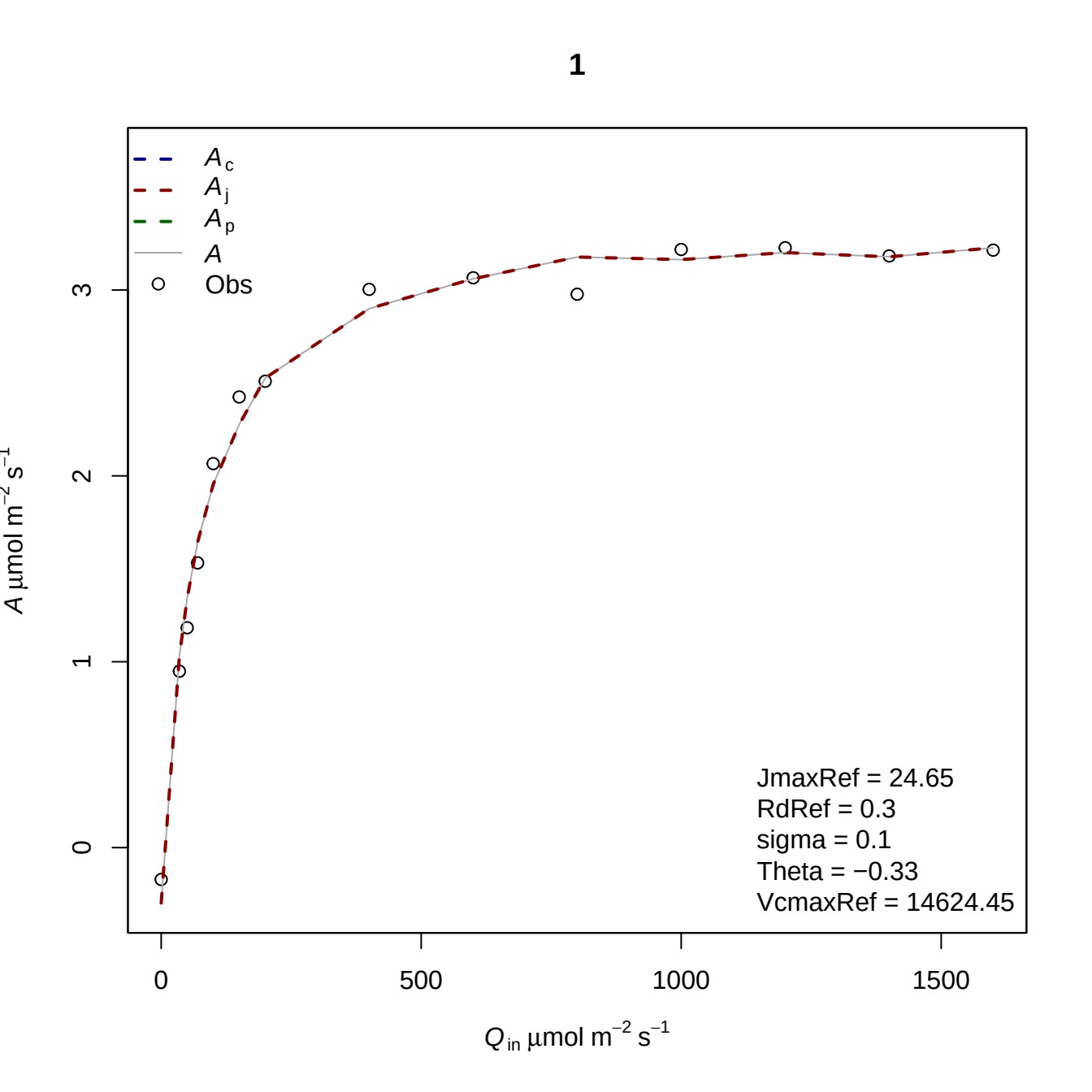
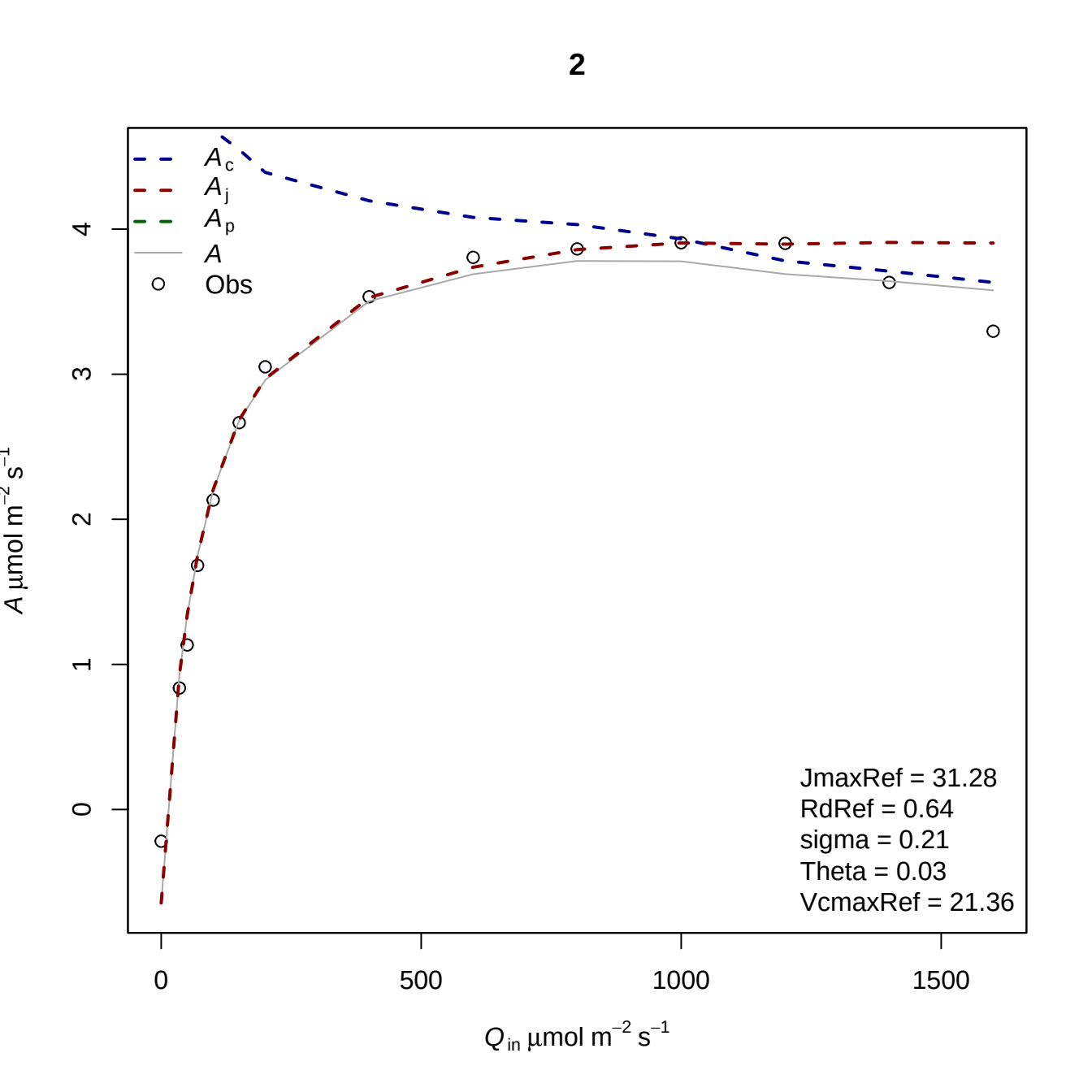
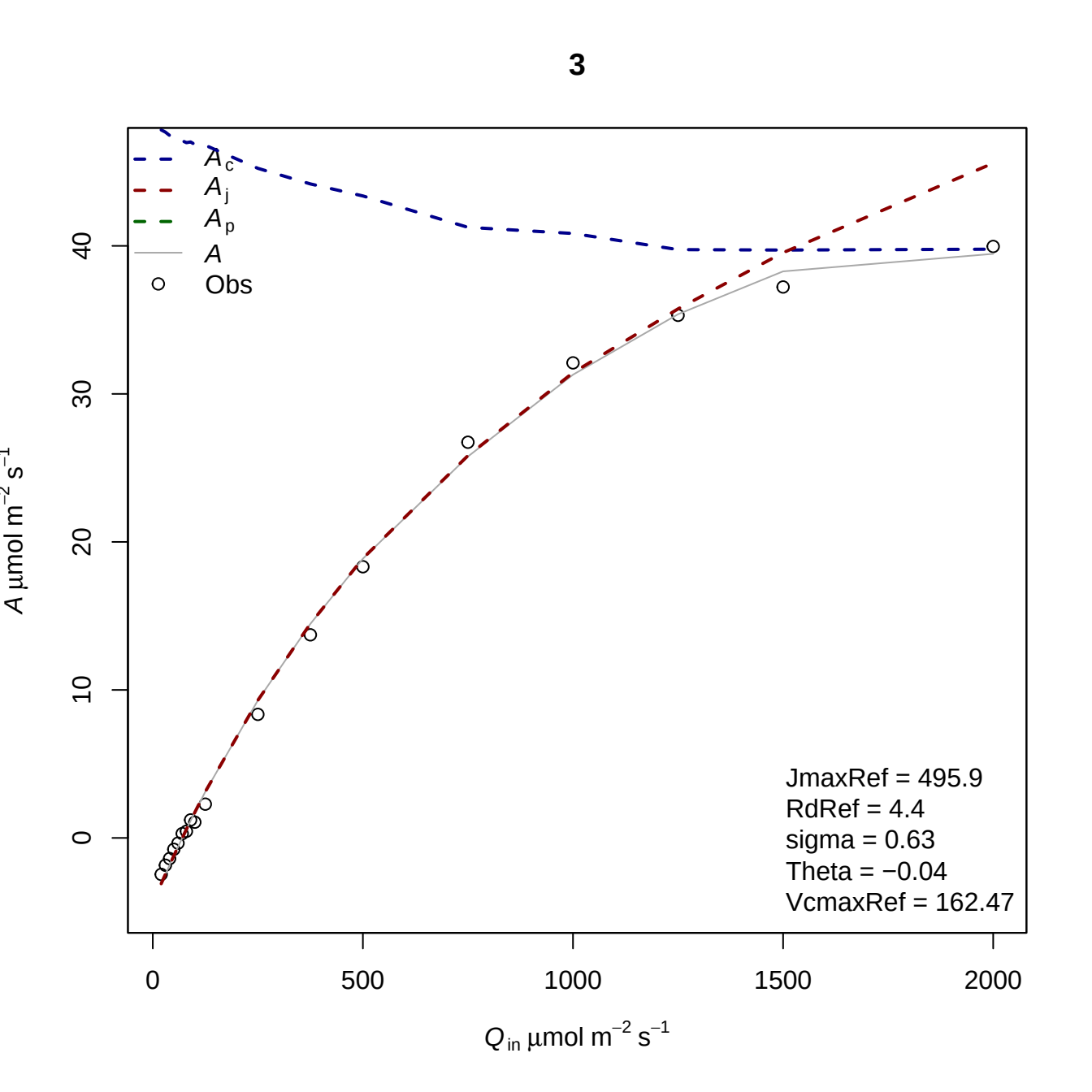
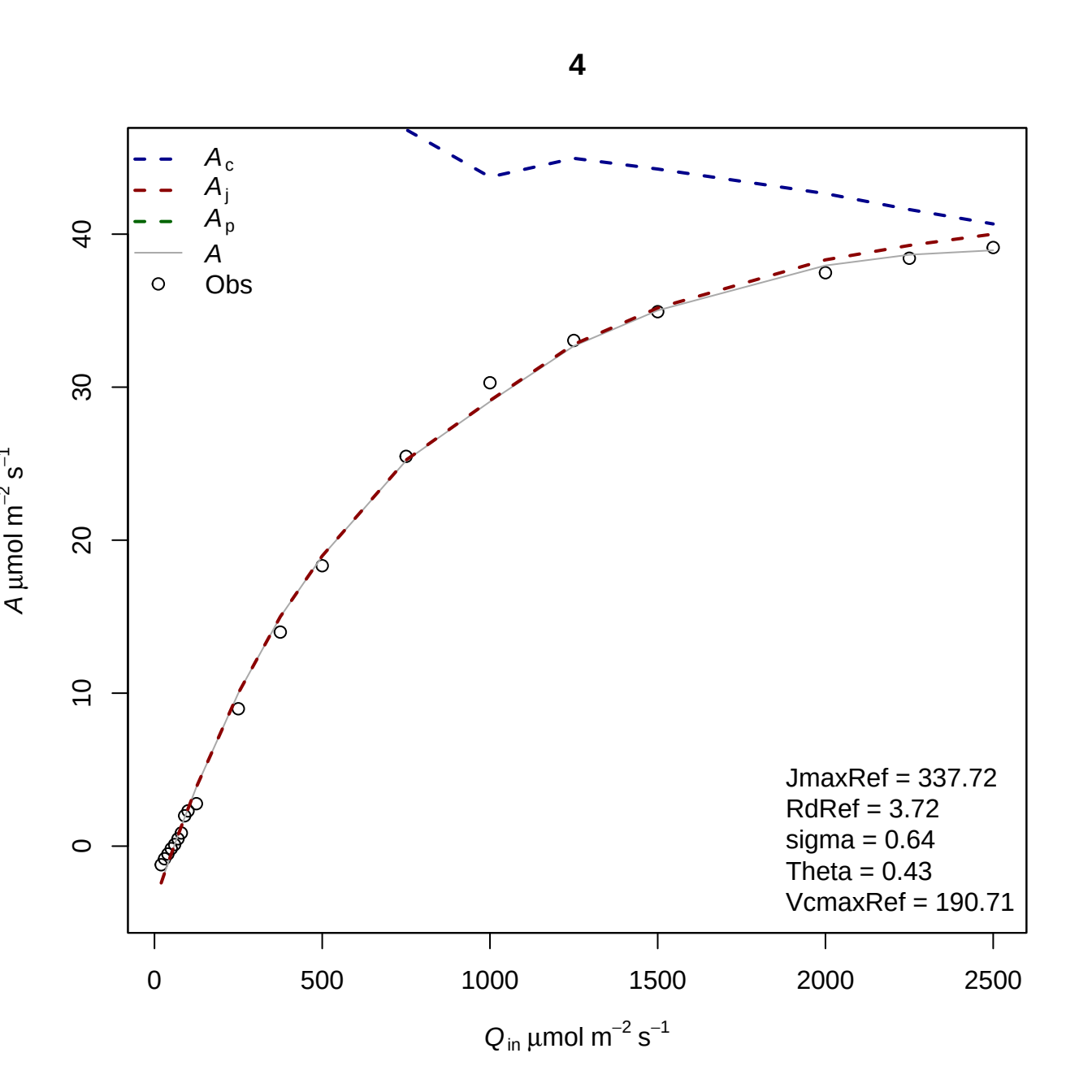
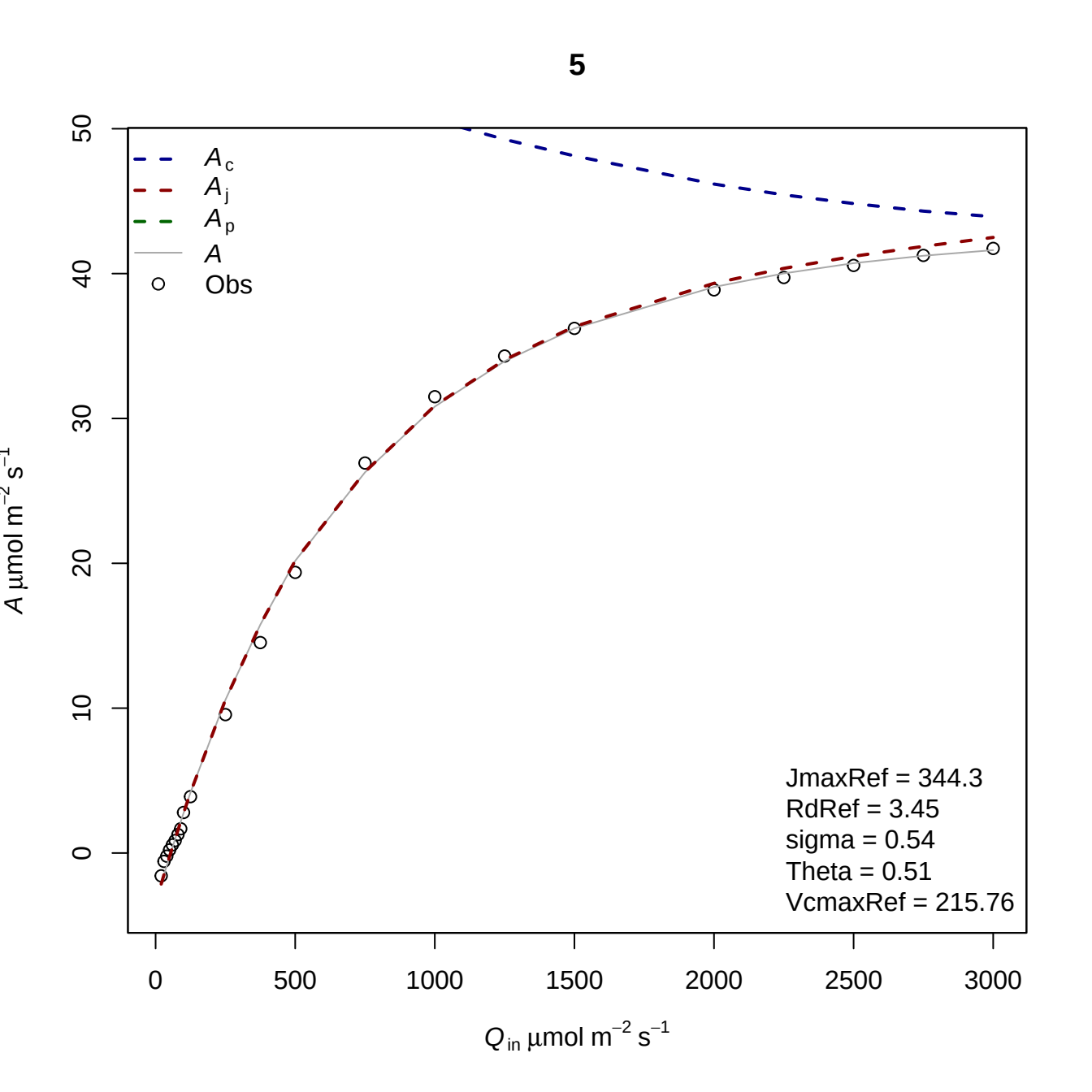
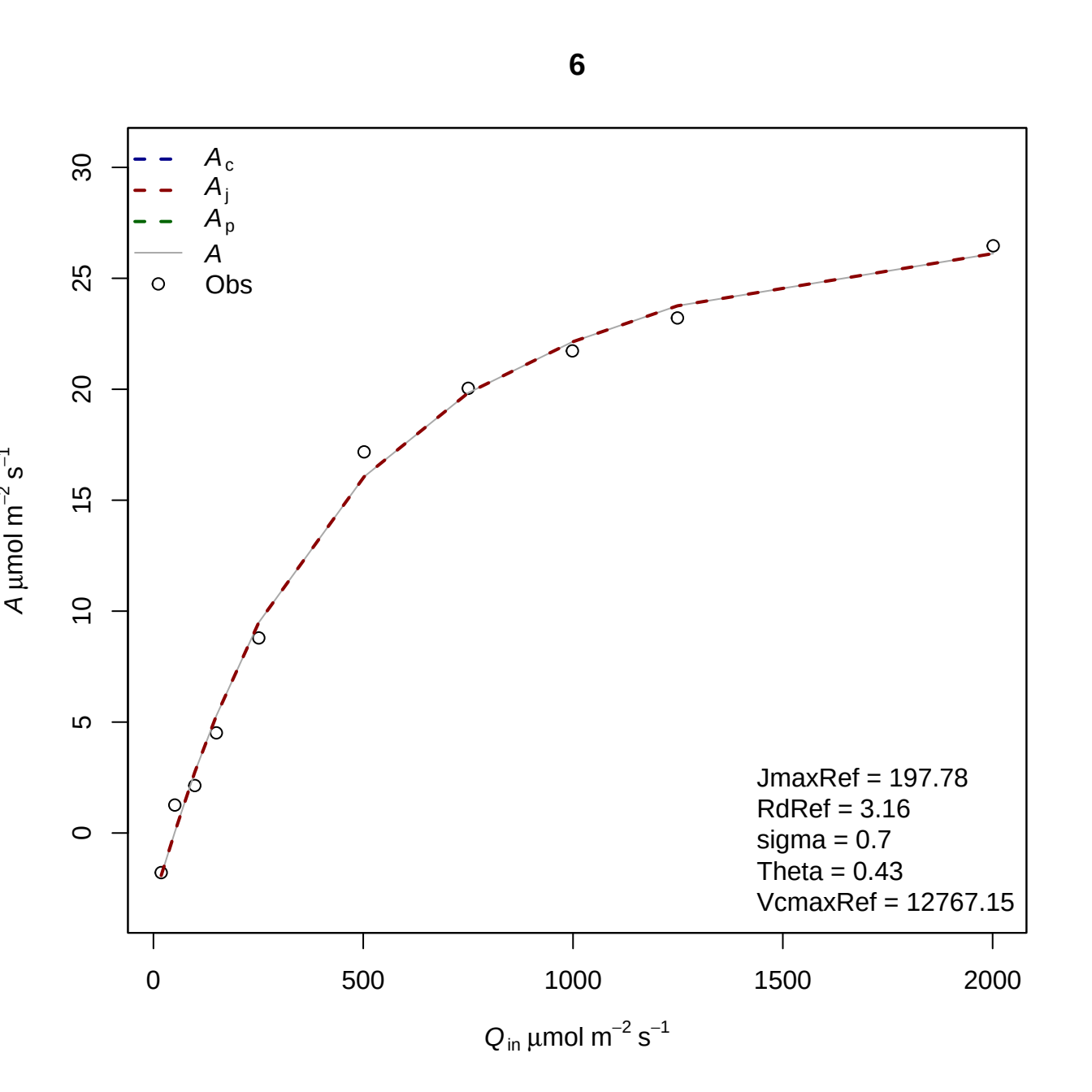

1
A_c
A_j
A_p
A
Obs
JmaxRef = 24.65
RdRef = 0.3
sigma = 0.1
Theta = −0.33
VcmaxRef = 14624.45
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

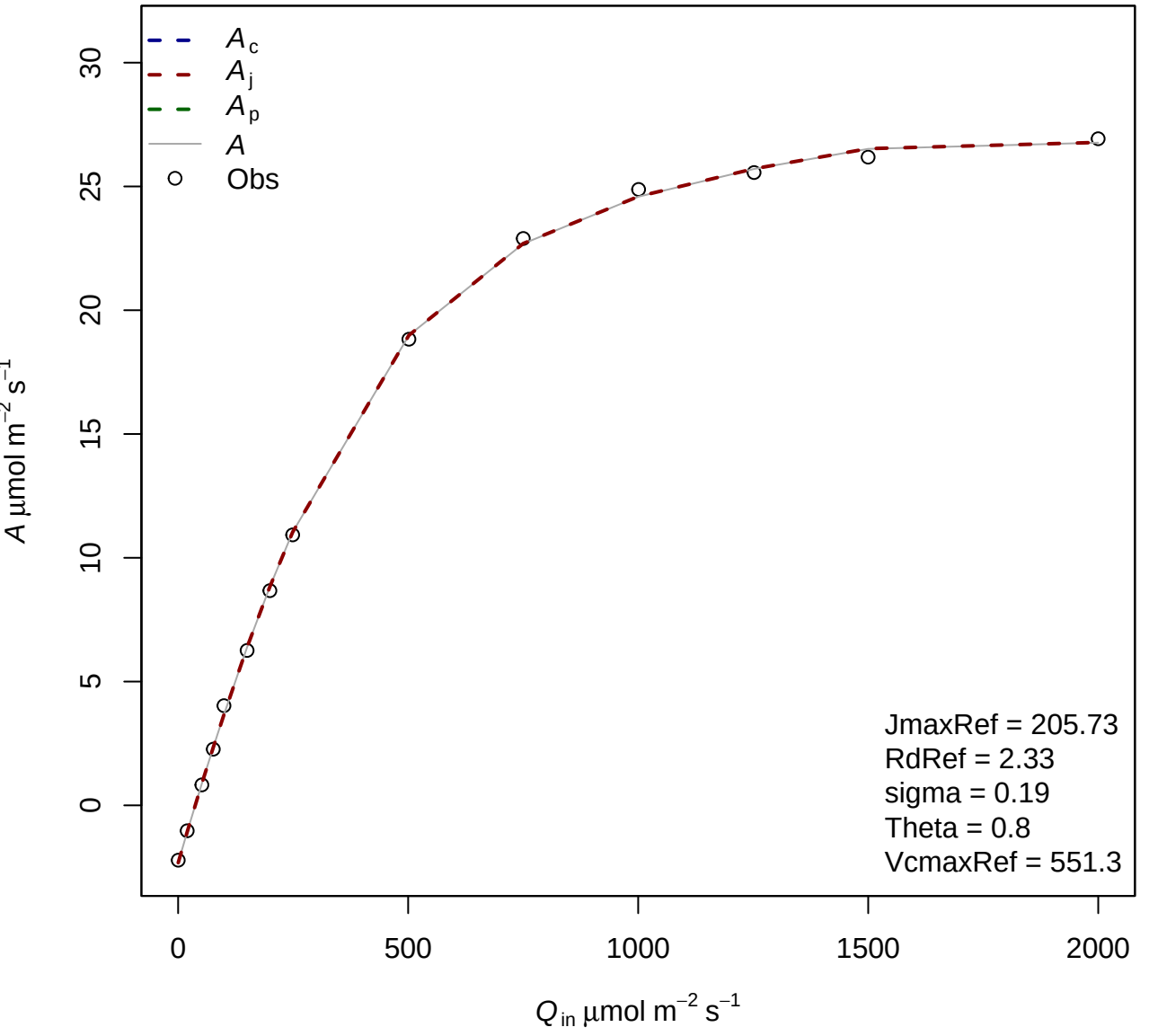

2
A_c
A_j
A_p
A
Obs
JmaxRef = 31.28
RdRef = 0.64
sigma = 0.21
Theta = 0.03
VcmaxRef = 21.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

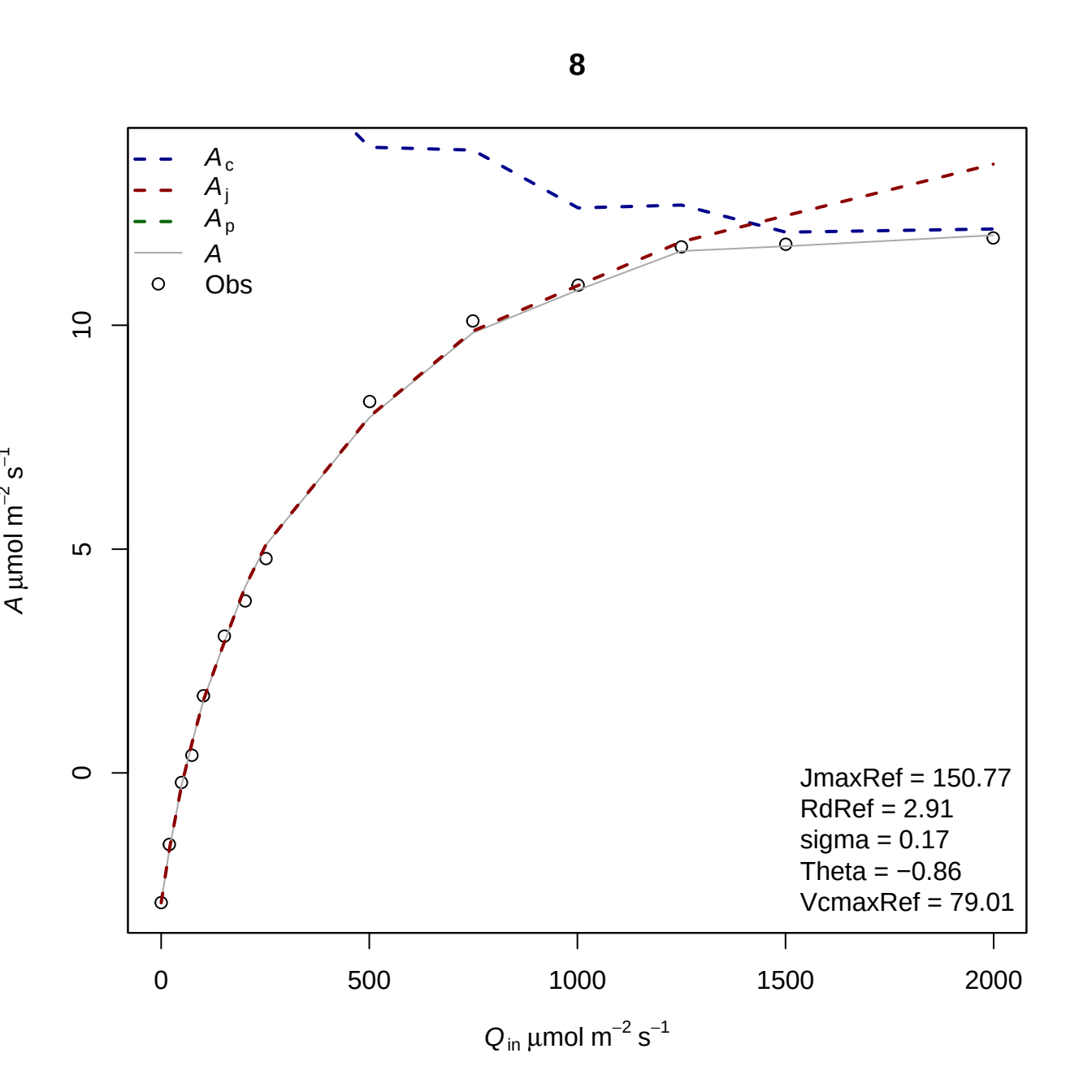

3
A_c
A_j
A_p
A
Obs
JmaxRef = 495.9
RdRef = 4.4
sigma = 0.63
Theta = −0.04
VcmaxRef = 162.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000

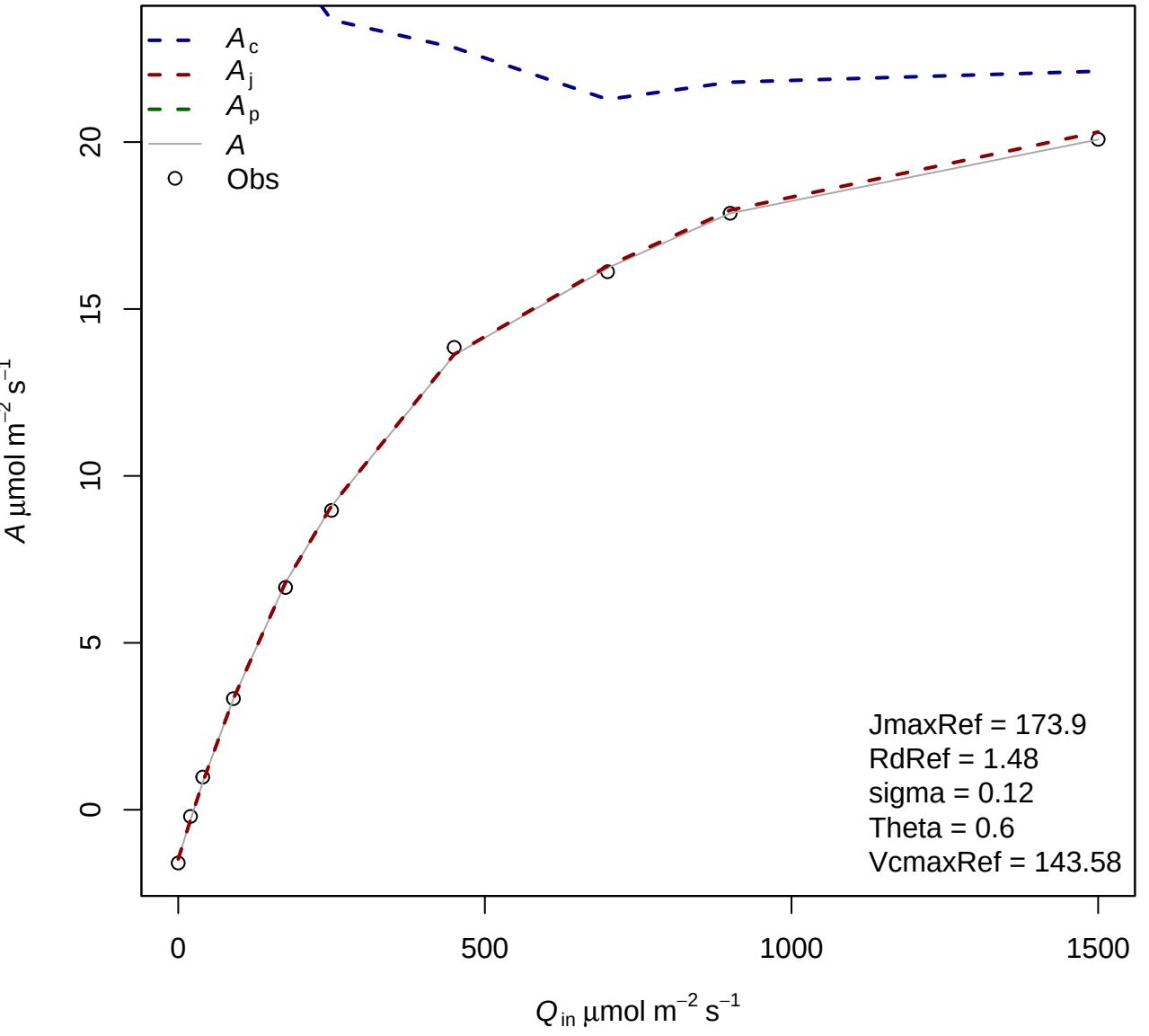

4
A_c
A_j
A_p
A
Obs
JmaxRef = 337.72
RdRef = 3.72
sigma = 0.64
Theta = 0.43
VcmaxRef = 190.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000
2500

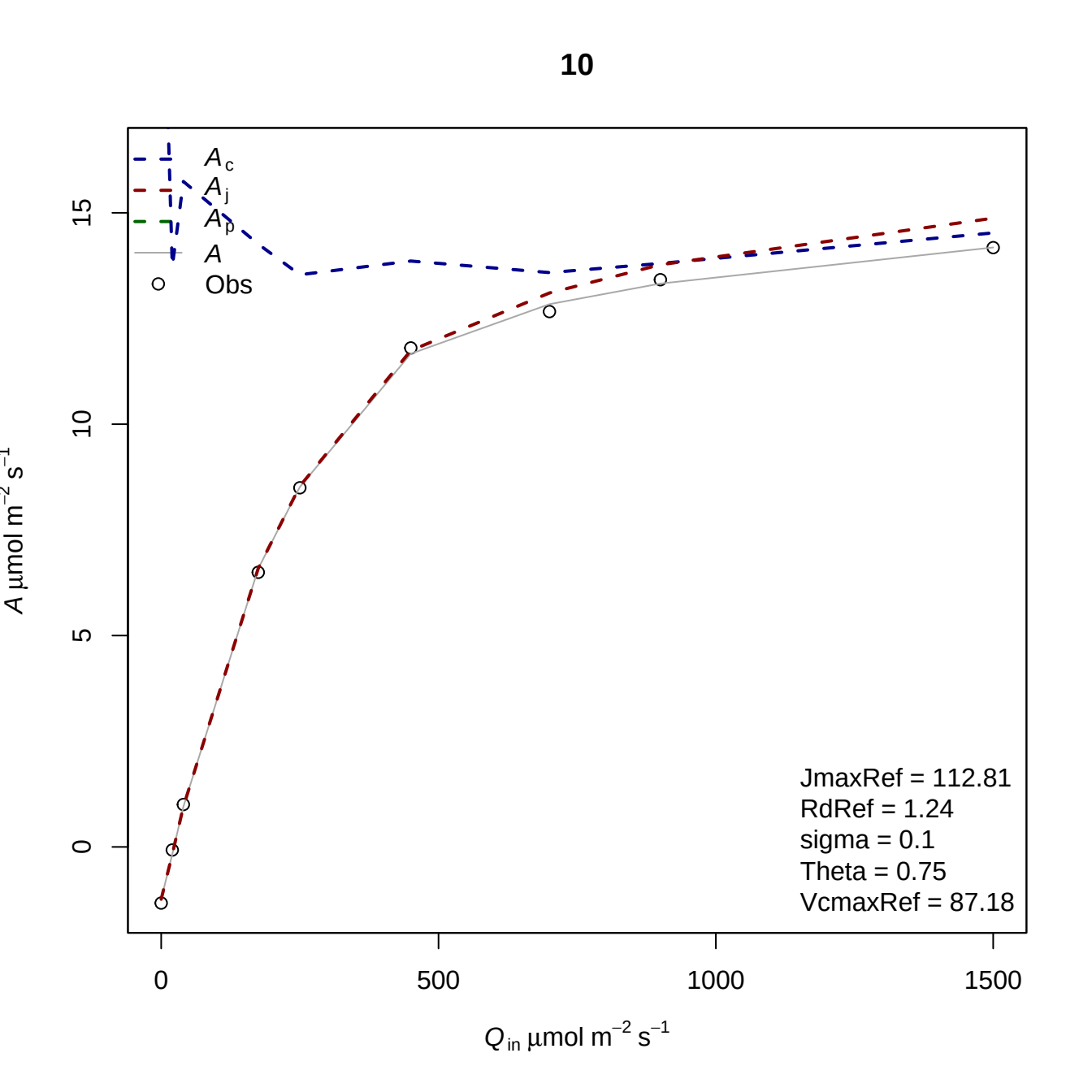

5
A_c
A_j
A_p
A
Obs
JmaxRef = 344.3
RdRef = 3.45
sigma = 0.54
Theta = 0.51
VcmaxRef = 215.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

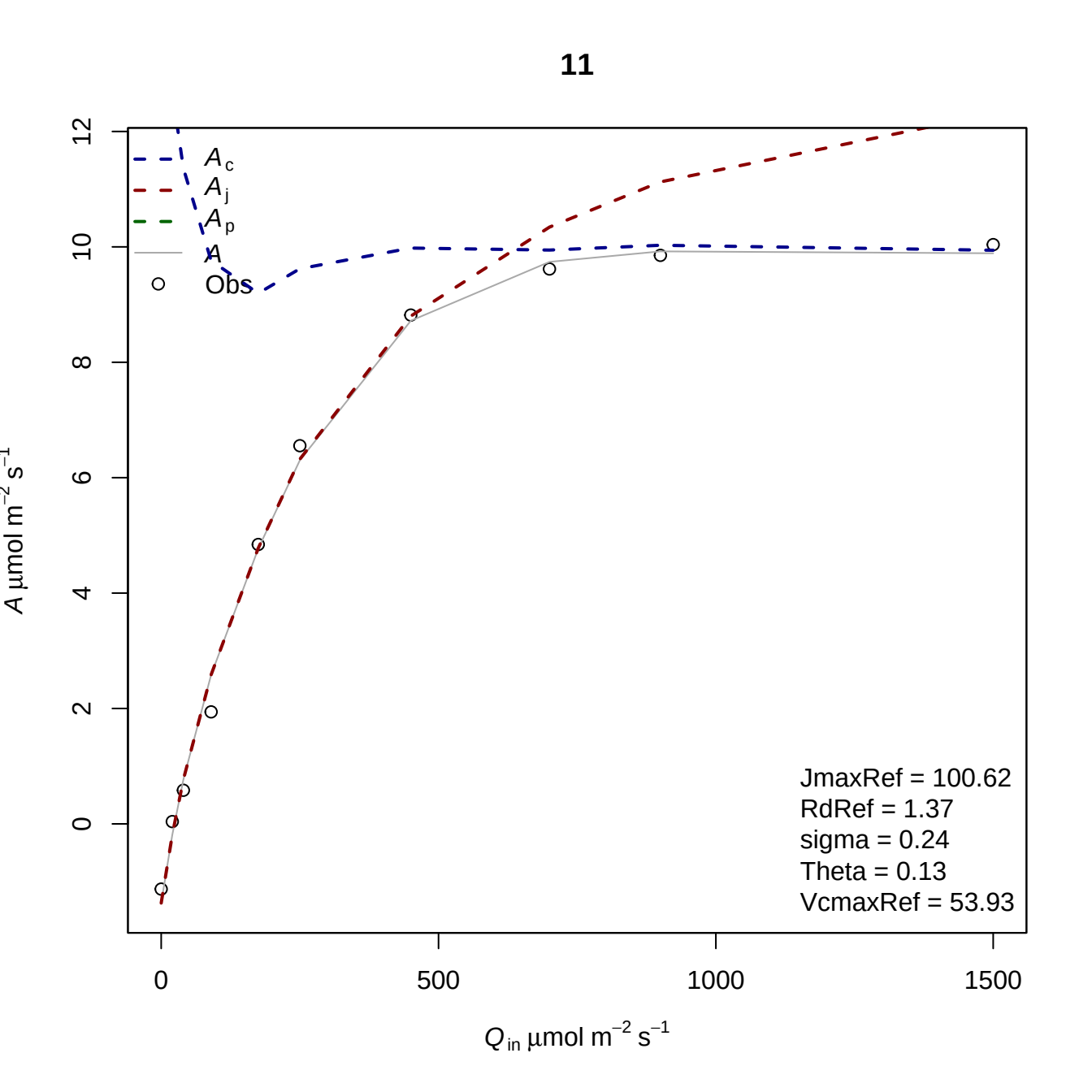

6
A_c
A_j
A_p
A
Obs
JmaxRef = 197.78
RdRef = 3.16
sigma = 0.7
Theta = 0.43
VcmaxRef = 12767.15
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

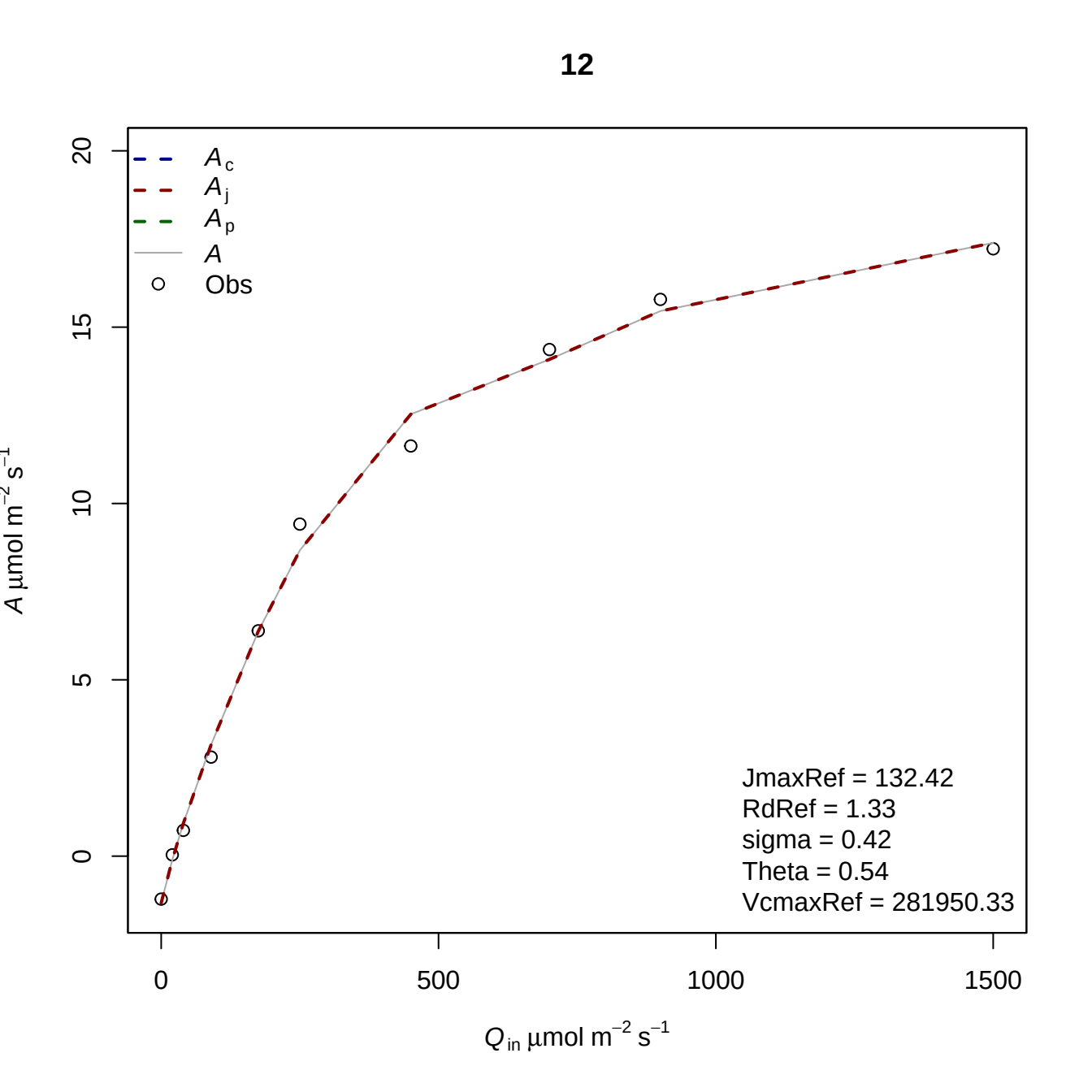

A_c
A_j
A_p
A
Obs
JmaxRef = 205.73
RdRef = 2.33
sigma = 0.19
Theta = 0.8
VcmaxRef = 551.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

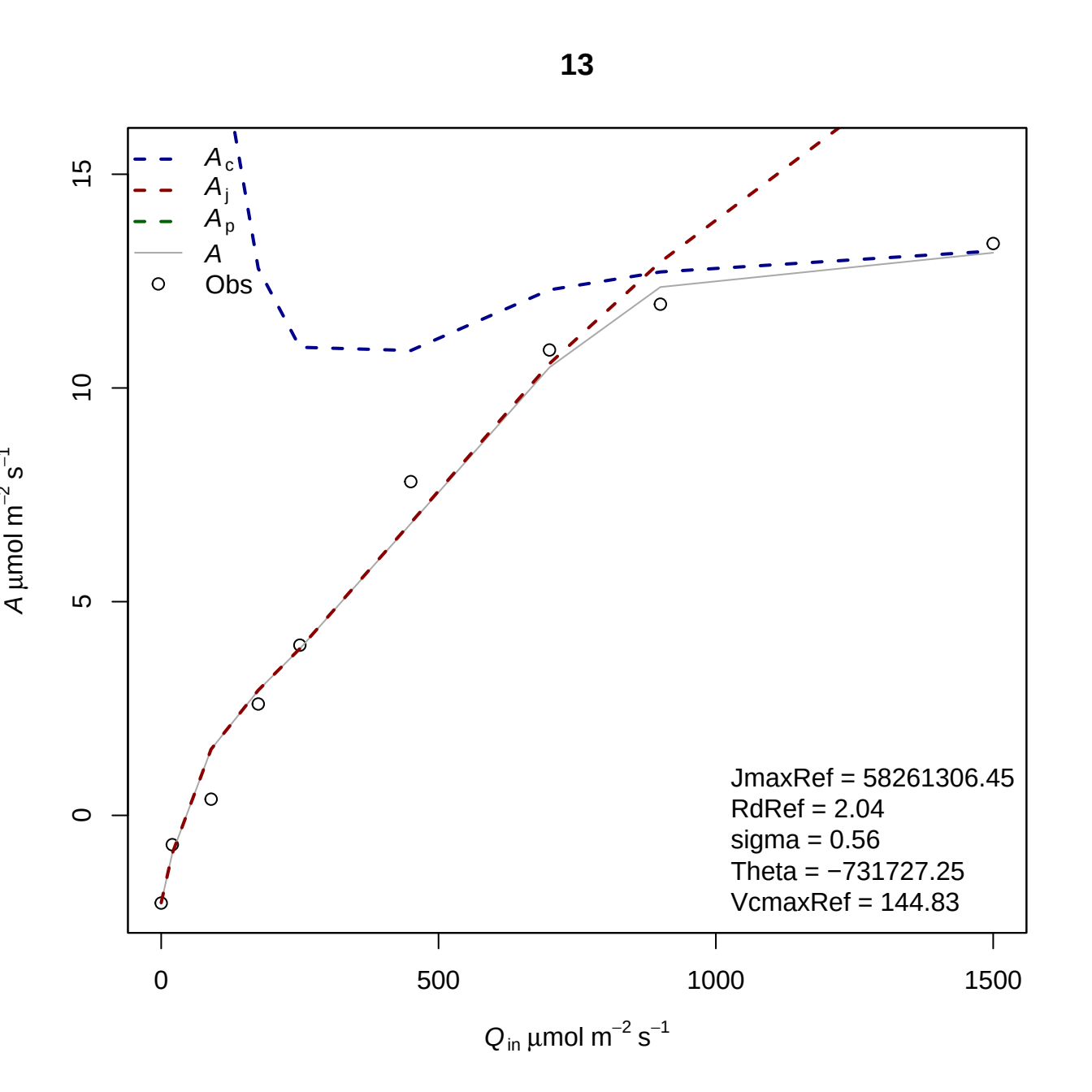

8
A_c
A_j
A_p
A
Obs
JmaxRef = 150.77
RdRef = 2.91
sigma = 0.17
Theta = −0.86
VcmaxRef = 79.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

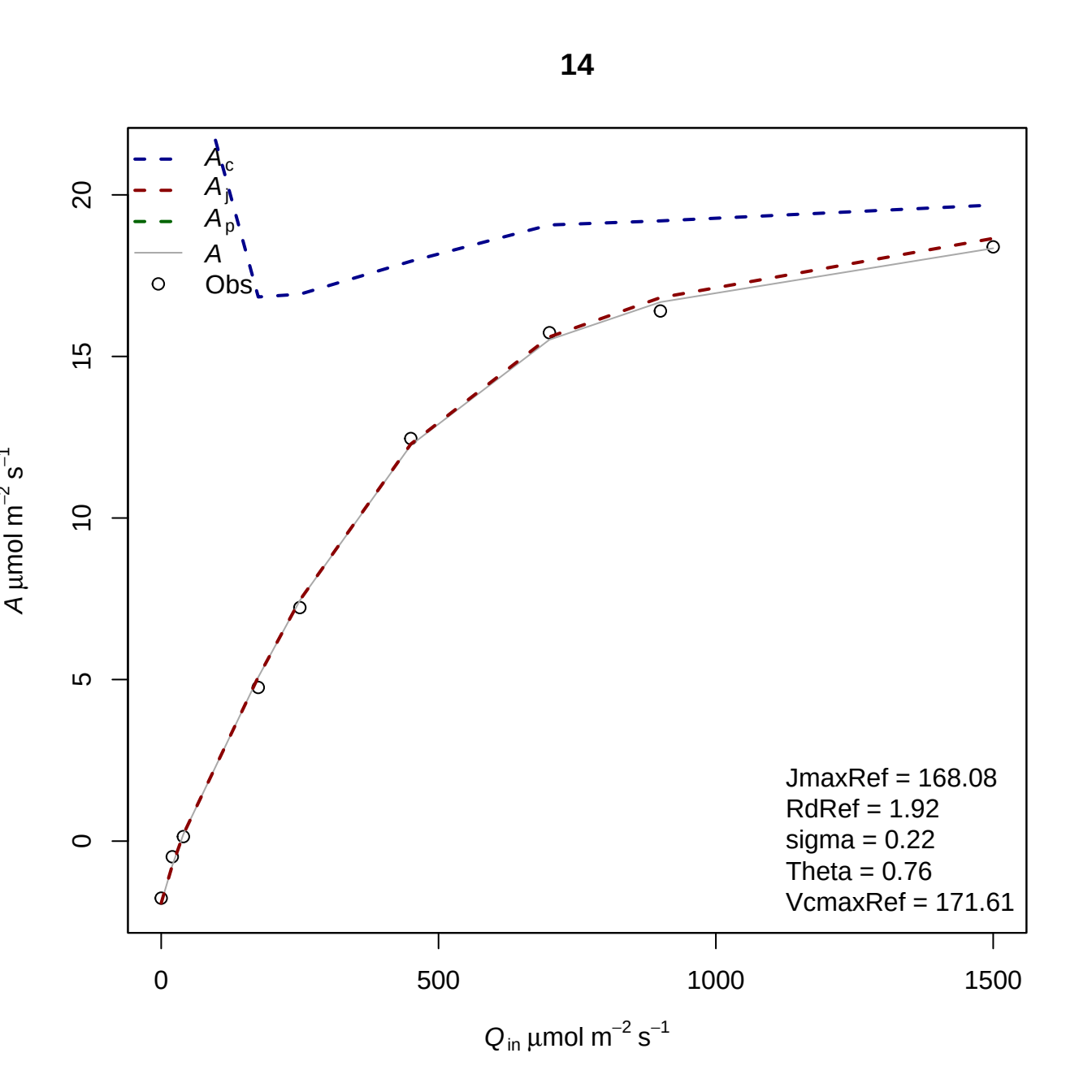

A_c
A_j
A_p
A
Obs
JmaxRef = 173.9
RdRef = 1.48
sigma = 0.12
Theta = 0.6
VcmaxRef = 143.58
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

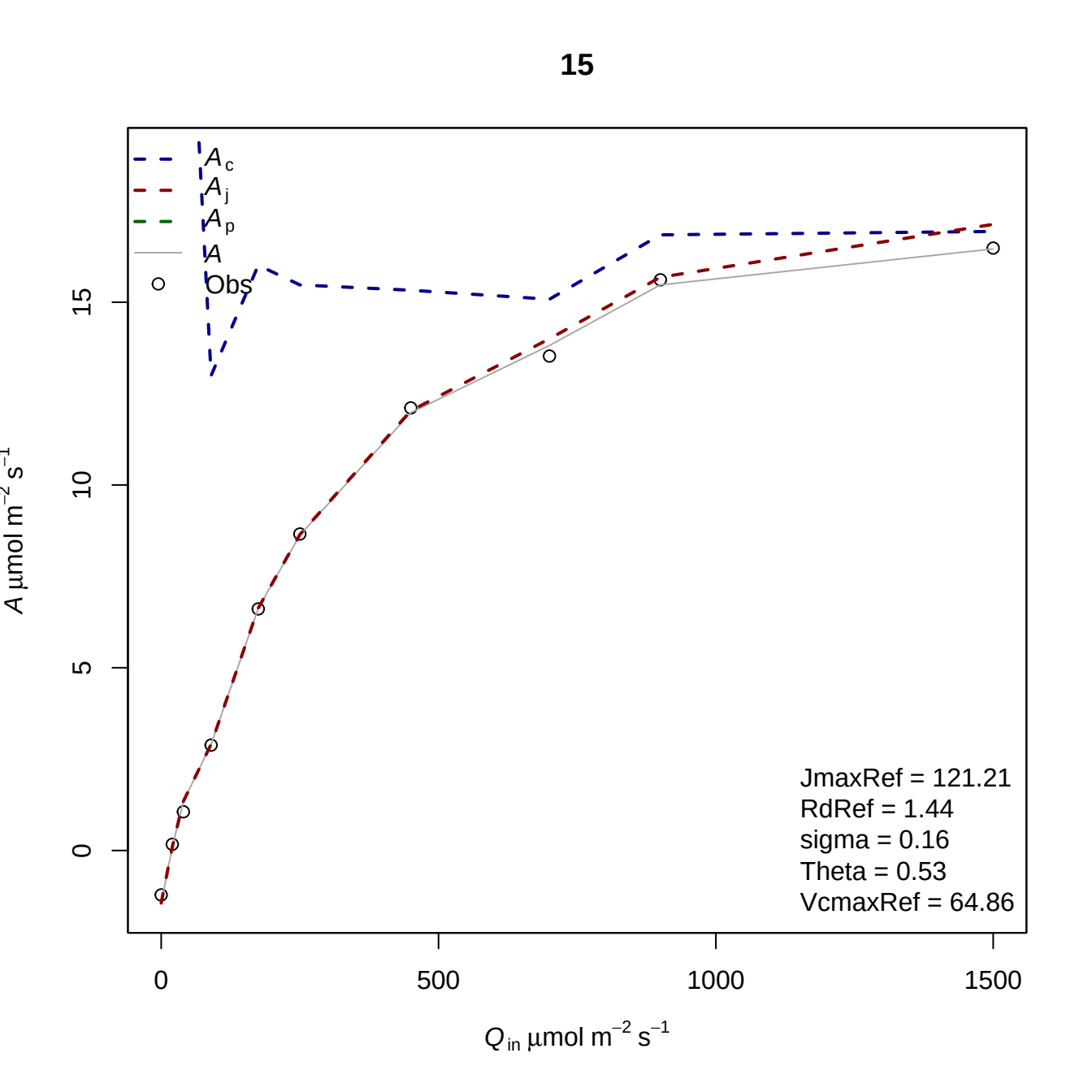

10
A_c
A_j
A_p
A
Obs
JmaxRef = 112.81
RdRef = 1.24
sigma = 0.1
Theta = 0.75
VcmaxRef = 87.18
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

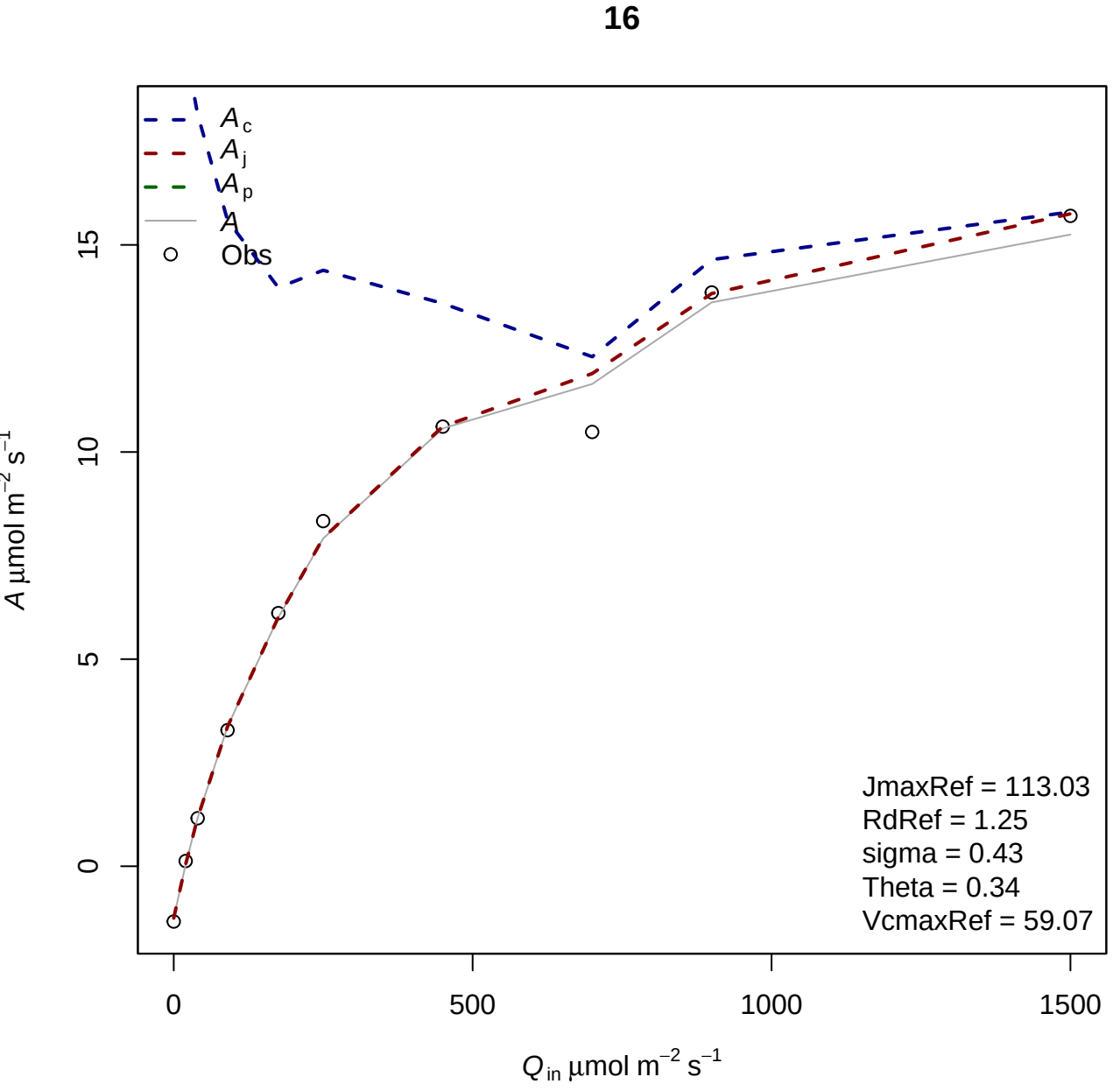

11
A_c
A_j
A_p
A
Obs
JmaxRef = 100.62
RdRef = 1.37
sigma = 0.24
Theta = 0.13
VcmaxRef = 53.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

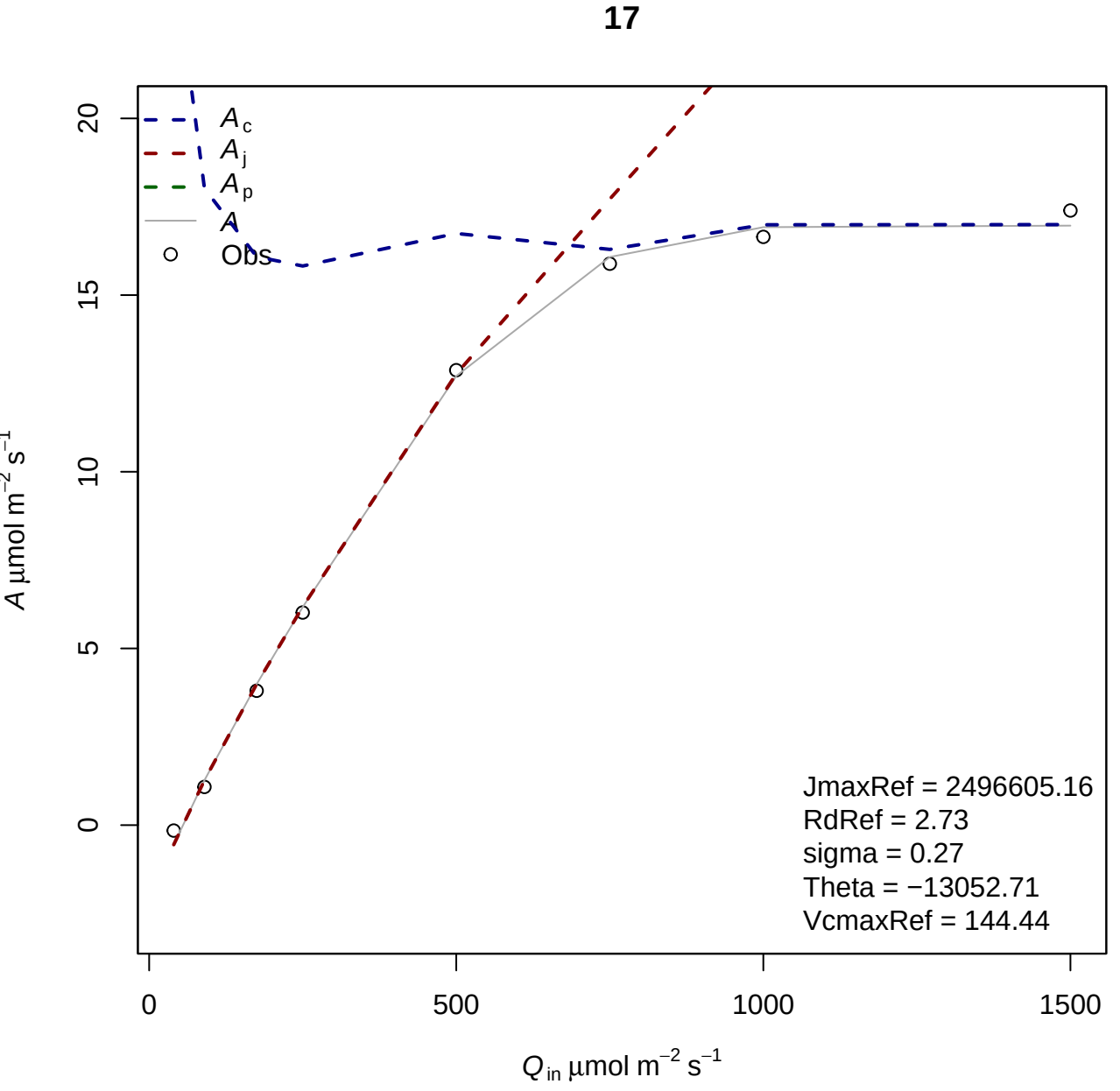

12
A_c
A_j
A_p
A
Obs
JmaxRef = 132.42
RdRef = 1.33
sigma = 0.42
Theta = 0.54
VcmaxRef = 281950.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

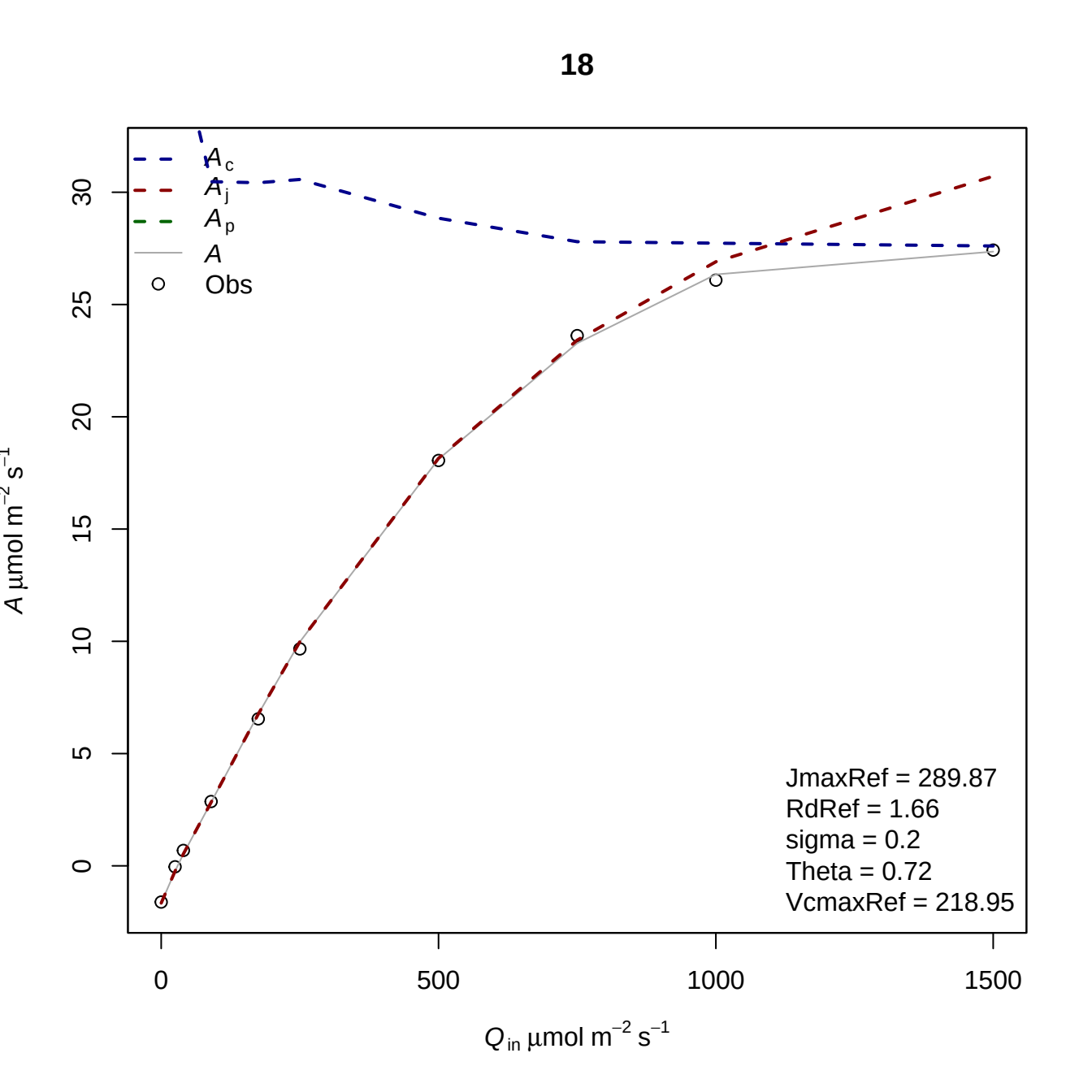

13
A_c
A_j
A_p
A
Obs
JmaxRef = 58261306.45
RdRef = 2.04
sigma = 0.56
Theta = −731727.25
VcmaxRef = 144.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

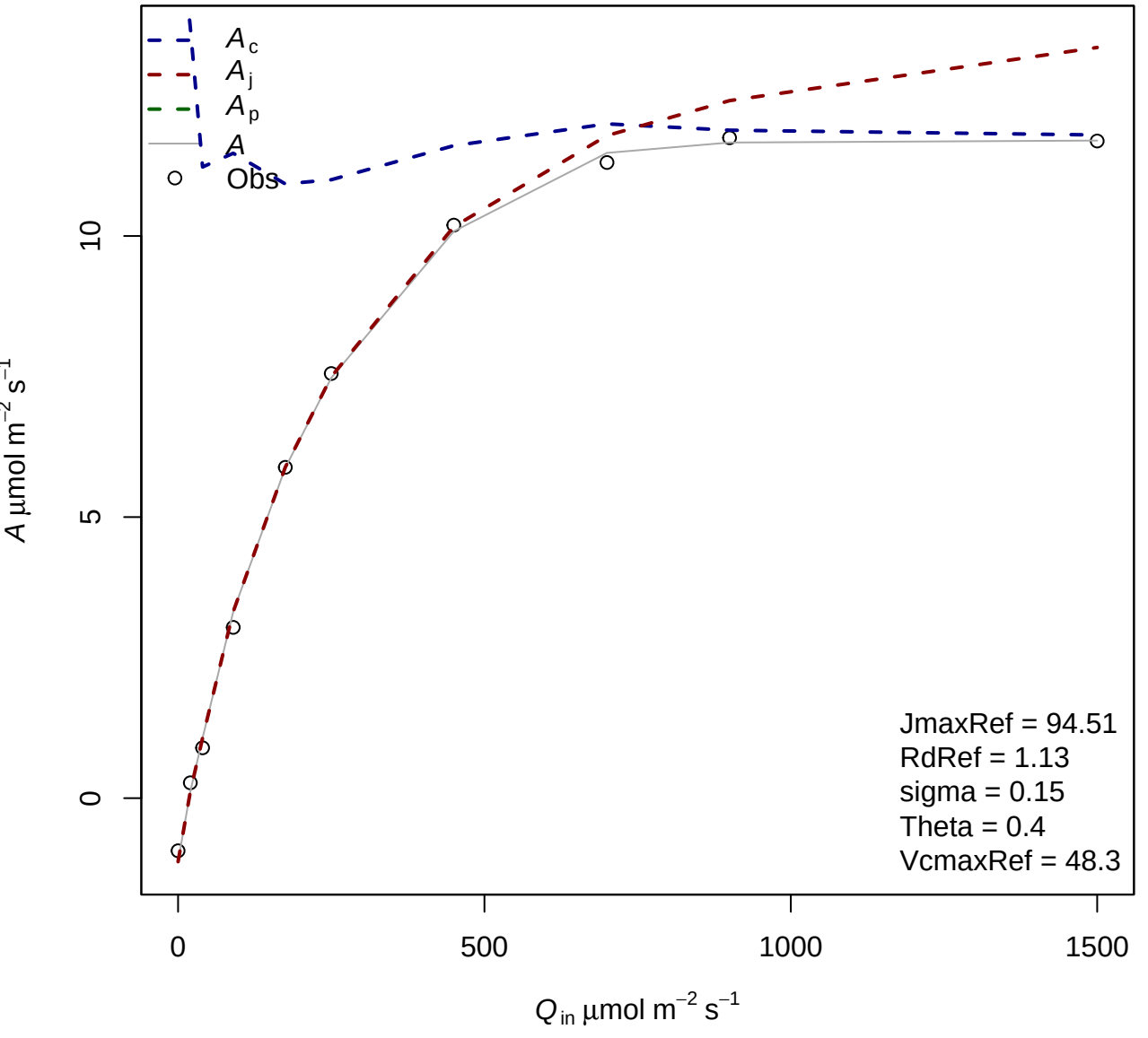

14
A_c
A_j
A_p
A
Obs
JmaxRef = 168.08
RdRef = 1.92
sigma = 0.22
Theta = 0.76
VcmaxRef = 171.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

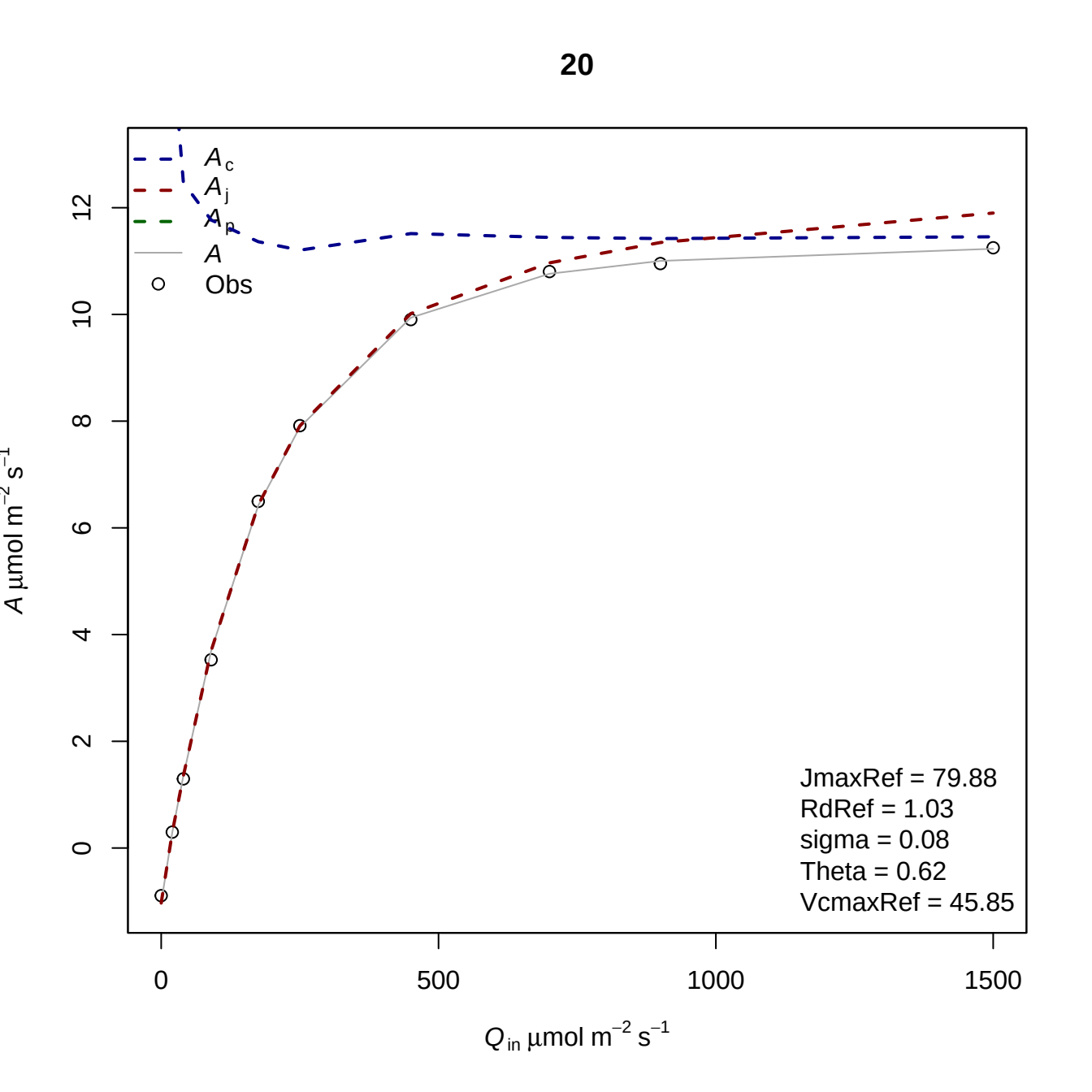

15
A_c
A_j
A_p
A
Obs
JmaxRef = 121.21
RdRef = 1.44
sigma = 0.16
Theta = 0.53
VcmaxRef = 64.86
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

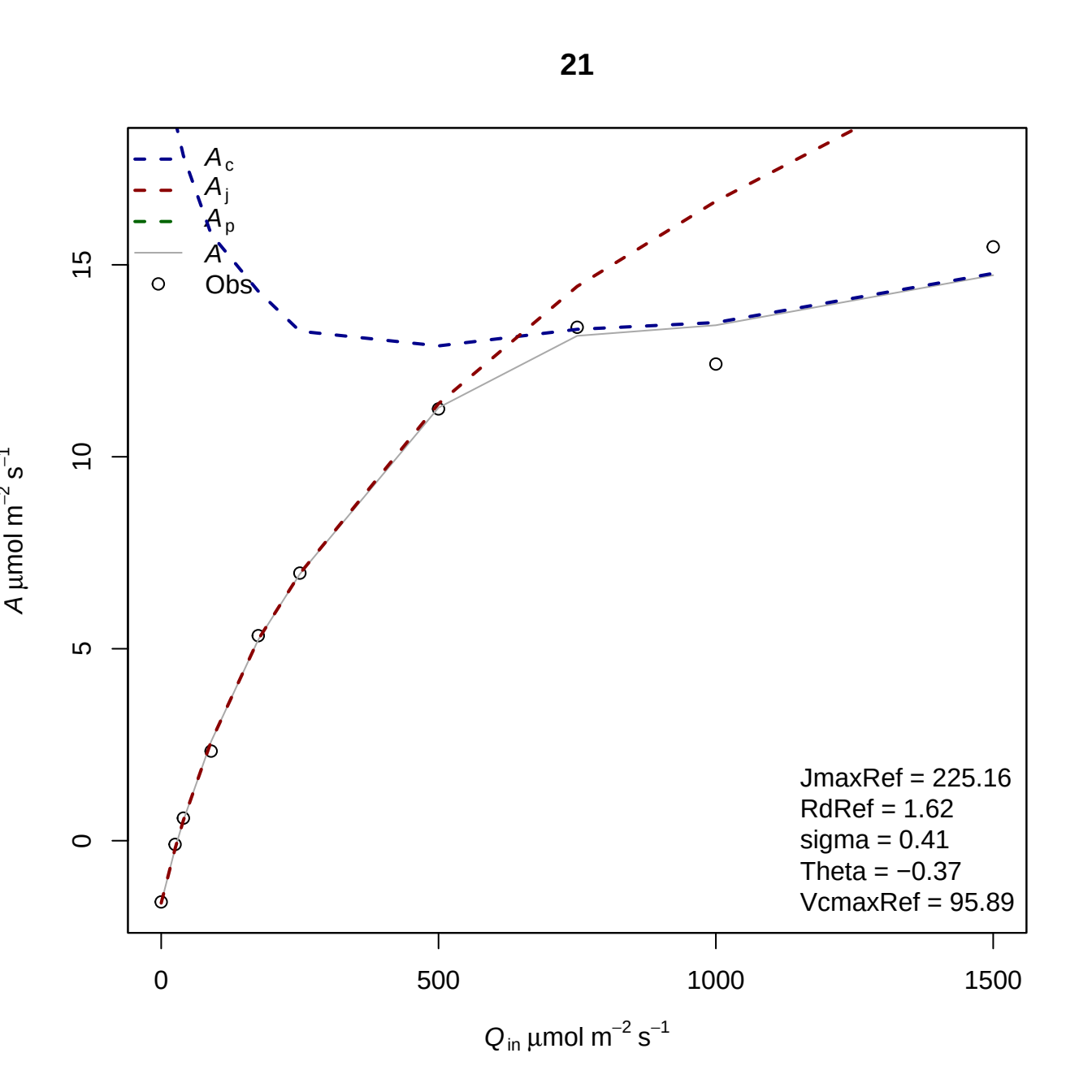

16
A_c
A_j
A_p
A
Obs
JmaxRef = 113.03
RdRef = 1.25
sigma = 0.43
Theta = 0.34
VcmaxRef = 59.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

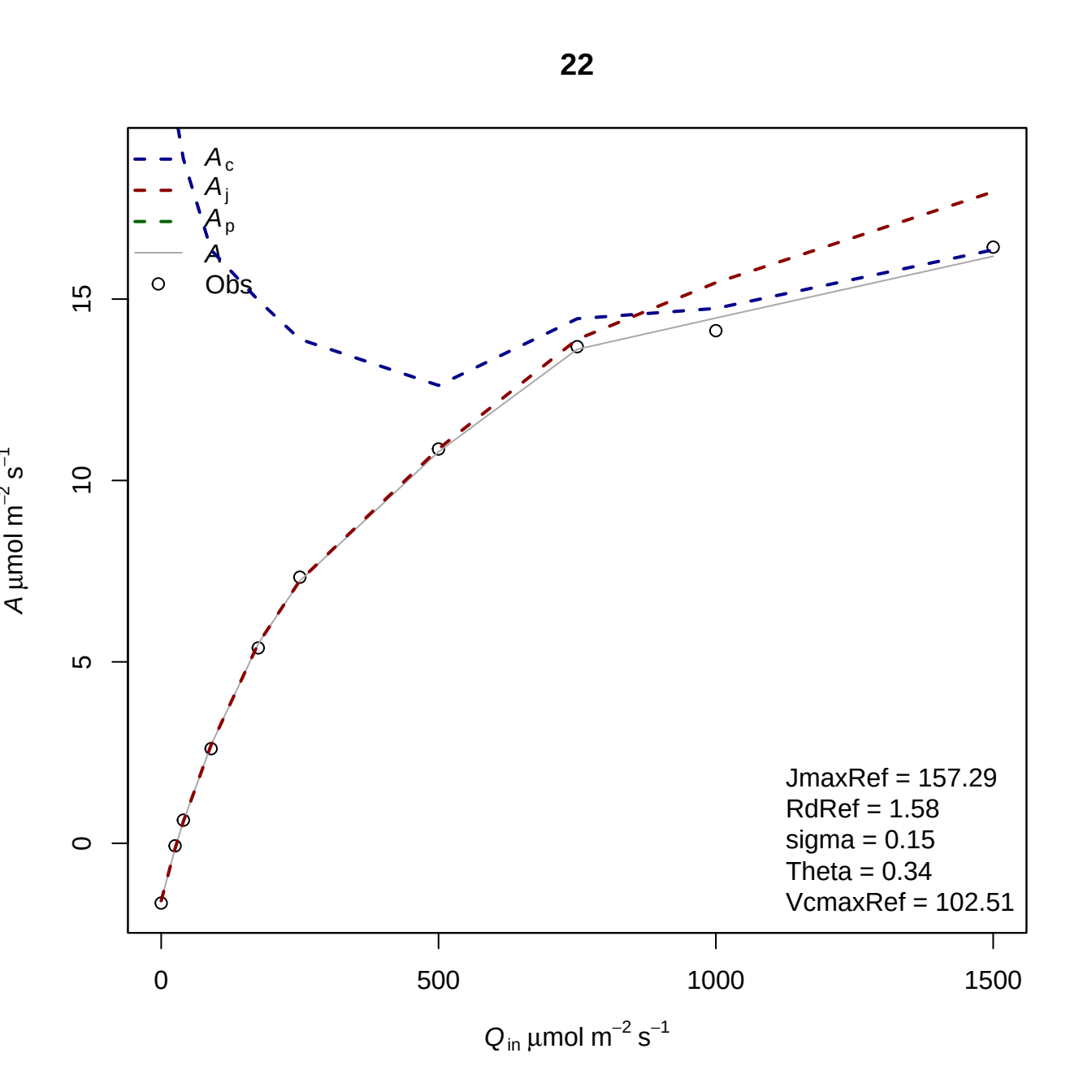

17
A_c
A_j
A_p
A
Obs
JmaxRef = 2496605.16
RdRef = 2.73
sigma = 0.27
Theta = −13052.71
VcmaxRef = 144.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

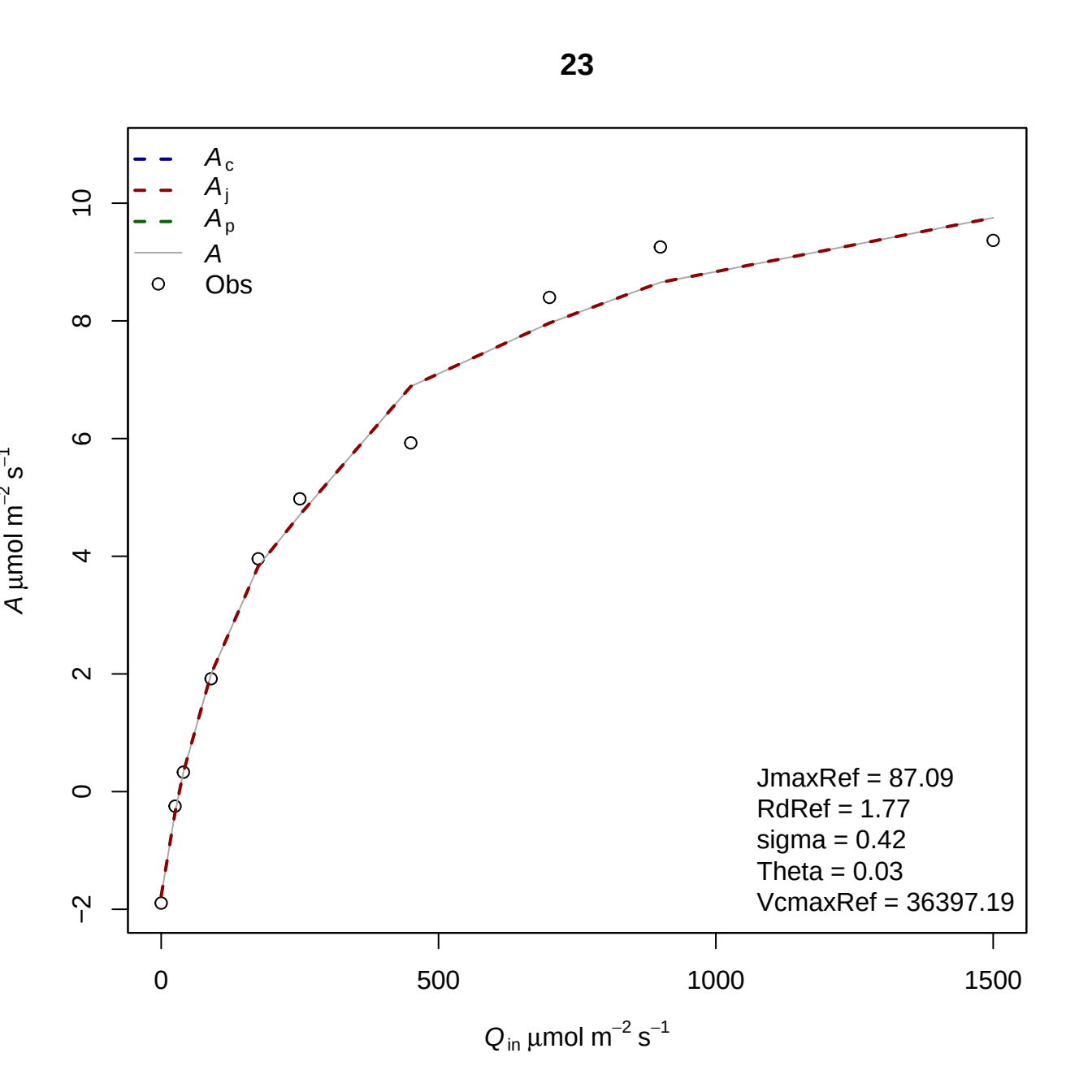

18
A_c
A_j
A_p
A
Obs
JmaxRef = 289.87
RdRef = 1.66
sigma = 0.2
Theta = 0.72
VcmaxRef = 218.95
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

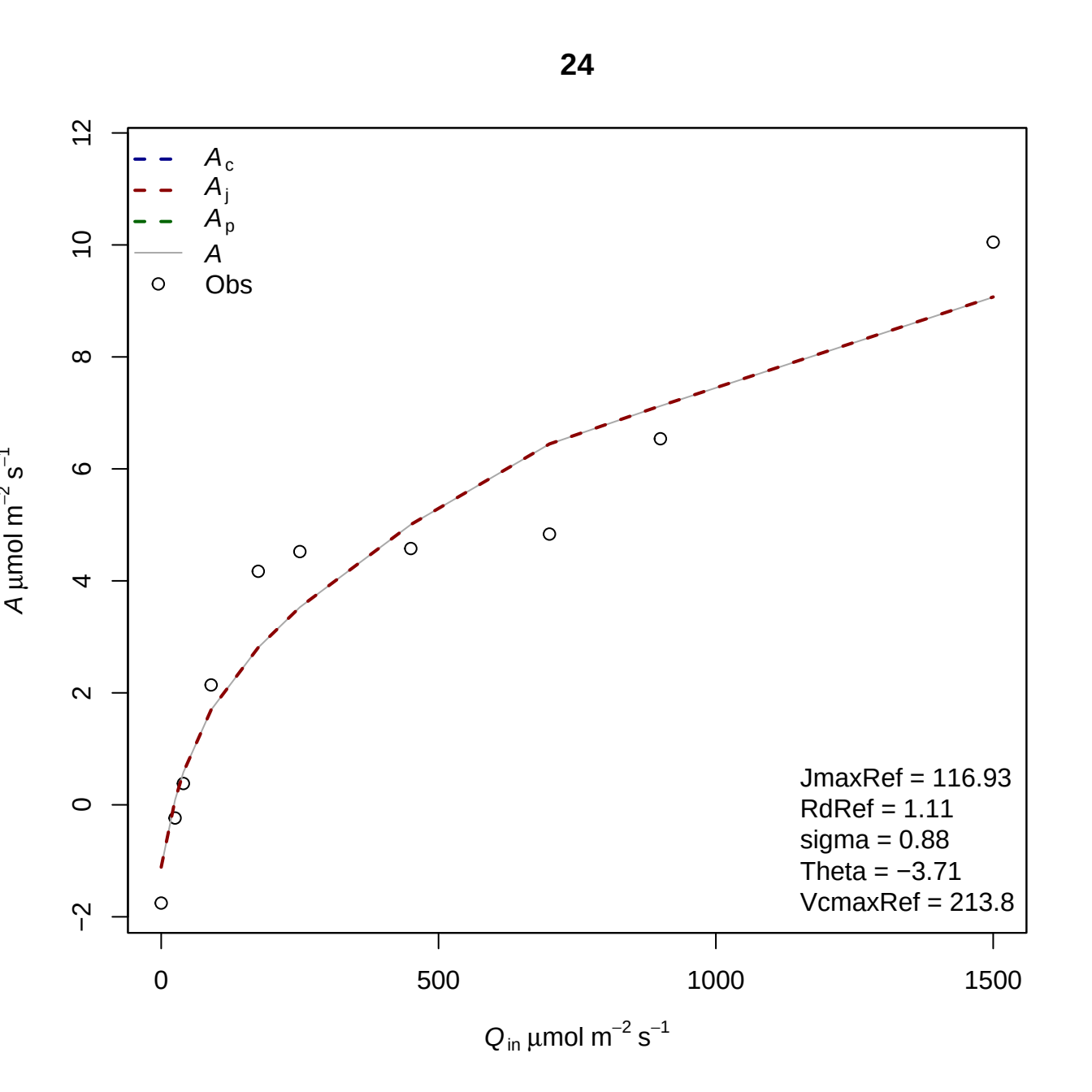

A_c
A_j
A_p
A
Obs
JmaxRef = 94.51
RdRef = 1.13
sigma = 0.15
Theta = 0.4
VcmaxRef = 48.3
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

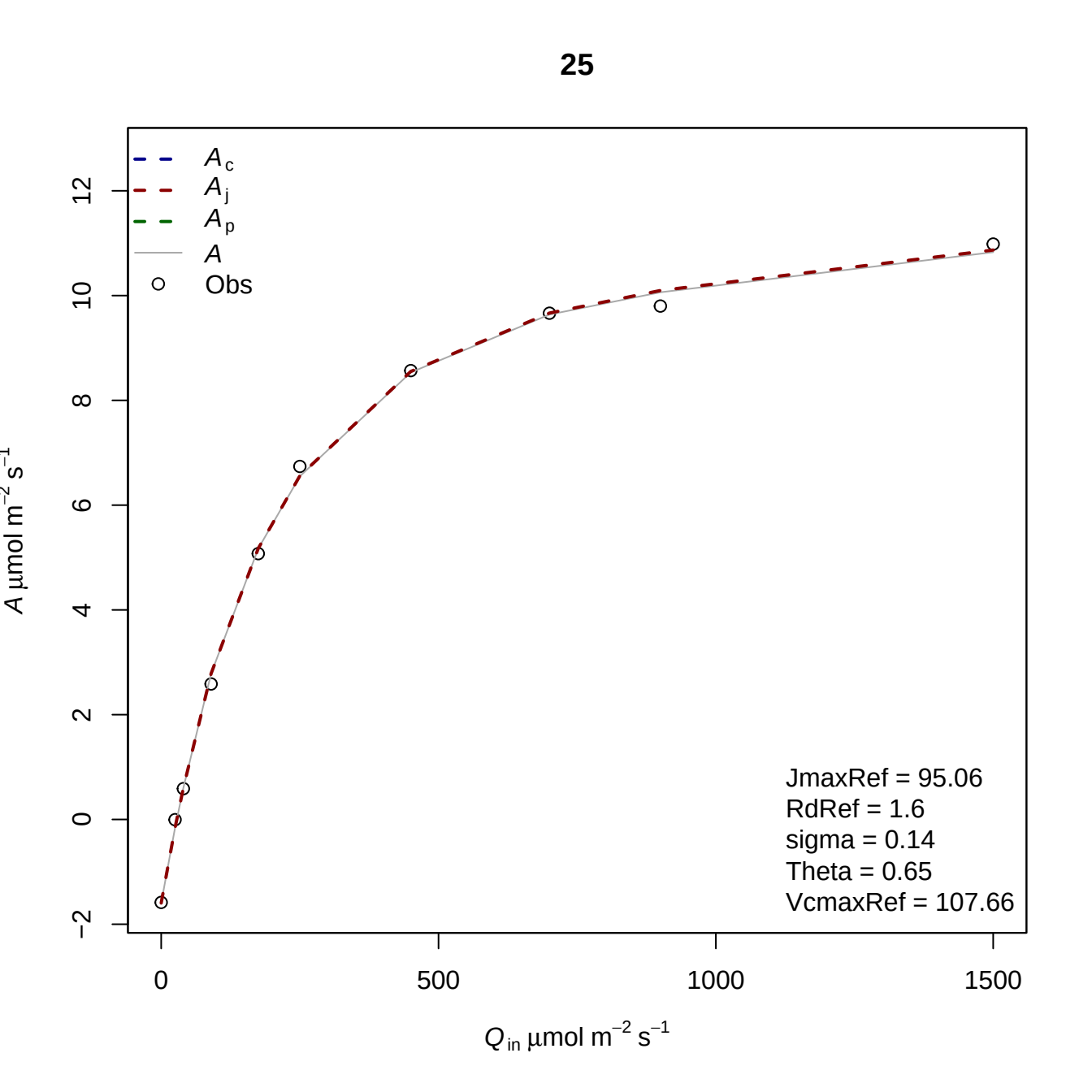

20
A_c
A_j
A_p
A
Obs
JmaxRef = 79.88
RdRef = 1.03
sigma = 0.08
Theta = 0.62
VcmaxRef = 45.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

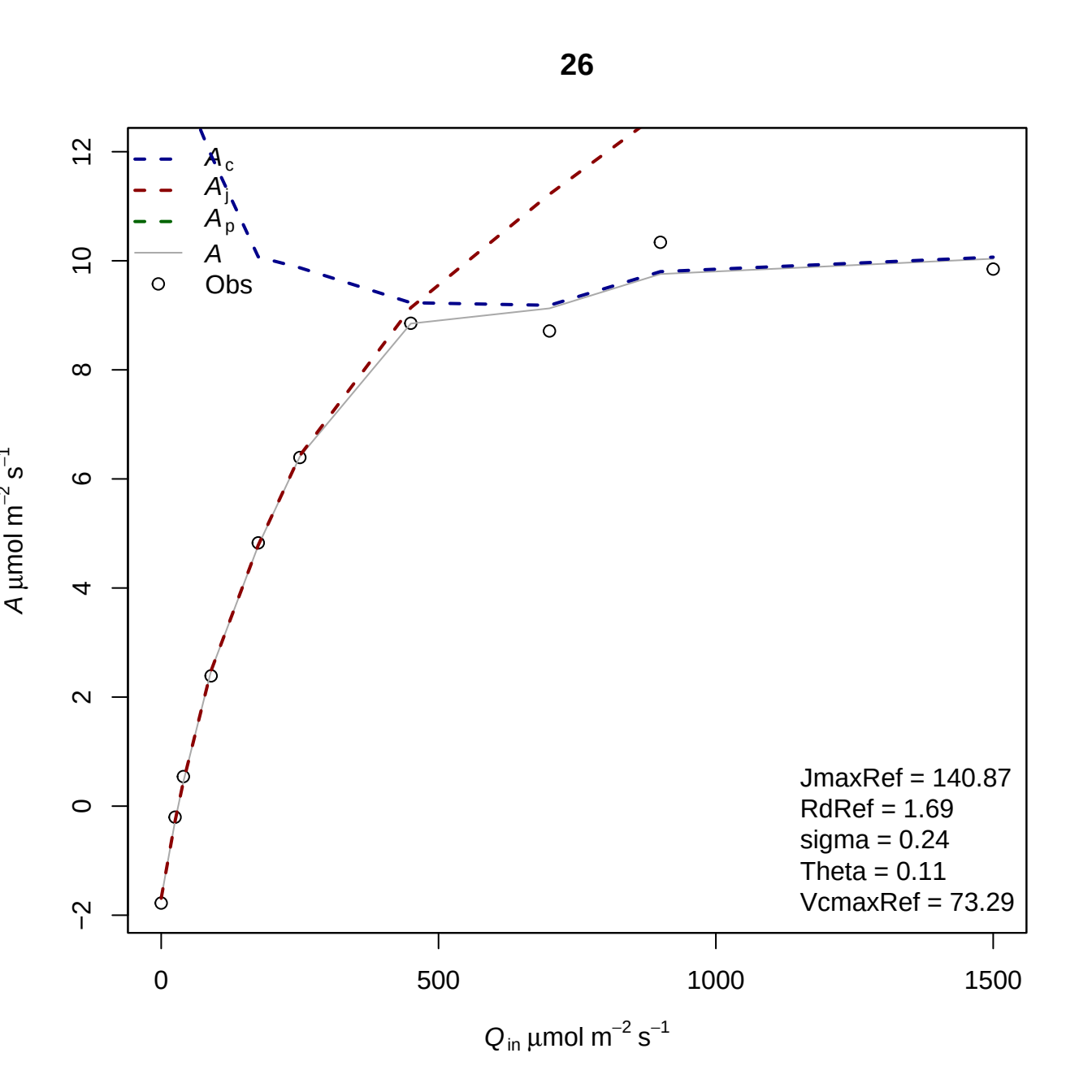

21
A_c
A_j
A_p
A
Obs
JmaxRef = 225.16
RdRef = 1.62
sigma = 0.41
Theta = −0.37
VcmaxRef = 95.89
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

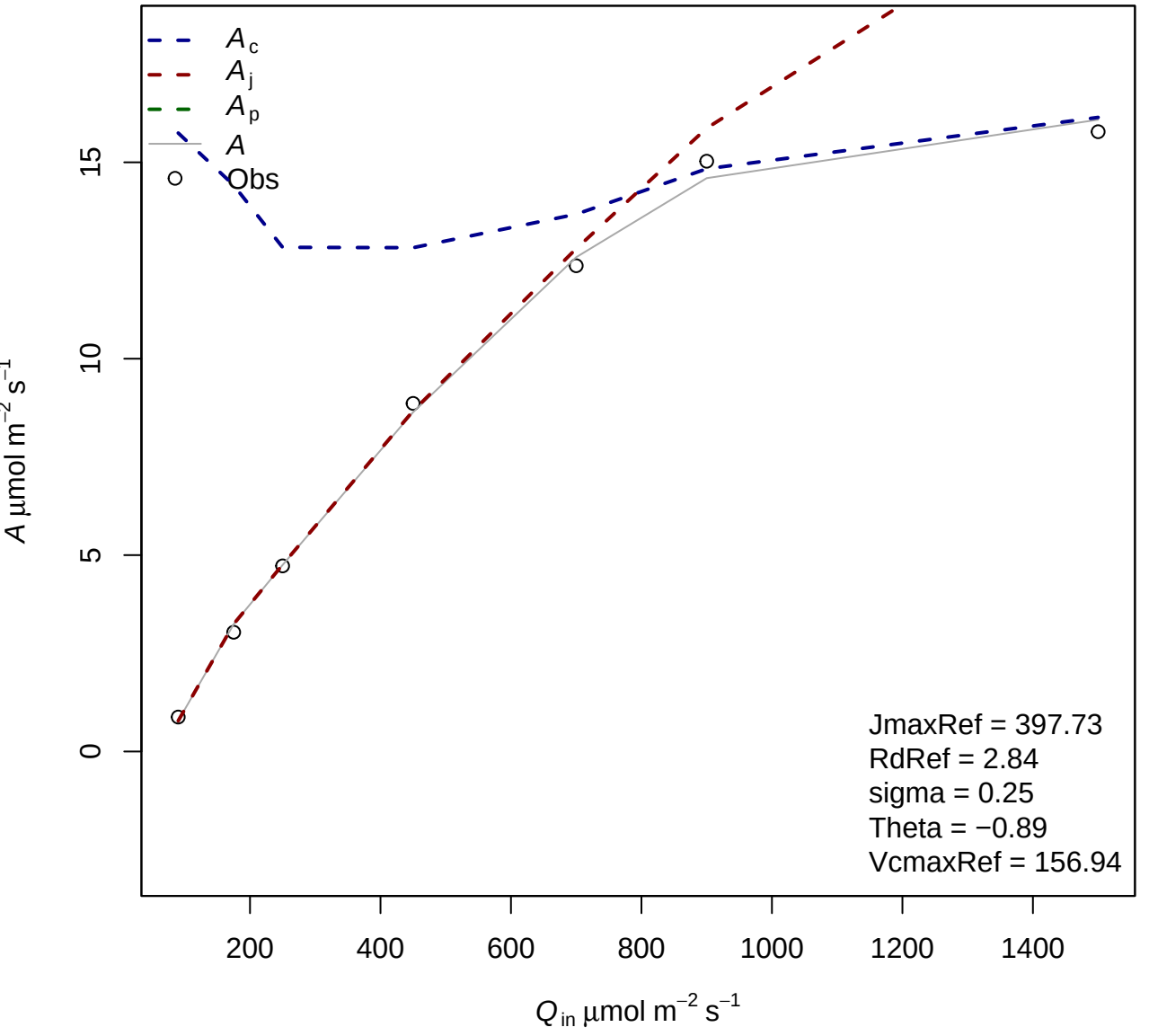

22
A_c
A_j
A_p
A
Obs
JmaxRef = 157.29
RdRef = 1.58
sigma = 0.15
Theta = 0.34
VcmaxRef = 102.51
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

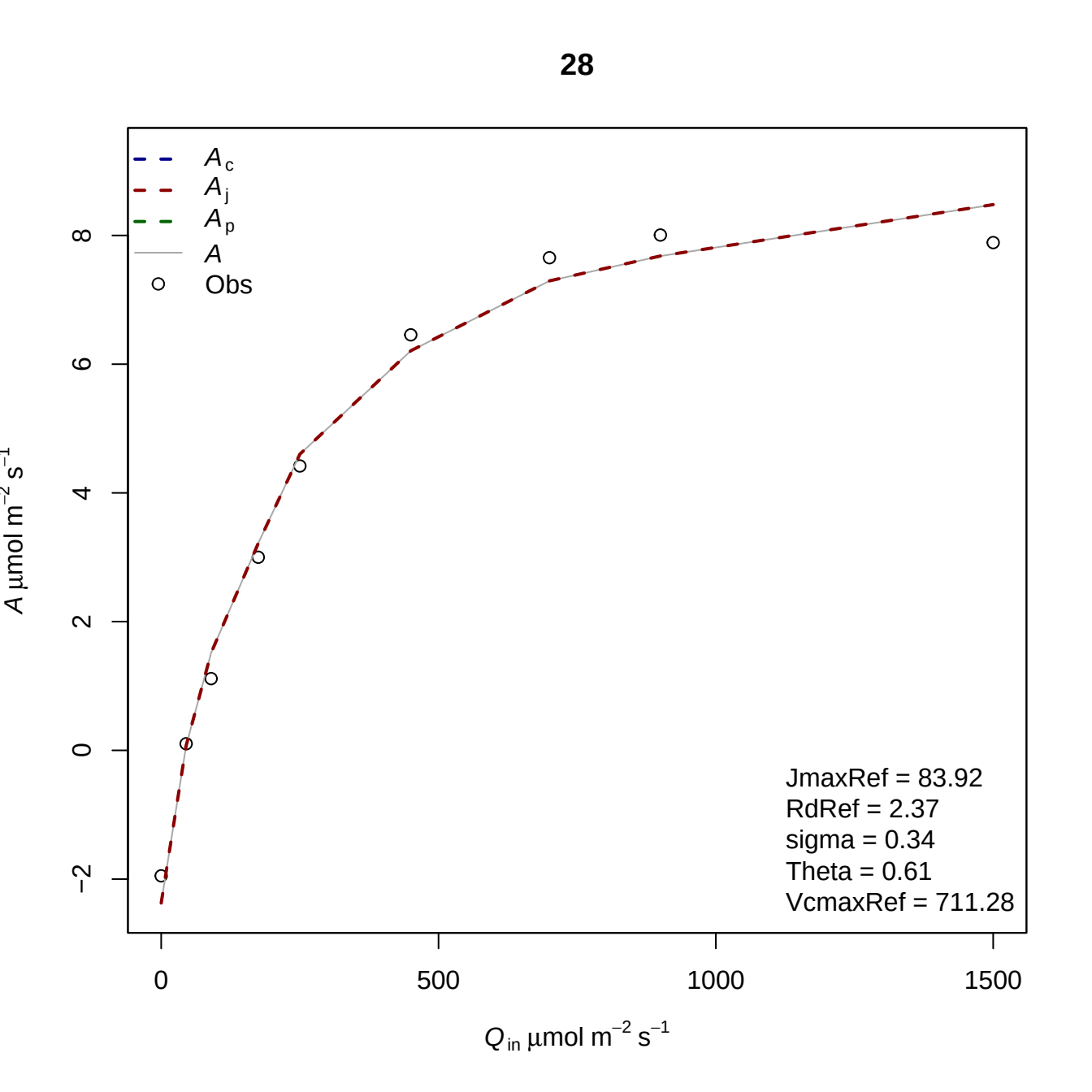

23
A_c
A_j
A_p
A
Obs
JmaxRef = 87.09
RdRef = 1.77
sigma = 0.42
Theta = 0.03
VcmaxRef = 36397.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

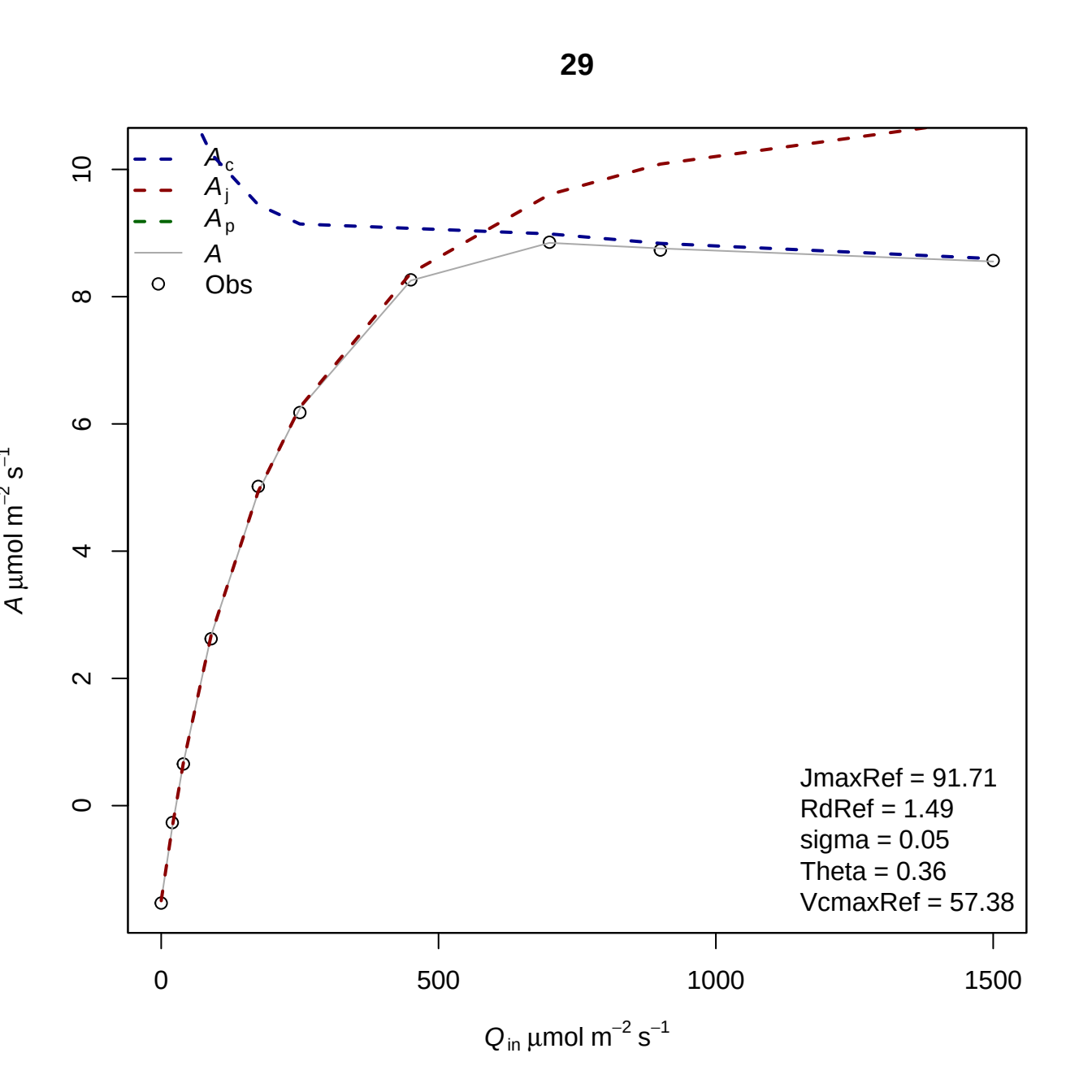

24
A_c
A_j
A_p
A
Obs
JmaxRef = 116.93
RdRef = 1.11
sigma = 0.88
Theta = −3.71
VcmaxRef = 213.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

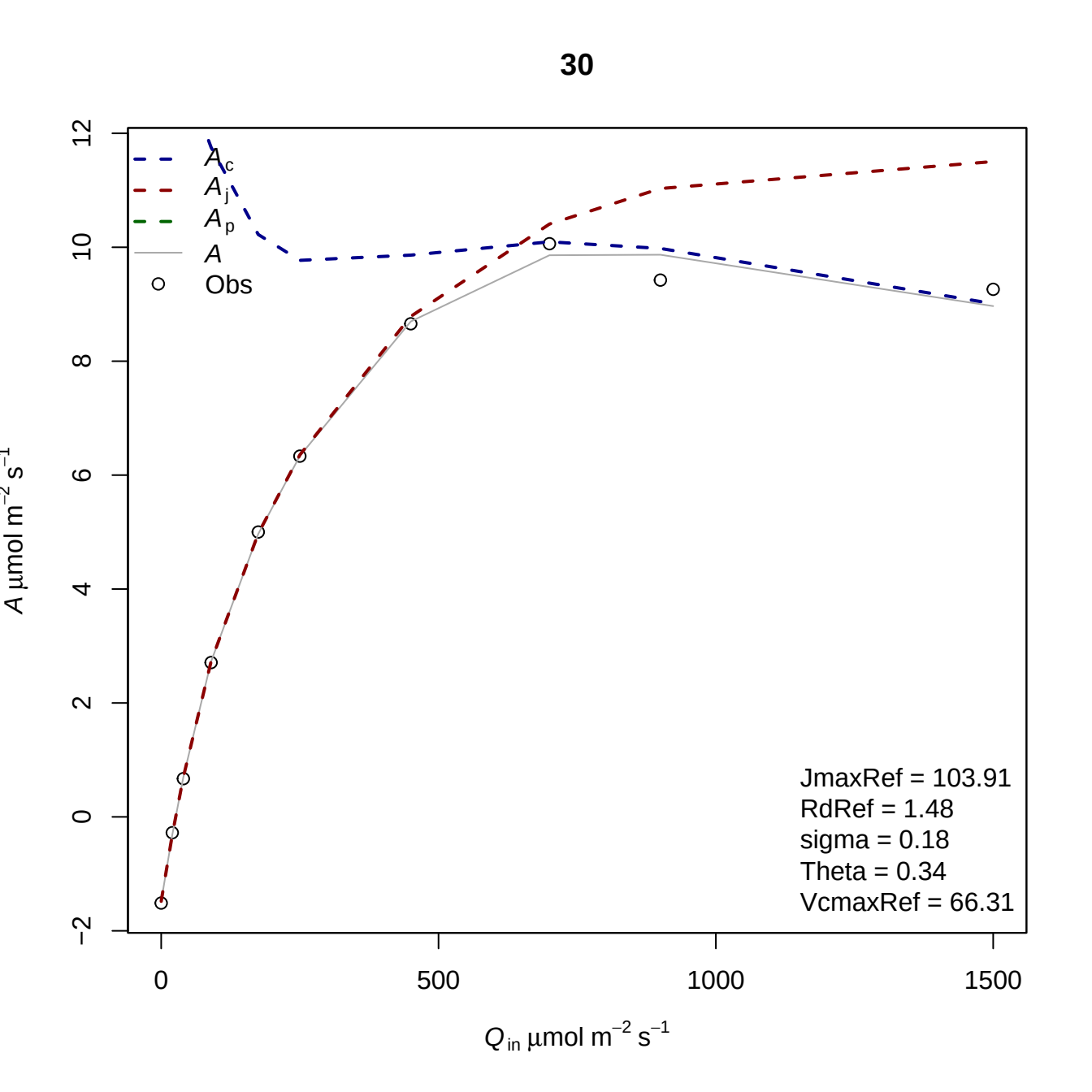

25
A_c
A_j
A_p
A
Obs
JmaxRef = 95.06
RdRef = 1.6
sigma = 0.14
Theta = 0.65
VcmaxRef = 107.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

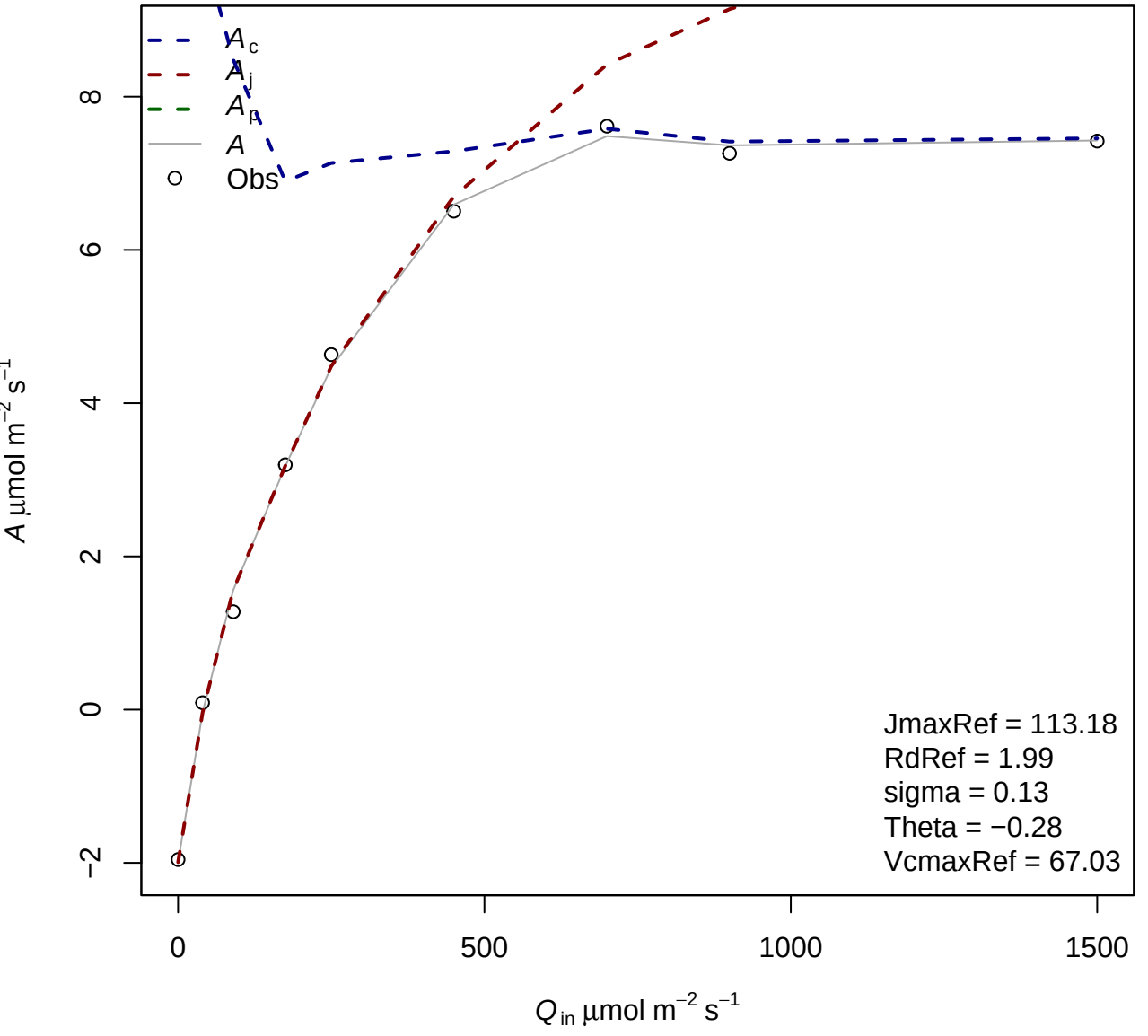

26
A_c
A_j
A_p
A
Obs
JmaxRef = 140.87
RdRef = 1.69
sigma = 0.24
Theta = 0.11
VcmaxRef = 73.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

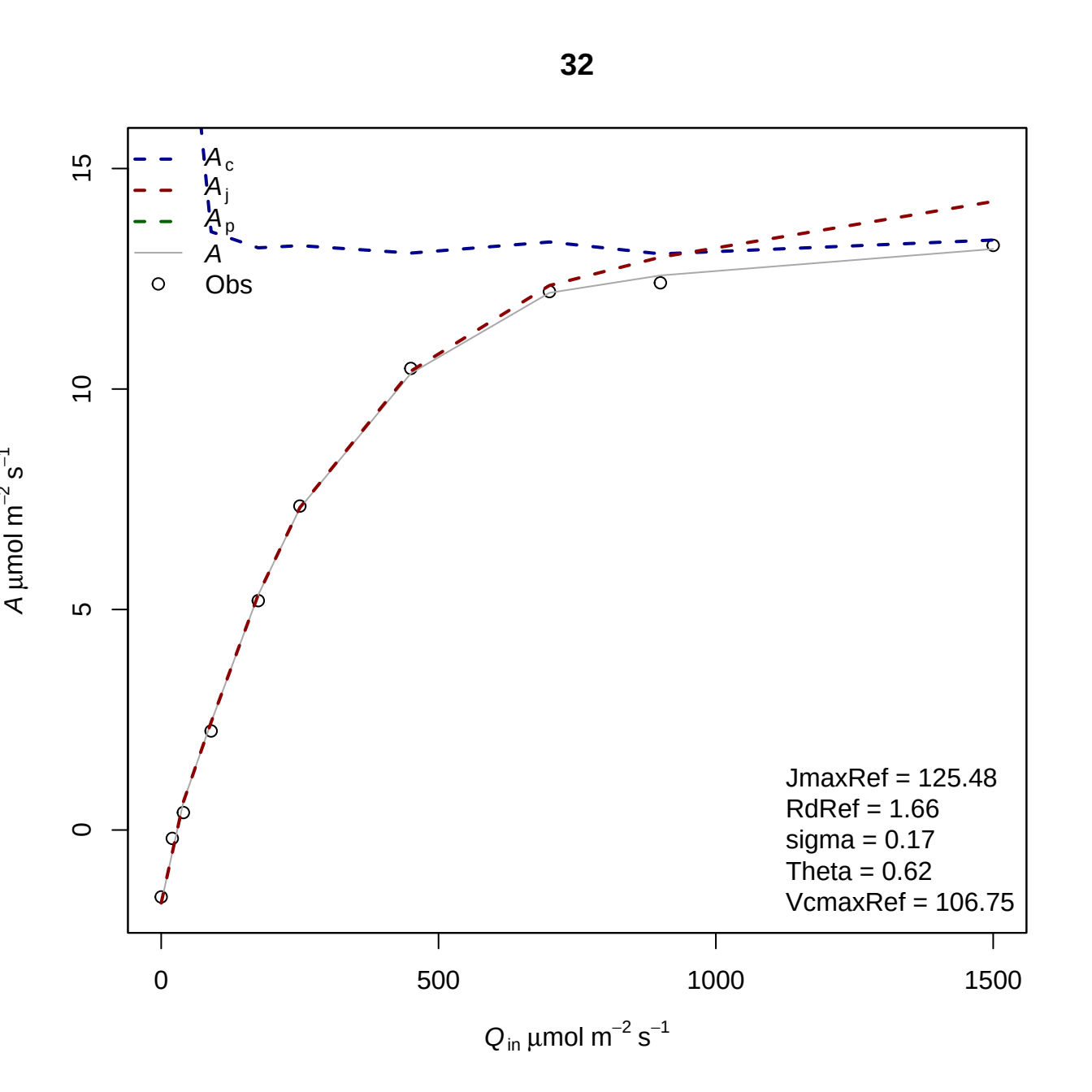

A_c
A_j
A_p
A
Obs
JmaxRef = 397.73
RdRef = 2.84
sigma = 0.25
Theta = −0.89
VcmaxRef = 156.94
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

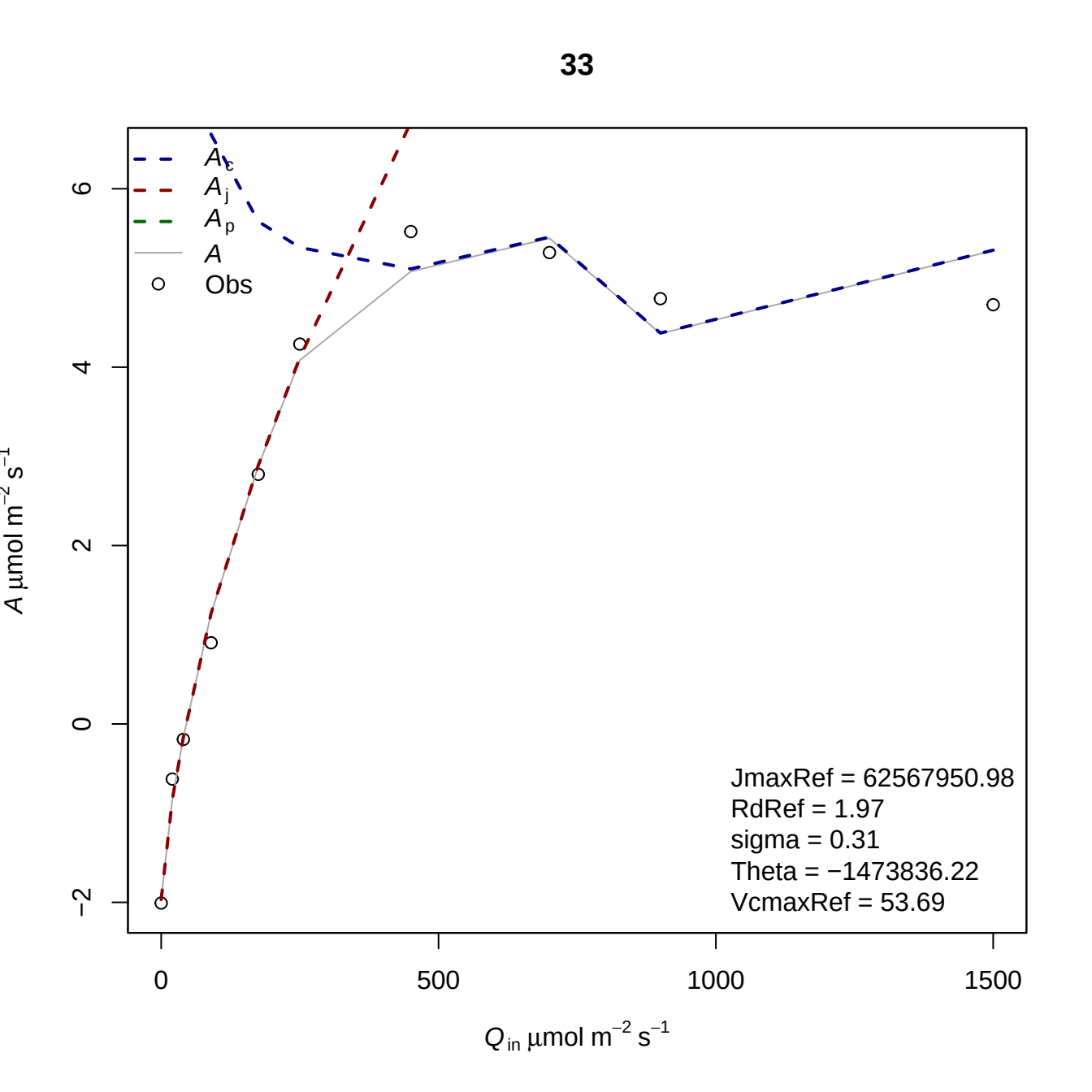

28
A_c
A_j
A_p
A
Obs
JmaxRef = 83.92
RdRef = 2.37
sigma = 0.34
Theta = 0.61
VcmaxRef = 711.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

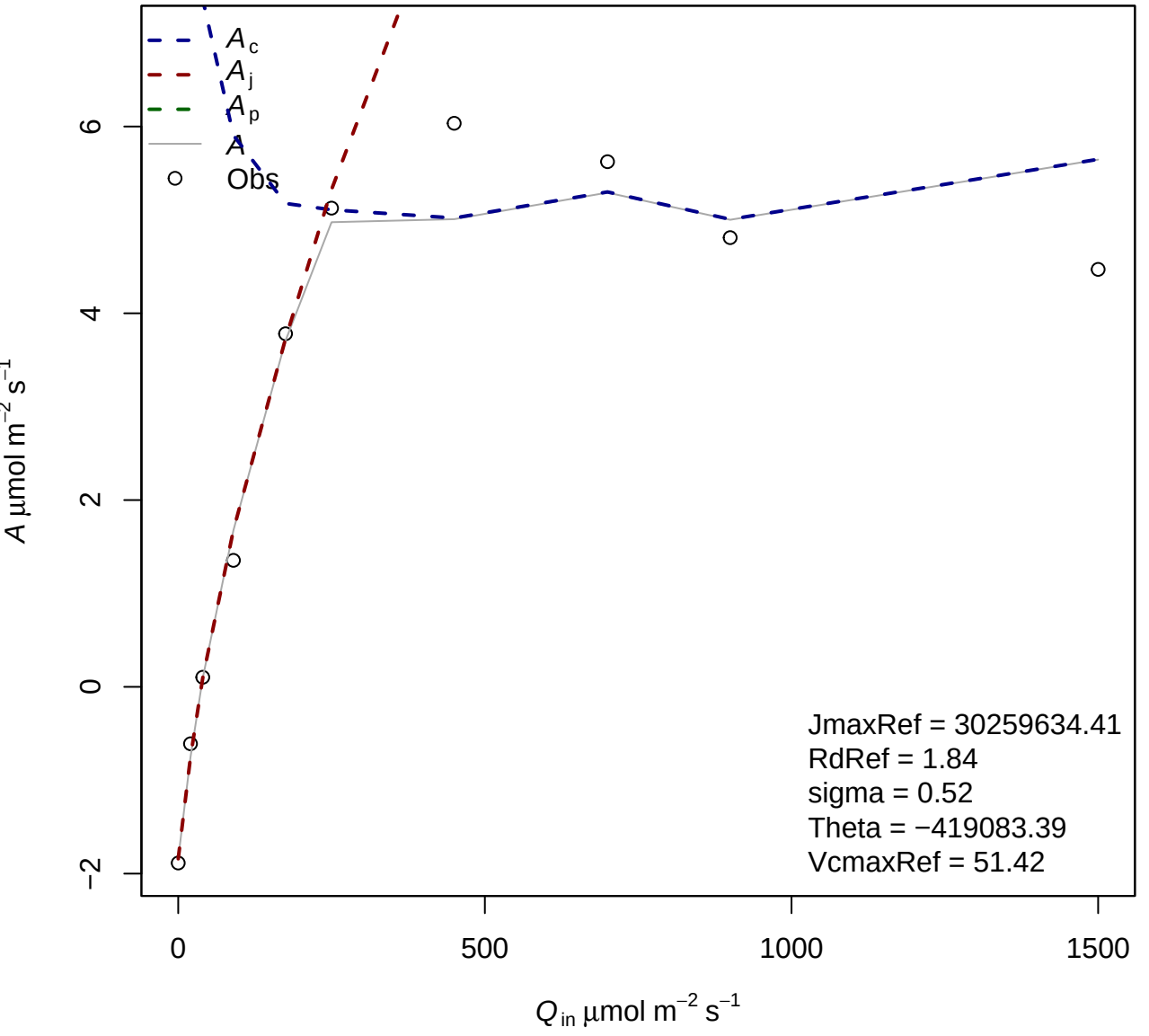

29
A_c
A_j
A_p
A
Obs
JmaxRef = 91.71
RdRef = 1.49
sigma = 0.05
Theta = 0.36
VcmaxRef = 57.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

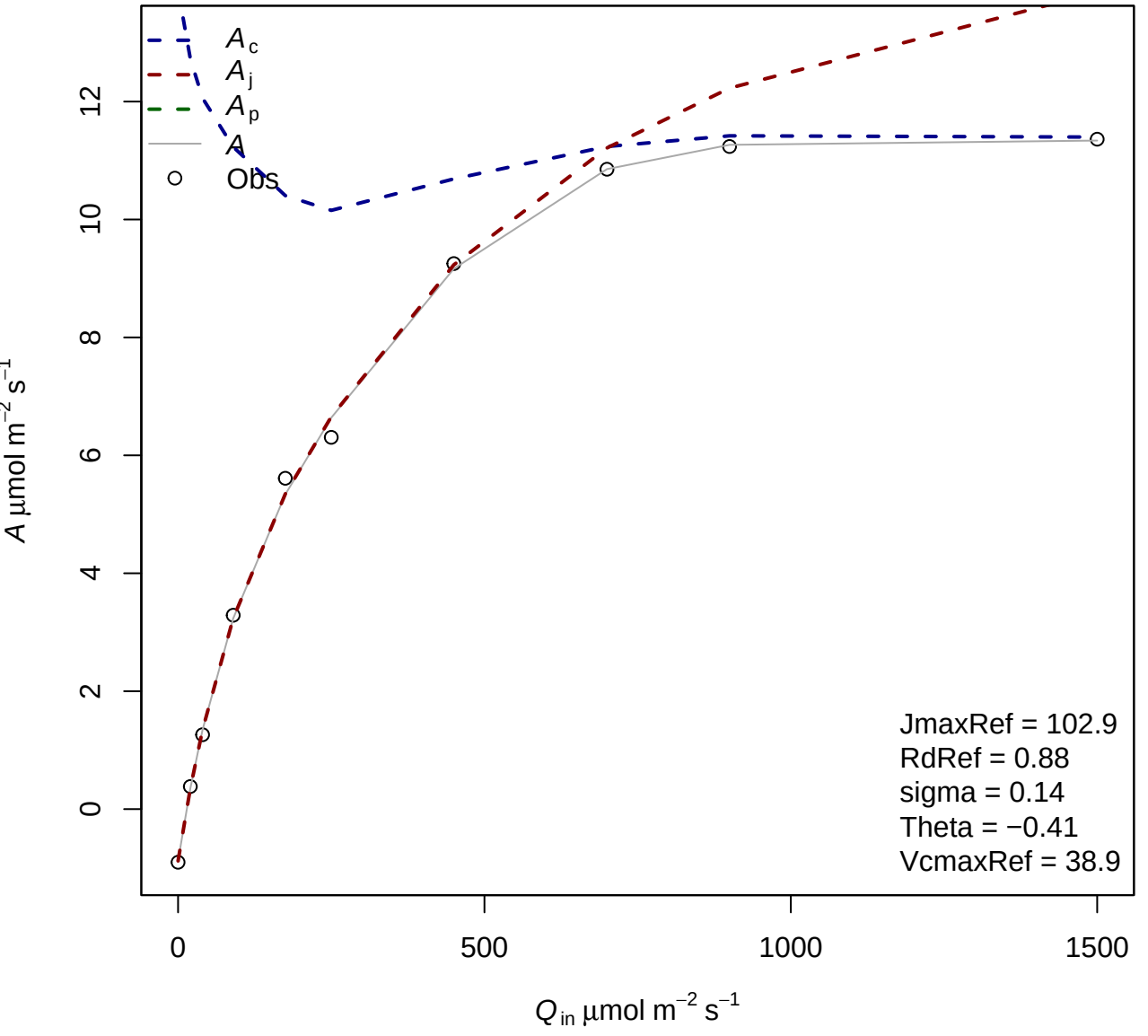

30
A_c
A_j
A_p
A
Obs
JmaxRef = 103.91
RdRef = 1.48
sigma = 0.18
Theta = 0.34
VcmaxRef = 66.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

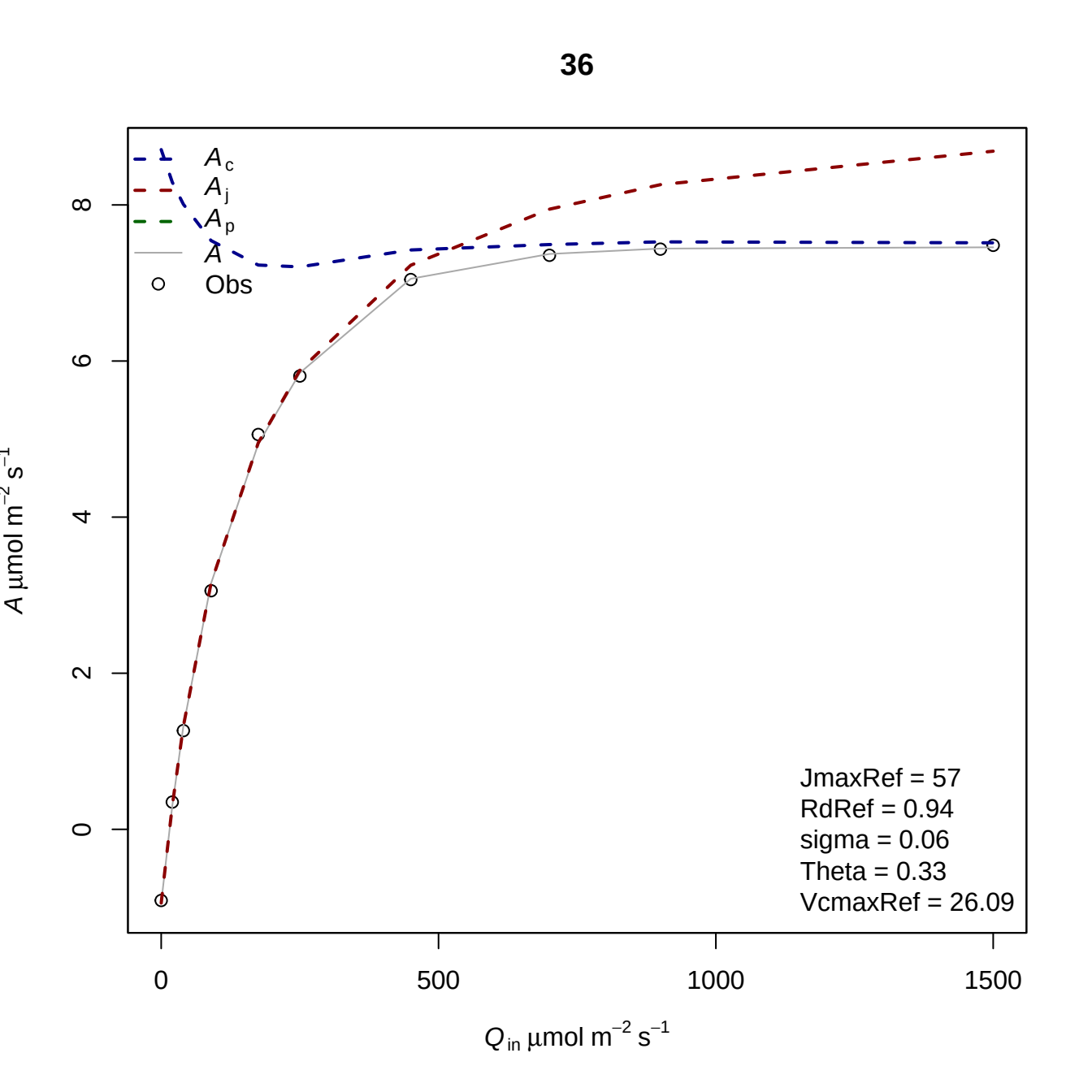

A_c
A_j
A_p
A
Obs
JmaxRef = 113.18
RdRef = 1.99
sigma = 0.13
Theta = −0.28
VcmaxRef = 67.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

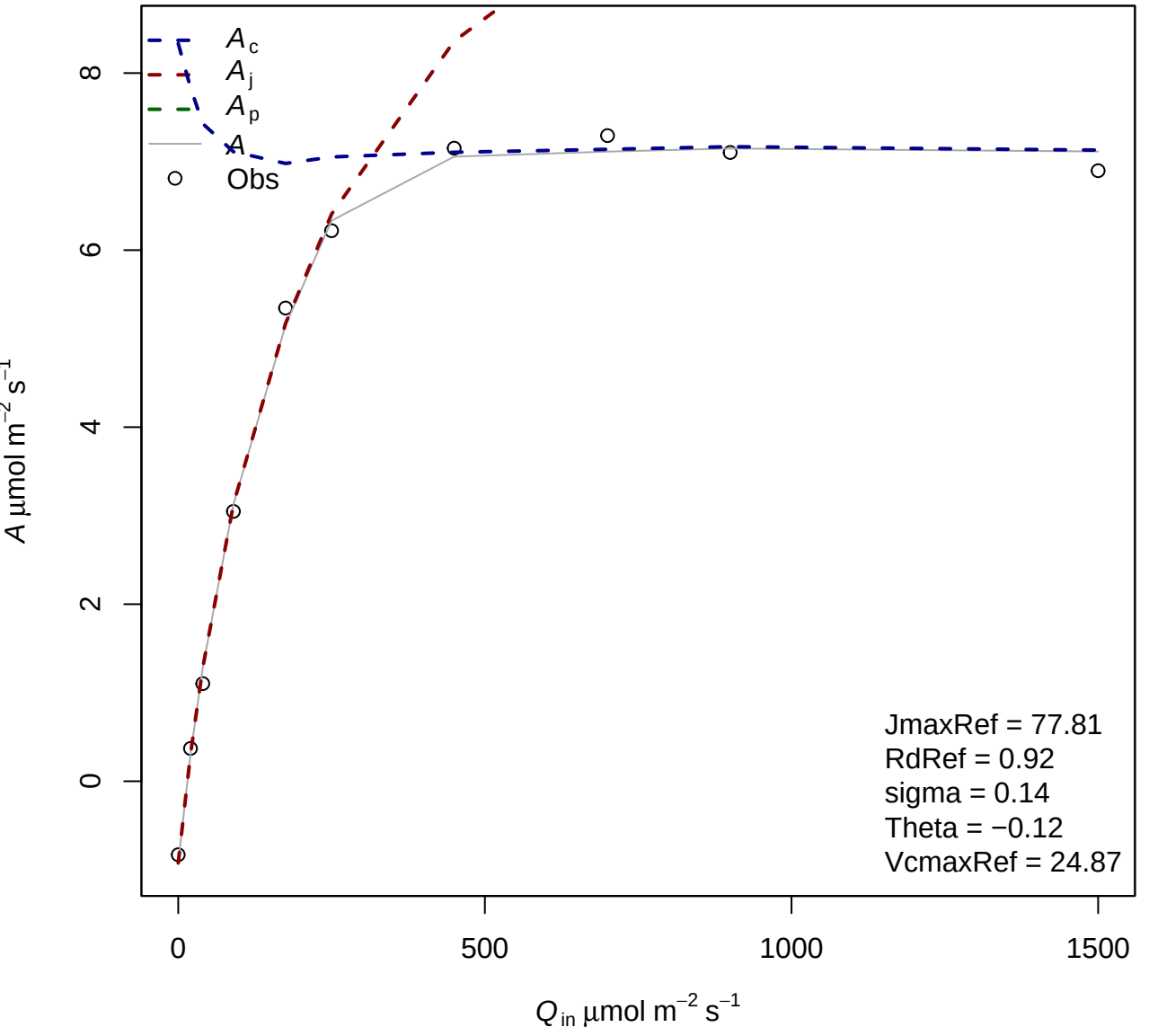

32
A_c
A_j
A_p
A
Obs
JmaxRef = 125.48
RdRef = 1.66
sigma = 0.17
Theta = 0.62
VcmaxRef = 106.75
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

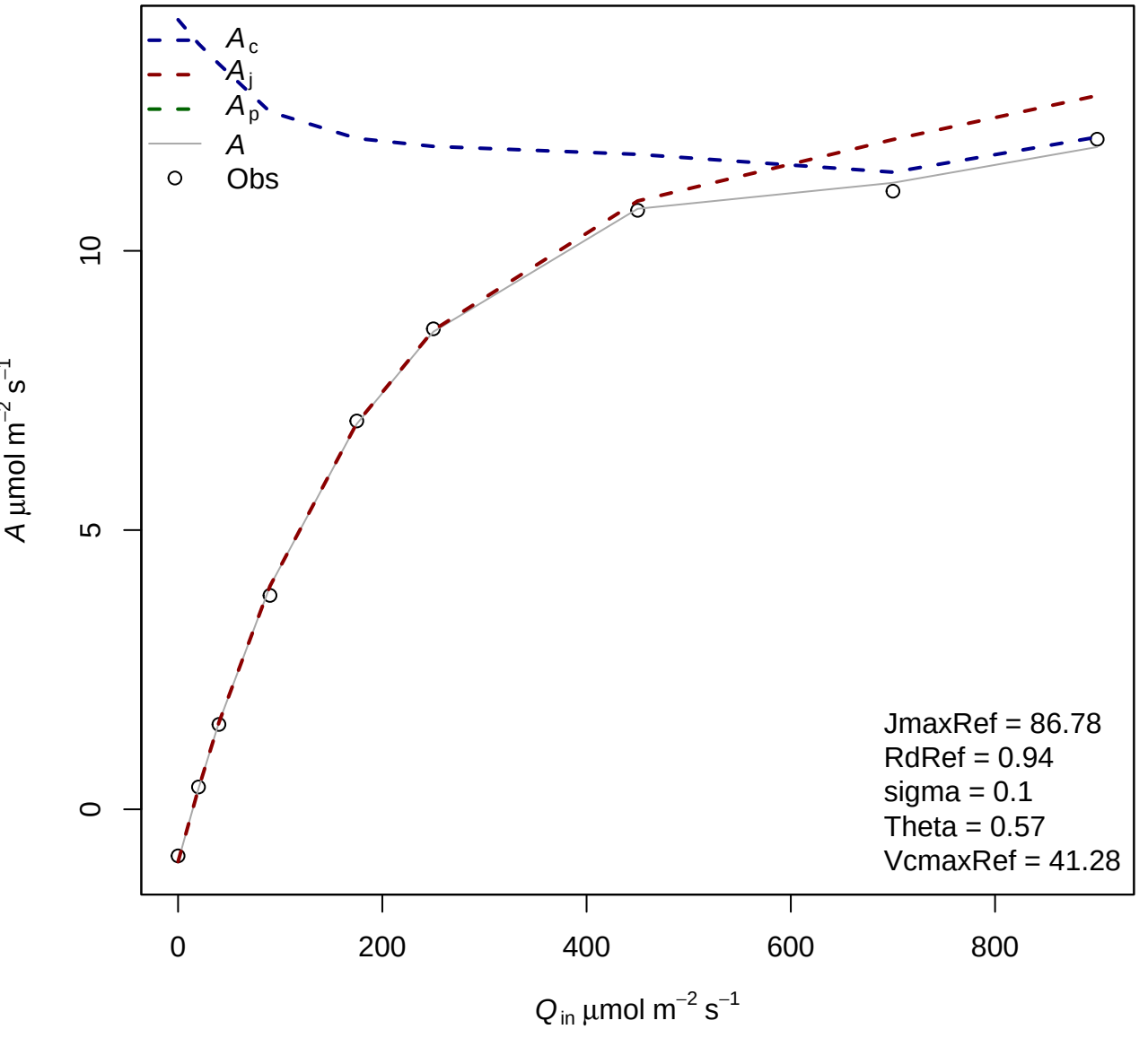

33
A_c
A_j
A_p
A
Obs
JmaxRef = 62567950.98
RdRef = 1.97
sigma = 0.31
Theta = −1473836.22
VcmaxRef = 53.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

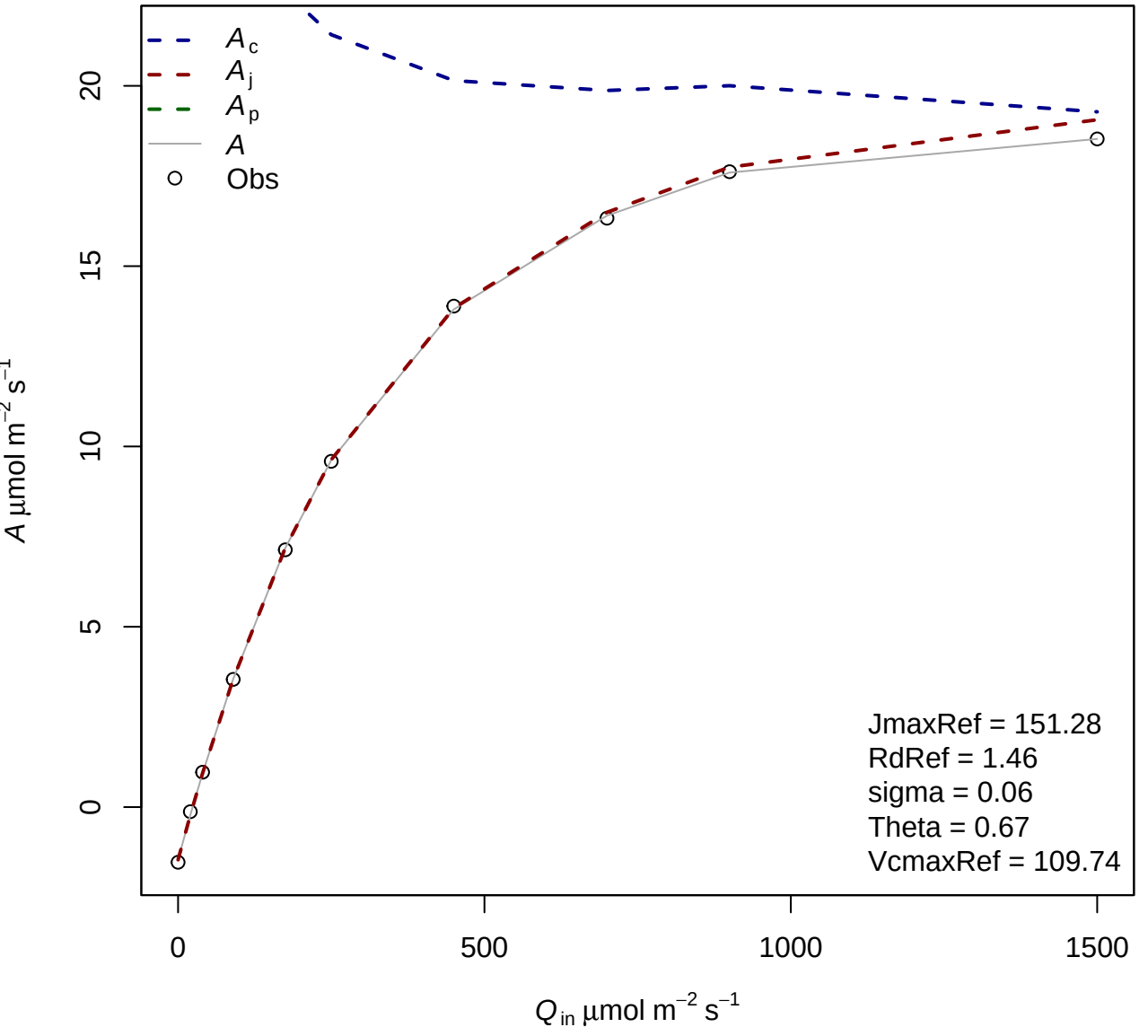

A_c
A_j
A_p
A
Obs
JmaxRef = 30259634.41
RdRef = 1.84
sigma = 0.52
Theta = −419083.39
VcmaxRef = 51.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

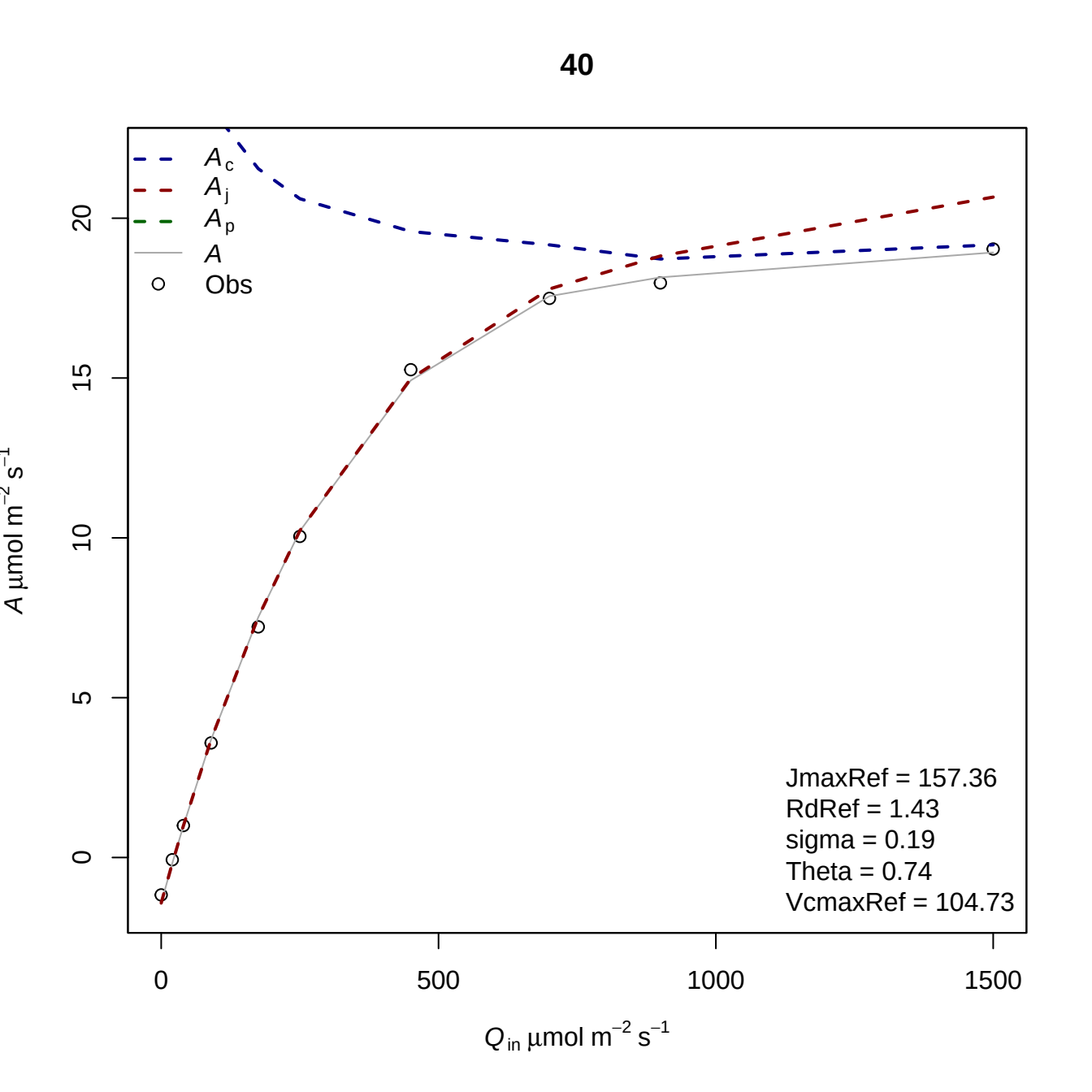

A_c
A_j
A_p
A
Obs
JmaxRef = 102.9
RdRef = 0.88
sigma = 0.14
Theta = −0.41
VcmaxRef = 38.9
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

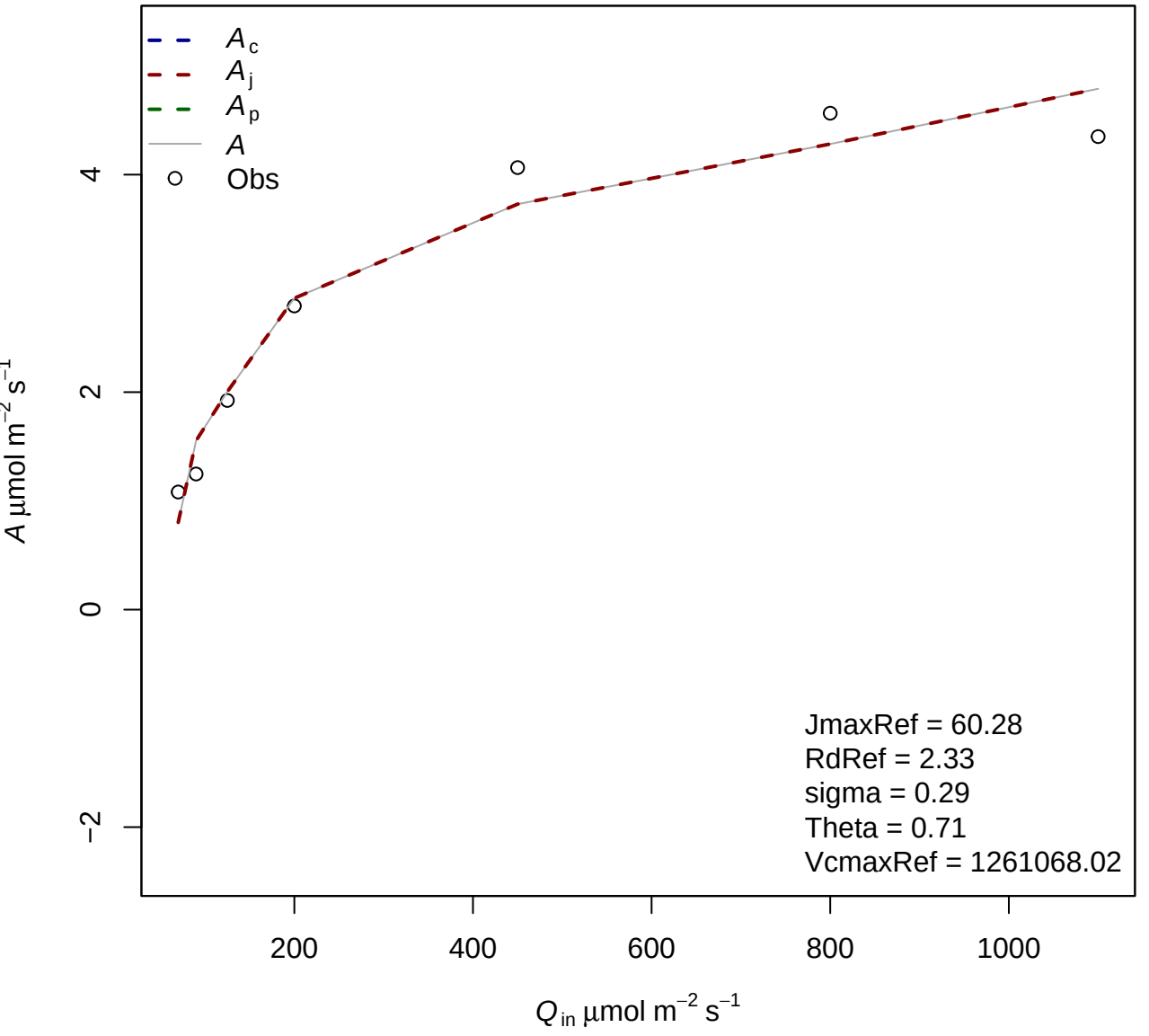

36
A_c
A_j
A_p
A
Obs
JmaxRef = 57
RdRef = 0.94
sigma = 0.06
Theta = 0.33
VcmaxRef = 26.09
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

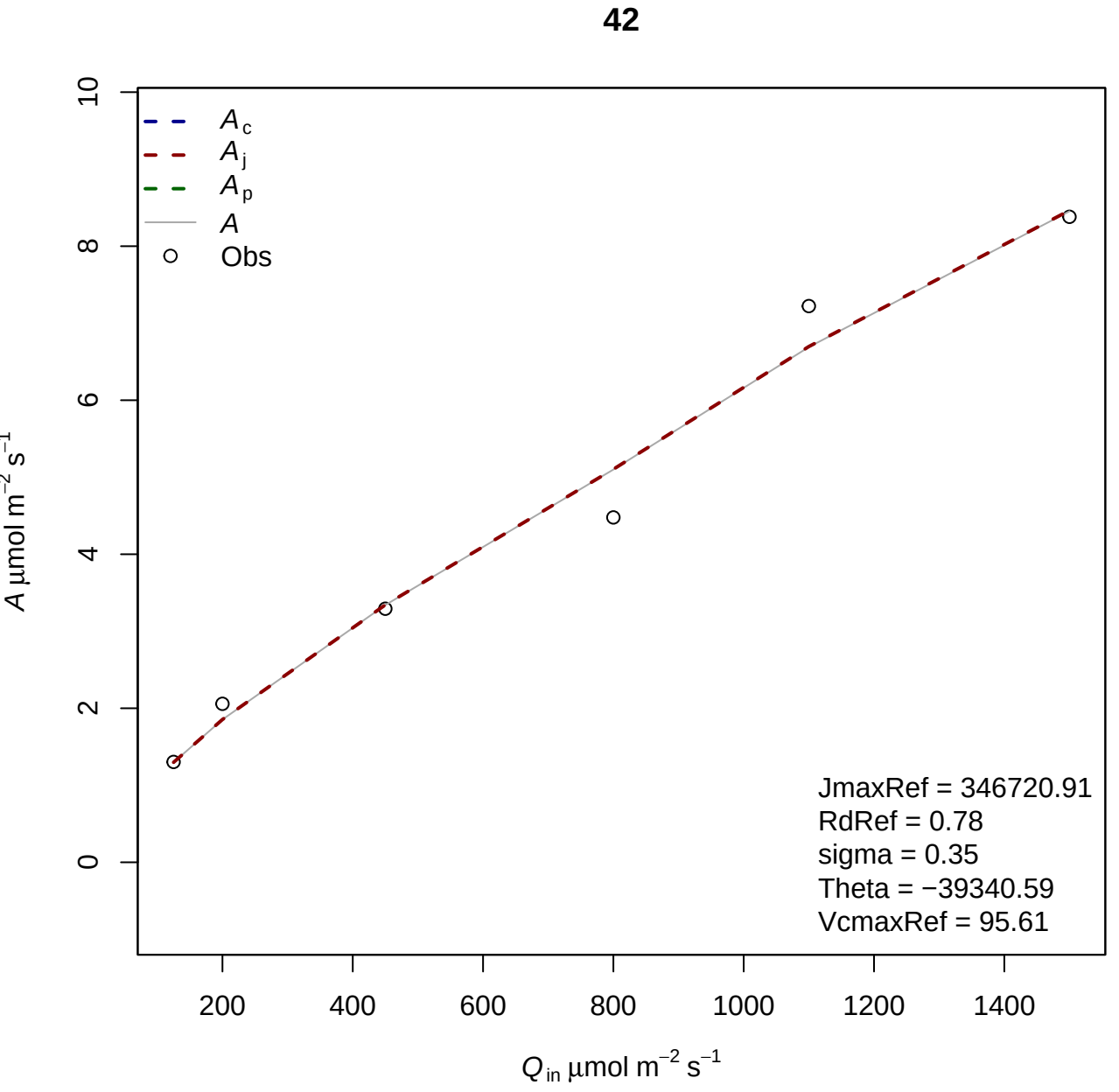

A_c
A_j
A_p
A
Obs
JmaxRef = 77.81
RdRef = 0.92
sigma = 0.14
Theta = −0.12
VcmaxRef = 24.87
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

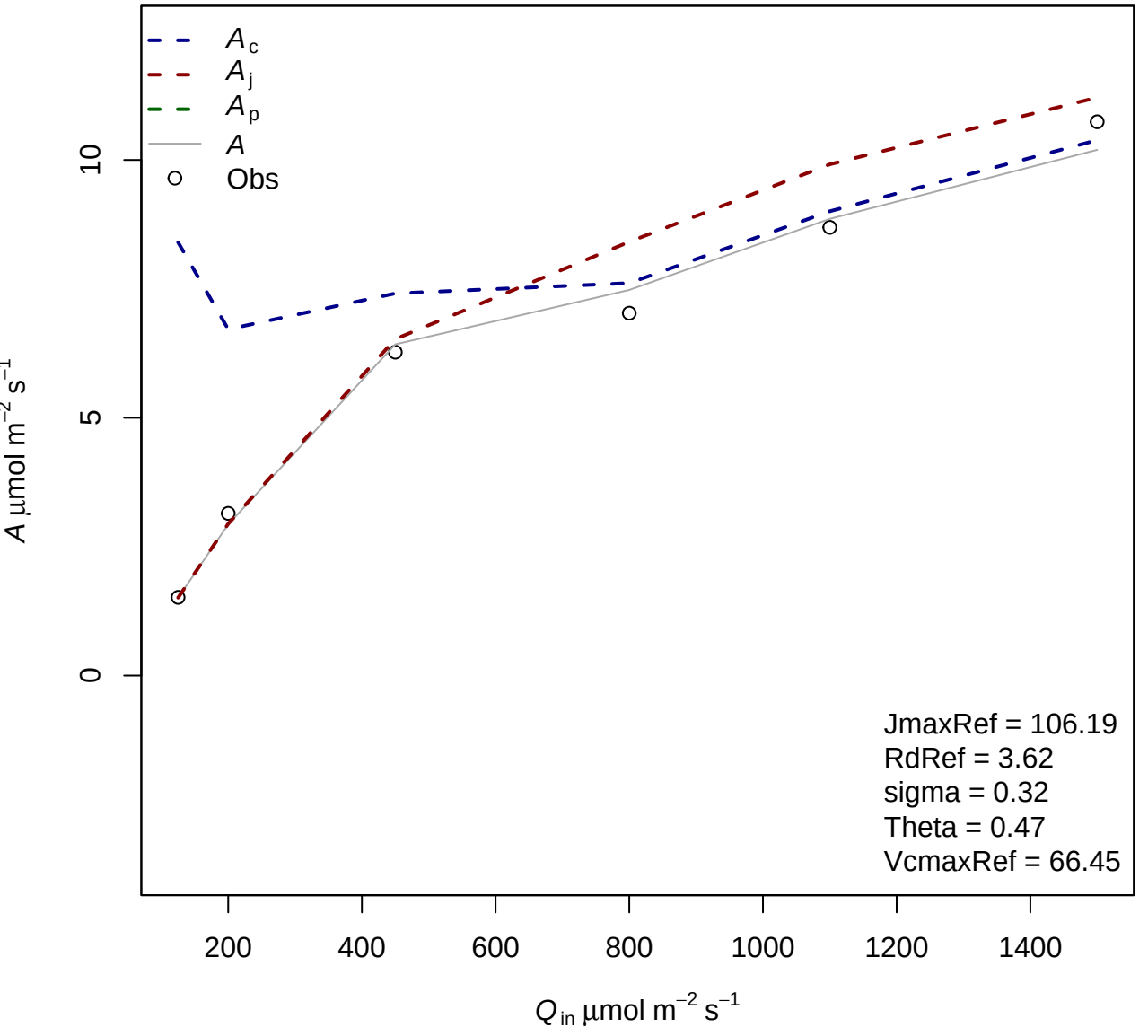

A_c
A_j
A_p
A
Obs
JmaxRef = 86.78
RdRef = 0.94
sigma = 0.1
Theta = 0.57
VcmaxRef = 41.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800

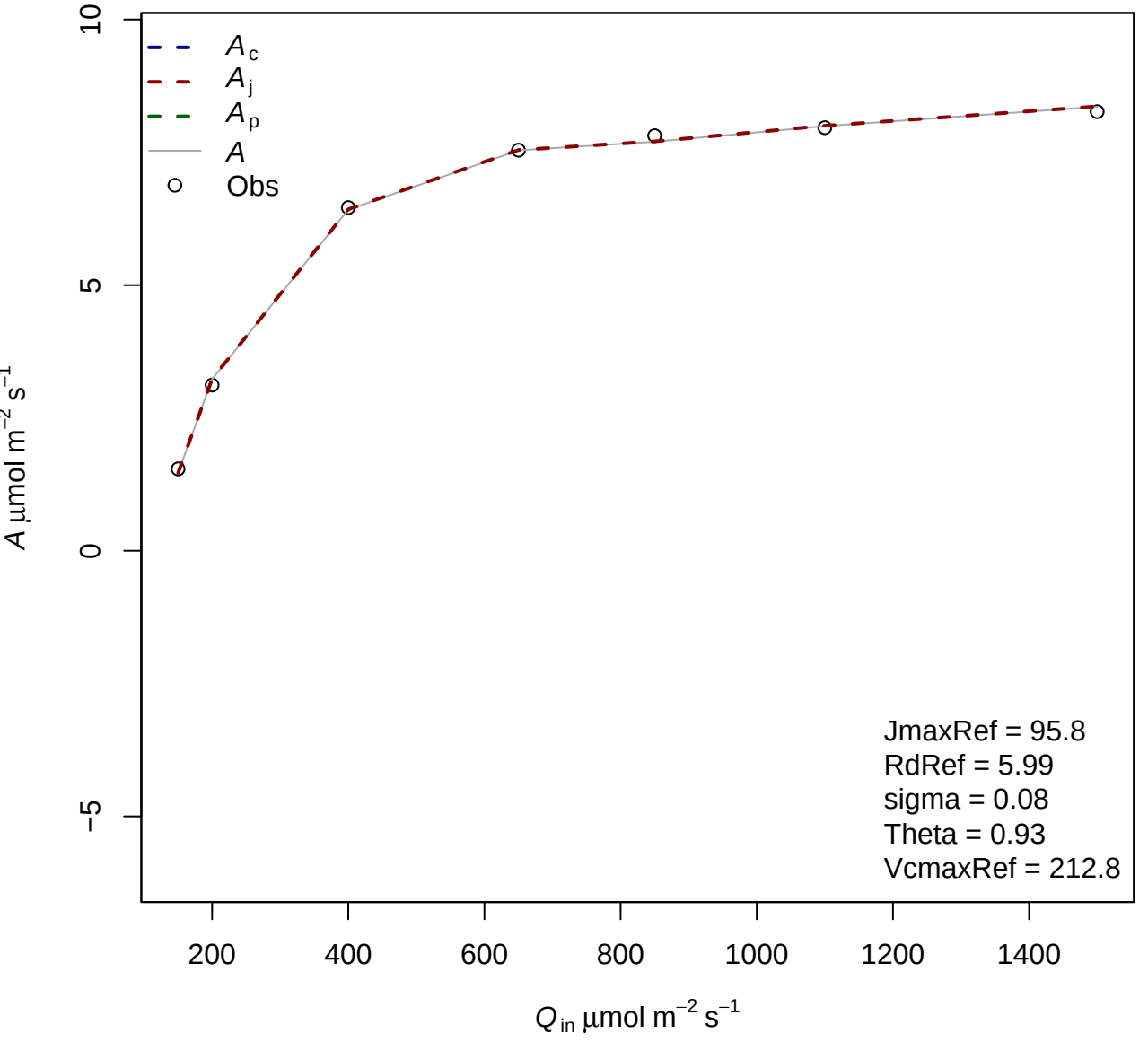

A_c
A_j
A_p
A
Obs
JmaxRef = 151.28
RdRef = 1.46
sigma = 0.06
Theta = 0.67
VcmaxRef = 109.74
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

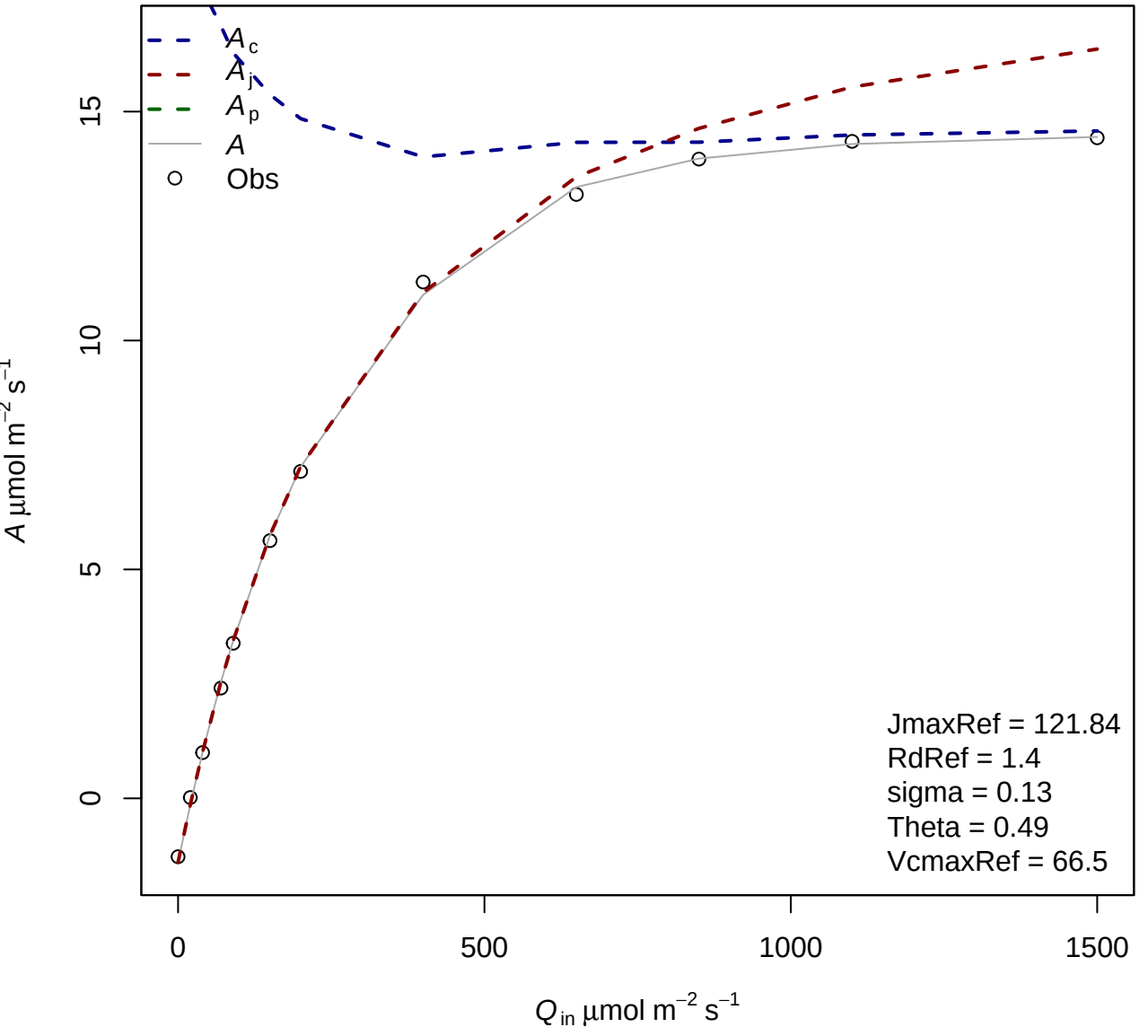

40
A_c
A_j
A_p
A
Obs
JmaxRef = 157.36
RdRef = 1.43
sigma = 0.19
Theta = 0.74
VcmaxRef = 104.73
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

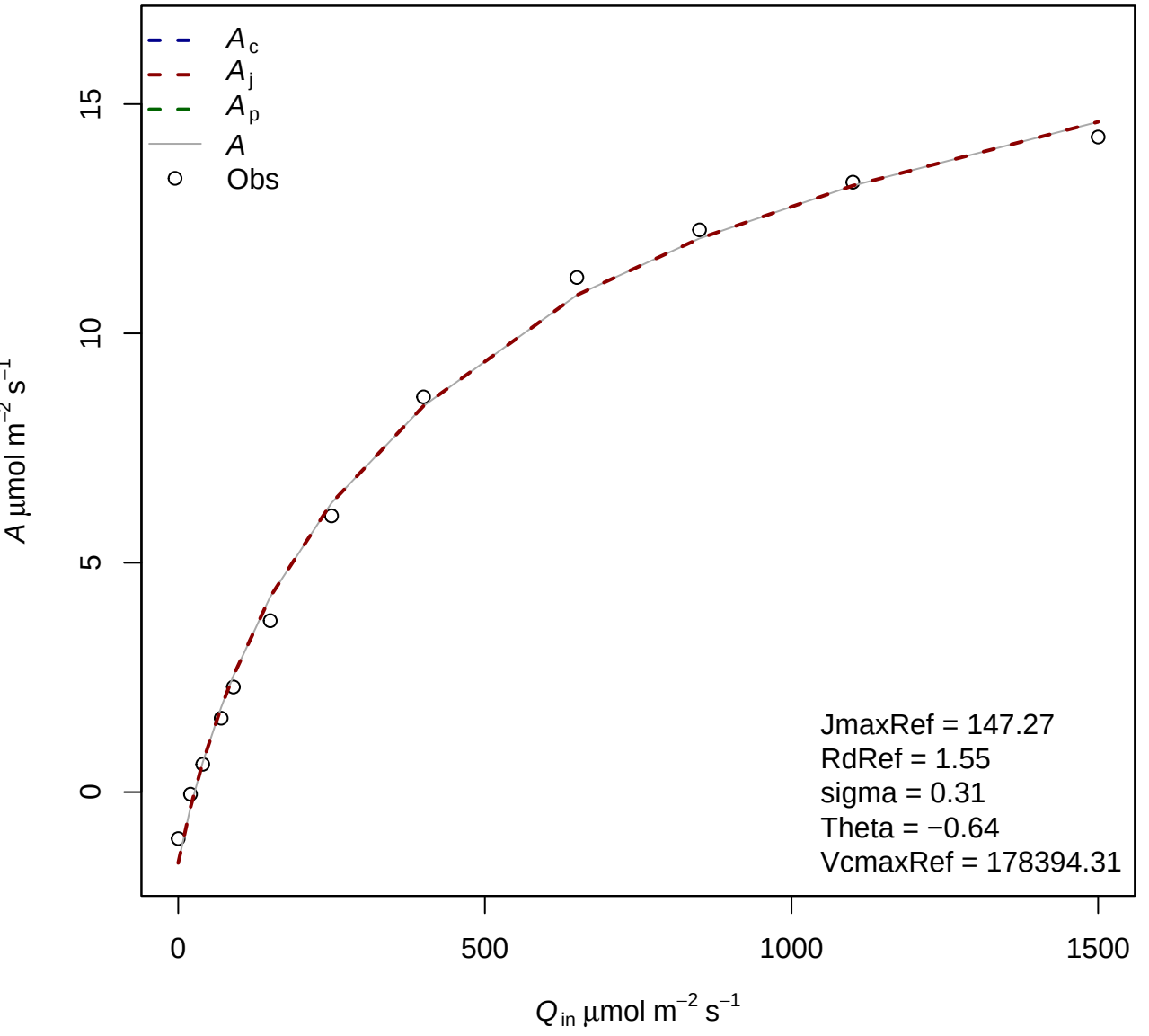

A_c
A_j
A_p
A
Obs
JmaxRef = 60.28
RdRef = 2.33
sigma = 0.29
Theta = 0.71
VcmaxRef = 1261068.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000

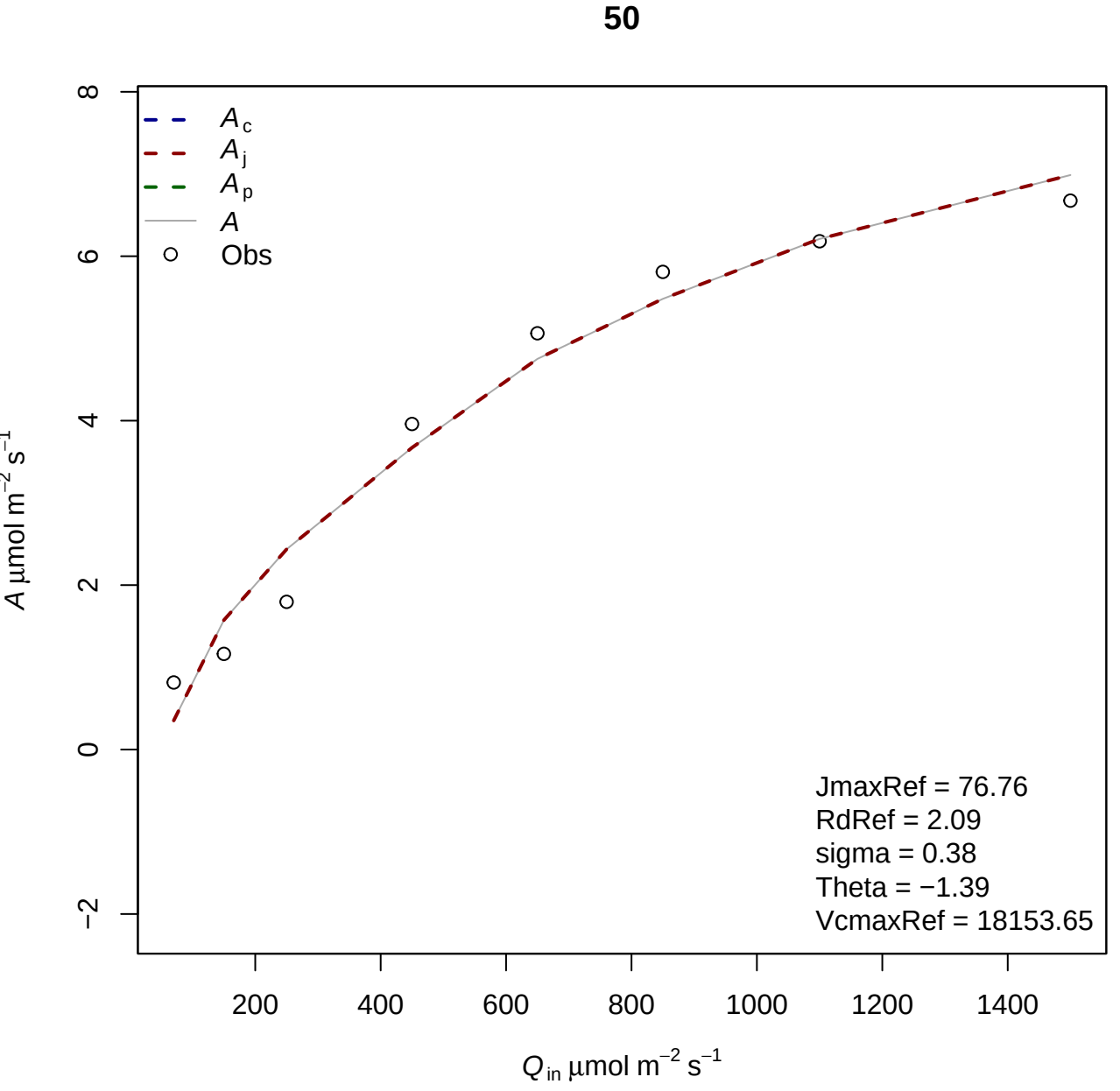

42
A_c
A_j
A_p
A
Obs
JmaxRef = 346720.91
RdRef = 0.78
sigma = 0.35
Theta = −39340.59
VcmaxRef = 95.61
0
2
4
6
8
10
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

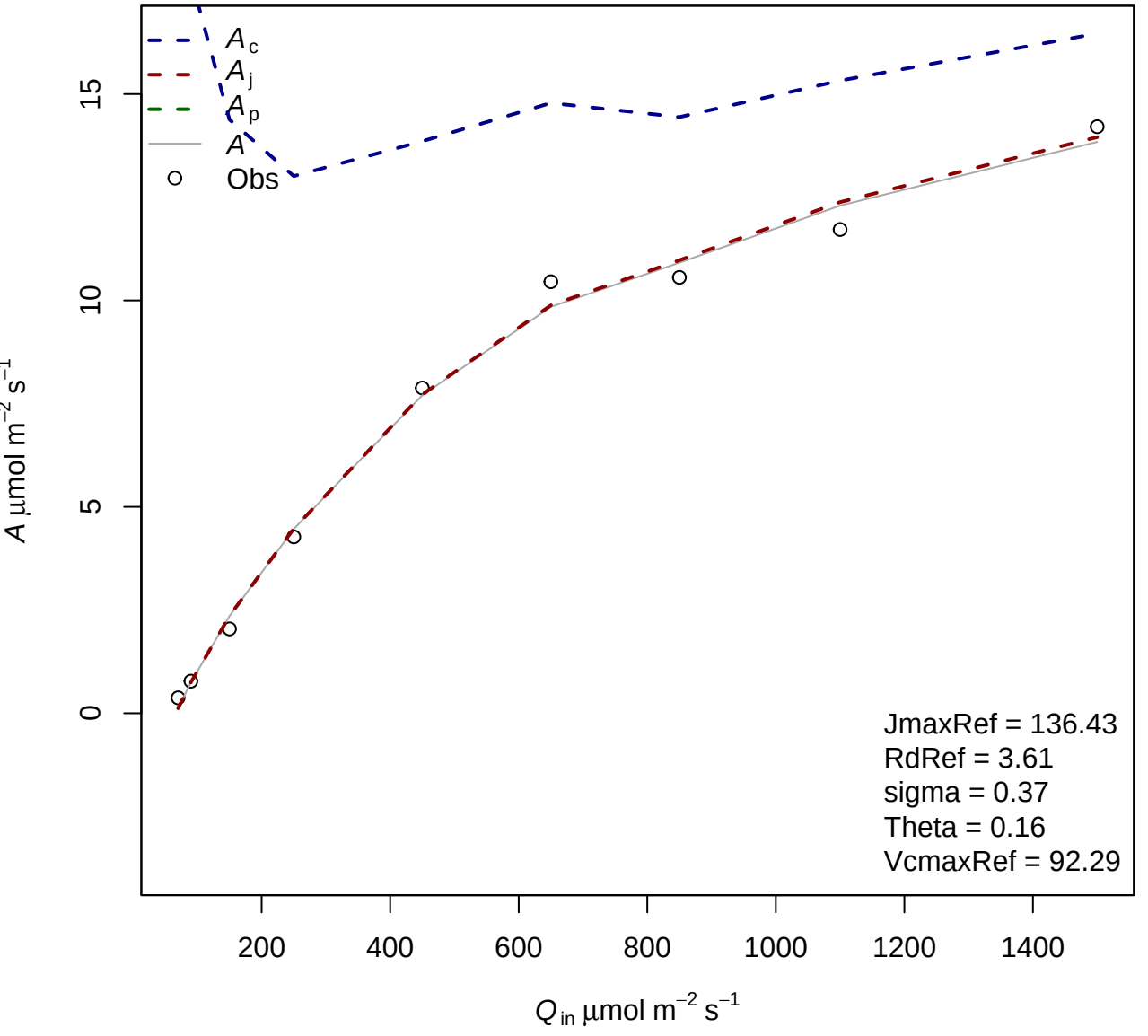

A_c
A_j
A_p
A
Obs
JmaxRef = 106.19
RdRef = 3.62
sigma = 0.32
Theta = 0.47
VcmaxRef = 66.45
0
5
10
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

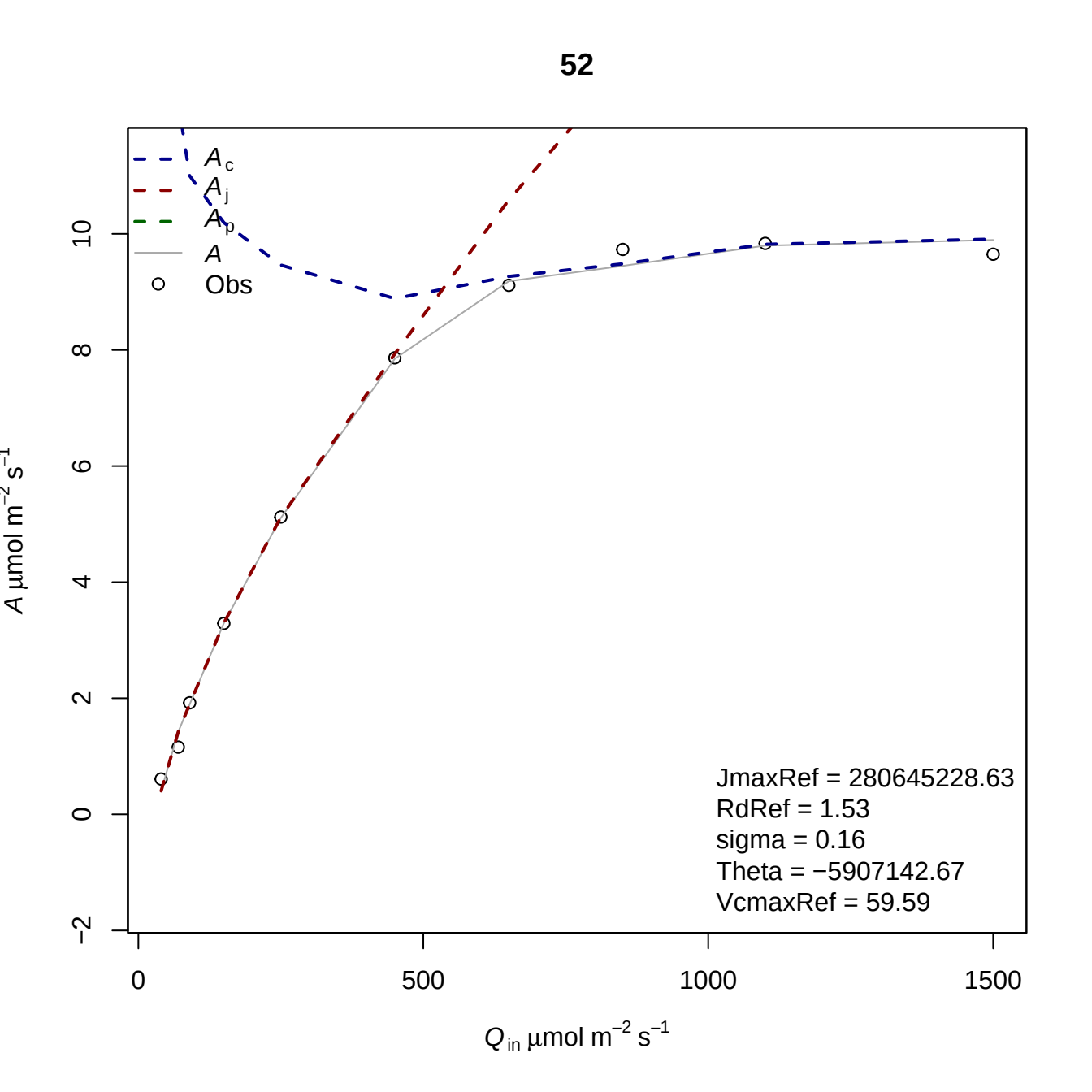

A_c
A_j
A_p
A
Obs
JmaxRef = 95.8
RdRef = 5.99
sigma = 0.08
Theta = 0.93
VcmaxRef = 212.8
10
5
0
−5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

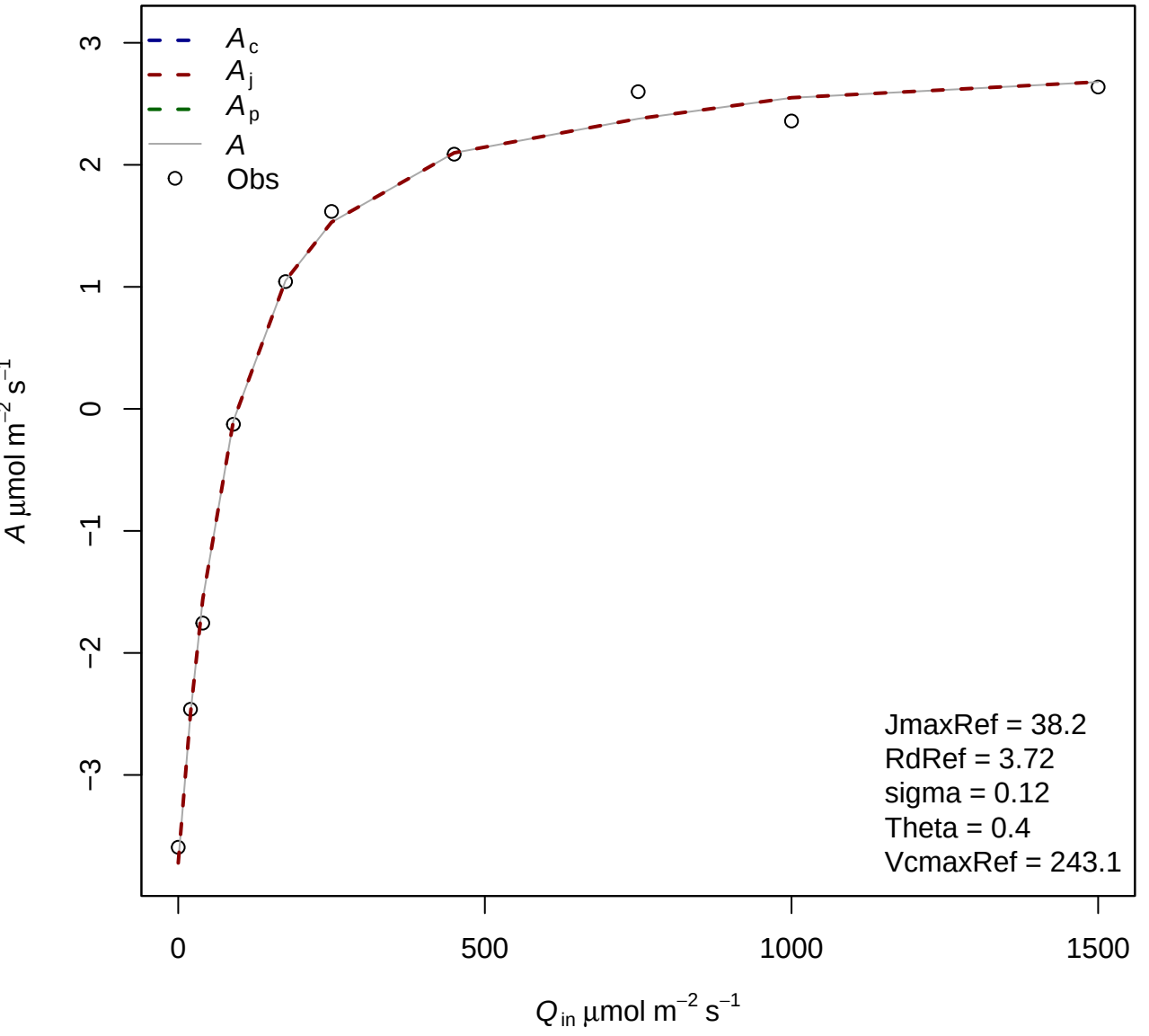

A_c
A_j
A_p
A
Obs
JmaxRef = 121.84
RdRef = 1.4
sigma = 0.13
Theta = 0.49
VcmaxRef = 66.5
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

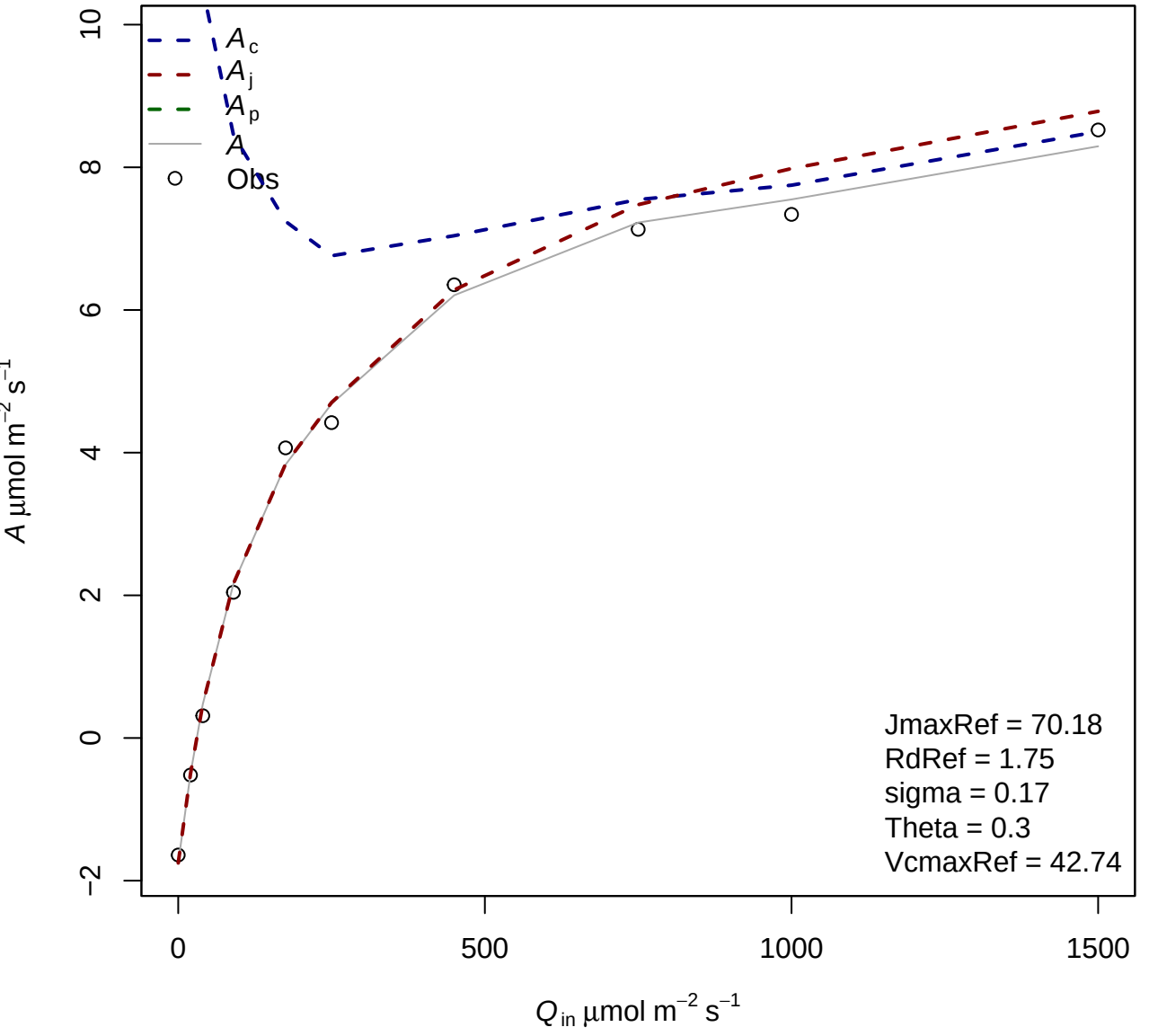

A_c
A_j
A_p
A
Obs
JmaxRef = 147.27
RdRef = 1.55
sigma = 0.31
Theta = −0.64
VcmaxRef = 178394.31
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

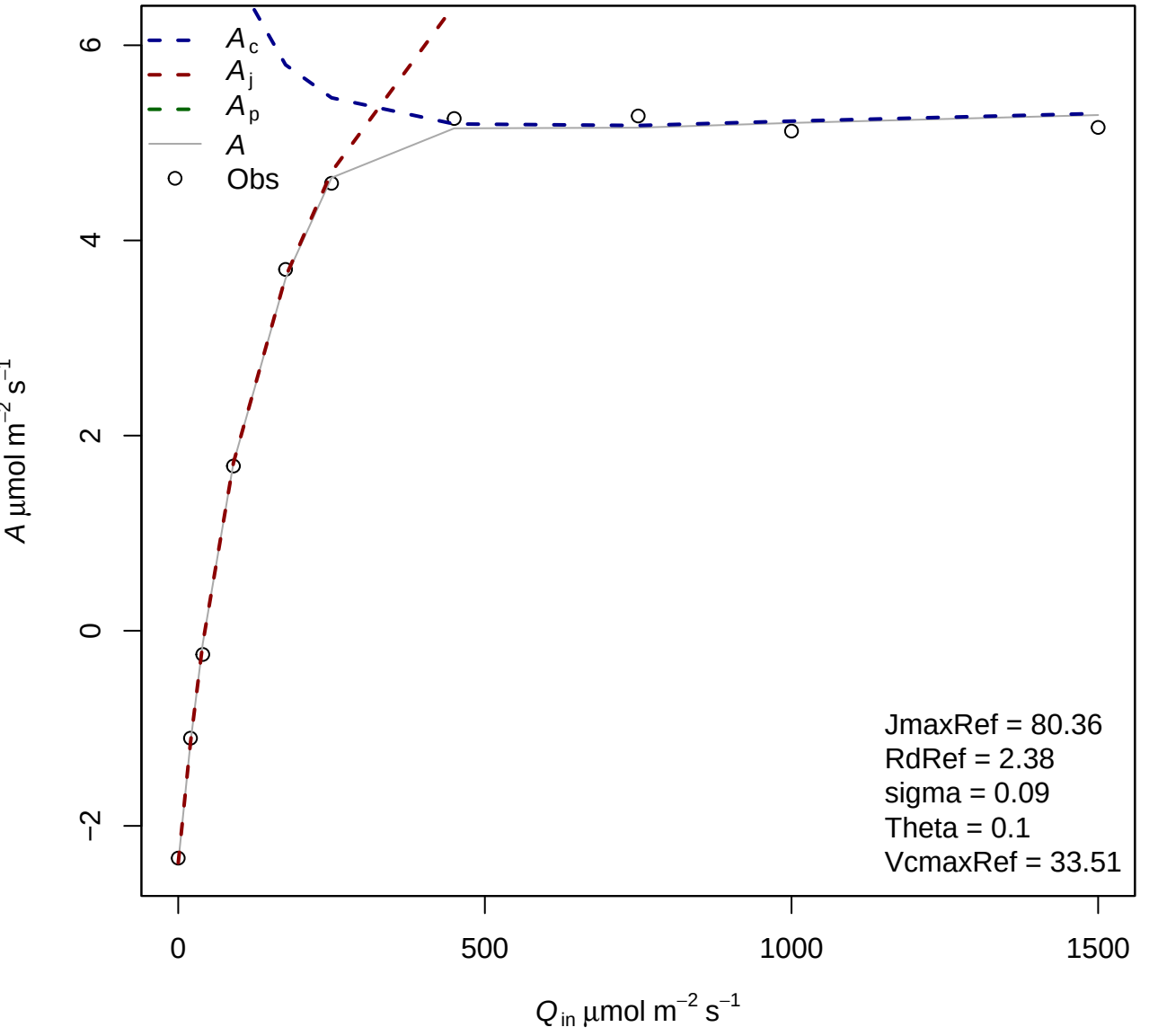

50
A_c
A_j
A_p
A
Obs
JmaxRef = 76.76
RdRef = 2.09
sigma = 0.38
Theta = −1.39
VcmaxRef = 18153.65
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

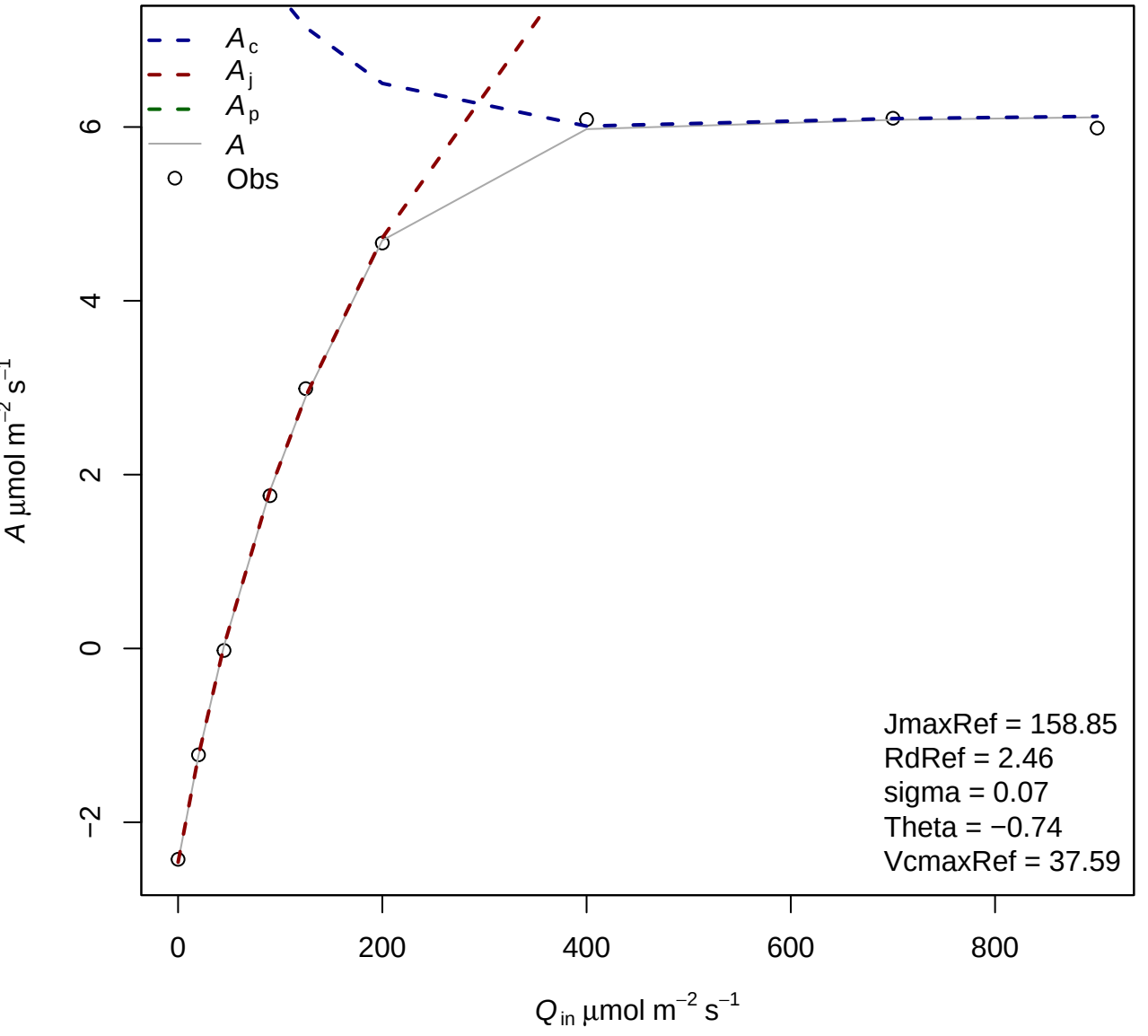

A_c
A_j
A_p
A
Obs
JmaxRef = 136.43
RdRef = 3.61
sigma = 0.37
Theta = 0.16
VcmaxRef = 92.29
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

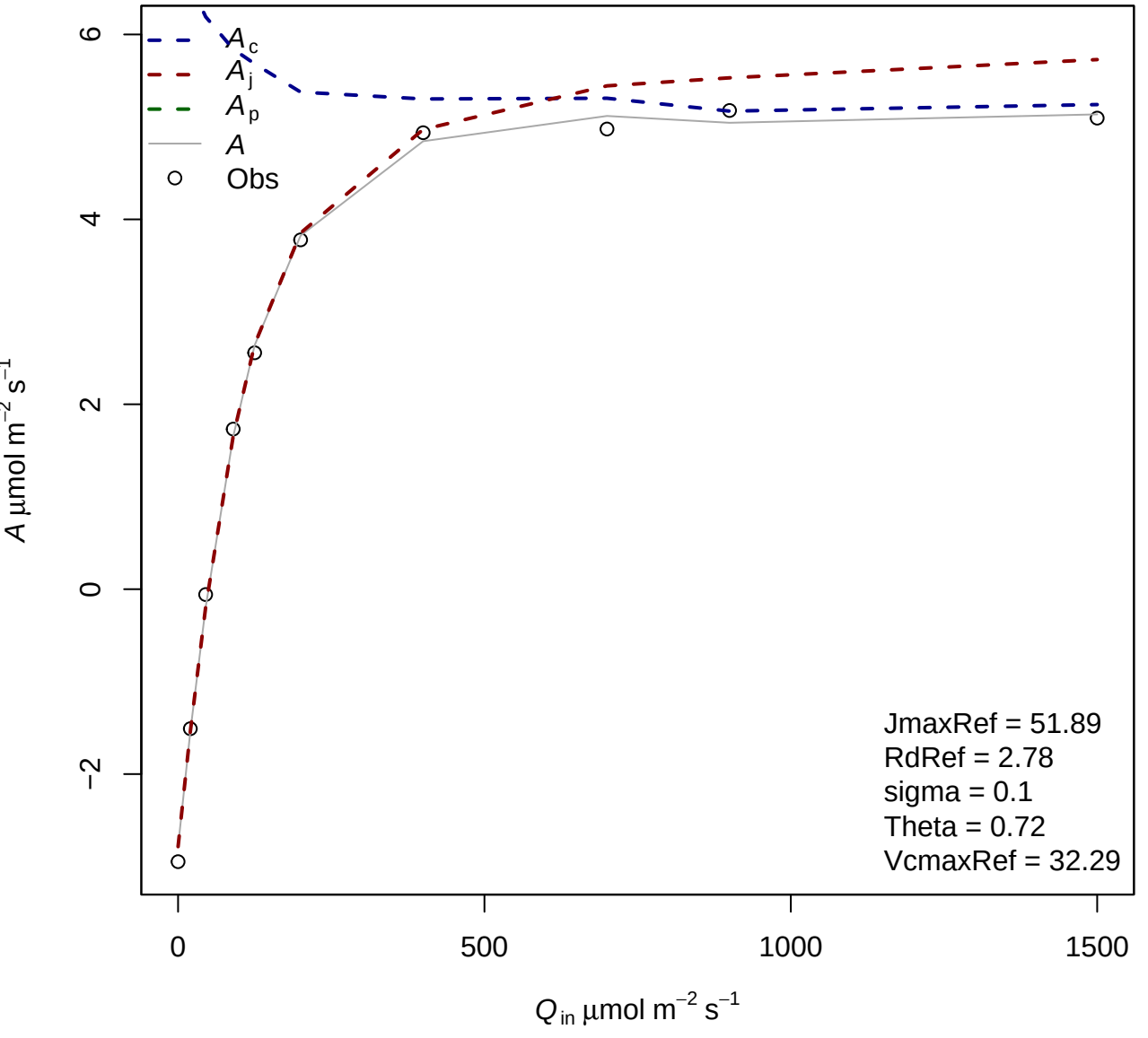

52
A_c
A_j
A_p
A
Obs
JmaxRef = 280645228.63
RdRef = 1.53
sigma = 0.16
Theta = −5907142.67
VcmaxRef = 59.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

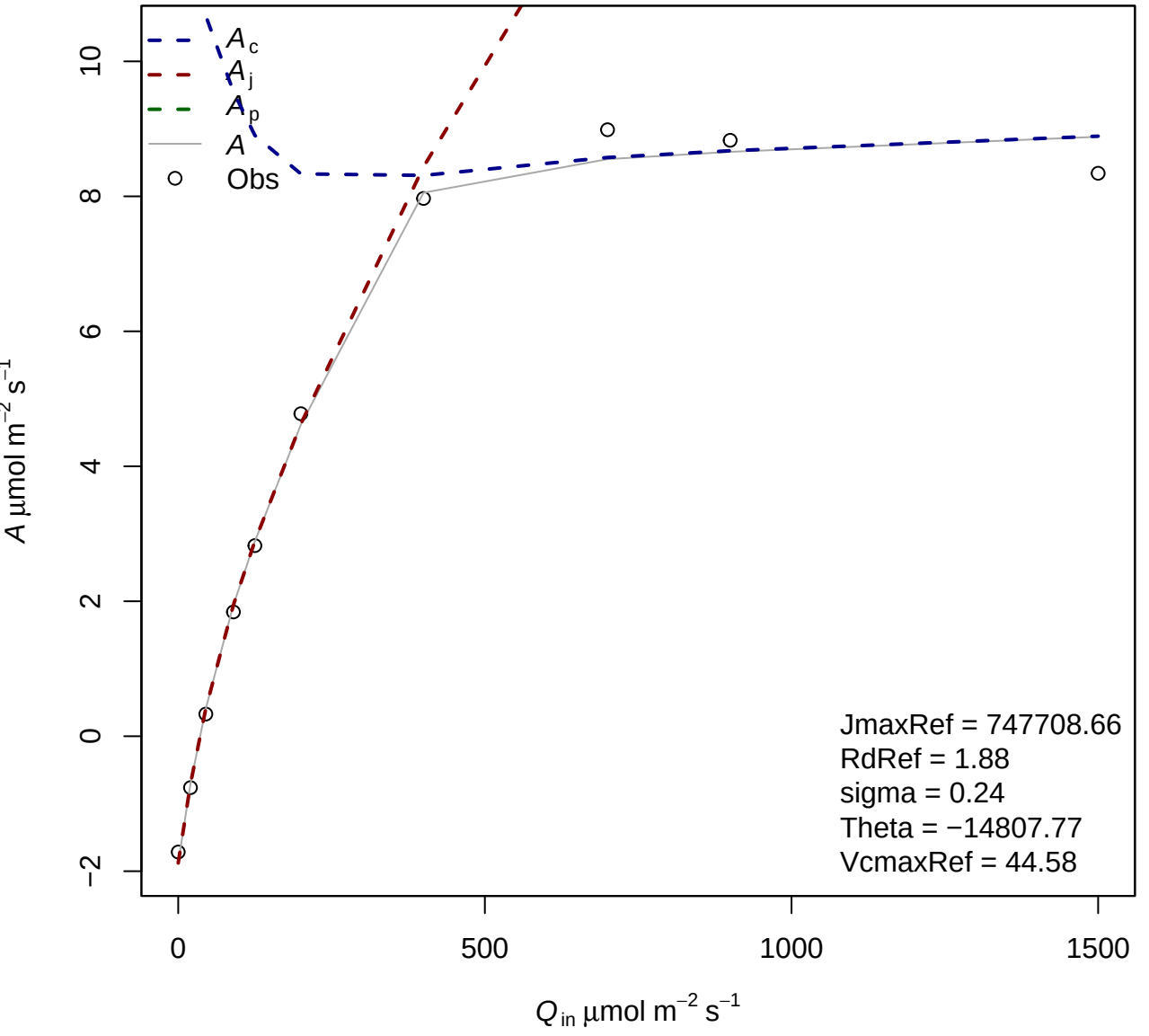

A_c
A_j
A_p
A
Obs
JmaxRef = 38.2
RdRef = 3.72
sigma = 0.12
Theta = 0.4
VcmaxRef = 243.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
500
1000
1500

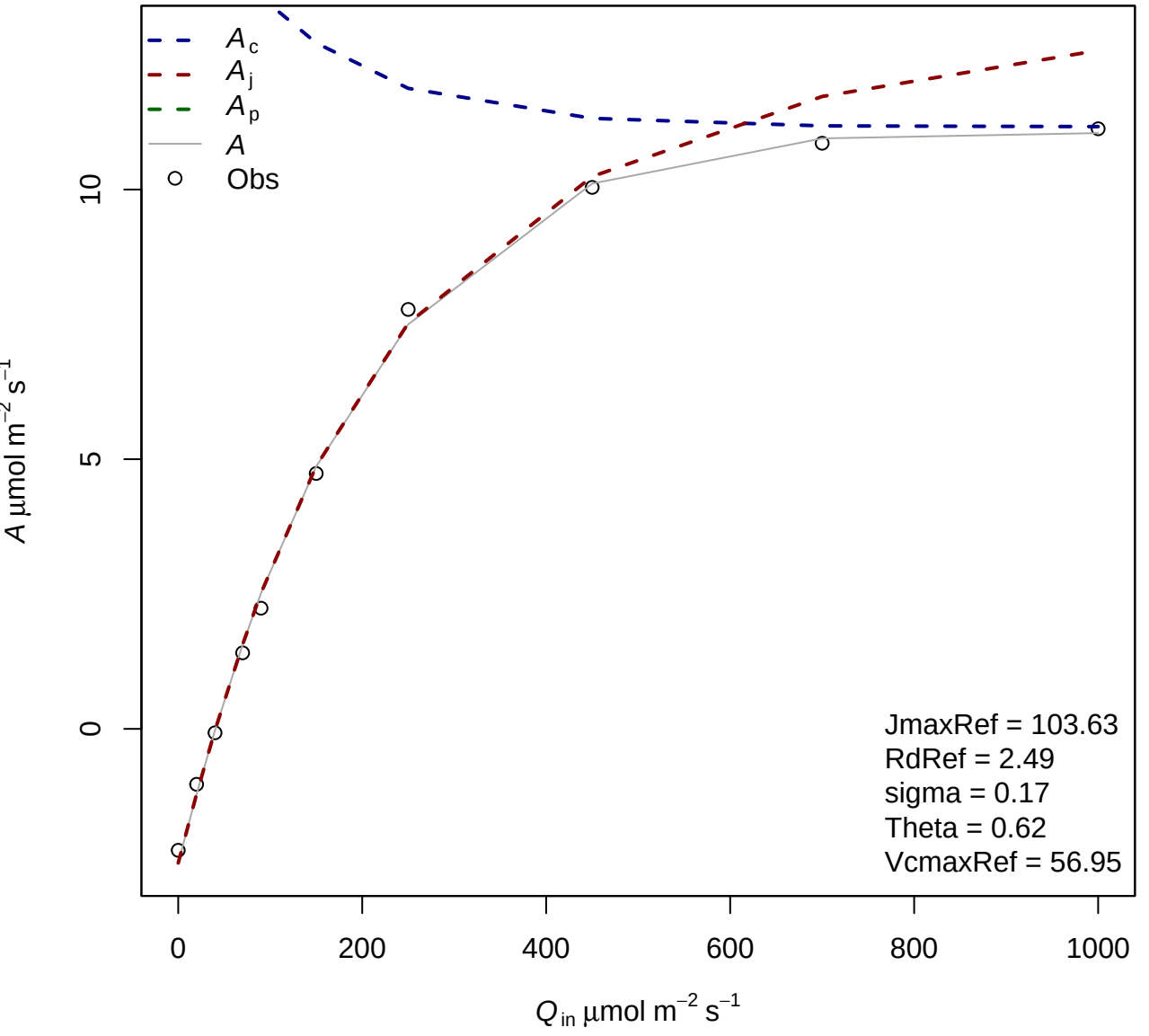

A_c
A_j
A_p
A
Obs
JmaxRef = 70.18
RdRef = 1.75
sigma = 0.17
Theta = 0.3
VcmaxRef = 42.74
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

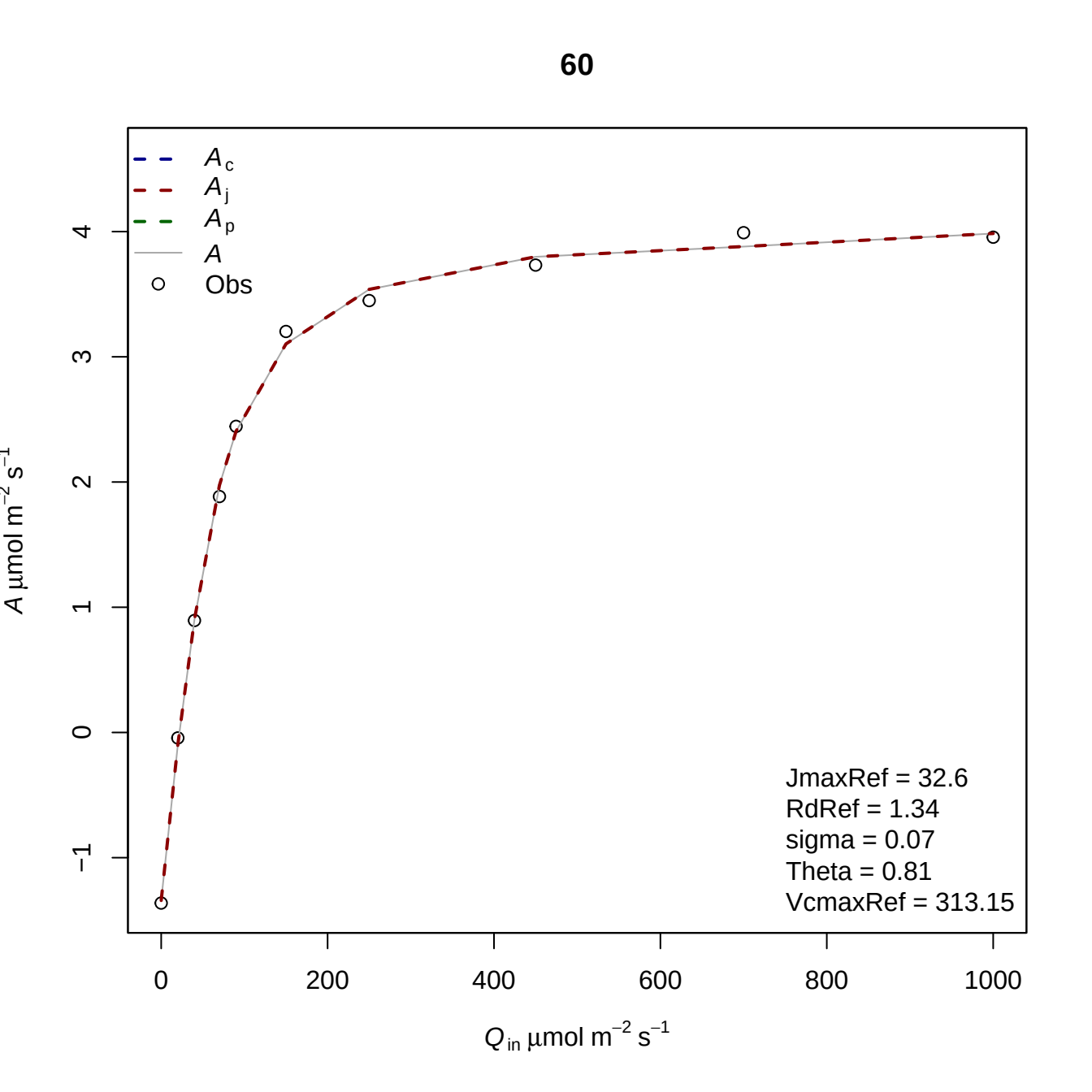

A_c
A_j
A_p
A
Obs
JmaxRef = 80.36
RdRef = 2.38
sigma = 0.09
Theta = 0.1
VcmaxRef = 33.51
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

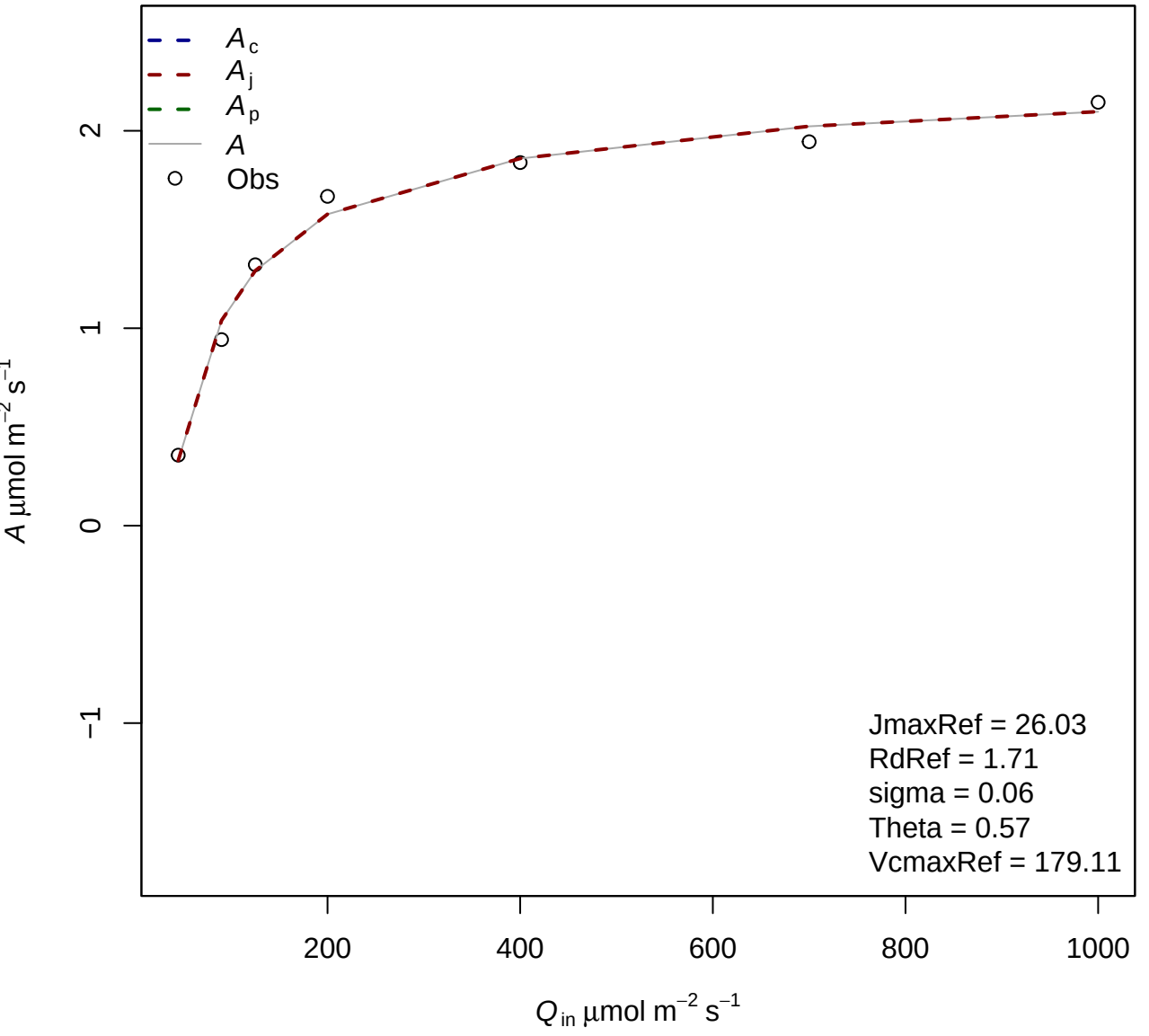

A_c
A_j
A_p
A
Obs
JmaxRef = 158.85
RdRef = 2.46
sigma = 0.07
Theta = −0.74
VcmaxRef = 37.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

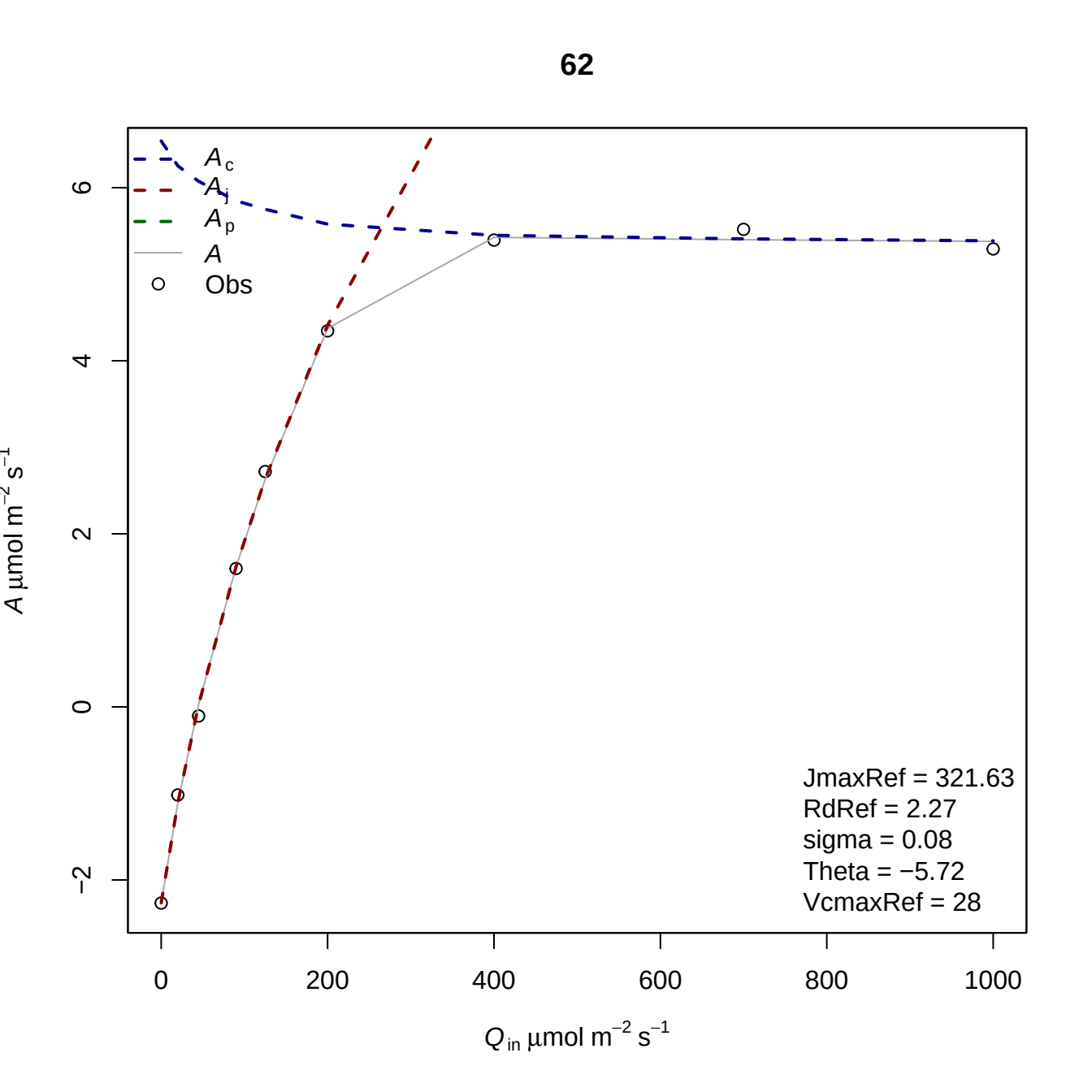

A_c
A_j
A_p
A
Obs
JmaxRef = 51.89
RdRef = 2.78
sigma = 0.1
Theta = 0.72
VcmaxRef = 32.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

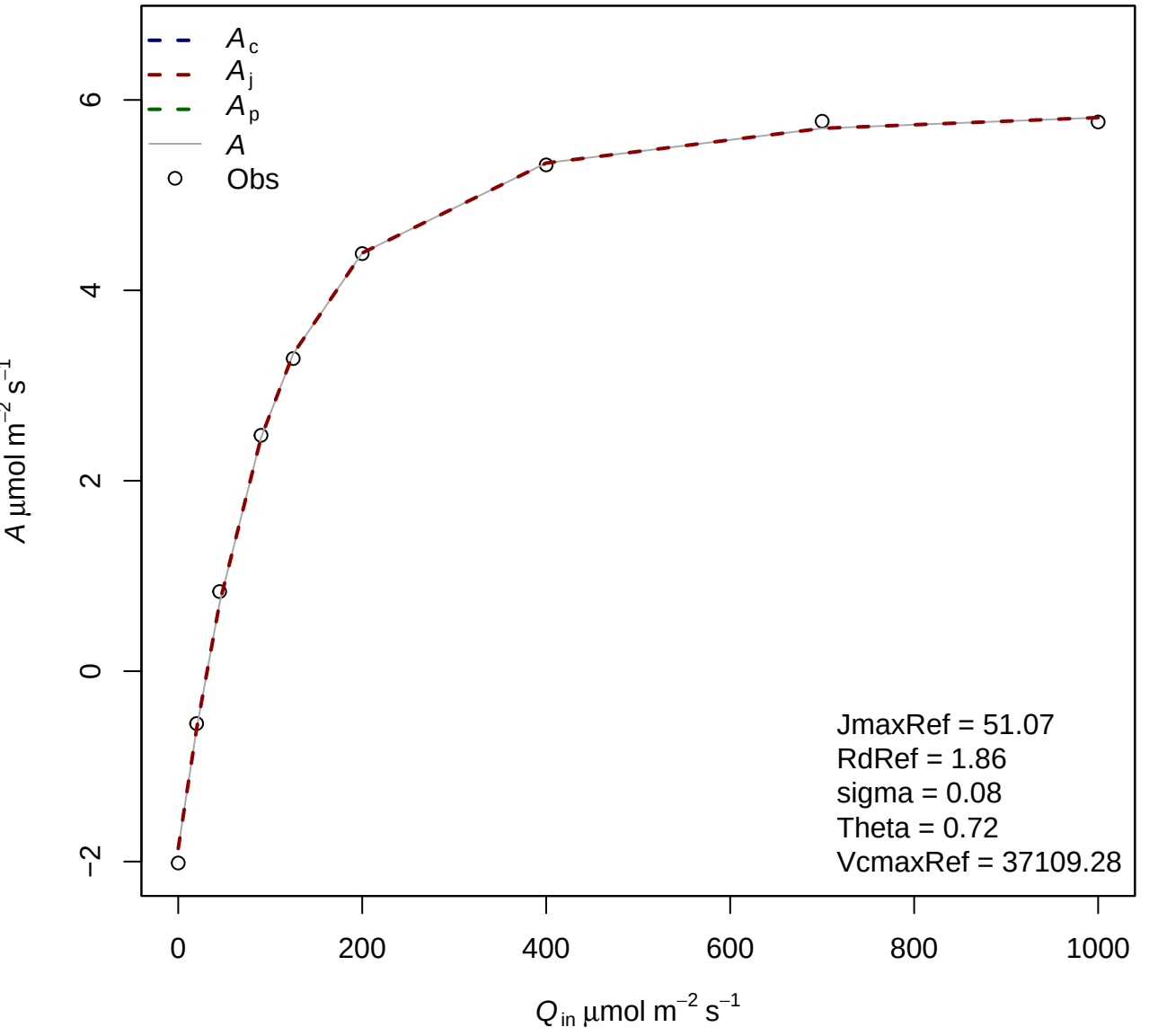

A_c
A_j
A_p
A
Obs
JmaxRef = 747708.66
RdRef = 1.88
sigma = 0.24
Theta = −14807.77
VcmaxRef = 44.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

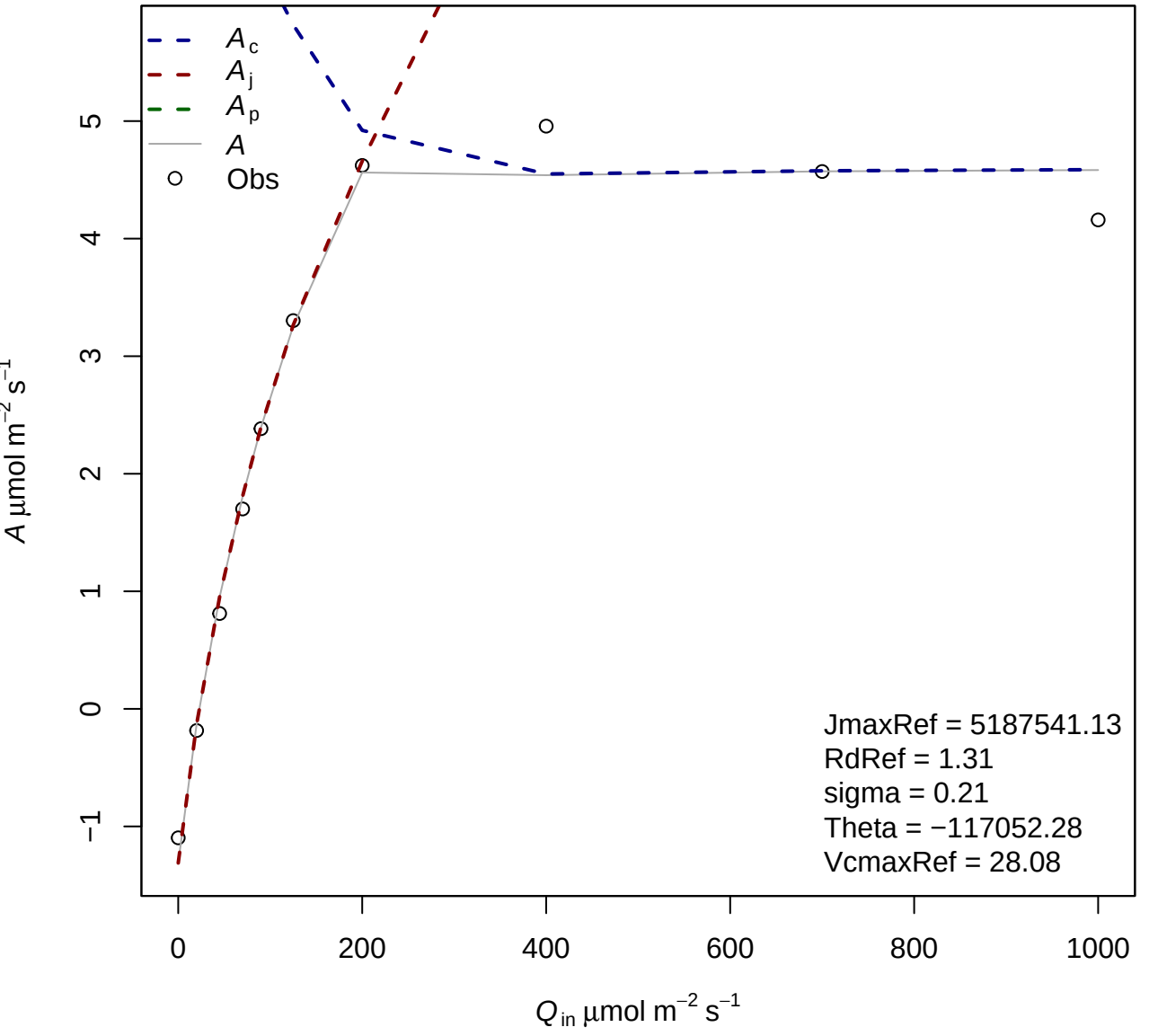

A_c
A_j
A_p
A
Obs
JmaxRef = 103.63
RdRef = 2.49
sigma = 0.17
Theta = 0.62
VcmaxRef = 56.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

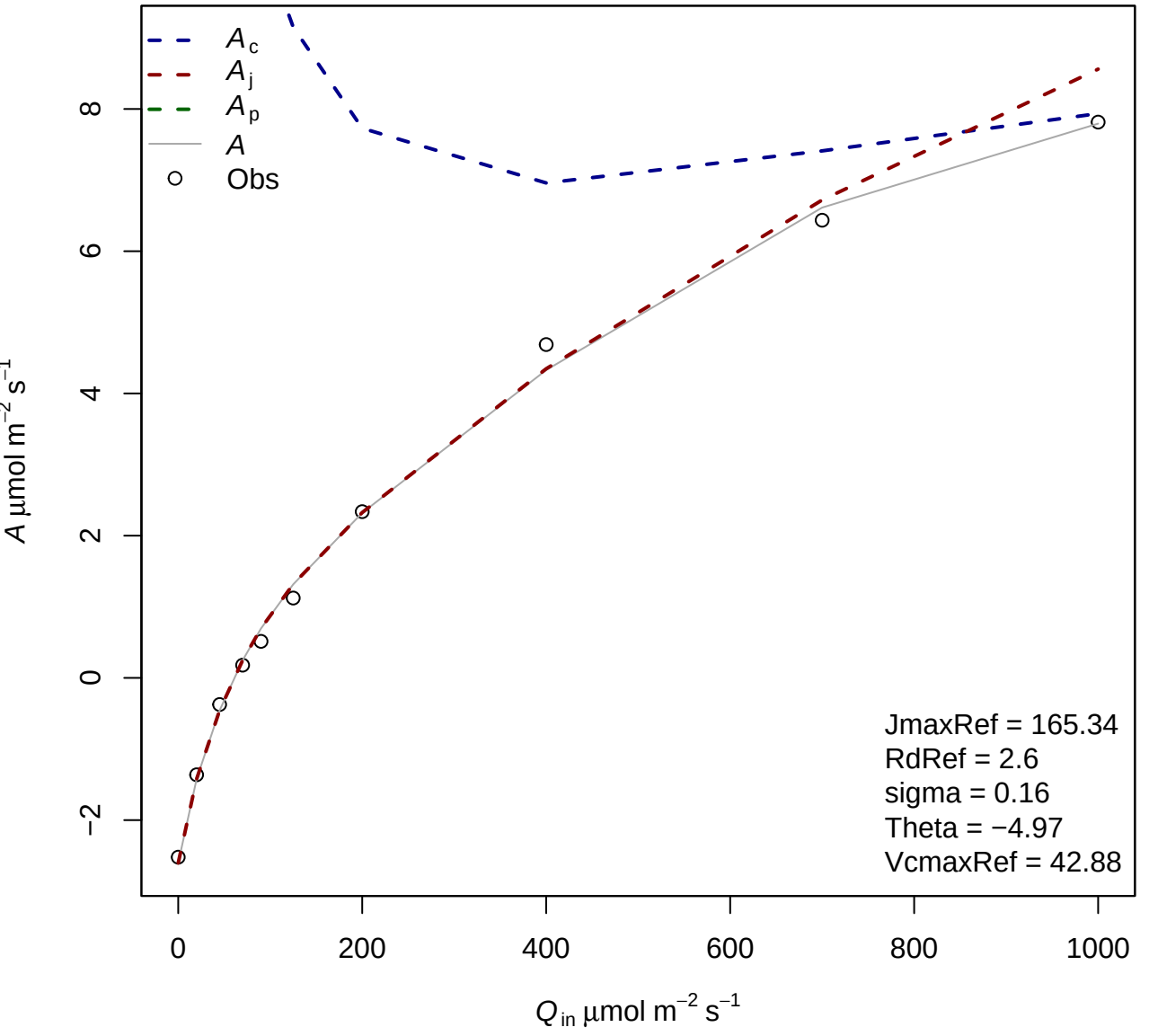

60
A_c
A_j
A_p
A
Obs
JmaxRef = 32.6
RdRef = 1.34
sigma = 0.07
Theta = 0.81
VcmaxRef = 313.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

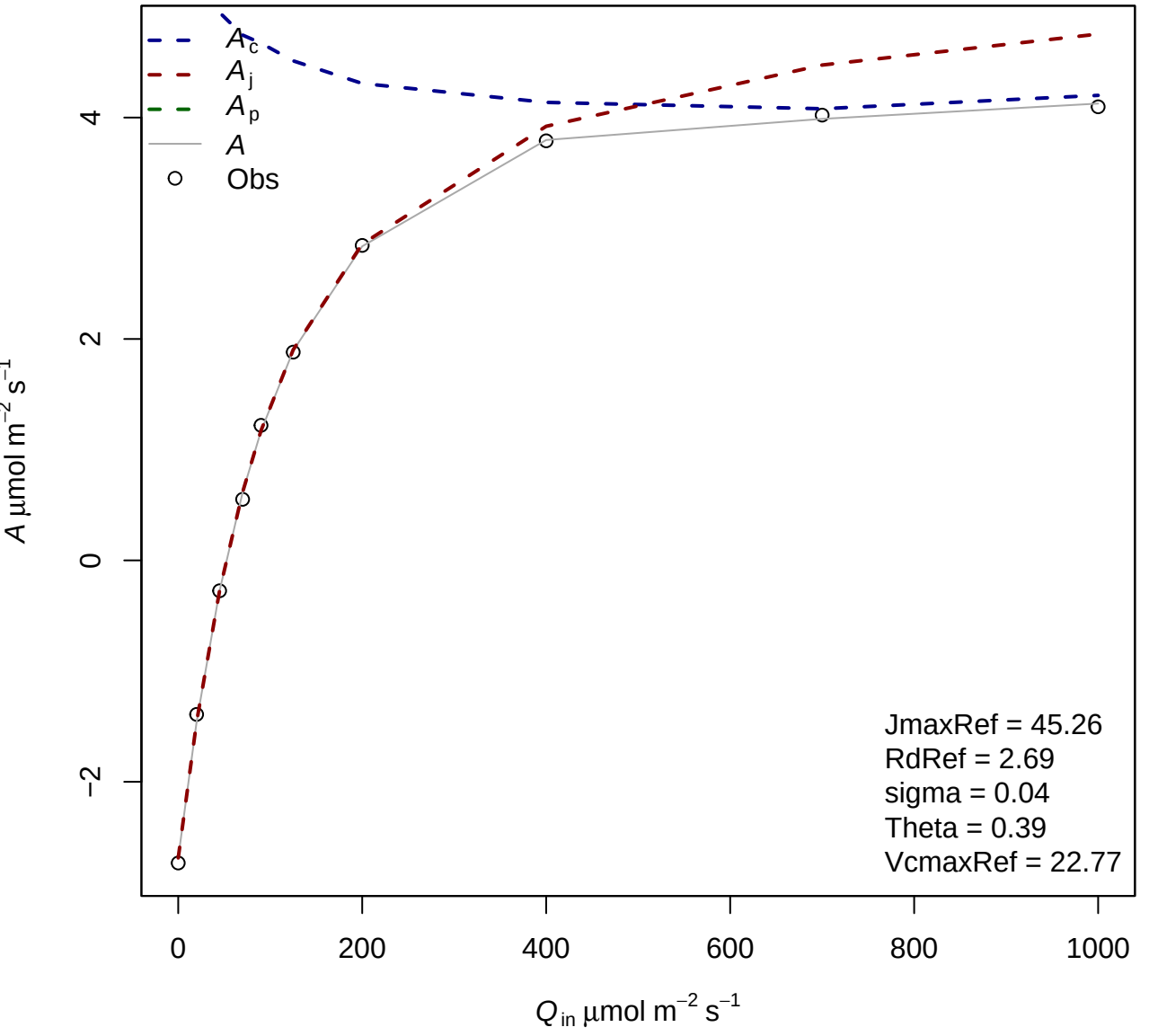

A_c
A_j
A_p
A
Obs
JmaxRef = 26.03
RdRef = 1.71
sigma = 0.06
Theta = 0.57
VcmaxRef = 179.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
200
400
600
800
1000

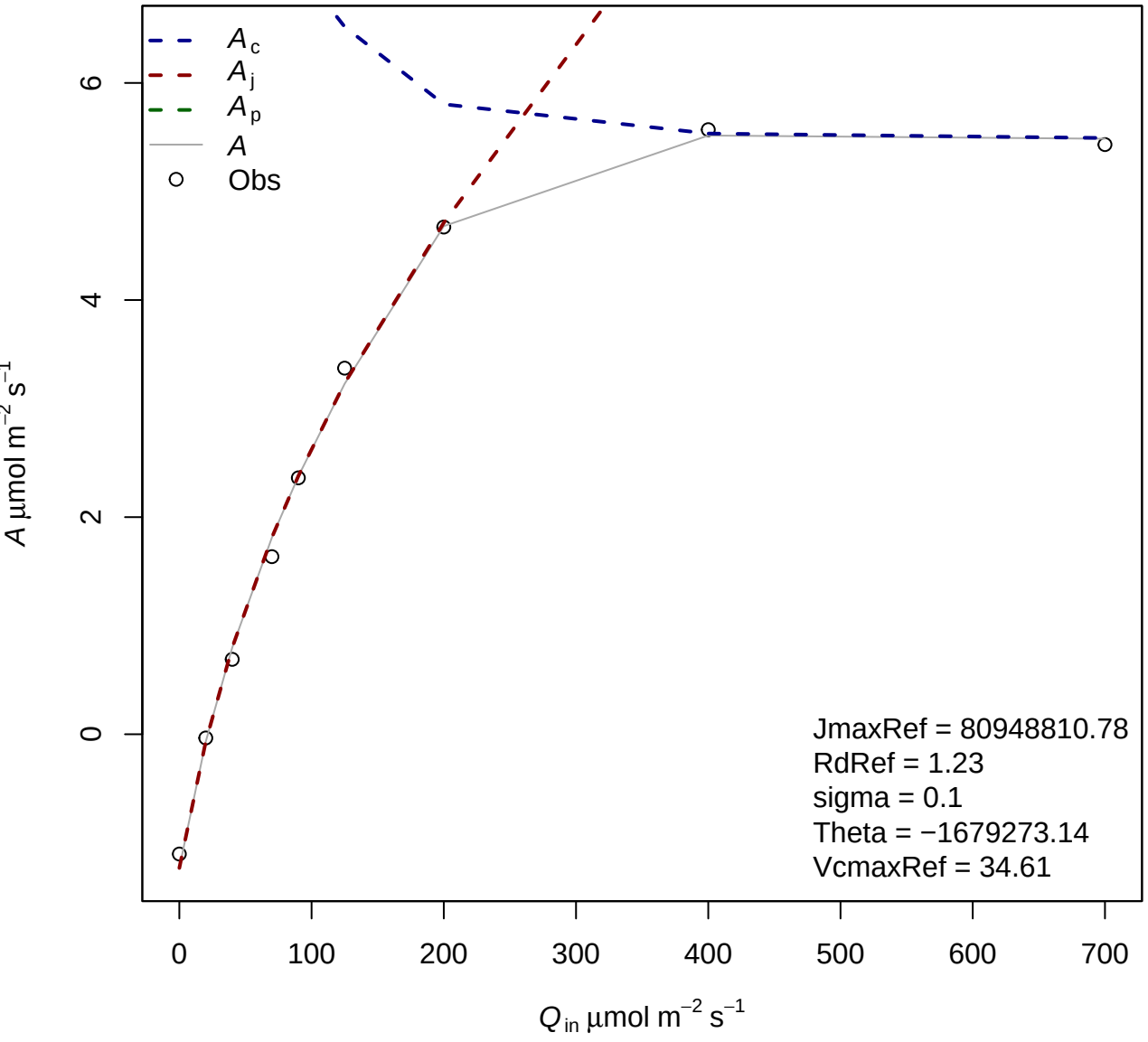

62
A_c
A_j
A_p
A
Obs
JmaxRef = 321.63
RdRef = 2.27
sigma = 0.08
Theta = −5.72
VcmaxRef = 28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

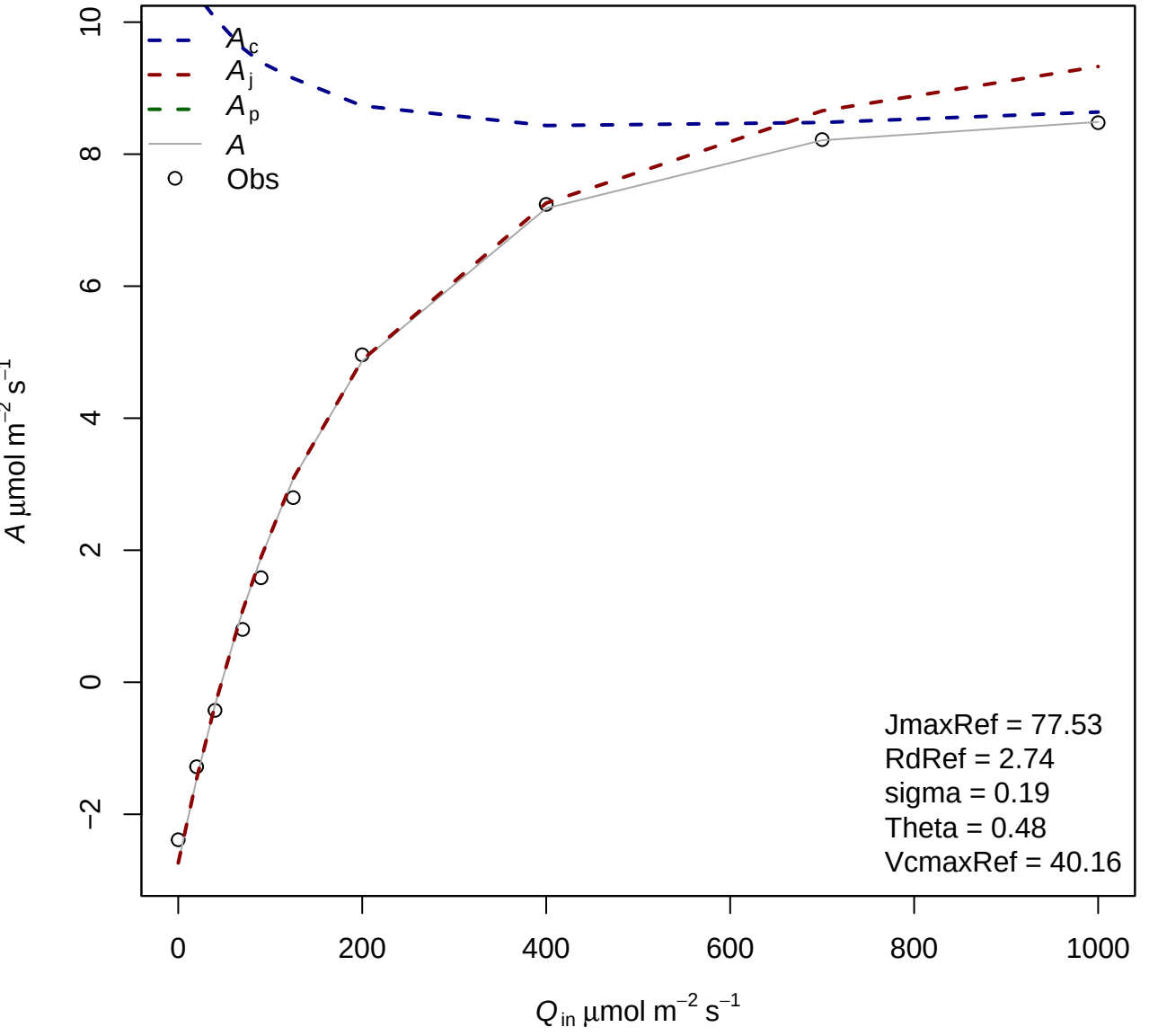

A_c
A_j
A_p
A
Obs
JmaxRef = 51.07
RdRef = 1.86
sigma = 0.08
Theta = 0.72
VcmaxRef = 37109.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

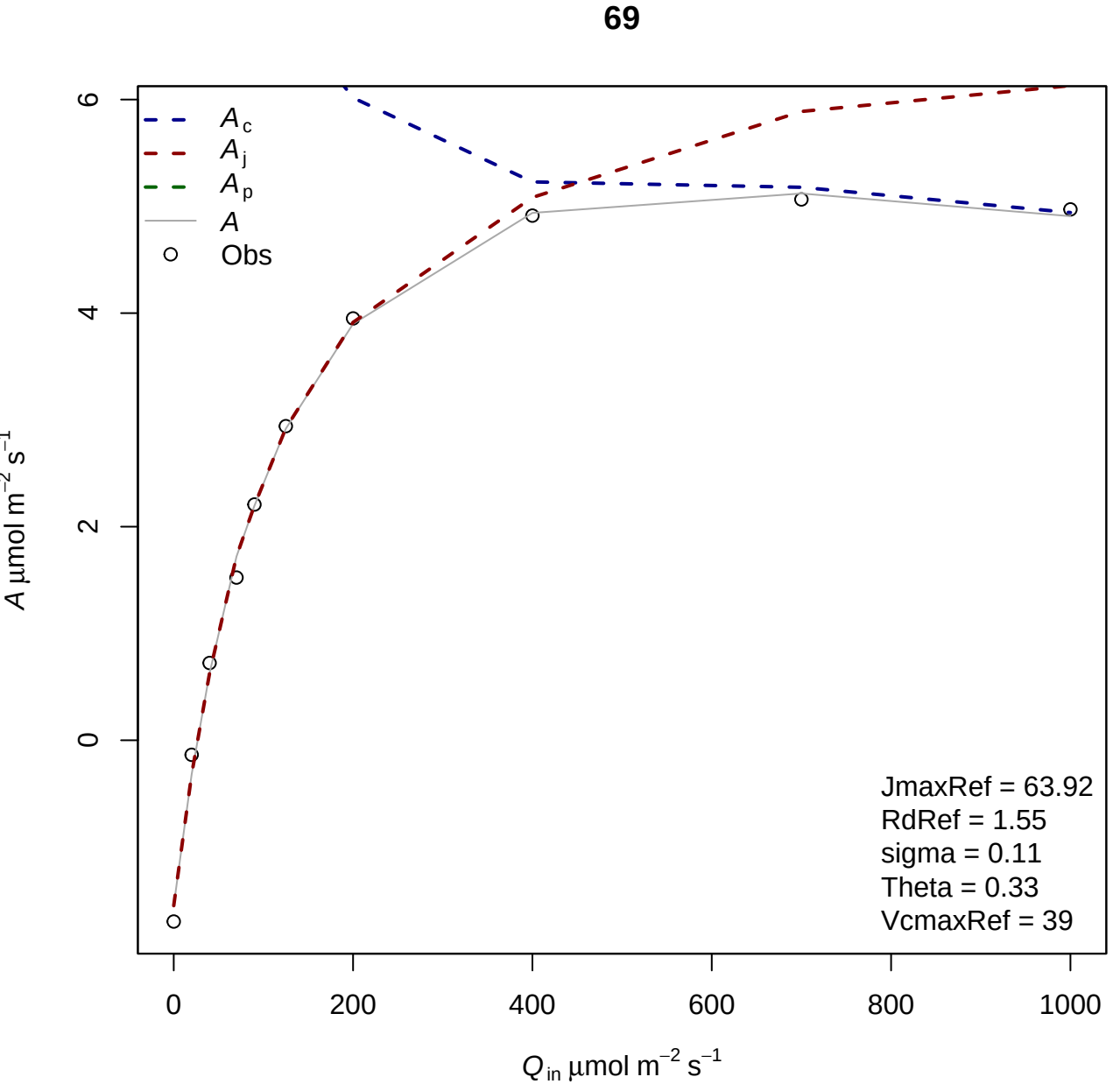

A_c
A_j
A_p
A
Obs
JmaxRef = 5187541.13
RdRef = 1.31
sigma = 0.21
Theta = −117052.28
VcmaxRef = 28.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

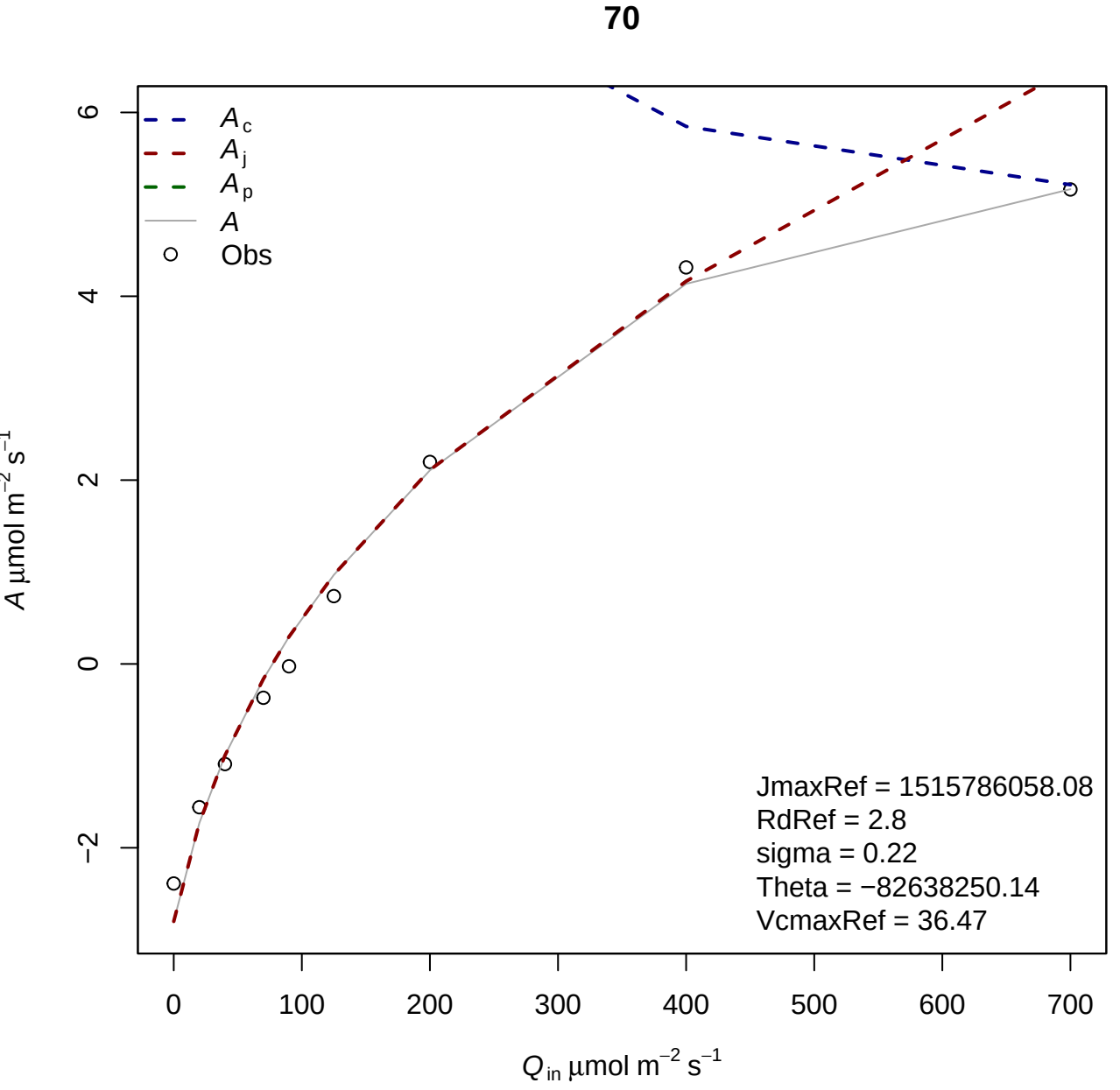

A_c
A_j
A_p
A
Obs
JmaxRef = 165.34
RdRef = 2.6
sigma = 0.16
Theta = −4.97
VcmaxRef = 42.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

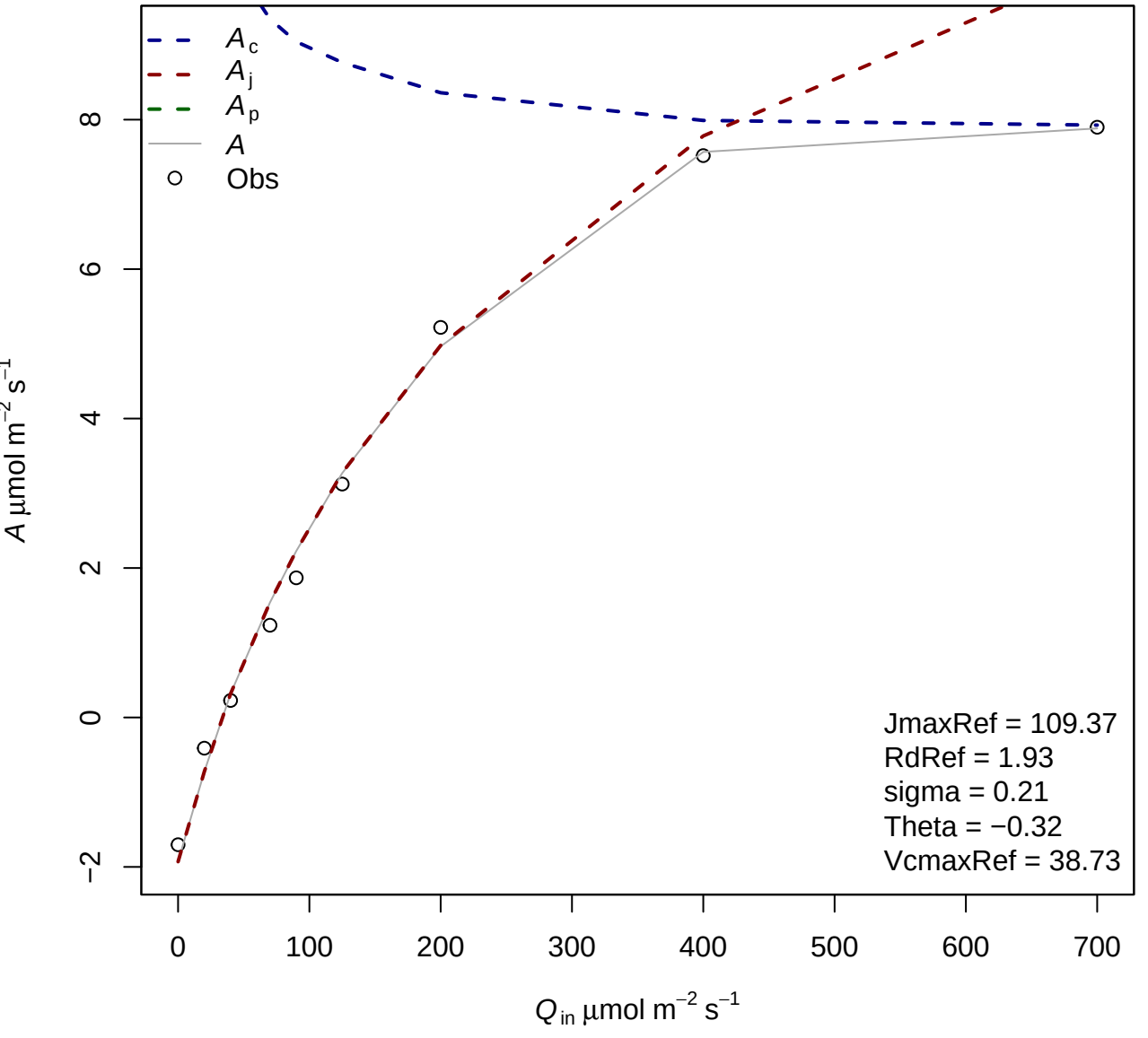

A_c
A_j
A_p
A
Obs
JmaxRef = 45.26
RdRef = 2.69
sigma = 0.04
Theta = 0.39
VcmaxRef = 22.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

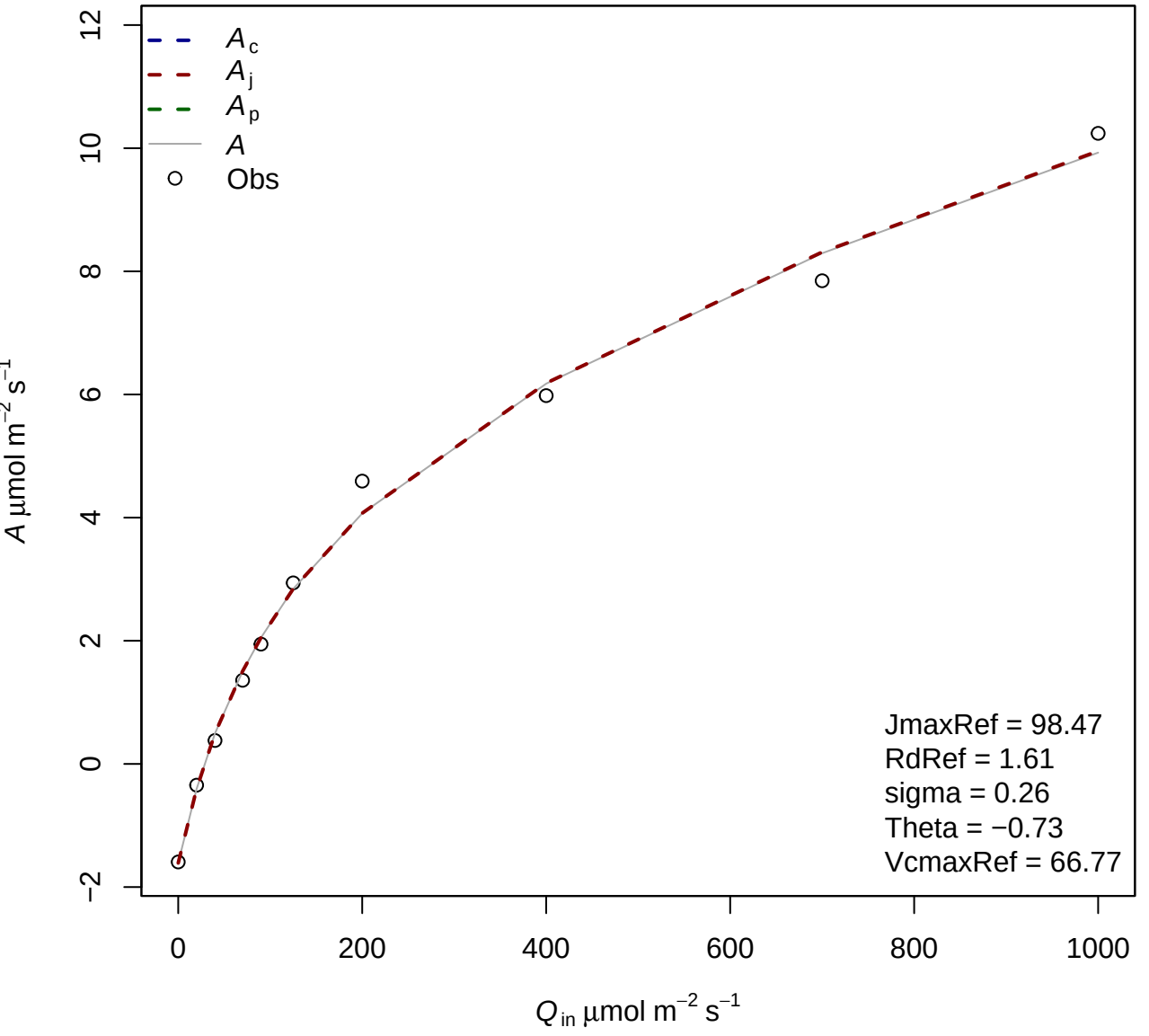

A_c
A_j
A_p
A
Obs
JmaxRef = 80948810.78
RdRef = 1.23
sigma = 0.1
Theta = −1679273.14
VcmaxRef = 34.61
0
2
4
6
0
100
200
300
400
500
600
700
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

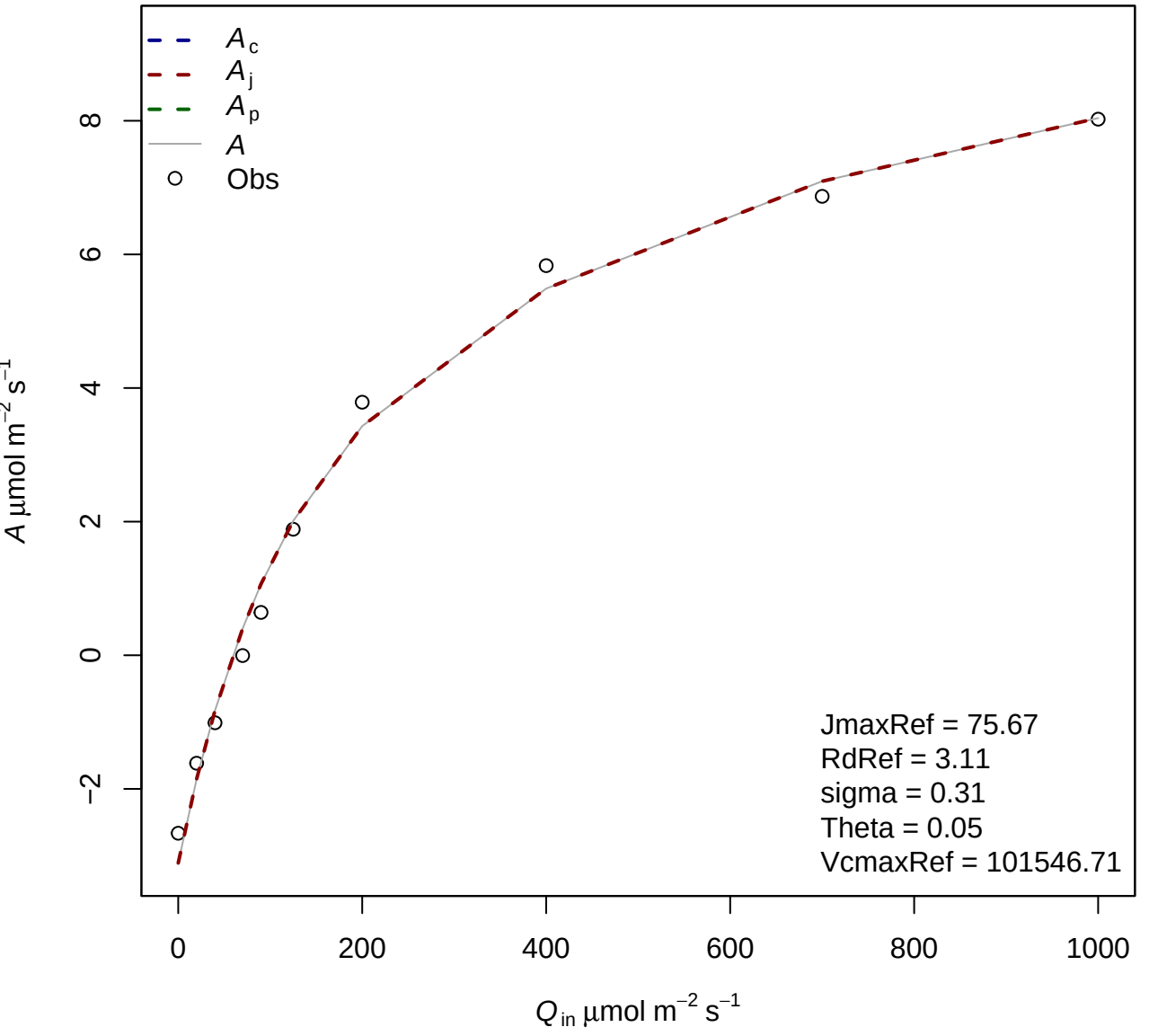

A_c
A_j
A_p
A
Obs
JmaxRef = 77.53
RdRef = 2.74
sigma = 0.19
Theta = 0.48
VcmaxRef = 40.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

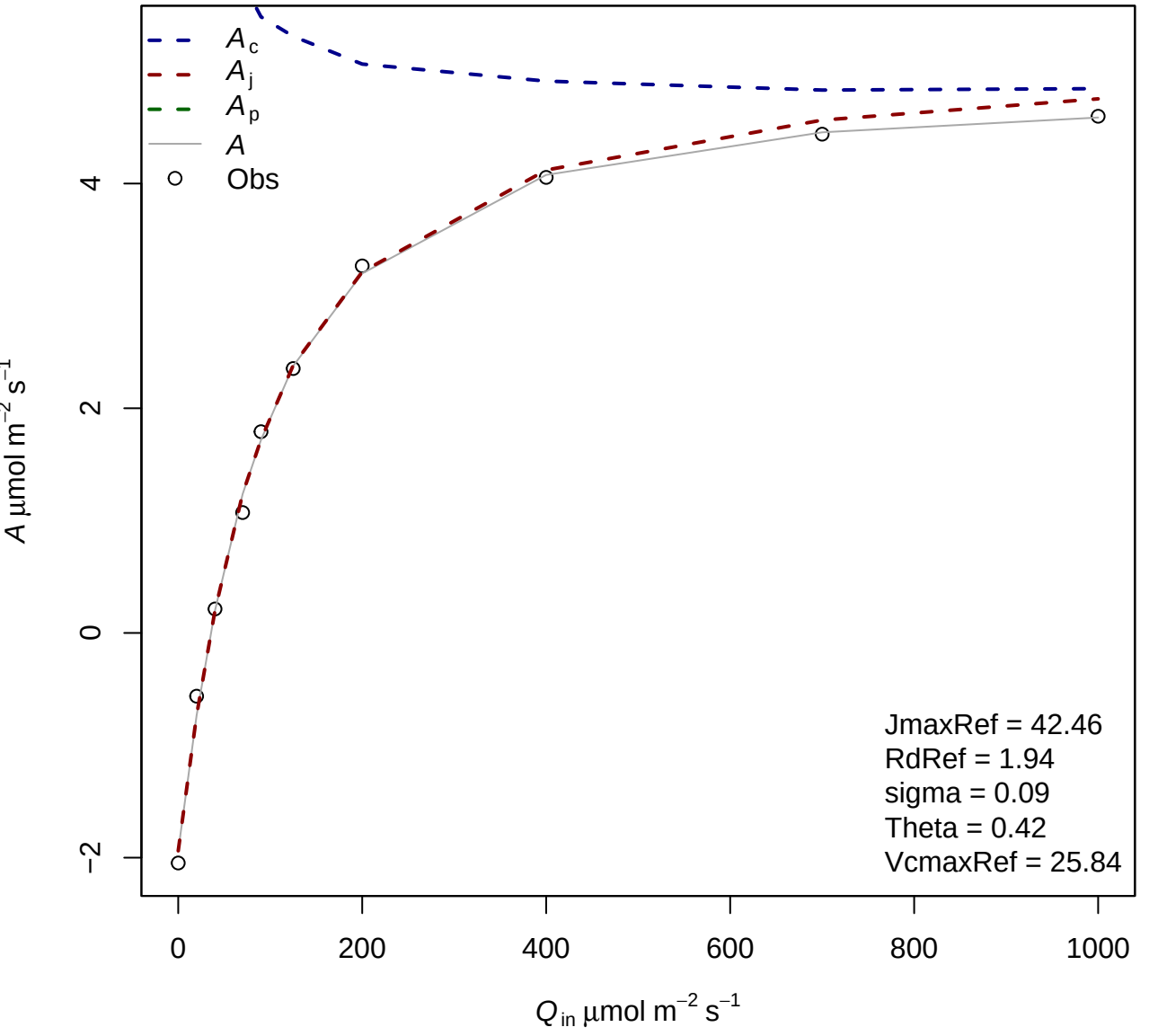

69
A_c
A_j
A_p
A
Obs
JmaxRef = 63.92
RdRef = 1.55
sigma = 0.11
Theta = 0.33
VcmaxRef = 39
0
2
4
6
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

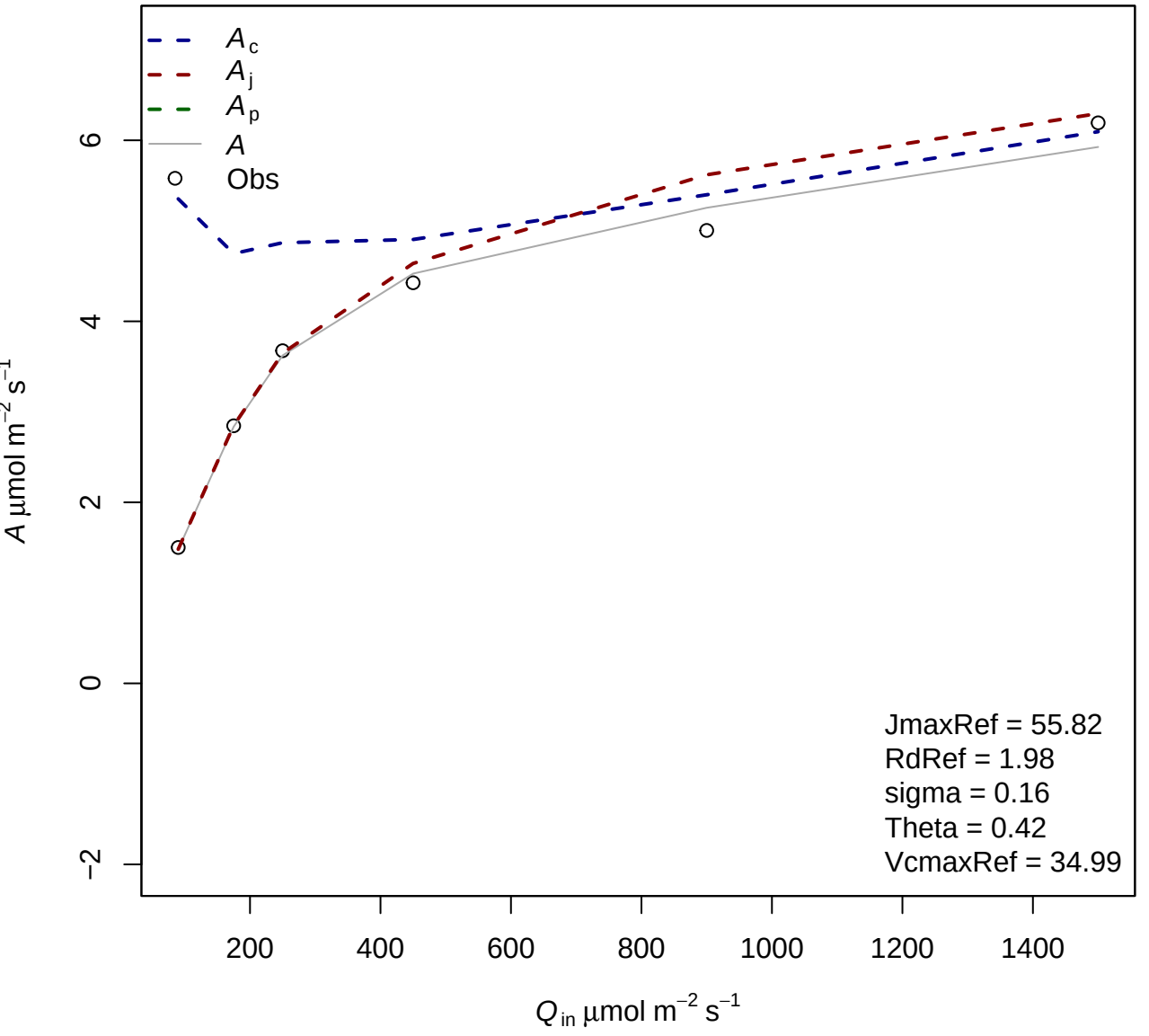

70
A_c
A_j
A_p
A
Obs
JmaxRef = 1515786058.08
RdRef = 2.8
sigma = 0.22
Theta = −82638250.14
VcmaxRef = 36.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
100
200
300
400
500
600
700

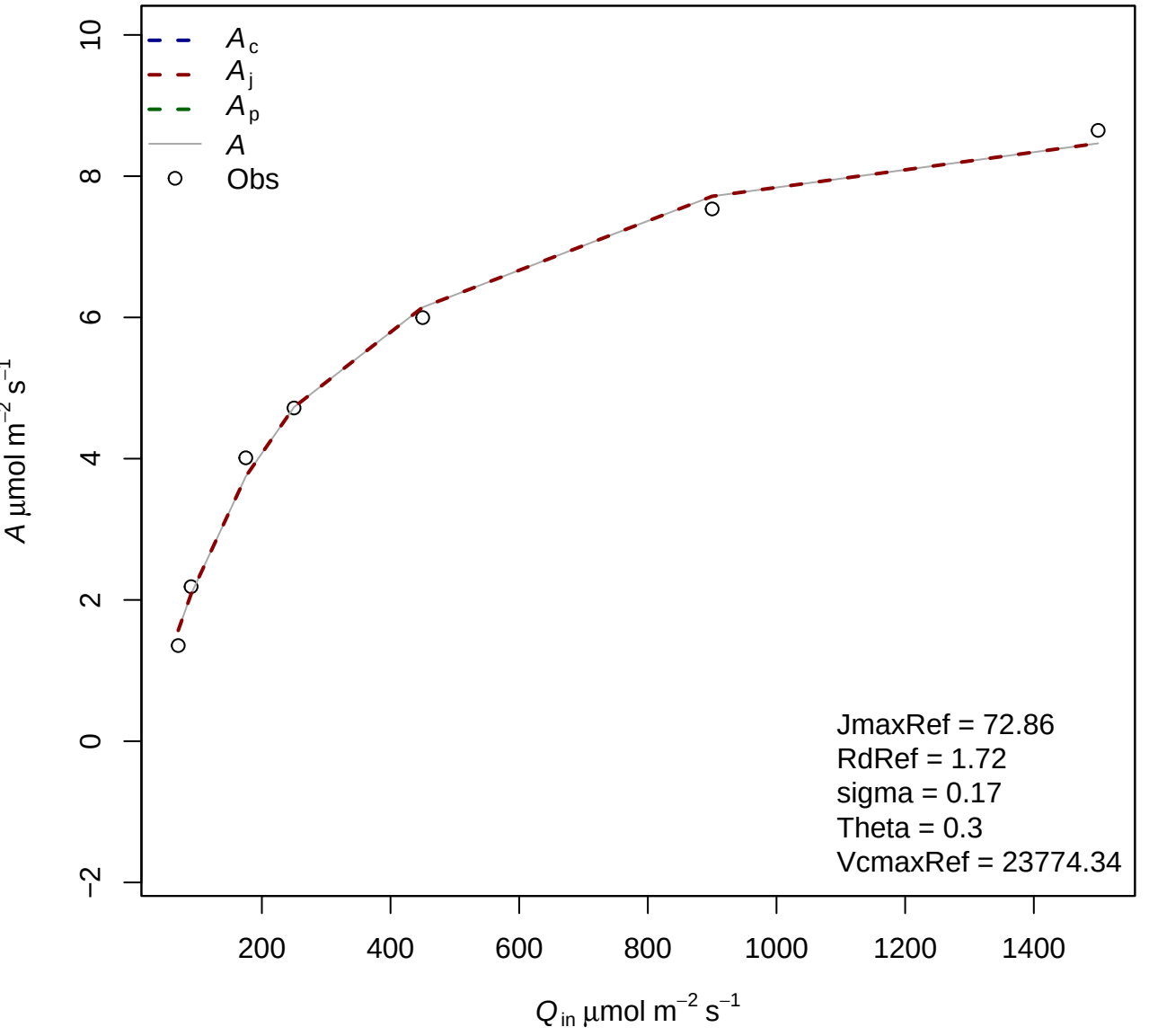

A_c
A_j
A_p
A
Obs
JmaxRef = 109.37
RdRef = 1.93
sigma = 0.21
Theta = −0.32
VcmaxRef = 38.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

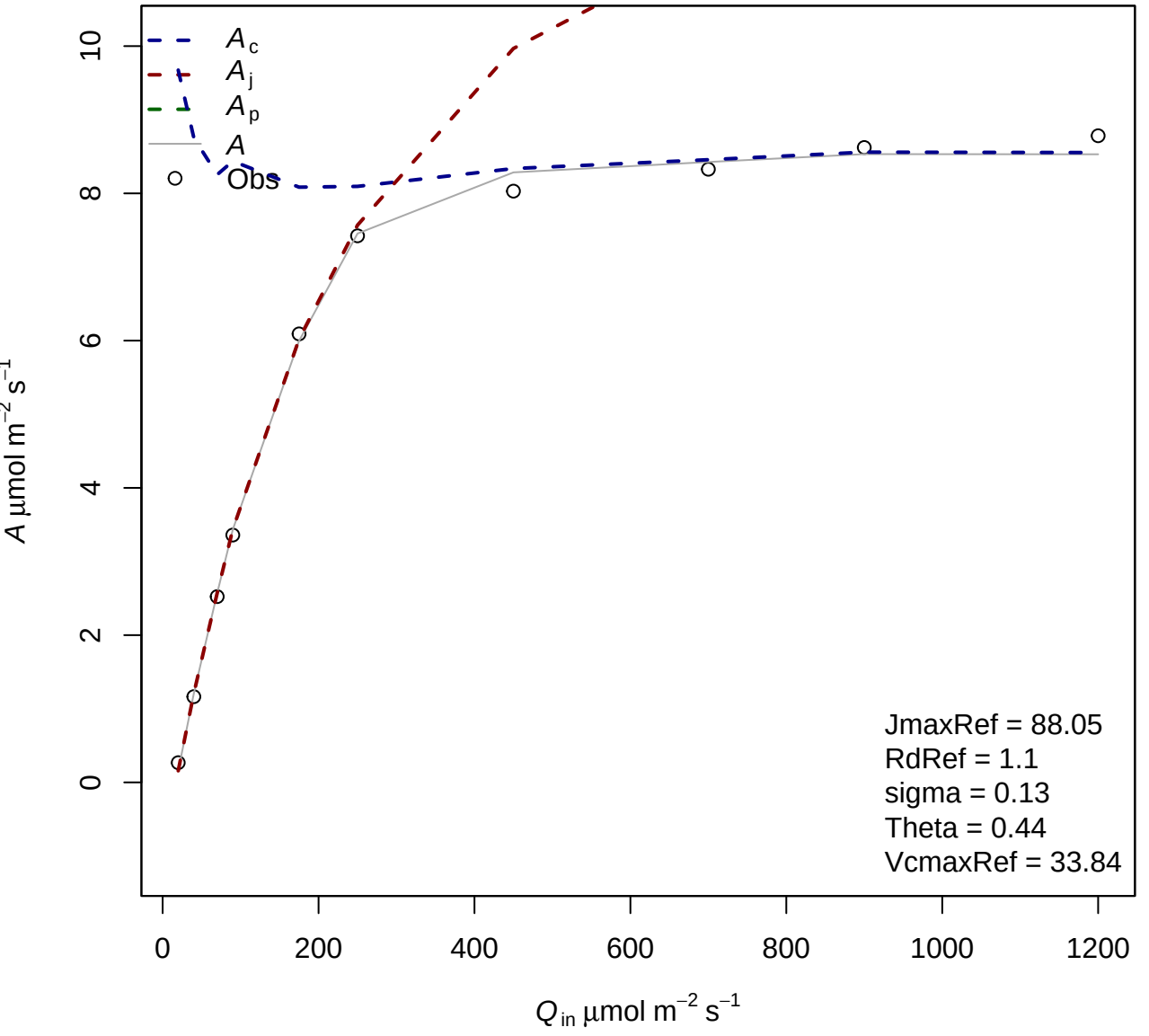

A_c
A_j
A_p
A
Obs
JmaxRef = 98.47
RdRef = 1.61
sigma = 0.26
Theta = −0.73
VcmaxRef = 66.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

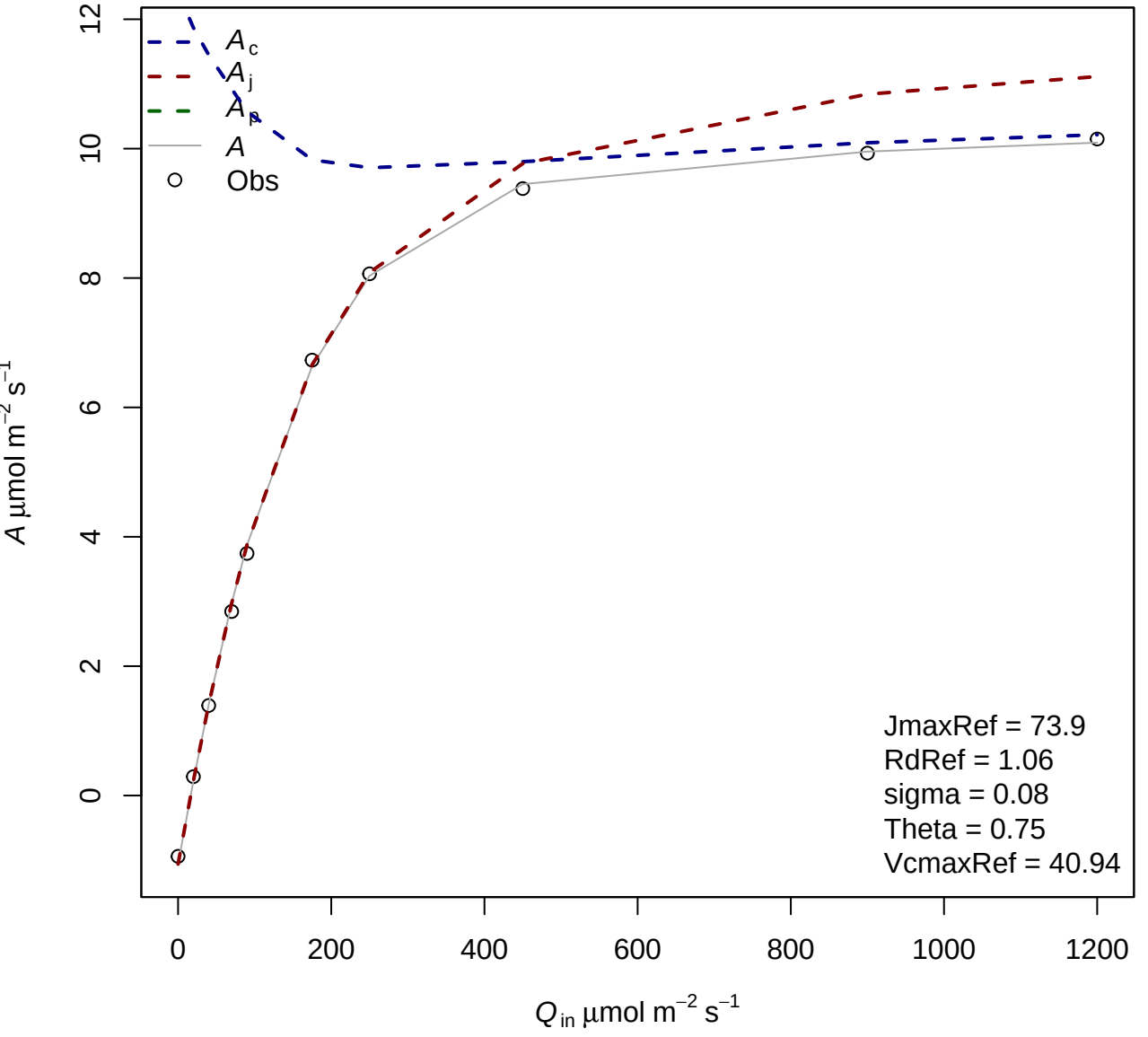

A_c
A_j
A_p
A
Obs
JmaxRef = 75.67
RdRef = 3.11
sigma = 0.31
Theta = 0.05
VcmaxRef = 101546.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

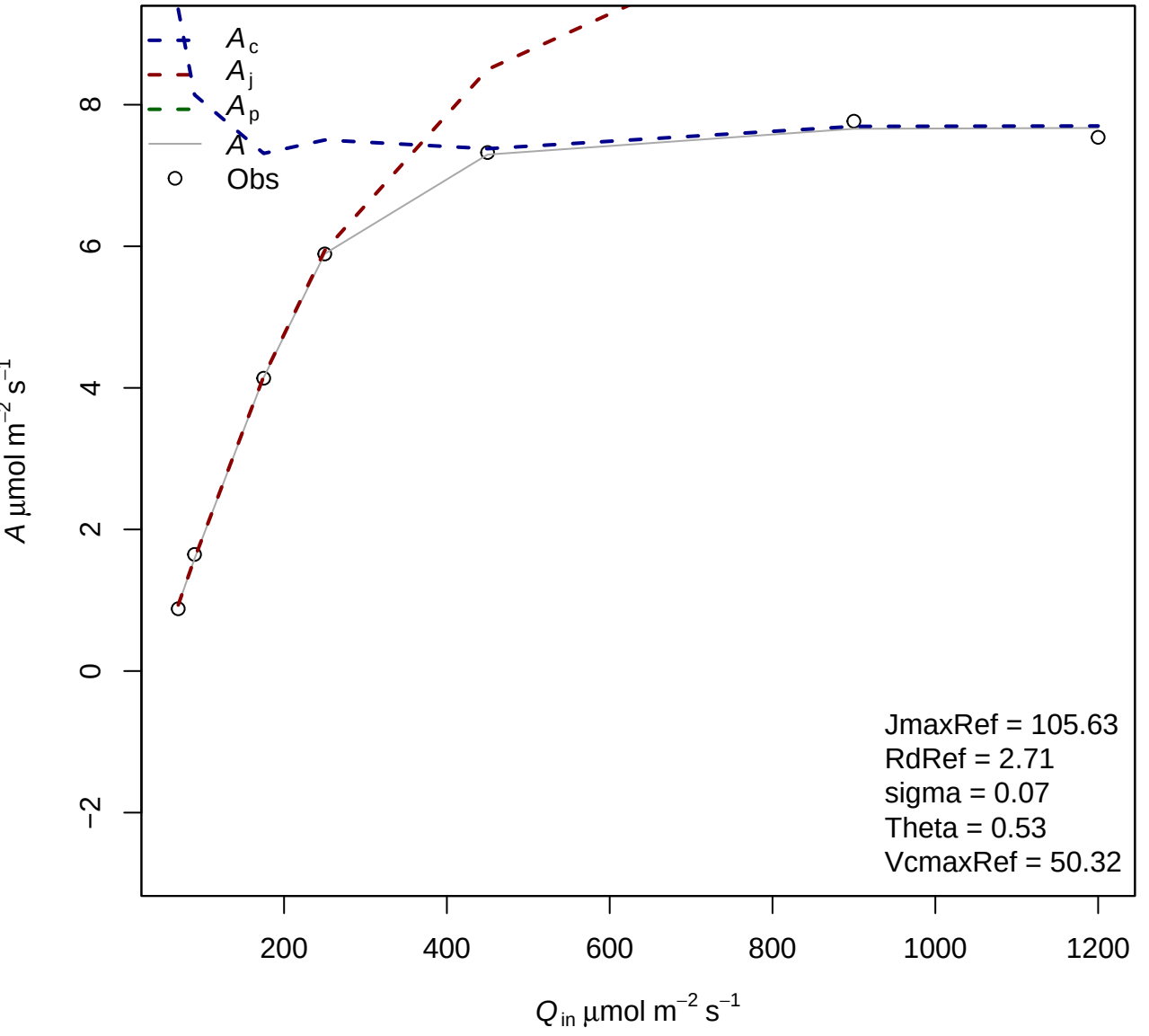

A_c
A_j
A_p
A
Obs
JmaxRef = 42.46
RdRef = 1.94
sigma = 0.09
Theta = 0.42
VcmaxRef = 25.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

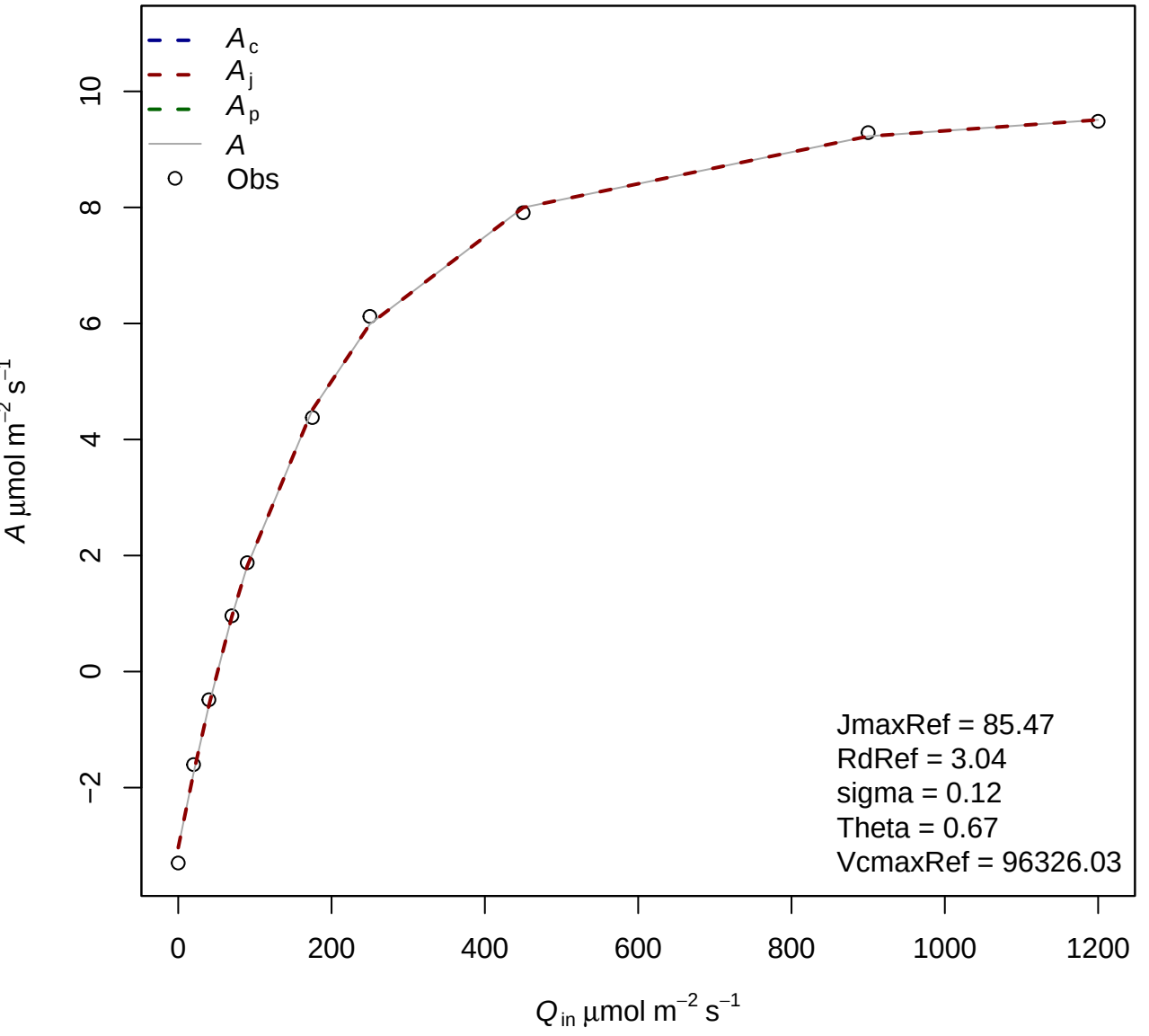

A_c
A_j
A_p
A
Obs
JmaxRef = 55.82
RdRef = 1.98
sigma = 0.16
Theta = 0.42
VcmaxRef = 34.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

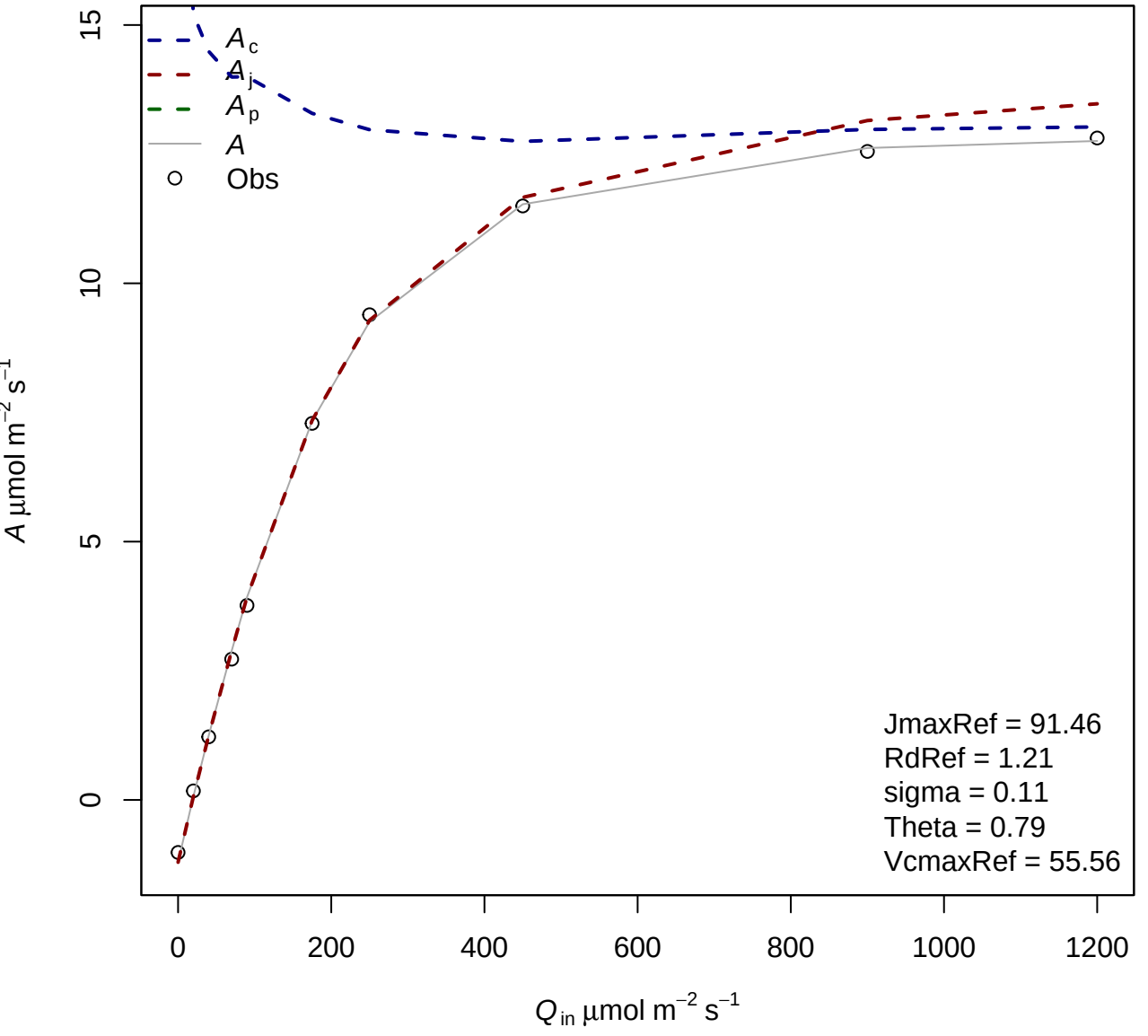

A_c
A_j
A_p
A
Obs
JmaxRef = 72.86
RdRef = 1.72
sigma = 0.17
Theta = 0.3
VcmaxRef = 23774.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

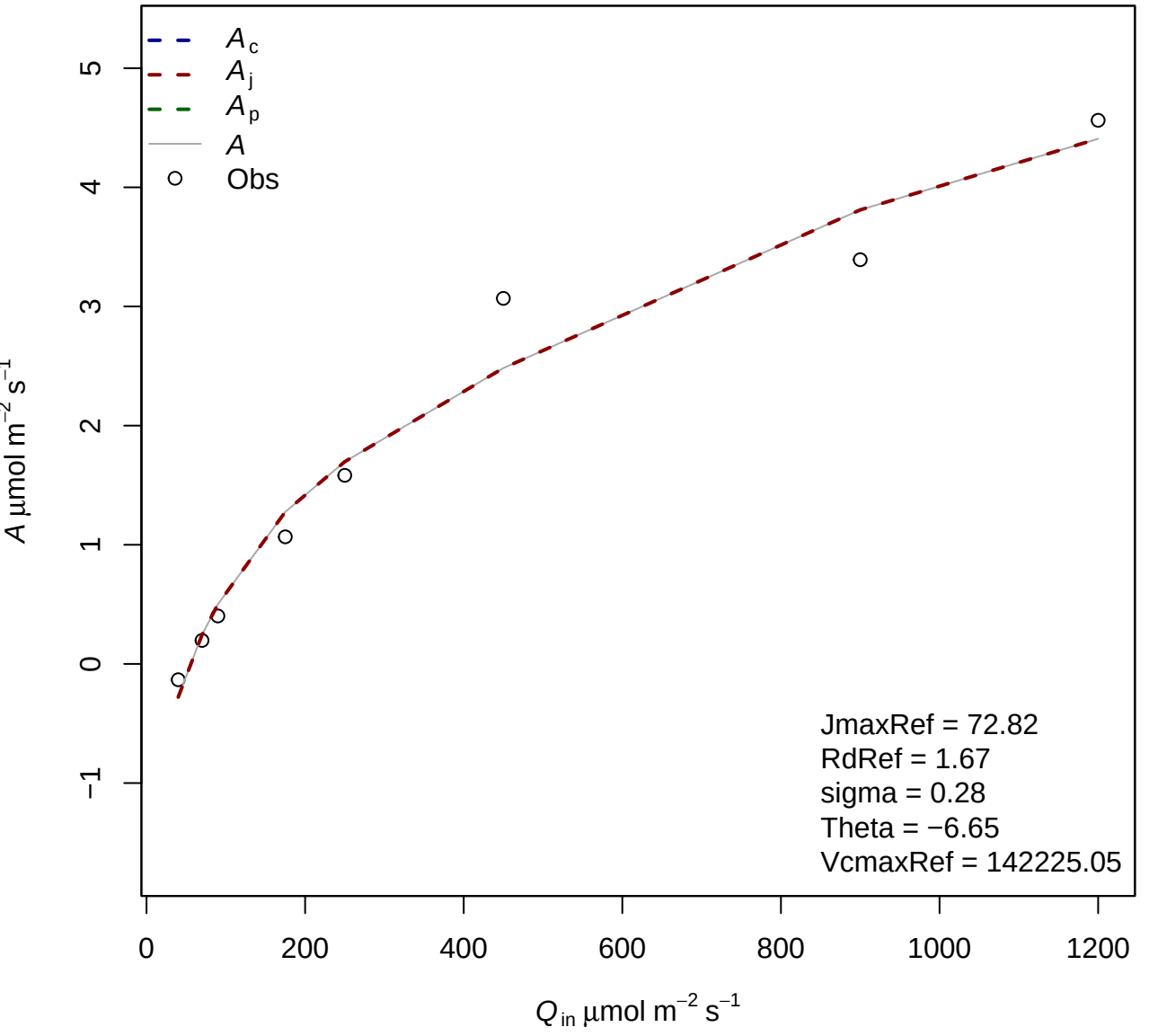

A_c
A_j
A_p
A
Obs
JmaxRef = 88.05
RdRef = 1.1
sigma = 0.13
Theta = 0.44
VcmaxRef = 33.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

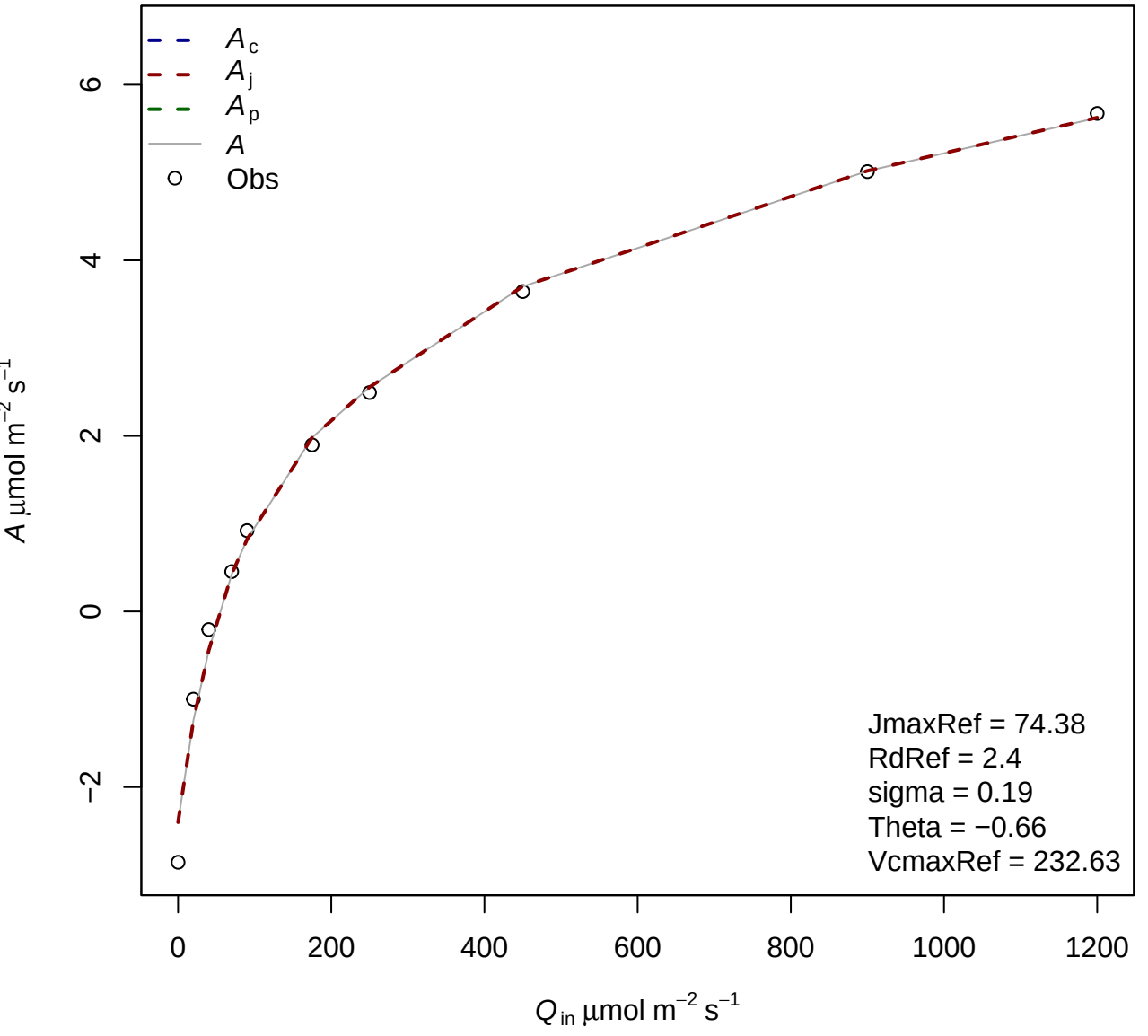

A_c
A_j
A_p
A
Obs
JmaxRef = 73.9
RdRef = 1.06
sigma = 0.08
Theta = 0.75
VcmaxRef = 40.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

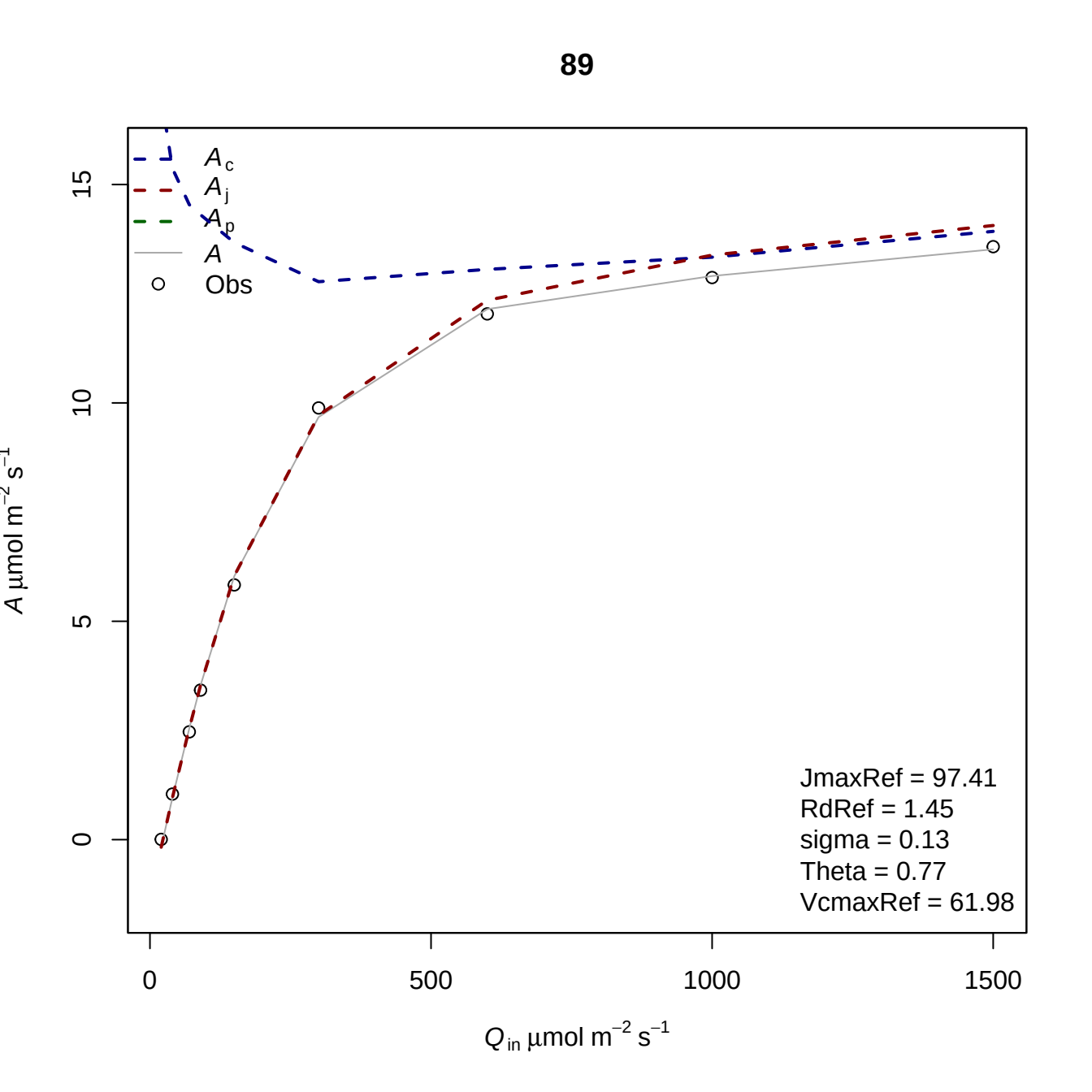

A_c
A_j
A_p
A
Obs
JmaxRef = 105.63
RdRef = 2.71
sigma = 0.07
Theta = 0.53
VcmaxRef = 50.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200

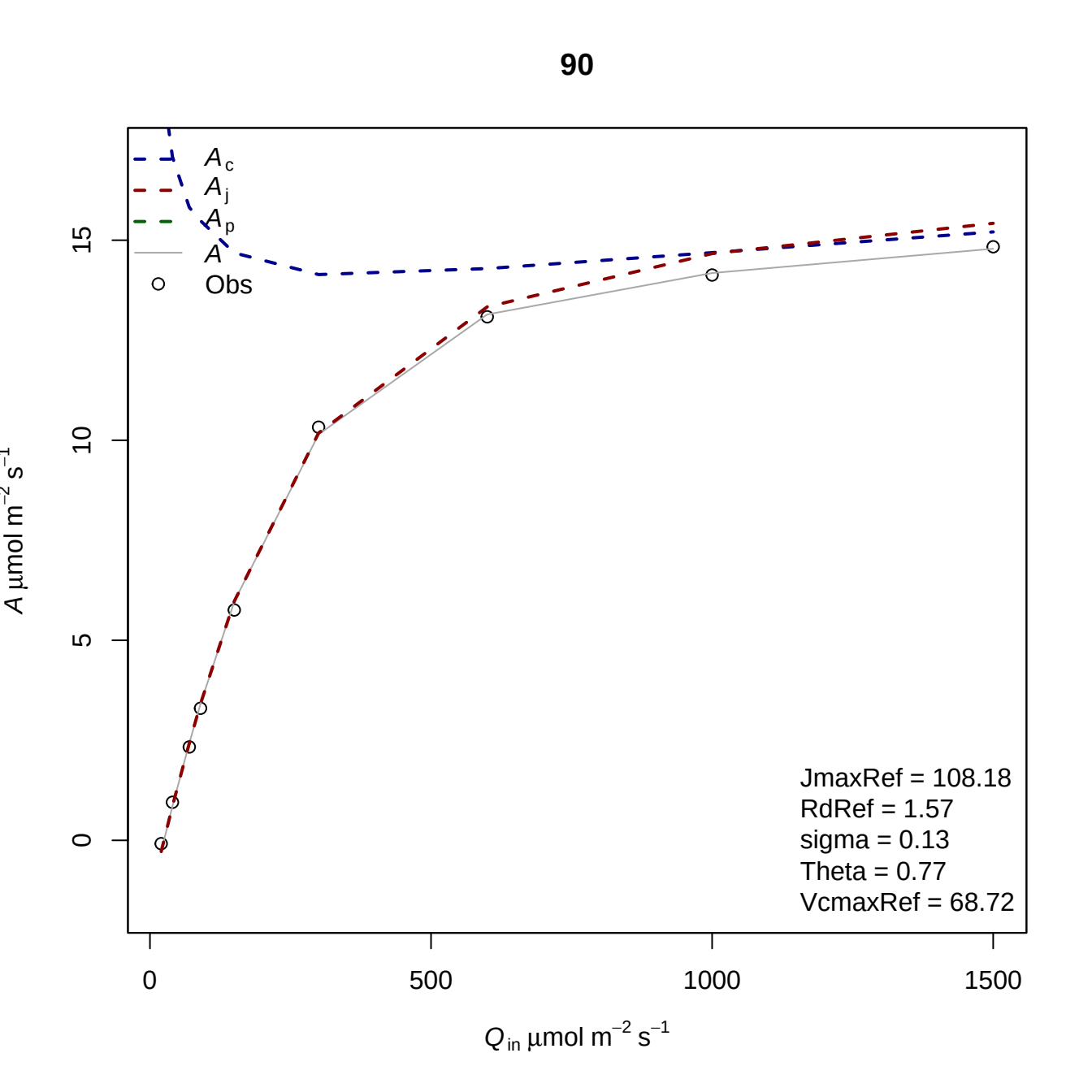

A_c
A_j
A_p
A
Obs
JmaxRef = 85.47
RdRef = 3.04
sigma = 0.12
Theta = 0.67
VcmaxRef = 96326.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

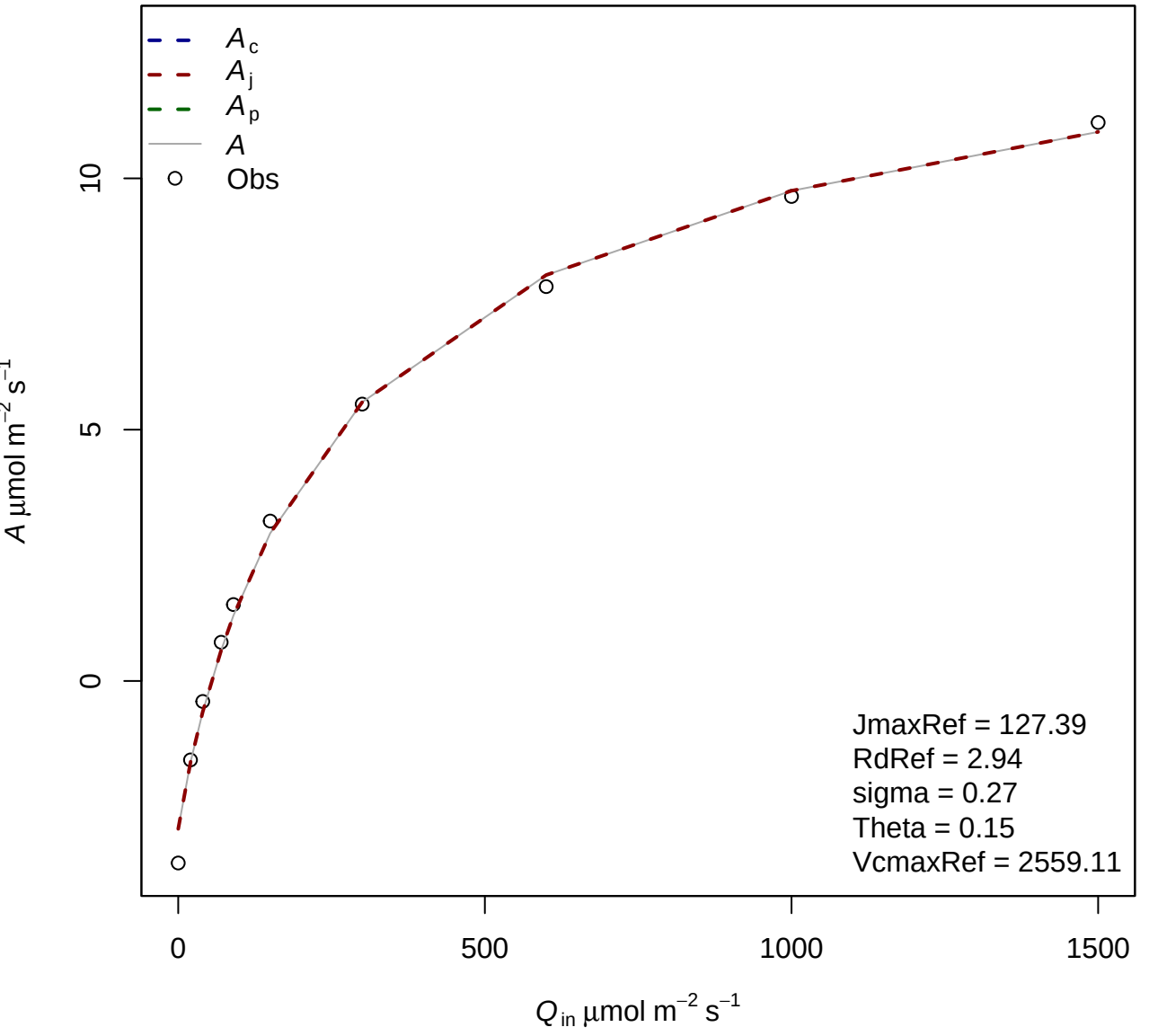

A_c
A_j
A_p
A
Obs
JmaxRef = 91.46
RdRef = 1.21
sigma = 0.11
Theta = 0.79
VcmaxRef = 55.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

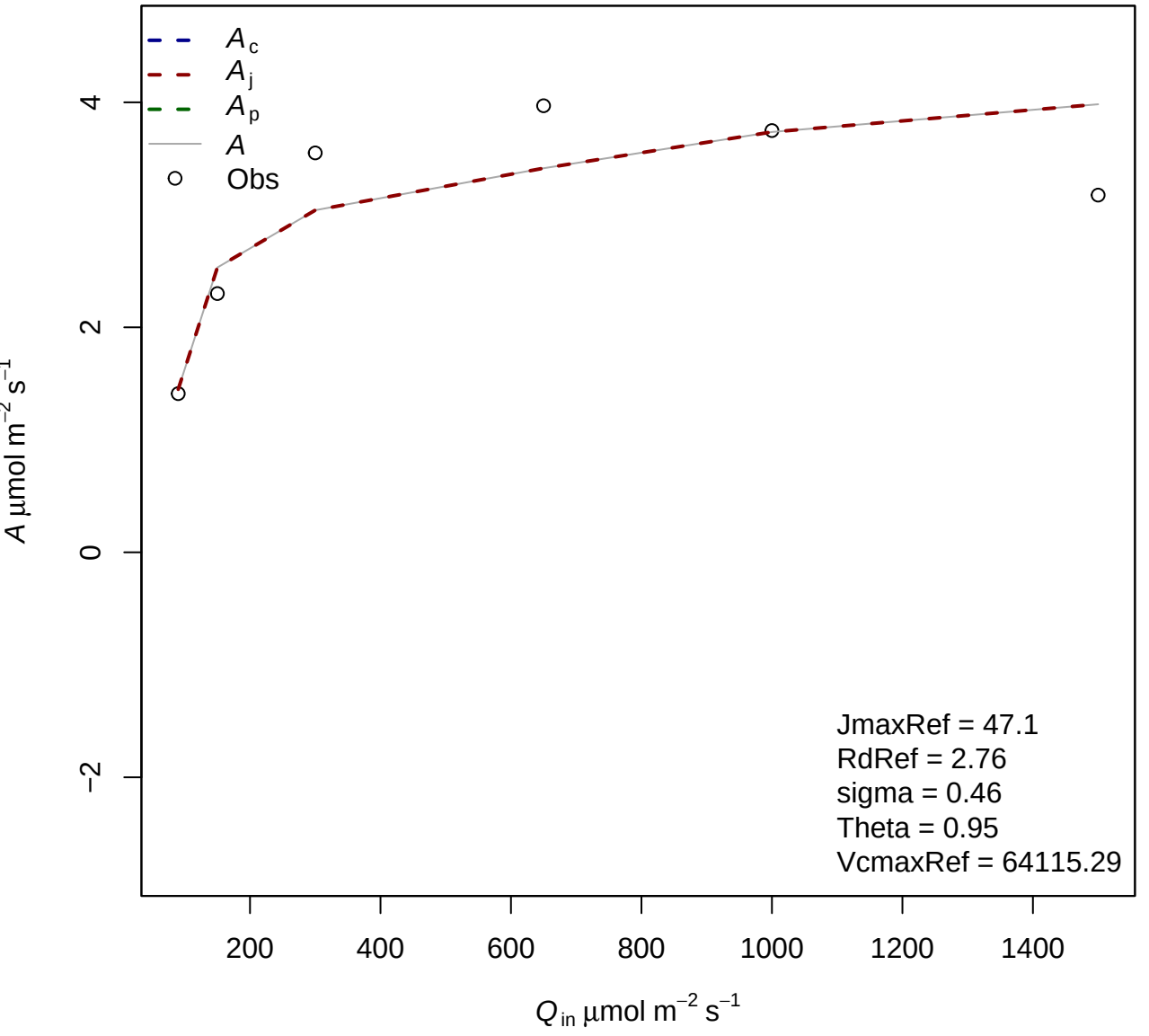

A_c
A_j
A_p
A
Obs
JmaxRef = 72.82
RdRef = 1.67
sigma = 0.28
Theta = −6.65
VcmaxRef = 142225.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

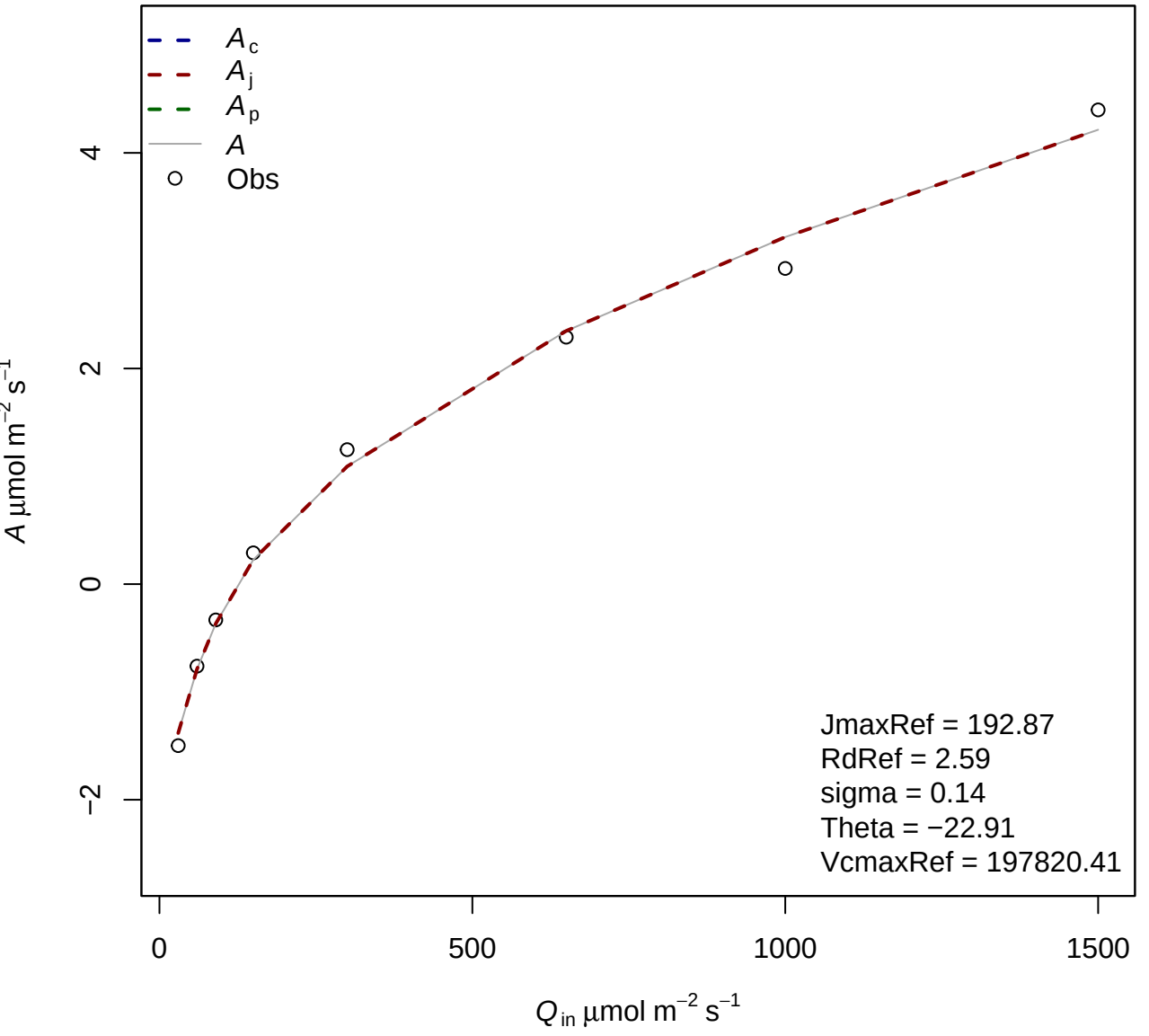

A_c
A_j
A_p
A
Obs
JmaxRef = 74.38
RdRef = 2.4
sigma = 0.19
Theta = −0.66
VcmaxRef = 232.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

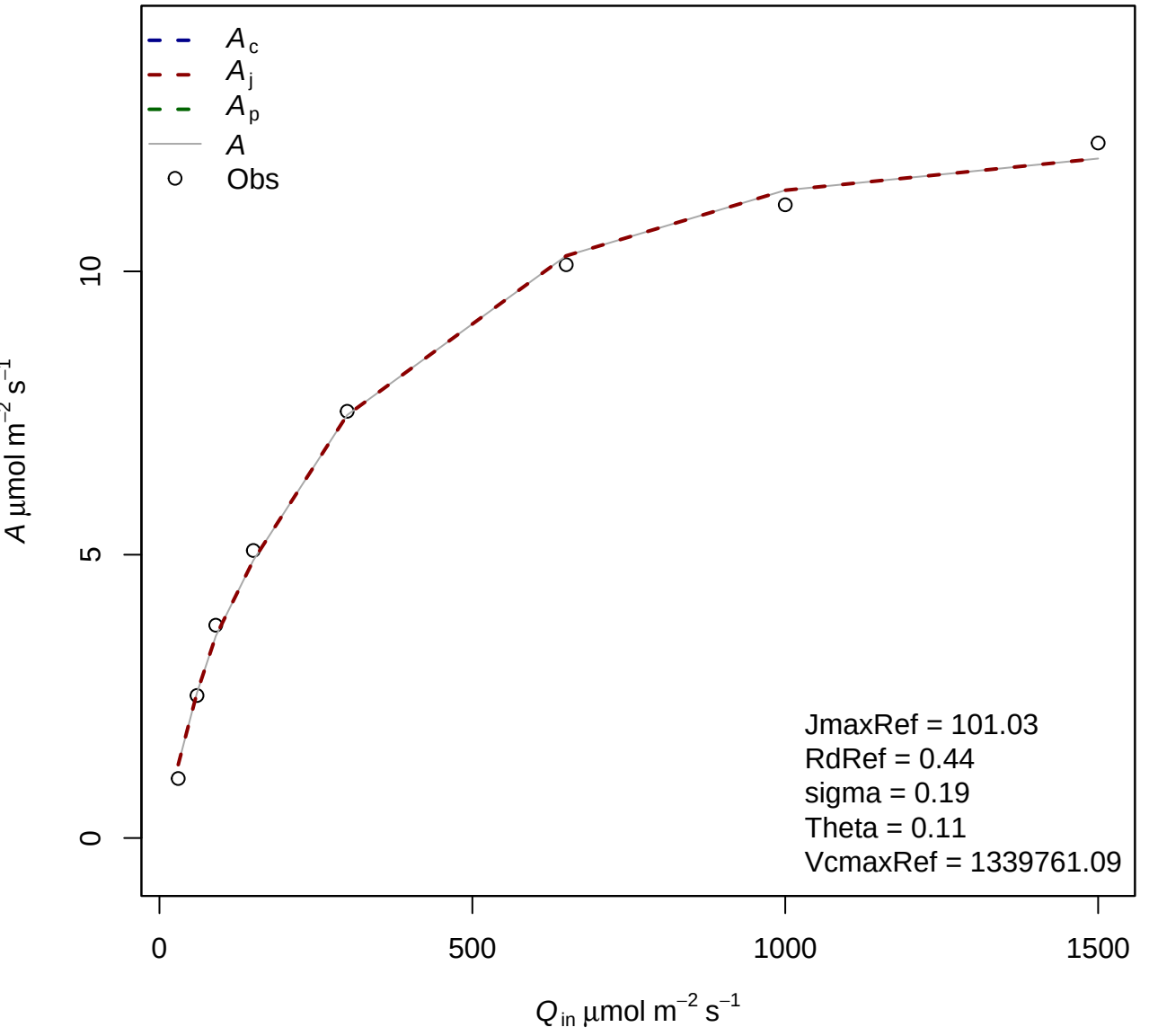

89
A_c
A_j
A_p
A
Obs
JmaxRef = 97.41
RdRef = 1.45
sigma = 0.13
Theta = 0.77
VcmaxRef = 61.98
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

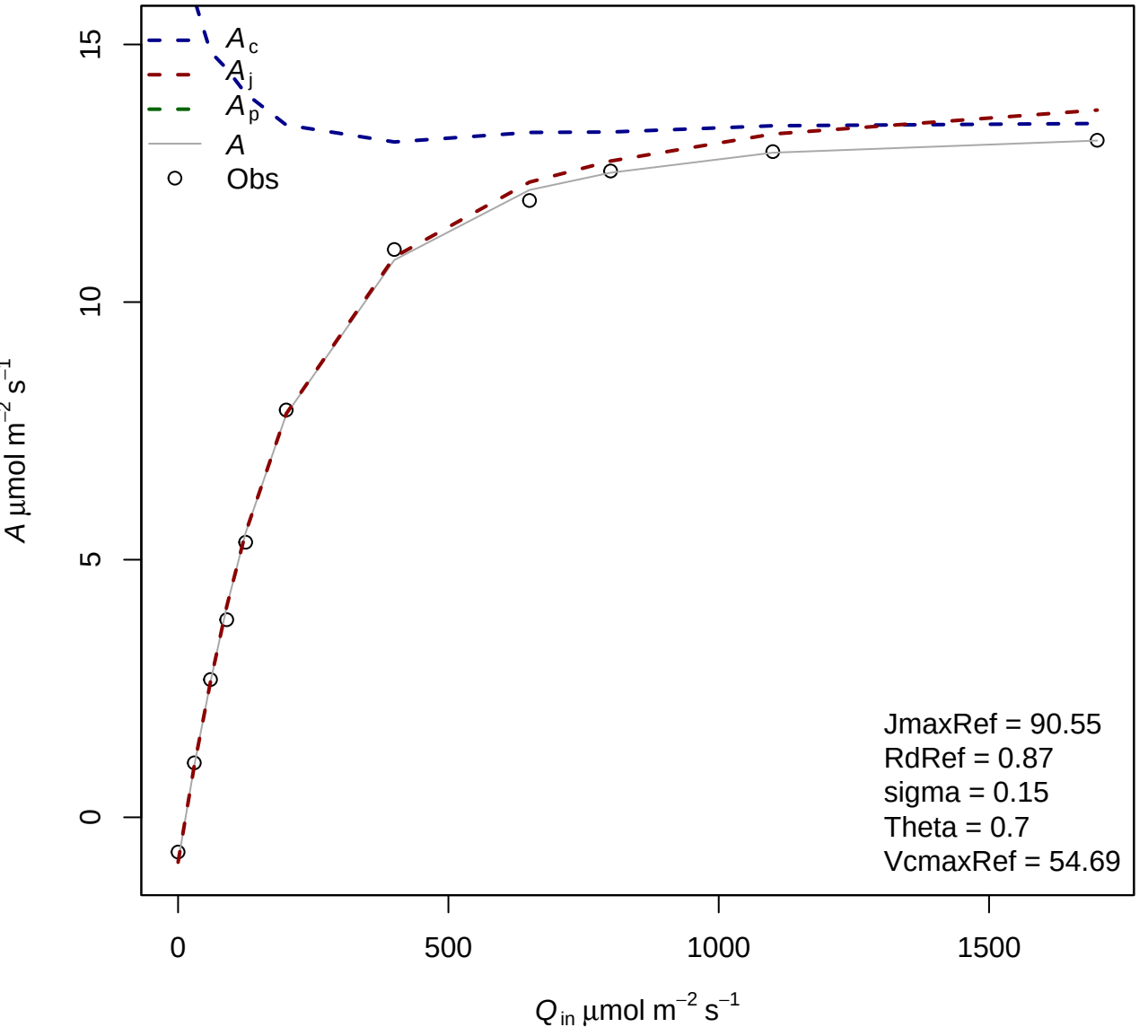

90
A_c
A_j
A_p
A
Obs
JmaxRef = 108.18
RdRef = 1.57
sigma = 0.13
Theta = 0.77
VcmaxRef = 68.72
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

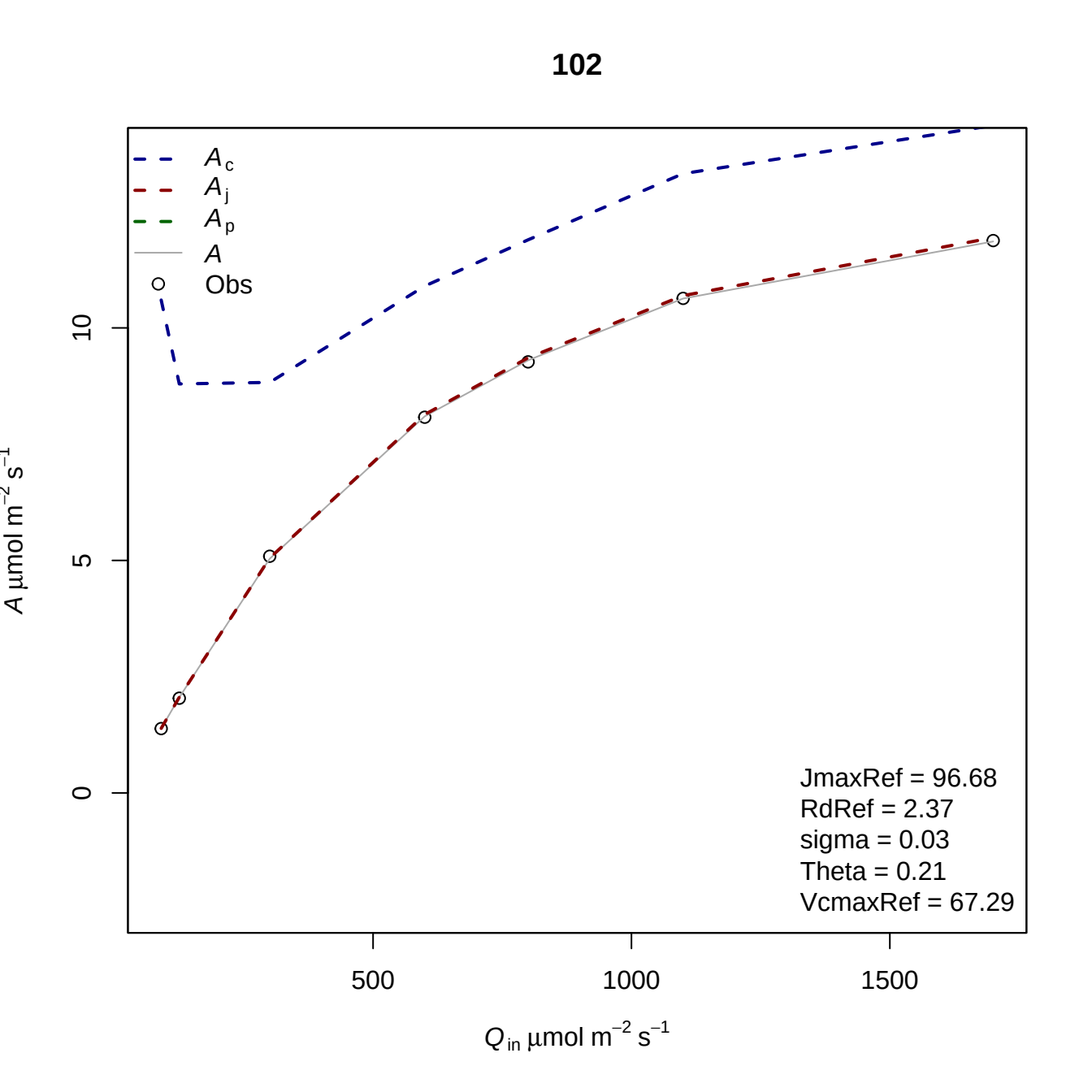

A_c
A_j
A_p
A
Obs
JmaxRef = 127.39
RdRef = 2.94
sigma = 0.27
Theta = 0.15
VcmaxRef = 2559.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

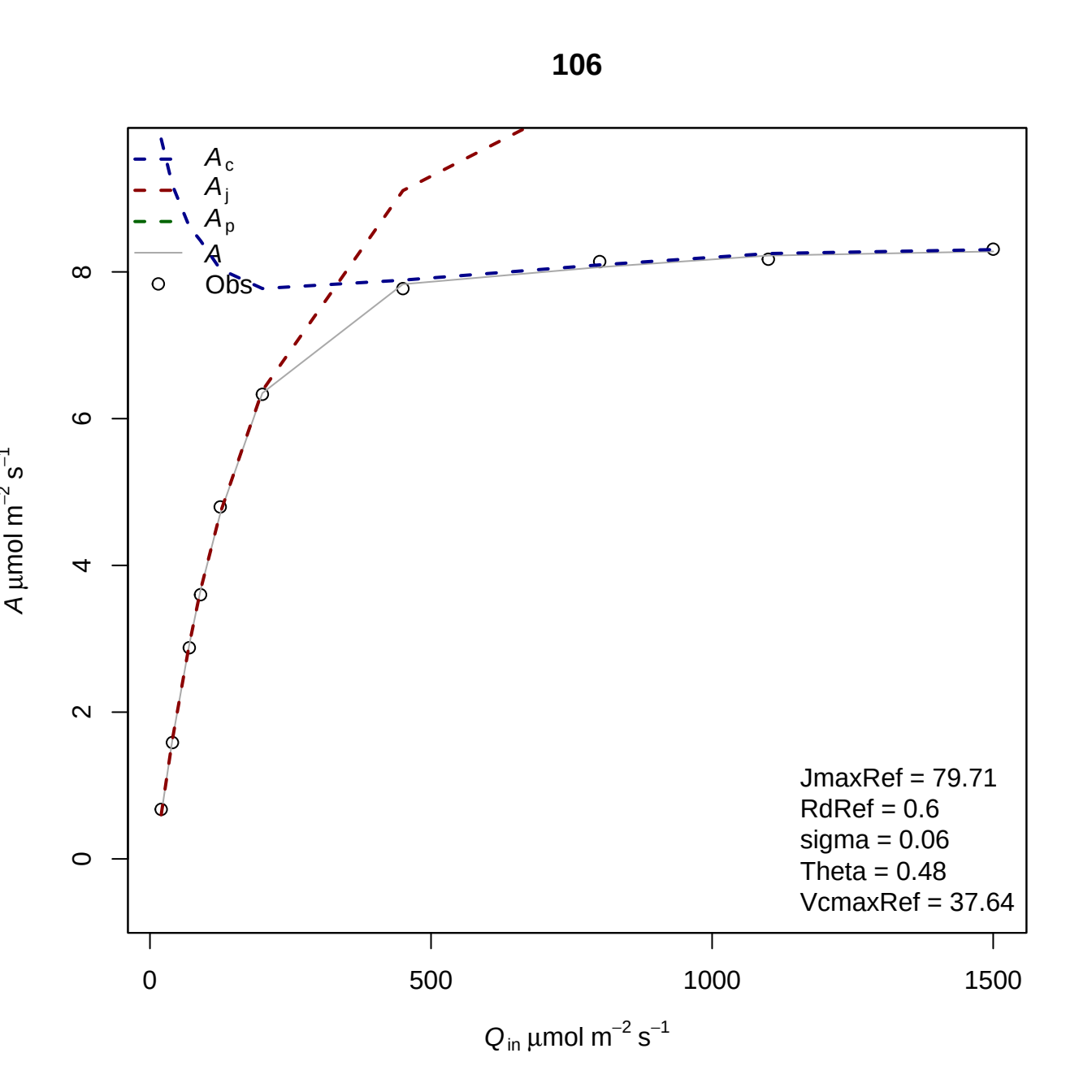

A_c
A_j
A_p
A
Obs
JmaxRef = 47.1
RdRef = 2.76
sigma = 0.46
Theta = 0.95
VcmaxRef = 64115.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

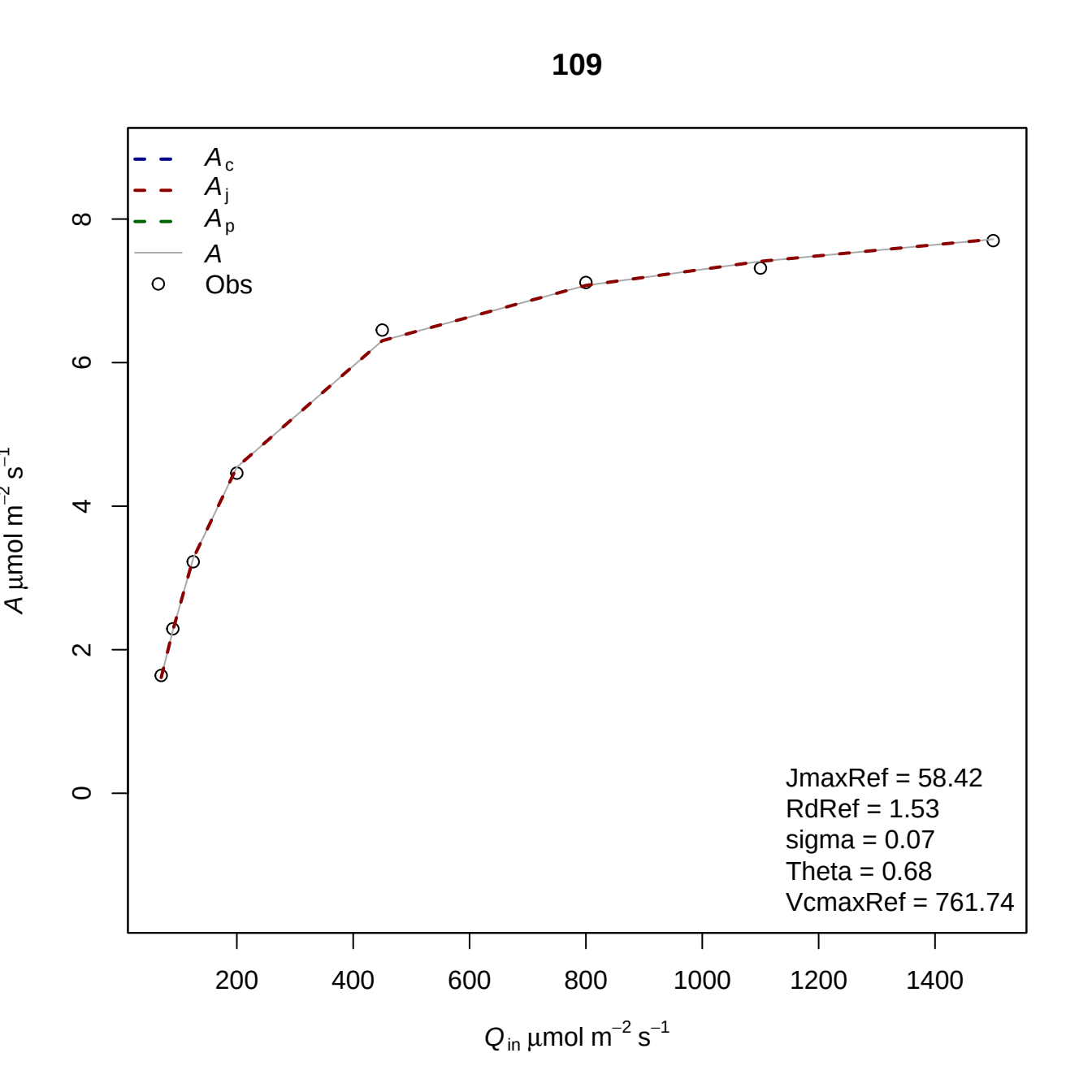

A_c
A_j
A_p
A
Obs
JmaxRef = 192.87
RdRef = 2.59
sigma = 0.14
Theta = −22.91
VcmaxRef = 197820.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

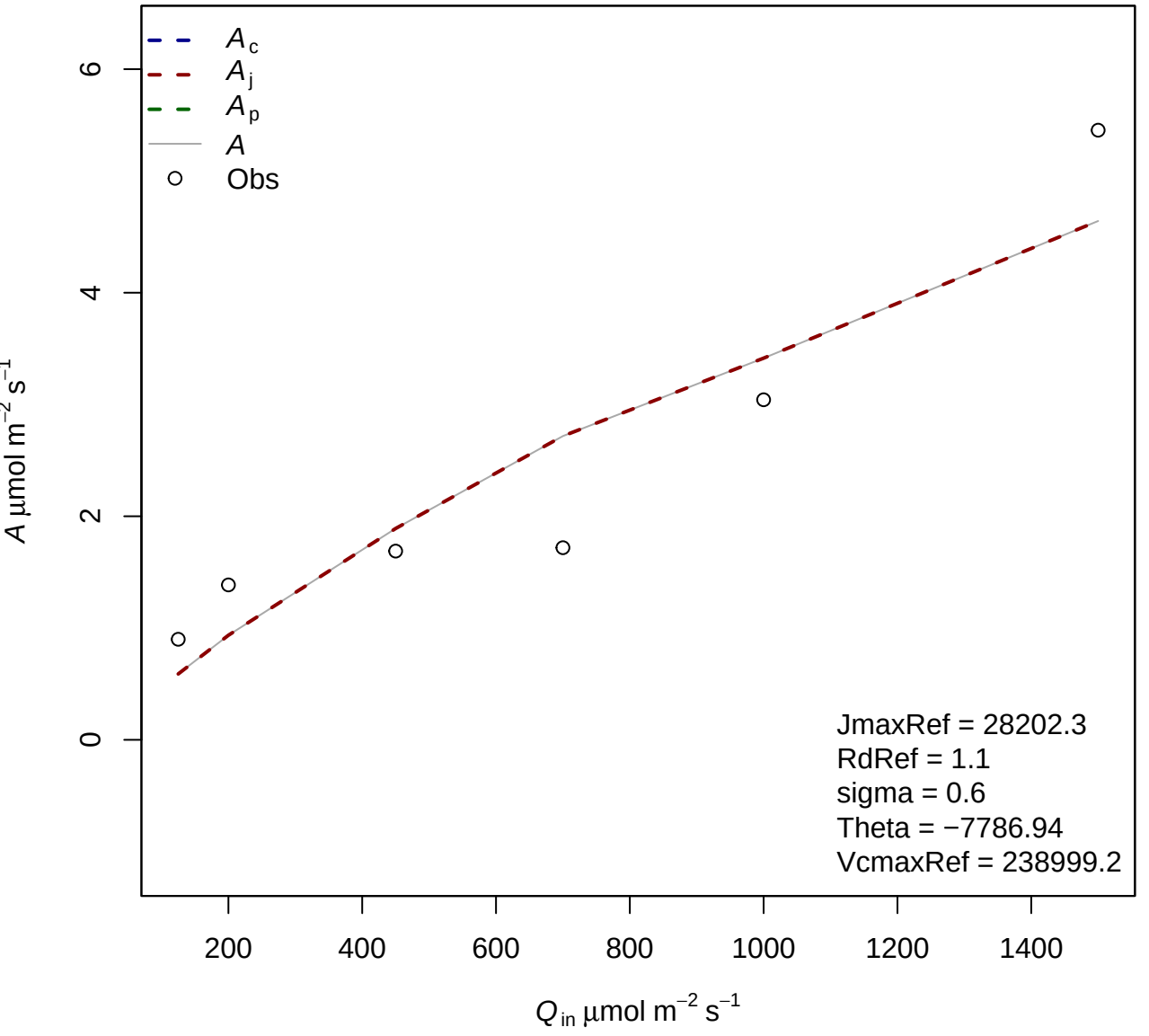

A_c
A_j
A_p
A
Obs
JmaxRef = 101.03
RdRef = 0.44
sigma = 0.19
Theta = 0.11
VcmaxRef = 1339761.09
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

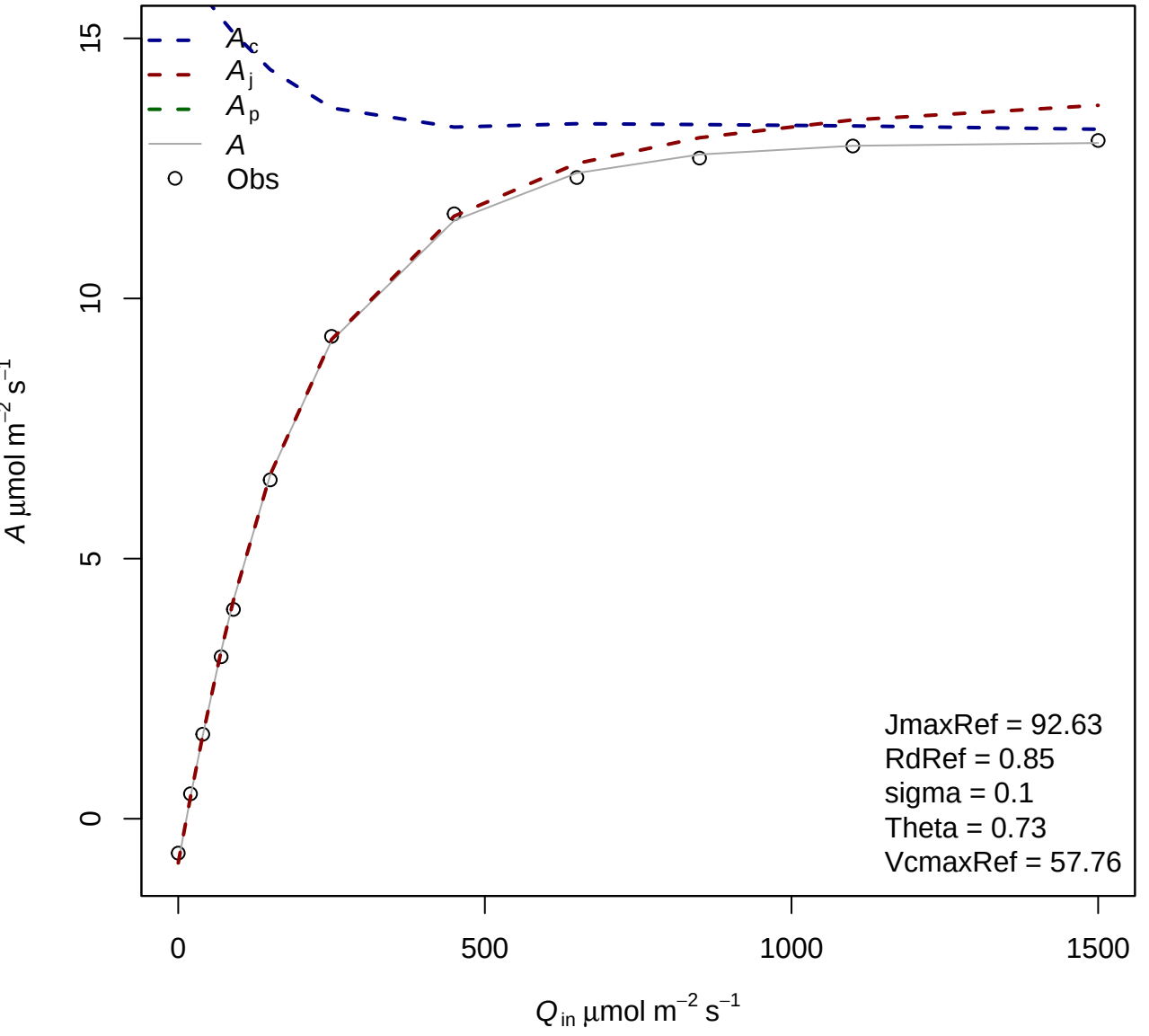

A_c
A_j
A_p
A
Obs
JmaxRef = 90.55
RdRef = 0.87
sigma = 0.15
Theta = 0.7
VcmaxRef = 54.69
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

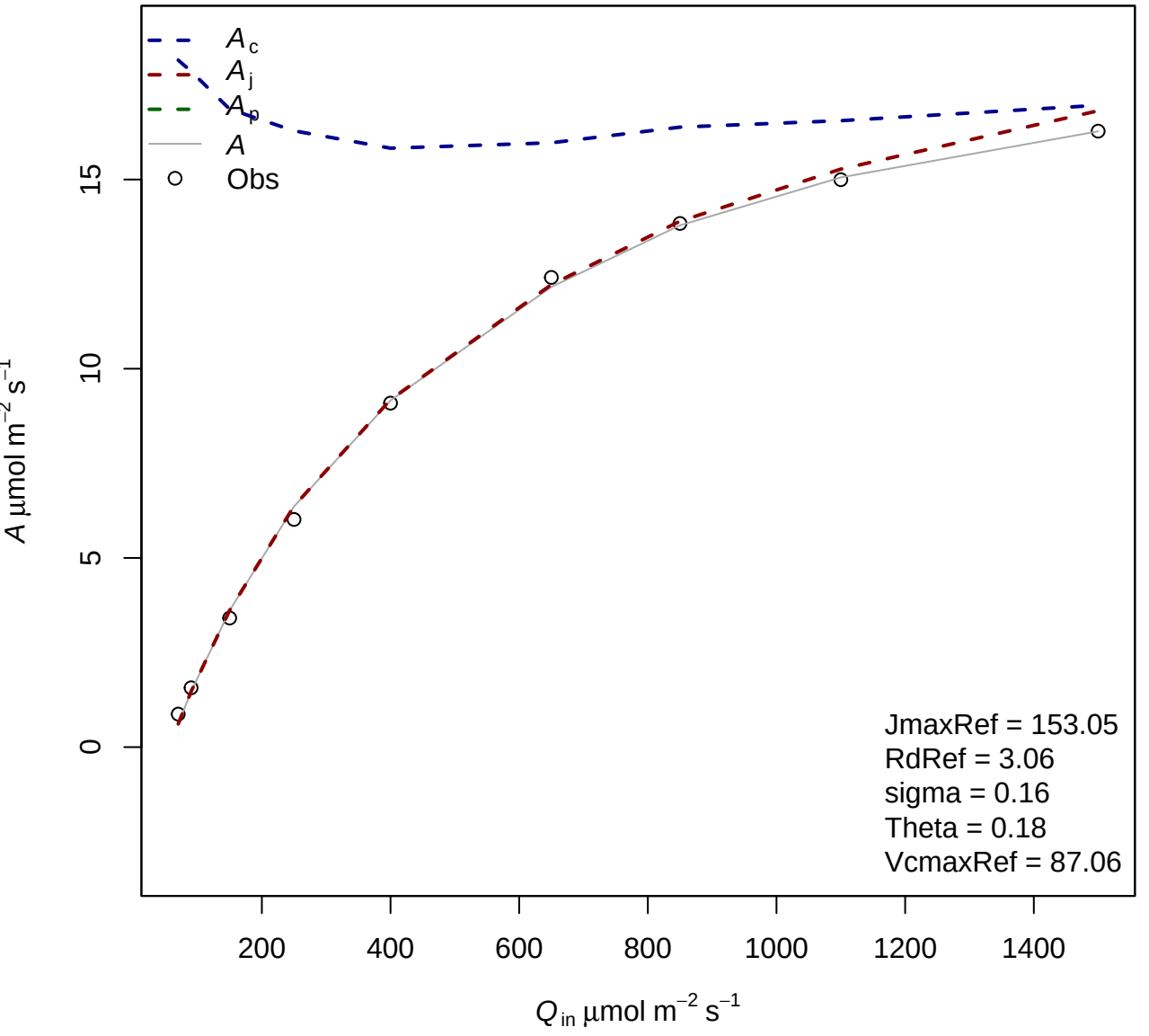

102
A_c
A_j
A_p
A
Obs
JmaxRef = 96.68
RdRef = 2.37
sigma = 0.03
Theta = 0.21
VcmaxRef = 67.29
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

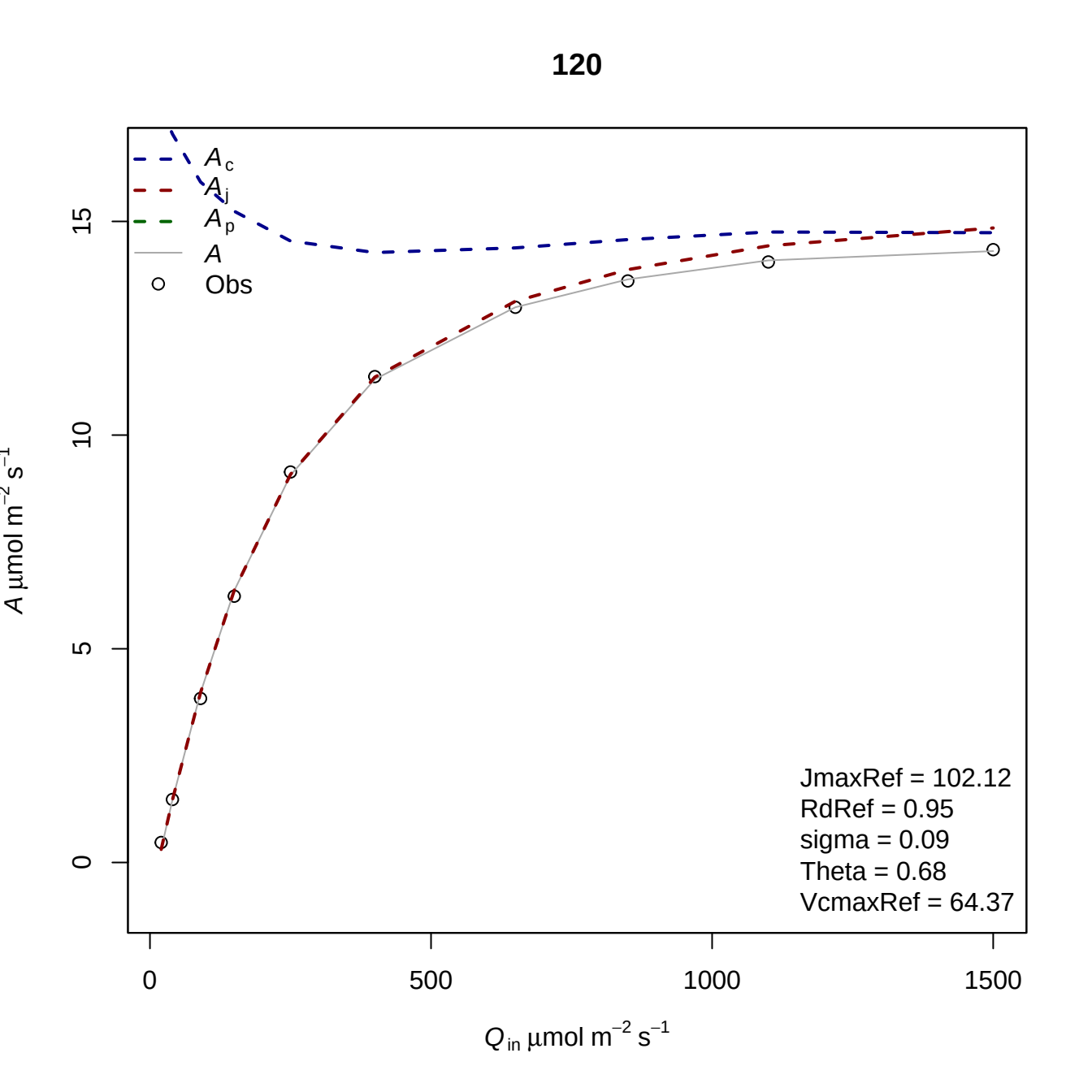

106
A_c
A_j
A_p
A
Obs
JmaxRef = 79.71
RdRef = 0.6
sigma = 0.06
Theta = 0.48
VcmaxRef = 37.64
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

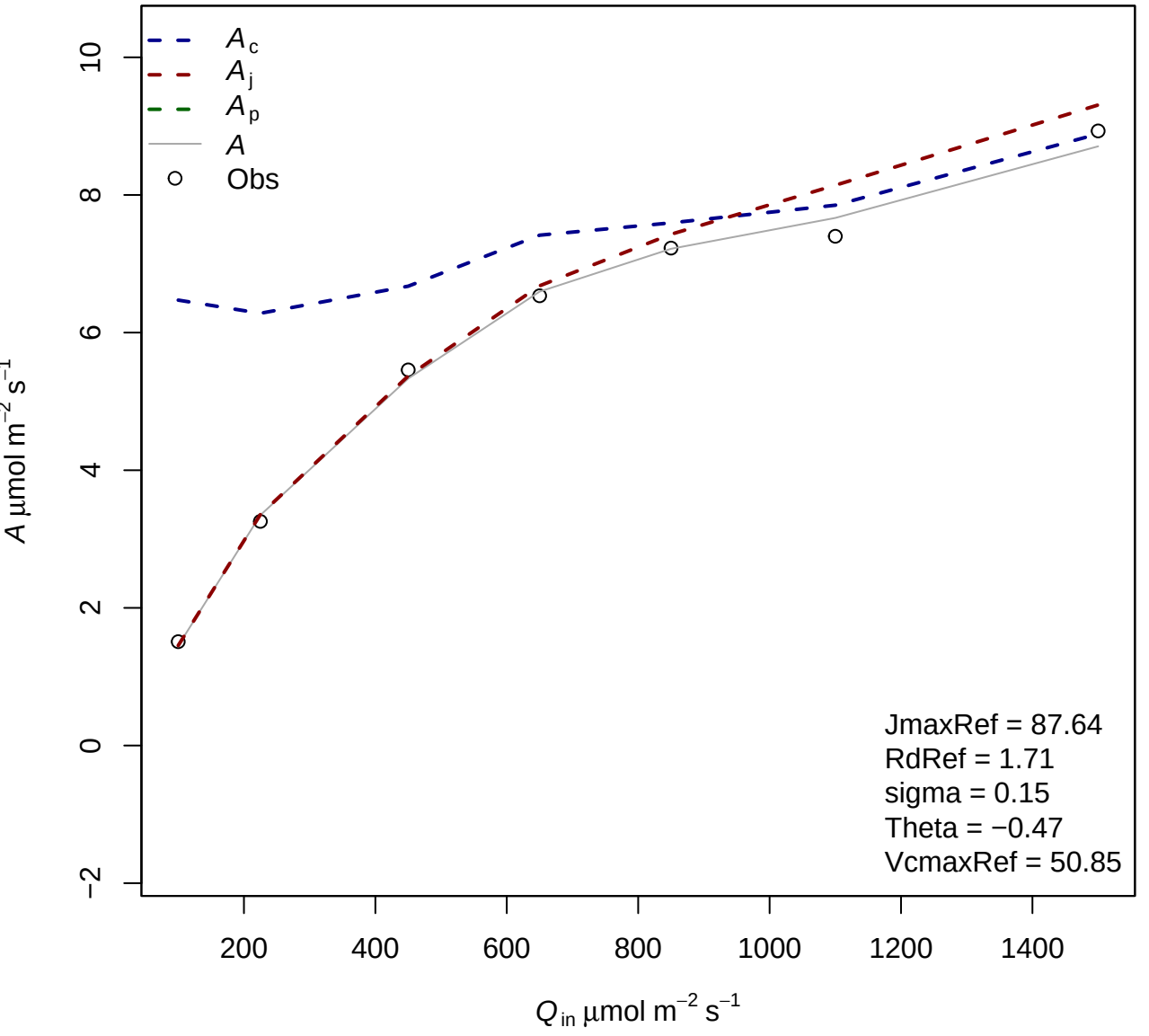

109
A_c
A_j
A_p
A
Obs
JmaxRef = 58.42
RdRef = 1.53
sigma = 0.07
Theta = 0.68
VcmaxRef = 761.74
0
2
4
6
8
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

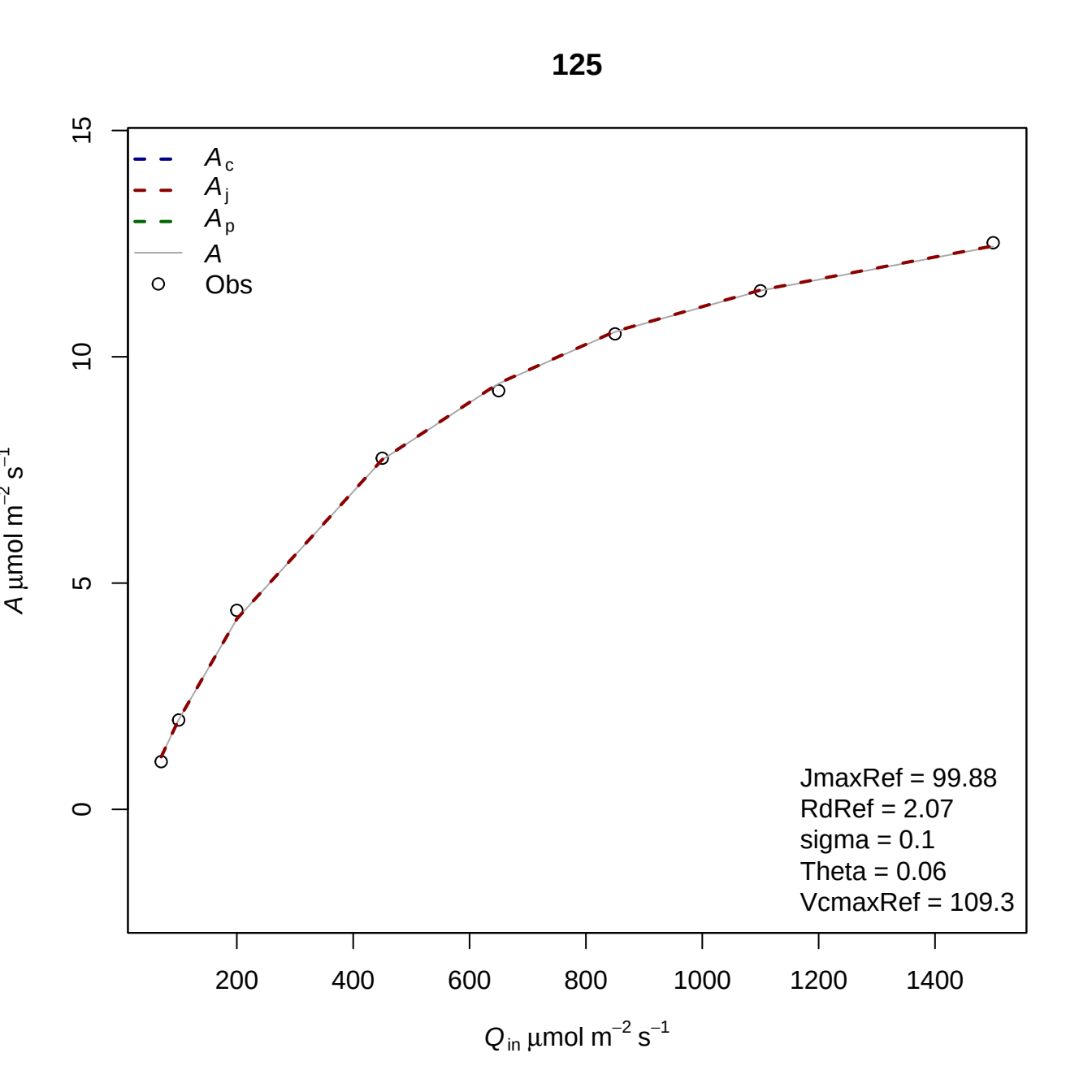

A_c
A_j
A_p
A
Obs
JmaxRef = 28202.3
RdRef = 1.1
sigma = 0.6
Theta = −7786.94
VcmaxRef = 238999.2
0
2
4
6
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

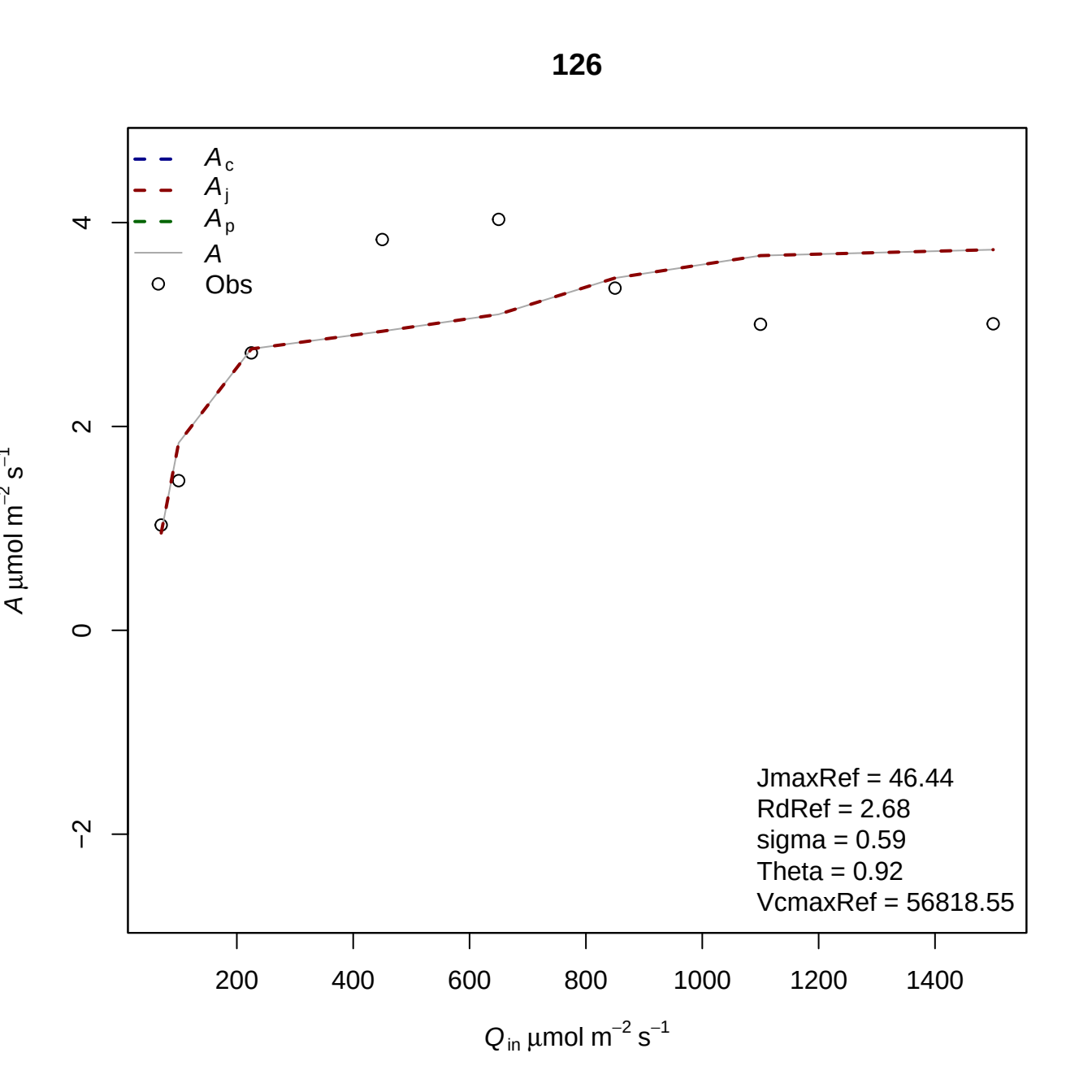

A_c
A_j
A_p
A
Obs
JmaxRef = 92.63
RdRef = 0.85
sigma = 0.1
Theta = 0.73
VcmaxRef = 57.76
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

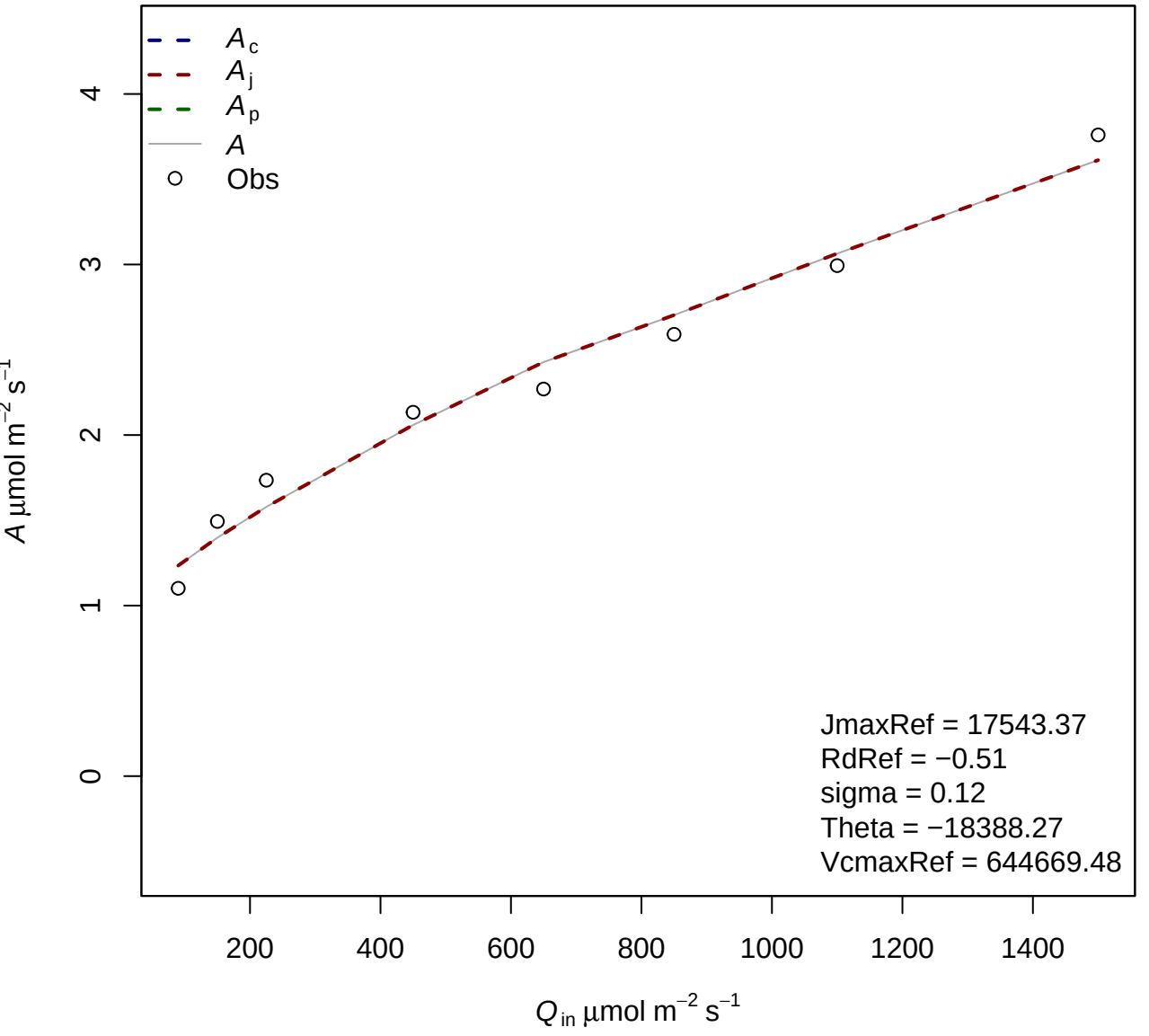

A_c
A_j
A_p
A
Obs
JmaxRef = 153.05
RdRef = 3.06
sigma = 0.16
Theta = 0.18
VcmaxRef = 87.06
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

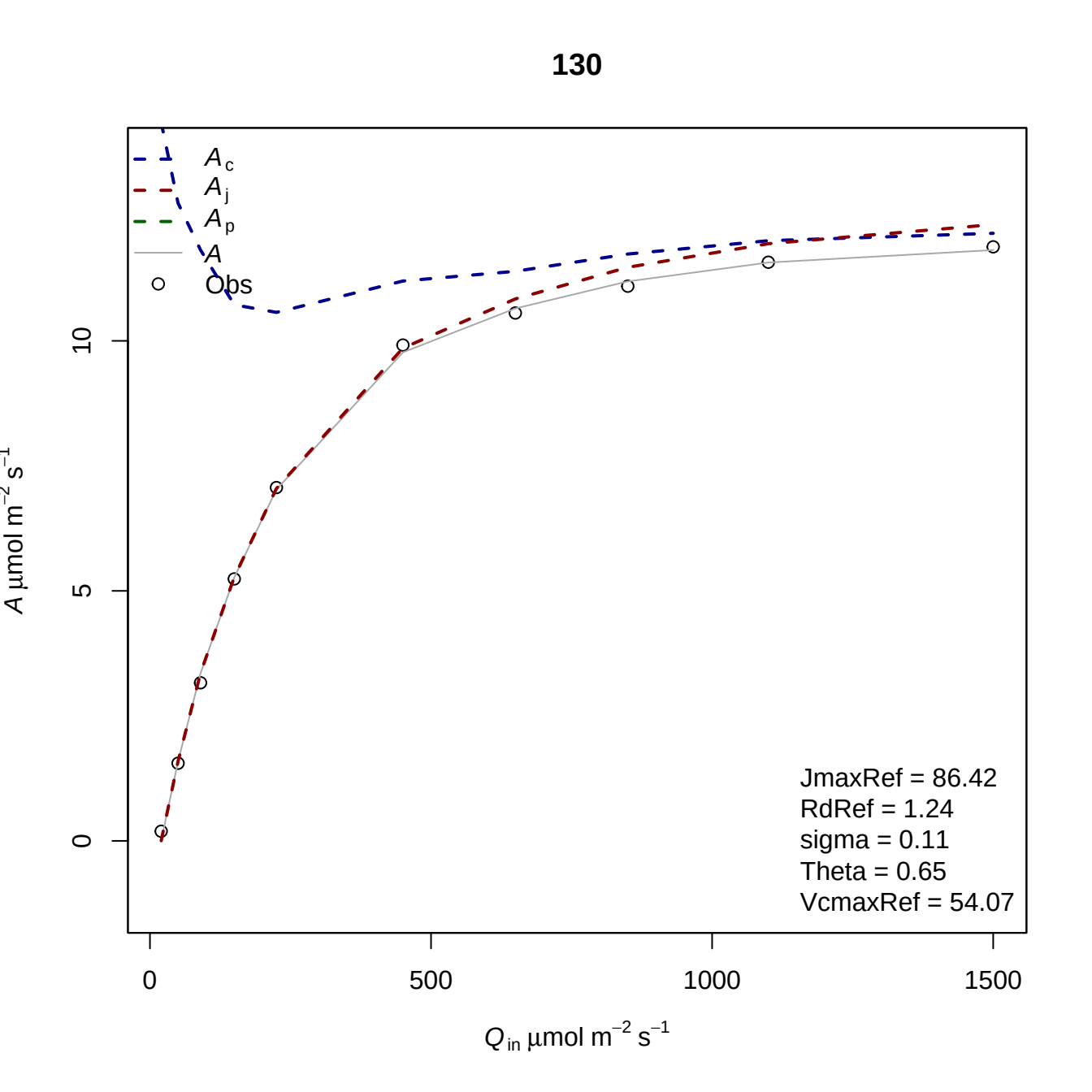

120
A_c
A_j
A_p
A
Obs
JmaxRef = 102.12
RdRef = 0.95
sigma = 0.09
Theta = 0.68
VcmaxRef = 64.37
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

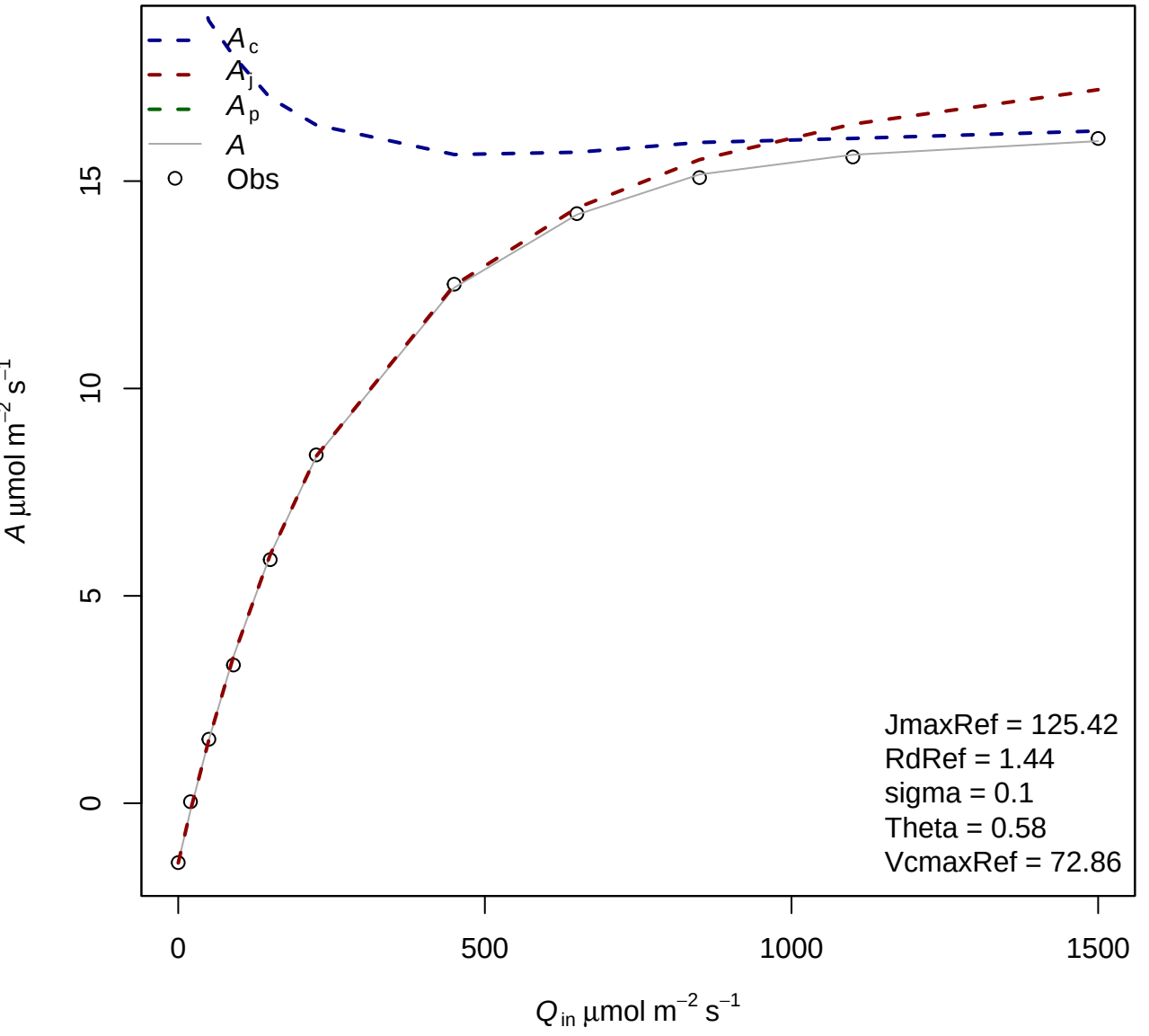

A_c
A_j
A_p
A
Obs
JmaxRef = 87.64
RdRef = 1.71
sigma = 0.15
Theta = −0.47
VcmaxRef = 50.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

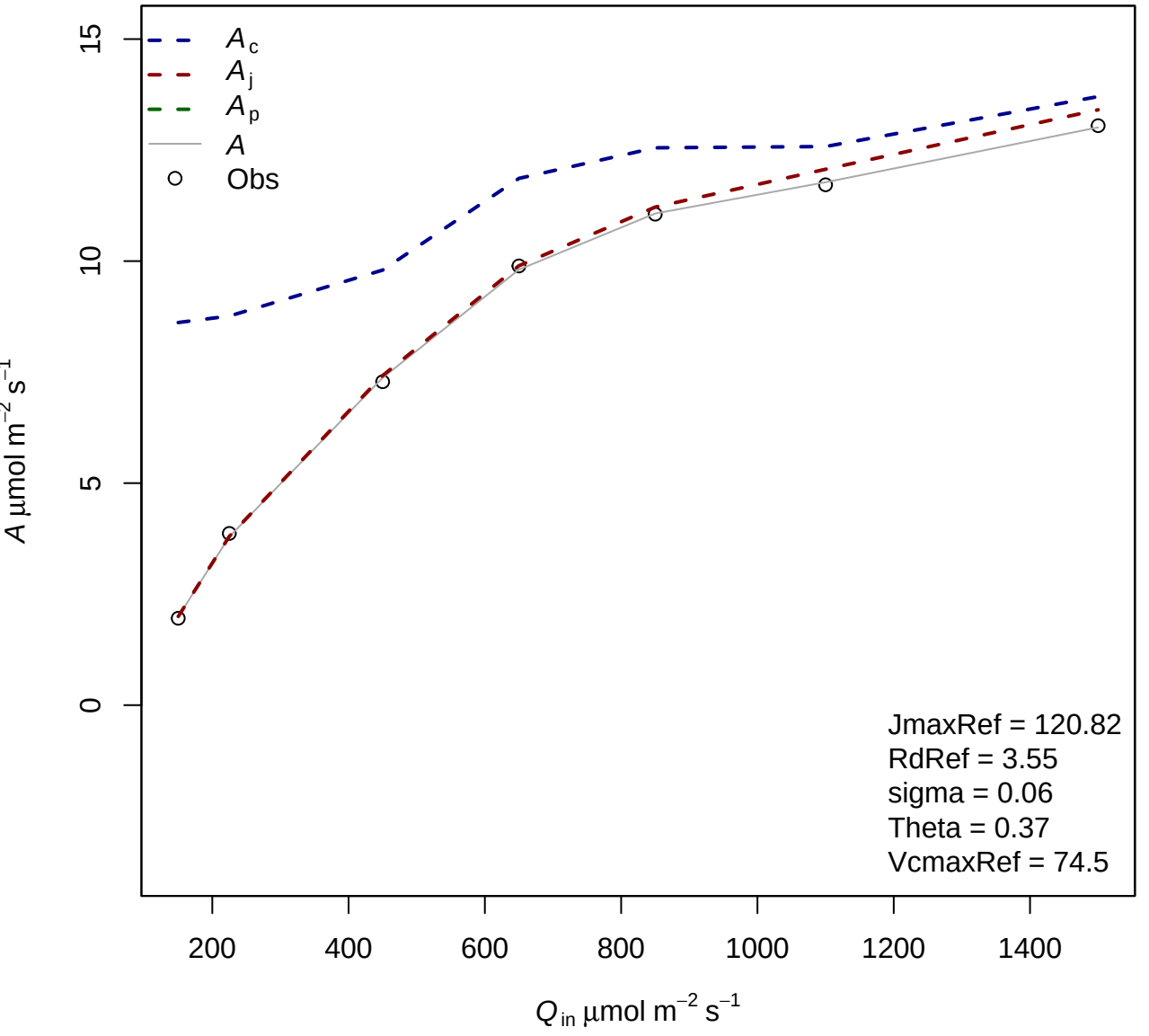

125
A_c
A_j
A_p
A
Obs
JmaxRef = 99.88
RdRef = 2.07
sigma = 0.1
Theta = 0.06
VcmaxRef = 109.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
200
400
600
800
1000
1200
1400

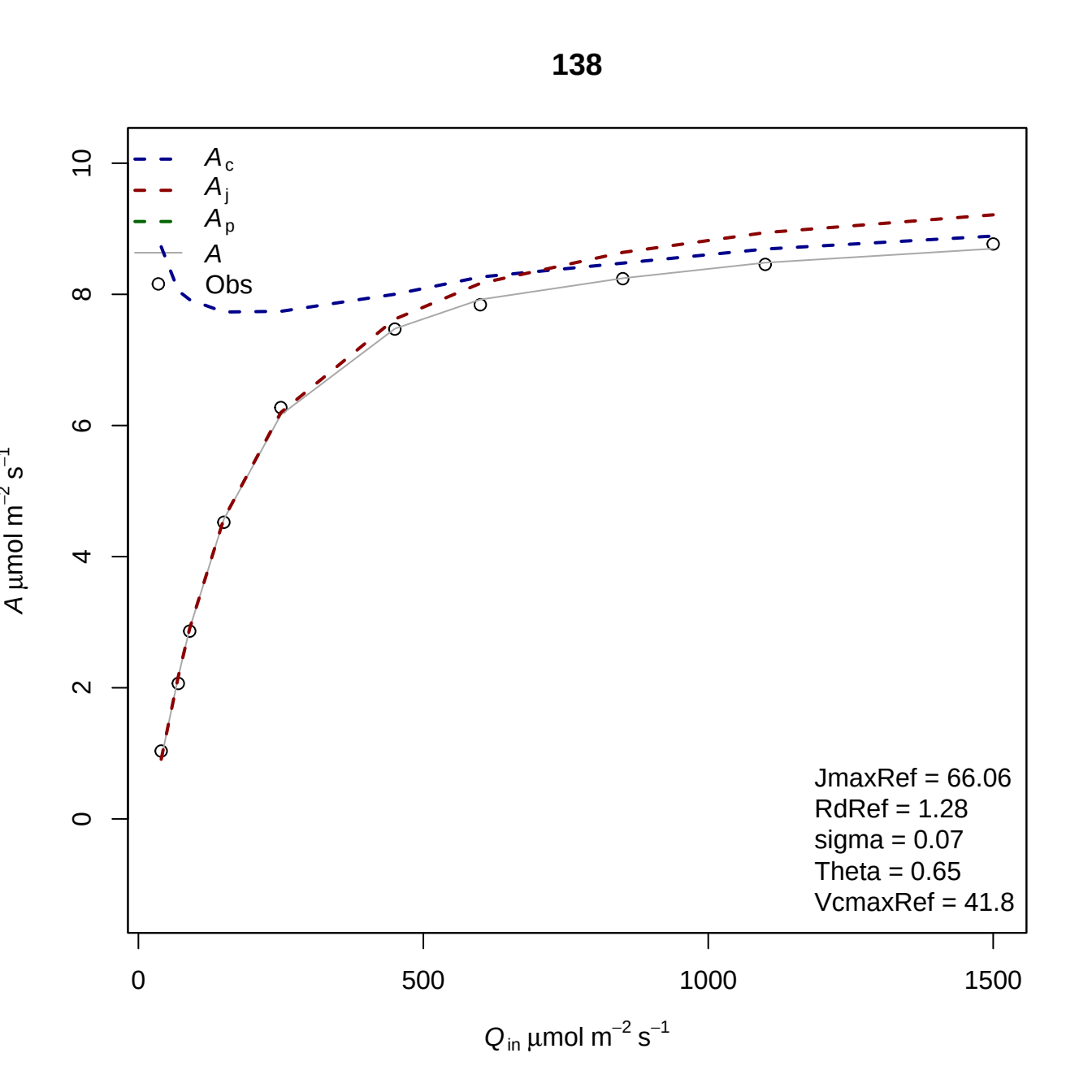

126
A_c
A_j
A_p
A
Obs
JmaxRef = 46.44
RdRef = 2.68
sigma = 0.59
Theta = 0.92
VcmaxRef = 56818.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

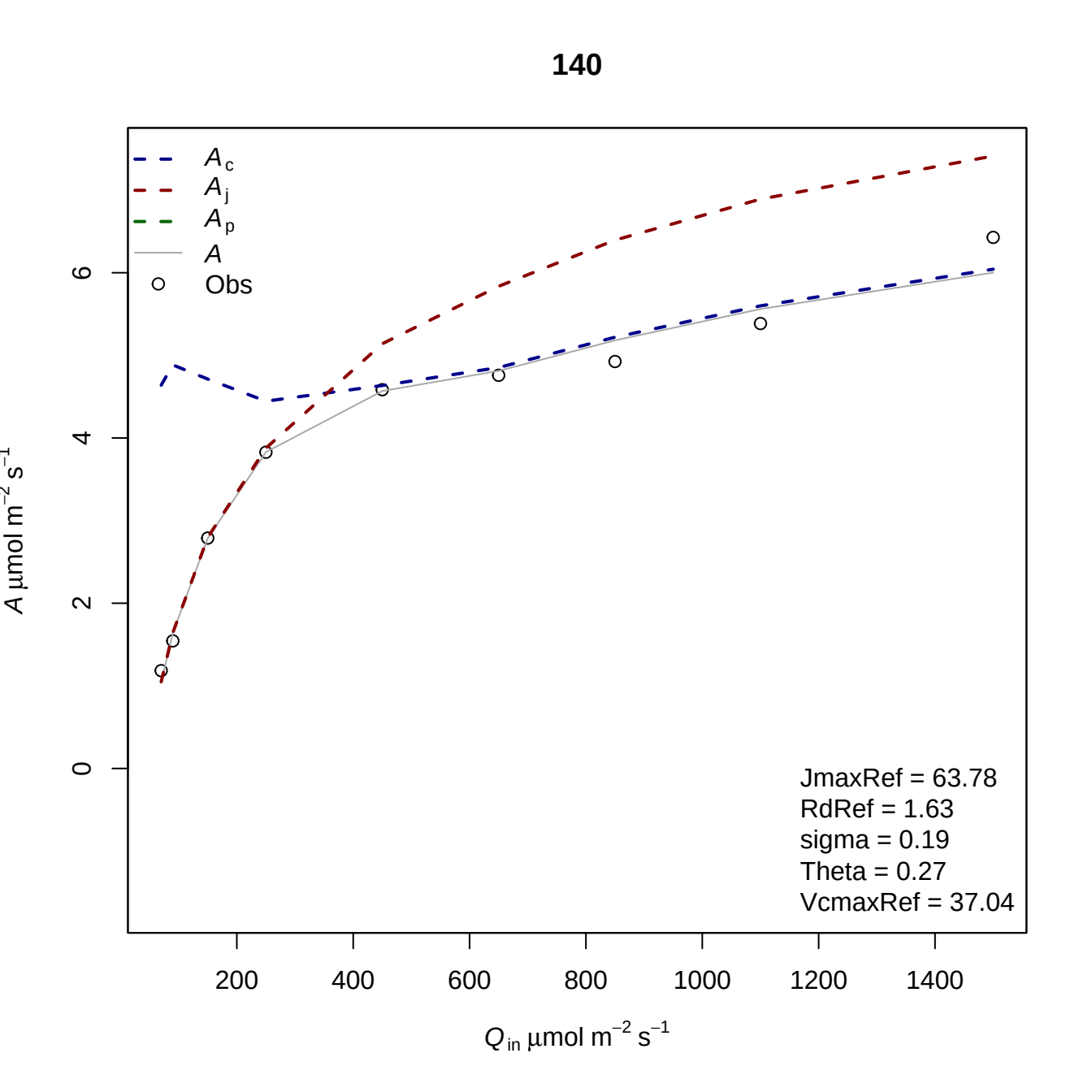

A_c
A_j
A_p
A
Obs
JmaxRef = 17543.37
RdRef = −0.51
sigma = 0.12
Theta = −18388.27
VcmaxRef = 644669.48
0
1
2
3
4
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

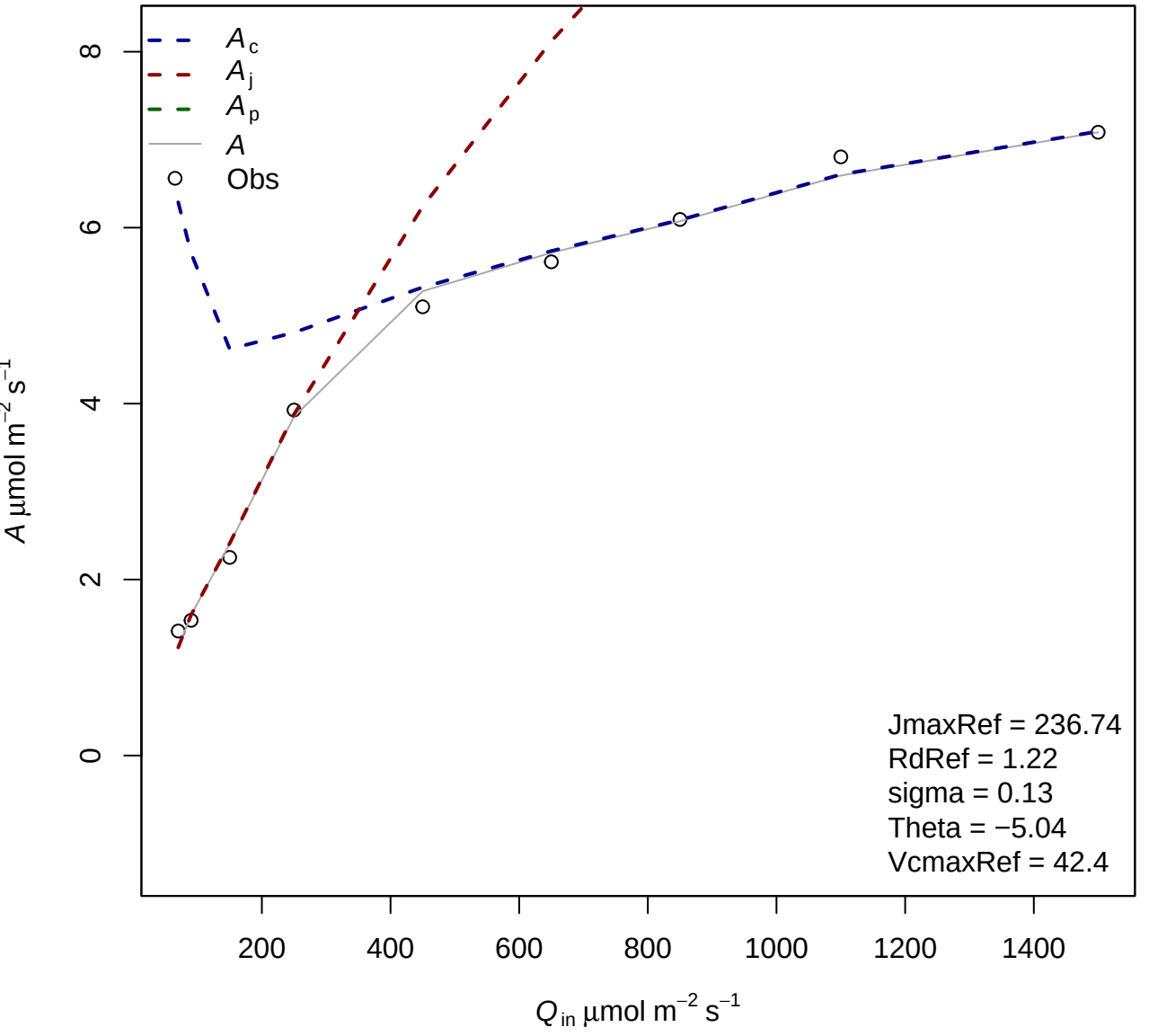

130
A_c
A_j
A_p
A
Obs
JmaxRef = 86.42
RdRef = 1.24
sigma = 0.11
Theta = 0.65
VcmaxRef = 54.07
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

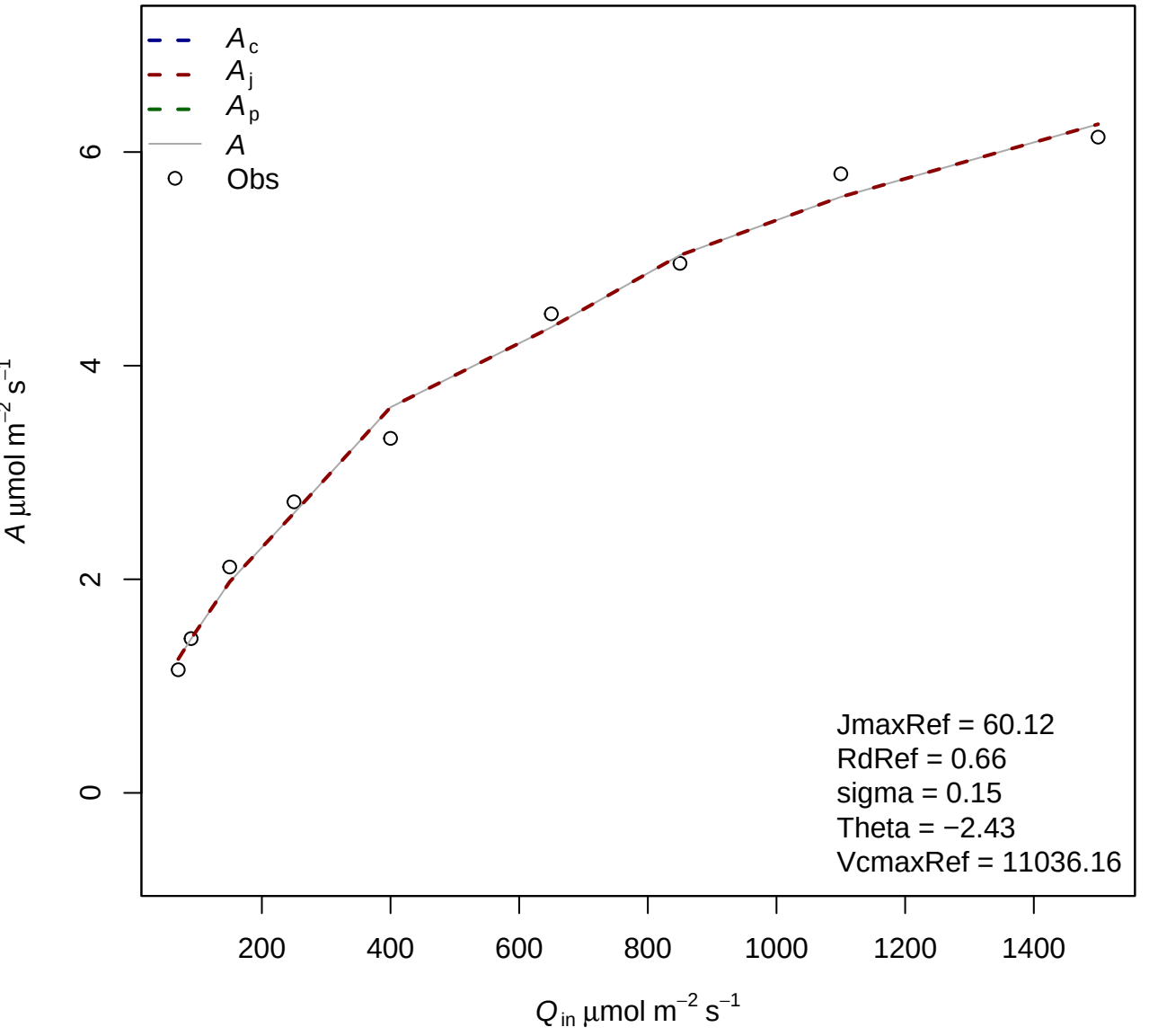

A_c
A_j
A_p
A
Obs
JmaxRef = 125.42
RdRef = 1.44
sigma = 0.1
Theta = 0.58
VcmaxRef = 72.86
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

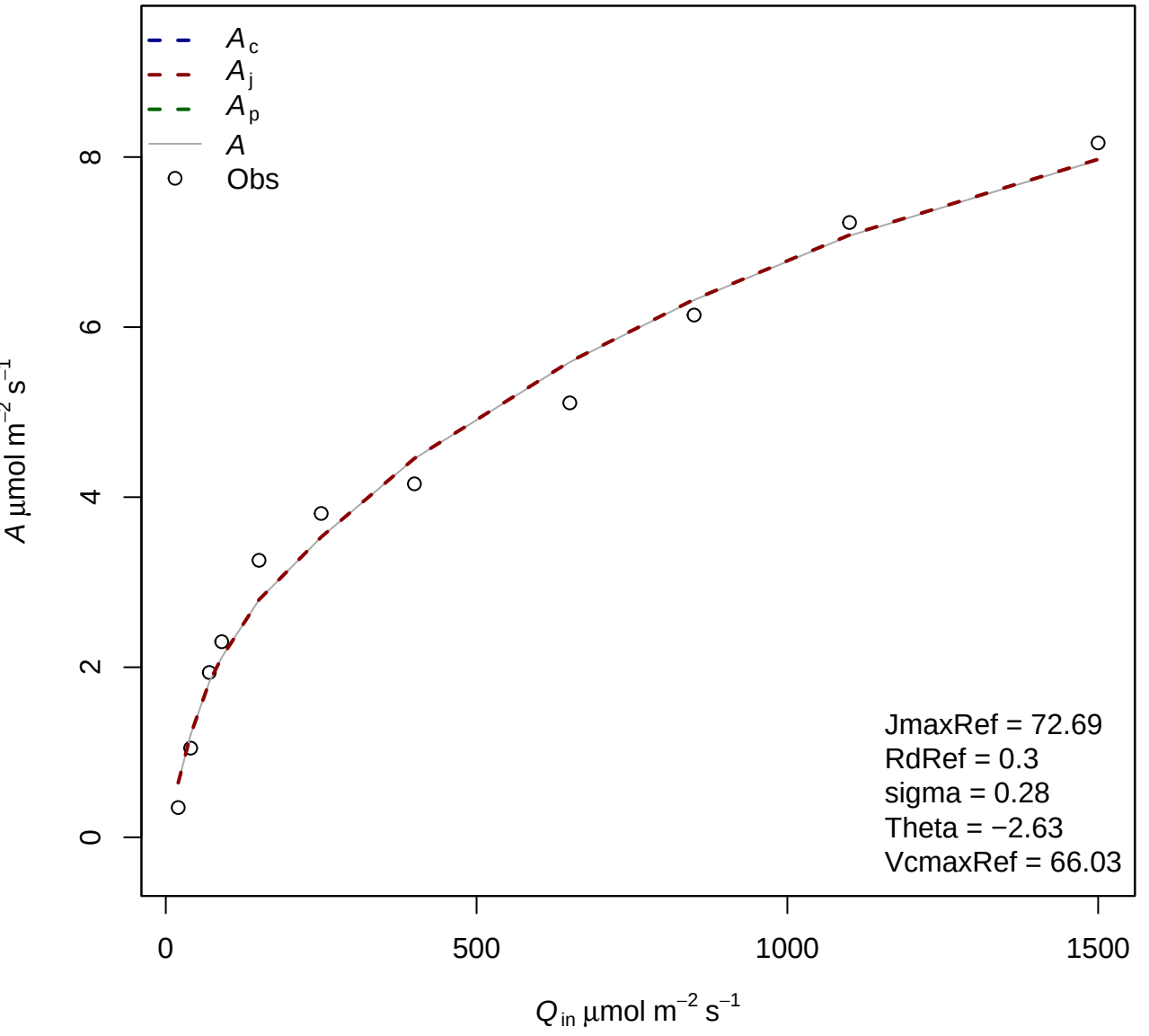

A_c
A_j
A_p
A
Obs
JmaxRef = 120.82
RdRef = 3.55
sigma = 0.06
Theta = 0.37
VcmaxRef = 74.5
0
5
10
15
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

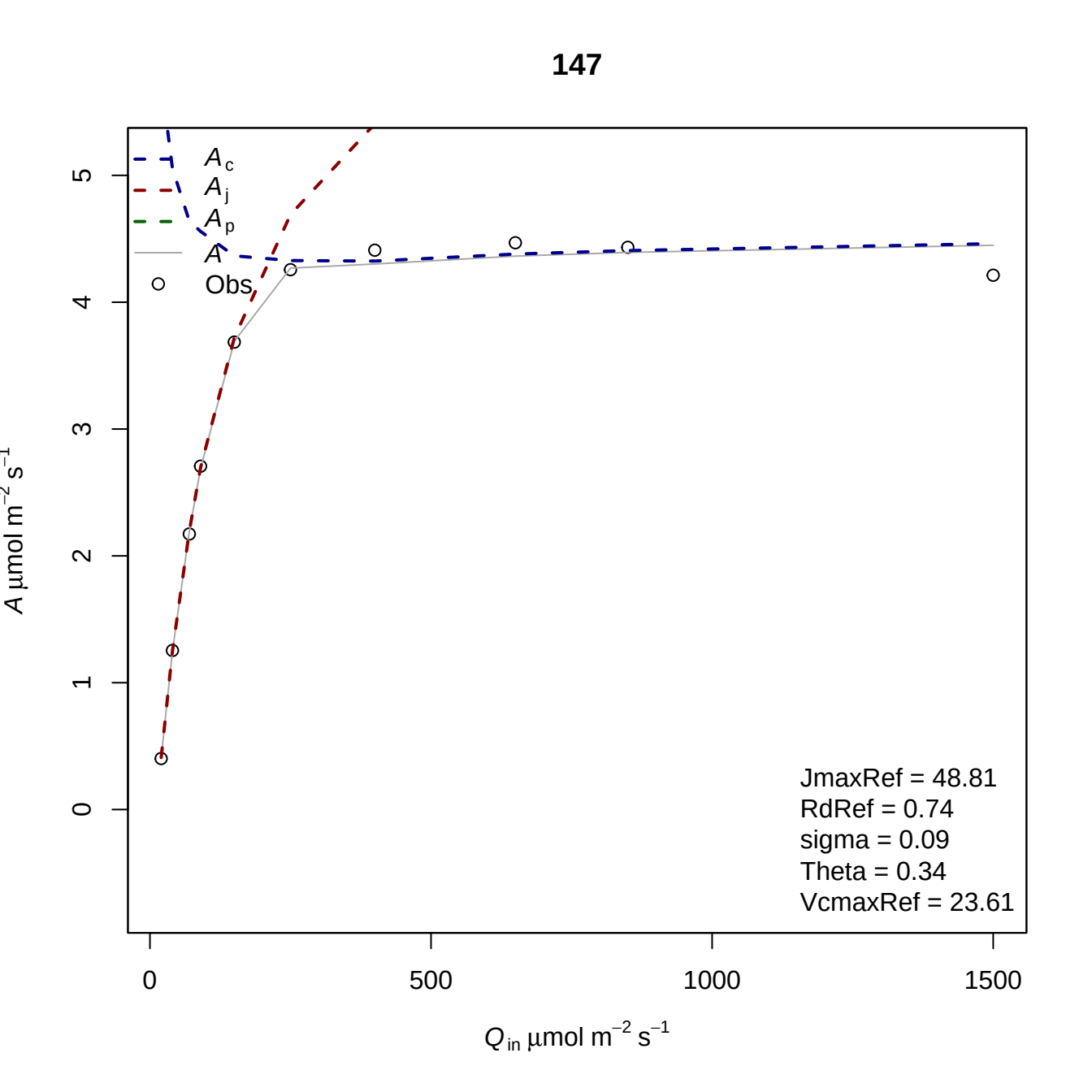

138
A_c
A_j
A_p
A
Obs
JmaxRef = 66.06
RdRef = 1.28
sigma = 0.07
Theta = 0.65
VcmaxRef = 41.8
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

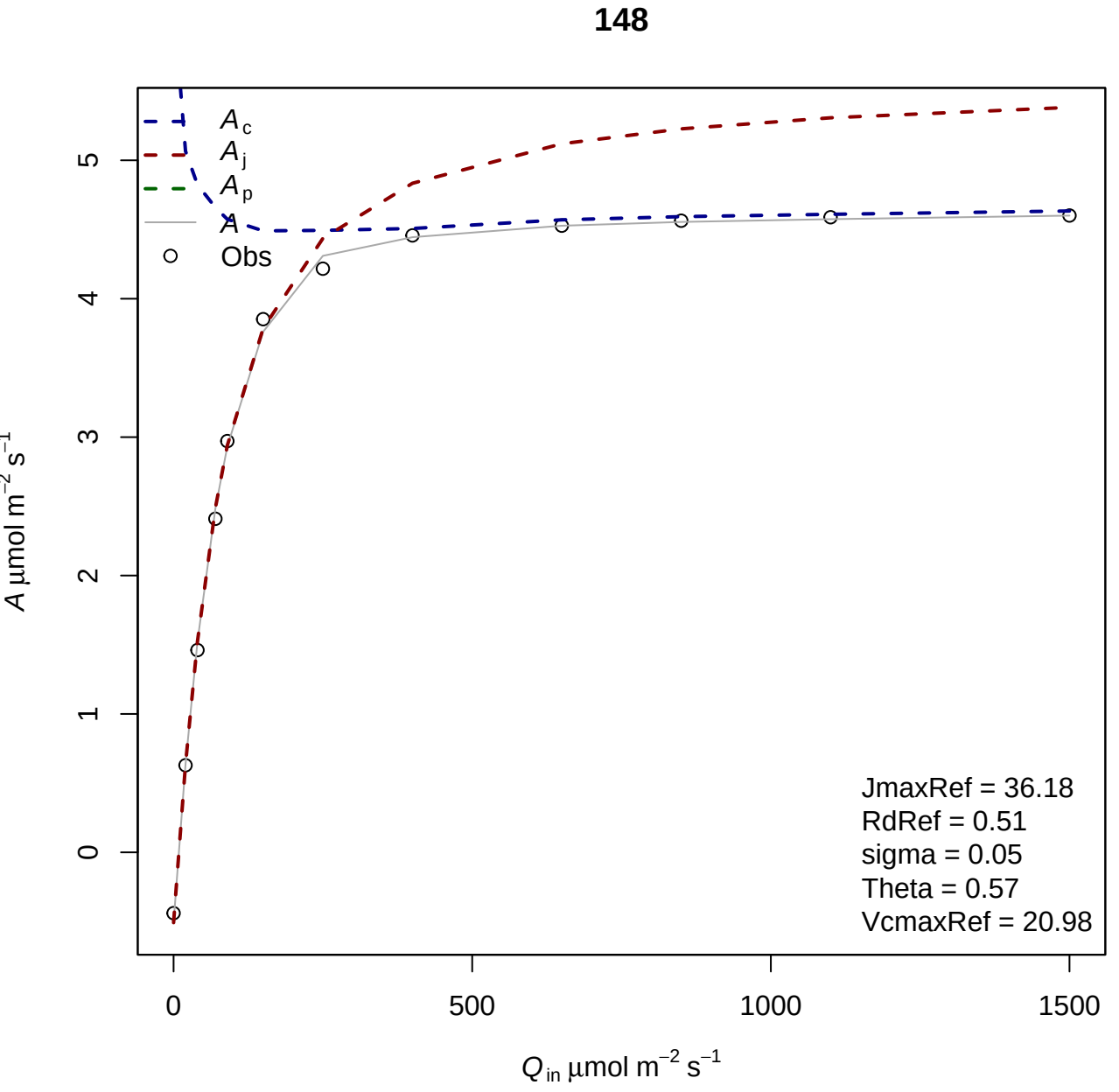

140
A_c
A_j
A_p
A
Obs
JmaxRef = 63.78
RdRef = 1.63
sigma = 0.19
Theta = 0.27
VcmaxRef = 37.04
0
2
4
6
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

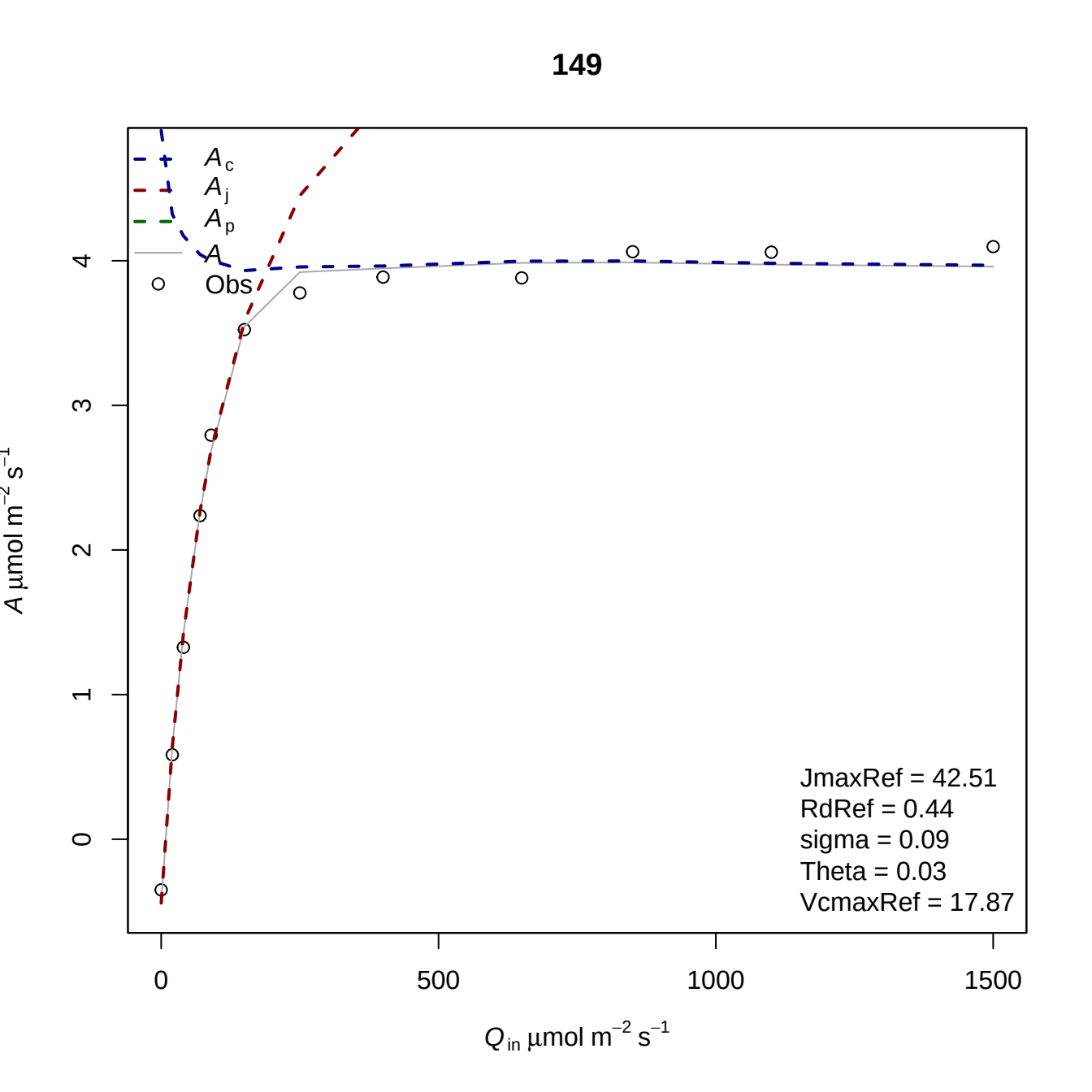

A_c
A_j
A_p
A
Obs
JmaxRef = 236.74
RdRef = 1.22
sigma = 0.13
Theta = −5.04
VcmaxRef = 42.4
0
2
4
6
8
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

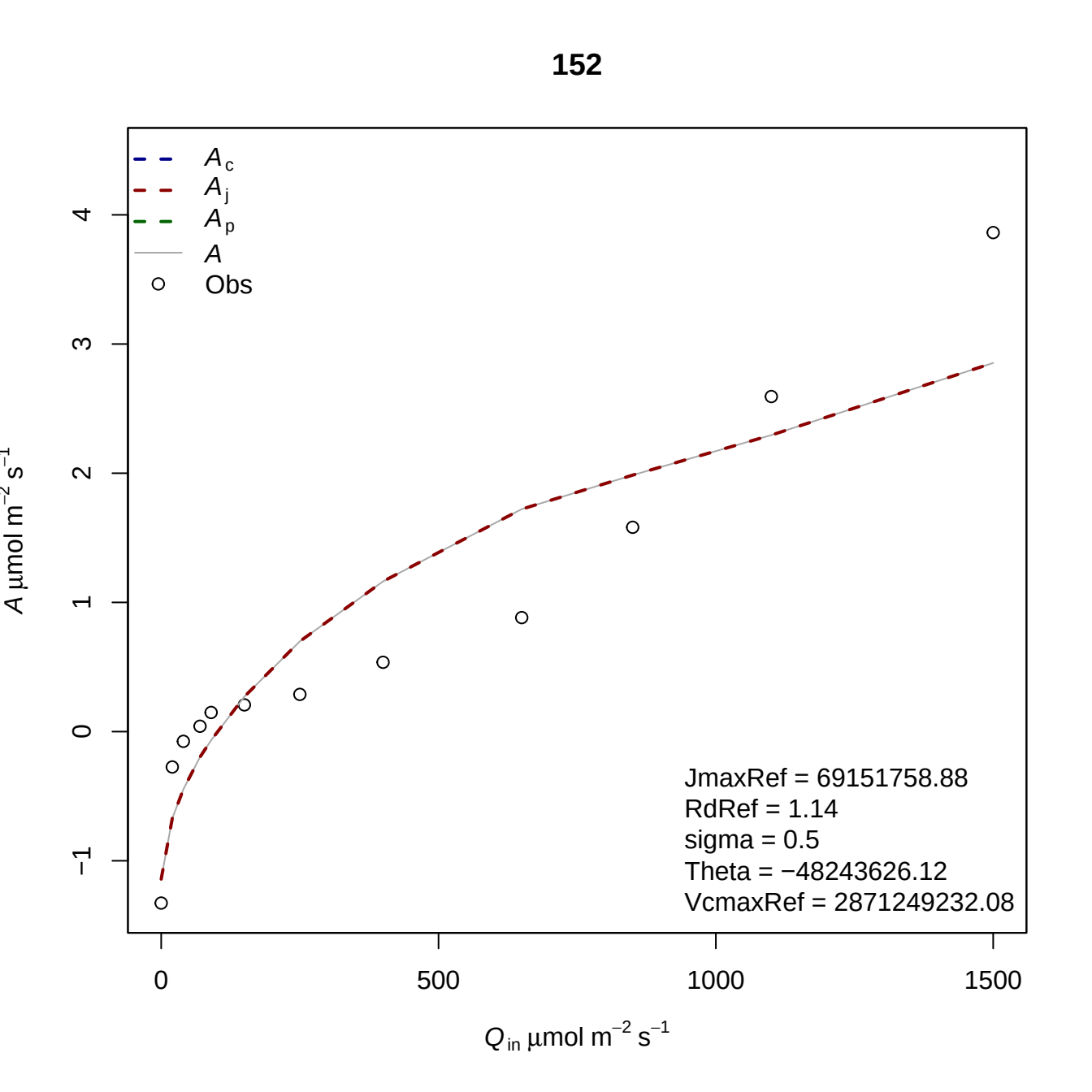

A_c
A_j
A_p
A
Obs
JmaxRef = 60.12
RdRef = 0.66
sigma = 0.15
Theta = −2.43
VcmaxRef = 11036.16
0
2
4
6
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

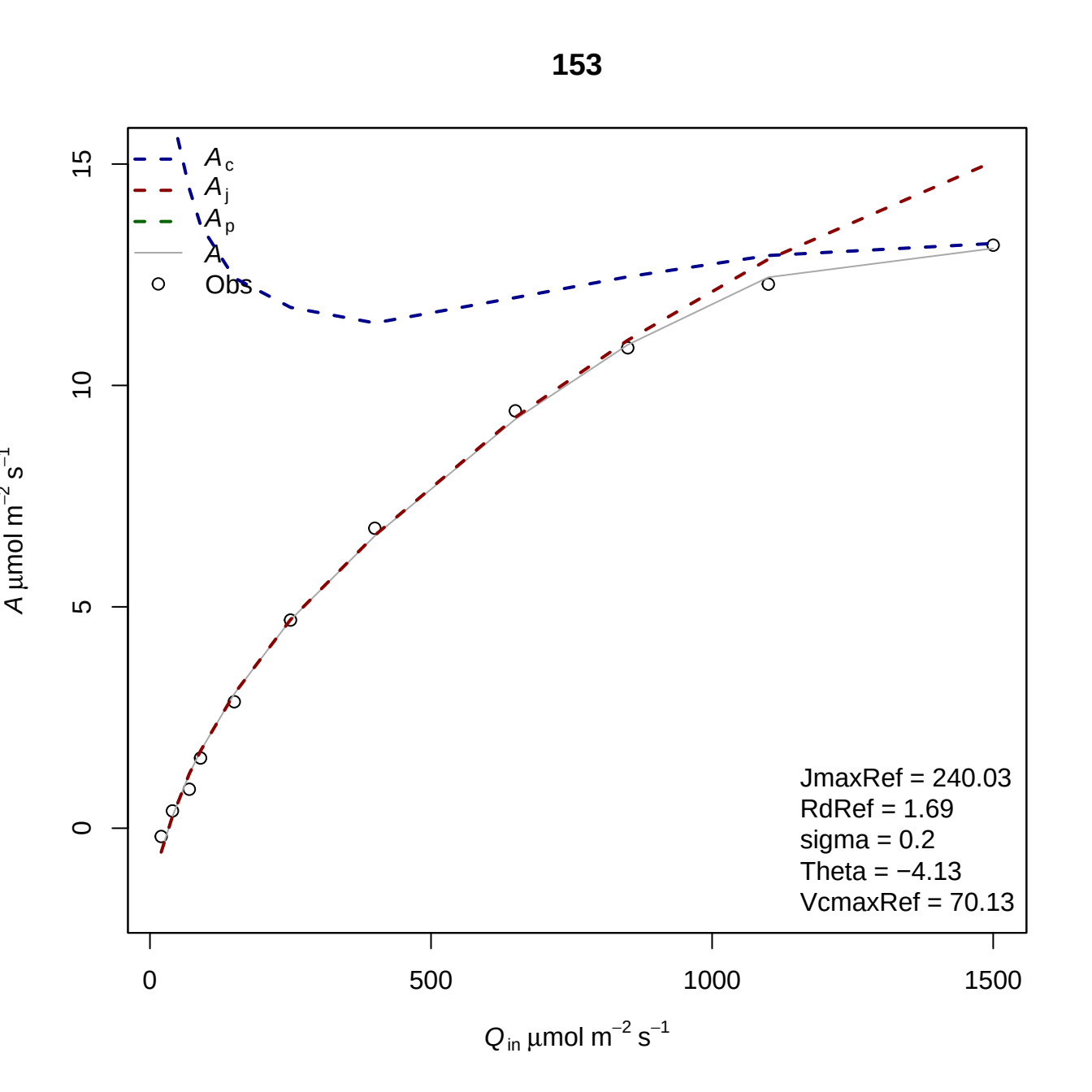

A_c
A_j
A_p
A
Obs
JmaxRef = 72.69
RdRef = 0.3
sigma = 0.28
Theta = −2.63
VcmaxRef = 66.03
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

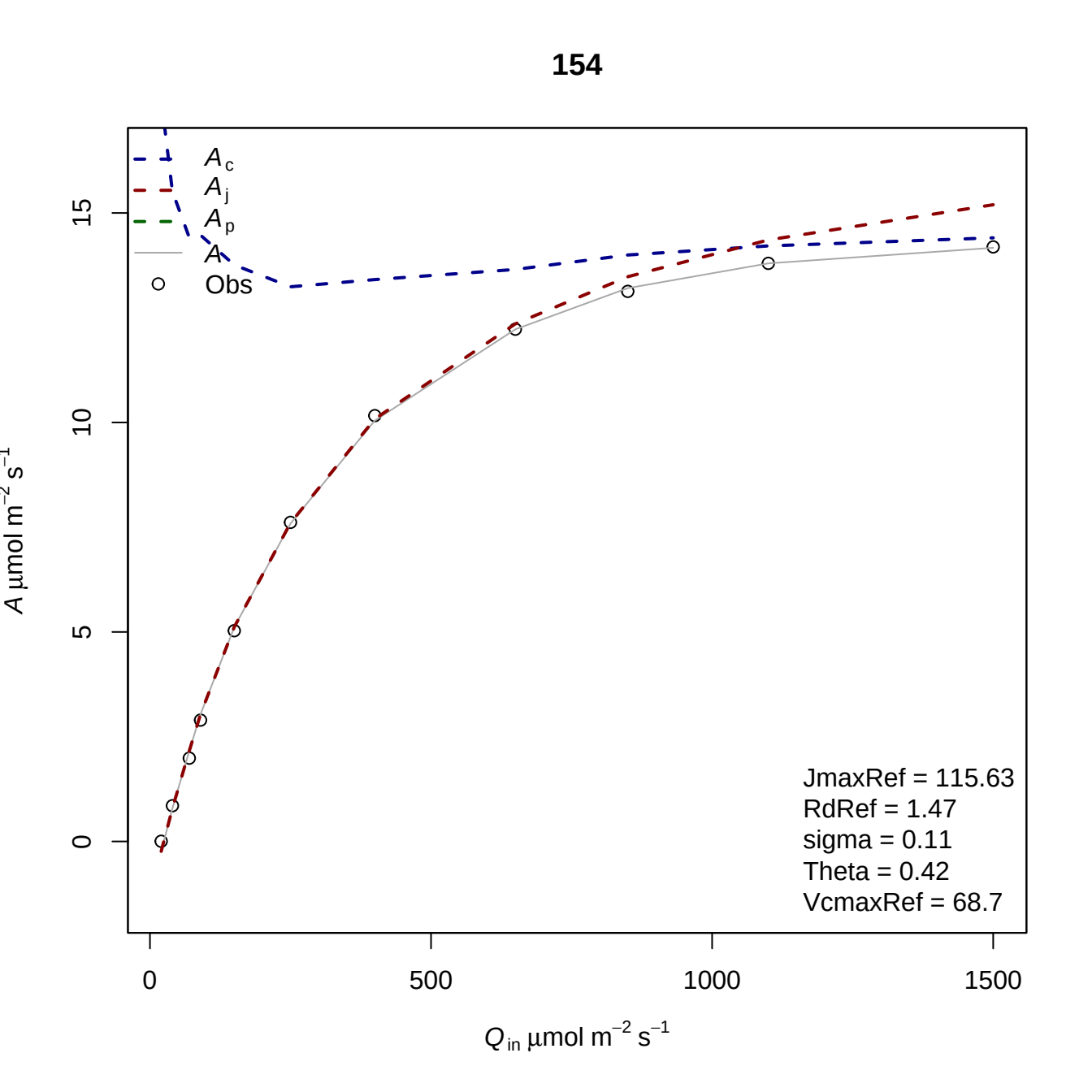

147
A_c
A_j
A_p
A
Obs
JmaxRef = 48.81
RdRef = 0.74
sigma = 0.09
Theta = 0.34
VcmaxRef = 23.61
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

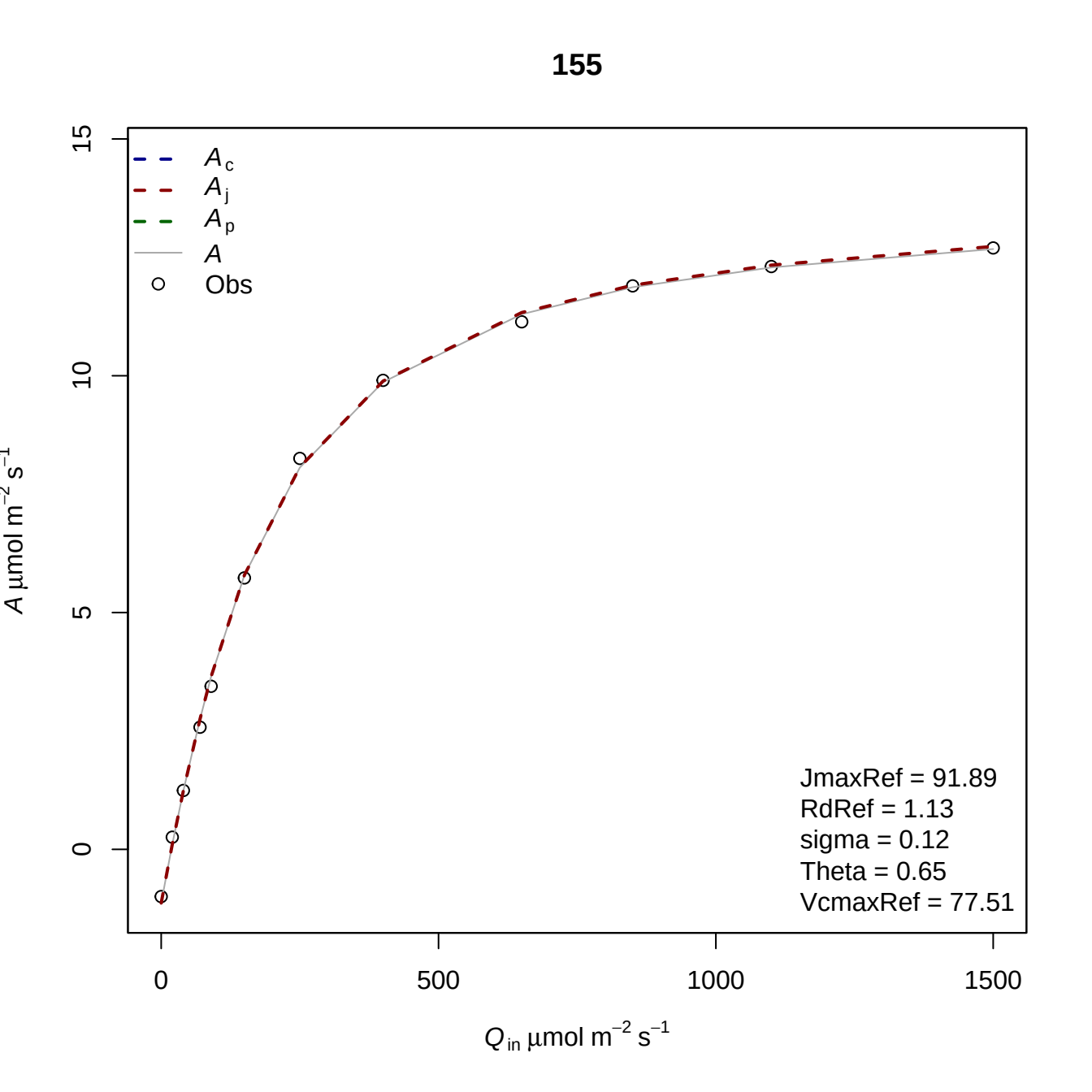

148
A_c
A_j
A_p
A
Obs
JmaxRef = 36.18
RdRef = 0.51
sigma = 0.05
Theta = 0.57
VcmaxRef = 20.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

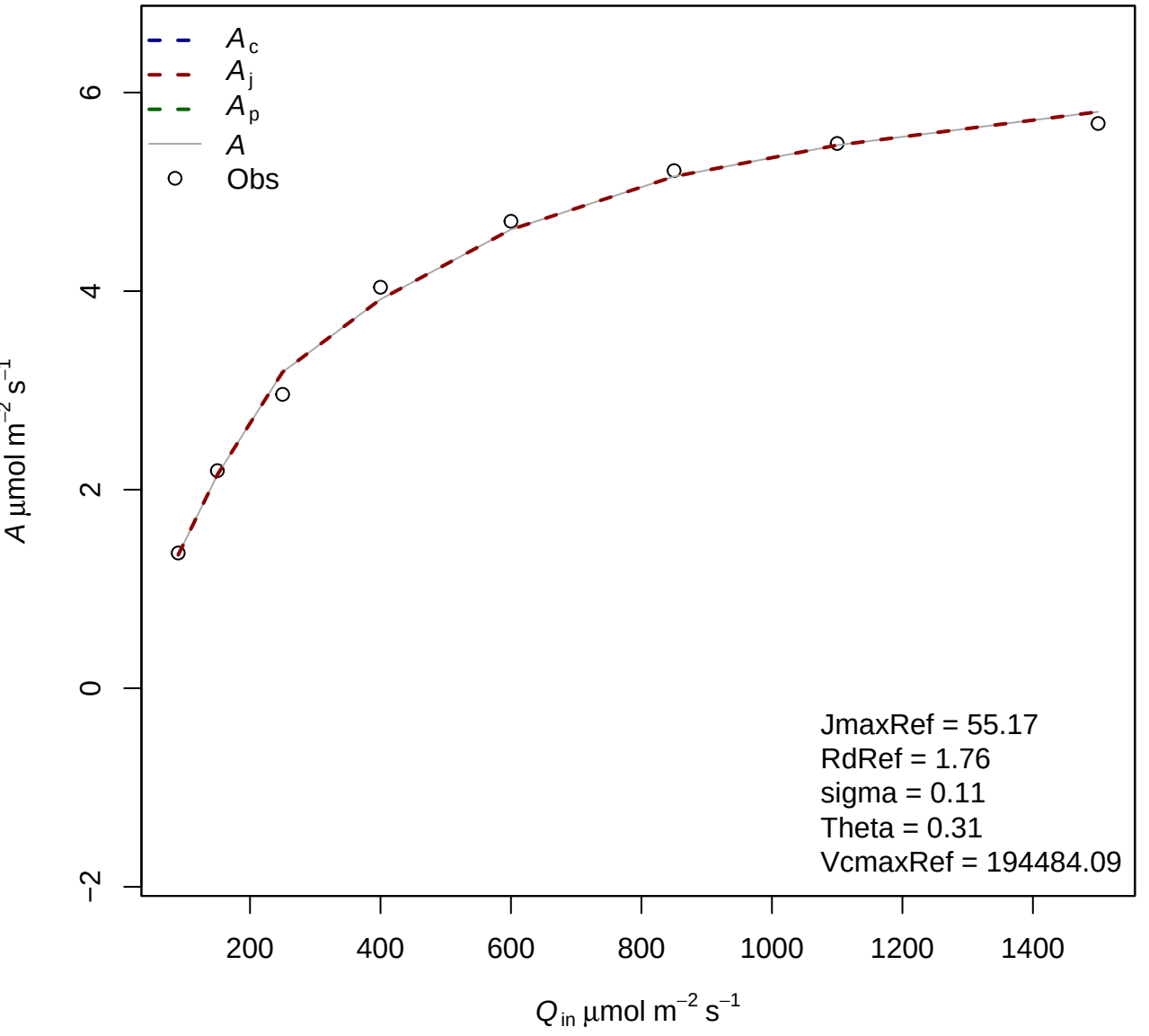

149
A_c
A_j
A_p
A
Obs
JmaxRef = 42.51
RdRef = 0.44
sigma = 0.09
Theta = 0.03
VcmaxRef = 17.87
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

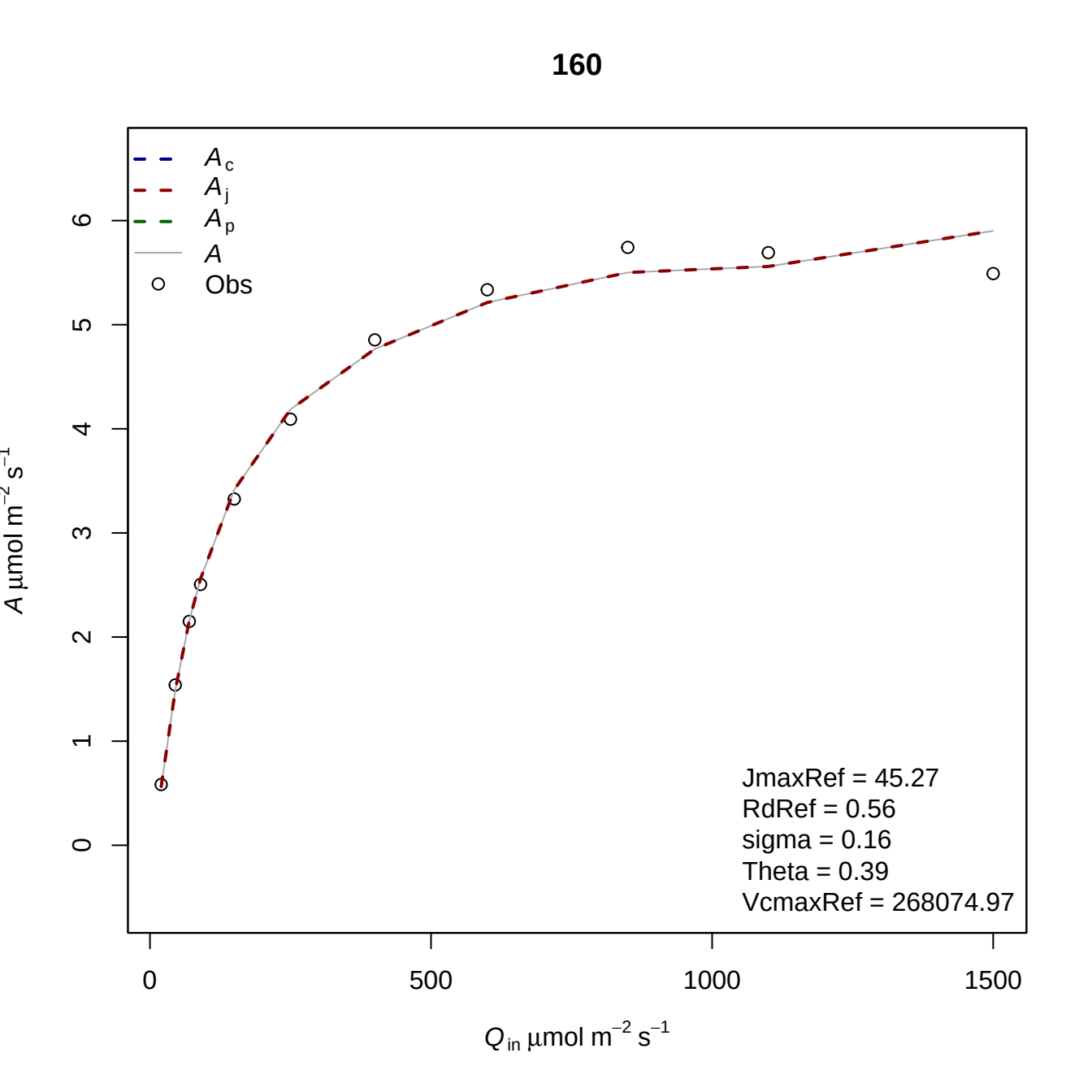

152
A_c
A_j
A_p
A
Obs
JmaxRef = 69151758.88
RdRef = 1.14
sigma = 0.5
Theta = −48243626.12
VcmaxRef = 2871249232.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

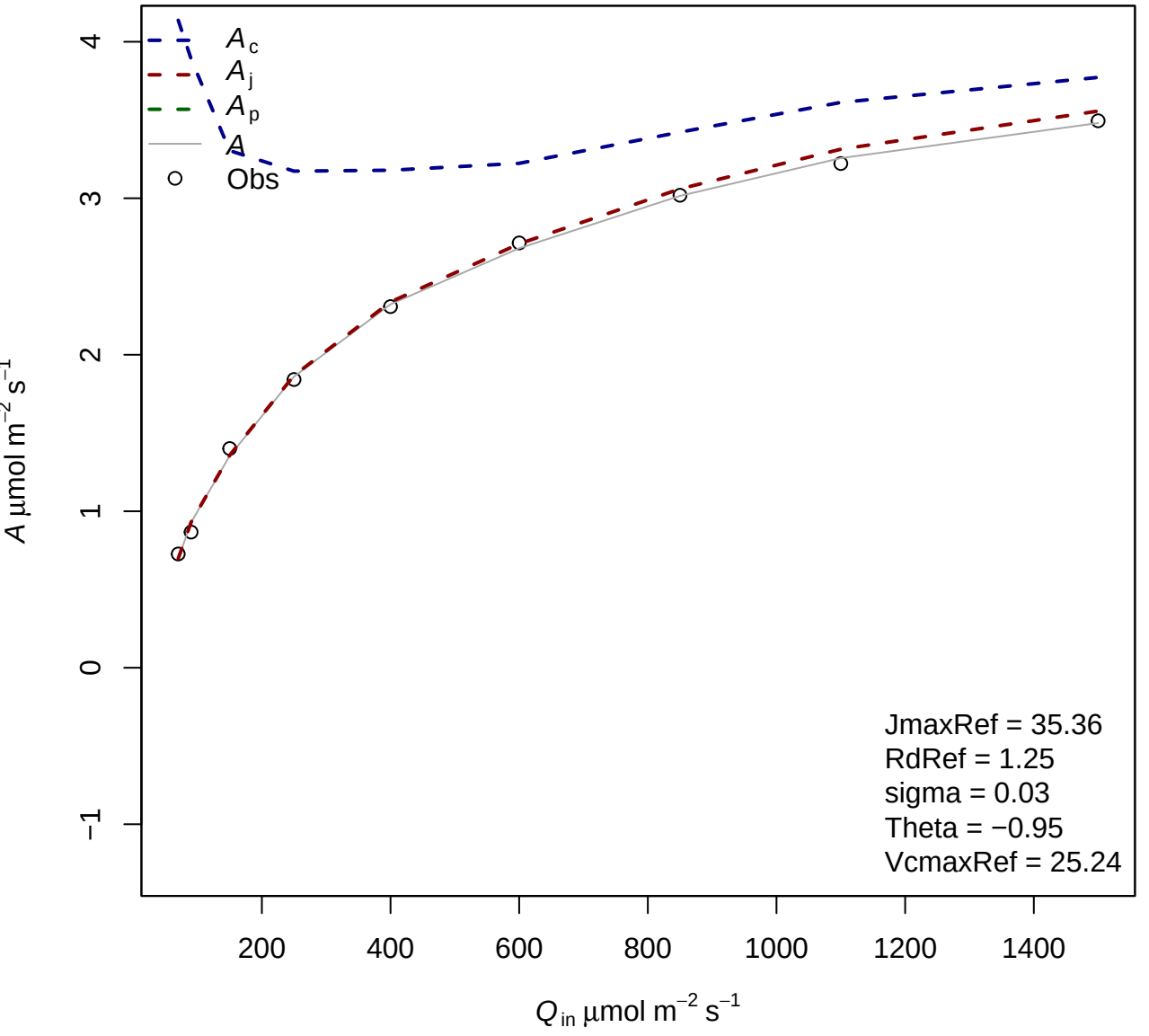

153
A_c
A_j
A_p
A
Obs
JmaxRef = 240.03
RdRef = 1.69
sigma = 0.2
Theta = −4.13
VcmaxRef = 70.13
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

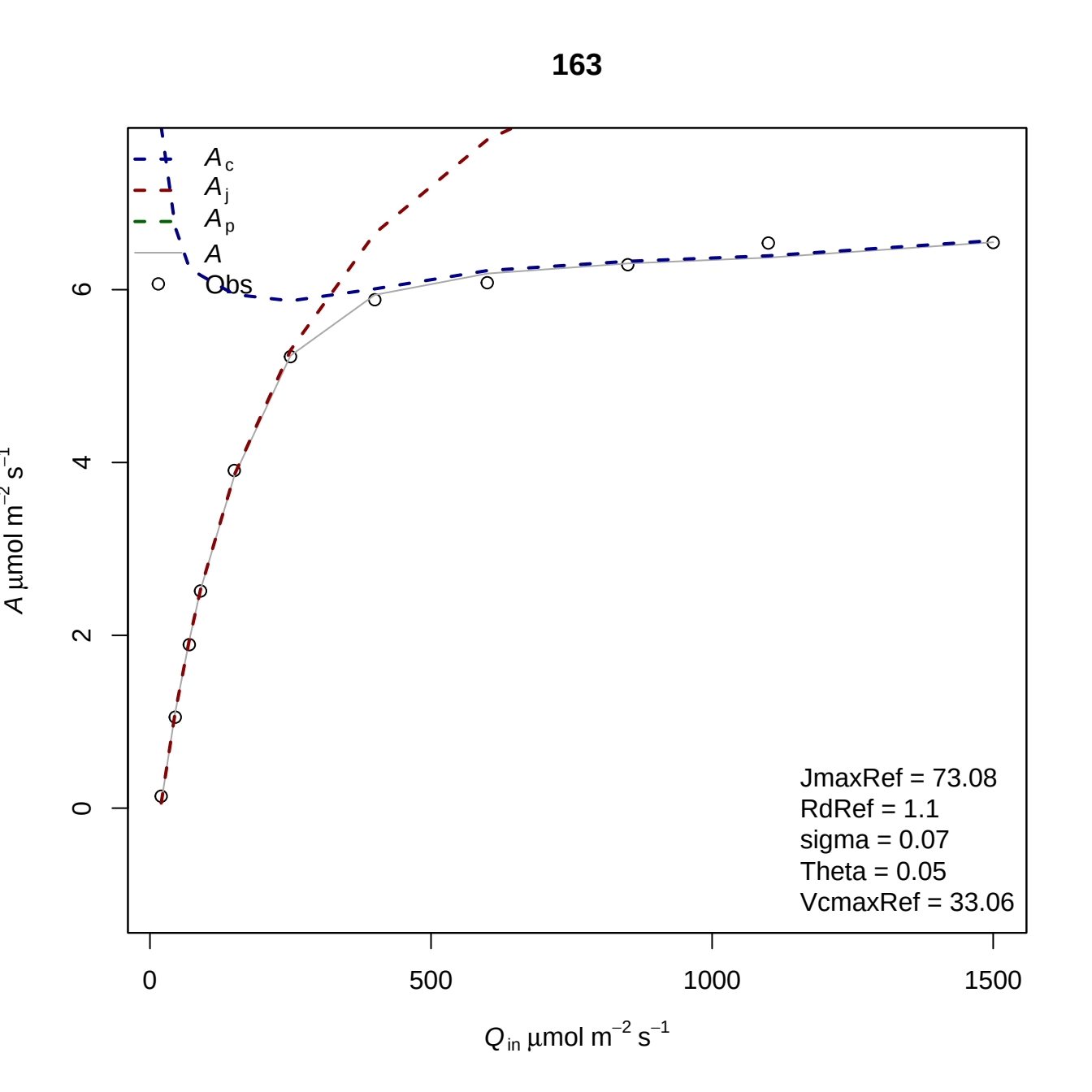

154
A_c
A_j
A_p
A
Obs
JmaxRef = 115.63
RdRef = 1.47
sigma = 0.11
Theta = 0.42
VcmaxRef = 68.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

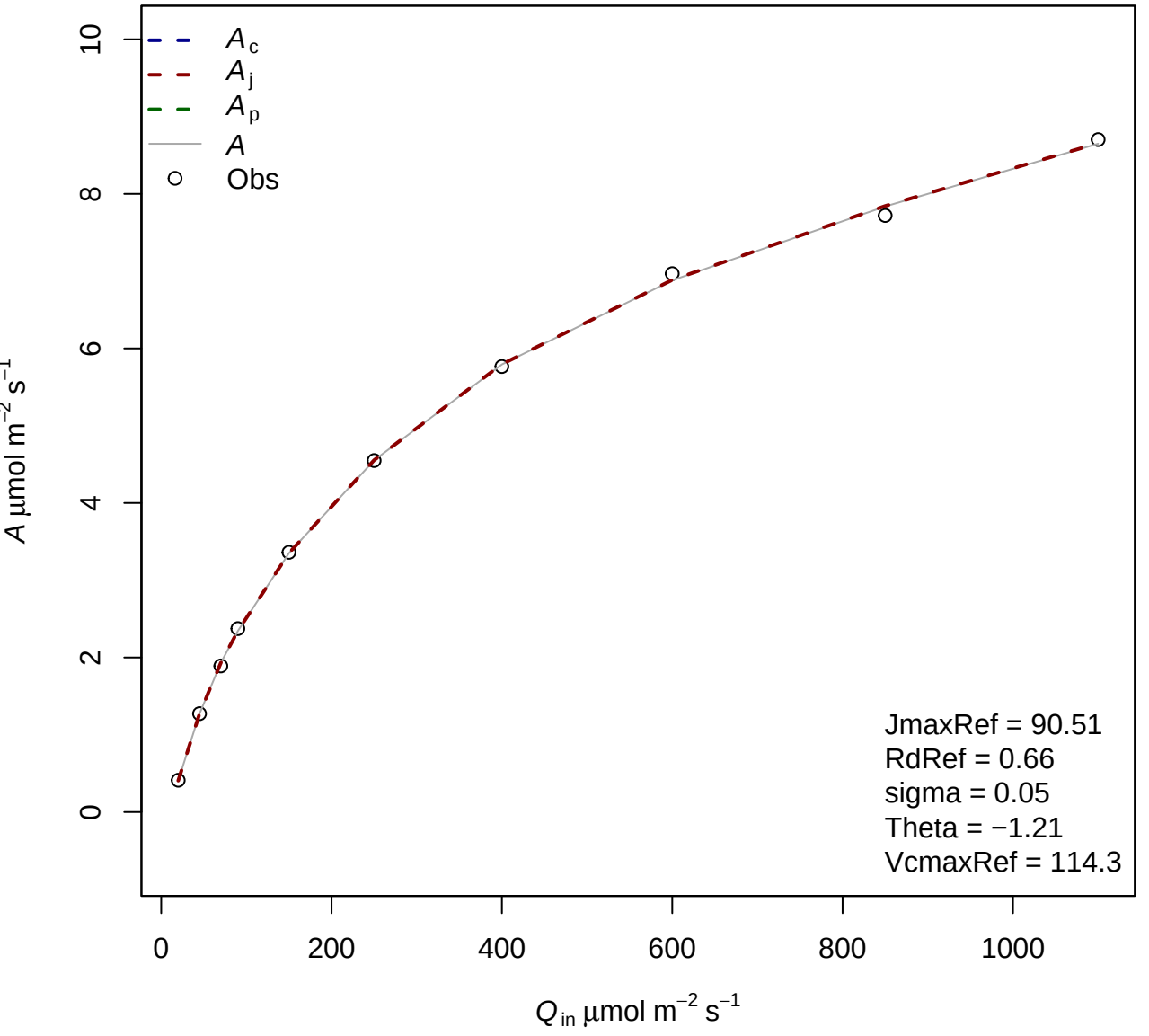

155
A_c
A_j
A_p
A
Obs
JmaxRef = 91.89
RdRef = 1.13
sigma = 0.12
Theta = 0.65
VcmaxRef = 77.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

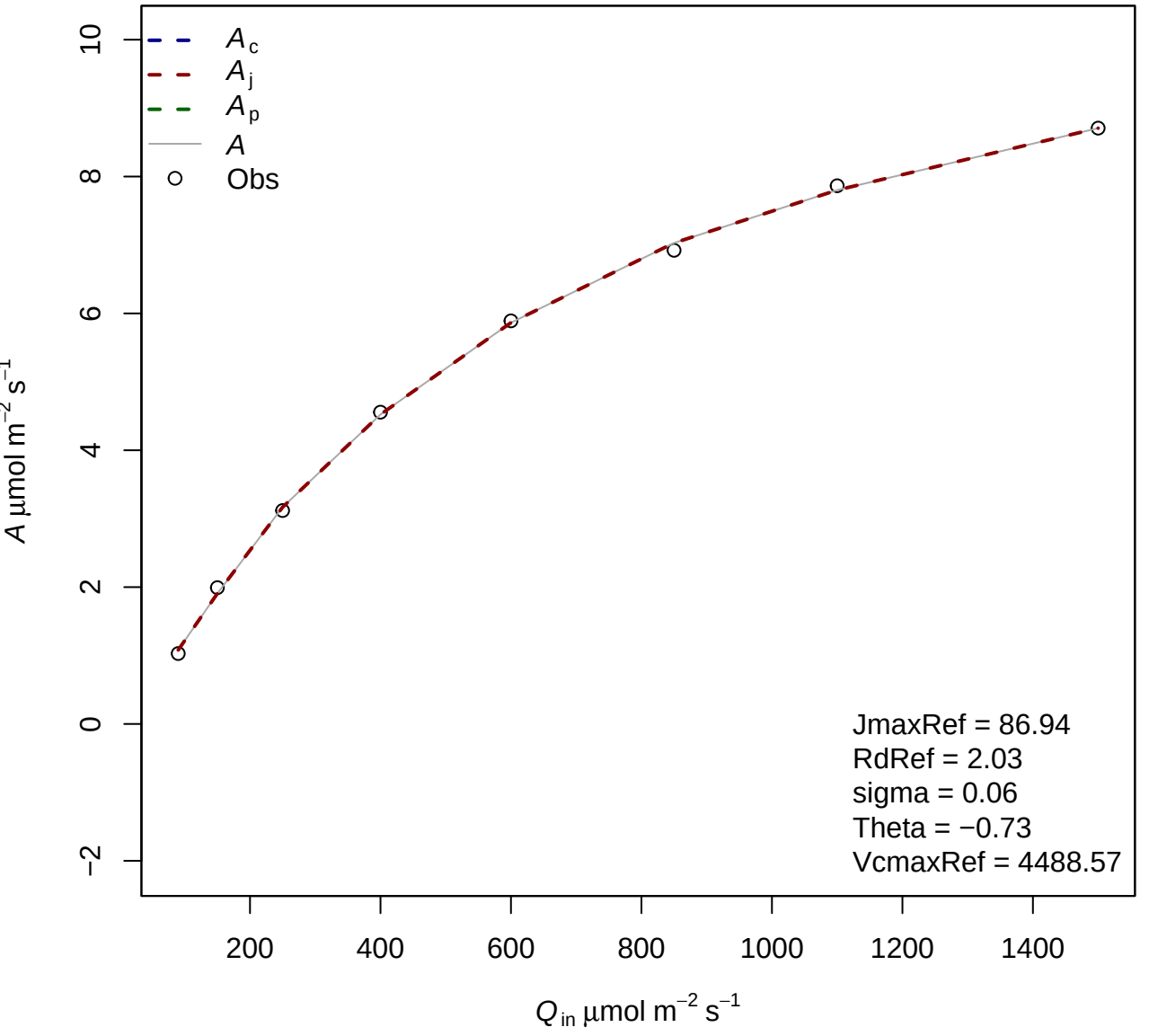

A_c
A_j
A_p
A
Obs
JmaxRef = 55.17
RdRef = 1.76
sigma = 0.11
Theta = 0.31
VcmaxRef = 194484.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000
1200
1400

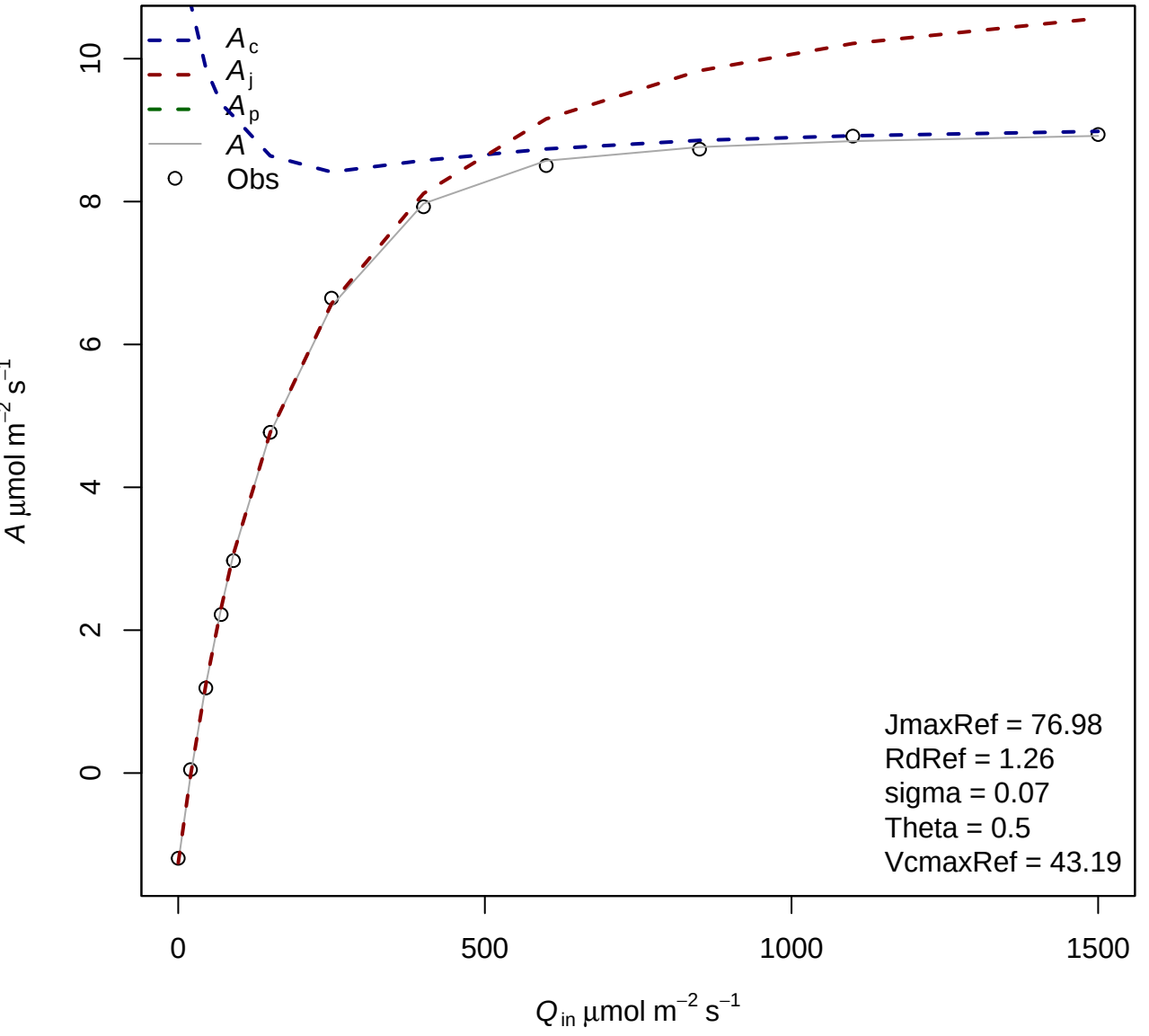

160
A_c
A_j
A_p
A
Obs
JmaxRef = 45.27
RdRef = 0.56
sigma = 0.16
Theta = 0.39
VcmaxRef = 268074.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

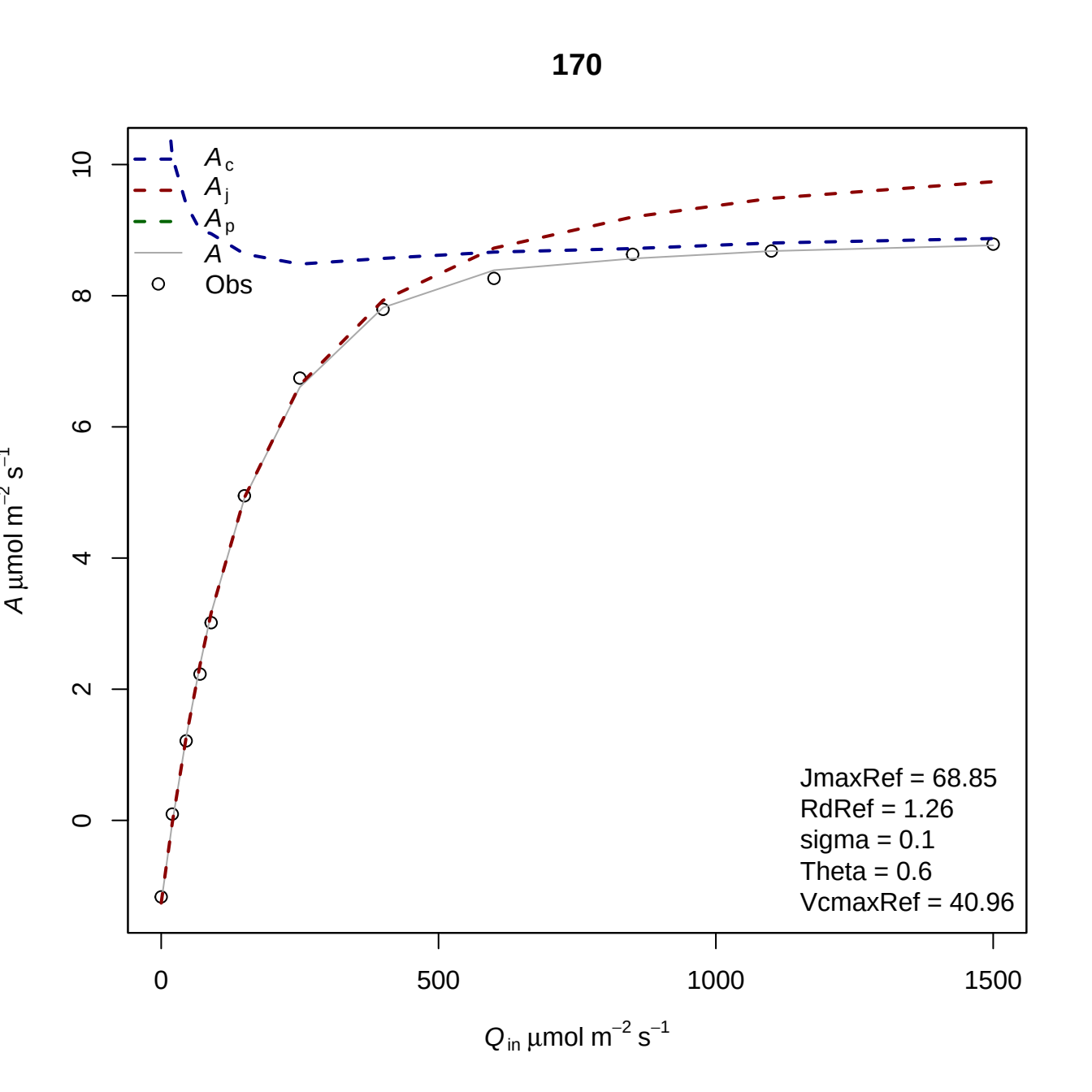

A_c
A_j
A_p
A
Obs
JmaxRef = 35.36
RdRef = 1.25
sigma = 0.03
Theta = −0.95
VcmaxRef = 25.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
200
400
600
800
1000
1200
1400

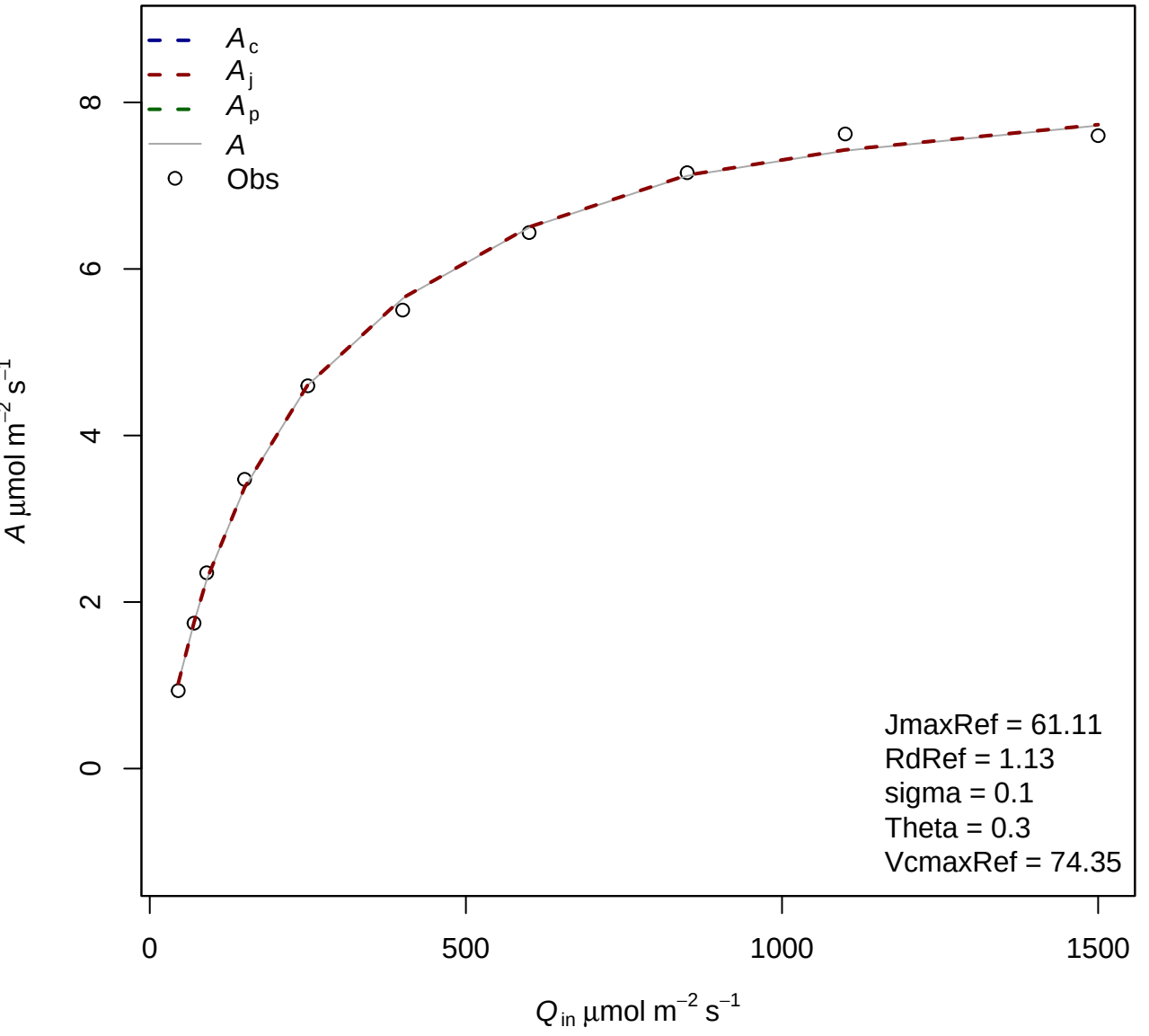

163
A_c
A_j
A_p
A
Obs
JmaxRef = 73.08
RdRef = 1.1
sigma = 0.07
Theta = 0.05
VcmaxRef = 33.06
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

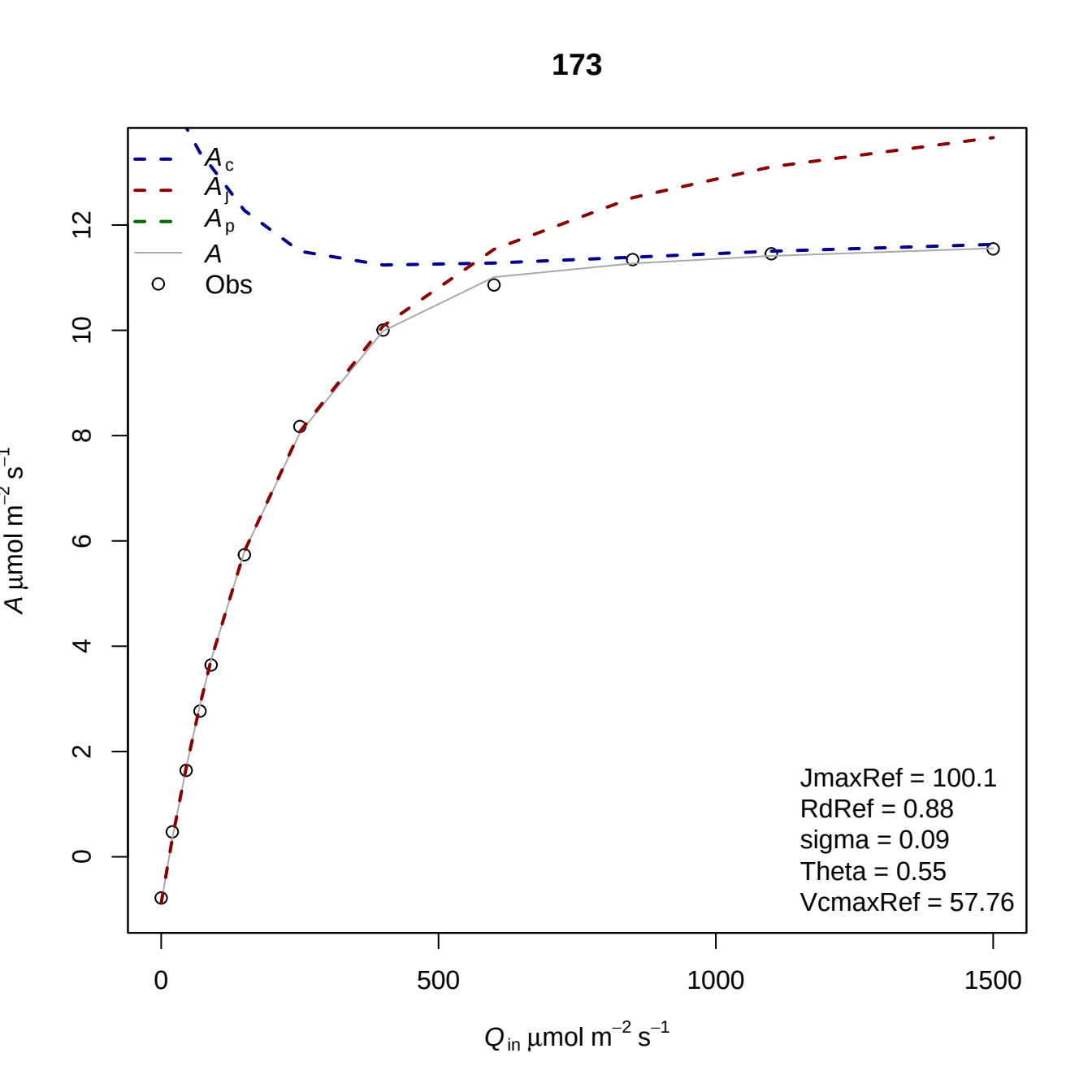

A_c
A_j
A_p
A
Obs
JmaxRef = 90.51
RdRef = 0.66
sigma = 0.05
Theta = −1.21
VcmaxRef = 114.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000

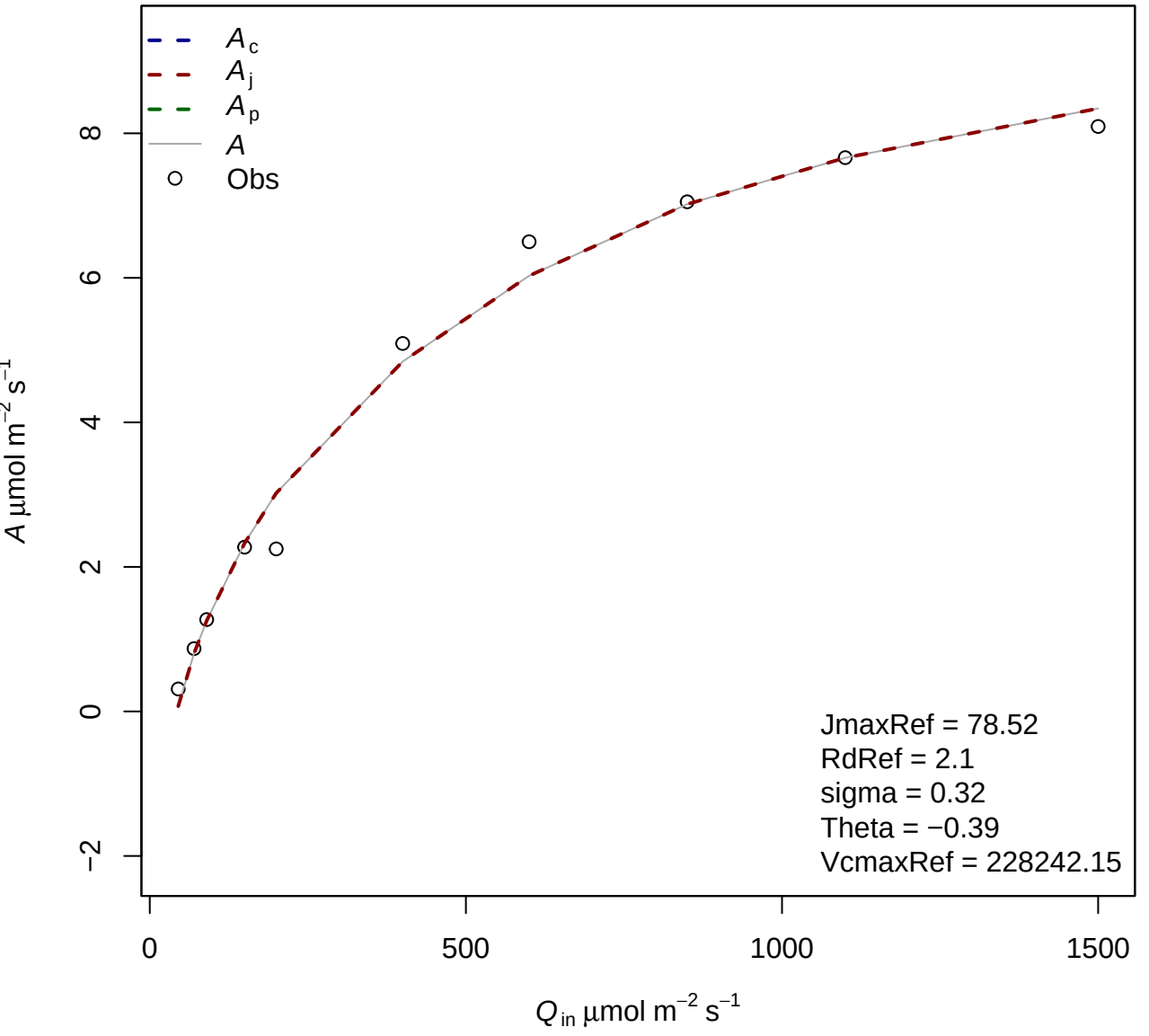

A_c
A_j
A_p
A
Obs
JmaxRef = 86.94
RdRef = 2.03
sigma = 0.06
Theta = −0.73
VcmaxRef = 4488.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

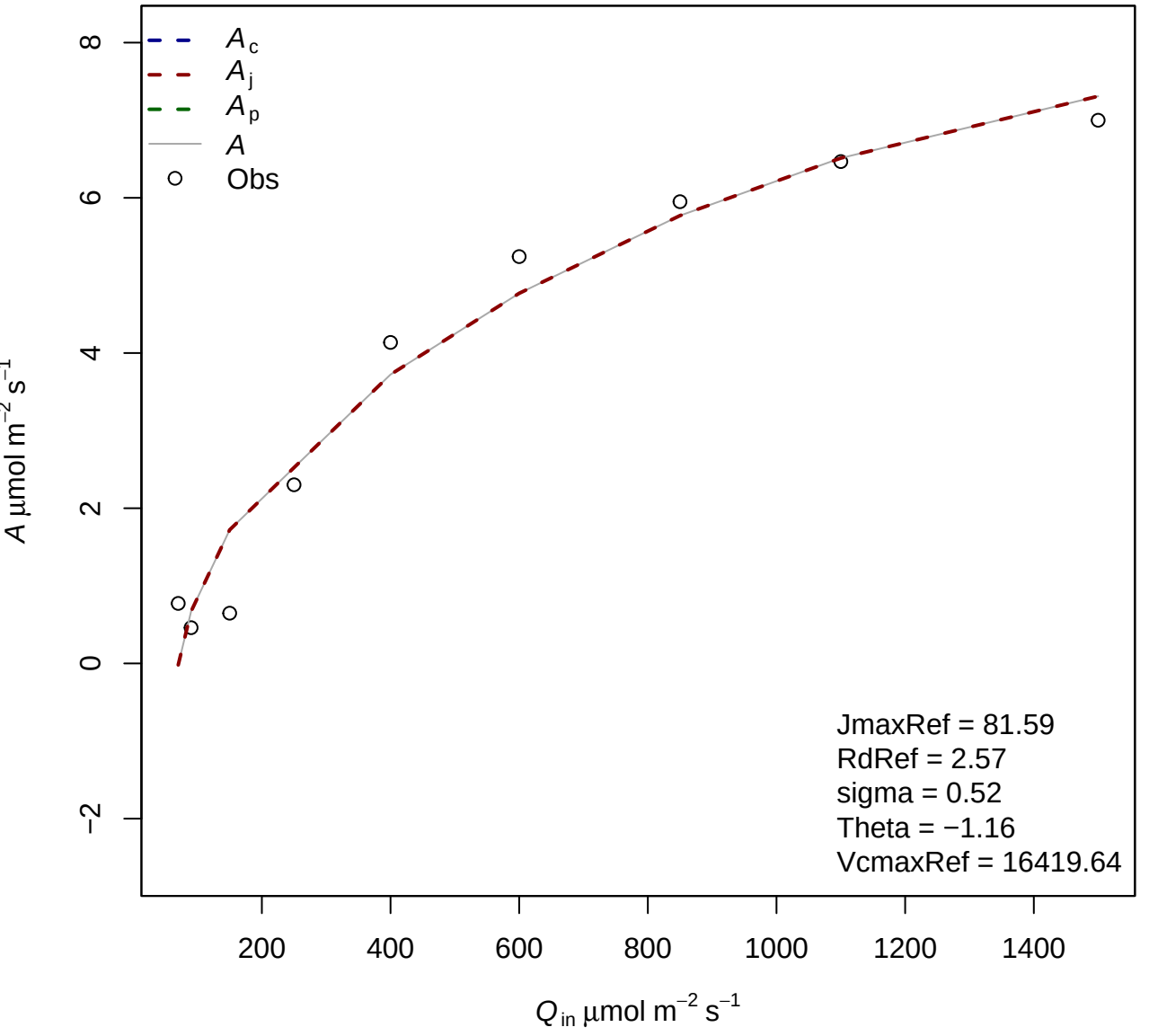

A_c
A_j
A_p
A
Obs
JmaxRef = 76.98
RdRef = 1.26
sigma = 0.07
Theta = 0.5
VcmaxRef = 43.19
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

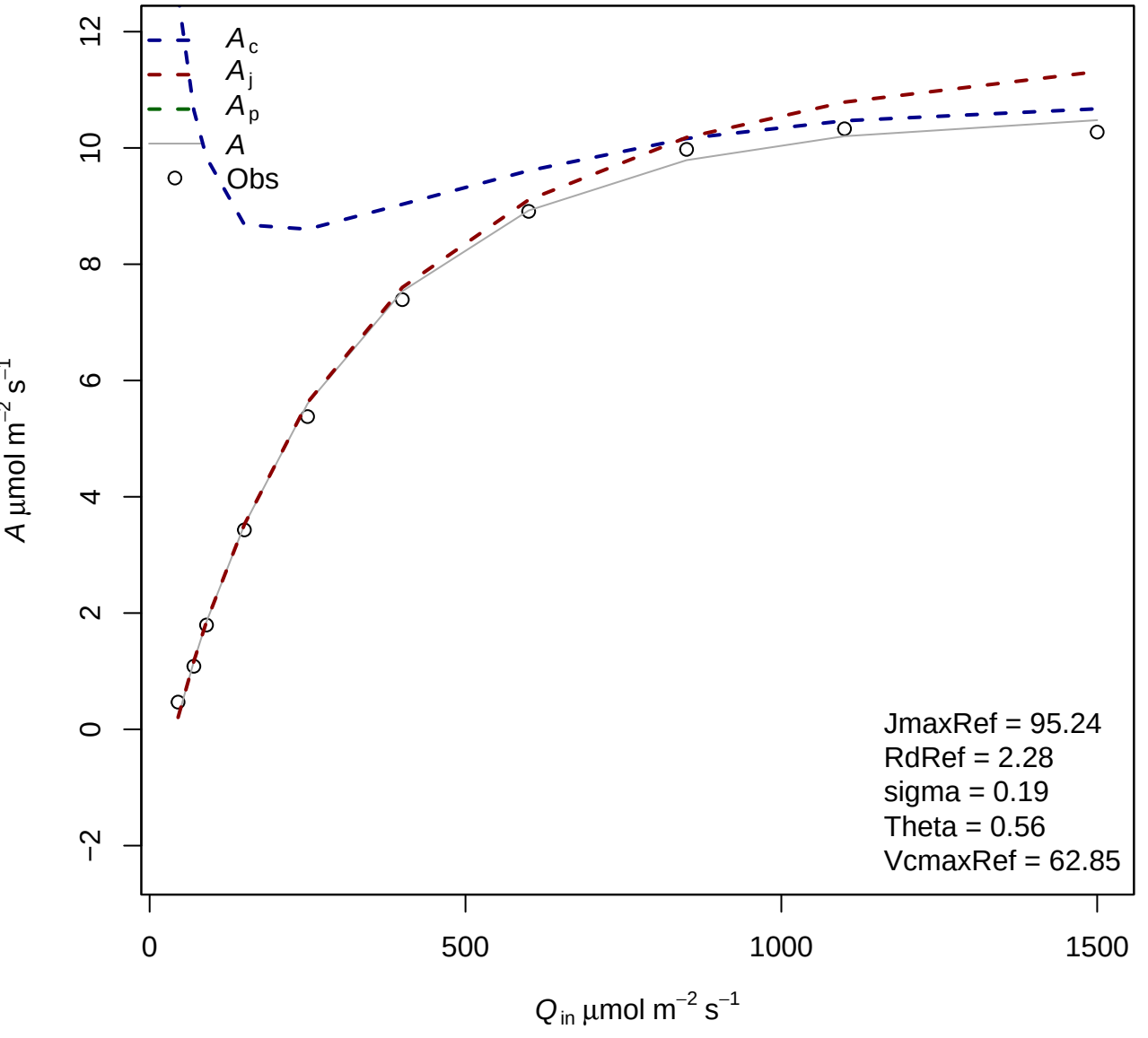

170
A_c
A_j
A_p
A
Obs
JmaxRef = 68.85
RdRef = 1.26
sigma = 0.1
Theta = 0.6
VcmaxRef = 40.96
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

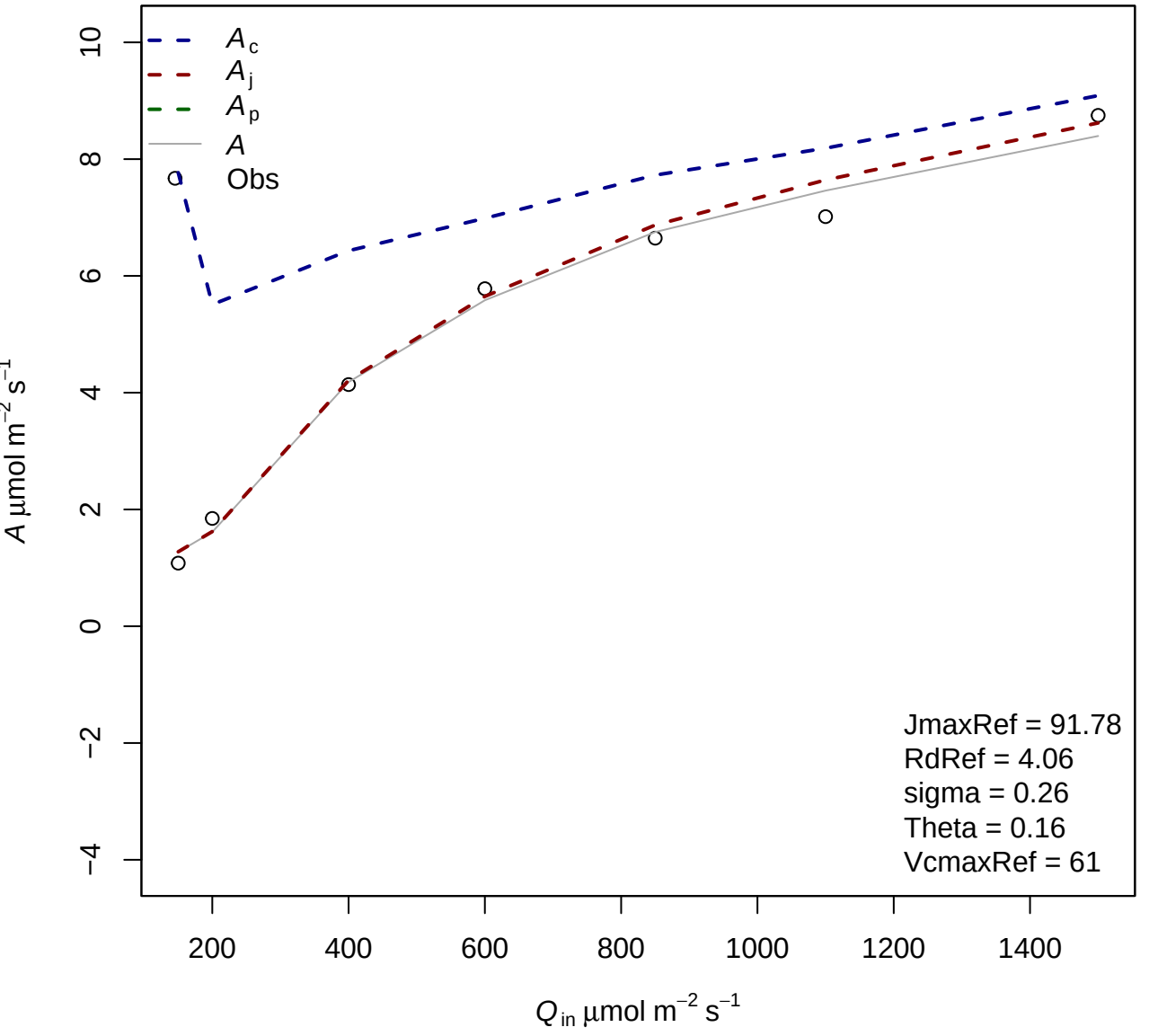

A_c
A_j
A_p
A
Obs
JmaxRef = 61.11
RdRef = 1.13
sigma = 0.1
Theta = 0.3
VcmaxRef = 74.35
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

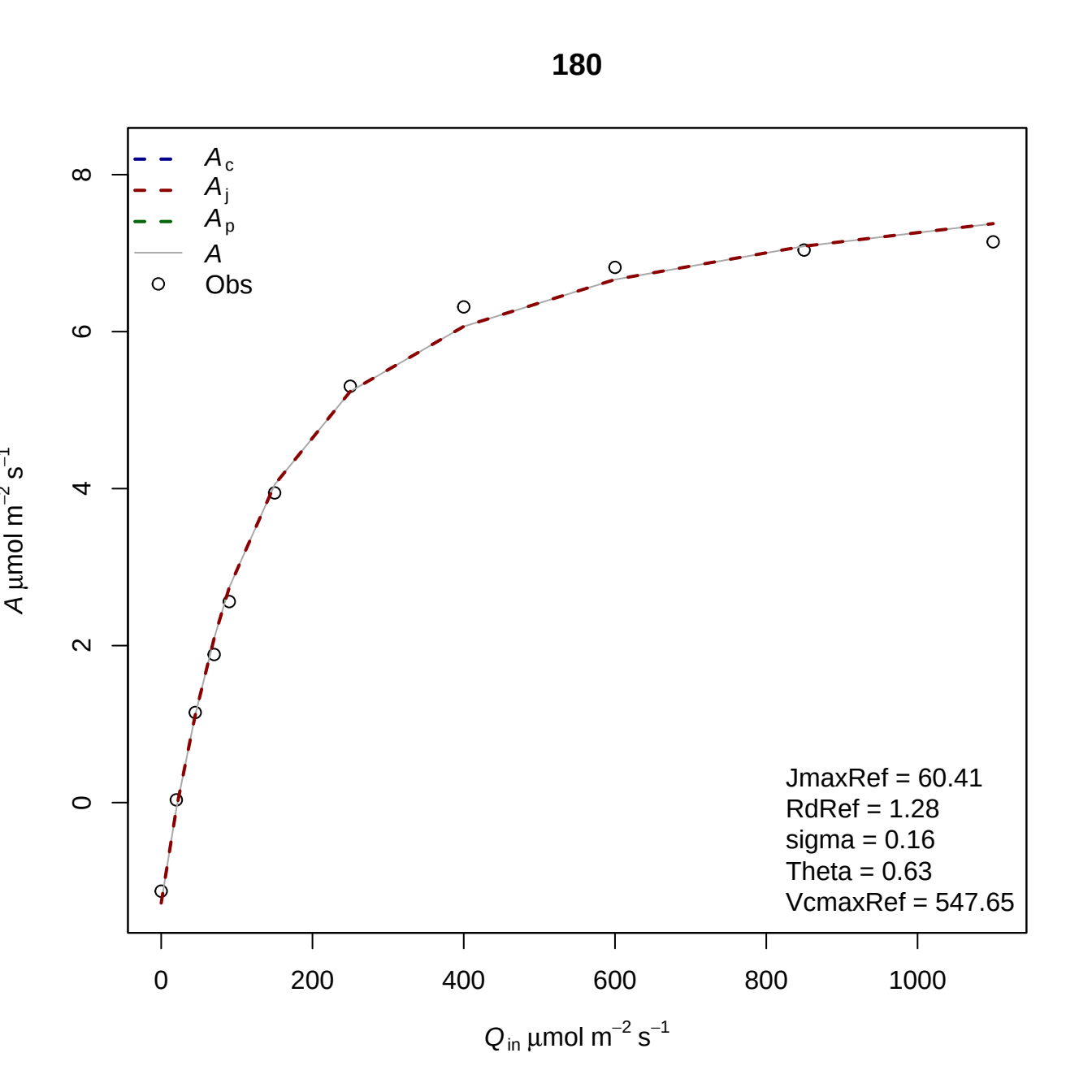

173
A_c
A_j
A_p
A
Obs
JmaxRef = 100.1
RdRef = 0.88
sigma = 0.09
Theta = 0.55
VcmaxRef = 57.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

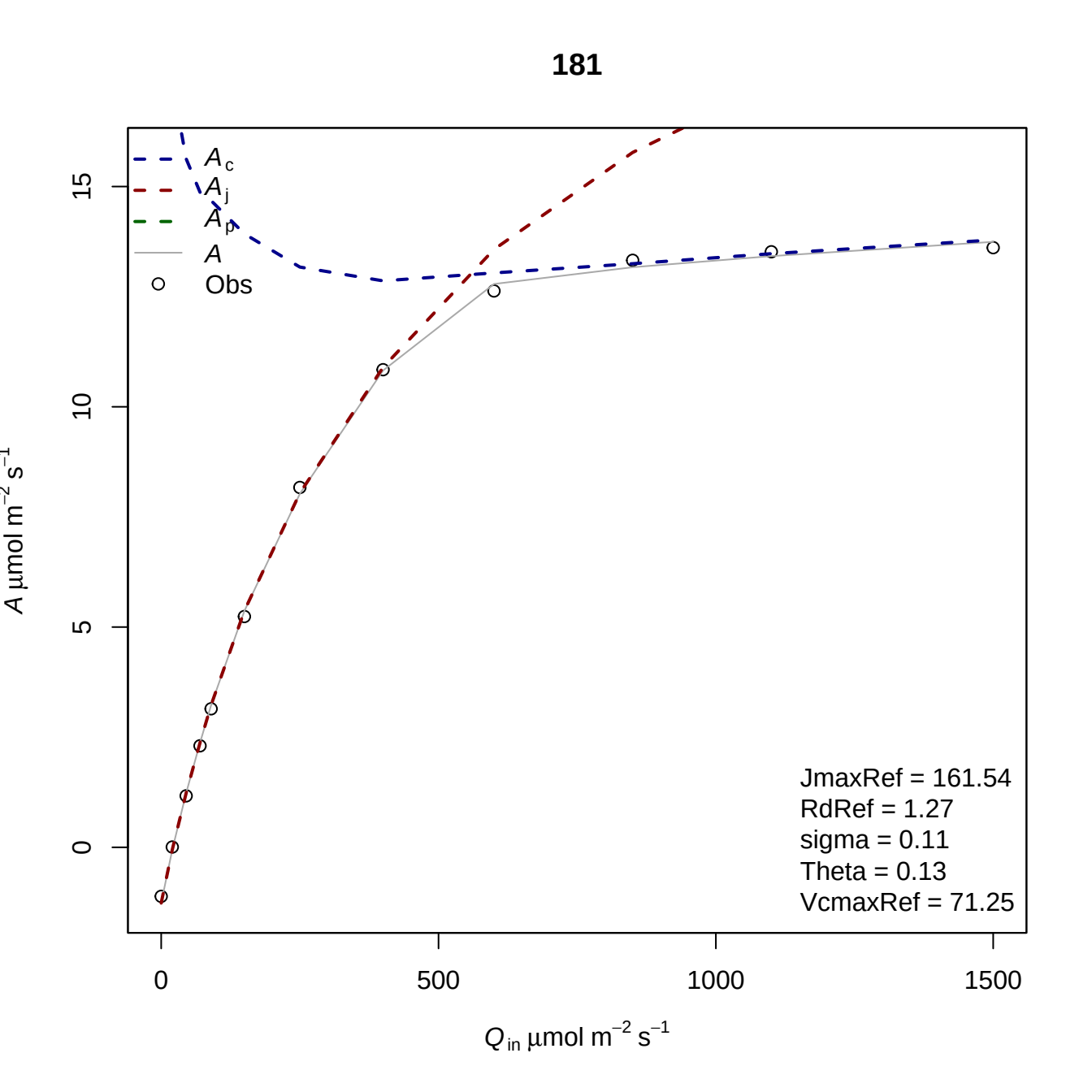

A_c
A_j
A_p
A
Obs
JmaxRef = 78.52
RdRef = 2.1
sigma = 0.32
Theta = −0.39
VcmaxRef = 228242.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

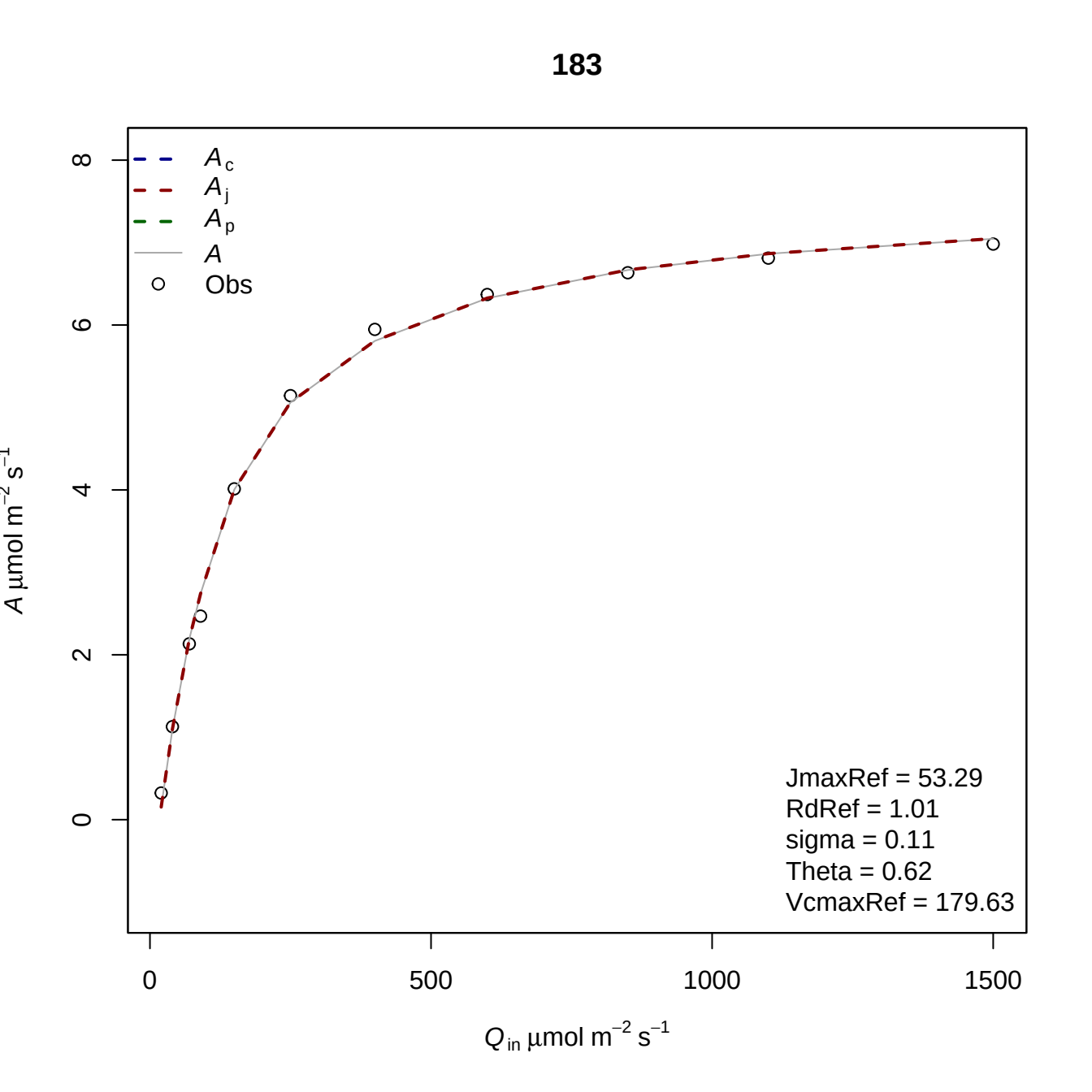

A_c
A_j
A_p
A
Obs
JmaxRef = 81.59
RdRef = 2.57
sigma = 0.52
Theta = −1.16
VcmaxRef = 16419.64
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

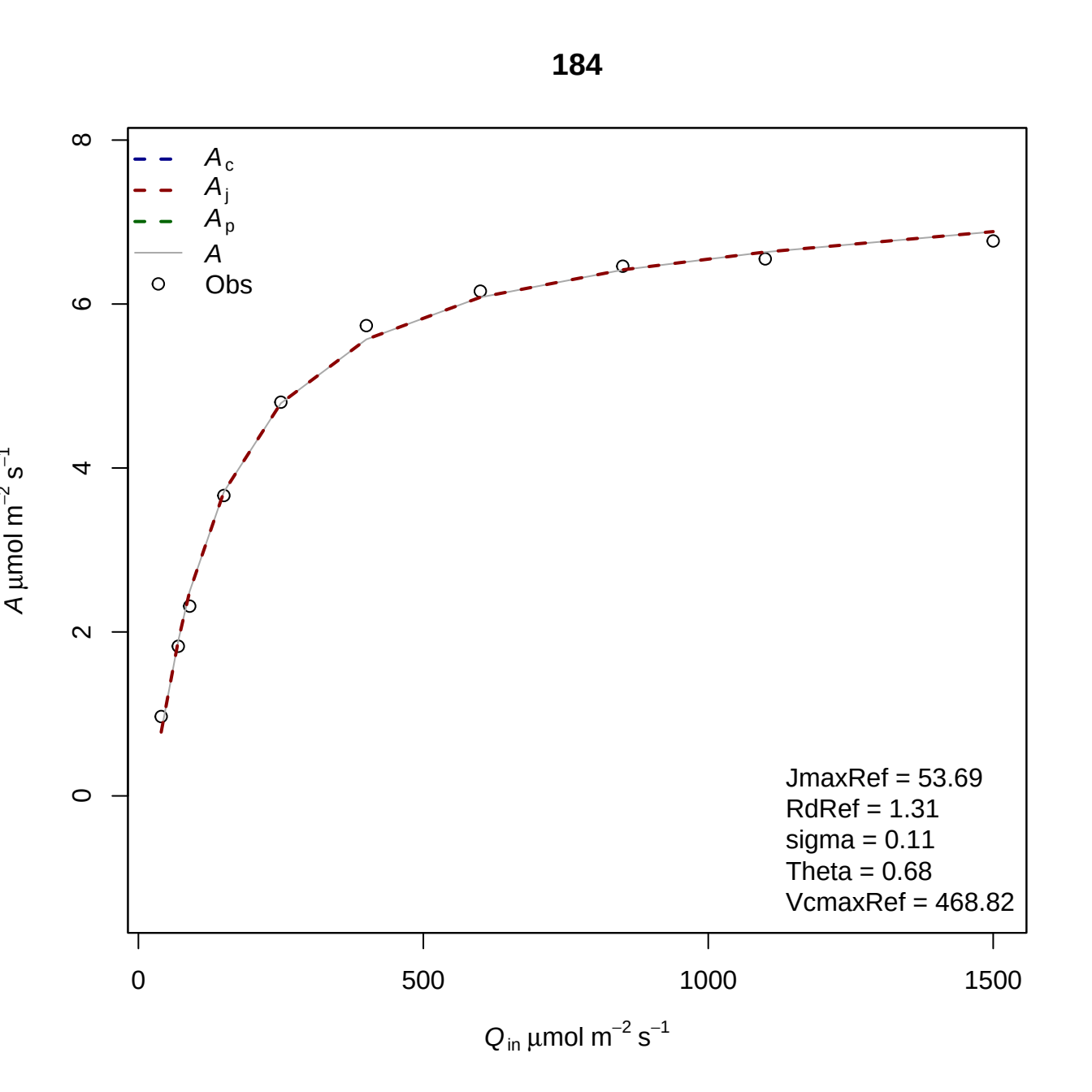

A_c
A_j
A_p
A
Obs
JmaxRef = 95.24
RdRef = 2.28
sigma = 0.19
Theta = 0.56
VcmaxRef = 62.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

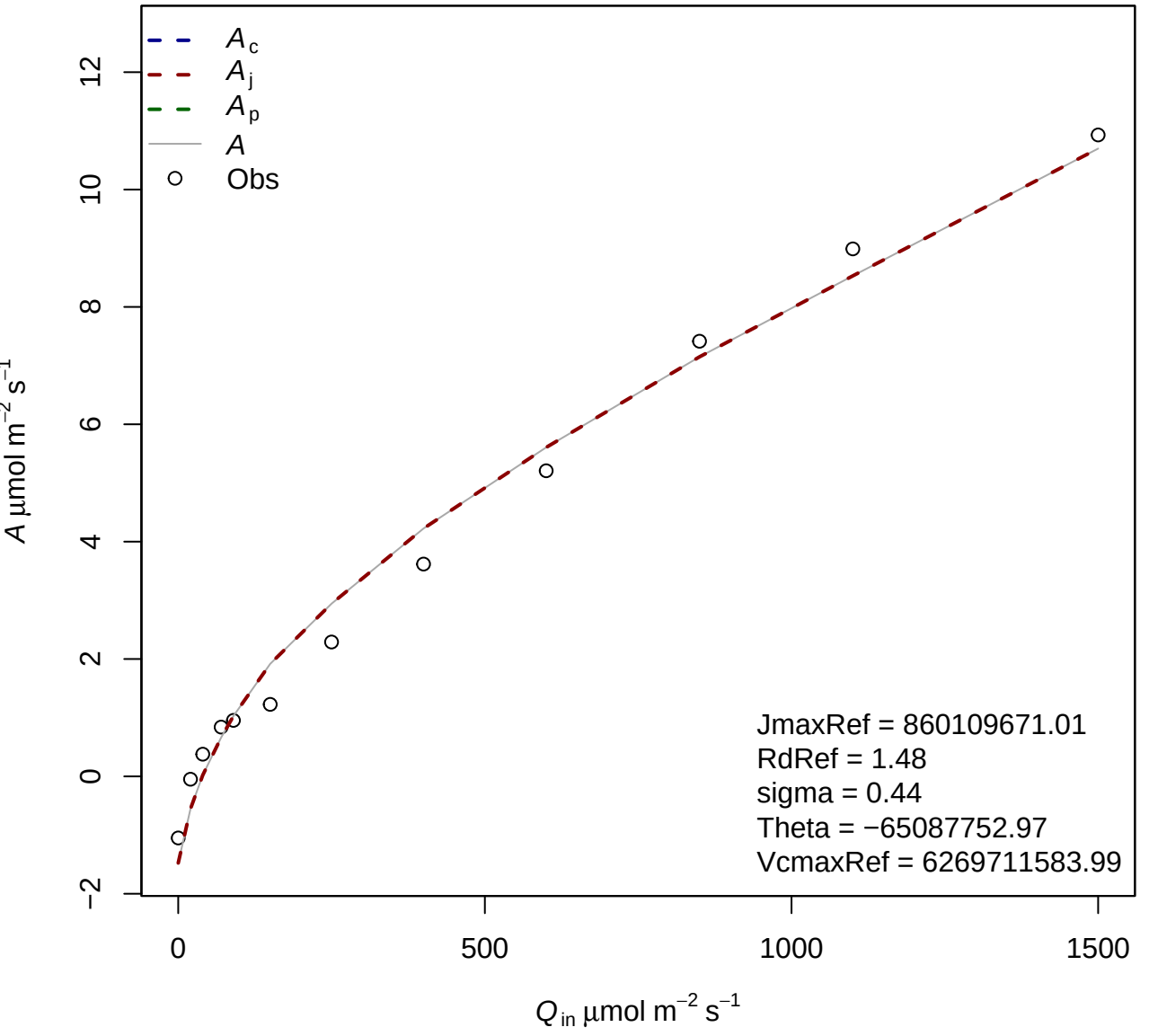

A_c
A_j
A_p
A
Obs
JmaxRef = 91.78
RdRef = 4.06
sigma = 0.26
Theta = 0.16
VcmaxRef = 61
10
8
6
4
2
0
−2
−4
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

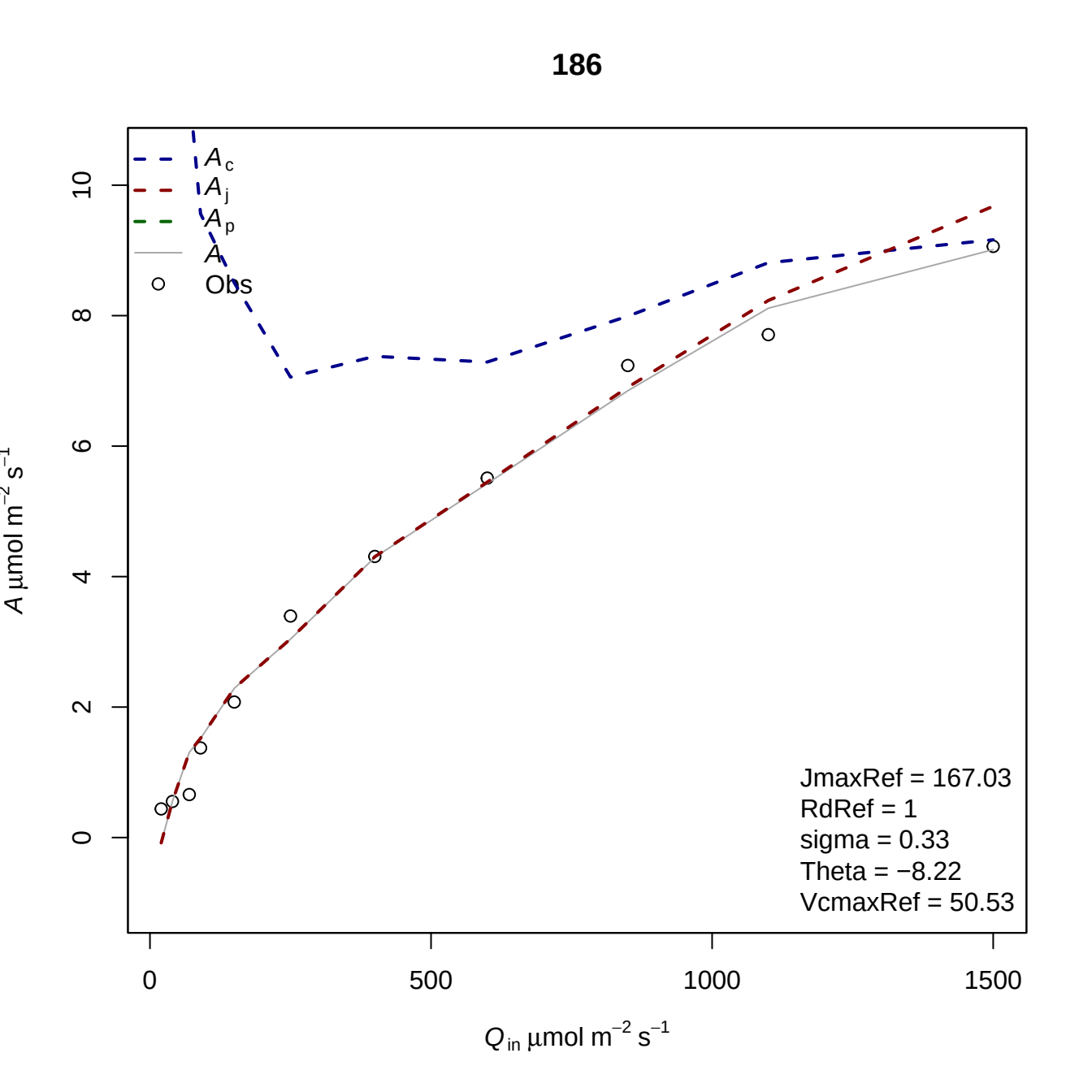

180
A_c
A_j
A_p
A
Obs
JmaxRef = 60.41
RdRef = 1.28
sigma = 0.16
Theta = 0.63
VcmaxRef = 547.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

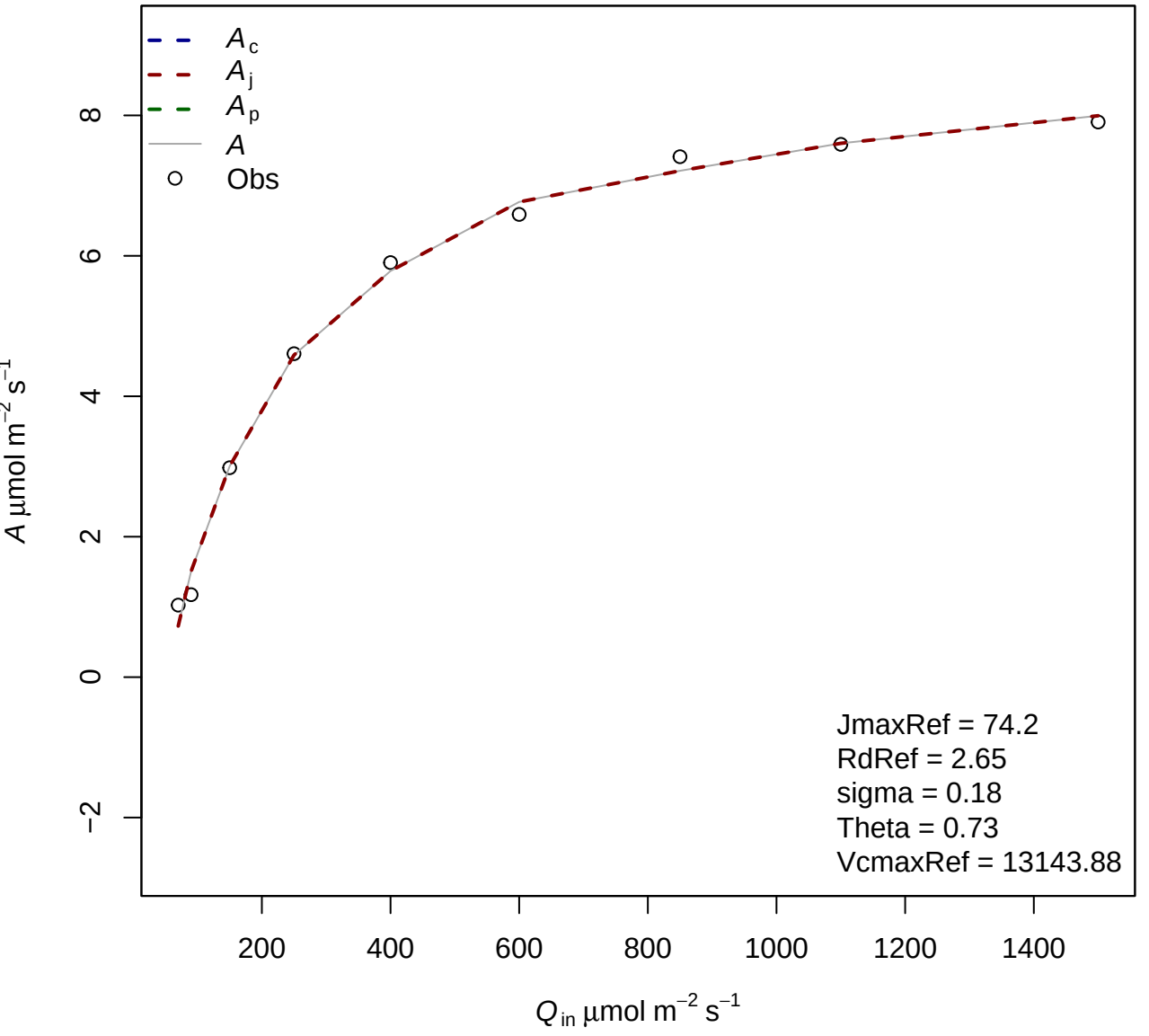

181
A_c
A_j
A_p
A
Obs
JmaxRef = 161.54
RdRef = 1.27
sigma = 0.11
Theta = 0.13
VcmaxRef = 71.25
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

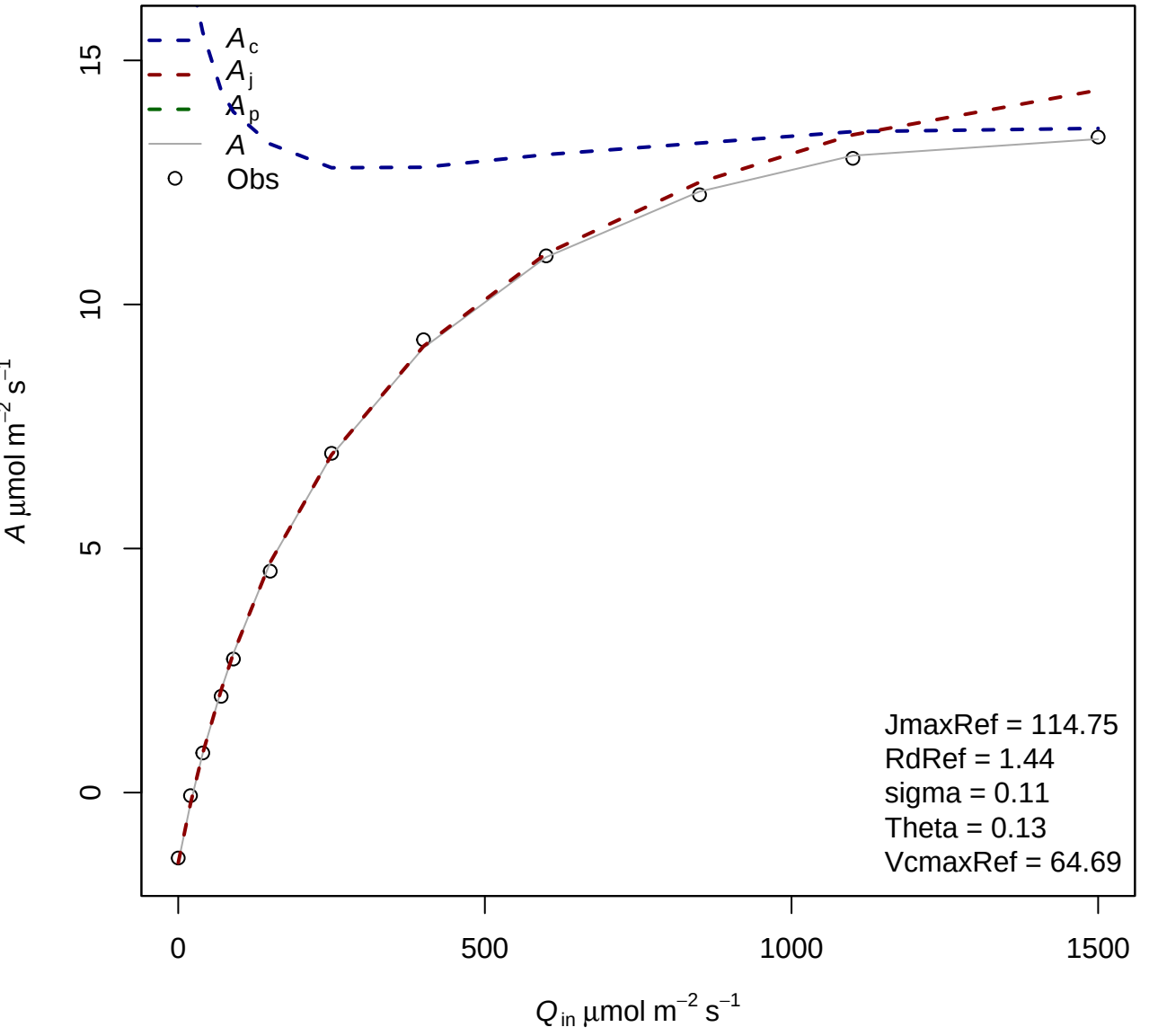

183
A_c
A_j
A_p
A
Obs
JmaxRef = 53.29
RdRef = 1.01
sigma = 0.11
Theta = 0.62
VcmaxRef = 179.63
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

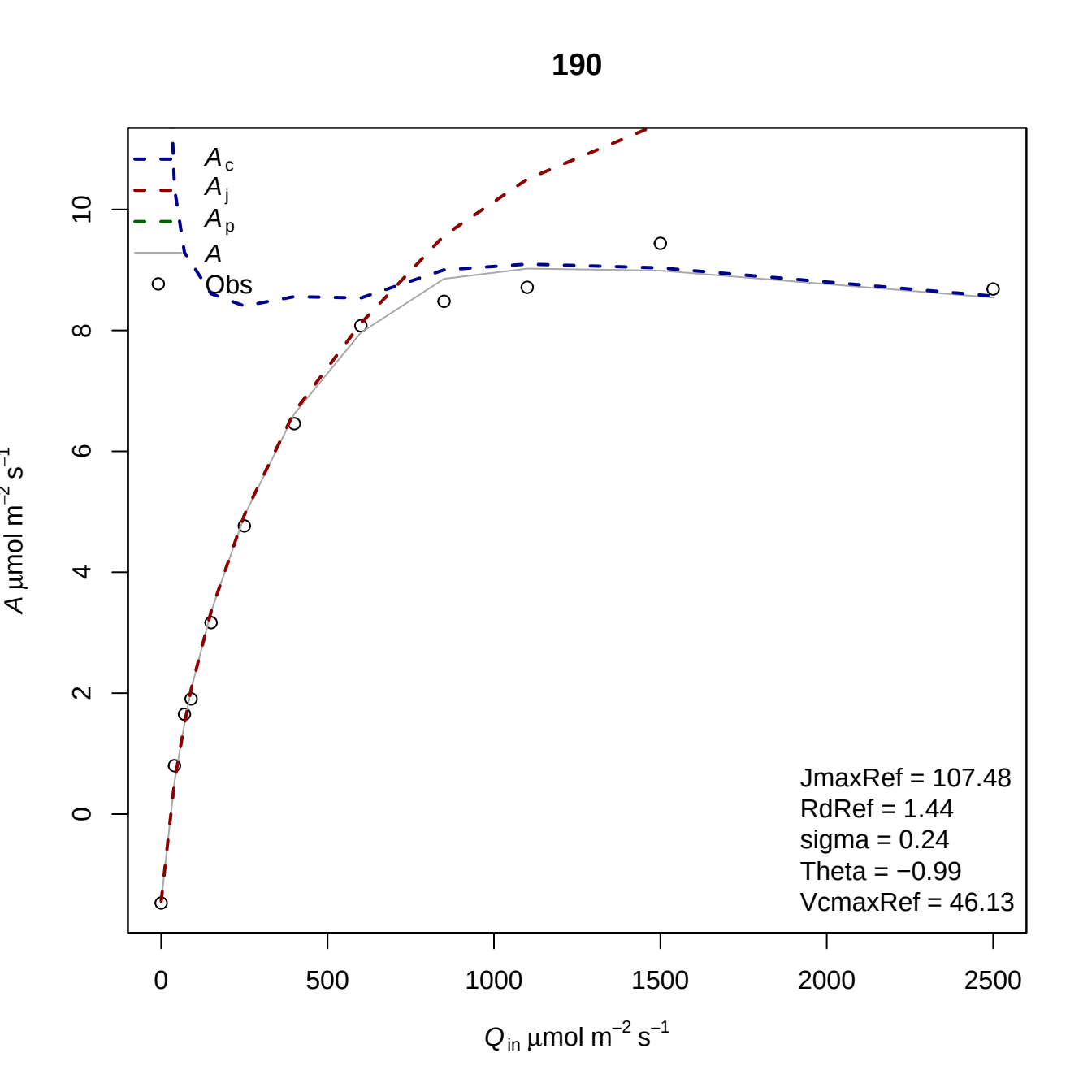

184
A_c
A_j
A_p
A
Obs
JmaxRef = 53.69
RdRef = 1.31
sigma = 0.11
Theta = 0.68
VcmaxRef = 468.82
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

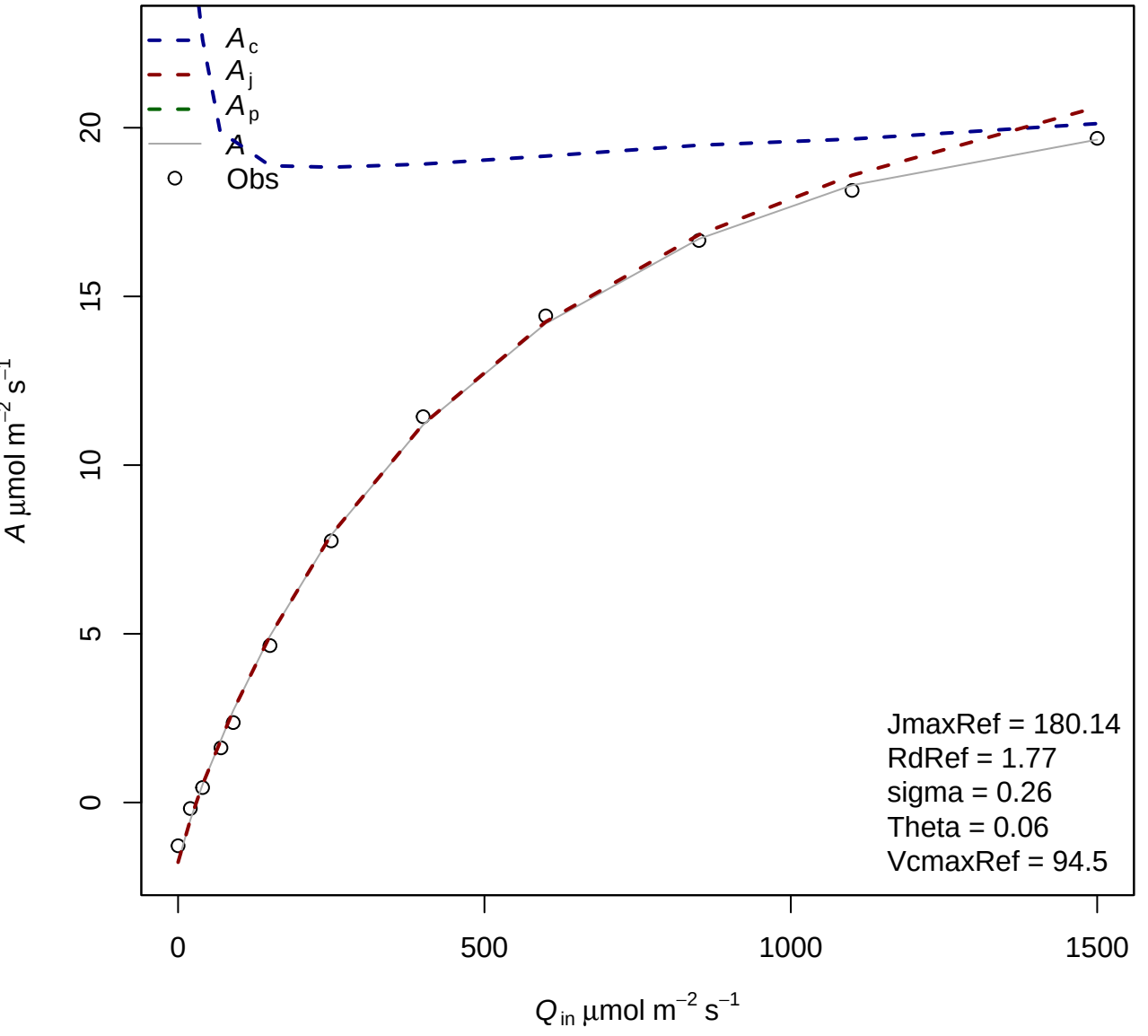

A_c
A_j
A_p
A
Obs
JmaxRef = 860109671.01
RdRef = 1.48
sigma = 0.44
Theta = −65087752.97
VcmaxRef = 6269711583.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

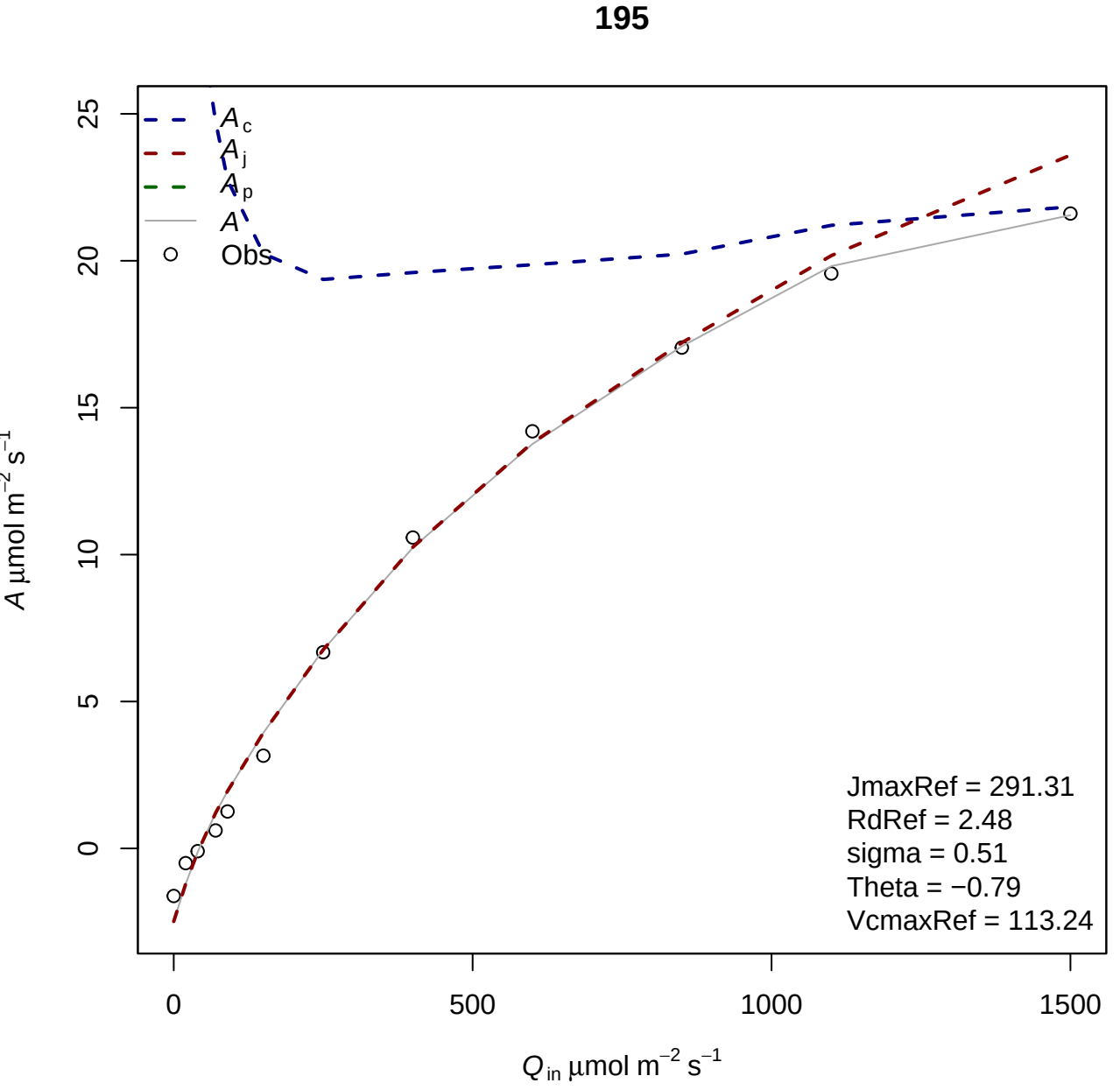

186
A_c
A_j
A_p
A
Obs
JmaxRef = 167.03
RdRef = 1
sigma = 0.33
Theta = −8.22
VcmaxRef = 50.53
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

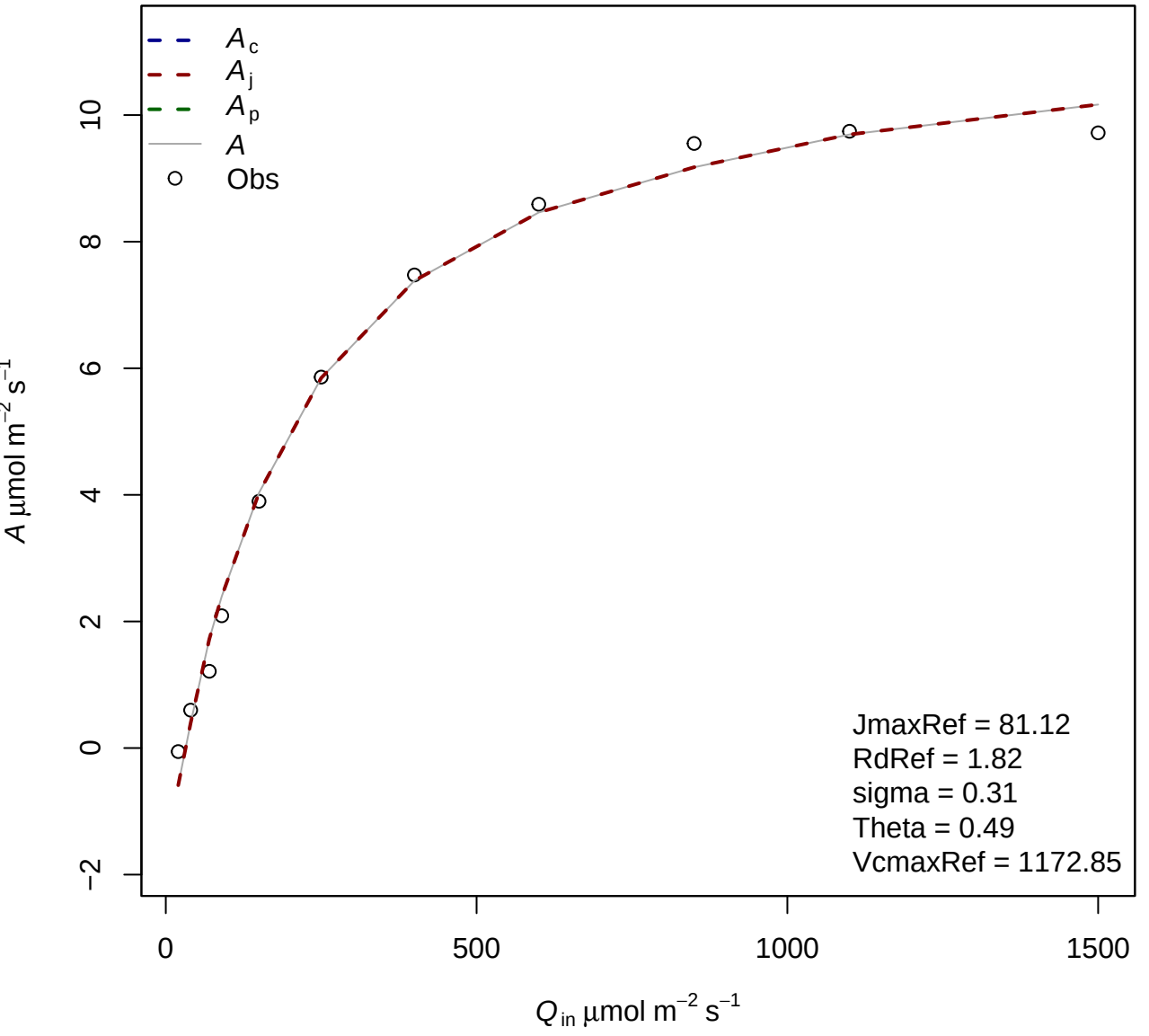

A_c
A_j
A_p
A
Obs
JmaxRef = 74.2
RdRef = 2.65
sigma = 0.18
Theta = 0.73
VcmaxRef = 13143.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

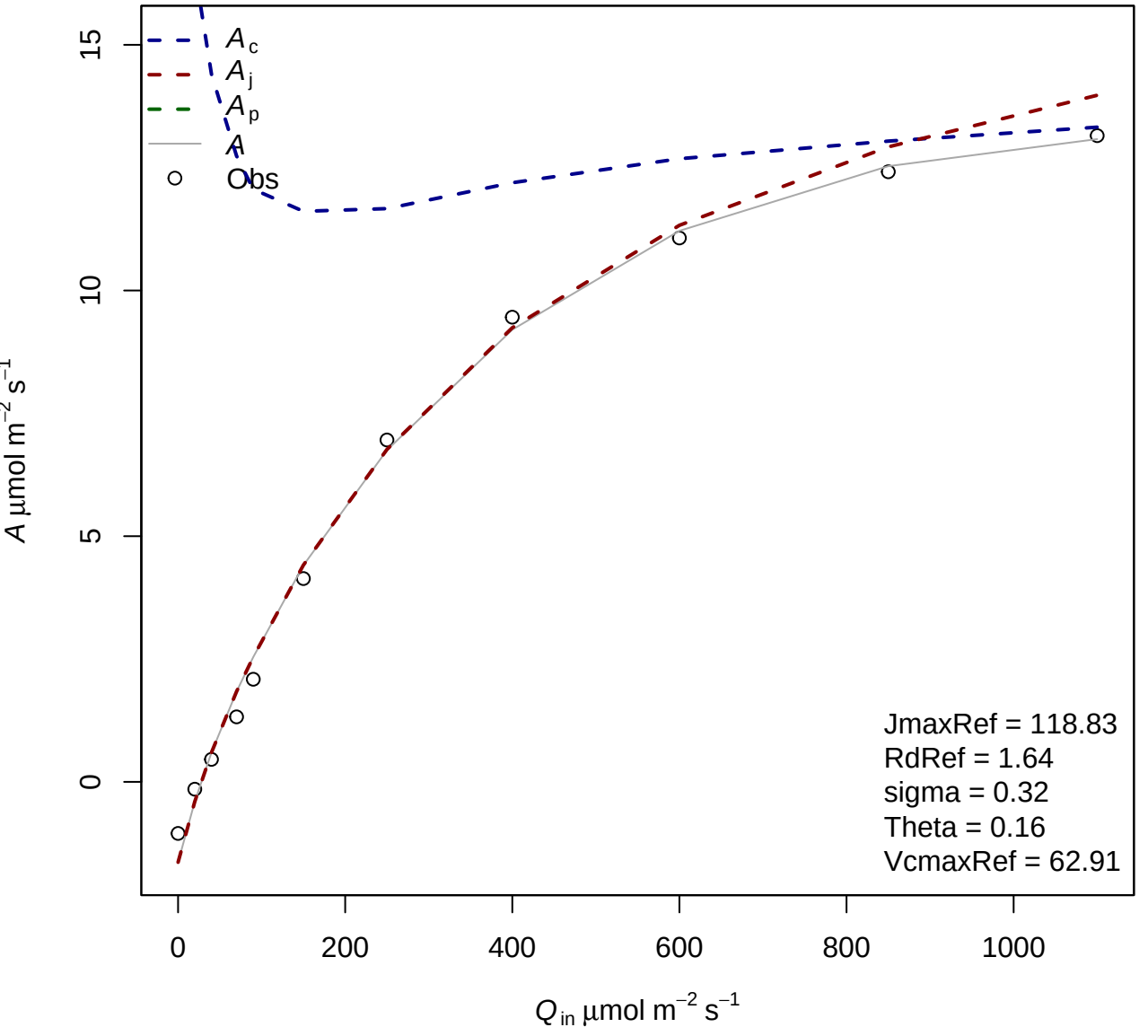

A_c
A_j
A_p
A
Obs
JmaxRef = 114.75
RdRef = 1.44
sigma = 0.11
Theta = 0.13
VcmaxRef = 64.69
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

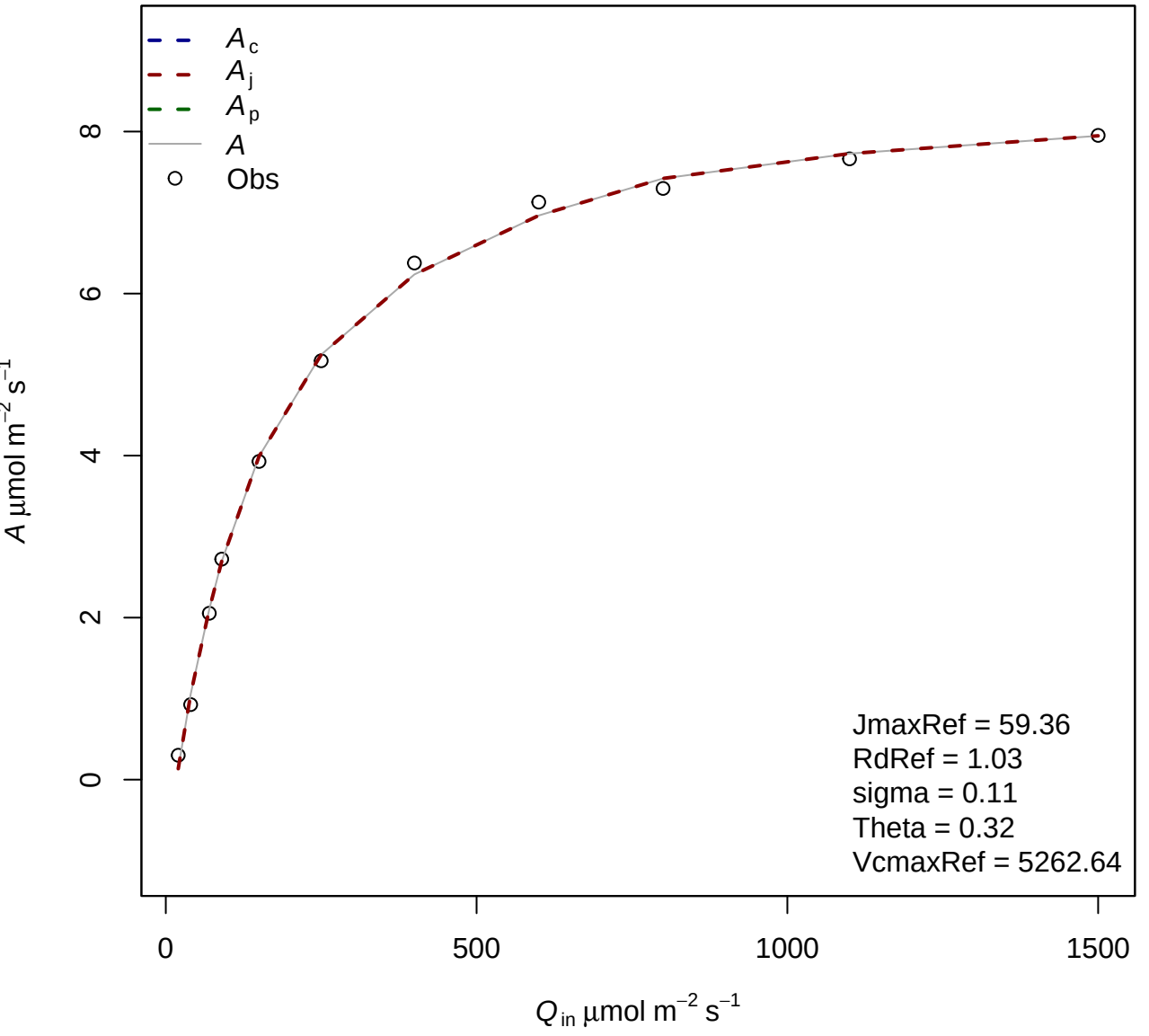

190
A_c
A_j
A_p
A
Obs
JmaxRef = 107.48
RdRef = 1.44
sigma = 0.24
Theta = −0.99
VcmaxRef = 46.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000
2500

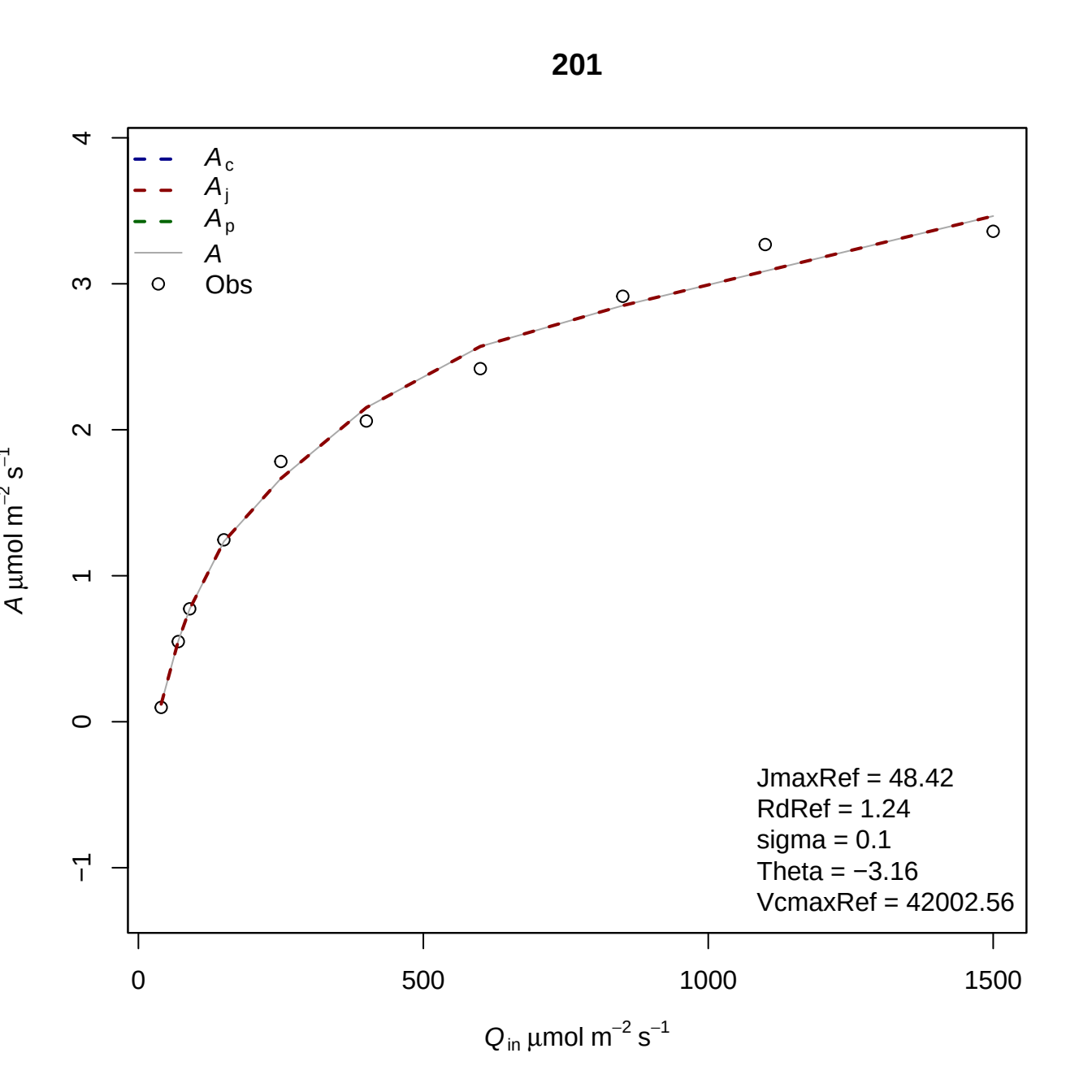

A_c
A_j
A_p
A
Obs
JmaxRef = 180.14
RdRef = 1.77
sigma = 0.26
Theta = 0.06
VcmaxRef = 94.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

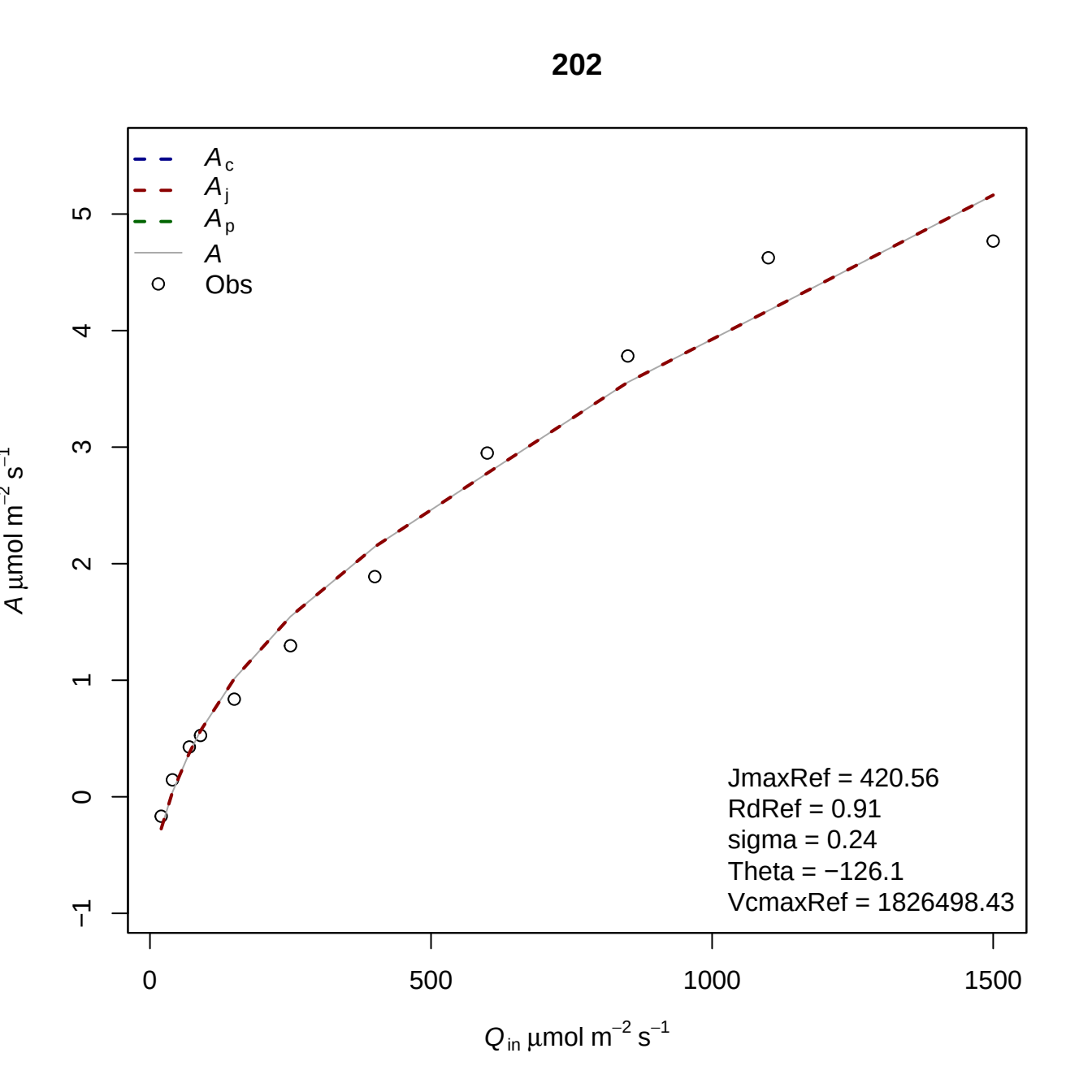

195
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
JmaxRef = 291.31
RdRef = 2.48
sigma = 0.51
Theta = −0.79
VcmaxRef = 113.24

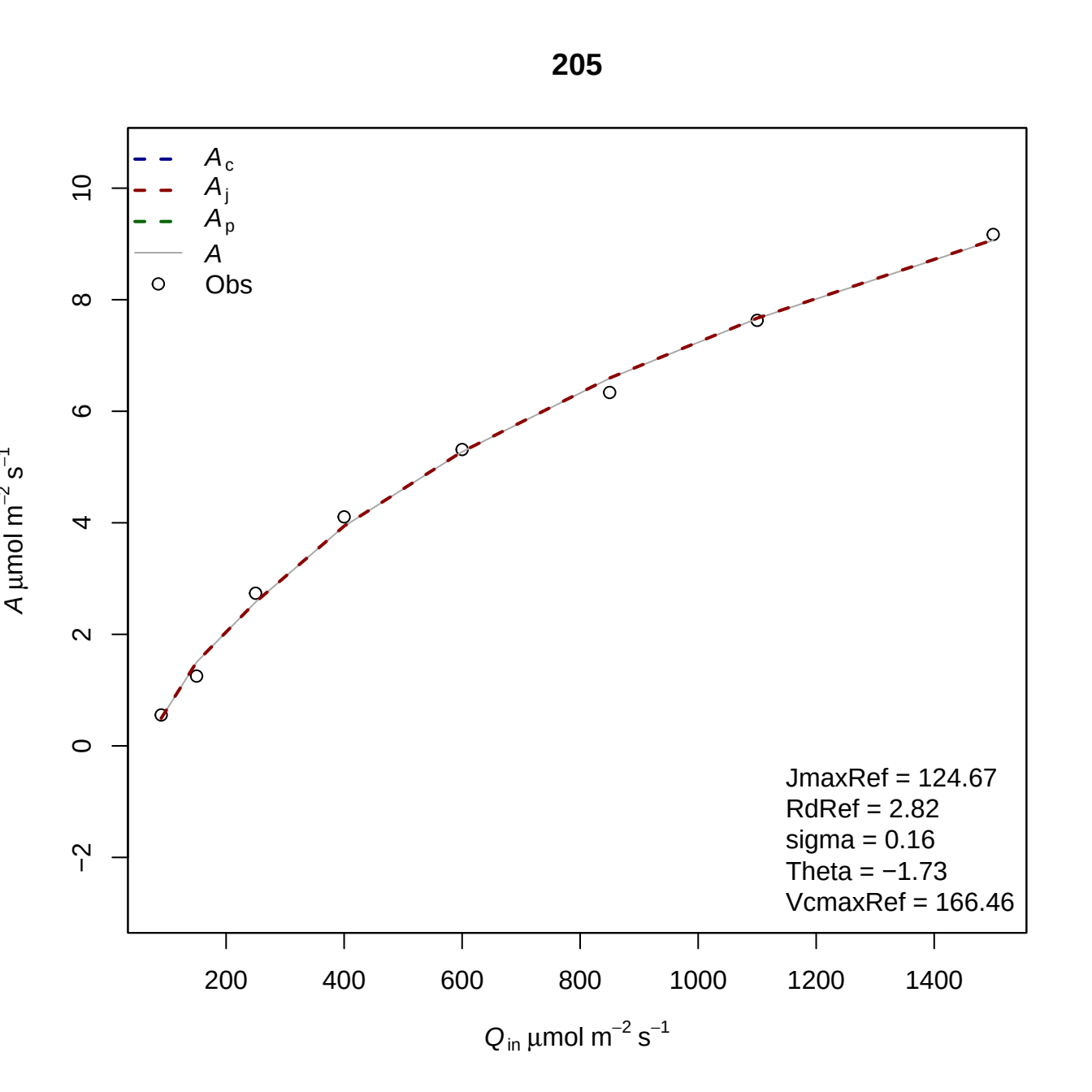

A_c
A_j
A_p
A
Obs
JmaxRef = 81.12
RdRef = 1.82
sigma = 0.31
Theta = 0.49
VcmaxRef = 1172.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

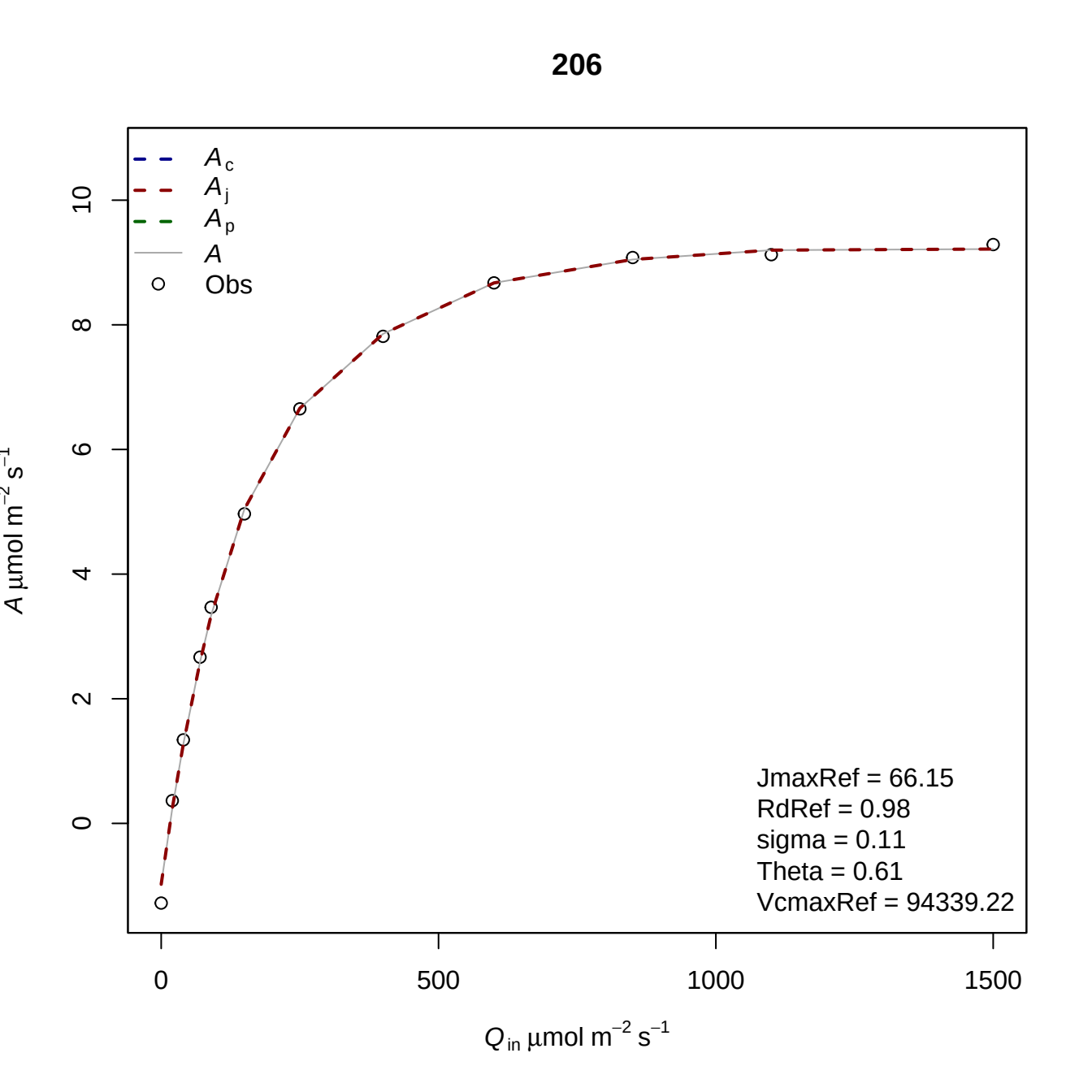

A_c
A_j
A_p
A
Obs
JmaxRef = 118.83
RdRef = 1.64
sigma = 0.32
Theta = 0.16
VcmaxRef = 62.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

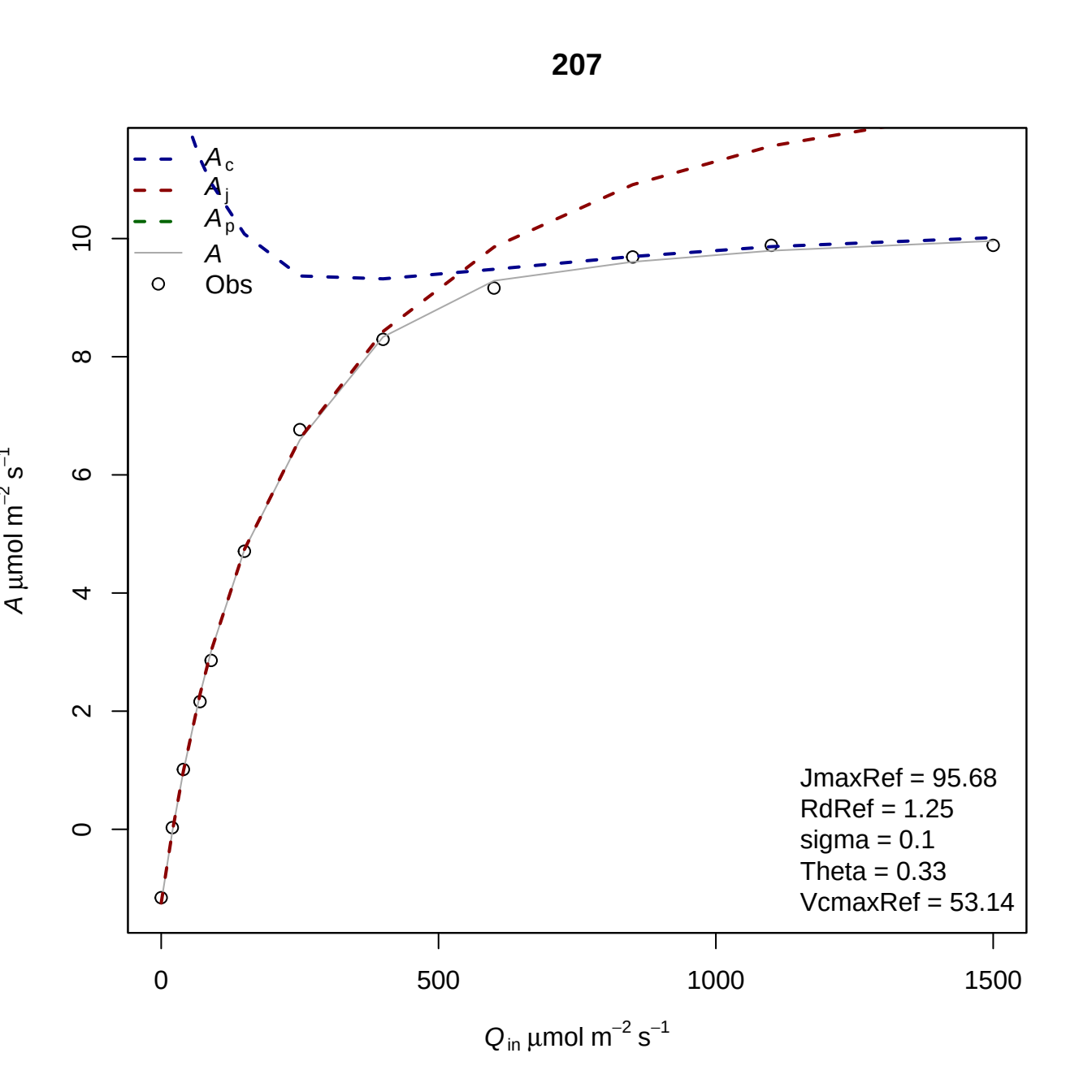

A_c
A_j
A_p
A
Obs
JmaxRef = 59.36
RdRef = 1.03
sigma = 0.11
Theta = 0.32
VcmaxRef = 5262.64
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

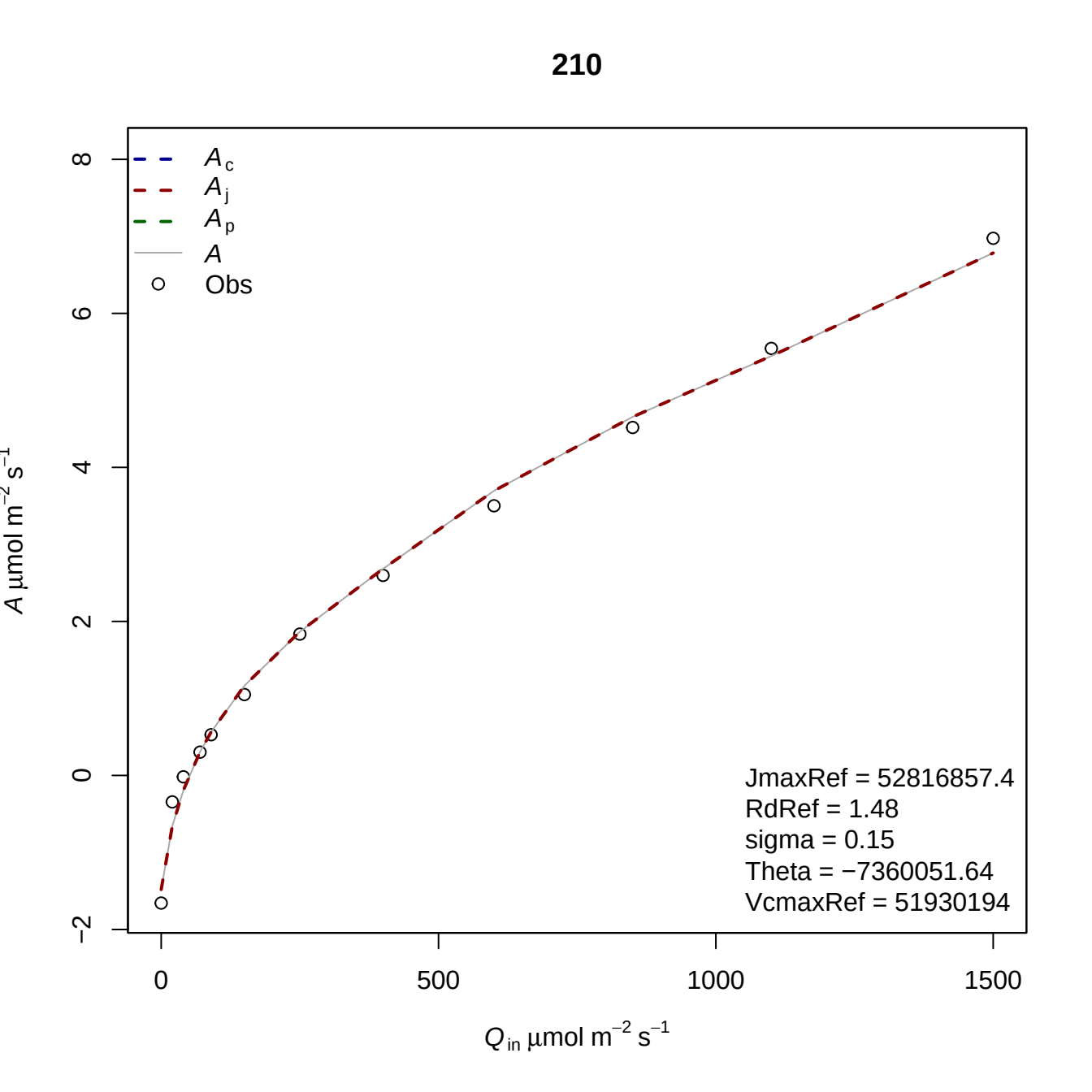

201
A_c
A_j
A_p
A
Obs
JmaxRef = 48.42
RdRef = 1.24
sigma = 0.1
Theta = −3.16
VcmaxRef = 42002.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

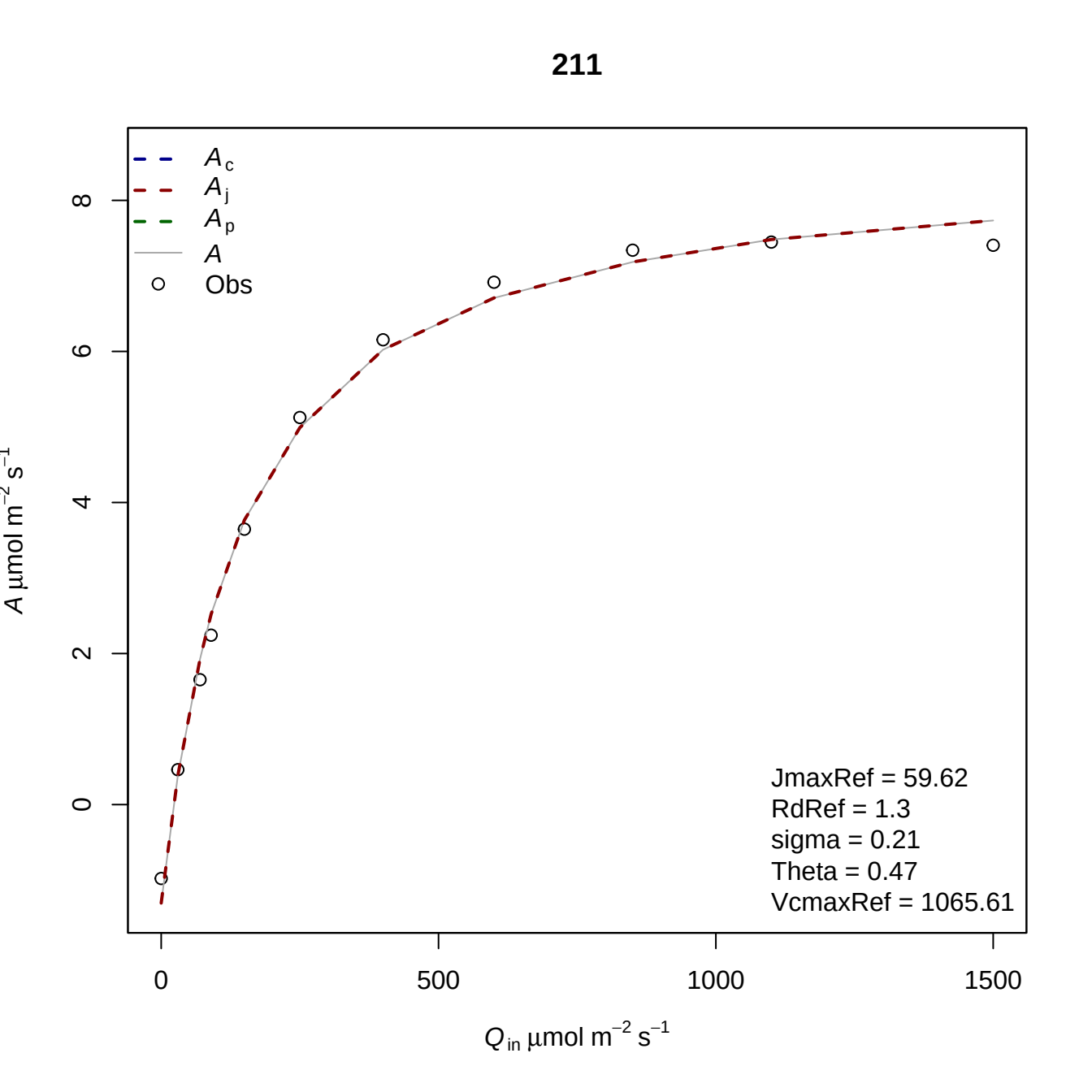

202
A_c
A_j
A_p
A
Obs
JmaxRef = 420.56
RdRef = 0.91
sigma = 0.24
Theta = −126.1
VcmaxRef = 1826498.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

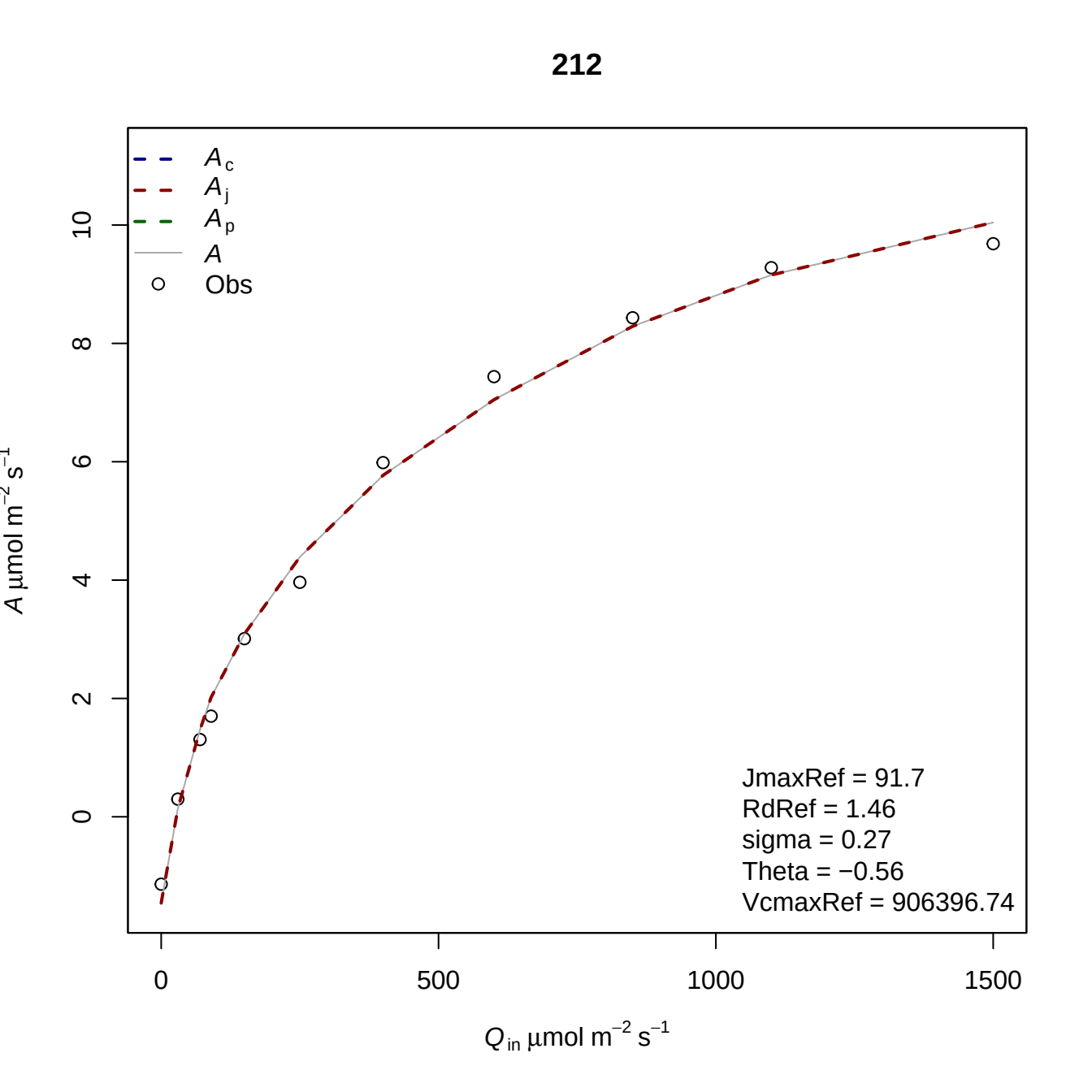

205
A_c
A_j
A_p
A
Obs
JmaxRef = 124.67
RdRef = 2.82
sigma = 0.16
Theta = −1.73
VcmaxRef = 166.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
200
400
600
800
1000
1200
1400

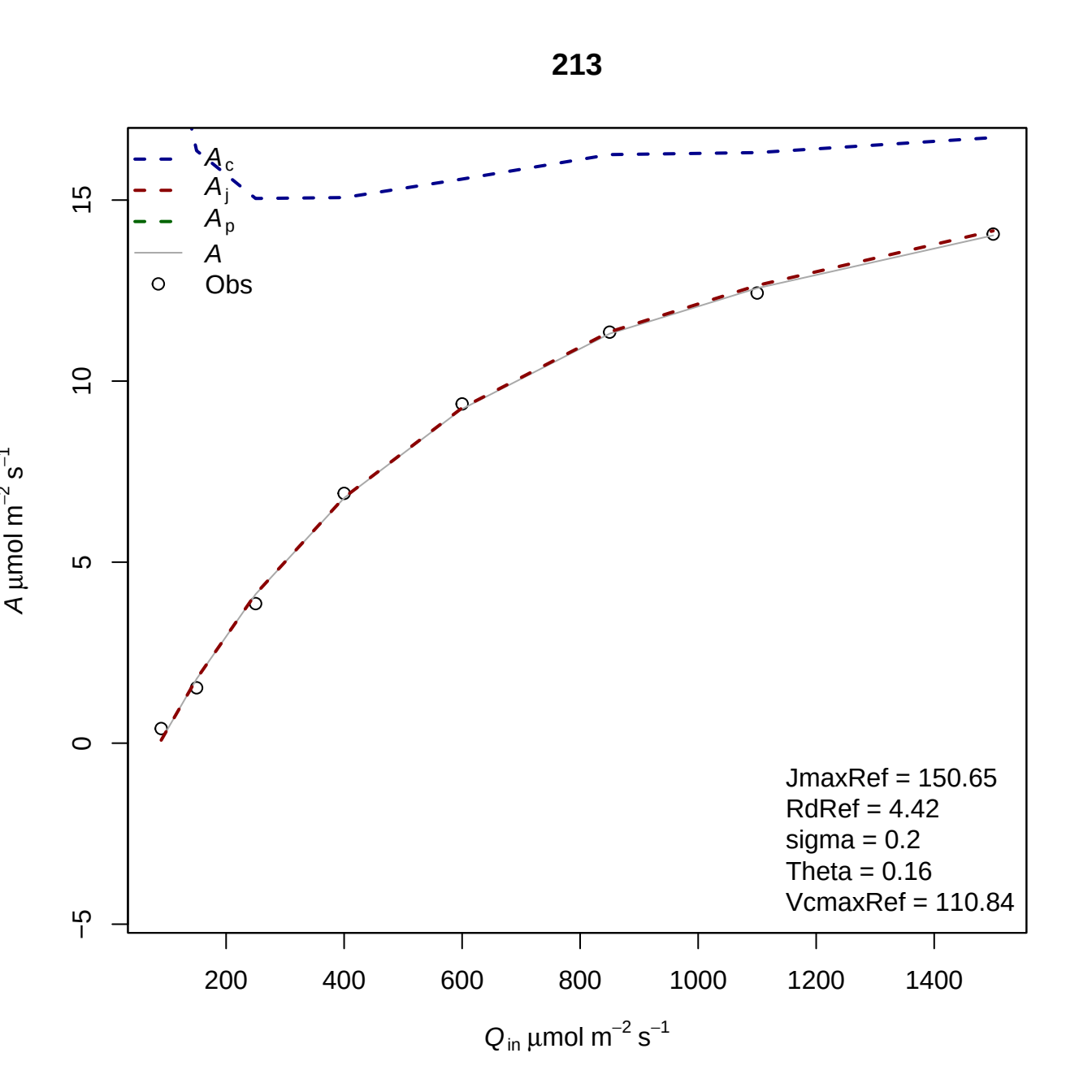

206
A_c
A_j
A_p
A
Obs
JmaxRef = 66.15
RdRef = 0.98
sigma = 0.11
Theta = 0.61
VcmaxRef = 94339.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

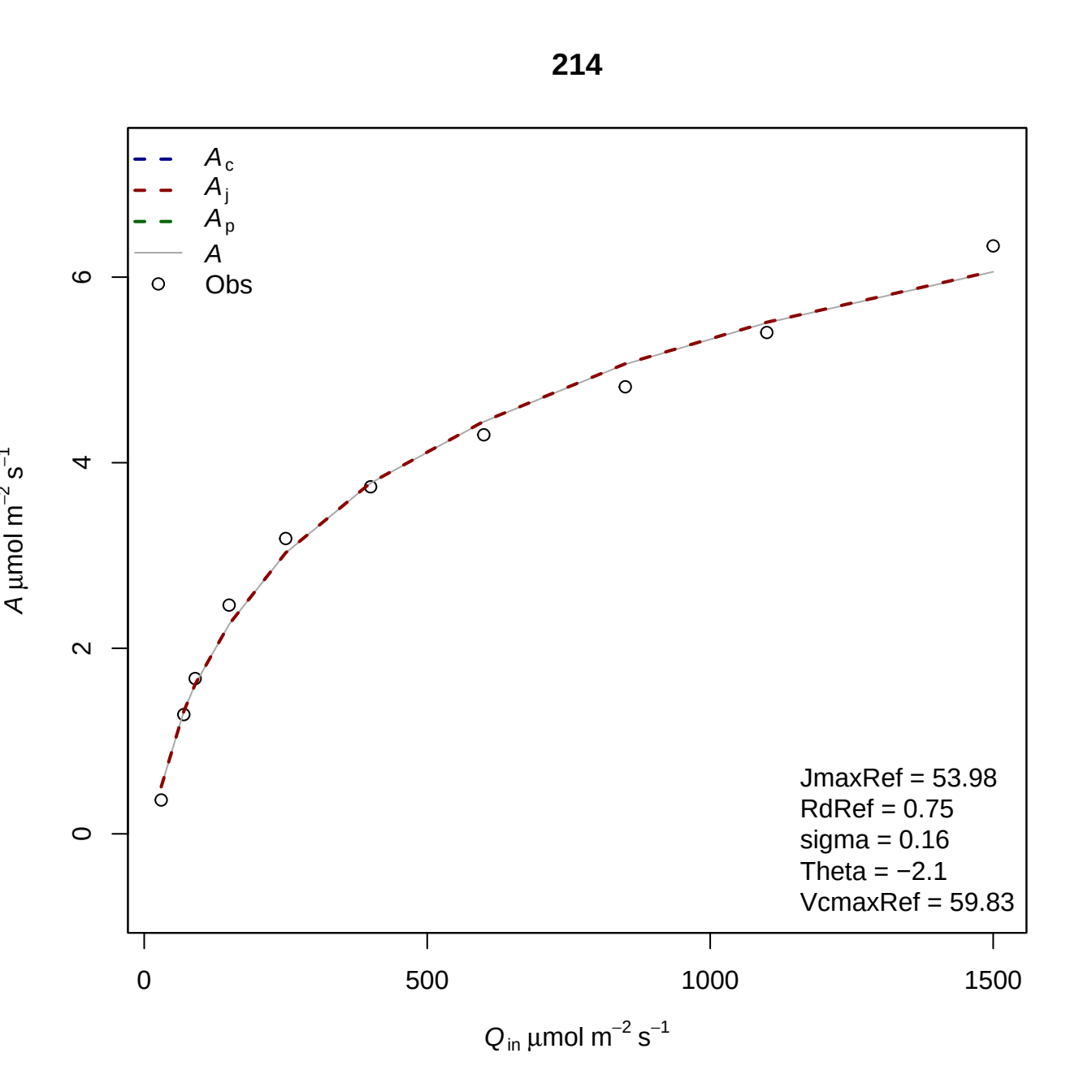

207
A_c
A_j
A_p
A
Obs
JmaxRef = 95.68
RdRef = 1.25
sigma = 0.1
Theta = 0.33
VcmaxRef = 53.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

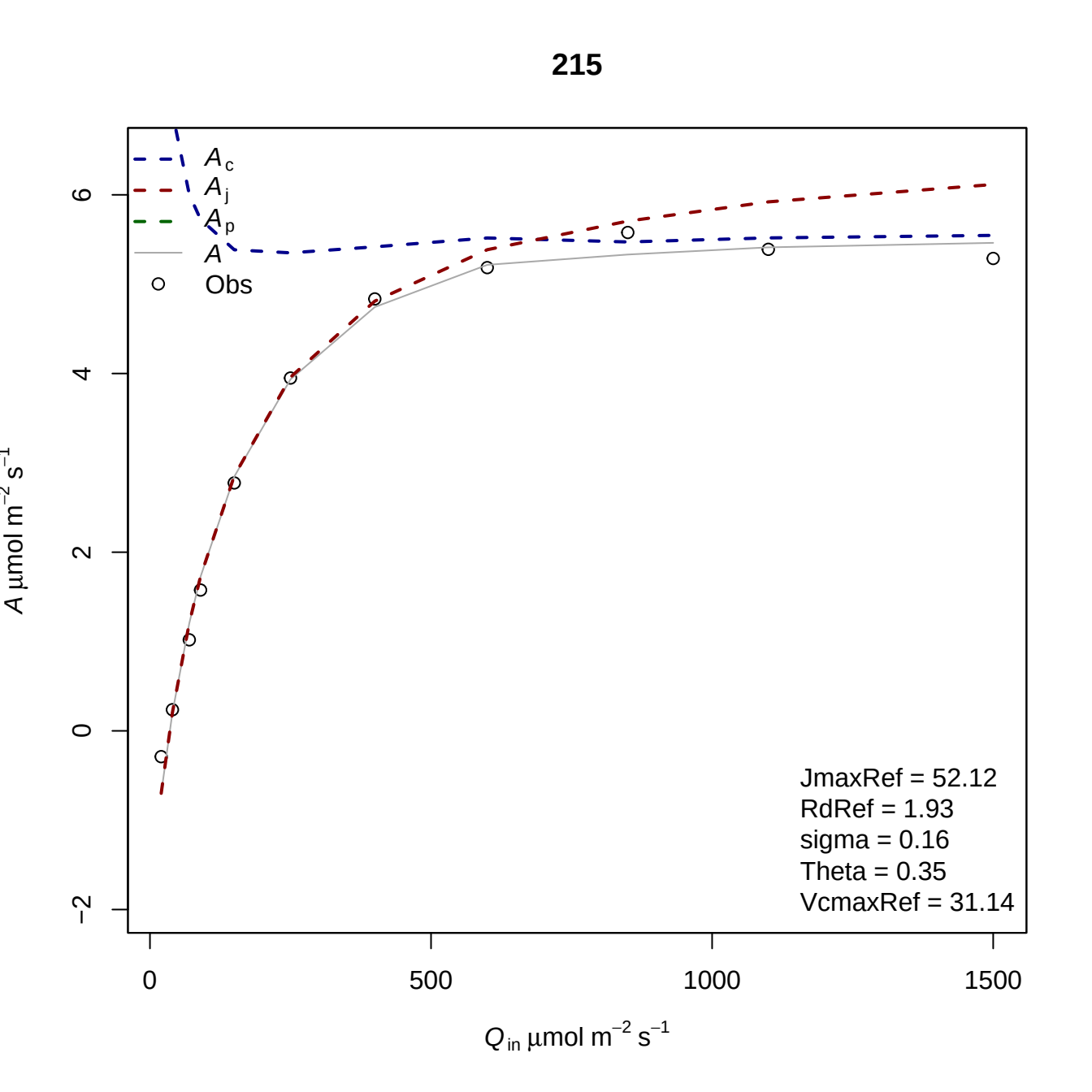

210
A_c
A_j
A_p
A
Obs
JmaxRef = 52816857.4
RdRef = 1.48
sigma = 0.15
Theta = −7360051.64
VcmaxRef = 51930194
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

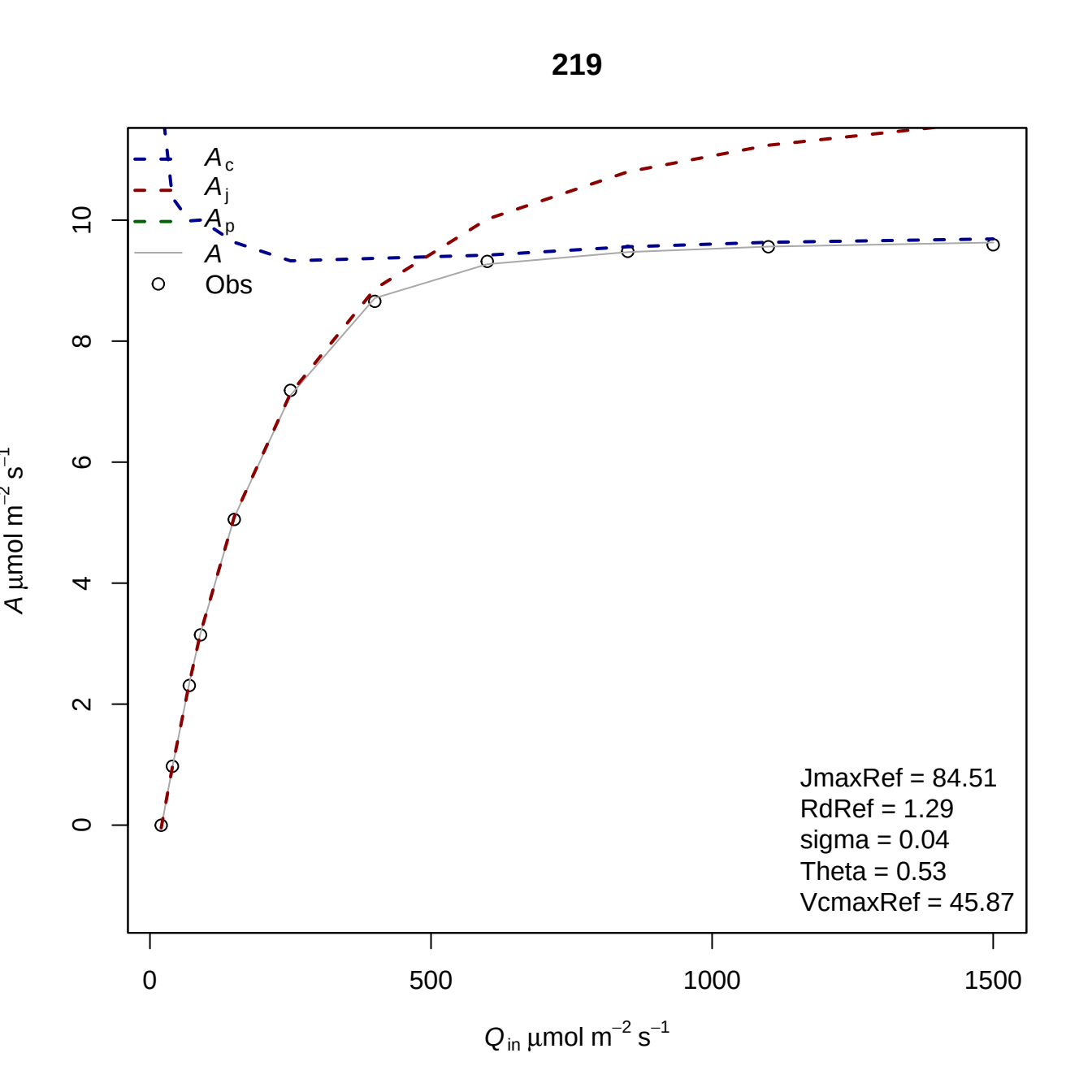

211
A_c
A_j
A_p
A
Obs
JmaxRef = 59.62
RdRef = 1.3
sigma = 0.21
Theta = 0.47
VcmaxRef = 1065.61
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

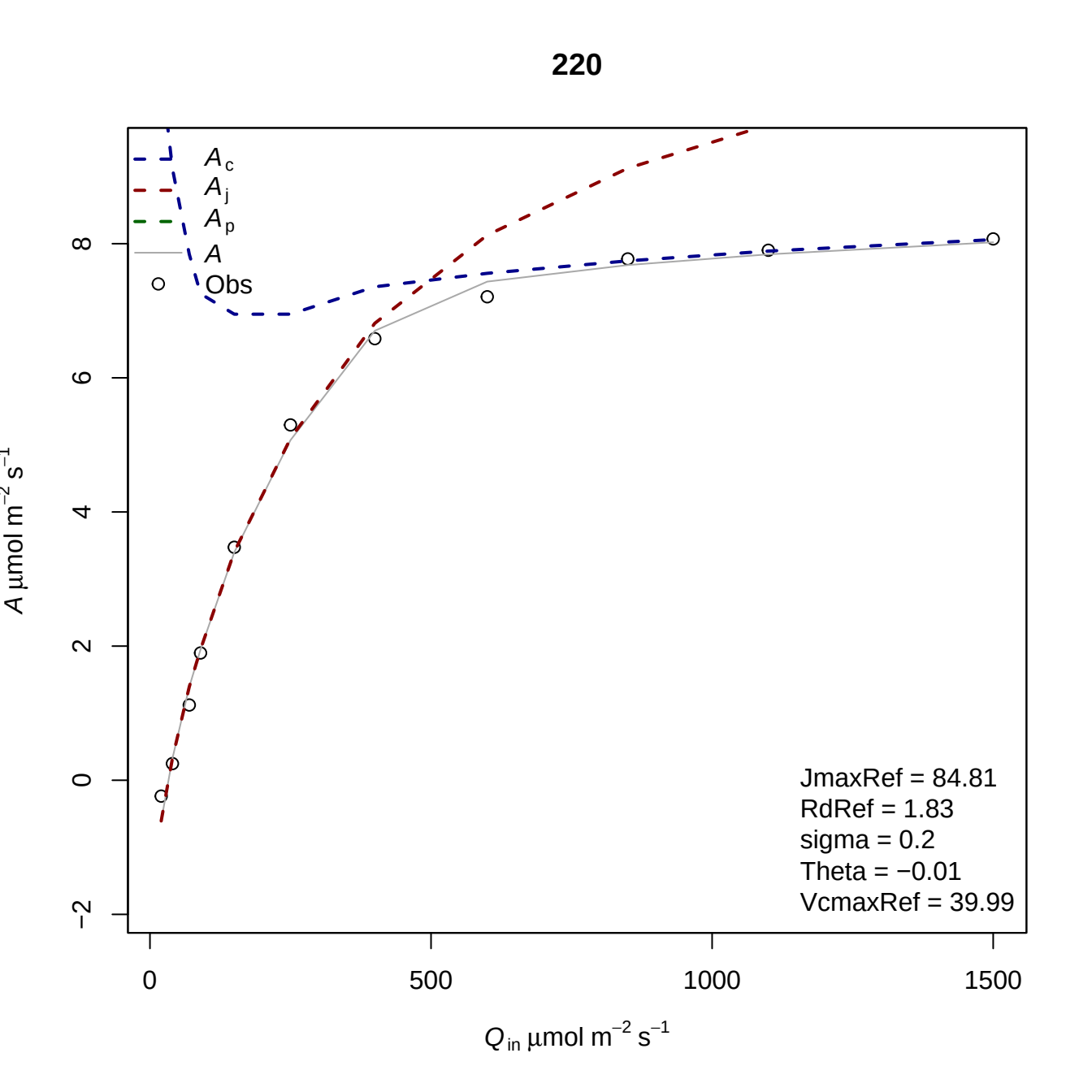

212
A_c
A_j
A_p
A
Obs
JmaxRef = 91.7
RdRef = 1.46
sigma = 0.27
Theta = −0.56
VcmaxRef = 906396.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

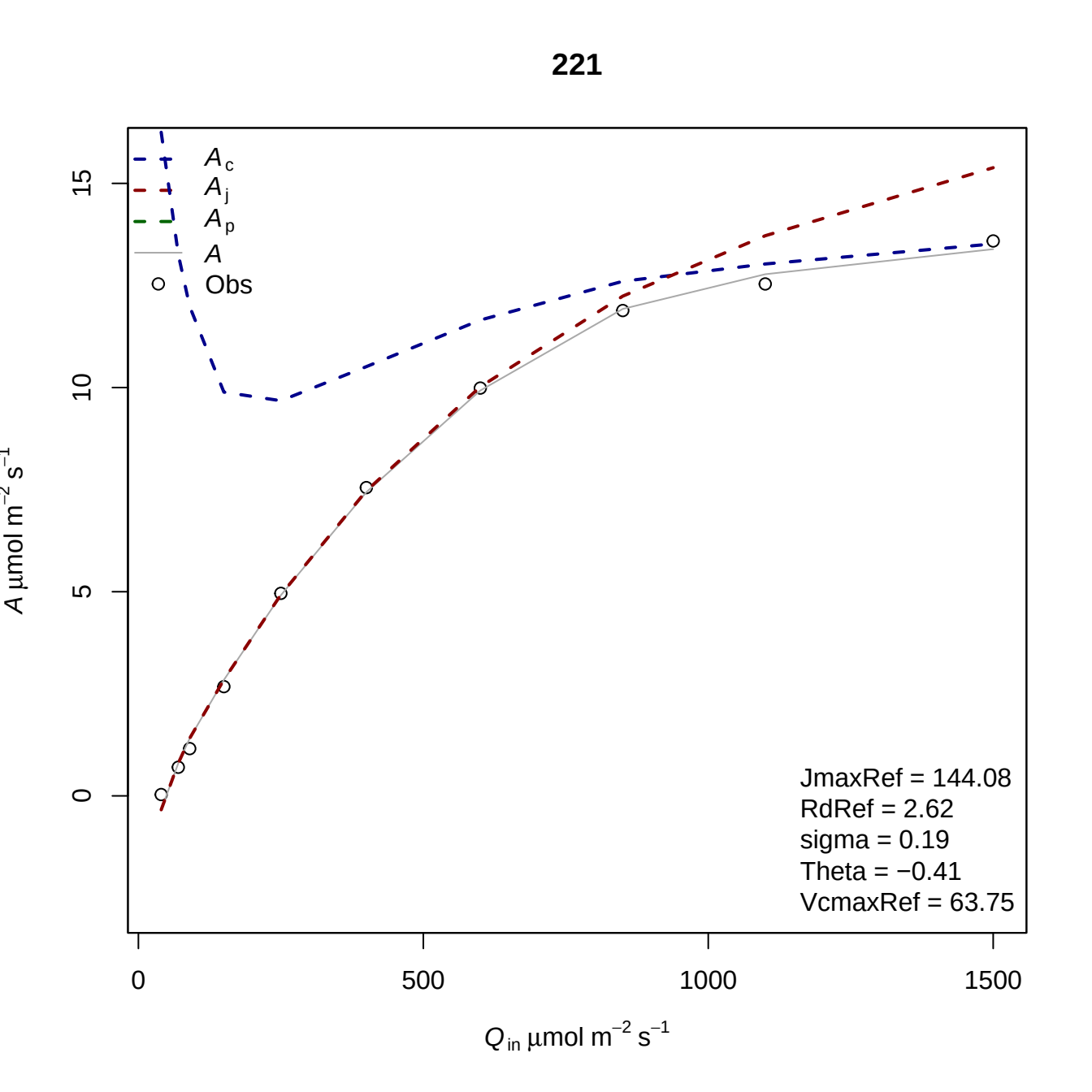

213
A_c
A_j
A_p
A
Obs
JmaxRef = 150.65
RdRef = 4.42
sigma = 0.2
Theta = 0.16
VcmaxRef = 110.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
200
400
600
800
1000
1200
1400

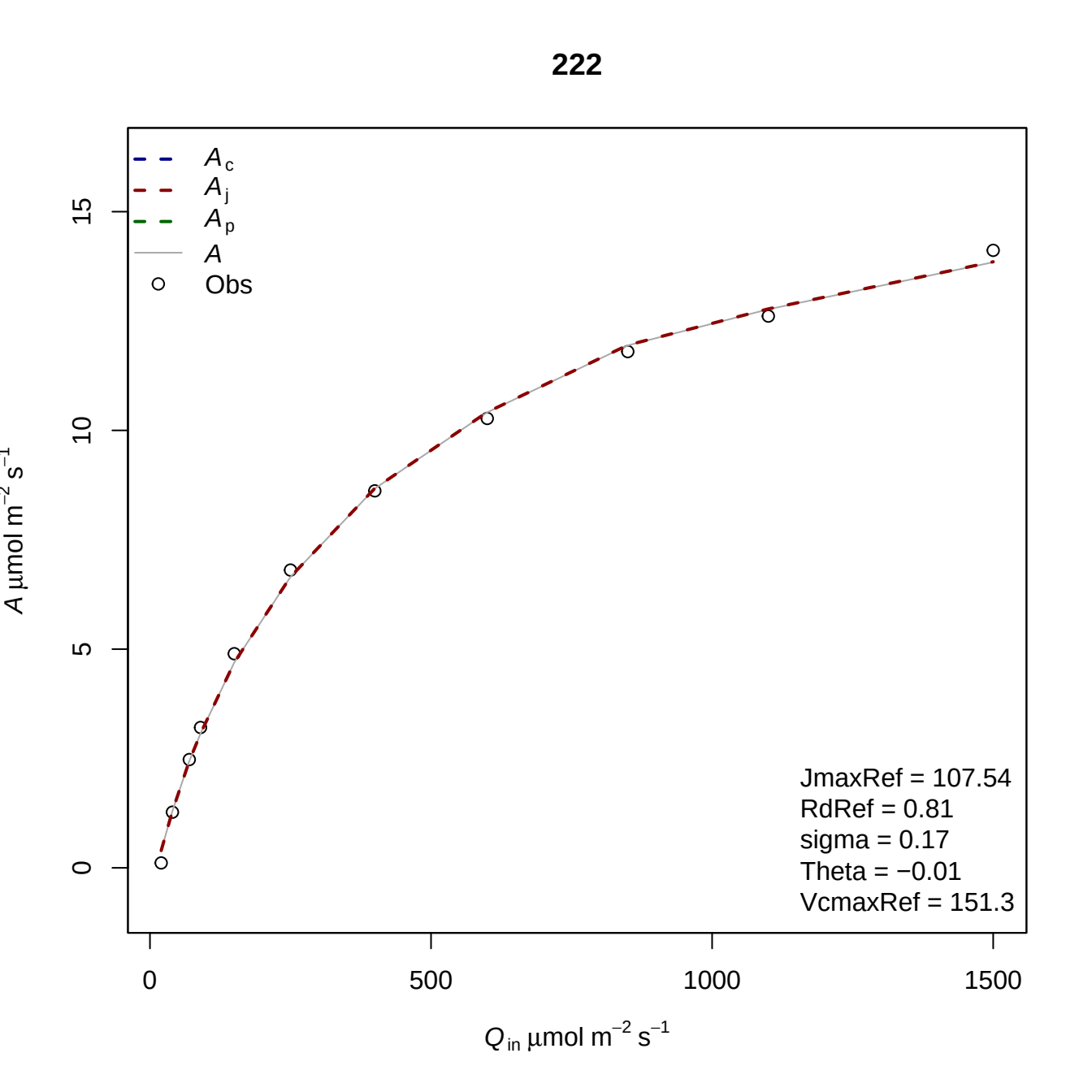

214
A_c
A_j
A_p
A
Obs
JmaxRef = 53.98
RdRef = 0.75
sigma = 0.16
Theta = −2.1
VcmaxRef = 59.83
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

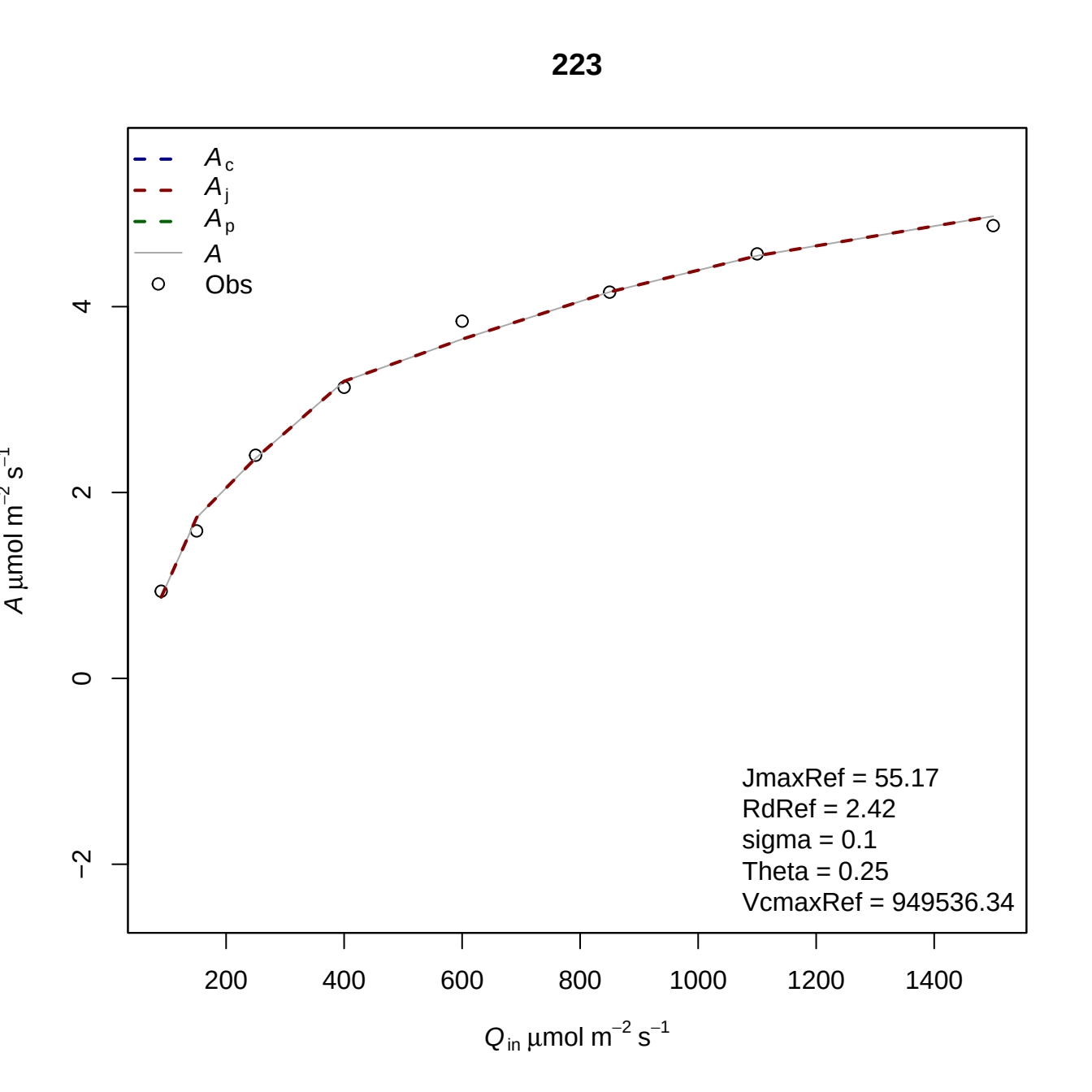

215
A_c
A_j
A_p
A
Obs
JmaxRef = 52.12
RdRef = 1.93
sigma = 0.16
Theta = 0.35
VcmaxRef = 31.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

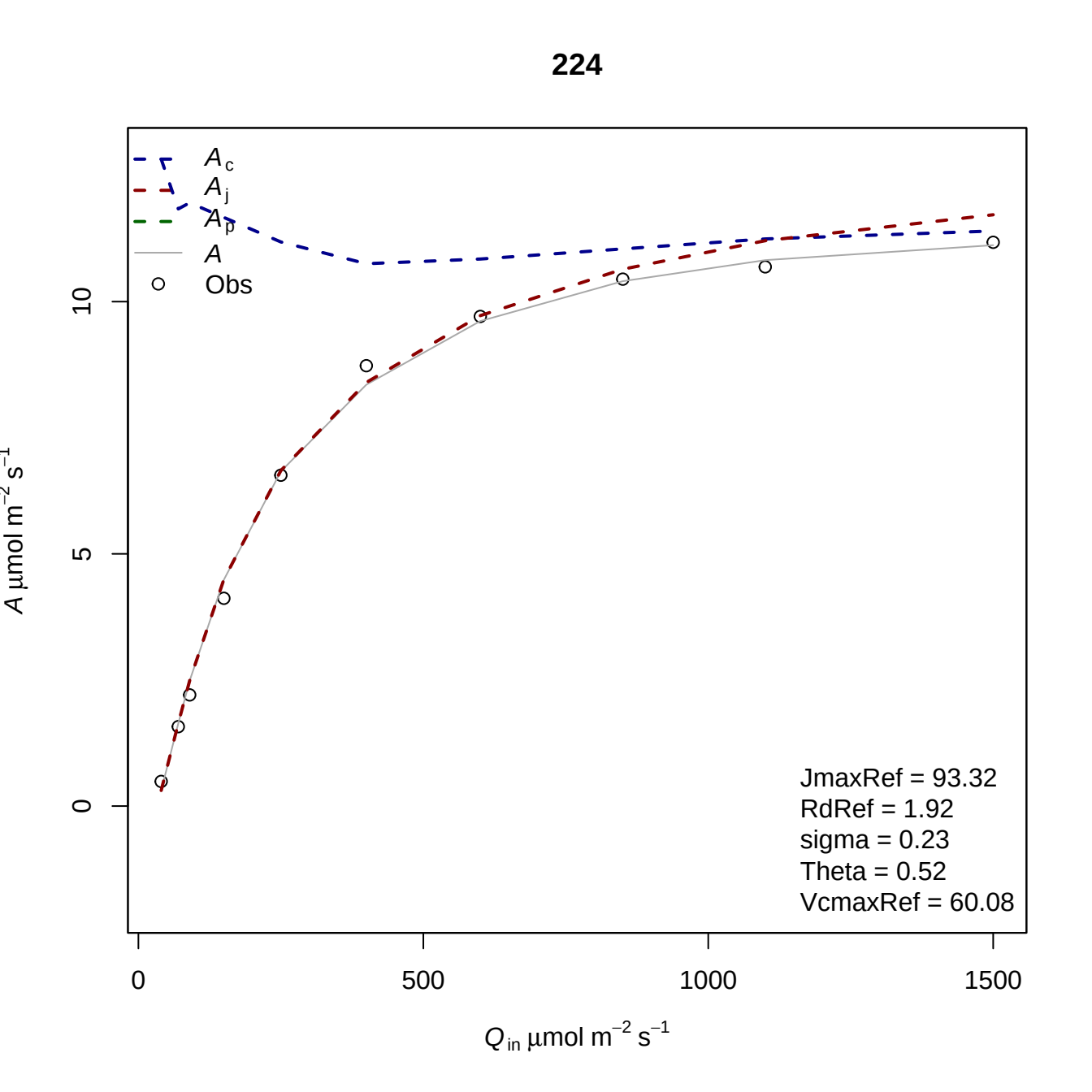

219
A_c
A_j
A_p
A
Obs
JmaxRef = 84.51
RdRef = 1.29
sigma = 0.04
Theta = 0.53
VcmaxRef = 45.87
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

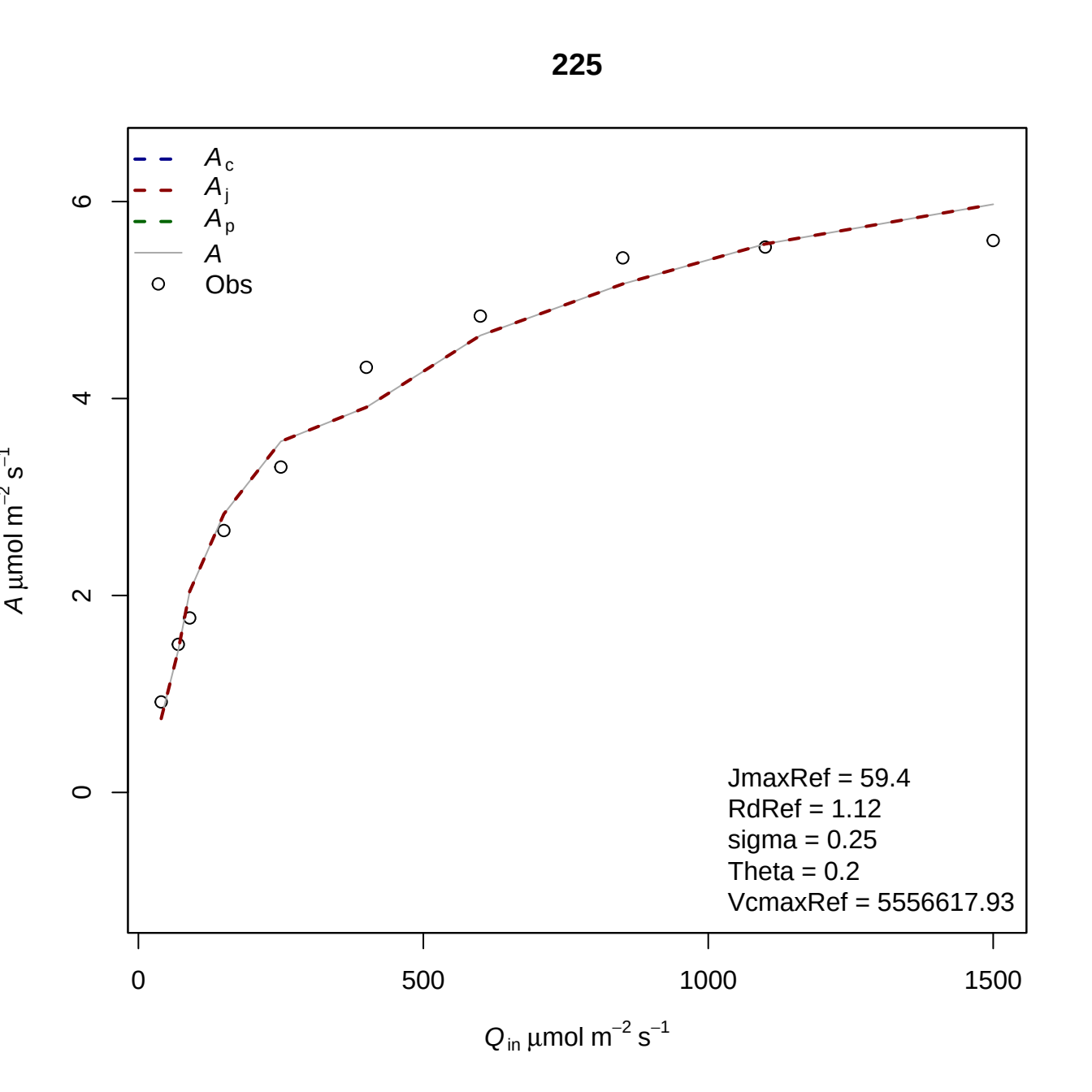

220
A_c
A_j
A_p
A
Obs
JmaxRef = 84.81
RdRef = 1.83
sigma = 0.2
Theta = −0.01
VcmaxRef = 39.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

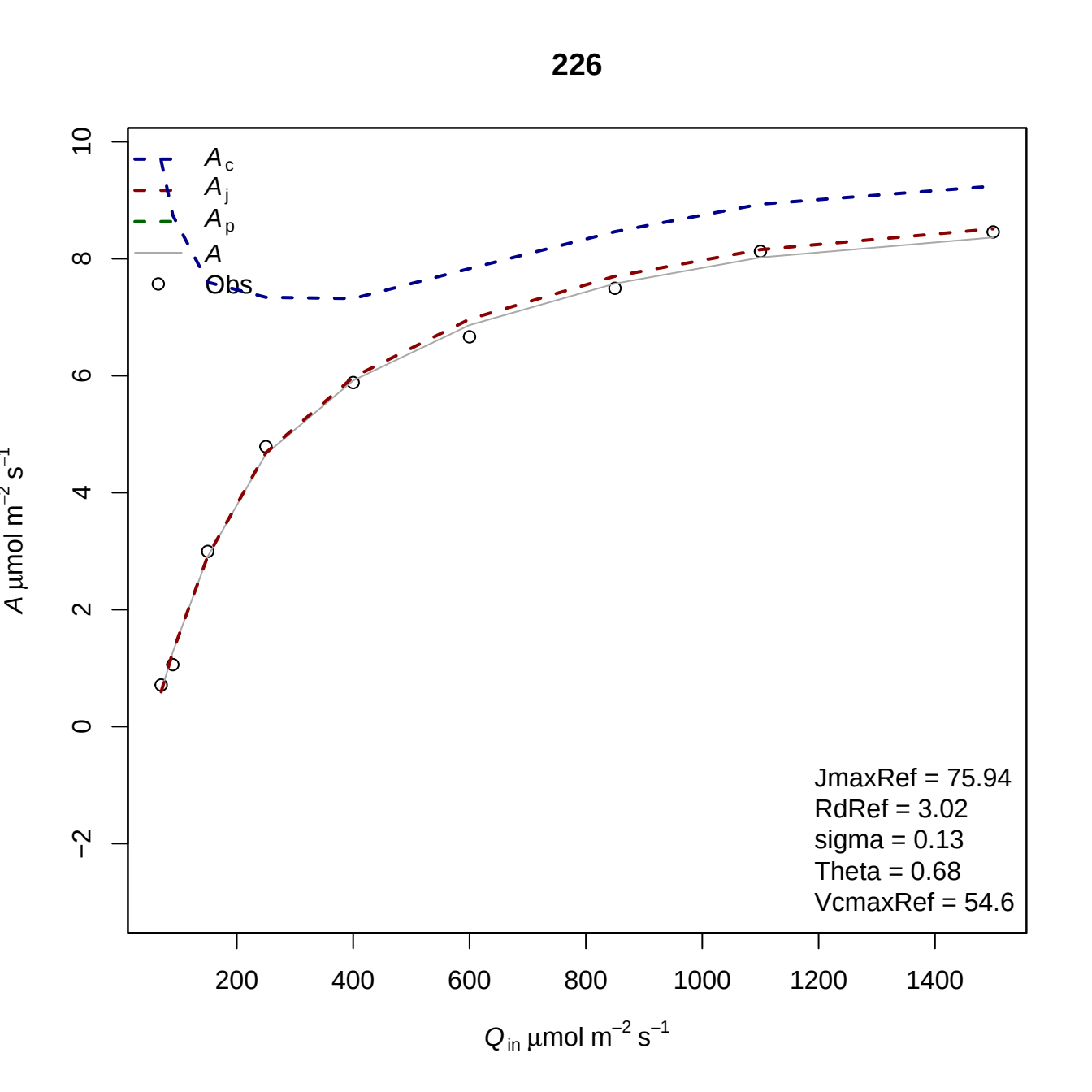

221
A_c
A_j
A_p
A
Obs
JmaxRef = 144.08
RdRef = 2.62
sigma = 0.19
Theta = −0.41
VcmaxRef = 63.75
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

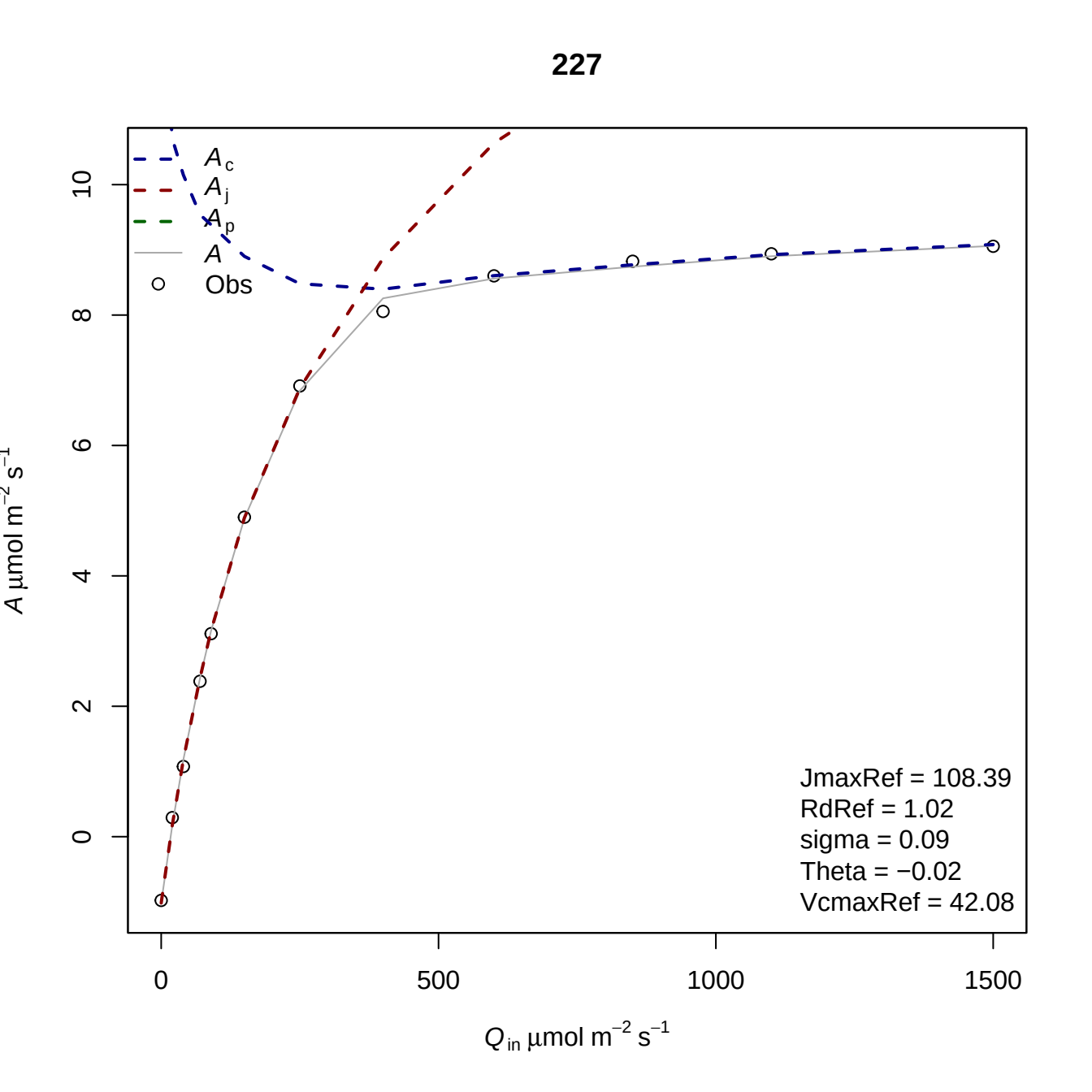

222
A_c
A_j
A_p
A
Obs
JmaxRef = 107.54
RdRef = 0.81
sigma = 0.17
Theta = −0.01
VcmaxRef = 151.3
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

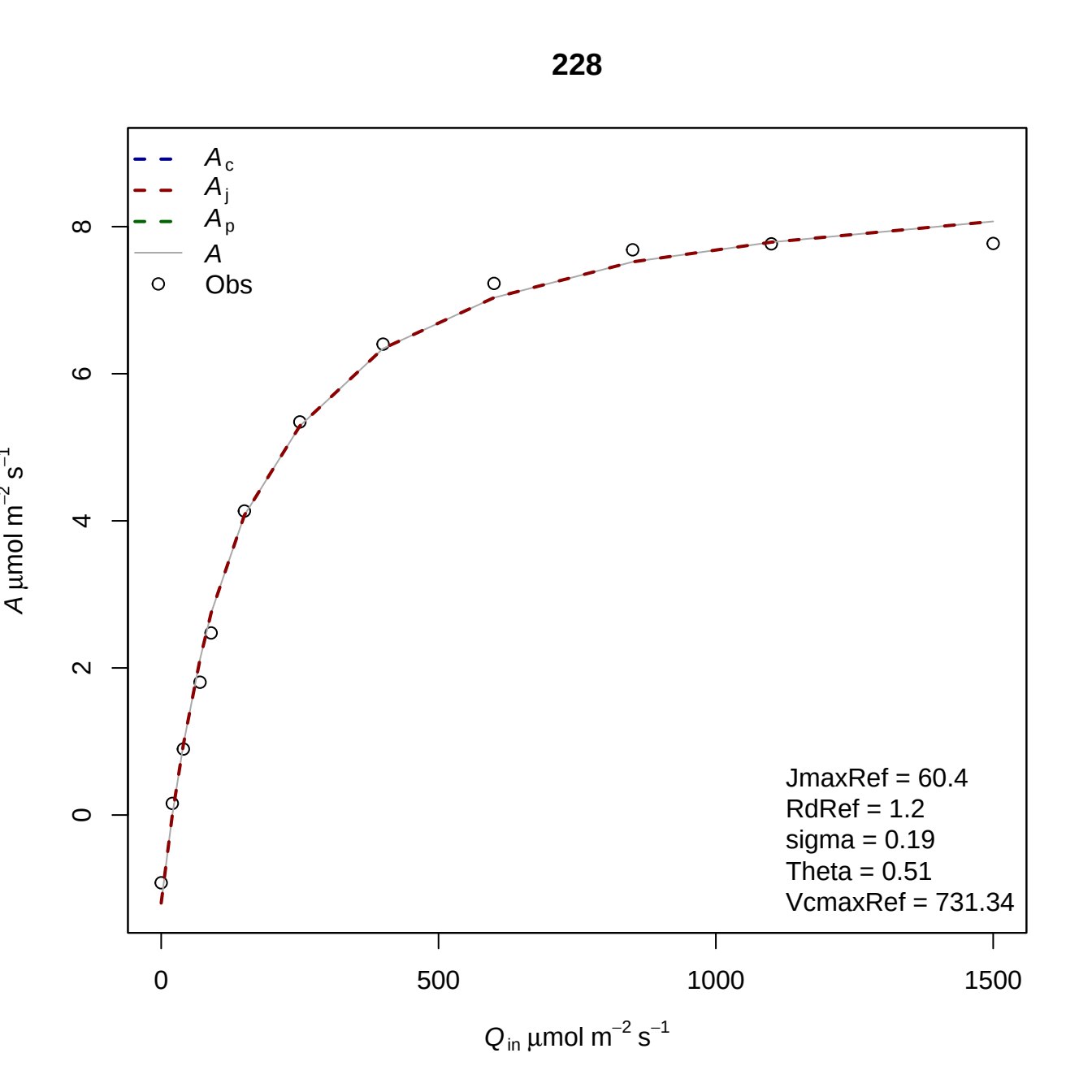

223
A_c
A_j
A_p
A
Obs
JmaxRef = 55.17
RdRef = 2.42
sigma = 0.1
Theta = 0.25
VcmaxRef = 949536.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

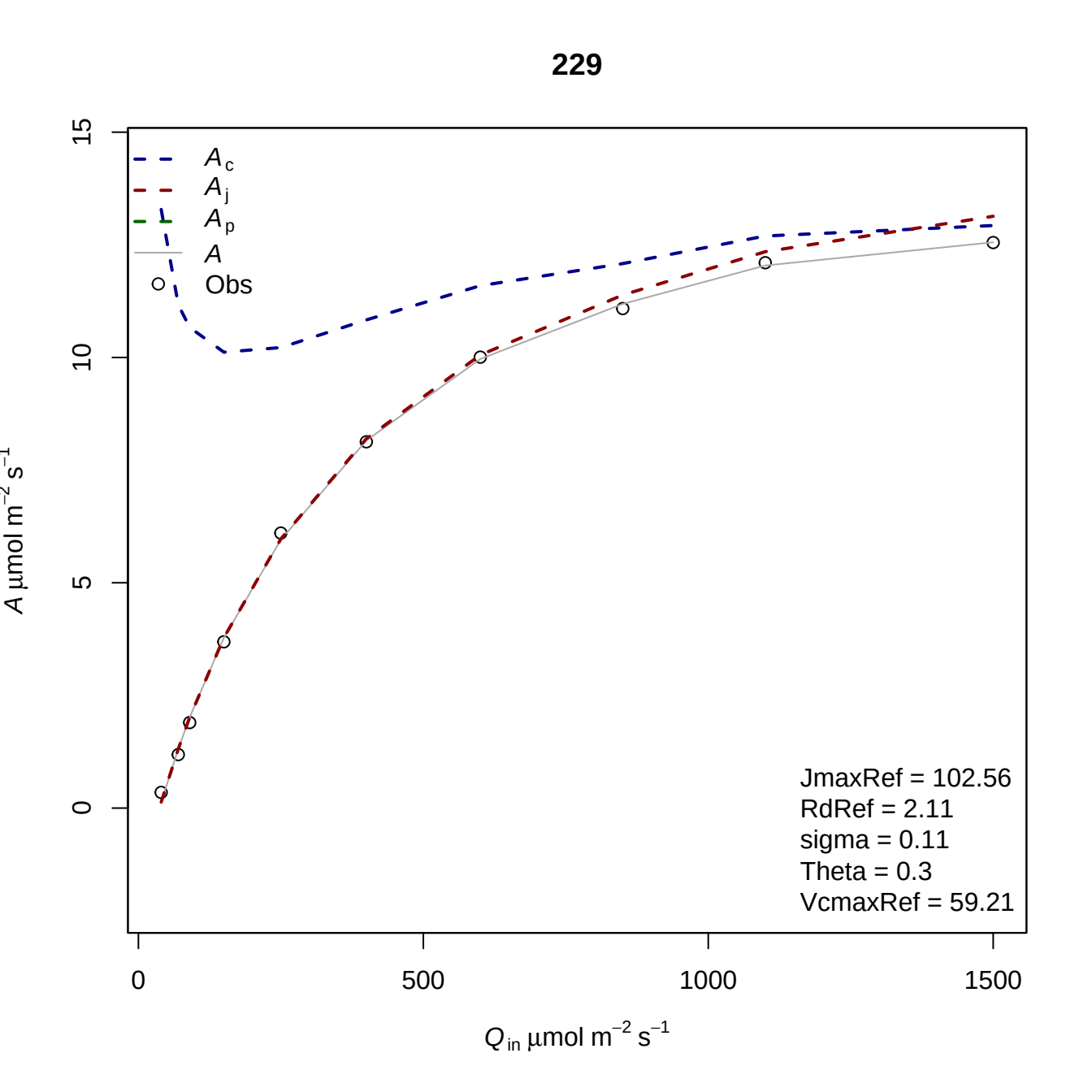

224
A_c
A_j
A_p
A
Obs
JmaxRef = 93.32
RdRef = 1.92
sigma = 0.23
Theta = 0.52
VcmaxRef = 60.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

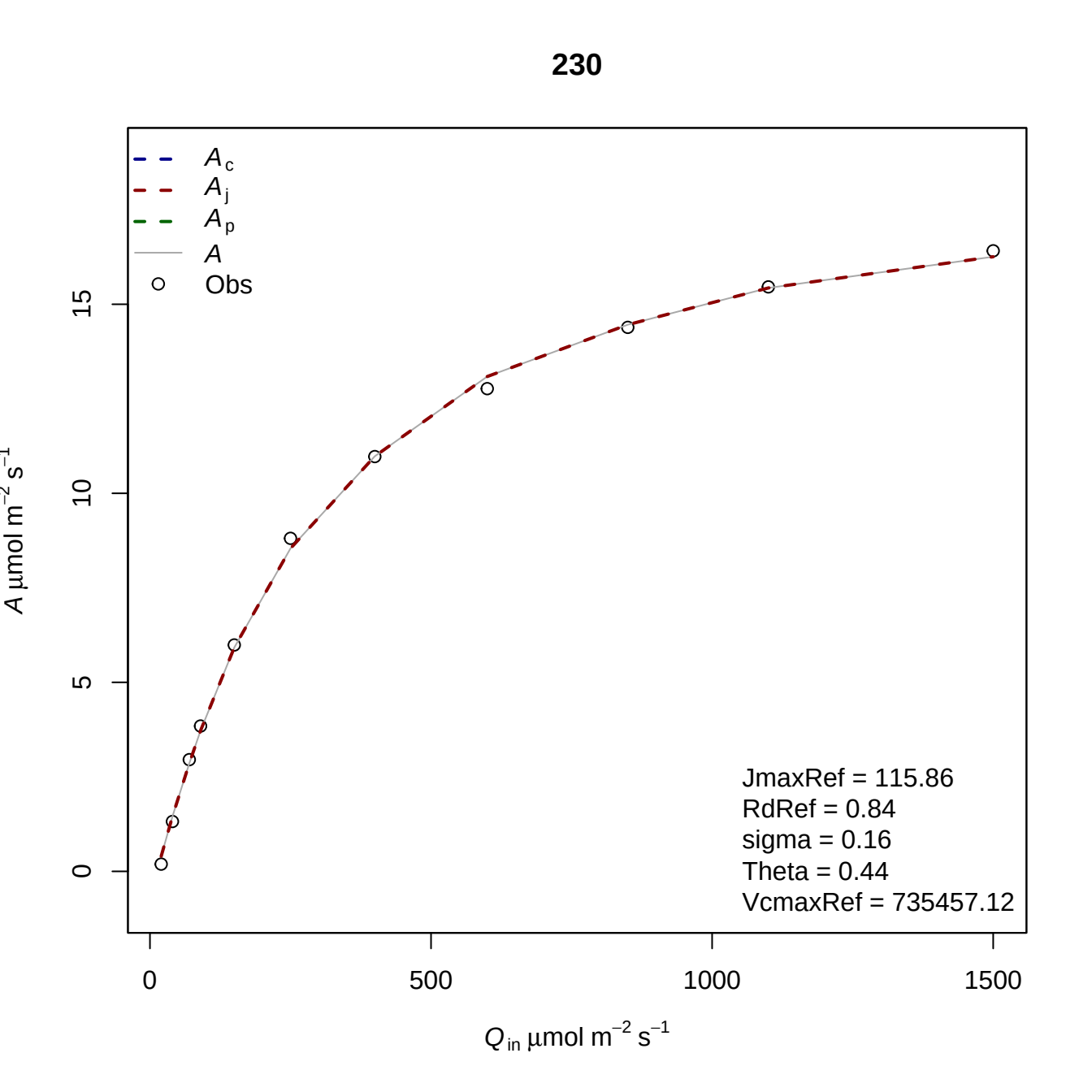

225
A_c
A_j
A_p
A
Obs
JmaxRef = 59.4
RdRef = 1.12
sigma = 0.25
Theta = 0.2
VcmaxRef = 5556617.93
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

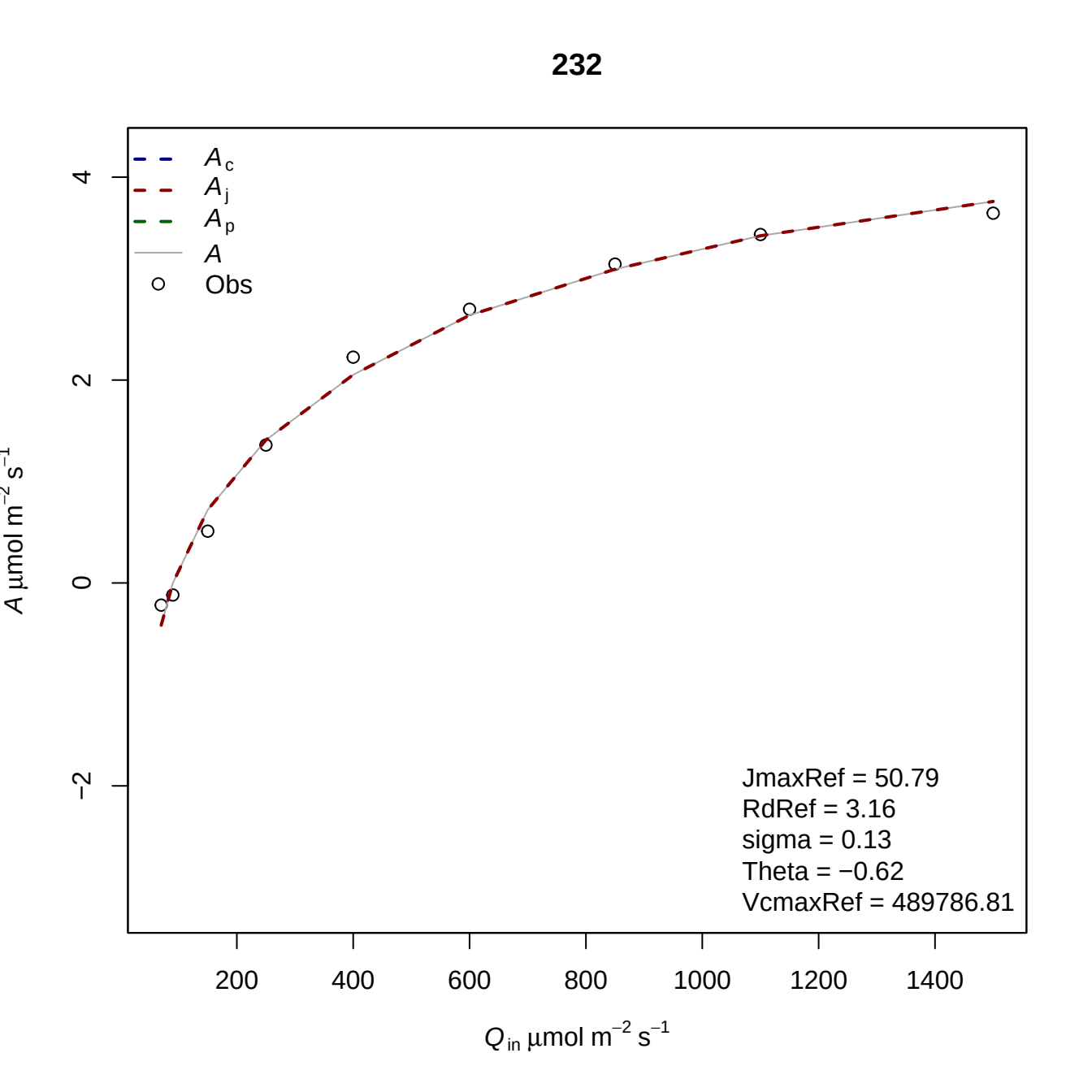

226
A_c
A_j
A_p
A
Obs
JmaxRef = 75.94
RdRef = 3.02
sigma = 0.13
Theta = 0.68
VcmaxRef = 54.6
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

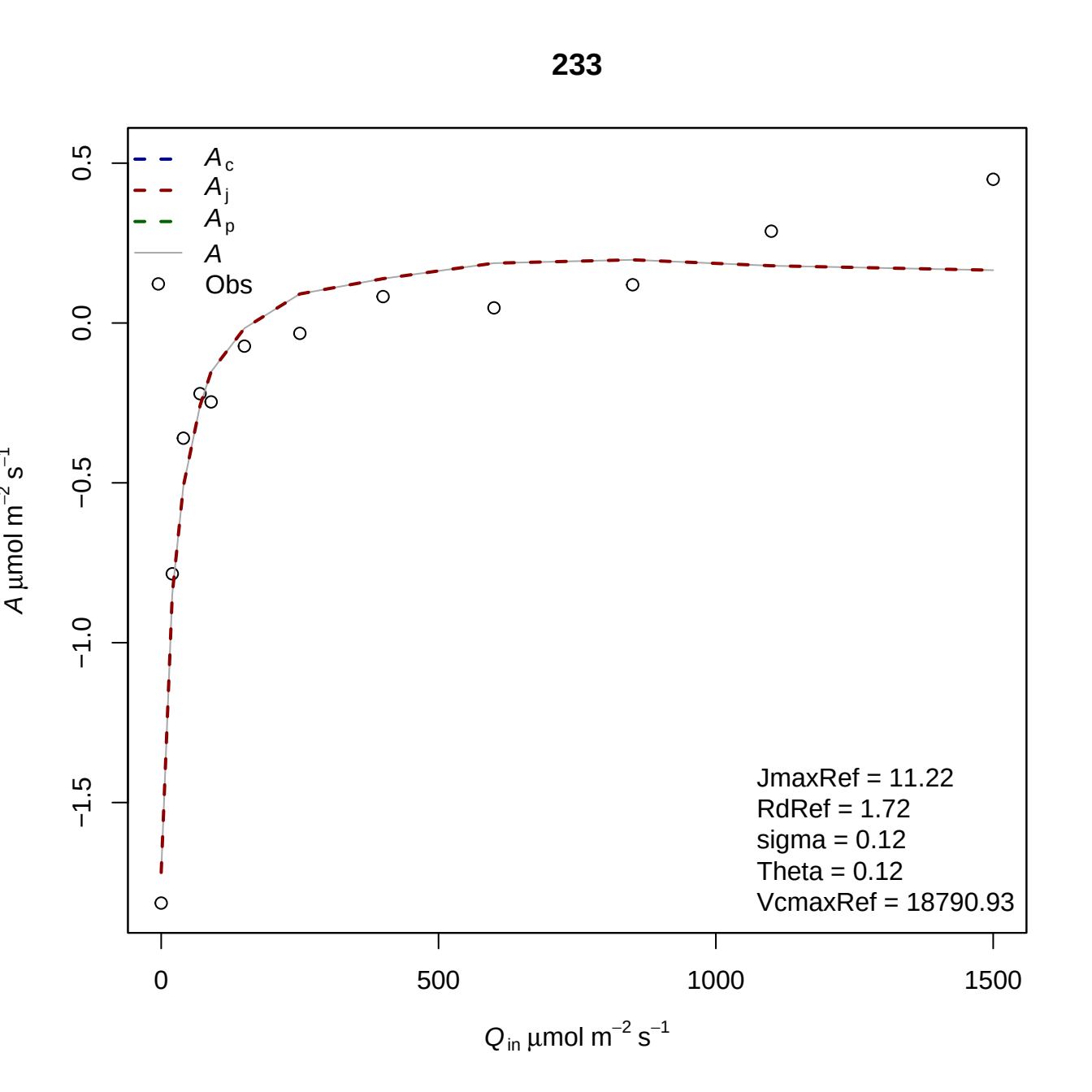

227
A_c
A_j
A_p
A
Obs
JmaxRef = 108.39
RdRef = 1.02
sigma = 0.09
Theta = −0.02
VcmaxRef = 42.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

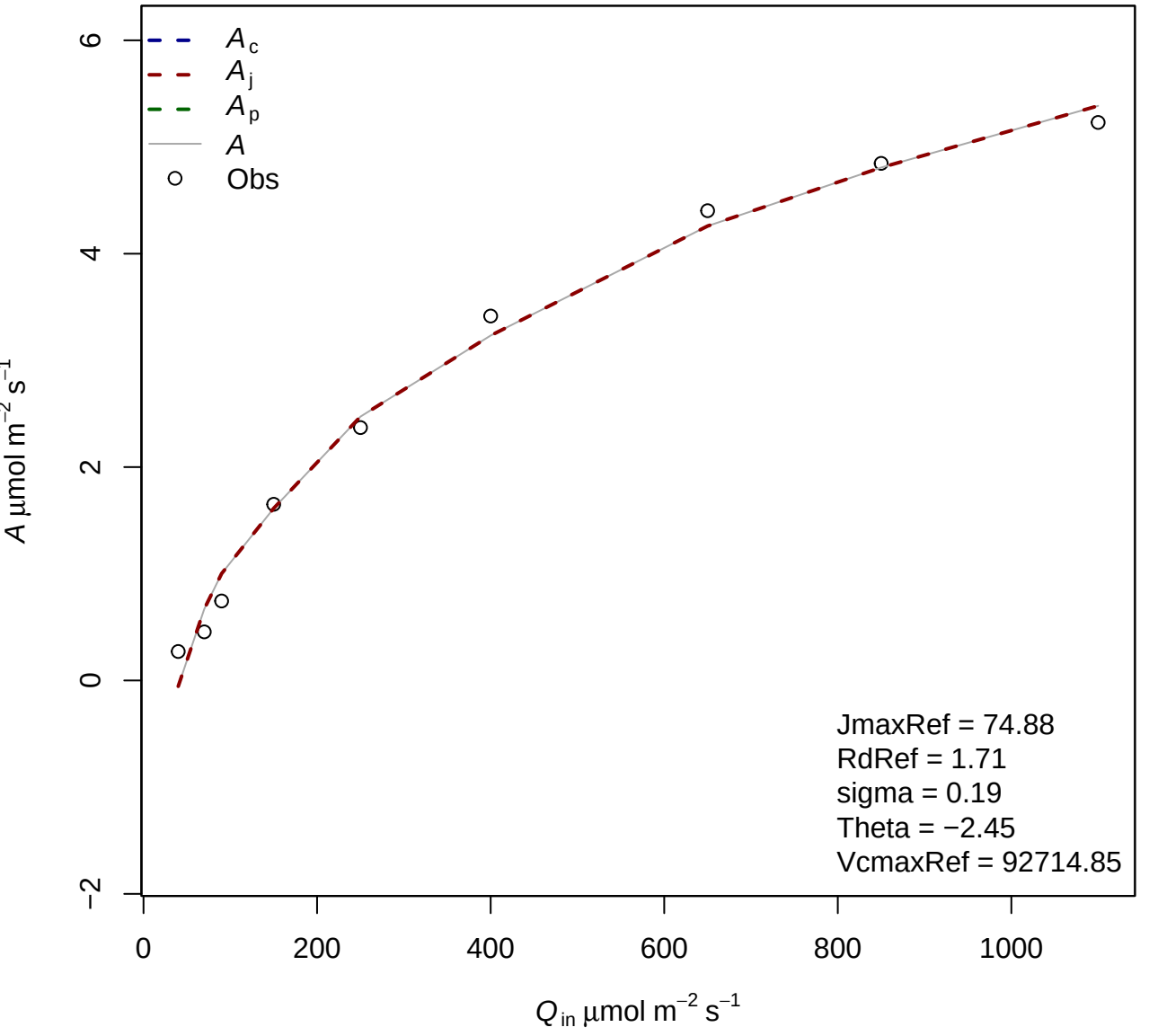

228
A_c
A_j
A_p
A
Obs
JmaxRef = 60.4
RdRef = 1.2
sigma = 0.19
Theta = 0.51
VcmaxRef = 731.34
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

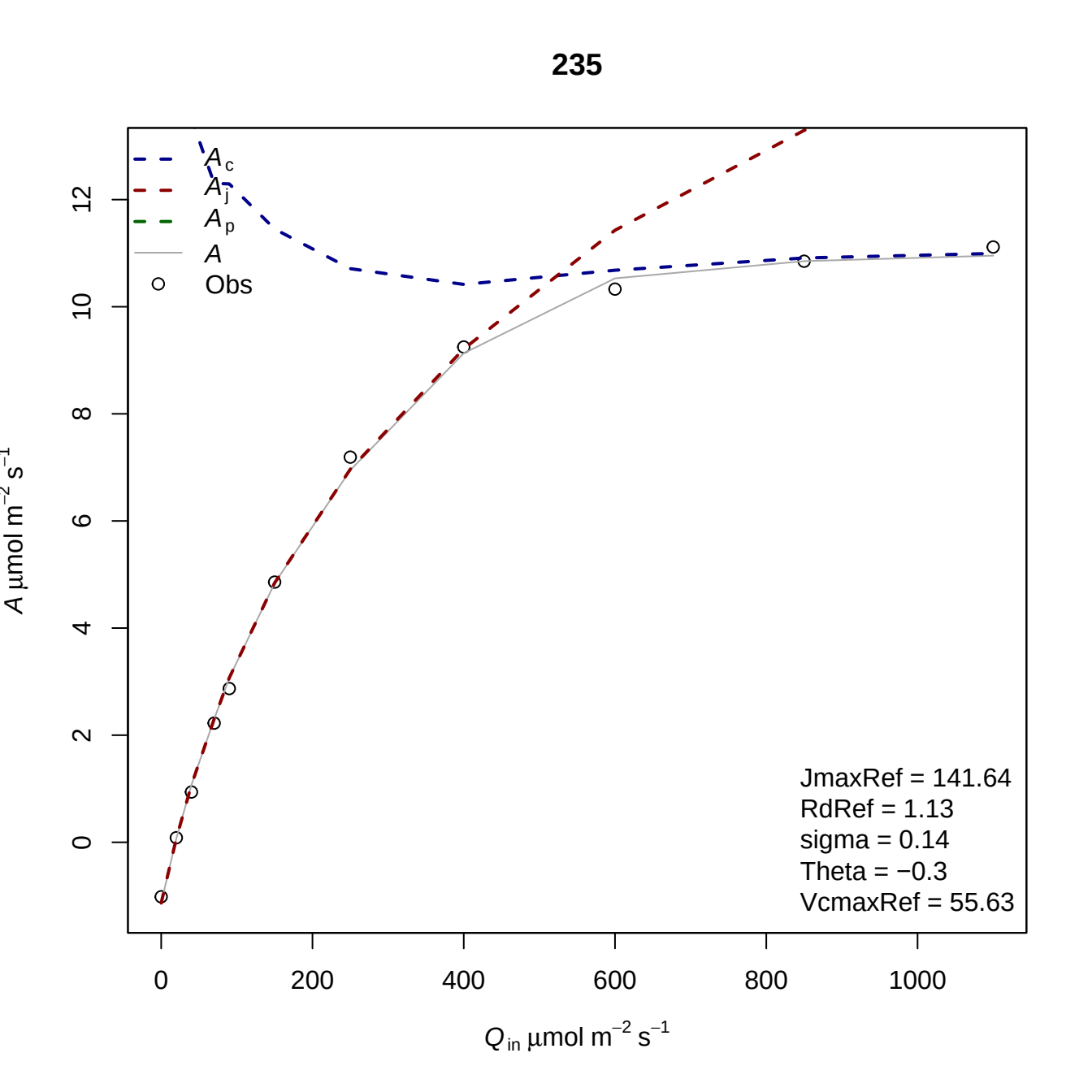

229
A_c
A_j
A_p
A
Obs
JmaxRef = 102.56
RdRef = 2.11
sigma = 0.11
Theta = 0.3
VcmaxRef = 59.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

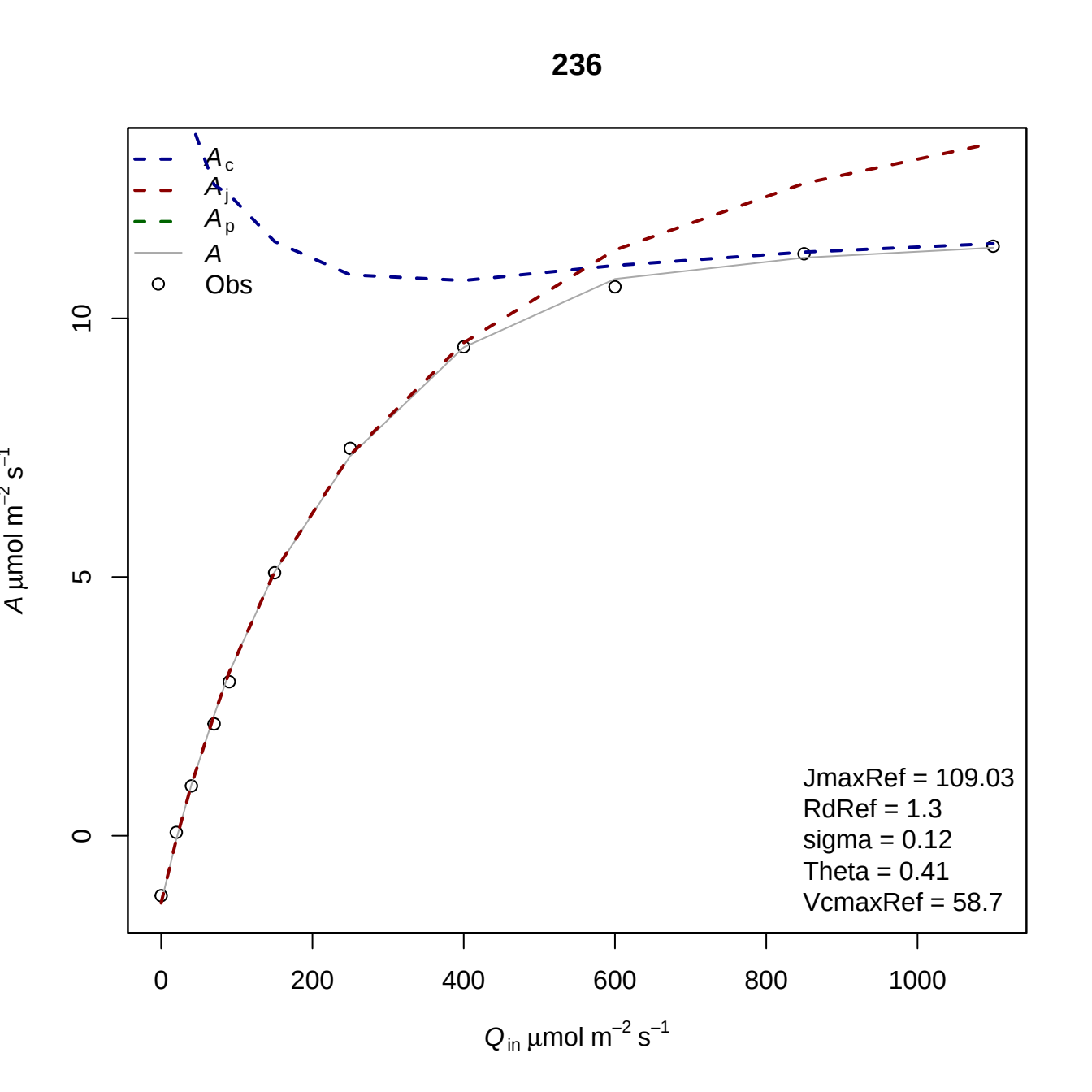

230
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 0.84
sigma = 0.16
Theta = 0.44
VcmaxRef = 735457.12
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

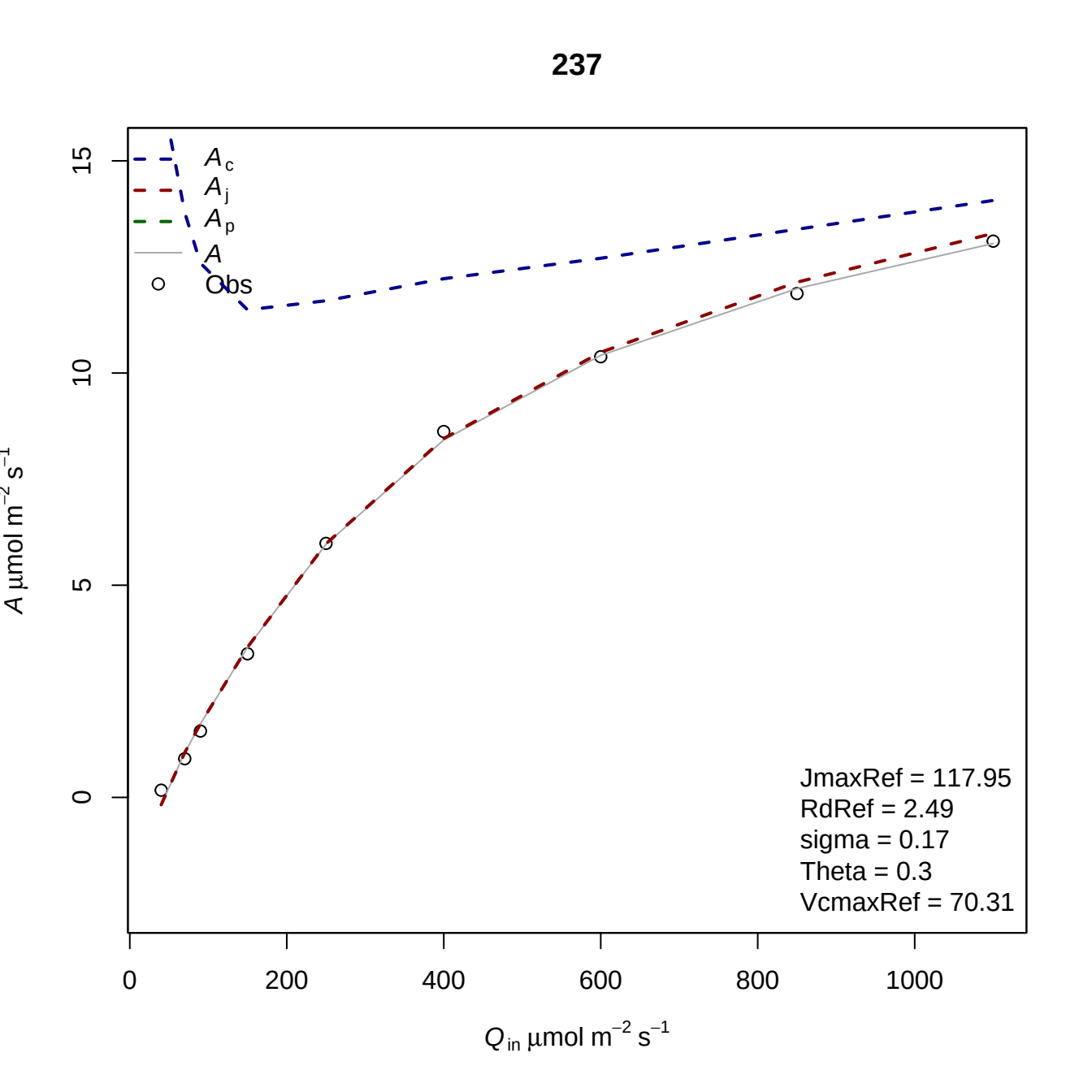

232
A_c
A_j
A_p
A
Obs
JmaxRef = 50.79
RdRef = 3.16
sigma = 0.13
Theta = −0.62
VcmaxRef = 489786.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
200
400
600
800
1000
1200
1400

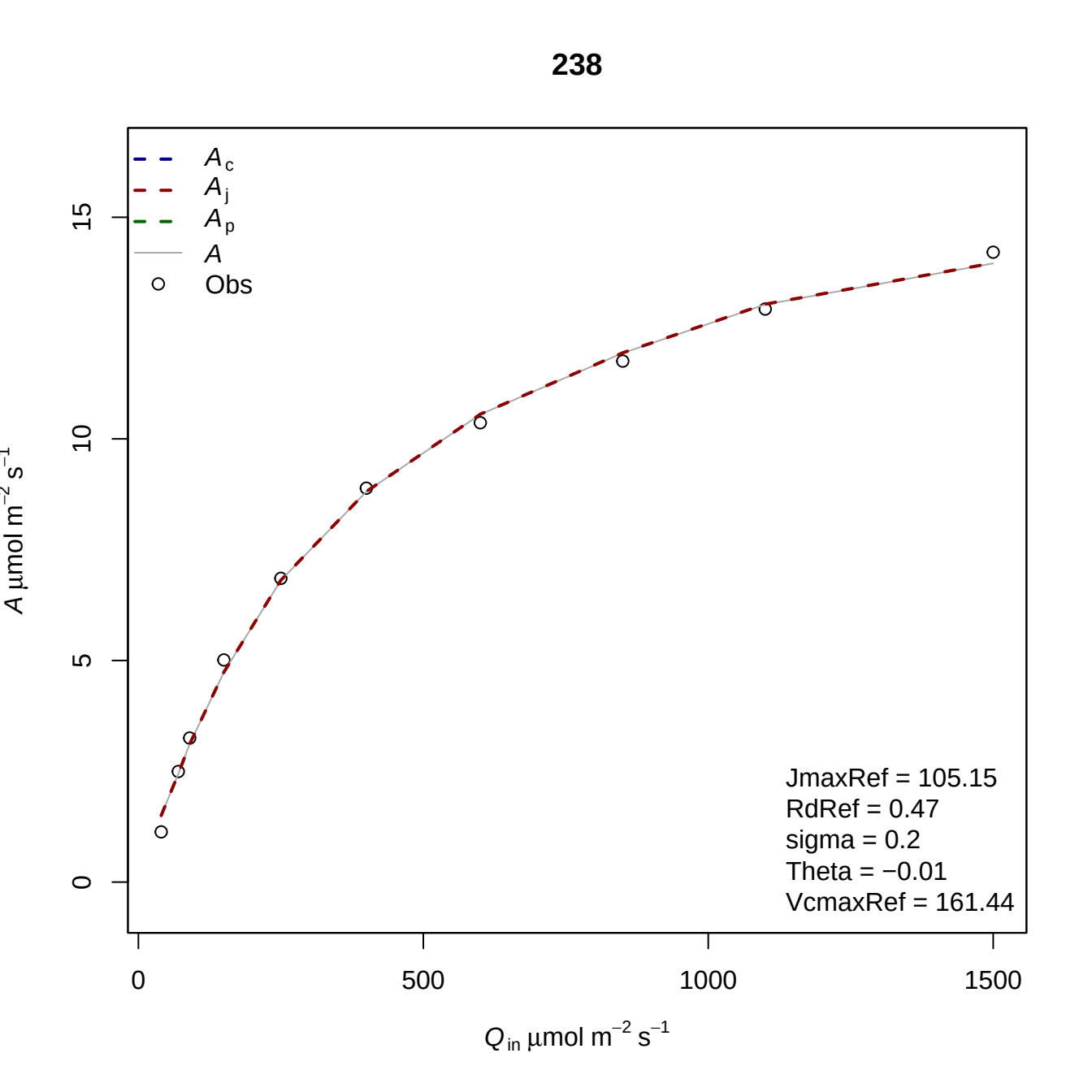

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.22
RdRef = 1.72
sigma = 0.12
Theta = 0.12
VcmaxRef = 18790.93
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

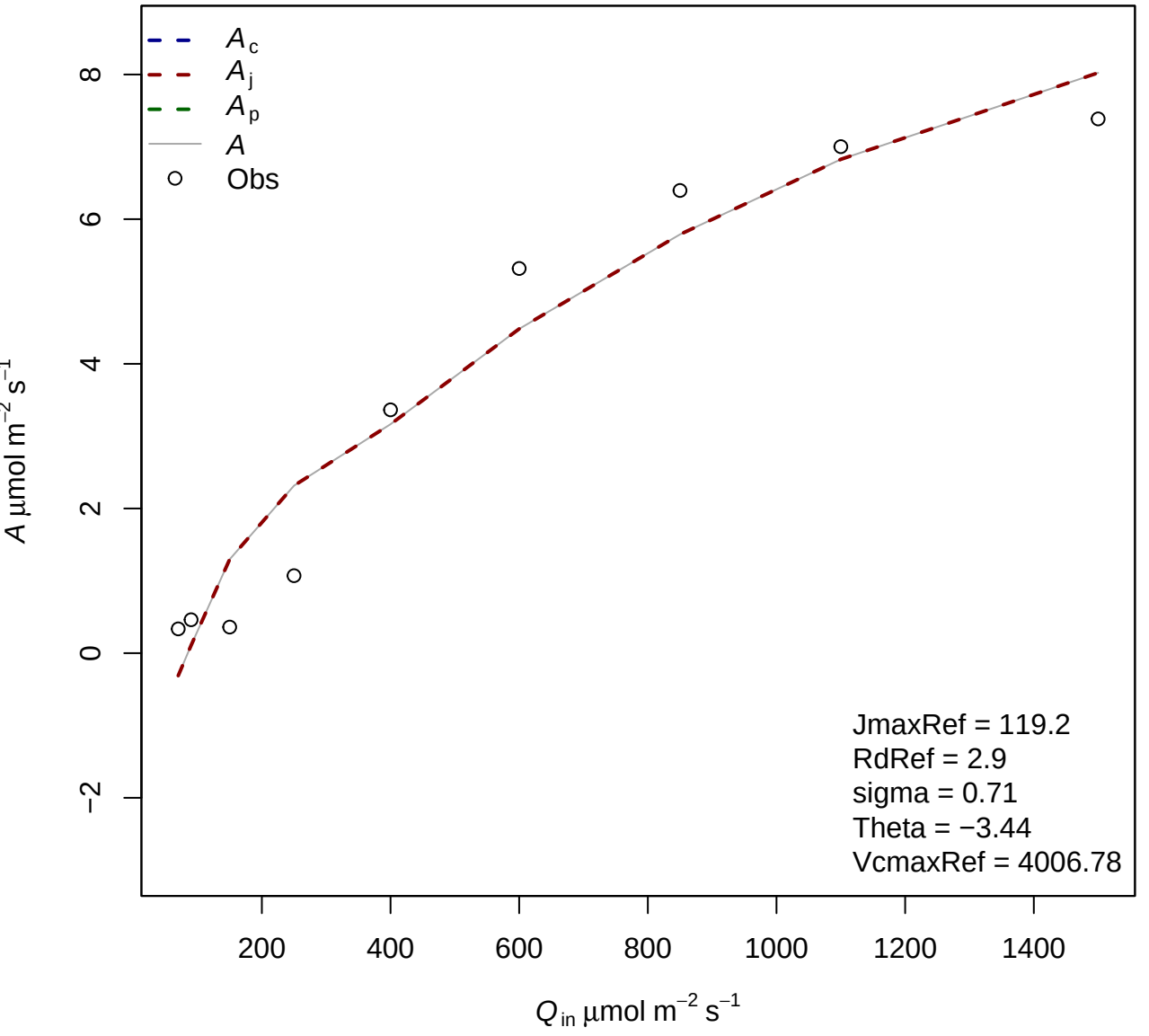

A_c
A_j
A_p
A
Obs
JmaxRef = 74.88
RdRef = 1.71
sigma = 0.19
Theta = −2.45
VcmaxRef = 92714.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

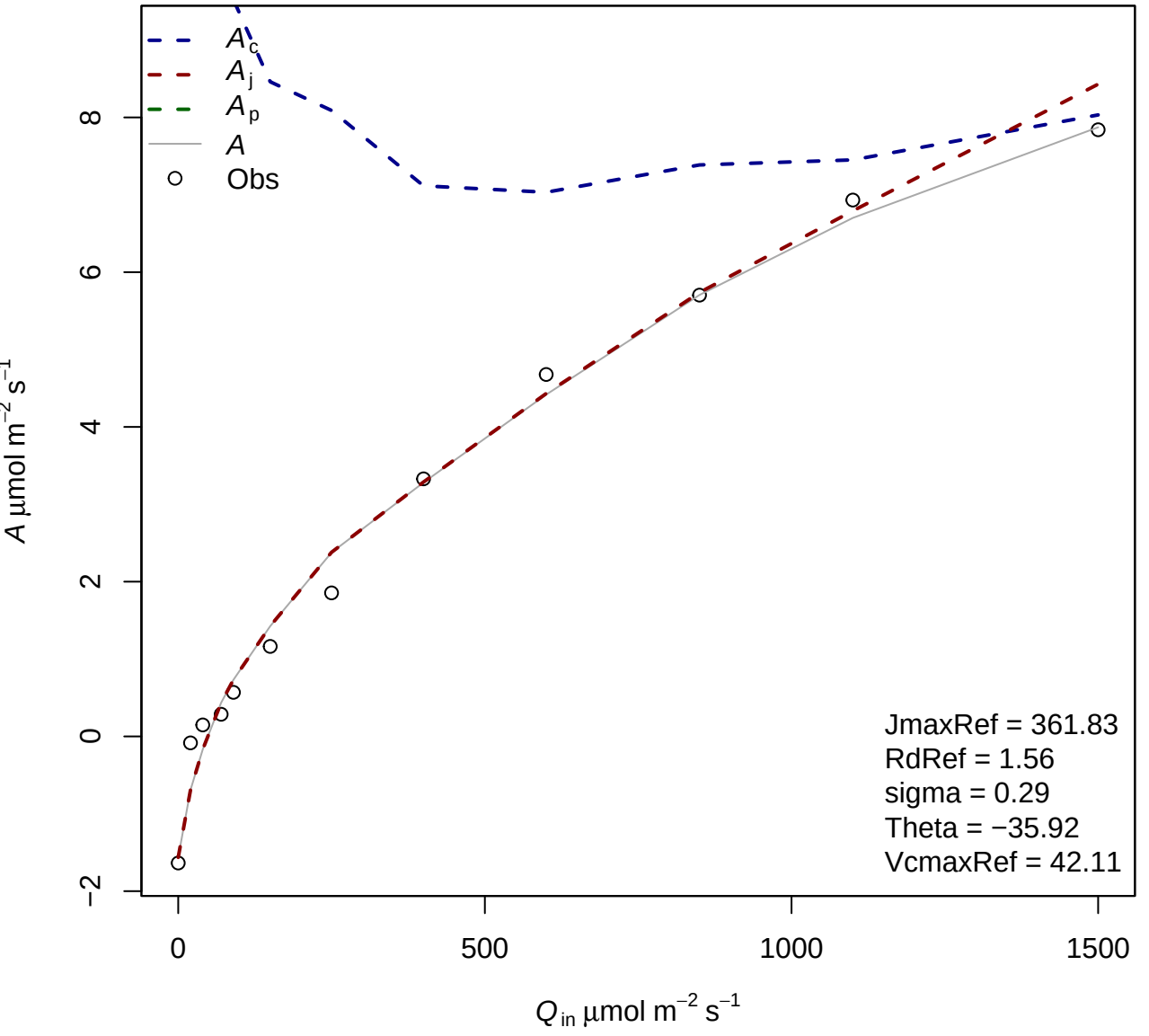

235
A_c
A_j
A_p
A
Obs
JmaxRef = 141.64
RdRef = 1.13
sigma = 0.14
Theta = −0.3
VcmaxRef = 55.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000

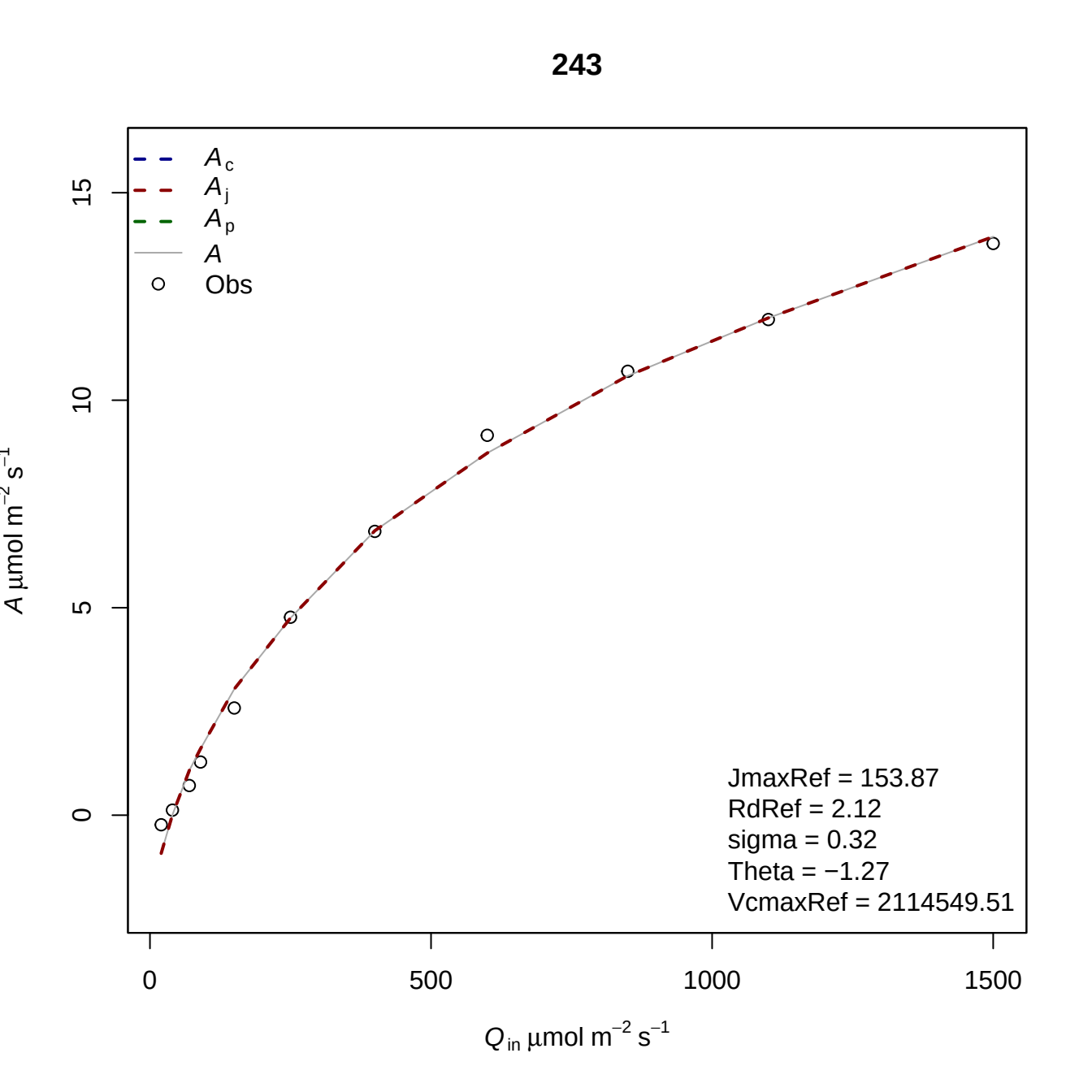

236
A_c
A_j
A_p
A
Obs
JmaxRef = 109.03
RdRef = 1.3
sigma = 0.12
Theta = 0.41
VcmaxRef = 58.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

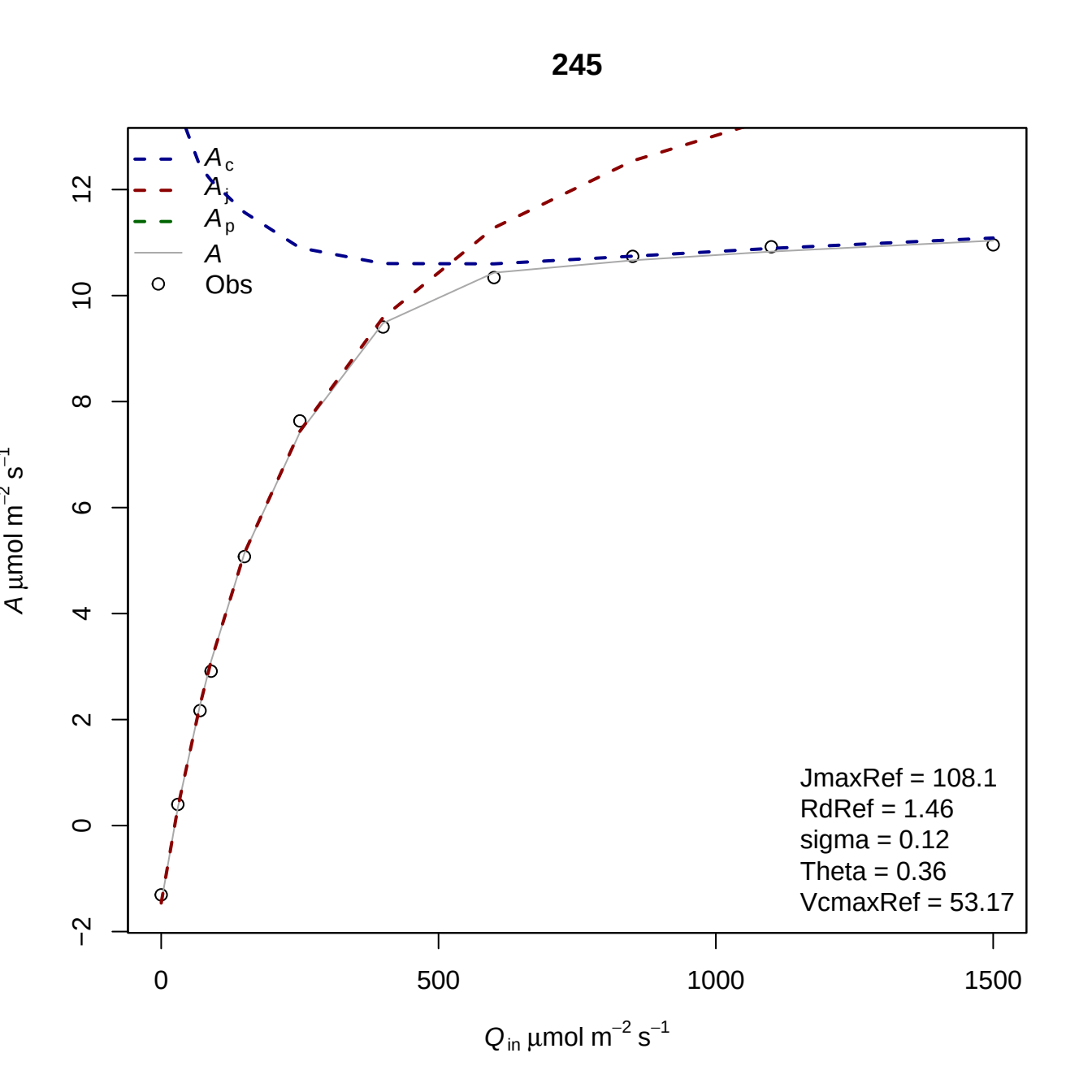

237
A_c
A_j
A_p
A
Obs
JmaxRef = 117.95
RdRef = 2.49
sigma = 0.17
Theta = 0.3
VcmaxRef = 70.31
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

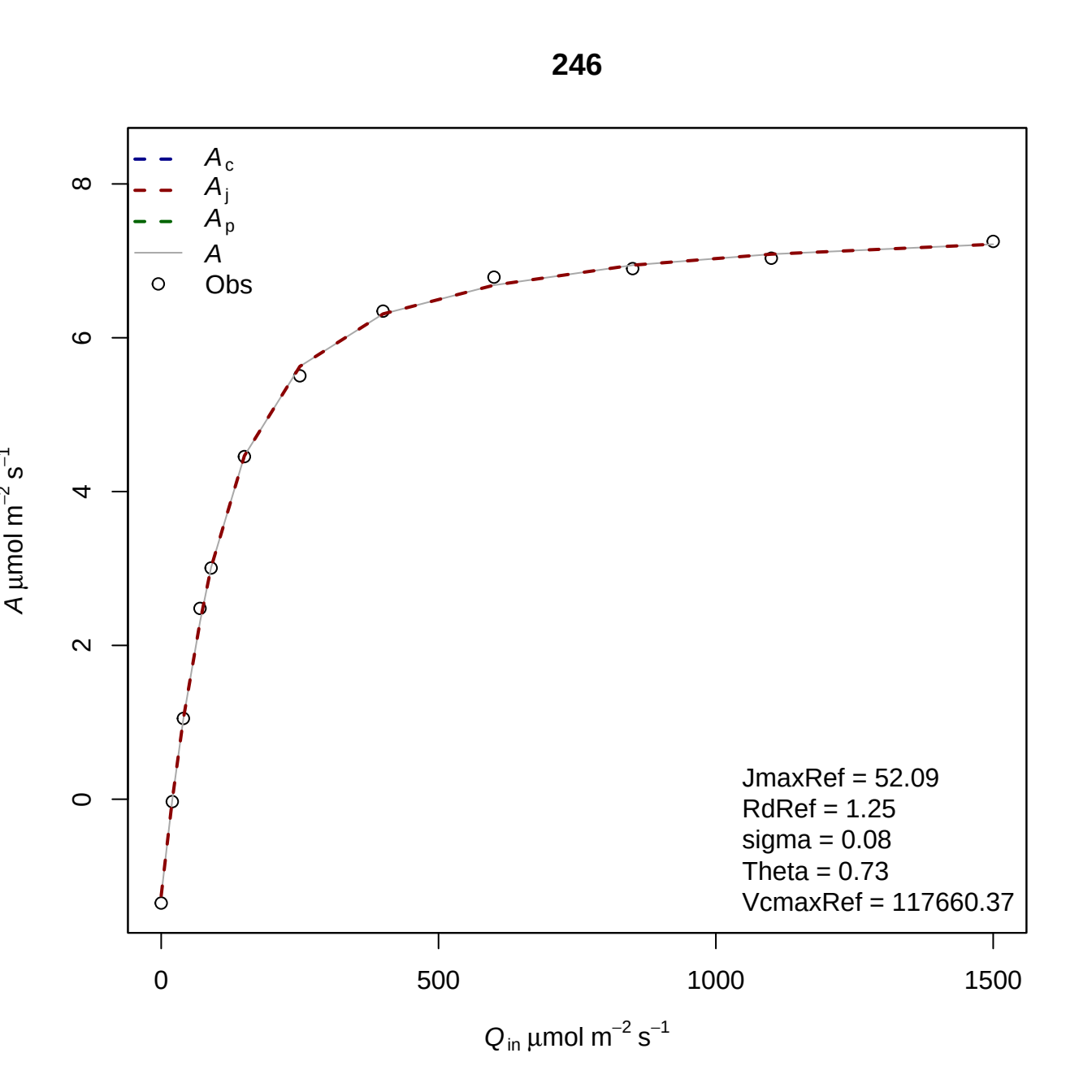

238
A_c
A_j
A_p
A
Obs
JmaxRef = 105.15
RdRef = 0.47
sigma = 0.2
Theta = −0.01
VcmaxRef = 161.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

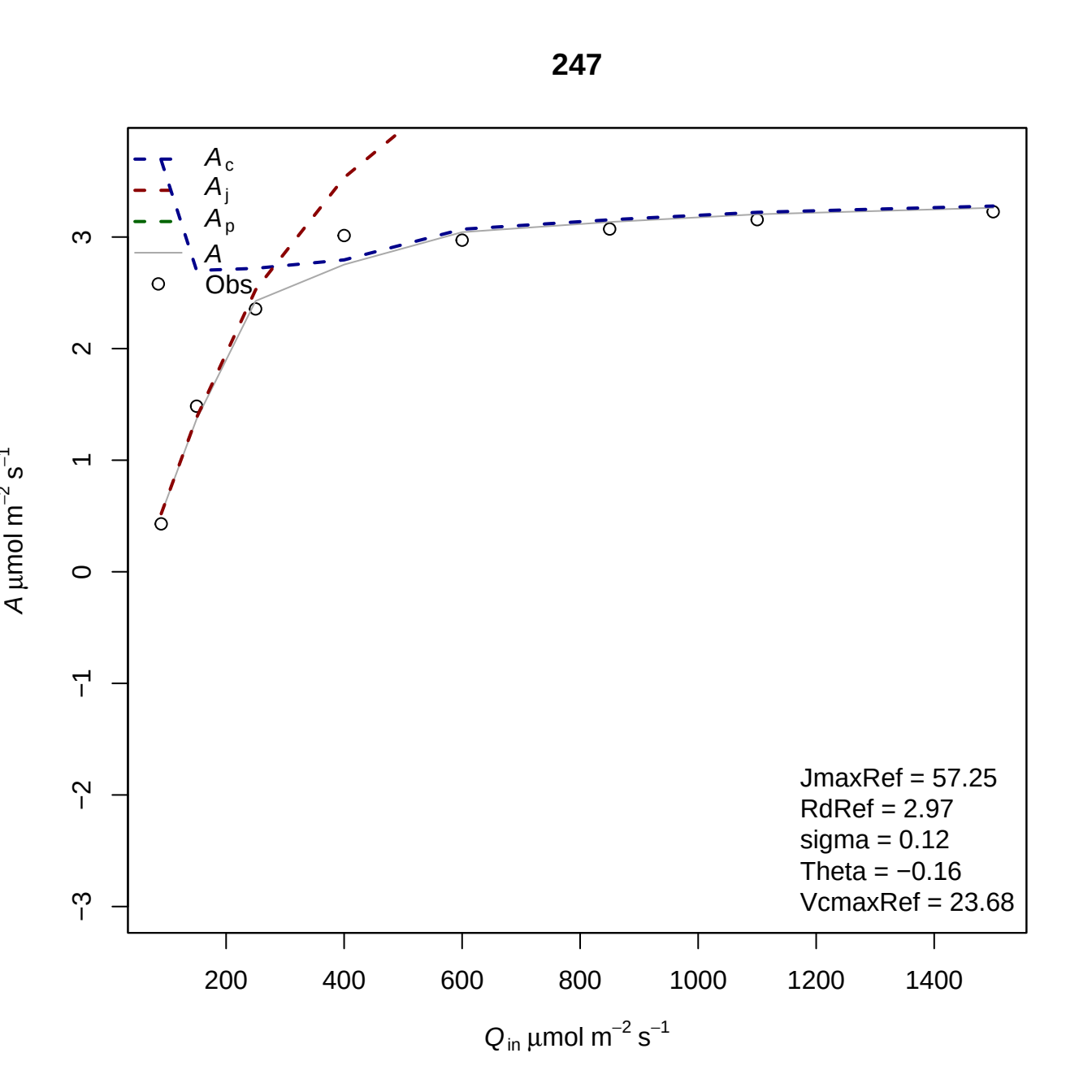

A_c
A_j
A_p
A
Obs
JmaxRef = 119.2
RdRef = 2.9
sigma = 0.71
Theta = −3.44
VcmaxRef = 4006.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

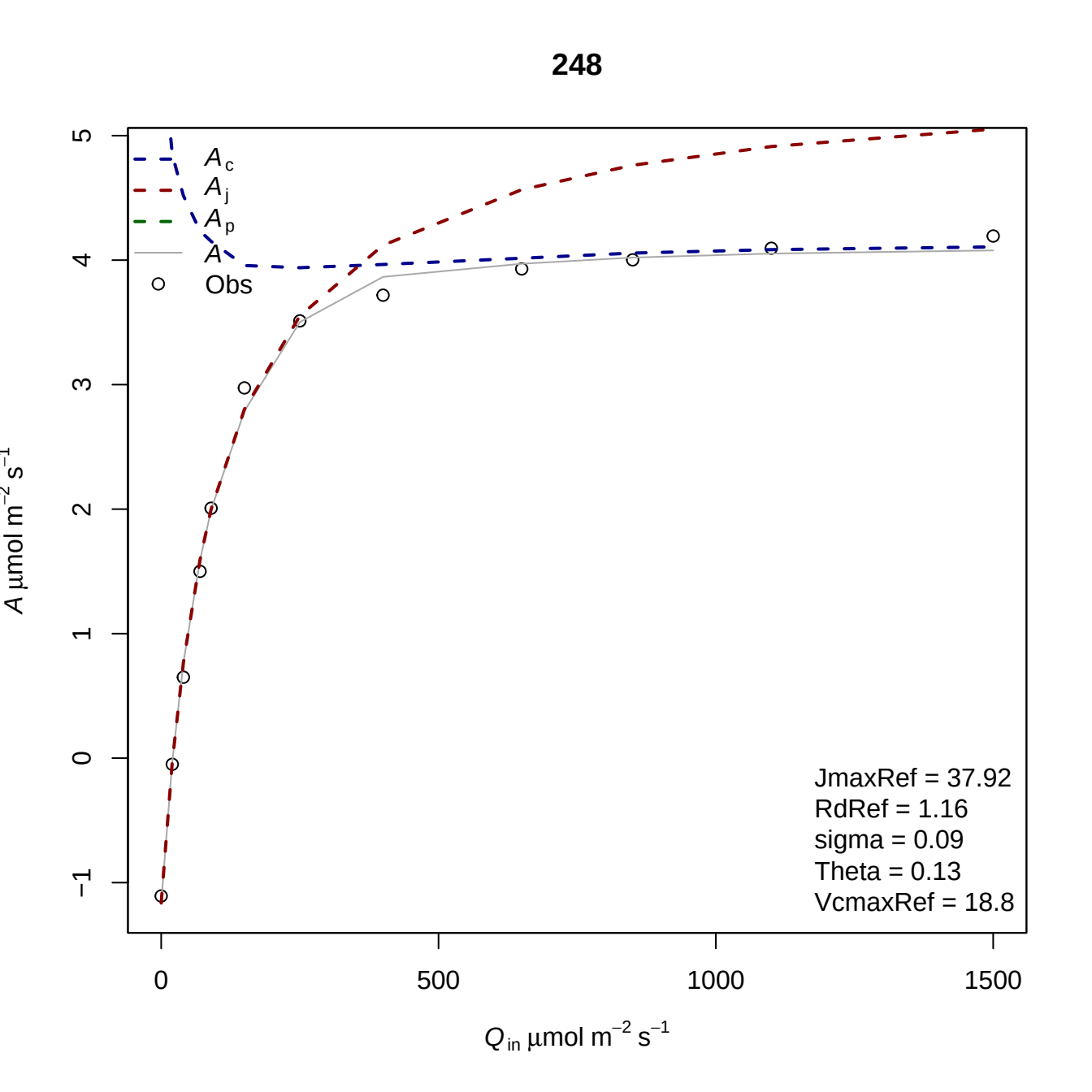

A_c
A_j
A_p
A
Obs
JmaxRef = 361.83
RdRef = 1.56
sigma = 0.29
Theta = −35.92
VcmaxRef = 42.11
−2
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

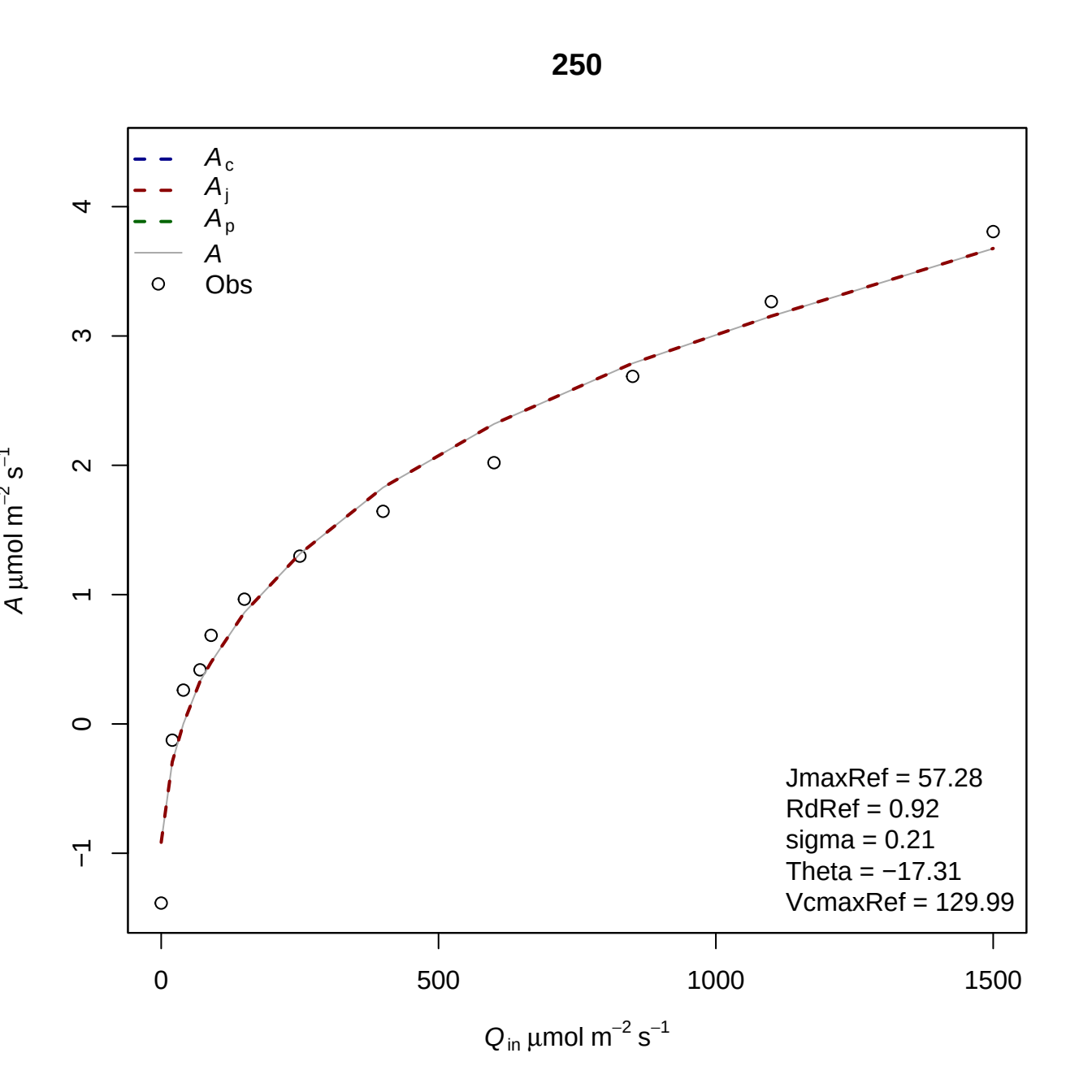

243
A_c
A_j
A_p
A
Obs
JmaxRef = 153.87
RdRef = 2.12
sigma = 0.32
Theta = −1.27
VcmaxRef = 2114549.51
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

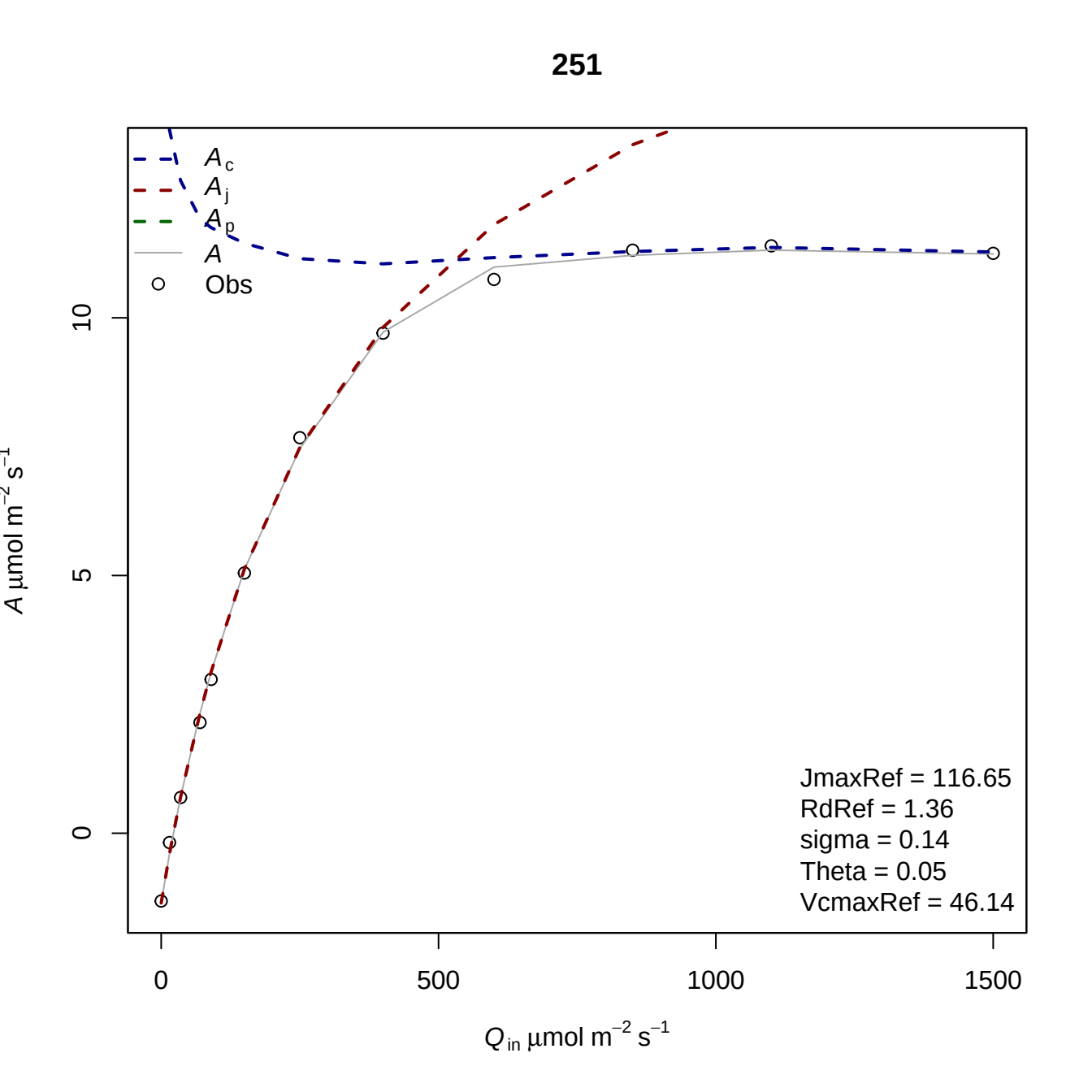

245
A_c
A_j
A_p
A
Obs
JmaxRef = 108.1
RdRef = 1.46
sigma = 0.12
Theta = 0.36
VcmaxRef = 53.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

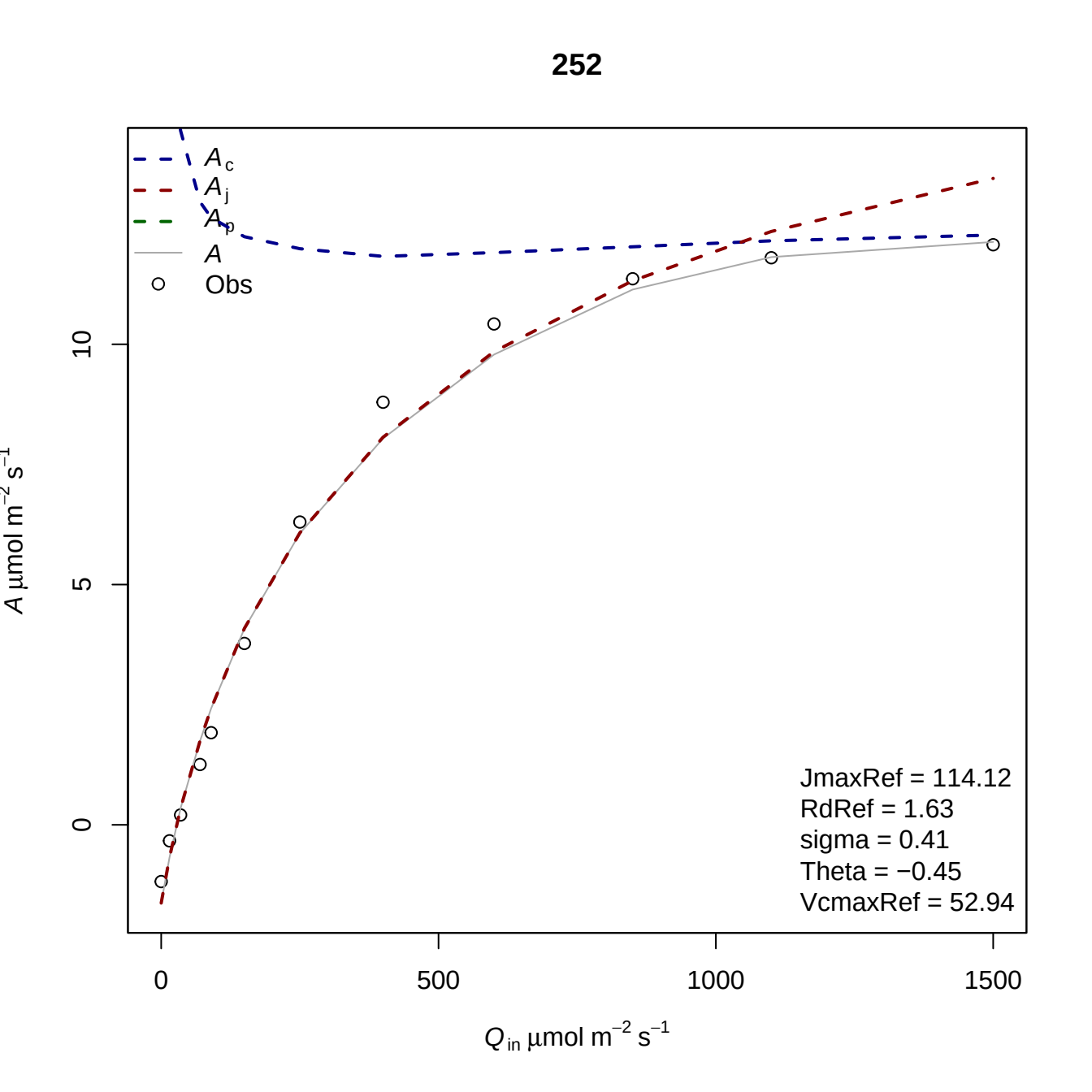

246
A_c
A_j
A_p
A
Obs
JmaxRef = 52.09
RdRef = 1.25
sigma = 0.08
Theta = 0.73
VcmaxRef = 117660.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

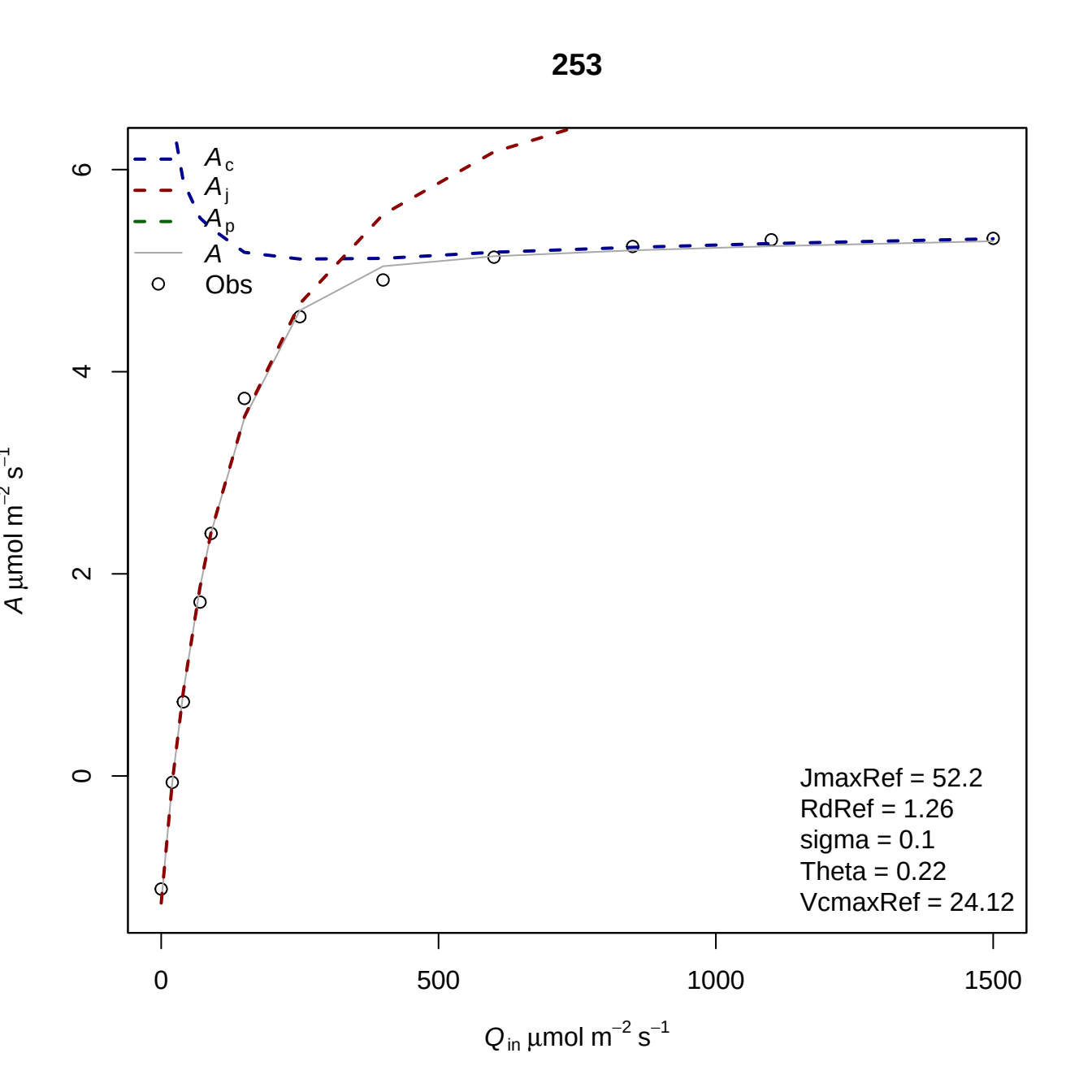

247
A_c
A_j
A_p
A
Obs
JmaxRef = 57.25
RdRef = 2.97
sigma = 0.12
Theta = −0.16
VcmaxRef = 23.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
200
400
600
800
1000
1200
1400

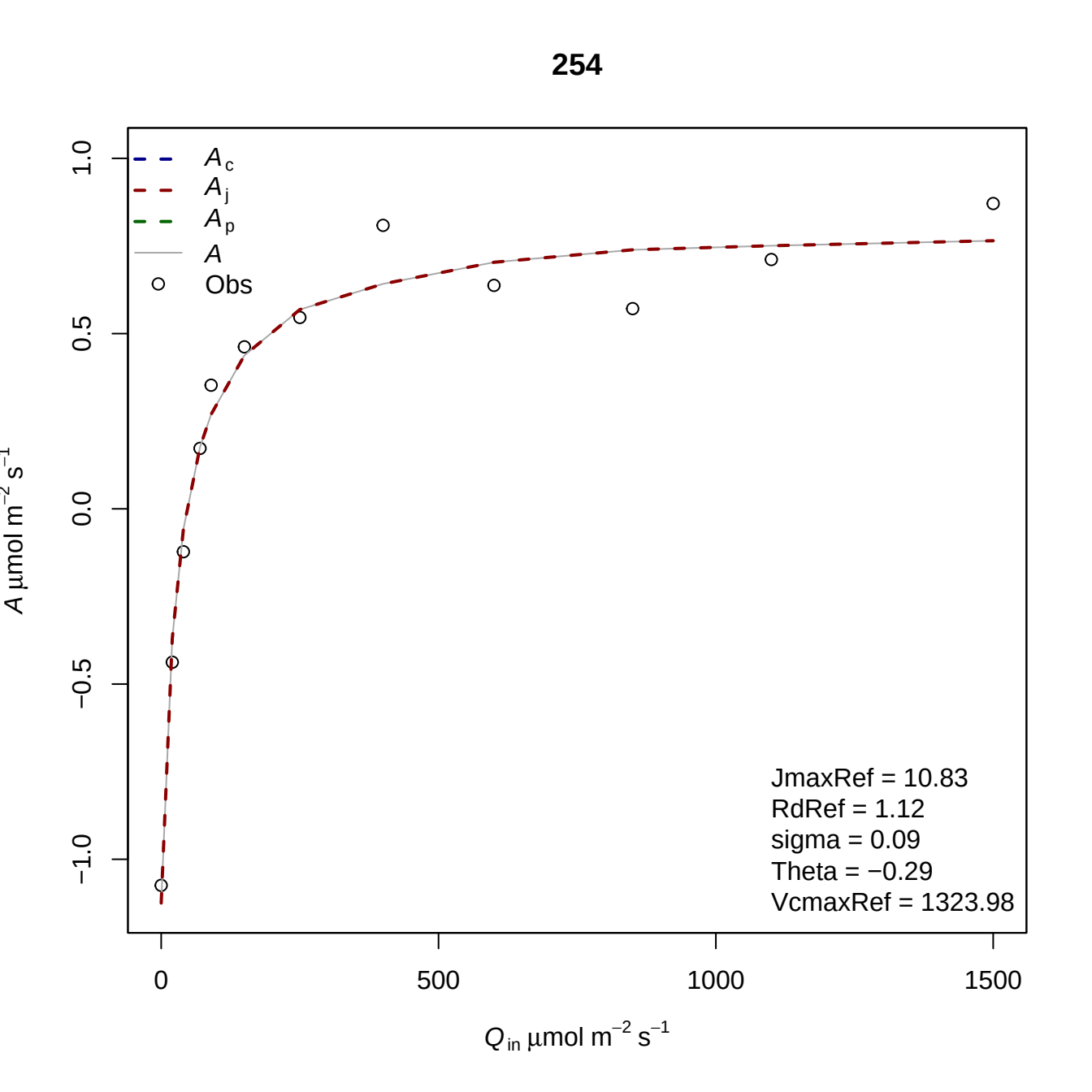

248
A_c
A_j
A_p
A
Obs
JmaxRef = 37.92
RdRef = 1.16
sigma = 0.09
Theta = 0.13
VcmaxRef = 18.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

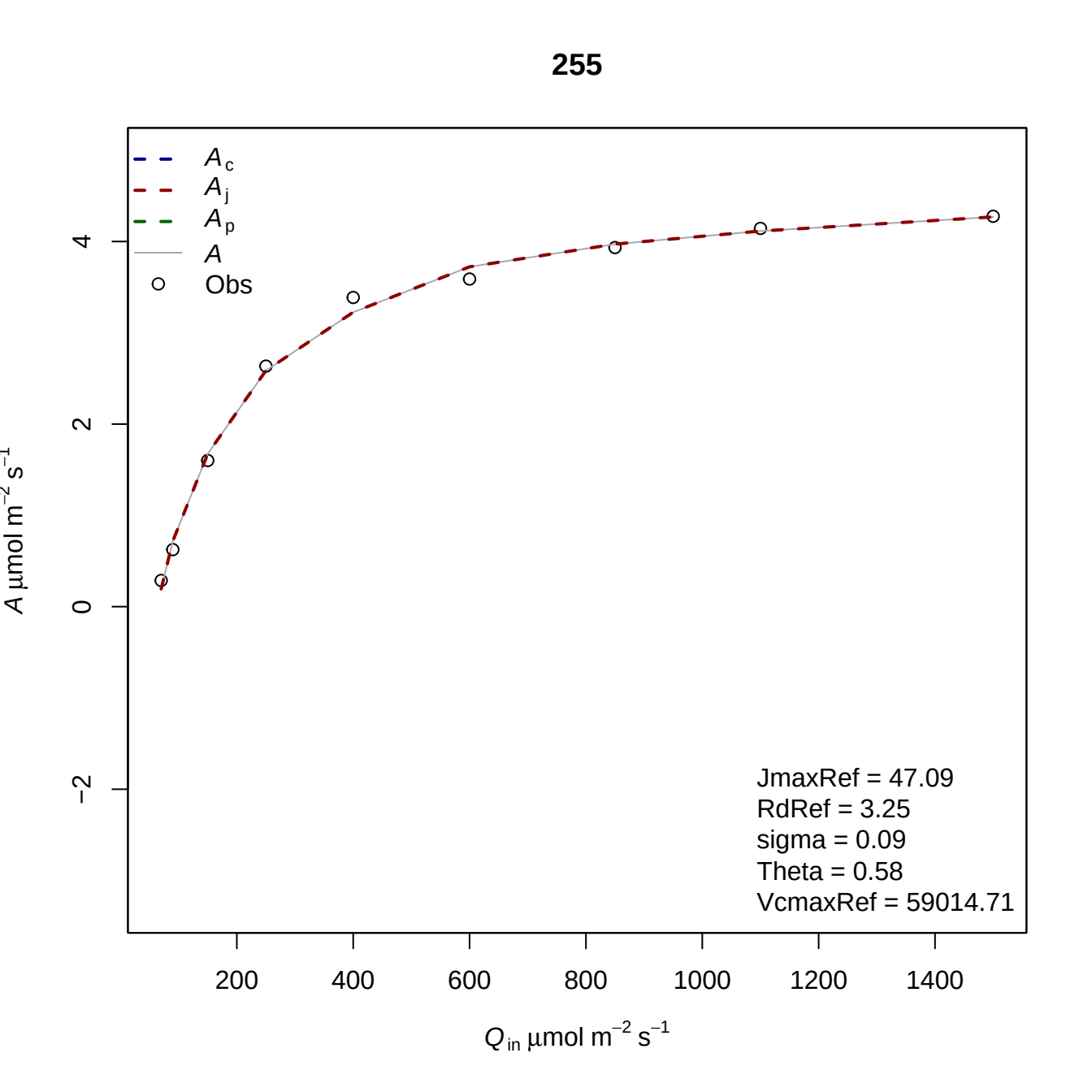

250
A_c
A_j
A_p
A
Obs
JmaxRef = 57.28
RdRef = 0.92
sigma = 0.21
Theta = −17.31
VcmaxRef = 129.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

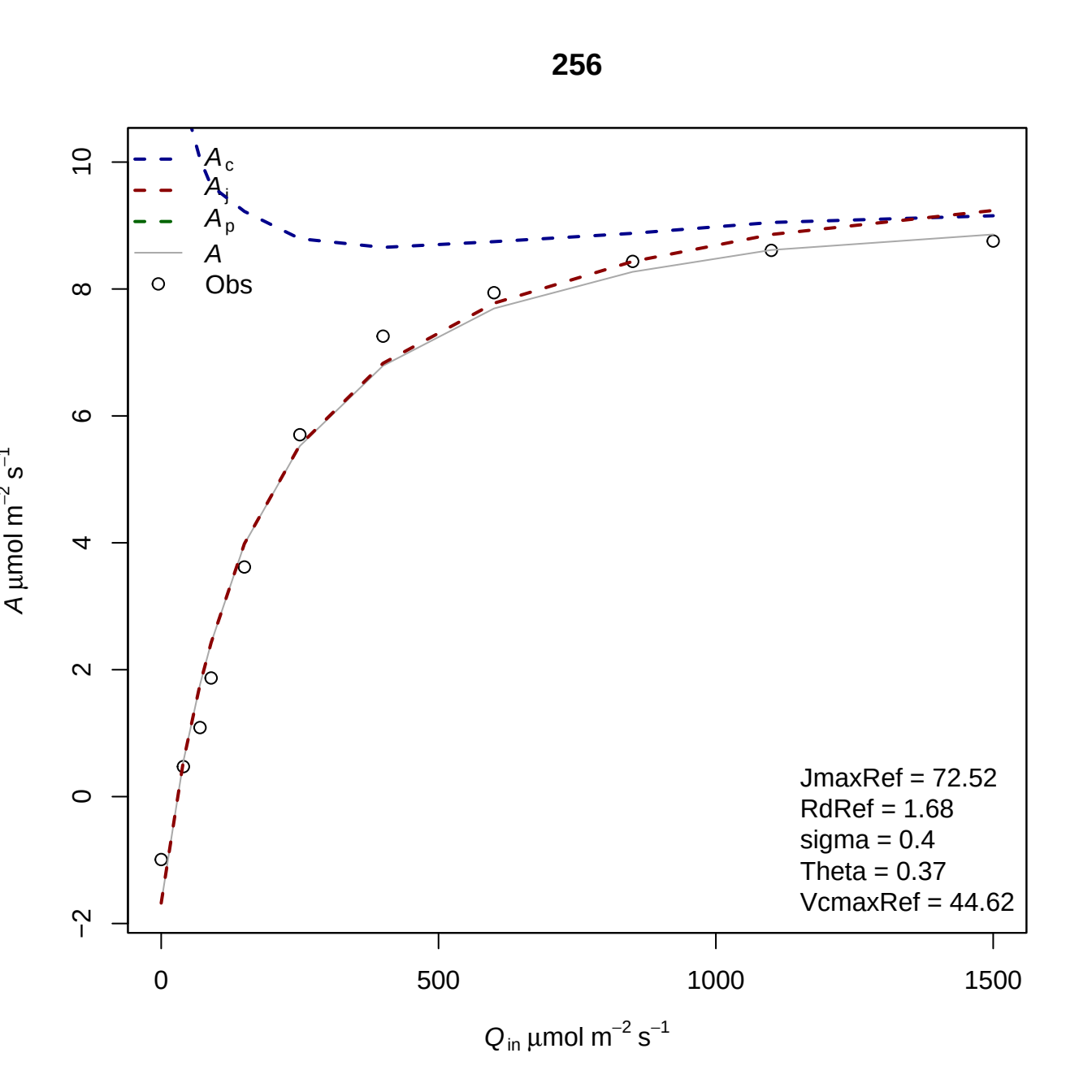

251
A_c
A_j
A_p
A
Obs
JmaxRef = 116.65
RdRef = 1.36
sigma = 0.14
Theta = 0.05
VcmaxRef = 46.14
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

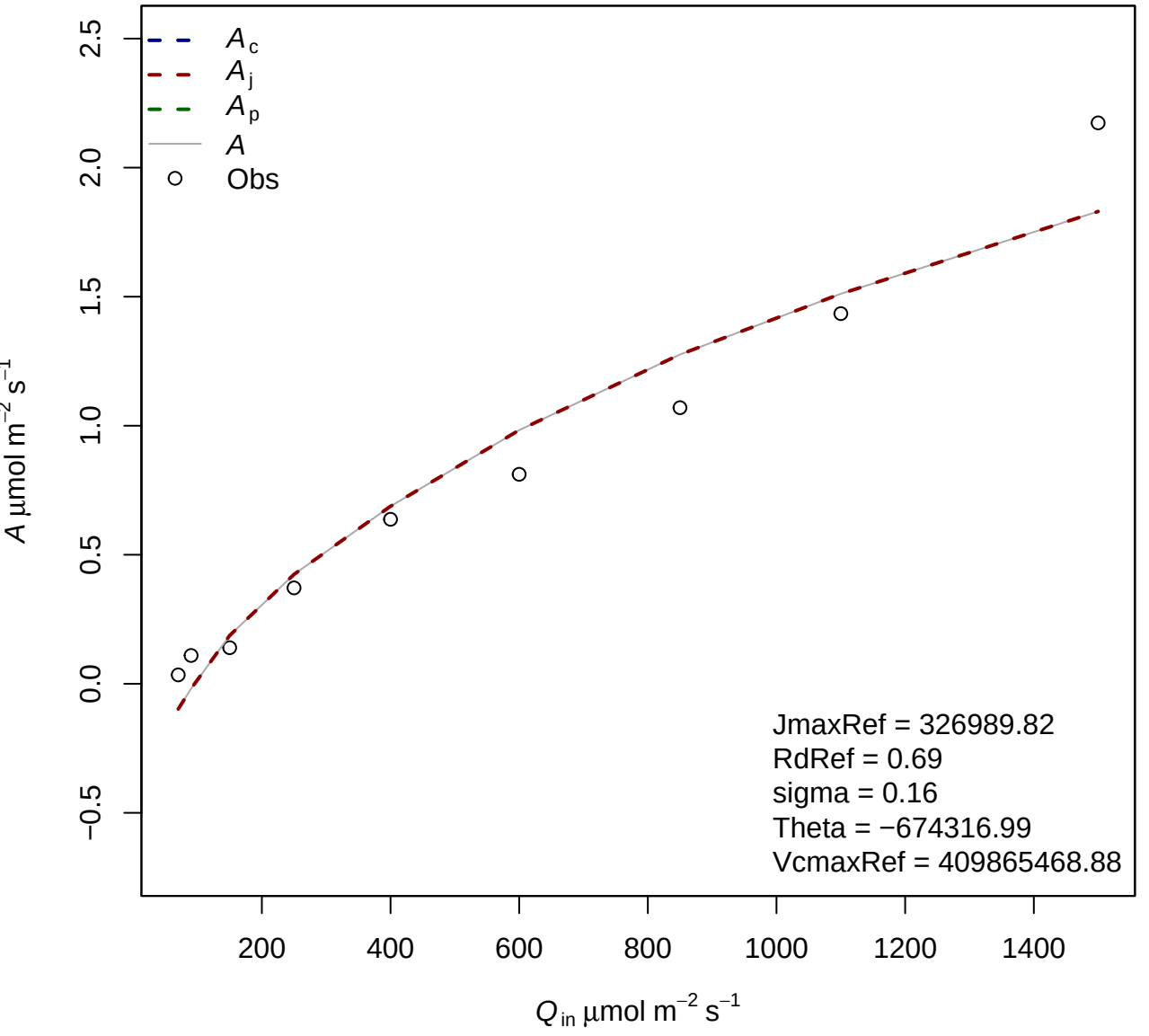

252
A_c
A_j
A_p
A
Obs
JmaxRef = 114.12
RdRef = 1.63
sigma = 0.41
Theta = −0.45
VcmaxRef = 52.94
0
5
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

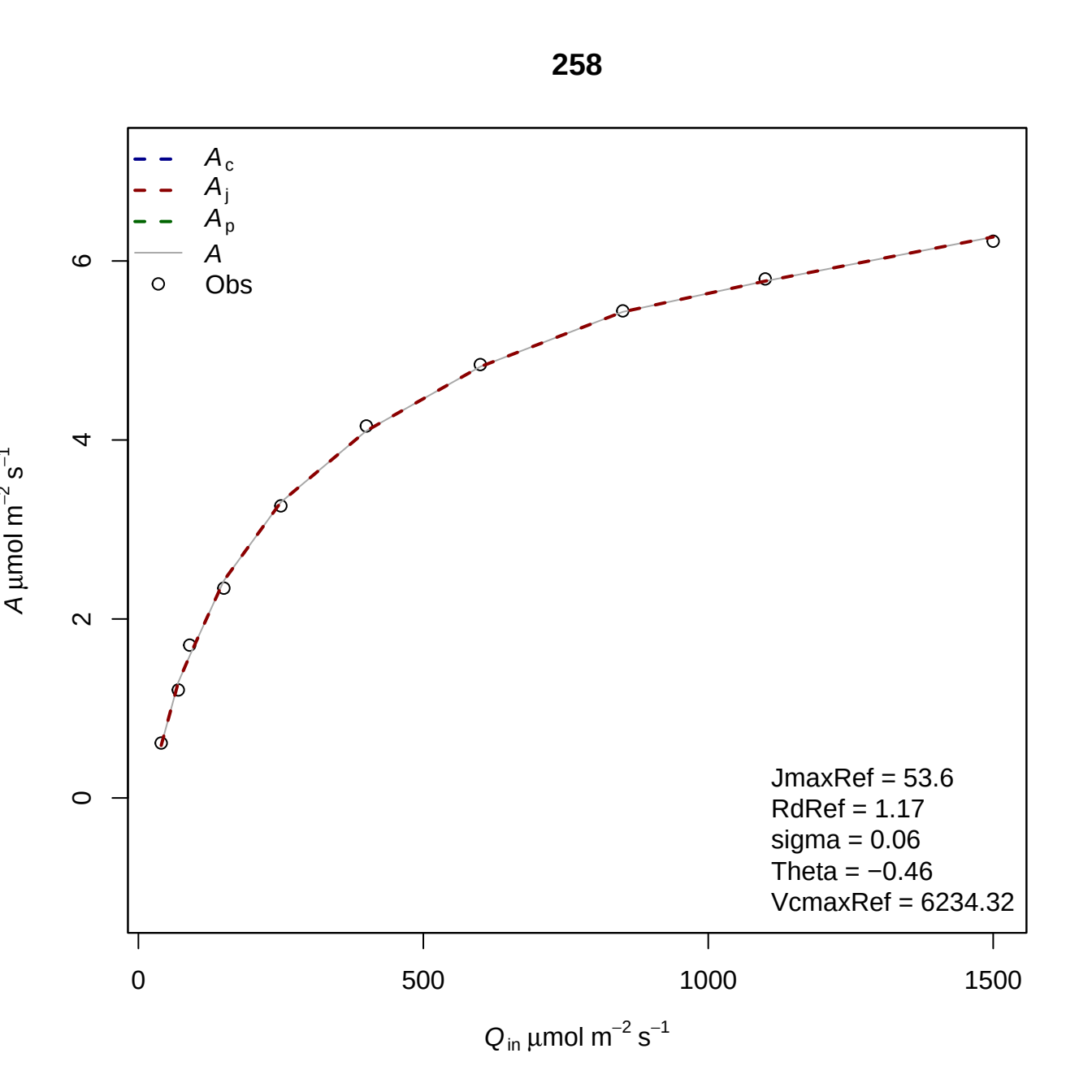

253
A_c
A_j
A_p
A
Obs
JmaxRef = 52.2
RdRef = 1.26
sigma = 0.1
Theta = 0.22
VcmaxRef = 24.12
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

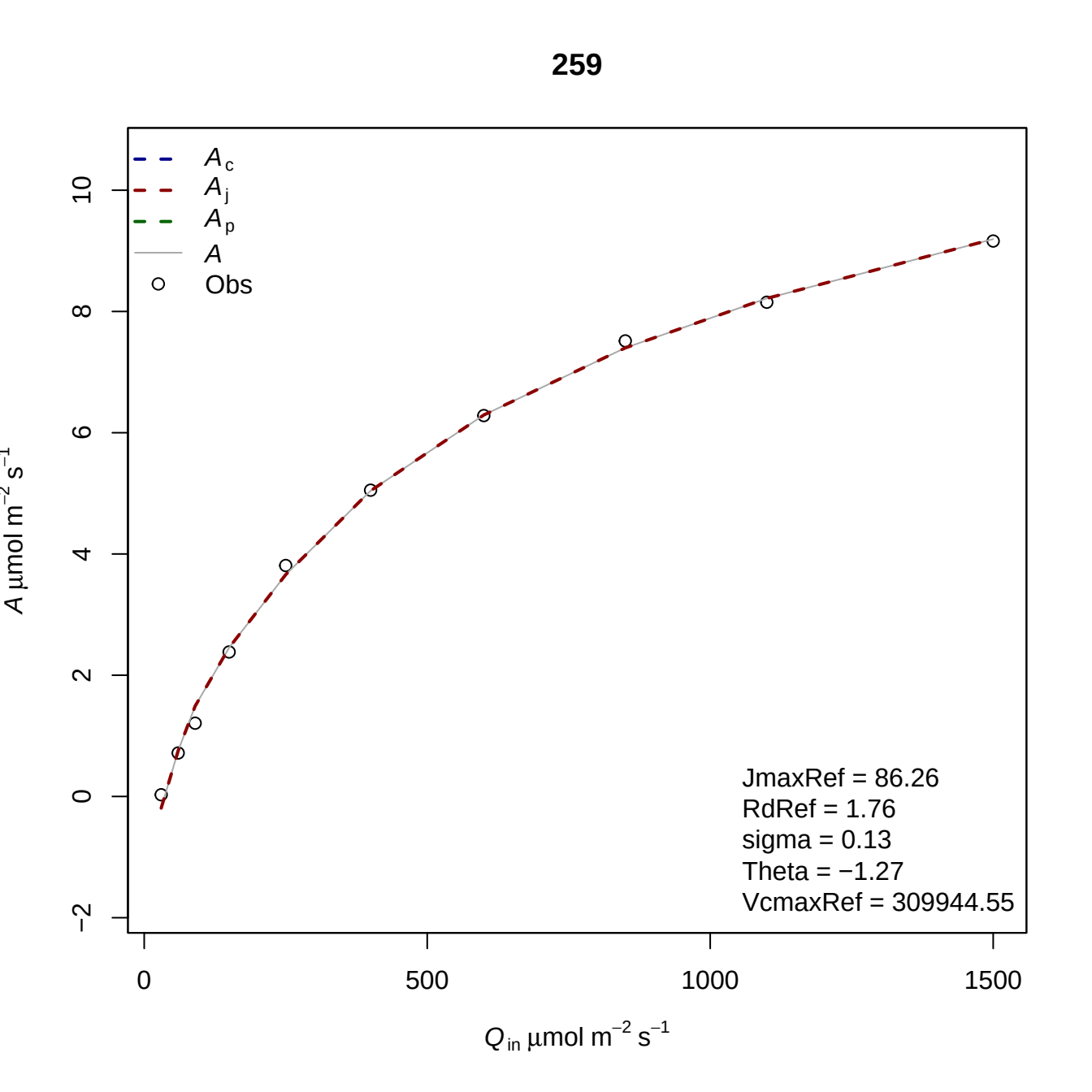

254
A_c
A_j
A_p
A
Obs
JmaxRef = 10.83
RdRef = 1.12
sigma = 0.09
Theta = −0.29
VcmaxRef = 1323.98
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

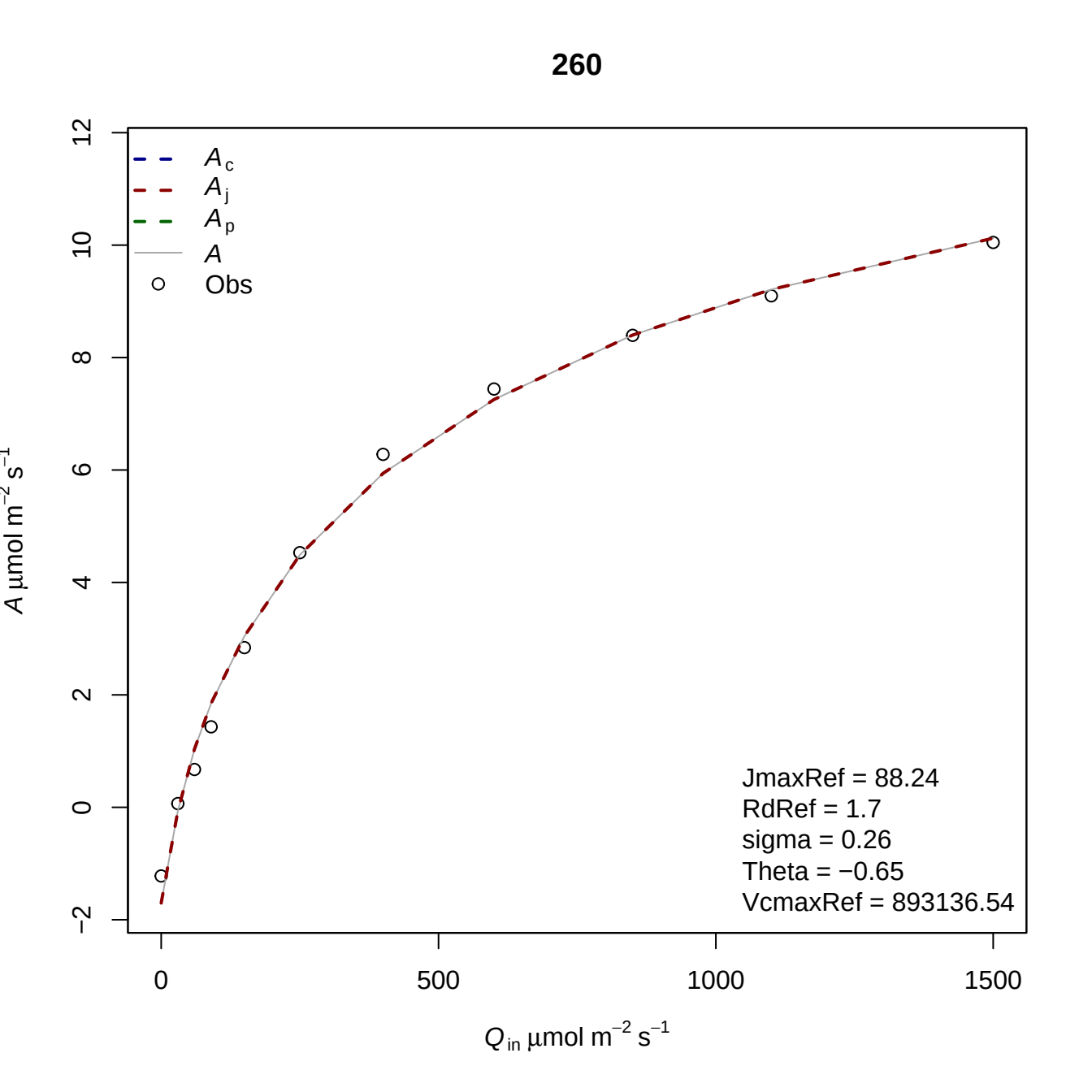

255
A_c
A_j
A_p
A
Obs
JmaxRef = 47.09
RdRef = 3.25
sigma = 0.09
Theta = 0.58
VcmaxRef = 59014.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
200
400
600
800
1000
1200
1400

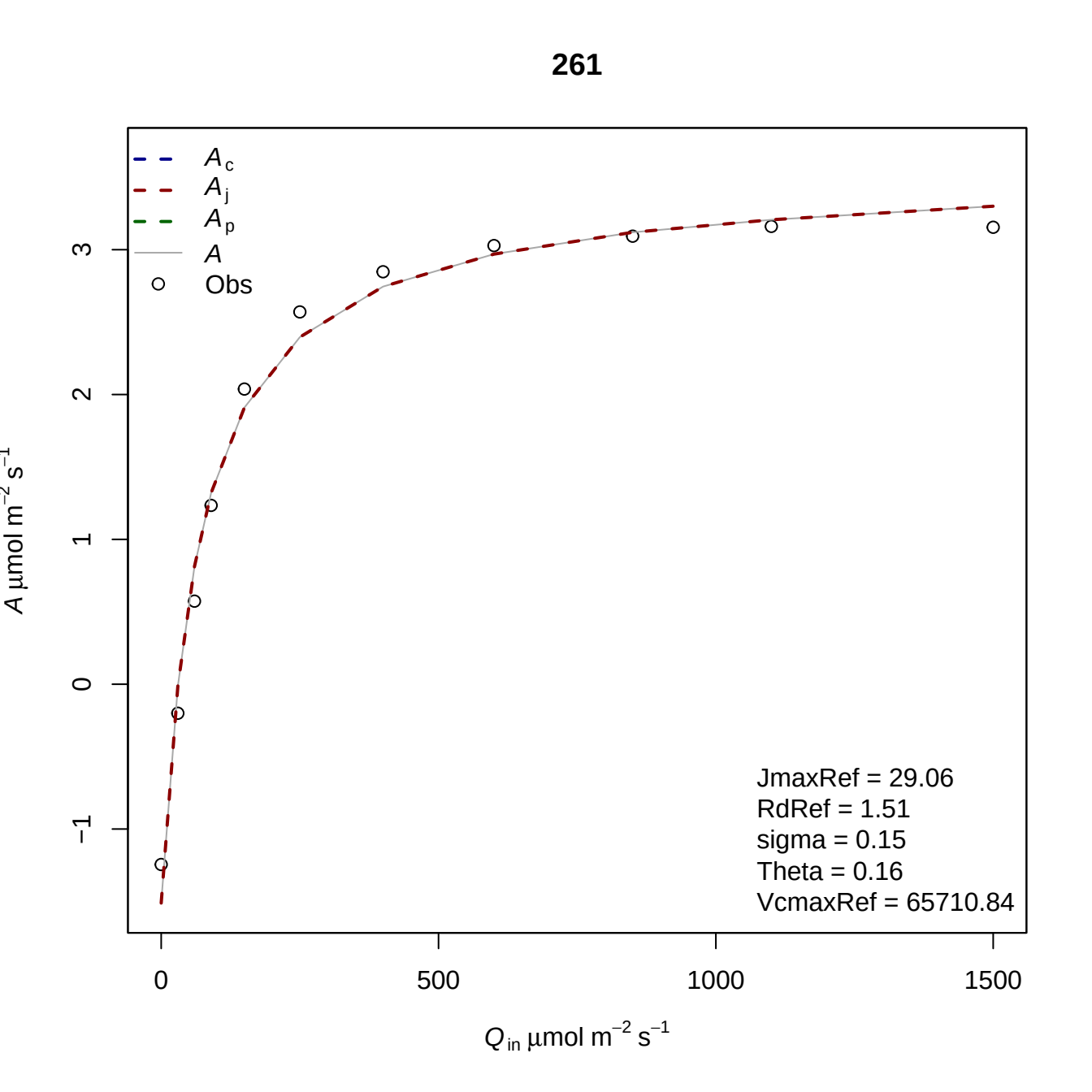

256
A_c
A_j
A_p
A
Obs
JmaxRef = 72.52
RdRef = 1.68
sigma = 0.4
Theta = 0.37
VcmaxRef = 44.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

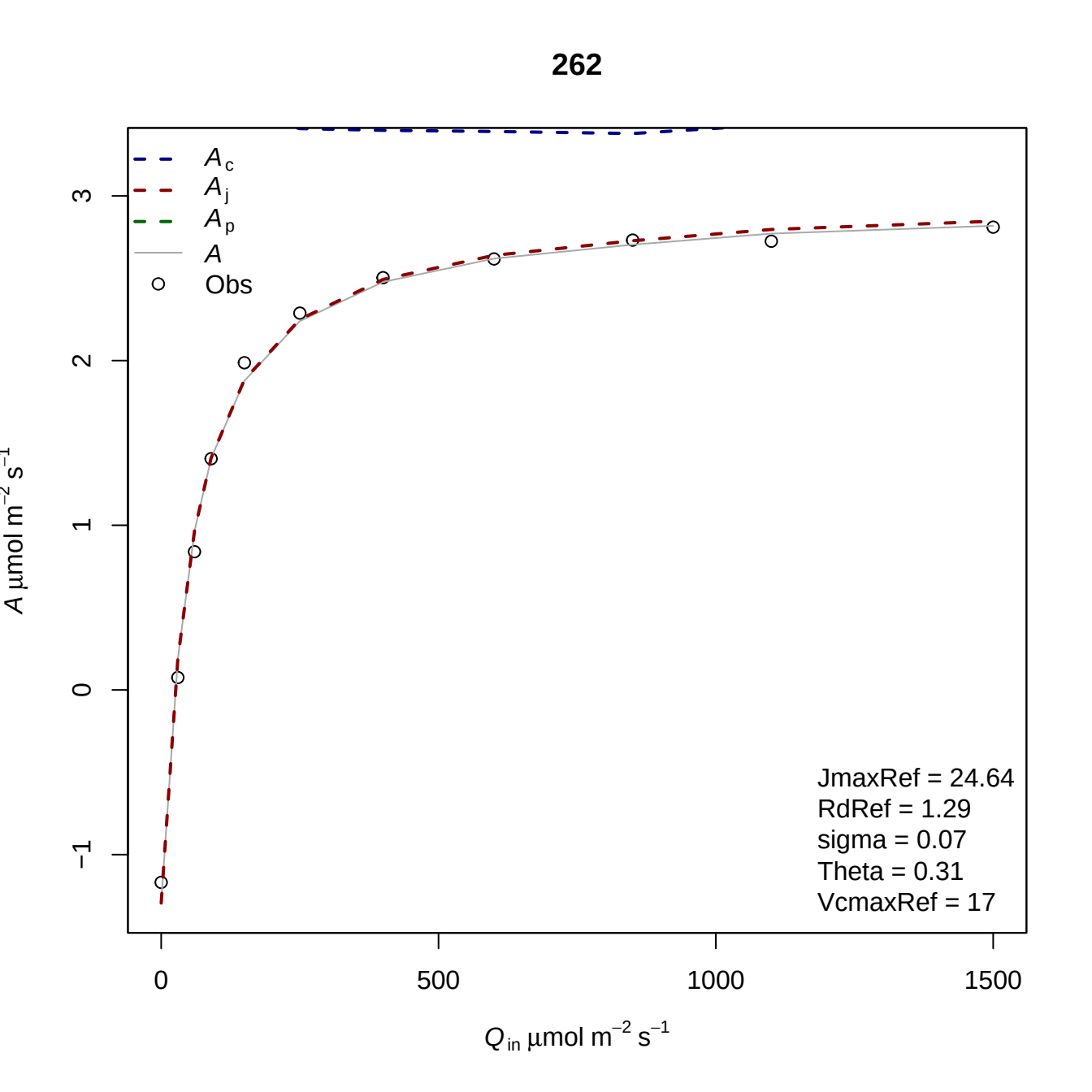

A_c
A_j
A_p
A
Obs
JmaxRef = 326989.82
RdRef = 0.69
sigma = 0.16
Theta = −674316.99
VcmaxRef = 409865468.88
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

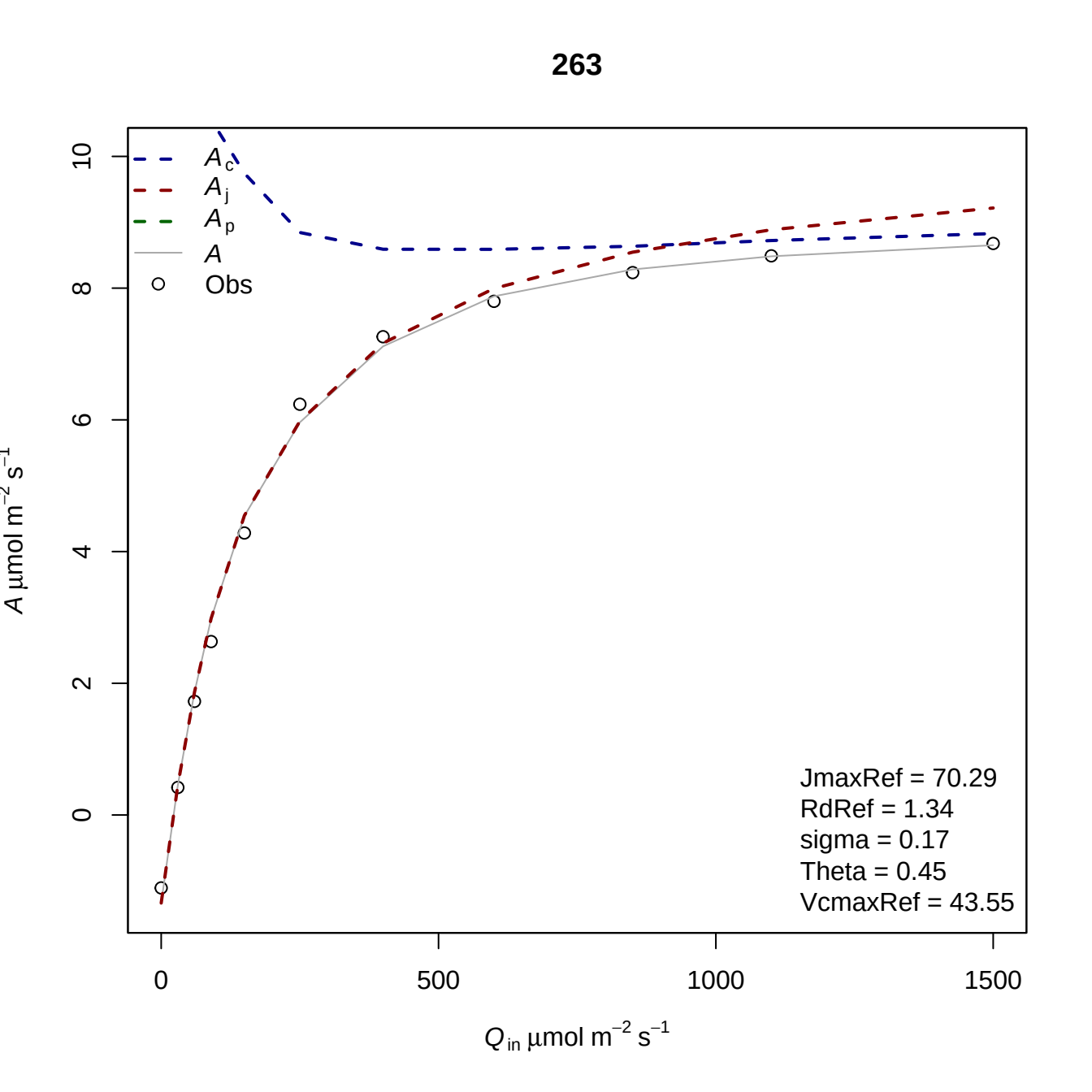

258
A_c
A_j
A_p
A
Obs
JmaxRef = 53.6
RdRef = 1.17
sigma = 0.06
Theta = −0.46
VcmaxRef = 6234.32
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

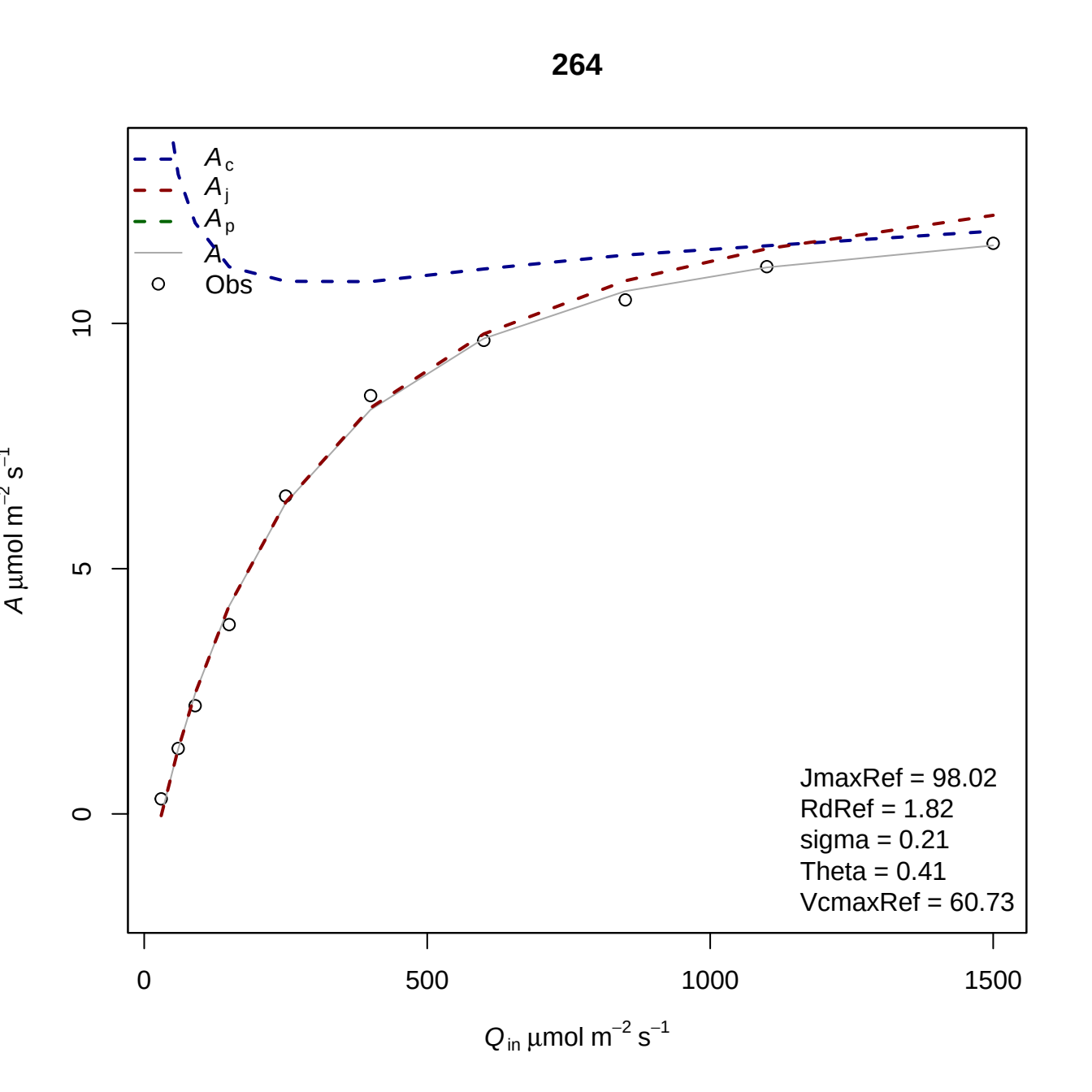

259
A_c
A_j
A_p
A
Obs
JmaxRef = 86.26
RdRef = 1.76
sigma = 0.13
Theta = −1.27
VcmaxRef = 309944.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

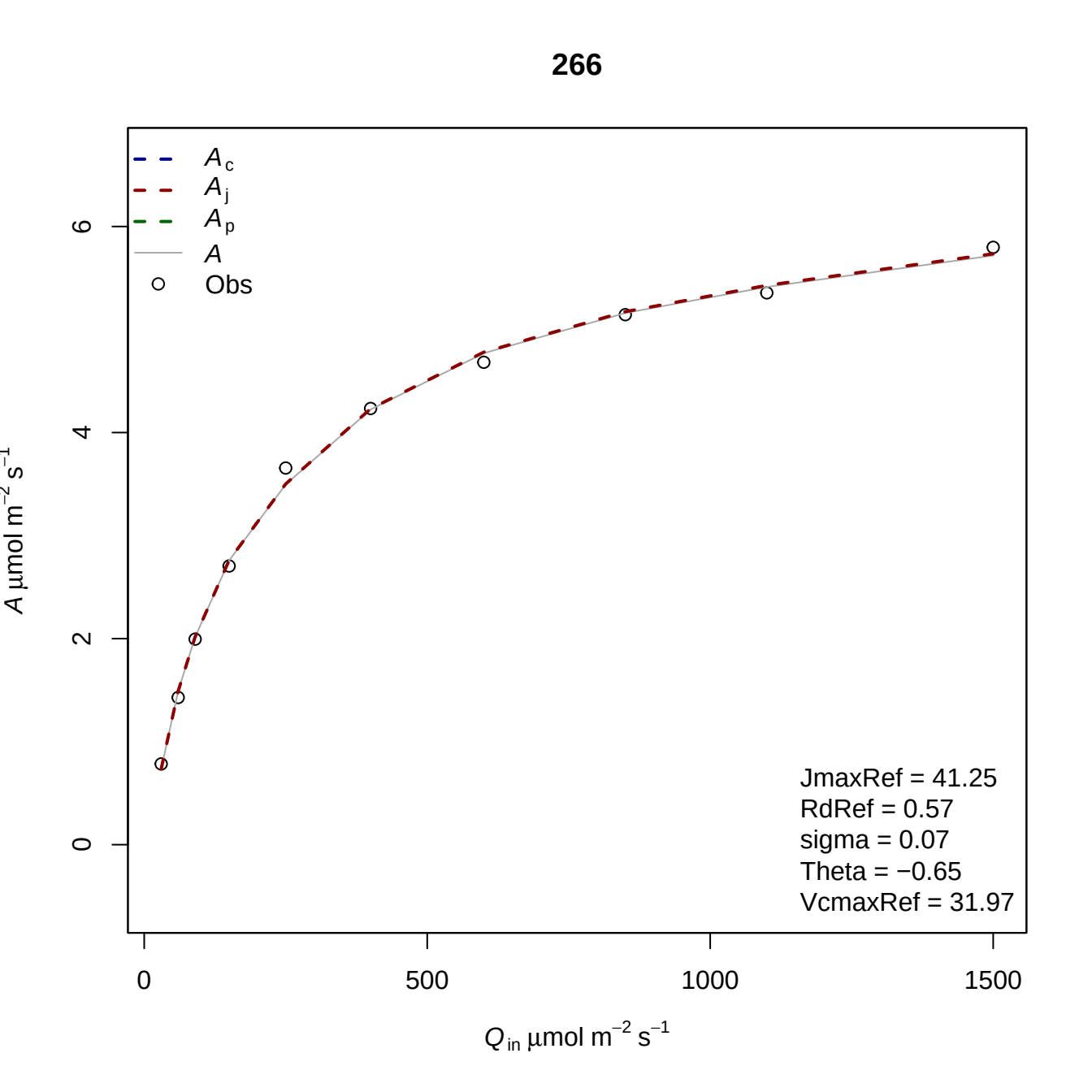

260
A_c
A_j
A_p
A
Obs
JmaxRef = 88.24
RdRef = 1.7
sigma = 0.26
Theta = −0.65
VcmaxRef = 893136.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

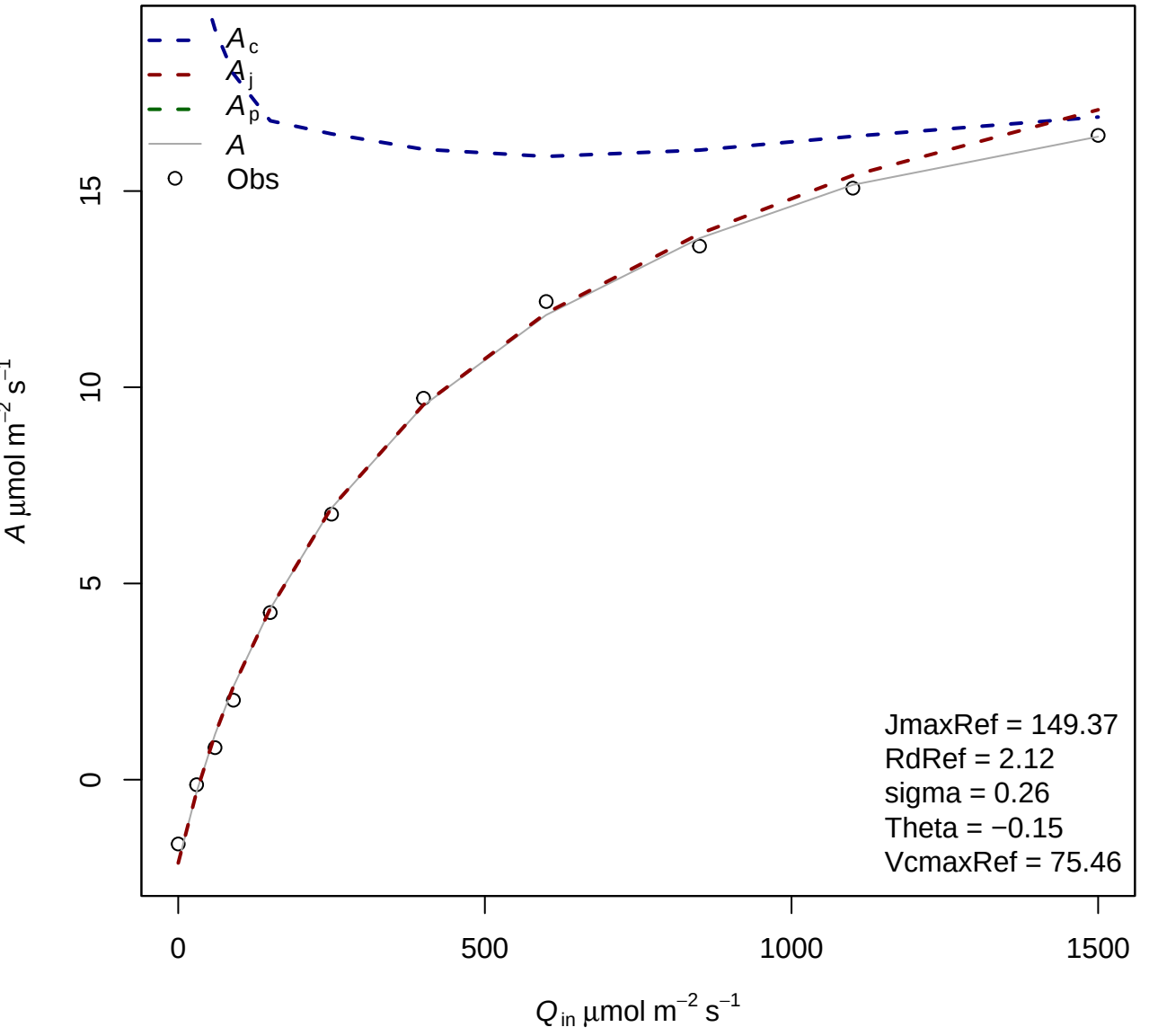

261
A_c
A_j
A_p
A
Obs
JmaxRef = 29.06
RdRef = 1.51
sigma = 0.15
Theta = 0.16
VcmaxRef = 65710.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

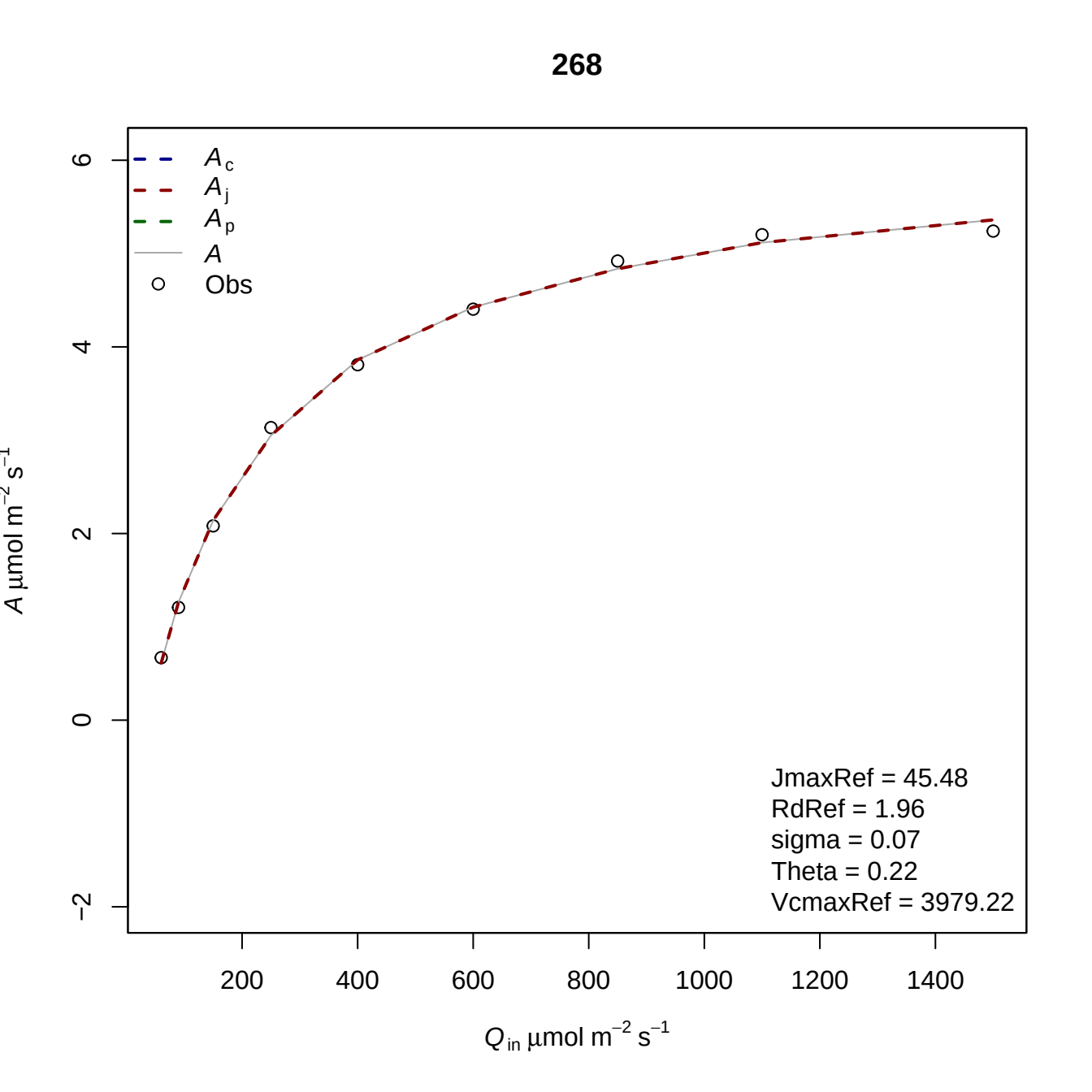

262
A_c
A_j
A_p
A
Obs
JmaxRef = 24.64
RdRef = 1.29
sigma = 0.07
Theta = 0.31
VcmaxRef = 17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

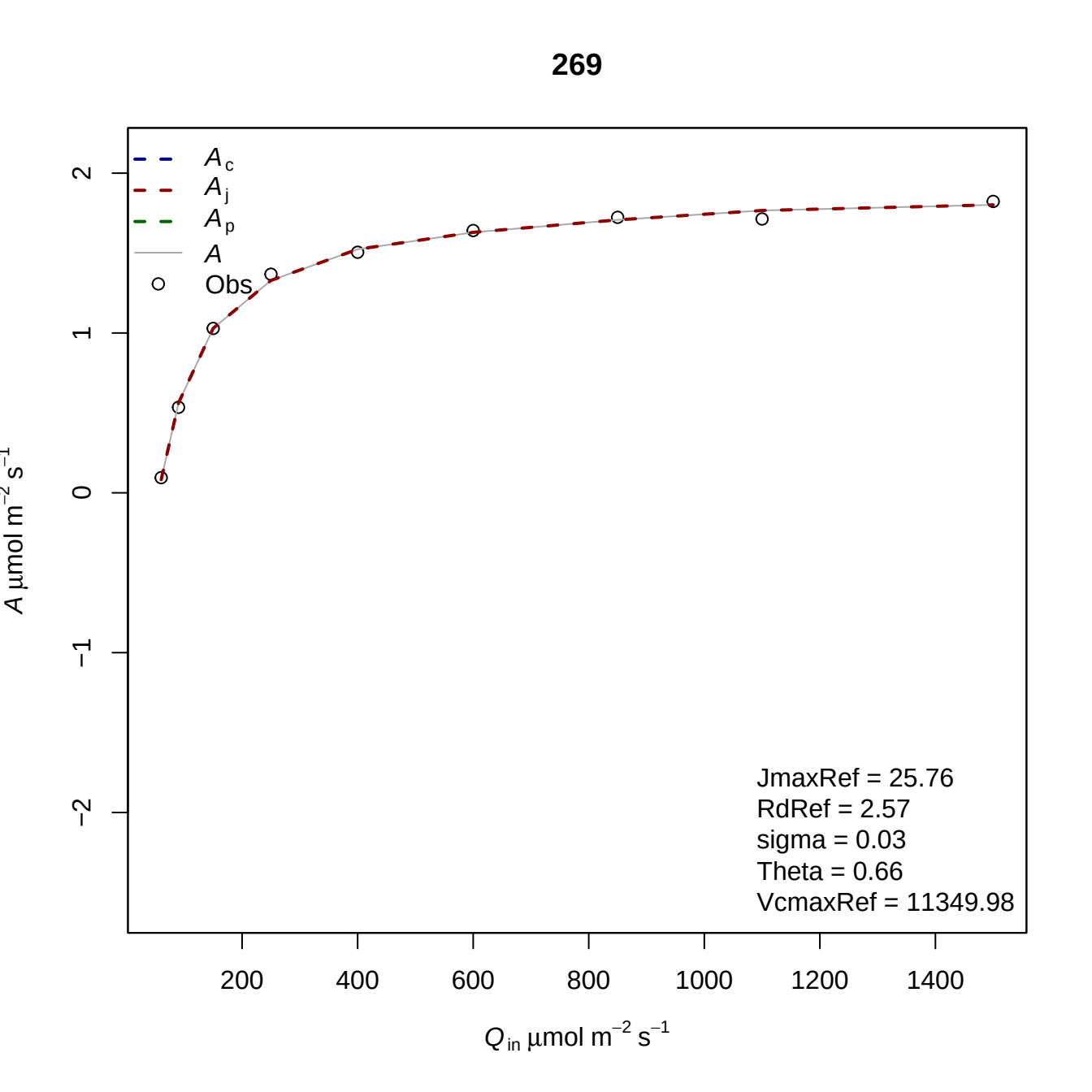

263
A_c
A_j
A_p
A
Obs
JmaxRef = 70.29
RdRef = 1.34
sigma = 0.17
Theta = 0.45
VcmaxRef = 43.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

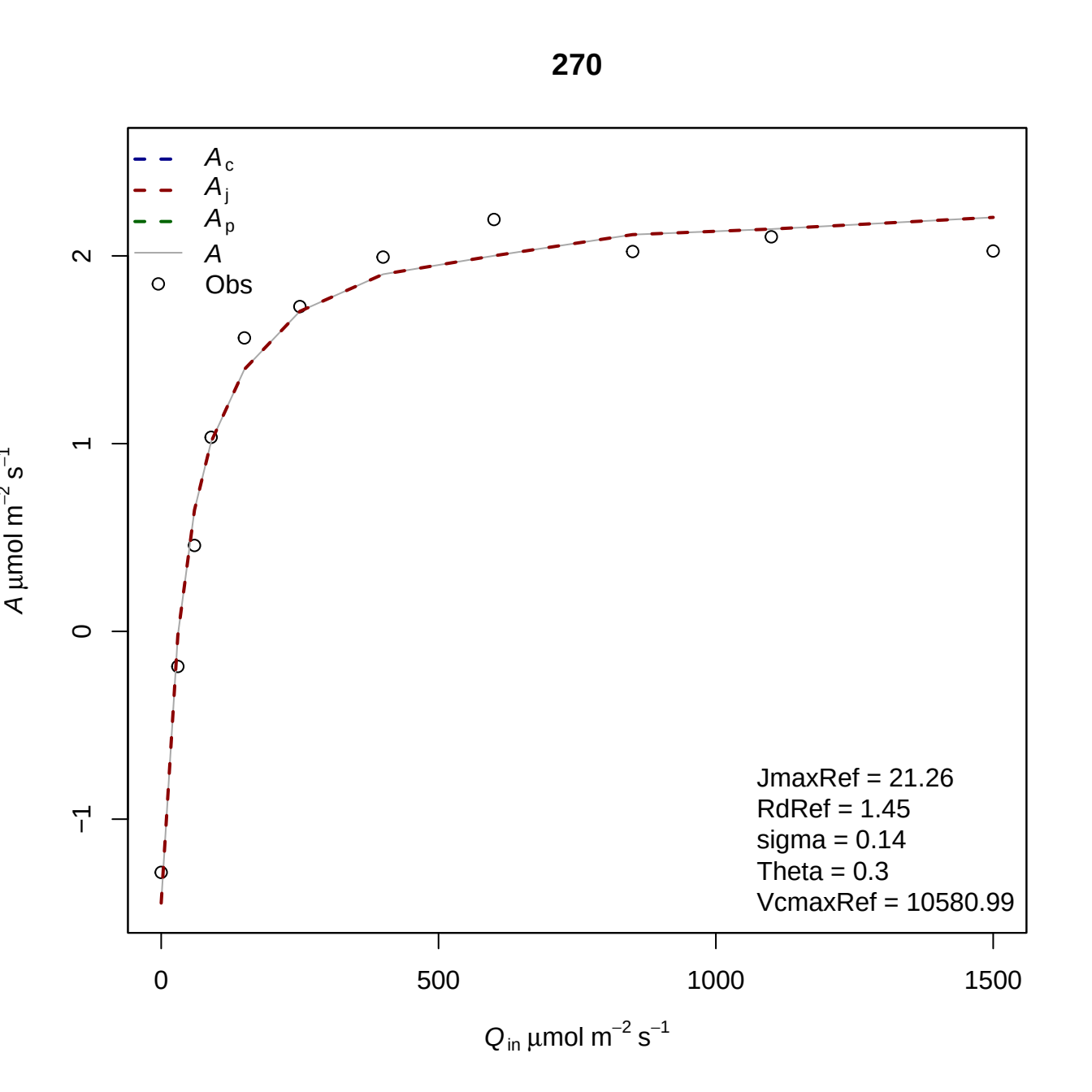

264
A_c
A_j
A_p
A
Obs
JmaxRef = 98.02
RdRef = 1.82
sigma = 0.21
Theta = 0.41
VcmaxRef = 60.73
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

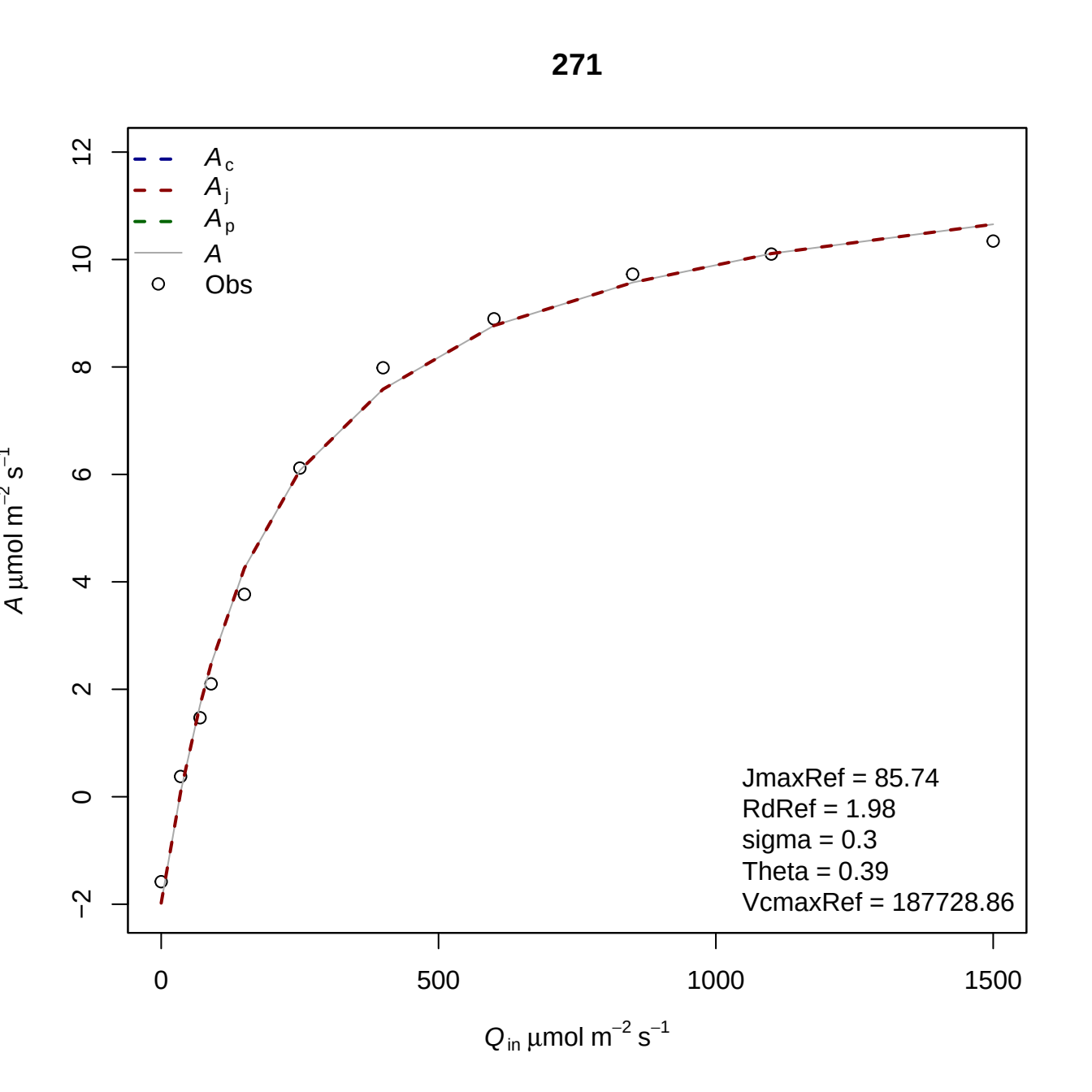

266
A_c
A_j
A_p
A
Obs
JmaxRef = 41.25
RdRef = 0.57
sigma = 0.07
Theta = −0.65
VcmaxRef = 31.97
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

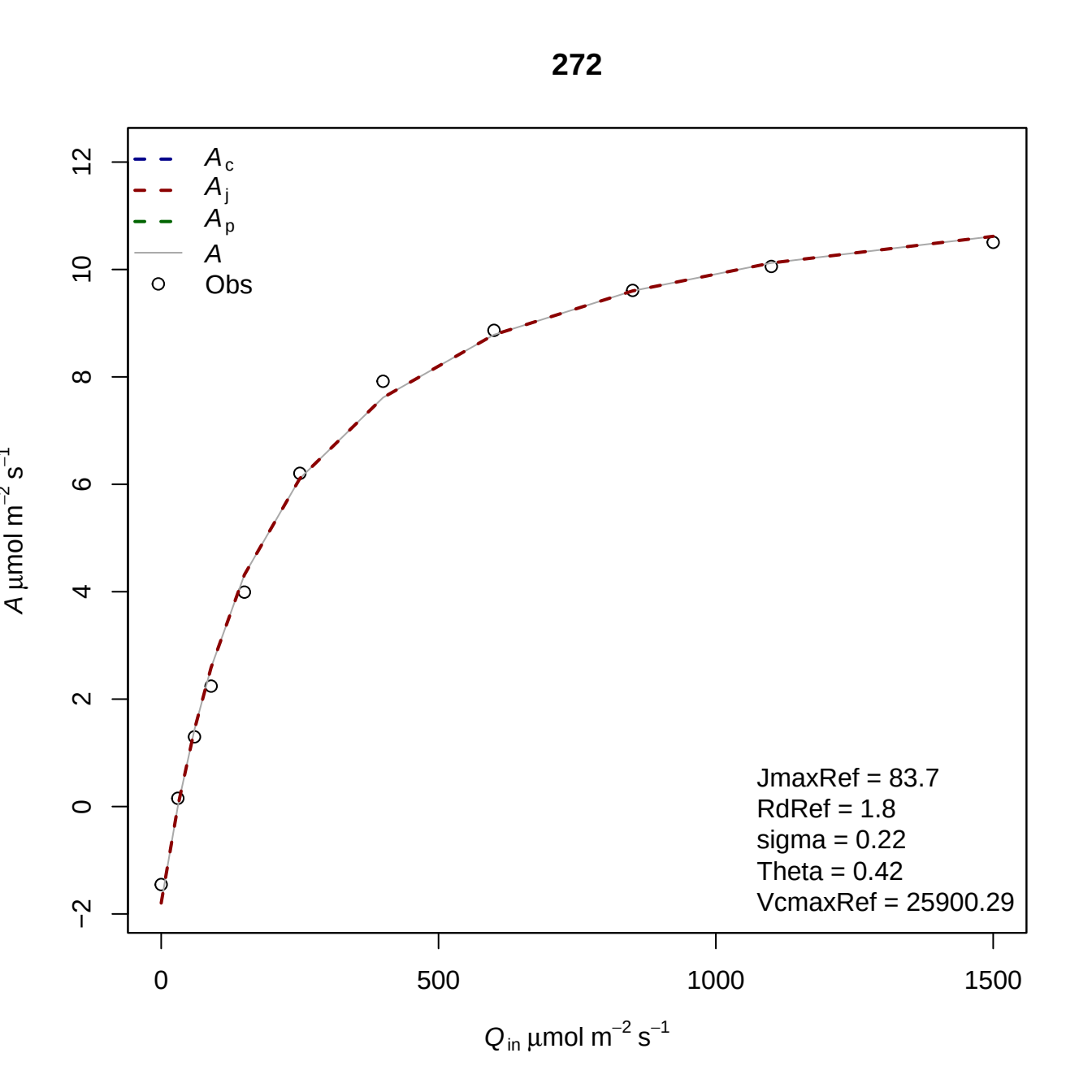

A_c
A_j
A_p
A
Obs
JmaxRef = 149.37
RdRef = 2.12
sigma = 0.26
Theta = −0.15
VcmaxRef = 75.46
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

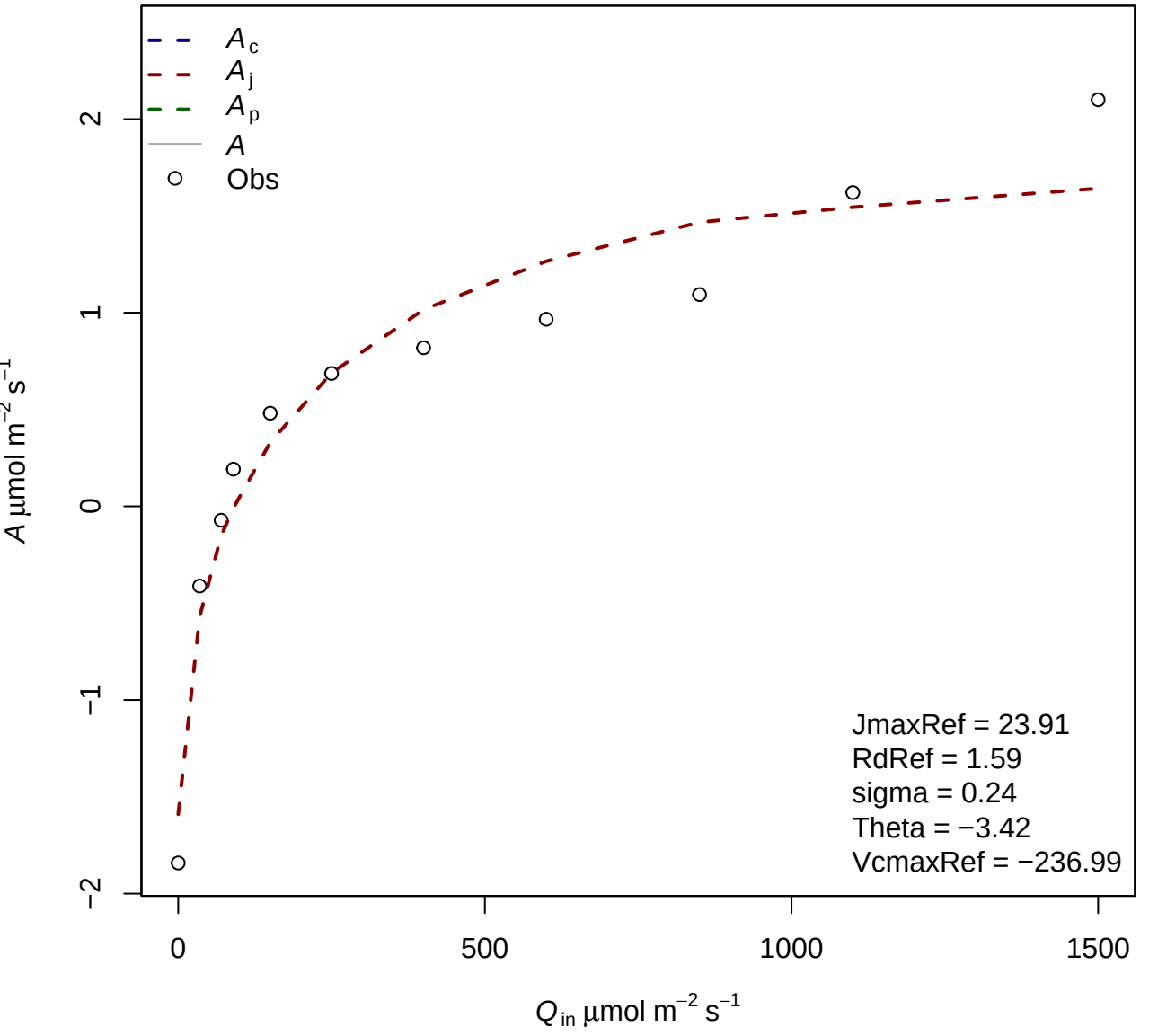

268
A_c
A_j
A_p
A
Obs
JmaxRef = 45.48
RdRef = 1.96
sigma = 0.07
Theta = 0.22
VcmaxRef = 3979.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
200
400
600
800
1000
1200
1400

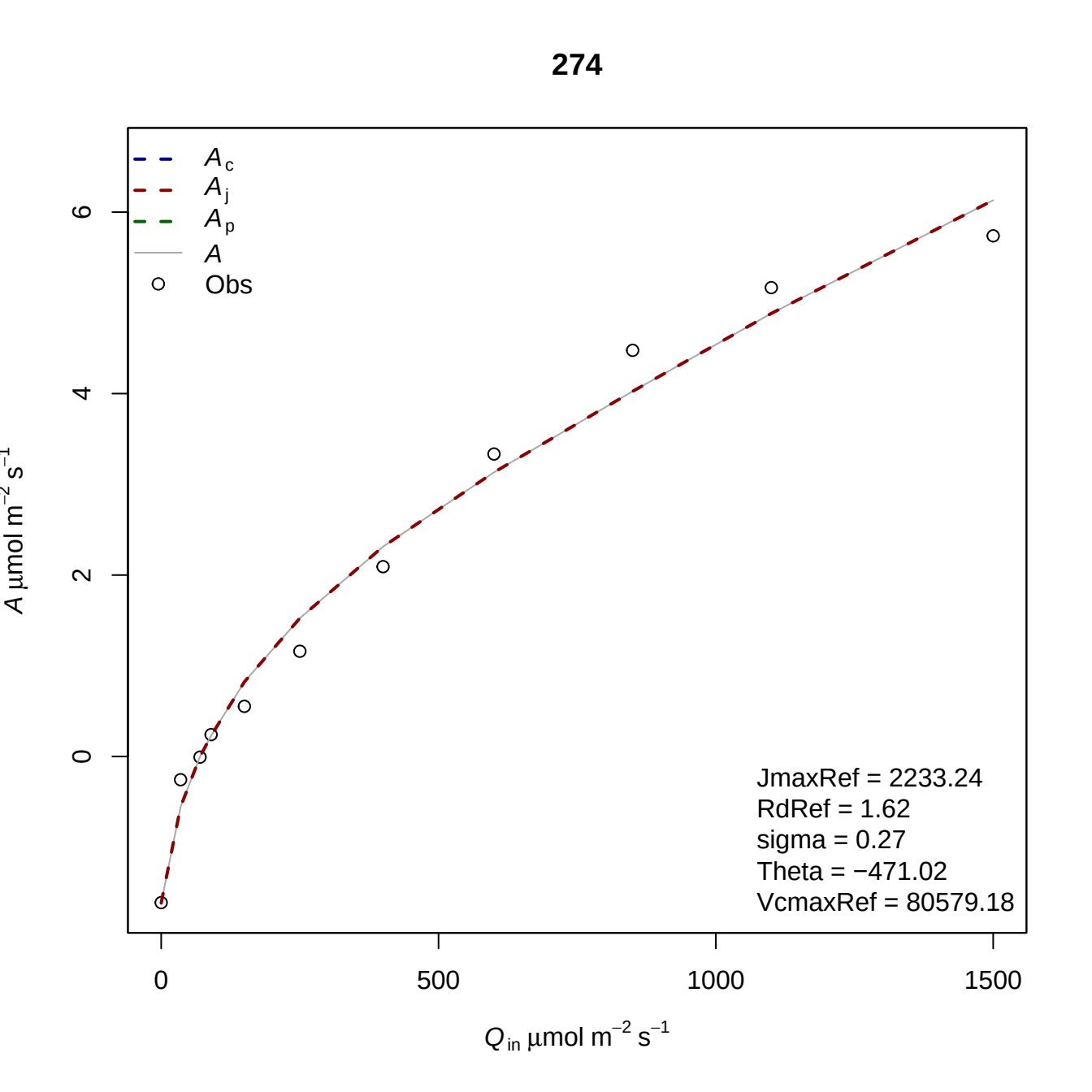

269
A_c
A_j
A_p
A
Obs
JmaxRef = 25.76
RdRef = 2.57
sigma = 0.03
Theta = 0.66
VcmaxRef = 11349.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
200
400
600
800
1000
1200
1400

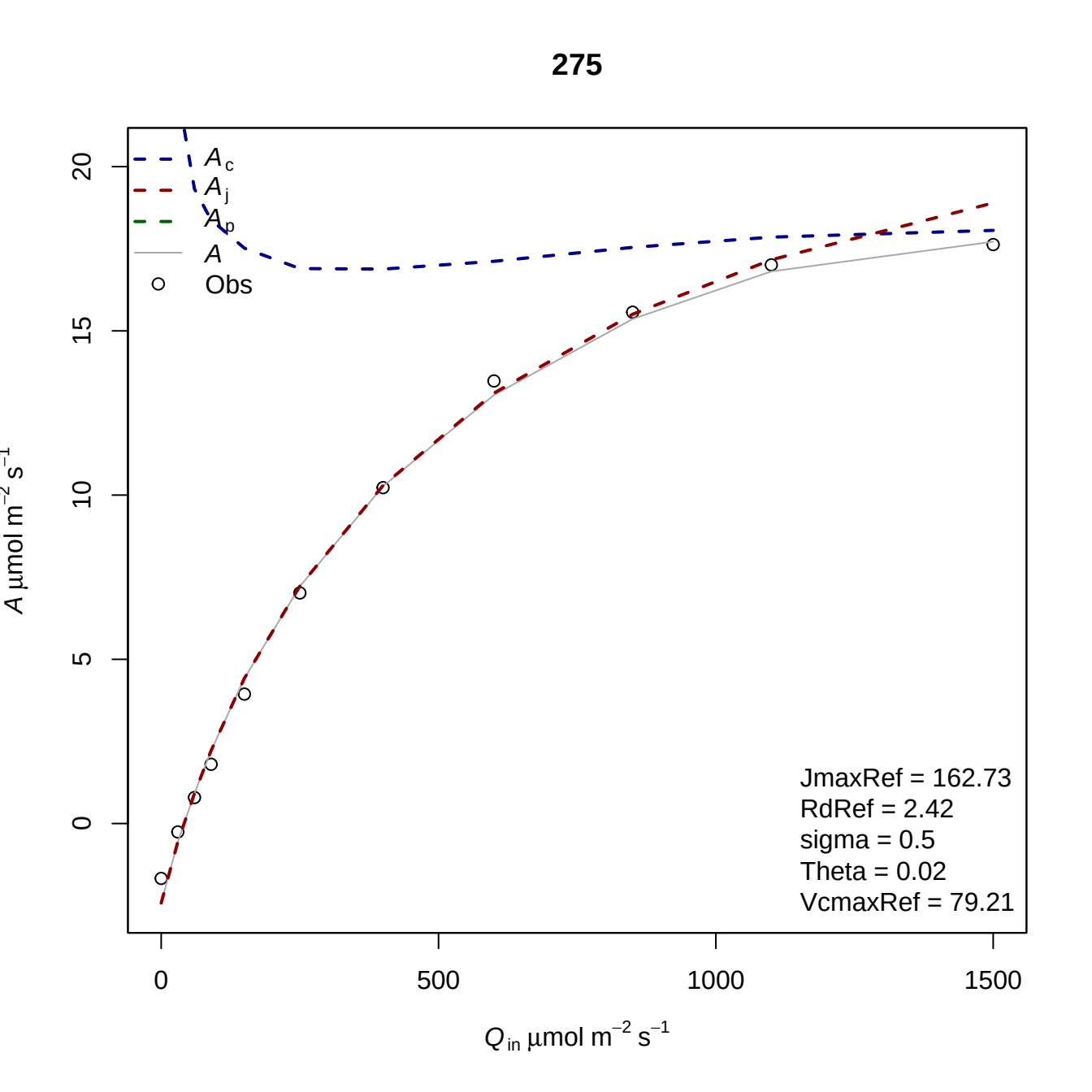

270
A_c
A_j
A_p
A
Obs
JmaxRef = 21.26
RdRef = 1.45
sigma = 0.14
Theta = 0.3
VcmaxRef = 10580.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

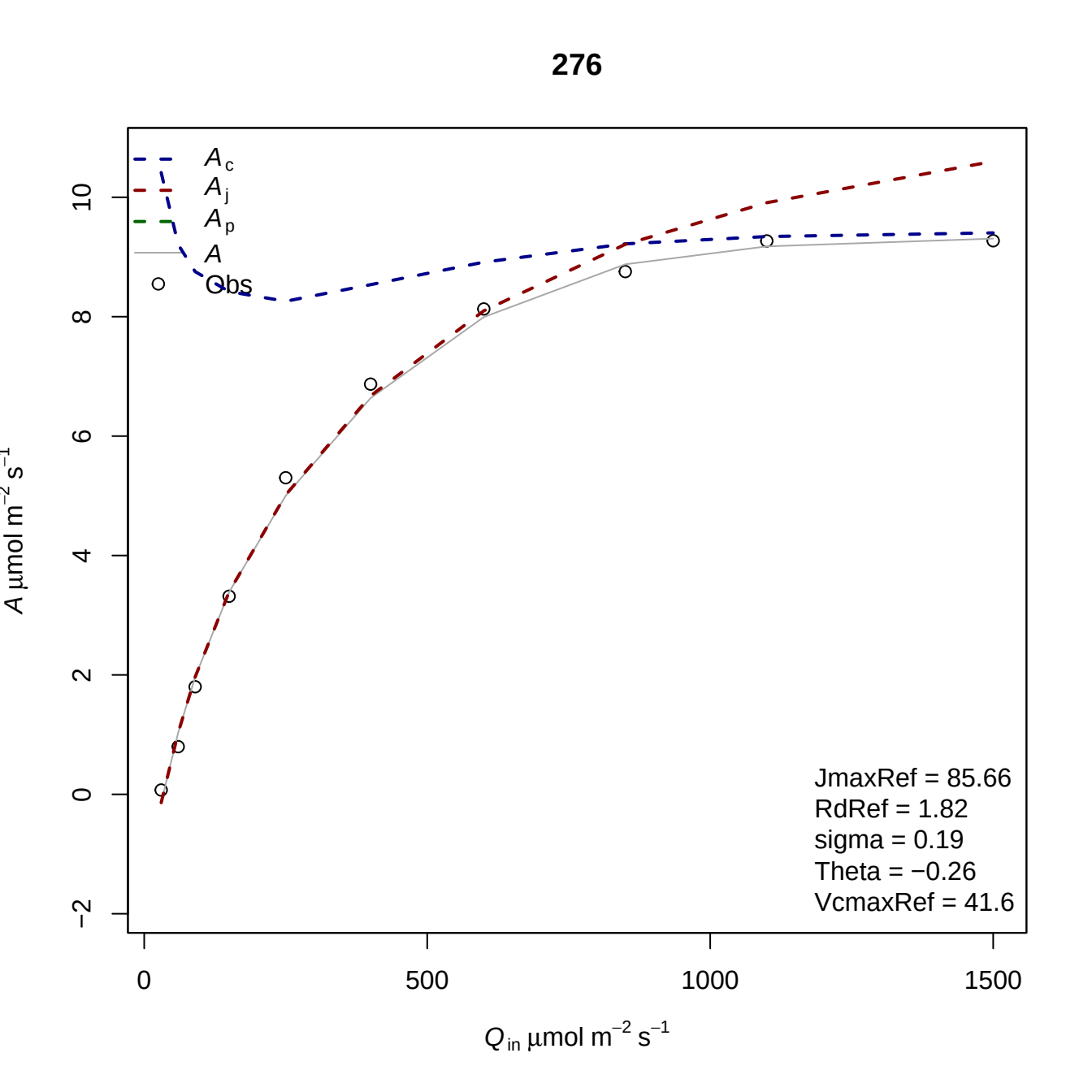

271
A_c
A_j
A_p
A
Obs
JmaxRef = 85.74
RdRef = 1.98
sigma = 0.3
Theta = 0.39
VcmaxRef = 187728.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

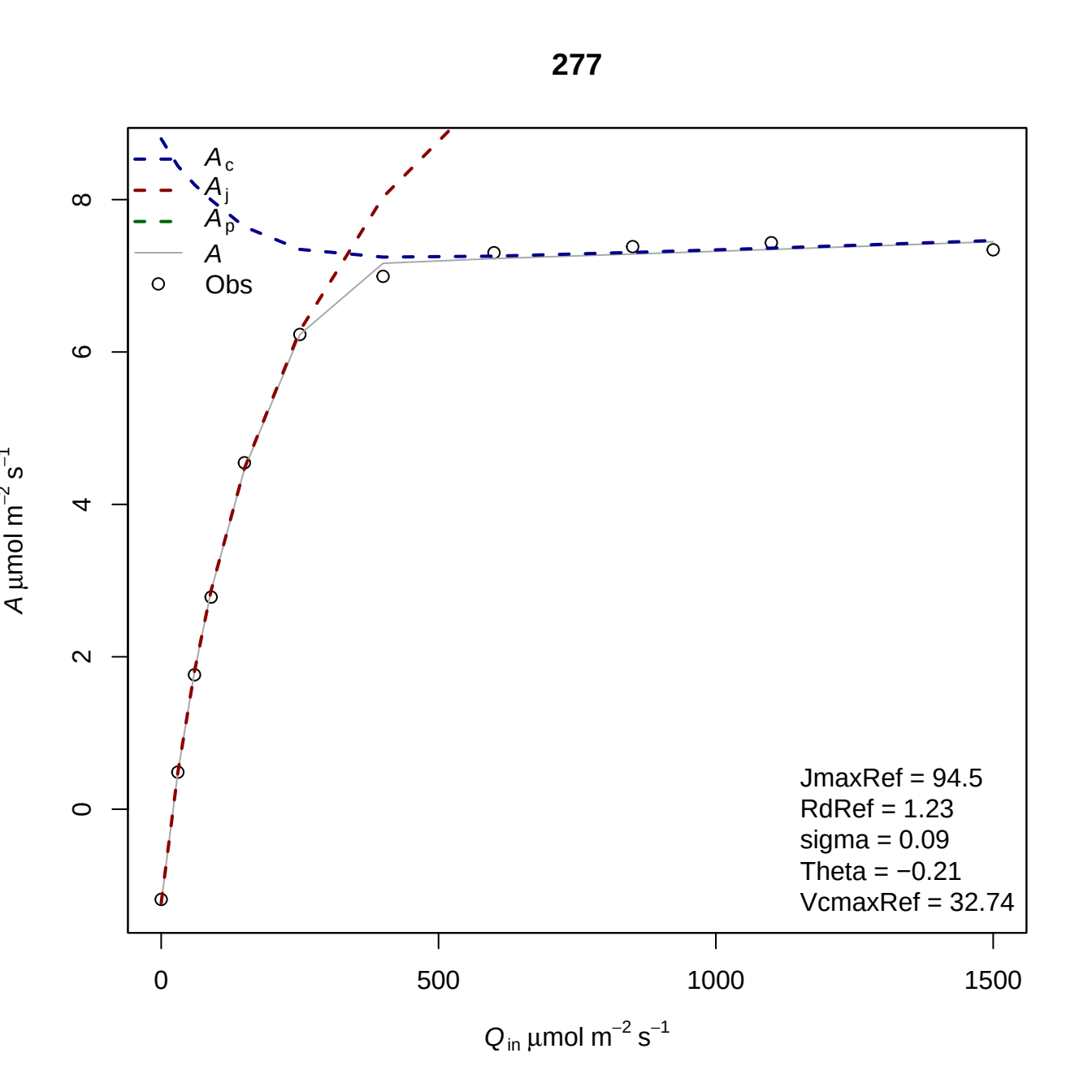

272
A_c
A_j
A_p
A
Obs
JmaxRef = 83.7
RdRef = 1.8
sigma = 0.22
Theta = 0.42
VcmaxRef = 25900.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

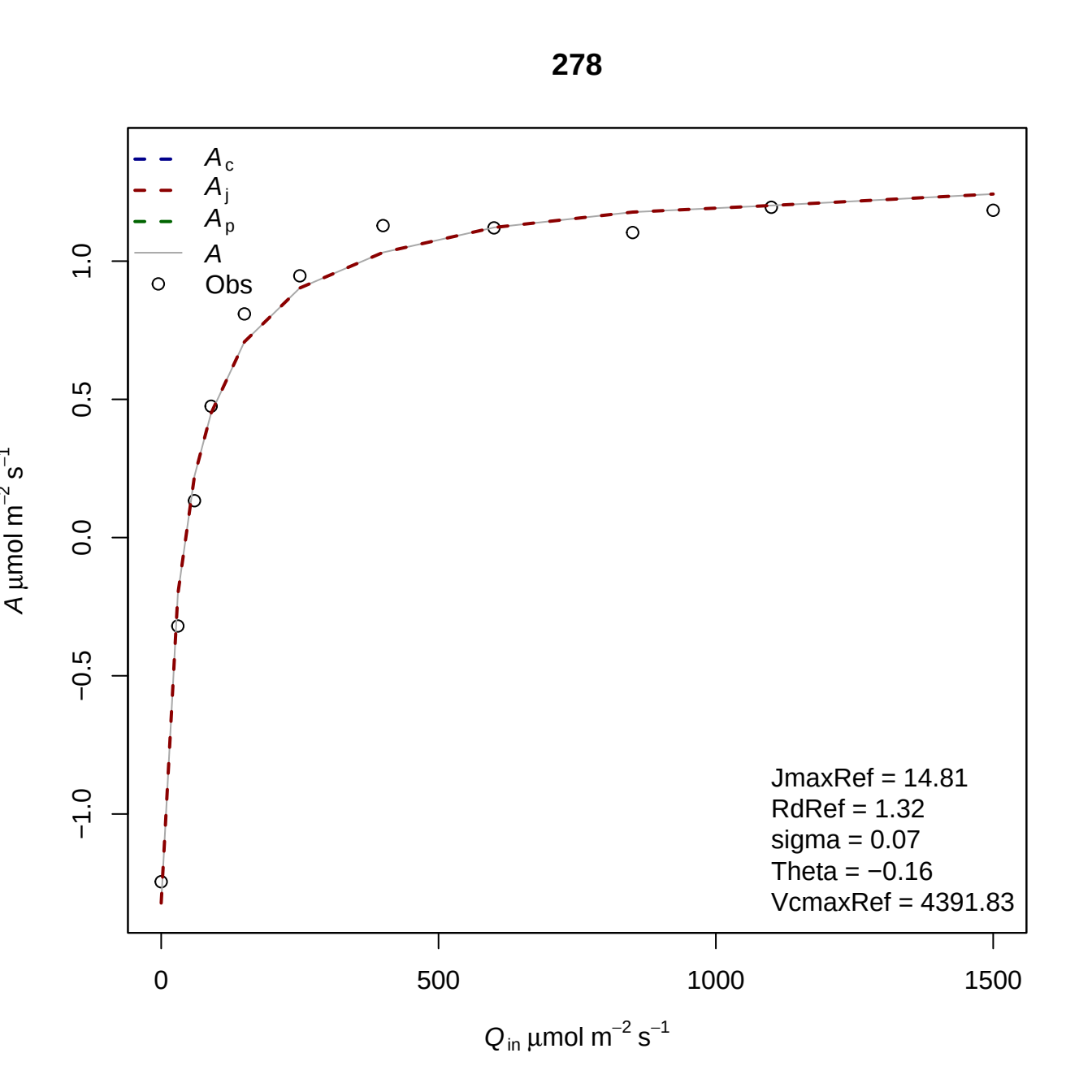

A_c
A_j
A_p
A
Obs
JmaxRef = 23.91
RdRef = 1.59
sigma = 0.24
Theta = −3.42
VcmaxRef = −236.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
0
500
1000
1500

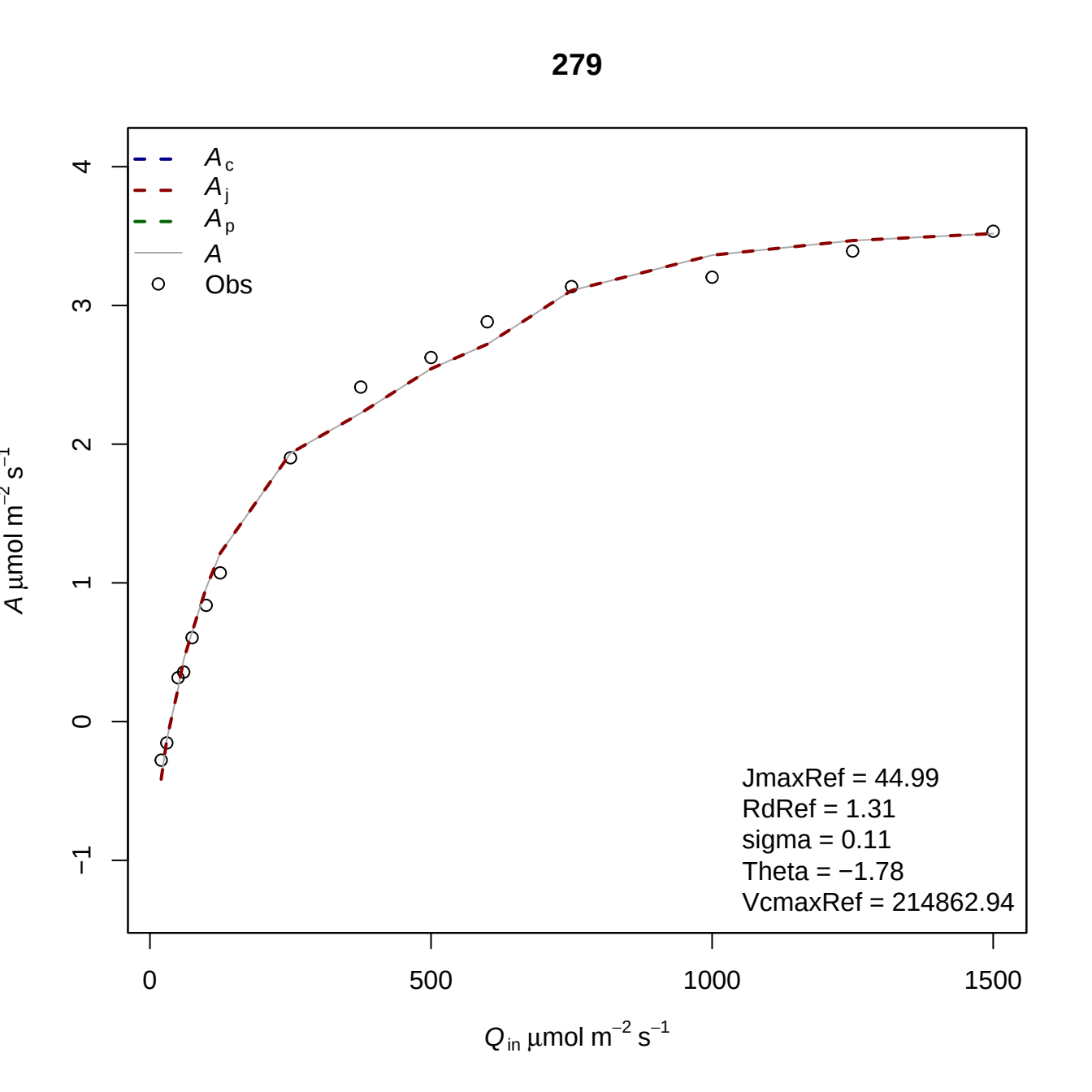

274
A_c
A_j
A_p
A
Obs
JmaxRef = 2233.24
RdRef = 1.62
sigma = 0.27
Theta = −471.02
VcmaxRef = 80579.18
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

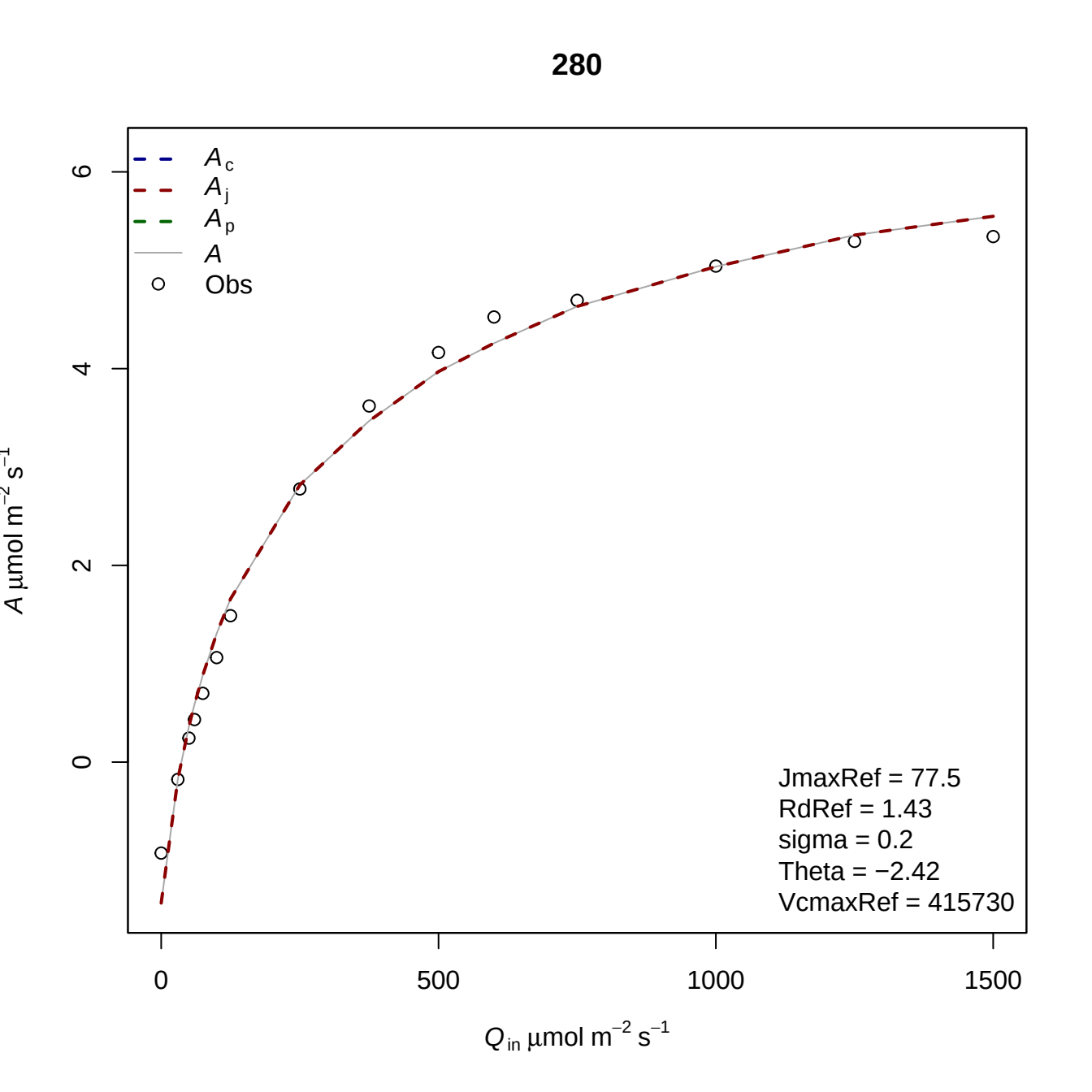

275
A_c
A_j
A_p
A
Obs
JmaxRef = 162.73
RdRef = 2.42
sigma = 0.5
Theta = 0.02
VcmaxRef = 79.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

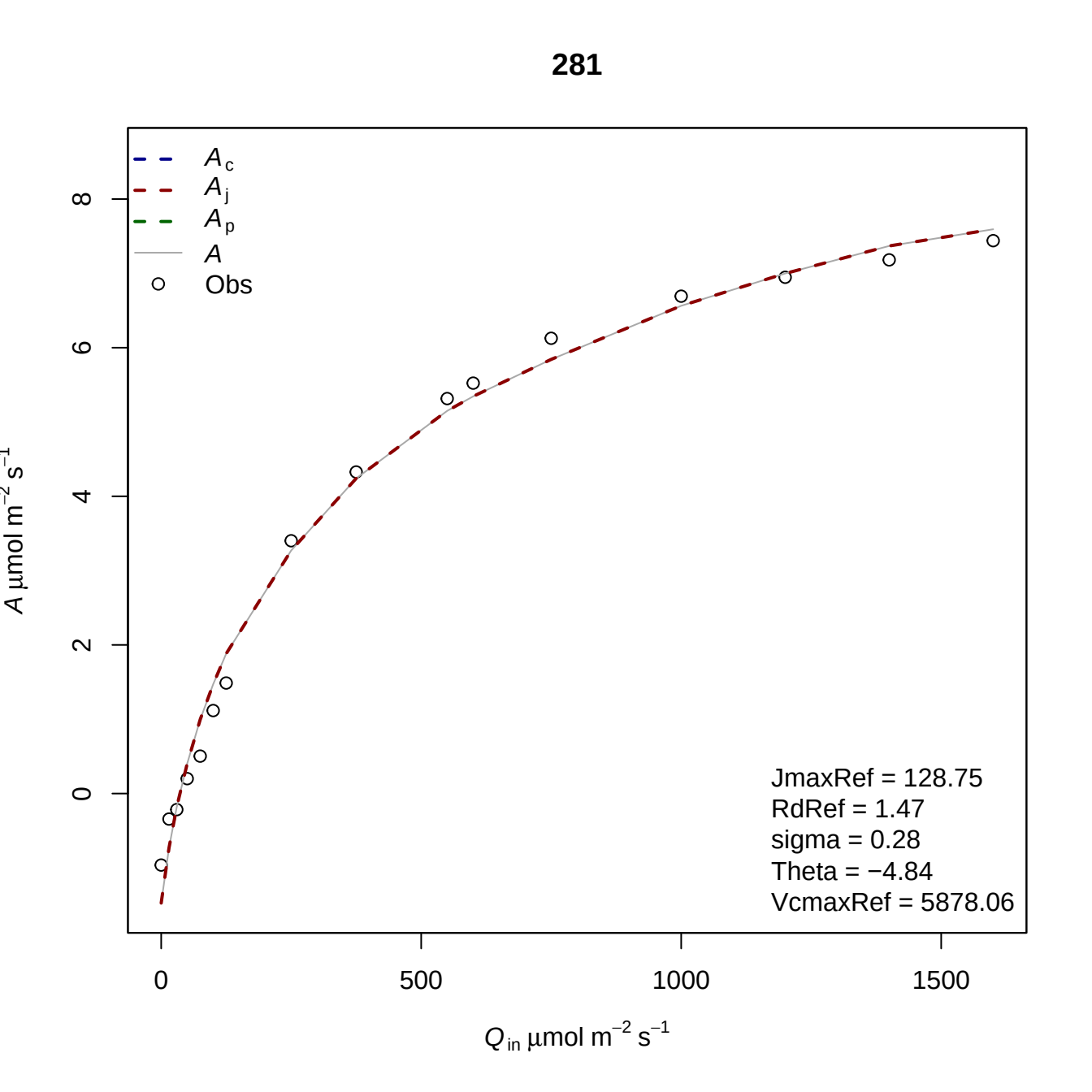

276
A_c
A_j
A_p
A
Obs
JmaxRef = 85.66
RdRef = 1.82
sigma = 0.19
Theta = −0.26
VcmaxRef = 41.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

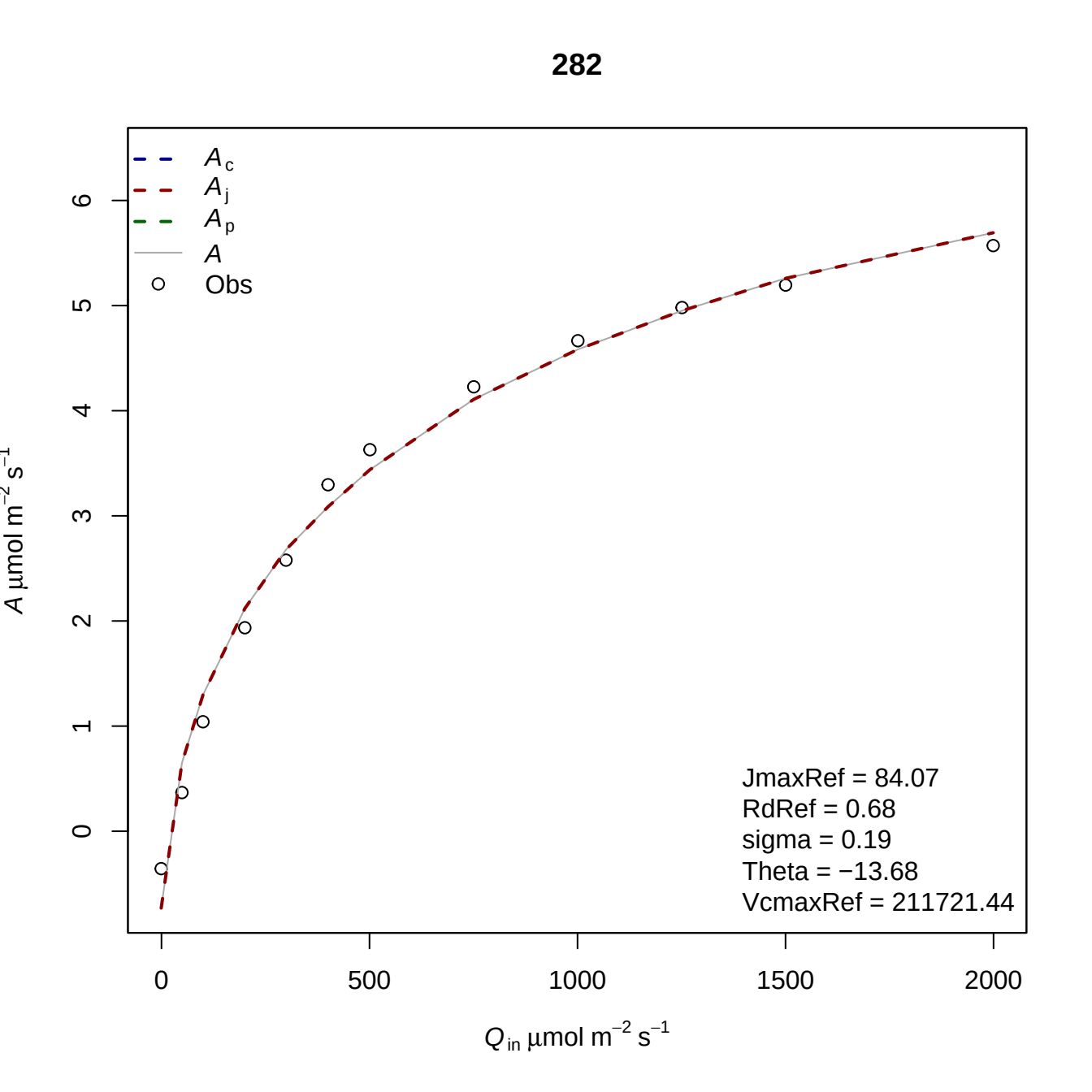

277
A_c
A_j
A_p
A
Obs
JmaxRef = 94.5
RdRef = 1.23
sigma = 0.09
Theta = −0.21
VcmaxRef = 32.74
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

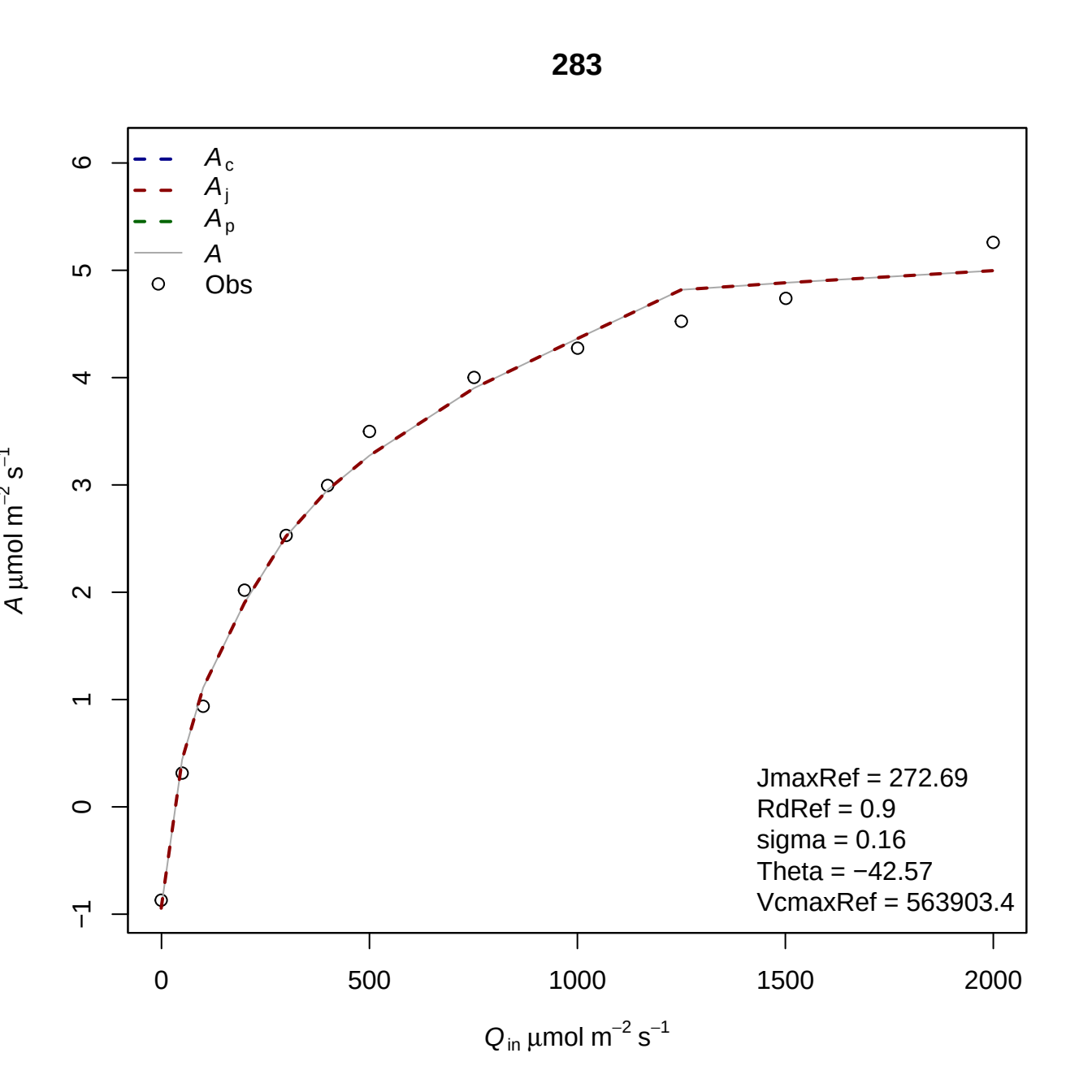

278
A_c
A_j
A_p
A
Obs
JmaxRef = 14.81
RdRef = 1.32
sigma = 0.07
Theta = −0.16
VcmaxRef = 4391.83
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

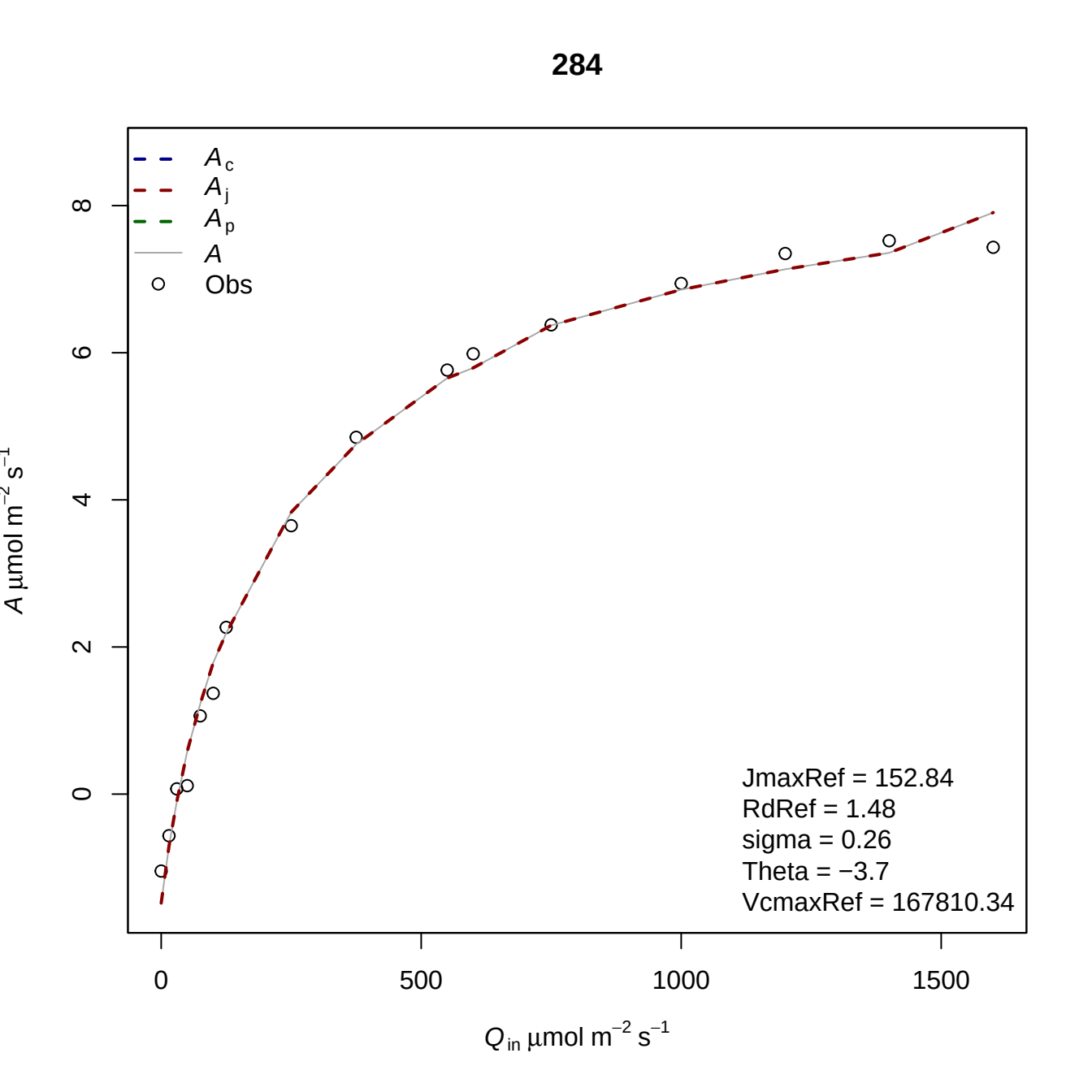

279
A_c
A_j
A_p
A
Obs
JmaxRef = 44.99
RdRef = 1.31
sigma = 0.11
Theta = −1.78
VcmaxRef = 214862.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

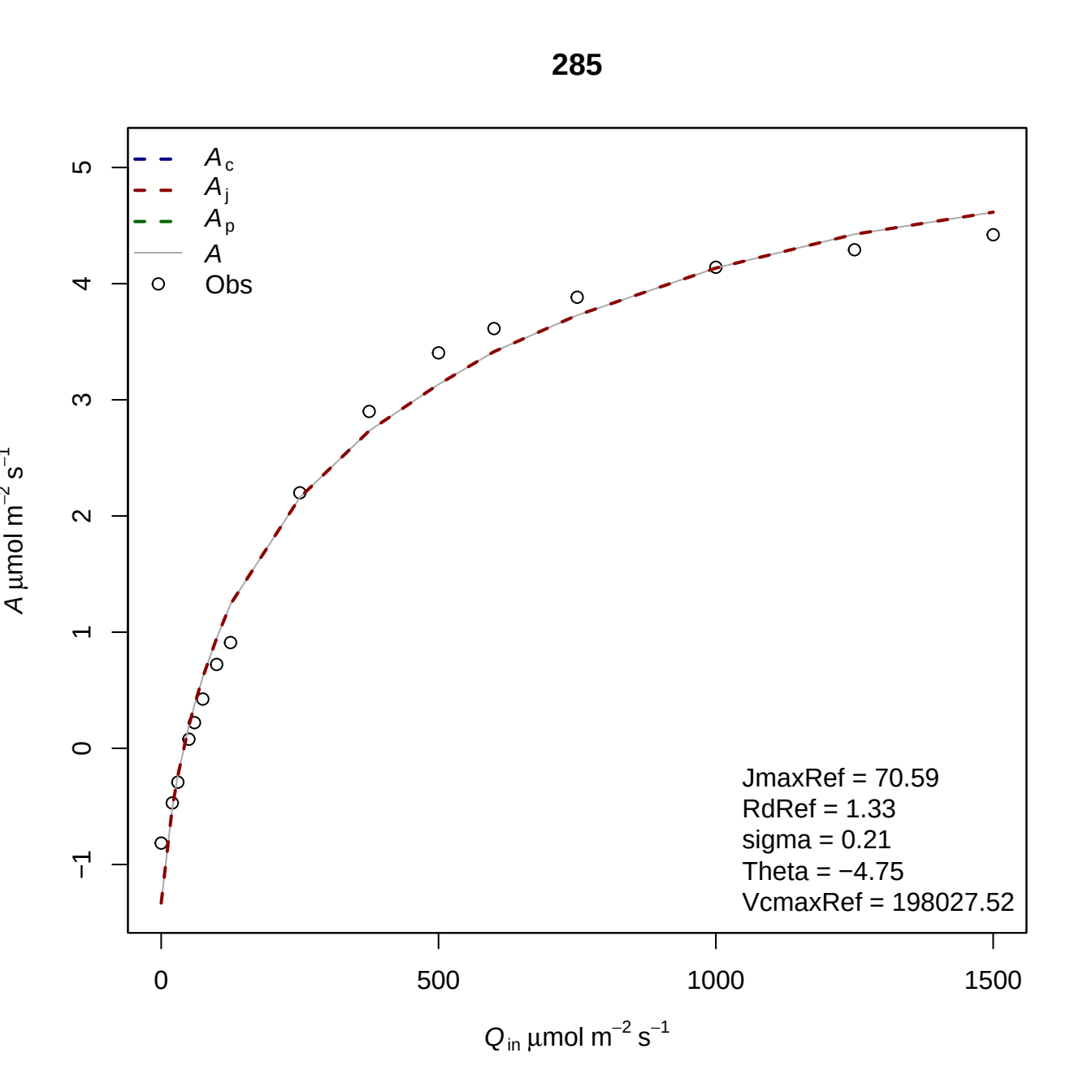

280
A_c
A_j
A_p
A
Obs
JmaxRef = 77.5
RdRef = 1.43
sigma = 0.2
Theta = −2.42
VcmaxRef = 415730
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

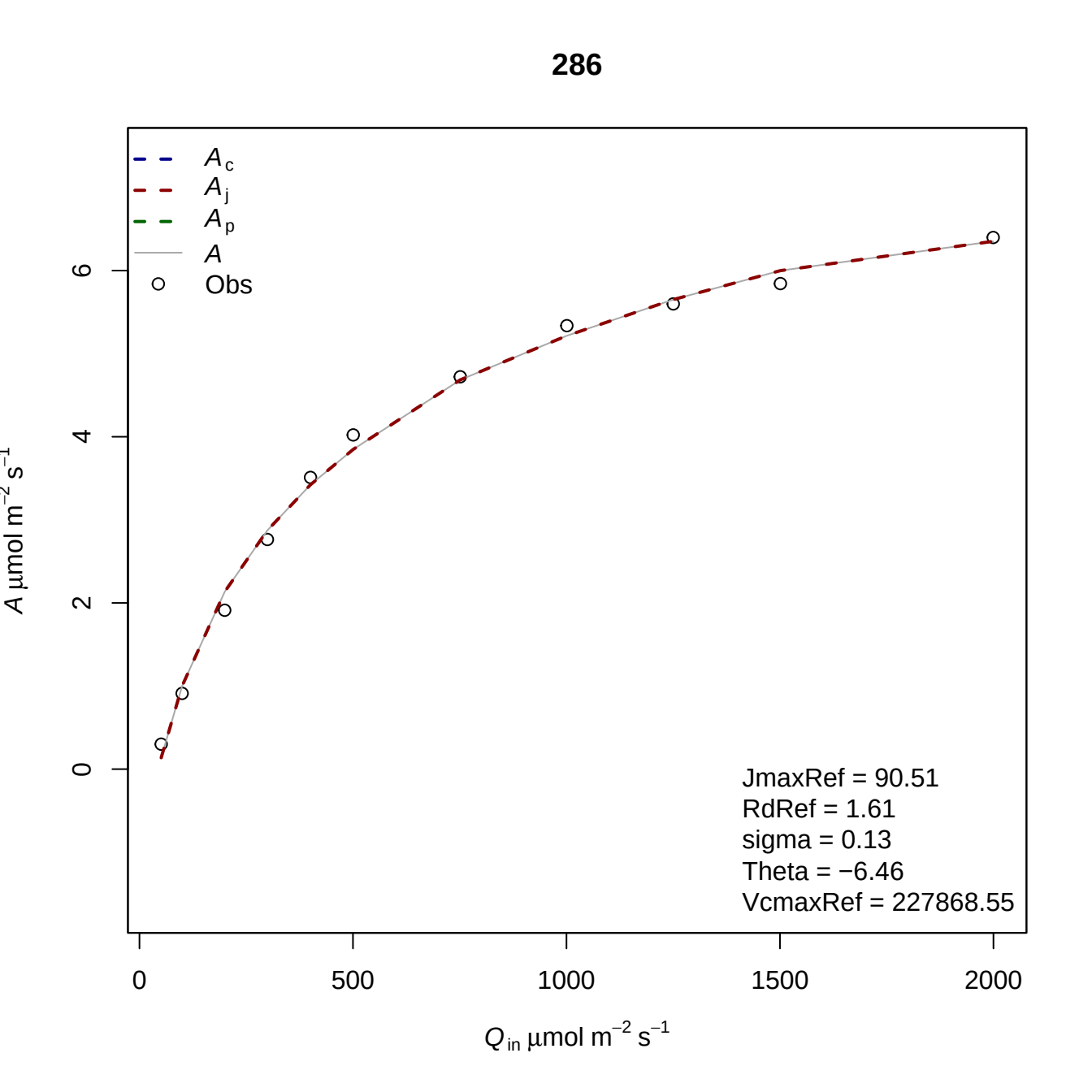

281
A_c
A_j
A_p
A
Obs
JmaxRef = 128.75
RdRef = 1.47
sigma = 0.28
Theta = −4.84
VcmaxRef = 5878.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

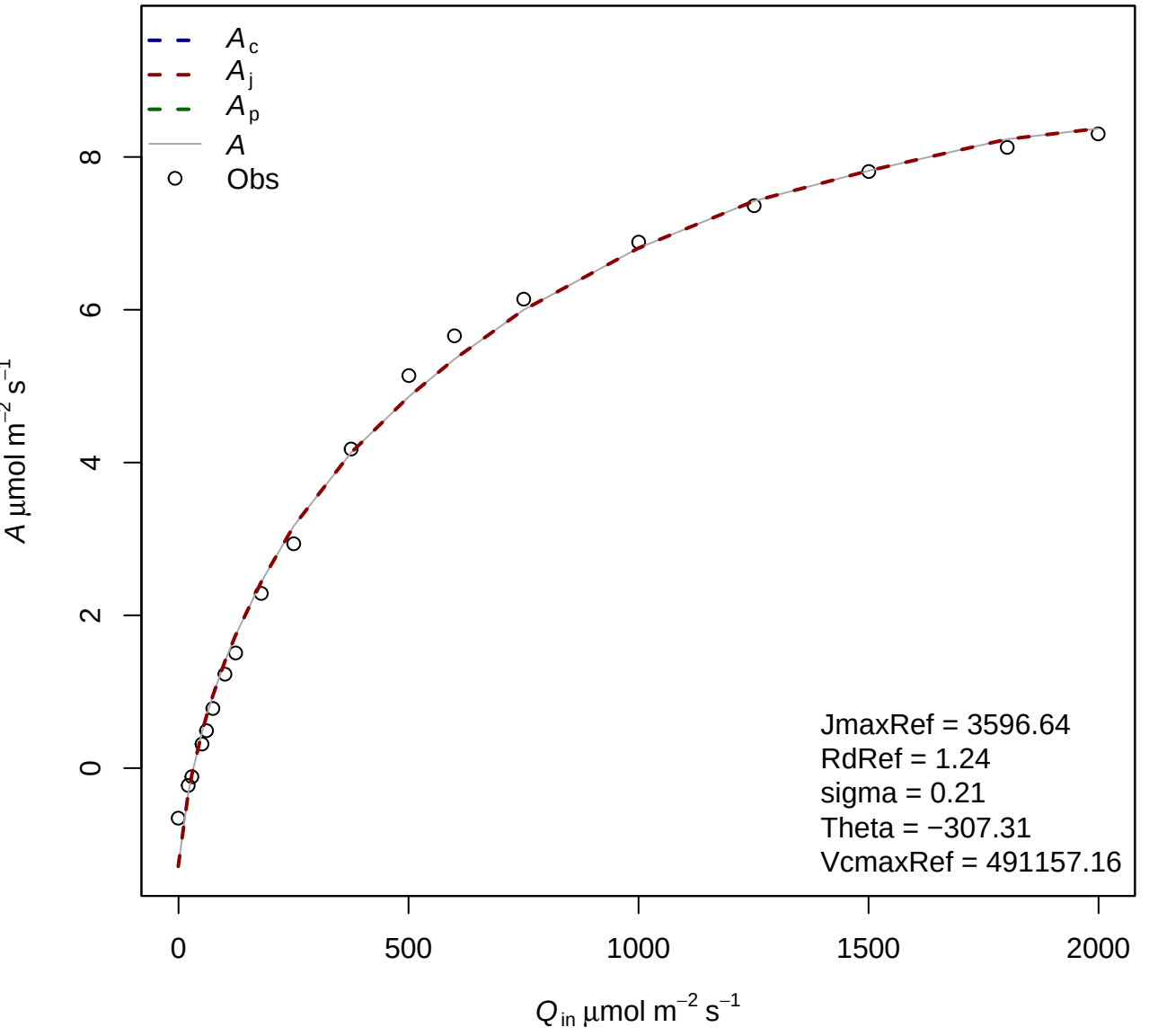

282
A_c
A_j
A_p
A
Obs
JmaxRef = 84.07
RdRef = 0.68
sigma = 0.19
Theta = −13.68
VcmaxRef = 211721.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

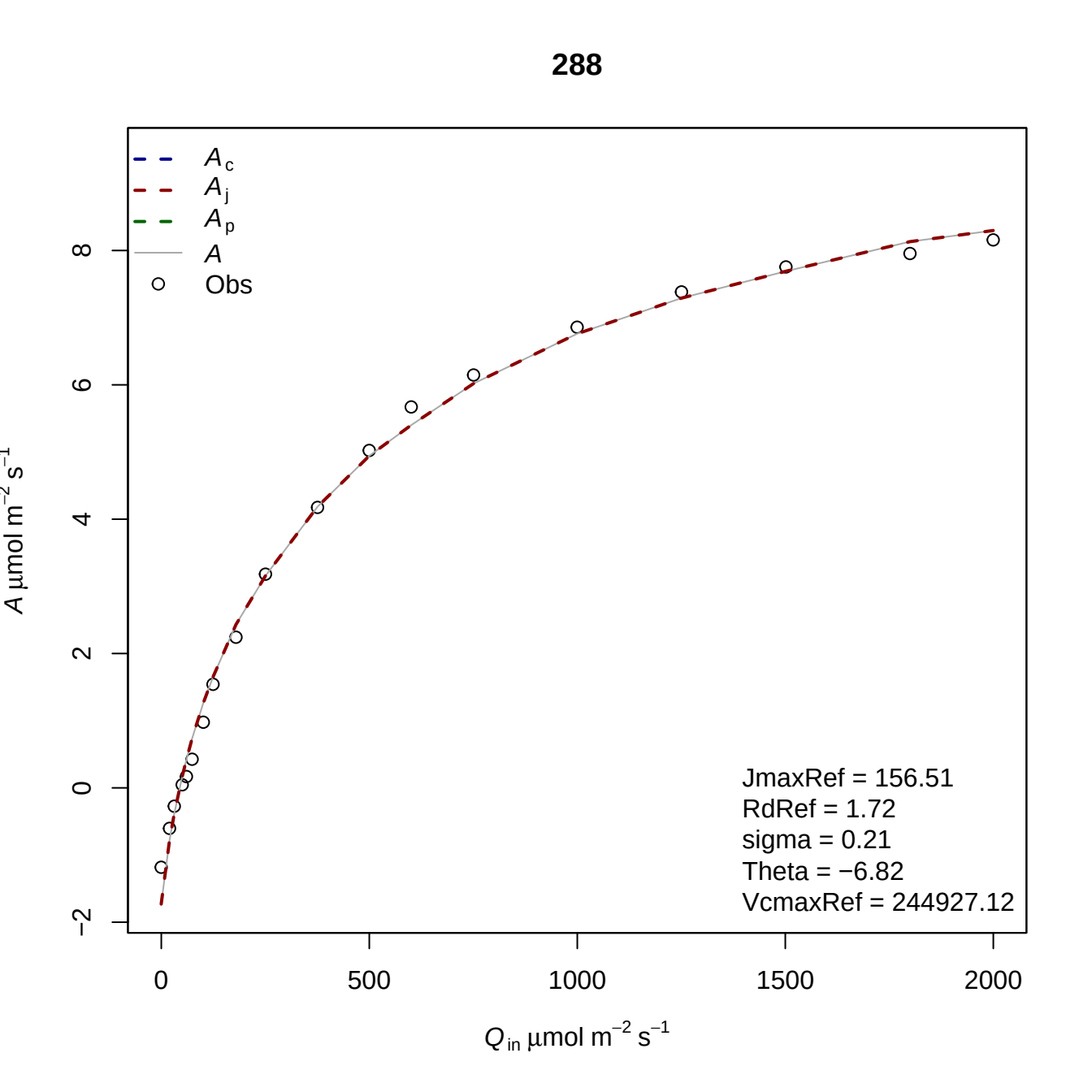

283
A_c
A_j
A_p
A
Obs
JmaxRef = 272.69
RdRef = 0.9
sigma = 0.16
Theta = −42.57
VcmaxRef = 563903.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

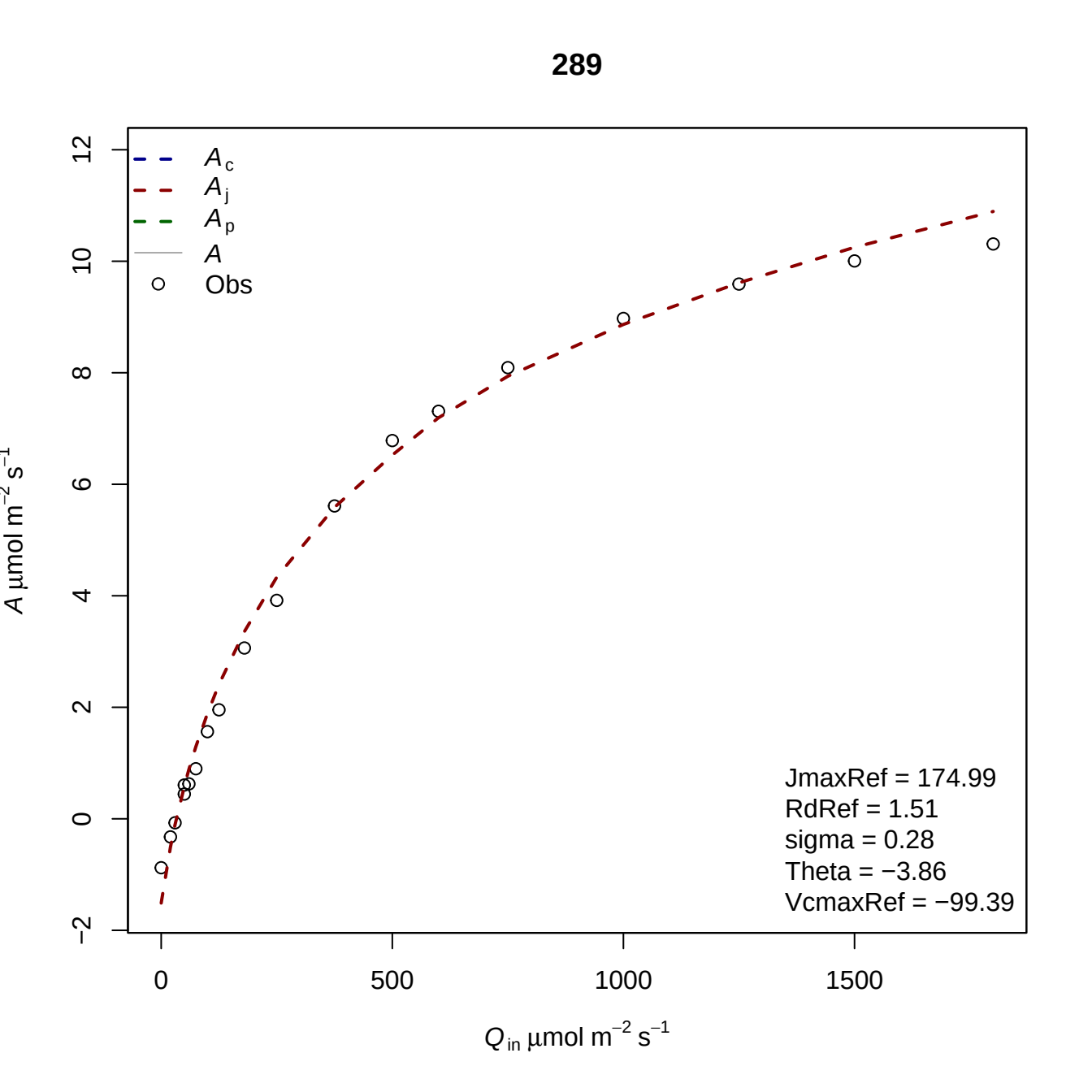

284
A_c
A_j
A_p
A
Obs
JmaxRef = 152.84
RdRef = 1.48
sigma = 0.26
Theta = −3.7
VcmaxRef = 167810.34
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

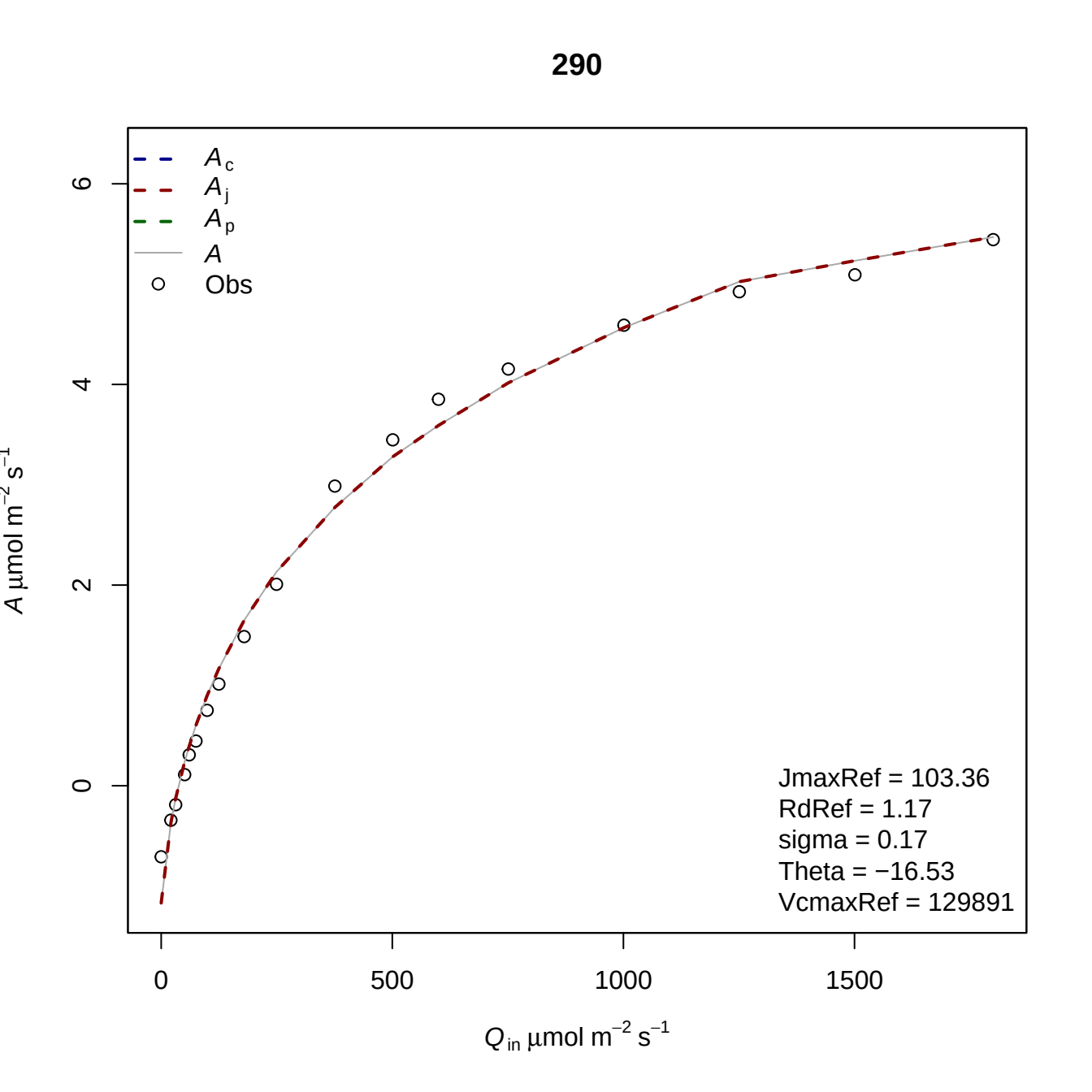

285
A_c
A_j
A_p
A
Obs
JmaxRef = 70.59
RdRef = 1.33
sigma = 0.21
Theta = −4.75
VcmaxRef = 198027.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

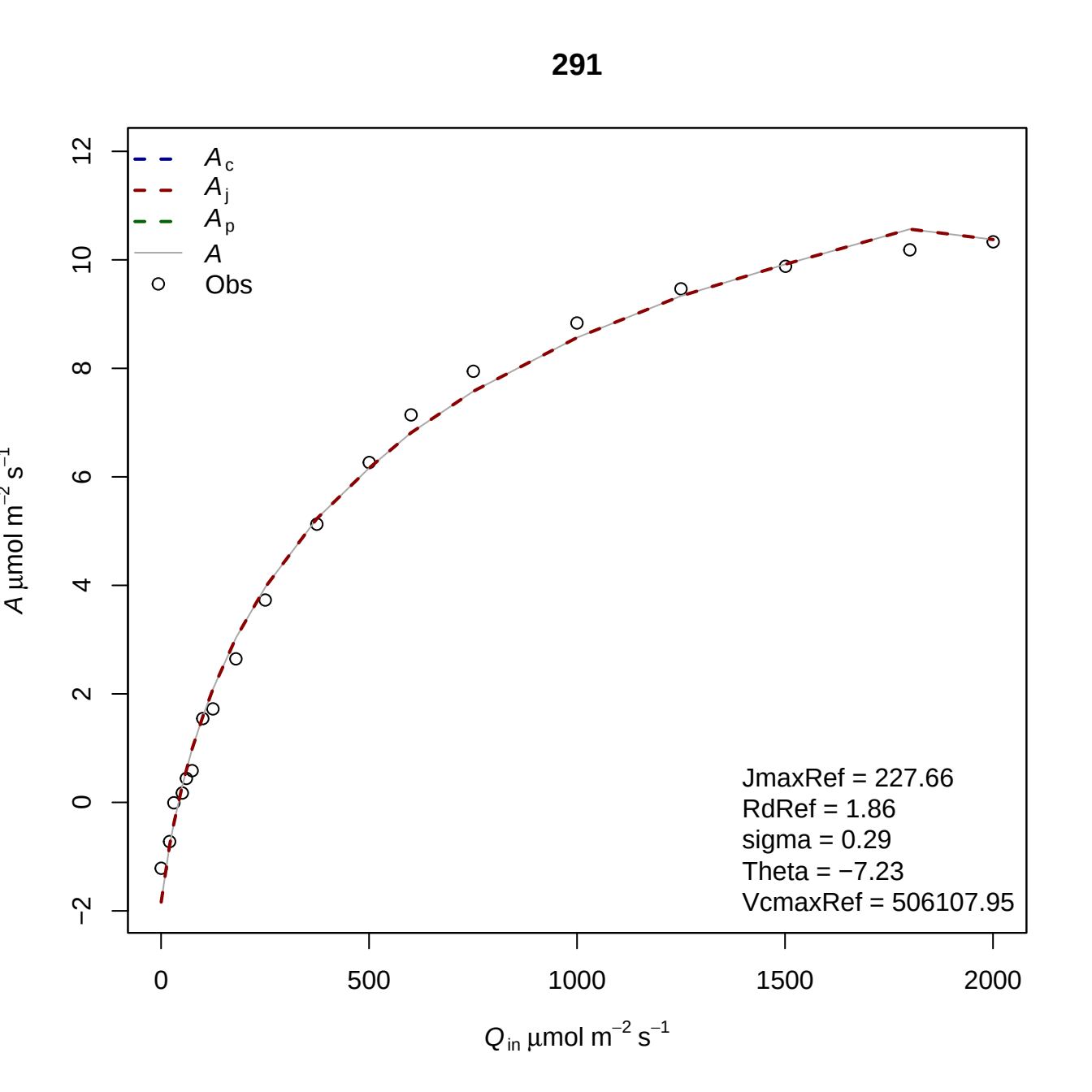

286
A_c
A_j
A_p
A
Obs
JmaxRef = 90.51
RdRef = 1.61
sigma = 0.13
Theta = −6.46
VcmaxRef = 227868.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

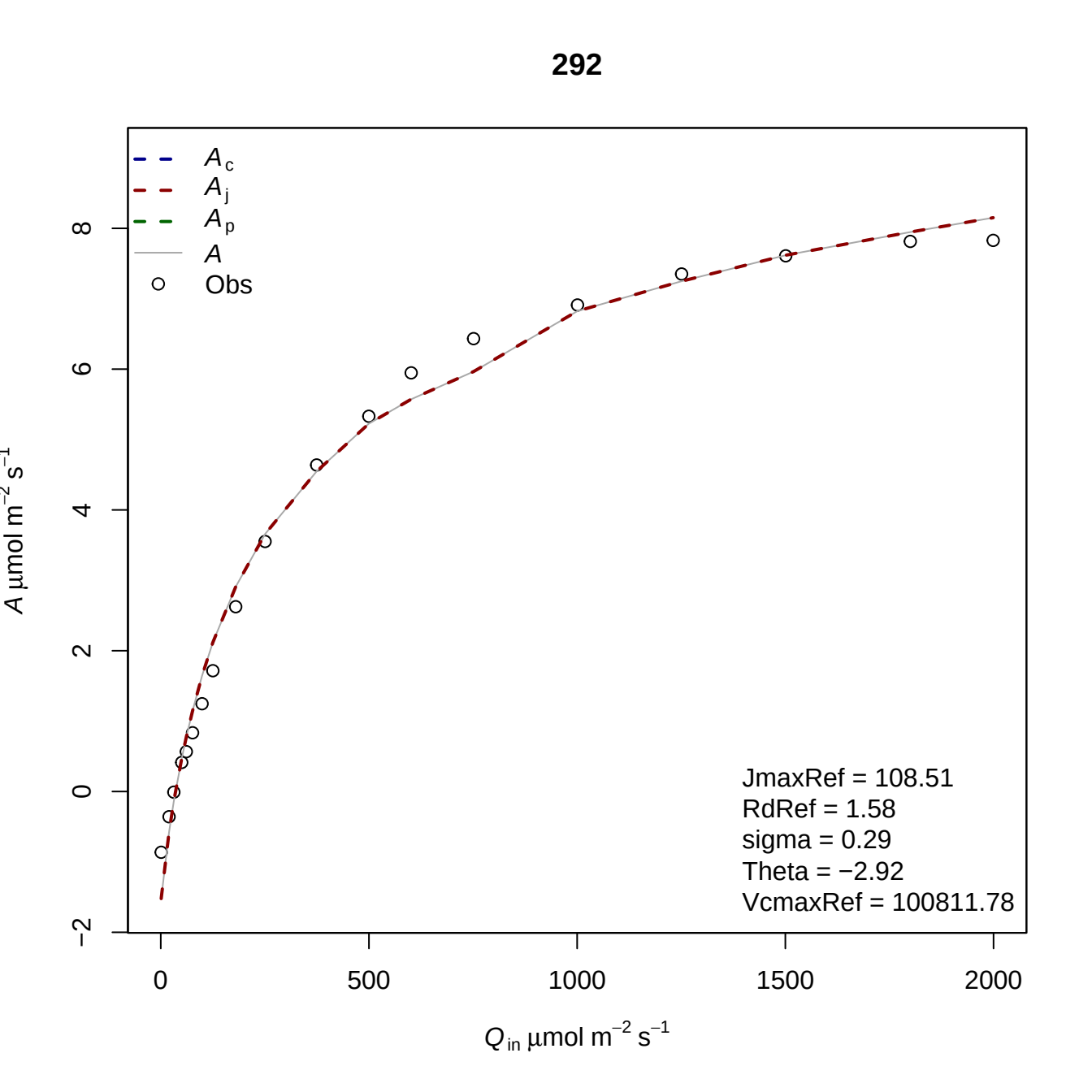

A_c
A_j
A_p
A
Obs
JmaxRef = 3596.64
RdRef = 1.24
sigma = 0.21
Theta = −307.31
VcmaxRef = 491157.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

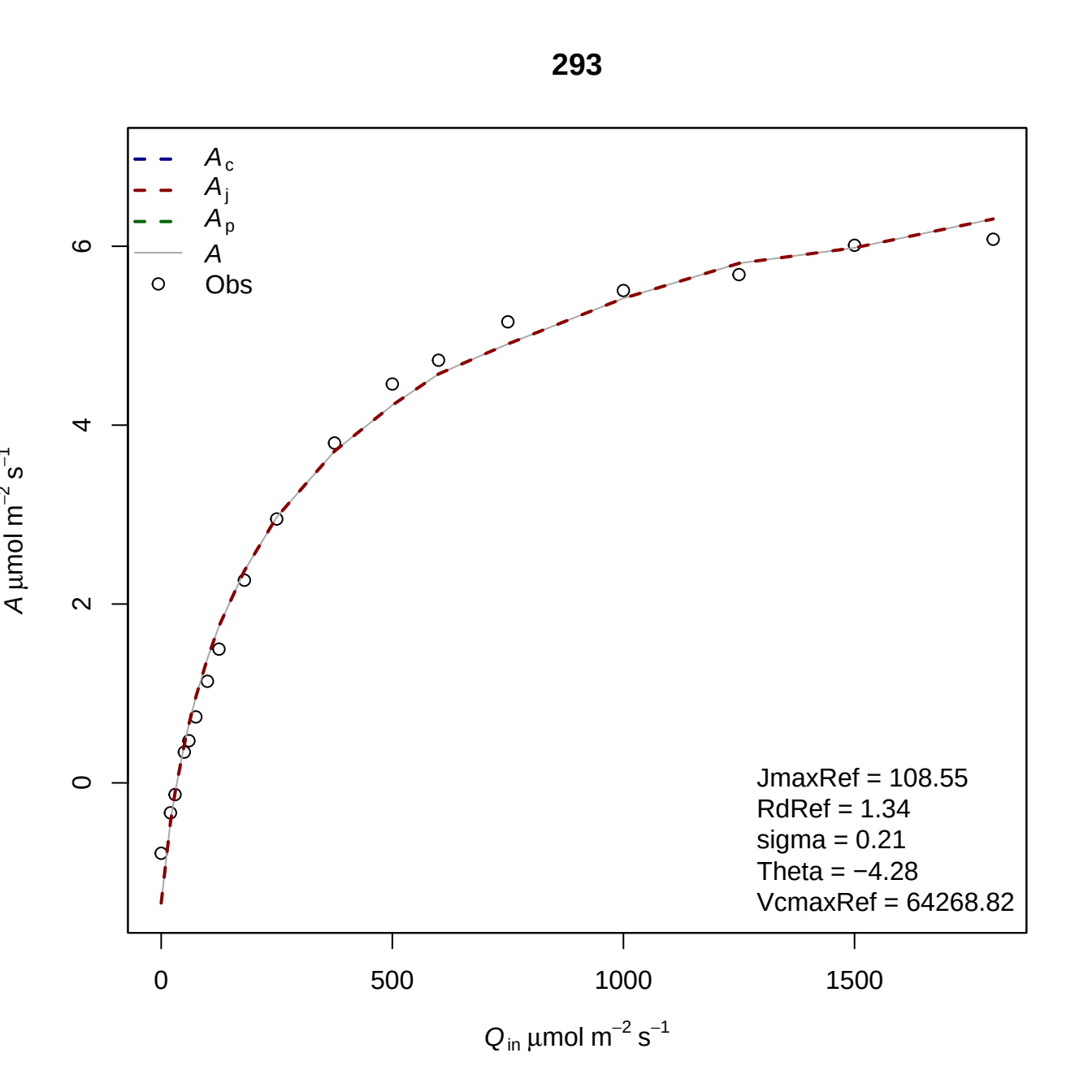

288
A_c
A_j
A_p
A
Obs
JmaxRef = 156.51
RdRef = 1.72
sigma = 0.21
Theta = −6.82
VcmaxRef = 244927.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

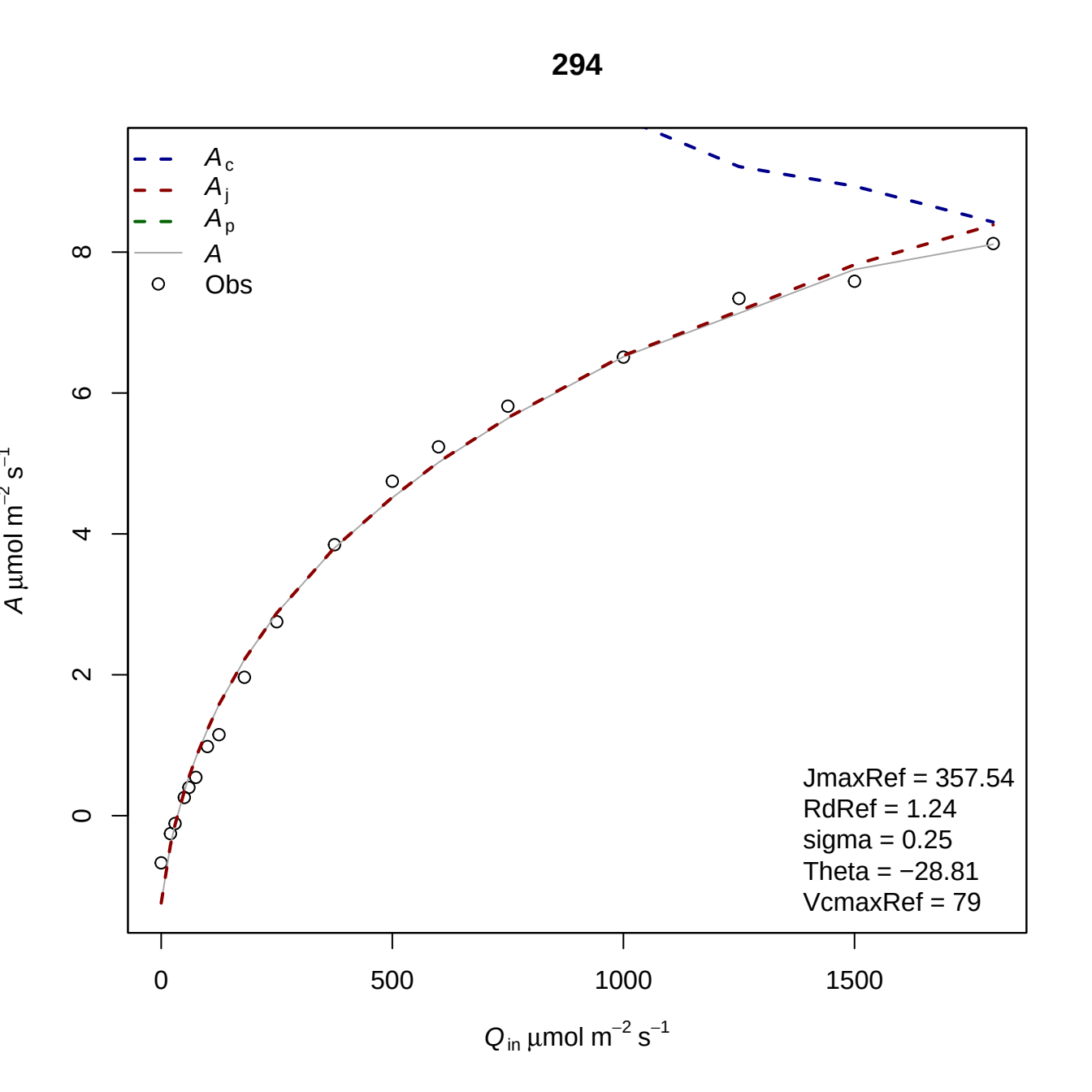

289
A_c
A_j
A_p
A
Obs
JmaxRef = 174.99
RdRef = 1.51
sigma = 0.28
Theta = −3.86
VcmaxRef = −99.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

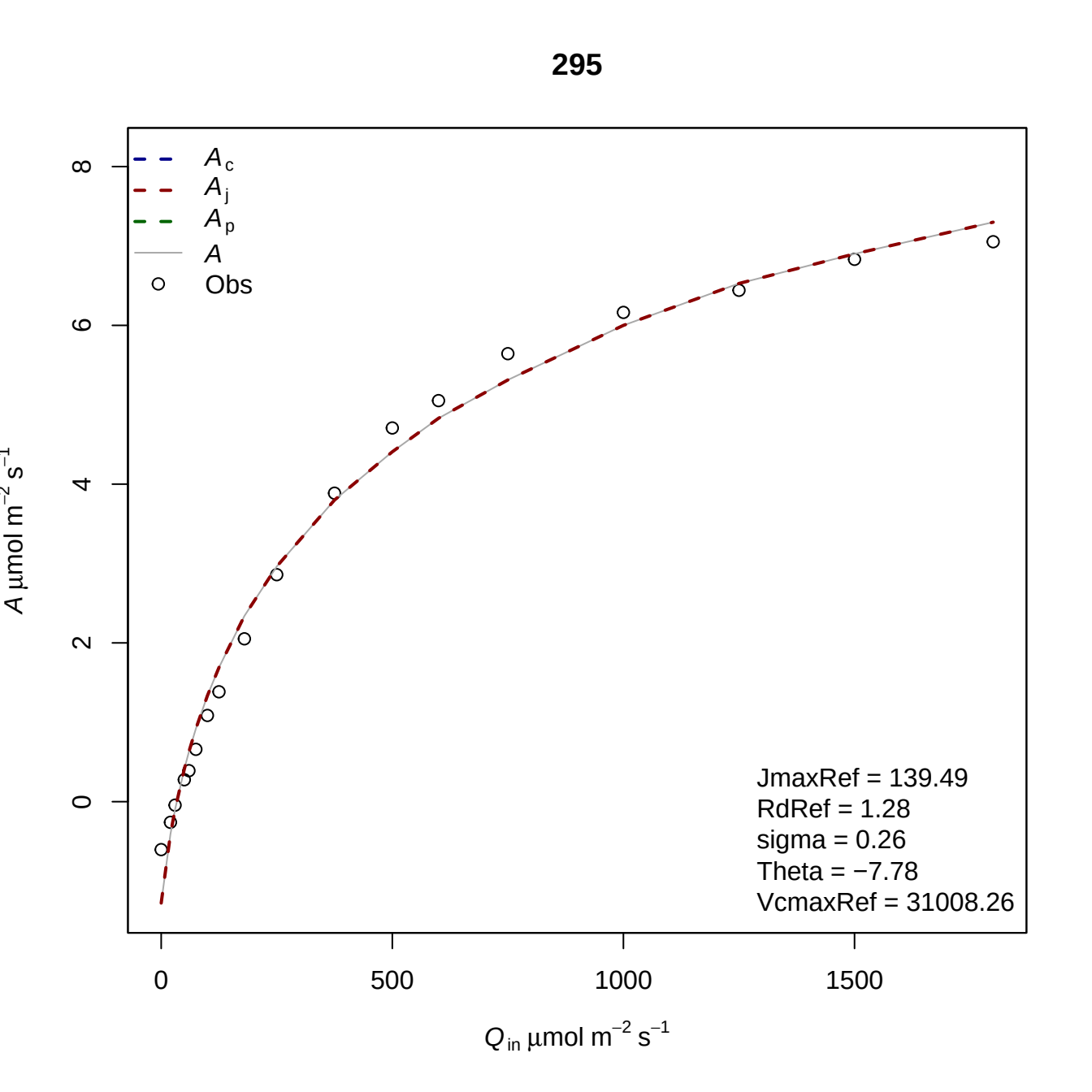

290
A_c
A_j
A_p
A
Obs
JmaxRef = 103.36
RdRef = 1.17
sigma = 0.17
Theta = −16.53
VcmaxRef = 129891
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

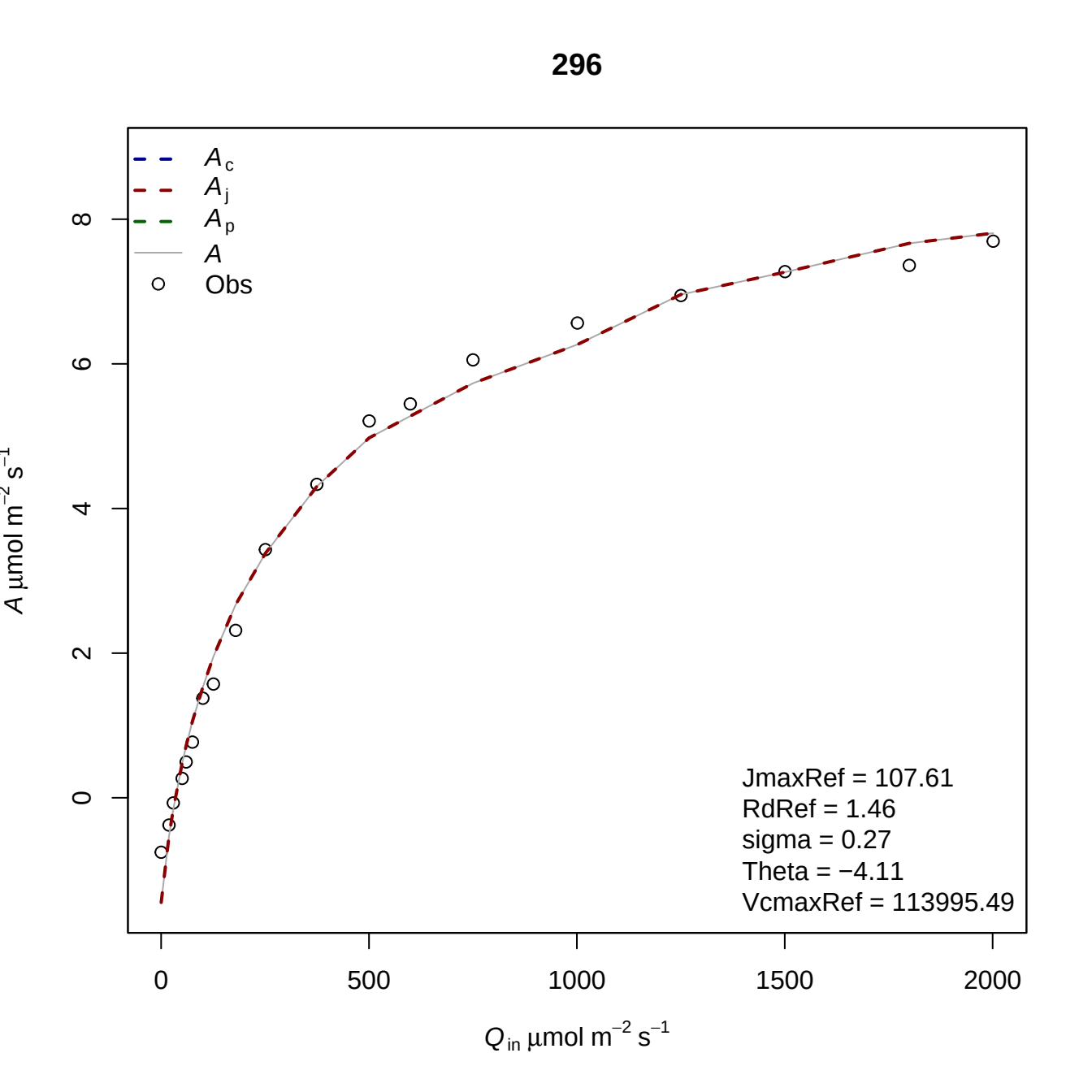

291
A_c
A_j
A_p
A
Obs
JmaxRef = 227.66
RdRef = 1.86
sigma = 0.29
Theta = −7.23
VcmaxRef = 506107.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

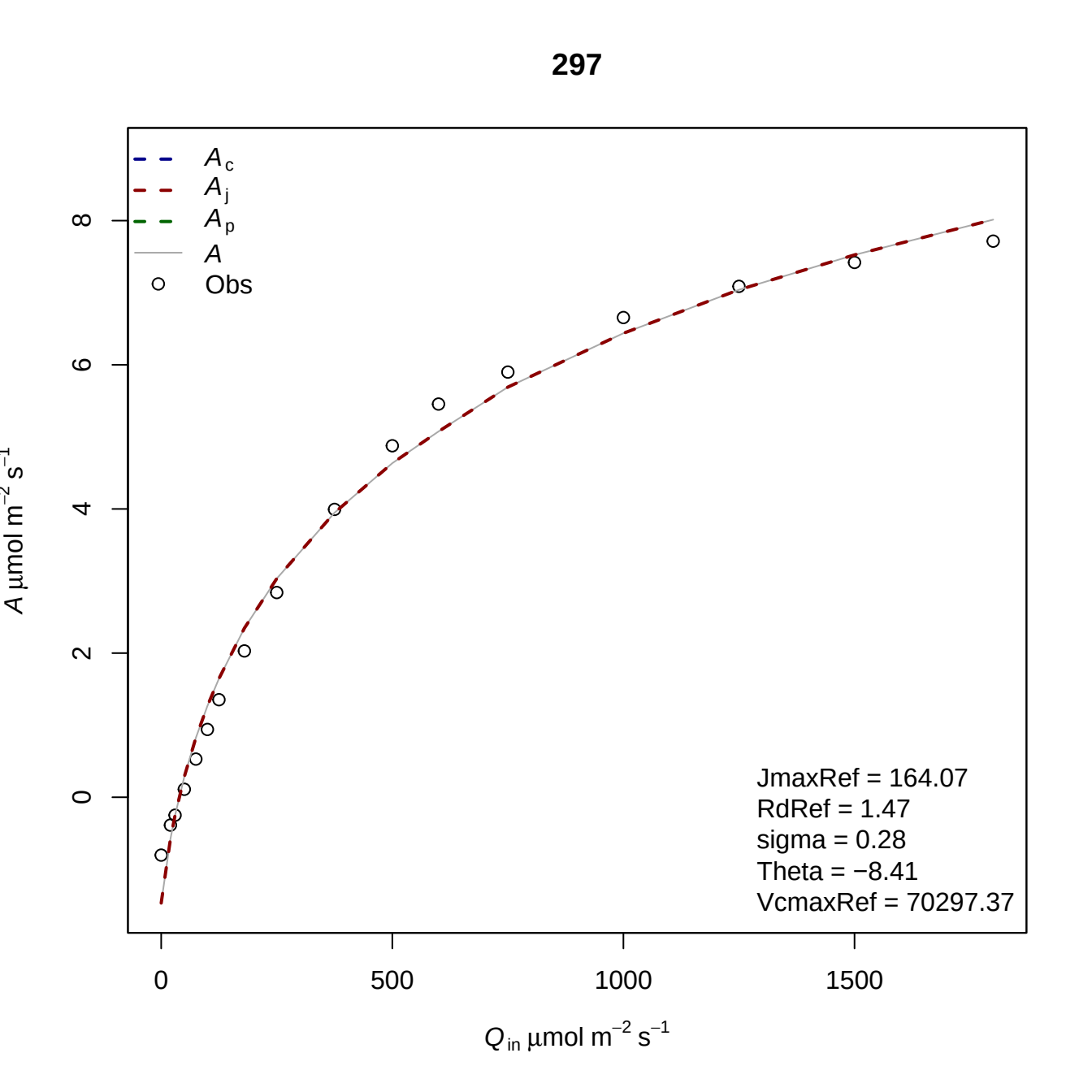

292
A_c
A_j
A_p
A
Obs
JmaxRef = 108.51
RdRef = 1.58
sigma = 0.29
Theta = −2.92
VcmaxRef = 100811.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

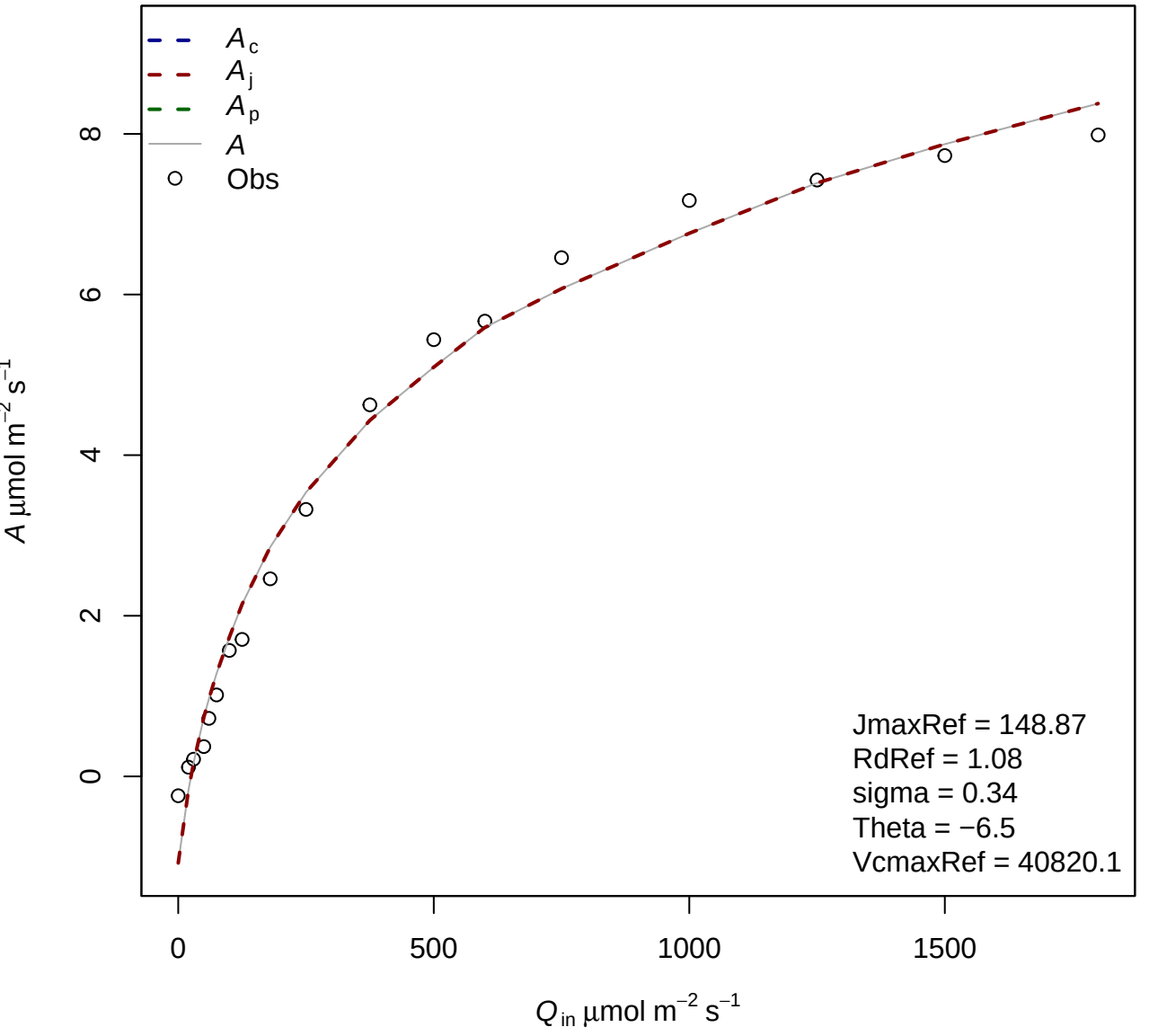

293
A_c
A_j
A_p
A
Obs
JmaxRef = 108.55
RdRef = 1.34
sigma = 0.21
Theta = −4.28
VcmaxRef = 64268.82
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

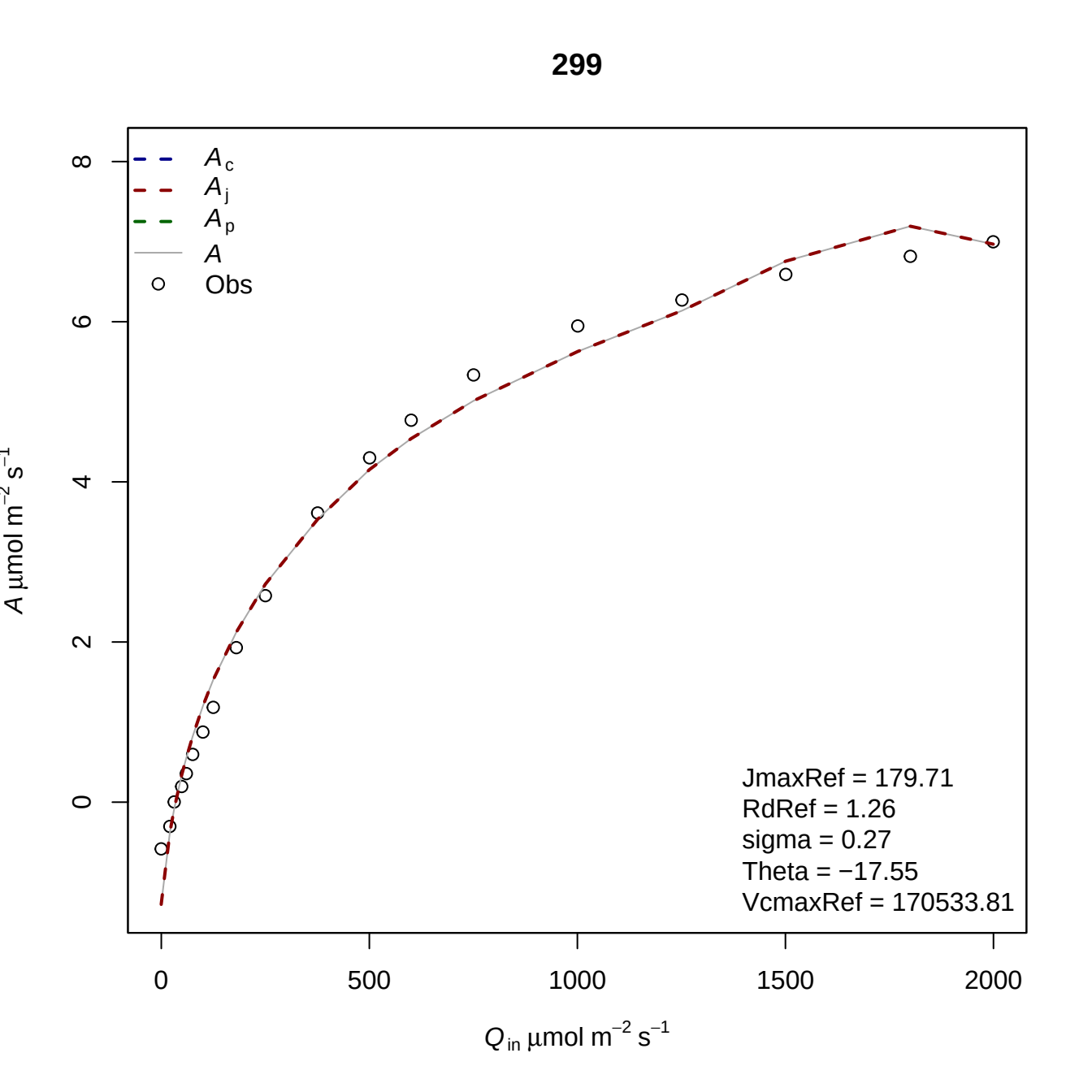

294
A_c
A_j
A_p
A
Obs
JmaxRef = 357.54
RdRef = 1.24
sigma = 0.25
Theta = −28.81
VcmaxRef = 79
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

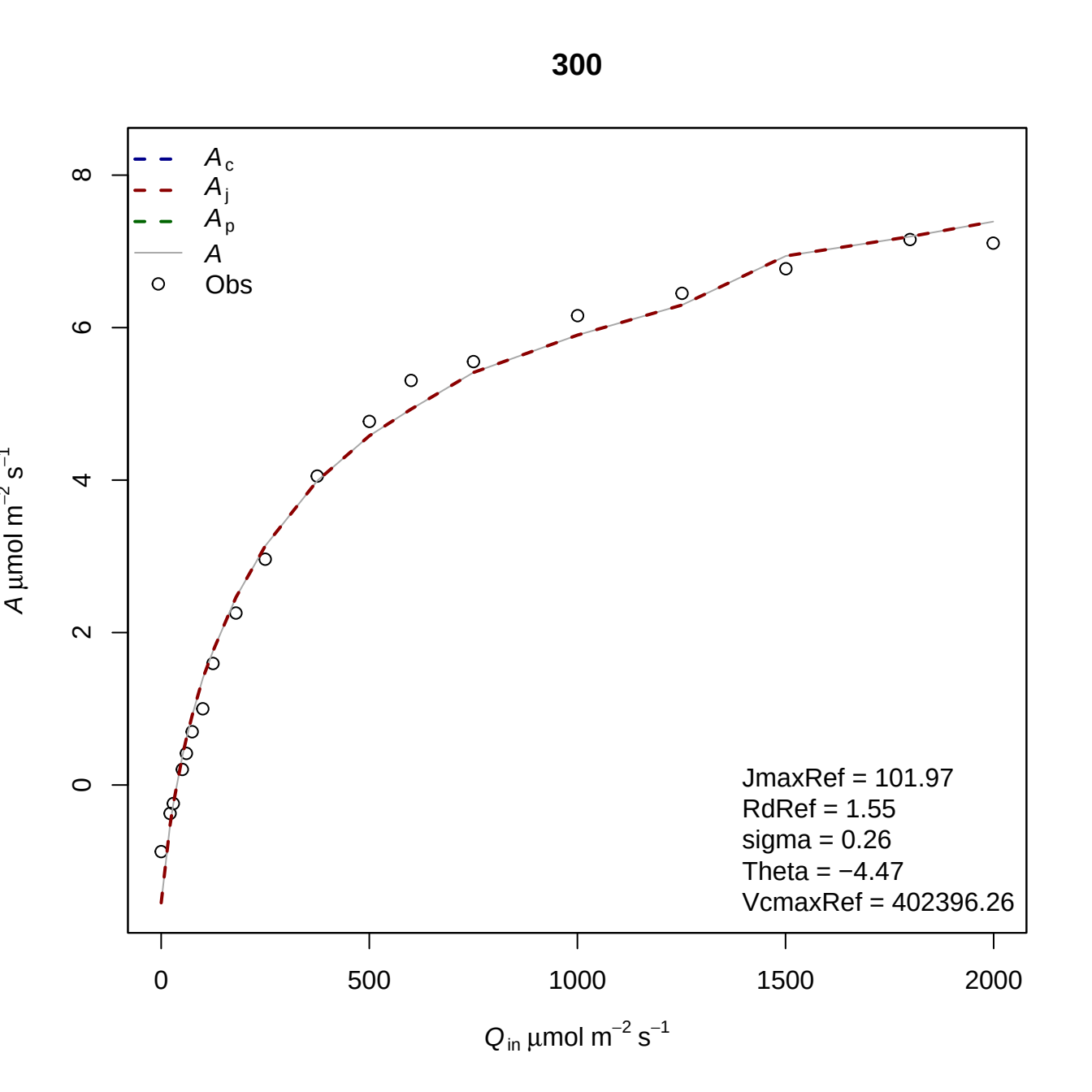

295
A_c
A_j
A_p
A
Obs
JmaxRef = 139.49
RdRef = 1.28
sigma = 0.26
Theta = −7.78
VcmaxRef = 31008.26
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

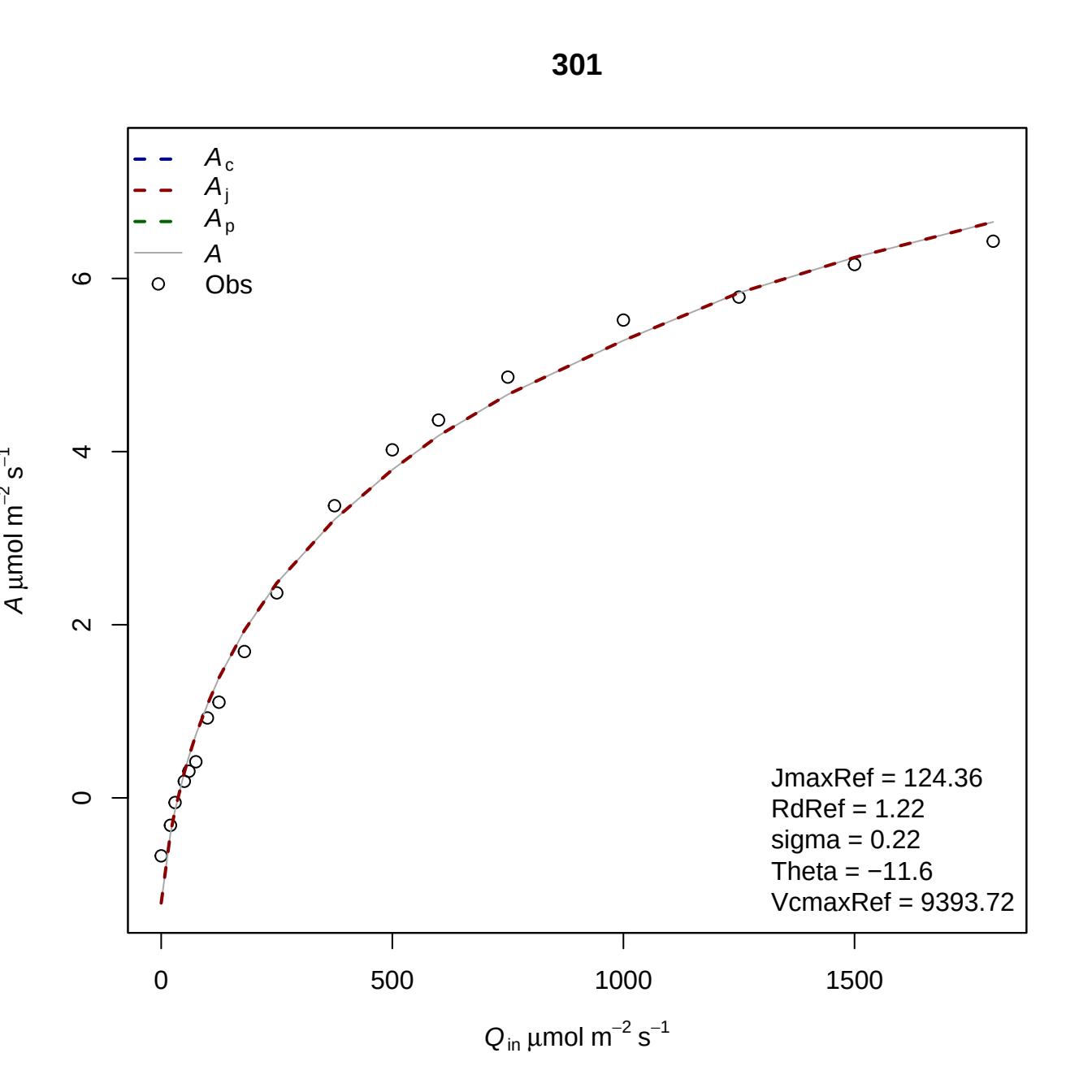

296
A_c
A_j
A_p
A
Obs
JmaxRef = 107.61
RdRef = 1.46
sigma = 0.27
Theta = −4.11
VcmaxRef = 113995.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

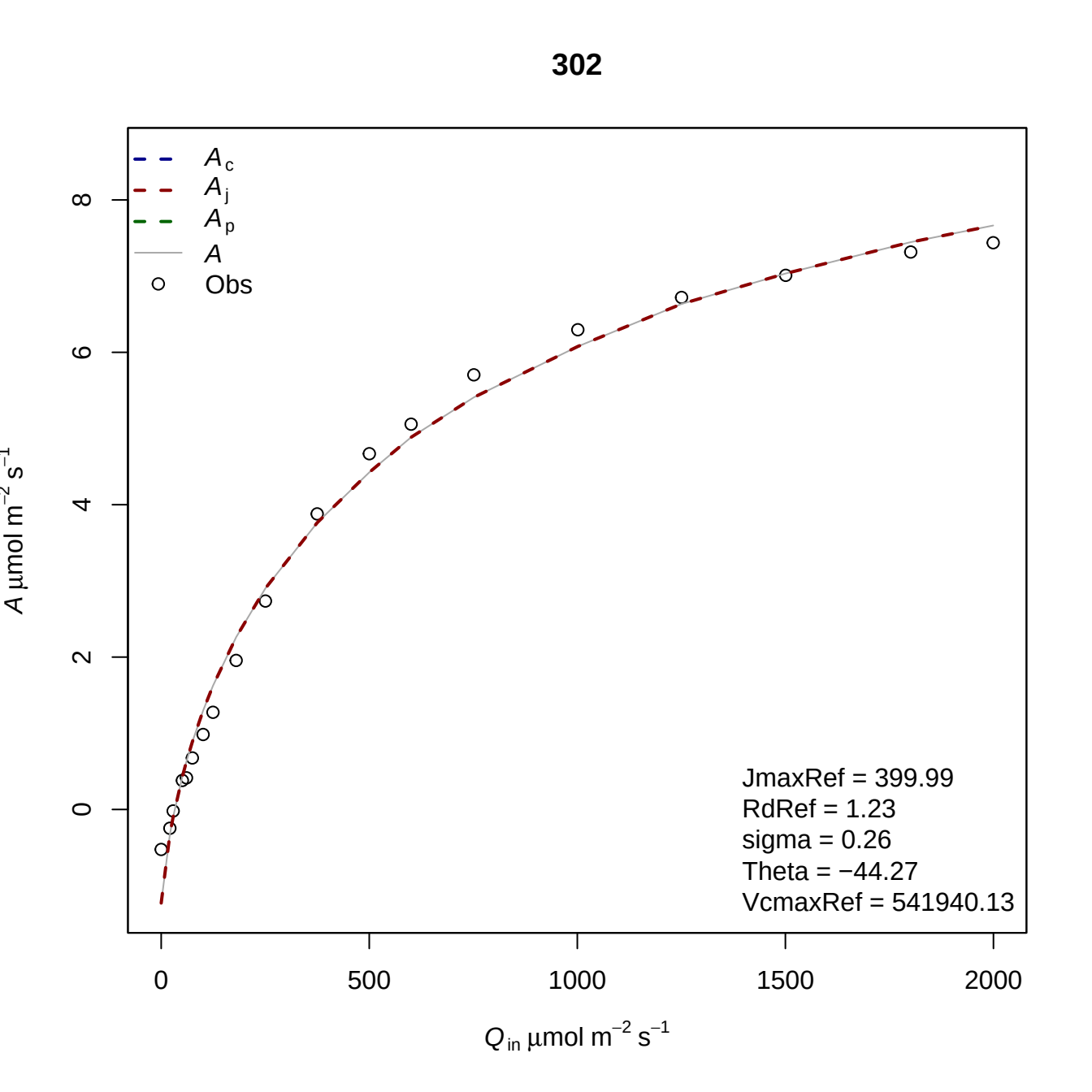

297
A_c
A_j
A_p
A
Obs
JmaxRef = 164.07
RdRef = 1.47
sigma = 0.28
Theta = −8.41
VcmaxRef = 70297.37
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

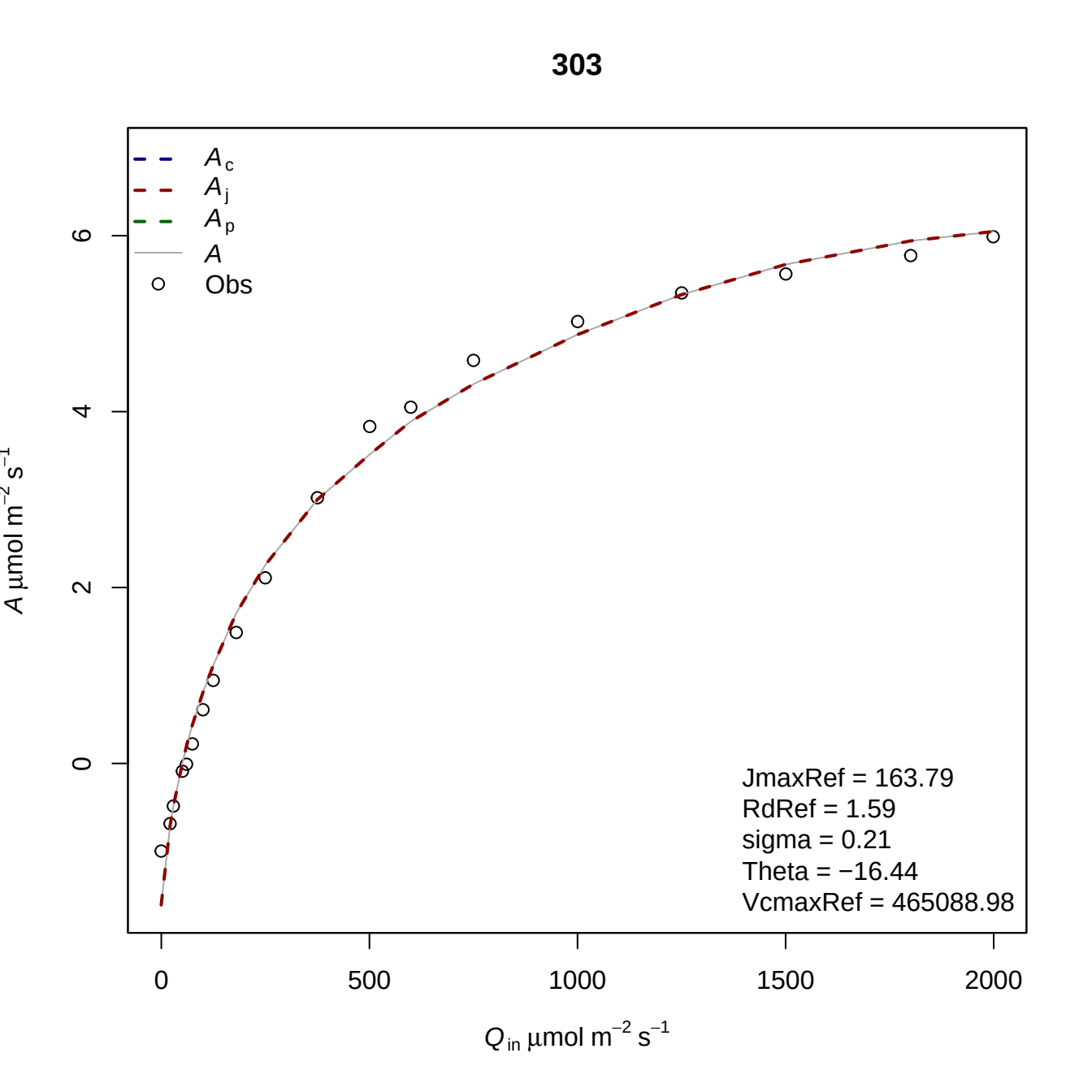

A_c
A_j
A_p
A
Obs
JmaxRef = 148.87
RdRef = 1.08
sigma = 0.34
Theta = −6.5
VcmaxRef = 40820.1
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

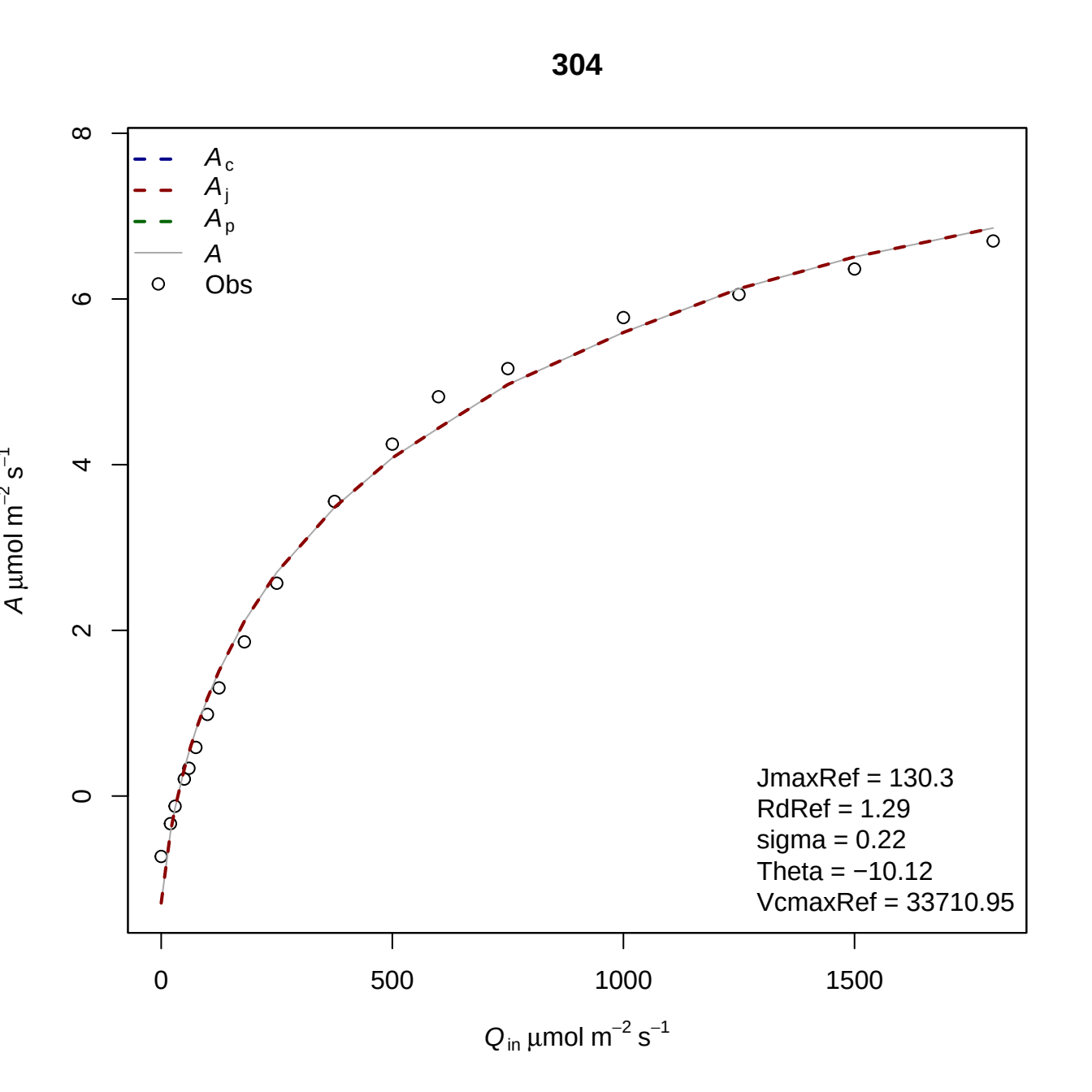

299
A_c
A_j
A_p
A
Obs
JmaxRef = 179.71
RdRef = 1.26
sigma = 0.27
Theta = −17.55
VcmaxRef = 170533.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

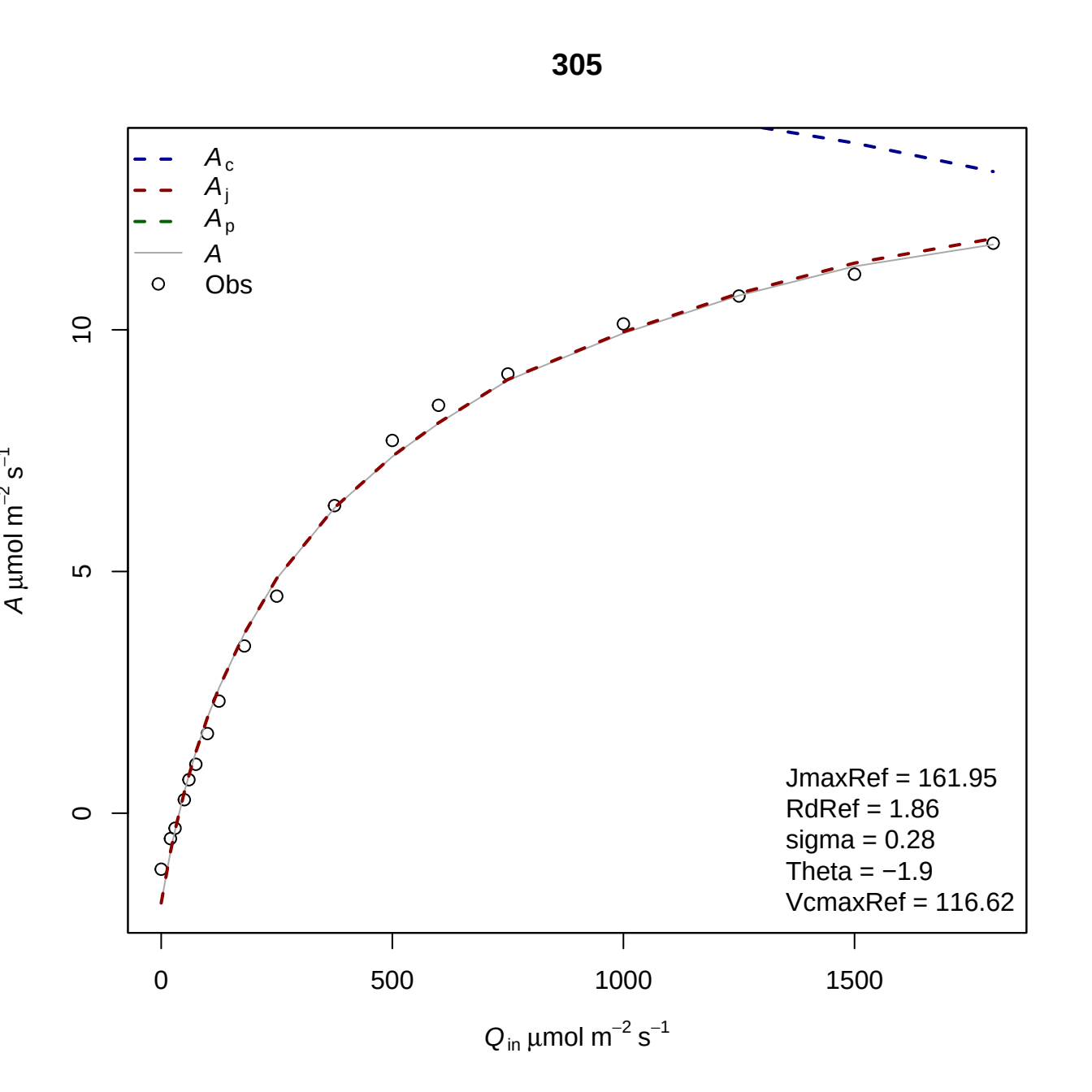

300
A_c
A_j
A_p
A
Obs
JmaxRef = 101.97
RdRef = 1.55
sigma = 0.26
Theta = −4.47
VcmaxRef = 402396.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

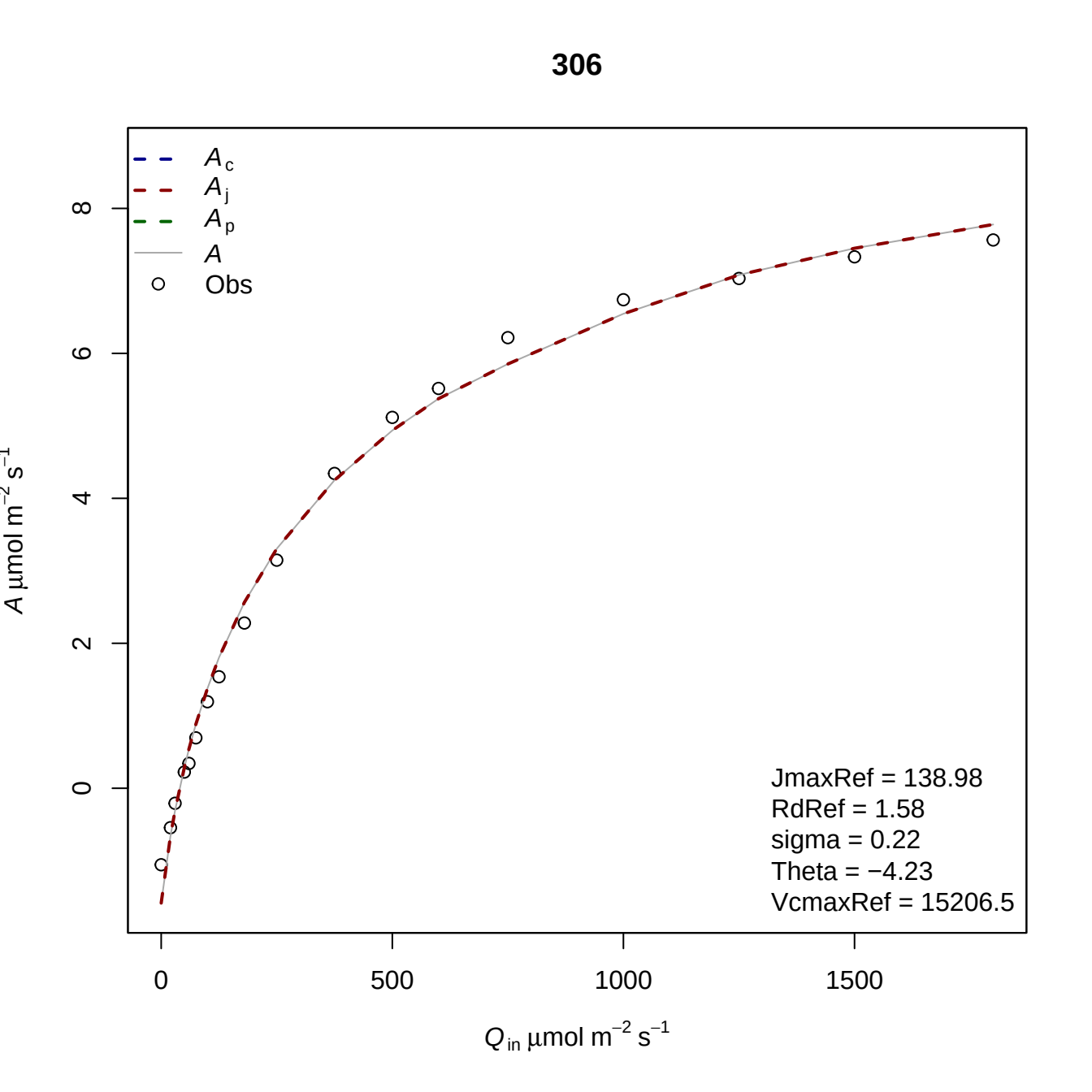

301
A_c
A_j
A_p
A
Obs
JmaxRef = 124.36
RdRef = 1.22
sigma = 0.22
Theta = −11.6
VcmaxRef = 9393.72
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

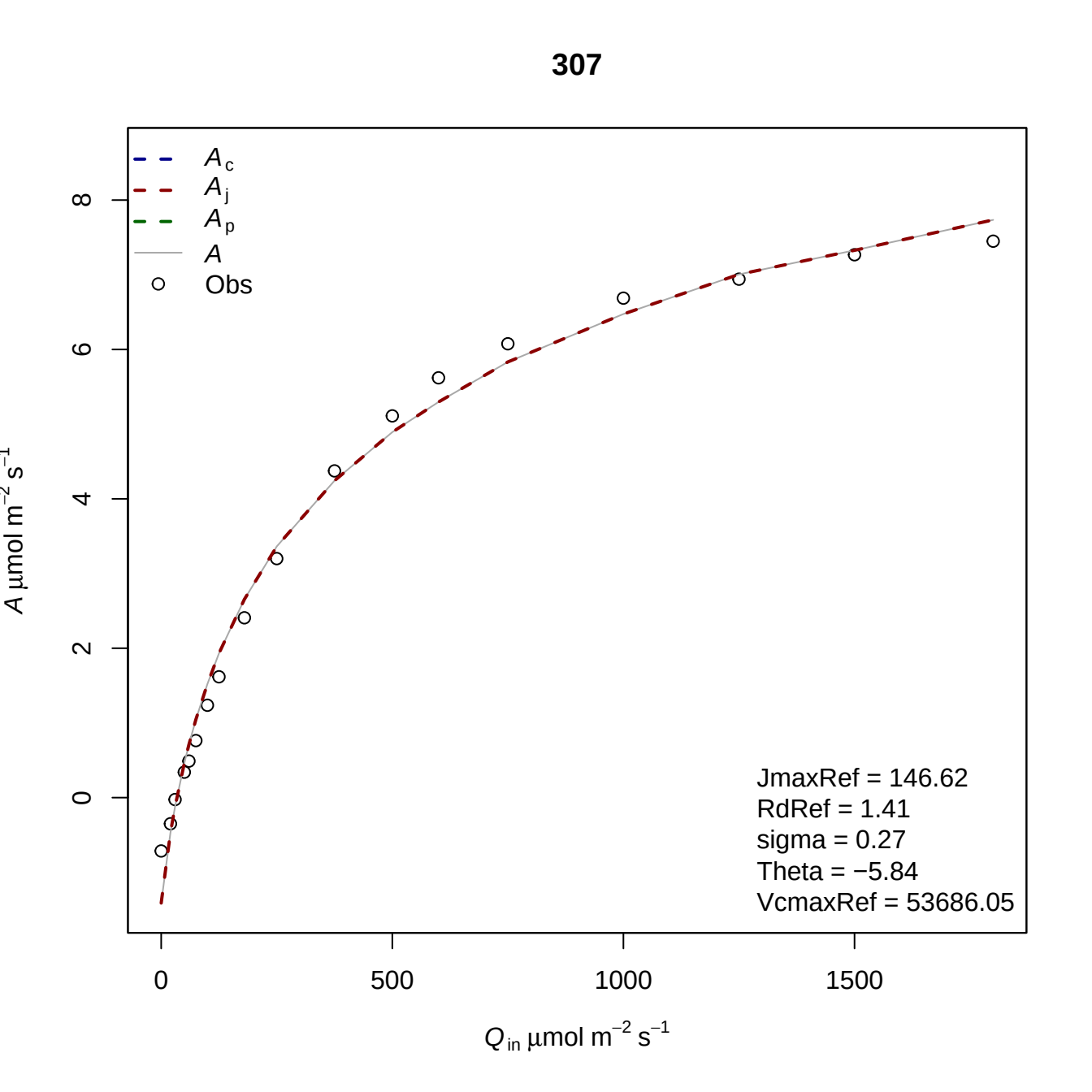

302
A_c
A_j
A_p
A
Obs
JmaxRef = 399.99
RdRef = 1.23
sigma = 0.26
Theta = −44.27
VcmaxRef = 541940.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

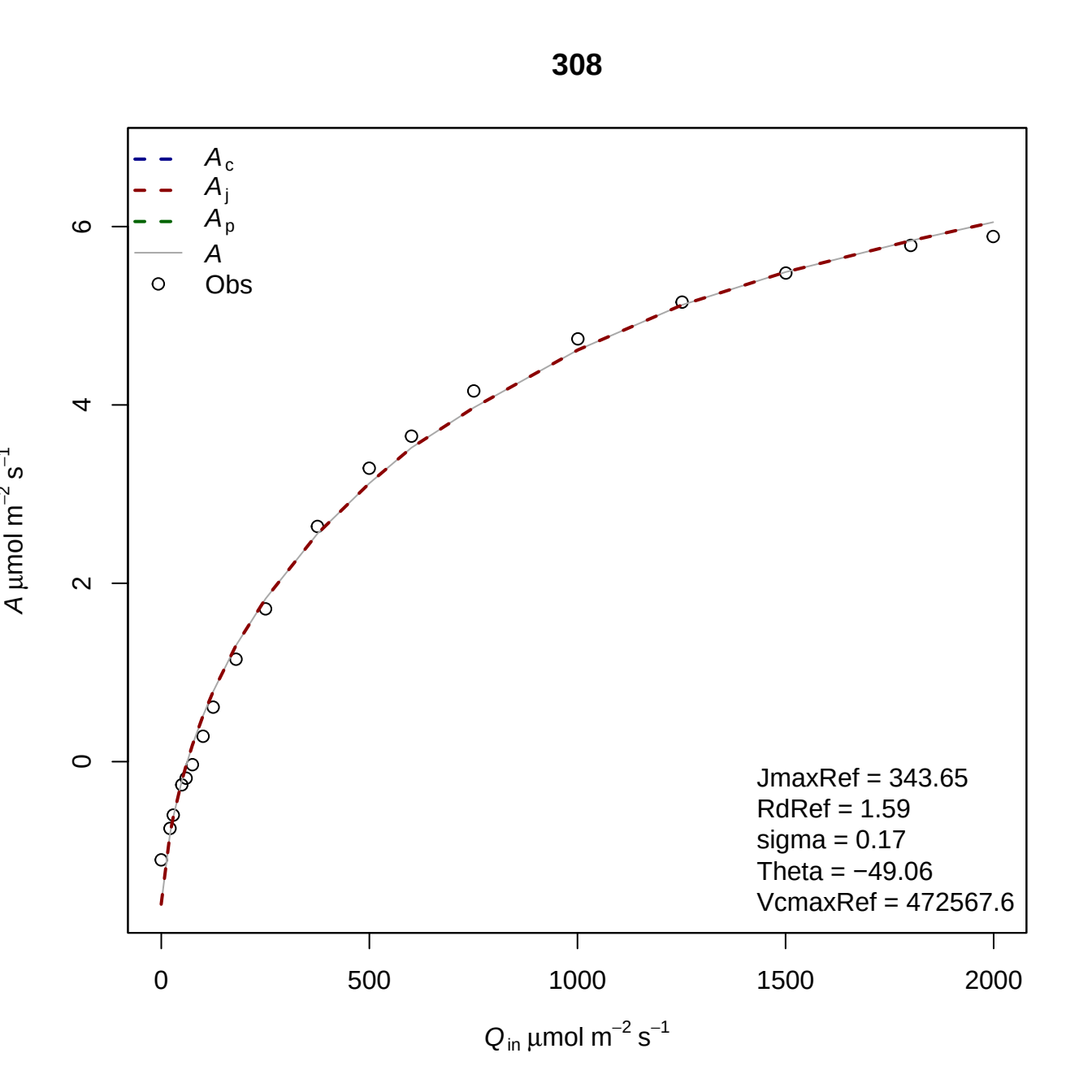

303
A_c
A_j
A_p
A
Obs
JmaxRef = 163.79
RdRef = 1.59
sigma = 0.21
Theta = −16.44
VcmaxRef = 465088.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

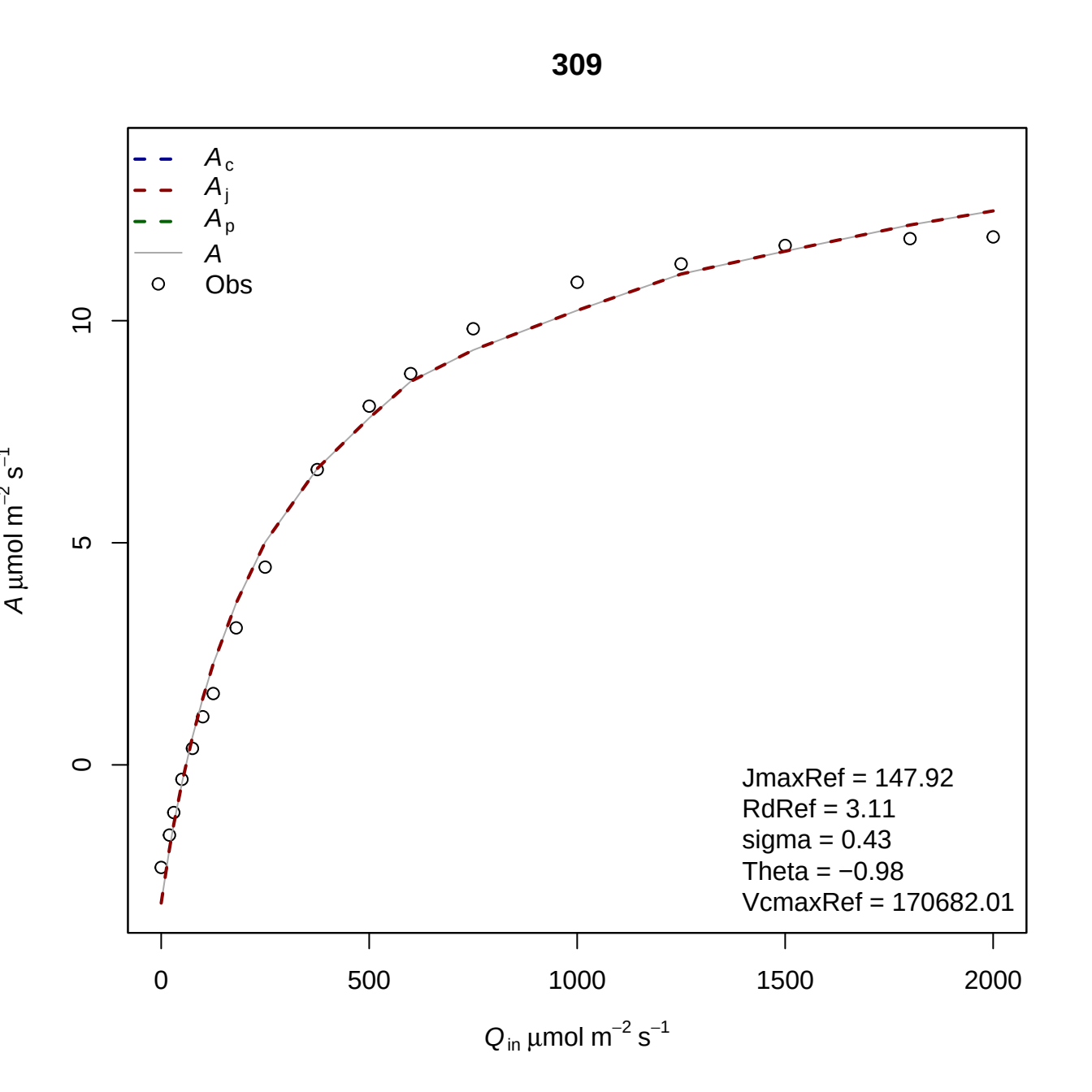

304
A_c
A_j
A_p
A
Obs
JmaxRef = 130.3
RdRef = 1.29
sigma = 0.22
Theta = −10.12
VcmaxRef = 33710.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

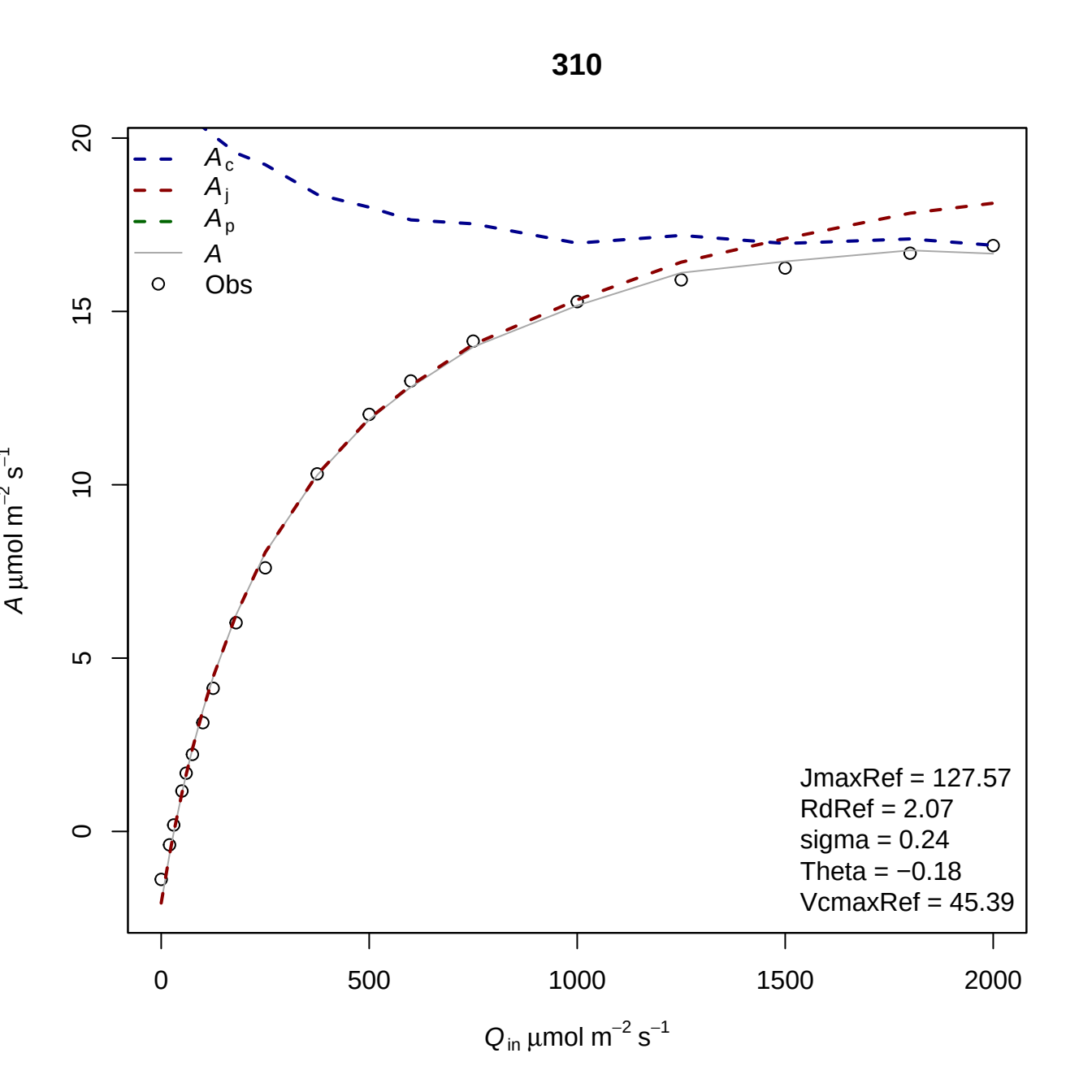

305
A_c
A_j
A_p
A
Obs
JmaxRef = 161.95
RdRef = 1.86
sigma = 0.28
Theta = −1.9
VcmaxRef = 116.62
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

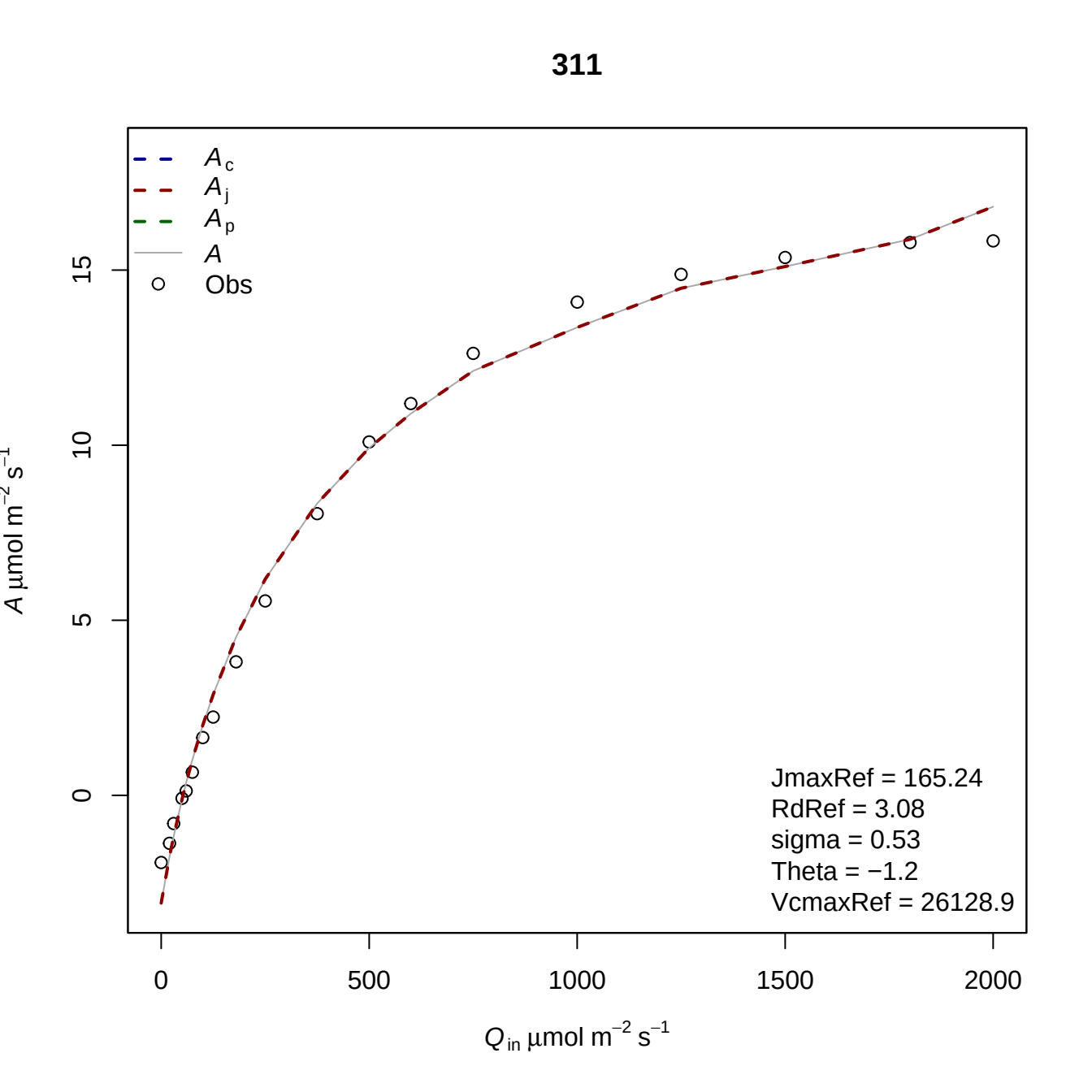

306
A_c
A_j
A_p
A
Obs
JmaxRef = 138.98
RdRef = 1.58
sigma = 0.22
Theta = −4.23
VcmaxRef = 15206.5
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

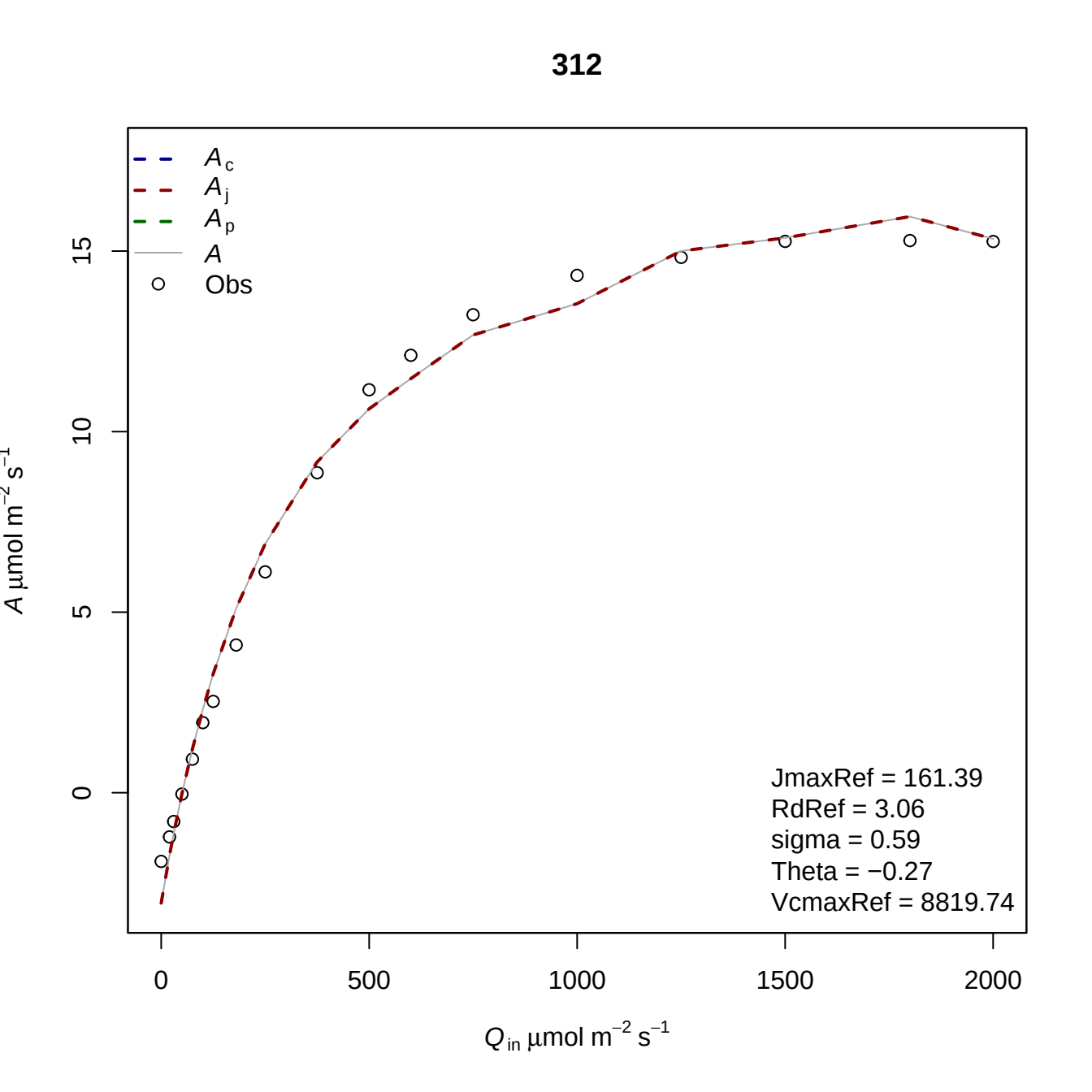

307
A_c
A_j
A_p
A
Obs
JmaxRef = 146.62
RdRef = 1.41
sigma = 0.27
Theta = −5.84
VcmaxRef = 53686.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

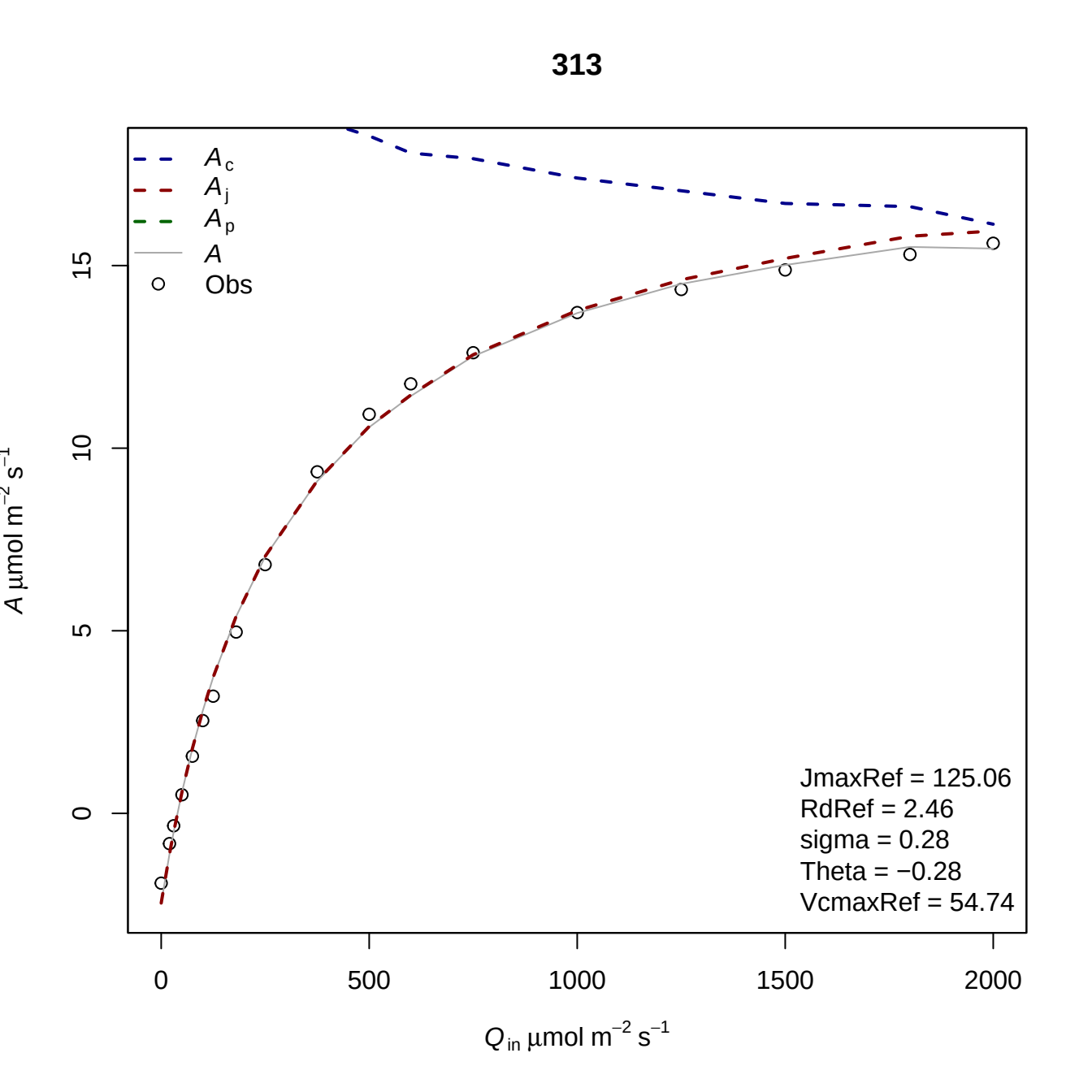

308
A_c
A_j
A_p
A
Obs
JmaxRef = 343.65
RdRef = 1.59
sigma = 0.17
Theta = −49.06
VcmaxRef = 472567.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

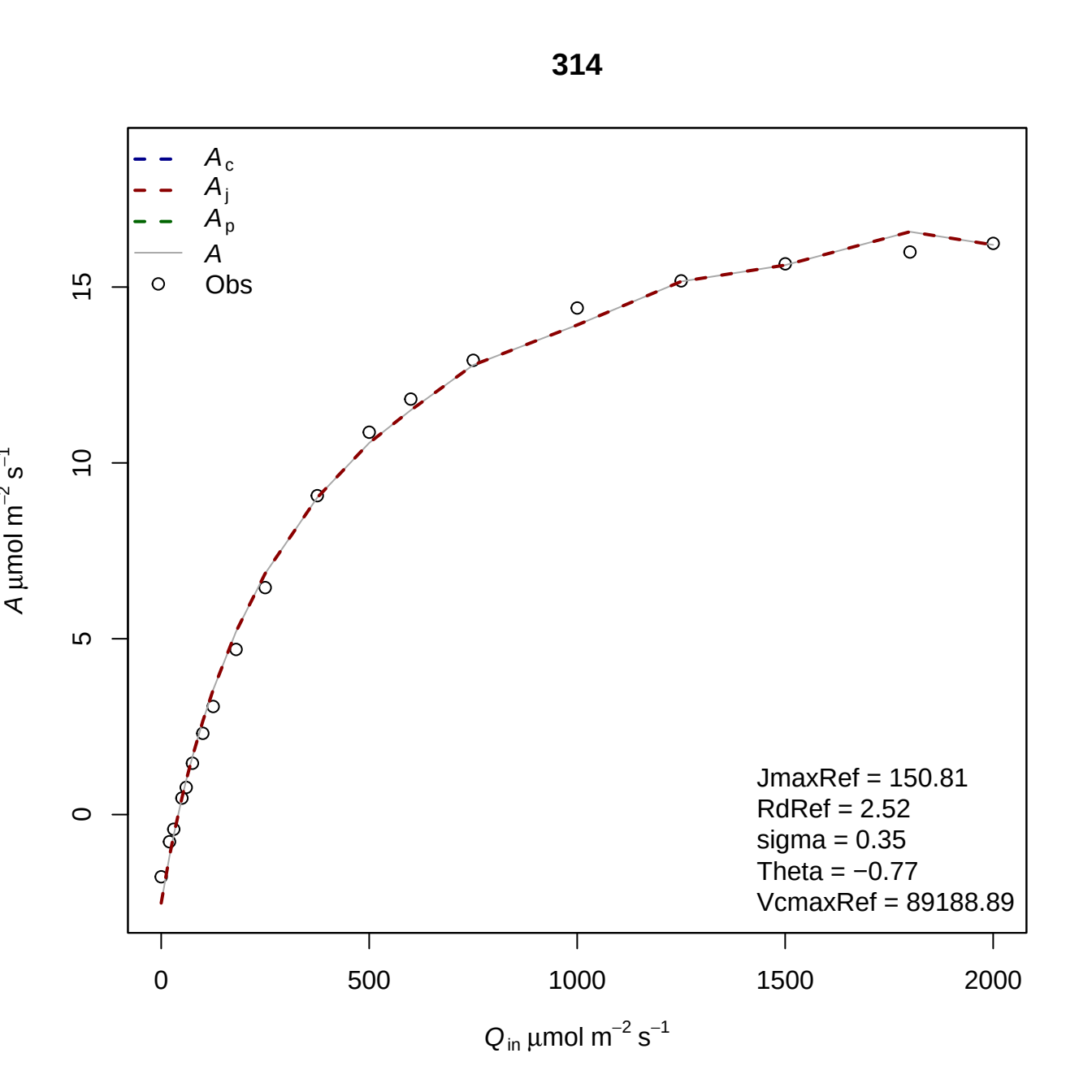

309
A_c
A_j
A_p
A
Obs
JmaxRef = 147.92
RdRef = 3.11
sigma = 0.43
Theta = −0.98
VcmaxRef = 170682.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

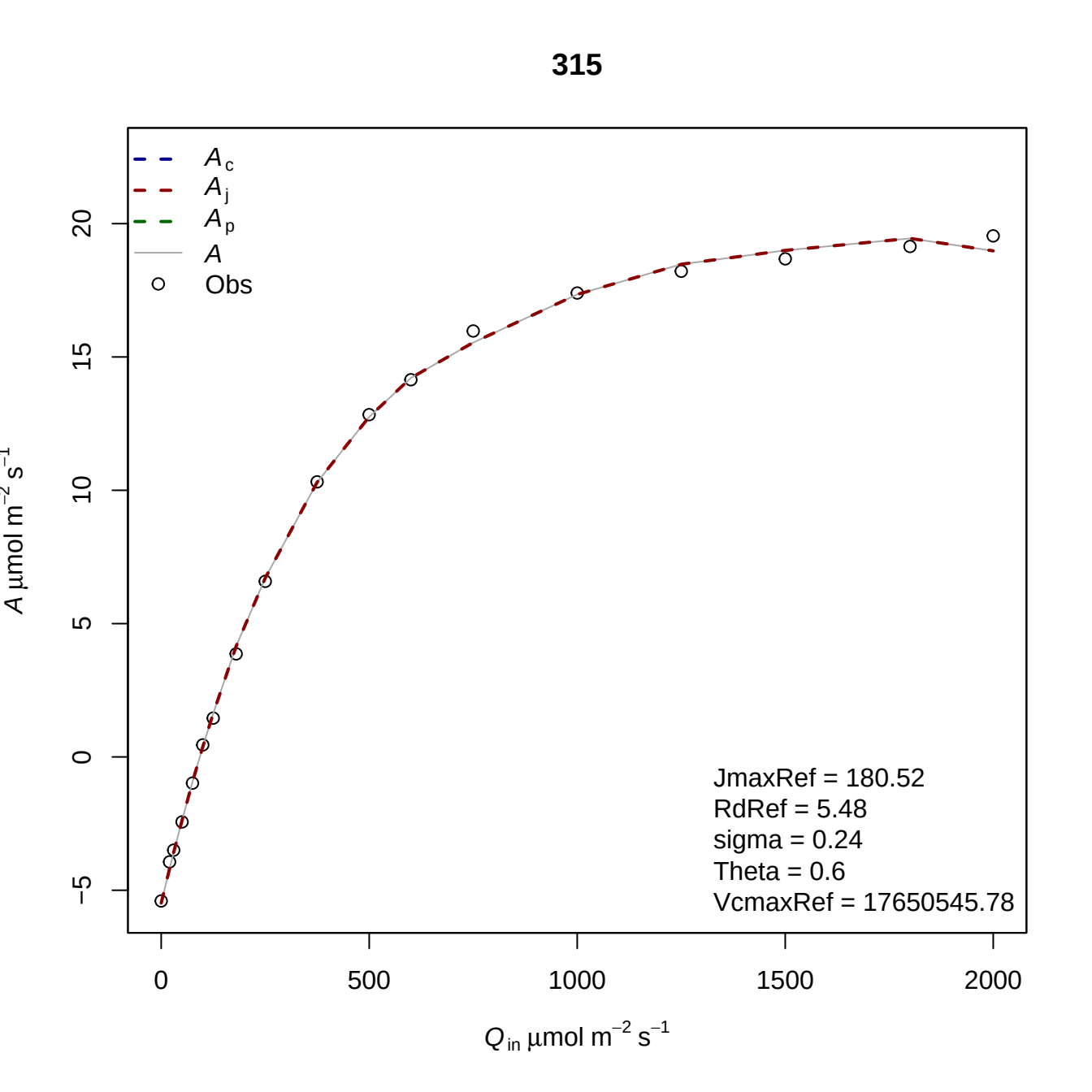

310
A_c
A_j
A_p
A
Obs
JmaxRef = 127.57
RdRef = 2.07
sigma = 0.24
Theta = −0.18
VcmaxRef = 45.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

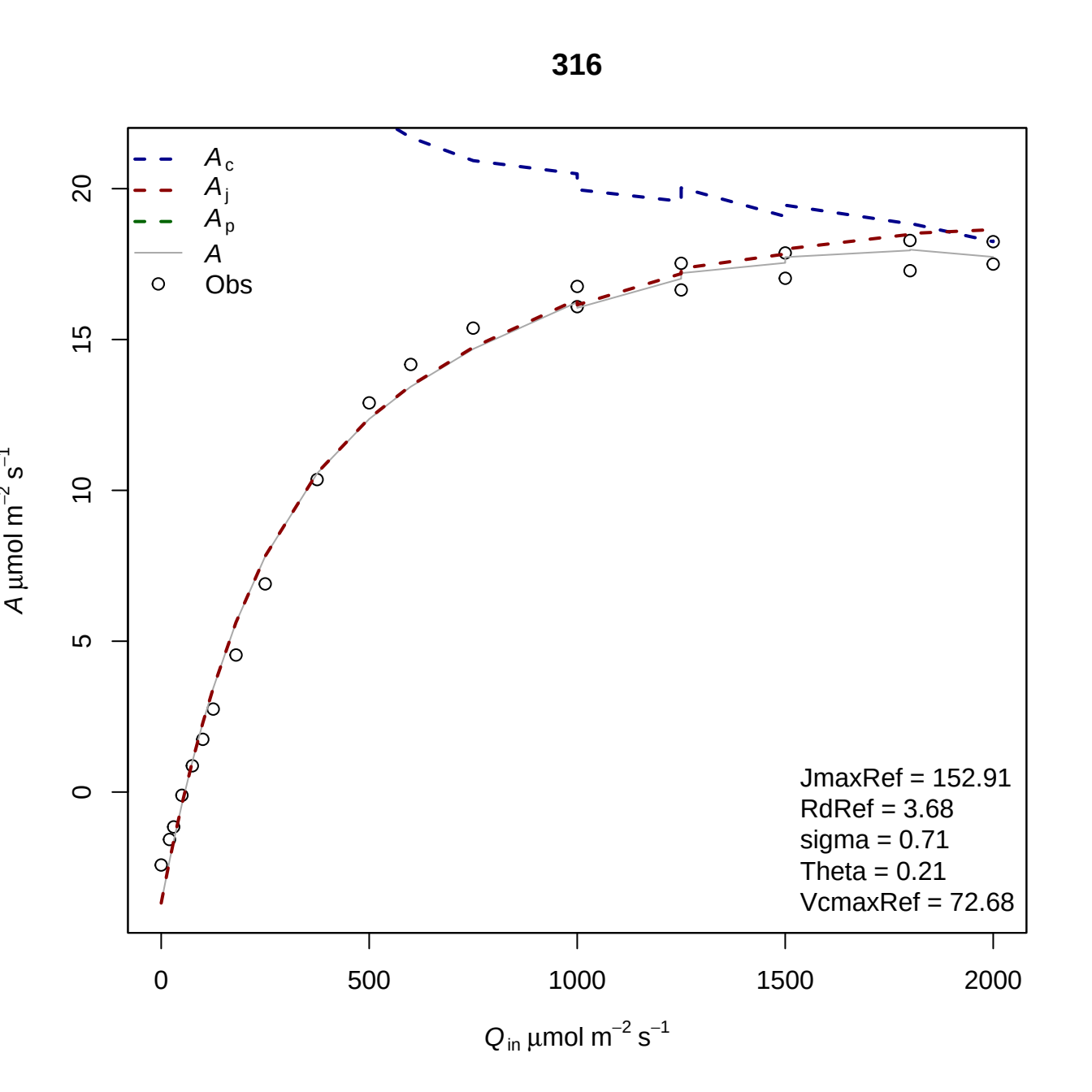

311
A_c
A_j
A_p
A
Obs
JmaxRef = 165.24
RdRef = 3.08
sigma = 0.53
Theta = −1.2
VcmaxRef = 26128.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

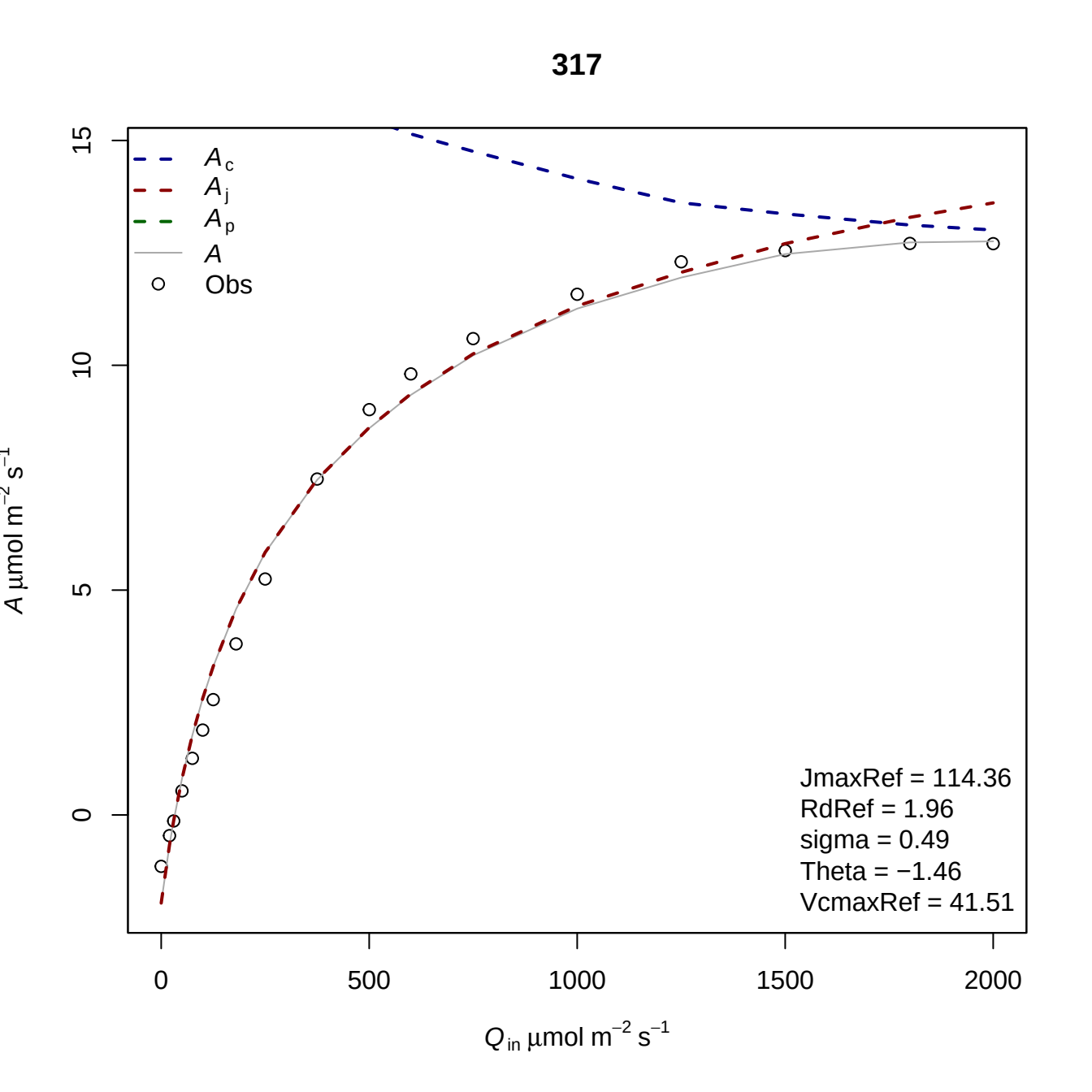

312
A_c
A_j
A_p
A
Obs
JmaxRef = 161.39
RdRef = 3.06
sigma = 0.59
Theta = −0.27
VcmaxRef = 8819.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

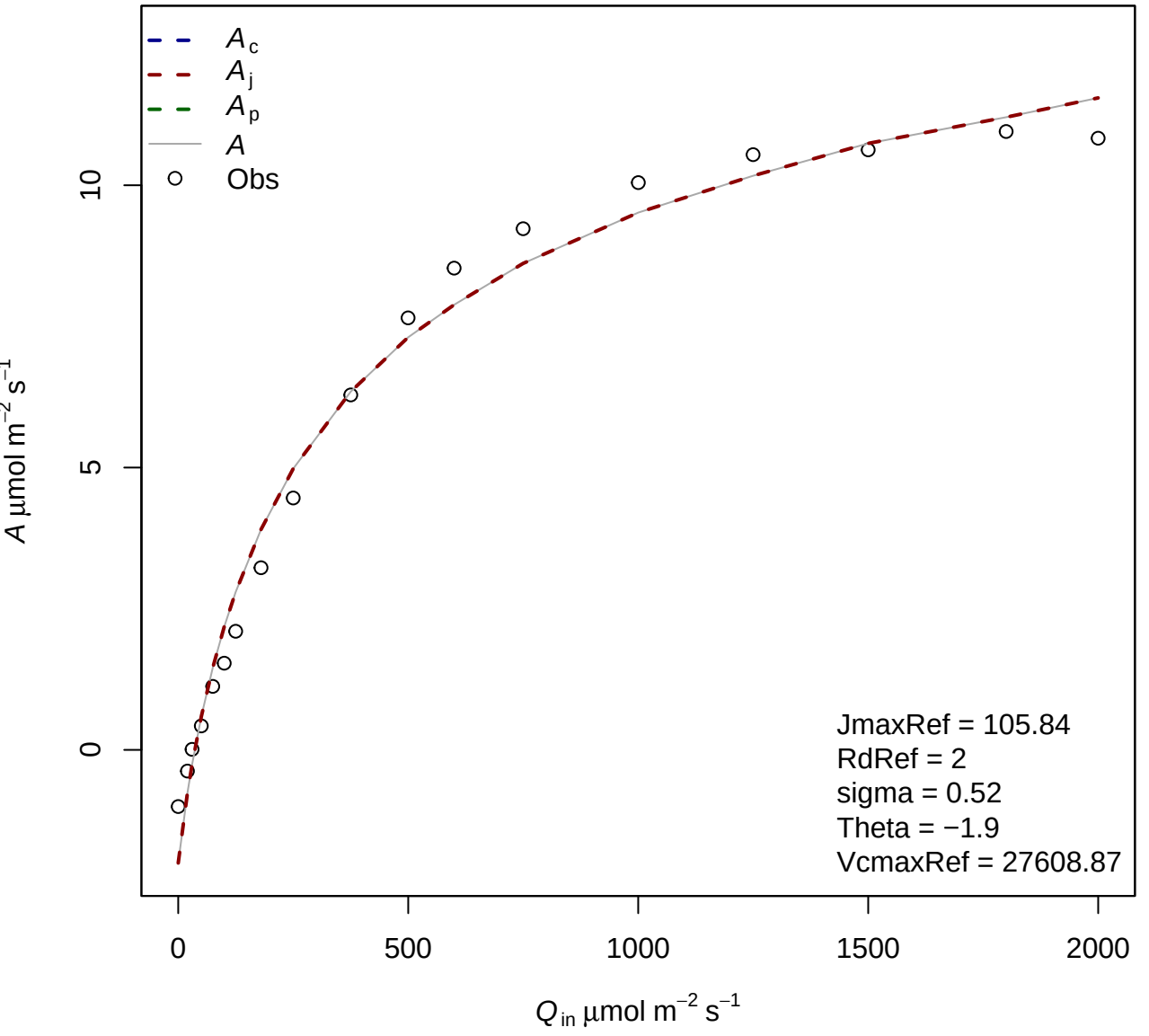

313
A_c
A_j
A_p
A
Obs
JmaxRef = 125.06
RdRef = 2.46
sigma = 0.28
Theta = −0.28
VcmaxRef = 54.74
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

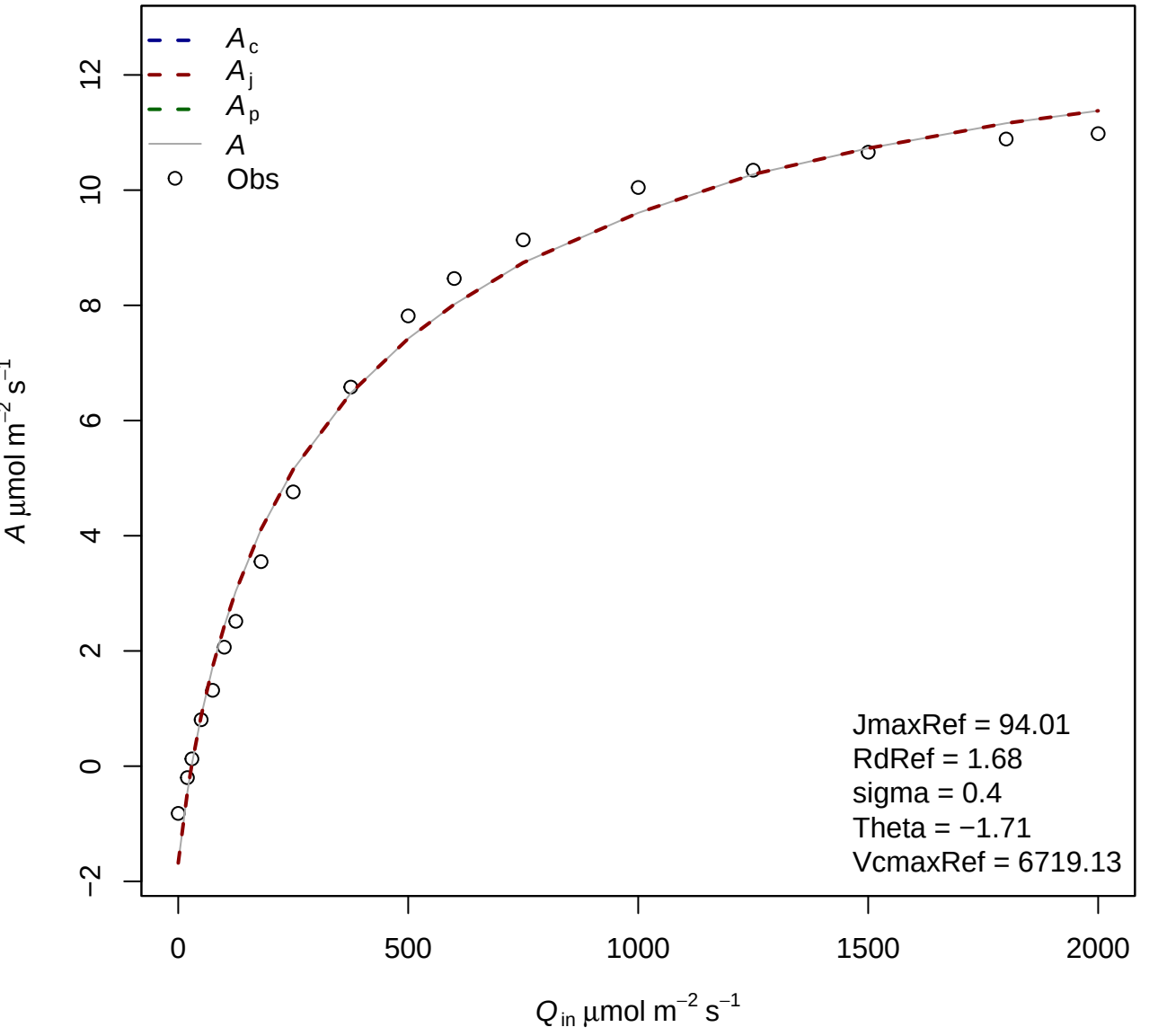

314
A_c
A_j
A_p
A
Obs
JmaxRef = 150.81
RdRef = 2.52
sigma = 0.35
Theta = −0.77
VcmaxRef = 89188.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

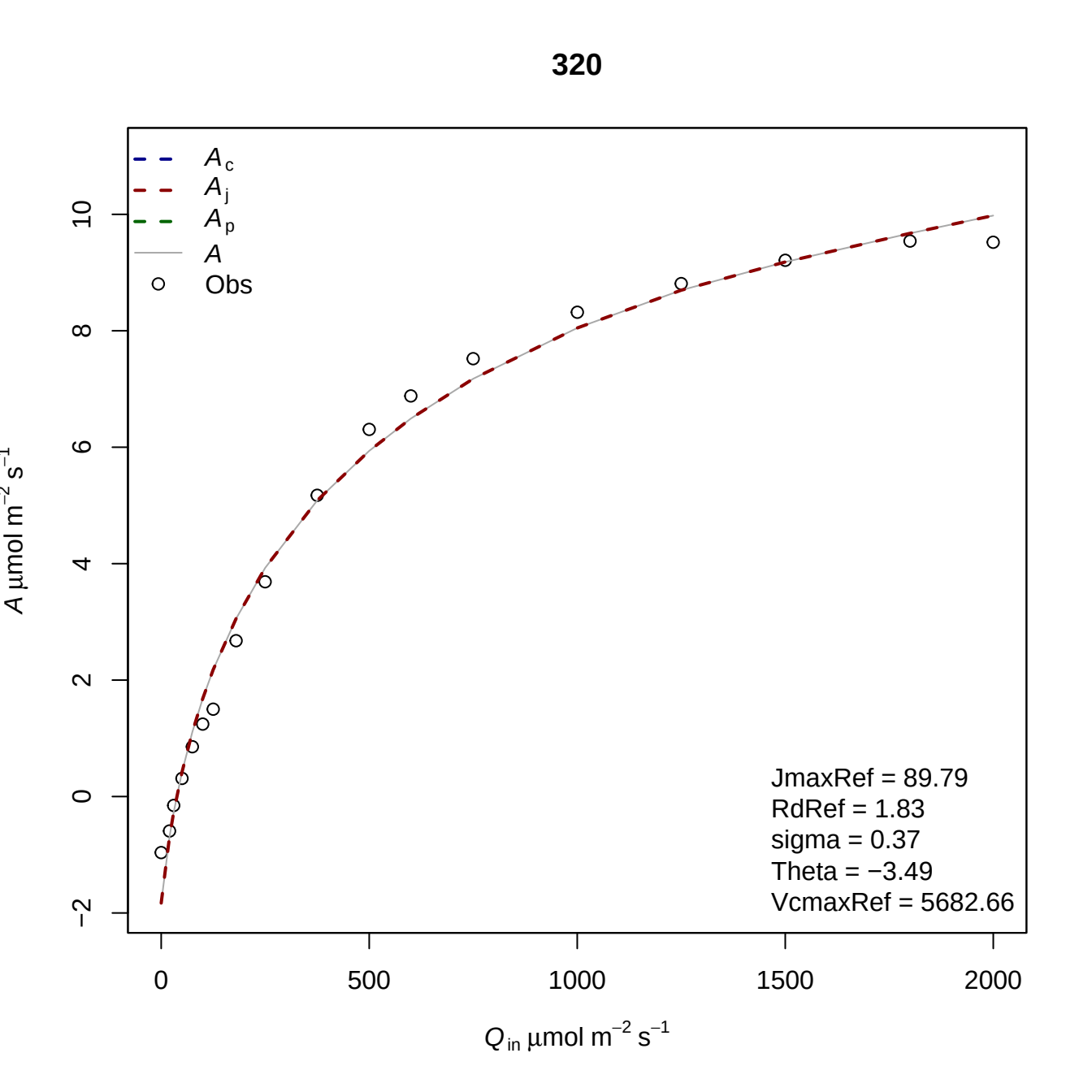

315
A_c
A_j
A_p
A
Obs
JmaxRef = 180.52
RdRef = 5.48
sigma = 0.24
Theta = 0.6
VcmaxRef = 17650545.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

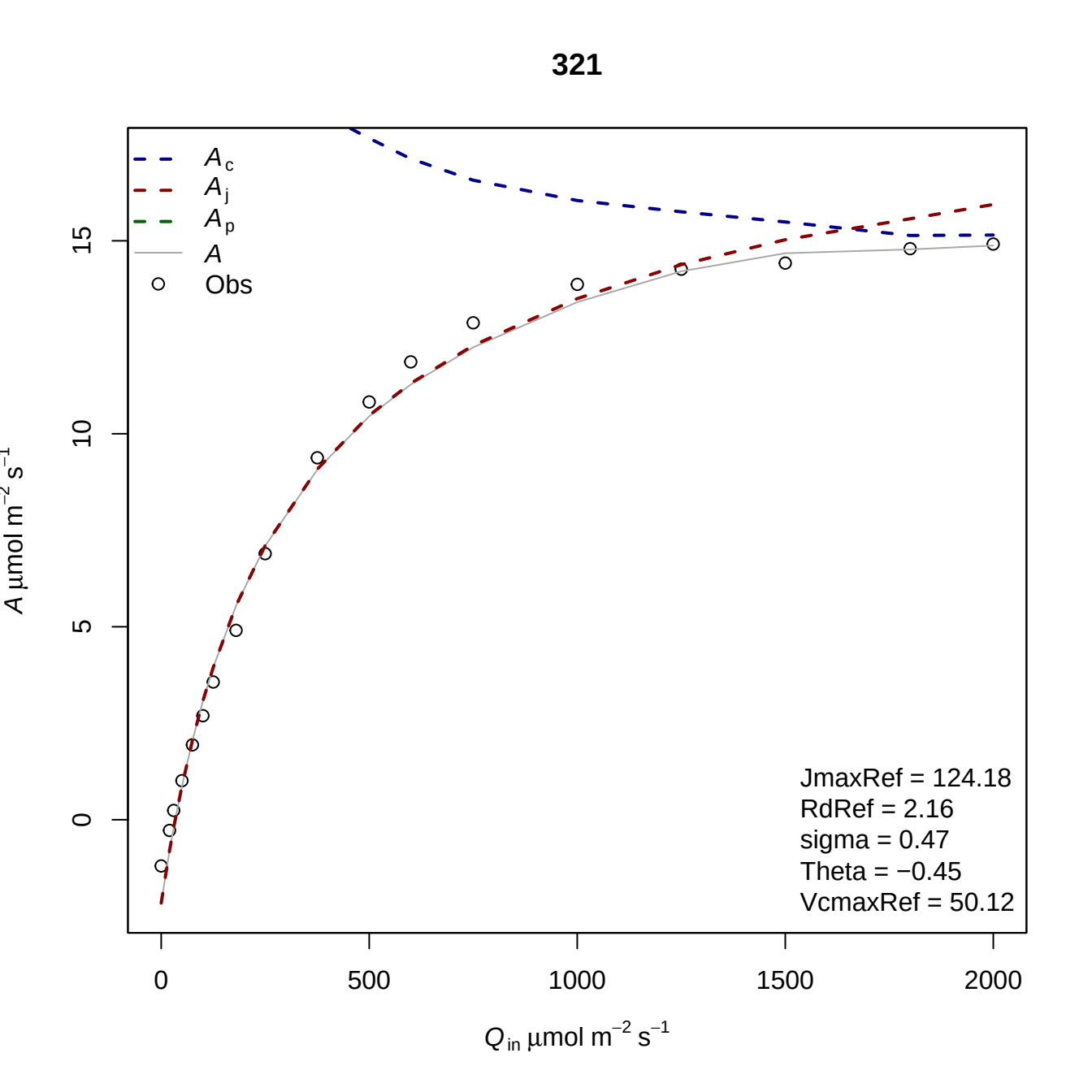

316
A_c
A_j
A_p
A
Obs
JmaxRef = 152.91
RdRef = 3.68
sigma = 0.71
Theta = 0.21
VcmaxRef = 72.68
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

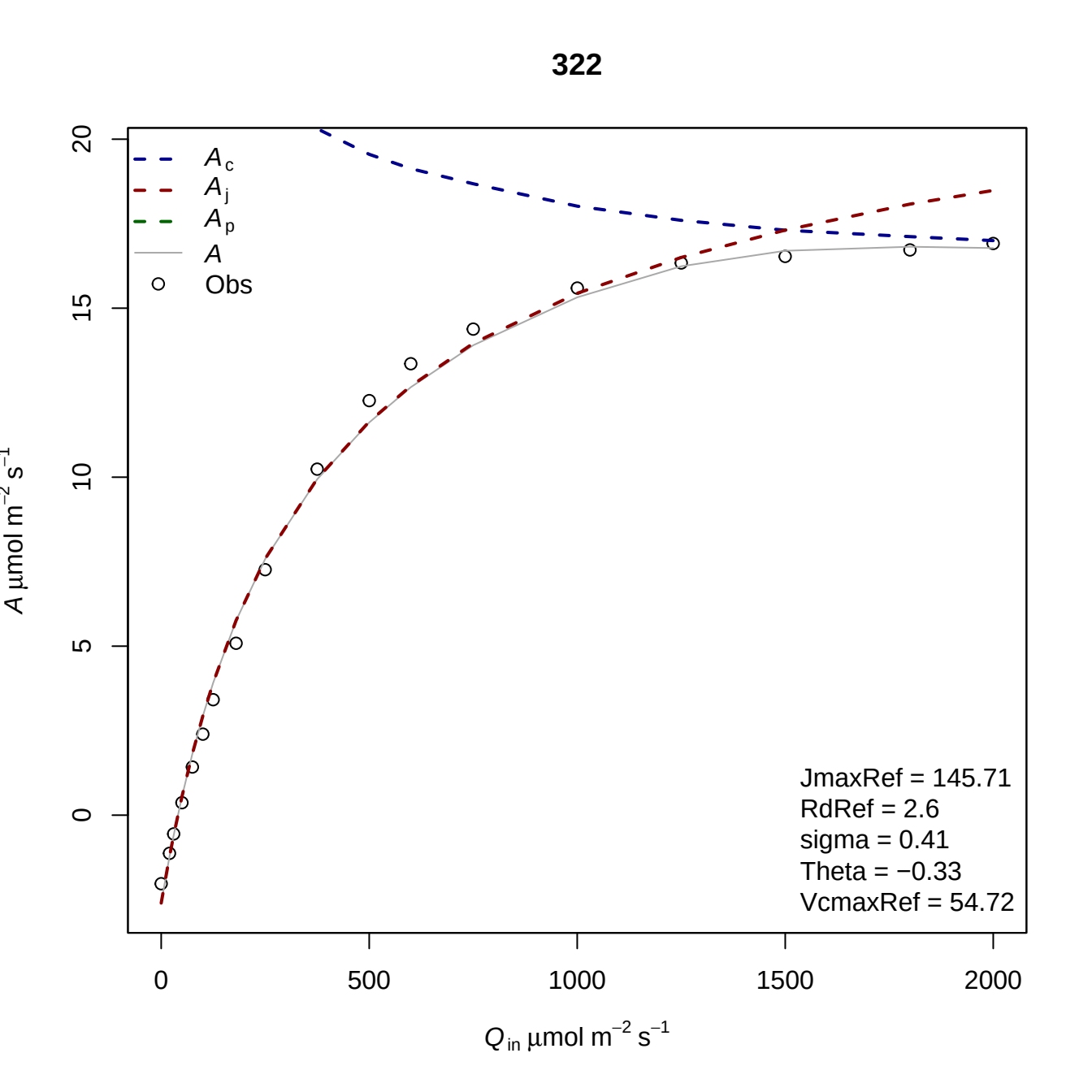

317
A_c
A_j
A_p
A
Obs
JmaxRef = 114.36
RdRef = 1.96
sigma = 0.49
Theta = −1.46
VcmaxRef = 41.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

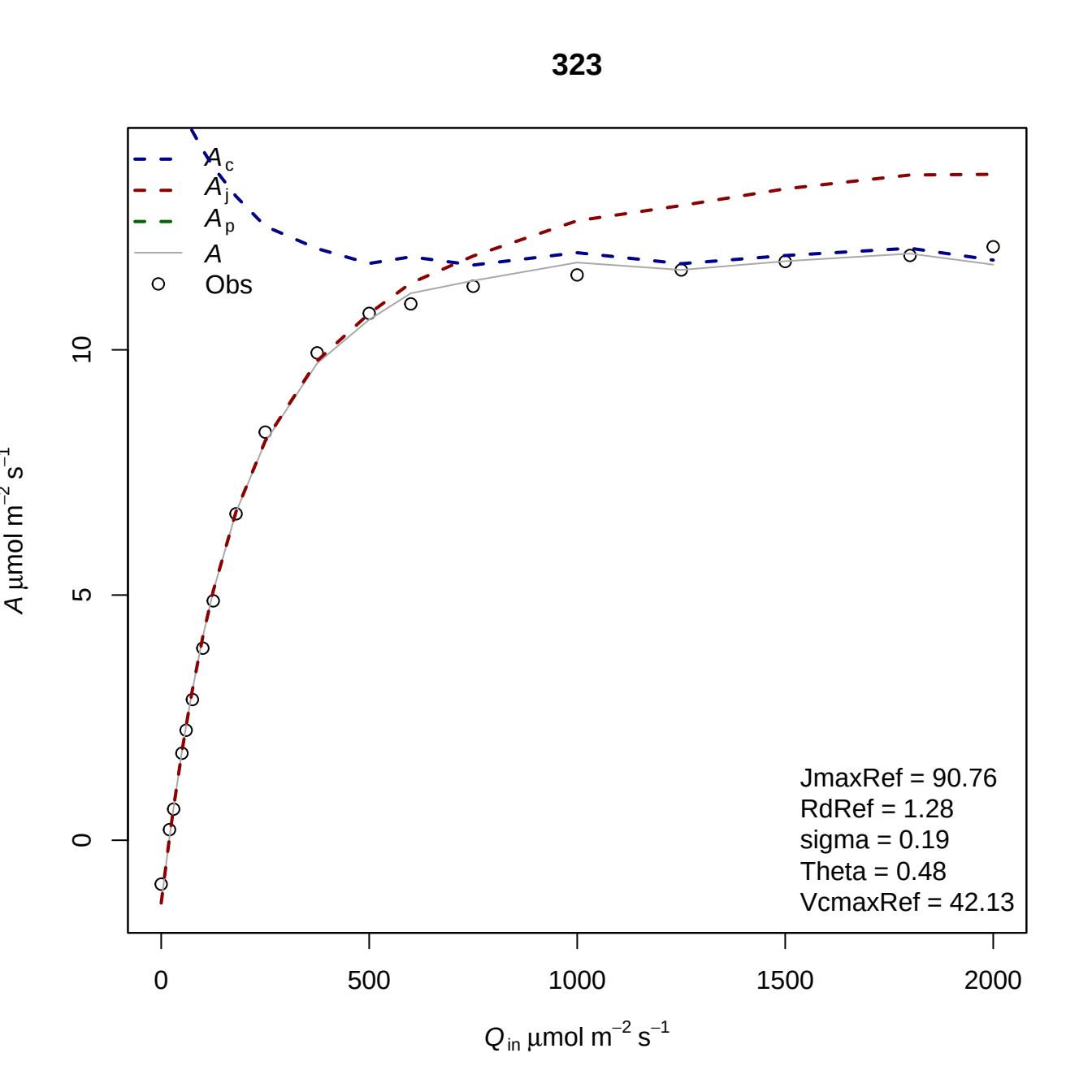

A_c
A_j
A_p
A
Obs
JmaxRef = 105.84
RdRef = 2
sigma = 0.52
Theta = −1.9
VcmaxRef = 27608.87
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

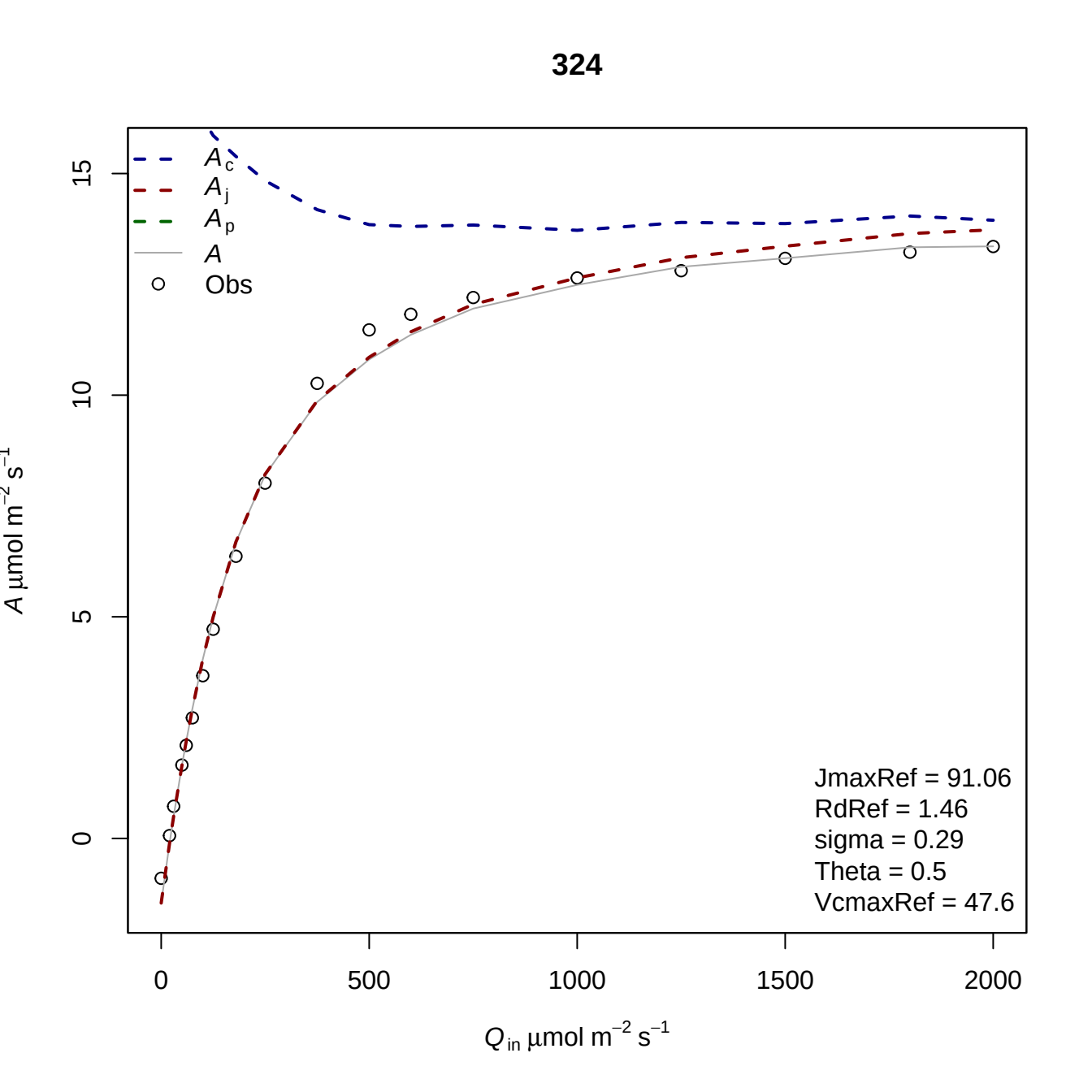

A_c
A_j
A_p
A
Obs
JmaxRef = 94.01
RdRef = 1.68
sigma = 0.4
Theta = −1.71
VcmaxRef = 6719.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

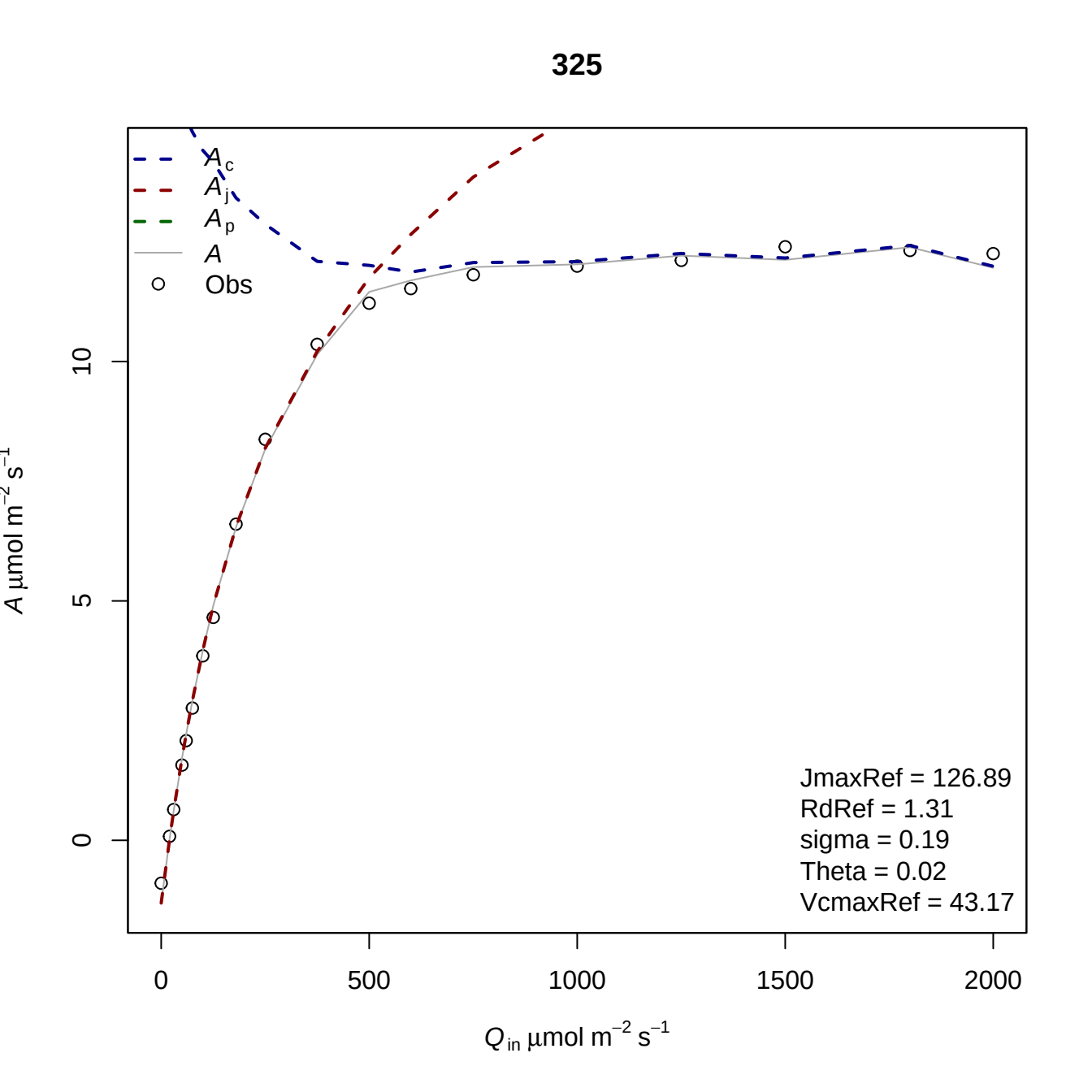

320
A_c
A_j
A_p
A
Obs
JmaxRef = 89.79
RdRef = 1.83
sigma = 0.37
Theta = −3.49
VcmaxRef = 5682.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

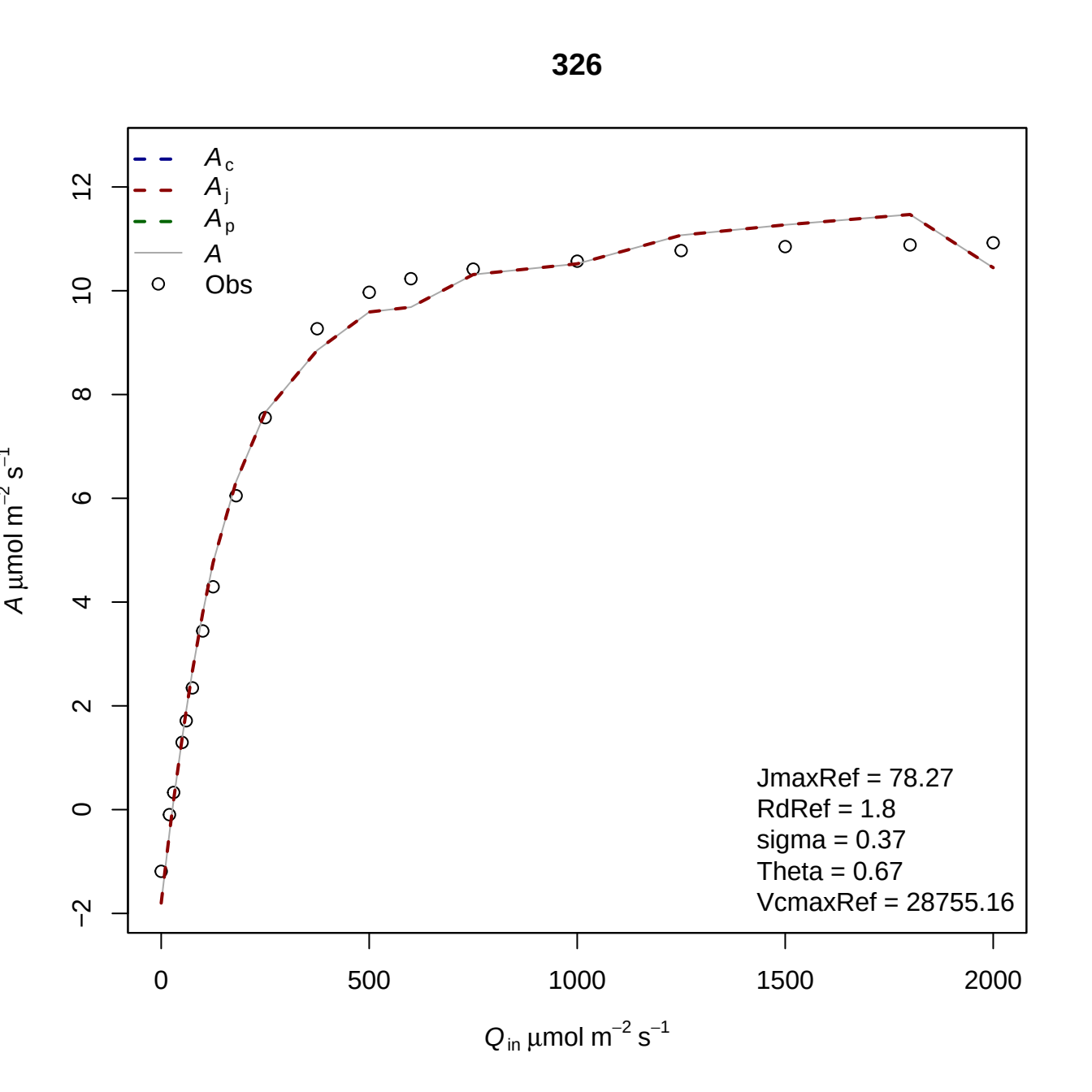

321
A_c
A_j
A_p
A
Obs
JmaxRef = 124.18
RdRef = 2.16
sigma = 0.47
Theta = −0.45
VcmaxRef = 50.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

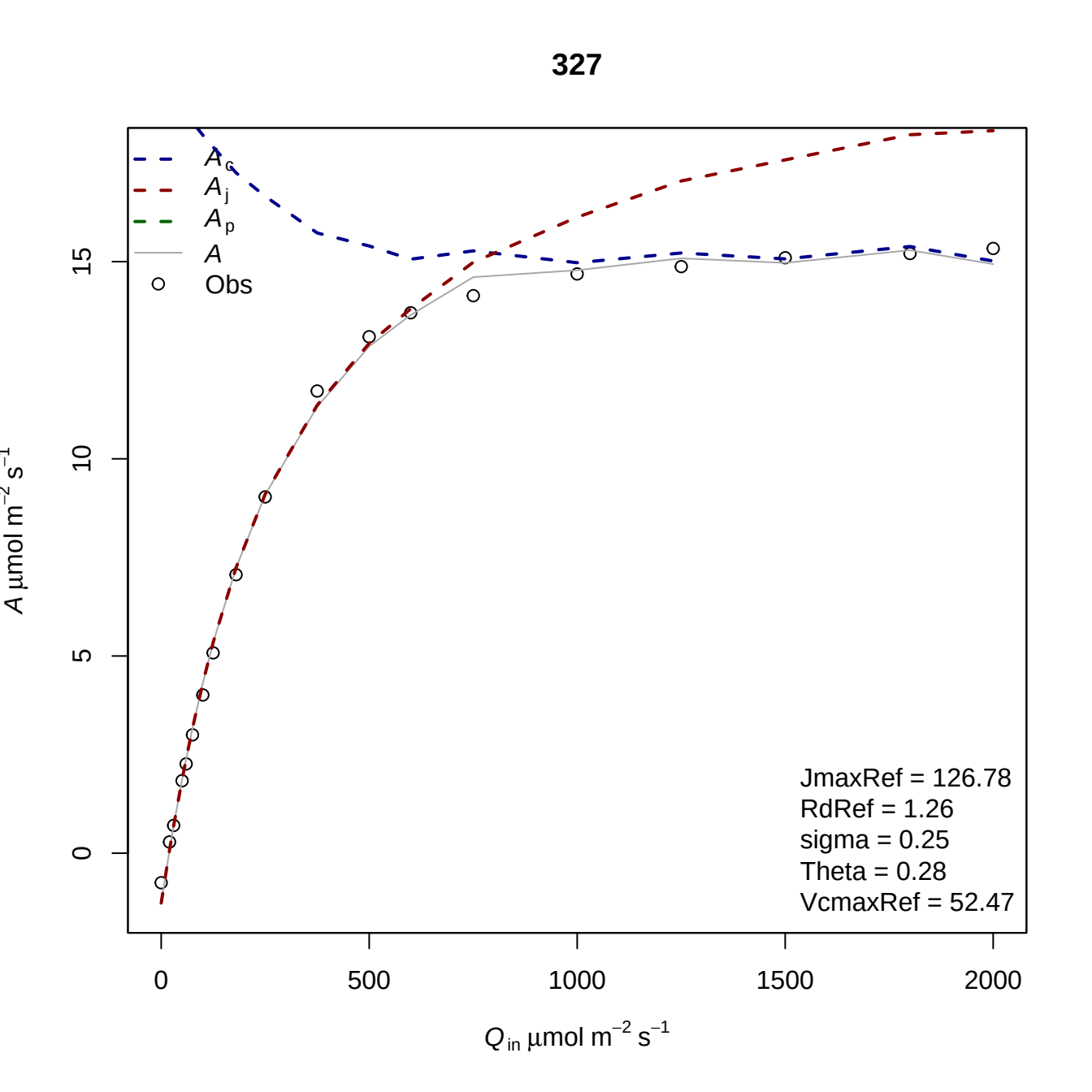

322
A_c
A_j
A_p
A
Obs
JmaxRef = 145.71
RdRef = 2.6
sigma = 0.41
Theta = −0.33
VcmaxRef = 54.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

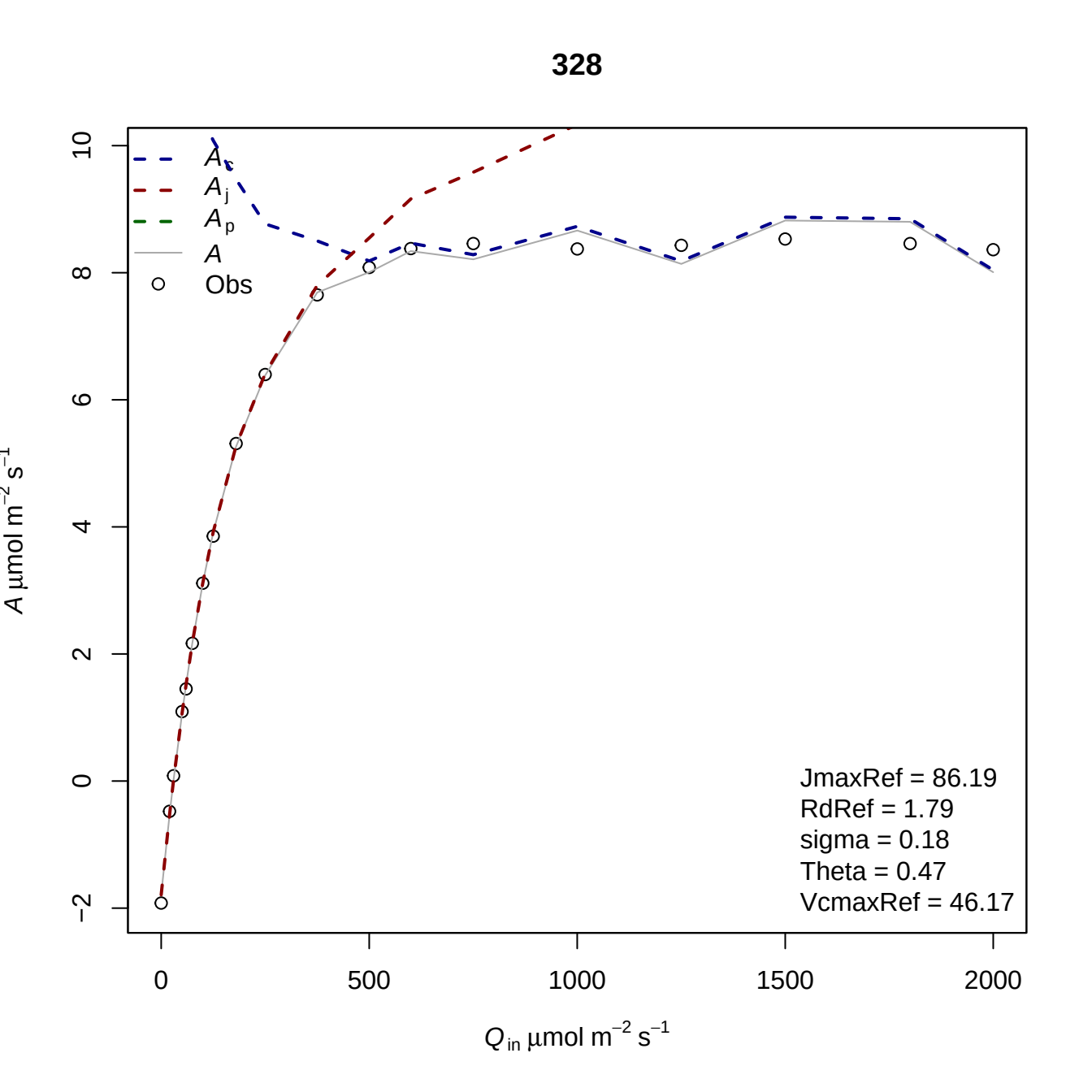

323
A_c
A_j
A_p
A
Obs
JmaxRef = 90.76
RdRef = 1.28
sigma = 0.19
Theta = 0.48
VcmaxRef = 42.13
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

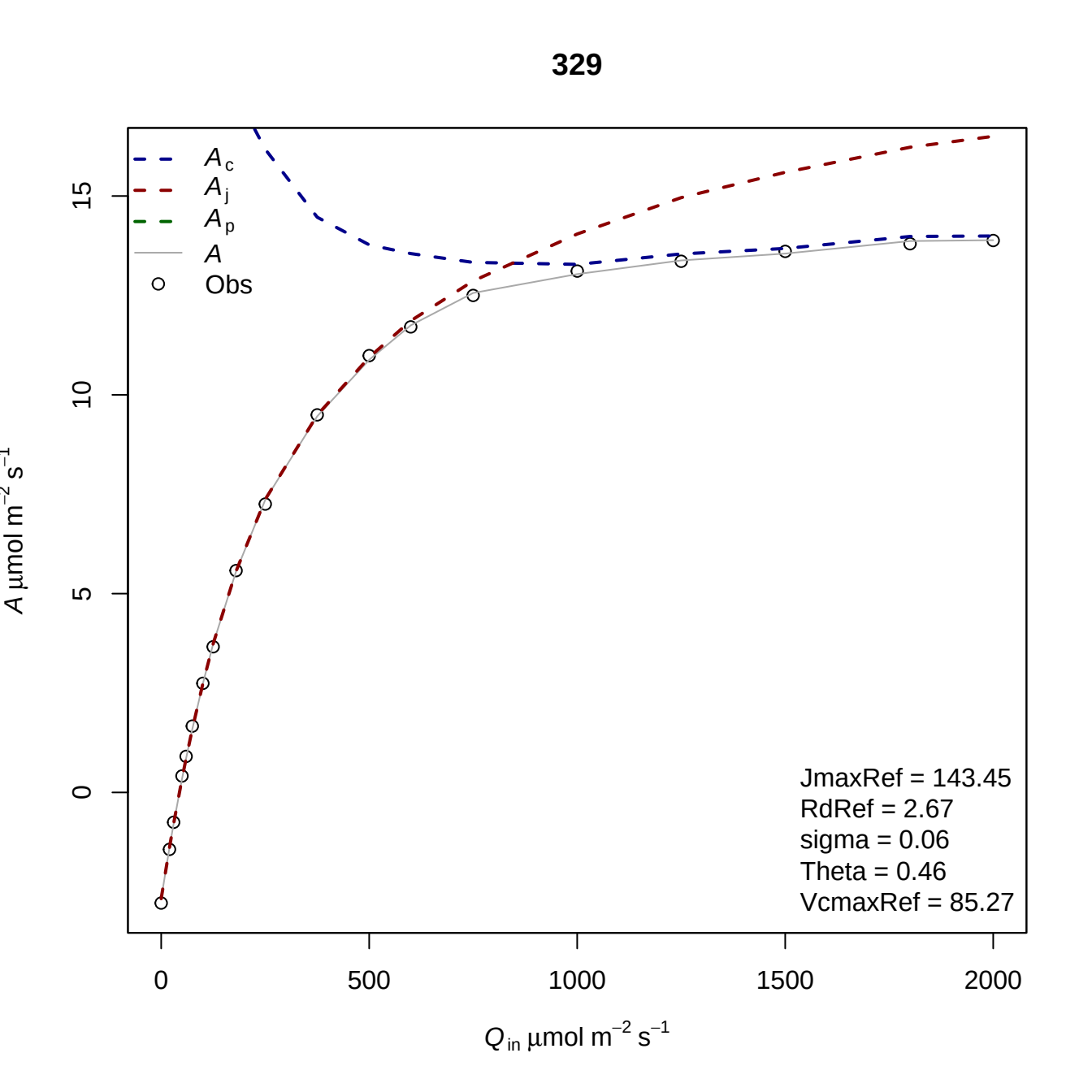

324
A_c
A_j
A_p
A
Obs
JmaxRef = 91.06
RdRef = 1.46
sigma = 0.29
Theta = 0.5
VcmaxRef = 47.6
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

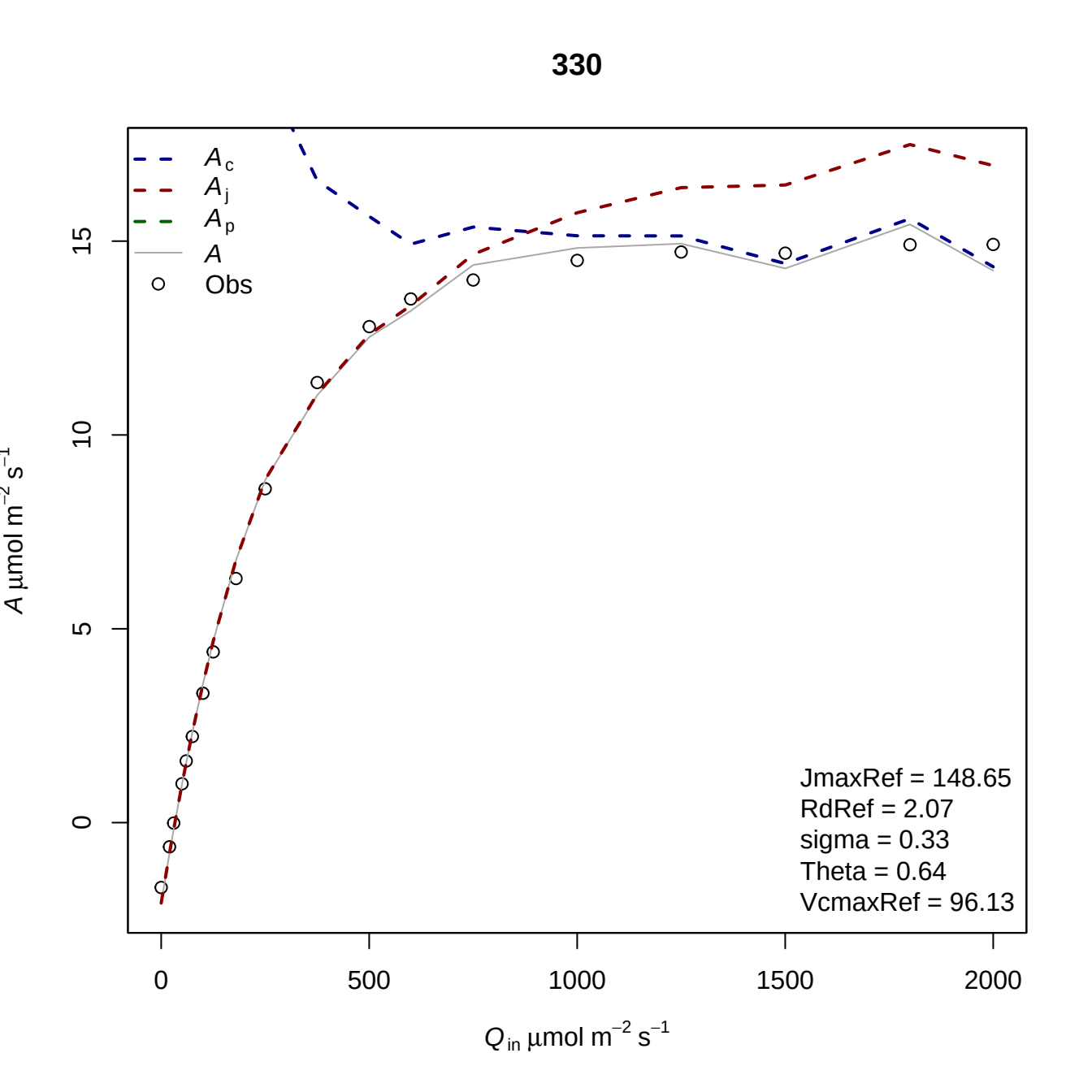

325
A_c
A_j
A_p
A
Obs
JmaxRef = 126.89
RdRef = 1.31
sigma = 0.19
Theta = 0.02
VcmaxRef = 43.17
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

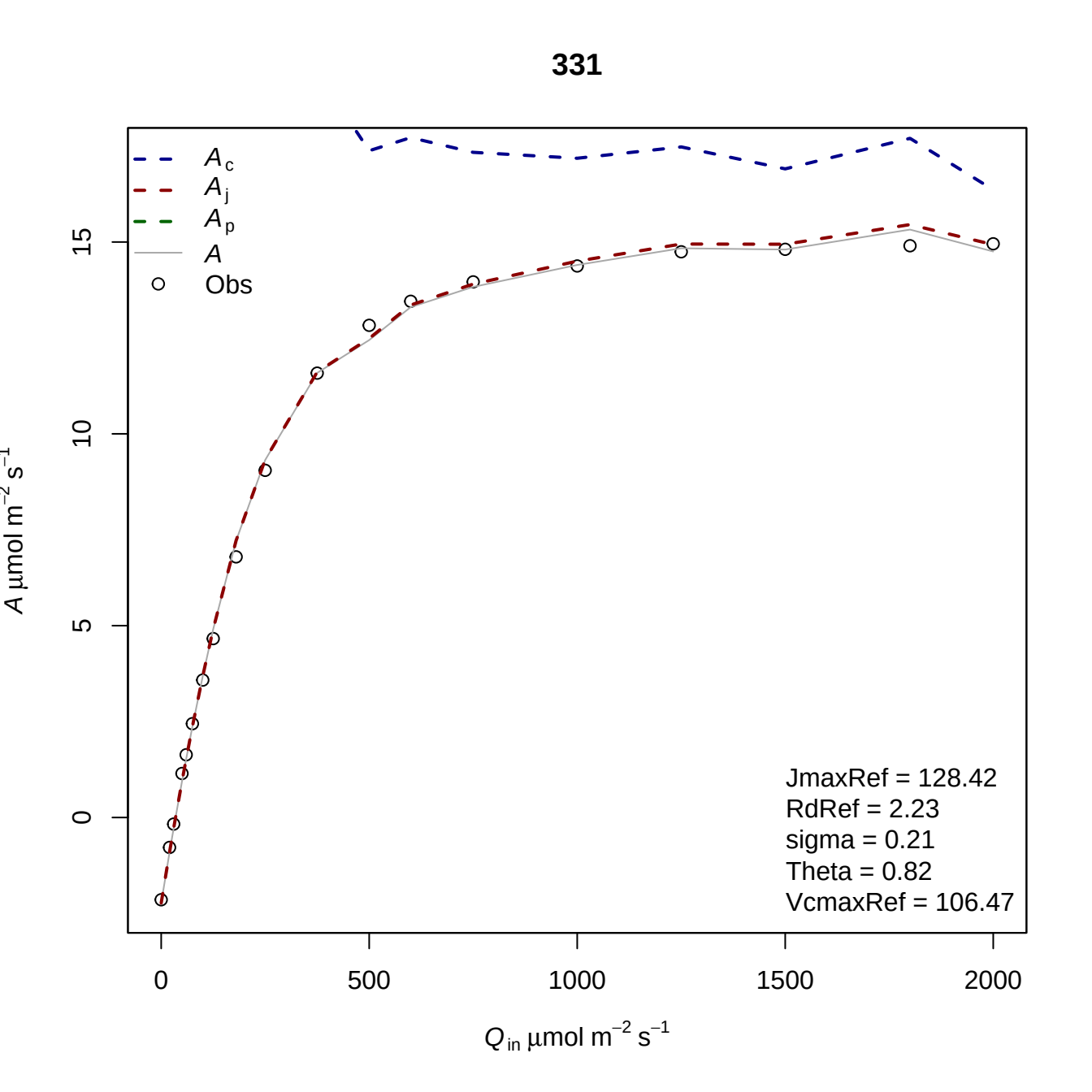

326
A_c
A_j
A_p
A
Obs
JmaxRef = 78.27
RdRef = 1.8
sigma = 0.37
Theta = 0.67
VcmaxRef = 28755.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

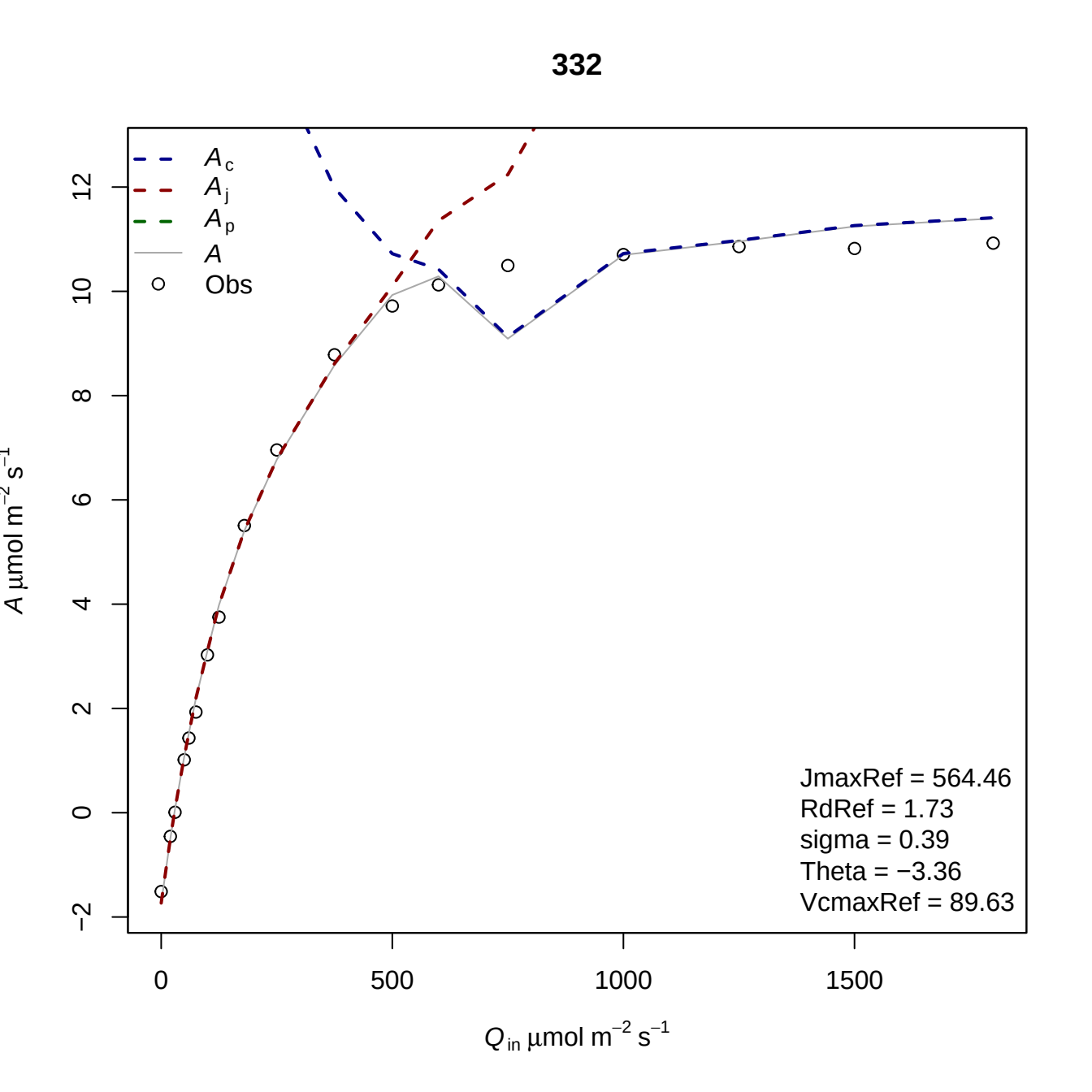

327
A_c
A_j
A_p
A
Obs
JmaxRef = 126.78
RdRef = 1.26
sigma = 0.25
Theta = 0.28
VcmaxRef = 52.47
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

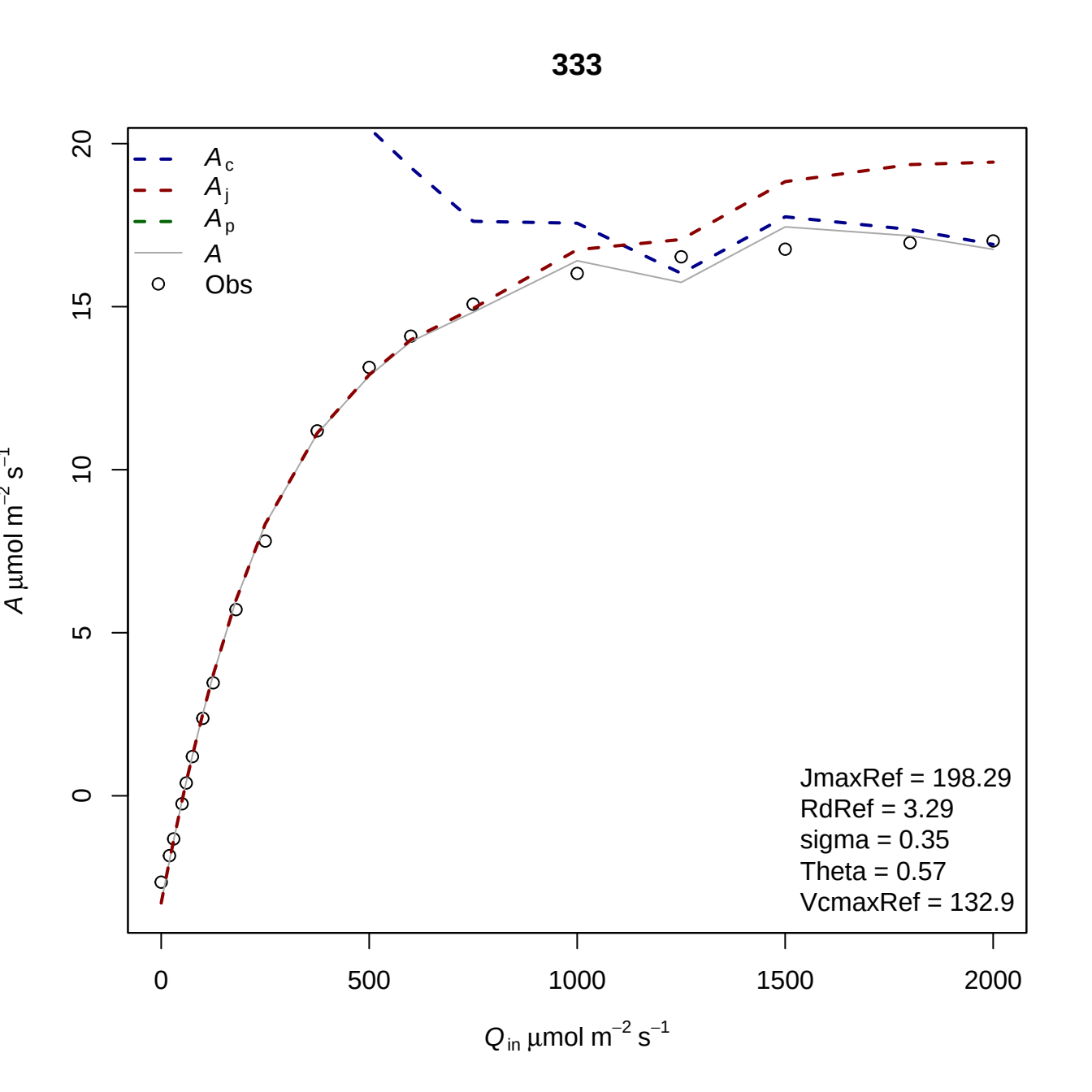

328
A_c
A_j
A_p
A
Obs
JmaxRef = 86.19
RdRef = 1.79
sigma = 0.18
Theta = 0.47
VcmaxRef = 46.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

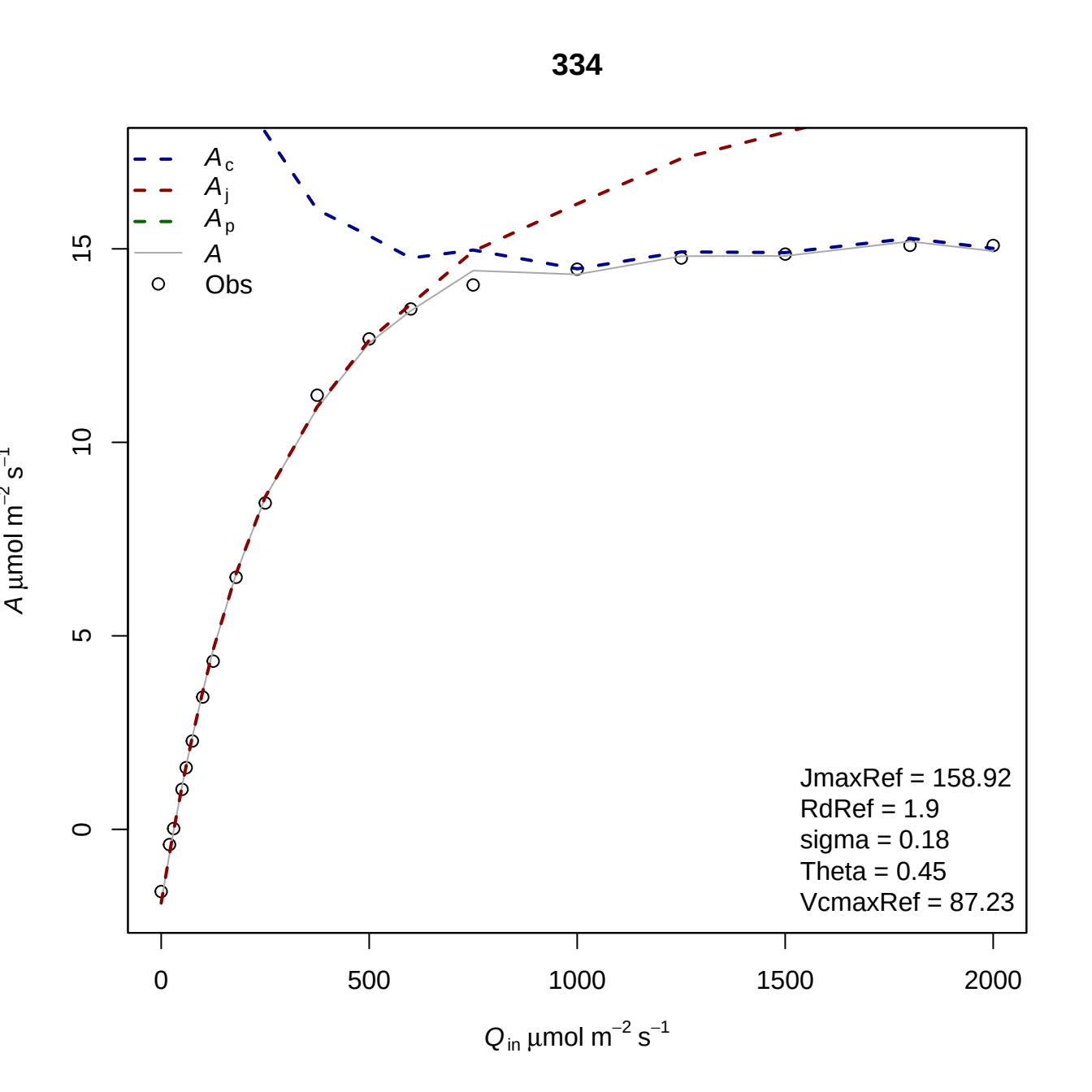

329
A_c
A_j
A_p
A
Obs
JmaxRef = 143.45
RdRef = 2.67
sigma = 0.06
Theta = 0.46
VcmaxRef = 85.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

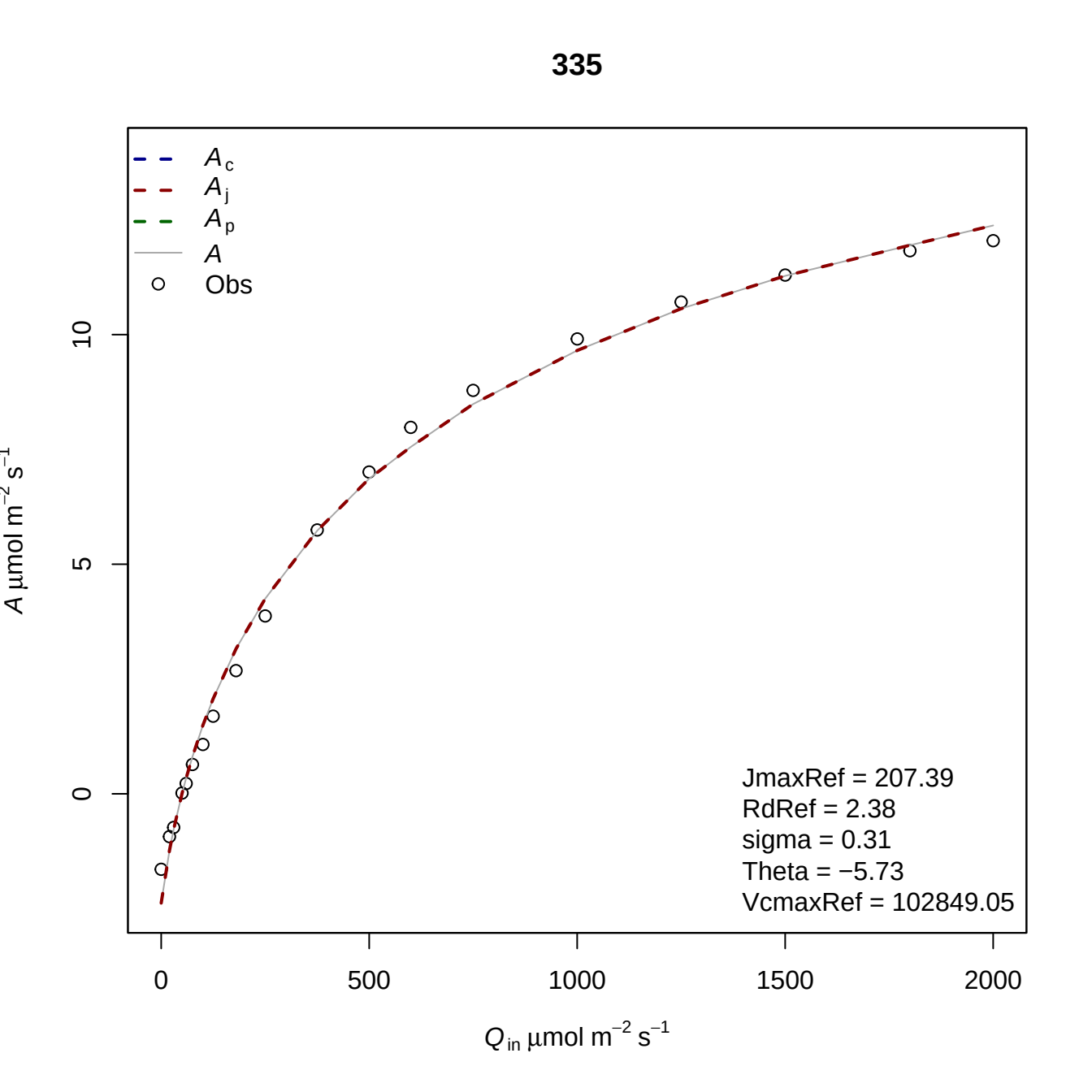

330
A_c
A_j
A_p
A
Obs
JmaxRef = 148.65
RdRef = 2.07
sigma = 0.33
Theta = 0.64
VcmaxRef = 96.13
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

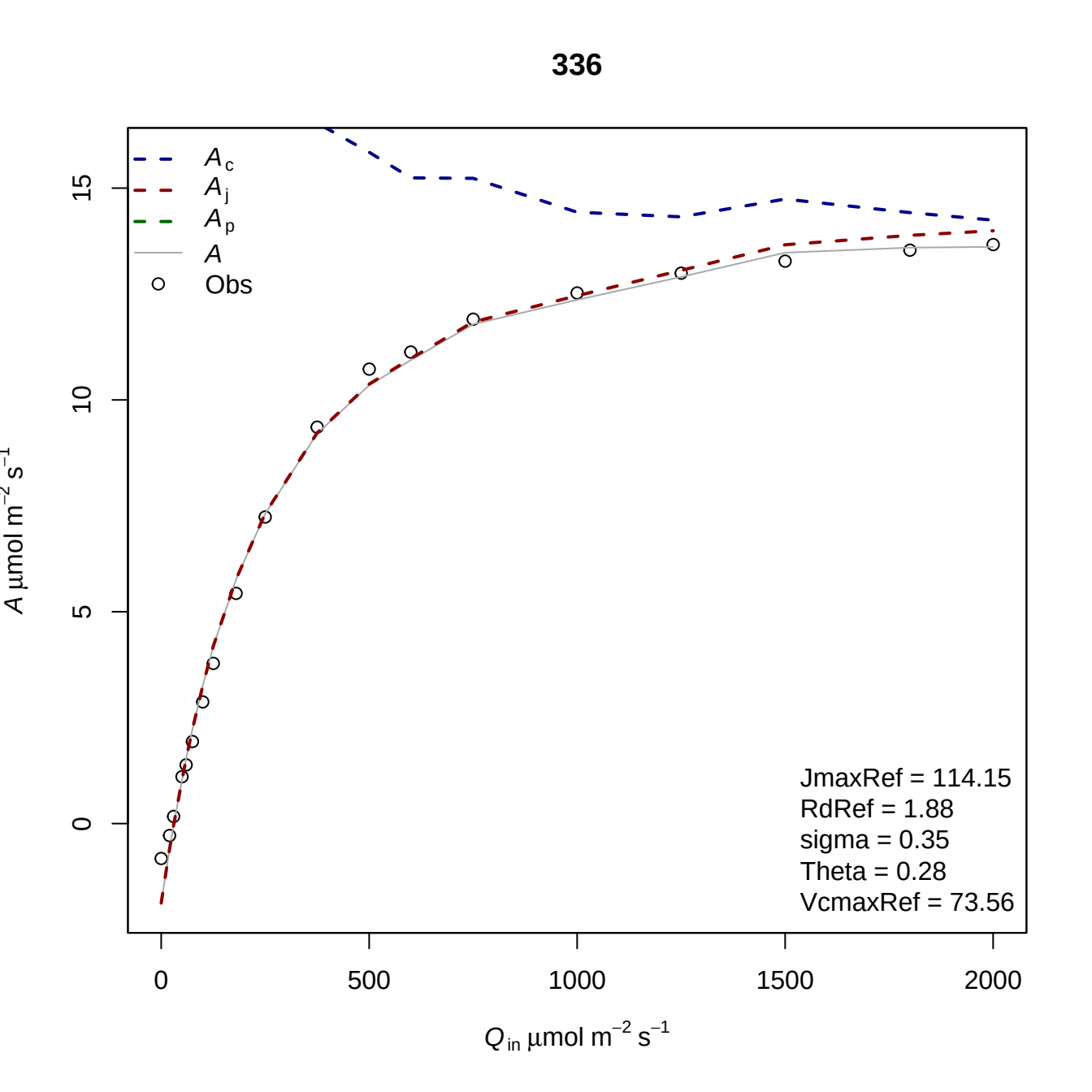

331
A_c
A_j
A_p
A
Obs
JmaxRef = 128.42
RdRef = 2.23
sigma = 0.21
Theta = 0.82
VcmaxRef = 106.47
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

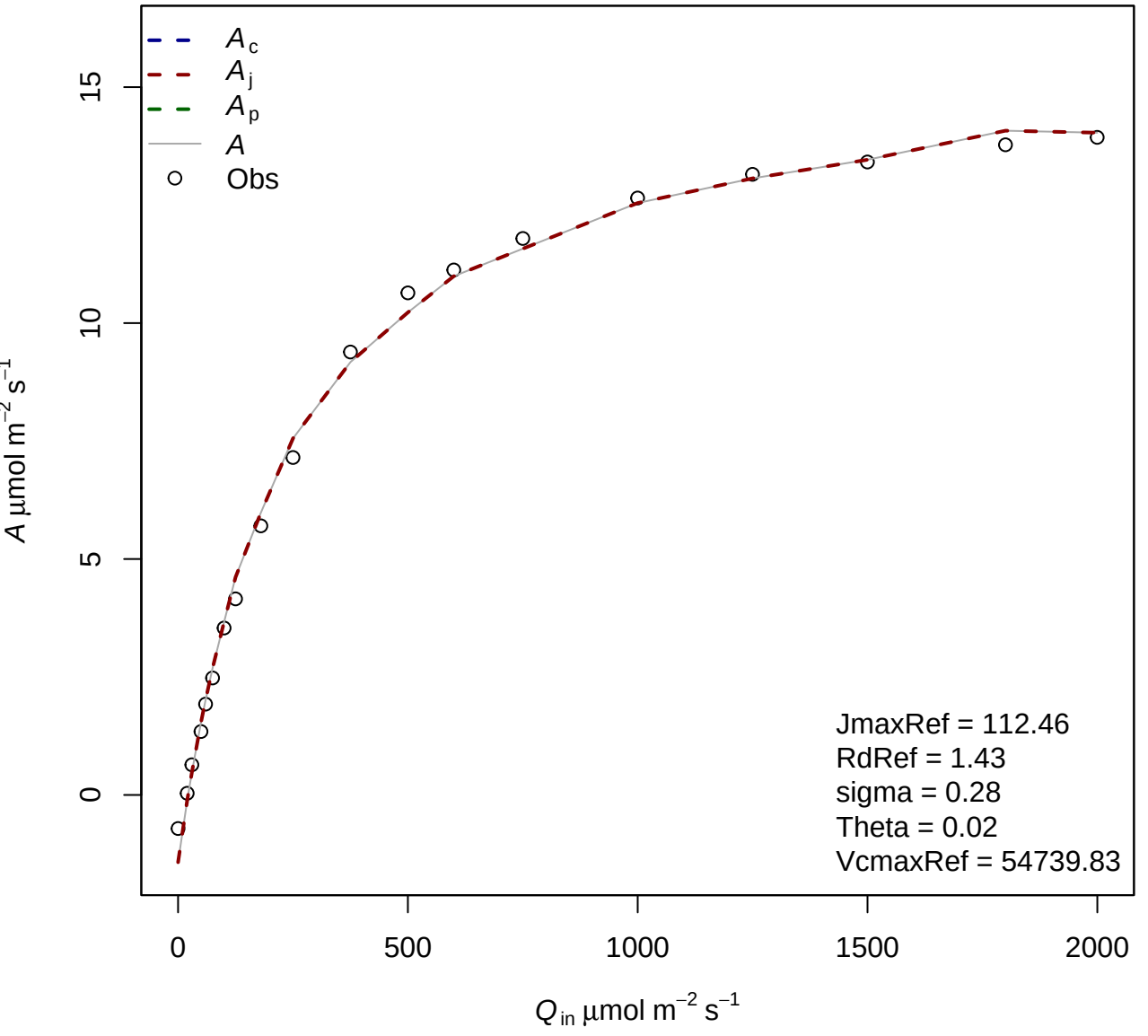

332
A_c
A_j
A_p
A
Obs
JmaxRef = 564.46
RdRef = 1.73
sigma = 0.39
Theta = −3.36
VcmaxRef = 89.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

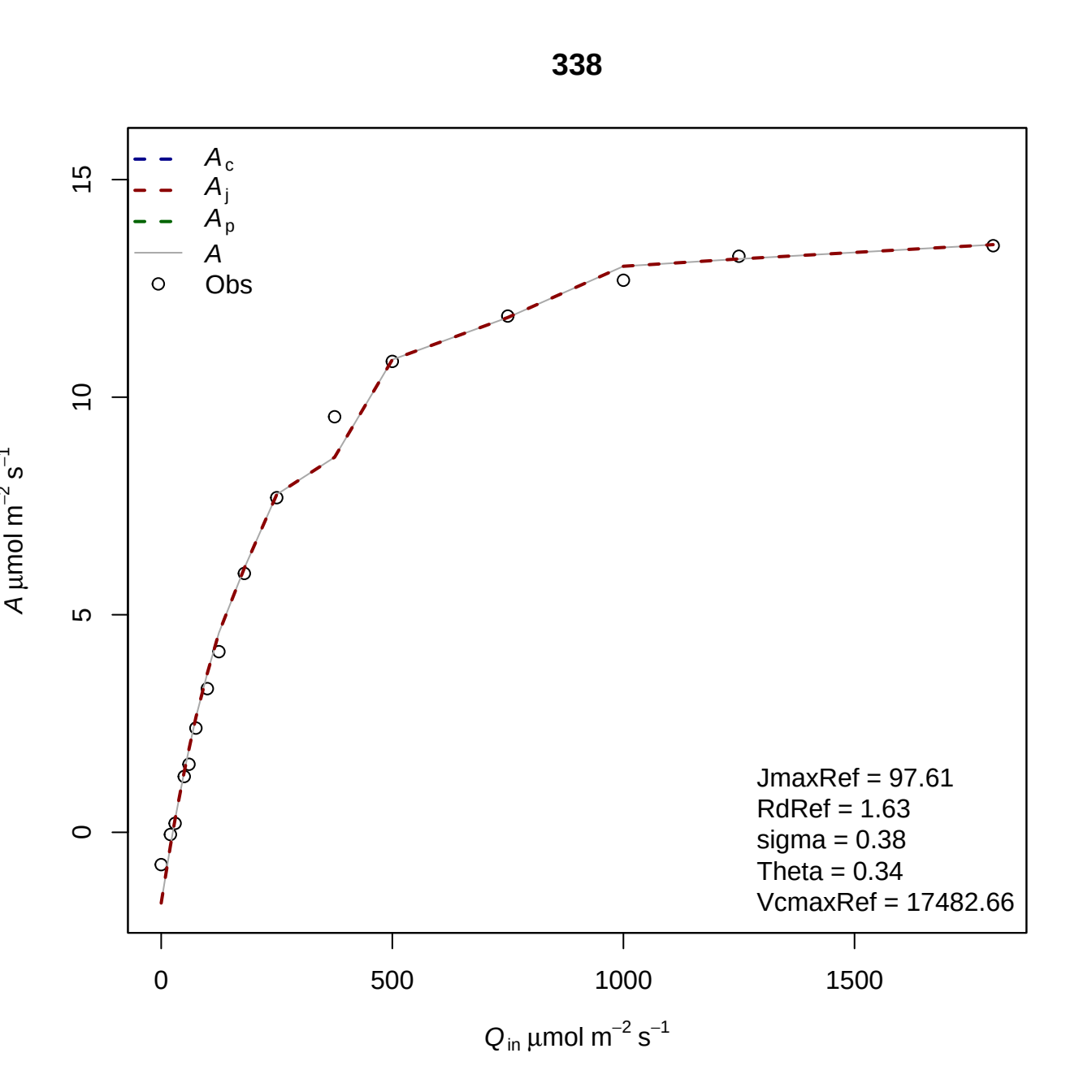

333
A_c
A_j
A_p
A
Obs
JmaxRef = 198.29
RdRef = 3.29
sigma = 0.35
Theta = 0.57
VcmaxRef = 132.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

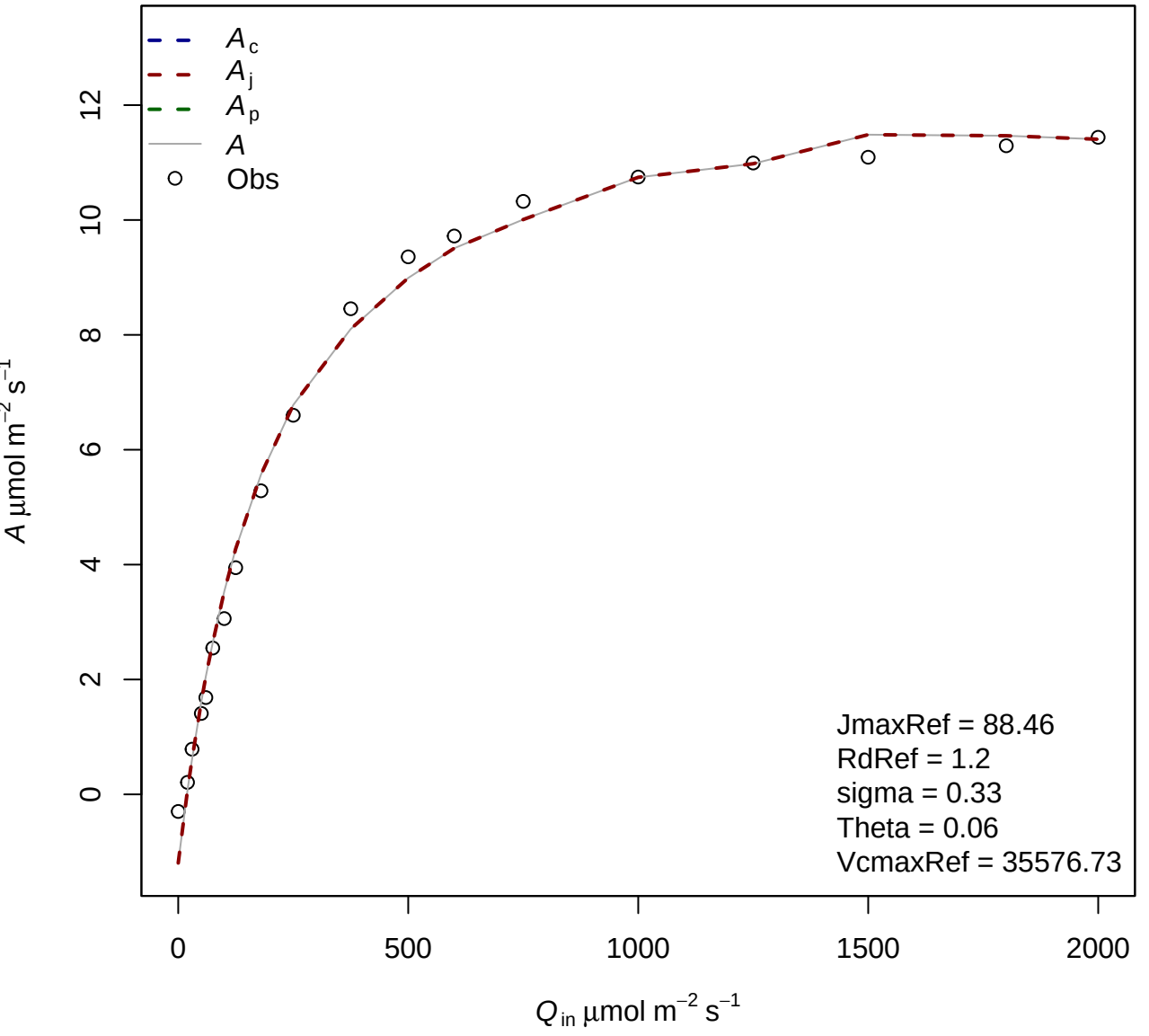

334
A_c
A_j
A_p
A
Obs
JmaxRef = 158.92
RdRef = 1.9
sigma = 0.18
Theta = 0.45
VcmaxRef = 87.23
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

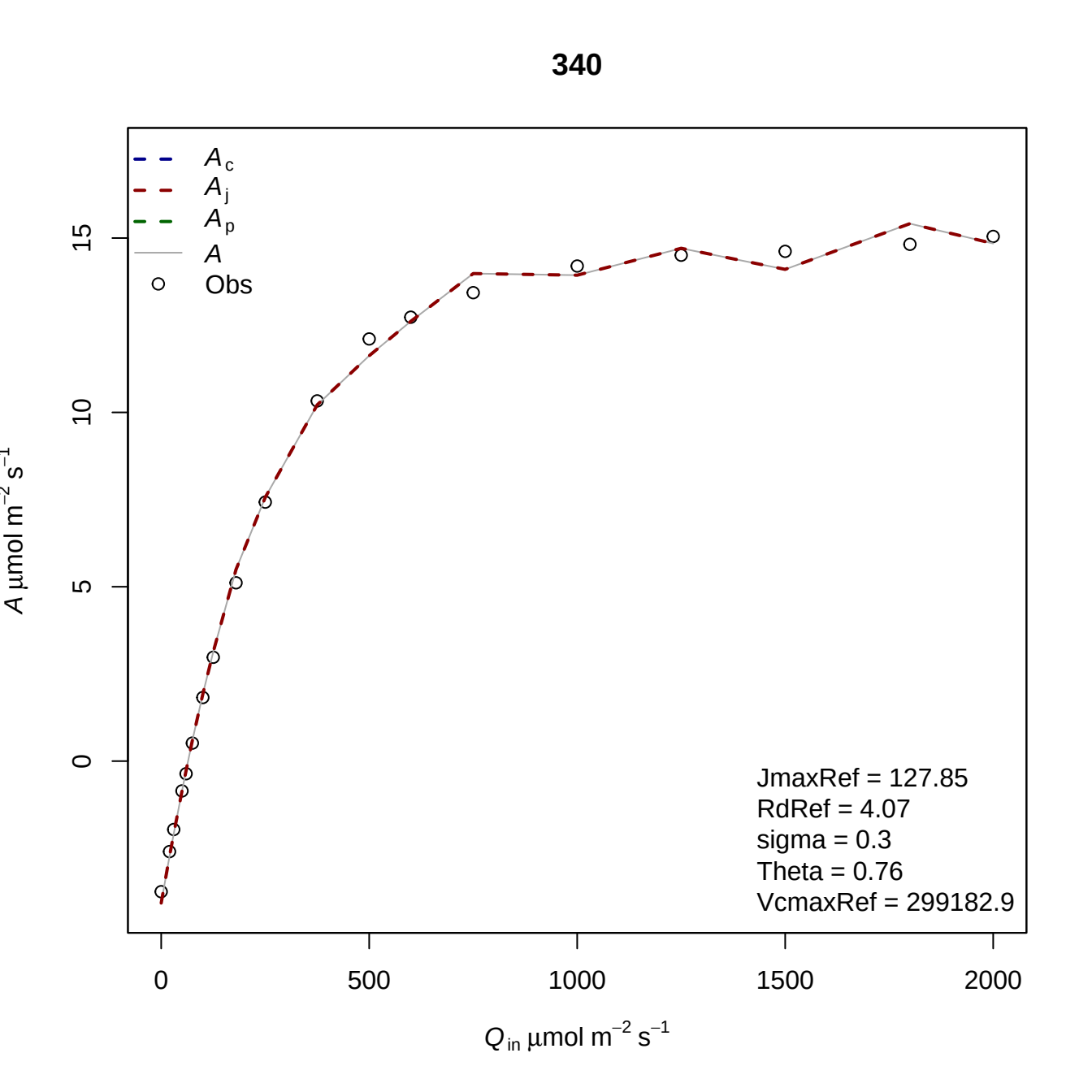

335
A_c
A_j
A_p
A
Obs
JmaxRef = 207.39
RdRef = 2.38
sigma = 0.31
Theta = −5.73
VcmaxRef = 102849.05
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

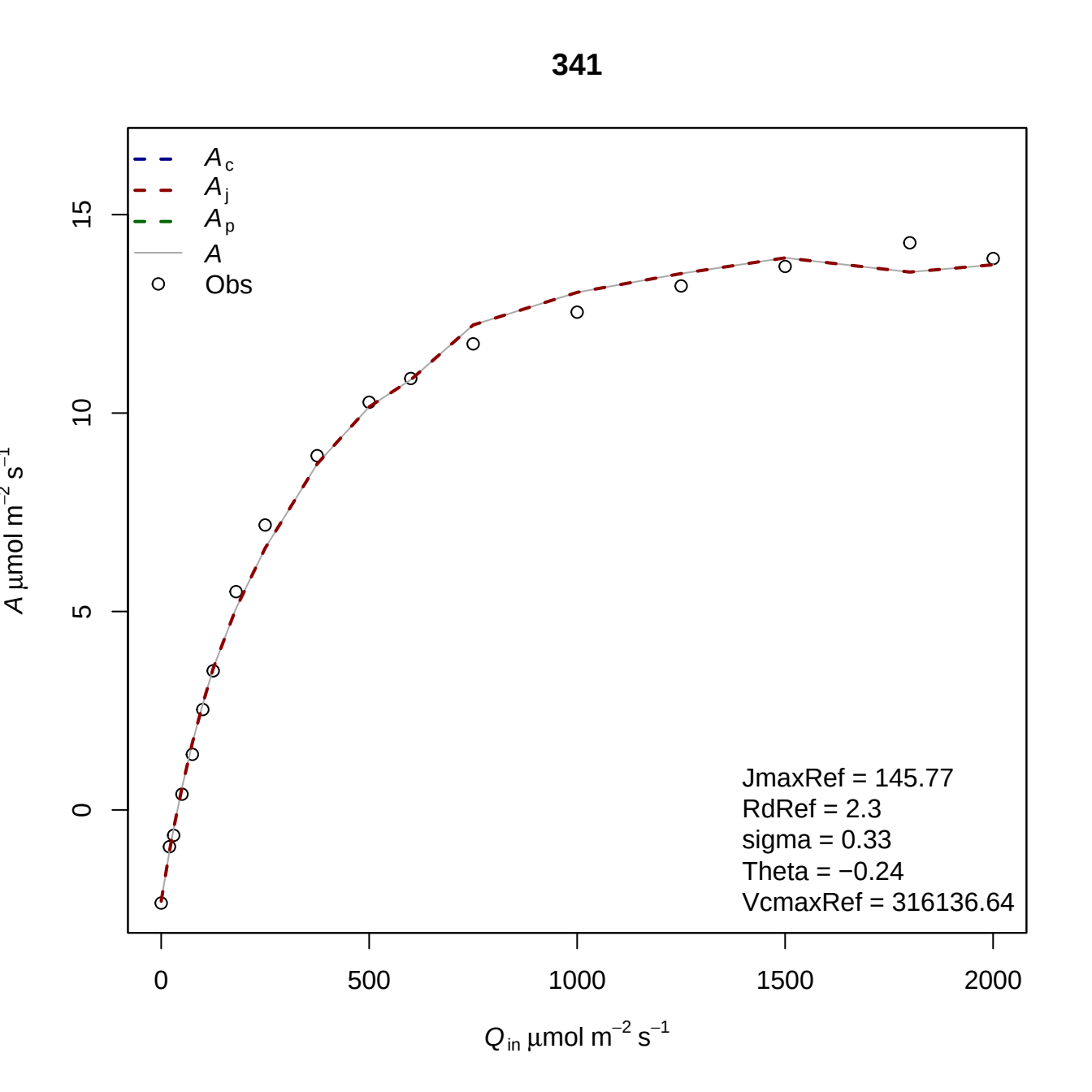

336
A_c
A_j
A_p
A
Obs
JmaxRef = 114.15
RdRef = 1.88
sigma = 0.35
Theta = 0.28
VcmaxRef = 73.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

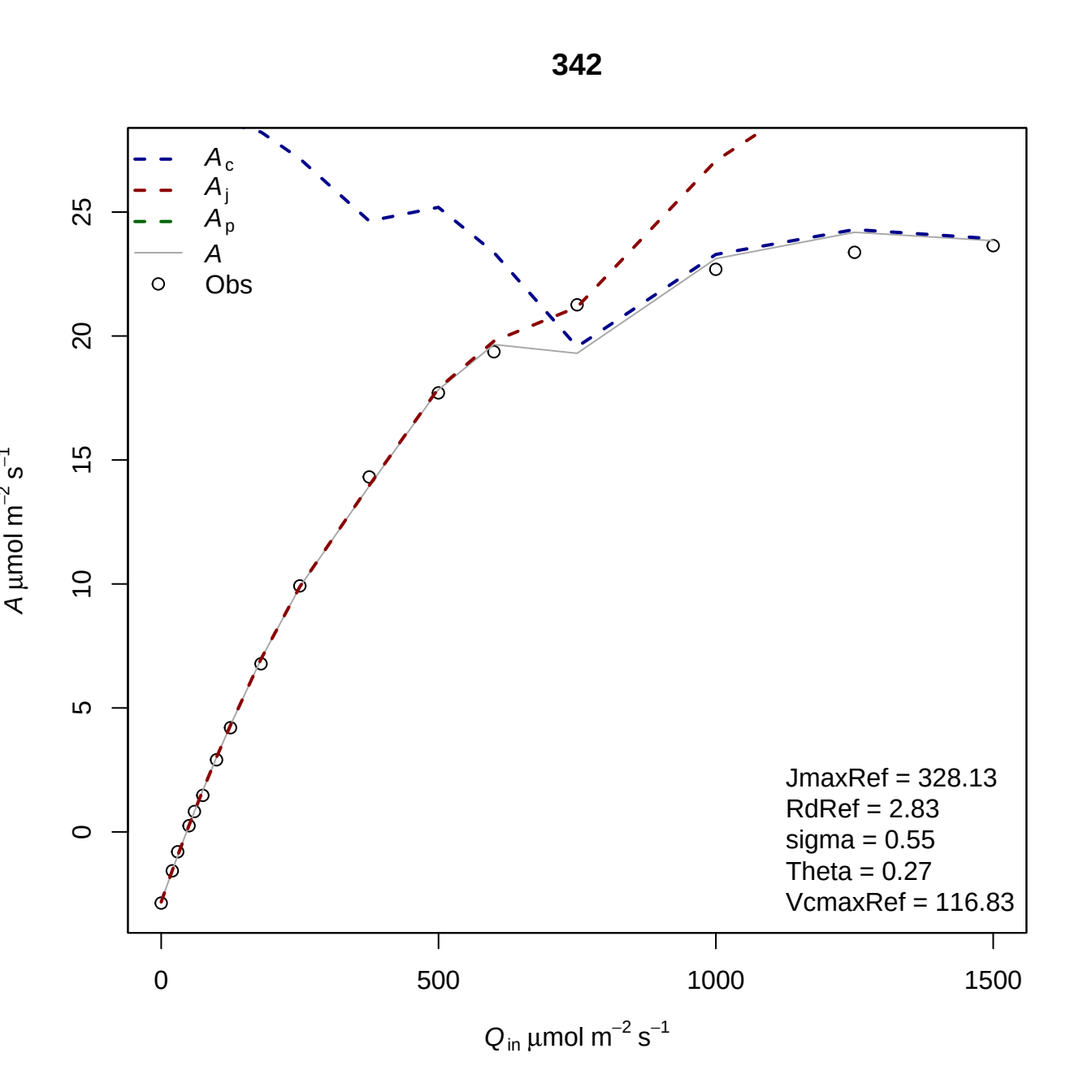

A_c
A_j
A_p
A
Obs
JmaxRef = 112.46
RdRef = 1.43
sigma = 0.28
Theta = 0.02
VcmaxRef = 54739.83
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

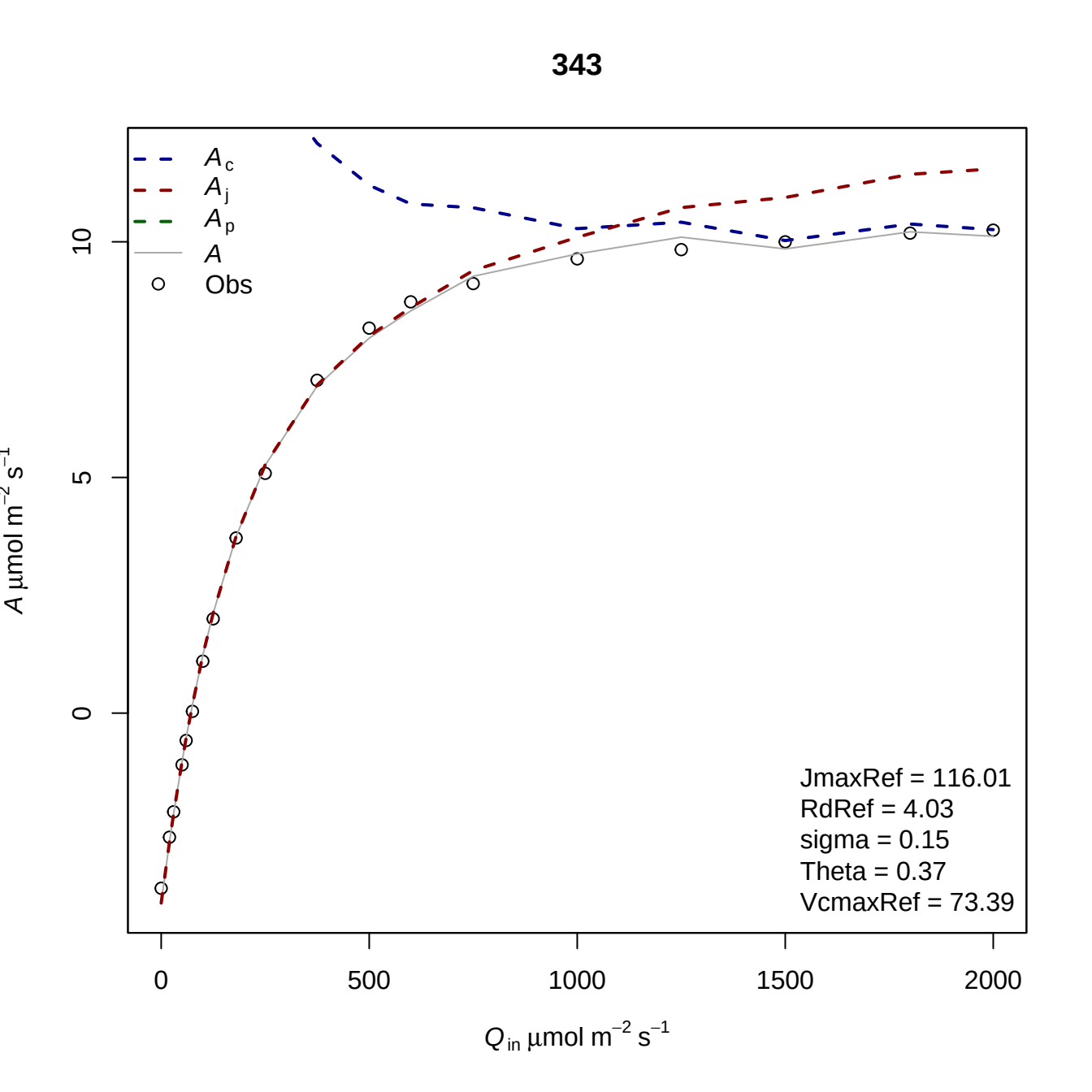

338
A_c
A_j
A_p
A
Obs
JmaxRef = 97.61
RdRef = 1.63
sigma = 0.38
Theta = 0.34
VcmaxRef = 17482.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

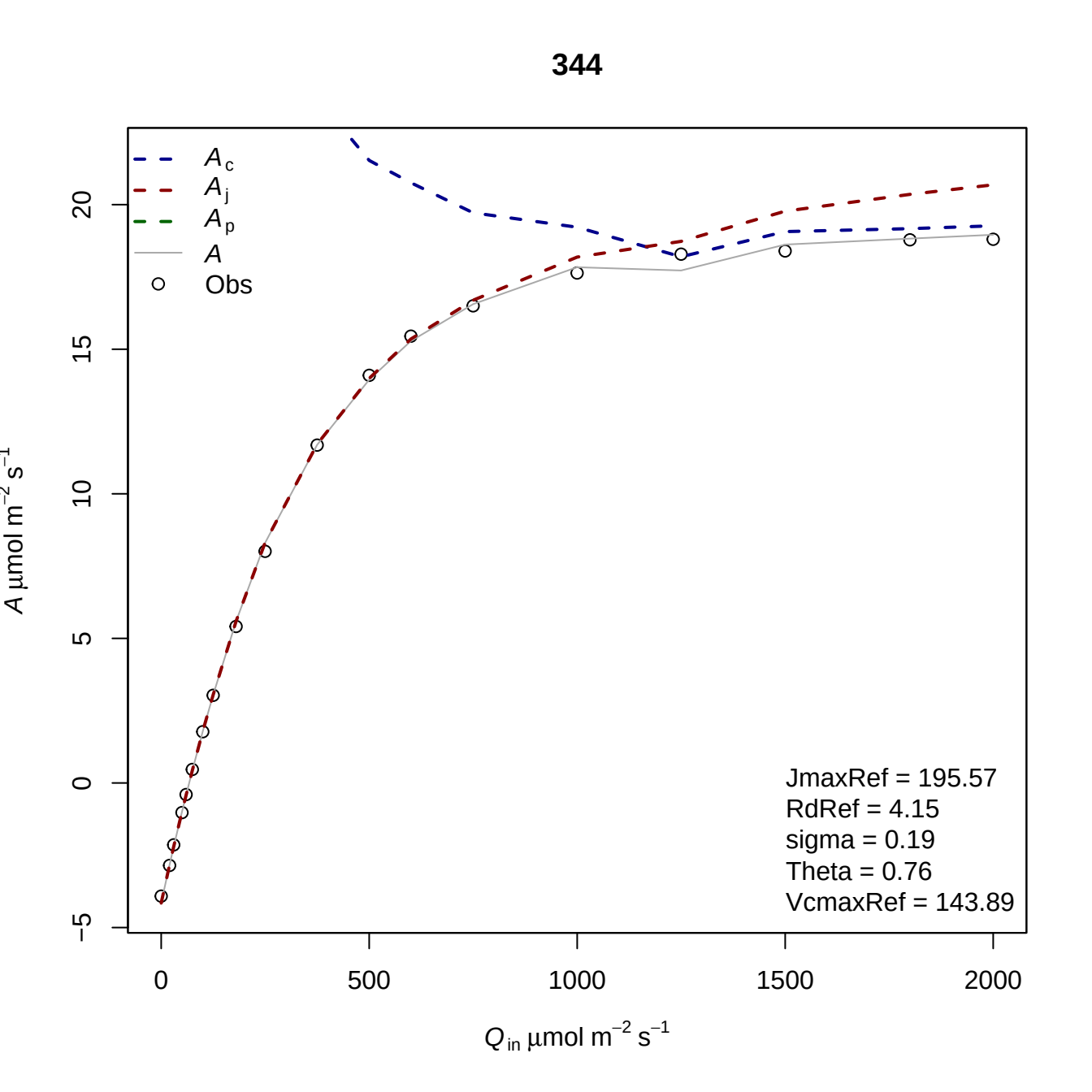

A_c
A_j
A_p
A
Obs
JmaxRef = 88.46
RdRef = 1.2
sigma = 0.33
Theta = 0.06
VcmaxRef = 35576.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

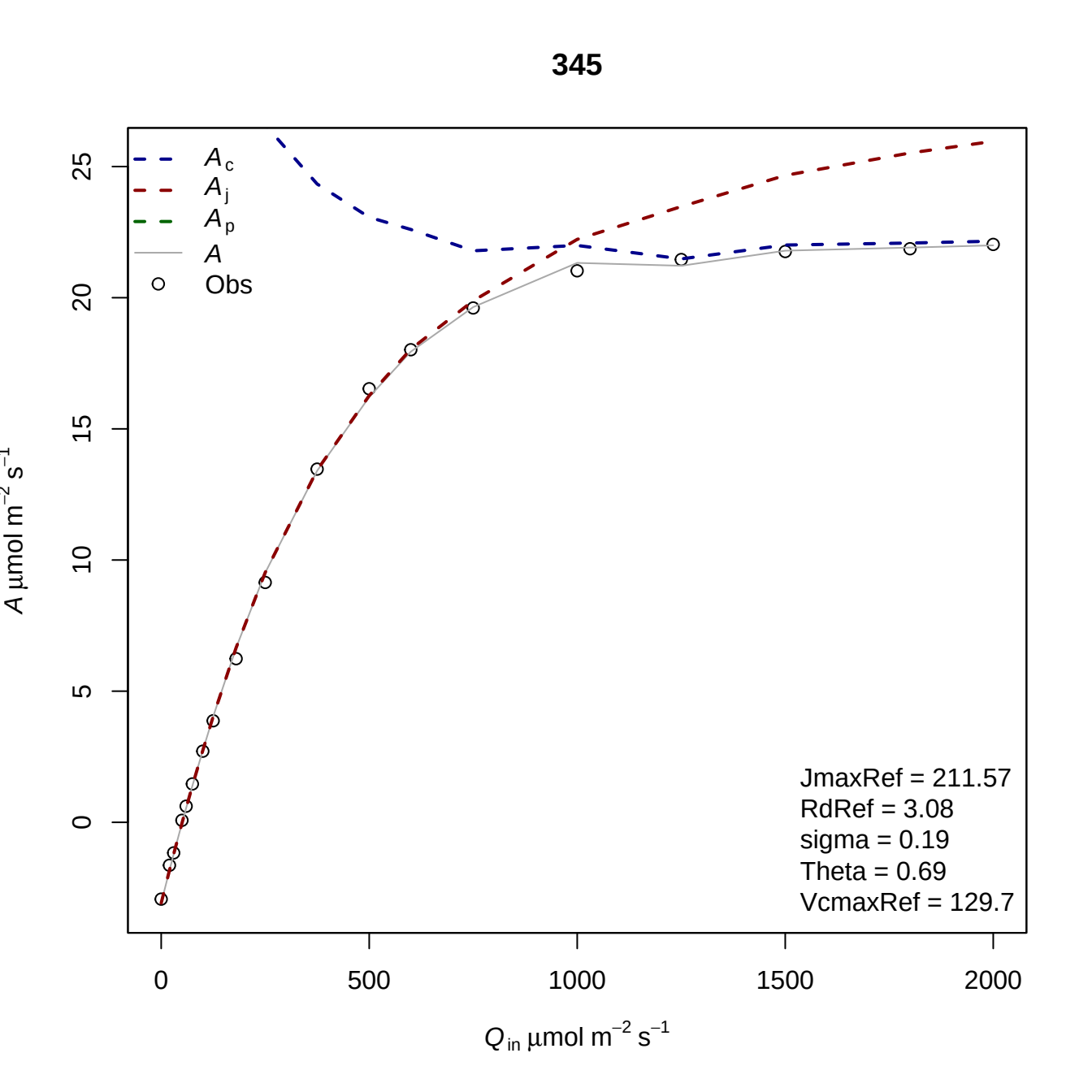

340
A_c
A_j
A_p
A
Obs
JmaxRef = 127.85
RdRef = 4.07
sigma = 0.3
Theta = 0.76
VcmaxRef = 299182.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

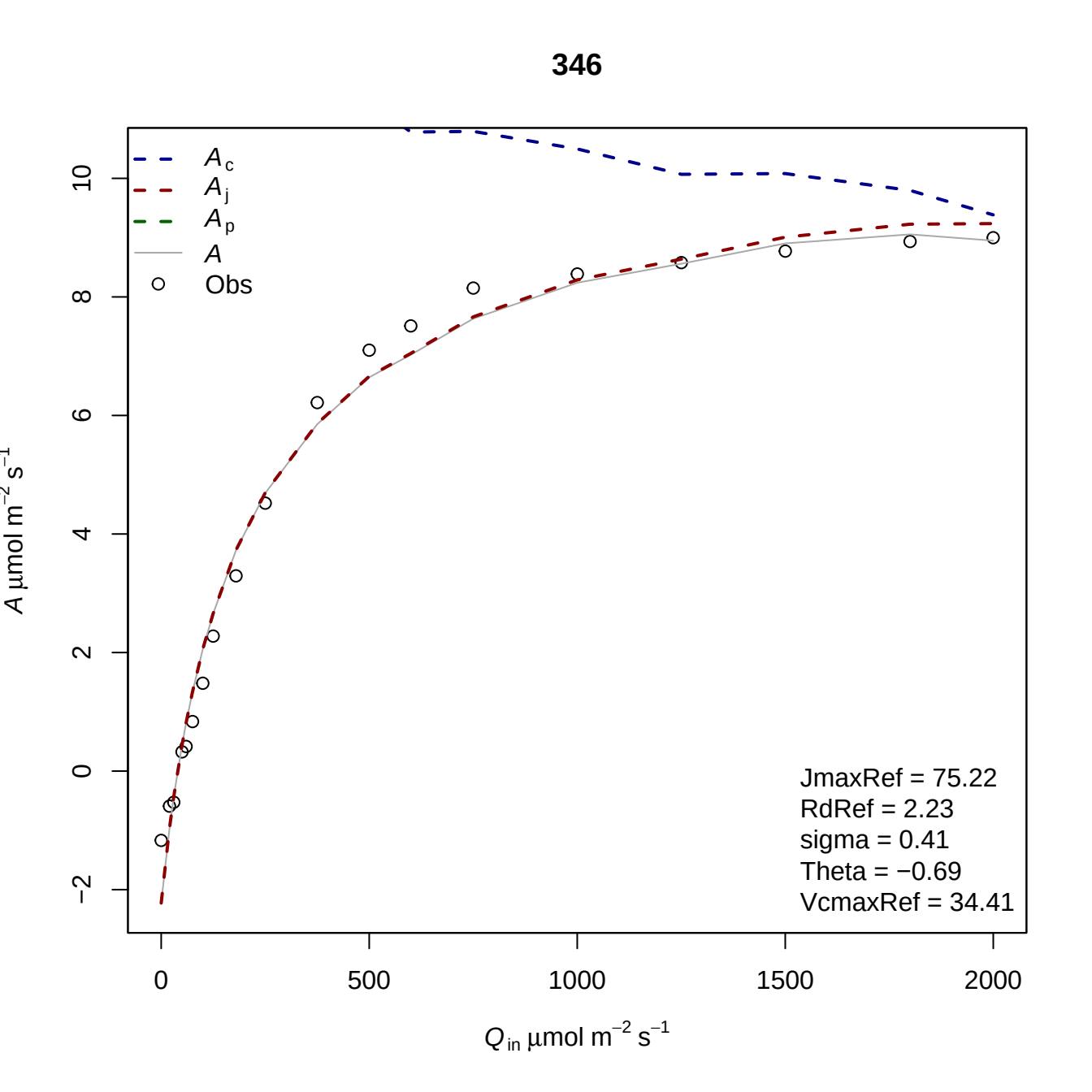

341
A_c
A_j
A_p
A
Obs
JmaxRef = 145.77
RdRef = 2.3
sigma = 0.33
Theta = −0.24
VcmaxRef = 316136.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

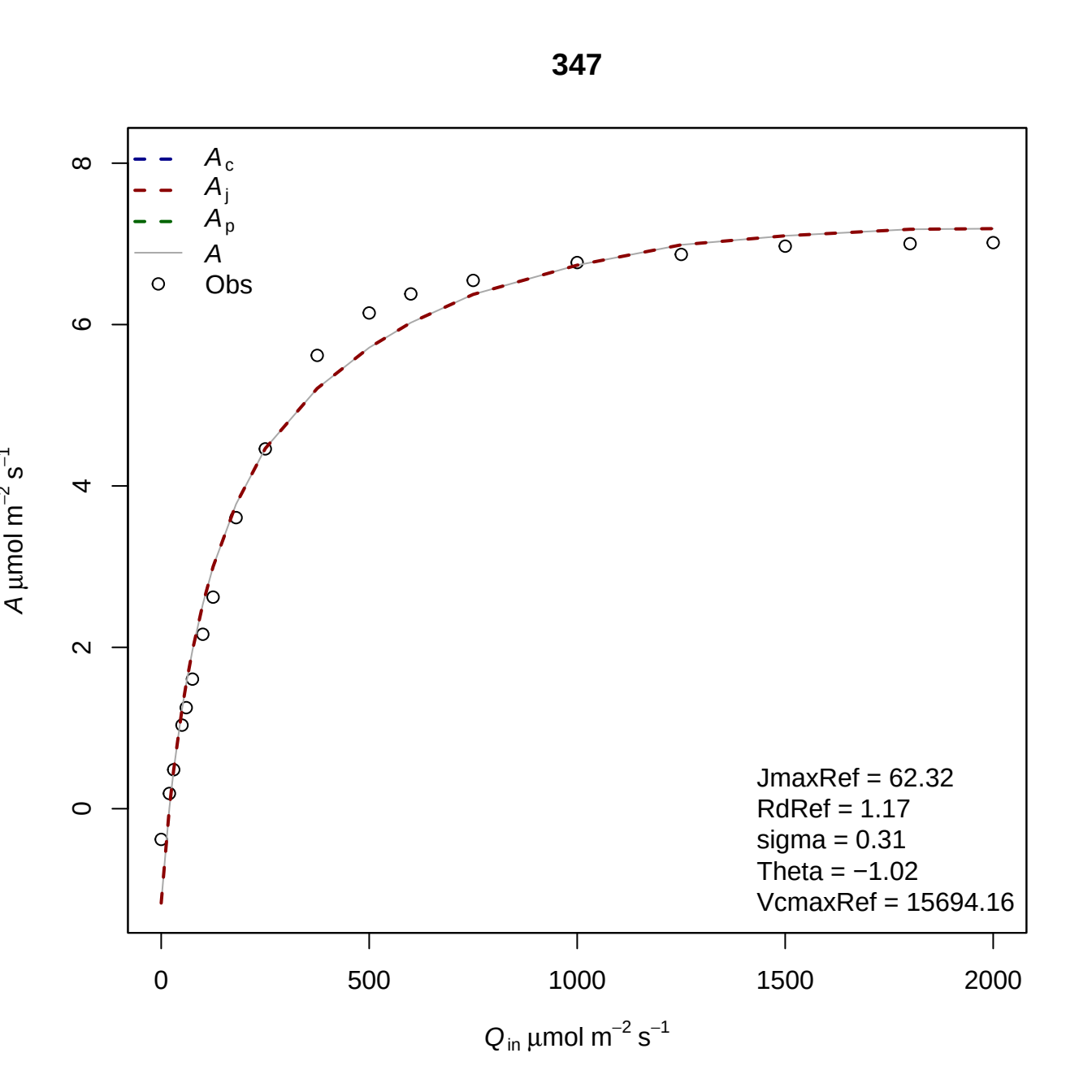

342
A_c
A_j
A_p
A
Obs
JmaxRef = 328.13
RdRef = 2.83
sigma = 0.55
Theta = 0.27
VcmaxRef = 116.83
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

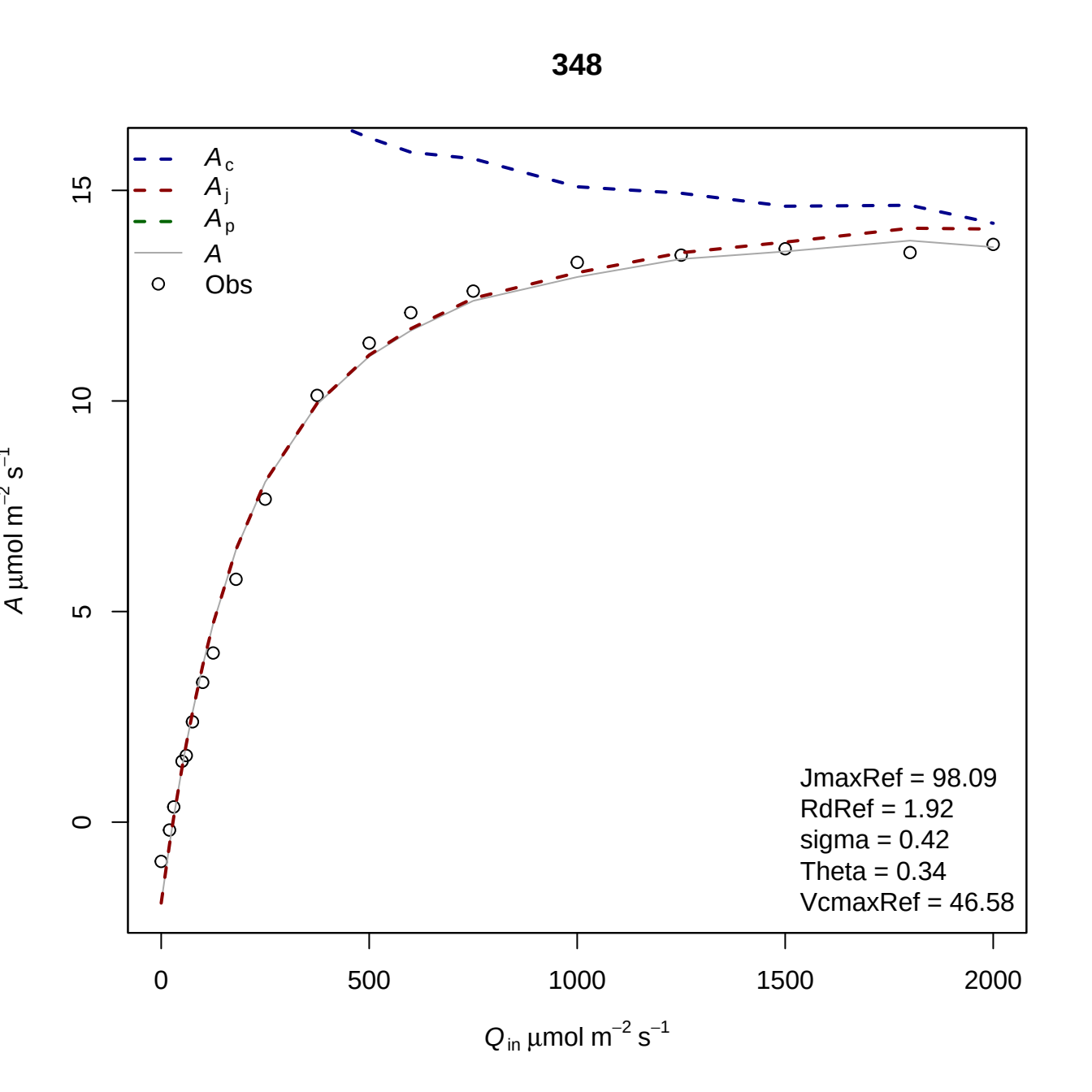

343
A_c
A_j
A_p
A
Obs
JmaxRef = 116.01
RdRef = 4.03
sigma = 0.15
Theta = 0.37
VcmaxRef = 73.39
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

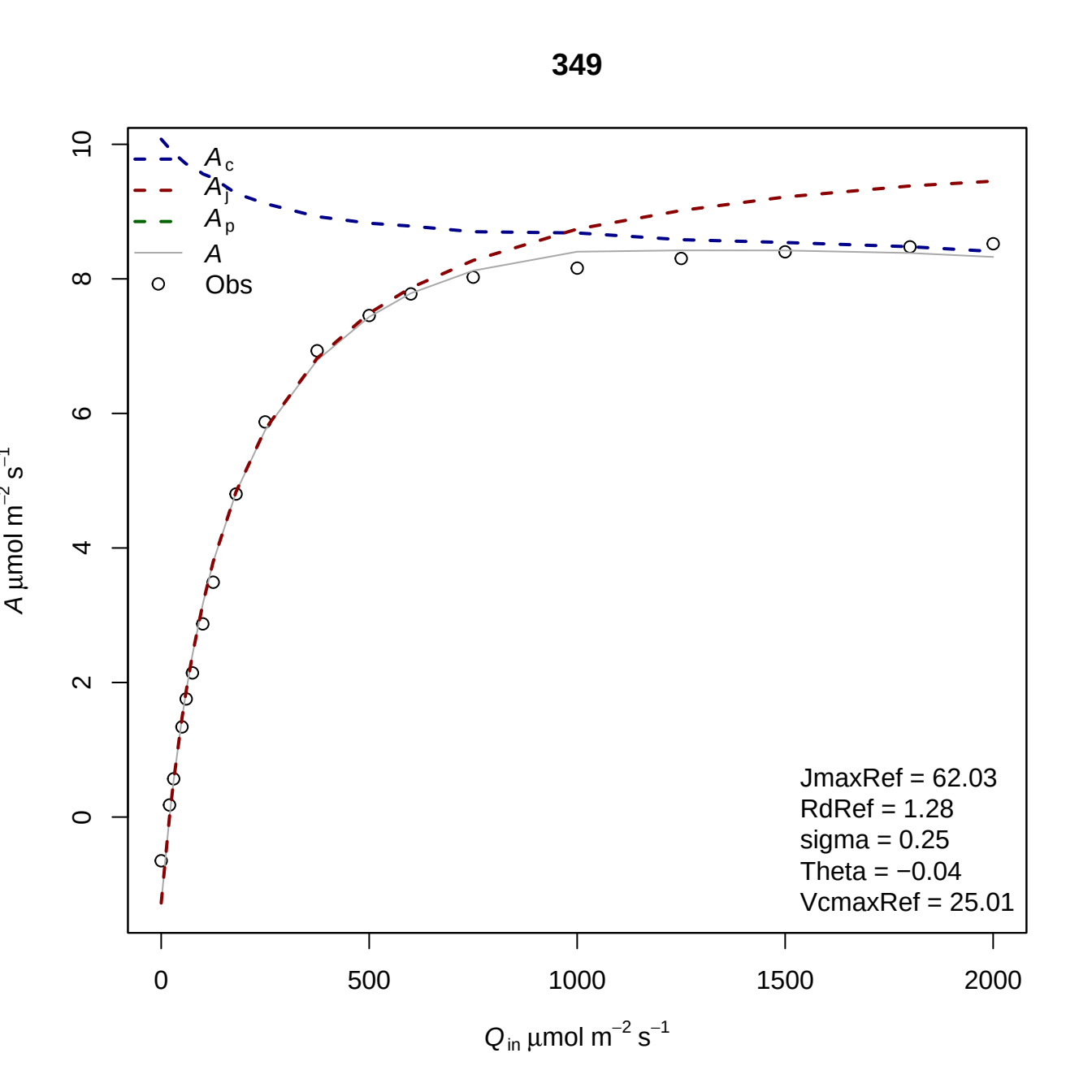

344
A_c
A_j
A_p
A
Obs
JmaxRef = 195.57
RdRef = 4.15
sigma = 0.19
Theta = 0.76
VcmaxRef = 143.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

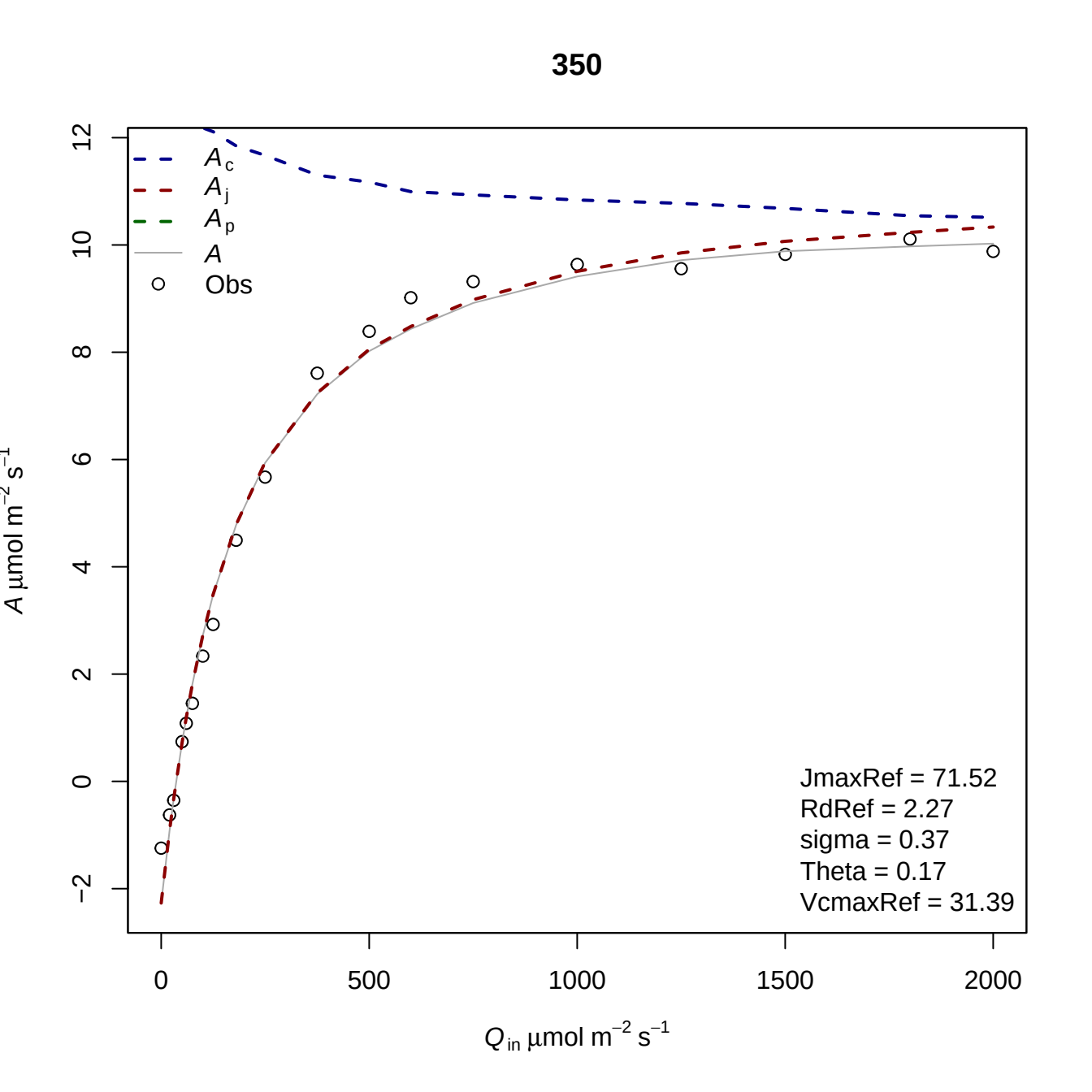

345
A_c
A_j
A_p
A
Obs
JmaxRef = 211.57
RdRef = 3.08
sigma = 0.19
Theta = 0.69
VcmaxRef = 129.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

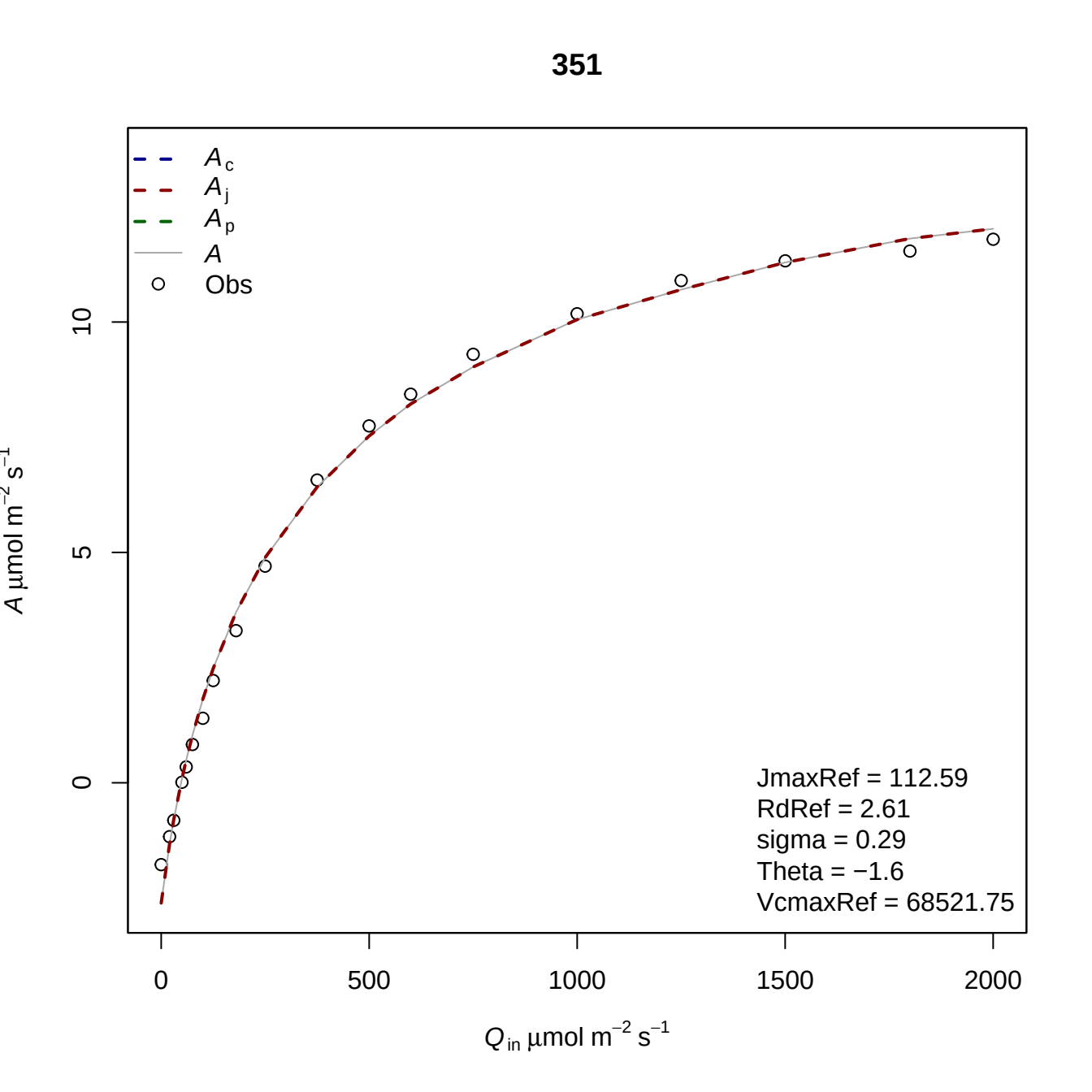

346
A_c
A_j
A_p
A
Obs
JmaxRef = 75.22
RdRef = 2.23
sigma = 0.41
Theta = −0.69
VcmaxRef = 34.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

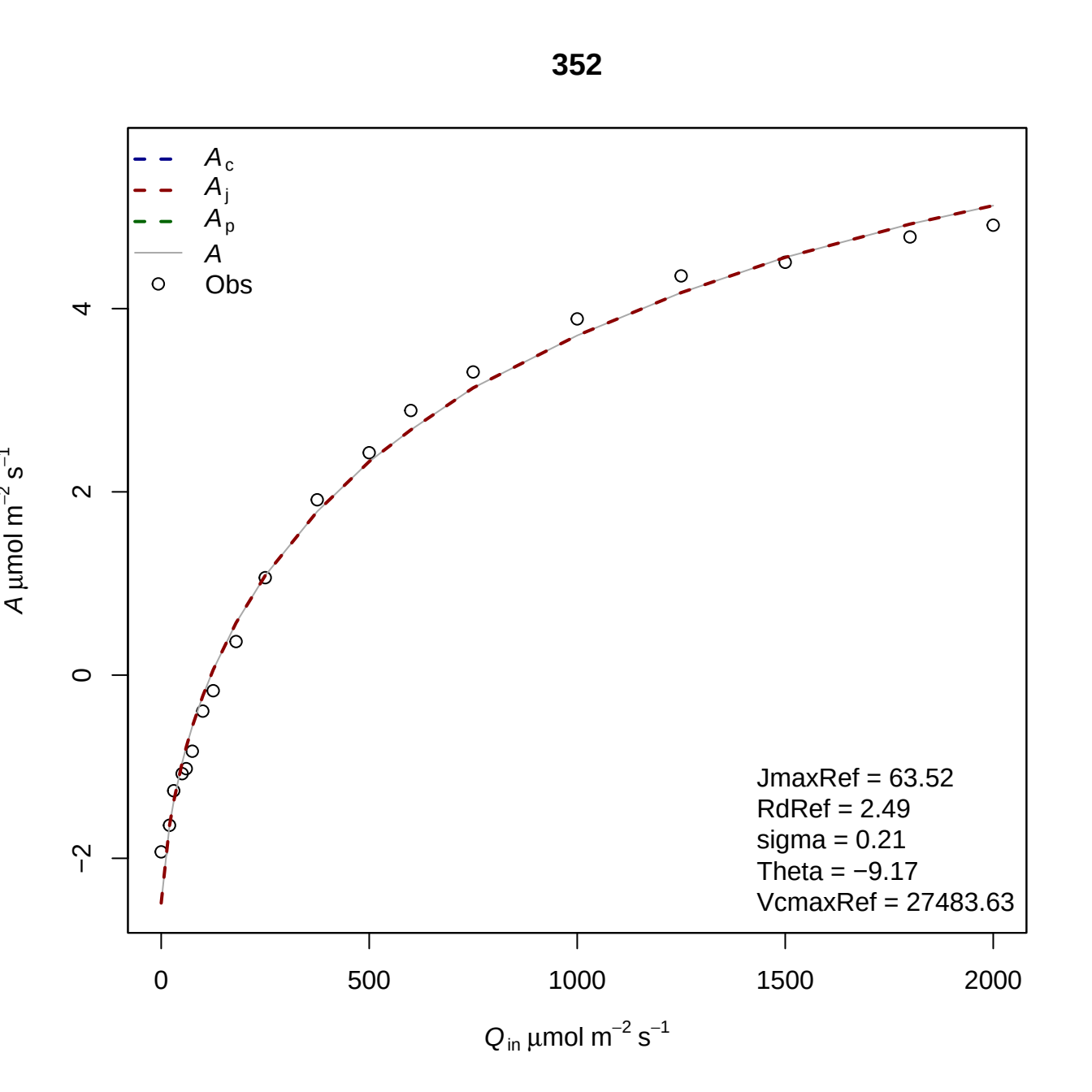

347
A_c
A_j
A_p
A
Obs
JmaxRef = 62.32
RdRef = 1.17
sigma = 0.31
Theta = −1.02
VcmaxRef = 15694.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

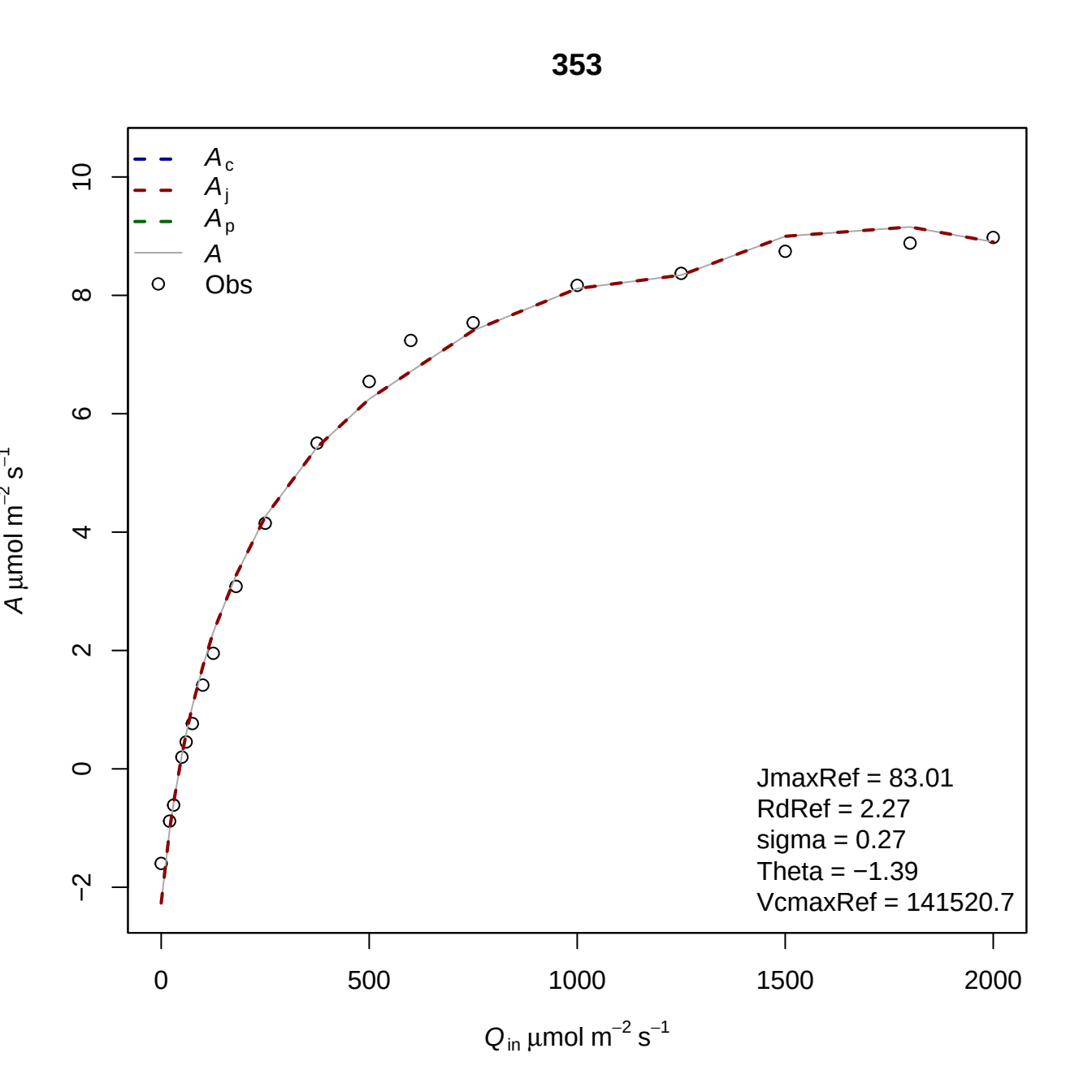

348
A_c
A_j
A_p
A
Obs
JmaxRef = 98.09
RdRef = 1.92
sigma = 0.42
Theta = 0.34
VcmaxRef = 46.58
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

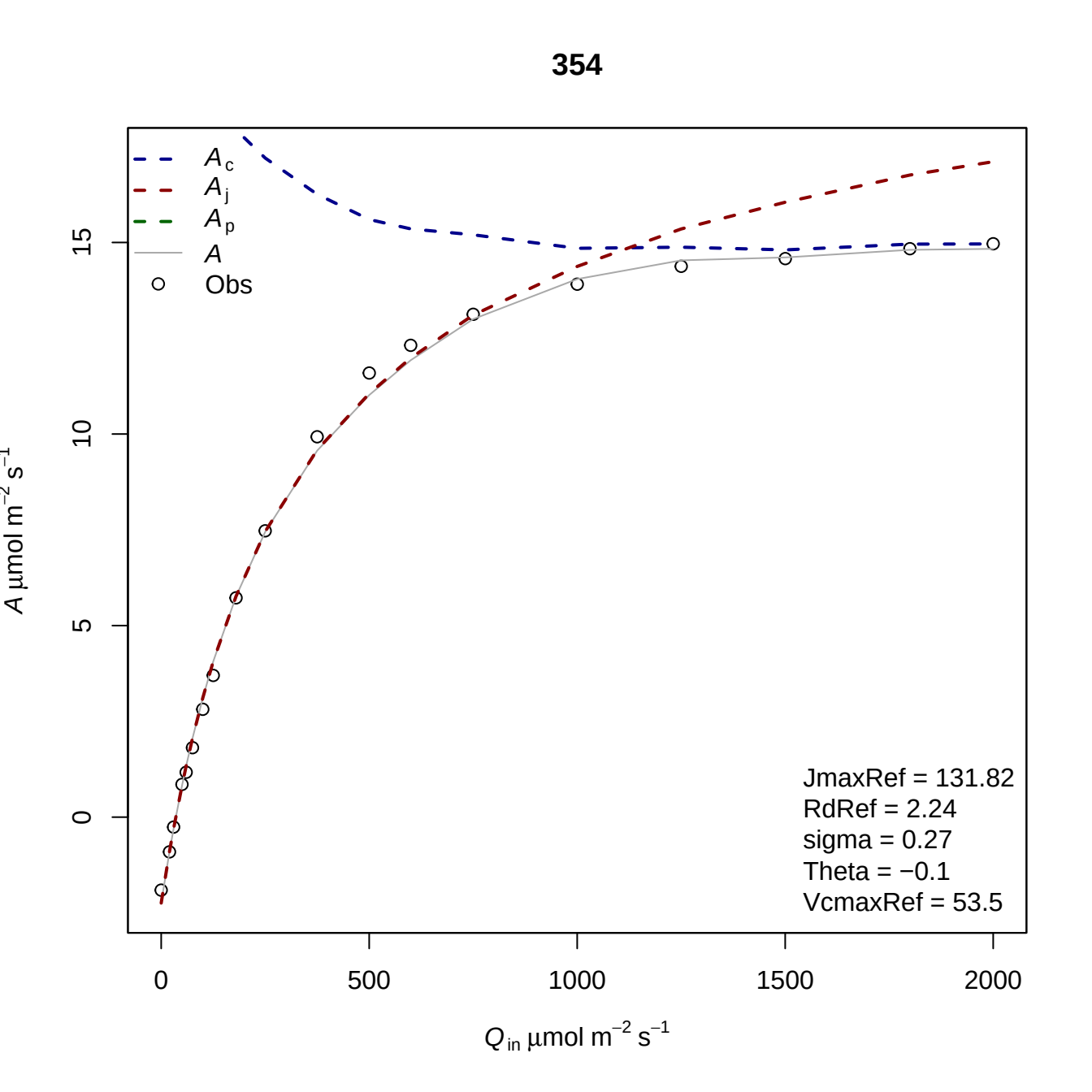

349
A_c
A_j
A_p
A
Obs
JmaxRef = 62.03
RdRef = 1.28
sigma = 0.25
Theta = −0.04
VcmaxRef = 25.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

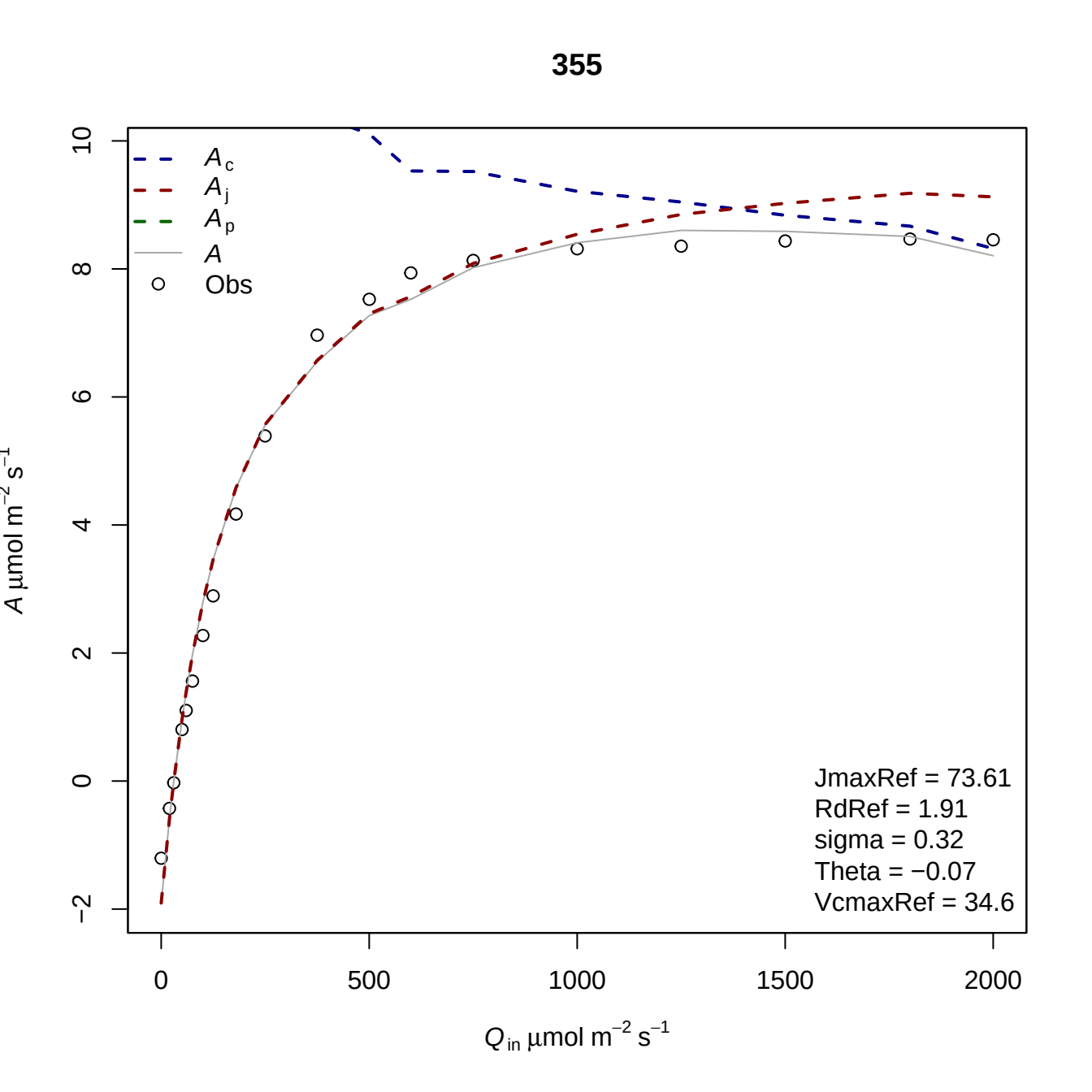

350
A_c
A_j
A_p
A
Obs
JmaxRef = 71.52
RdRef = 2.27
sigma = 0.37
Theta = 0.17
VcmaxRef = 31.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

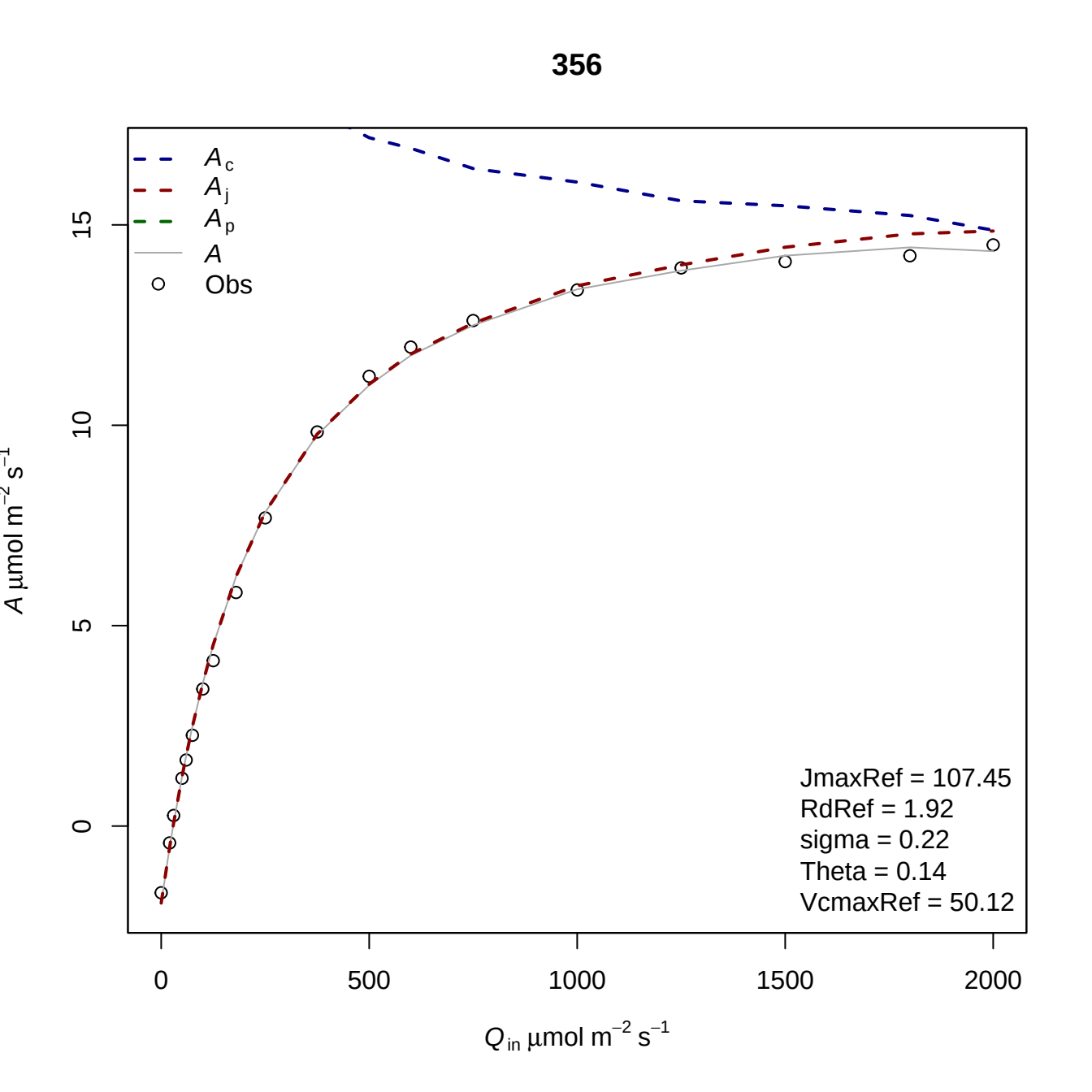

351
A_c
A_j
A_p
A
Obs
JmaxRef = 112.59
RdRef = 2.61
sigma = 0.29
Theta = −1.6
VcmaxRef = 68521.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

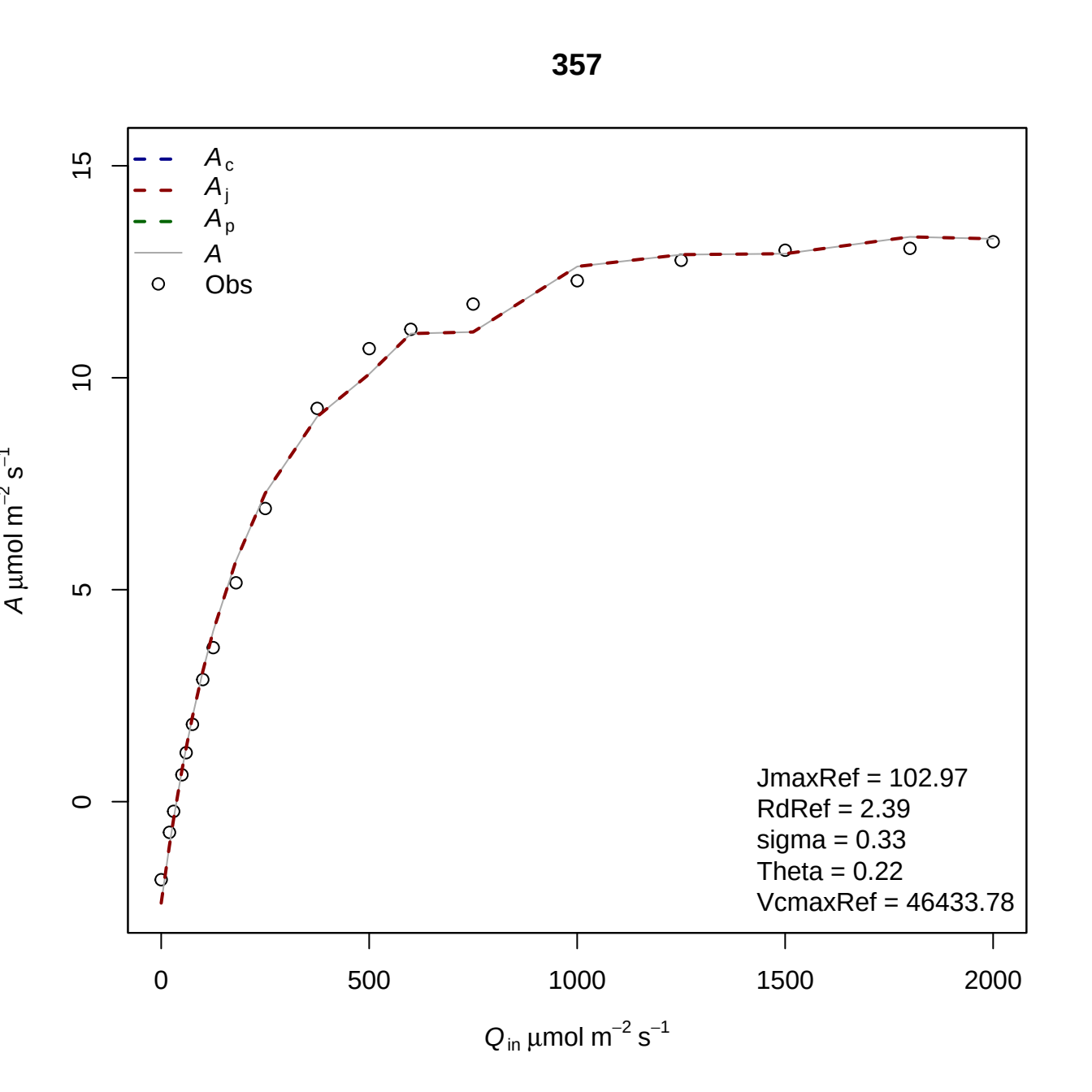

352
A_c
A_j
A_p
A
Obs
JmaxRef = 63.52
RdRef = 2.49
sigma = 0.21
Theta = −9.17
VcmaxRef = 27483.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

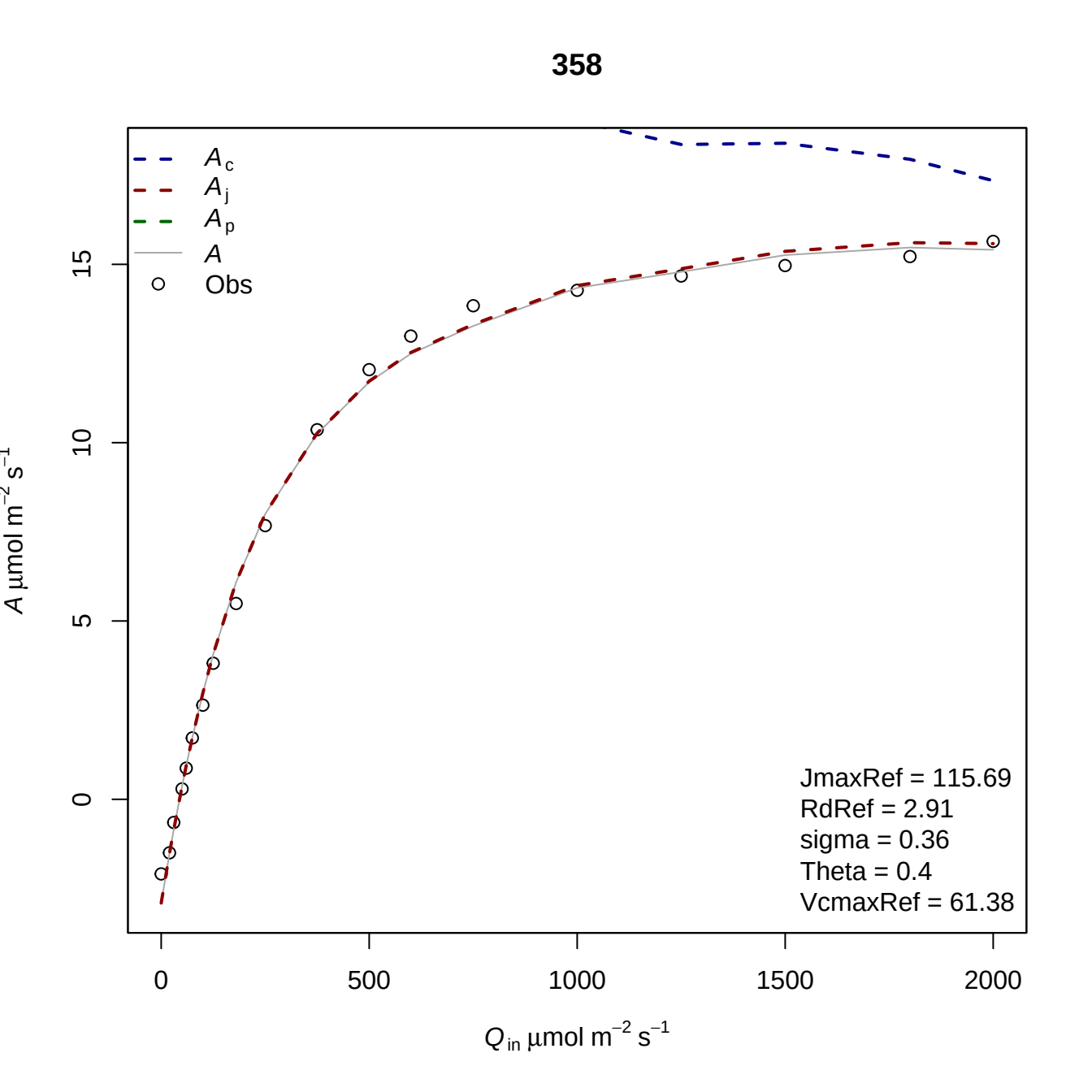

353
A_c
A_j
A_p
A
Obs
JmaxRef = 83.01
RdRef = 2.27
sigma = 0.27
Theta = −1.39
VcmaxRef = 141520.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

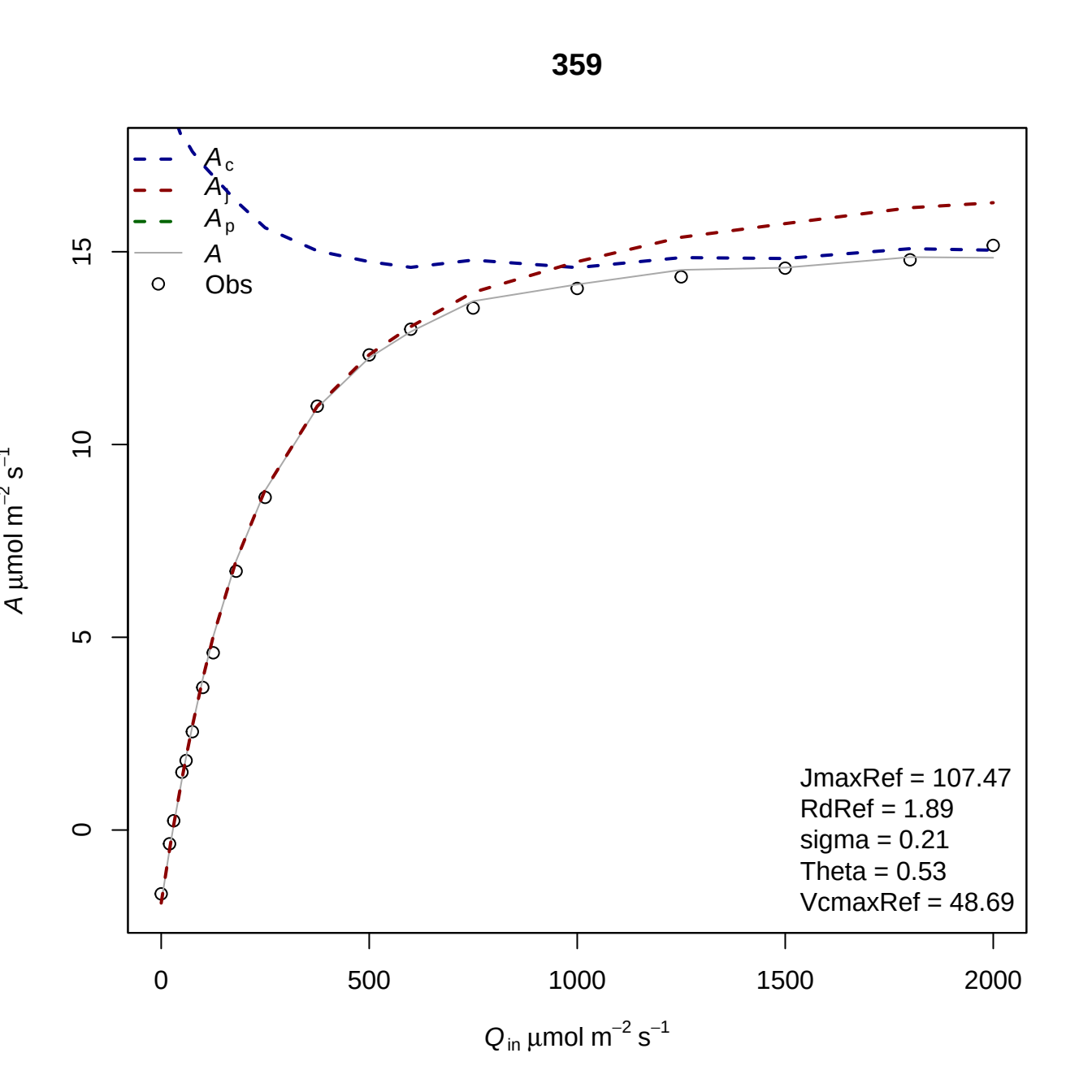

354
A_c
A_j
A_p
A
Obs
JmaxRef = 131.82
RdRef = 2.24
sigma = 0.27
Theta = −0.1
VcmaxRef = 53.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

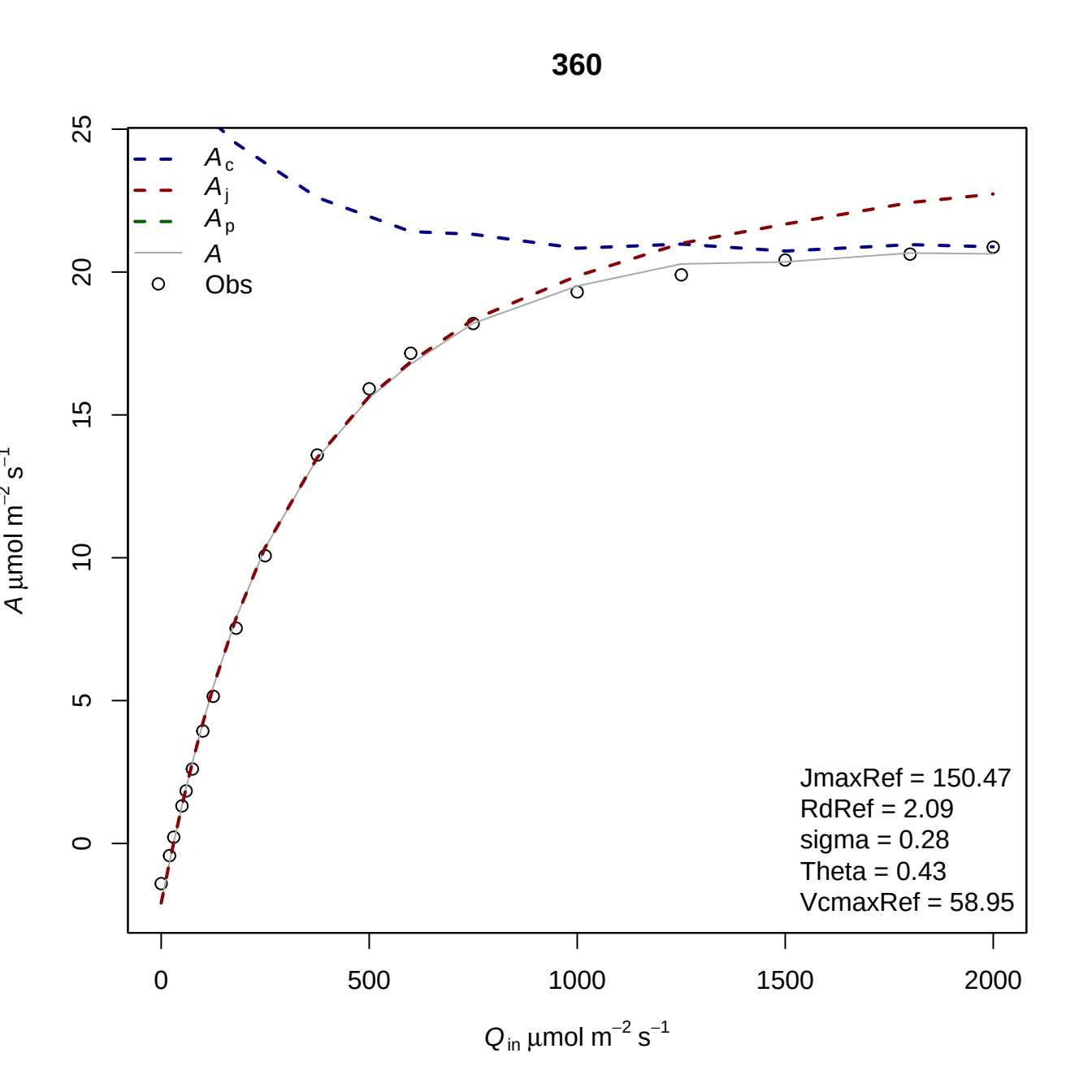

355
A_c
A_j
A_p
A
Obs
JmaxRef = 73.61
RdRef = 1.91
sigma = 0.32
Theta = −0.07
VcmaxRef = 34.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

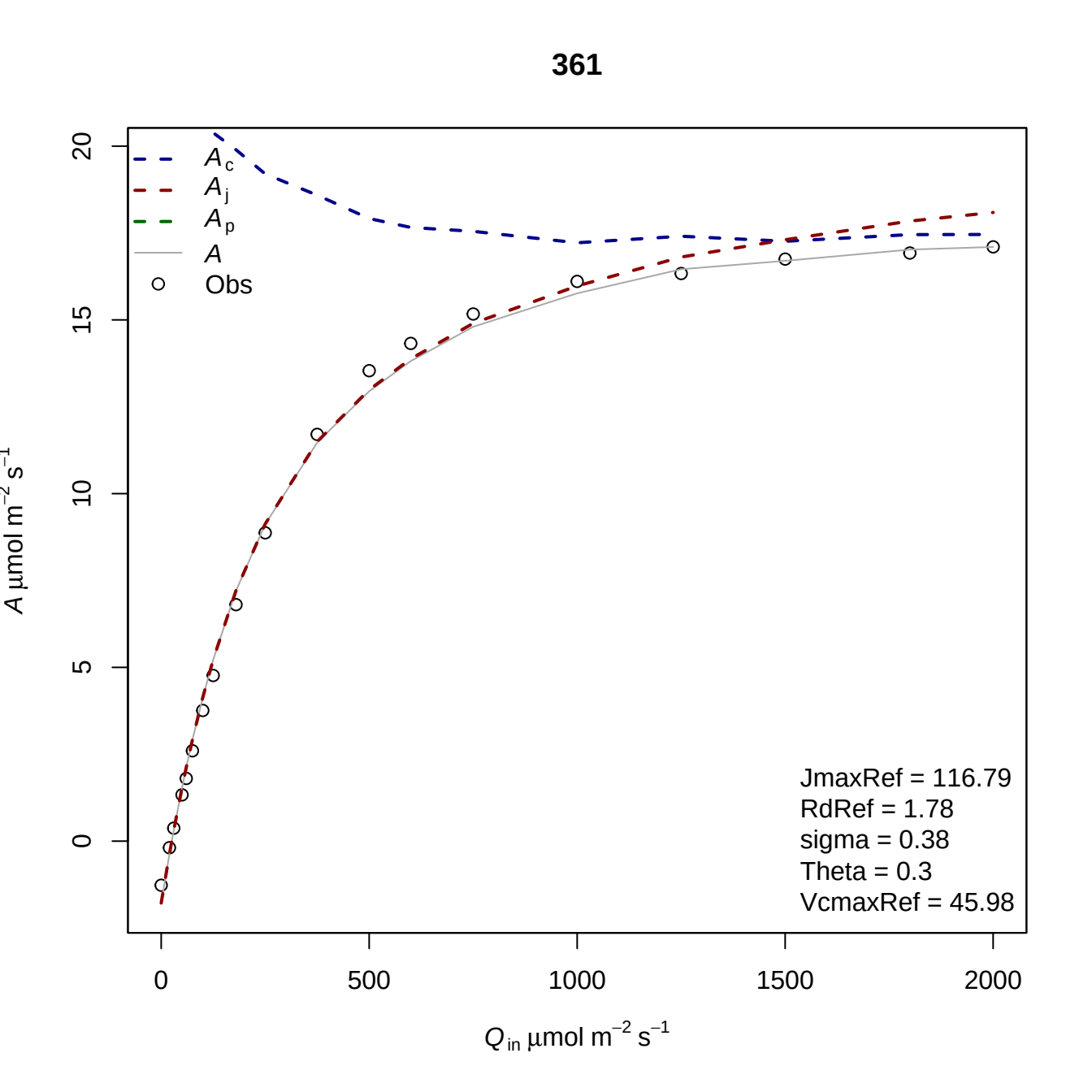

356
A_c
A_j
A_p
A
Obs
JmaxRef = 107.45
RdRef = 1.92
sigma = 0.22
Theta = 0.14
VcmaxRef = 50.12
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

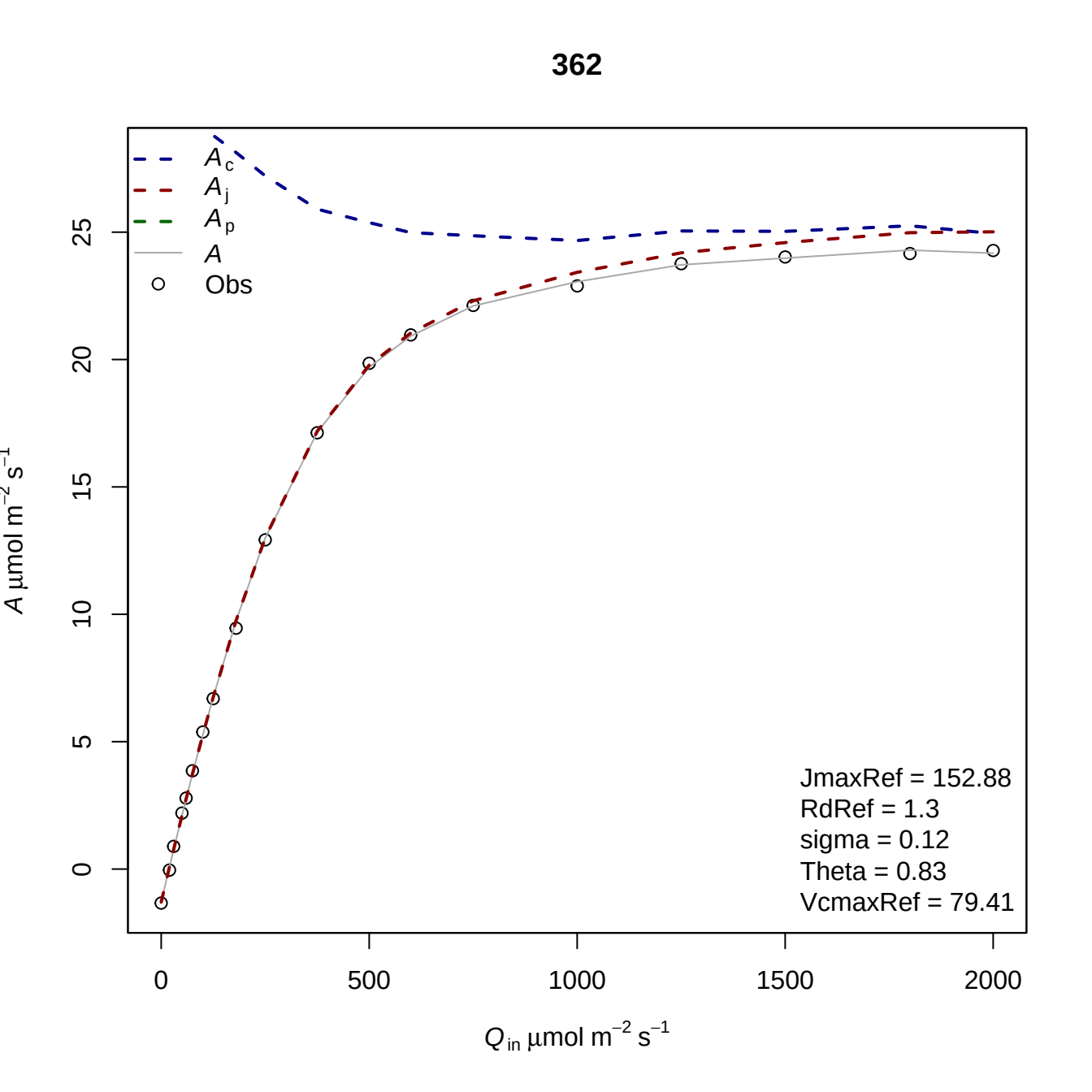

357
A_c
A_j
A_p
A
Obs
JmaxRef = 102.97
RdRef = 2.39
sigma = 0.33
Theta = 0.22
VcmaxRef = 46433.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

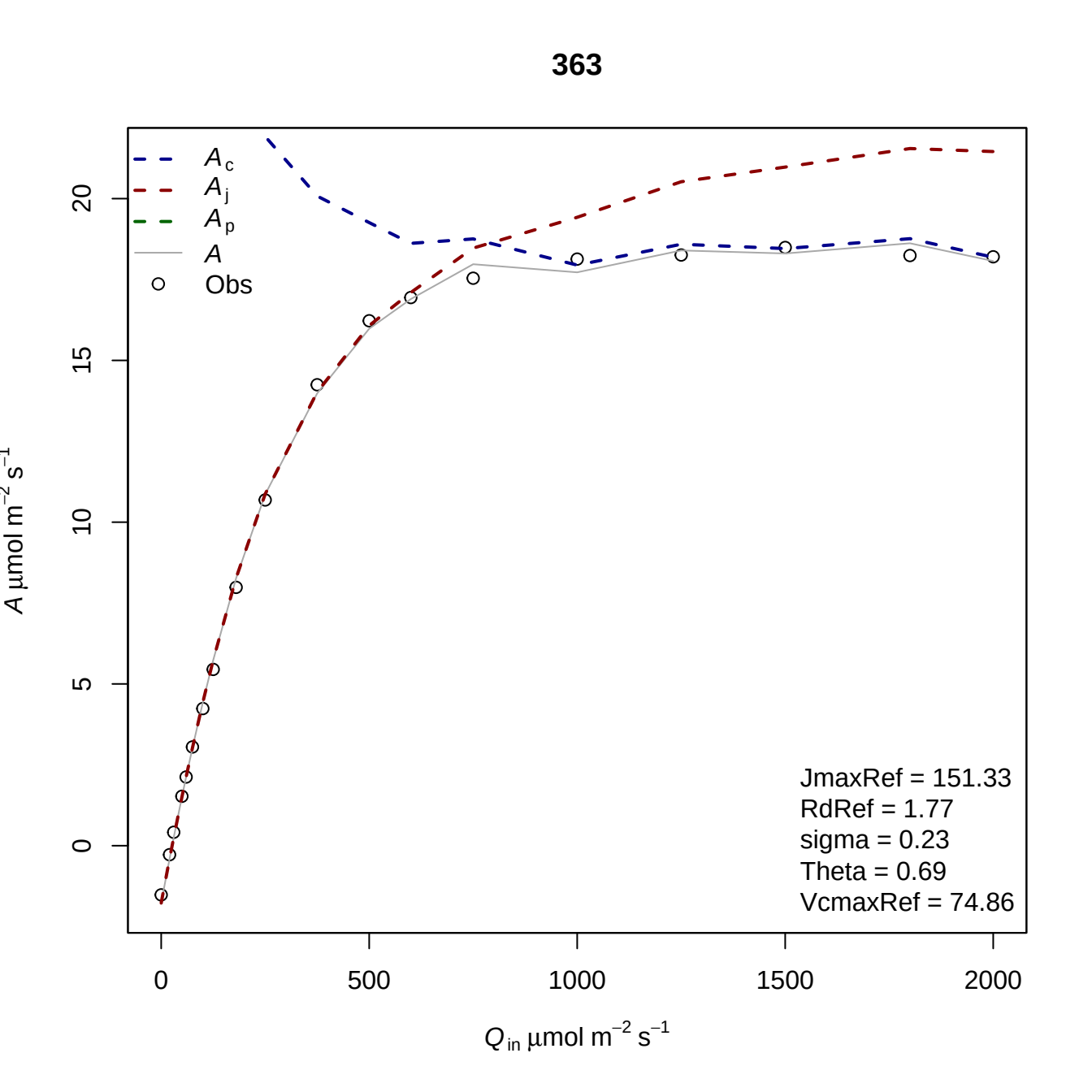

358
A_c
A_j
A_p
A
Obs
JmaxRef = 115.69
RdRef = 2.91
sigma = 0.36
Theta = 0.4
VcmaxRef = 61.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

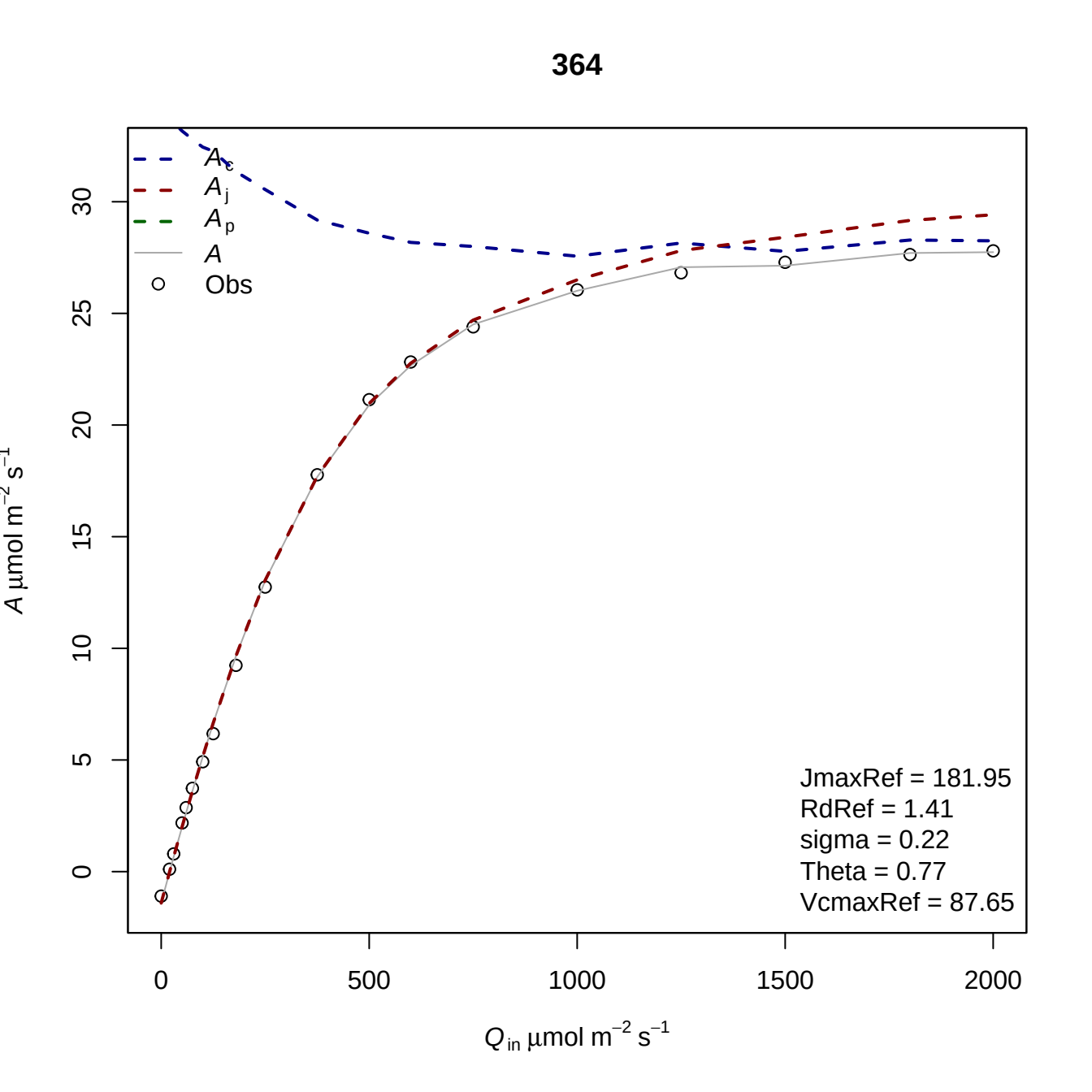

359
A_c
A_j
A_p
A
Obs
JmaxRef = 107.47
RdRef = 1.89
sigma = 0.21
Theta = 0.53
VcmaxRef = 48.69
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

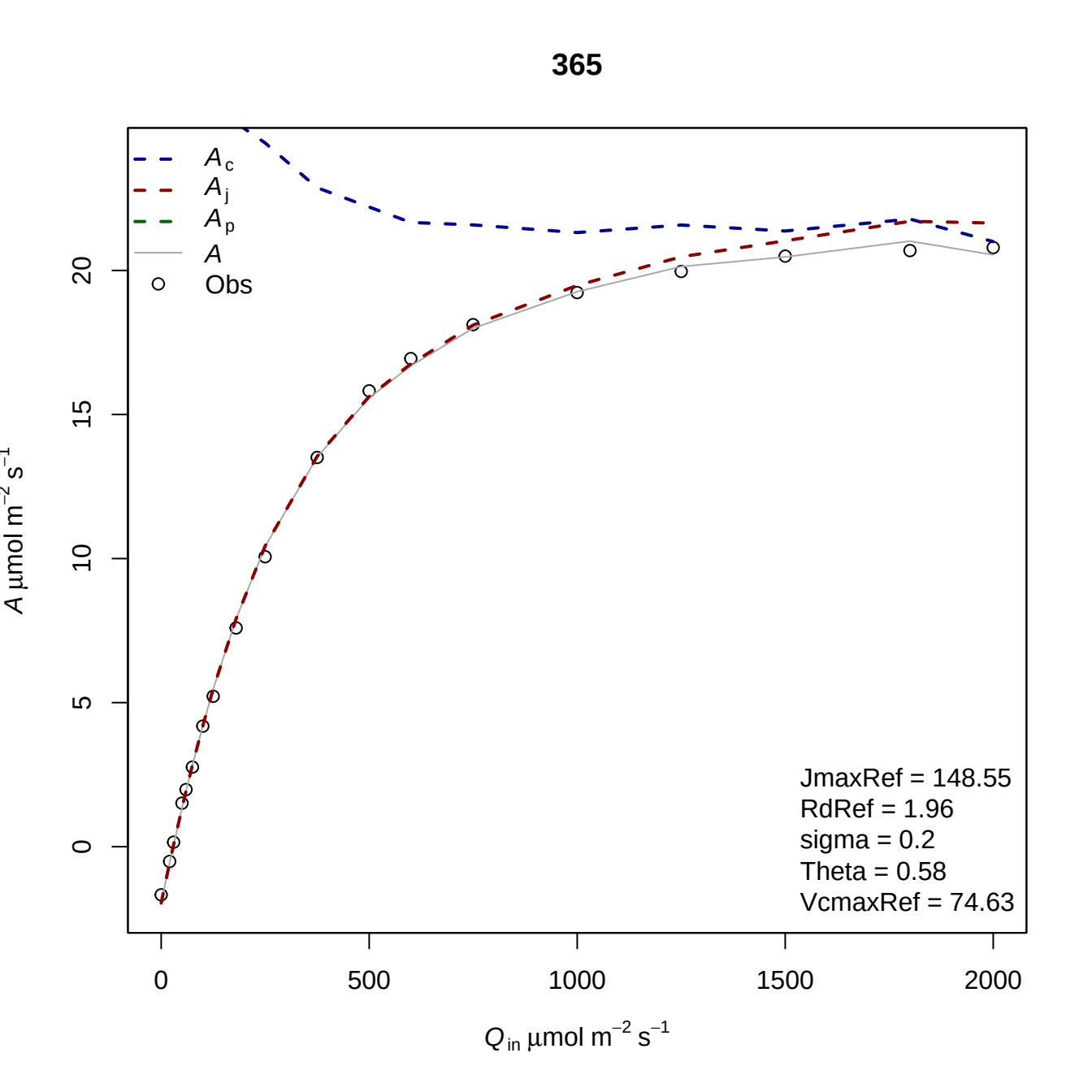

360
A_c
A_j
A_p
A
Obs
JmaxRef = 150.47
RdRef = 2.09
sigma = 0.28
Theta = 0.43
VcmaxRef = 58.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

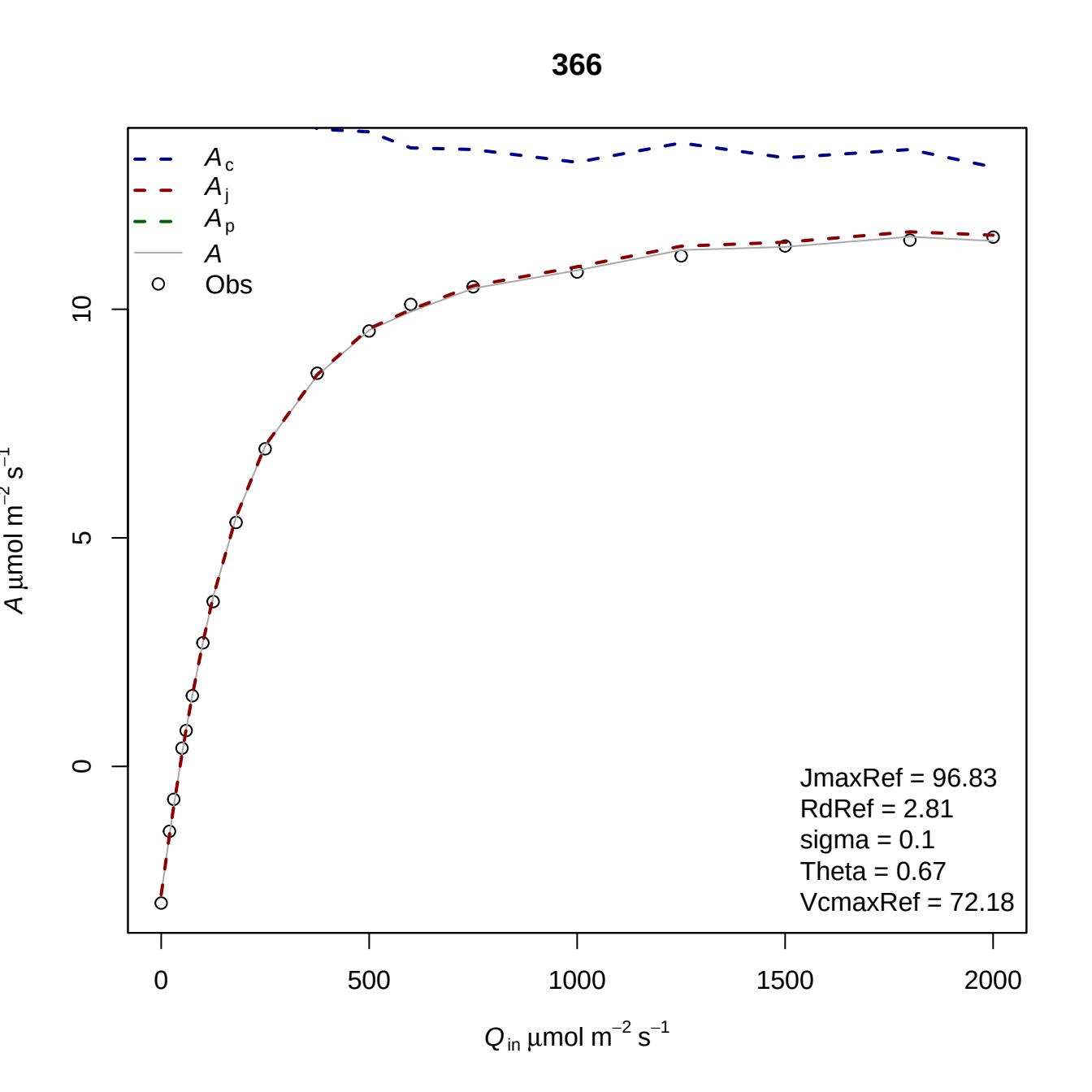

361
A_c
A_j
A_p
A
Obs
JmaxRef = 116.79
RdRef = 1.78
sigma = 0.38
Theta = 0.3
VcmaxRef = 45.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

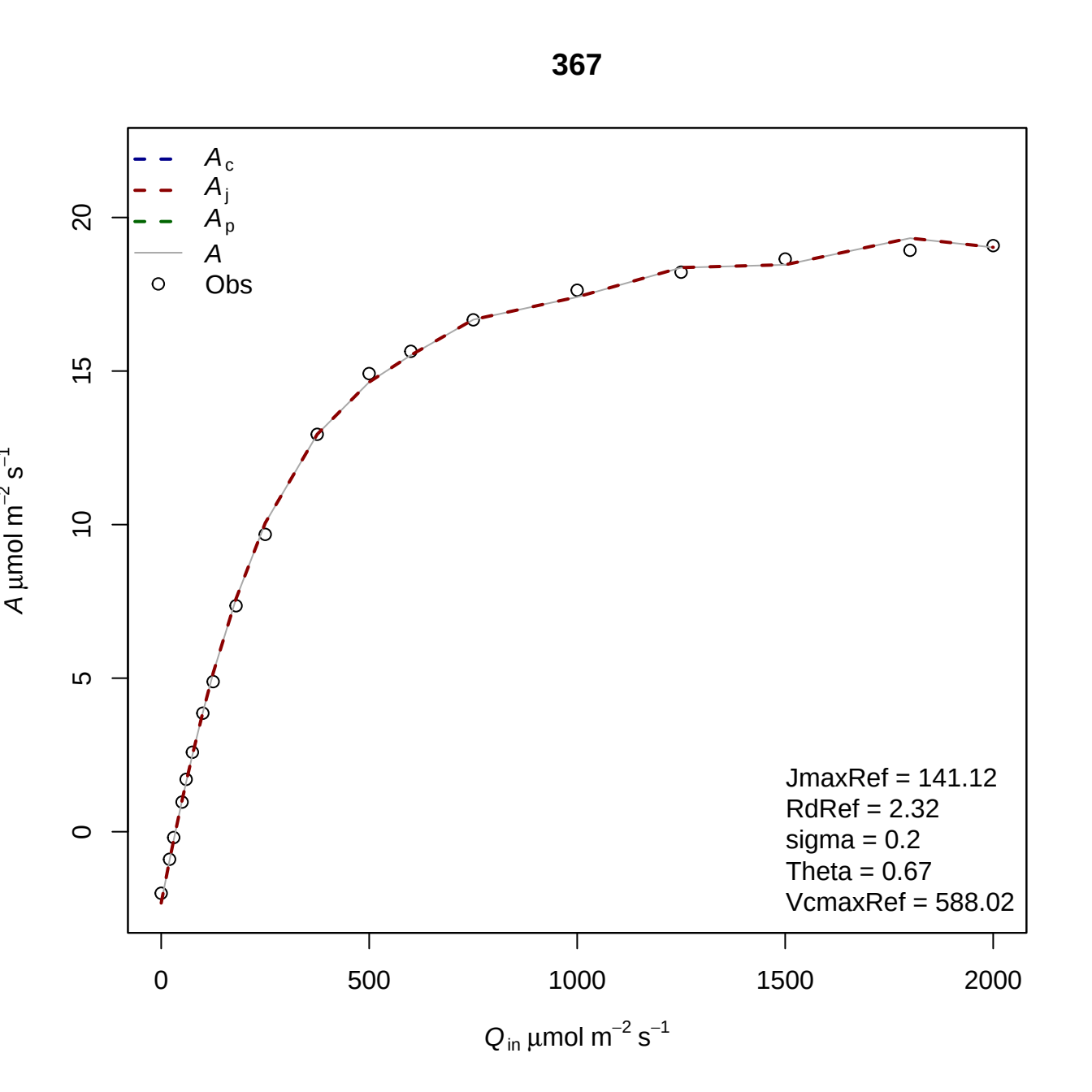

362
A_c
A_j
A_p
A
Obs
JmaxRef = 152.88
RdRef = 1.3
sigma = 0.12
Theta = 0.83
VcmaxRef = 79.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

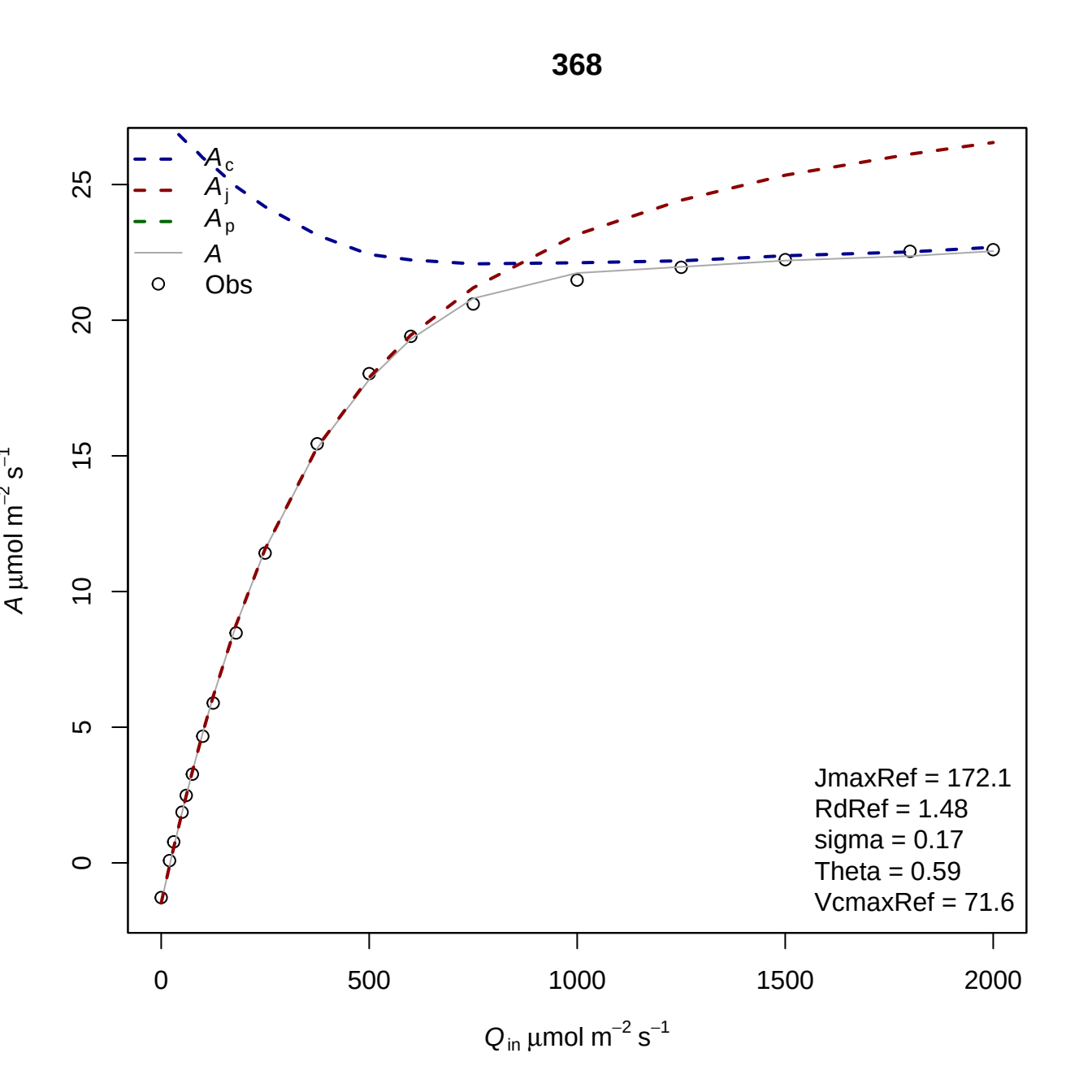

363
A_c
A_j
A_p
A
Obs
JmaxRef = 151.33
RdRef = 1.77
sigma = 0.23
Theta = 0.69
VcmaxRef = 74.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

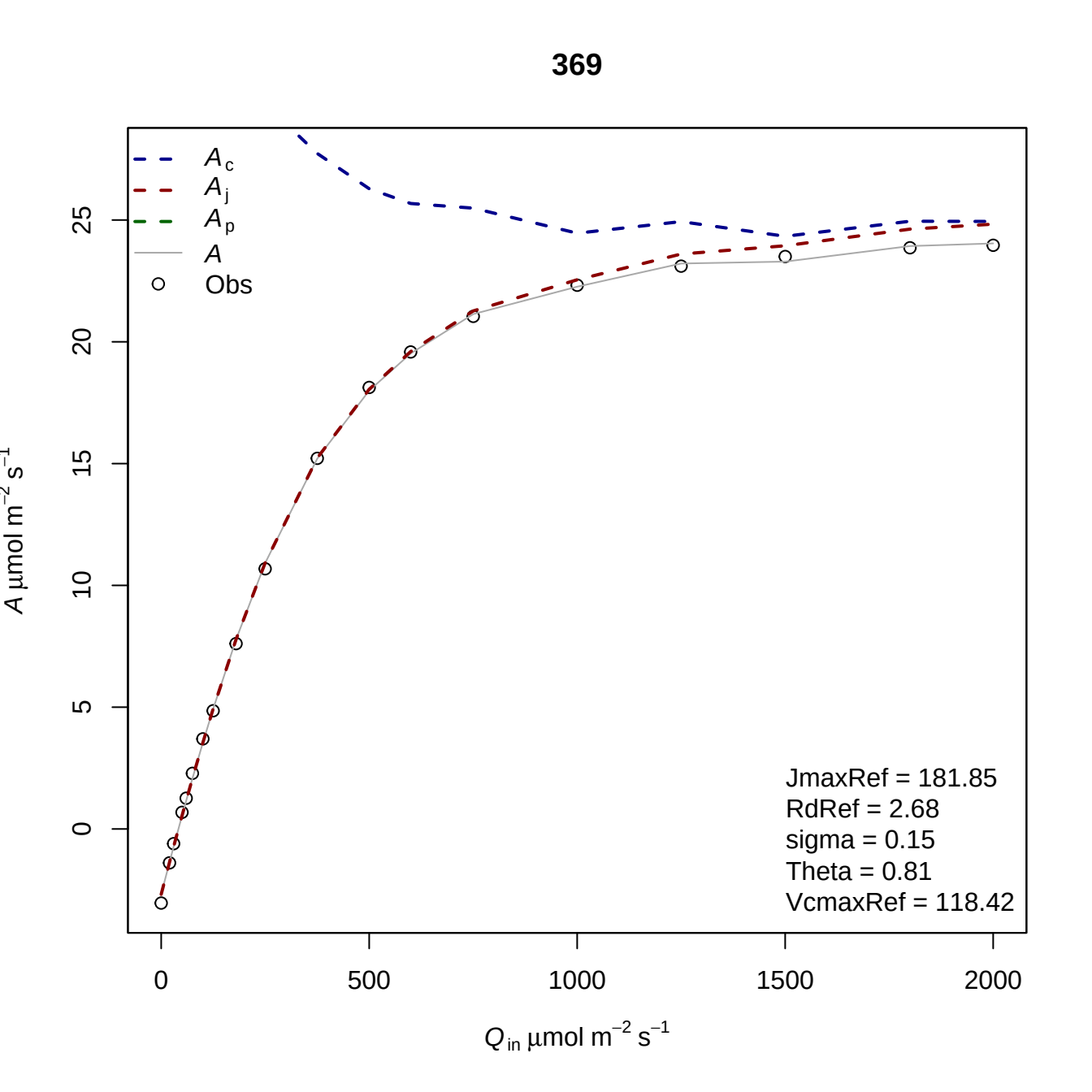

364
A_c
A_j
A_p
A
Obs
JmaxRef = 181.95
RdRef = 1.41
sigma = 0.22
Theta = 0.77
VcmaxRef = 87.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

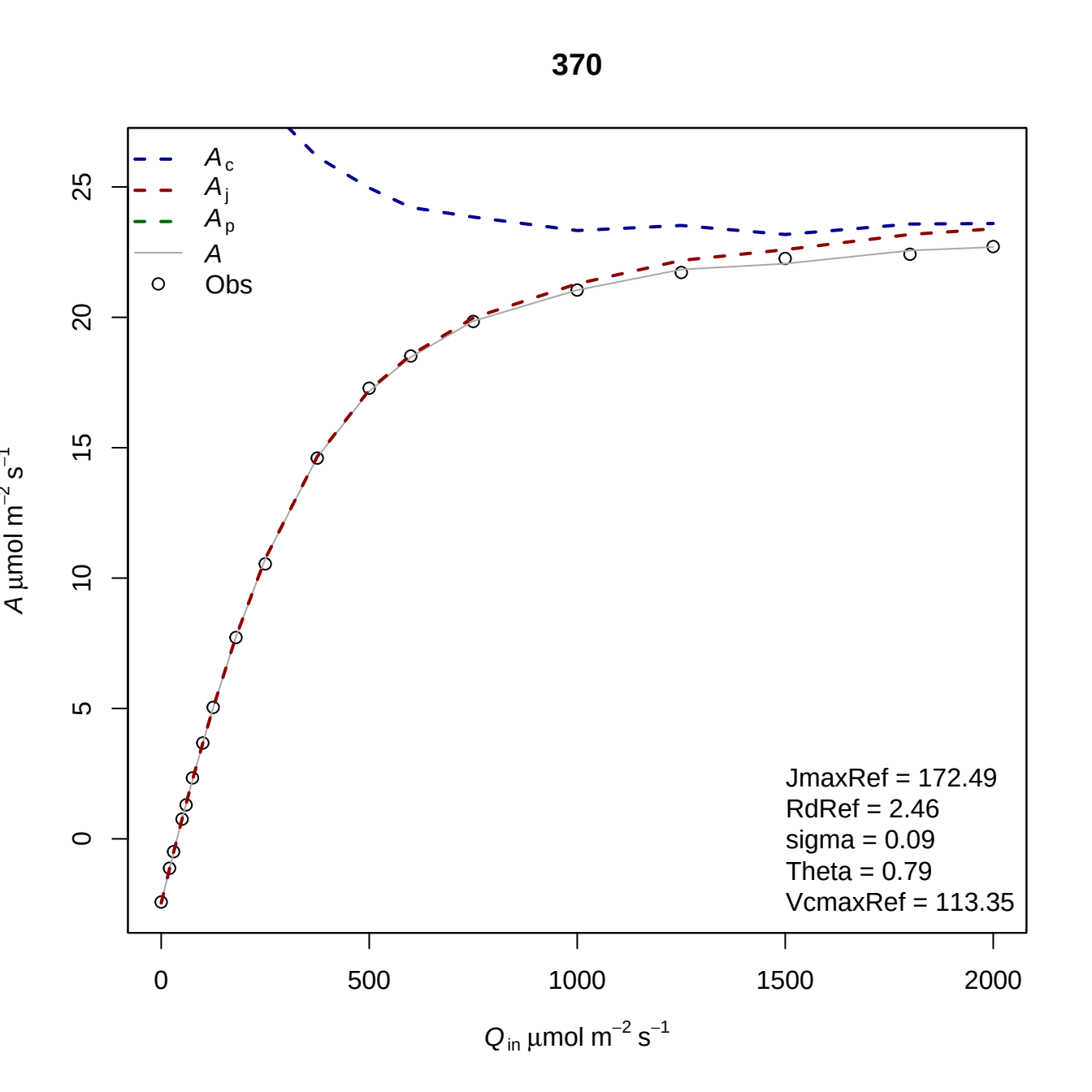

365
A_c
A_j
A_p
A
Obs
JmaxRef = 148.55
RdRef = 1.96
sigma = 0.2
Theta = 0.58
VcmaxRef = 74.63
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

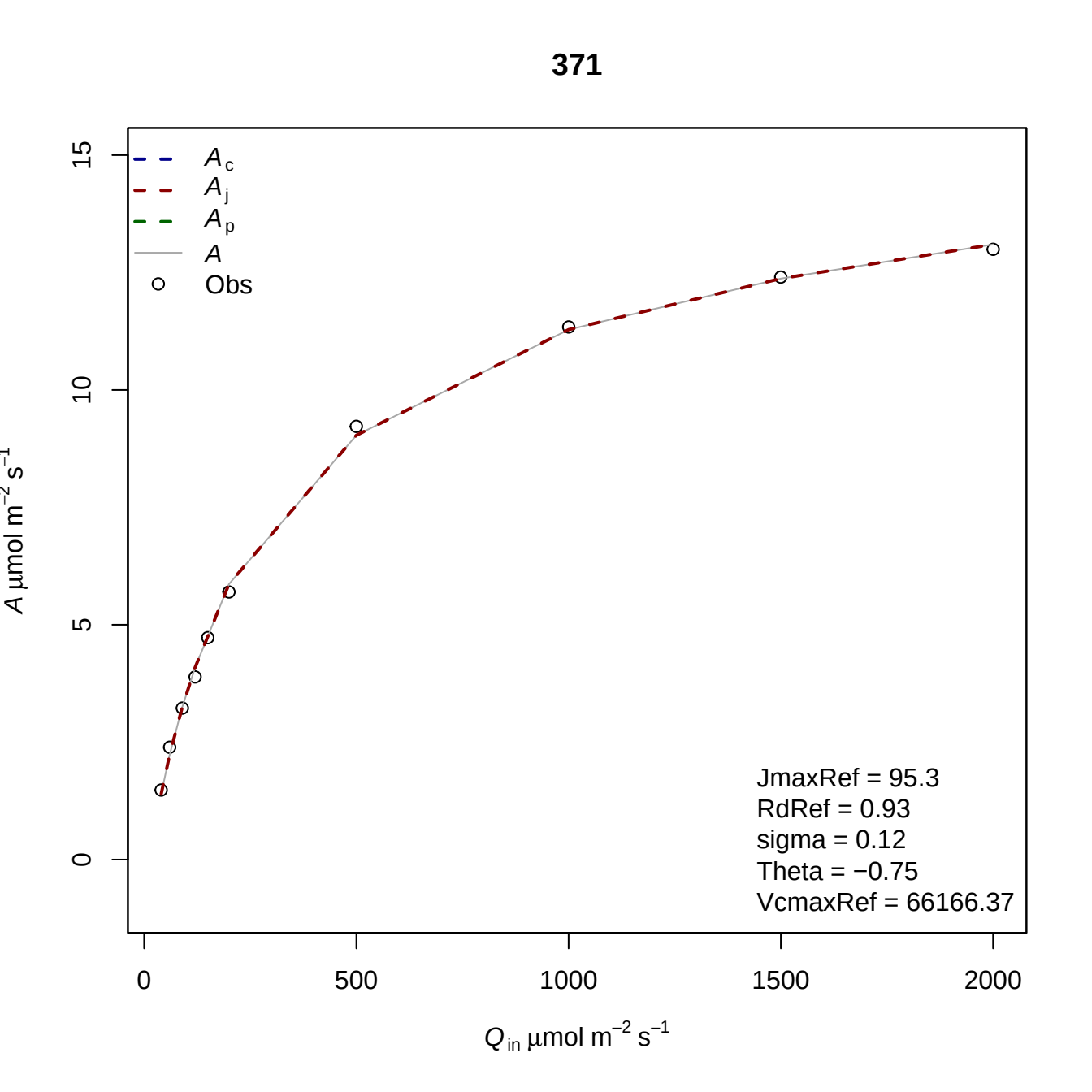

366
A_c
A_j
A_p
A
Obs
JmaxRef = 96.83
RdRef = 2.81
sigma = 0.1
Theta = 0.67
VcmaxRef = 72.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

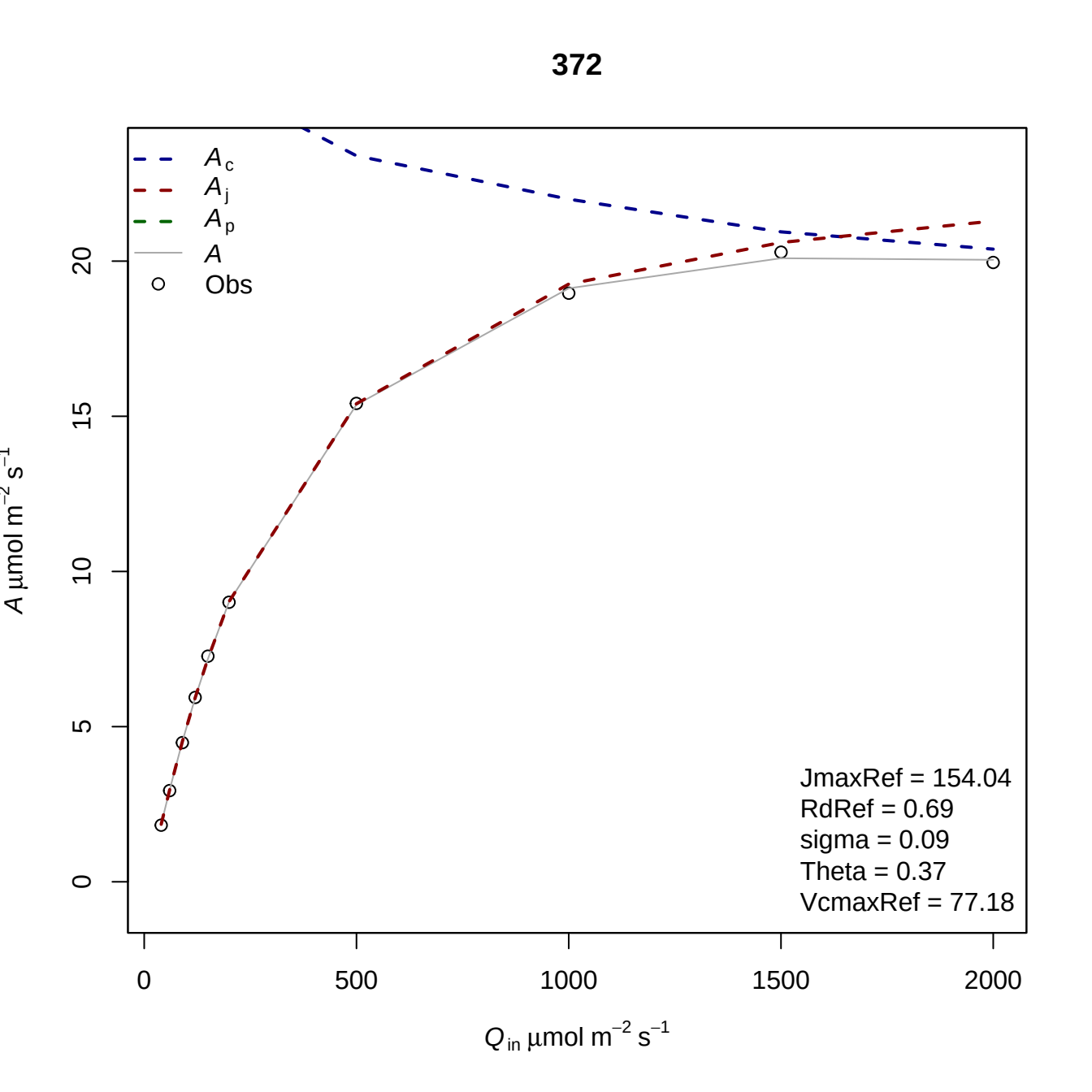

367
A_c
A_j
A_p
A
Obs
JmaxRef = 141.12
RdRef = 2.32
sigma = 0.2
Theta = 0.67
VcmaxRef = 588.02
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

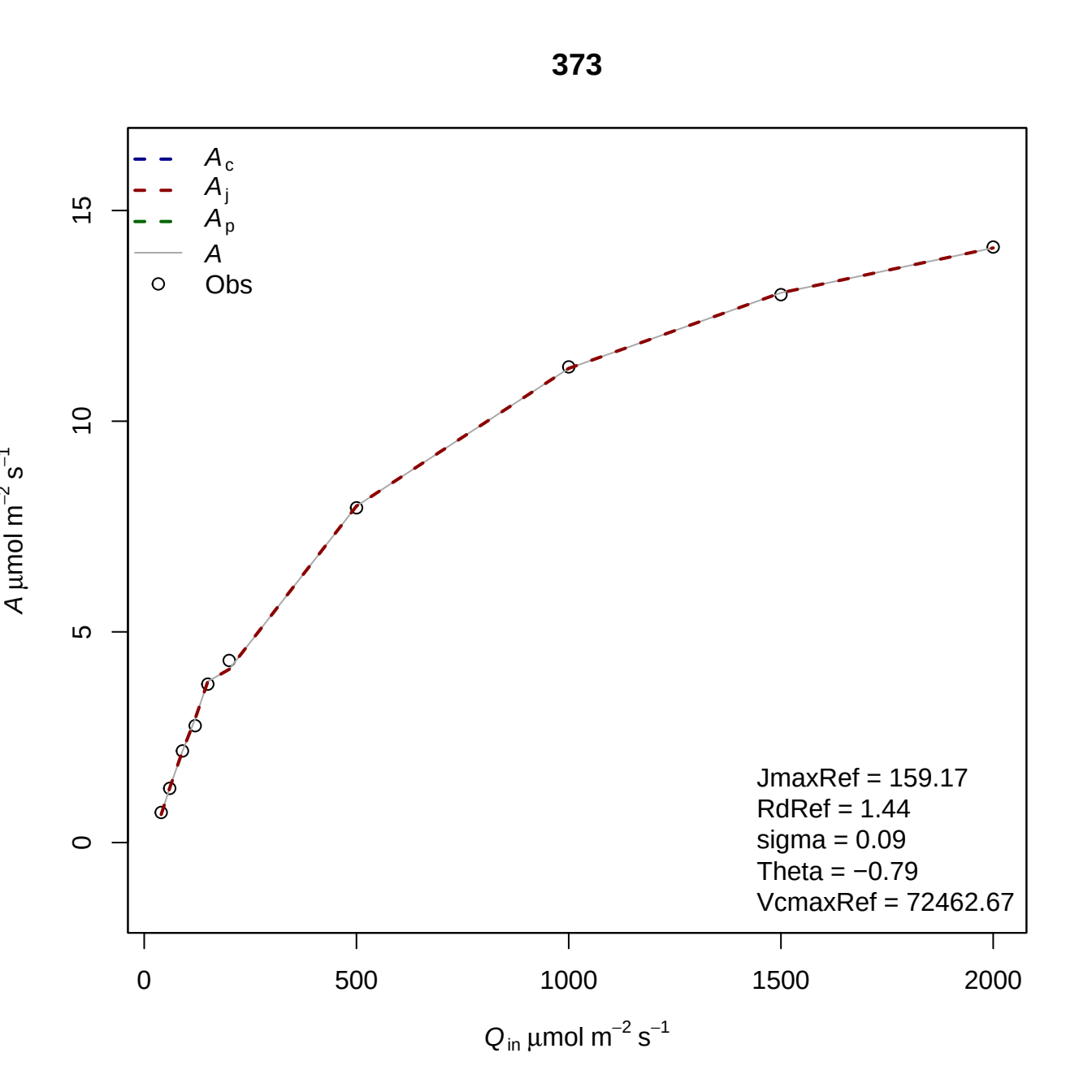

368
A_c
A_j
A_p
A
Obs
JmaxRef = 172.1
RdRef = 1.48
sigma = 0.17
Theta = 0.59
VcmaxRef = 71.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

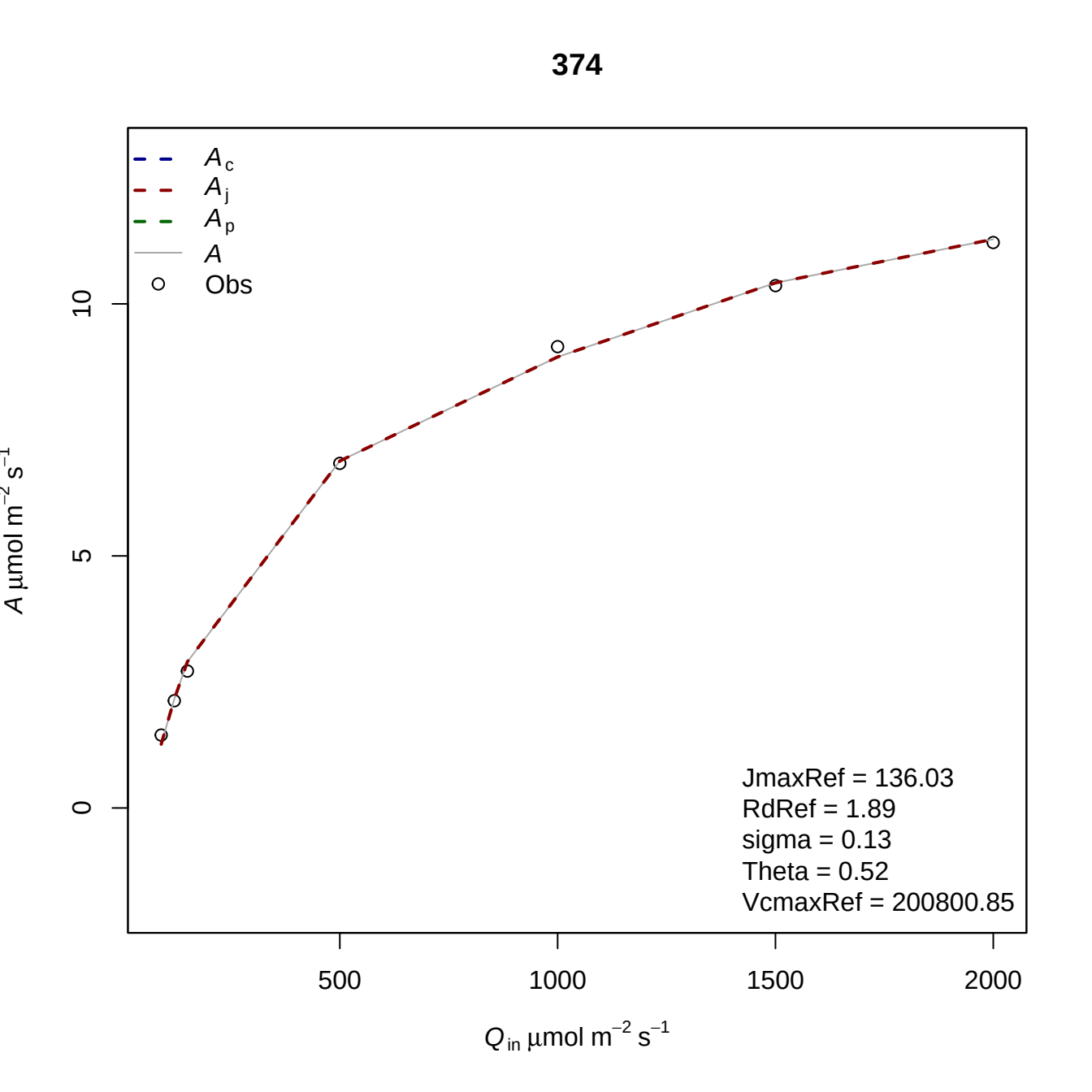

369
A_c
A_j
A_p
A
Obs
JmaxRef = 181.85
RdRef = 2.68
sigma = 0.15
Theta = 0.81
VcmaxRef = 118.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

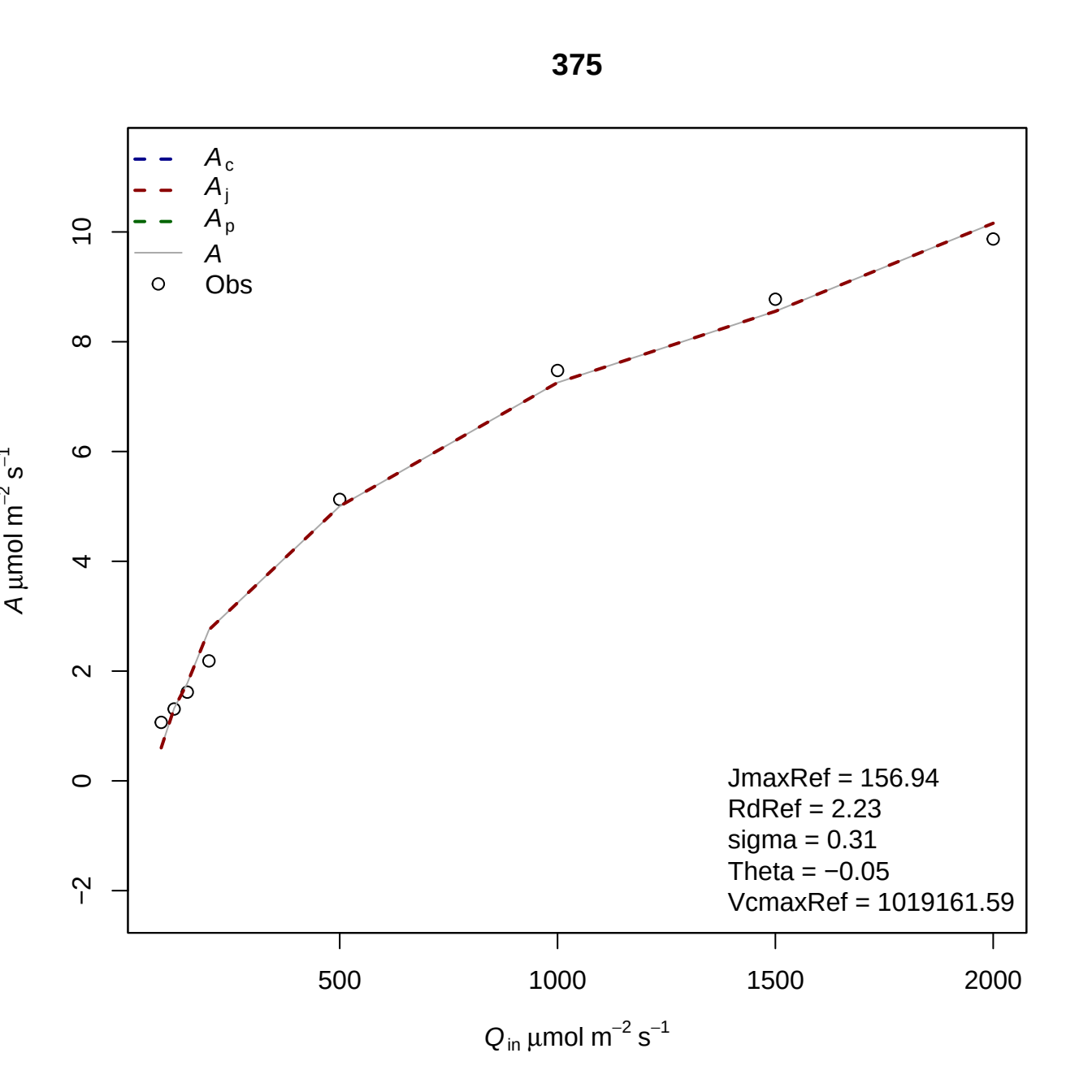

370
A_c
A_j
A_p
A
Obs
JmaxRef = 172.49
RdRef = 2.46
sigma = 0.09
Theta = 0.79
VcmaxRef = 113.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

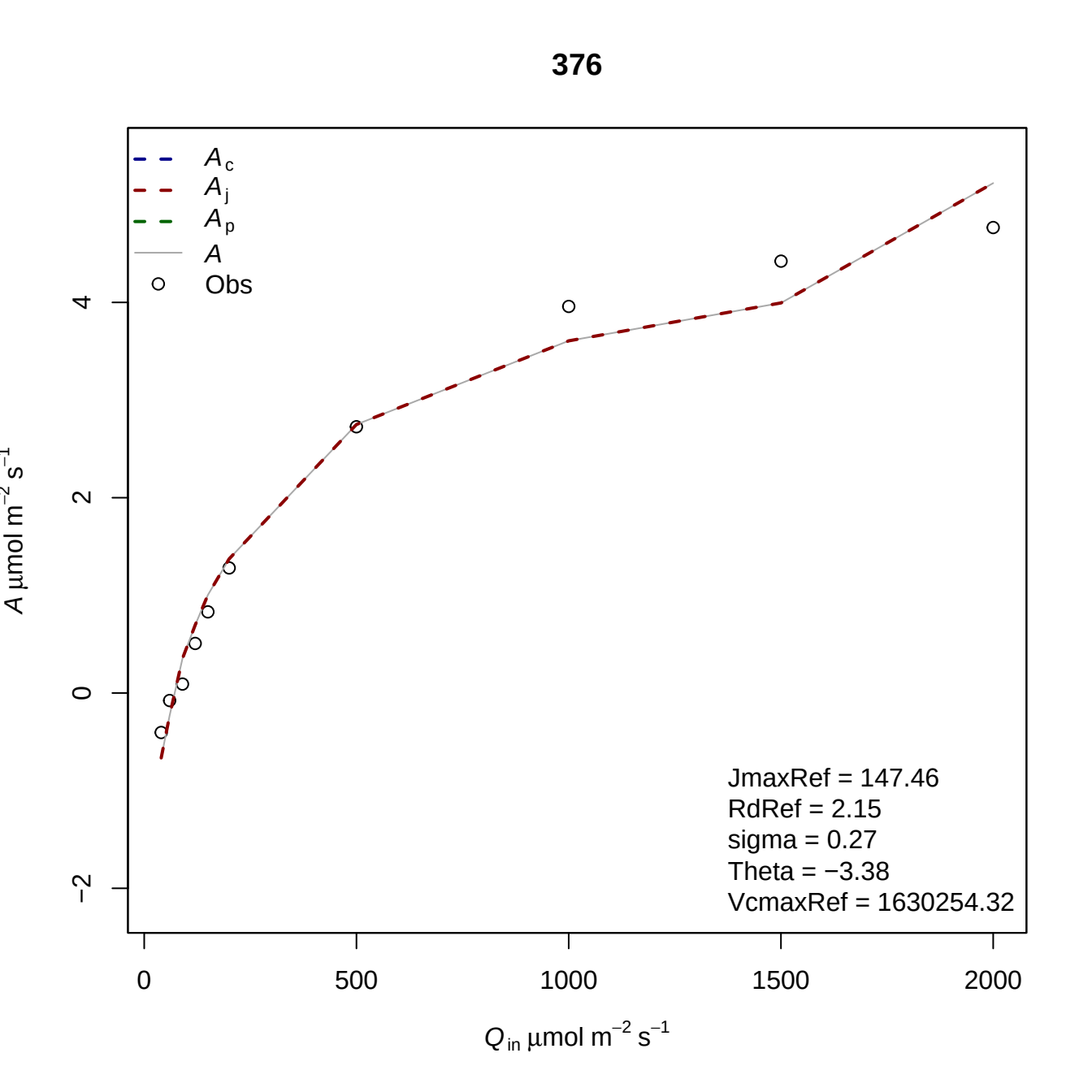

371
A_c
A_j
A_p
A
Obs
JmaxRef = 95.3
RdRef = 0.93
sigma = 0.12
Theta = −0.75
VcmaxRef = 66166.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

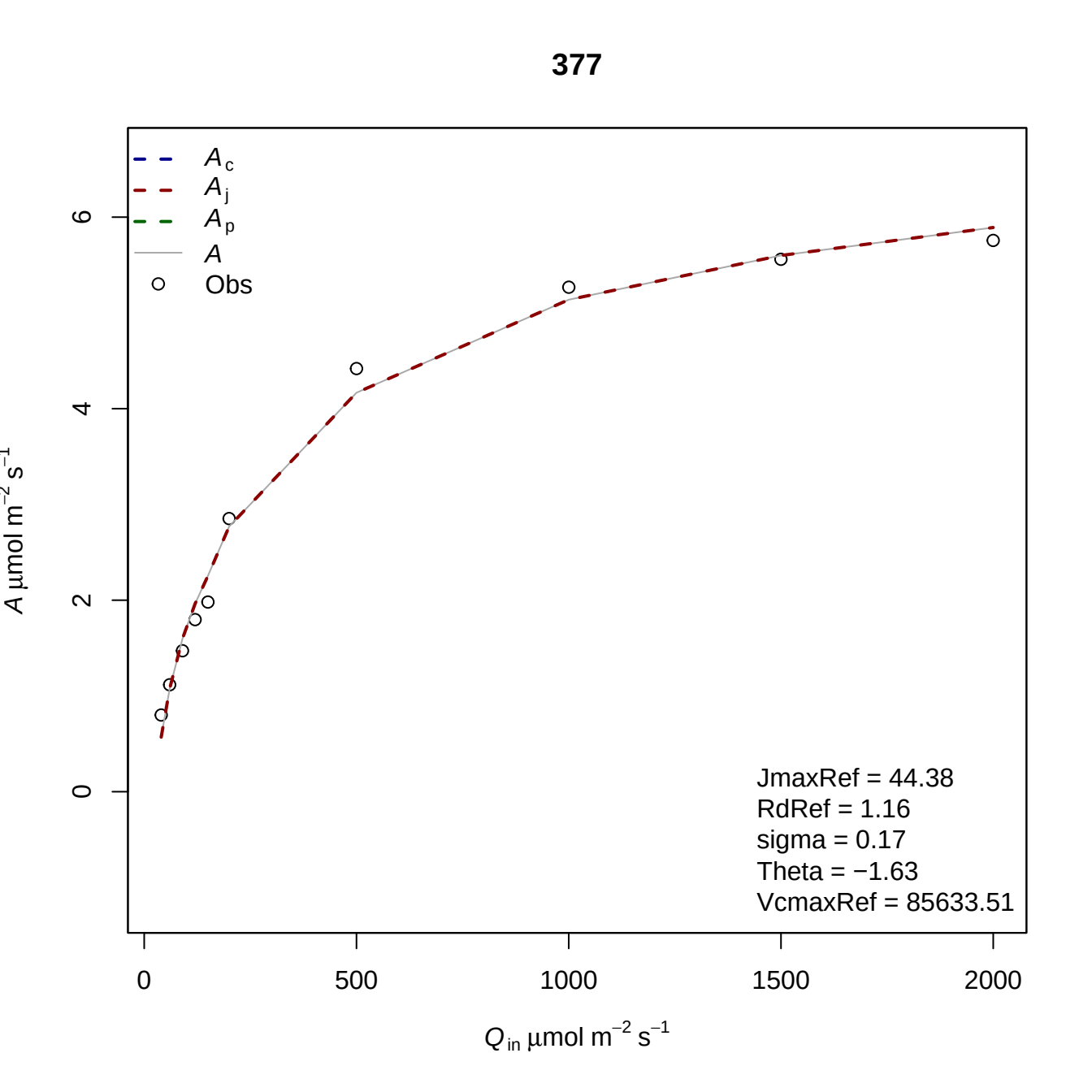

372
A_c
A_j
A_p
A
Obs
JmaxRef = 154.04
RdRef = 0.69
sigma = 0.09
Theta = 0.37
VcmaxRef = 77.18
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

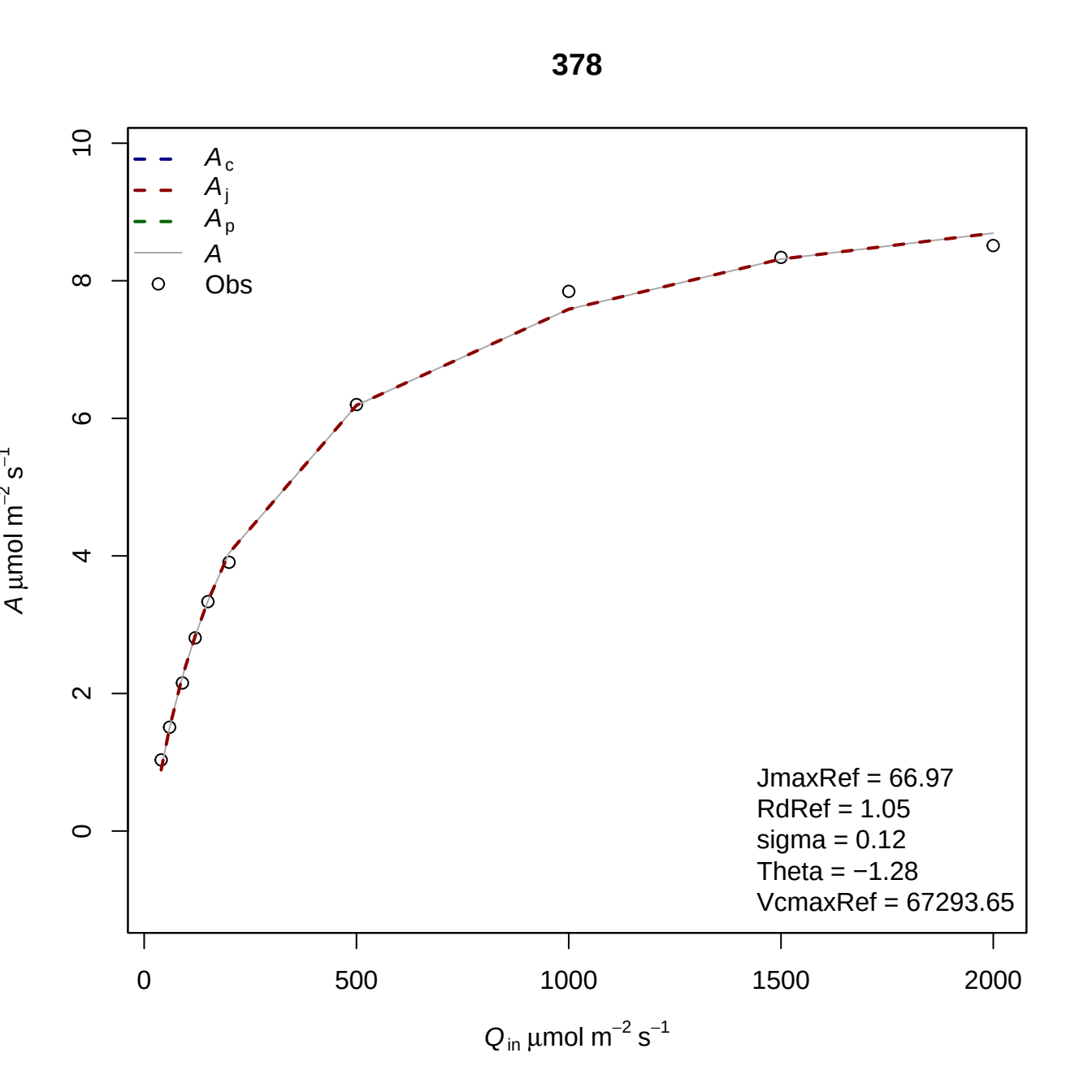

373
A_c
A_j
A_p
A
Obs
JmaxRef = 159.17
RdRef = 1.44
sigma = 0.09
Theta = −0.79
VcmaxRef = 72462.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

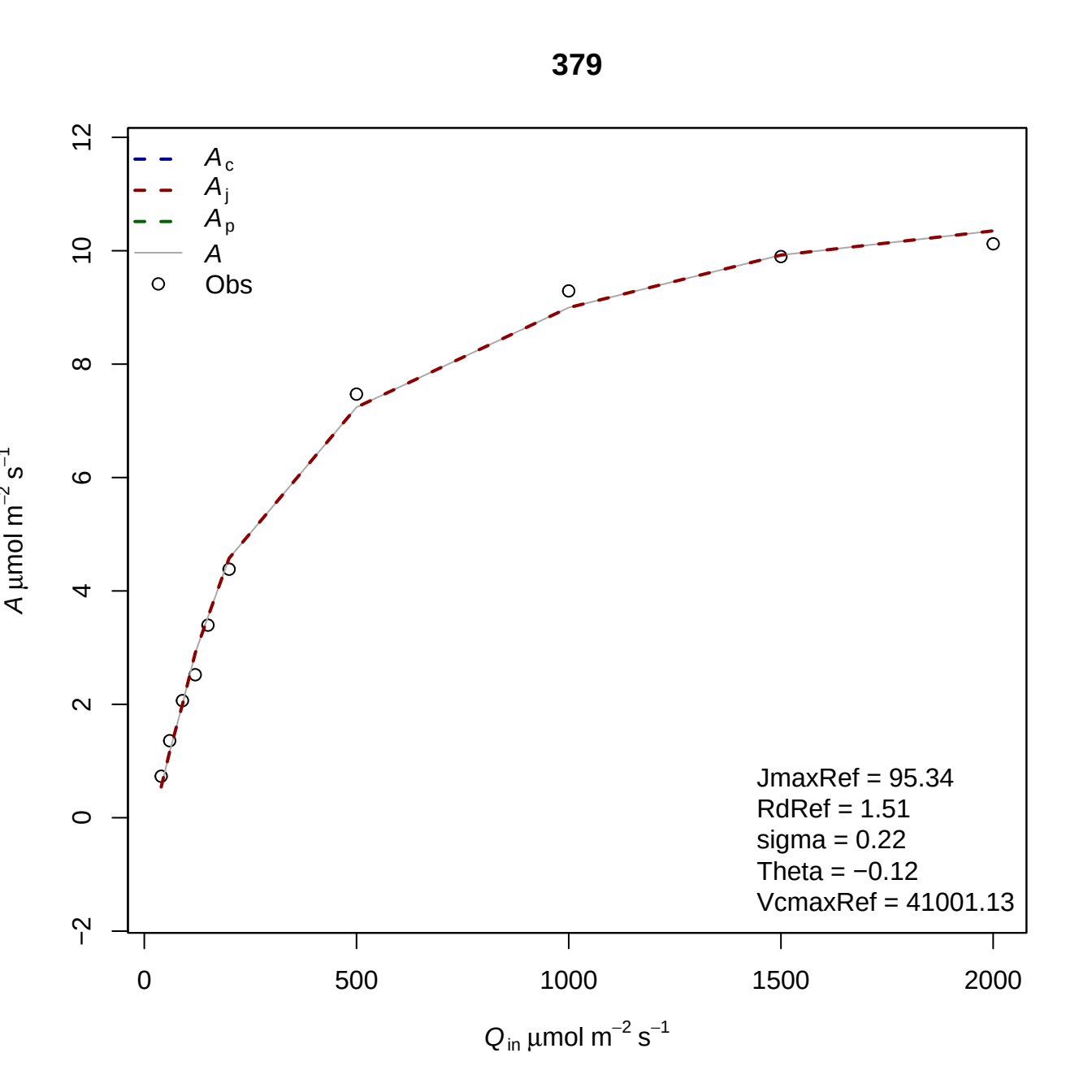

374
A_c
A_j
A_p
A
Obs
JmaxRef = 136.03
RdRef = 1.89
sigma = 0.13
Theta = 0.52
VcmaxRef = 200800.85
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

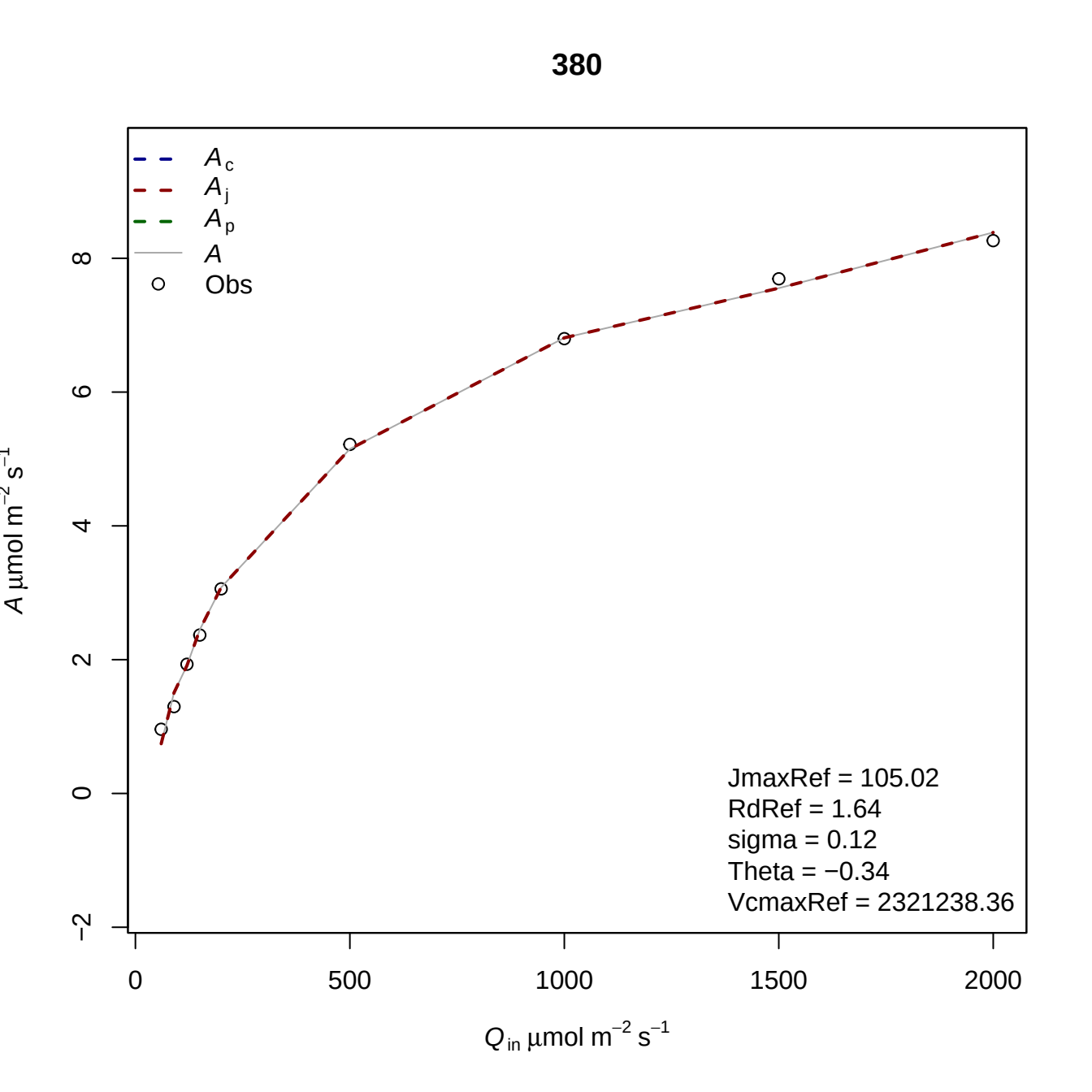

375
A_c
A_j
A_p
A
Obs
JmaxRef = 156.94
RdRef = 2.23
sigma = 0.31
Theta = −0.05
VcmaxRef = 1019161.59
−2
0
2
4
6
8
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

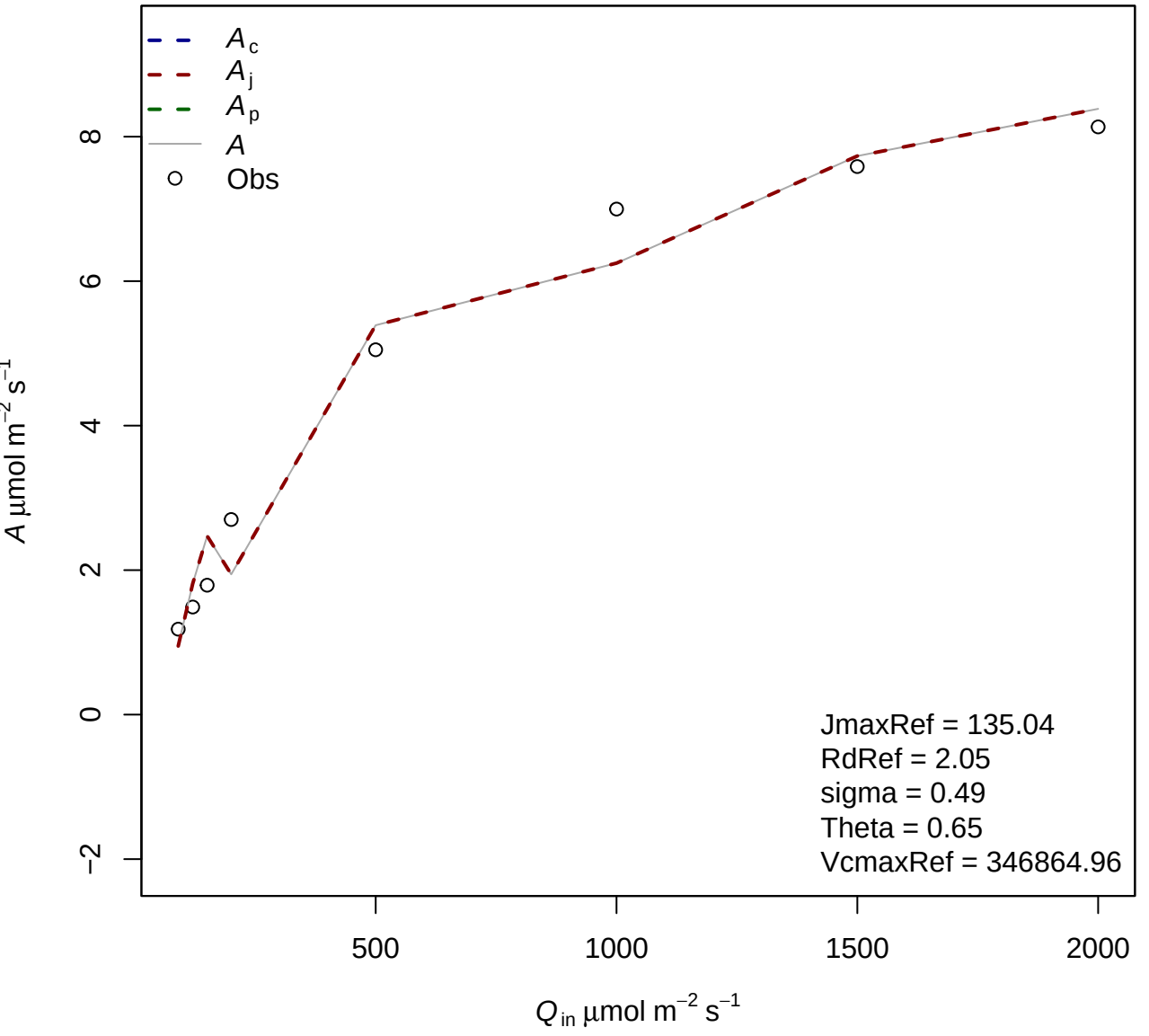

376
A_c
A_j
A_p
A
Obs
JmaxRef = 147.46
RdRef = 2.15
sigma = 0.27
Theta = −3.38
VcmaxRef = 1630254.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

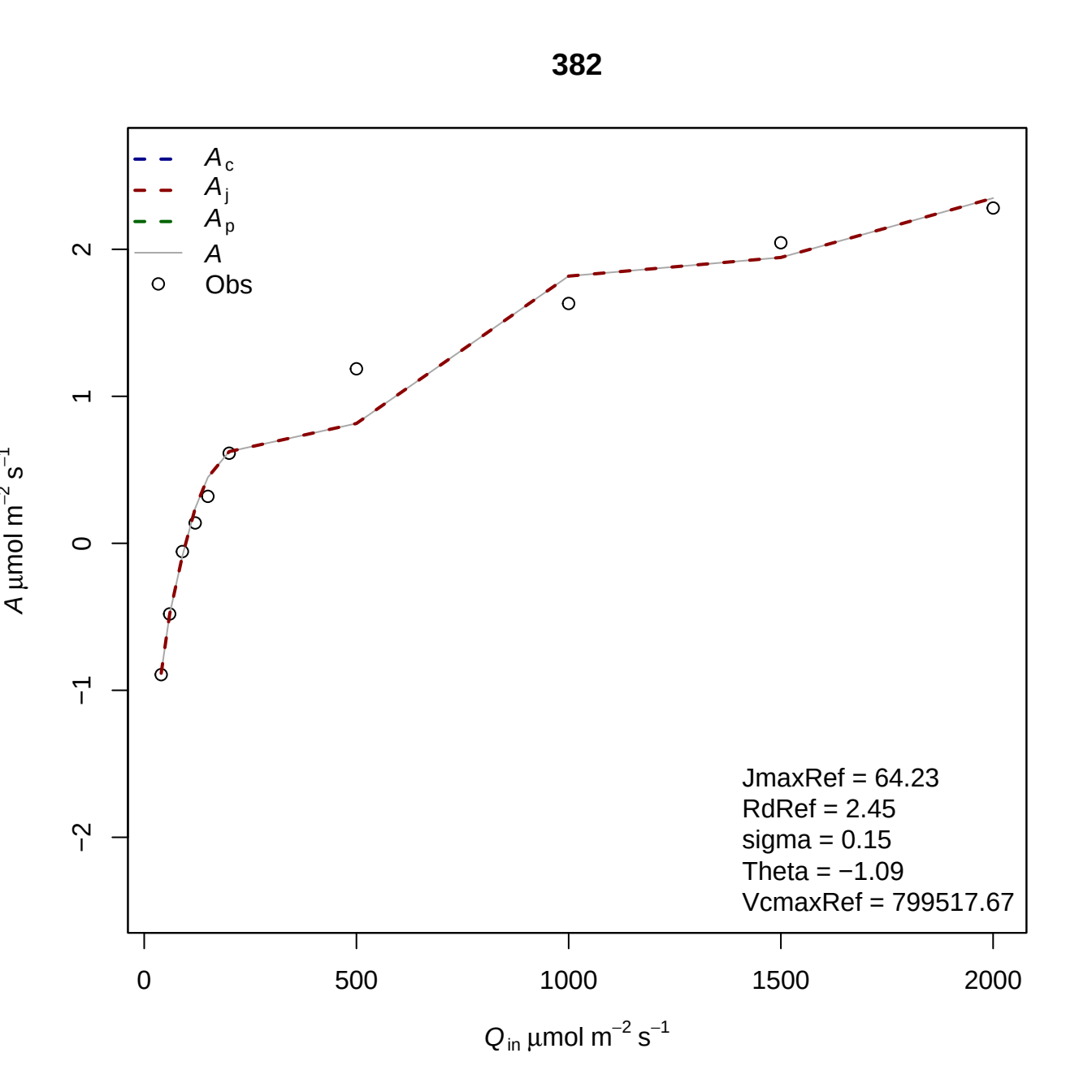

377
A_c
A_j
A_p
A
Obs
JmaxRef = 44.38
RdRef = 1.16
sigma = 0.17
Theta = −1.63
VcmaxRef = 85633.51
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

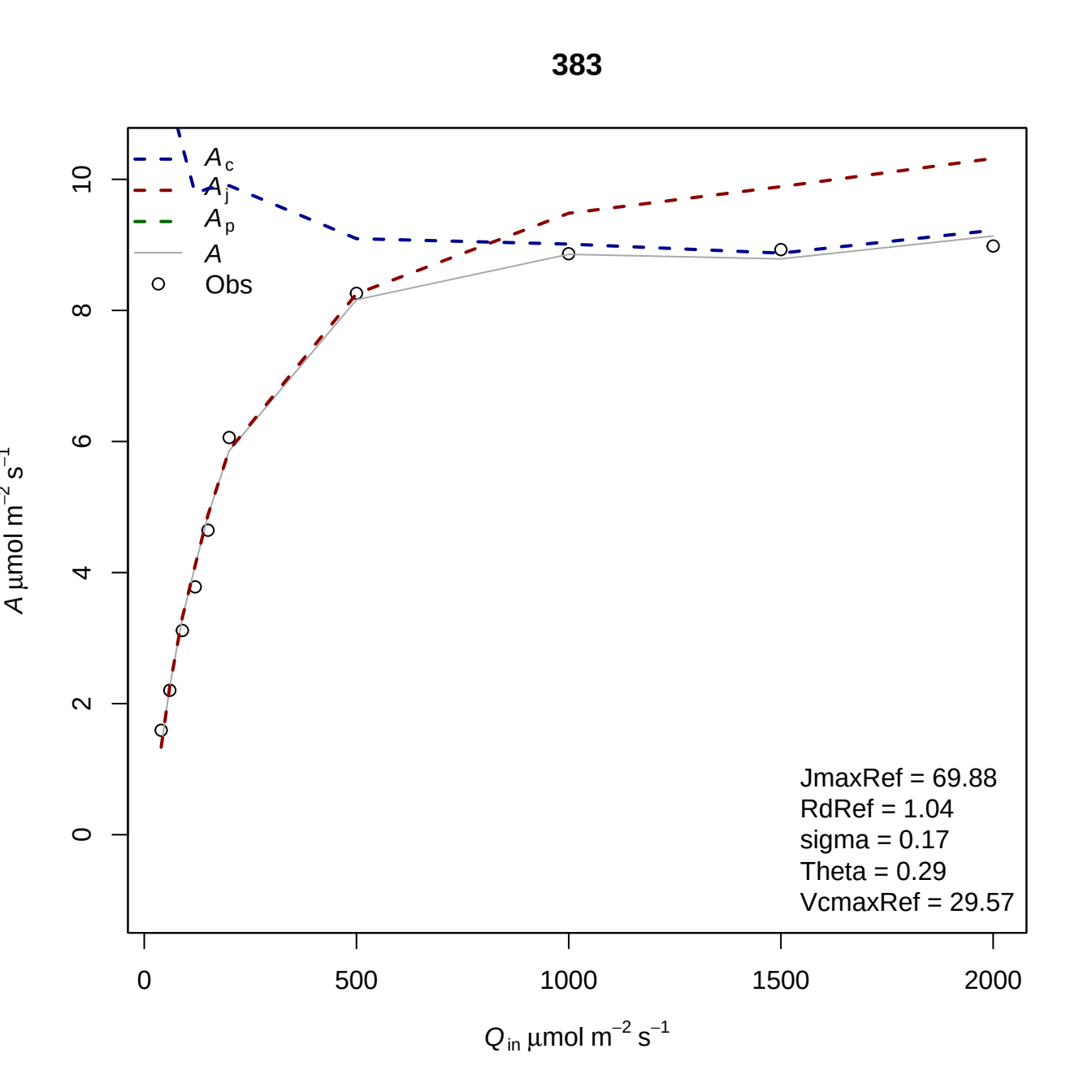

378
A_c
A_j
A_p
A
Obs
JmaxRef = 66.97
RdRef = 1.05
sigma = 0.12
Theta = −1.28
VcmaxRef = 67293.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

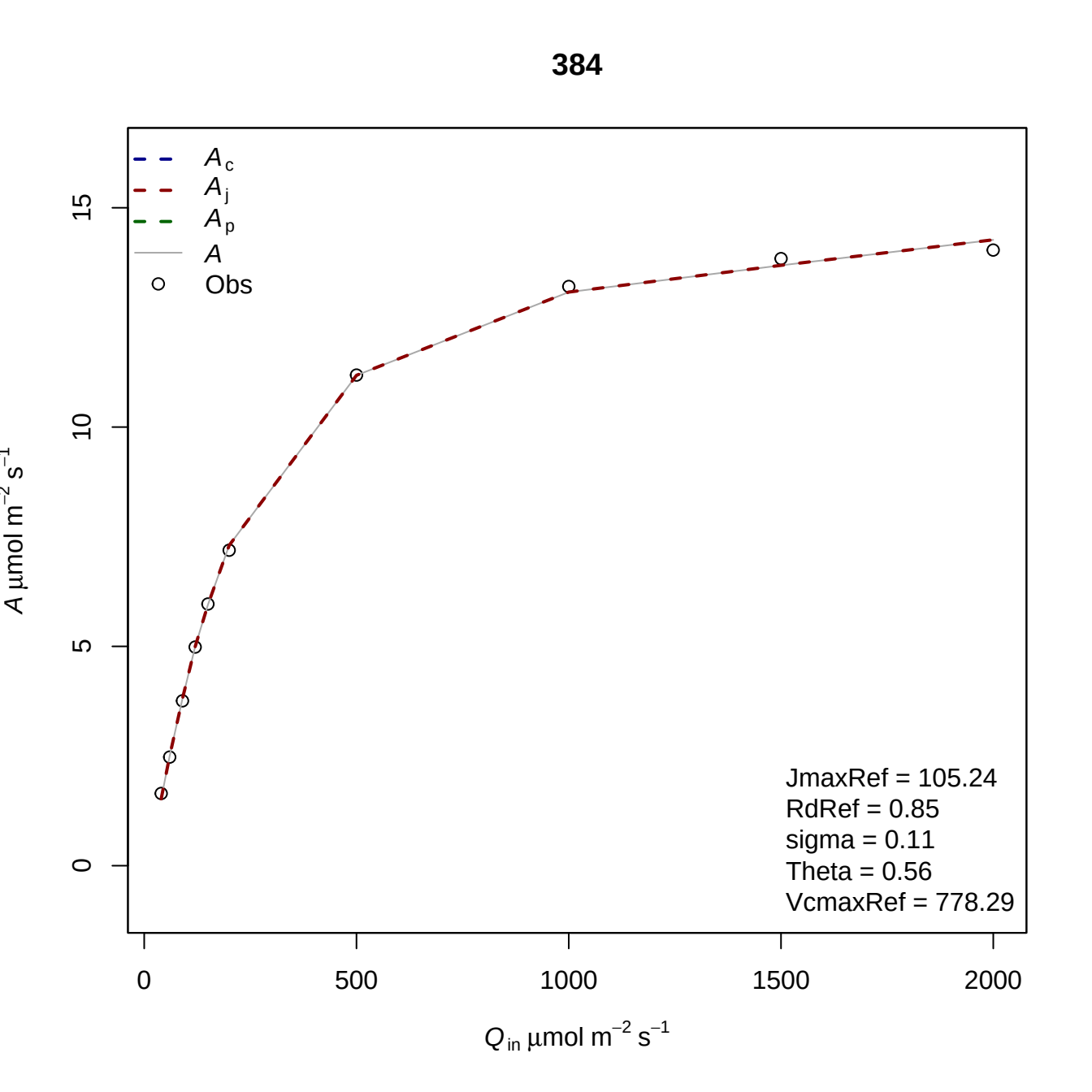

379
A_c
A_j
A_p
A
Obs
JmaxRef = 95.34
RdRef = 1.51
sigma = 0.22
Theta = −0.12
VcmaxRef = 41001.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

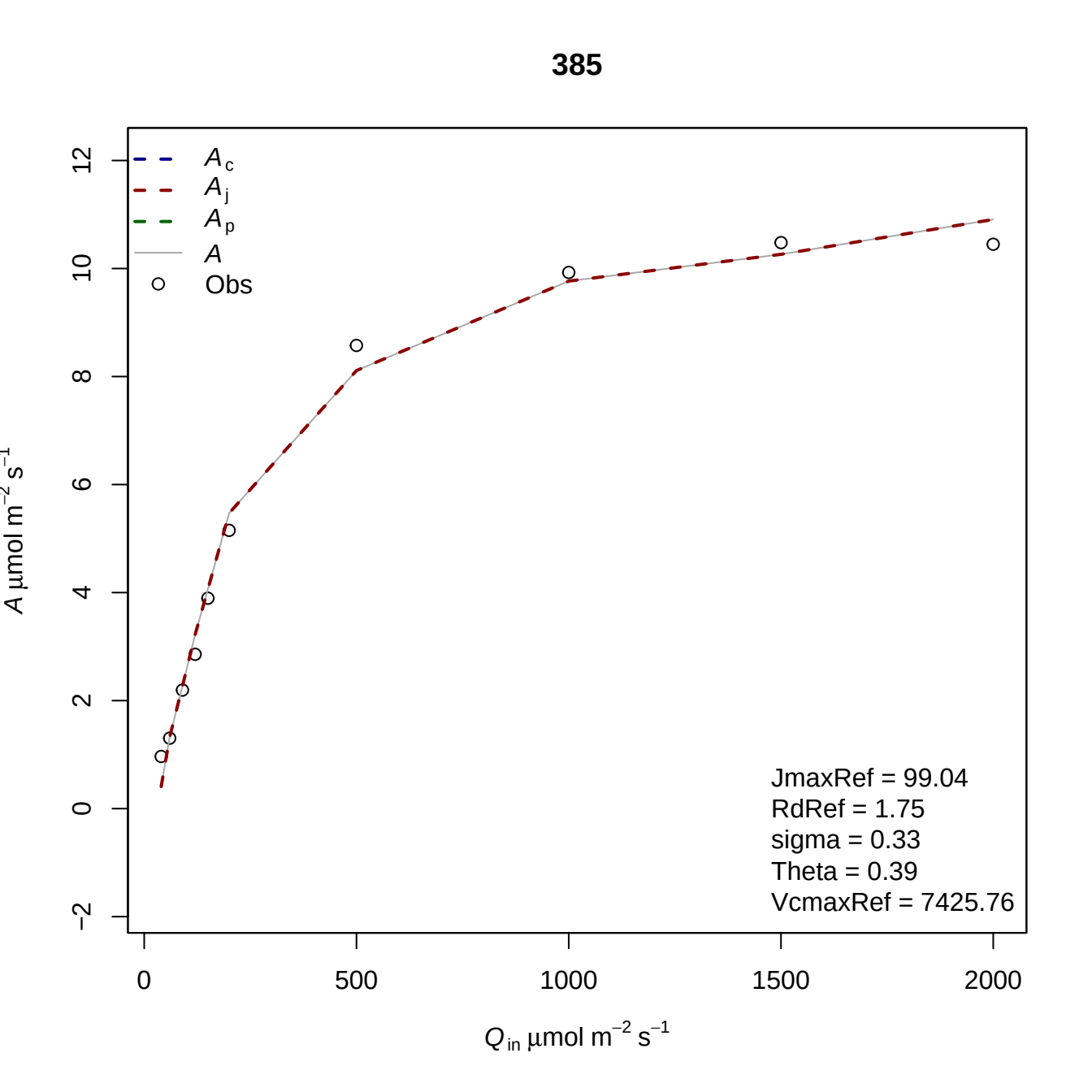

380
A_c
A_j
A_p
A
Obs
JmaxRef = 105.02
RdRef = 1.64
sigma = 0.12
Theta = −0.34
VcmaxRef = 2321238.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

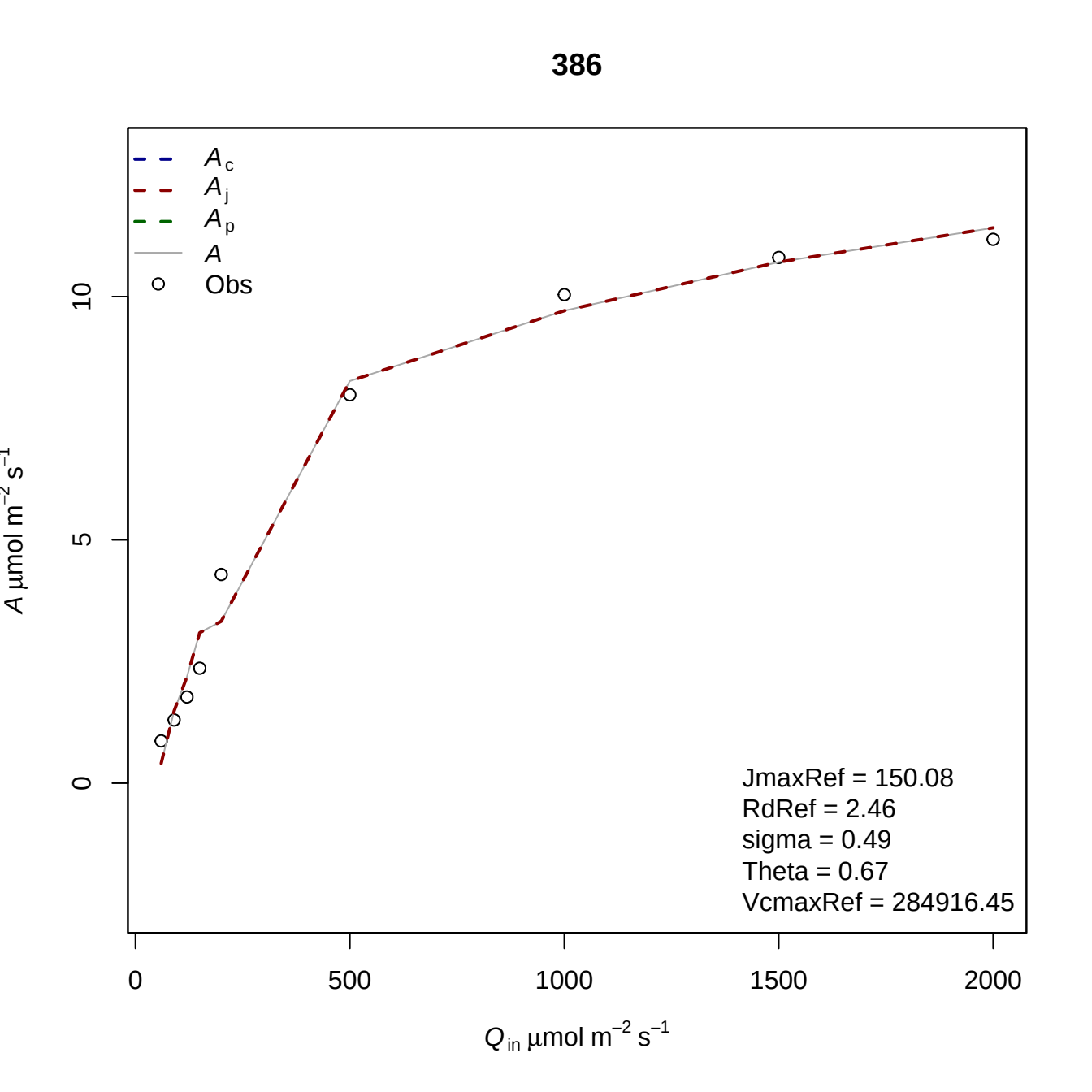

A_c
A_j
A_p
A
Obs
JmaxRef = 135.04
RdRef = 2.05
sigma = 0.49
Theta = 0.65
VcmaxRef = 346864.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
500
1000
1500
2000

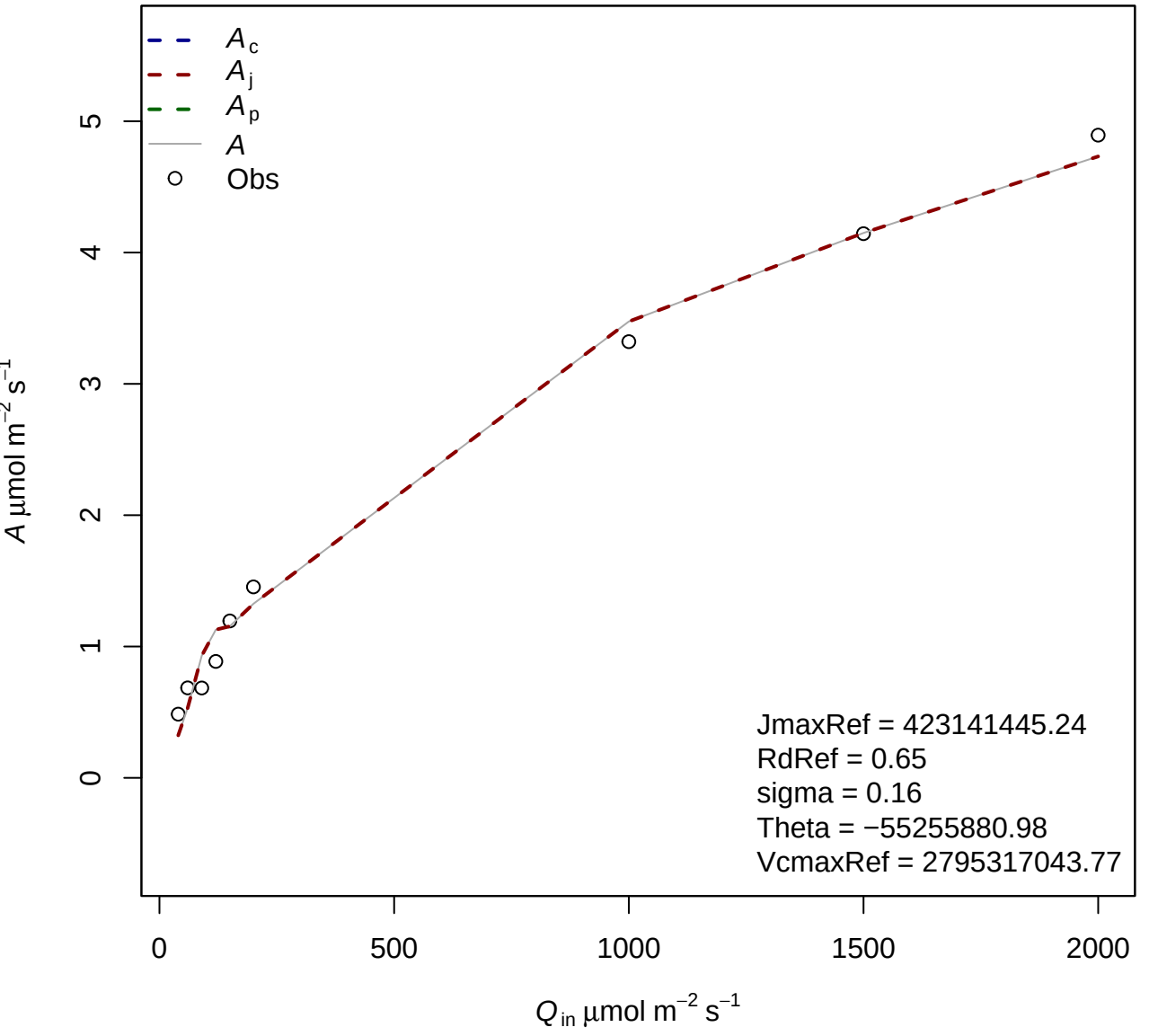

382
A_c
A_j
A_p
A
Obs
JmaxRef = 64.23
RdRef = 2.45
sigma = 0.15
Theta = −1.09
VcmaxRef = 799517.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500
2000

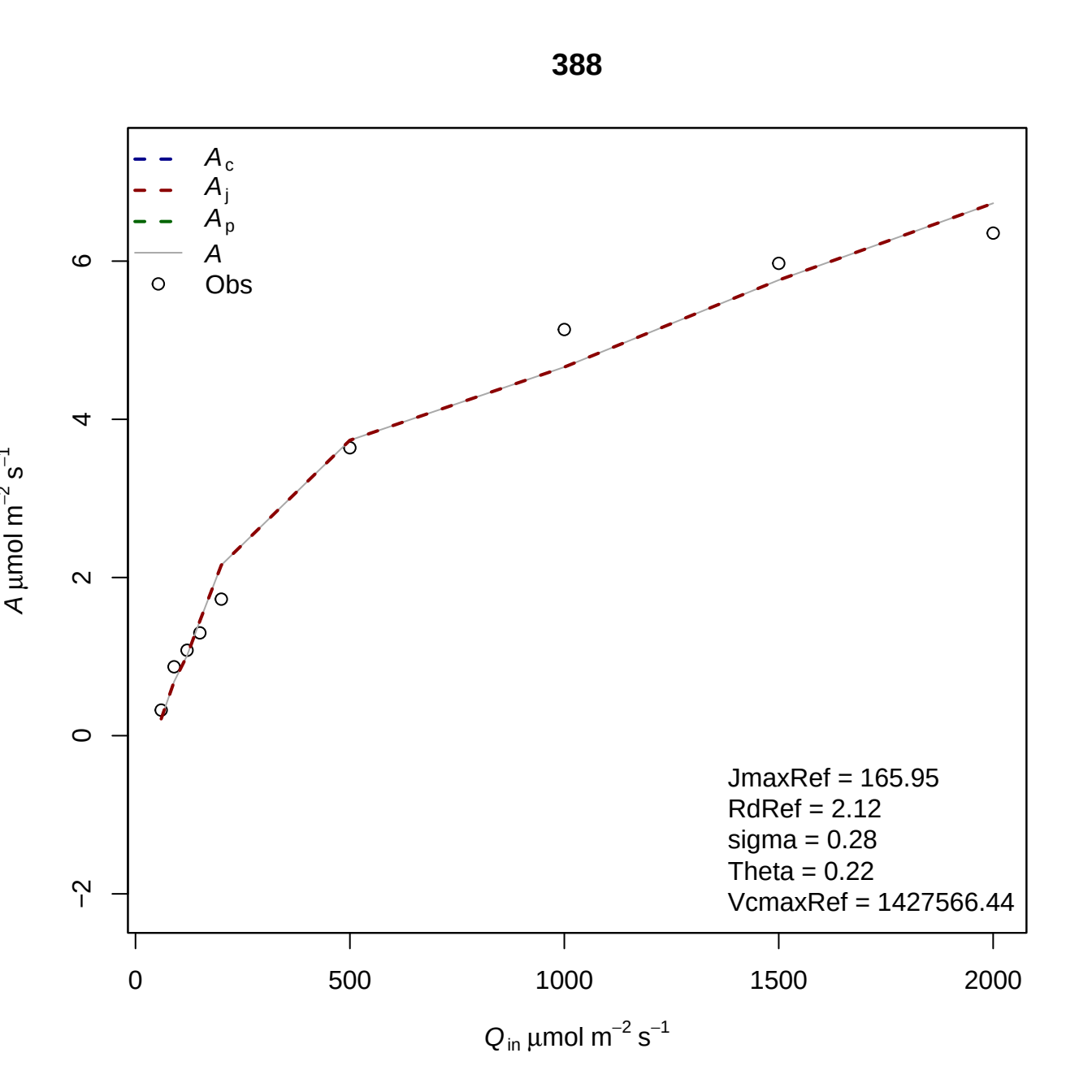

383
A_c
A_j
A_p
A
Obs
JmaxRef = 69.88
RdRef = 1.04
sigma = 0.17
Theta = 0.29
VcmaxRef = 29.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

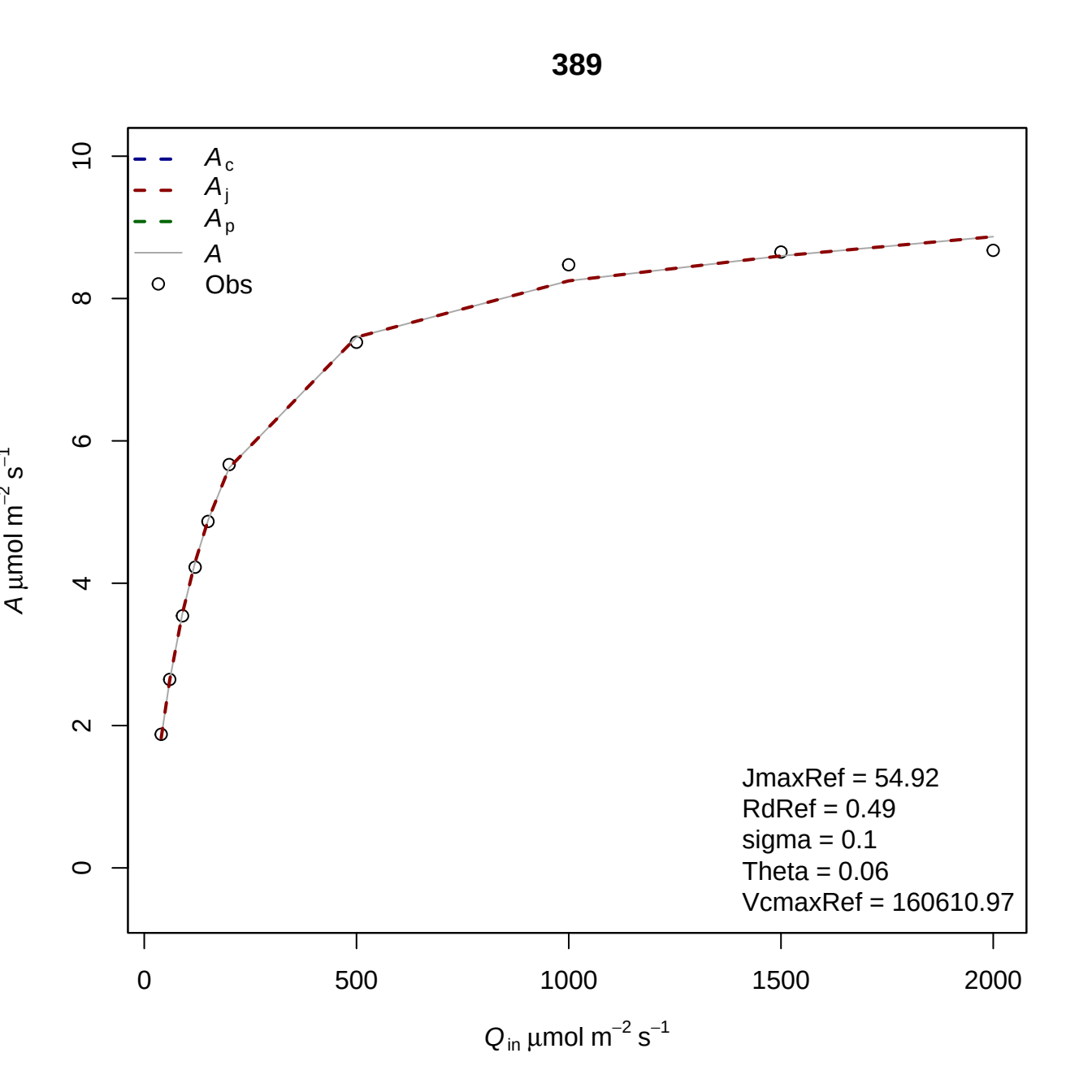

384
A_c
A_j
A_p
A
Obs
JmaxRef = 105.24
RdRef = 0.85
sigma = 0.11
Theta = 0.56
VcmaxRef = 778.29
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

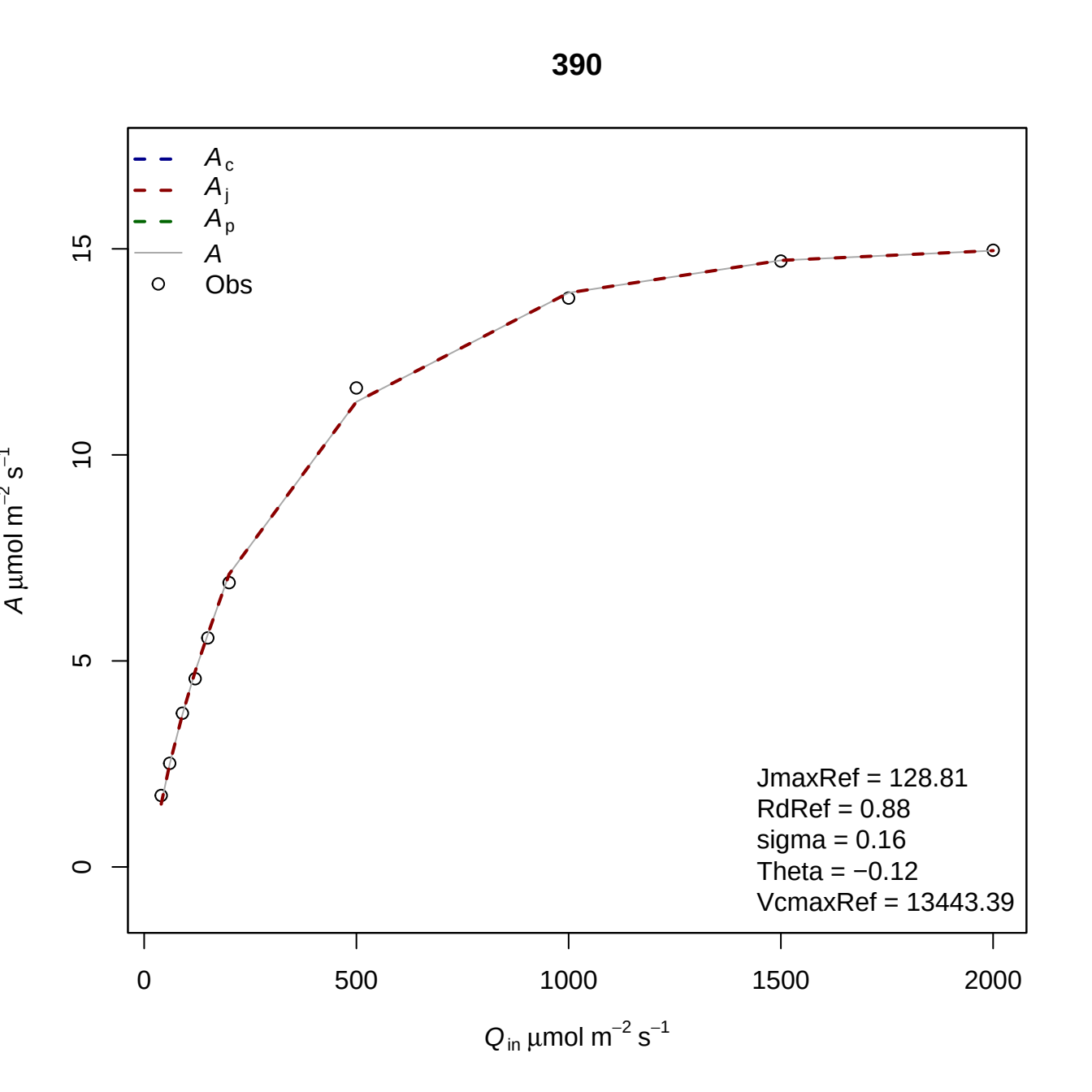

385
A_c
A_j
A_p
A
Obs
JmaxRef = 99.04
RdRef = 1.75
sigma = 0.33
Theta = 0.39
VcmaxRef = 7425.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

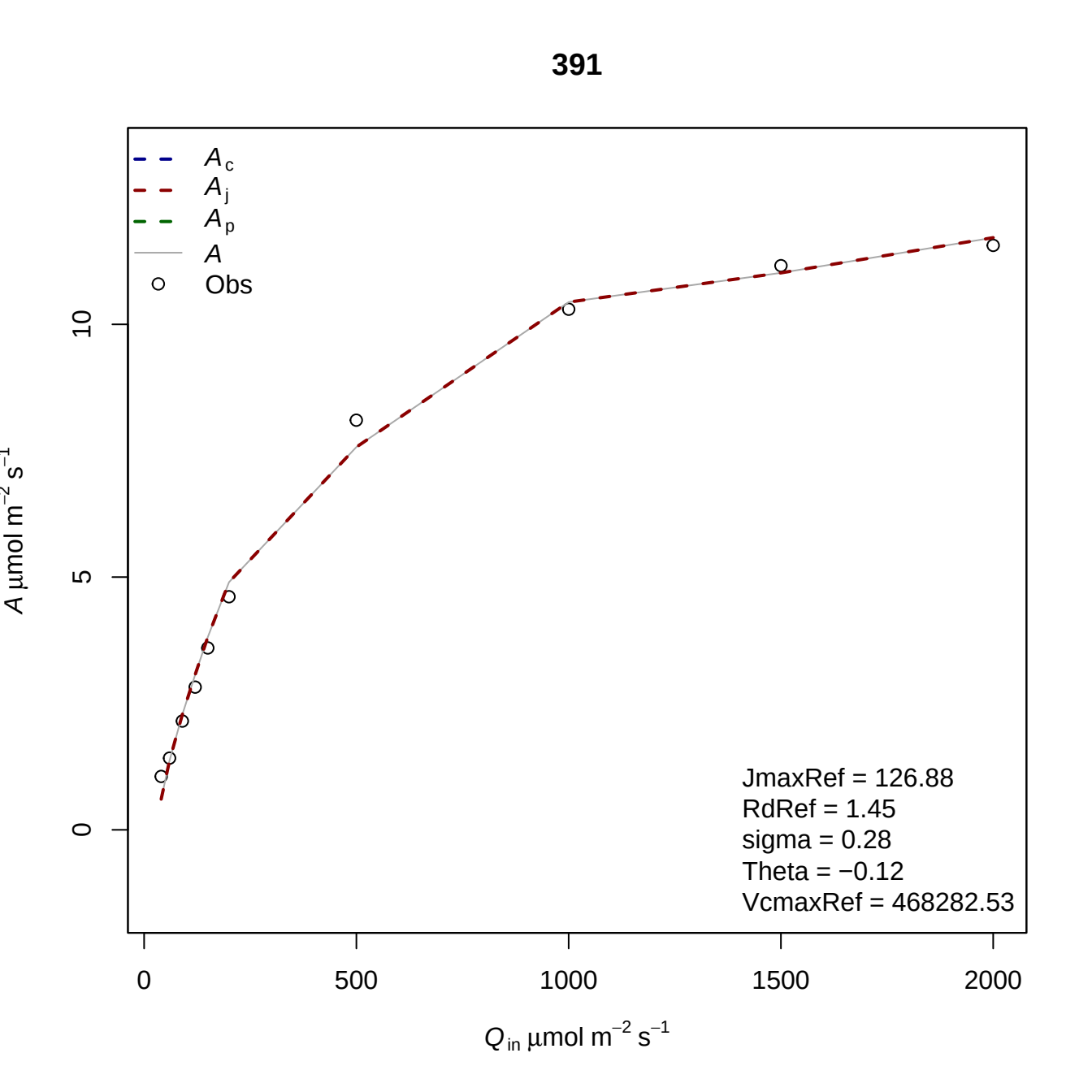

386
A_c
A_j
A_p
A
Obs
JmaxRef = 150.08
RdRef = 2.46
sigma = 0.49
Theta = 0.67
VcmaxRef = 284916.45
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

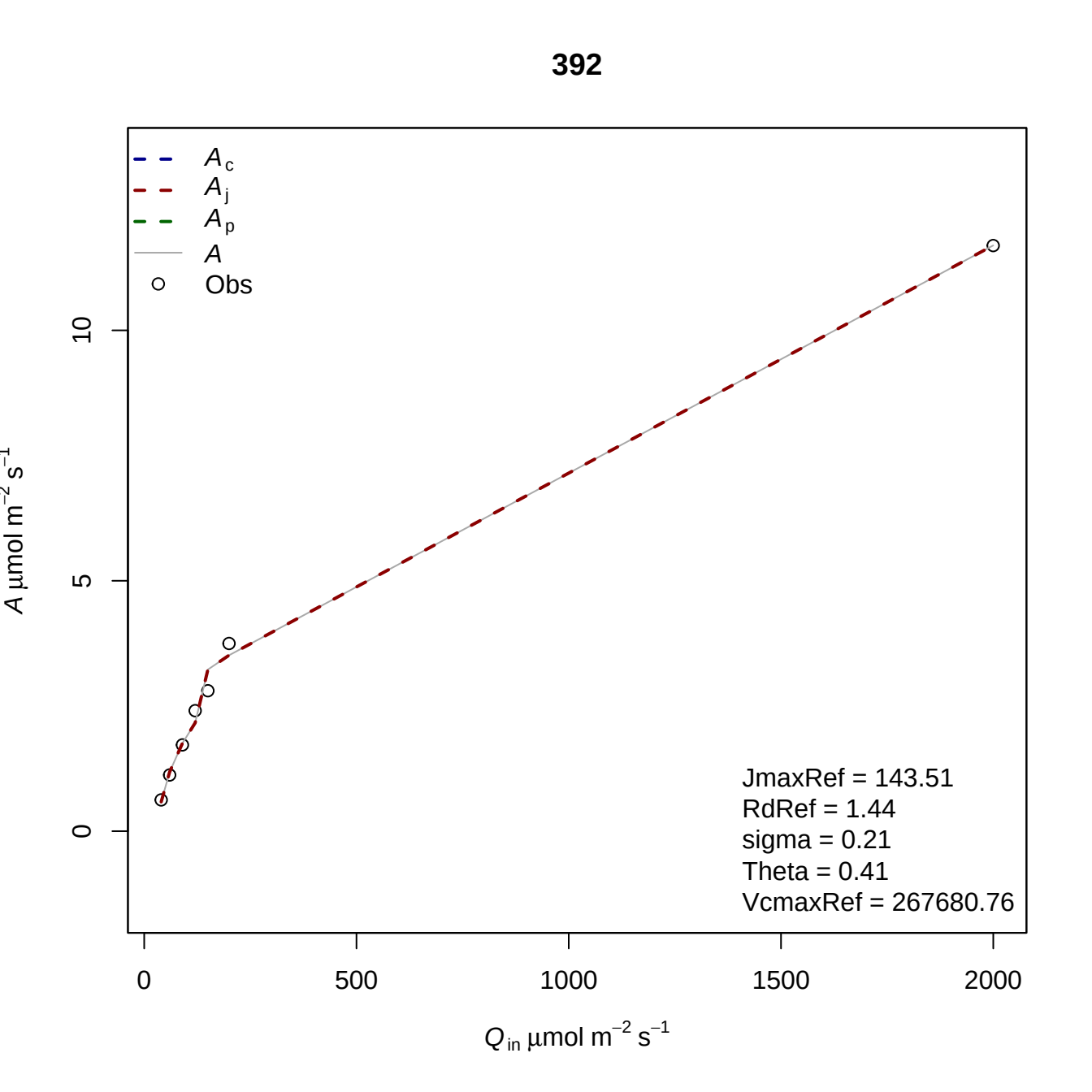

A_c
A_j
A_p
A
Obs
JmaxRef = 423141445.24
RdRef = 0.65
sigma = 0.16
Theta = −55255880.98
VcmaxRef = 2795317043.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

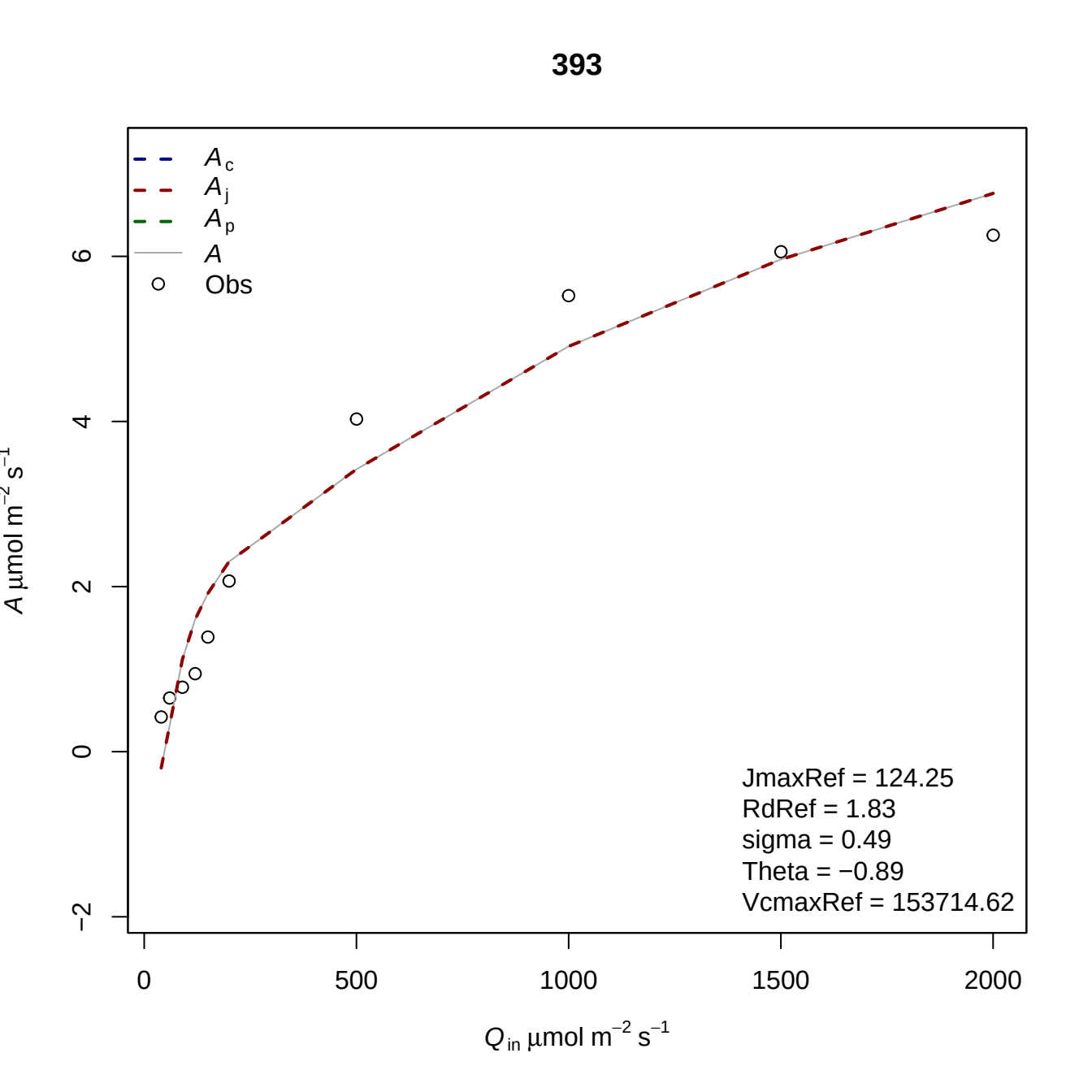

388
A_c
A_j
A_p
A
Obs
JmaxRef = 165.95
RdRef = 2.12
sigma = 0.28
Theta = 0.22
VcmaxRef = 1427566.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

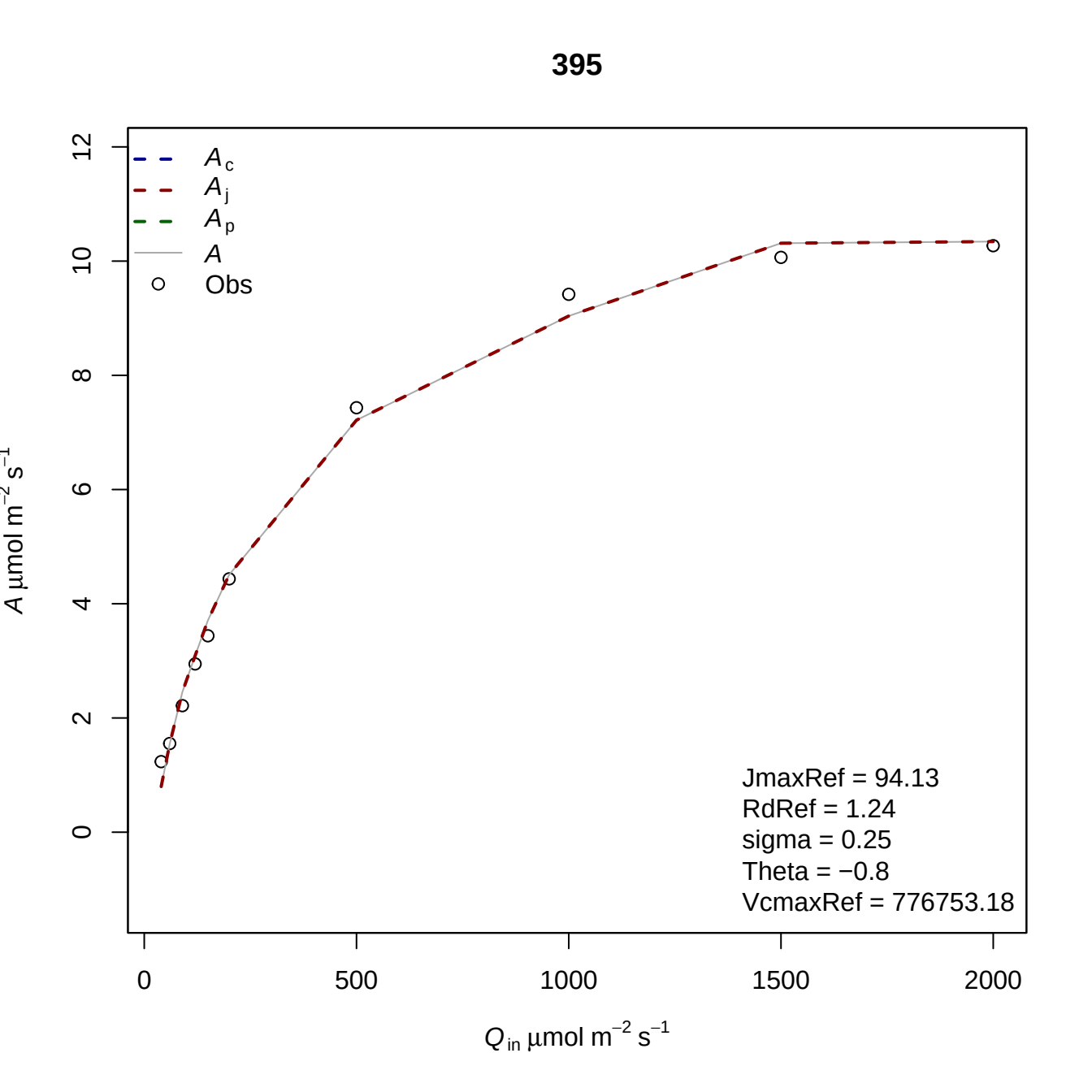

389
A_c
A_j
A_p
A
Obs
JmaxRef = 54.92
RdRef = 0.49
sigma = 0.1
Theta = 0.06
VcmaxRef = 160610.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

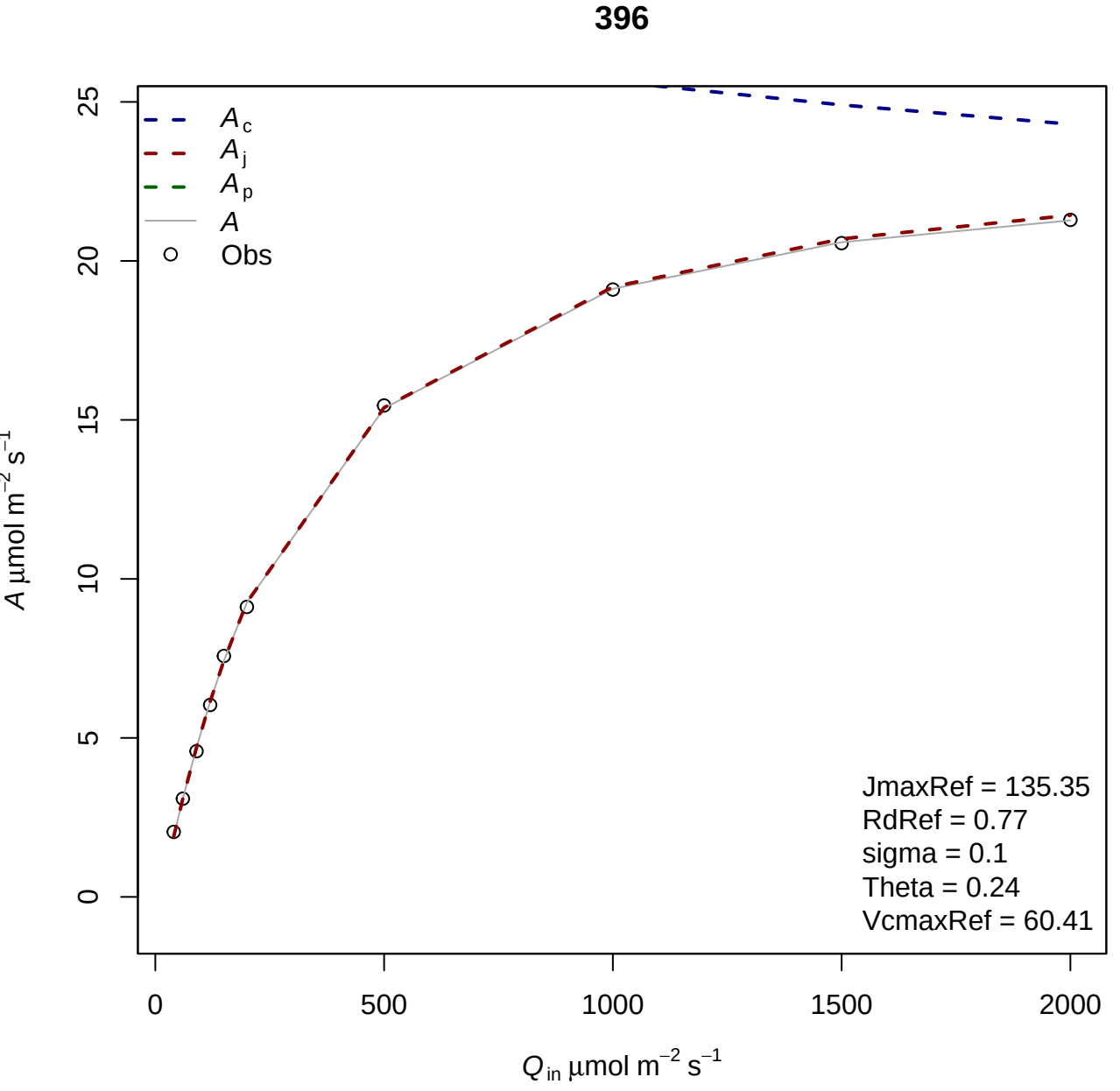

390
A_c
A_j
A_p
A
Obs
JmaxRef = 128.81
RdRef = 0.88
sigma = 0.16
Theta = −0.12
VcmaxRef = 13443.39
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

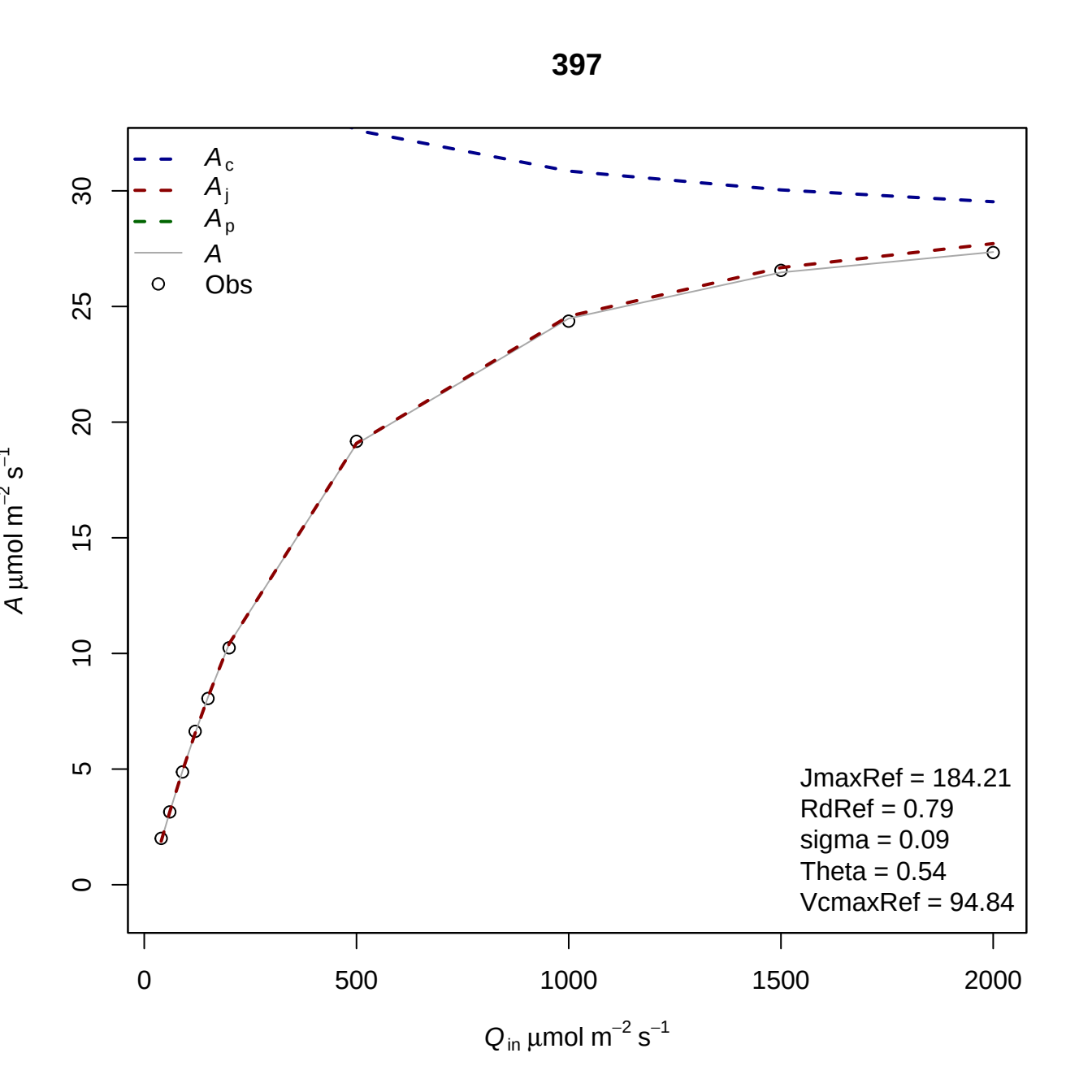

391
A_c
A_j
A_p
A
Obs
JmaxRef = 126.88
RdRef = 1.45
sigma = 0.28
Theta = −0.12
VcmaxRef = 468282.53
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

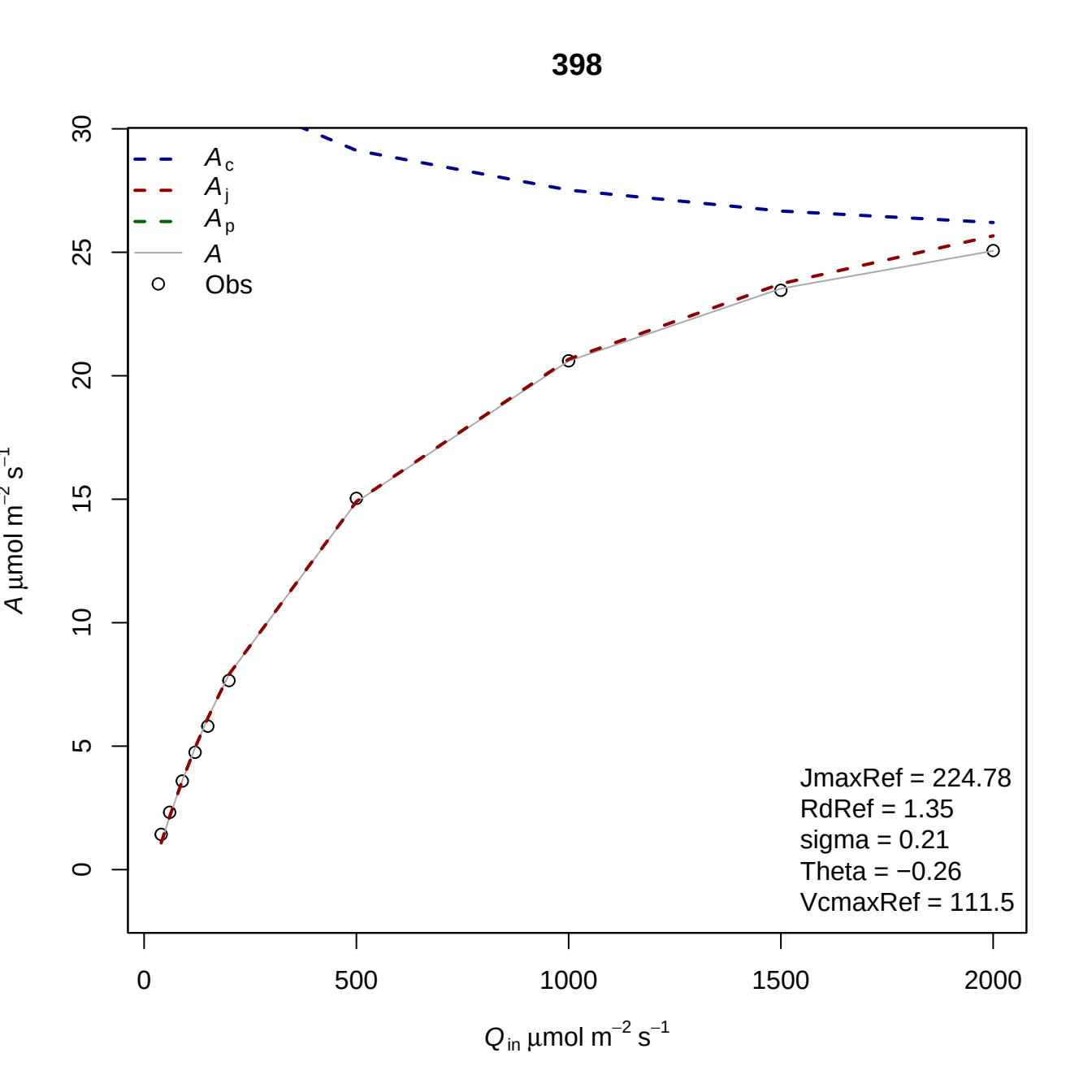

392
A_c
A_j
A_p
A
Obs
JmaxRef = 143.51
RdRef = 1.44
sigma = 0.21
Theta = 0.41
VcmaxRef = 267680.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

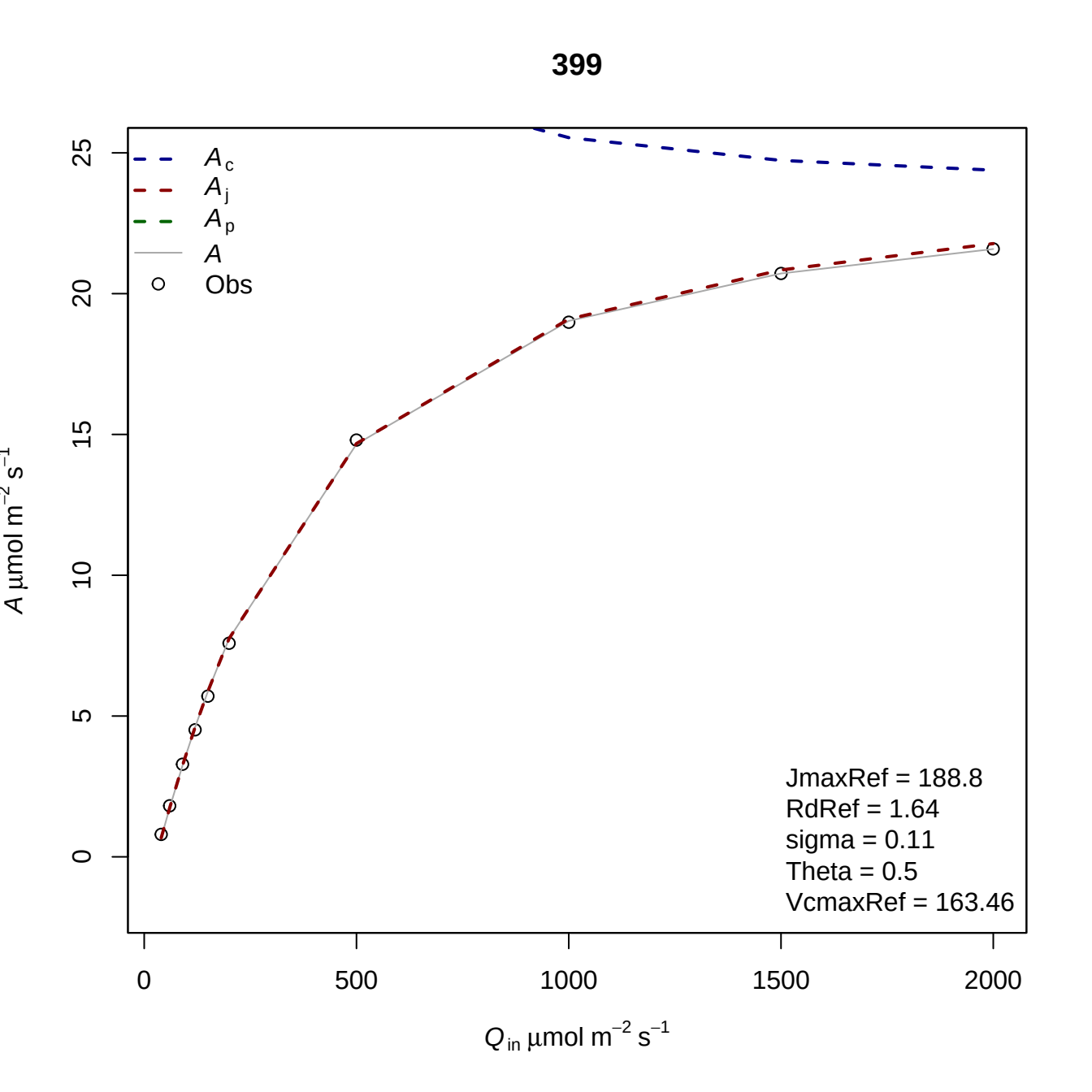

393
A_c
A_j
A_p
A
Obs
JmaxRef = 124.25
RdRef = 1.83
sigma = 0.49
Theta = −0.89
VcmaxRef = 153714.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

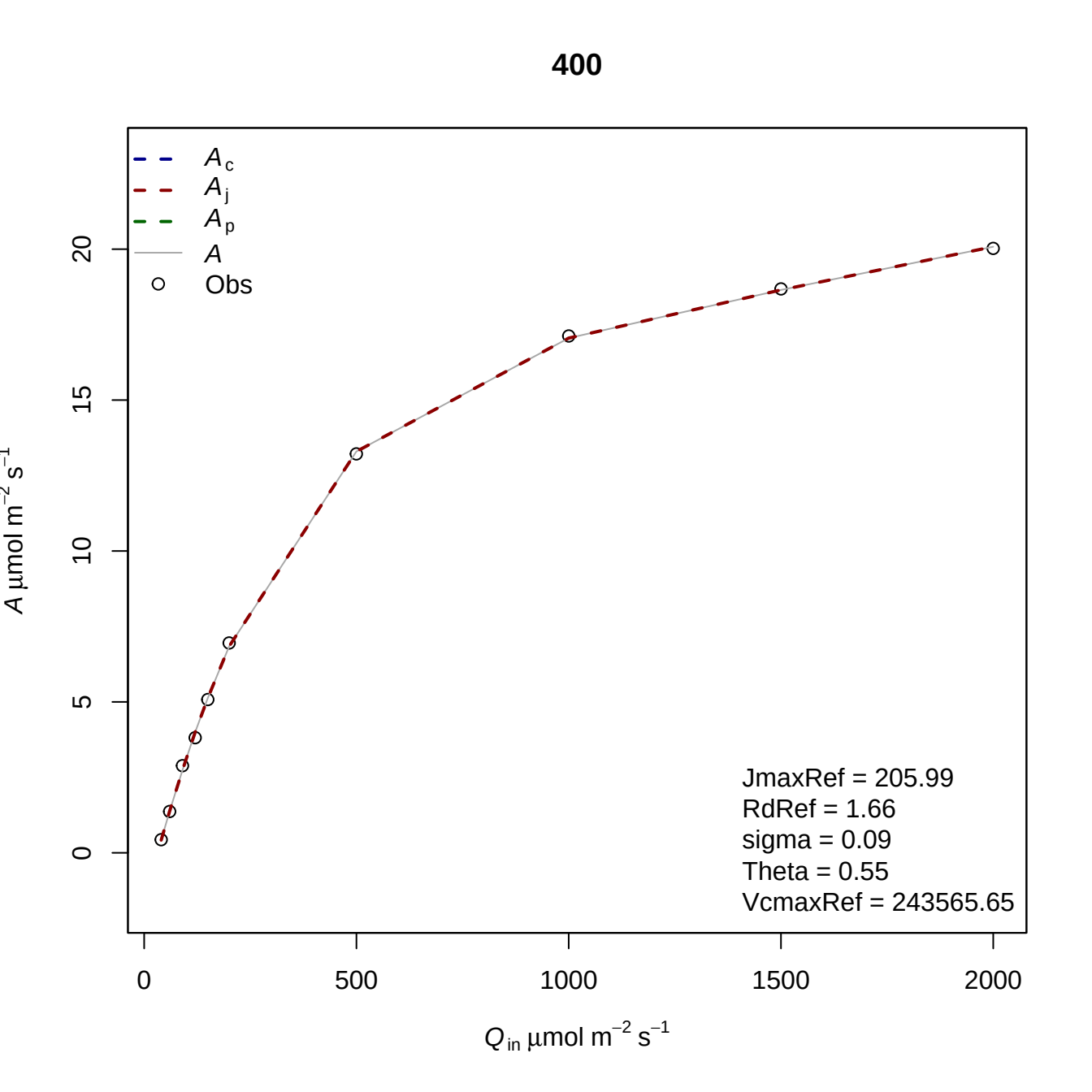

395
A_c
A_j
A_p
A
Obs
JmaxRef = 94.13
RdRef = 1.24
sigma = 0.25
Theta = −0.8
VcmaxRef = 776753.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

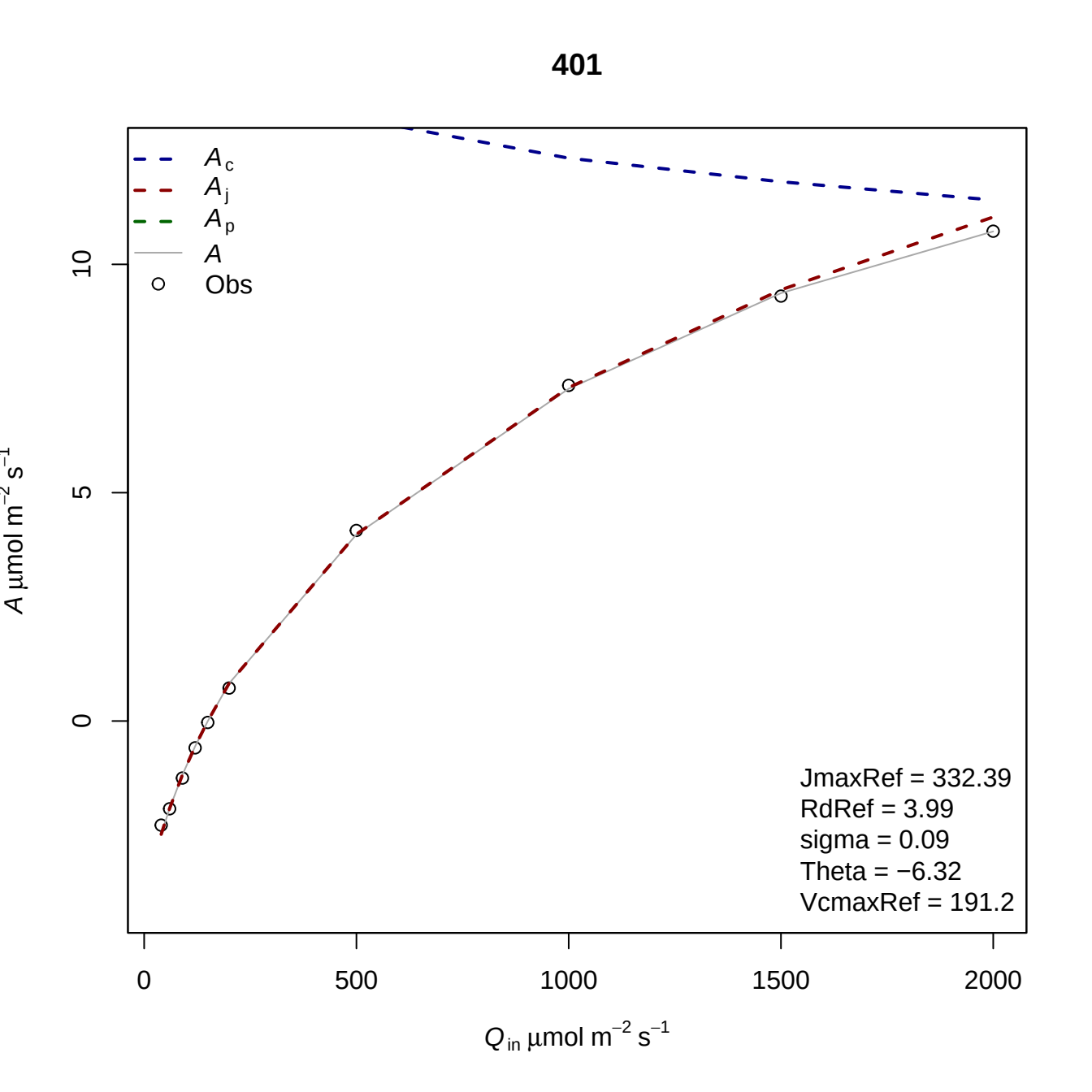

396
A_c
A_j
A_p
A
Obs
JmaxRef = 135.35
RdRef = 0.77
sigma = 0.1
Theta = 0.24
VcmaxRef = 60.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

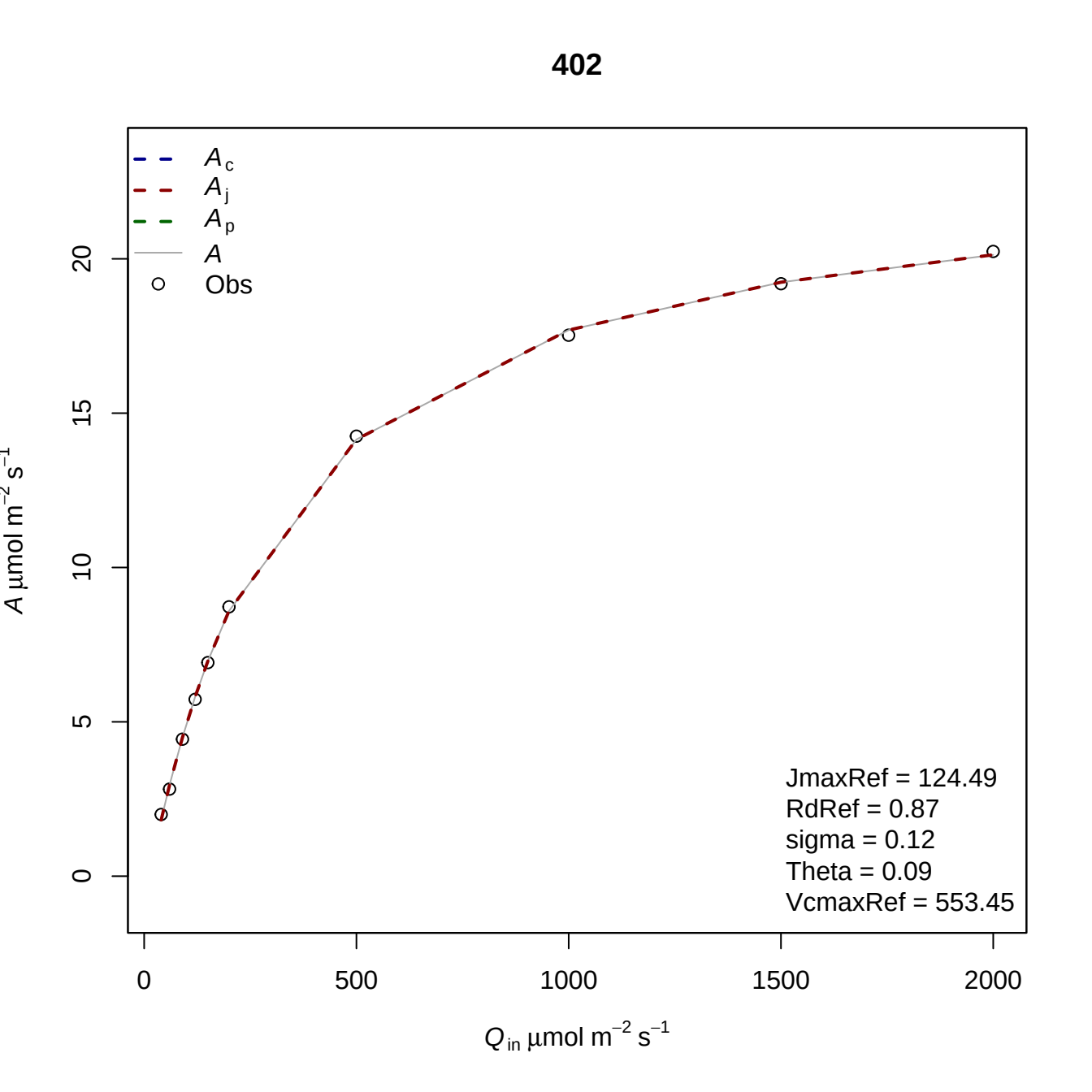

397
A_c
A_j
A_p
A
Obs
JmaxRef = 184.21
RdRef = 0.79
sigma = 0.09
Theta = 0.54
VcmaxRef = 94.84
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

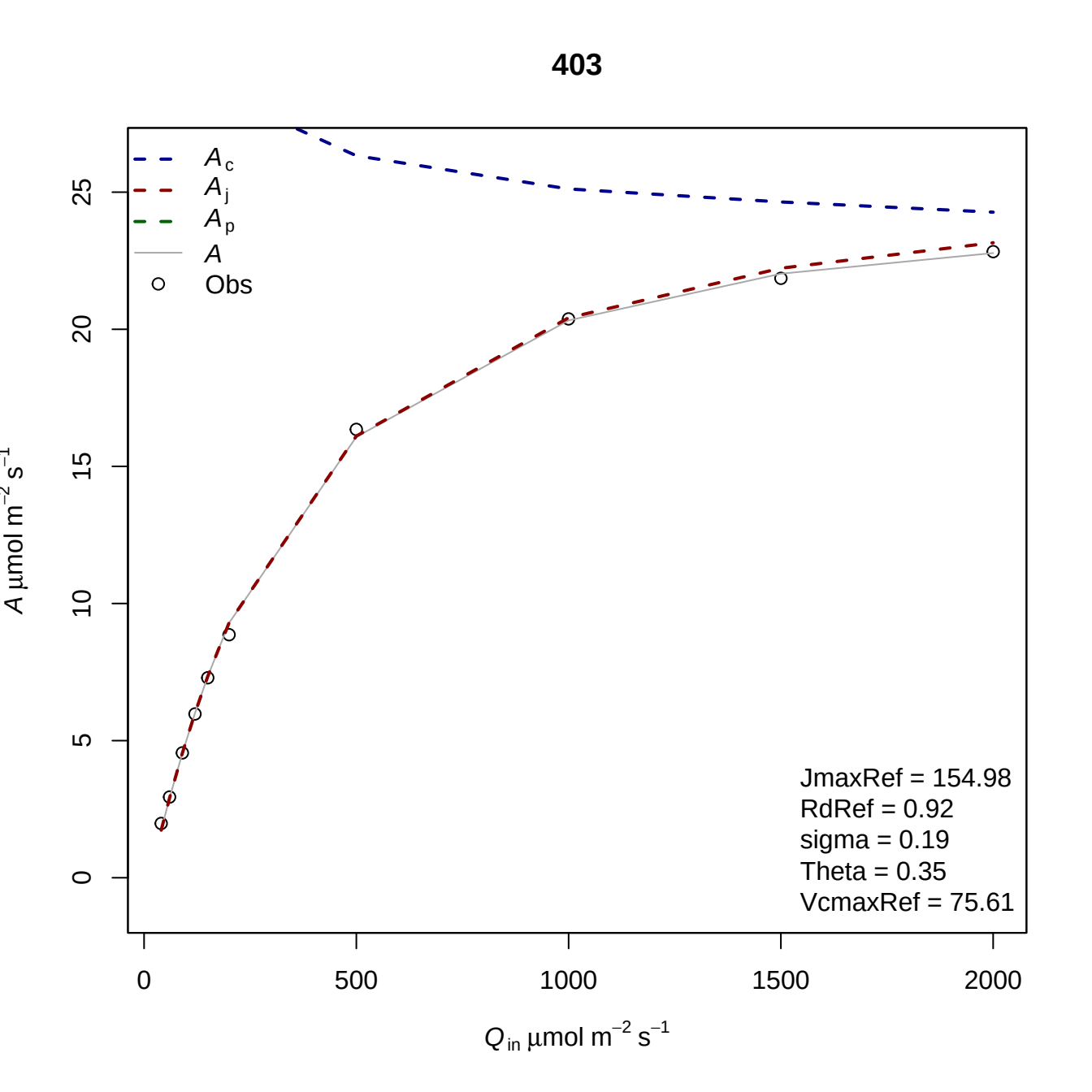

398
A_c
A_j
A_p
A
Obs
JmaxRef = 224.78
RdRef = 1.35
sigma = 0.21
Theta = −0.26
VcmaxRef = 111.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

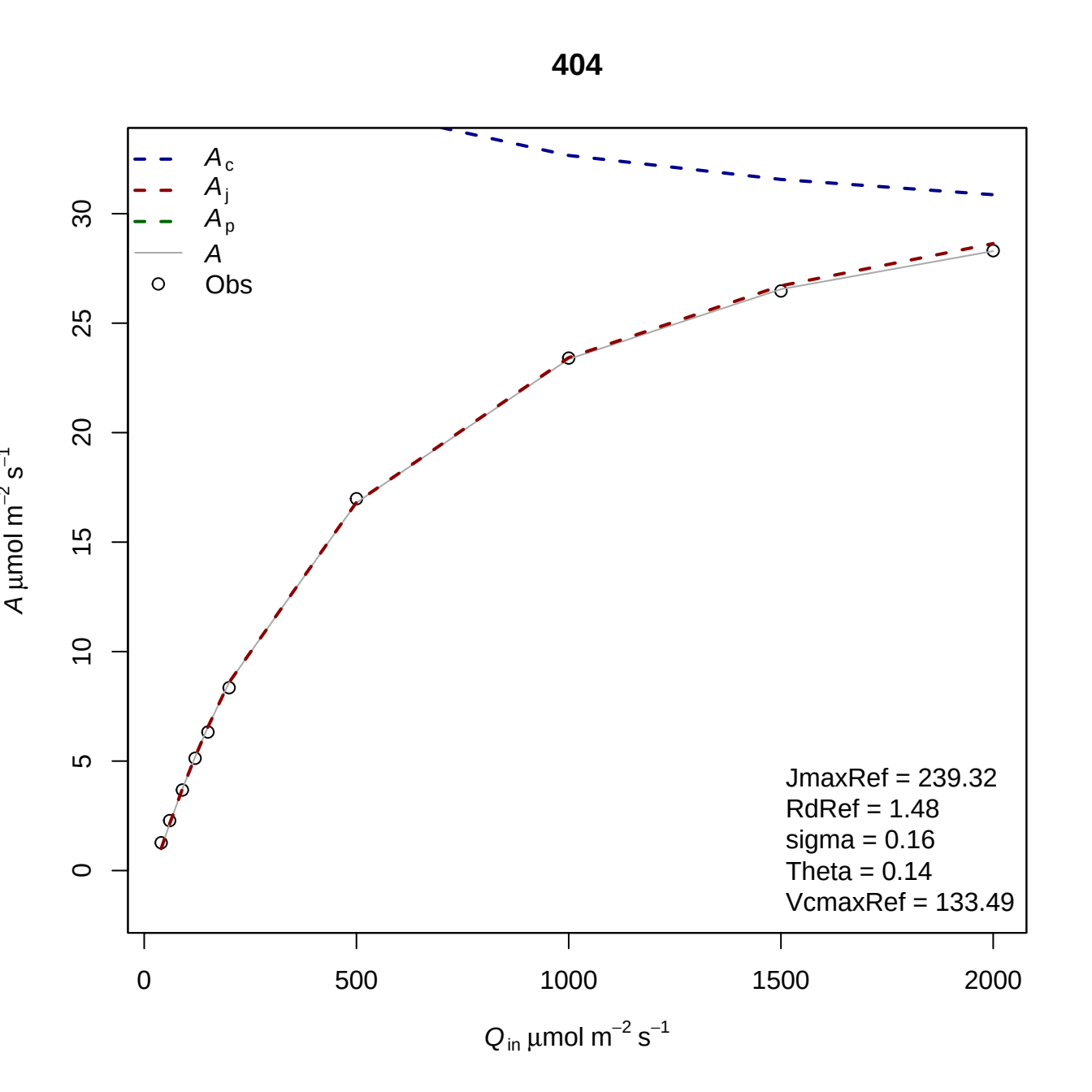

399
A_c
A_j
A_p
A
Obs
JmaxRef = 188.8
RdRef = 1.64
sigma = 0.11
Theta = 0.5
VcmaxRef = 163.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

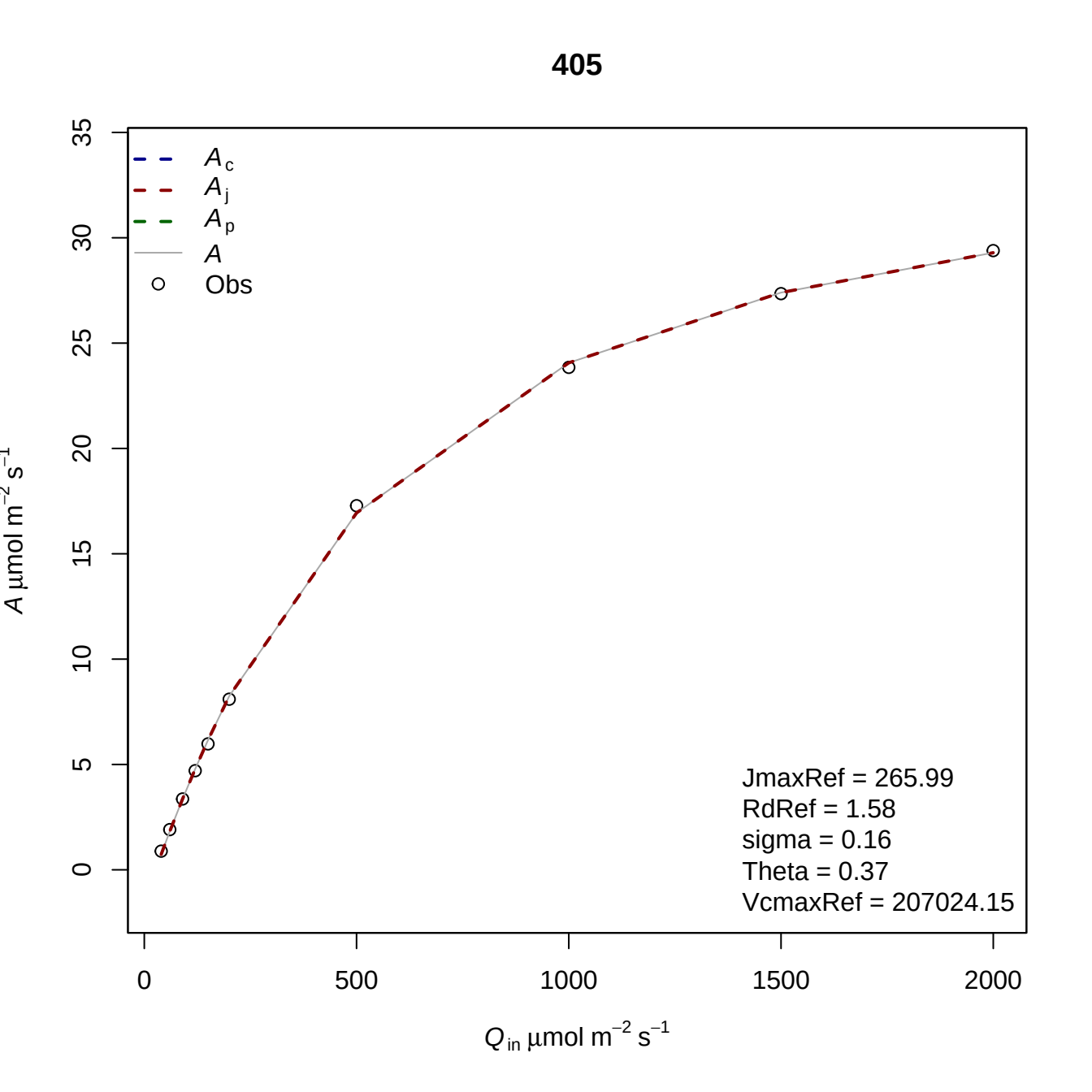

400
A_c
A_j
A_p
A
Obs
JmaxRef = 205.99
RdRef = 1.66
sigma = 0.09
Theta = 0.55
VcmaxRef = 243565.65
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

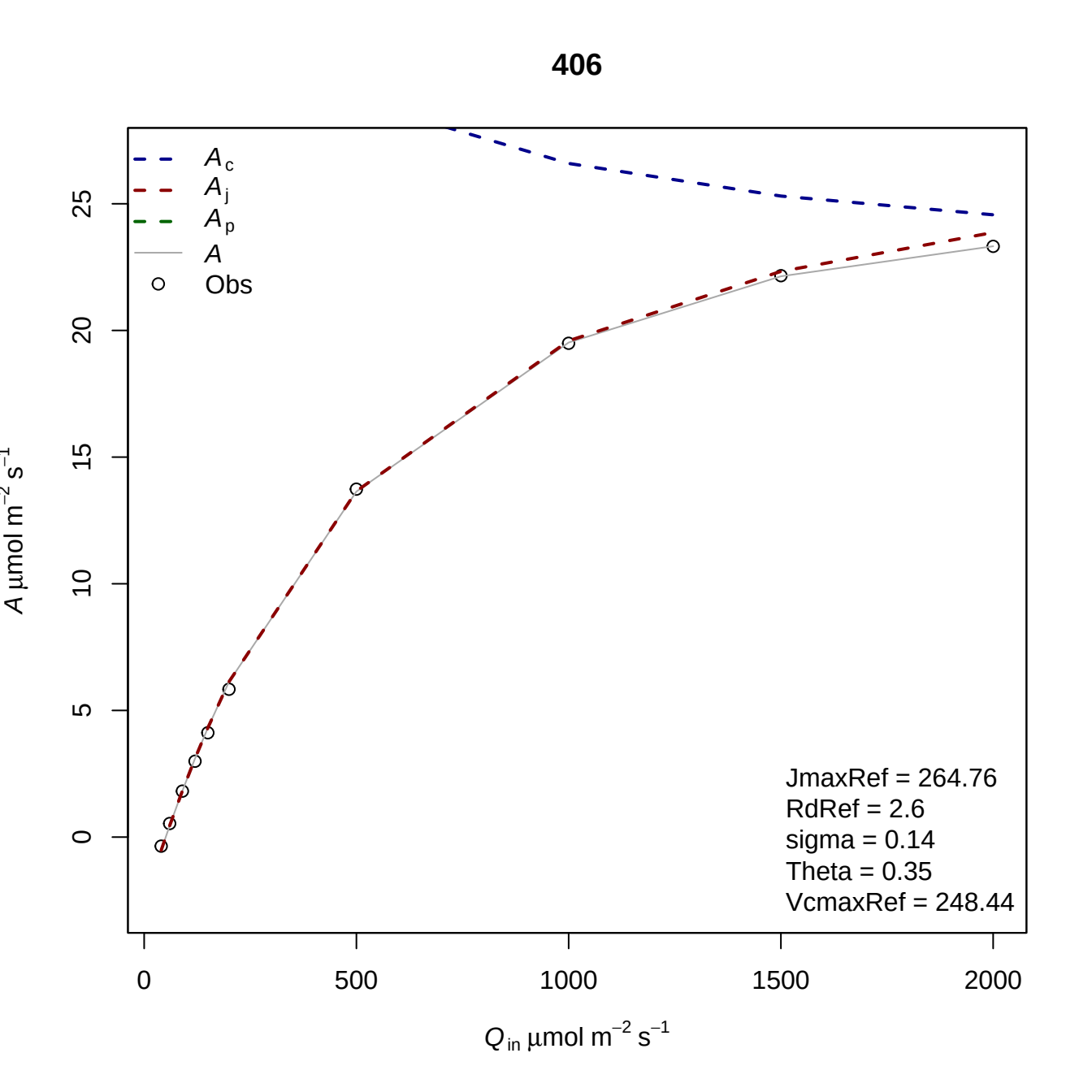

401
A_c
A_j
A_p
A
Obs
JmaxRef = 332.39
RdRef = 3.99
sigma = 0.09
Theta = −6.32
VcmaxRef = 191.2
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

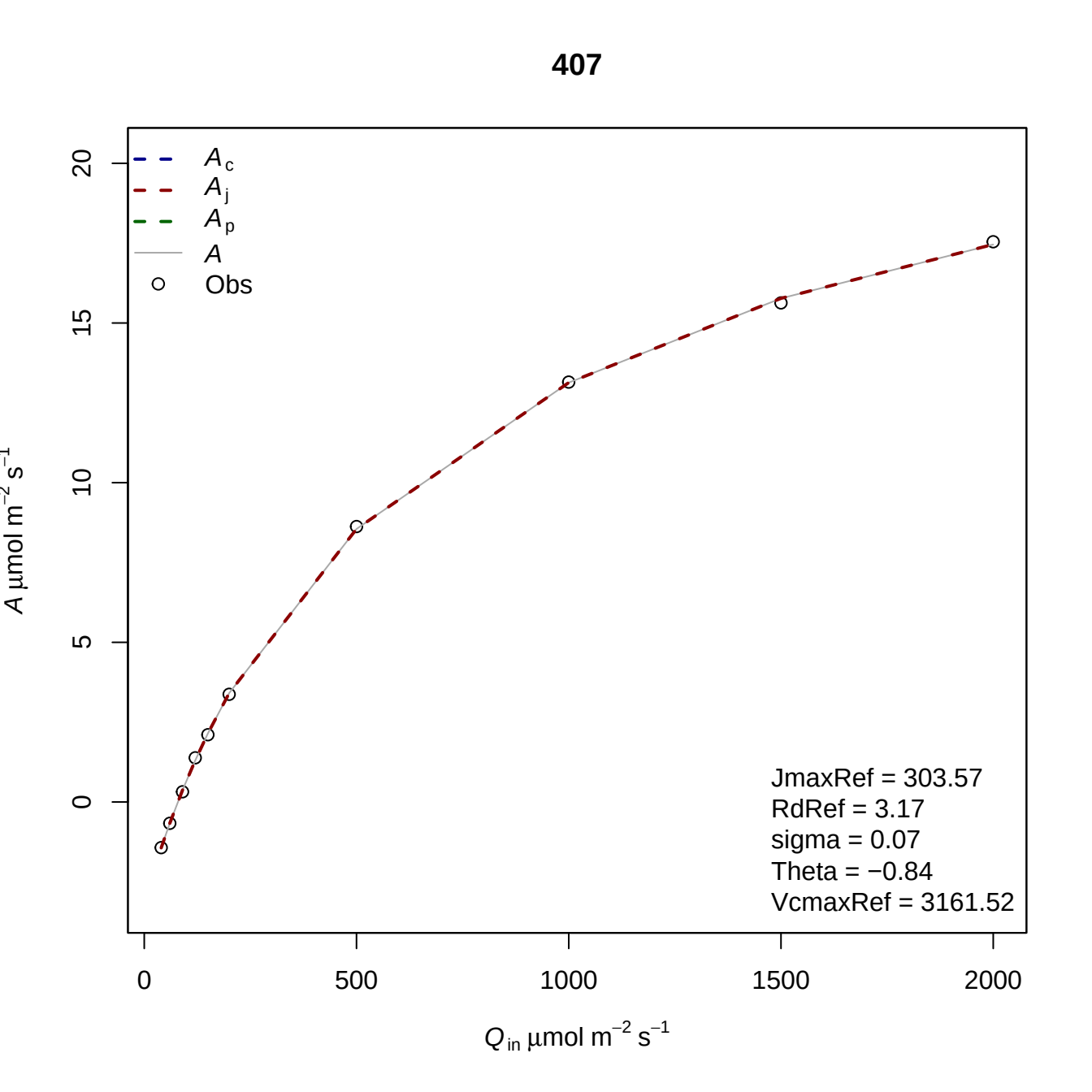

402
A_c
A_j
A_p
A
Obs
JmaxRef = 124.49
RdRef = 0.87
sigma = 0.12
Theta = 0.09
VcmaxRef = 553.45
A µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

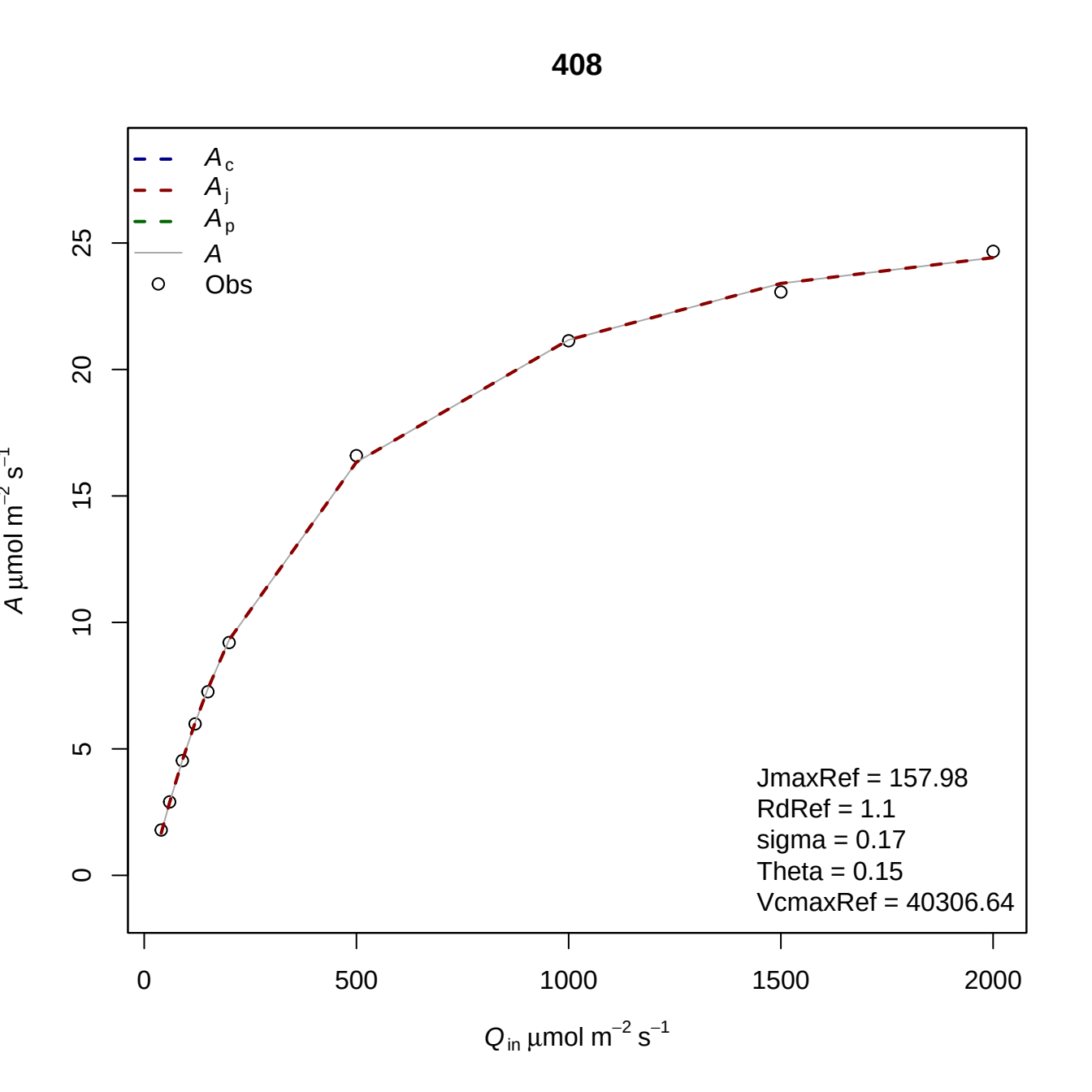

403
A_c
A_j
A_p
A
Obs
JmaxRef = 154.98
RdRef = 0.92
sigma = 0.19
Theta = 0.35
VcmaxRef = 75.61
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

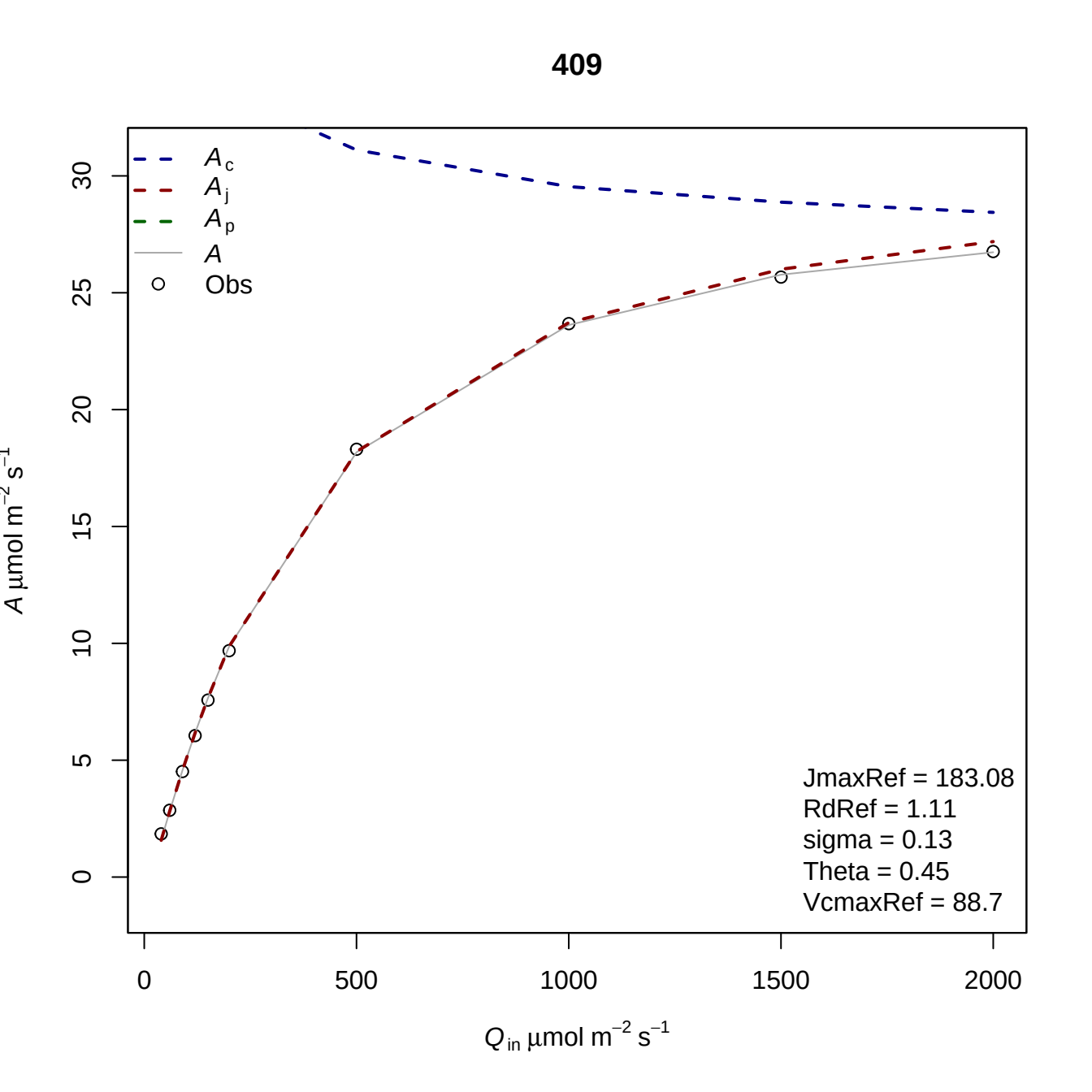

404
A_c
A_j
A_p
A
Obs
JmaxRef = 239.32
RdRef = 1.48
sigma = 0.16
Theta = 0.14
VcmaxRef = 133.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

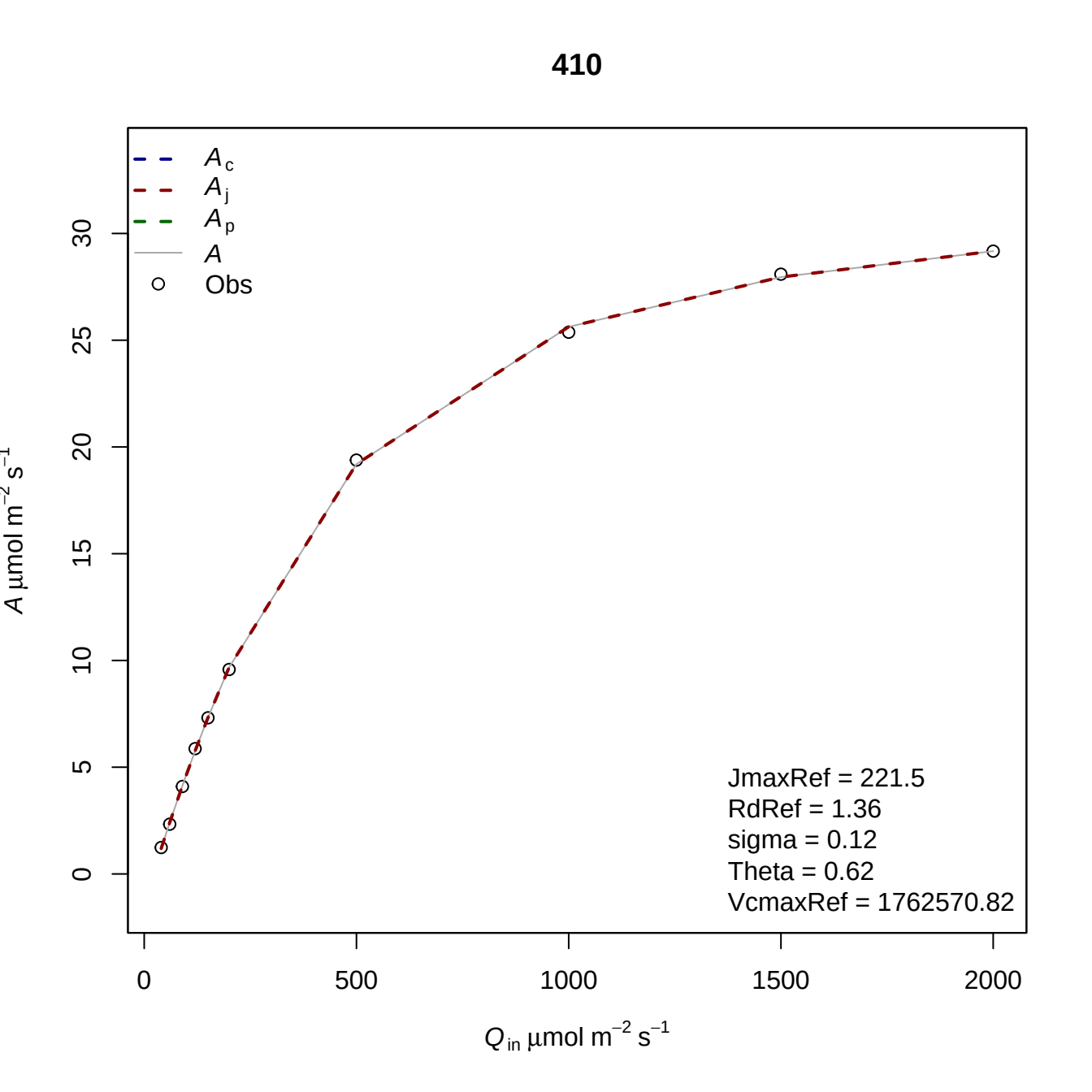

405
A_c
A_j
A_p
A
Obs
JmaxRef = 265.99
RdRef = 1.58
sigma = 0.16
Theta = 0.37
VcmaxRef = 207024.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

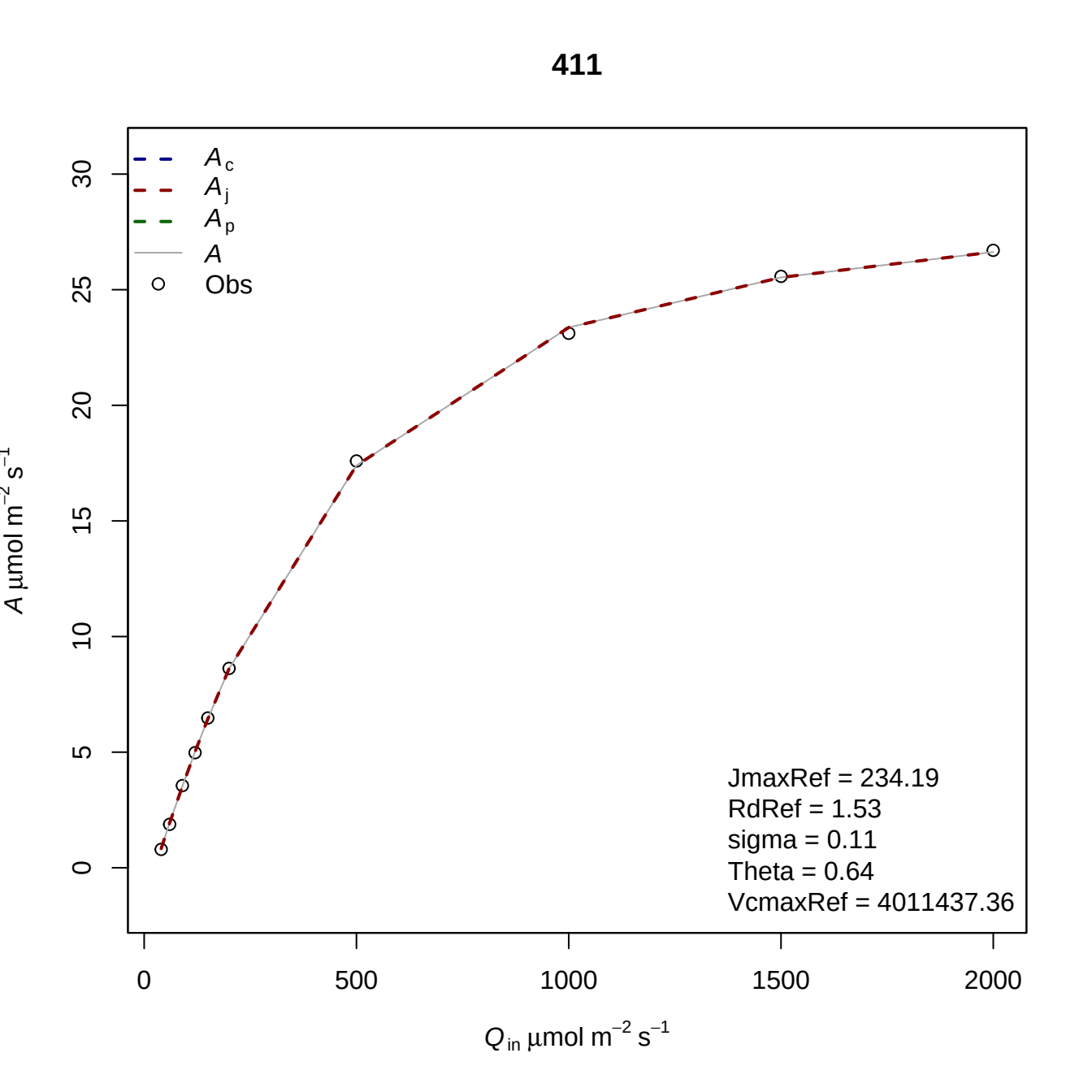

406
A_c
A_j
A_p
A
Obs
JmaxRef = 264.76
RdRef = 2.6
sigma = 0.14
Theta = 0.35
VcmaxRef = 248.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

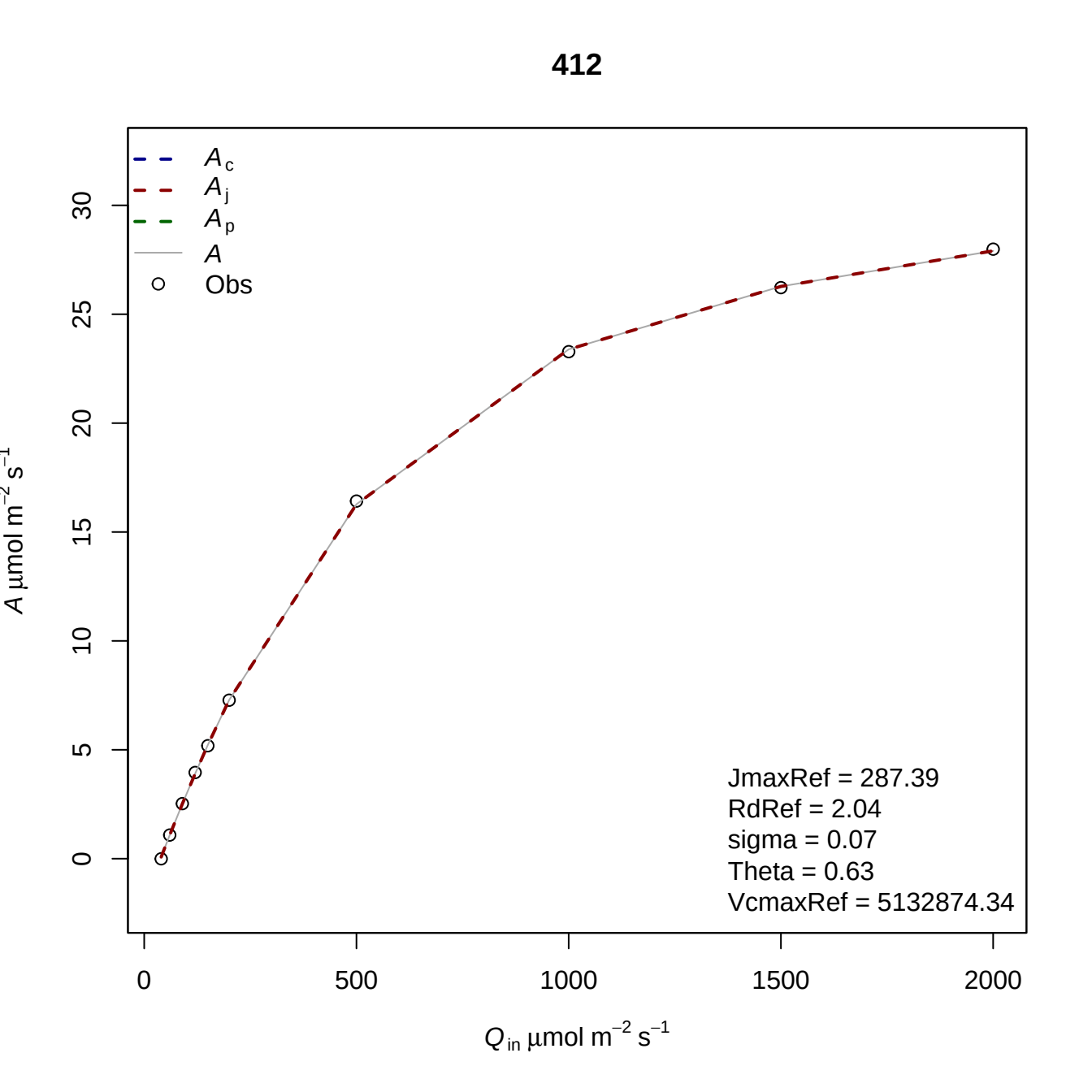

407
A_c
A_j
A_p
A
Obs
JmaxRef = 303.57
RdRef = 3.17
sigma = 0.07
Theta = −0.84
VcmaxRef = 3161.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

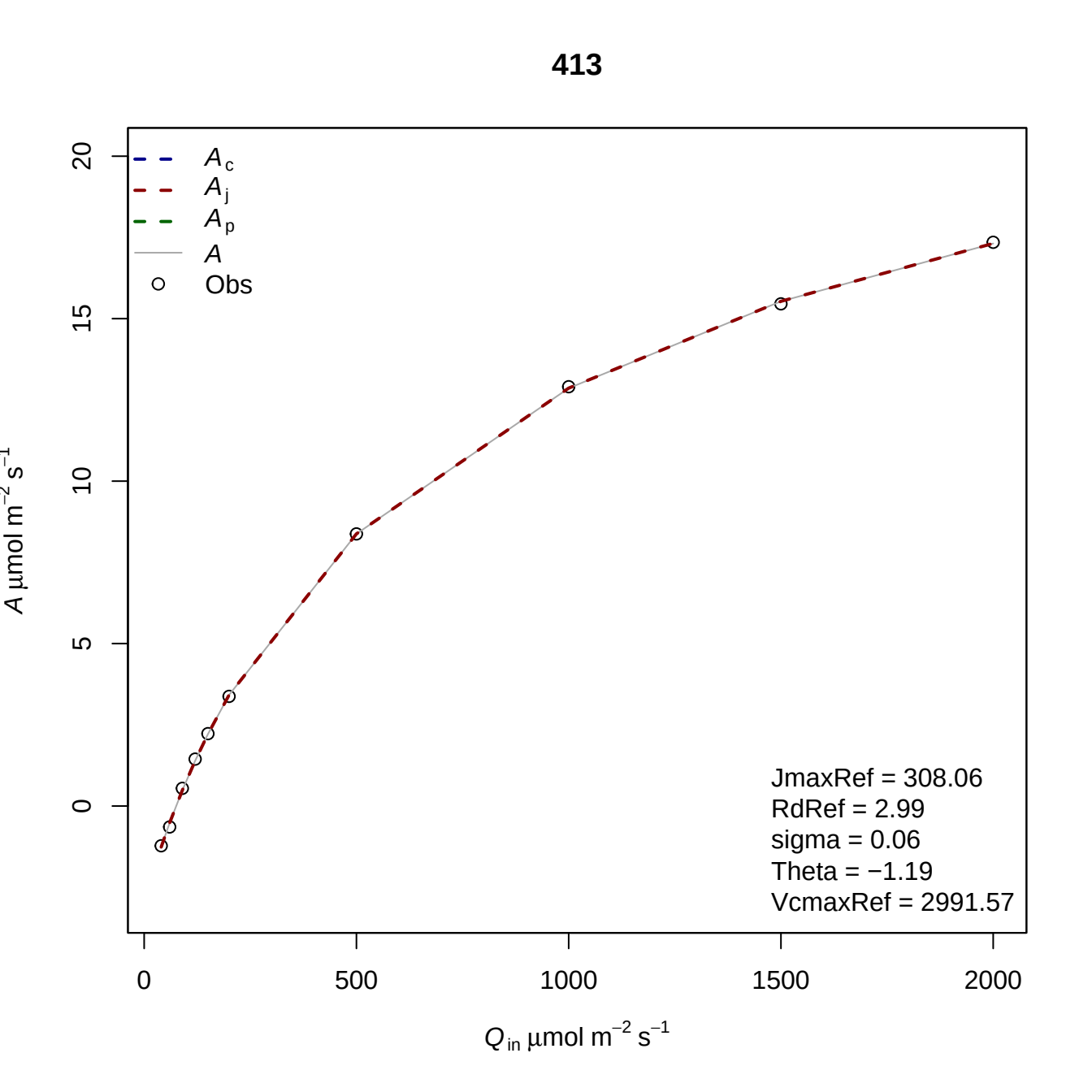

408
A_c
A_j
A_p
A
Obs
JmaxRef = 157.98
RdRef = 1.1
sigma = 0.17
Theta = 0.15
VcmaxRef = 40306.64
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

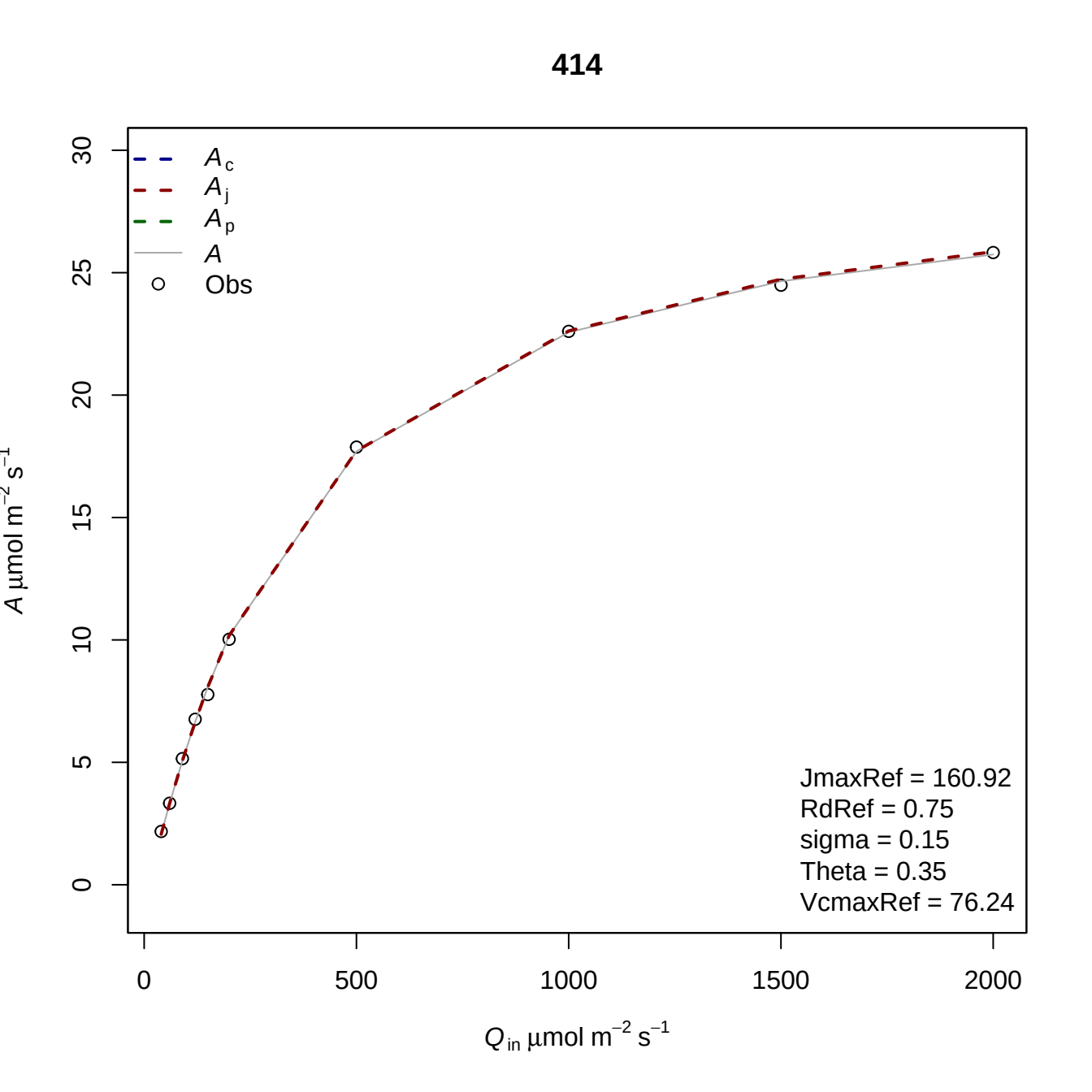

409
A_c
A_j
A_p
A
Obs
JmaxRef = 183.08
RdRef = 1.11
sigma = 0.13
Theta = 0.45
VcmaxRef = 88.7
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

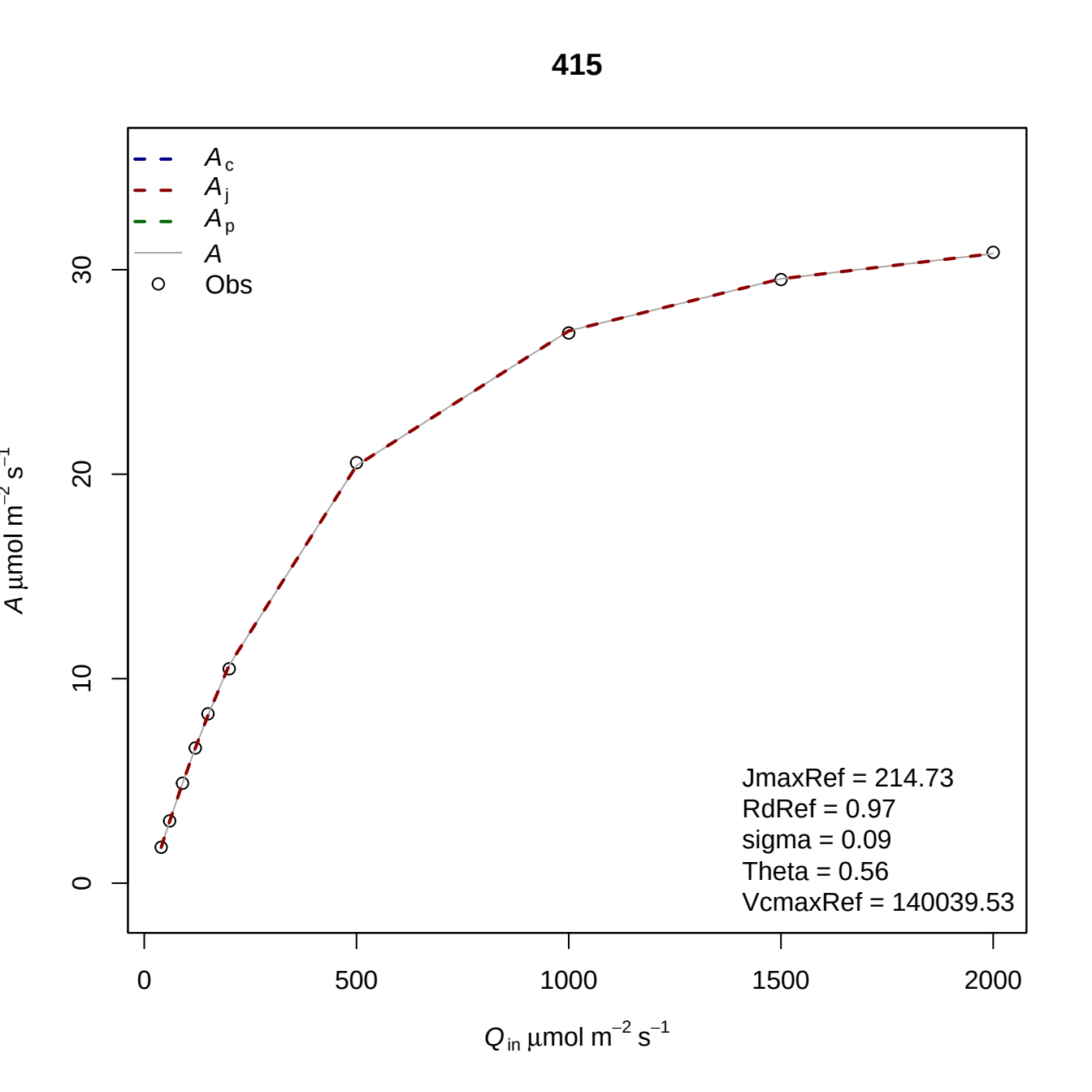

410
A_c
A_j
A_p
A
Obs
JmaxRef = 221.5
RdRef = 1.36
sigma = 0.12
Theta = 0.62
VcmaxRef = 1762570.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

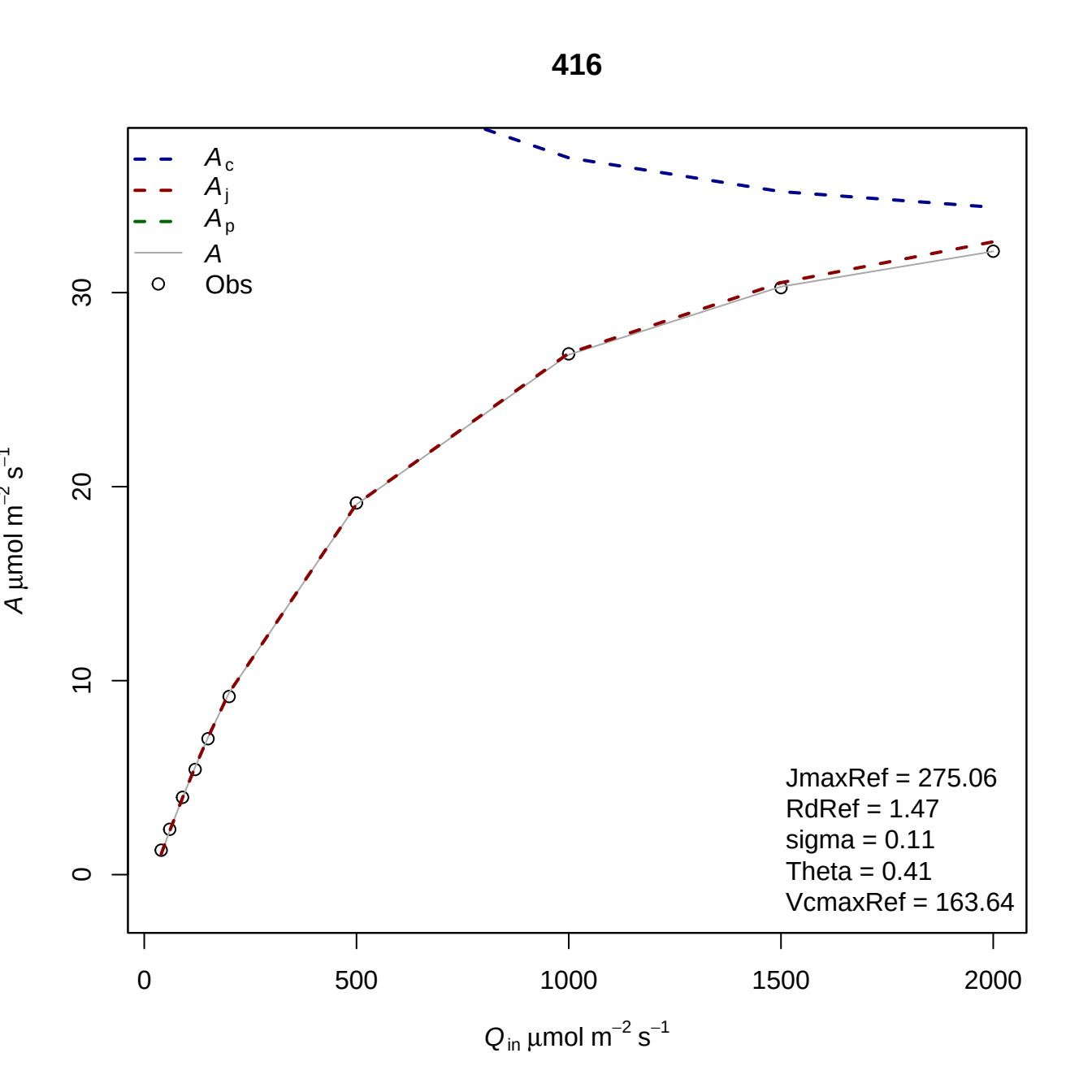

411
A_c
A_j
A_p
A
Obs
JmaxRef = 234.19
RdRef = 1.53
sigma = 0.11
Theta = 0.64
VcmaxRef = 4011437.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

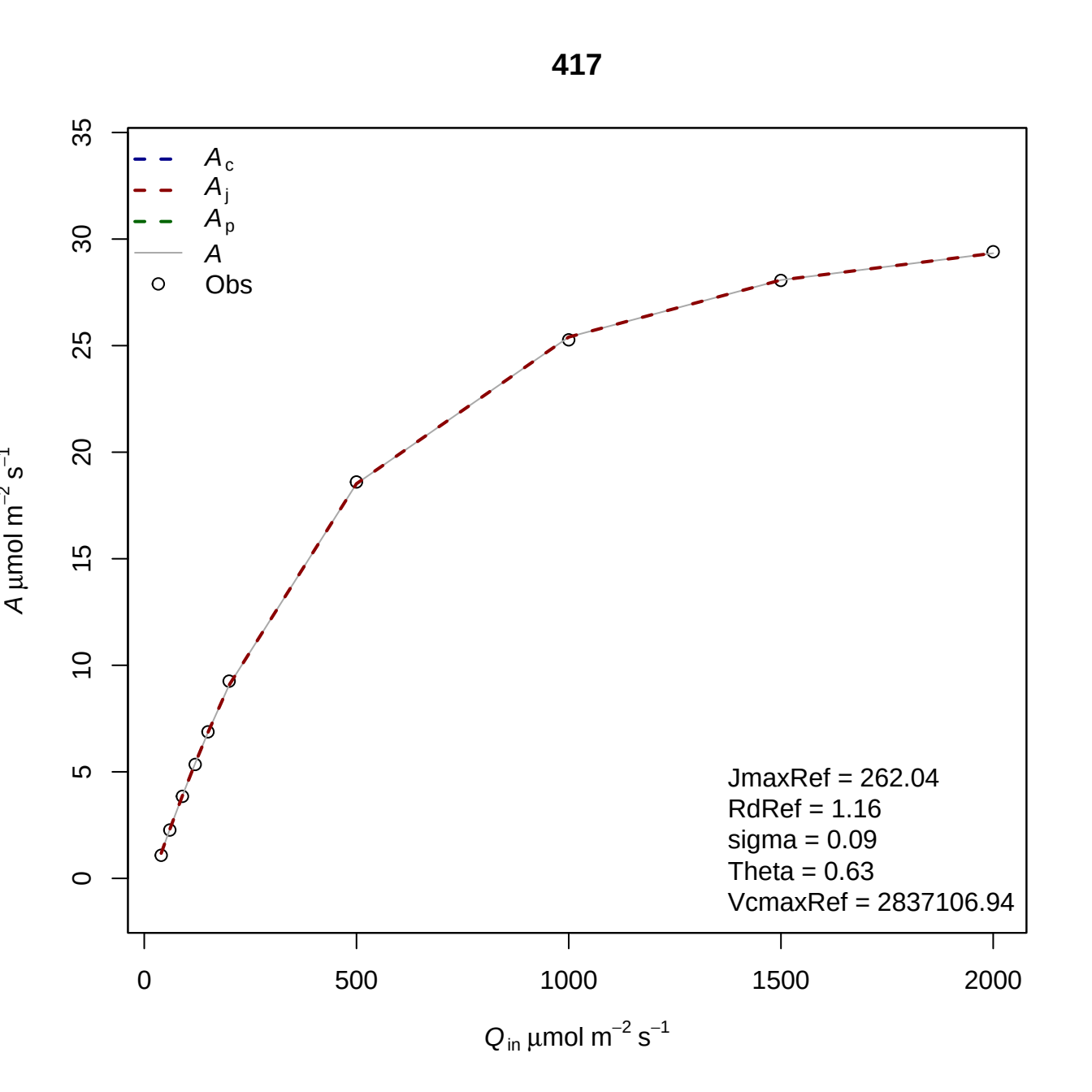

412
A_c
A_j
A_p
A
Obs
JmaxRef = 287.39
RdRef = 2.04
sigma = 0.07
Theta = 0.63
VcmaxRef = 5132874.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

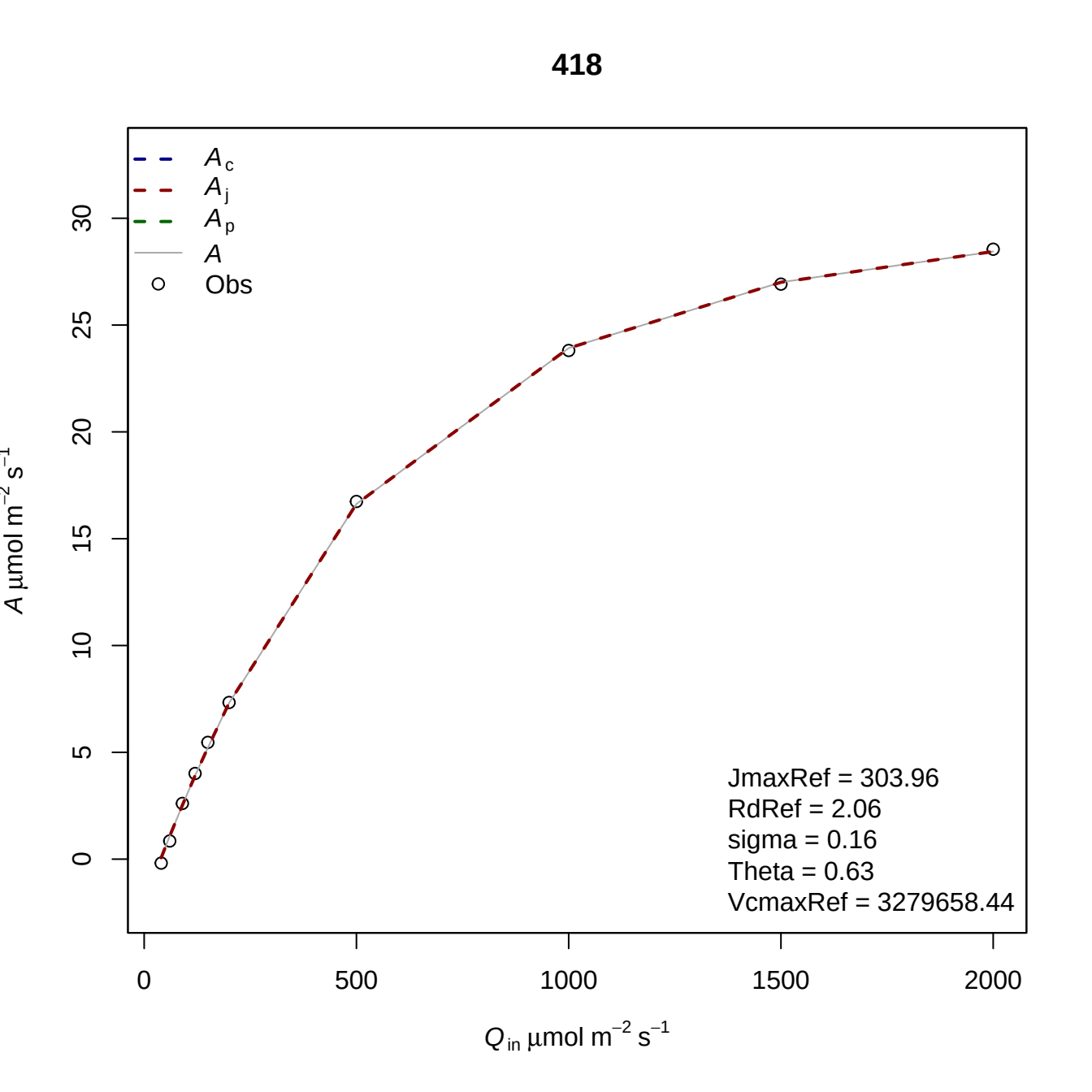

413
A_c
A_j
A_p
A
Obs
JmaxRef = 308.06
RdRef = 2.99
sigma = 0.06
Theta = −1.19
VcmaxRef = 2991.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

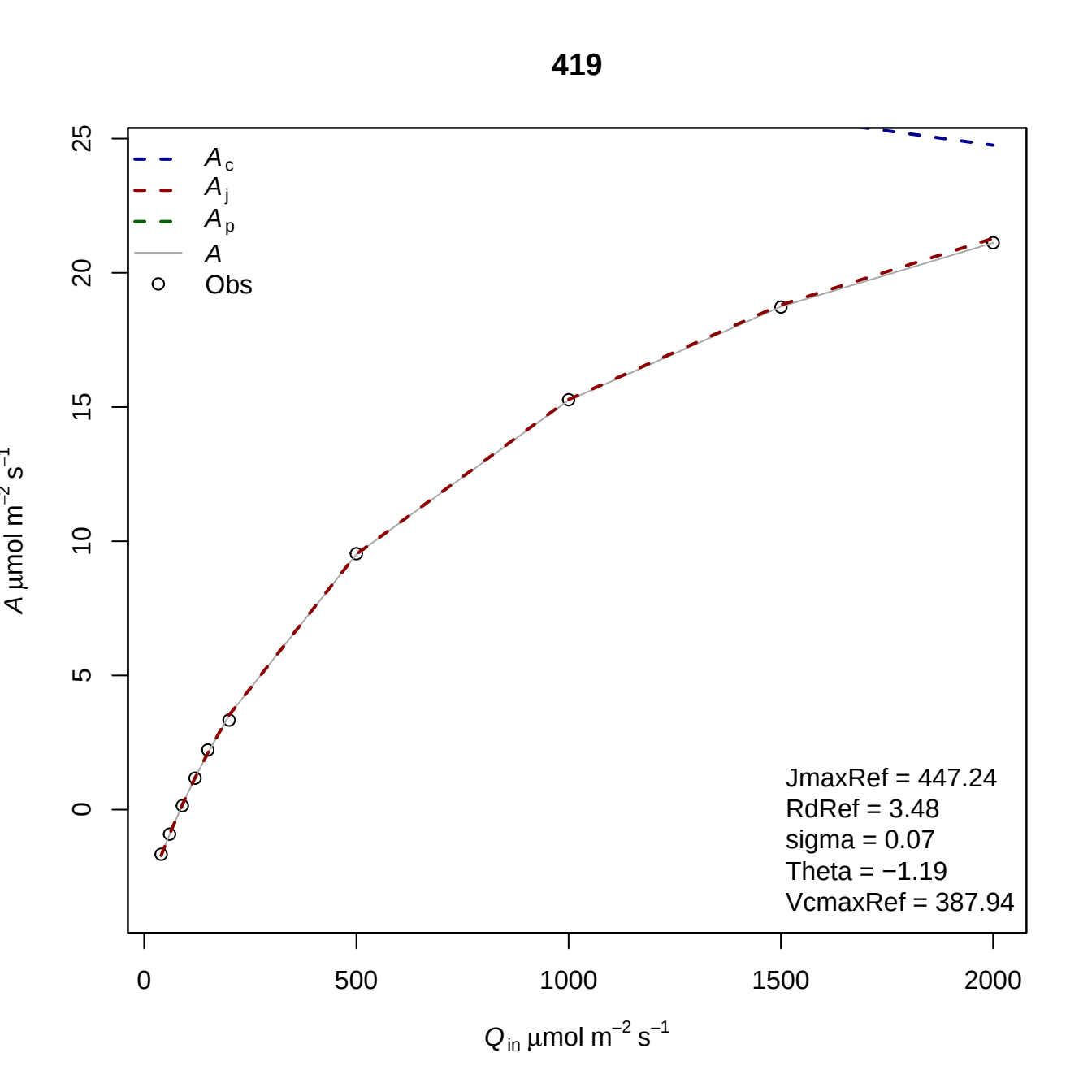

414
A_c
A_j
A_p
A
Obs
JmaxRef = 160.92
RdRef = 0.75
sigma = 0.15
Theta = 0.35
VcmaxRef = 76.24
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

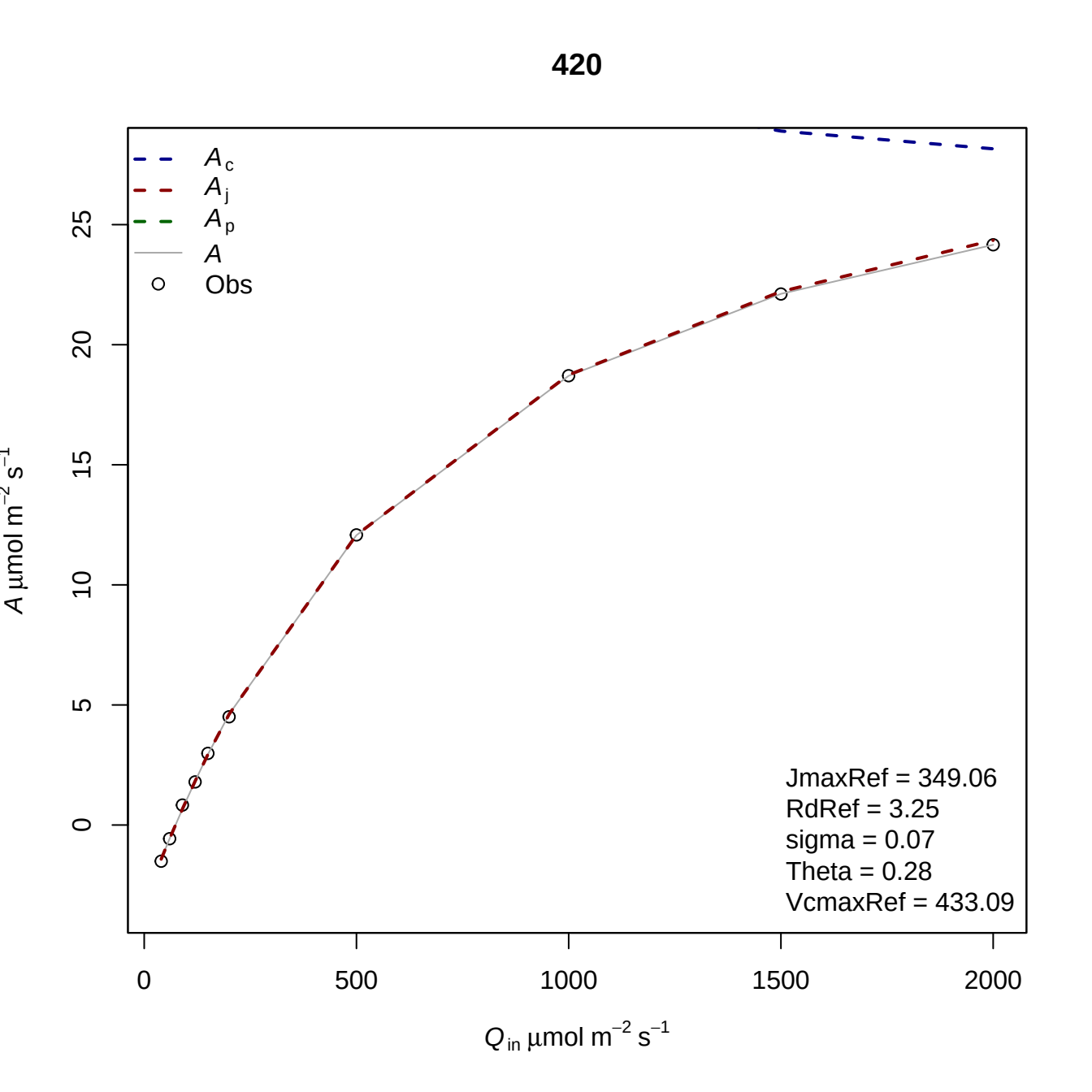

415
A_c
A_j
A_p
A
Obs
JmaxRef = 214.73
RdRef = 0.97
sigma = 0.09
Theta = 0.56
VcmaxRef = 140039.53
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

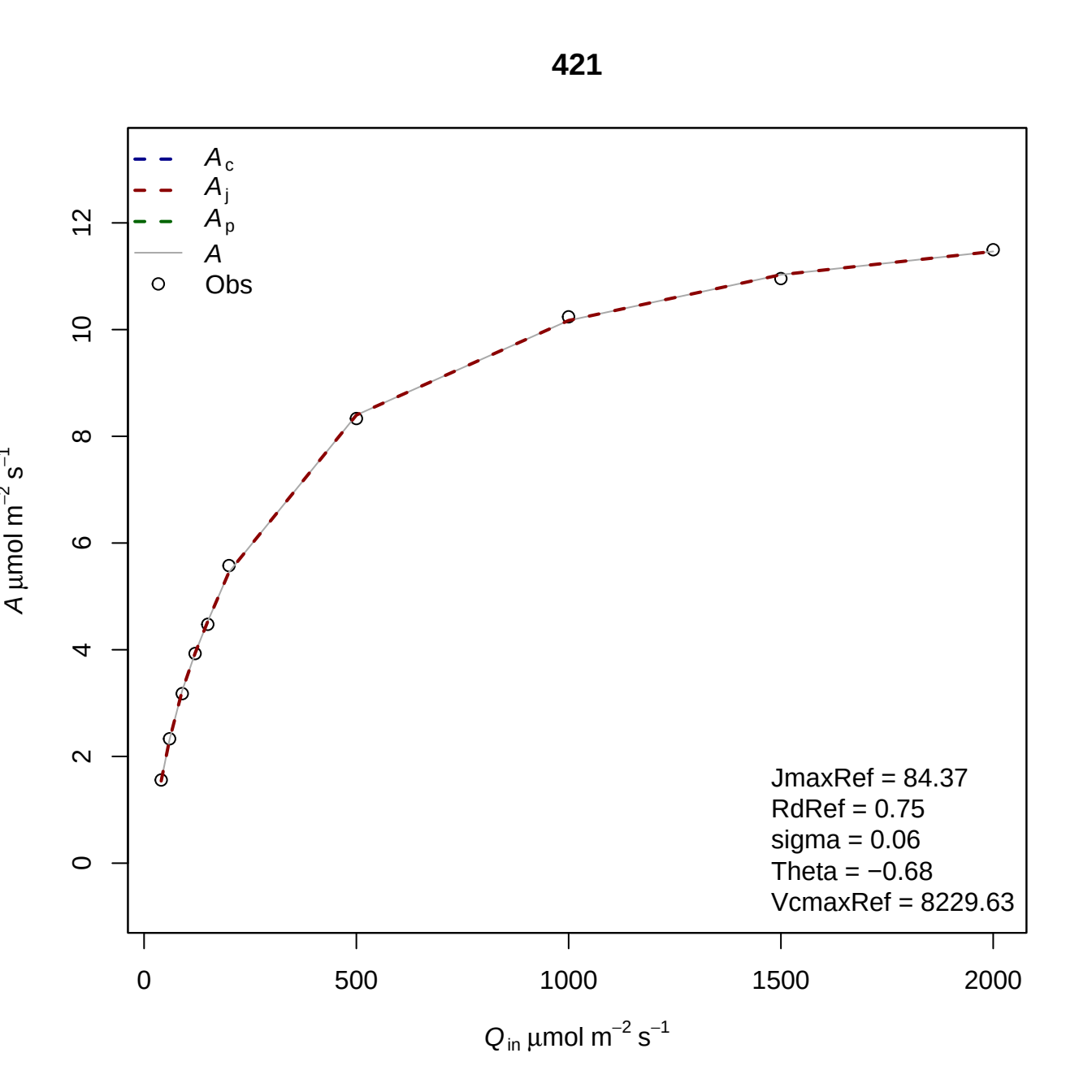

416
A_c
A_j
A_p
A
Obs
JmaxRef = 275.06
RdRef = 1.47
sigma = 0.11
Theta = 0.41
VcmaxRef = 163.64
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

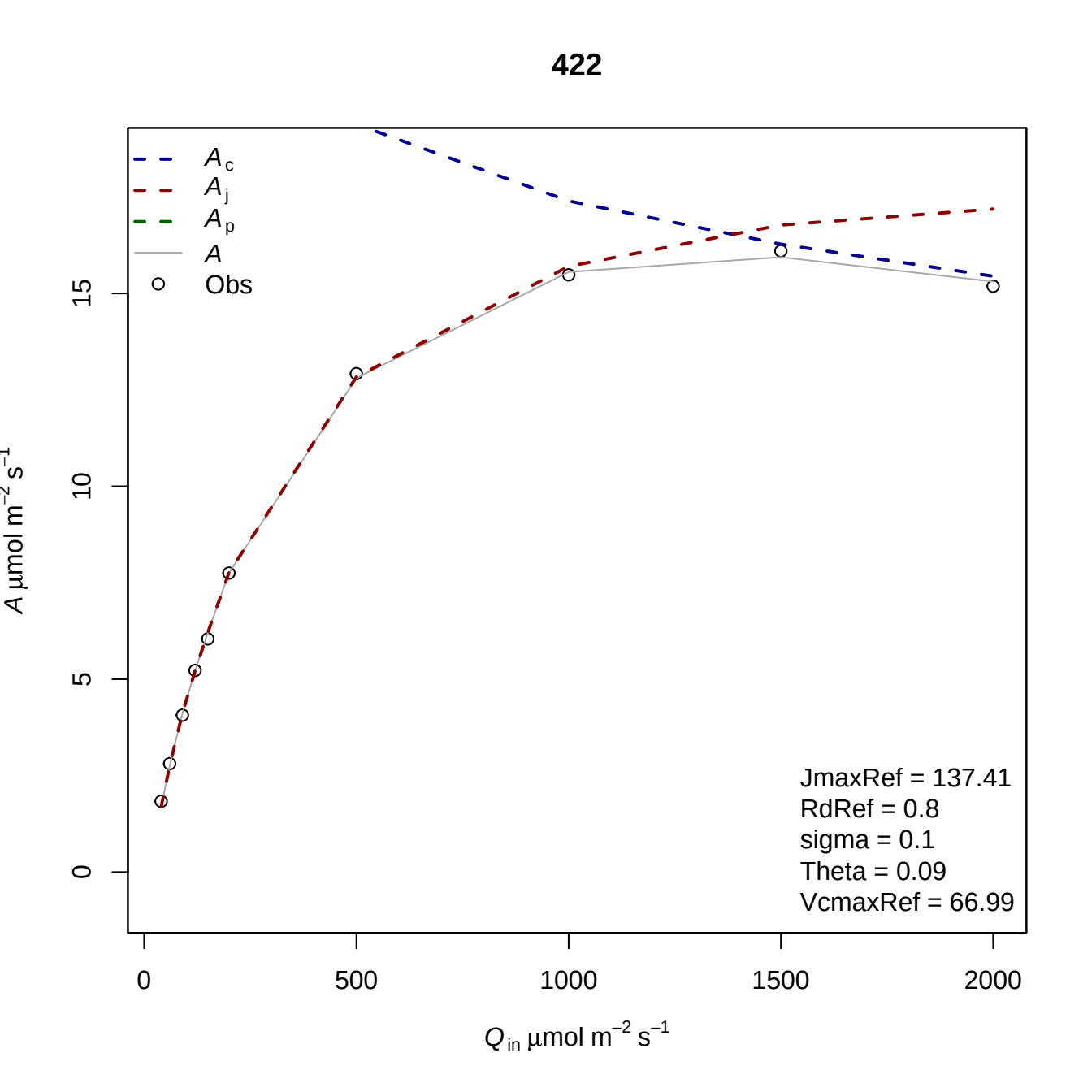

417
A_c
A_j
A_p
A
Obs
JmaxRef = 262.04
RdRef = 1.16
sigma = 0.09
Theta = 0.63
VcmaxRef = 2837106.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

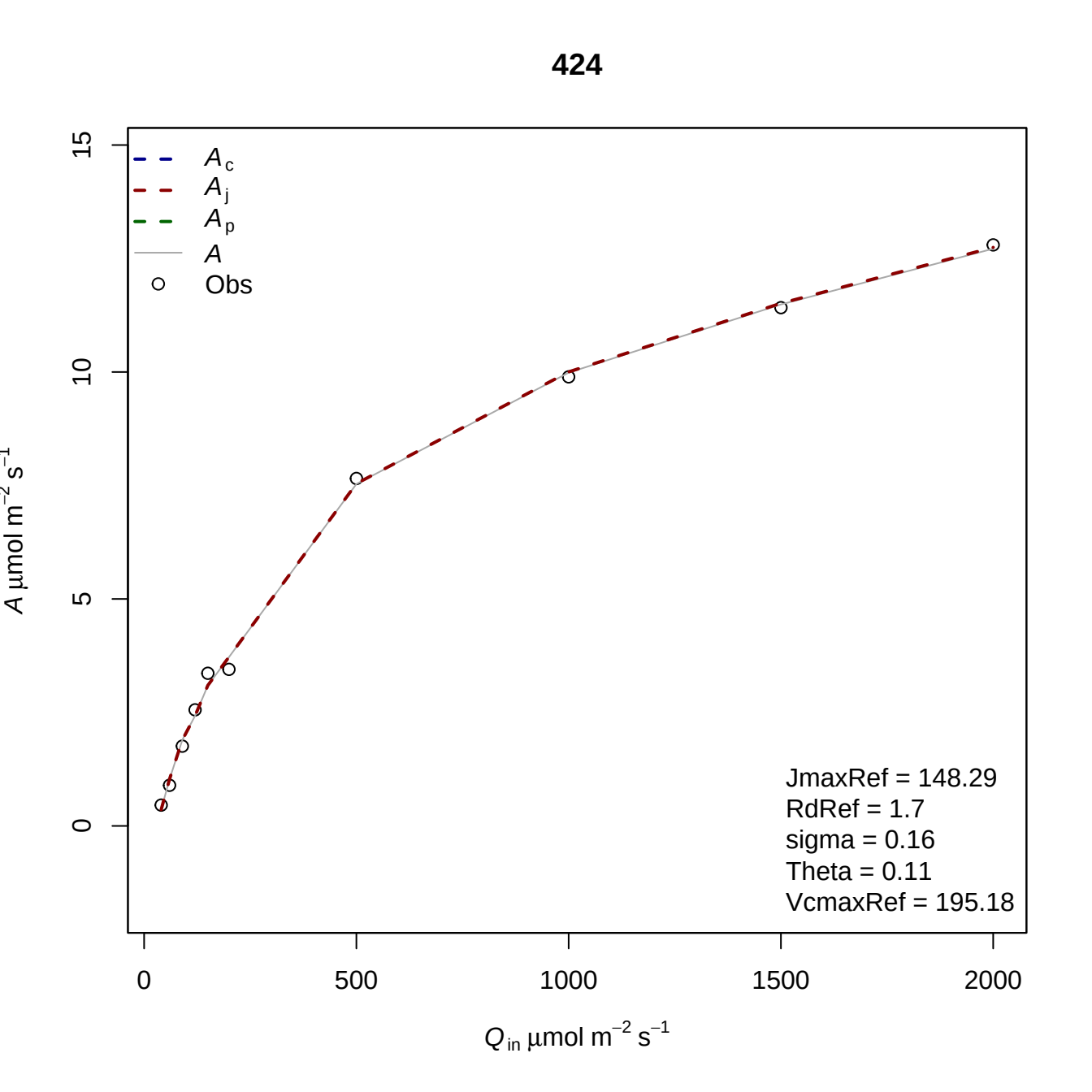

418
A_c
A_j
A_p
A
Obs
JmaxRef = 303.96
RdRef = 2.06
sigma = 0.16
Theta = 0.63
VcmaxRef = 3279658.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

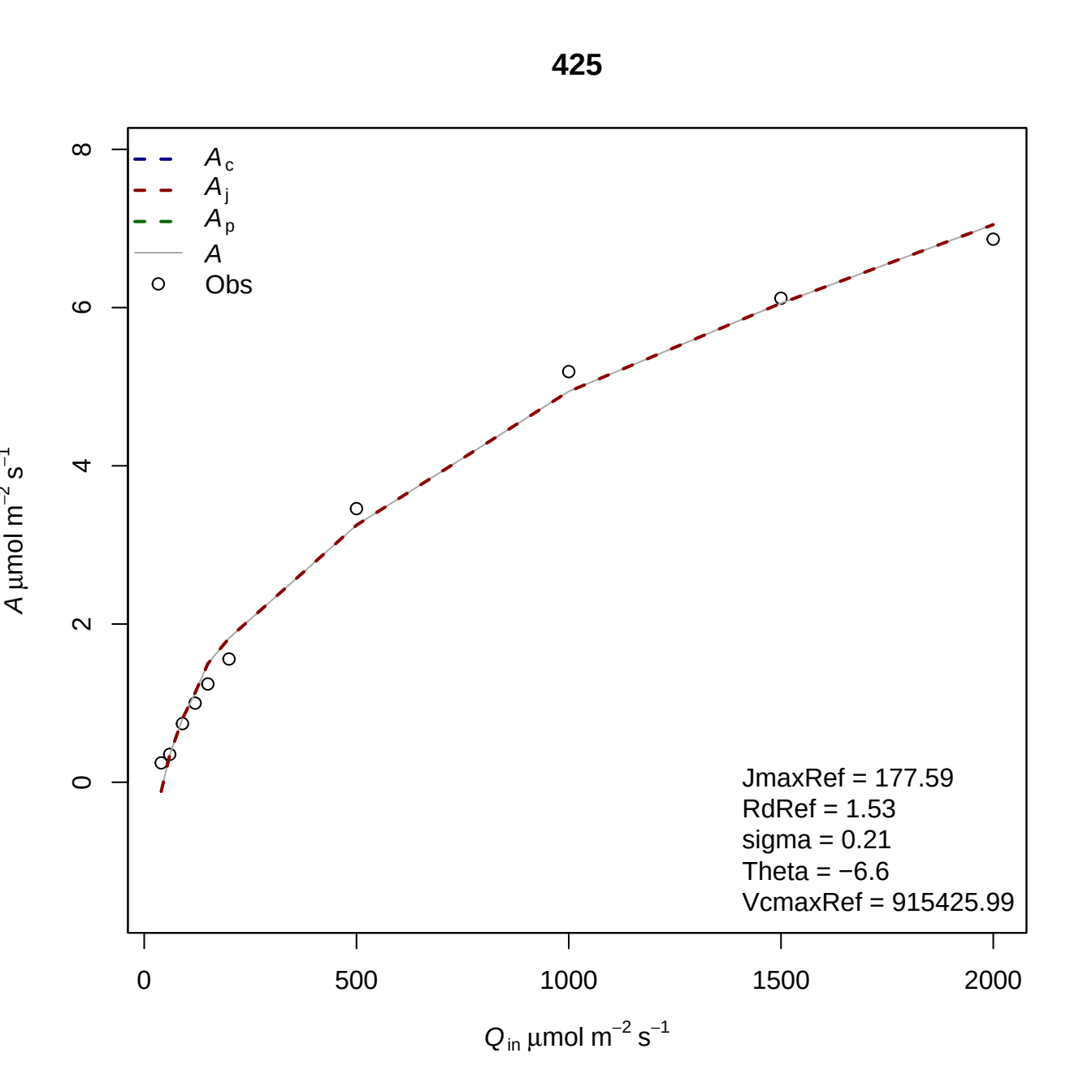

419
A_c
A_j
A_p
A
Obs
JmaxRef = 447.24
RdRef = 3.48
sigma = 0.07
Theta = −1.19
VcmaxRef = 387.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

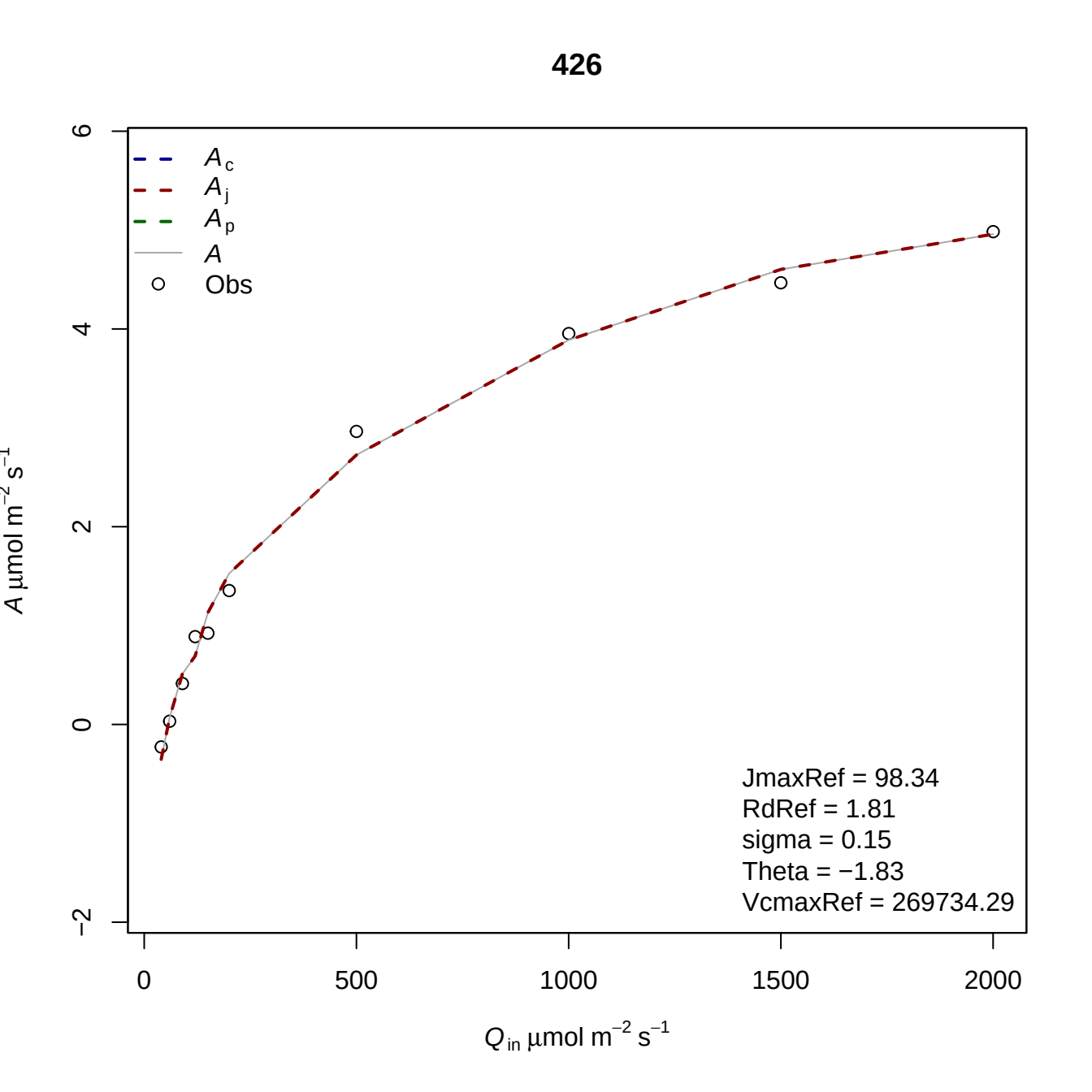

420
A_c
A_j
A_p
A
Obs
JmaxRef = 349.06
RdRef = 3.25
sigma = 0.07
Theta = 0.28
VcmaxRef = 433.09
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

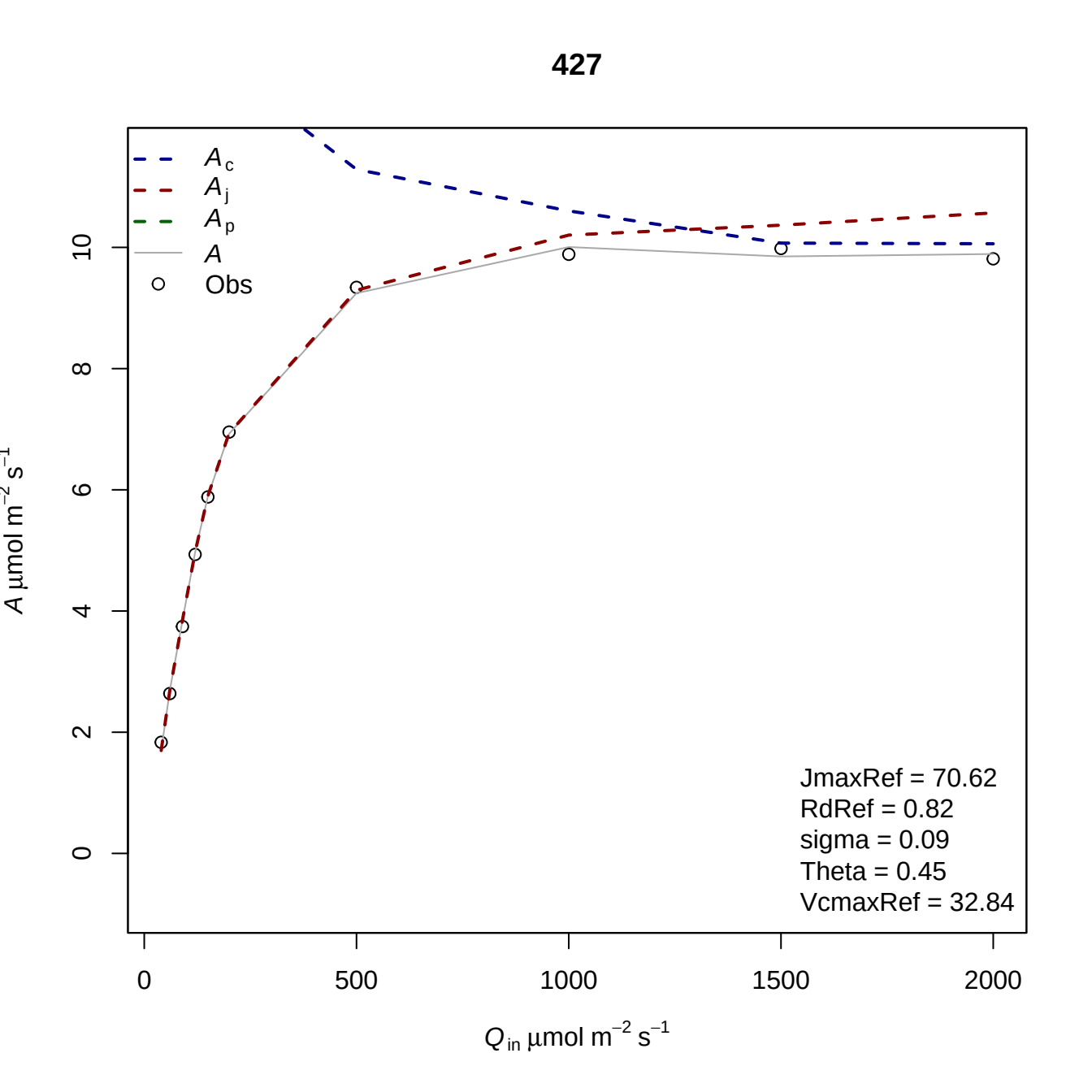

421
A_c
A_j
A_p
A
Obs
JmaxRef = 84.37
RdRef = 0.75
sigma = 0.06
Theta = −0.68
VcmaxRef = 8229.63
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

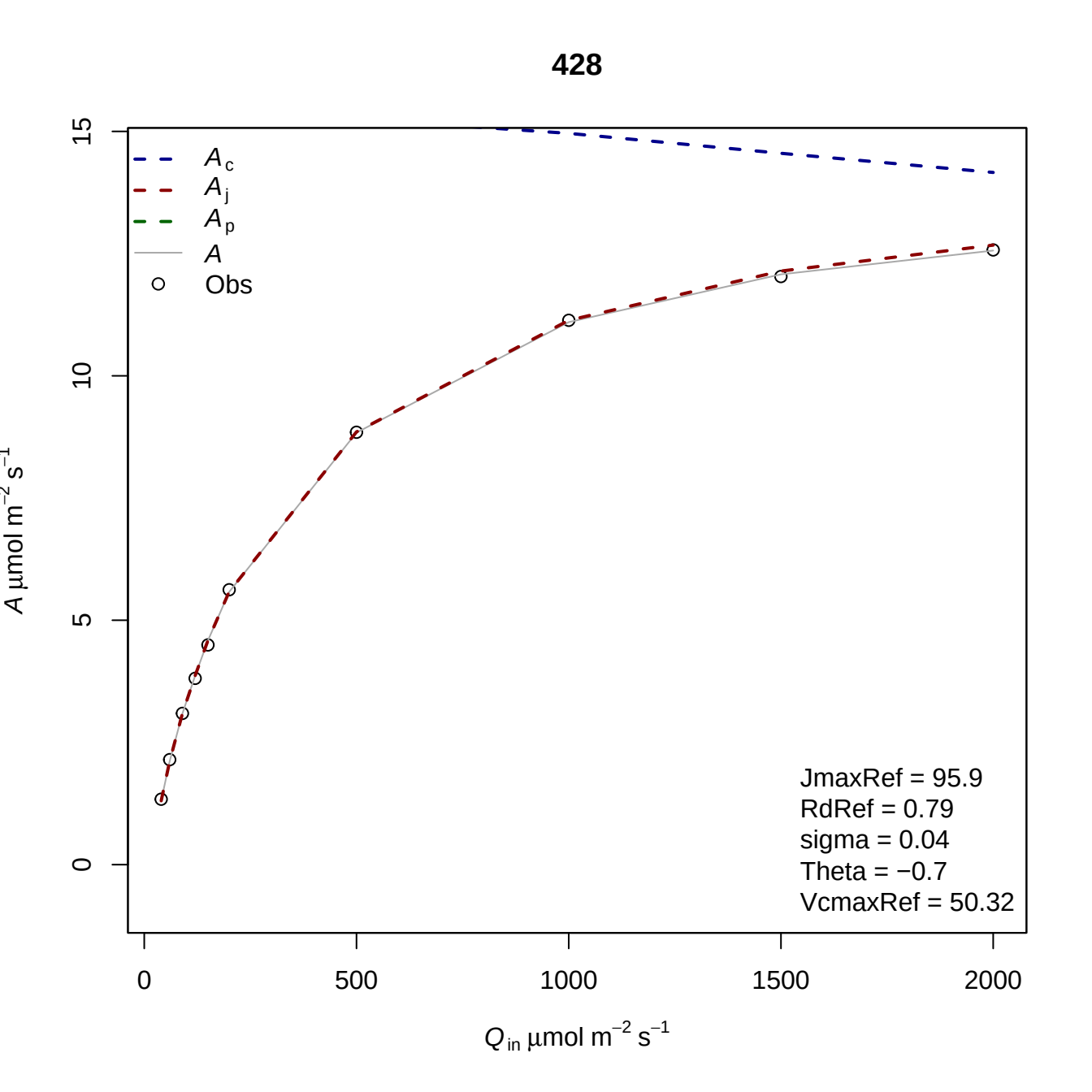

422
A_c
A_j
A_p
A
Obs
JmaxRef = 137.41
RdRef = 0.8
sigma = 0.1
Theta = 0.09
VcmaxRef = 66.99
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

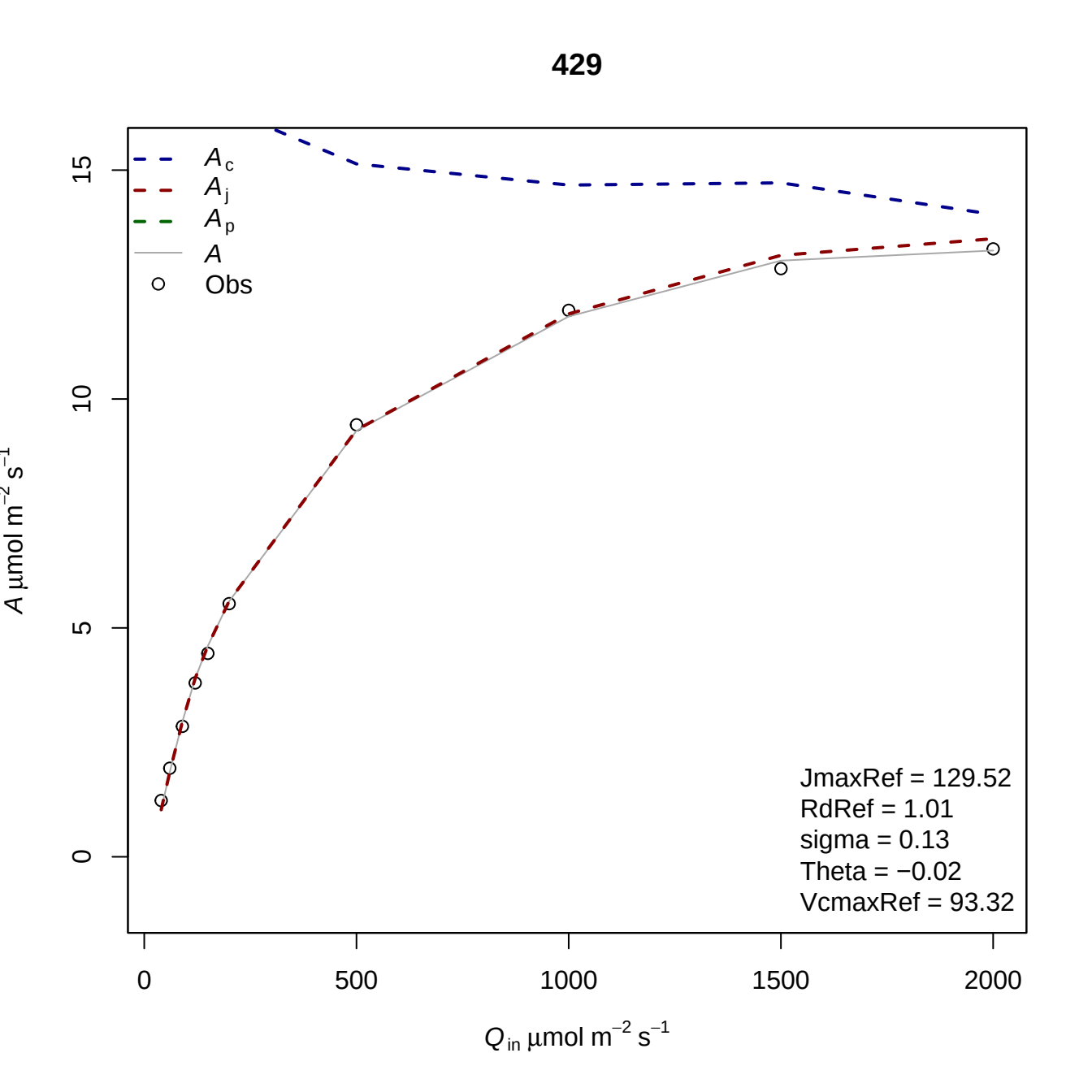

424
A_c
A_j
A_p
A
Obs
JmaxRef = 148.29
RdRef = 1.7
sigma = 0.16
Theta = 0.11
VcmaxRef = 195.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

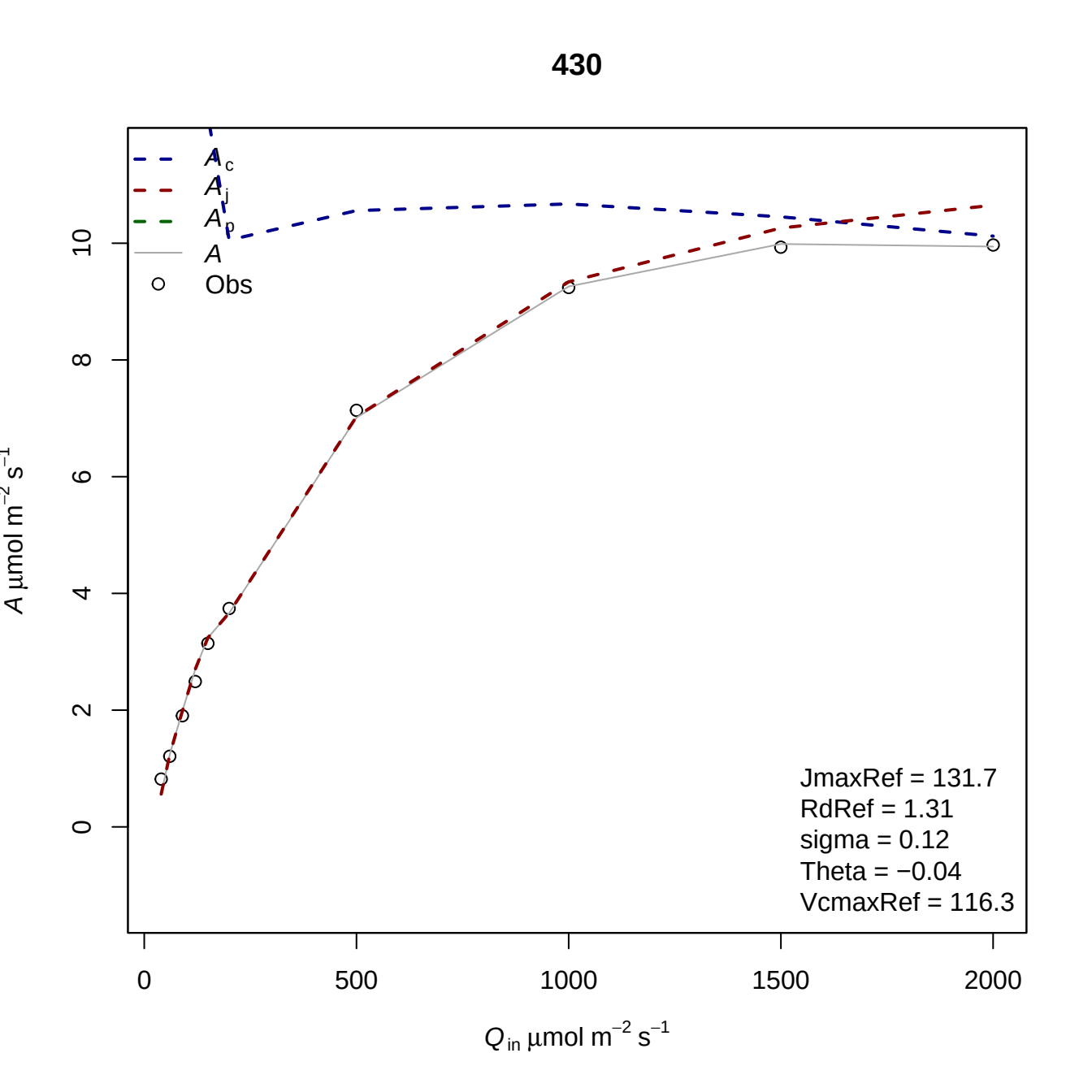

425
A_c
A_j
A_p
A
Obs
JmaxRef = 177.59
RdRef = 1.53
sigma = 0.21
Theta = −6.6
VcmaxRef = 915425.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

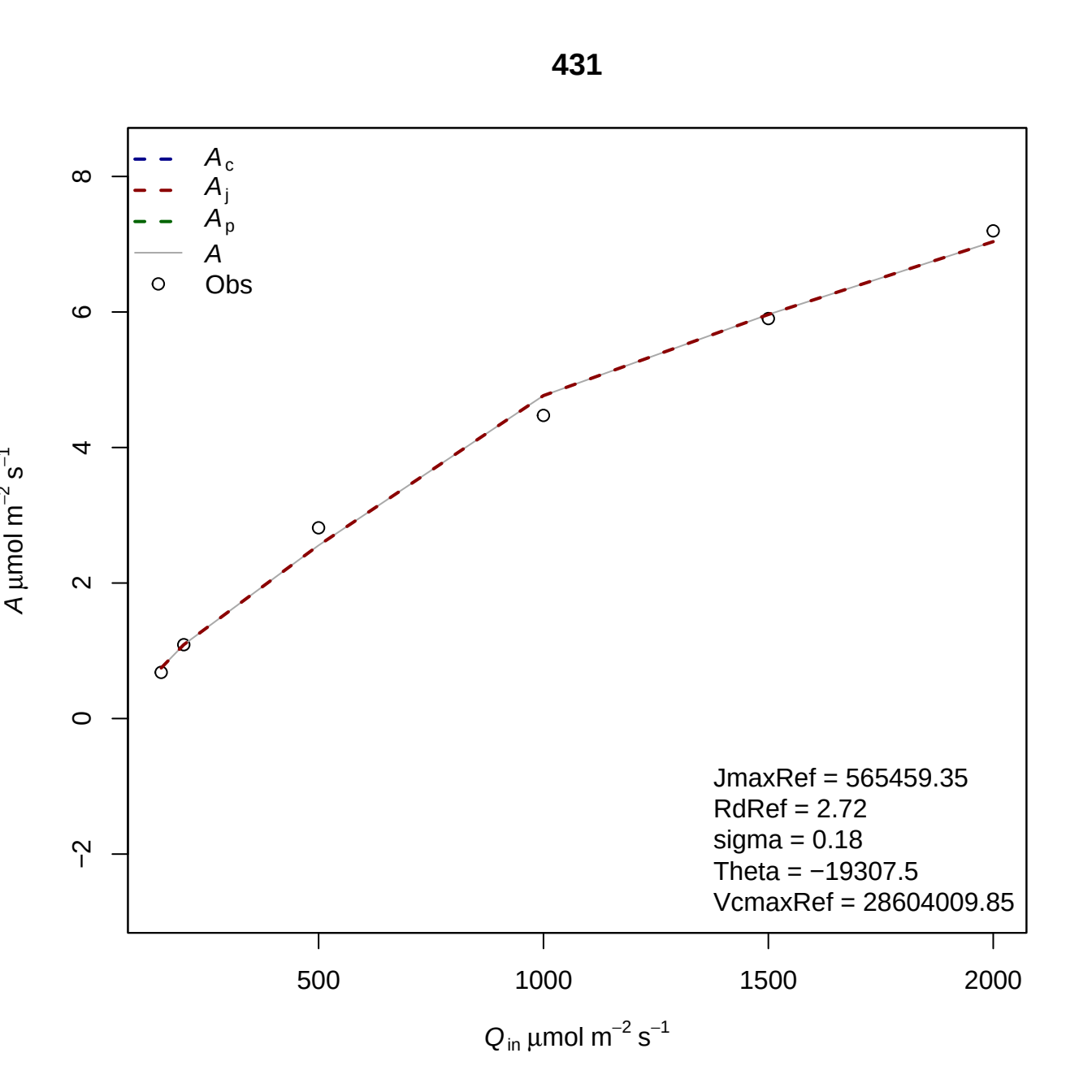

426
A_c
A_j
A_p
A
Obs
JmaxRef = 98.34
RdRef = 1.81
sigma = 0.15
Theta = −1.83
VcmaxRef = 269734.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

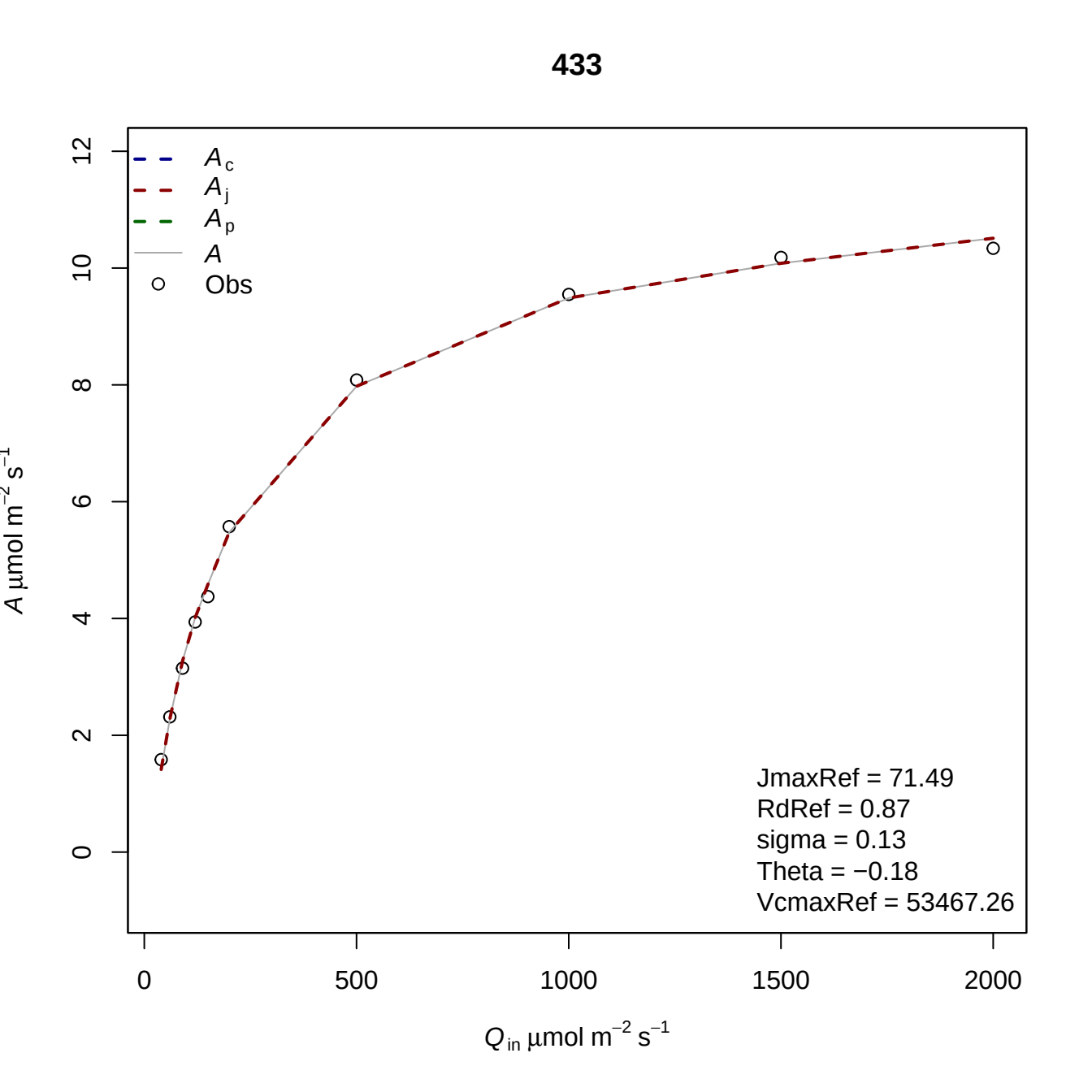

427
A_c
A_j
A_p
A
Obs
JmaxRef = 70.62
RdRef = 0.82
sigma = 0.09
Theta = 0.45
VcmaxRef = 32.84
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

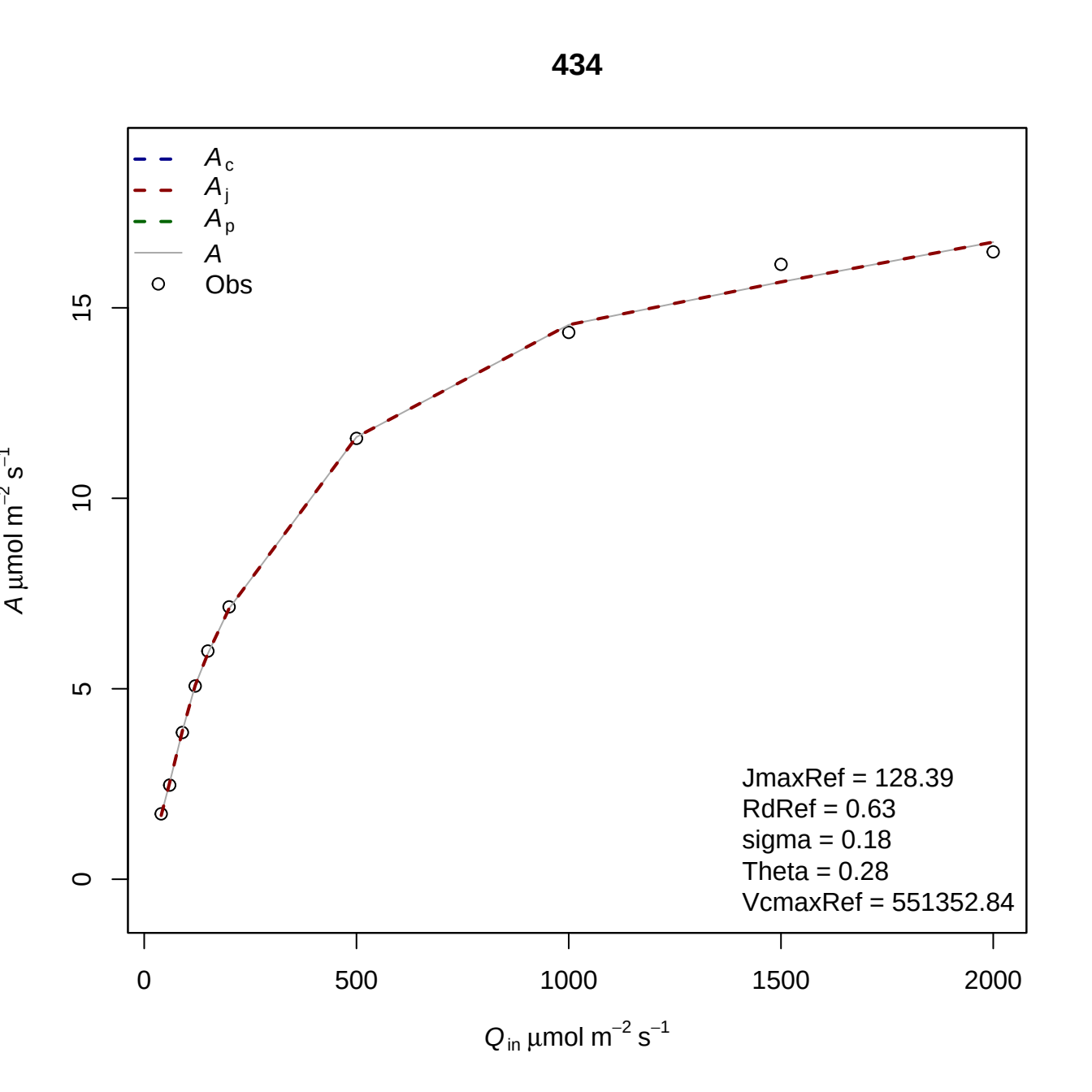

428
A_c
A_j
A_p
A
Obs
JmaxRef = 95.9
RdRef = 0.79
sigma = 0.04
Theta = −0.7
VcmaxRef = 50.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

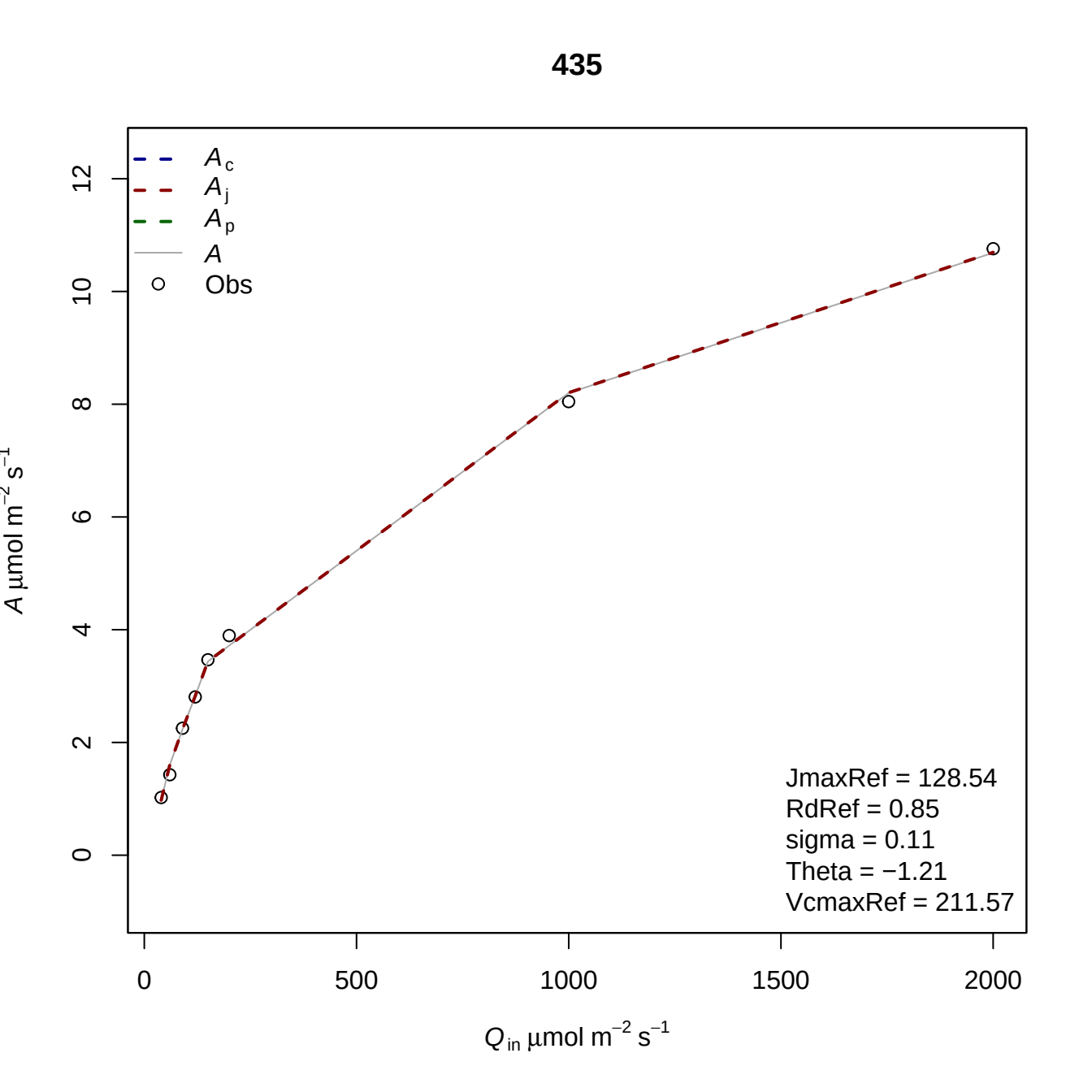

429
A_c
A_j
A_p
A
Obs
JmaxRef = 129.52
RdRef = 1.01
sigma = 0.13
Theta = −0.02
VcmaxRef = 93.32
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

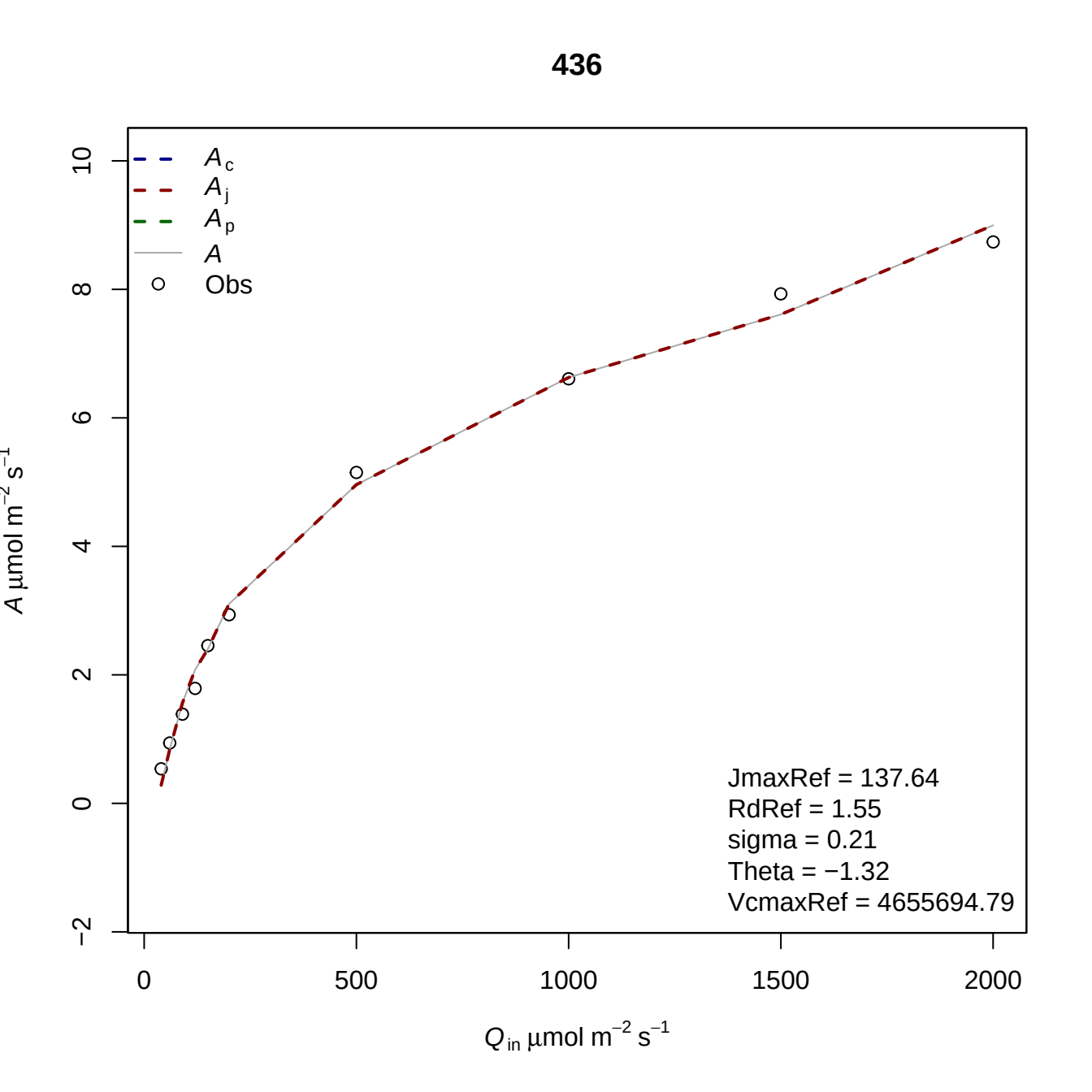

430
A_c
A_j
A_p
A
Obs
JmaxRef = 131.7
RdRef = 1.31
sigma = 0.12
Theta = −0.04
VcmaxRef = 116.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

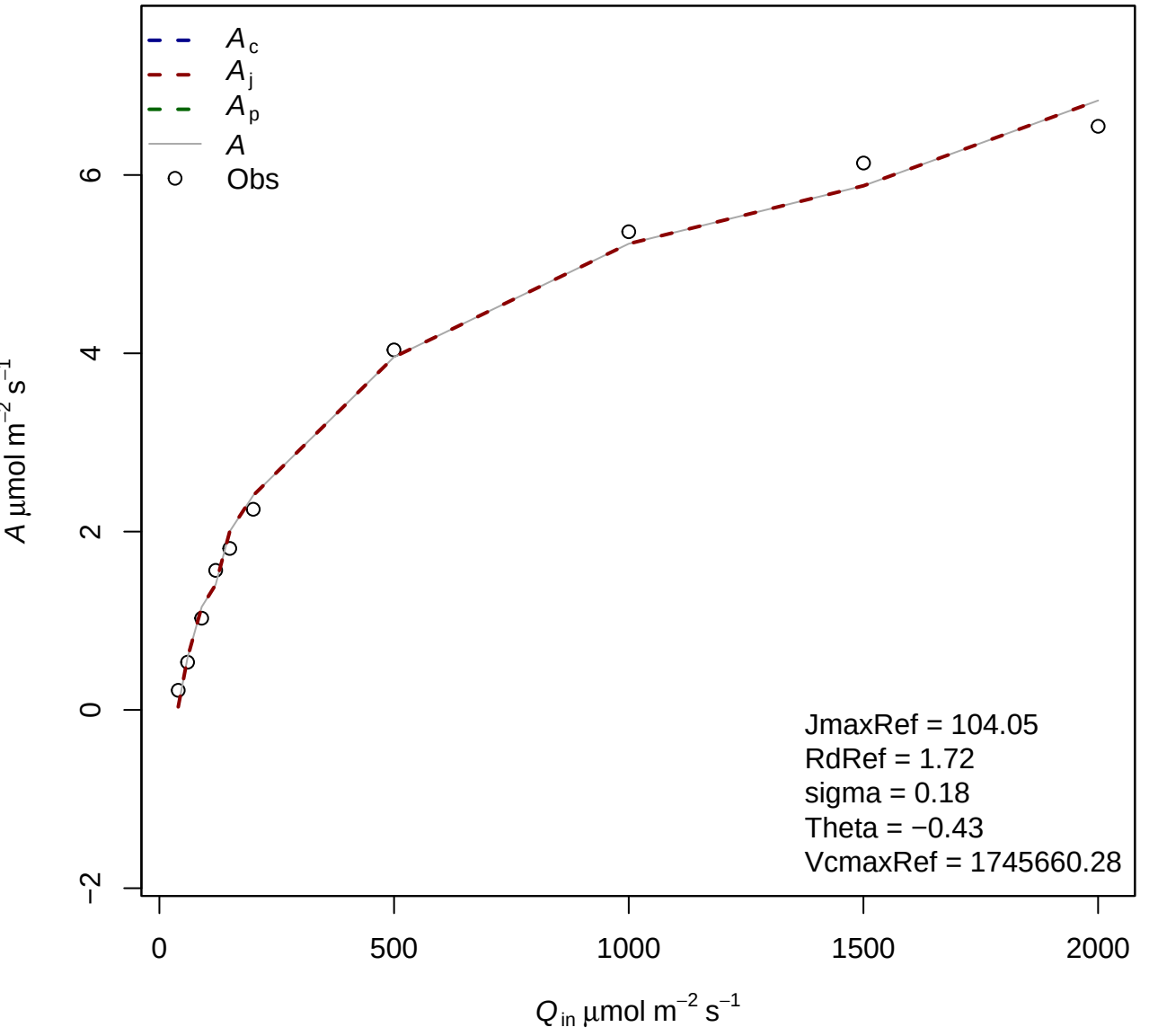

431
A_c
A_j
A_p
A
Obs
JmaxRef = 565459.35
RdRef = 2.72
sigma = 0.18
Theta = −19307.5
VcmaxRef = 28604009.85
8
6
4
2
0
−2
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

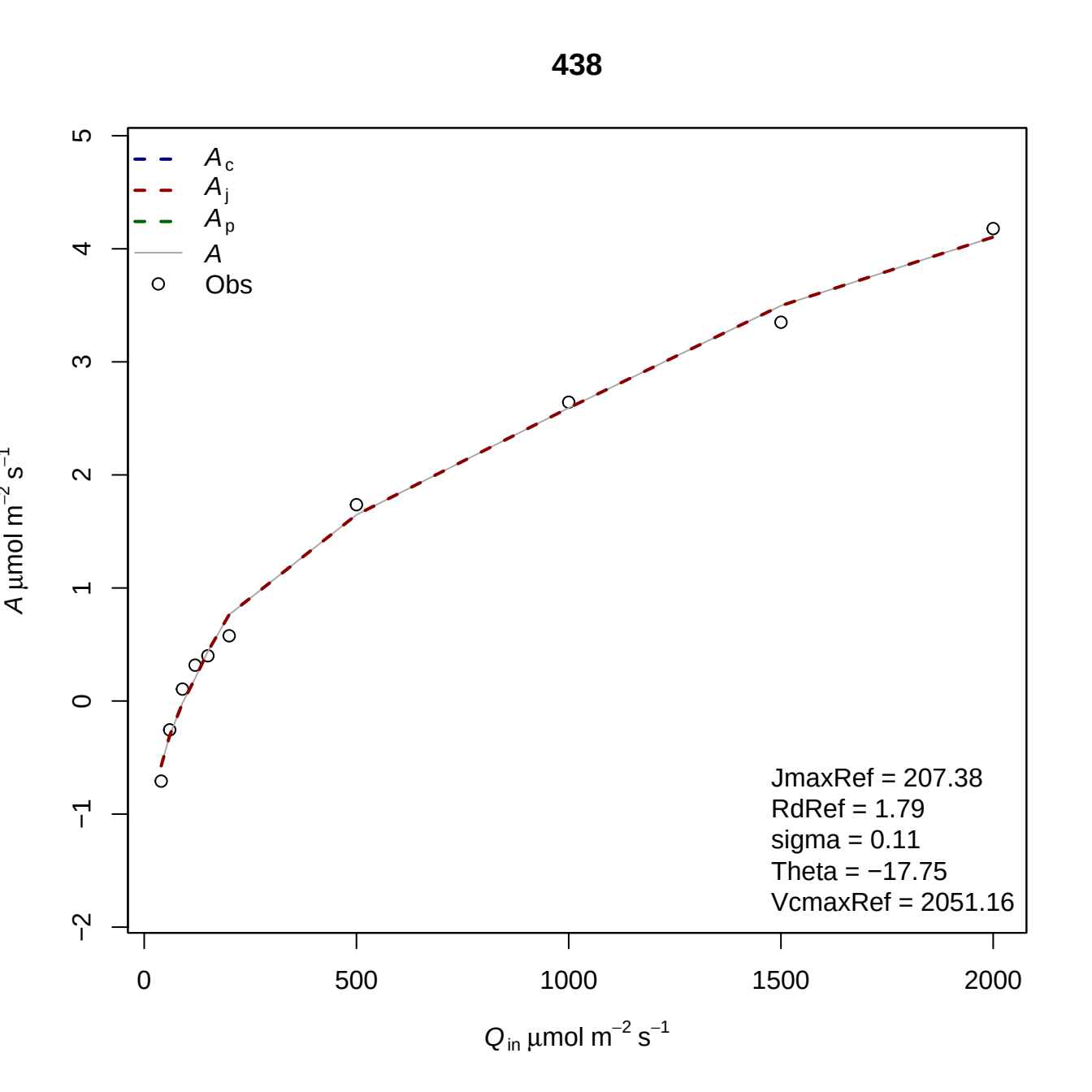

433
A_c
A_j
A_p
A
Obs
JmaxRef = 71.49
RdRef = 0.87
sigma = 0.13
Theta = −0.18
VcmaxRef = 53467.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

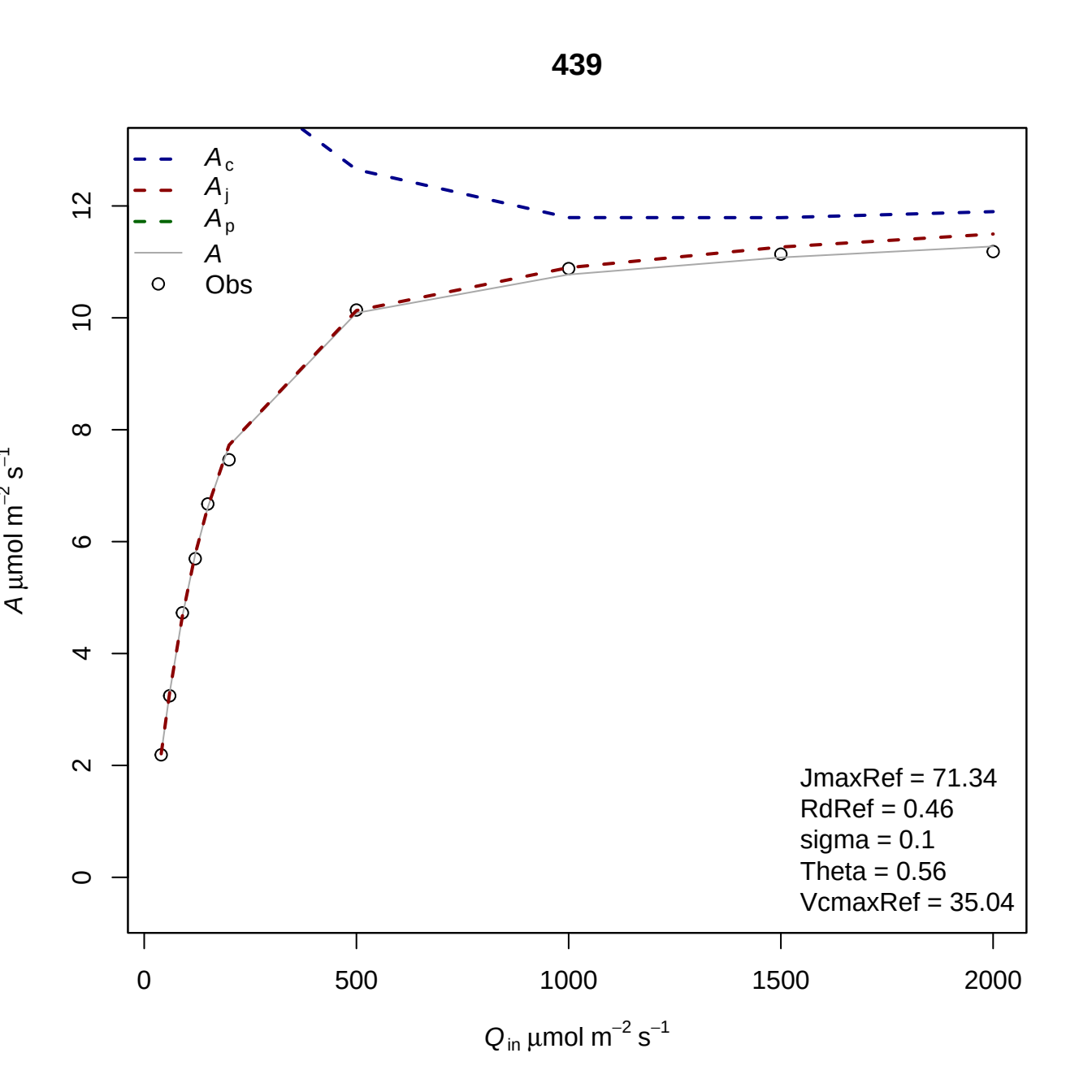

434
A_c
A_j
A_p
A
Obs
JmaxRef = 128.39
RdRef = 0.63
sigma = 0.18
Theta = 0.28
VcmaxRef = 551352.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

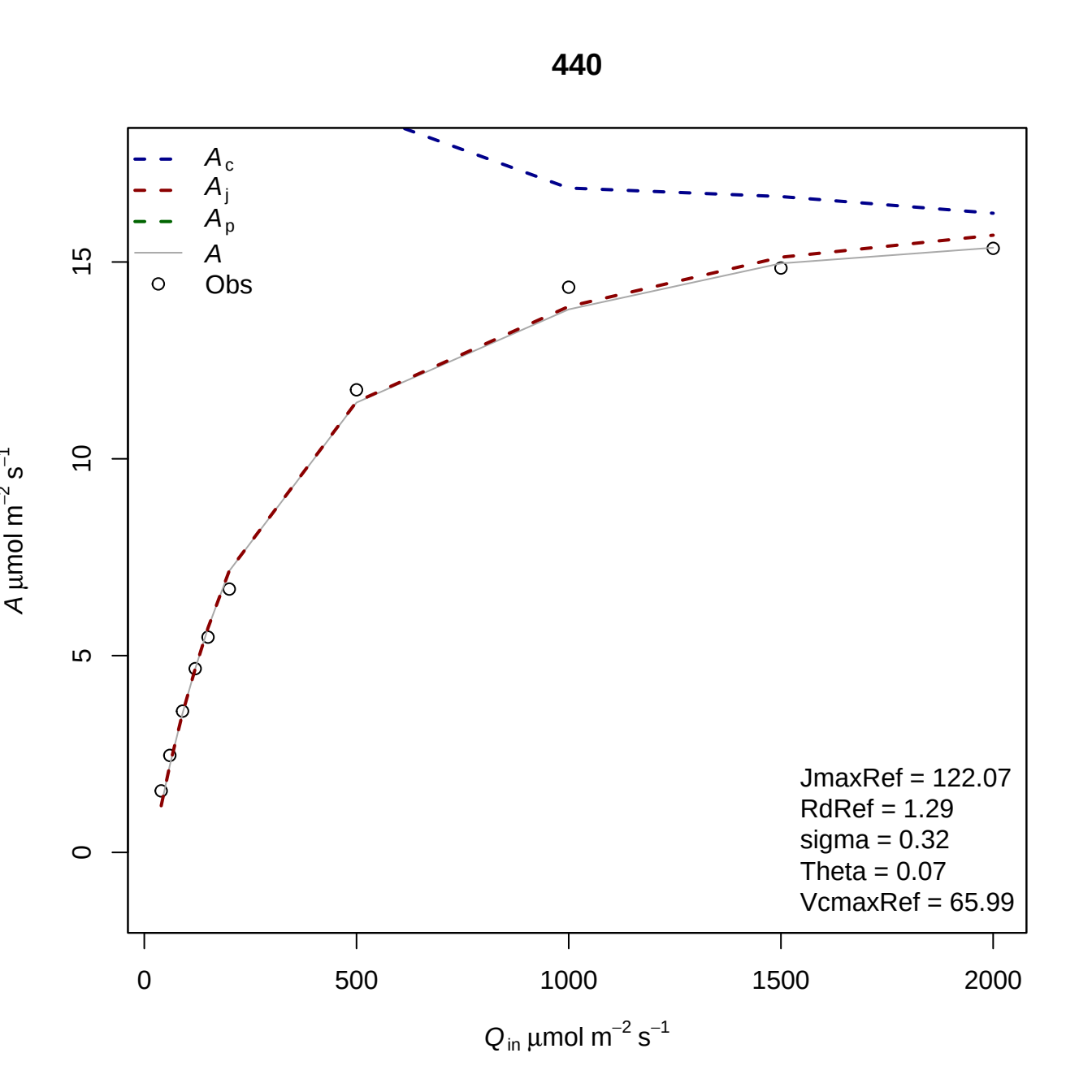

435
A_c
A_j
A_p
A
Obs
JmaxRef = 128.54
RdRef = 0.85
sigma = 0.11
Theta = −1.21
VcmaxRef = 211.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

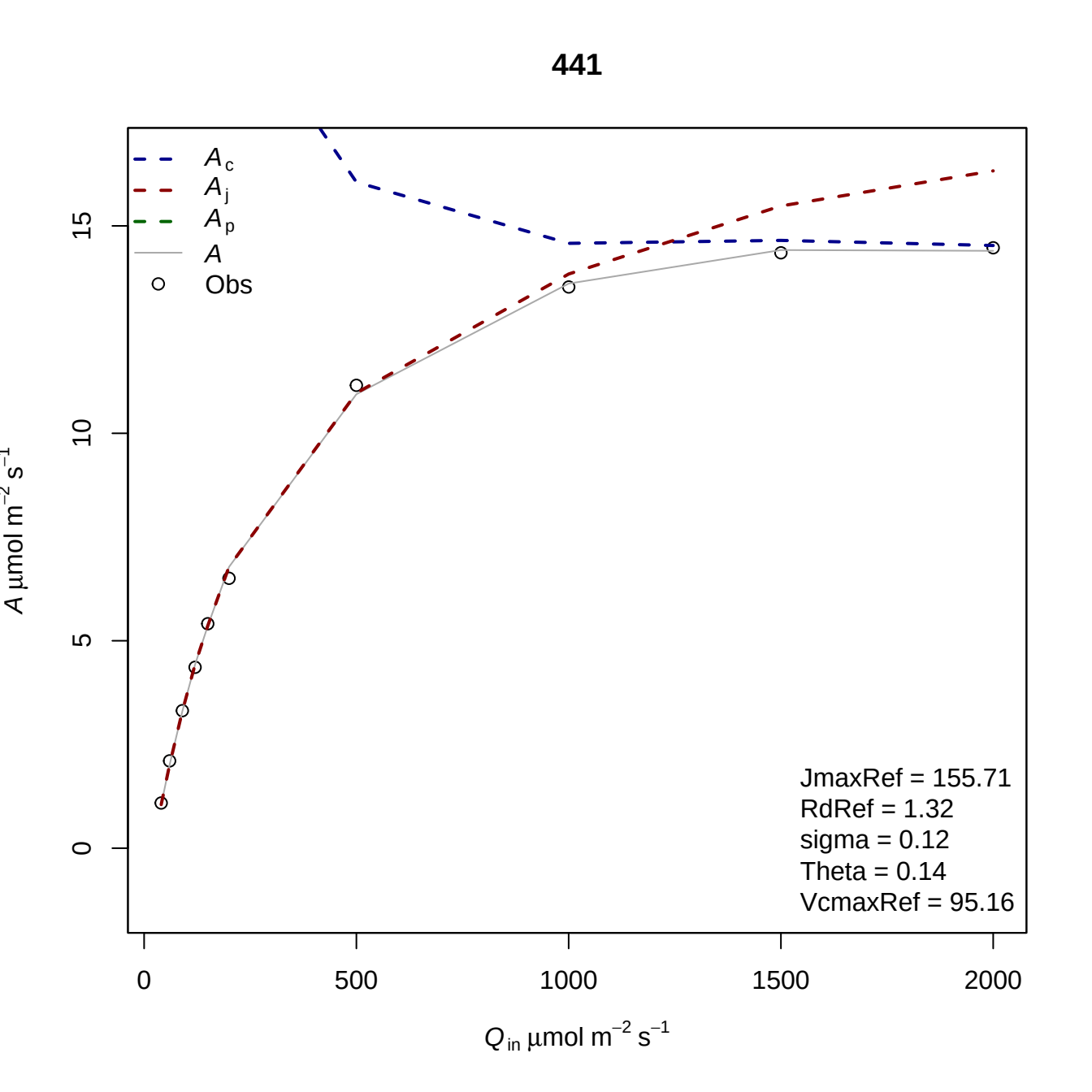

436
A_c
A_j
A_p
A
Obs
JmaxRef = 137.64
RdRef = 1.55
sigma = 0.21
Theta = −1.32
VcmaxRef = 4655694.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

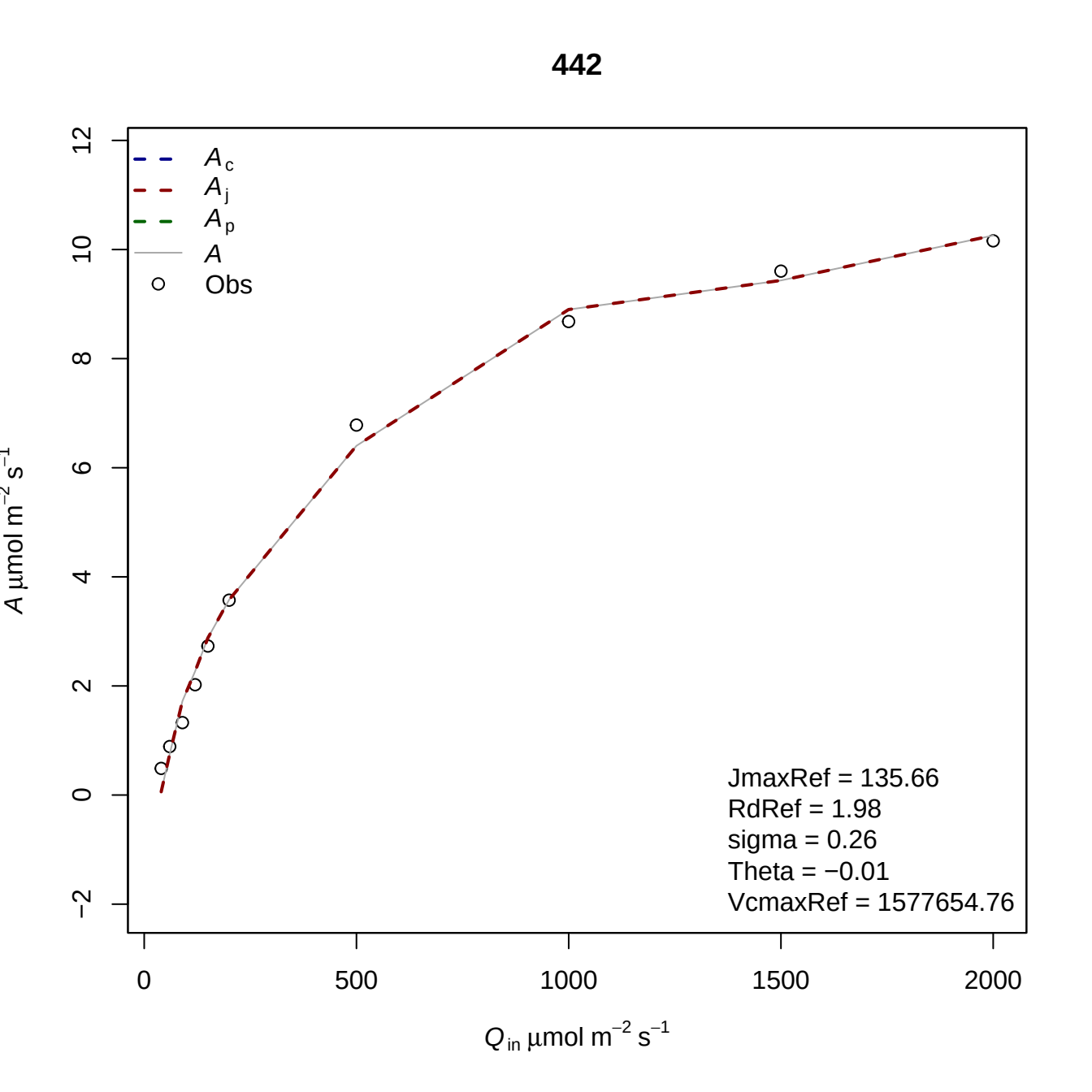

A_c
A_j
A_p
A
Obs
JmaxRef = 104.05
RdRef = 1.72
sigma = 0.18
Theta = −0.43
VcmaxRef = 1745660.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

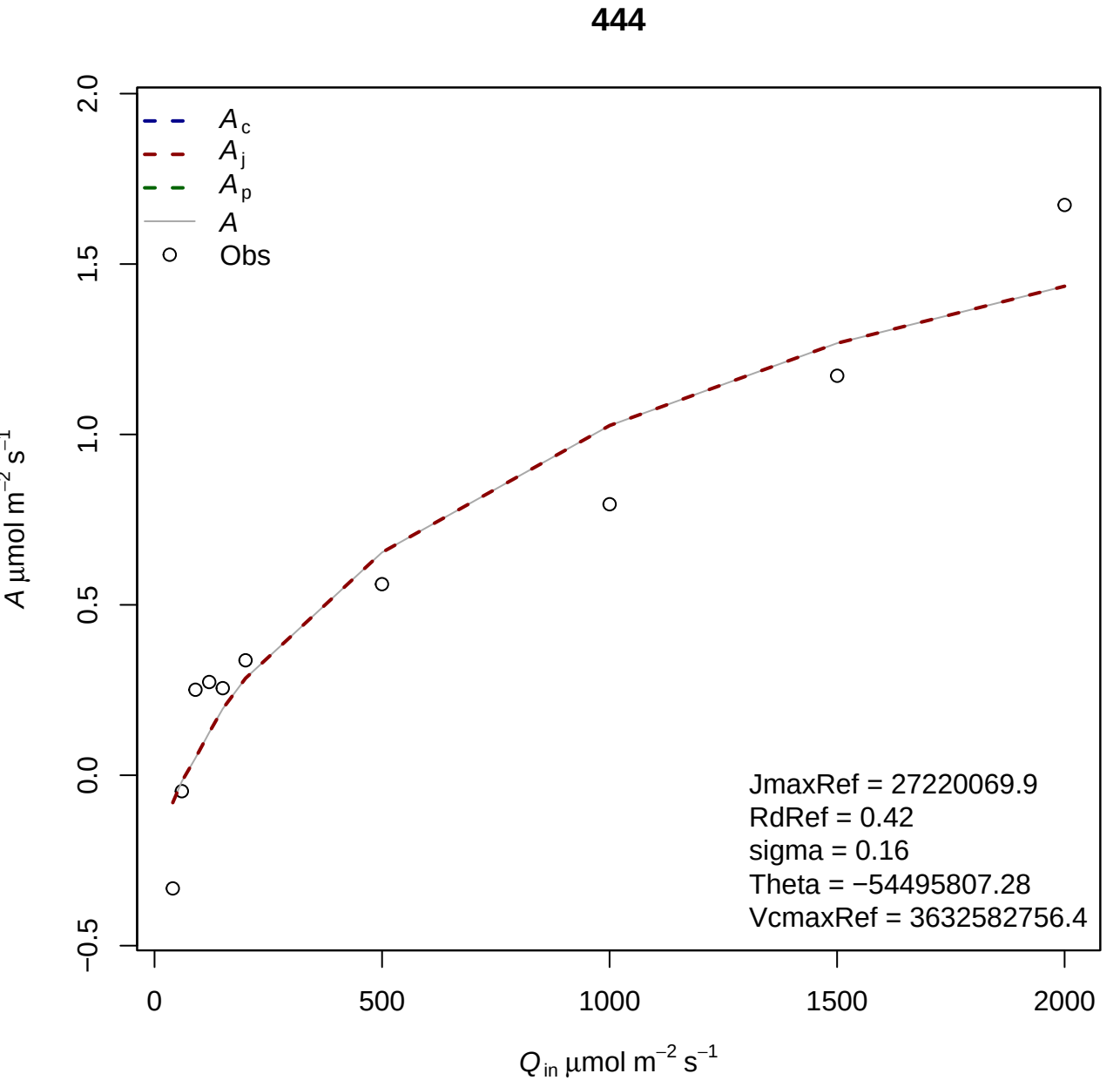

438
A_c
A_j
A_p
A
Obs
JmaxRef = 207.38
RdRef = 1.79
sigma = 0.11
Theta = −17.75
VcmaxRef = 2051.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

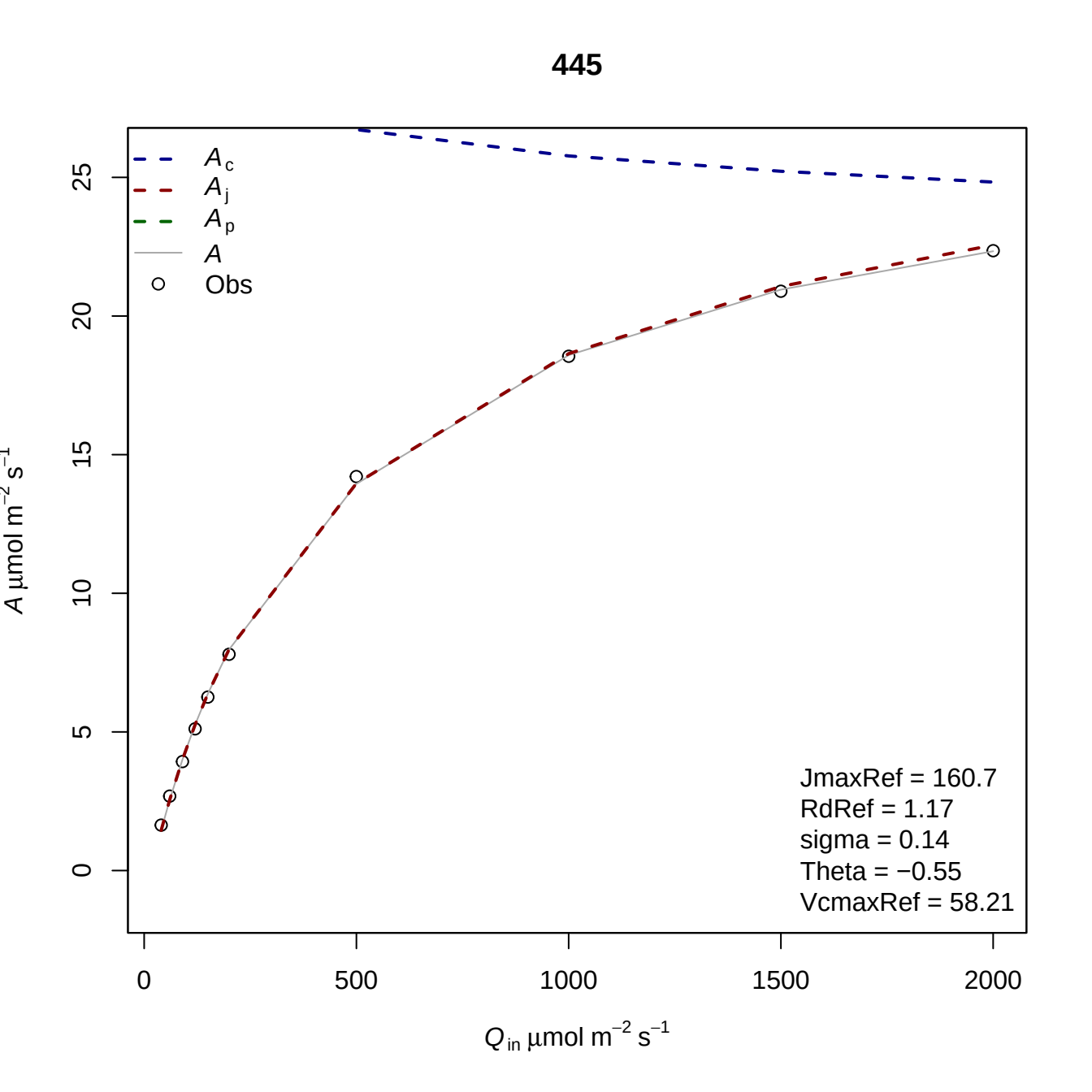

439
A_c
A_j
A_p
A
Obs
JmaxRef = 71.34
RdRef = 0.46
sigma = 0.1
Theta = 0.56
VcmaxRef = 35.04
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

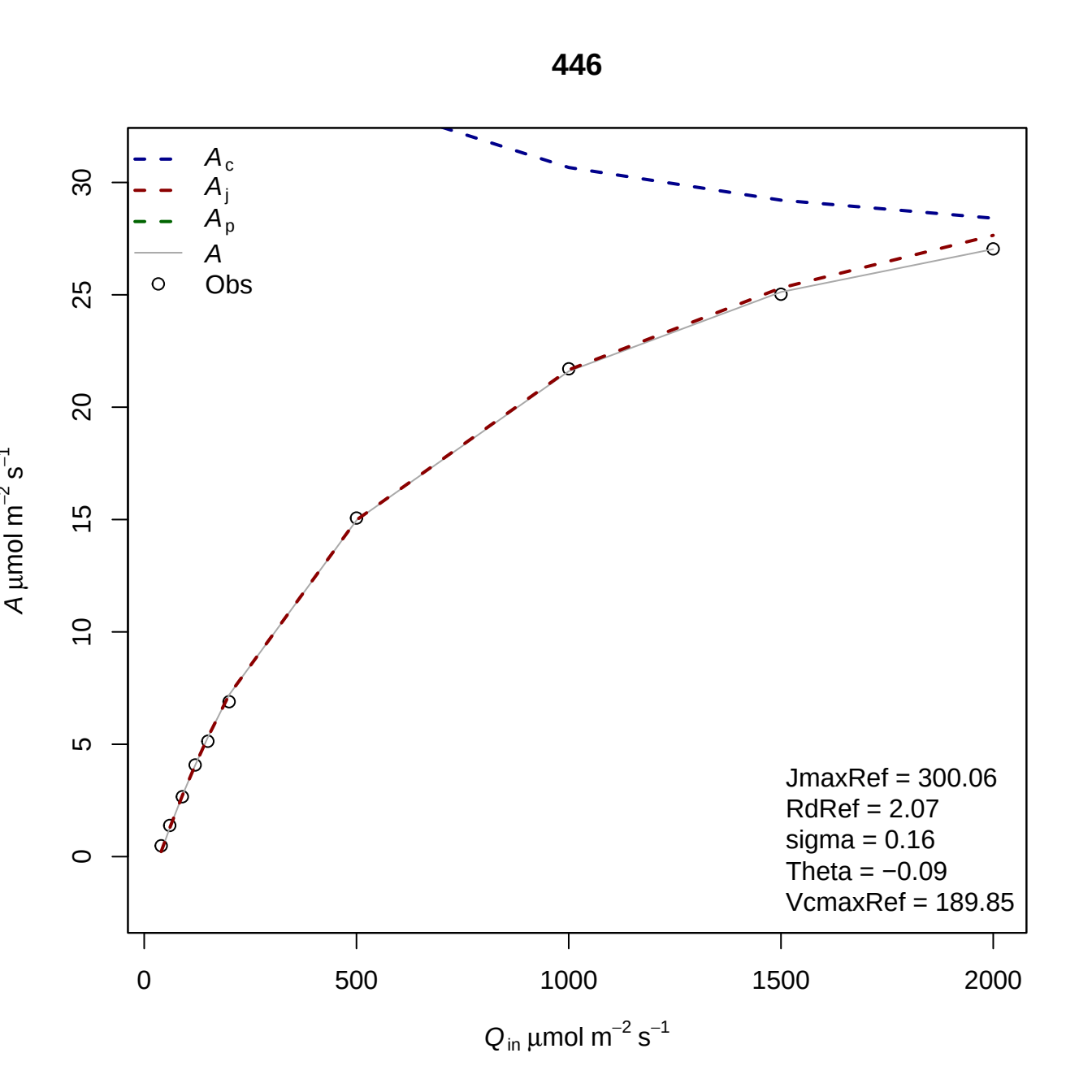

440
A_c
A_j
A_p
A
Obs
JmaxRef = 122.07
RdRef = 1.29
sigma = 0.32
Theta = 0.07
VcmaxRef = 65.99
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

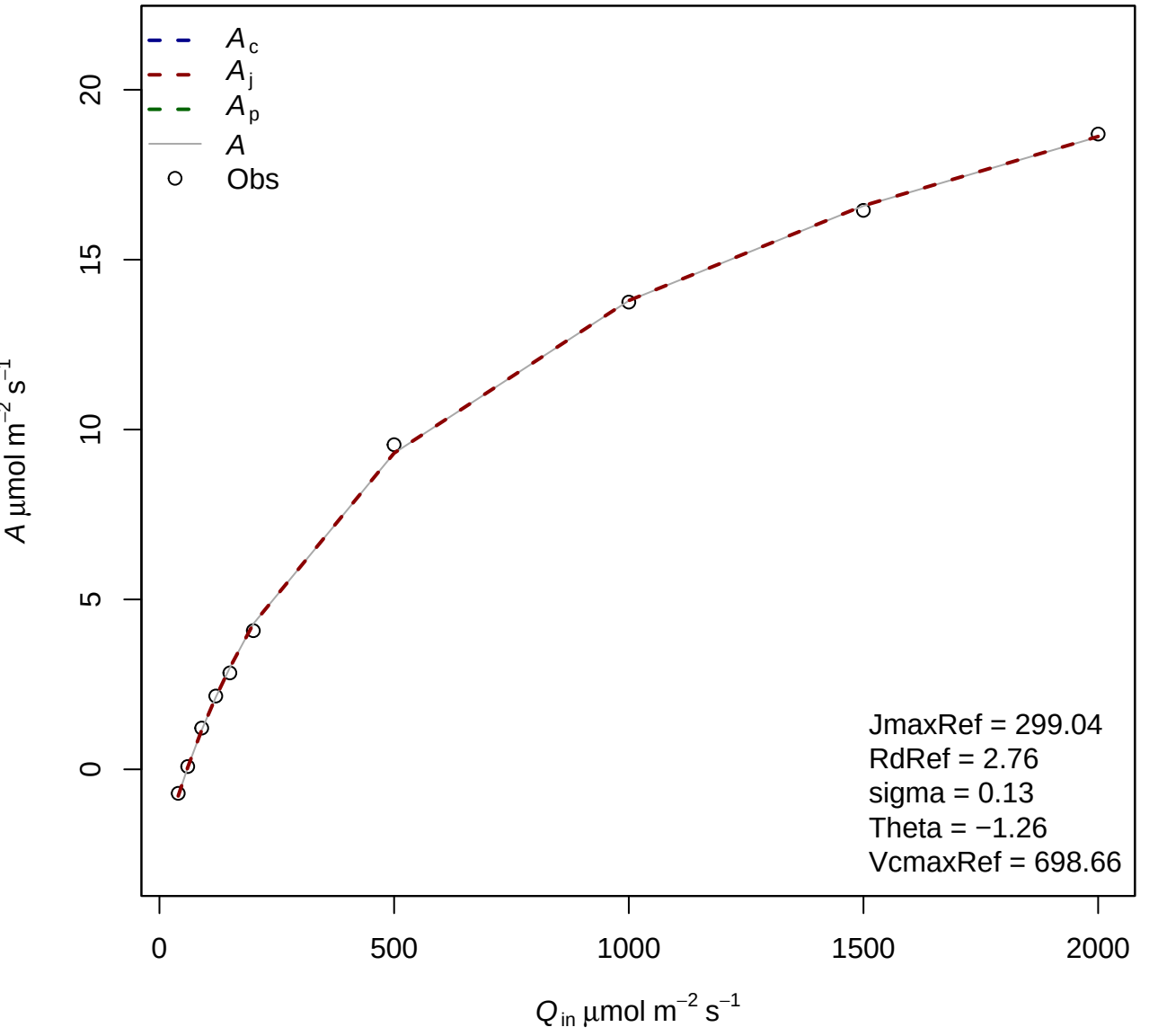

441
A_c
A_j
A_p
A
Obs
JmaxRef = 155.71
RdRef = 1.32
sigma = 0.12
Theta = 0.14
VcmaxRef = 95.16
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

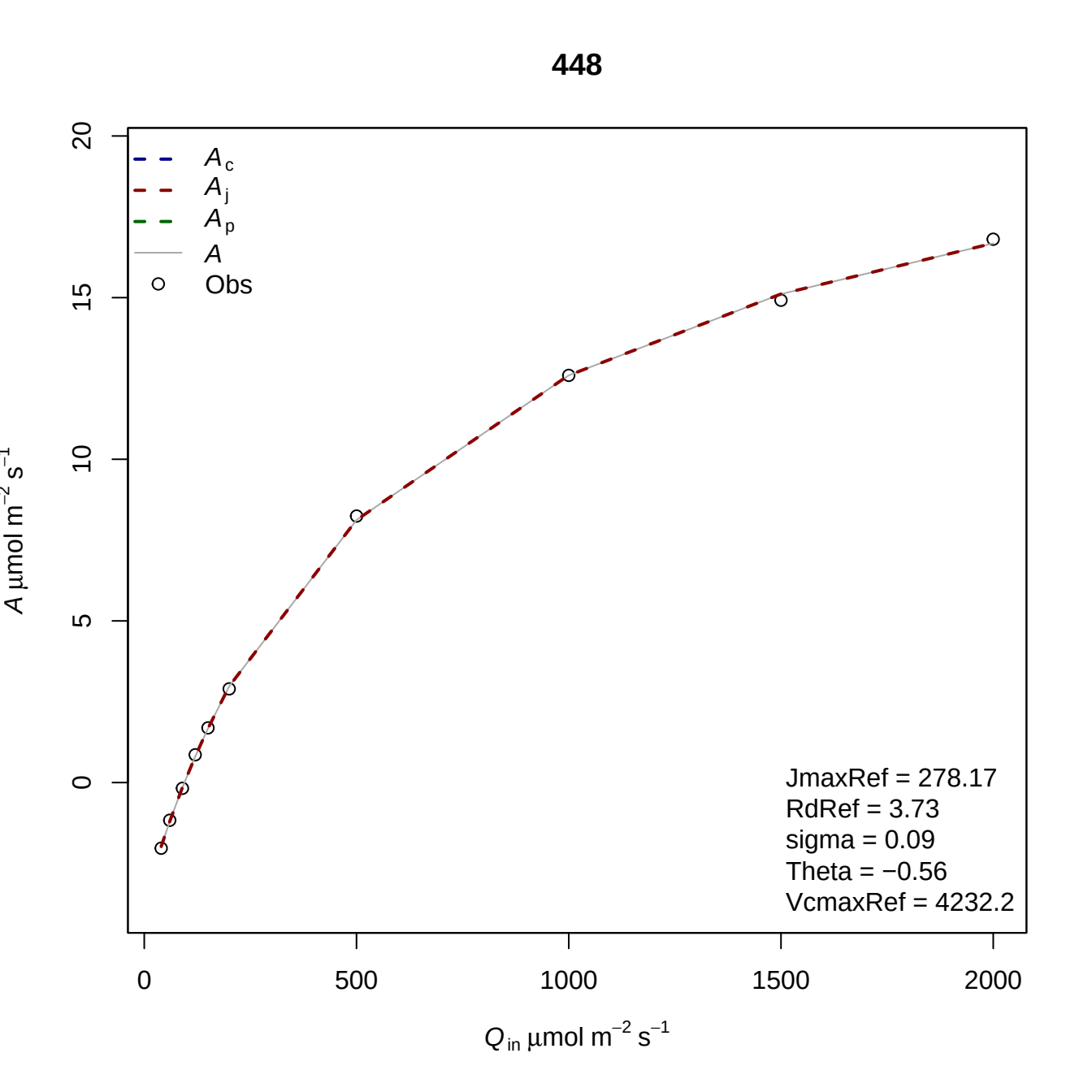

442
A_c
A_j
A_p
A
Obs
JmaxRef = 135.66
RdRef = 1.98
sigma = 0.26
Theta = −0.01
VcmaxRef = 1577654.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

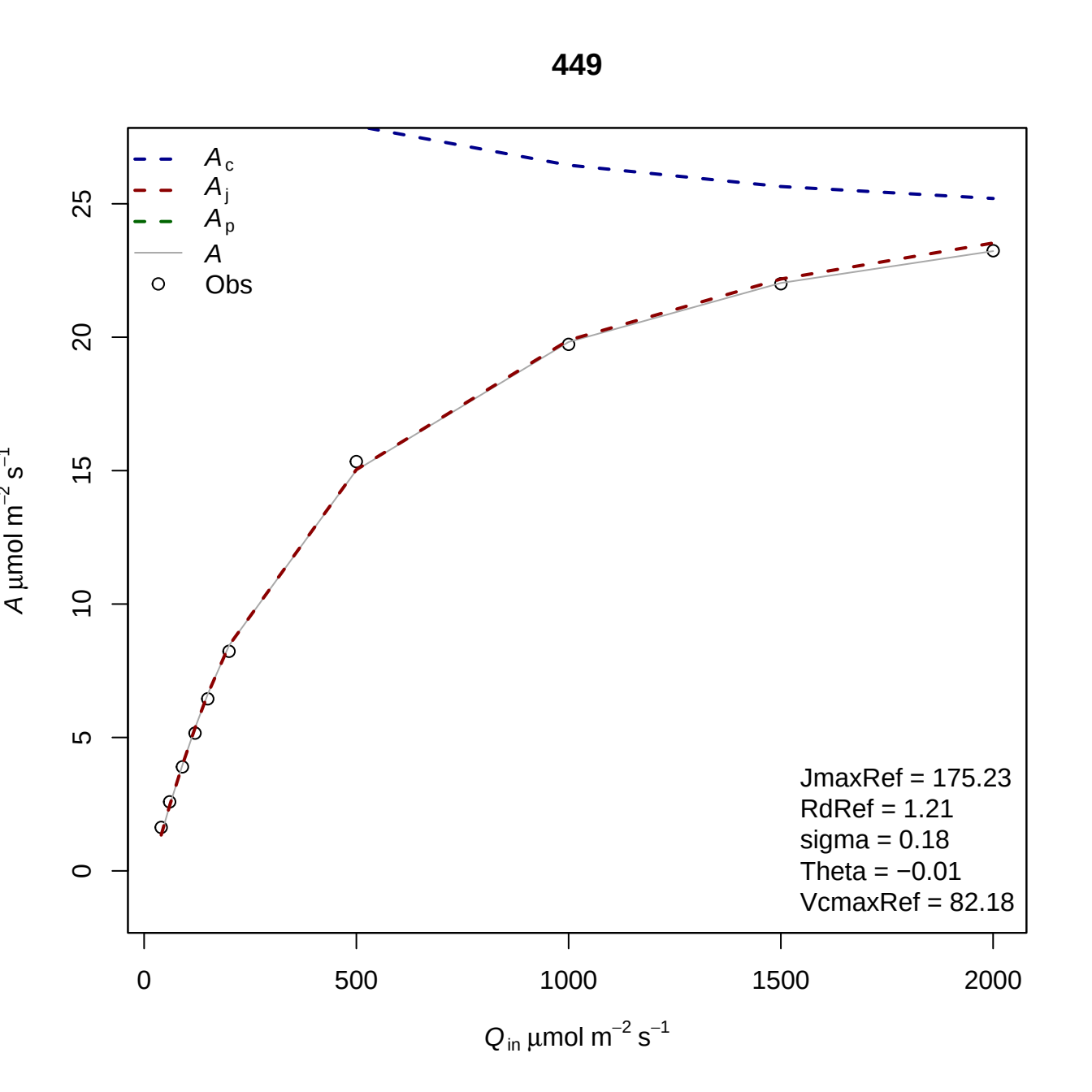

444
A_c
A_j
A_p
A
Obs
JmaxRef = 27220069.9
RdRef = 0.42
sigma = 0.16
Theta = −54495807.28
VcmaxRef = 3632582756.4
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

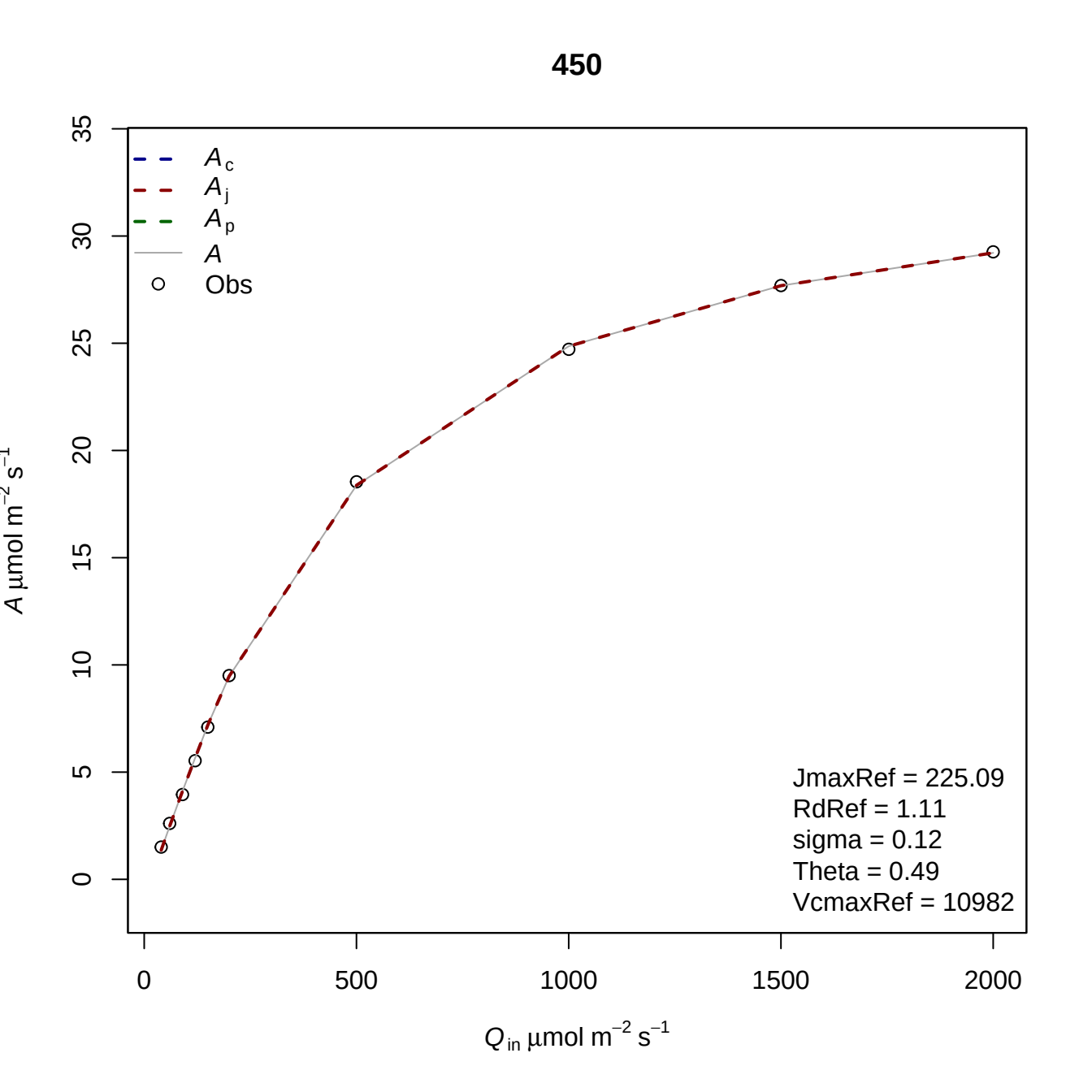

445
A_c
A_j
A_p
A
Obs
JmaxRef = 160.7
RdRef = 1.17
sigma = 0.14
Theta = −0.55
VcmaxRef = 58.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

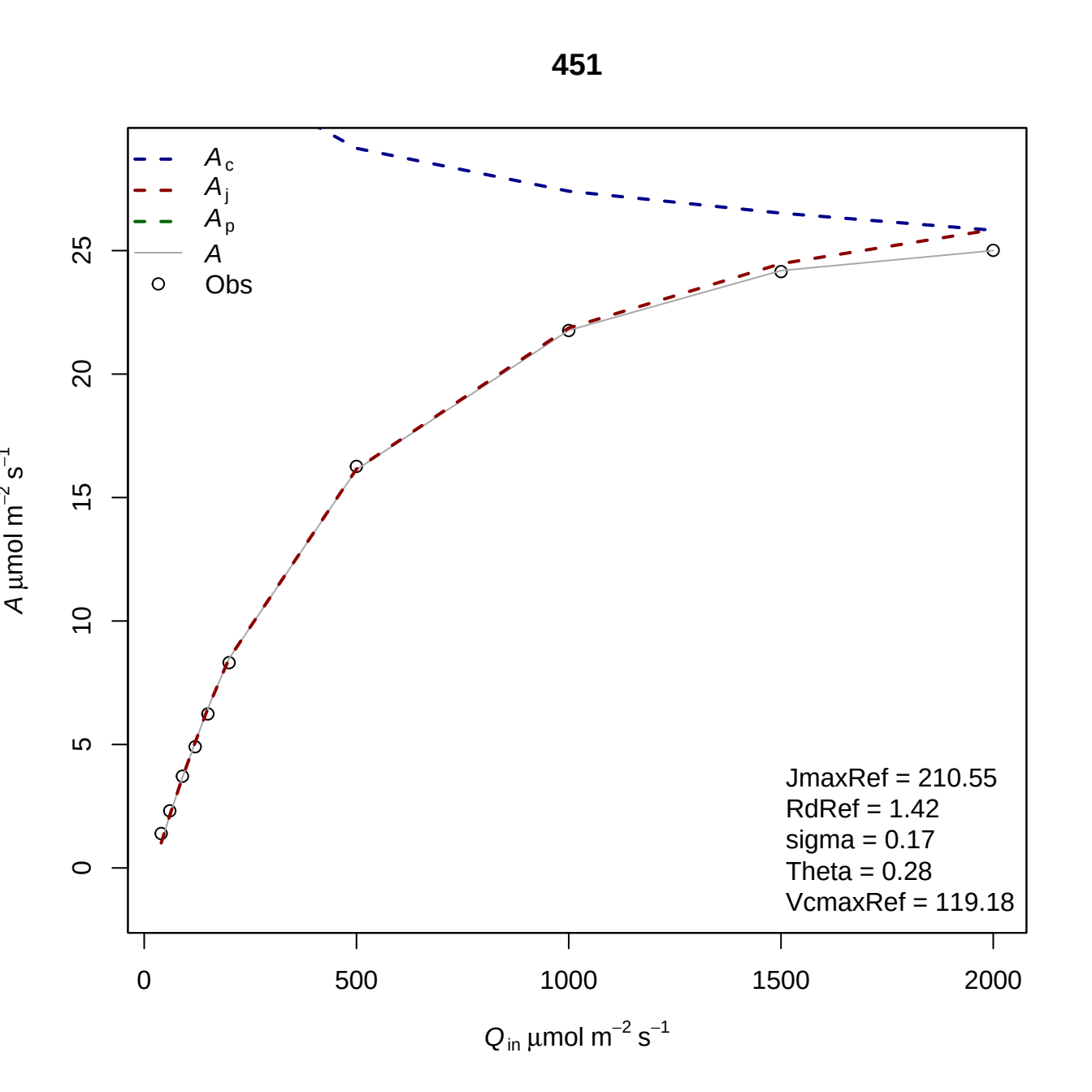

446
A_c
A_j
A_p
A
Obs
JmaxRef = 300.06
RdRef = 2.07
sigma = 0.16
Theta = −0.09
VcmaxRef = 189.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

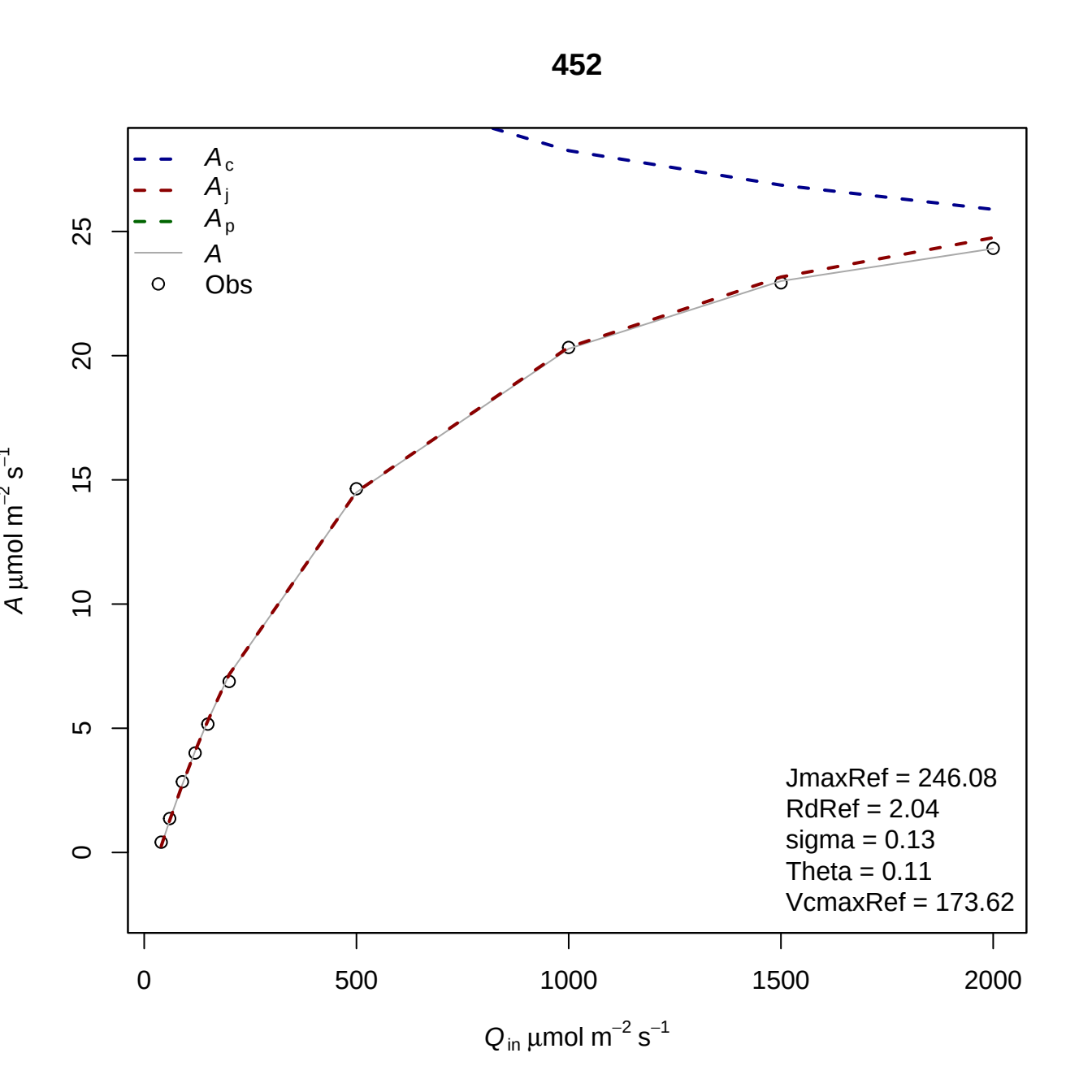

A_c
A_j
A_p
A
Obs
JmaxRef = 299.04
RdRef = 2.76
sigma = 0.13
Theta = −1.26
VcmaxRef = 698.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

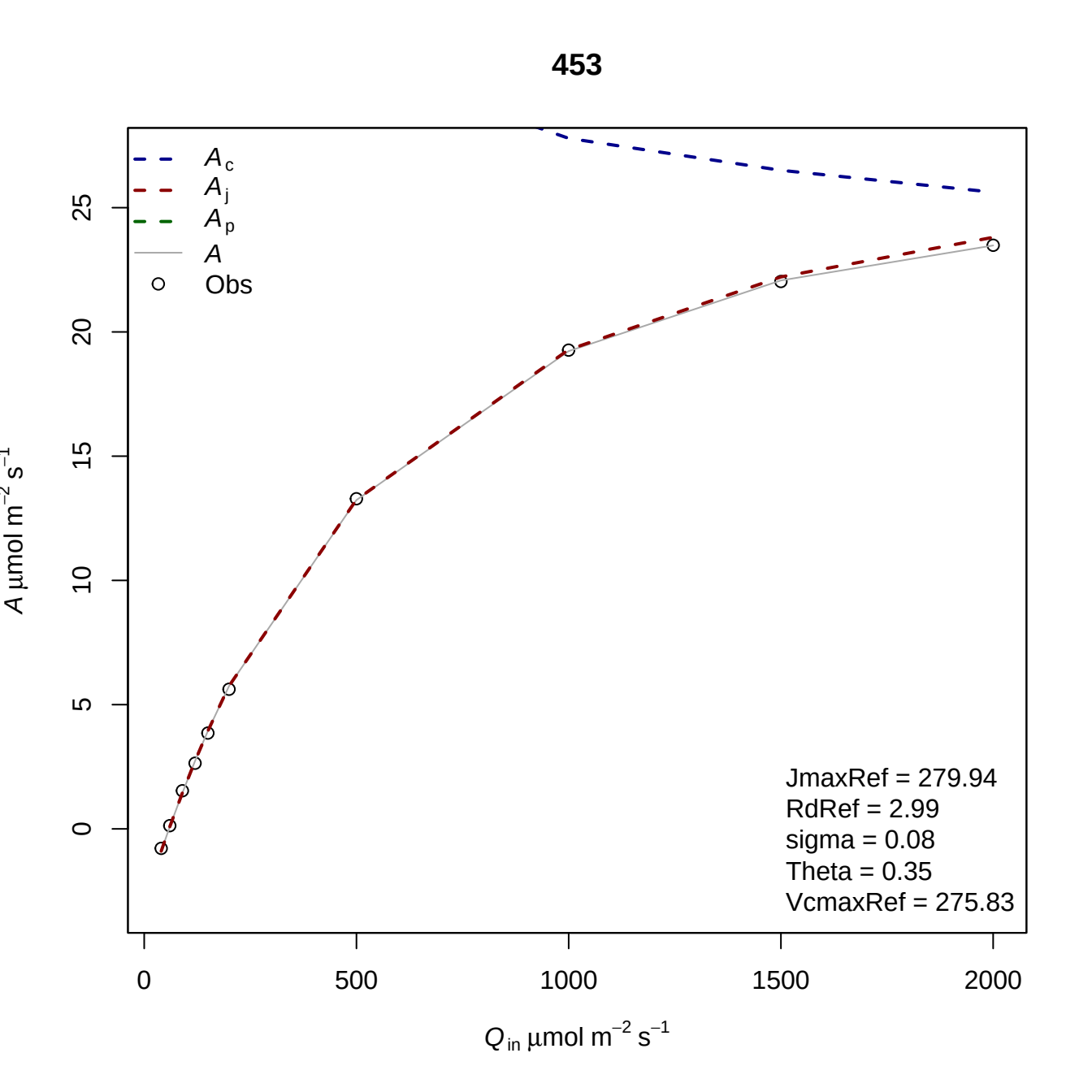

448
A_c
A_j
A_p
A
Obs
JmaxRef = 278.17
RdRef = 3.73
sigma = 0.09
Theta = −0.56
VcmaxRef = 4232.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

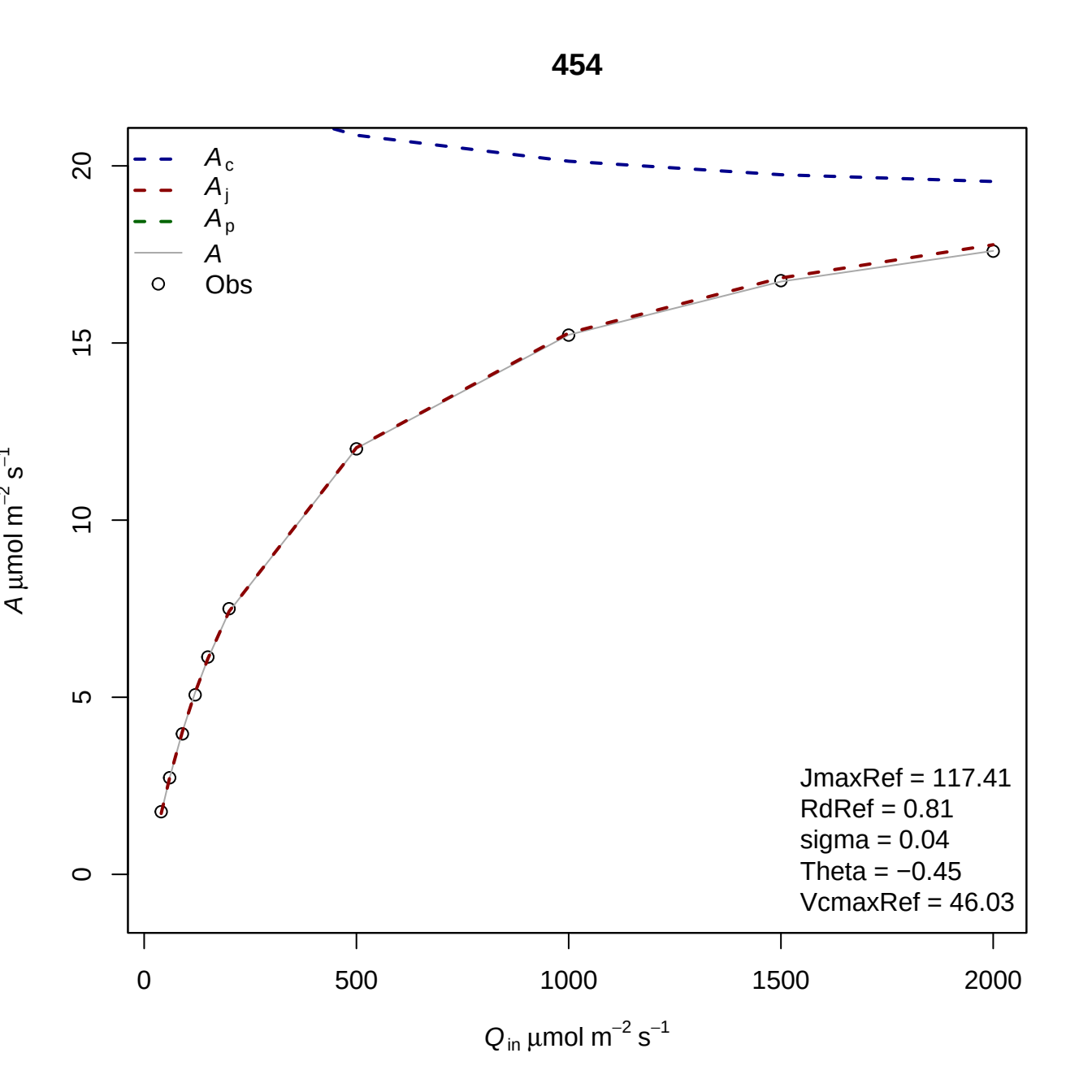

449
A_c
A_j
A_p
A
Obs
JmaxRef = 175.23
RdRef = 1.21
sigma = 0.18
Theta = −0.01
VcmaxRef = 82.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

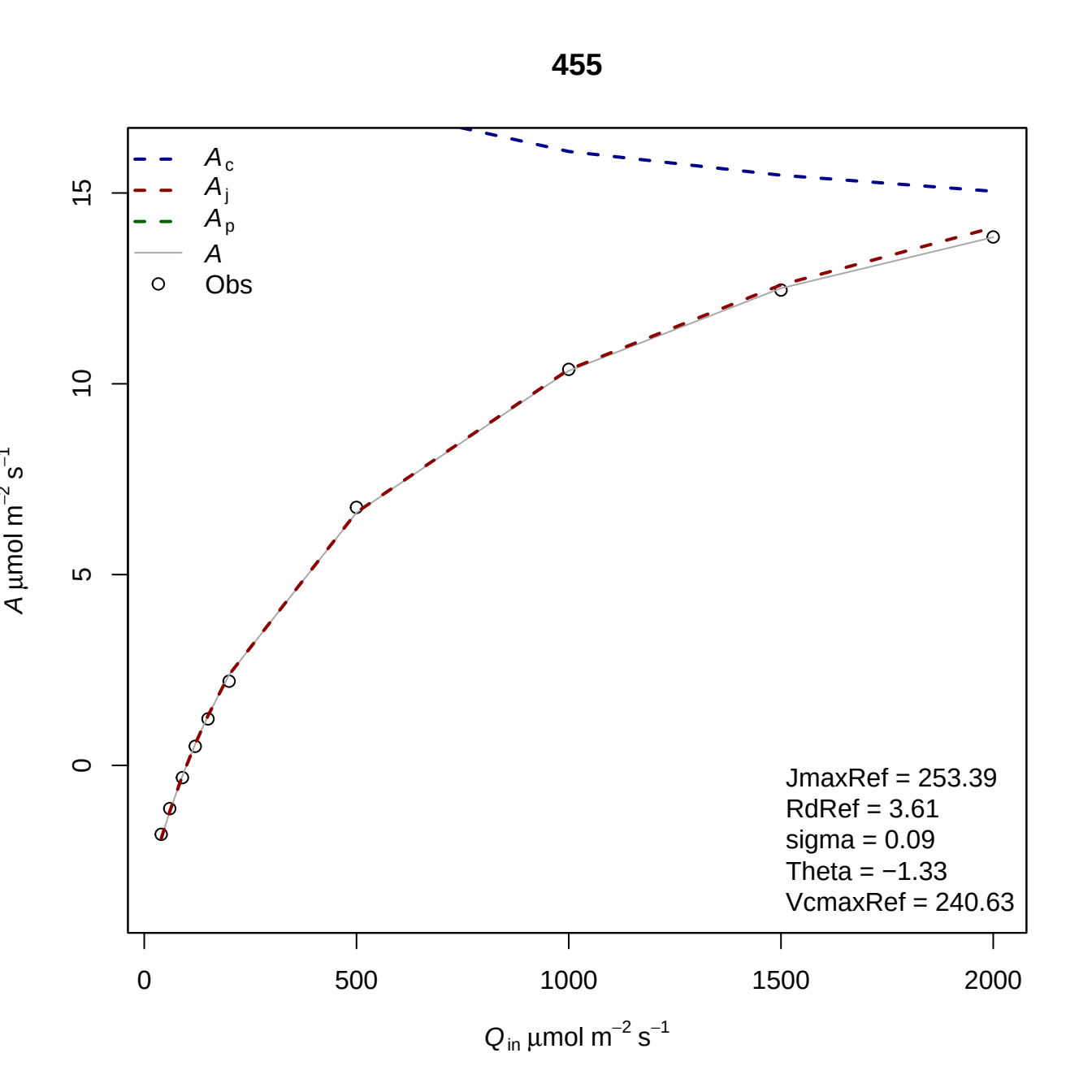

450
A_c
A_j
A_p
A
Obs
JmaxRef = 225.09
RdRef = 1.11
sigma = 0.12
Theta = 0.49
VcmaxRef = 10982
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

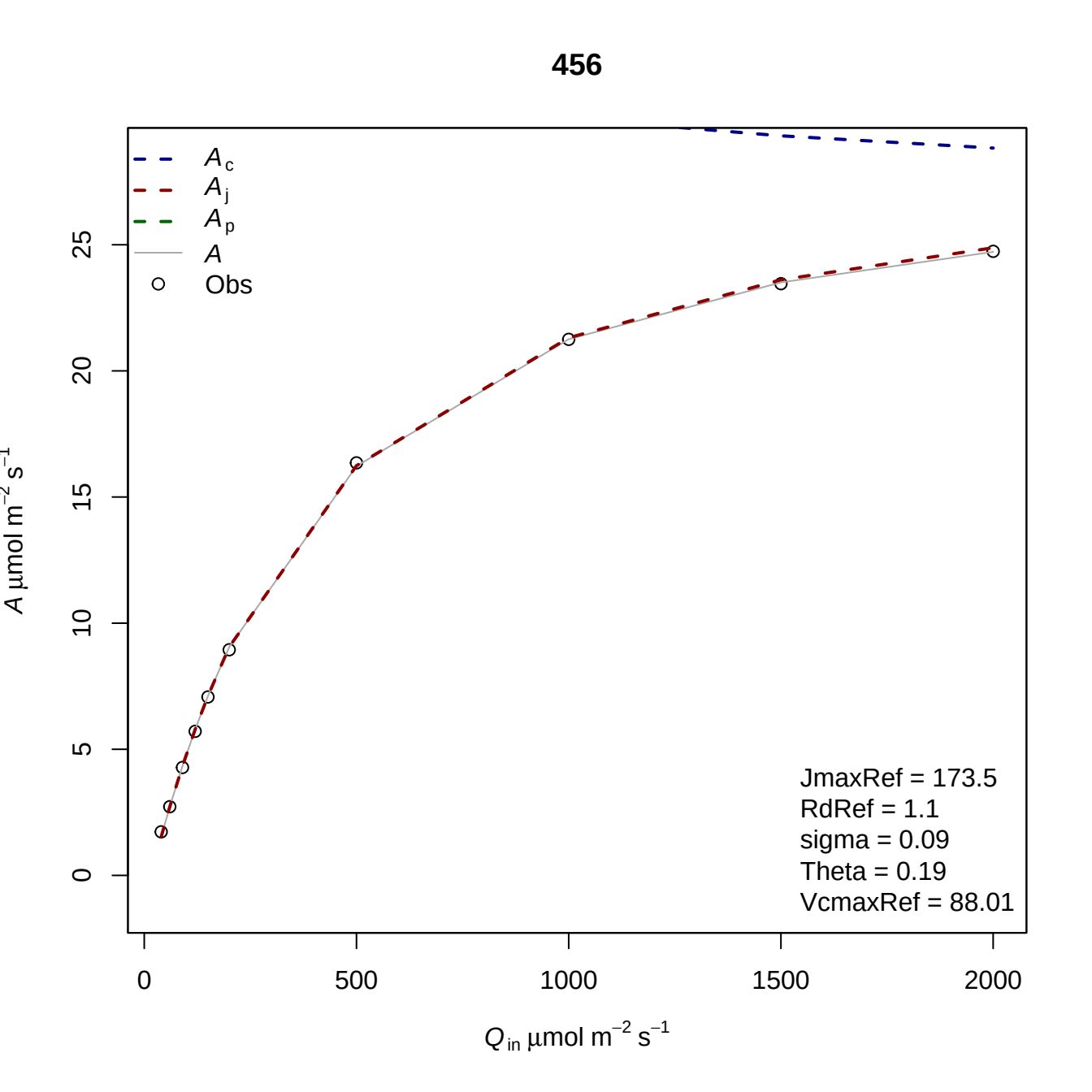

451
A_c
A_j
A_p
A
Obs
JmaxRef = 210.55
RdRef = 1.42
sigma = 0.17
Theta = 0.28
VcmaxRef = 119.18
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

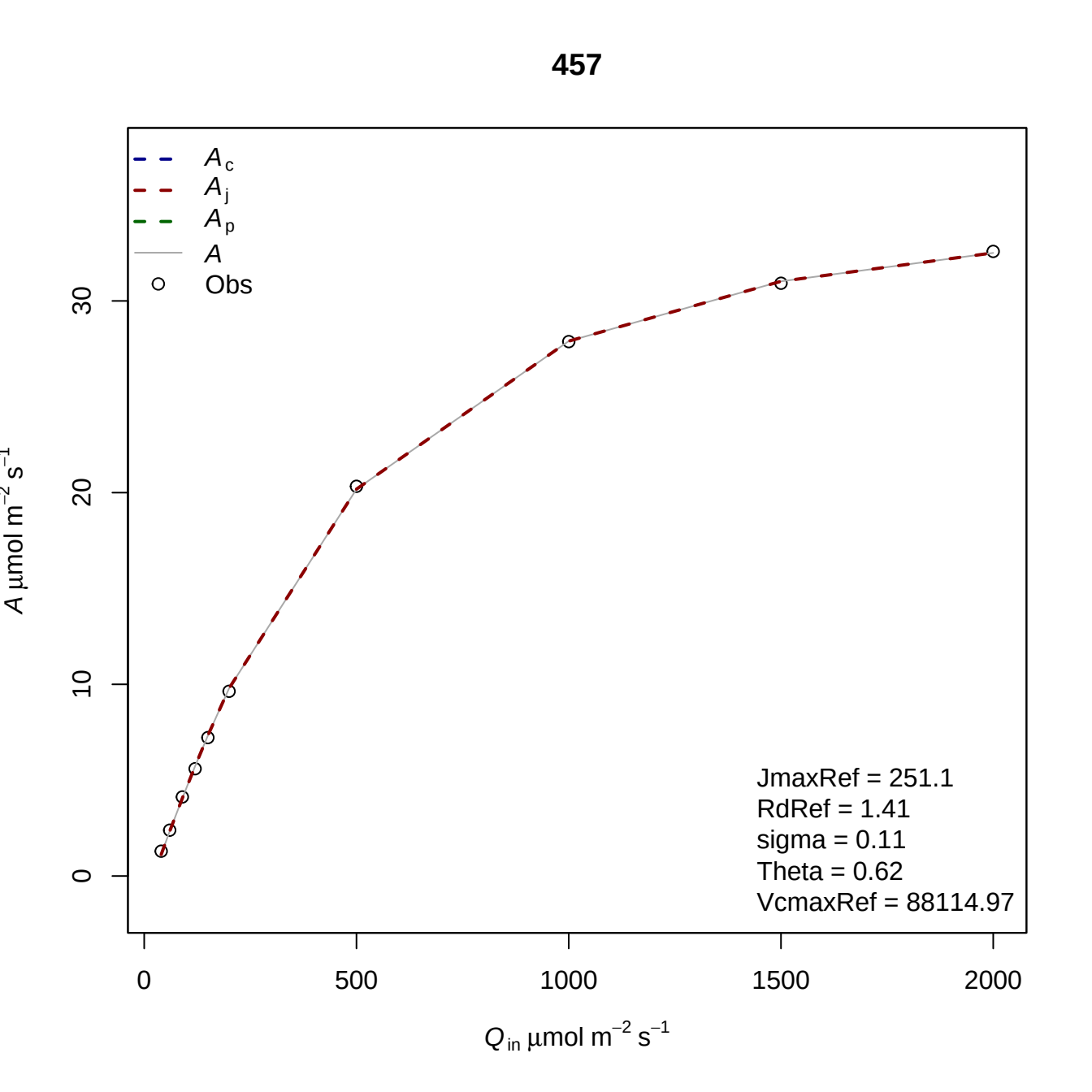

452
A_c
A_j
A_p
A
Obs
JmaxRef = 246.08
RdRef = 2.04
sigma = 0.13
Theta = 0.11
VcmaxRef = 173.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

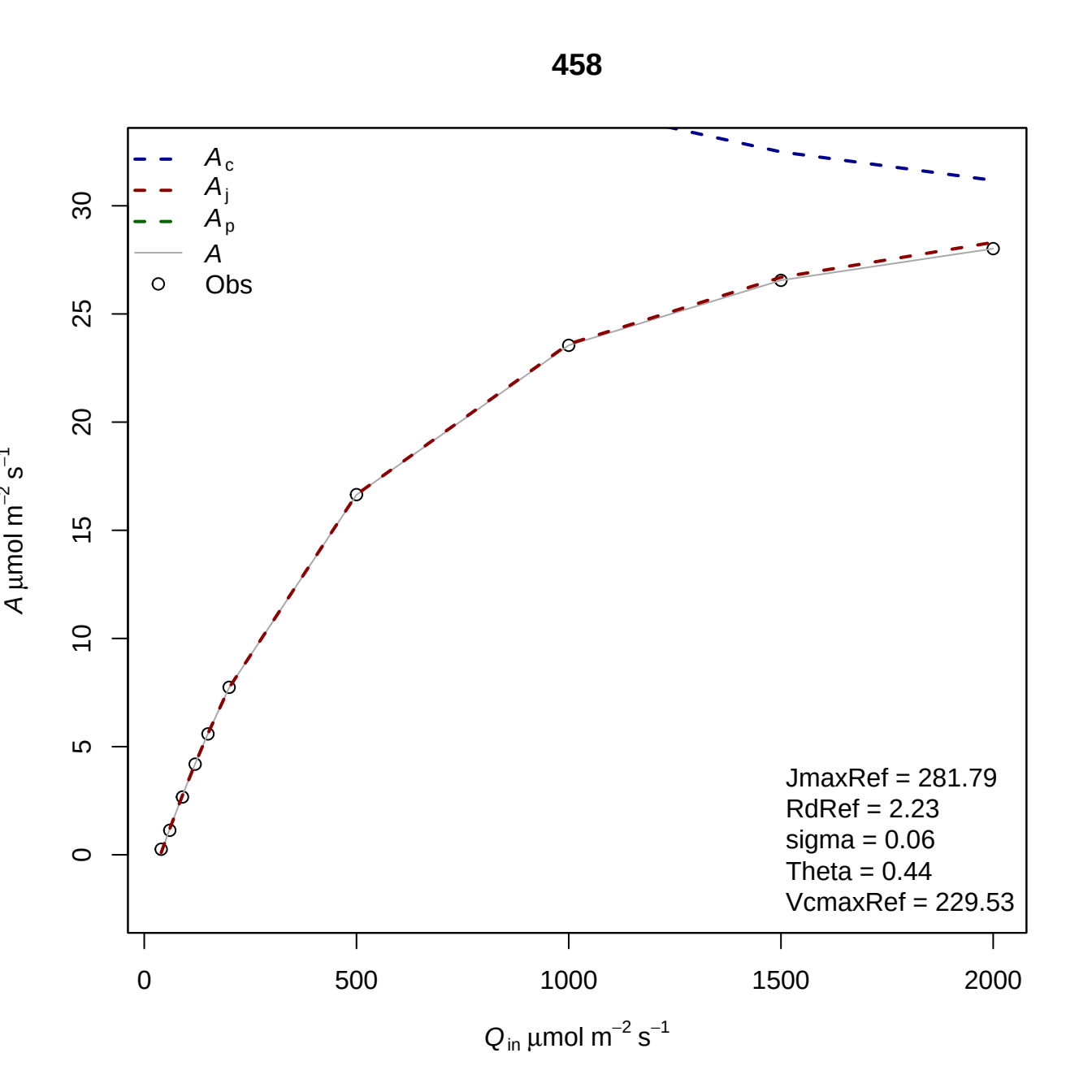

453
A_c
A_j
A_p
A
Obs
JmaxRef = 279.94
RdRef = 2.99
sigma = 0.08
Theta = 0.35
VcmaxRef = 275.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

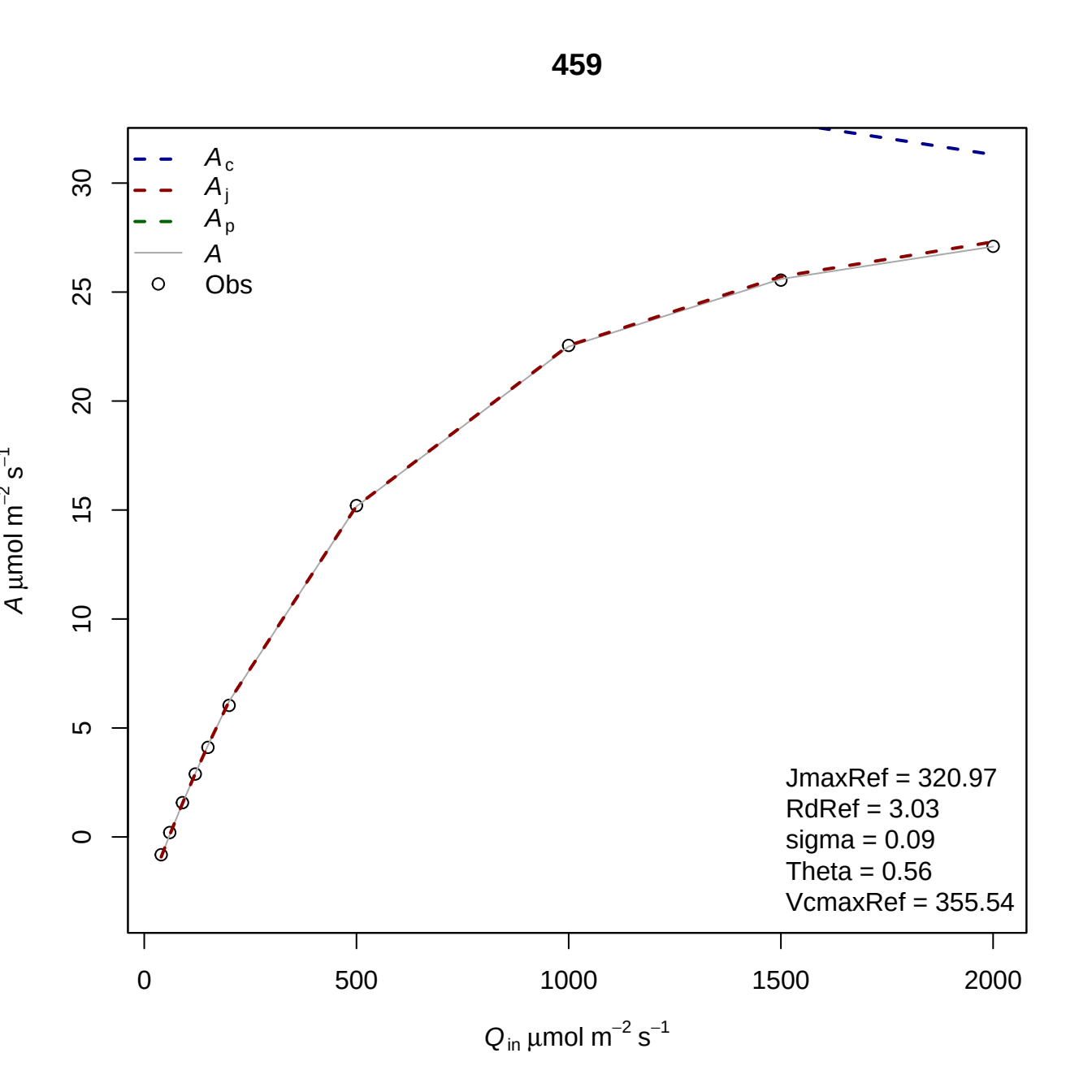

454
A_c
A_j
A_p
A
Obs
JmaxRef = 117.41
RdRef = 0.81
sigma = 0.04
Theta = −0.45
VcmaxRef = 46.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

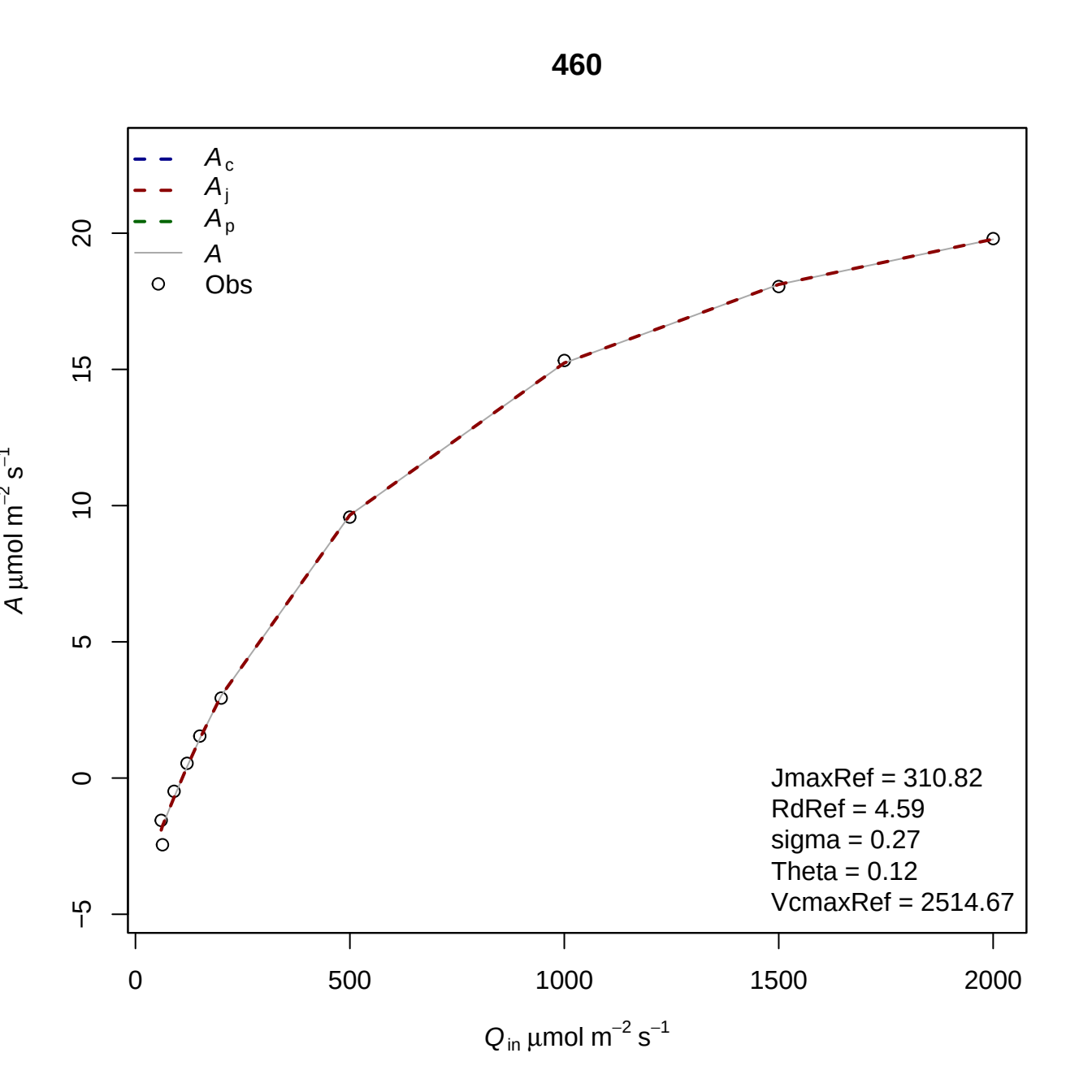

455
A_c
A_j
A_p
A
Obs
JmaxRef = 253.39
RdRef = 3.61
sigma = 0.09
Theta = −1.33
VcmaxRef = 240.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

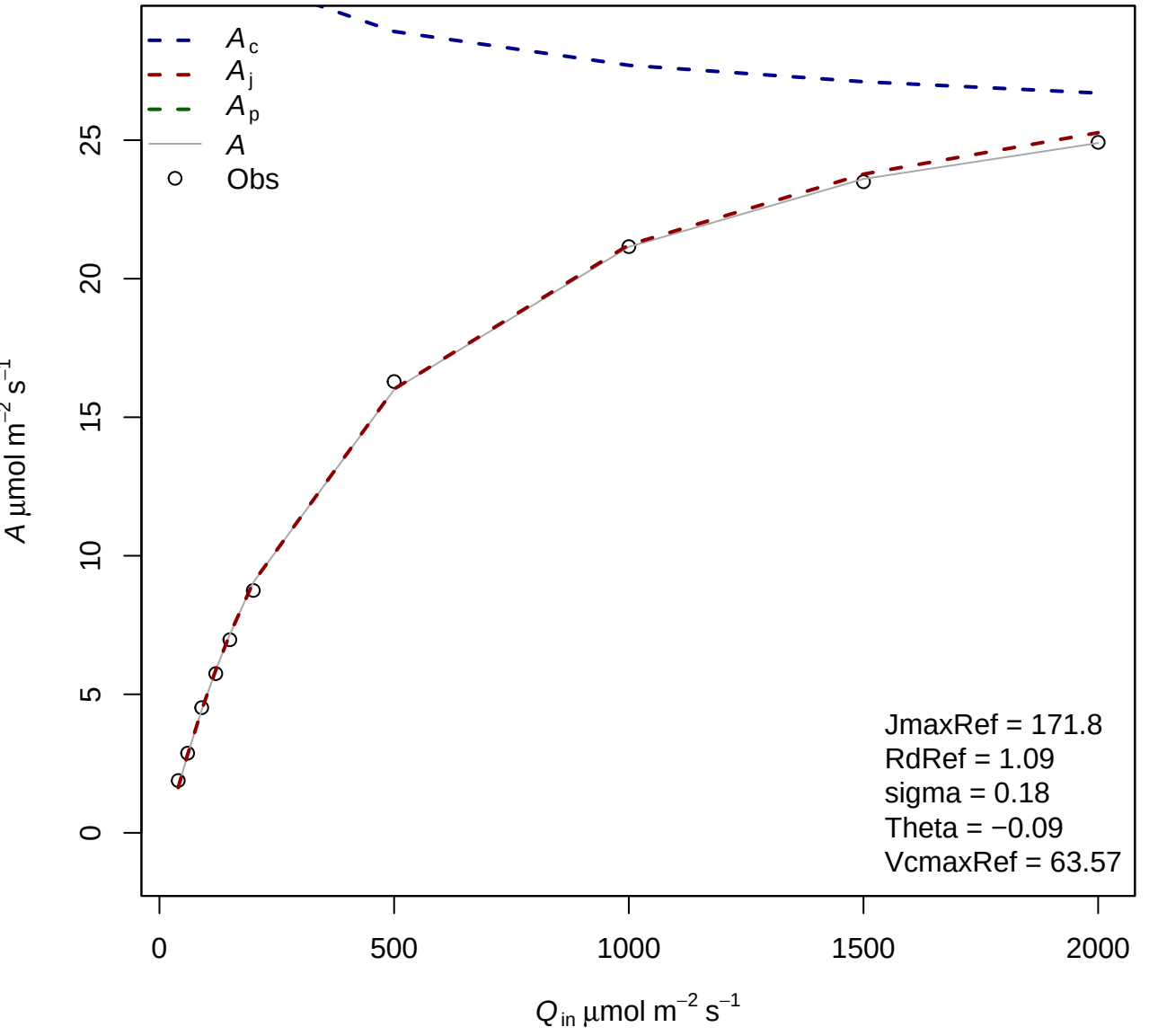

456
A_c
A_j
A_p
A
Obs
JmaxRef = 173.5
RdRef = 1.1
sigma = 0.09
Theta = 0.19
VcmaxRef = 88.01
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

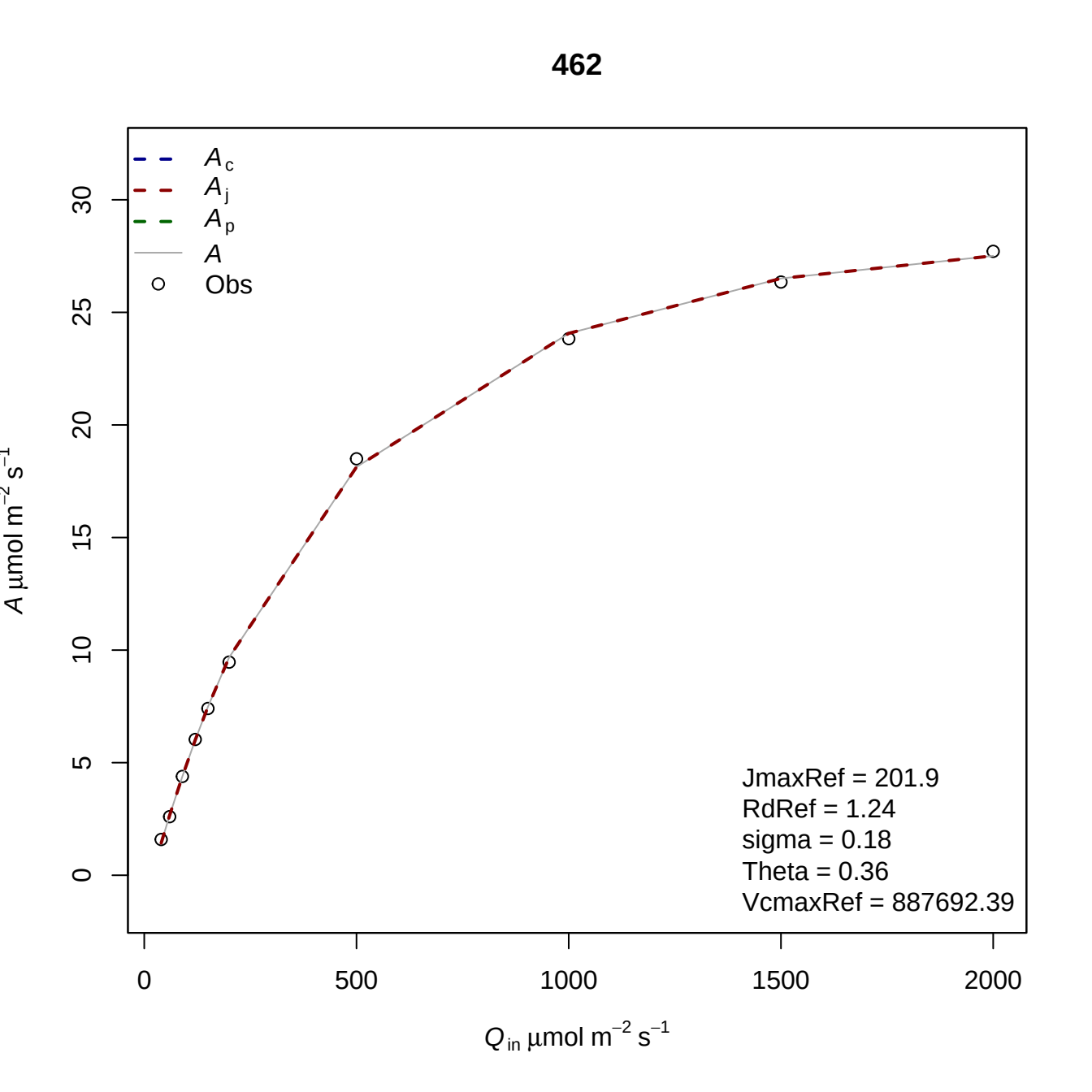

457
A_c
A_j
A_p
A
Obs
JmaxRef = 251.1
RdRef = 1.41
sigma = 0.11
Theta = 0.62
VcmaxRef = 88114.97
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

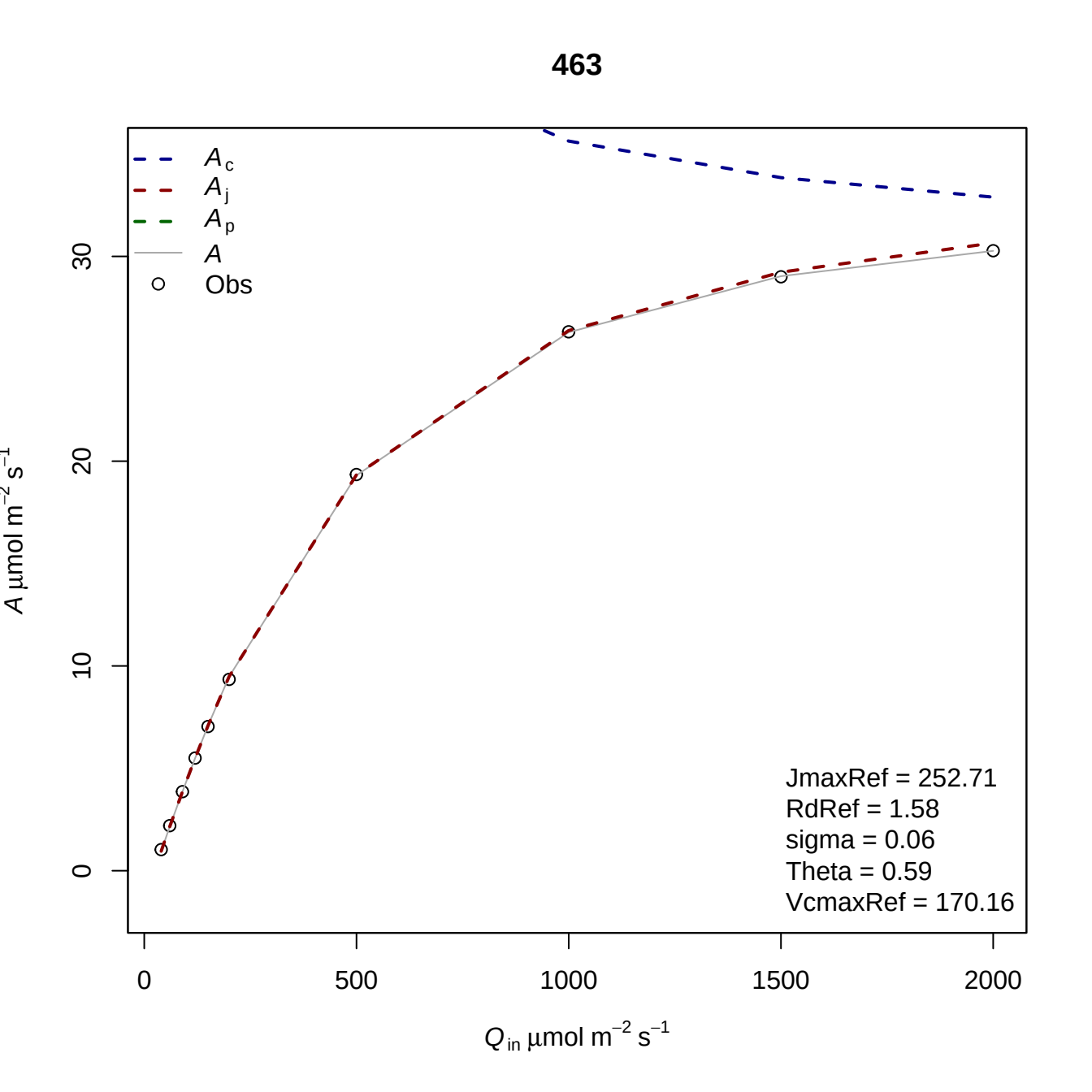

458
A_c
A_j
A_p
A
Obs
JmaxRef = 281.79
RdRef = 2.23
sigma = 0.06
Theta = 0.44
VcmaxRef = 229.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

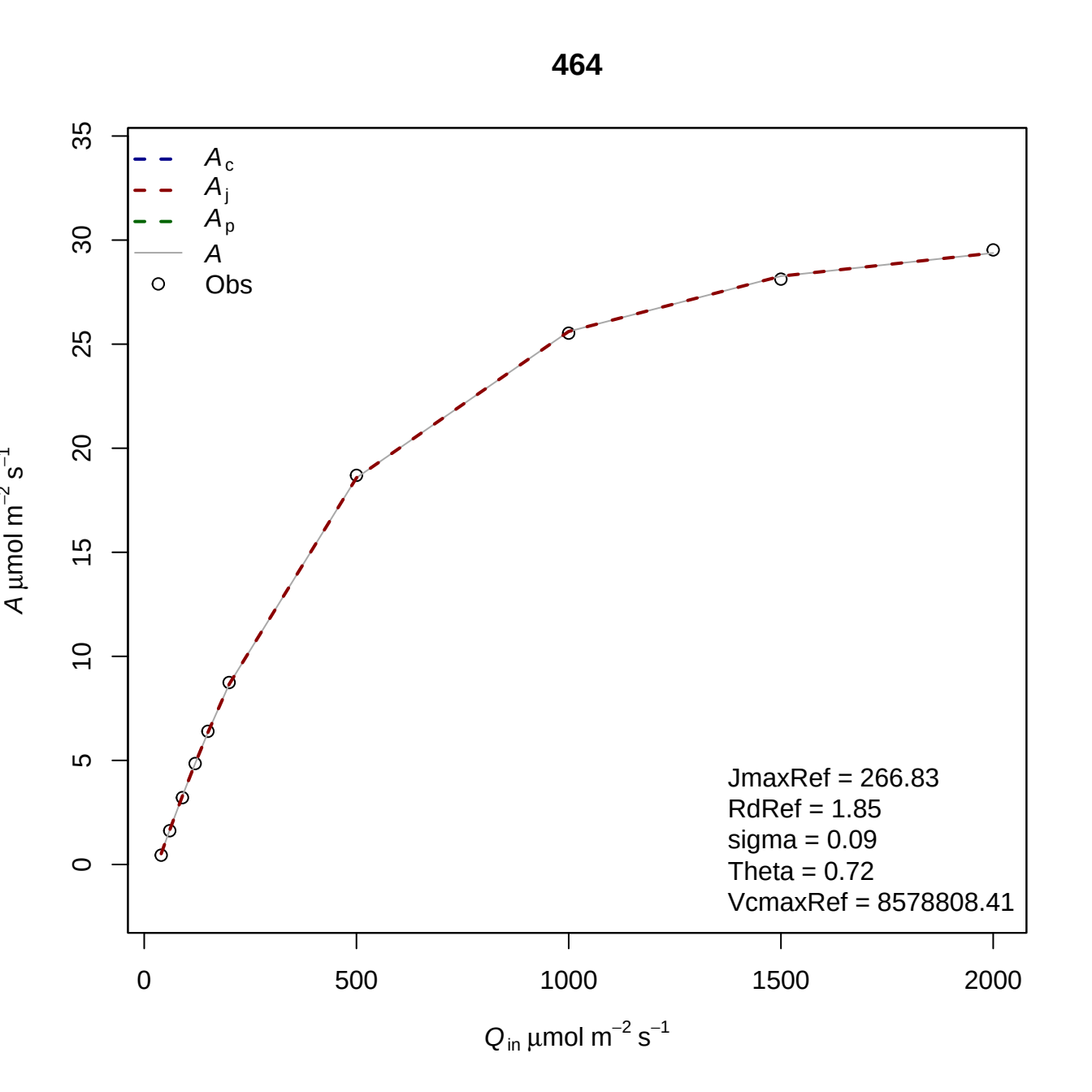

459
A_c
A_j
A_p
A
Obs
JmaxRef = 320.97
RdRef = 3.03
sigma = 0.09
Theta = 0.56
VcmaxRef = 355.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

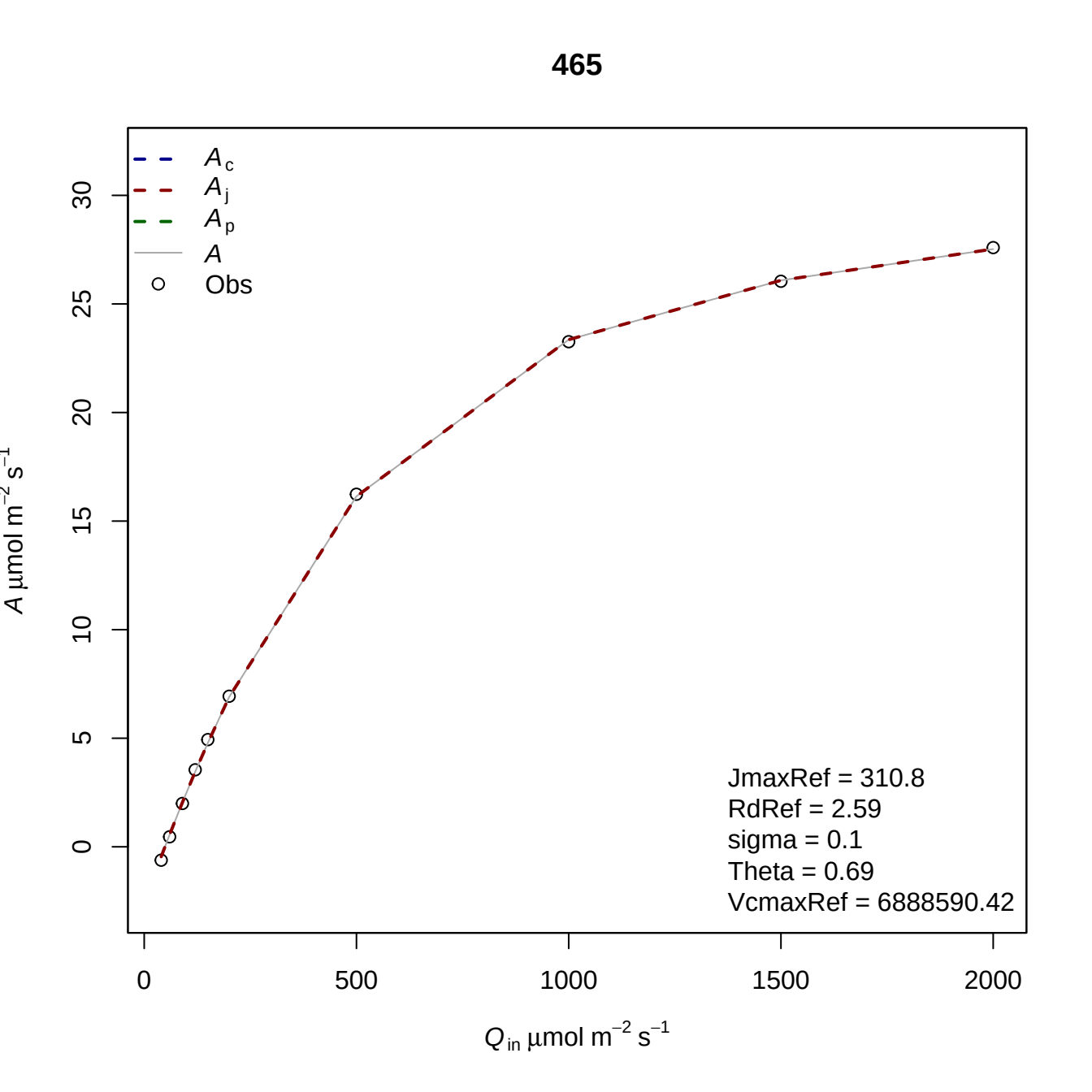

460
A_c
A_j
A_p
A
Obs
JmaxRef = 310.82
RdRef = 4.59
sigma = 0.27
Theta = 0.12
VcmaxRef = 2514.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

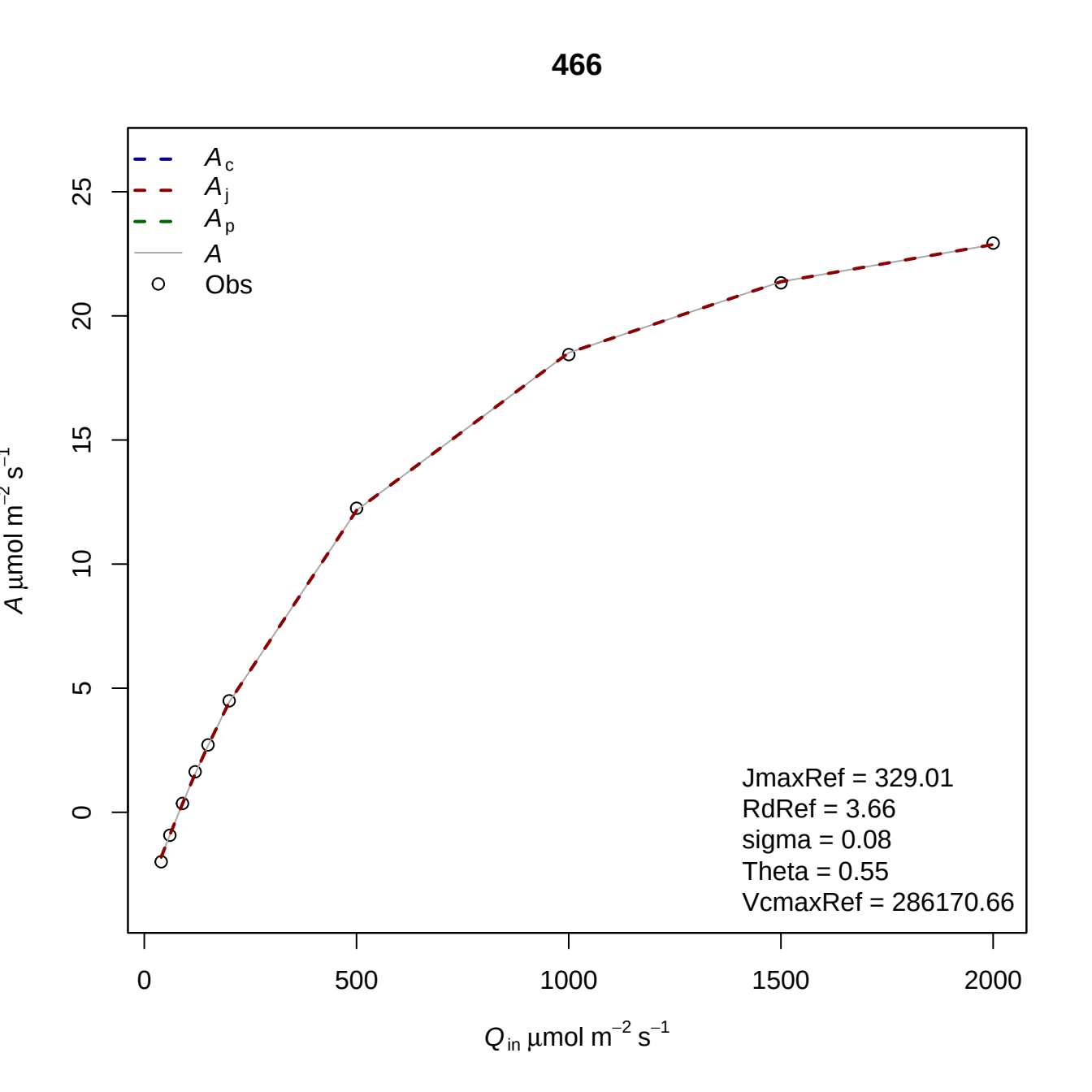

A_c
A_j
A_p
A
Obs
JmaxRef = 171.8
RdRef = 1.09
sigma = 0.18
Theta = −0.09
VcmaxRef = 63.57
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

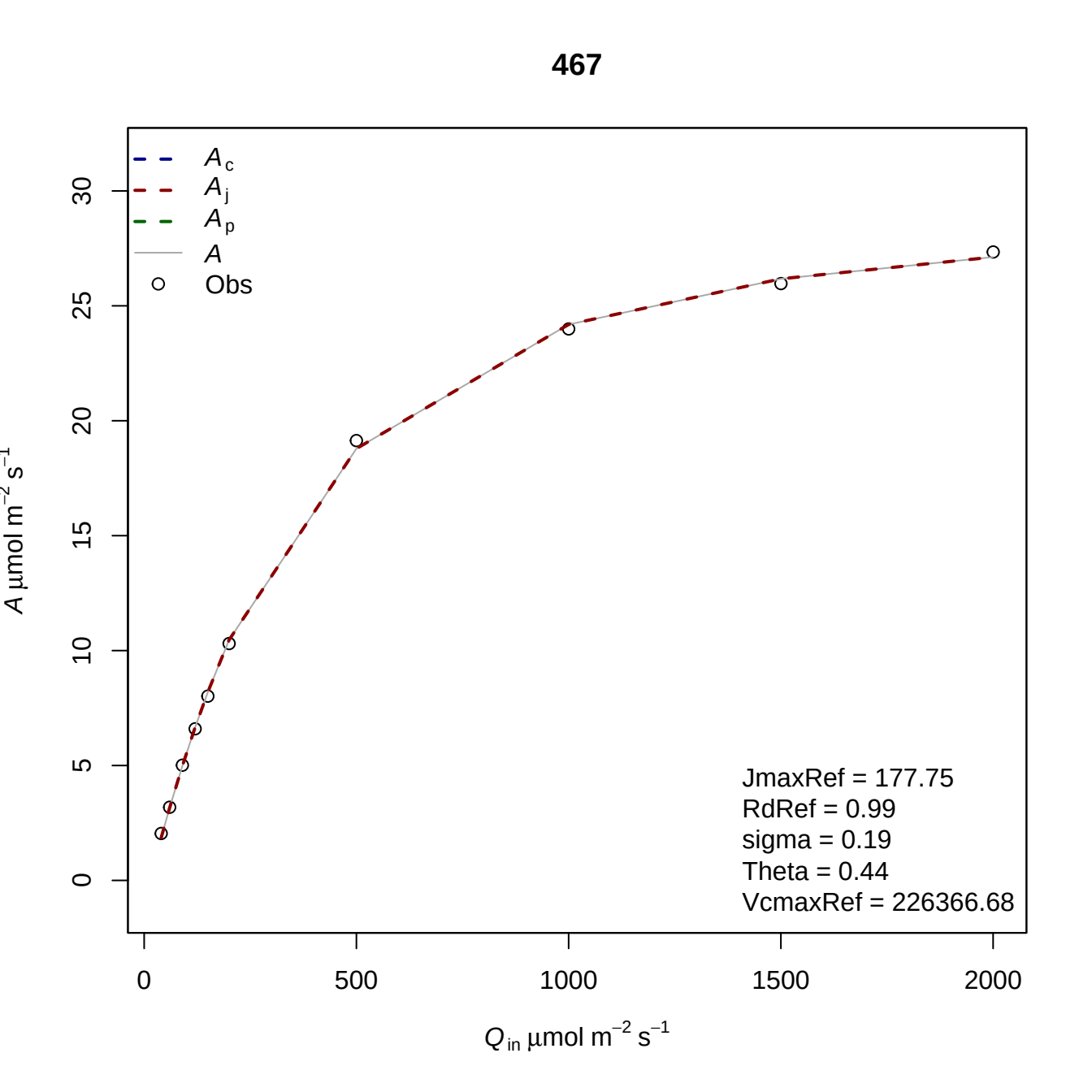

462
A_c
A_j
A_p
A
Obs
JmaxRef = 201.9
RdRef = 1.24
sigma = 0.18
Theta = 0.36
VcmaxRef = 887692.39
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

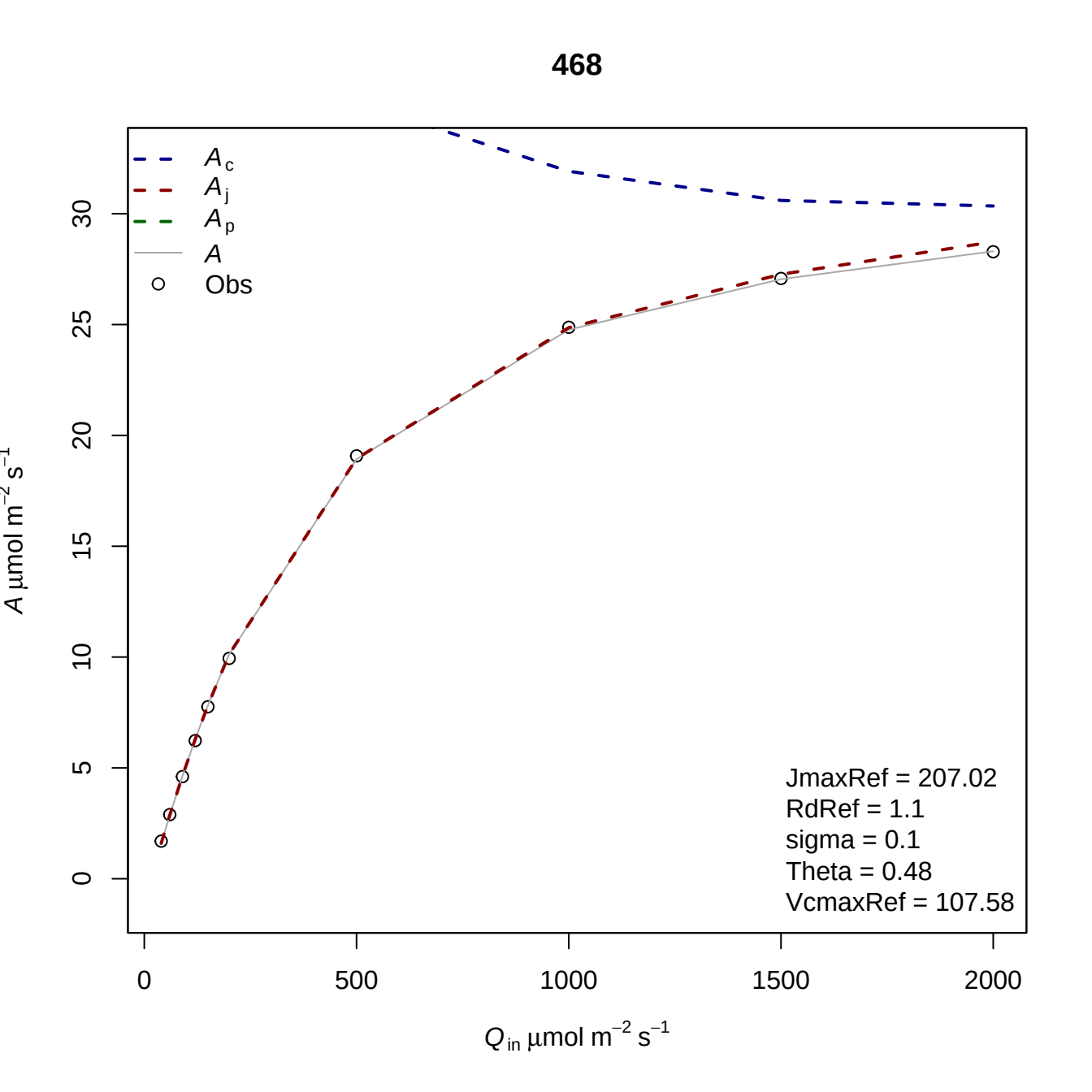

463
A_c
A_j
A_p
A
Obs
JmaxRef = 252.71
RdRef = 1.58
sigma = 0.06
Theta = 0.59
VcmaxRef = 170.16
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

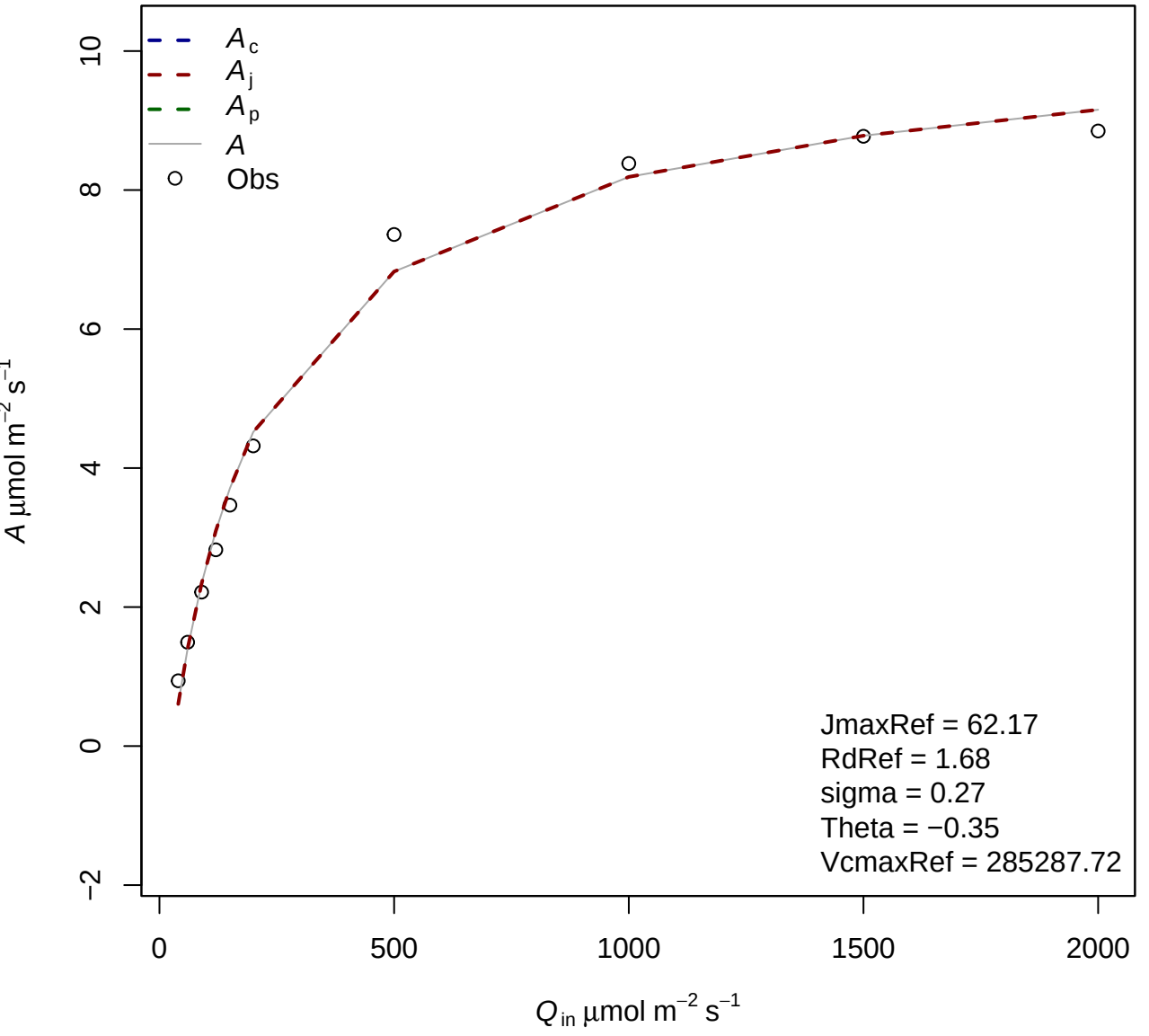

464
A_c
A_j
A_p
A
Obs
JmaxRef = 266.83
RdRef = 1.85
sigma = 0.09
Theta = 0.72
VcmaxRef = 8578808.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

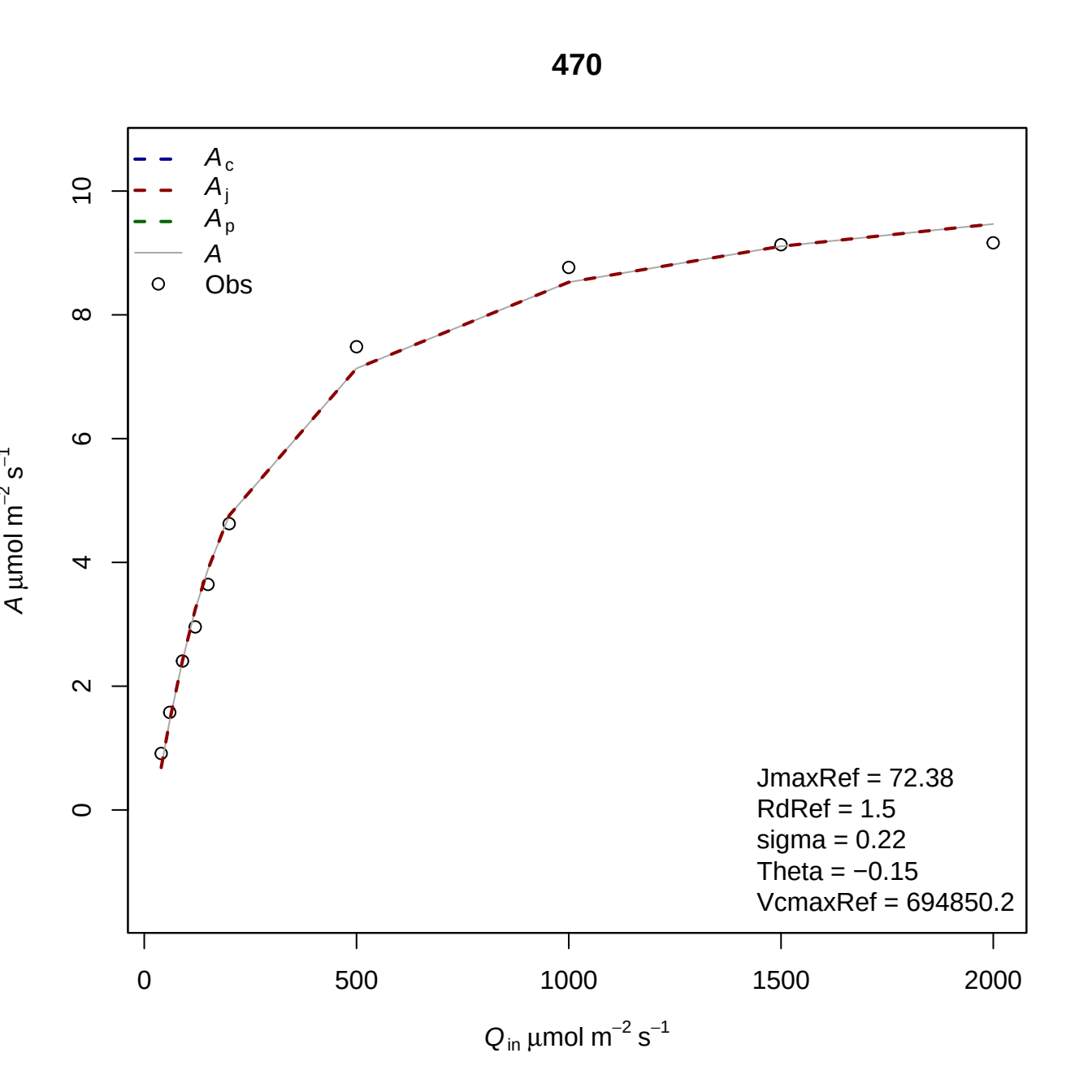

465
A_c
A_j
A_p
A
Obs
JmaxRef = 310.8
RdRef = 2.59
sigma = 0.1
Theta = 0.69
VcmaxRef = 6888590.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

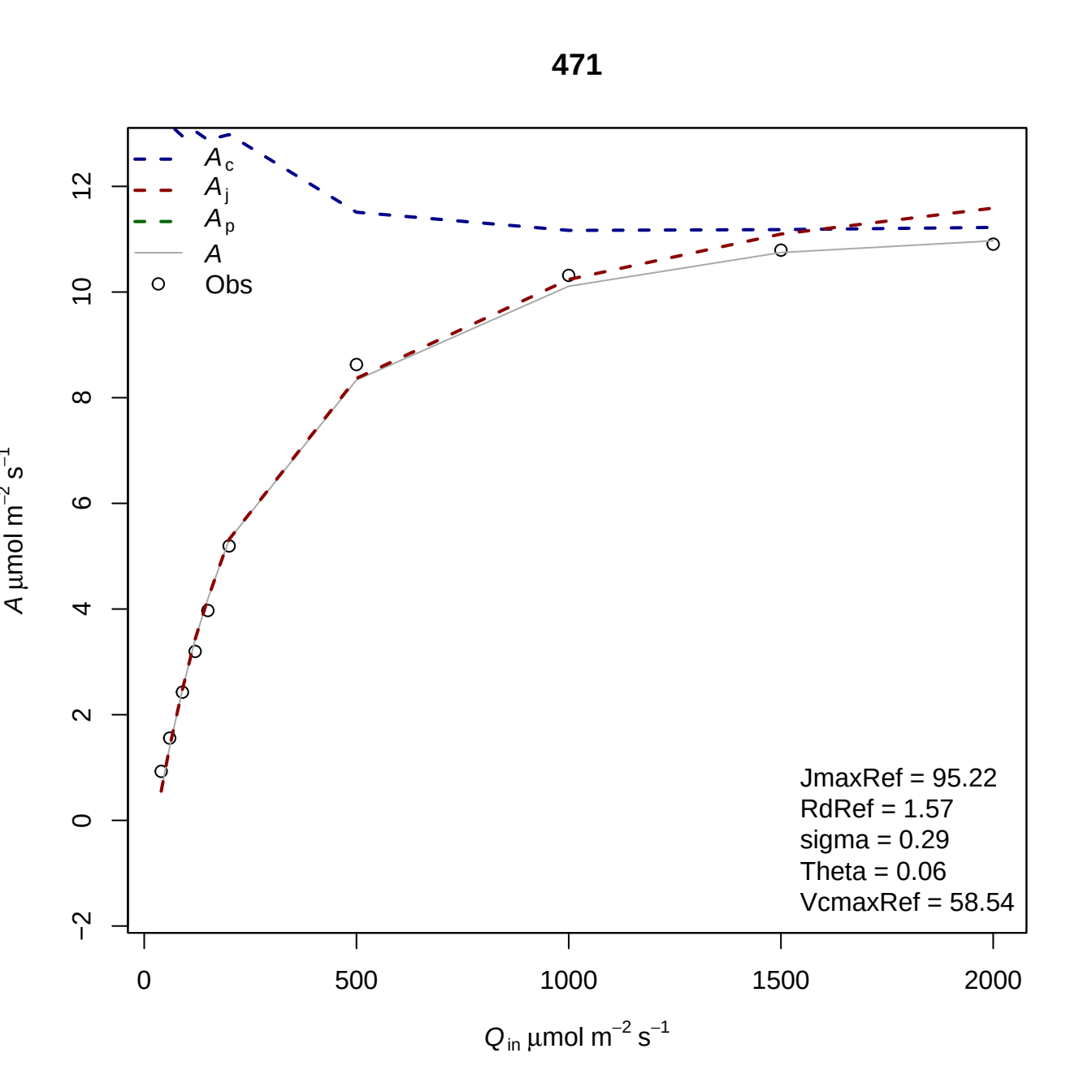

466
A_c
A_j
A_p
A
Obs
JmaxRef = 329.01
RdRef = 3.66
sigma = 0.08
Theta = 0.55
VcmaxRef = 286170.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

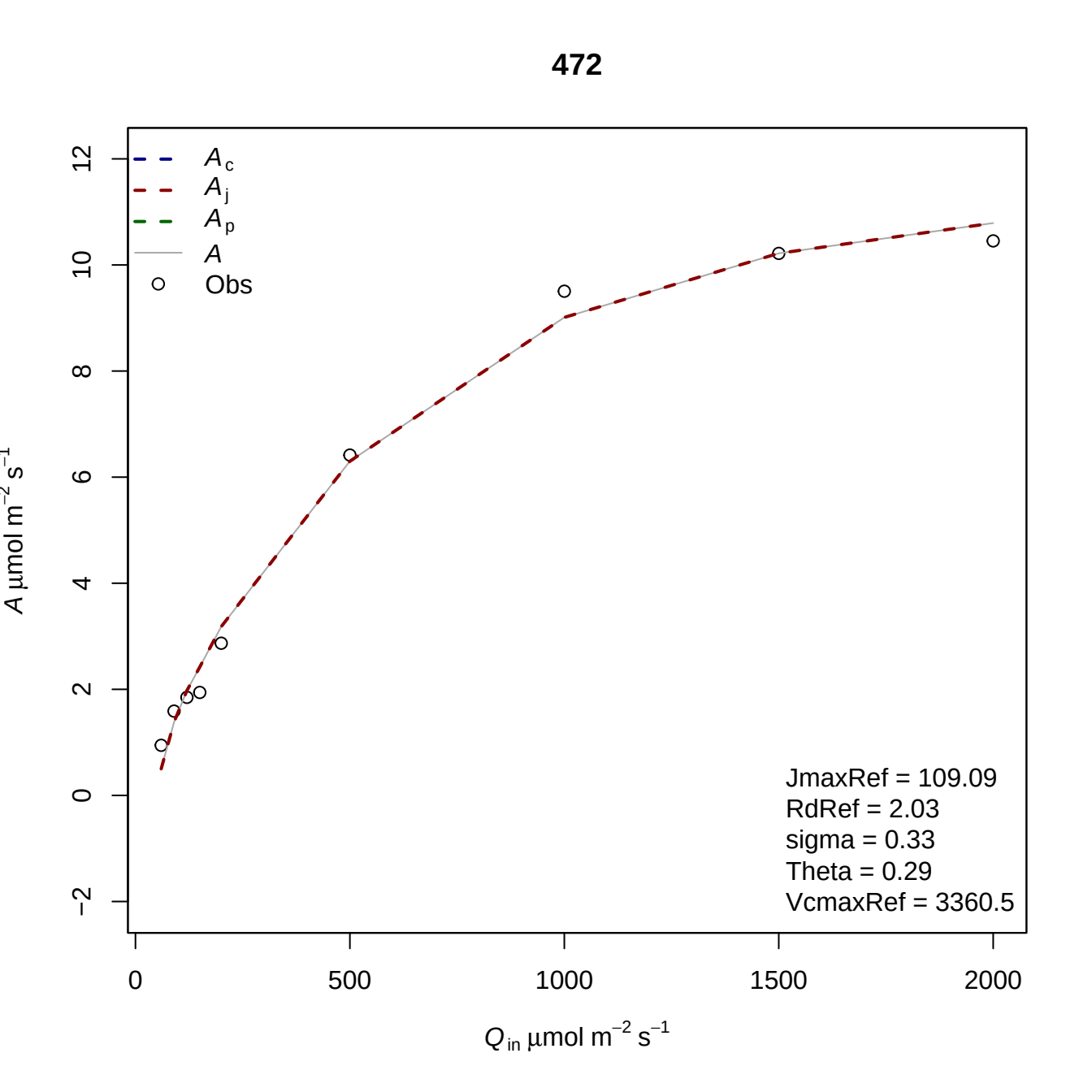

467
A_c
A_j
A_p
A
Obs
JmaxRef = 177.75
RdRef = 0.99
sigma = 0.19
Theta = 0.44
VcmaxRef = 226366.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

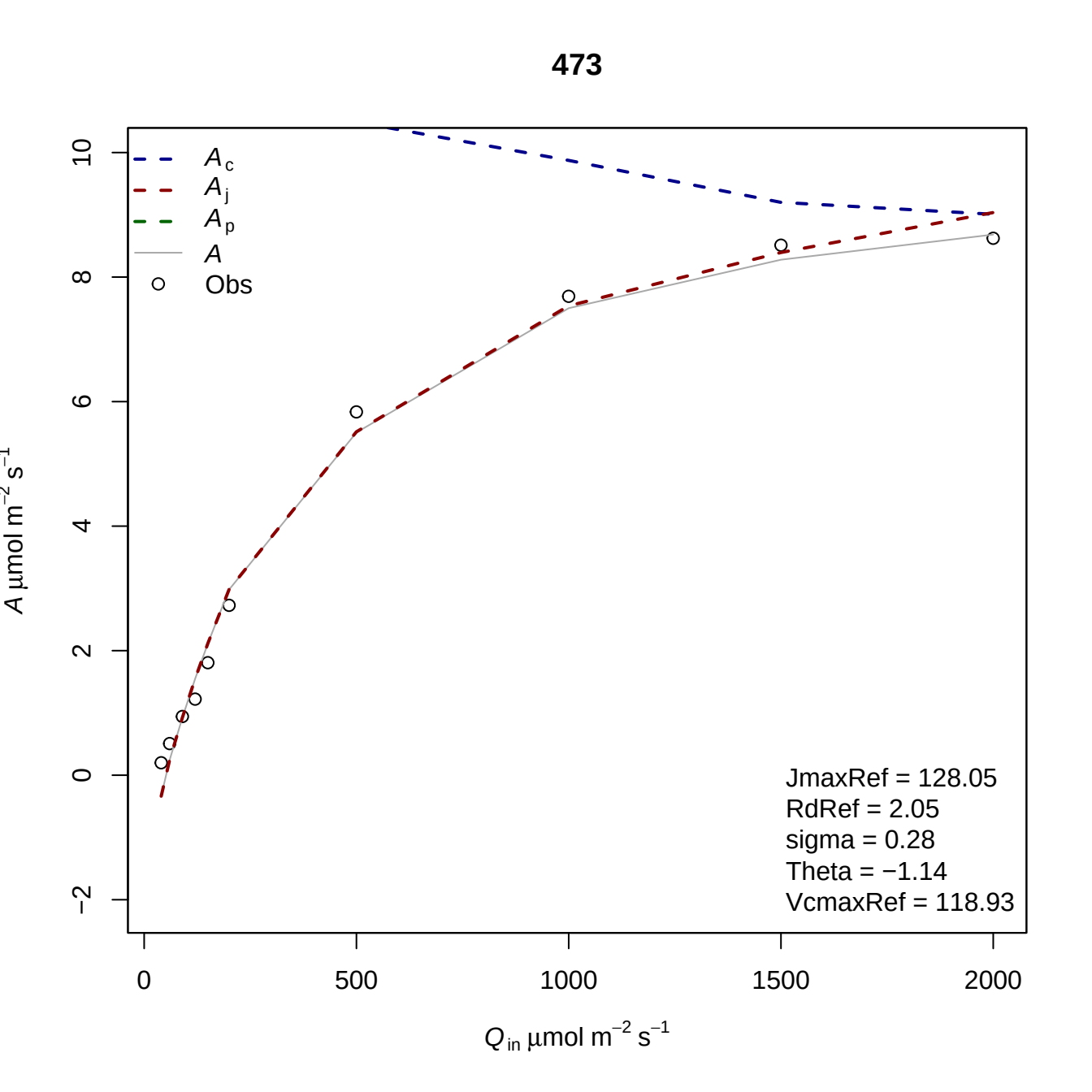

468
A_c
A_j
A_p
A
Obs
JmaxRef = 207.02
RdRef = 1.1
sigma = 0.1
Theta = 0.48
VcmaxRef = 107.58
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

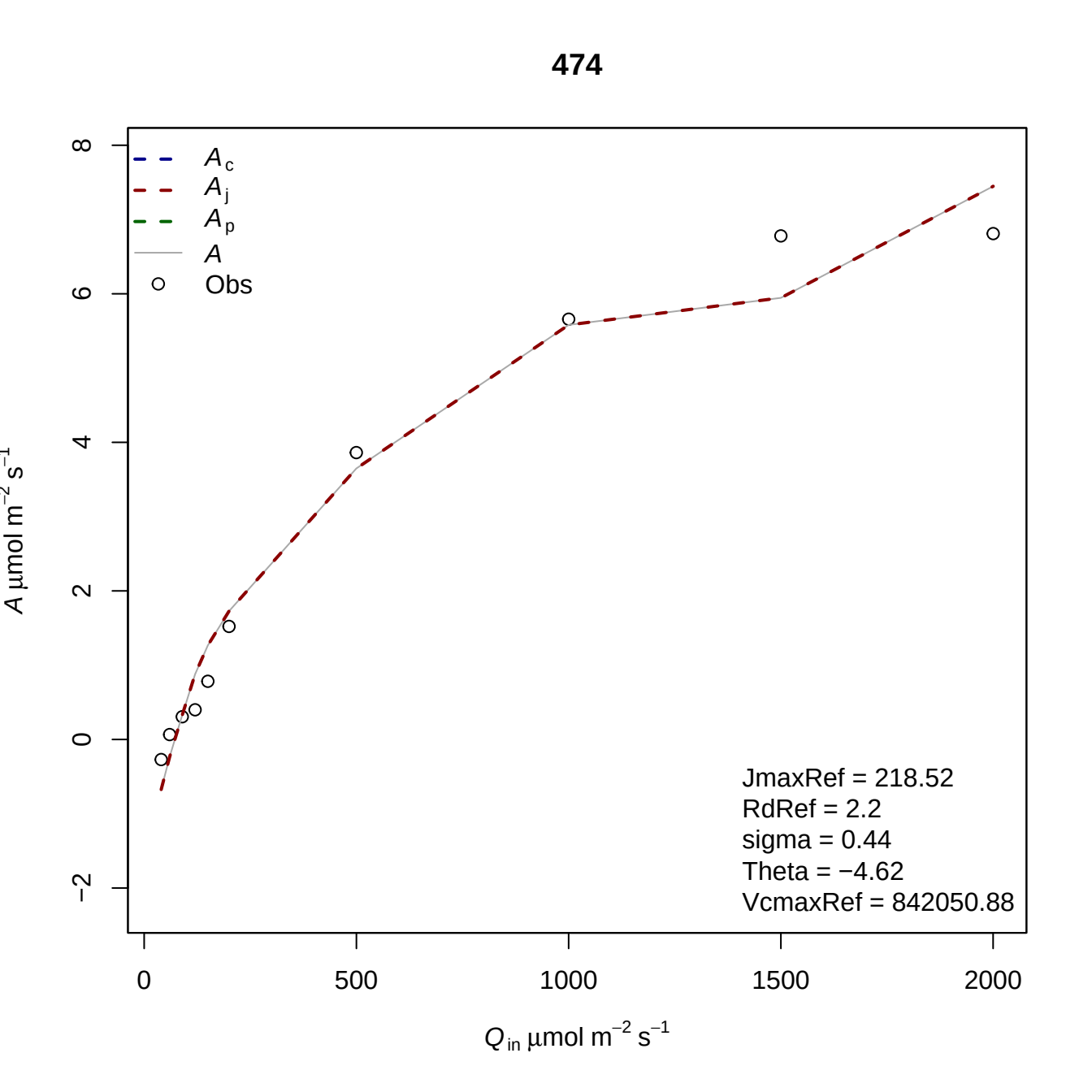

A_c
A_j
A_p
A
Obs
JmaxRef = 62.17
RdRef = 1.68
sigma = 0.27
Theta = −0.35
VcmaxRef = 285287.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

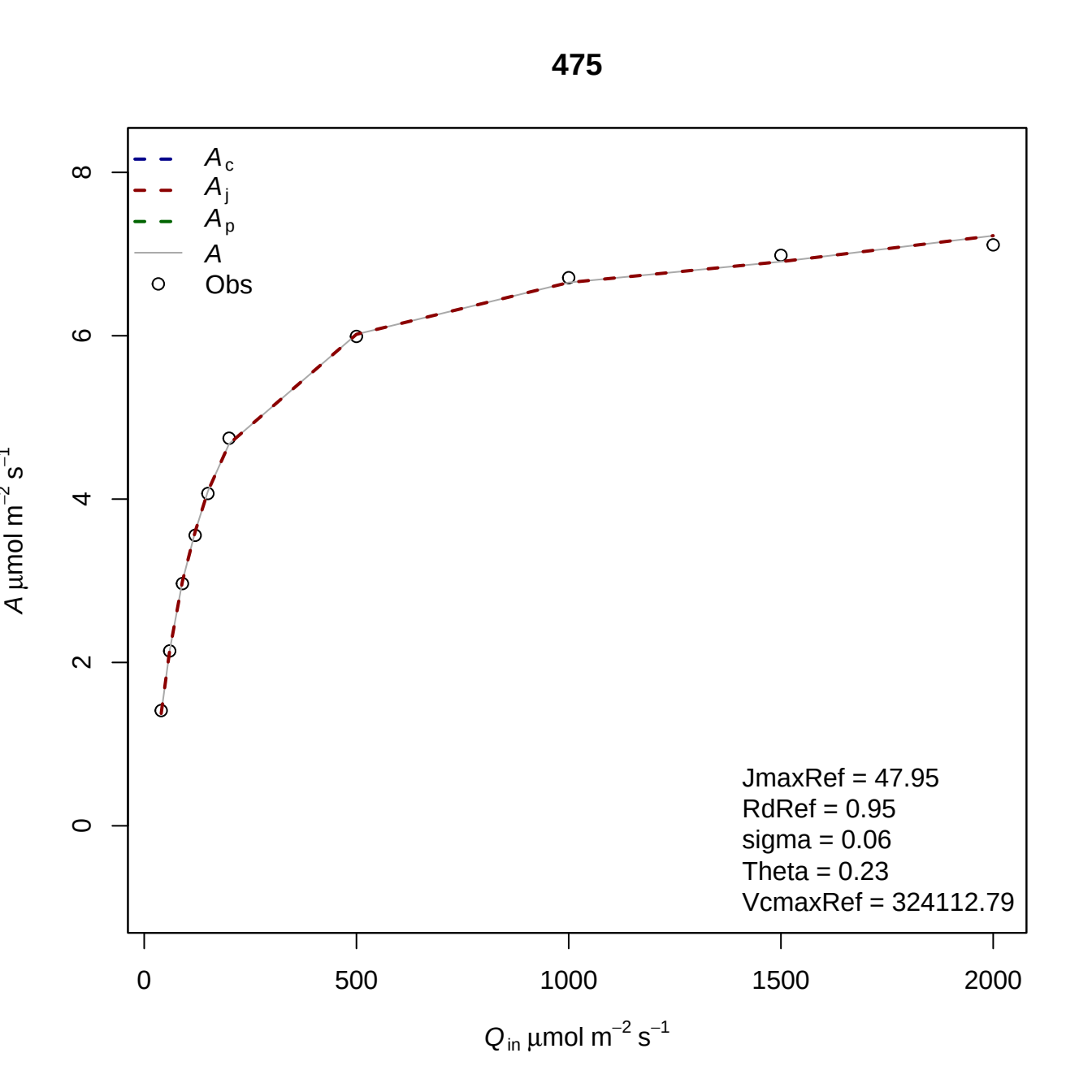

470
A_c
A_j
A_p
A
Obs
JmaxRef = 72.38
RdRef = 1.5
sigma = 0.22
Theta = −0.15
VcmaxRef = 694850.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

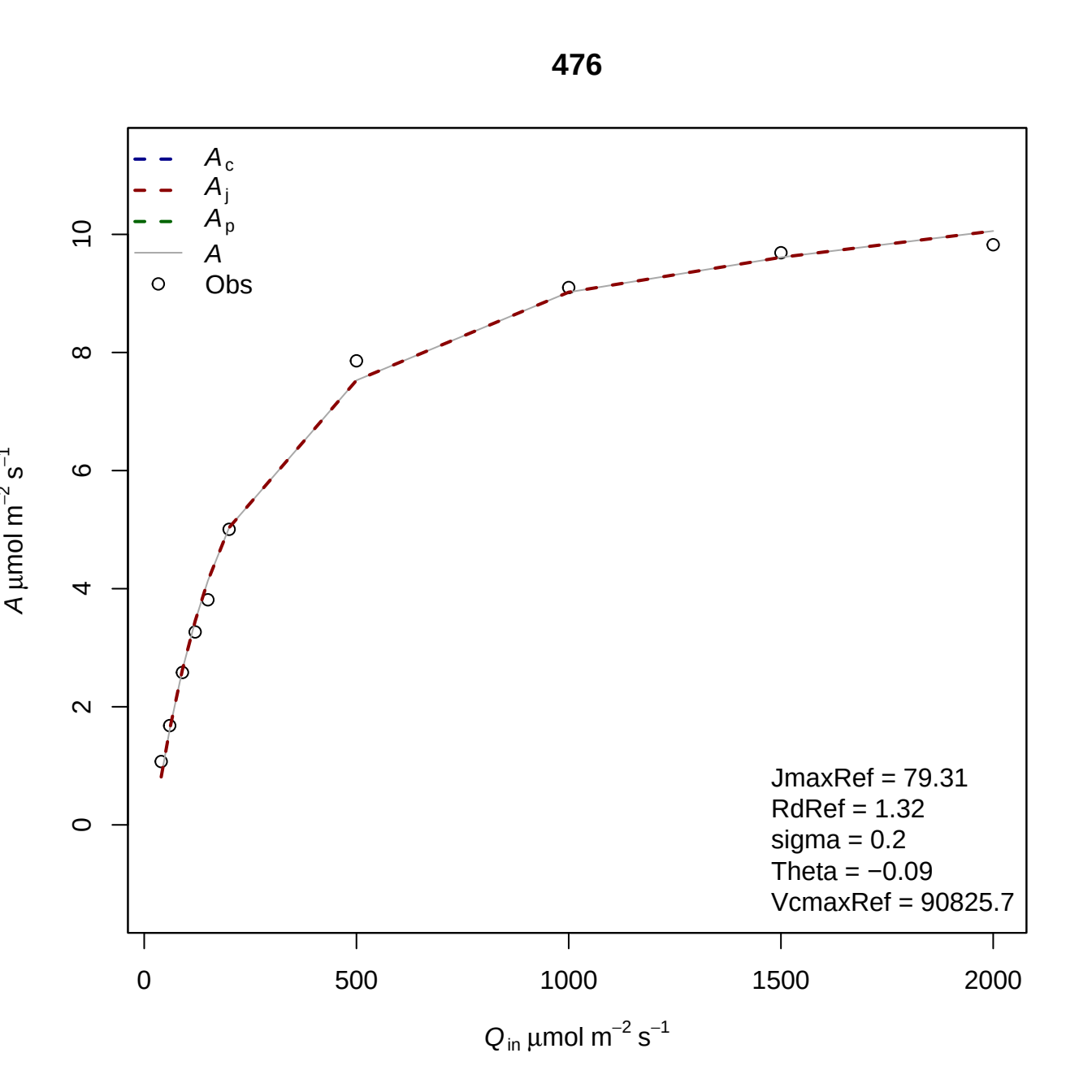

471
A_c
A_j
A_p
A
Obs
JmaxRef = 95.22
RdRef = 1.57
sigma = 0.29
Theta = 0.06
VcmaxRef = 58.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

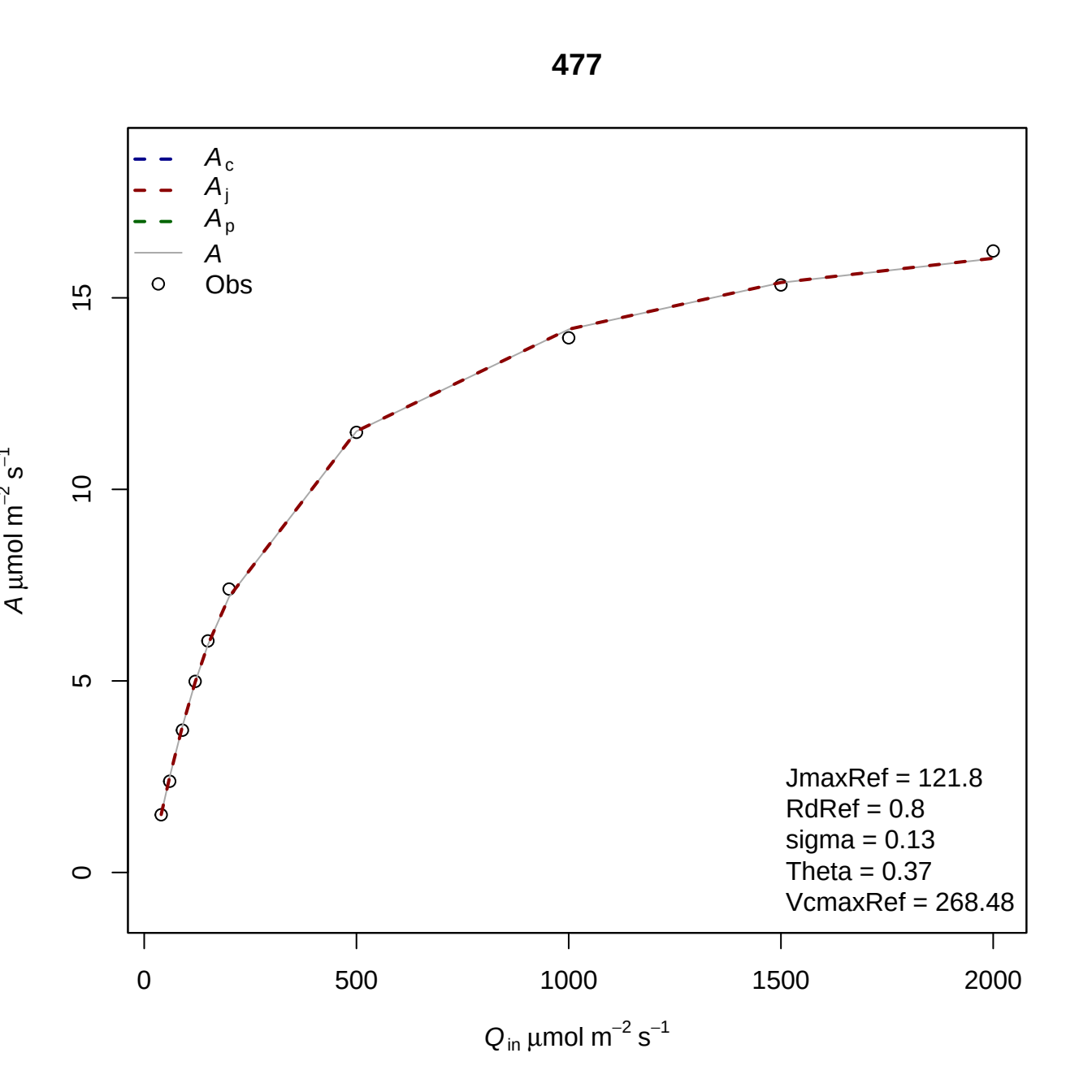

472
A_c
A_j
A_p
A
Obs
JmaxRef = 109.09
RdRef = 2.03
sigma = 0.33
Theta = 0.29
VcmaxRef = 3360.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

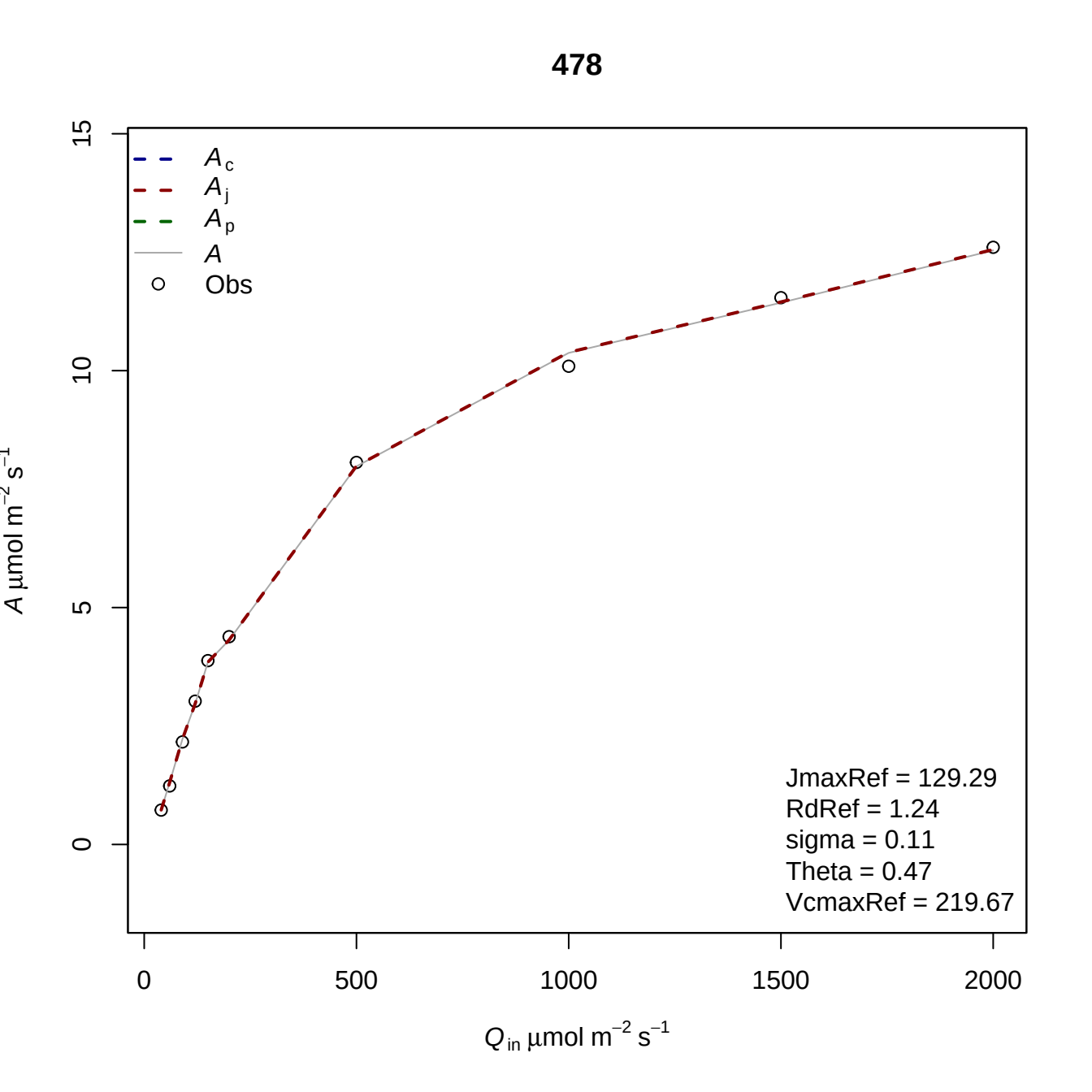

473
A_c
A_j
A_p
A
Obs
JmaxRef = 128.05
RdRef = 2.05
sigma = 0.28
Theta = −1.14
VcmaxRef = 118.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

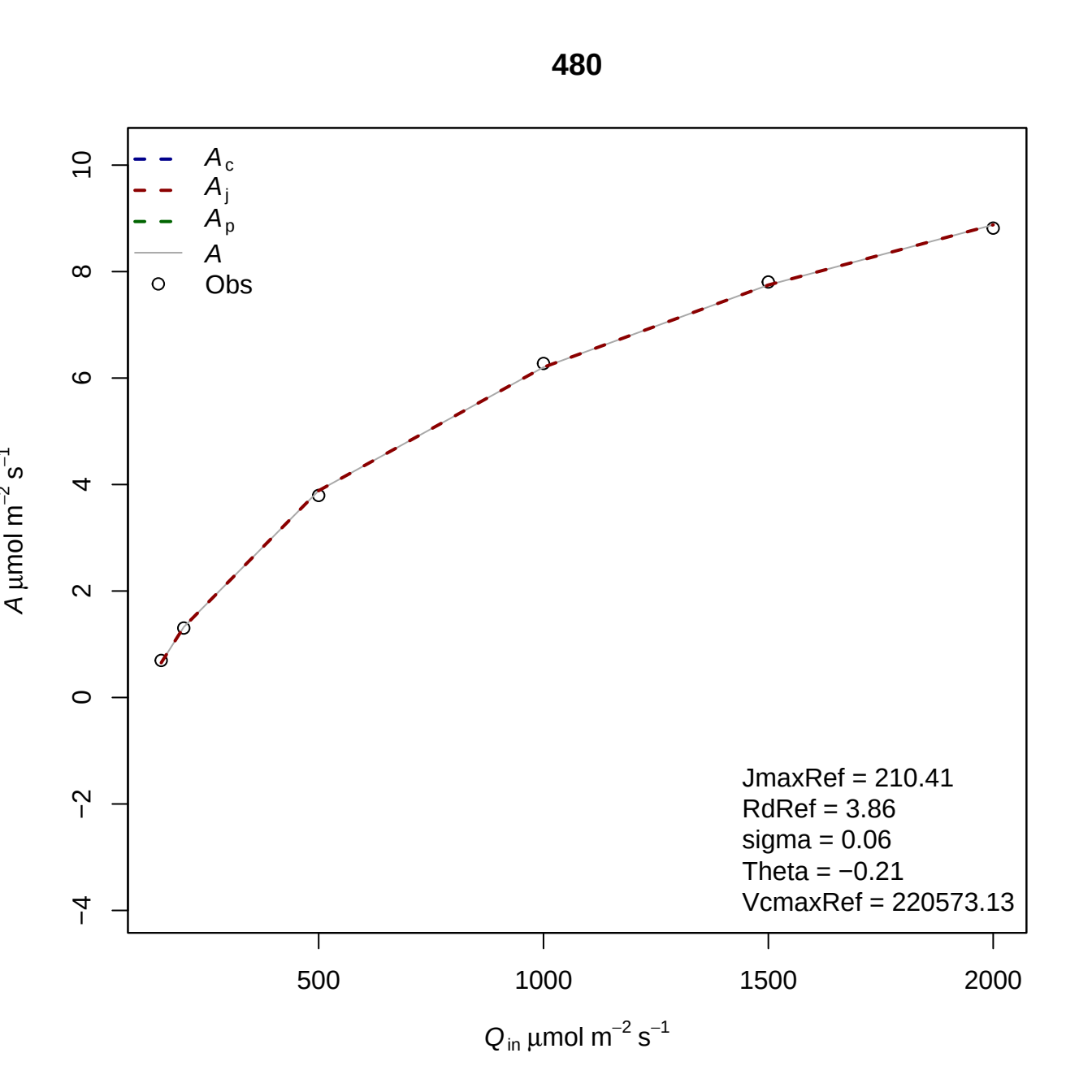

474
A_c
A_j
A_p
A
Obs
JmaxRef = 218.52
RdRef = 2.2
sigma = 0.44
Theta = −4.62
VcmaxRef = 842050.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

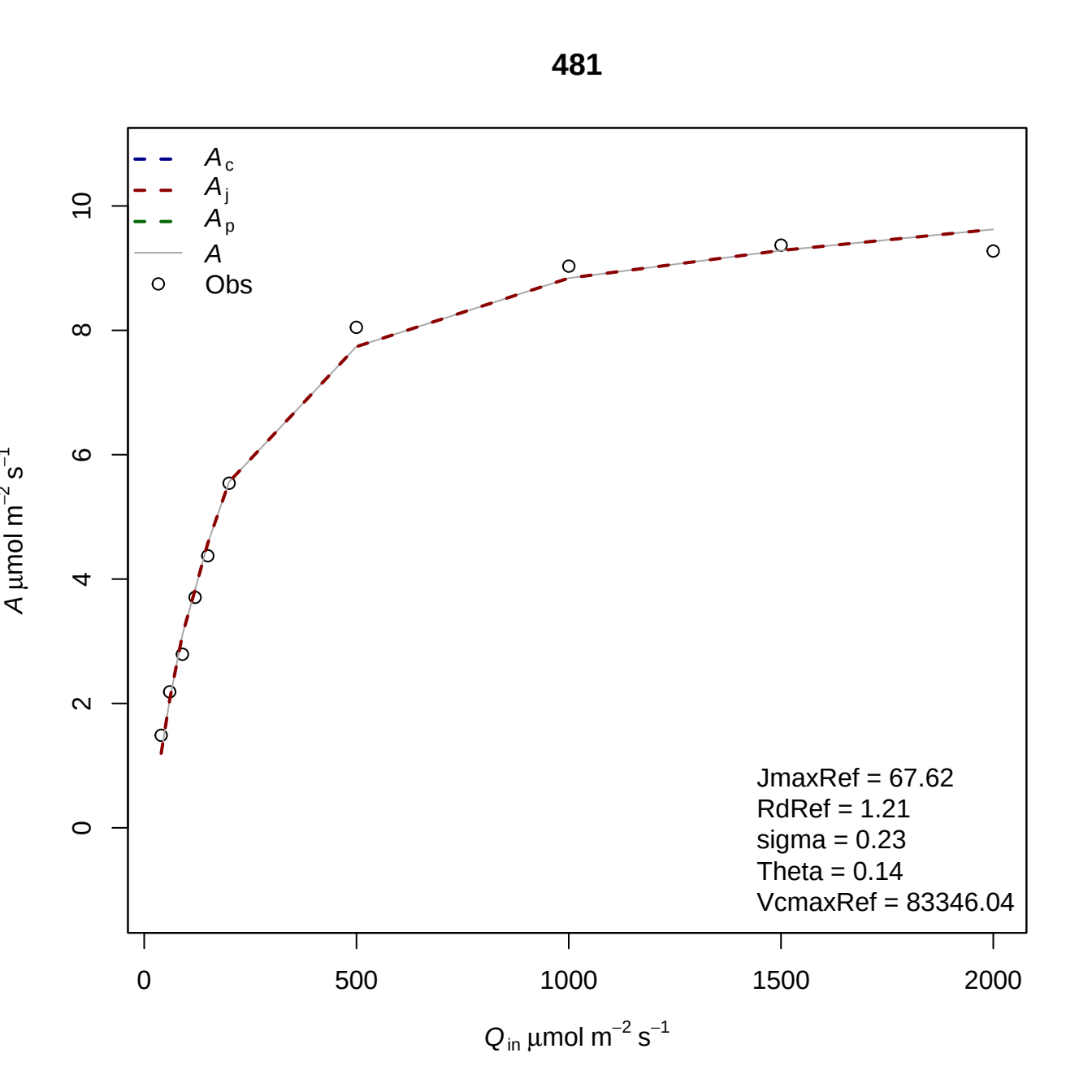

475
A_c
A_j
A_p
A
Obs
JmaxRef = 47.95
RdRef = 0.95
sigma = 0.06
Theta = 0.23
VcmaxRef = 324112.79
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

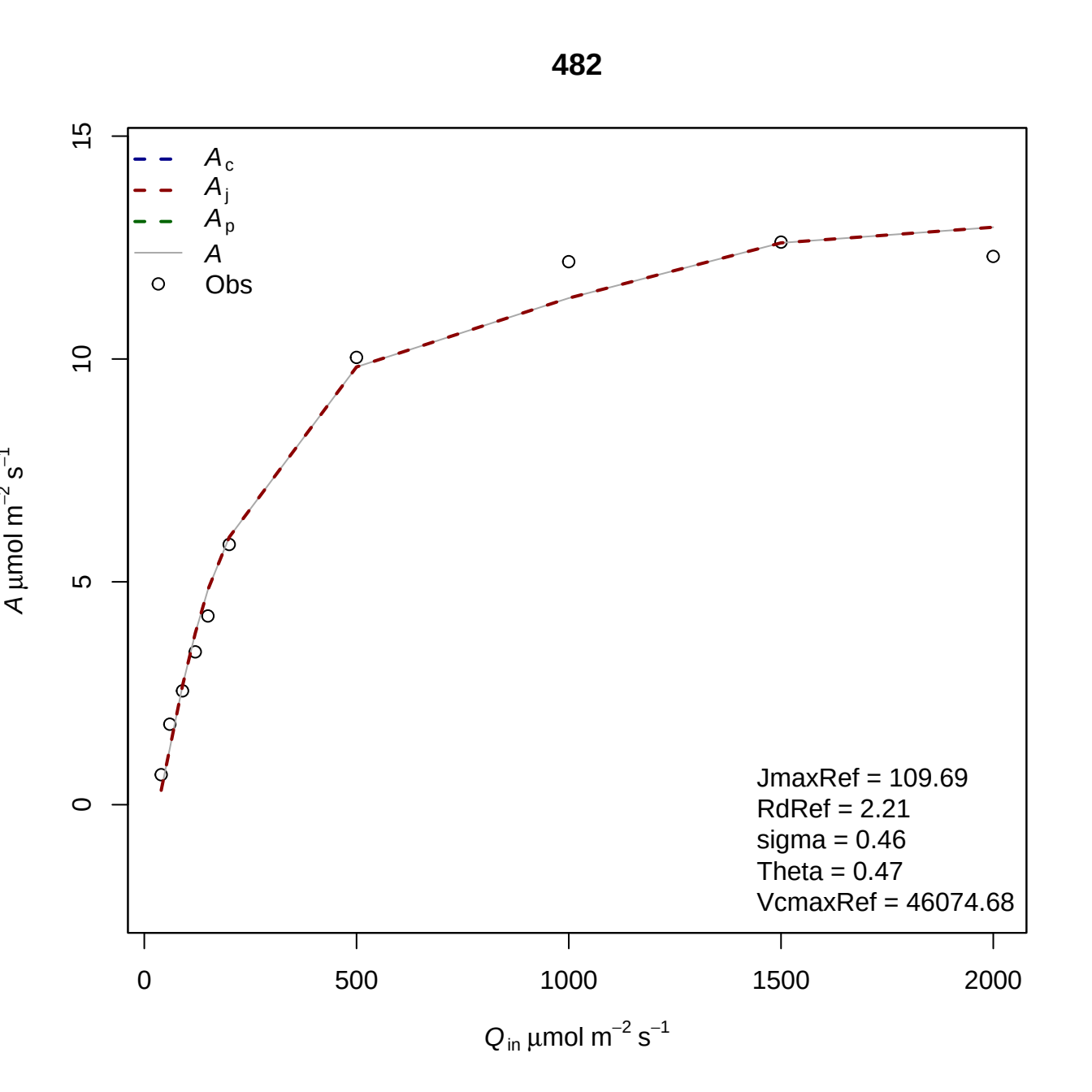

476
A_c
A_j
A_p
A
Obs
JmaxRef = 79.31
RdRef = 1.32
sigma = 0.2
Theta = −0.09
VcmaxRef = 90825.7
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

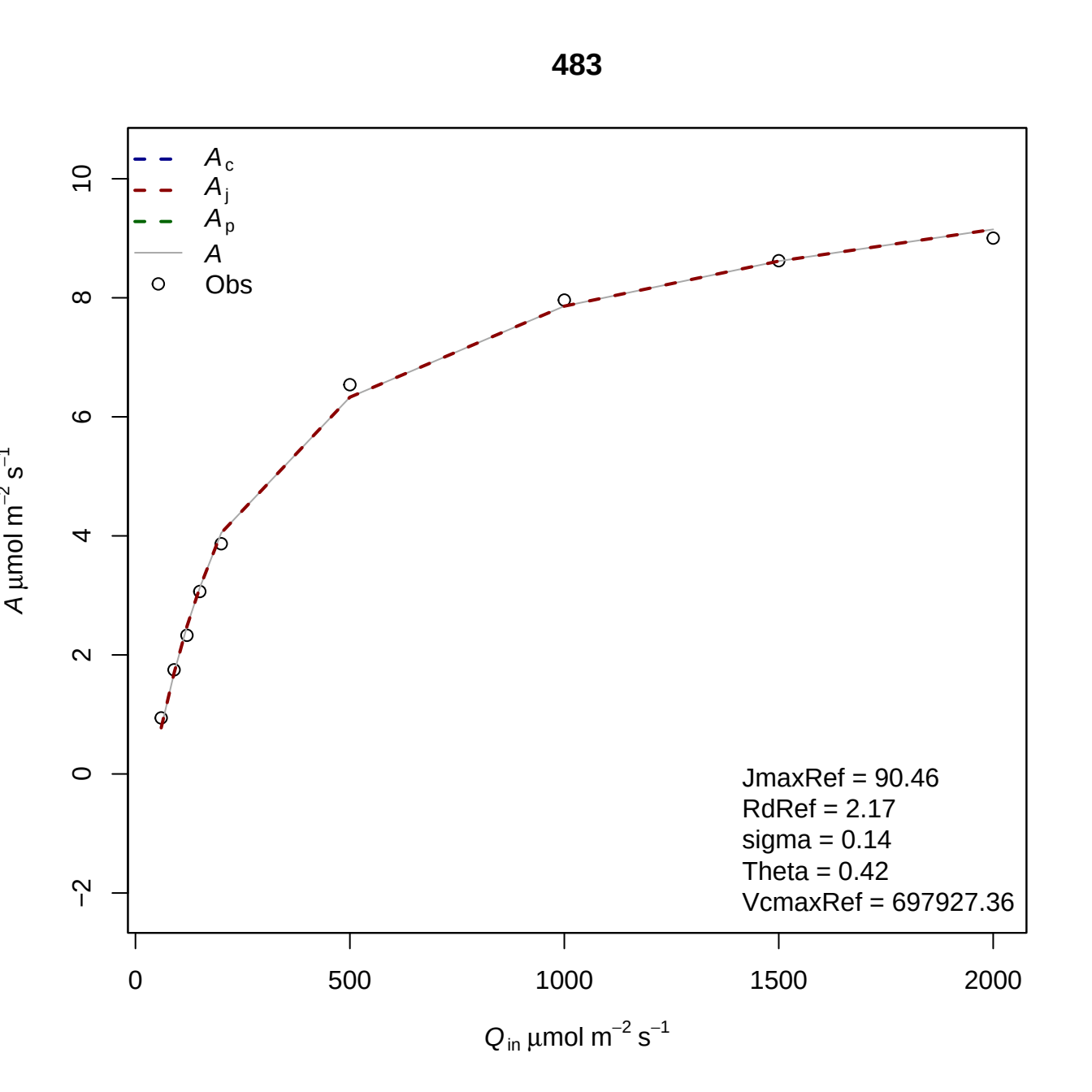

477
A_c
A_j
A_p
A
Obs
JmaxRef = 121.8
RdRef = 0.8
sigma = 0.13
Theta = 0.37
VcmaxRef = 268.48
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

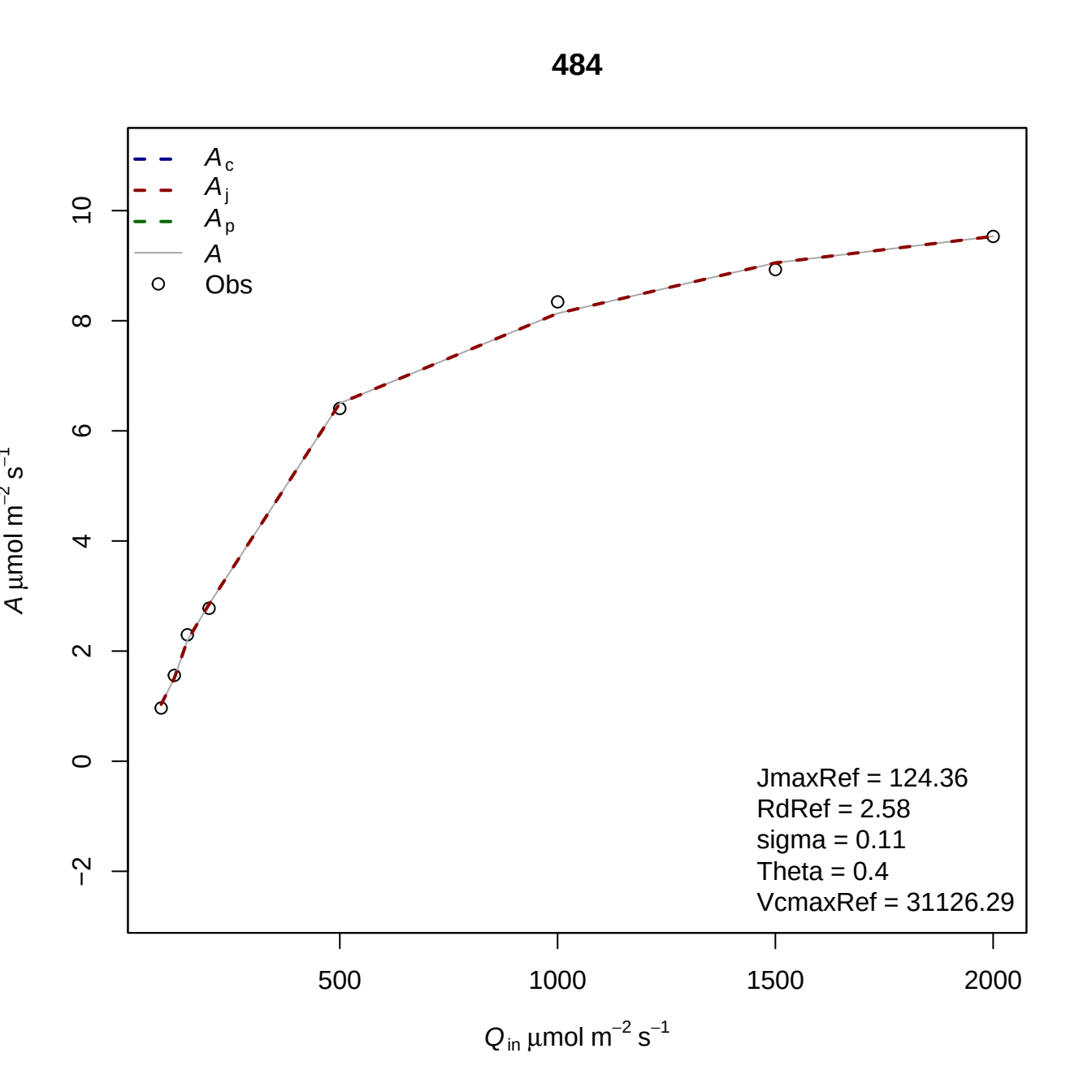

478
A_c
A_j
A_p
A
Obs
JmaxRef = 129.29
RdRef = 1.24
sigma = 0.11
Theta = 0.47
VcmaxRef = 219.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

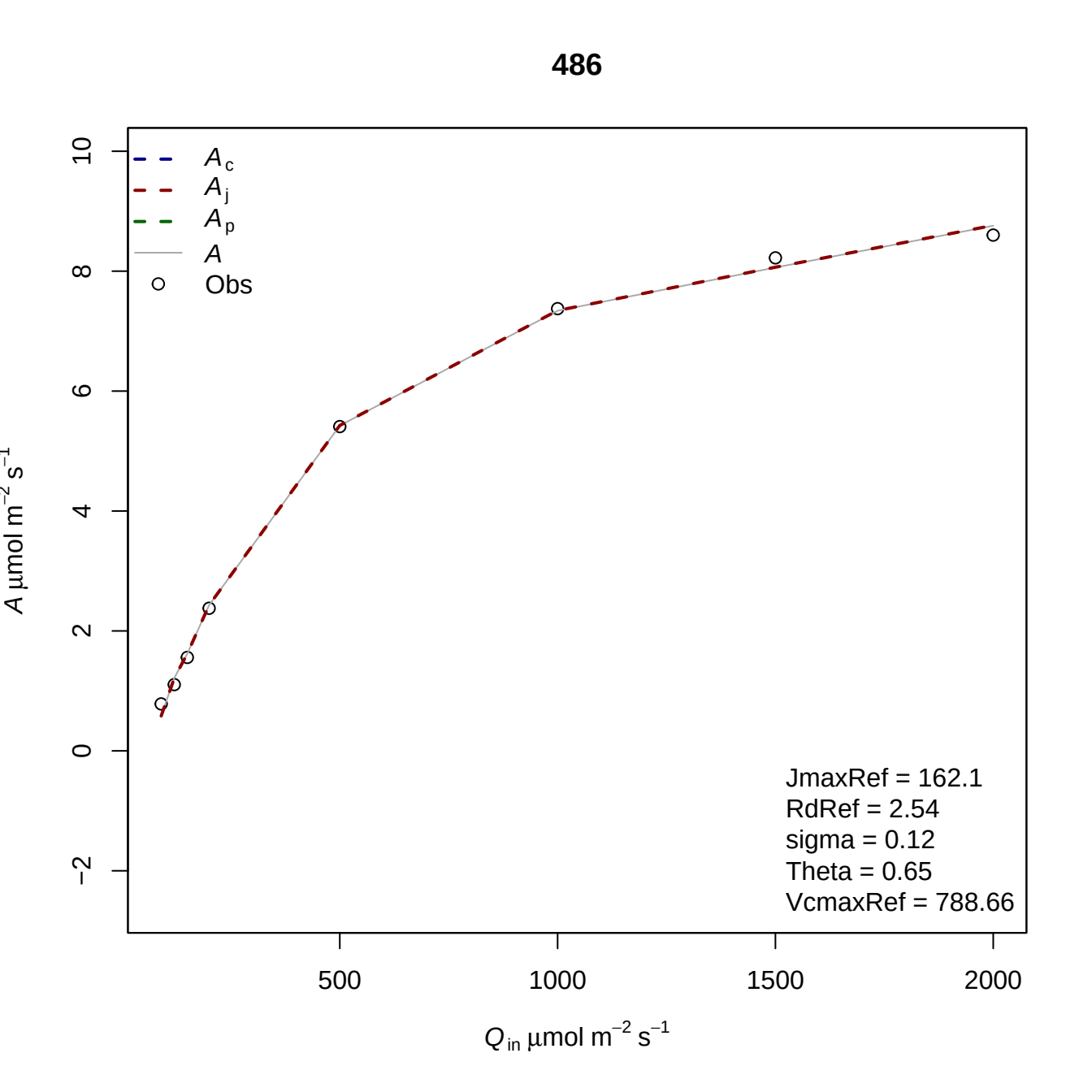

480
A_c
A_j
A_p
A
Obs
JmaxRef = 210.41
RdRef = 3.86
sigma = 0.06
Theta = −0.21
VcmaxRef = 220573.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
500
1000
1500
2000

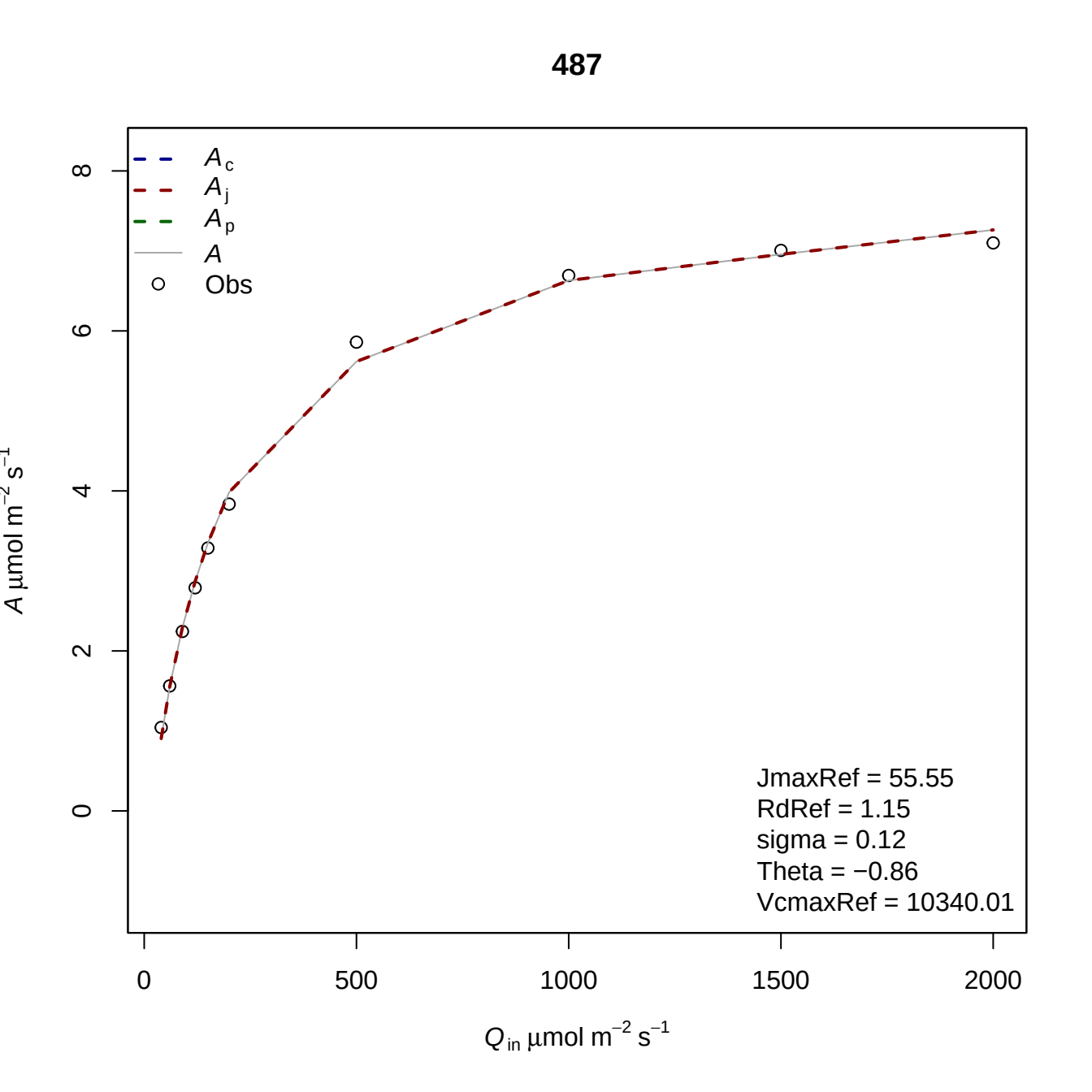

481
A_c
A_j
A_p
A
Obs
JmaxRef = 67.62
RdRef = 1.21
sigma = 0.23
Theta = 0.14
VcmaxRef = 83346.04
0
2
4
6
8
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

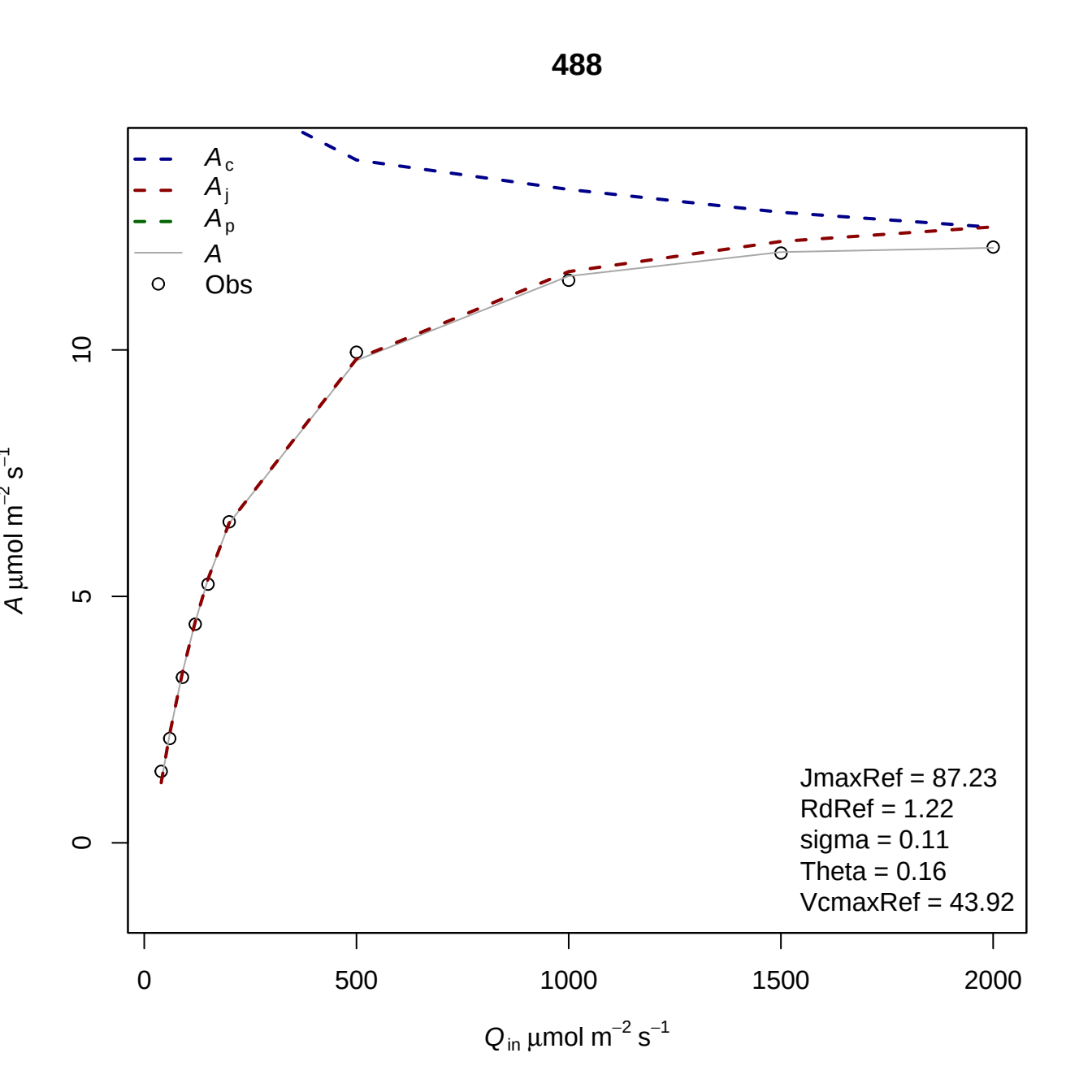

482
A_c
A_j
A_p
A
Obs
JmaxRef = 109.69
RdRef = 2.21
sigma = 0.46
Theta = 0.47
VcmaxRef = 46074.68
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

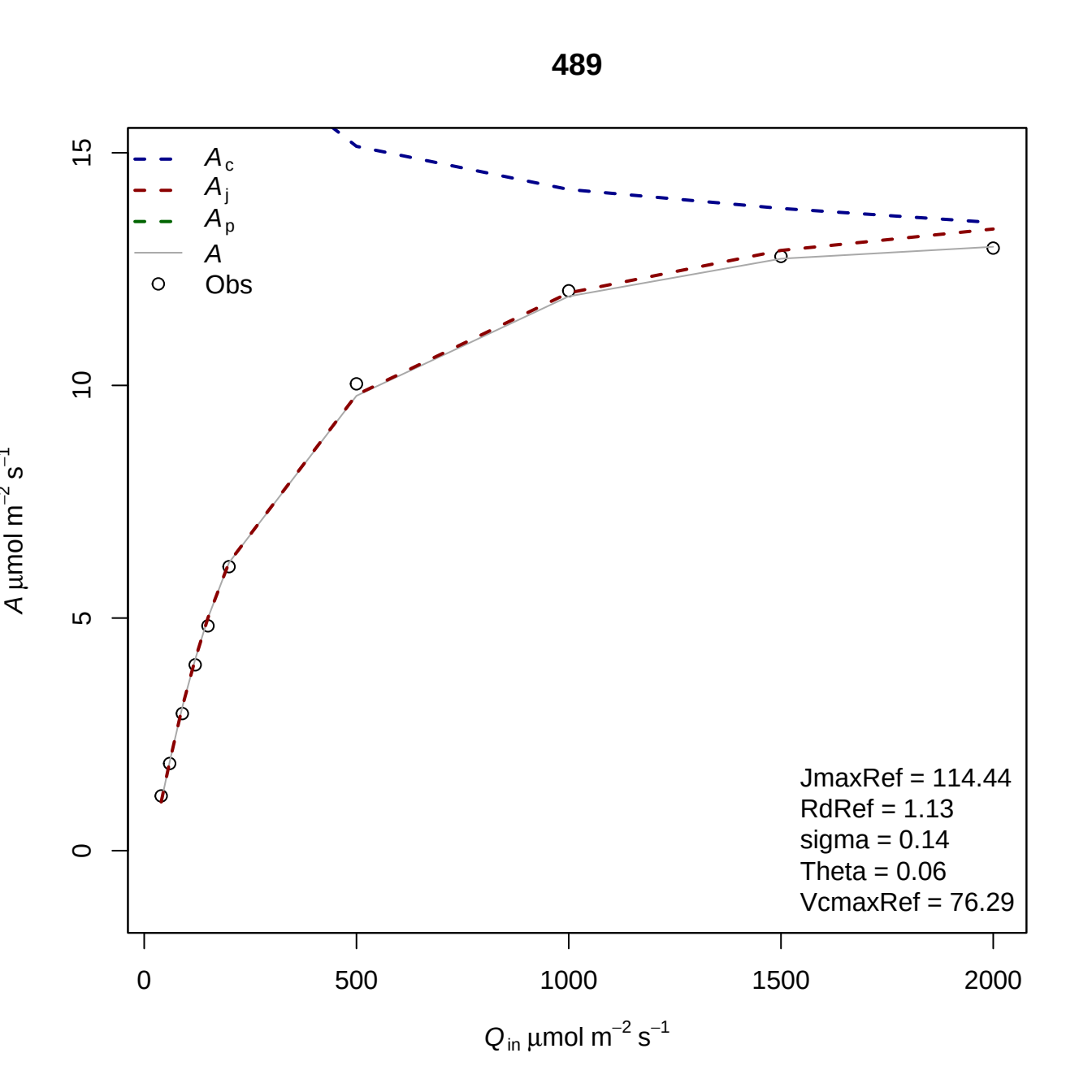

483
A_c
A_j
A_p
A
Obs
JmaxRef = 90.46
RdRef = 2.17
sigma = 0.14
Theta = 0.42
VcmaxRef = 697927.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

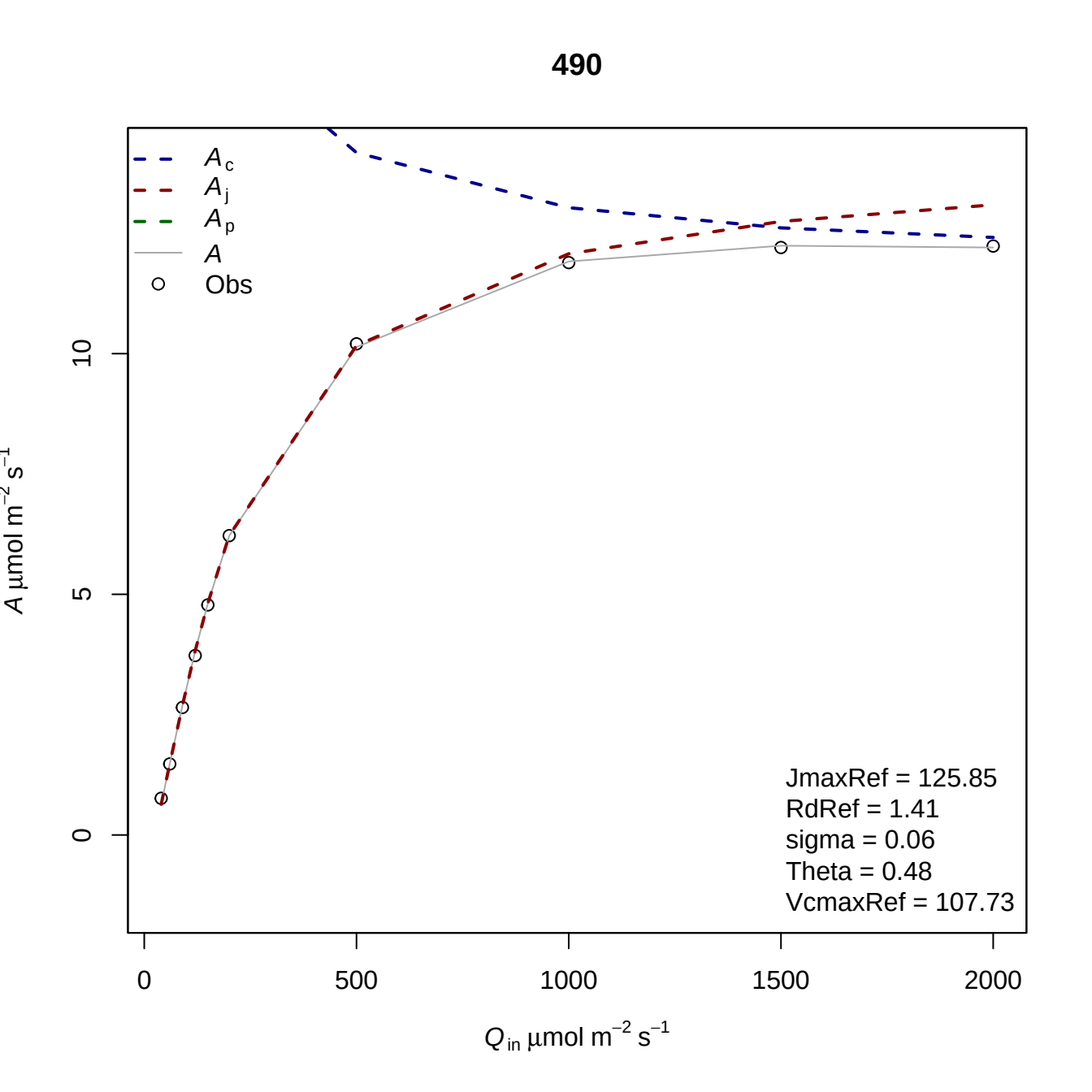

484
A_c
A_j
A_p
A
Obs
JmaxRef = 124.36
RdRef = 2.58
sigma = 0.11
Theta = 0.4
VcmaxRef = 31126.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

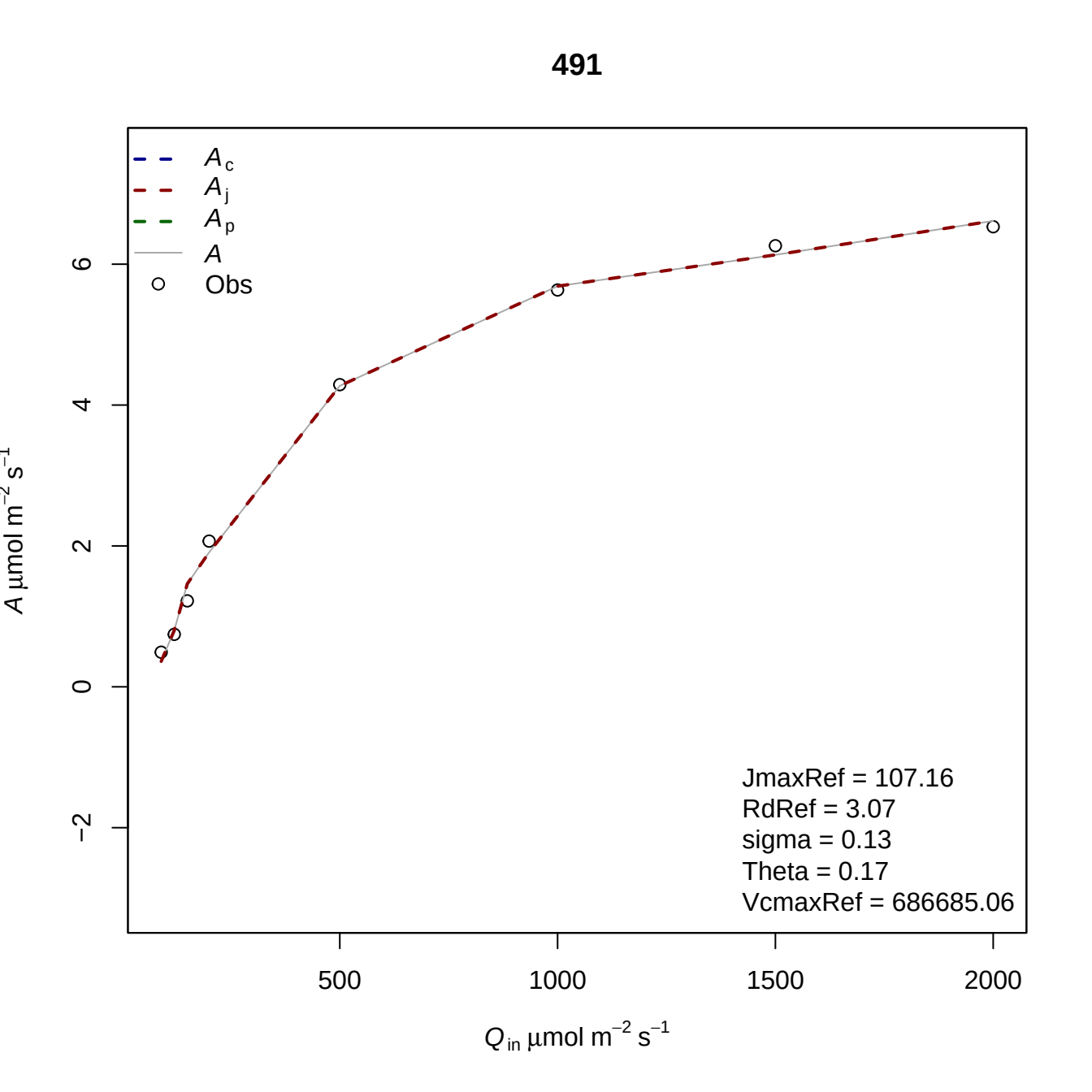

486
A_c
A_j
A_p
A
Obs
JmaxRef = 162.1
RdRef = 2.54
sigma = 0.12
Theta = 0.65
VcmaxRef = 788.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
500
1000
1500
2000

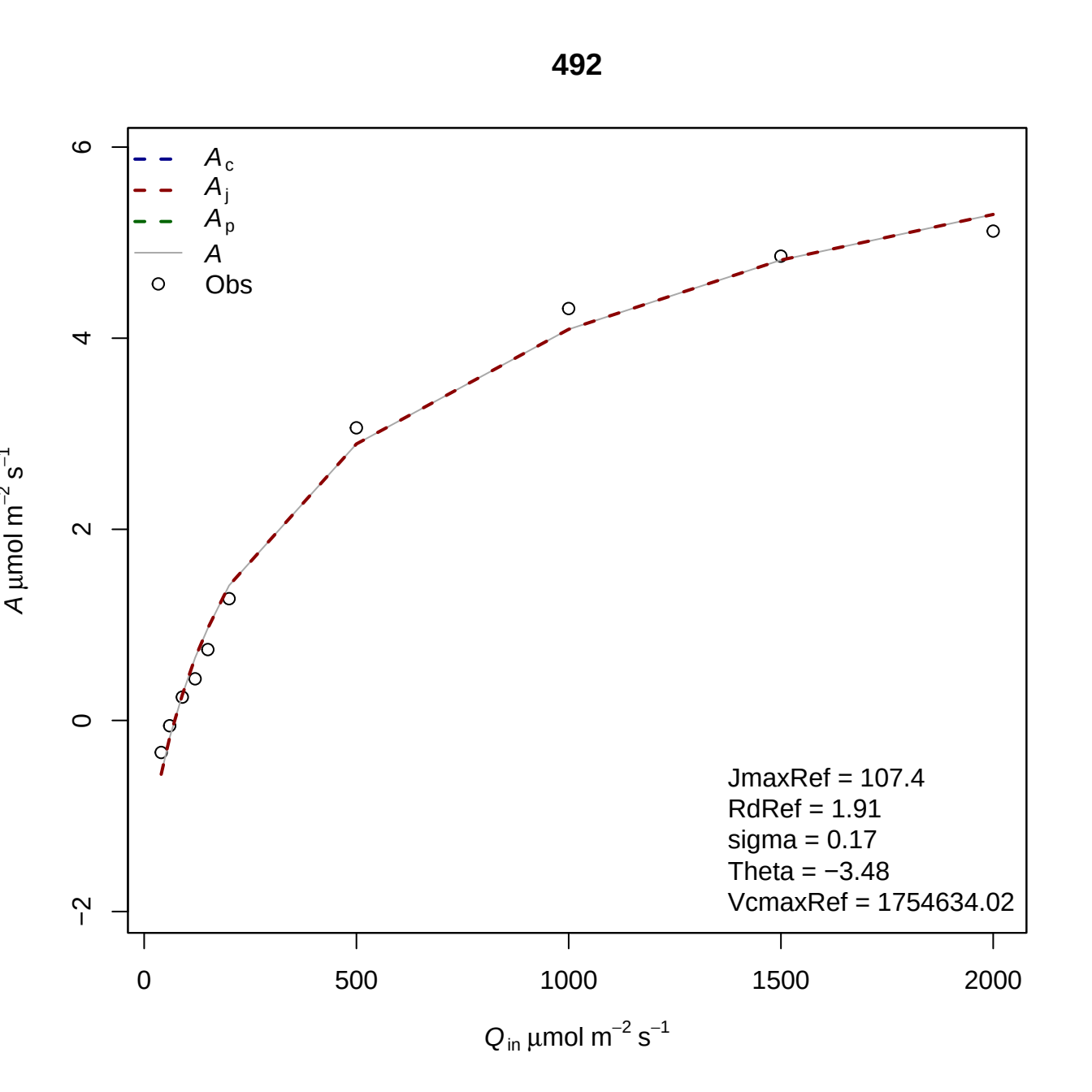

487
A_c
A_j
A_p
A
Obs
JmaxRef = 55.55
RdRef = 1.15
sigma = 0.12
Theta = −0.86
VcmaxRef = 10340.01
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

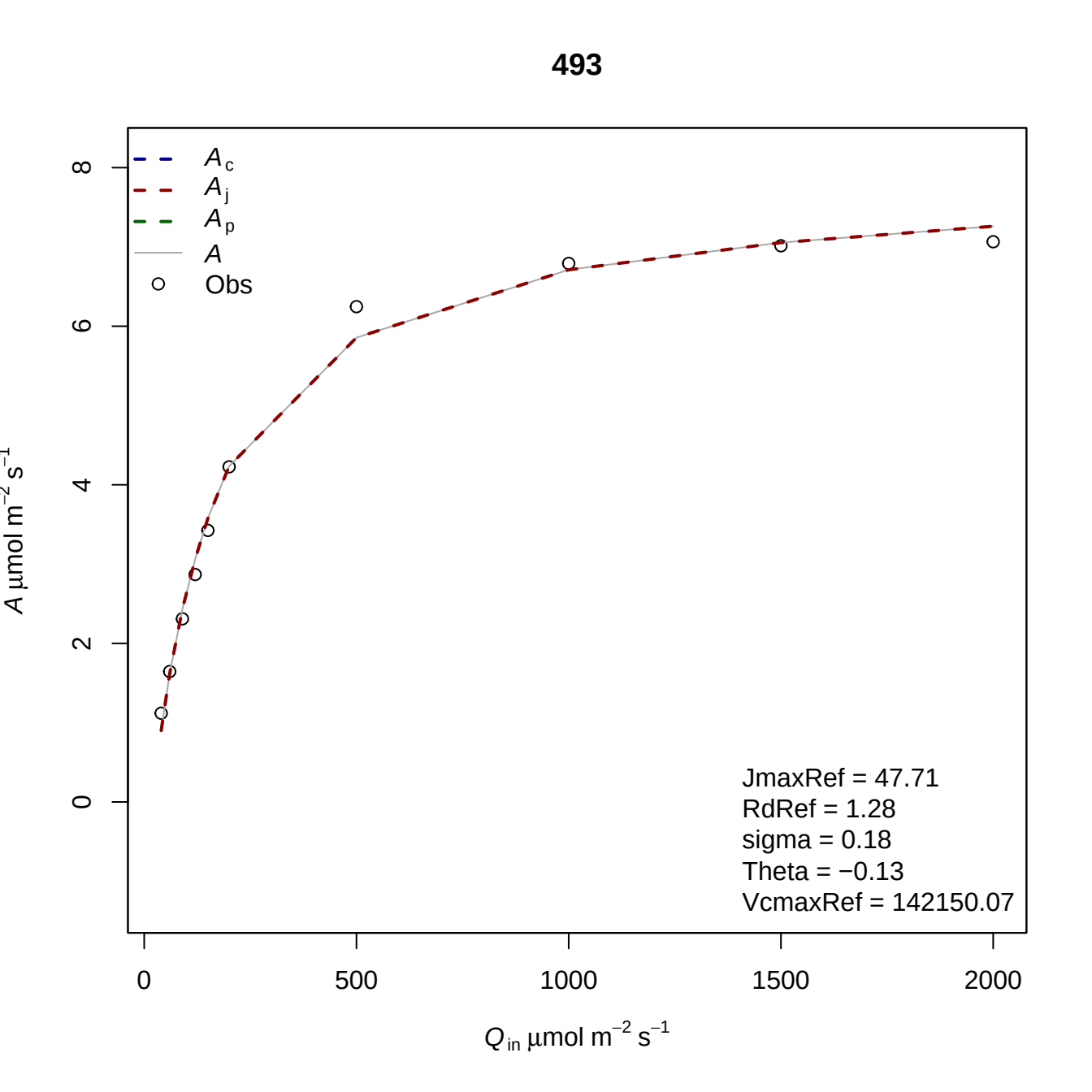

488
A_c
A_j
A_p
A
Obs
JmaxRef = 87.23
RdRef = 1.22
sigma = 0.11
Theta = 0.16
VcmaxRef = 43.92
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

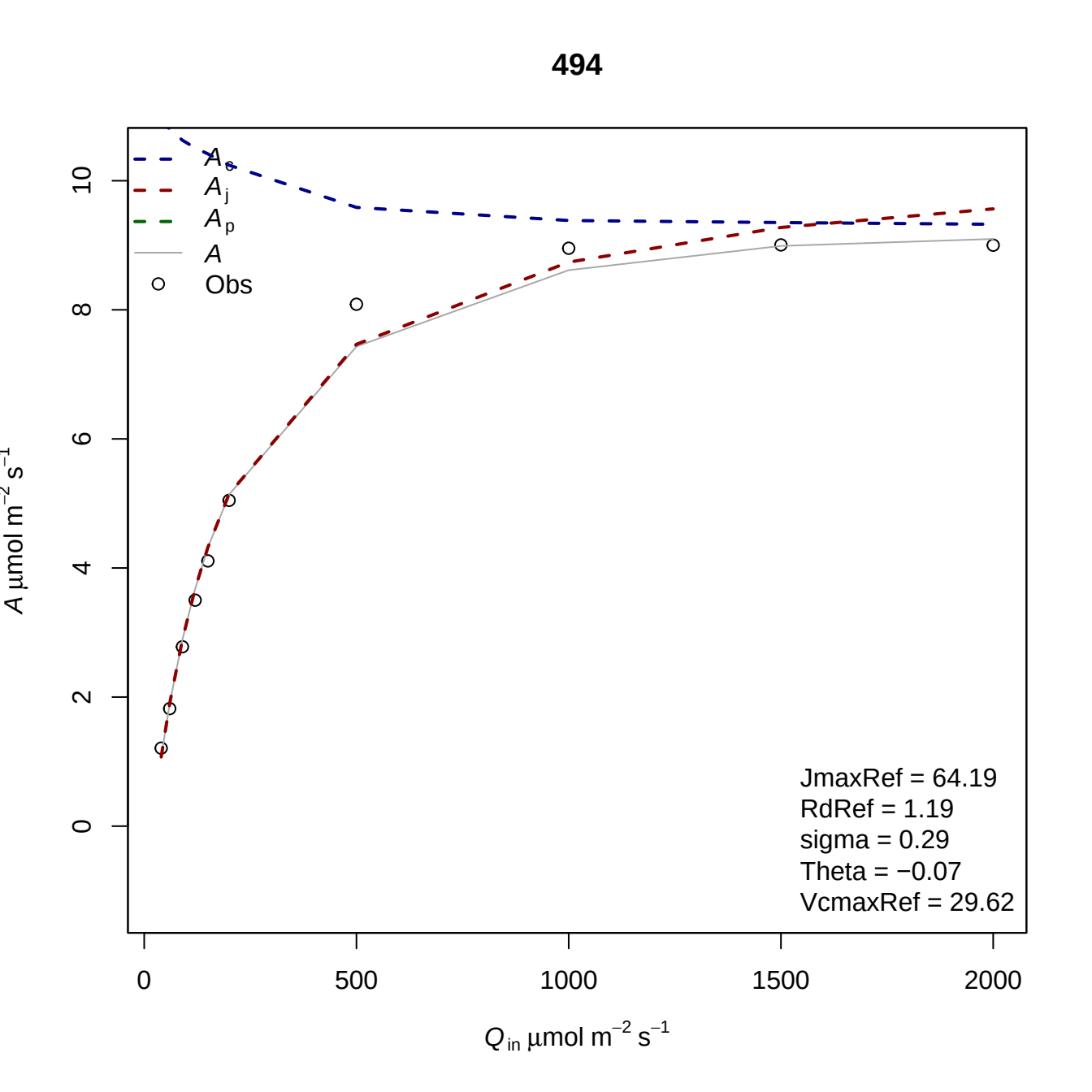

489
A_c
A_j
A_p
A
Obs
JmaxRef = 114.44
RdRef = 1.13
sigma = 0.14
Theta = 0.06
VcmaxRef = 76.29
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

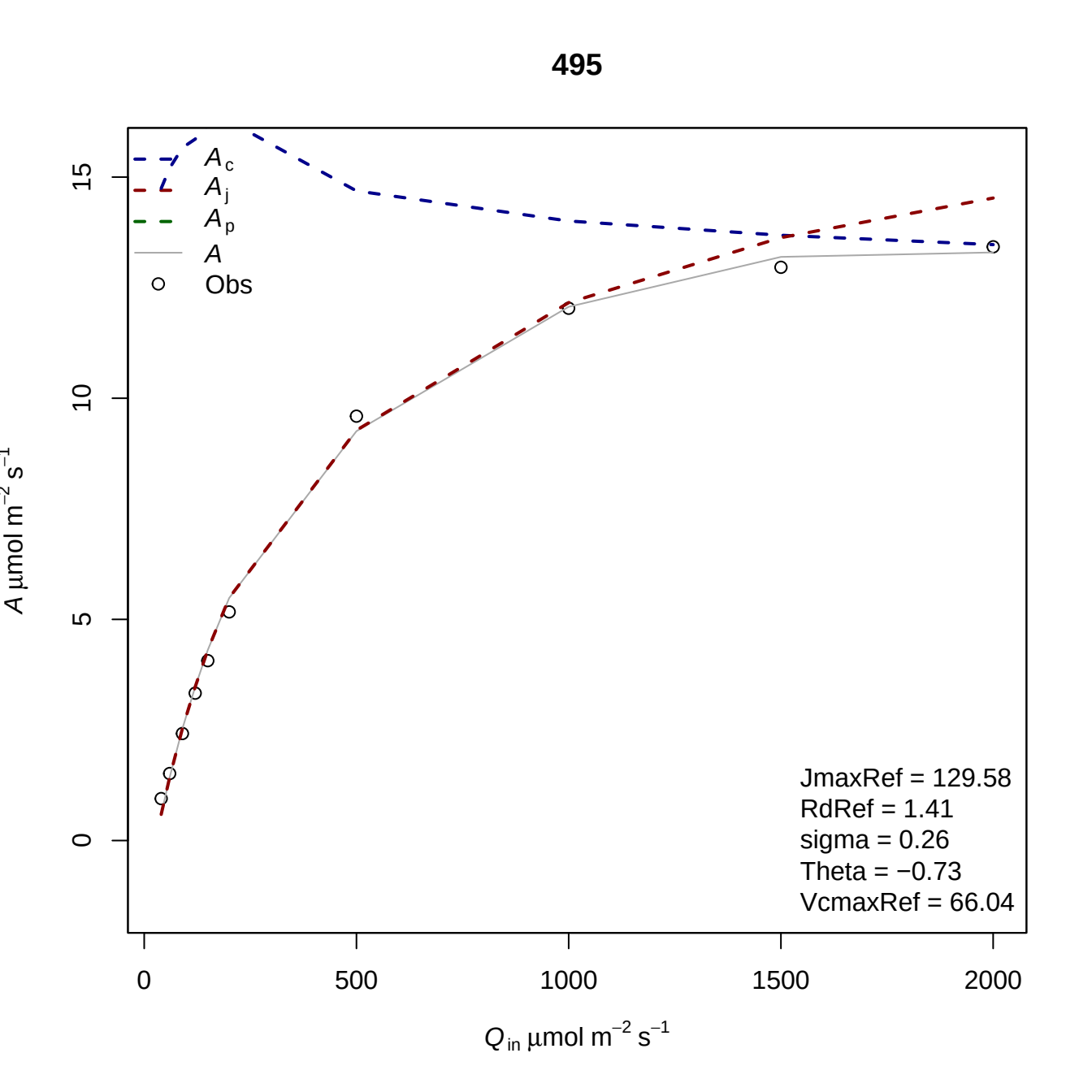

490
A_c
A_j
A_p
A
Obs
JmaxRef = 125.85
RdRef = 1.41
sigma = 0.06
Theta = 0.48
VcmaxRef = 107.73
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

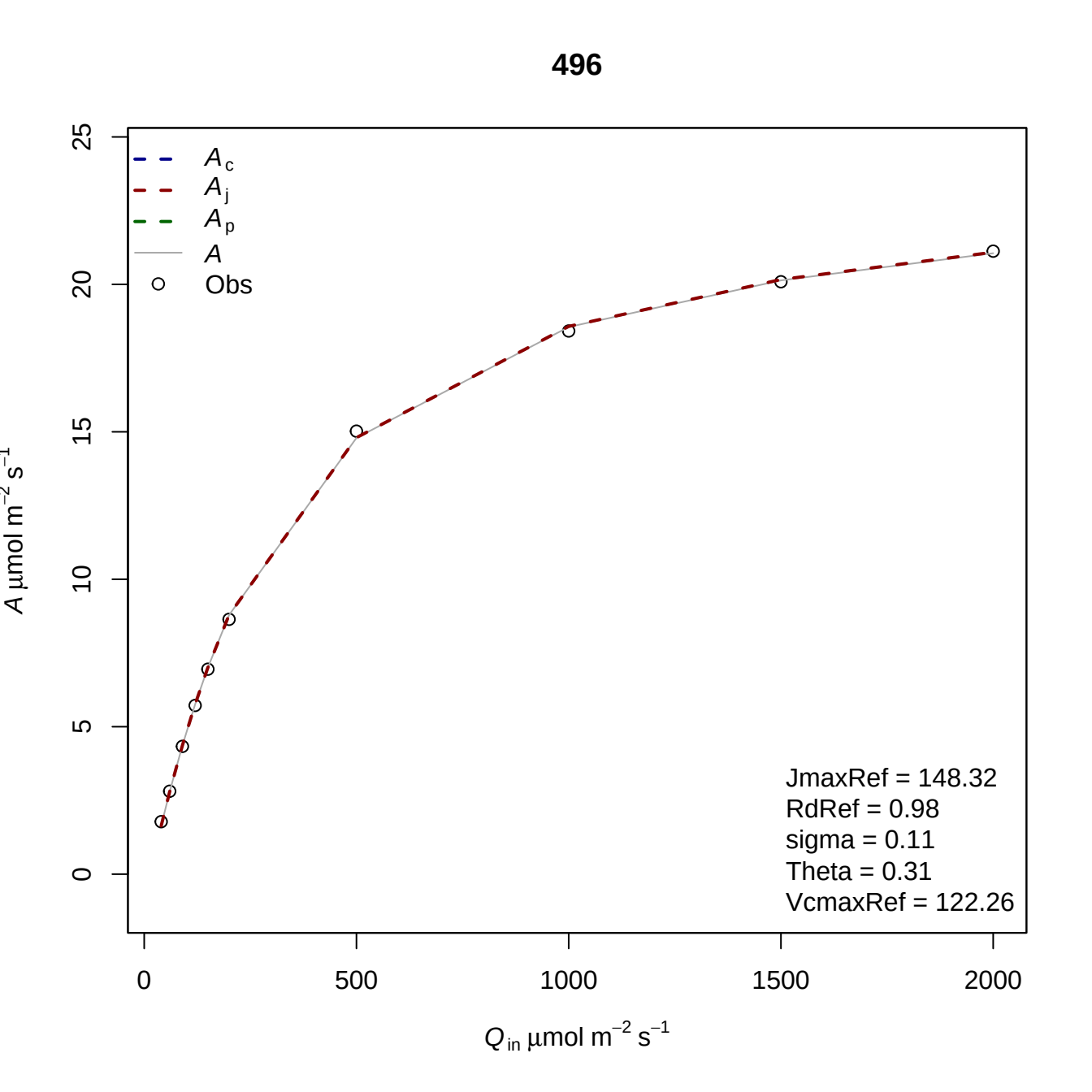

491
A_c
A_j
A_p
A
Obs
JmaxRef = 107.16
RdRef = 3.07
sigma = 0.13
Theta = 0.17
VcmaxRef = 686685.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
500
1000
1500
2000

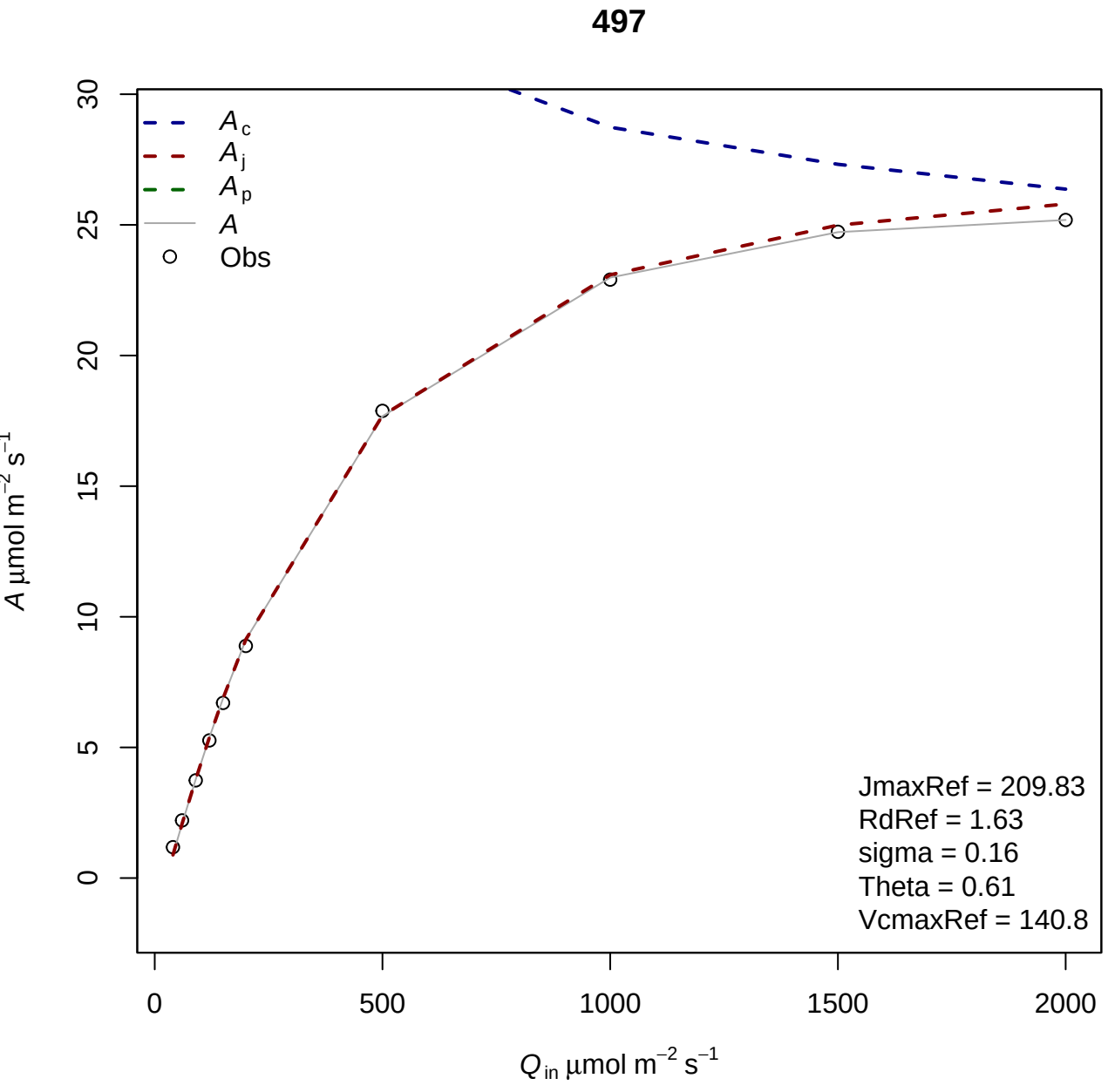

492
A_c
A_j
A_p
A
Obs
JmaxRef = 107.4
RdRef = 1.91
sigma = 0.17
Theta = −3.48
VcmaxRef = 1754634.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

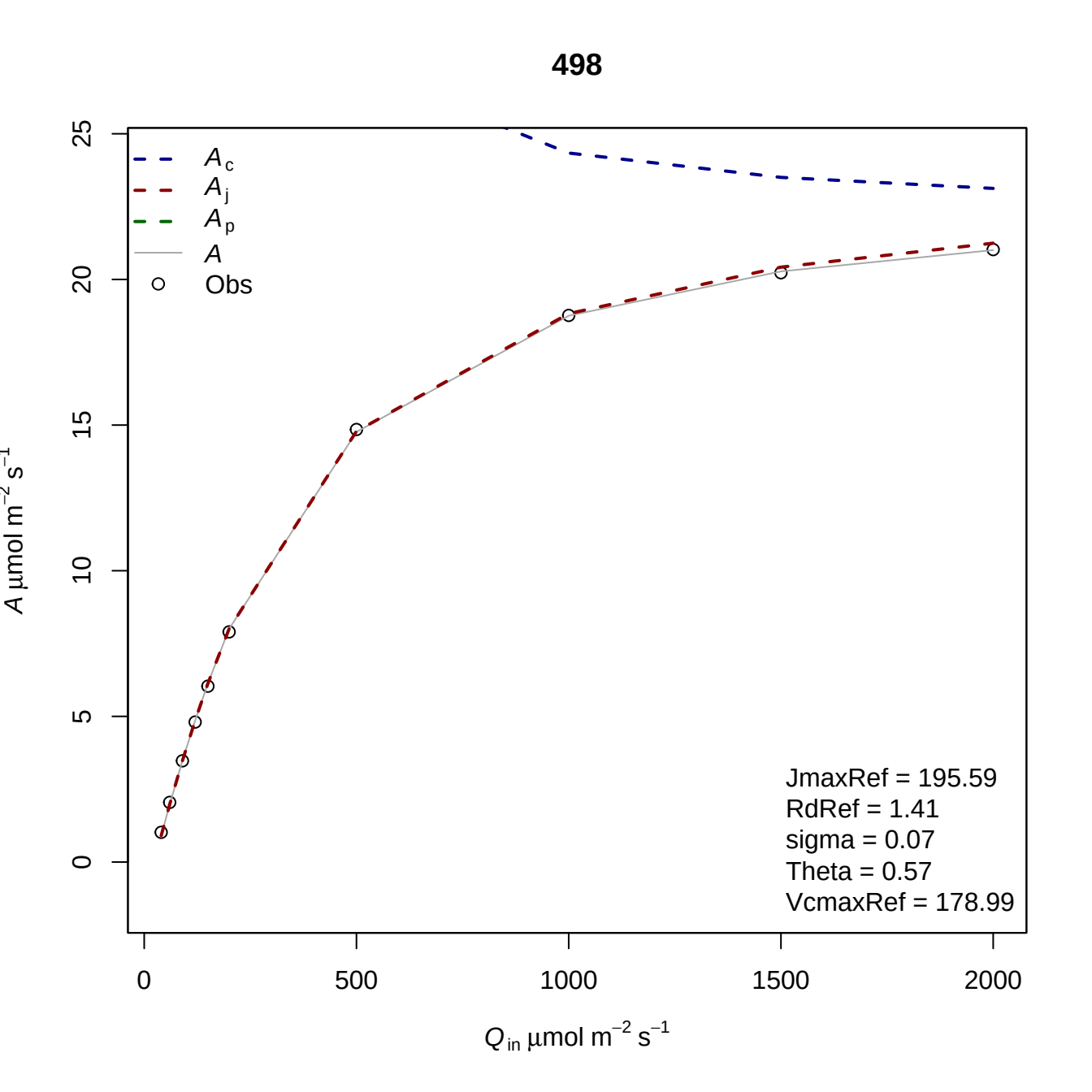

493
A_c
A_j
A_p
A
Obs
JmaxRef = 47.71
RdRef = 1.28
sigma = 0.18
Theta = −0.13
VcmaxRef = 142150.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

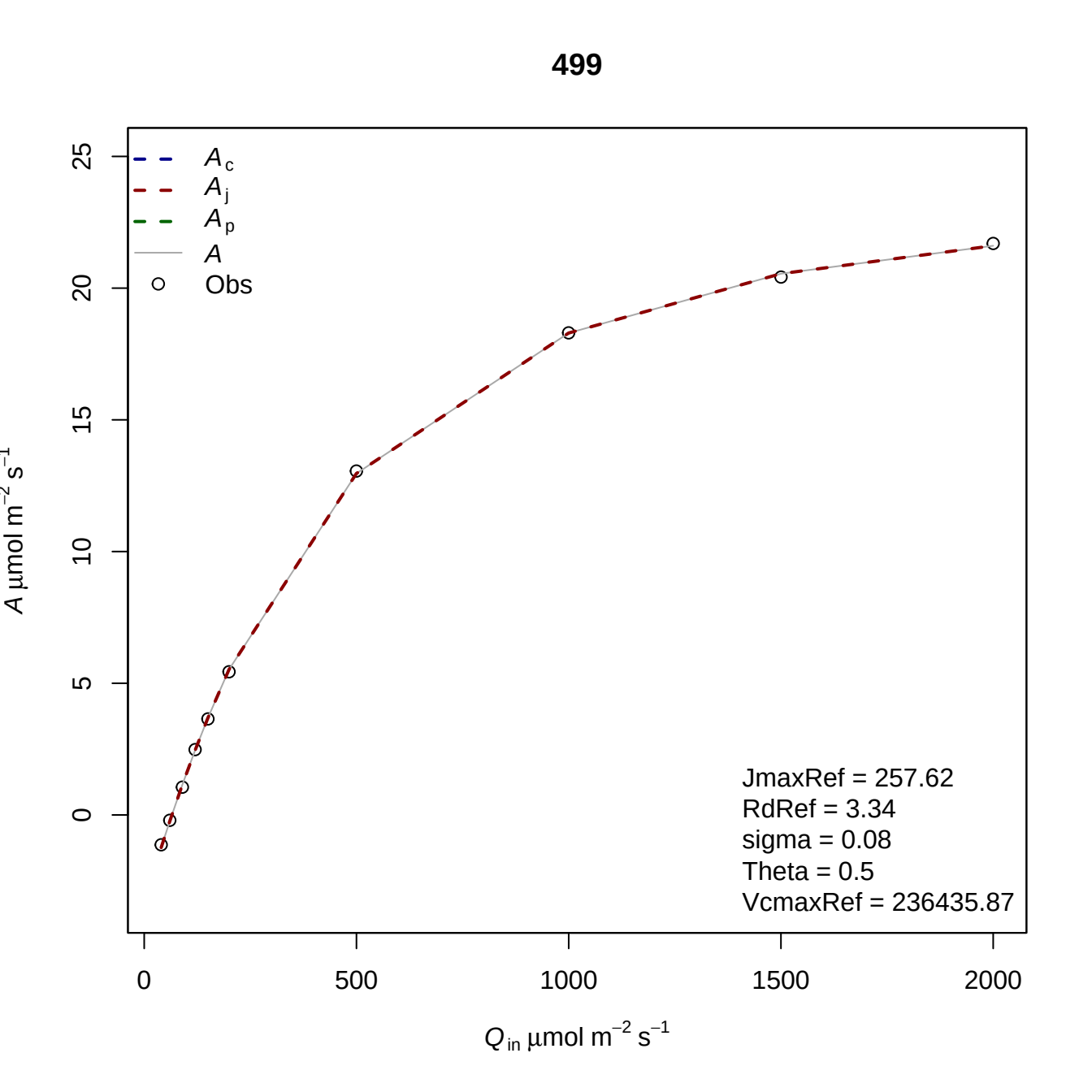

494
A_c
A_j
A_p
A
Obs
JmaxRef = 64.19
RdRef = 1.19
sigma = 0.29
Theta = −0.07
VcmaxRef = 29.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

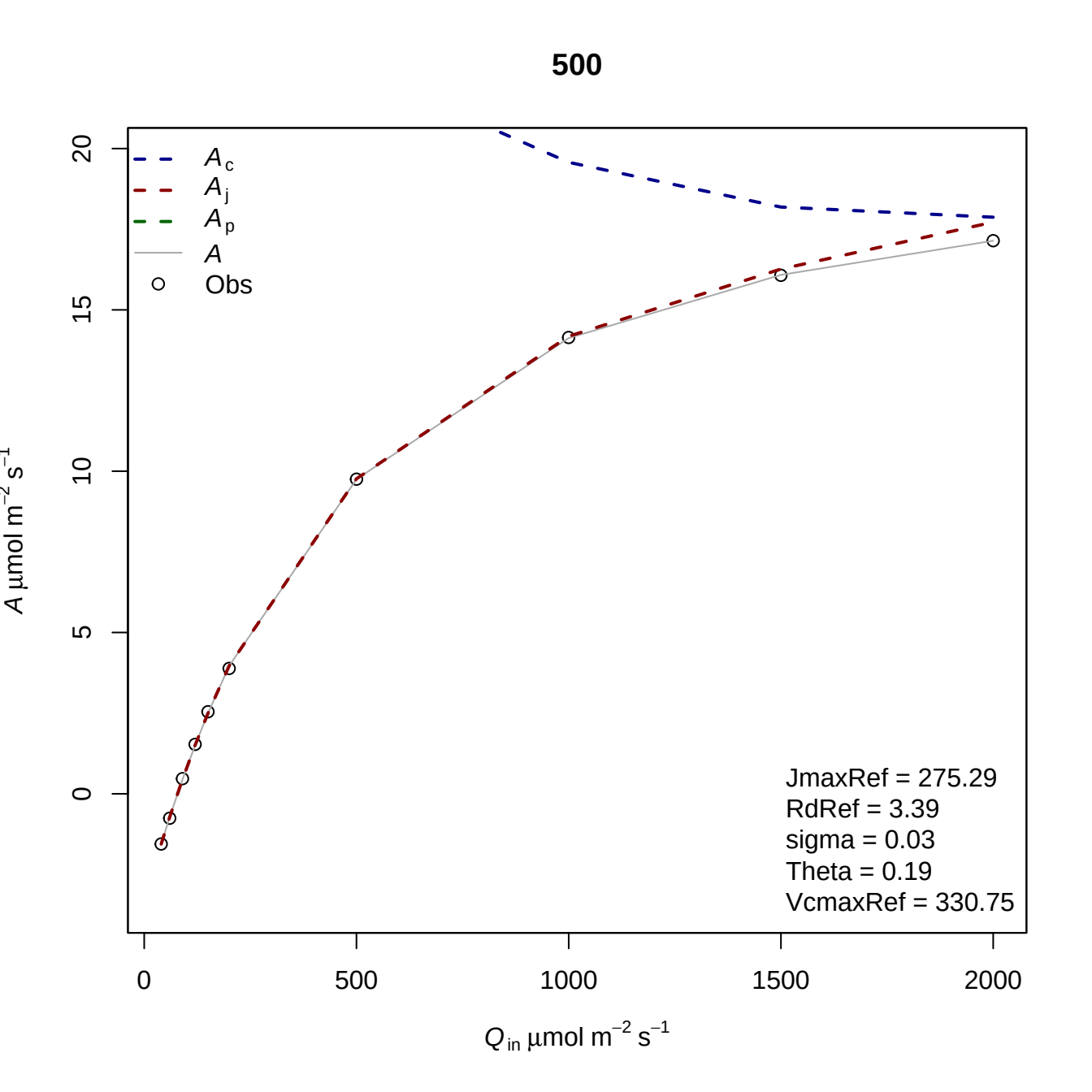

495
A_c
A_j
A_p
A
Obs
JmaxRef = 129.58
RdRef = 1.41
sigma = 0.26
Theta = −0.73
VcmaxRef = 66.04
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

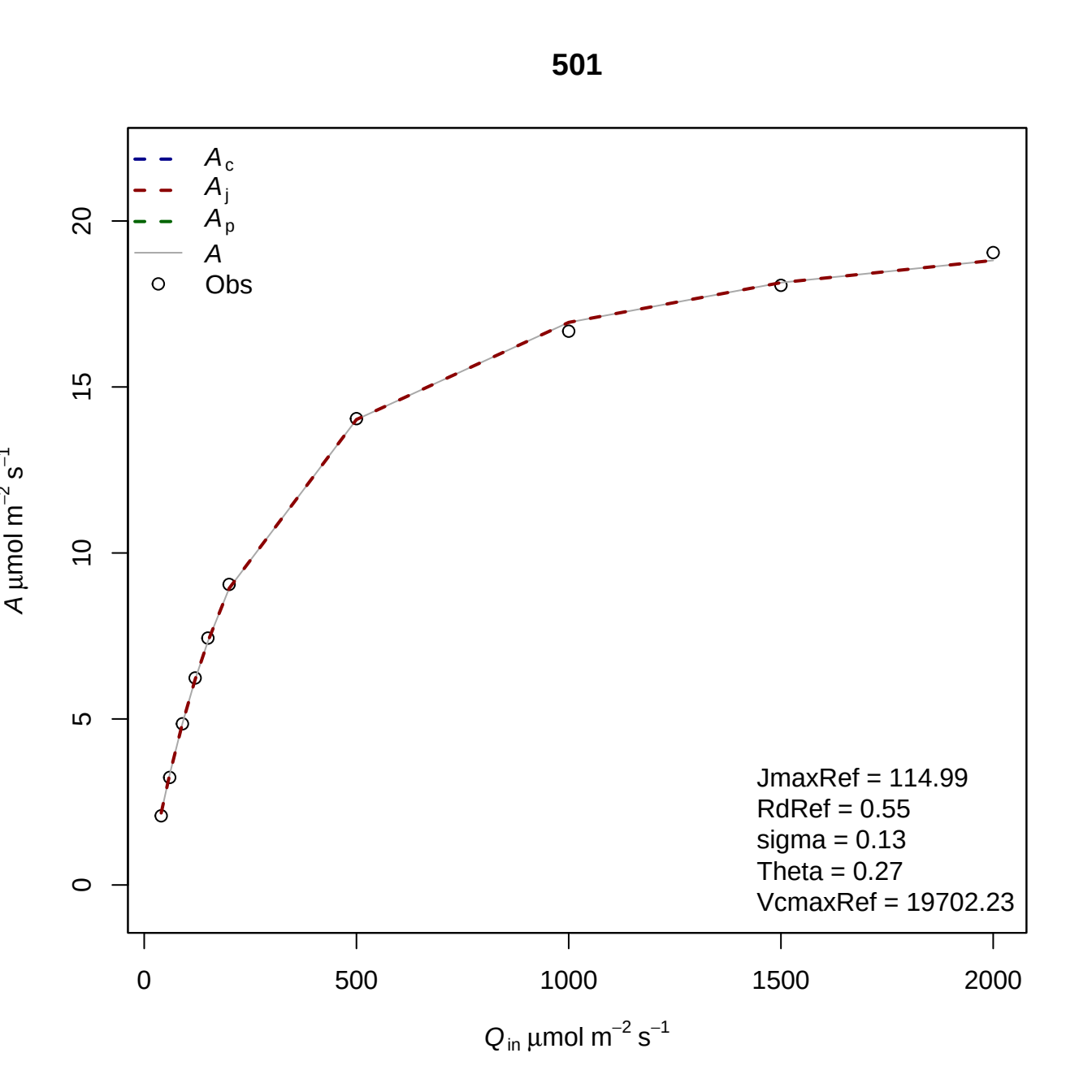

496
A_c
A_j
A_p
A
Obs
JmaxRef = 148.32
RdRef = 0.98
sigma = 0.11
Theta = 0.31
VcmaxRef = 122.26
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

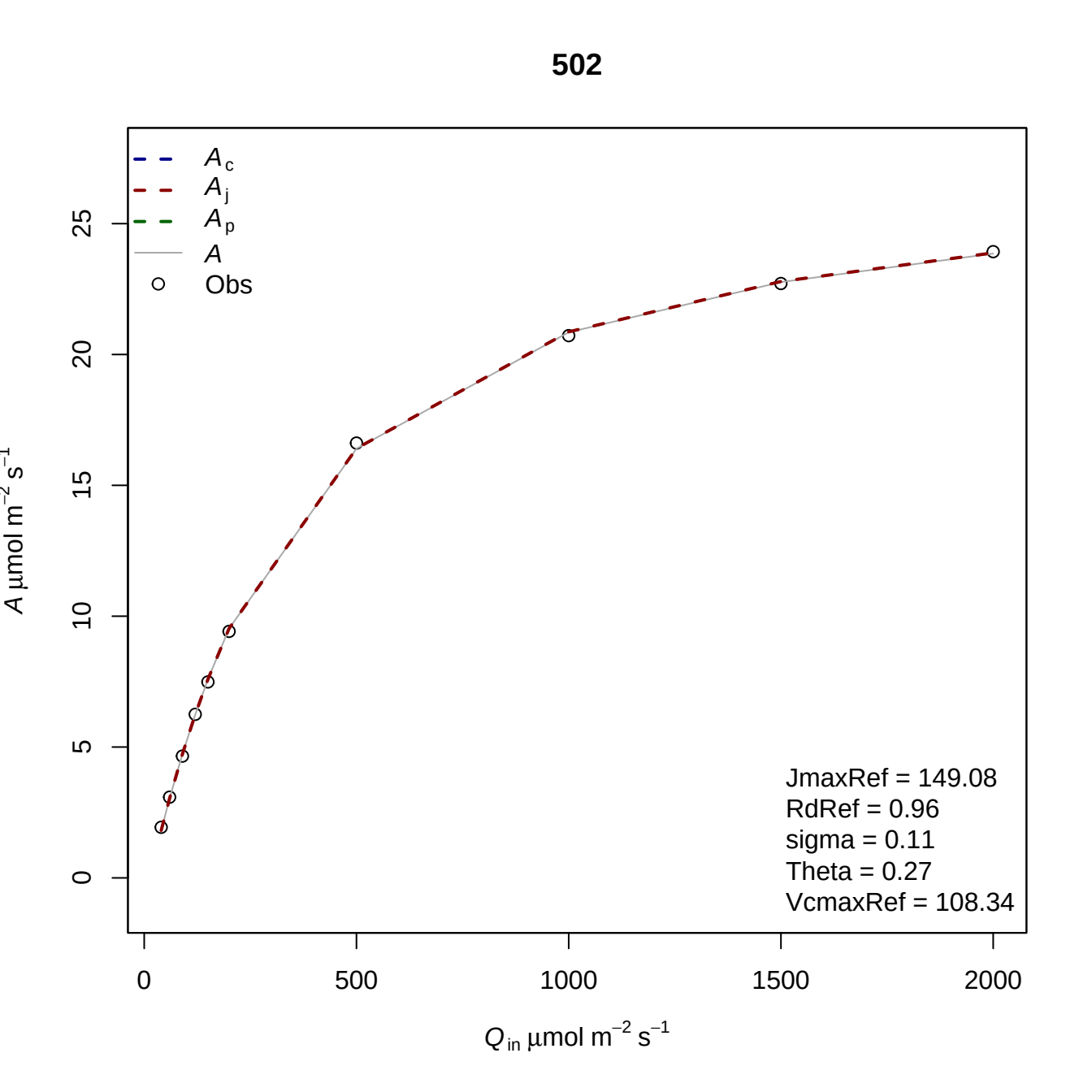

497
A_c
A_j
A_p
A
Obs
JmaxRef = 209.83
RdRef = 1.63
sigma = 0.16
Theta = 0.61
VcmaxRef = 140.8
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

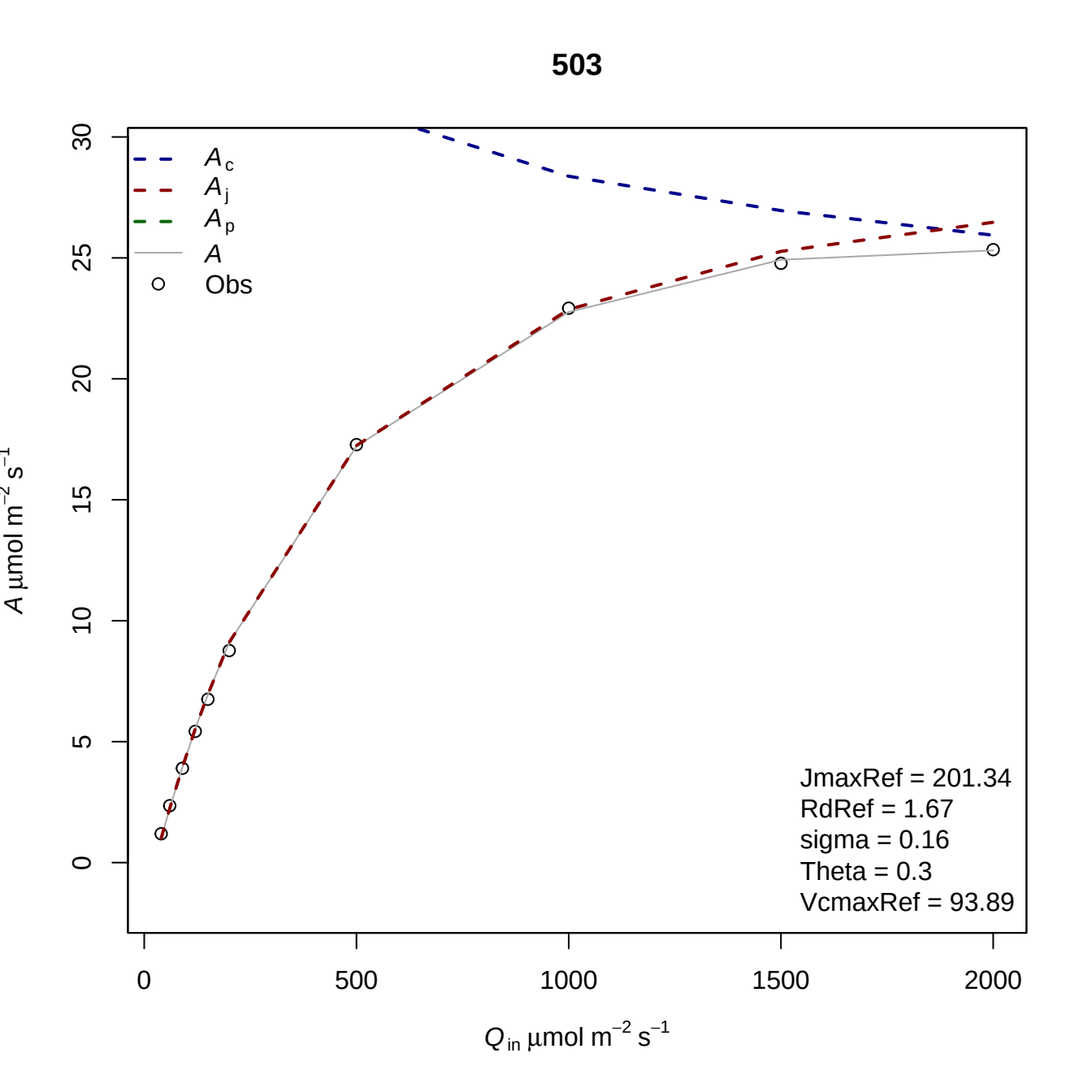

498
A_c
A_j
A_p
A
Obs
JmaxRef = 195.59
RdRef = 1.41
sigma = 0.07
Theta = 0.57
VcmaxRef = 178.99
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

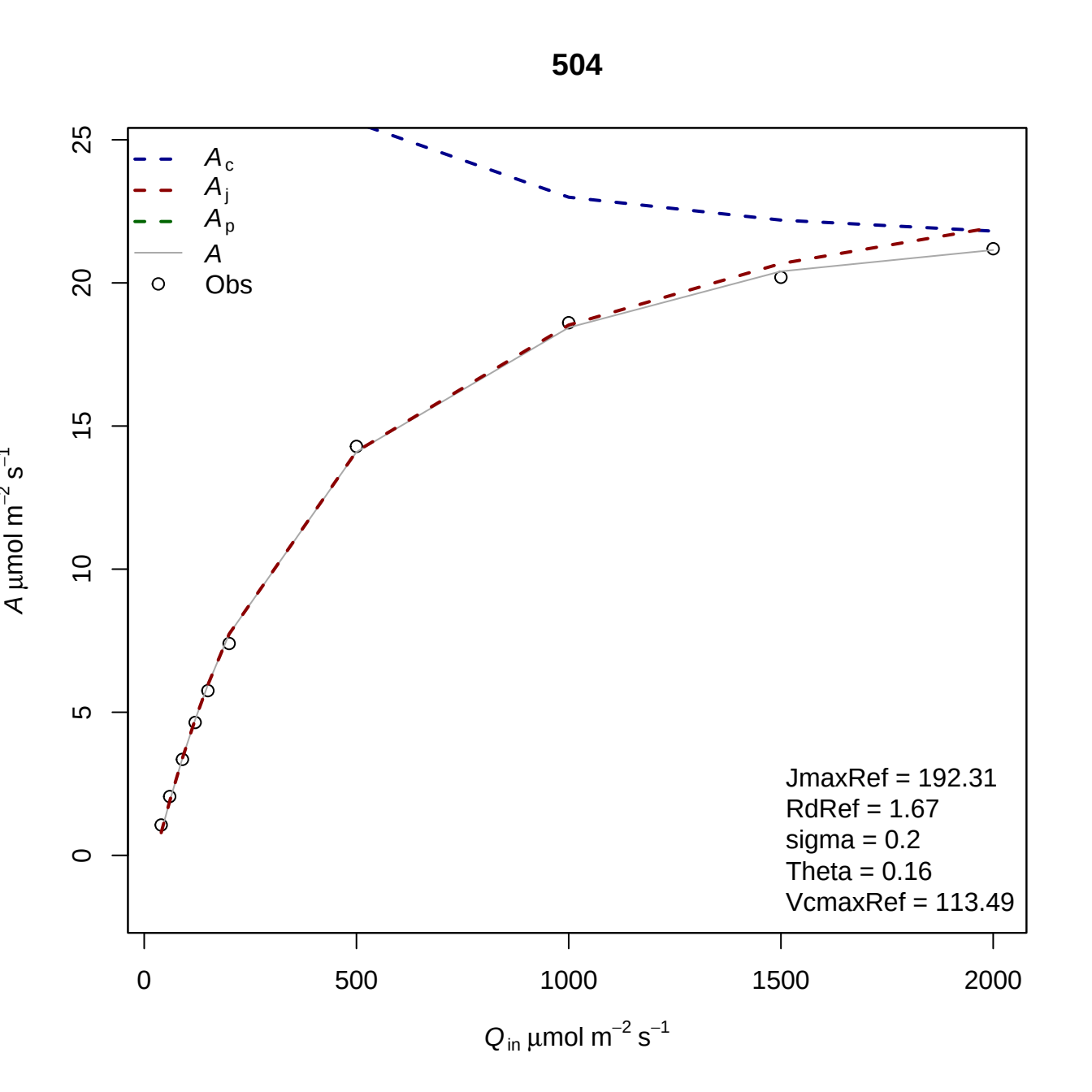

499
A_c
A_j
A_p
A
Obs
JmaxRef = 257.62
RdRef = 3.34
sigma = 0.08
Theta = 0.5
VcmaxRef = 236435.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

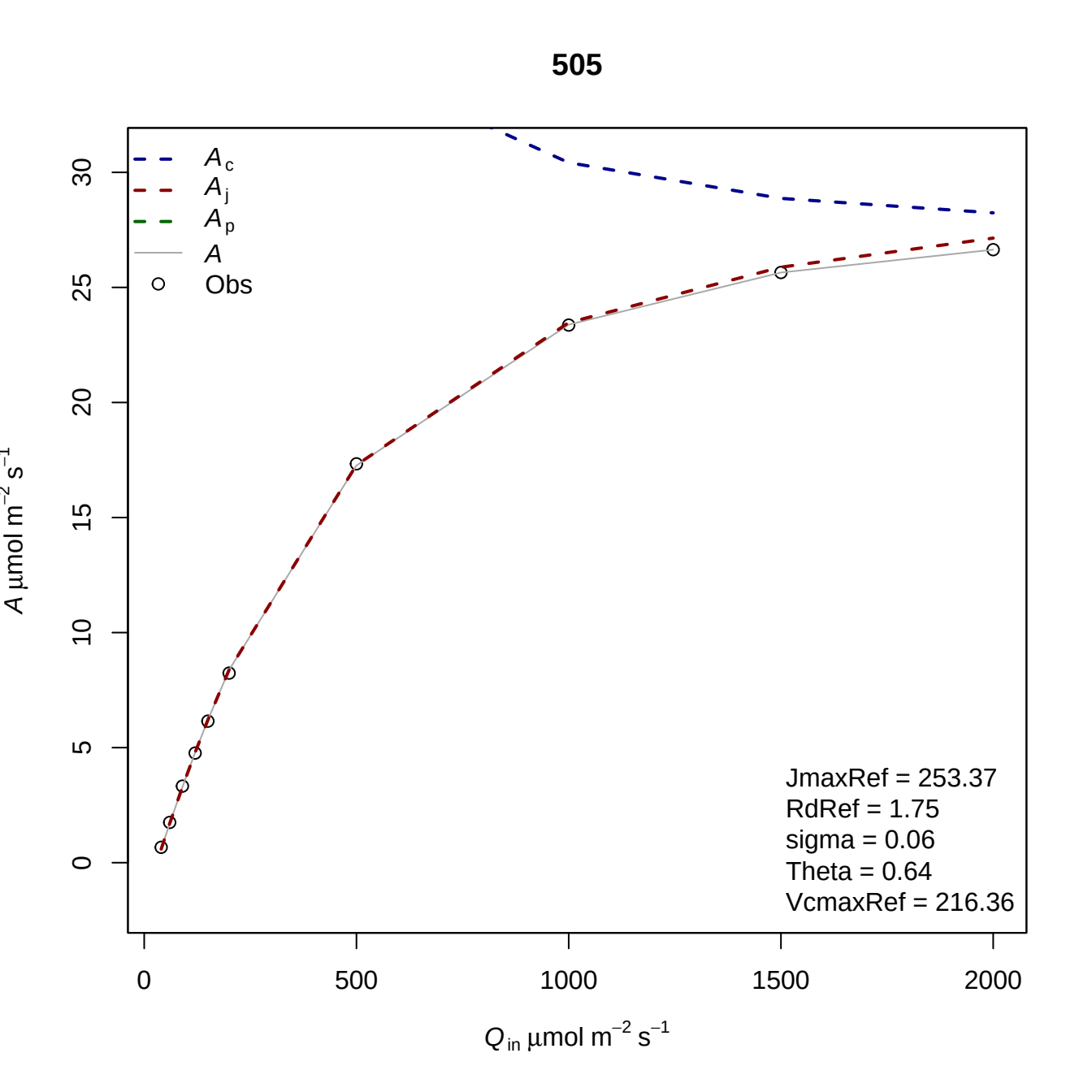

500
A_c
A_j
A_p
A
Obs
JmaxRef = 275.29
RdRef = 3.39
sigma = 0.03
Theta = 0.19
VcmaxRef = 330.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

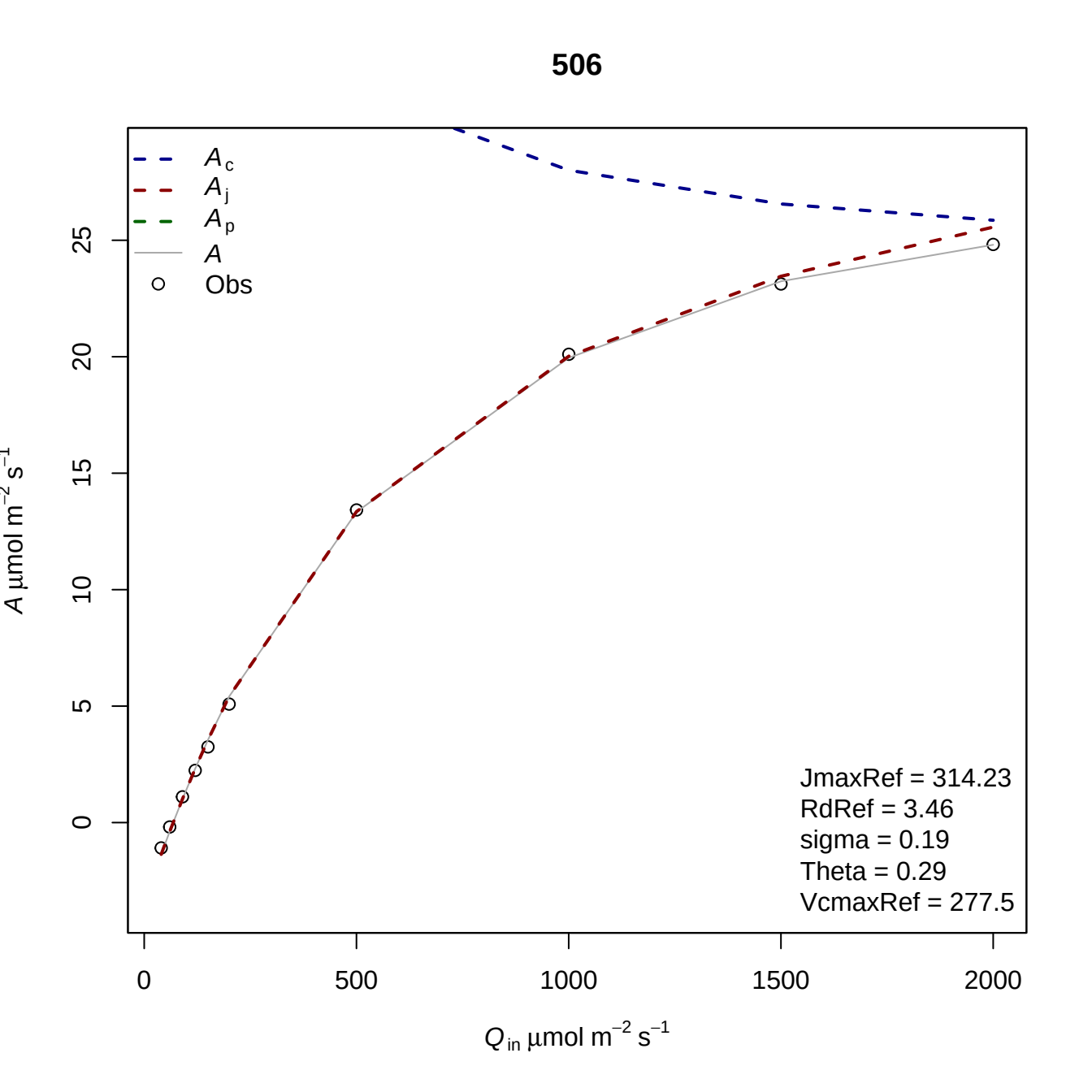

501
A_c
A_j
A_p
A
Obs
JmaxRef = 114.99
RdRef = 0.55
sigma = 0.13
Theta = 0.27
VcmaxRef = 19702.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

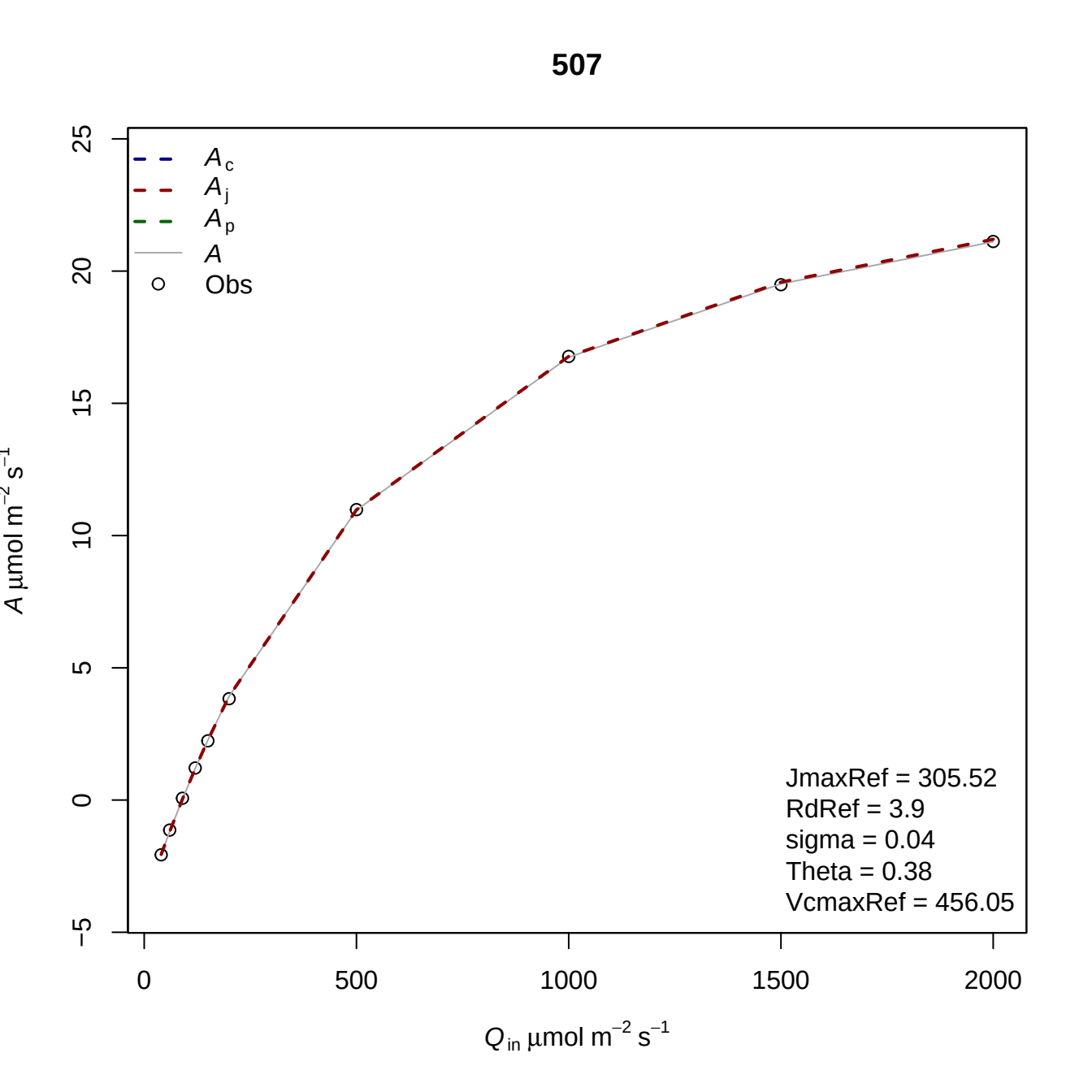

502
A_c
A_j
A_p
A
Obs
JmaxRef = 149.08
RdRef = 0.96
sigma = 0.11
Theta = 0.27
VcmaxRef = 108.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

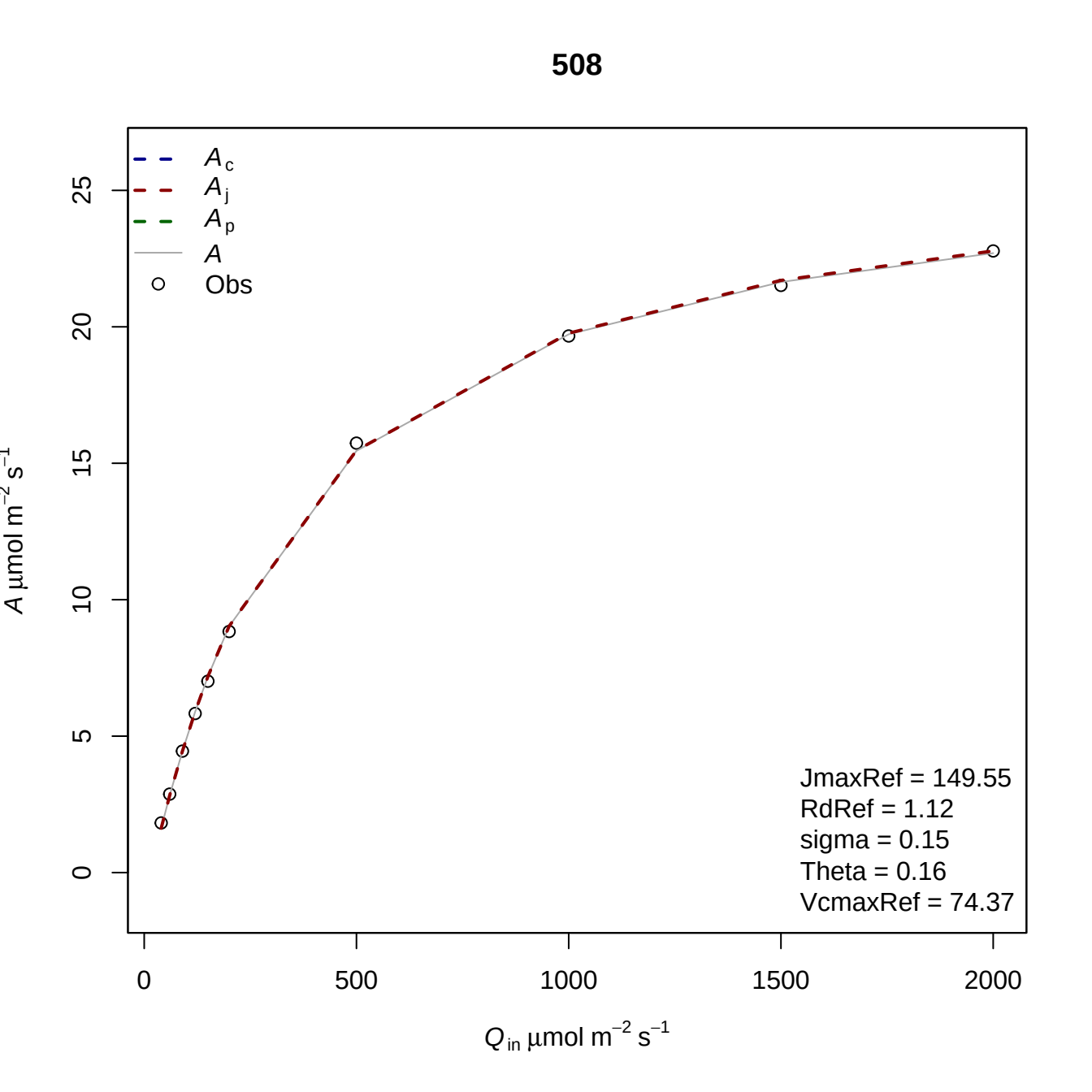

503
A_c
A_j
A_p
A
Obs
JmaxRef = 201.34
RdRef = 1.67
sigma = 0.16
Theta = 0.3
VcmaxRef = 93.89
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

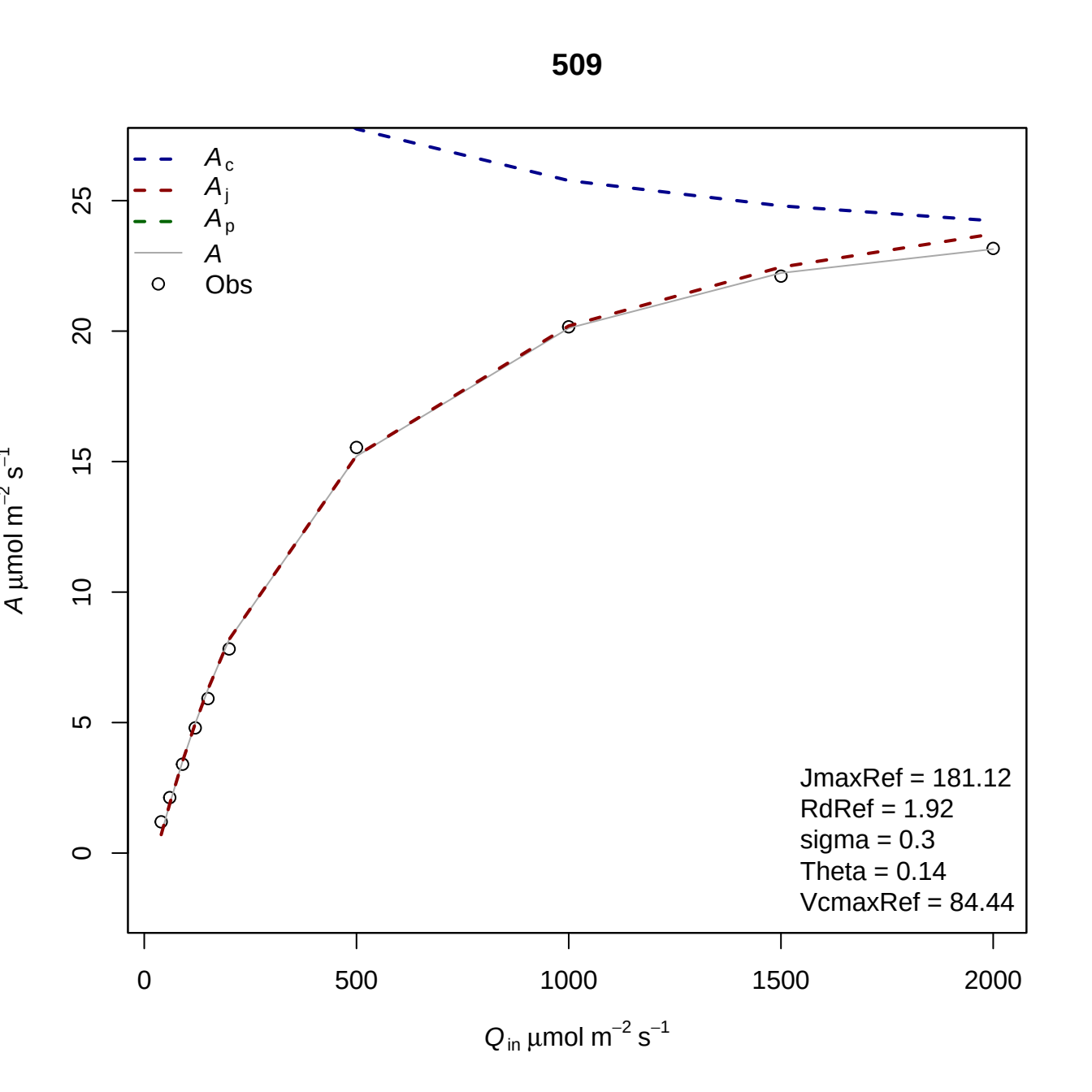

504
A_c
A_j
A_p
A
Obs
JmaxRef = 192.31
RdRef = 1.67
sigma = 0.2
Theta = 0.16
VcmaxRef = 113.49
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

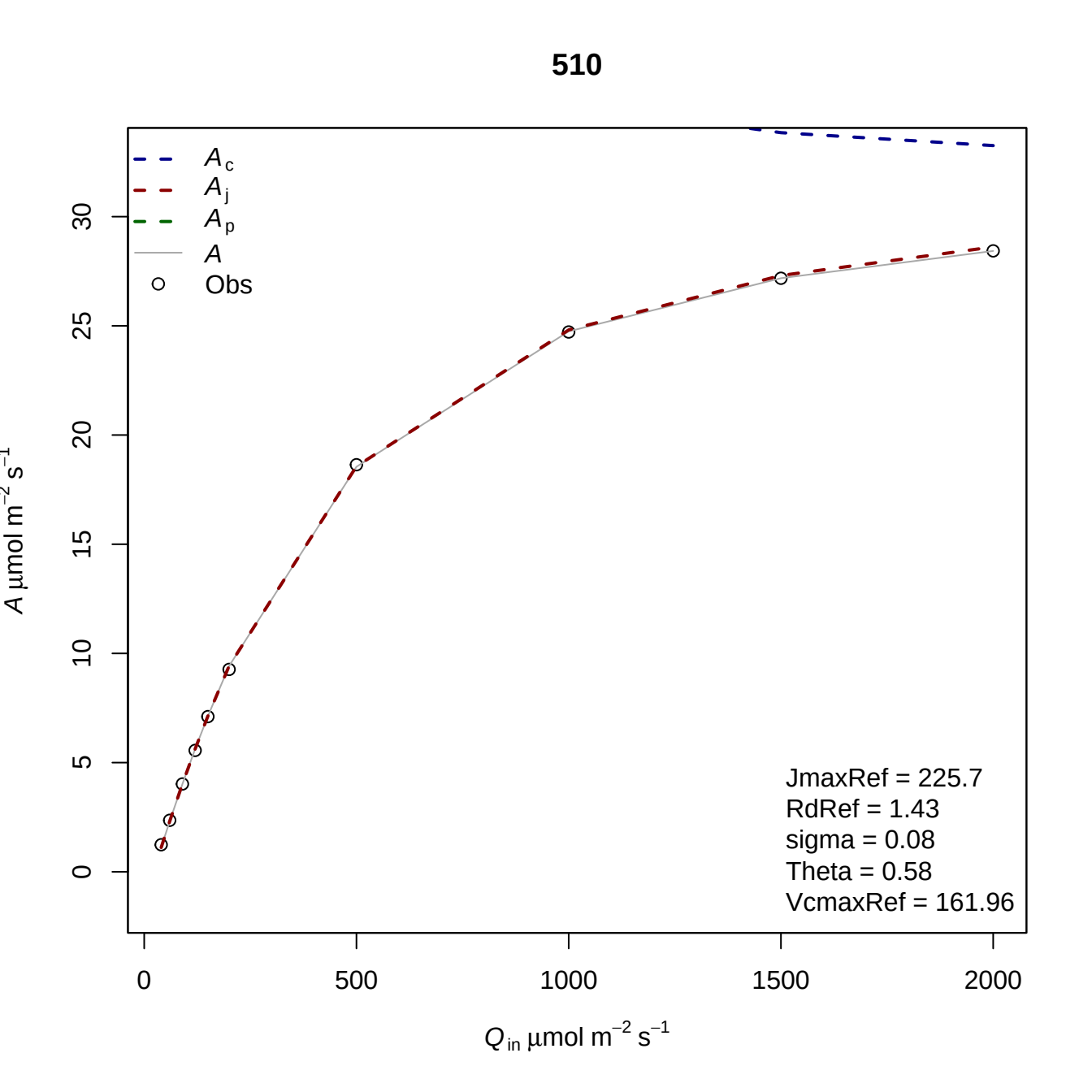

505
A_c
A_j
A_p
A
Obs
JmaxRef = 253.37
RdRef = 1.75
sigma = 0.06
Theta = 0.64
VcmaxRef = 216.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

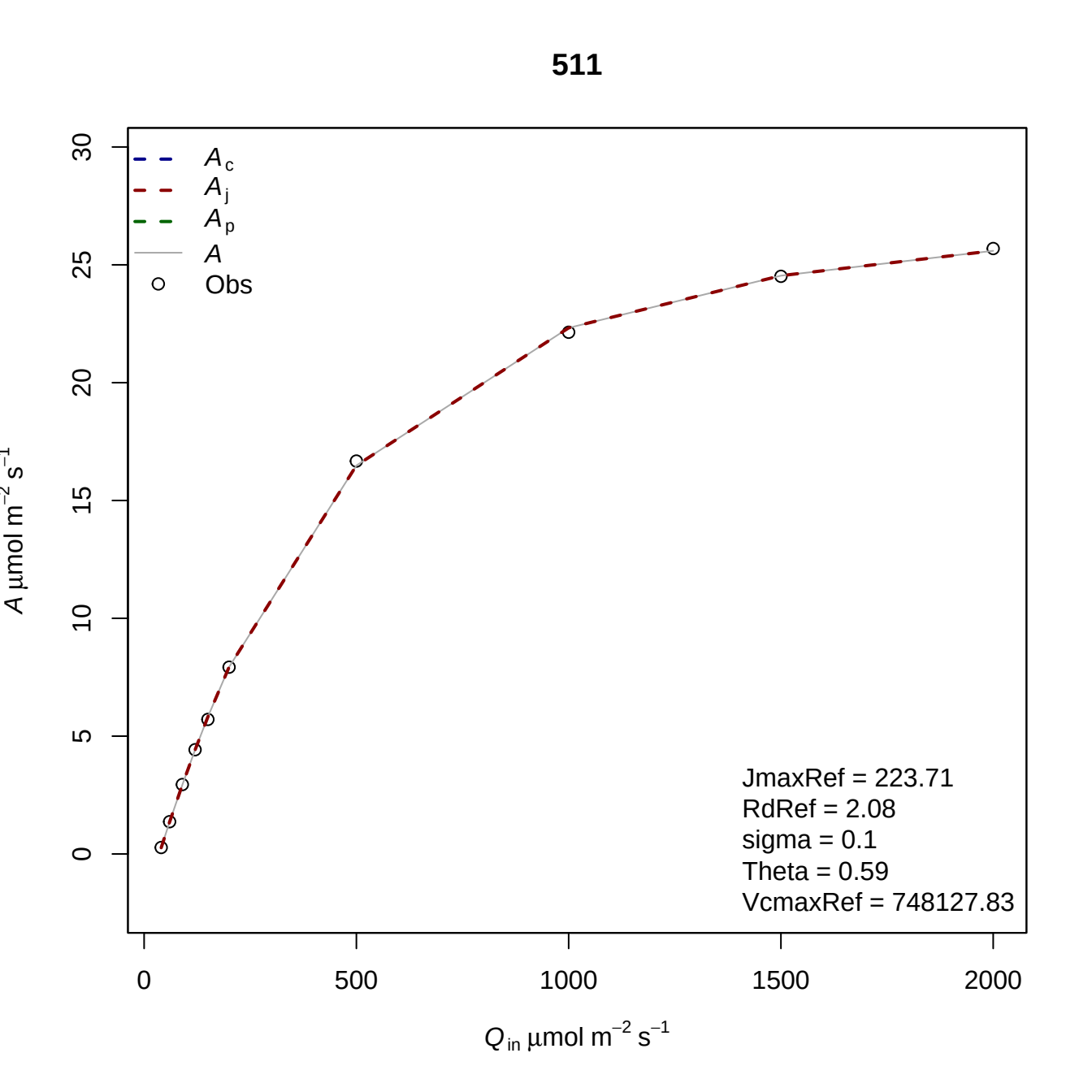

506
A_c
A_j
A_p
A
Obs
JmaxRef = 314.23
RdRef = 3.46
sigma = 0.19
Theta = 0.29
VcmaxRef = 277.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

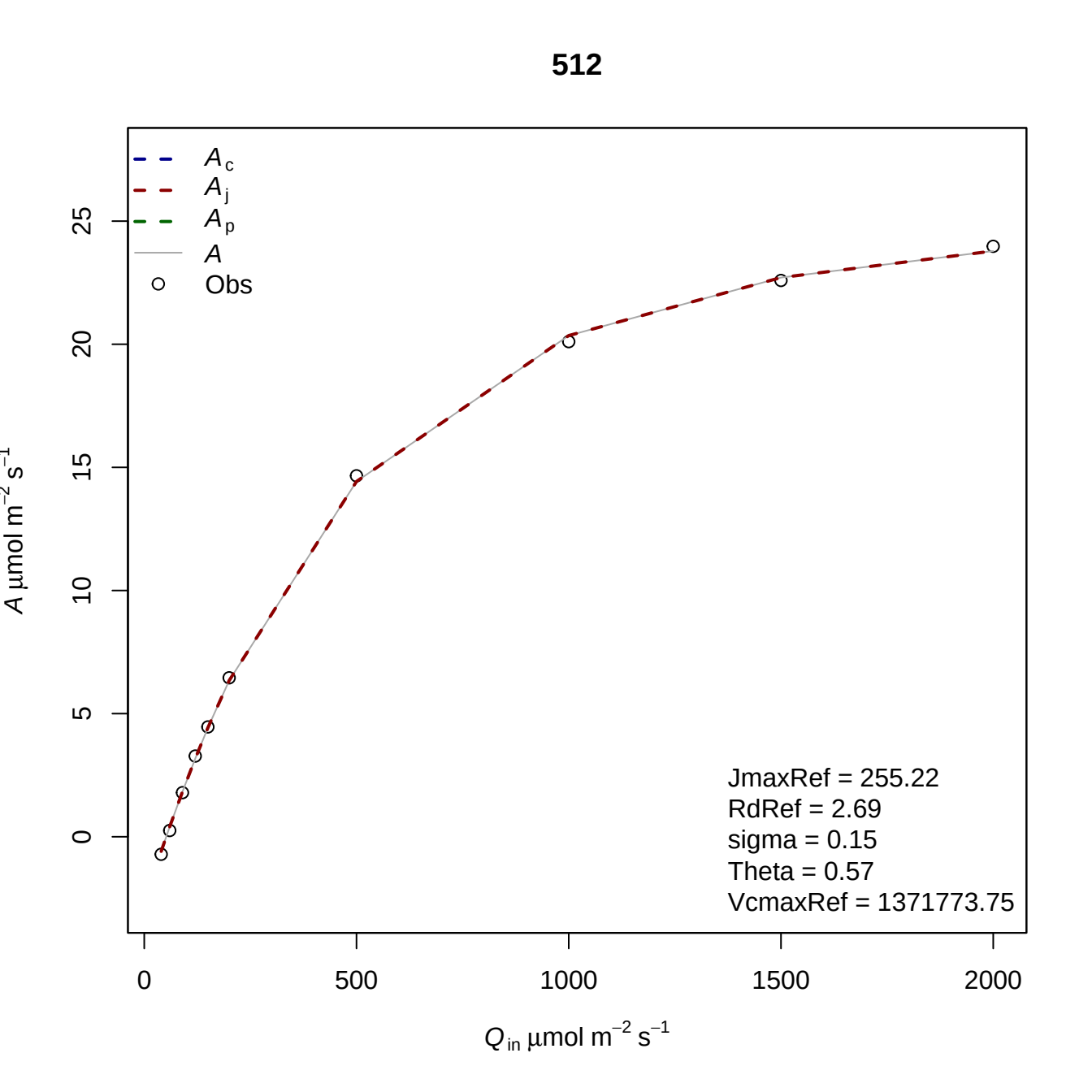

507
A_c
A_j
A_p
A
Obs
JmaxRef = 305.52
RdRef = 3.9
sigma = 0.04
Theta = 0.38
VcmaxRef = 456.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

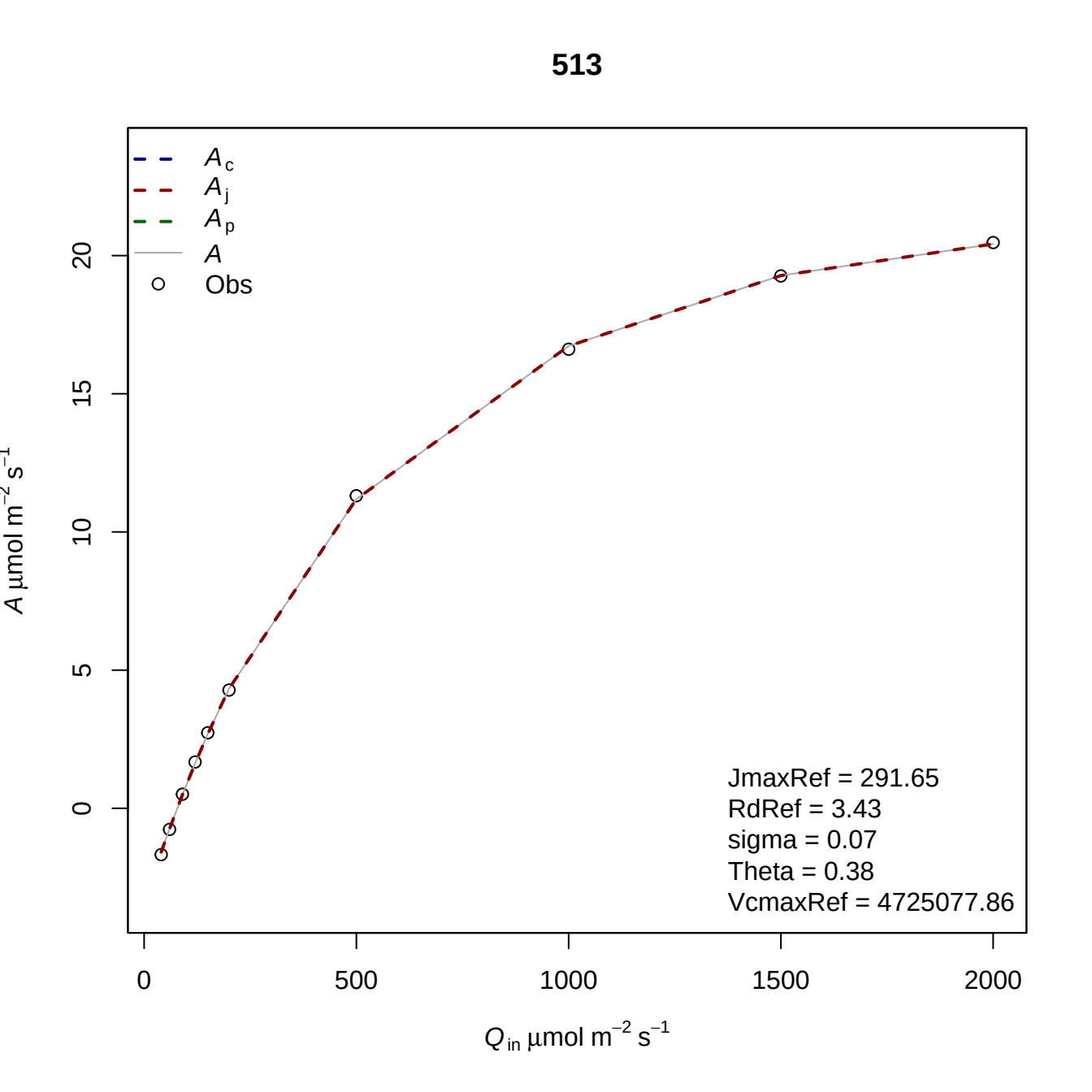

508
A_c
A_j
A_p
A
Obs
JmaxRef = 149.55
RdRef = 1.12
sigma = 0.15
Theta = 0.16
VcmaxRef = 74.37
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

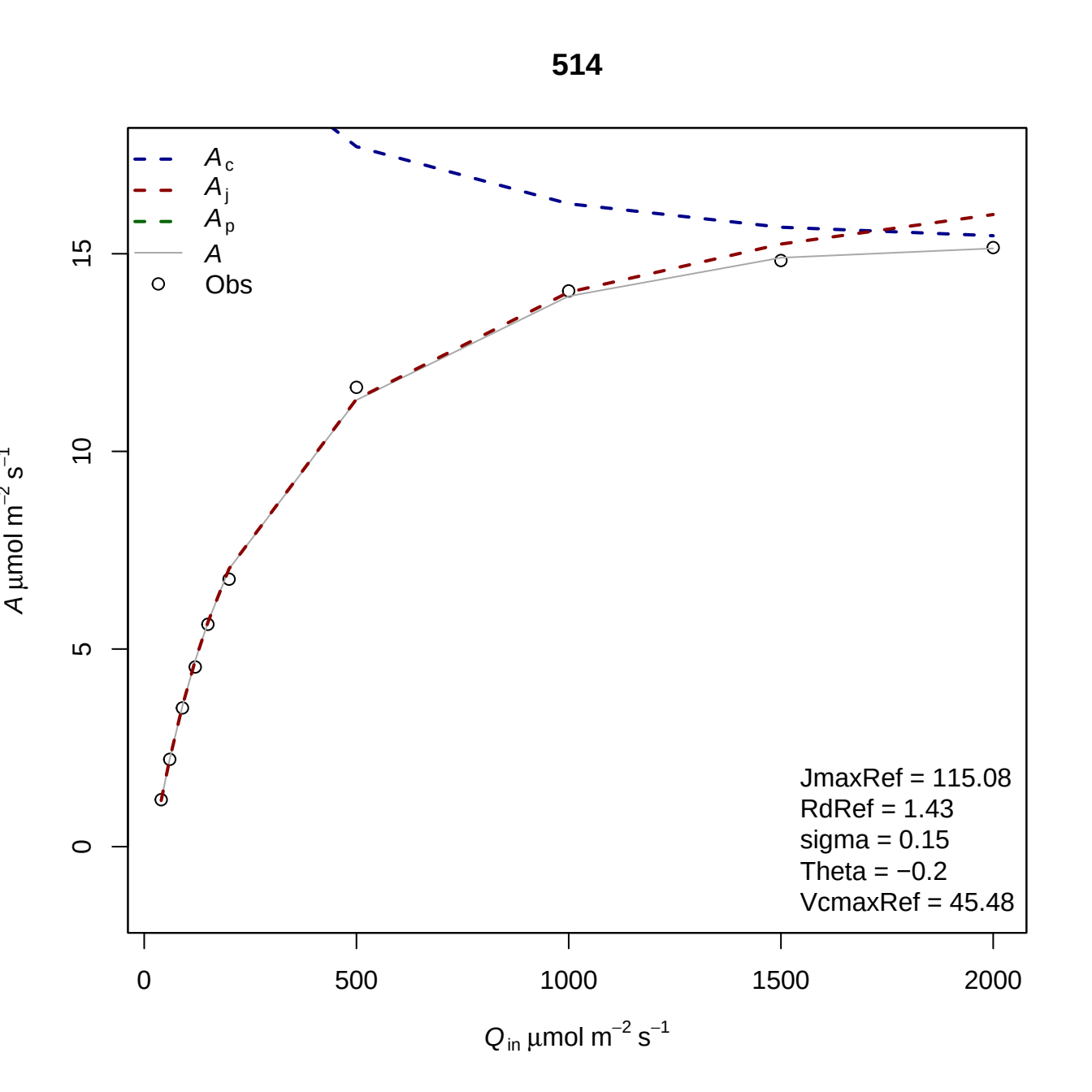

509
A_c
A_j
A_p
A
Obs
JmaxRef = 181.12
RdRef = 1.92
sigma = 0.3
Theta = 0.14
VcmaxRef = 84.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

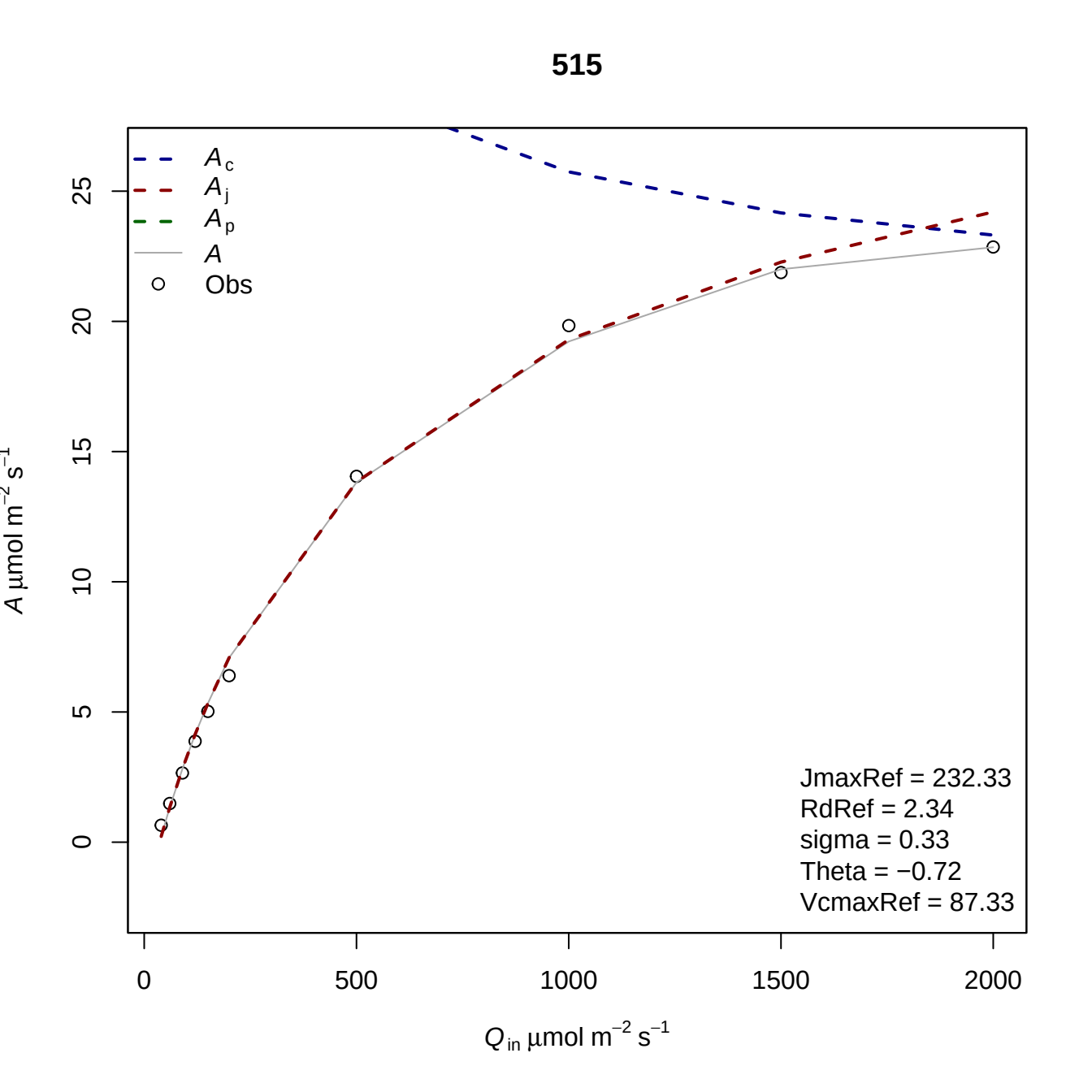

510
A_c
A_j
A_p
A
Obs
JmaxRef = 225.7
RdRef = 1.43
sigma = 0.08
Theta = 0.58
VcmaxRef = 161.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

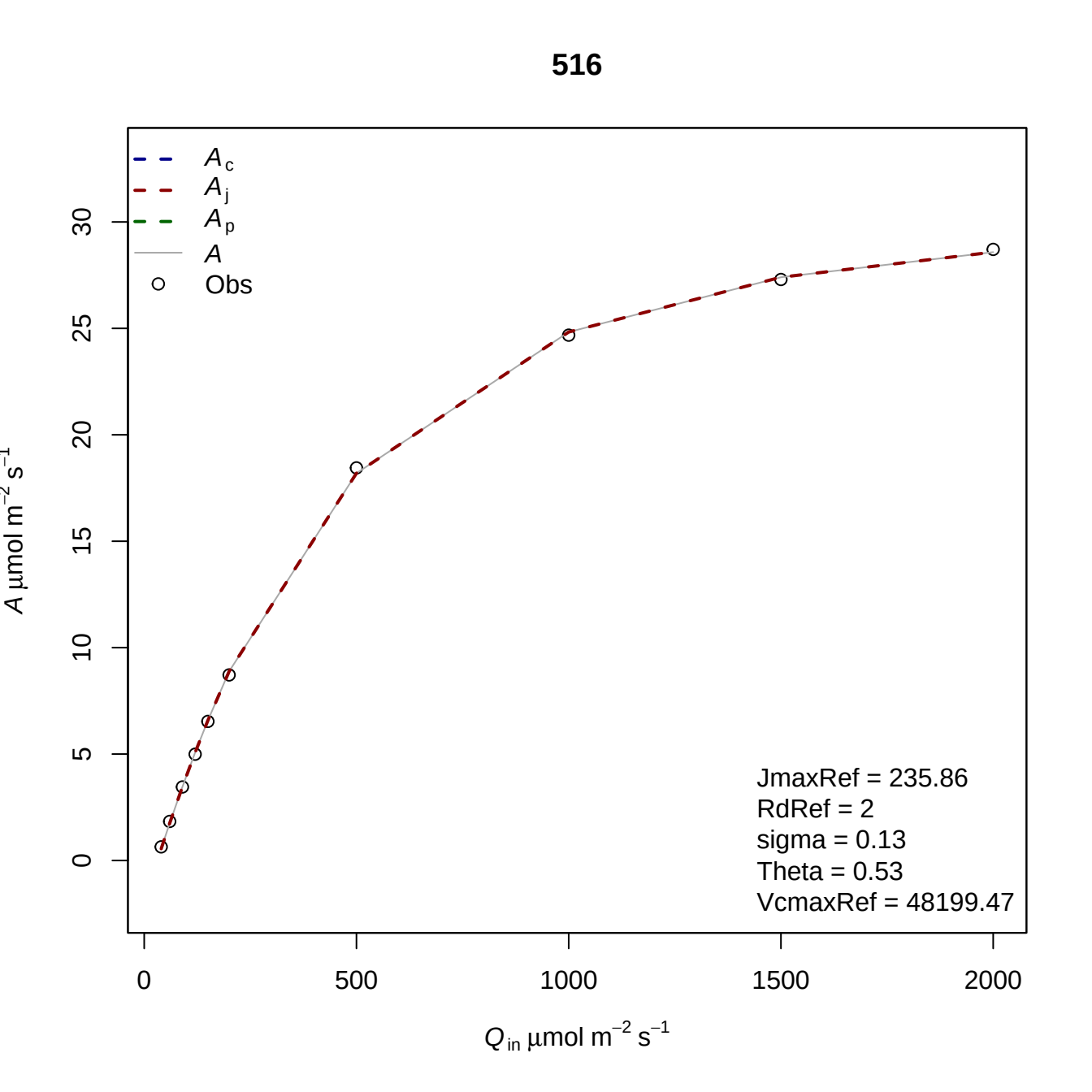

511
A_c
A_j
A_p
A
Obs
JmaxRef = 223.71
RdRef = 2.08
sigma = 0.1
Theta = 0.59
VcmaxRef = 748127.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

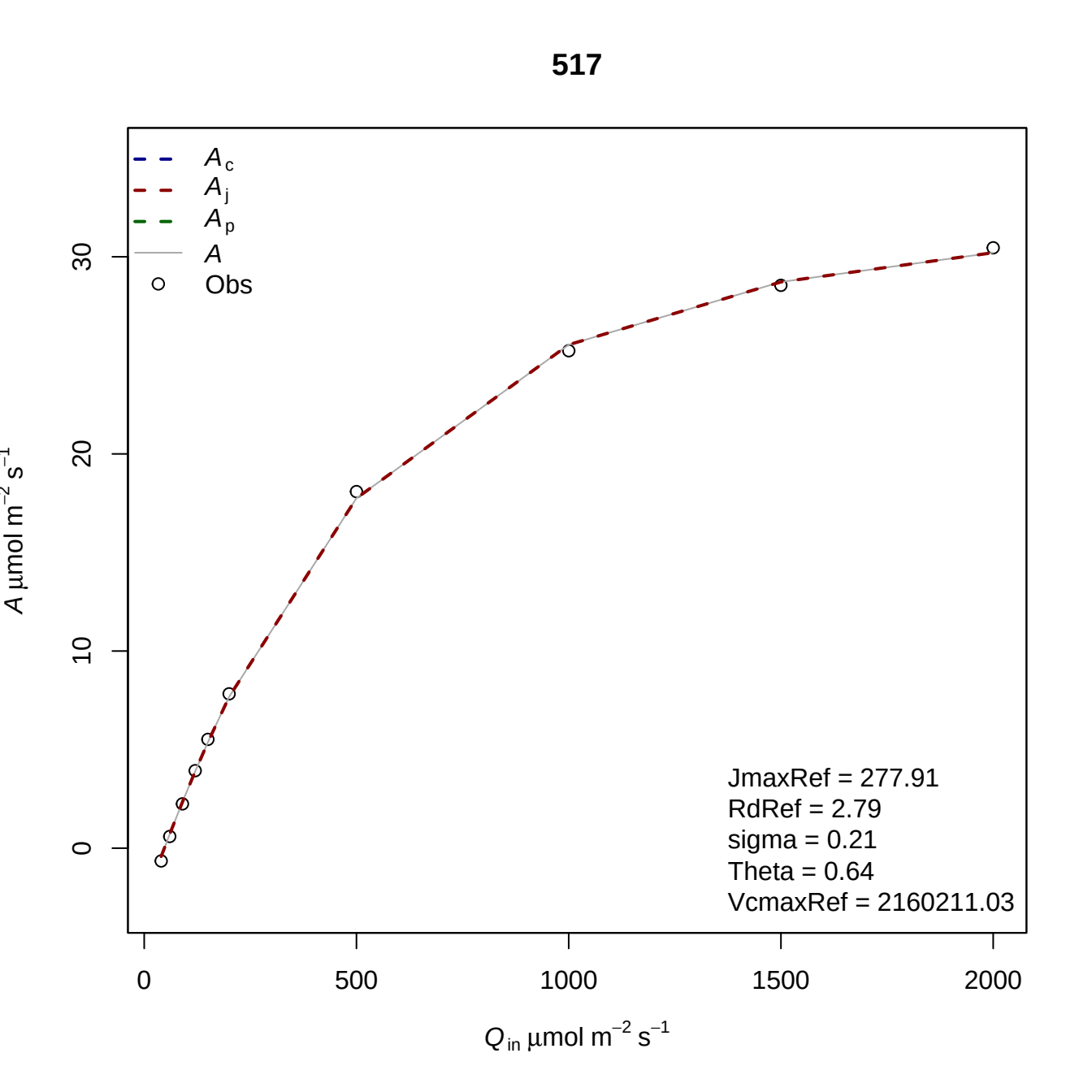

512
A_c
A_j
A_p
A
Obs
JmaxRef = 255.22
RdRef = 2.69
sigma = 0.15
Theta = 0.57
VcmaxRef = 1371773.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

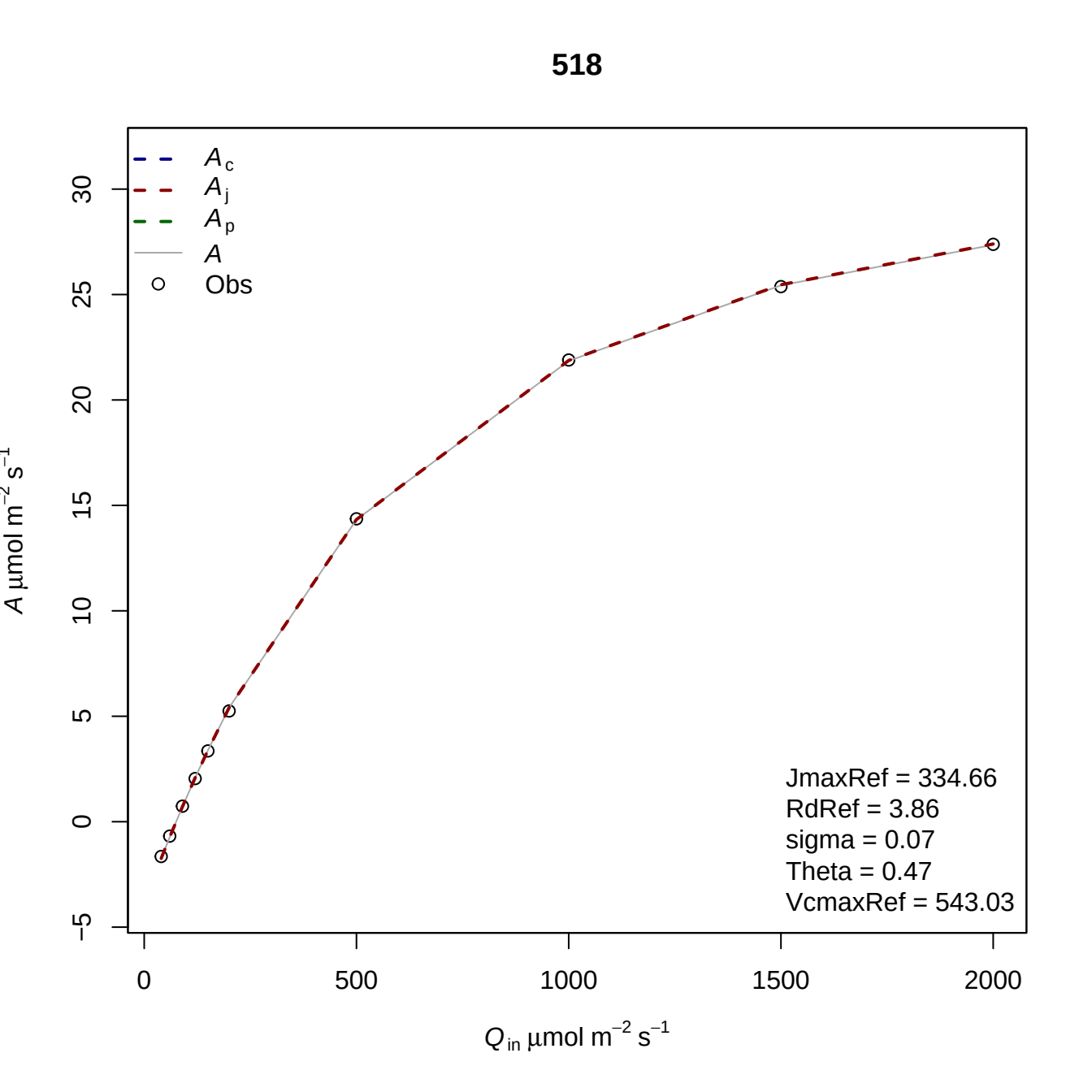

513
A_c
A_j
A_p
A
Obs
JmaxRef = 291.65
RdRef = 3.43
sigma = 0.07
Theta = 0.38
VcmaxRef = 4725077.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

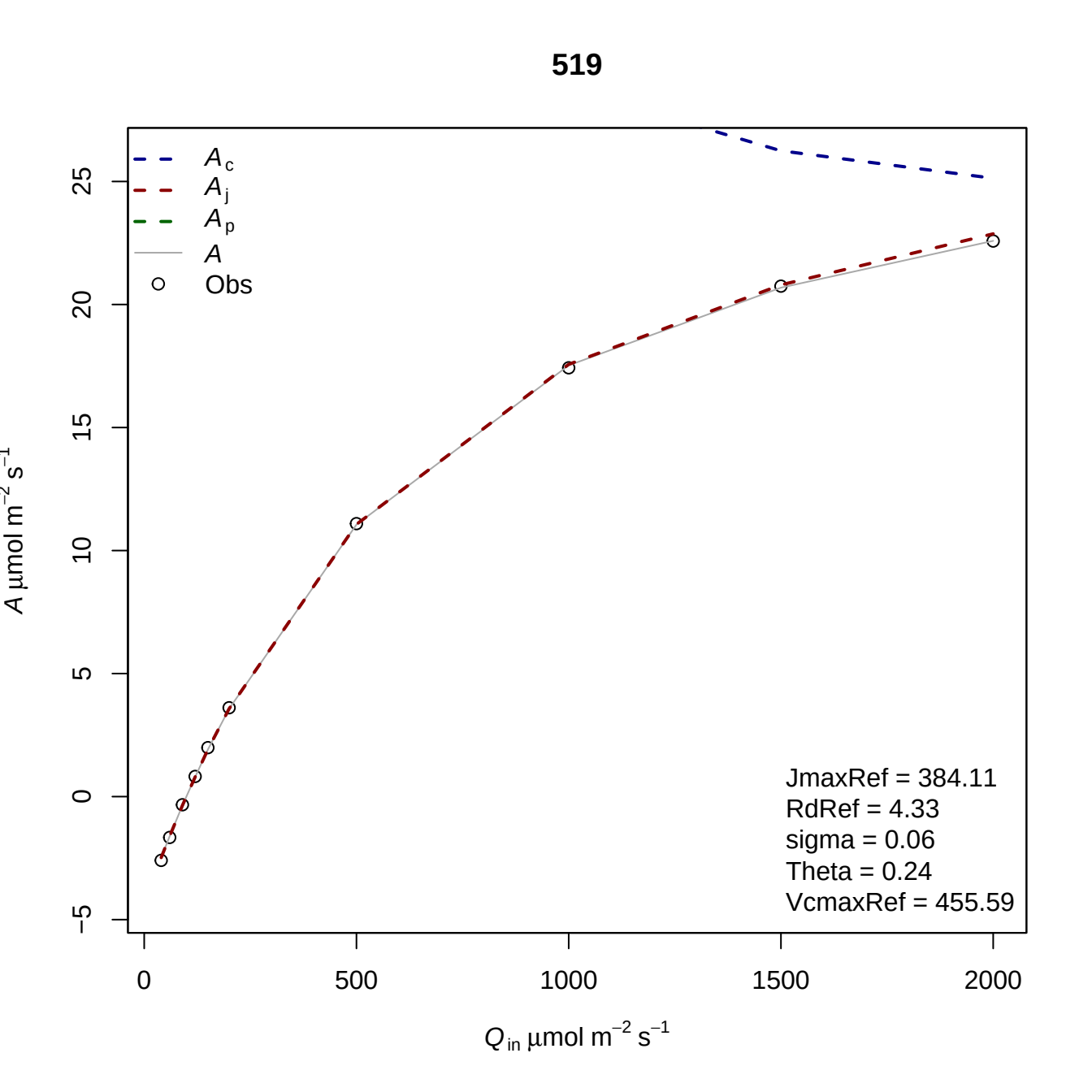

514
A_c
A_j
A_p
A
Obs
JmaxRef = 115.08
RdRef = 1.43
sigma = 0.15
Theta = −0.2
VcmaxRef = 45.48
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

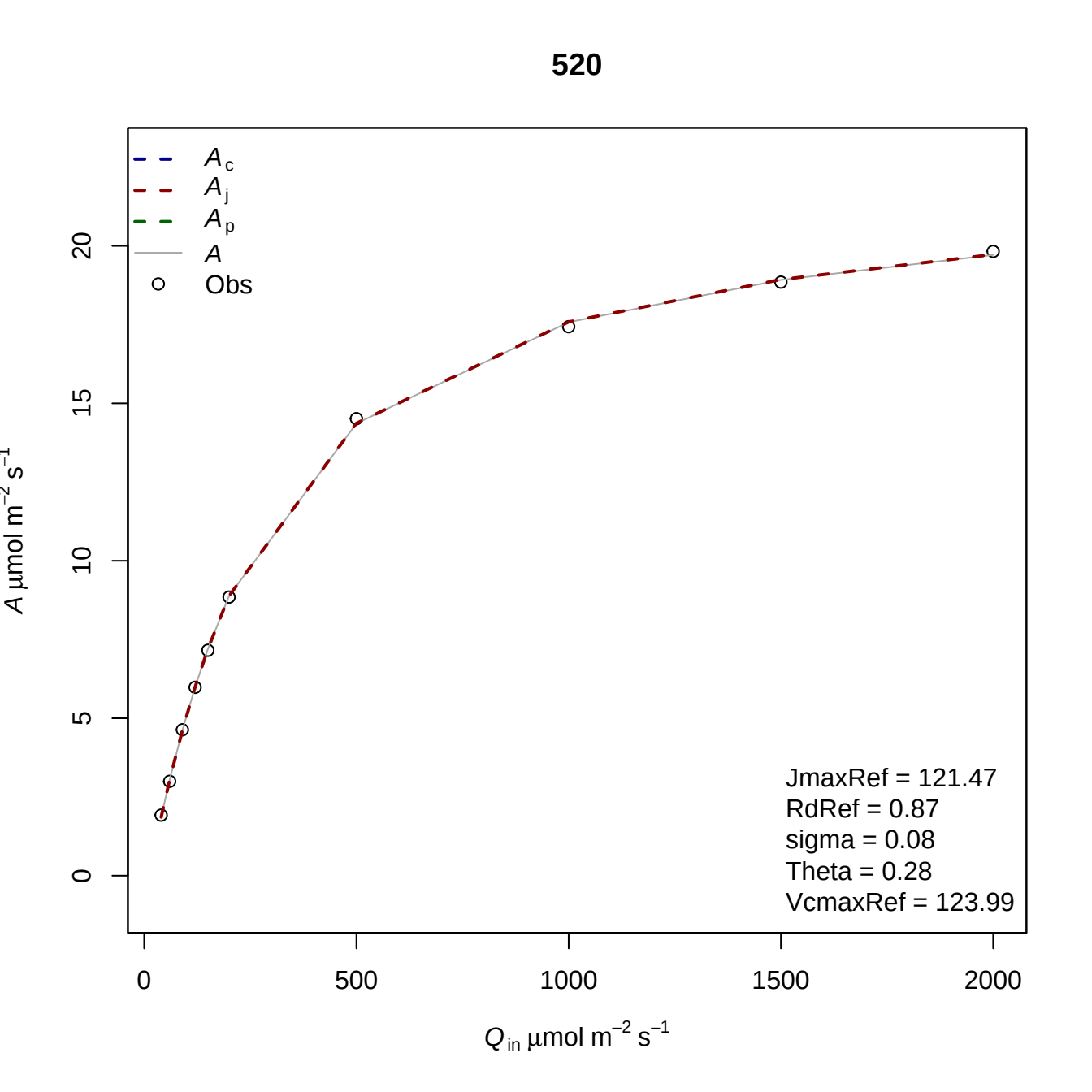

515
A_c
A_j
A_p
A
Obs
JmaxRef = 232.33
RdRef = 2.34
sigma = 0.33
Theta = −0.72
VcmaxRef = 87.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

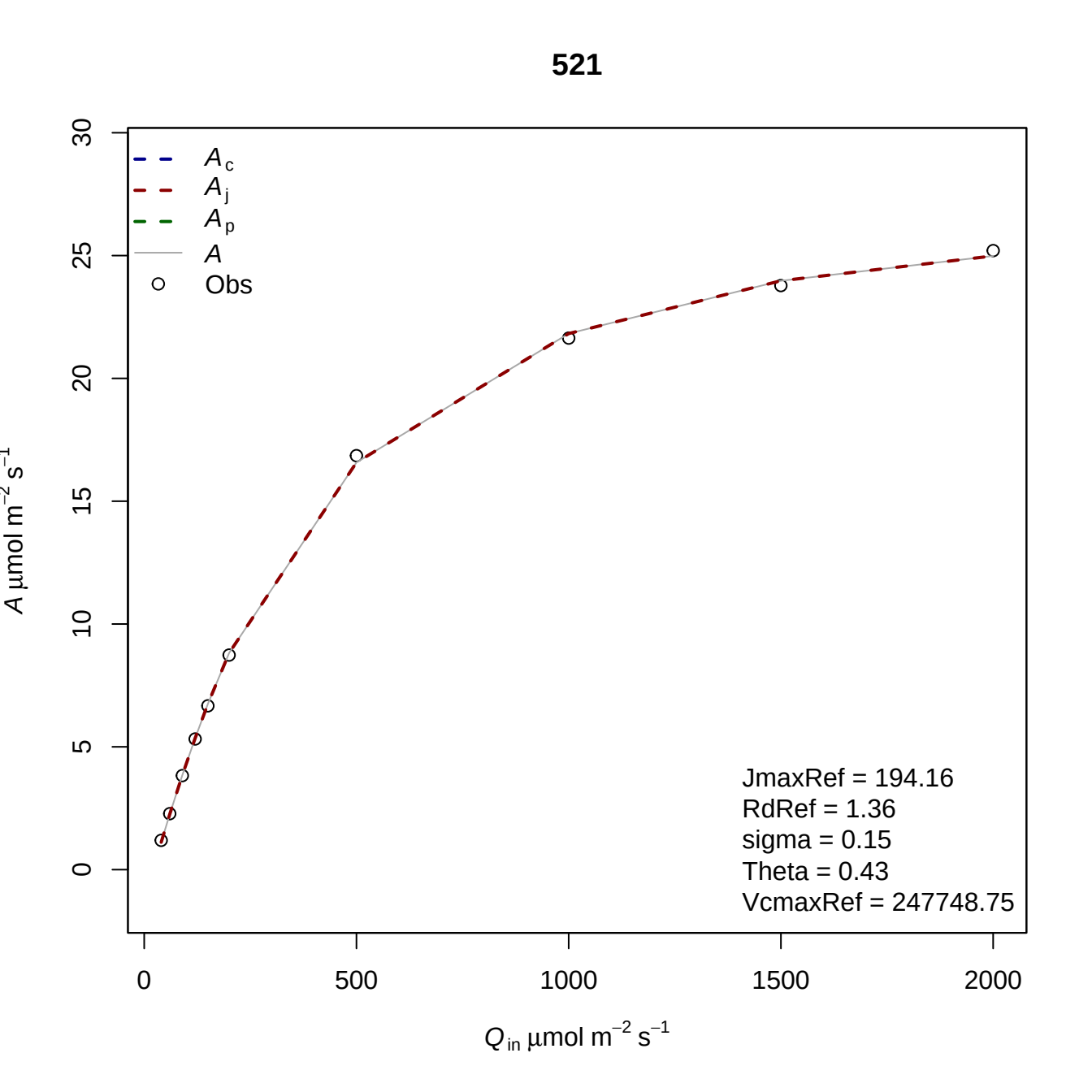

516
A_c
A_j
A_p
A
Obs
JmaxRef = 235.86
RdRef = 2
sigma = 0.13
Theta = 0.53
VcmaxRef = 48199.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

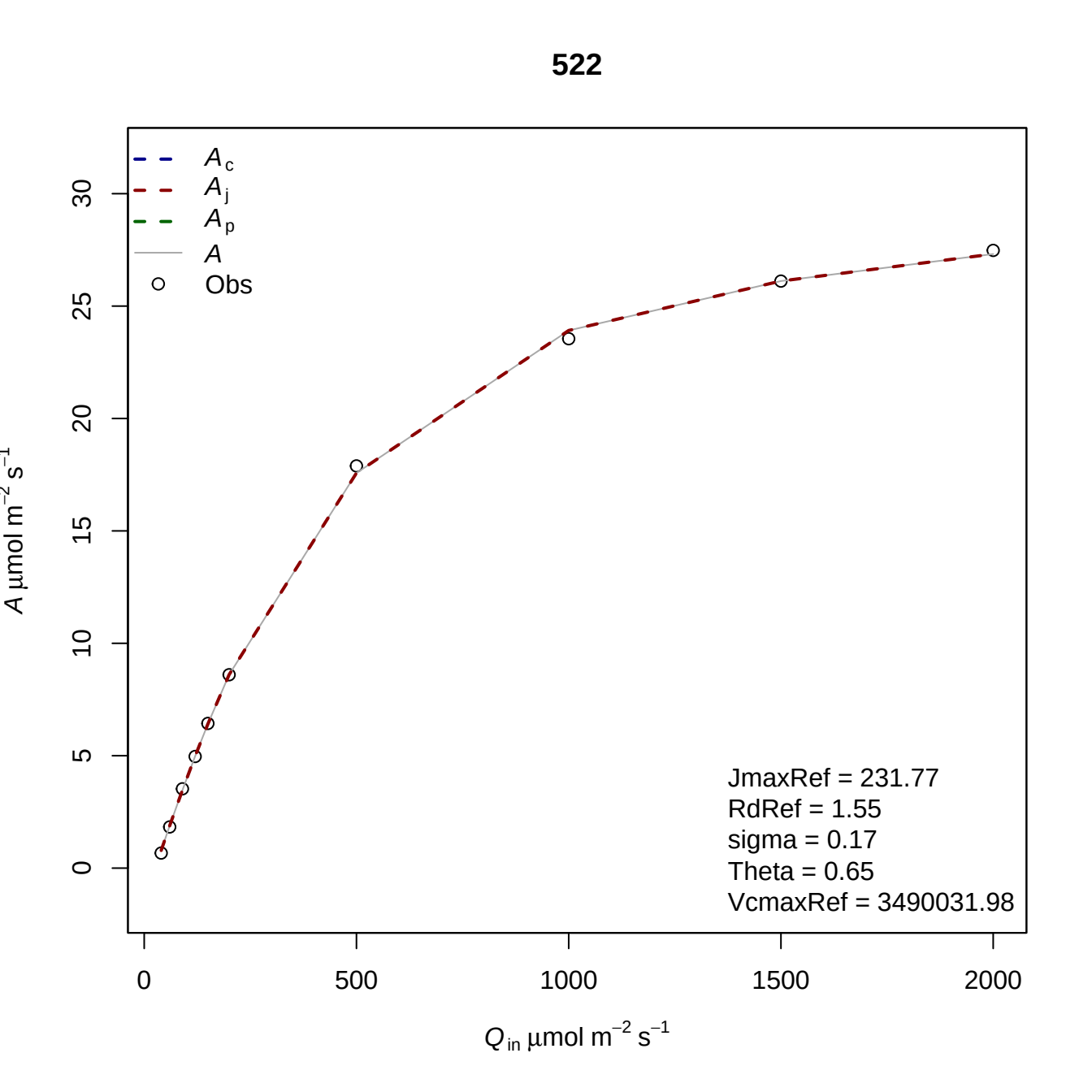

517
A_c
A_j
A_p
A
Obs
JmaxRef = 277.91
RdRef = 2.79
sigma = 0.21
Theta = 0.64
VcmaxRef = 2160211.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

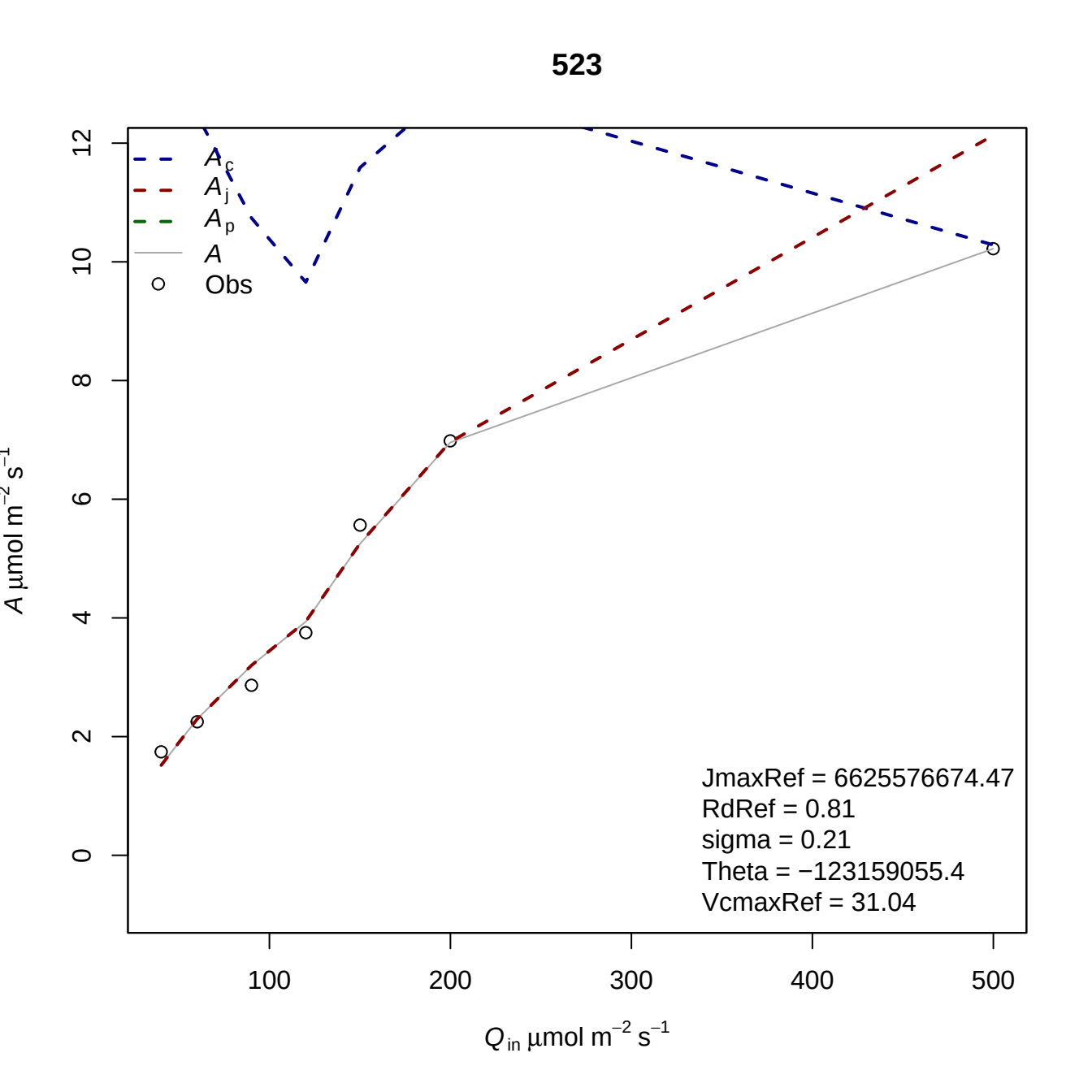

518
A_c
A_j
A_p
A
Obs
JmaxRef = 334.66
RdRef = 3.86
sigma = 0.07
Theta = 0.47
VcmaxRef = 543.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

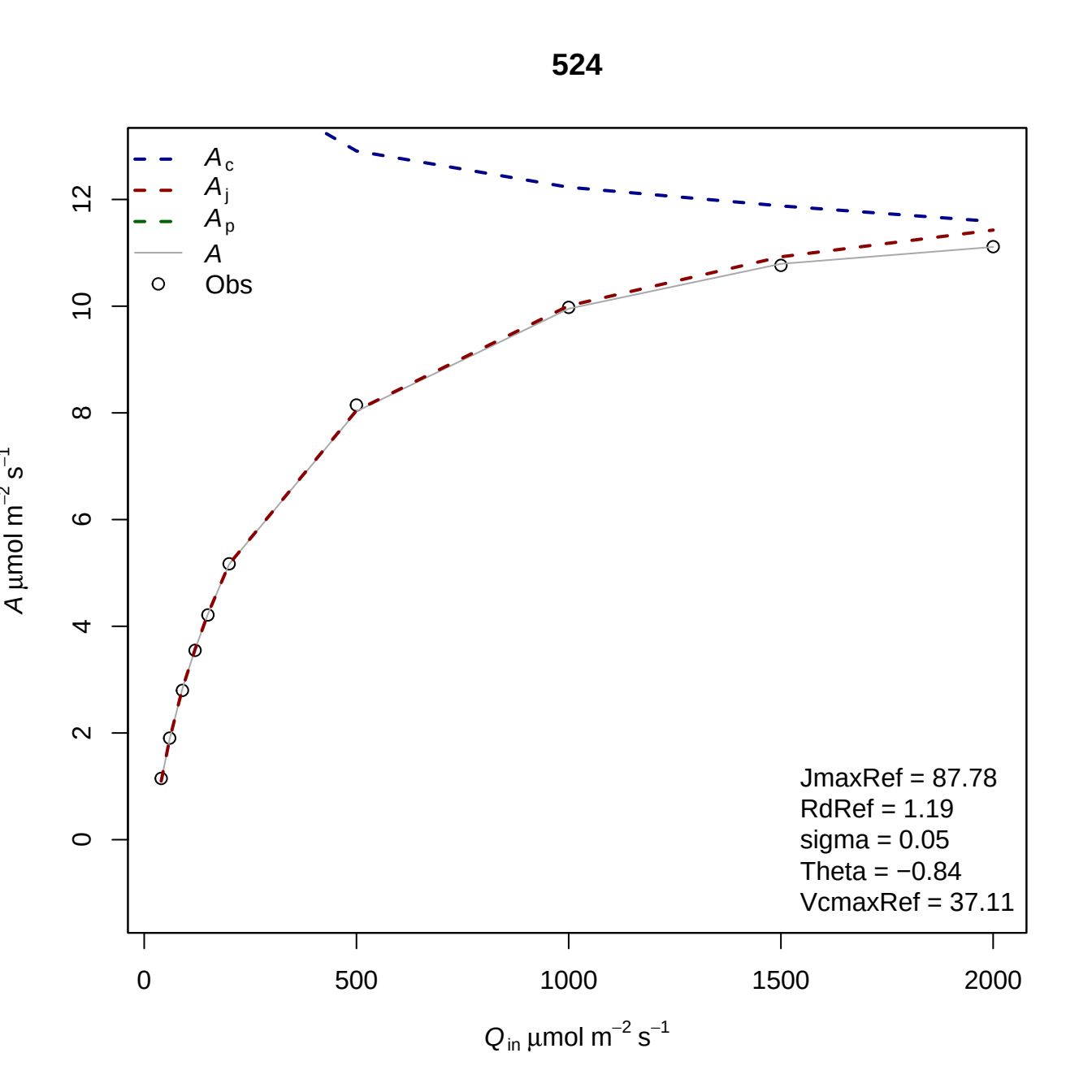

519
A_c
A_j
A_p
A
Obs
JmaxRef = 384.11
RdRef = 4.33
sigma = 0.06
Theta = 0.24
VcmaxRef = 455.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

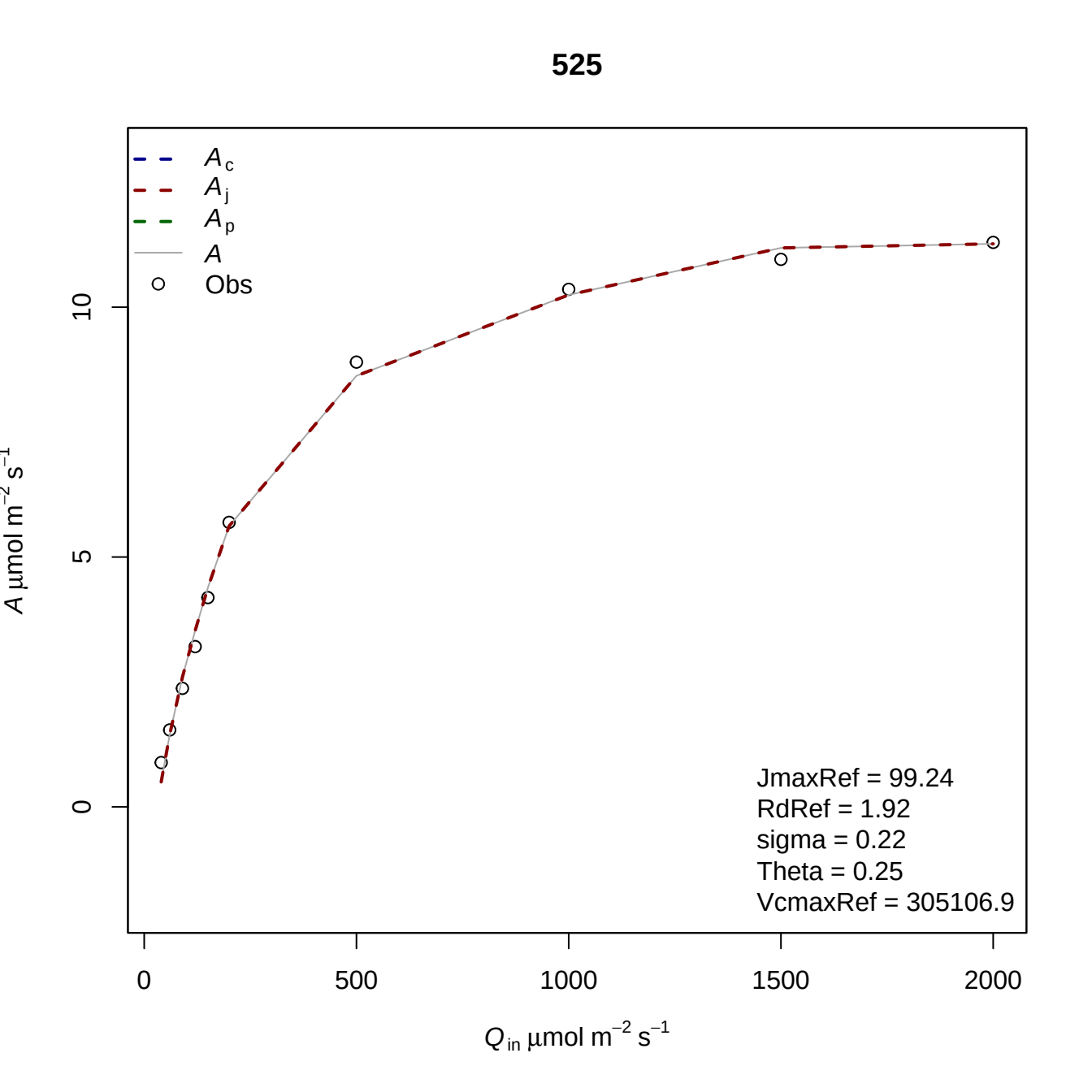

520
A_c
A_j
A_p
A
Obs
JmaxRef = 121.47
RdRef = 0.87
sigma = 0.08
Theta = 0.28
VcmaxRef = 123.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

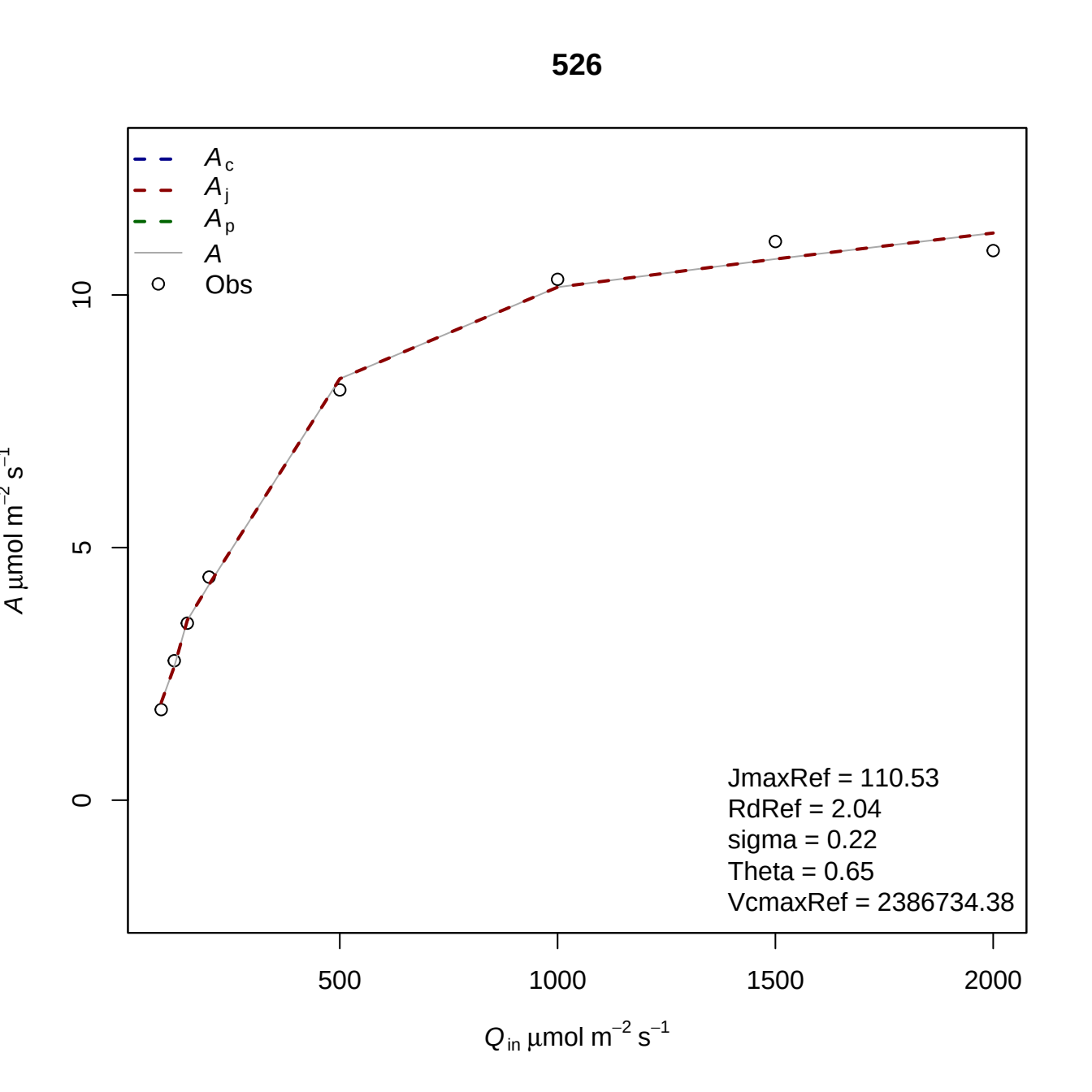

521
A_c
A_j
A_p
A
Obs
JmaxRef = 194.16
RdRef = 1.36
sigma = 0.15
Theta = 0.43
VcmaxRef = 247748.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

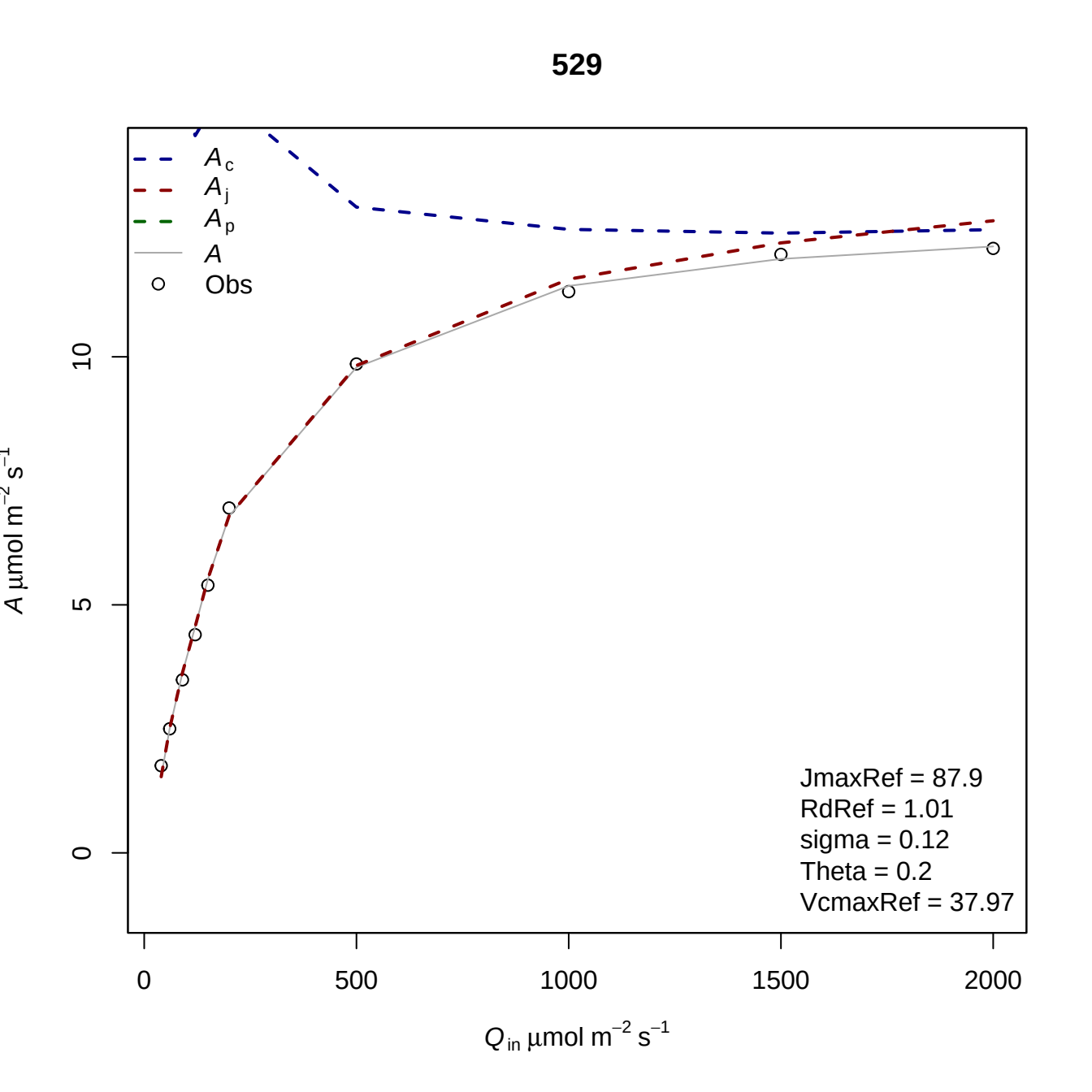

522
A_c
A_j
A_p
A
Obs
JmaxRef = 231.77
RdRef = 1.55
sigma = 0.17
Theta = 0.65
VcmaxRef = 3490031.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

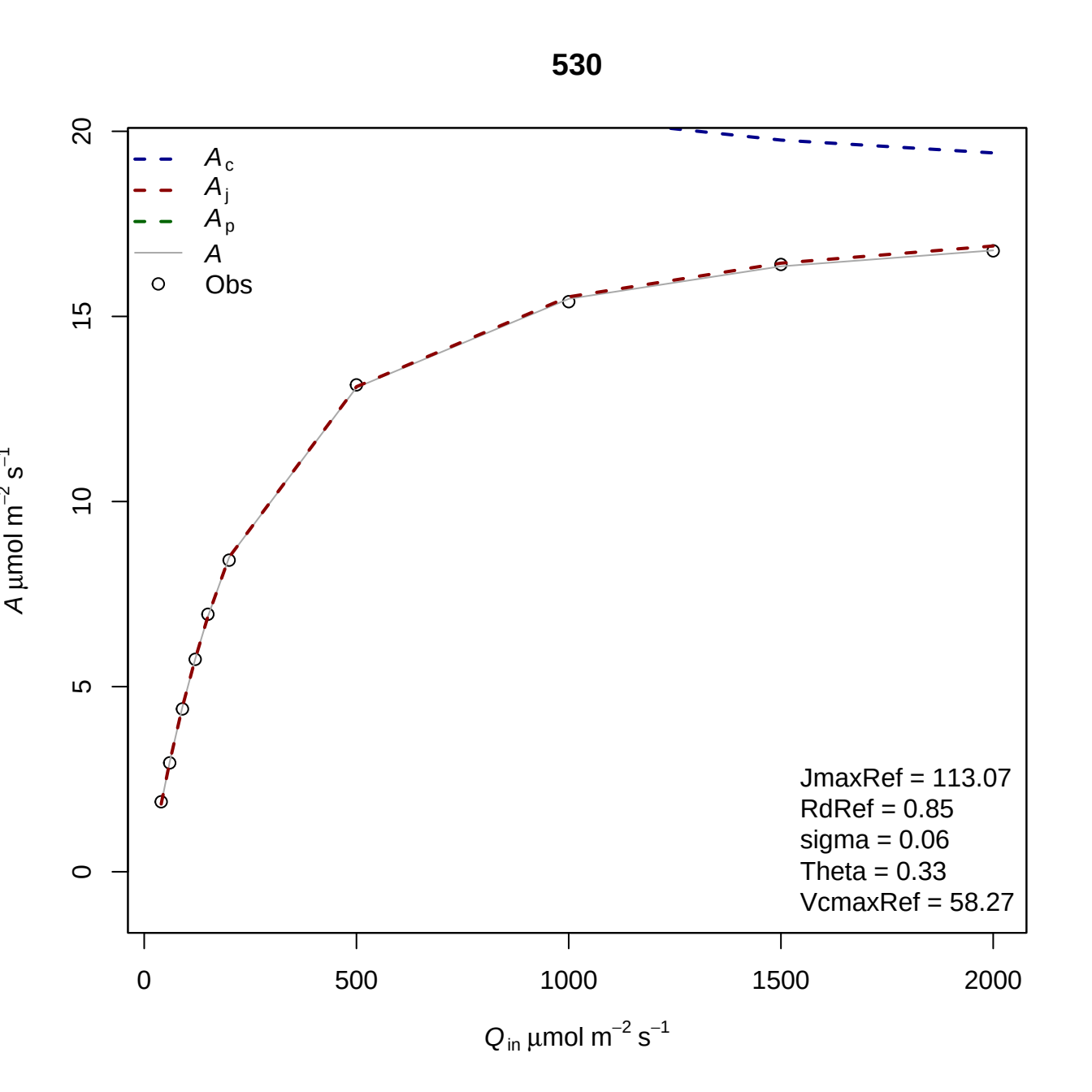

523
A_c
A_j
A_p
A
Obs
JmaxRef = 6625576674.47
RdRef = 0.81
sigma = 0.21
Theta = −123159055.4
VcmaxRef = 31.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
100
200
300
400
500

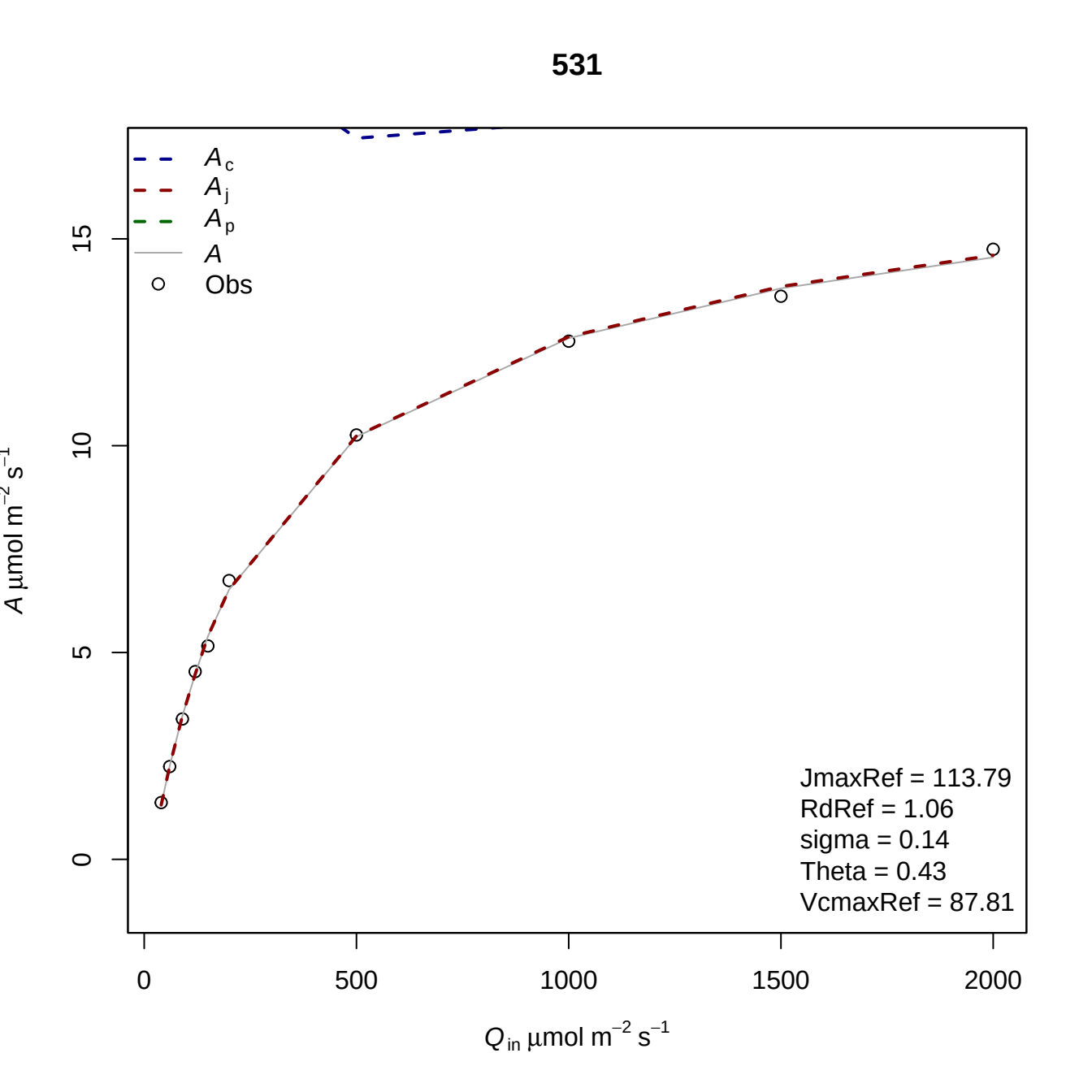

524
A_c
A_j
A_p
A
Obs
JmaxRef = 87.78
RdRef = 1.19
sigma = 0.05
Theta = −0.84
VcmaxRef = 37.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

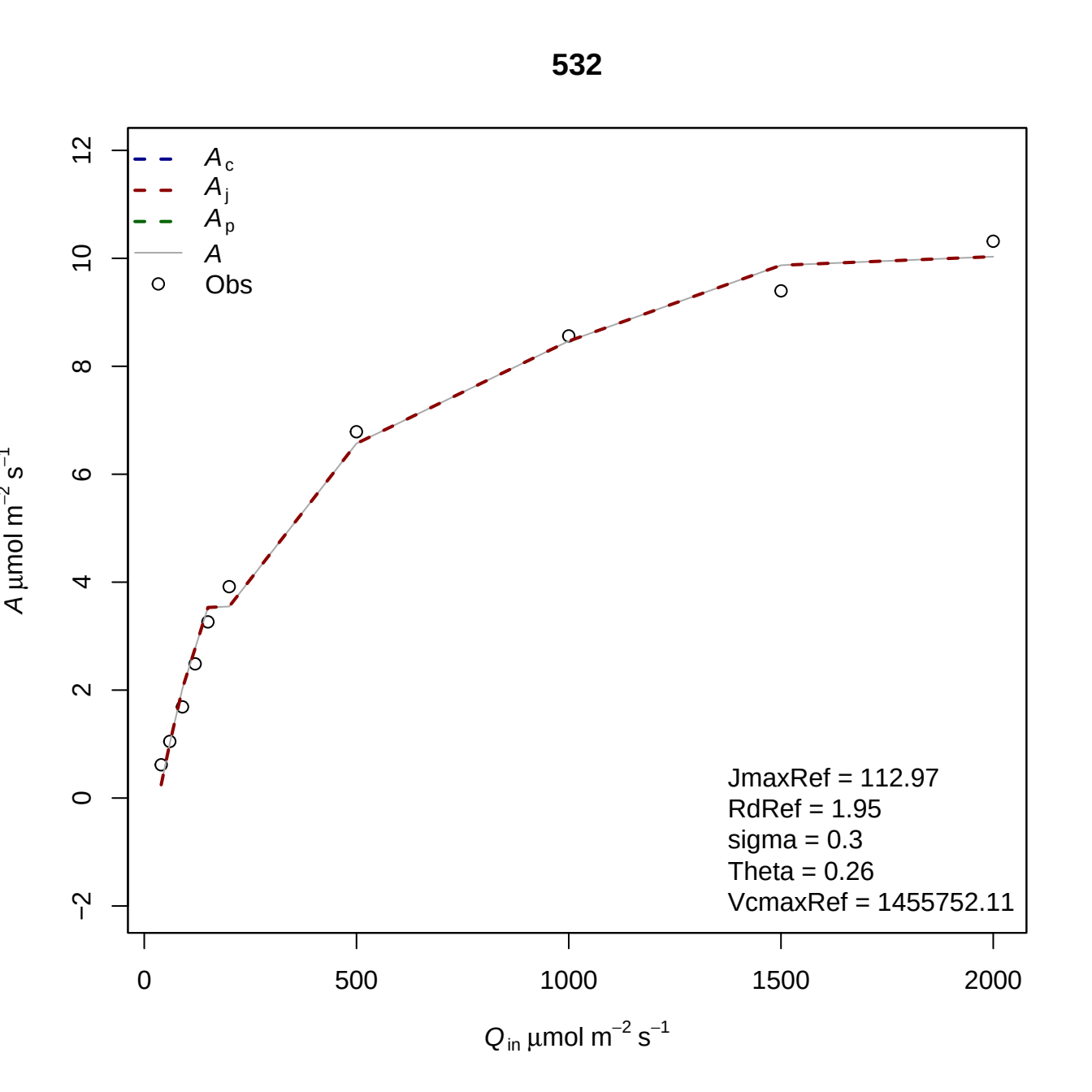

525
A_c
A_j
A_p
A
Obs
JmaxRef = 99.24
RdRef = 1.92
sigma = 0.22
Theta = 0.25
VcmaxRef = 305106.9
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

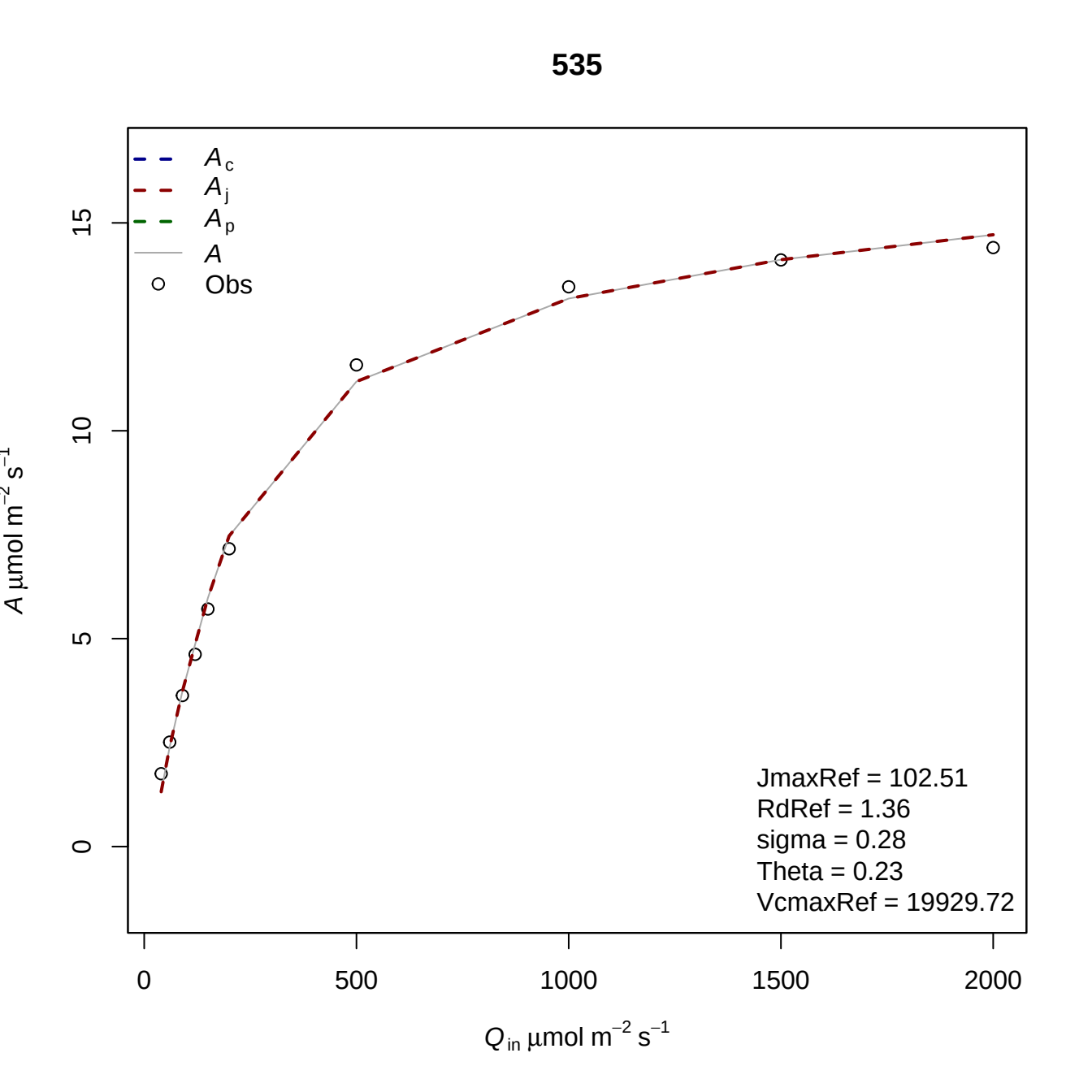

526
A_c
A_j
A_p
A
Obs
JmaxRef = 110.53
RdRef = 2.04
sigma = 0.22
Theta = 0.65
VcmaxRef = 2386734.38
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

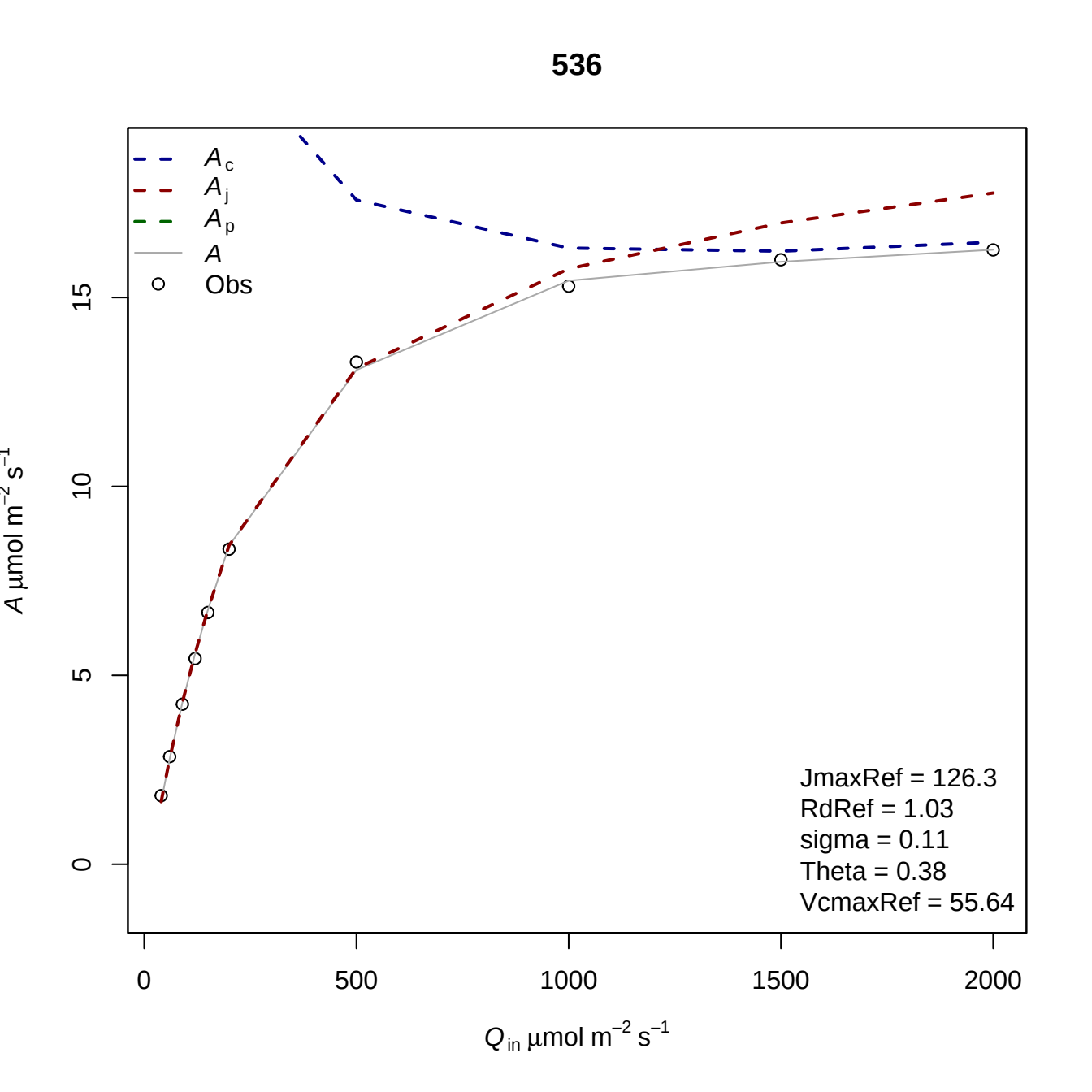

529
A_c
A_j
A_p
A
Obs
JmaxRef = 87.9
RdRef = 1.01
sigma = 0.12
Theta = 0.2
VcmaxRef = 37.97
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

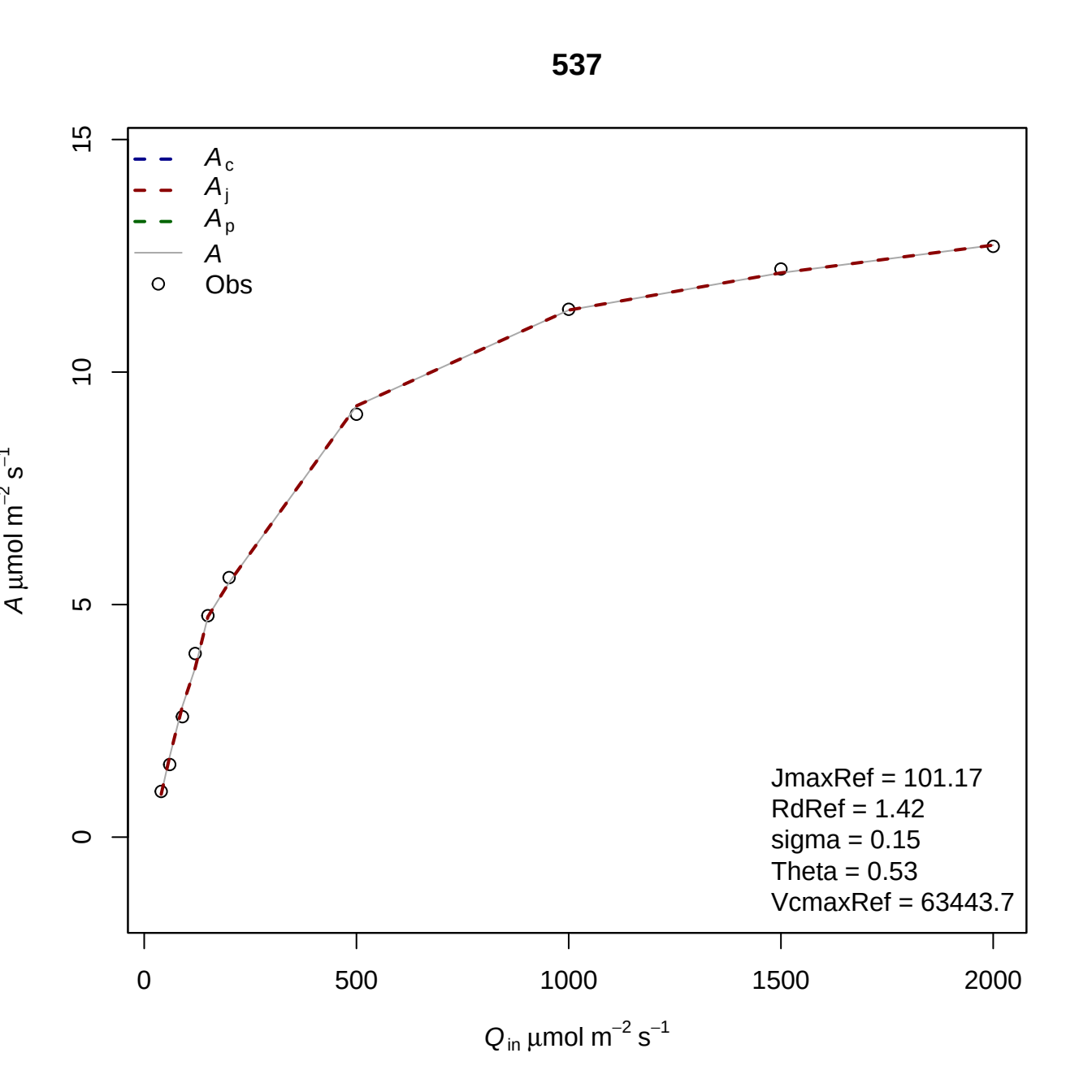

530
A_c
A_j
A_p
A
Obs
JmaxRef = 113.07
RdRef = 0.85
sigma = 0.06
Theta = 0.33
VcmaxRef = 58.27
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

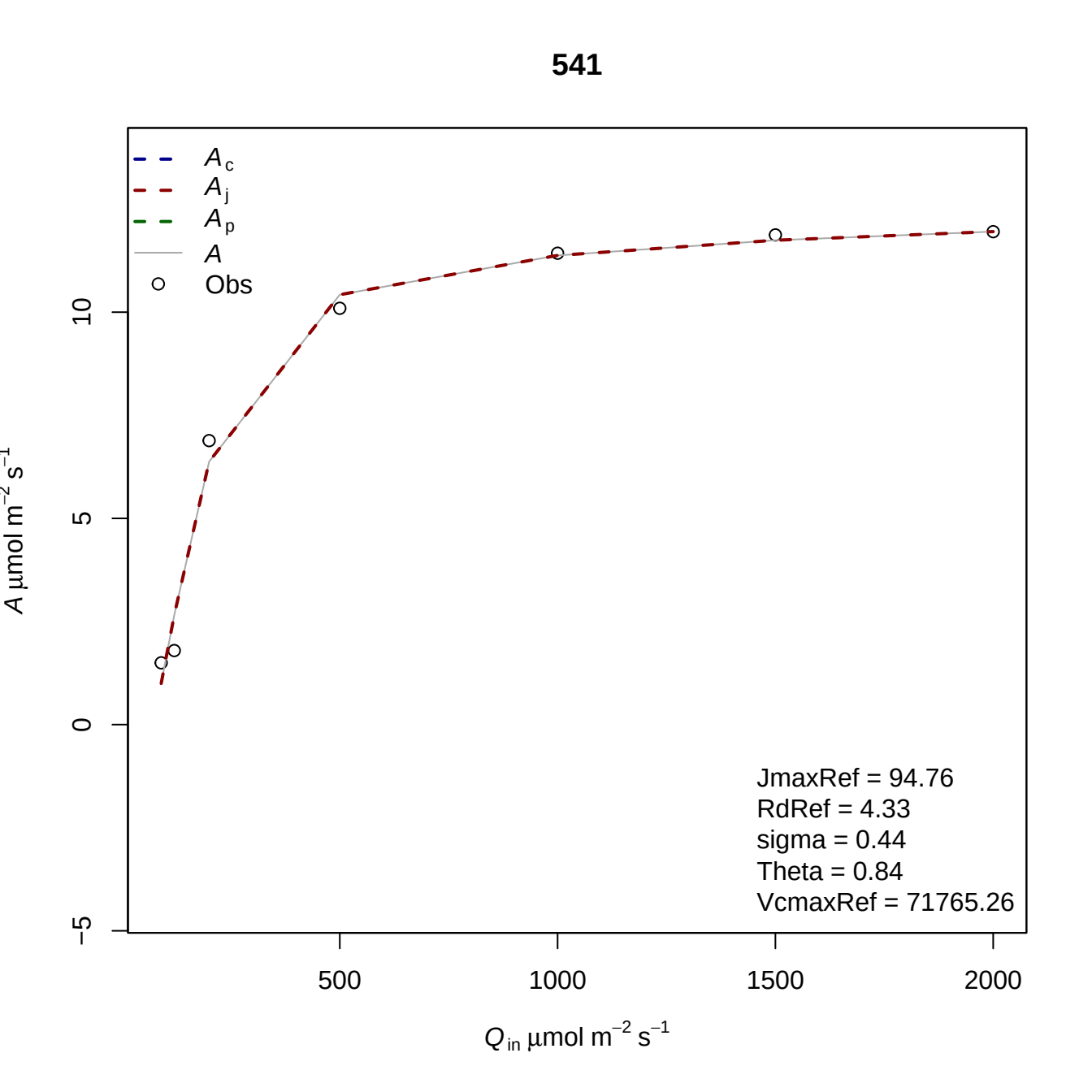

531
A_c
A_j
A_p
A
Obs
JmaxRef = 113.79
RdRef = 1.06
sigma = 0.14
Theta = 0.43
VcmaxRef = 87.81
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

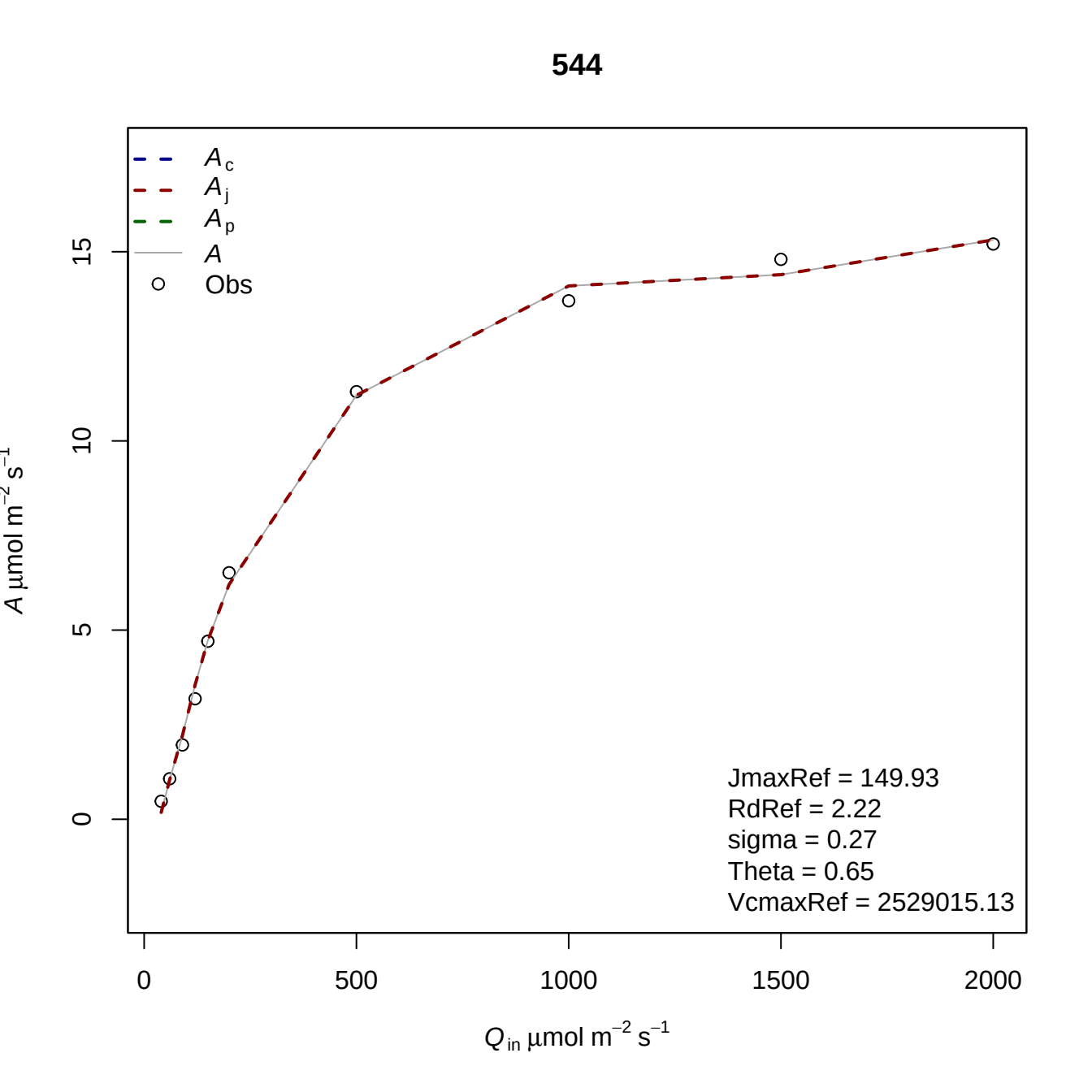

532
A_c
A_j
A_p
A
Obs
JmaxRef = 112.97
RdRef = 1.95
sigma = 0.3
Theta = 0.26
VcmaxRef = 1455752.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

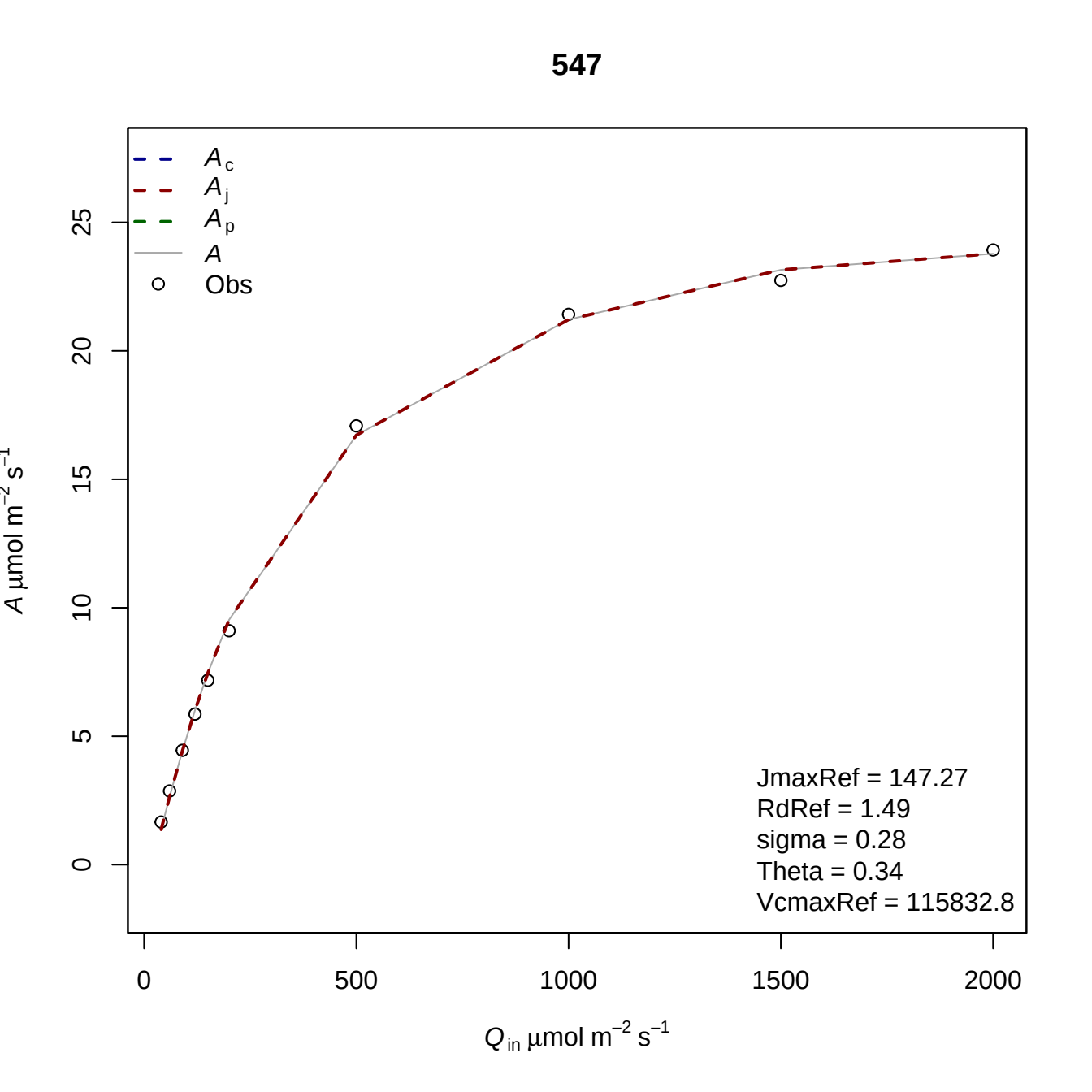

535
A_c
A_j
A_p
A
Obs
JmaxRef = 102.51
RdRef = 1.36
sigma = 0.28
Theta = 0.23
VcmaxRef = 19929.72
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

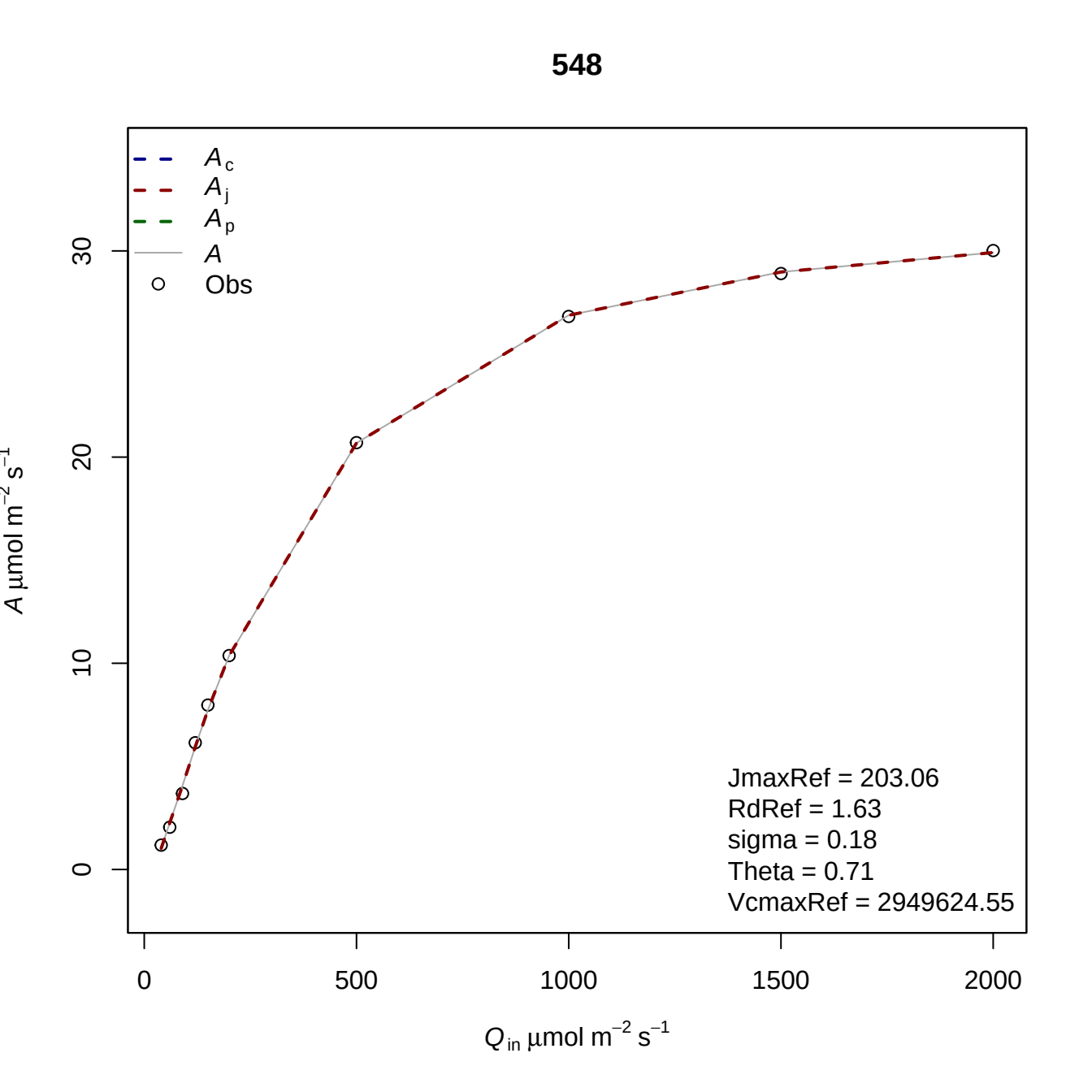

536
A_c
A_j
A_p
A
Obs
JmaxRef = 126.3
RdRef = 1.03
sigma = 0.11
Theta = 0.38
VcmaxRef = 55.64
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

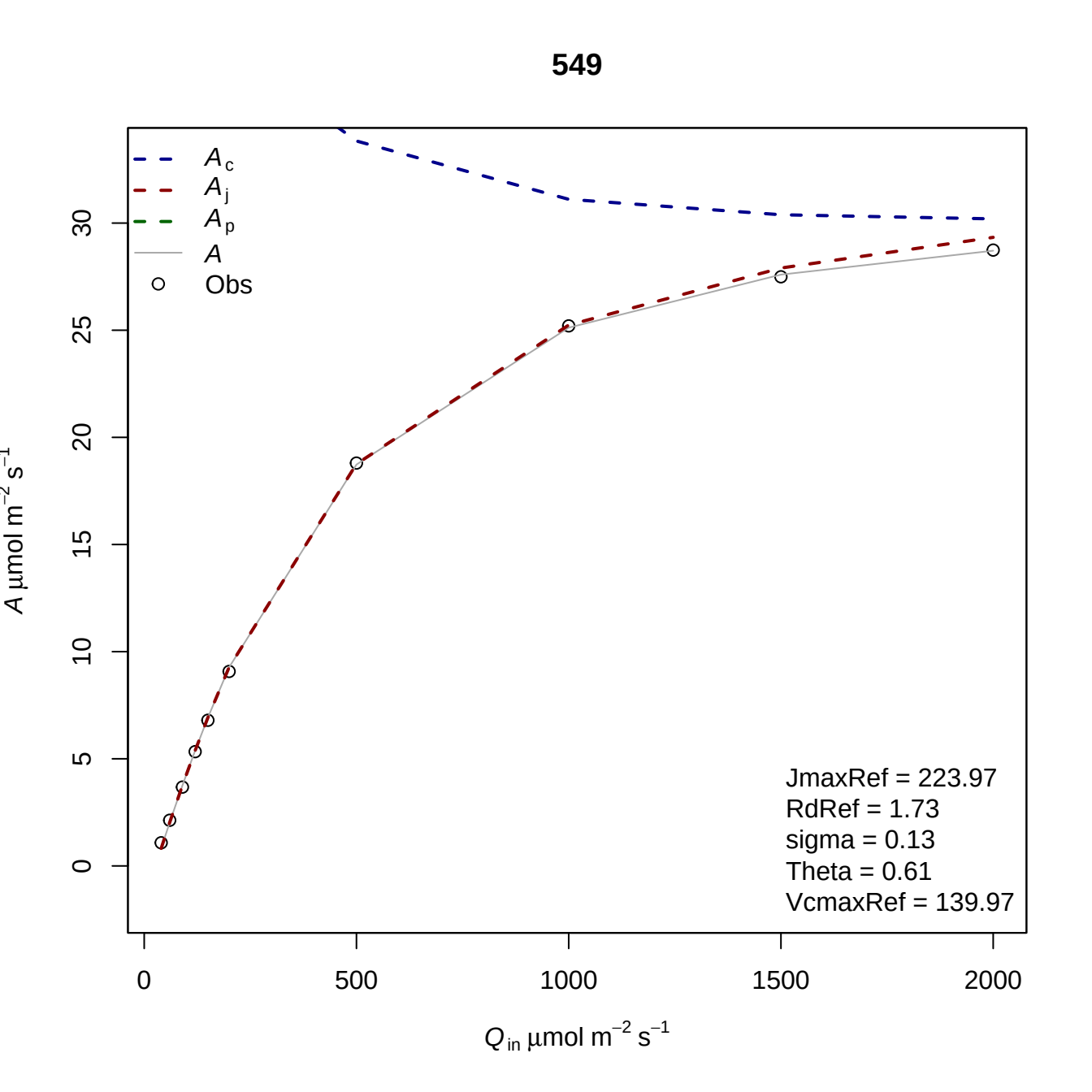

537
A_c
A_j
A_p
A
Obs
JmaxRef = 101.17
RdRef = 1.42
sigma = 0.15
Theta = 0.53
VcmaxRef = 63443.7
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

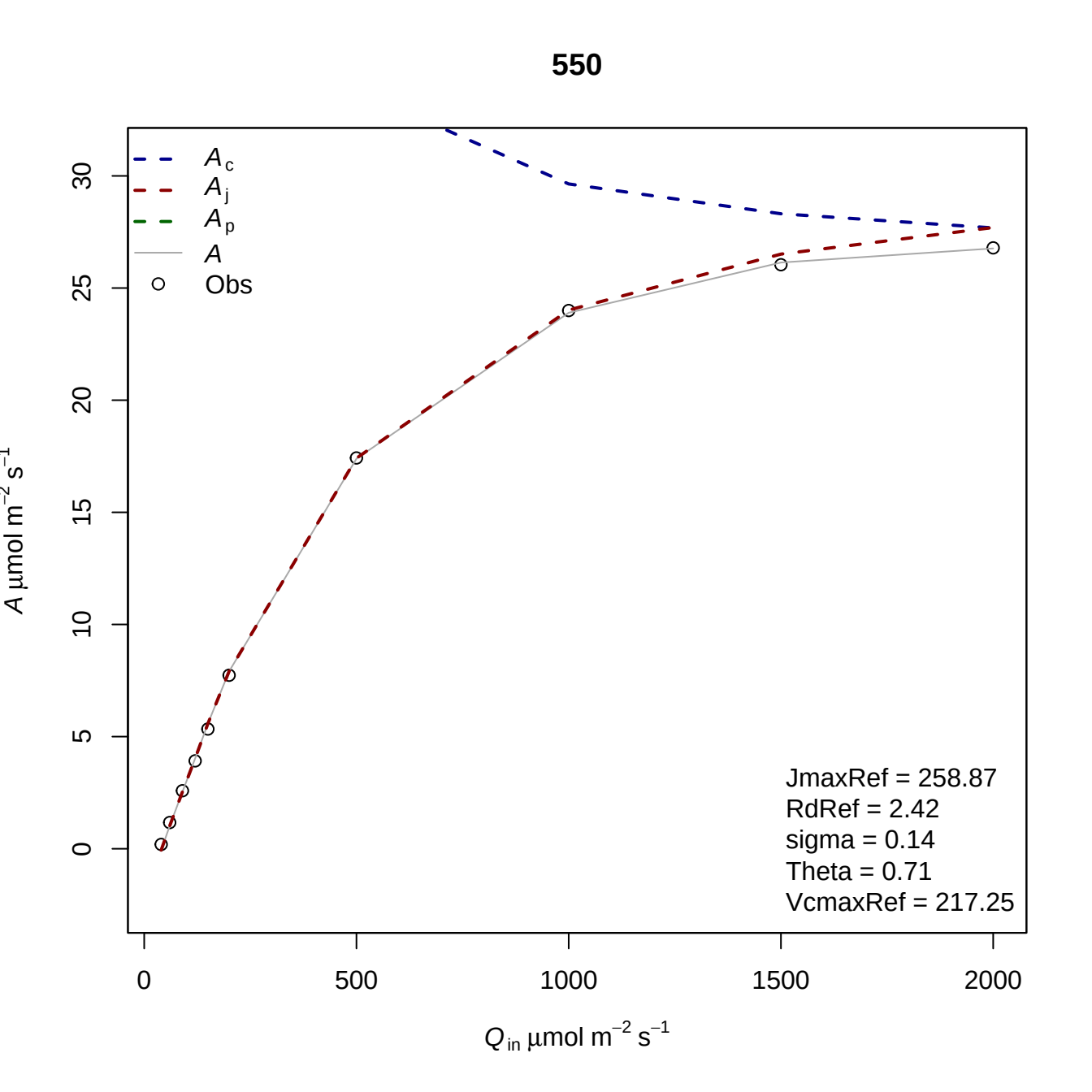

541
A_c
A_j
A_p
A
Obs
JmaxRef = 94.76
RdRef = 4.33
sigma = 0.44
Theta = 0.84
VcmaxRef = 71765.26
−5
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

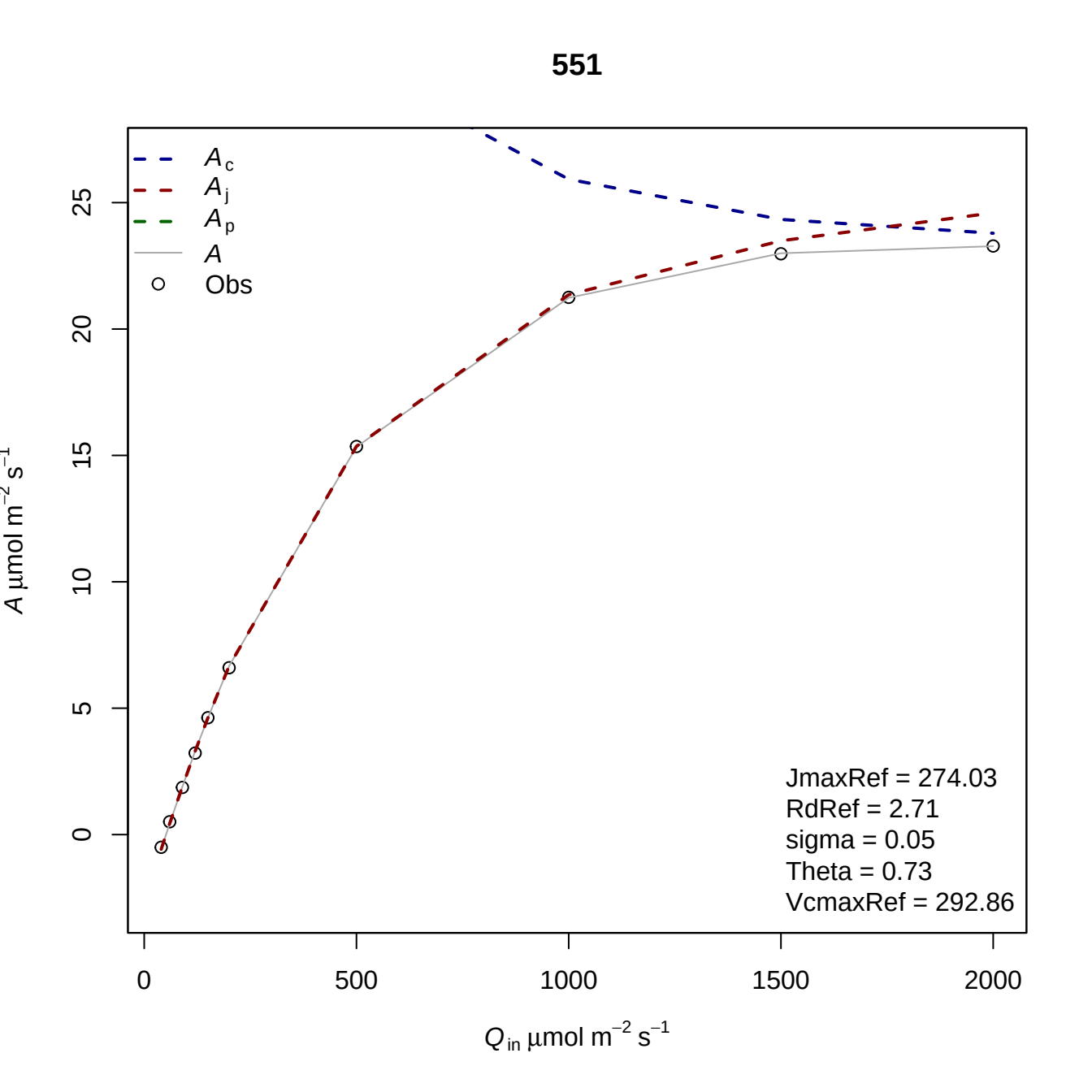

544
A_c
A_j
A_p
A
Obs
JmaxRef = 149.93
RdRef = 2.22
sigma = 0.27
Theta = 0.65
VcmaxRef = 2529015.13
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

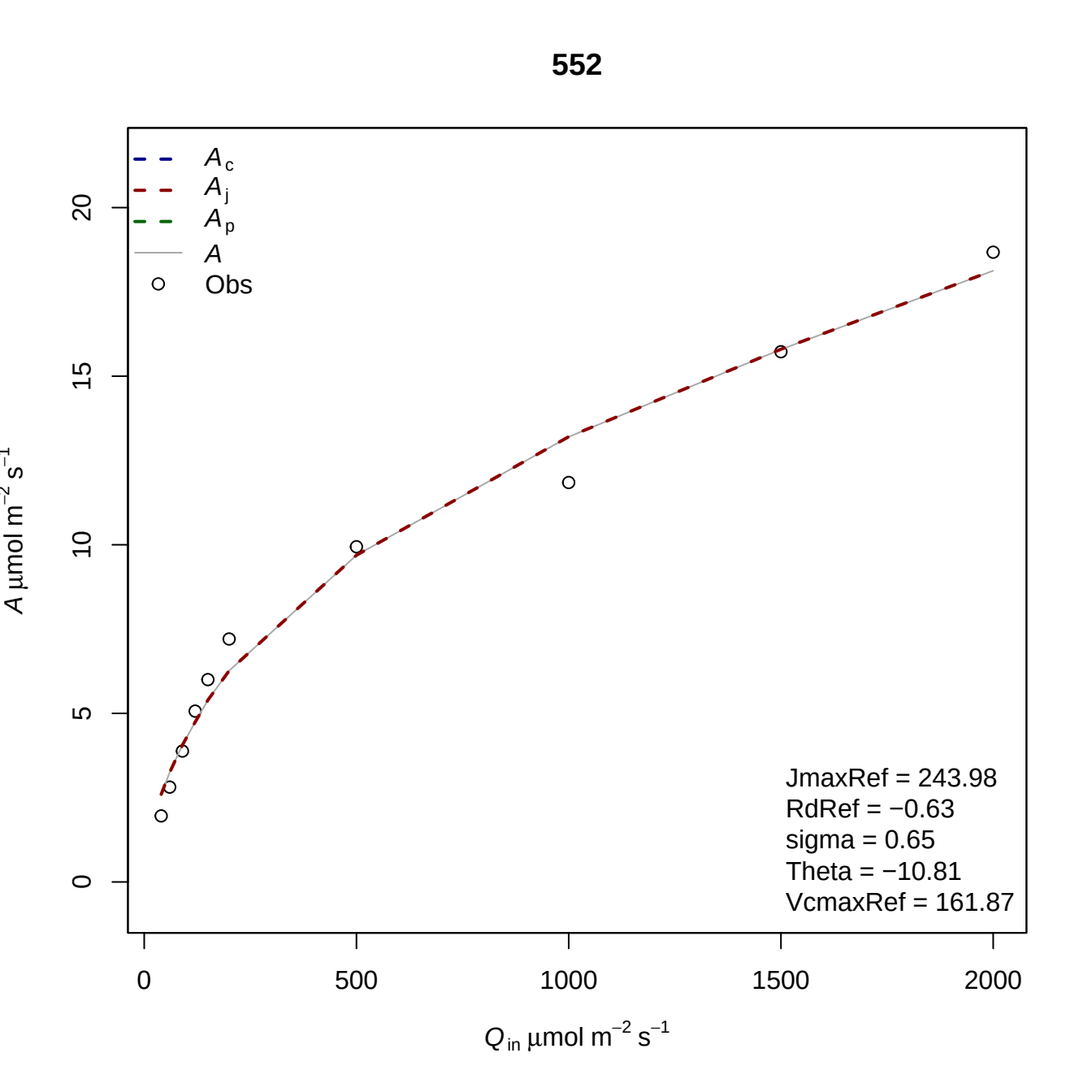

547
A_c
A_j
A_p
A
Obs
JmaxRef = 147.27
RdRef = 1.49
sigma = 0.28
Theta = 0.34
VcmaxRef = 115832.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

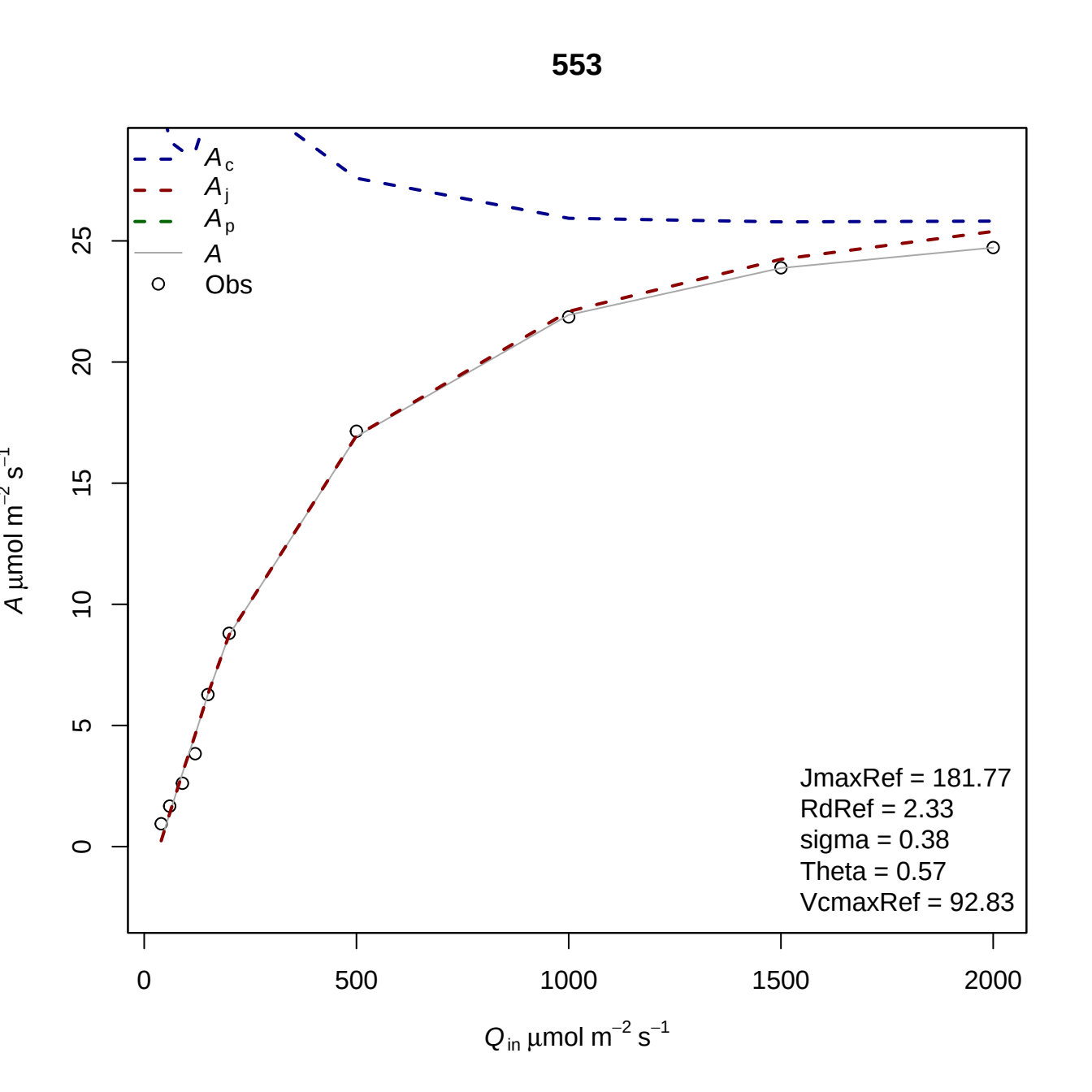

548
A_c
A_j
A_p
A
Obs
JmaxRef = 203.06
RdRef = 1.63
sigma = 0.18
Theta = 0.71
VcmaxRef = 2949624.55
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

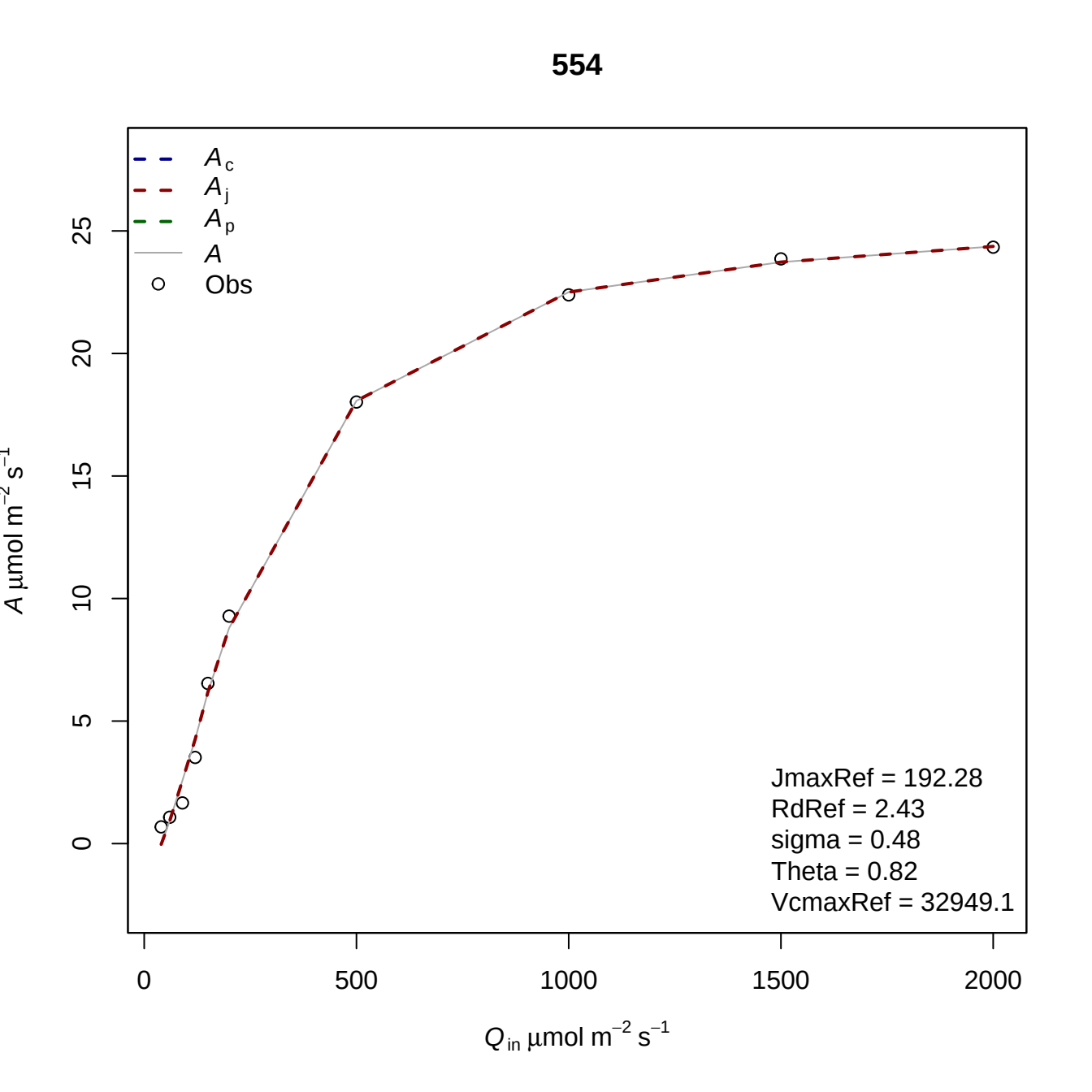

549
A_c
A_j
A_p
A
Obs
JmaxRef = 223.97
RdRef = 1.73
sigma = 0.13
Theta = 0.61
VcmaxRef = 139.97
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

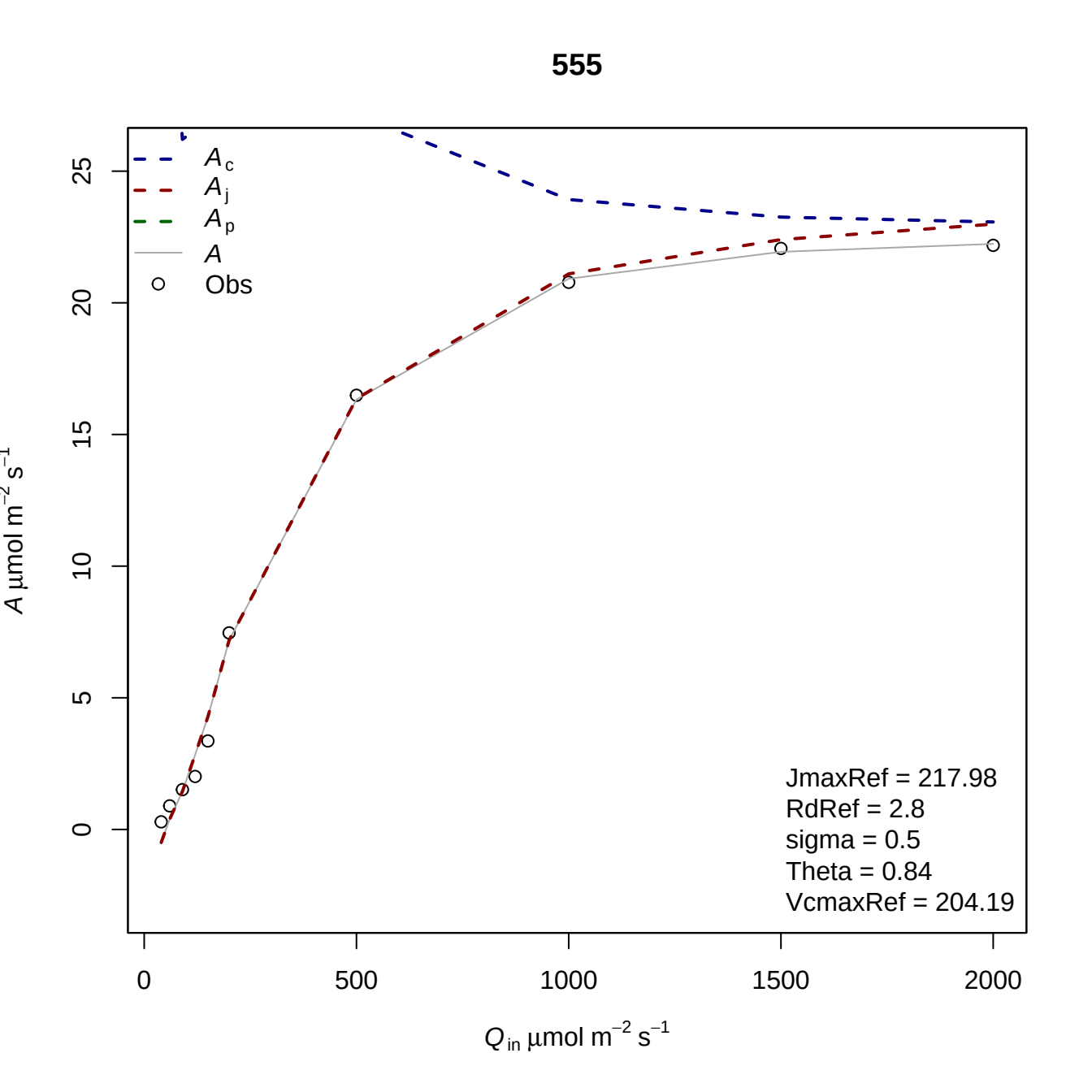

550
A_c
A_j
A_p
A
Obs
JmaxRef = 258.87
RdRef = 2.42
sigma = 0.14
Theta = 0.71
VcmaxRef = 217.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

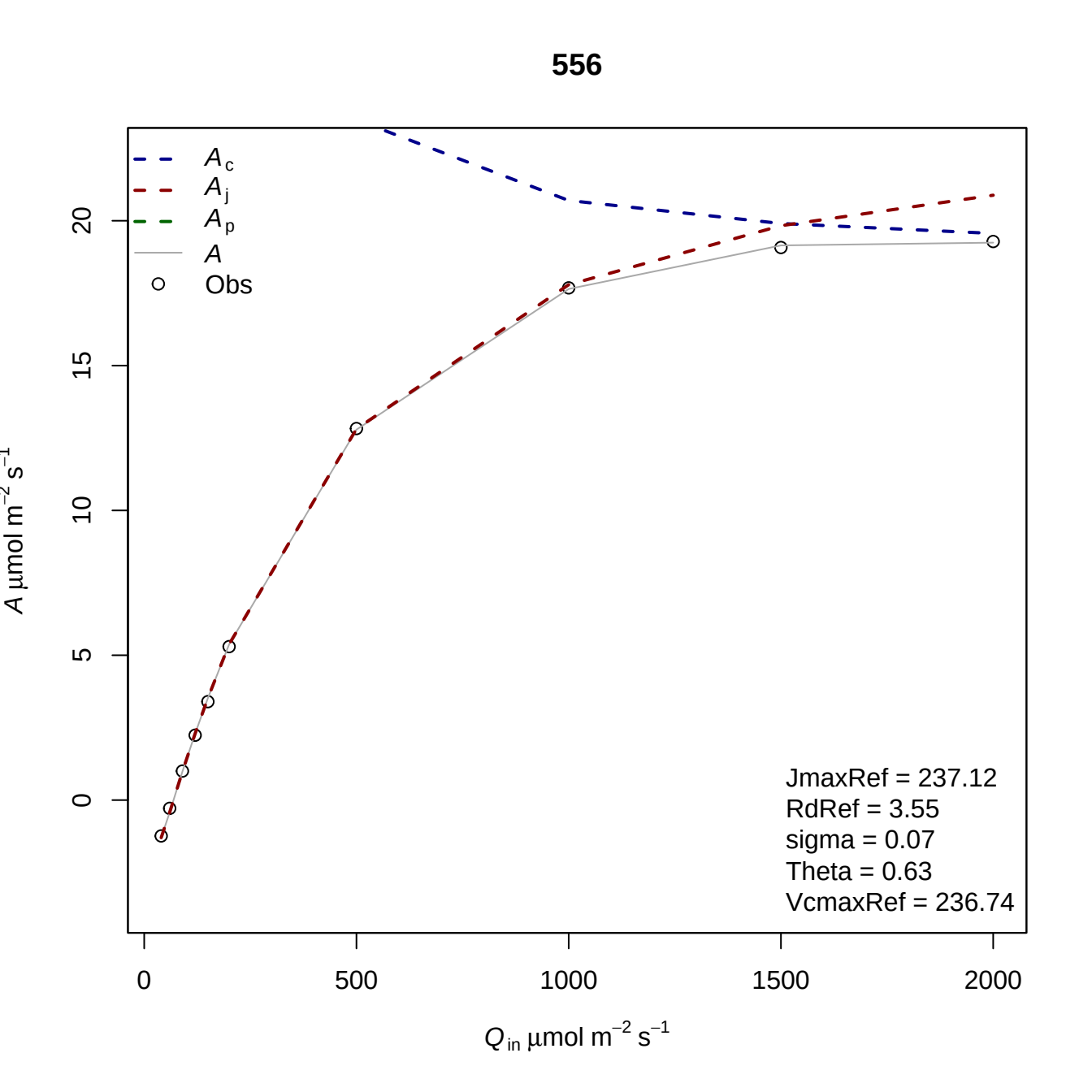

551
A_c
A_j
A_p
A
Obs
JmaxRef = 274.03
RdRef = 2.71
sigma = 0.05
Theta = 0.73
VcmaxRef = 292.86
A µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

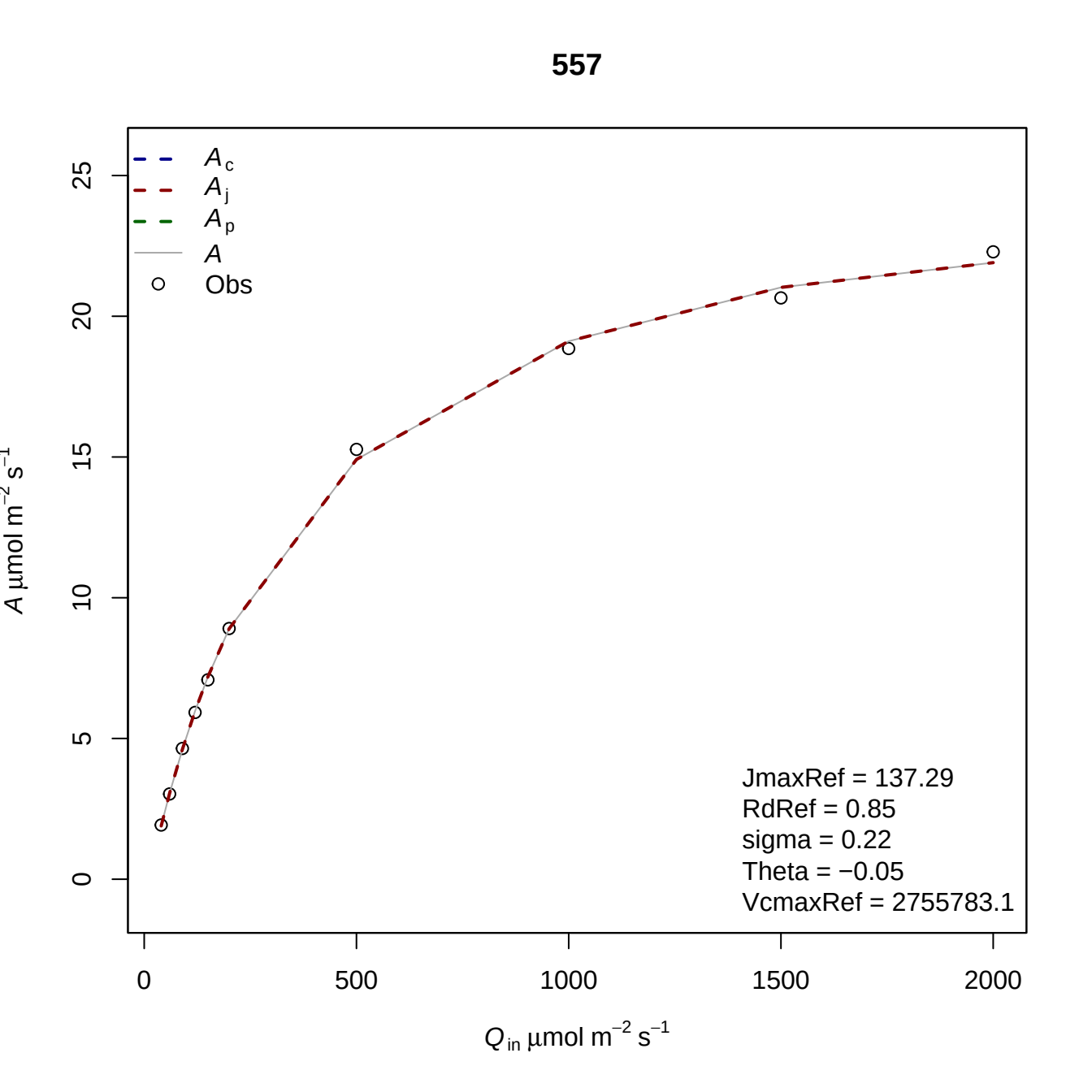

552
A_c
A_j
A_p
A
Obs
JmaxRef = 243.98
RdRef = −0.63
sigma = 0.65
Theta = −10.81
VcmaxRef = 161.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

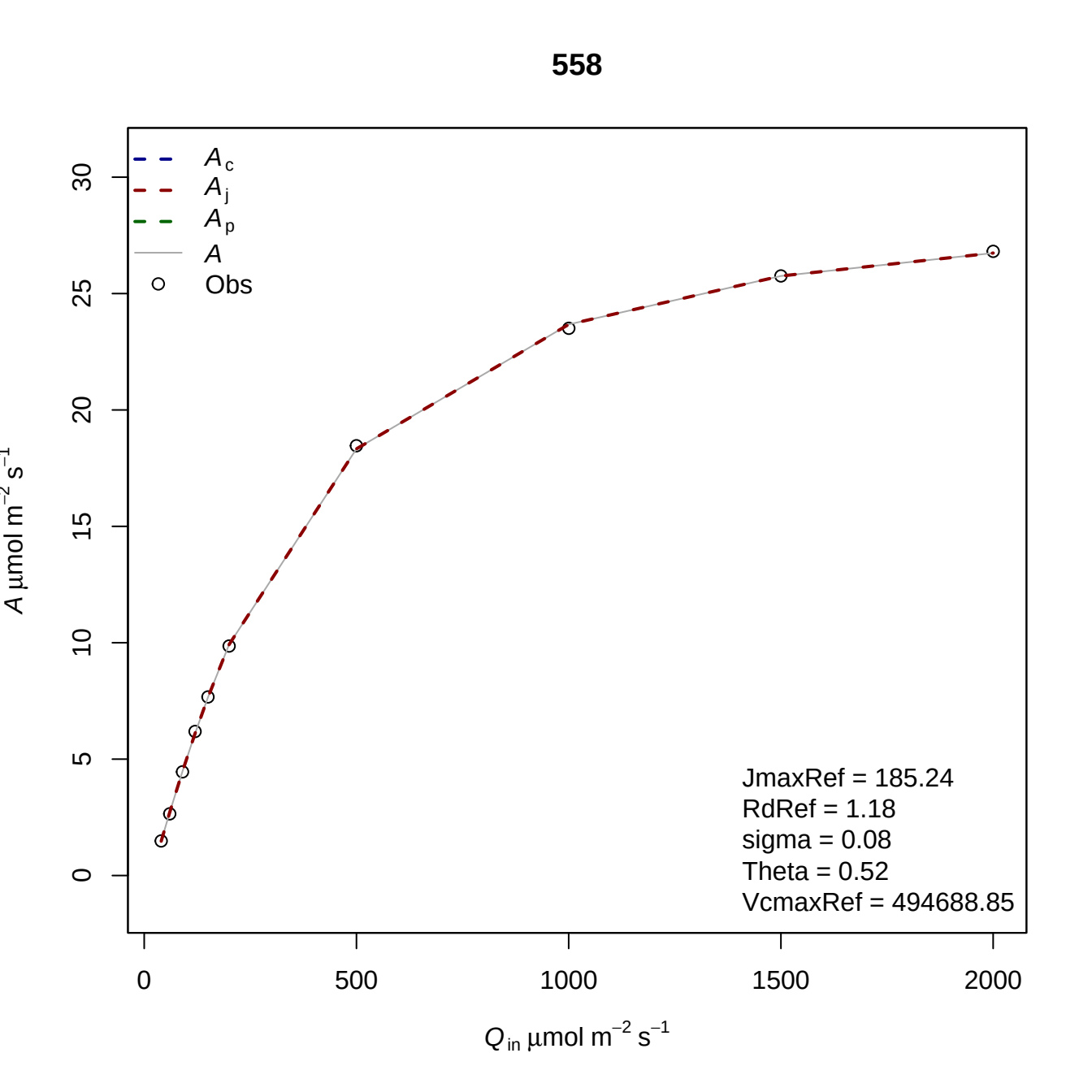

553
A_c
A_j
A_p
A
Obs
JmaxRef = 181.77
RdRef = 2.33
sigma = 0.38
Theta = 0.57
VcmaxRef = 92.83
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

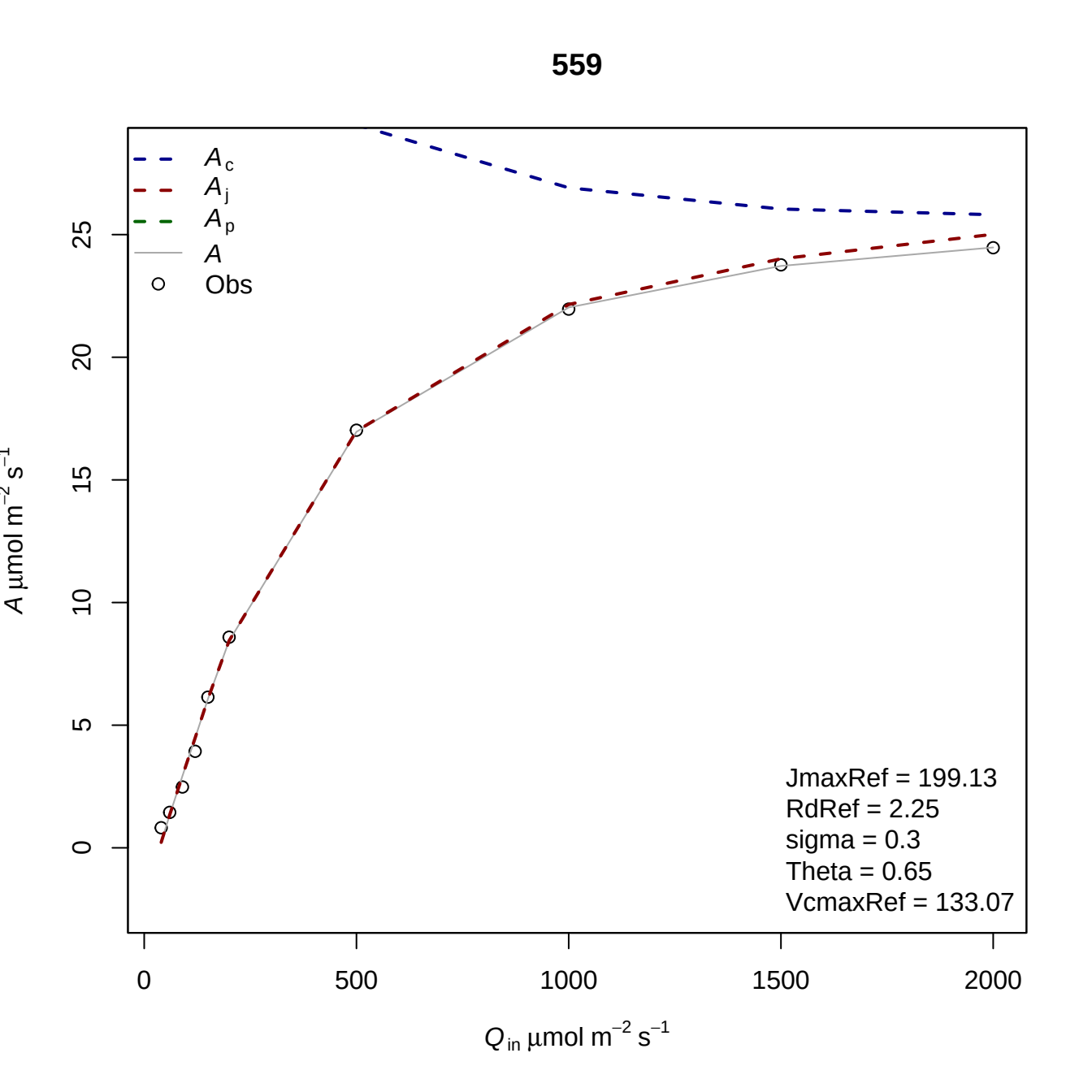

554
A_c
A_j
A_p
A
Obs
JmaxRef = 192.28
RdRef = 2.43
sigma = 0.48
Theta = 0.82
VcmaxRef = 32949.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

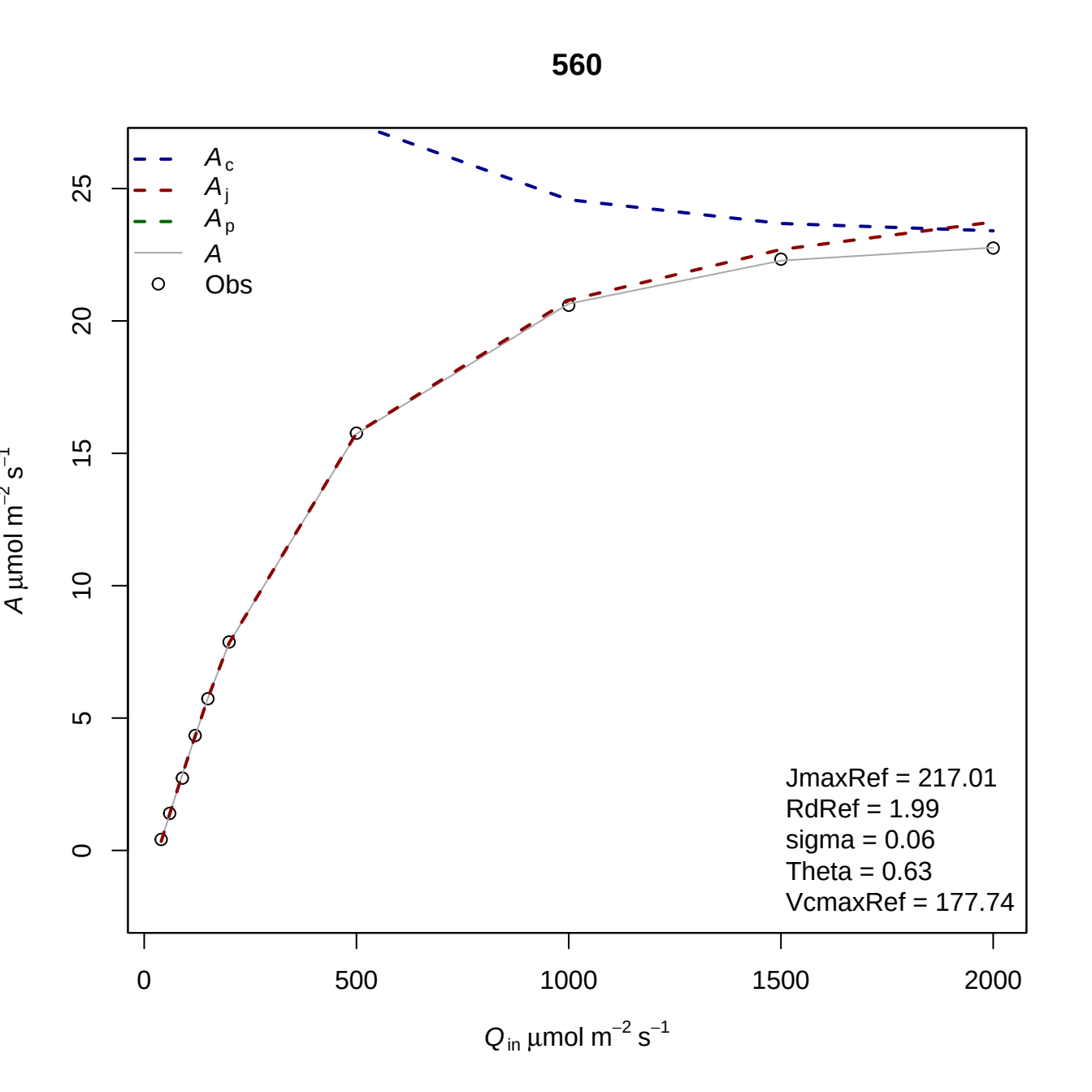

555
A_c
A_j
A_p
A
Obs
JmaxRef = 217.98
RdRef = 2.8
sigma = 0.5
Theta = 0.84
VcmaxRef = 204.19
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

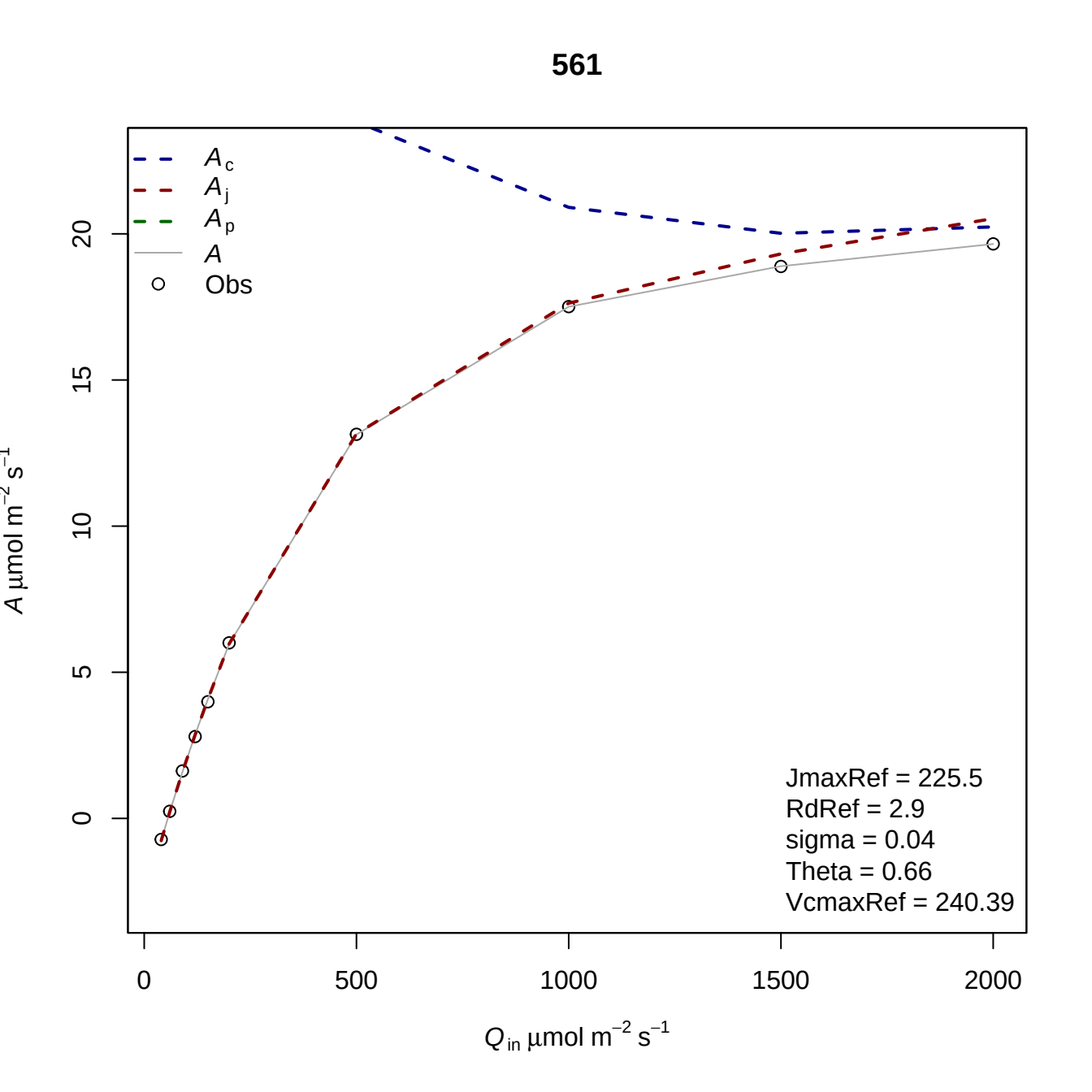

556
A_c
A_j
A_p
A
Obs
JmaxRef = 237.12
RdRef = 3.55
sigma = 0.07
Theta = 0.63
VcmaxRef = 236.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

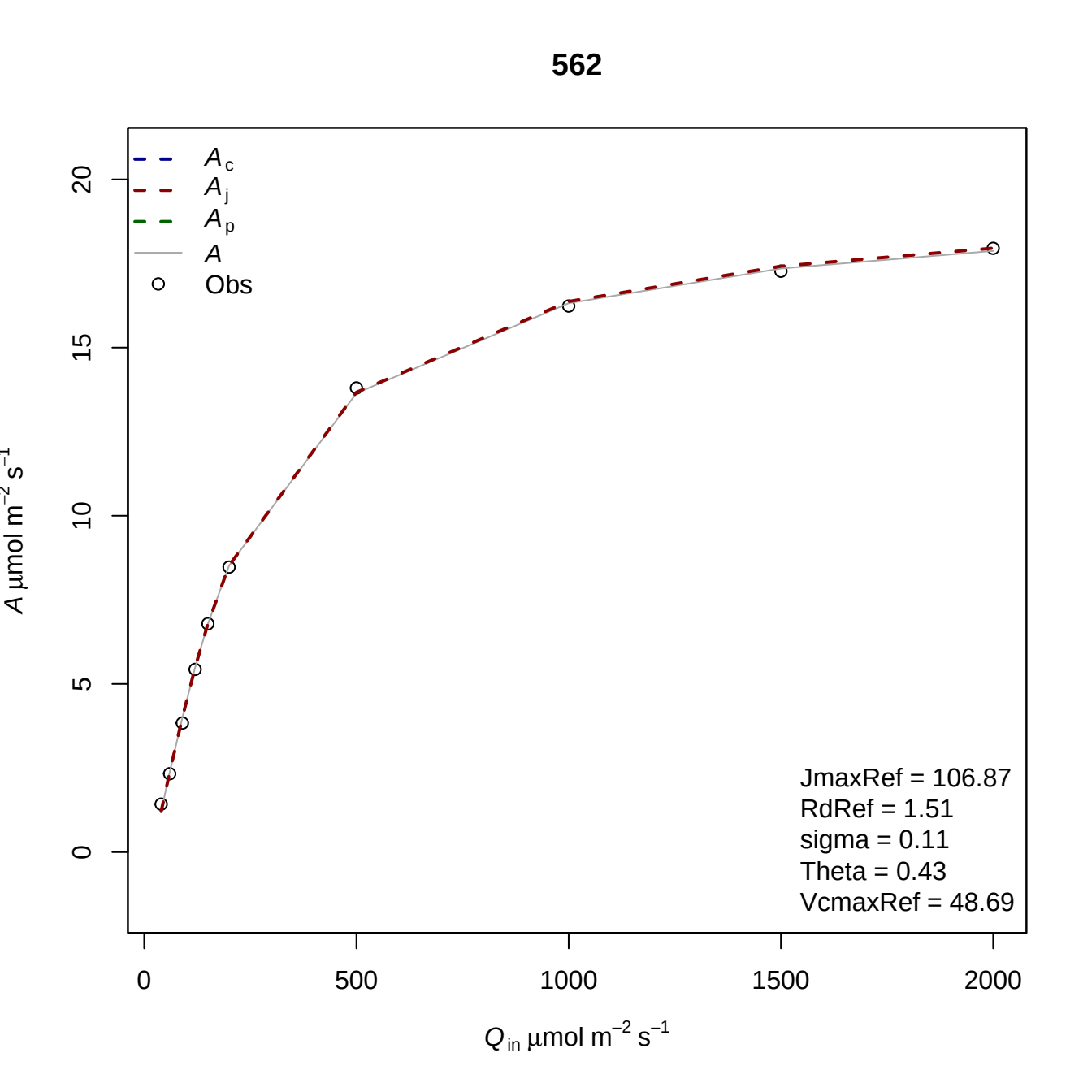

557
A_c
A_j
A_p
A
Obs
JmaxRef = 137.29
RdRef = 0.85
sigma = 0.22
Theta = −0.05
VcmaxRef = 2755783.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

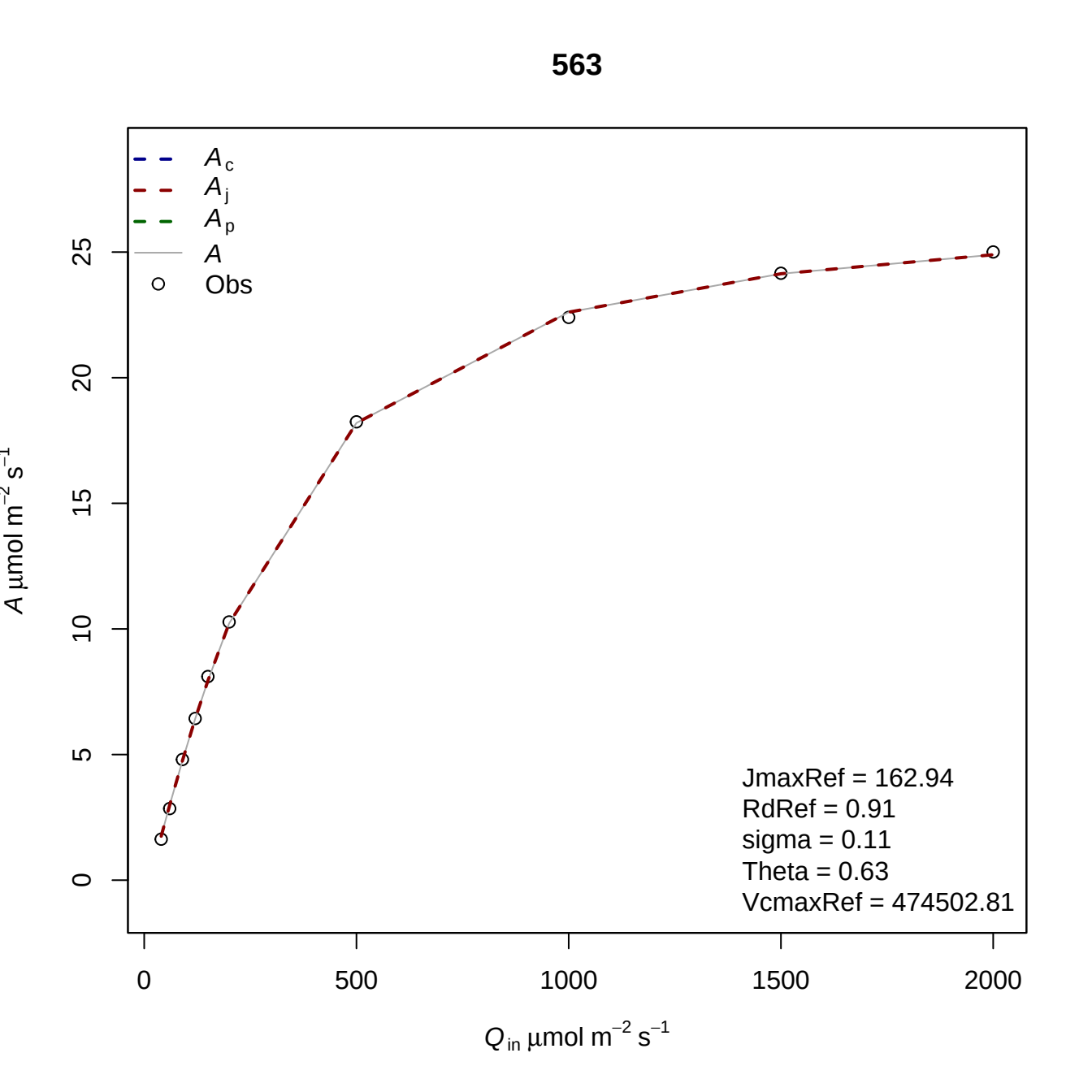

558
A_c
A_j
A_p
A
Obs
JmaxRef = 185.24
RdRef = 1.18
sigma = 0.08
Theta = 0.52
VcmaxRef = 494688.85
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

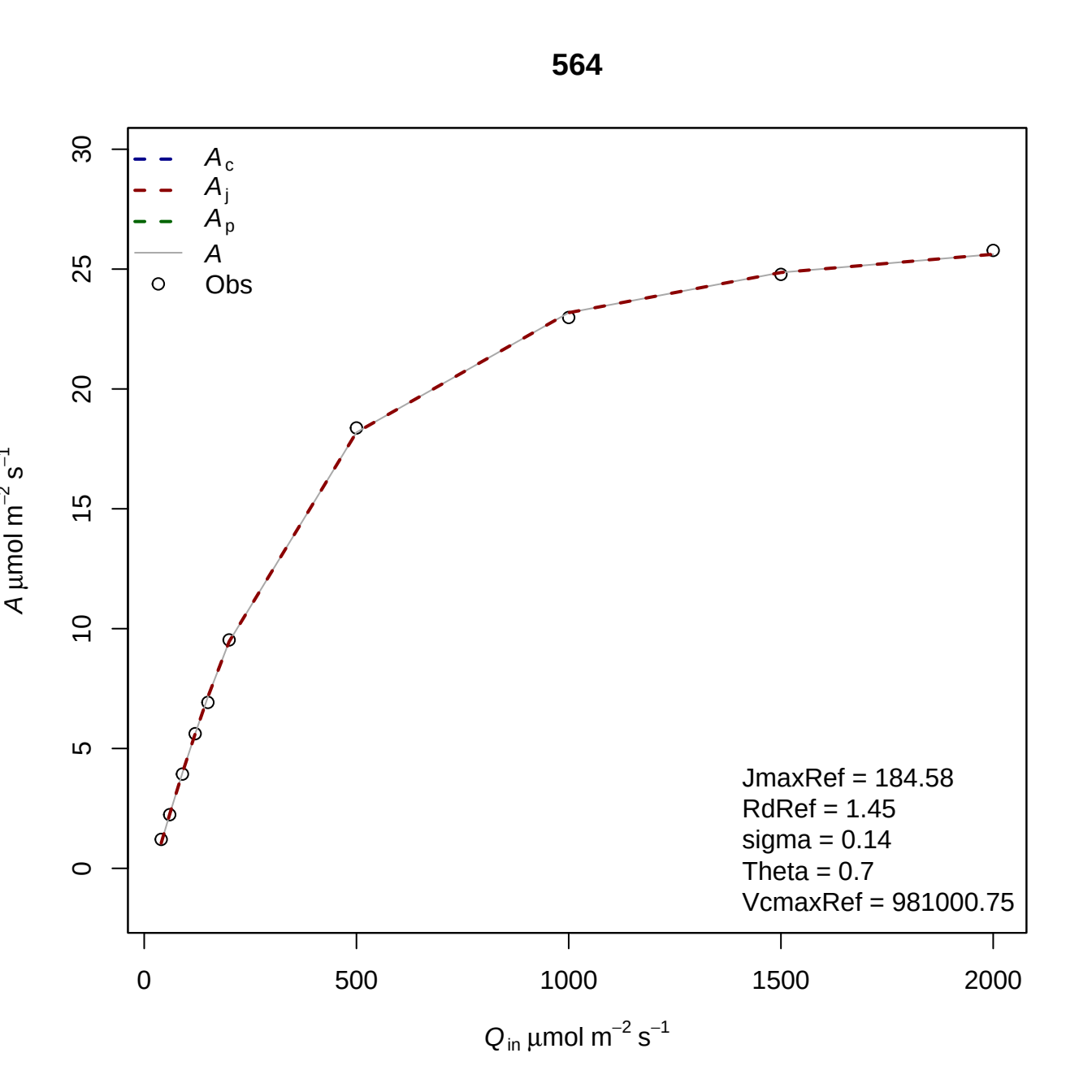

559
A_c
A_j
A_p
A
Obs
JmaxRef = 199.13
RdRef = 2.25
sigma = 0.3
Theta = 0.65
VcmaxRef = 133.07
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

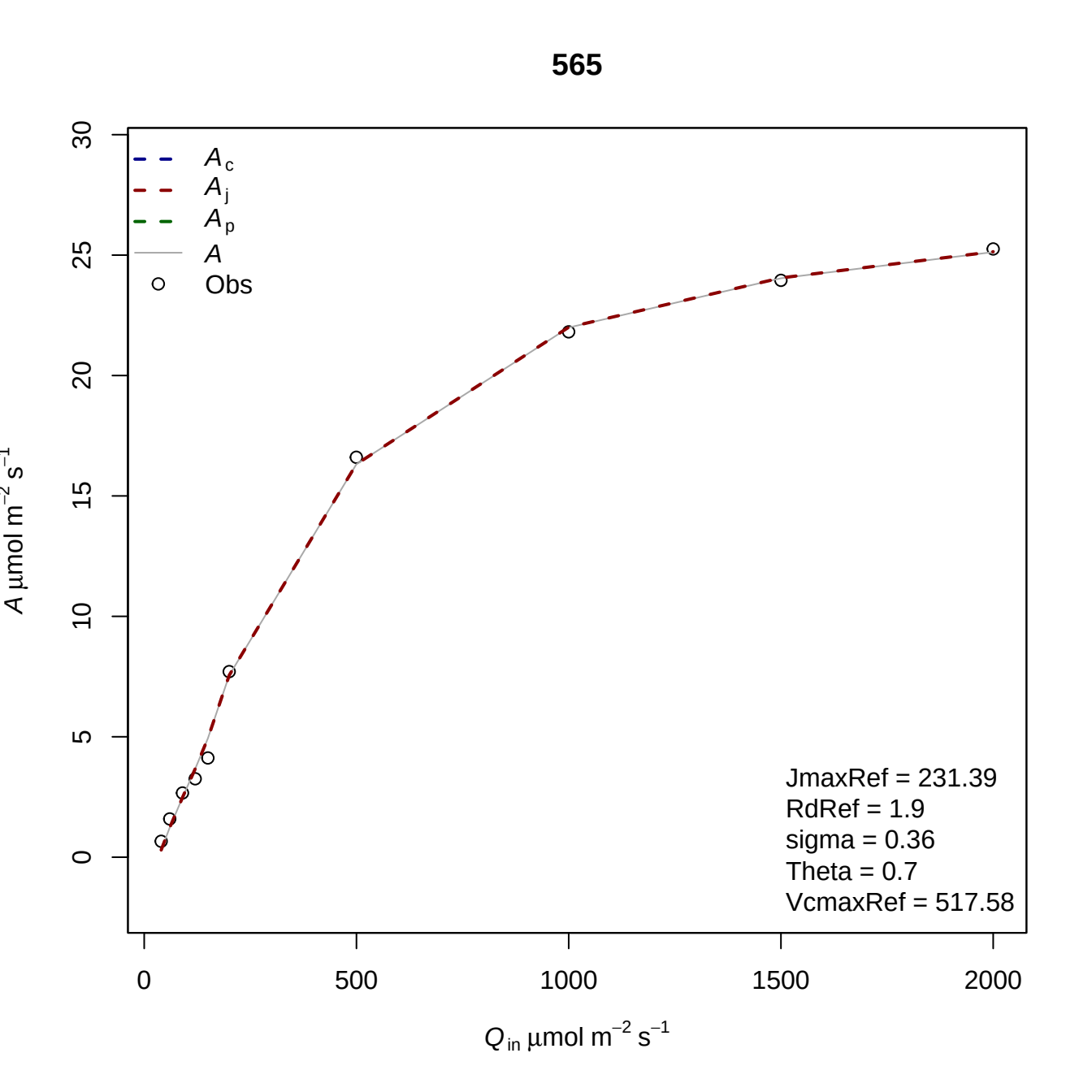

560
A_c
A_j
A_p
A
Obs
JmaxRef = 217.01
RdRef = 1.99
sigma = 0.06
Theta = 0.63
VcmaxRef = 177.74
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

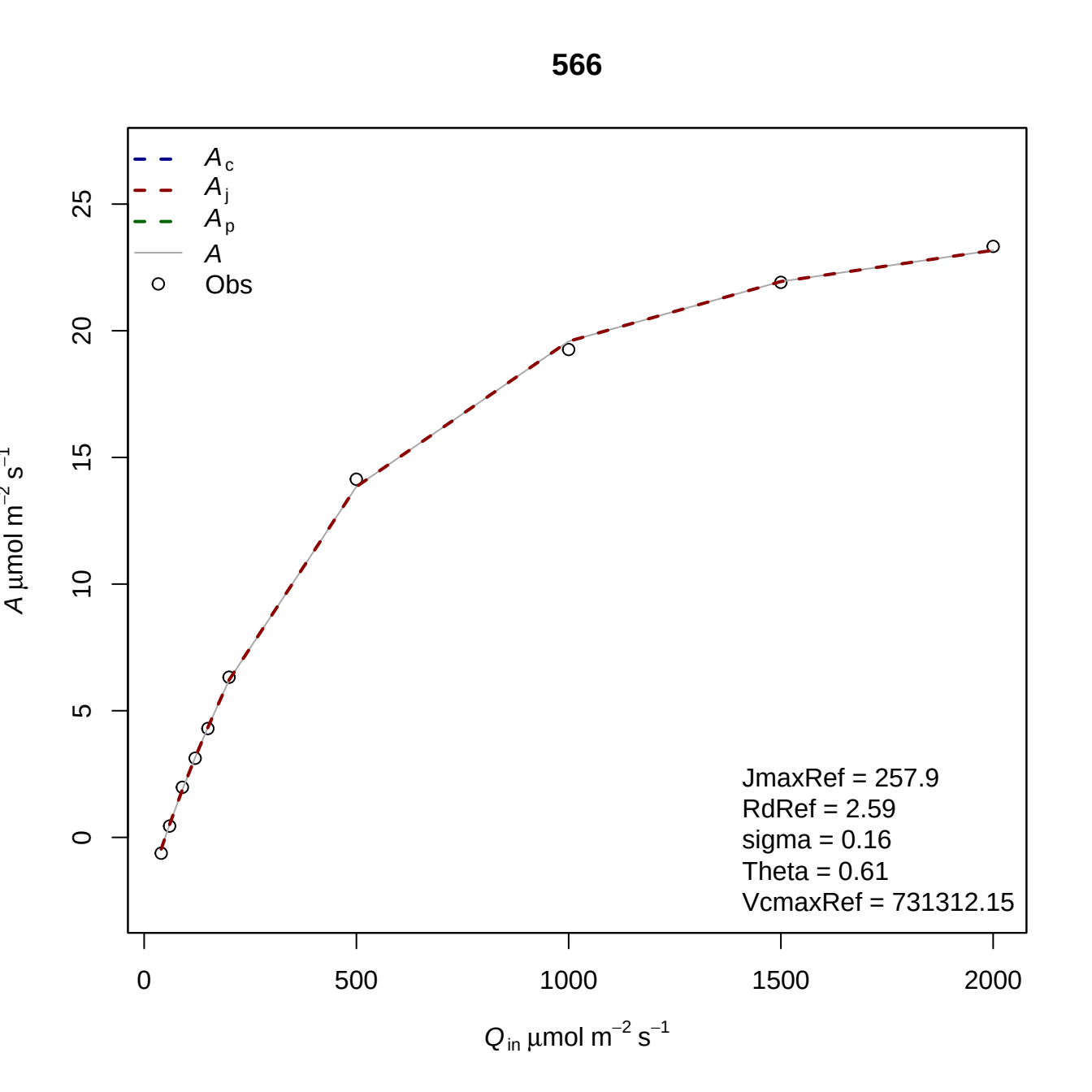

561
A_c
A_j
A_p
A
Obs
JmaxRef = 225.5
RdRef = 2.9
sigma = 0.04
Theta = 0.66
VcmaxRef = 240.39
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

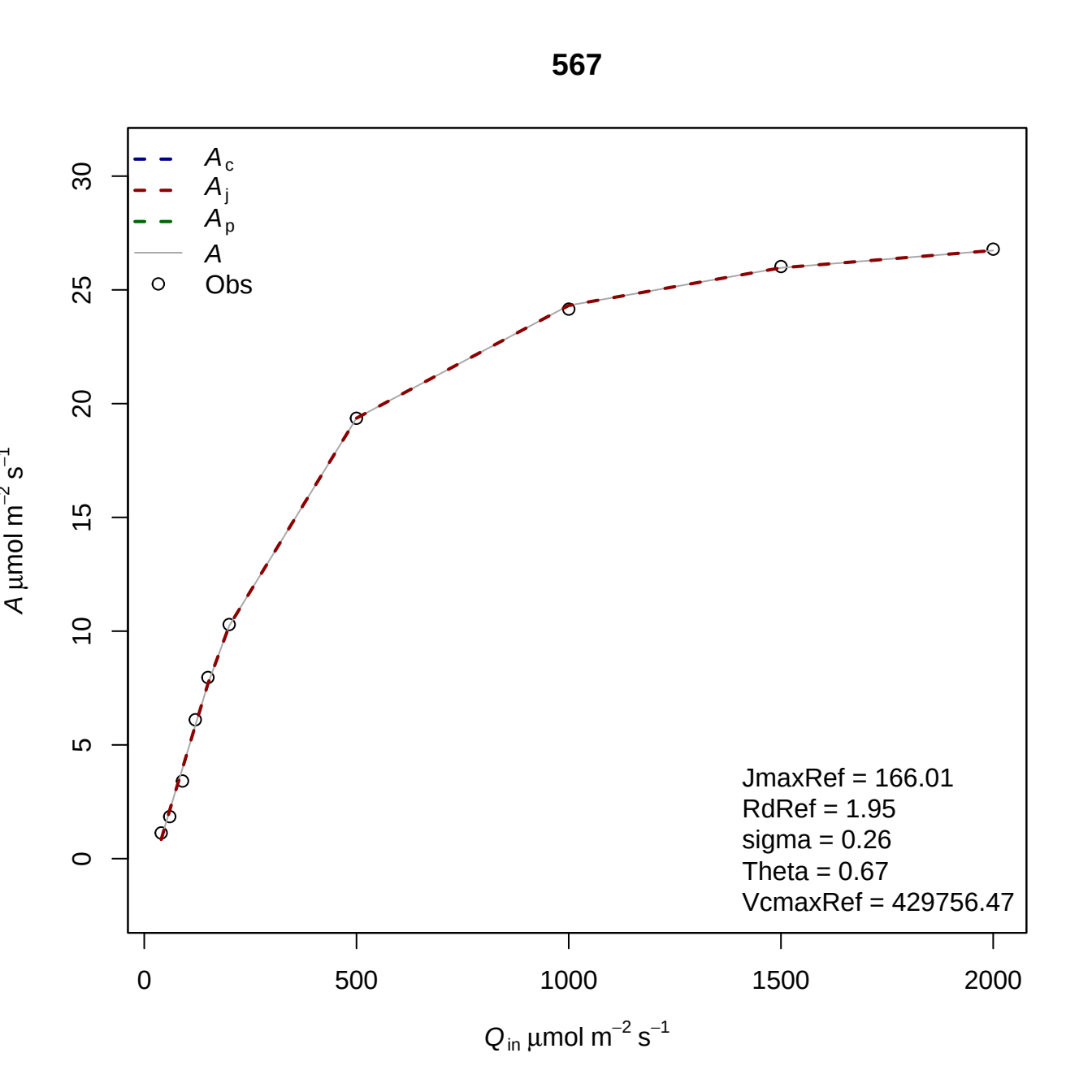

562
A_c
A_j
A_p
A
Obs
JmaxRef = 106.87
RdRef = 1.51
sigma = 0.11
Theta = 0.43
VcmaxRef = 48.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

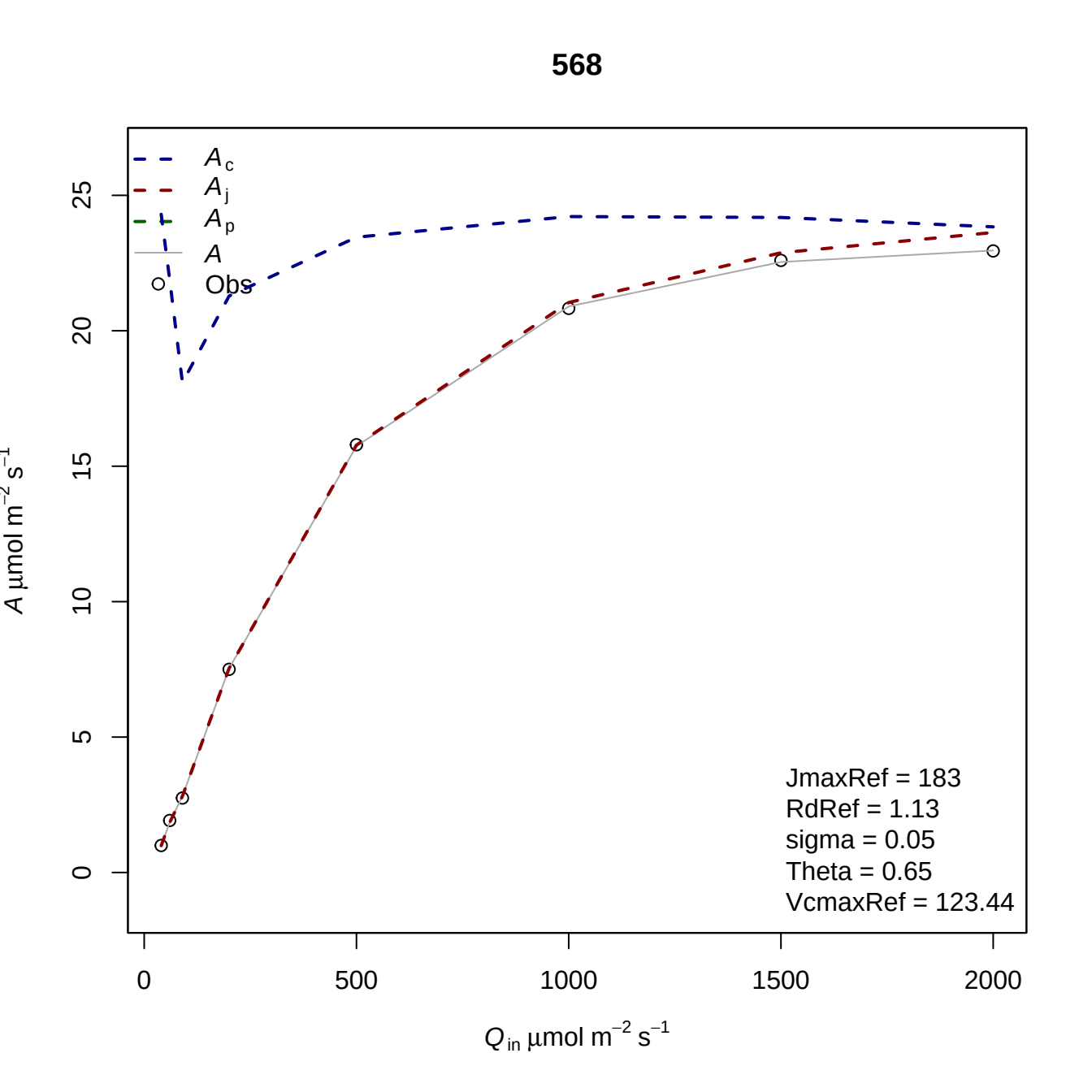

563
A_c
A_j
A_p
A
Obs
JmaxRef = 162.94
RdRef = 0.91
sigma = 0.11
Theta = 0.63
VcmaxRef = 474502.81
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

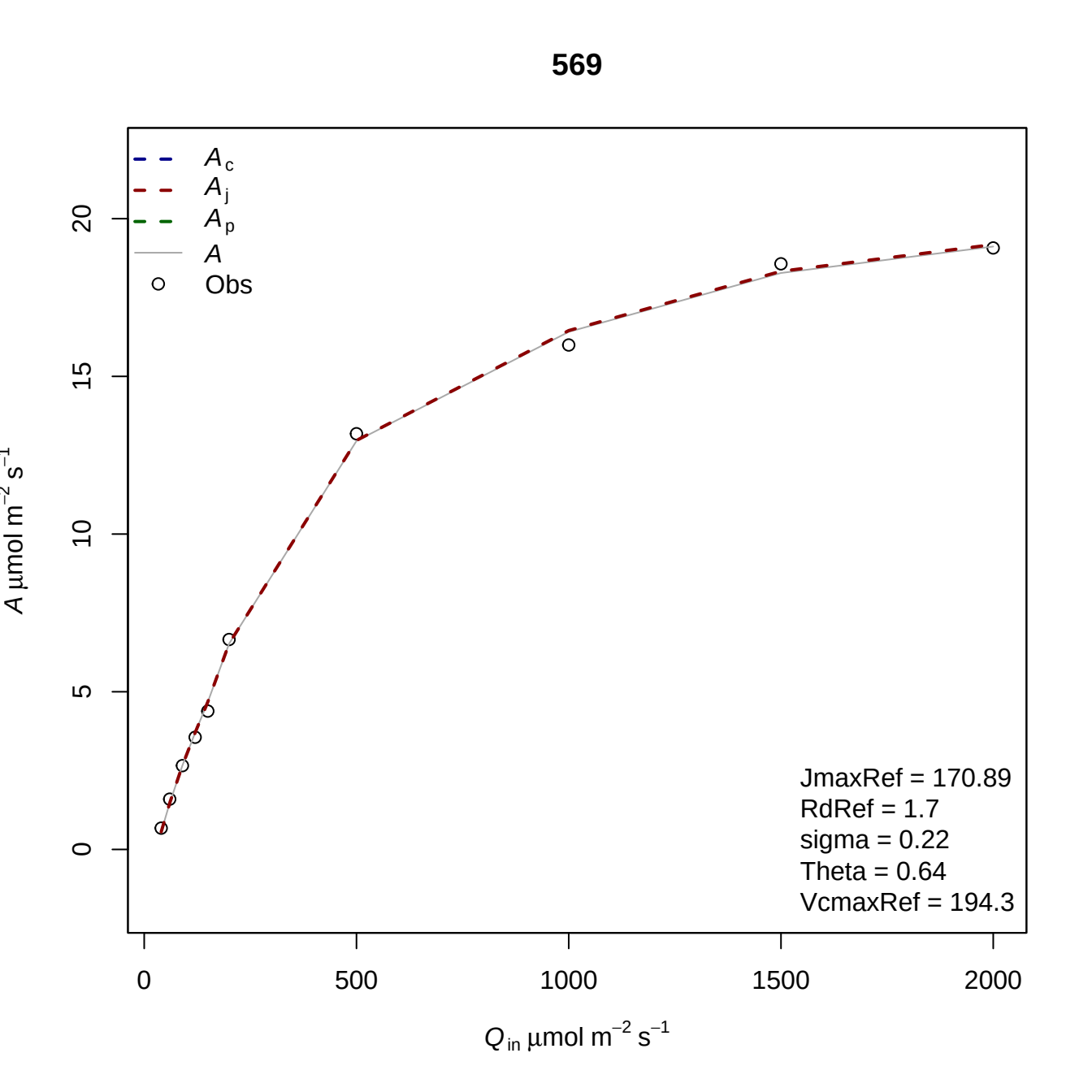

564
A_c
A_j
A_p
A
Obs
JmaxRef = 184.58
RdRef = 1.45
sigma = 0.14
Theta = 0.7
VcmaxRef = 981000.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

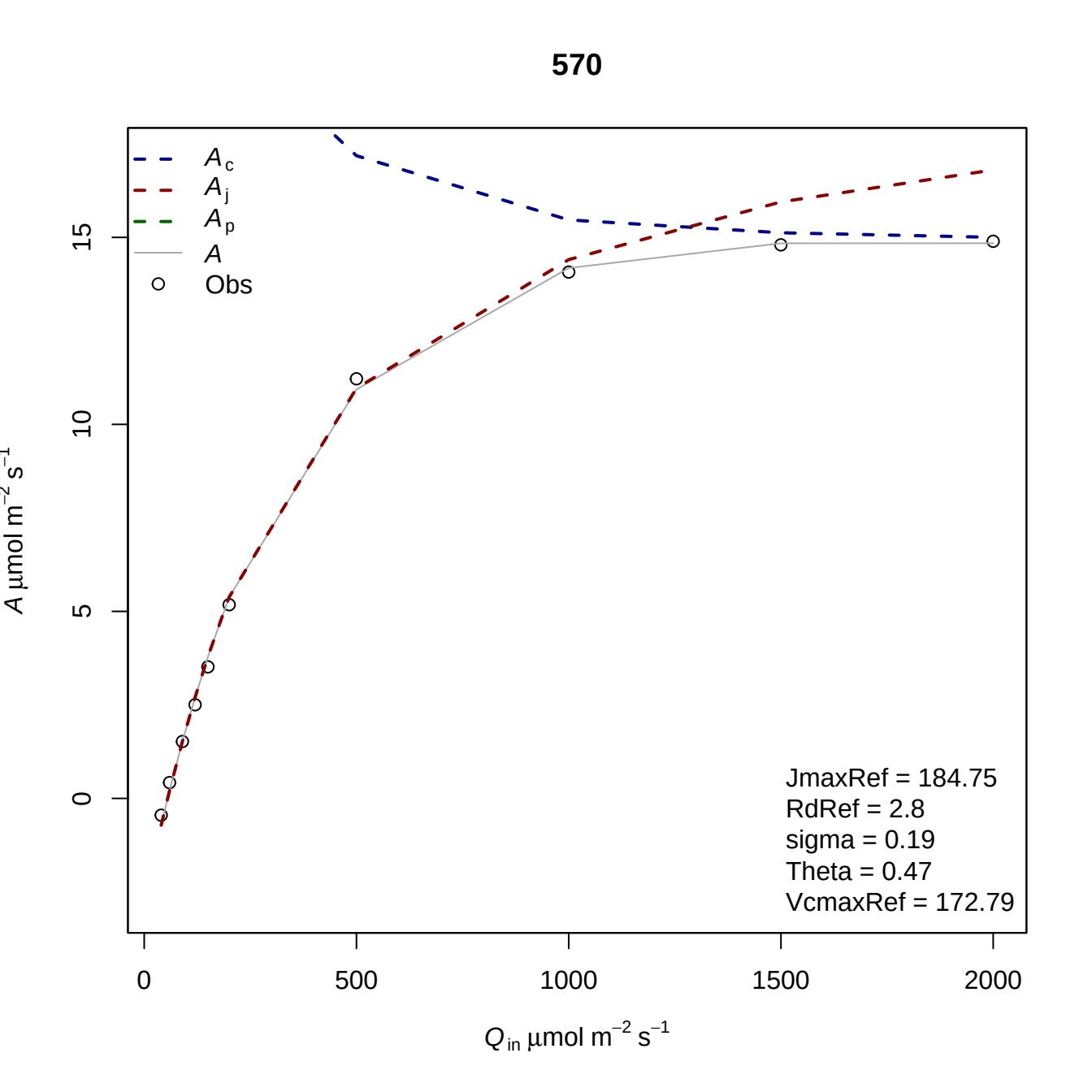

565
A_c
A_j
A_p
A
Obs
JmaxRef = 231.39
RdRef = 1.9
sigma = 0.36
Theta = 0.7
VcmaxRef = 517.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

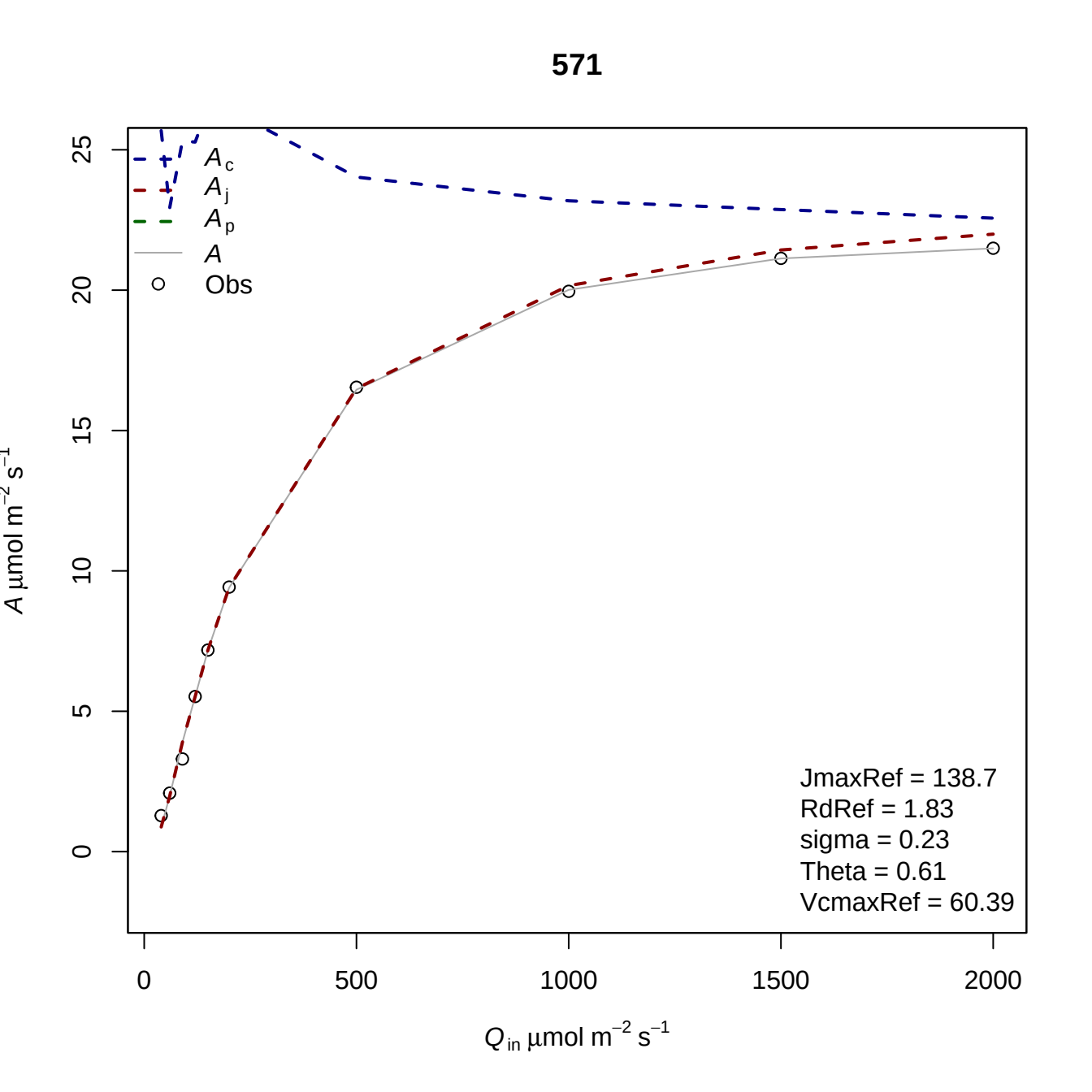

566
A_c
A_j
A_p
A
Obs
JmaxRef = 257.9
RdRef = 2.59
sigma = 0.16
Theta = 0.61
VcmaxRef = 731312.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

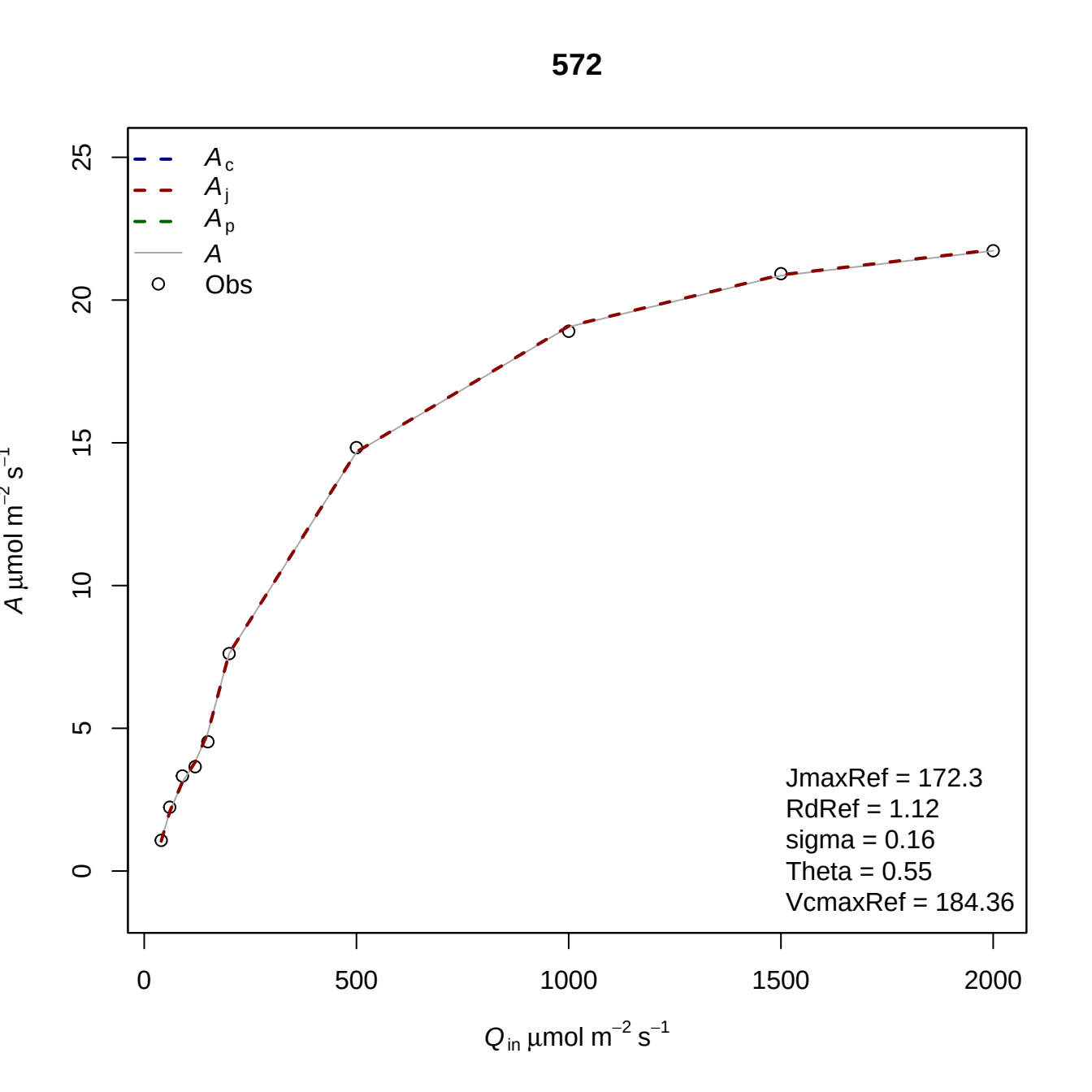

567
A_c
A_j
A_p
A
Obs
JmaxRef = 166.01
RdRef = 1.95
sigma = 0.26
Theta = 0.67
VcmaxRef = 429756.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

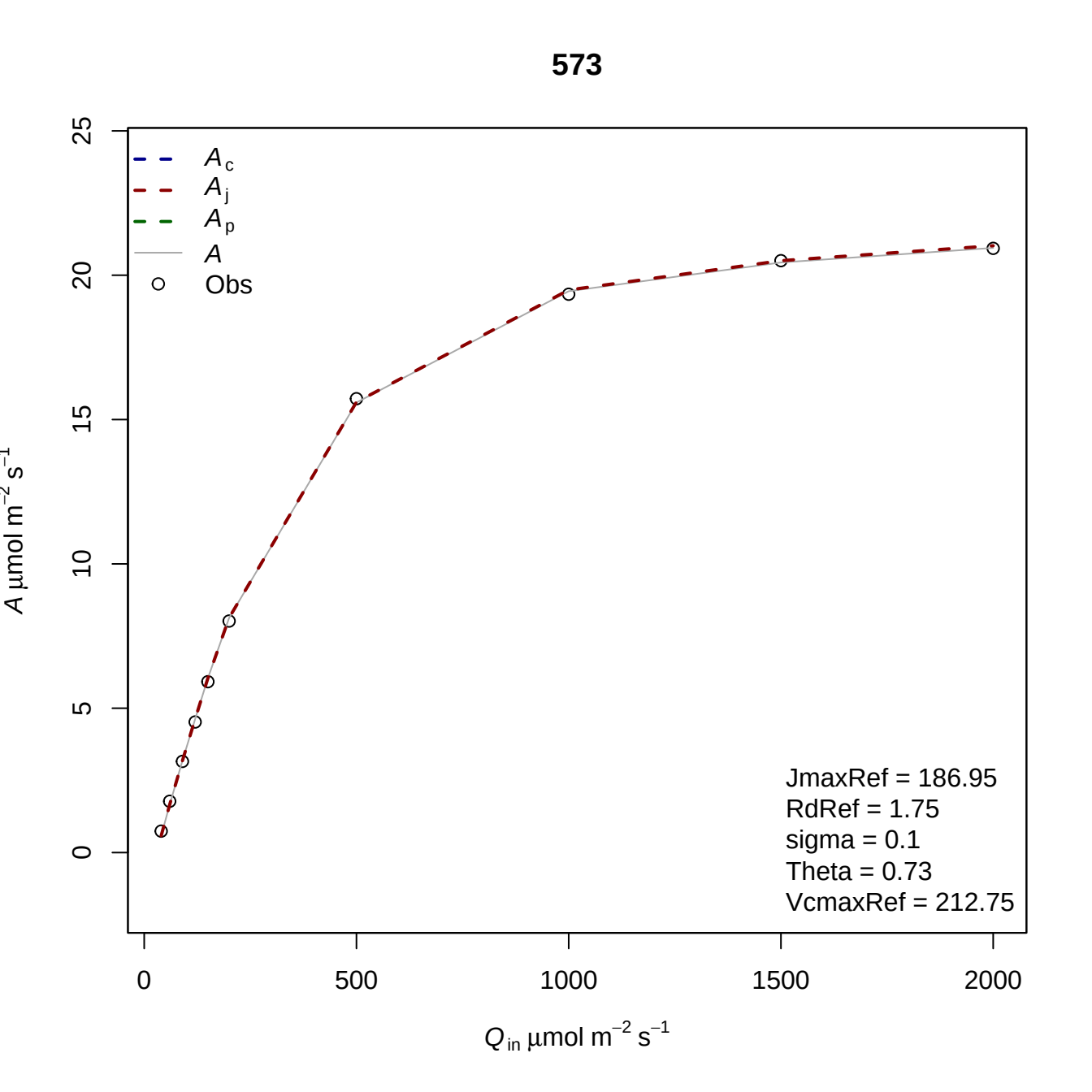

568
A_c
A_j
A_p
A
Obs
JmaxRef = 183
RdRef = 1.13
sigma = 0.05
Theta = 0.65
VcmaxRef = 123.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

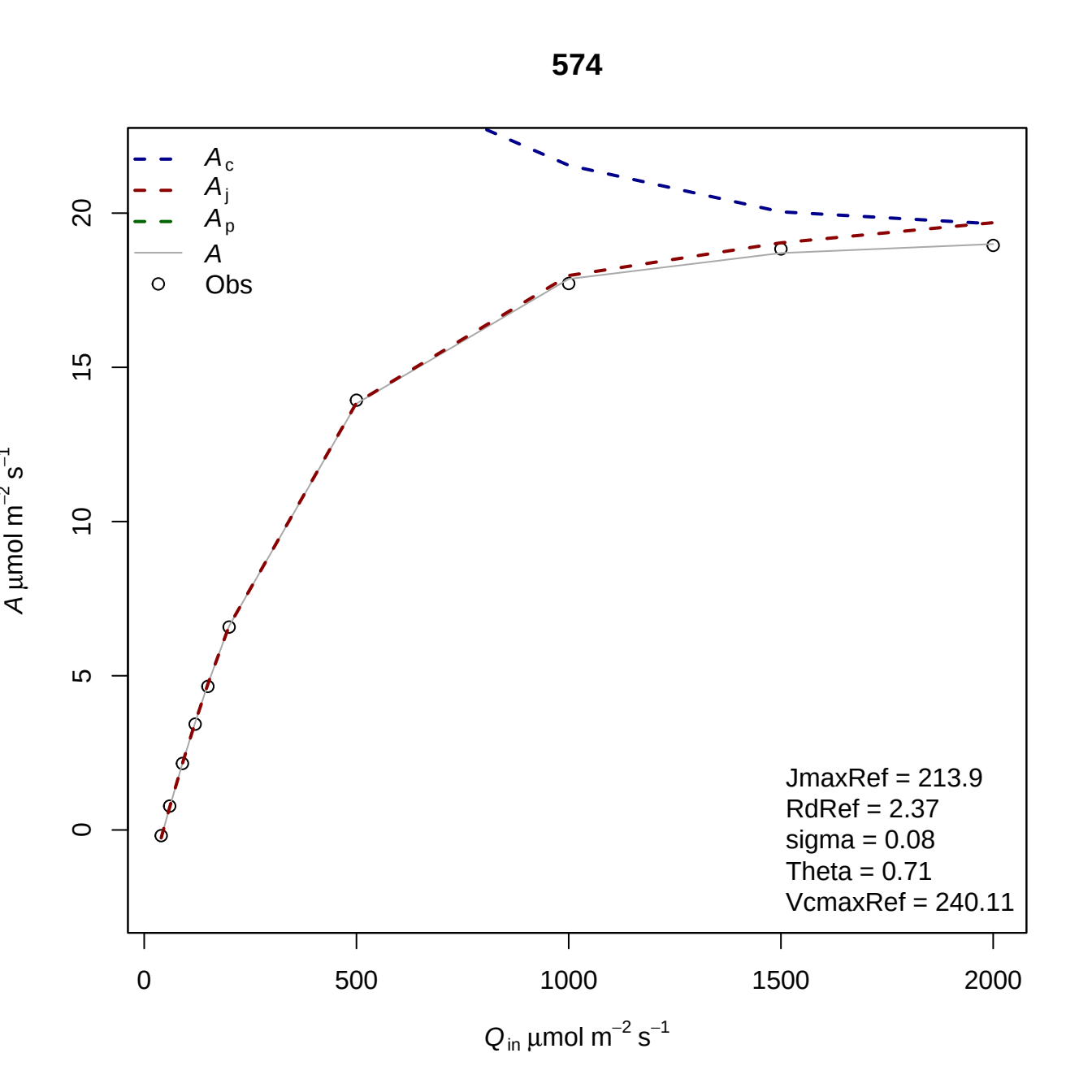

569
A_c
A_j
A_p
A
Obs
JmaxRef = 170.89
RdRef = 1.7
sigma = 0.22
Theta = 0.64
VcmaxRef = 194.3
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

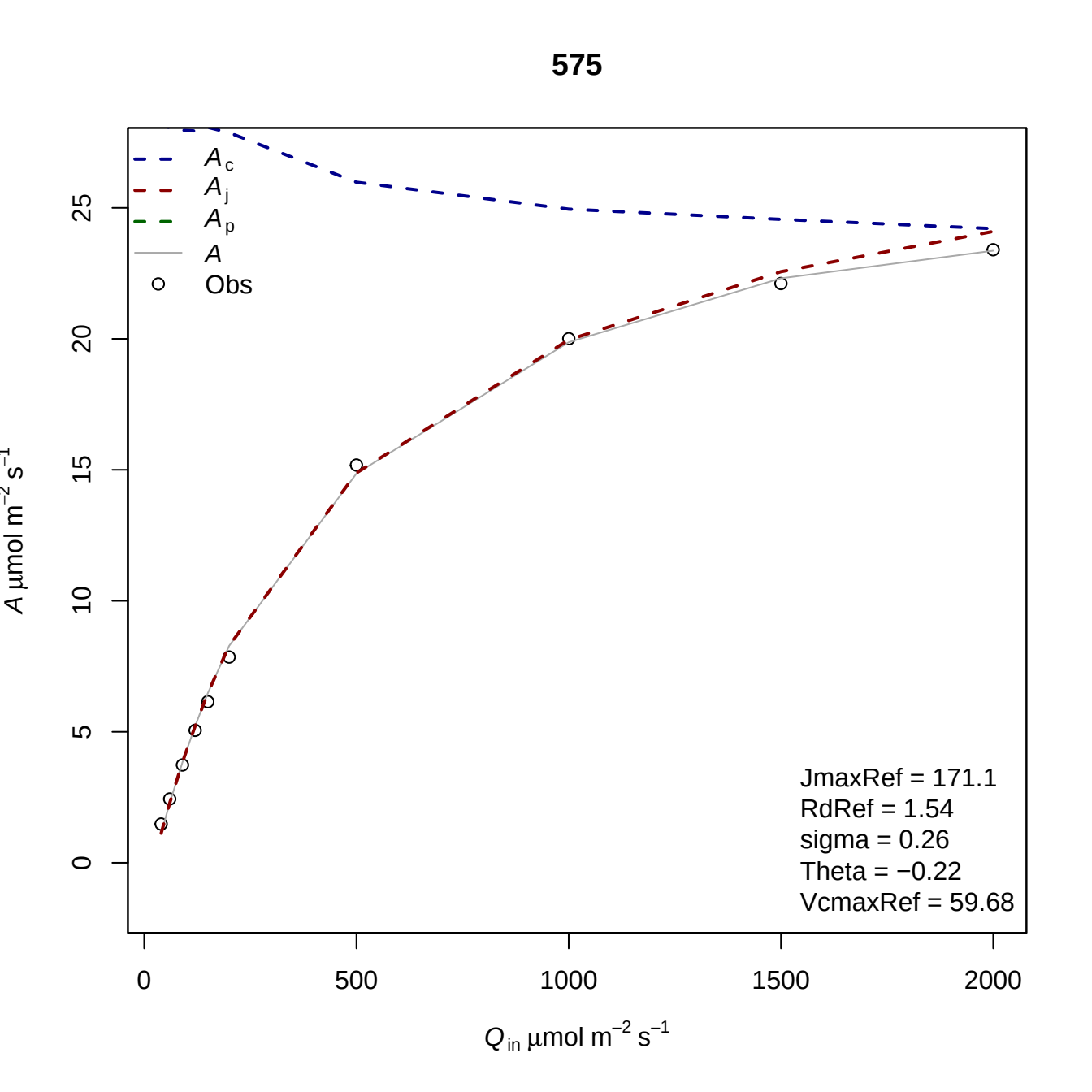

570
A_c
A_j
A_p
A
Obs
JmaxRef = 184.75
RdRef = 2.8
sigma = 0.19
Theta = 0.47
VcmaxRef = 172.79
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

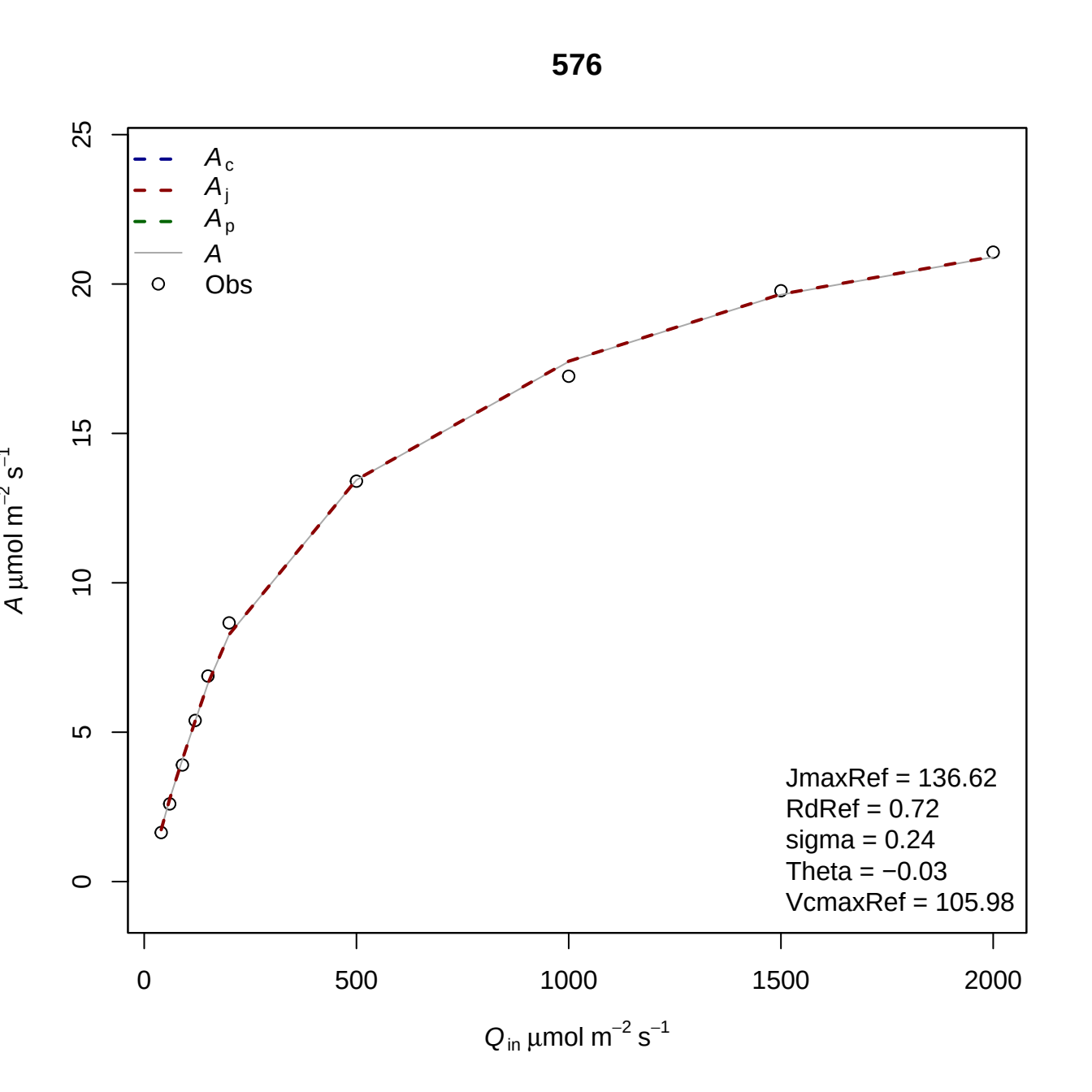

571
A_c
A_j
A_p
A
Obs
JmaxRef = 138.7
RdRef = 1.83
sigma = 0.23
Theta = 0.61
VcmaxRef = 60.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

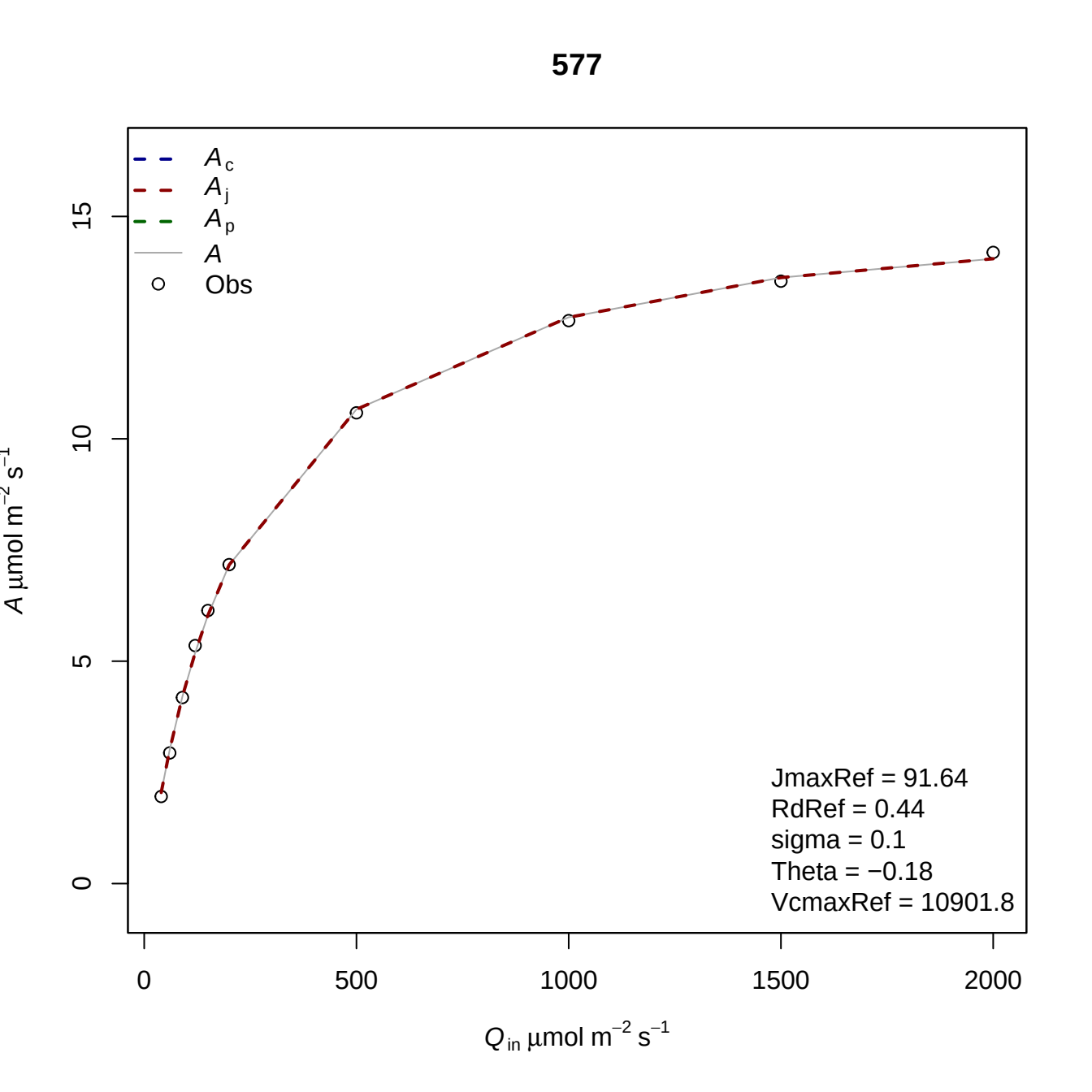

572
A_c
A_j
A_p
A
Obs
JmaxRef = 172.3
RdRef = 1.12
sigma = 0.16
Theta = 0.55
VcmaxRef = 184.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

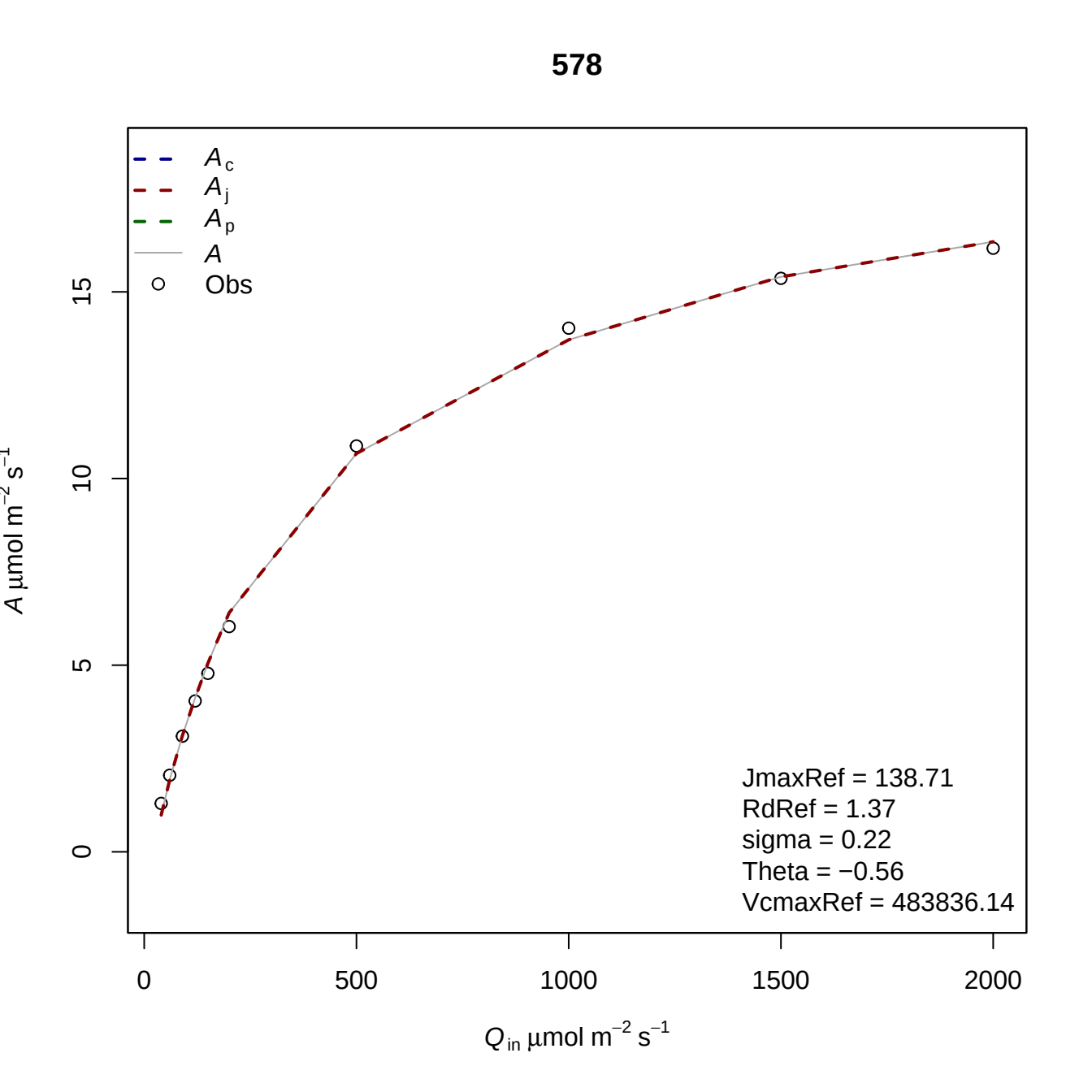

573
A_c
A_j
A_p
A
Obs
JmaxRef = 186.95
RdRef = 1.75
sigma = 0.1
Theta = 0.73
VcmaxRef = 212.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

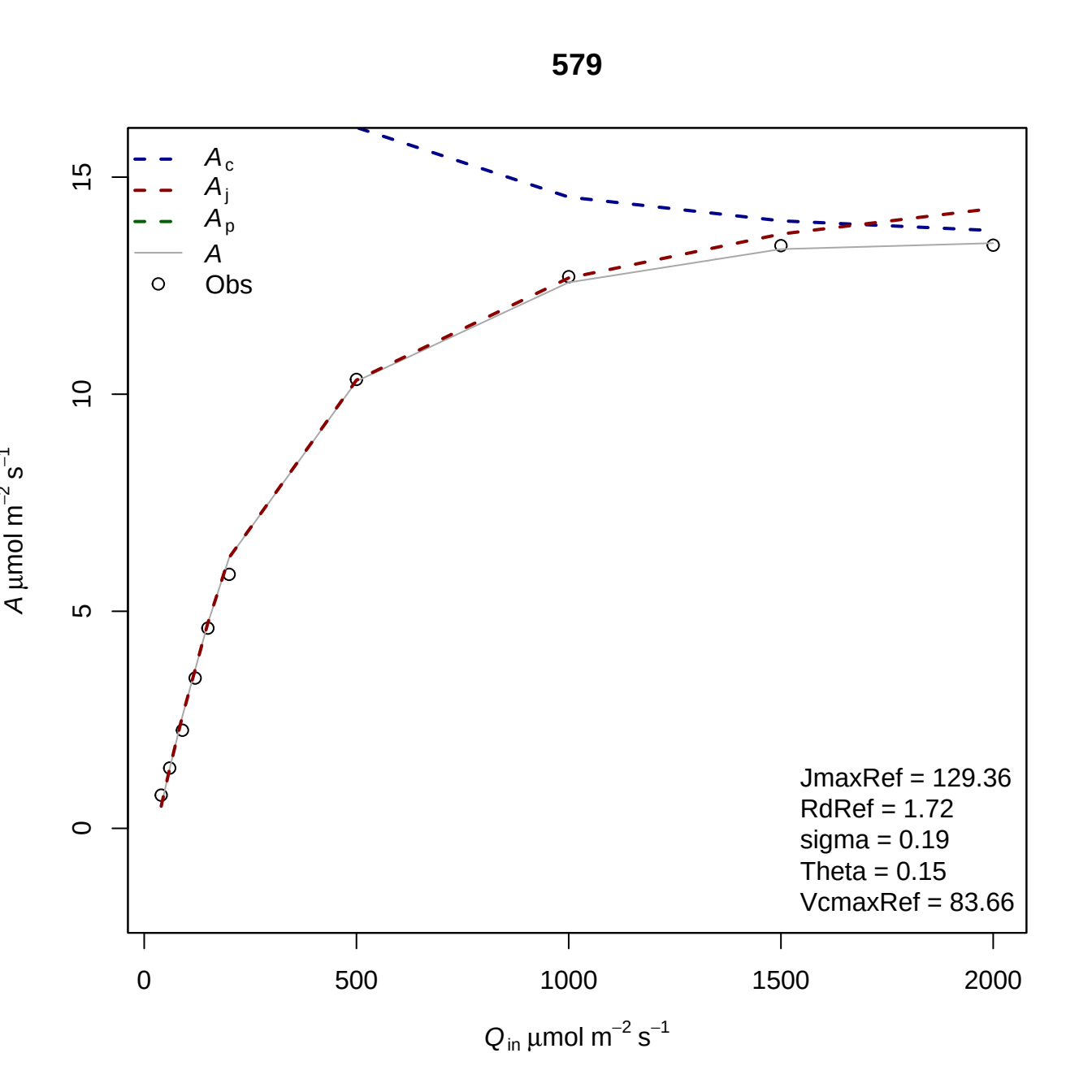

574
A_c
A_j
A_p
A
Obs
JmaxRef = 213.9
RdRef = 2.37
sigma = 0.08
Theta = 0.71
VcmaxRef = 240.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

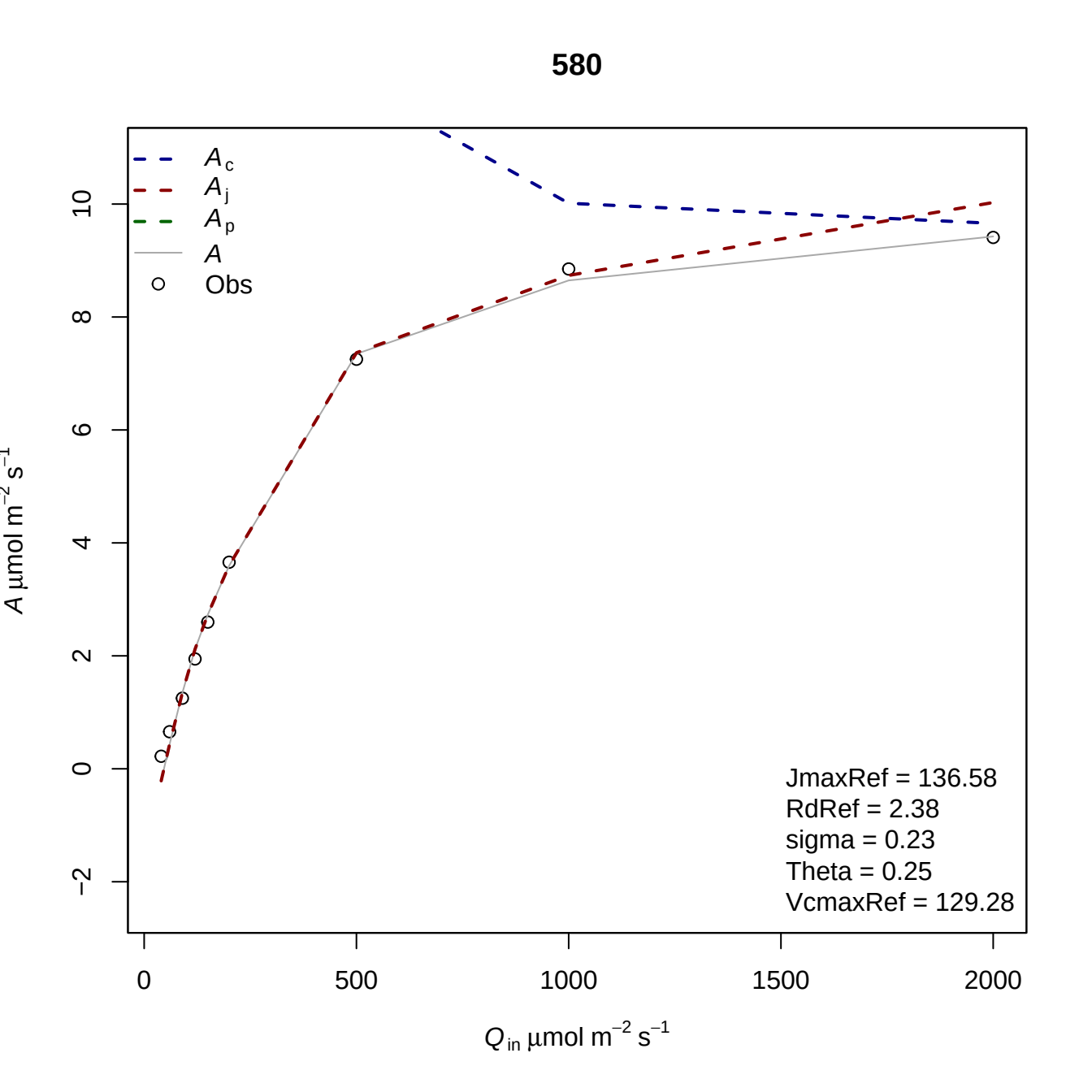

575
A_c
A_j
A_p
A
Obs
JmaxRef = 171.1
RdRef = 1.54
sigma = 0.26
Theta = −0.22
VcmaxRef = 59.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

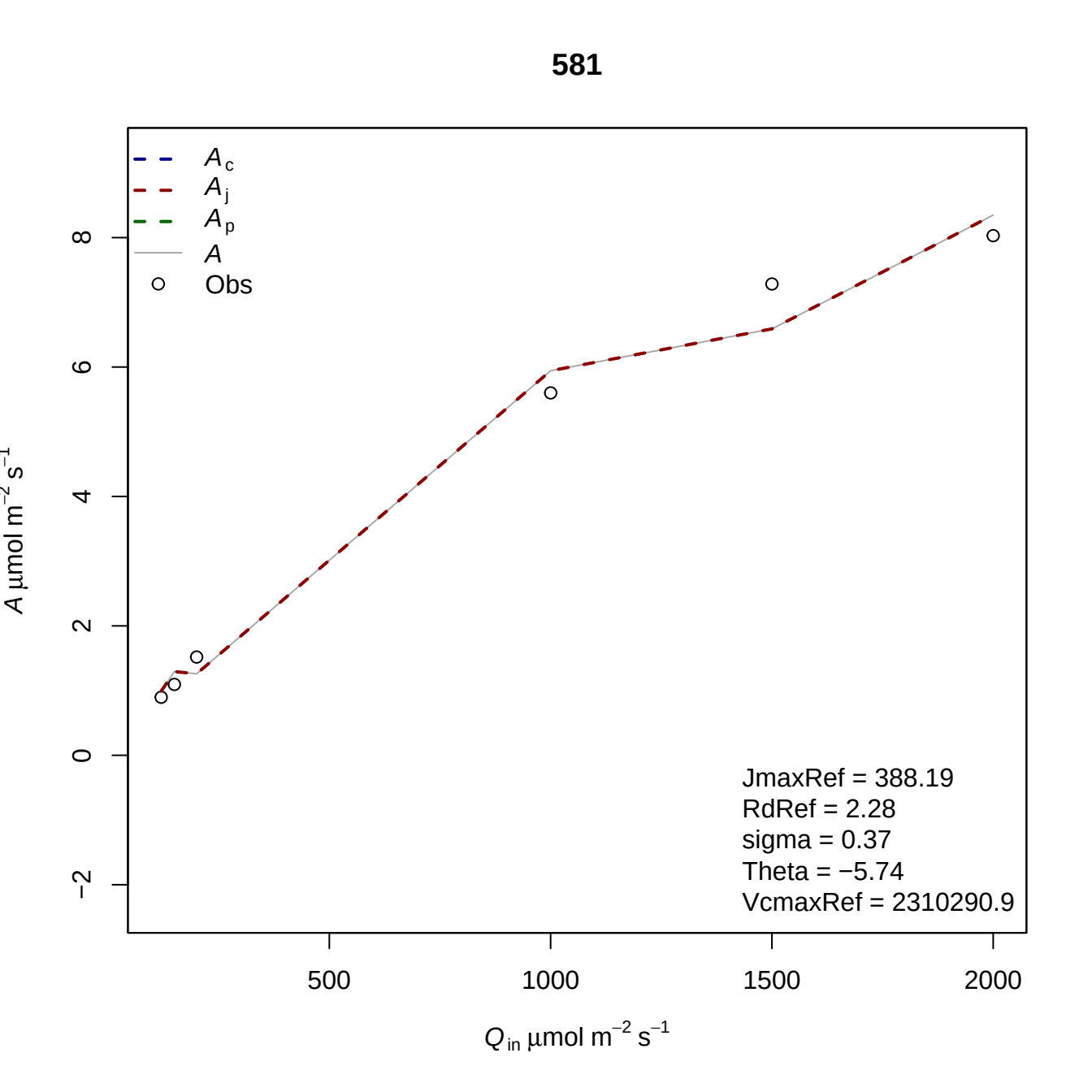

576
A_c
A_j
A_p
A
Obs
JmaxRef = 136.62
RdRef = 0.72
sigma = 0.24
Theta = −0.03
VcmaxRef = 105.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

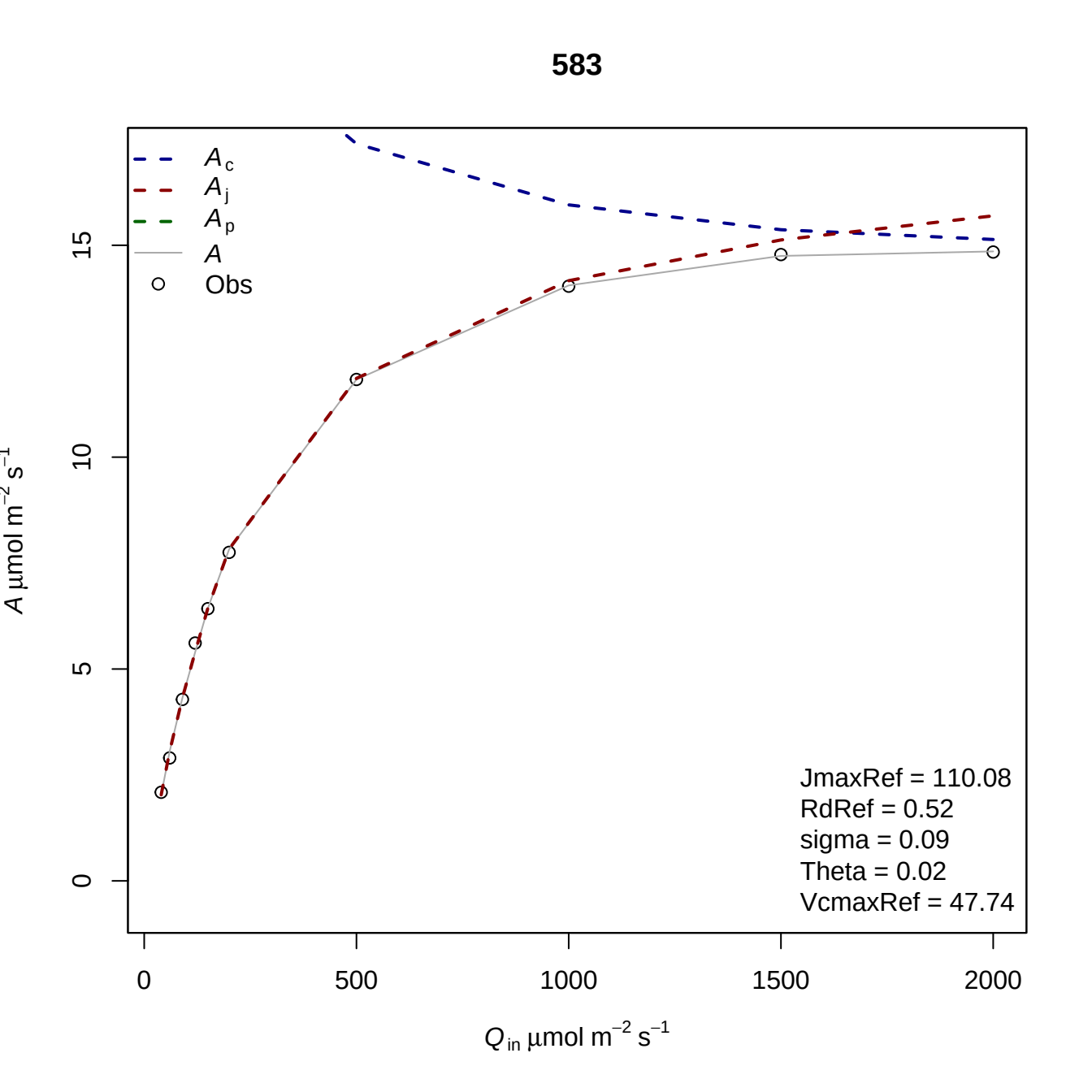

577
A_c
A_j
A_p
A
Obs
JmaxRef = 91.64
RdRef = 0.44
sigma = 0.1
Theta = −0.18
VcmaxRef = 10901.8
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

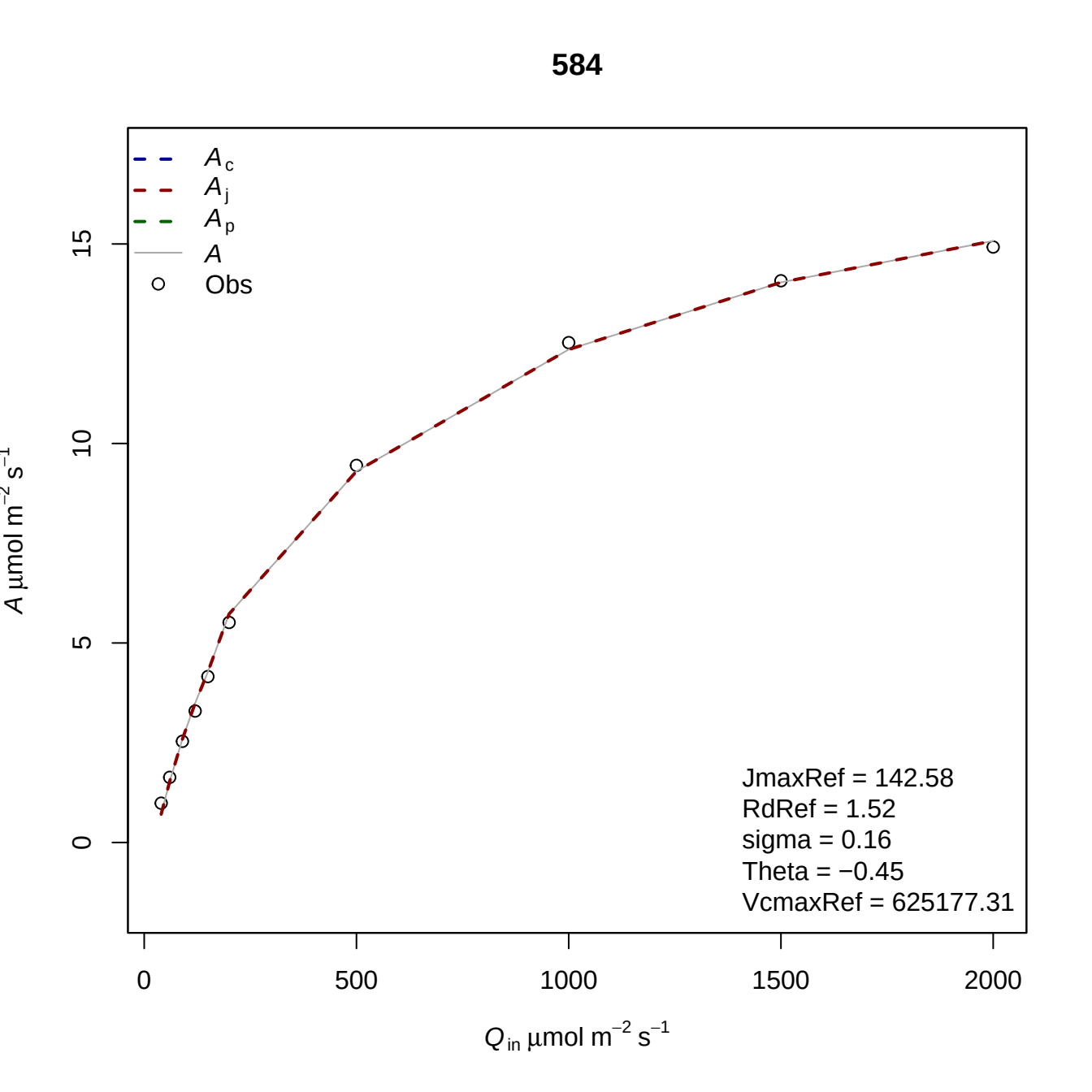

578
A_c
A_j
A_p
A
Obs
JmaxRef = 138.71
RdRef = 1.37
sigma = 0.22
Theta = −0.56
VcmaxRef = 483836.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

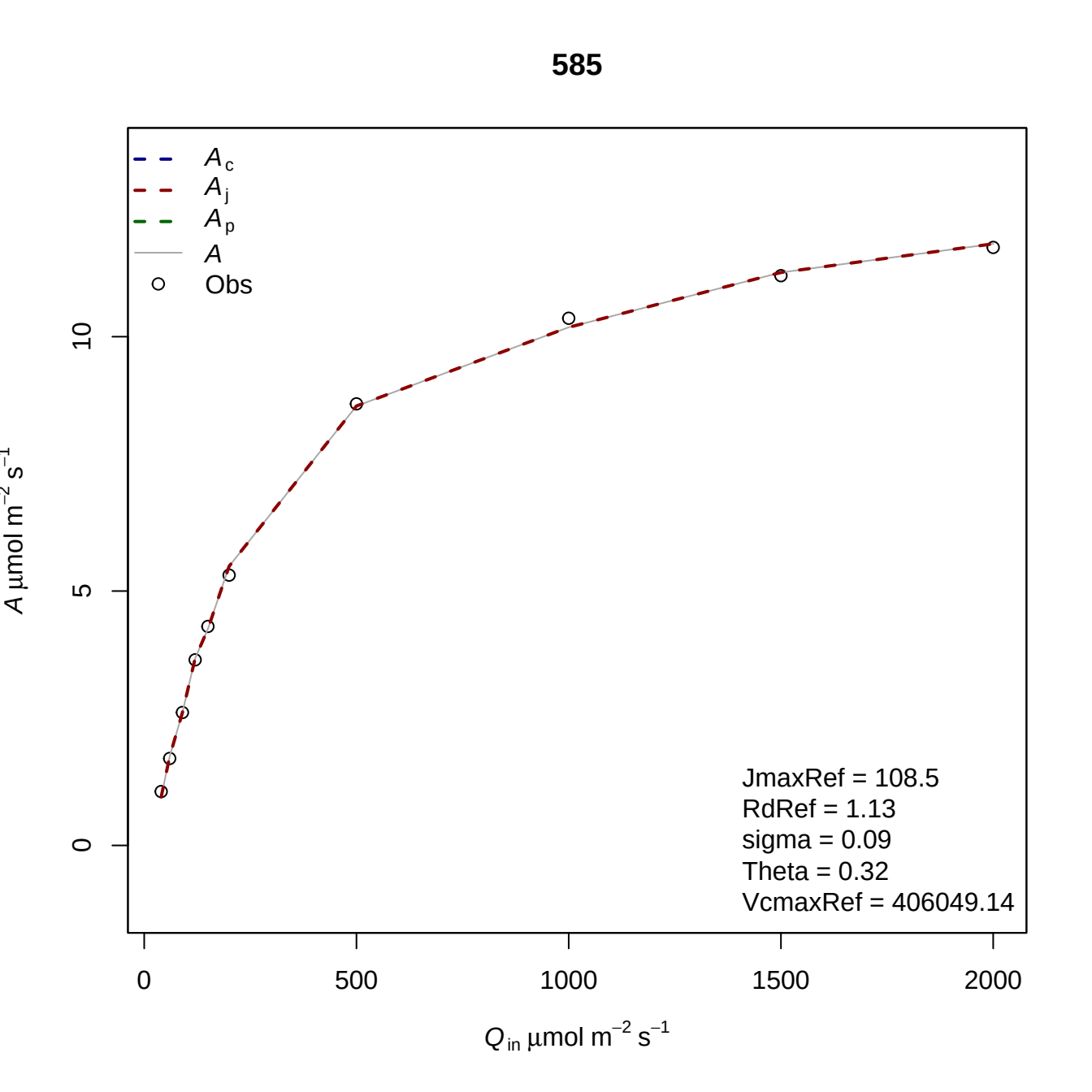

579
A_c
A_j
A_p
A
Obs
JmaxRef = 129.36
RdRef = 1.72
sigma = 0.19
Theta = 0.15
VcmaxRef = 83.66
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

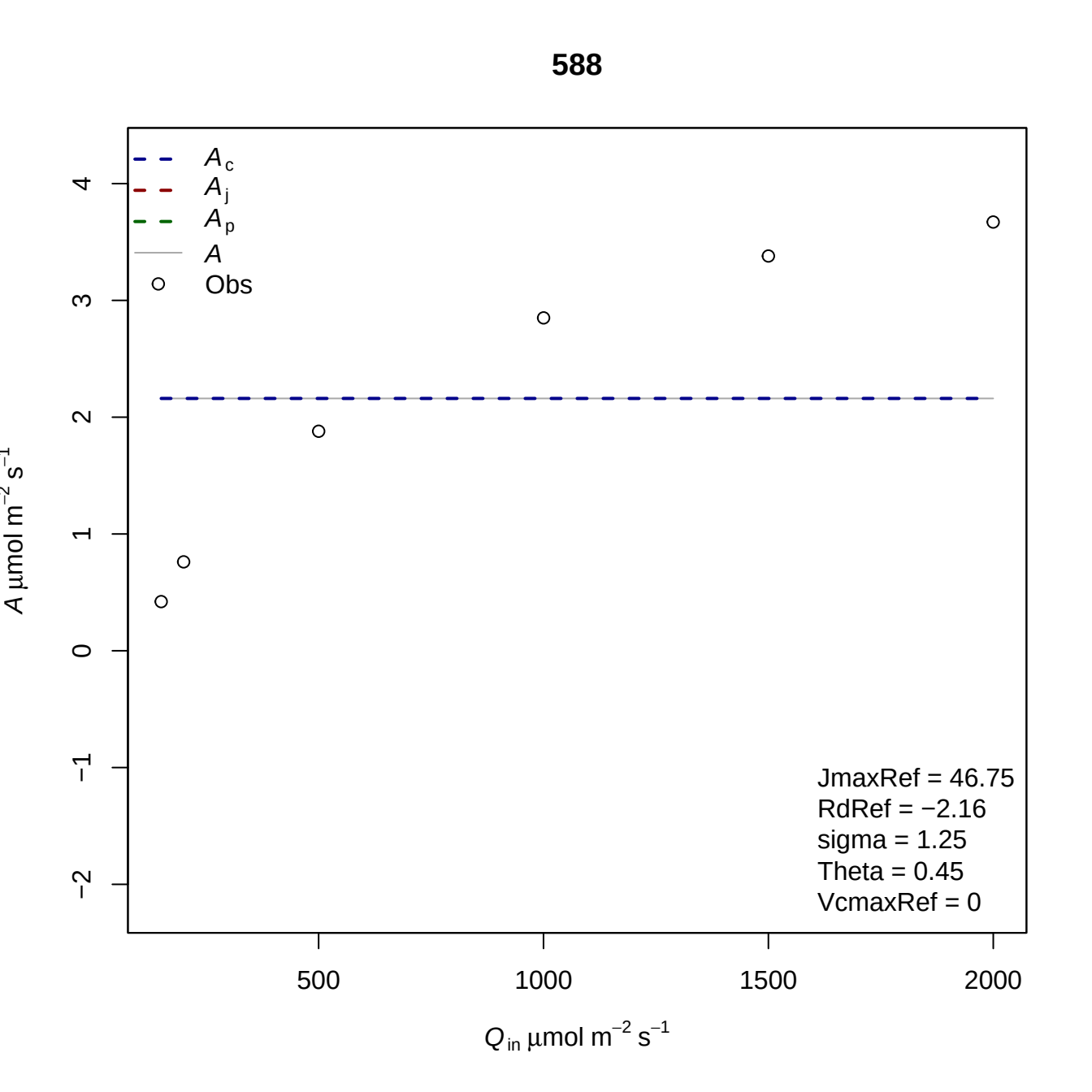

580
A_c
A_j
A_p
A
Obs
JmaxRef = 136.58
RdRef = 2.38
sigma = 0.23
Theta = 0.25
VcmaxRef = 129.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

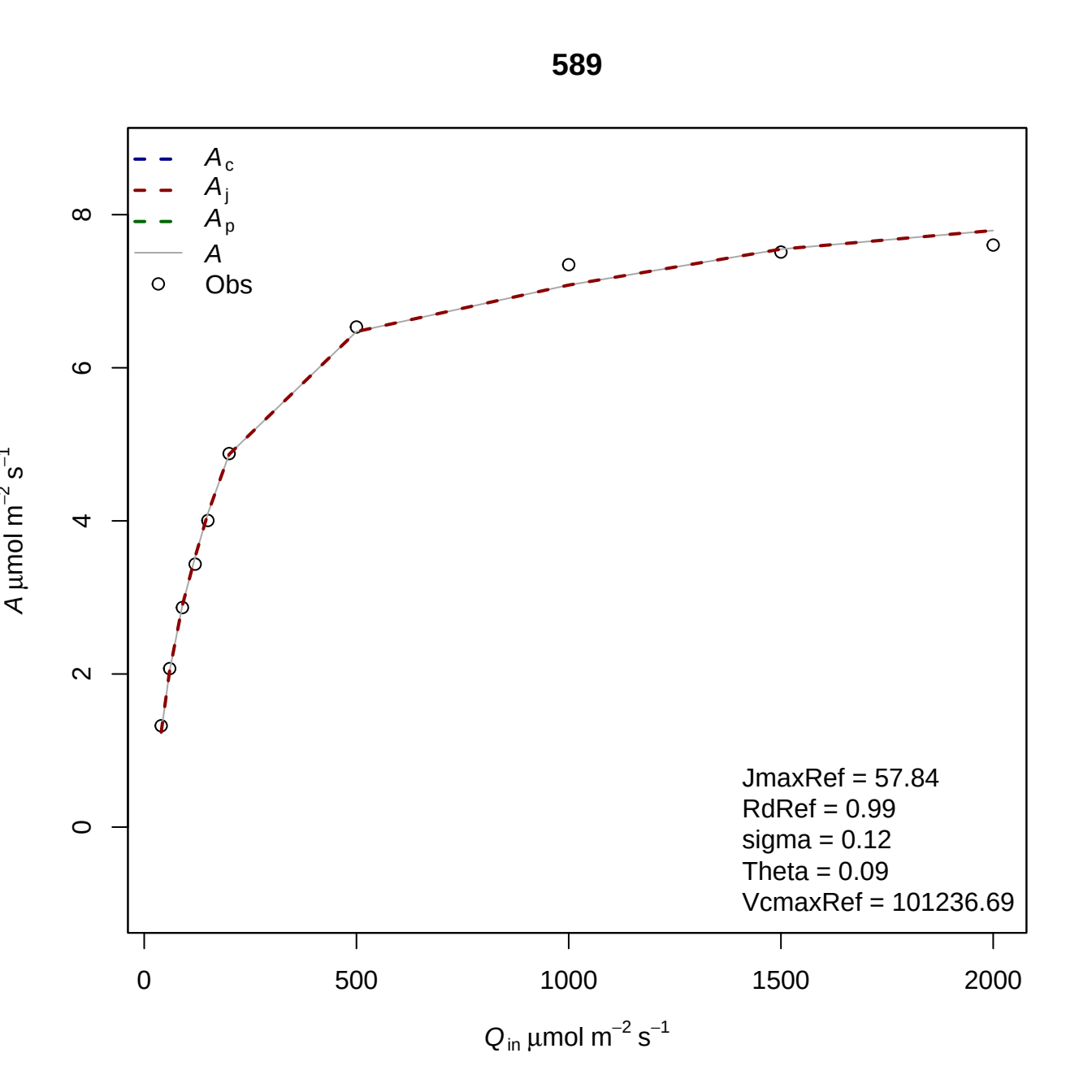

581
A_c
A_j
A_p
A
Obs
JmaxRef = 388.19
RdRef = 2.28
sigma = 0.37
Theta = −5.74
VcmaxRef = 2310290.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
500
1000
1500
2000

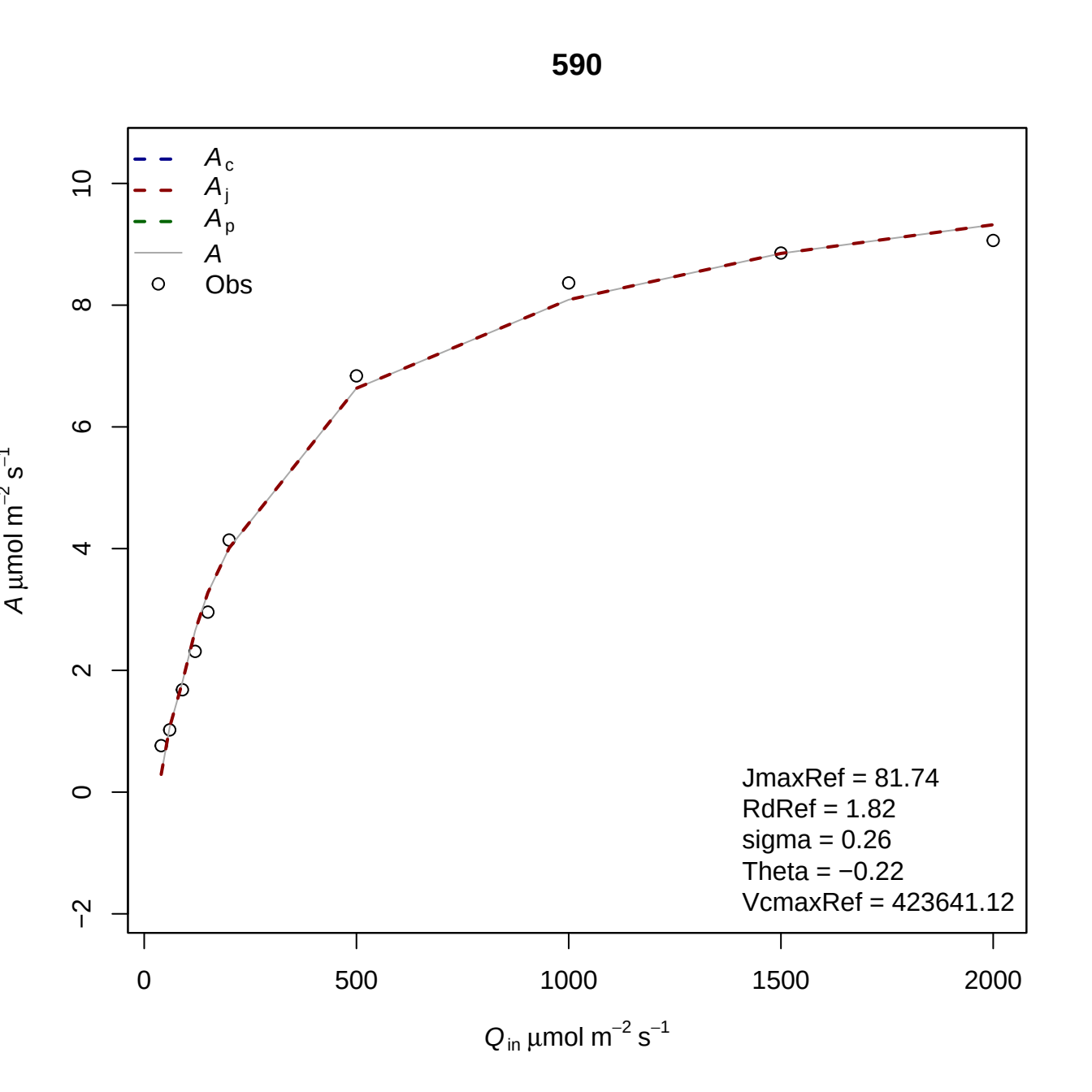

583
A_c
A_j
A_p
A
Obs
JmaxRef = 110.08
RdRef = 0.52
sigma = 0.09
Theta = 0.02
VcmaxRef = 47.74
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

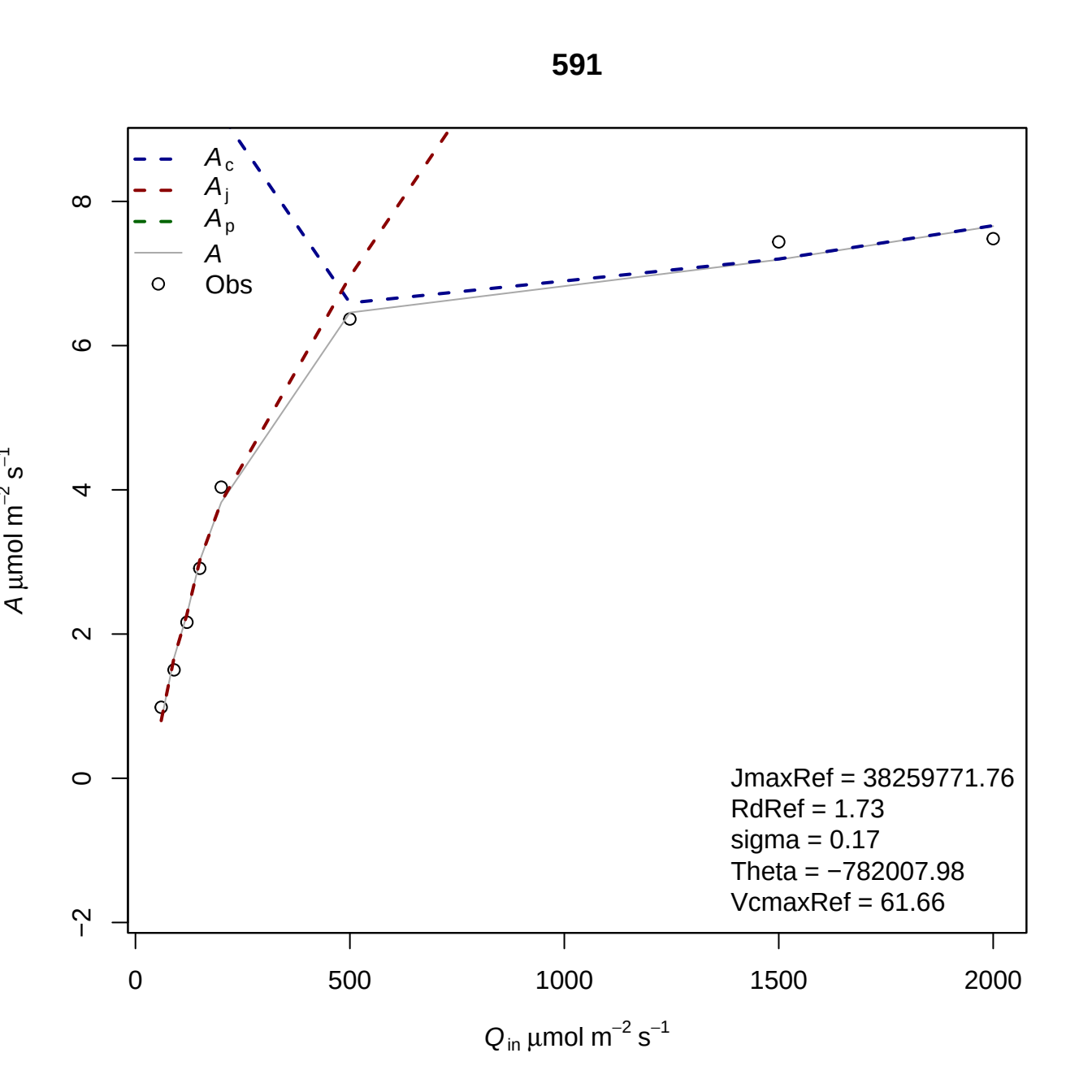

584
A_c
A_j
A_p
A
Obs
JmaxRef = 142.58
RdRef = 1.52
sigma = 0.16
Theta = −0.45
VcmaxRef = 625177.31
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

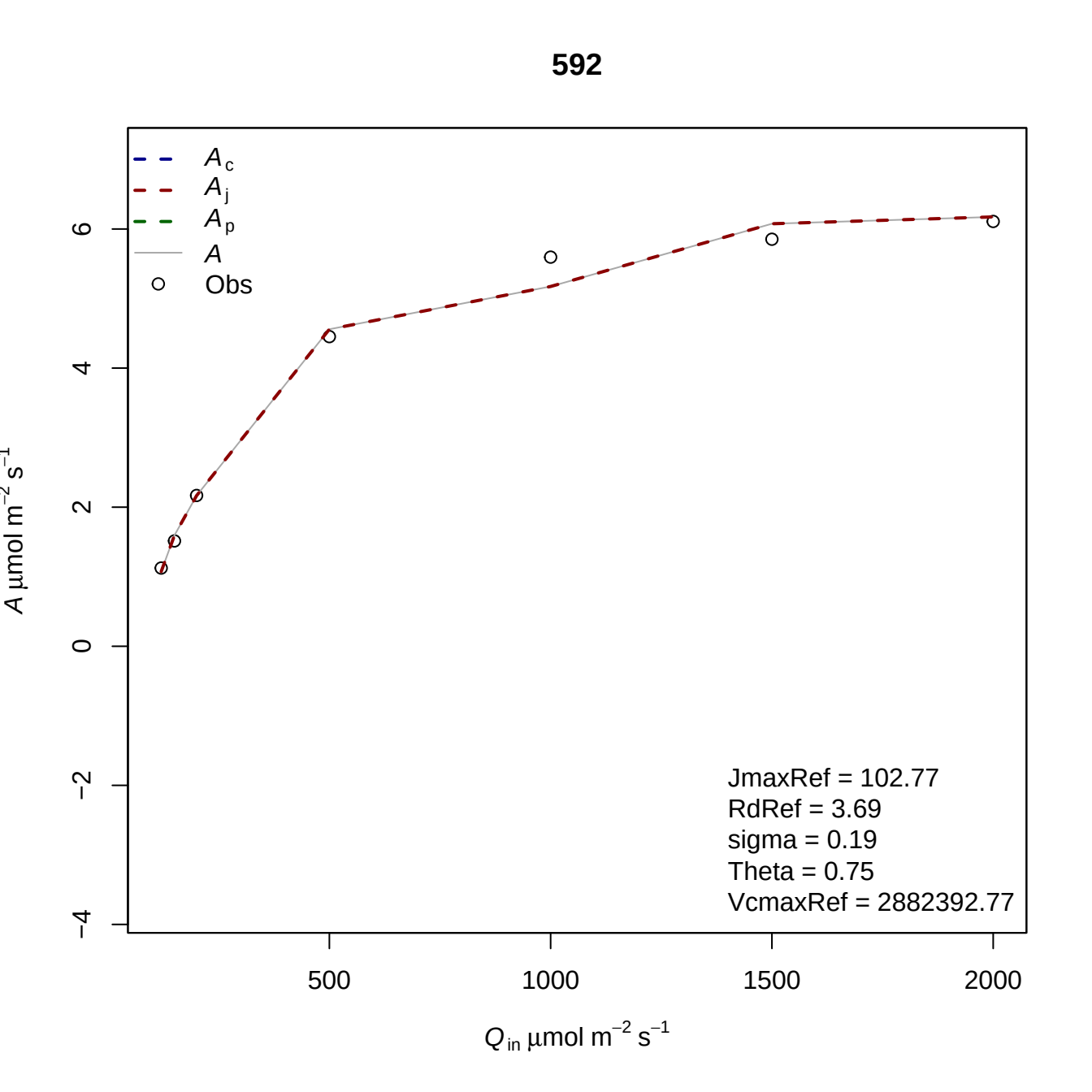

585
A_c
A_j
A_p
A
Obs
JmaxRef = 108.5
RdRef = 1.13
sigma = 0.09
Theta = 0.32
VcmaxRef = 406049.14
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

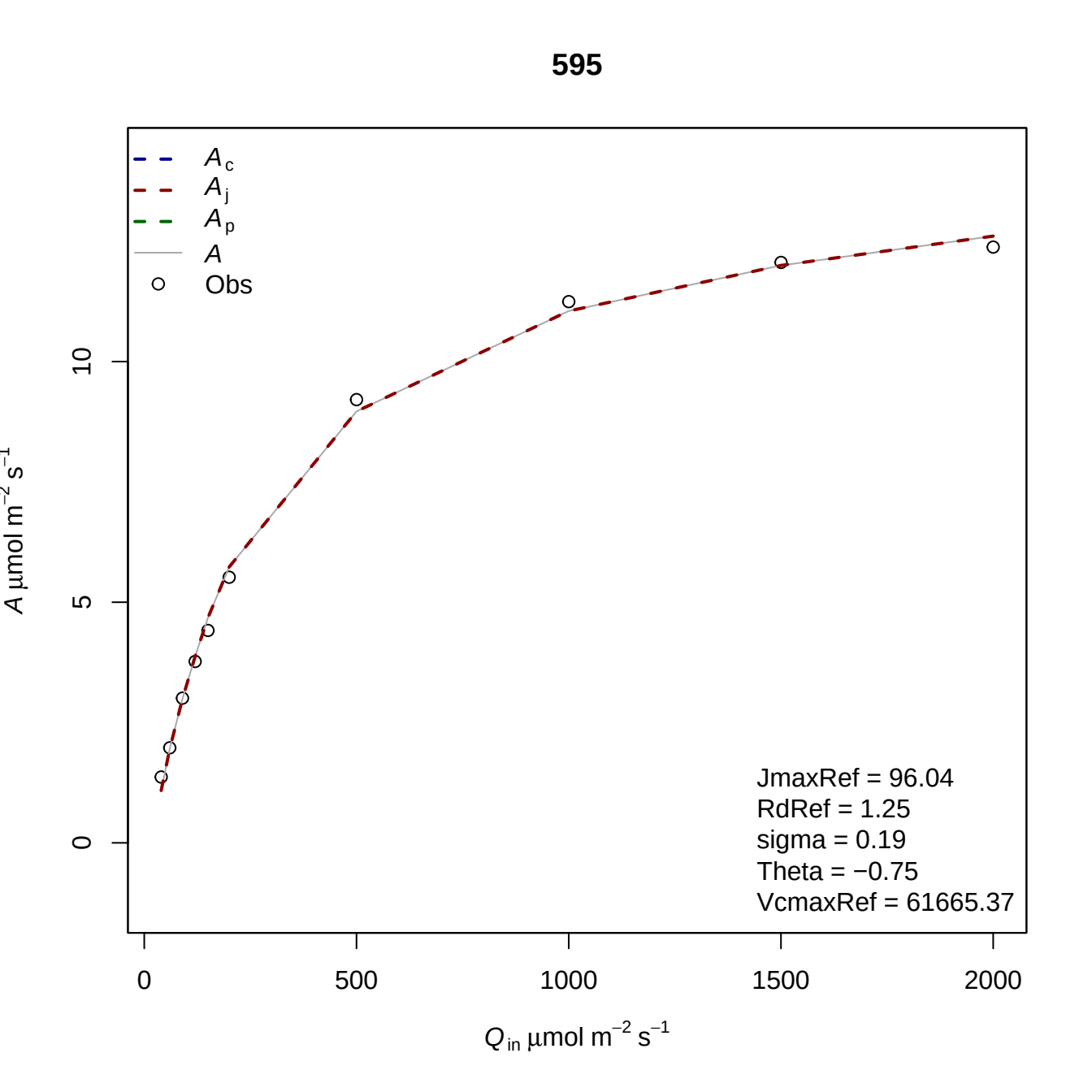

588
A_c
A_j
A_p
A
Obs
JmaxRef = 46.75
RdRef = −2.16
sigma = 1.25
Theta = 0.45
VcmaxRef = 0
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
500
1000
1500
2000

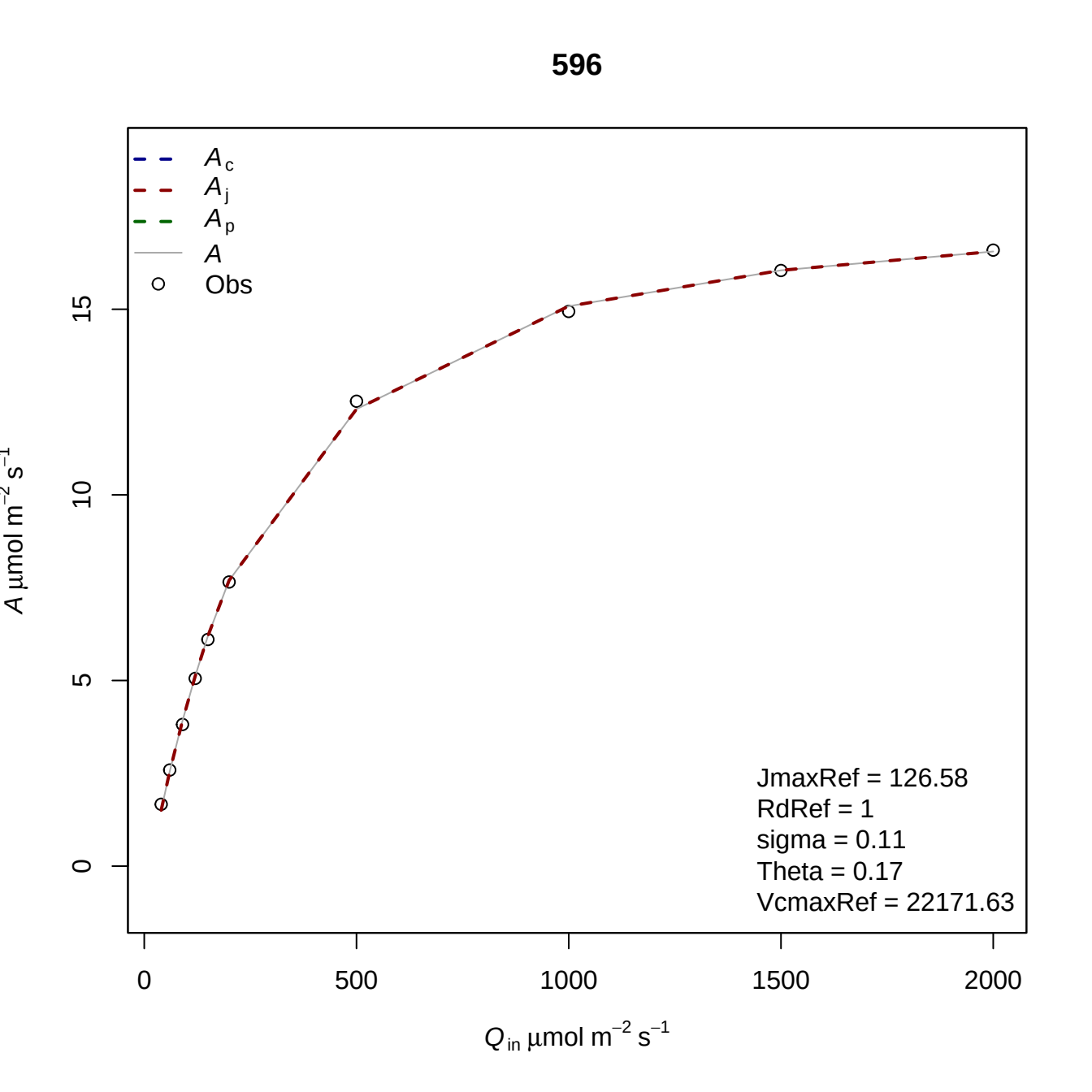

589
A_c
A_j
A_p
A
Obs
JmaxRef = 57.84
RdRef = 0.99
sigma = 0.12
Theta = 0.09
VcmaxRef = 101236.69
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

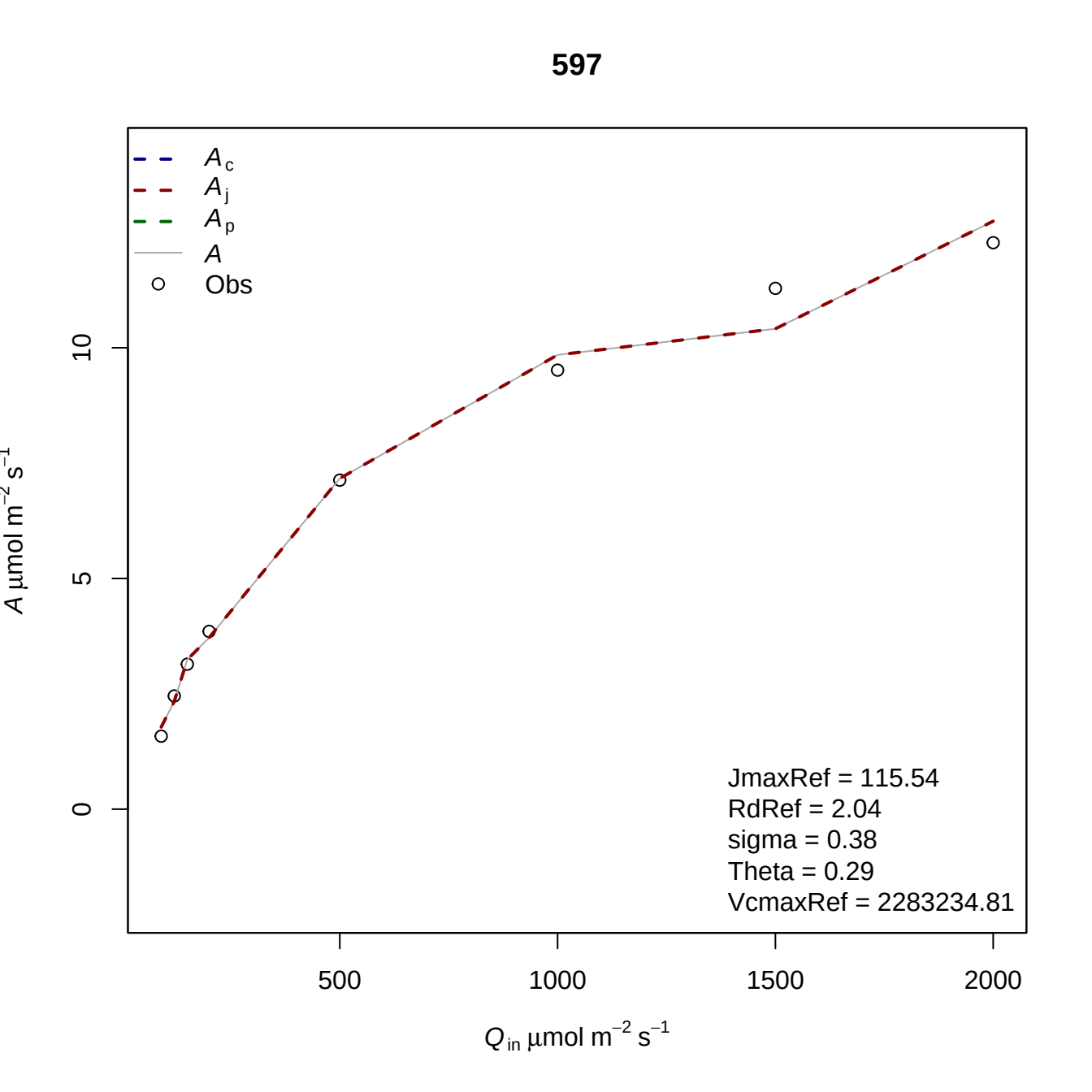

590
A_c
A_j
A_p
A
Obs
JmaxRef = 81.74
RdRef = 1.82
sigma = 0.26
Theta = −0.22
VcmaxRef = 423641.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

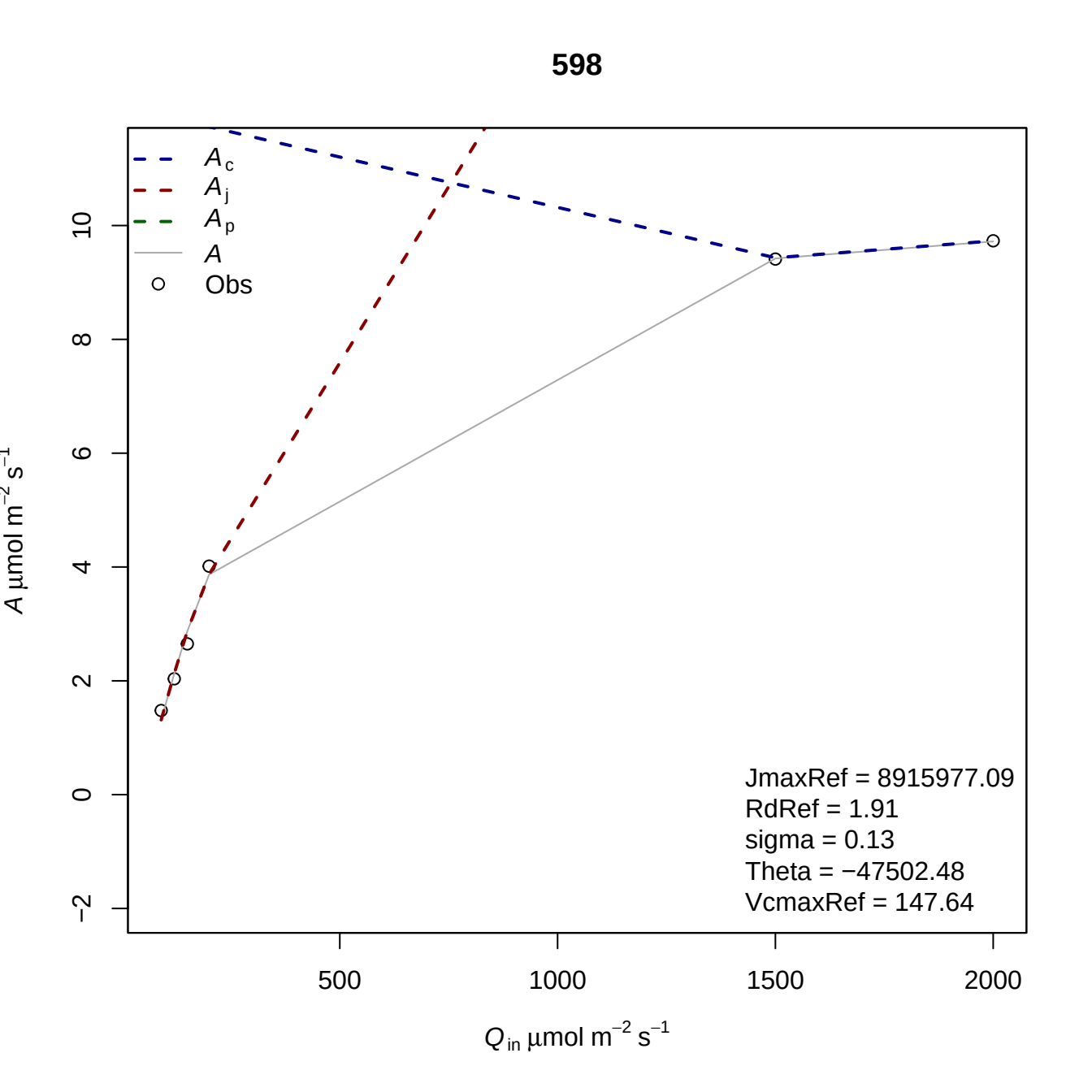

591
A_c
A_j
A_p
A
Obs
JmaxRef = 38259771.76
RdRef = 1.73
sigma = 0.17
Theta = −782007.98
VcmaxRef = 61.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

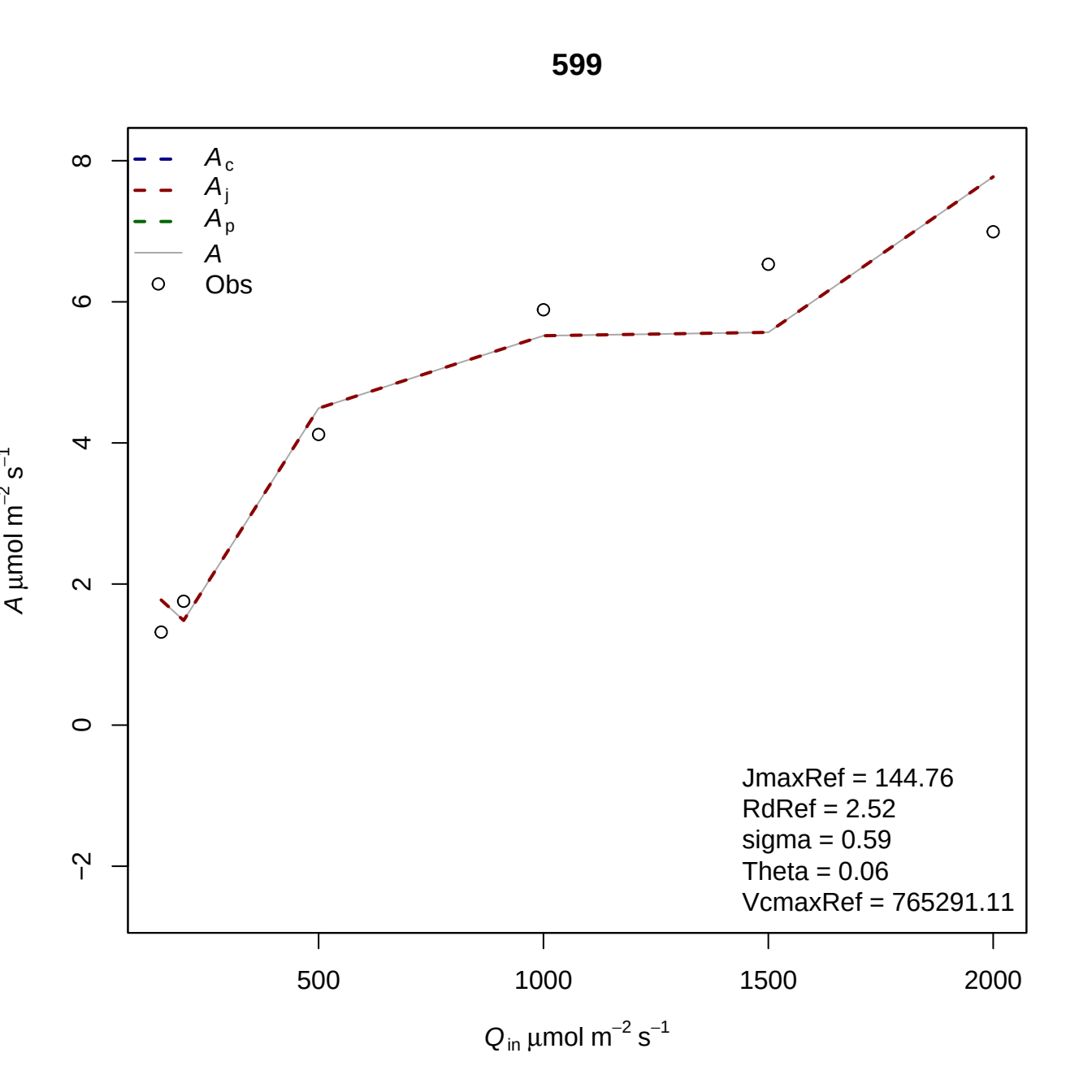

592
A_c
A_j
A_p
A
Obs
JmaxRef = 102.77
RdRef = 3.69
sigma = 0.19
Theta = 0.75
VcmaxRef = 2882392.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
500
1000
1500
2000

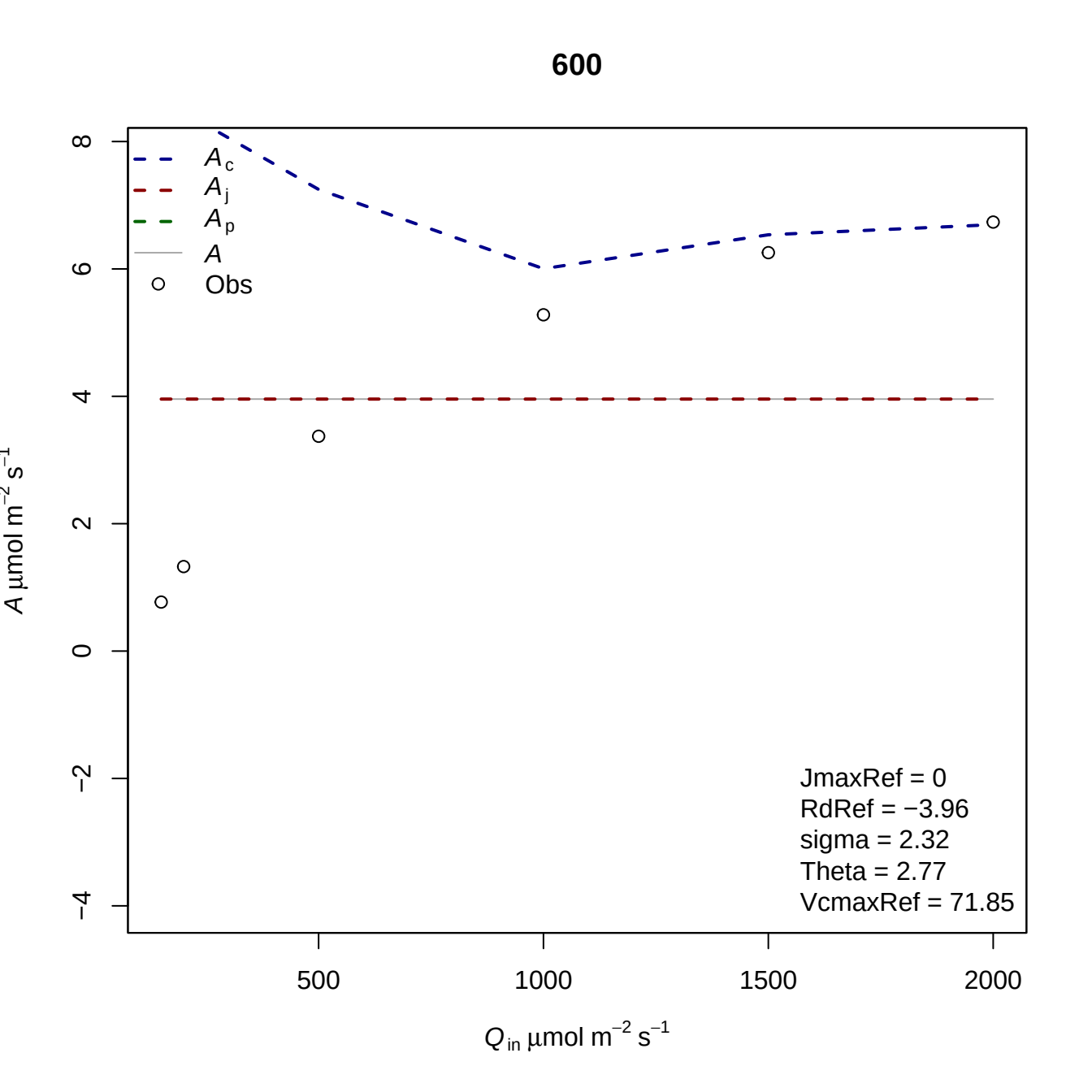

595
A_c
A_j
A_p
A
Obs
JmaxRef = 96.04
RdRef = 1.25
sigma = 0.19
Theta = −0.75
VcmaxRef = 61665.37
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

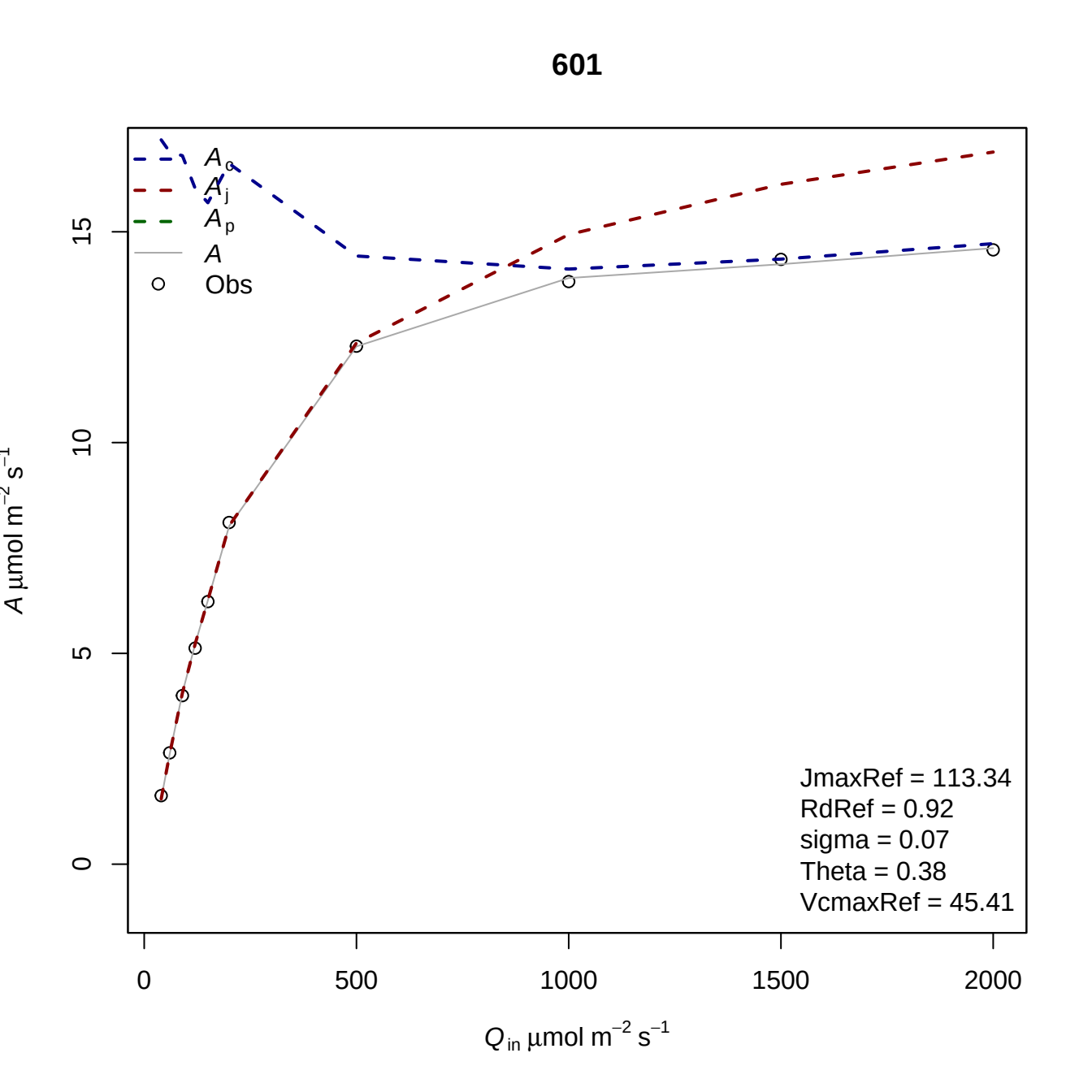

596
A_c
A_j
A_p
A
Obs
JmaxRef = 126.58
RdRef = 1
sigma = 0.11
Theta = 0.17
VcmaxRef = 22171.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

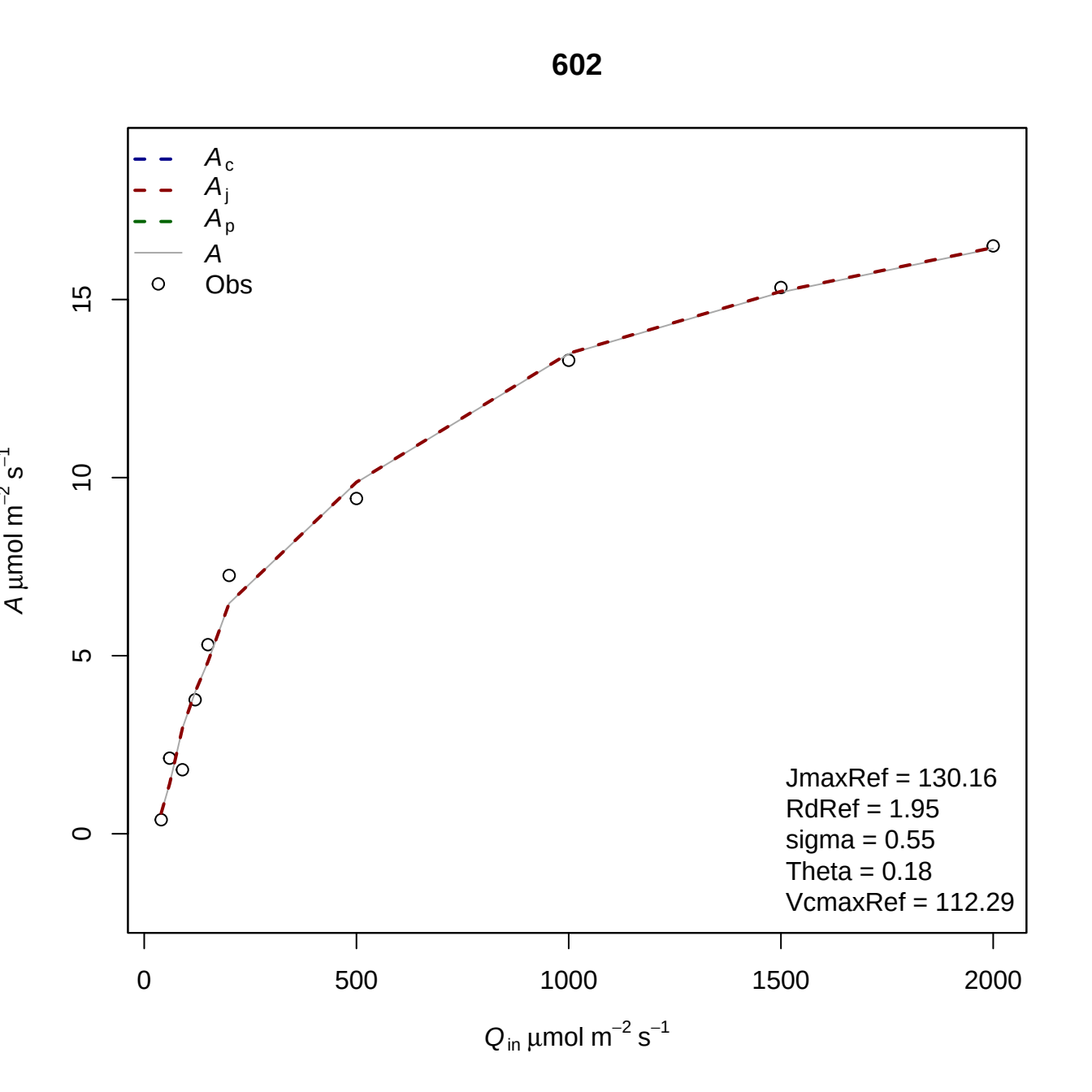

597
A_c
A_j
A_p
A
Obs
JmaxRef = 115.54
RdRef = 2.04
sigma = 0.38
Theta = 0.29
VcmaxRef = 2283234.81
0
5
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

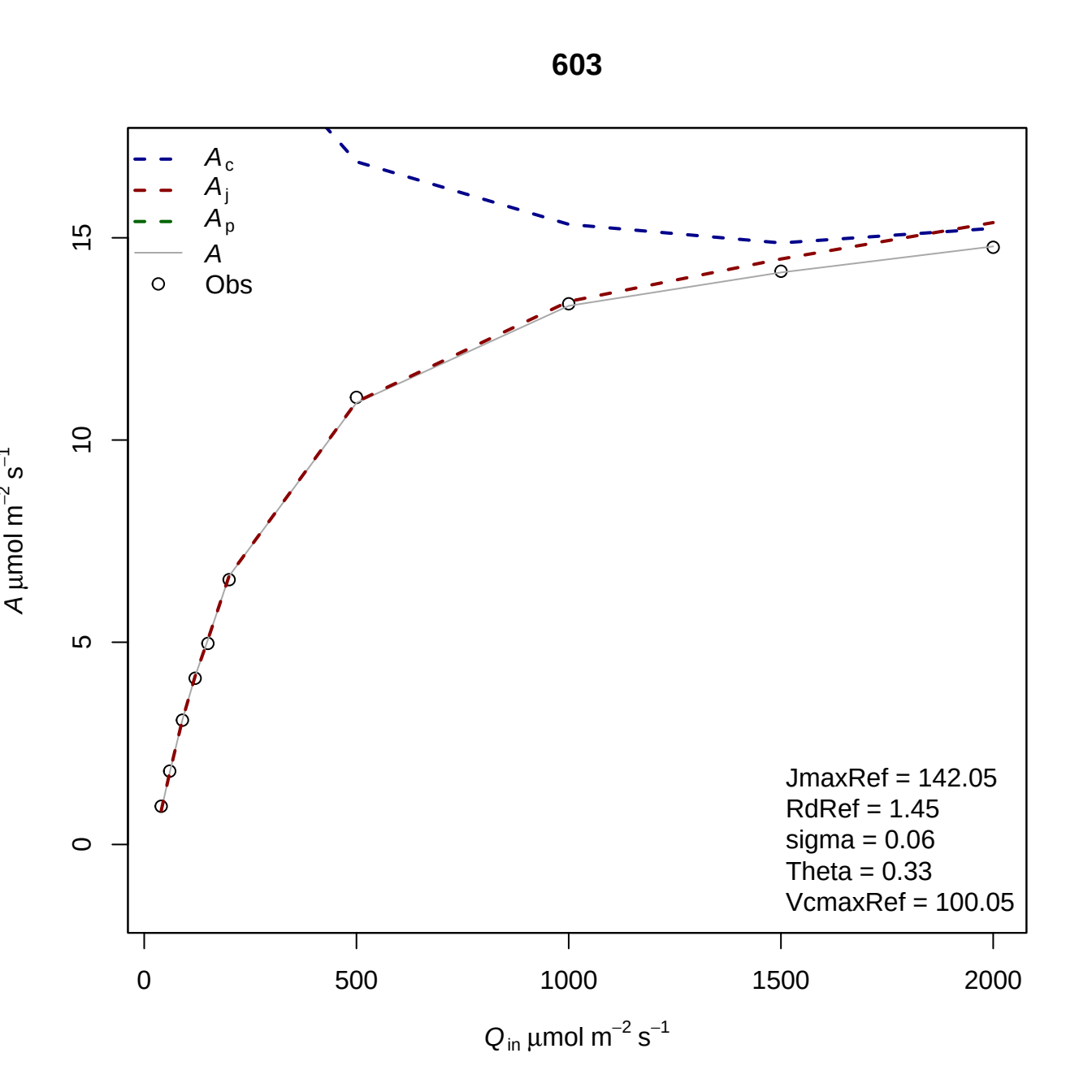

598
A_c
A_j
A_p
A
Obs
JmaxRef = 8915977.09
RdRef = 1.91
sigma = 0.13
Theta = −47502.48
VcmaxRef = 147.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

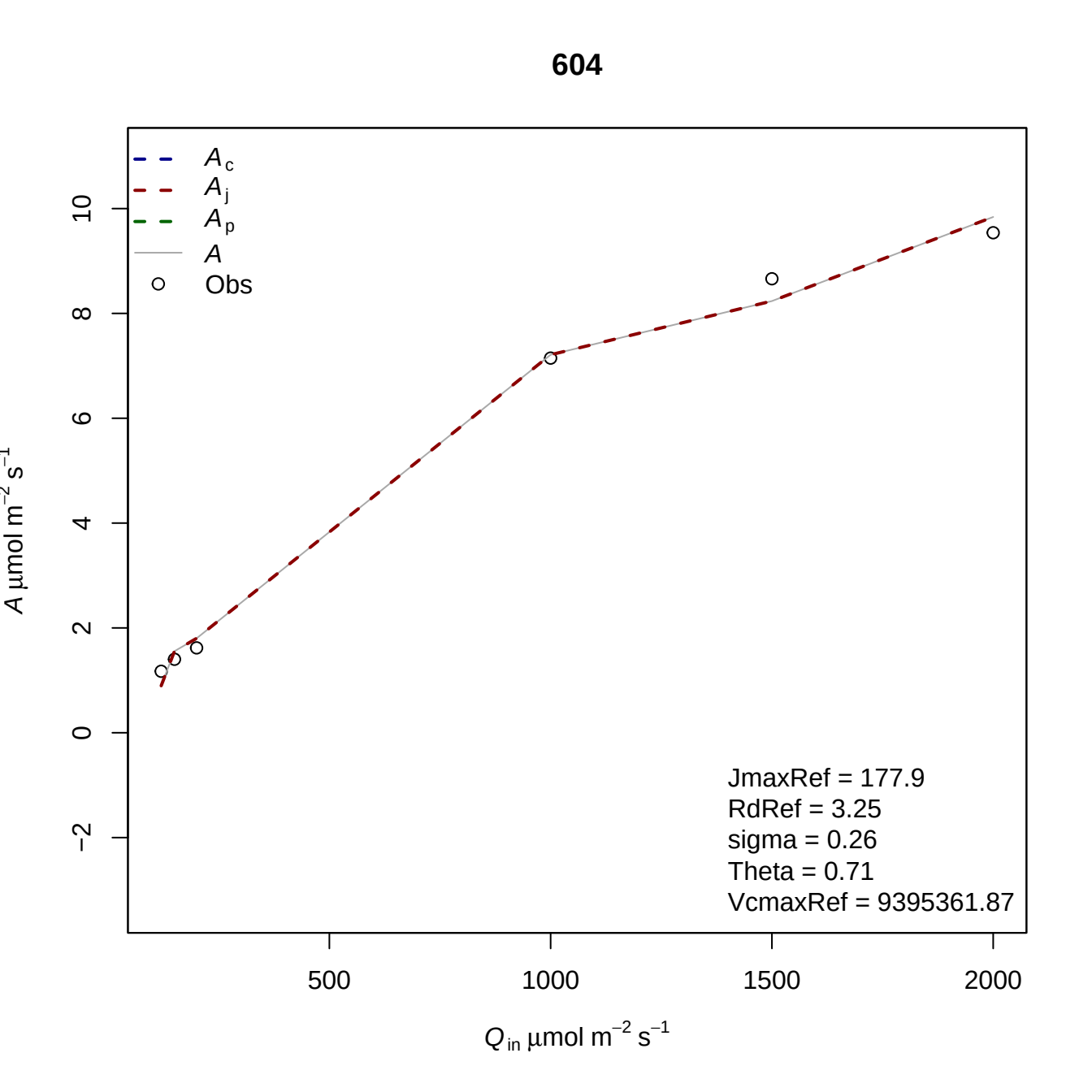

599
A_c
A_j
A_p
A
Obs
JmaxRef = 144.76
RdRef = 2.52
sigma = 0.59
Theta = 0.06
VcmaxRef = 765291.11
8
6
4
2
0
−2
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

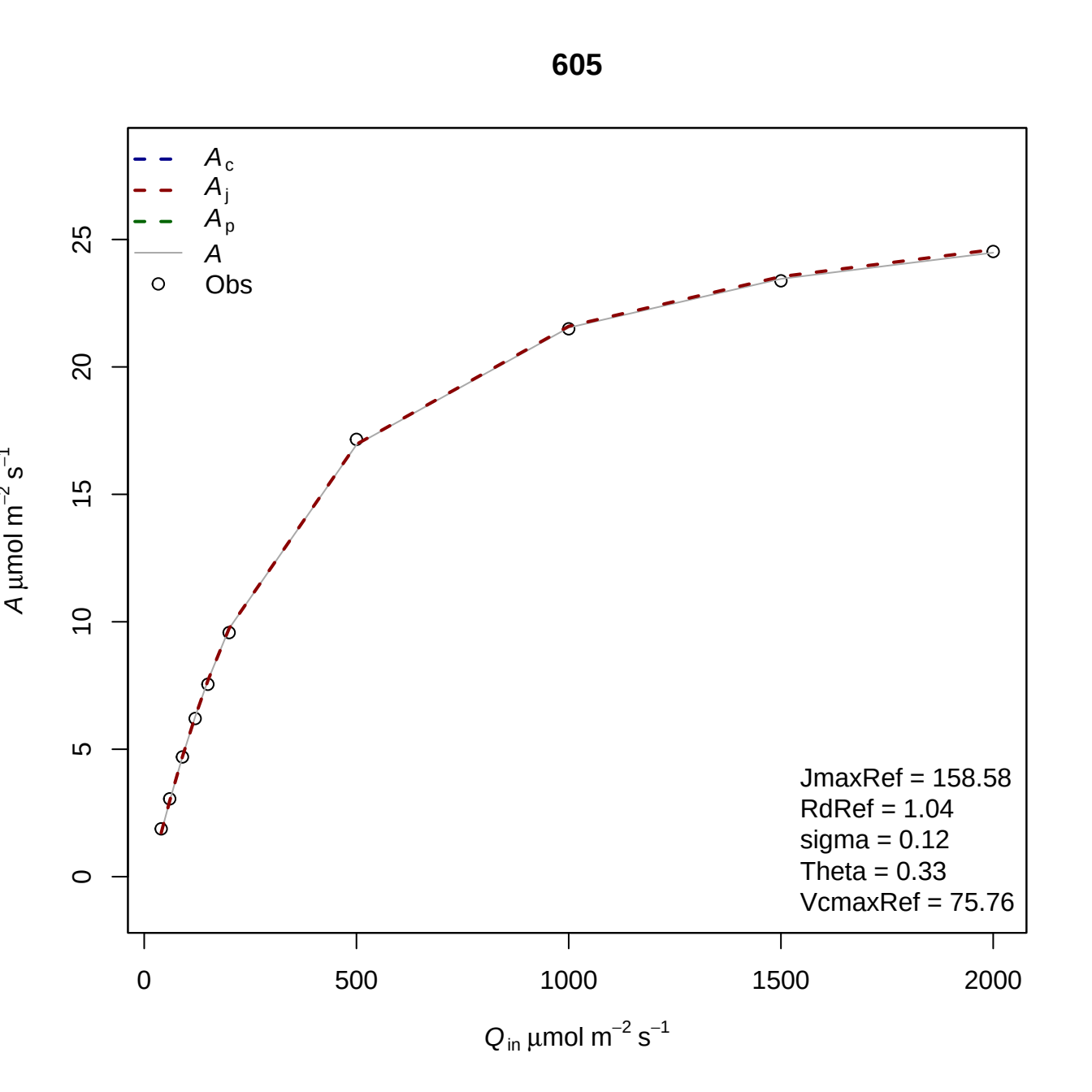

600
A_c
A_j
A_p
A
Obs
JmaxRef = 0
RdRef = −3.96
sigma = 2.32
Theta = 2.77
VcmaxRef = 71.85
8
6
4
2
0
−2
−4
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

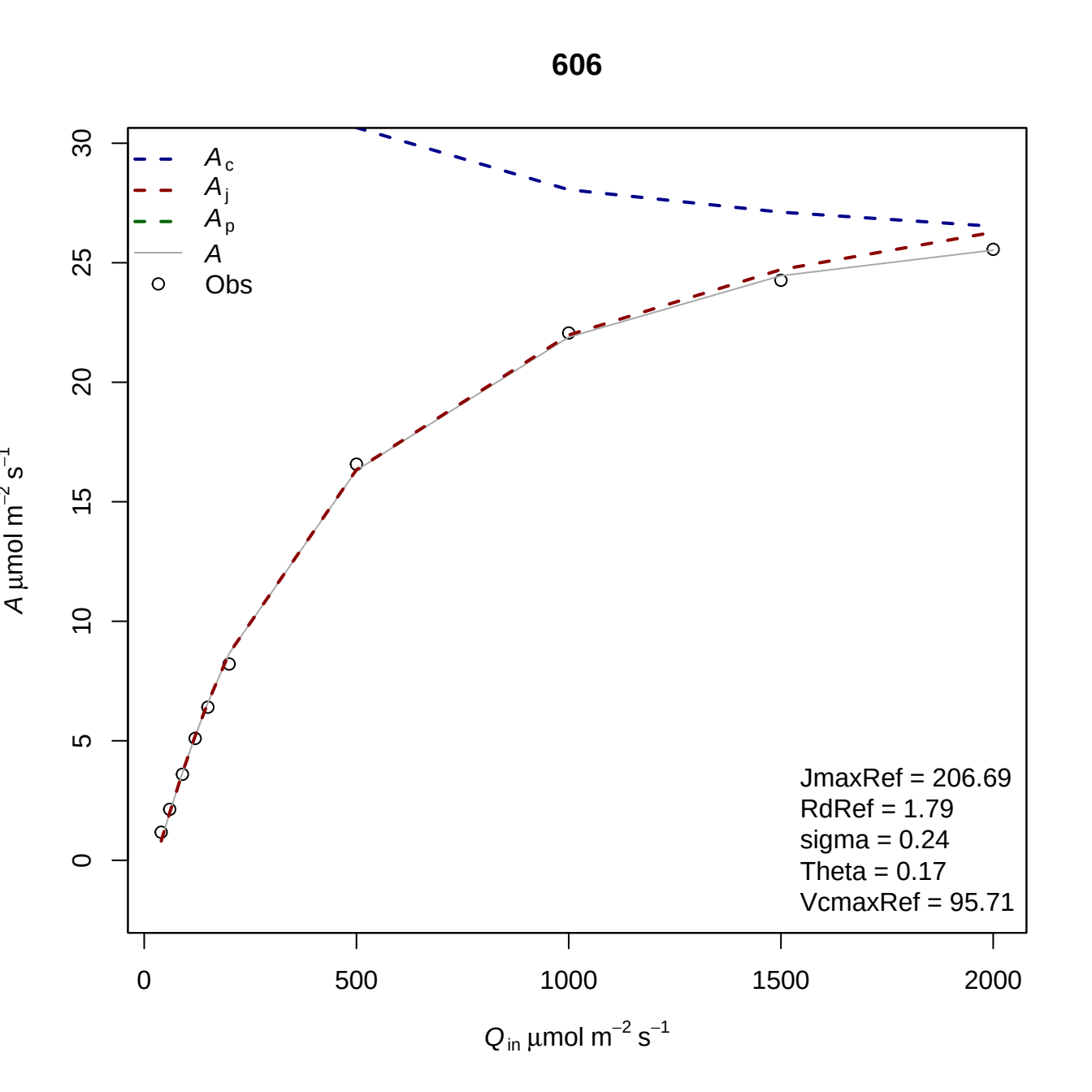

601
A_c
A_j
A_p
A
Obs
JmaxRef = 113.34
RdRef = 0.92
sigma = 0.07
Theta = 0.38
VcmaxRef = 45.41
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

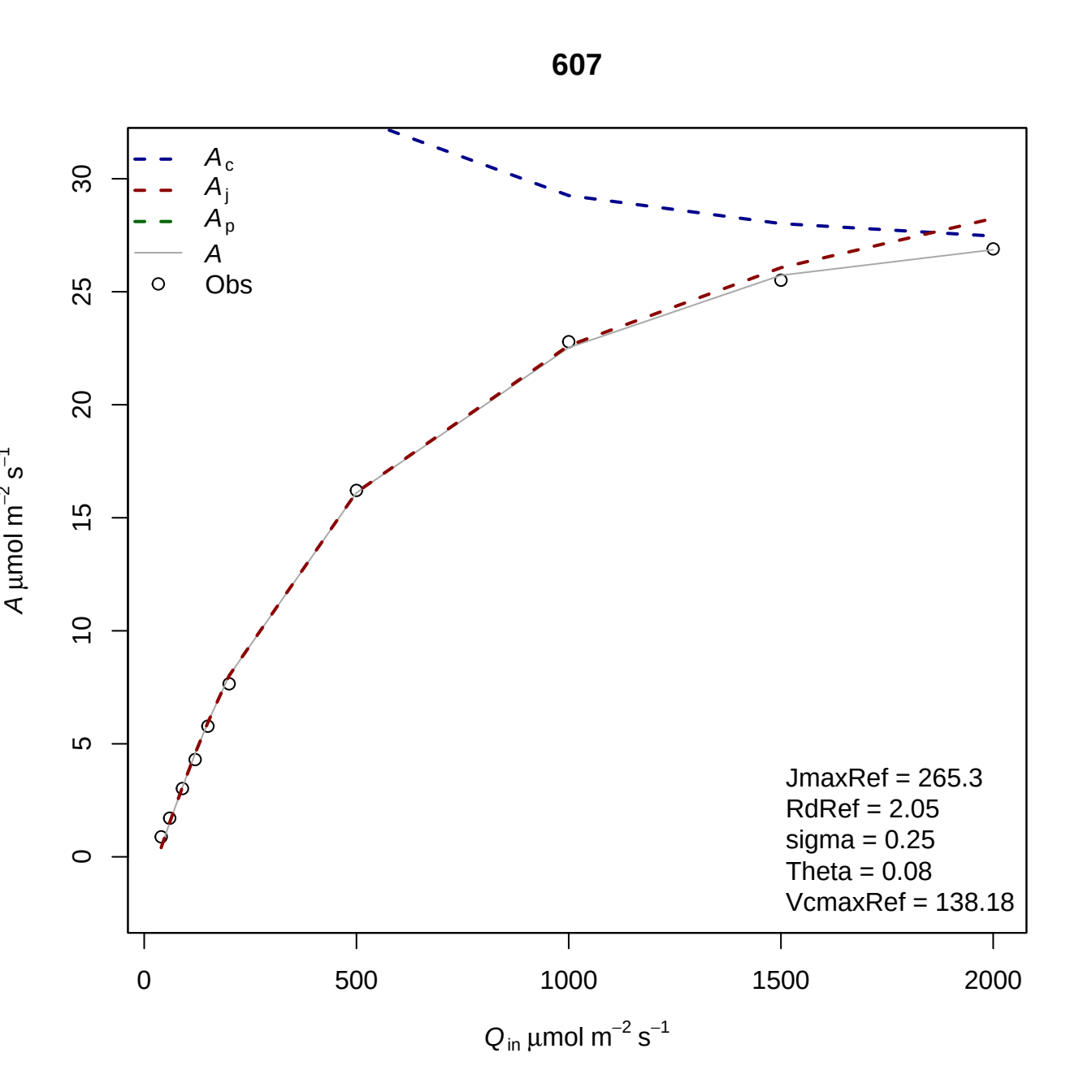

602
A_c
A_j
A_p
A
Obs
JmaxRef = 130.16
RdRef = 1.95
sigma = 0.55
Theta = 0.18
VcmaxRef = 112.29
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

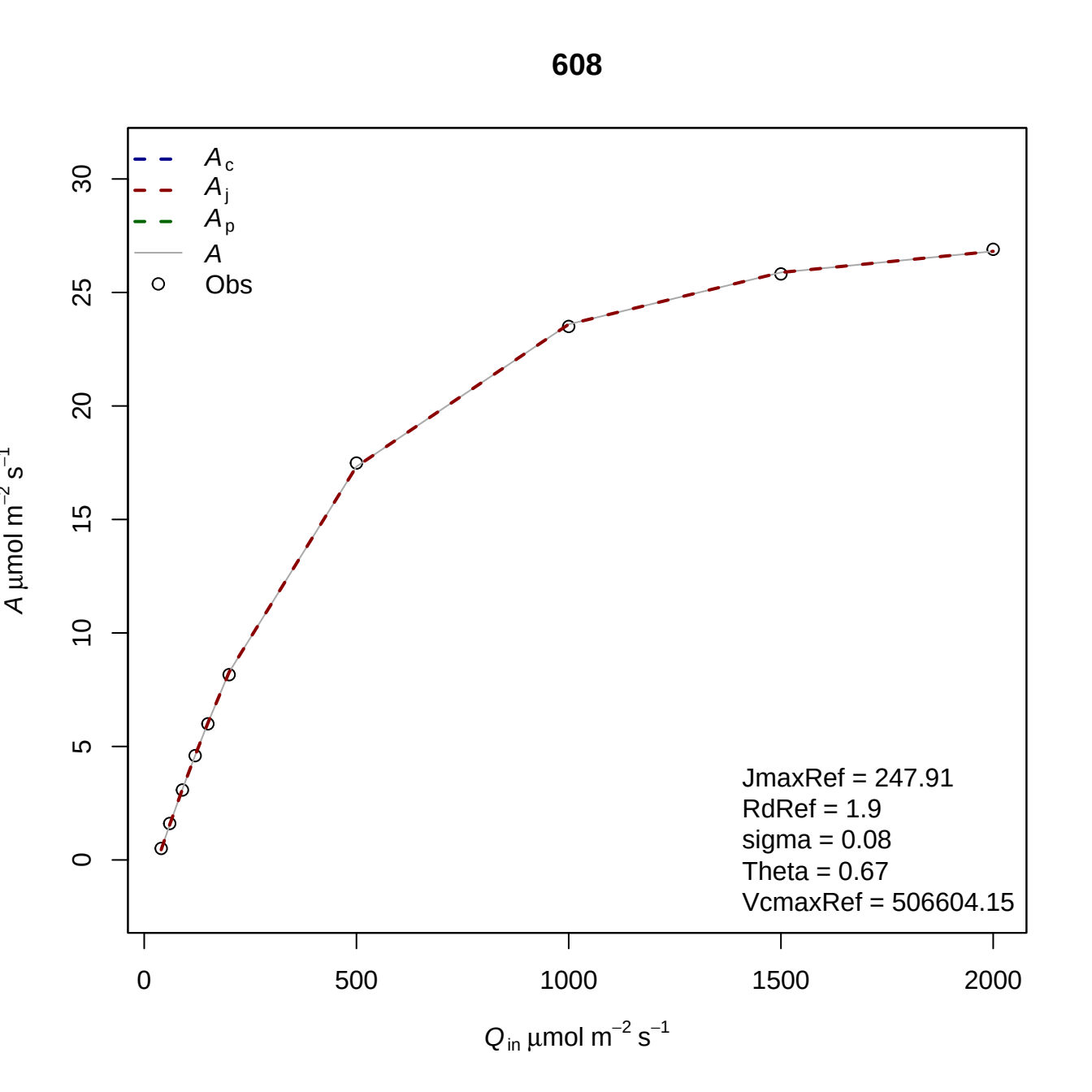

603
A_c
A_j
A_p
A
Obs
JmaxRef = 142.05
RdRef = 1.45
sigma = 0.06
Theta = 0.33
VcmaxRef = 100.05
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

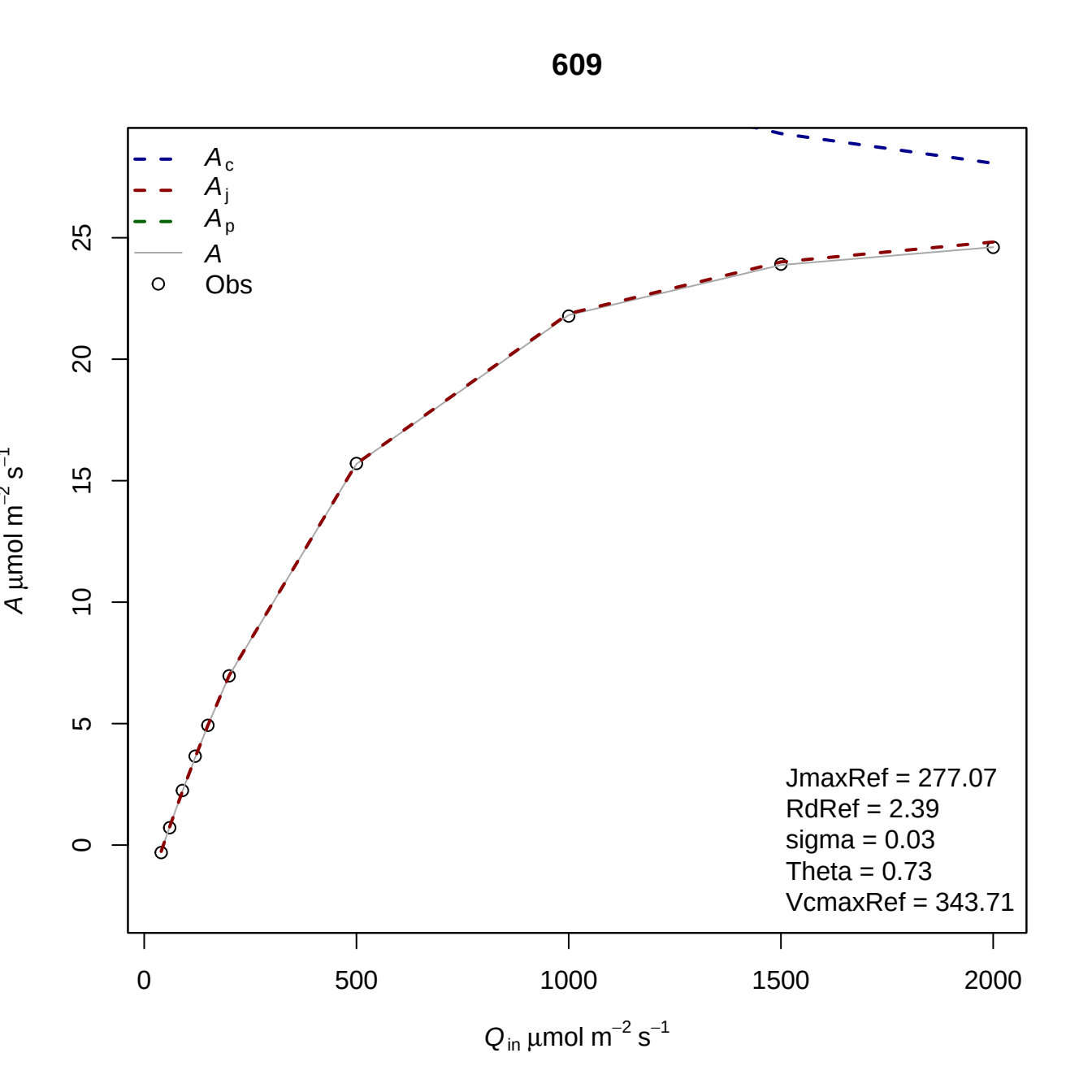

604
A_c
A_j
A_p
A
Obs
JmaxRef = 177.9
RdRef = 3.25
sigma = 0.26
Theta = 0.71
VcmaxRef = 9395361.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
500
1000
1500
2000

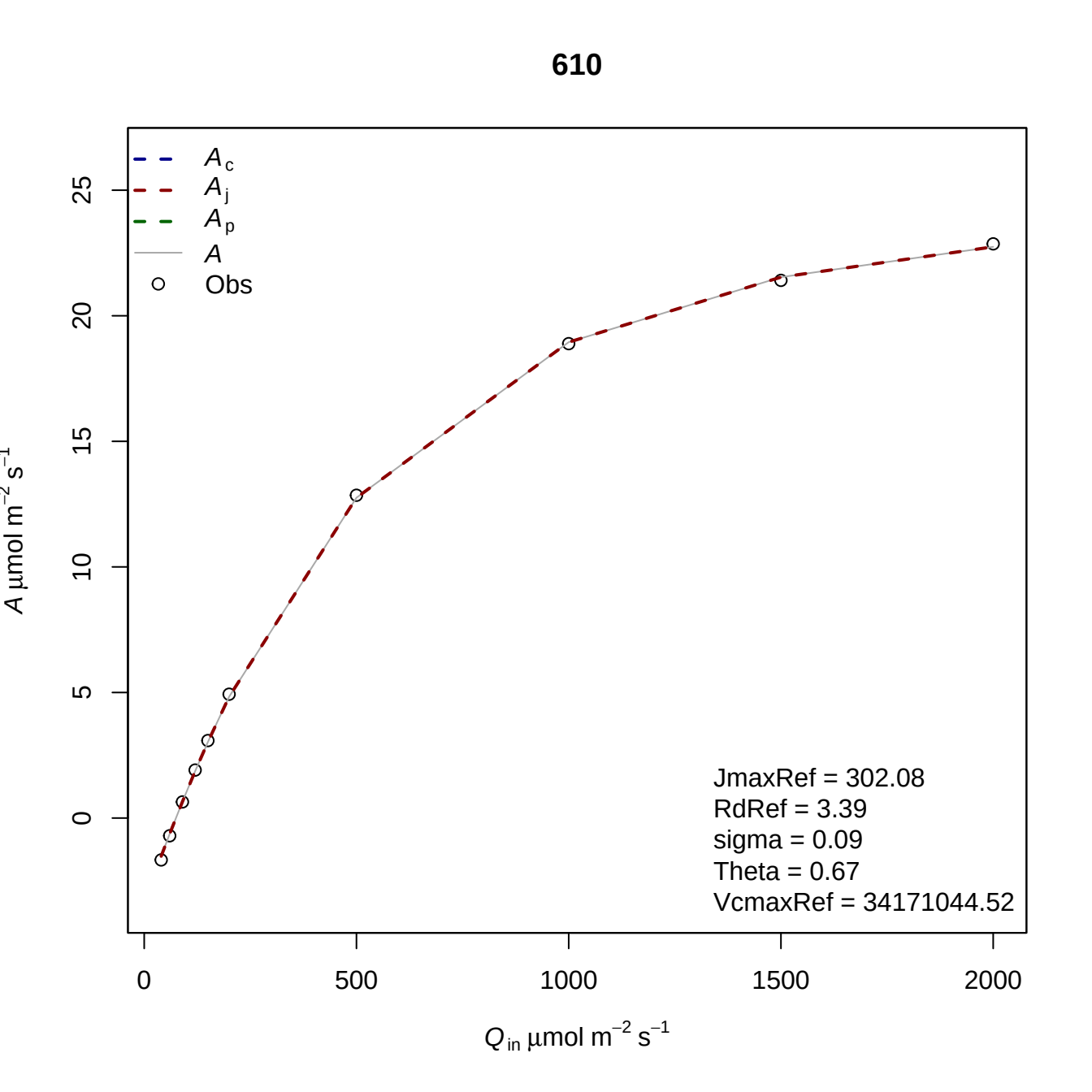

605
A_c
A_j
A_p
A
Obs
JmaxRef = 158.58
RdRef = 1.04
sigma = 0.12
Theta = 0.33
VcmaxRef = 75.76
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

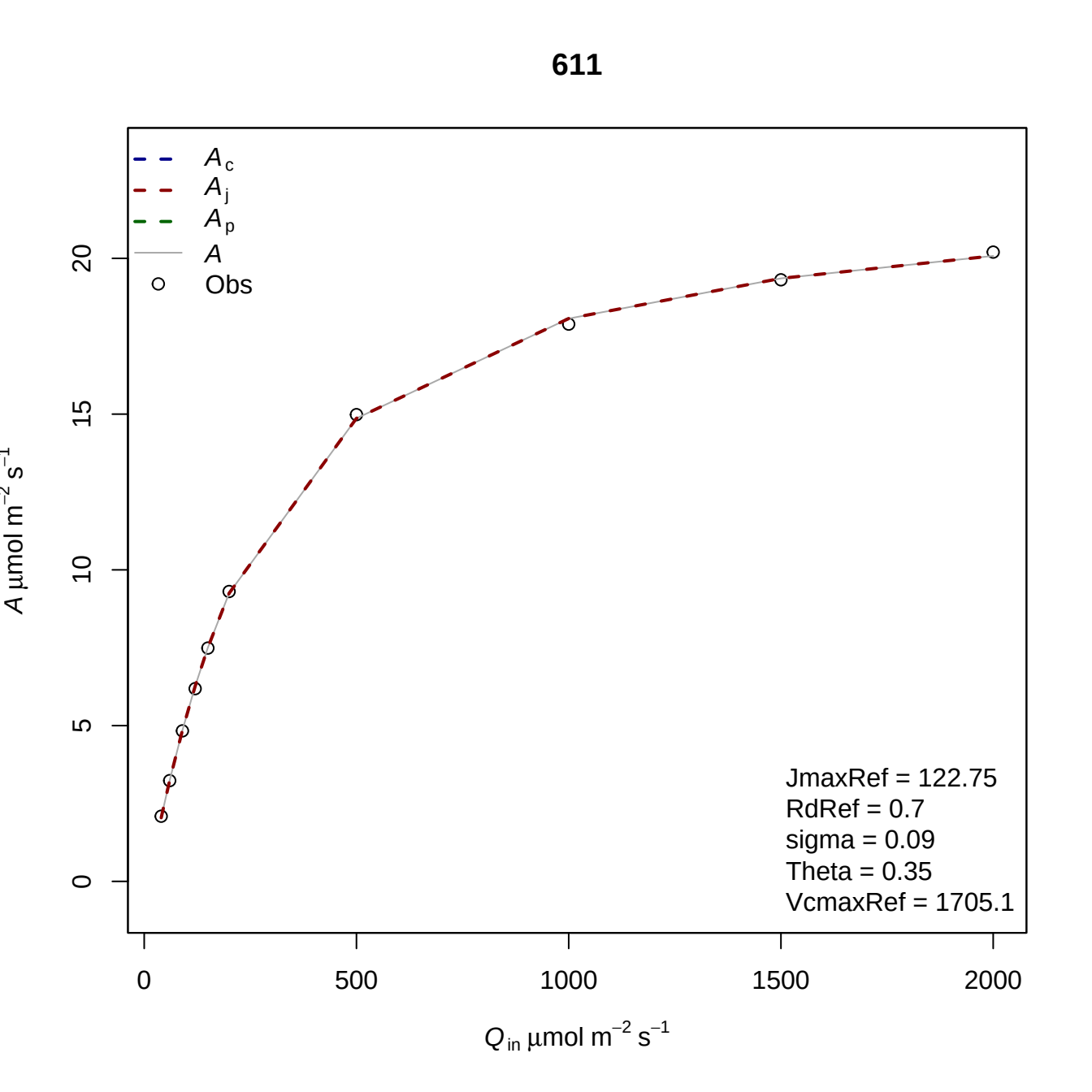

606
A_c
A_j
A_p
A
Obs
JmaxRef = 206.69
RdRef = 1.79
sigma = 0.24
Theta = 0.17
VcmaxRef = 95.71
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

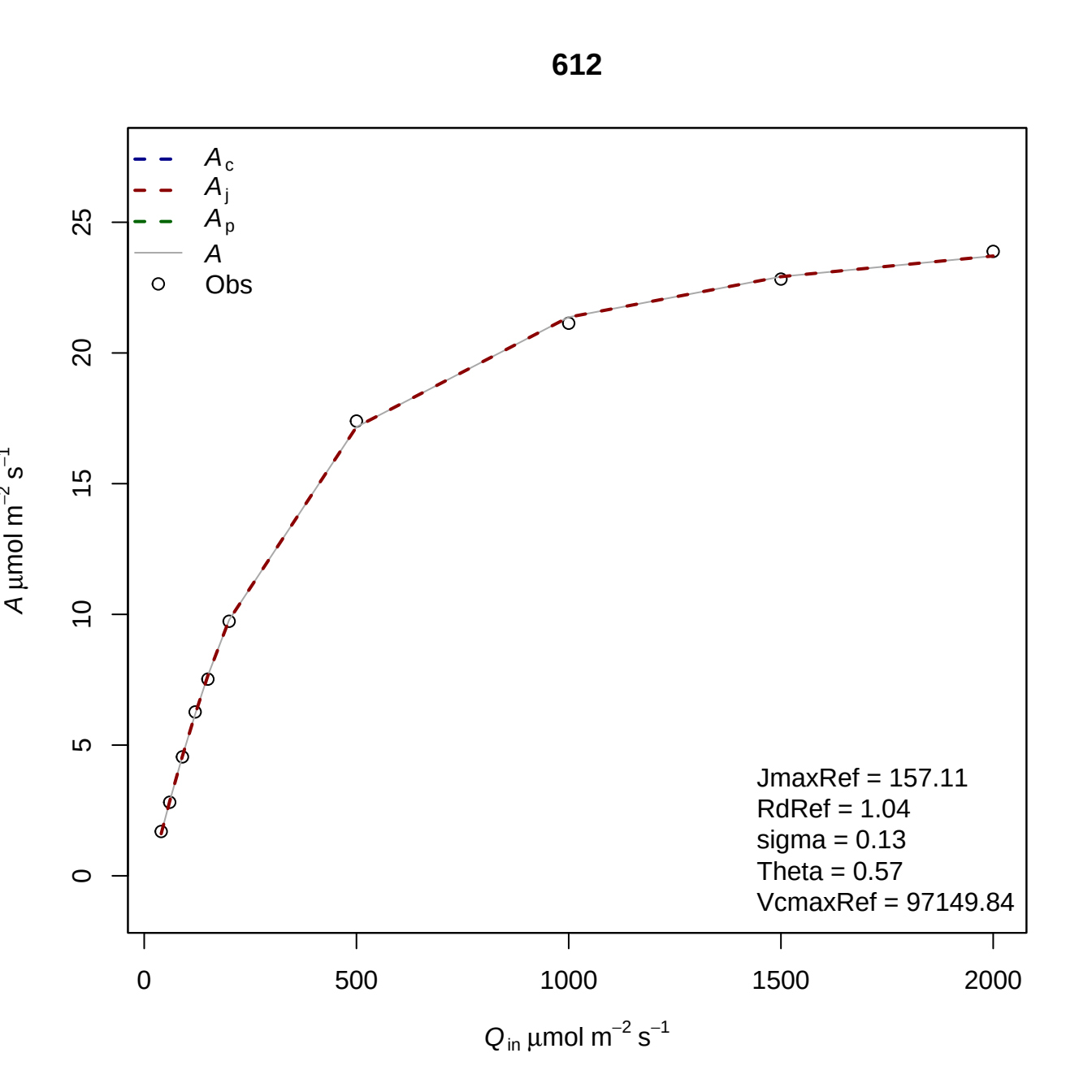

607
A_c
A_j
A_p
A
Obs
JmaxRef = 265.3
RdRef = 2.05
sigma = 0.25
Theta = 0.08
VcmaxRef = 138.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

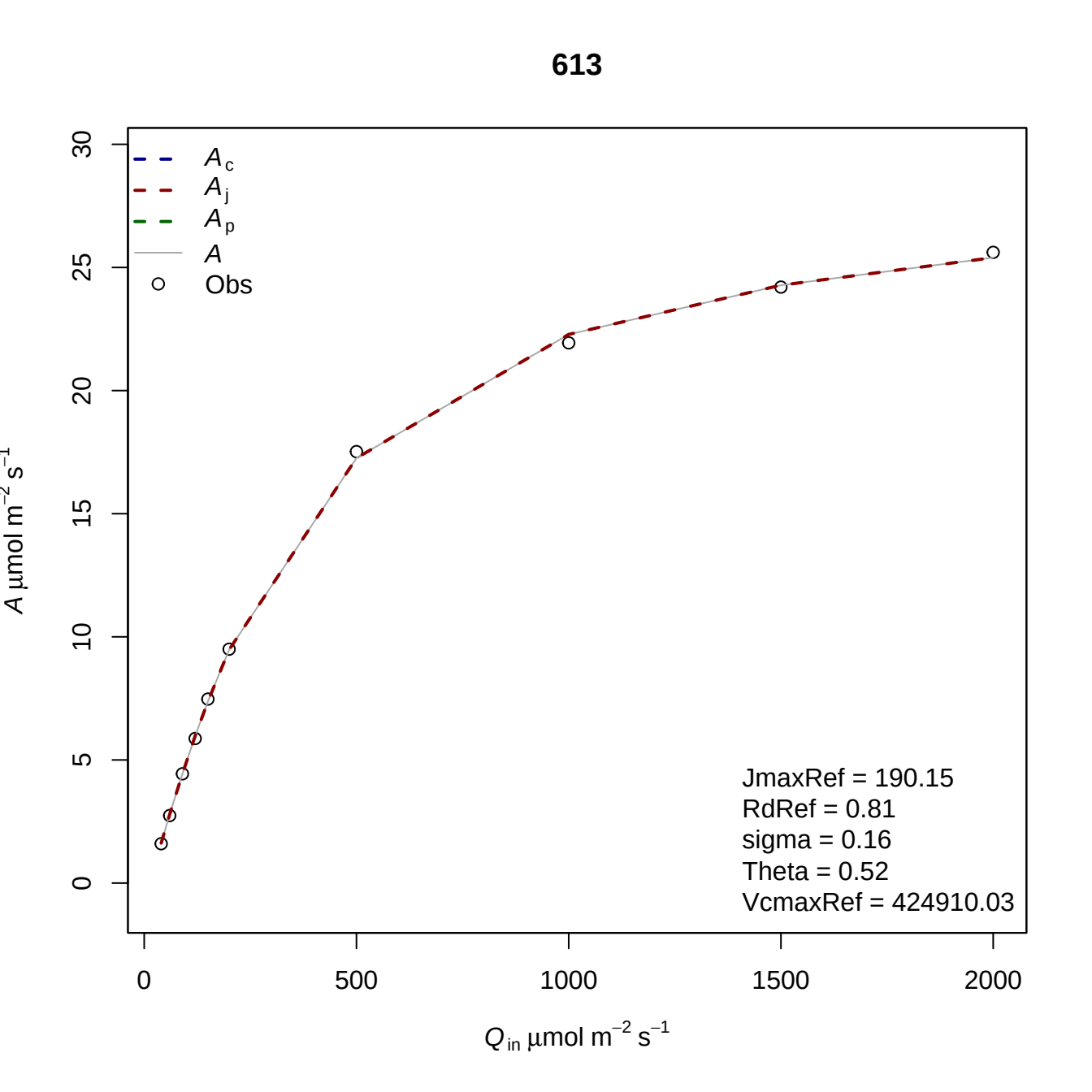

608
A_c
A_j
A_p
A
Obs
JmaxRef = 247.91
RdRef = 1.9
sigma = 0.08
Theta = 0.67
VcmaxRef = 506604.15
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

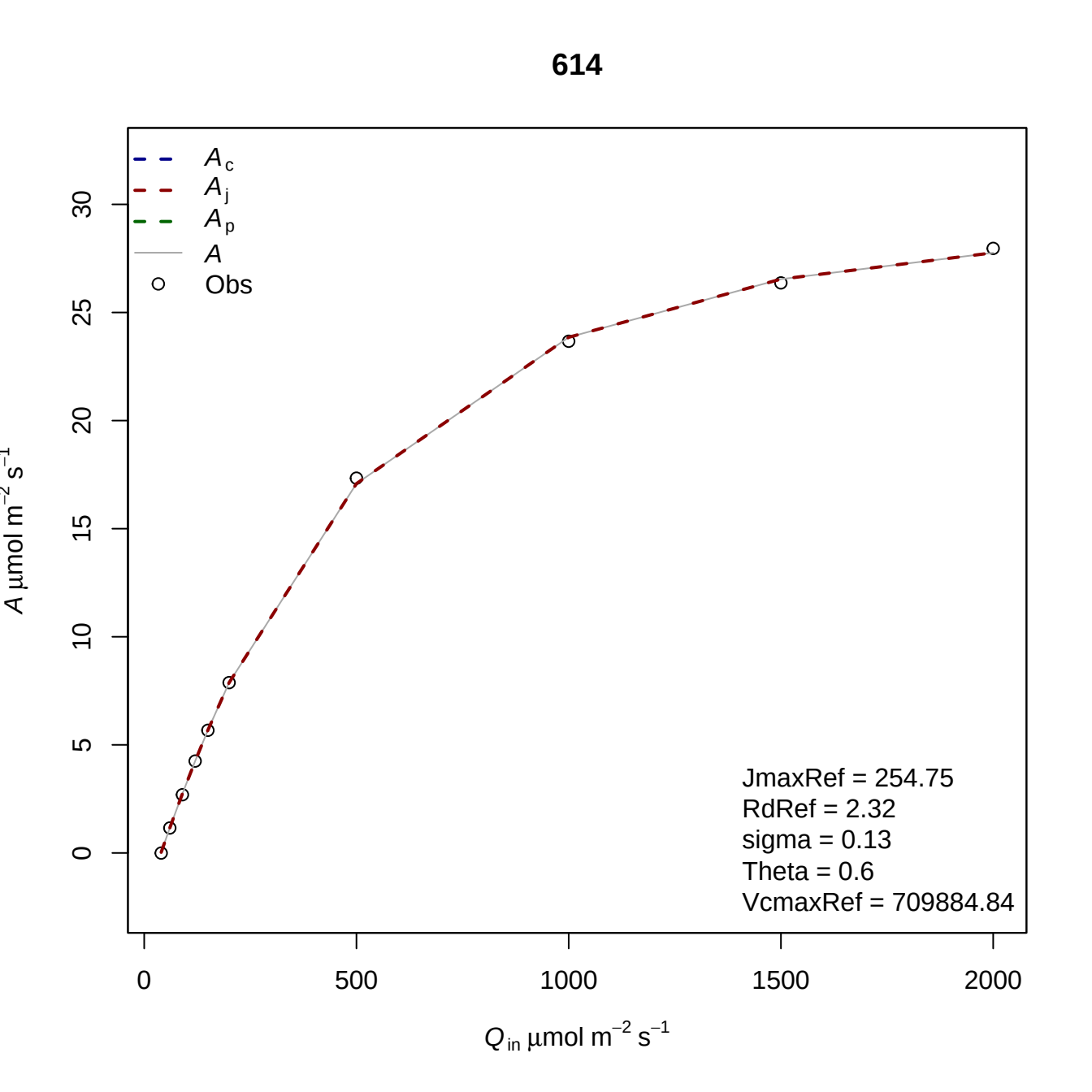

609
A_c
A_j
A_p
A
Obs
JmaxRef = 277.07
RdRef = 2.39
sigma = 0.03
Theta = 0.73
VcmaxRef = 343.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

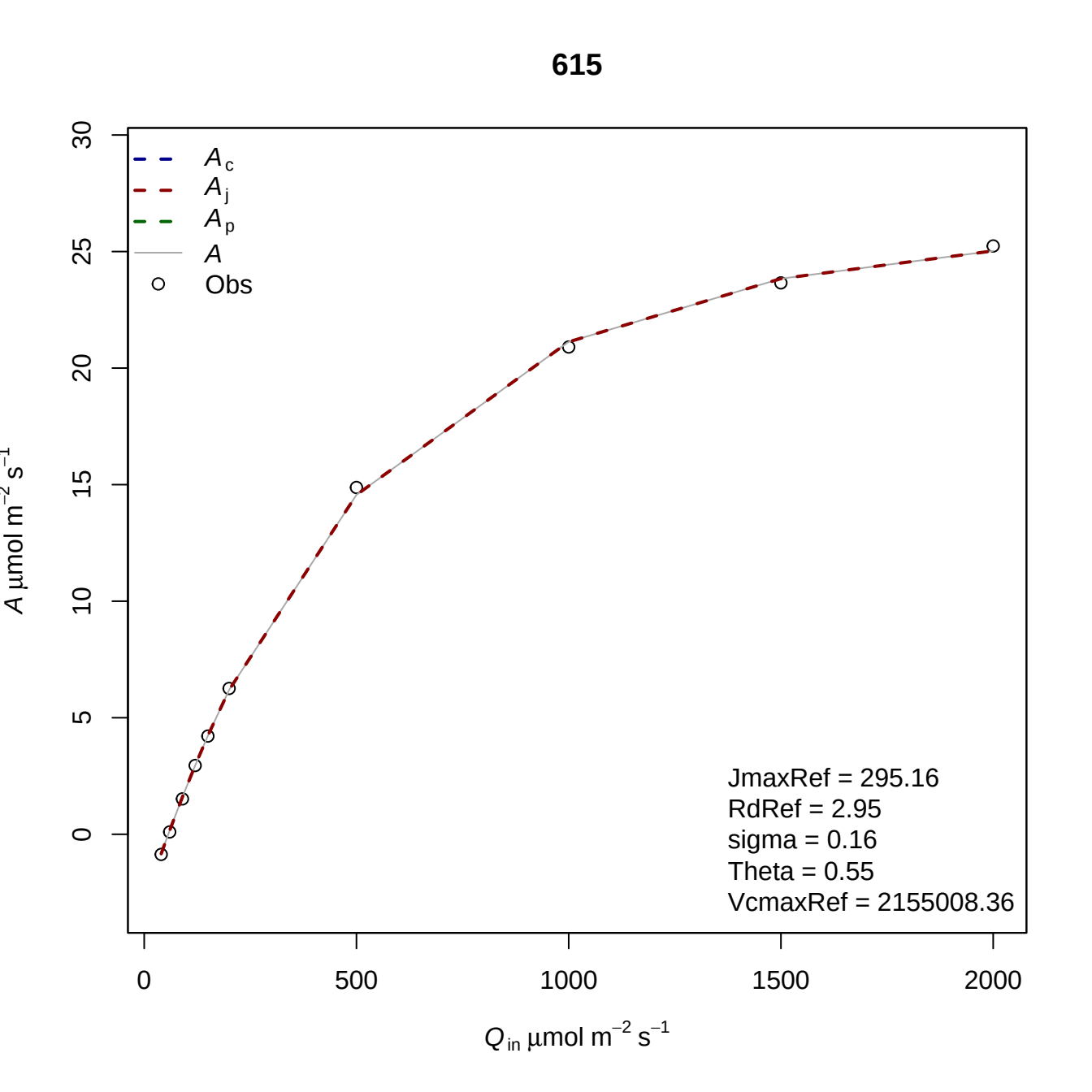

610
A_c
A_j
A_p
A
Obs
JmaxRef = 302.08
RdRef = 3.39
sigma = 0.09
Theta = 0.67
VcmaxRef = 34171044.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

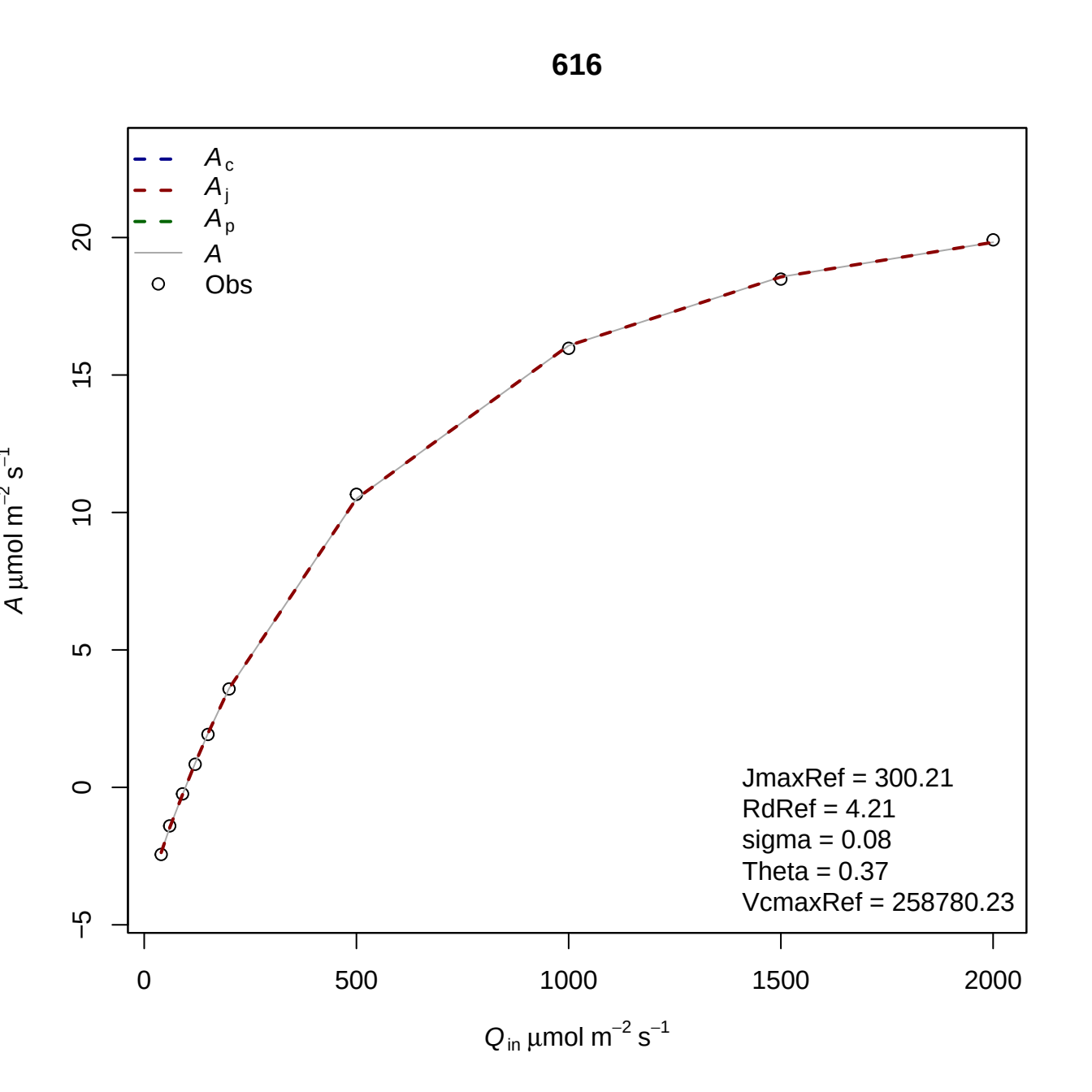

611
A_c
A_j
A_p
A
Obs
JmaxRef = 122.75
RdRef = 0.7
sigma = 0.09
Theta = 0.35
VcmaxRef = 1705.1
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

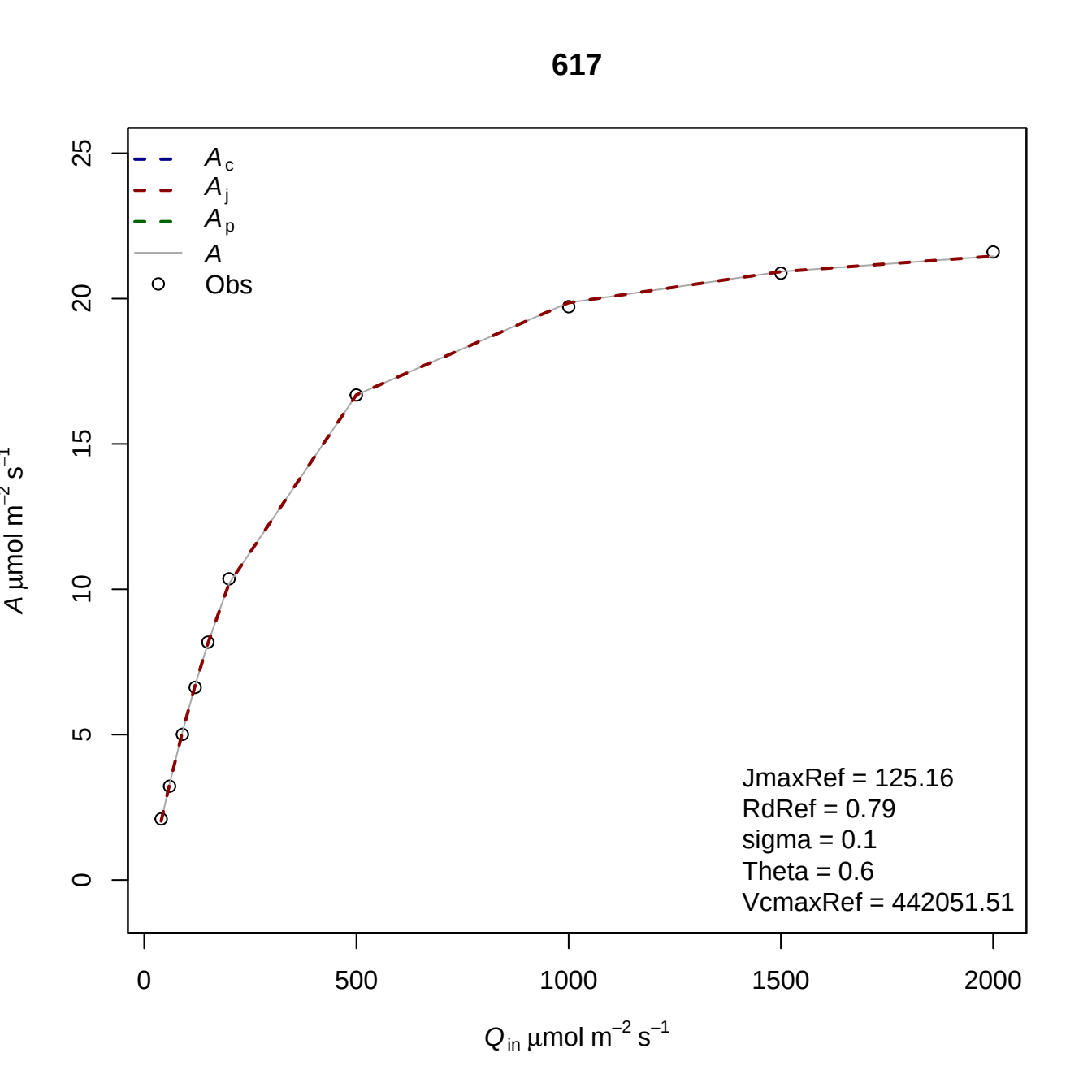

612
A_c
A_j
A_p
A
Obs
JmaxRef = 157.11
RdRef = 1.04
sigma = 0.13
Theta = 0.57
VcmaxRef = 97149.84
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

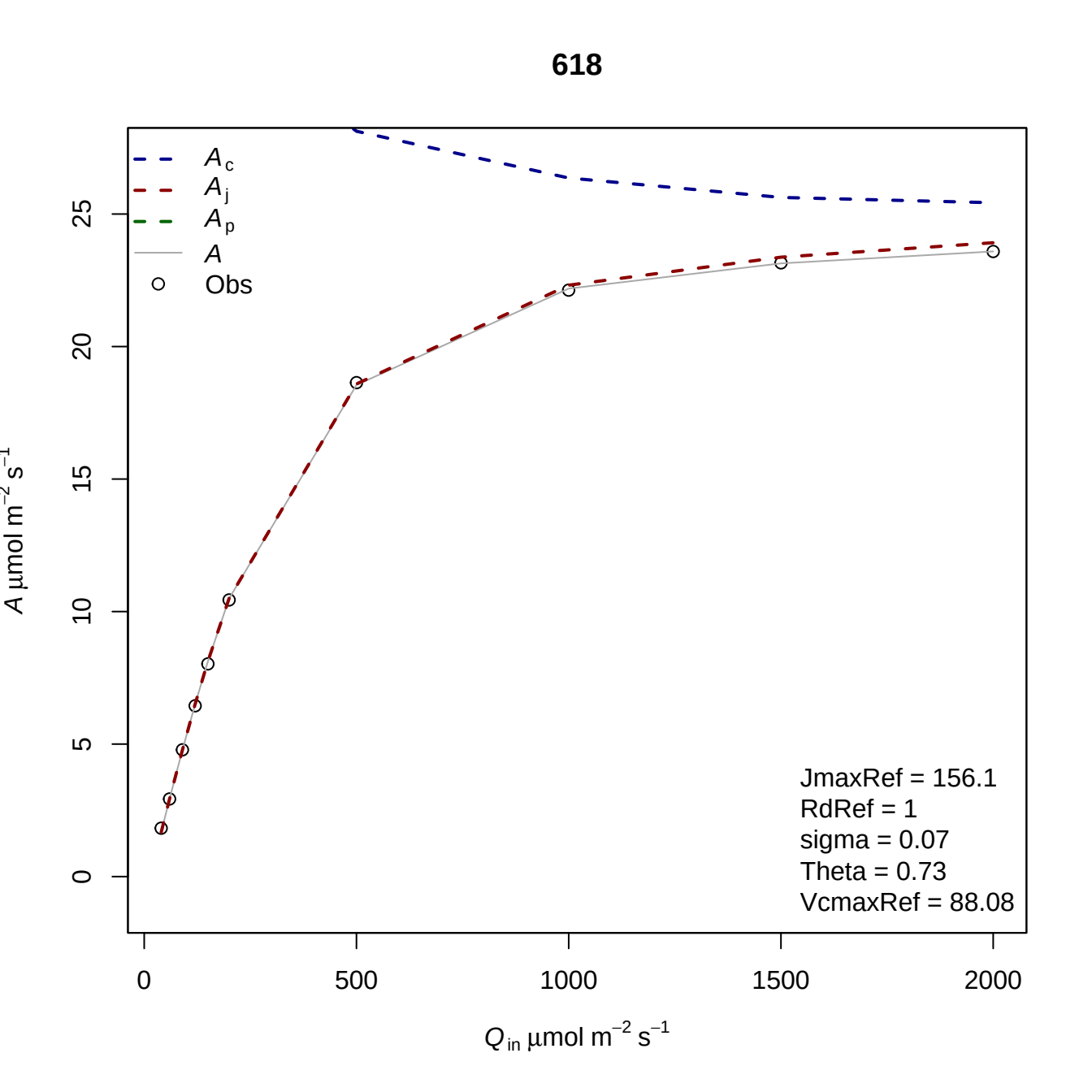

613
A_c
A_j
A_p
A
Obs
JmaxRef = 190.15
RdRef = 0.81
sigma = 0.16
Theta = 0.52
VcmaxRef = 424910.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

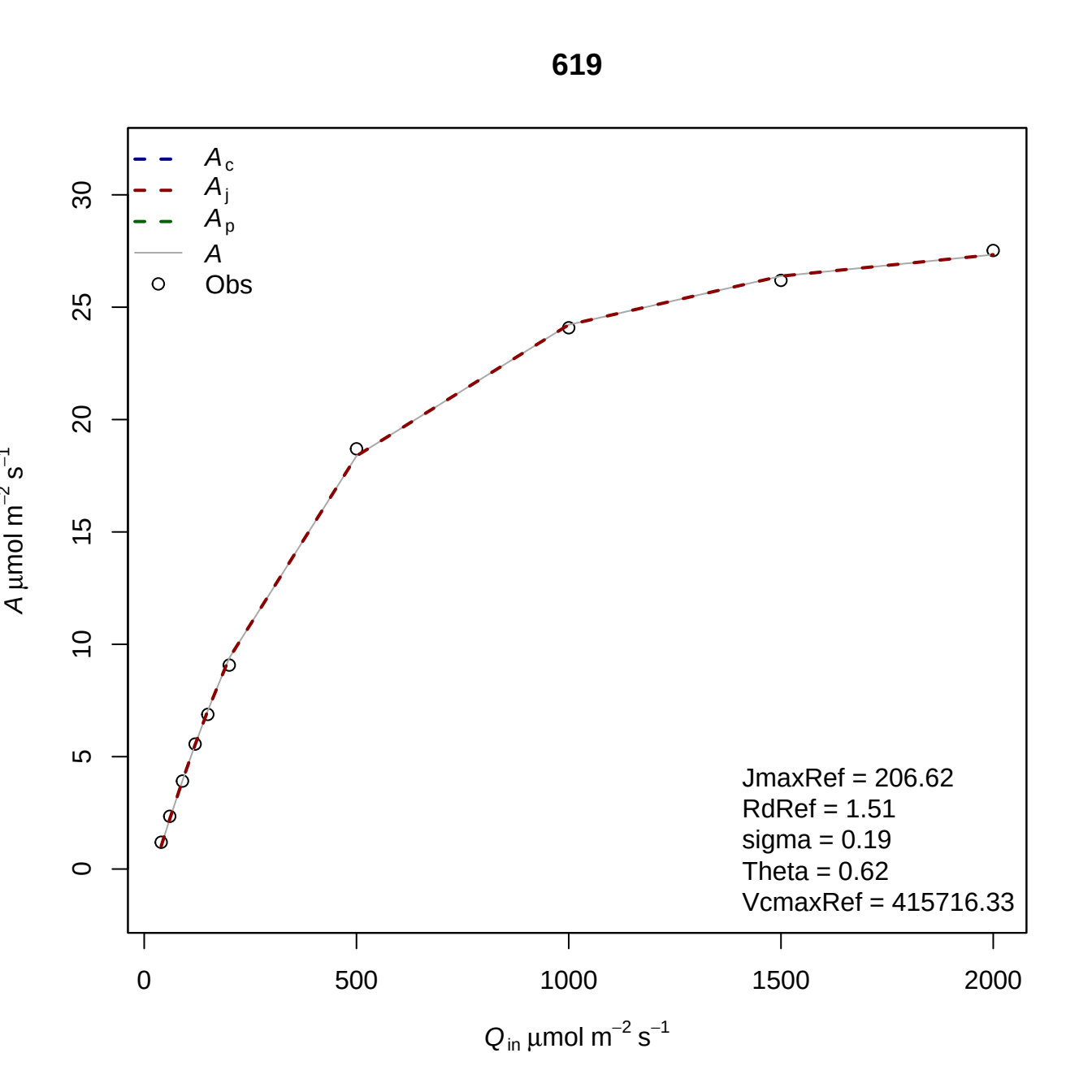

614
A_c
A_j
A_p
A
Obs
JmaxRef = 254.75
RdRef = 2.32
sigma = 0.13
Theta = 0.6
VcmaxRef = 709884.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

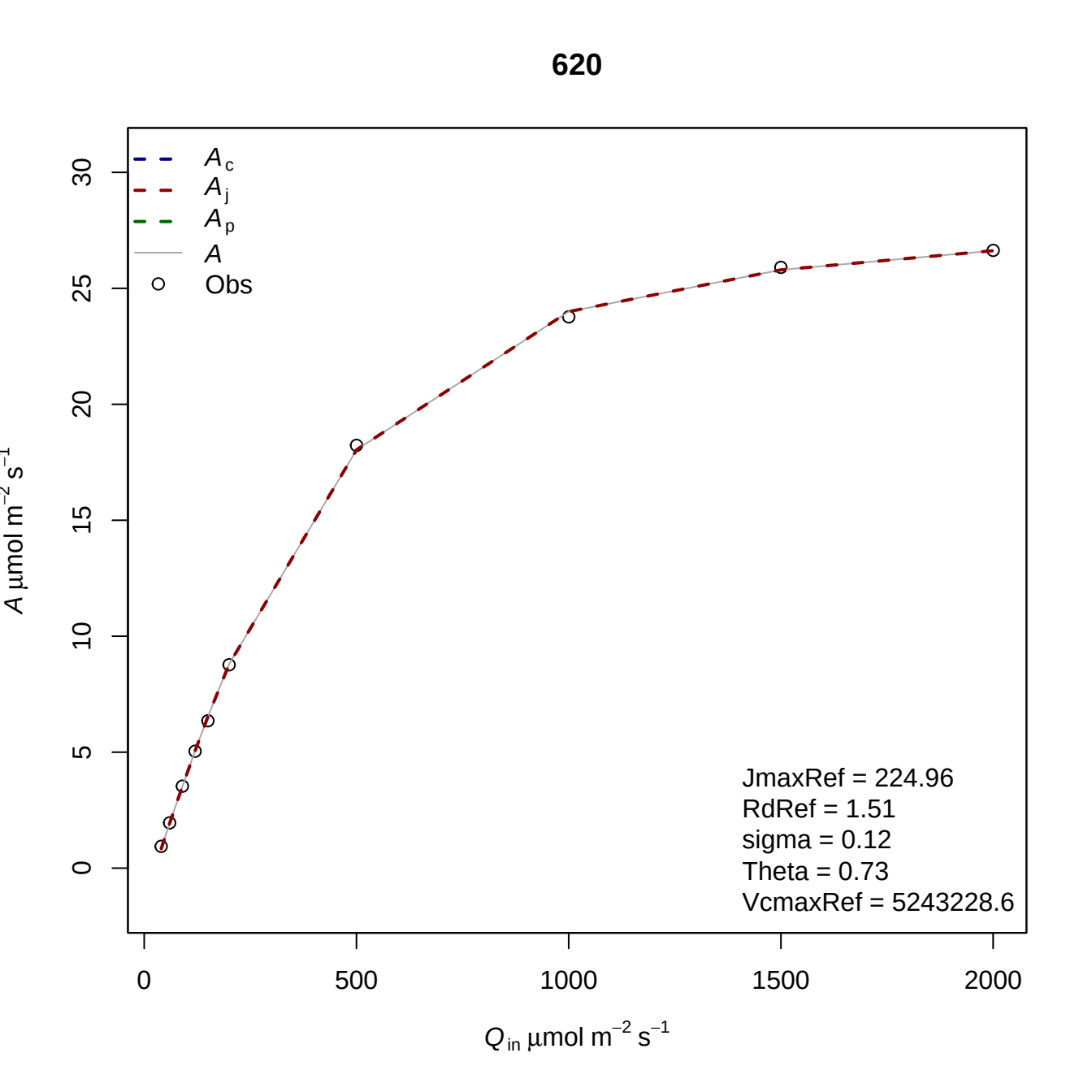

615
A_c
A_j
A_p
A
Obs
JmaxRef = 295.16
RdRef = 2.95
sigma = 0.16
Theta = 0.55
VcmaxRef = 2155008.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

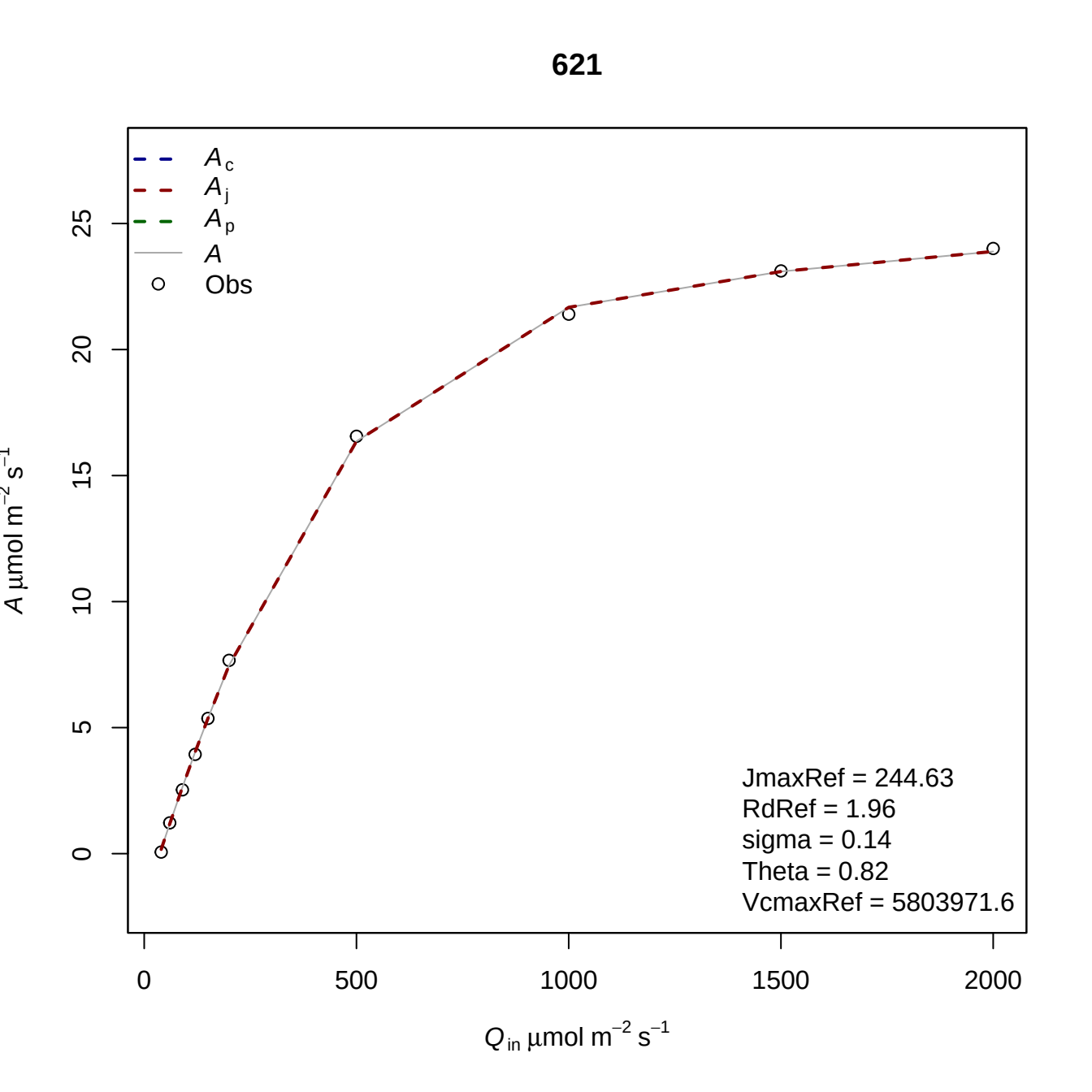

616
A_c
A_j
A_p
A
Obs
JmaxRef = 300.21
RdRef = 4.21
sigma = 0.08
Theta = 0.37
VcmaxRef = 258780.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

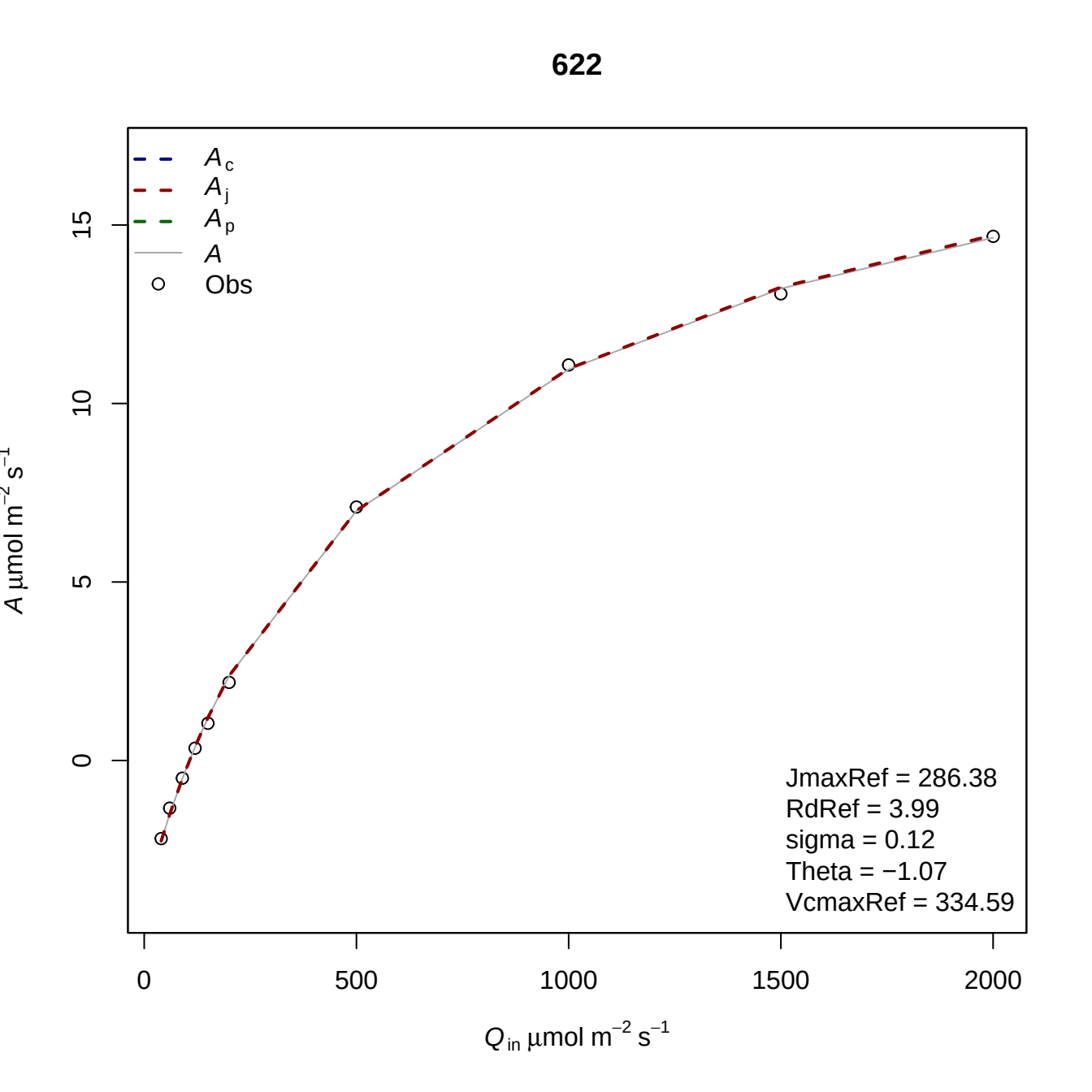

617
A_c
A_j
A_p
A
Obs
JmaxRef = 125.16
RdRef = 0.79
sigma = 0.1
Theta = 0.6
VcmaxRef = 442051.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

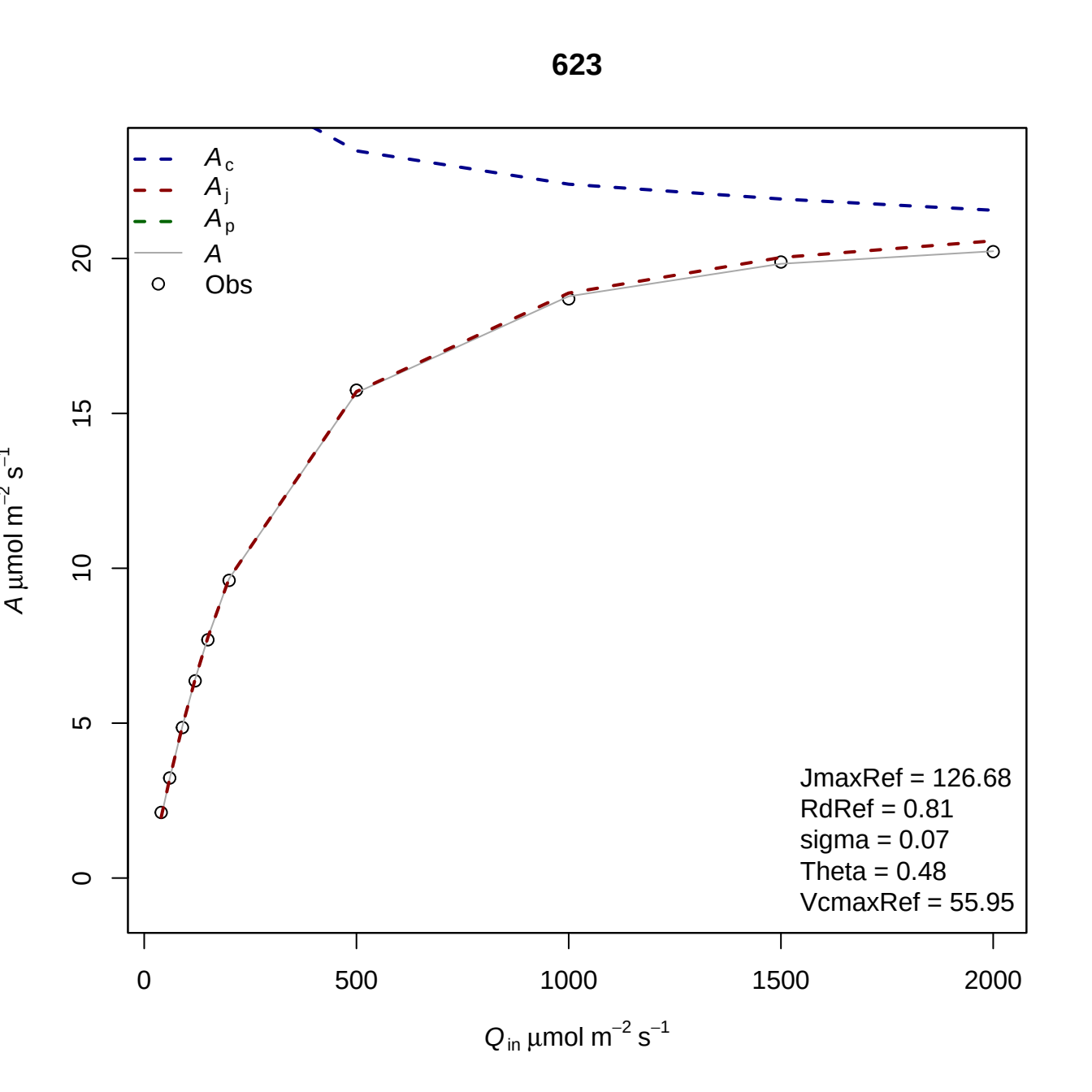

618
A_c
A_j
A_p
A
Obs
JmaxRef = 156.1
RdRef = 1
sigma = 0.07
Theta = 0.73
VcmaxRef = 88.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

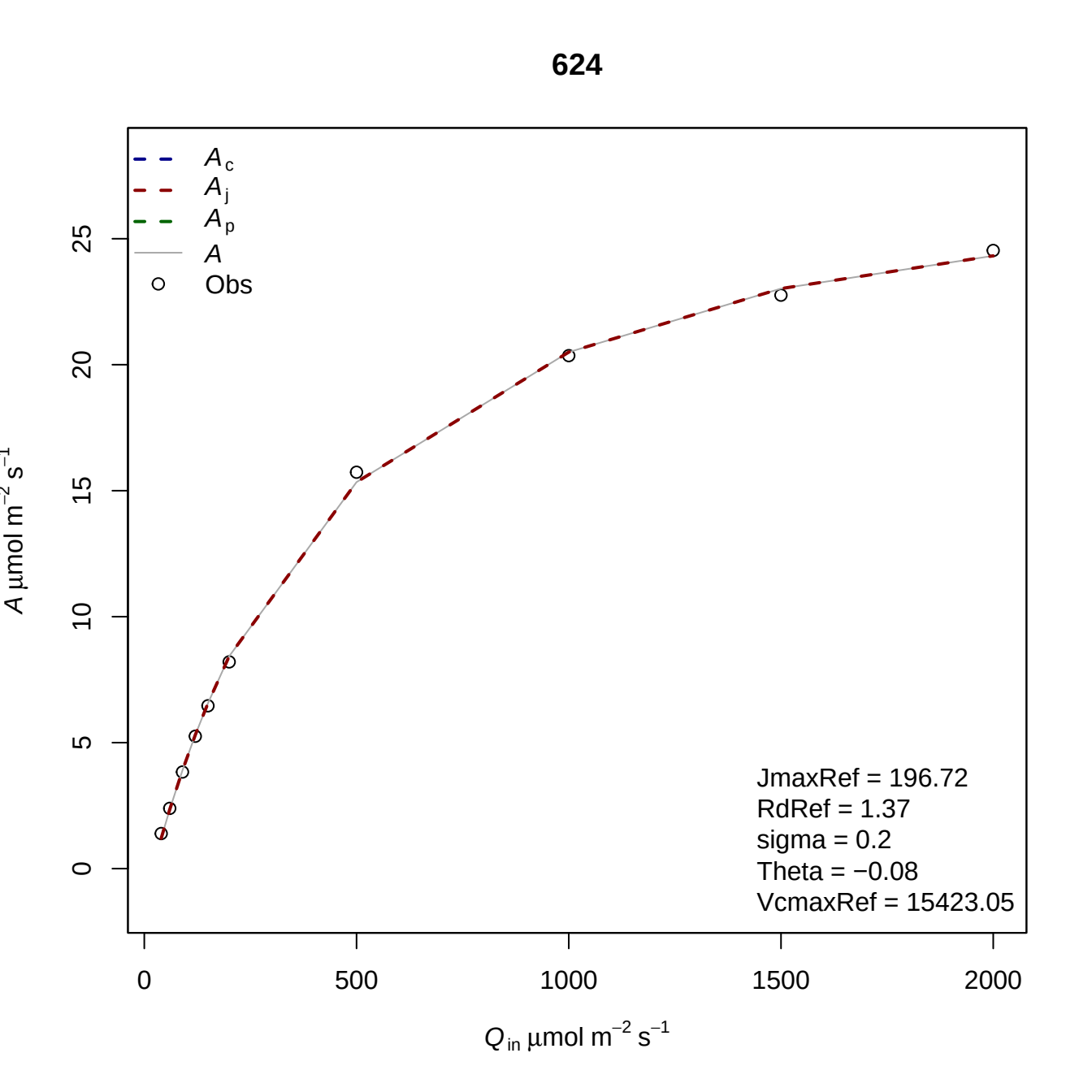

619
A_c
A_j
A_p
A
Obs
JmaxRef = 206.62
RdRef = 1.51
sigma = 0.19
Theta = 0.62
VcmaxRef = 415716.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

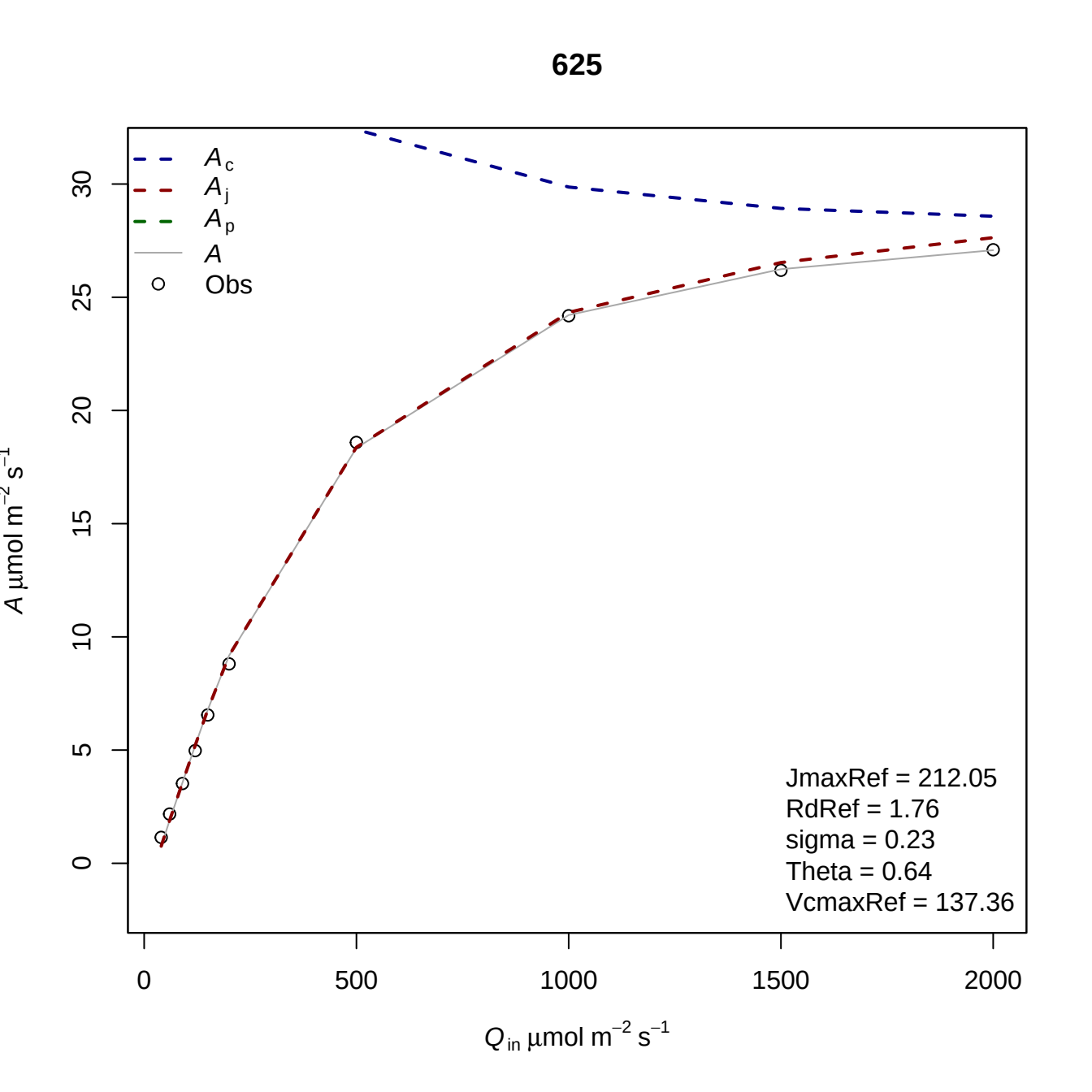

620
A_c
A_j
A_p
A
Obs
JmaxRef = 224.96
RdRef = 1.51
sigma = 0.12
Theta = 0.73
VcmaxRef = 5243228.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

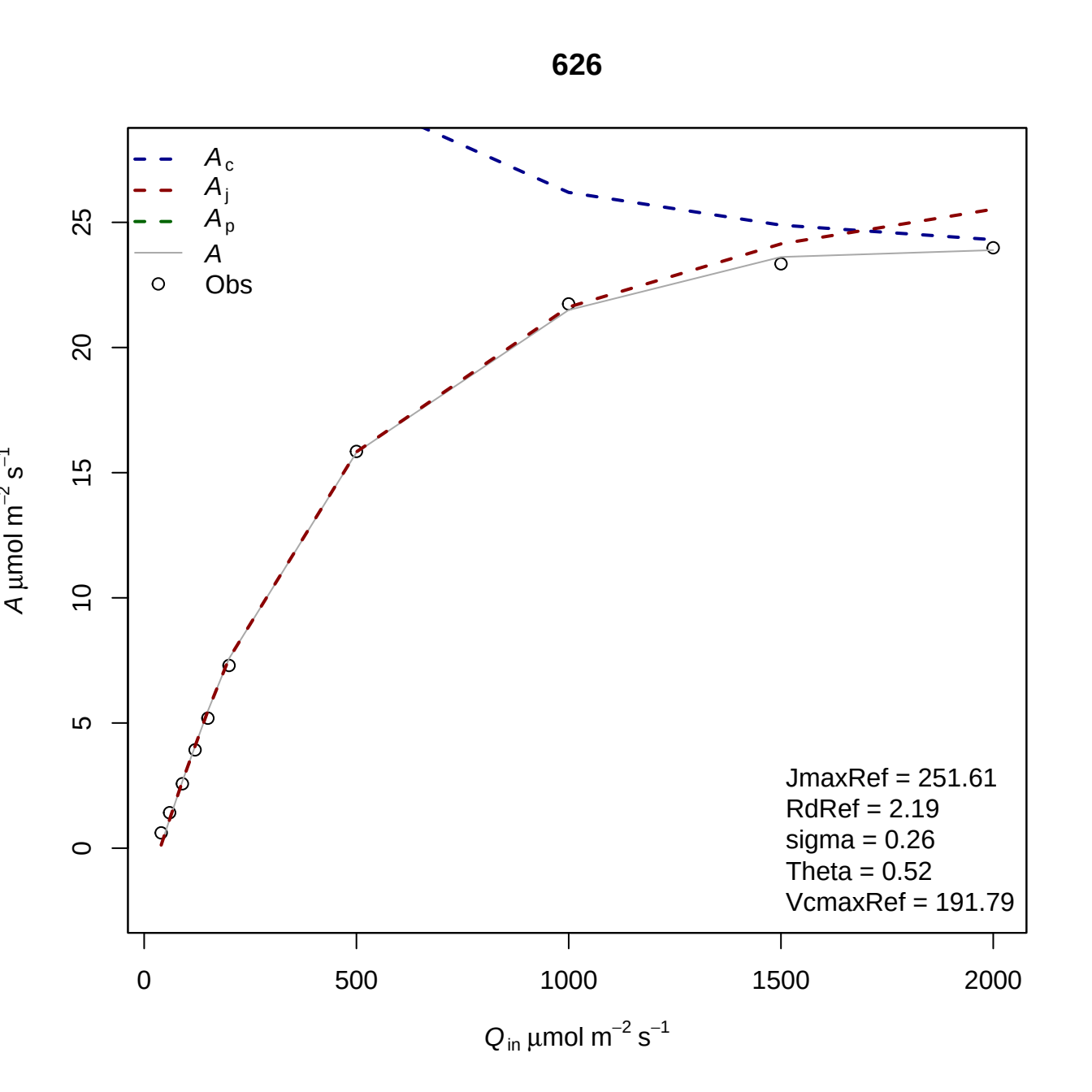

621
A_c
A_j
A_p
A
Obs
JmaxRef = 244.63
RdRef = 1.96
sigma = 0.14
Theta = 0.82
VcmaxRef = 5803971.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

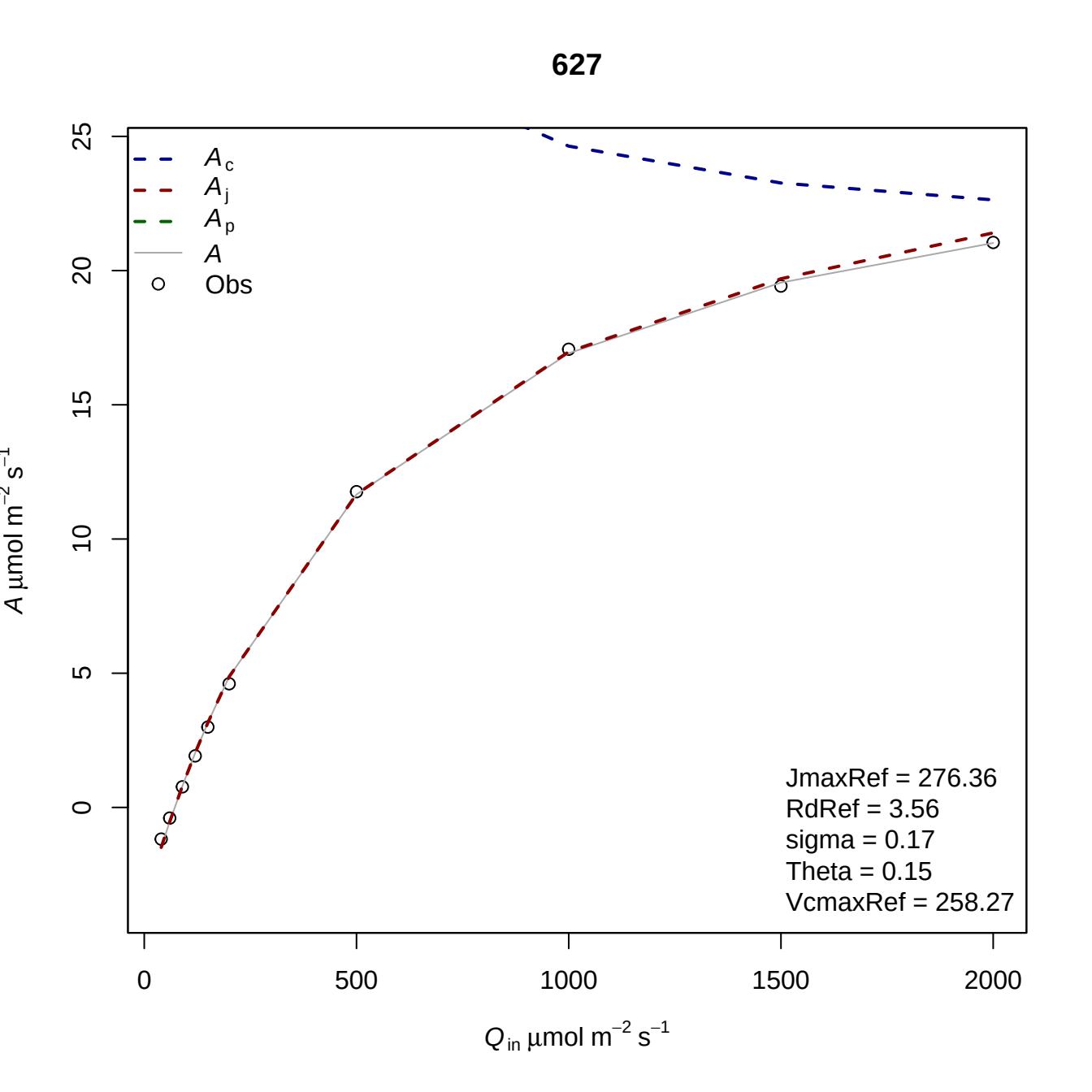

622
A_c
A_j
A_p
A
Obs
JmaxRef = 286.38
RdRef = 3.99
sigma = 0.12
Theta = −1.07
VcmaxRef = 334.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

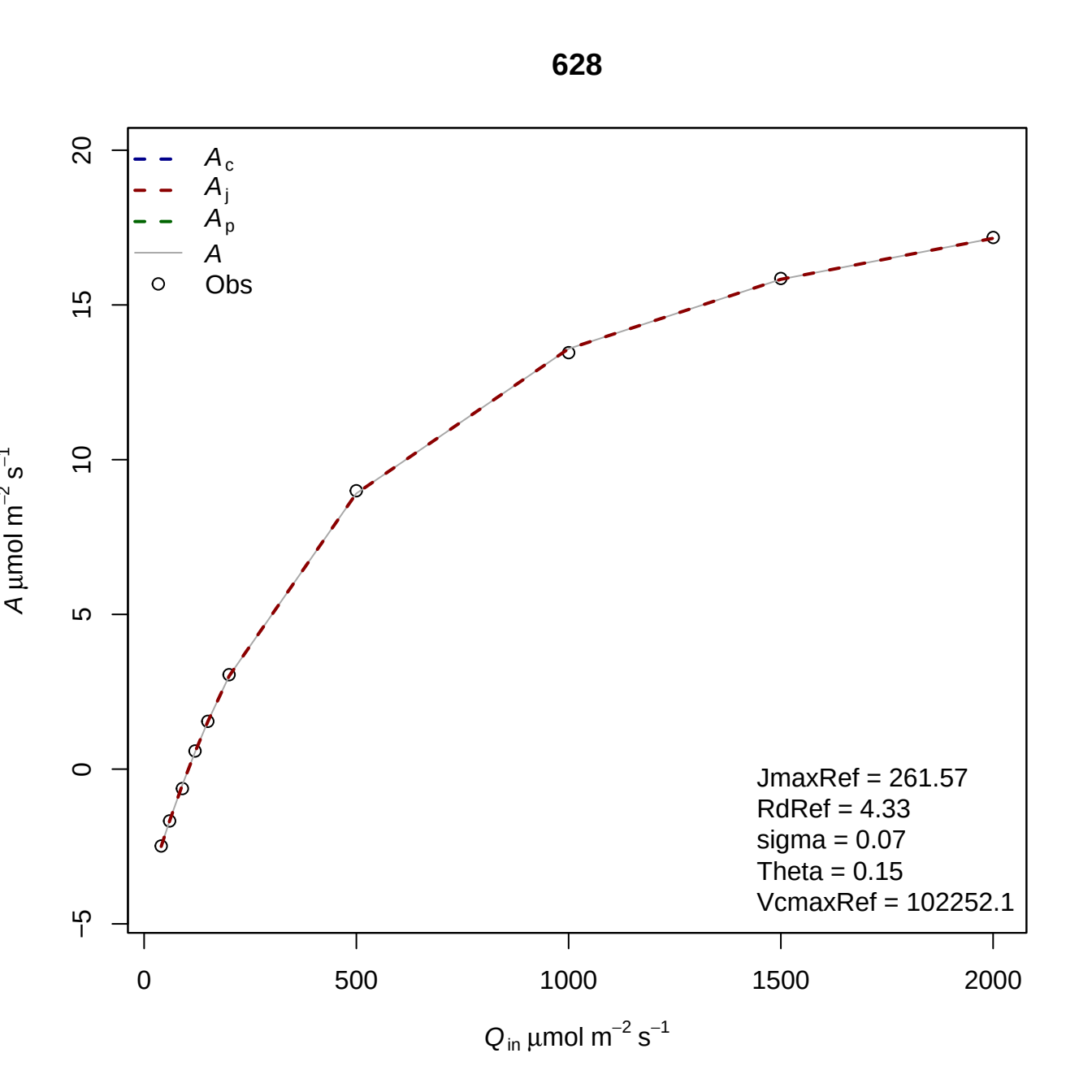

623
A_c
A_j
A_p
A
Obs
JmaxRef = 126.68
RdRef = 0.81
sigma = 0.07
Theta = 0.48
VcmaxRef = 55.95
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

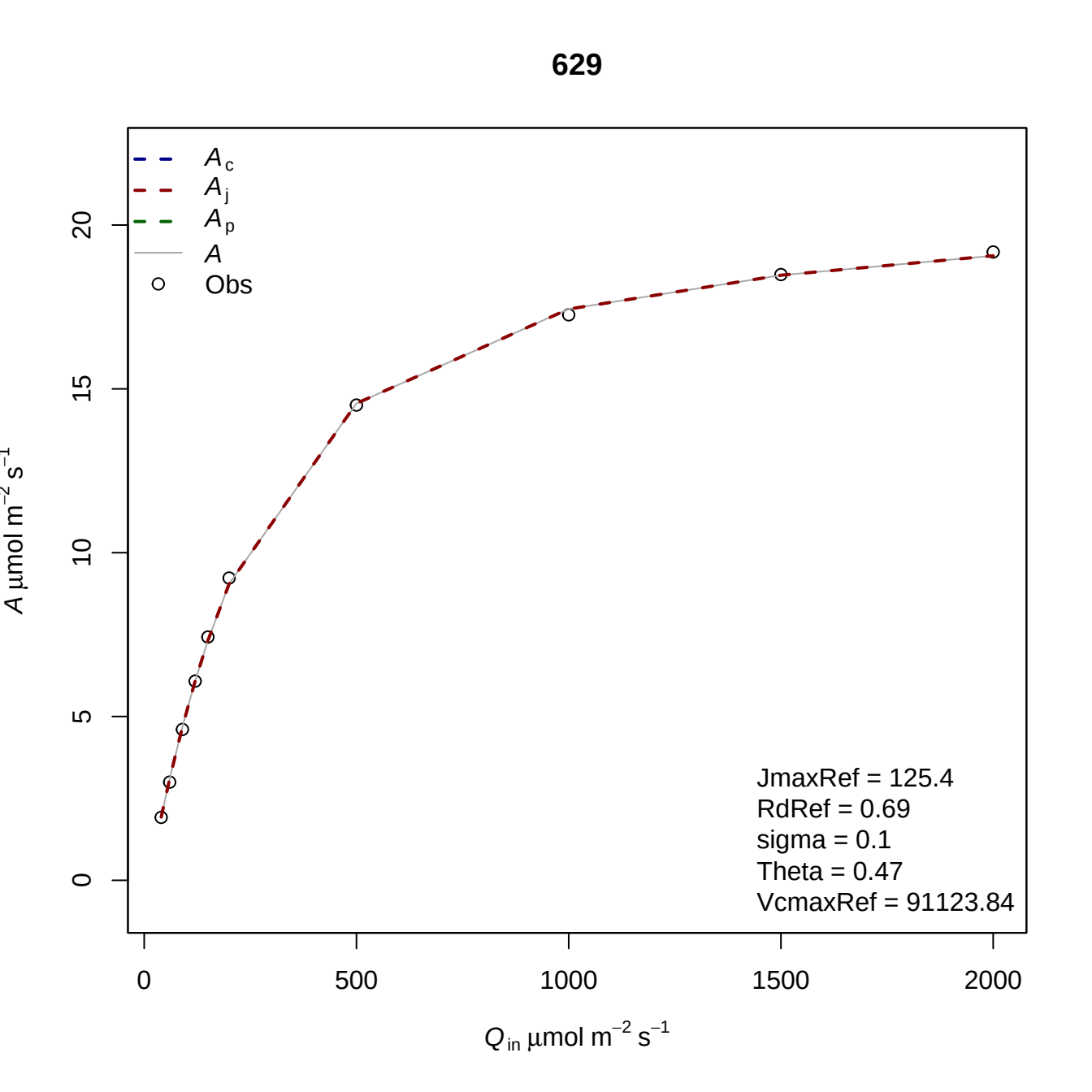

624
A_c
A_j
A_p
A
Obs
JmaxRef = 196.72
RdRef = 1.37
sigma = 0.2
Theta = −0.08
VcmaxRef = 15423.05
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

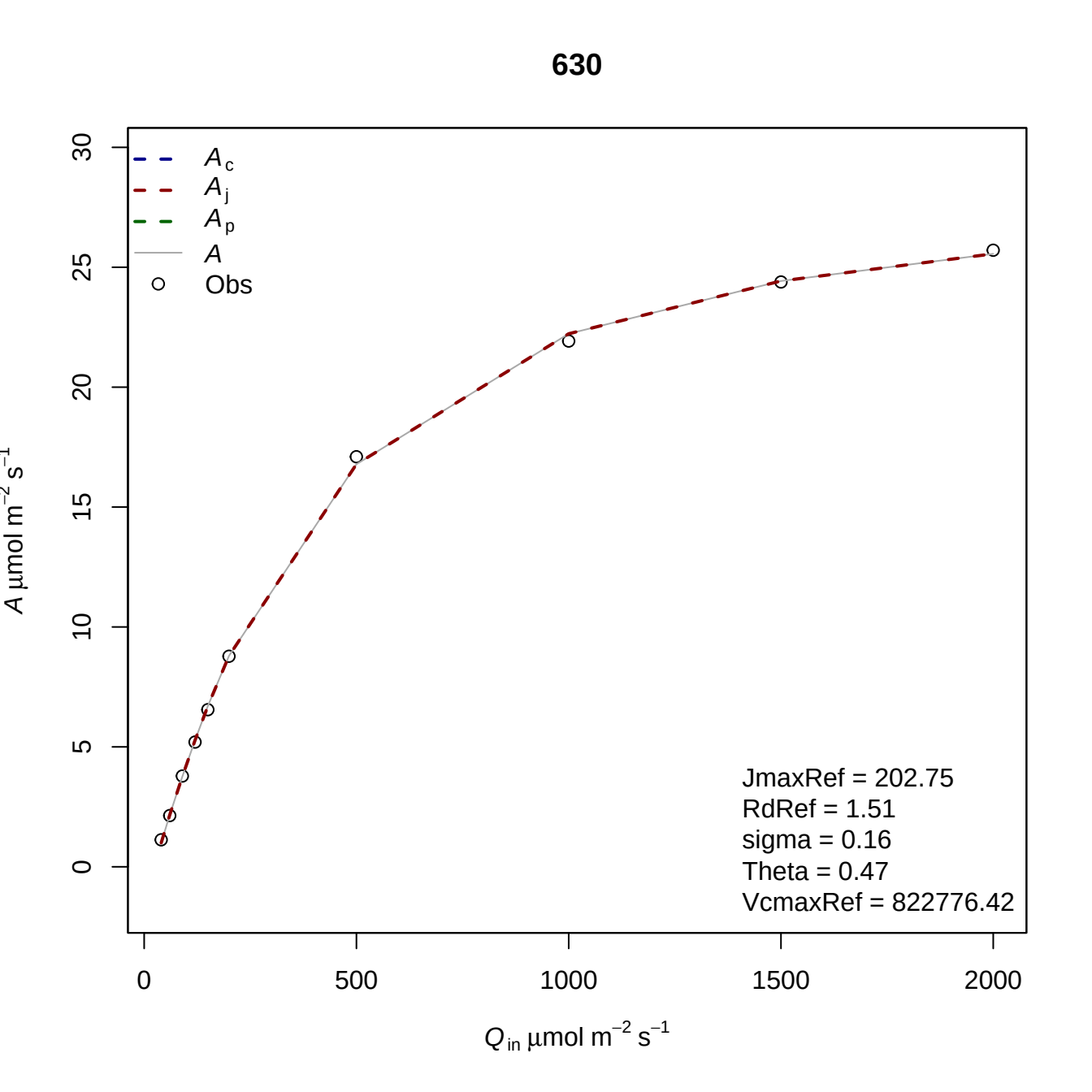

625
A_c
A_j
A_p
A
Obs
JmaxRef = 212.05
RdRef = 1.76
sigma = 0.23
Theta = 0.64
VcmaxRef = 137.36
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

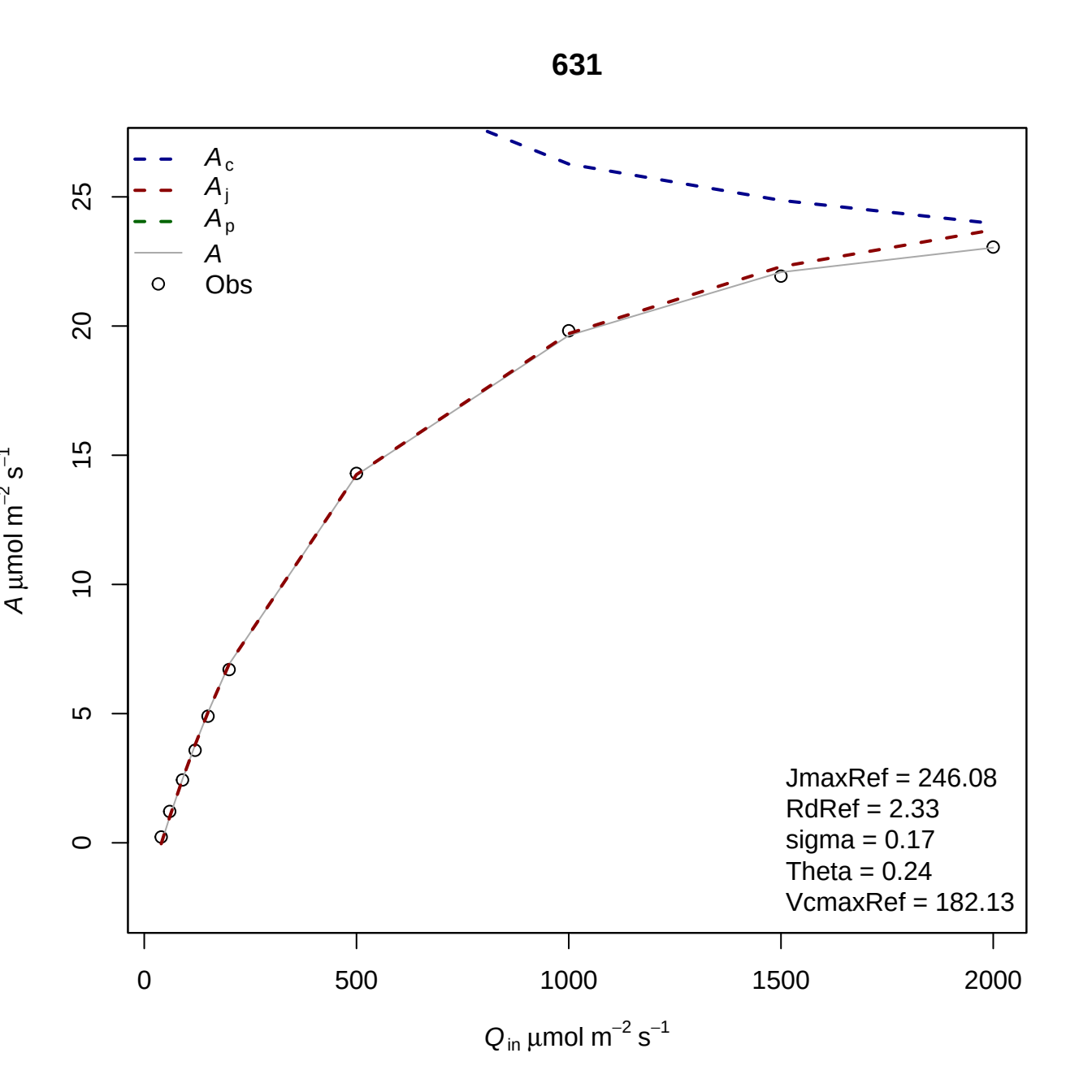

626
A_c
A_j
A_p
A
Obs
JmaxRef = 251.61
RdRef = 2.19
sigma = 0.26
Theta = 0.52
VcmaxRef = 191.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

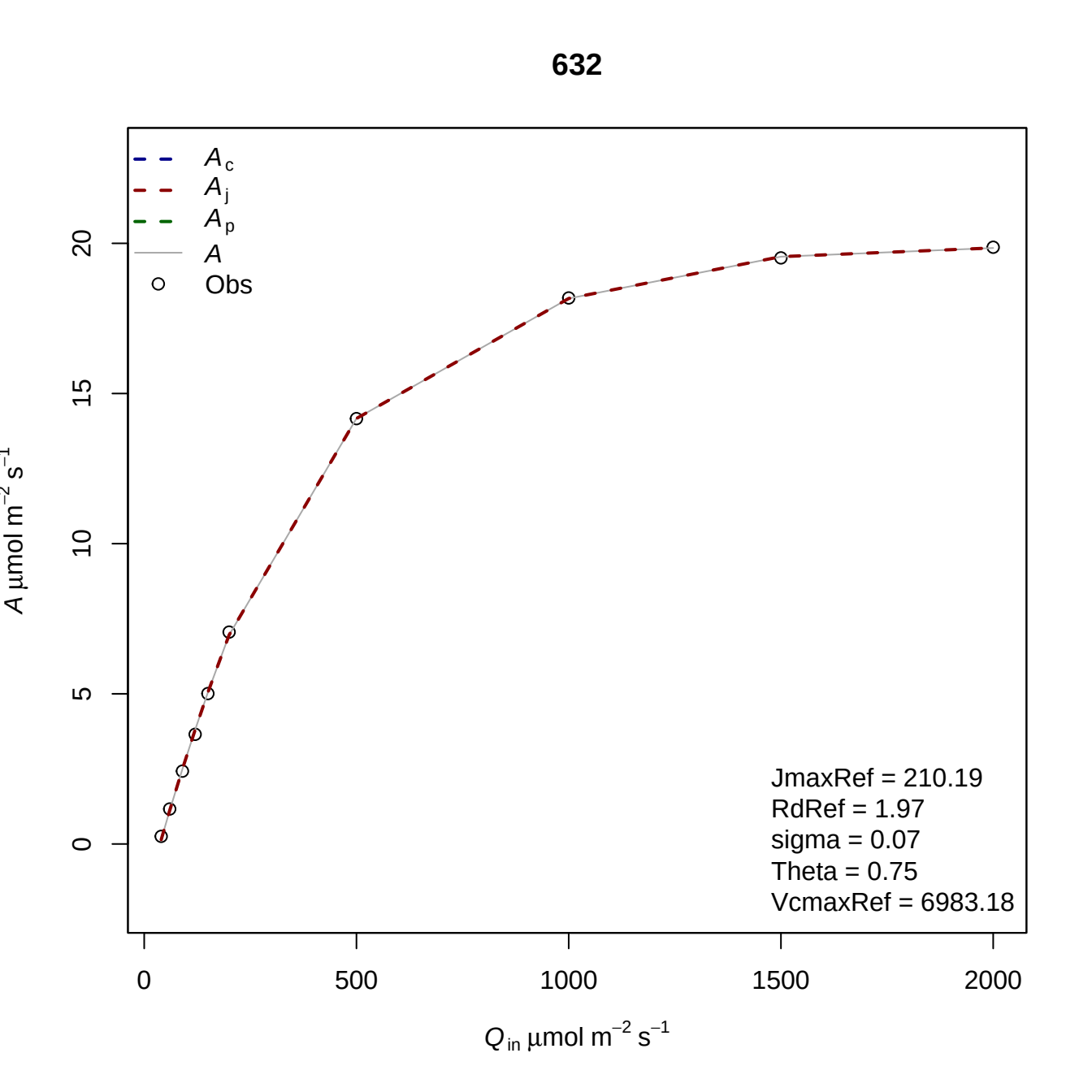

627
A_c
A_j
A_p
A
Obs
JmaxRef = 276.36
RdRef = 3.56
sigma = 0.17
Theta = 0.15
VcmaxRef = 258.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

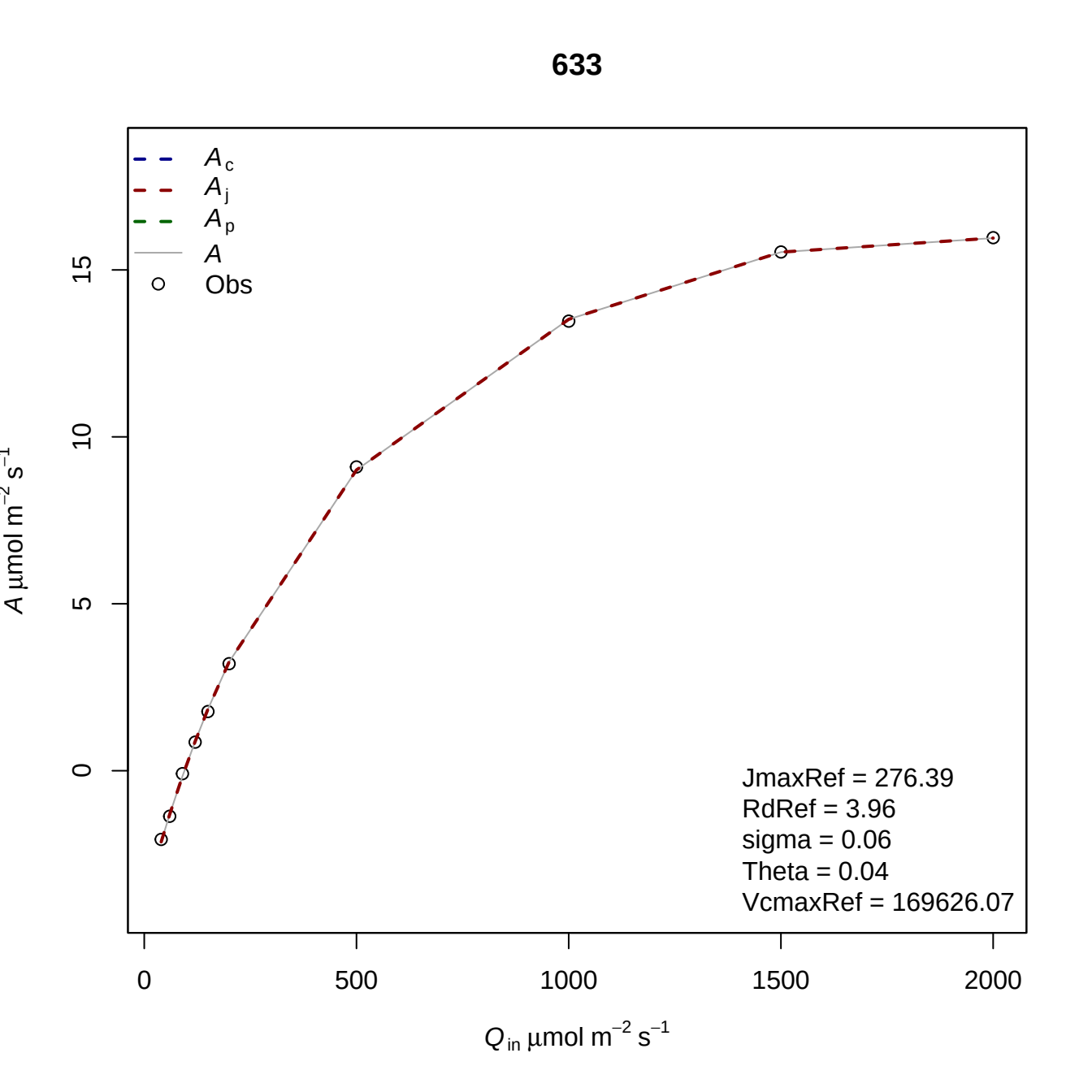

628
A_c
A_j
A_p
A
Obs
JmaxRef = 261.57
RdRef = 4.33
sigma = 0.07
Theta = 0.15
VcmaxRef = 102252.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

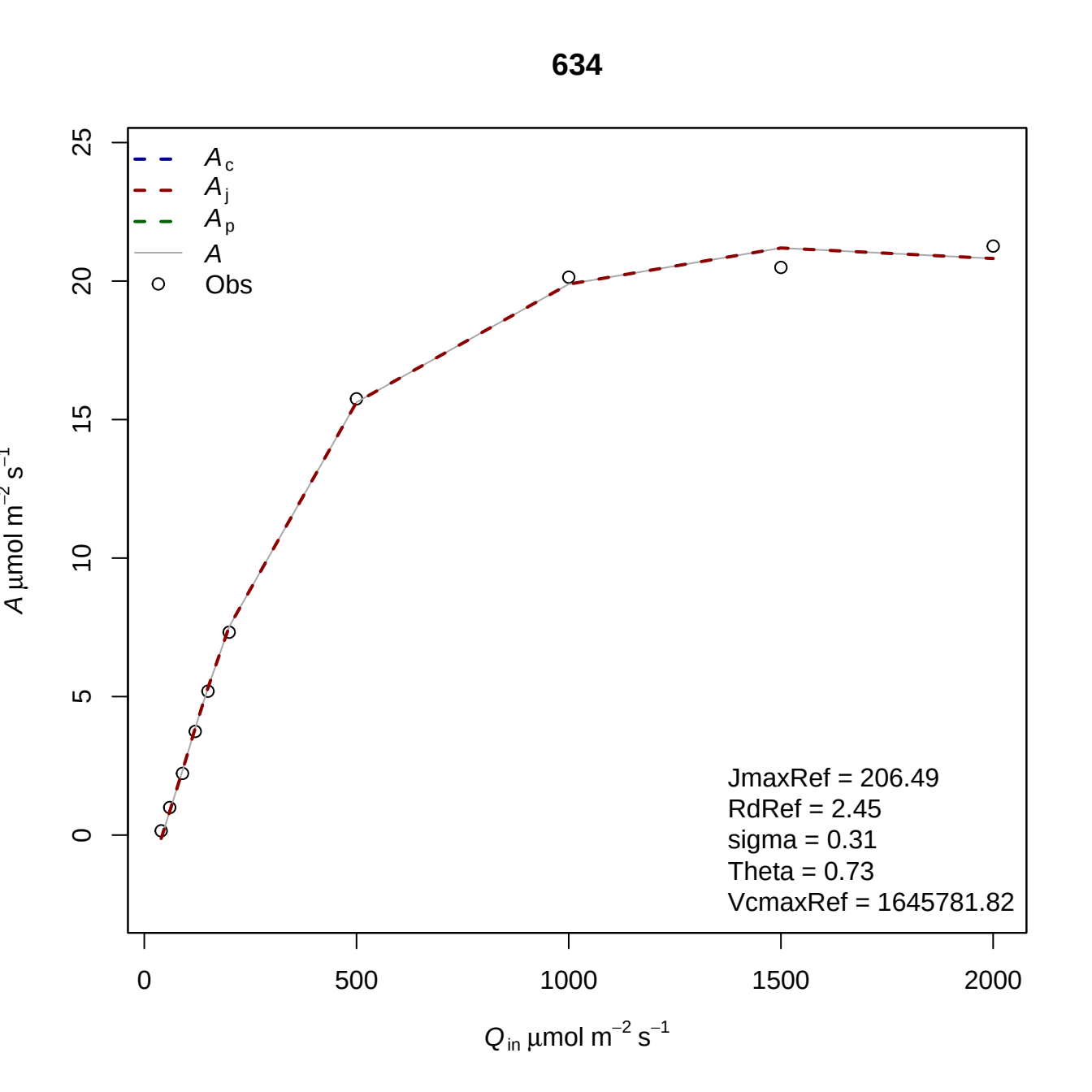

629
A_c
A_j
A_p
A
Obs
JmaxRef = 125.4
RdRef = 0.69
sigma = 0.1
Theta = 0.47
VcmaxRef = 91123.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

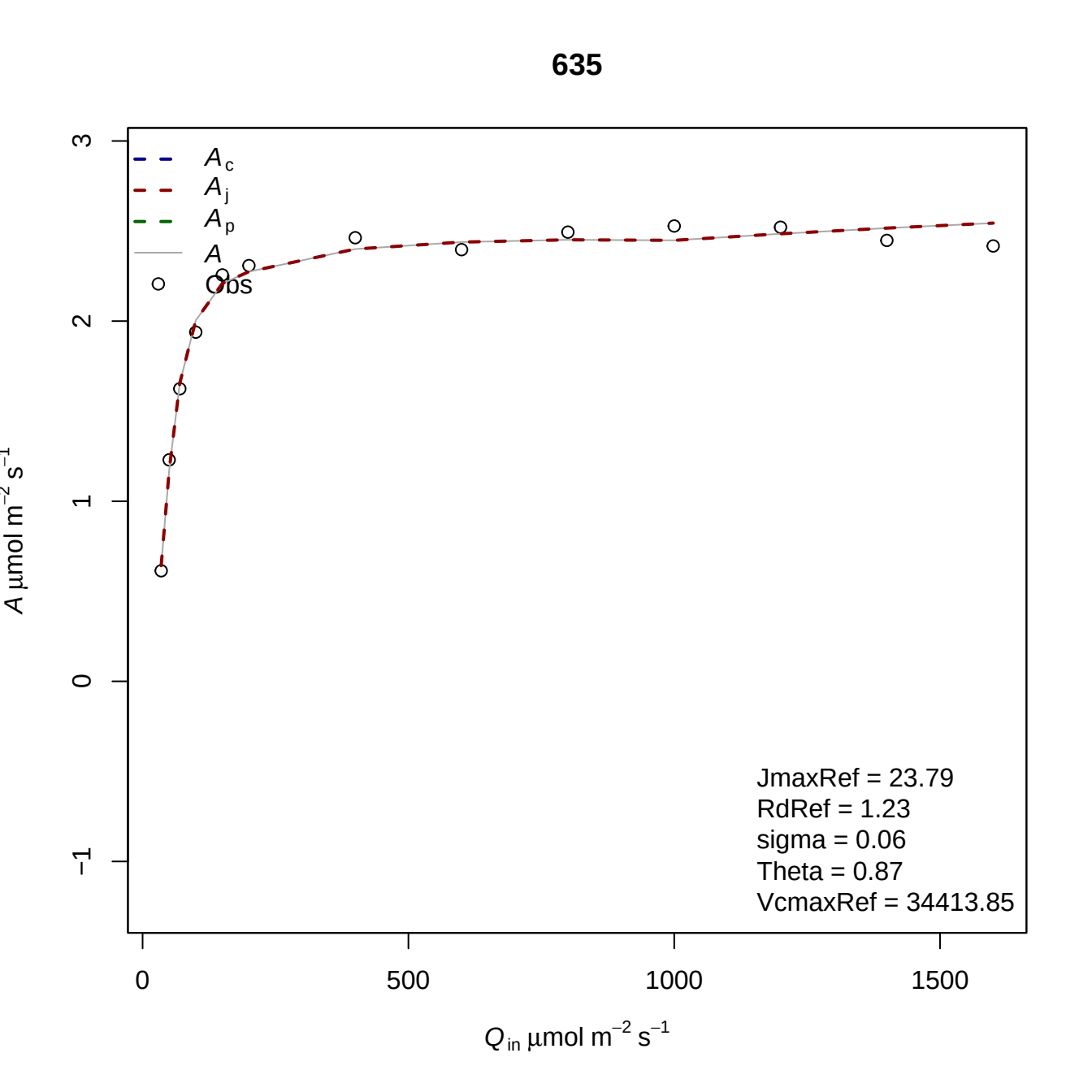

630
A_c
A_j
A_p
A
Obs
JmaxRef = 202.75
RdRef = 1.51
sigma = 0.16
Theta = 0.47
VcmaxRef = 822776.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

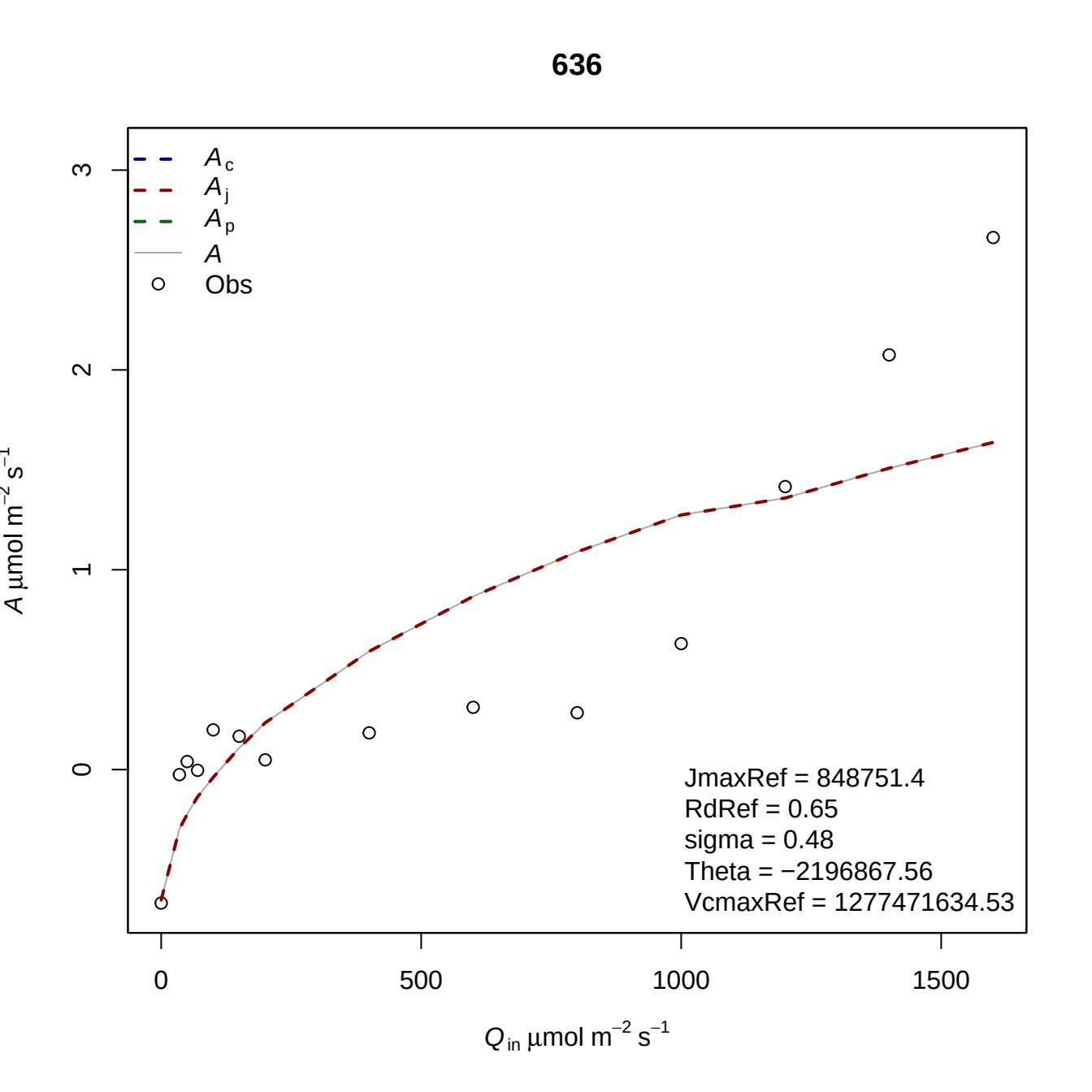

631
A_c
A_j
A_p
A
Obs
JmaxRef = 246.08
RdRef = 2.33
sigma = 0.17
Theta = 0.24
VcmaxRef = 182.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

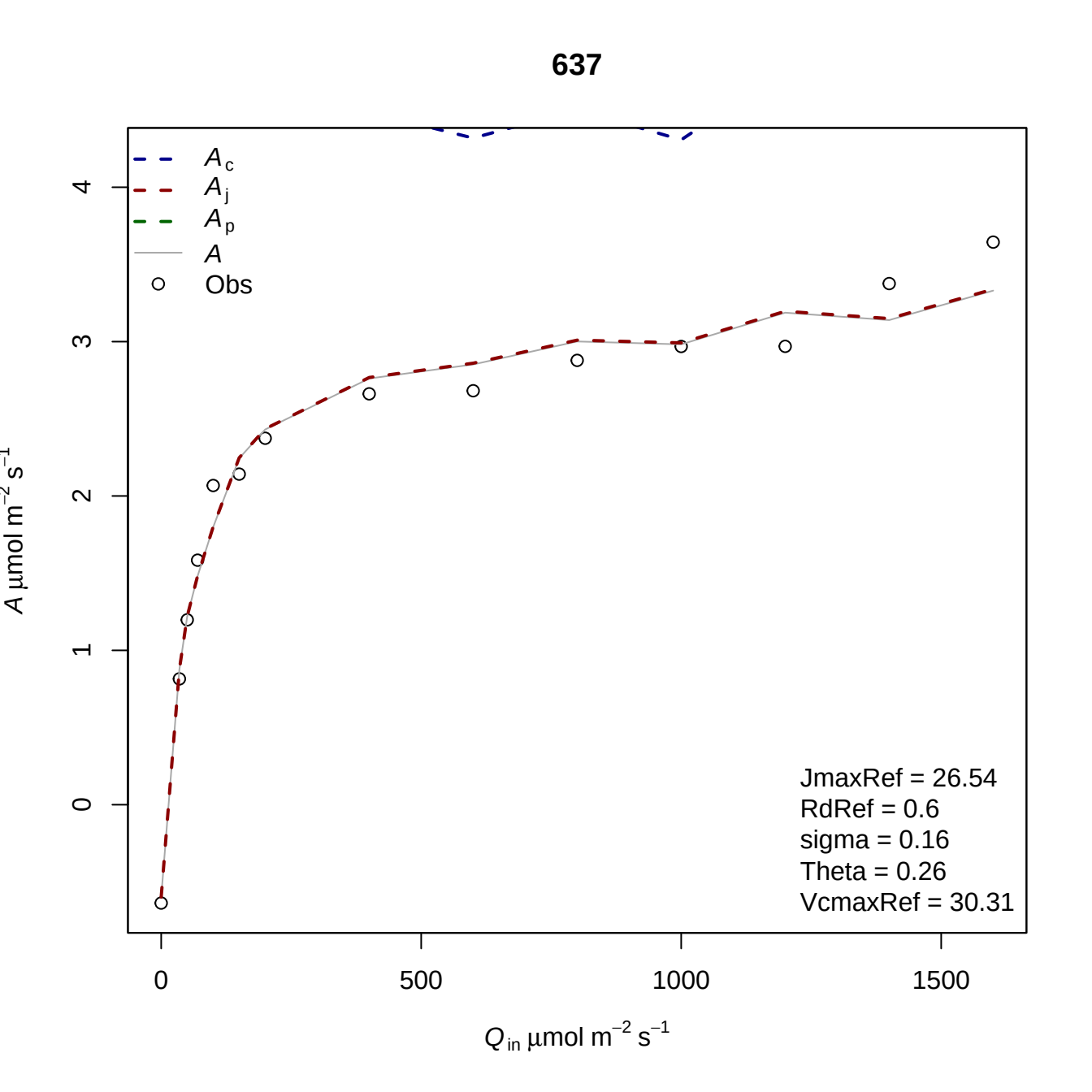

632
A_c
A_j
A_p
A
Obs
JmaxRef = 210.19
RdRef = 1.97
sigma = 0.07
Theta = 0.75
VcmaxRef = 6983.18
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

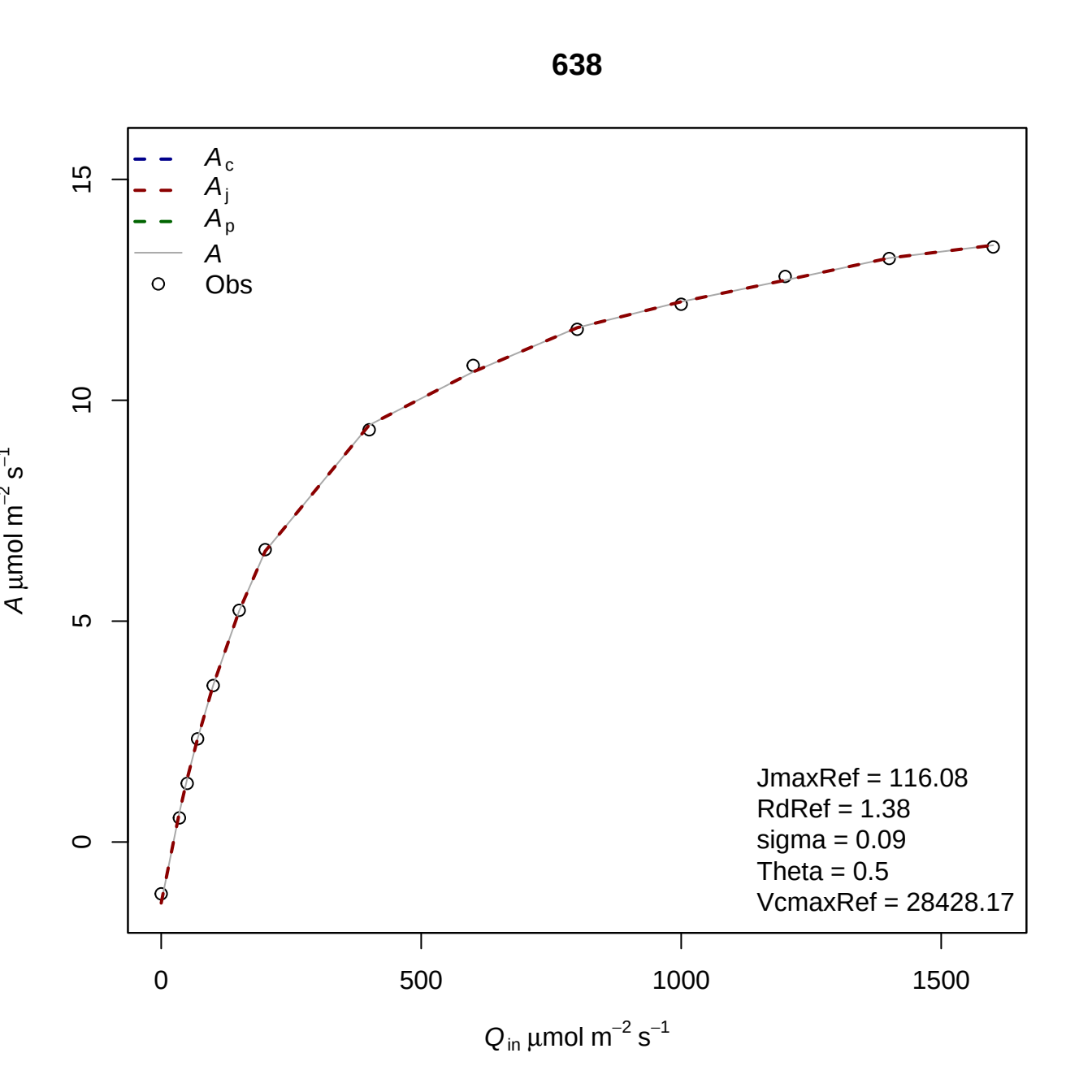

633
A_c
A_j
A_p
A
Obs
JmaxRef = 276.39
RdRef = 3.96
sigma = 0.06
Theta = 0.04
VcmaxRef = 169626.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

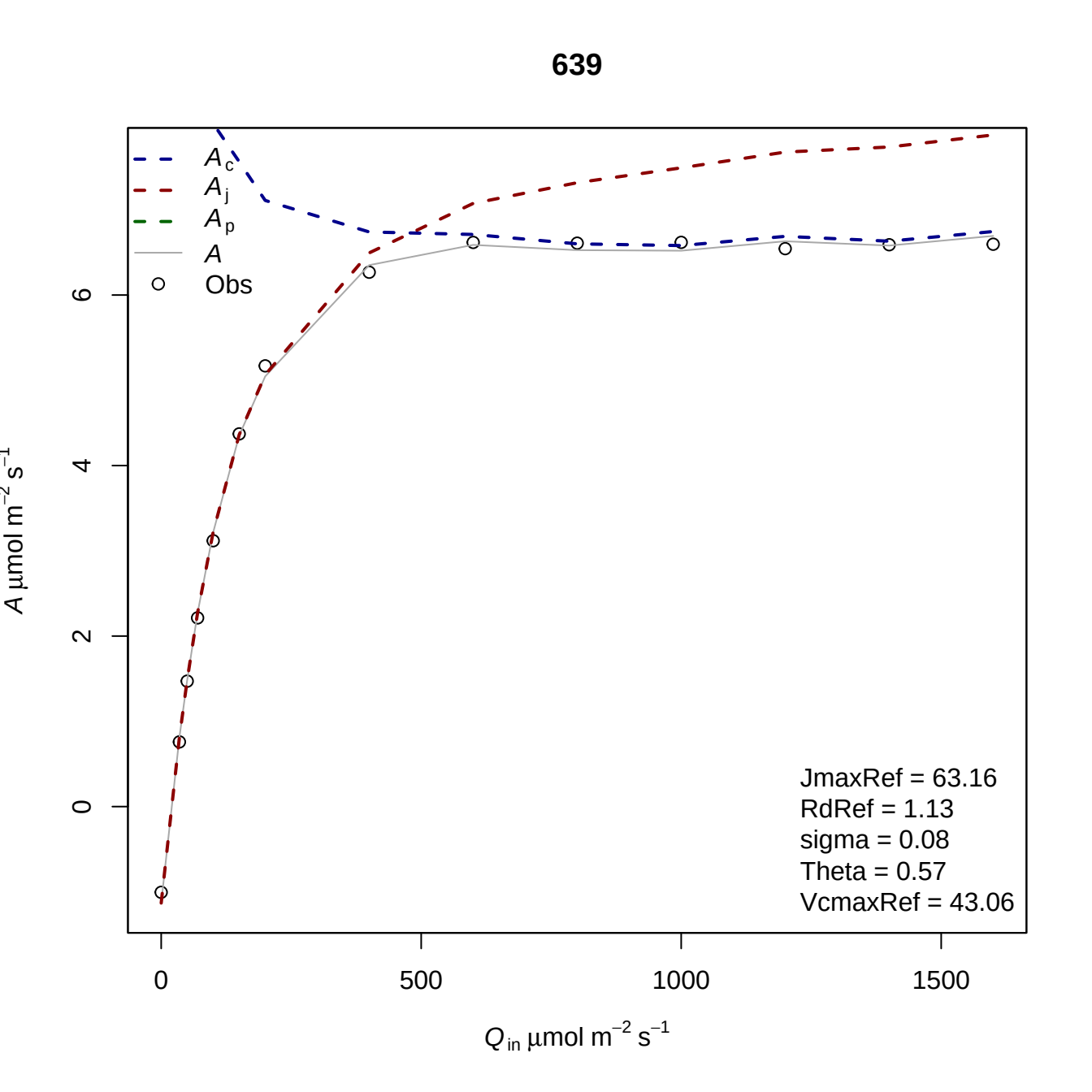

634
A_c
A_j
A_p
A
Obs
JmaxRef = 206.49
RdRef = 2.45
sigma = 0.31
Theta = 0.73
VcmaxRef = 1645781.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

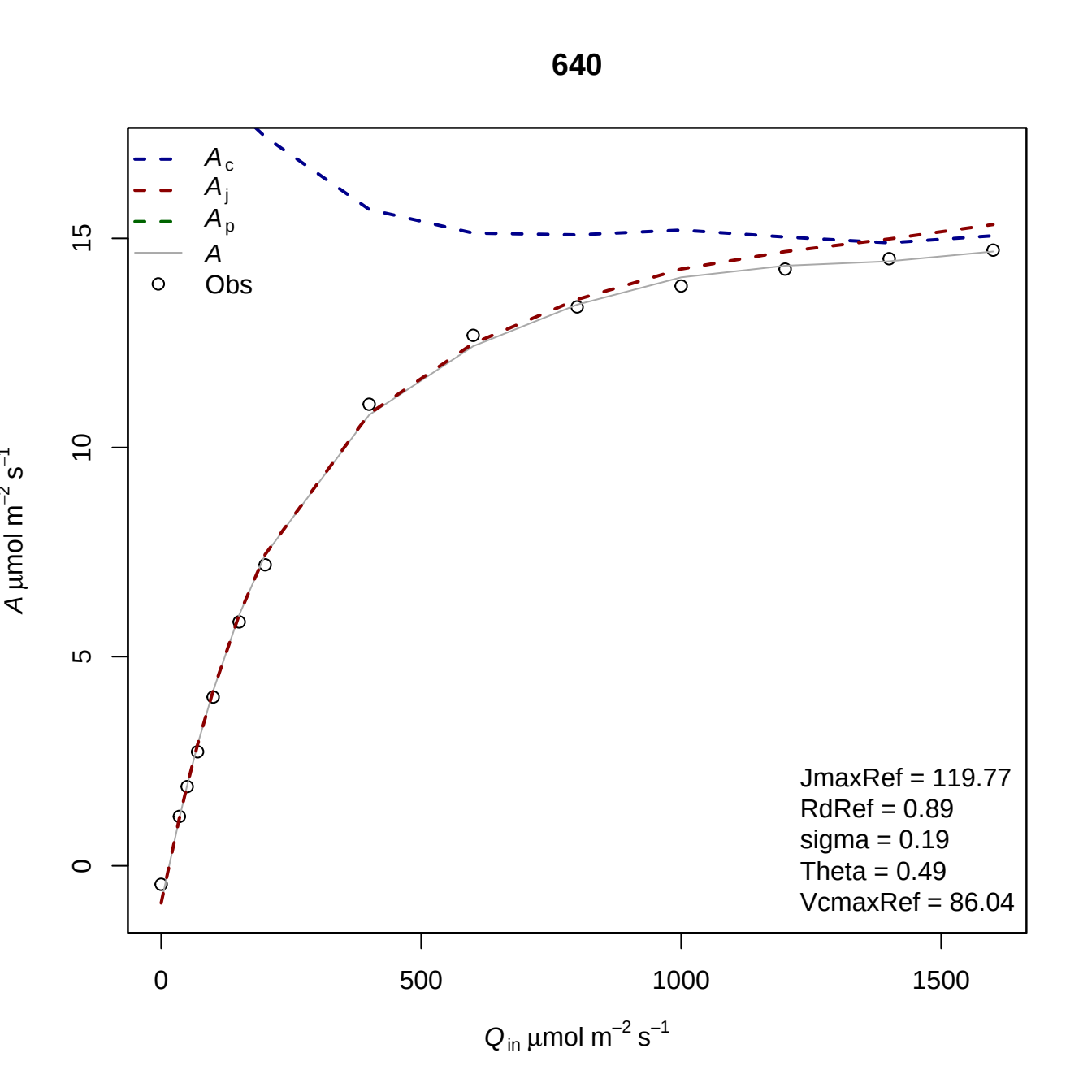

635
A_c
A_j
A_p
A
Obs
JmaxRef = 23.79
RdRef = 1.23
sigma = 0.06
Theta = 0.87
VcmaxRef = 34413.85
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

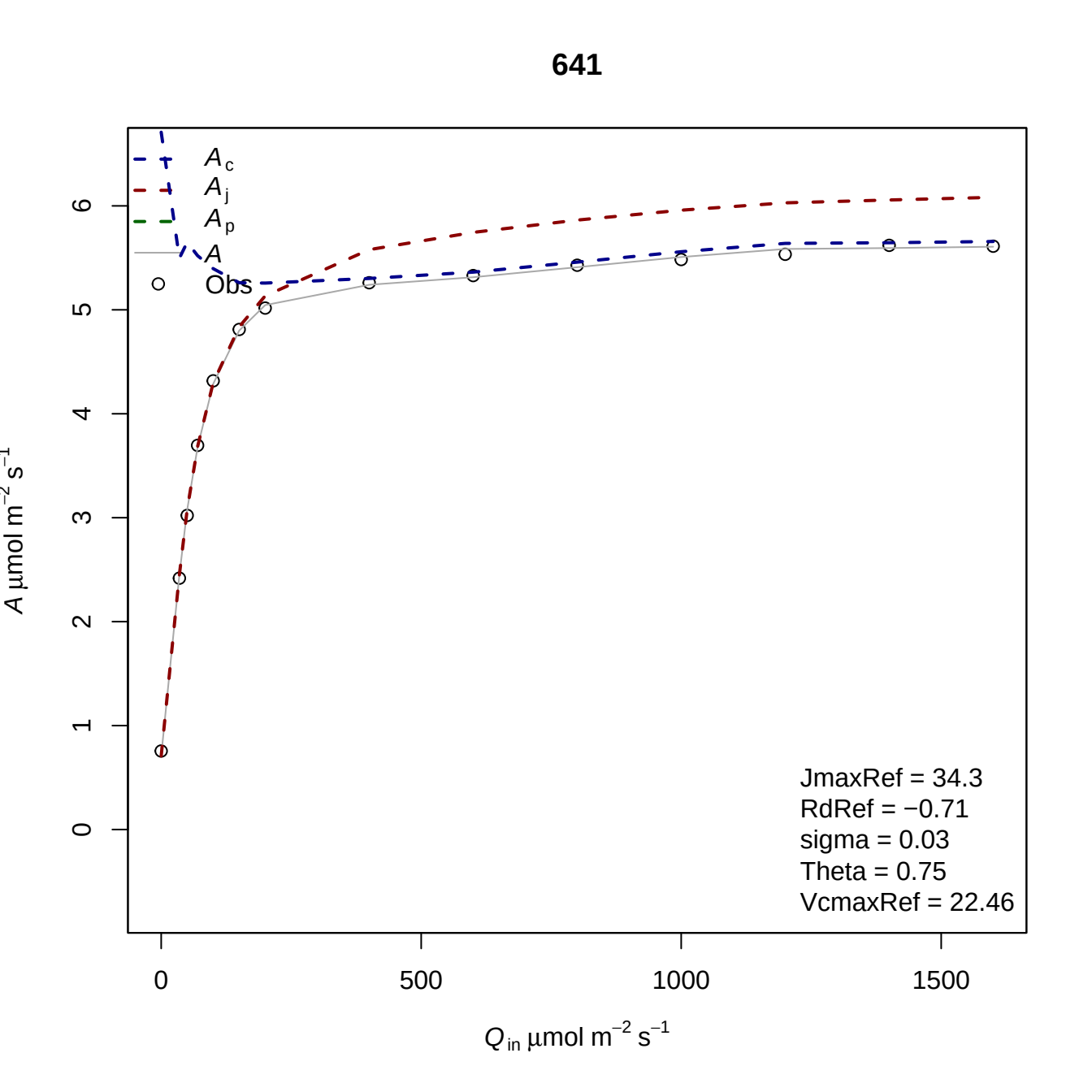

636
A_c
A_j
A_p
A
Obs
JmaxRef = 848751.4
RdRef = 0.65
sigma = 0.48
Theta = −2196867.56
VcmaxRef = 1277471634.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

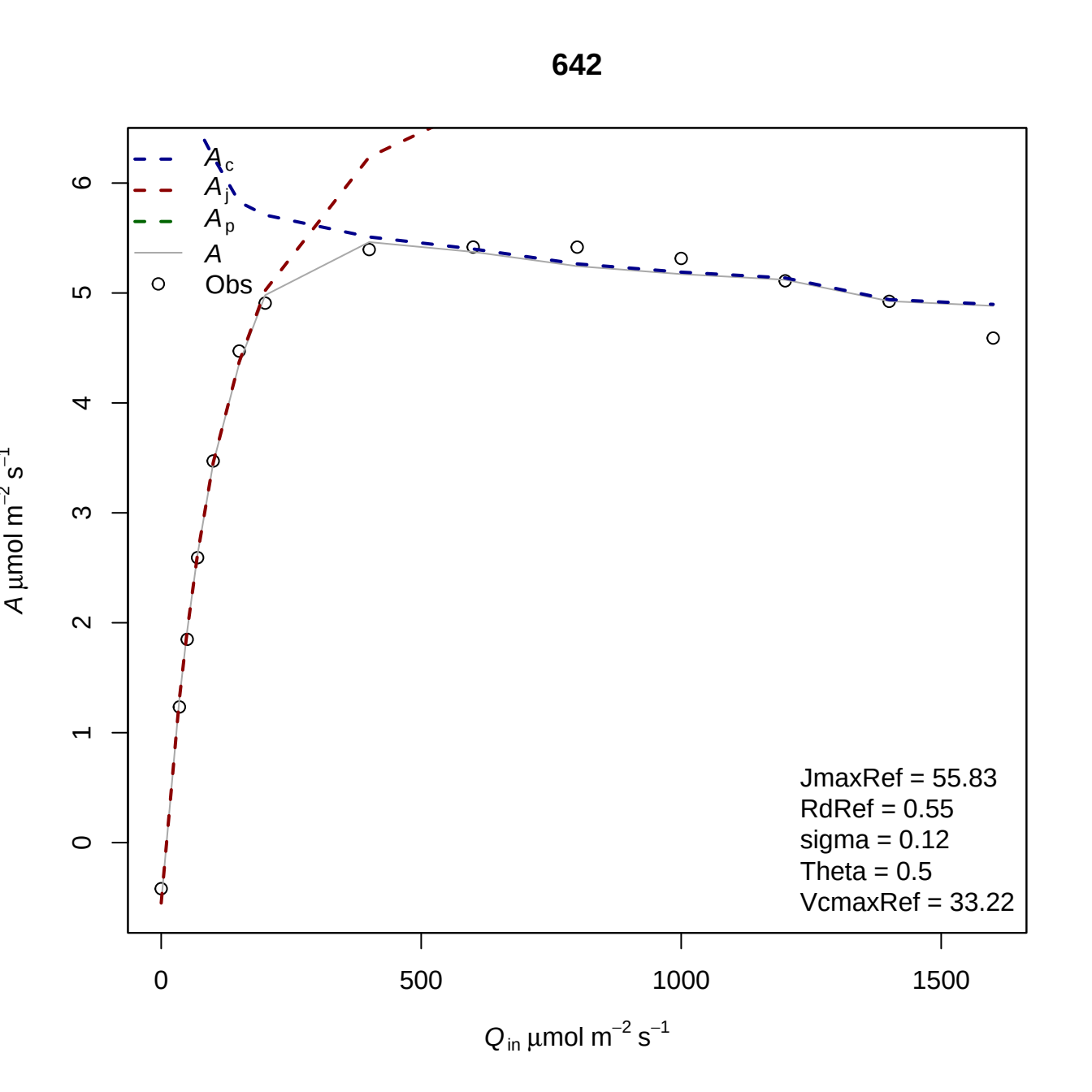

637
A_c
A_j
A_p
A
Obs
JmaxRef = 26.54
RdRef = 0.6
sigma = 0.16
Theta = 0.26
VcmaxRef = 30.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

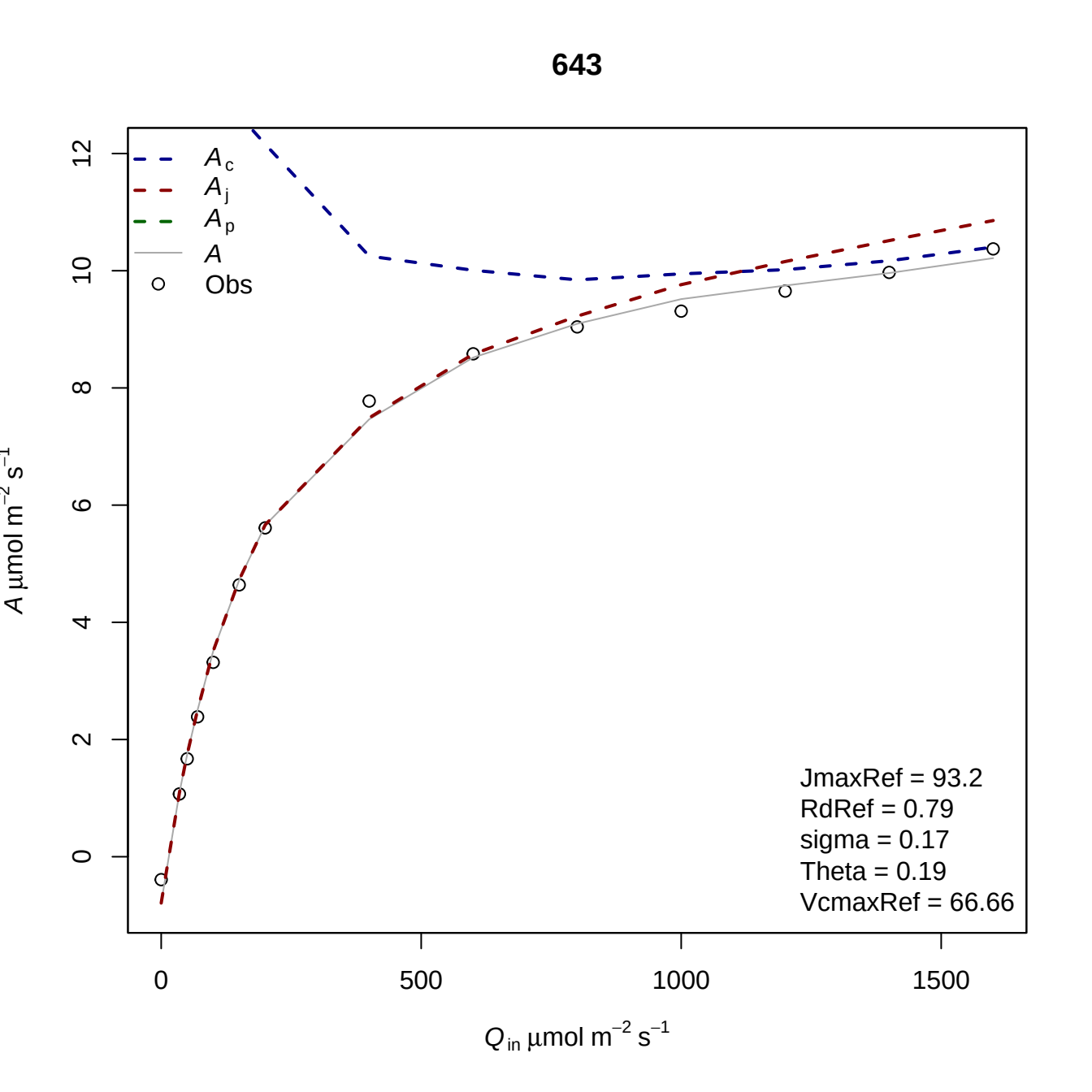

638
A_c
A_j
A_p
A
Obs
JmaxRef = 116.08
RdRef = 1.38
sigma = 0.09
Theta = 0.5
VcmaxRef = 28428.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

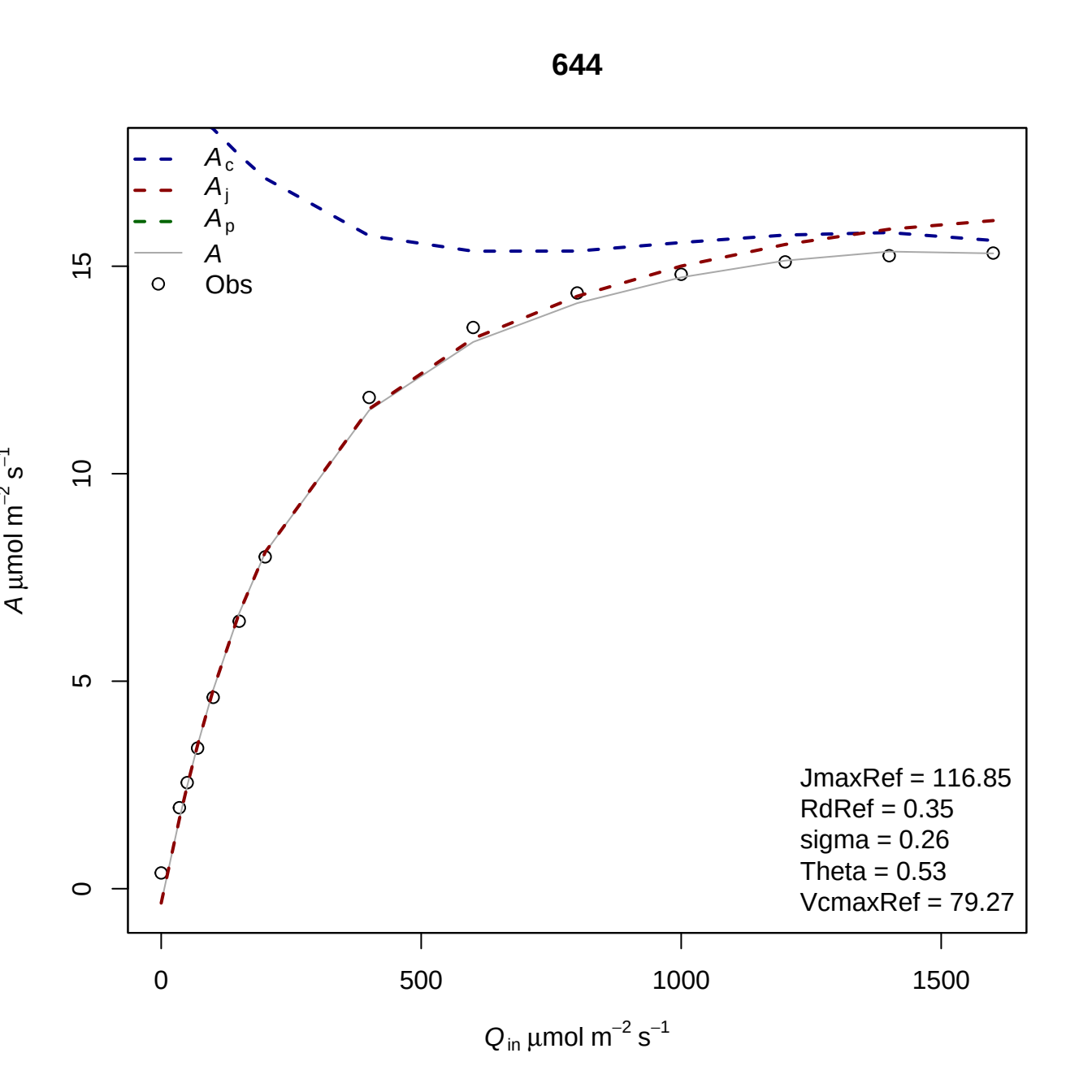

639
A_c
A_j
A_p
A
Obs
JmaxRef = 63.16
RdRef = 1.13
sigma = 0.08
Theta = 0.57
VcmaxRef = 43.06
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

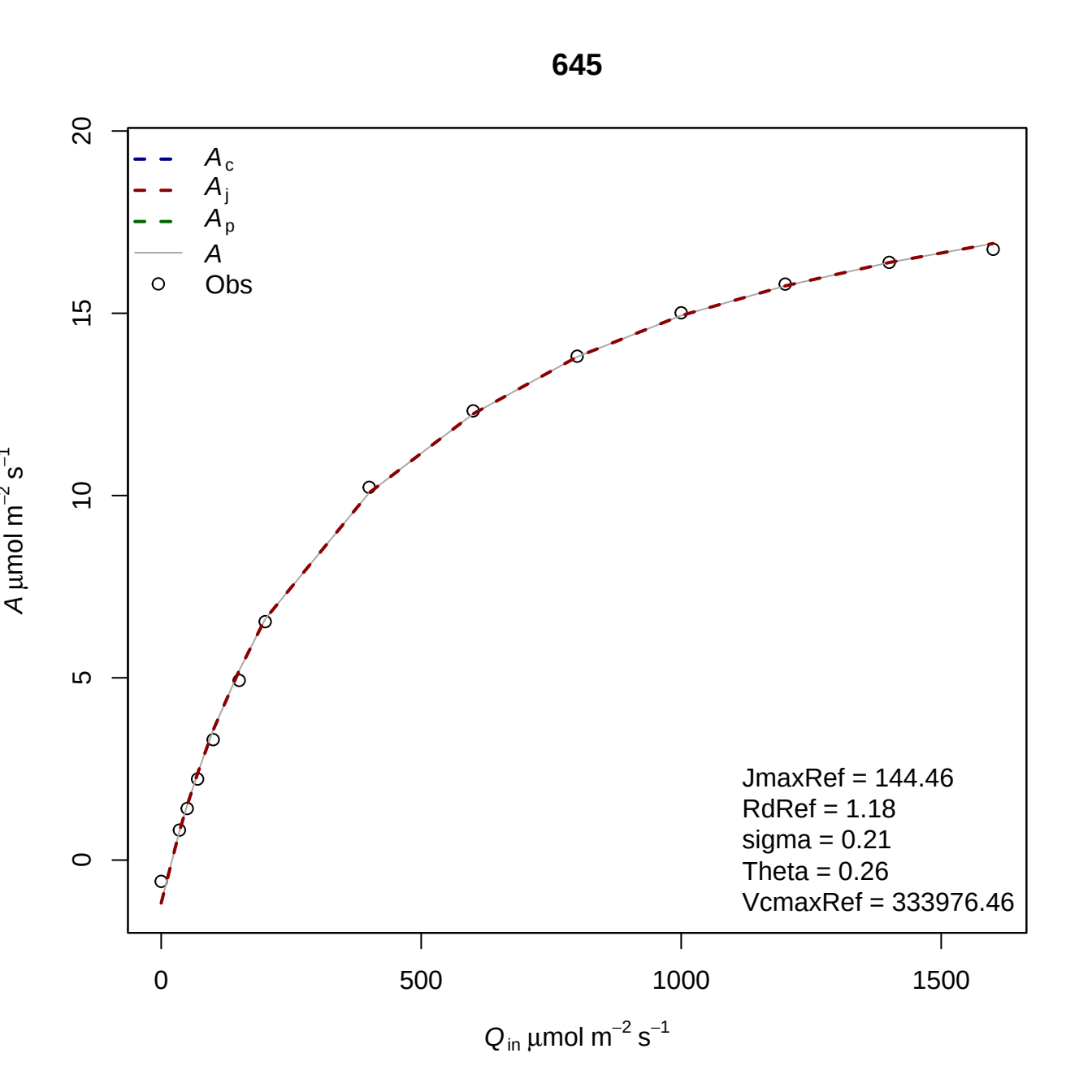

640
A_c
A_j
A_p
A
Obs
JmaxRef = 119.77
RdRef = 0.89
sigma = 0.19
Theta = 0.49
VcmaxRef = 86.04
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

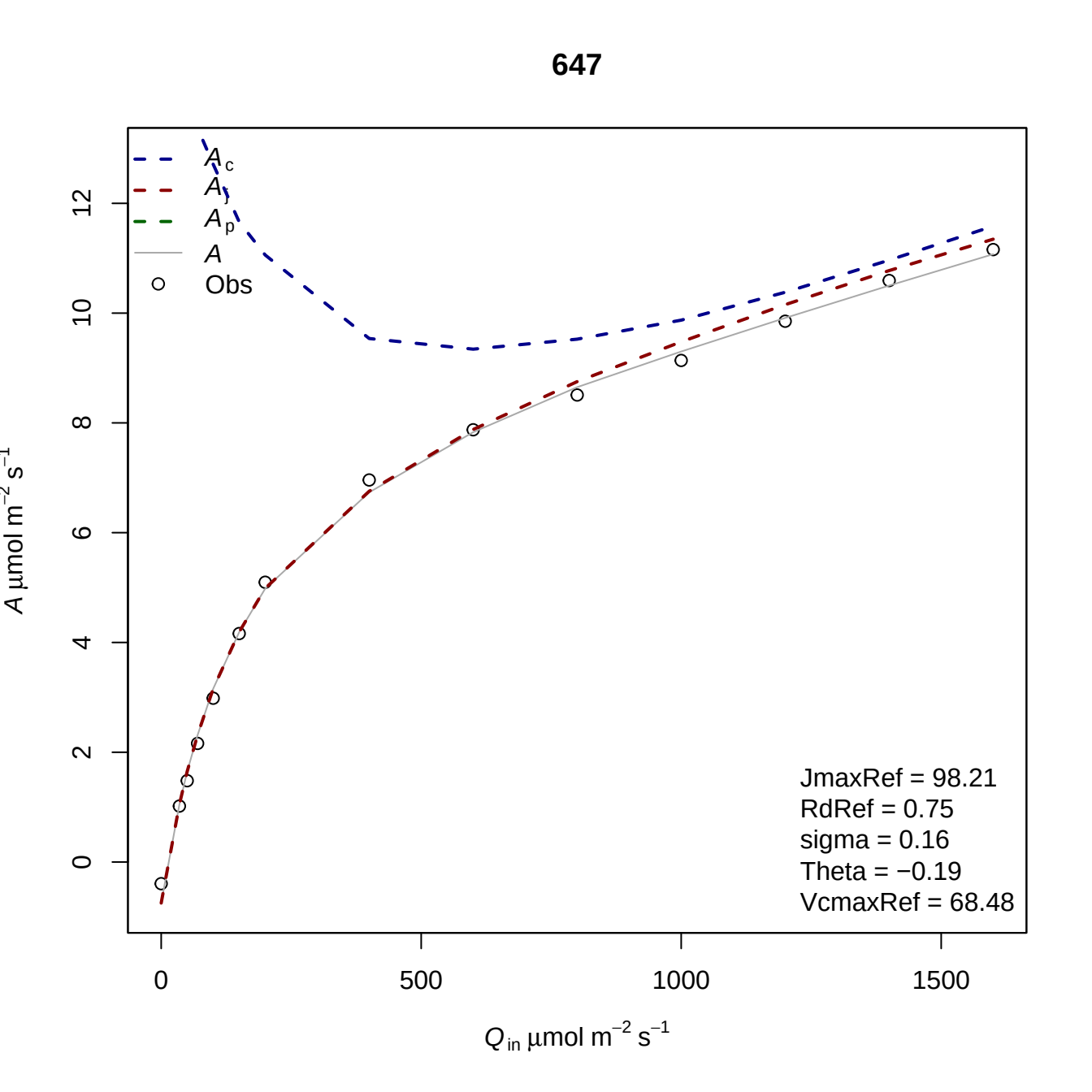

641
A_c
A_j
A_p
A
Obs
JmaxRef = 34.3
RdRef = −0.71
sigma = 0.03
Theta = 0.75
VcmaxRef = 22.46
0
1
2
3
4
5
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

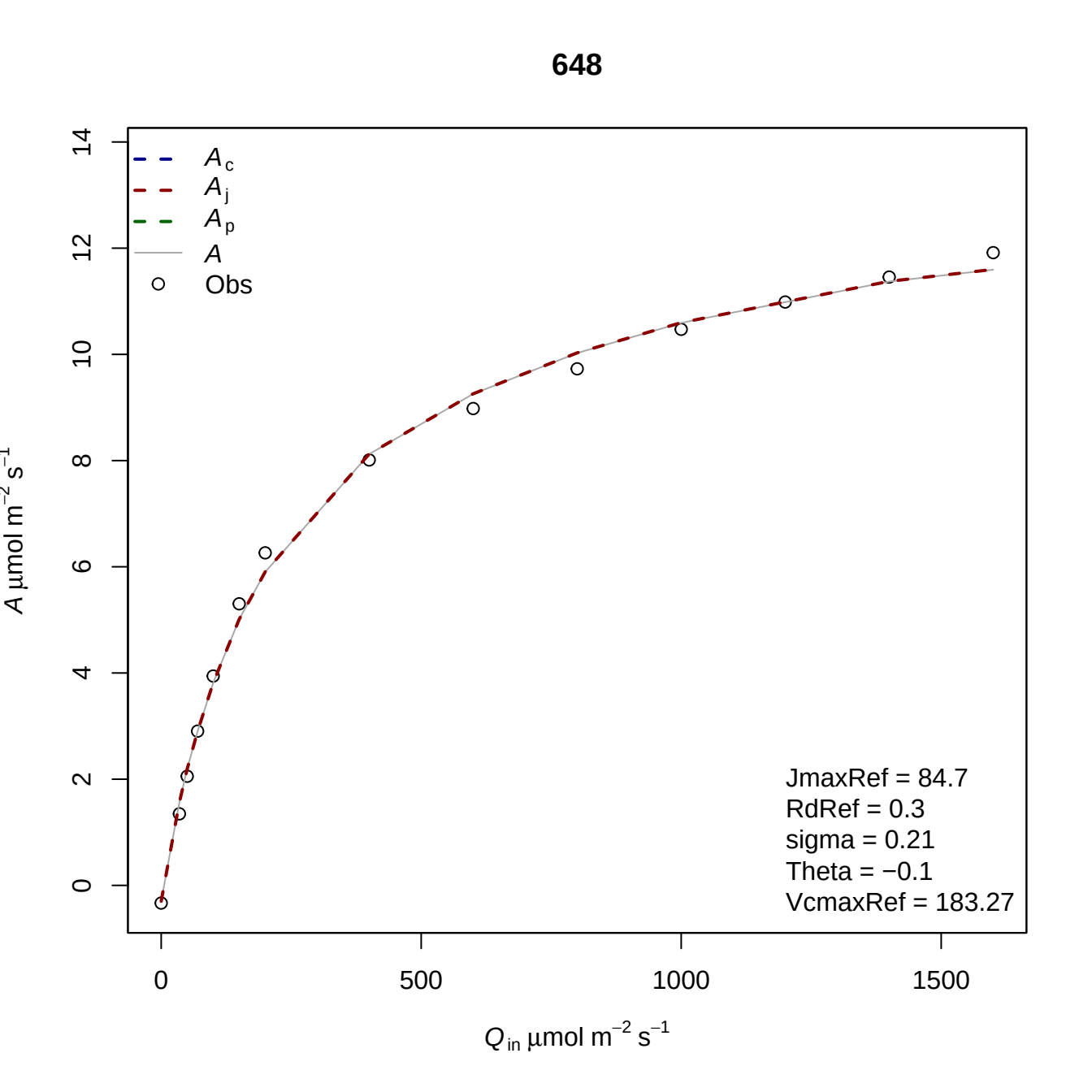

642
A_c
A_j
A_p
A
Obs
JmaxRef = 55.83
RdRef = 0.55
sigma = 0.12
Theta = 0.5
VcmaxRef = 33.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

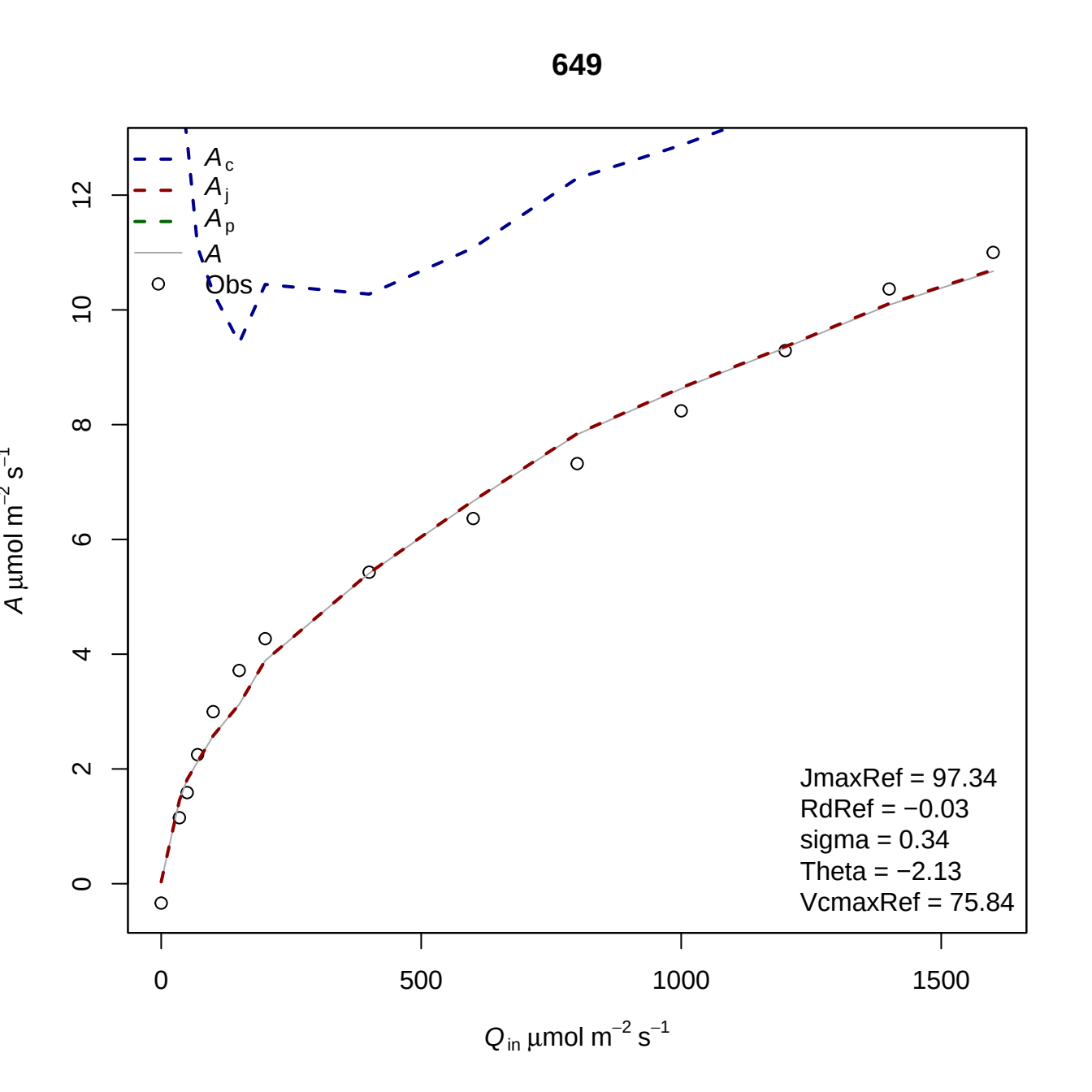

643
A_c
A_j
A_p
A
Obs
JmaxRef = 93.2
RdRef = 0.79
sigma = 0.17
Theta = 0.19
VcmaxRef = 66.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

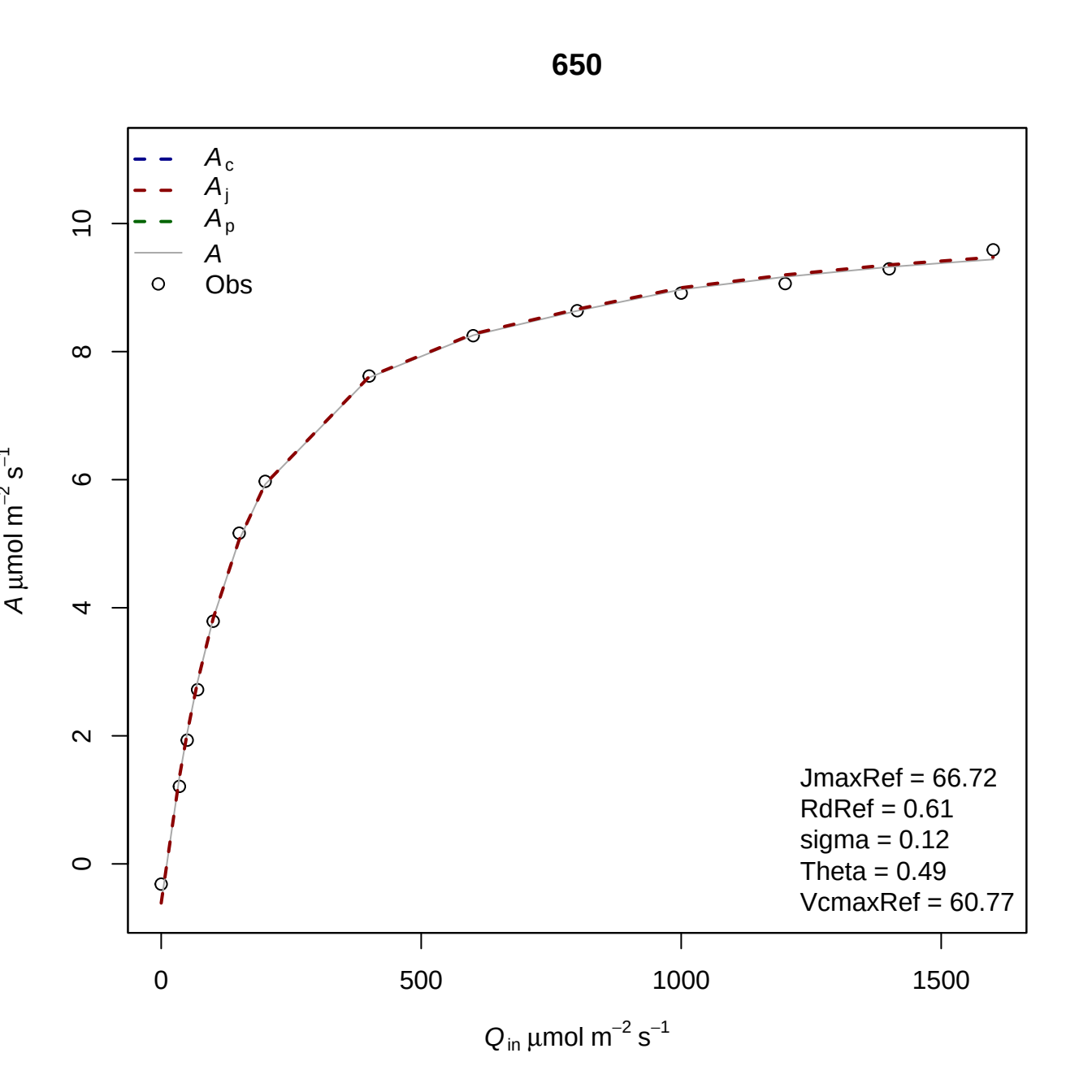

644
A_c
A_j
A_p
A
Obs
JmaxRef = 116.85
RdRef = 0.35
sigma = 0.26
Theta = 0.53
VcmaxRef = 79.27
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

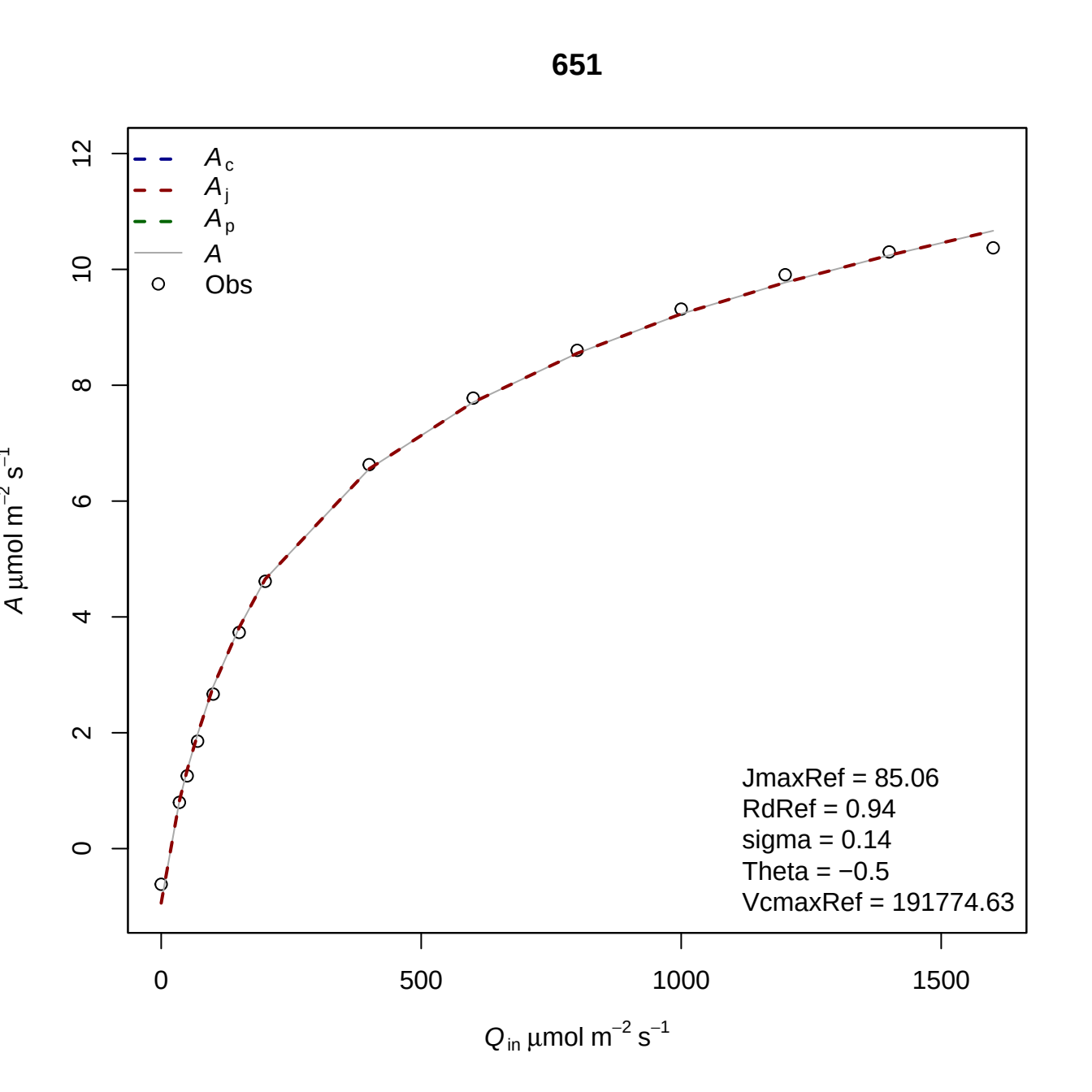

645
A_c
A_j
A_p
A
Obs
JmaxRef = 144.46
RdRef = 1.18
sigma = 0.21
Theta = 0.26
VcmaxRef = 333976.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

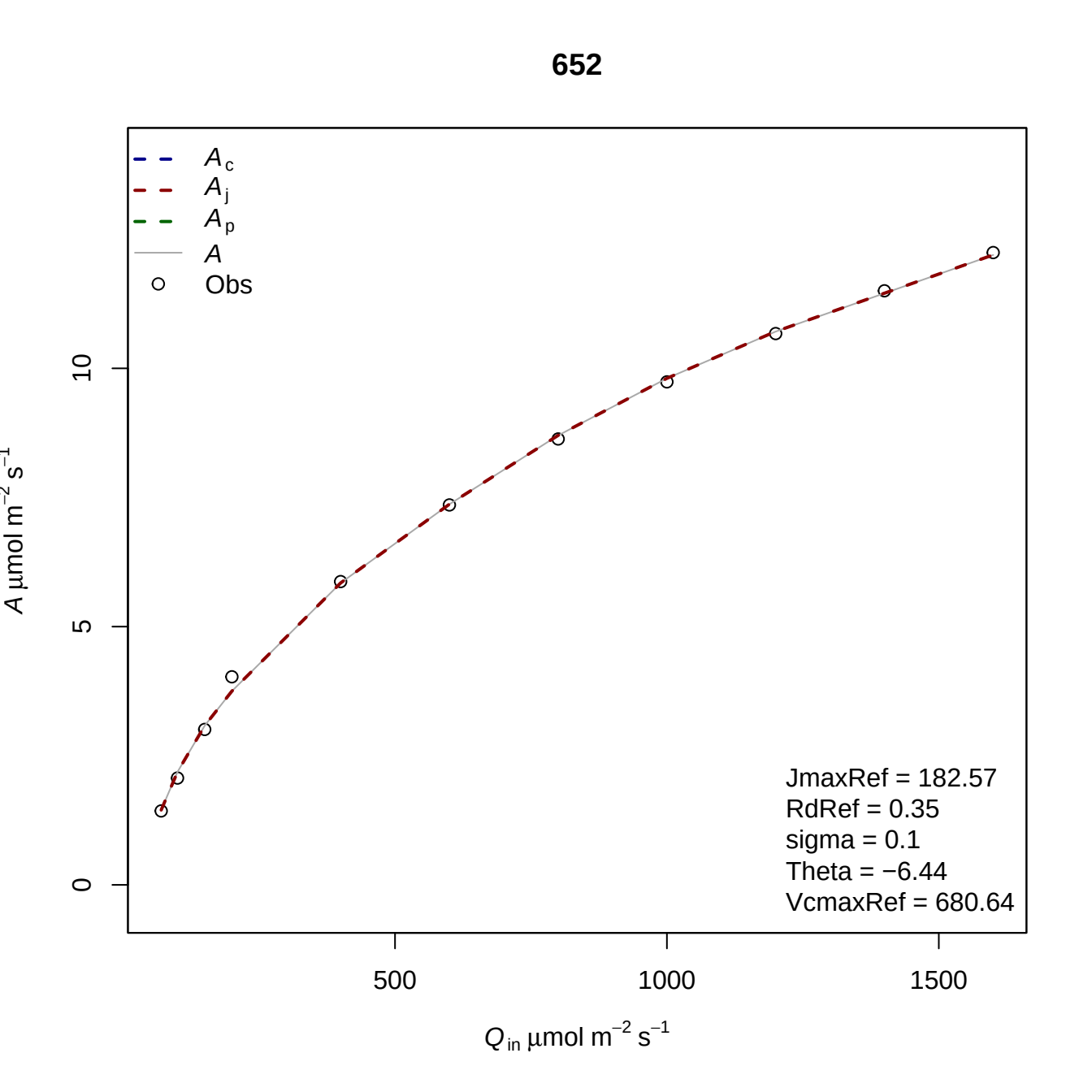

647
A_c
A_j
A_p
A
Obs
JmaxRef = 98.21
RdRef = 0.75
sigma = 0.16
Theta = −0.19
VcmaxRef = 68.48
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

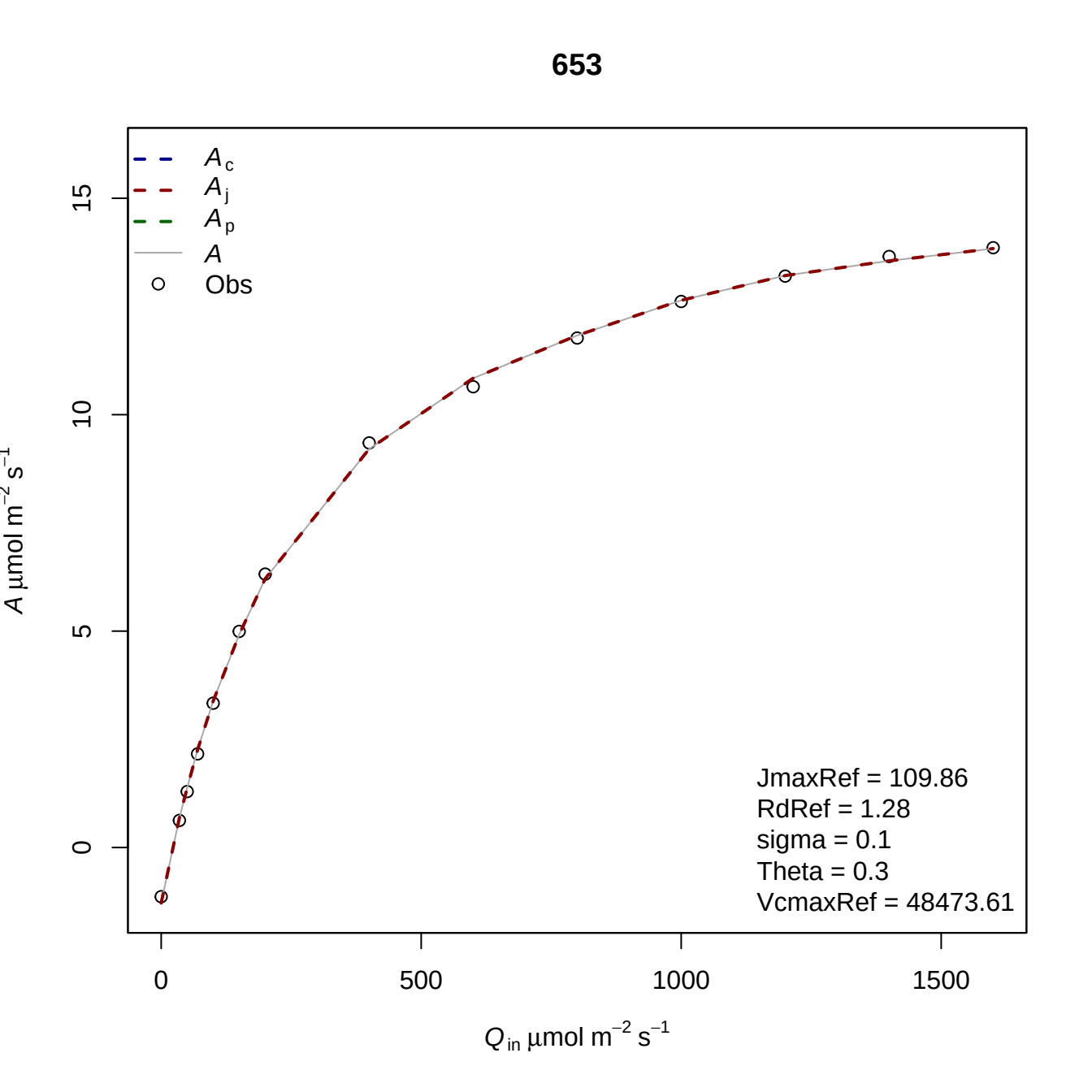

648
A_c
A_j
A_p
A
Obs
JmaxRef = 84.7
RdRef = 0.3
sigma = 0.21
Theta = −0.1
VcmaxRef = 183.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

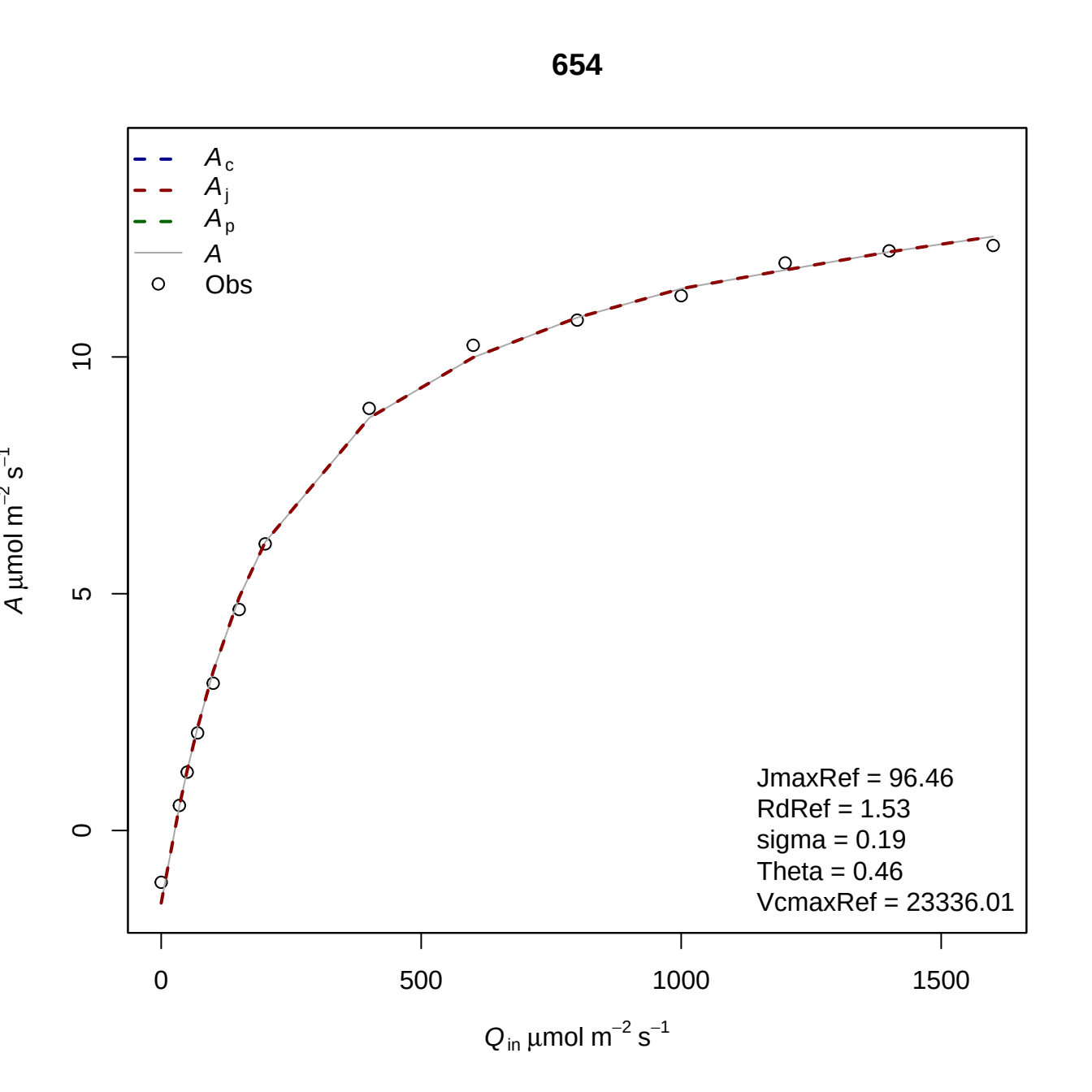

649
A_c
A_j
A_p
A
Obs
JmaxRef = 97.34
RdRef = −0.03
sigma = 0.34
Theta = −2.13
VcmaxRef = 75.84
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

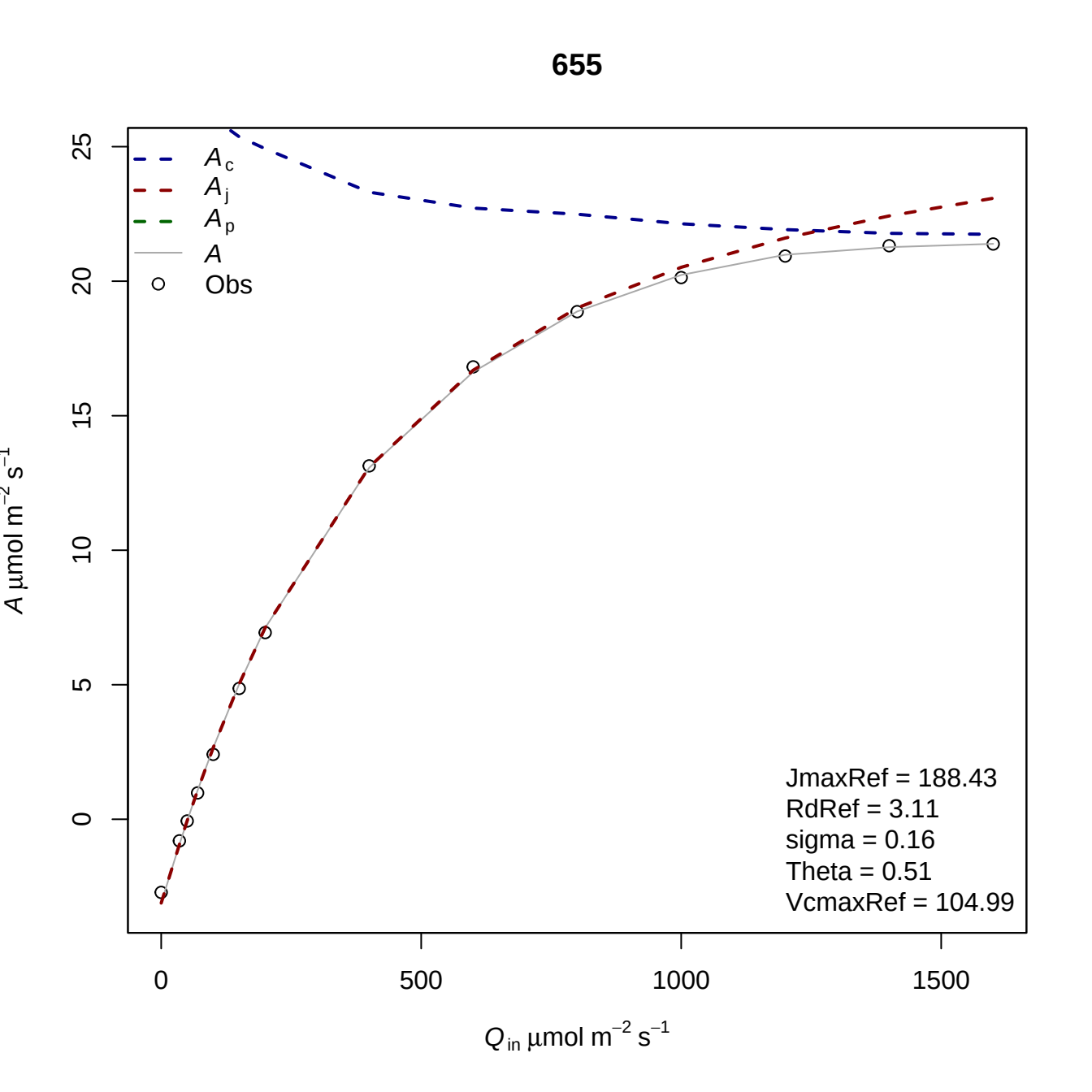

650
A_c
A_j
A_p
A
Obs
JmaxRef = 66.72
RdRef = 0.61
sigma = 0.12
Theta = 0.49
VcmaxRef = 60.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

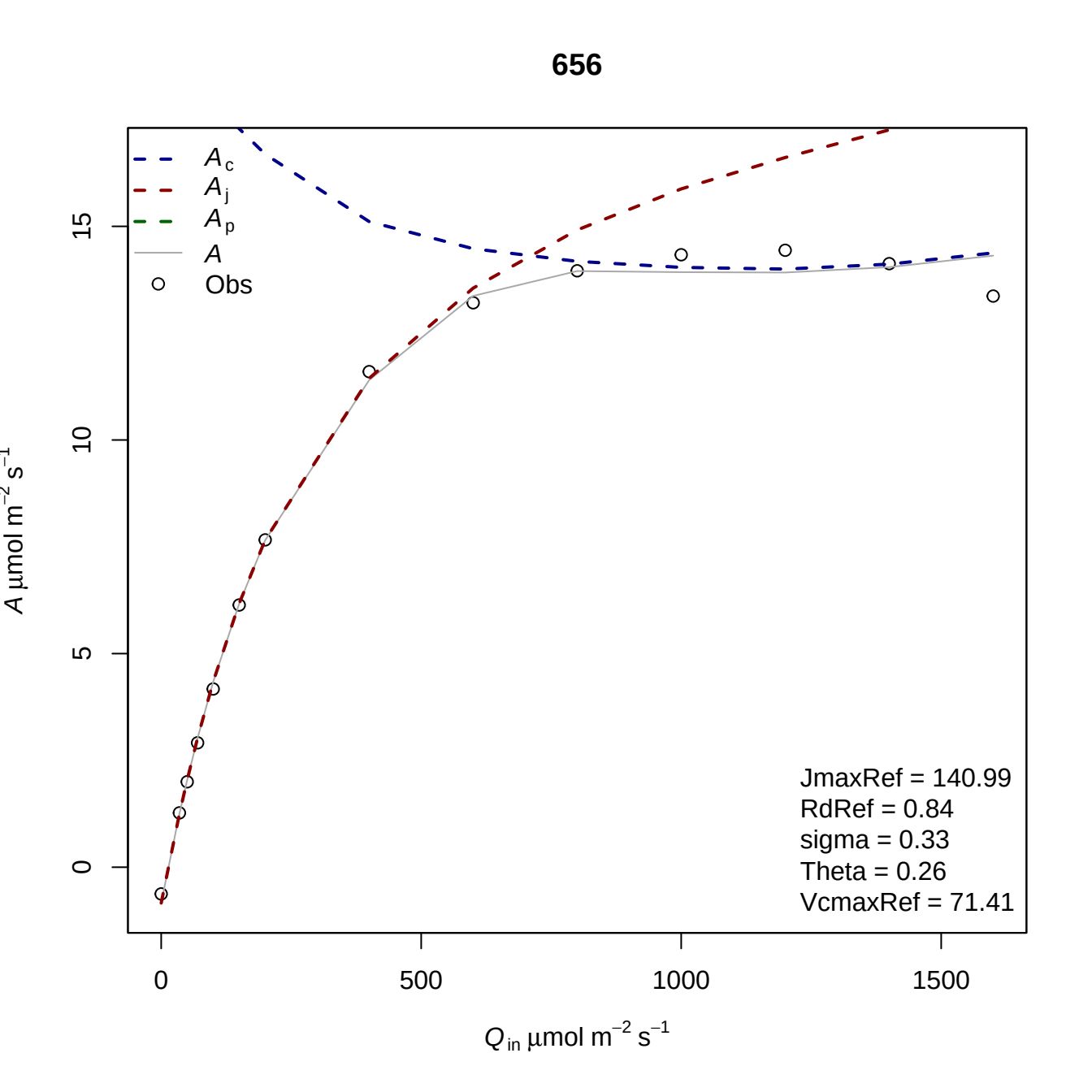

651
A_c
A_j
A_p
A
Obs
JmaxRef = 85.06
RdRef = 0.94
sigma = 0.14
Theta = −0.5
VcmaxRef = 191774.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

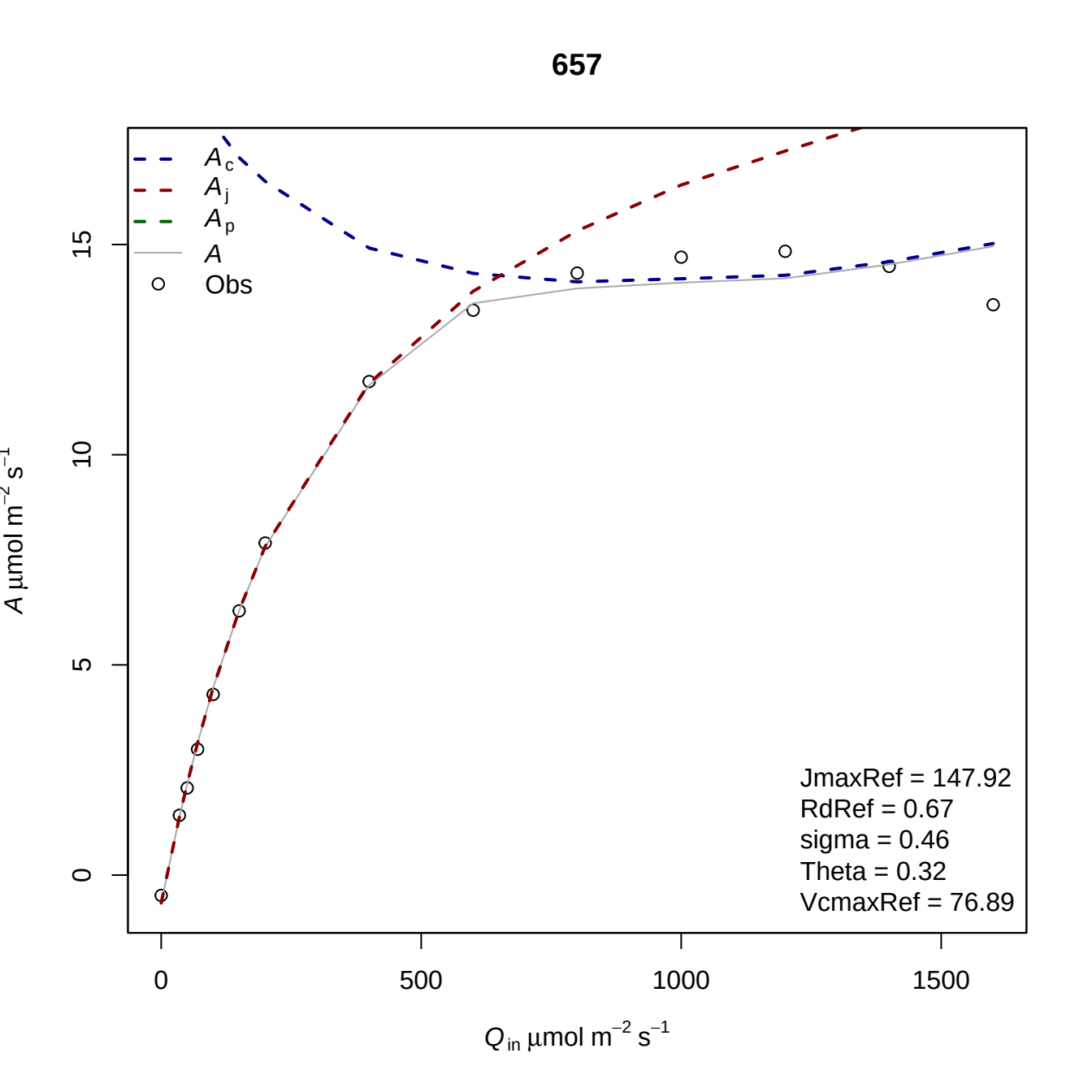

652
A_c
A_j
A_p
A
Obs
JmaxRef = 182.57
RdRef = 0.35
sigma = 0.1
Theta = −6.44
VcmaxRef = 680.64
0
5
10
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

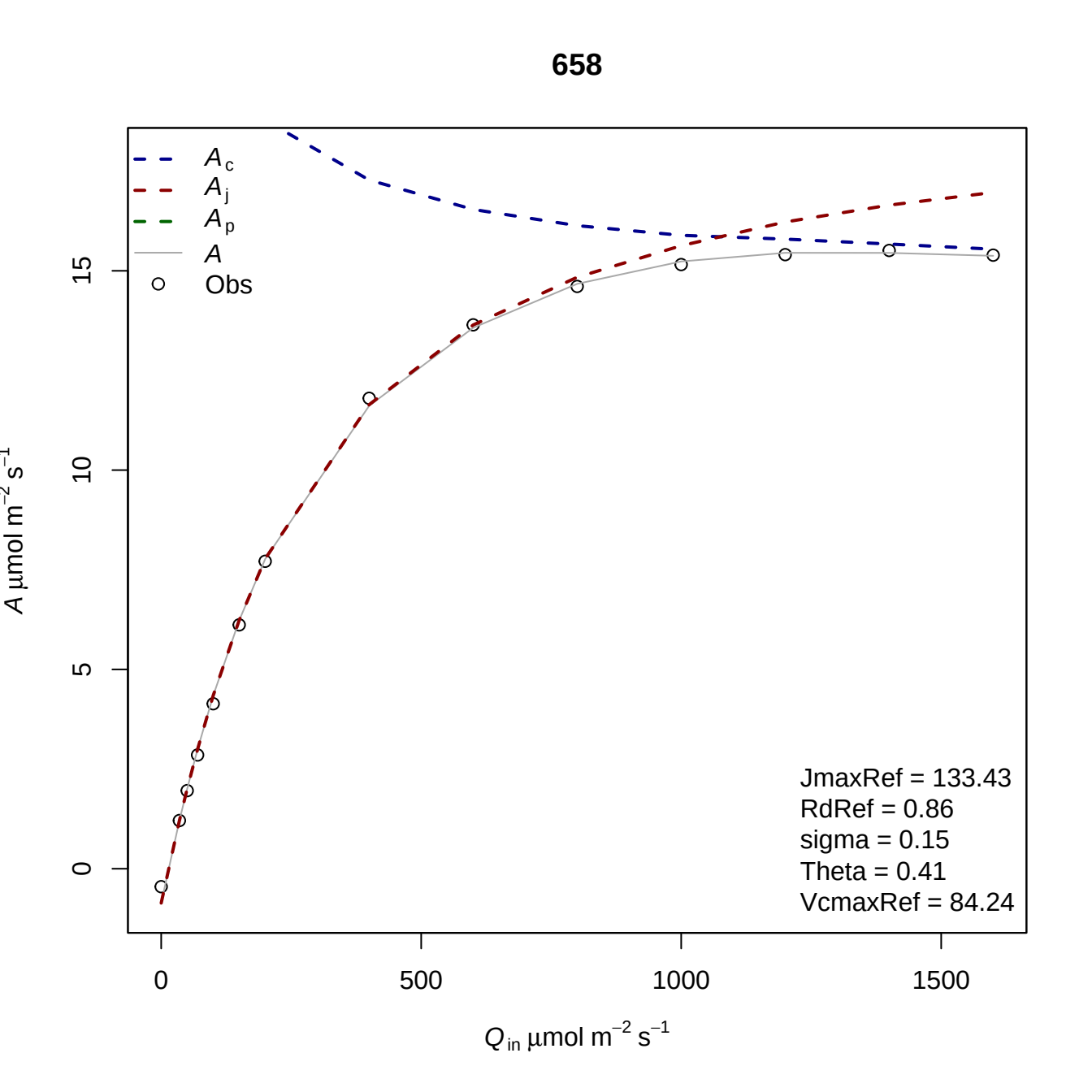

653
A_c
A_j
A_p
A
Obs
JmaxRef = 109.86
RdRef = 1.28
sigma = 0.1
Theta = 0.3
VcmaxRef = 48473.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

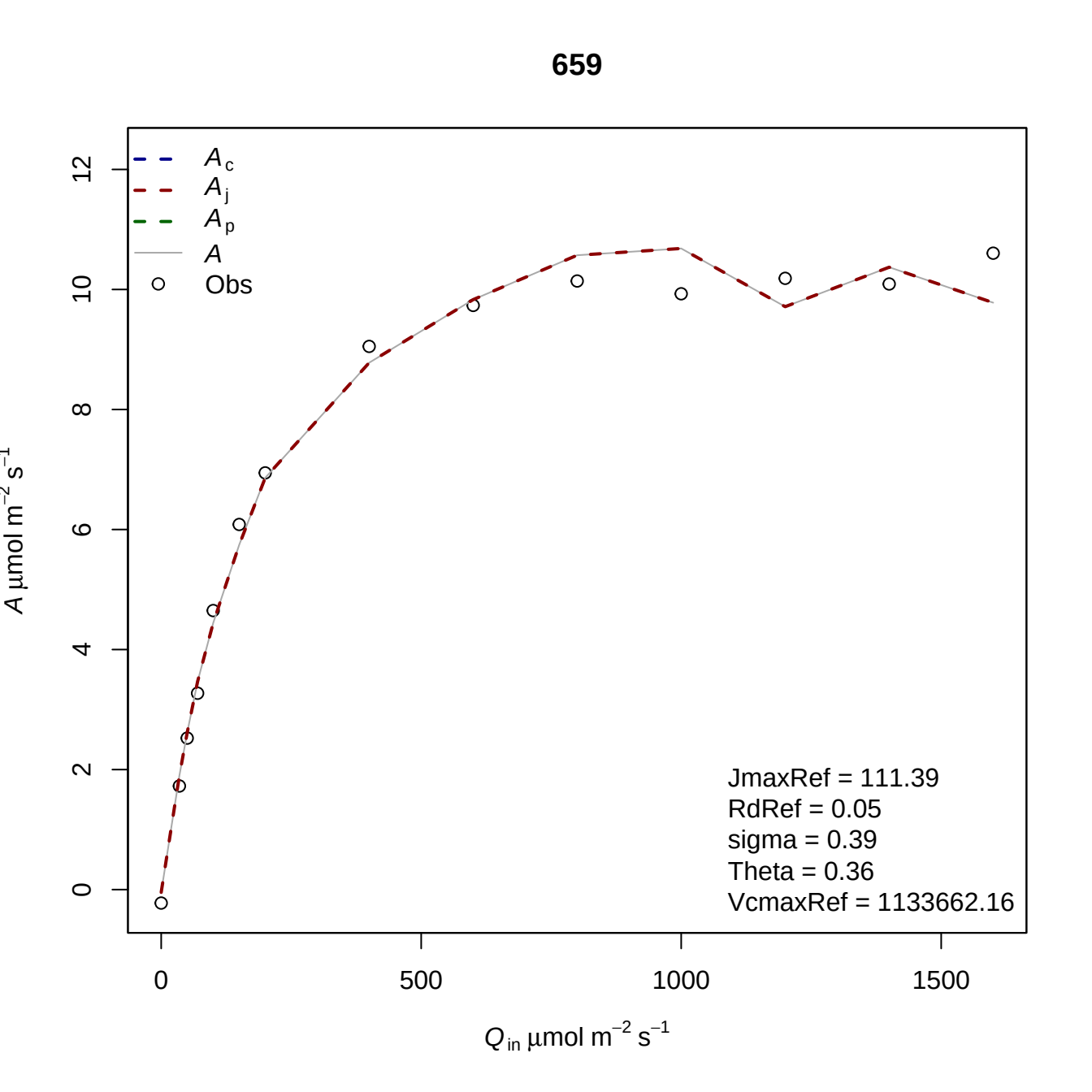

654
A_c
A_j
A_p
A
Obs
JmaxRef = 96.46
RdRef = 1.53
sigma = 0.19
Theta = 0.46
VcmaxRef = 23336.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

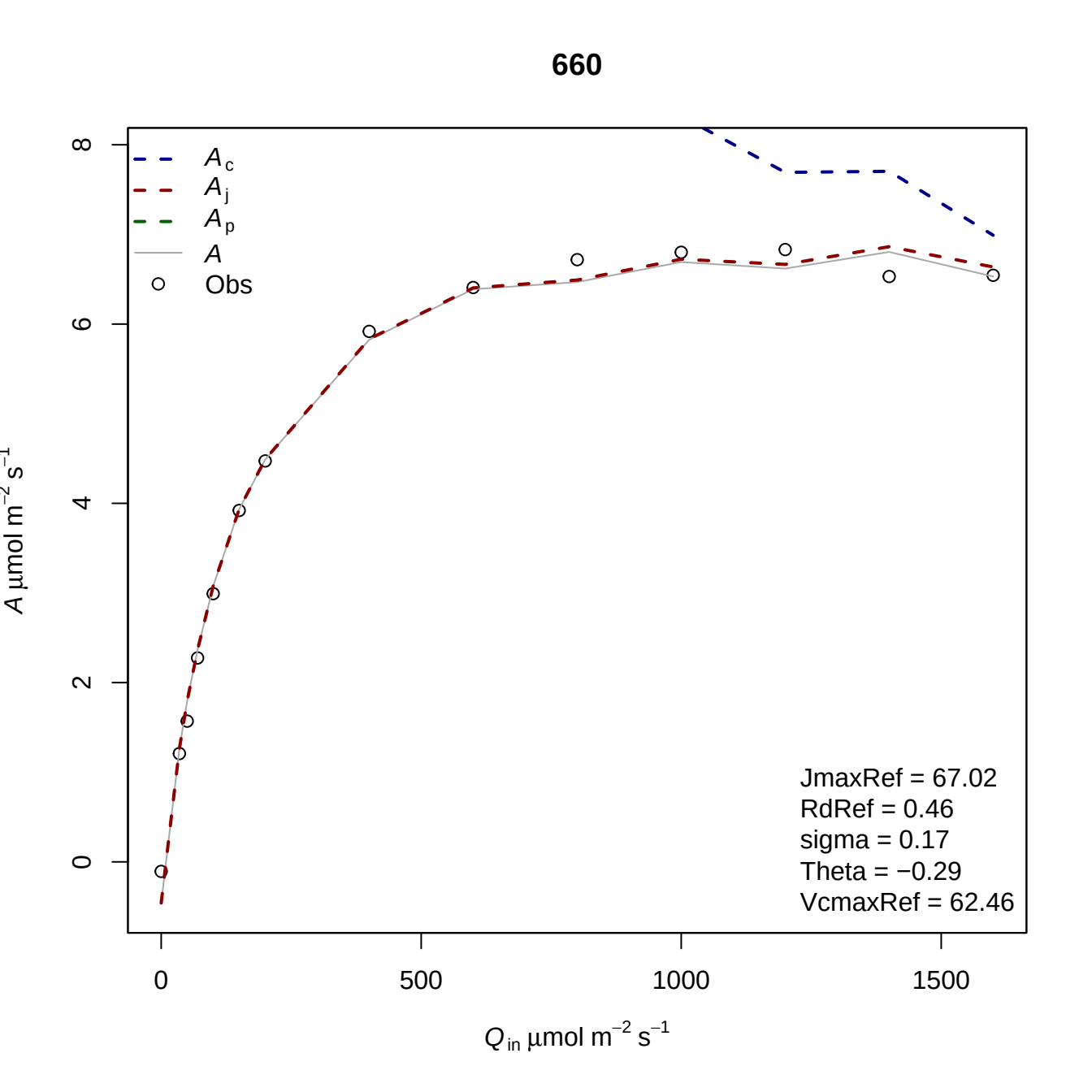

655
A_c
A_j
A_p
A
Obs
JmaxRef = 188.43
RdRef = 3.11
sigma = 0.16
Theta = 0.51
VcmaxRef = 104.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

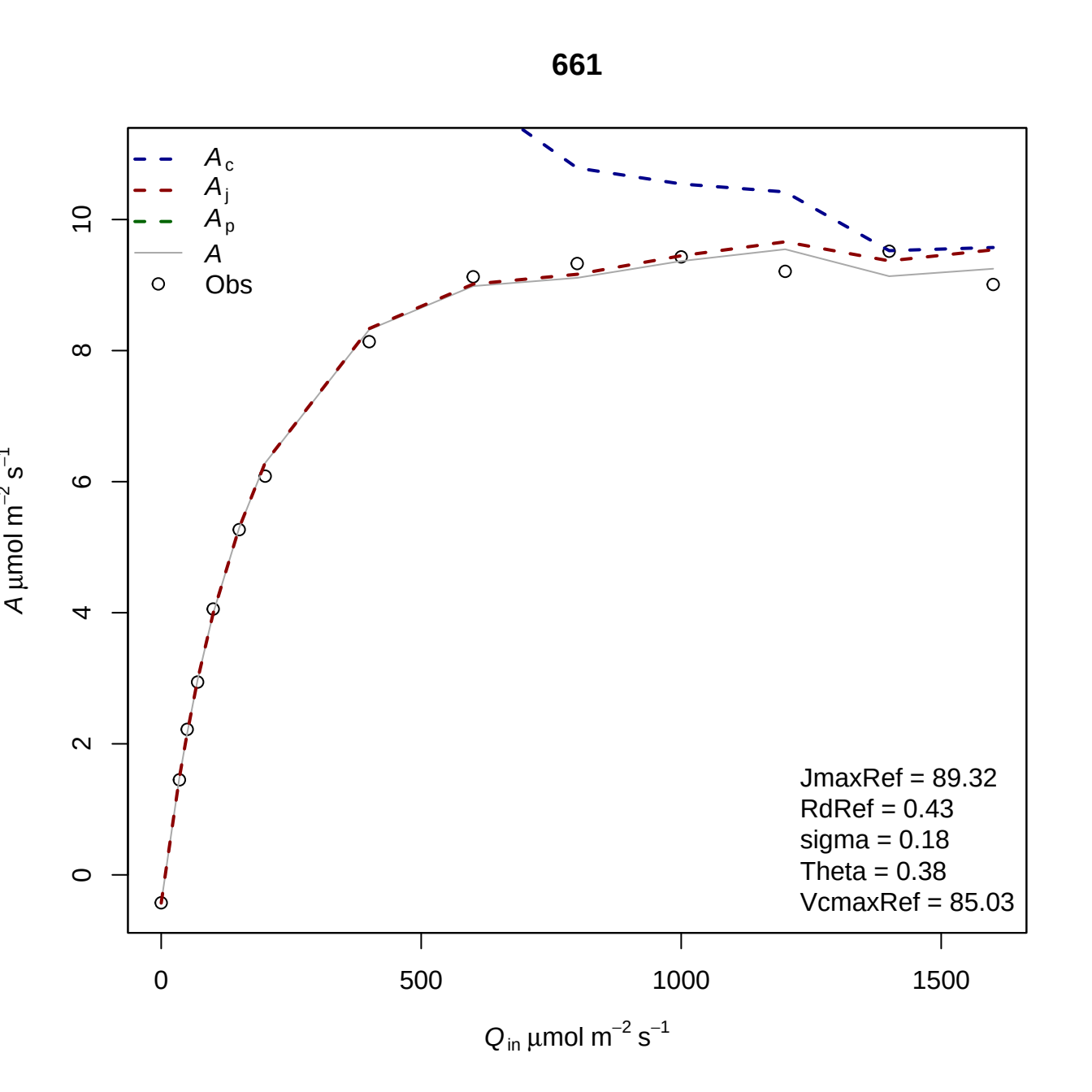

656
A_c
A_j
A_p
A
Obs
JmaxRef = 140.99
RdRef = 0.84
sigma = 0.33
Theta = 0.26
VcmaxRef = 71.41
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

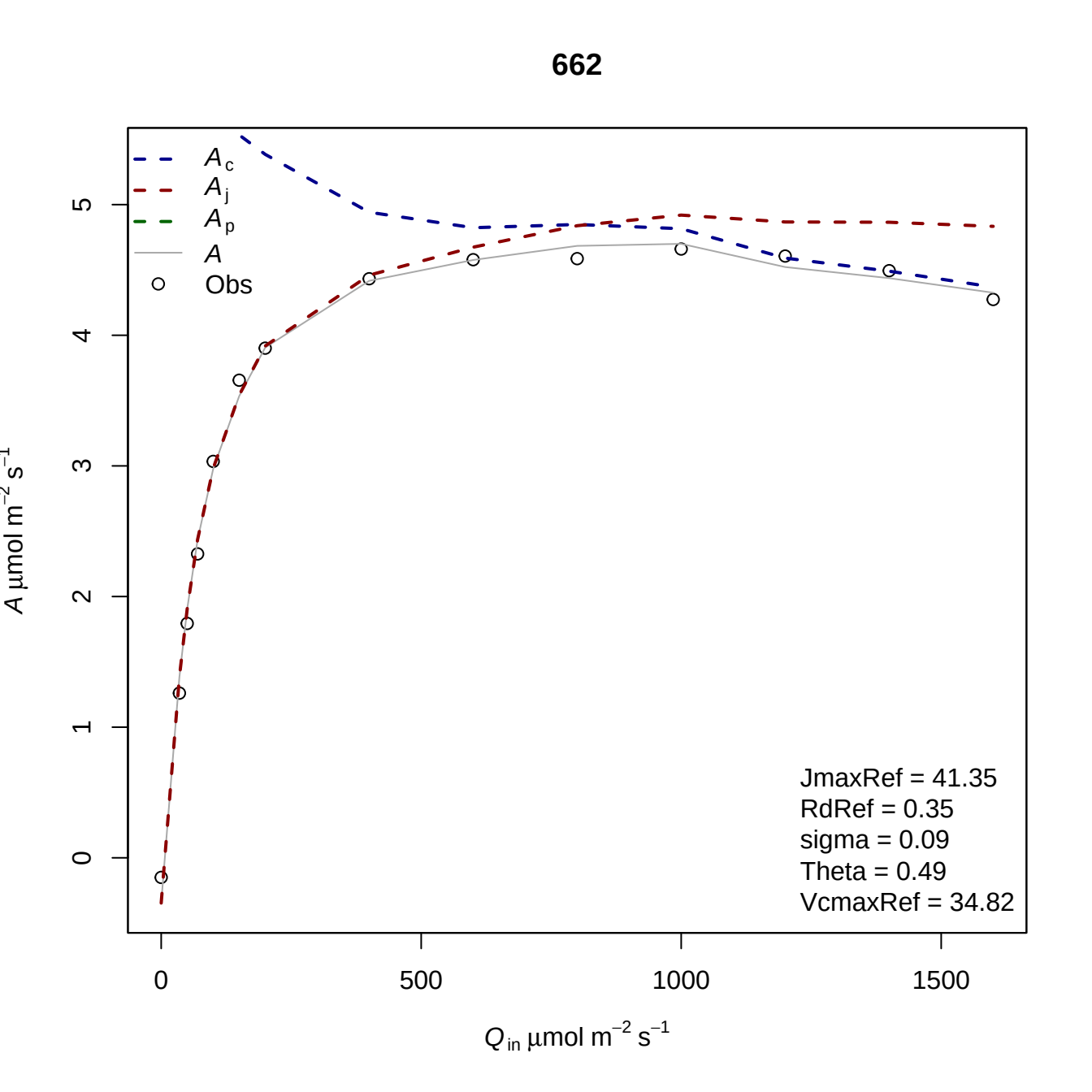

657
A_c
A_j
A_p
A
Obs
JmaxRef = 147.92
RdRef = 0.67
sigma = 0.46
Theta = 0.32
VcmaxRef = 76.89
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

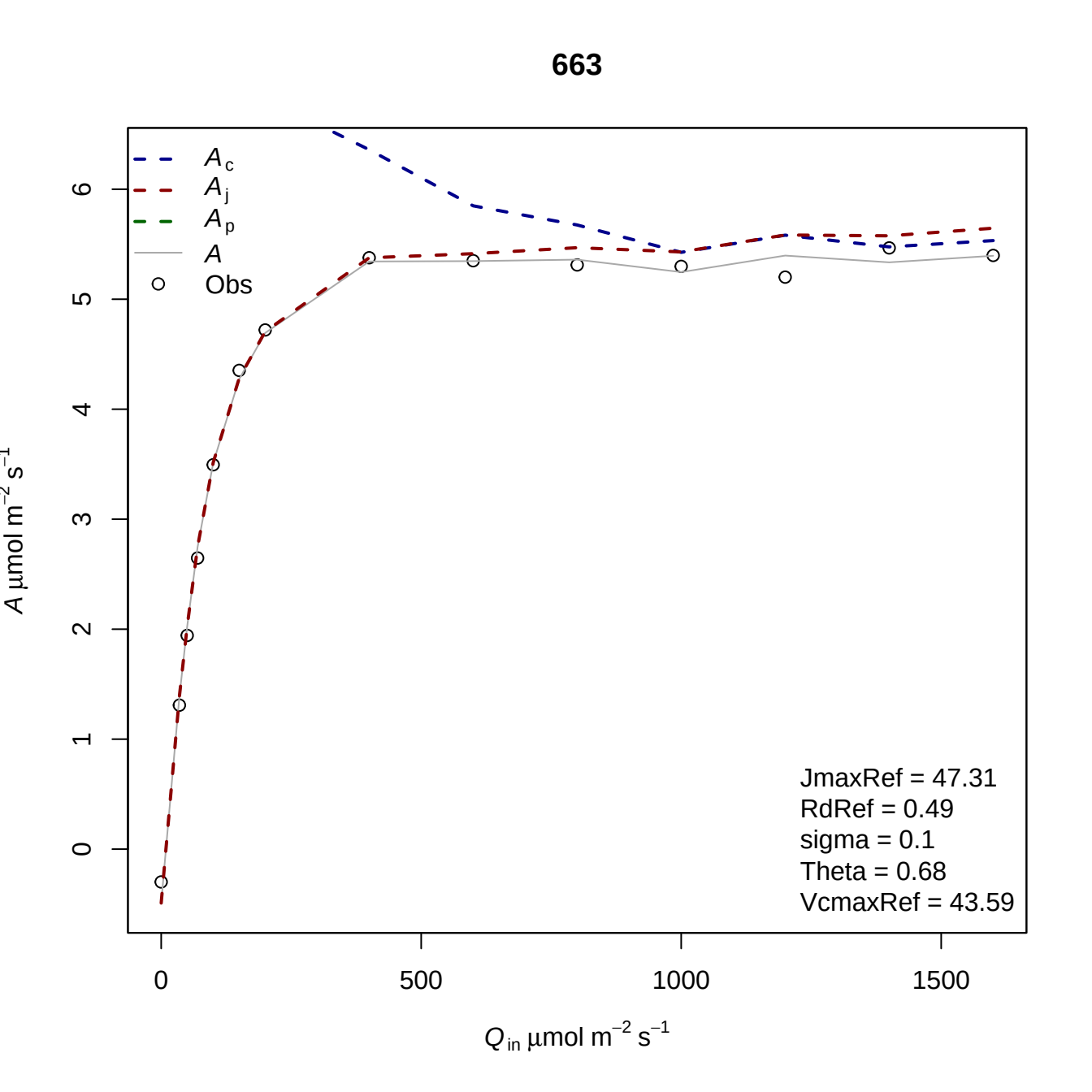

658
A_c
A_j
A_p
A
Obs
JmaxRef = 133.43
RdRef = 0.86
sigma = 0.15
Theta = 0.41
VcmaxRef = 84.24
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

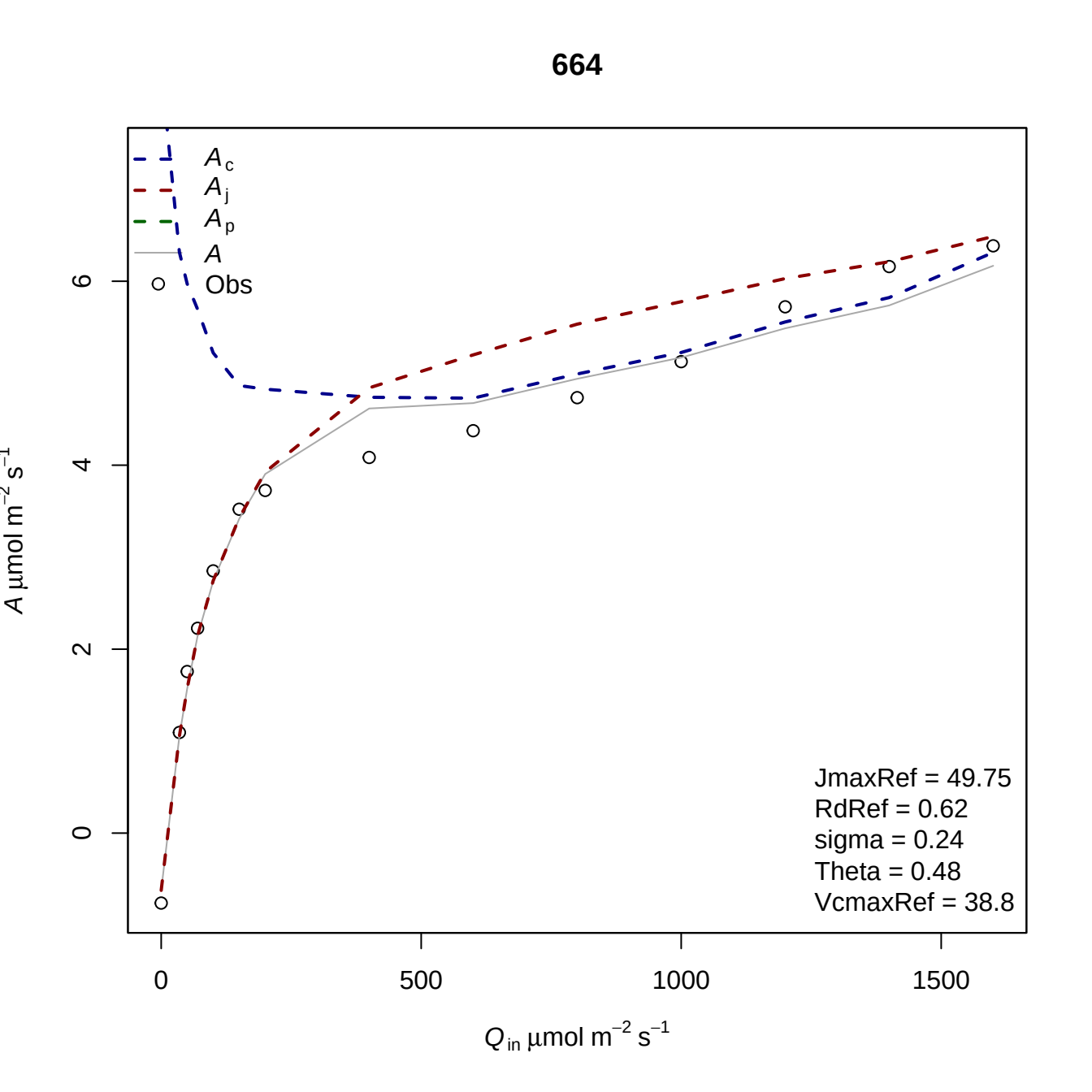

659
A_c
A_j
A_p
A
Obs
JmaxRef = 111.39
RdRef = 0.05
sigma = 0.39
Theta = 0.36
VcmaxRef = 1133662.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

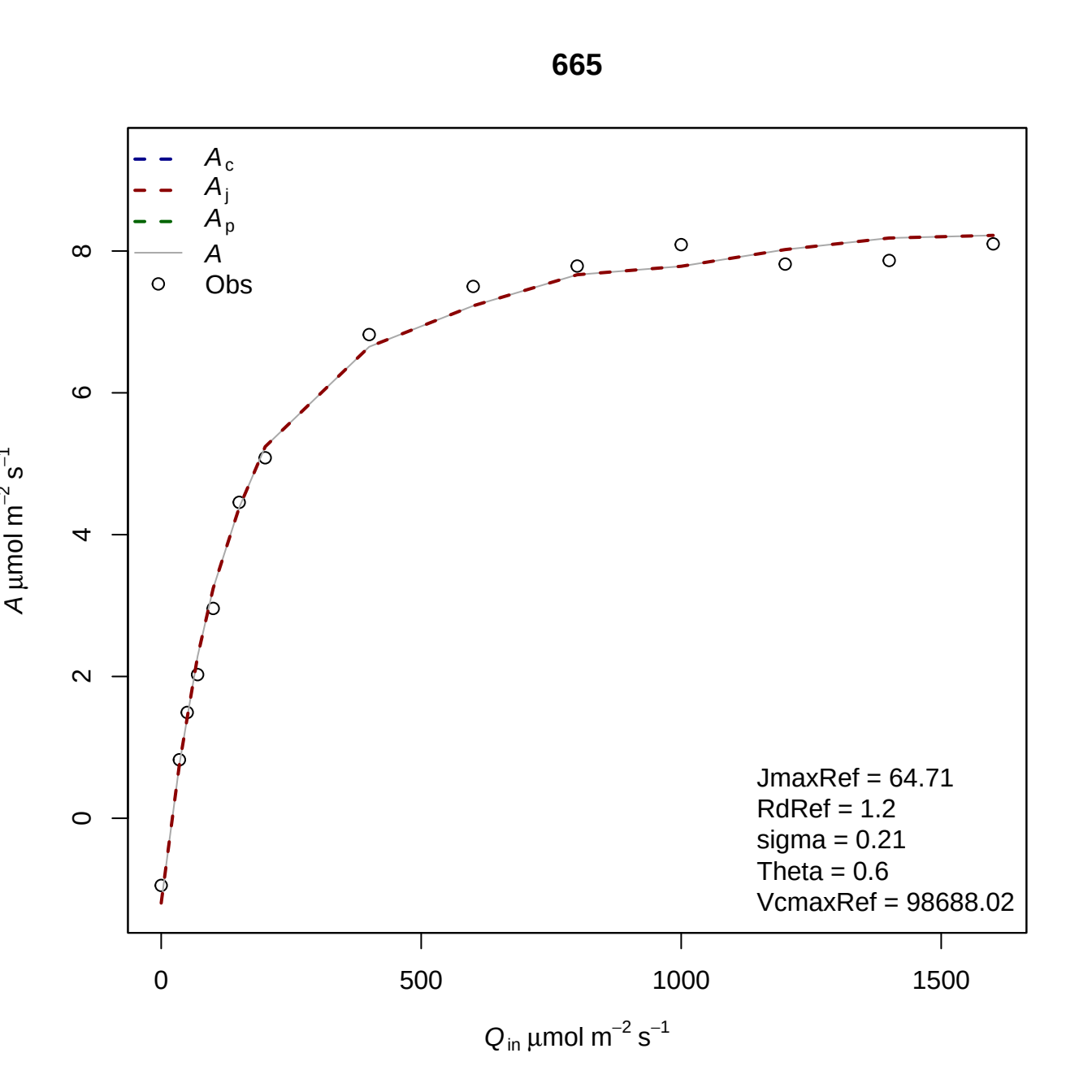

660
A_c
A_j
A_p
A
Obs
JmaxRef = 67.02
RdRef = 0.46
sigma = 0.17
Theta = −0.29
VcmaxRef = 62.46
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

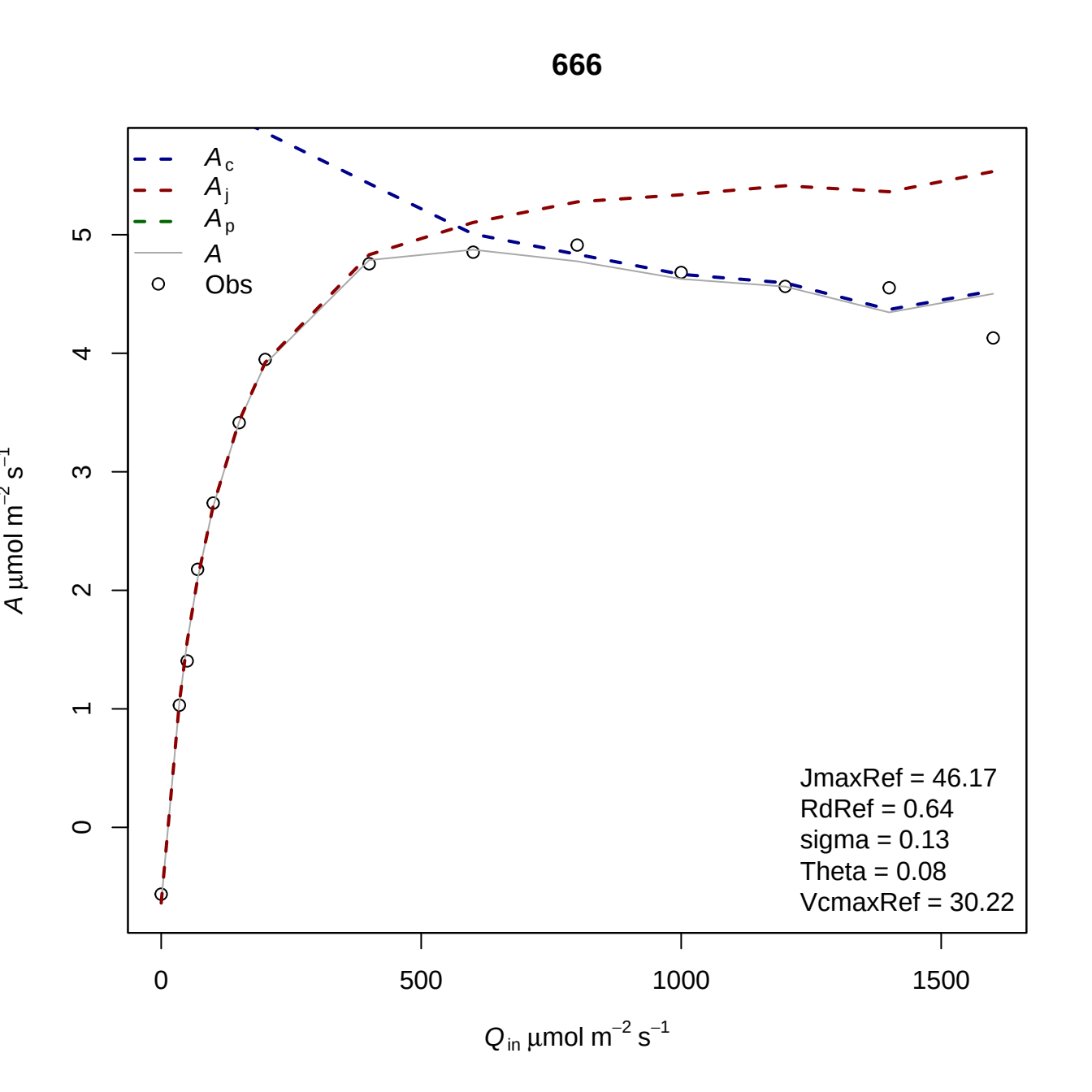

661
A_c
A_j
A_p
A
Obs
JmaxRef = 89.32
RdRef = 0.43
sigma = 0.18
Theta = 0.38
VcmaxRef = 85.03
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

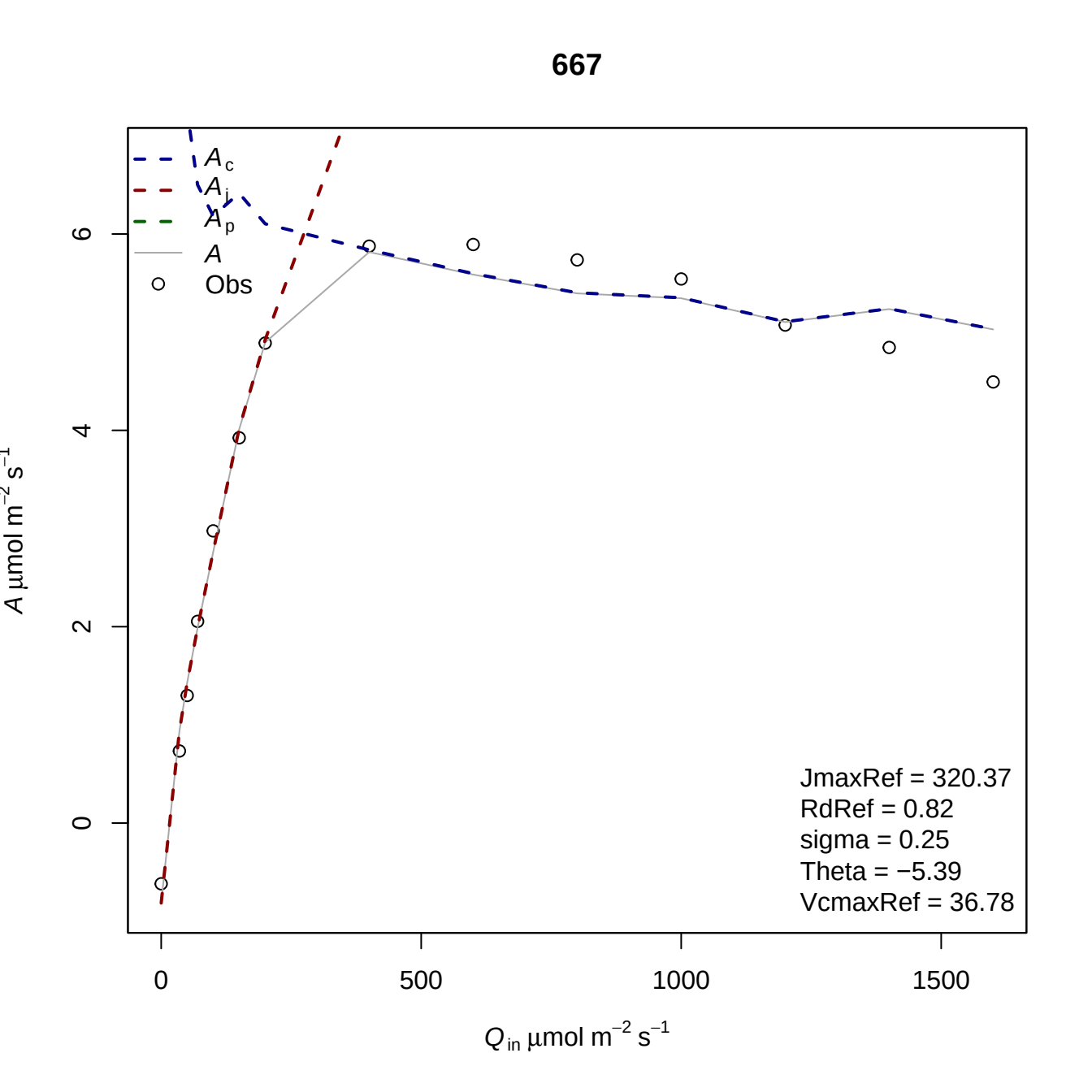

662
A_c
A_j
A_p
A
Obs
JmaxRef = 41.35
RdRef = 0.35
sigma = 0.09
Theta = 0.49
VcmaxRef = 34.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

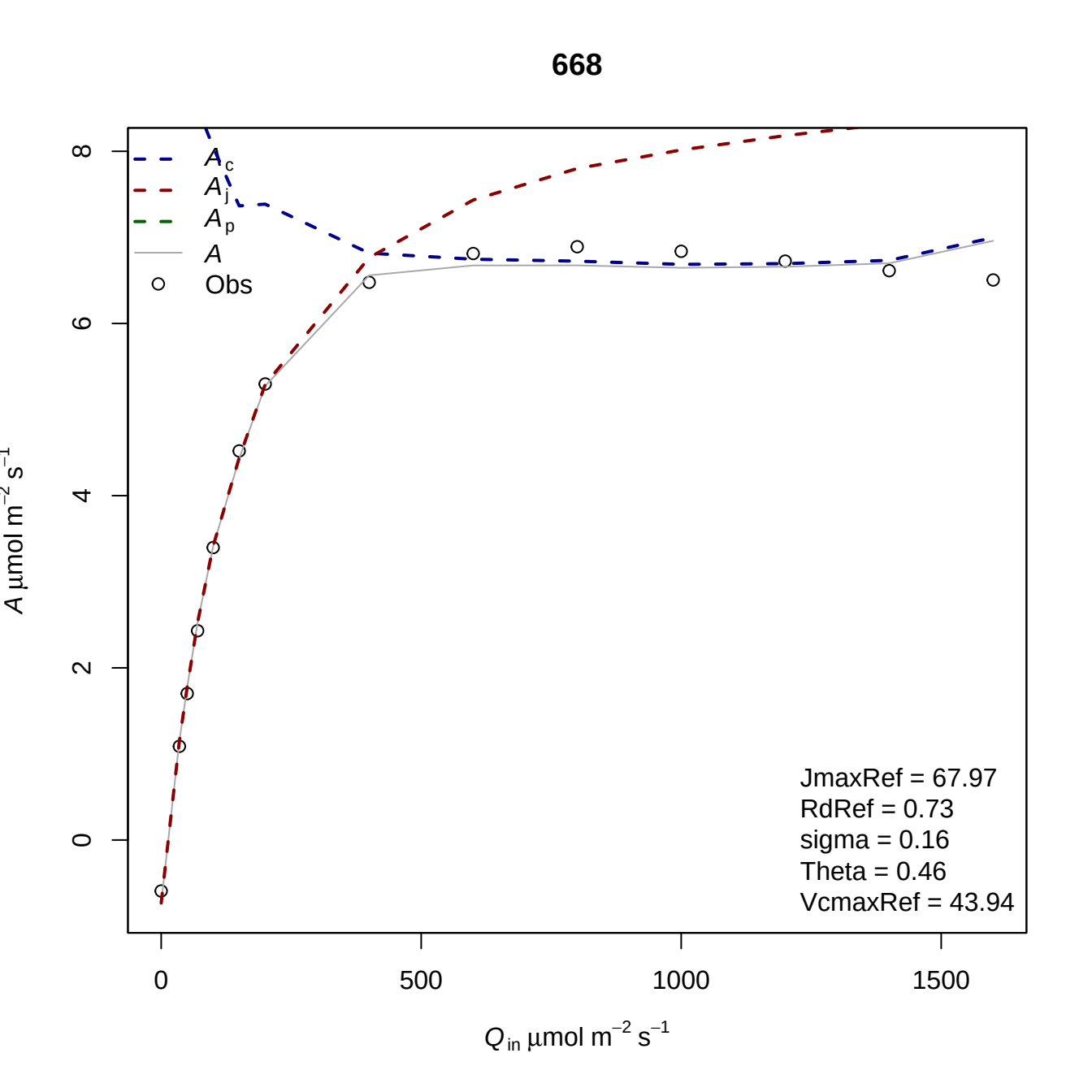

663
A_c
A_j
A_p
A
Obs
JmaxRef = 47.31
RdRef = 0.49
sigma = 0.1
Theta = 0.68
VcmaxRef = 43.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

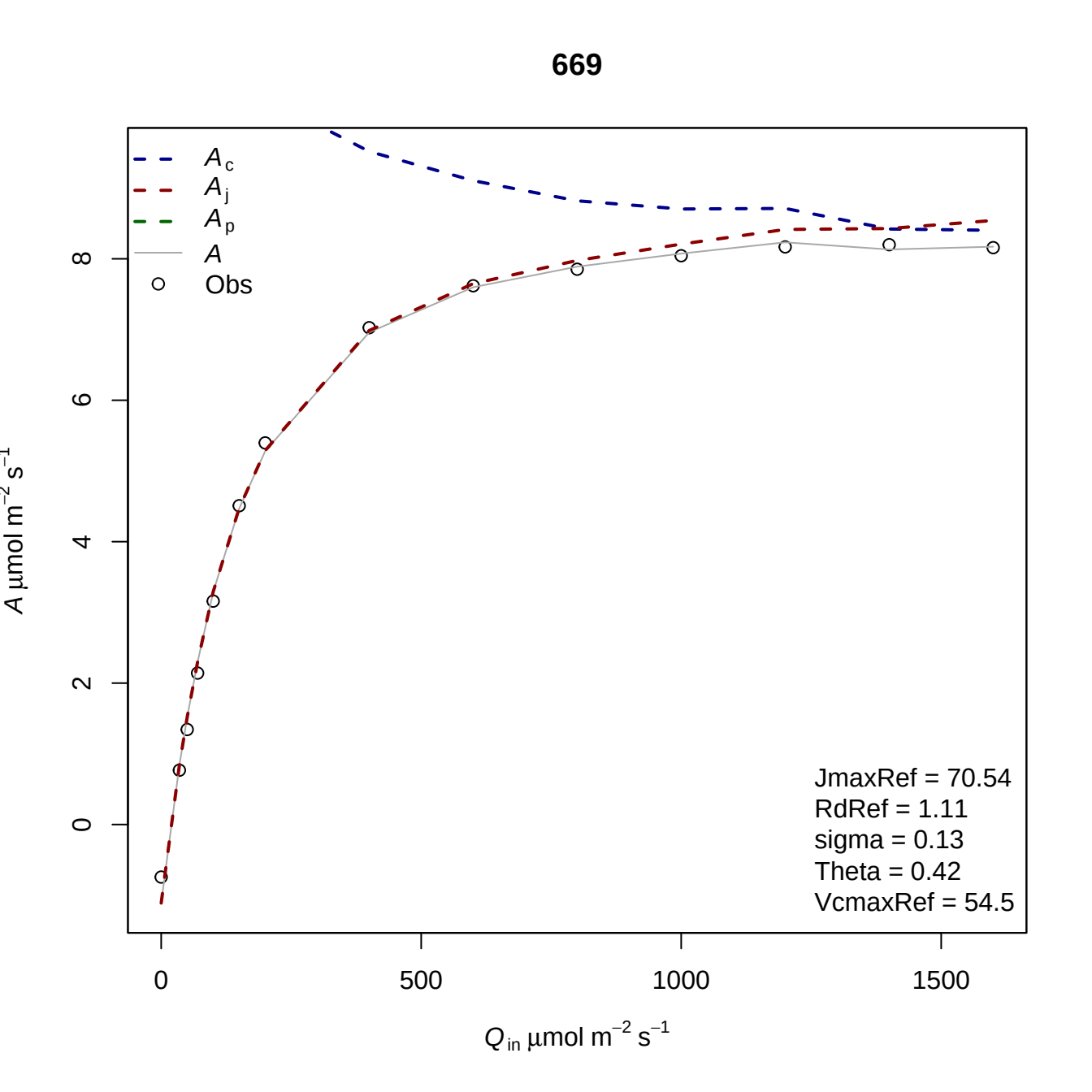

664
A_c
A_j
A_p
A
Obs
JmaxRef = 49.75
RdRef = 0.62
sigma = 0.24
Theta = 0.48
VcmaxRef = 38.8
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

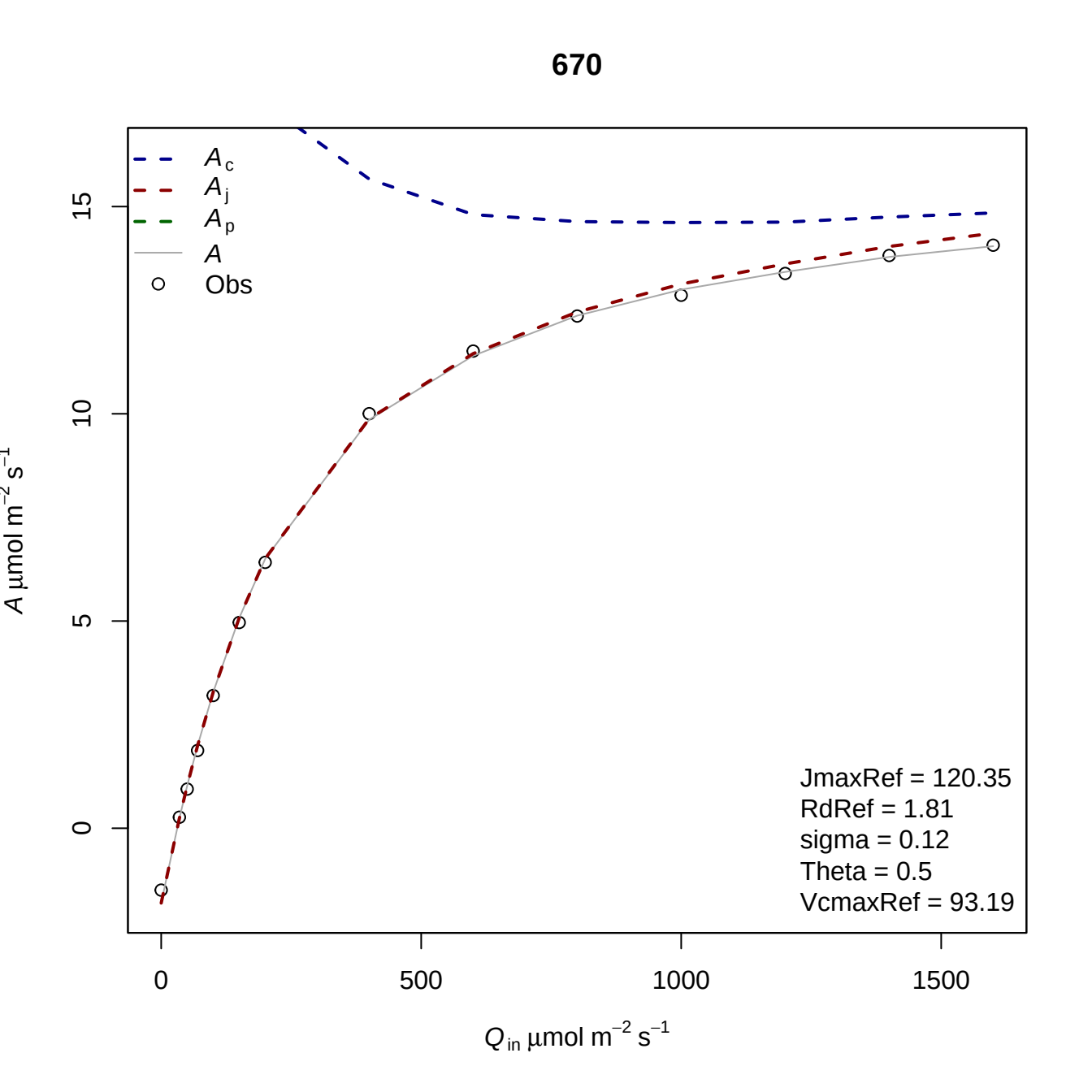

665
A_c
A_j
A_p
A
Obs
JmaxRef = 64.71
RdRef = 1.2
sigma = 0.21
Theta = 0.6
VcmaxRef = 98688.02
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

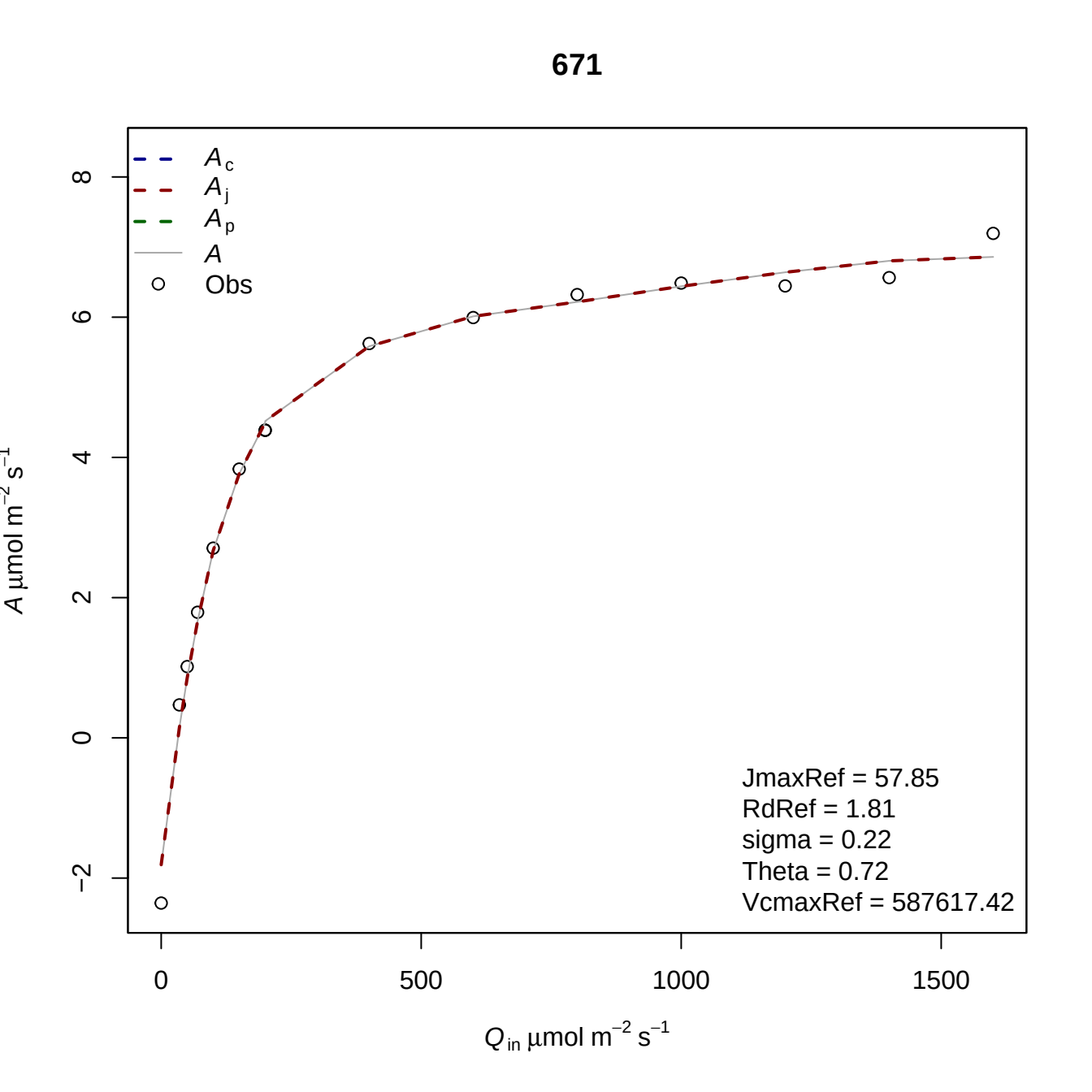

666
A_c
A_j
A_p
A
Obs
JmaxRef = 46.17
RdRef = 0.64
sigma = 0.13
Theta = 0.08
VcmaxRef = 30.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

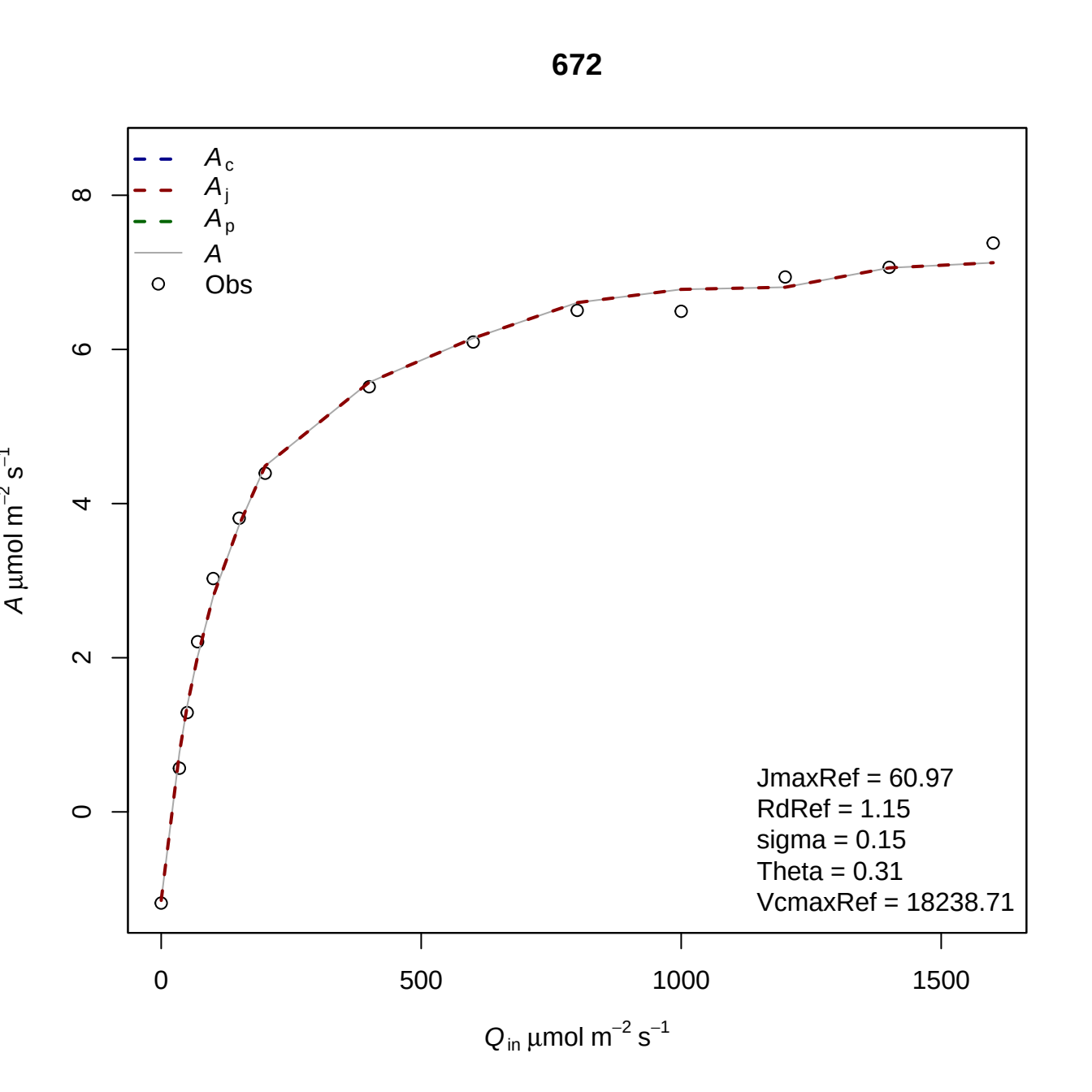

667
A_c
A_j
A_p
A
Obs
JmaxRef = 320.37
RdRef = 0.82
sigma = 0.25
Theta = −5.39
VcmaxRef = 36.78
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

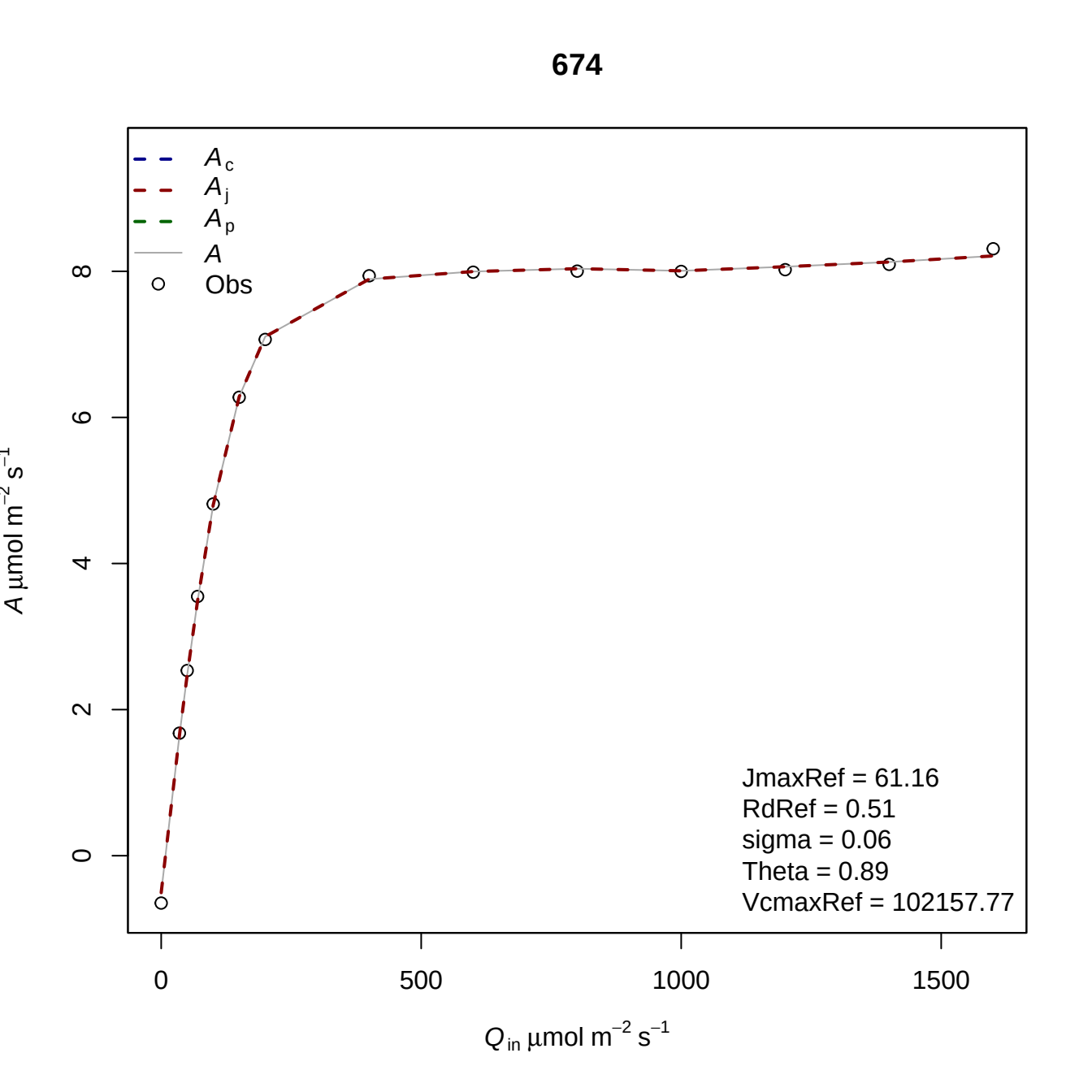

668
A_c
A_j
A_p
A
Obs
JmaxRef = 67.97
RdRef = 0.73
sigma = 0.16
Theta = 0.46
VcmaxRef = 43.94
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

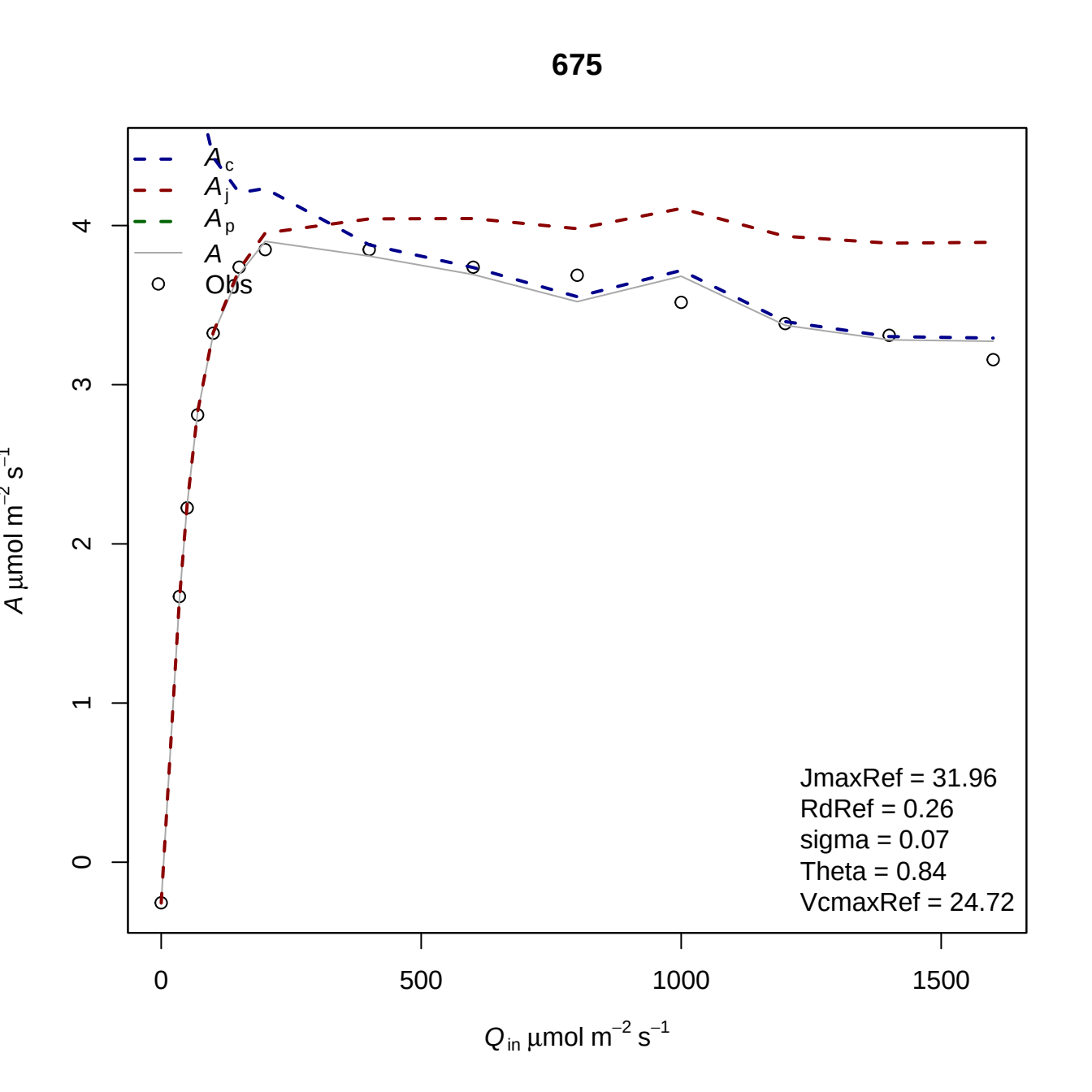

669
A_c
A_j
A_p
A
Obs
JmaxRef = 70.54
RdRef = 1.11
sigma = 0.13
Theta = 0.42
VcmaxRef = 54.5
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

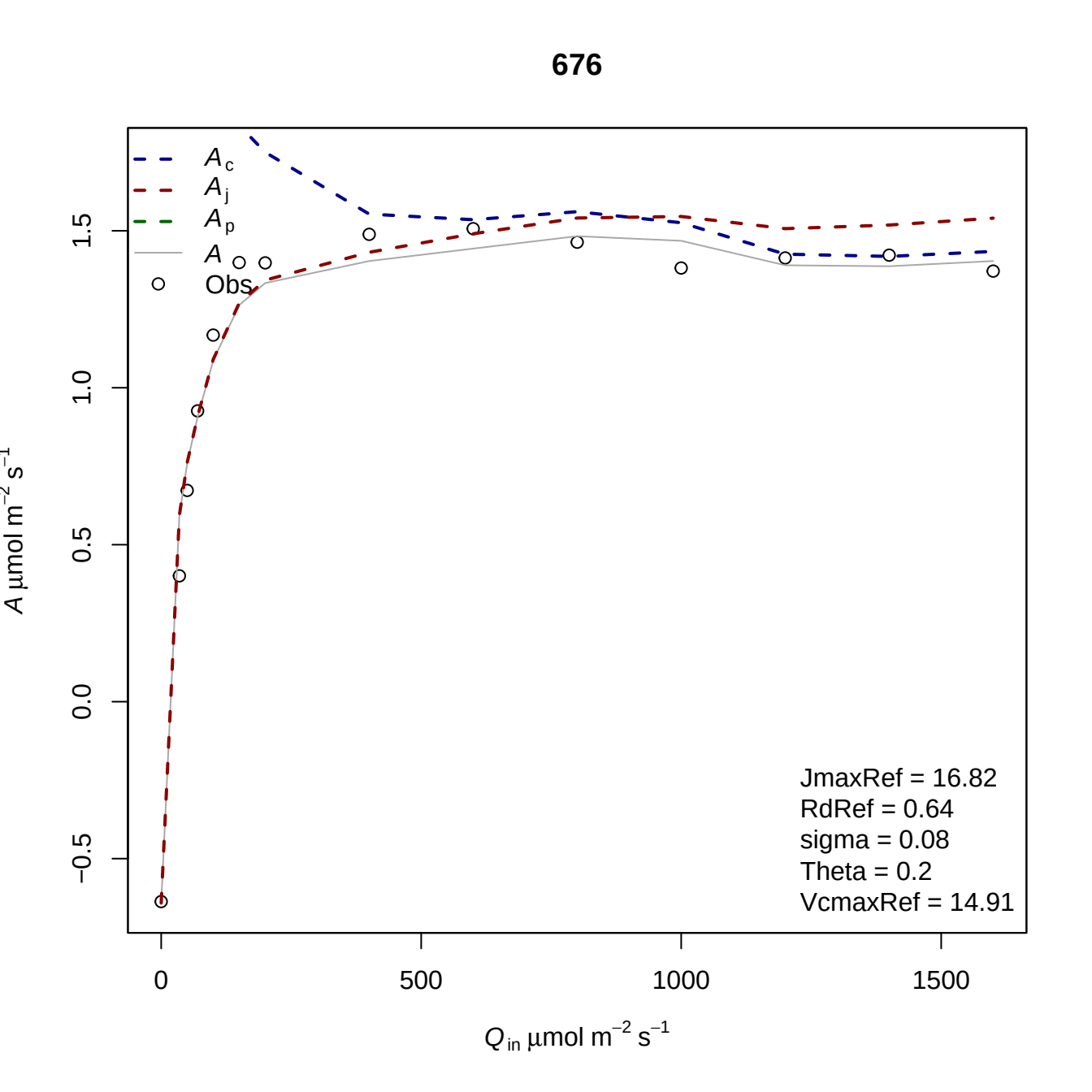

670
A_c
A_j
A_p
A
Obs
JmaxRef = 120.35
RdRef = 1.81
sigma = 0.12
Theta = 0.5
VcmaxRef = 93.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

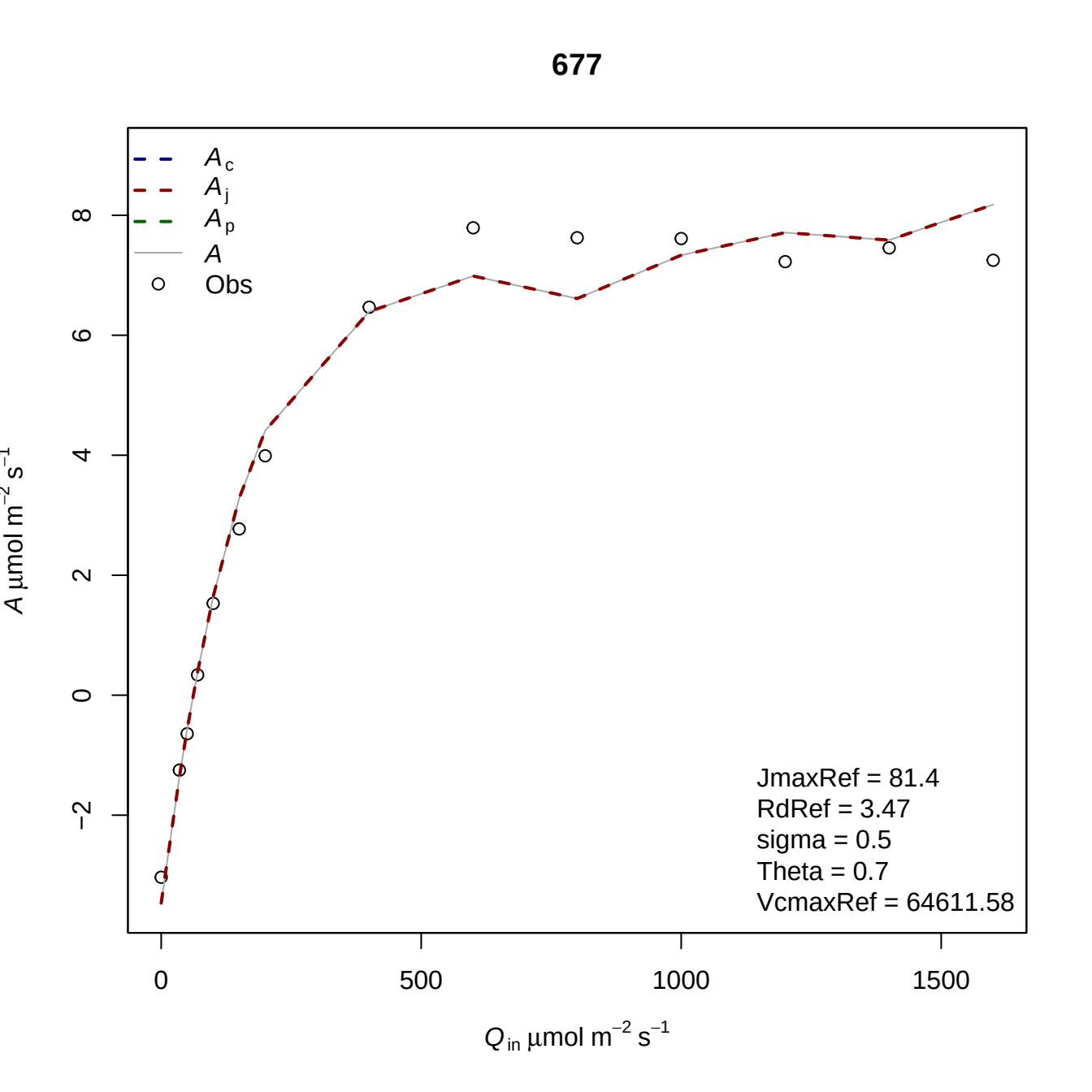

671
A_c
A_j
A_p
A
Obs
JmaxRef = 57.85
RdRef = 1.81
sigma = 0.22
Theta = 0.72
VcmaxRef = 587617.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

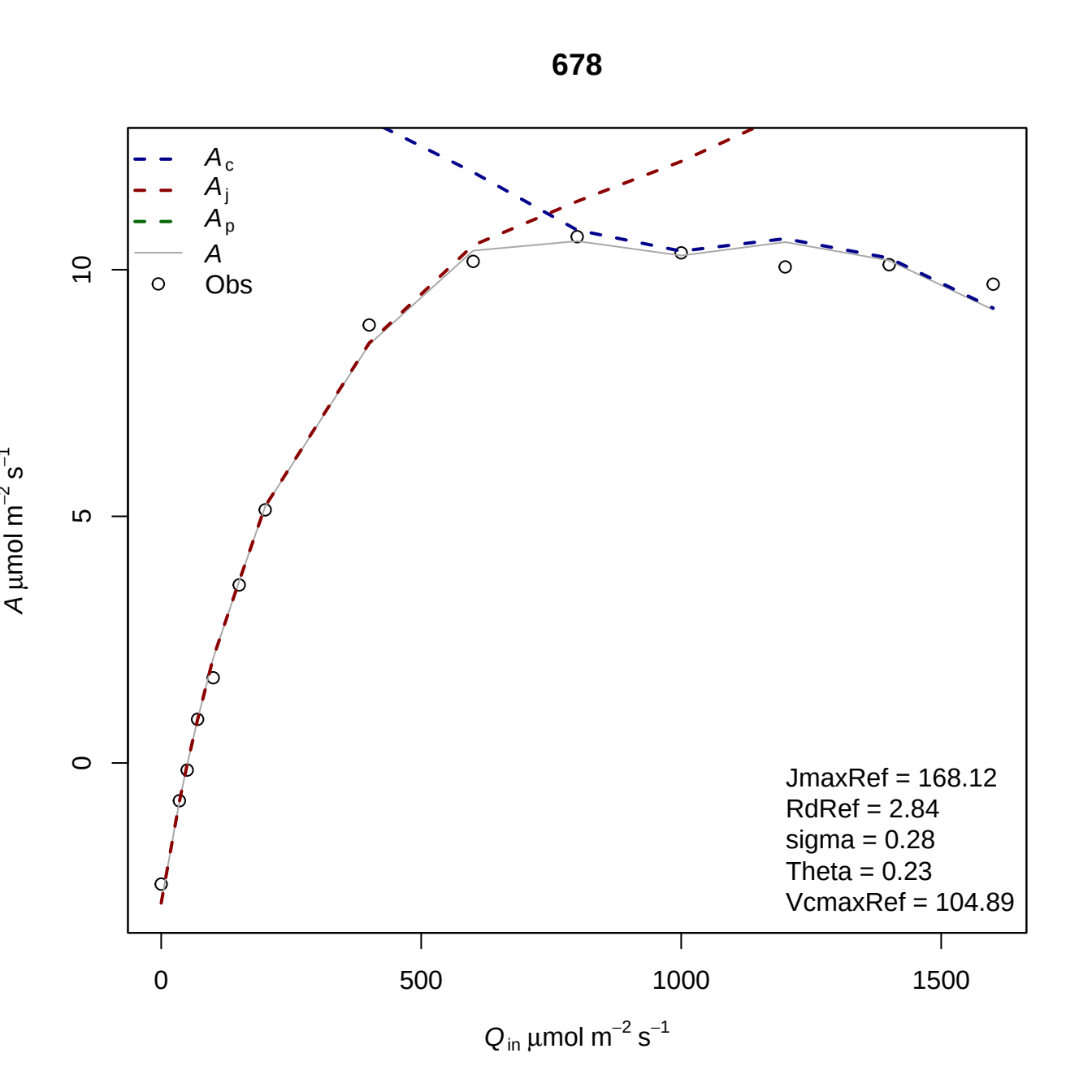

672
A_c
A_j
A_p
A
Obs
JmaxRef = 60.97
RdRef = 1.15
sigma = 0.15
Theta = 0.31
VcmaxRef = 18238.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

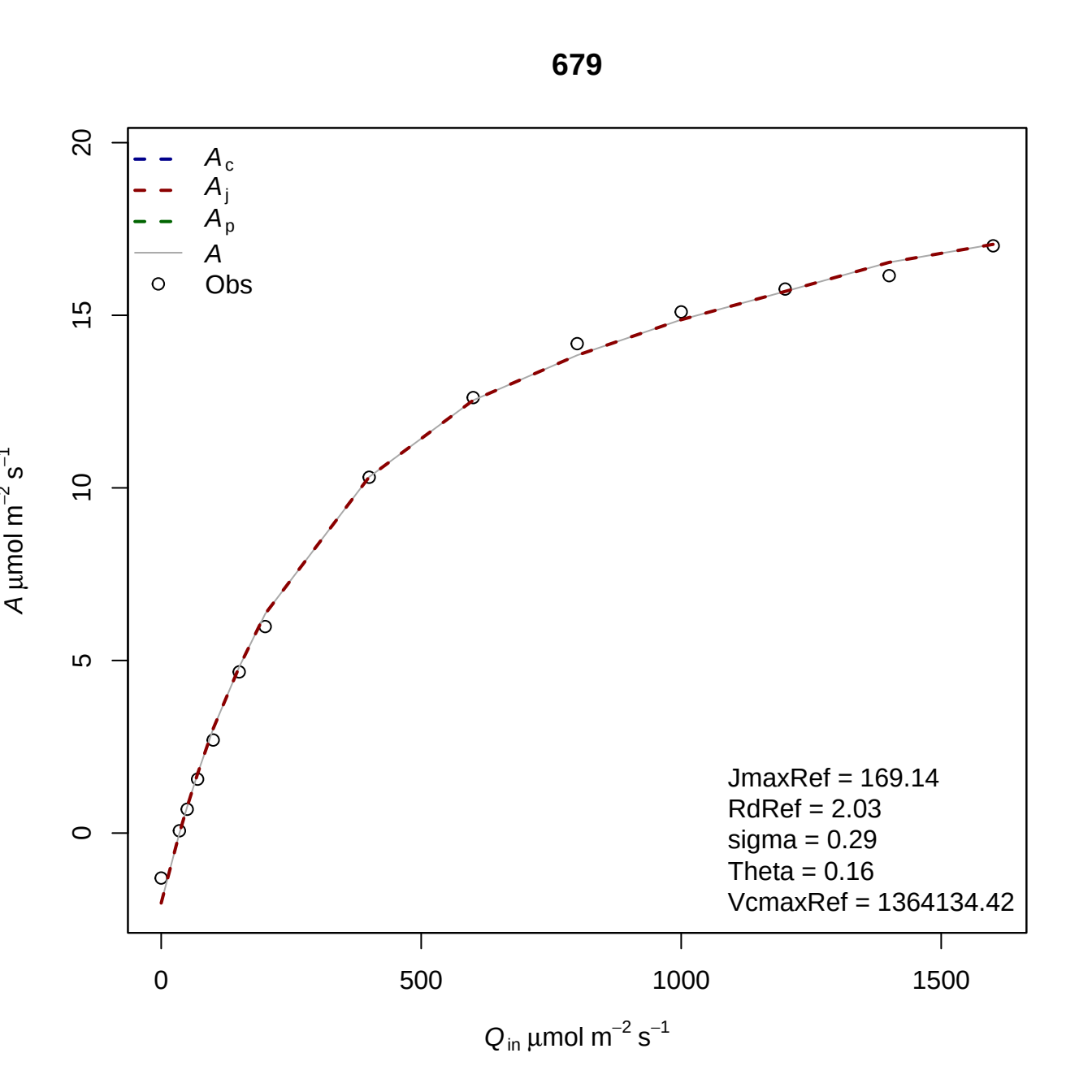

674
A_c
A_j
A_p
A
Obs
JmaxRef = 61.16
RdRef = 0.51
sigma = 0.06
Theta = 0.89
VcmaxRef = 102157.77
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

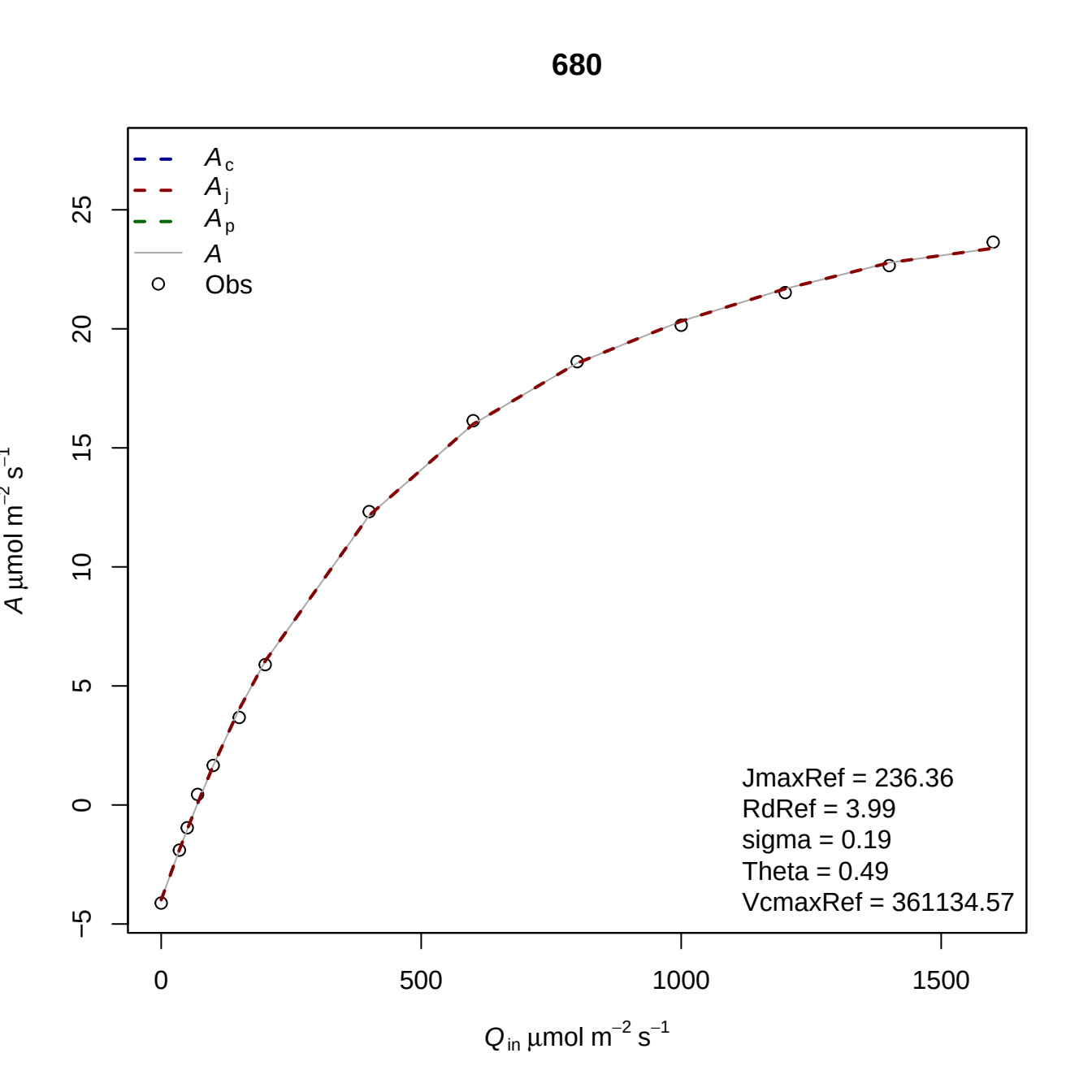

675
A_c
A_j
A_p
A
Obs
JmaxRef = 31.96
RdRef = 0.26
sigma = 0.07
Theta = 0.84
VcmaxRef = 24.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

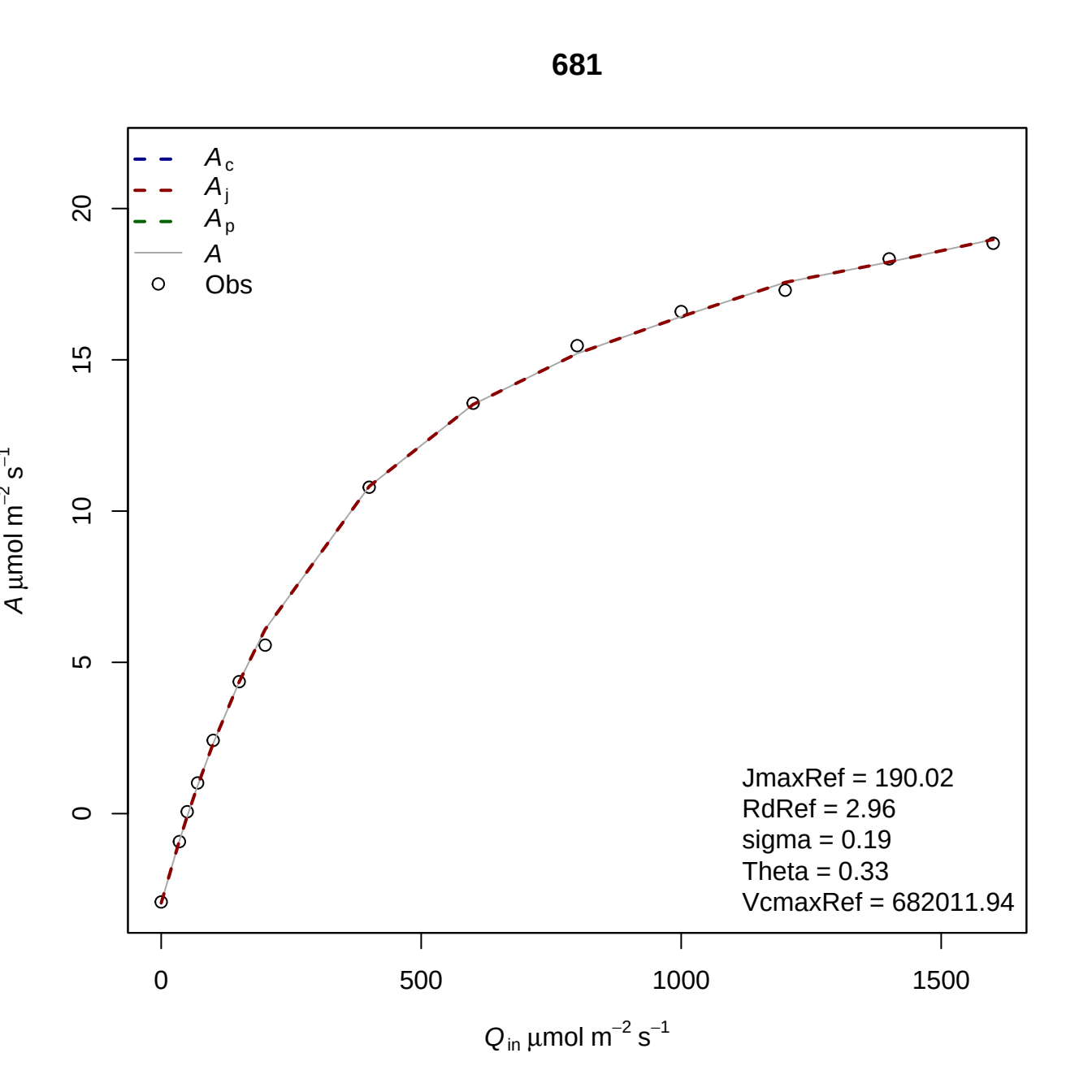

676
A_c
A_j
A_p
A
Obs
JmaxRef = 16.82
RdRef = 0.64
sigma = 0.08
Theta = 0.2
VcmaxRef = 14.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500

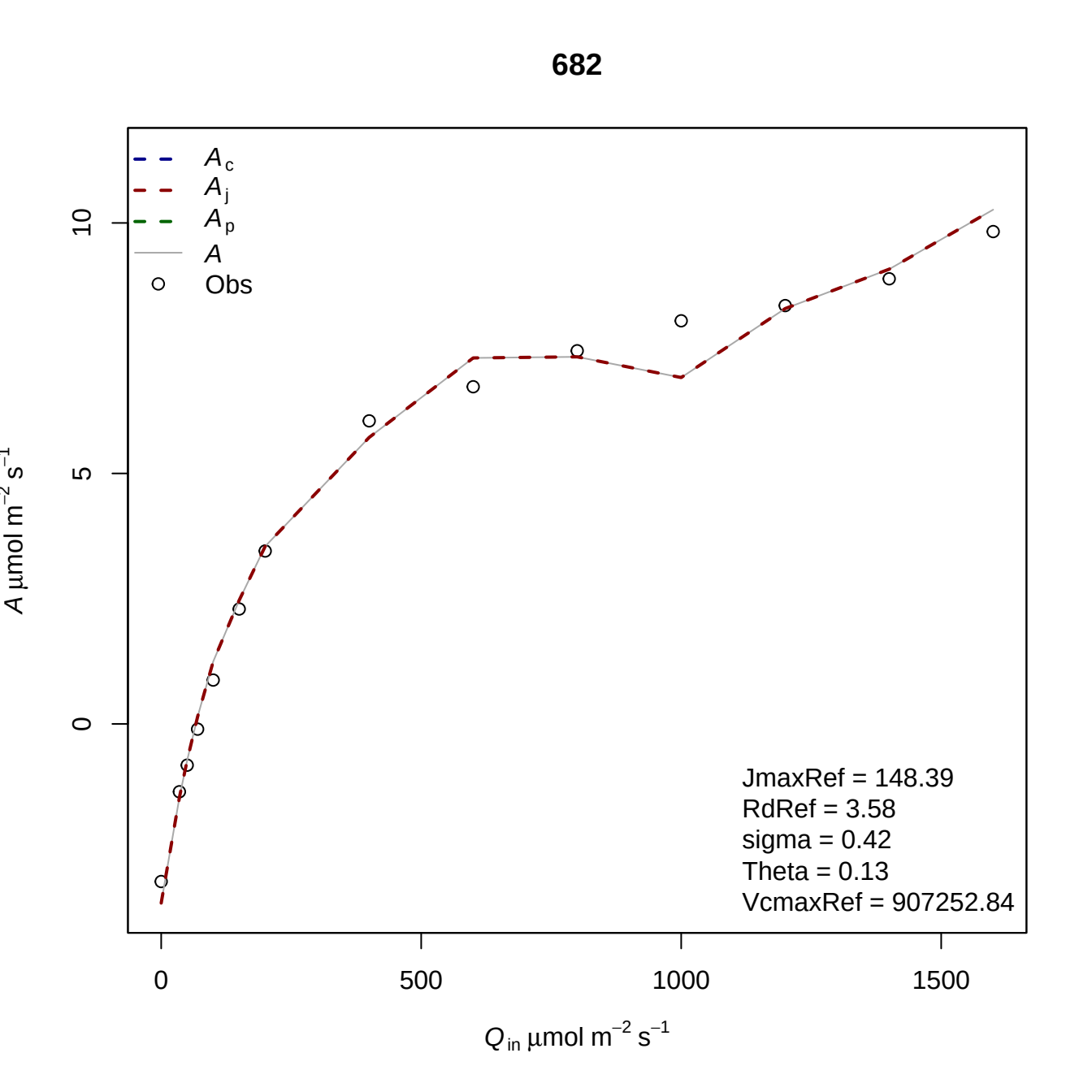

677
A_c
A_j
A_p
A
Obs
JmaxRef = 81.4
RdRef = 3.47
sigma = 0.5
Theta = 0.7
VcmaxRef = 64611.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

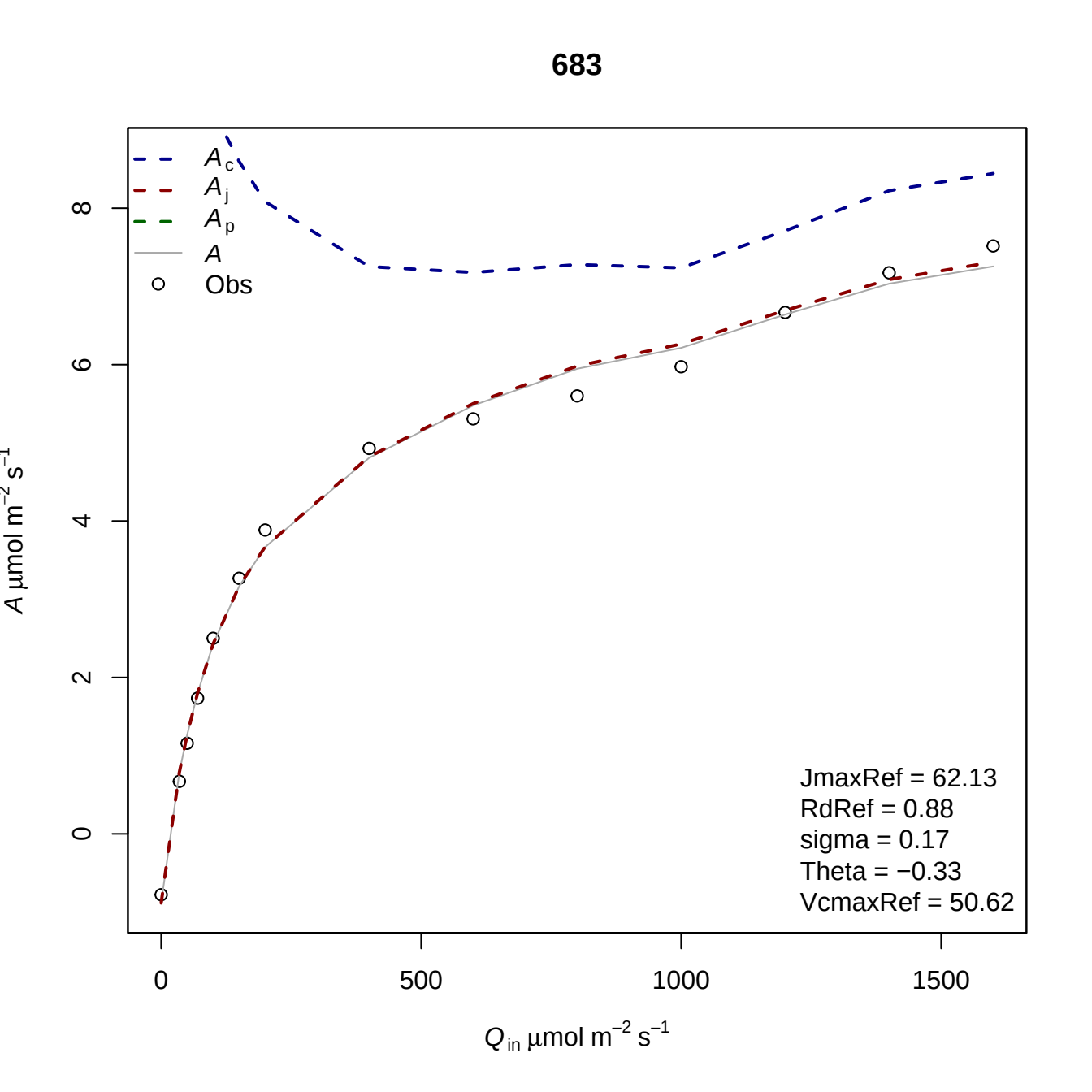

678
A_c
A_j
A_p
A
Obs
JmaxRef = 168.12
RdRef = 2.84
sigma = 0.28
Theta = 0.23
VcmaxRef = 104.89
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

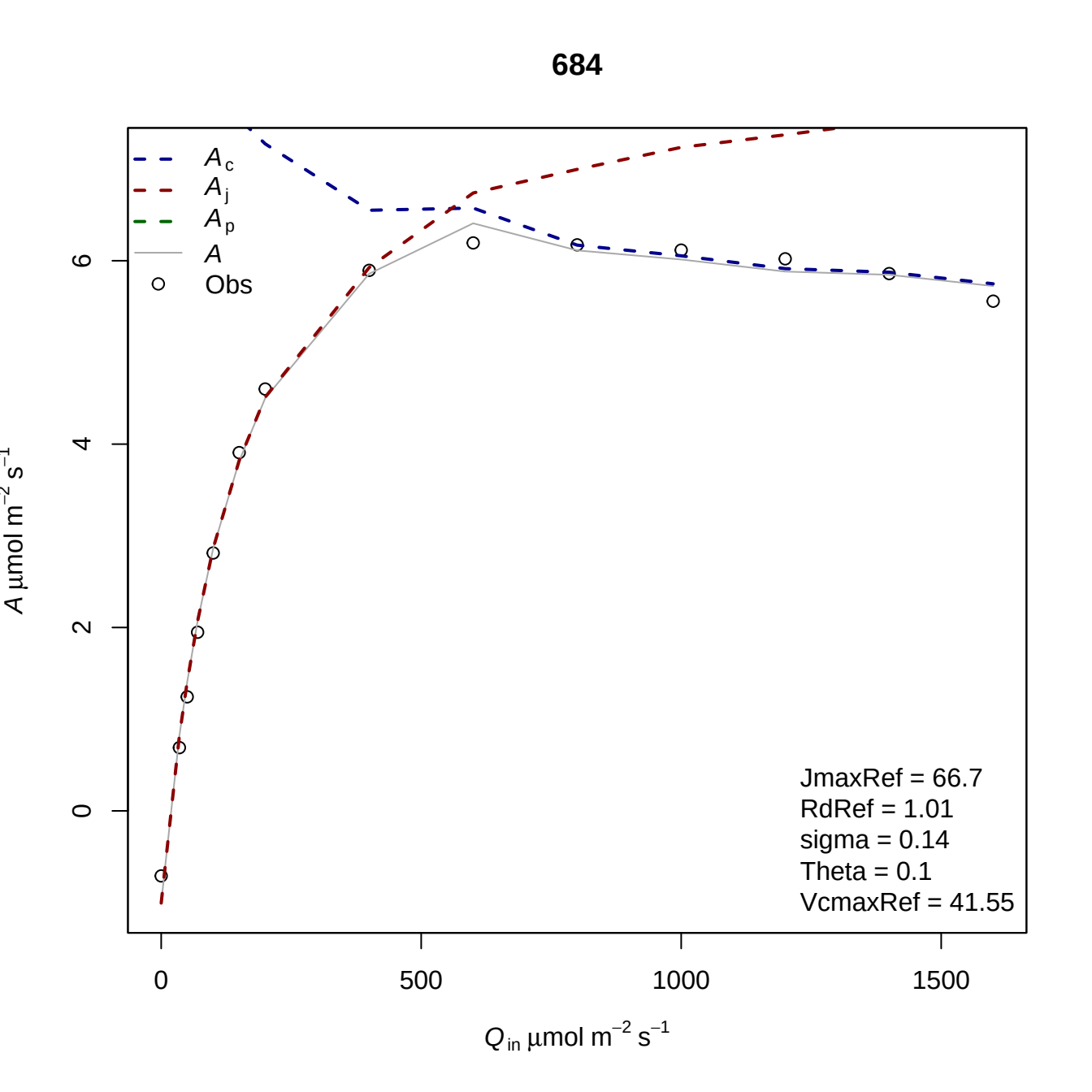

679
A_c
A_j
A_p
A
Obs
JmaxRef = 169.14
RdRef = 2.03
sigma = 0.29
Theta = 0.16
VcmaxRef = 1364134.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

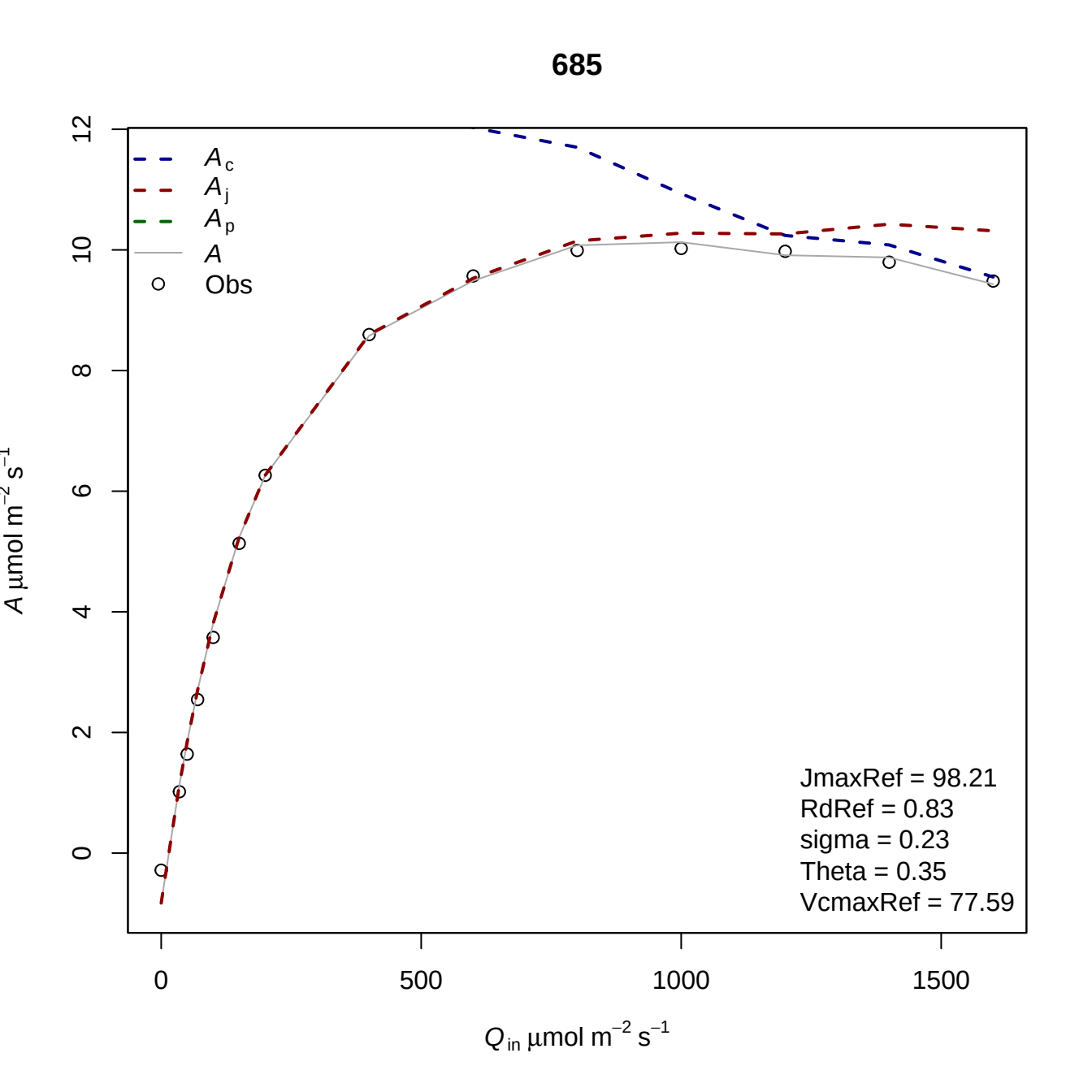

680
A_c
A_j
A_p
A
Obs
JmaxRef = 236.36
RdRef = 3.99
sigma = 0.19
Theta = 0.49
VcmaxRef = 361134.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

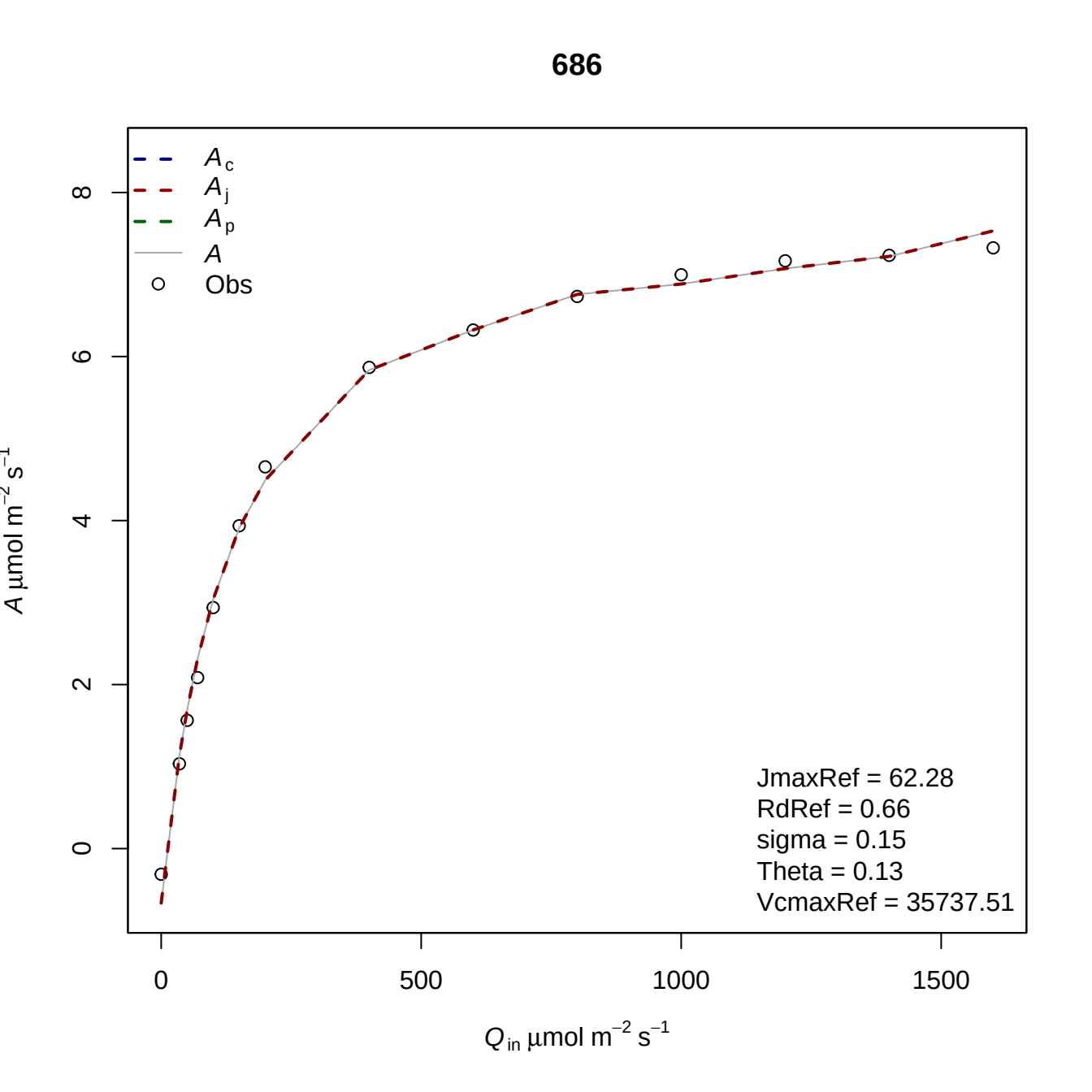

681
A_c
A_j
A_p
A
Obs
JmaxRef = 190.02
RdRef = 2.96
sigma = 0.19
Theta = 0.33
VcmaxRef = 682011.94
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

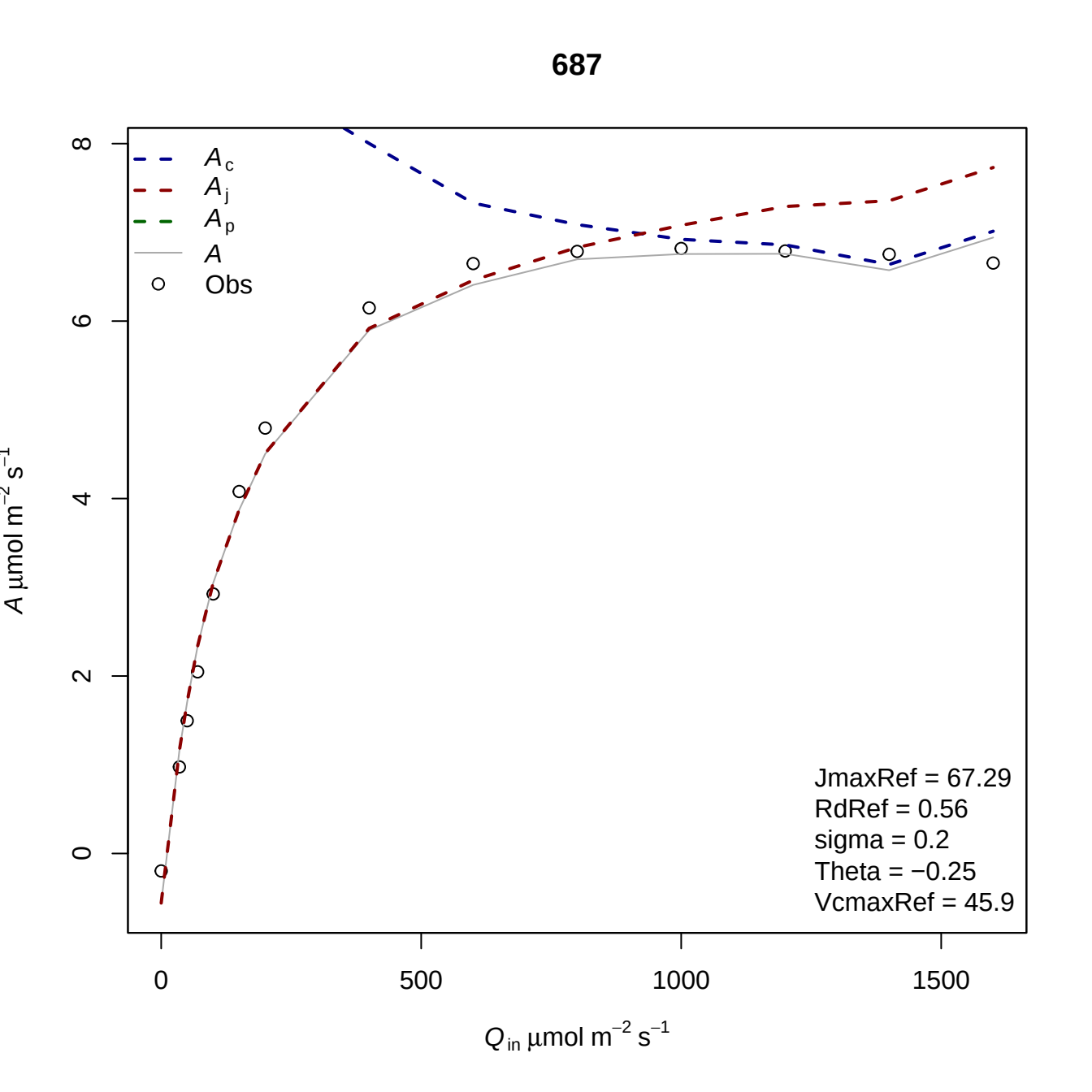

682
A_c
A_j
A_p
A
Obs
JmaxRef = 148.39
RdRef = 3.58
sigma = 0.42
Theta = 0.13
VcmaxRef = 907252.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

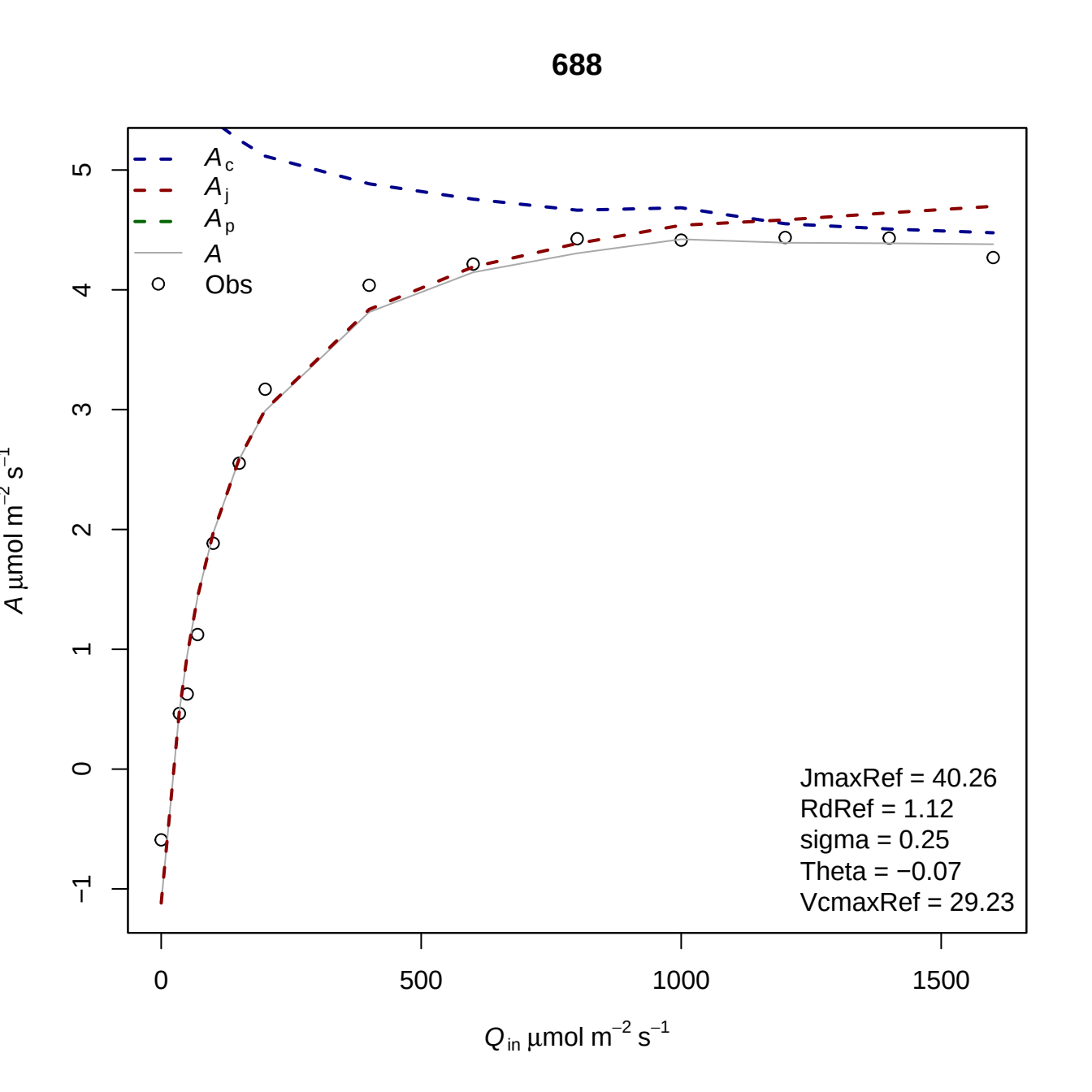

683
A_c
A_j
A_p
A
Obs
JmaxRef = 62.13
RdRef = 0.88
sigma = 0.17
Theta = −0.33
VcmaxRef = 50.62
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

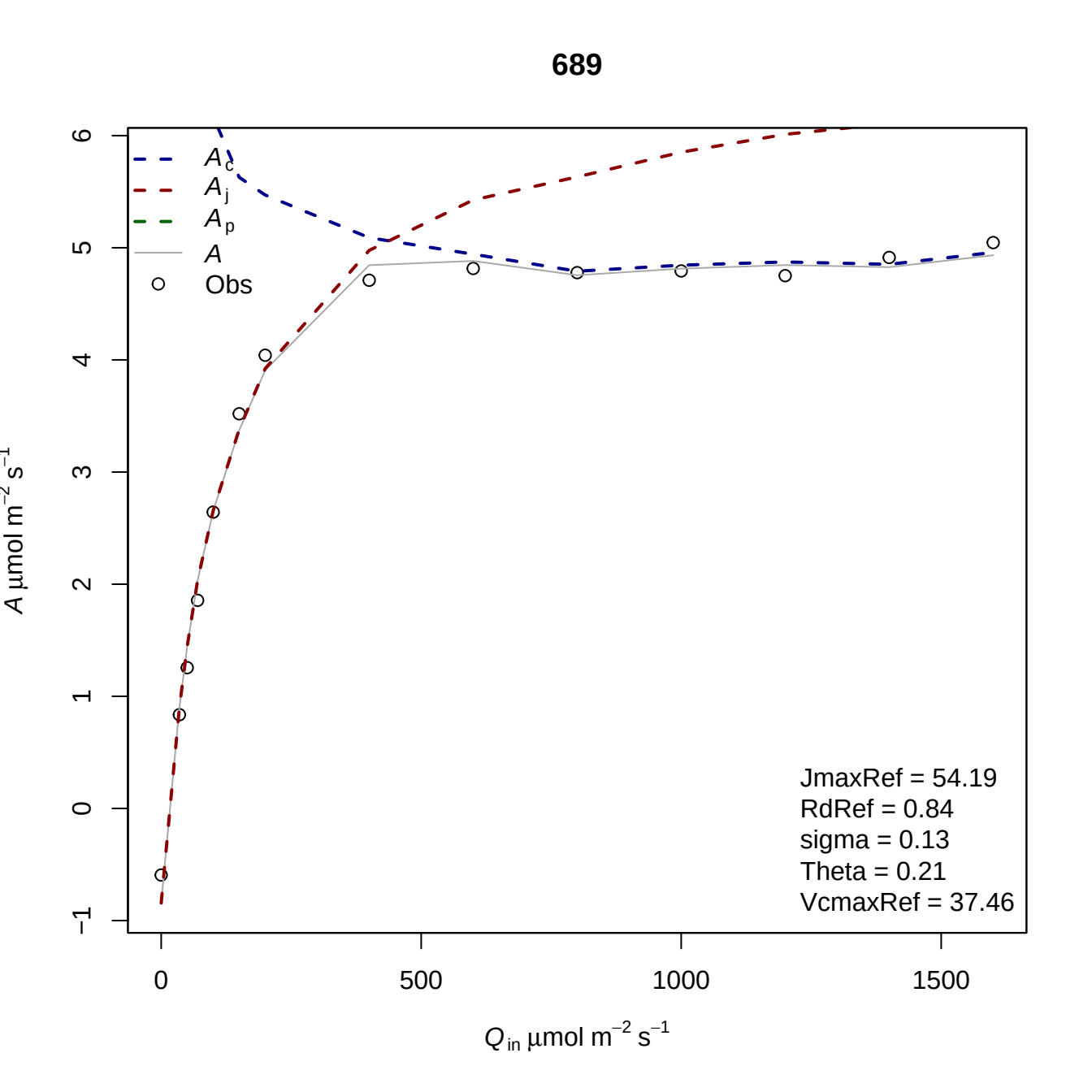

684
A_c
A_j
A_p
A
Obs
JmaxRef = 66.7
RdRef = 1.01
sigma = 0.14
Theta = 0.1
VcmaxRef = 41.55
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

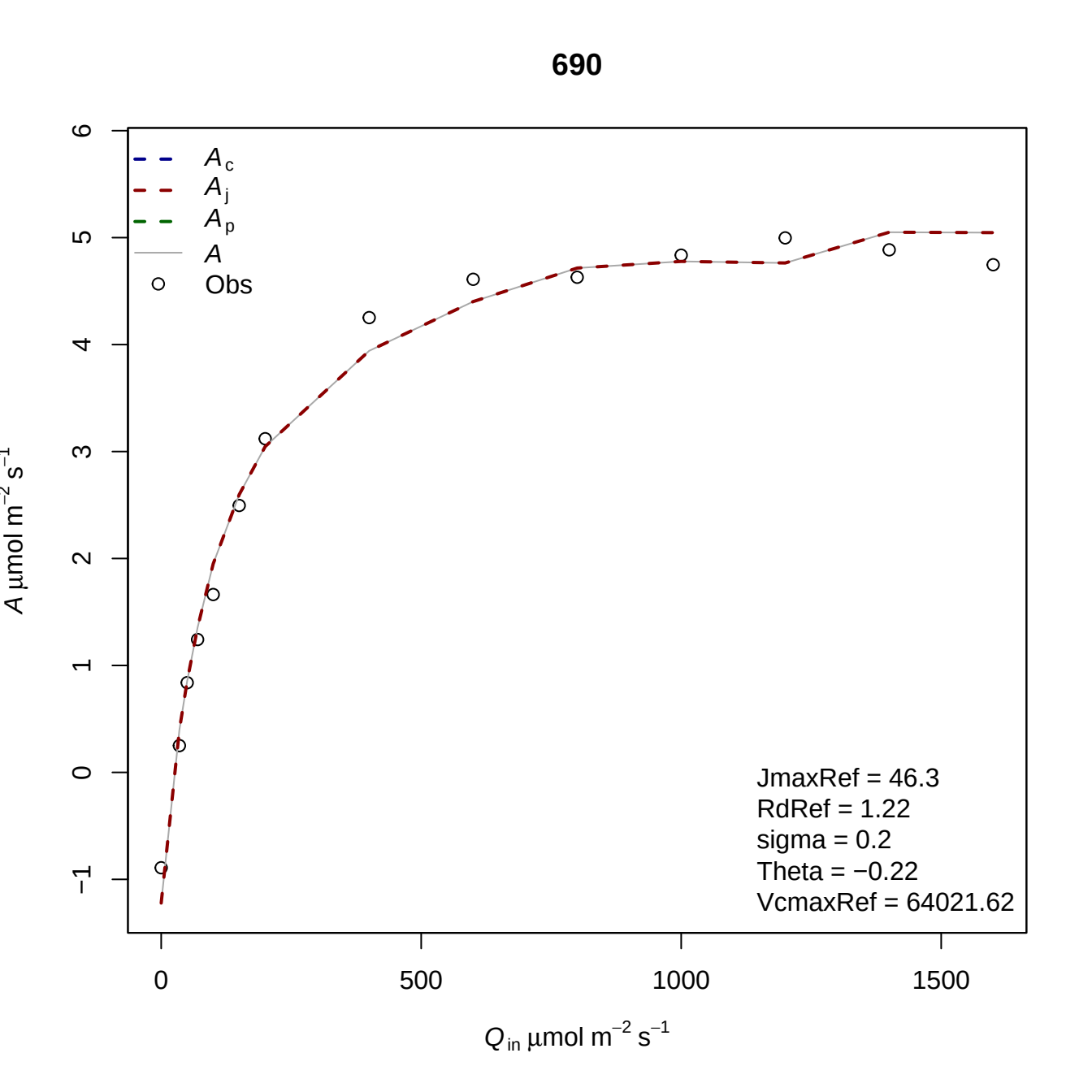

685
A_c
A_j
A_p
A
Obs
JmaxRef = 98.21
RdRef = 0.83
sigma = 0.23
Theta = 0.35
VcmaxRef = 77.59
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

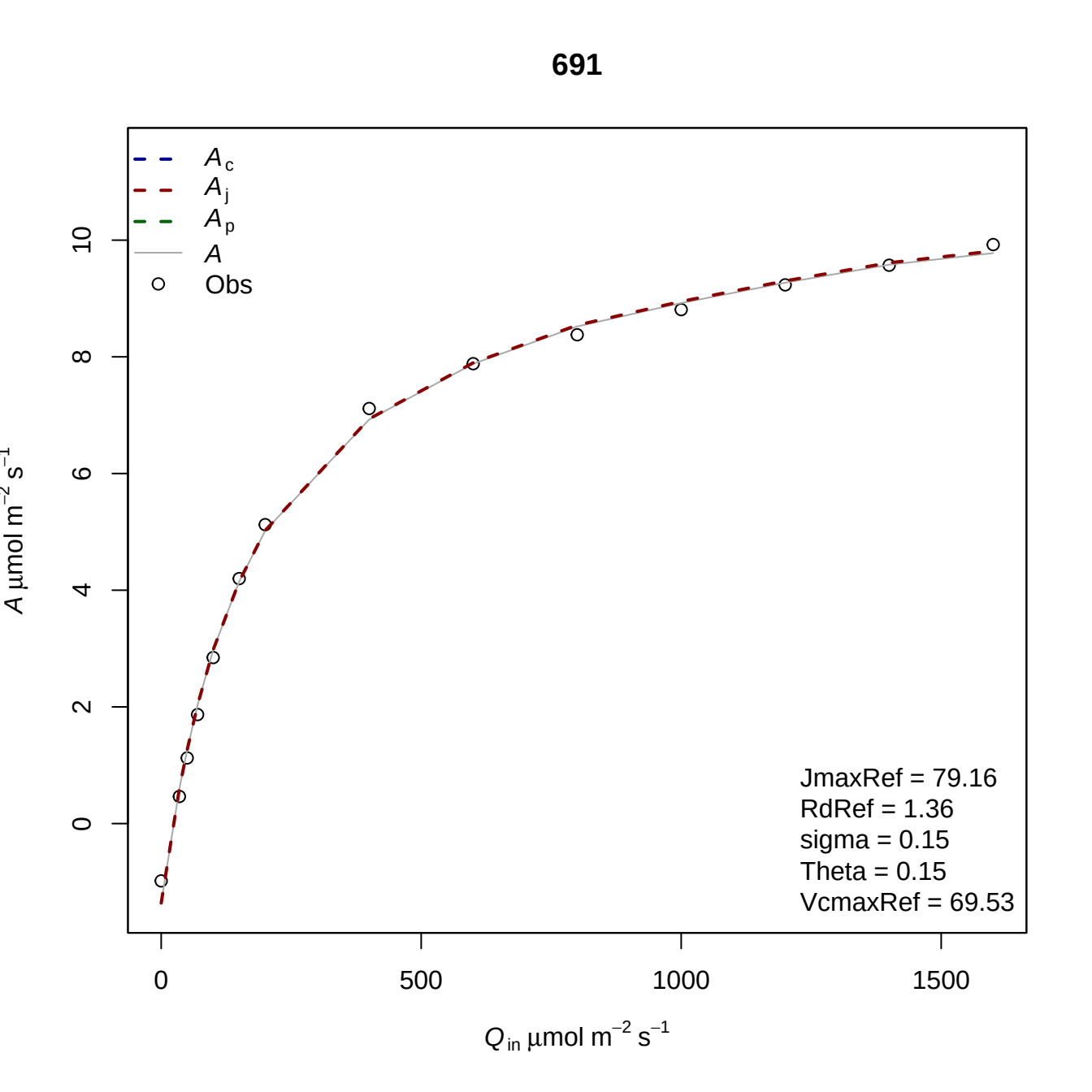

686
A_c
A_j
A_p
A
Obs
JmaxRef = 62.28
RdRef = 0.66
sigma = 0.15
Theta = 0.13
VcmaxRef = 35737.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

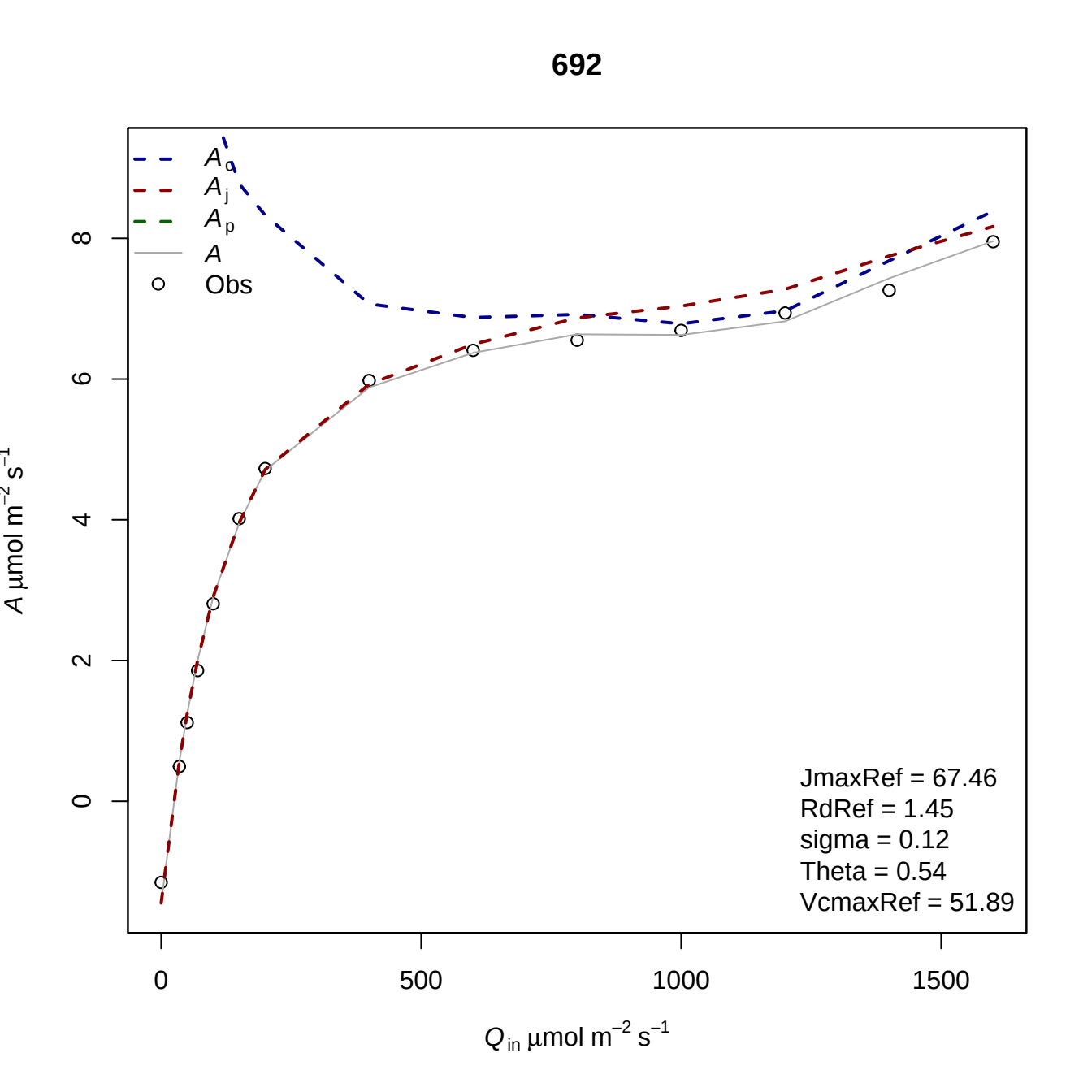

687
A_c
A_j
A_p
A
Obs
JmaxRef = 67.29
RdRef = 0.56
sigma = 0.2
Theta = −0.25
VcmaxRef = 45.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

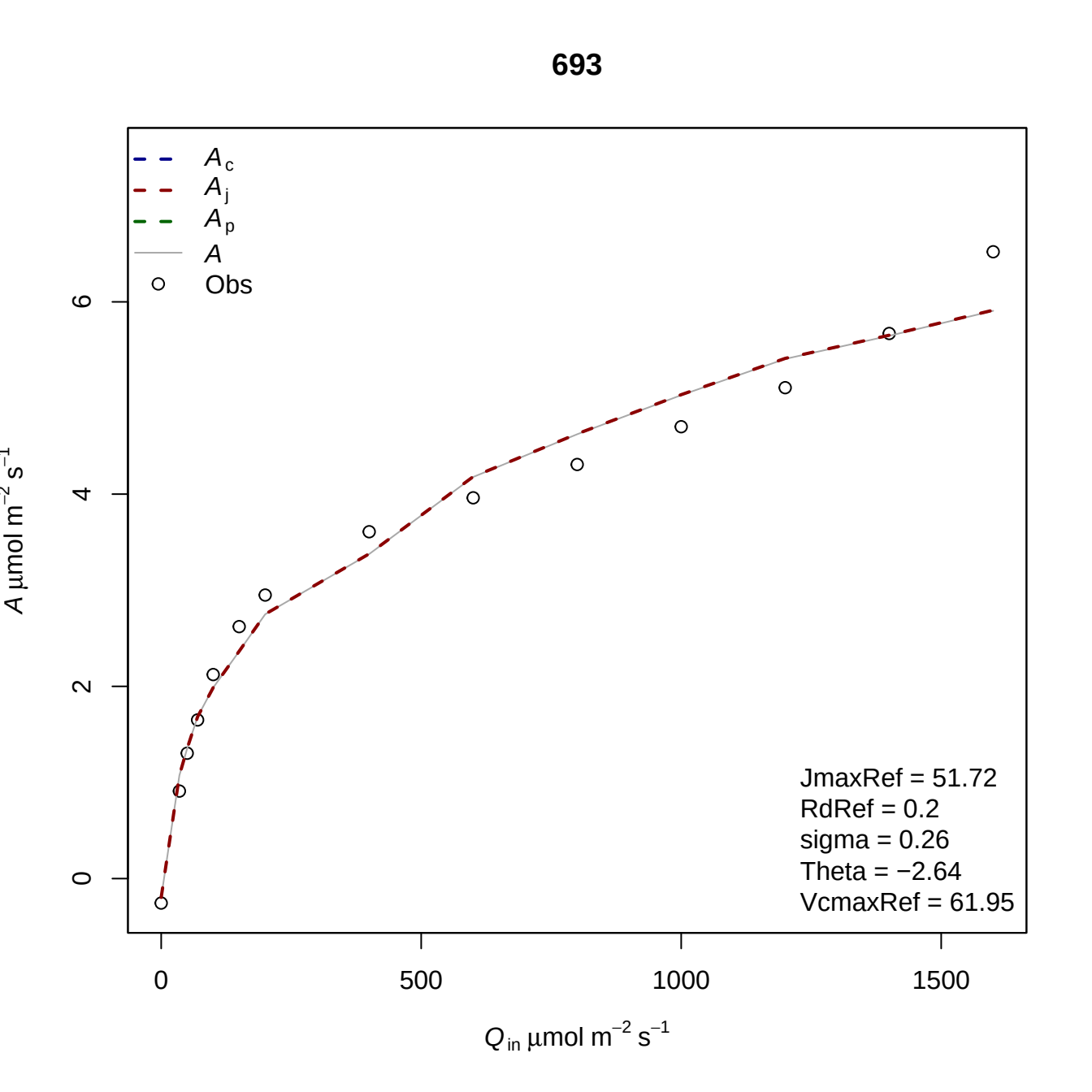

688
A_c
A_j
A_p
A
Obs
JmaxRef = 40.26
RdRef = 1.12
sigma = 0.25
Theta = −0.07
VcmaxRef = 29.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

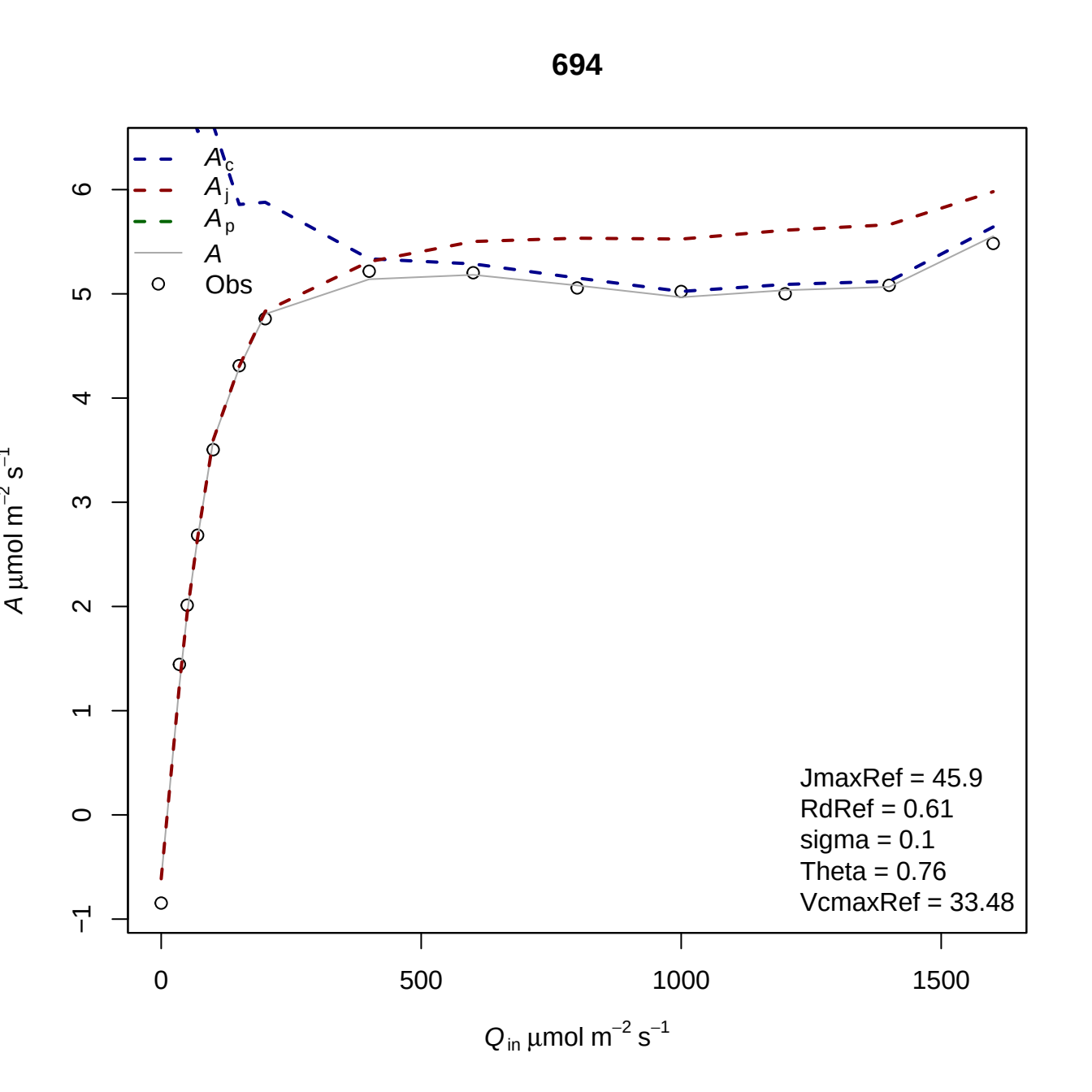

689
A_c
A_j
A_p
A
Obs
JmaxRef = 54.19
RdRef = 0.84
sigma = 0.13
Theta = 0.21
VcmaxRef = 37.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

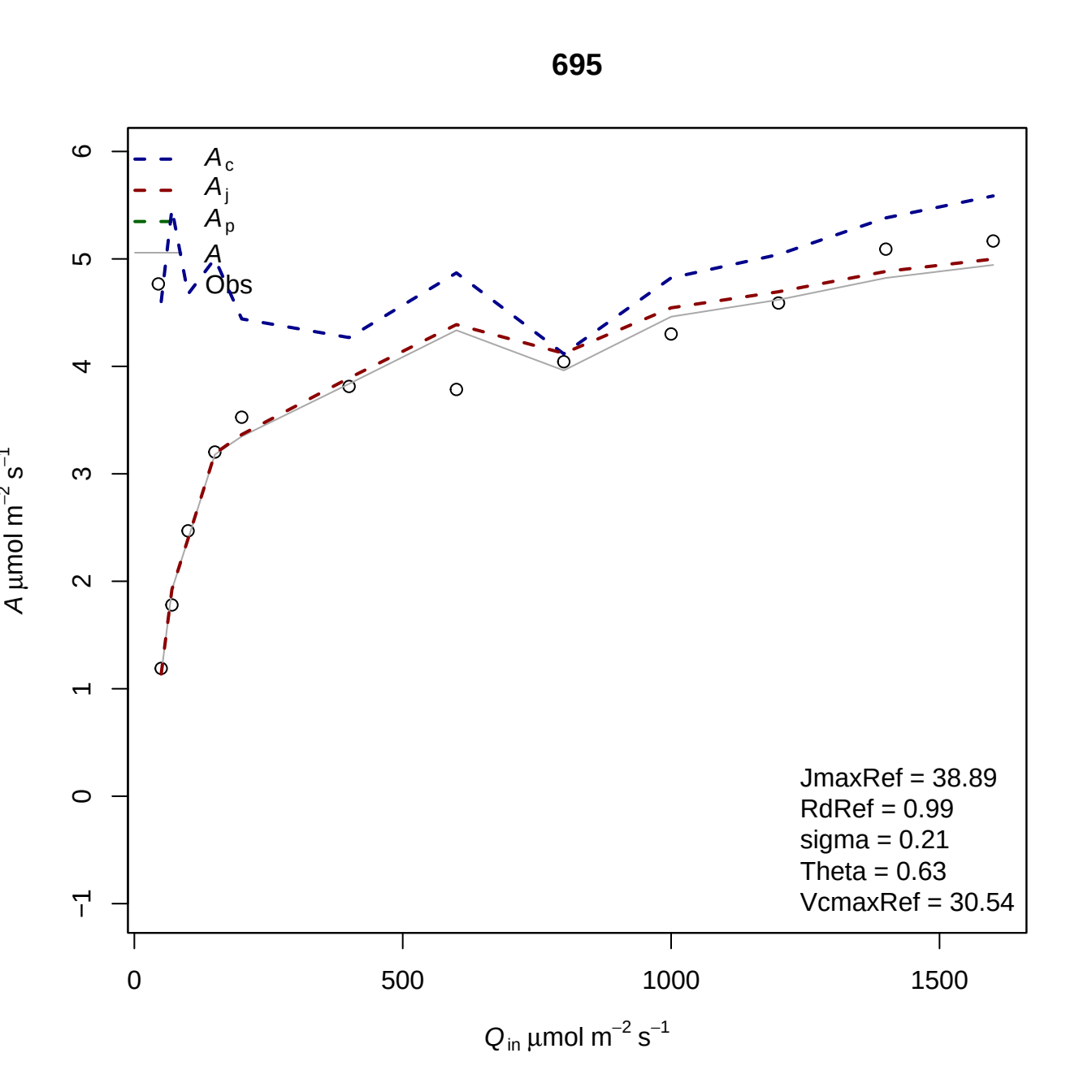

690
A_c
A_j
A_p
A
Obs
JmaxRef = 46.3
RdRef = 1.22
sigma = 0.2
Theta = −0.22
VcmaxRef = 64021.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

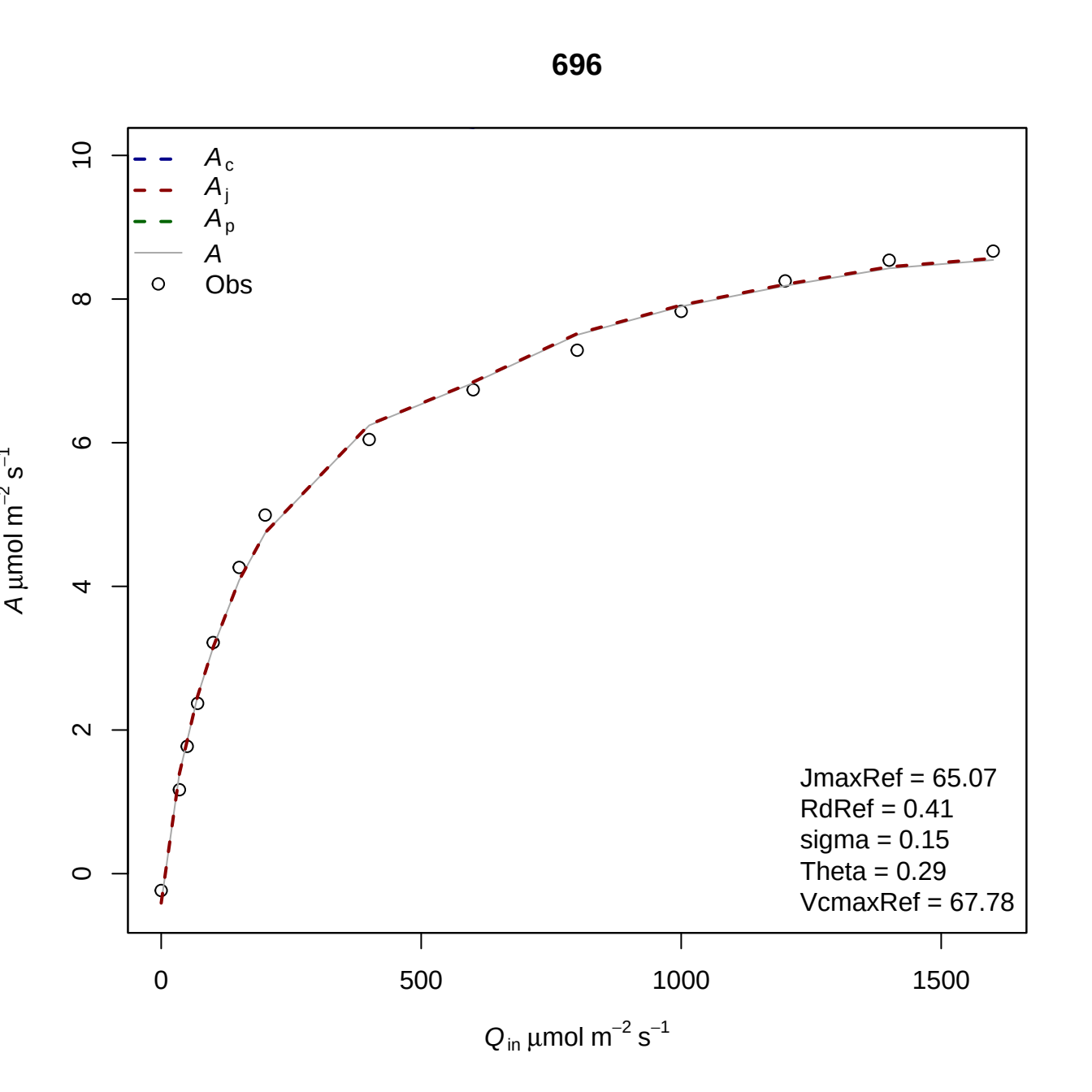

691
A_c
A_j
A_p
A
Obs
JmaxRef = 79.16
RdRef = 1.36
sigma = 0.15
Theta = 0.15
VcmaxRef = 69.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

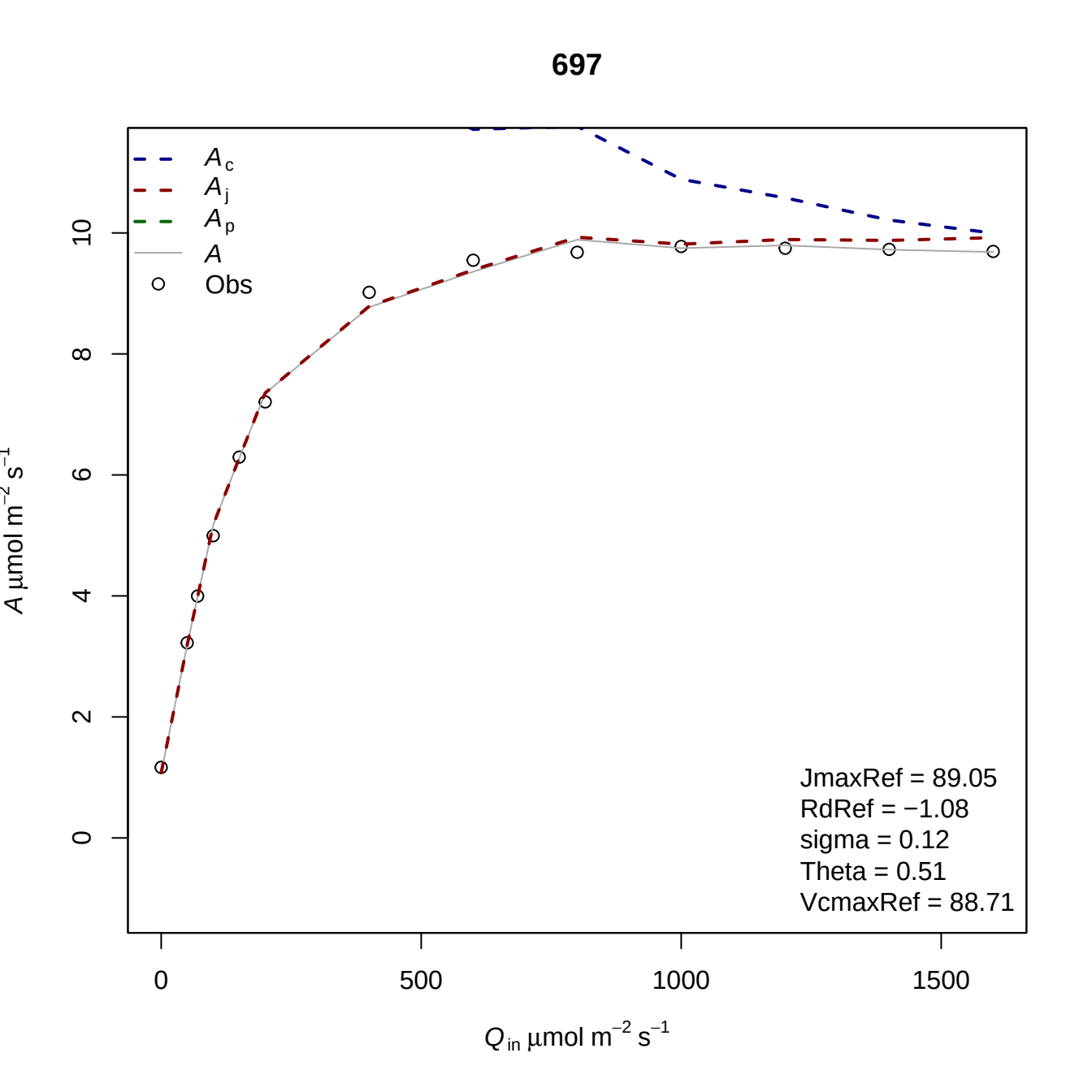

692
A_c
A_j
A_p
A
Obs
JmaxRef = 67.46
RdRef = 1.45
sigma = 0.12
Theta = 0.54
VcmaxRef = 51.89
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

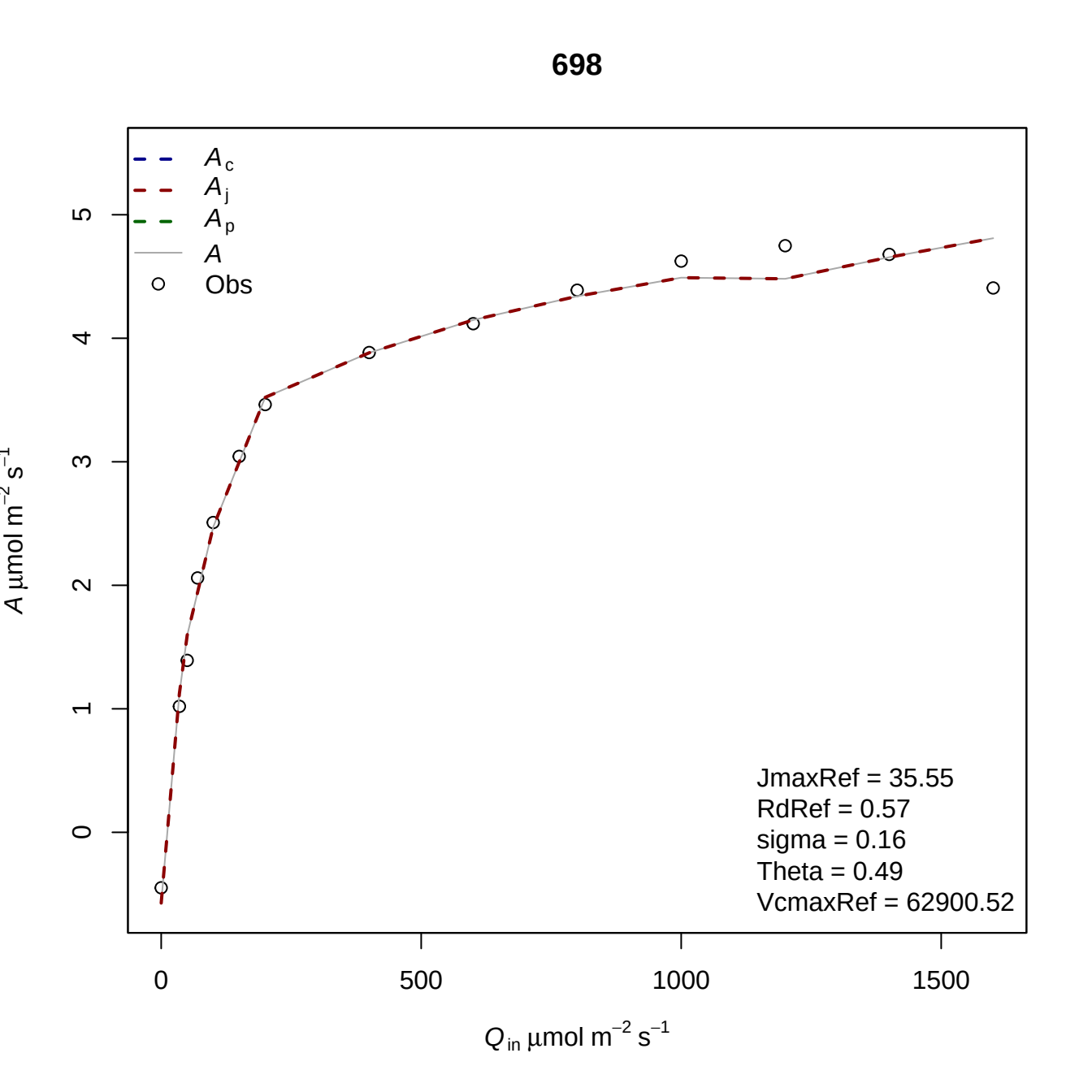

693
A_c
A_j
A_p
A
Obs
JmaxRef = 51.72
RdRef = 0.2
sigma = 0.26
Theta = −2.64
VcmaxRef = 61.95
A µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

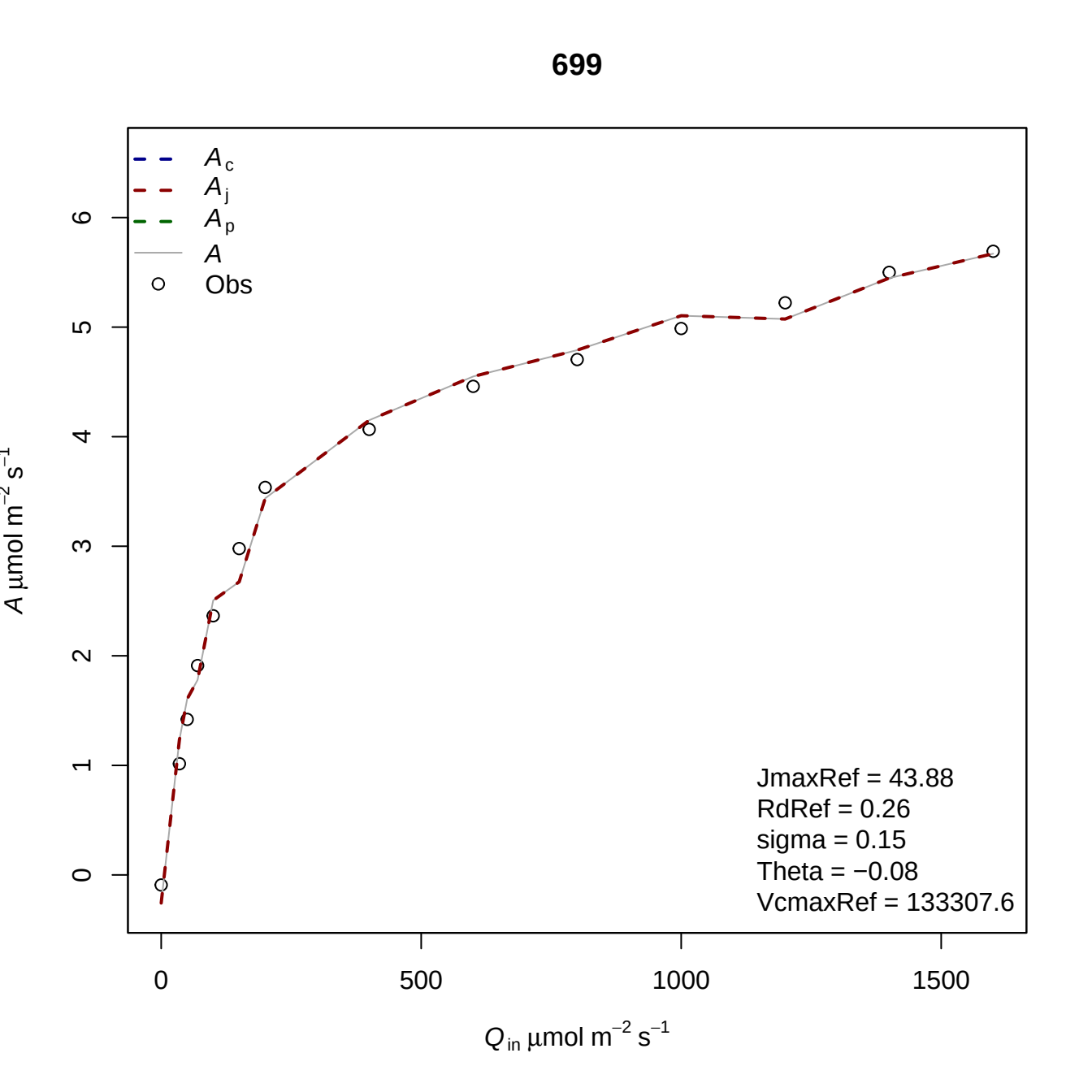

694
A_c
A_j
A_p
A
Obs
JmaxRef = 45.9
RdRef = 0.61
sigma = 0.1
Theta = 0.76
VcmaxRef = 33.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

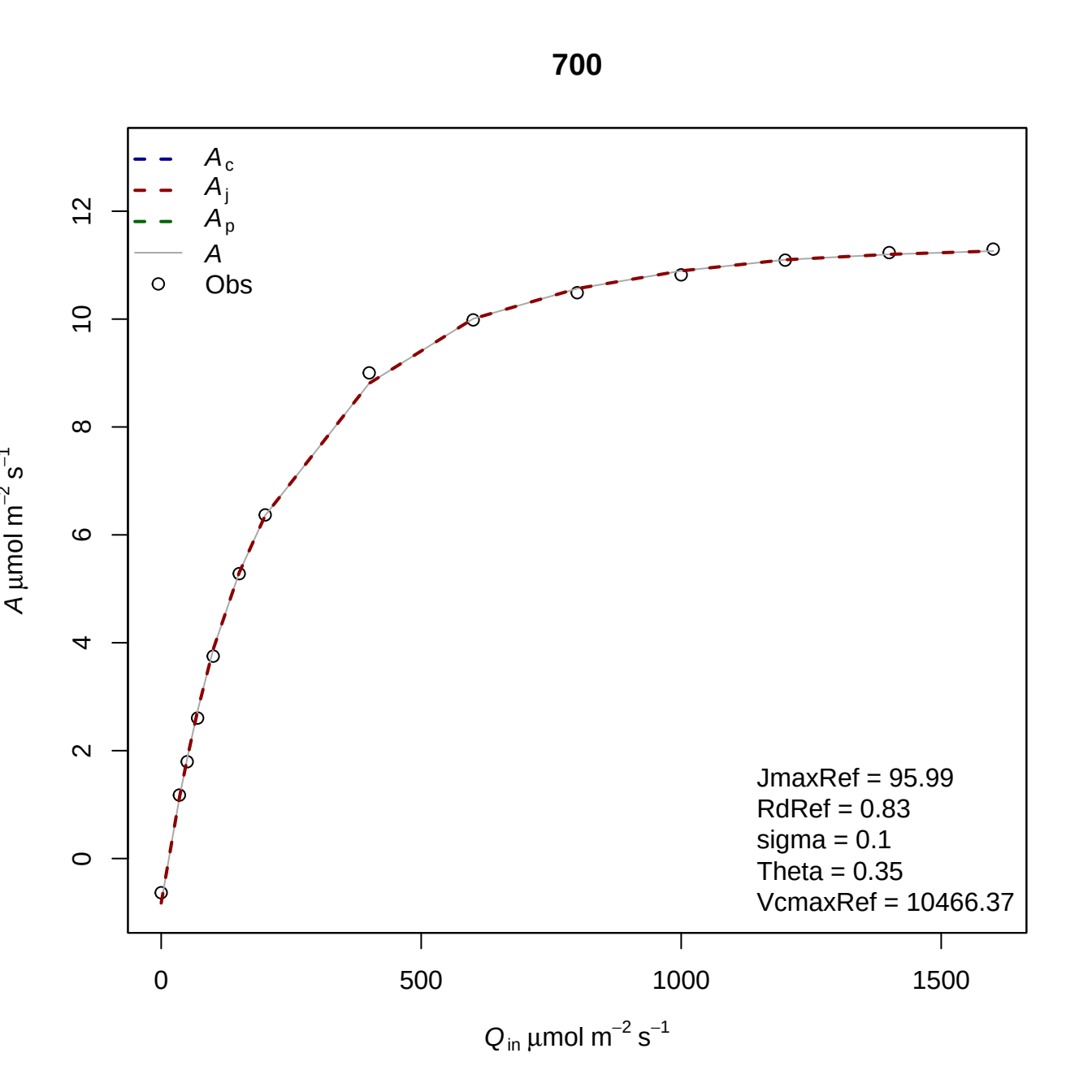

695
A_c
A_j
A_p
A
Obs
JmaxRef = 38.89
RdRef = 0.99
sigma = 0.21
Theta = 0.63
VcmaxRef = 30.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

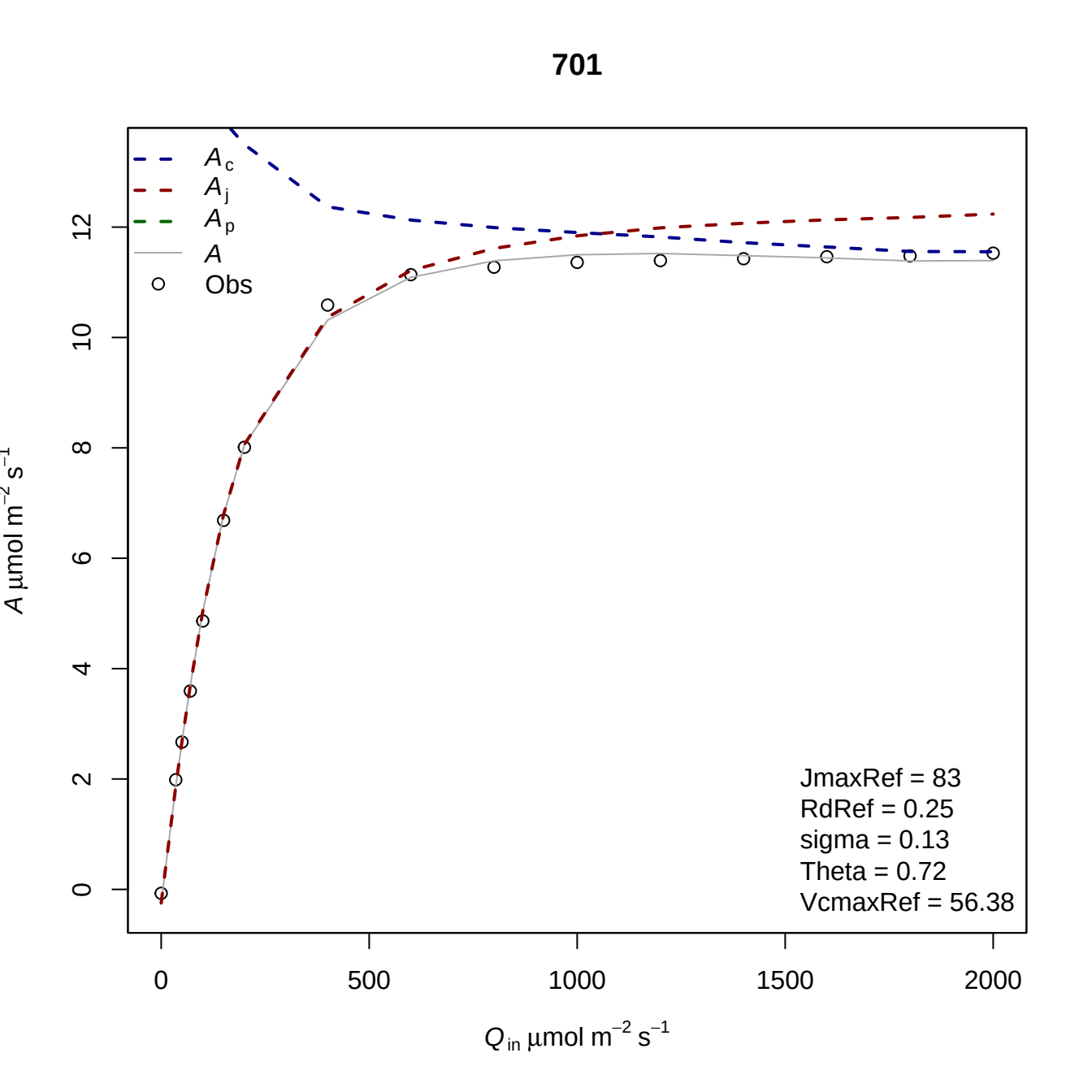

696
A_c
A_j
A_p
A
Obs
JmaxRef = 65.07
RdRef = 0.41
sigma = 0.15
Theta = 0.29
VcmaxRef = 67.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

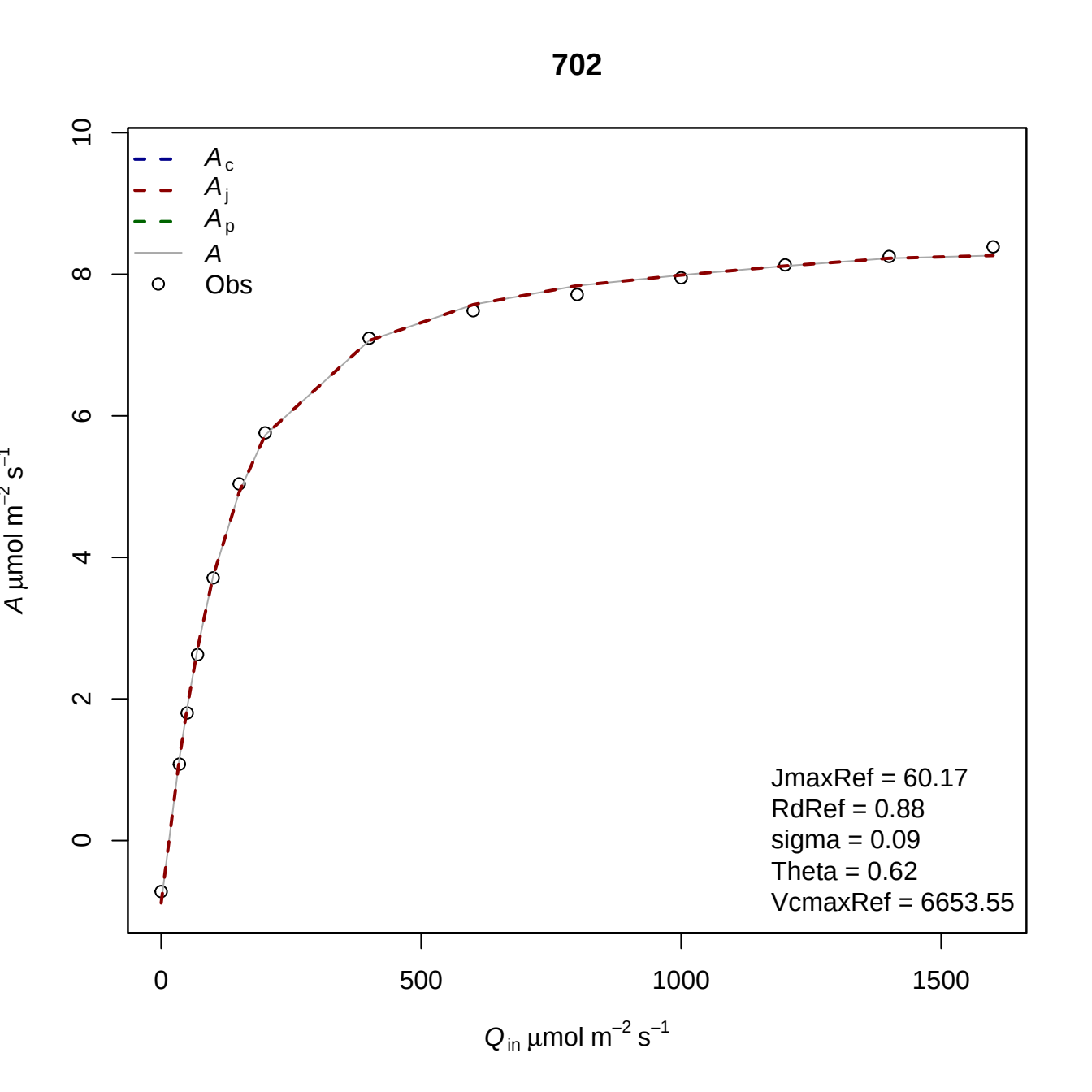

697
A_c
A_j
A_p
A
Obs
JmaxRef = 89.05
RdRef = −1.08
sigma = 0.12
Theta = 0.51
VcmaxRef = 88.71
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

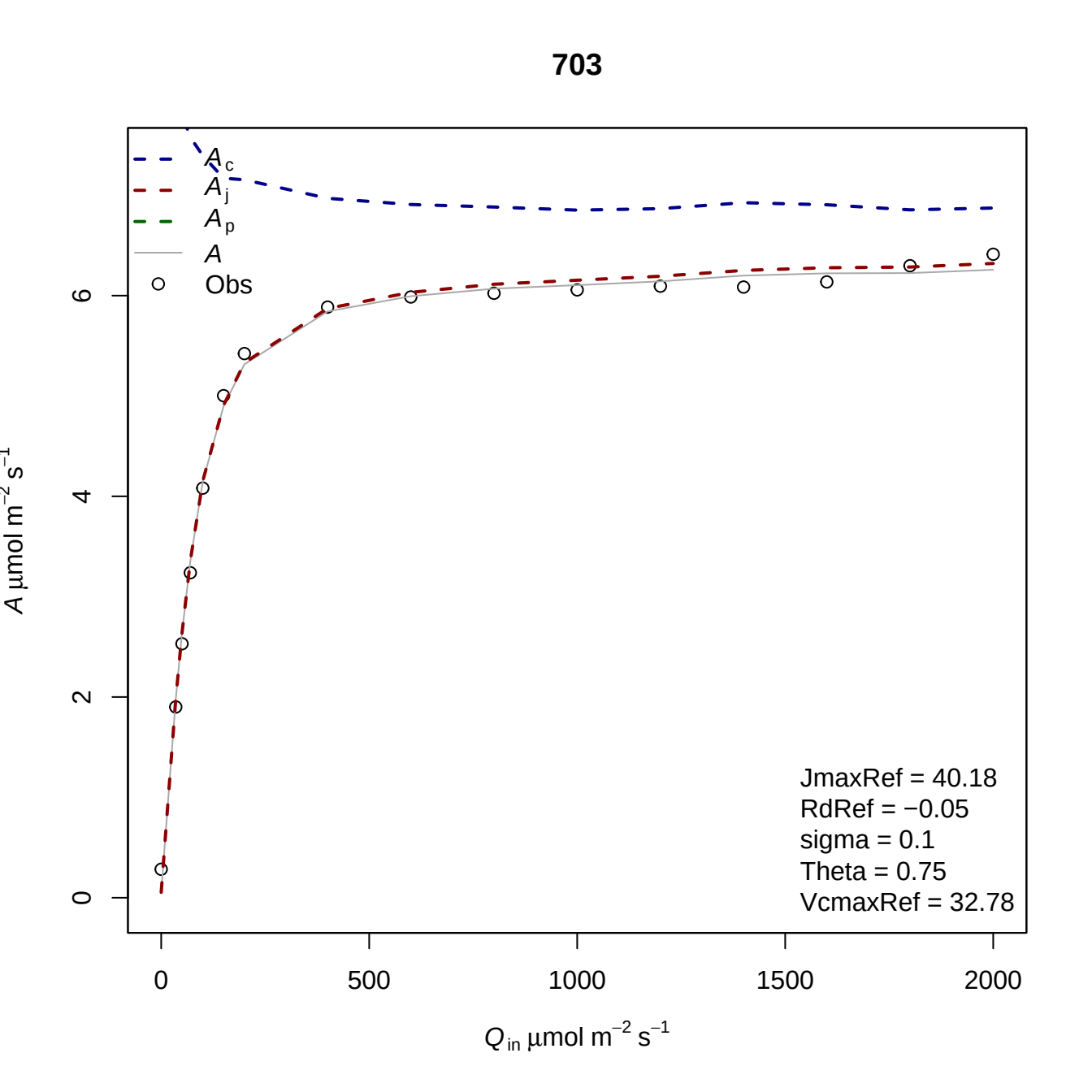

698
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 0.57
sigma = 0.16
Theta = 0.49
VcmaxRef = 62900.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

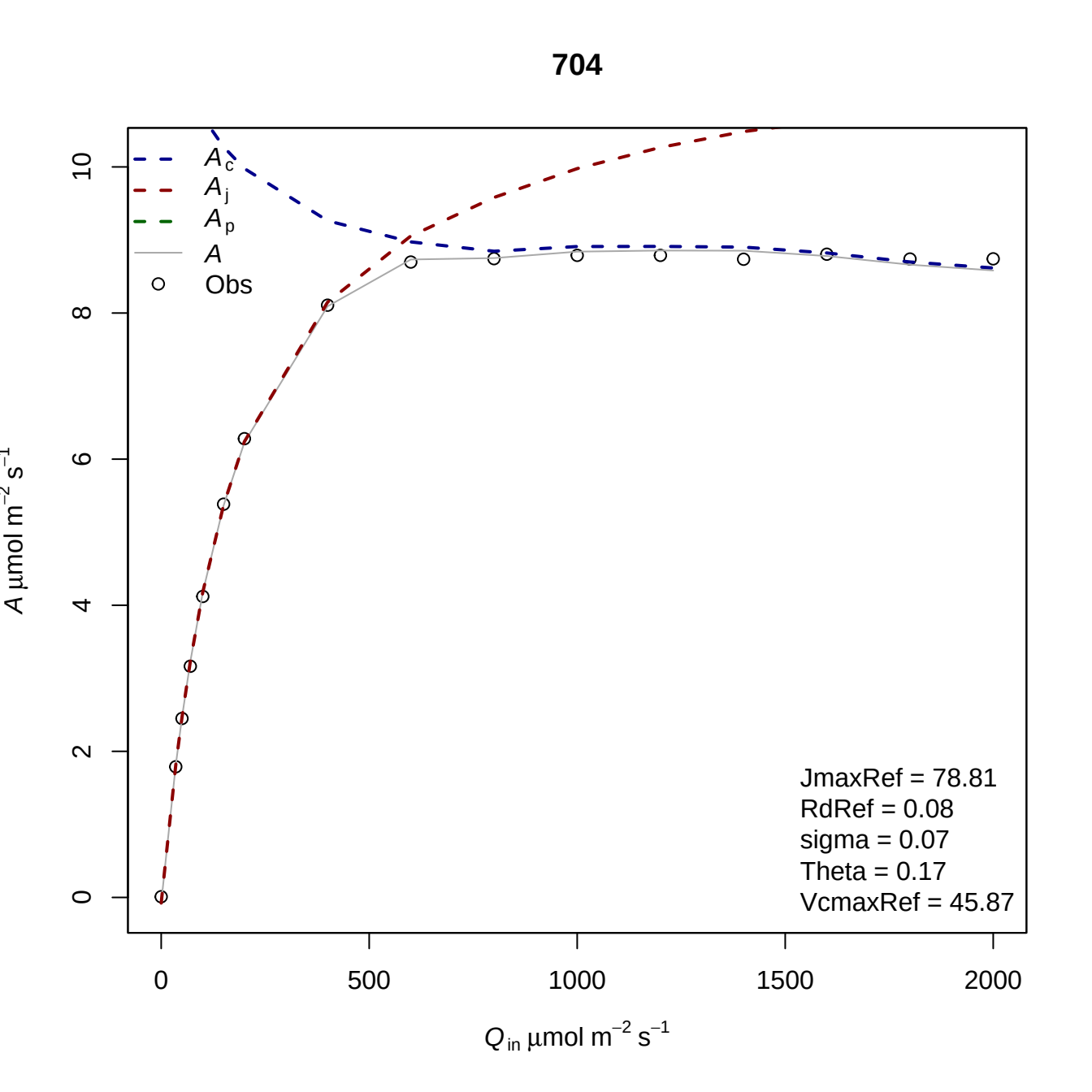

699
A_c
A_j
A_p
A
Obs
JmaxRef = 43.88
RdRef = 0.26
sigma = 0.15
Theta = −0.08
VcmaxRef = 133307.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

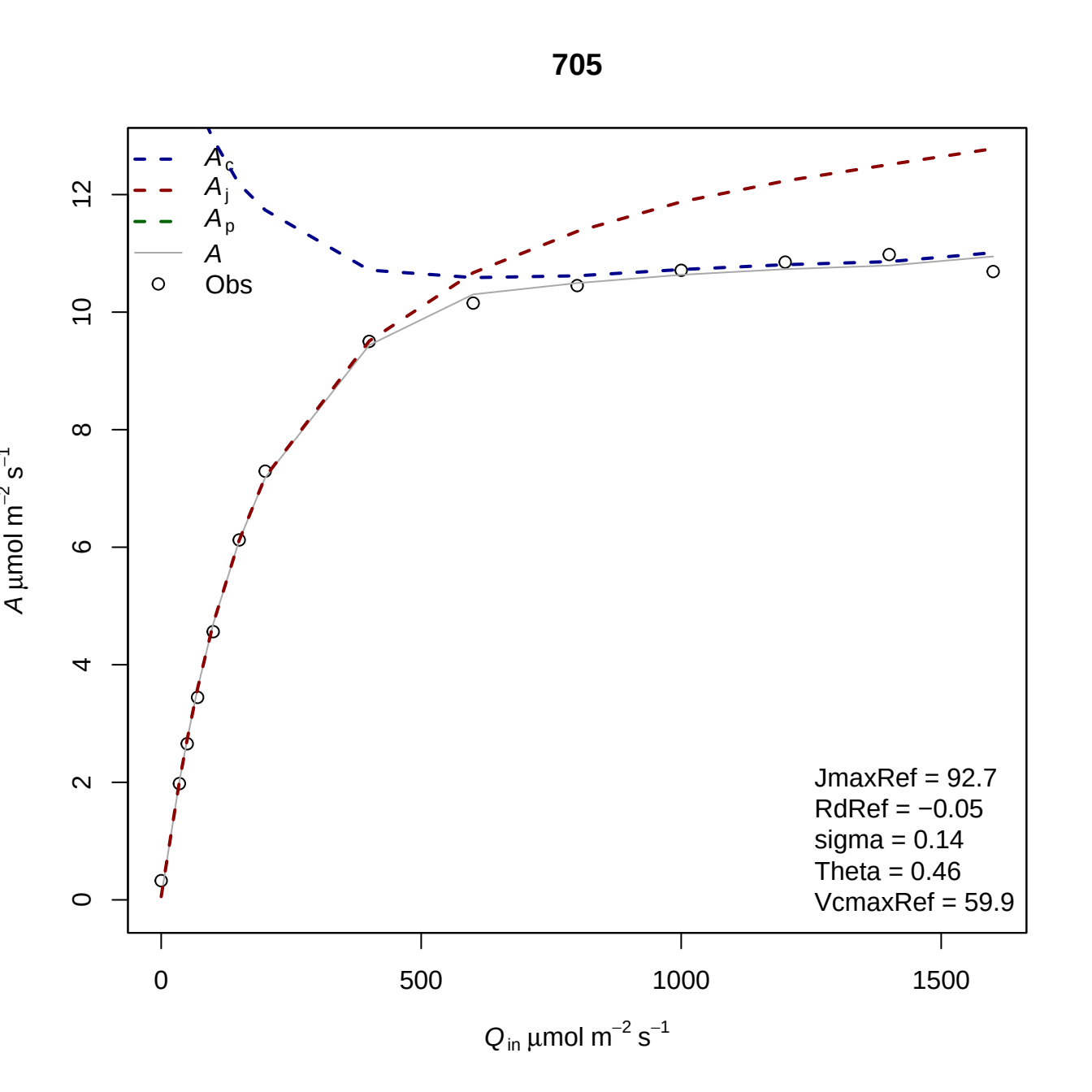

700
A_c
A_j
A_p
A
Obs
JmaxRef = 95.99
RdRef = 0.83
sigma = 0.1
Theta = 0.35
VcmaxRef = 10466.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

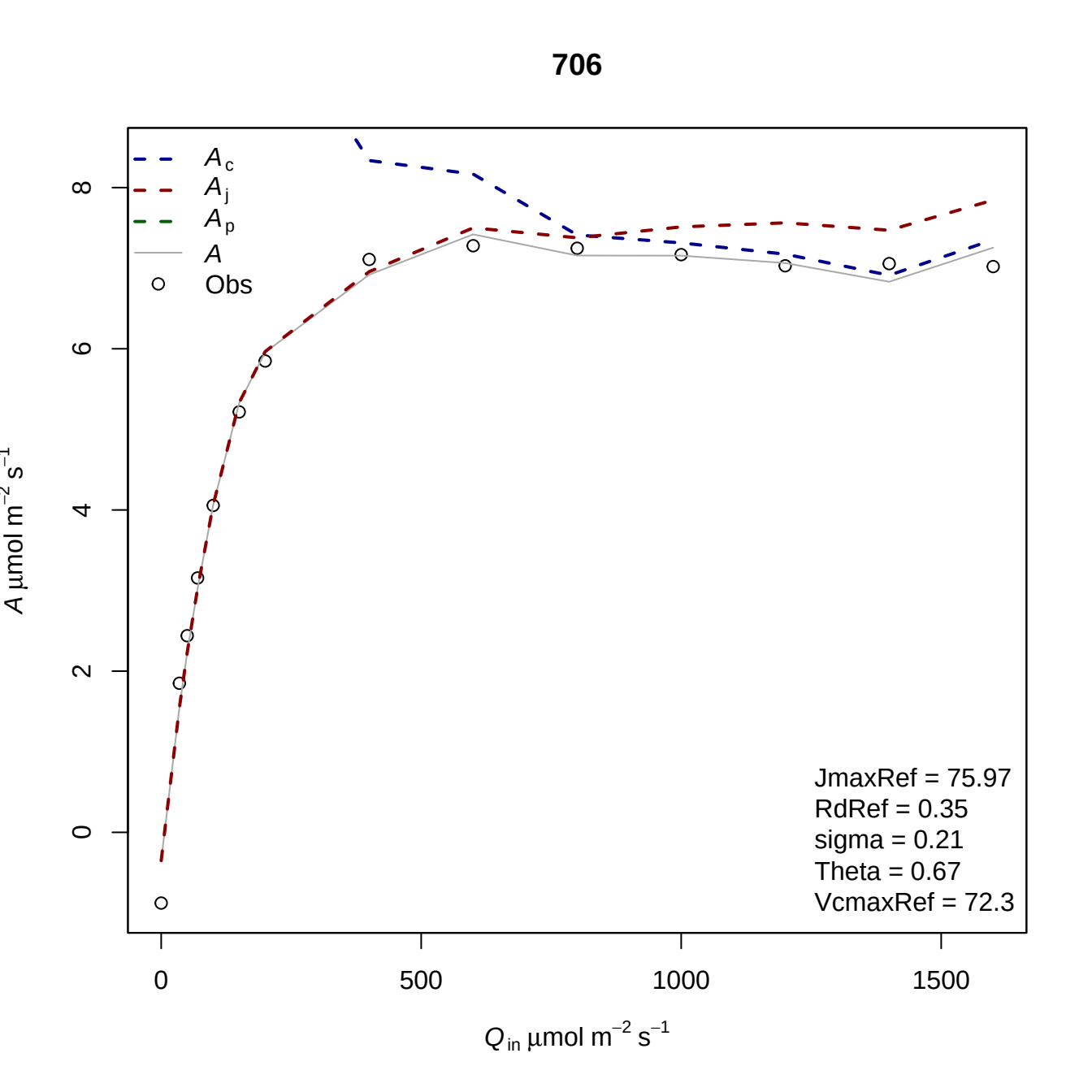

701
A_c
A_j
A_p
A
Obs
JmaxRef = 83
RdRef = 0.25
sigma = 0.13
Theta = 0.72
VcmaxRef = 56.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

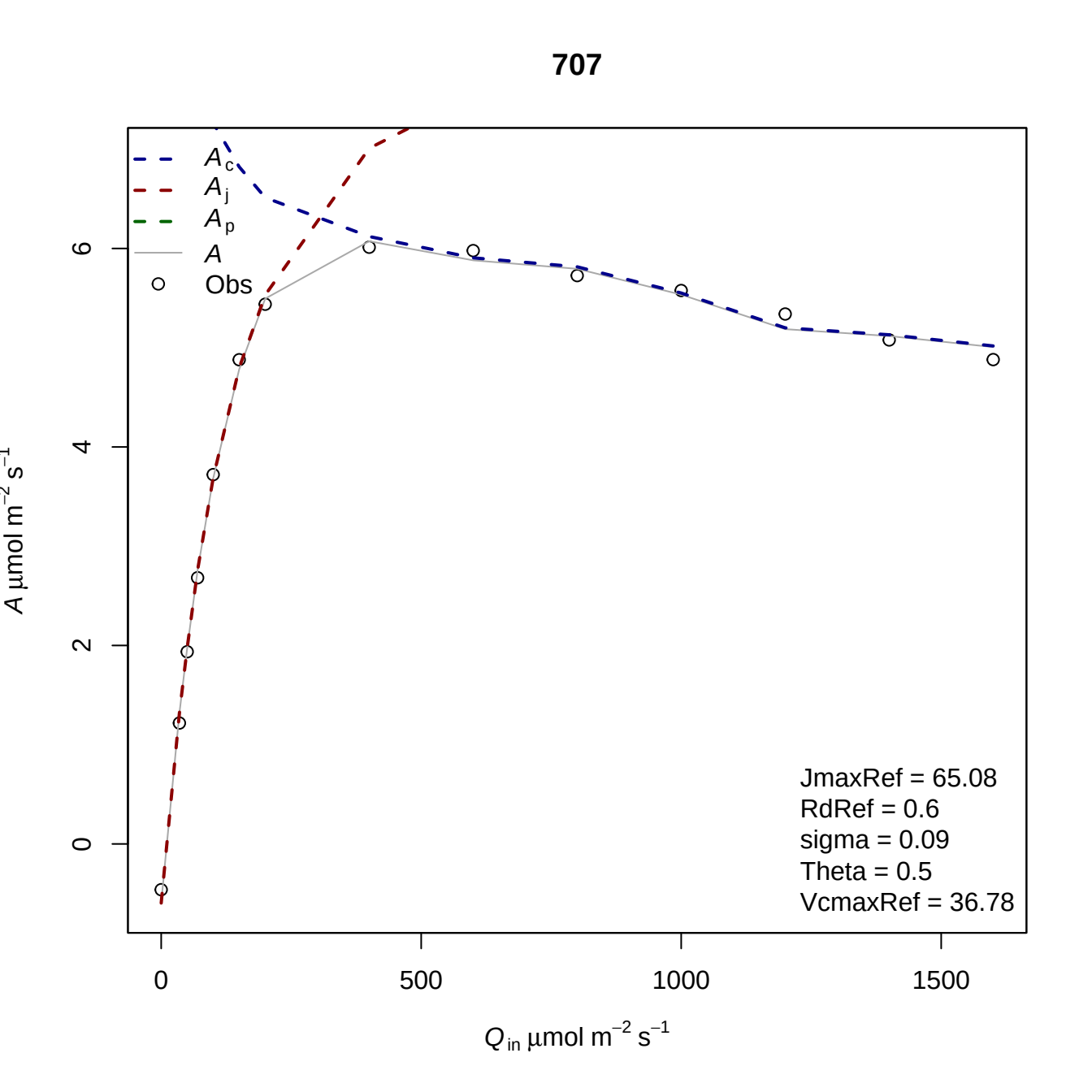

702
A_c
A_j
A_p
A
Obs
JmaxRef = 60.17
RdRef = 0.88
sigma = 0.09
Theta = 0.62
VcmaxRef = 6653.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

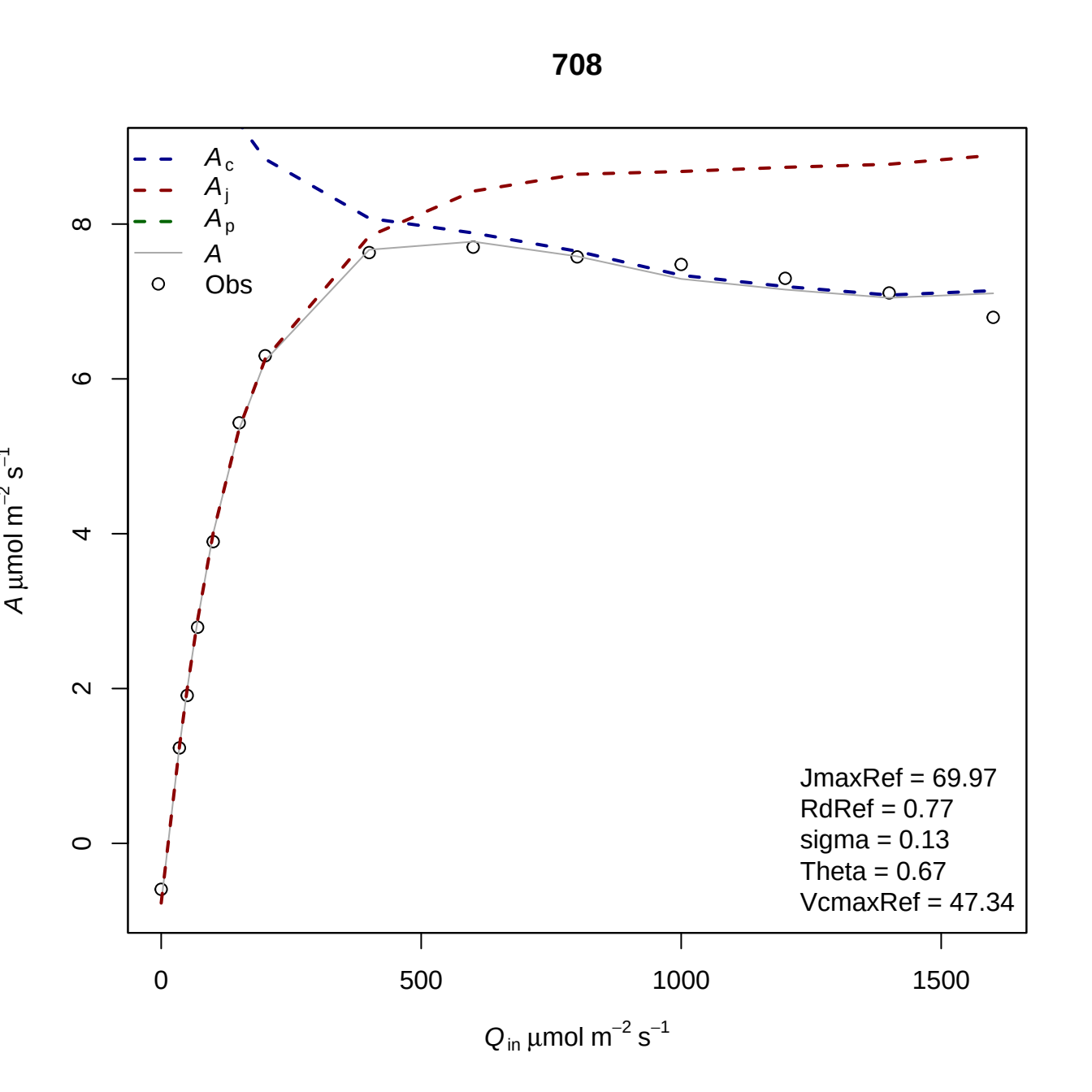

703
A_c
A_j
A_p
A
Obs
JmaxRef = 40.18
RdRef = −0.05
sigma = 0.1
Theta = 0.75
VcmaxRef = 32.78
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

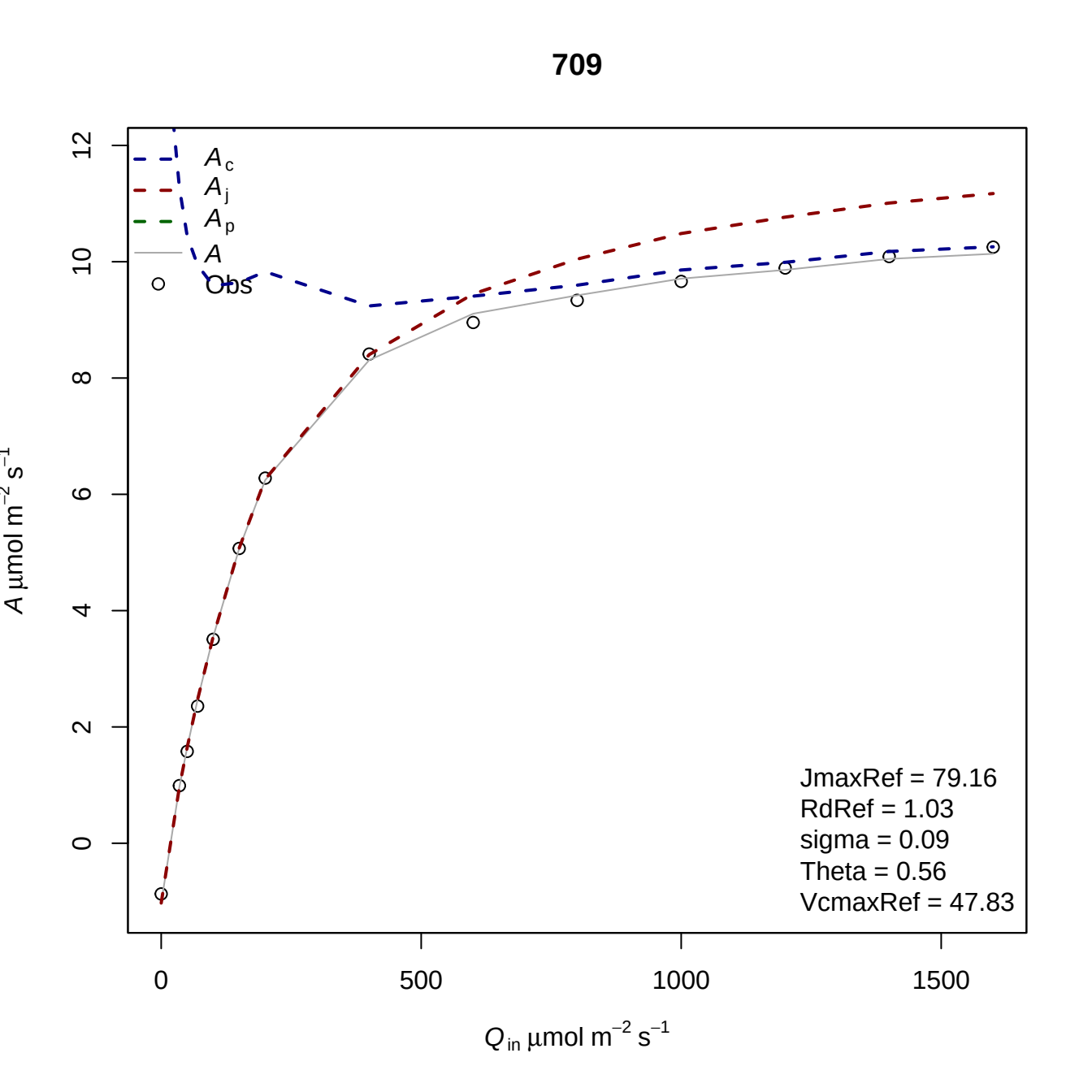

704
A_c
A_j
A_p
A
Obs
JmaxRef = 78.81
RdRef = 0.08
sigma = 0.07
Theta = 0.17
VcmaxRef = 45.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

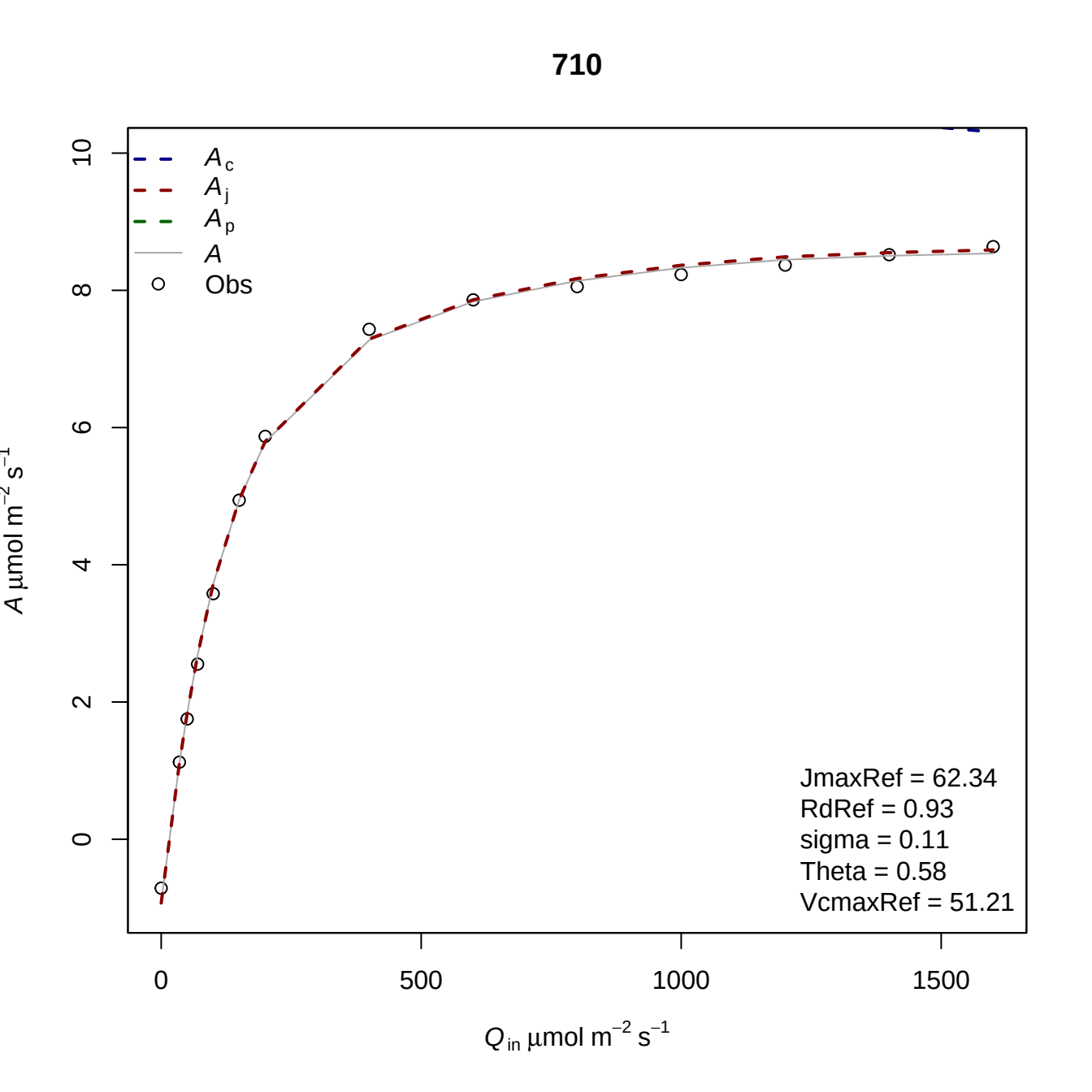

705
A_c
A_j
A_p
A
Obs
JmaxRef = 92.7
RdRef = −0.05
sigma = 0.14
Theta = 0.46
VcmaxRef = 59.9
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

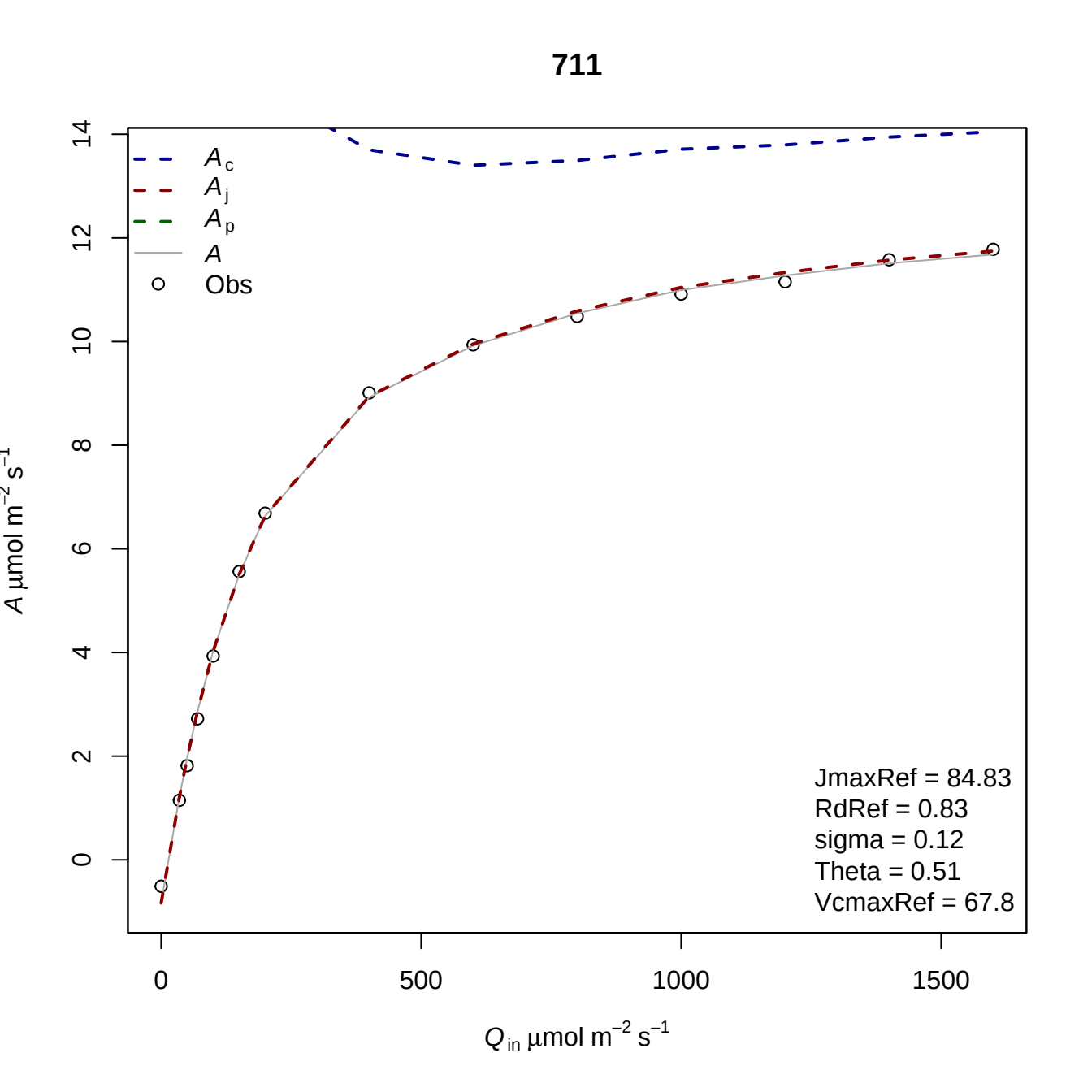

706
A_c
A_j
A_p
A
Obs
JmaxRef = 75.97
RdRef = 0.35
sigma = 0.21
Theta = 0.67
VcmaxRef = 72.3
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

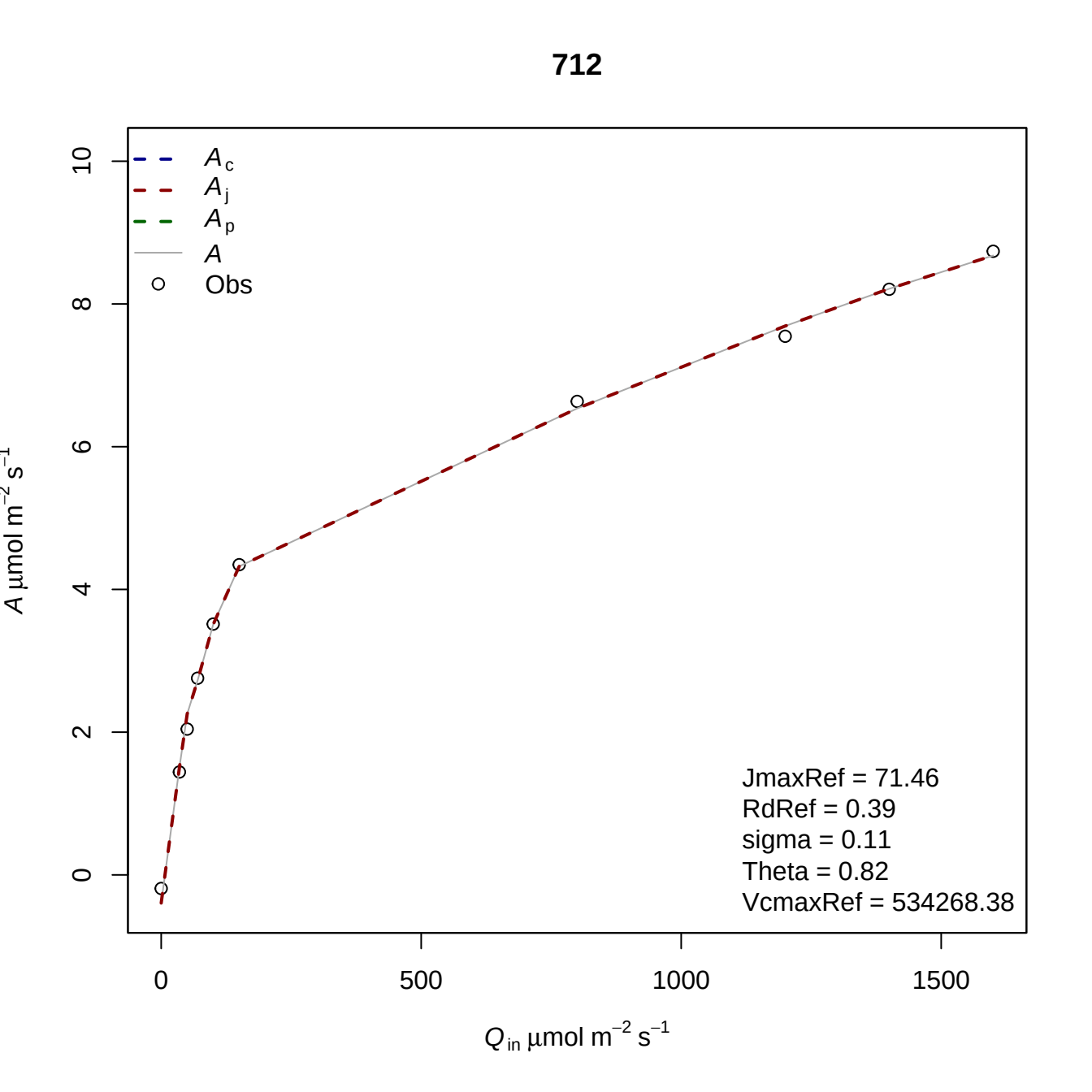

707
A_c
A_j
A_p
A
Obs
JmaxRef = 65.08
RdRef = 0.6
sigma = 0.09
Theta = 0.5
VcmaxRef = 36.78
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

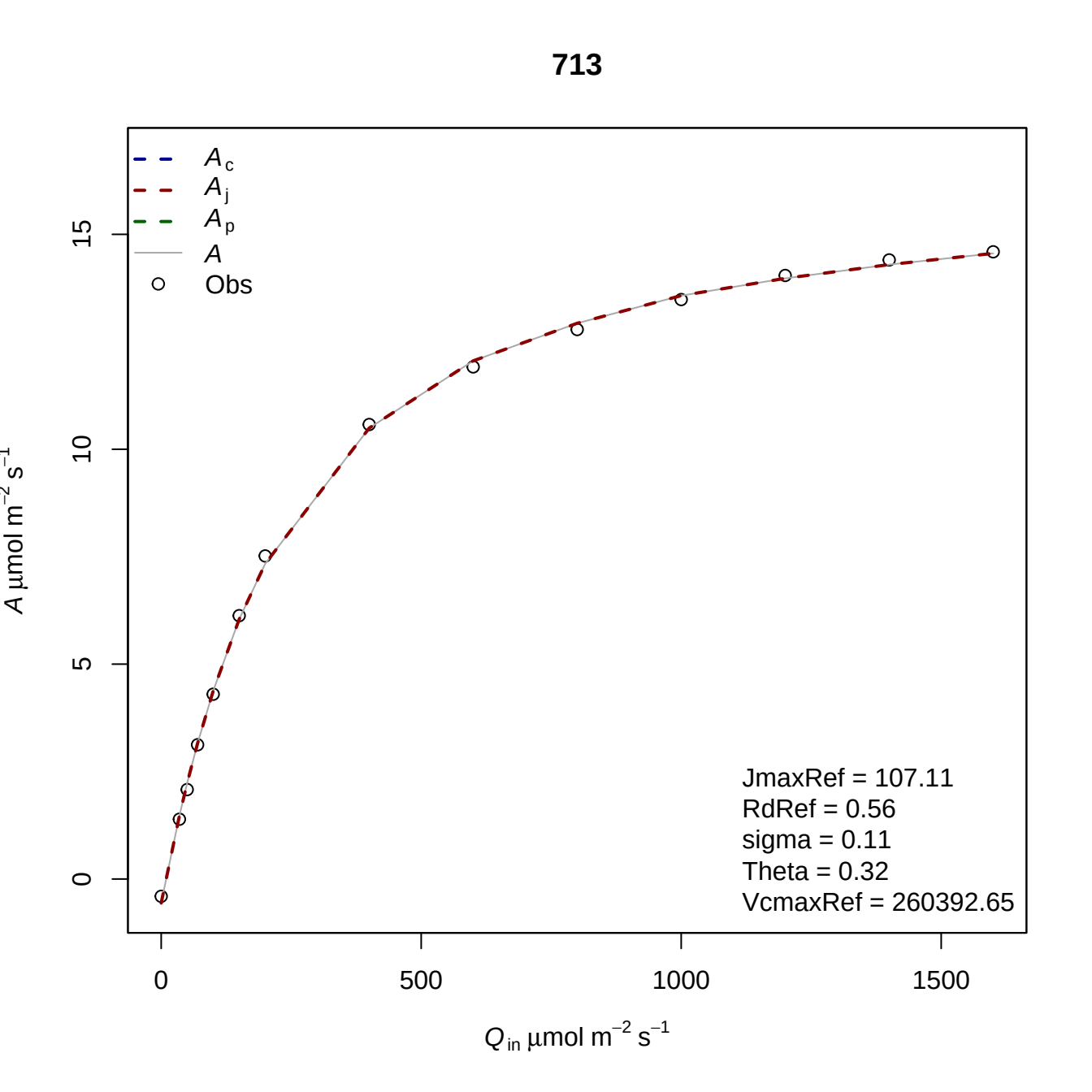

708
A_c
A_j
A_p
A
Obs
JmaxRef = 69.97
RdRef = 0.77
sigma = 0.13
Theta = 0.67
VcmaxRef = 47.34
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

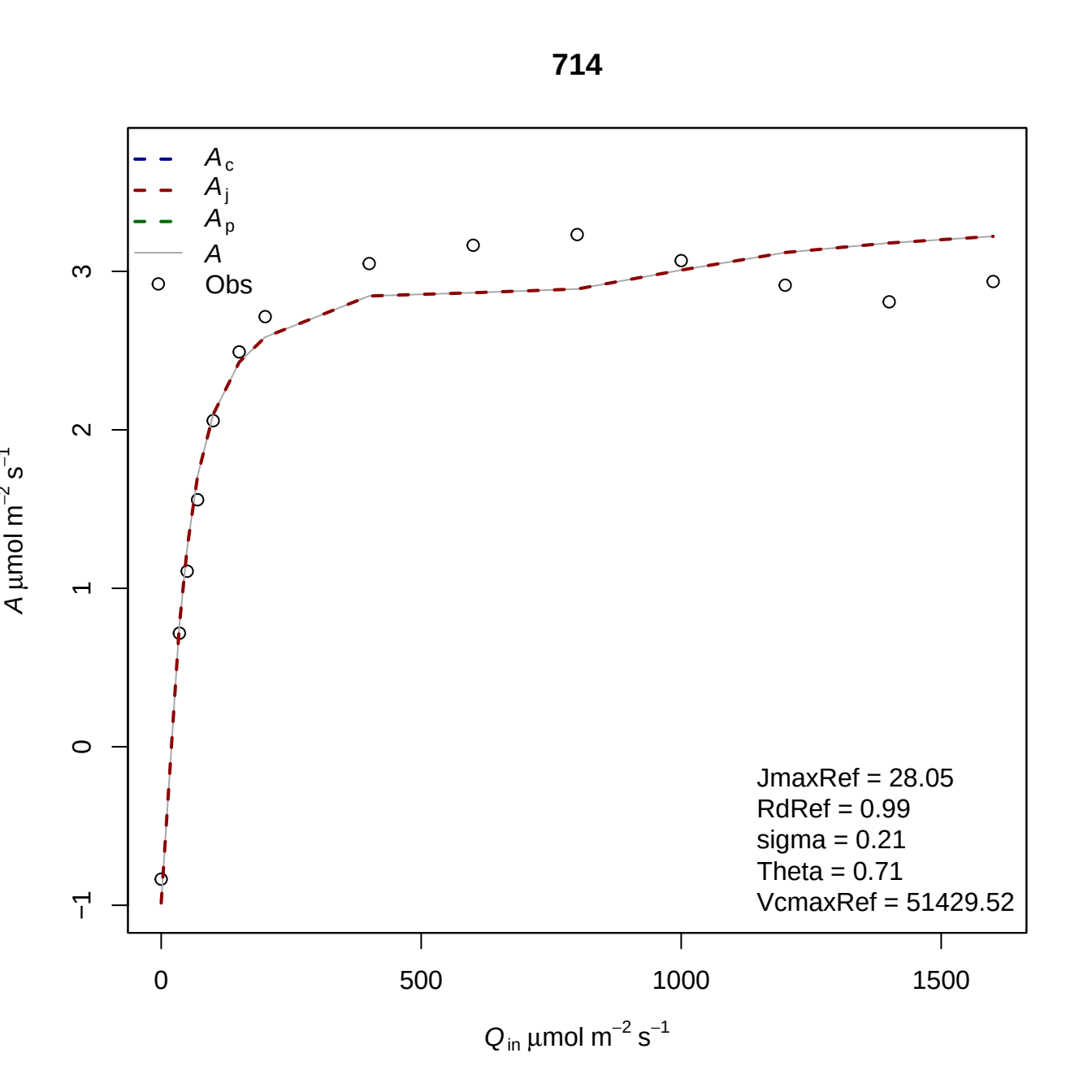

709
A_c
A_j
A_p
A
Obs
JmaxRef = 79.16
RdRef = 1.03
sigma = 0.09
Theta = 0.56
VcmaxRef = 47.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

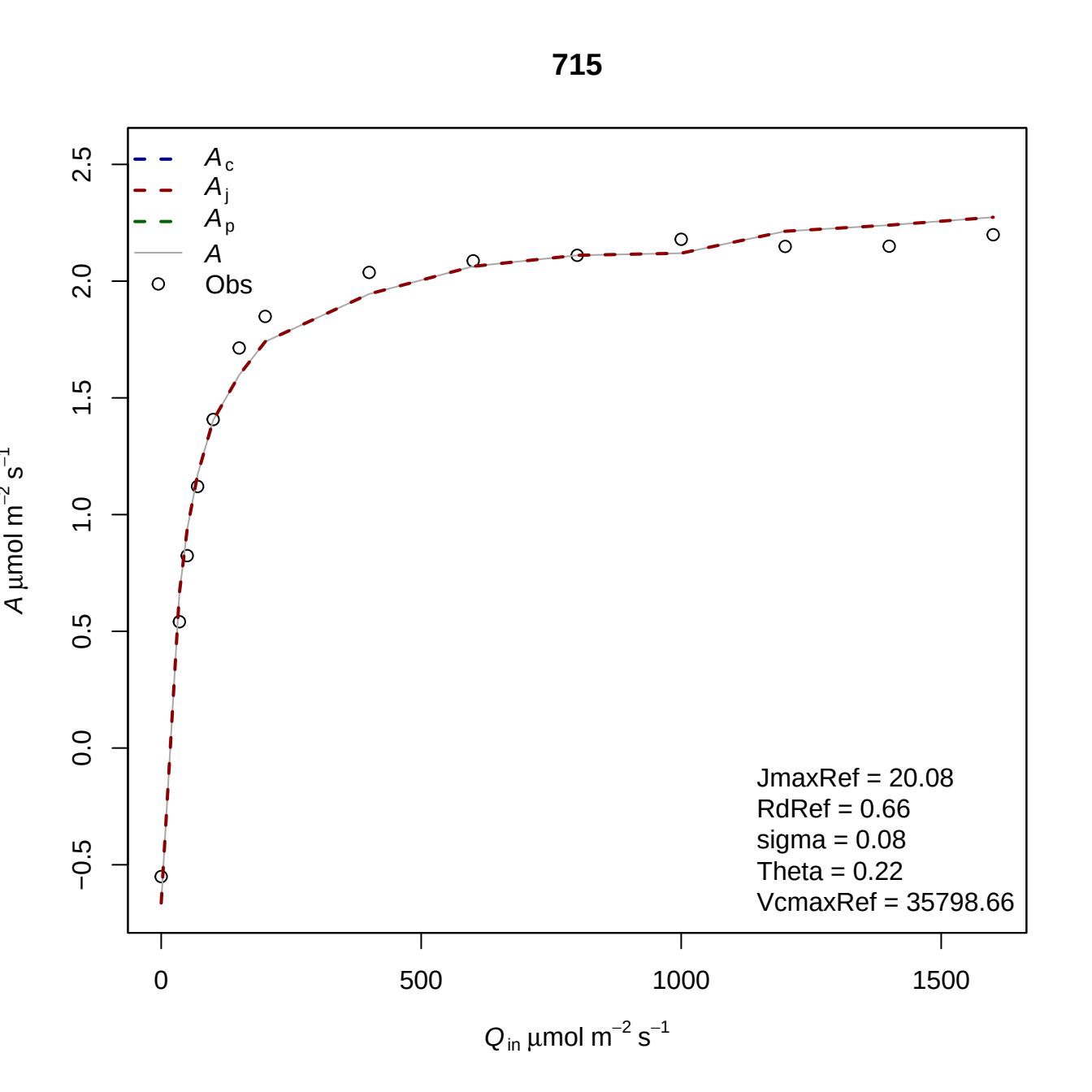

710
A_c
A_j
A_p
A
Obs
JmaxRef = 62.34
RdRef = 0.93
sigma = 0.11
Theta = 0.58
VcmaxRef = 51.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

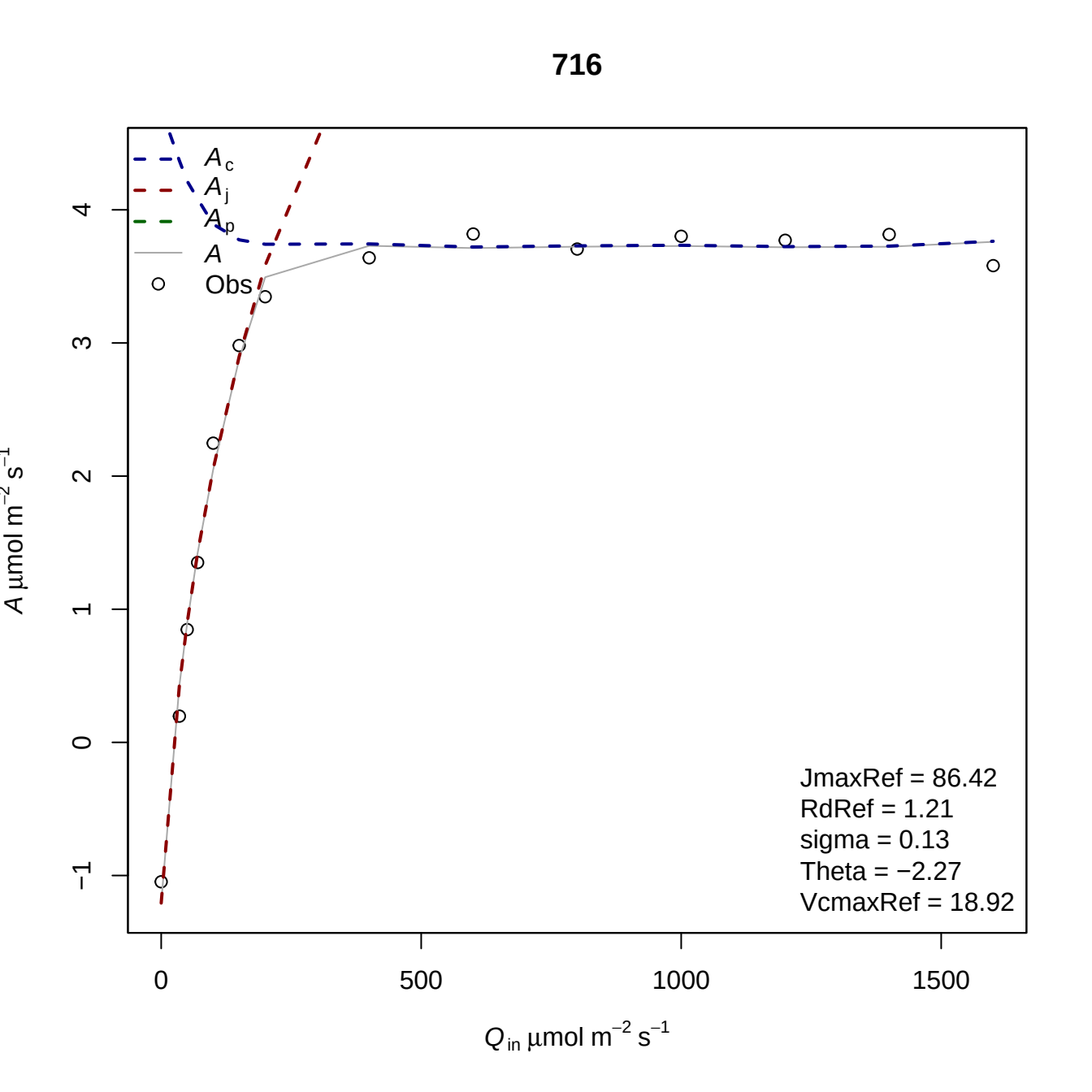

711
A_c
A_j
A_p
A
Obs
JmaxRef = 84.83
RdRef = 0.83
sigma = 0.12
Theta = 0.51
VcmaxRef = 67.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

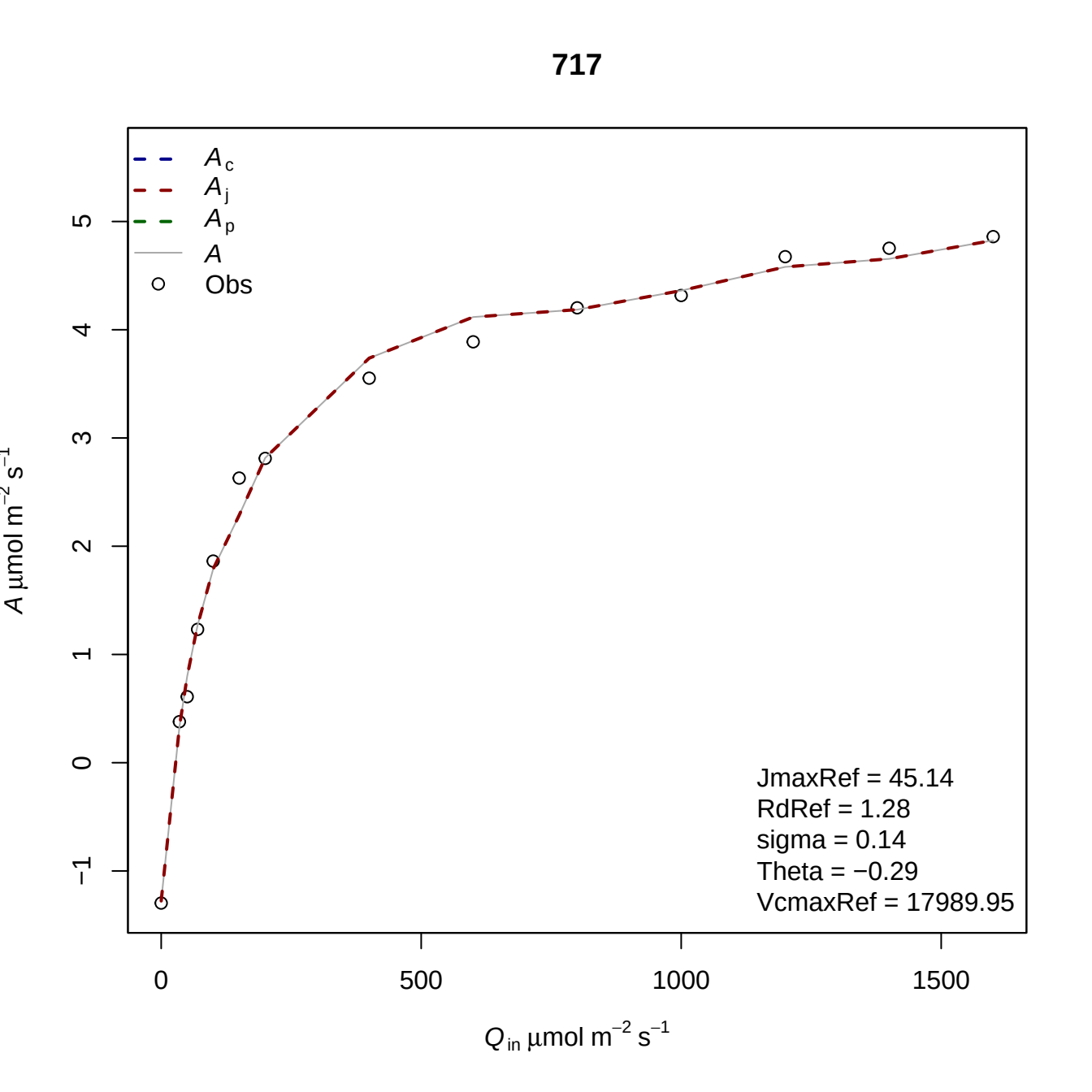

712
A_c
A_j
A_p
A
Obs
JmaxRef = 71.46
RdRef = 0.39
sigma = 0.11
Theta = 0.82
VcmaxRef = 534268.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

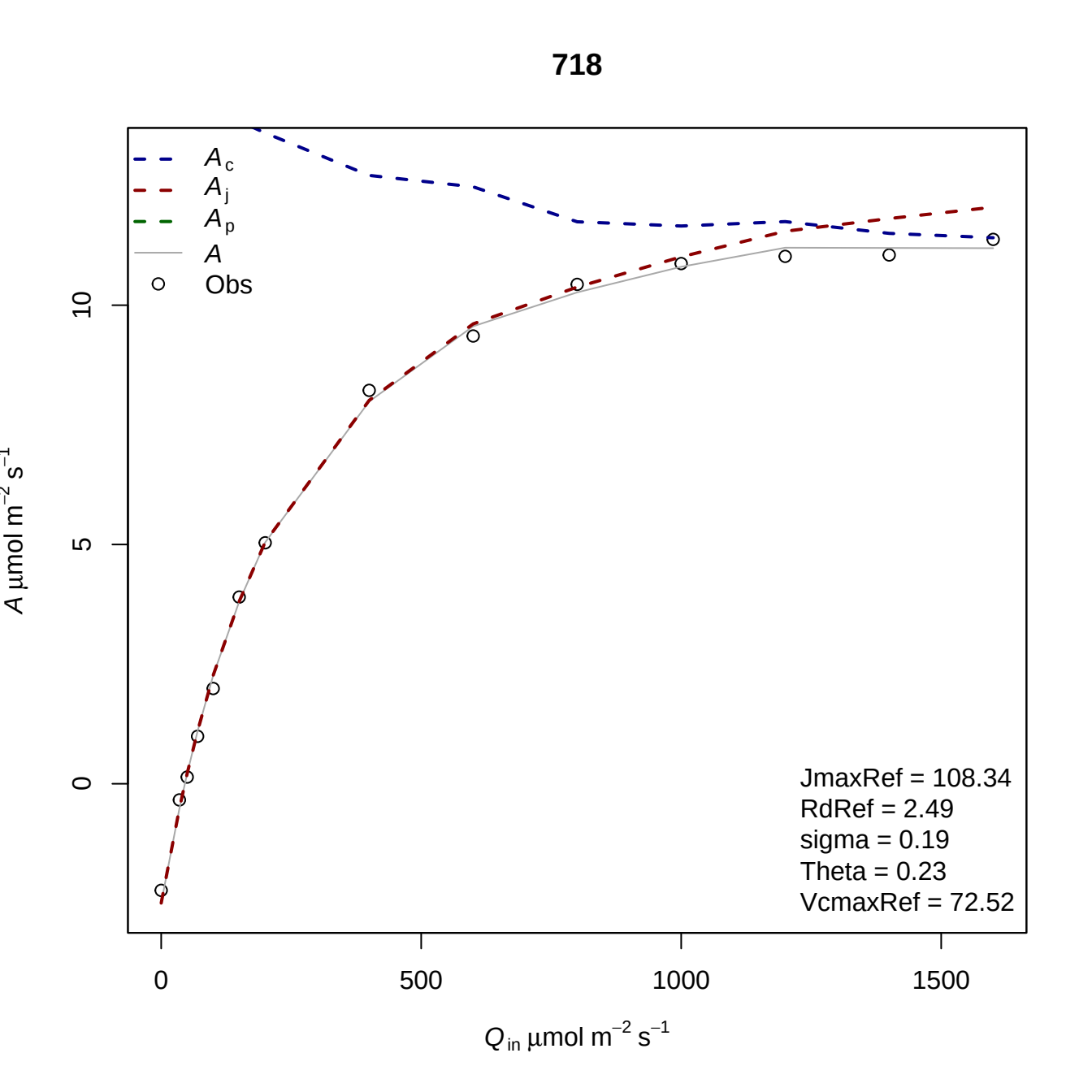

713
A_c
A_j
A_p
A
Obs
JmaxRef = 107.11
RdRef = 0.56
sigma = 0.11
Theta = 0.32
VcmaxRef = 260392.65
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

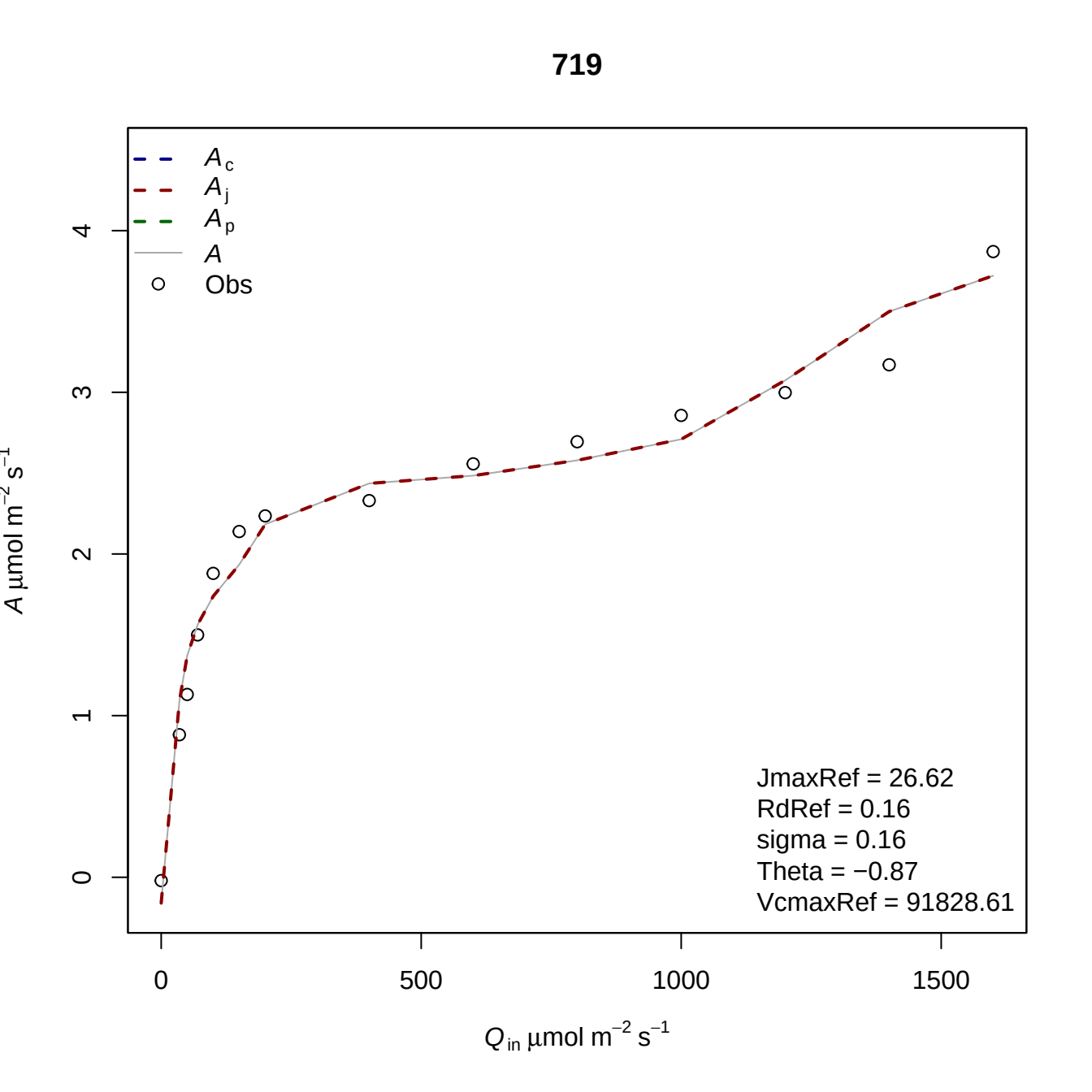

714
A_c
A_j
A_p
A
Obs
JmaxRef = 28.05
RdRef = 0.99
sigma = 0.21
Theta = 0.71
VcmaxRef = 51429.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

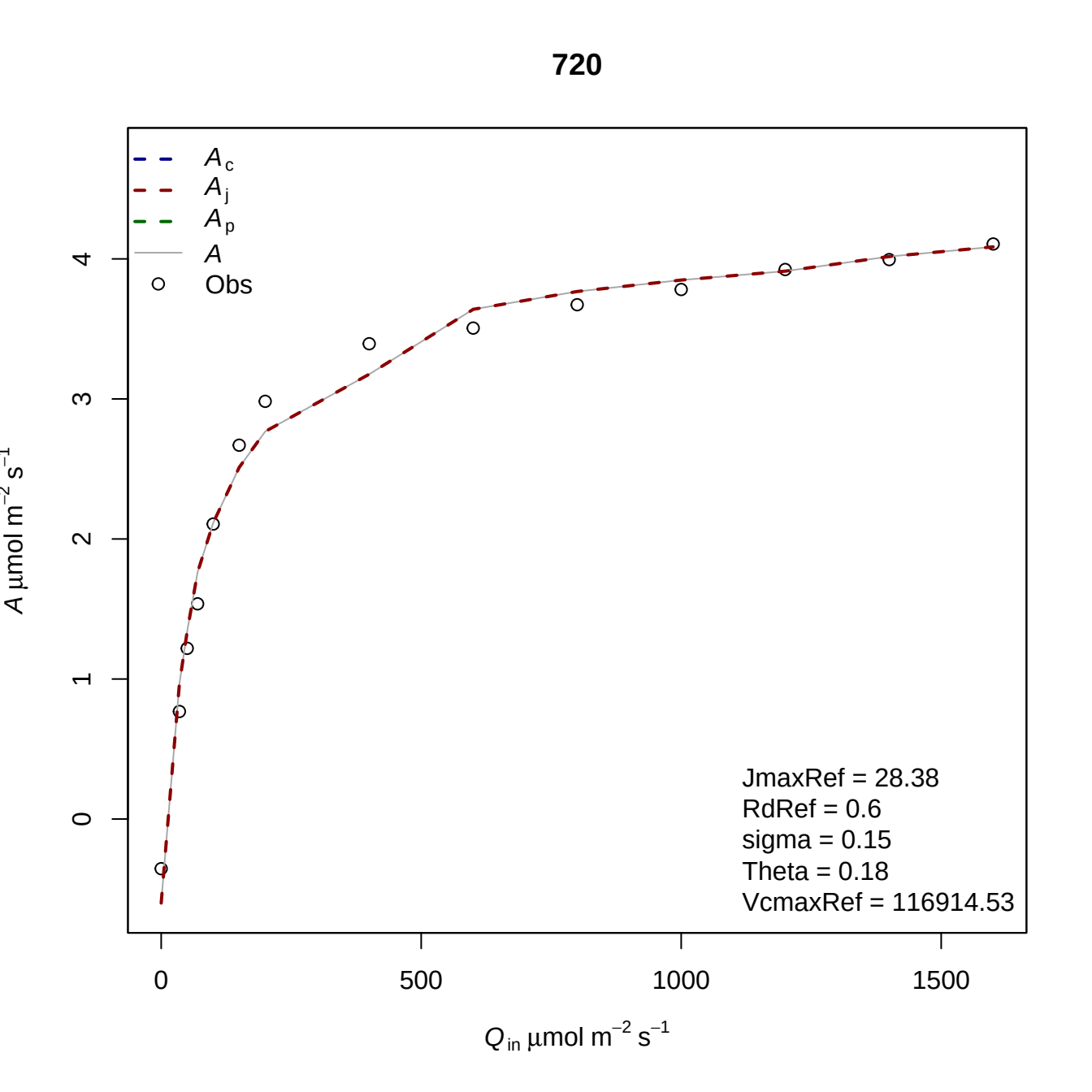

715
A_c
A_j
A_p
A
Obs
JmaxRef = 20.08
RdRef = 0.66
sigma = 0.08
Theta = 0.22
VcmaxRef = 35798.66
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

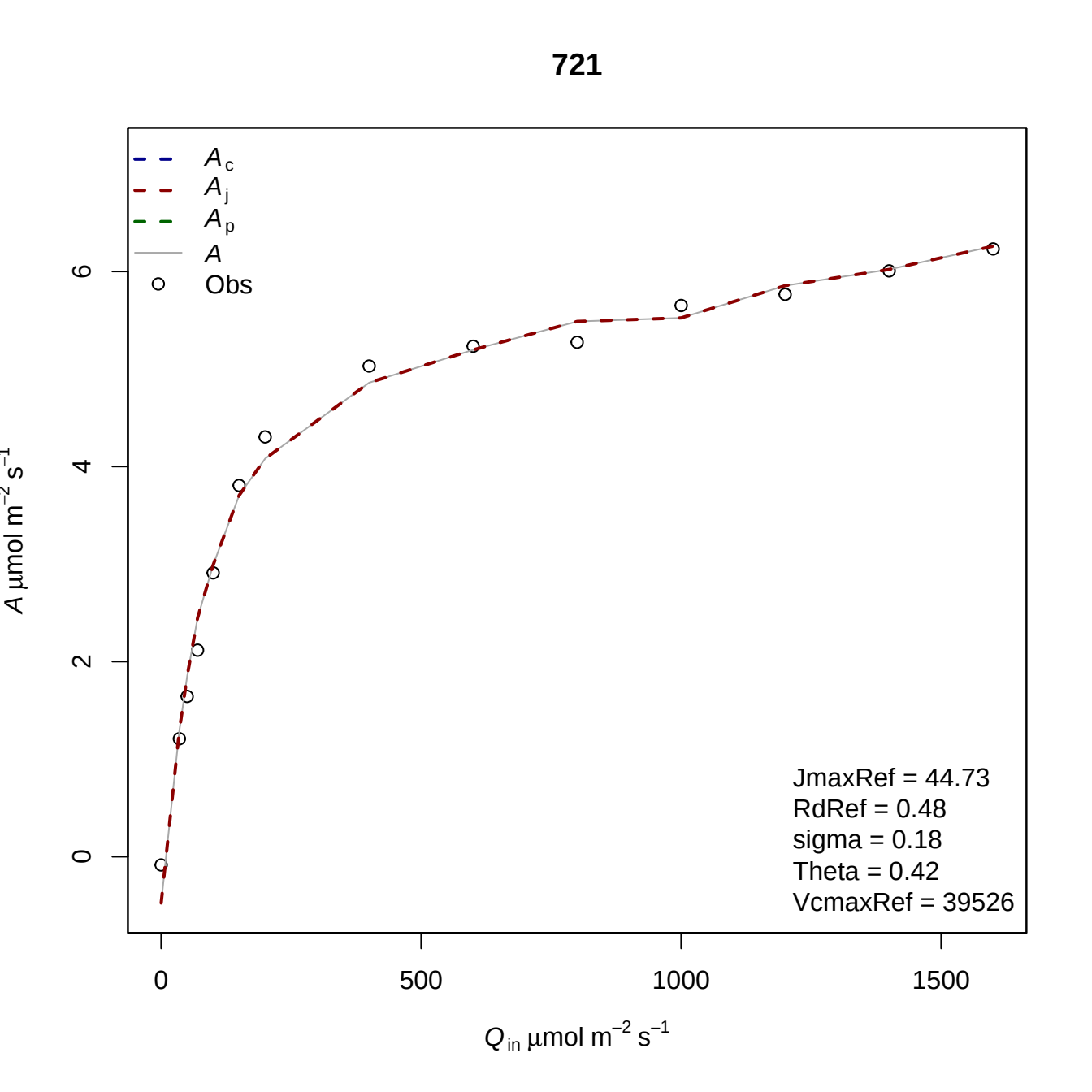

716
A_c
A_j
A_p
A
Obs
JmaxRef = 86.42
RdRef = 1.21
sigma = 0.13
Theta = −2.27
VcmaxRef = 18.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

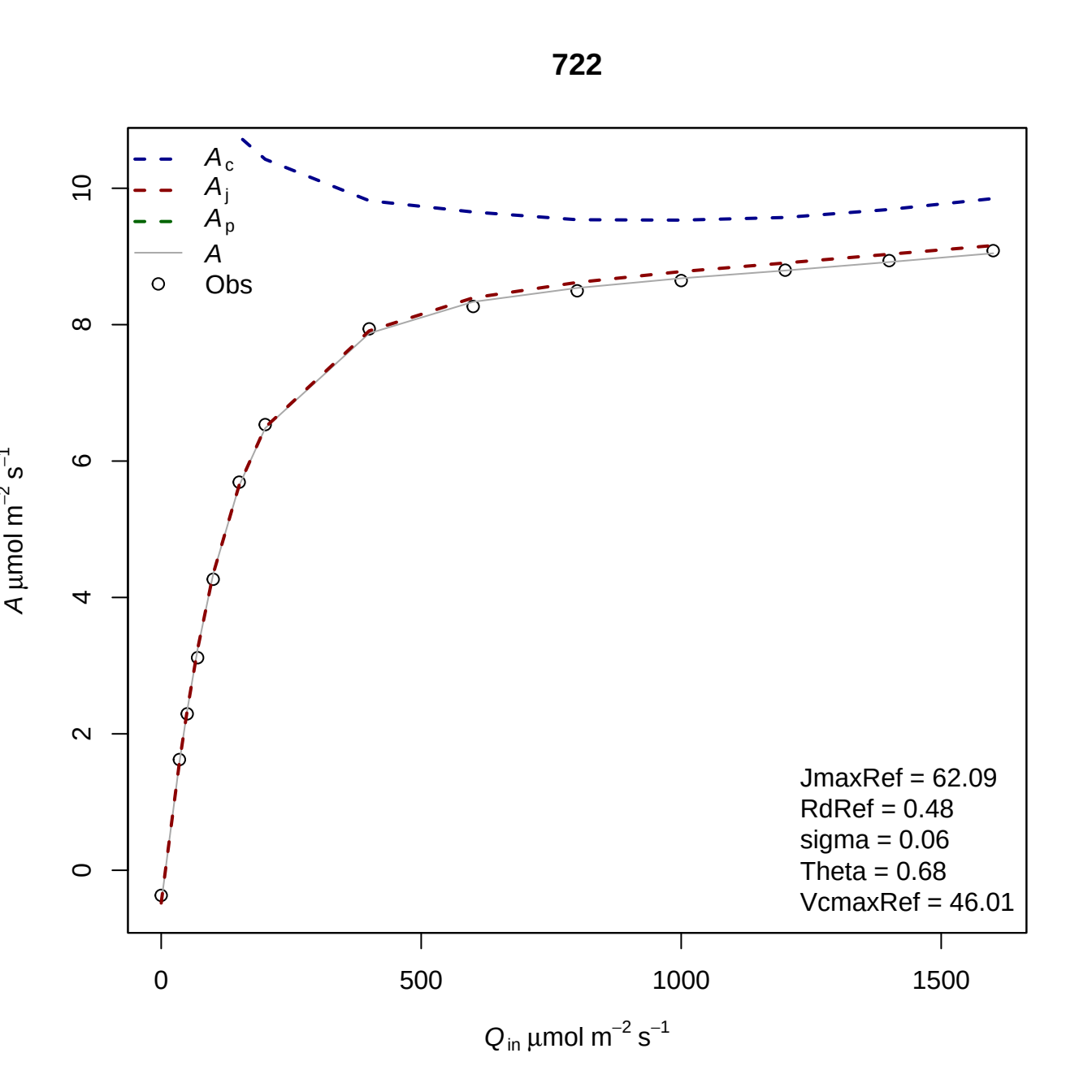

717
A_c
A_j
A_p
A
Obs
JmaxRef = 45.14
RdRef = 1.28
sigma = 0.14
Theta = −0.29
VcmaxRef = 17989.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

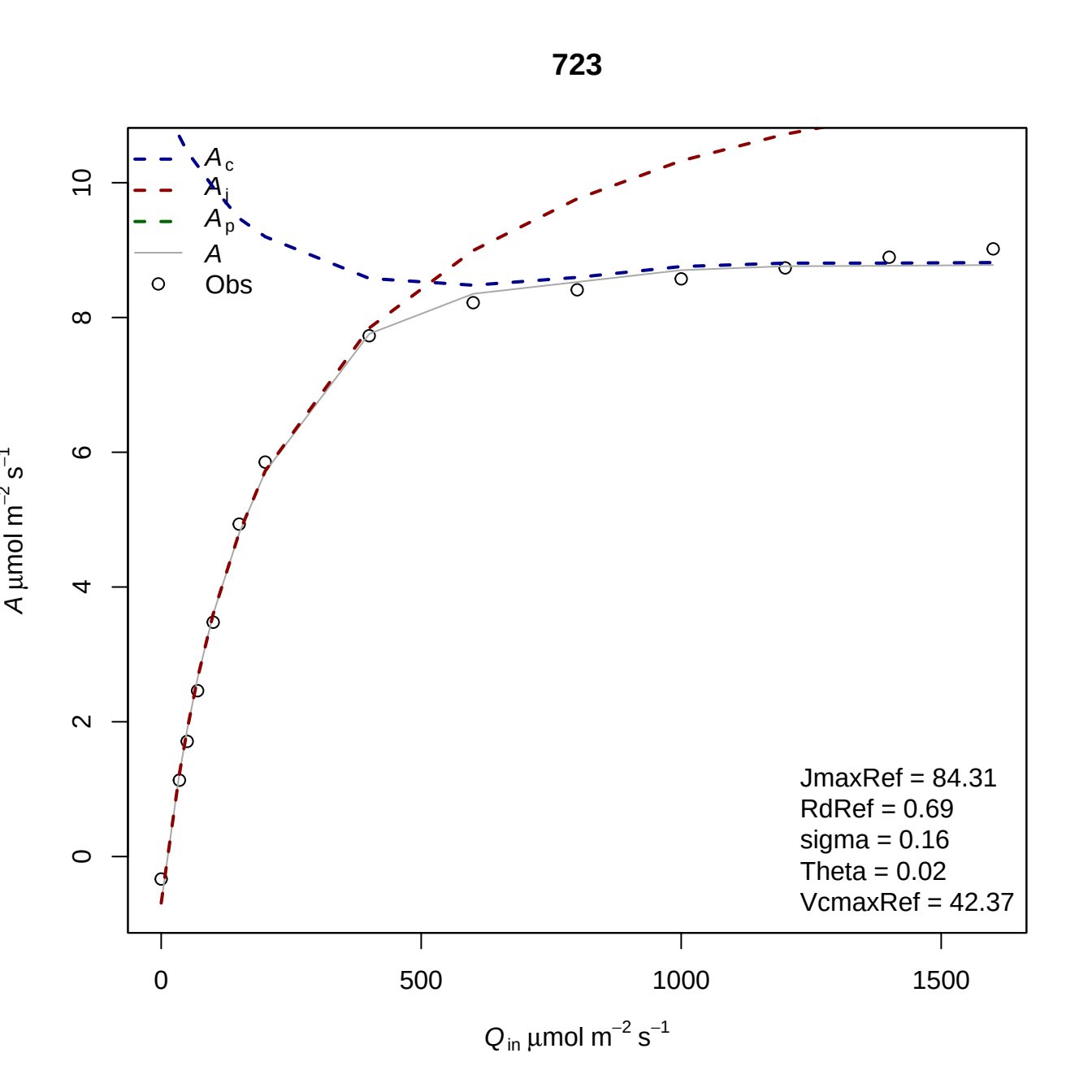

718
A_c
A_j
A_p
A
Obs
JmaxRef = 108.34
RdRef = 2.49
sigma = 0.19
Theta = 0.23
VcmaxRef = 72.52
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

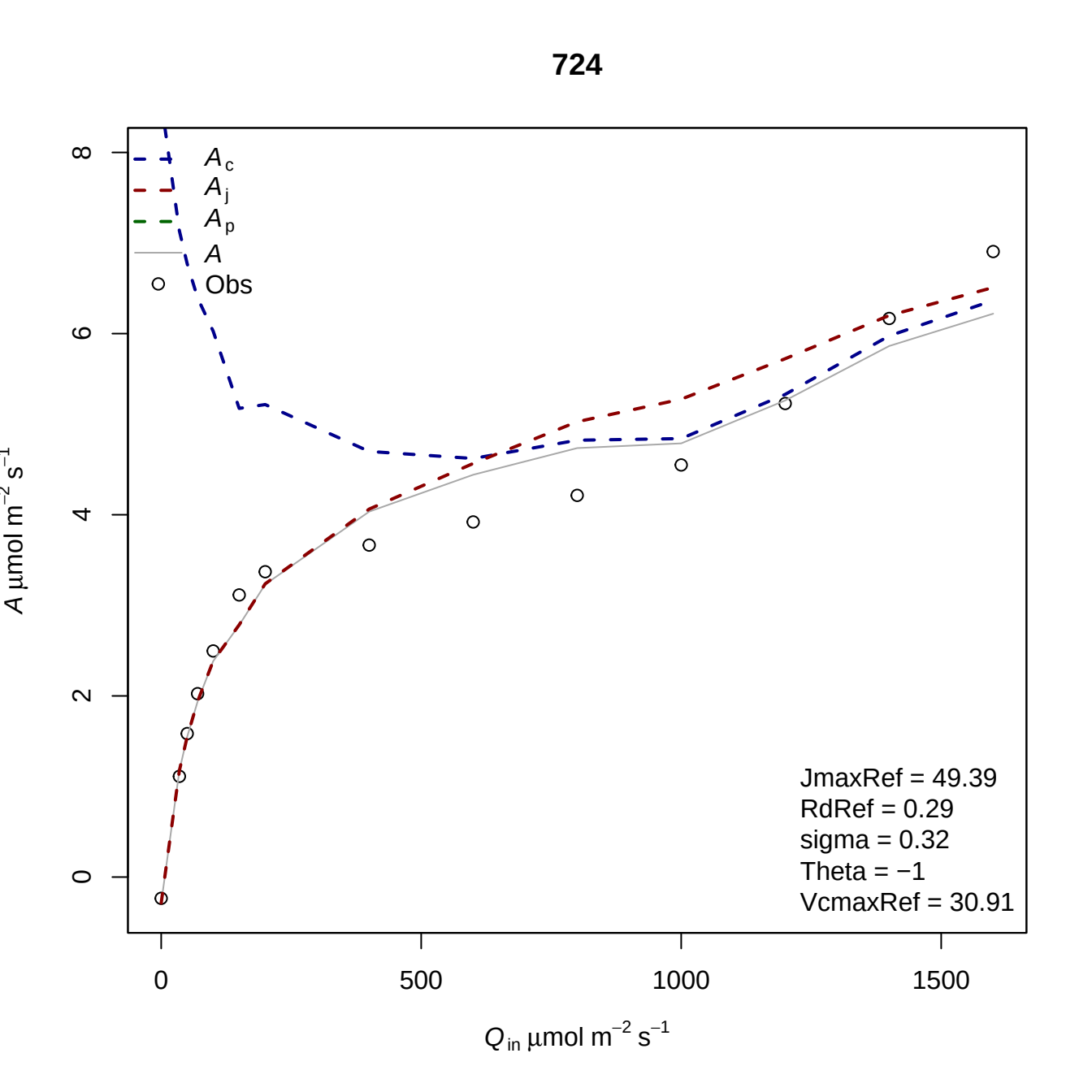

719
A_c
A_j
A_p
A
Obs
JmaxRef = 26.62
RdRef = 0.16
sigma = 0.16
Theta = −0.87
VcmaxRef = 91828.61
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

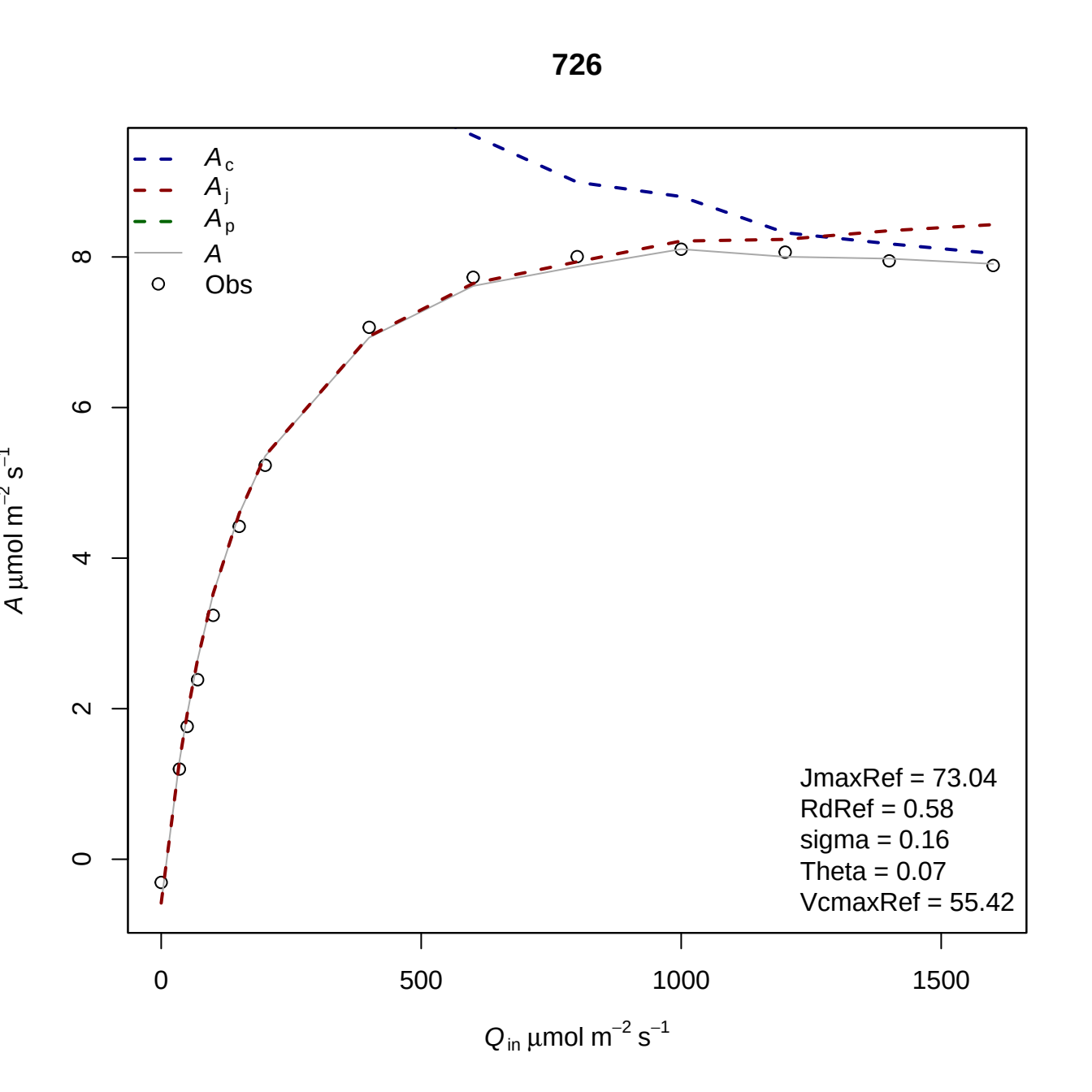

720
A_c
A_j
A_p
A
Obs
JmaxRef = 28.38
RdRef = 0.6
sigma = 0.15
Theta = 0.18
VcmaxRef = 116914.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

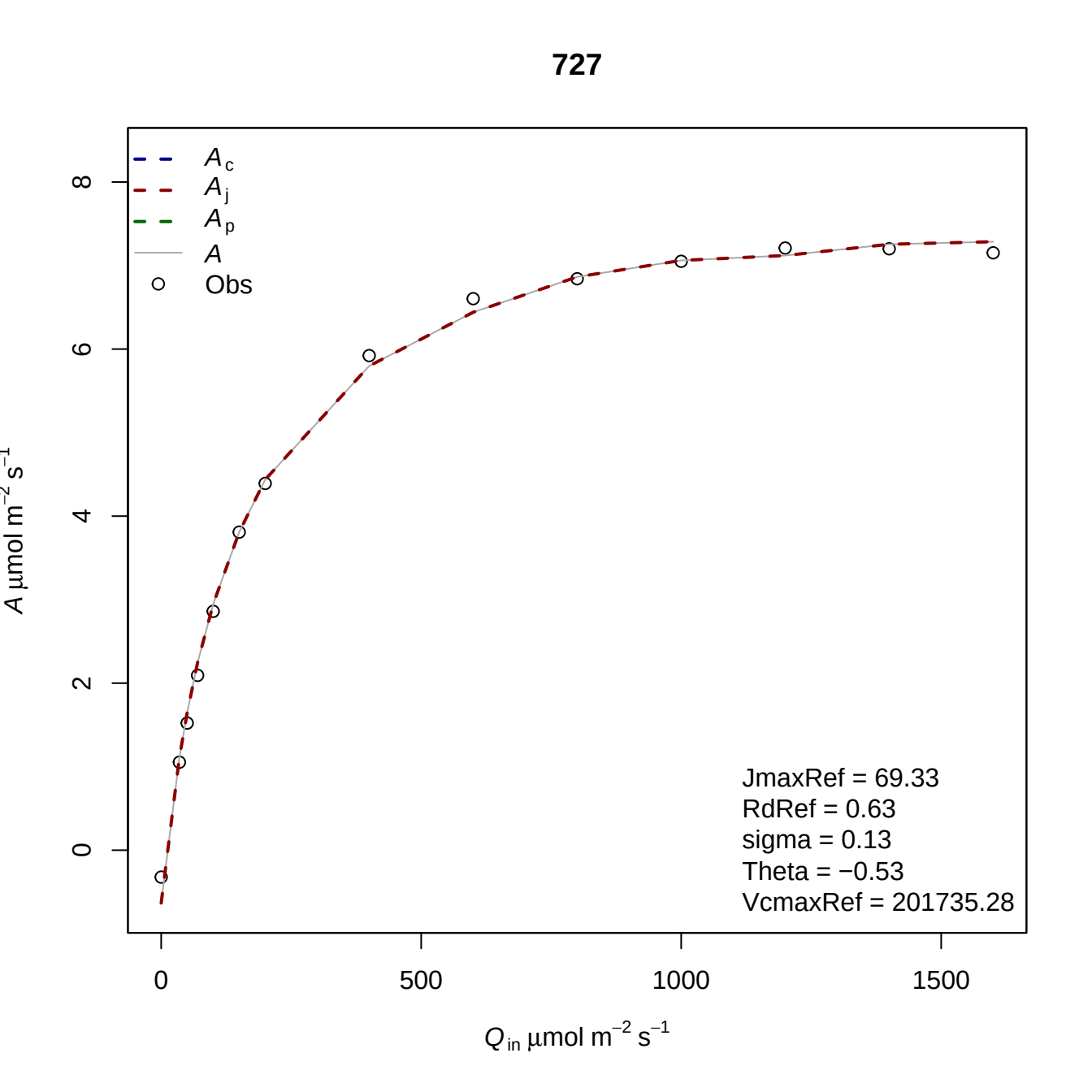

721
A_c
A_j
A_p
A
Obs
JmaxRef = 44.73
RdRef = 0.48
sigma = 0.18
Theta = 0.42
VcmaxRef = 39526
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

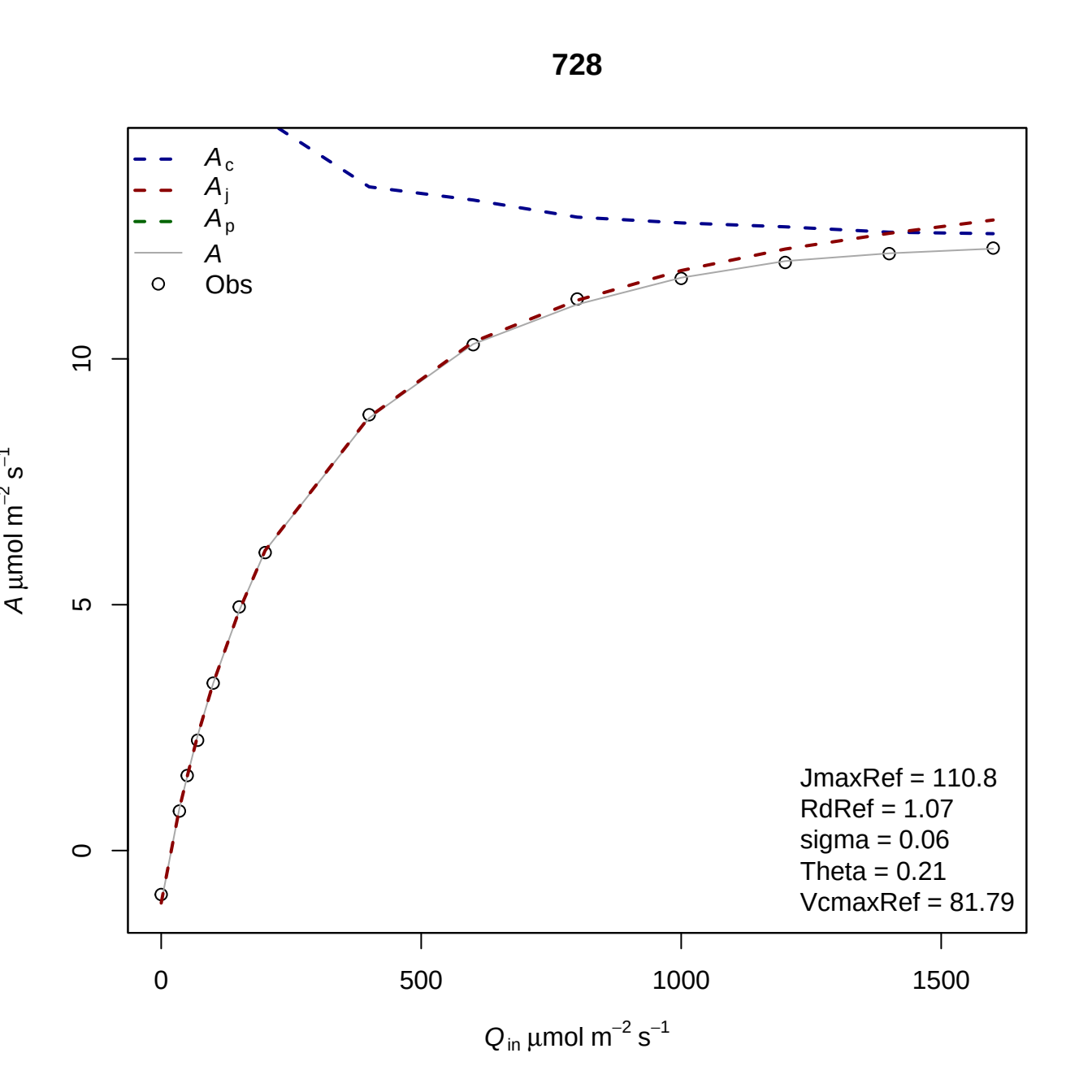

722
A_c
A_j
A_p
A
Obs
JmaxRef = 62.09
RdRef = 0.48
sigma = 0.06
Theta = 0.68
VcmaxRef = 46.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

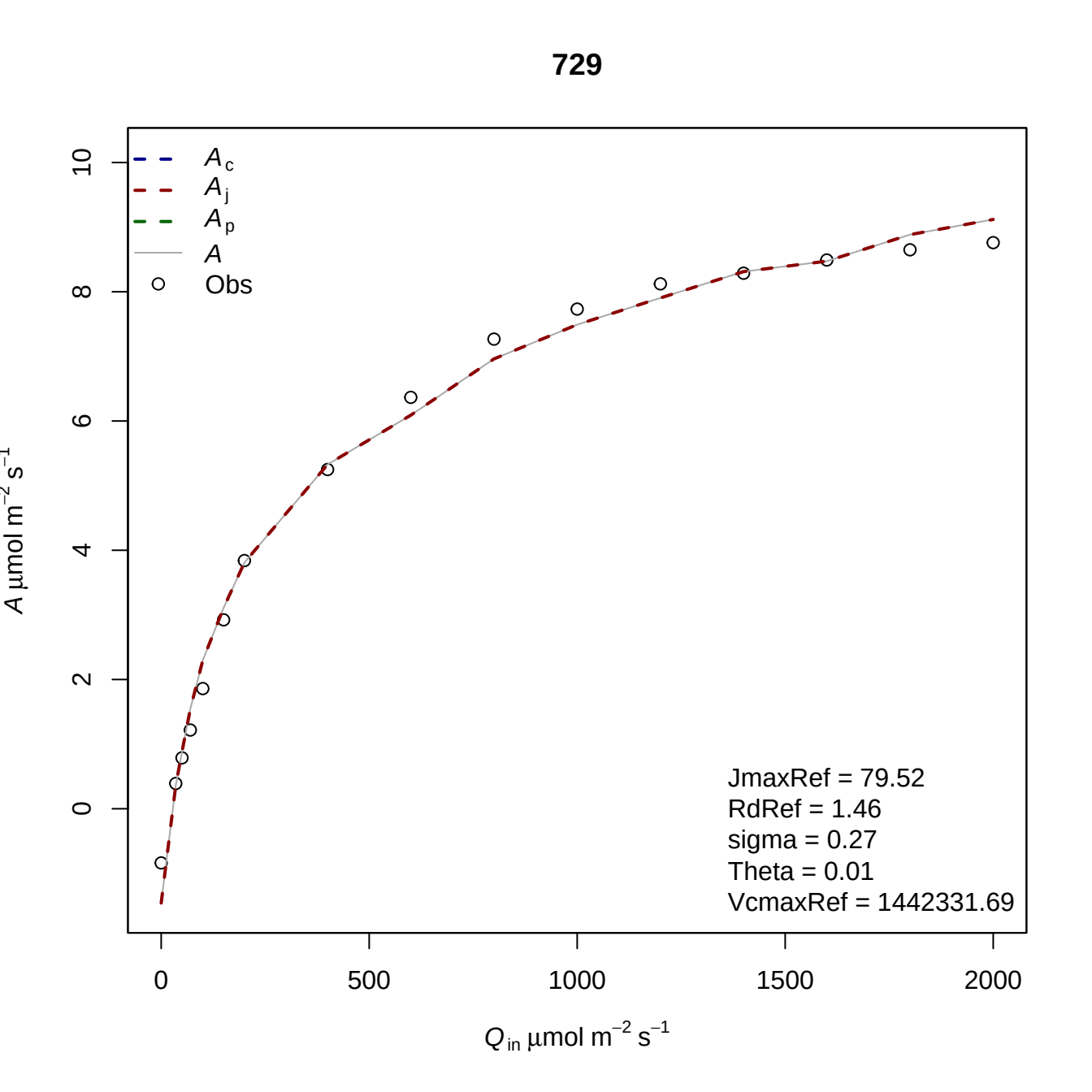

723
A_c
A_j
A_p
A
Obs
JmaxRef = 84.31
RdRef = 0.69
sigma = 0.16
Theta = 0.02
VcmaxRef = 42.37
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

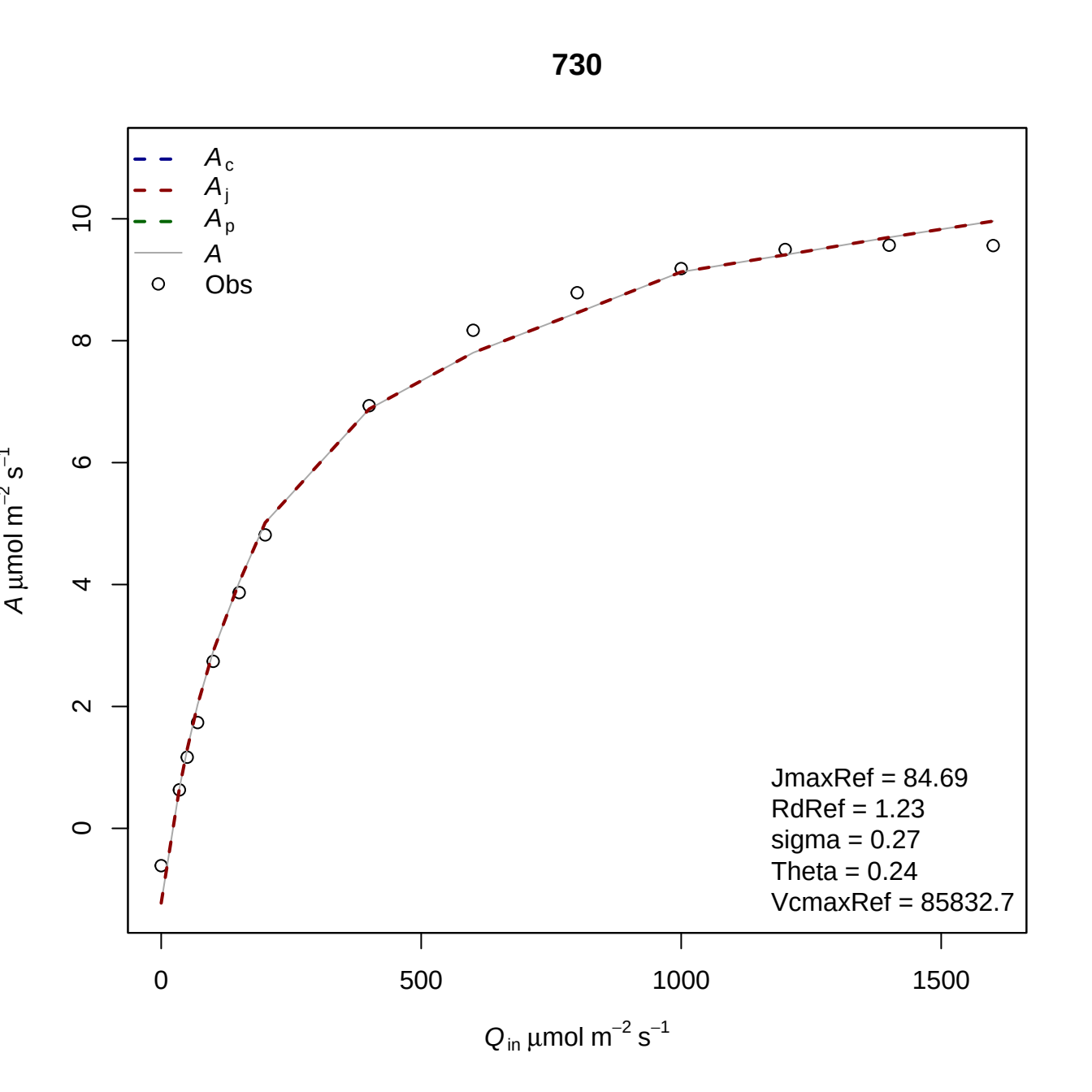

724
A_c
A_j
A_p
A
Obs
JmaxRef = 49.39
RdRef = 0.29
sigma = 0.32
Theta = −1
VcmaxRef = 30.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

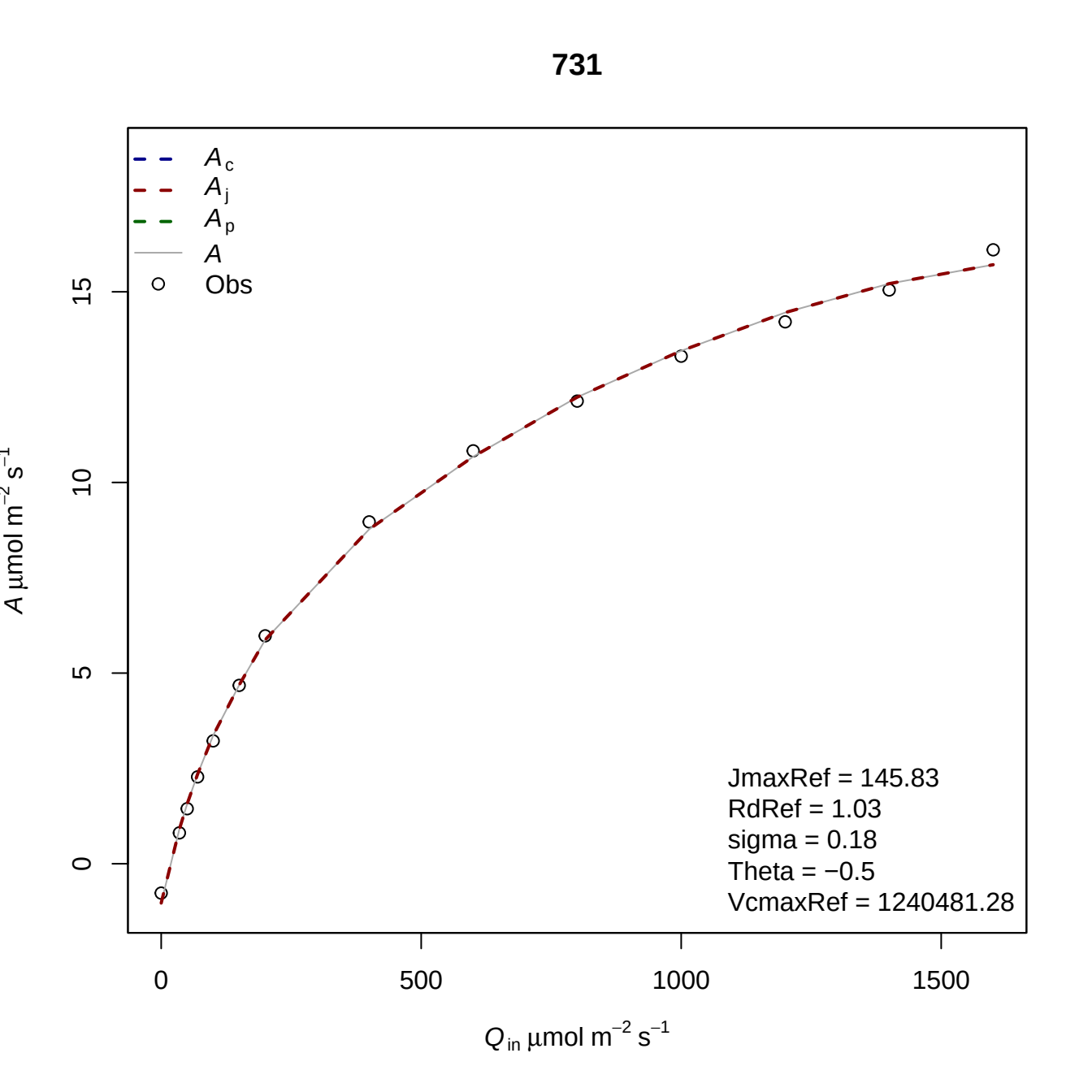

726
A_c
A_j
A_p
A
Obs
JmaxRef = 73.04
RdRef = 0.58
sigma = 0.16
Theta = 0.07
VcmaxRef = 55.42
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

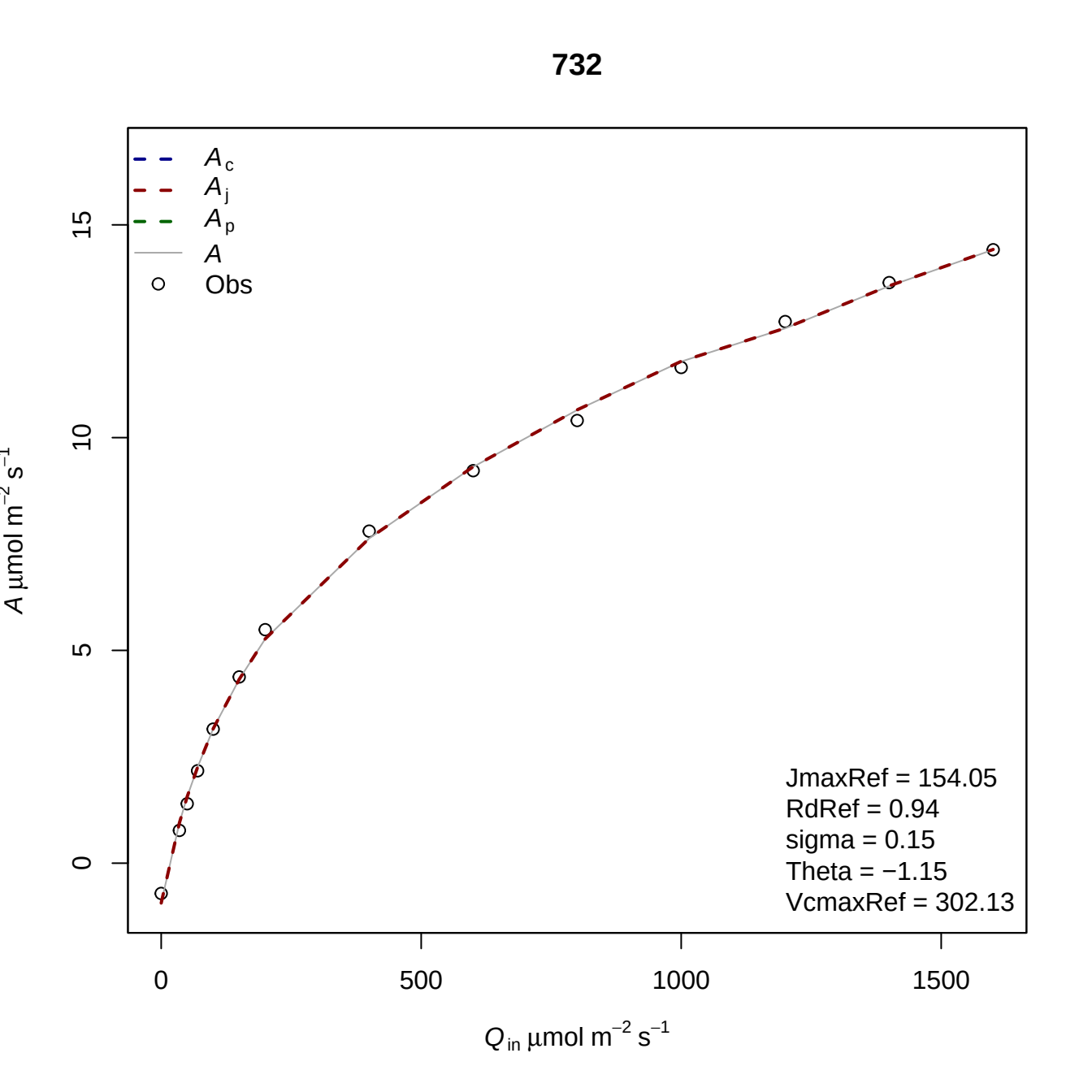

727
A_c
A_j
A_p
A
Obs
JmaxRef = 69.33
RdRef = 0.63
sigma = 0.13
Theta = −0.53
VcmaxRef = 201735.28
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

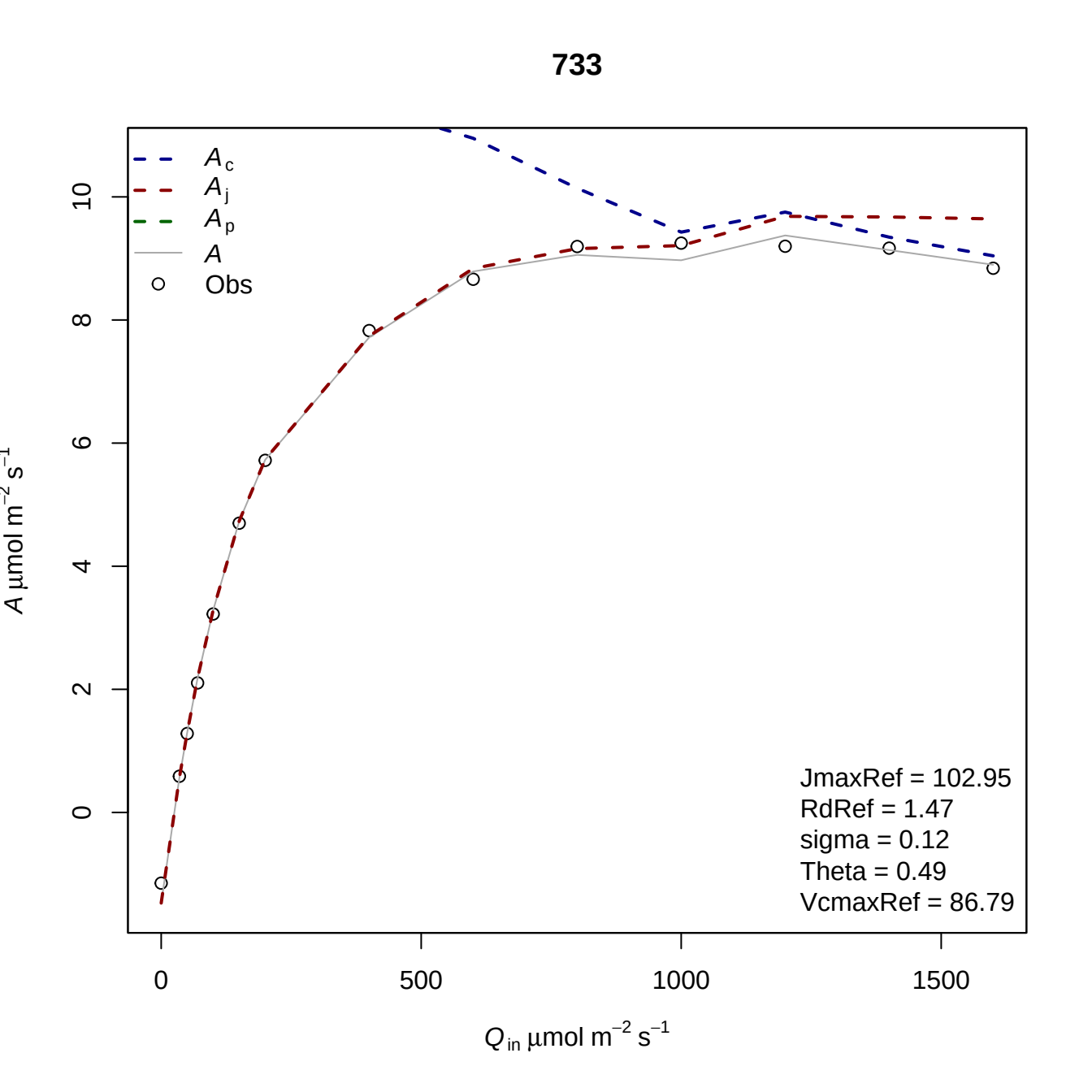

728
A_c
A_j
A_p
A
Obs
JmaxRef = 110.8
RdRef = 1.07
sigma = 0.06
Theta = 0.21
VcmaxRef = 81.79
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

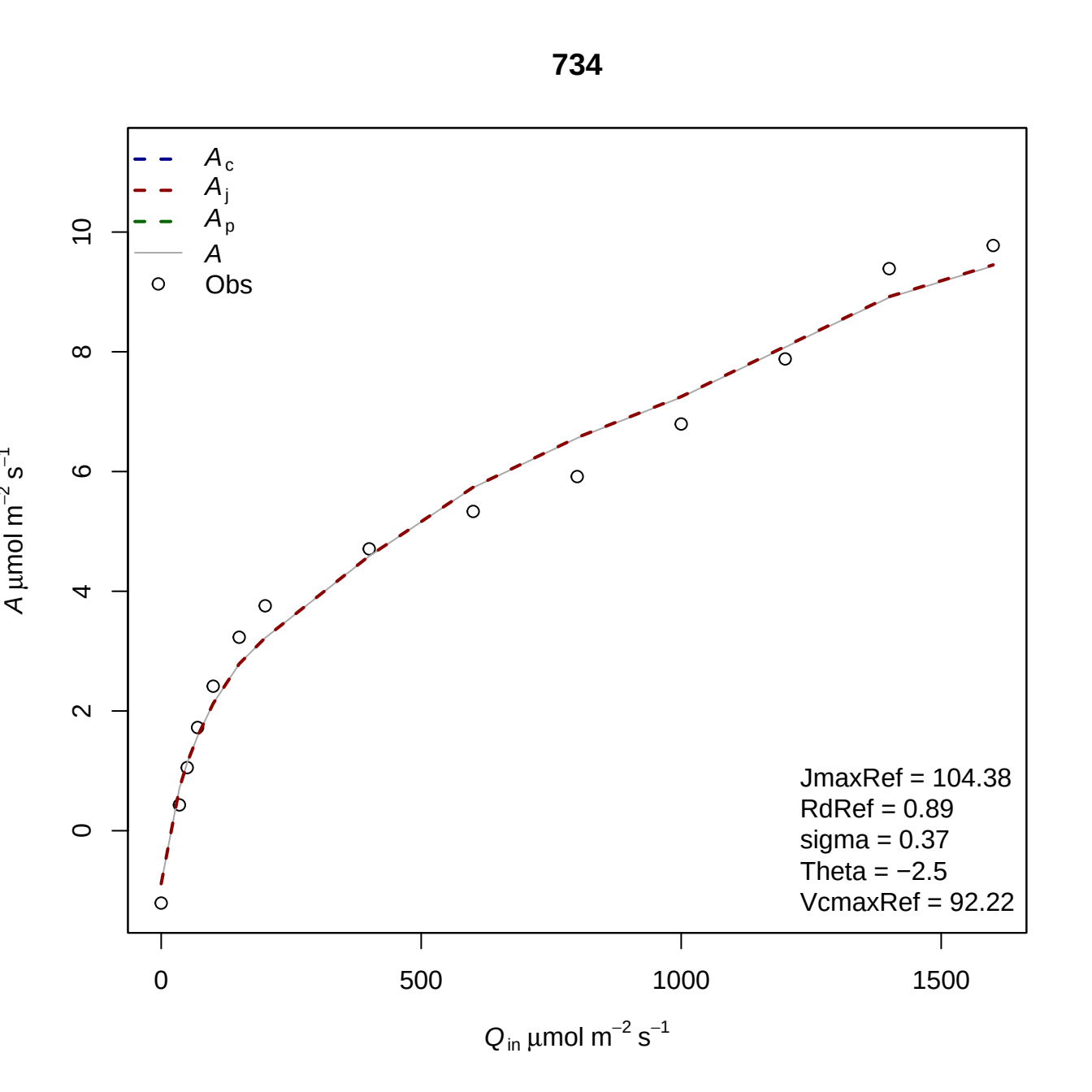

729
A_c
A_j
A_p
A
Obs
JmaxRef = 79.52
RdRef = 1.46
sigma = 0.27
Theta = 0.01
VcmaxRef = 1442331.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

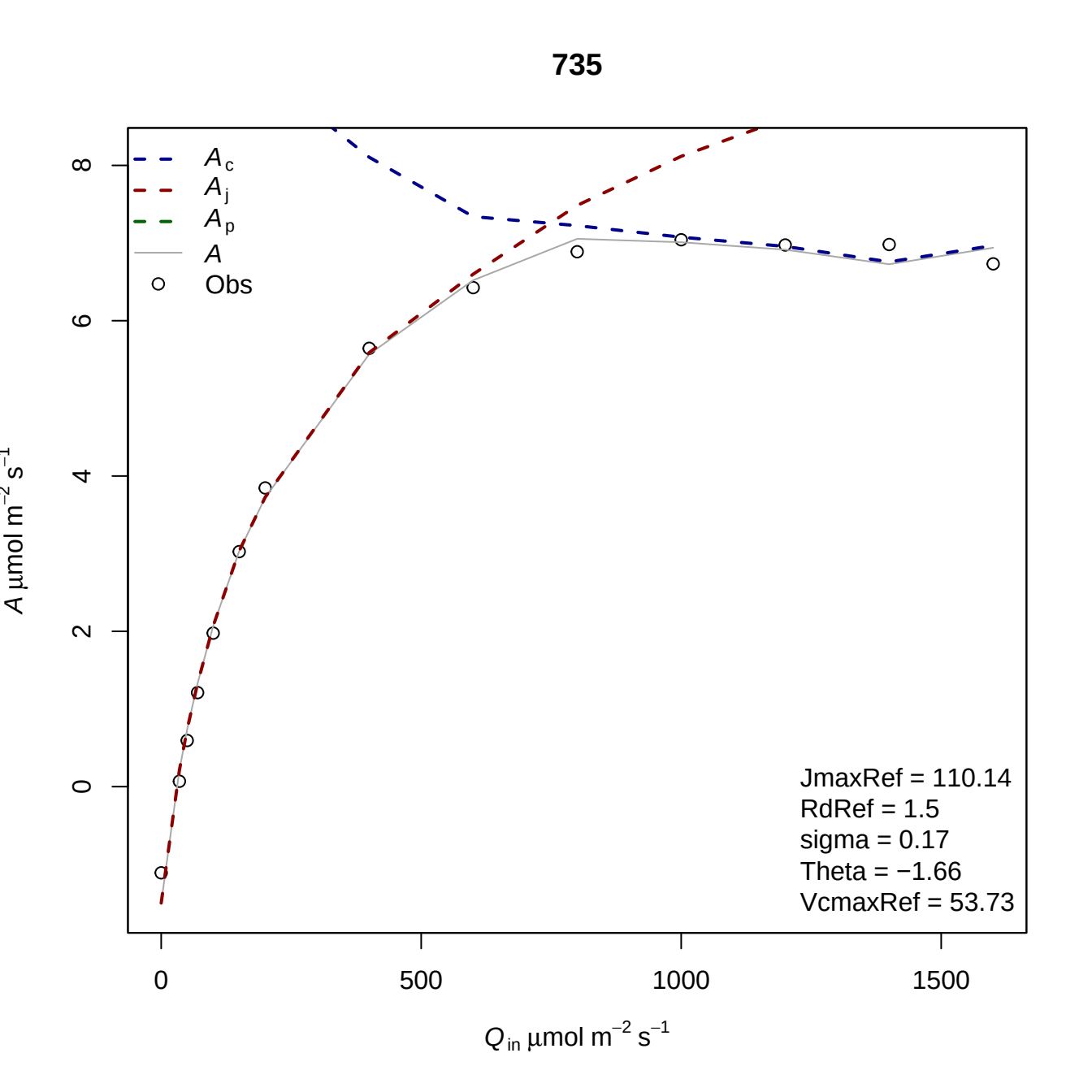

730
A_c
A_j
A_p
A
Obs
JmaxRef = 84.69
RdRef = 1.23
sigma = 0.27
Theta = 0.24
VcmaxRef = 85832.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

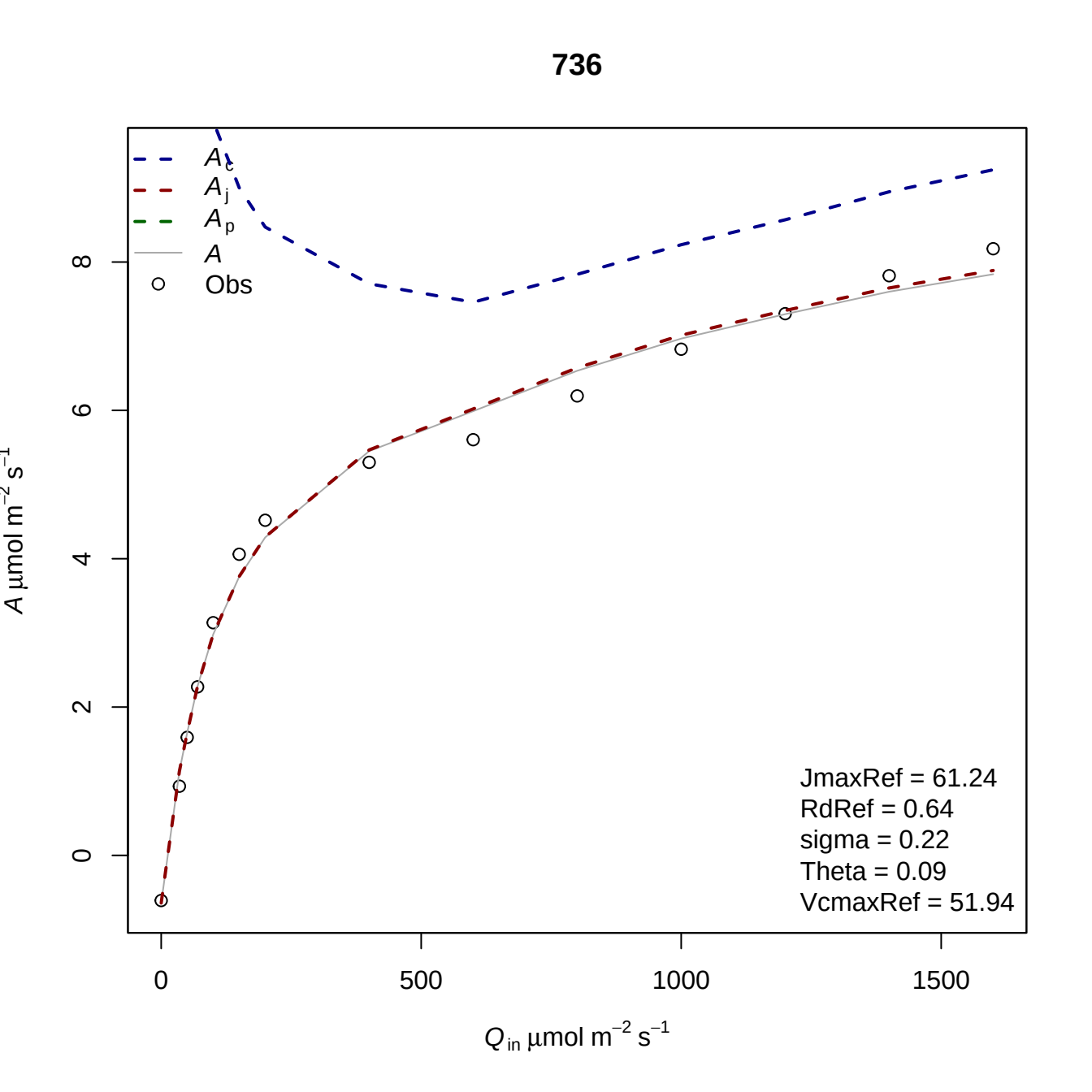

731
A_c
A_j
A_p
A
Obs
JmaxRef = 145.83
RdRef = 1.03
sigma = 0.18
Theta = −0.5
VcmaxRef = 1240481.28
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

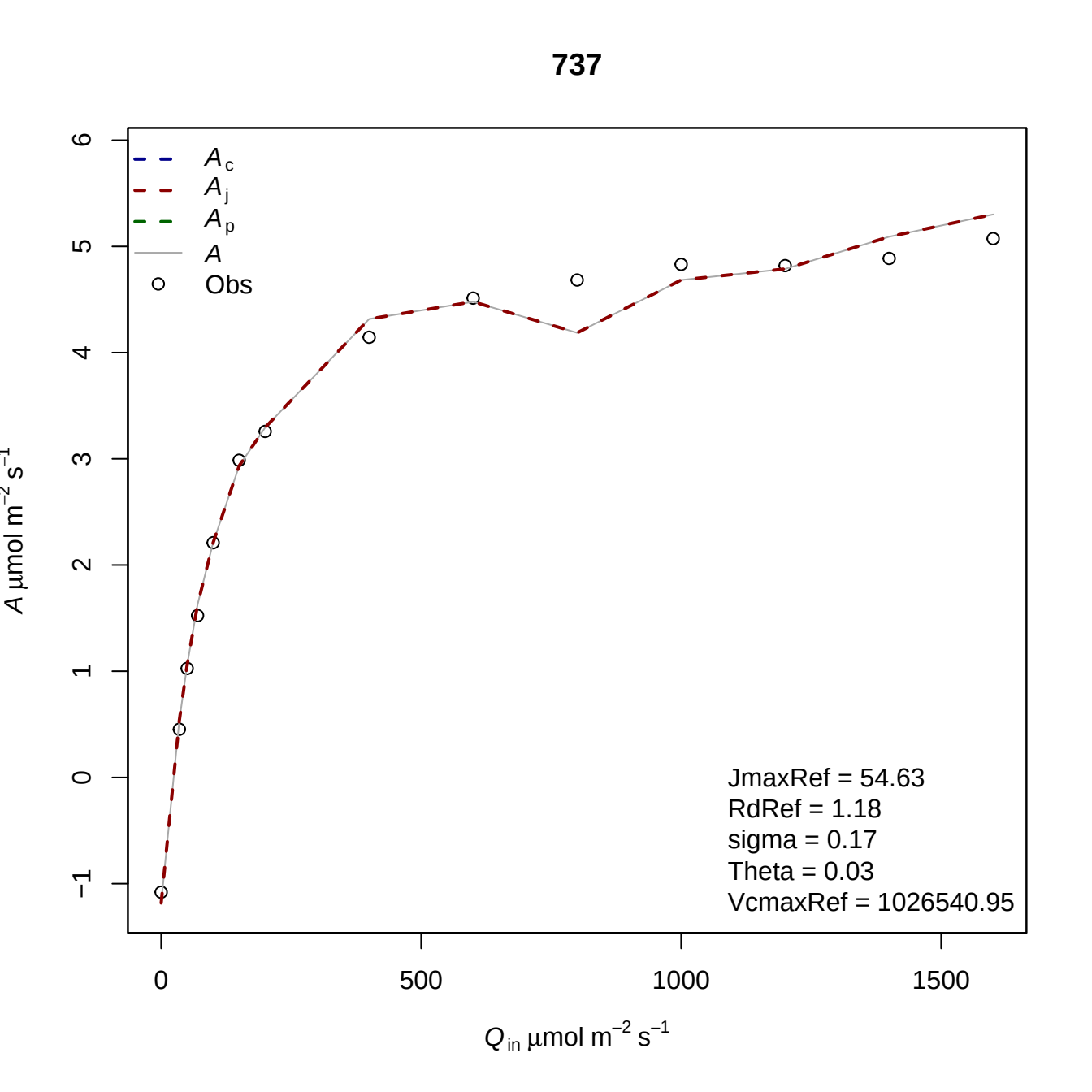

732
A_c
A_j
A_p
A
Obs
JmaxRef = 154.05
RdRef = 0.94
sigma = 0.15
Theta = −1.15
VcmaxRef = 302.13
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

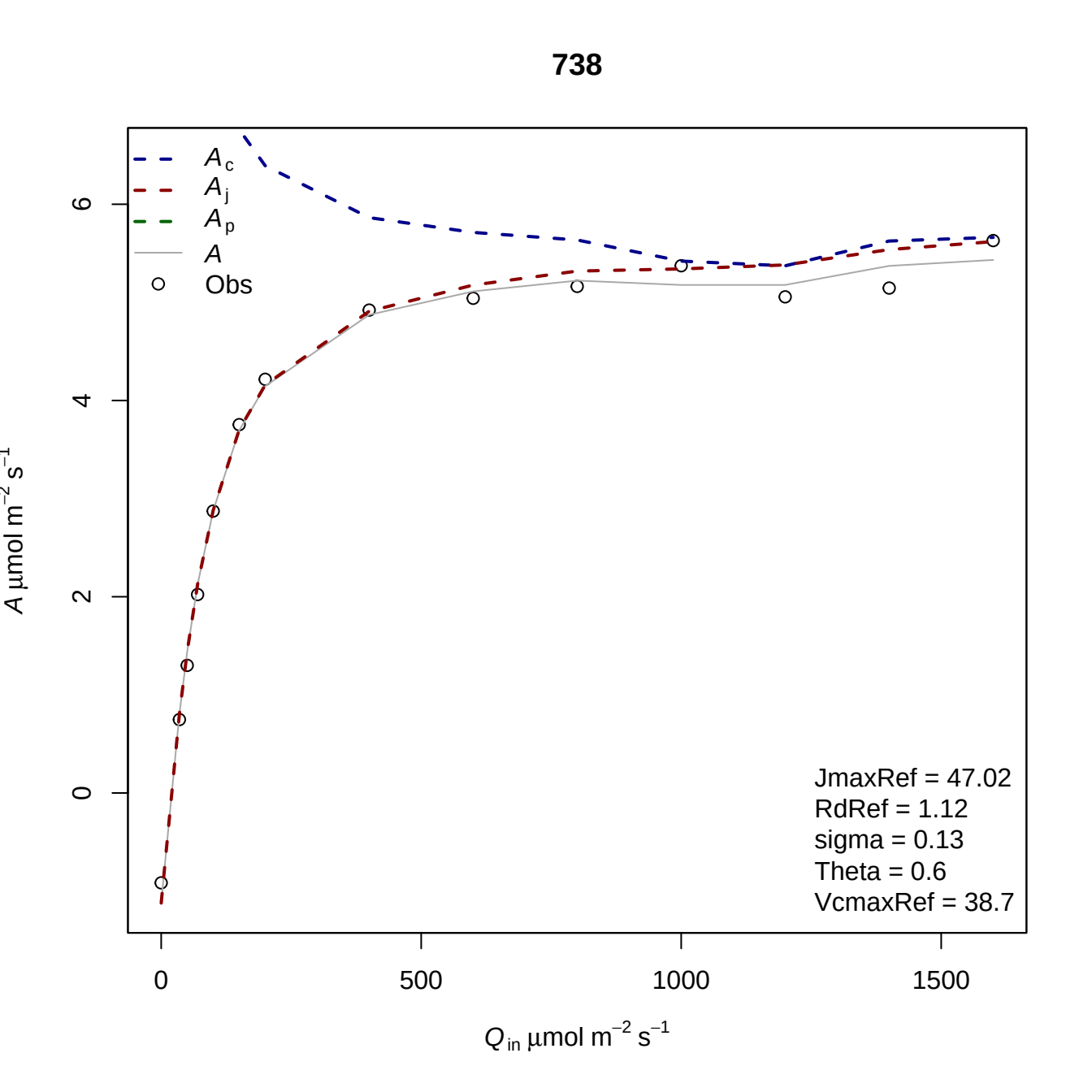

733
A_c
A_j
A_p
A
Obs
JmaxRef = 102.95
RdRef = 1.47
sigma = 0.12
Theta = 0.49
VcmaxRef = 86.79
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

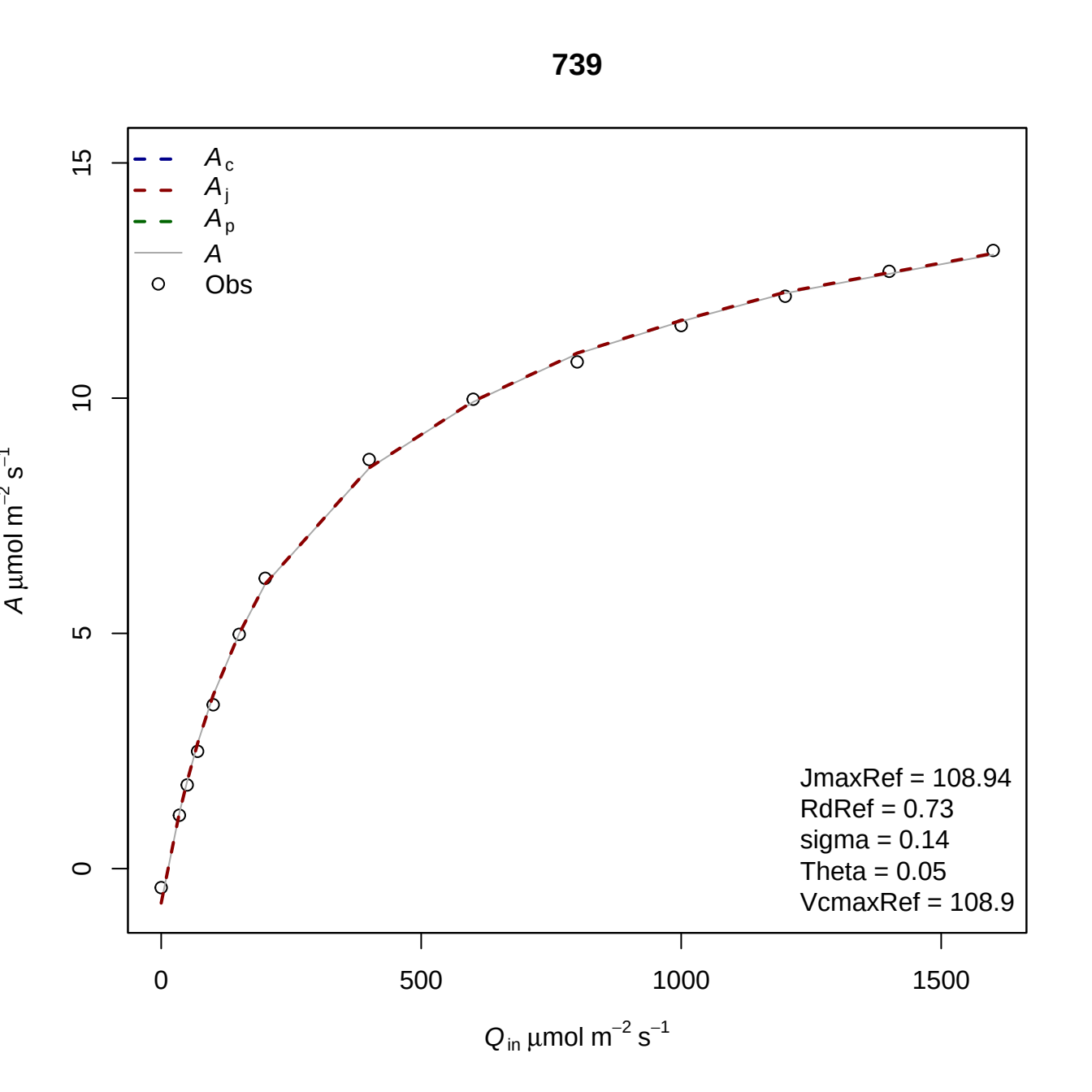

734
A_c
A_j
A_p
A
Obs
JmaxRef = 104.38
RdRef = 0.89
sigma = 0.37
Theta = −2.5
VcmaxRef = 92.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

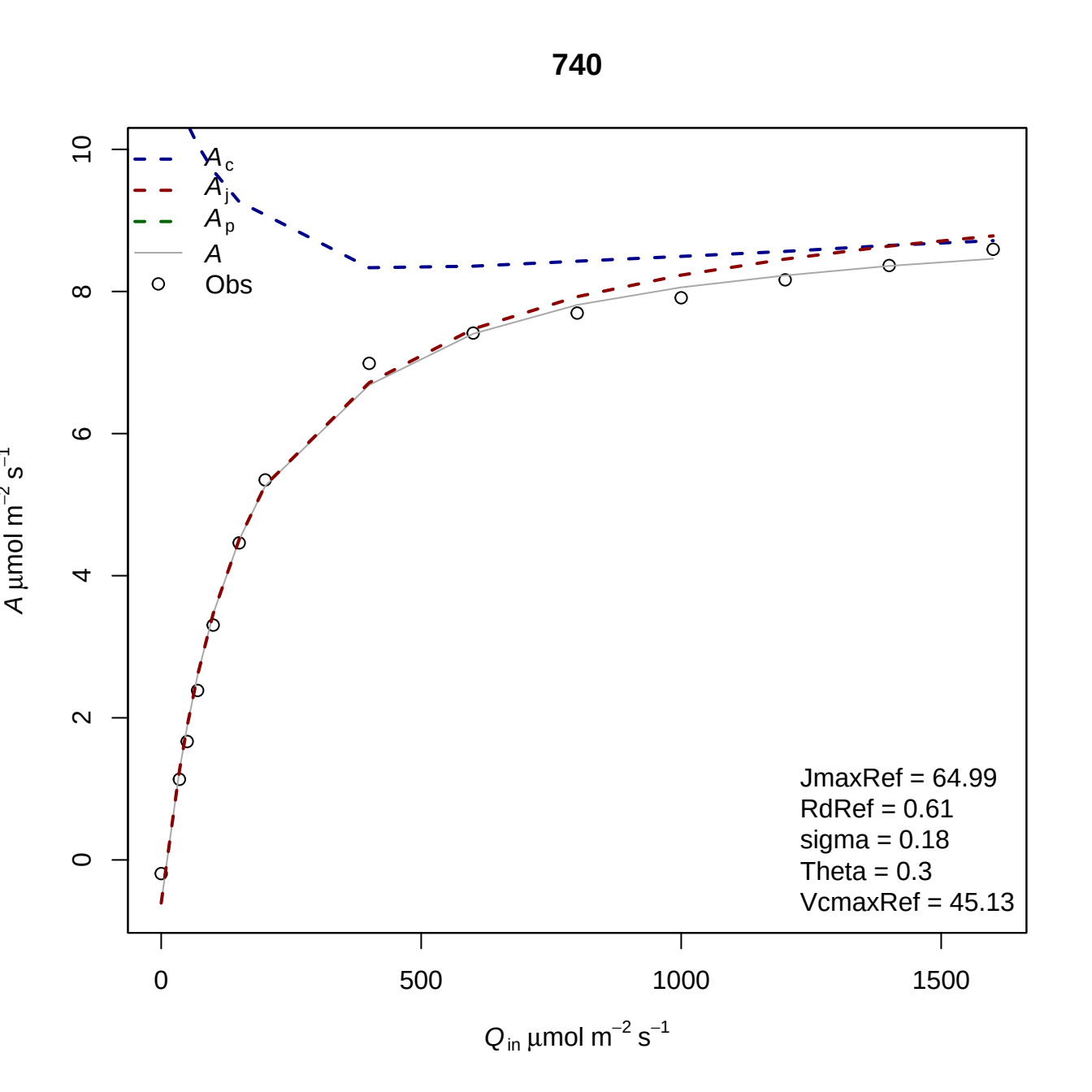

735
A_c
A_j
A_p
A
Obs
JmaxRef = 110.14
RdRef = 1.5
sigma = 0.17
Theta = −1.66
VcmaxRef = 53.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

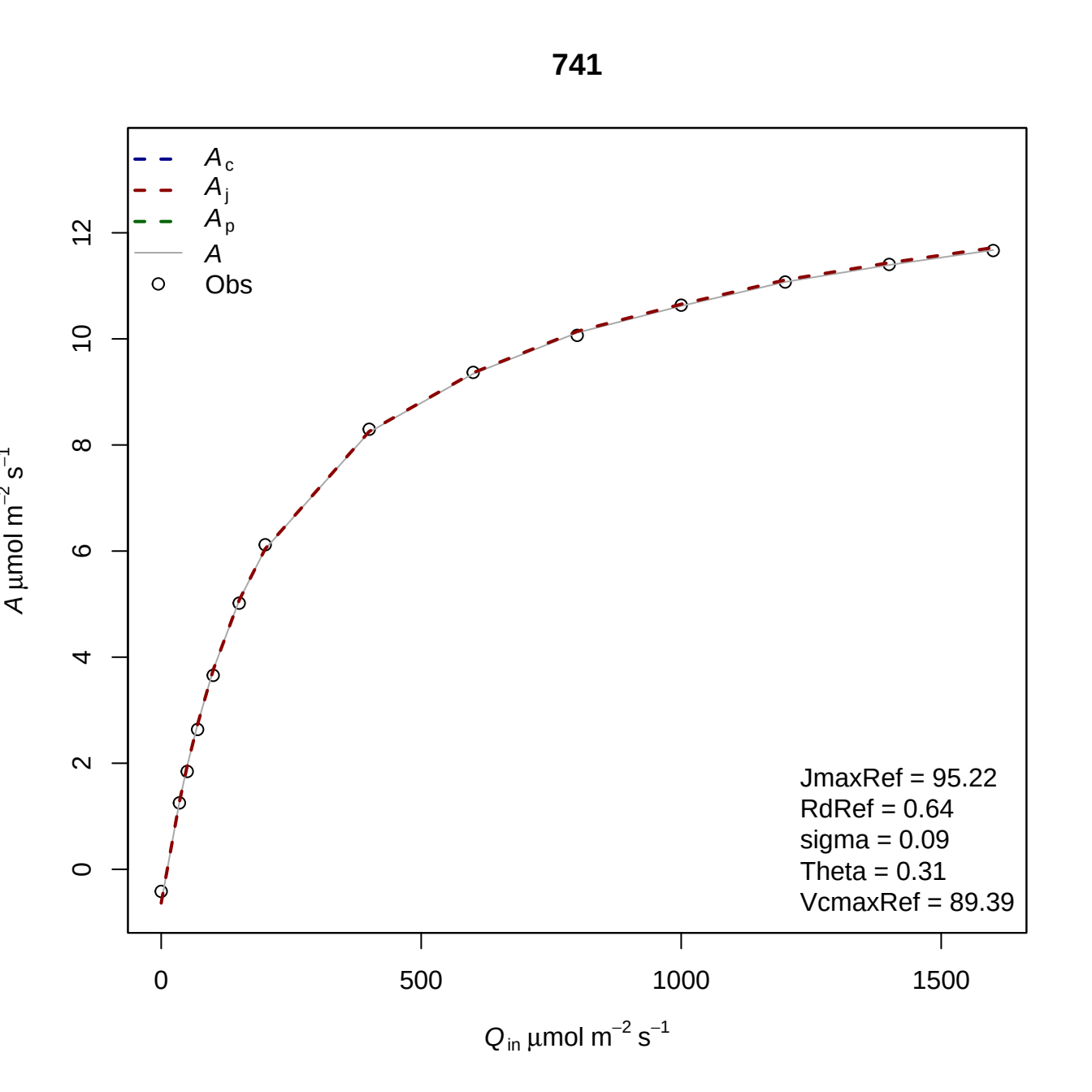

736
A_c
A_j
A_p
A
Obs
JmaxRef = 61.24
RdRef = 0.64
sigma = 0.22
Theta = 0.09
VcmaxRef = 51.94
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

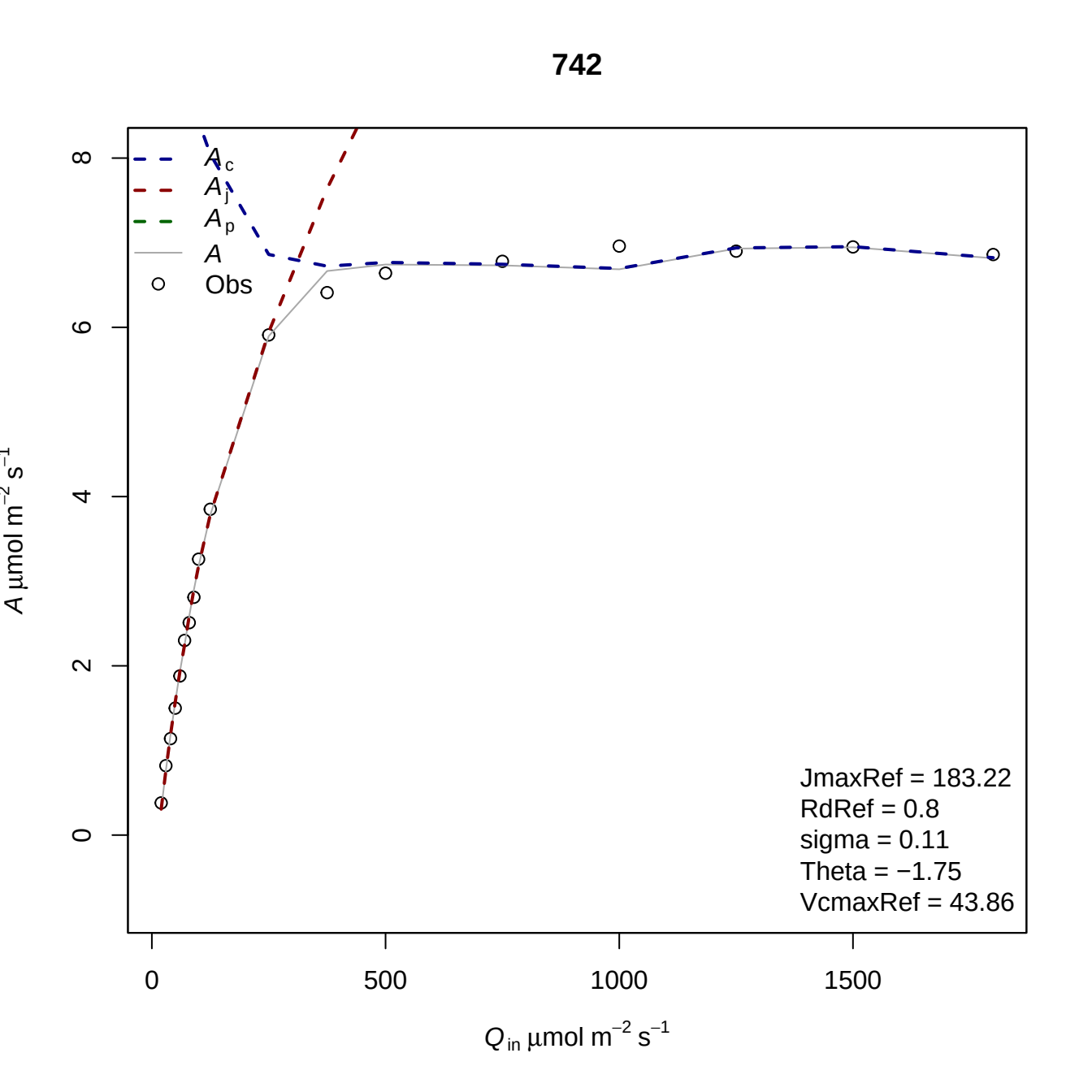

737
A_c
A_j
A_p
A
Obs
JmaxRef = 54.63
RdRef = 1.18
sigma = 0.17
Theta = 0.03
VcmaxRef = 1026540.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

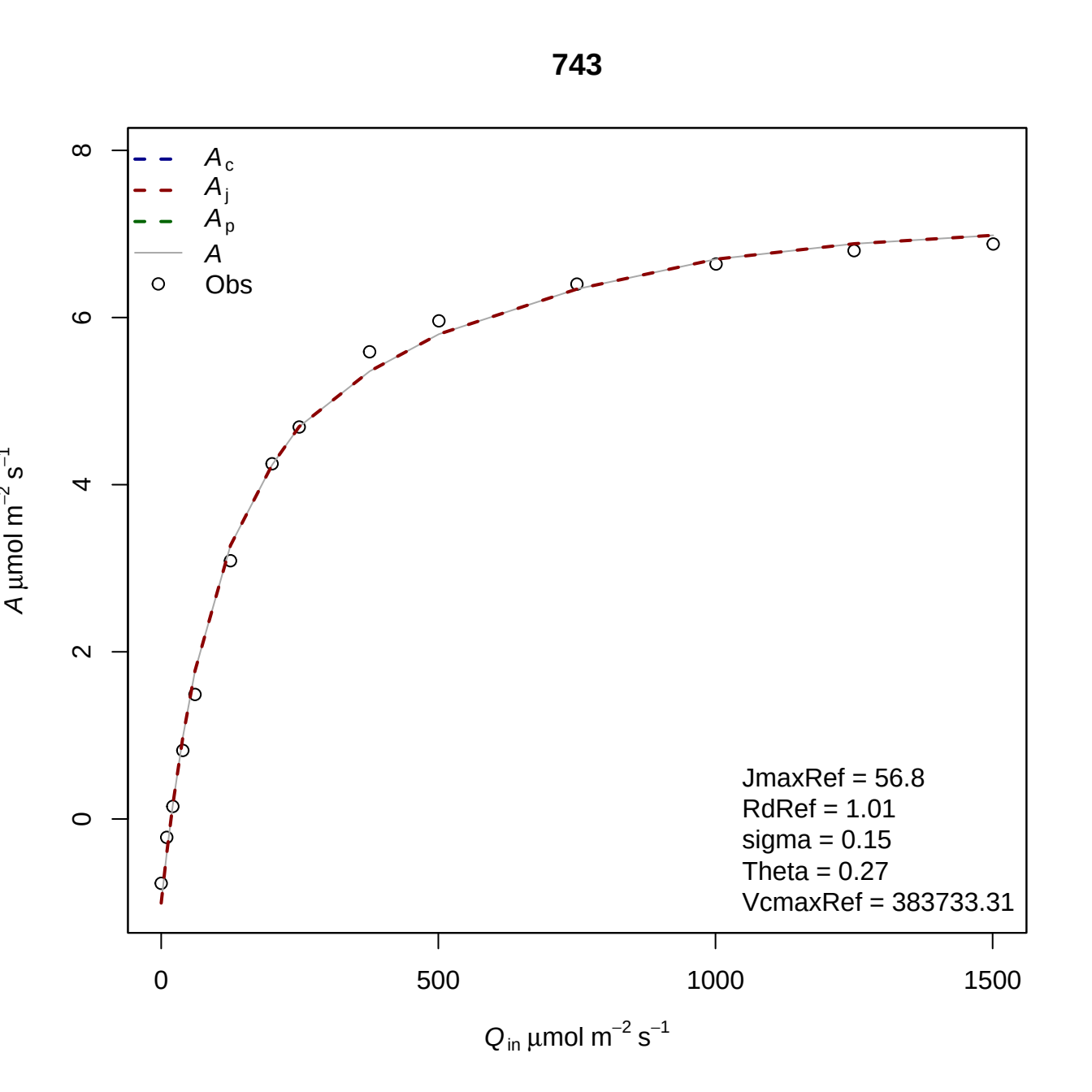

738
A_c
A_j
A_p
A
Obs
JmaxRef = 47.02
RdRef = 1.12
sigma = 0.13
Theta = 0.6
VcmaxRef = 38.7
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

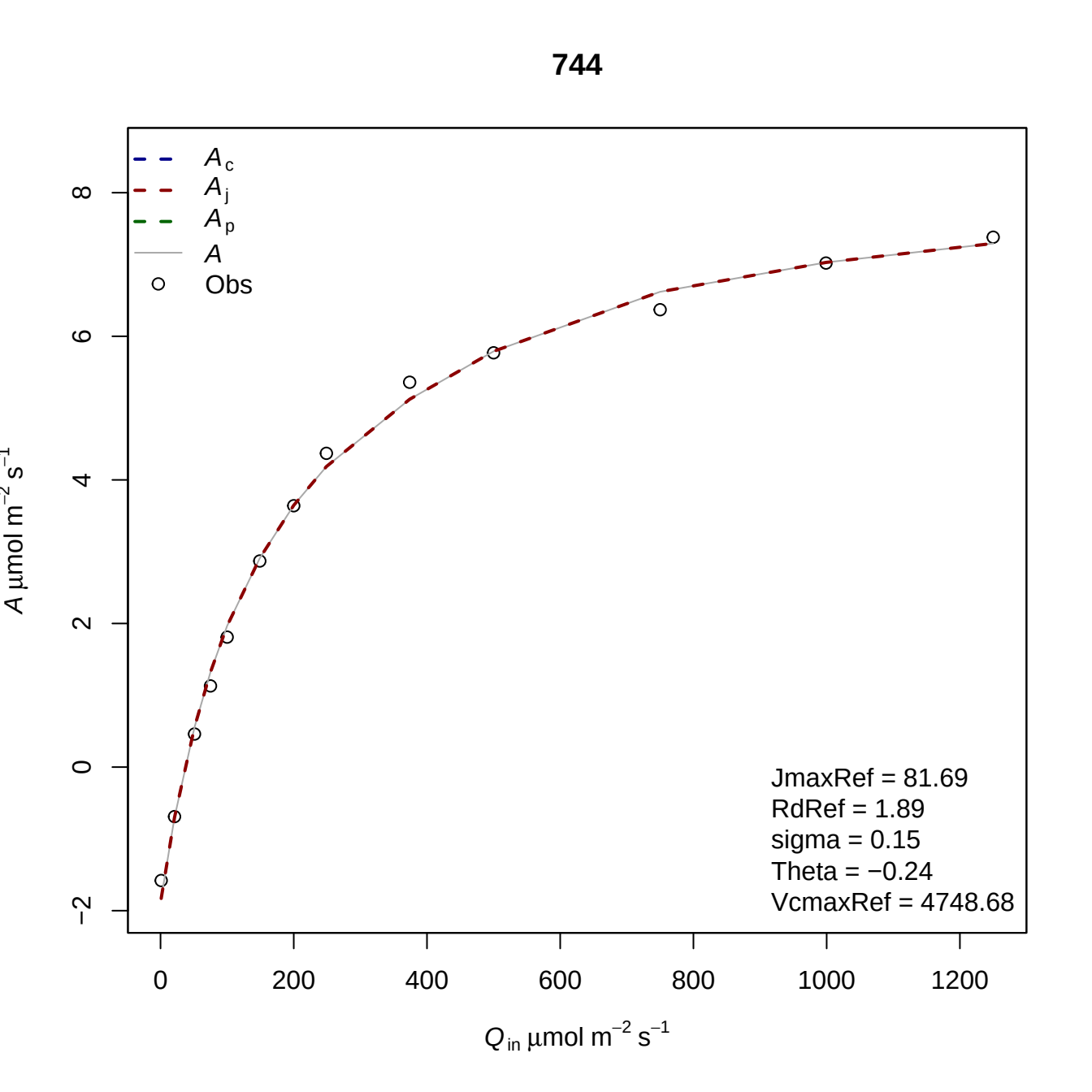

739
A_c
A_j
A_p
A
Obs
JmaxRef = 108.94
RdRef = 0.73
sigma = 0.14
Theta = 0.05
VcmaxRef = 108.9
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

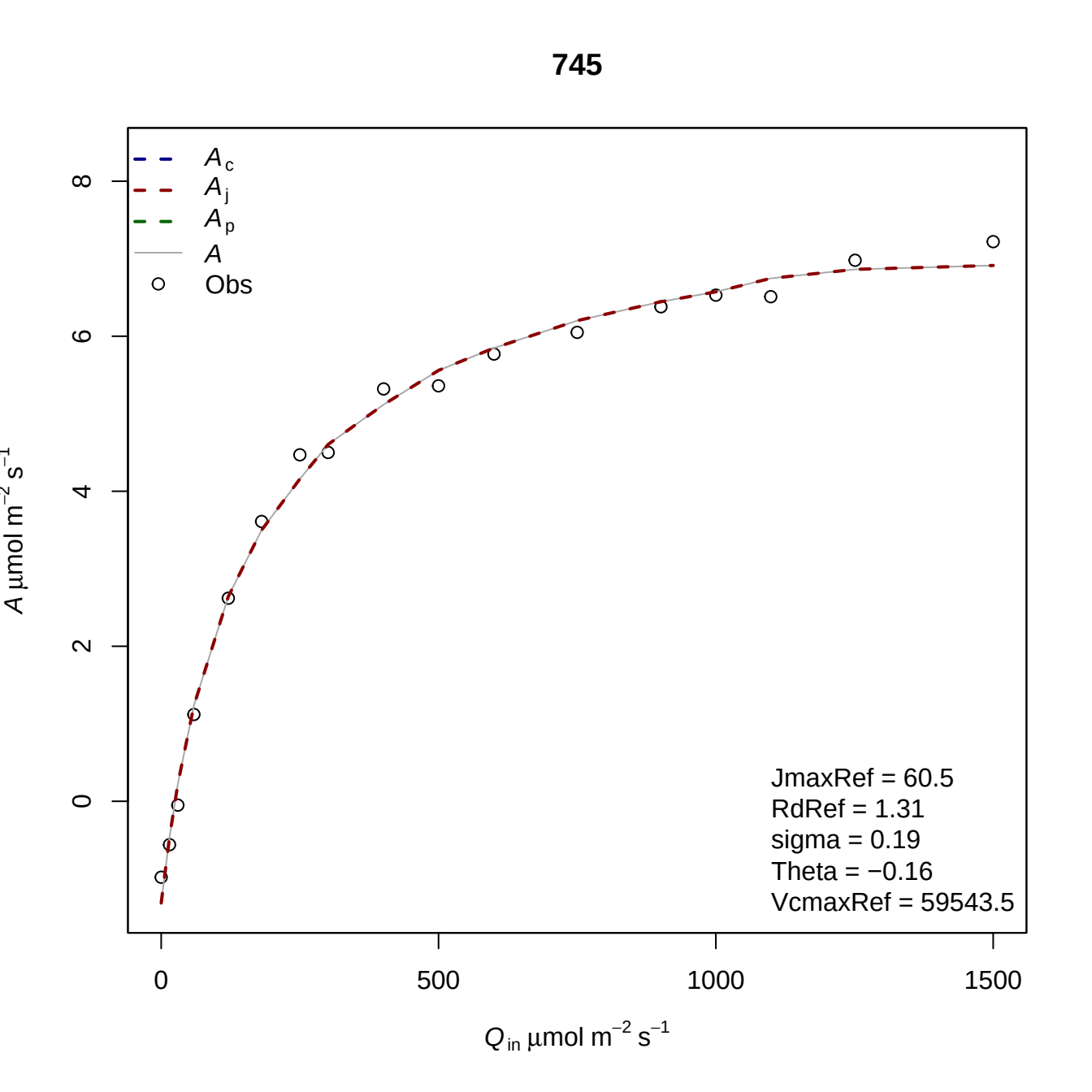

740
A_c
A_j
A_p
A
Obs
JmaxRef = 64.99
RdRef = 0.61
sigma = 0.18
Theta = 0.3
VcmaxRef = 45.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

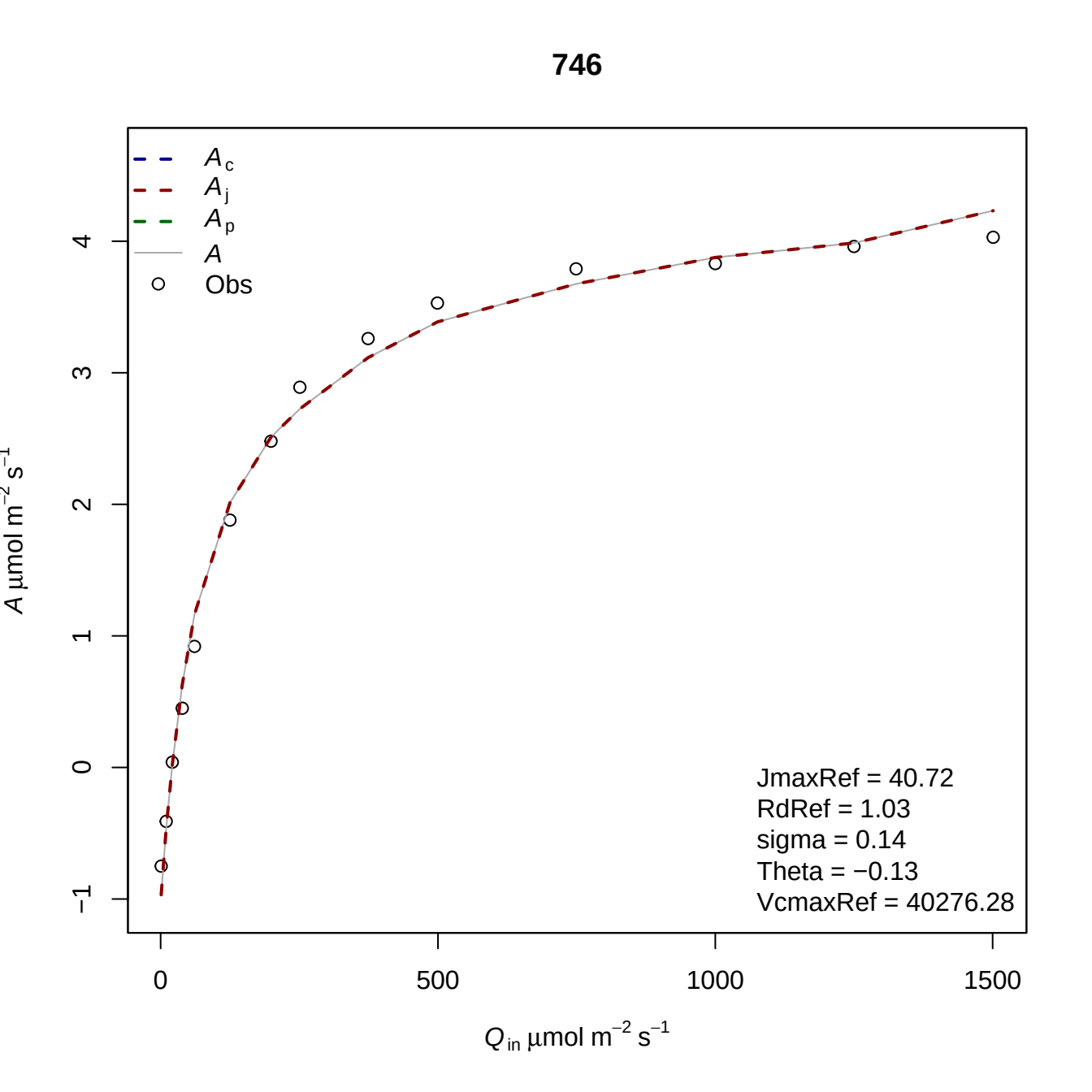

741
A_c
A_j
A_p
A
Obs
JmaxRef = 95.22
RdRef = 0.64
sigma = 0.09
Theta = 0.31
VcmaxRef = 89.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

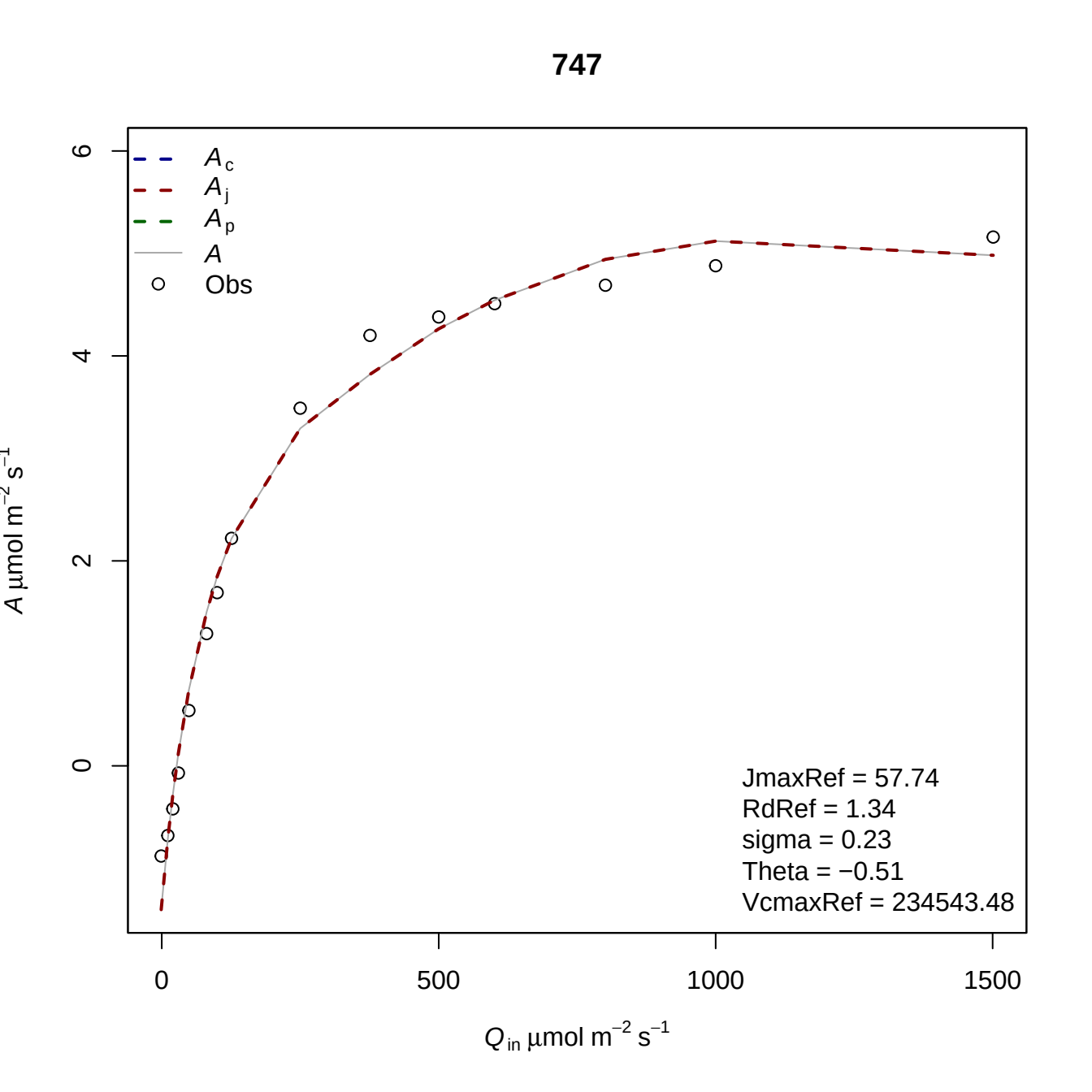

742
A_c
A_j
A_p
A
Obs
JmaxRef = 183.22
RdRef = 0.8
sigma = 0.11
Theta = −1.75
VcmaxRef = 43.86
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

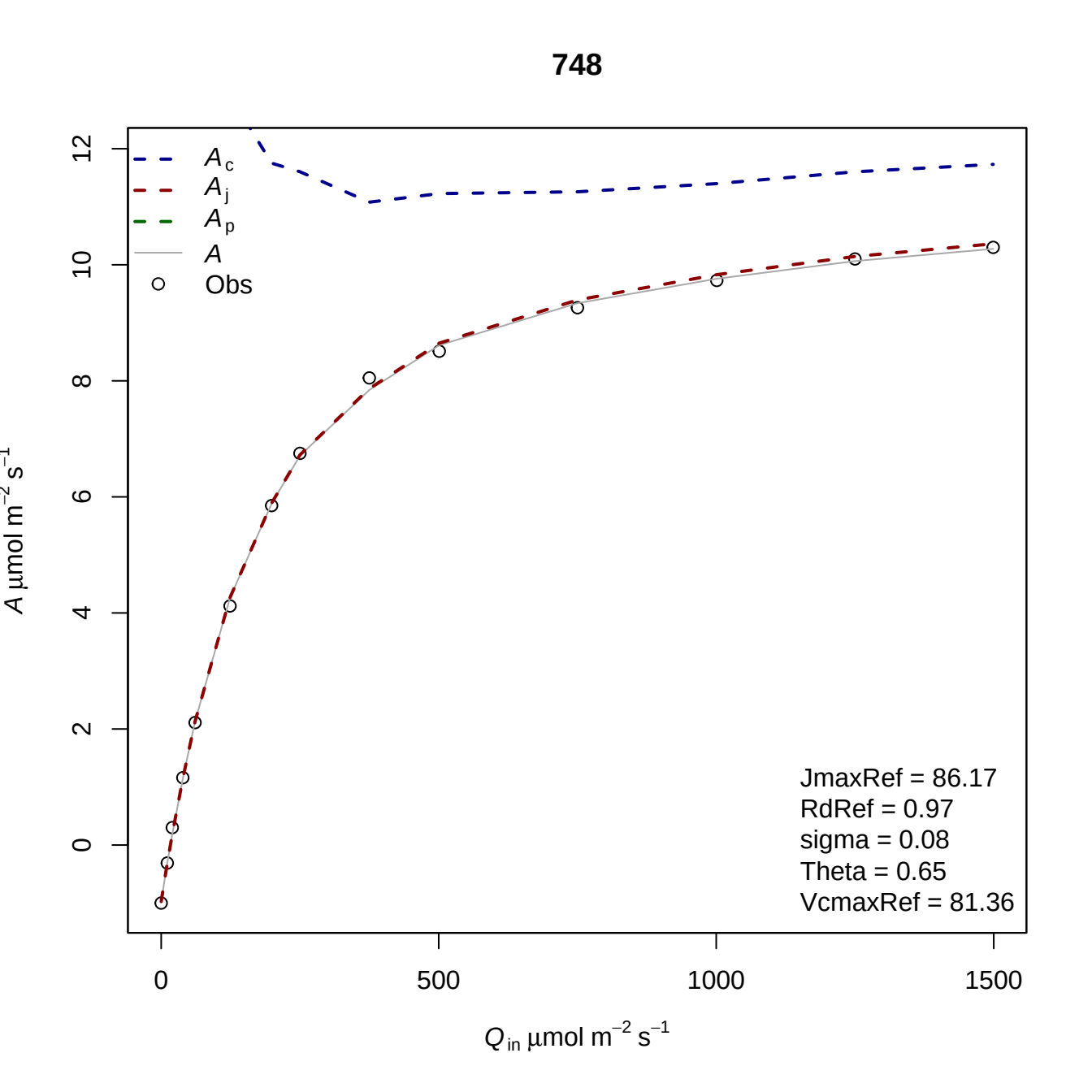

743
A_c
A_j
A_p
A
Obs
JmaxRef = 56.8
RdRef = 1.01
sigma = 0.15
Theta = 0.27
VcmaxRef = 383733.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

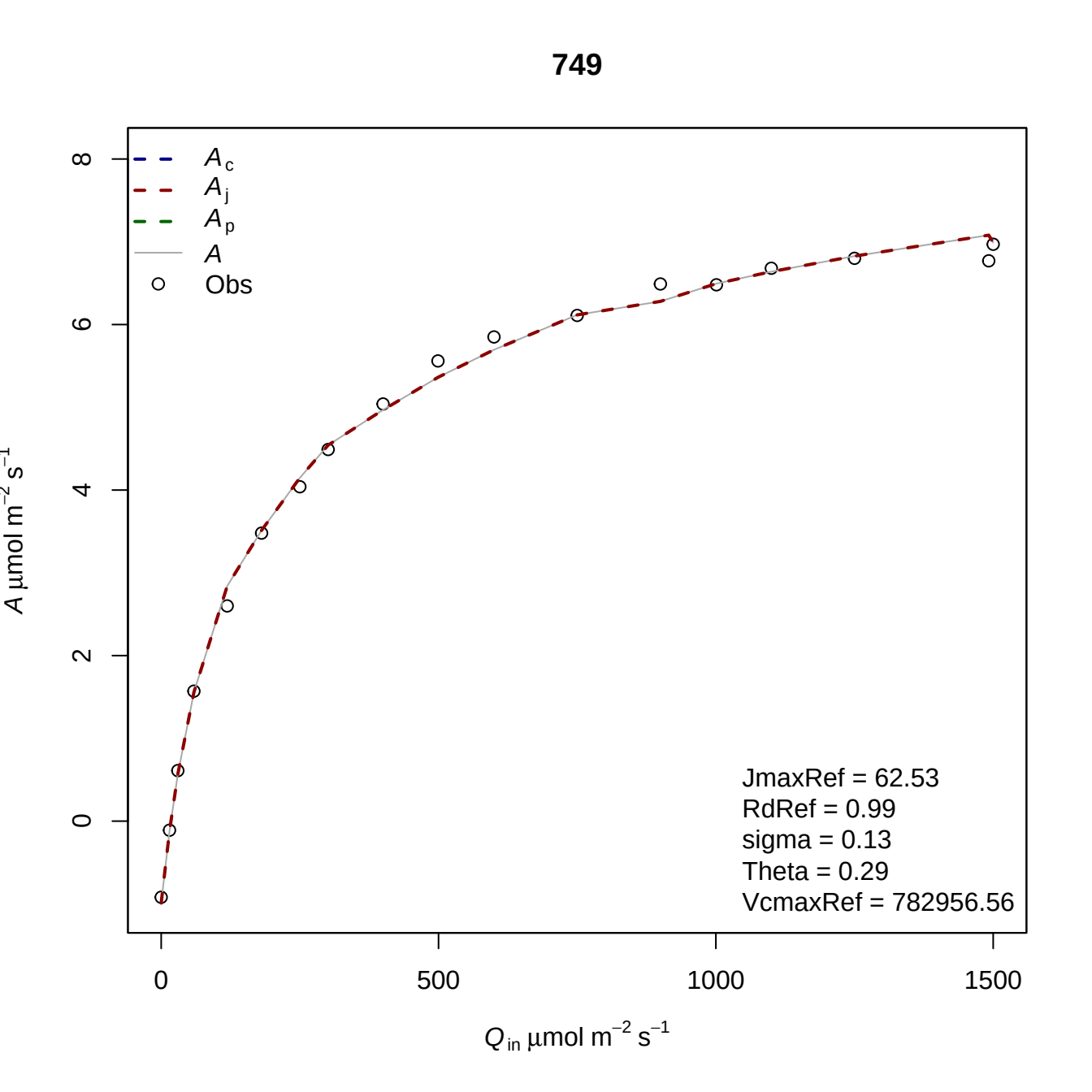

744
A_c
A_j
A_p
A
Obs
JmaxRef = 81.69
RdRef = 1.89
sigma = 0.15
Theta = −0.24
VcmaxRef = 4748.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

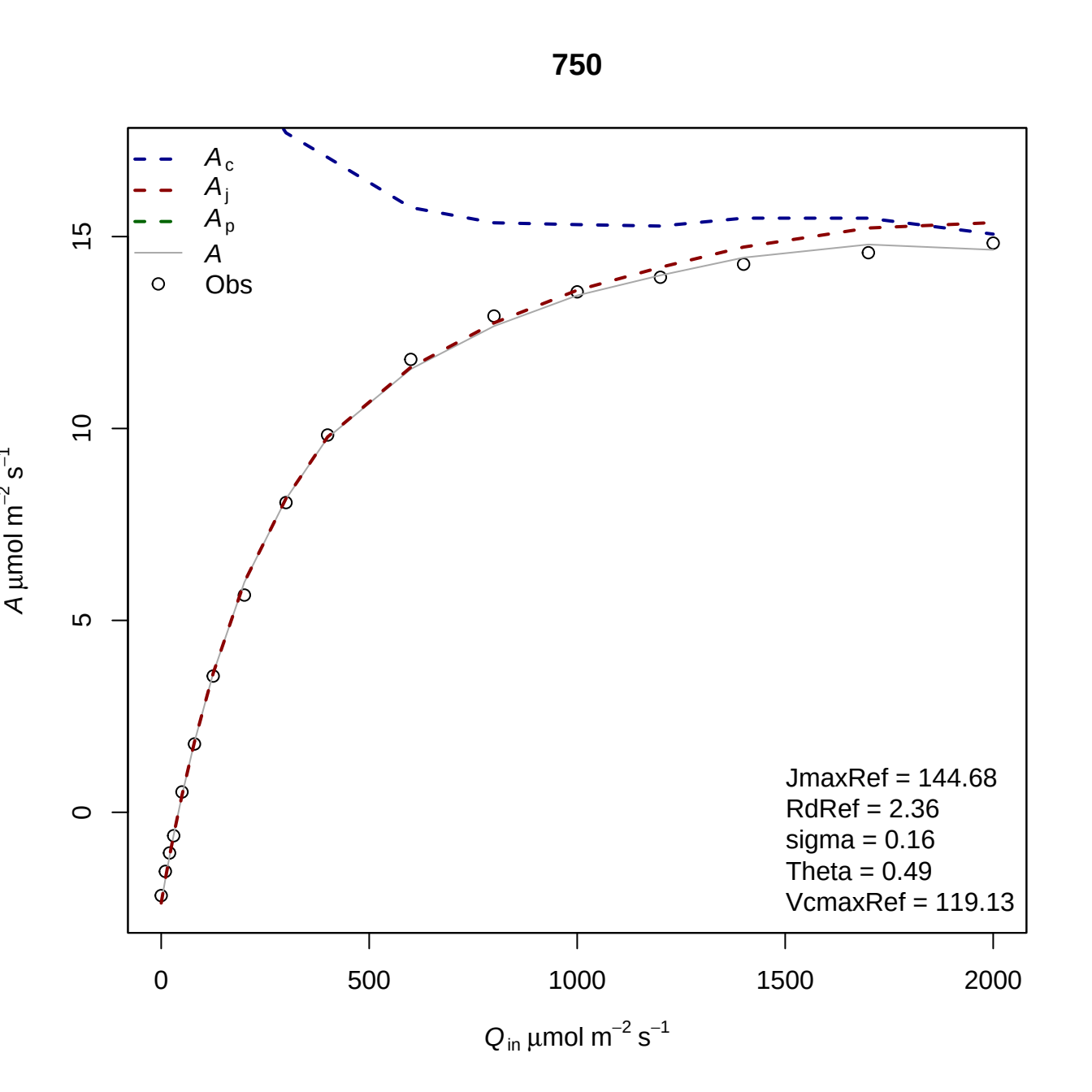

745
A_c
A_j
A_p
A
Obs
JmaxRef = 60.5
RdRef = 1.31
sigma = 0.19
Theta = −0.16
VcmaxRef = 59543.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

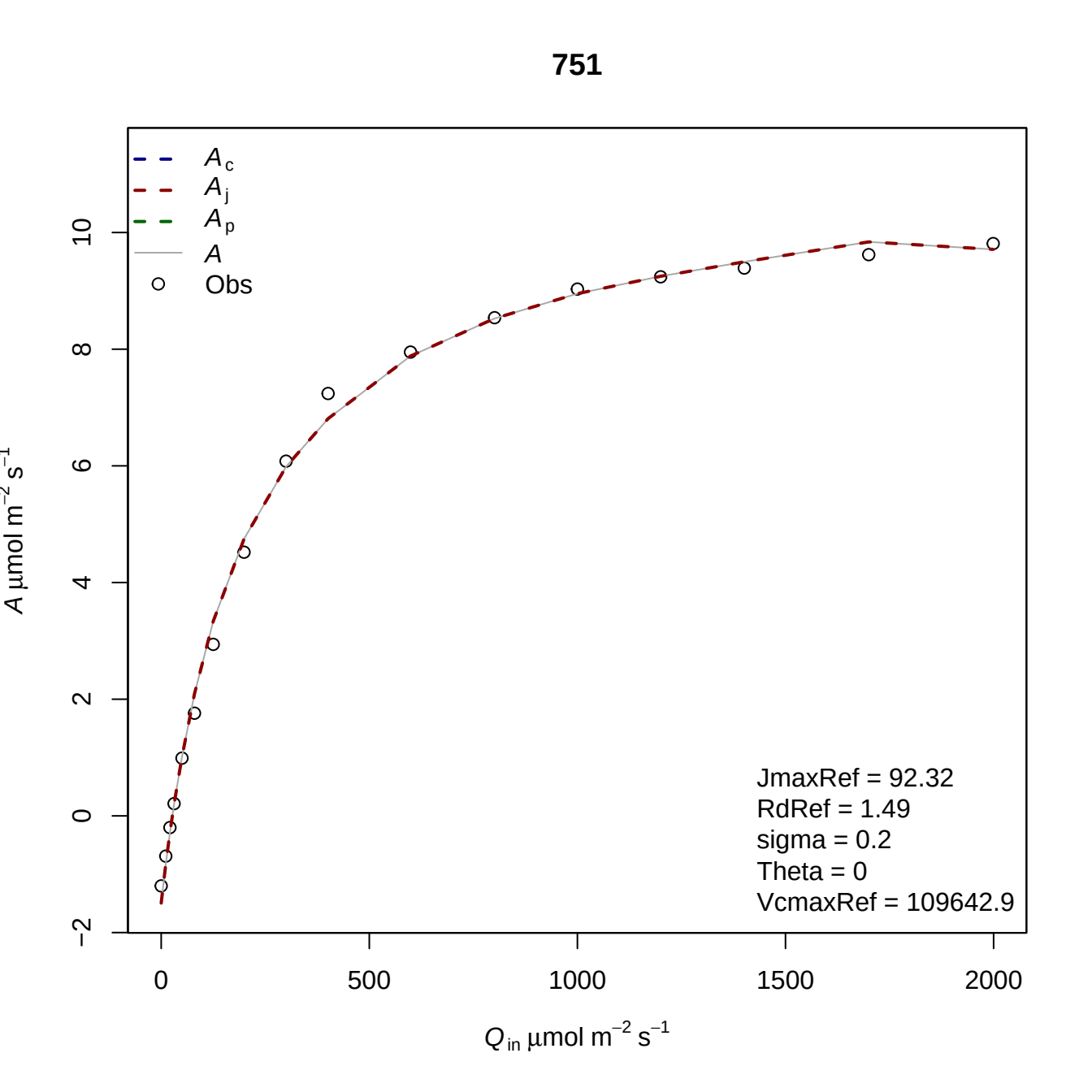

746
A_c
A_j
A_p
A
Obs
JmaxRef = 40.72
RdRef = 1.03
sigma = 0.14
Theta = −0.13
VcmaxRef = 40276.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

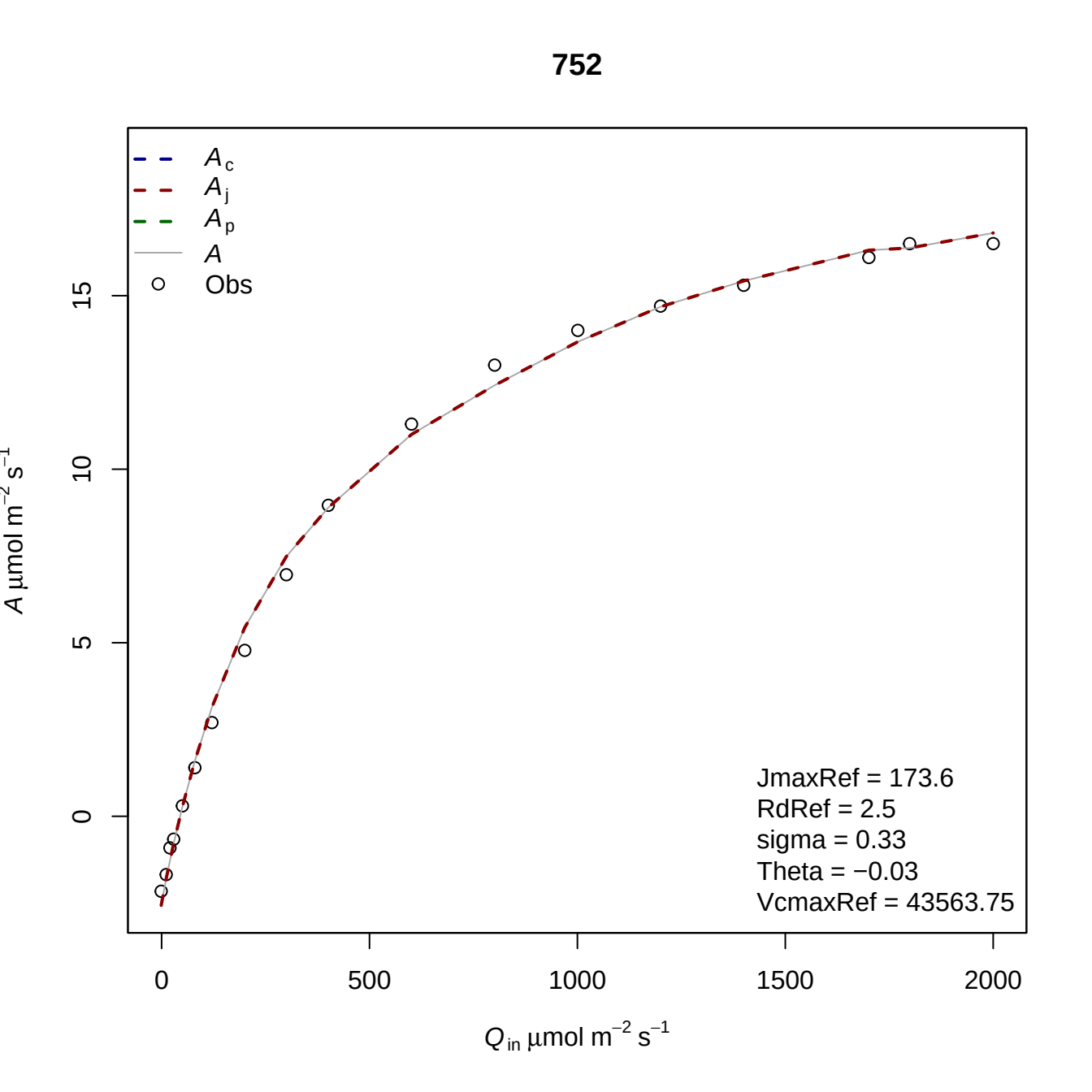

747
A_c
A_j
A_p
A
Obs
JmaxRef = 57.74
RdRef = 1.34
sigma = 0.23
Theta = −0.51
VcmaxRef = 234543.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

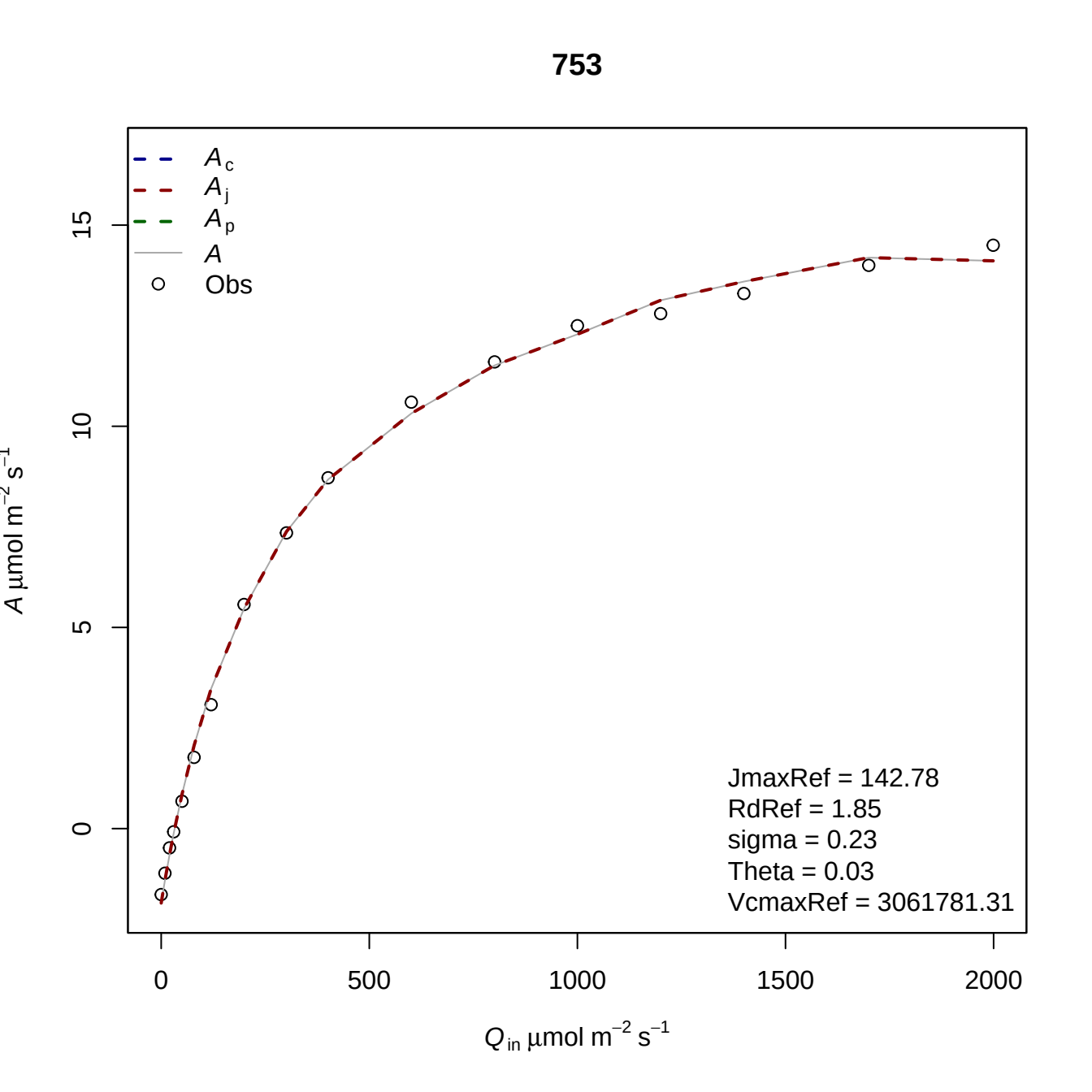

748
A_c
A_j
A_p
A
Obs
JmaxRef = 86.17
RdRef = 0.97
sigma = 0.08
Theta = 0.65
VcmaxRef = 81.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

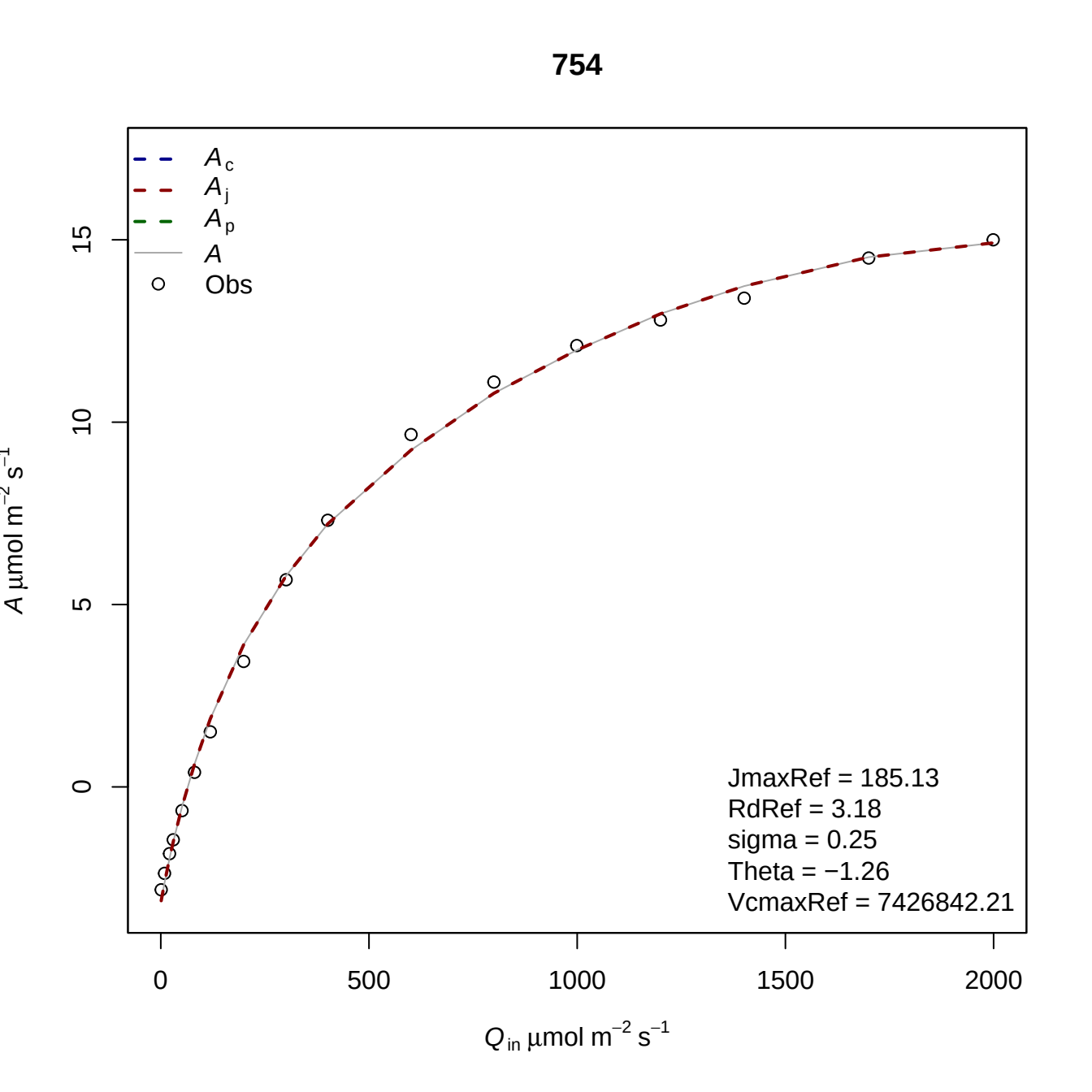

749
A_c
A_j
A_p
A
Obs
JmaxRef = 62.53
RdRef = 0.99
sigma = 0.13
Theta = 0.29
VcmaxRef = 782956.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

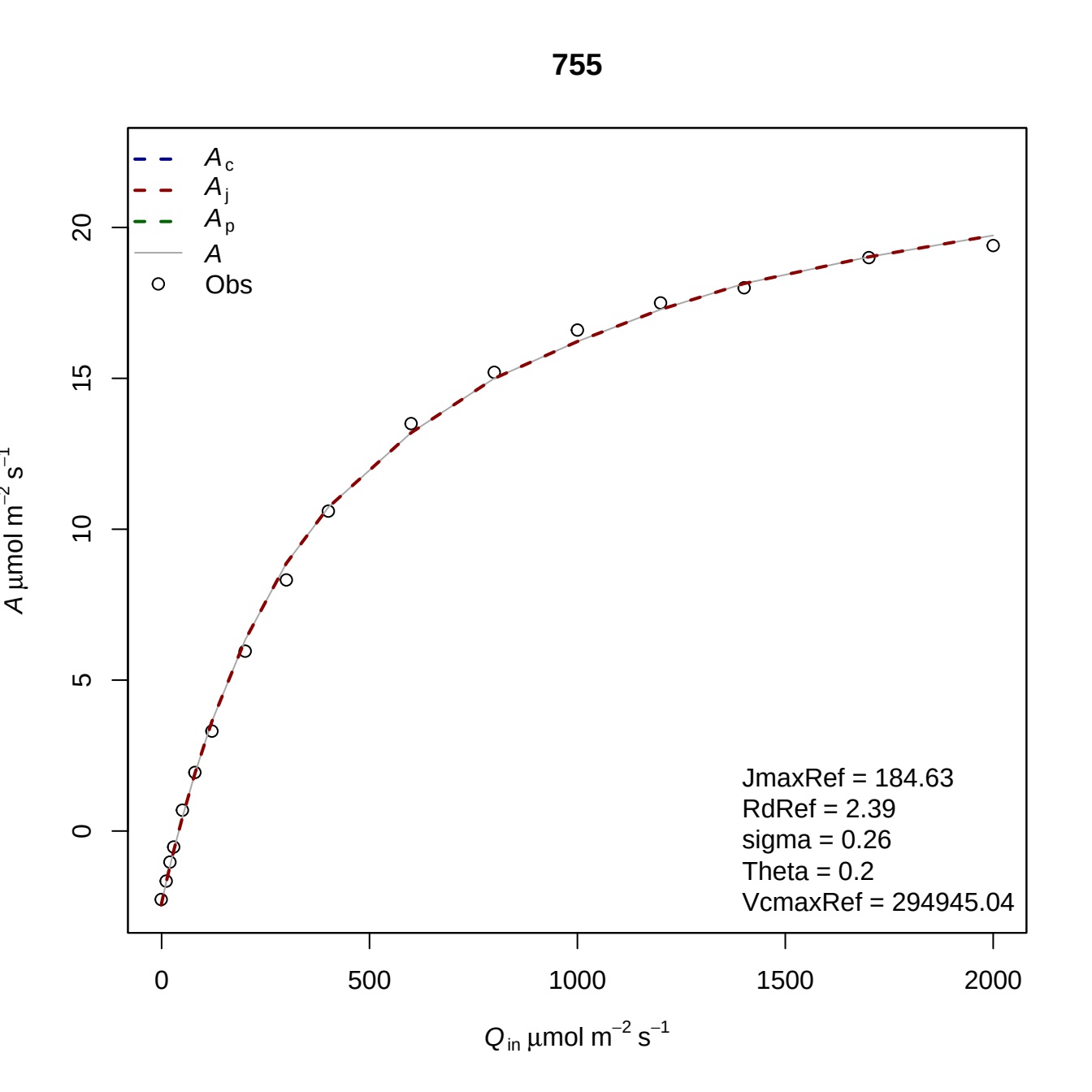

750
A_c
A_j
A_p
A
Obs
JmaxRef = 144.68
RdRef = 2.36
sigma = 0.16
Theta = 0.49
VcmaxRef = 119.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

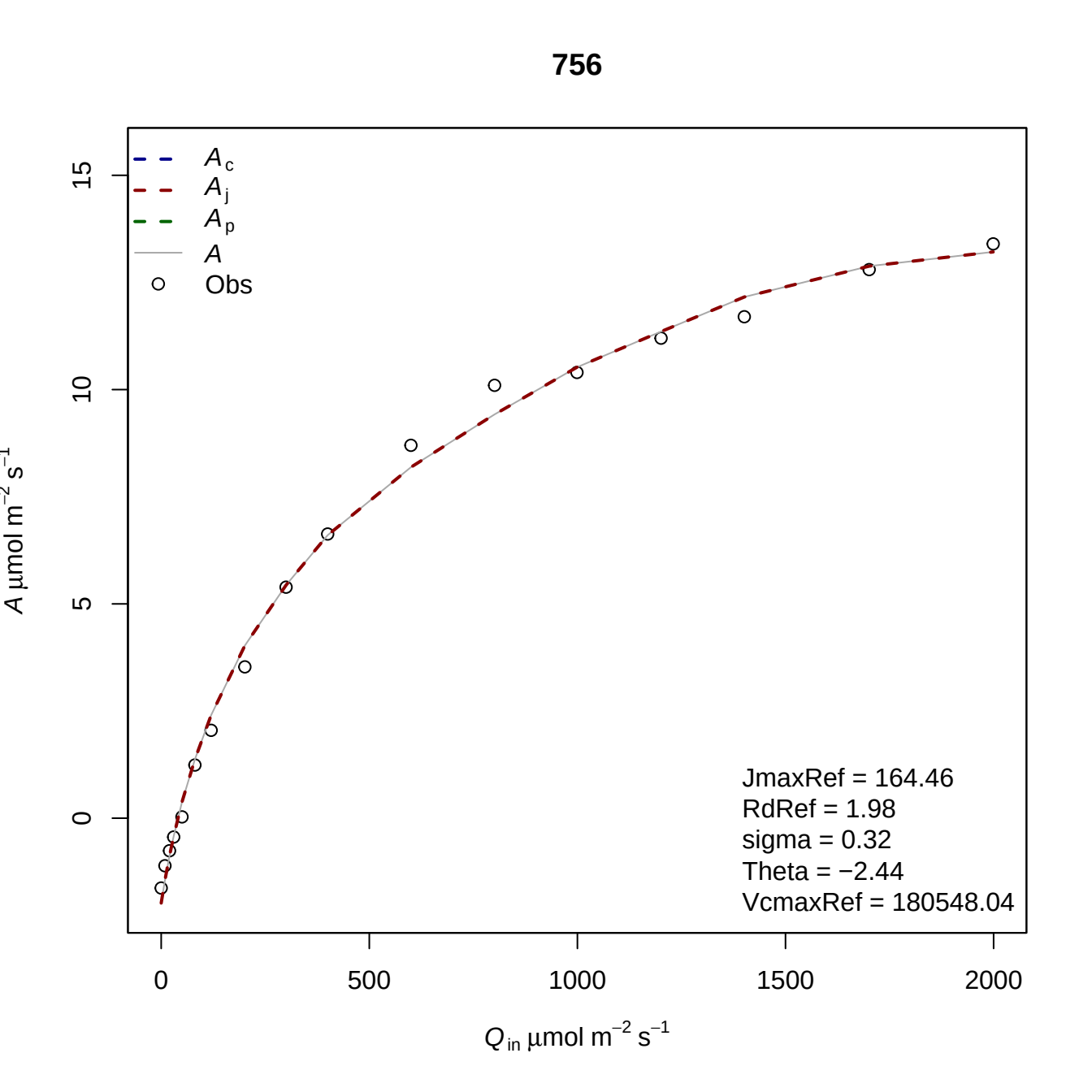

751
A_c
A_j
A_p
A
Obs
JmaxRef = 92.32
RdRef = 1.49
sigma = 0.2
Theta = 0
VcmaxRef = 109642.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

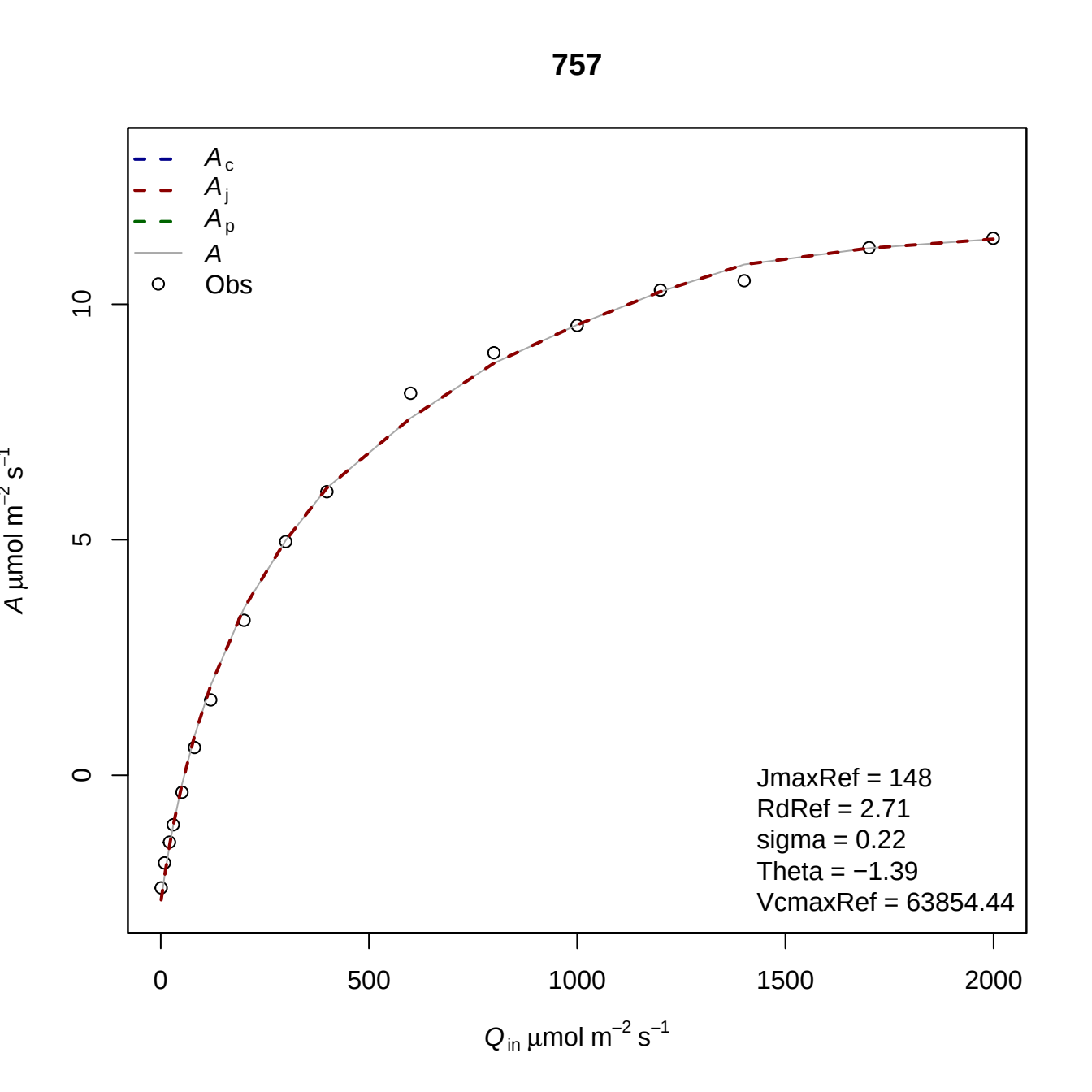

752
A_c
A_j
A_p
A
Obs
JmaxRef = 173.6
RdRef = 2.5
sigma = 0.33
Theta = −0.03
VcmaxRef = 43563.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

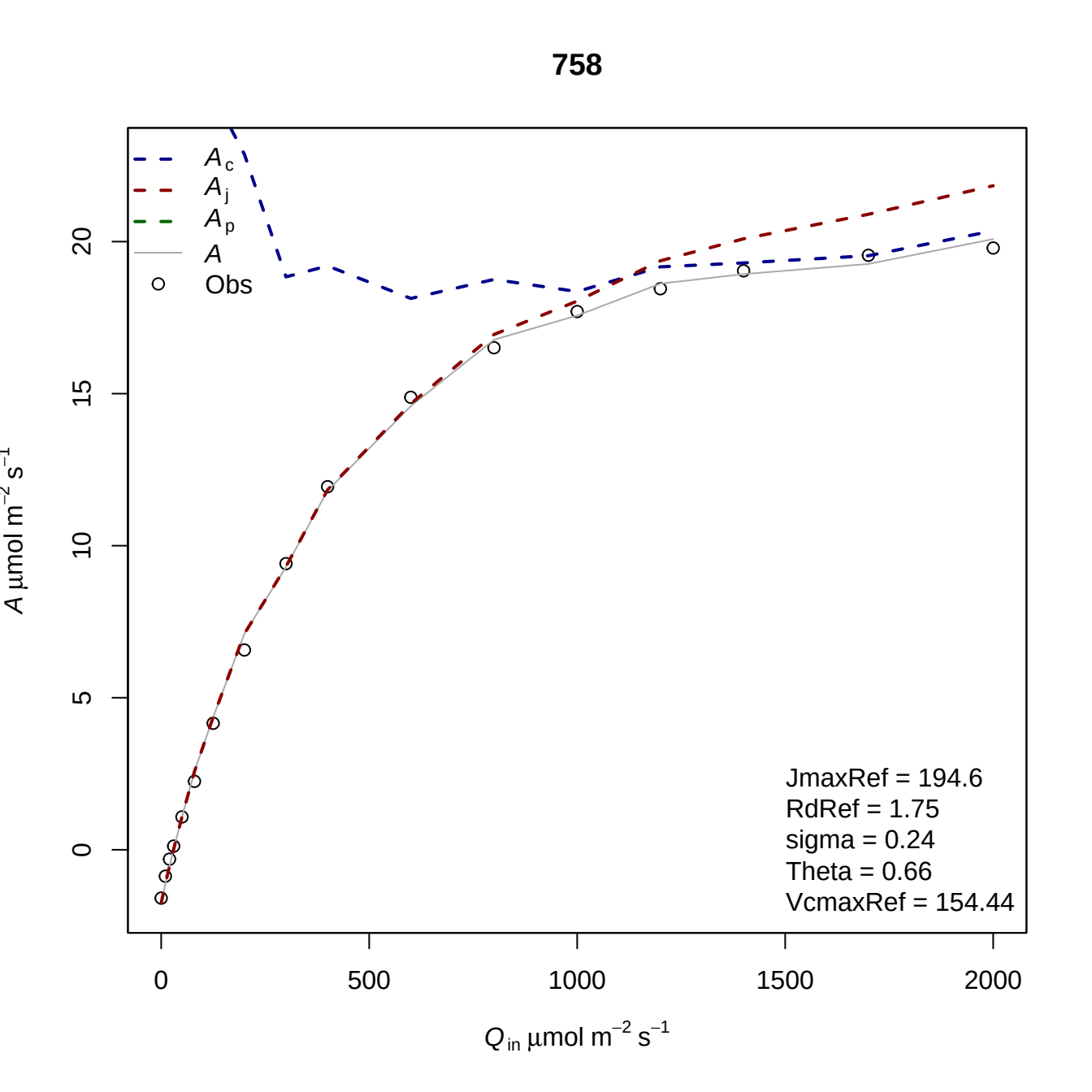

753
A_c
A_j
A_p
A
Obs
JmaxRef = 142.78
RdRef = 1.85
sigma = 0.23
Theta = 0.03
VcmaxRef = 3061781.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

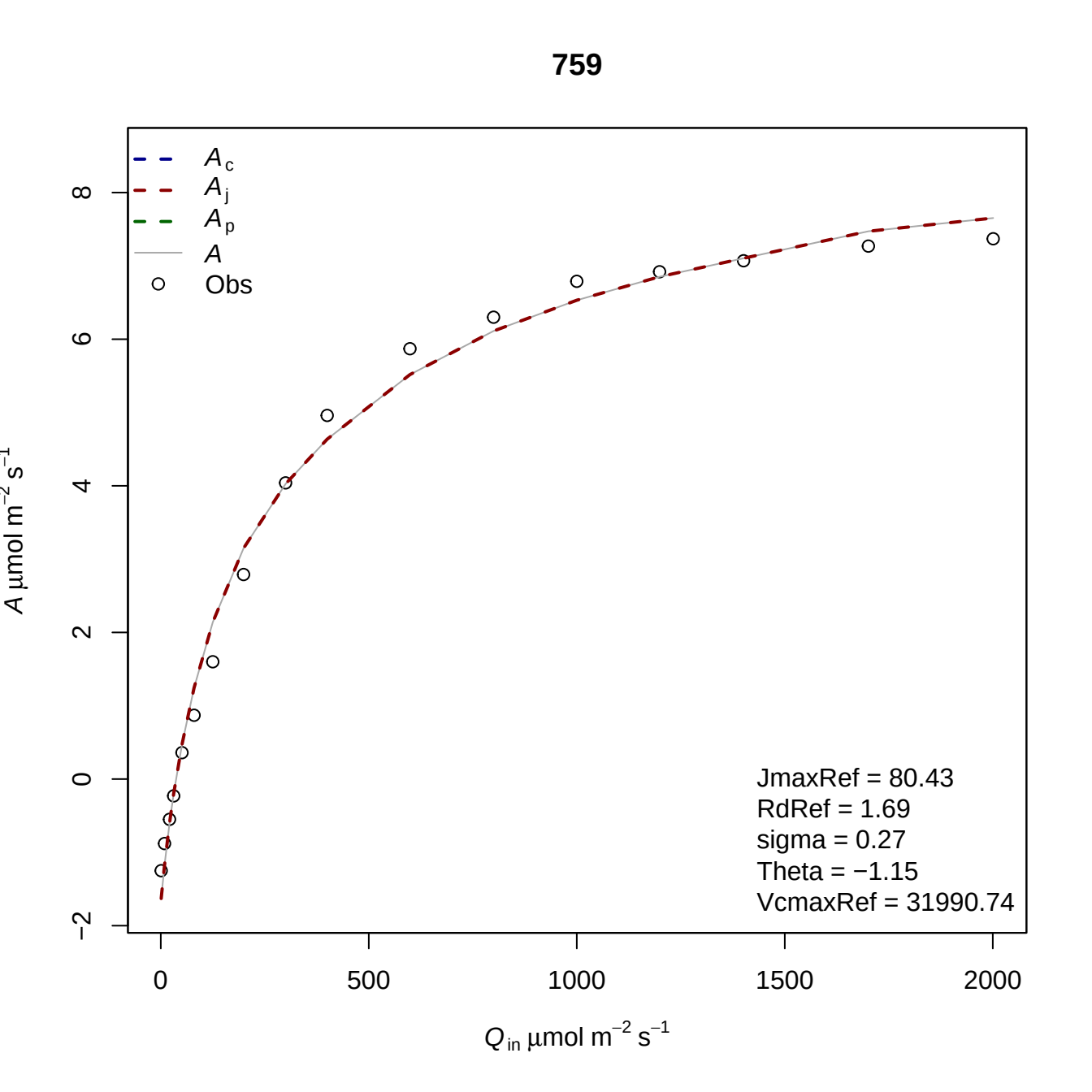

754
A_c
A_j
A_p
A
Obs
JmaxRef = 185.13
RdRef = 3.18
sigma = 0.25
Theta = −1.26
VcmaxRef = 7426842.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

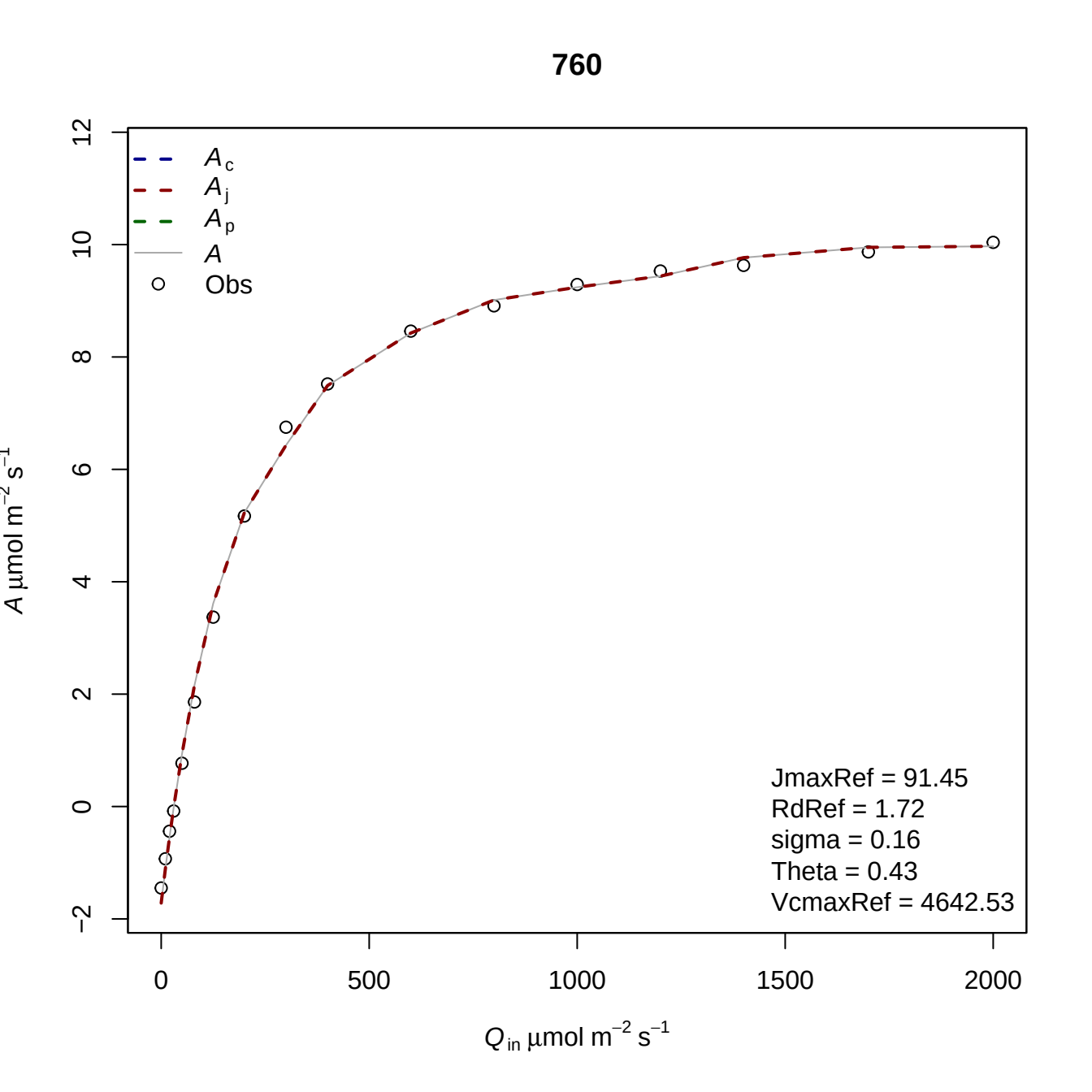

755
A_c
A_j
A_p
A
Obs
JmaxRef = 184.63
RdRef = 2.39
sigma = 0.26
Theta = 0.2
VcmaxRef = 294945.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

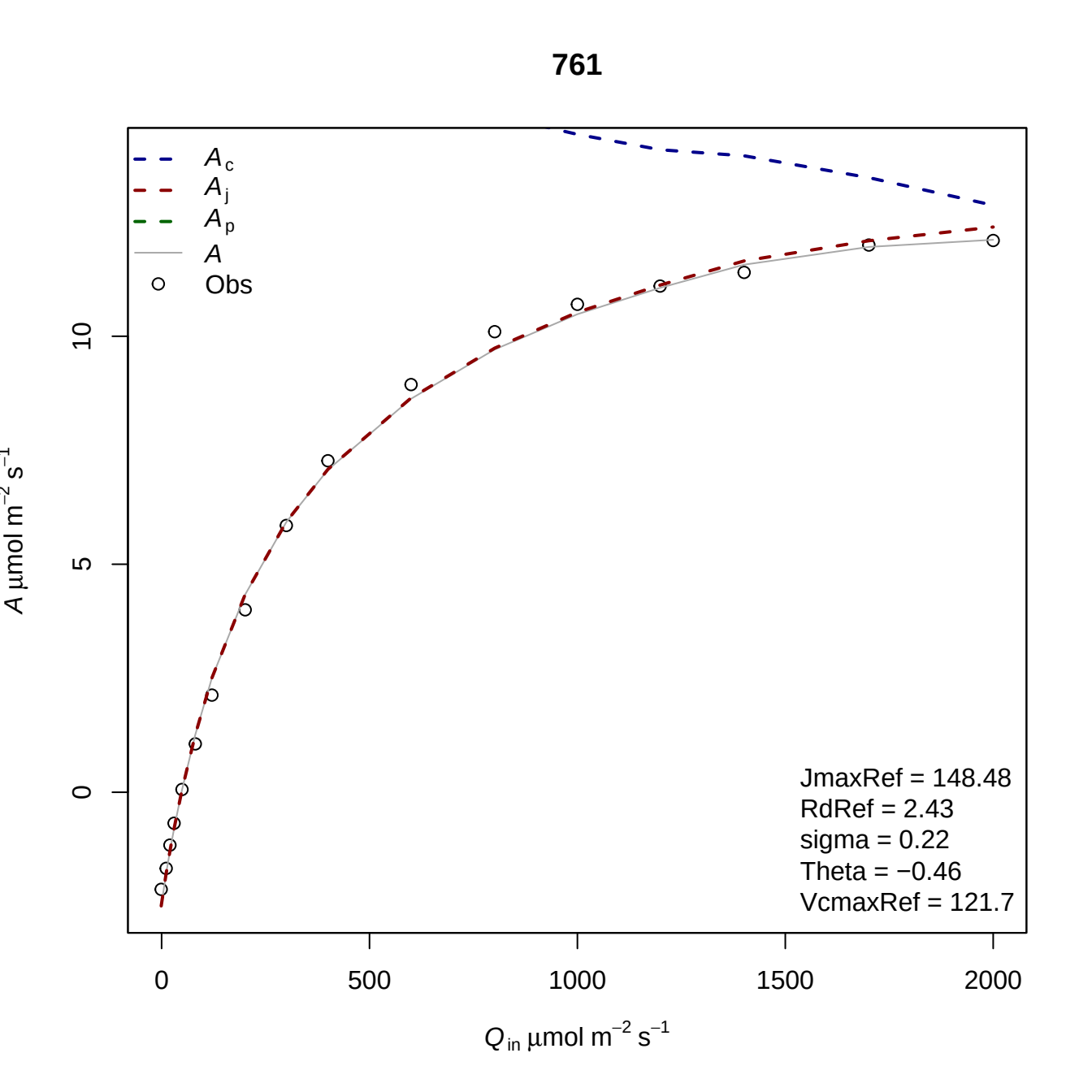

756
A_c
A_j
A_p
A
Obs
JmaxRef = 164.46
RdRef = 1.98
sigma = 0.32
Theta = −2.44
VcmaxRef = 180548.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

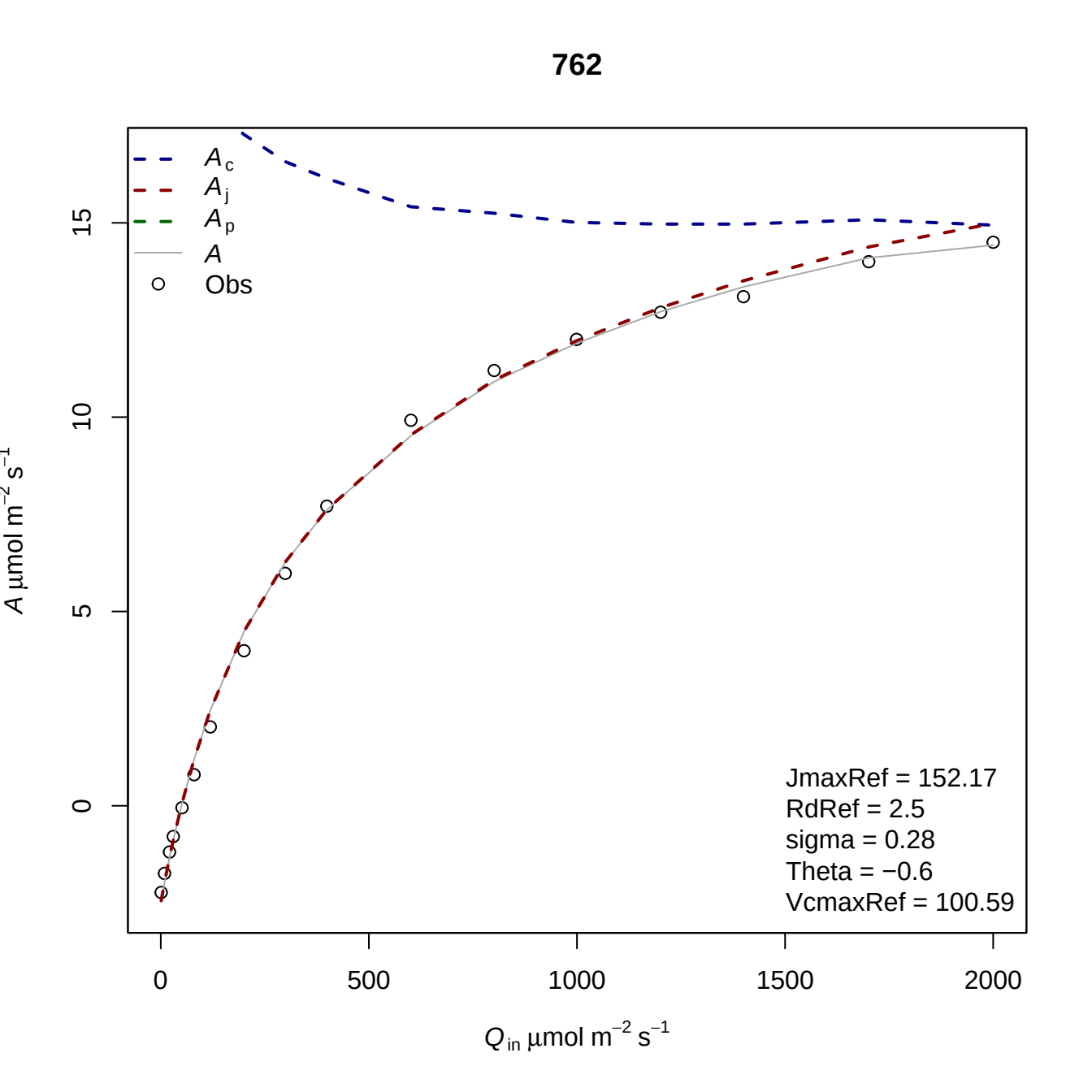

757
A_c
A_j
A_p
A
Obs
JmaxRef = 148
RdRef = 2.71
sigma = 0.22
Theta = −1.39
VcmaxRef = 63854.44
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

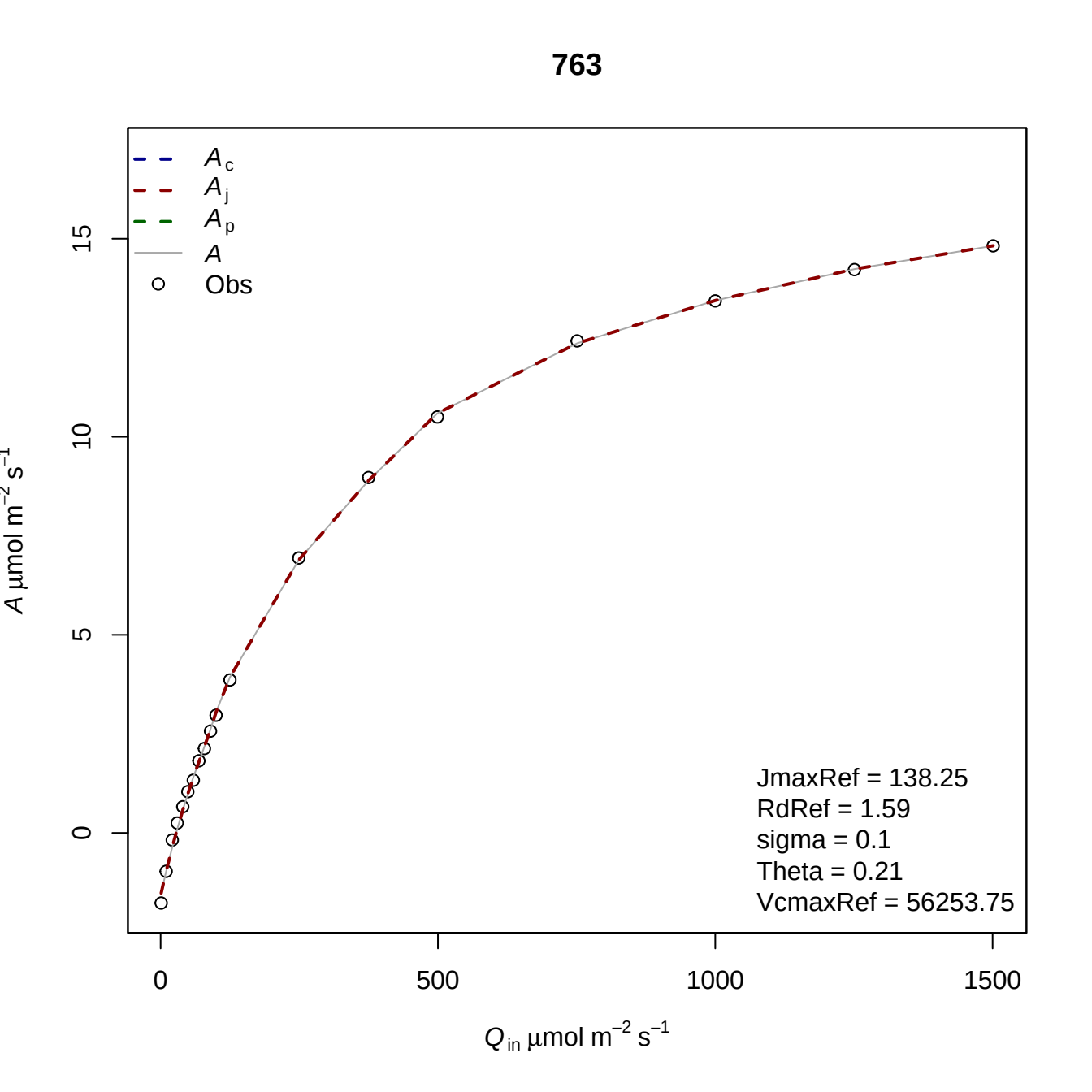

758
A_c
A_j
A_p
A
Obs
JmaxRef = 194.6
RdRef = 1.75
sigma = 0.24
Theta = 0.66
VcmaxRef = 154.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

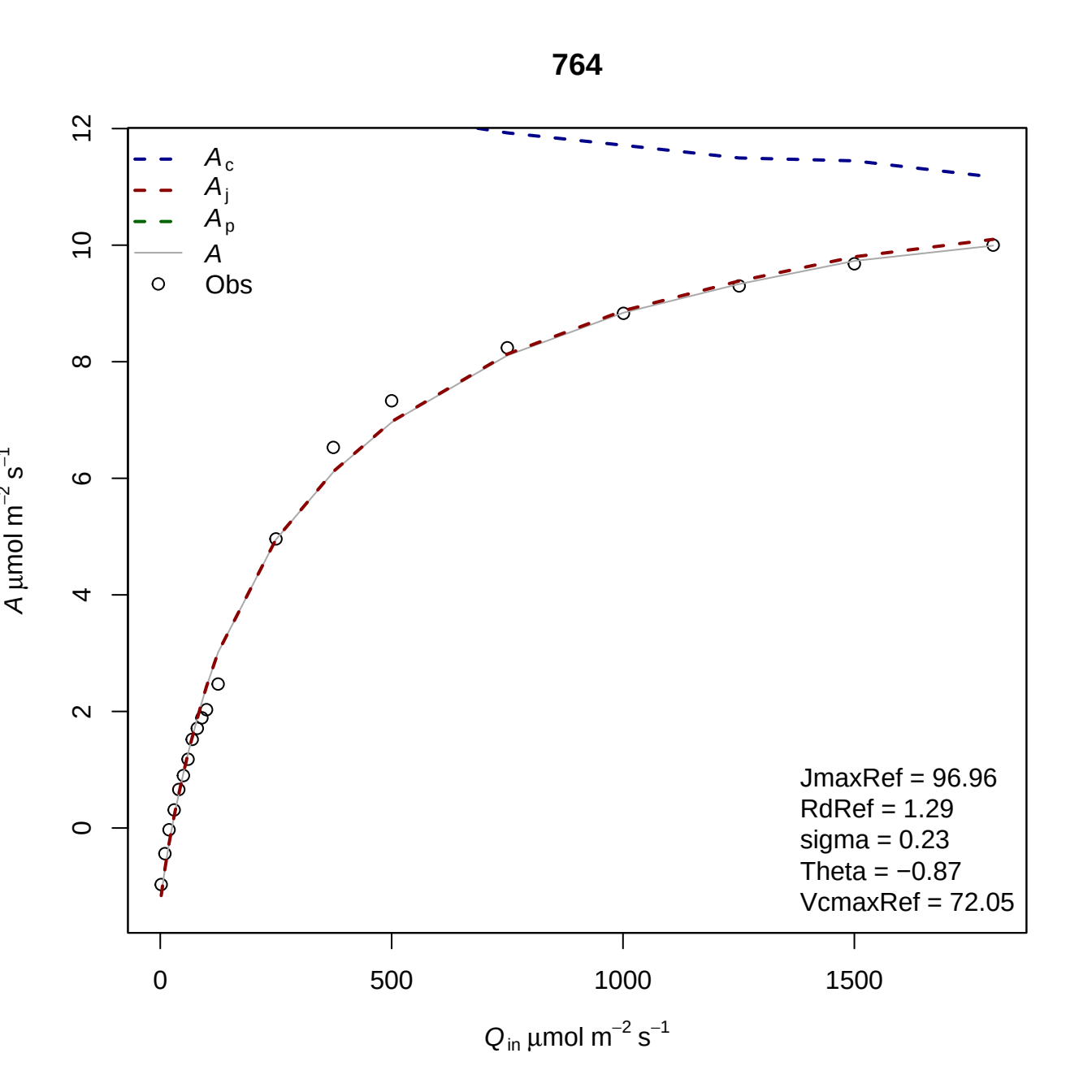

759
A_c
A_j
A_p
A
Obs
JmaxRef = 80.43
RdRef = 1.69
sigma = 0.27
Theta = −1.15
VcmaxRef = 31990.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

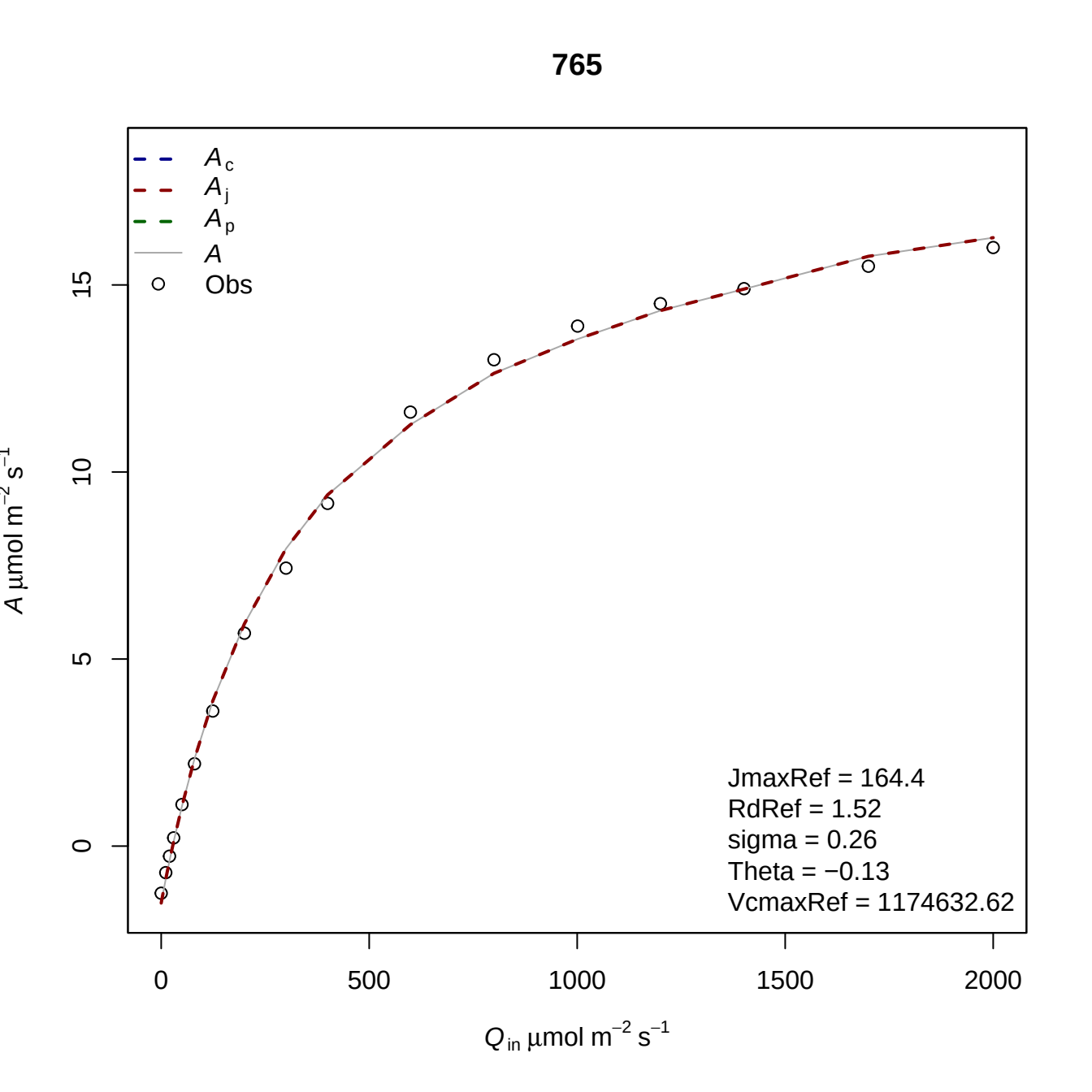

760
A_c
A_j
A_p
A
Obs
JmaxRef = 91.45
RdRef = 1.72
sigma = 0.16
Theta = 0.43
VcmaxRef = 4642.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

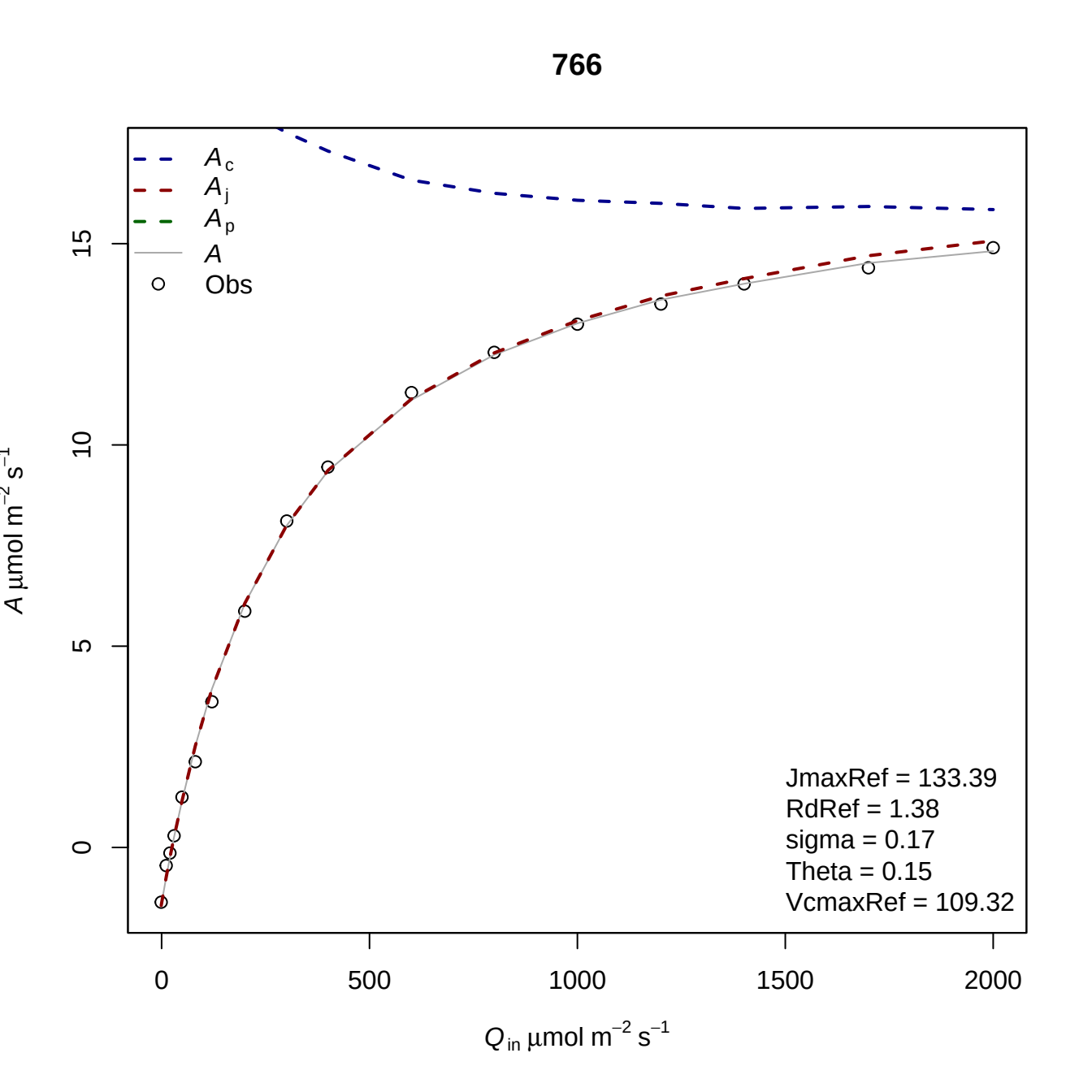

761
A_c
A_j
A_p
A
Obs
JmaxRef = 148.48
RdRef = 2.43
sigma = 0.22
Theta = −0.46
VcmaxRef = 121.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

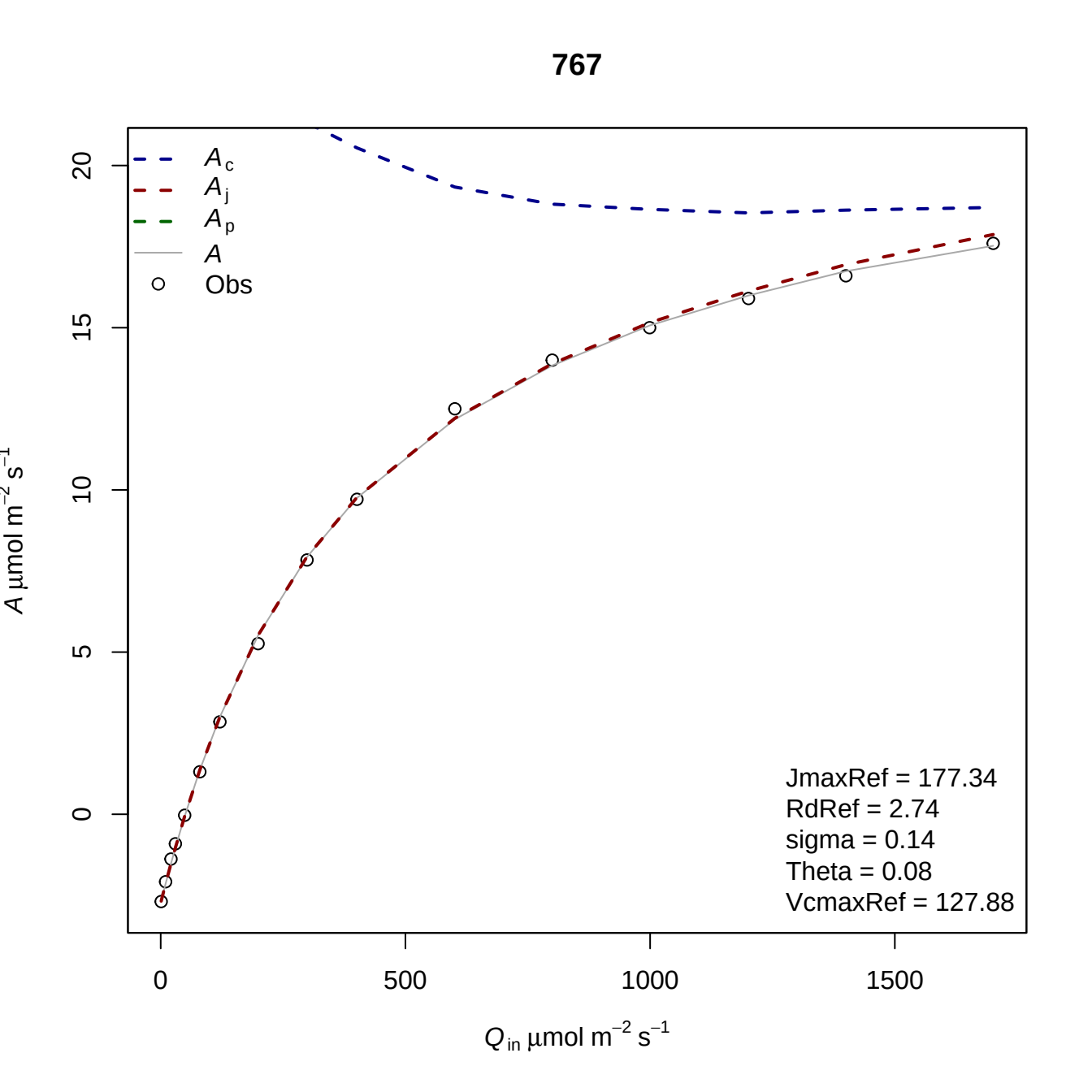

762
A_c
A_j
A_p
A
Obs
JmaxRef = 152.17
RdRef = 2.5
sigma = 0.28
Theta = −0.6
VcmaxRef = 100.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

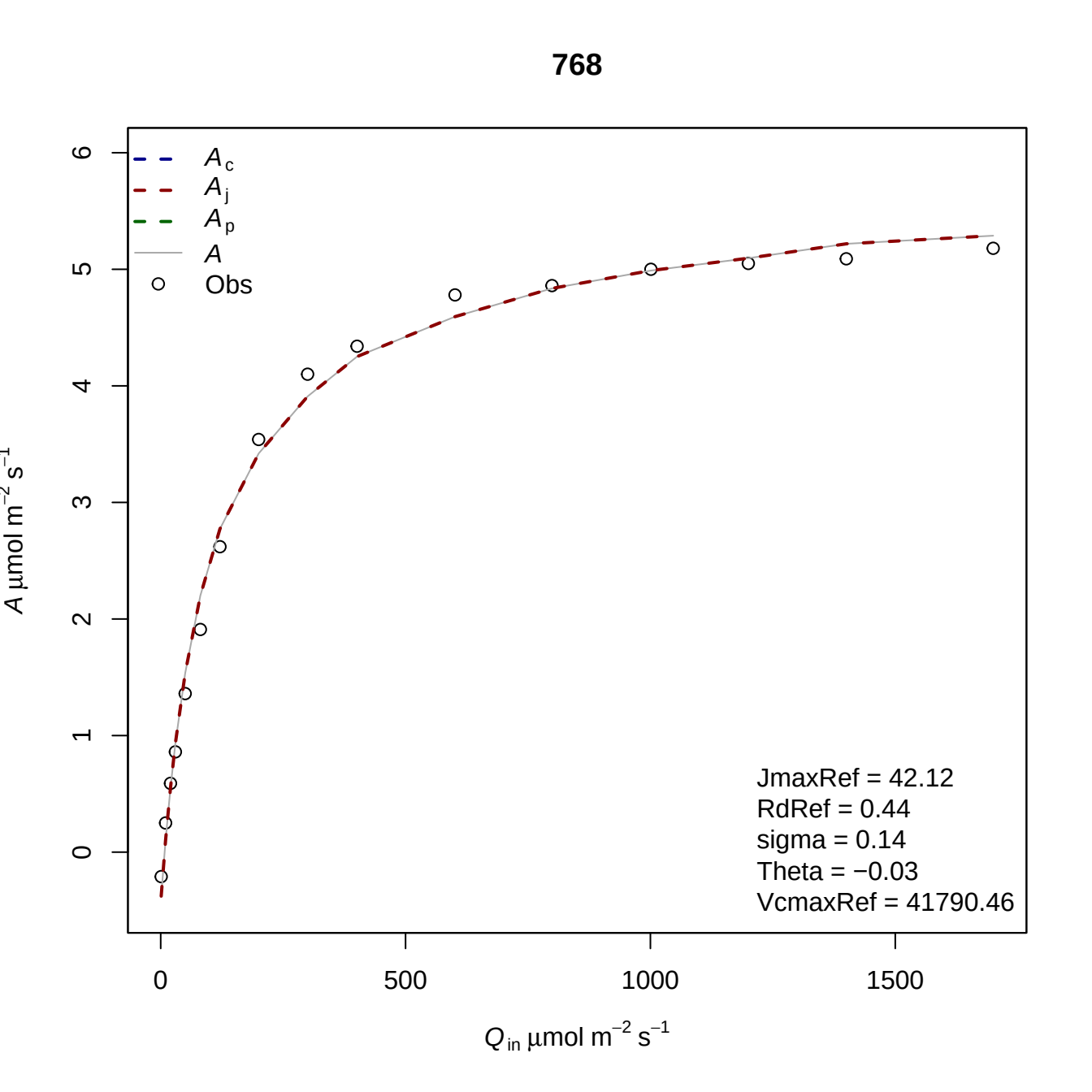

763
A_c
A_j
A_p
A
Obs
JmaxRef = 138.25
RdRef = 1.59
sigma = 0.1
Theta = 0.21
VcmaxRef = 56253.75
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

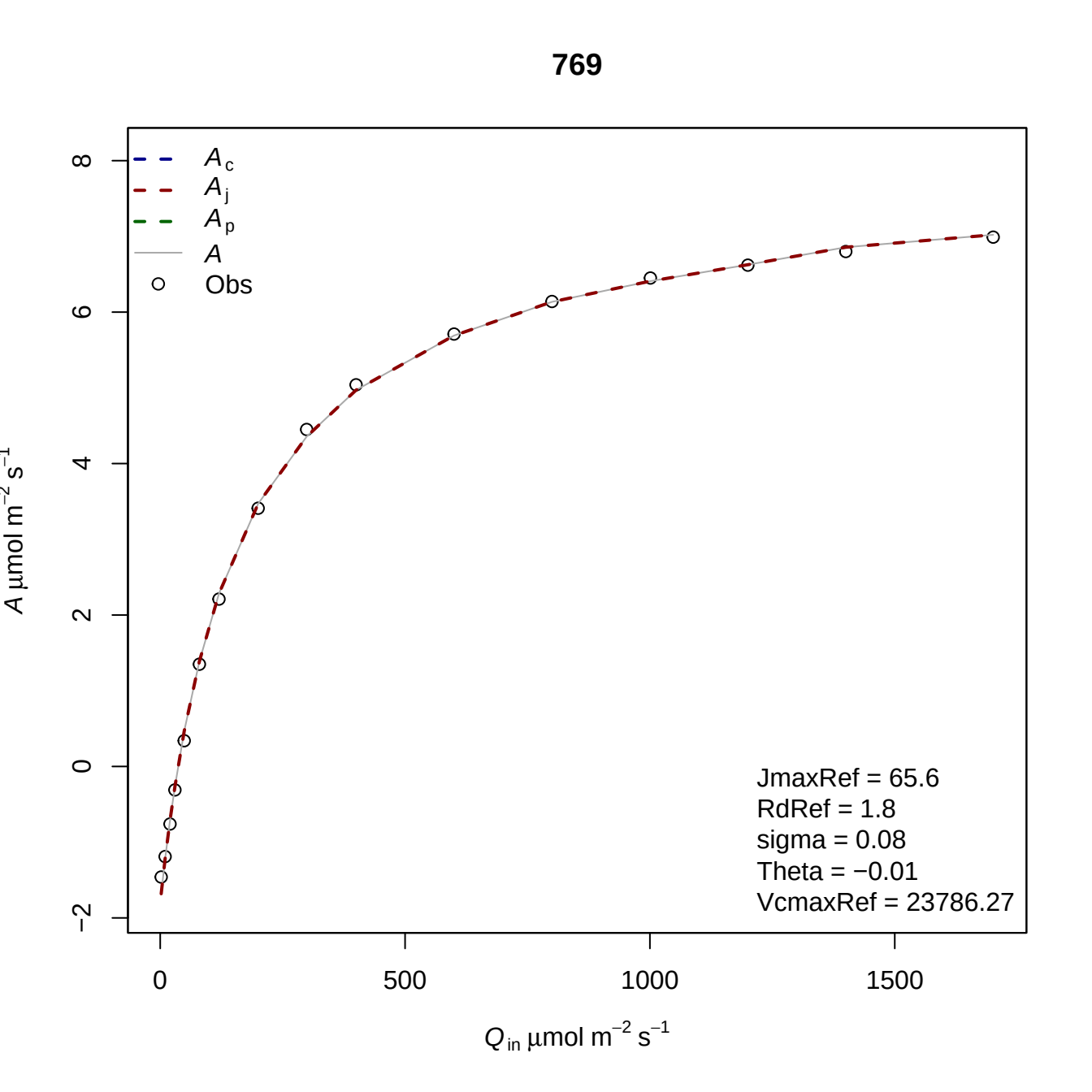

764
A_c
A_j
A_p
A
Obs
JmaxRef = 96.96
RdRef = 1.29
sigma = 0.23
Theta = −0.87
VcmaxRef = 72.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

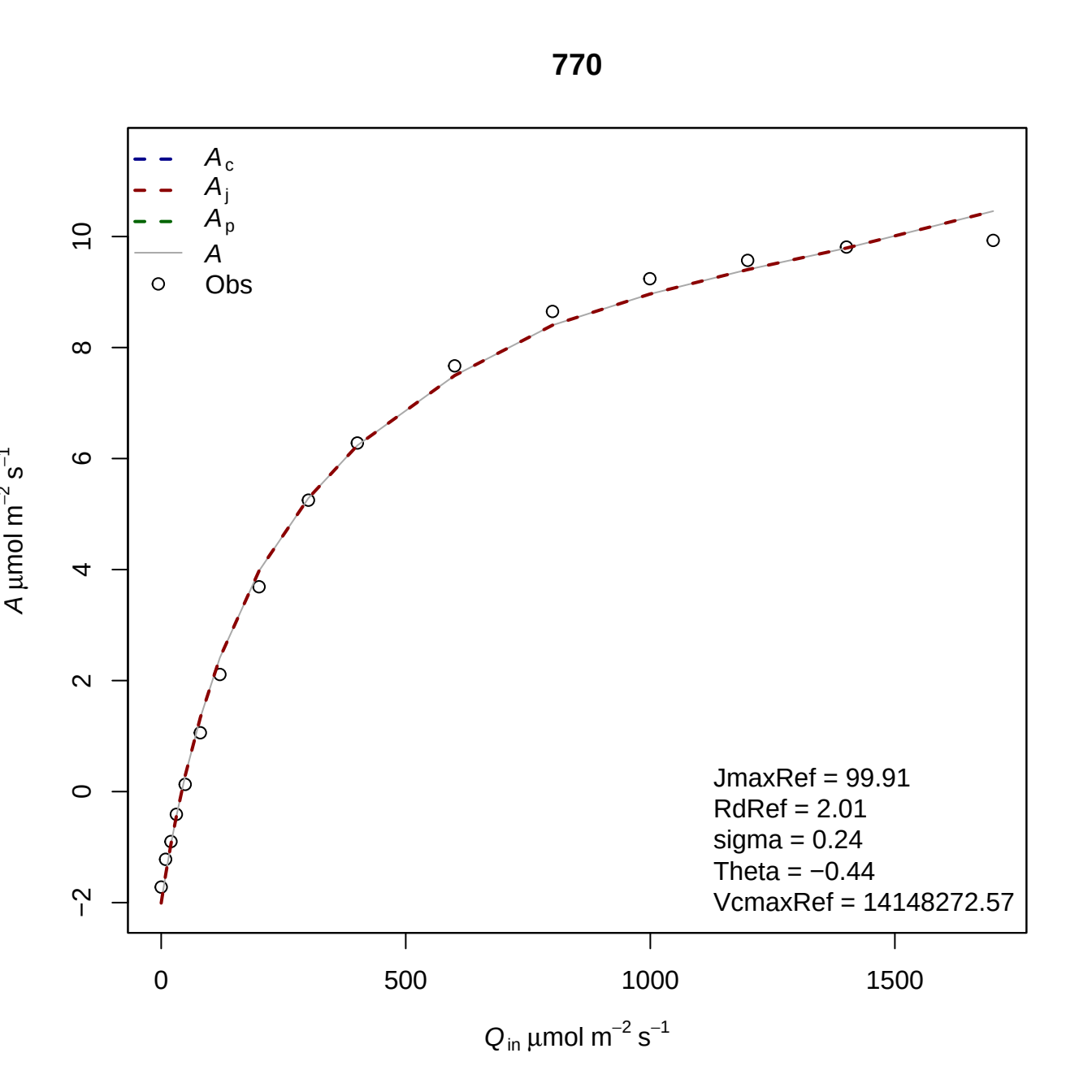

765
A_c
A_j
A_p
A
Obs
JmaxRef = 164.4
RdRef = 1.52
sigma = 0.26
Theta = −0.13
VcmaxRef = 1174632.62
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

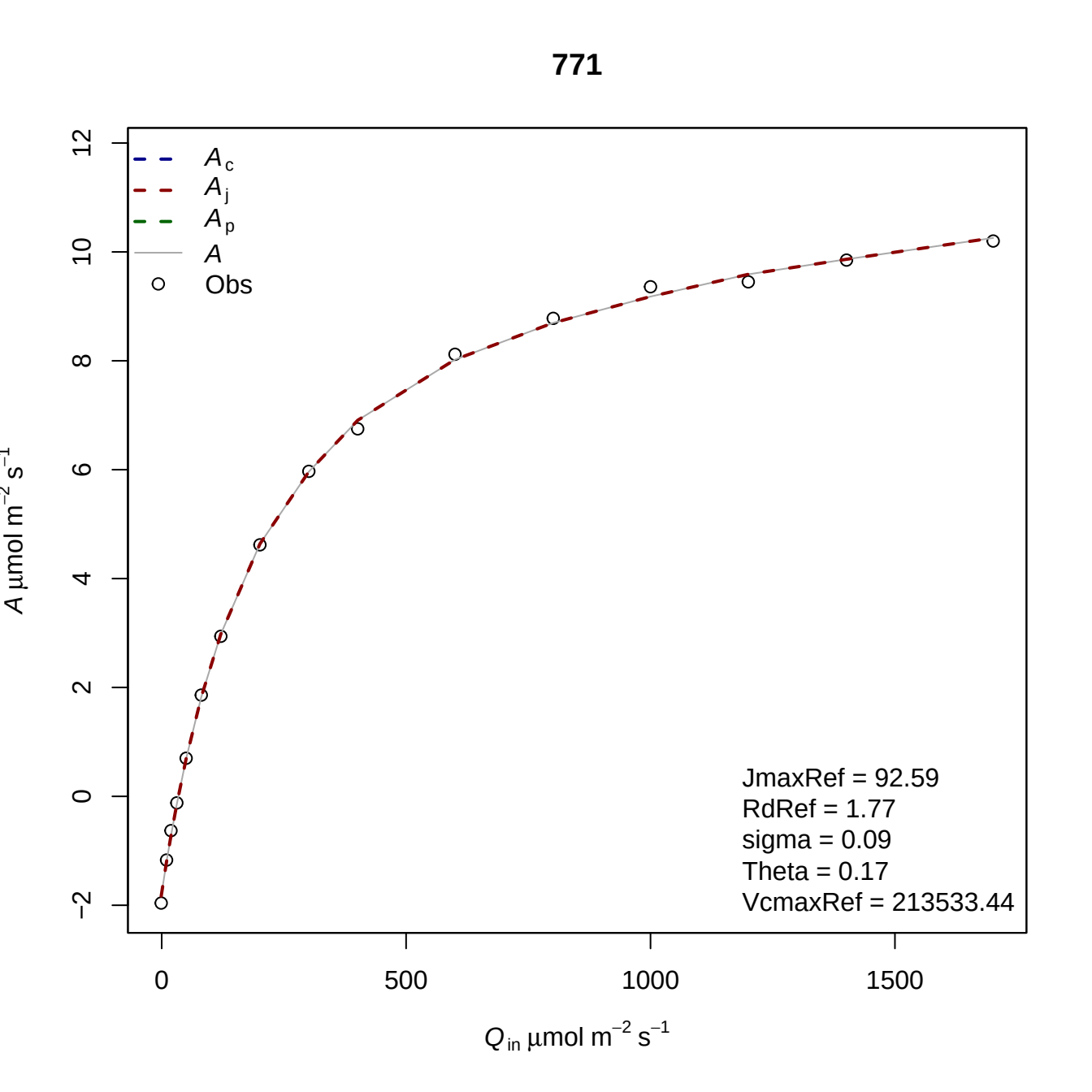

766
A_c
A_j
A_p
A
Obs
JmaxRef = 133.39
RdRef = 1.38
sigma = 0.17
Theta = 0.15
VcmaxRef = 109.32
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

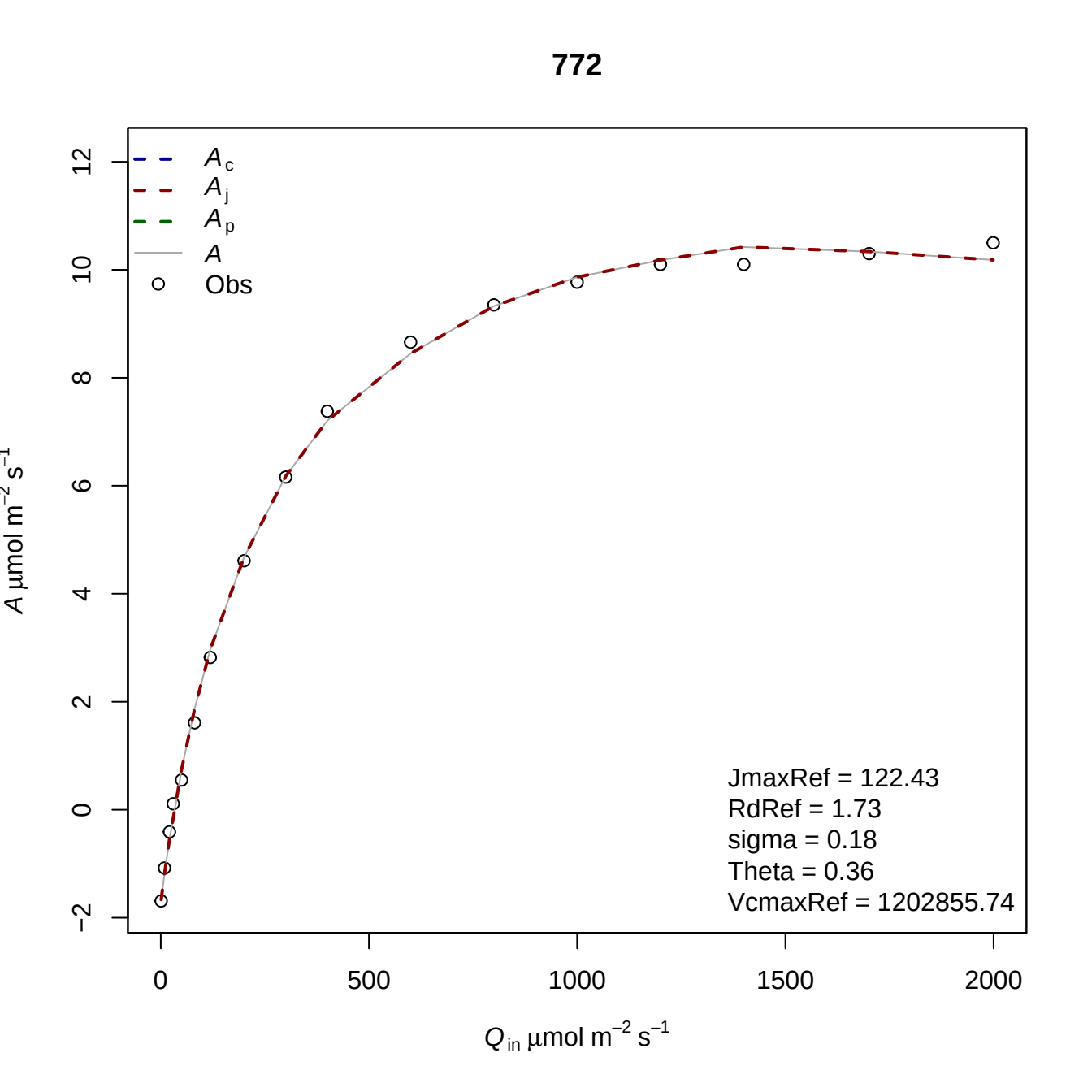

767
A_c
A_j
A_p
A
Obs
JmaxRef = 177.34
RdRef = 2.74
sigma = 0.14
Theta = 0.08
VcmaxRef = 127.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

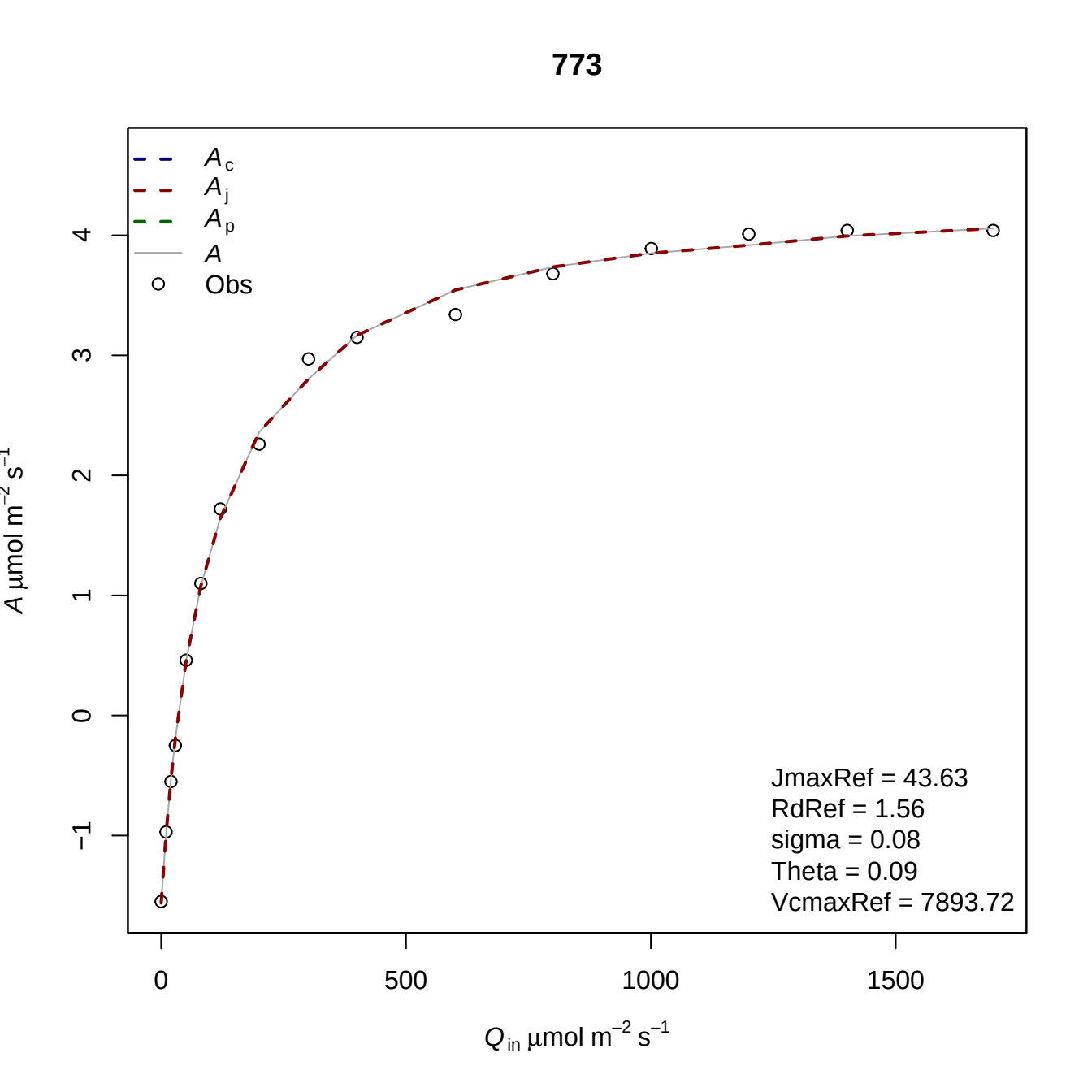

768
A_c
A_j
A_p
A
Obs
JmaxRef = 42.12
RdRef = 0.44
sigma = 0.14
Theta = −0.03
VcmaxRef = 41790.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

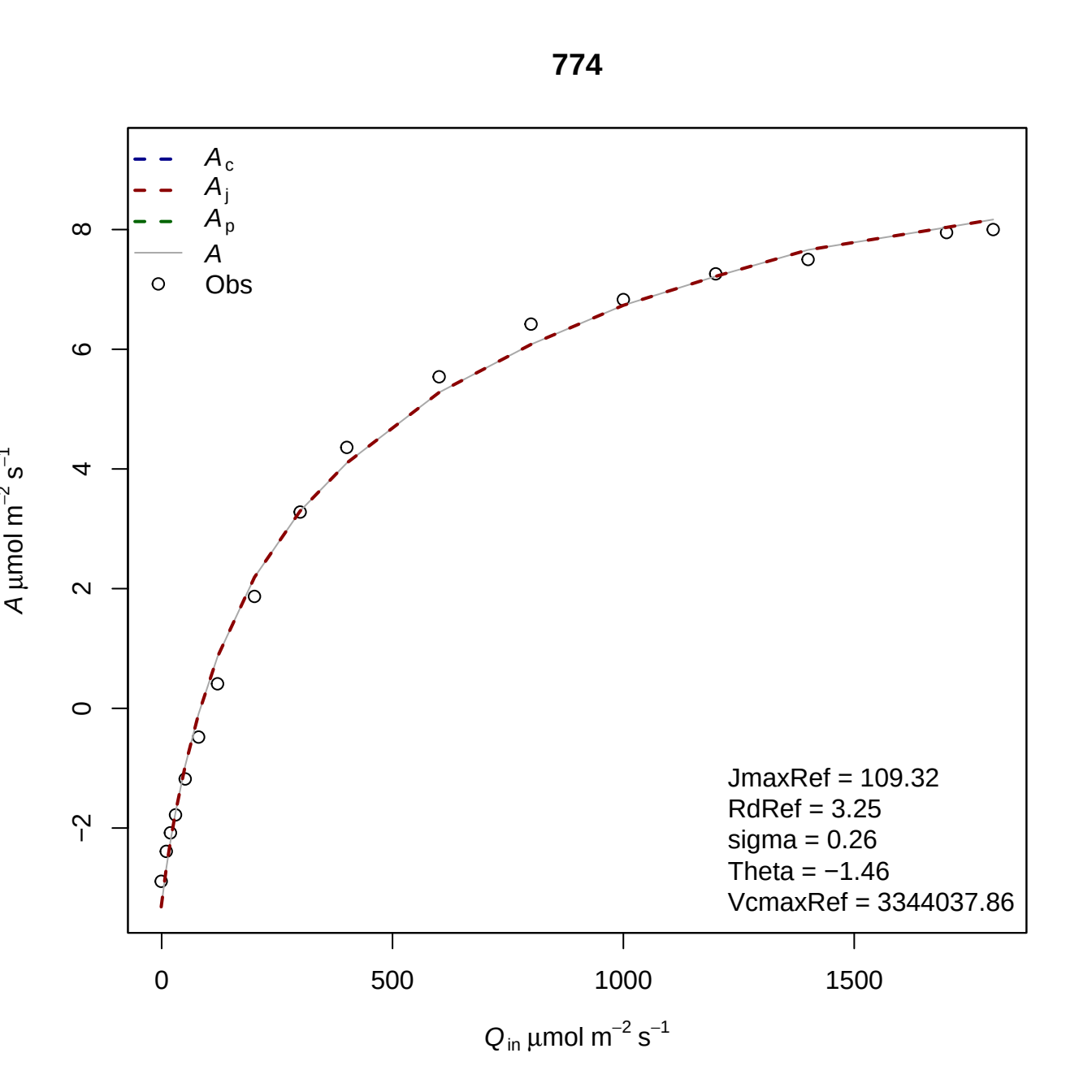

769
A_c
A_j
A_p
A
Obs
JmaxRef = 65.6
RdRef = 1.8
sigma = 0.08
Theta = −0.01
VcmaxRef = 23786.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

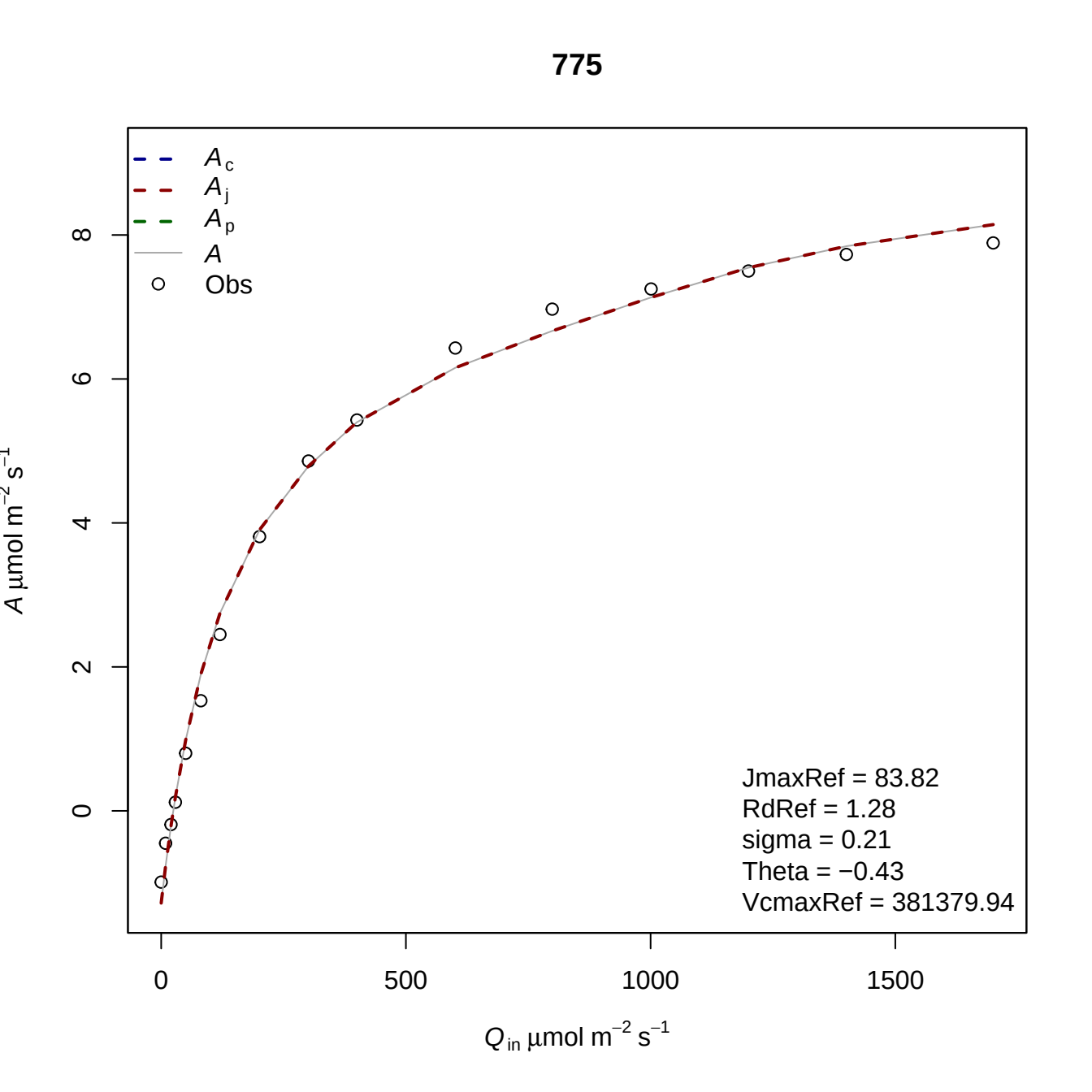

770
A_c
A_j
A_p
A
Obs
JmaxRef = 99.91
RdRef = 2.01
sigma = 0.24
Theta = −0.44
VcmaxRef = 14148272.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

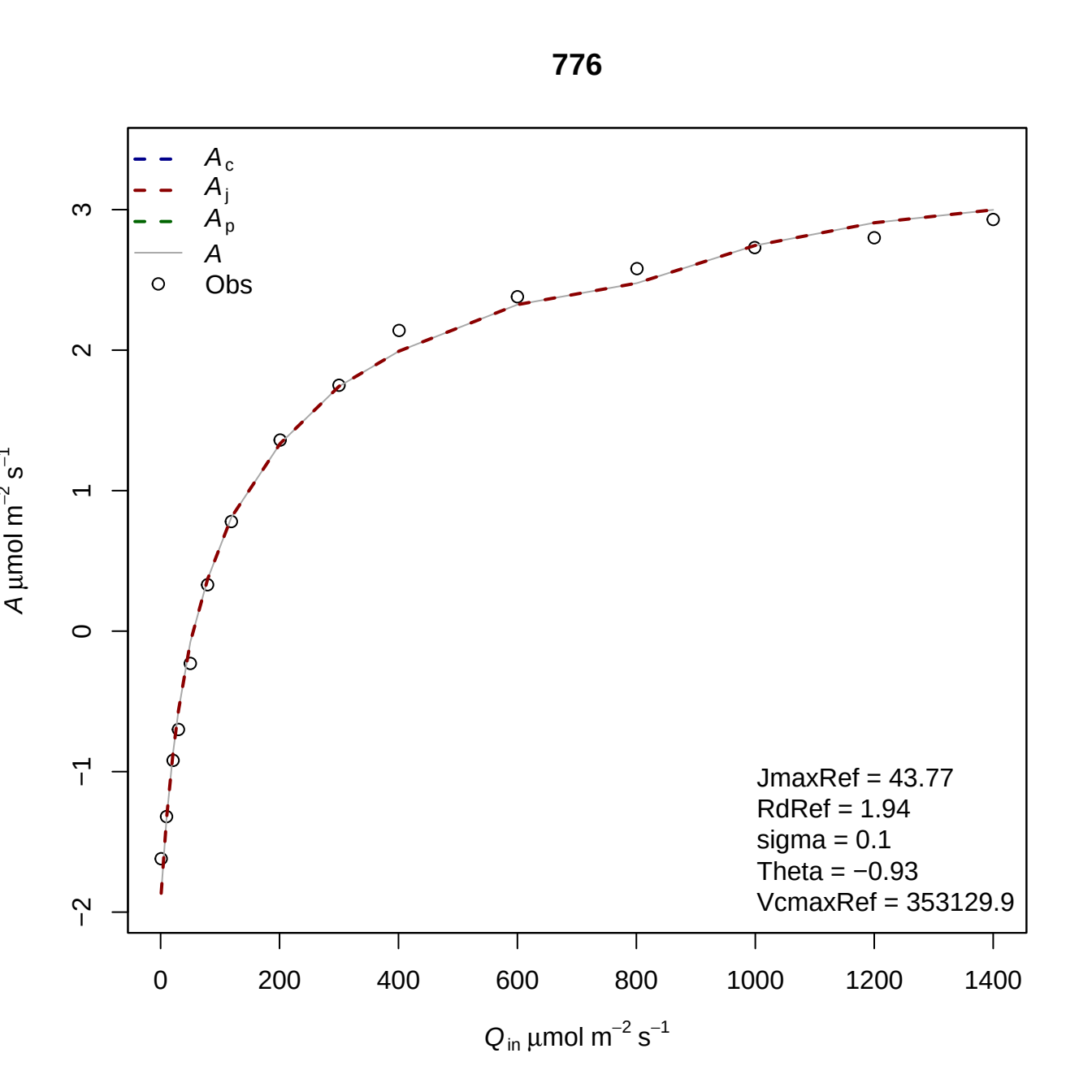

771
A_c
A_j
A_p
A
Obs
JmaxRef = 92.59
RdRef = 1.77
sigma = 0.09
Theta = 0.17
VcmaxRef = 213533.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

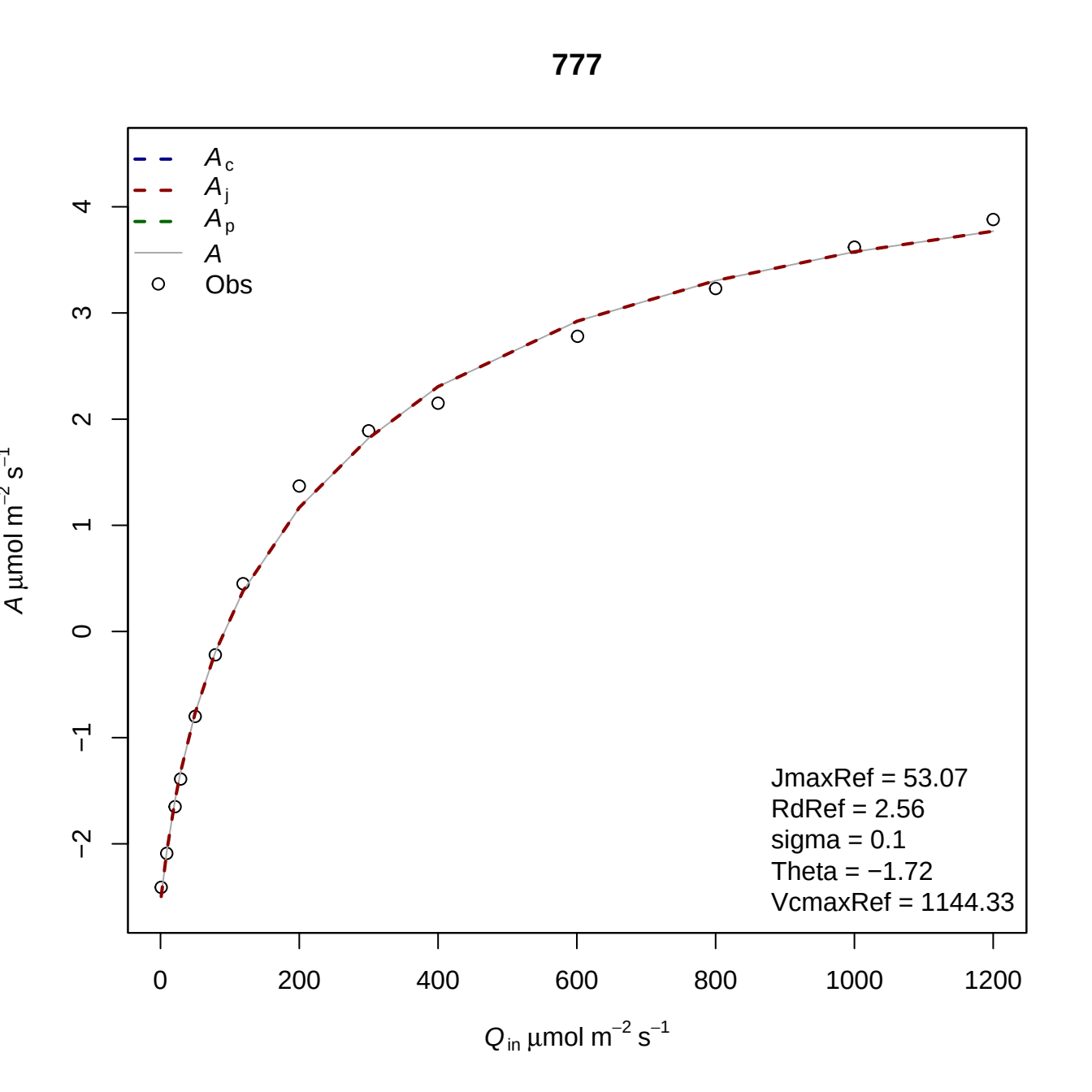

772
A_c
A_j
A_p
A
Obs
JmaxRef = 122.43
RdRef = 1.73
sigma = 0.18
Theta = 0.36
VcmaxRef = 1202855.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

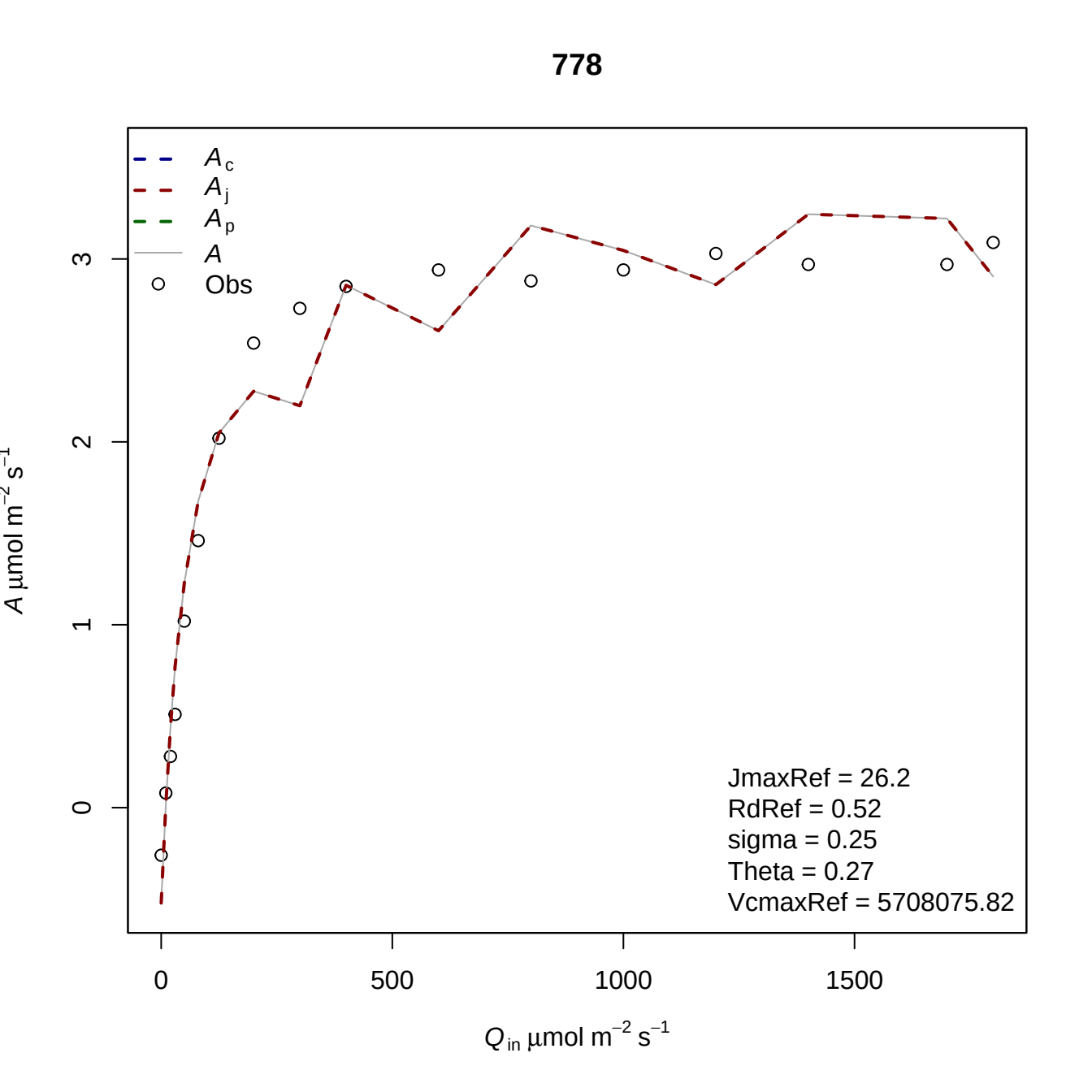

773
A_c
A_j
A_p
A
Obs
JmaxRef = 43.63
RdRef = 1.56
sigma = 0.08
Theta = 0.09
VcmaxRef = 7893.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

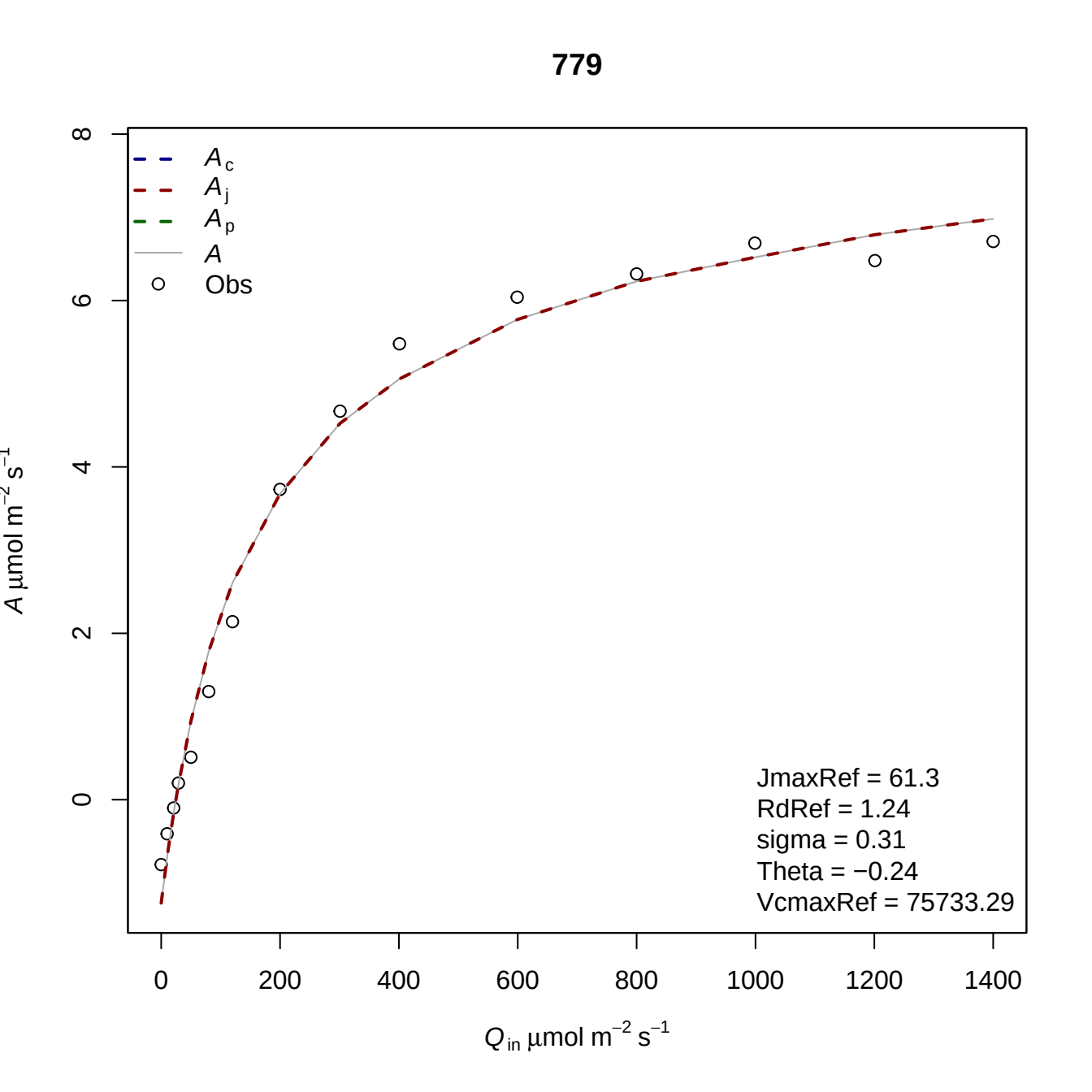

774
A_c
A_j
A_p
A
Obs
JmaxRef = 109.32
RdRef = 3.25
sigma = 0.26
Theta = −1.46
VcmaxRef = 3344037.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

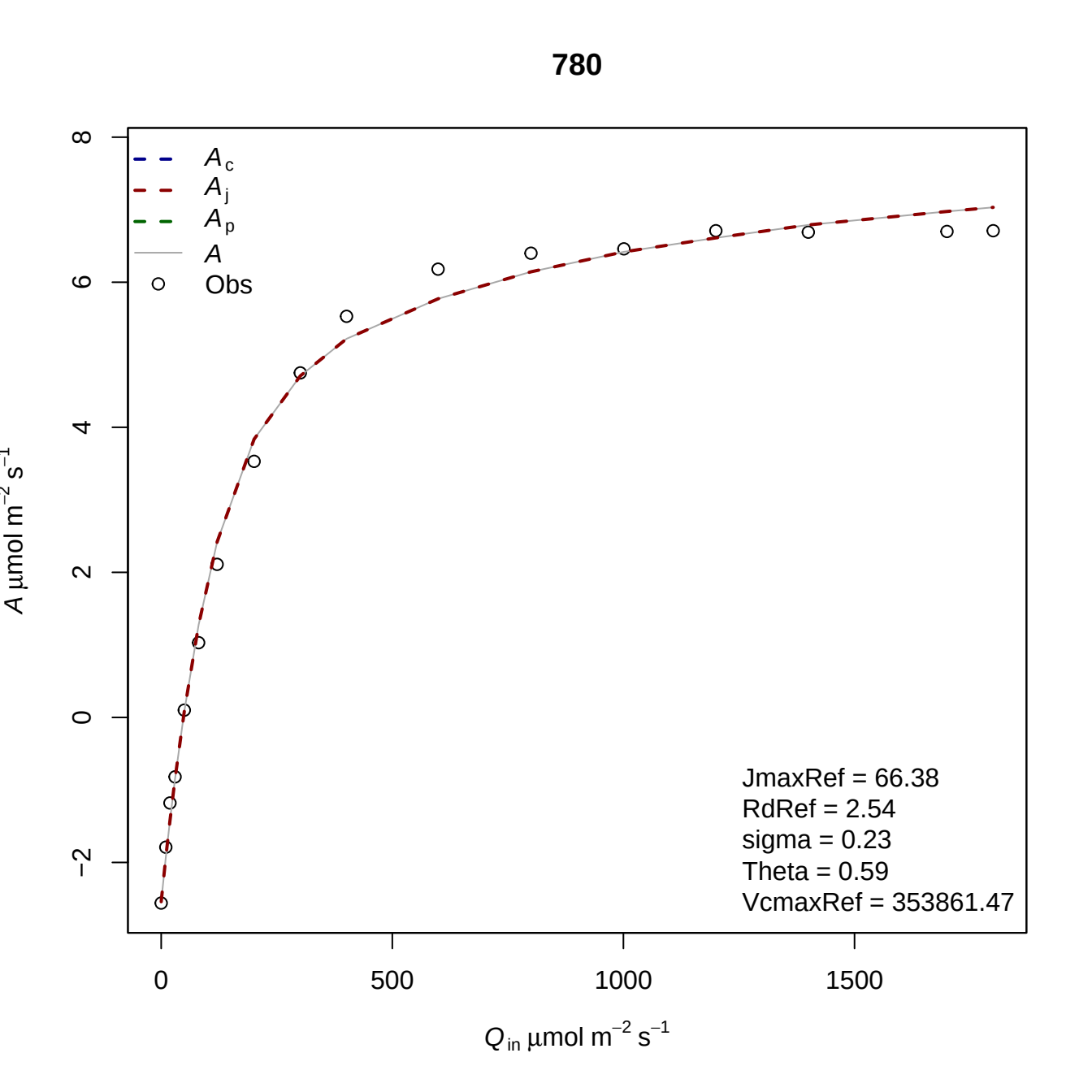

775
A_c
A_j
A_p
A
Obs
JmaxRef = 83.82
RdRef = 1.28
sigma = 0.21
Theta = −0.43
VcmaxRef = 381379.94
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

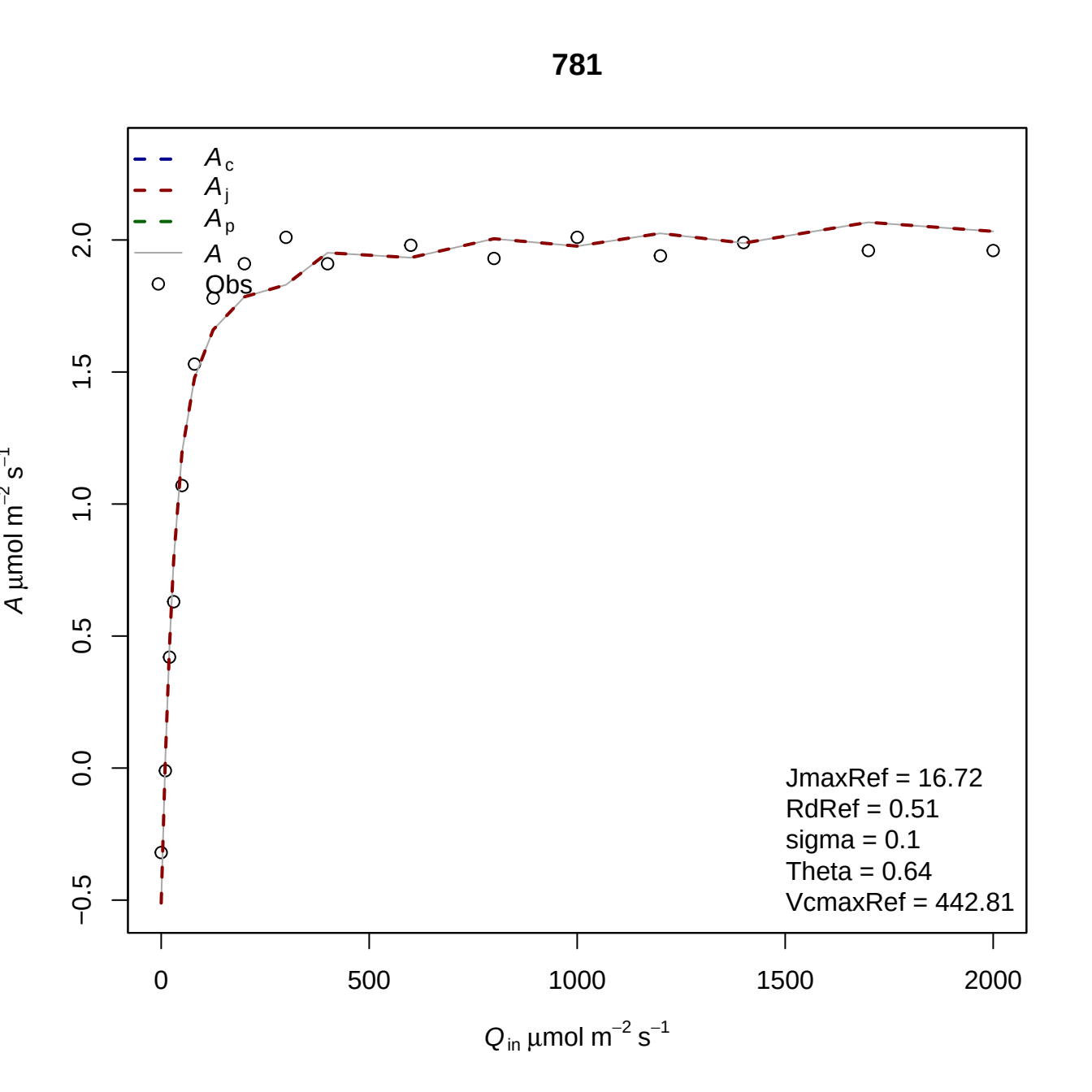

776
A_c
A_j
A_p
A
Obs
JmaxRef = 43.77
RdRef = 1.94
sigma = 0.1
Theta = −0.93
VcmaxRef = 353129.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

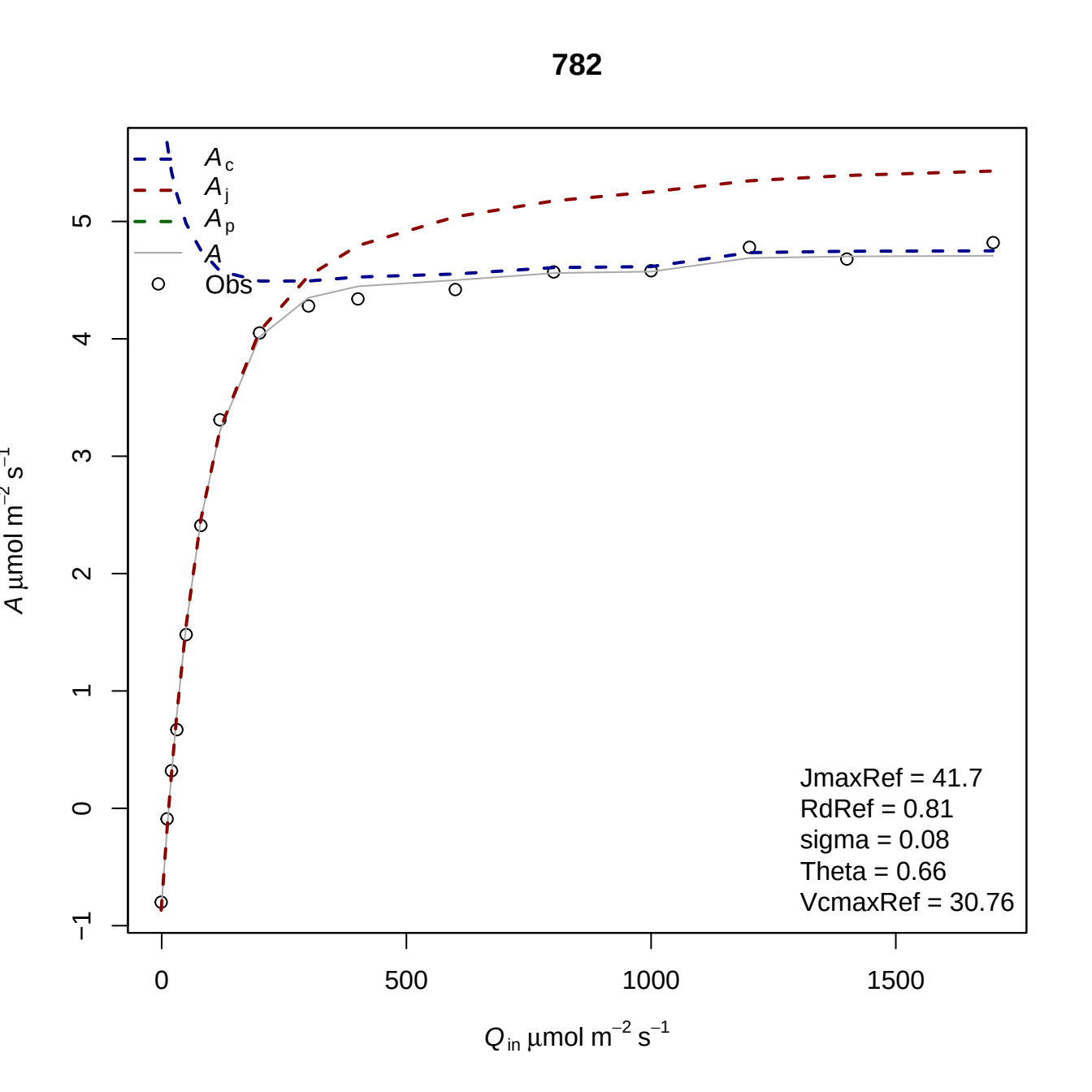

777
A_c
A_j
A_p
A
Obs
JmaxRef = 53.07
RdRef = 2.56
sigma = 0.1
Theta = −1.72
VcmaxRef = 1144.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200

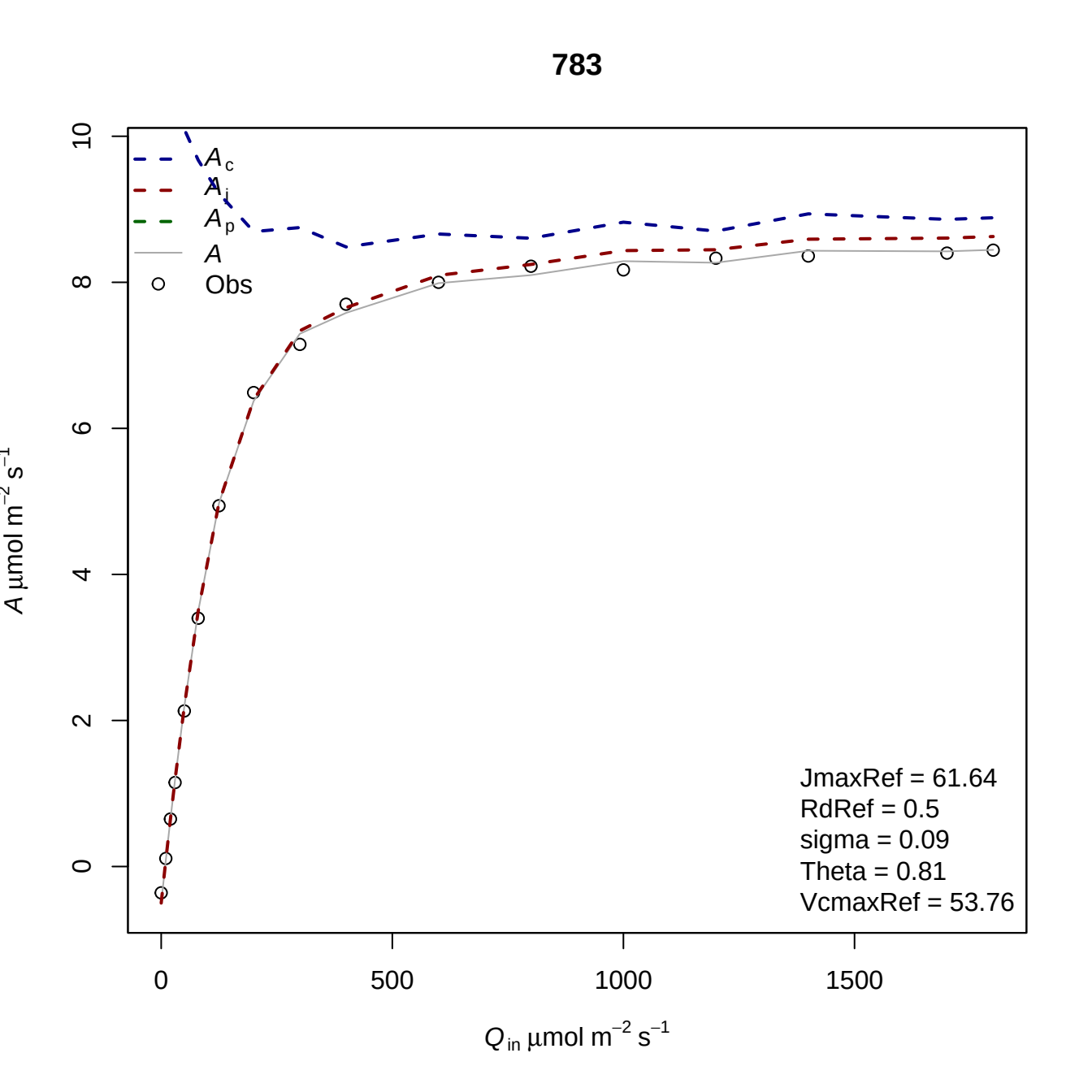

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.2
RdRef = 0.52
sigma = 0.25
Theta = 0.27
VcmaxRef = 5708075.82
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

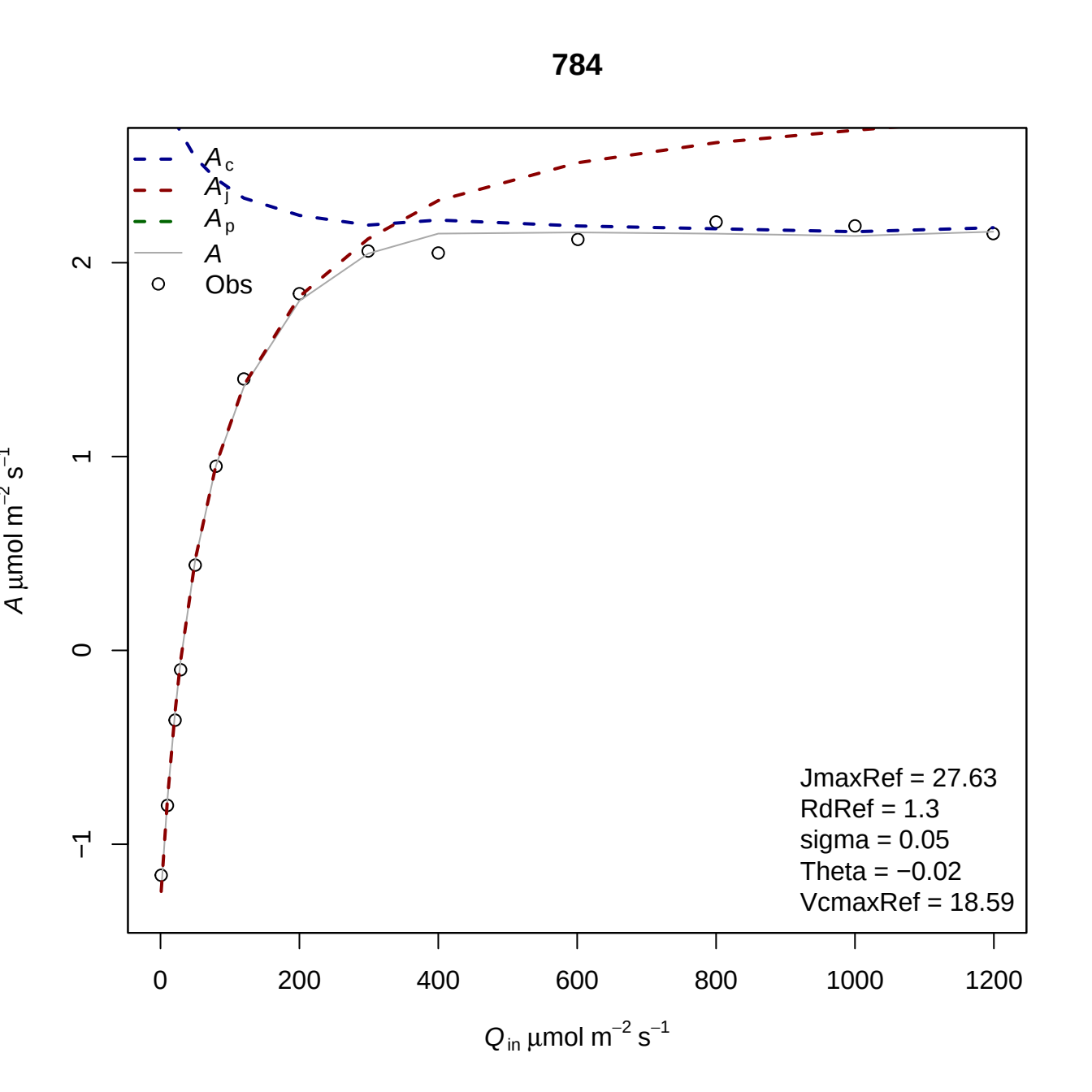

779
A_c
A_j
A_p
A
Obs
JmaxRef = 61.3
RdRef = 1.24
sigma = 0.31
Theta = −0.24
VcmaxRef = 75733.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

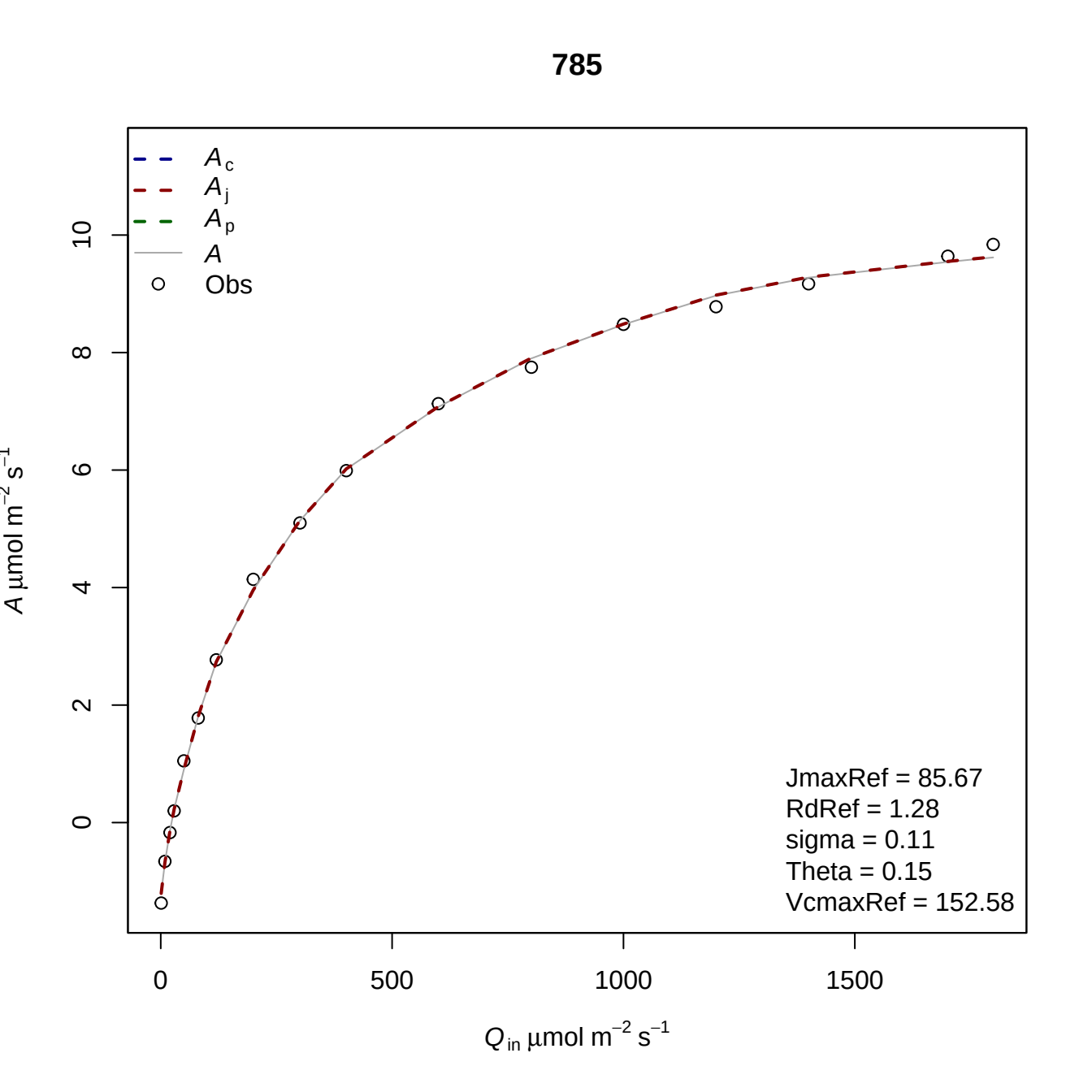

780
A_c
A_j
A_p
A
Obs
JmaxRef = 66.38
RdRef = 2.54
sigma = 0.23
Theta = 0.59
VcmaxRef = 353861.47
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

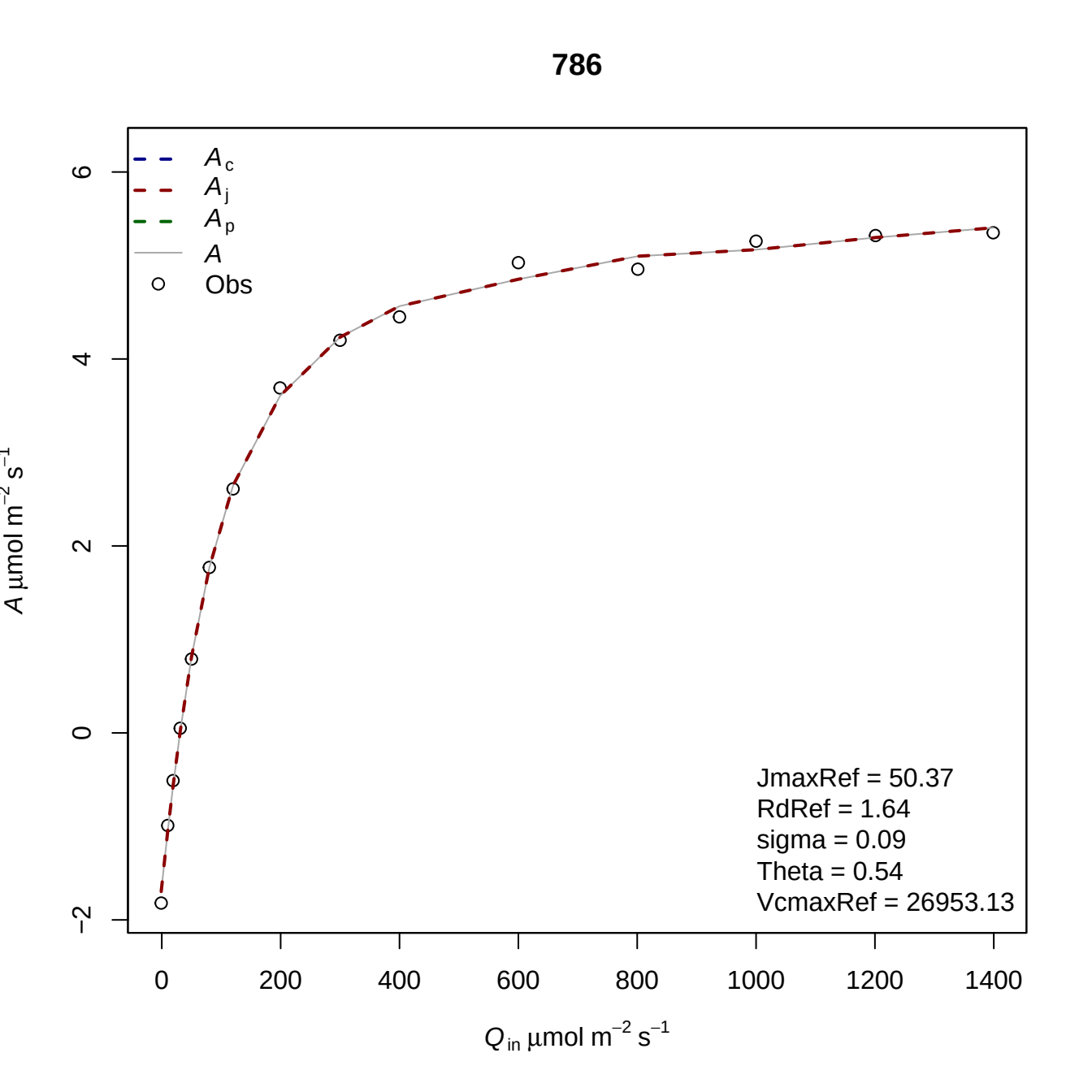

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.72
RdRef = 0.51
sigma = 0.1
Theta = 0.64
VcmaxRef = 442.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

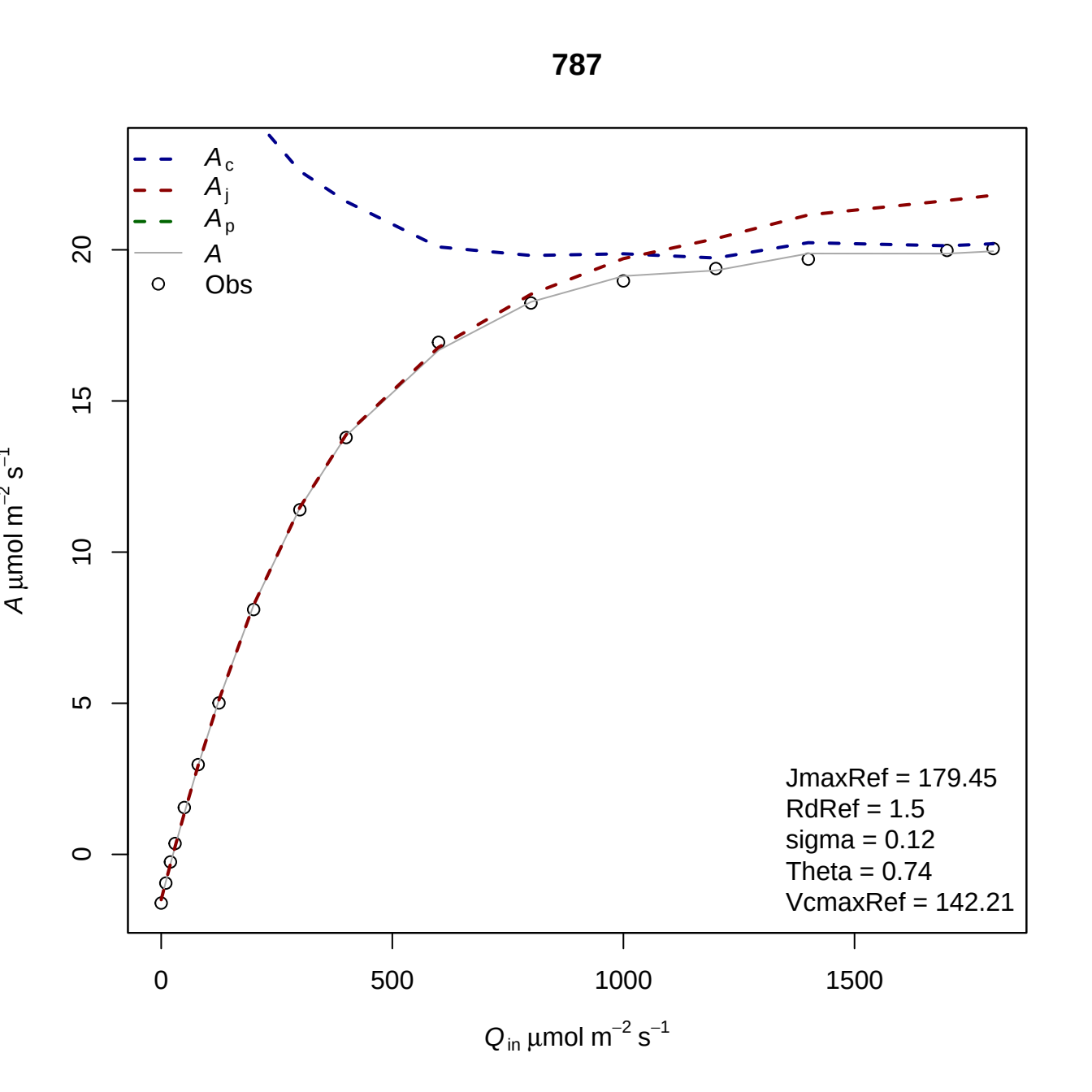

782
A_c
A_j
A_p
A
Obs
JmaxRef = 41.7
RdRef = 0.81
sigma = 0.08
Theta = 0.66
VcmaxRef = 30.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

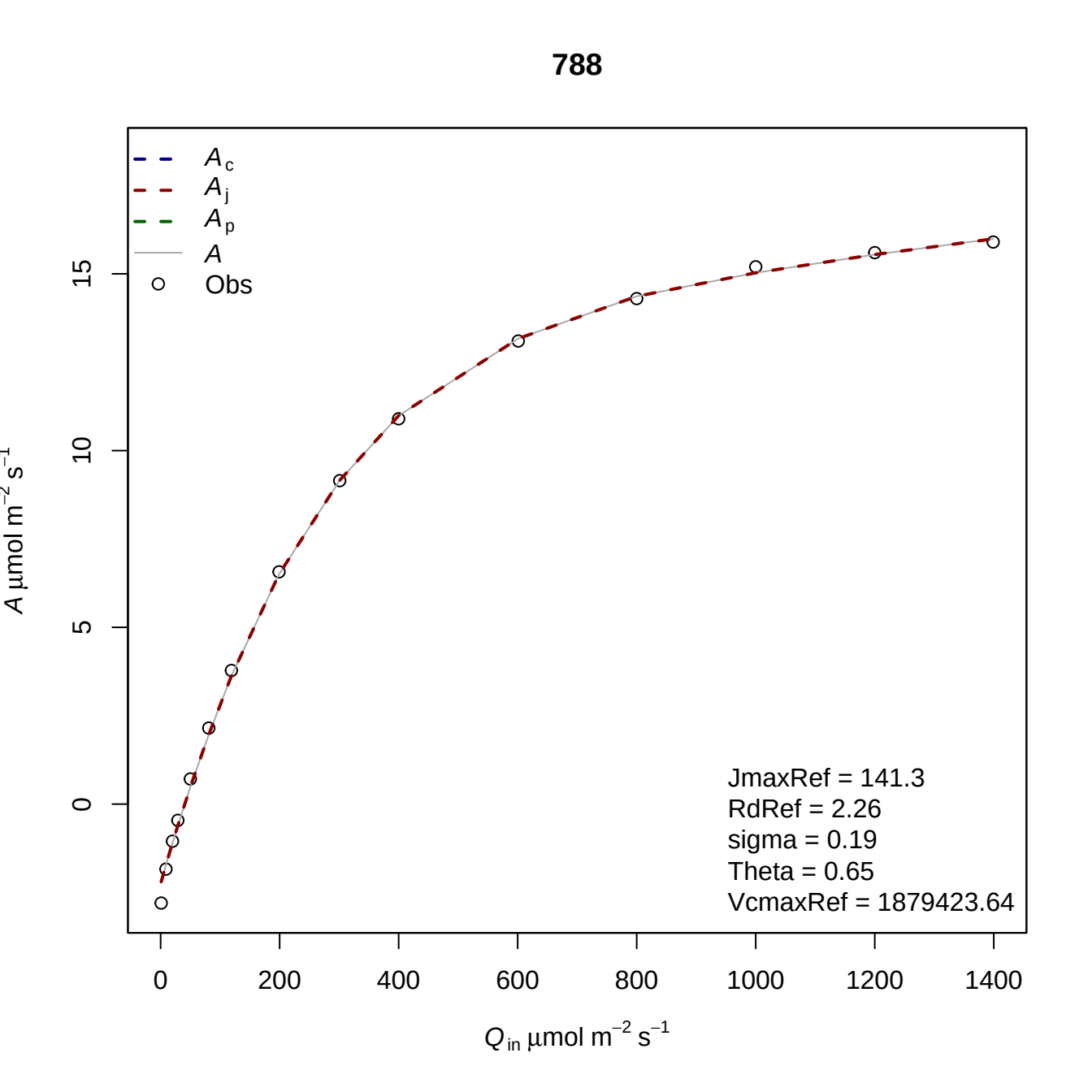

783
A_c
A_j
A_p
A
Obs
JmaxRef = 61.64
RdRef = 0.5
sigma = 0.09
Theta = 0.81
VcmaxRef = 53.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

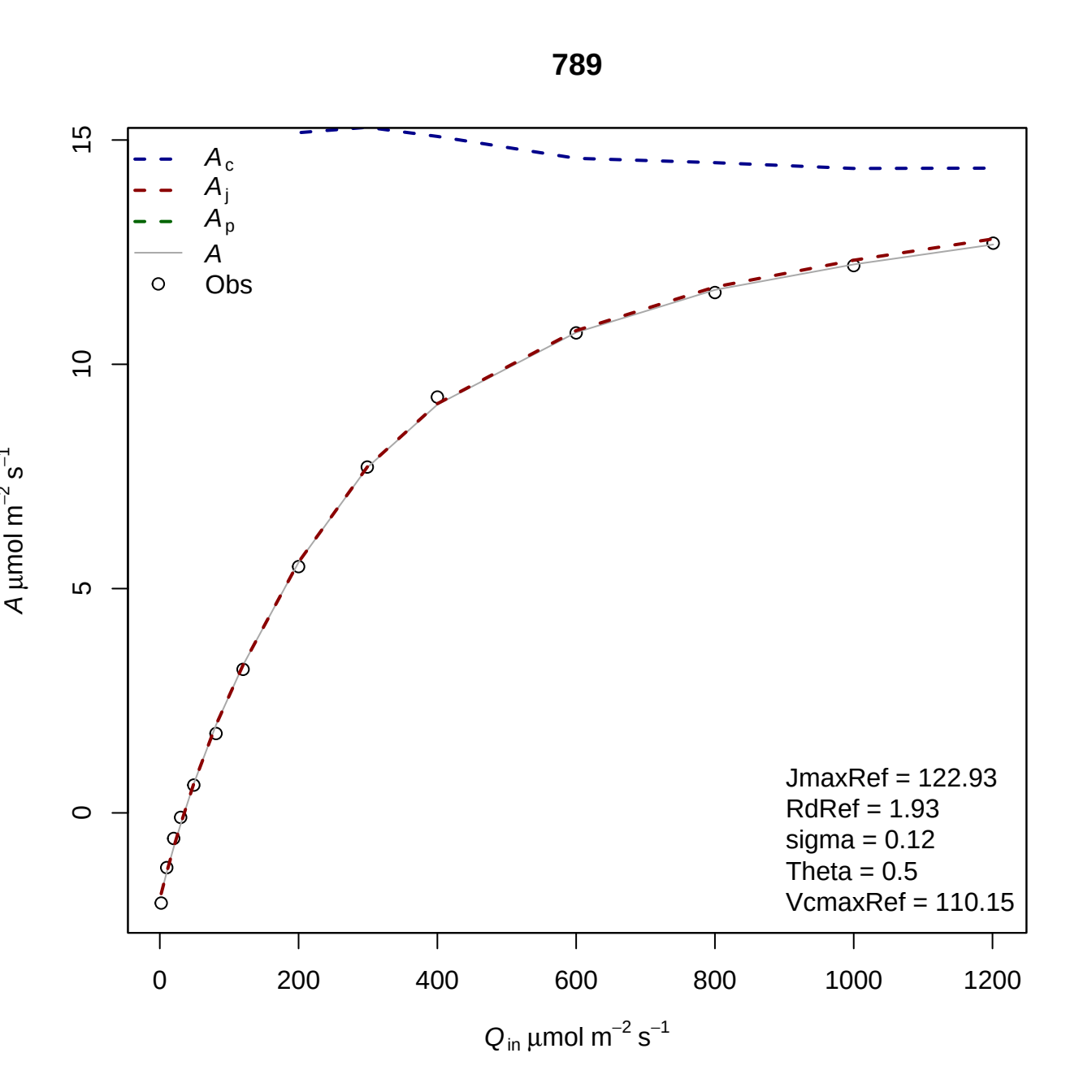

784
A_c
A_j
A_p
A
Obs
JmaxRef = 27.63
RdRef = 1.3
sigma = 0.05
Theta = −0.02
VcmaxRef = 18.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

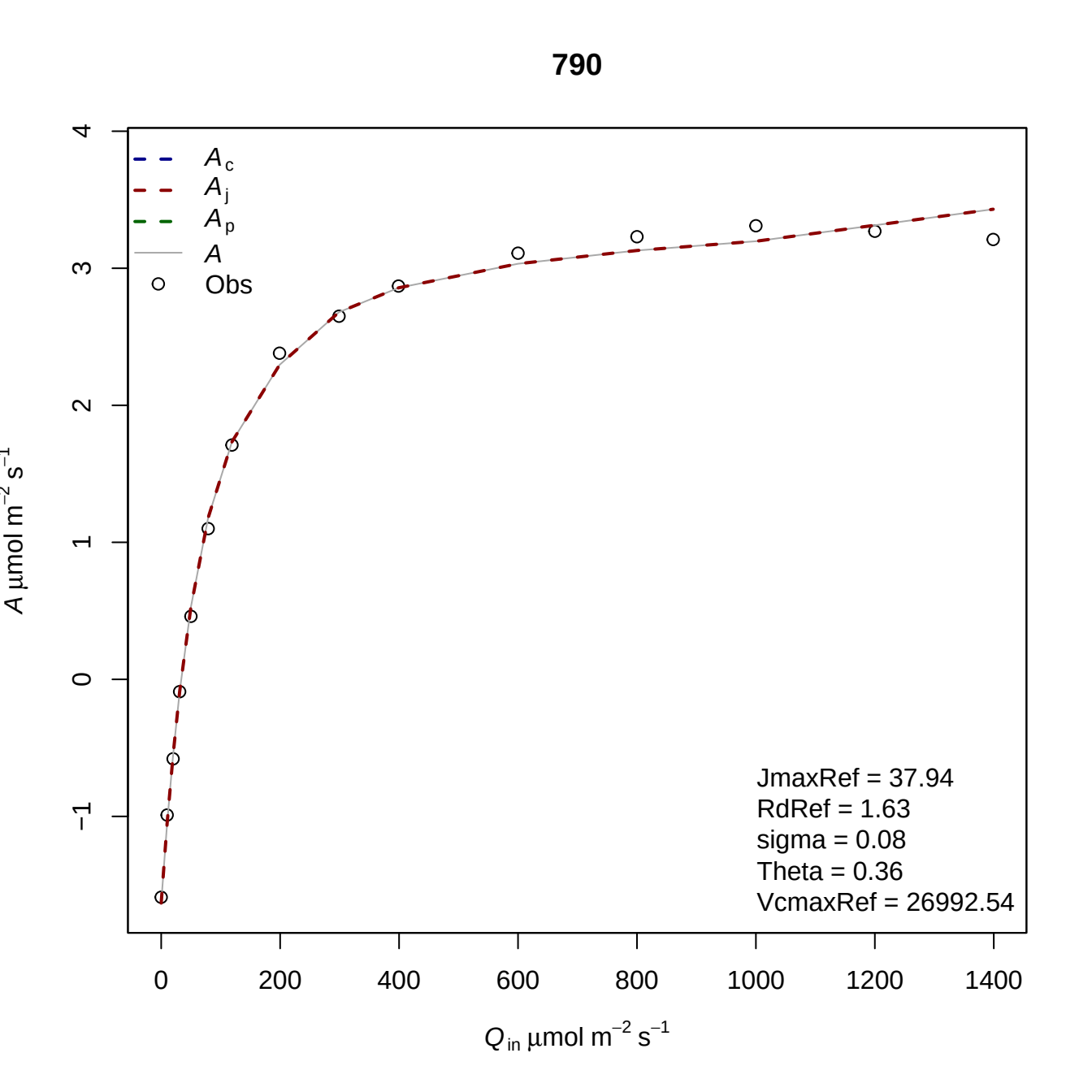

785
A_c
A_j
A_p
A
Obs
JmaxRef = 85.67
RdRef = 1.28
sigma = 0.11
Theta = 0.15
VcmaxRef = 152.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

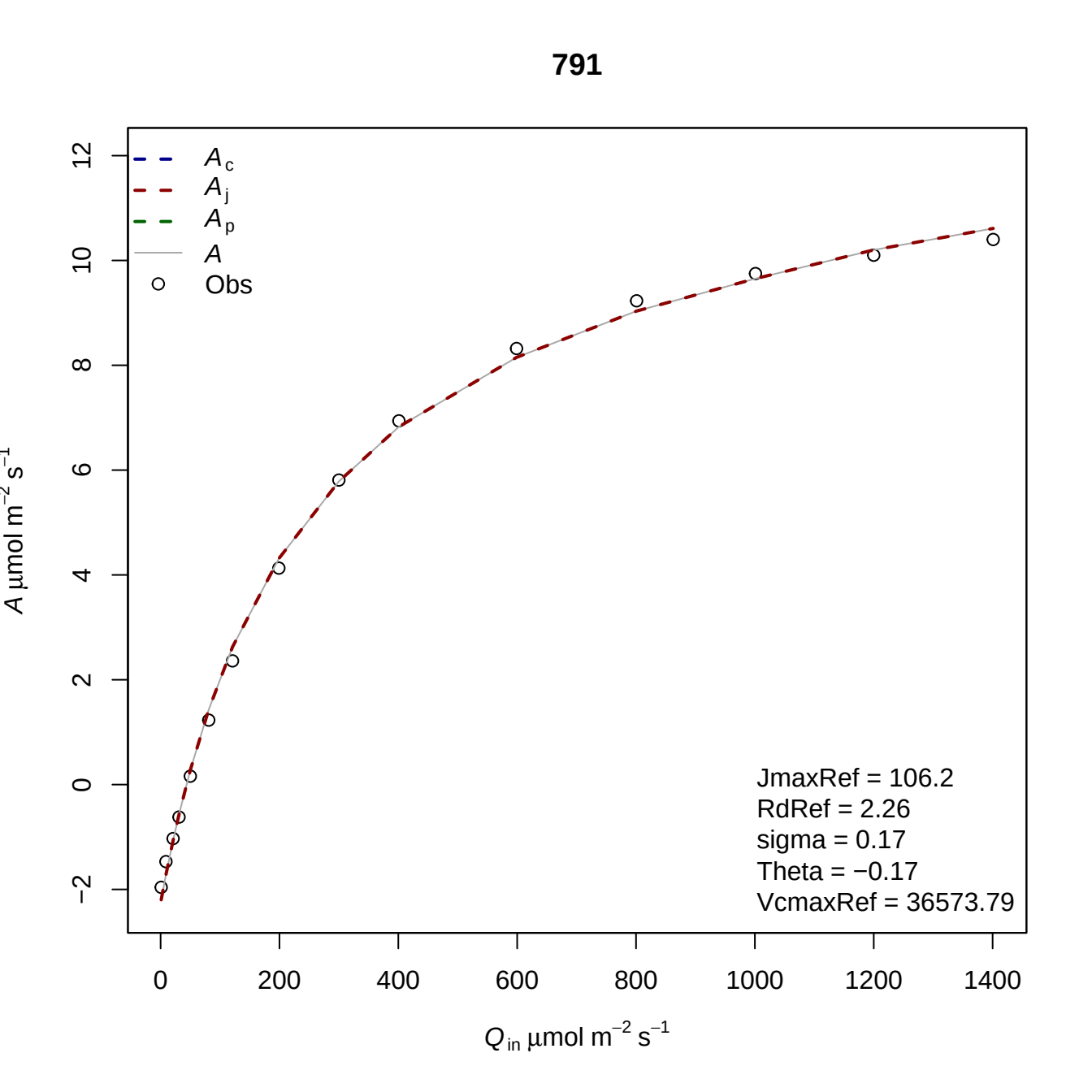

786
A_c
A_j
A_p
A
Obs
JmaxRef = 50.37
RdRef = 1.64
sigma = 0.09
Theta = 0.54
VcmaxRef = 26953.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

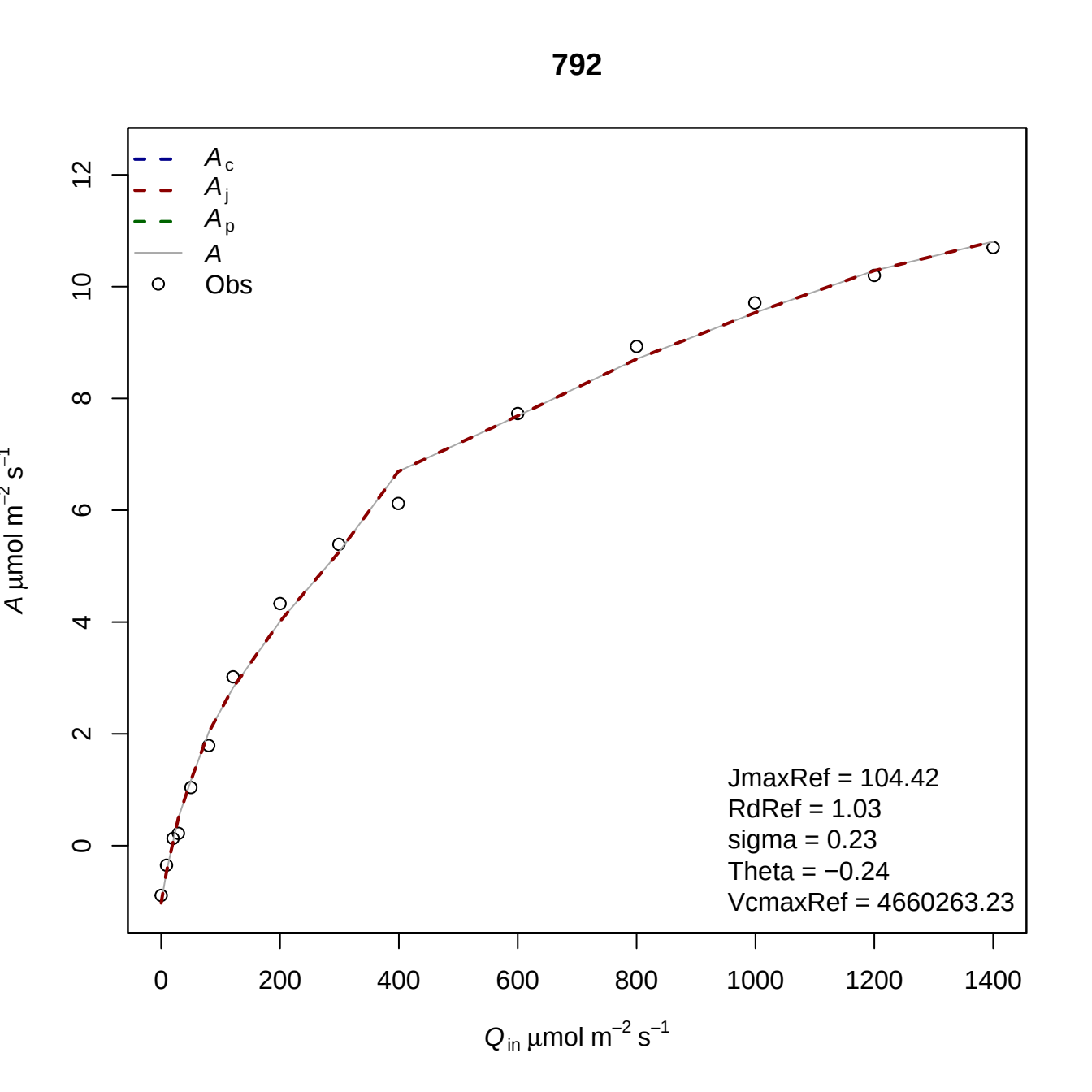

787
A_c
A_j
A_p
A
Obs
JmaxRef = 179.45
RdRef = 1.5
sigma = 0.12
Theta = 0.74
VcmaxRef = 142.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

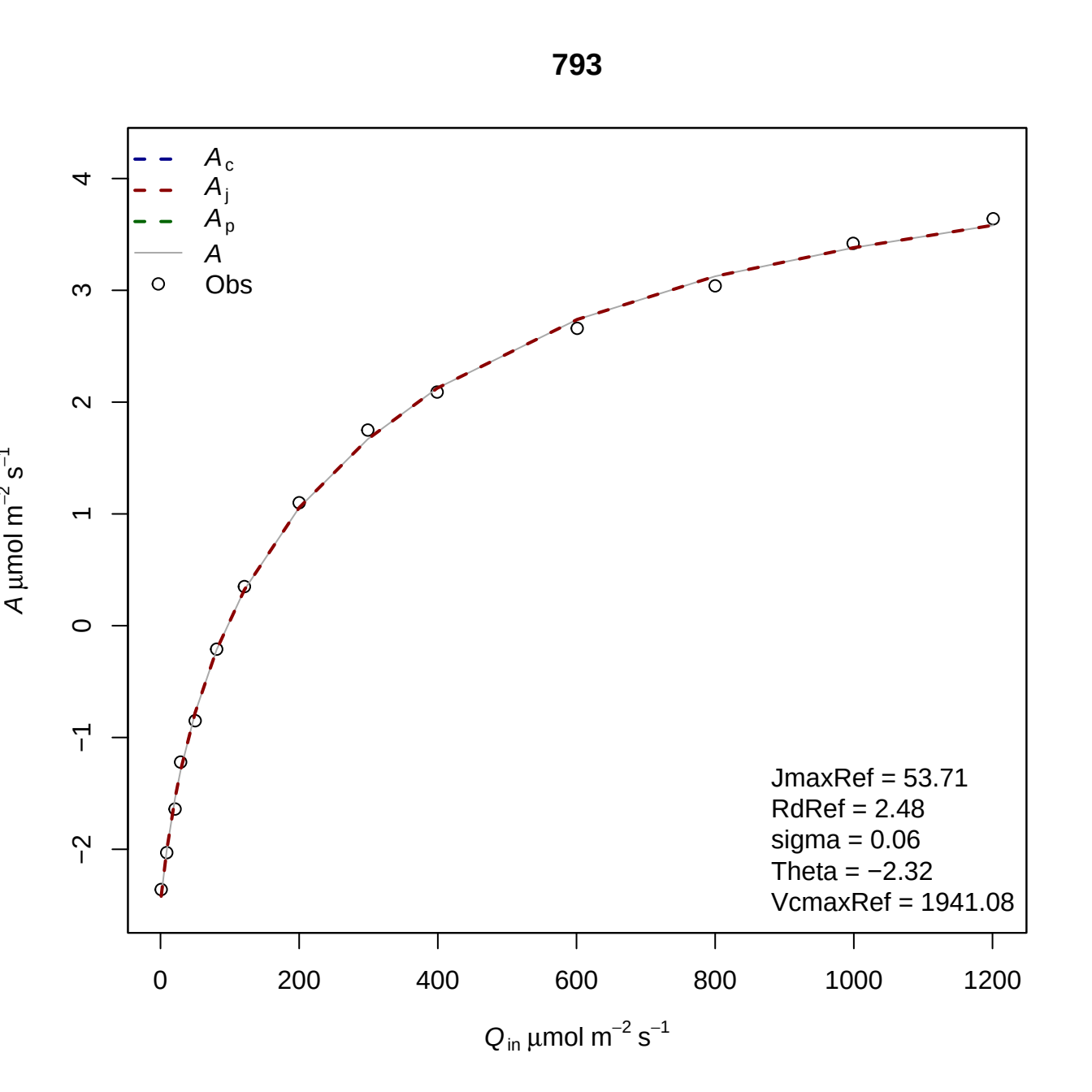

788
A_c
A_j
A_p
A
Obs
JmaxRef = 141.3
RdRef = 2.26
sigma = 0.19
Theta = 0.65
VcmaxRef = 1879423.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

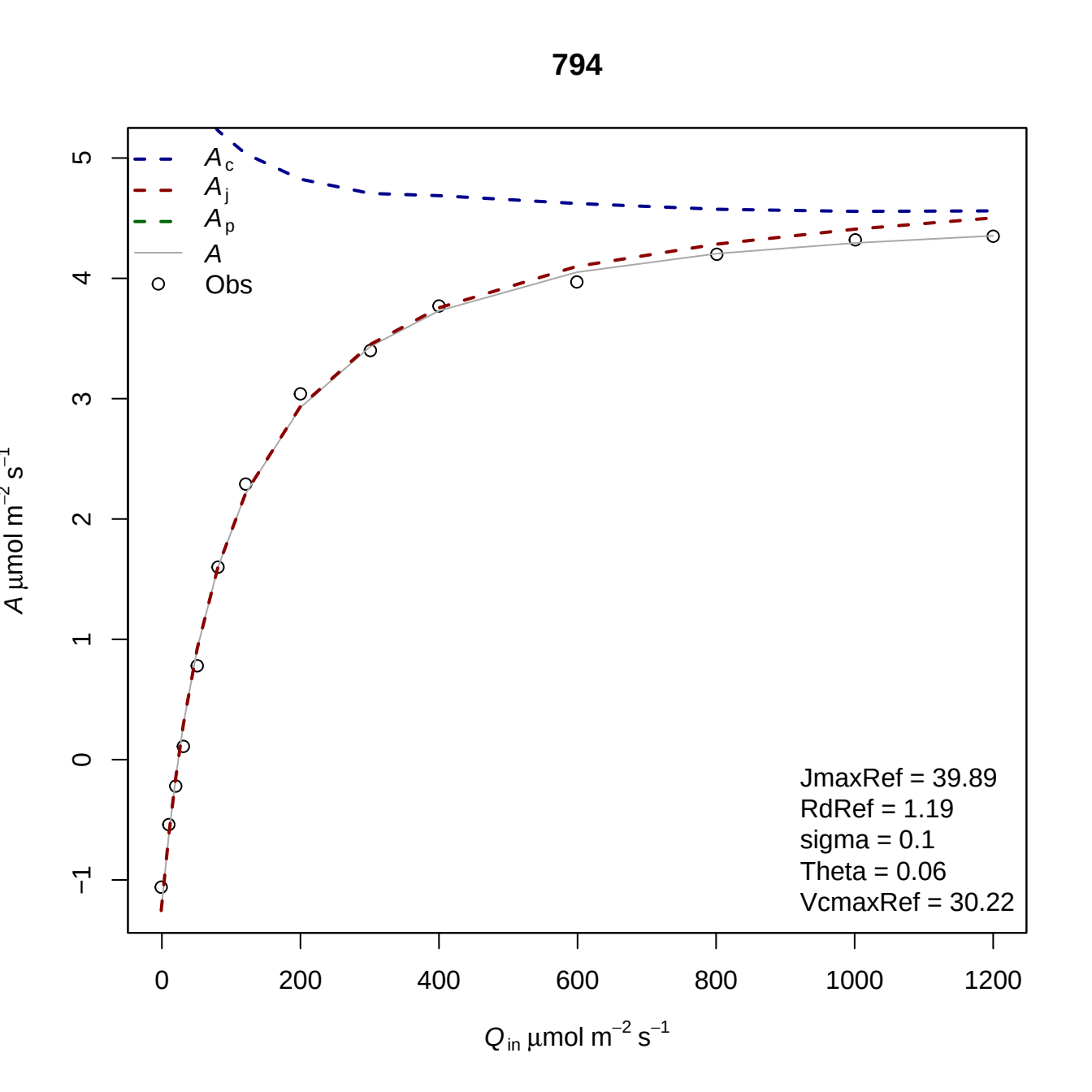

789
A_c
A_j
A_p
A
Obs
JmaxRef = 122.93
RdRef = 1.93
sigma = 0.12
Theta = 0.5
VcmaxRef = 110.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

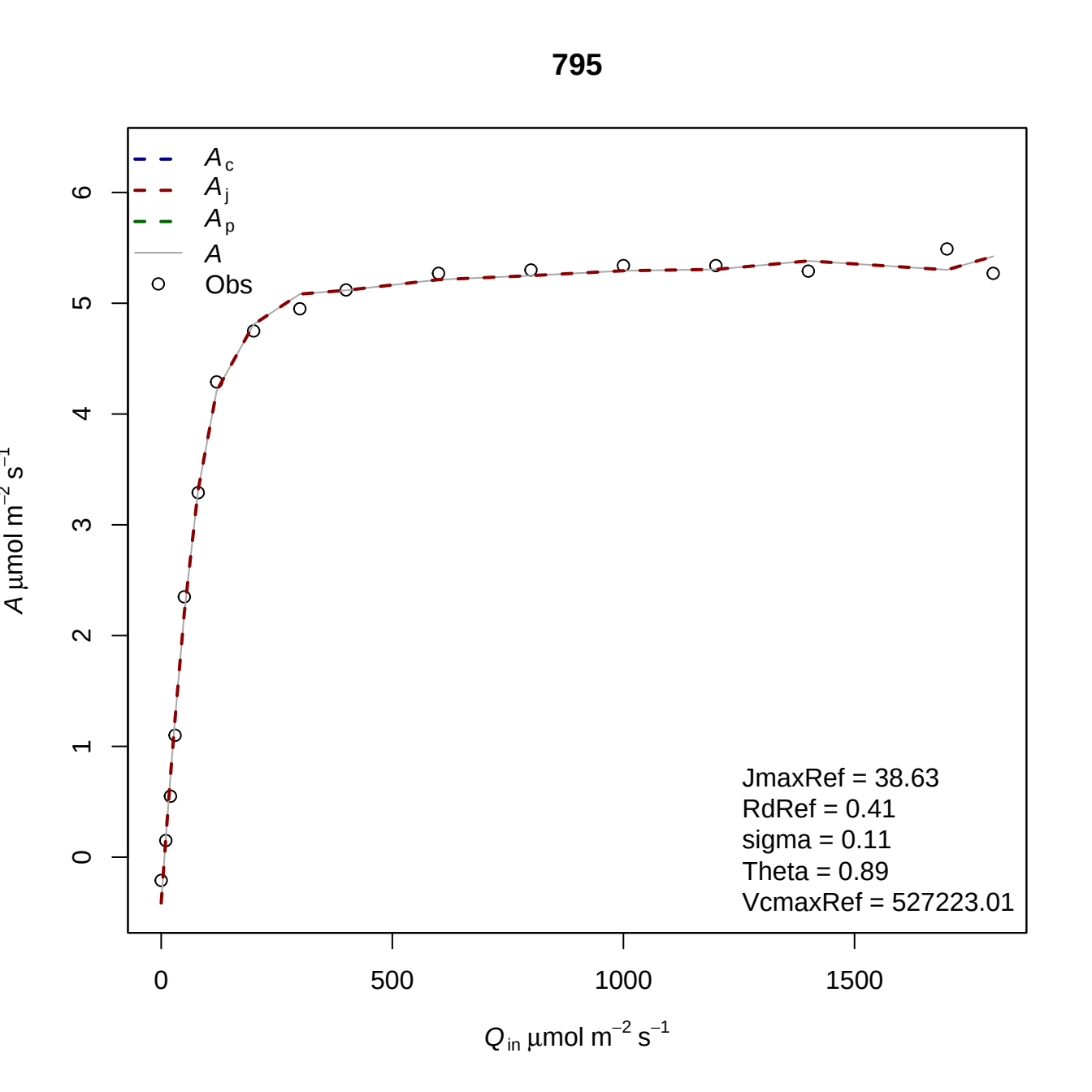

790
A_c
A_j
A_p
A
Obs
JmaxRef = 37.94
RdRef = 1.63
sigma = 0.08
Theta = 0.36
VcmaxRef = 26992.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

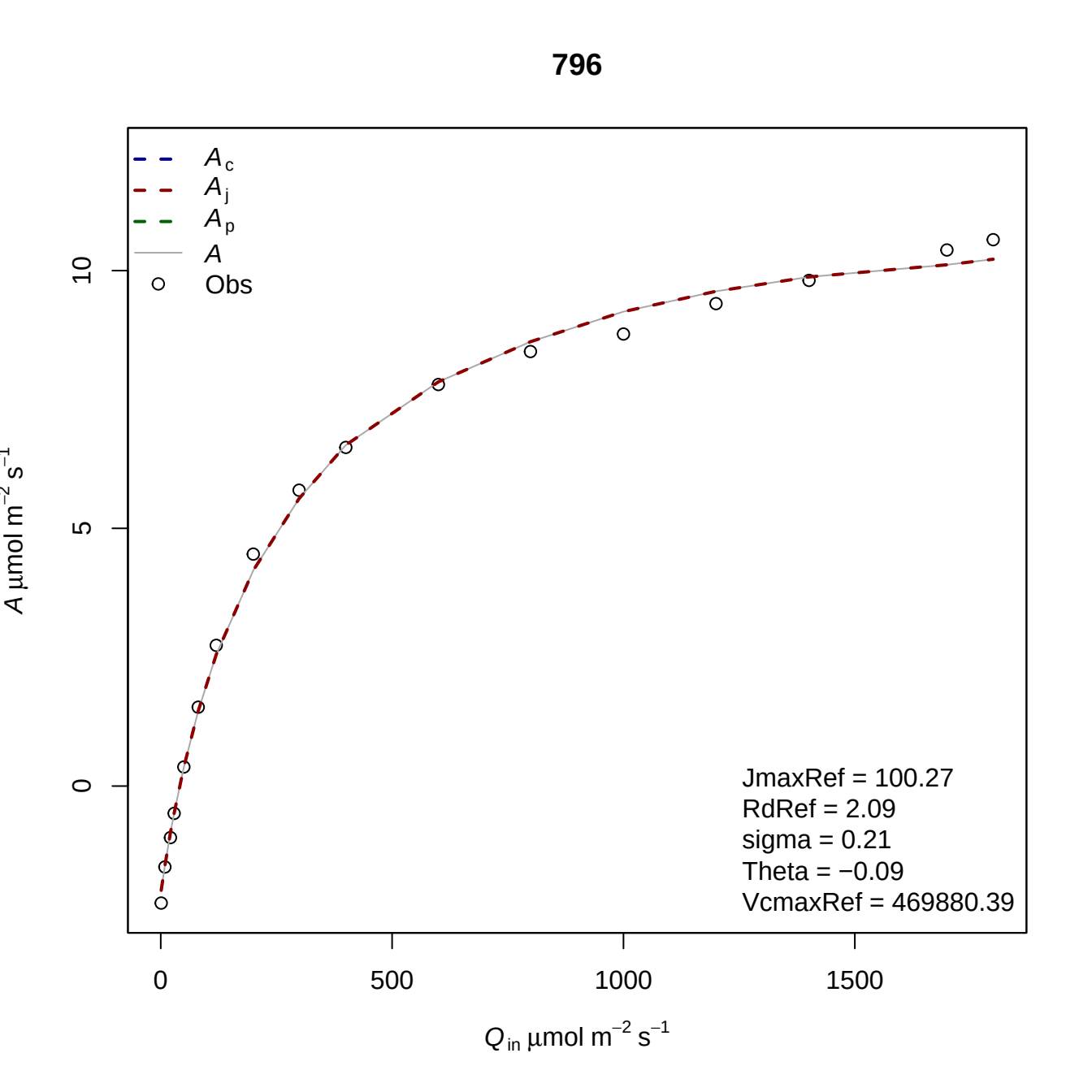

791
A_c
A_j
A_p
A
Obs
JmaxRef = 106.2
RdRef = 2.26
sigma = 0.17
Theta = −0.17
VcmaxRef = 36573.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

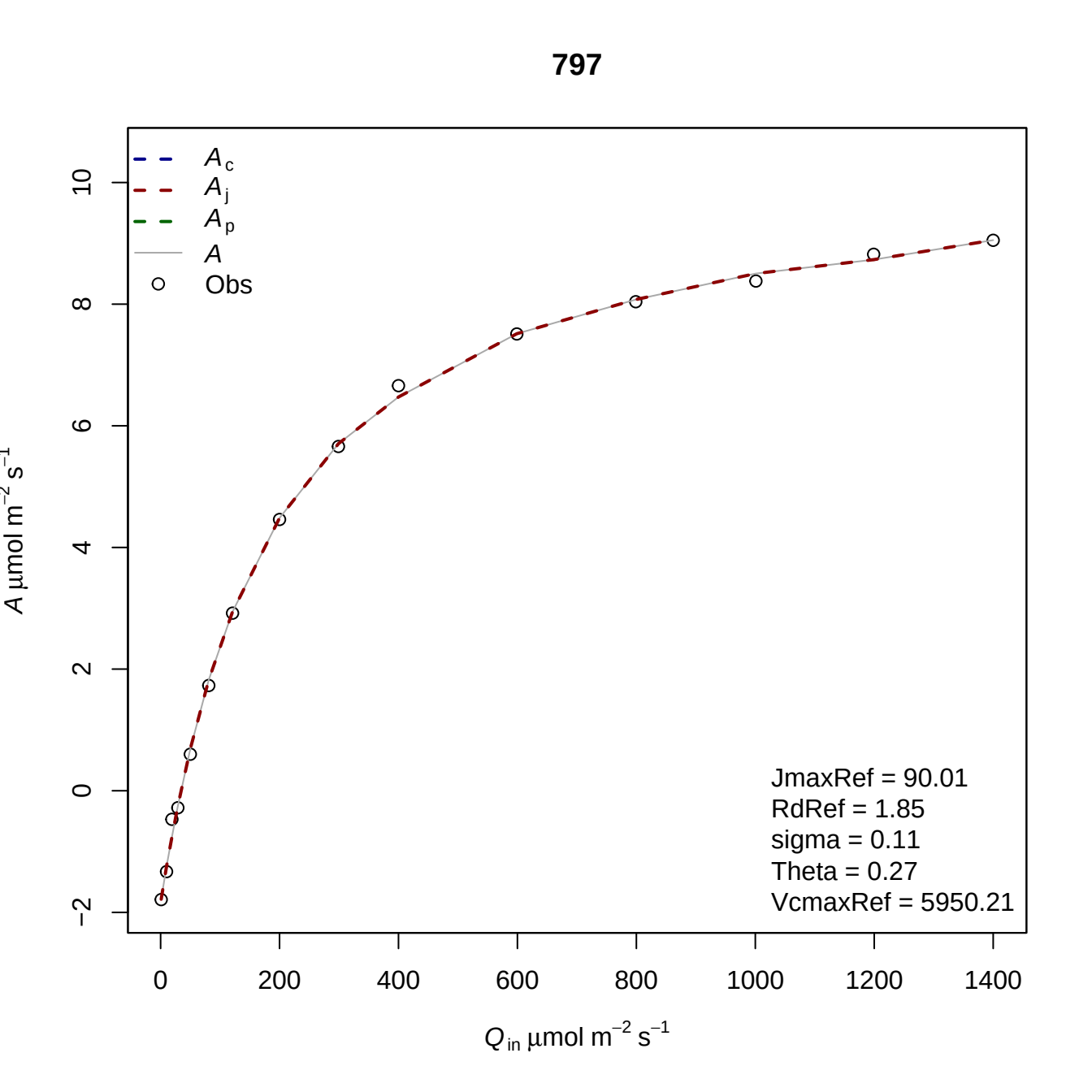

792
A_c
A_j
A_p
A
Obs
JmaxRef = 104.42
RdRef = 1.03
sigma = 0.23
Theta = −0.24
VcmaxRef = 4660263.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

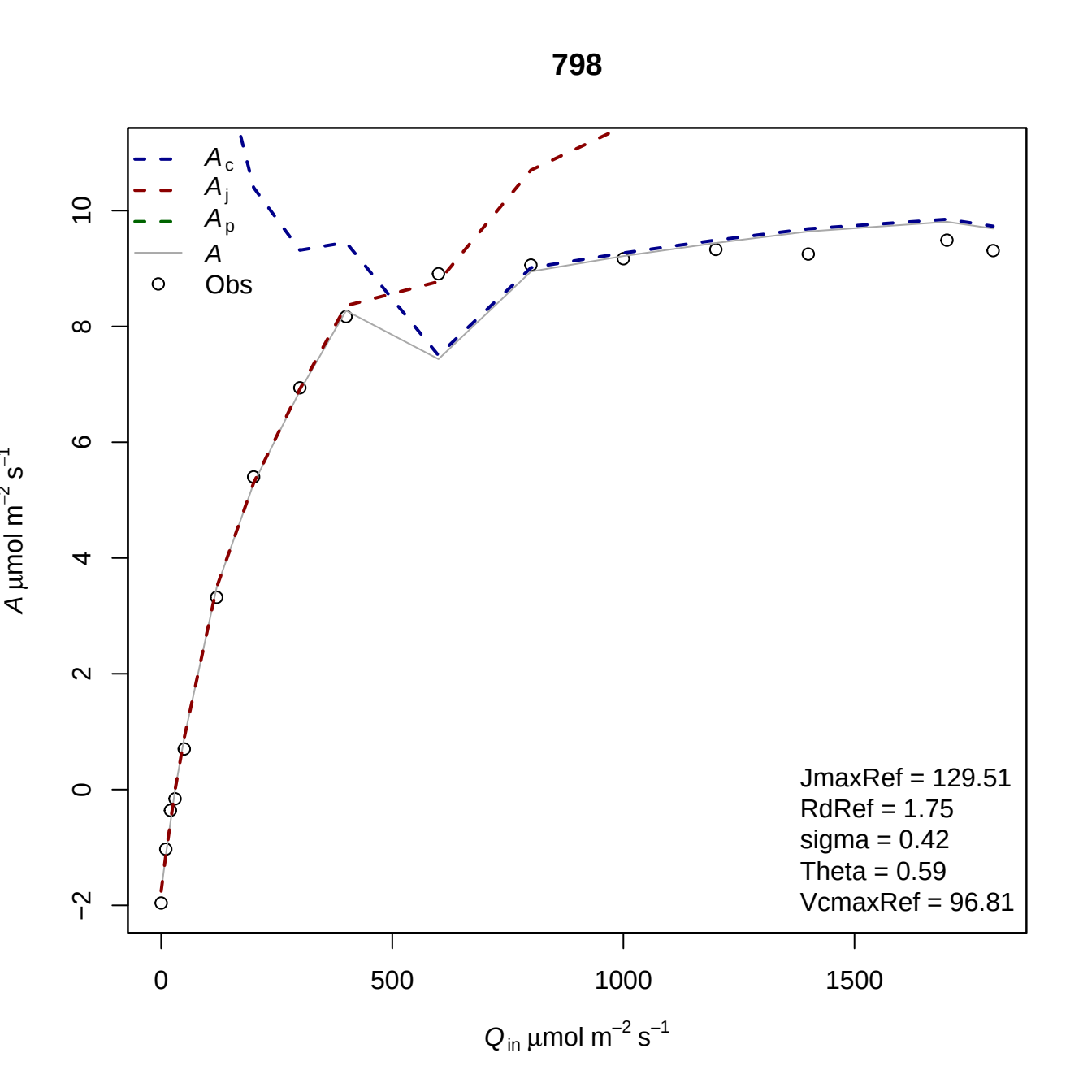

793
A_c
A_j
A_p
A
Obs
JmaxRef = 53.71
RdRef = 2.48
sigma = 0.06
Theta = −2.32
VcmaxRef = 1941.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200

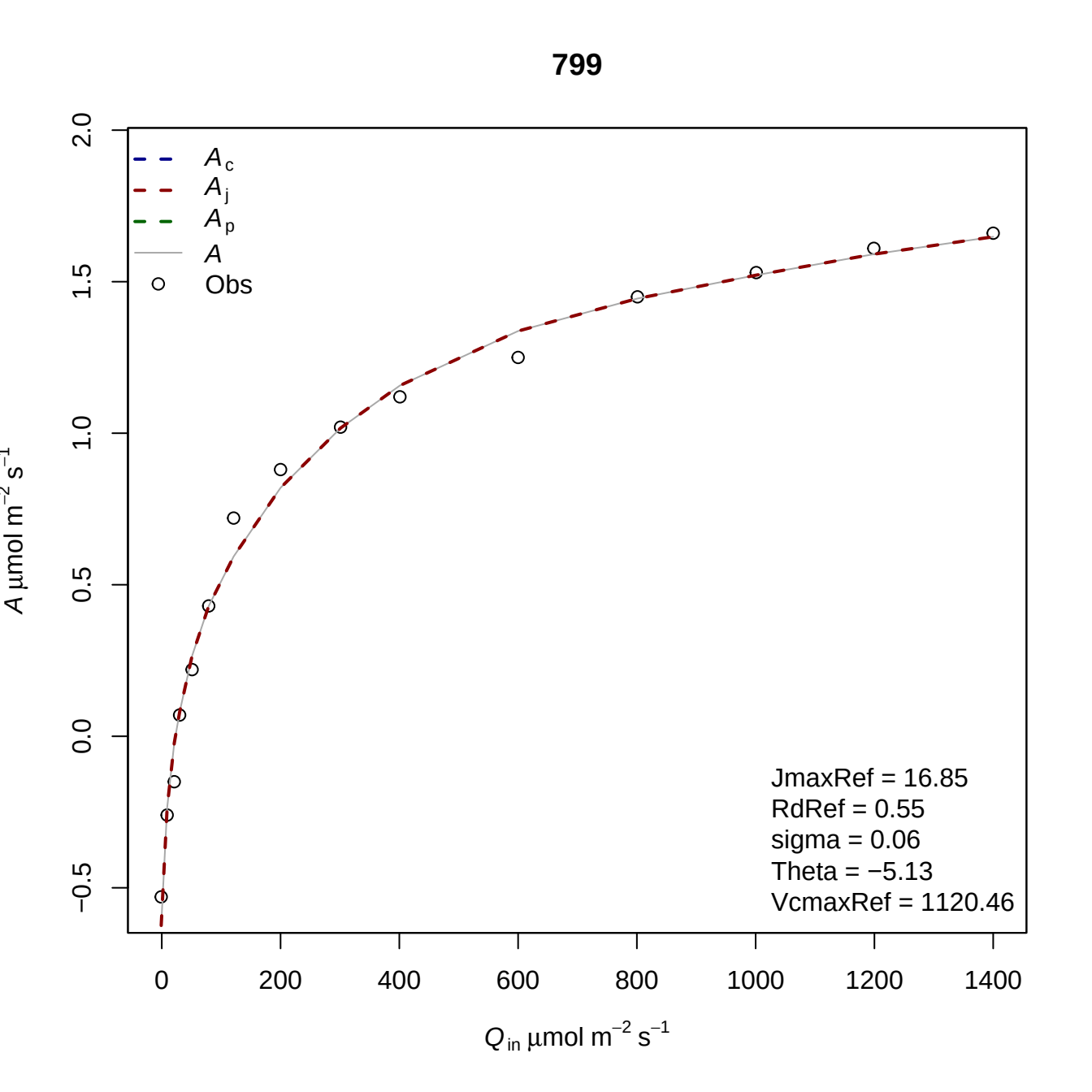

794
A_c
A_j
A_p
A
Obs
JmaxRef = 39.89
RdRef = 1.19
sigma = 0.1
Theta = 0.06
VcmaxRef = 30.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200

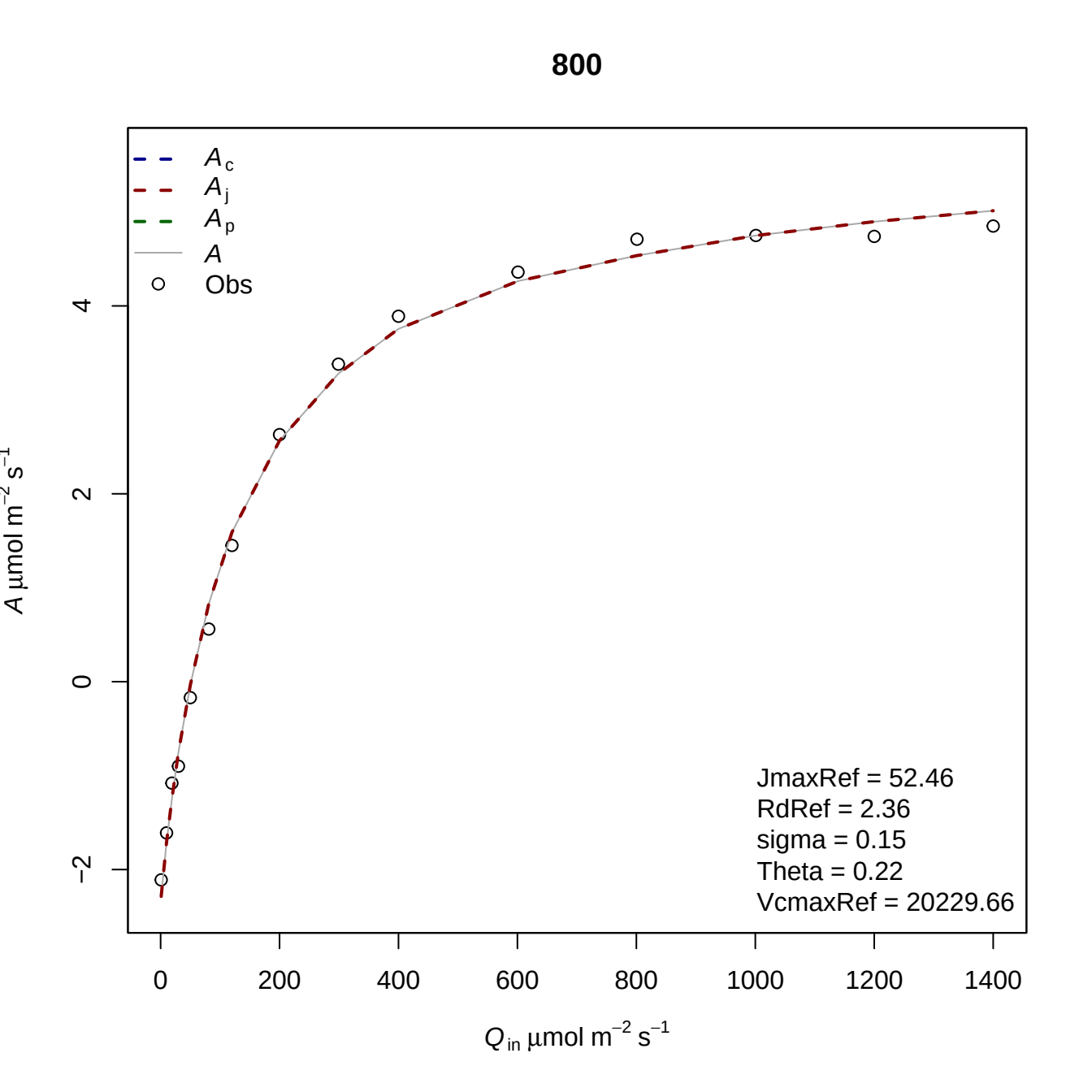

795
A_c
A_j
A_p
A
Obs
JmaxRef = 38.63
RdRef = 0.41
sigma = 0.11
Theta = 0.89
VcmaxRef = 527223.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

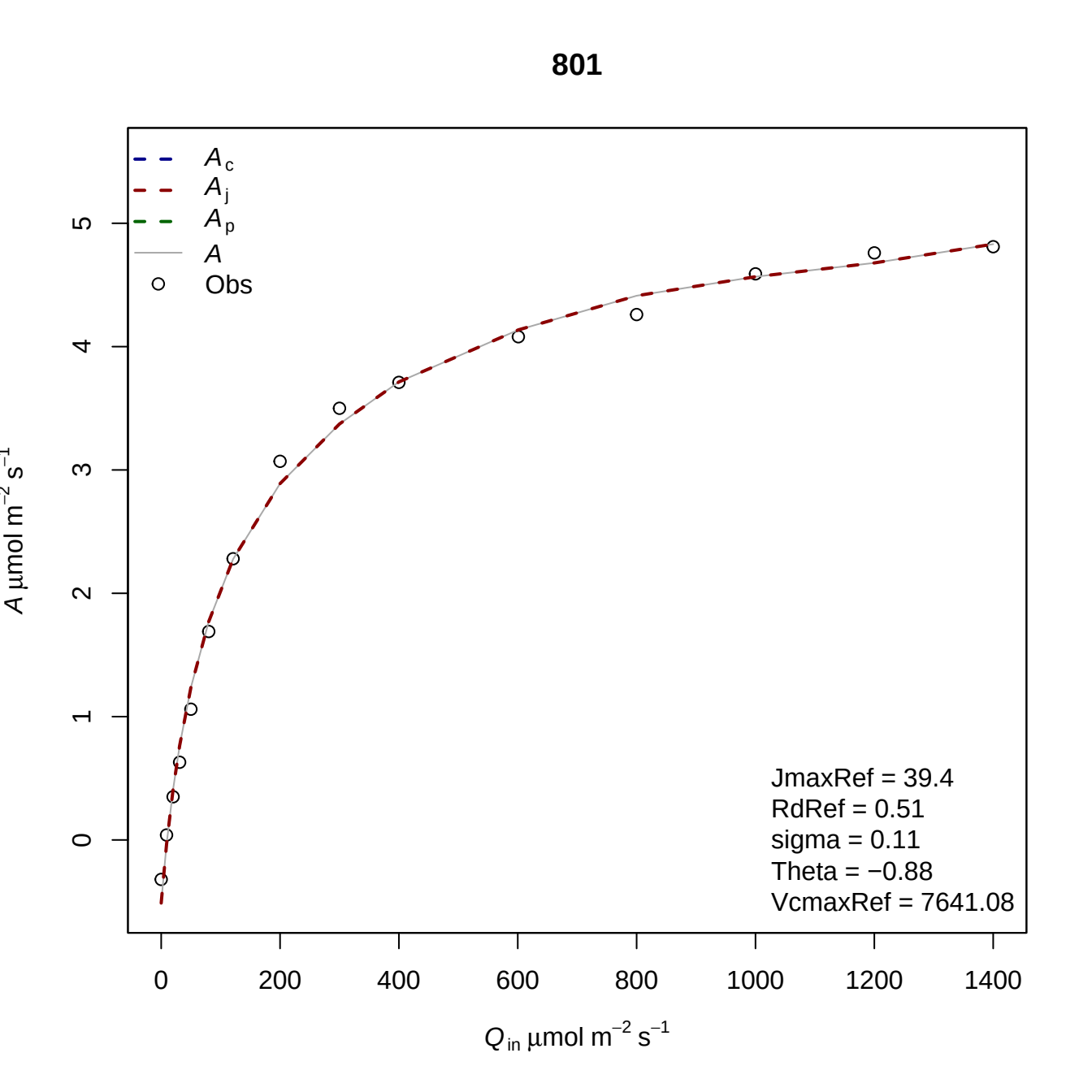

796
A_c
A_j
A_p
A
Obs
JmaxRef = 100.27
RdRef = 2.09
sigma = 0.21
Theta = −0.09
VcmaxRef = 469880.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

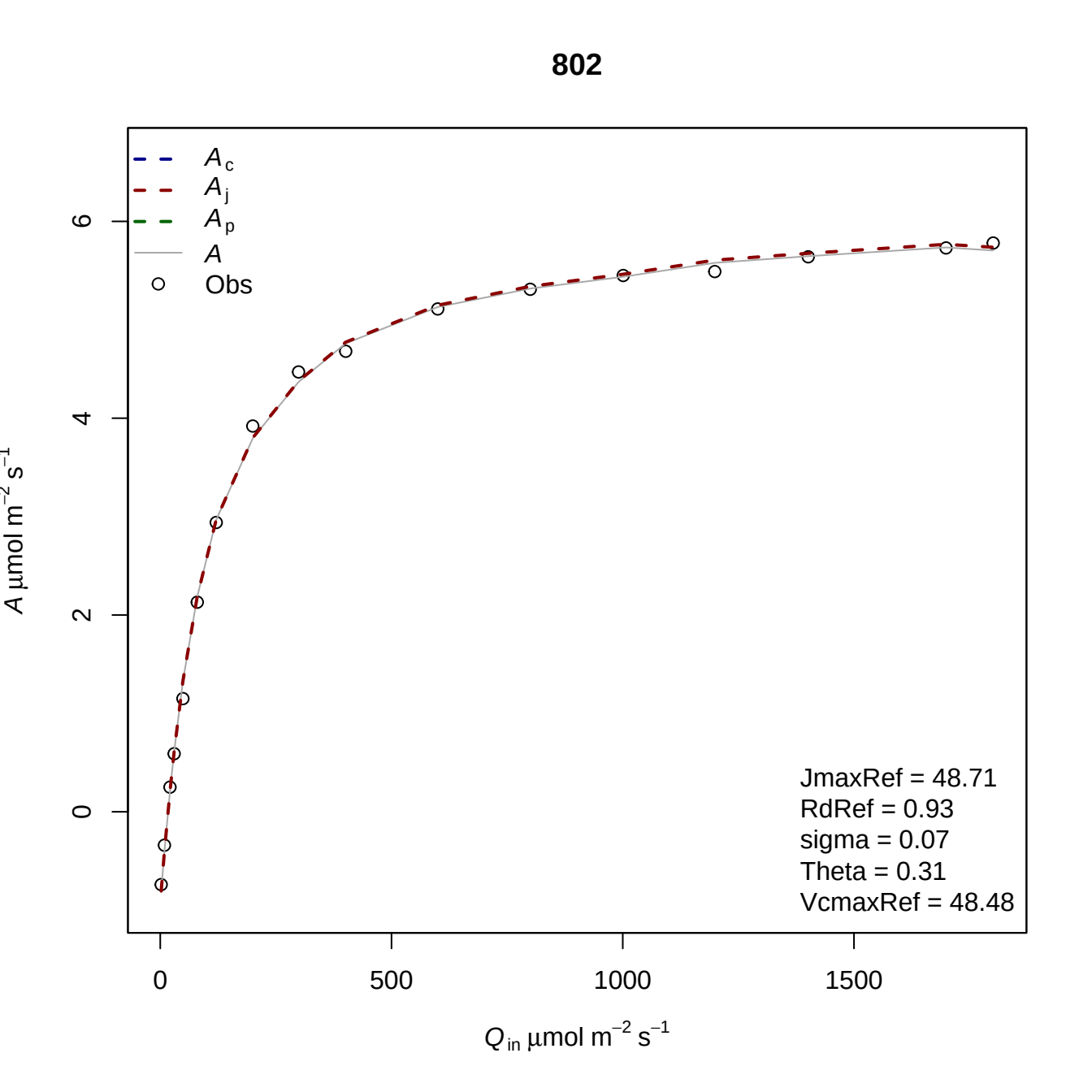

797
A_c
A_j
A_p
A
Obs
JmaxRef = 90.01
RdRef = 1.85
sigma = 0.11
Theta = 0.27
VcmaxRef = 5950.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

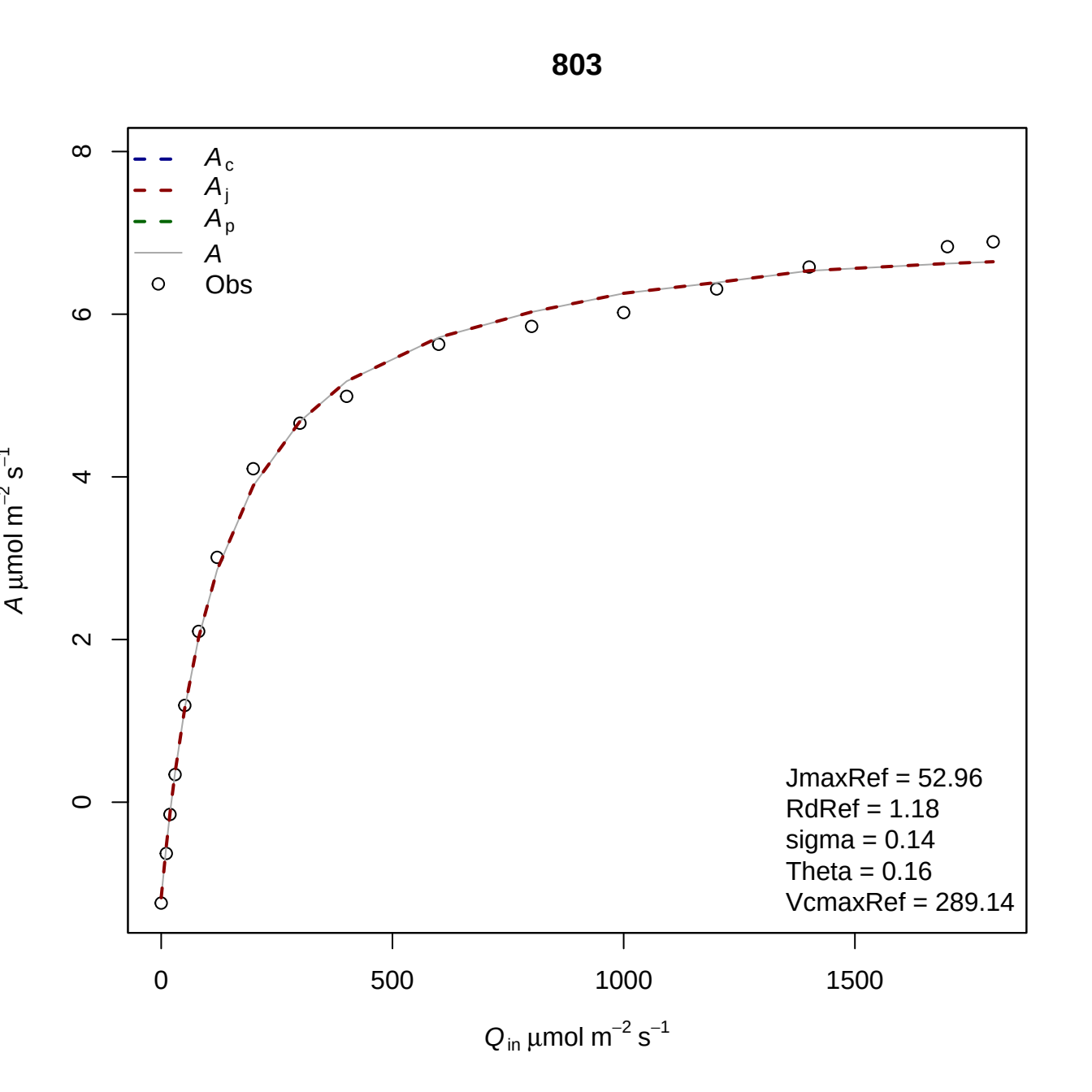

798
A_c
A_j
A_p
A
Obs
JmaxRef = 129.51
RdRef = 1.75
sigma = 0.42
Theta = 0.59
VcmaxRef = 96.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

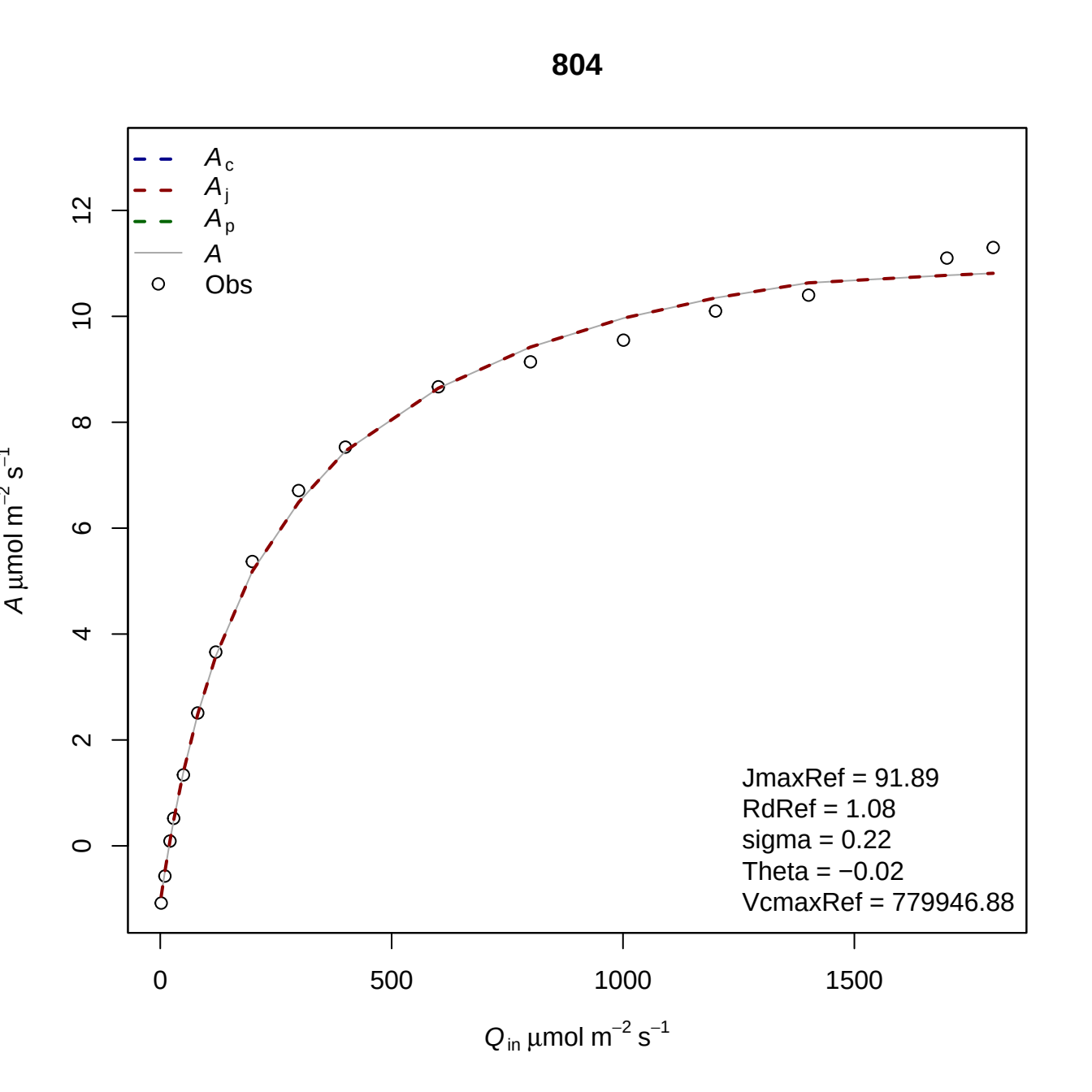

799
A_c
A_j
A_p
A
Obs
JmaxRef = 16.85
RdRef = 0.55
sigma = 0.06
Theta = −5.13
VcmaxRef = 1120.46
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

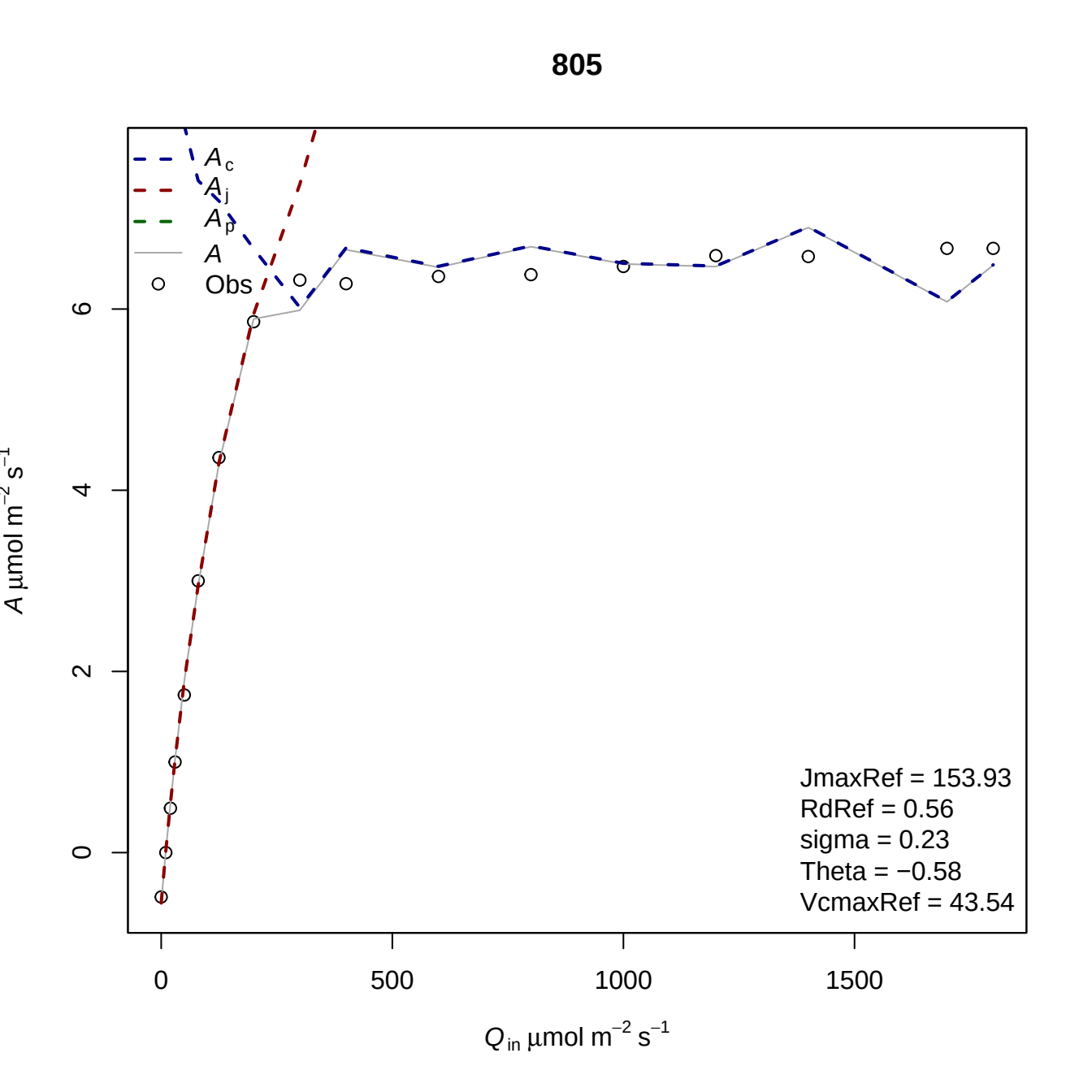

800
A_c
A_j
A_p
A
Obs
JmaxRef = 52.46
RdRef = 2.36
sigma = 0.15
Theta = 0.22
VcmaxRef = 20229.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

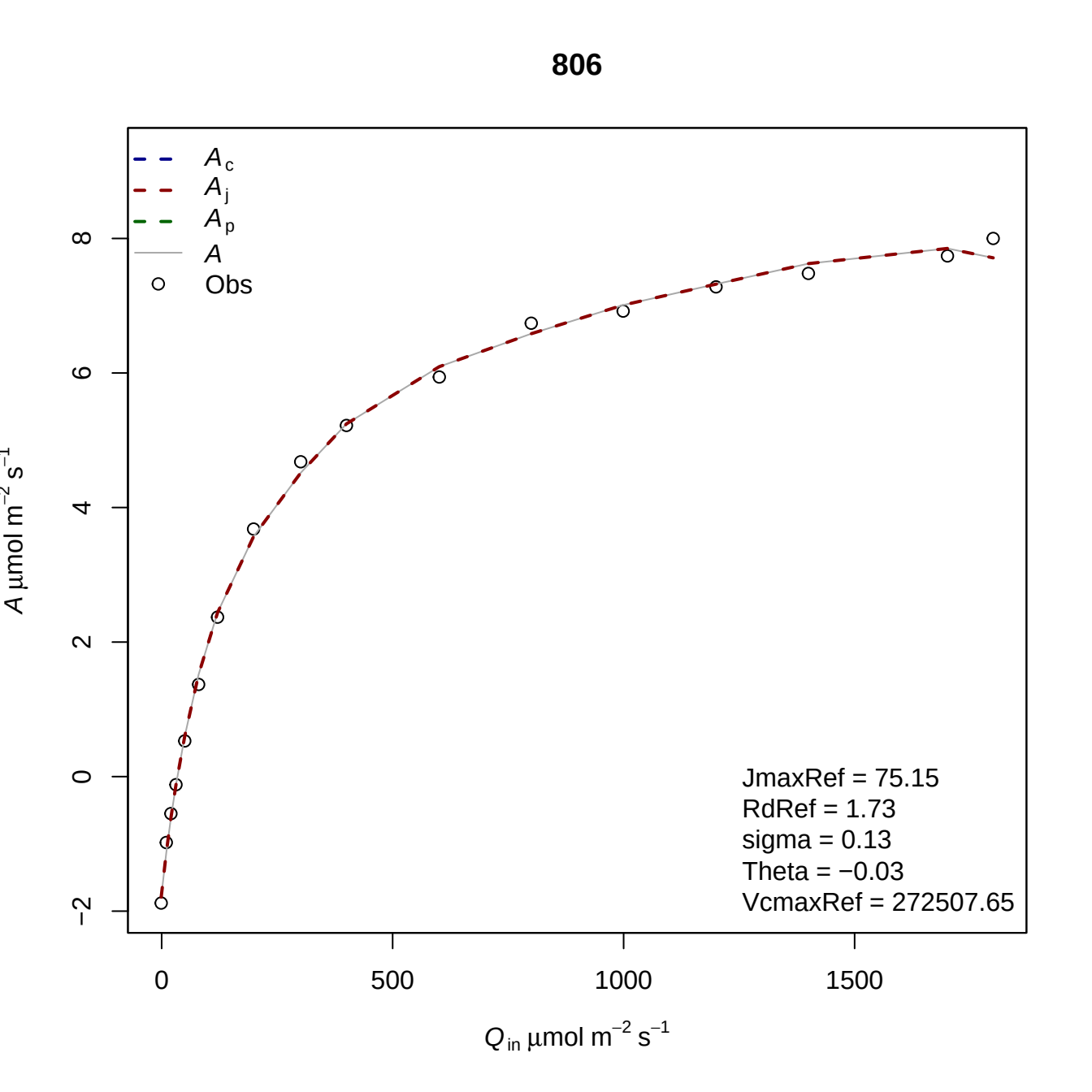

801
A_c
A_j
A_p
A
Obs
JmaxRef = 39.4
RdRef = 0.51
sigma = 0.11
Theta = −0.88
VcmaxRef = 7641.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

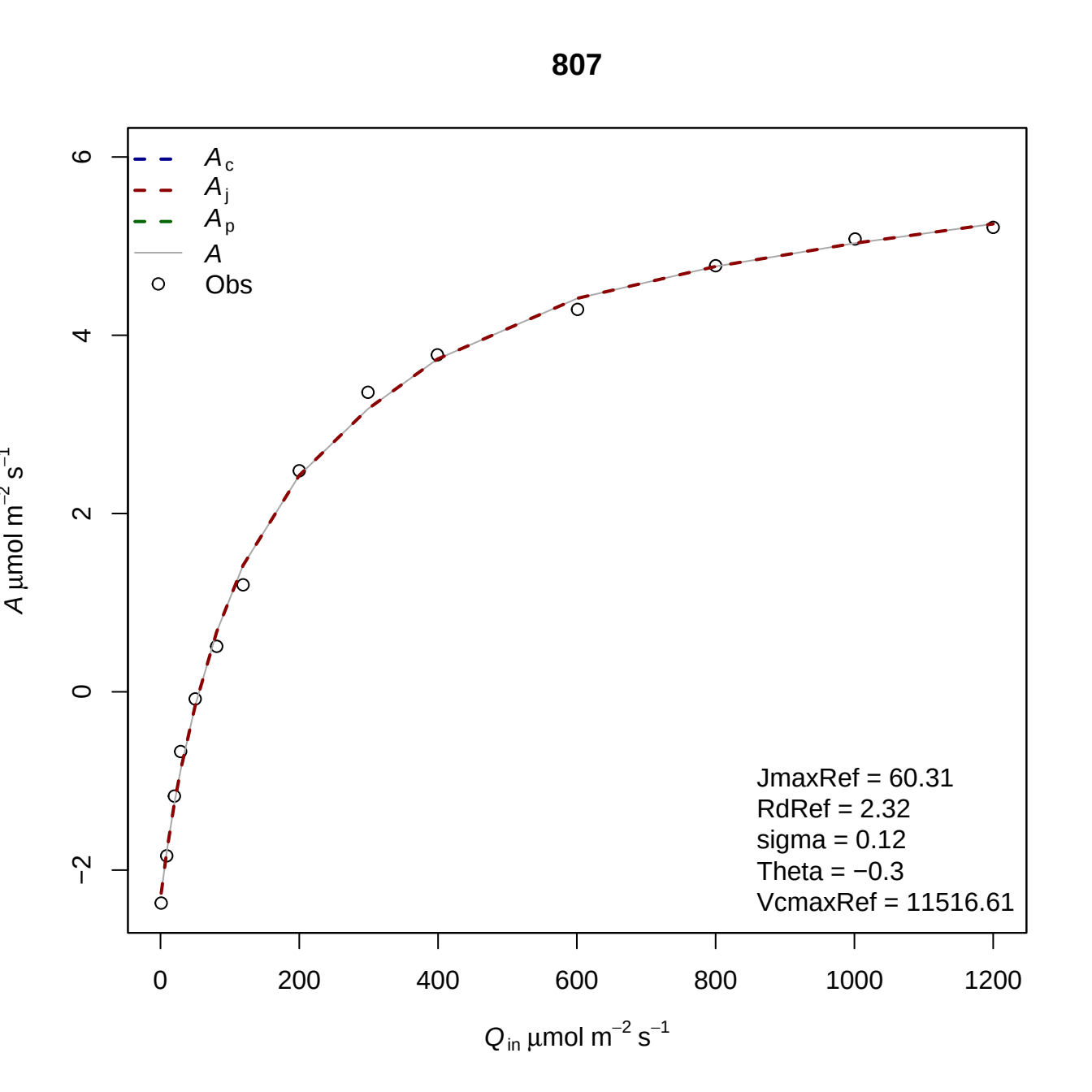

802
A_c
A_j
A_p
A
Obs
JmaxRef = 48.71
RdRef = 0.93
sigma = 0.07
Theta = 0.31
VcmaxRef = 48.48
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

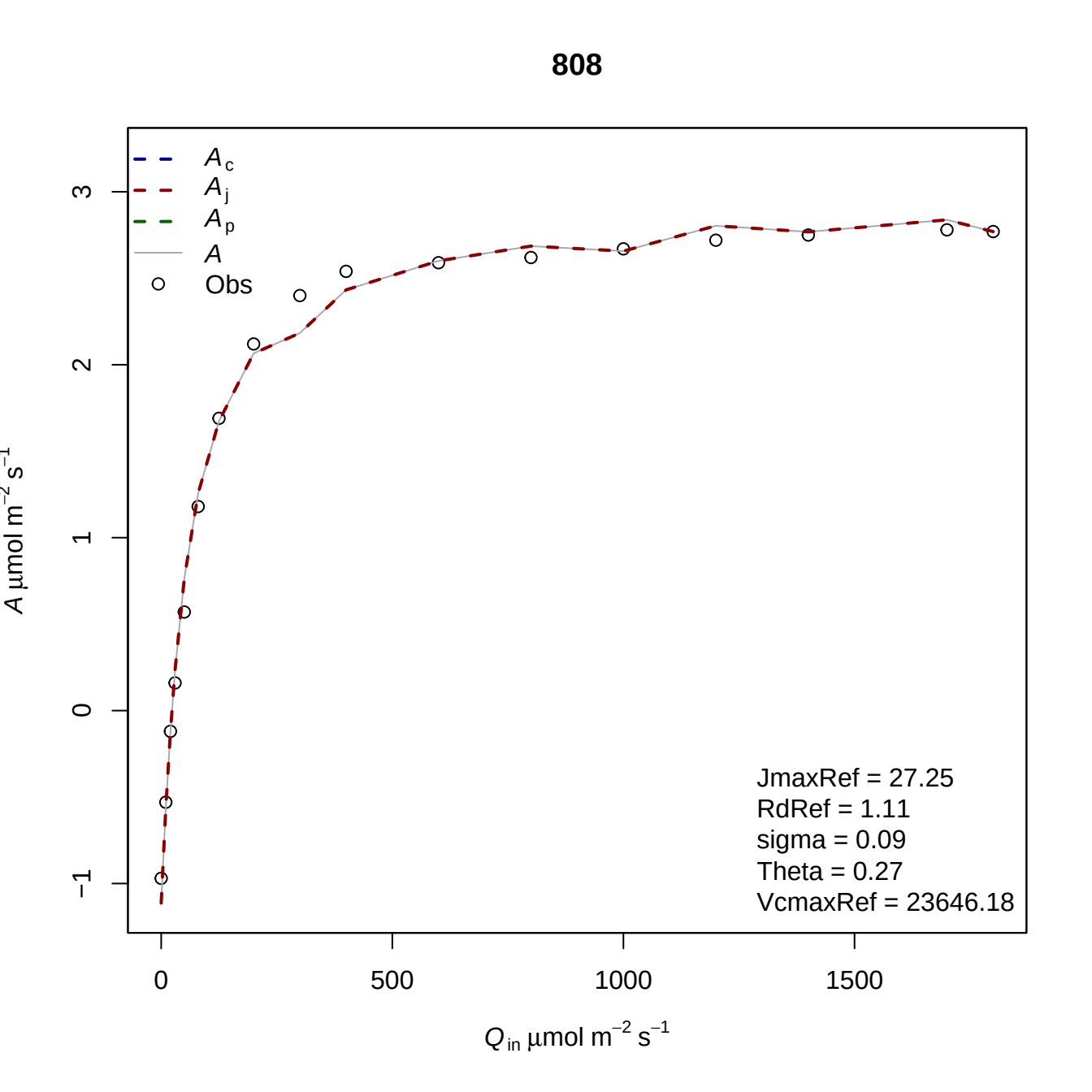

803
A_c
A_j
A_p
A
Obs
JmaxRef = 52.96
RdRef = 1.18
sigma = 0.14
Theta = 0.16
VcmaxRef = 289.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

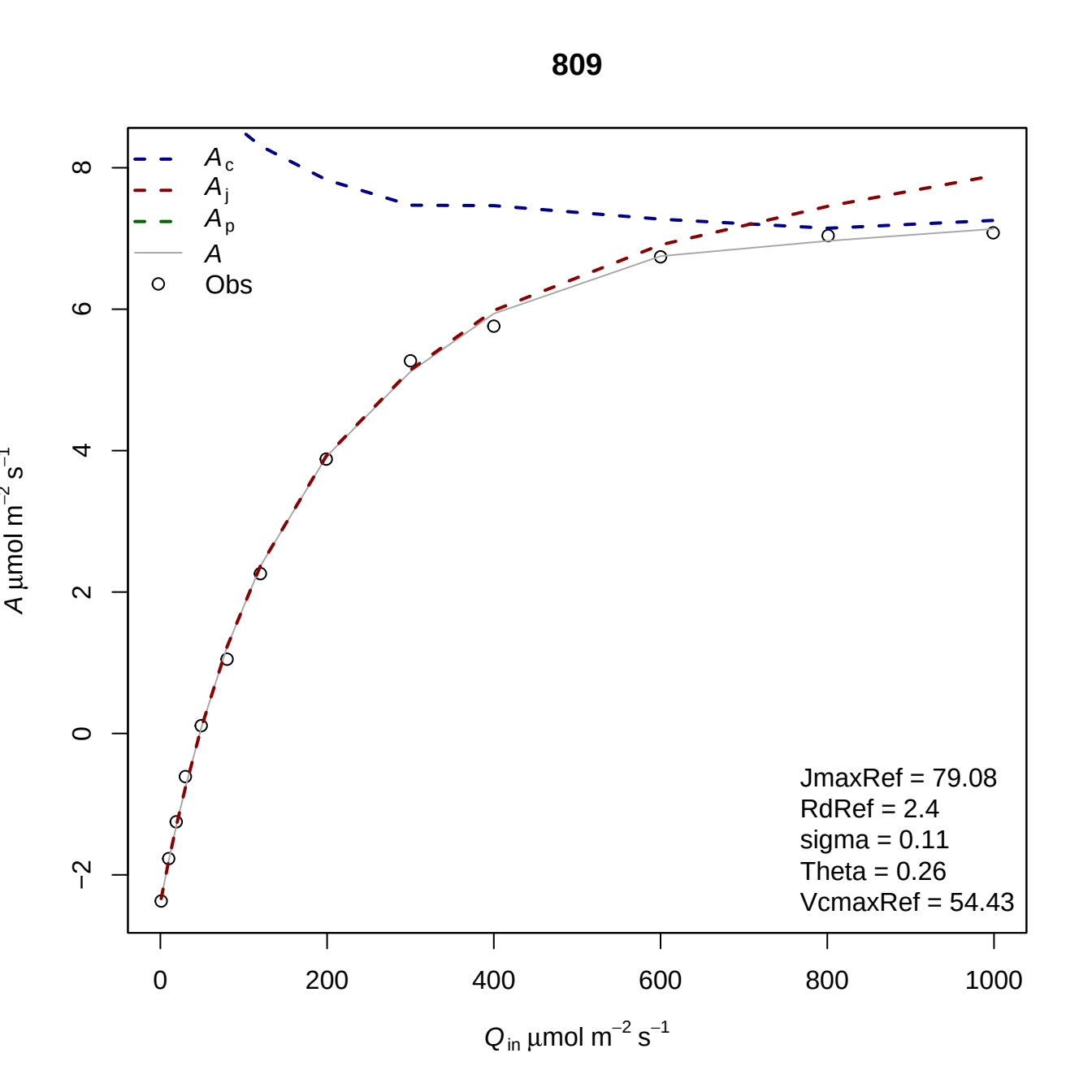

804
A_c
A_j
A_p
A
Obs
JmaxRef = 91.89
RdRef = 1.08
sigma = 0.22
Theta = −0.02
VcmaxRef = 779946.88
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

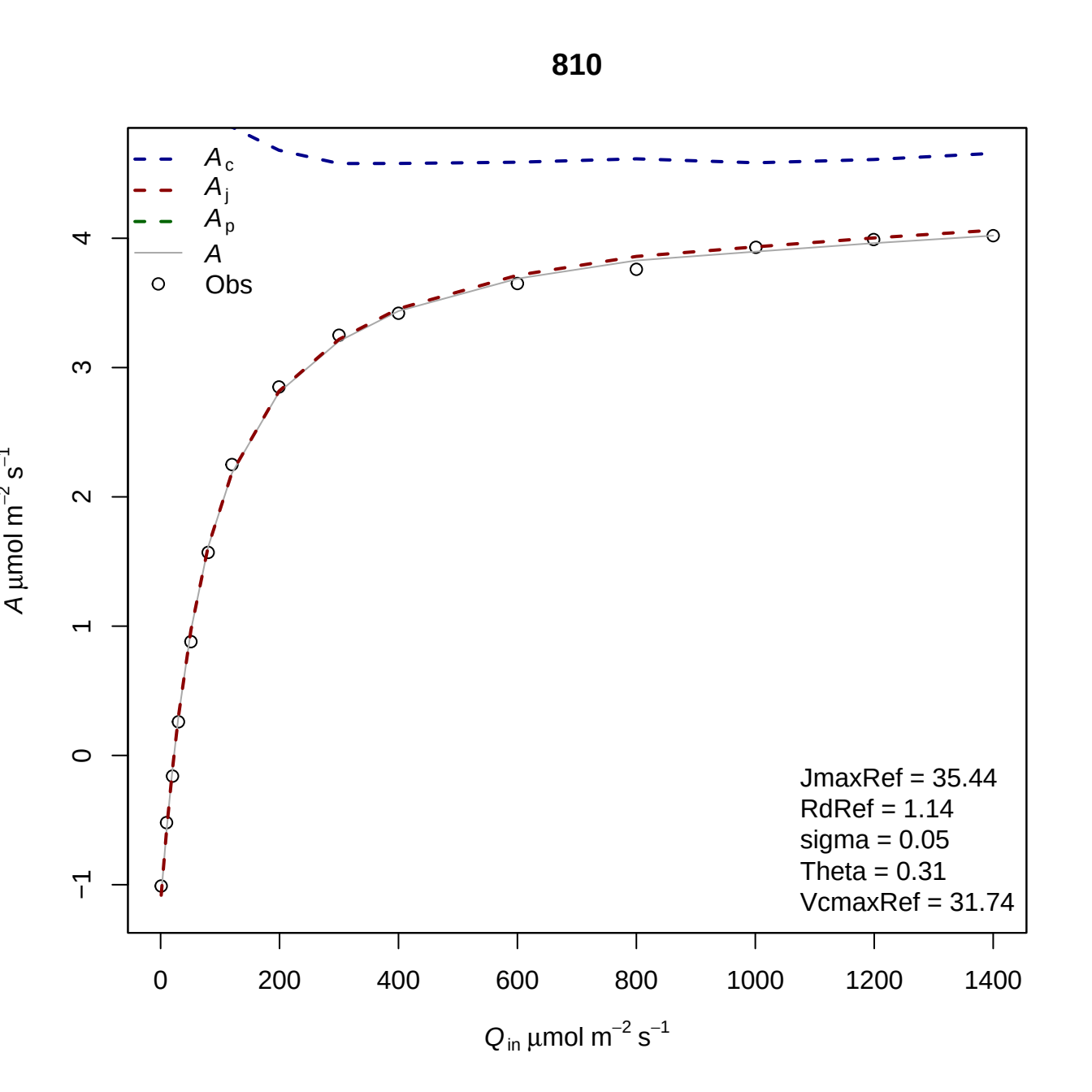

805
A_c
A_j
A_p
A
Obs
JmaxRef = 153.93
RdRef = 0.56
sigma = 0.23
Theta = −0.58
VcmaxRef = 43.54
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

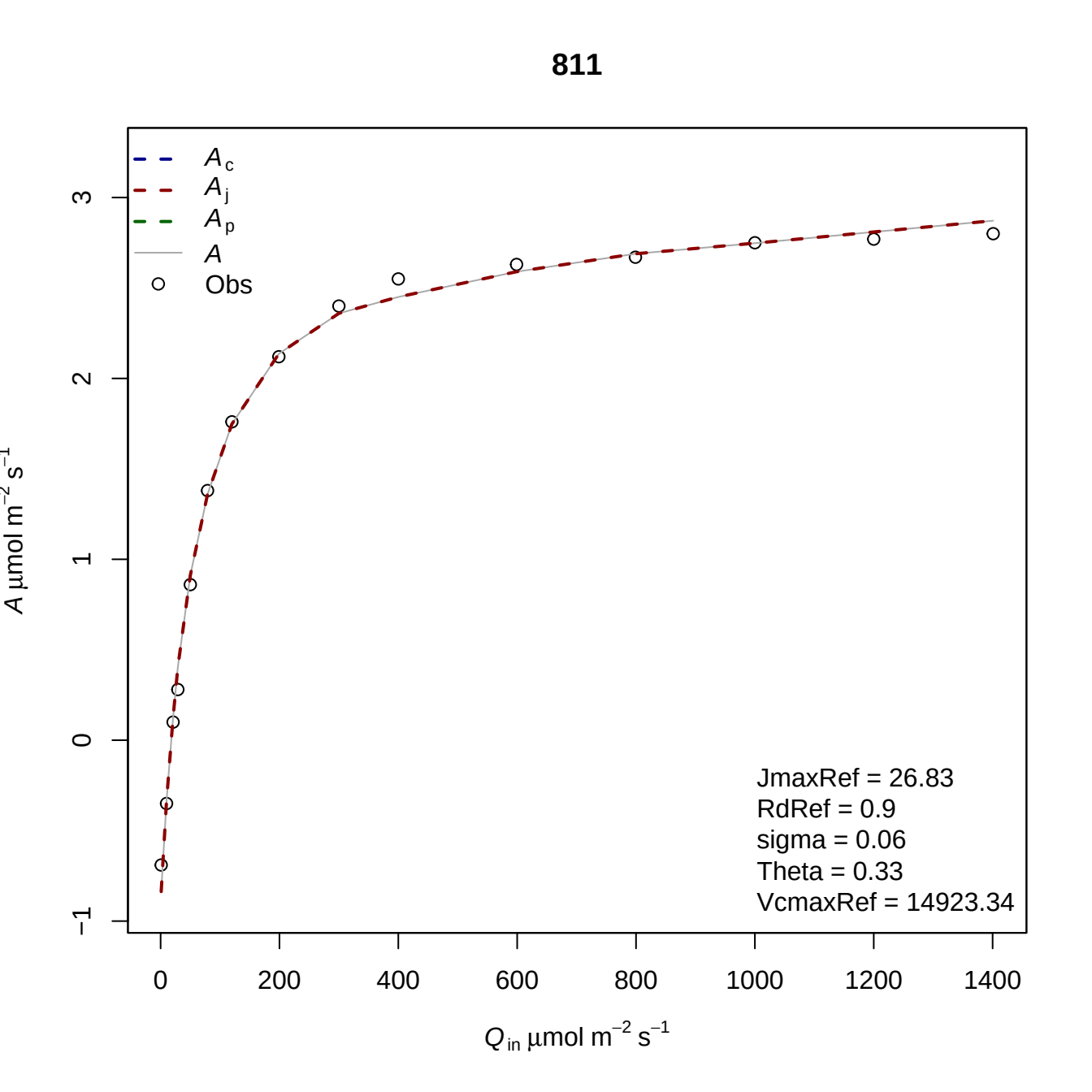

806
A_c
A_j
A_p
A
Obs
JmaxRef = 75.15
RdRef = 1.73
sigma = 0.13
Theta = −0.03
VcmaxRef = 272507.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

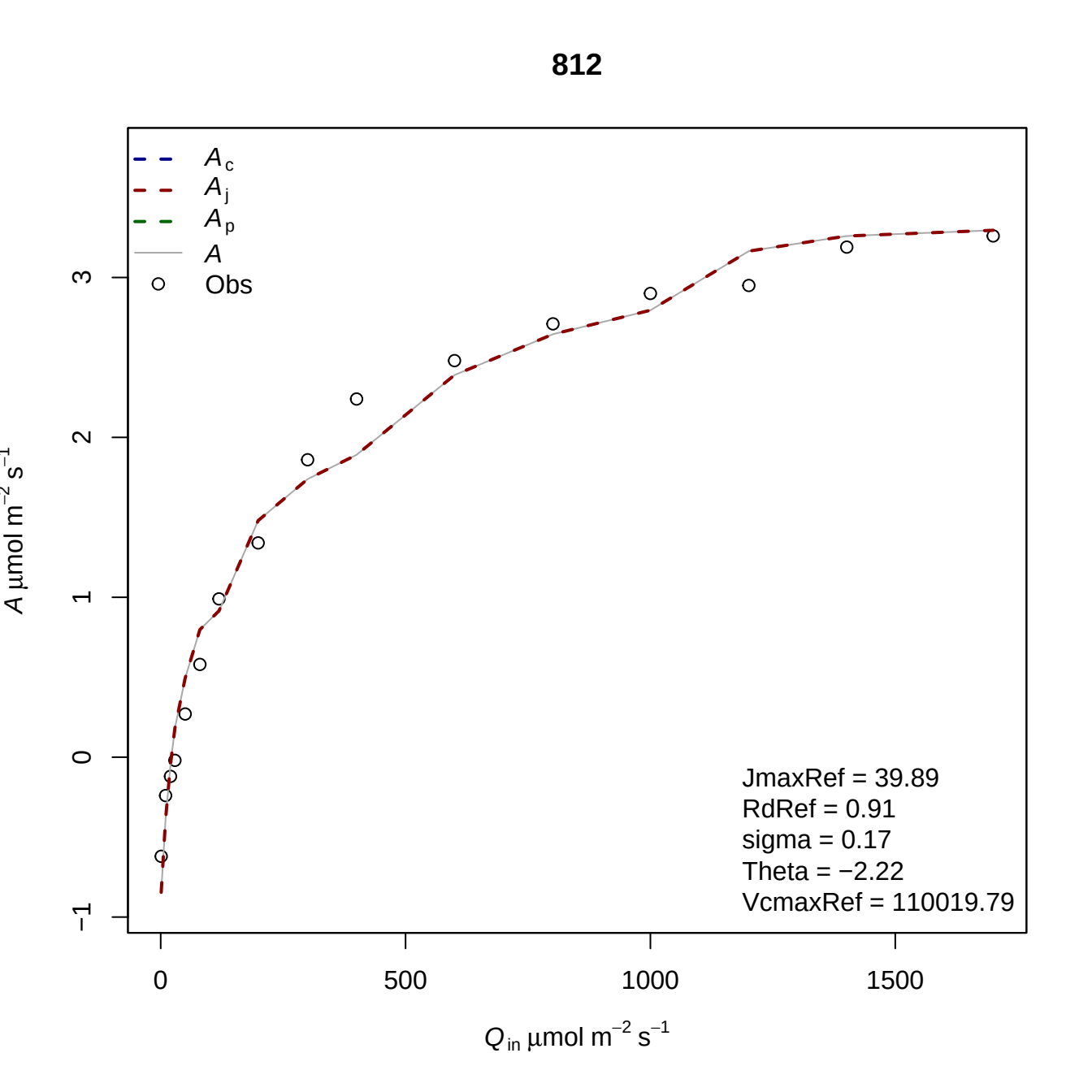

807
A_c
A_j
A_p
A
Obs
JmaxRef = 60.31
RdRef = 2.32
sigma = 0.12
Theta = −0.3
VcmaxRef = 11516.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

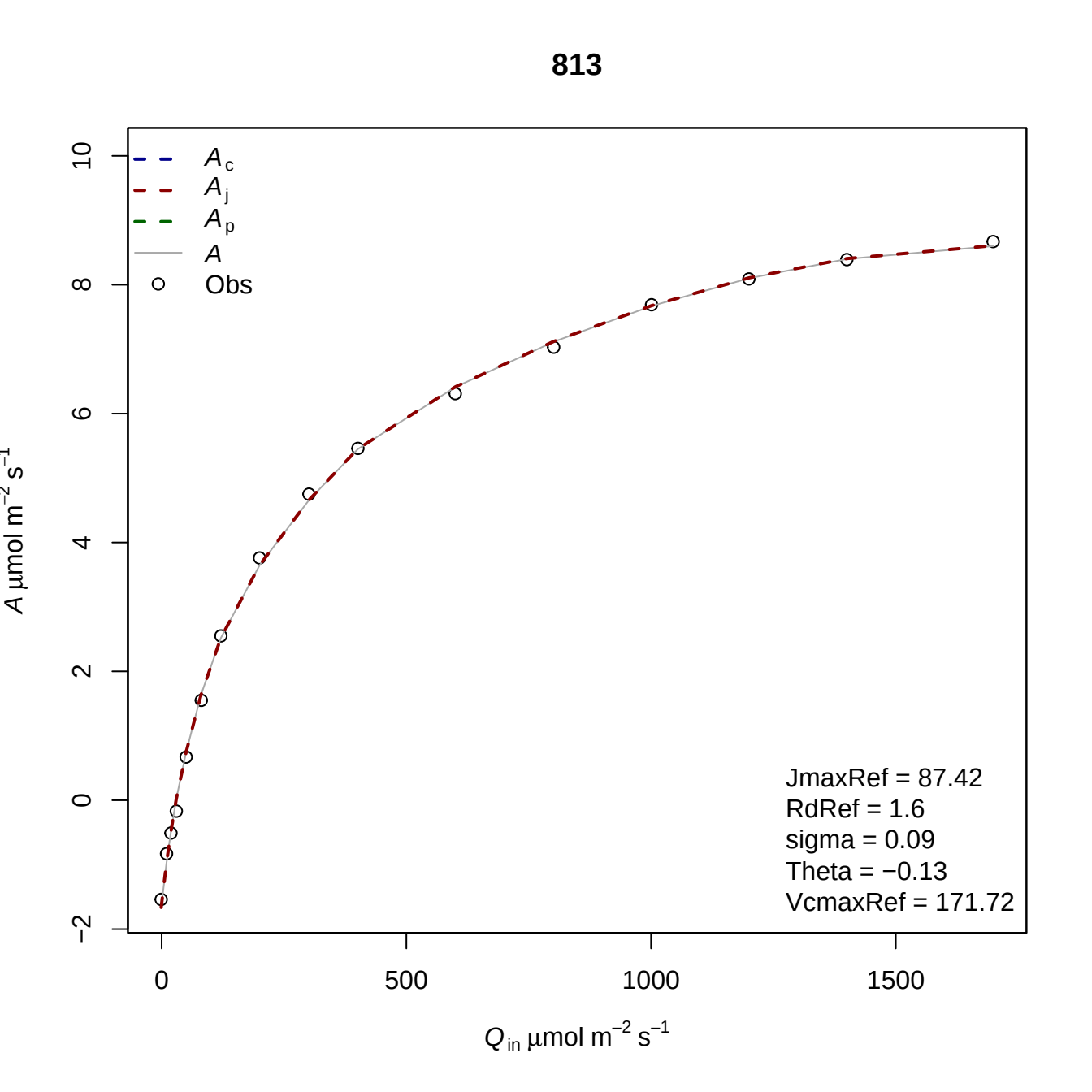

808
A_c
A_j
A_p
A
Obs
JmaxRef = 27.25
RdRef = 1.11
sigma = 0.09
Theta = 0.27
VcmaxRef = 23646.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

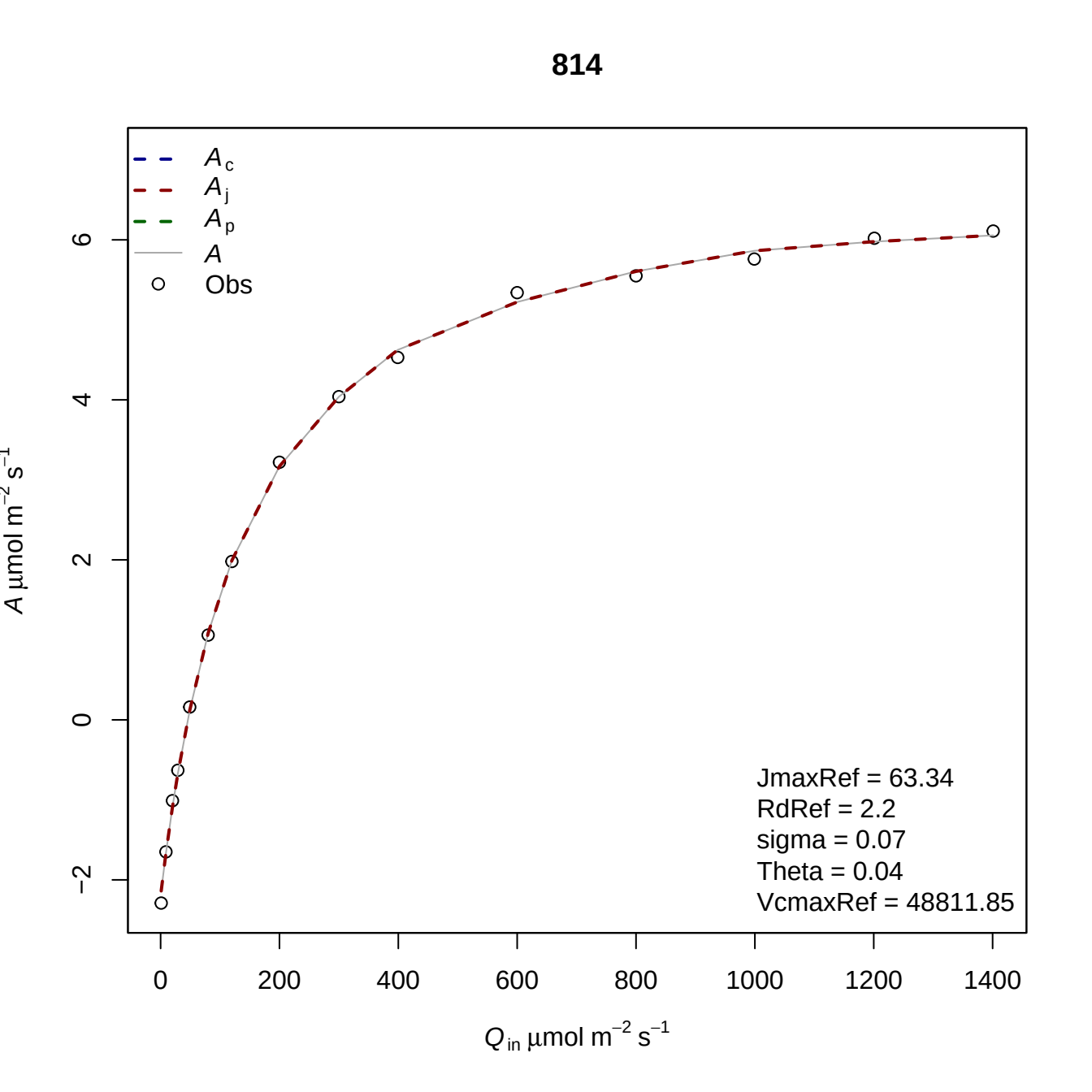

809
A_c
A_j
A_p
A
Obs
JmaxRef = 79.08
RdRef = 2.4
sigma = 0.11
Theta = 0.26
VcmaxRef = 54.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

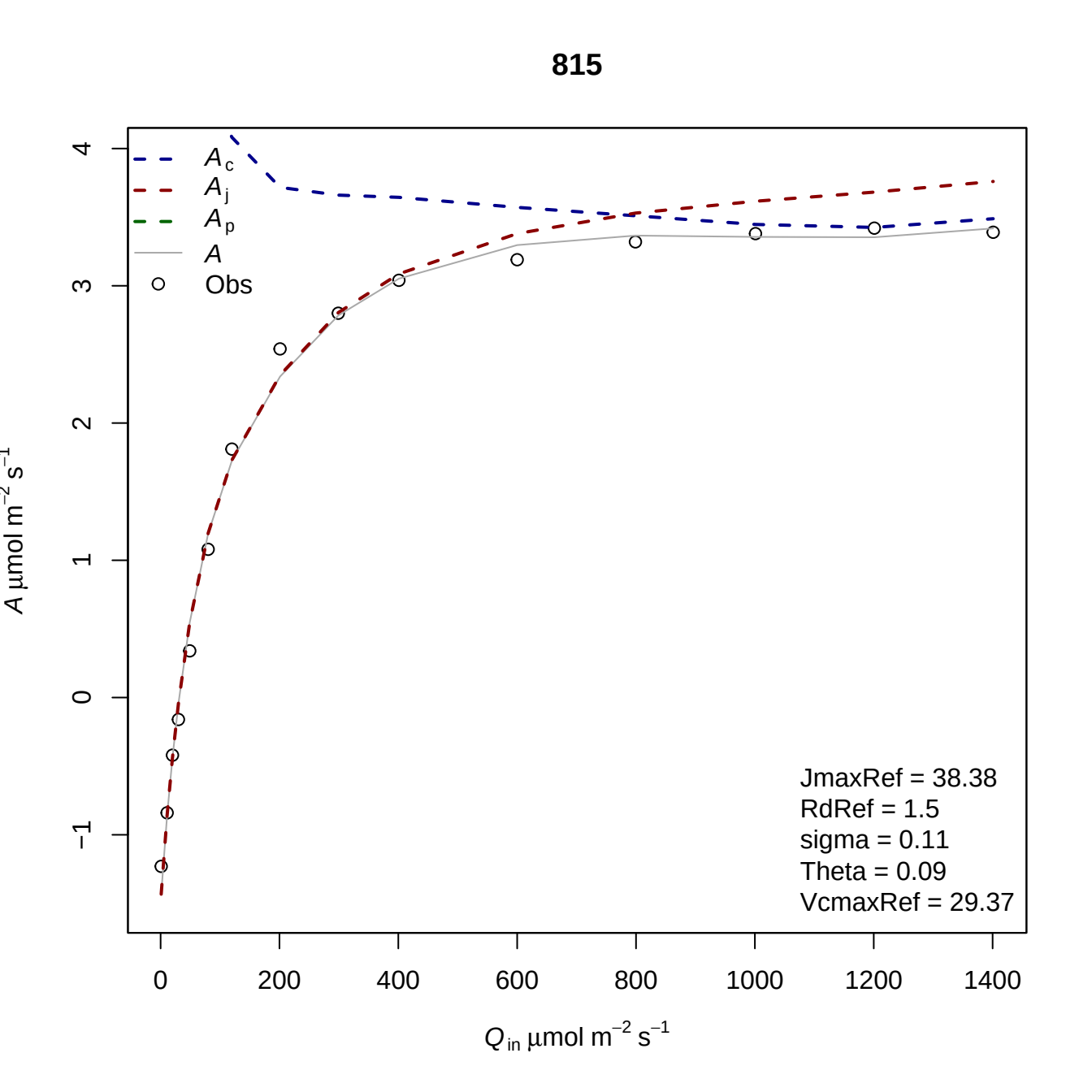

810
A_c
A_j
A_p
A
Obs
JmaxRef = 35.44
RdRef = 1.14
sigma = 0.05
Theta = 0.31
VcmaxRef = 31.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

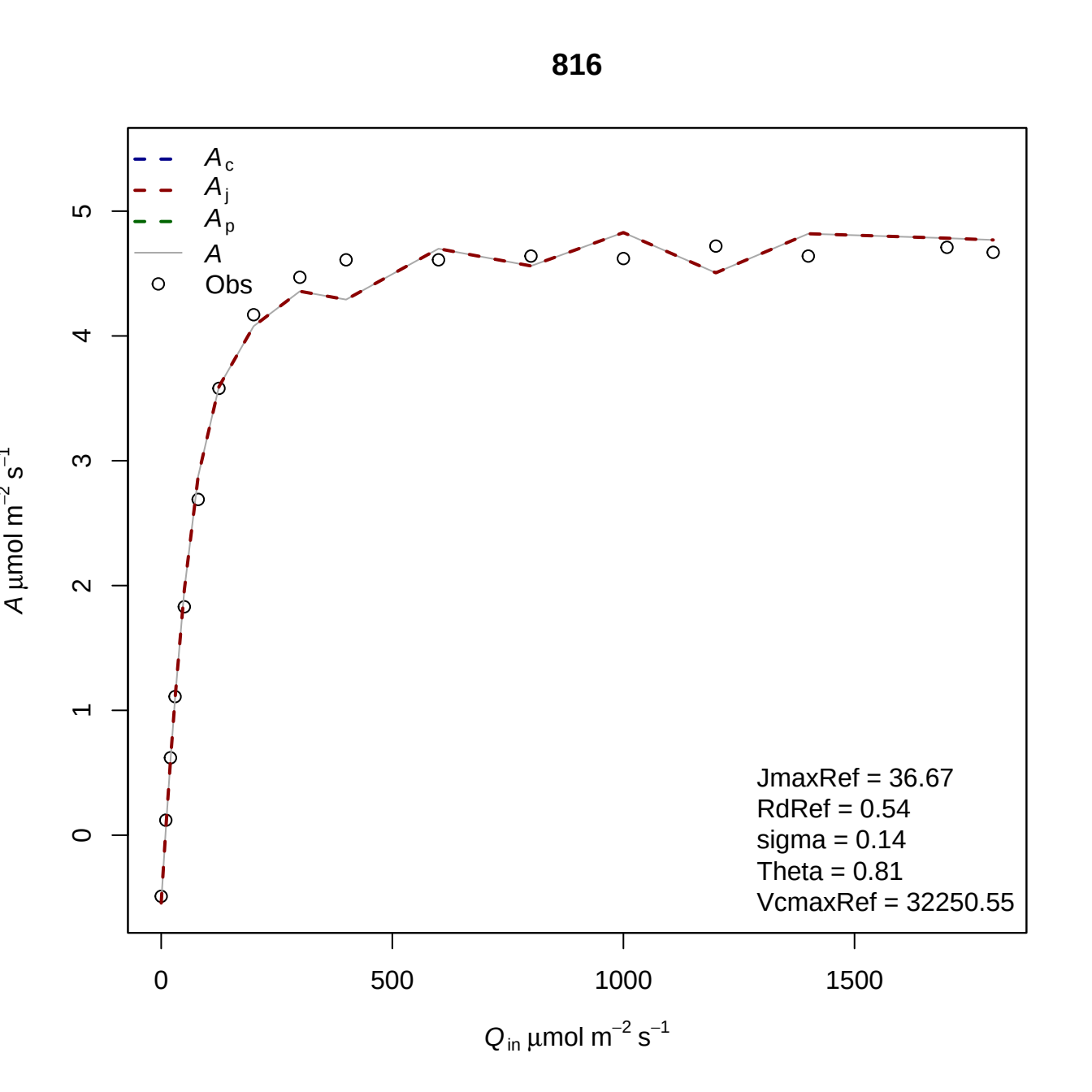

811
A_c
A_j
A_p
A
Obs
JmaxRef = 26.83
RdRef = 0.9
sigma = 0.06
Theta = 0.33
VcmaxRef = 14923.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

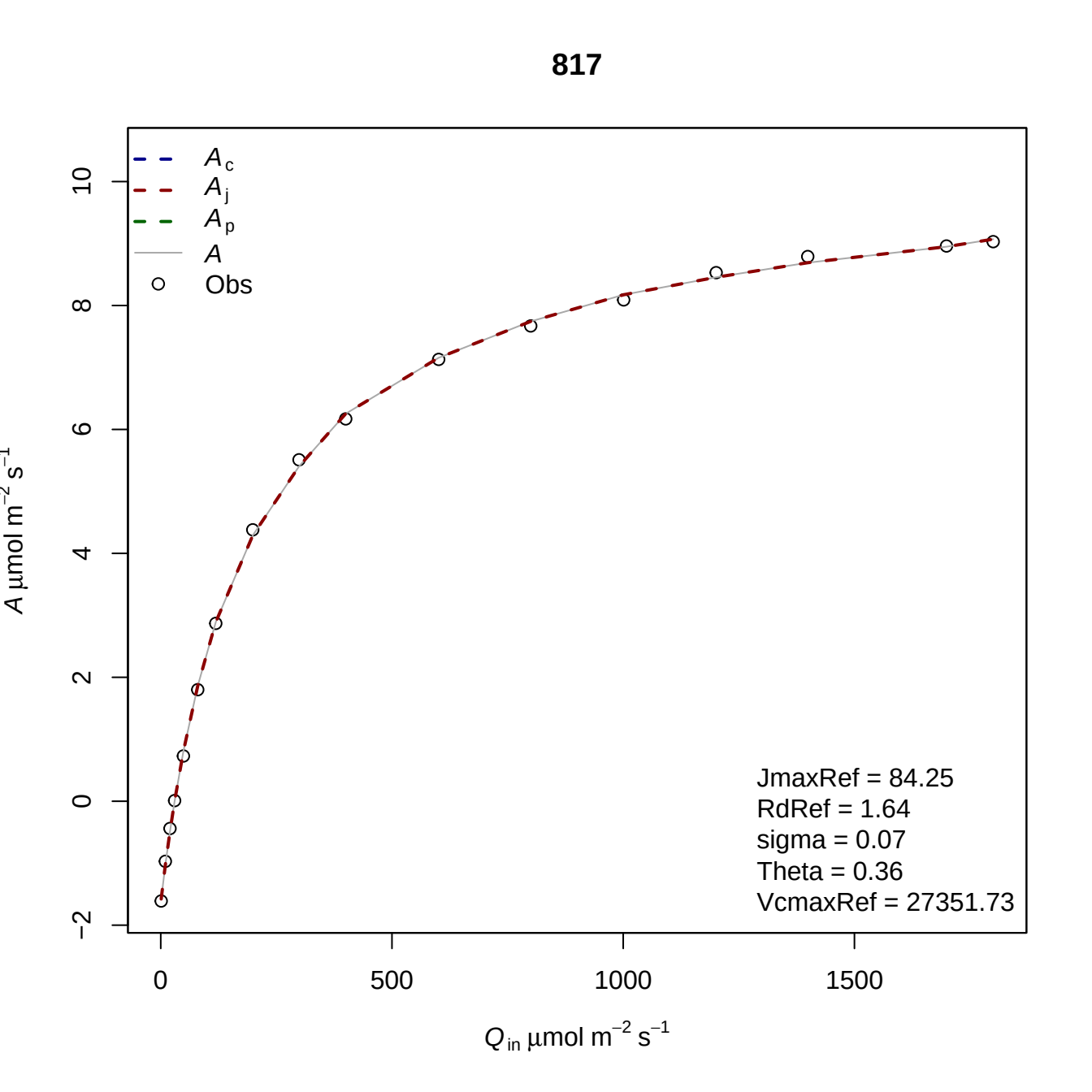

812
A_c
A_j
A_p
A
Obs
JmaxRef = 39.89
RdRef = 0.91
sigma = 0.17
Theta = −2.22
VcmaxRef = 110019.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

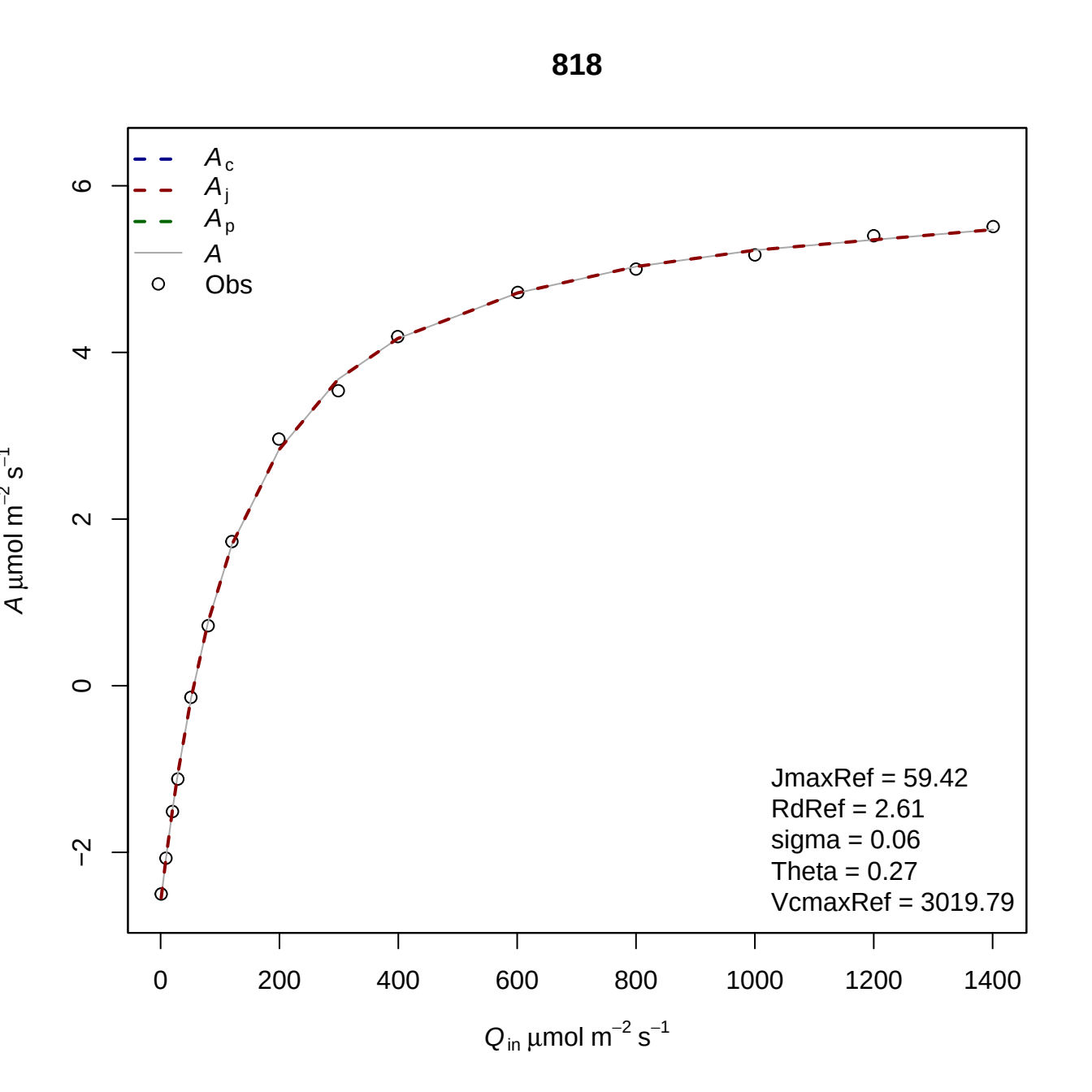

813
A_c
A_j
A_p
A
Obs
JmaxRef = 87.42
RdRef = 1.6
sigma = 0.09
Theta = −0.13
VcmaxRef = 171.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

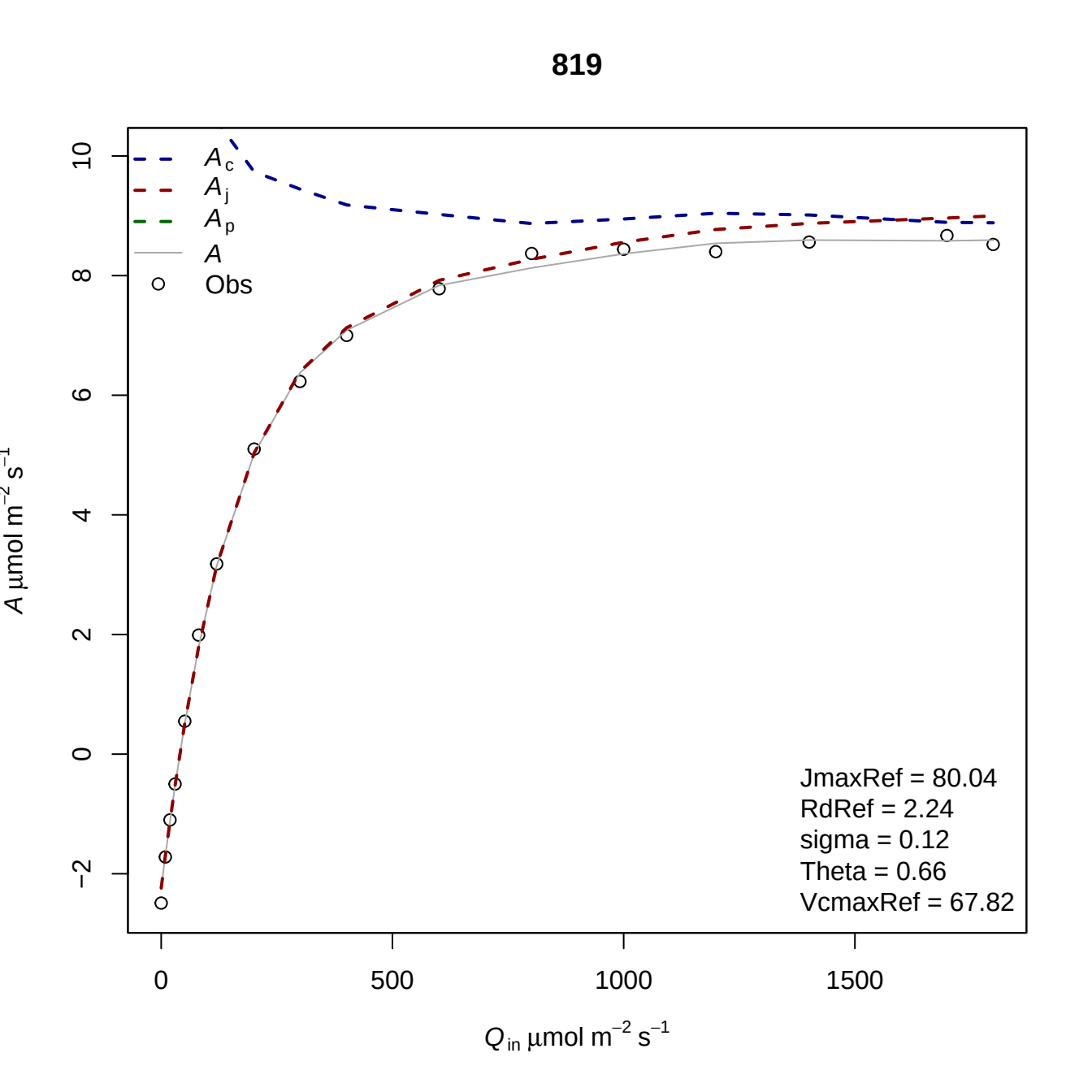

814
A_c
A_j
A_p
A
Obs
JmaxRef = 63.34
RdRef = 2.2
sigma = 0.07
Theta = 0.04
VcmaxRef = 48811.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

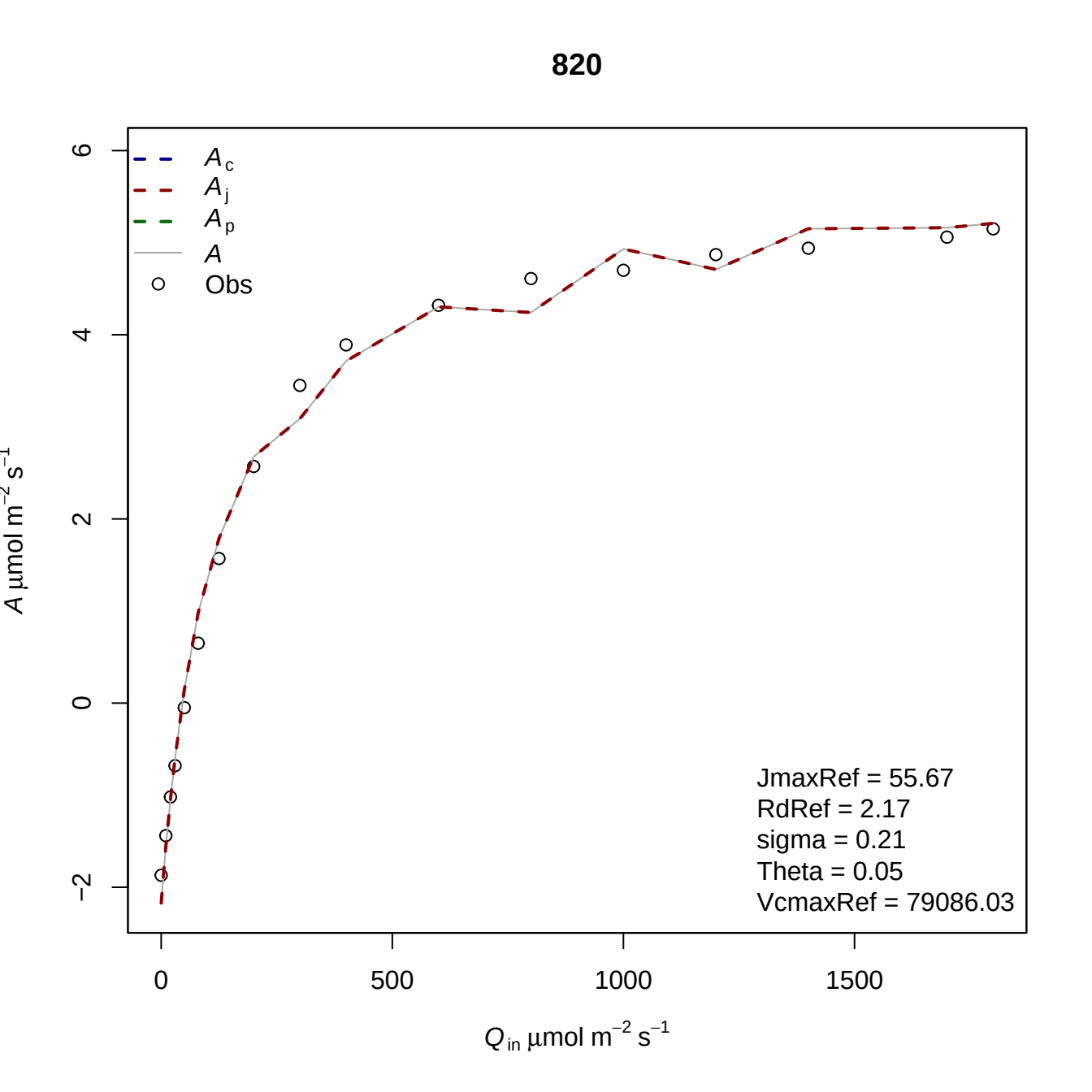

815
A_c
A_j
A_p
A
Obs
JmaxRef = 38.38
RdRef = 1.5
sigma = 0.11
Theta = 0.09
VcmaxRef = 29.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

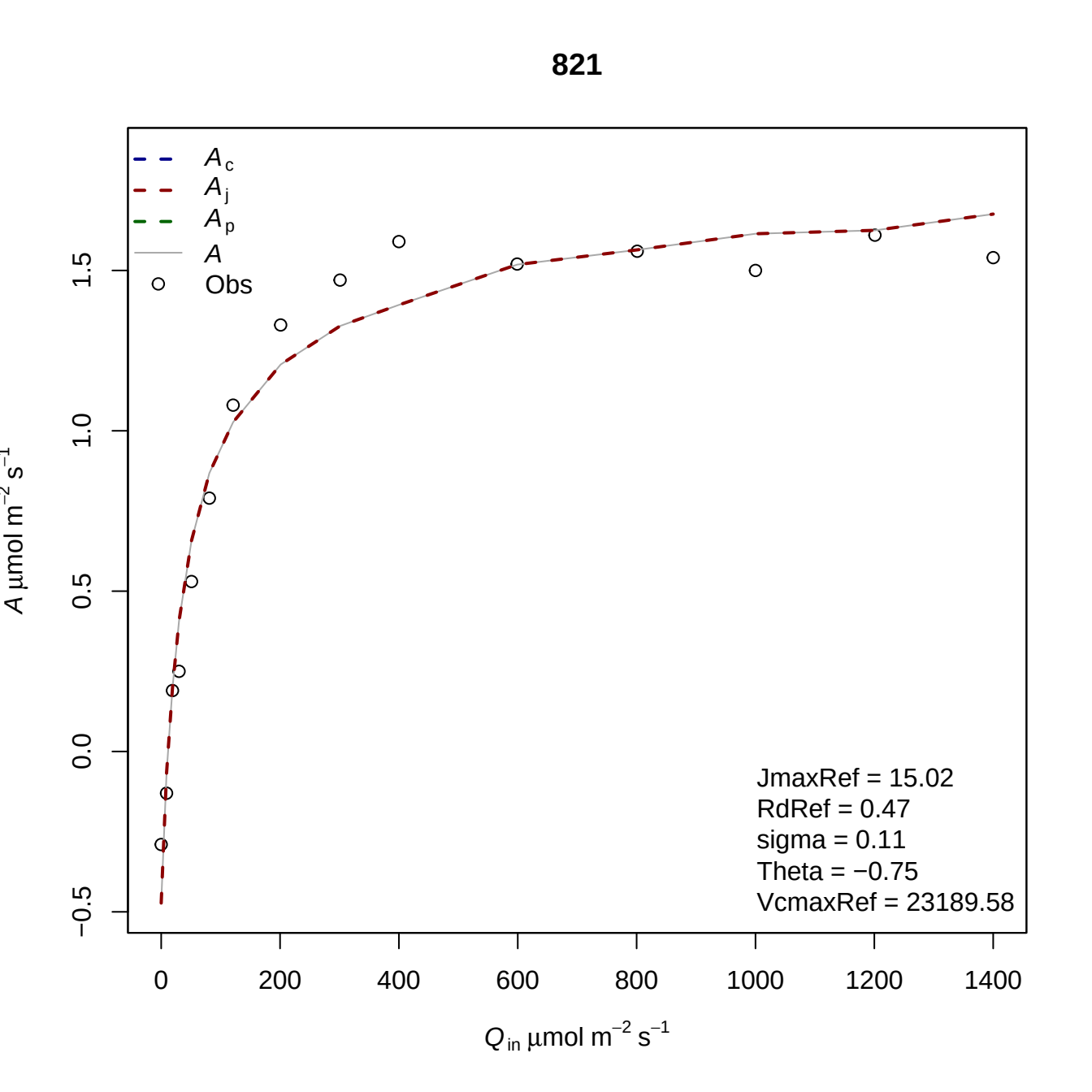

816
A_c
A_j
A_p
A
Obs
JmaxRef = 36.67
RdRef = 0.54
sigma = 0.14
Theta = 0.81
VcmaxRef = 32250.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

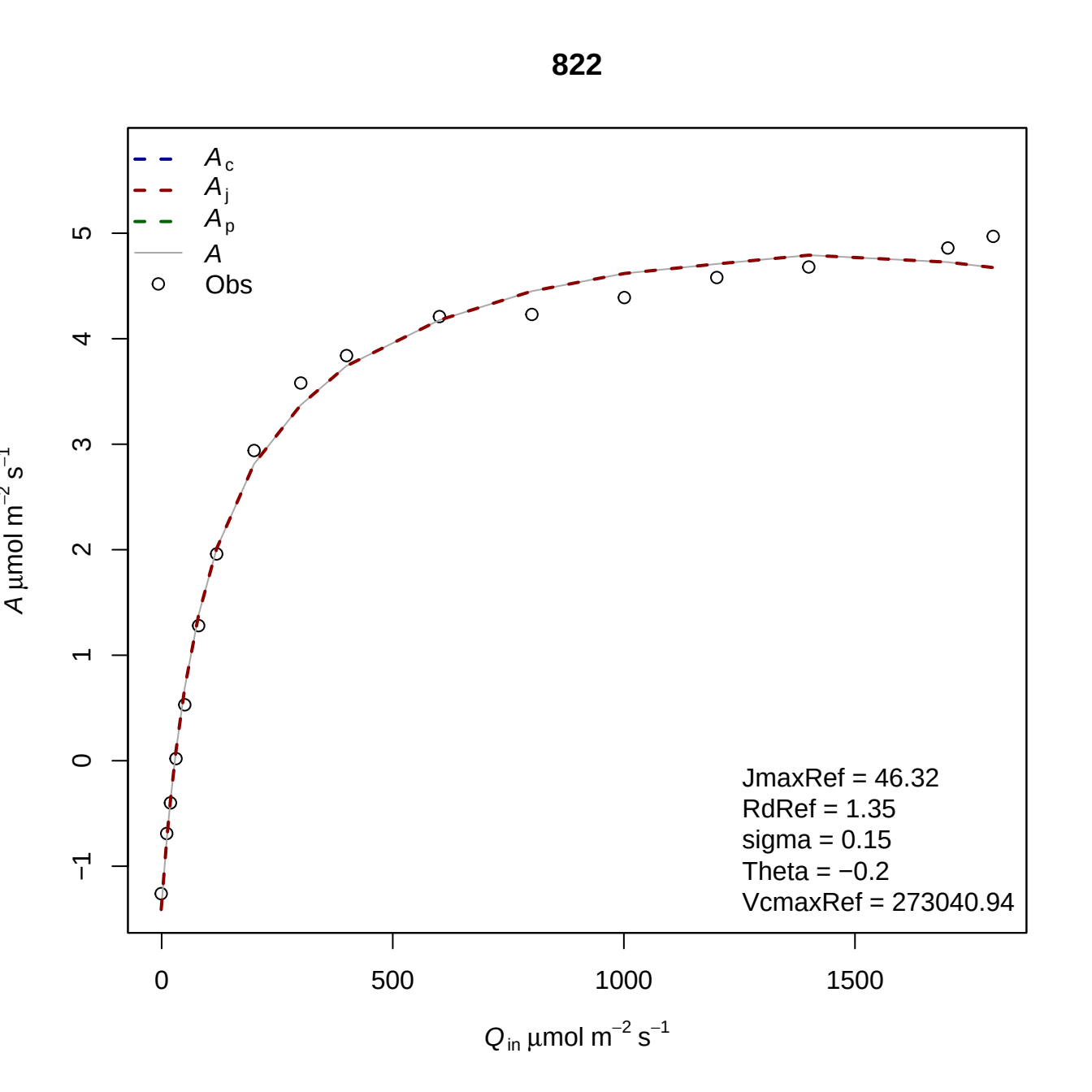

817
A_c
A_j
A_p
A
Obs
JmaxRef = 84.25
RdRef = 1.64
sigma = 0.07
Theta = 0.36
VcmaxRef = 27351.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

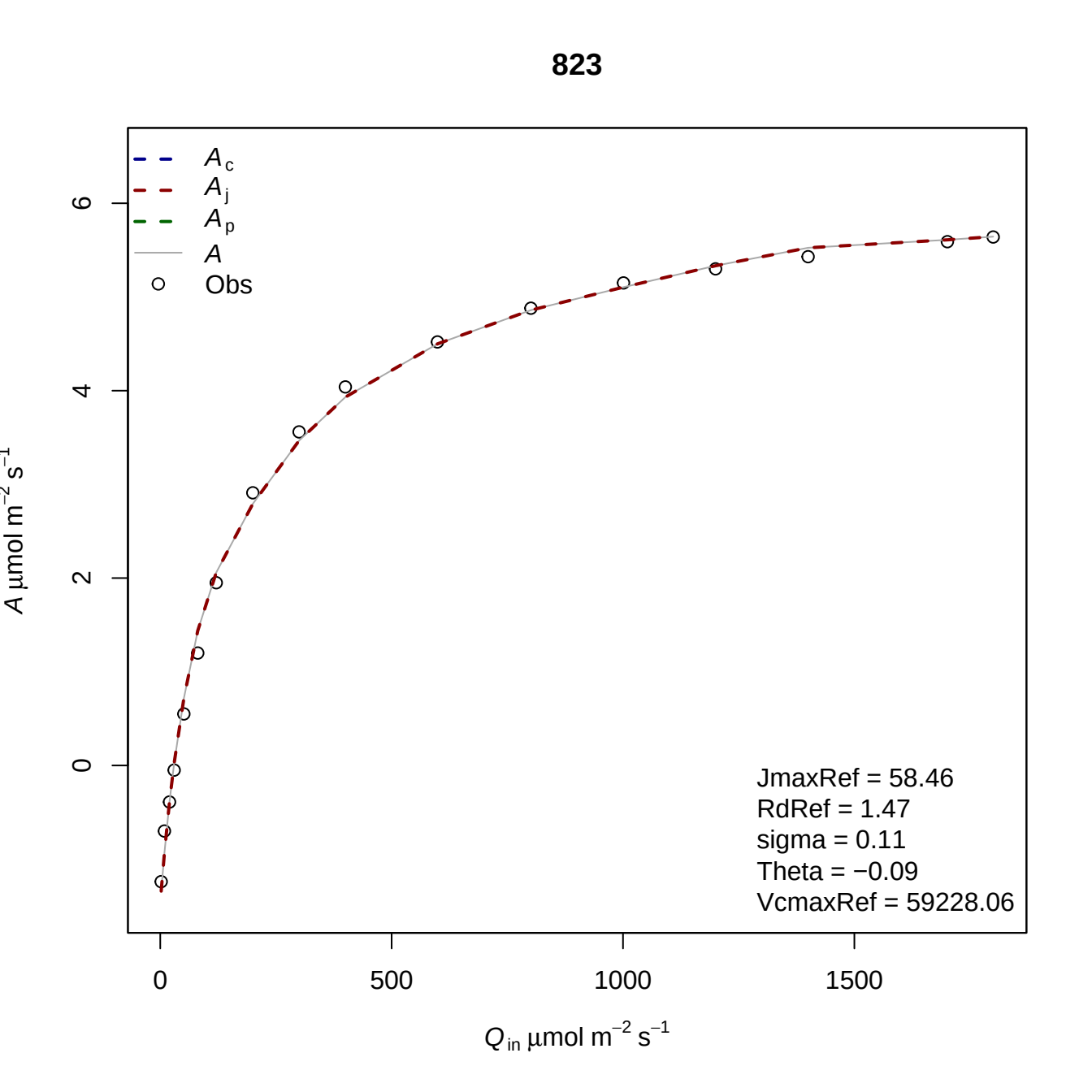

818
A_c
A_j
A_p
A
Obs
JmaxRef = 59.42
RdRef = 2.61
sigma = 0.06
Theta = 0.27
VcmaxRef = 3019.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

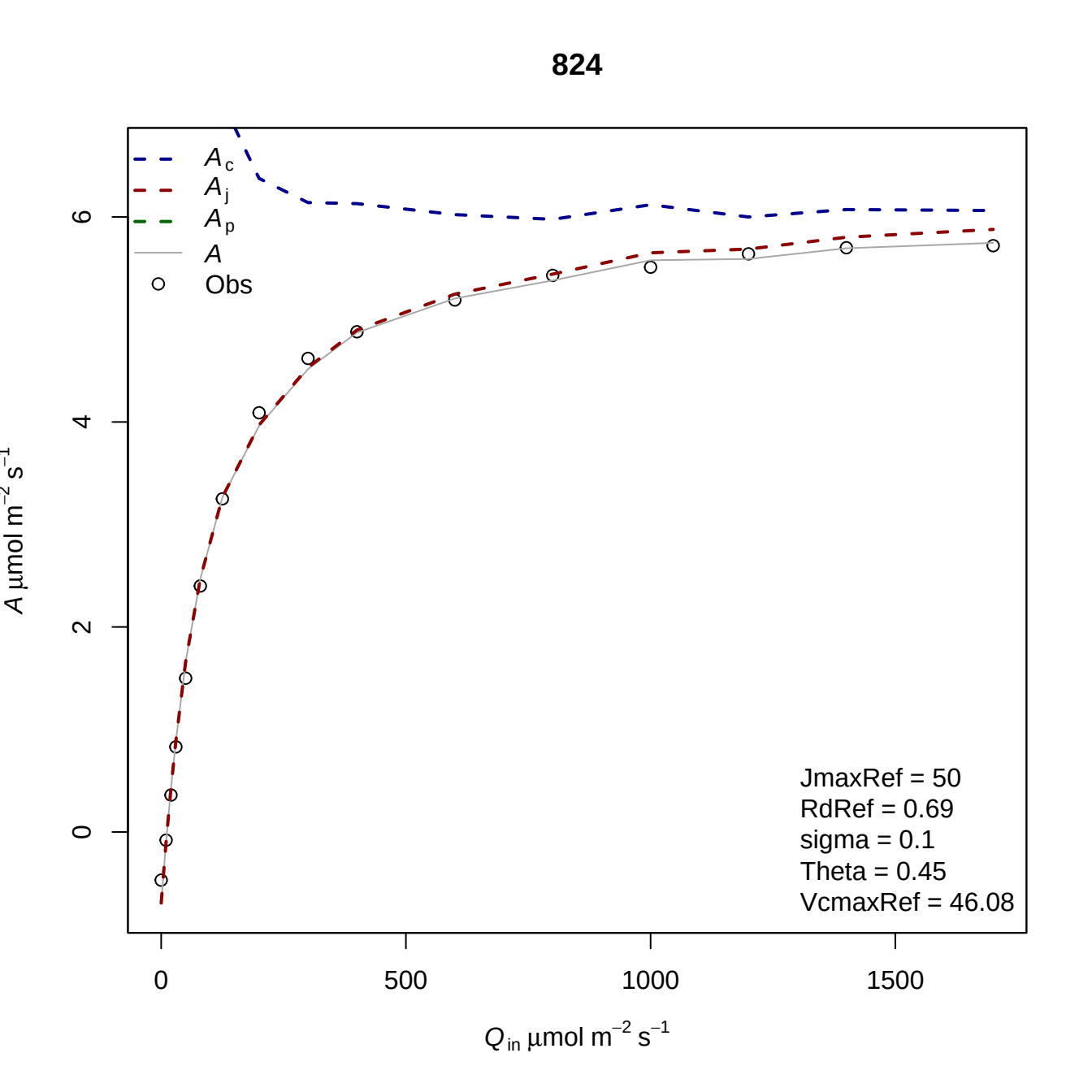

819
A_c
A_j
A_p
A
Obs
JmaxRef = 80.04
RdRef = 2.24
sigma = 0.12
Theta = 0.66
VcmaxRef = 67.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

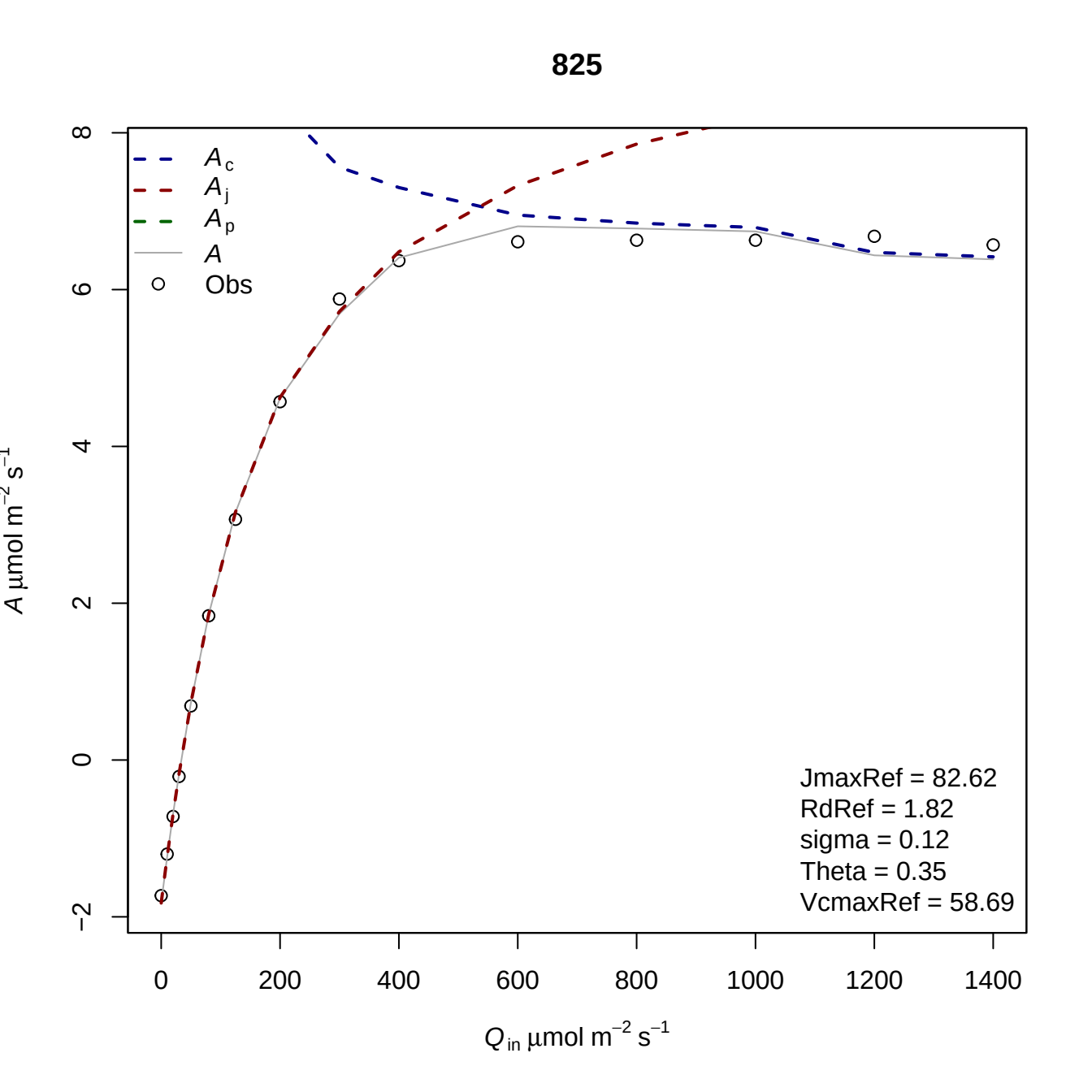

820
A_c
A_j
A_p
A
Obs
JmaxRef = 55.67
RdRef = 2.17
sigma = 0.21
Theta = 0.05
VcmaxRef = 79086.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

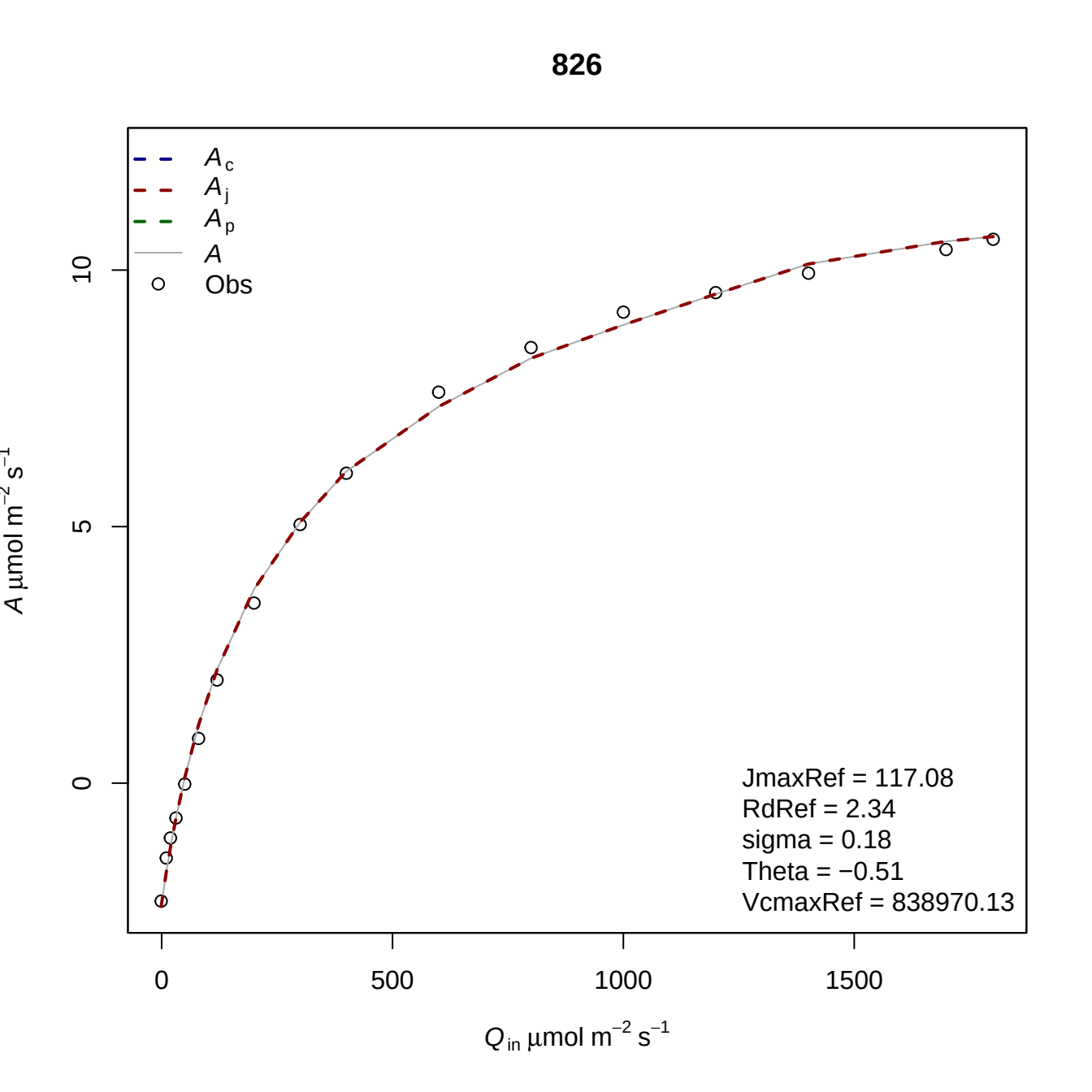

821
A_c
A_j
A_p
A
Obs
JmaxRef = 15.02
RdRef = 0.47
sigma = 0.11
Theta = −0.75
VcmaxRef = 23189.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400

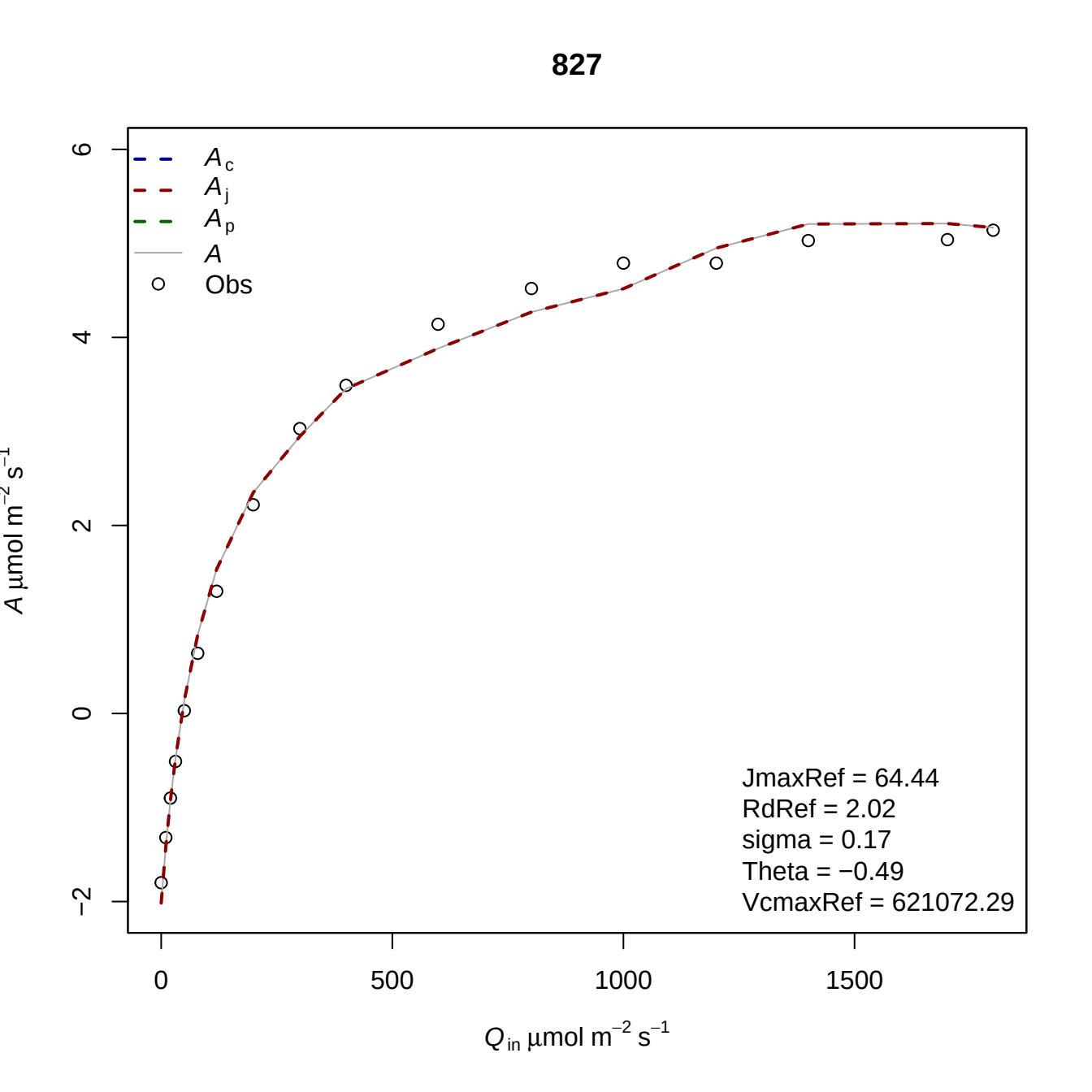

822
A_c
A_j
A_p
A
Obs
JmaxRef = 46.32
RdRef = 1.35
sigma = 0.15
Theta = −0.2
VcmaxRef = 273040.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

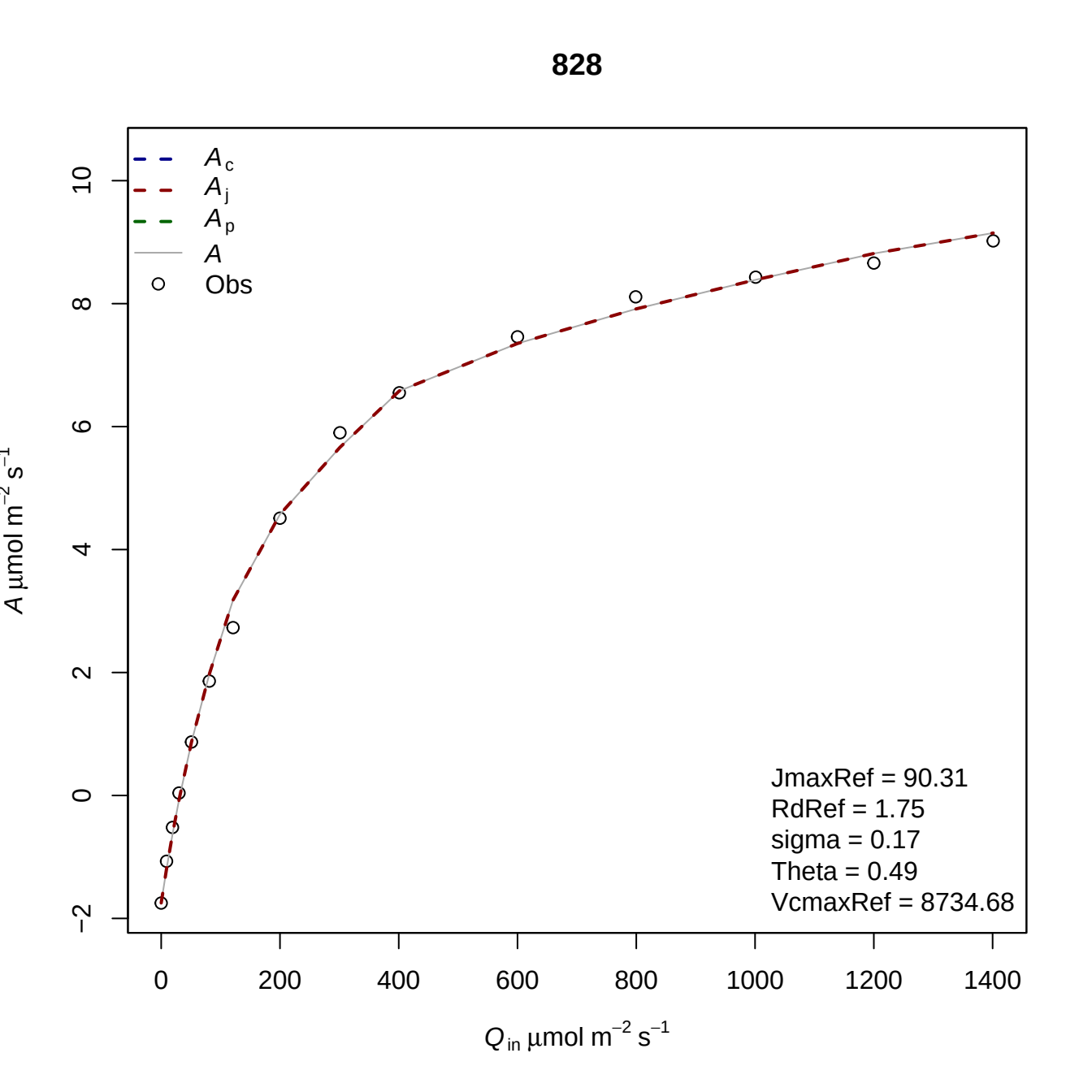

823
A_c
A_j
A_p
A
Obs
JmaxRef = 58.46
RdRef = 1.47
sigma = 0.11
Theta = −0.09
VcmaxRef = 59228.06
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

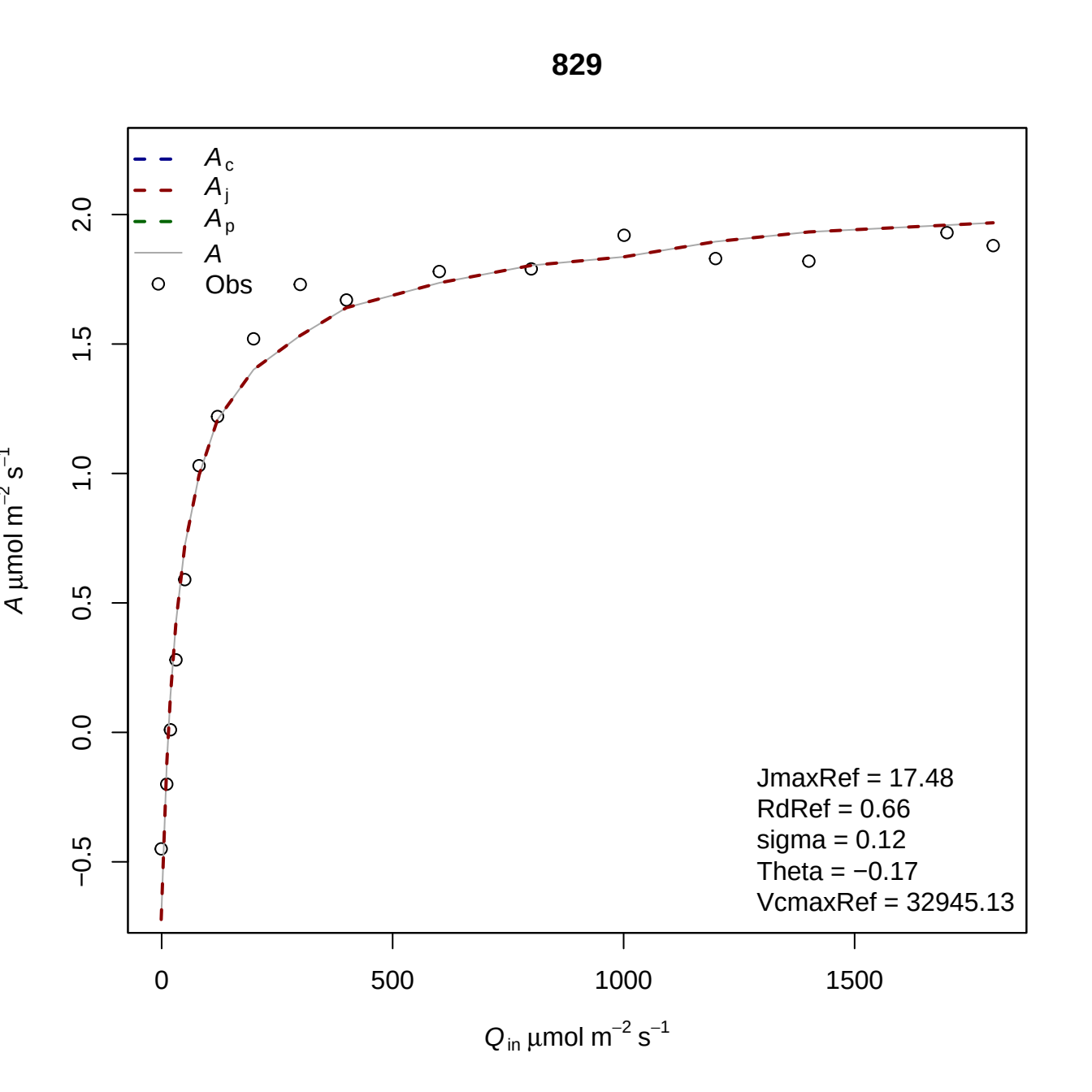

824
A_c
A_j
A_p
A
Obs
JmaxRef = 50
RdRef = 0.69
sigma = 0.1
Theta = 0.45
VcmaxRef = 46.08
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

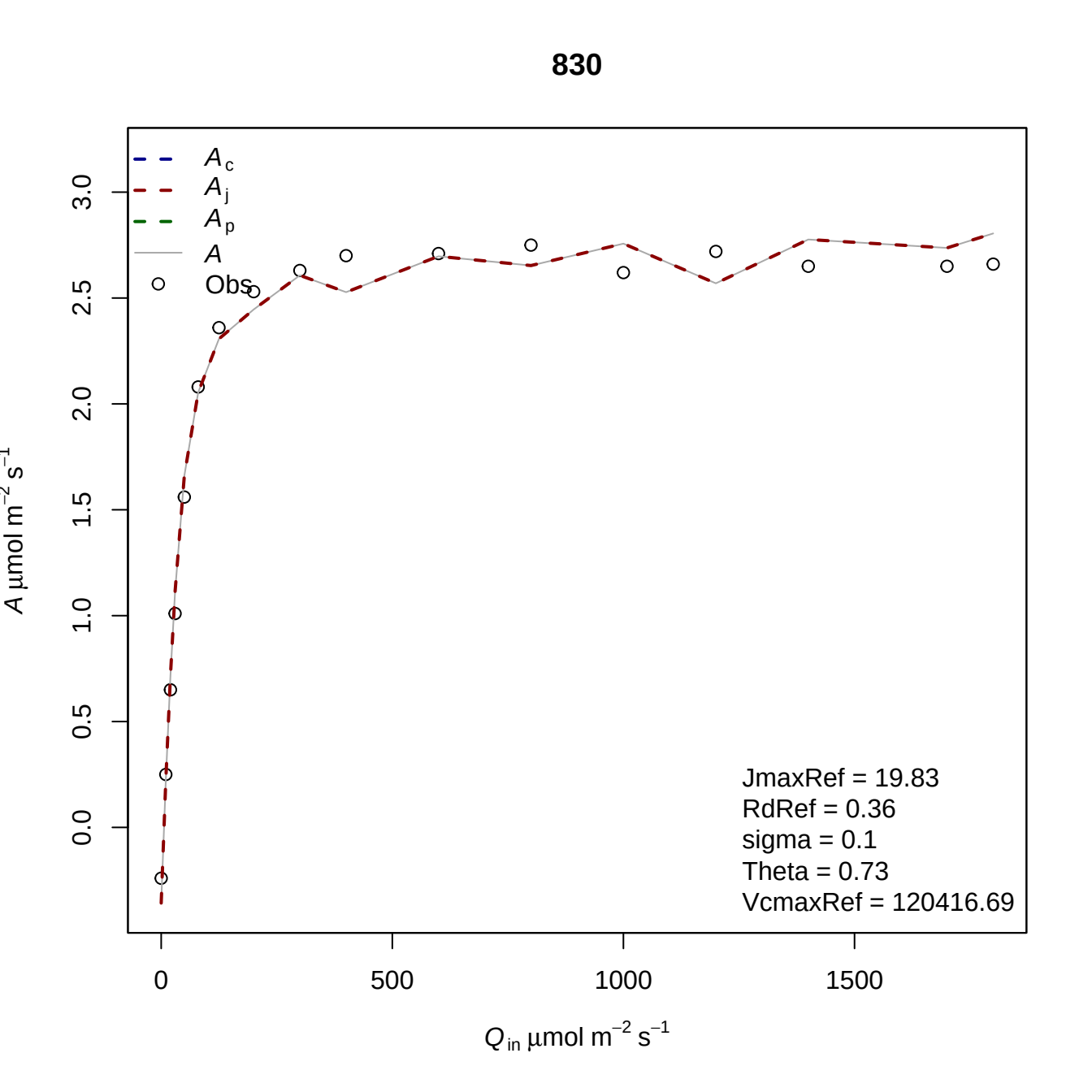

825
A_c
A_j
A_p
A
Obs
JmaxRef = 82.62
RdRef = 1.82
sigma = 0.12
Theta = 0.35
VcmaxRef = 58.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

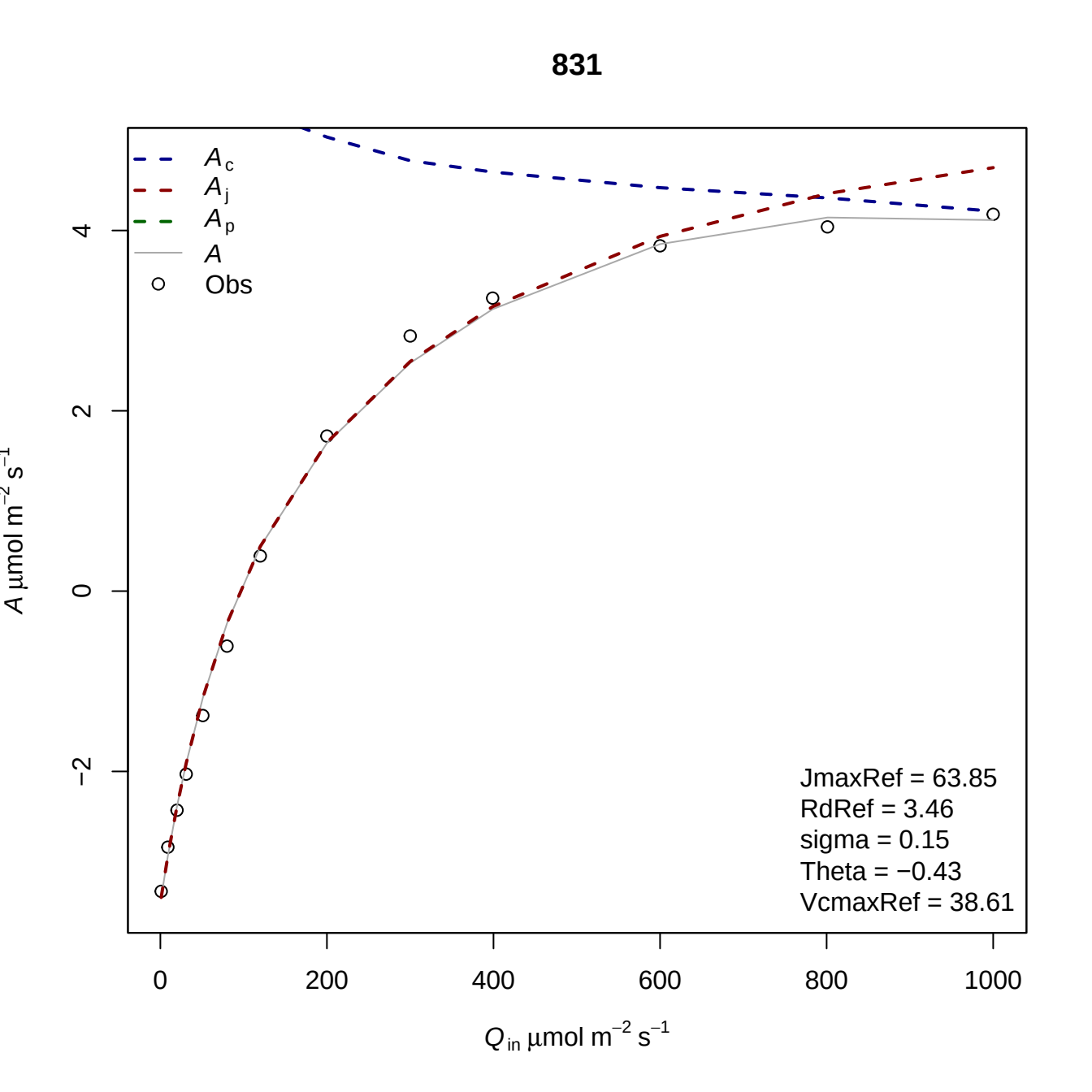

826
A_c
A_j
A_p
A
Obs
JmaxRef = 117.08
RdRef = 2.34
sigma = 0.18
Theta = −0.51
VcmaxRef = 838970.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

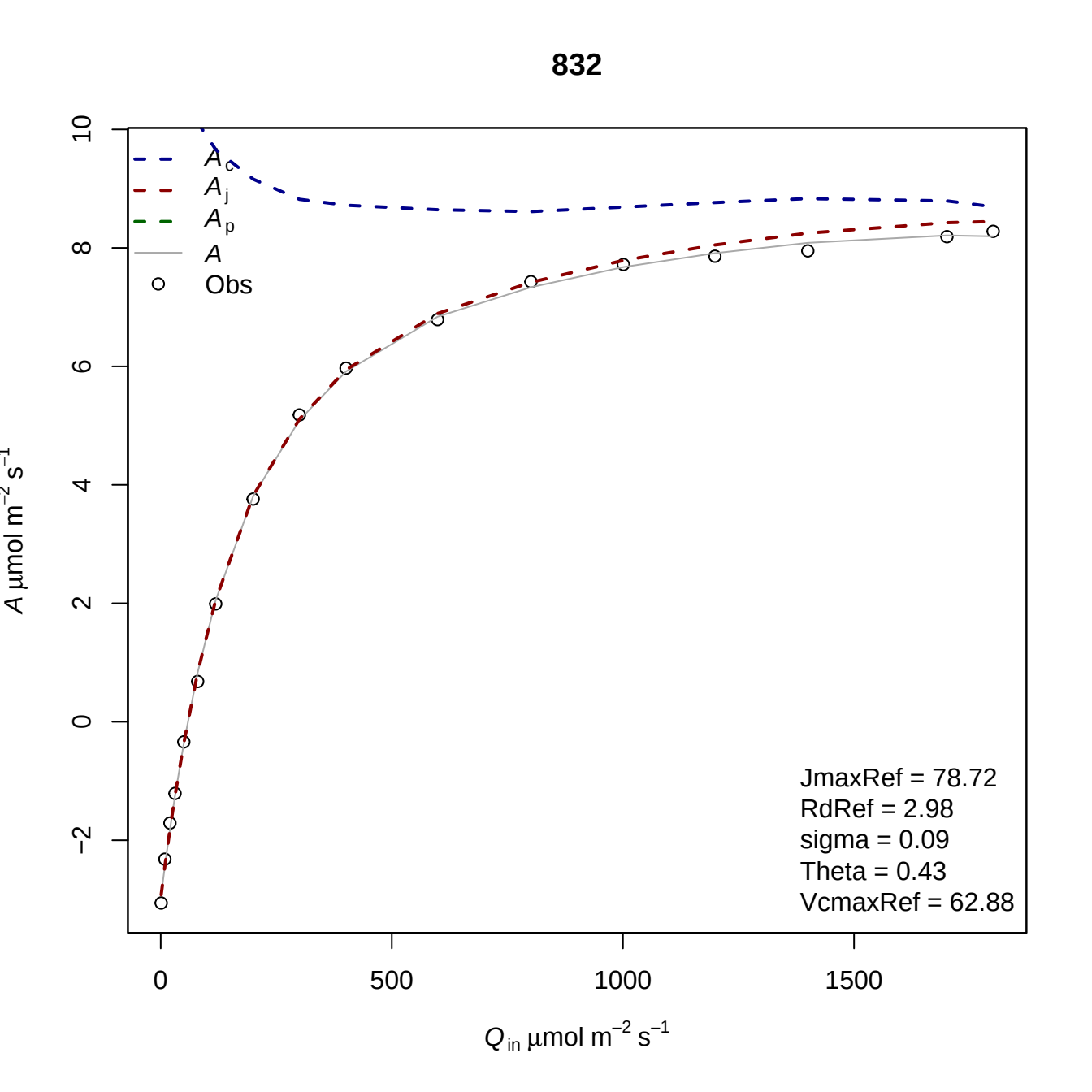

827
A_c
A_j
A_p
A
Obs
JmaxRef = 64.44
RdRef = 2.02
sigma = 0.17
Theta = −0.49
VcmaxRef = 621072.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

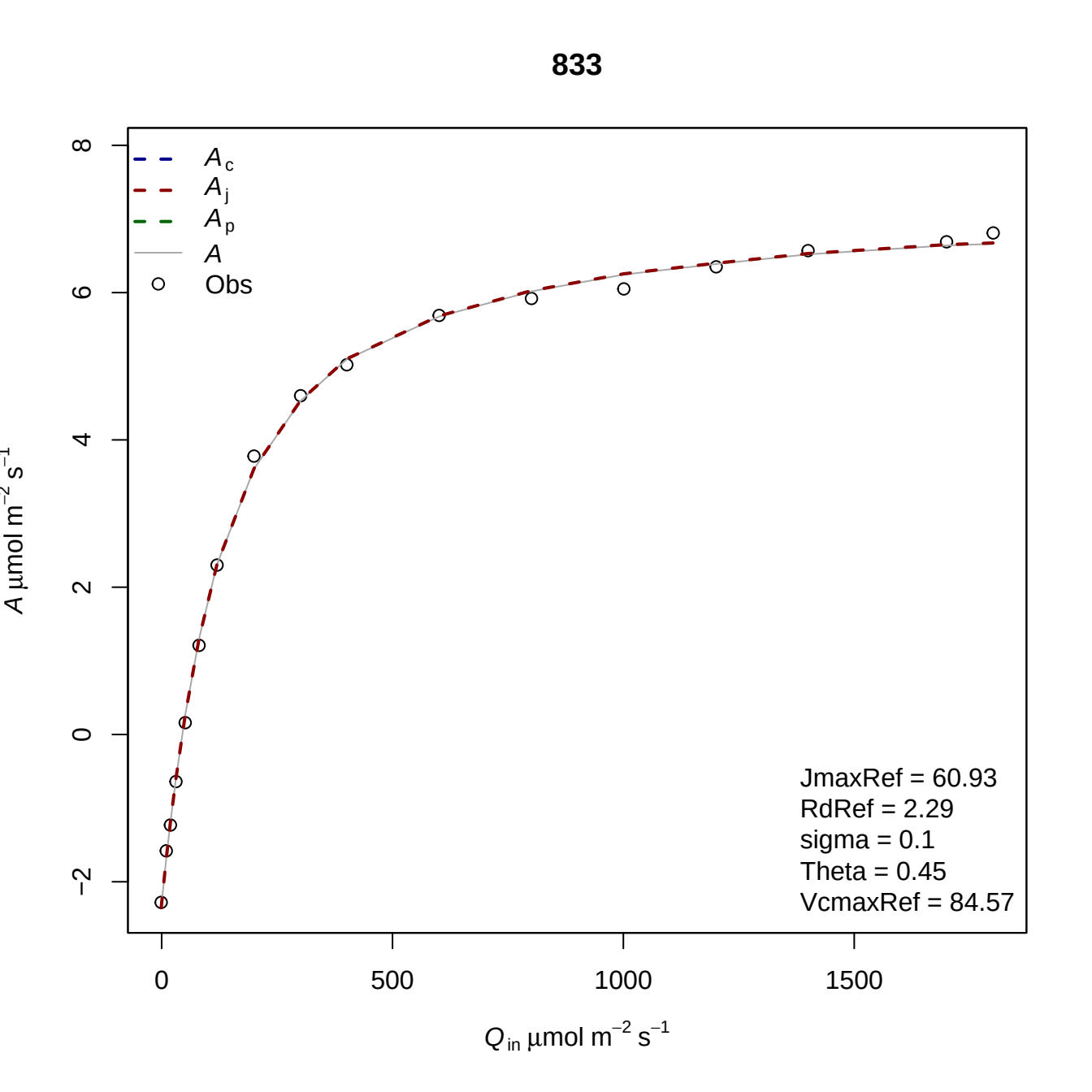

828
A_c
A_j
A_p
A
Obs
JmaxRef = 90.31
RdRef = 1.75
sigma = 0.17
Theta = 0.49
VcmaxRef = 8734.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

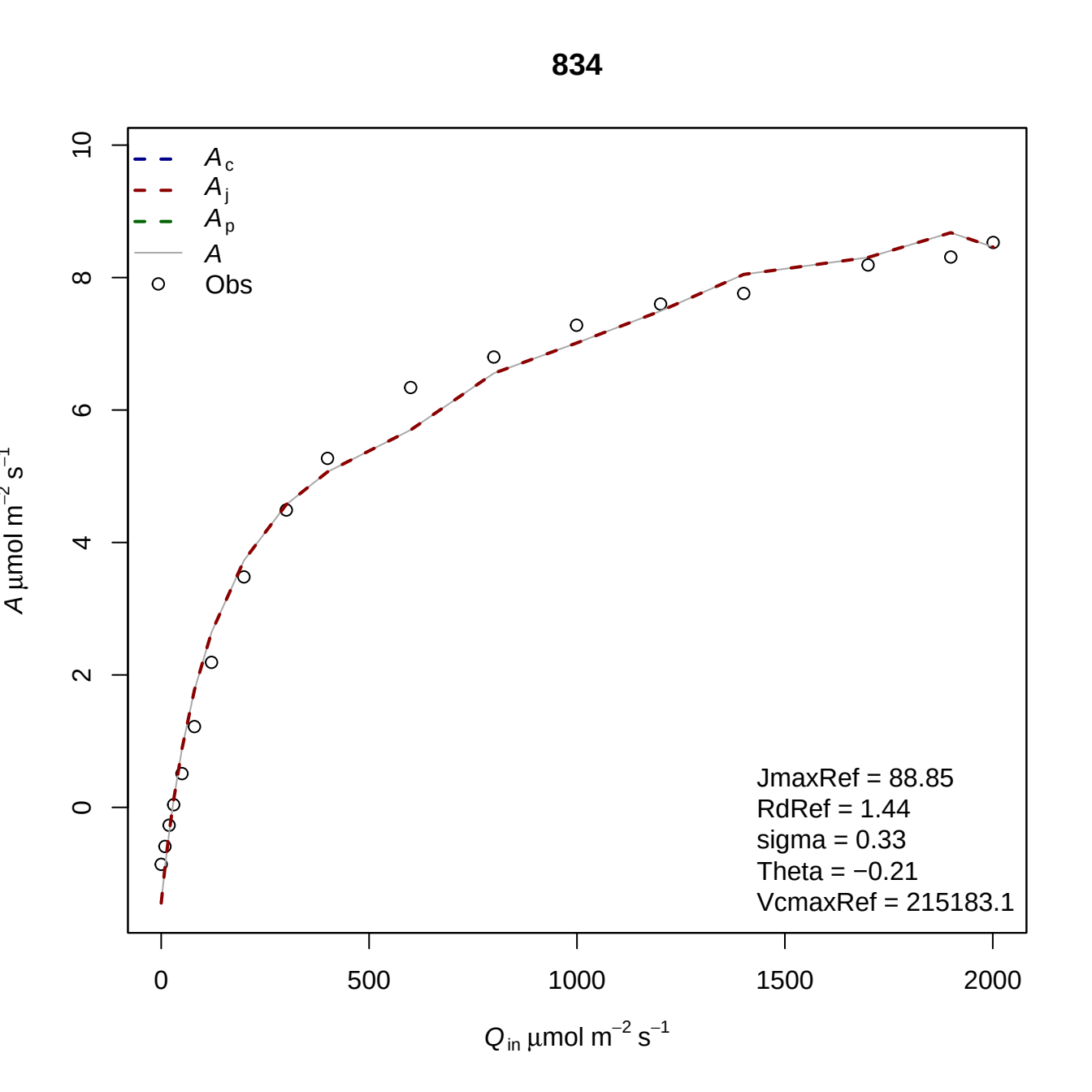

829
A_c
A_j
A_p
A
Obs
JmaxRef = 17.48
RdRef = 0.66
sigma = 0.12
Theta = −0.17
VcmaxRef = 32945.13
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

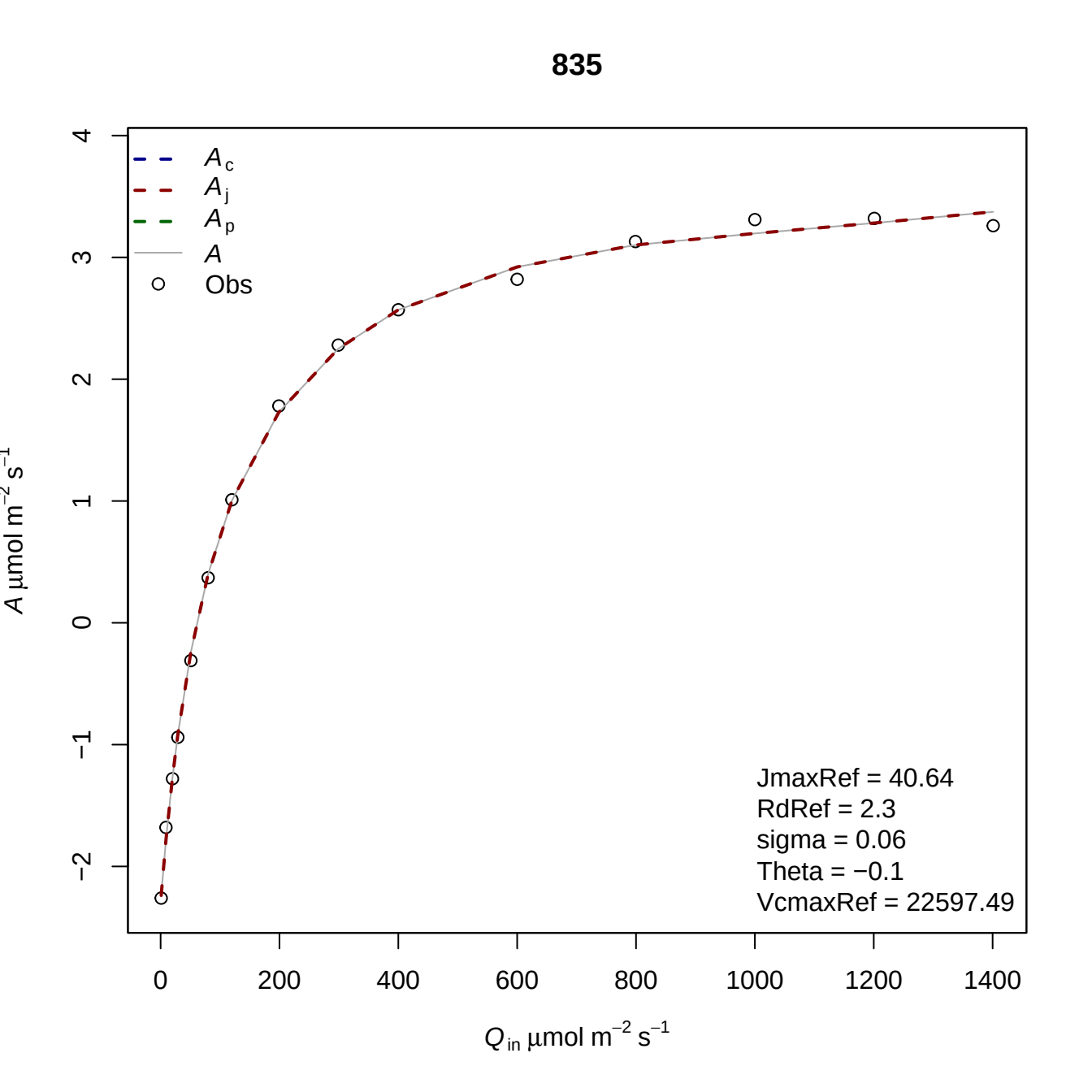

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.83
RdRef = 0.36
sigma = 0.1
Theta = 0.73
VcmaxRef = 120416.69
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

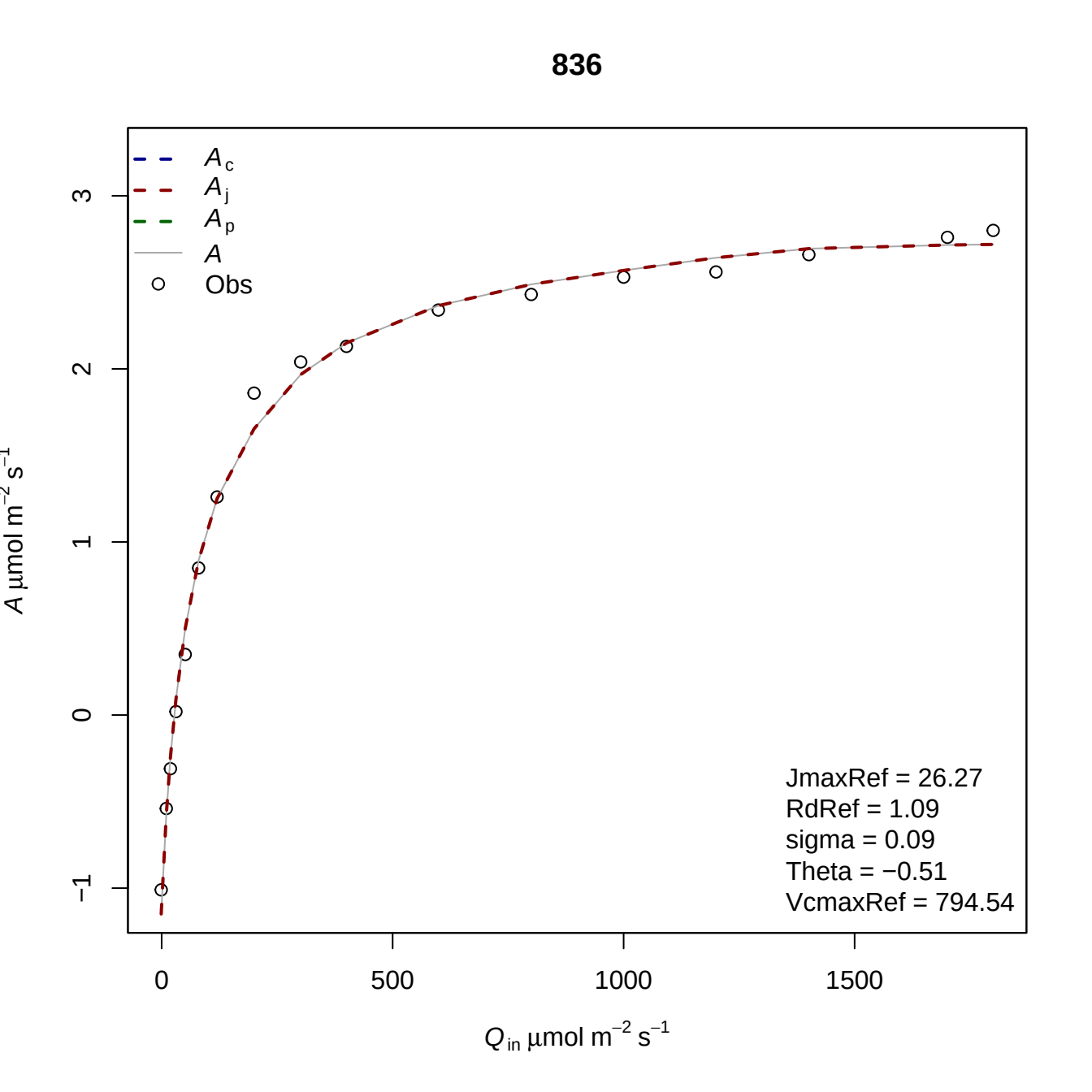

831
A_c
A_j
A_p
A
Obs
JmaxRef = 63.85
RdRef = 3.46
sigma = 0.15
Theta = −0.43
VcmaxRef = 38.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

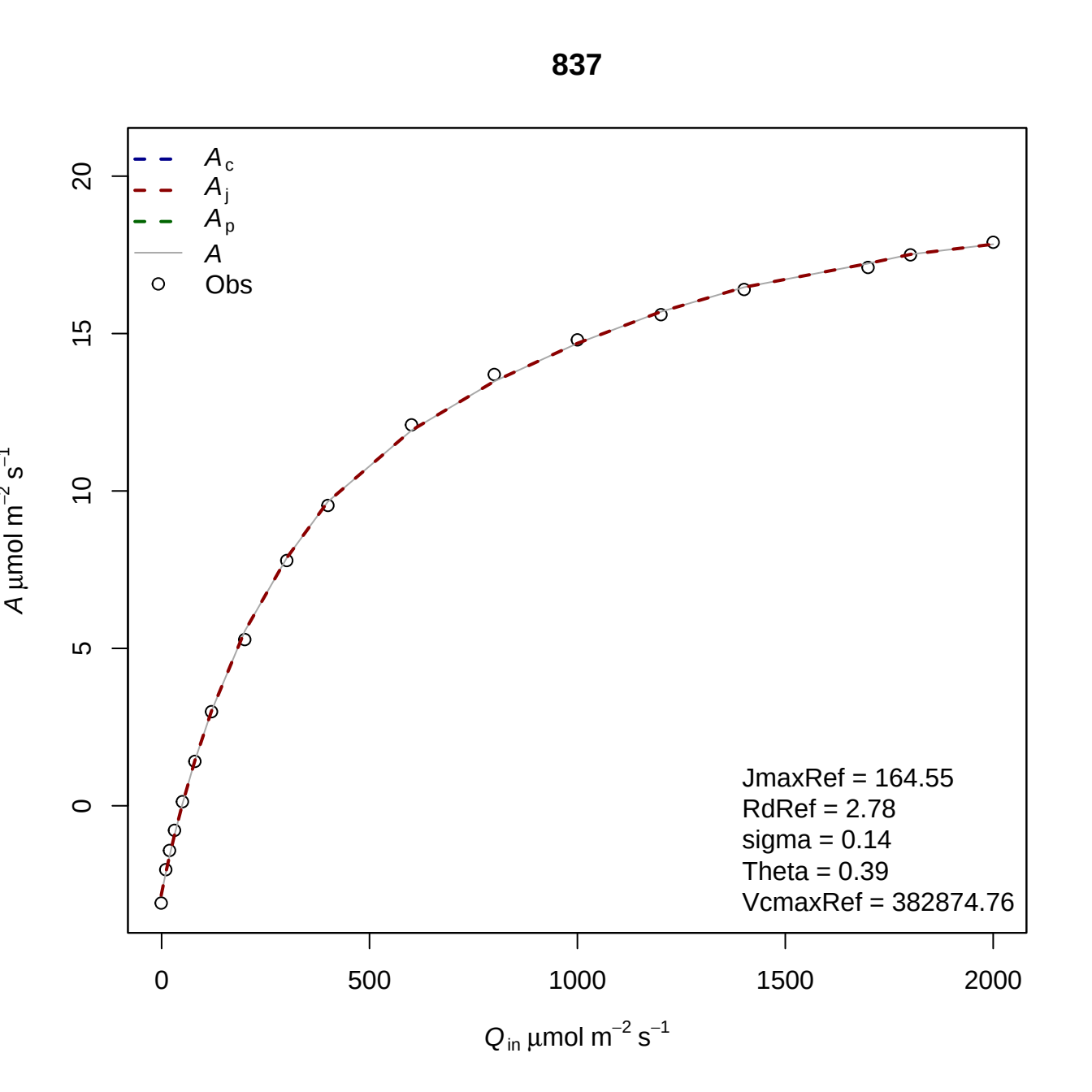

832
A_c
A_j
A_p
A
Obs
JmaxRef = 78.72
RdRef = 2.98
sigma = 0.09
Theta = 0.43
VcmaxRef = 62.88
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

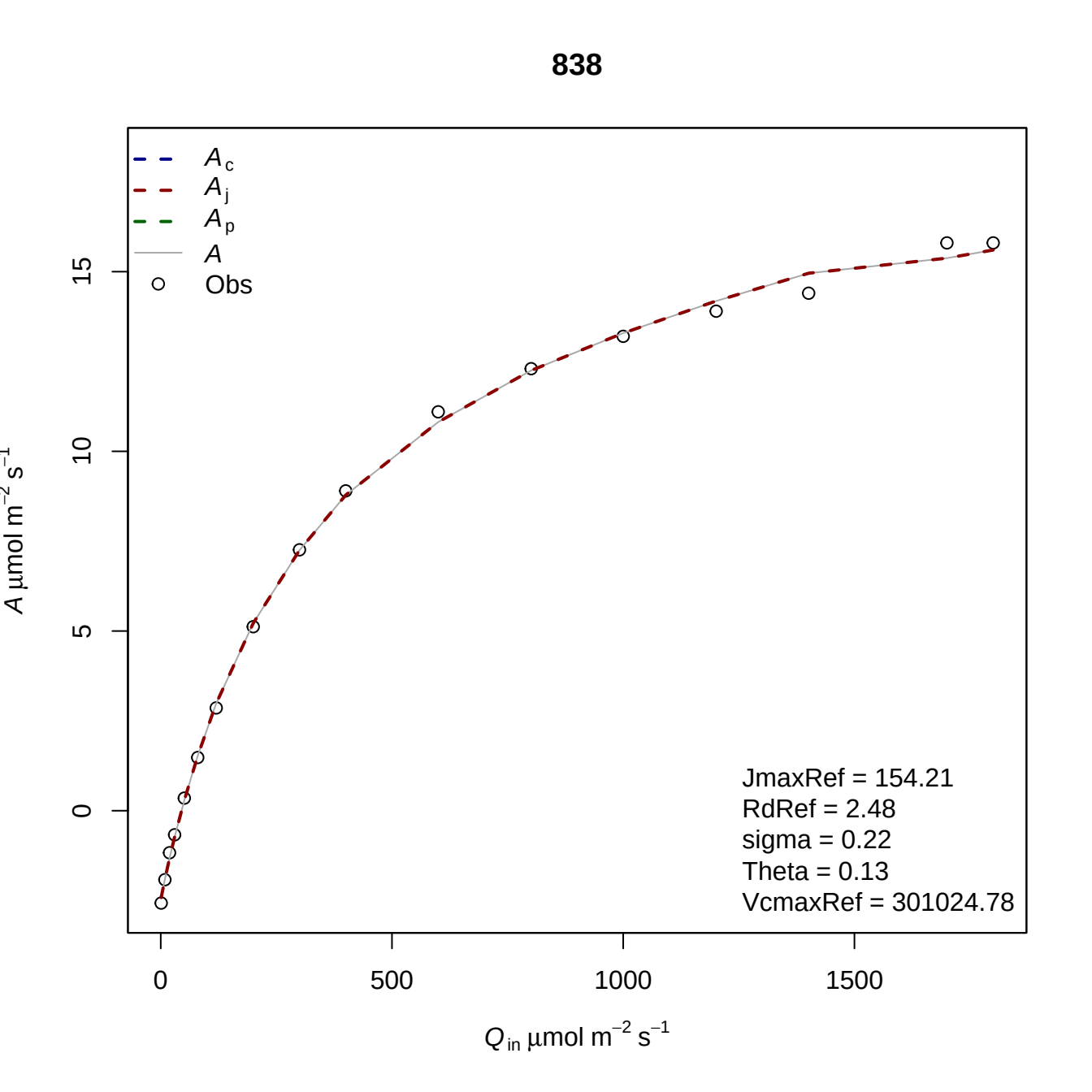

833
A_c
A_j
A_p
A
Obs
JmaxRef = 60.93
RdRef = 2.29
sigma = 0.1
Theta = 0.45
VcmaxRef = 84.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

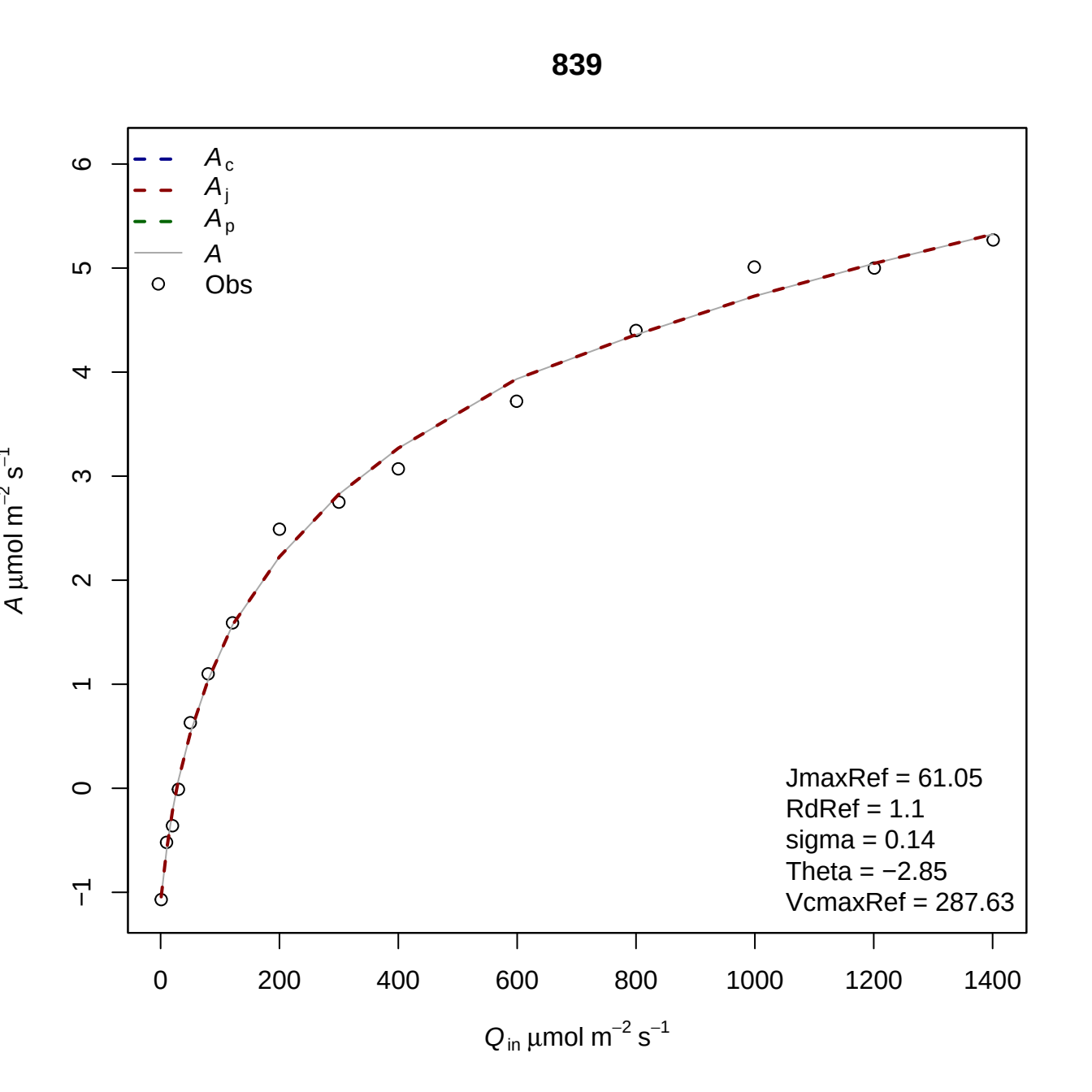

834
A_c
A_j
A_p
A
Obs
JmaxRef = 88.85
RdRef = 1.44
sigma = 0.33
Theta = −0.21
VcmaxRef = 215183.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

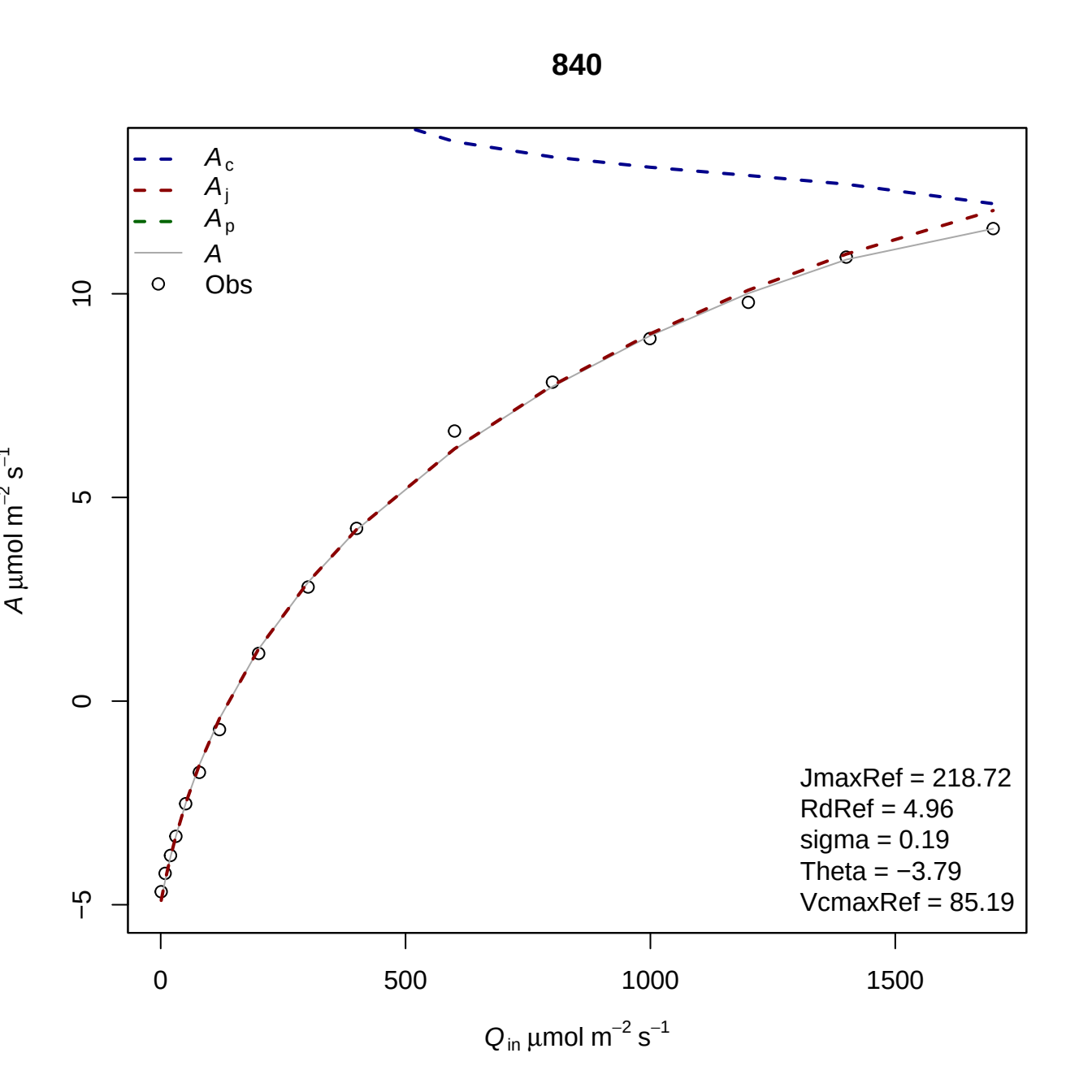

835
A_c
A_j
A_p
A
Obs
JmaxRef = 40.64
RdRef = 2.3
sigma = 0.06
Theta = −0.1
VcmaxRef = 22597.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

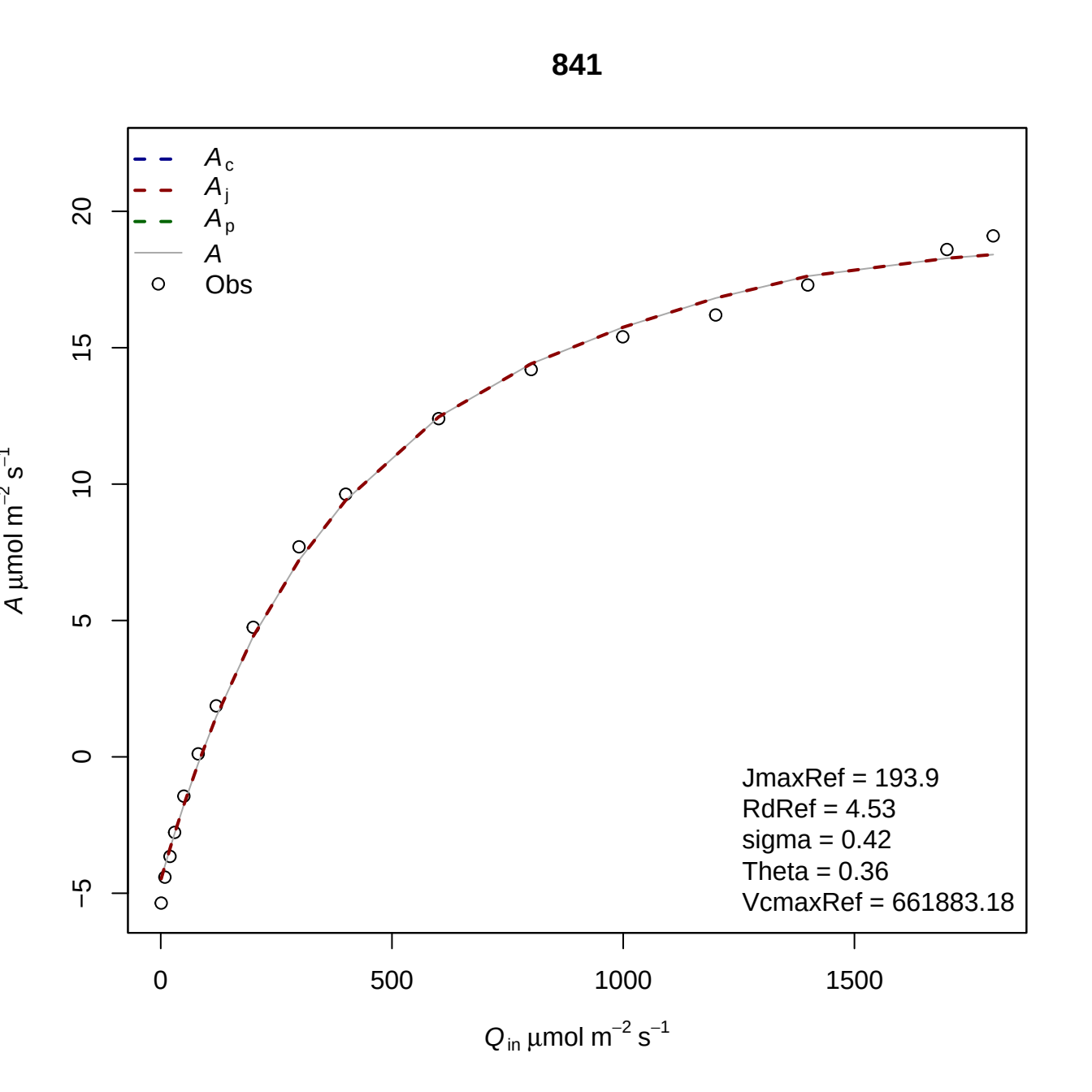

836
A_c
A_j
A_p
A
Obs
JmaxRef = 26.27
RdRef = 1.09
sigma = 0.09
Theta = −0.51
VcmaxRef = 794.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

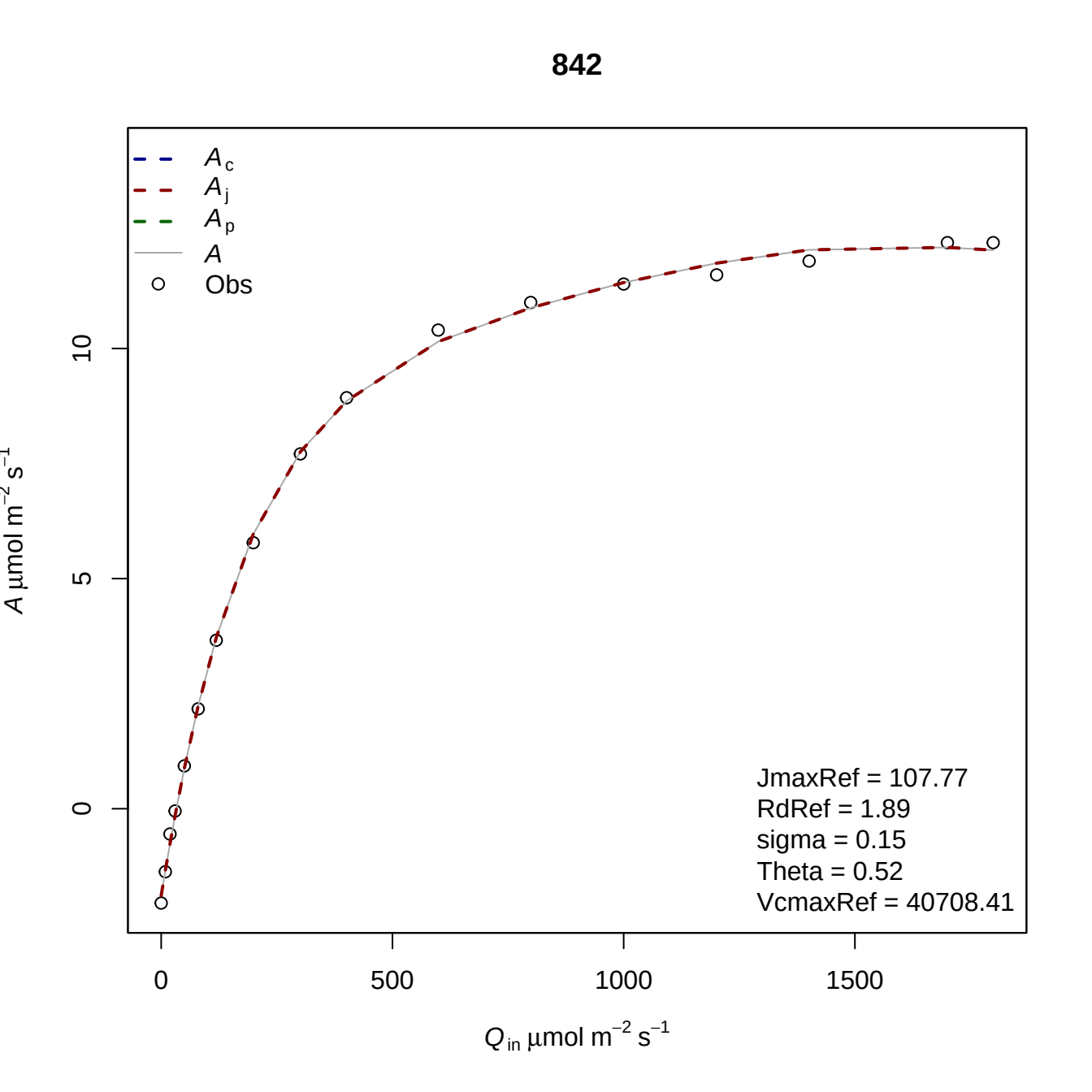

837
A_c
A_j
A_p
A
Obs
JmaxRef = 164.55
RdRef = 2.78
sigma = 0.14
Theta = 0.39
VcmaxRef = 382874.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

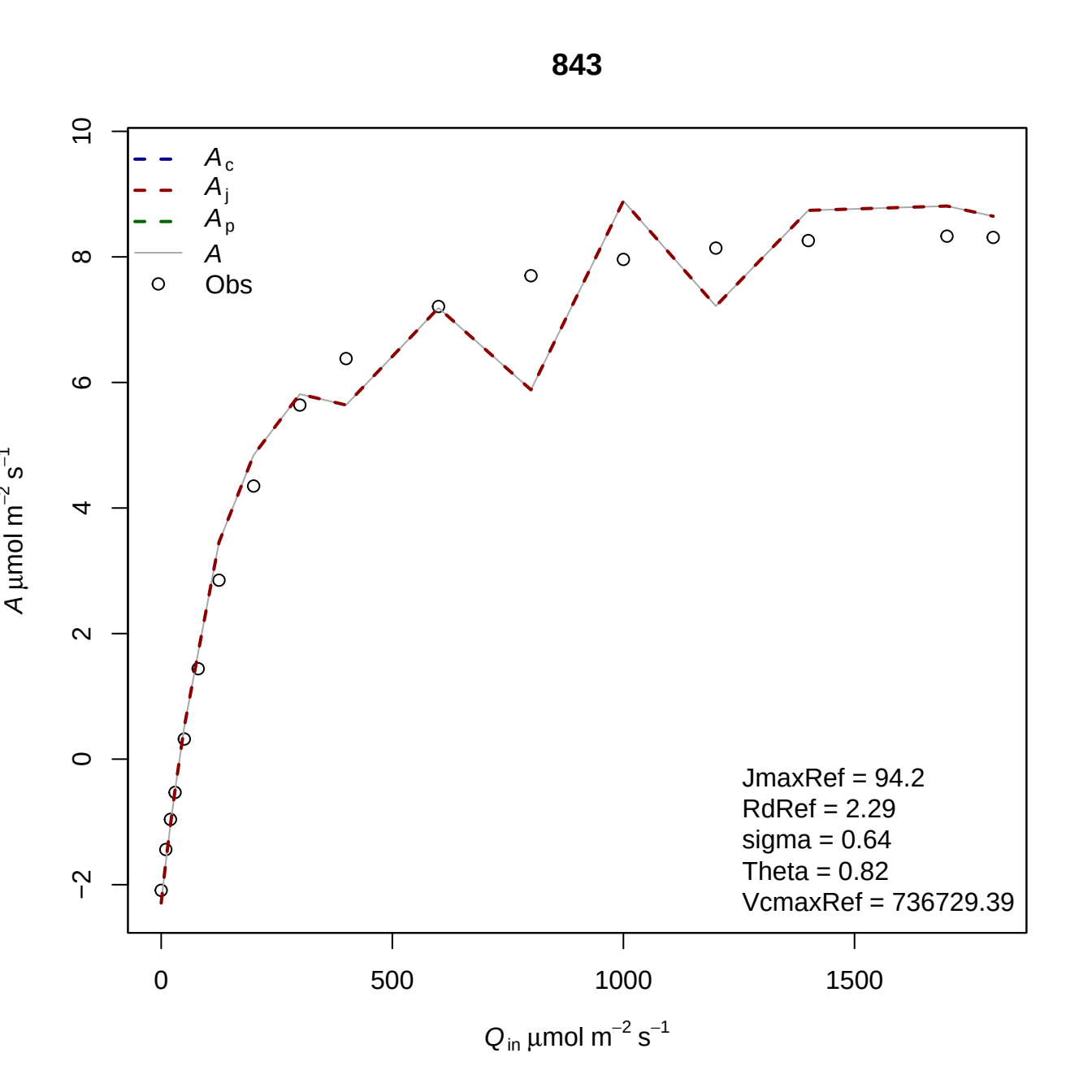

838
A_c
A_j
A_p
A
Obs
JmaxRef = 154.21
RdRef = 2.48
sigma = 0.22
Theta = 0.13
VcmaxRef = 301024.78
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

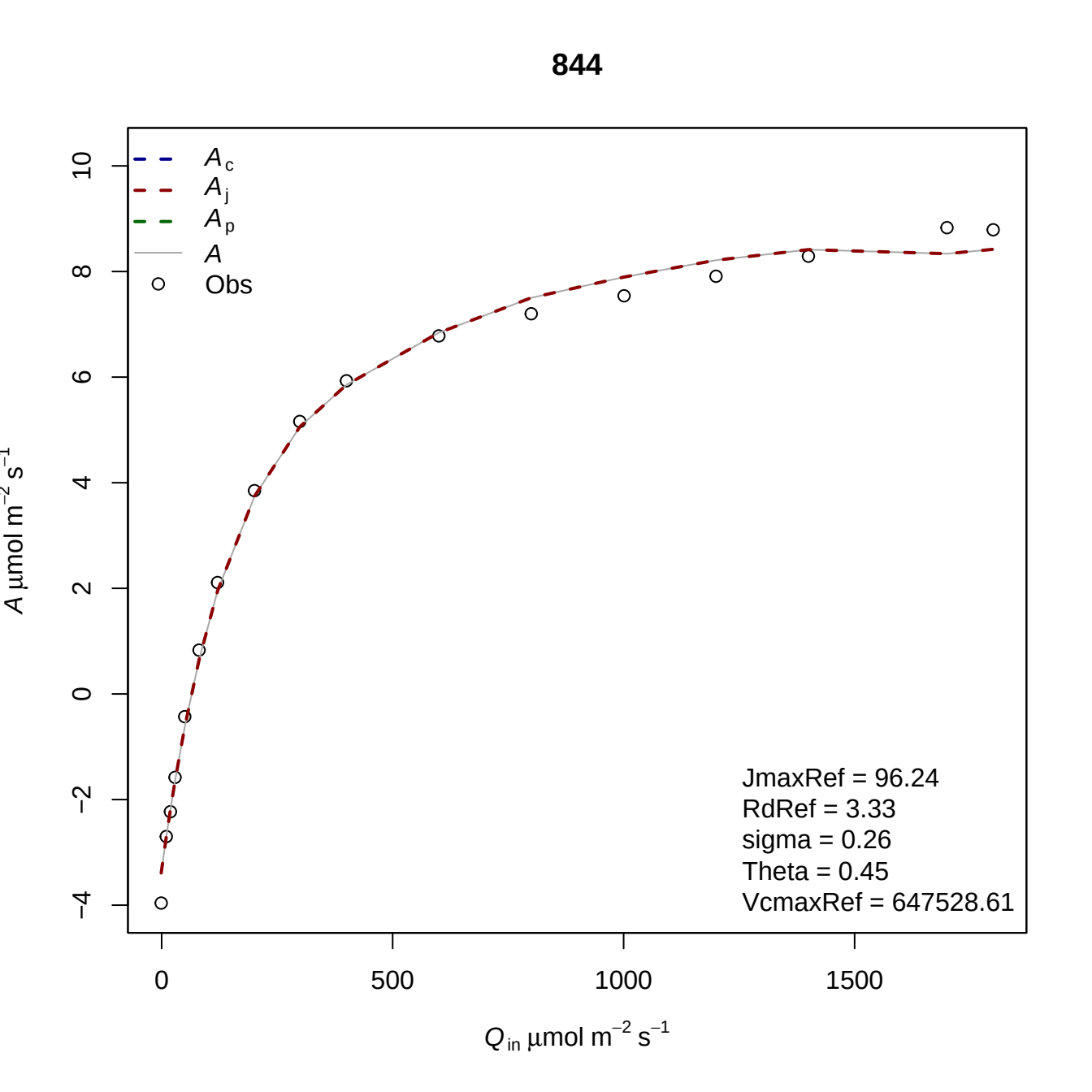

839
A_c
A_j
A_p
A
Obs
JmaxRef = 61.05
RdRef = 1.1
sigma = 0.14
Theta = −2.85
VcmaxRef = 287.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
1400

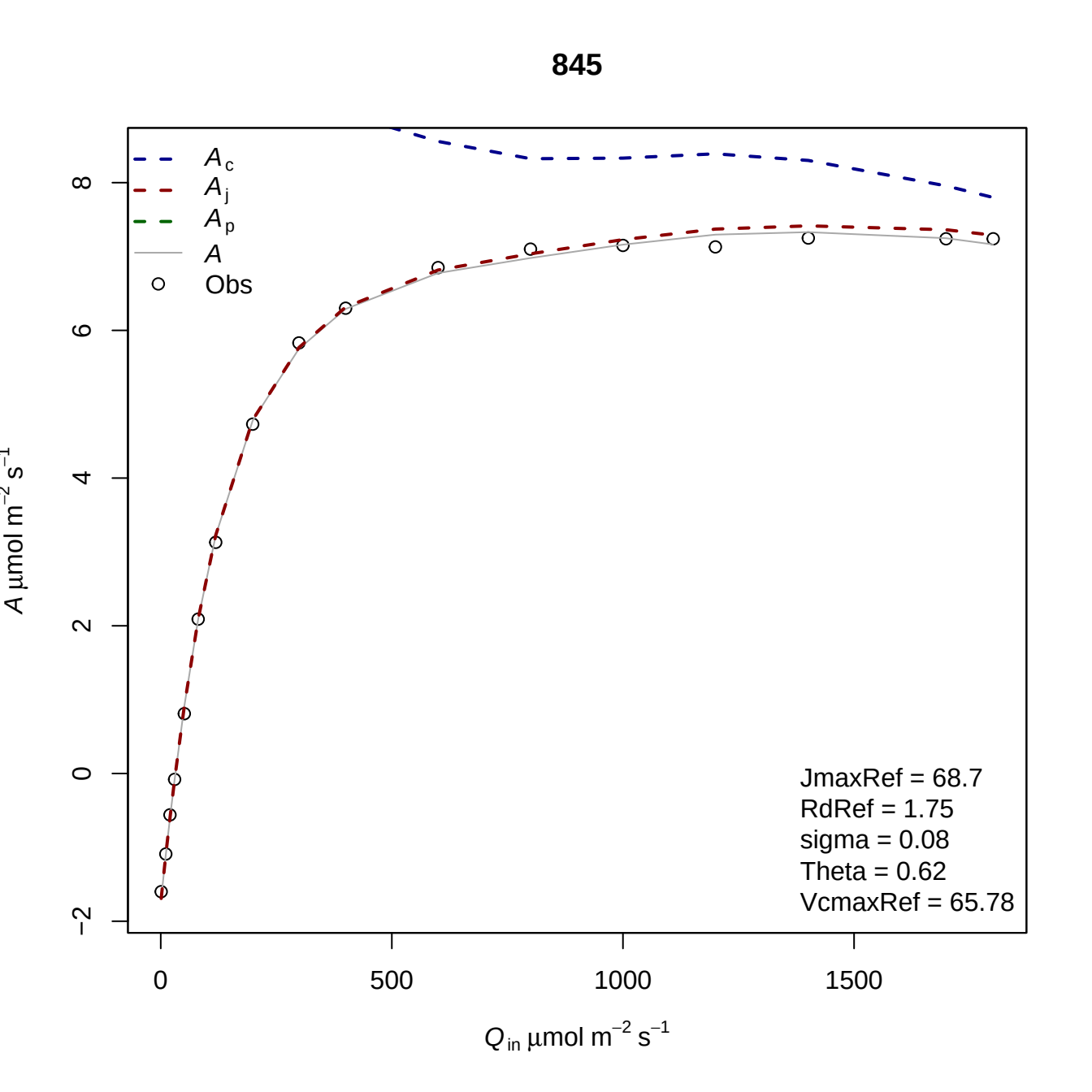

840
A_c
A_j
A_p
A
Obs
JmaxRef = 218.72
RdRef = 4.96
sigma = 0.19
Theta = −3.79
VcmaxRef = 85.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

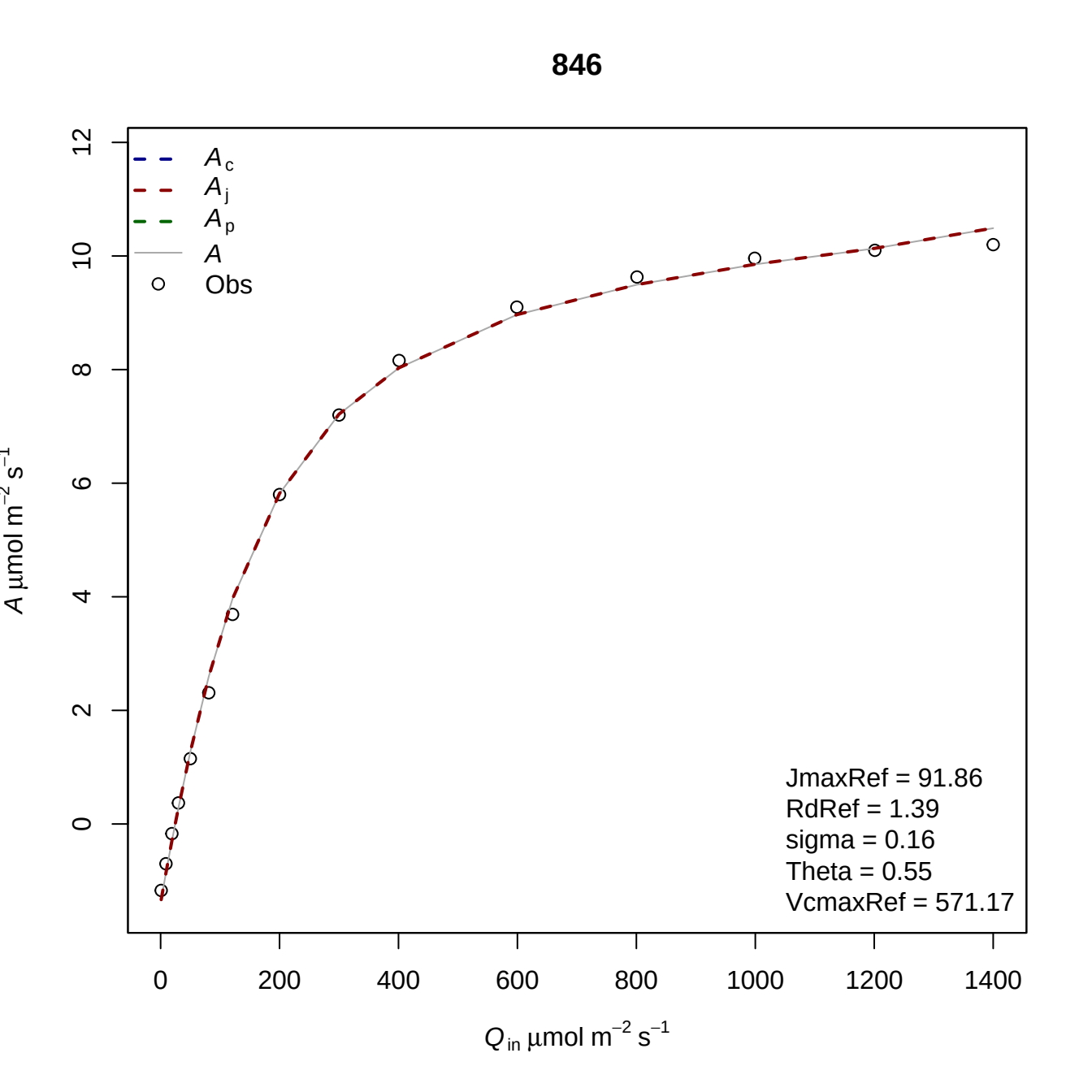

841
A_c
A_j
A_p
A
Obs
JmaxRef = 193.9
RdRef = 4.53
sigma = 0.42
Theta = 0.36
VcmaxRef = 661883.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

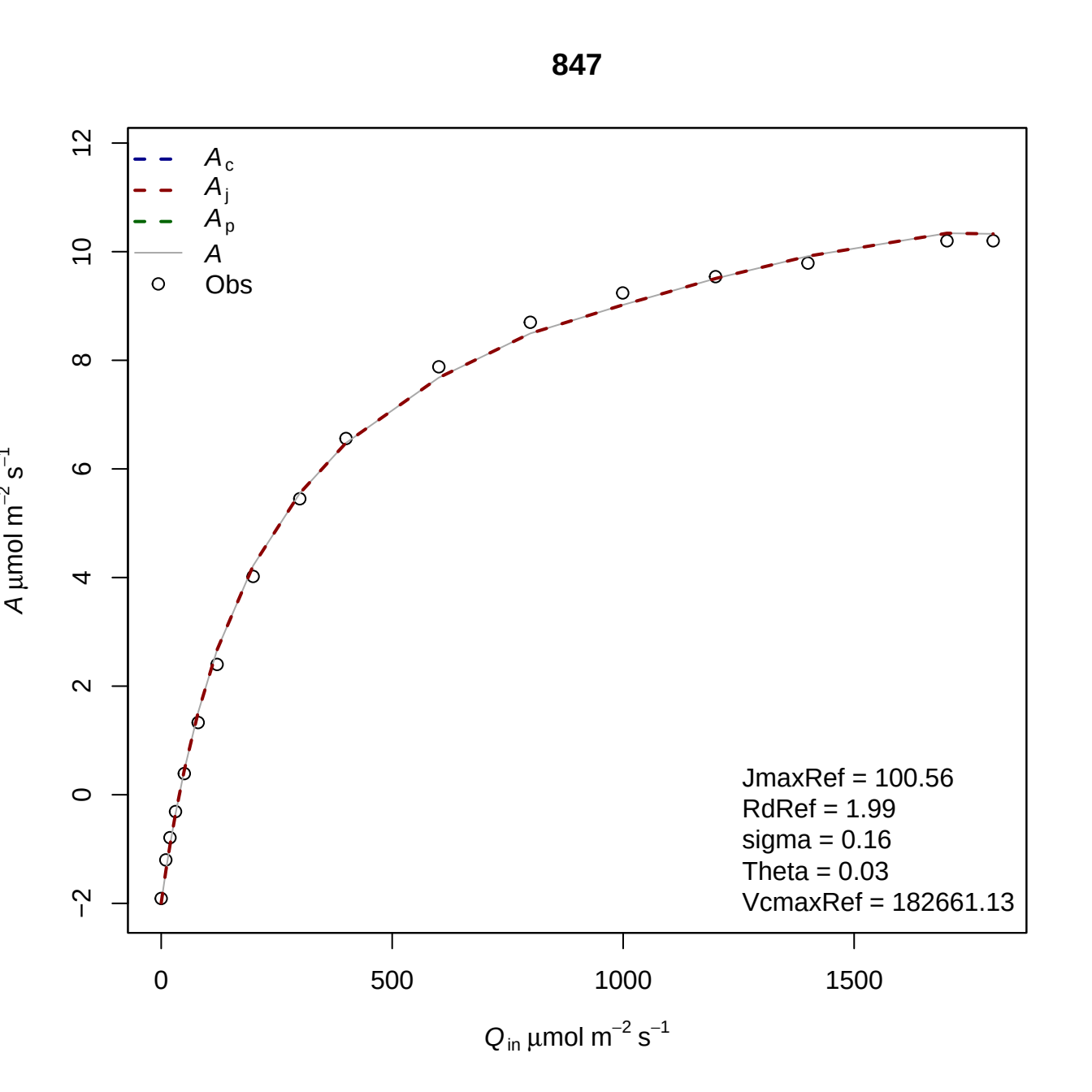

842
A_c
A_j
A_p
A
Obs
JmaxRef = 107.77
RdRef = 1.89
sigma = 0.15
Theta = 0.52
VcmaxRef = 40708.41
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

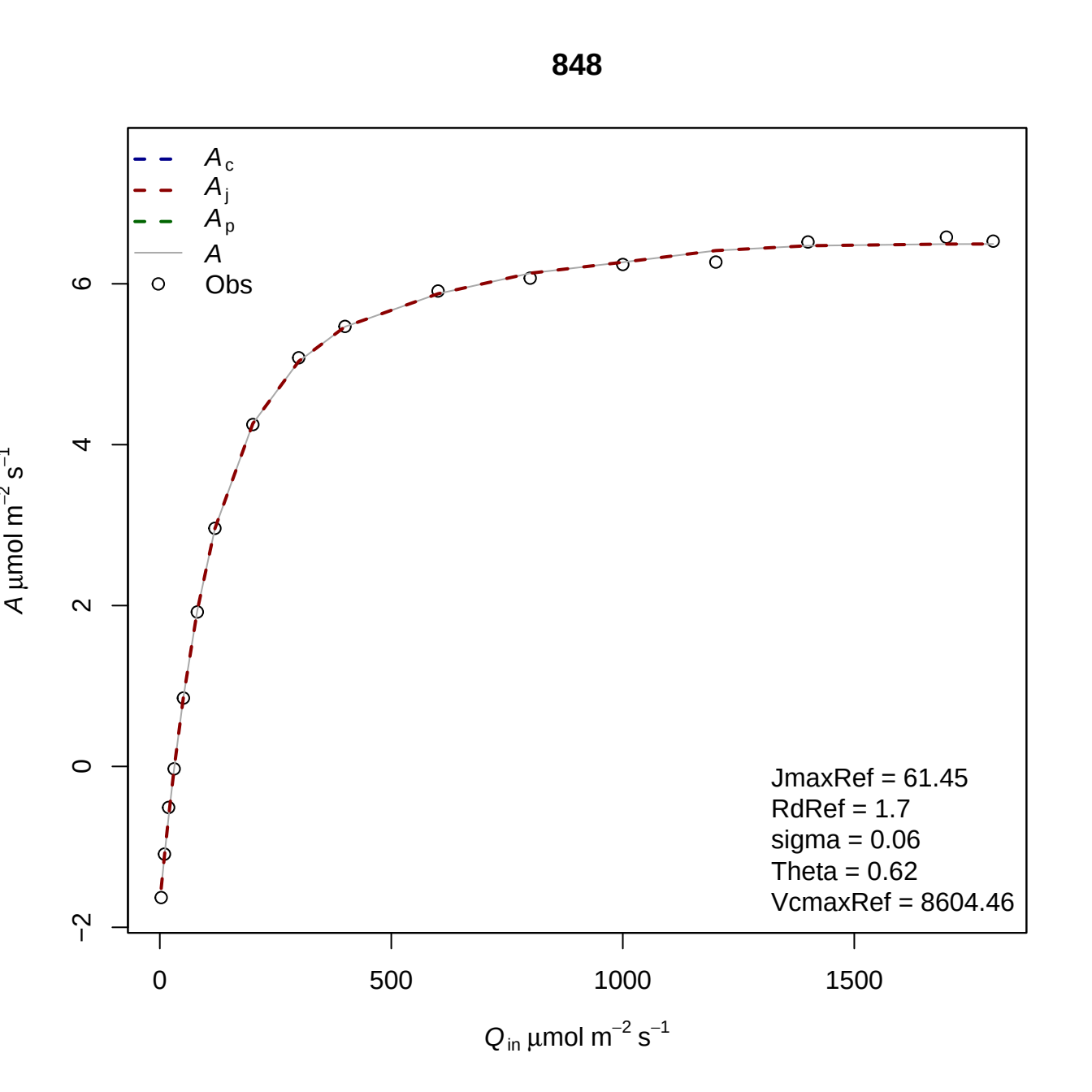

843
A_c
A_j
A_p
A
Obs
JmaxRef = 94.2
RdRef = 2.29
sigma = 0.64
Theta = 0.82
VcmaxRef = 736729.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

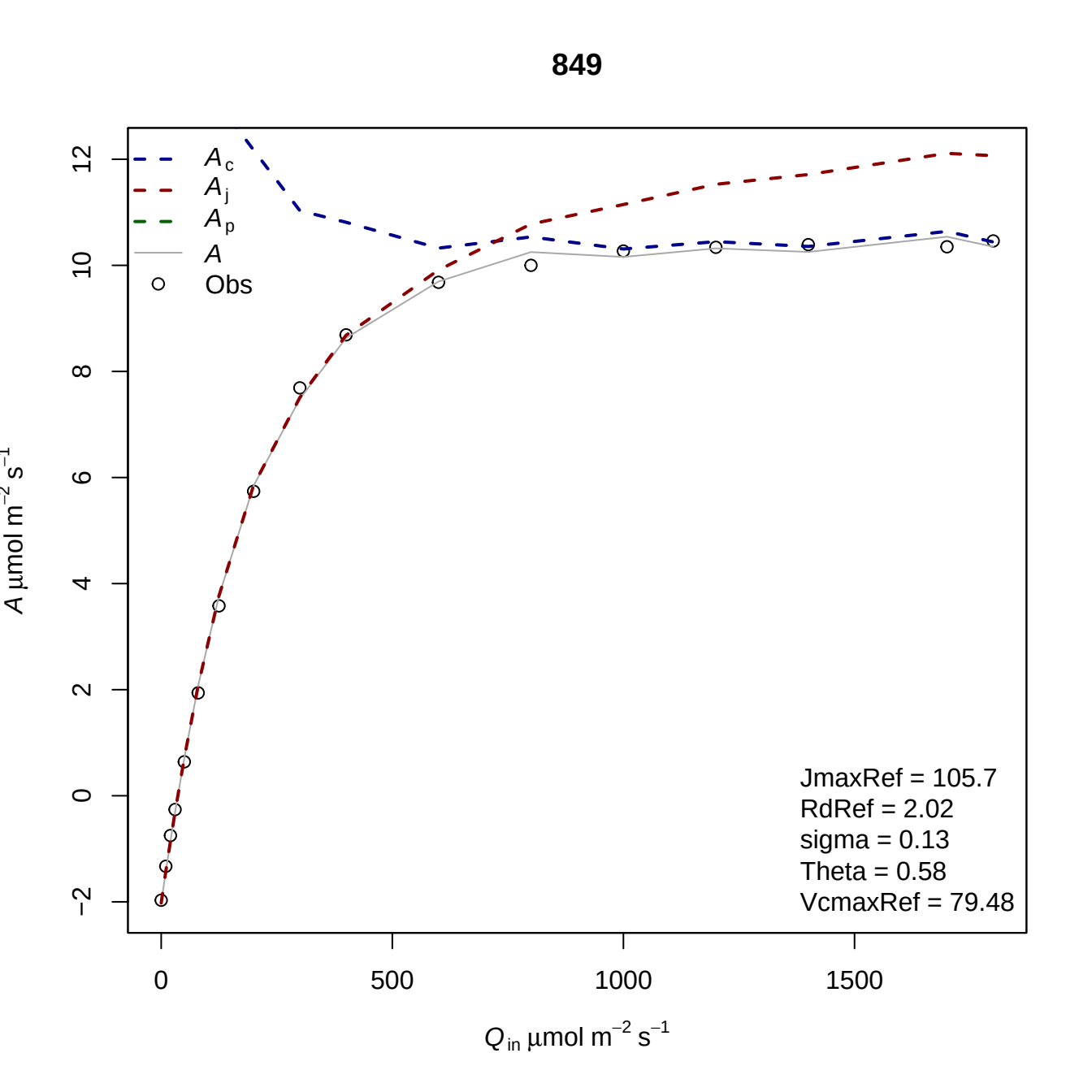

844
A_c
A_j
A_p
A
Obs
JmaxRef = 96.24
RdRef = 3.33
sigma = 0.26
Theta = 0.45
VcmaxRef = 647528.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

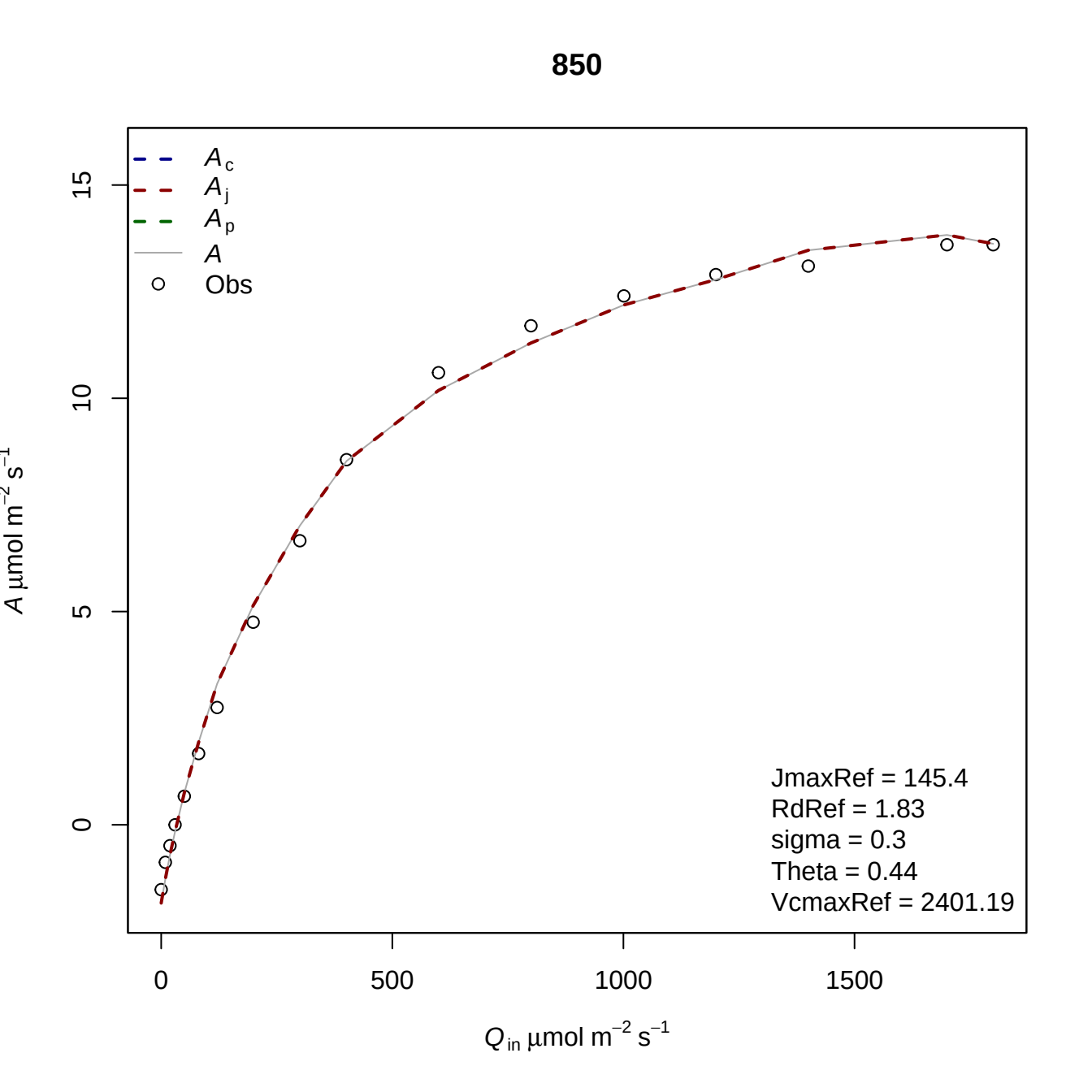

845
A_c
A_j
A_p
A
Obs
JmaxRef = 68.7
RdRef = 1.75
sigma = 0.08
Theta = 0.62
VcmaxRef = 65.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

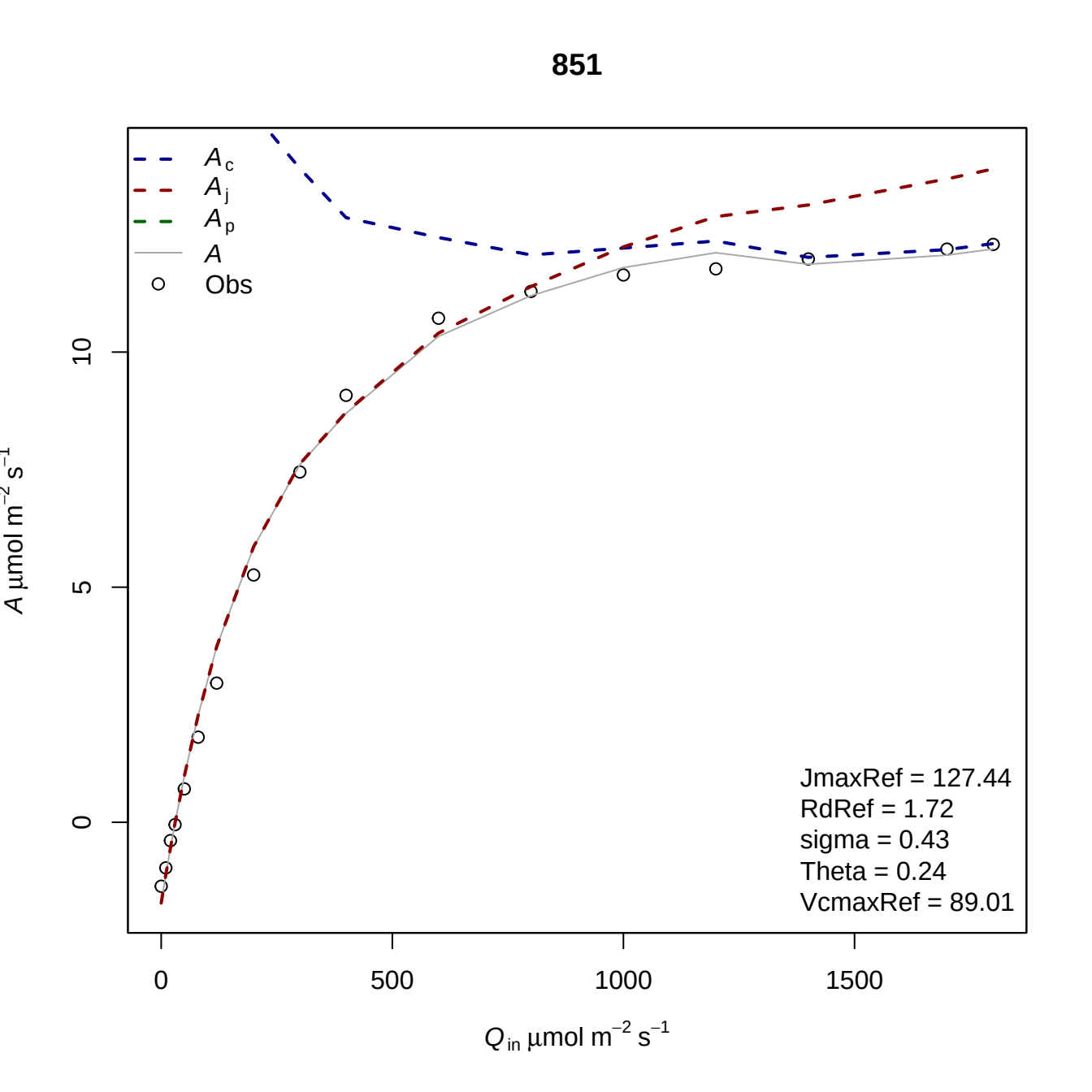

846
A_c
A_j
A_p
A
Obs
JmaxRef = 91.86
RdRef = 1.39
sigma = 0.16
Theta = 0.55
VcmaxRef = 571.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

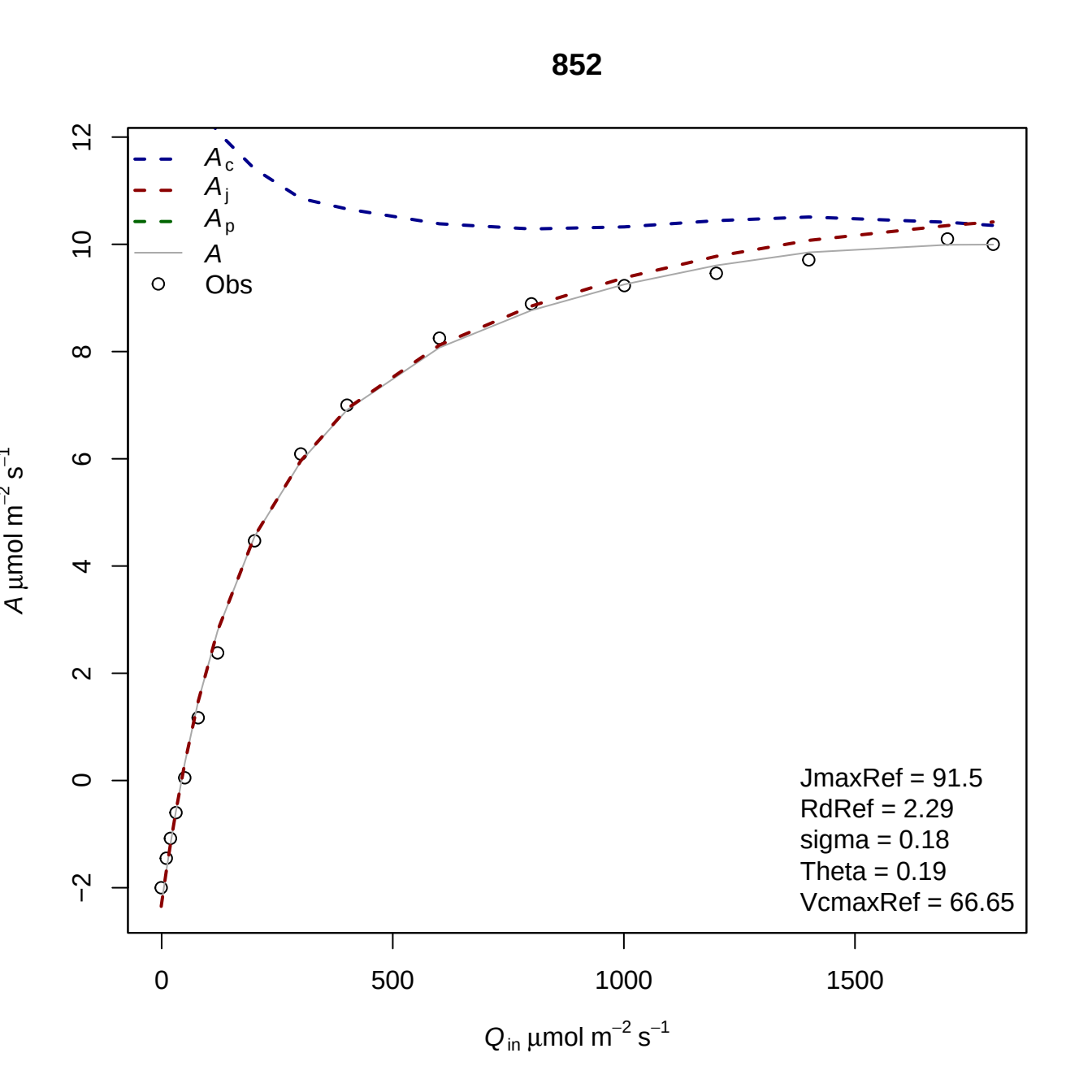

847
A_c
A_j
A_p
A
Obs
JmaxRef = 100.56
RdRef = 1.99
sigma = 0.16
Theta = 0.03
VcmaxRef = 182661.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

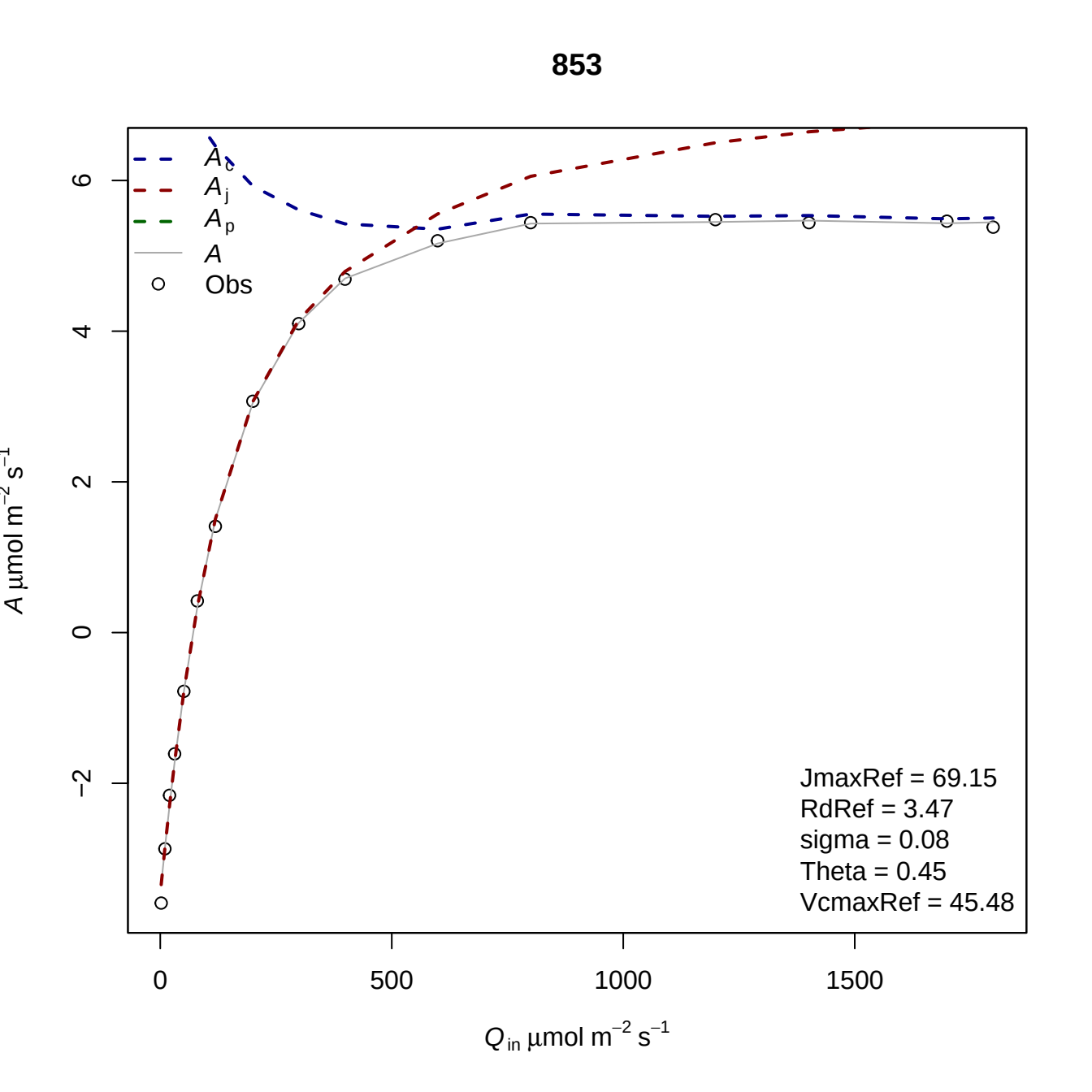

848
A_c
A_j
A_p
A
Obs
JmaxRef = 61.45
RdRef = 1.7
sigma = 0.06
Theta = 0.62
VcmaxRef = 8604.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

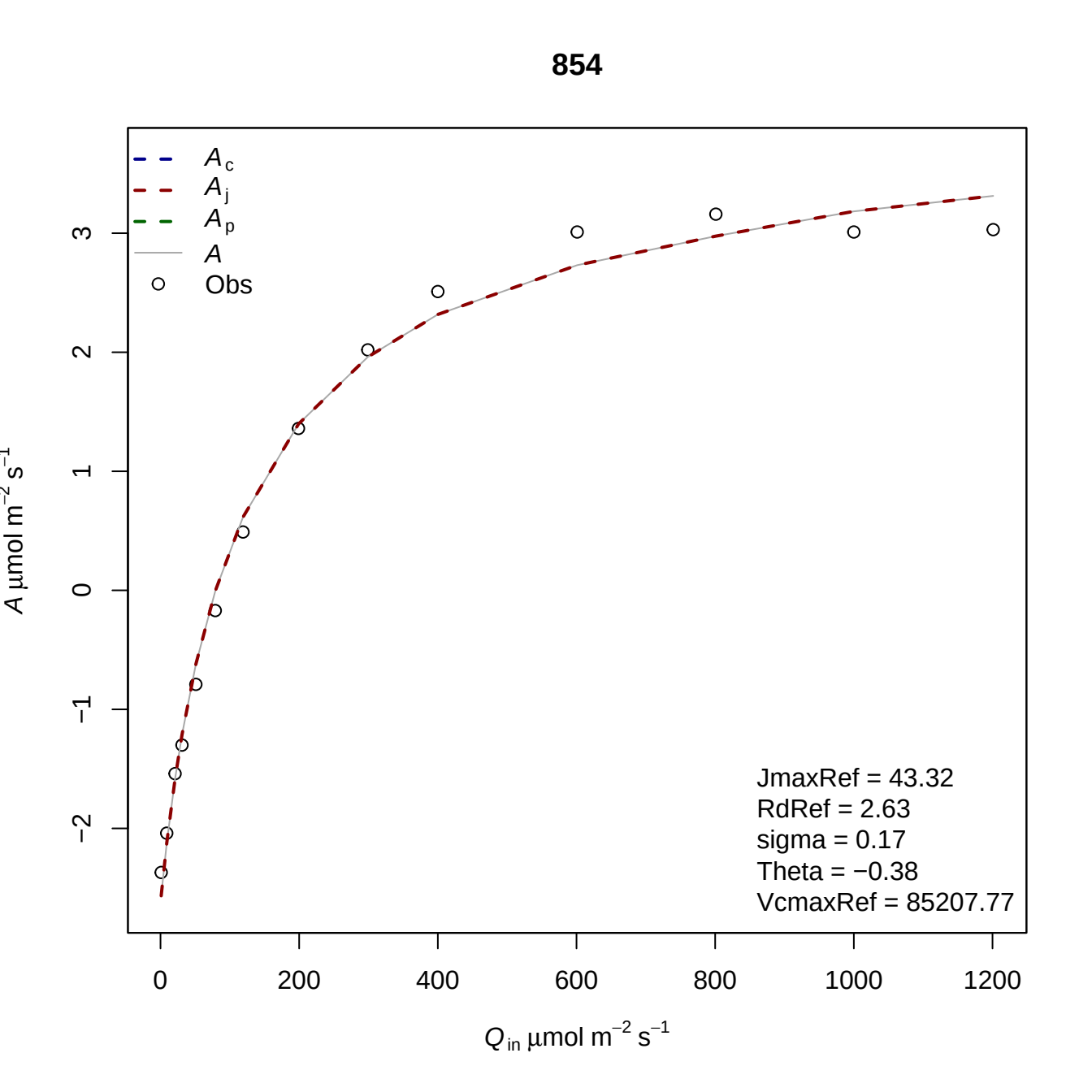

849
A_c
A_j
A_p
A
Obs
JmaxRef = 105.7
RdRef = 2.02
sigma = 0.13
Theta = 0.58
VcmaxRef = 79.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

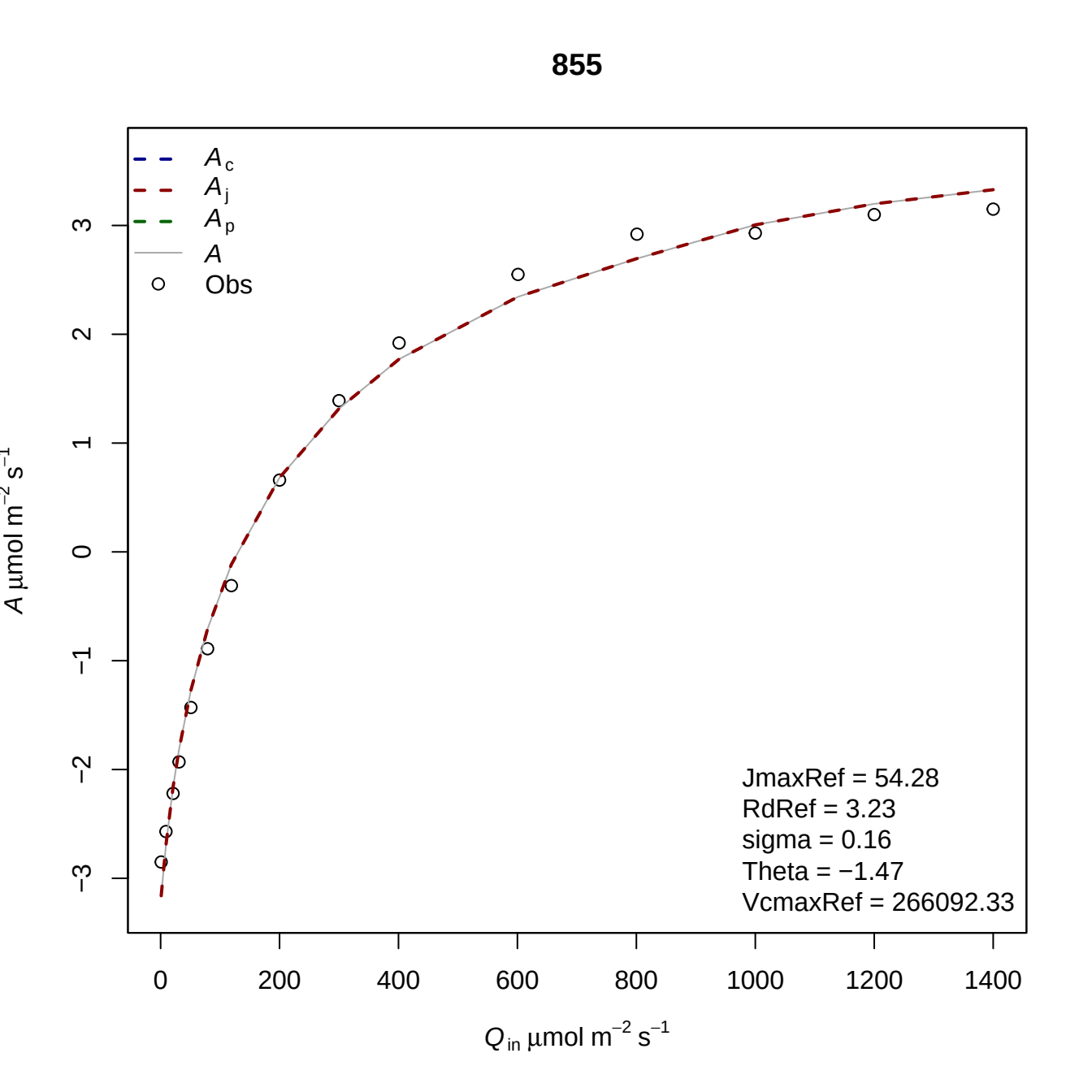

850
A_c
A_j
A_p
A
Obs
JmaxRef = 145.4
RdRef = 1.83
sigma = 0.3
Theta = 0.44
VcmaxRef = 2401.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

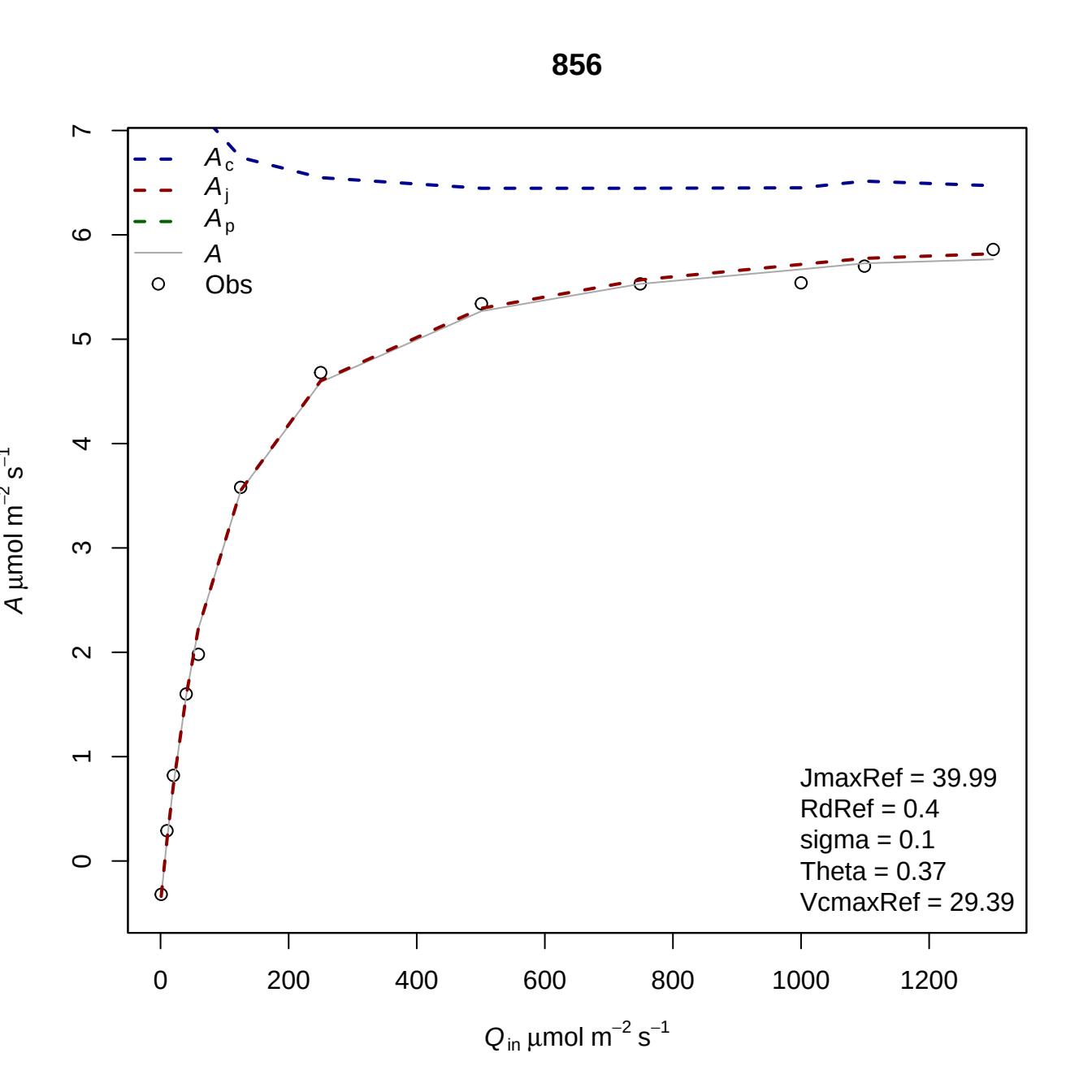

851
A_c
A_j
A_p
A
Obs
JmaxRef = 127.44
RdRef = 1.72
sigma = 0.43
Theta = 0.24
VcmaxRef = 89.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

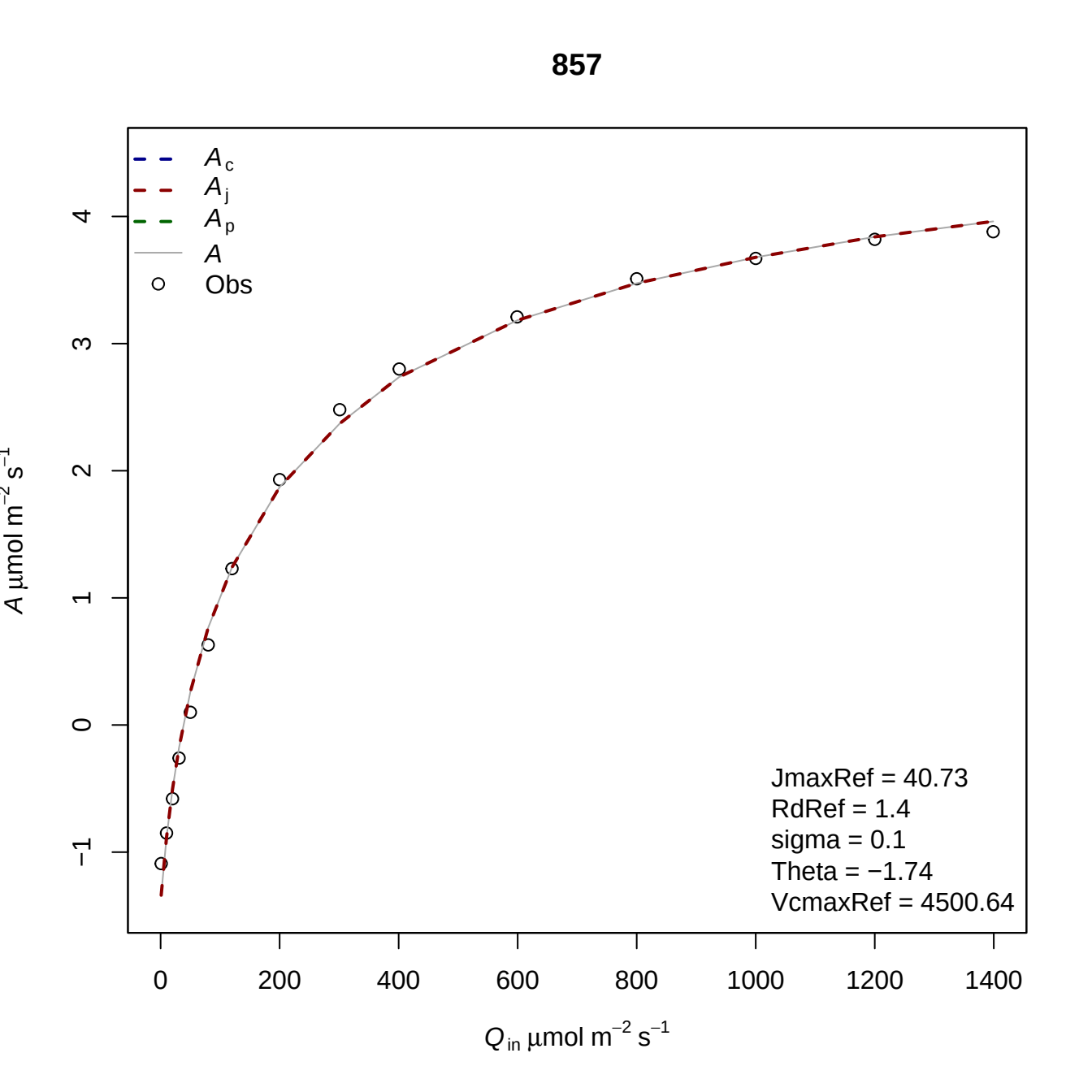

852
A_c
A_j
A_p
A
Obs
JmaxRef = 91.5
RdRef = 2.29
sigma = 0.18
Theta = 0.19
VcmaxRef = 66.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

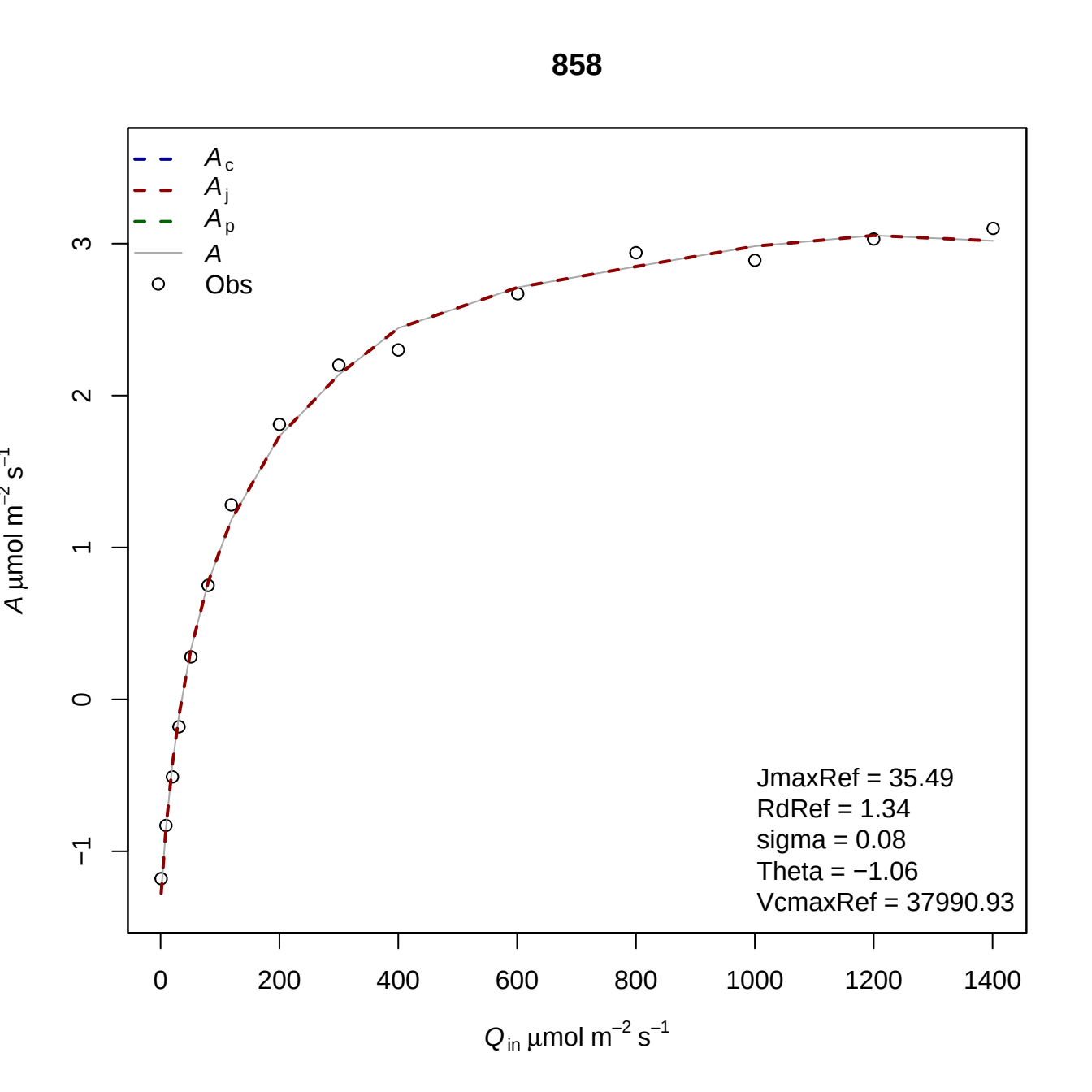

853
A_c
A_j
A_p
A
Obs
JmaxRef = 69.15
RdRef = 3.47
sigma = 0.08
Theta = 0.45
VcmaxRef = 45.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

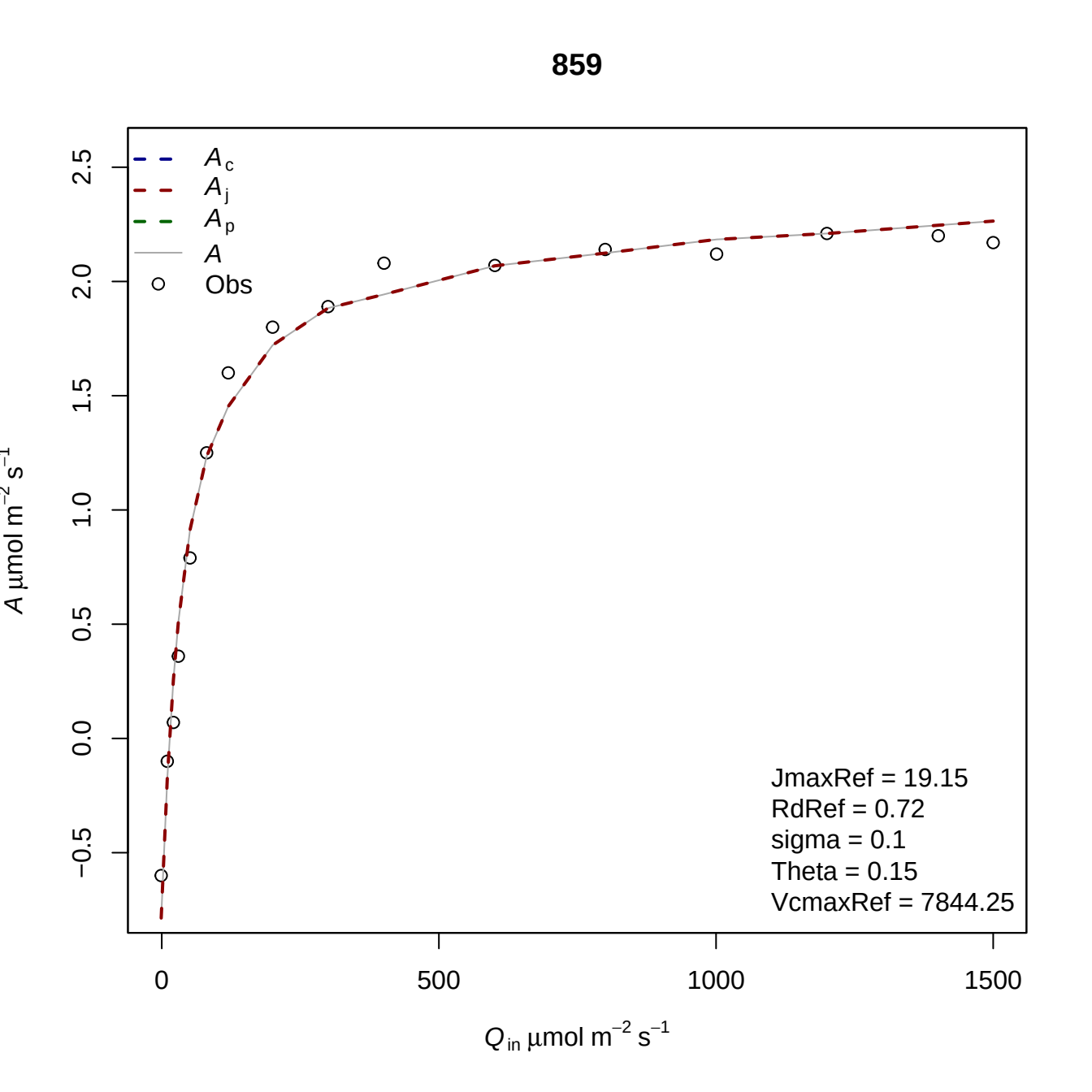

854
A_c
A_j
A_p
A
Obs
JmaxRef = 43.32
RdRef = 2.63
sigma = 0.17
Theta = −0.38
VcmaxRef = 85207.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

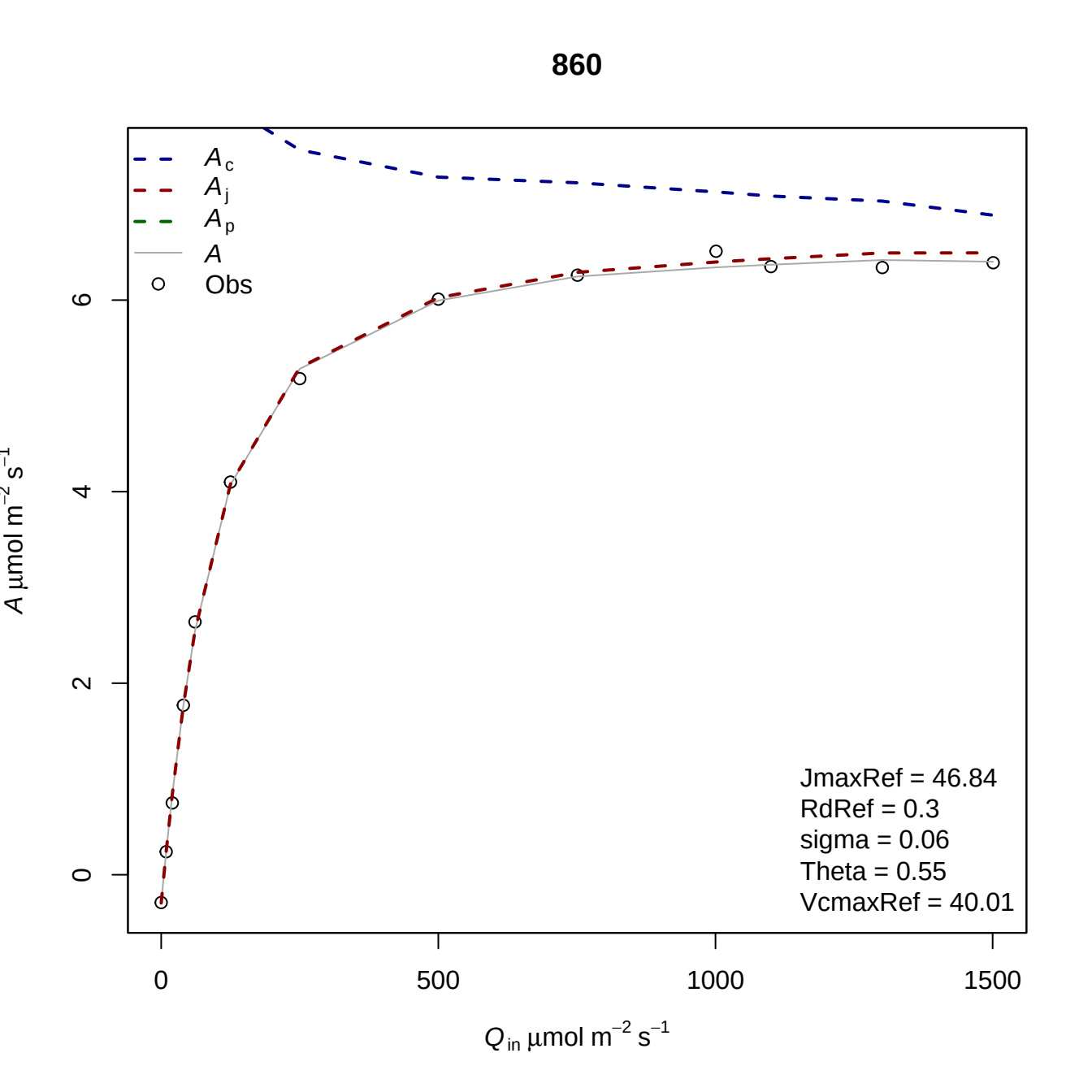

855
A_c
A_j
A_p
A
Obs
JmaxRef = 54.28
RdRef = 3.23
sigma = 0.16
Theta = −1.47
VcmaxRef = 266092.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

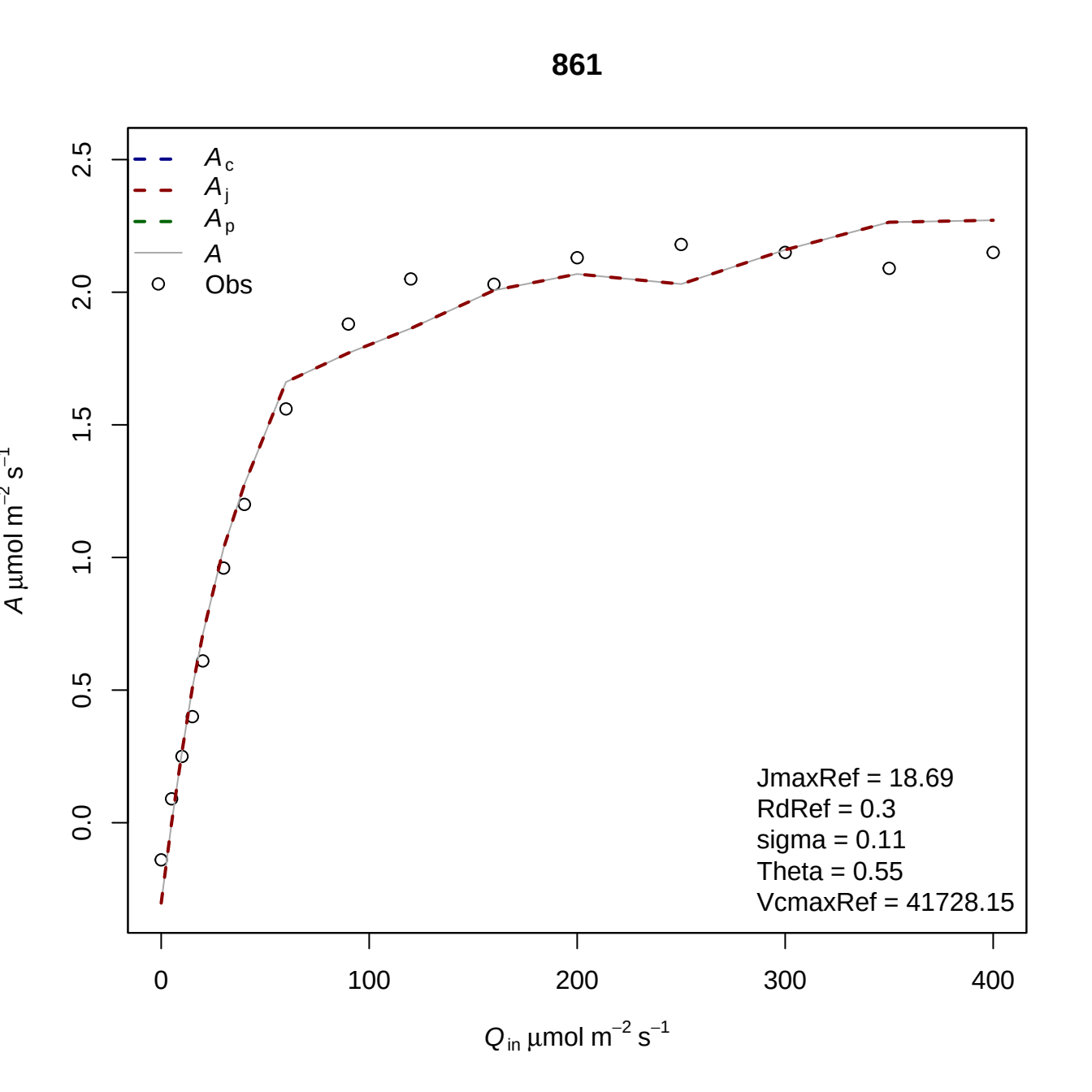

856
A_c
A_j
A_p
A
Obs
JmaxRef = 39.99
RdRef = 0.4
sigma = 0.1
Theta = 0.37
VcmaxRef = 29.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200

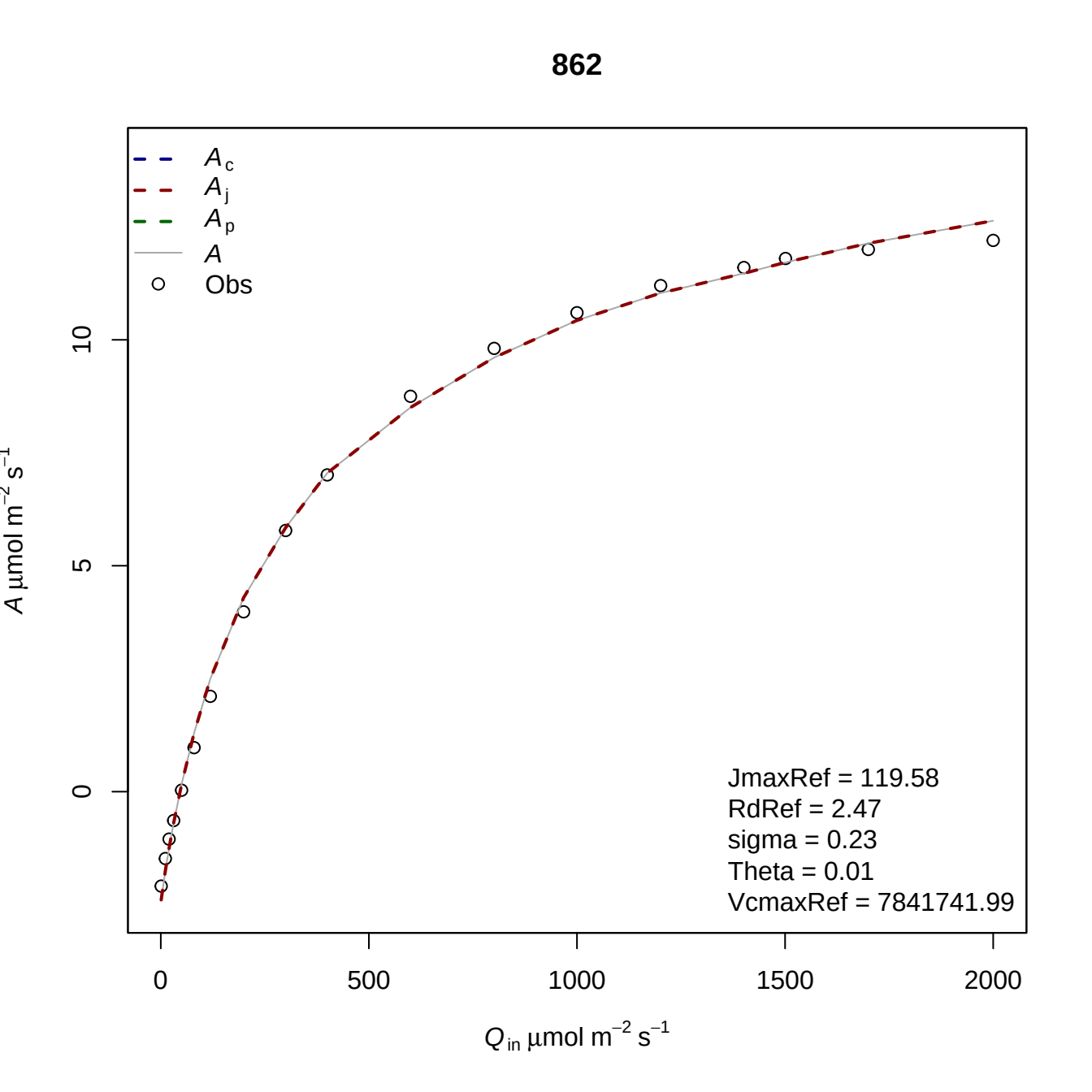

857
A_c
A_j
A_p
A
Obs
JmaxRef = 40.73
RdRef = 1.4
sigma = 0.1
Theta = −1.74
VcmaxRef = 4500.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

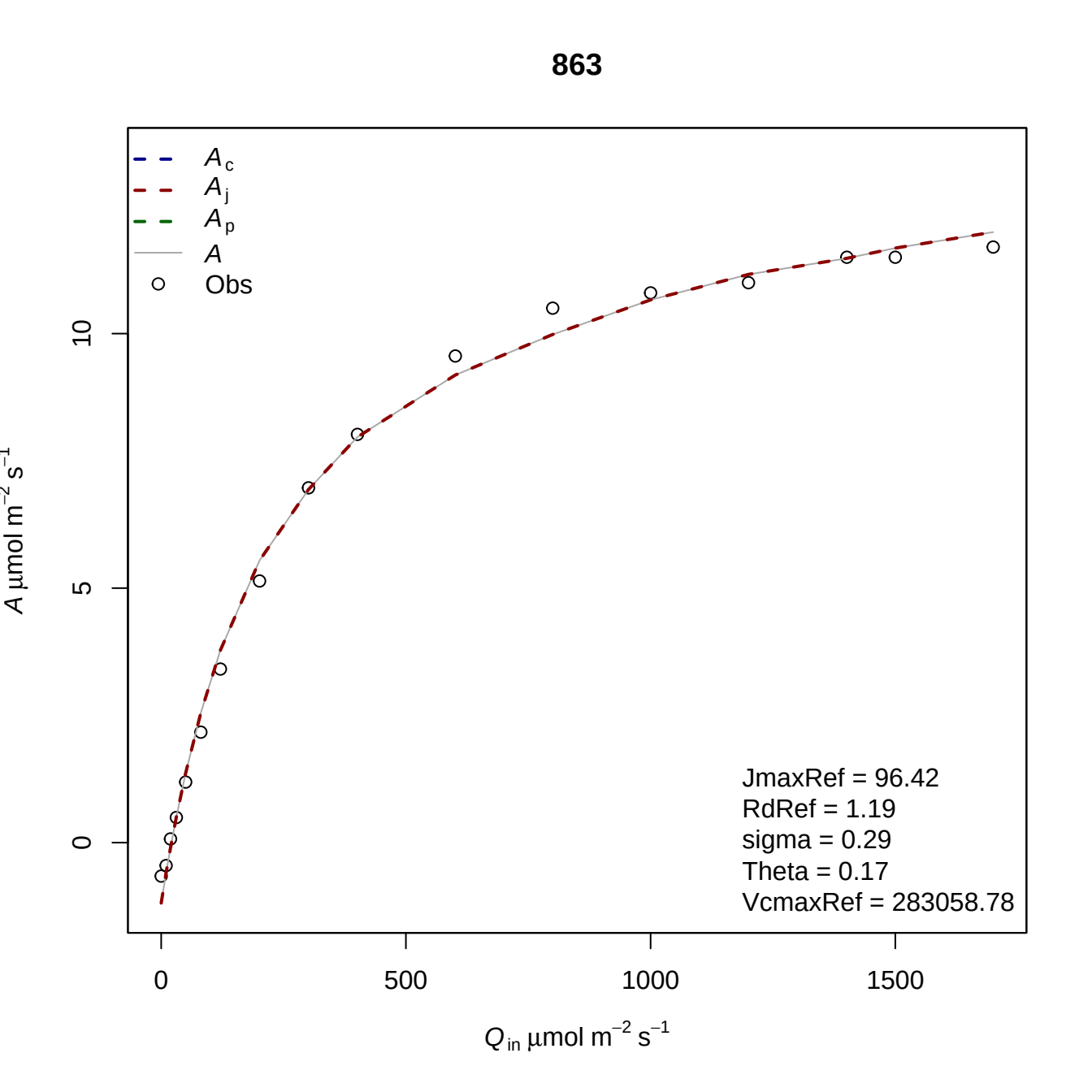

858
A_c
A_j
A_p
A
Obs
JmaxRef = 35.49
RdRef = 1.34
sigma = 0.08
Theta = −1.06
VcmaxRef = 37990.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

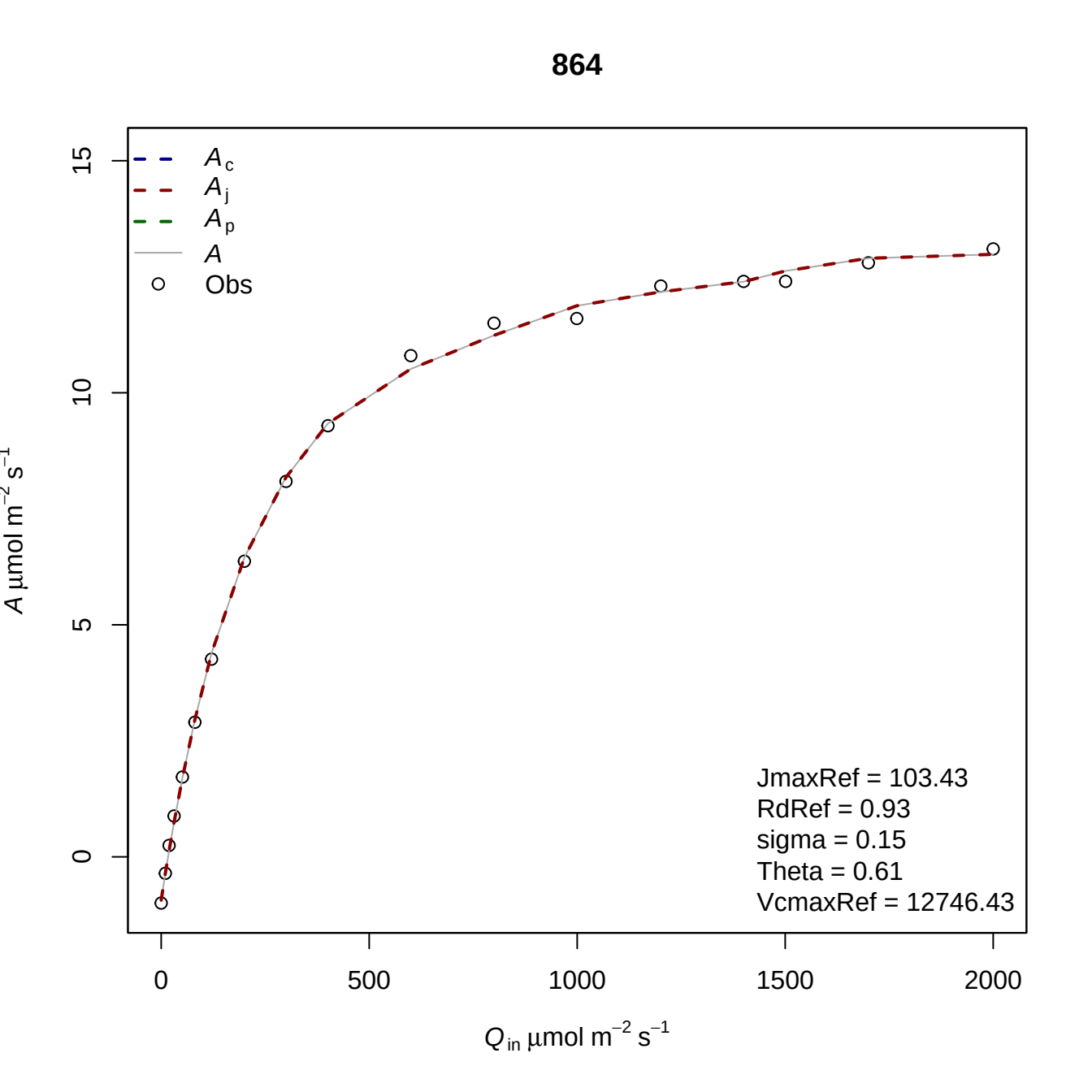

859
A_c
A_j
A_p
A
Obs
JmaxRef = 19.15
RdRef = 0.72
sigma = 0.1
Theta = 0.15
VcmaxRef = 7844.25
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

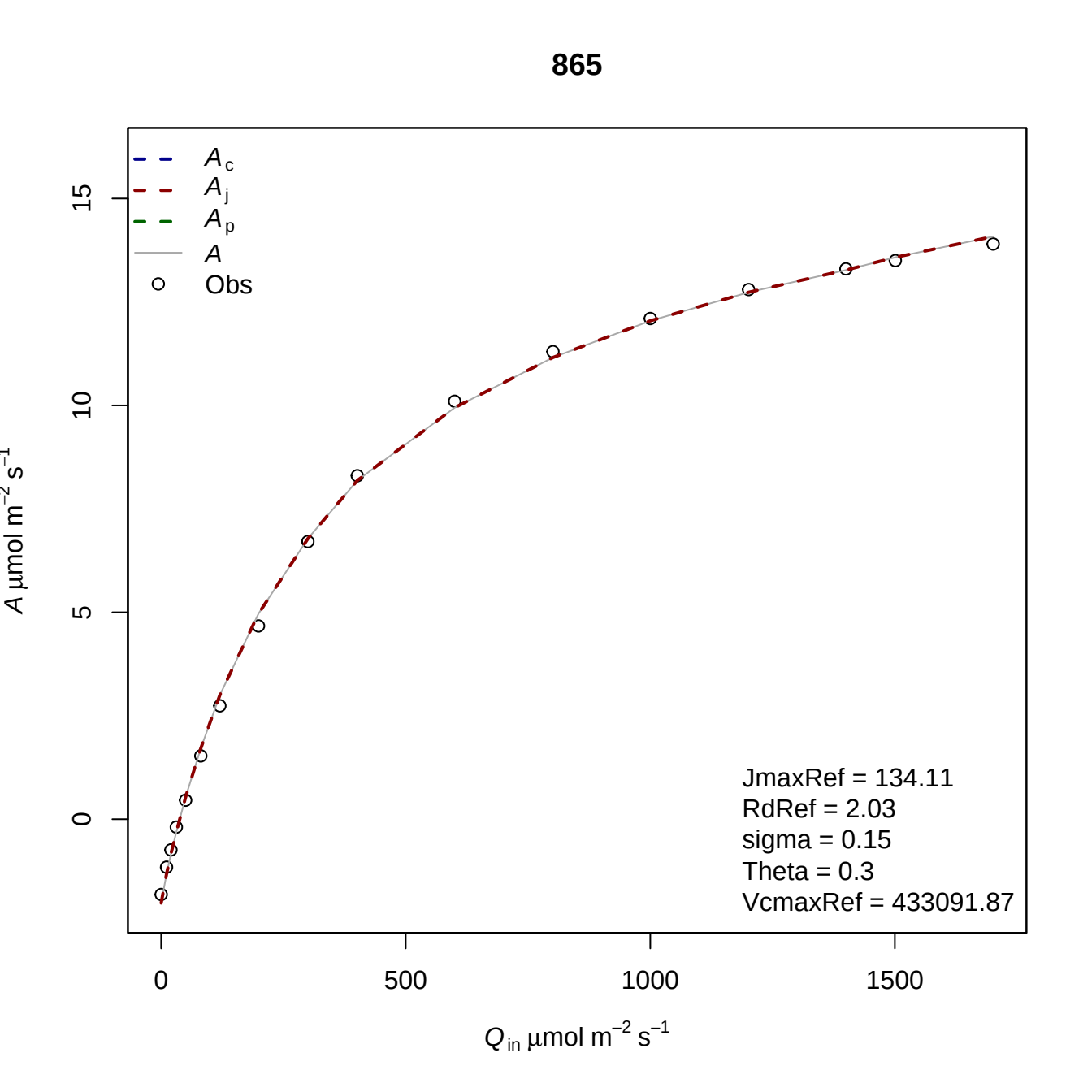

860
A_c
A_j
A_p
A
Obs
JmaxRef = 46.84
RdRef = 0.3
sigma = 0.06
Theta = 0.55
VcmaxRef = 40.01
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

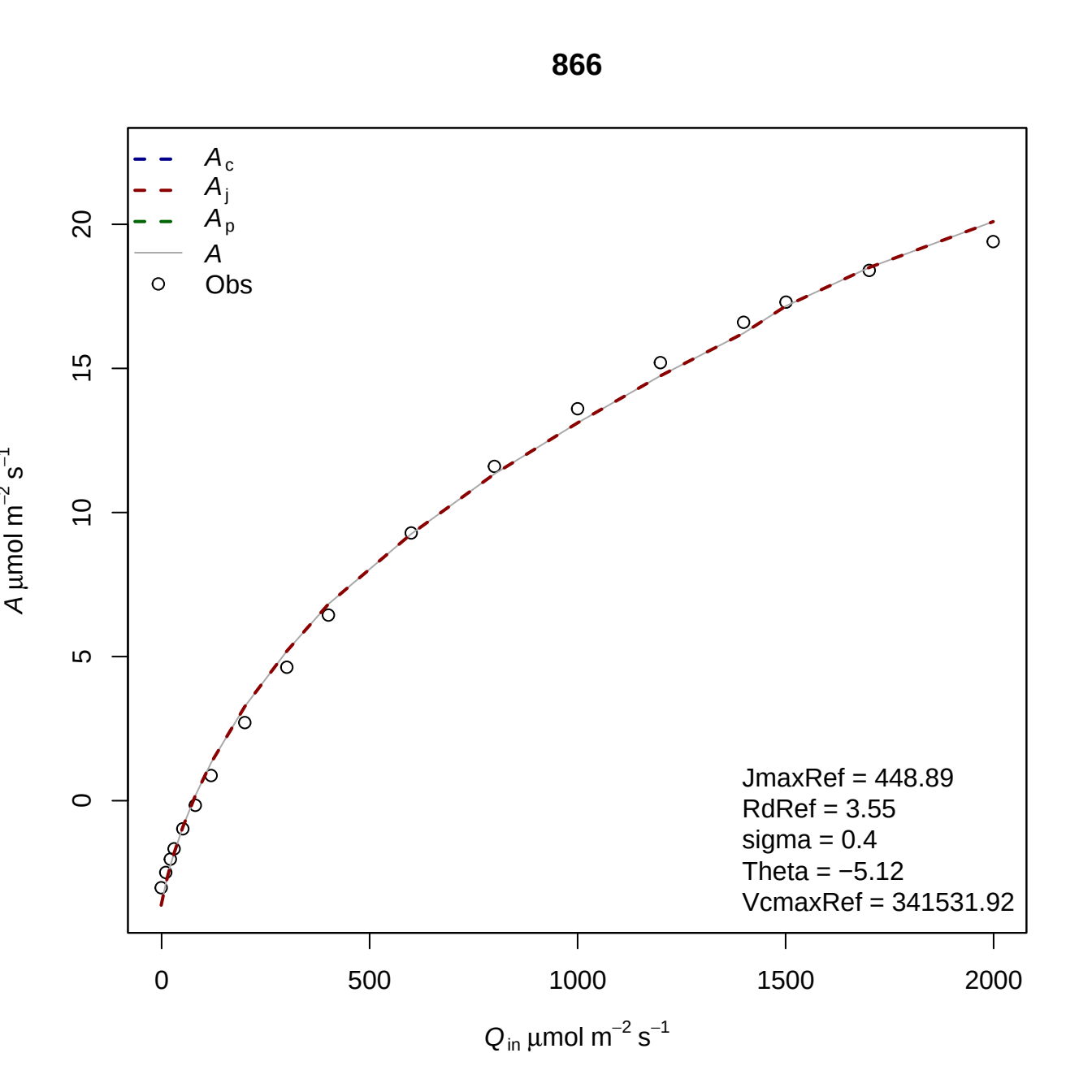

861
A_c
A_j
A_p
A
Obs
JmaxRef = 18.69
RdRef = 0.3
sigma = 0.11
Theta = 0.55
VcmaxRef = 41728.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

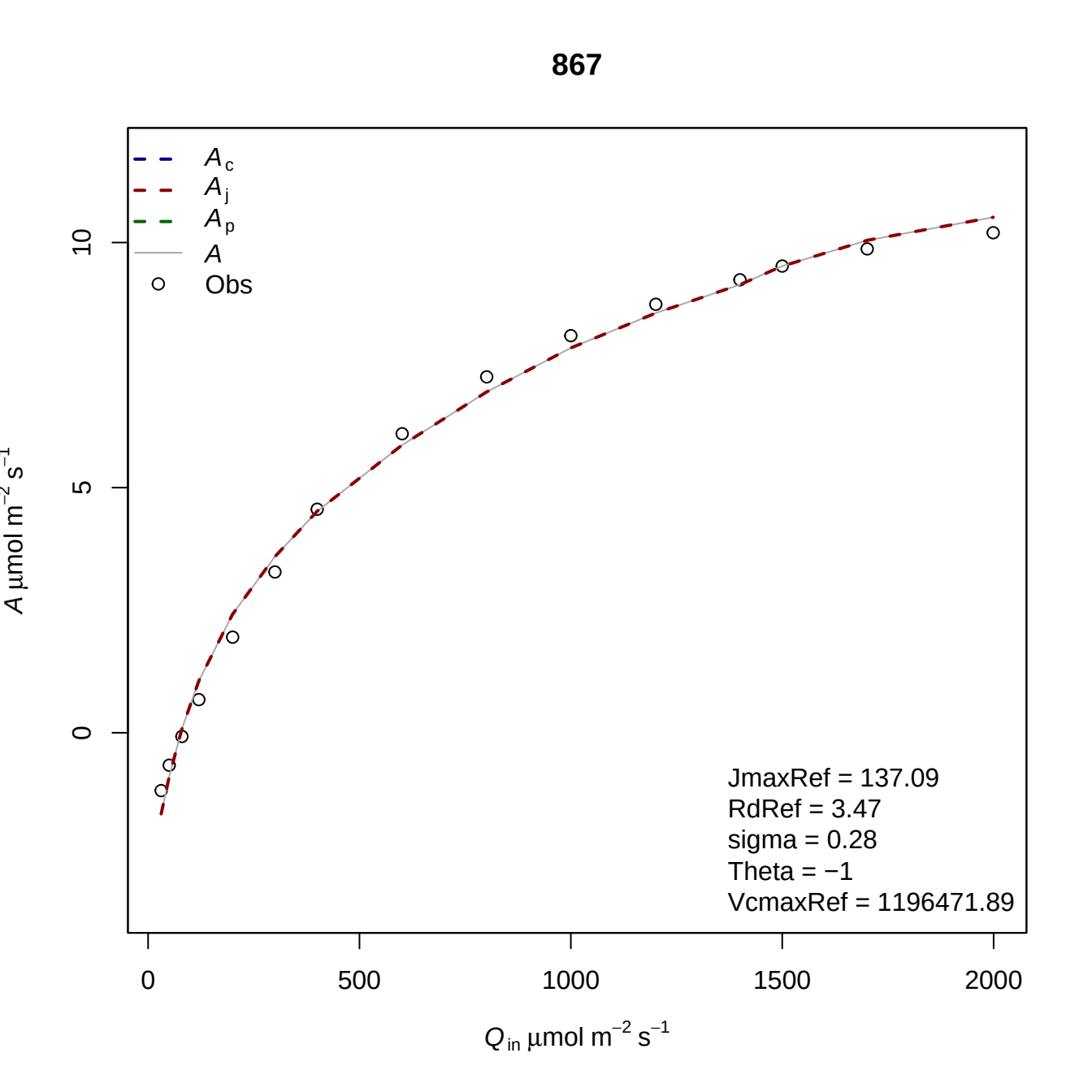

862
A_c
A_j
A_p
A
Obs
JmaxRef = 119.58
RdRef = 2.47
sigma = 0.23
Theta = 0.01
VcmaxRef = 7841741.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

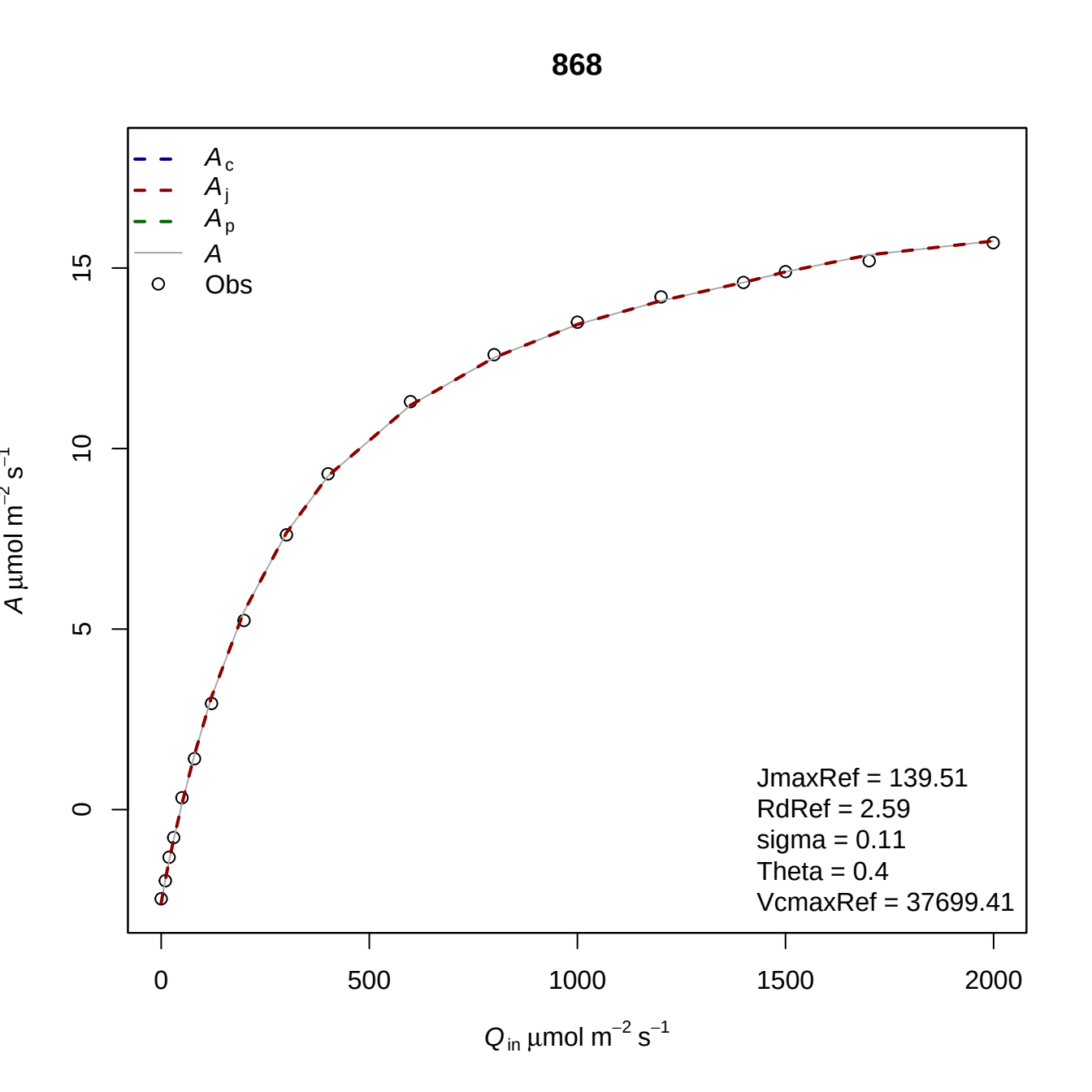

863
A_c
A_j
A_p
A
Obs
JmaxRef = 96.42
RdRef = 1.19
sigma = 0.29
Theta = 0.17
VcmaxRef = 283058.78
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

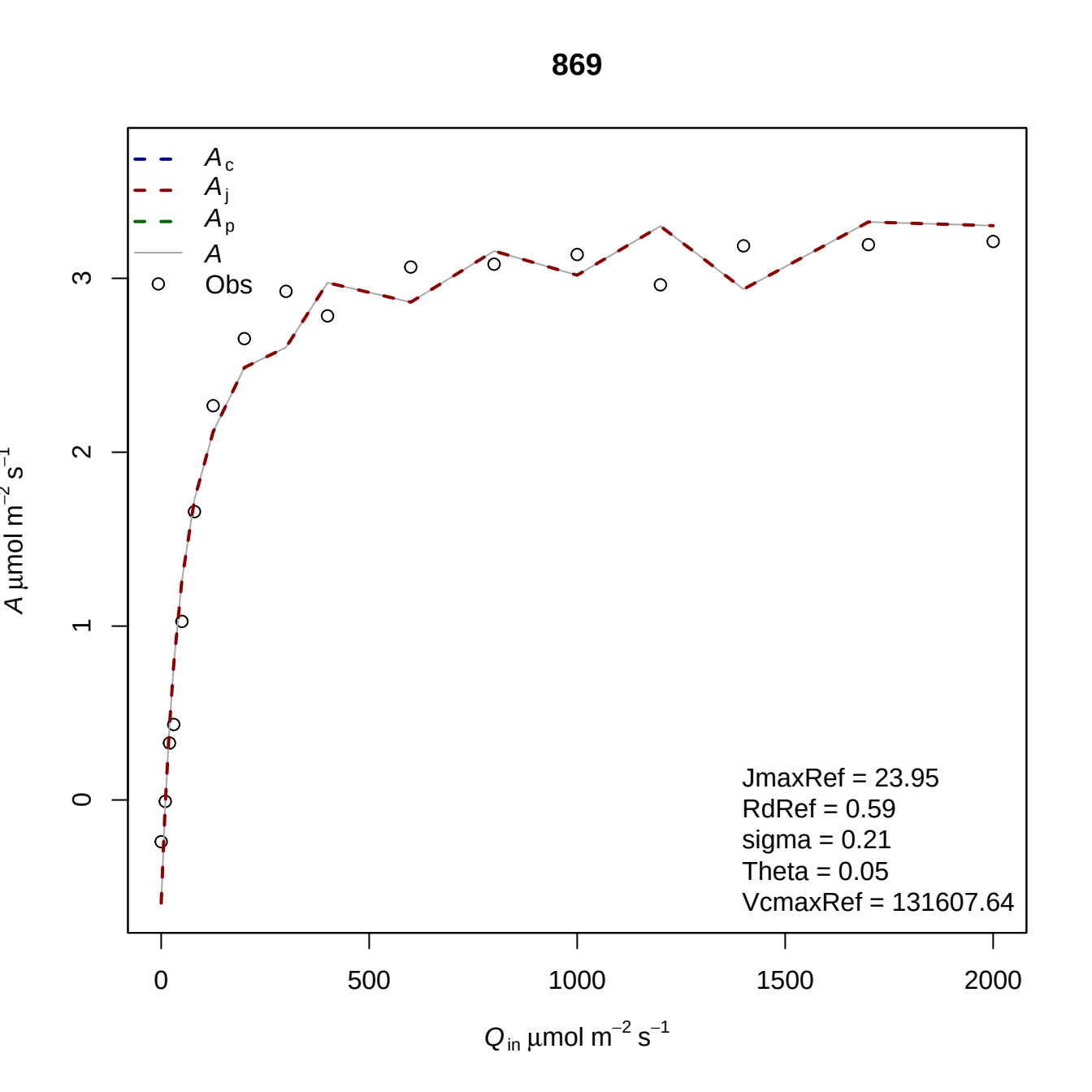

864
A_c
A_j
A_p
A
Obs
JmaxRef = 103.43
RdRef = 0.93
sigma = 0.15
Theta = 0.61
VcmaxRef = 12746.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

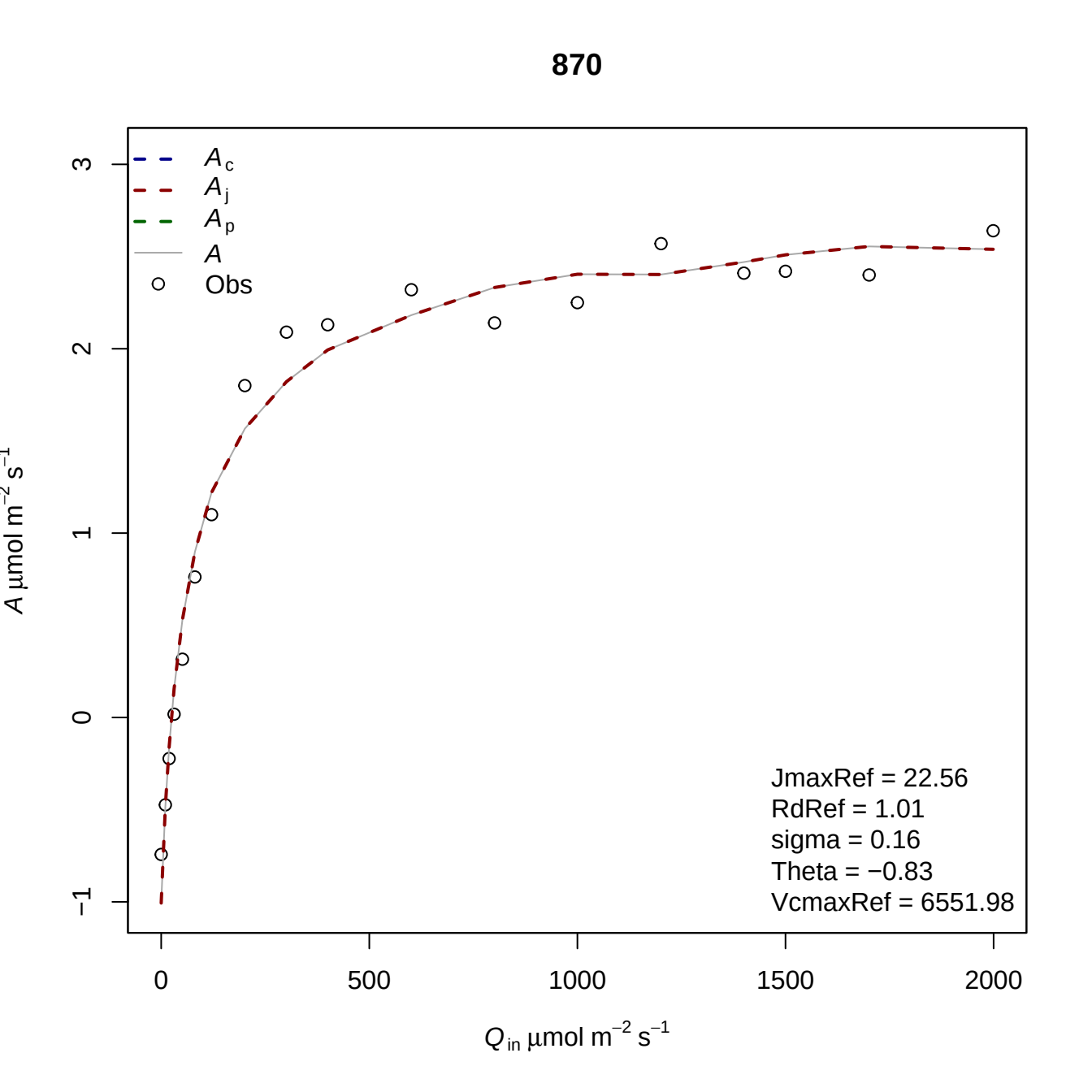

865
A_c
A_j
A_p
A
Obs
JmaxRef = 134.11
RdRef = 2.03
sigma = 0.15
Theta = 0.3
VcmaxRef = 433091.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

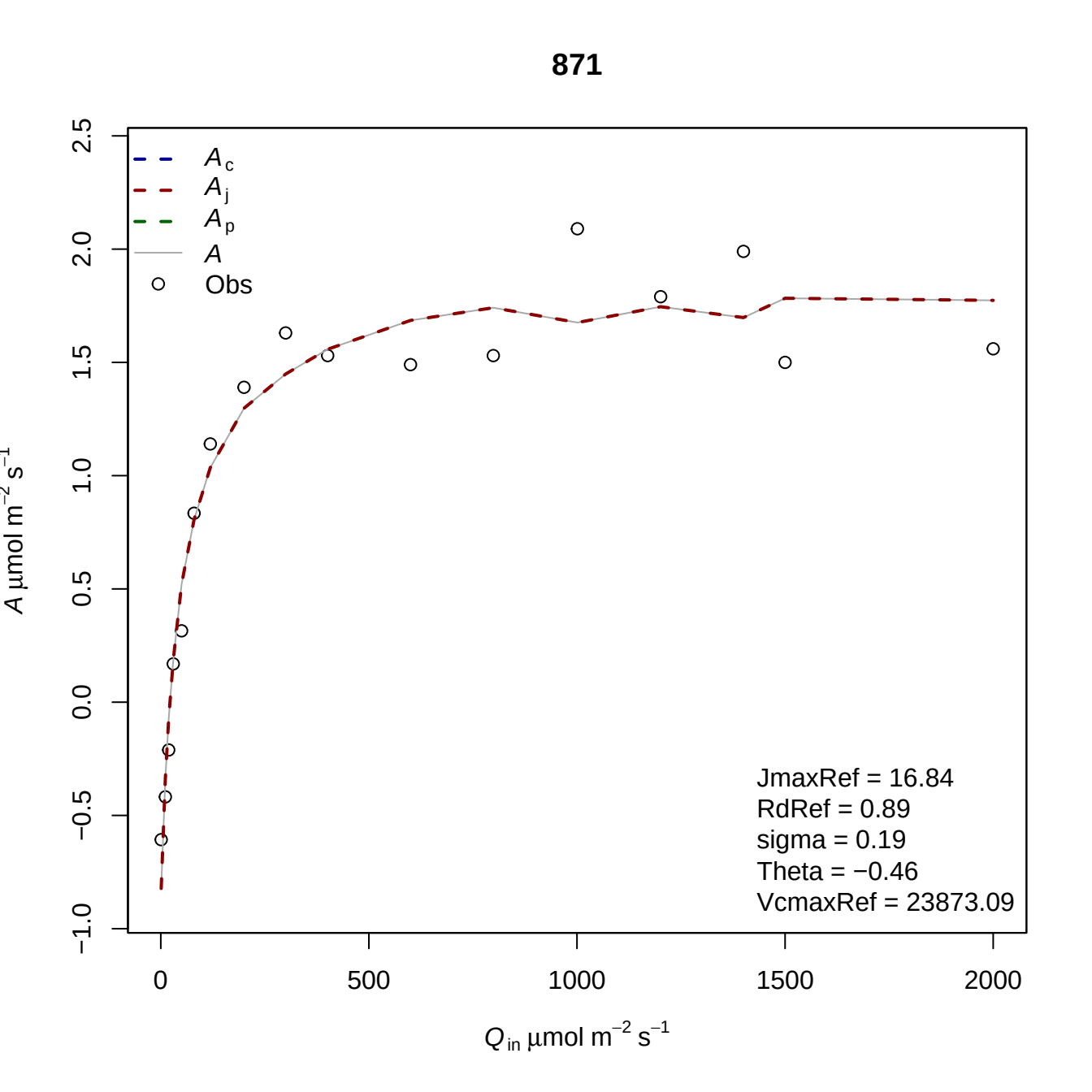

866
A_c
A_j
A_p
A
Obs
JmaxRef = 448.89
RdRef = 3.55
sigma = 0.4
Theta = −5.12
VcmaxRef = 341531.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

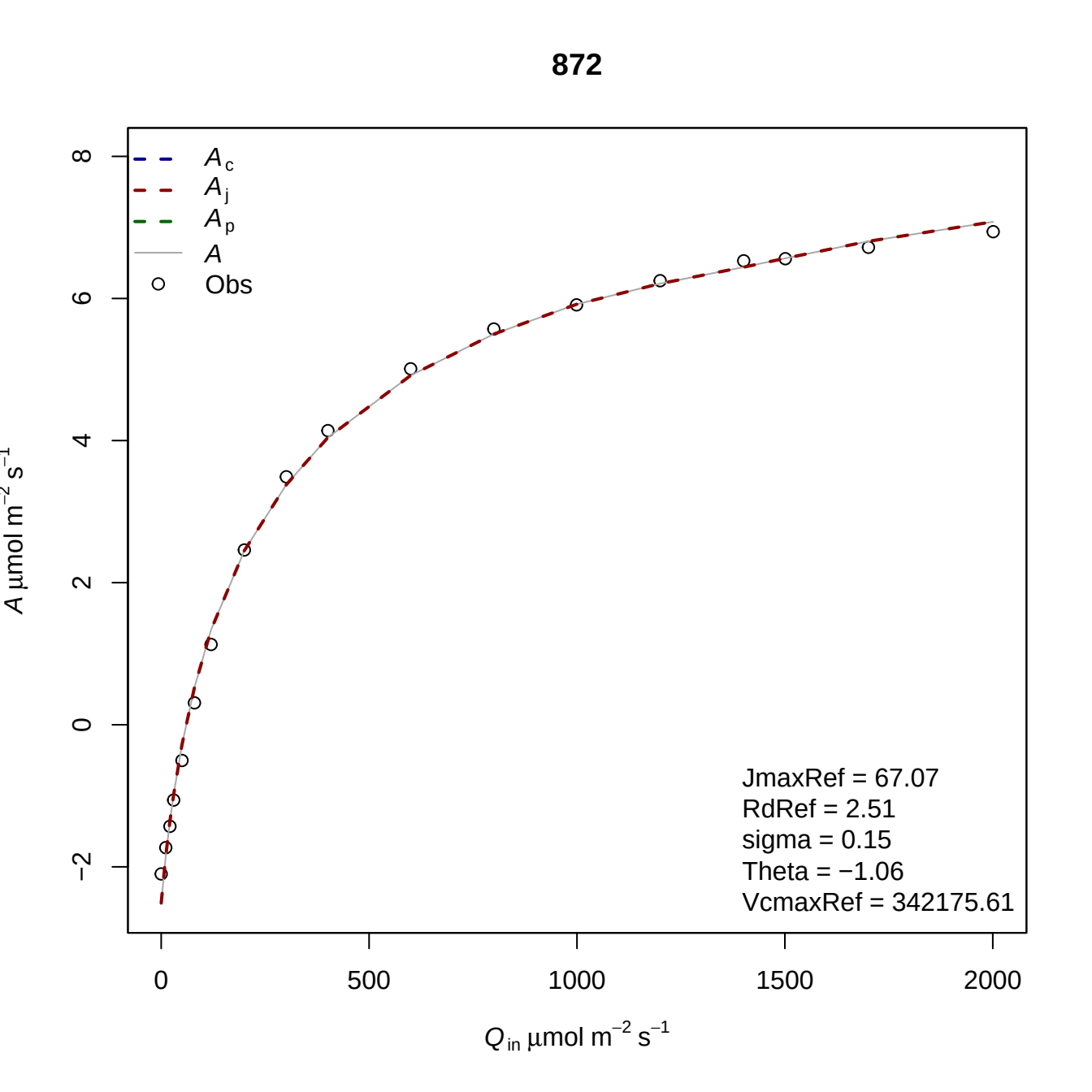

867
A_c
A_j
A_p
A
Obs
JmaxRef = 137.09
RdRef = 3.47
sigma = 0.28
Theta = −1
VcmaxRef = 1196471.89
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

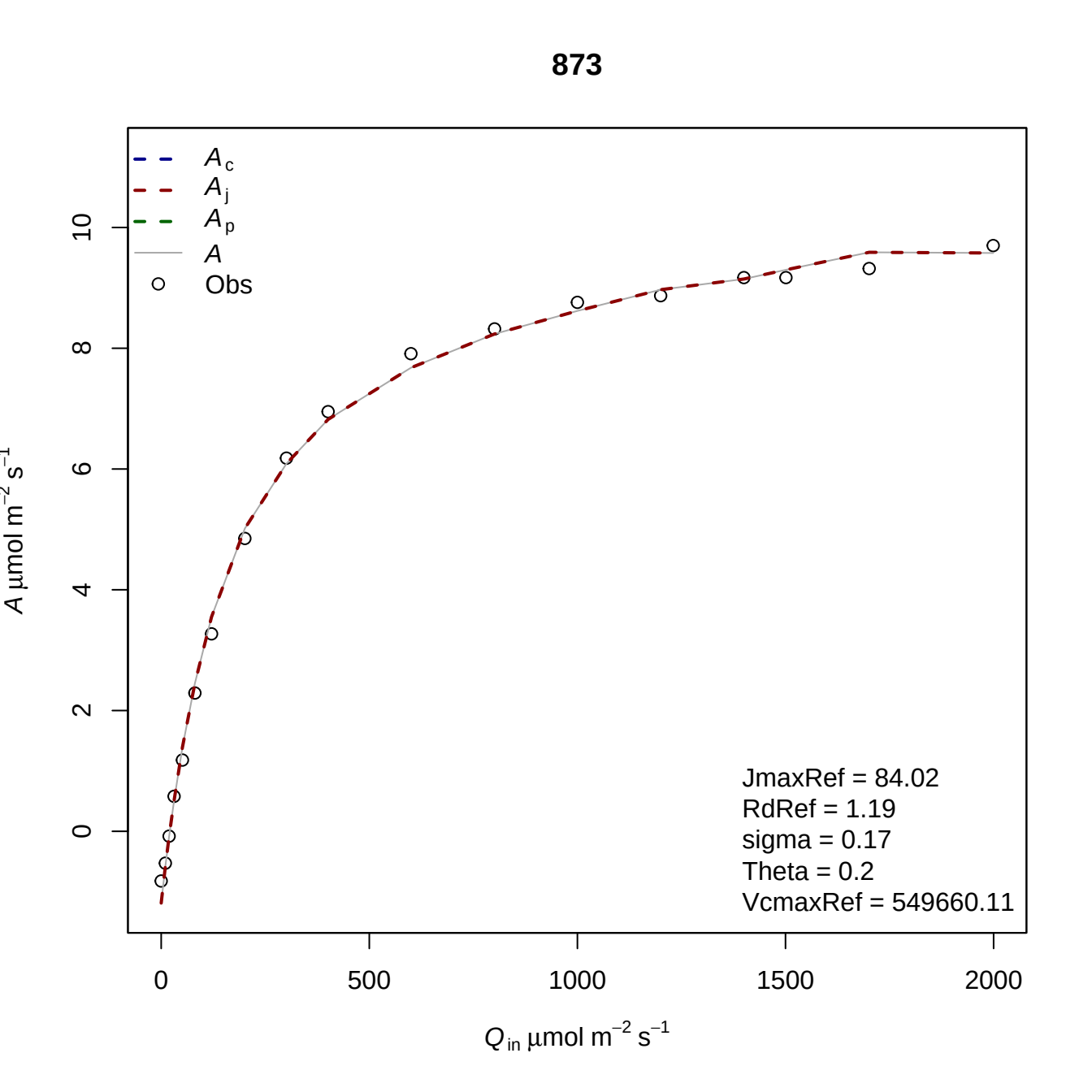

868
A_c
A_j
A_p
A
Obs
JmaxRef = 139.51
RdRef = 2.59
sigma = 0.11
Theta = 0.4
VcmaxRef = 37699.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

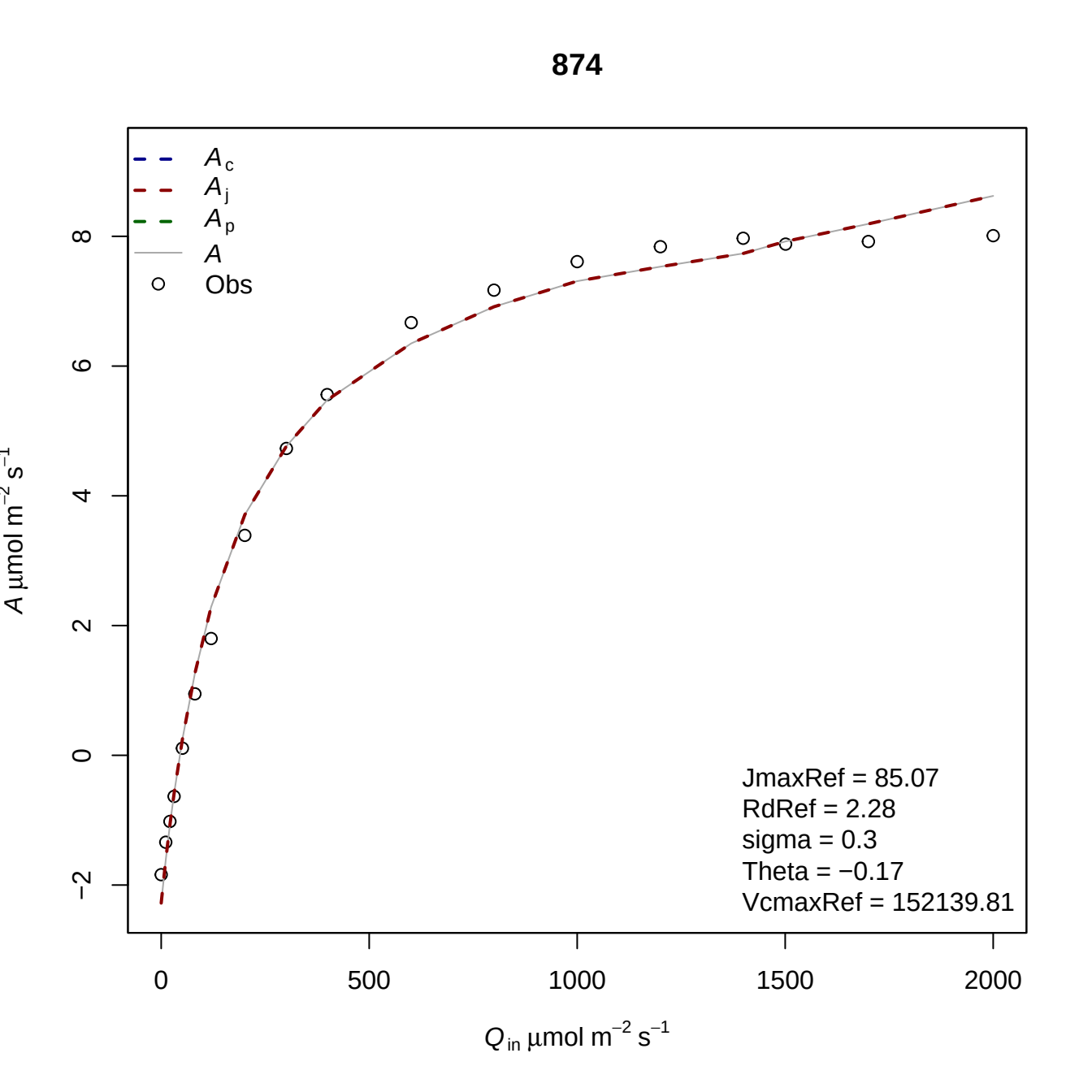

869
A_c
A_j
A_p
A
Obs
JmaxRef = 23.95
RdRef = 0.59
sigma = 0.21
Theta = 0.05
VcmaxRef = 131607.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

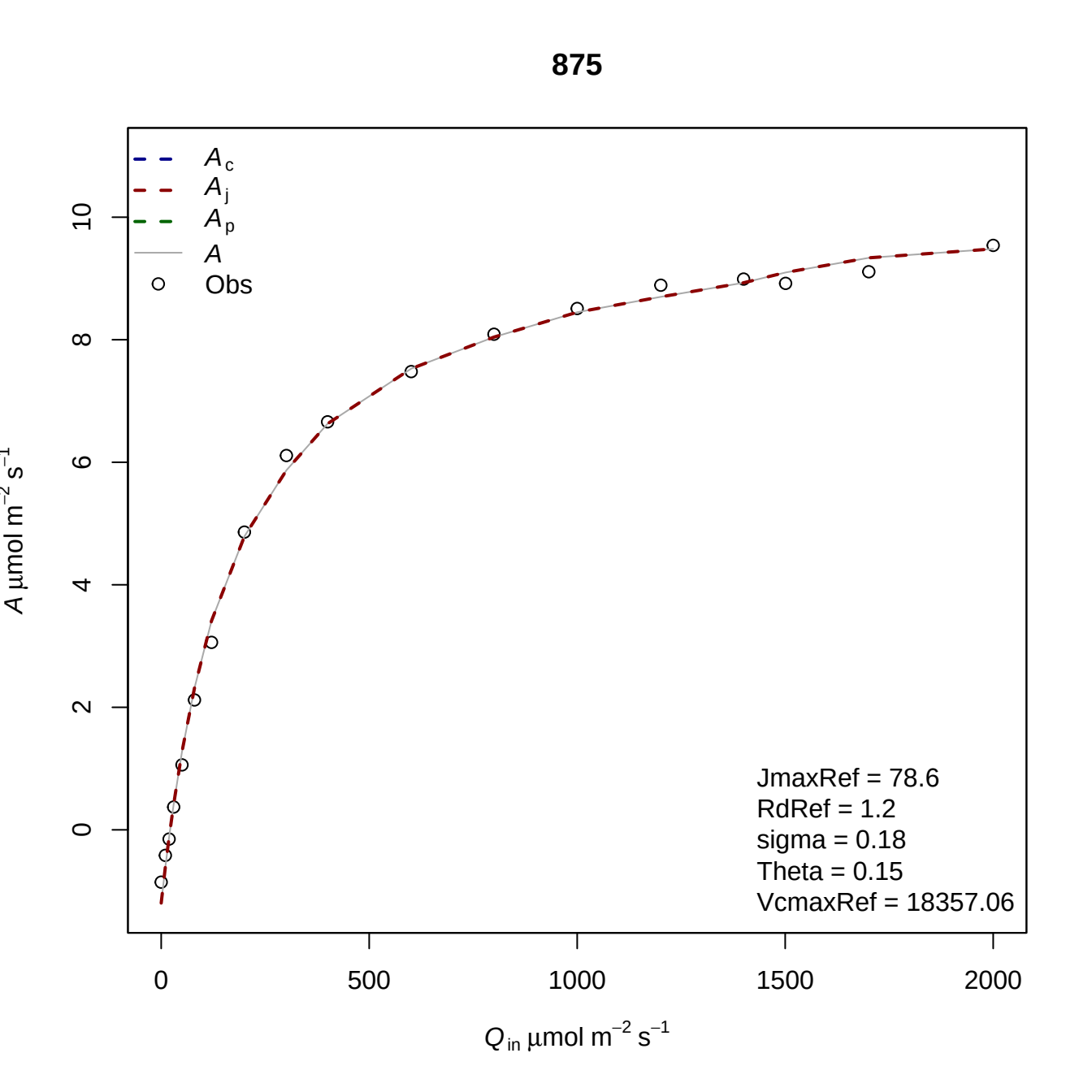

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.56
RdRef = 1.01
sigma = 0.16
Theta = −0.83
VcmaxRef = 6551.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

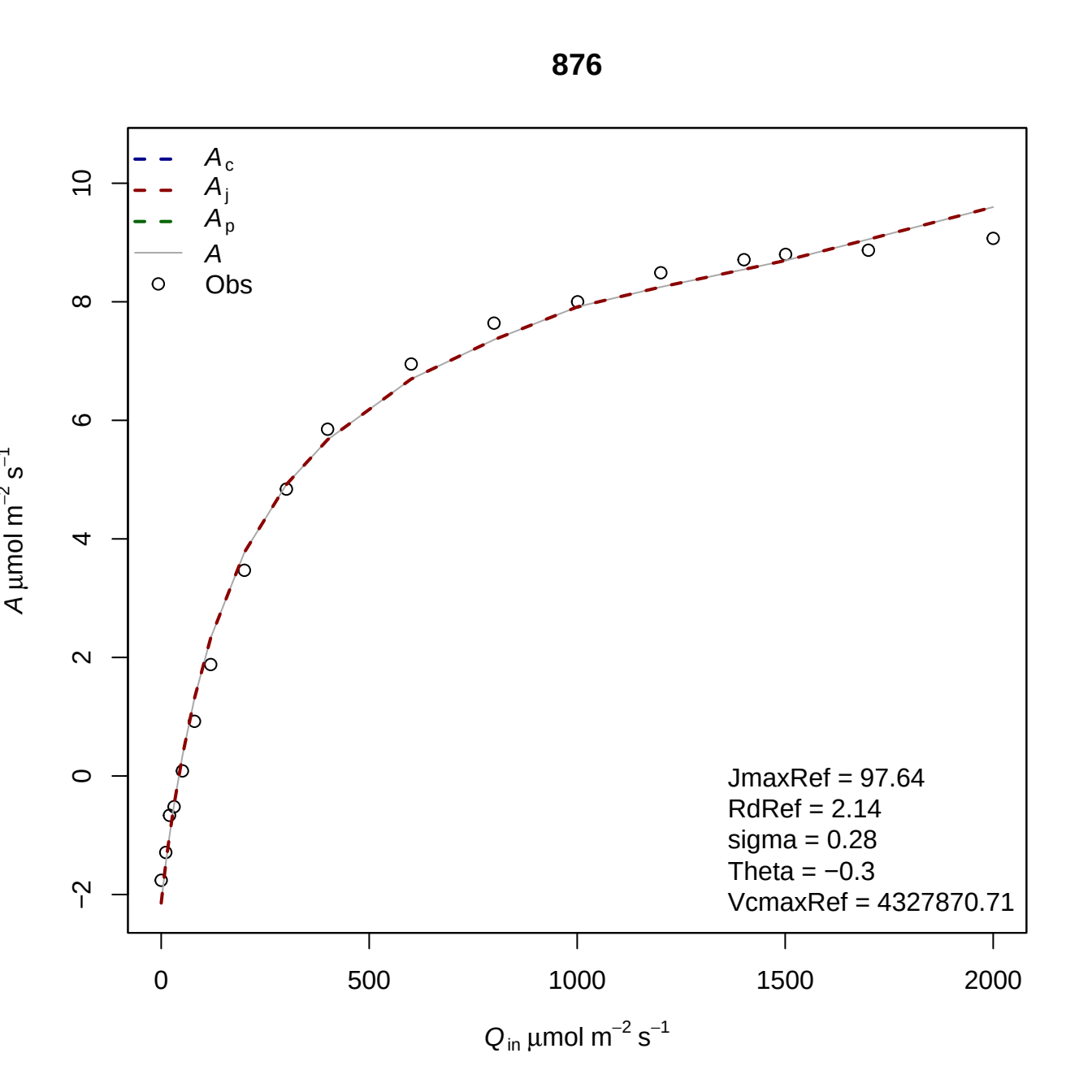

871
A_c
A_j
A_p
A
Obs
JmaxRef = 16.84
RdRef = 0.89
sigma = 0.19
Theta = −0.46
VcmaxRef = 23873.09
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

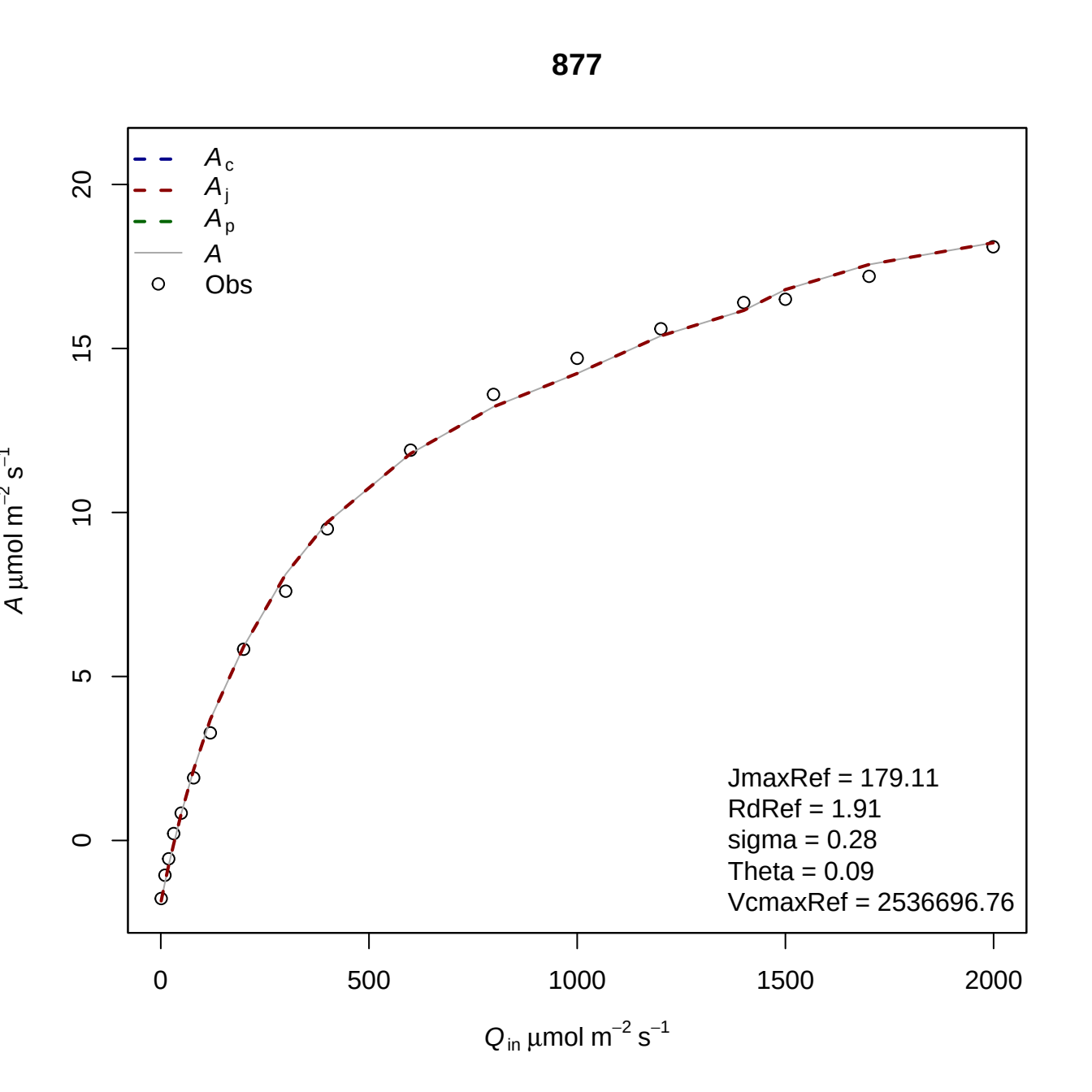

872
A_c
A_j
A_p
A
Obs
JmaxRef = 67.07
RdRef = 2.51
sigma = 0.15
Theta = −1.06
VcmaxRef = 342175.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

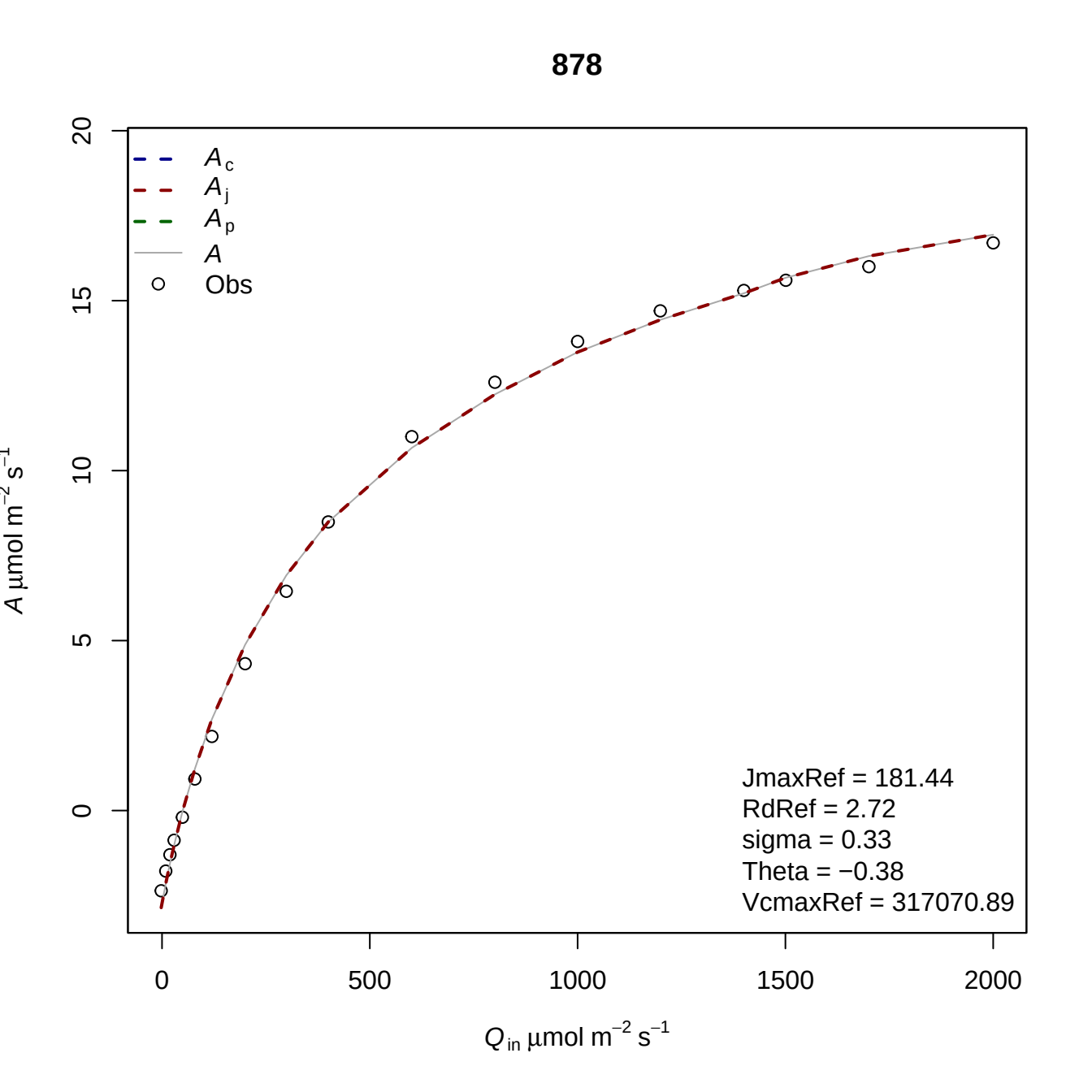

873
A_c
A_j
A_p
A
Obs
JmaxRef = 84.02
RdRef = 1.19
sigma = 0.17
Theta = 0.2
VcmaxRef = 549660.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

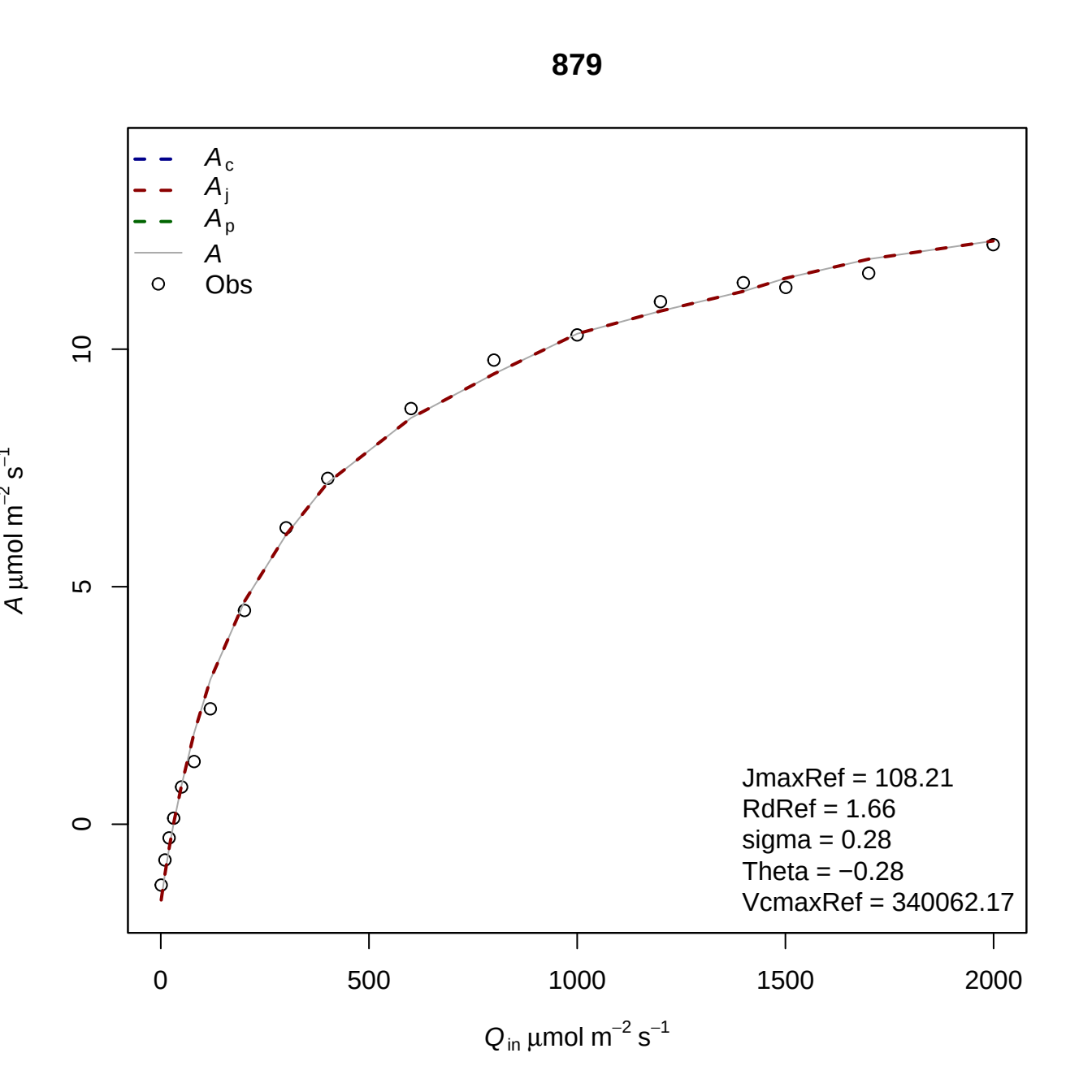

874
A_c
A_j
A_p
A
Obs
JmaxRef = 85.07
RdRef = 2.28
sigma = 0.3
Theta = −0.17
VcmaxRef = 152139.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

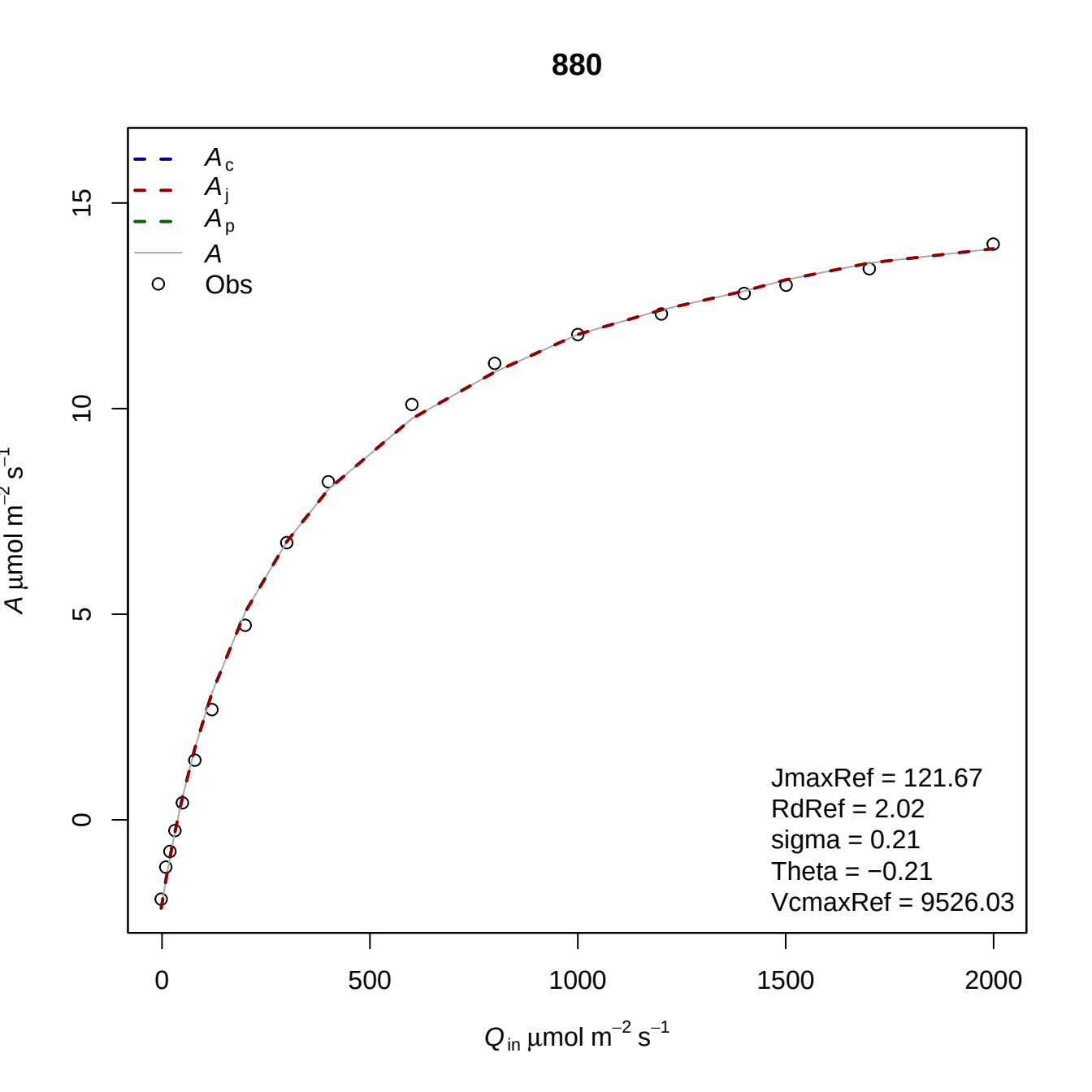

875
A_c
A_j
A_p
A
Obs
JmaxRef = 78.6
RdRef = 1.2
sigma = 0.18
Theta = 0.15
VcmaxRef = 18357.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

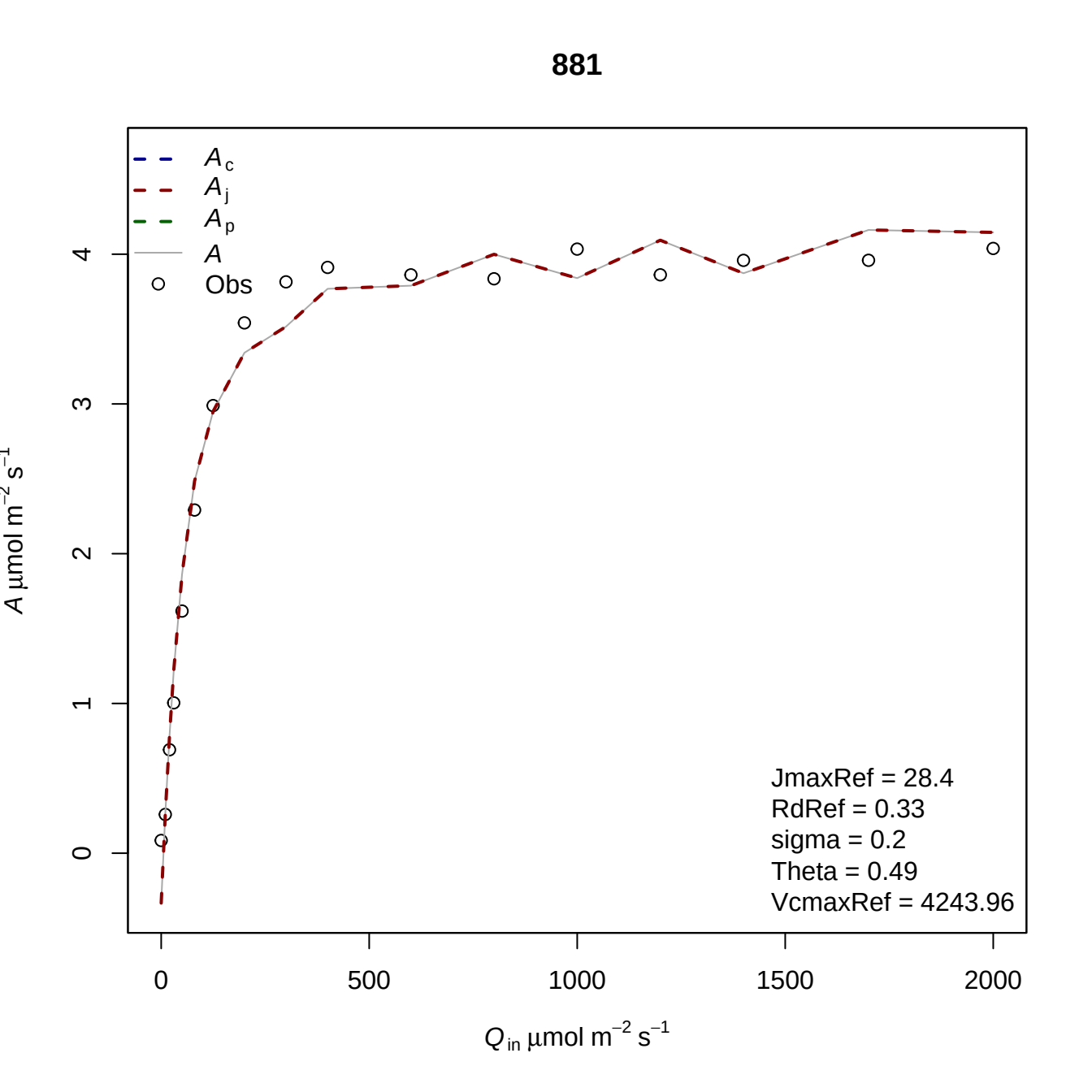

876
A_c
A_j
A_p
A
Obs
JmaxRef = 97.64
RdRef = 2.14
sigma = 0.28
Theta = −0.3
VcmaxRef = 4327870.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

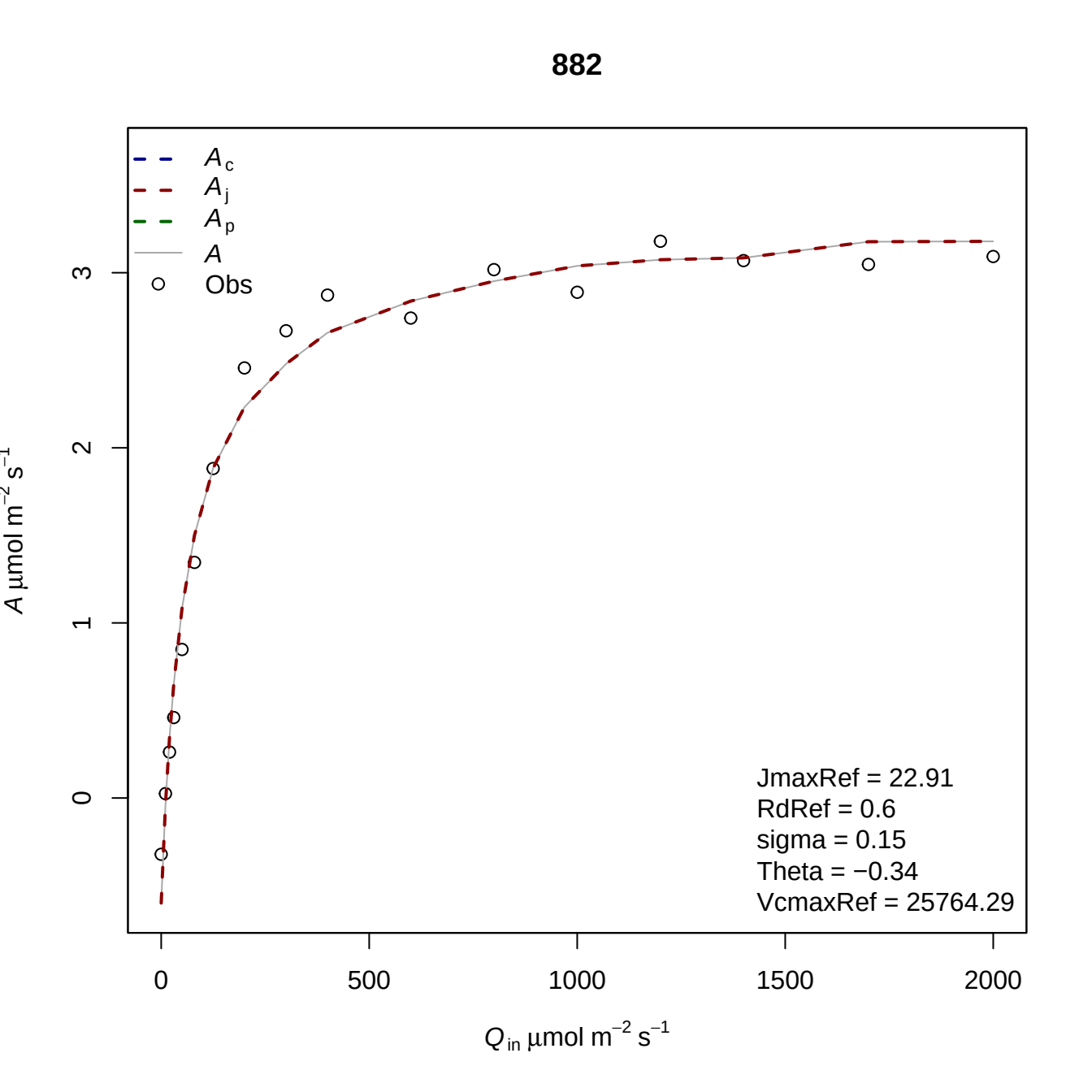

877
A_c
A_j
A_p
A
Obs
JmaxRef = 179.11
RdRef = 1.91
sigma = 0.28
Theta = 0.09
VcmaxRef = 2536696.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

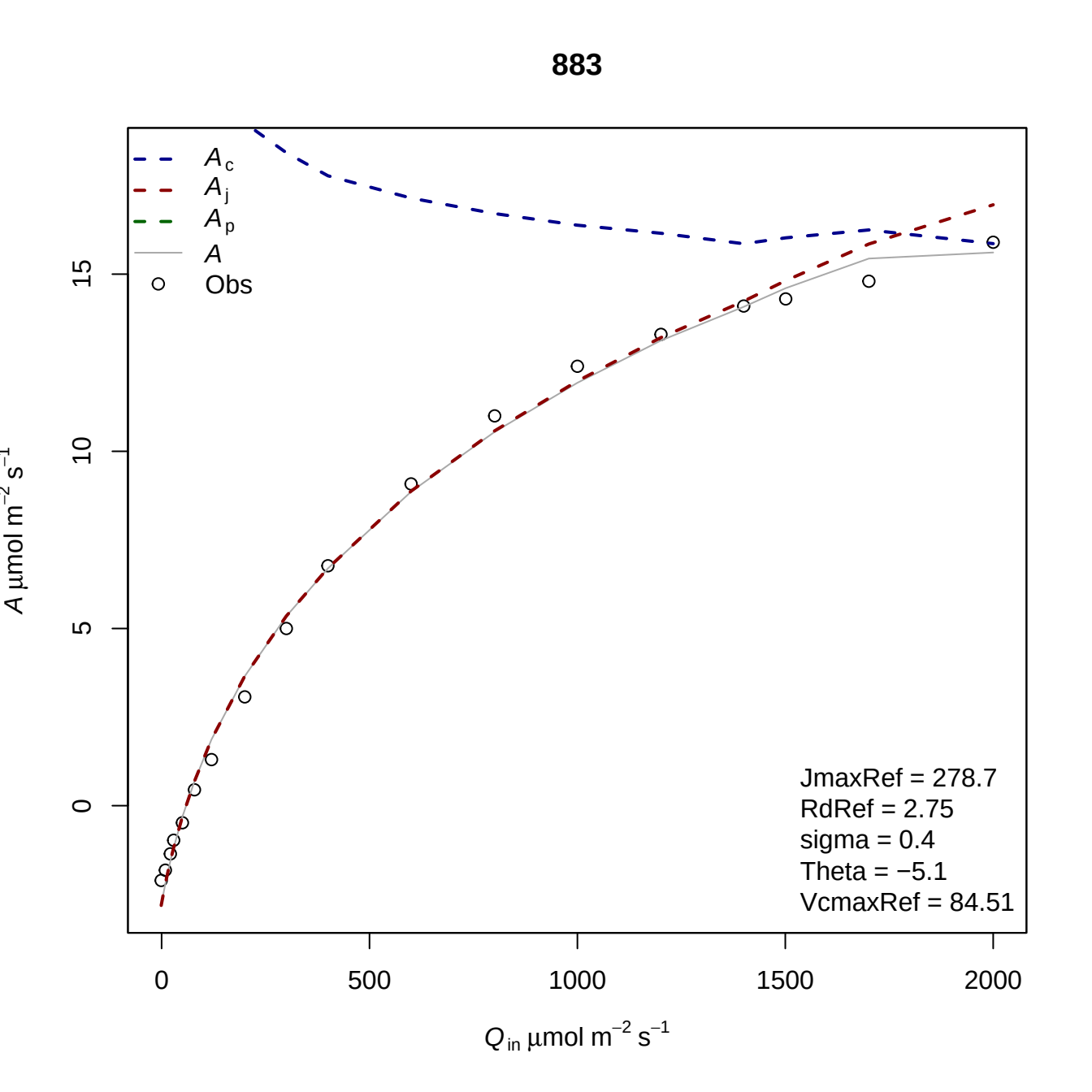

878
A_c
A_j
A_p
A
Obs
JmaxRef = 181.44
RdRef = 2.72
sigma = 0.33
Theta = −0.38
VcmaxRef = 317070.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

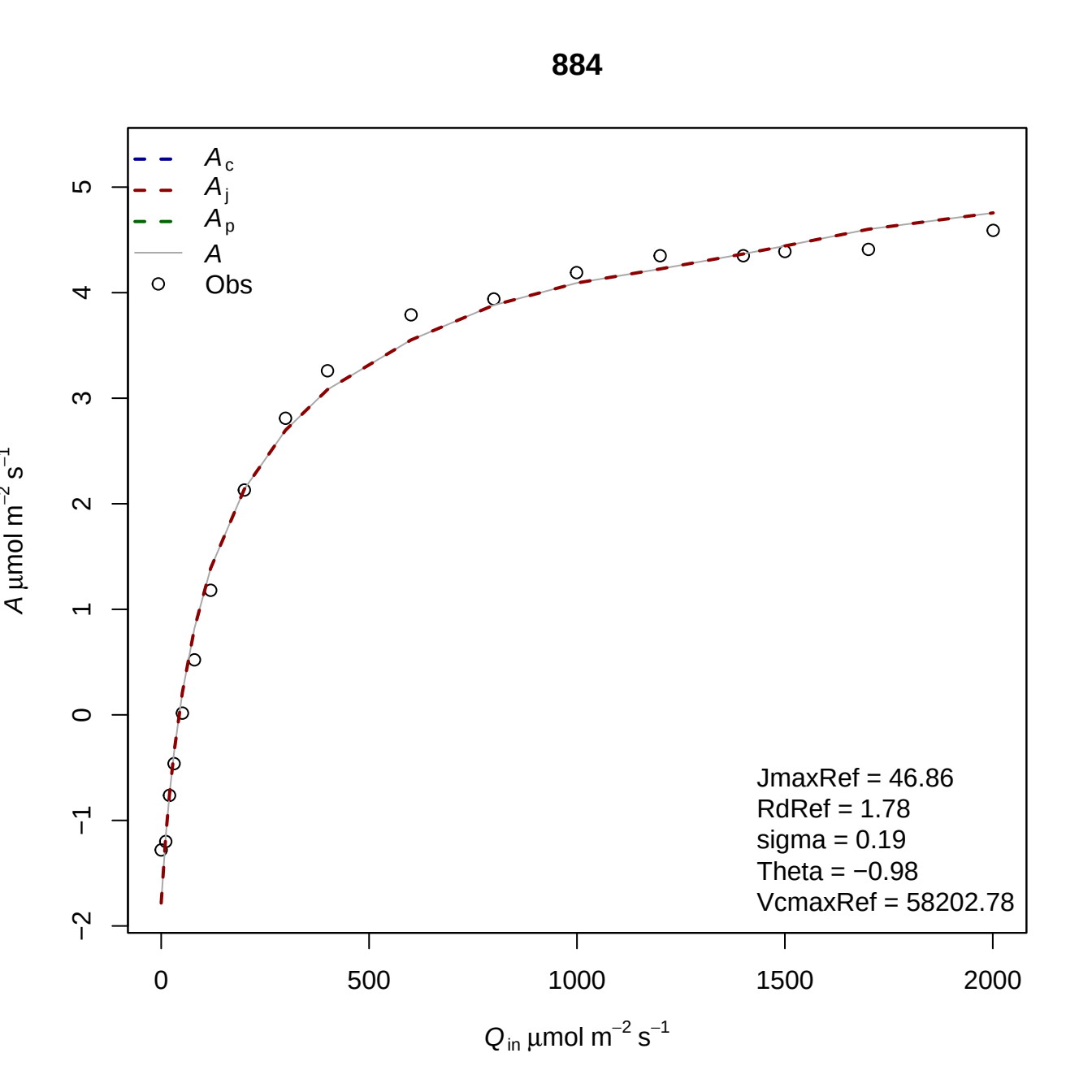

879
A_c
A_j
A_p
A
Obs
JmaxRef = 108.21
RdRef = 1.66
sigma = 0.28
Theta = −0.28
VcmaxRef = 340062.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

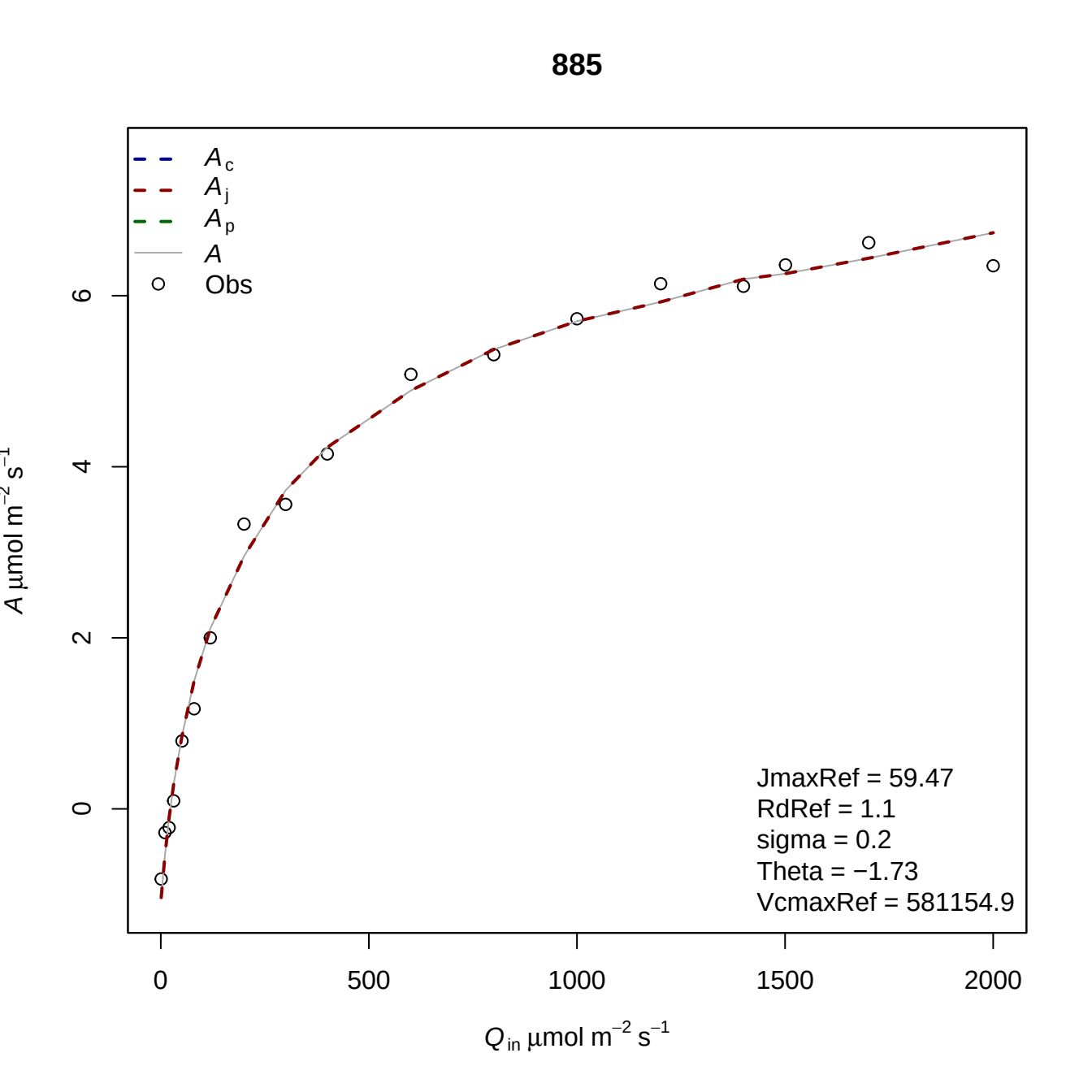

880
A_c
A_j
A_p
A
Obs
JmaxRef = 121.67
RdRef = 2.02
sigma = 0.21
Theta = −0.21
VcmaxRef = 9526.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

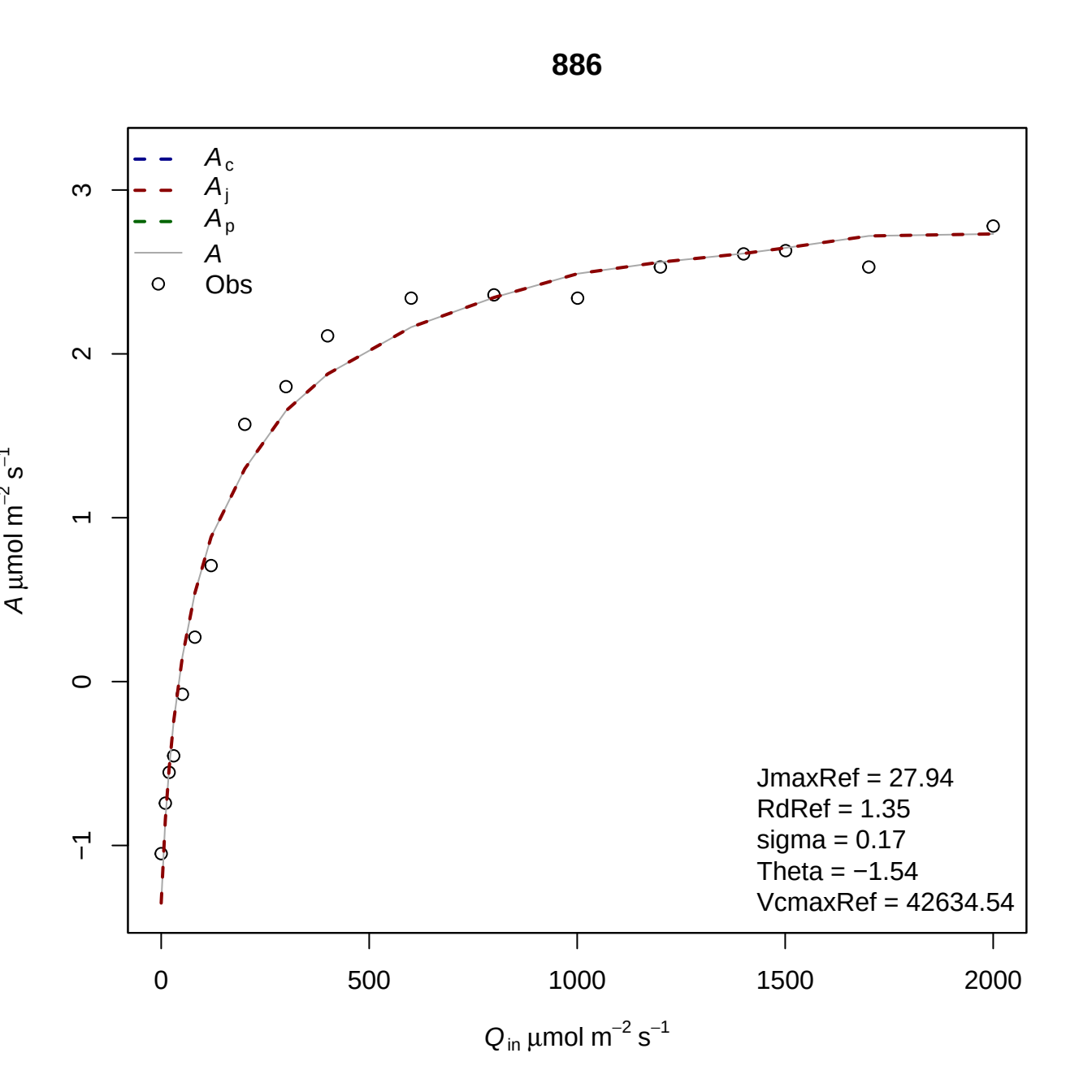

881
A_c
A_j
A_p
A
Obs
JmaxRef = 28.4
RdRef = 0.33
sigma = 0.2
Theta = 0.49
VcmaxRef = 4243.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

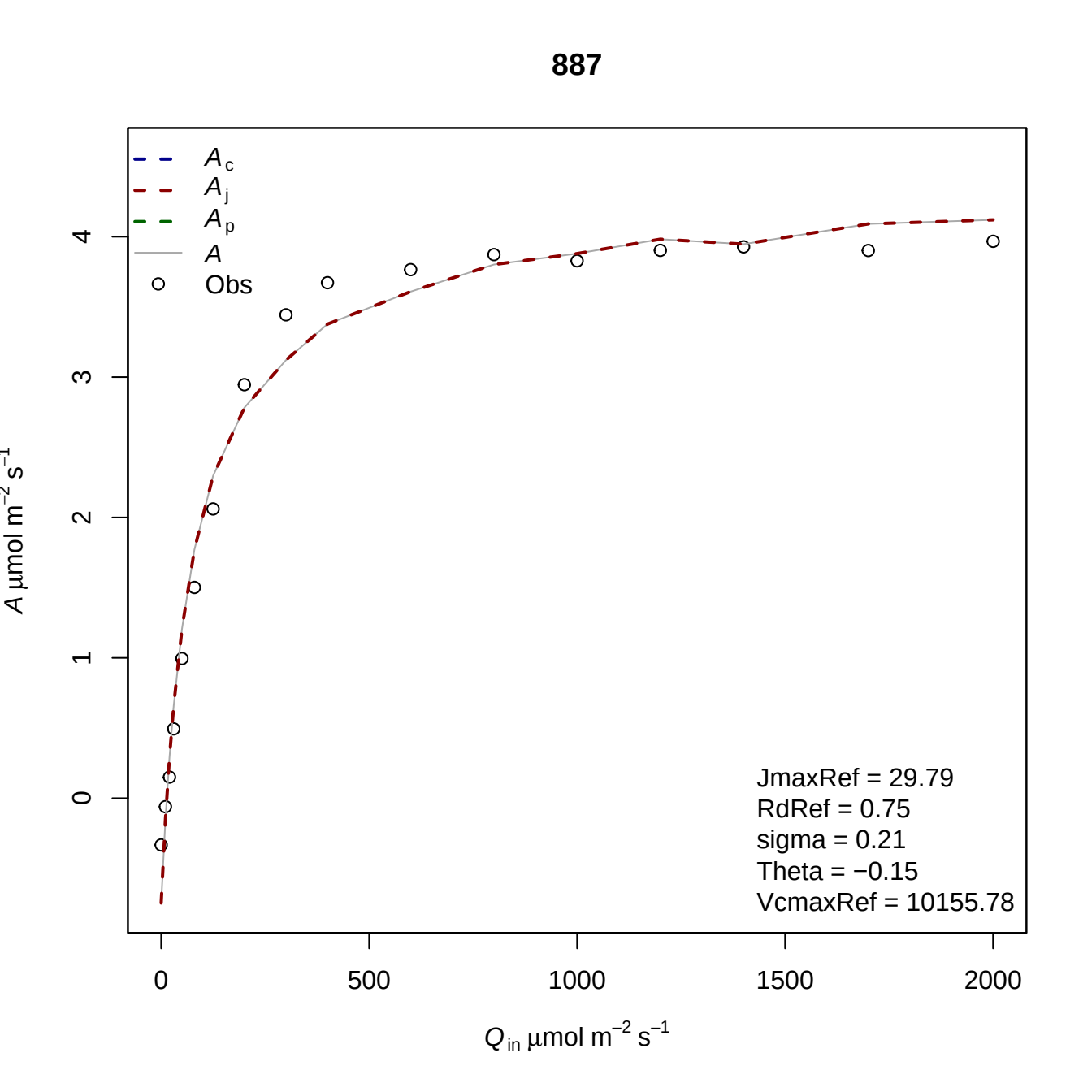

882
A_c
A_j
A_p
A
Obs
JmaxRef = 22.91
RdRef = 0.6
sigma = 0.15
Theta = −0.34
VcmaxRef = 25764.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

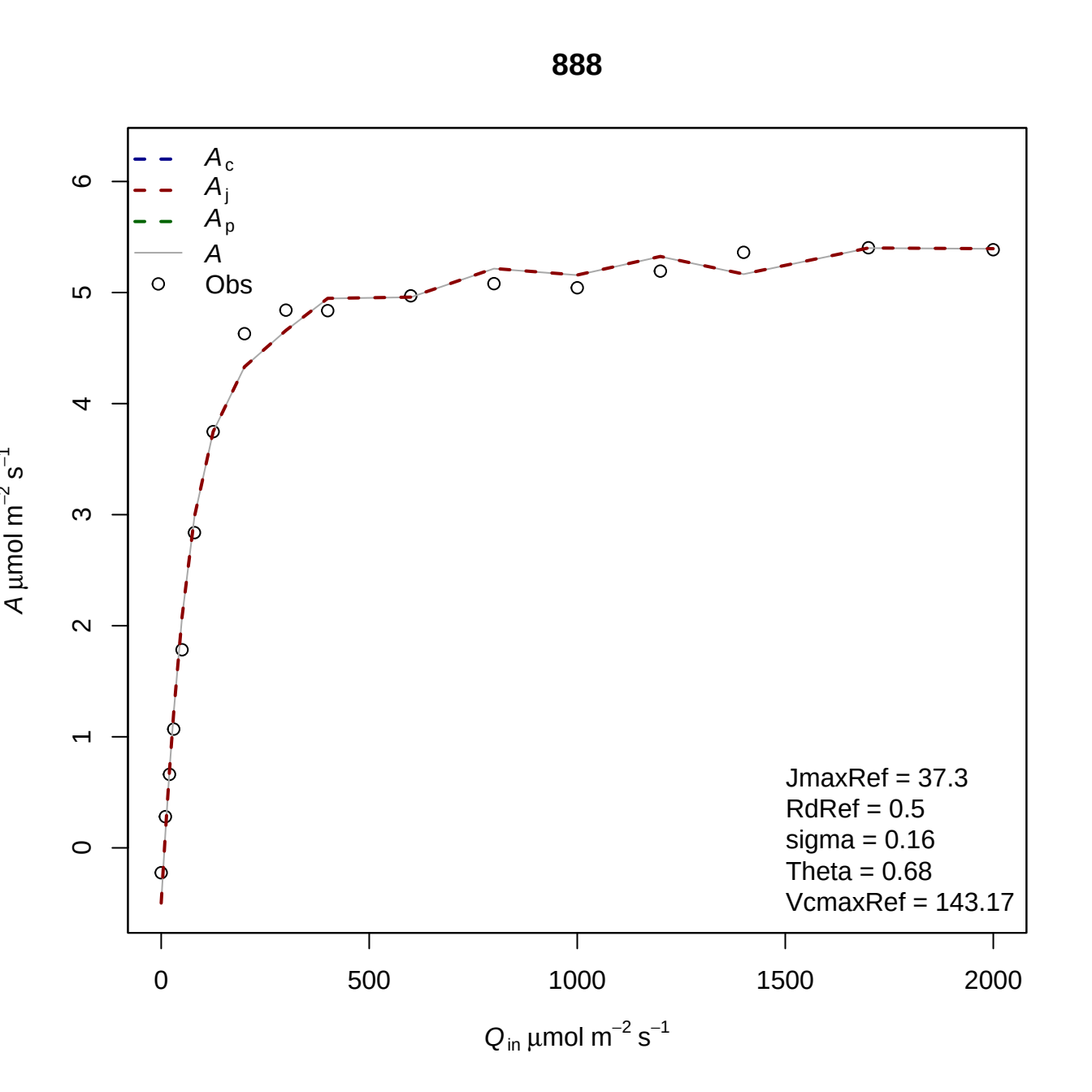

883
A_c
A_j
A_p
A
Obs
JmaxRef = 278.7
RdRef = 2.75
sigma = 0.4
Theta = −5.1
VcmaxRef = 84.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

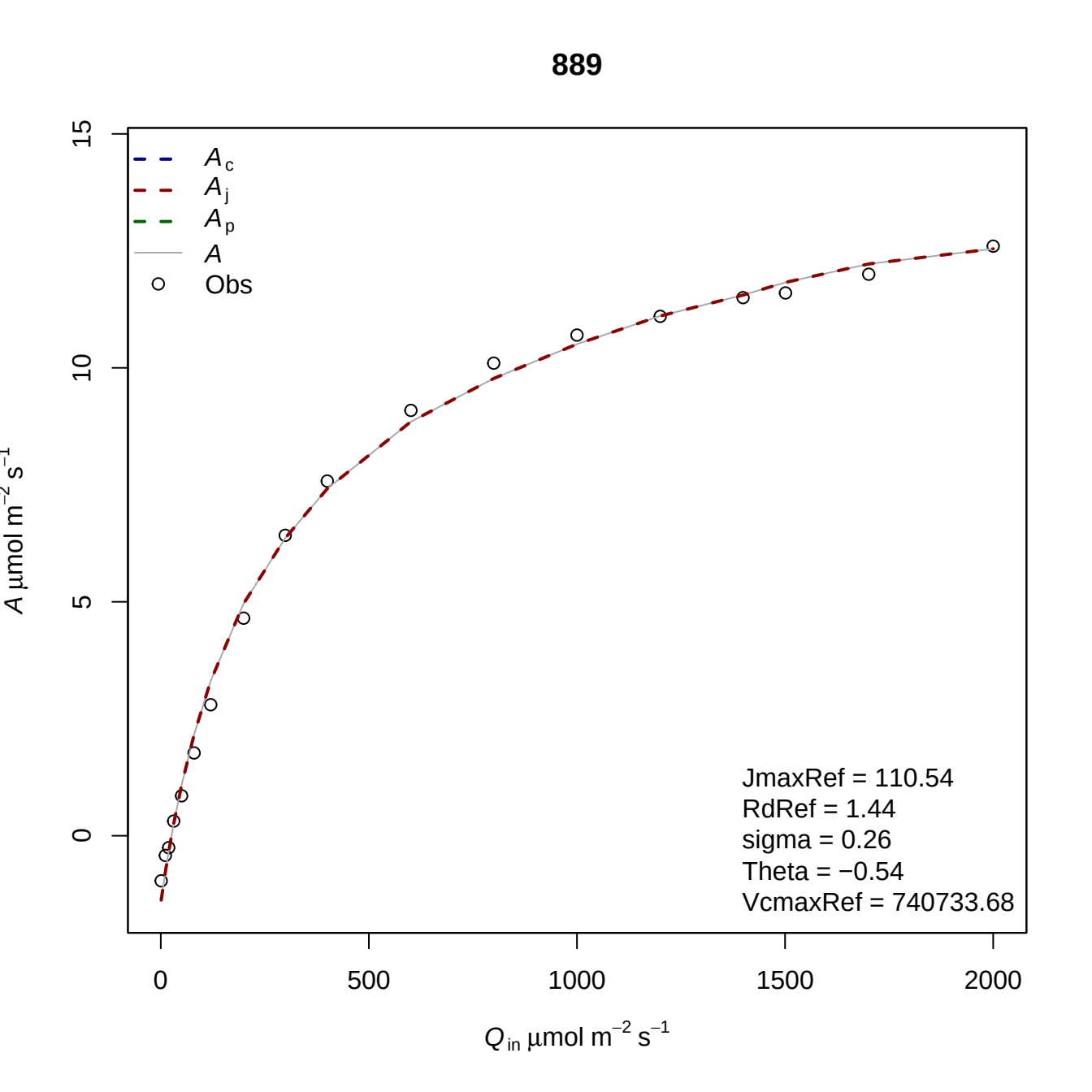

884
A_c
A_j
A_p
A
Obs
JmaxRef = 46.86
RdRef = 1.78
sigma = 0.19
Theta = −0.98
VcmaxRef = 58202.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
5
0
500
1000
1500
2000

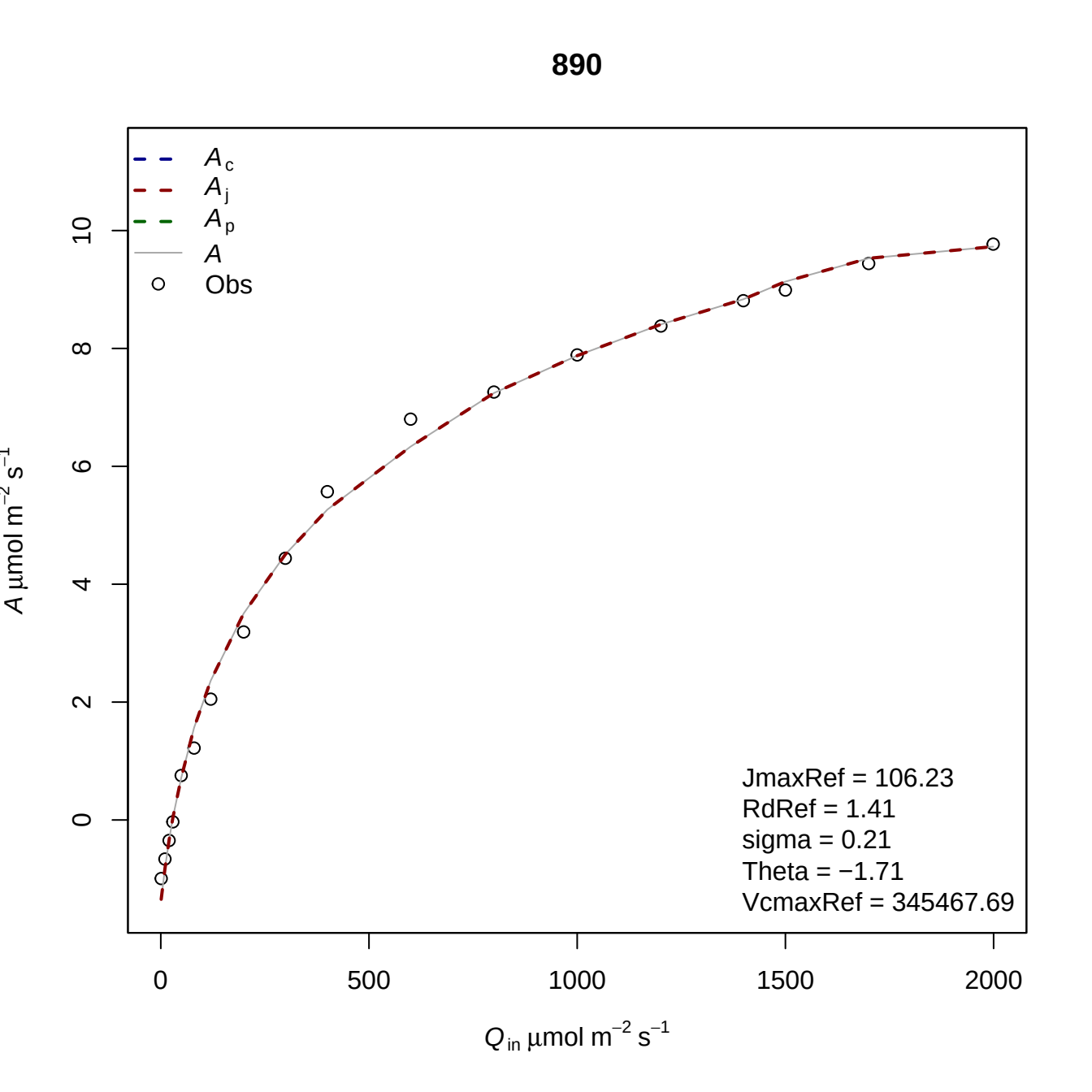

885
A_c
A_j
A_p
A
Obs
JmaxRef = 59.47
RdRef = 1.1
sigma = 0.2
Theta = −1.73
VcmaxRef = 581154.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

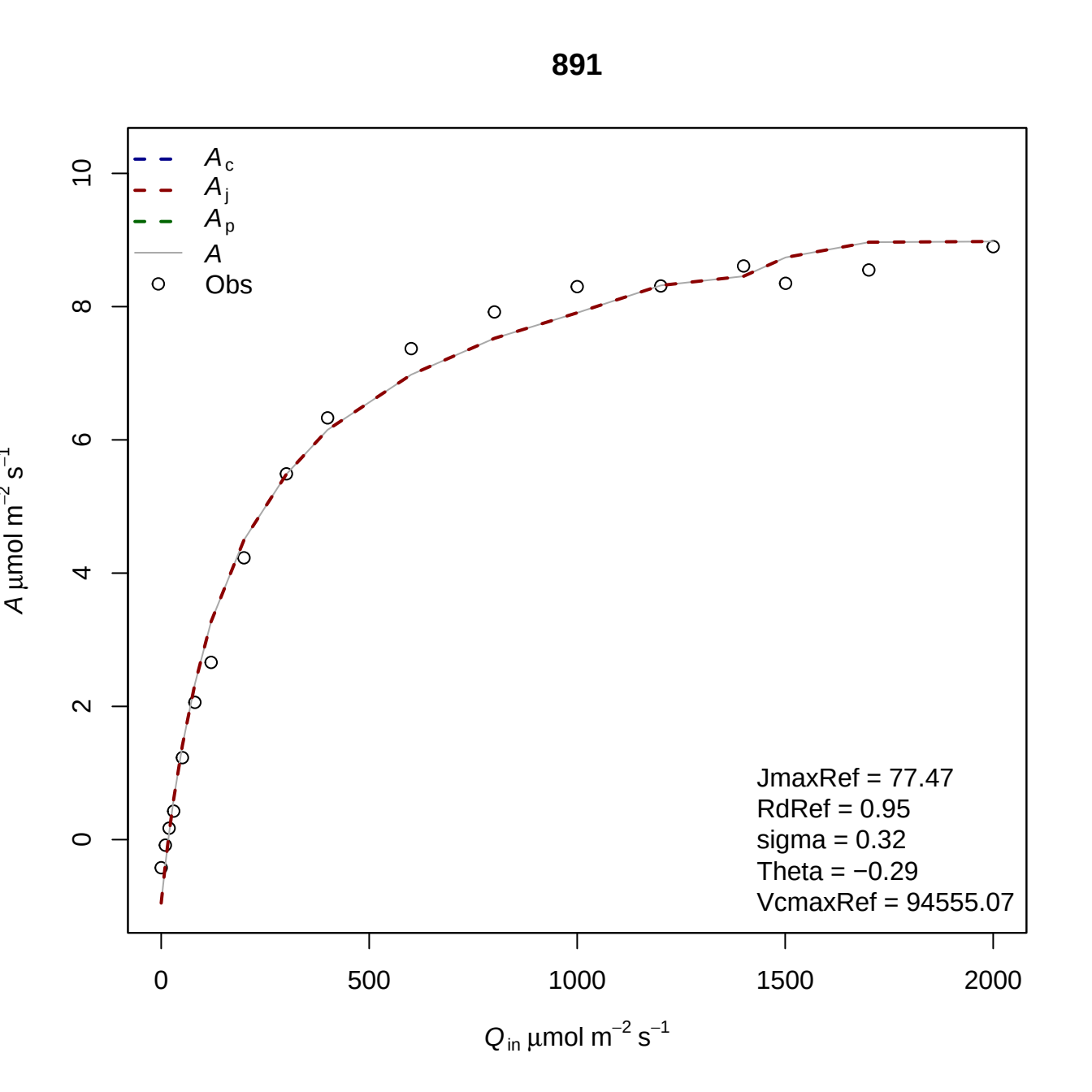

886
A_c
A_j
A_p
A
Obs
JmaxRef = 27.94
RdRef = 1.35
sigma = 0.17
Theta = −1.54
VcmaxRef = 42634.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

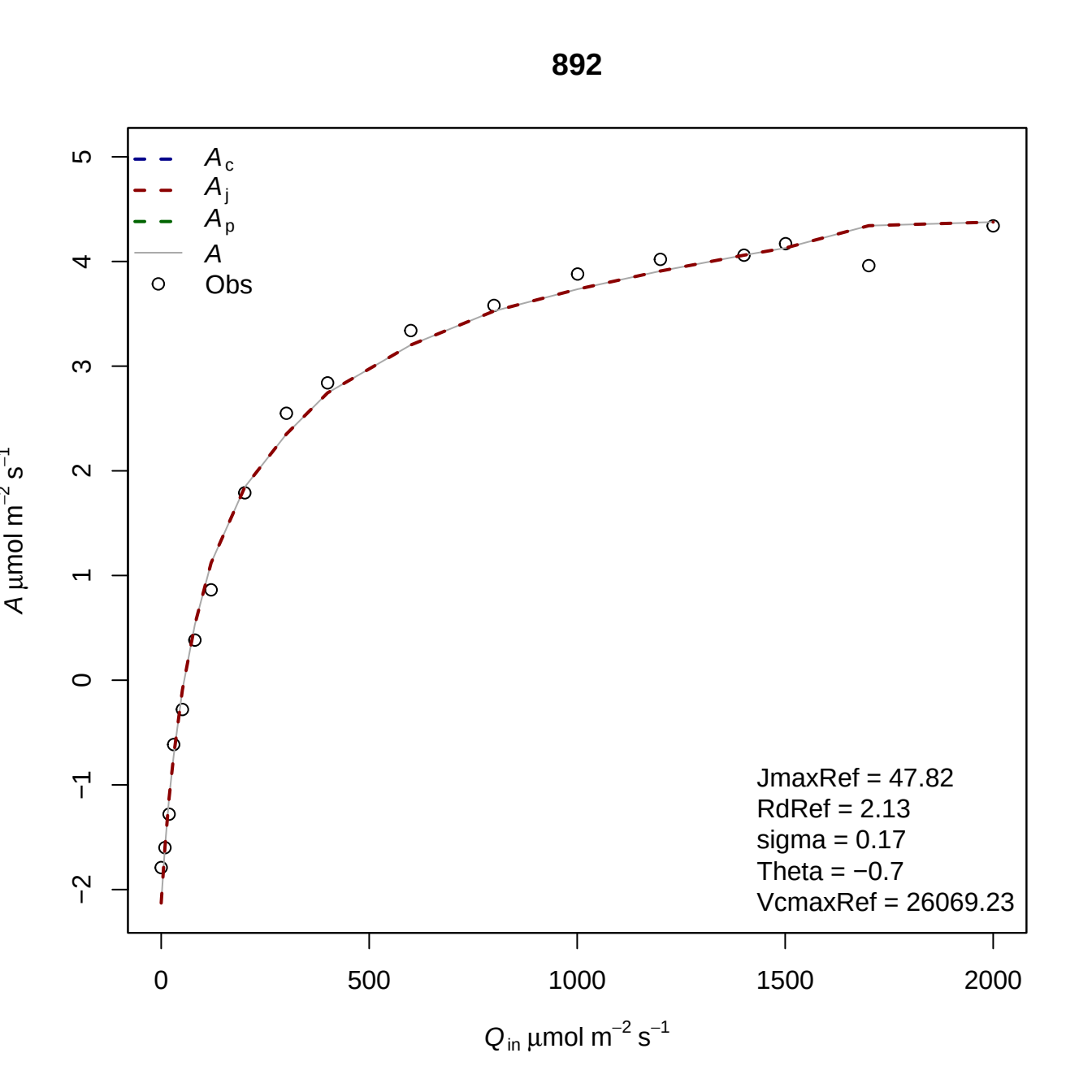

887
A_c
A_j
A_p
A
Obs
JmaxRef = 29.79
RdRef = 0.75
sigma = 0.21
Theta = −0.15
VcmaxRef = 10155.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

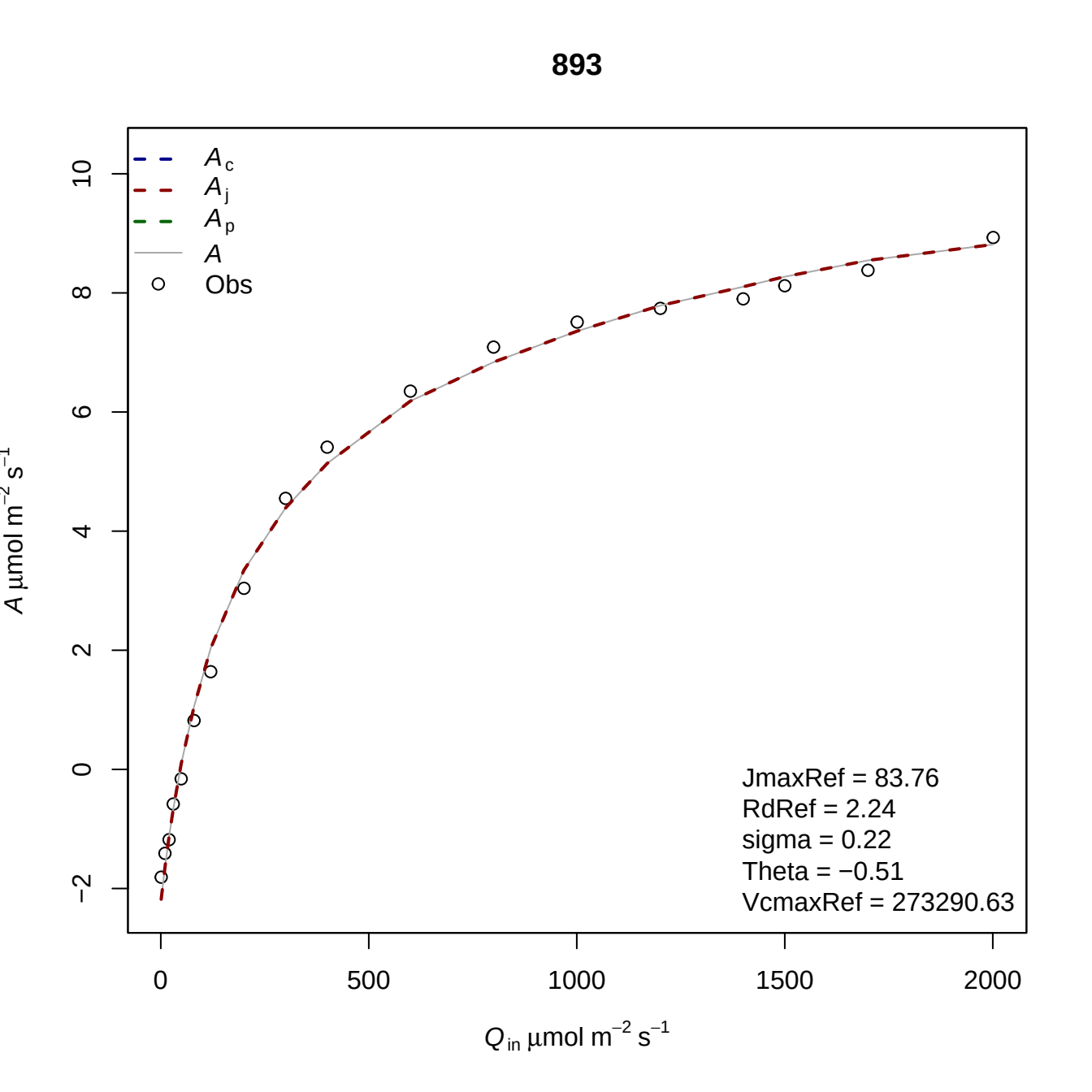

888
A_c
A_j
A_p
A
Obs
JmaxRef = 37.3
RdRef = 0.5
sigma = 0.16
Theta = 0.68
VcmaxRef = 143.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

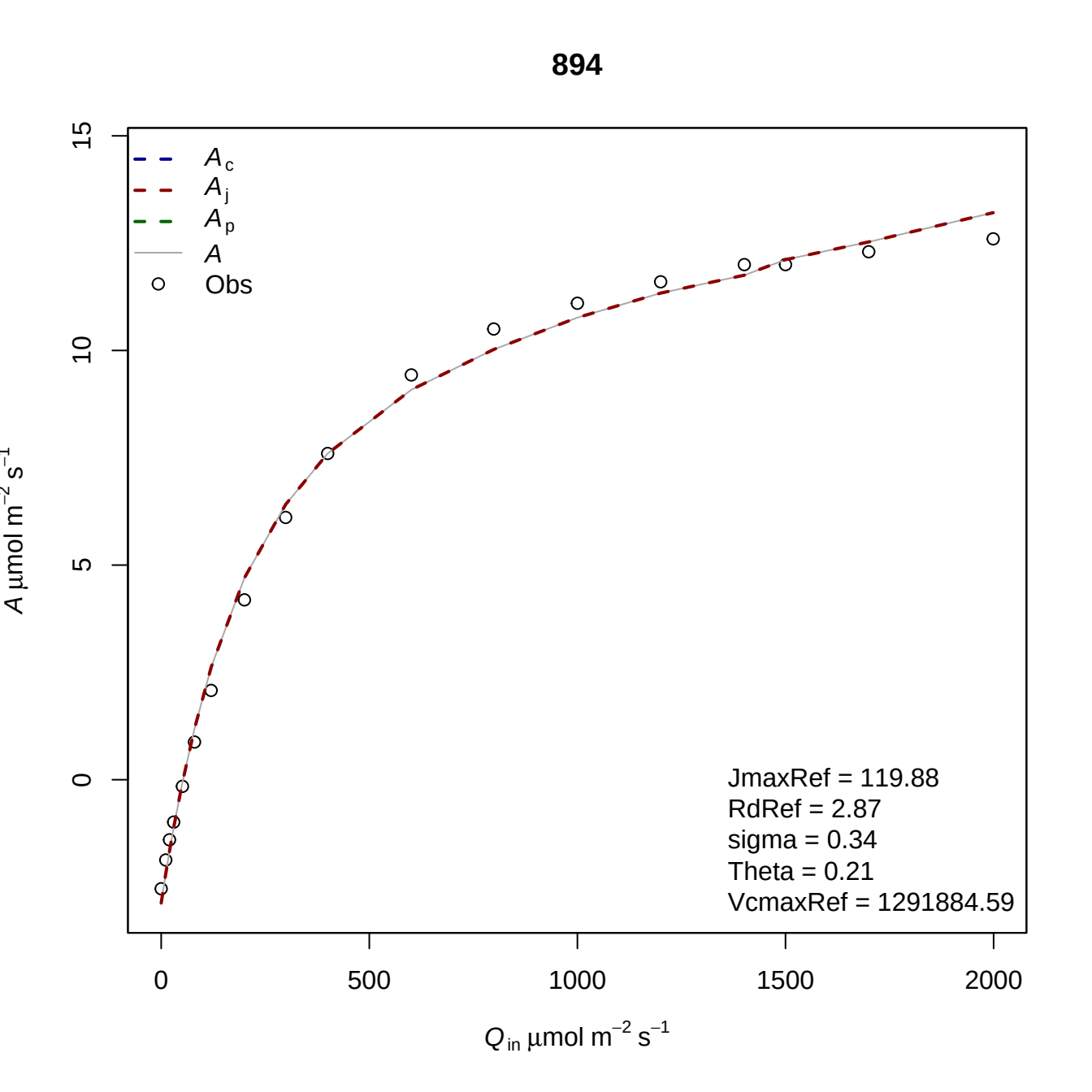

889
A_c
A_j
A_p
A
Obs
JmaxRef = 110.54
RdRef = 1.44
sigma = 0.26
Theta = −0.54
VcmaxRef = 740733.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

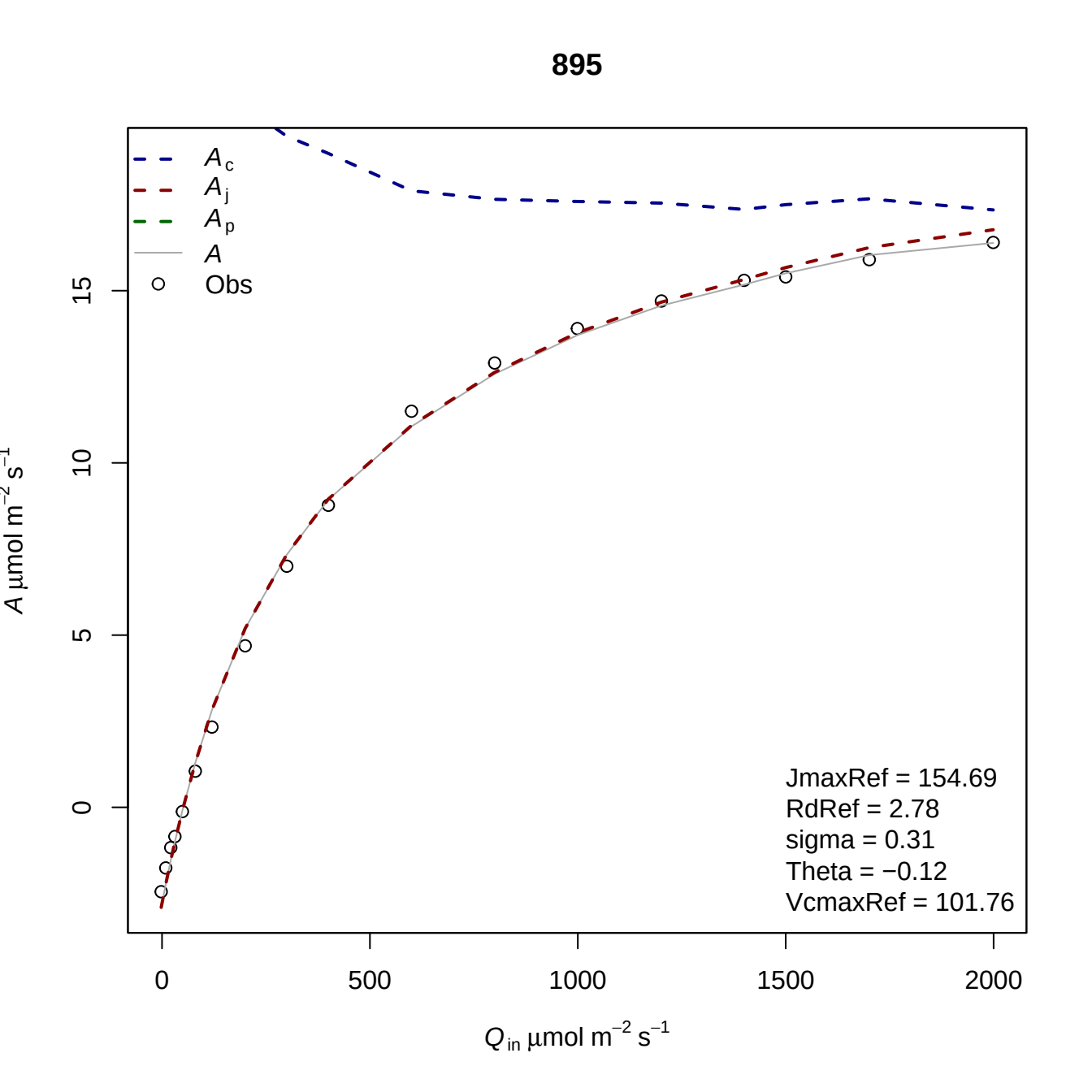

890
A_c
A_j
A_p
A
Obs
JmaxRef = 106.23
RdRef = 1.41
sigma = 0.21
Theta = −1.71
VcmaxRef = 345467.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

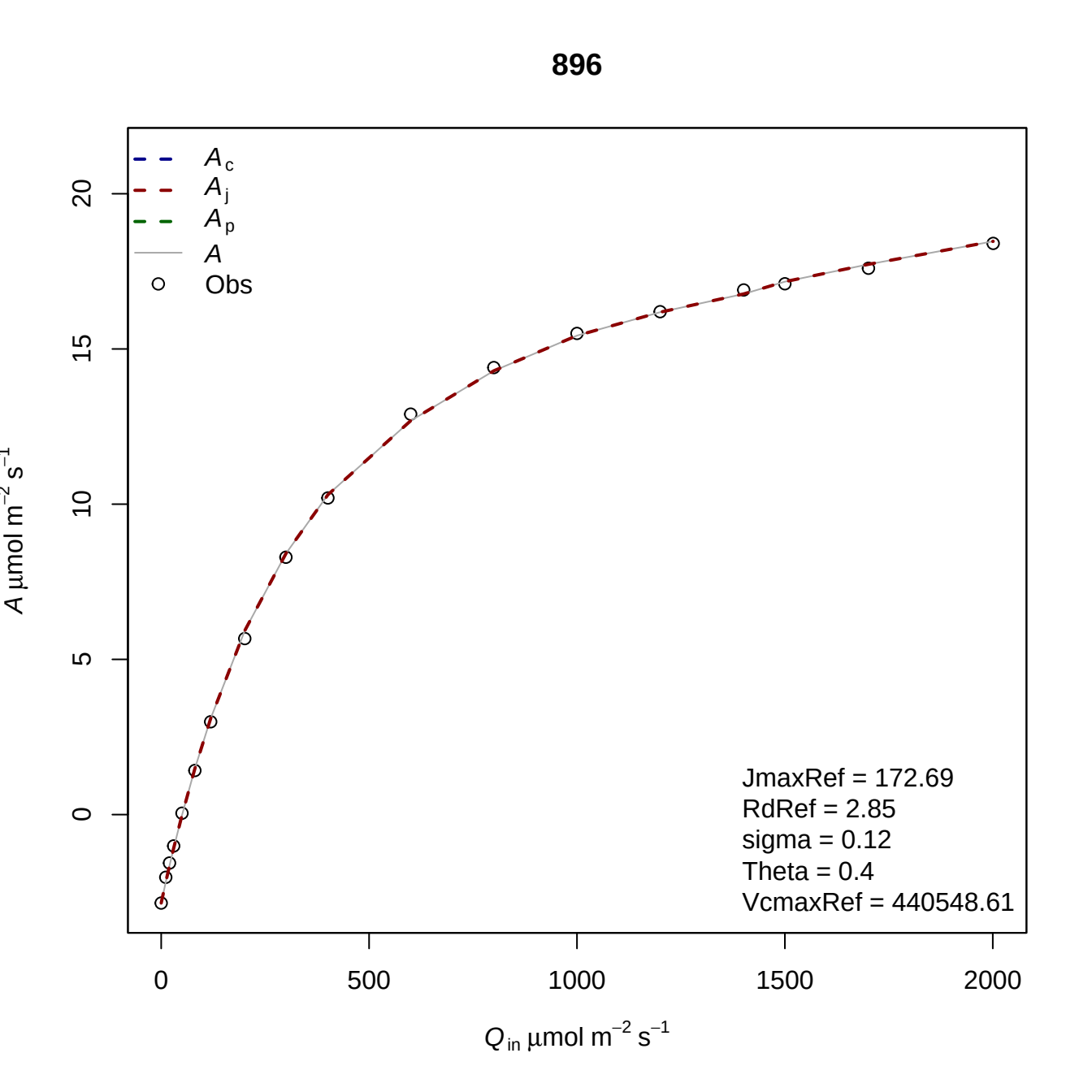

891
A_c
A_j
A_p
A
Obs
JmaxRef = 77.47
RdRef = 0.95
sigma = 0.32
Theta = −0.29
VcmaxRef = 94555.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

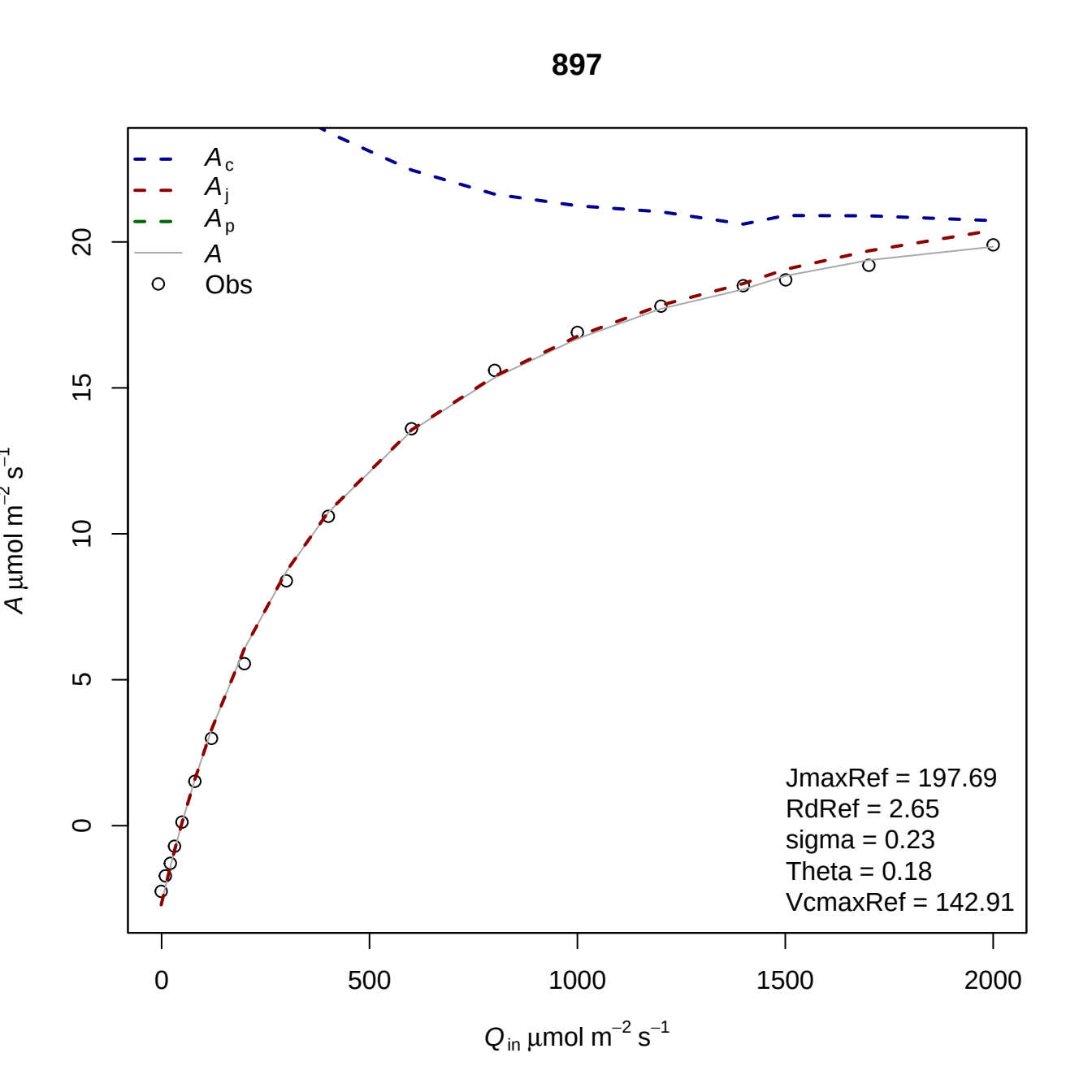

892
A_c
A_j
A_p
A
Obs
JmaxRef = 47.82
RdRef = 2.13
sigma = 0.17
Theta = −0.7
VcmaxRef = 26069.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

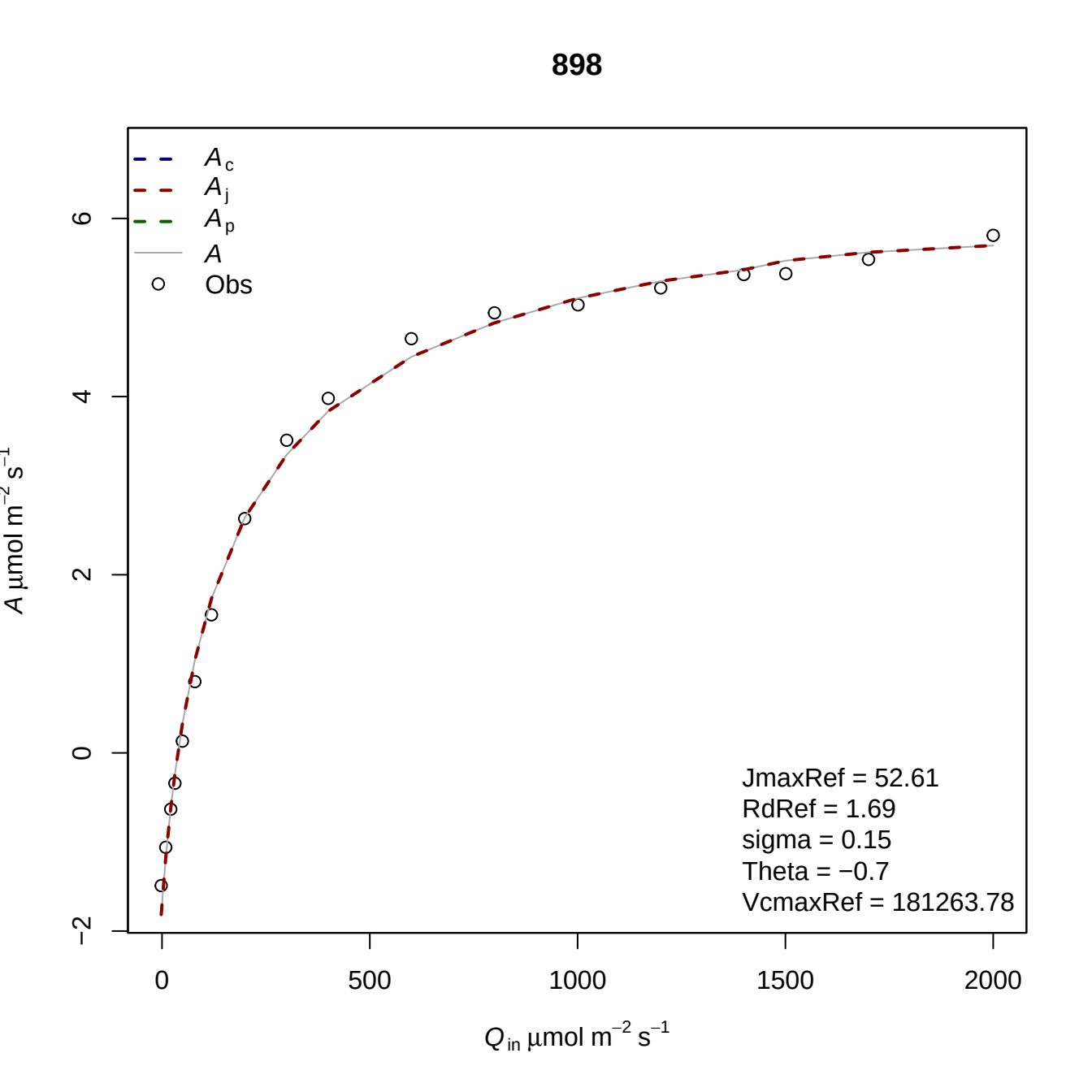

893
A_c
A_j
A_p
A
Obs
JmaxRef = 83.76
RdRef = 2.24
sigma = 0.22
Theta = −0.51
VcmaxRef = 273290.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

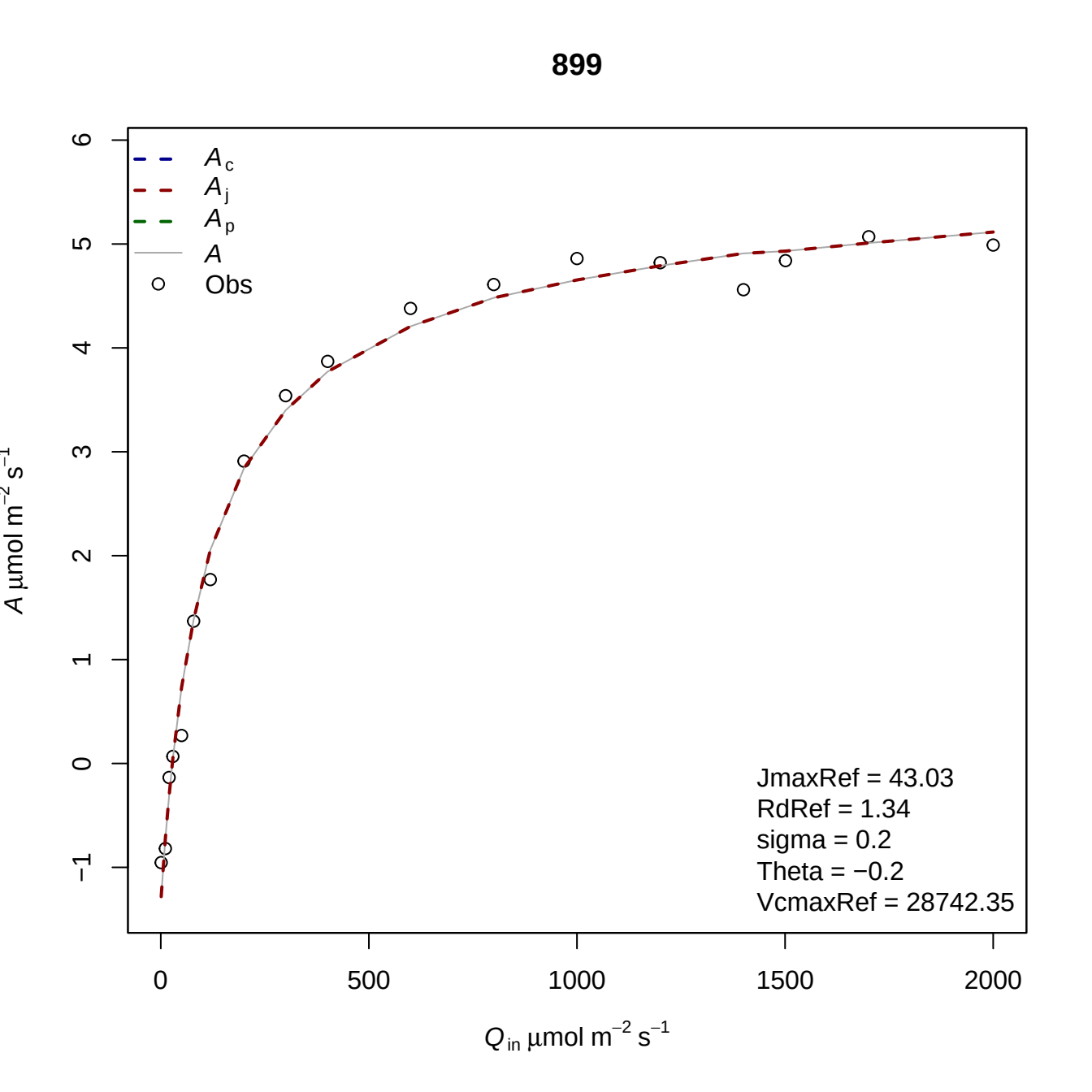

894
A_c
A_j
A_p
A
Obs
JmaxRef = 119.88
RdRef = 2.87
sigma = 0.34
Theta = 0.21
VcmaxRef = 1291884.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

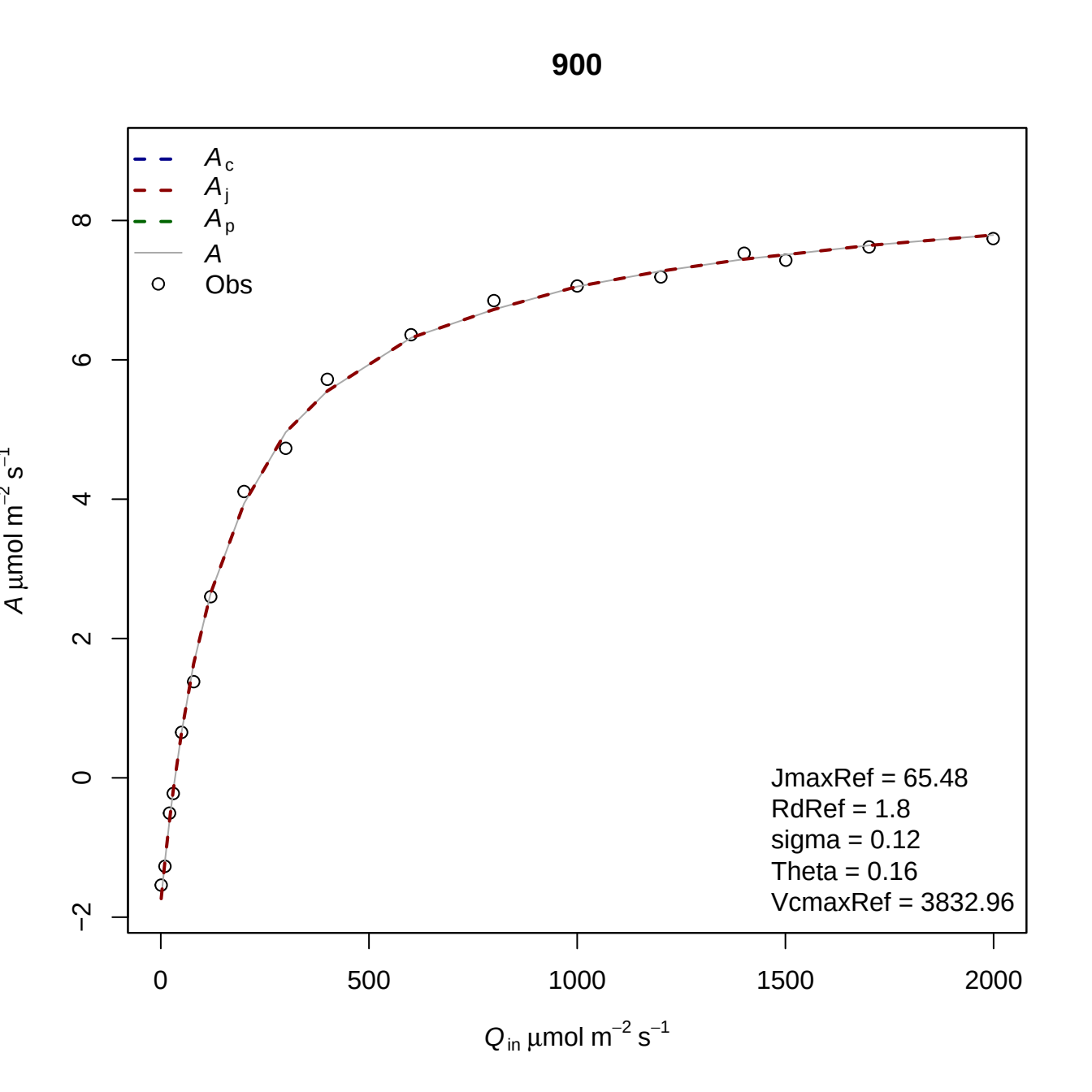

895
A_c
A_j
A_p
A
Obs
JmaxRef = 154.69
RdRef = 2.78
sigma = 0.31
Theta = −0.12
VcmaxRef = 101.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

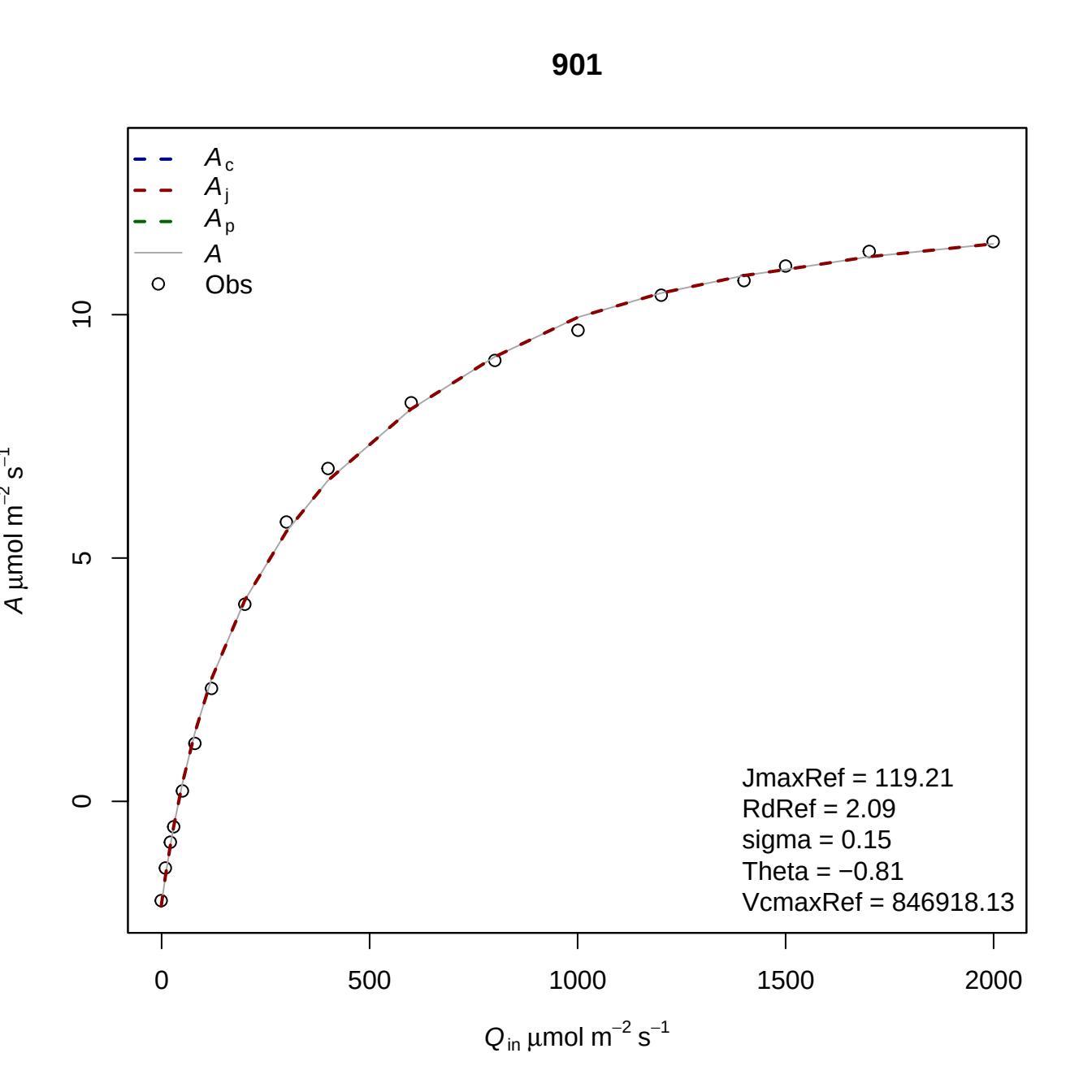

896
A_c
A_j
A_p
A
Obs
JmaxRef = 172.69
RdRef = 2.85
sigma = 0.12
Theta = 0.4
VcmaxRef = 440548.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

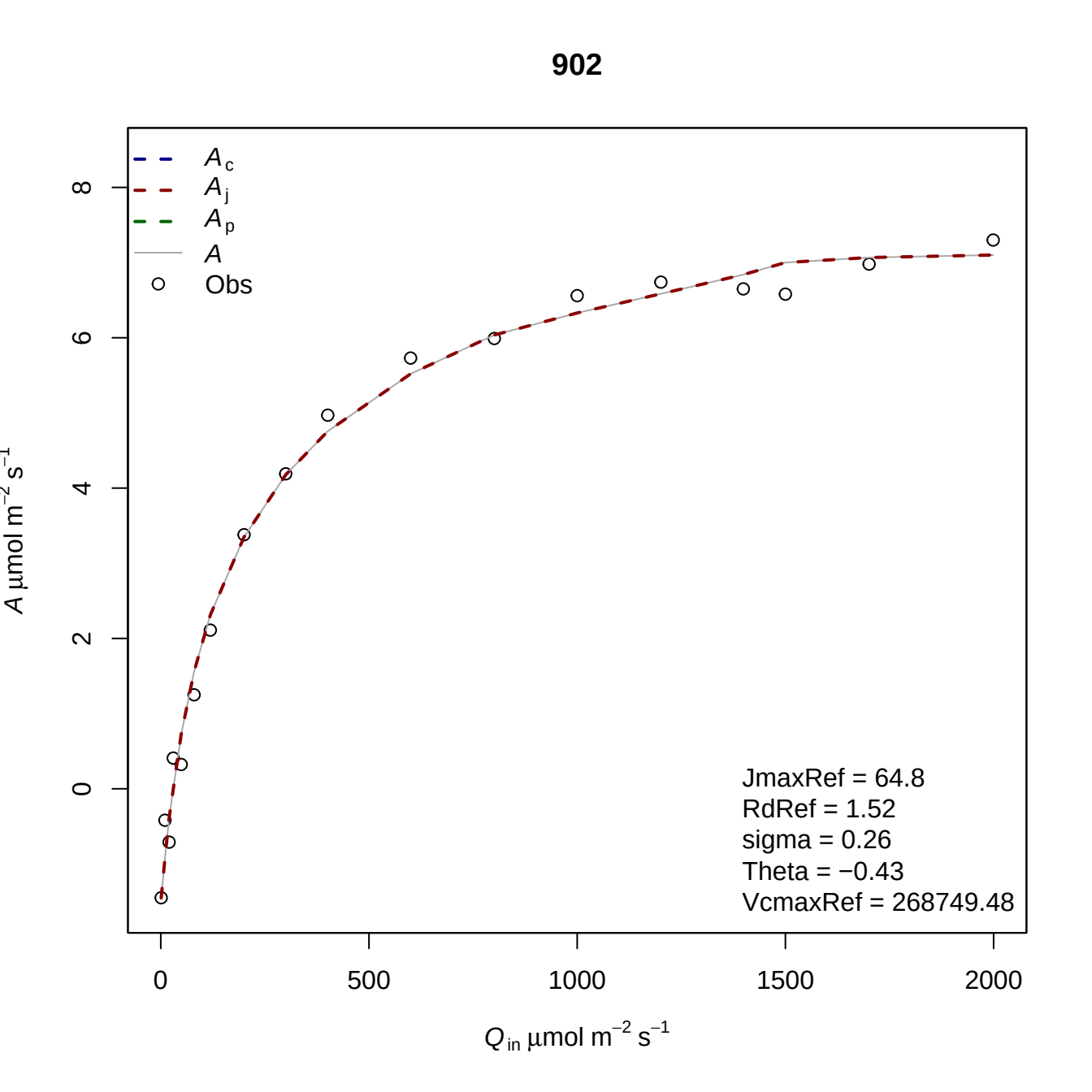

897
A_c
A_j
A_p
A
Obs
JmaxRef = 197.69
RdRef = 2.65
sigma = 0.23
Theta = 0.18
VcmaxRef = 142.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

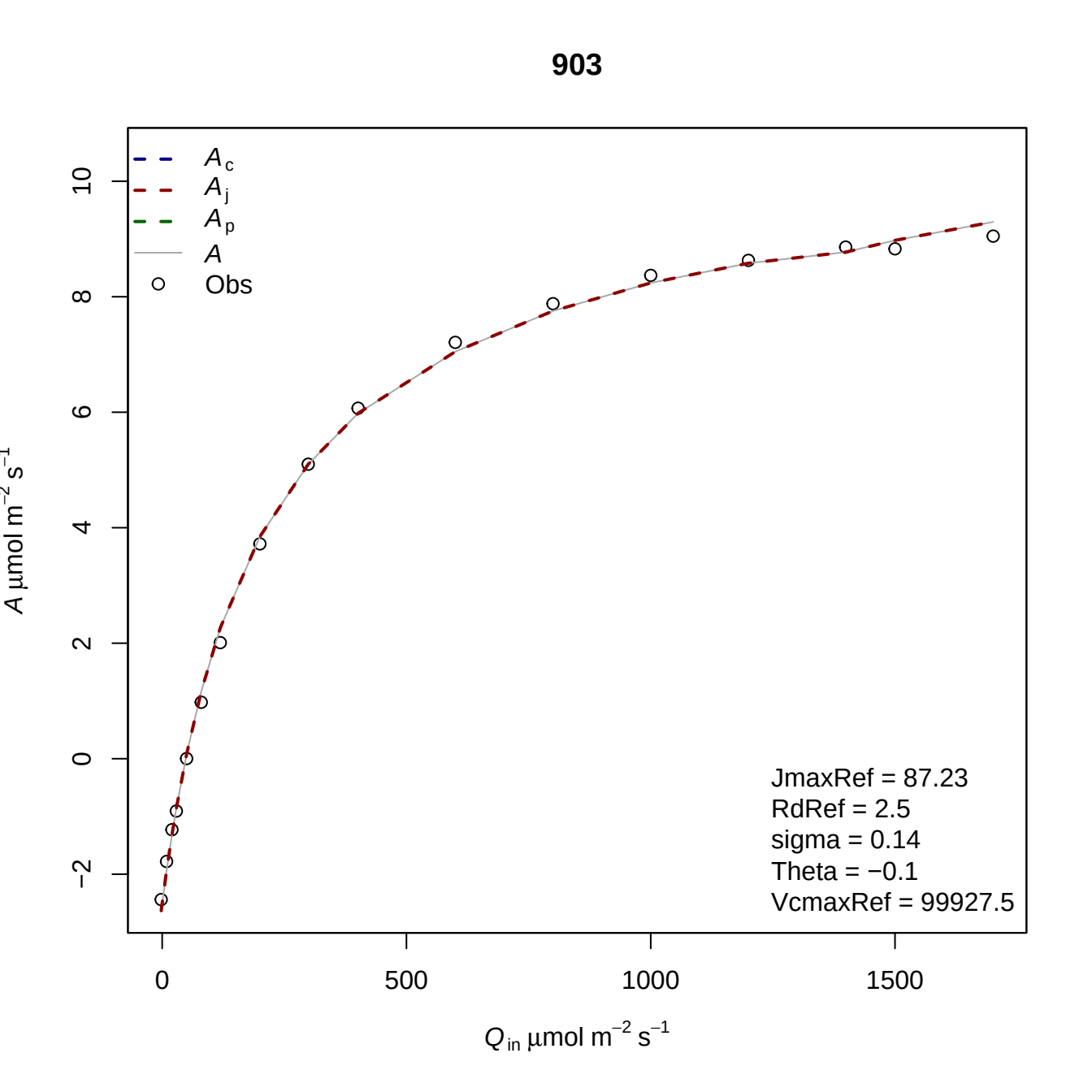

898
A_c
A_j
A_p
A
Obs
JmaxRef = 52.61
RdRef = 1.69
sigma = 0.15
Theta = −0.7
VcmaxRef = 181263.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

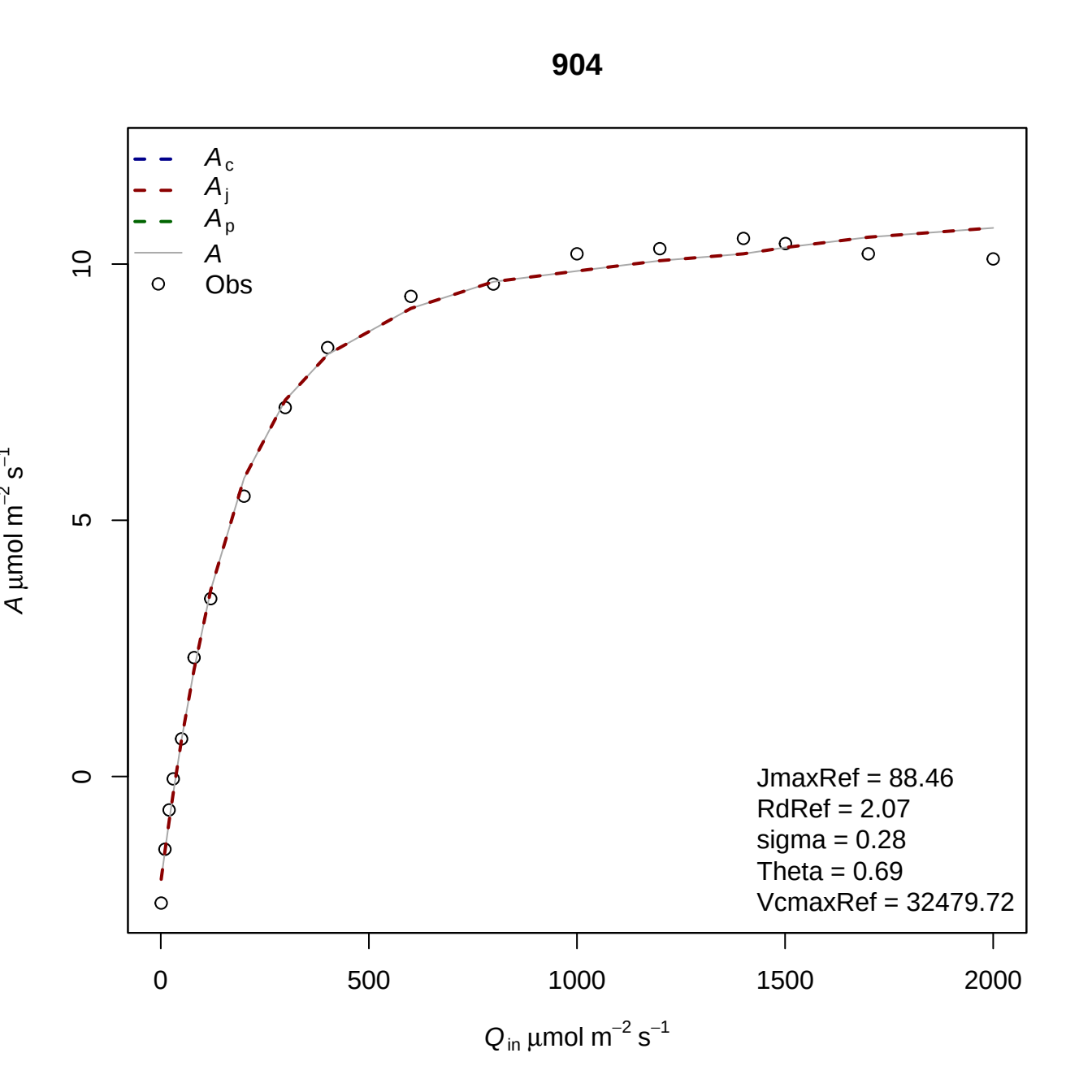

899
A_c
A_j
A_p
A
Obs
JmaxRef = 43.03
RdRef = 1.34
sigma = 0.2
Theta = −0.2
VcmaxRef = 28742.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

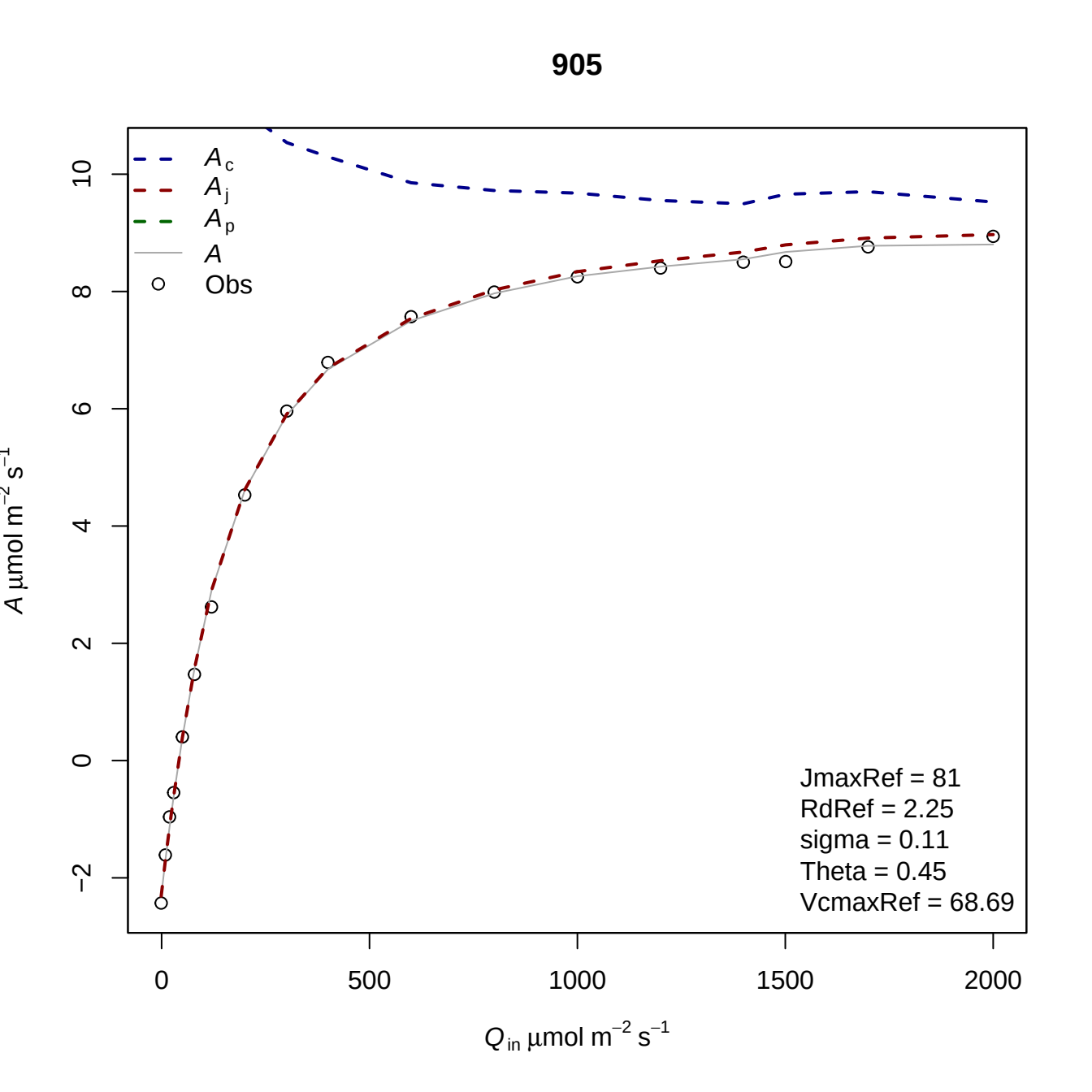

900
A_c
A_j
A_p
A
Obs
JmaxRef = 65.48
RdRef = 1.8
sigma = 0.12
Theta = 0.16
VcmaxRef = 3832.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

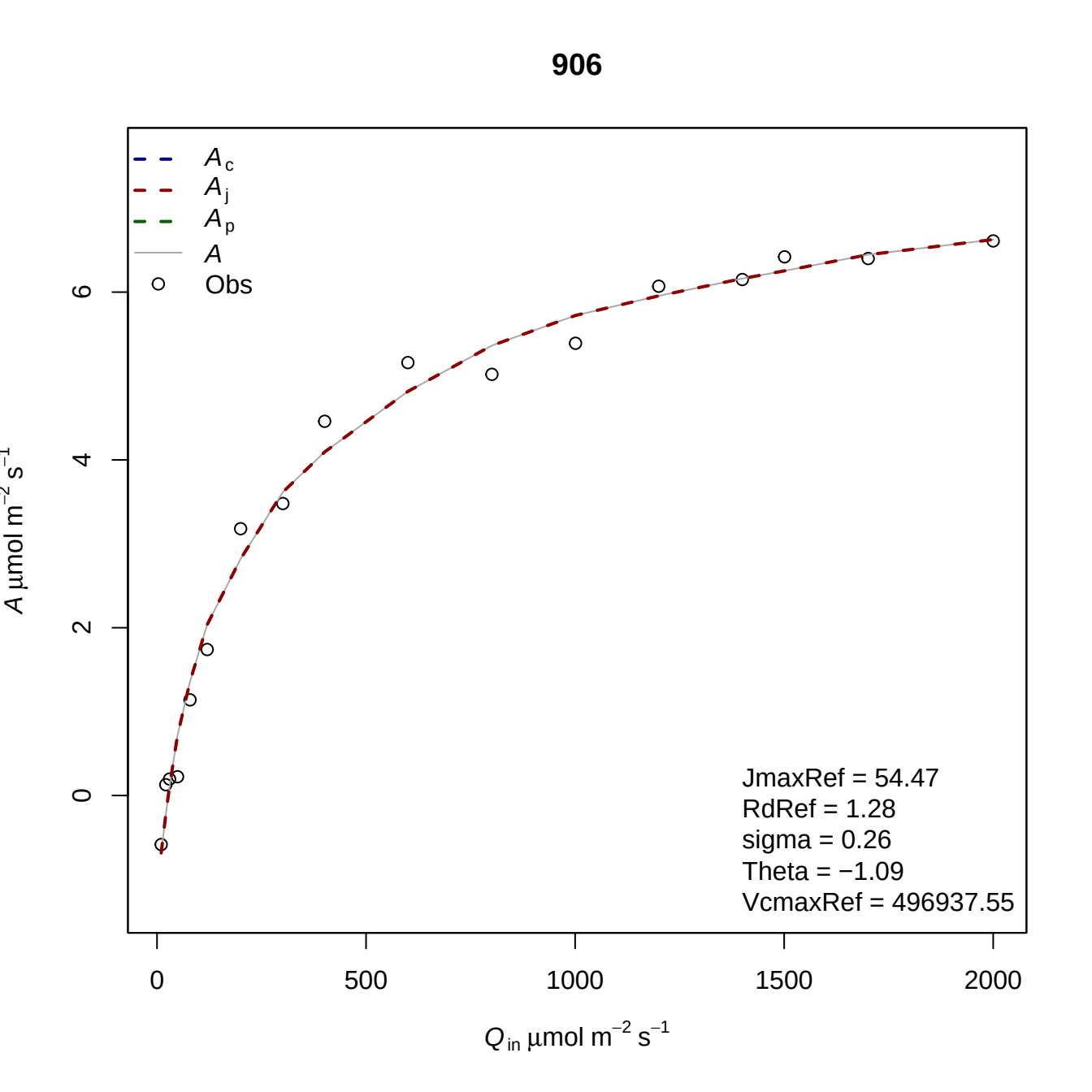

901
A_c
A_j
A_p
A
Obs
JmaxRef = 119.21
RdRef = 2.09
sigma = 0.15
Theta = −0.81
VcmaxRef = 846918.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

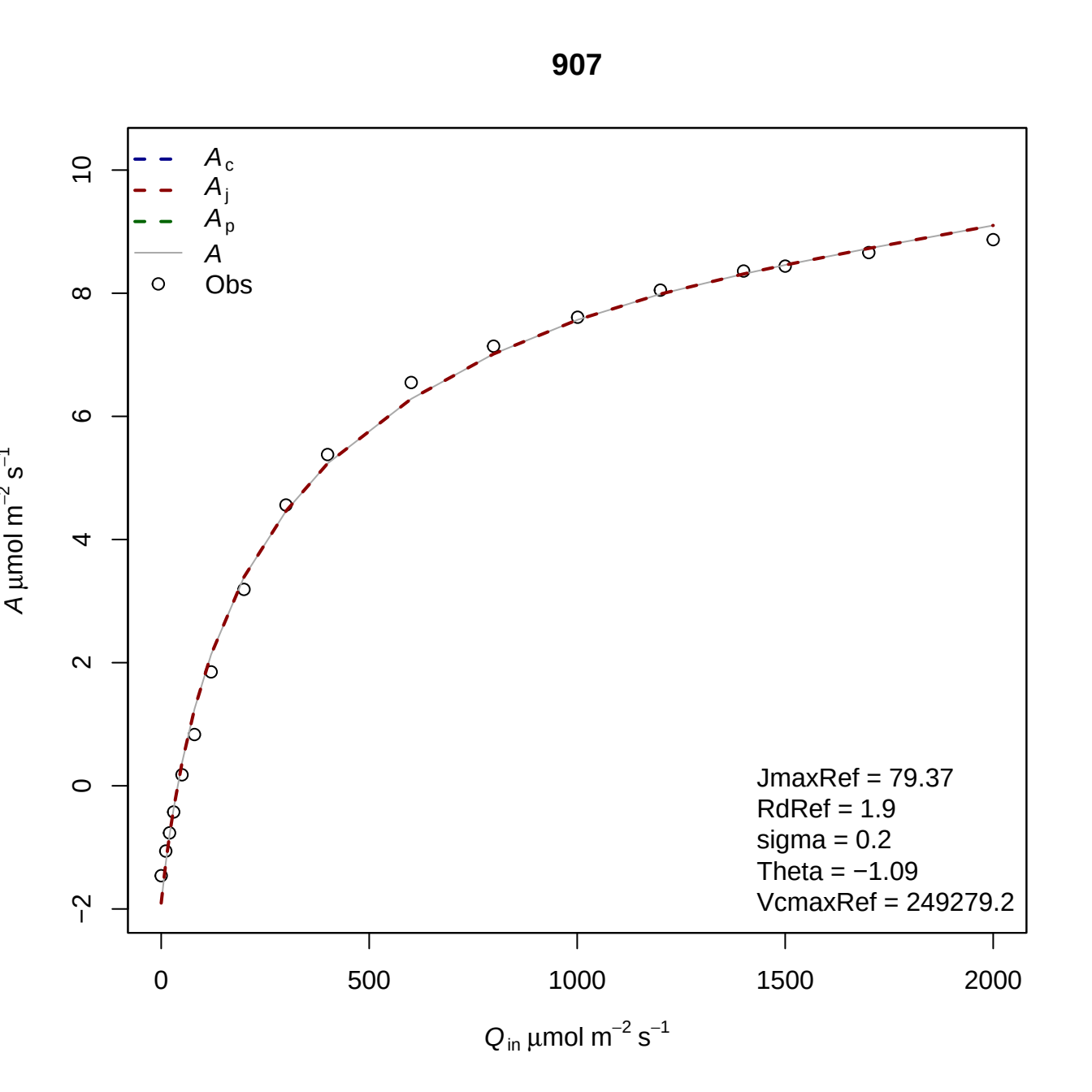

902
A_c
A_j
A_p
A
Obs
JmaxRef = 64.8
RdRef = 1.52
sigma = 0.26
Theta = −0.43
VcmaxRef = 268749.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

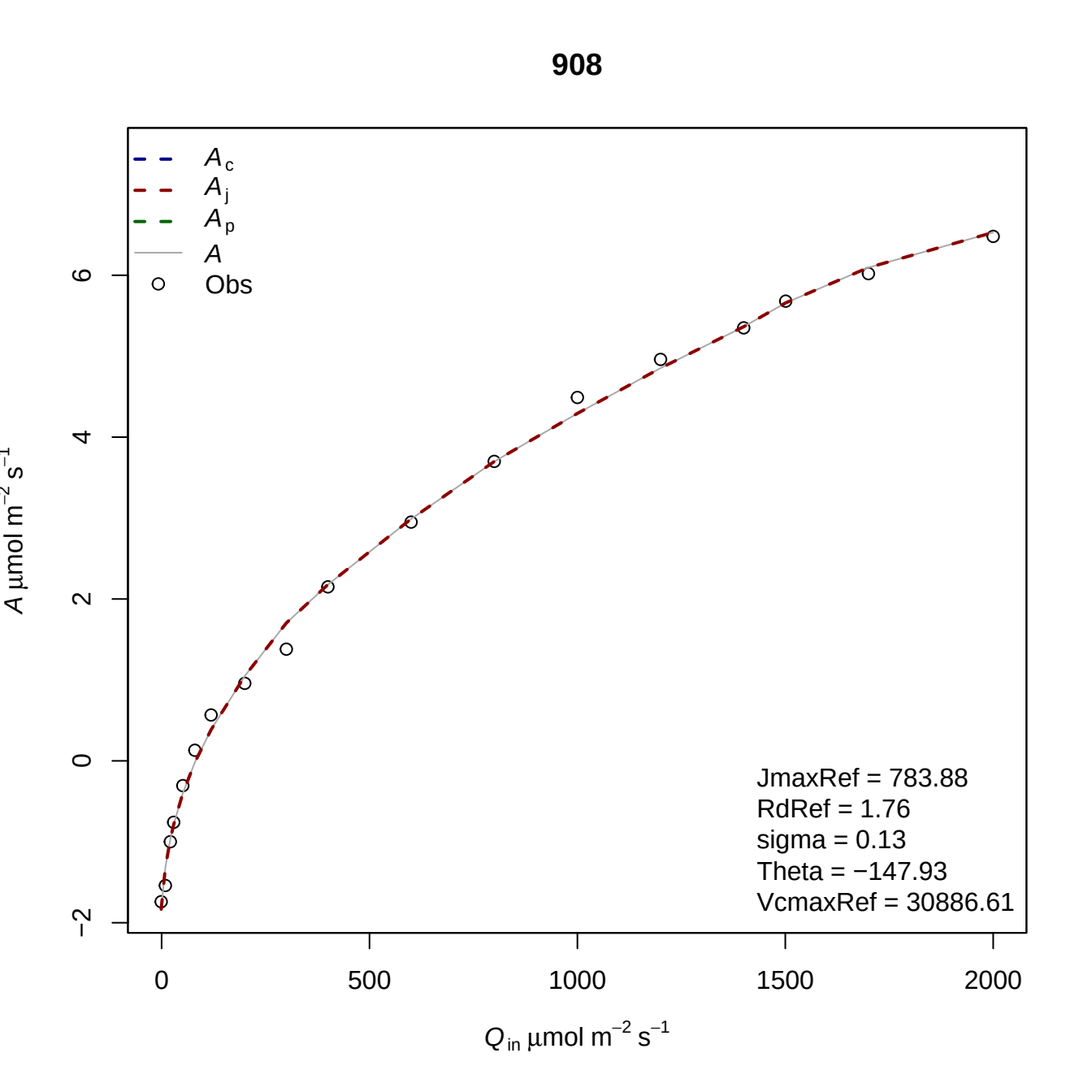

903
A_c
A_j
A_p
A
Obs
JmaxRef = 87.23
RdRef = 2.5
sigma = 0.14
Theta = −0.1
VcmaxRef = 99927.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

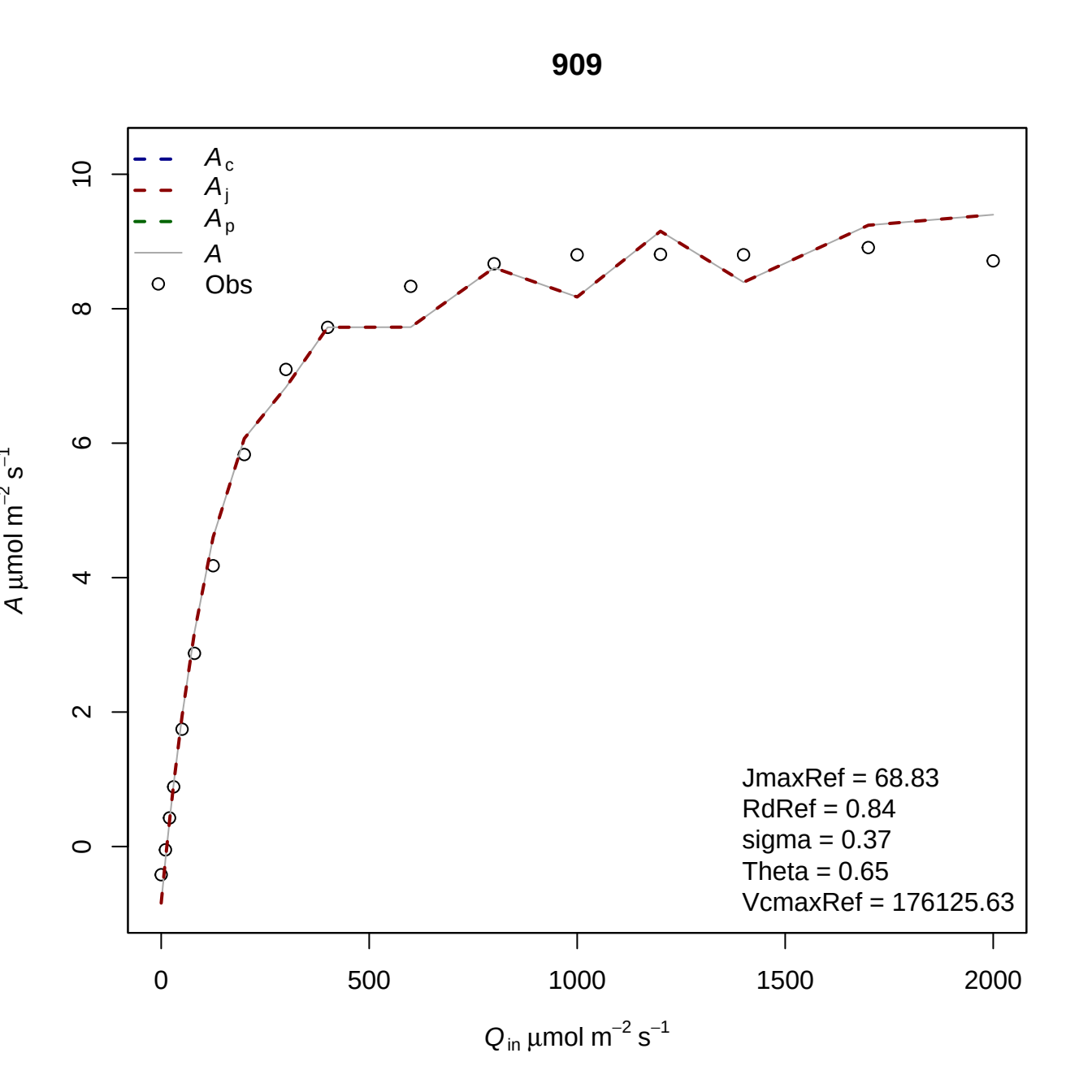

904
A_c
A_j
A_p
A
Obs
JmaxRef = 88.46
RdRef = 2.07
sigma = 0.28
Theta = 0.69
VcmaxRef = 32479.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

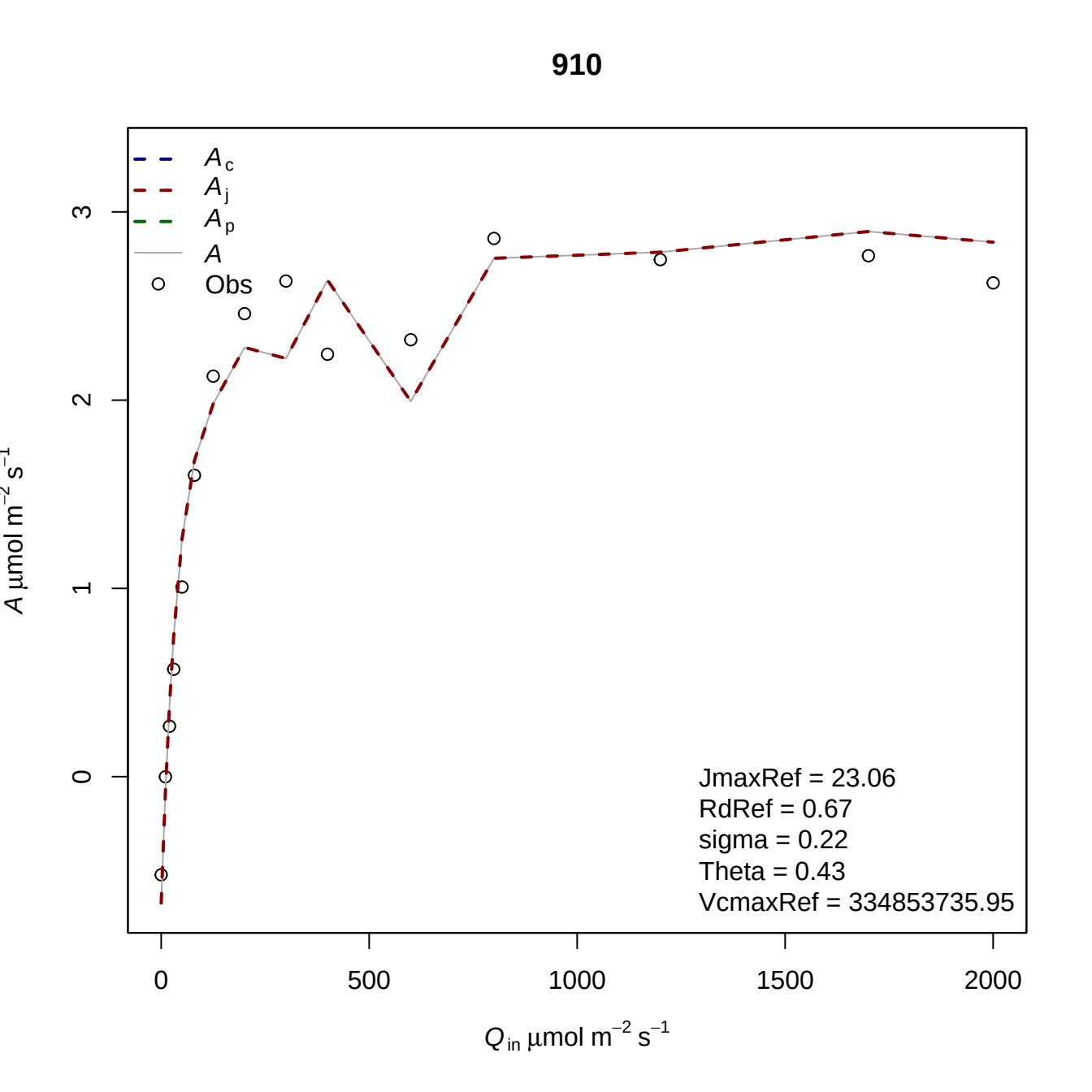

905
A_c
A_j
A_p
A
Obs
JmaxRef = 81
RdRef = 2.25
sigma = 0.11
Theta = 0.45
VcmaxRef = 68.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

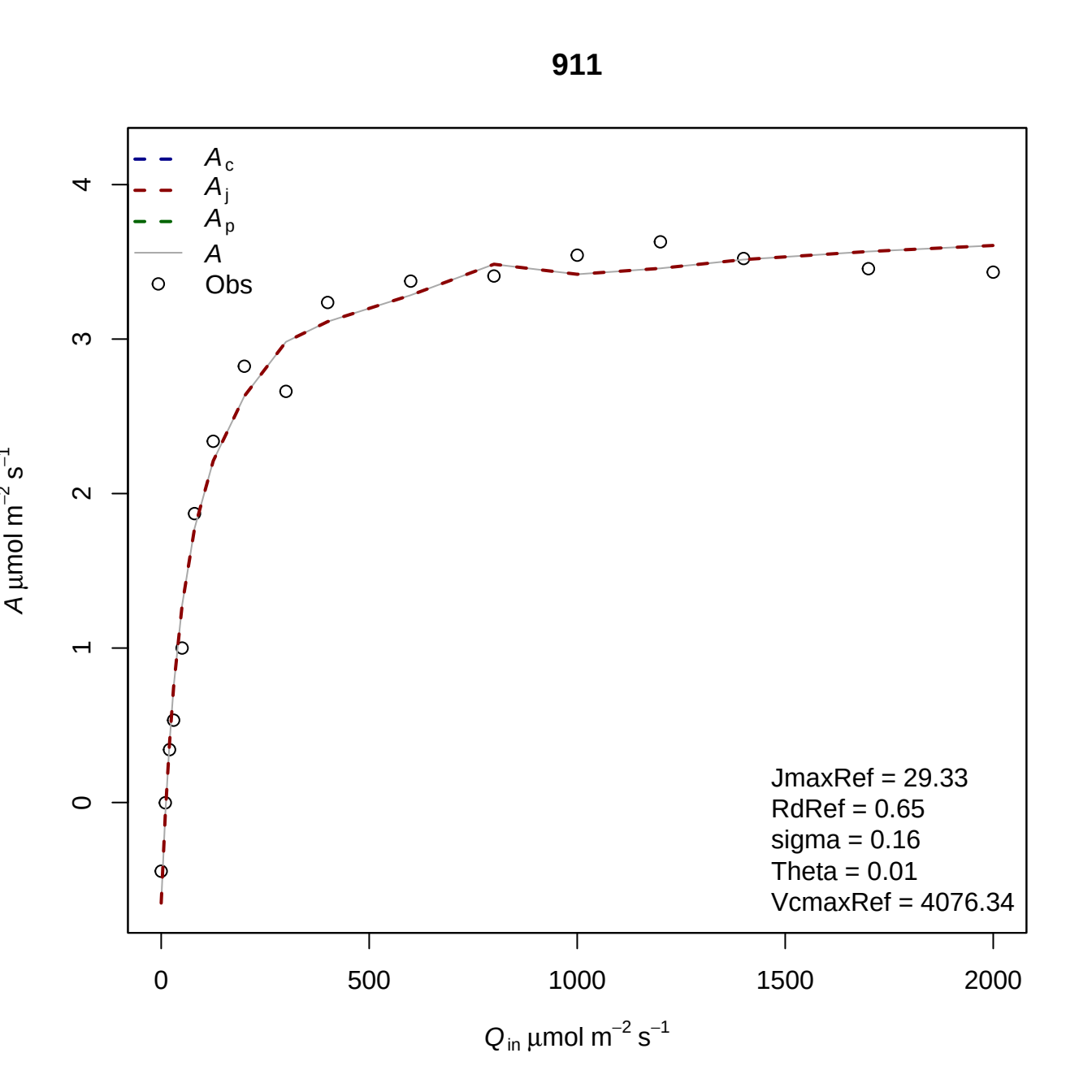

906
A_c
A_j
A_p
A
Obs
JmaxRef = 54.47
RdRef = 1.28
sigma = 0.26
Theta = −1.09
VcmaxRef = 496937.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

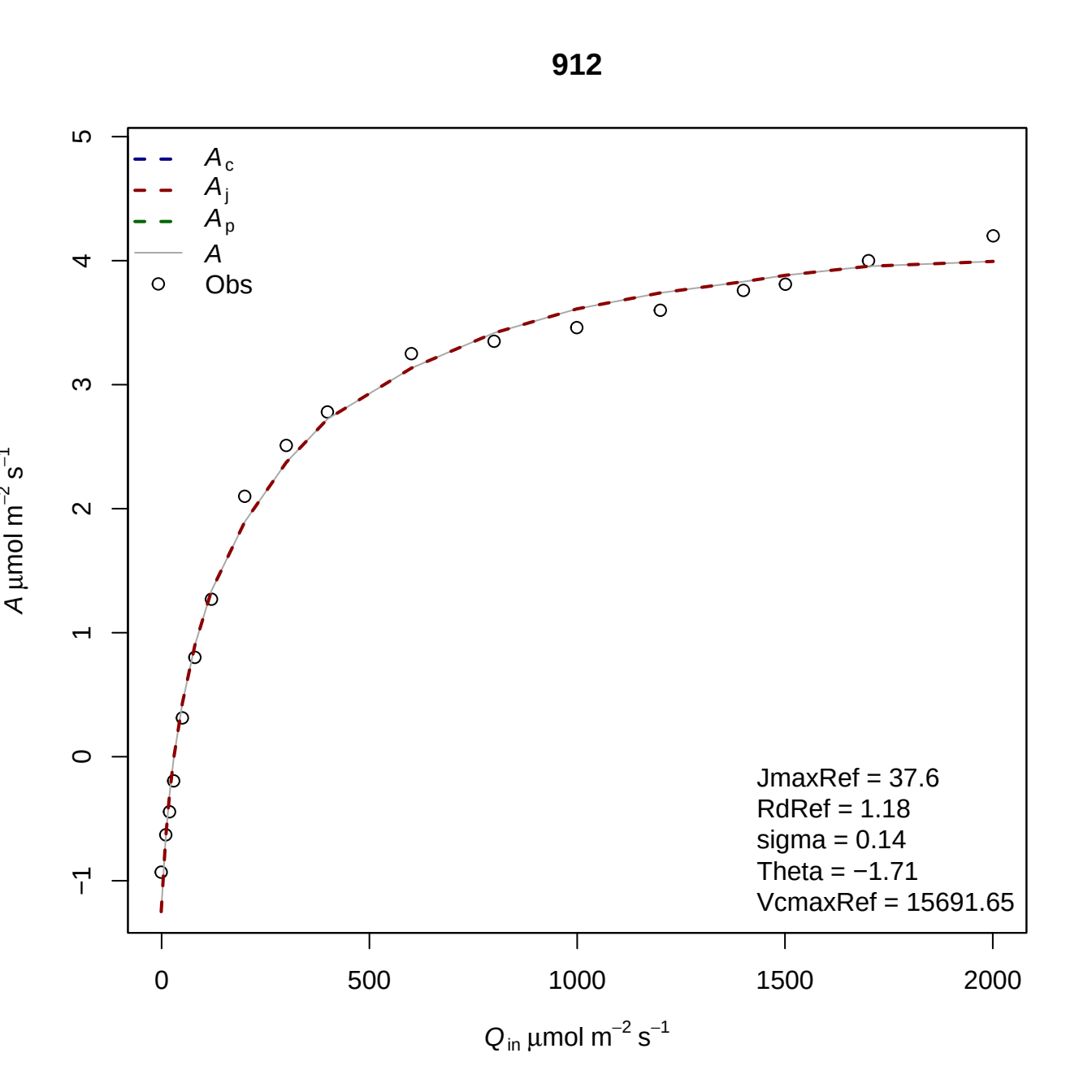

907
A_c
A_j
A_p
A
Obs
JmaxRef = 79.37
RdRef = 1.9
sigma = 0.2
Theta = −1.09
VcmaxRef = 249279.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

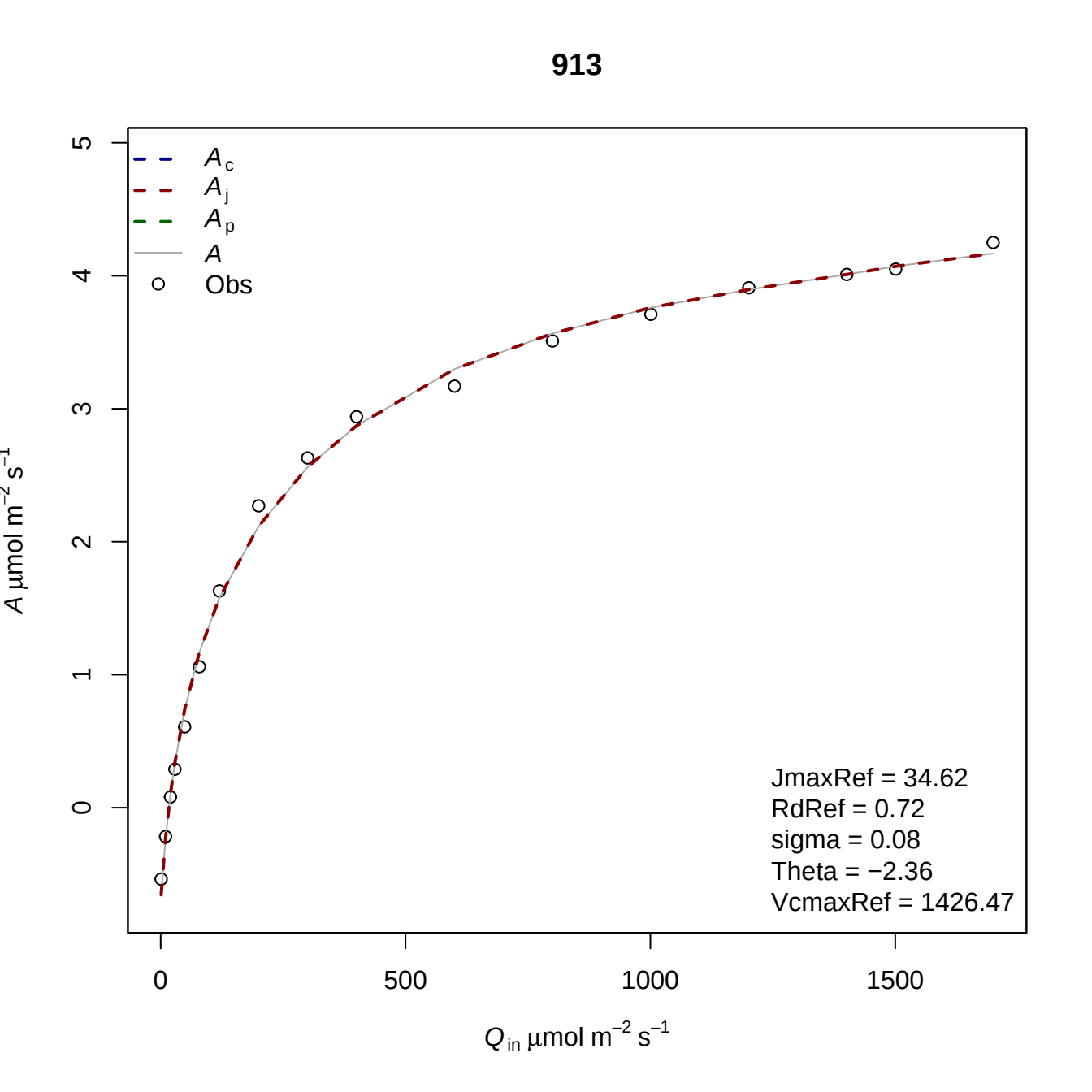

908
A_c
A_j
A_p
A
Obs
JmaxRef = 783.88
RdRef = 1.76
sigma = 0.13
Theta = −147.93
VcmaxRef = 30886.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

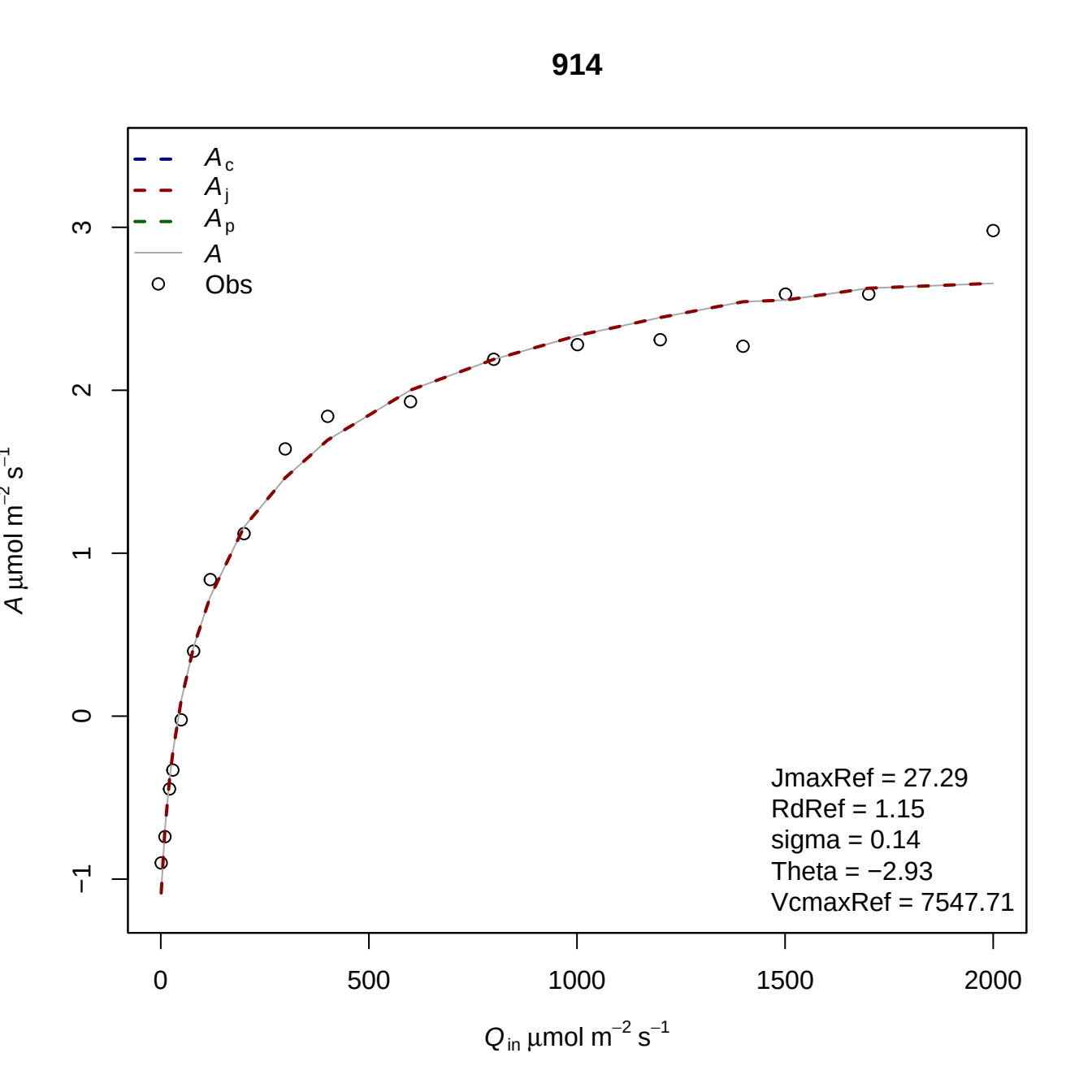

909
A_c
A_j
A_p
A
Obs
JmaxRef = 68.83
RdRef = 0.84
sigma = 0.37
Theta = 0.65
VcmaxRef = 176125.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

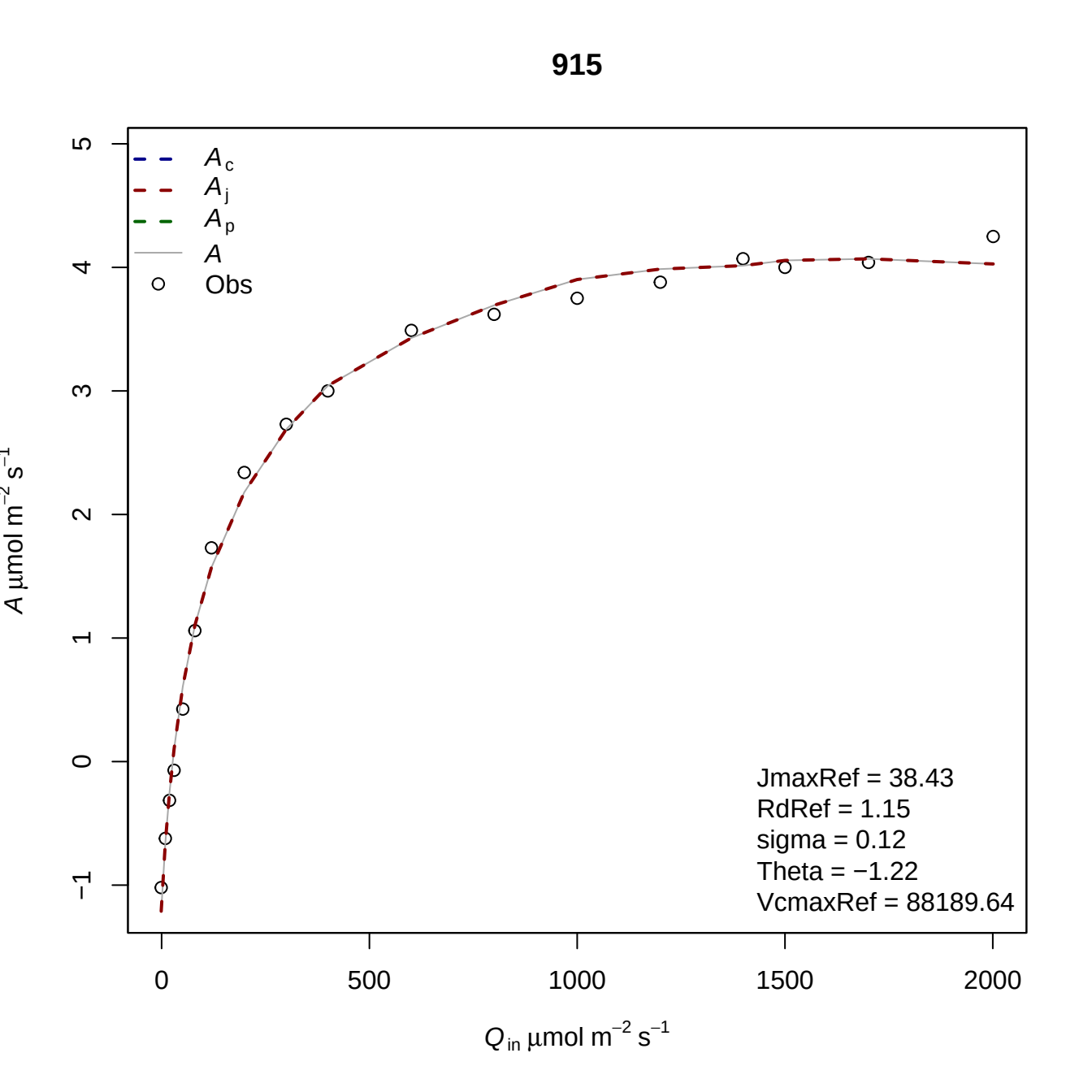

910
A_c
A_j
A_p
A
Obs
JmaxRef = 23.06
RdRef = 0.67
sigma = 0.22
Theta = 0.43
VcmaxRef = 334853735.95
0
1
2
3
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

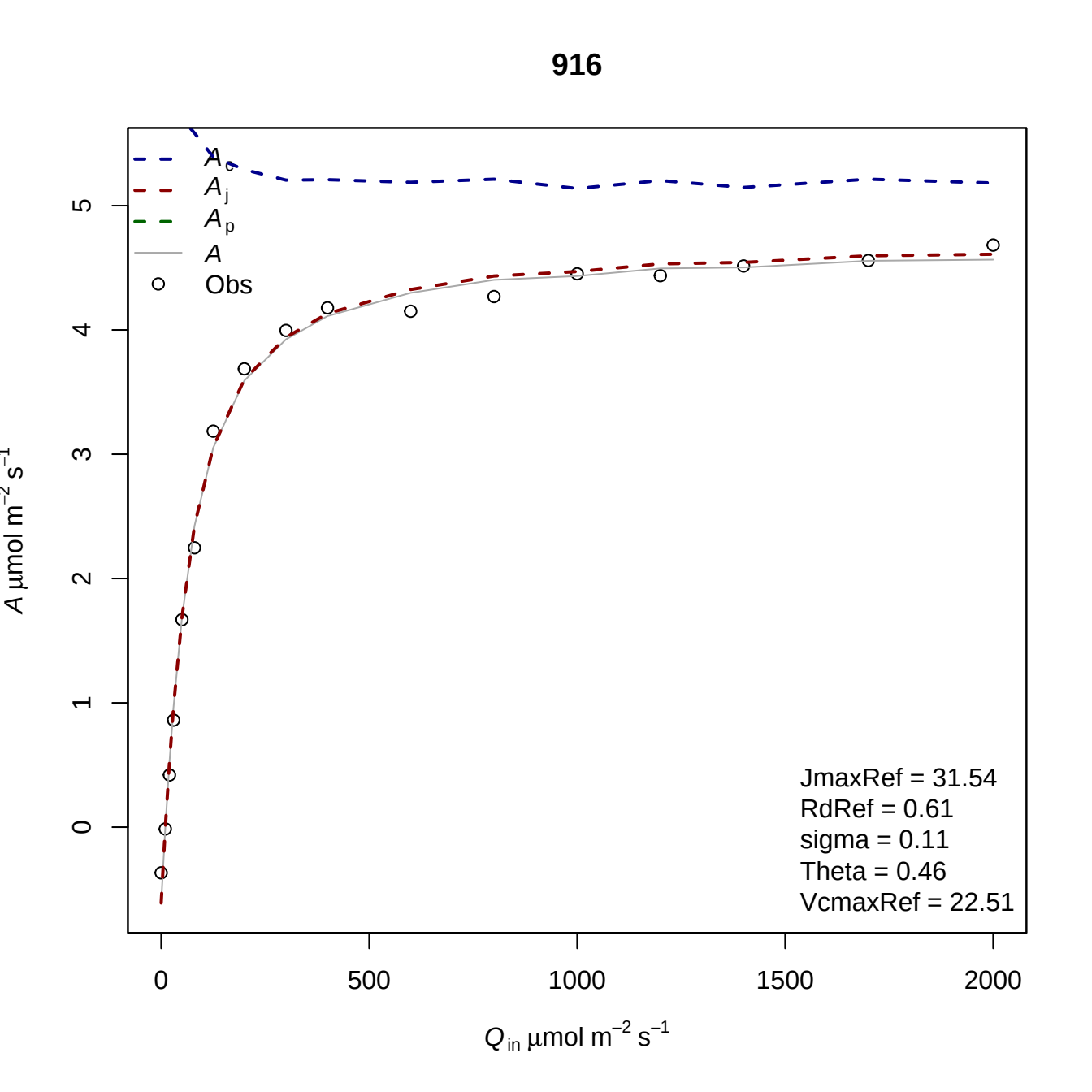

911
A_c
A_j
A_p
A
Obs
JmaxRef = 29.33
RdRef = 0.65
sigma = 0.16
Theta = 0.01
VcmaxRef = 4076.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

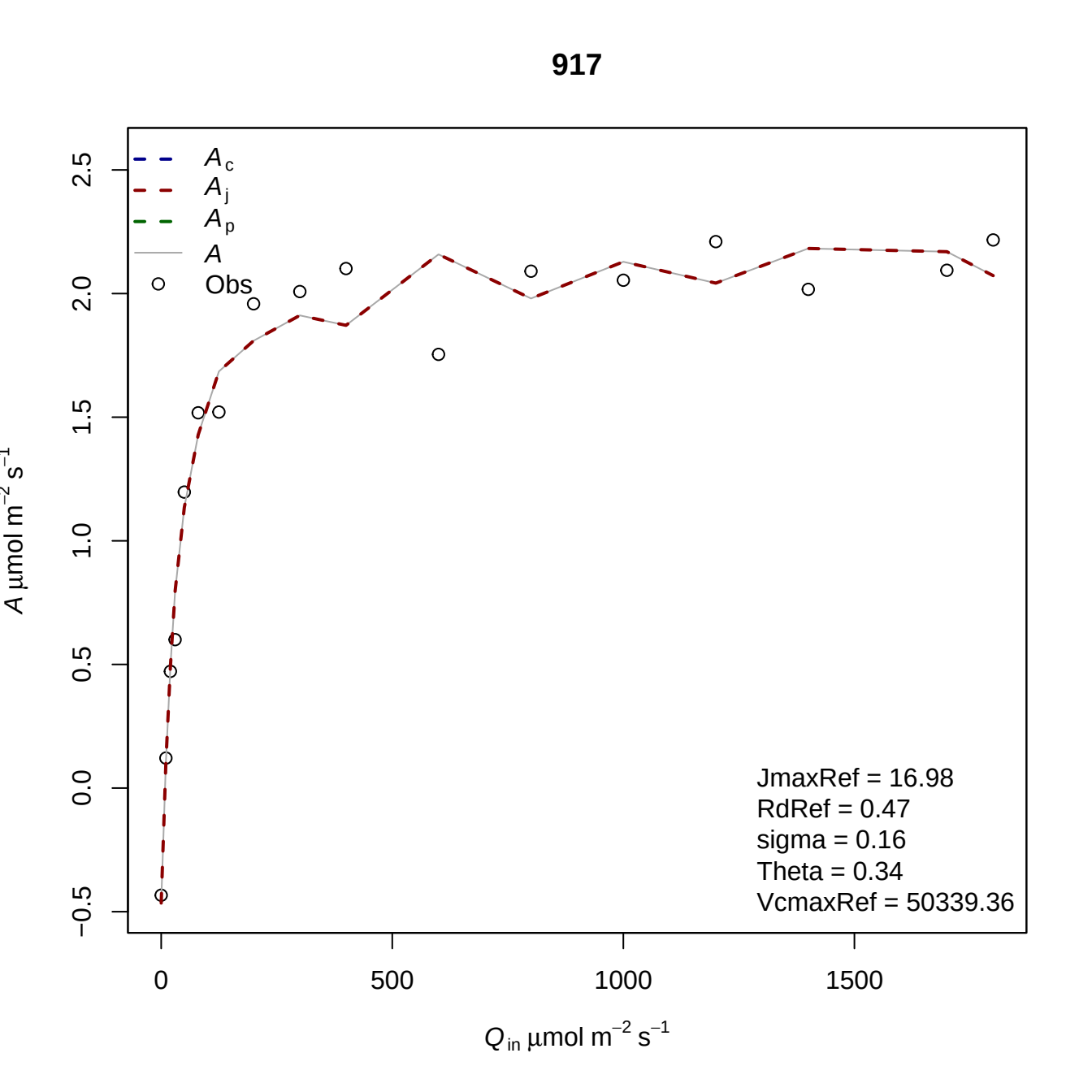

912
A_c
A_j
A_p
A
Obs
JmaxRef = 37.6
RdRef = 1.18
sigma = 0.14
Theta = −1.71
VcmaxRef = 15691.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

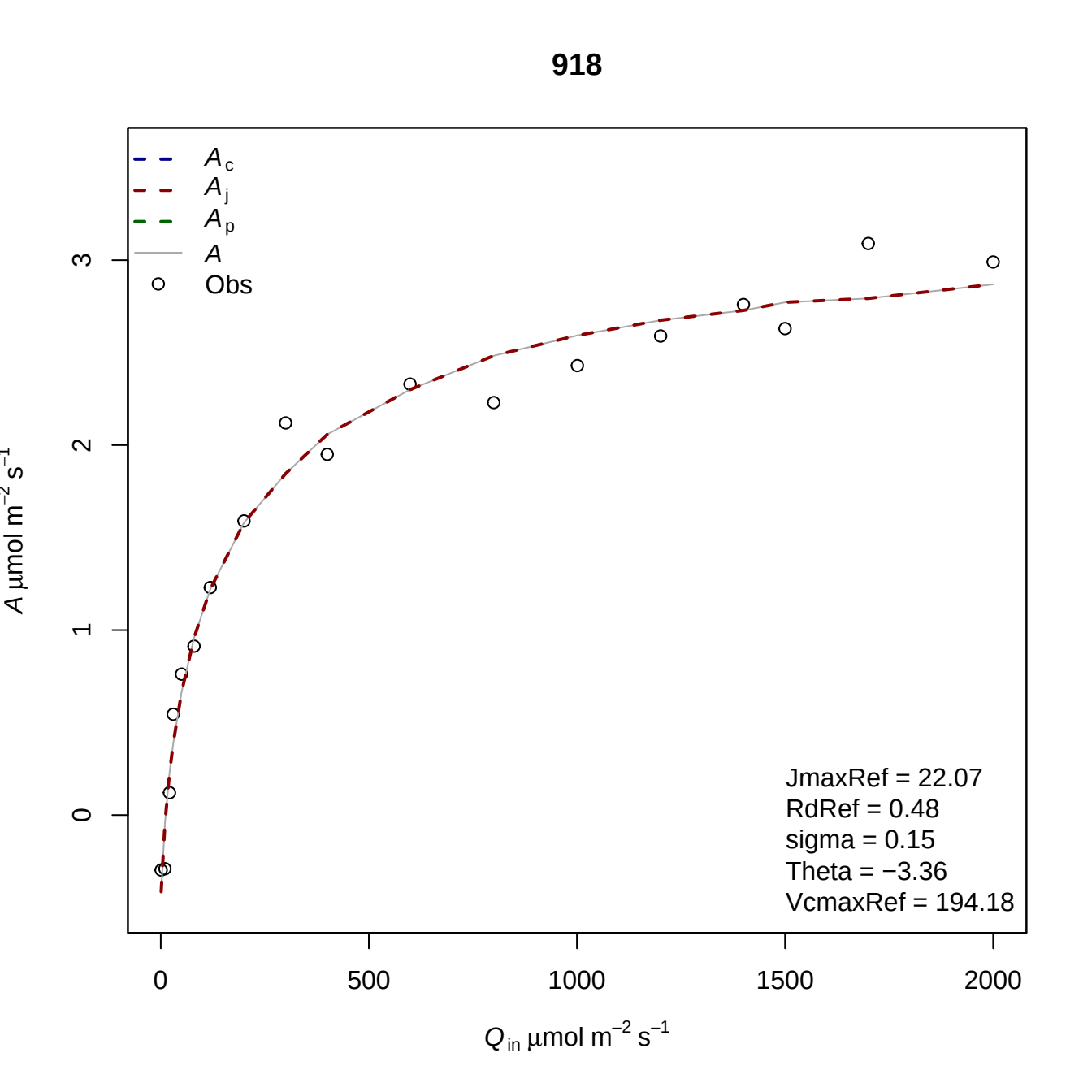

913
A_c
A_j
A_p
A
Obs
JmaxRef = 34.62
RdRef = 0.72
sigma = 0.08
Theta = −2.36
VcmaxRef = 1426.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

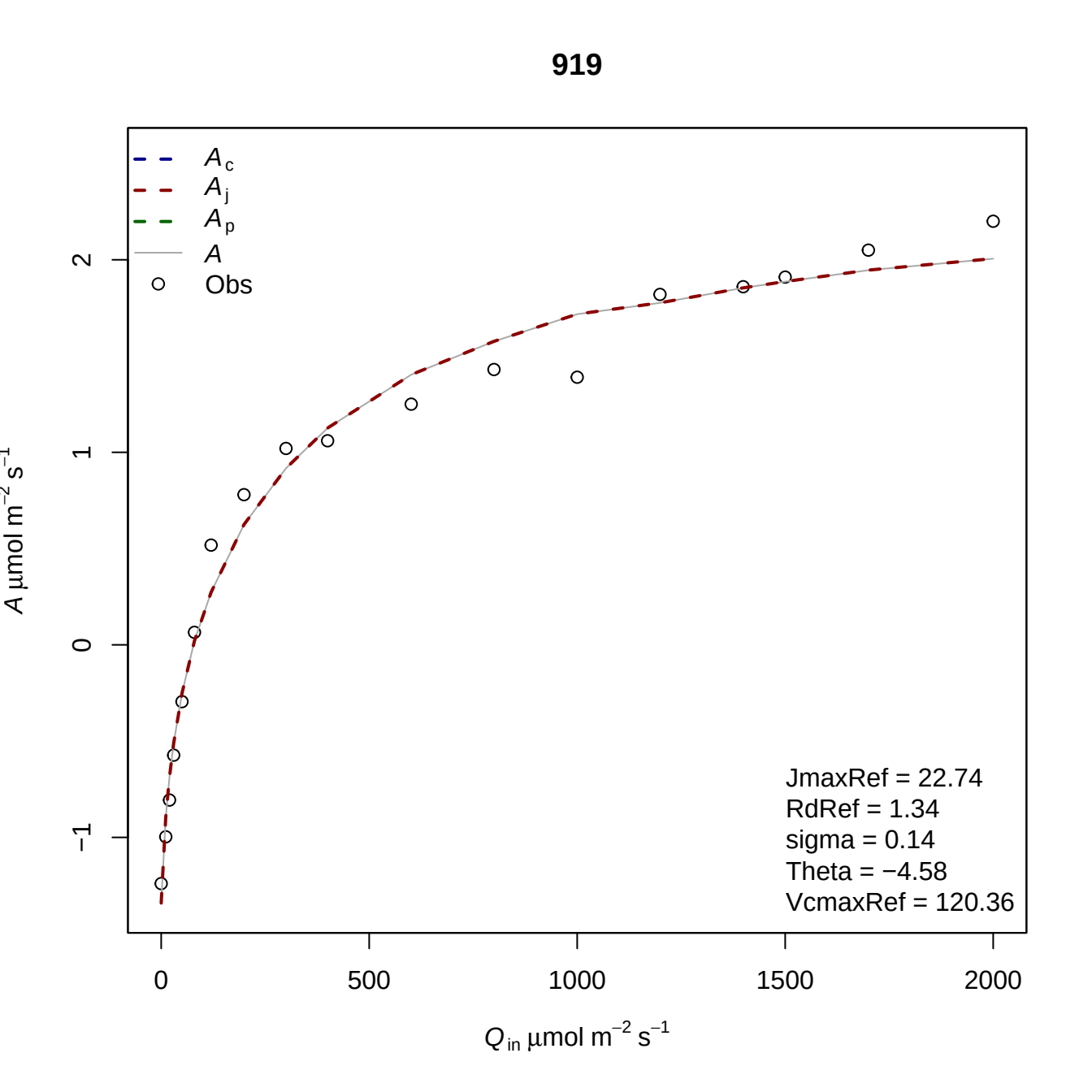

914
A_c
A_j
A_p
A
Obs
JmaxRef = 27.29
RdRef = 1.15
sigma = 0.14
Theta = −2.93
VcmaxRef = 7547.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

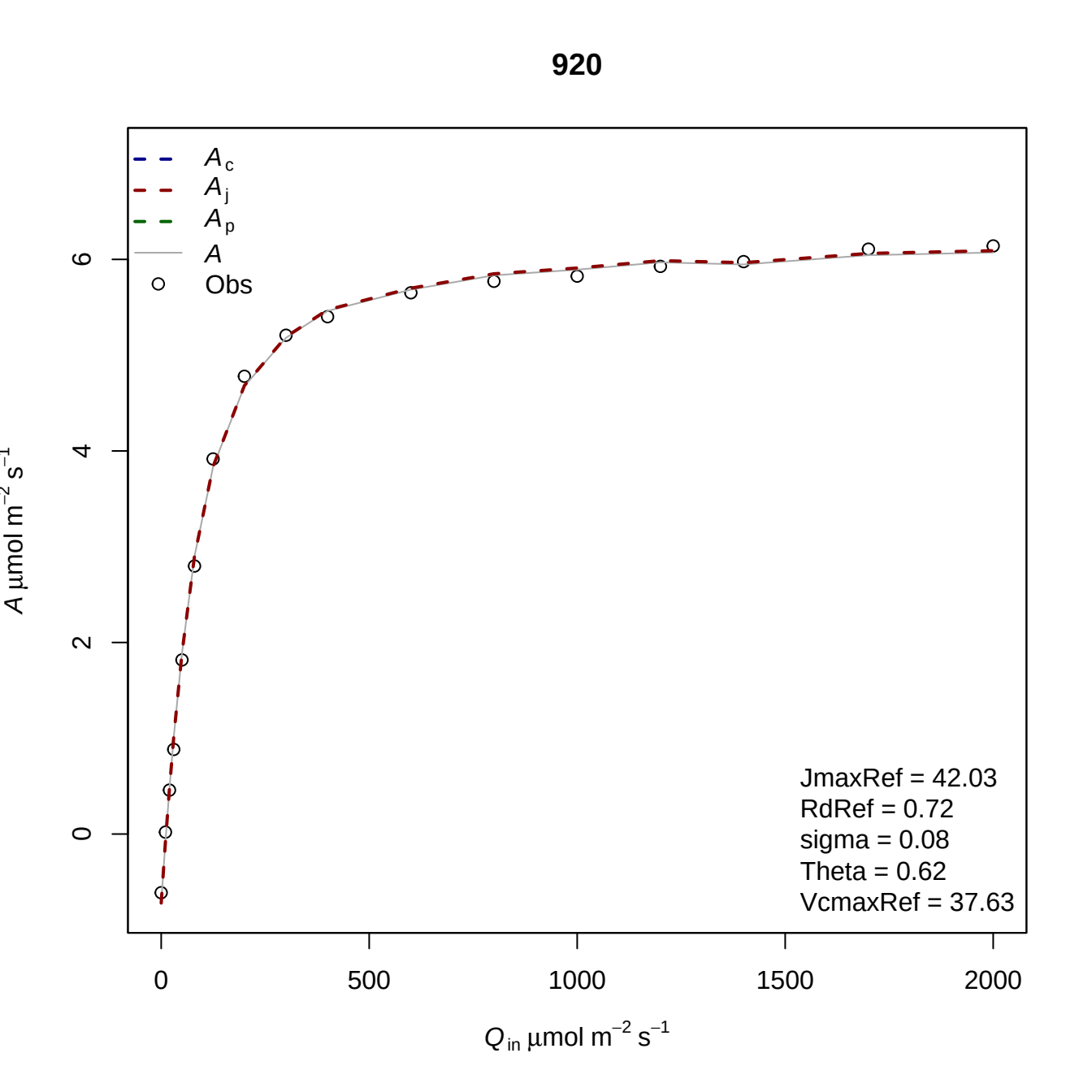

915
A_c
A_j
A_p
A
Obs
JmaxRef = 38.43
RdRef = 1.15
sigma = 0.12
Theta = −1.22
VcmaxRef = 88189.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

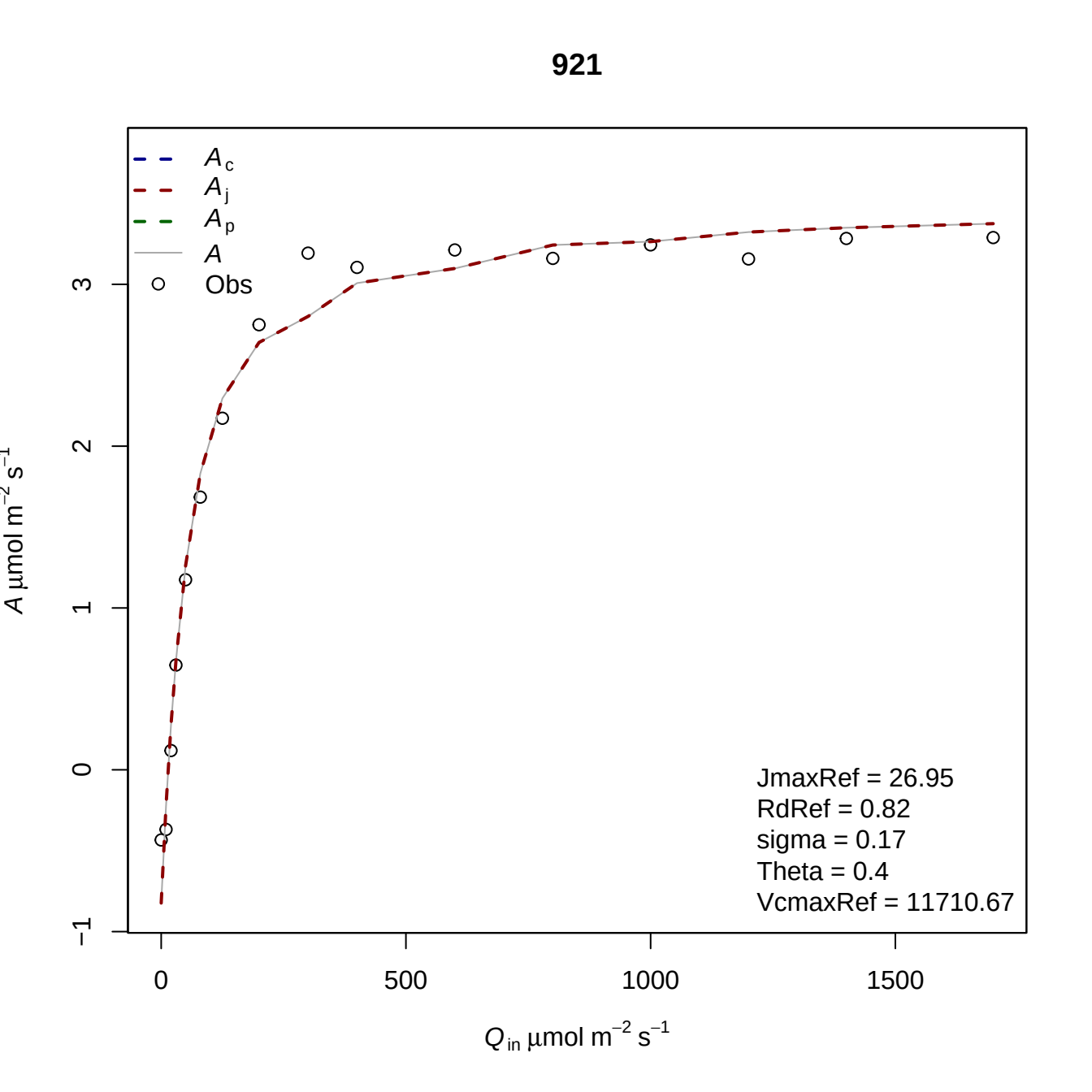

916
A_c
A_j
A_p
A
Obs
JmaxRef = 31.54
RdRef = 0.61
sigma = 0.11
Theta = 0.46
VcmaxRef = 22.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

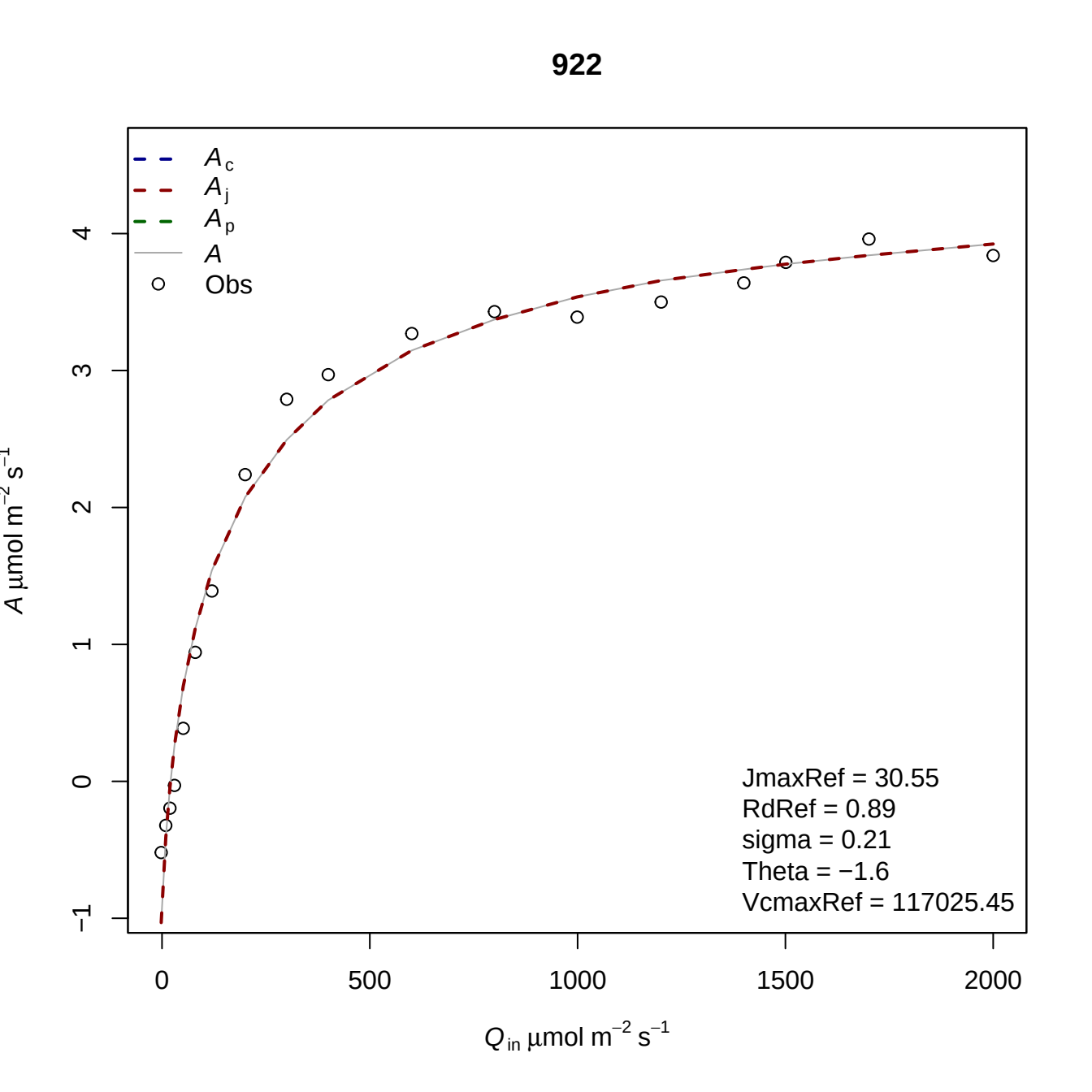

917
A_c
A_j
A_p
A
Obs
JmaxRef = 16.98
RdRef = 0.47
sigma = 0.16
Theta = 0.34
VcmaxRef = 50339.36
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

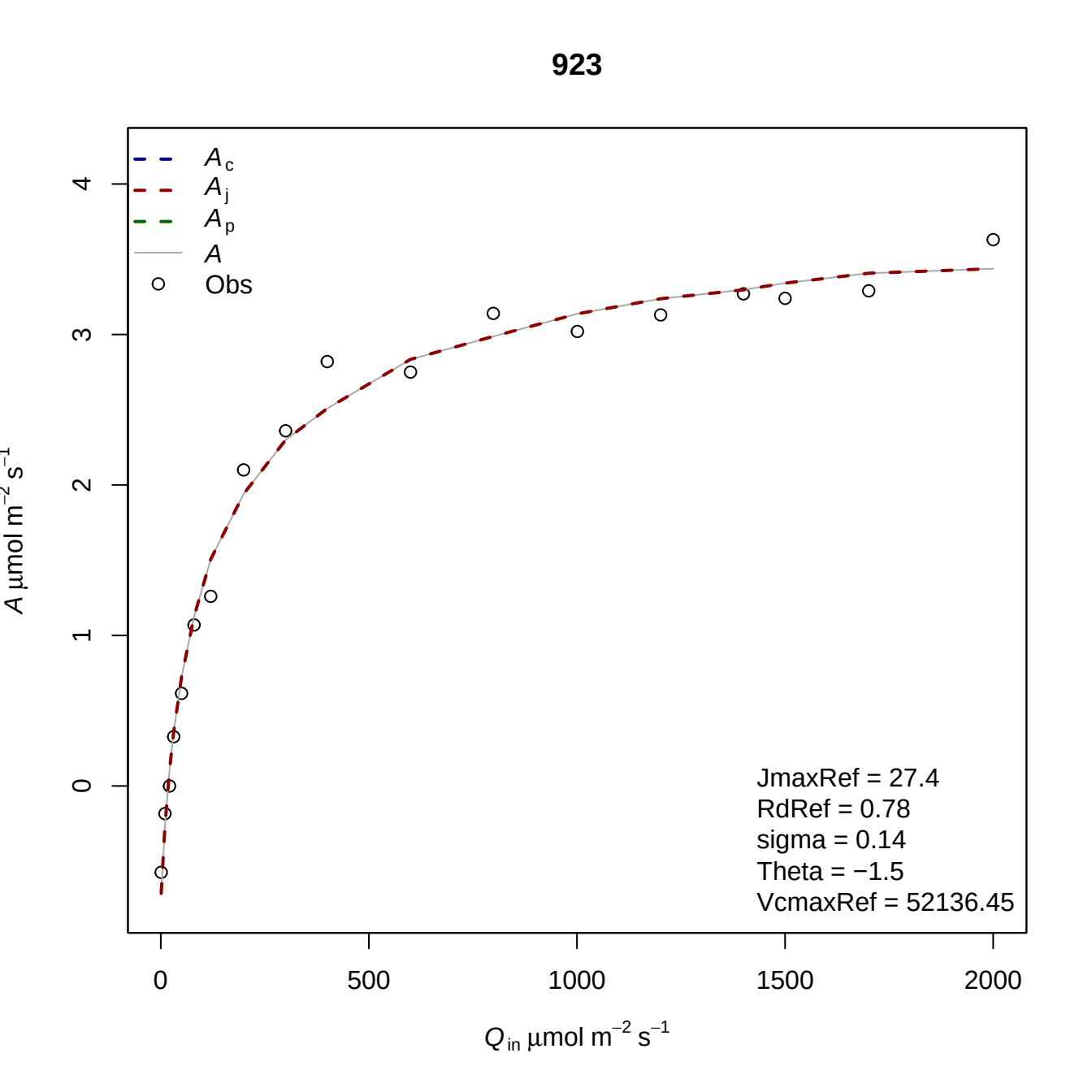

918
A_c
A_j
A_p
A
Obs
JmaxRef = 22.07
RdRef = 0.48
sigma = 0.15
Theta = −3.36
VcmaxRef = 194.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

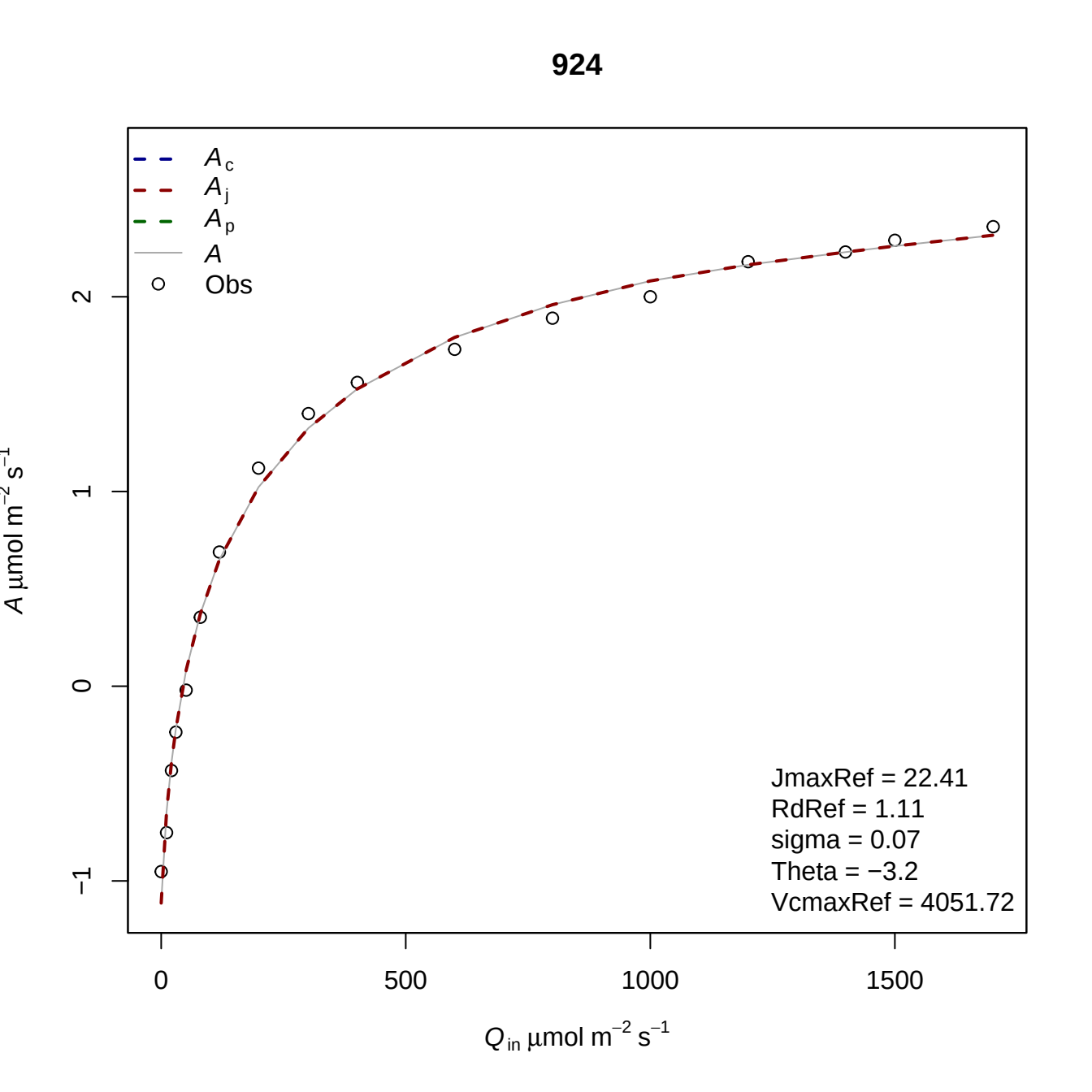

919
A_c
A_j
A_p
A
Obs
JmaxRef = 22.74
RdRef = 1.34
sigma = 0.14
Theta = −4.58
VcmaxRef = 120.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

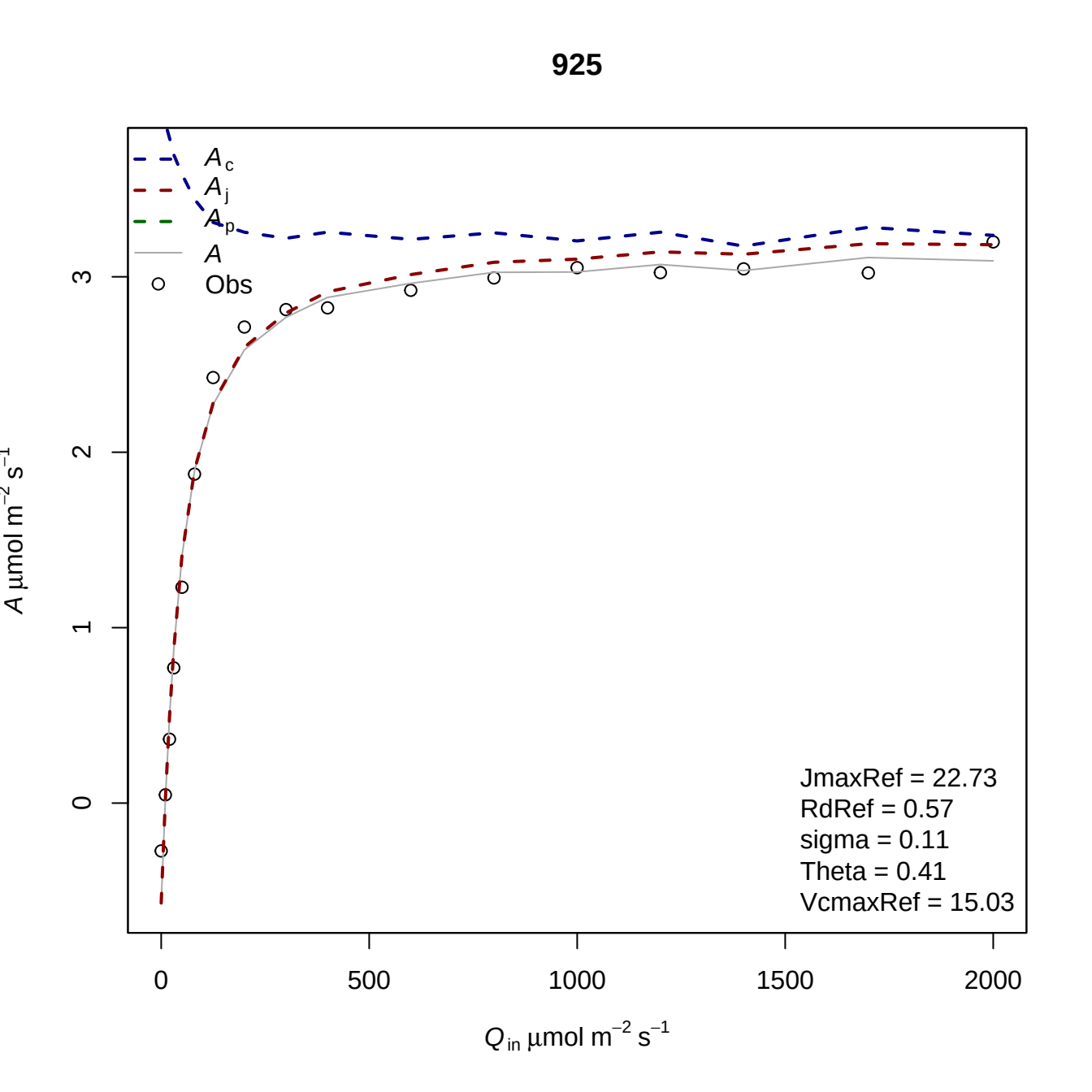

920
A_c
A_j
A_p
A
Obs
JmaxRef = 42.03
RdRef = 0.72
sigma = 0.08
Theta = 0.62
VcmaxRef = 37.63
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

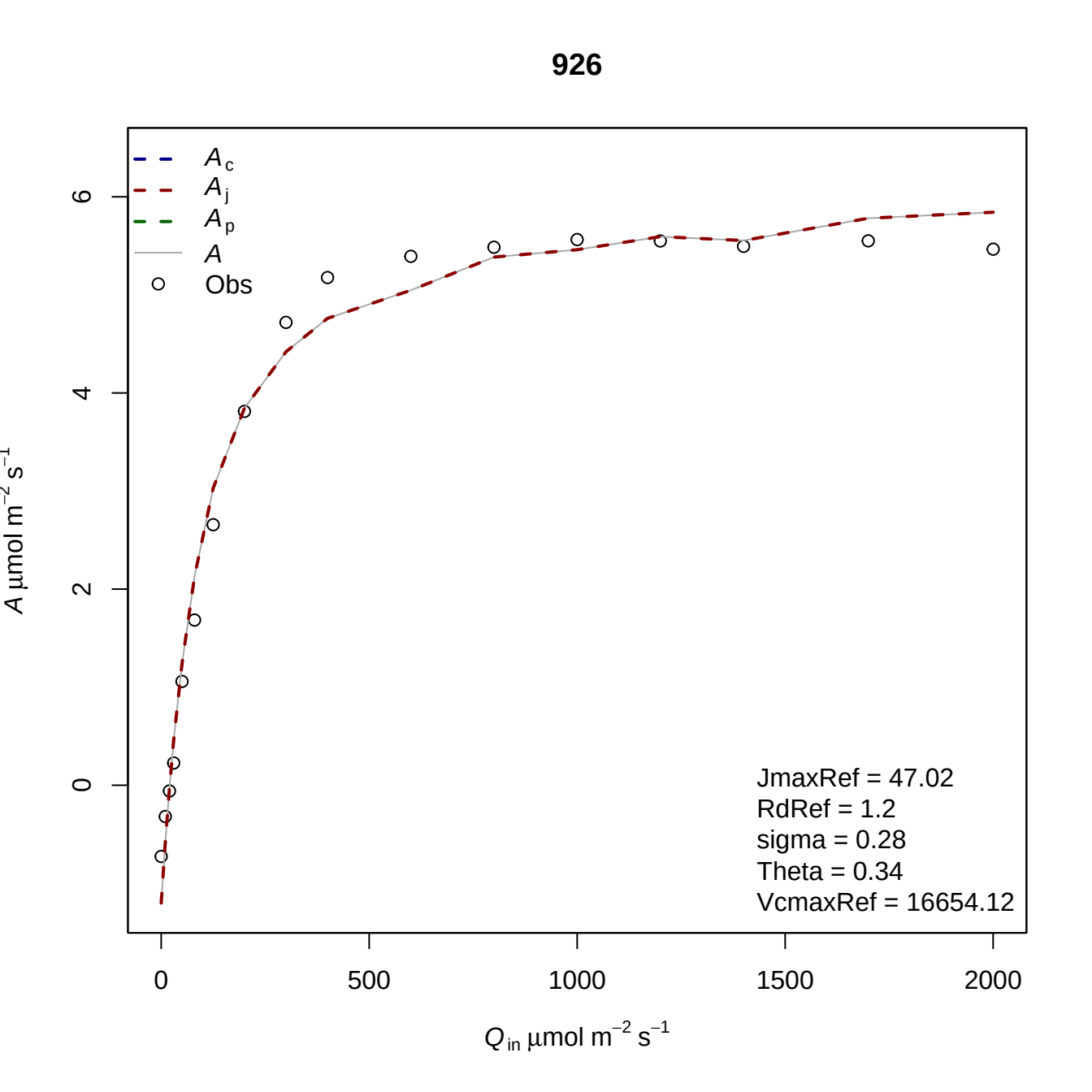

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.95
RdRef = 0.82
sigma = 0.17
Theta = 0.4
VcmaxRef = 11710.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

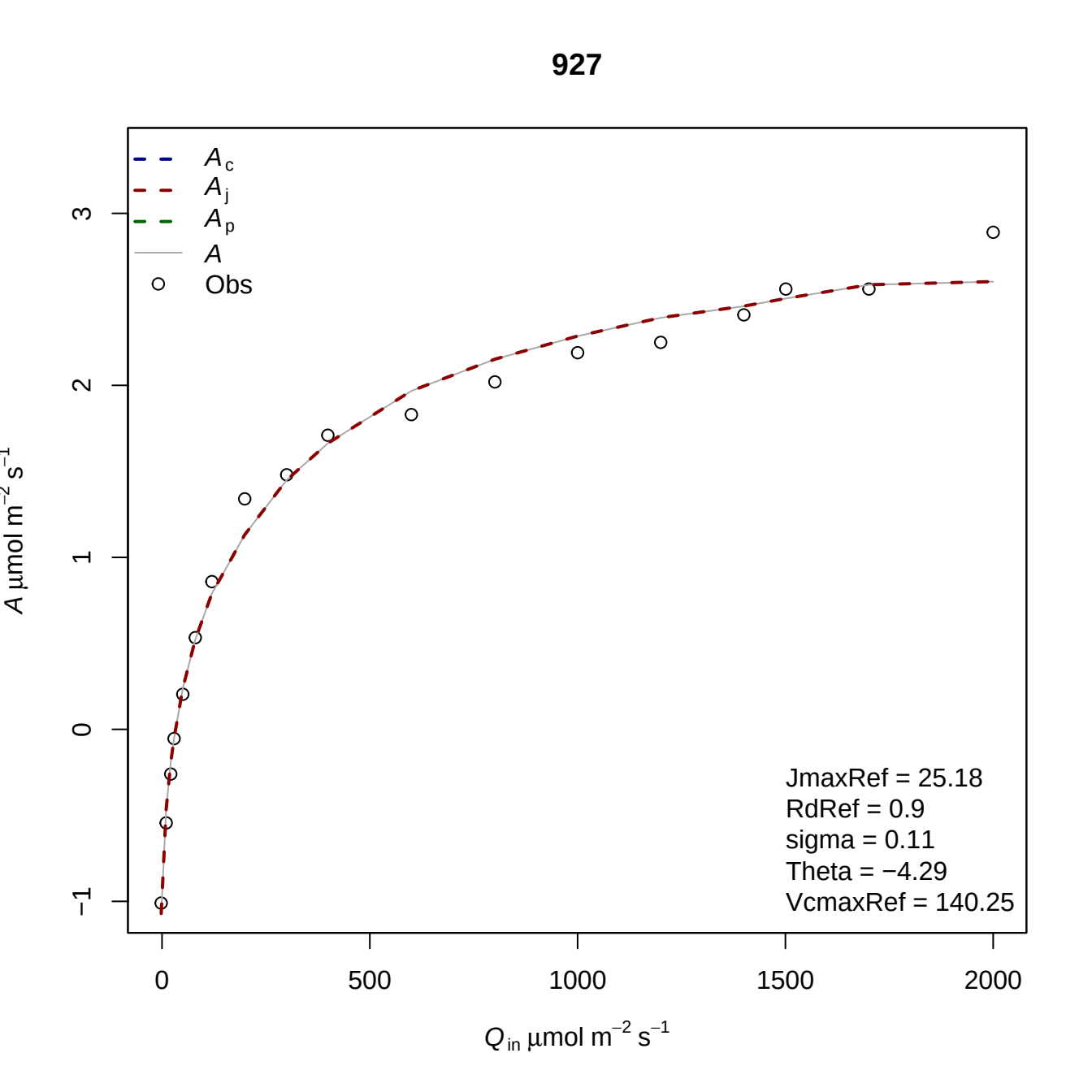

922
A_c
A_j
A_p
A
Obs
JmaxRef = 30.55
RdRef = 0.89
sigma = 0.21
Theta = −1.6
VcmaxRef = 117025.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

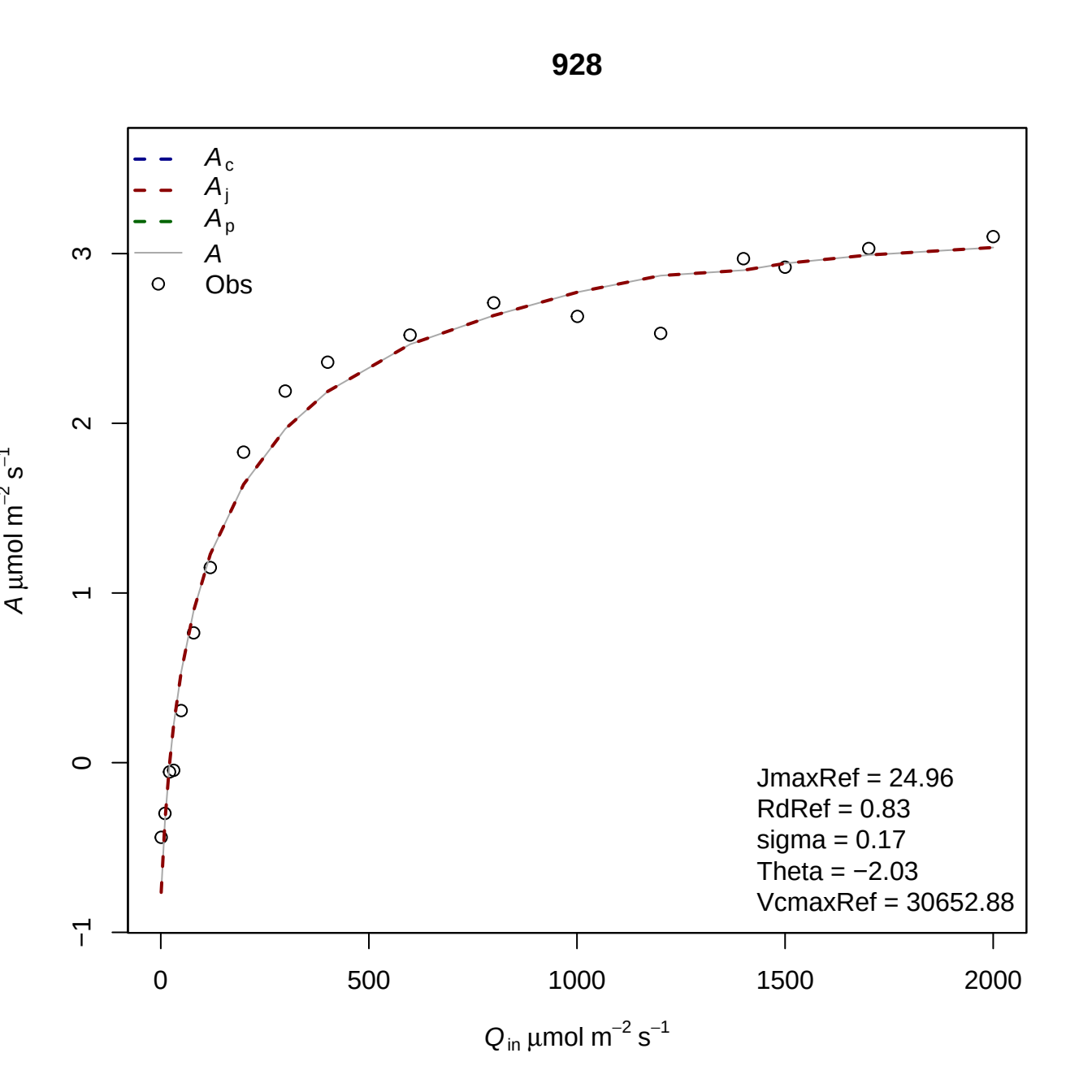

923
A_c
A_j
A_p
A
Obs
JmaxRef = 27.4
RdRef = 0.78
sigma = 0.14
Theta = −1.5
VcmaxRef = 52136.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

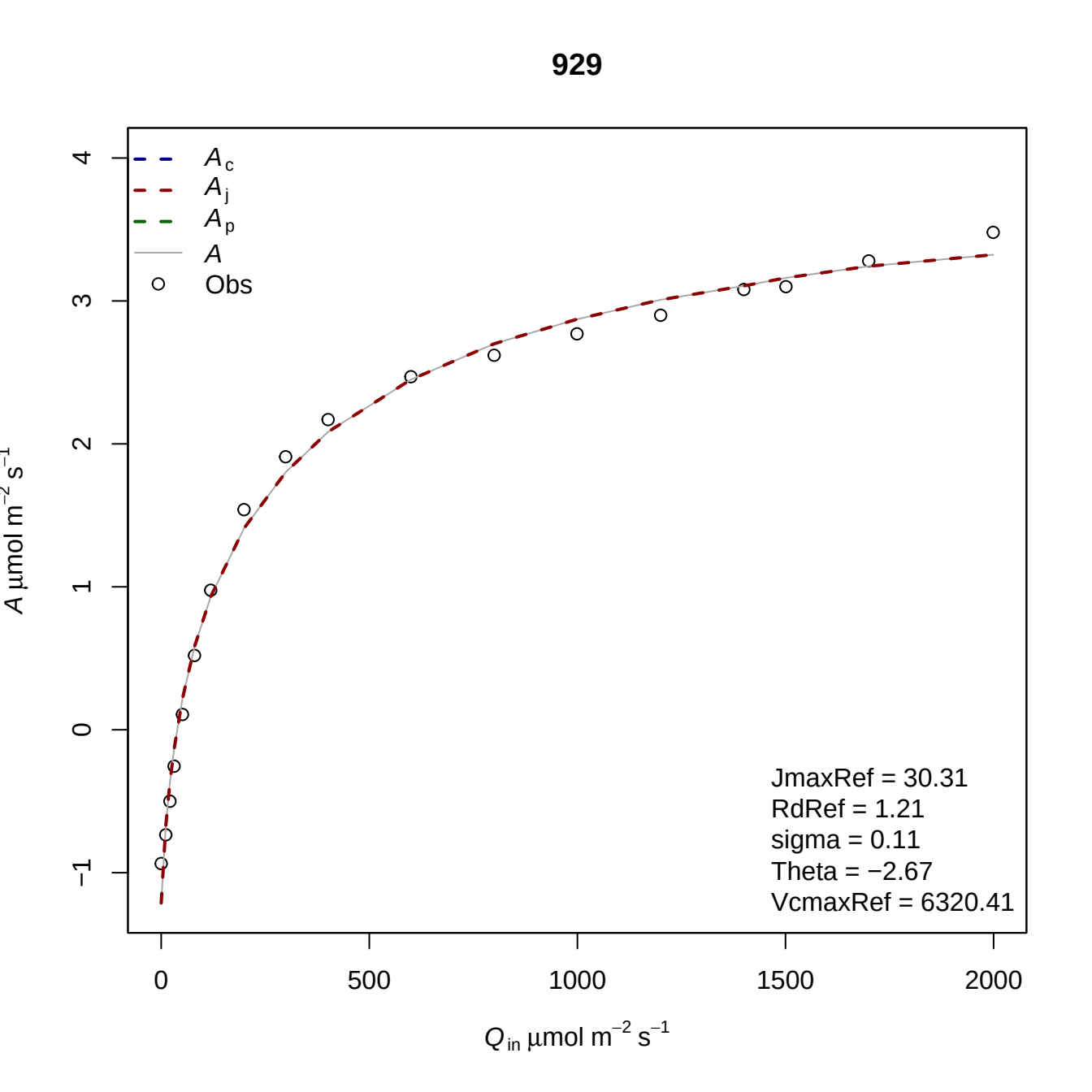

924
A_c
A_j
A_p
A
Obs
JmaxRef = 22.41
RdRef = 1.11
sigma = 0.07
Theta = −3.2
VcmaxRef = 4051.72
A µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

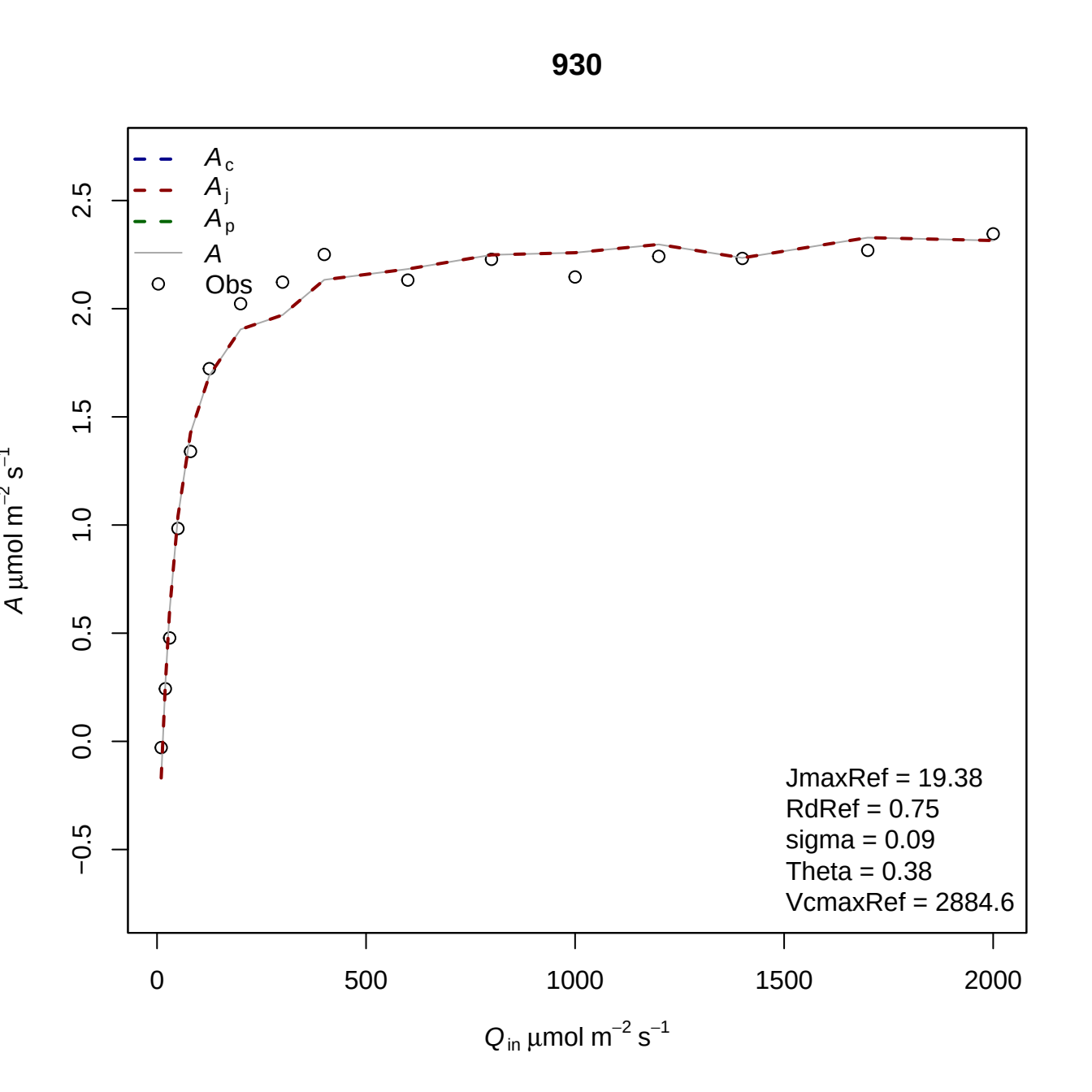

925
A_c
A_j
A_p
A
Obs
JmaxRef = 22.73
RdRef = 0.57
sigma = 0.11
Theta = 0.41
VcmaxRef = 15.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

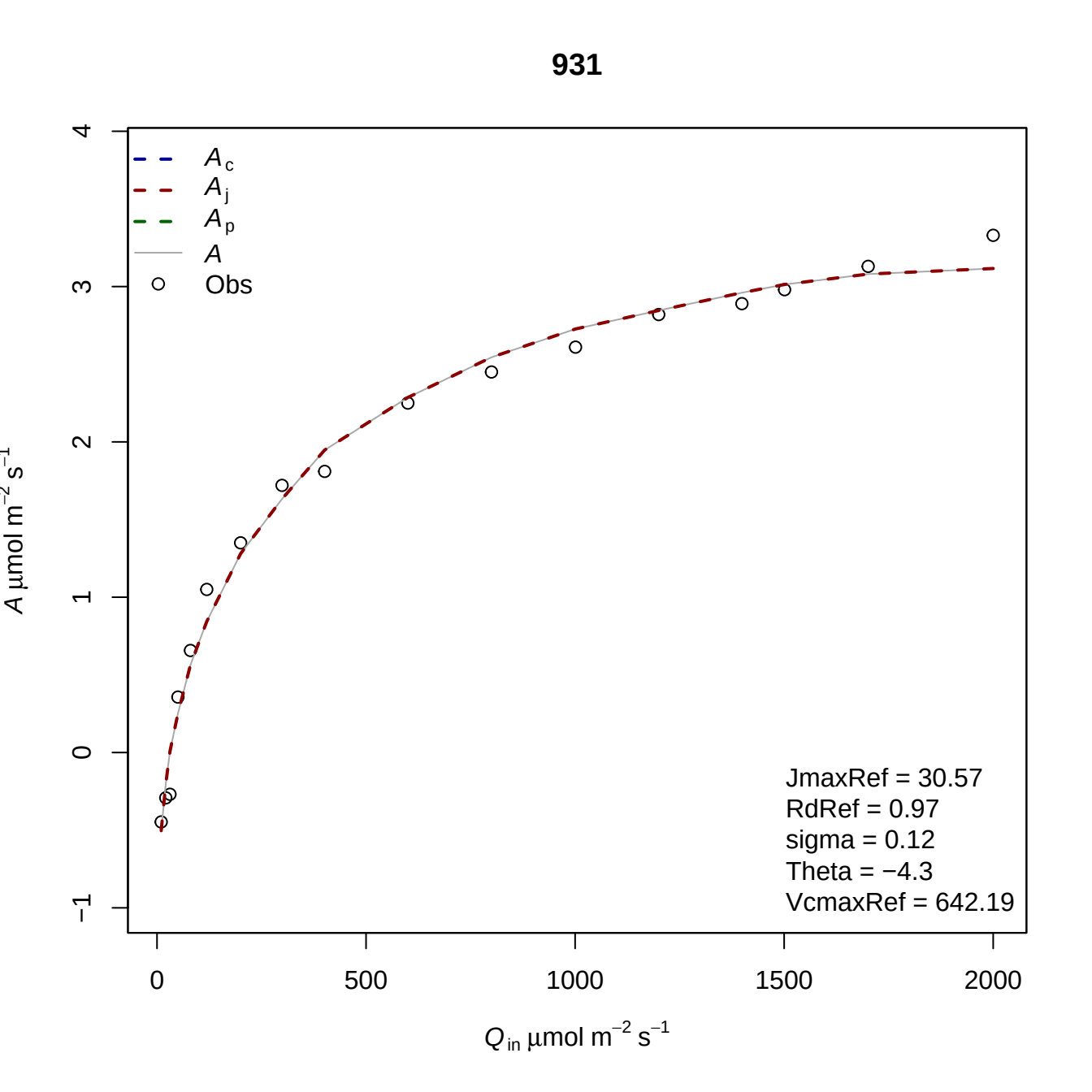

926
A_c
A_j
A_p
A
Obs
JmaxRef = 47.02
RdRef = 1.2
sigma = 0.28
Theta = 0.34
VcmaxRef = 16654.12
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

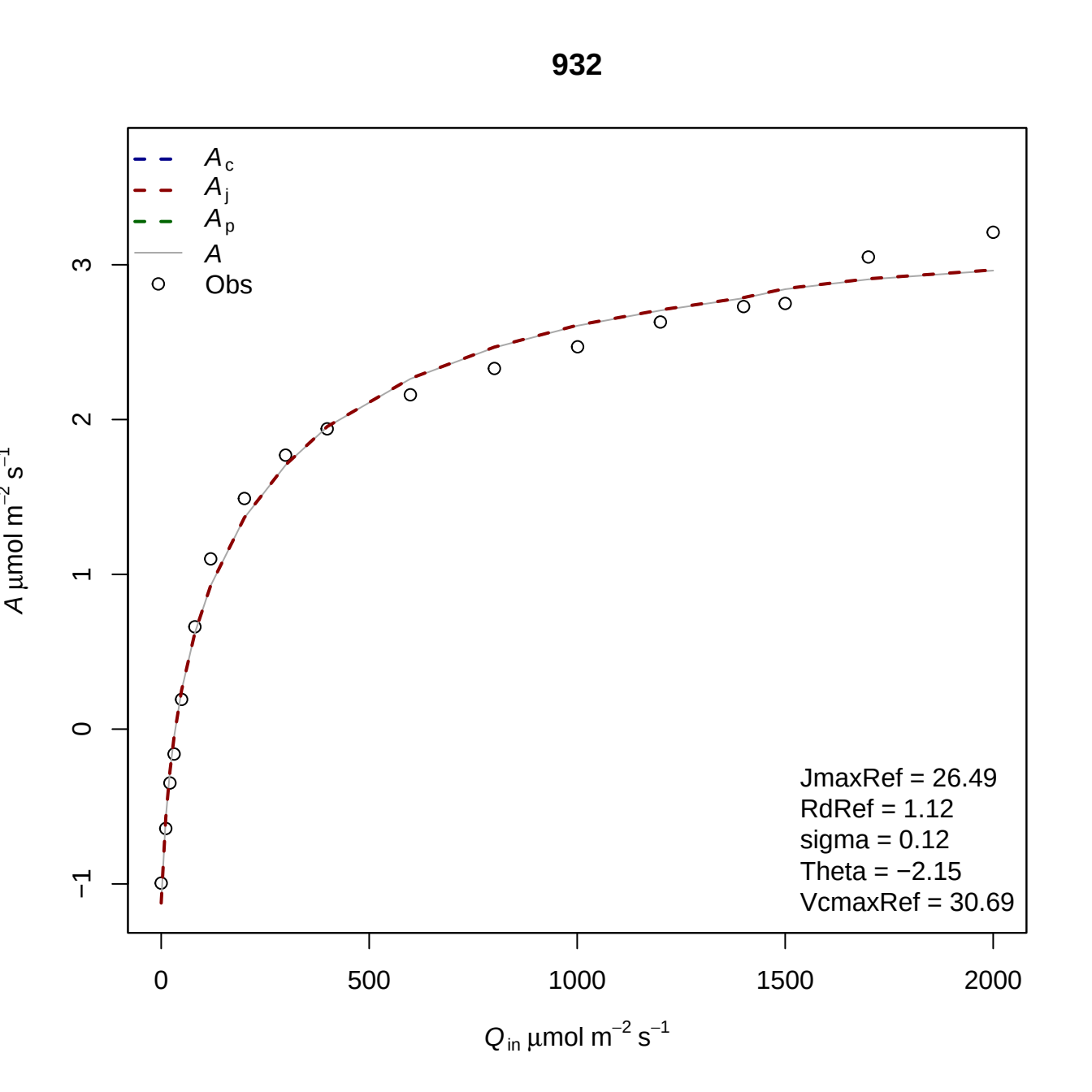

927
A_c
A_j
A_p
A
Obs
JmaxRef = 25.18
RdRef = 0.9
sigma = 0.11
Theta = −4.29
VcmaxRef = 140.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

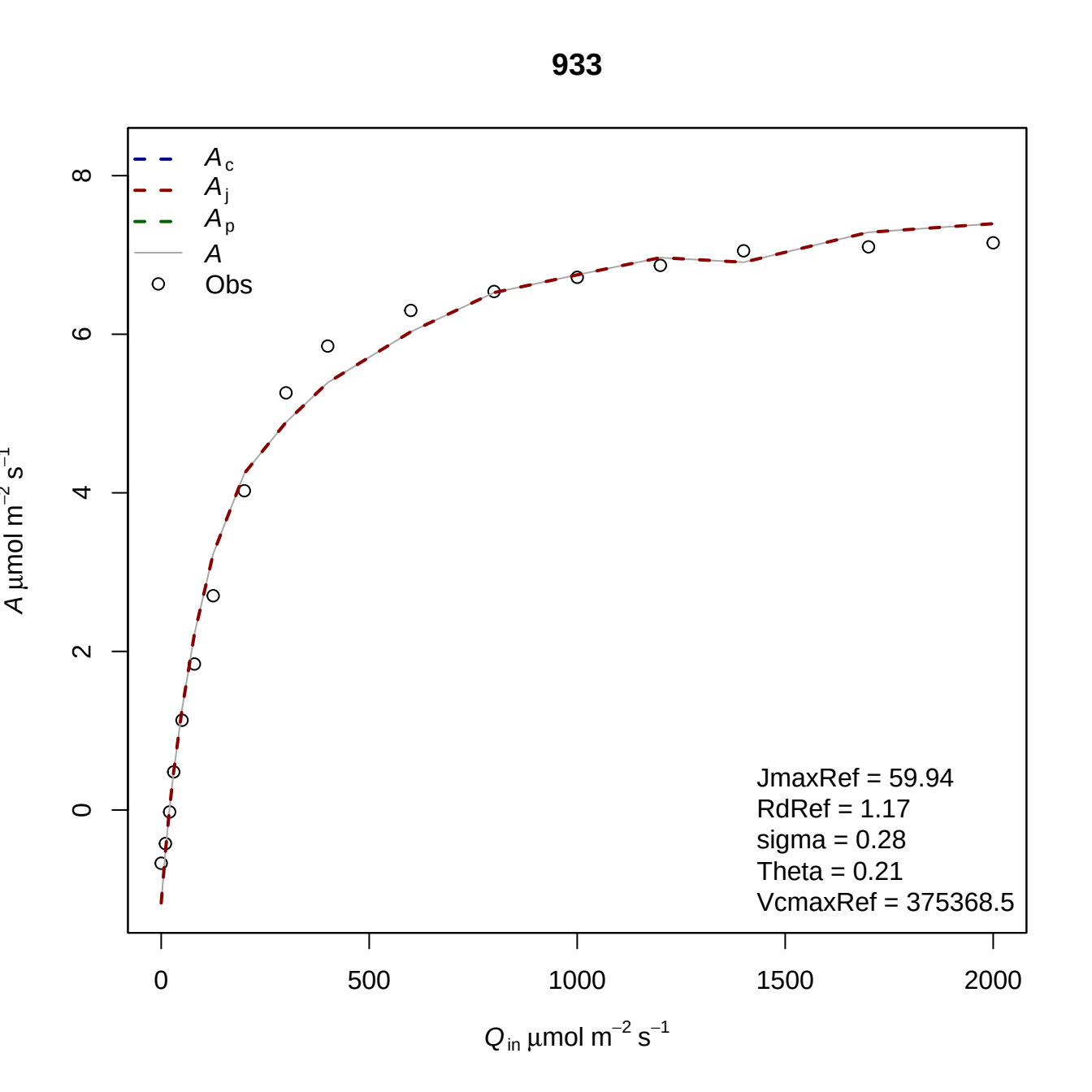

928
A_c
A_j
A_p
A
Obs
JmaxRef = 24.96
RdRef = 0.83
sigma = 0.17
Theta = −2.03
VcmaxRef = 30652.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

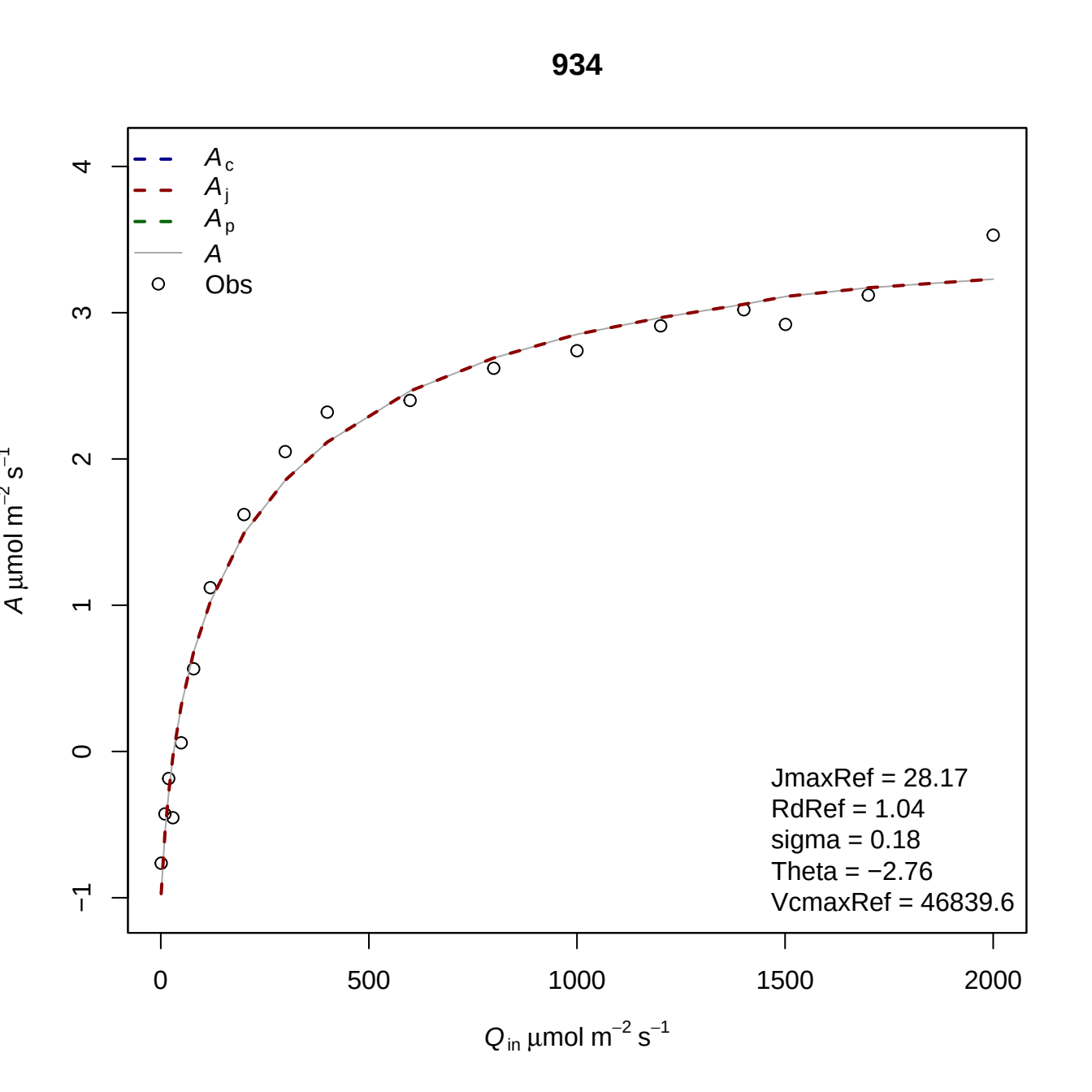

929
A_c
A_j
A_p
A
Obs
JmaxRef = 30.31
RdRef = 1.21
sigma = 0.11
Theta = −2.67
VcmaxRef = 6320.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500
2000

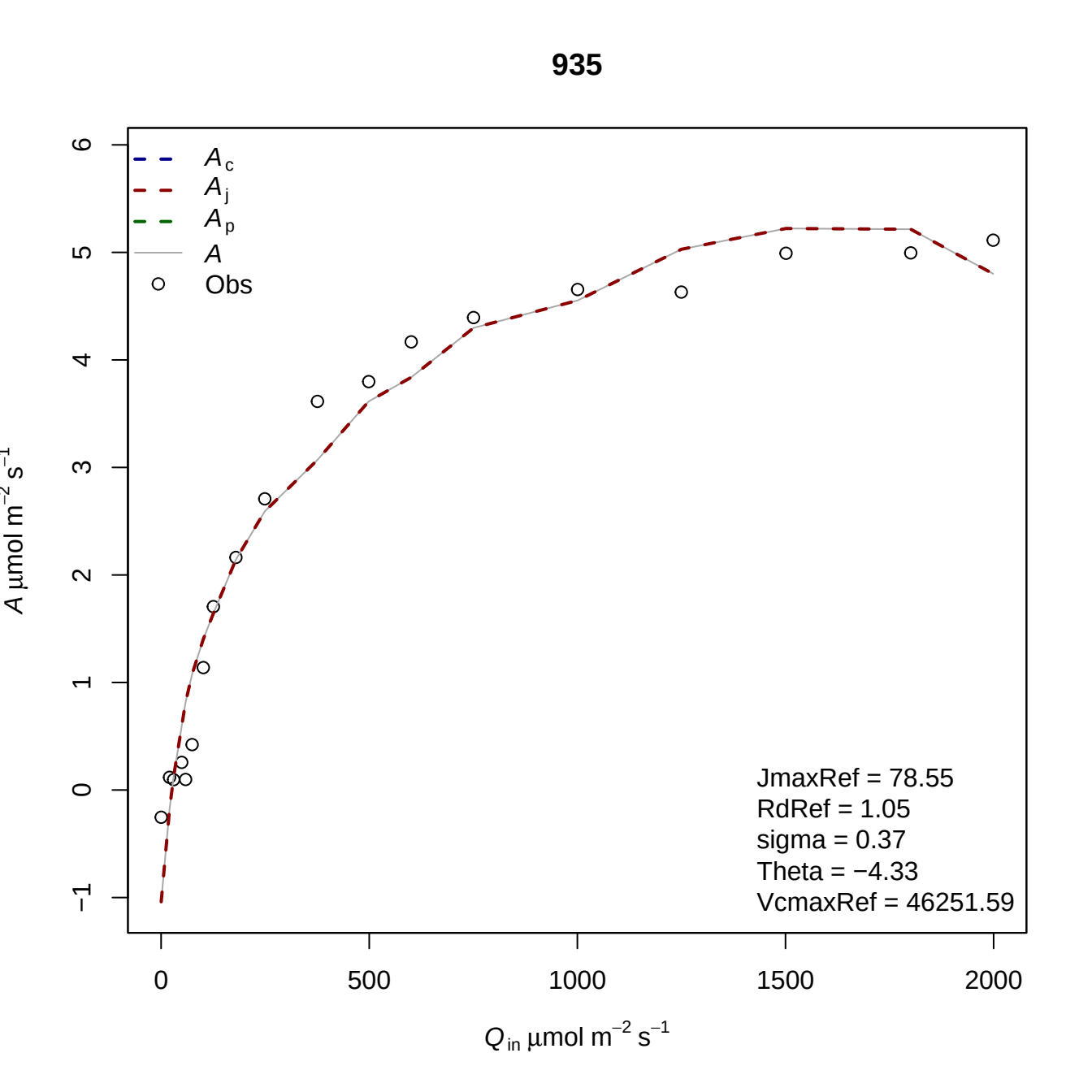

930
A_c
A_j
A_p
A
Obs
JmaxRef = 19.38
RdRef = 0.75
sigma = 0.09
Theta = 0.38
VcmaxRef = 2884.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
2000

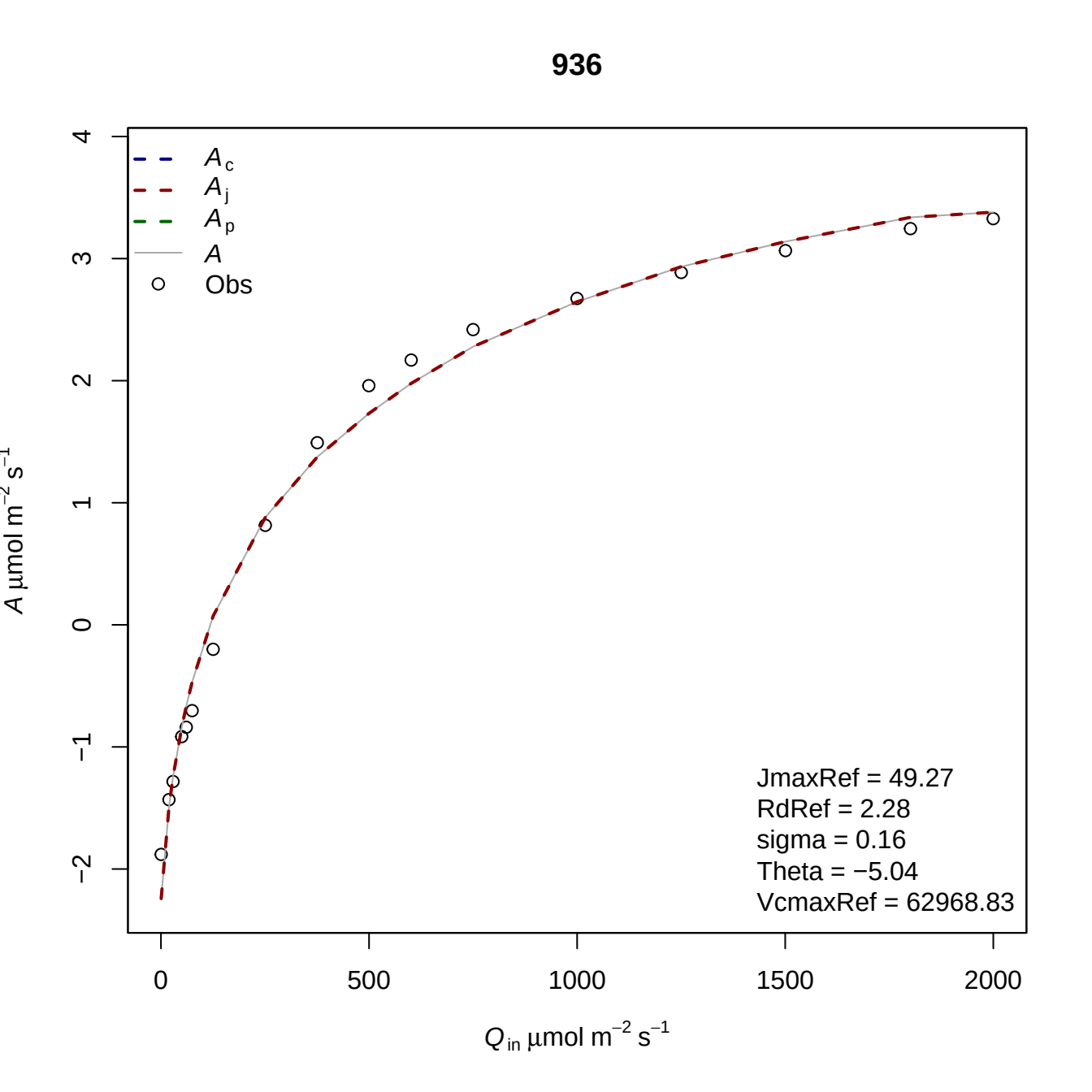

931
A_c
A_j
A_p
A
Obs
JmaxRef = 30.57
RdRef = 0.97
sigma = 0.12
Theta = −4.3
VcmaxRef = 642.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

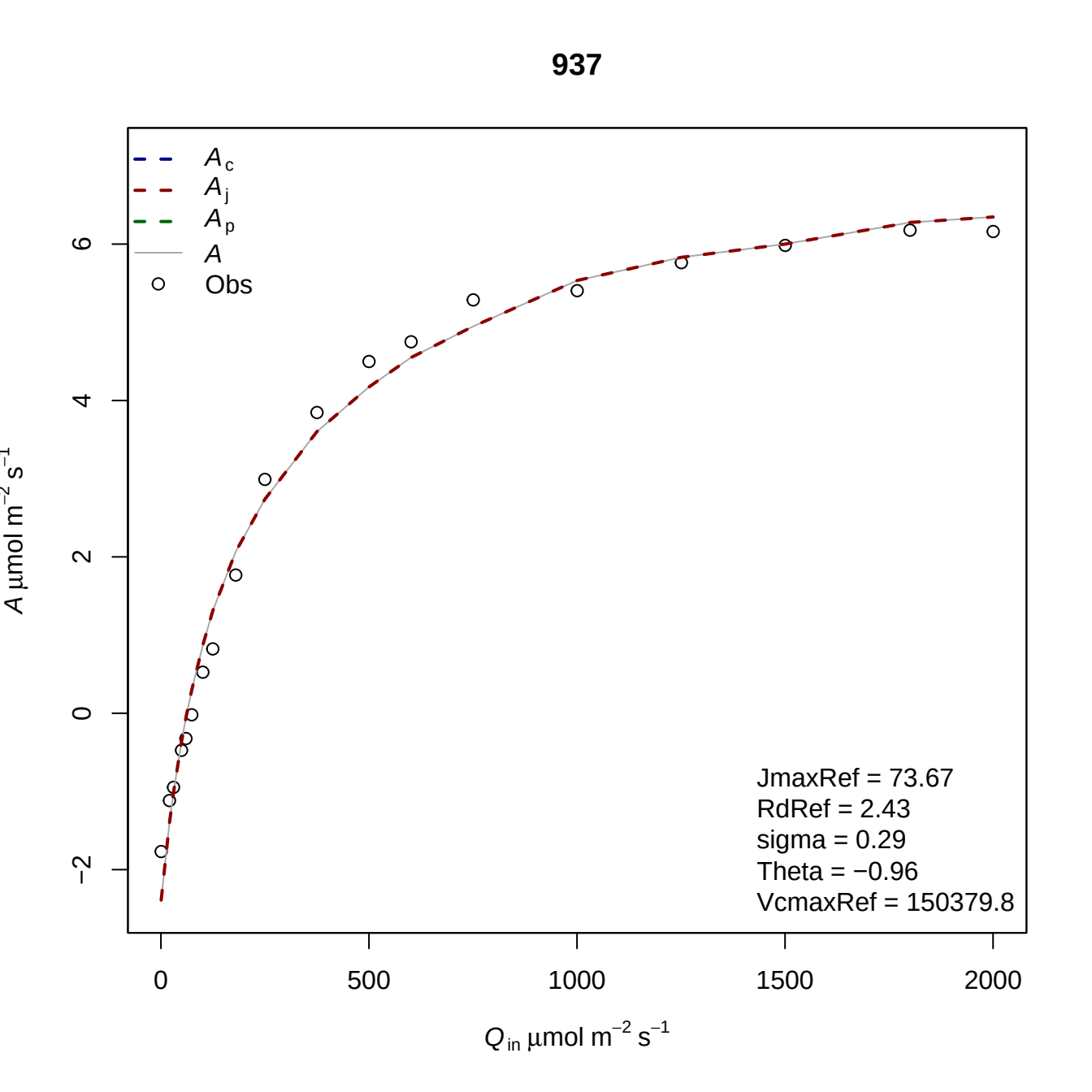

932
A_c
A_j
A_p
A
Obs
JmaxRef = 26.49
RdRef = 1.12
sigma = 0.12
Theta = −2.15
VcmaxRef = 30.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

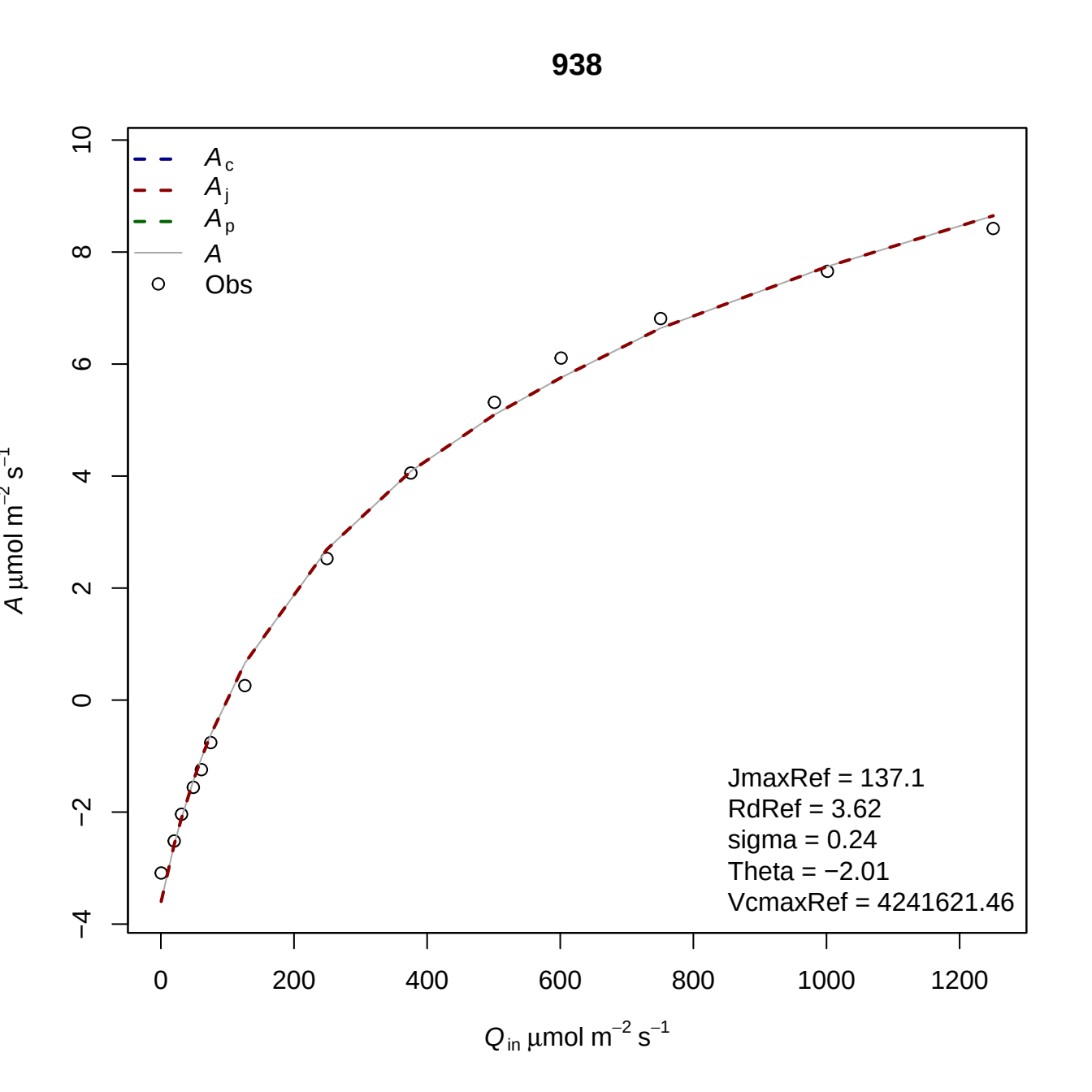

933
A_c
A_j
A_p
A
Obs
JmaxRef = 59.94
RdRef = 1.17
sigma = 0.28
Theta = 0.21
VcmaxRef = 375368.5
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

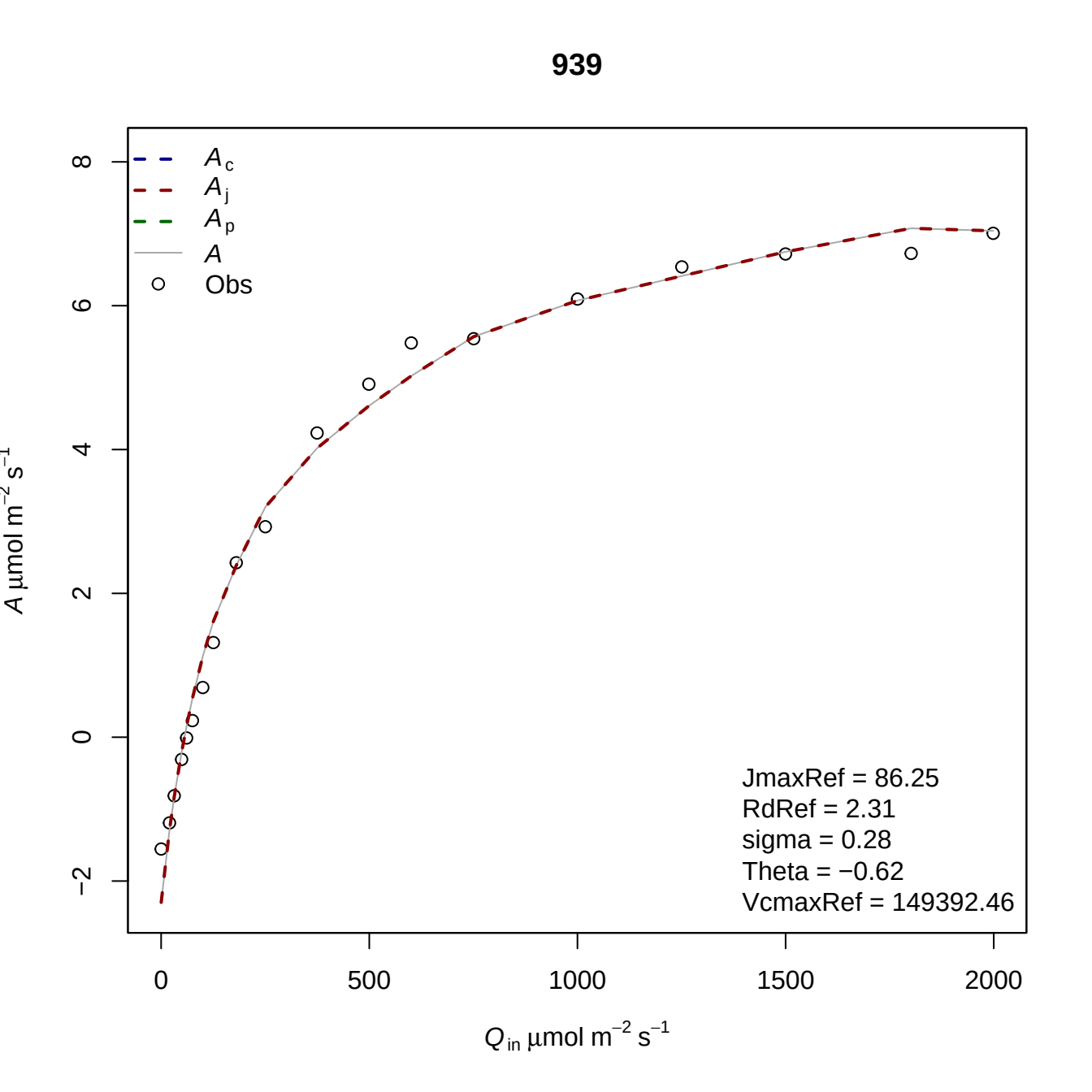

934
A_c
A_j
A_p
A
Obs
JmaxRef = 28.17
RdRef = 1.04
sigma = 0.18
Theta = −2.76
VcmaxRef = 46839.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

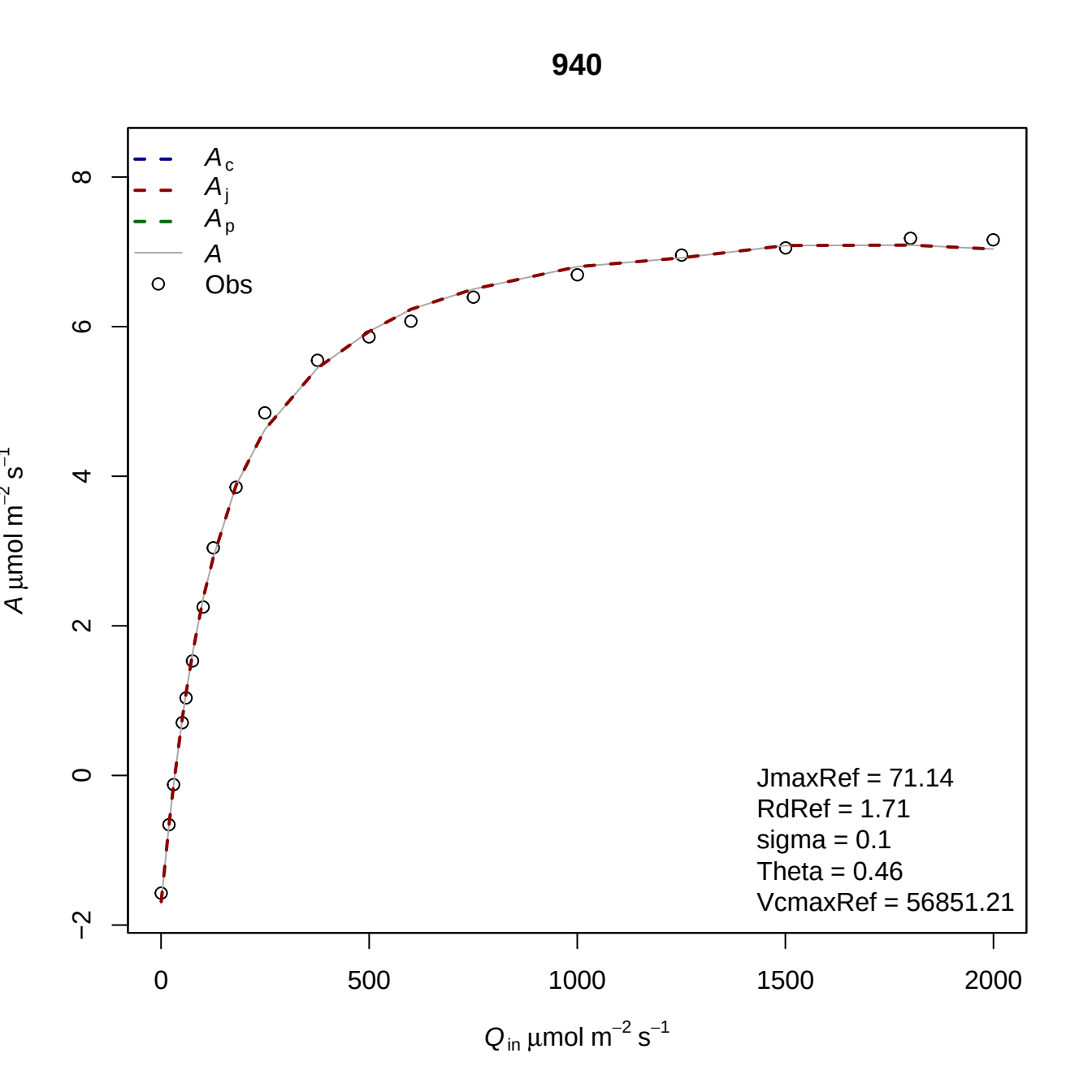

935
A_c
A_j
A_p
A
Obs
JmaxRef = 78.55
RdRef = 1.05
sigma = 0.37
Theta = −4.33
VcmaxRef = 46251.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

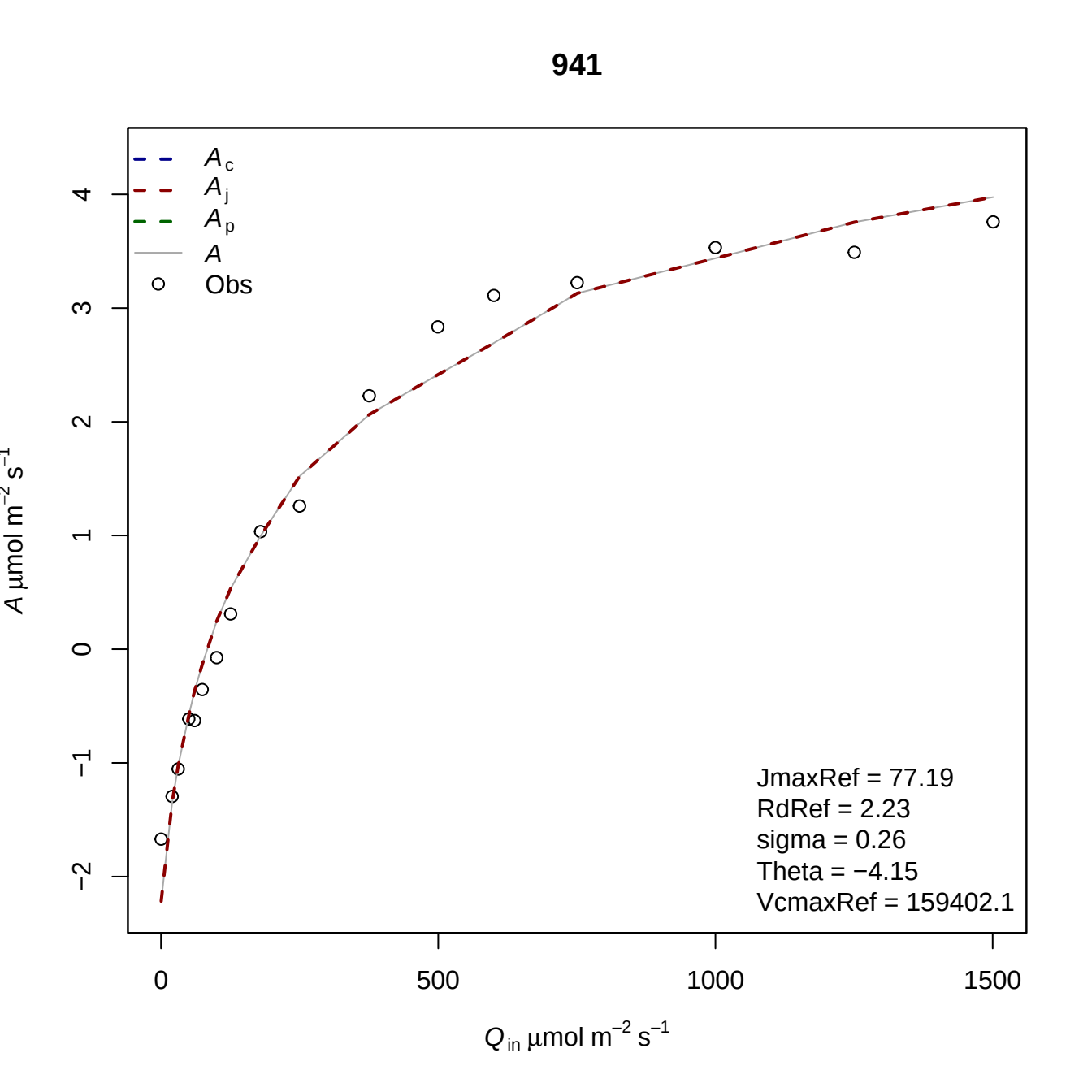

936
A_c
A_j
A_p
A
Obs
JmaxRef = 49.27
RdRef = 2.28
sigma = 0.16
Theta = −5.04
VcmaxRef = 62968.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

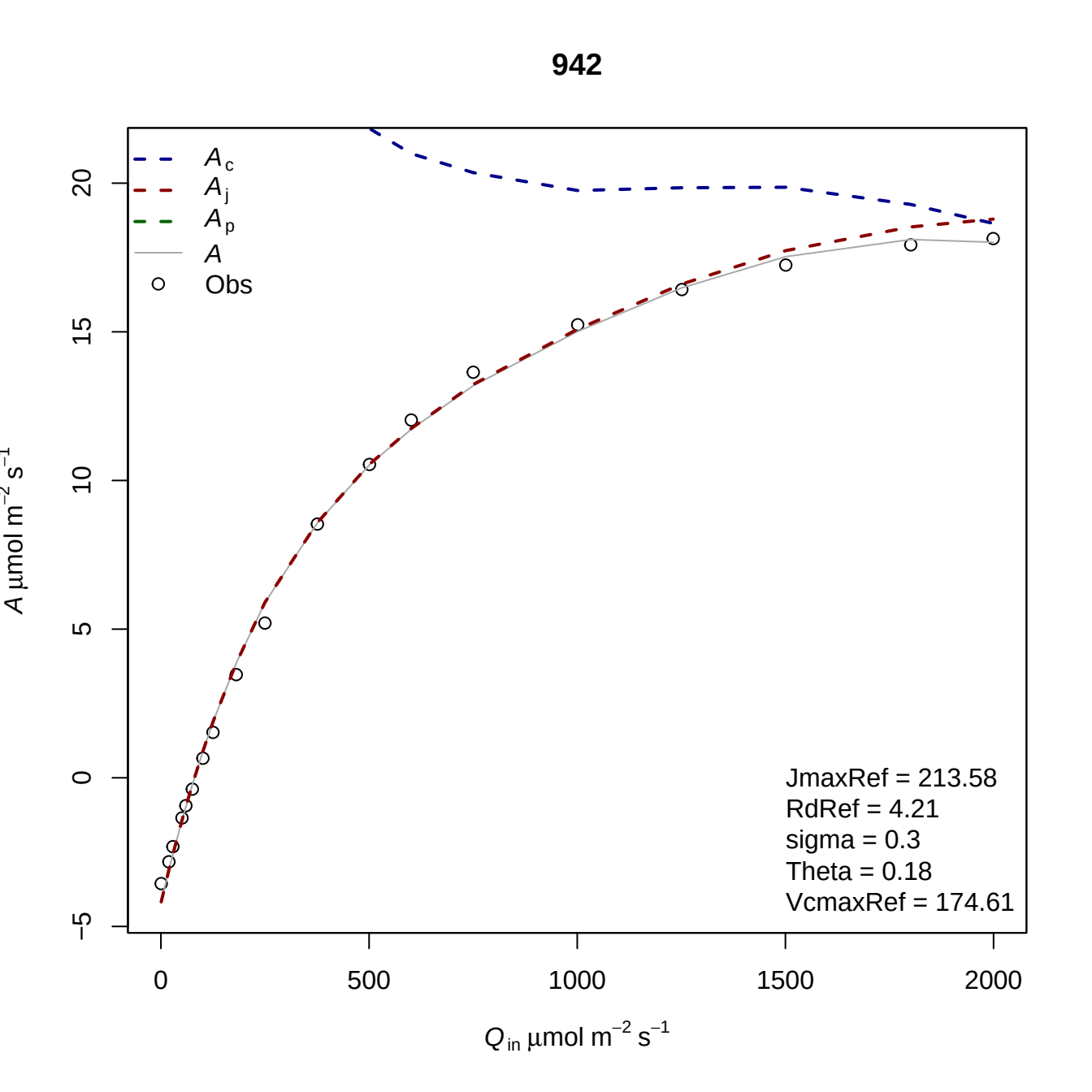

937
A_c
A_j
A_p
A
Obs
JmaxRef = 73.67
RdRef = 2.43
sigma = 0.29
Theta = −0.96
VcmaxRef = 150379.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

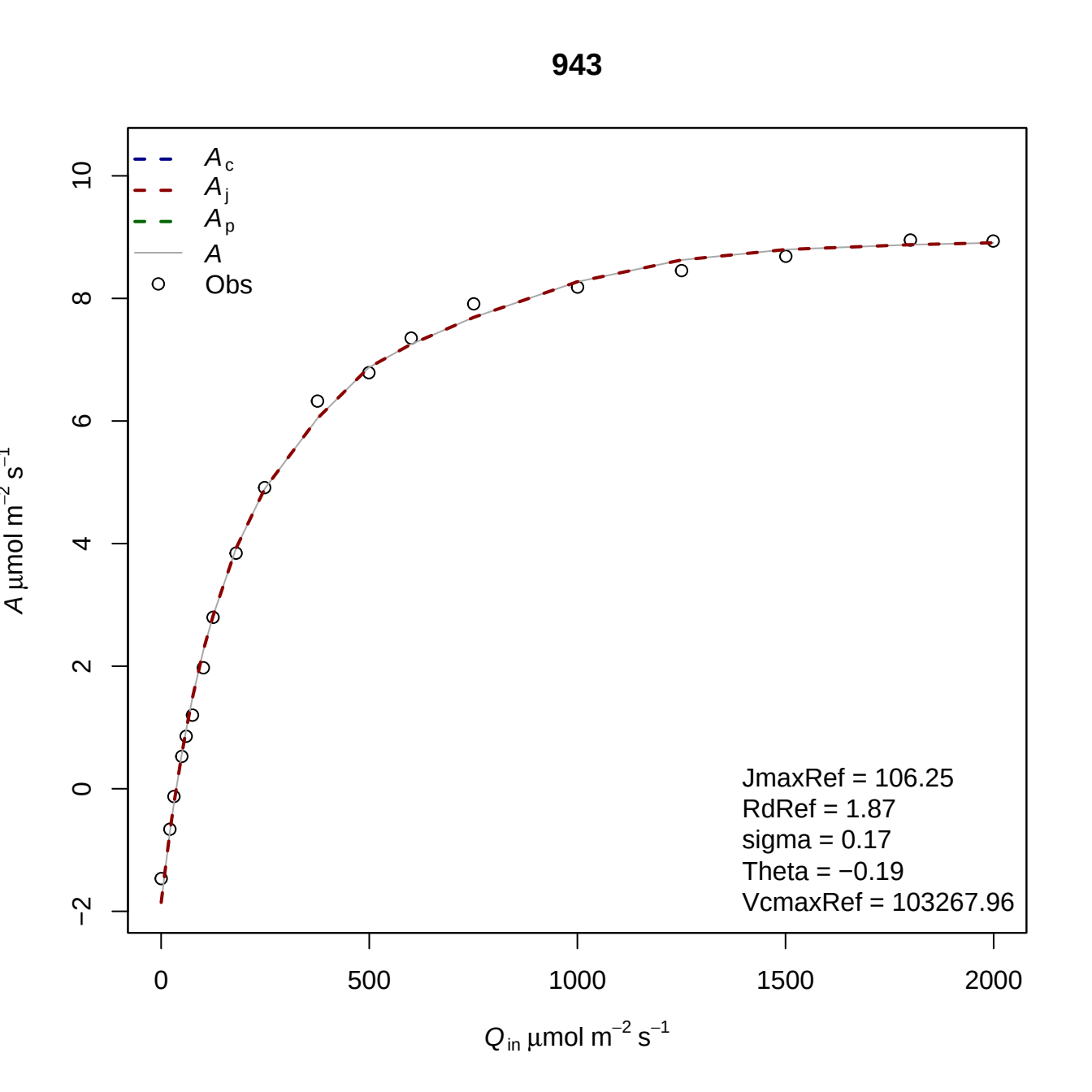

938
A_c
A_j
A_p
A
Obs
JmaxRef = 137.1
RdRef = 3.62
sigma = 0.24
Theta = −2.01
VcmaxRef = 4241621.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

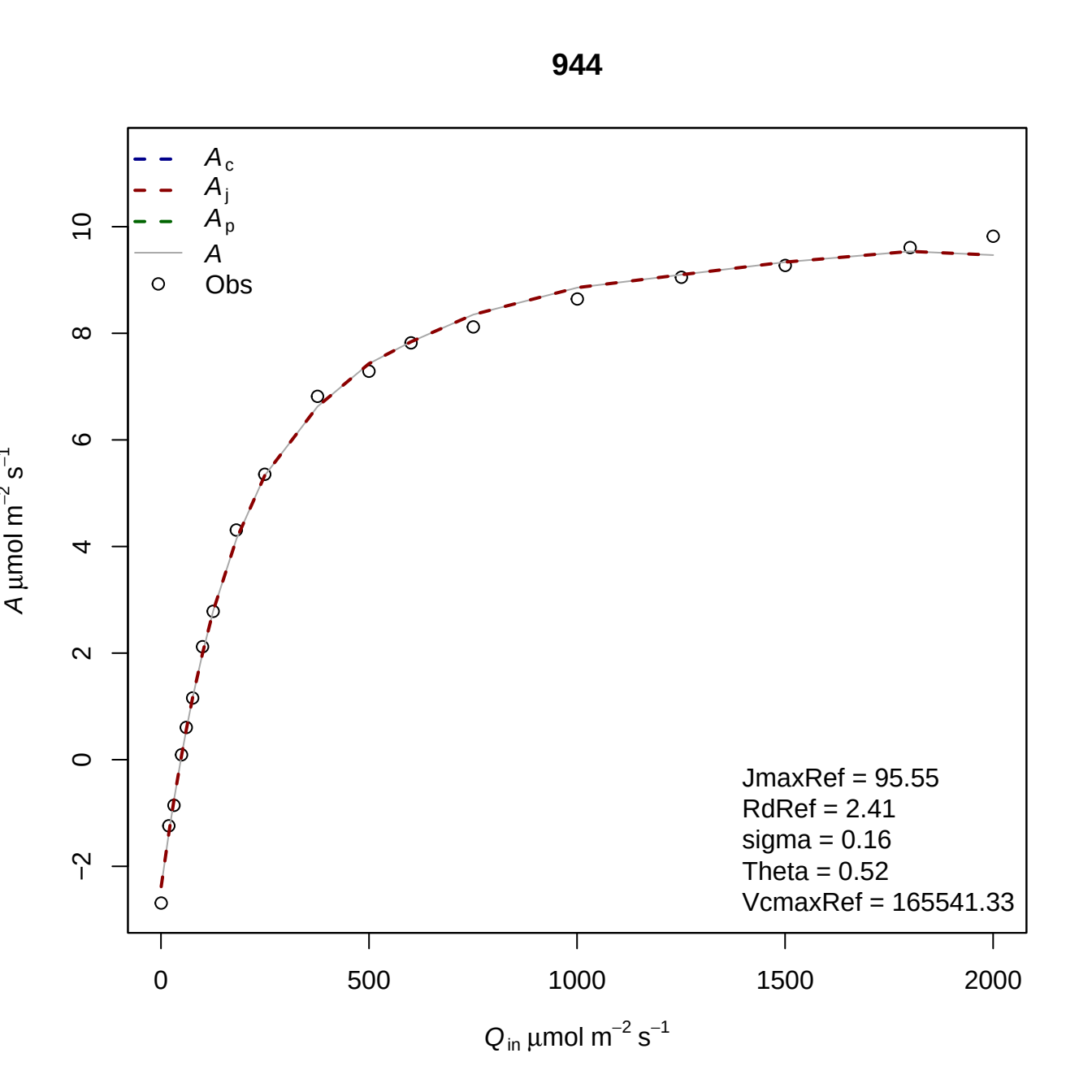

939
A_c
A_j
A_p
A
Obs
JmaxRef = 86.25
RdRef = 2.31
sigma = 0.28
Theta = −0.62
VcmaxRef = 149392.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

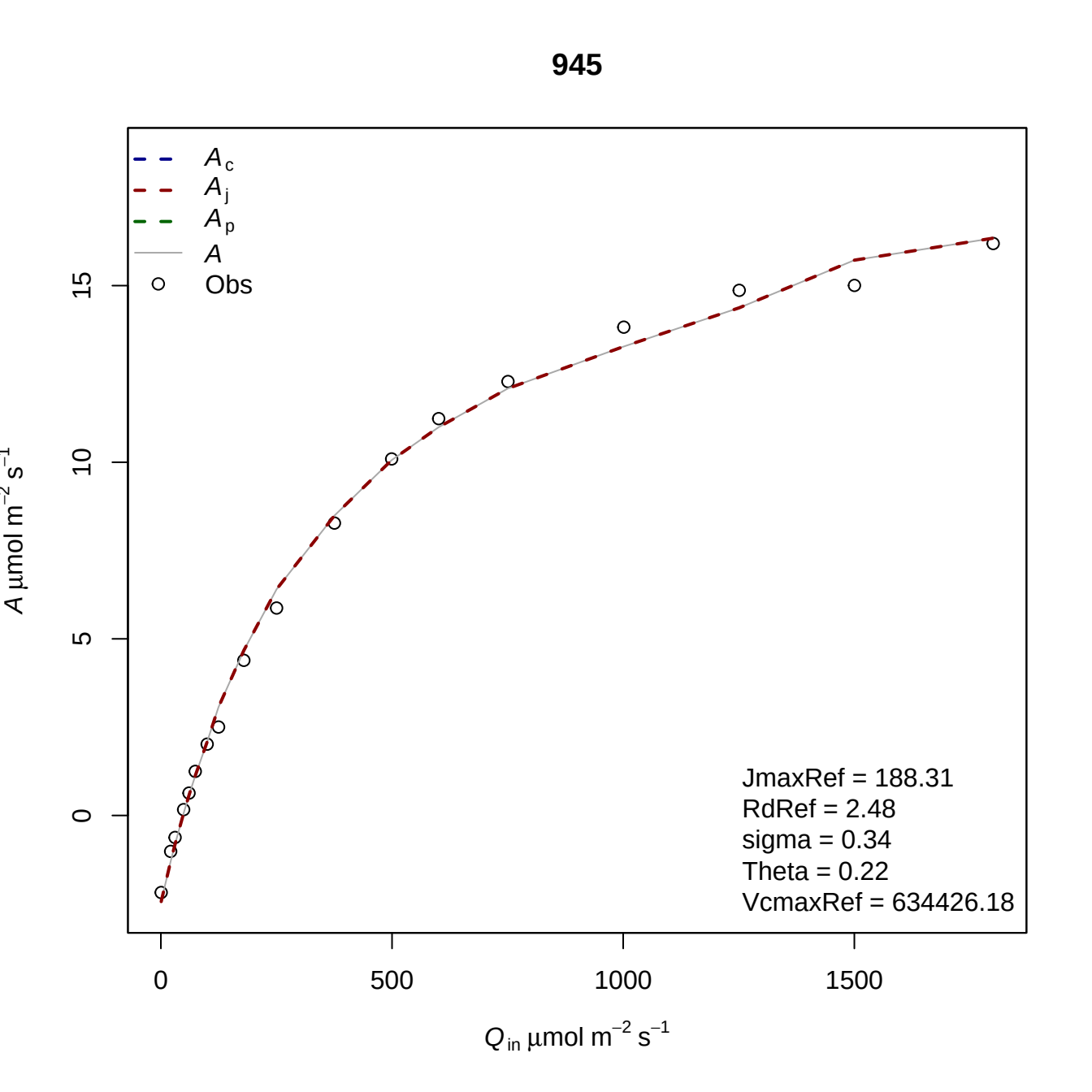

940
A_c
A_j
A_p
A
Obs
JmaxRef = 71.14
RdRef = 1.71
sigma = 0.1
Theta = 0.46
VcmaxRef = 56851.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

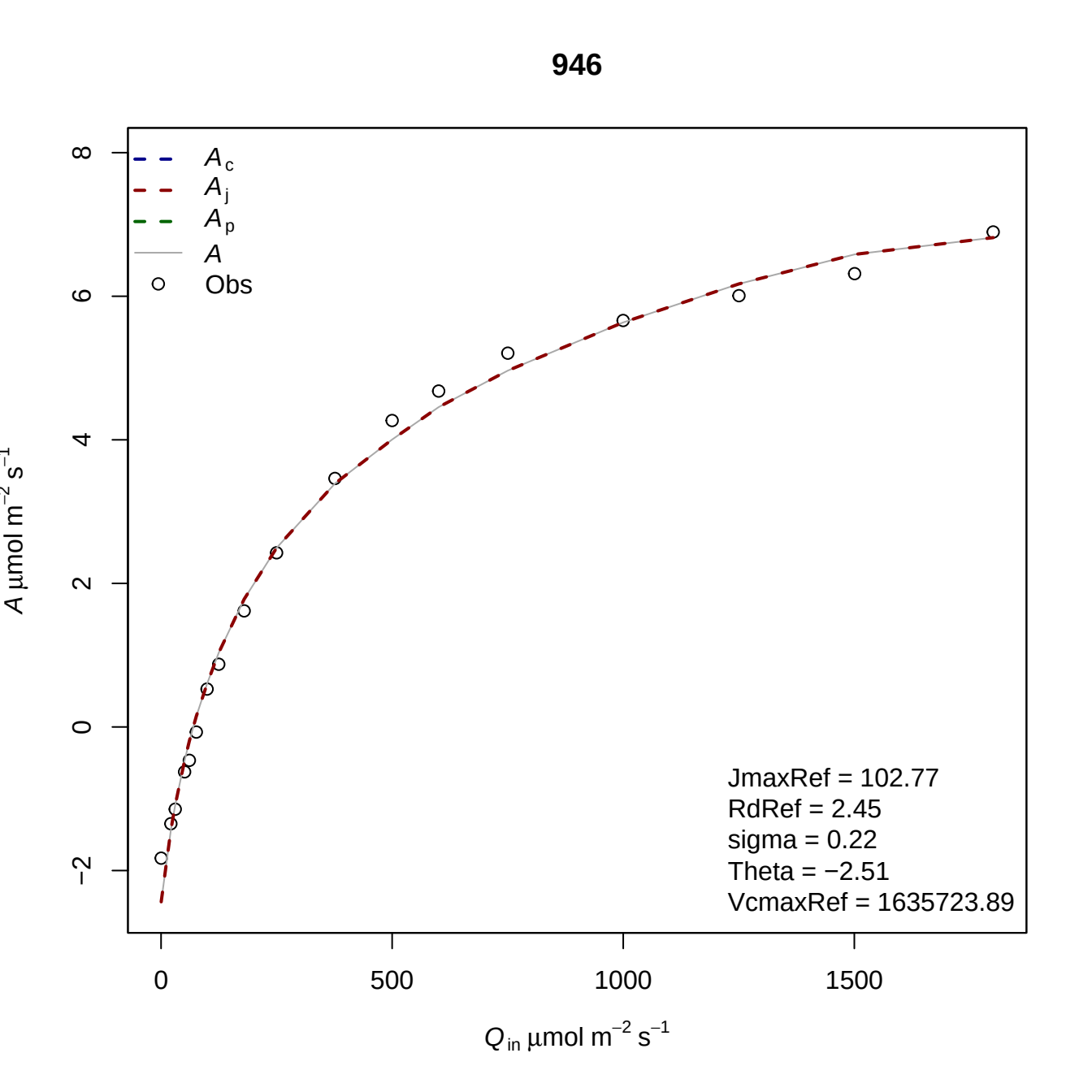

941
A_c
A_j
A_p
A
Obs
JmaxRef = 77.19
RdRef = 2.23
sigma = 0.26
Theta = −4.15
VcmaxRef = 159402.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

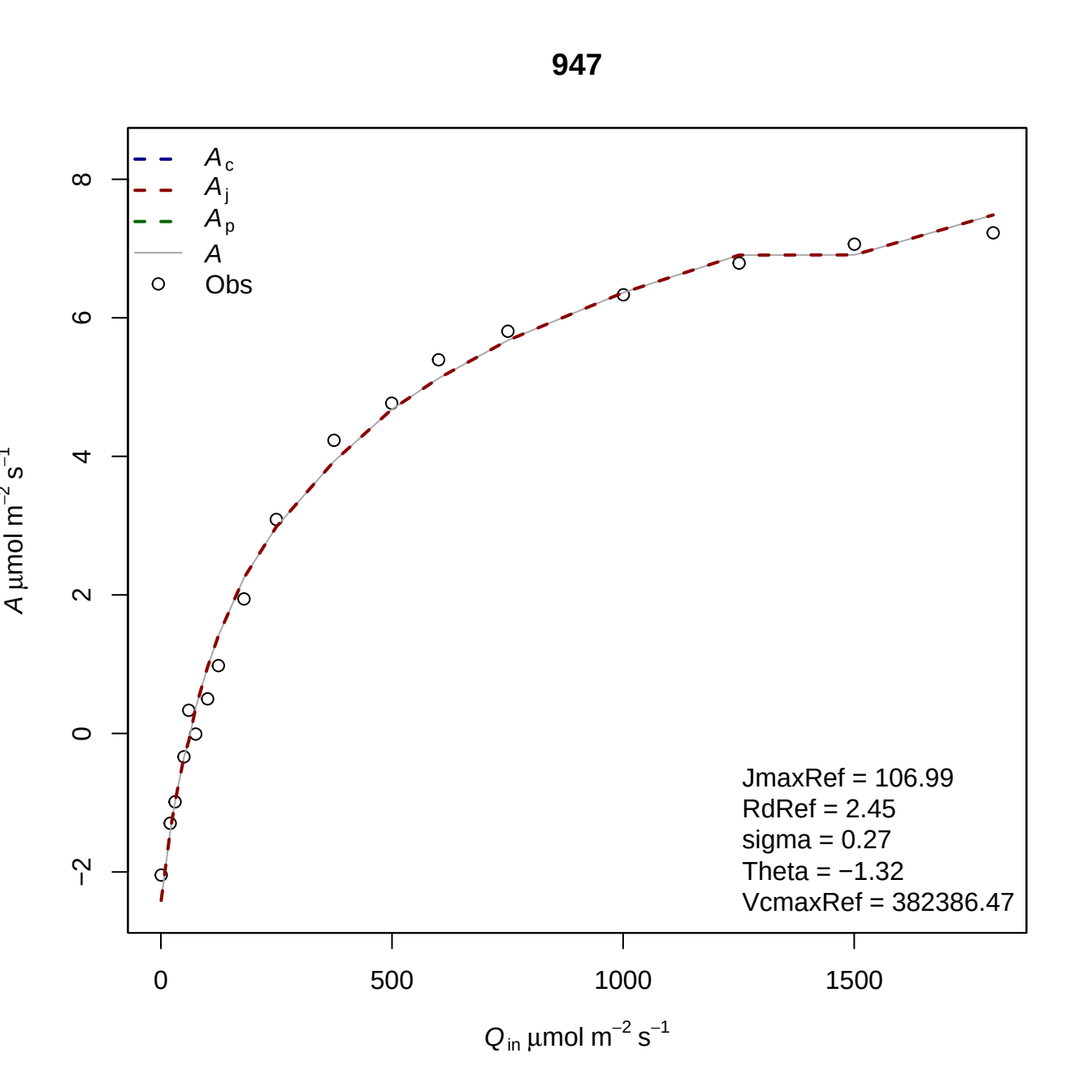

942
A_c
A_j
A_p
A
Obs
JmaxRef = 213.58
RdRef = 4.21
sigma = 0.3
Theta = 0.18
VcmaxRef = 174.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

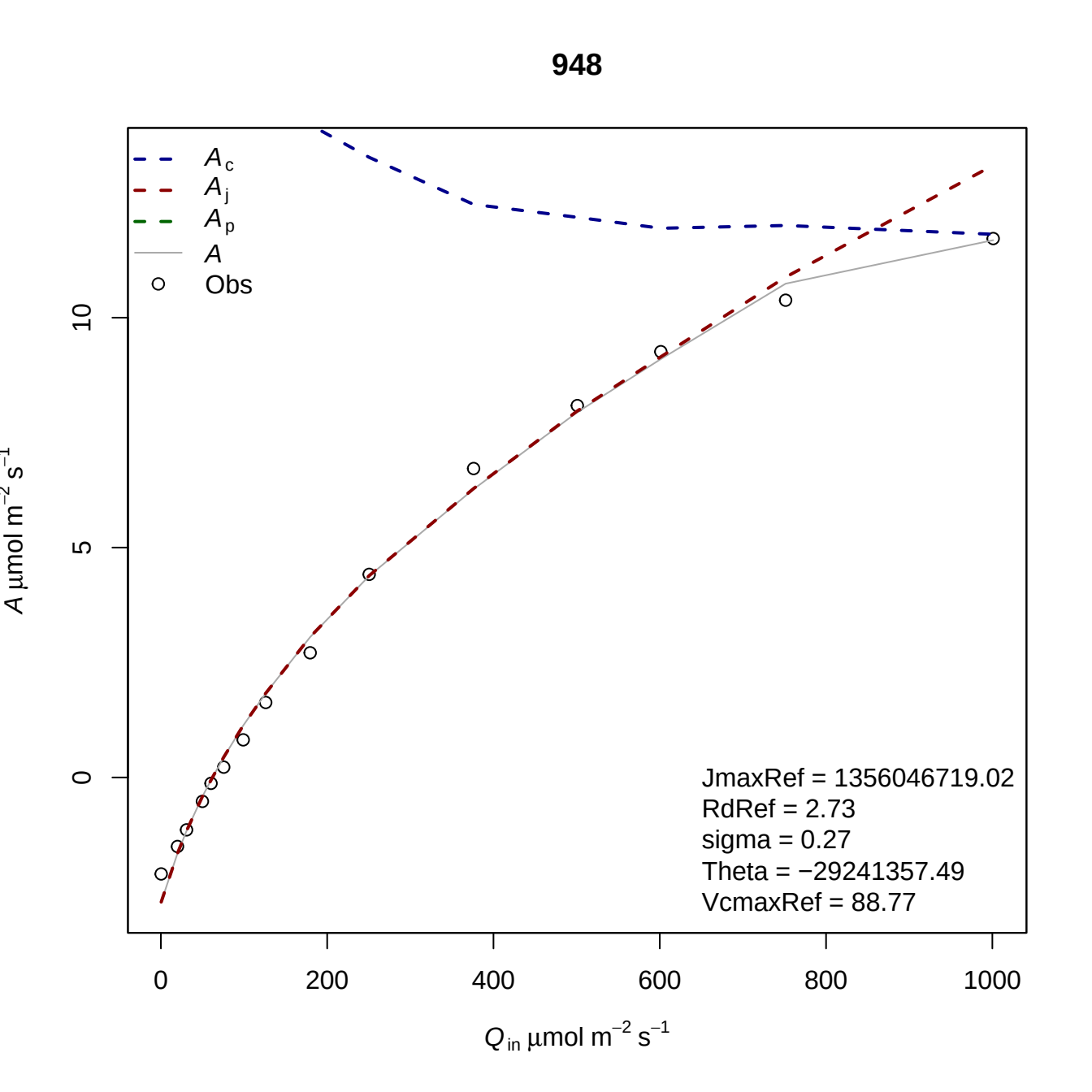

943
A_c
A_j
A_p
A
Obs
JmaxRef = 106.25
RdRef = 1.87
sigma = 0.17
Theta = −0.19
VcmaxRef = 103267.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

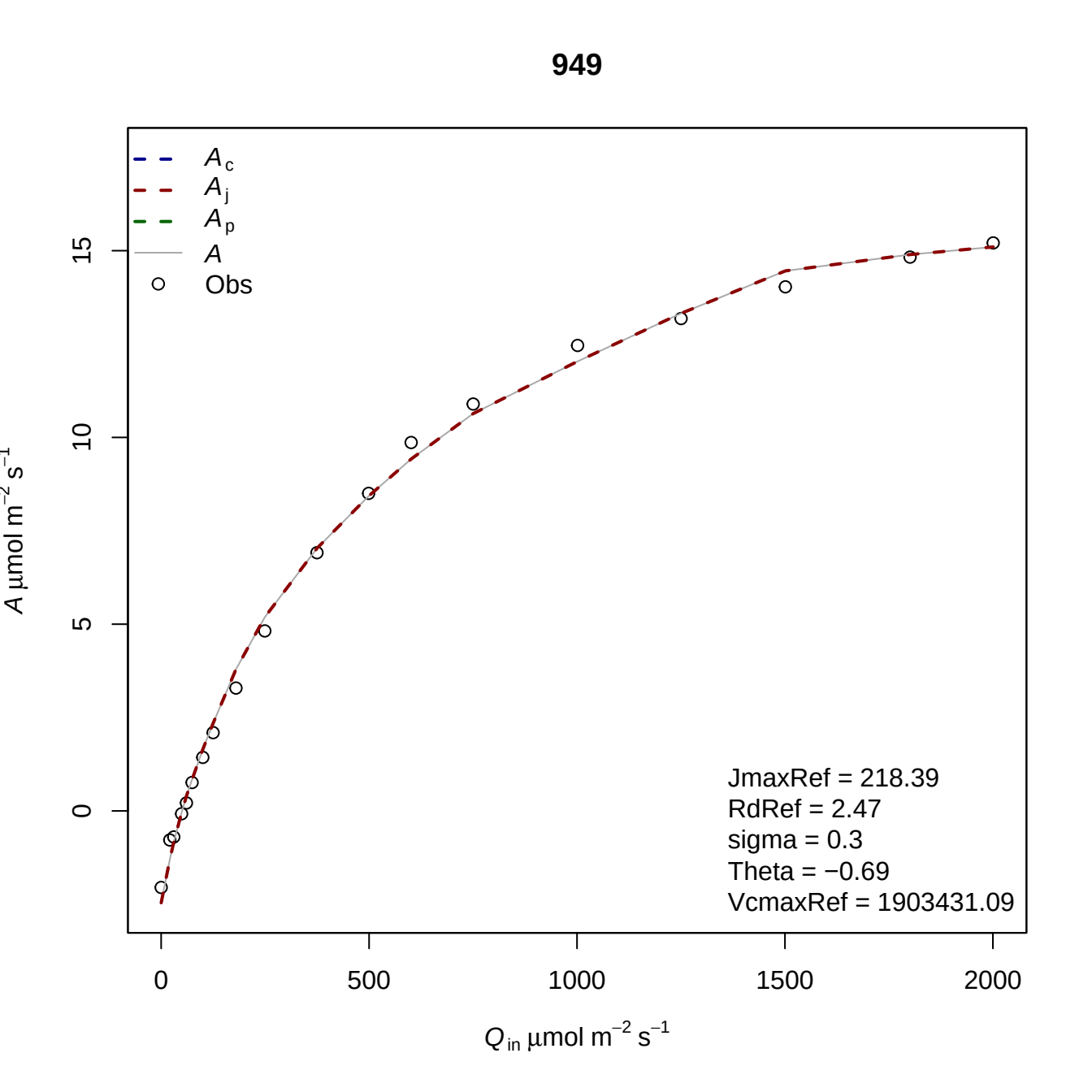

944
A_c
A_j
A_p
A
Obs
JmaxRef = 95.55
RdRef = 2.41
sigma = 0.16
Theta = 0.52
VcmaxRef = 165541.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

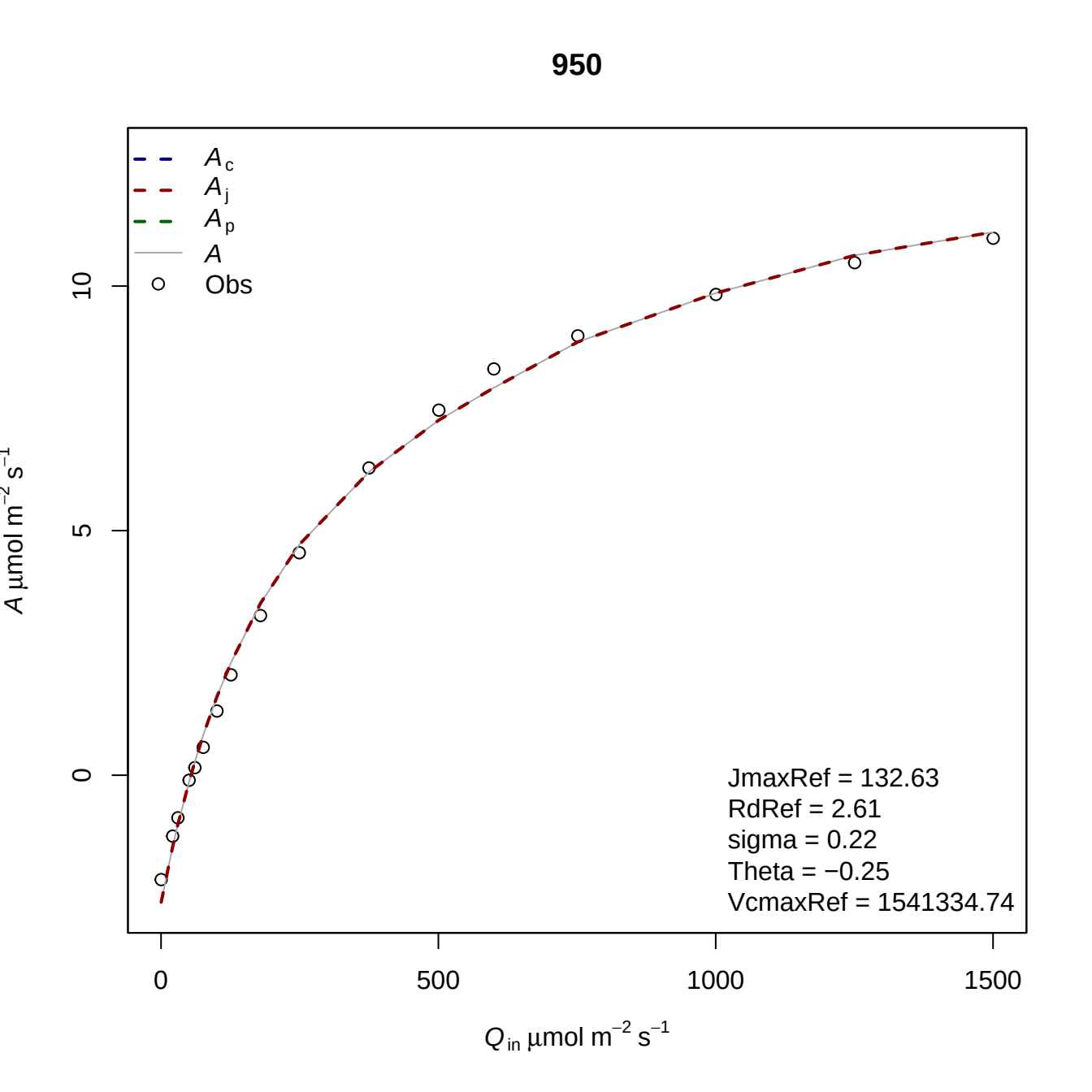

945
A_c
A_j
A_p
A
Obs
JmaxRef = 188.31
RdRef = 2.48
sigma = 0.34
Theta = 0.22
VcmaxRef = 634426.18
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

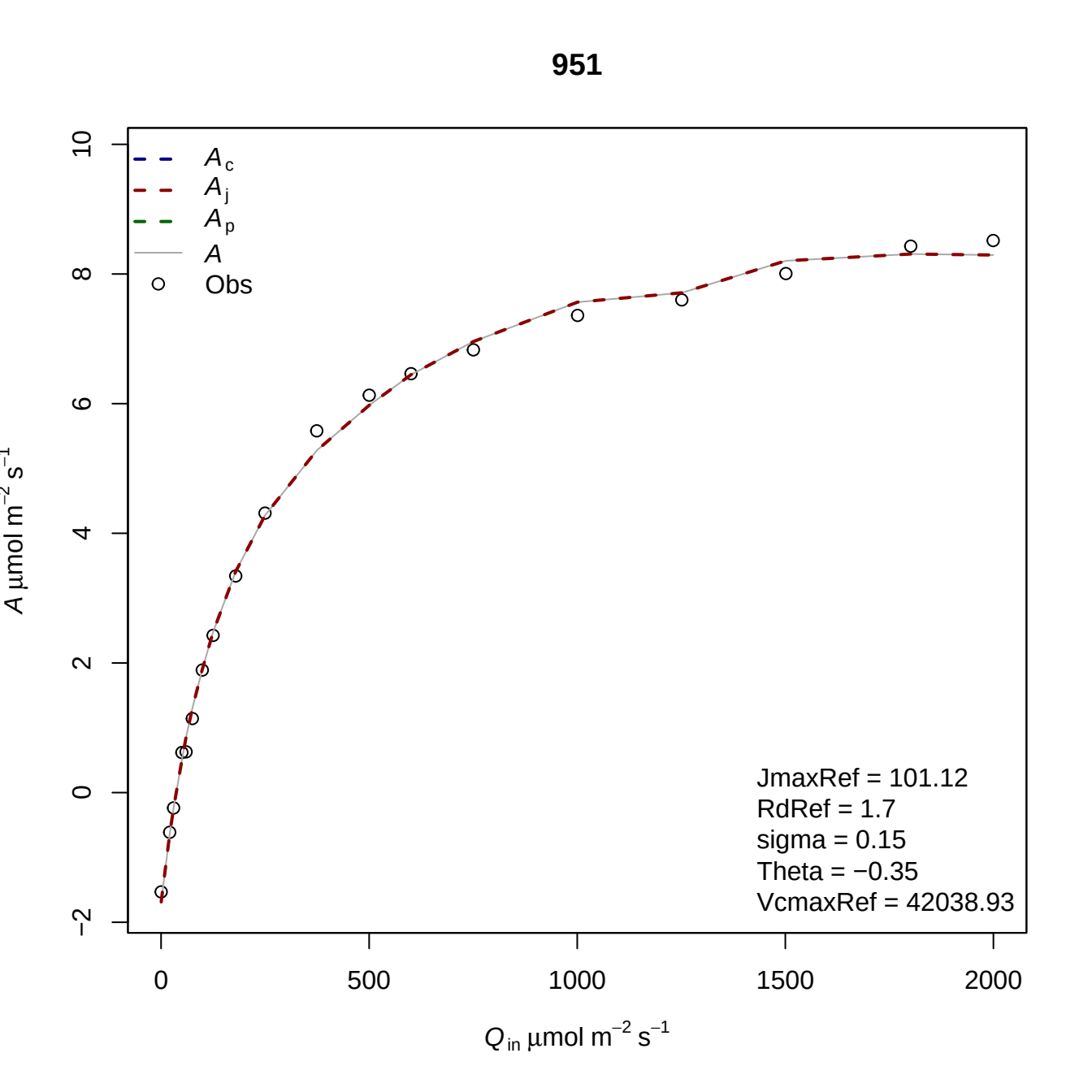

946
A_c
A_j
A_p
A
Obs
JmaxRef = 102.77
RdRef = 2.45
sigma = 0.22
Theta = −2.51
VcmaxRef = 1635723.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

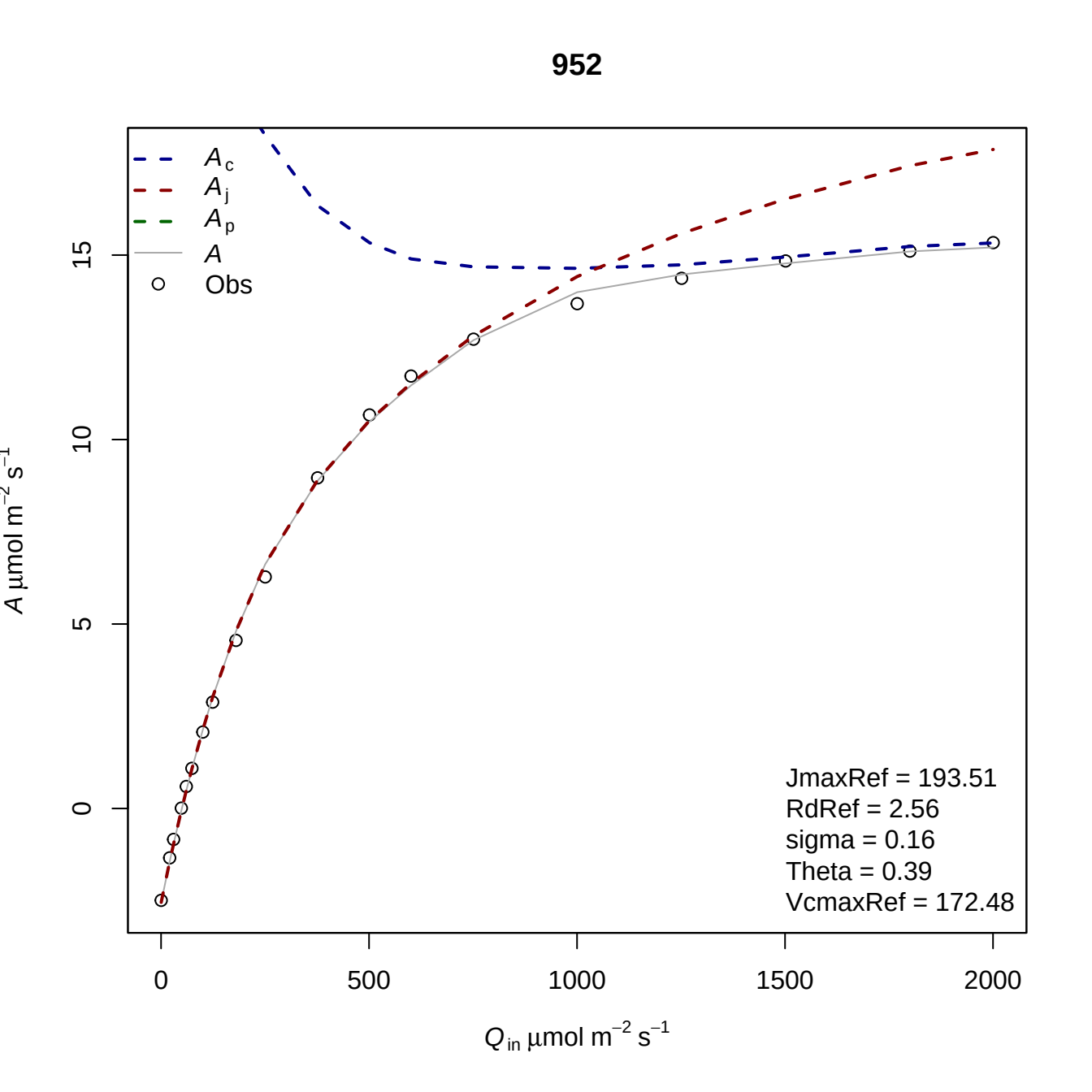

947
A_c
A_j
A_p
A
Obs
JmaxRef = 106.99
RdRef = 2.45
sigma = 0.27
Theta = −1.32
VcmaxRef = 382386.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

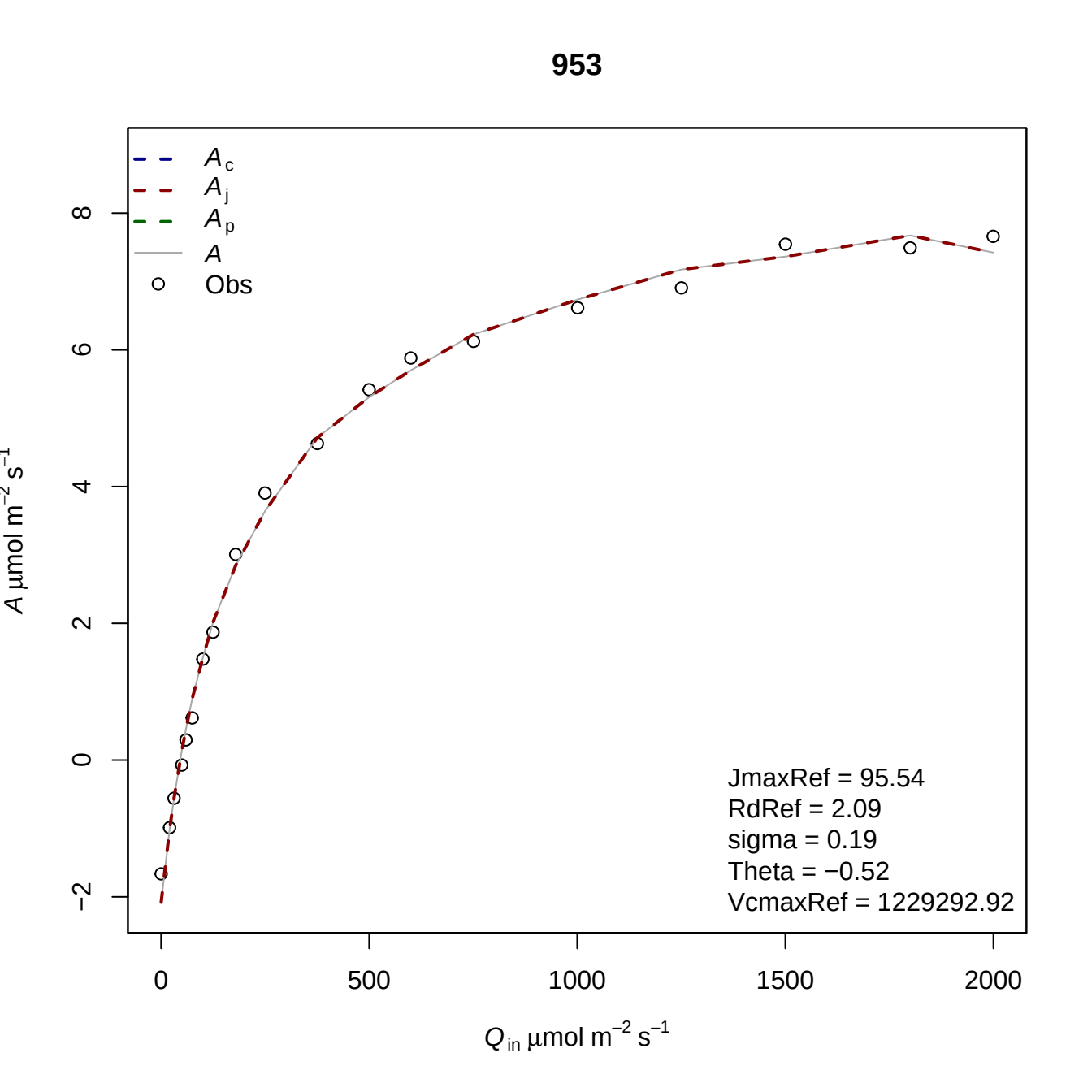

948
A_c
A_j
A_p
A
Obs
JmaxRef = 1356046719.02
RdRef = 2.73
sigma = 0.27
Theta = −29241357.49
VcmaxRef = 88.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

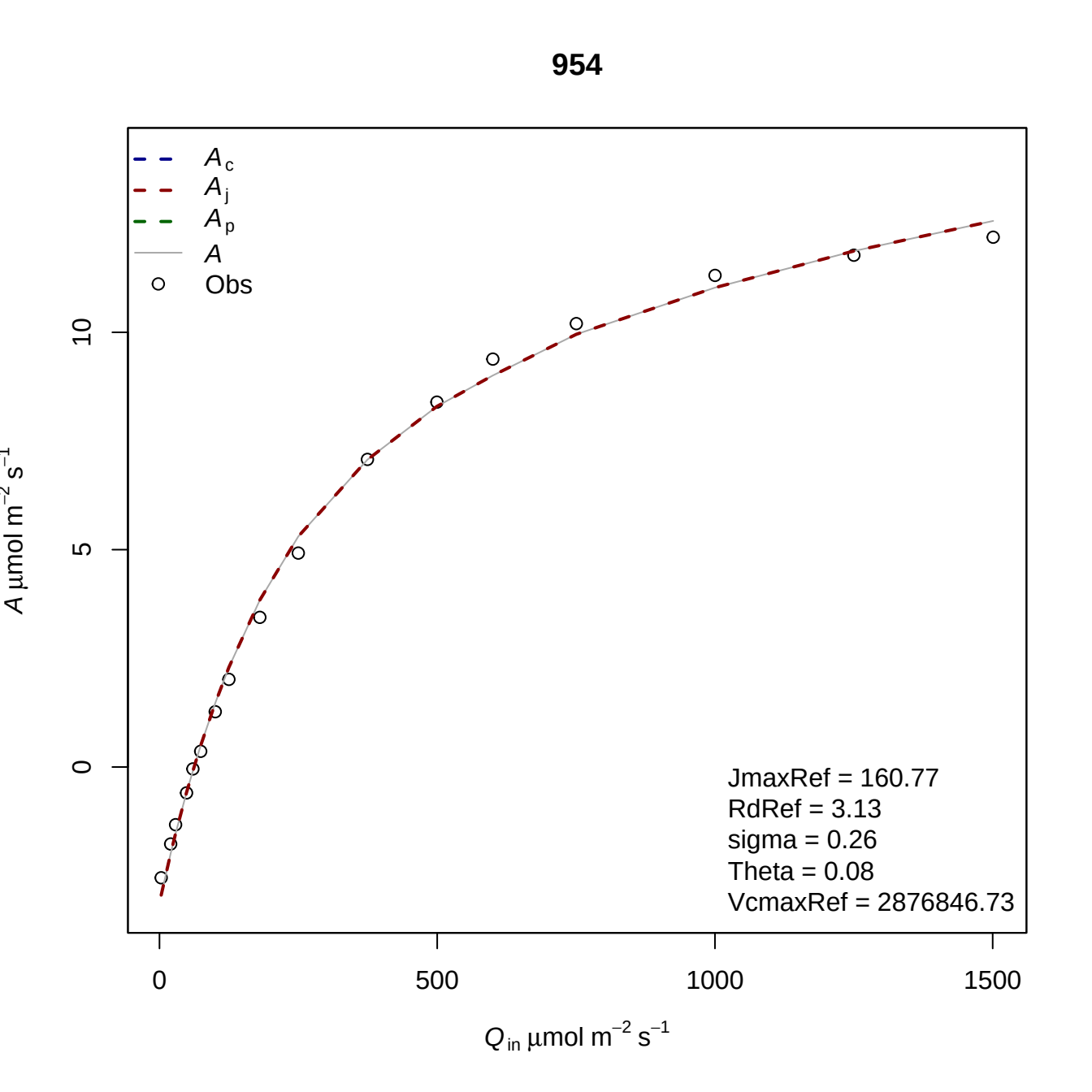

949
A_c
A_j
A_p
A
Obs
JmaxRef = 218.39
RdRef = 2.47
sigma = 0.3
Theta = −0.69
VcmaxRef = 1903431.09
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

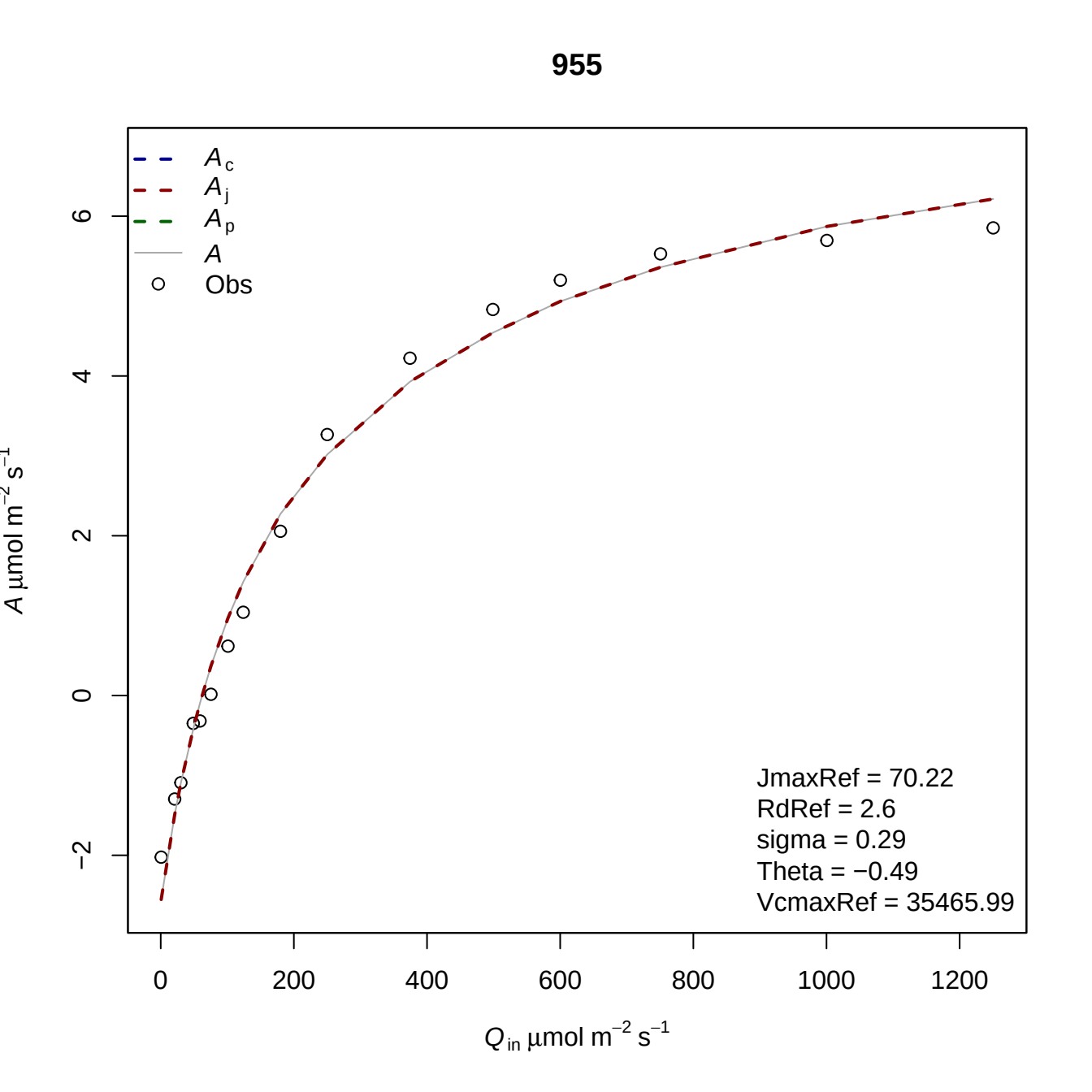

950
A_c
A_j
A_p
A
Obs
JmaxRef = 132.63
RdRef = 2.61
sigma = 0.22
Theta = −0.25
VcmaxRef = 1541334.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

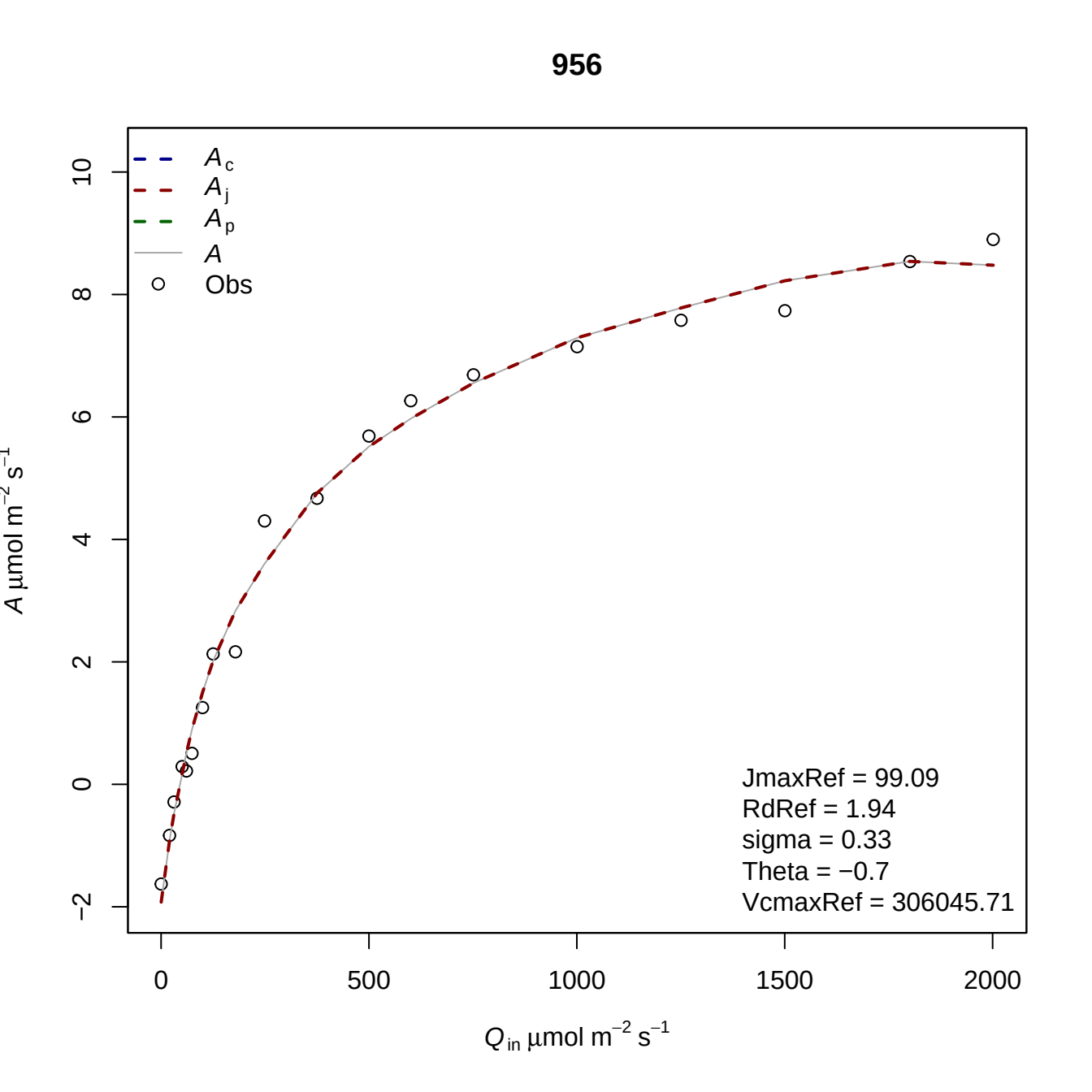

951
A_c
A_j
A_p
A
Obs
JmaxRef = 101.12
RdRef = 1.7
sigma = 0.15
Theta = −0.35
VcmaxRef = 42038.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

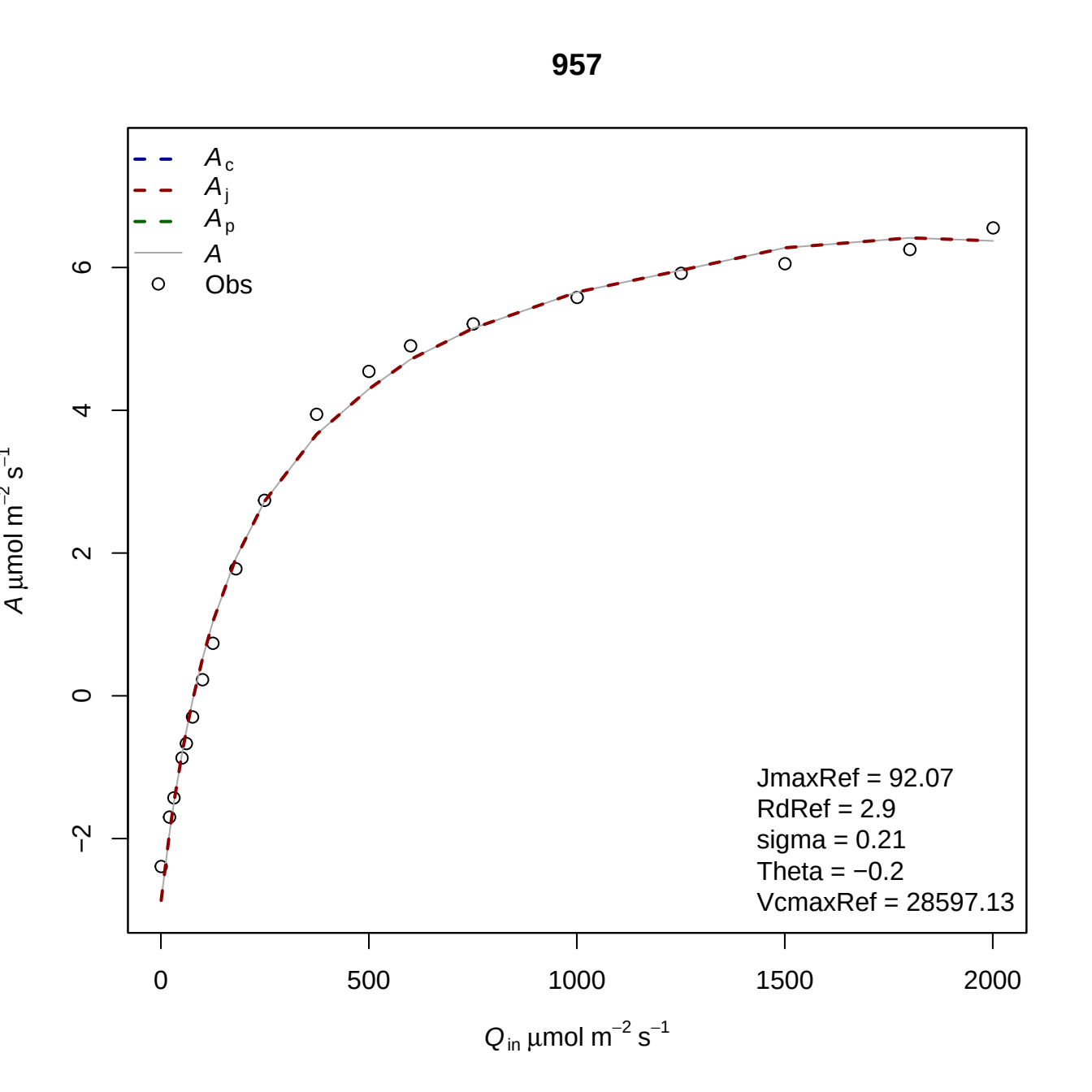

952
A_c
A_j
A_p
A
Obs
JmaxRef = 193.51
RdRef = 2.56
sigma = 0.16
Theta = 0.39
VcmaxRef = 172.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

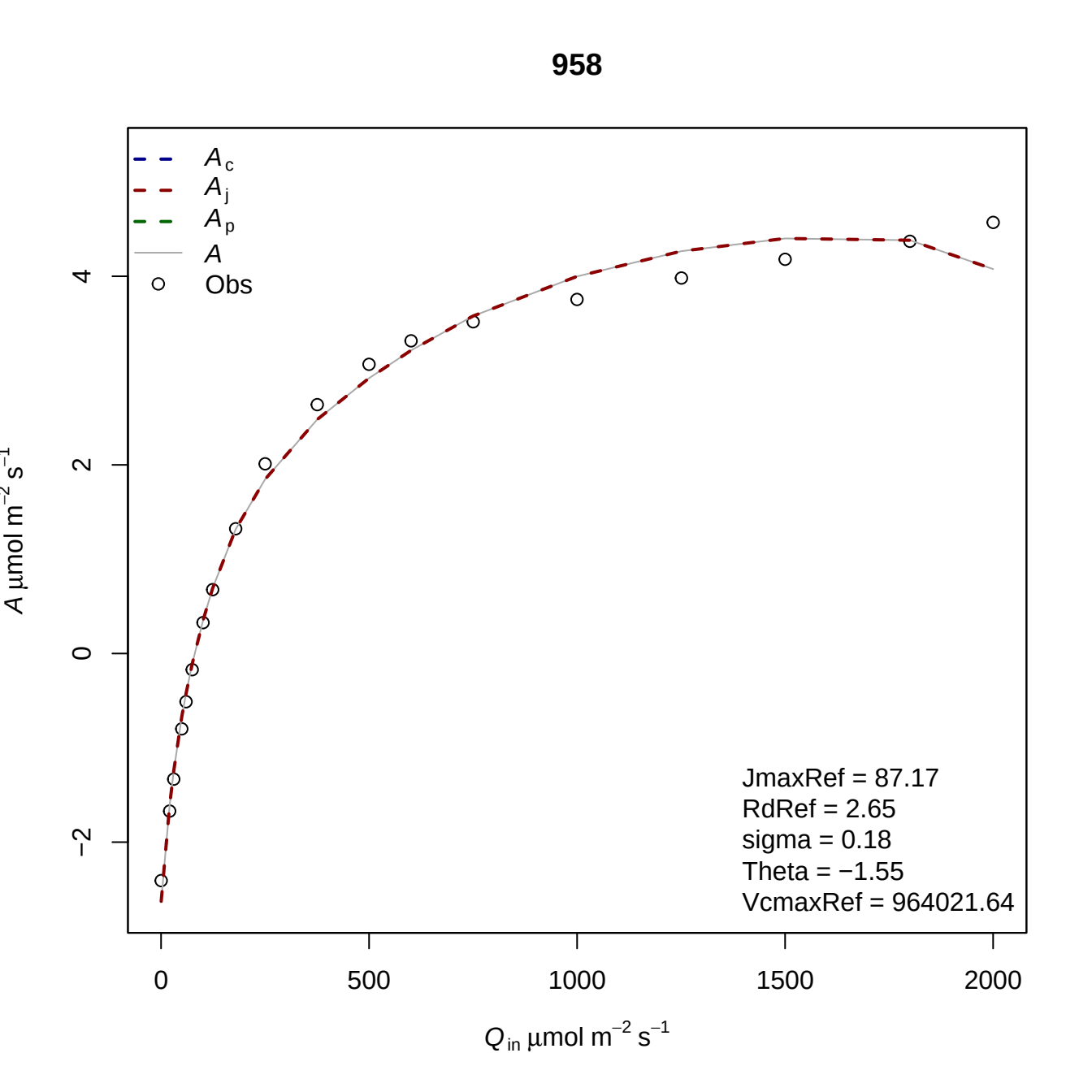

953
A_c
A_j
A_p
A
Obs
JmaxRef = 95.54
RdRef = 2.09
sigma = 0.19
Theta = −0.52
VcmaxRef = 1229292.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

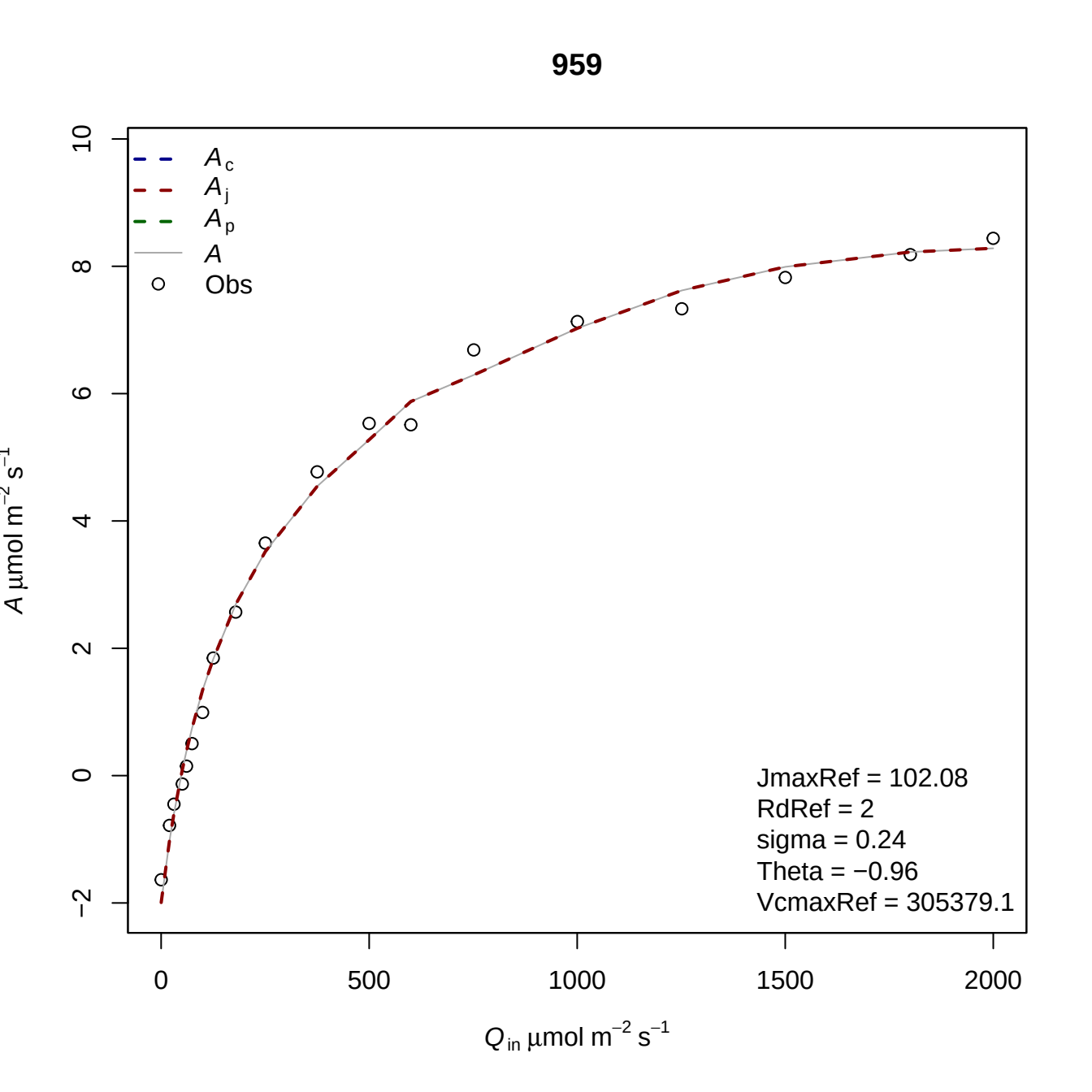

954
A_c
A_j
A_p
A
Obs
JmaxRef = 160.77
RdRef = 3.13
sigma = 0.26
Theta = 0.08
VcmaxRef = 2876846.73
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

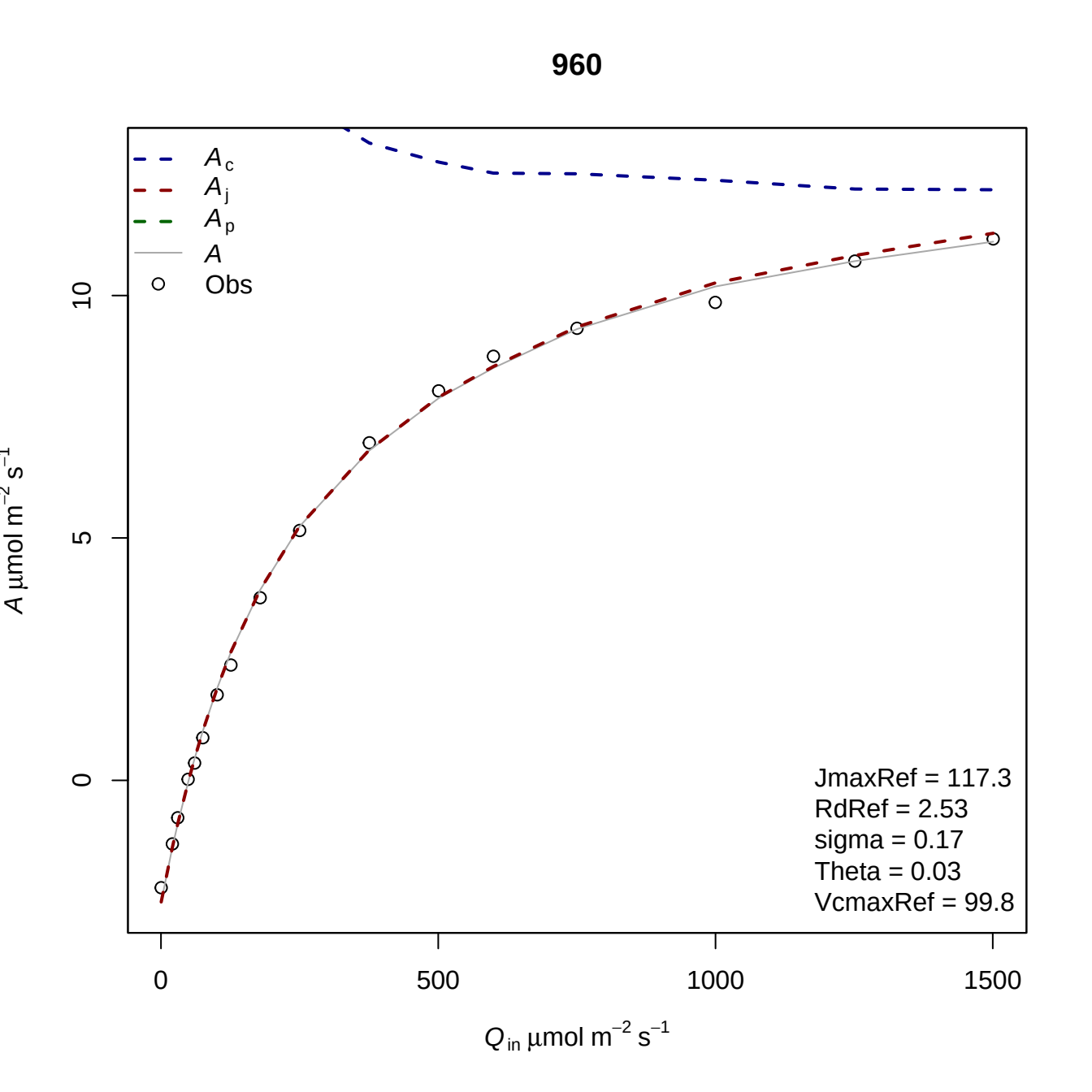

955
A_c
A_j
A_p
A
Obs
JmaxRef = 70.22
RdRef = 2.6
sigma = 0.29
Theta = −0.49
VcmaxRef = 35465.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

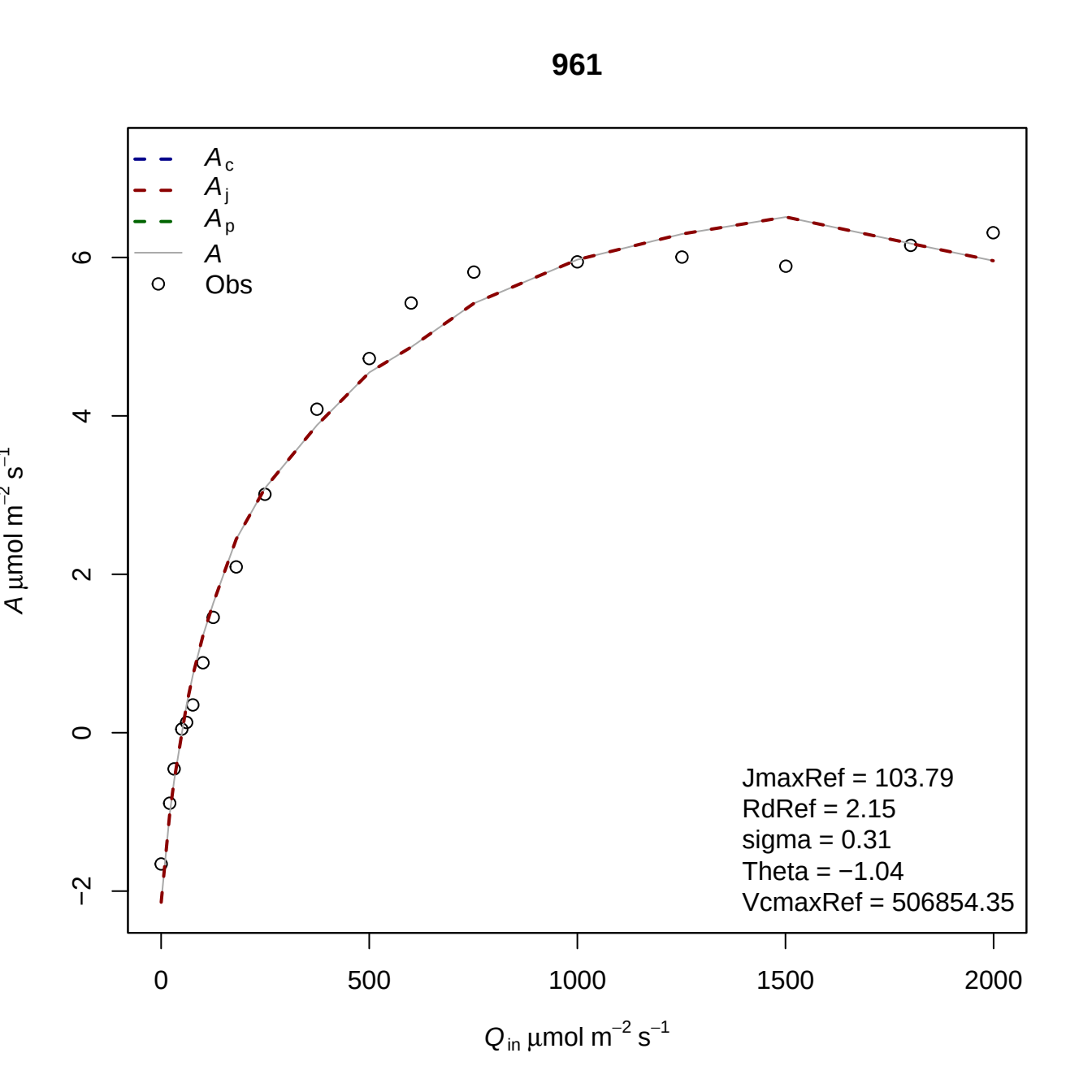

956
A_c
A_j
A_p
A
Obs
JmaxRef = 99.09
RdRef = 1.94
sigma = 0.33
Theta = −0.7
VcmaxRef = 306045.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

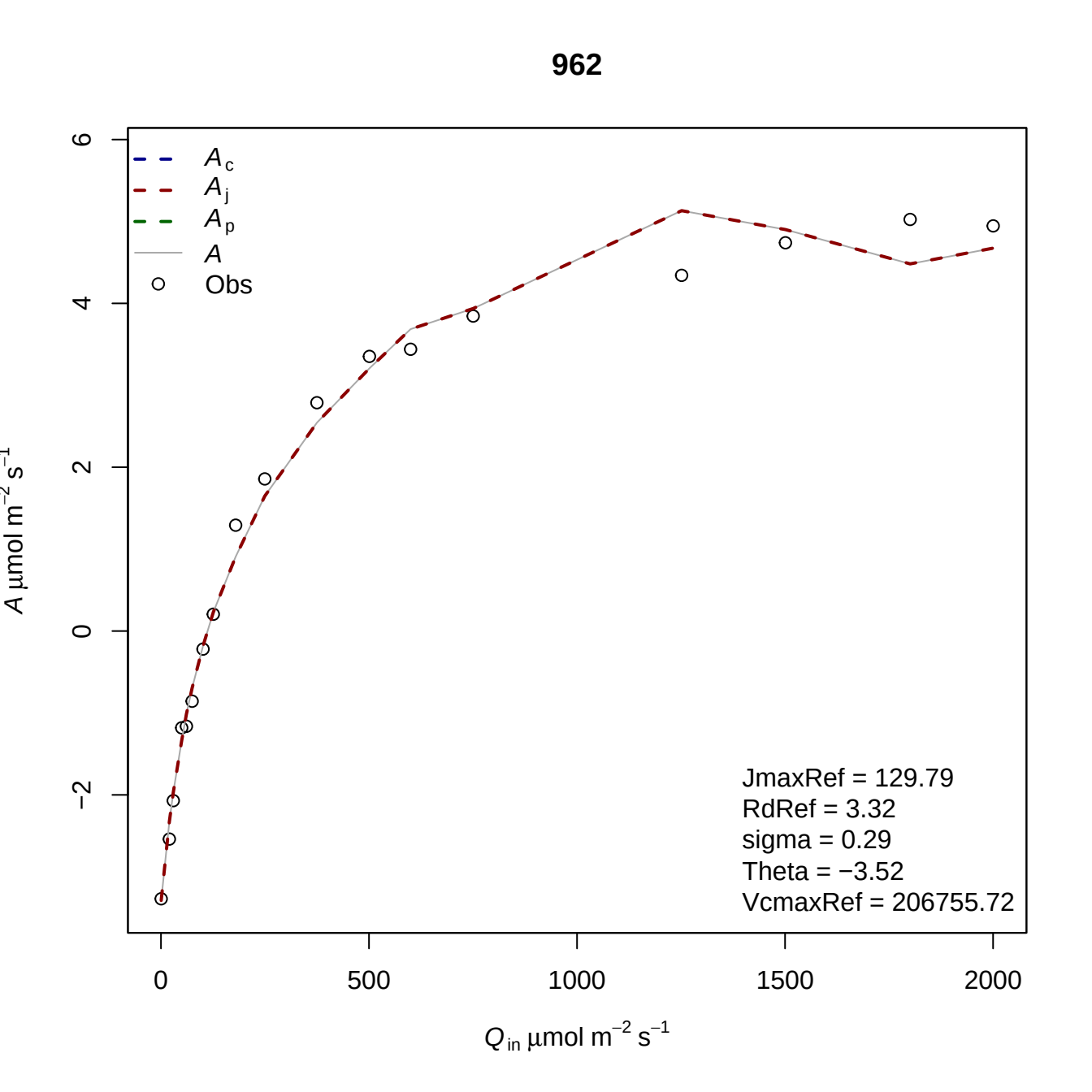

957
A_c
A_j
A_p
A
Obs
JmaxRef = 92.07
RdRef = 2.9
sigma = 0.21
Theta = −0.2
VcmaxRef = 28597.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

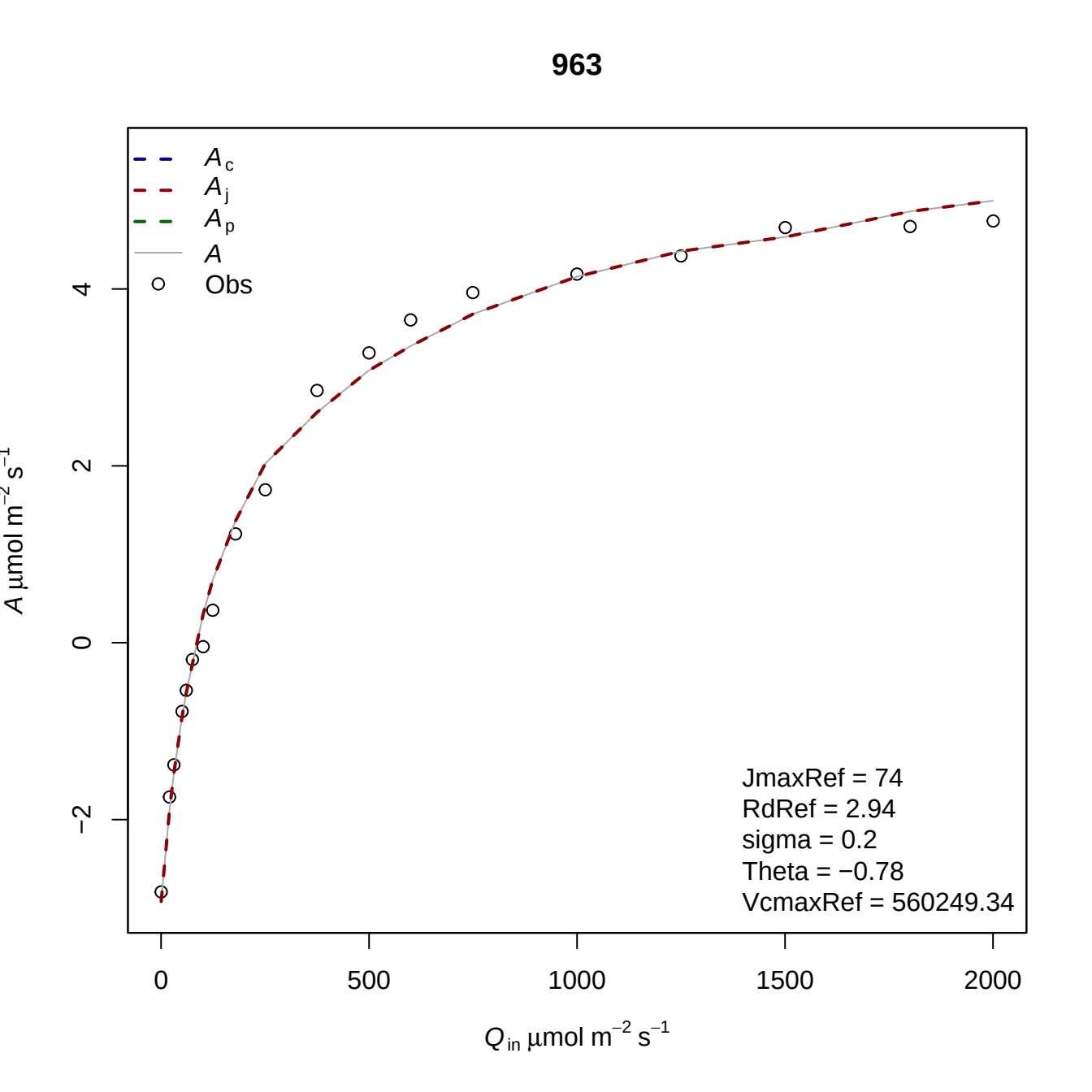

958
A_c
A_j
A_p
A
Obs
JmaxRef = 87.17
RdRef = 2.65
sigma = 0.18
Theta = −1.55
VcmaxRef = 964021.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

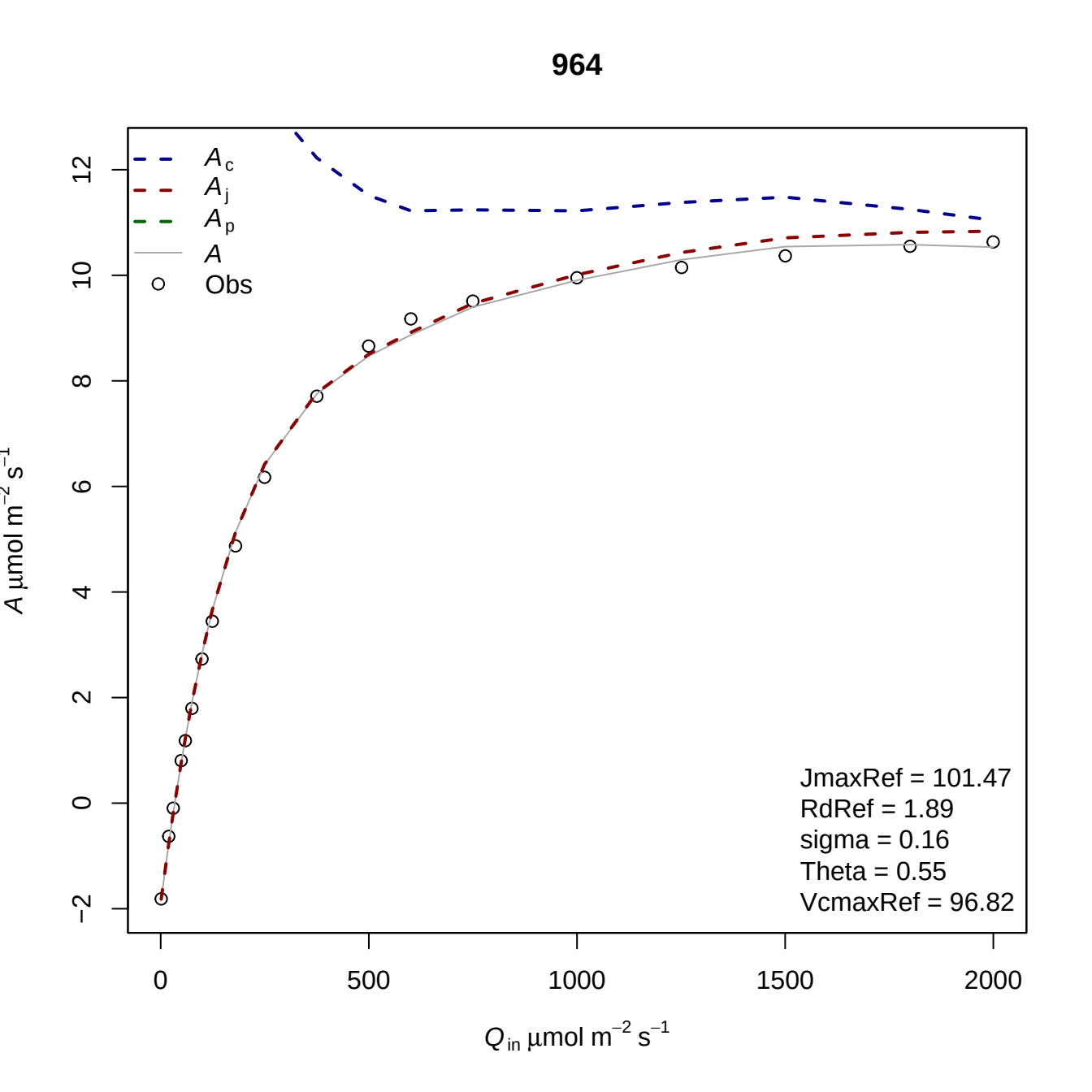

959
A_c
A_j
A_p
A
Obs
JmaxRef = 102.08
RdRef = 2
sigma = 0.24
Theta = −0.96
VcmaxRef = 305379.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

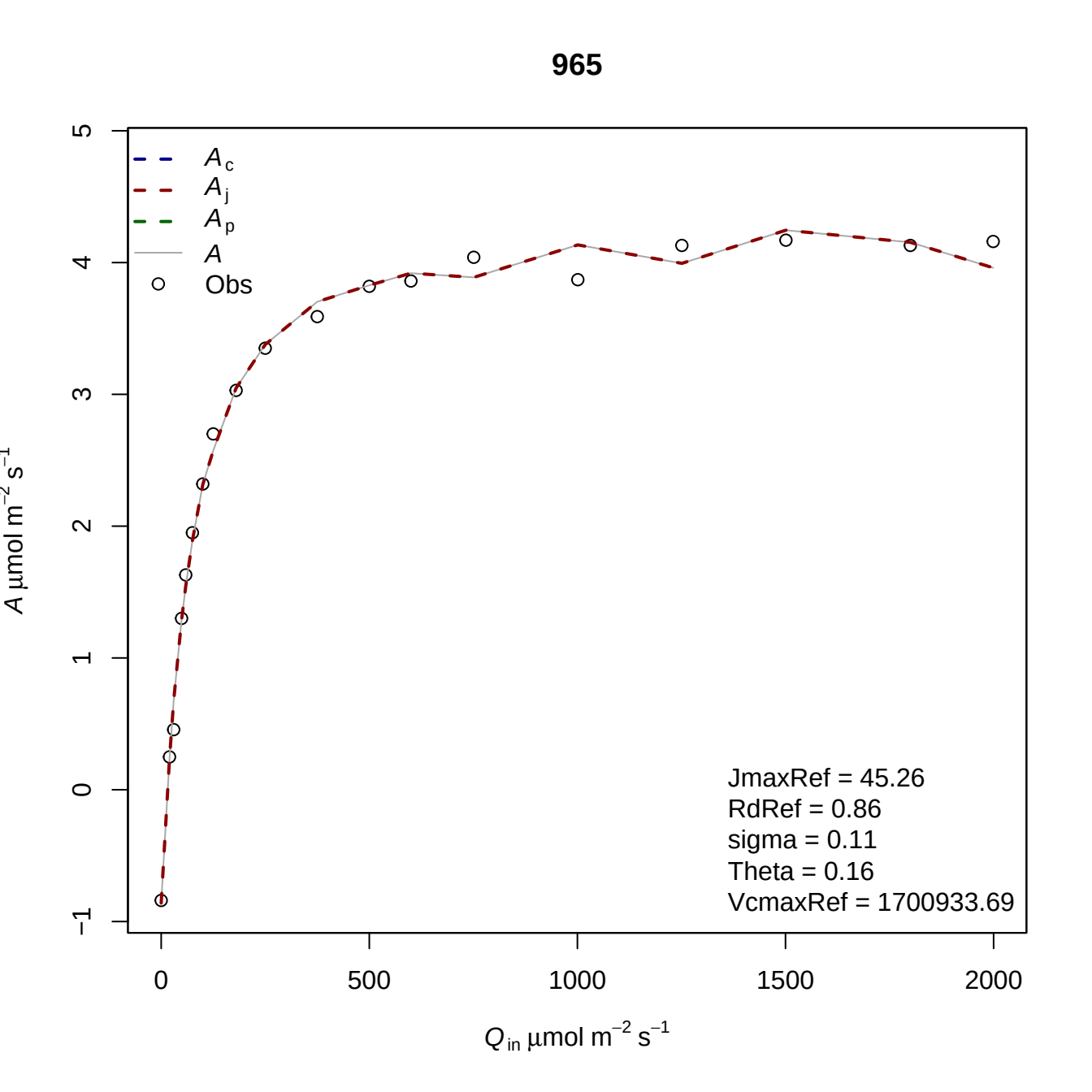

960
A_c
A_j
A_p
A
Obs
JmaxRef = 117.3
RdRef = 2.53
sigma = 0.17
Theta = 0.03
VcmaxRef = 99.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

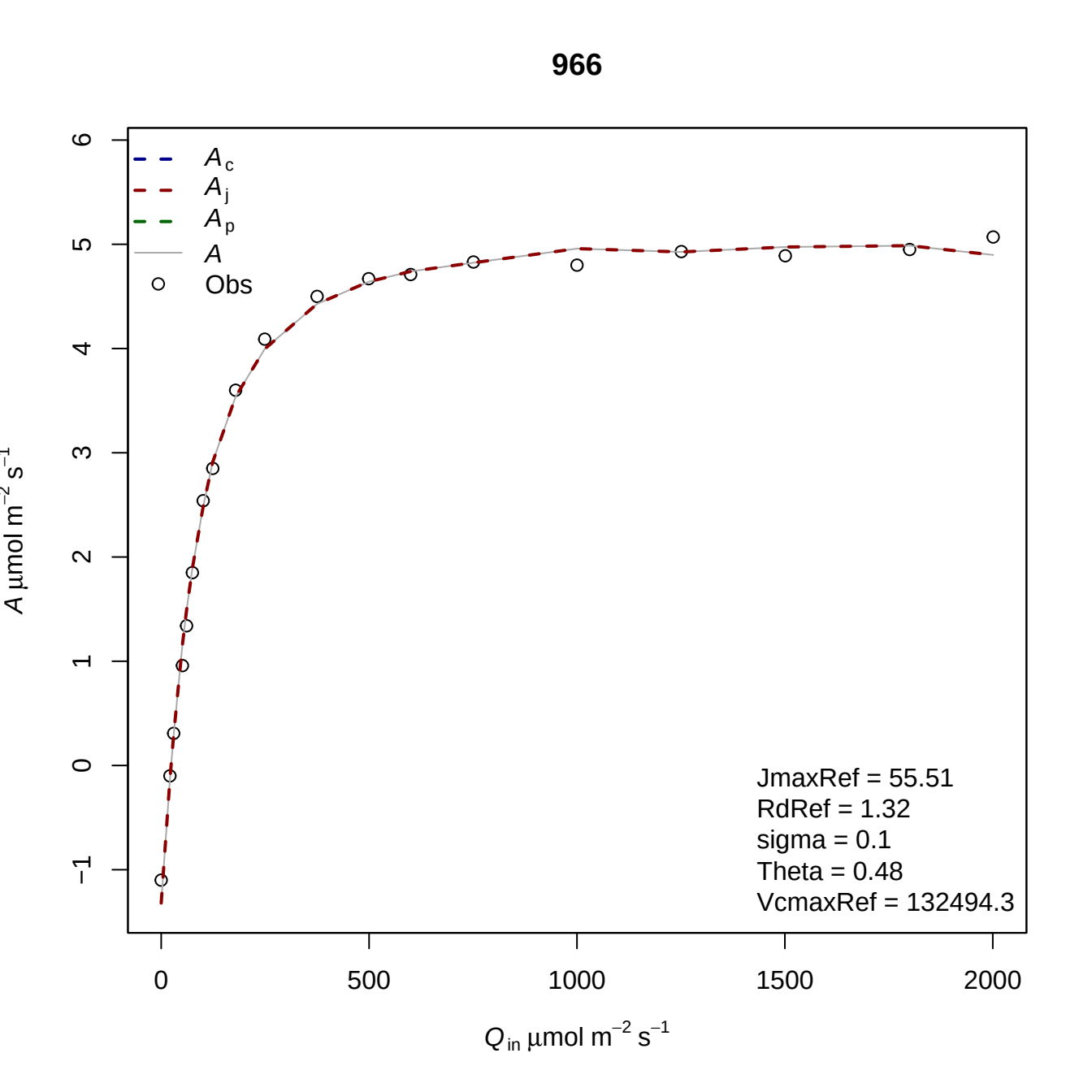

961
A_c
A_j
A_p
A
Obs
JmaxRef = 103.79
RdRef = 2.15
sigma = 0.31
Theta = −1.04
VcmaxRef = 506854.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

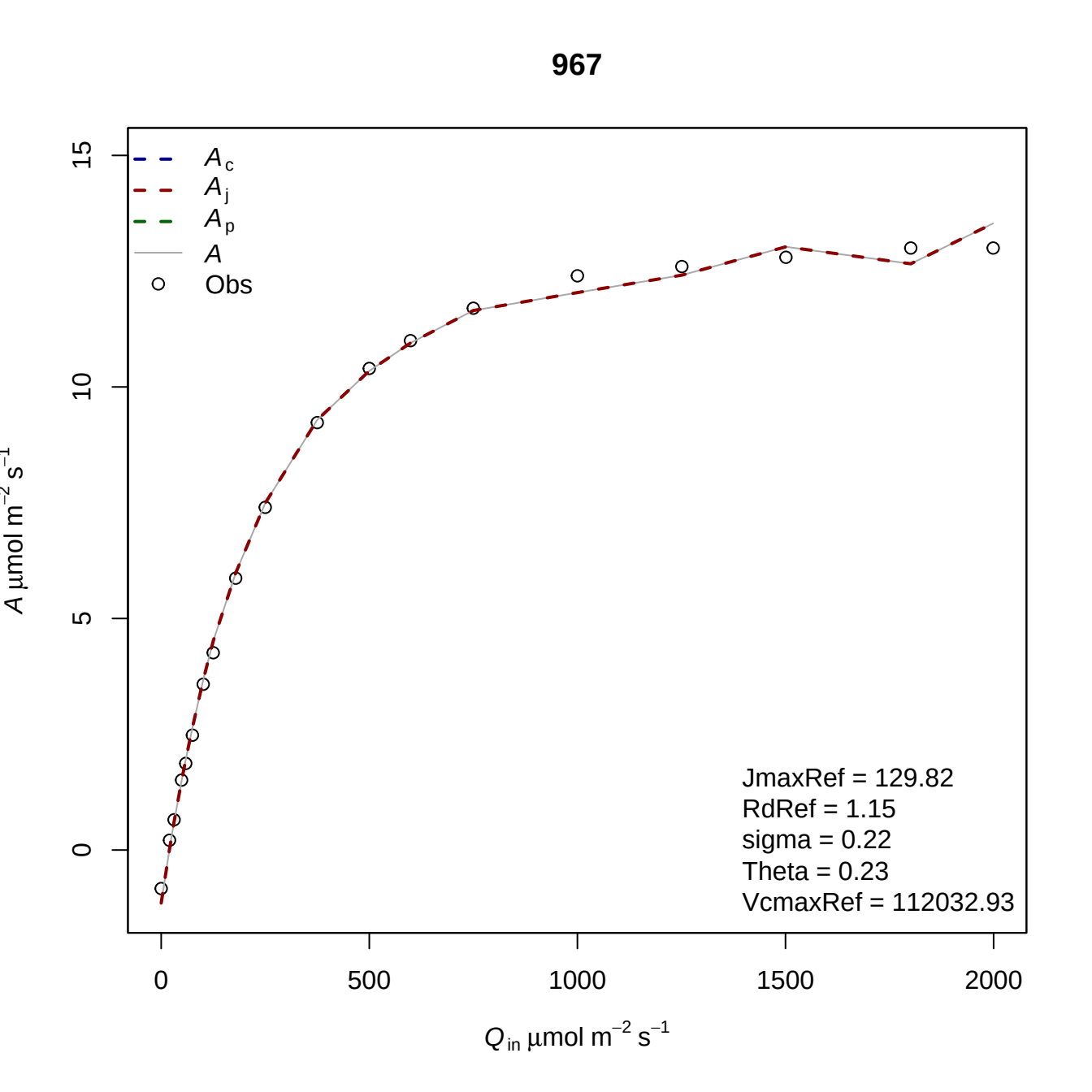

962
A_c
A_j
A_p
A
Obs
JmaxRef = 129.79
RdRef = 3.32
sigma = 0.29
Theta = −3.52
VcmaxRef = 206755.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

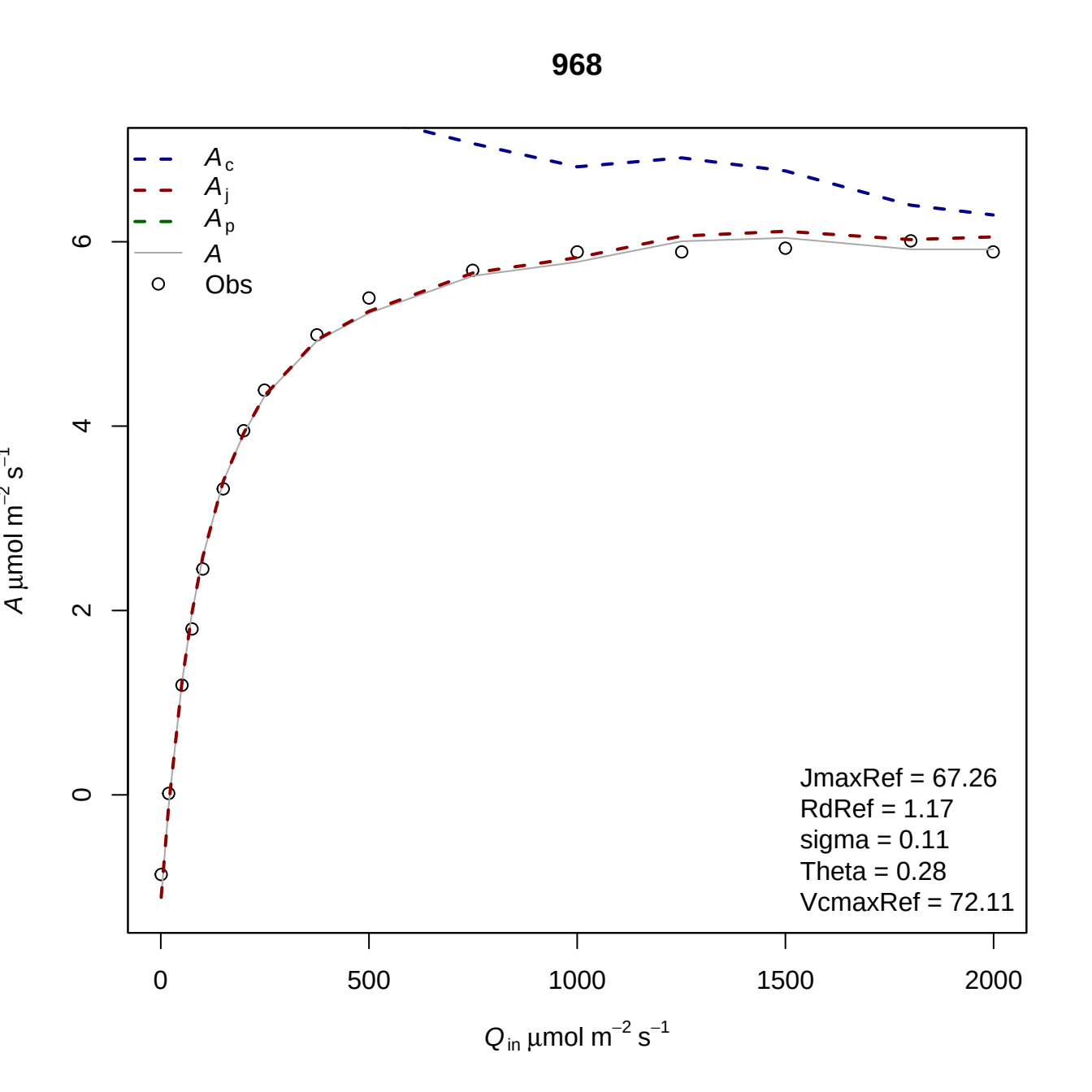

963
A_c
A_j
A_p
A
Obs
JmaxRef = 74
RdRef = 2.94
sigma = 0.2
Theta = −0.78
VcmaxRef = 560249.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

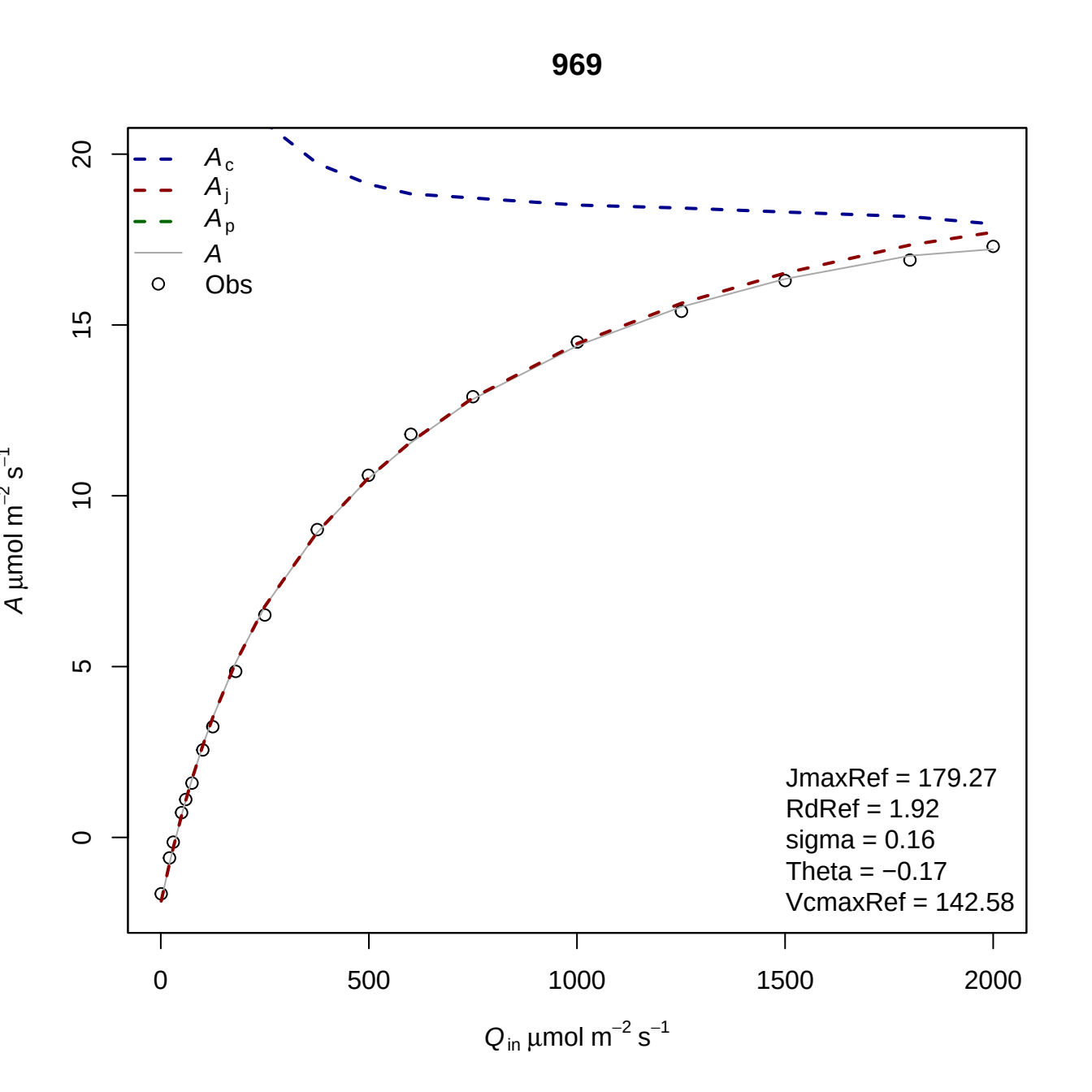

964
A_c
A_j
A_p
A
Obs
JmaxRef = 101.47
RdRef = 1.89
sigma = 0.16
Theta = 0.55
VcmaxRef = 96.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

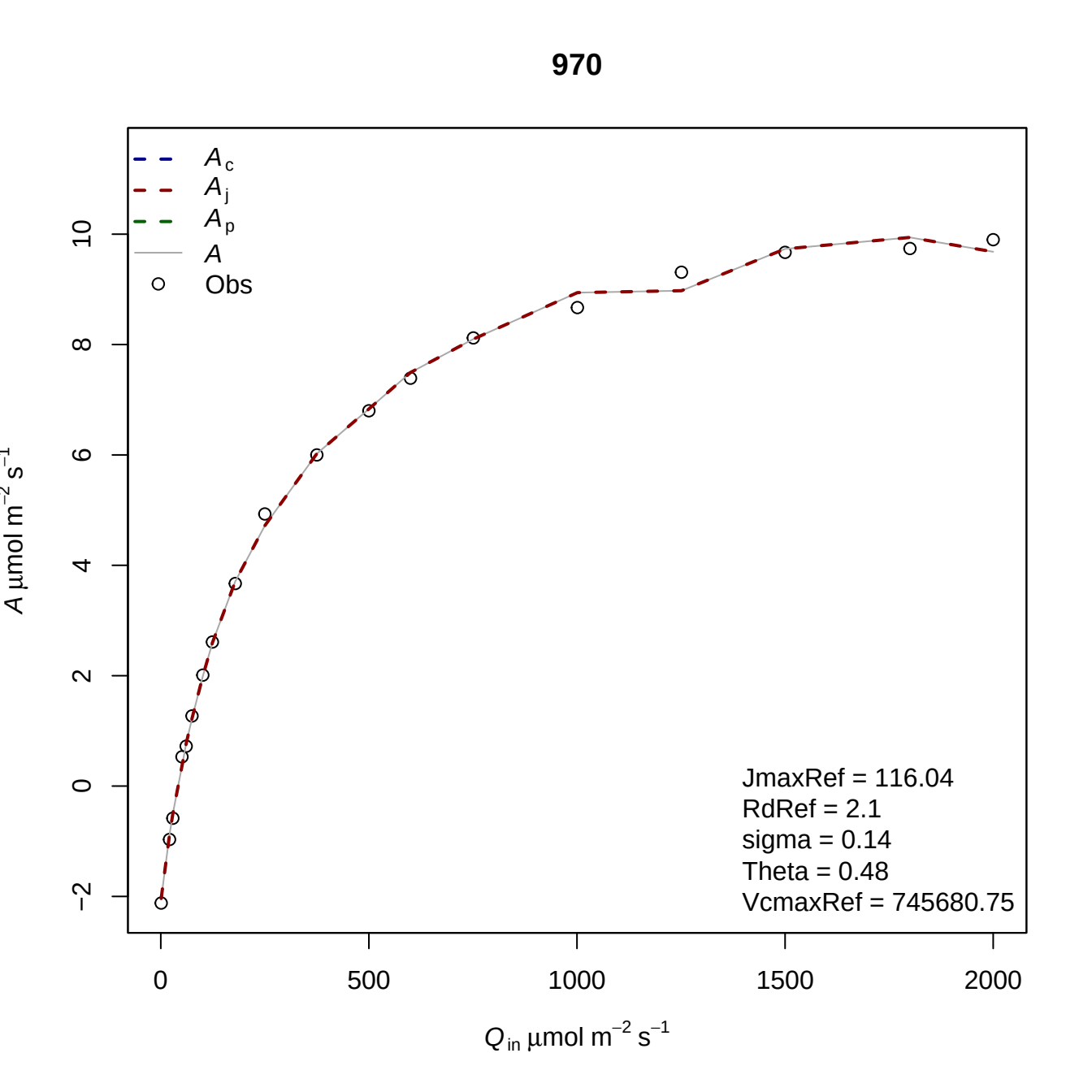

965
A_c
A_j
A_p
A
Obs
JmaxRef = 45.26
RdRef = 0.86
sigma = 0.11
Theta = 0.16
VcmaxRef = 1700933.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

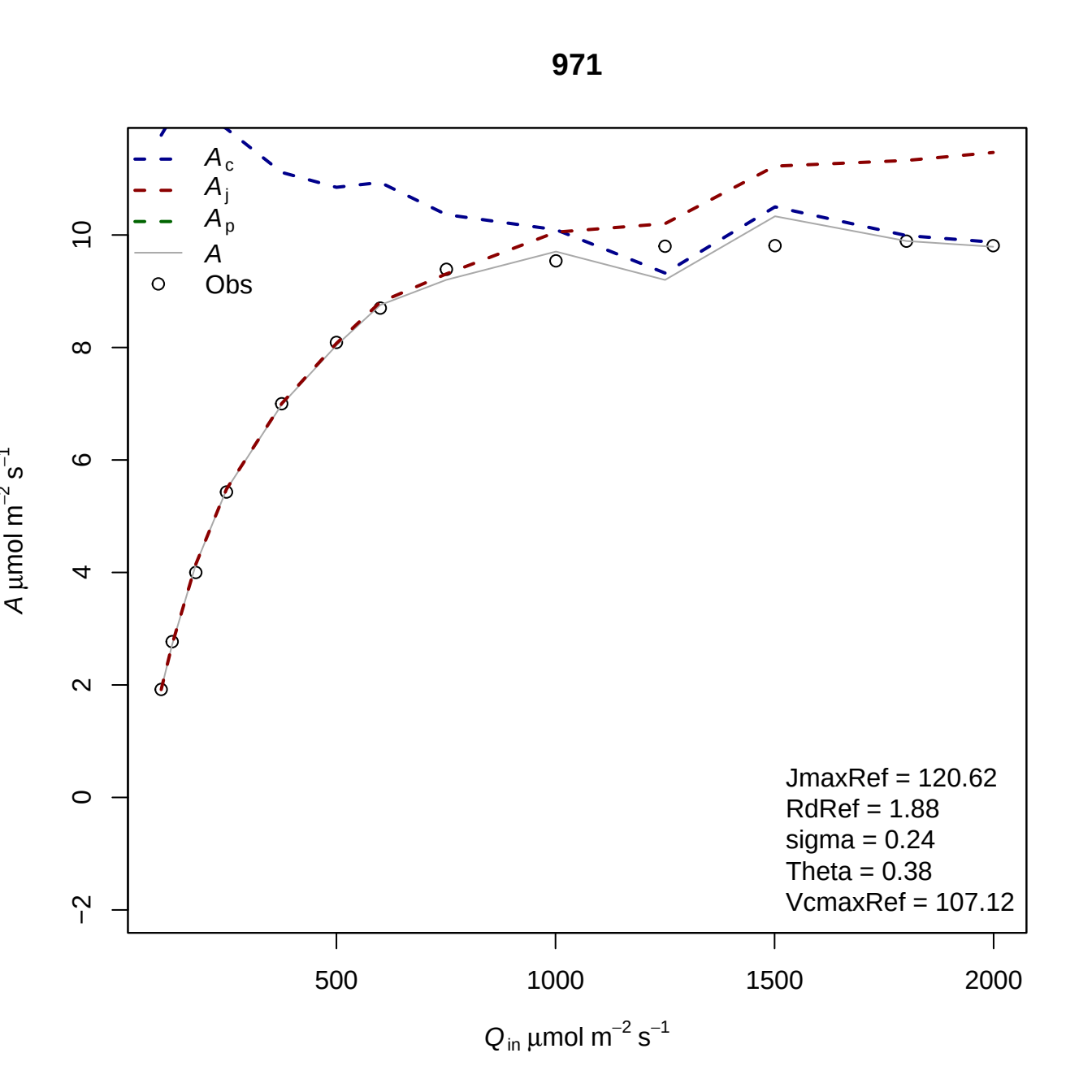

966
A_c
A_j
A_p
A
Obs
JmaxRef = 55.51
RdRef = 1.32
sigma = 0.1
Theta = 0.48
VcmaxRef = 132494.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

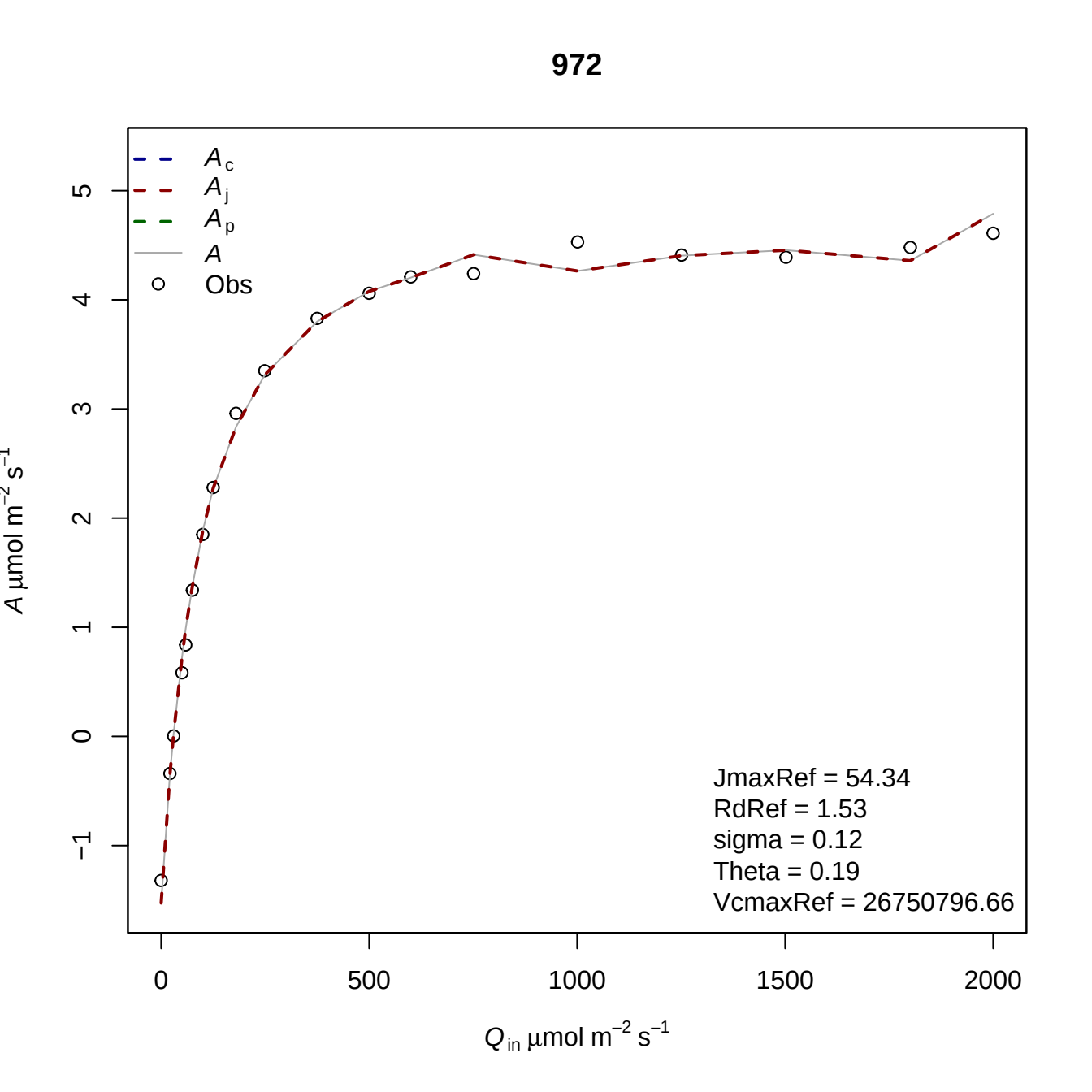

967
A_c
A_j
A_p
A
Obs
JmaxRef = 129.82
RdRef = 1.15
sigma = 0.22
Theta = 0.23
VcmaxRef = 112032.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

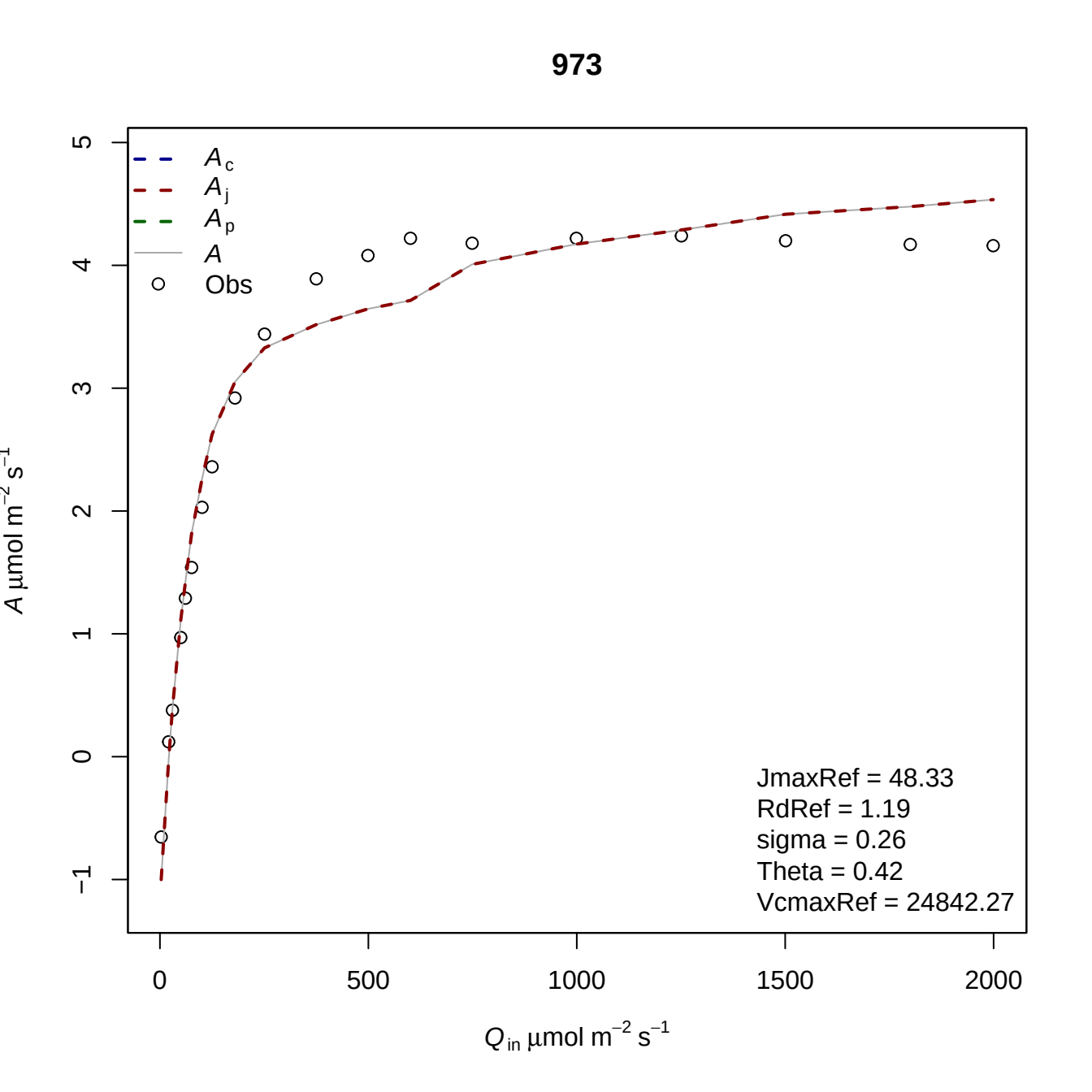

968
A_c
A_j
A_p
A
Obs
JmaxRef = 67.26
RdRef = 1.17
sigma = 0.11
Theta = 0.28
VcmaxRef = 72.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

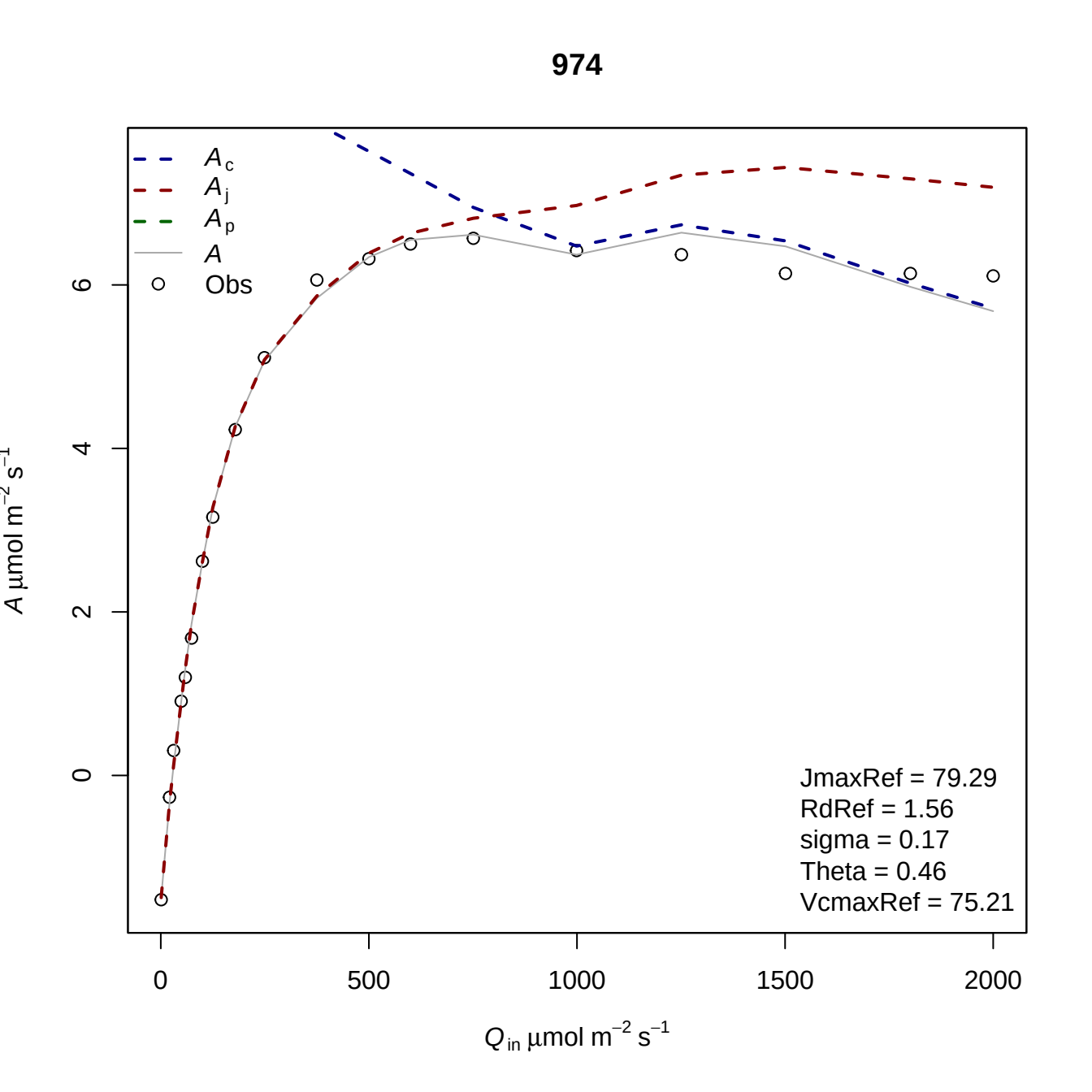

969
A_c
A_j
A_p
A
Obs
JmaxRef = 179.27
RdRef = 1.92
sigma = 0.16
Theta = −0.17
VcmaxRef = 142.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

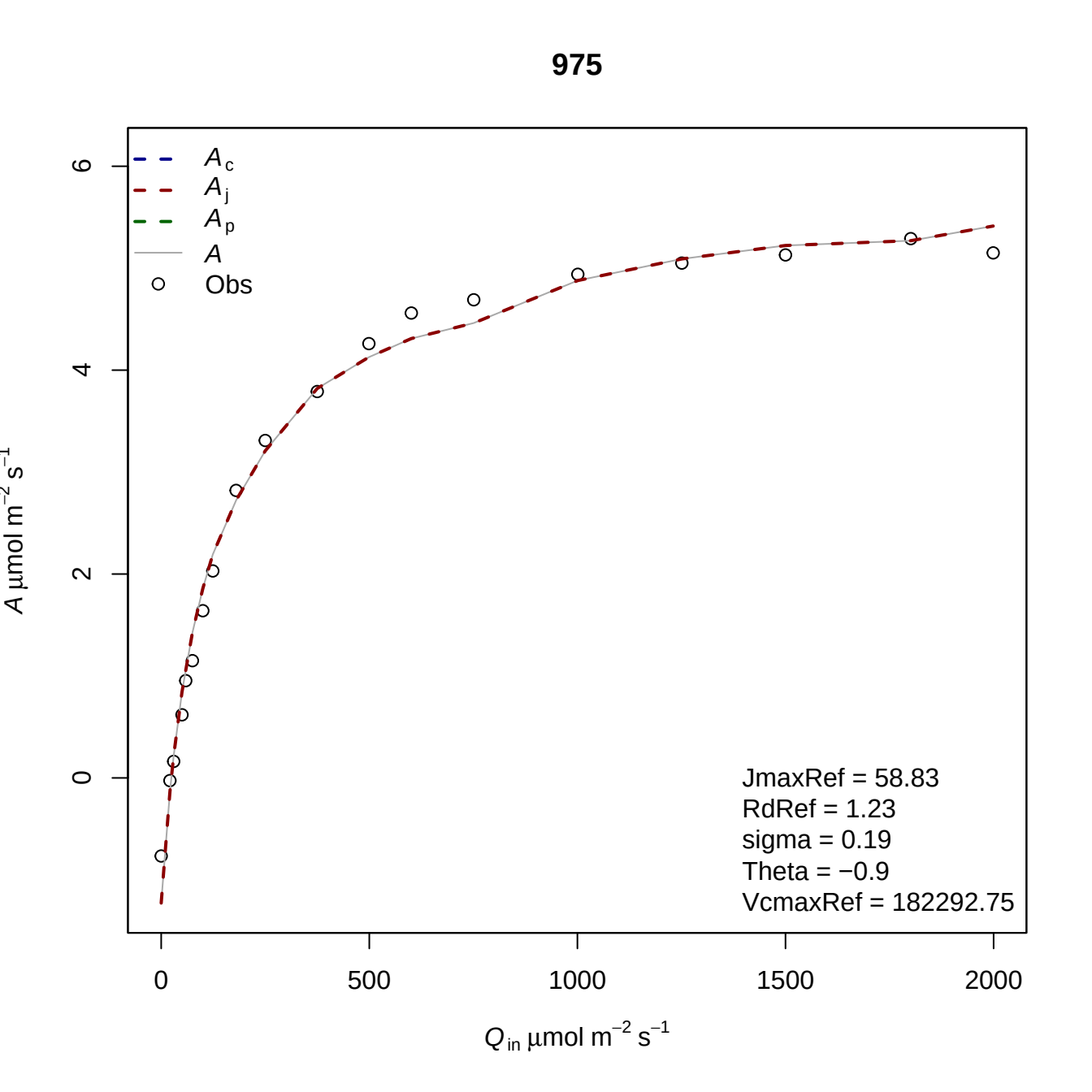

970
A_c
A_j
A_p
A
Obs
JmaxRef = 116.04
RdRef = 2.1
sigma = 0.14
Theta = 0.48
VcmaxRef = 745680.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

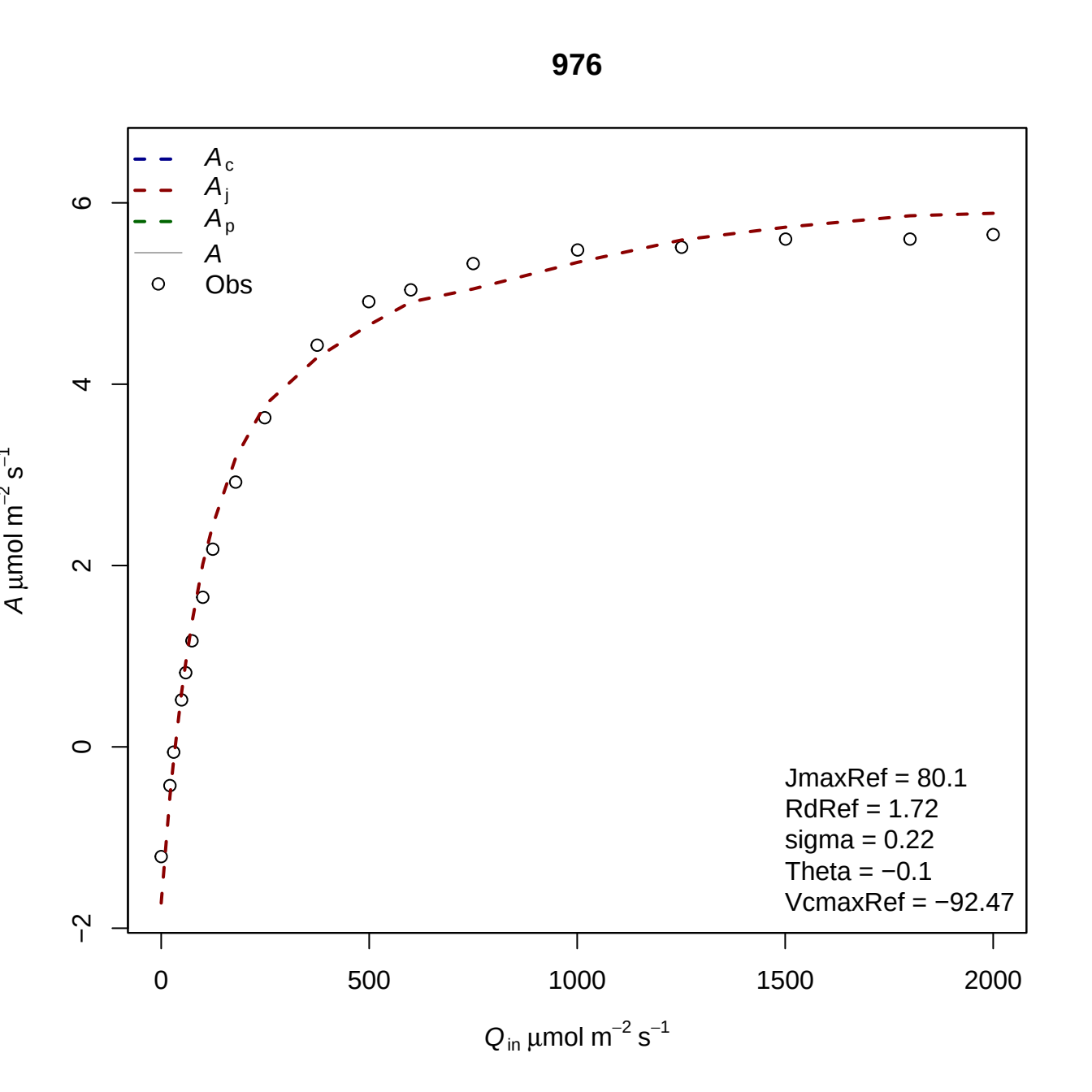

971
A_c
A_j
A_p
A
Obs
JmaxRef = 120.62
RdRef = 1.88
sigma = 0.24
Theta = 0.38
VcmaxRef = 107.12
−2
0
2
4
6
8
10
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

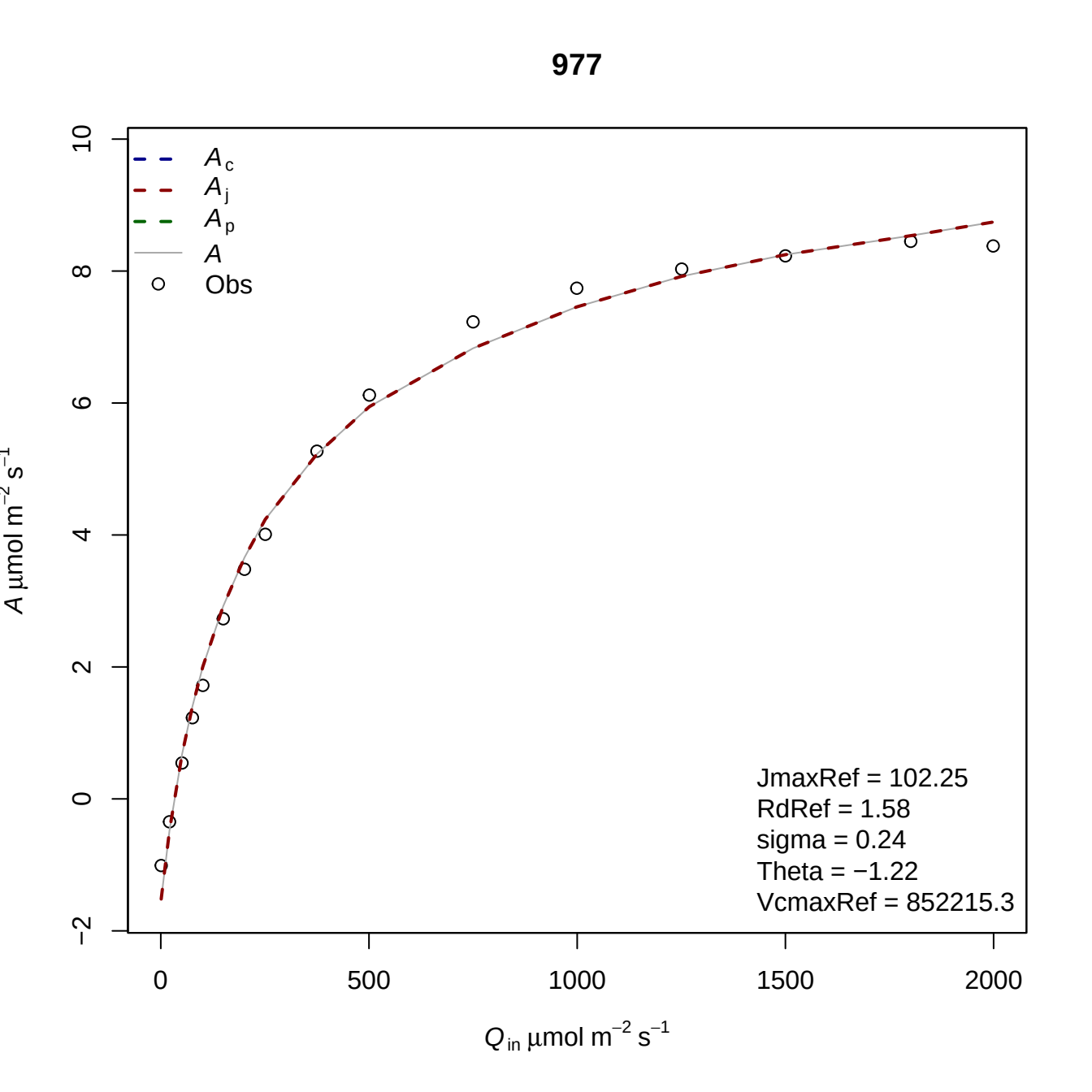

972
A_c
A_j
A_p
A
Obs
JmaxRef = 54.34
RdRef = 1.53
sigma = 0.12
Theta = 0.19
VcmaxRef = 26750796.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

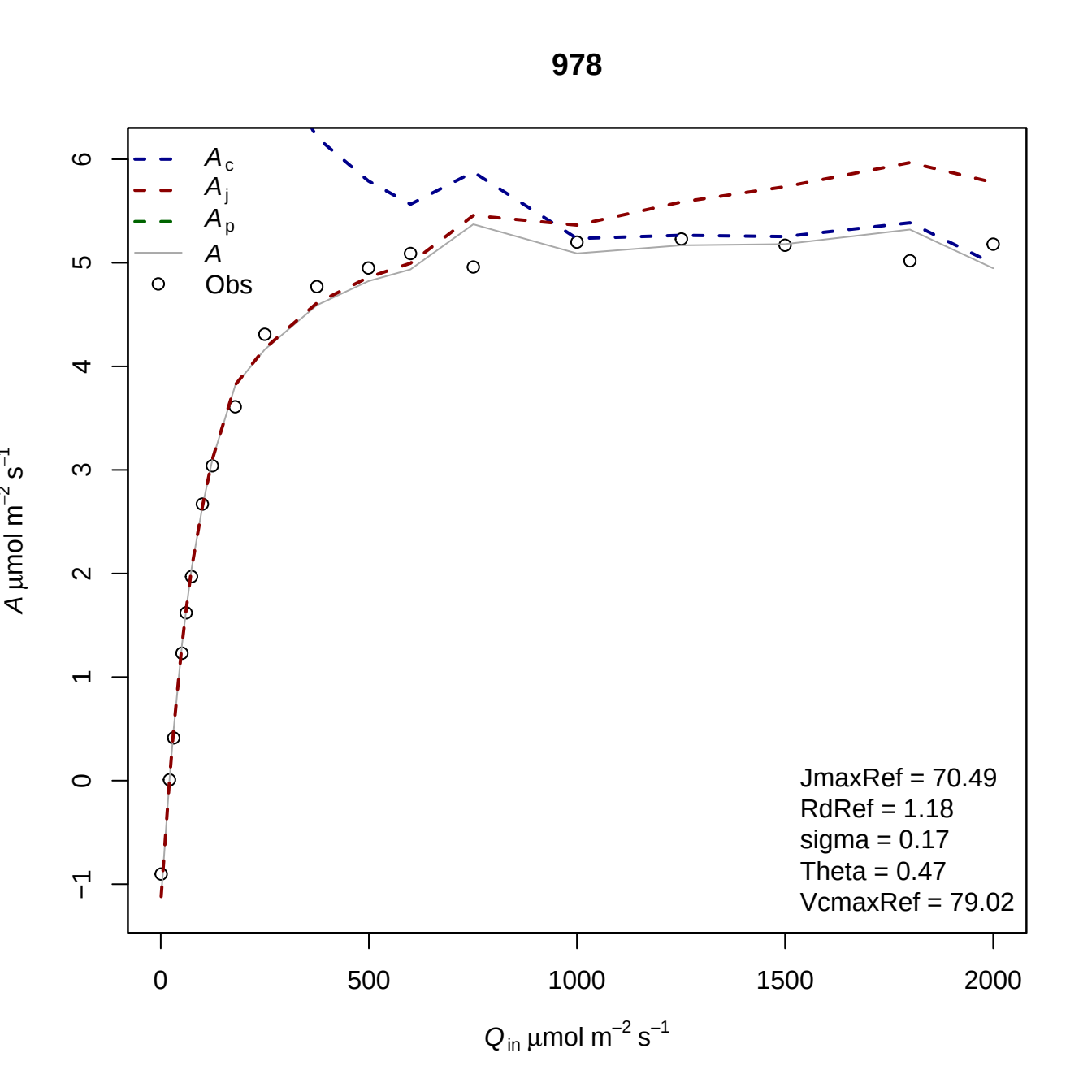

973
A_c
A_j
A_p
A
Obs
JmaxRef = 48.33
RdRef = 1.19
sigma = 0.26
Theta = 0.42
VcmaxRef = 24842.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

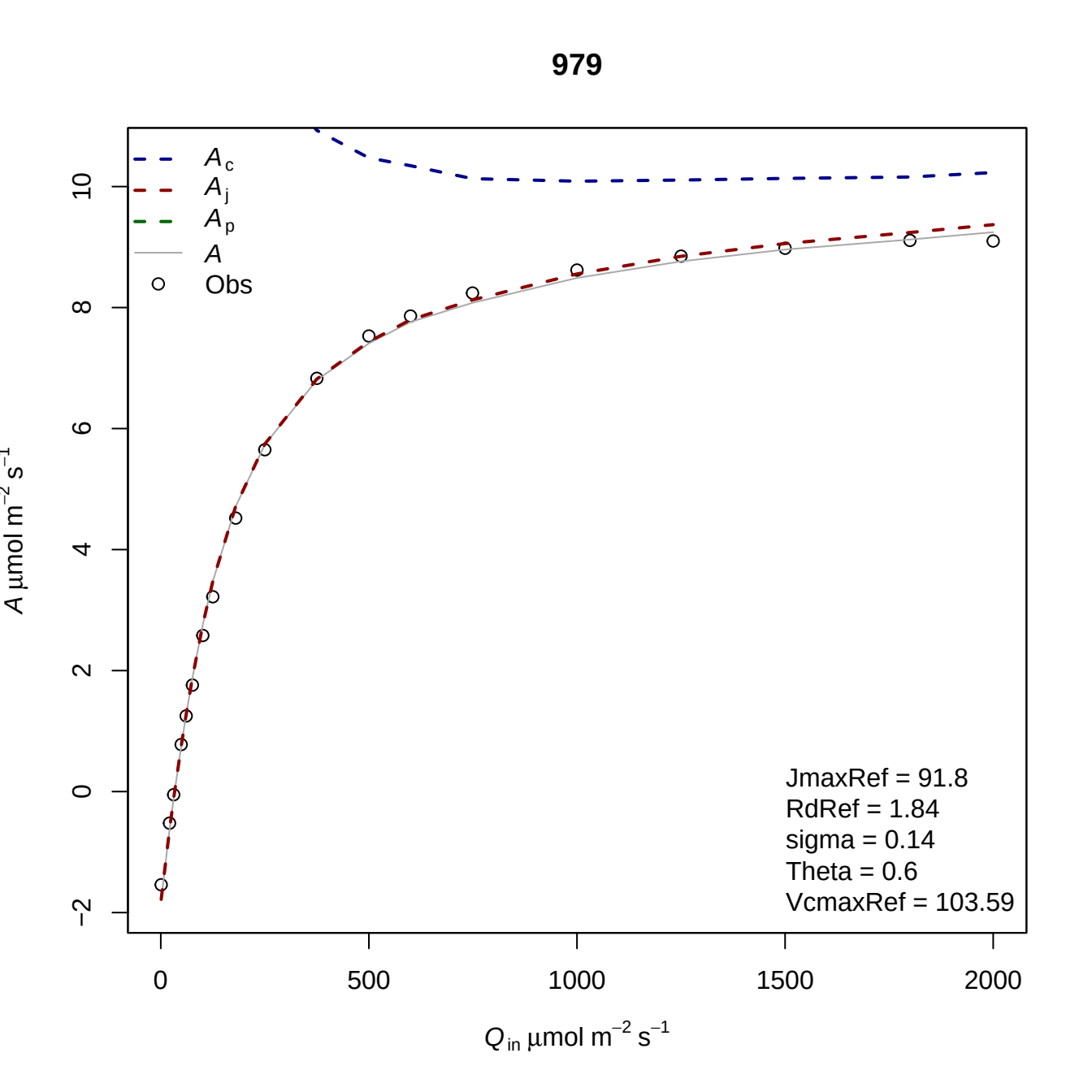

974
A_c
A_j
A_p
A
Obs
JmaxRef = 79.29
RdRef = 1.56
sigma = 0.17
Theta = 0.46
VcmaxRef = 75.21
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

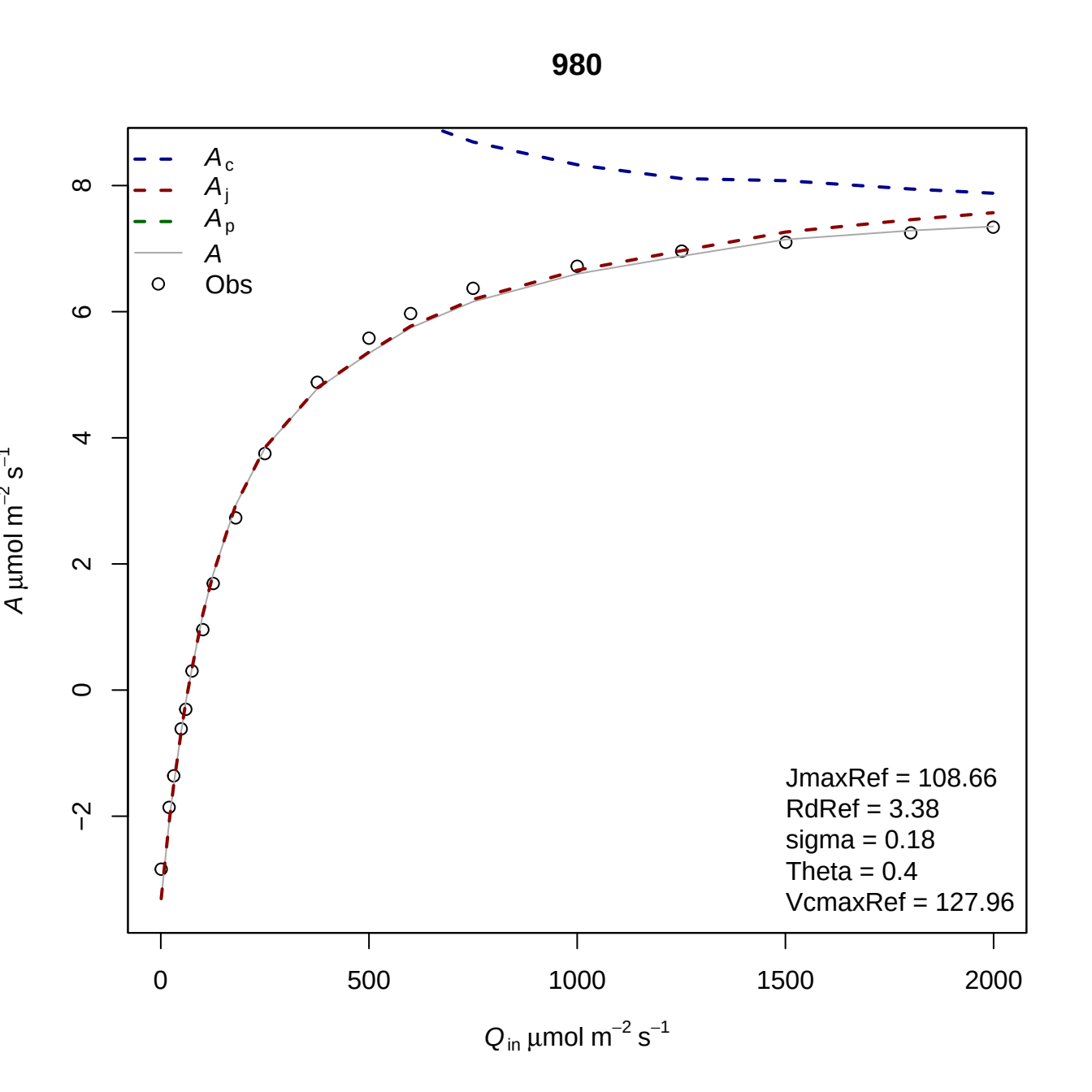

975
A_c
A_j
A_p
A
Obs
JmaxRef = 58.83
RdRef = 1.23
sigma = 0.19
Theta = −0.9
VcmaxRef = 182292.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

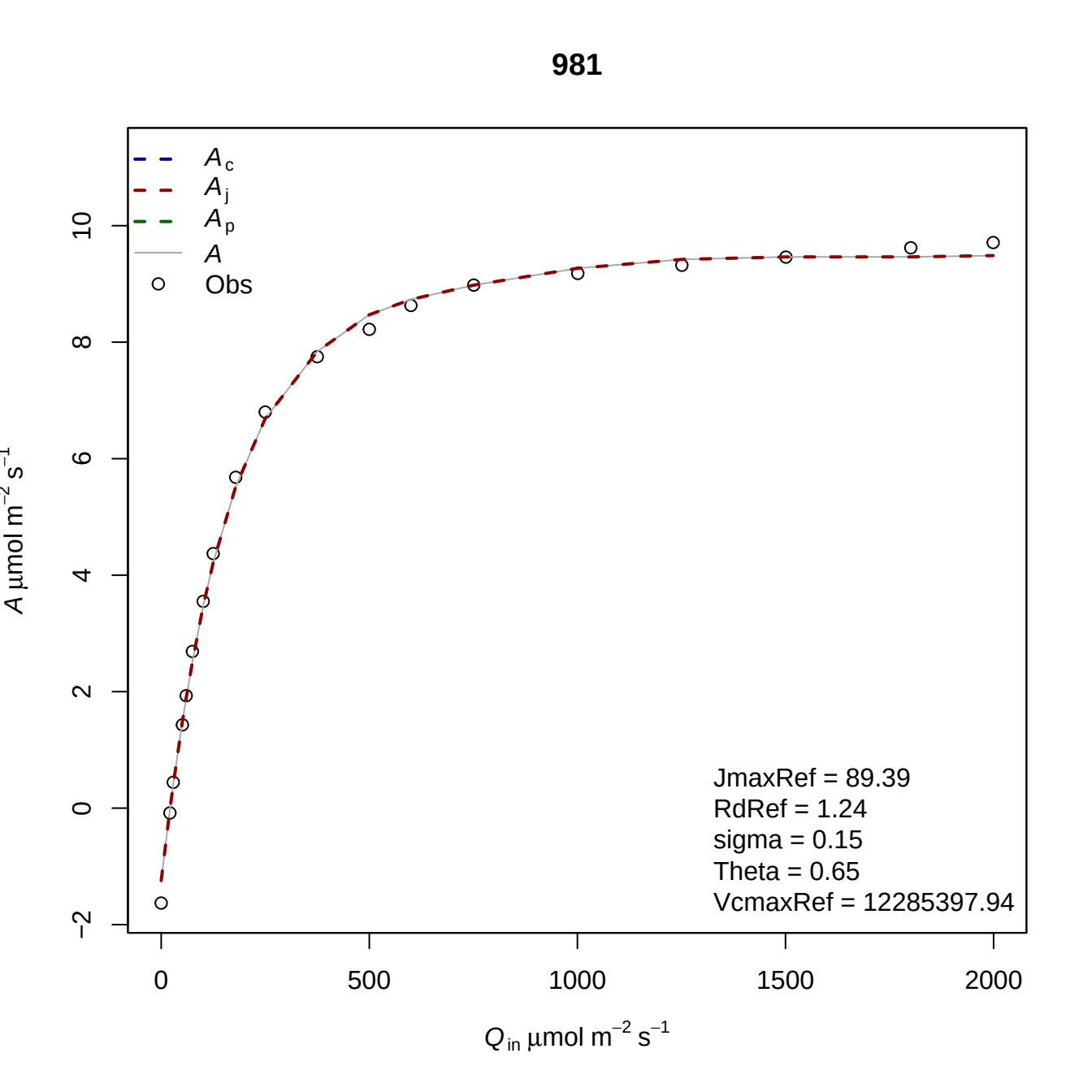

976
A_c
A_j
A_p
A
Obs
JmaxRef = 80.1
RdRef = 1.72
sigma = 0.22
Theta = −0.1
VcmaxRef = −92.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

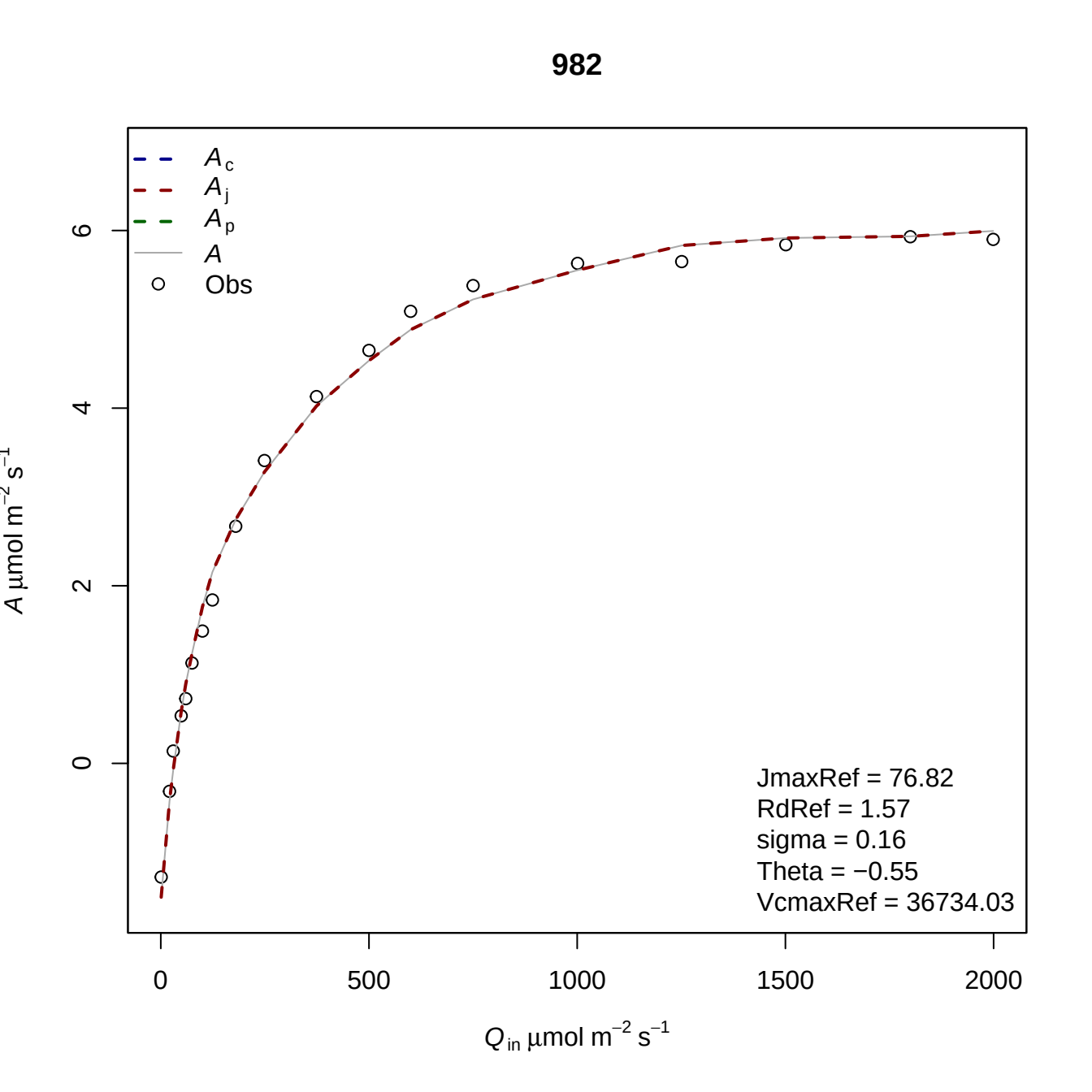

977
A_c
A_j
A_p
A
Obs
JmaxRef = 102.25
RdRef = 1.58
sigma = 0.24
Theta = −1.22
VcmaxRef = 852215.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

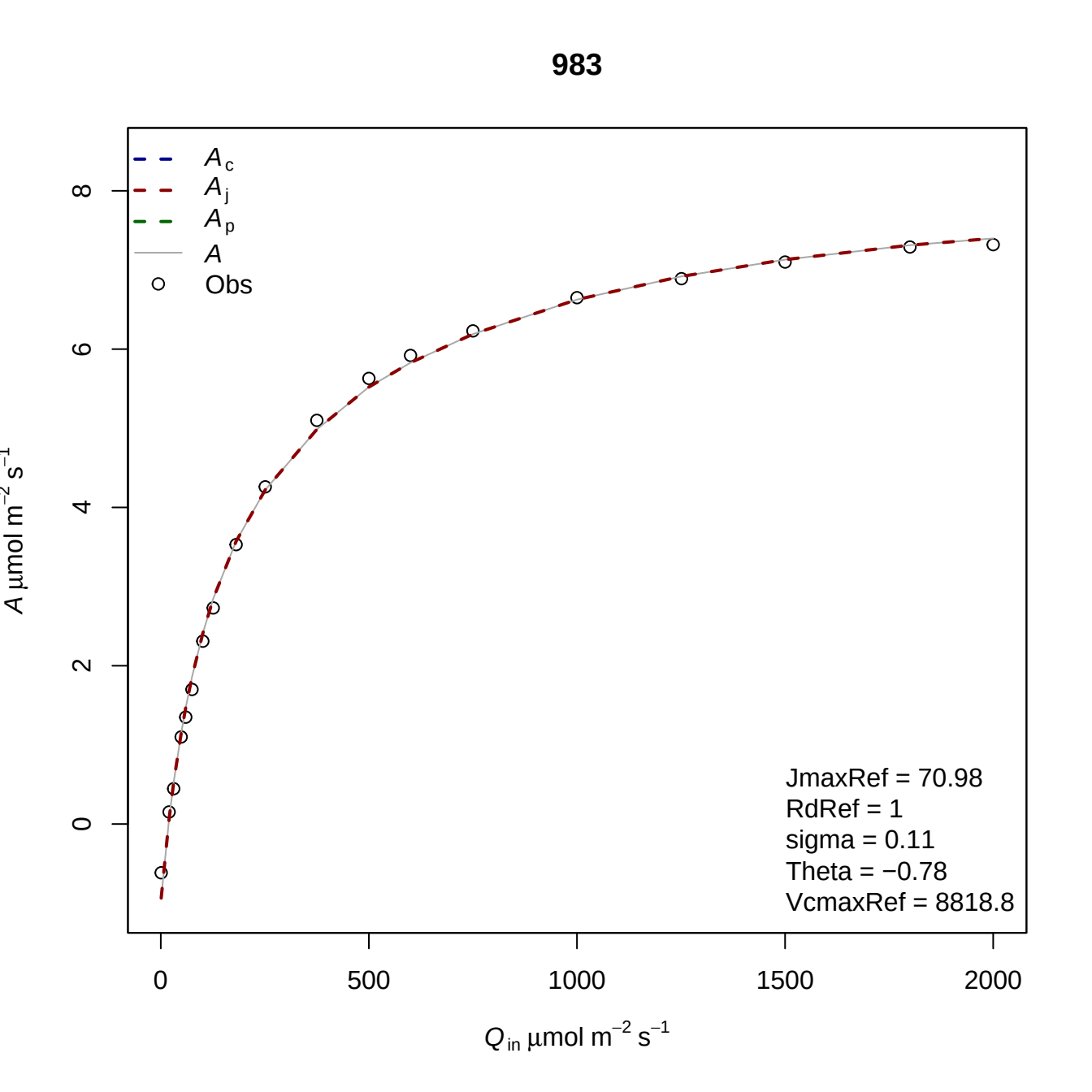

978
A_c
A_j
A_p
A
Obs
JmaxRef = 70.49
RdRef = 1.18
sigma = 0.17
Theta = 0.47
VcmaxRef = 79.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

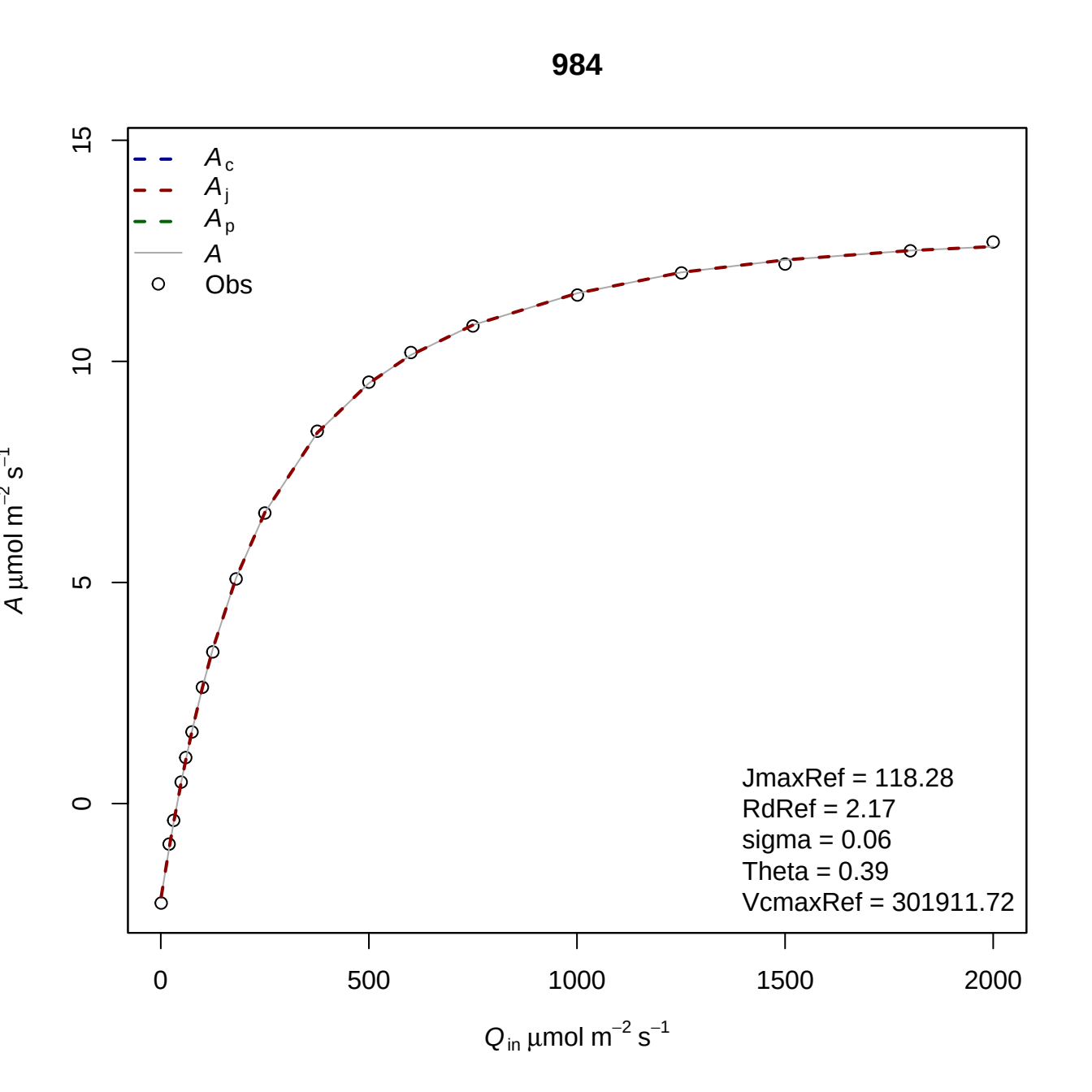

979
A_c
A_j
A_p
A
Obs
JmaxRef = 91.8
RdRef = 1.84
sigma = 0.14
Theta = 0.6
VcmaxRef = 103.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

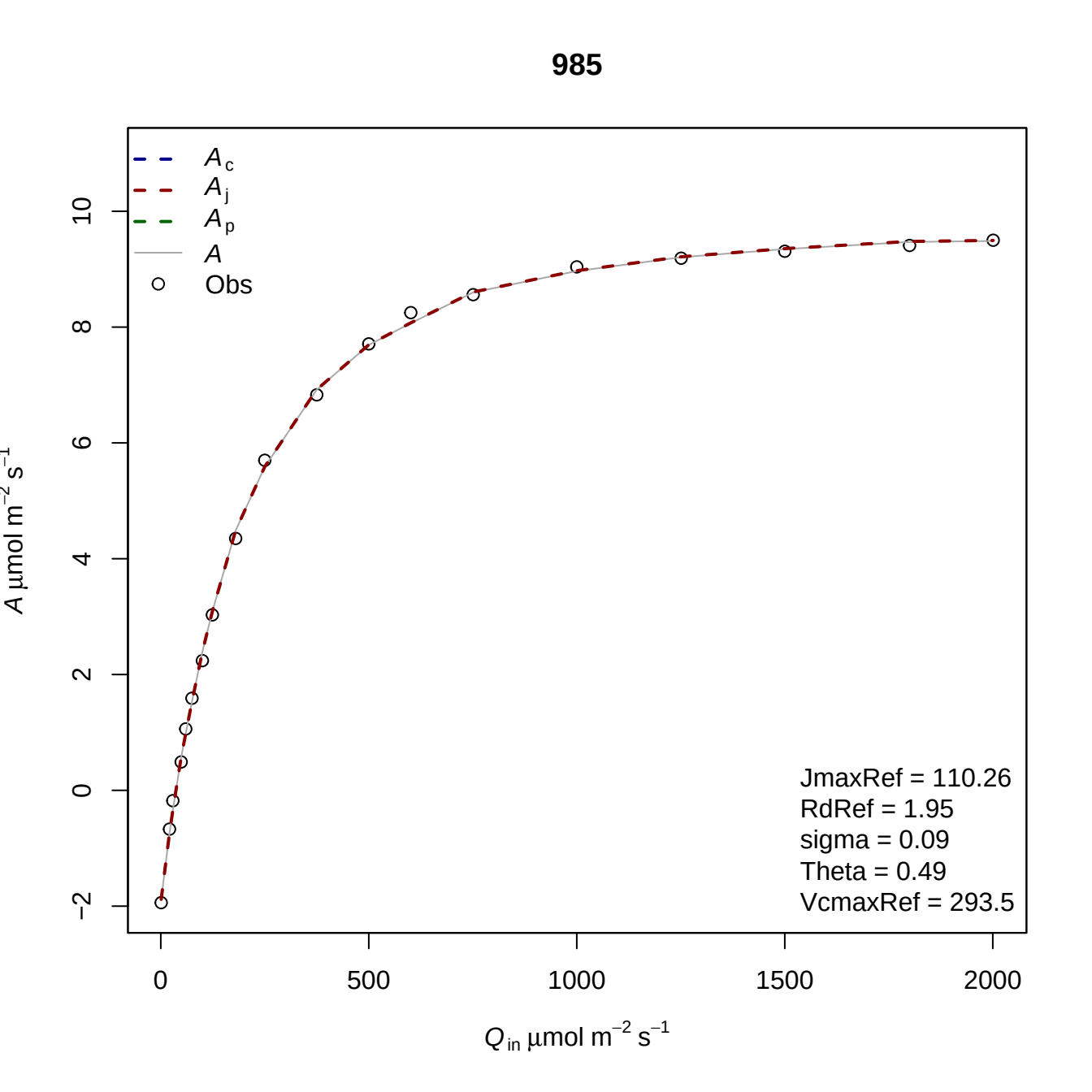

980
A_c
A_j
A_p
A
Obs
JmaxRef = 108.66
RdRef = 3.38
sigma = 0.18
Theta = 0.4
VcmaxRef = 127.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

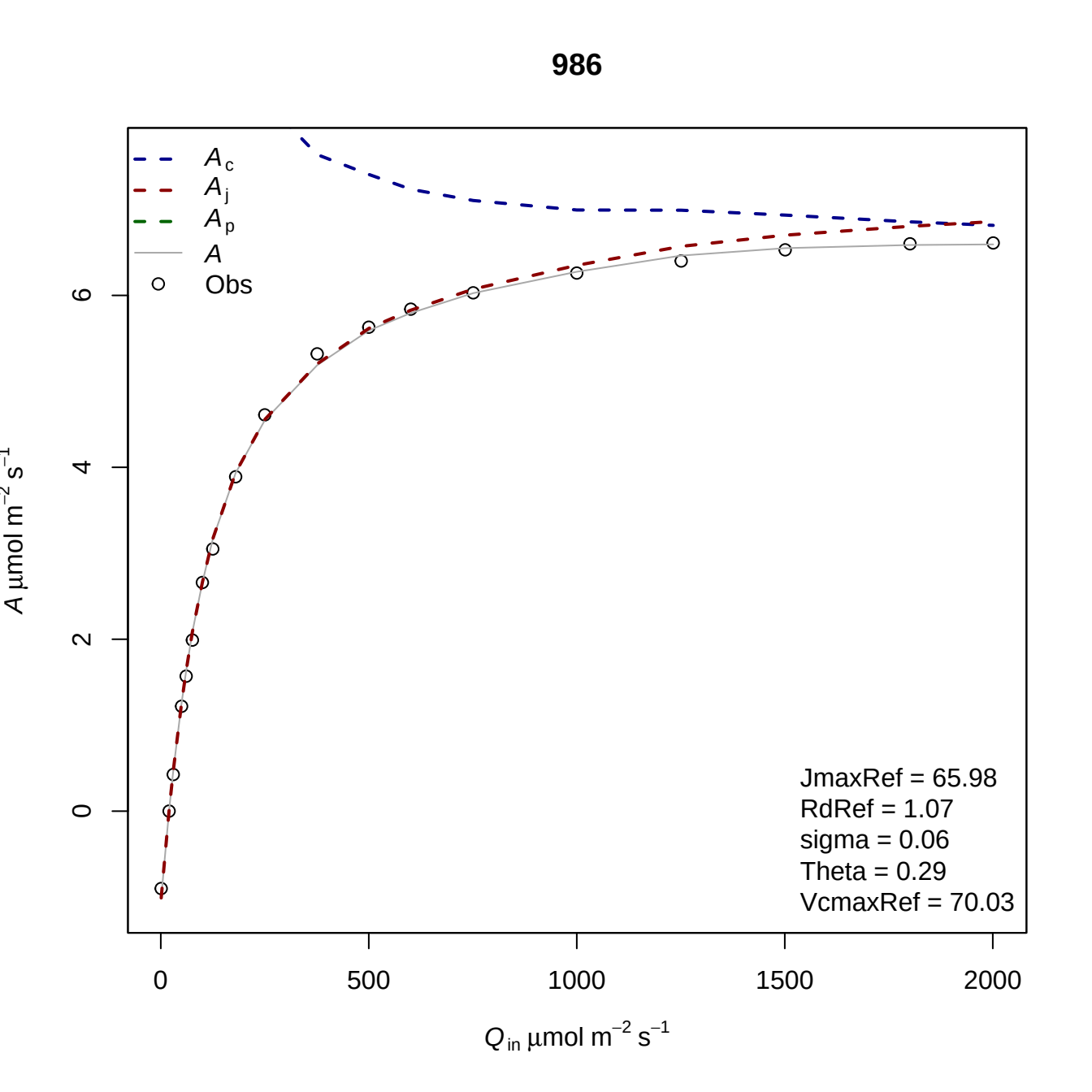

981
A_c
A_j
A_p
A
Obs
JmaxRef = 89.39
RdRef = 1.24
sigma = 0.15
Theta = 0.65
VcmaxRef = 12285397.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

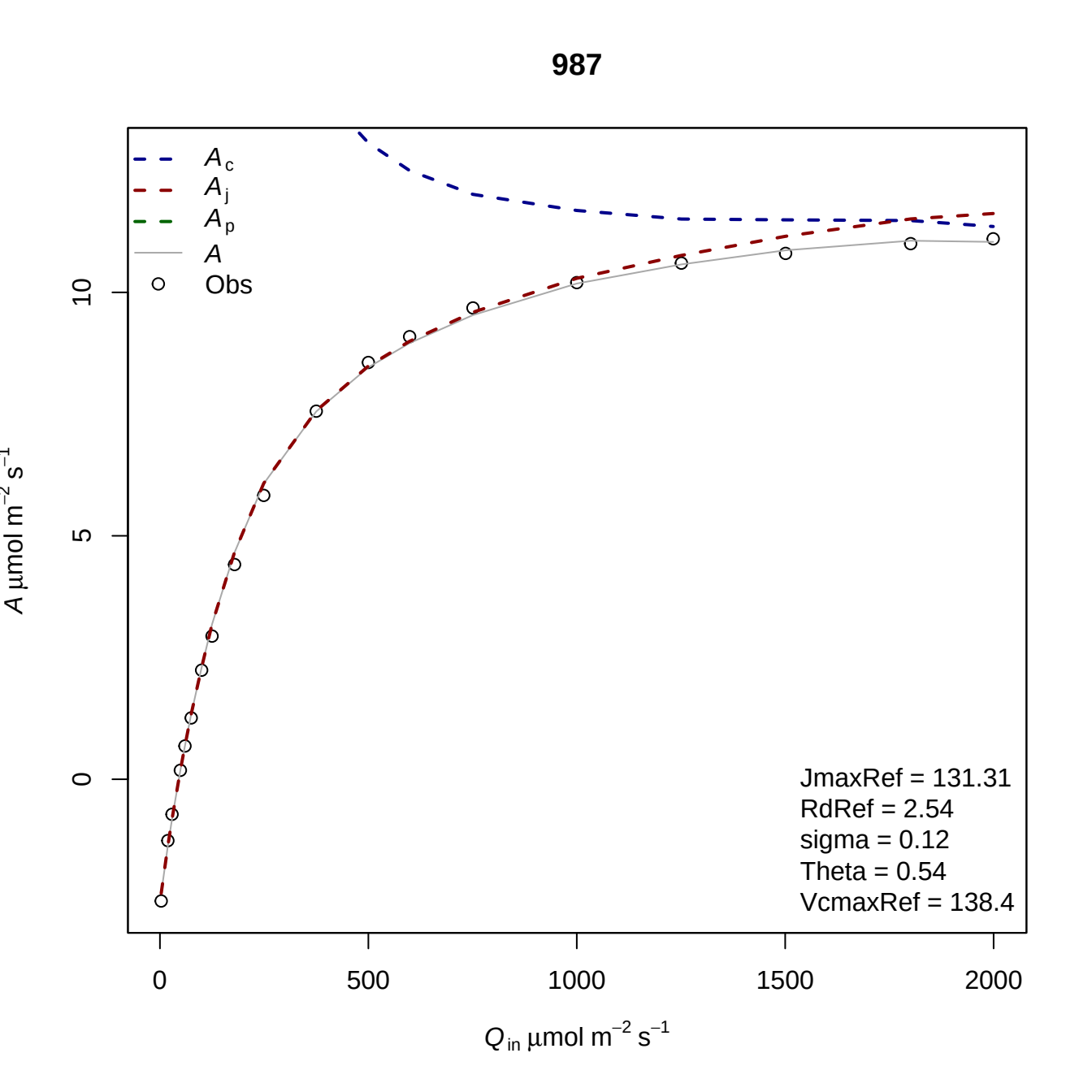

982
A_c
A_j
A_p
A
Obs
JmaxRef = 76.82
RdRef = 1.57
sigma = 0.16
Theta = −0.55
VcmaxRef = 36734.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

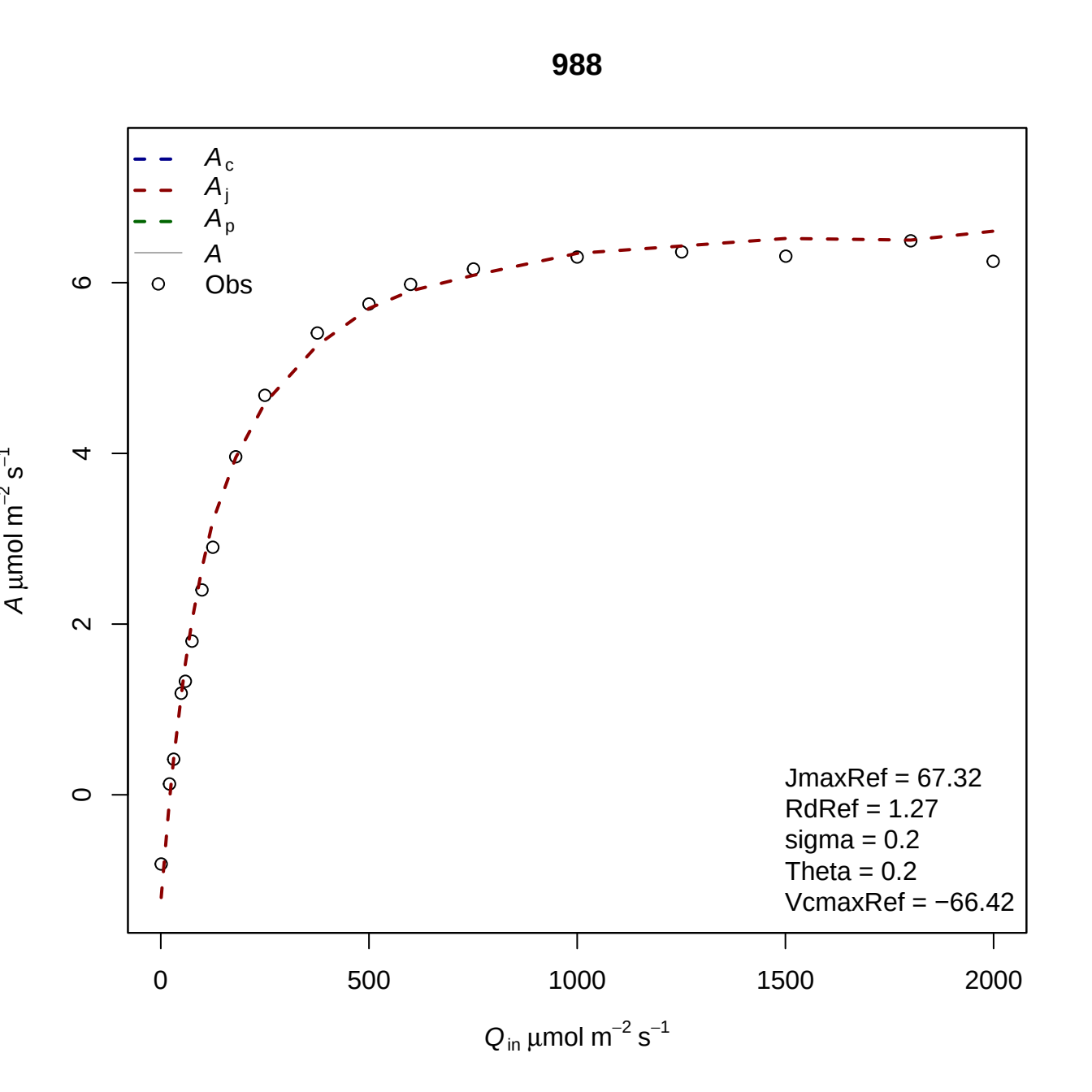

983
A_c
A_j
A_p
A
Obs
JmaxRef = 70.98
RdRef = 1
sigma = 0.11
Theta = −0.78
VcmaxRef = 8818.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

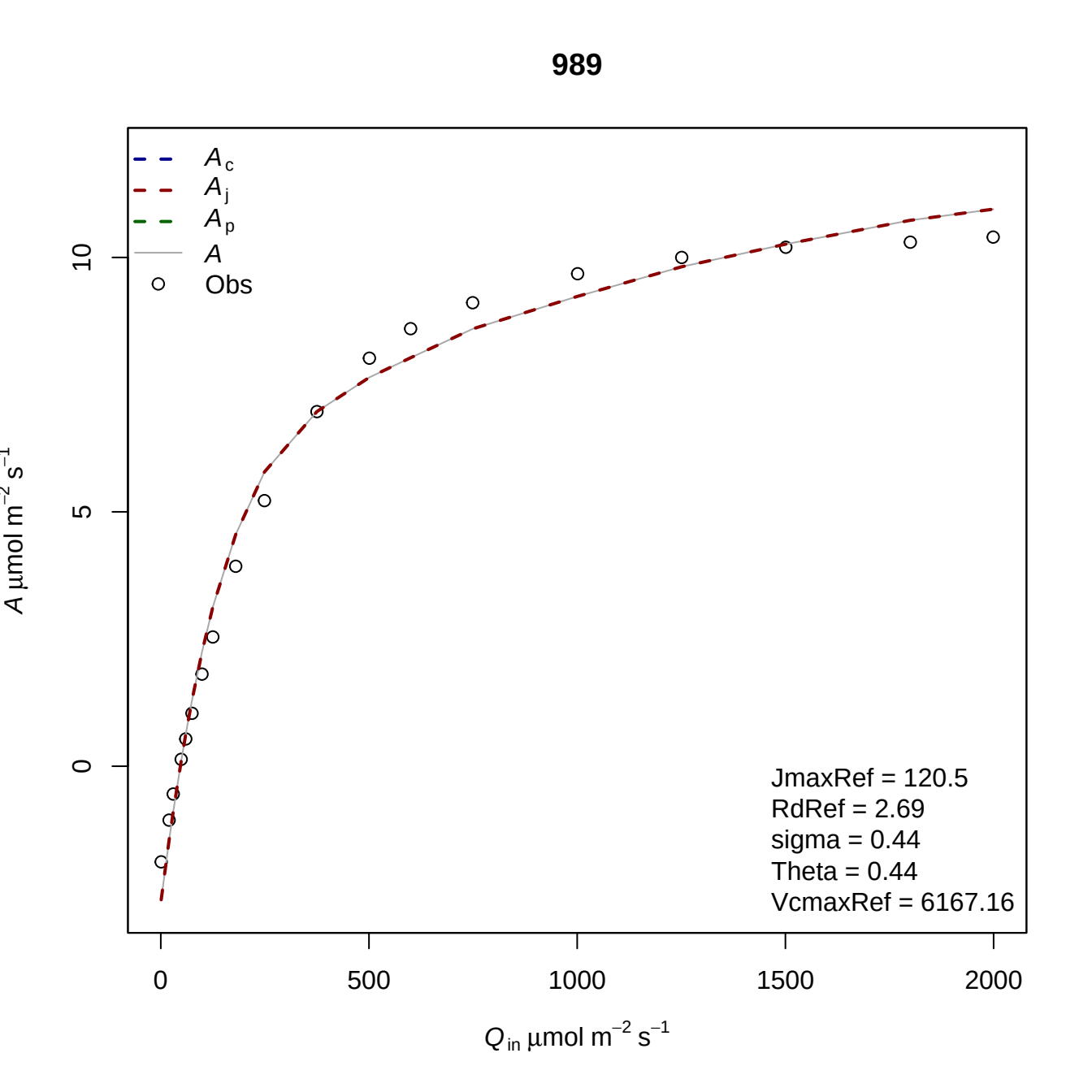

984
A_c
A_j
A_p
A
Obs
JmaxRef = 118.28
RdRef = 2.17
sigma = 0.06
Theta = 0.39
VcmaxRef = 301911.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

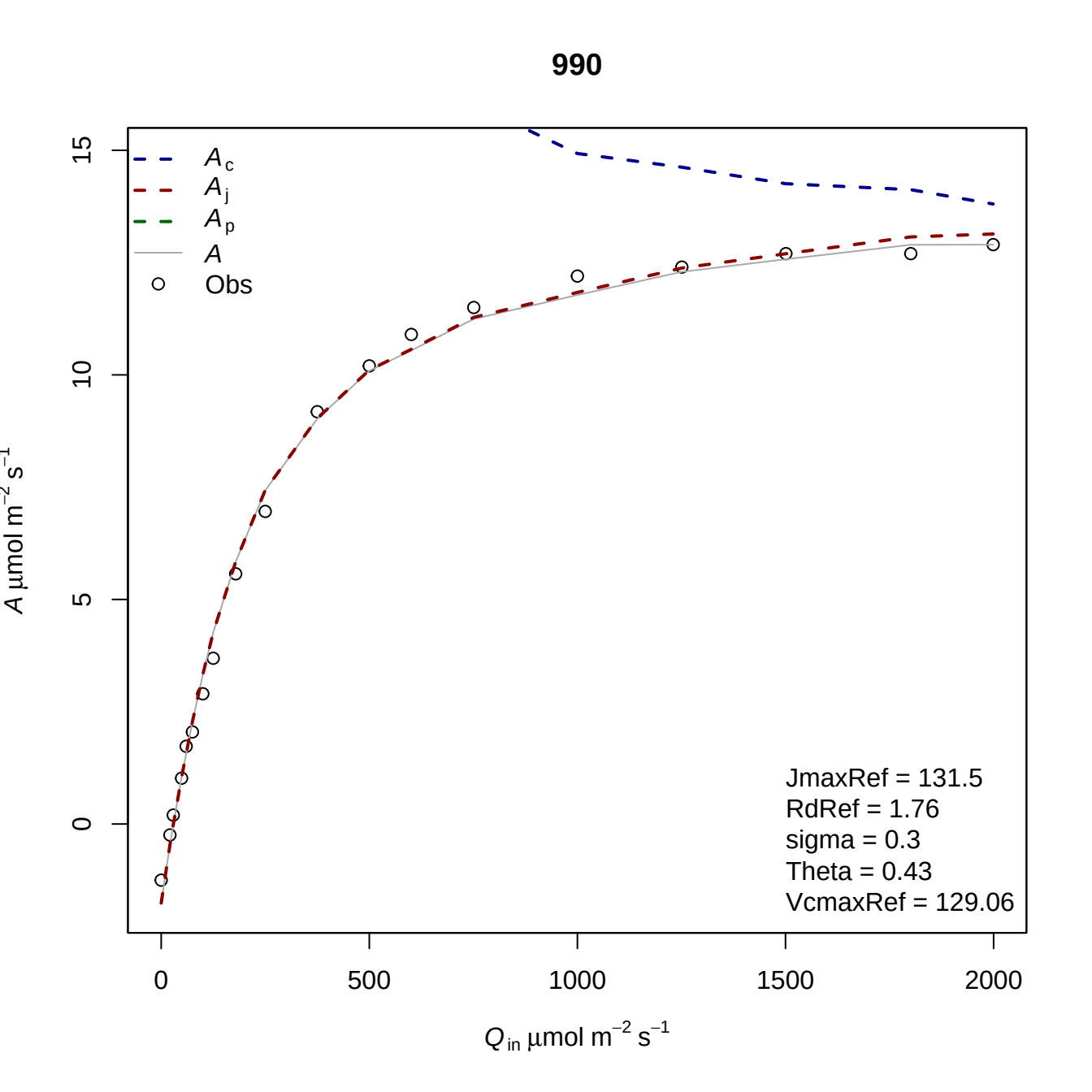

985
A_c
A_j
A_p
A
Obs
JmaxRef = 110.26
RdRef = 1.95
sigma = 0.09
Theta = 0.49
VcmaxRef = 293.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

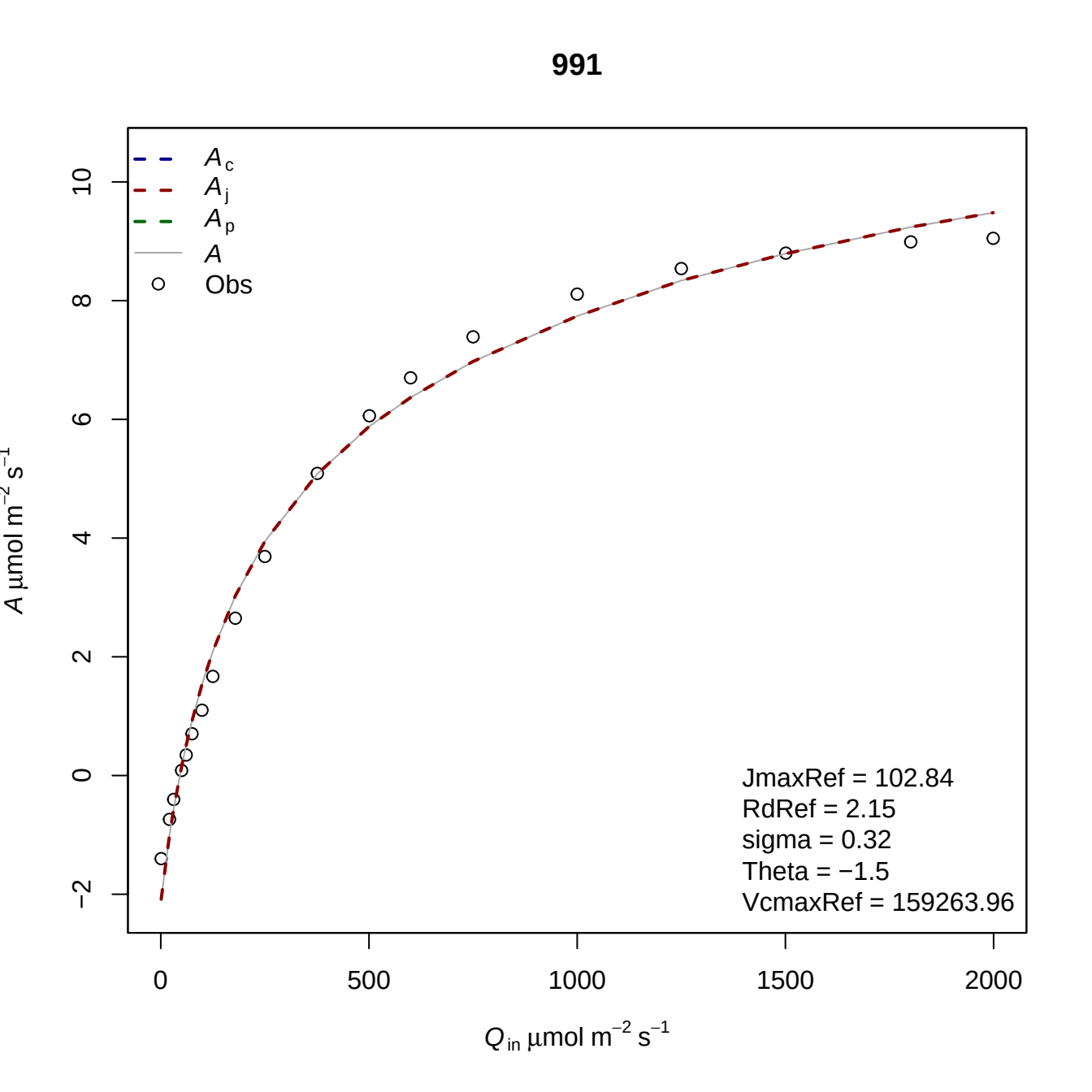

986
A_c
A_j
A_p
A
Obs
JmaxRef = 65.98
RdRef = 1.07
sigma = 0.06
Theta = 0.29
VcmaxRef = 70.03
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

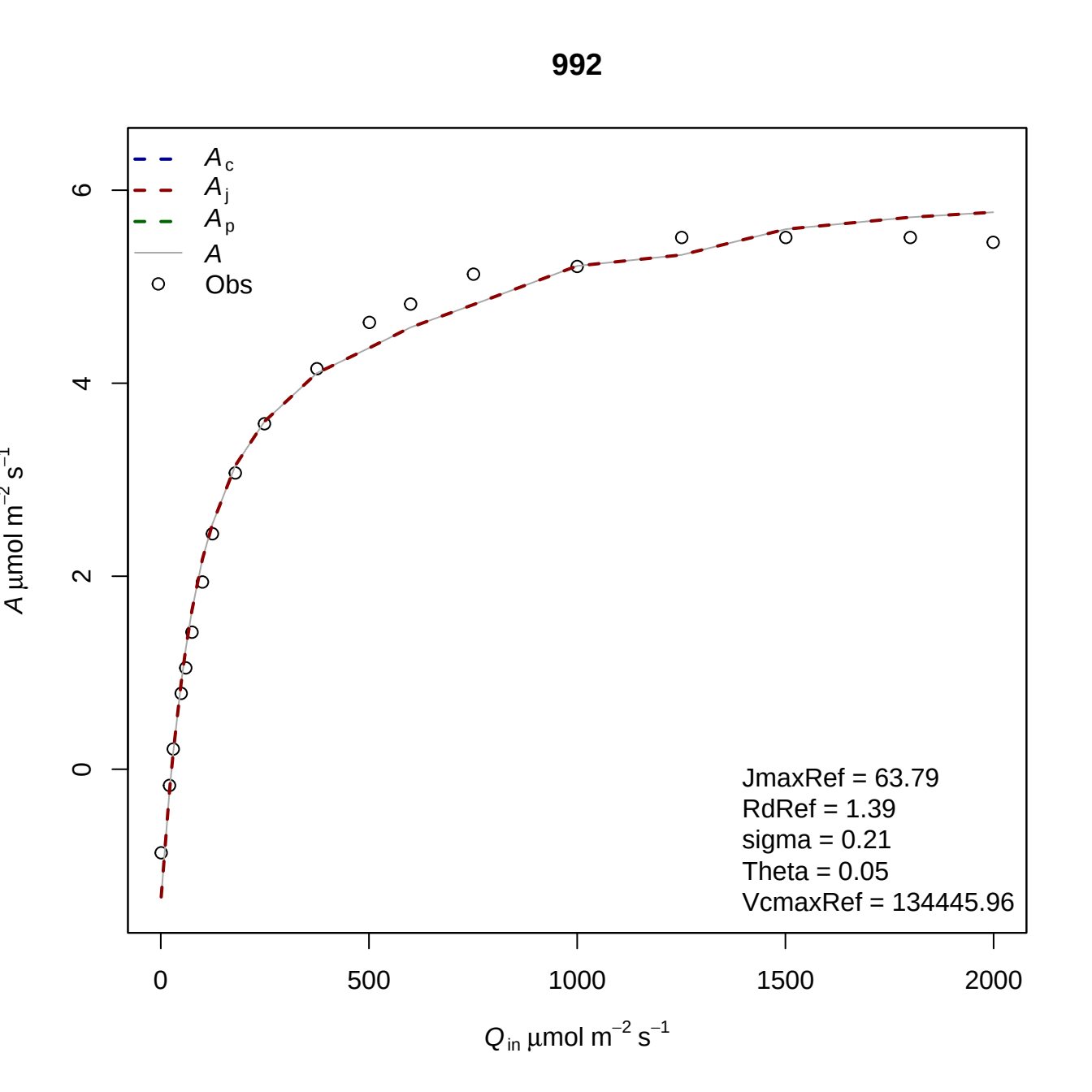

987
A_c
A_j
A_p
A
Obs
JmaxRef = 131.31
RdRef = 2.54
sigma = 0.12
Theta = 0.54
VcmaxRef = 138.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

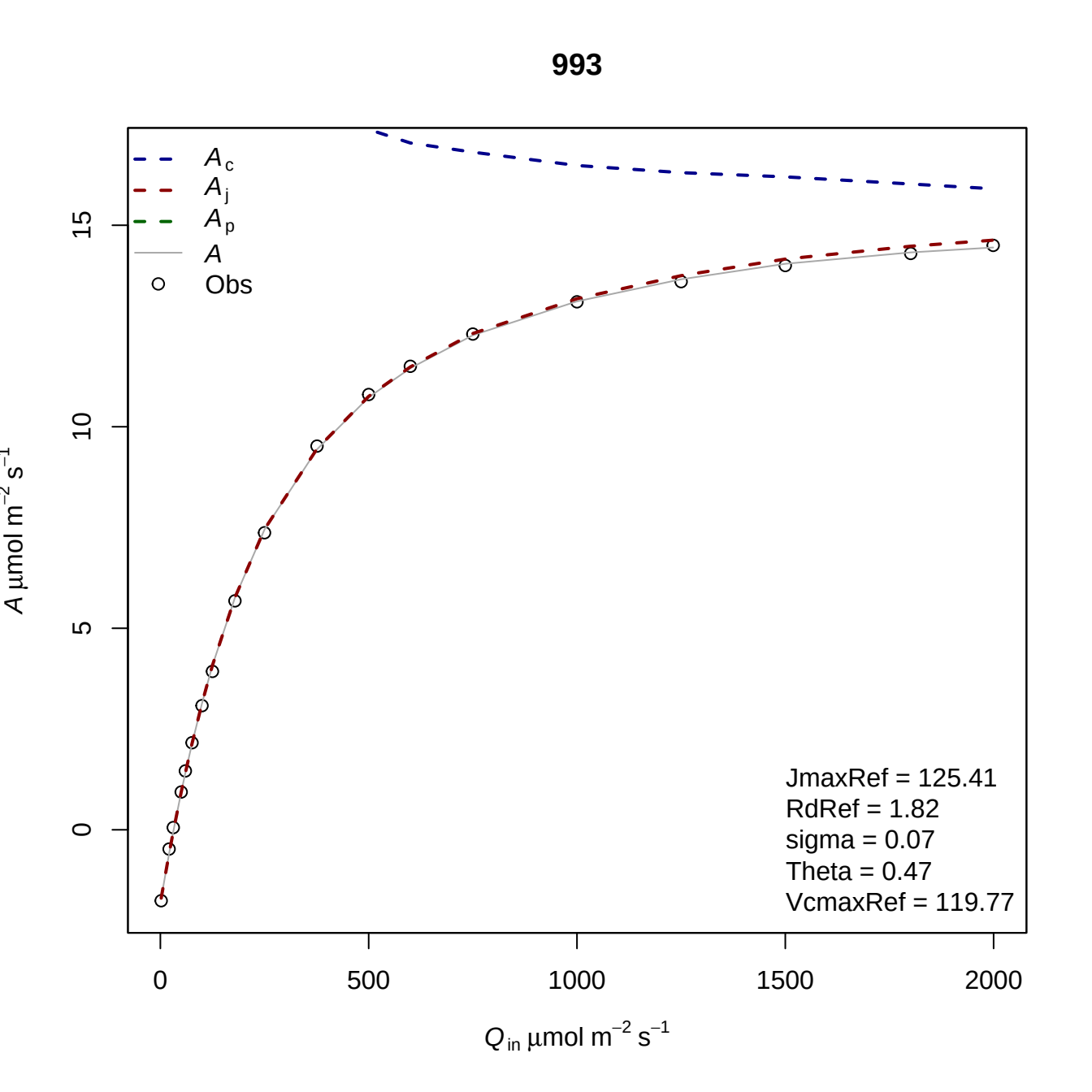

988
A_c
A_j
A_p
A
Obs
JmaxRef = 67.32
RdRef = 1.27
sigma = 0.2
Theta = 0.2
VcmaxRef = −66.42
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

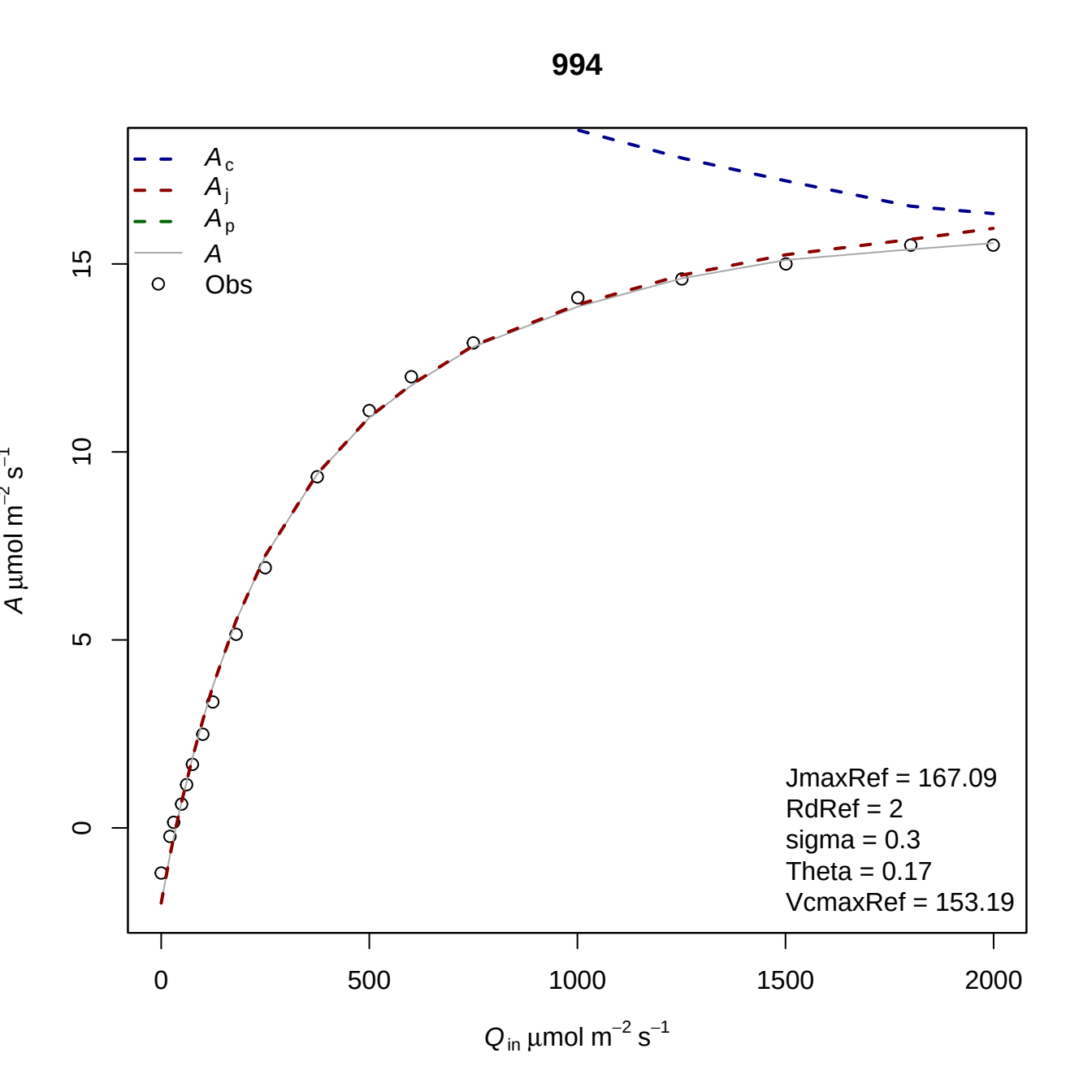

989
A_c
A_j
A_p
A
Obs
JmaxRef = 120.5
RdRef = 2.69
sigma = 0.44
Theta = 0.44
VcmaxRef = 6167.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

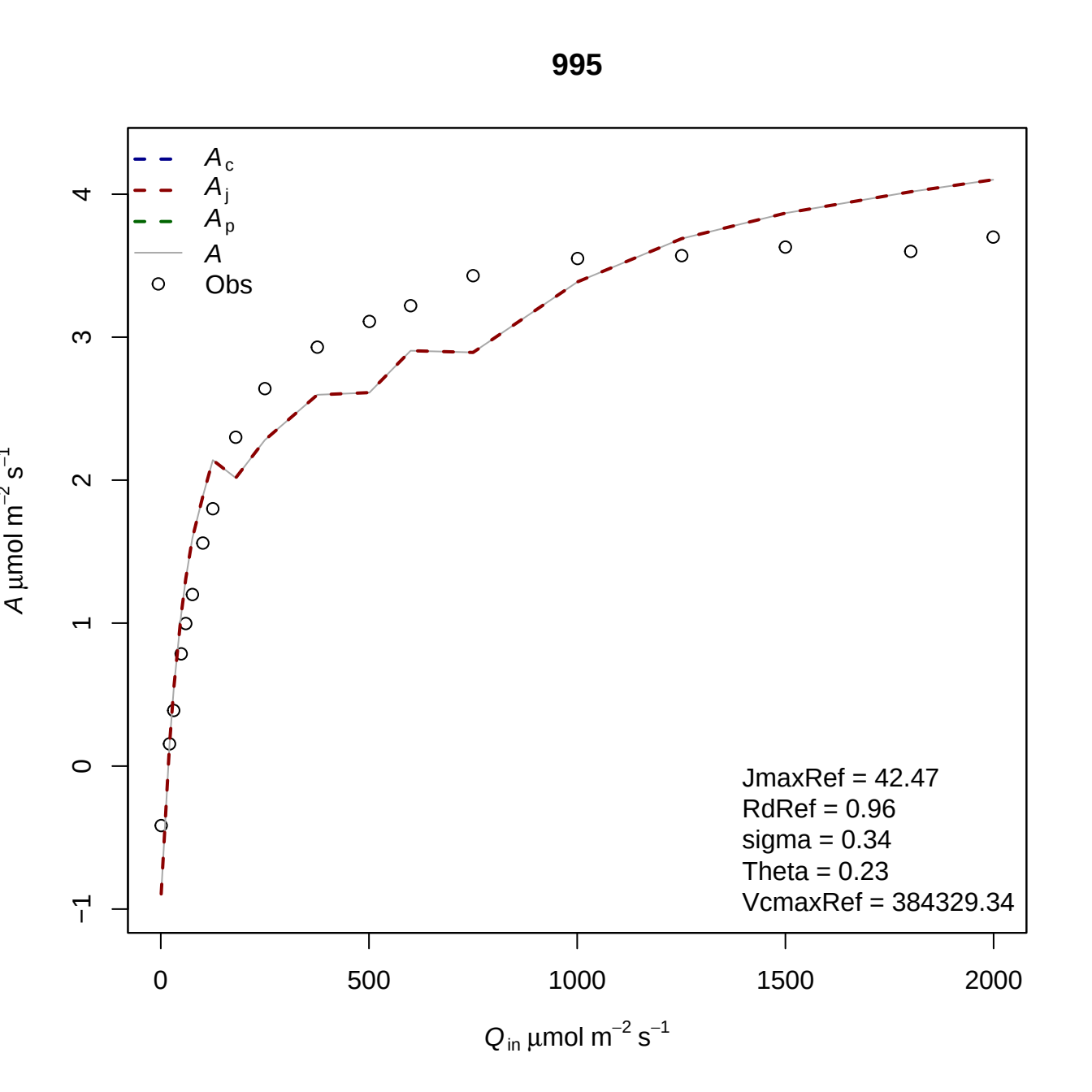

990
A_c
A_j
A_p
A
Obs
JmaxRef = 131.5
RdRef = 1.76
sigma = 0.3
Theta = 0.43
VcmaxRef = 129.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

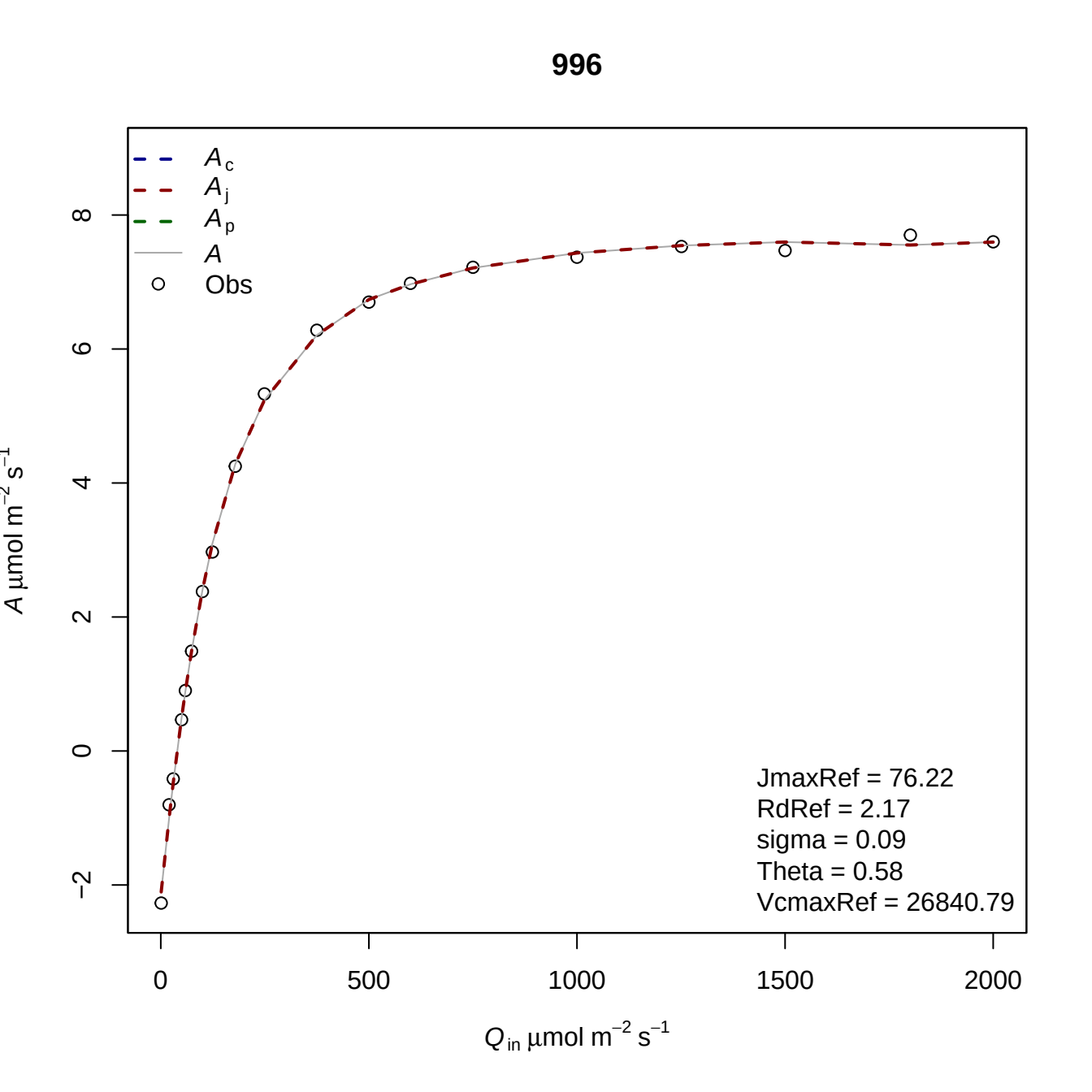

991
A_c
A_j
A_p
A
Obs
JmaxRef = 102.84
RdRef = 2.15
sigma = 0.32
Theta = −1.5
VcmaxRef = 159263.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

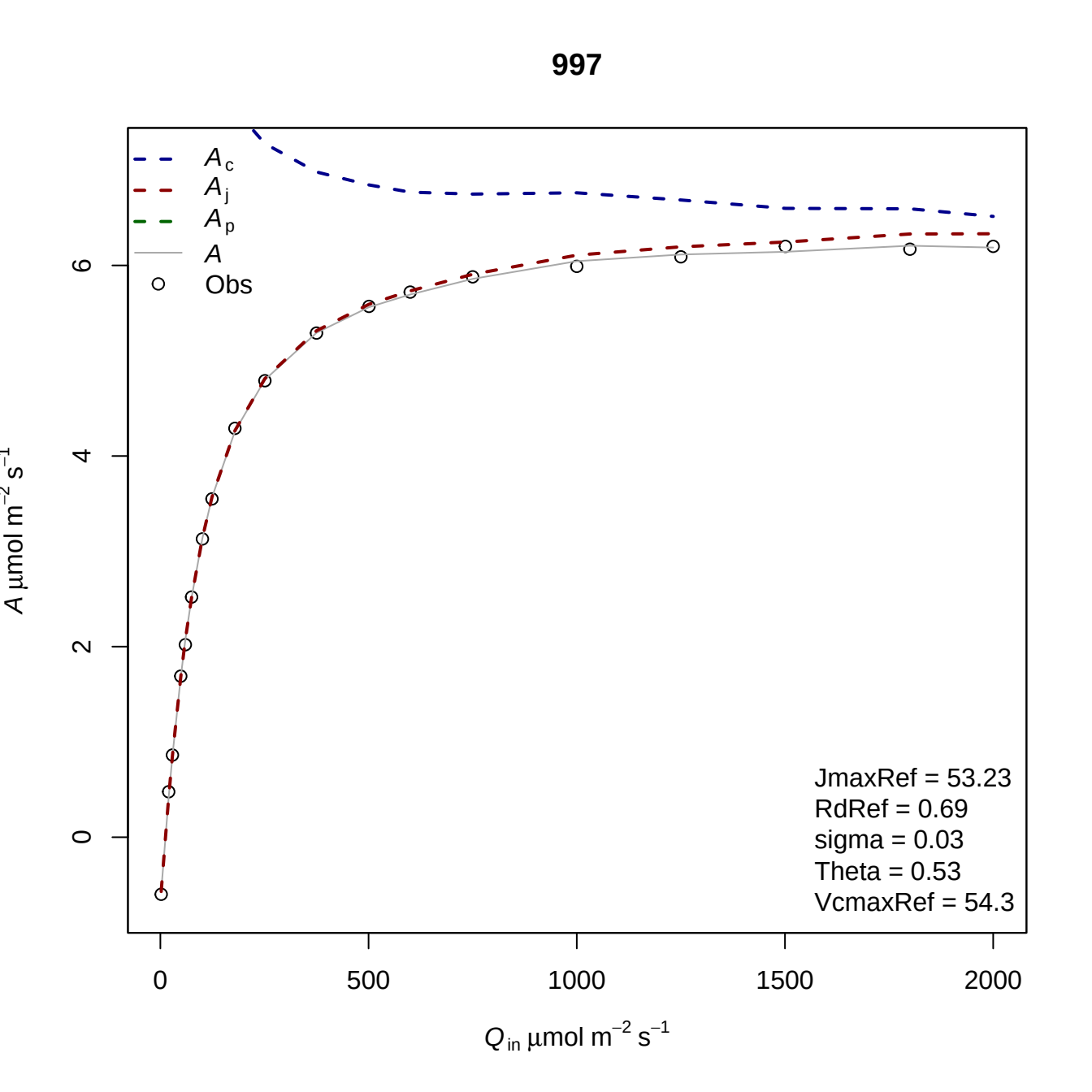

992
A_c
A_j
A_p
A
Obs
JmaxRef = 63.79
RdRef = 1.39
sigma = 0.21
Theta = 0.05
VcmaxRef = 134445.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

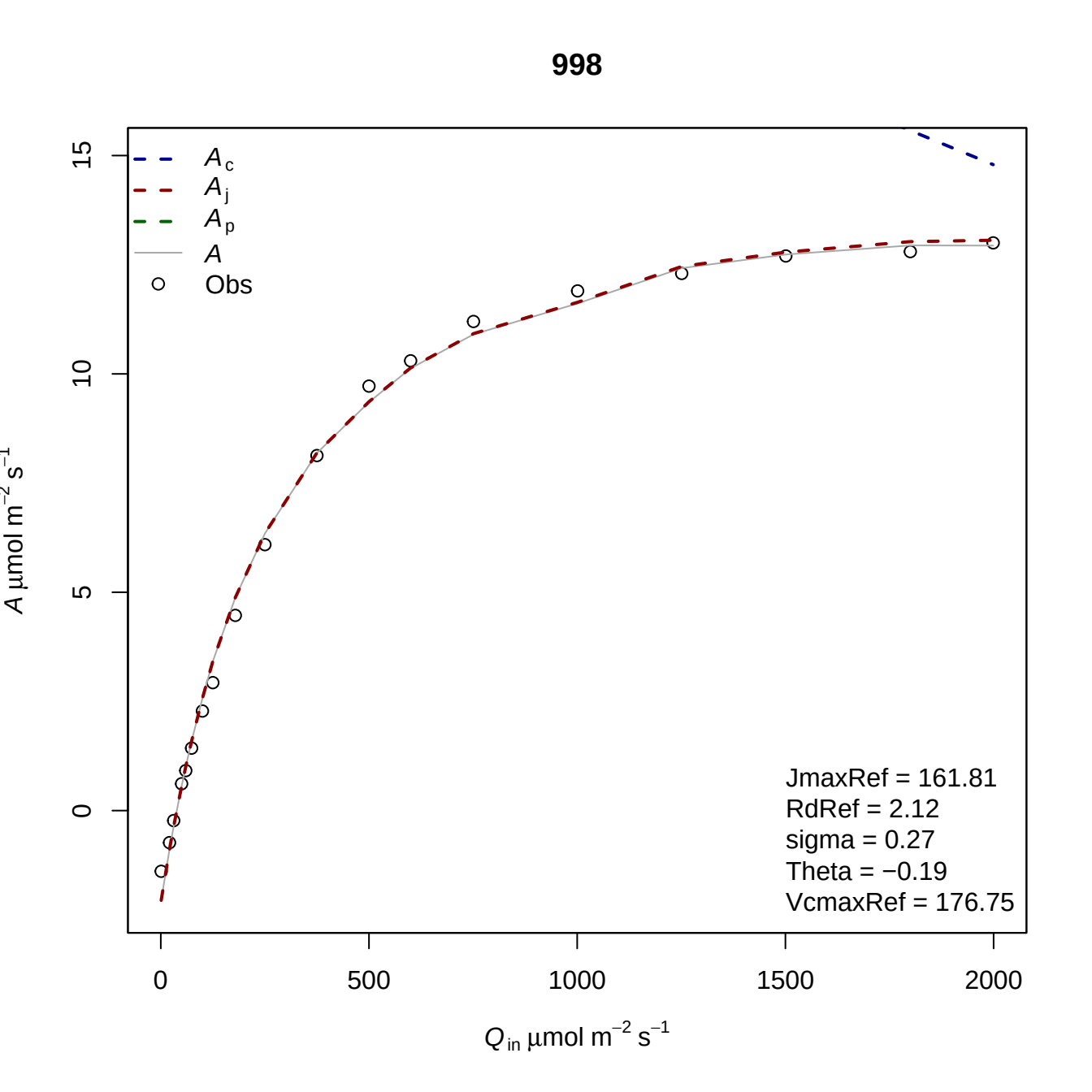

993
A_c
A_j
A_p
A
Obs
JmaxRef = 125.41
RdRef = 1.82
sigma = 0.07
Theta = 0.47
VcmaxRef = 119.77
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

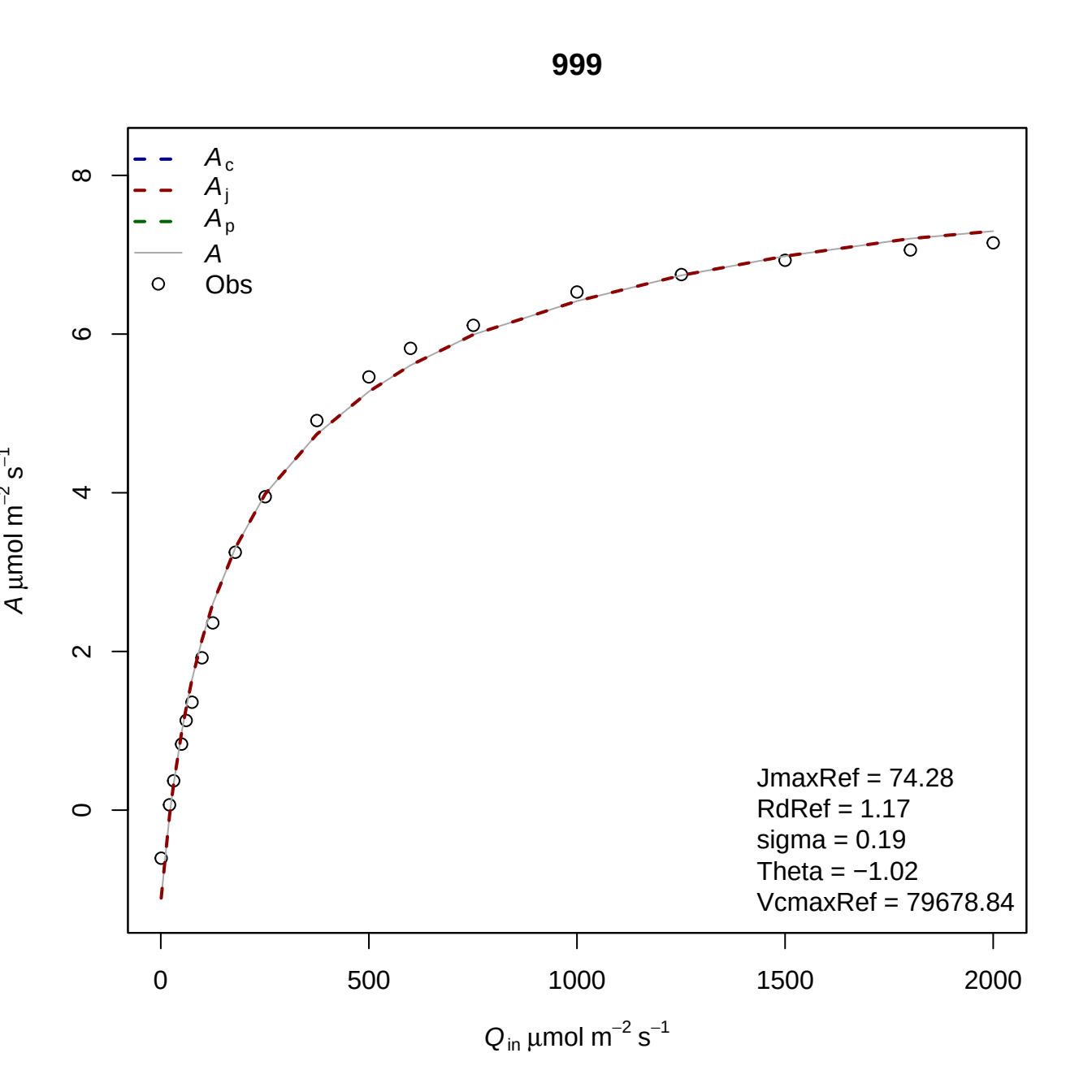

994
A_c
A_j
A_p
A
Obs
JmaxRef = 167.09
RdRef = 2
sigma = 0.3
Theta = 0.17
VcmaxRef = 153.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

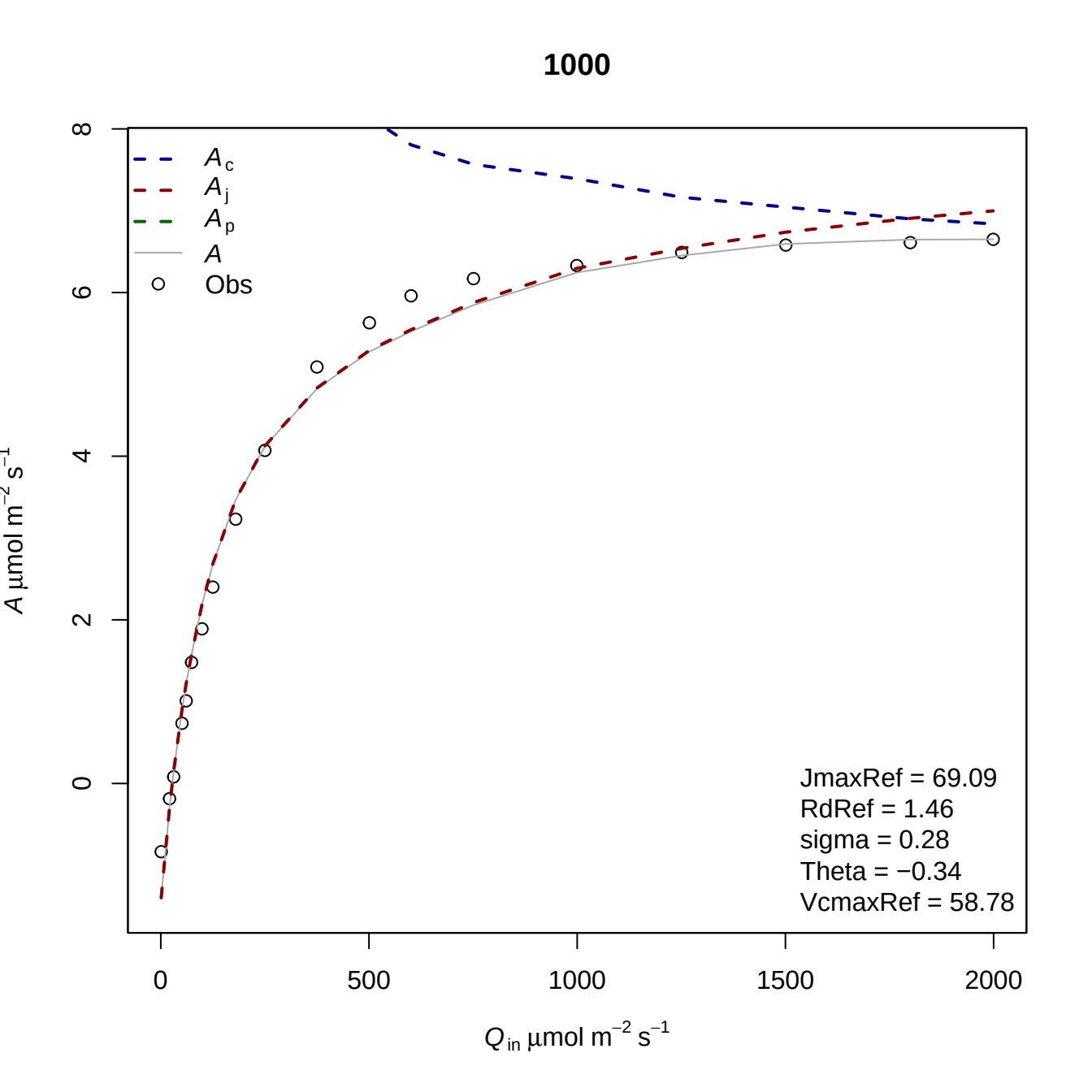

995
A_c
A_j
A_p
A
Obs
JmaxRef = 42.47
RdRef = 0.96
sigma = 0.34
Theta = 0.23
VcmaxRef = 384329.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

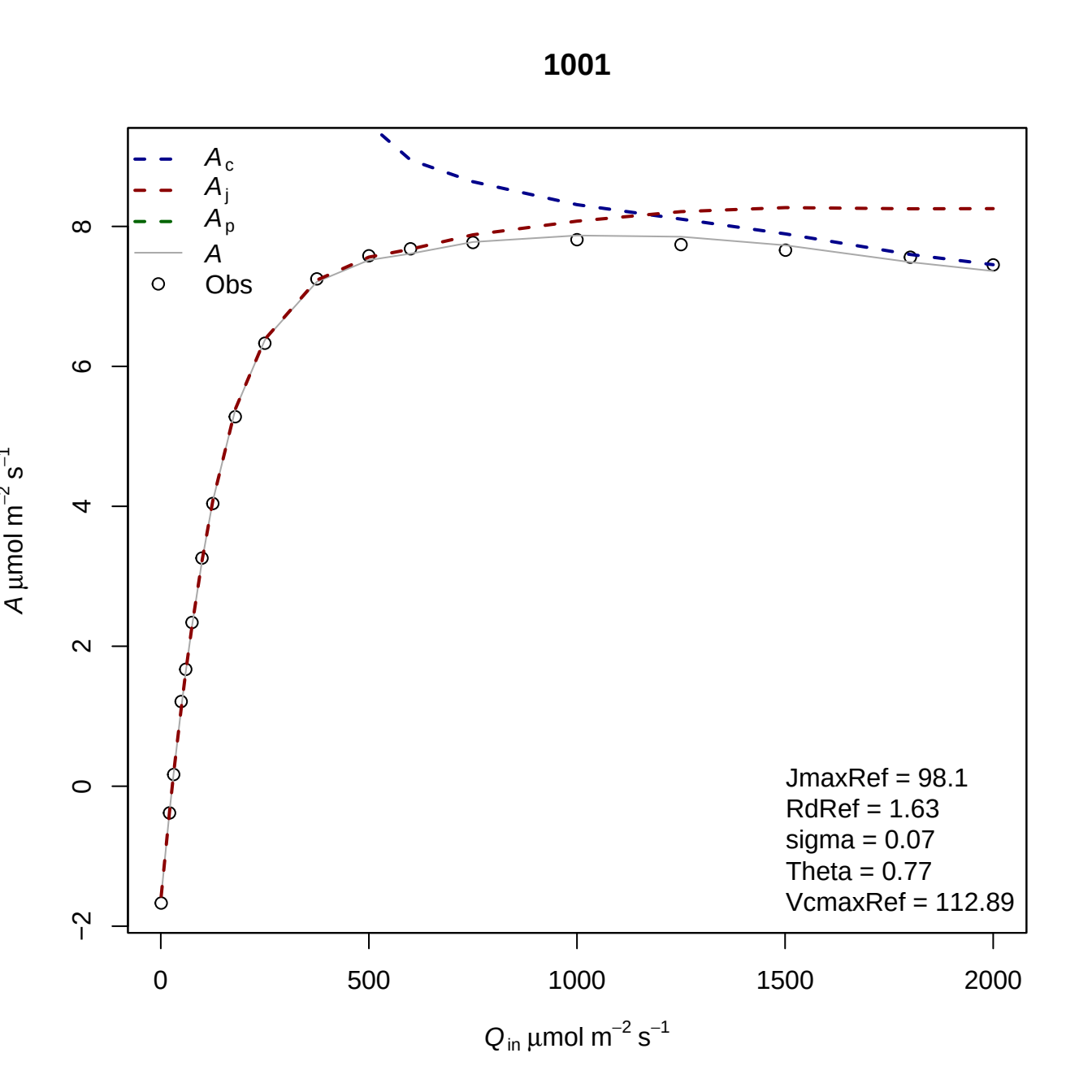

996
A_c
A_j
A_p
A
Obs
JmaxRef = 76.22
RdRef = 2.17
sigma = 0.09
Theta = 0.58
VcmaxRef = 26840.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

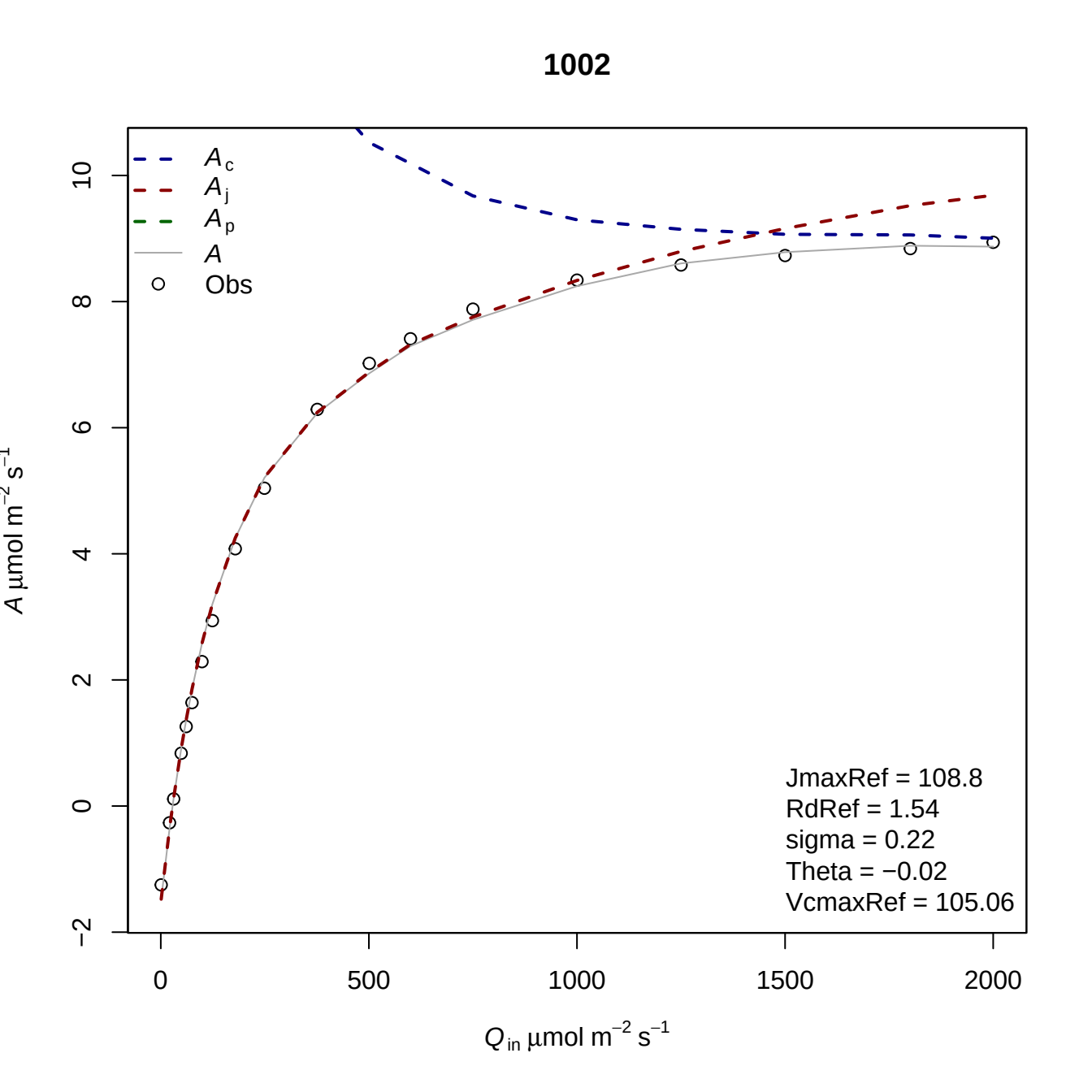

997
A_c
A_j
A_p
A
Obs
JmaxRef = 53.23
RdRef = 0.69
sigma = 0.03
Theta = 0.53
VcmaxRef = 54.3
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

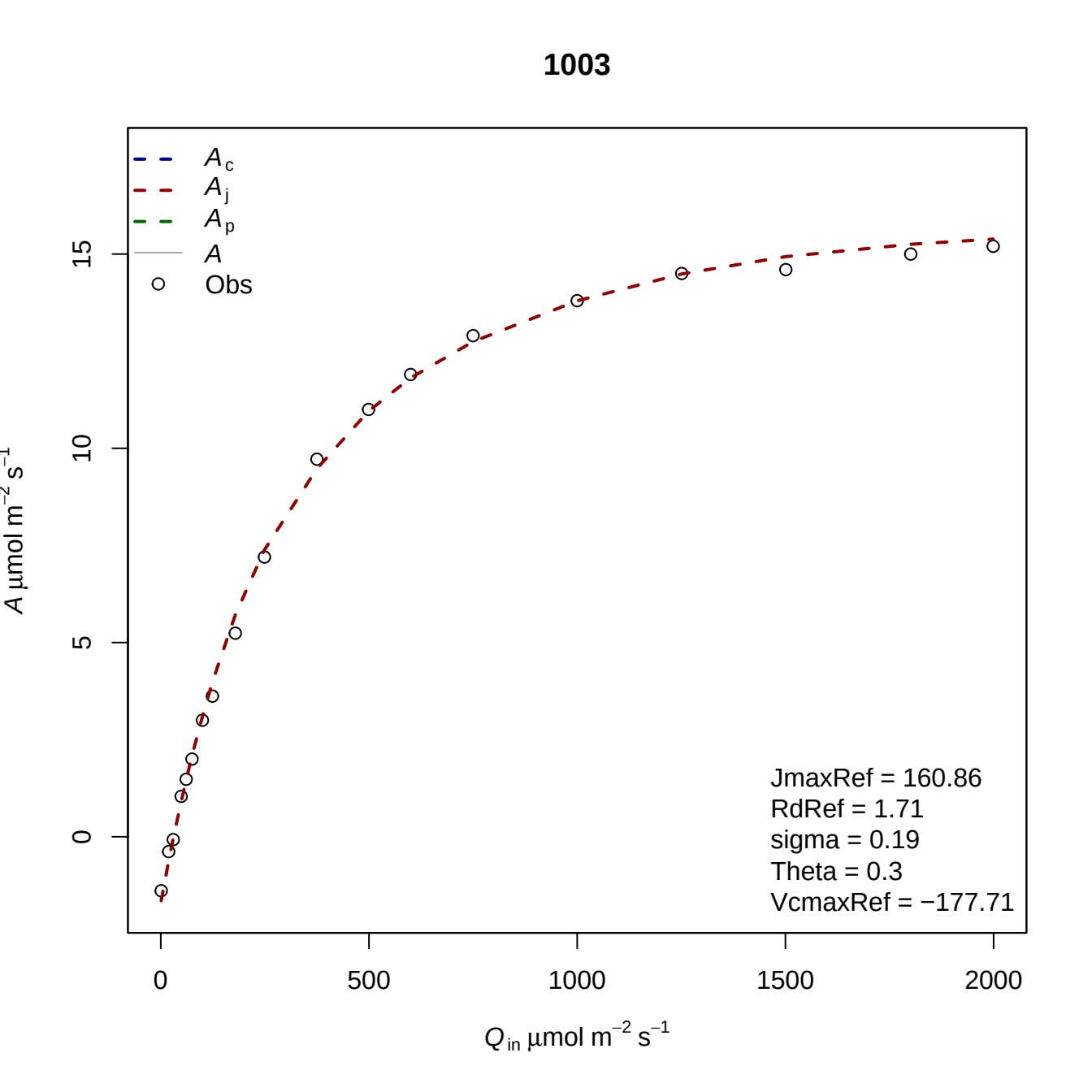

998
A_c
A_j
A_p
A
Obs
JmaxRef = 161.81
RdRef = 2.12
sigma = 0.27
Theta = −0.19
VcmaxRef = 176.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

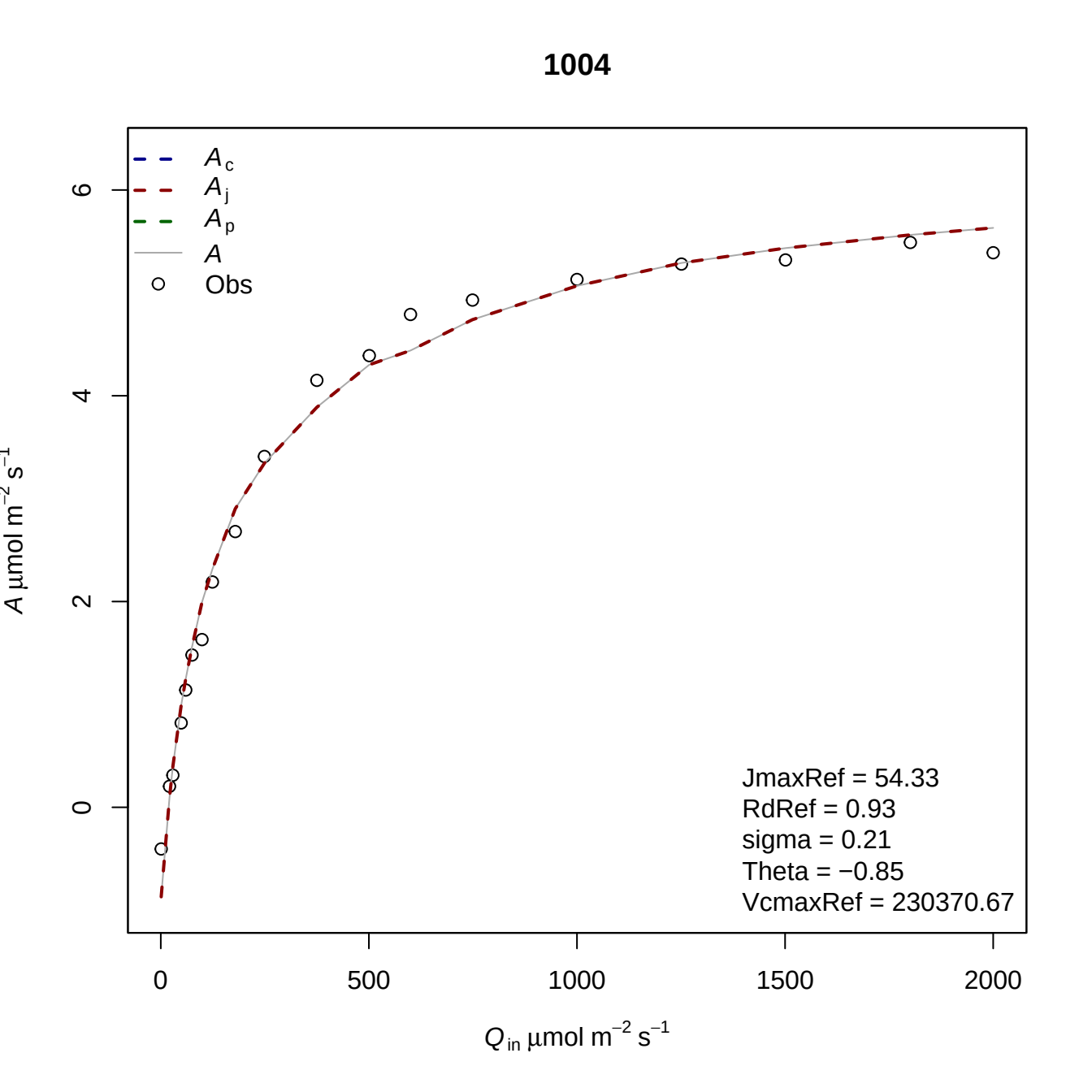

999
A_c
A_j
A_p
A
Obs
JmaxRef = 74.28
RdRef = 1.17
sigma = 0.19
Theta = −1.02
VcmaxRef = 79678.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

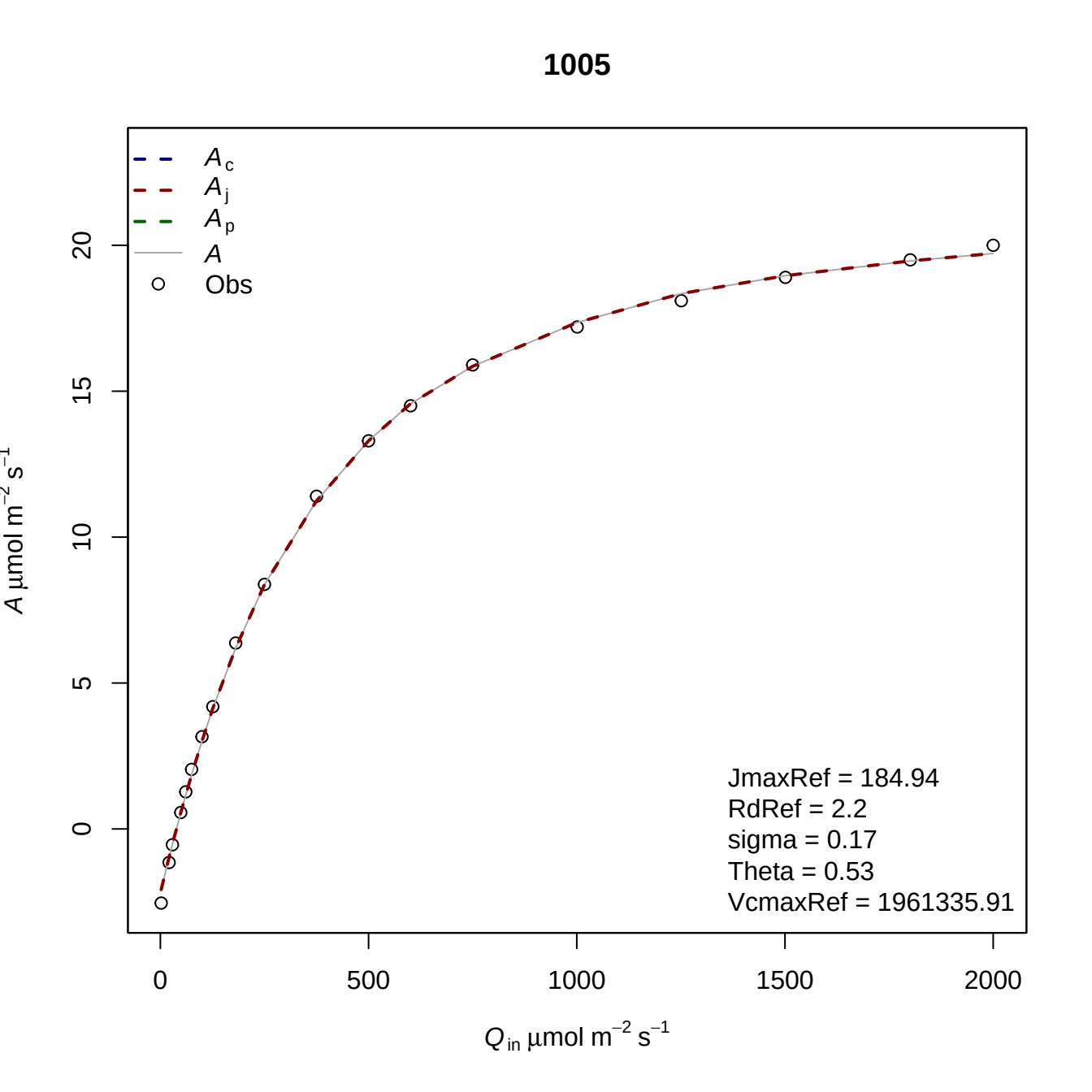

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 69.09
RdRef = 1.46
sigma = 0.28
Theta = −0.34
VcmaxRef = 58.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

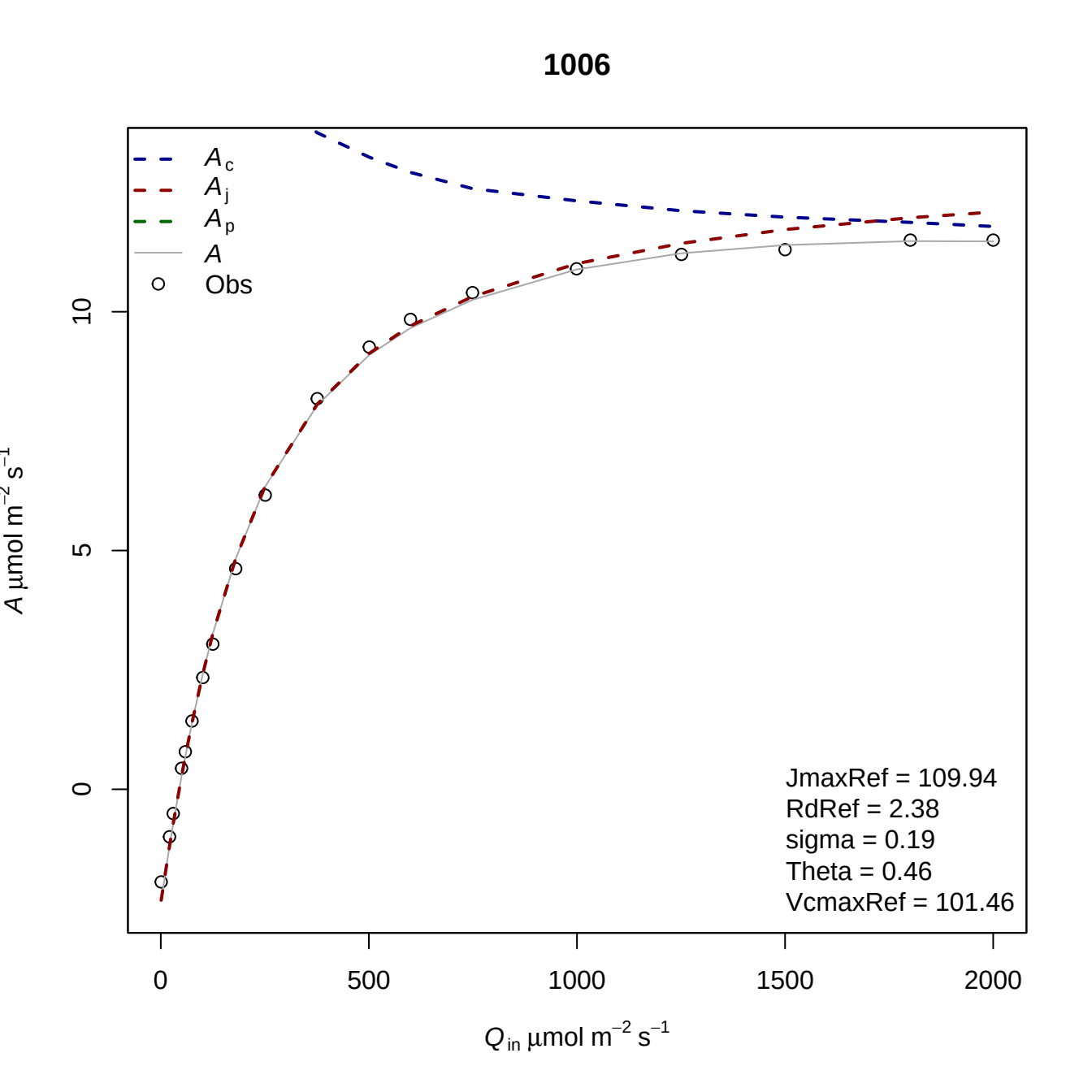

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 98.1
RdRef = 1.63
sigma = 0.07
Theta = 0.77
VcmaxRef = 112.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

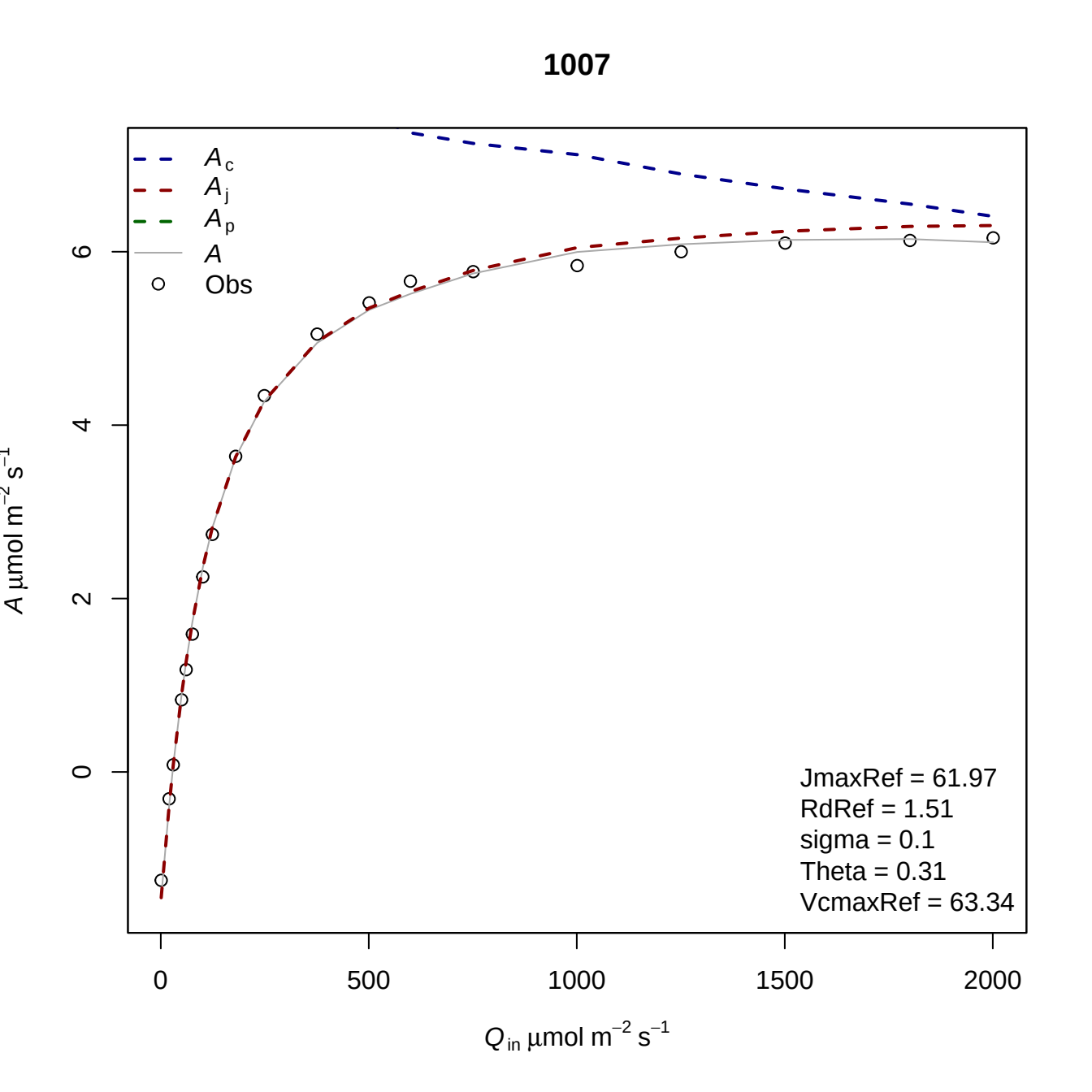

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 108.8
RdRef = 1.54
sigma = 0.22
Theta = −0.02
VcmaxRef = 105.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

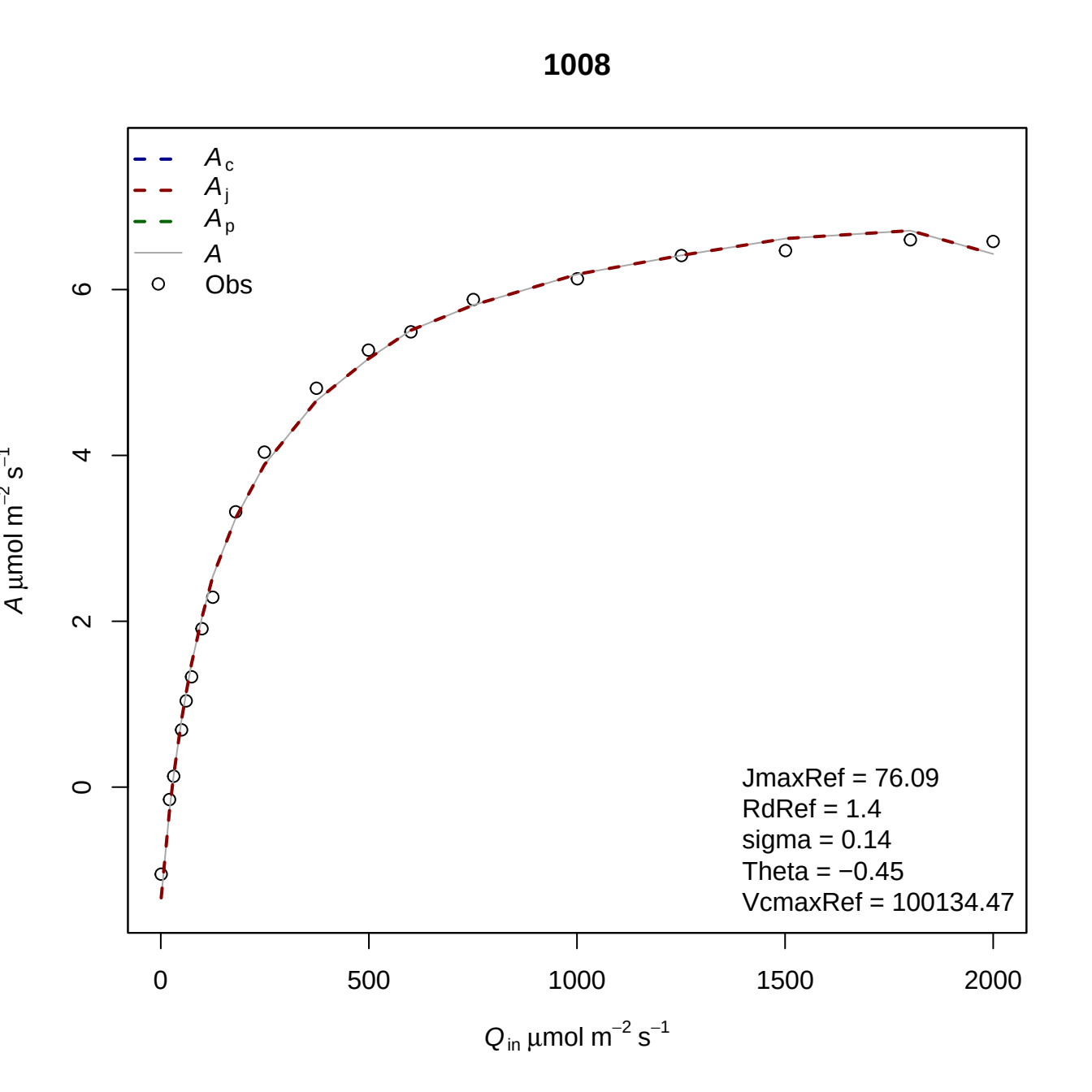

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 160.86
RdRef = 1.71
sigma = 0.19
Theta = 0.3
VcmaxRef = −177.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

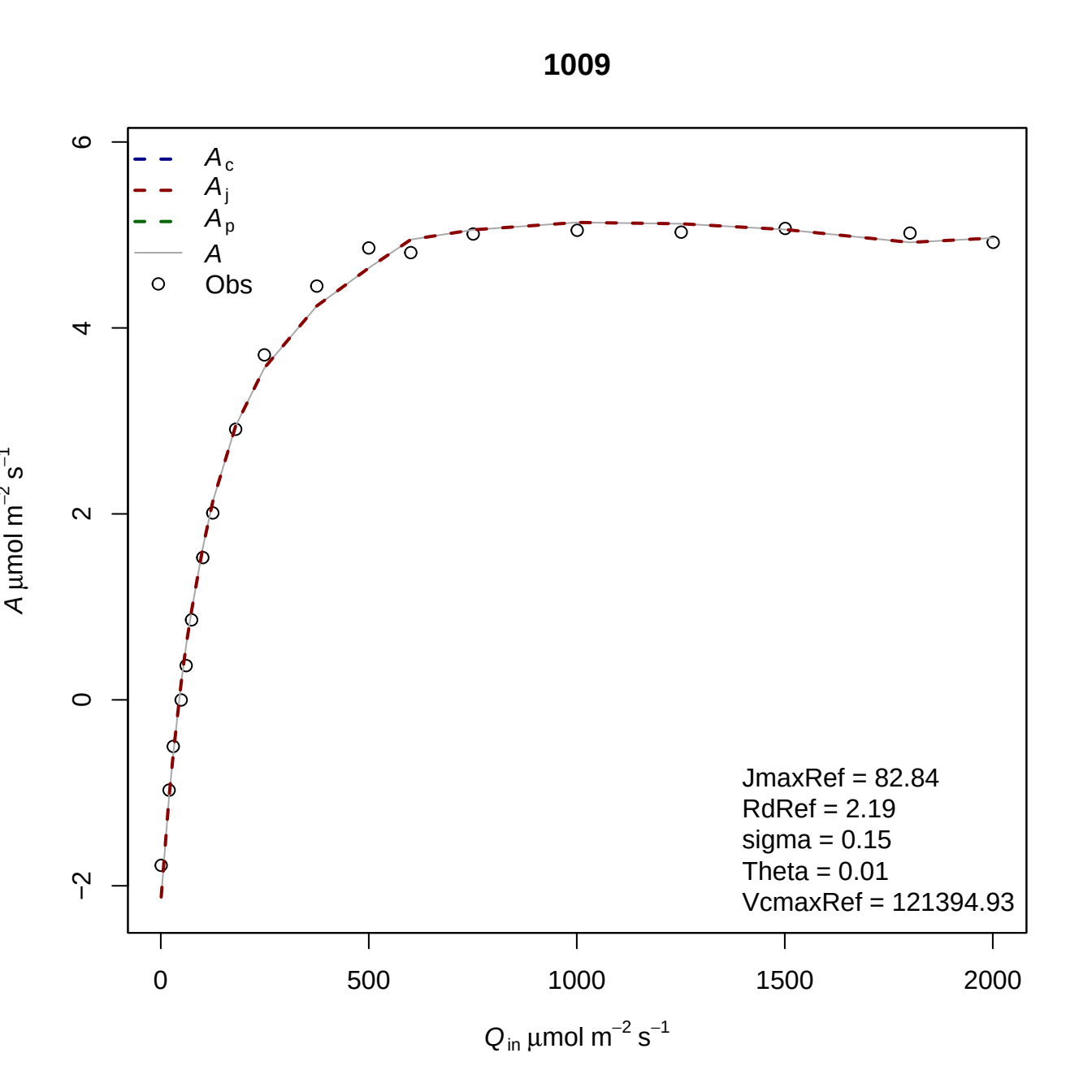

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 54.33
RdRef = 0.93
sigma = 0.21
Theta = −0.85
VcmaxRef = 230370.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

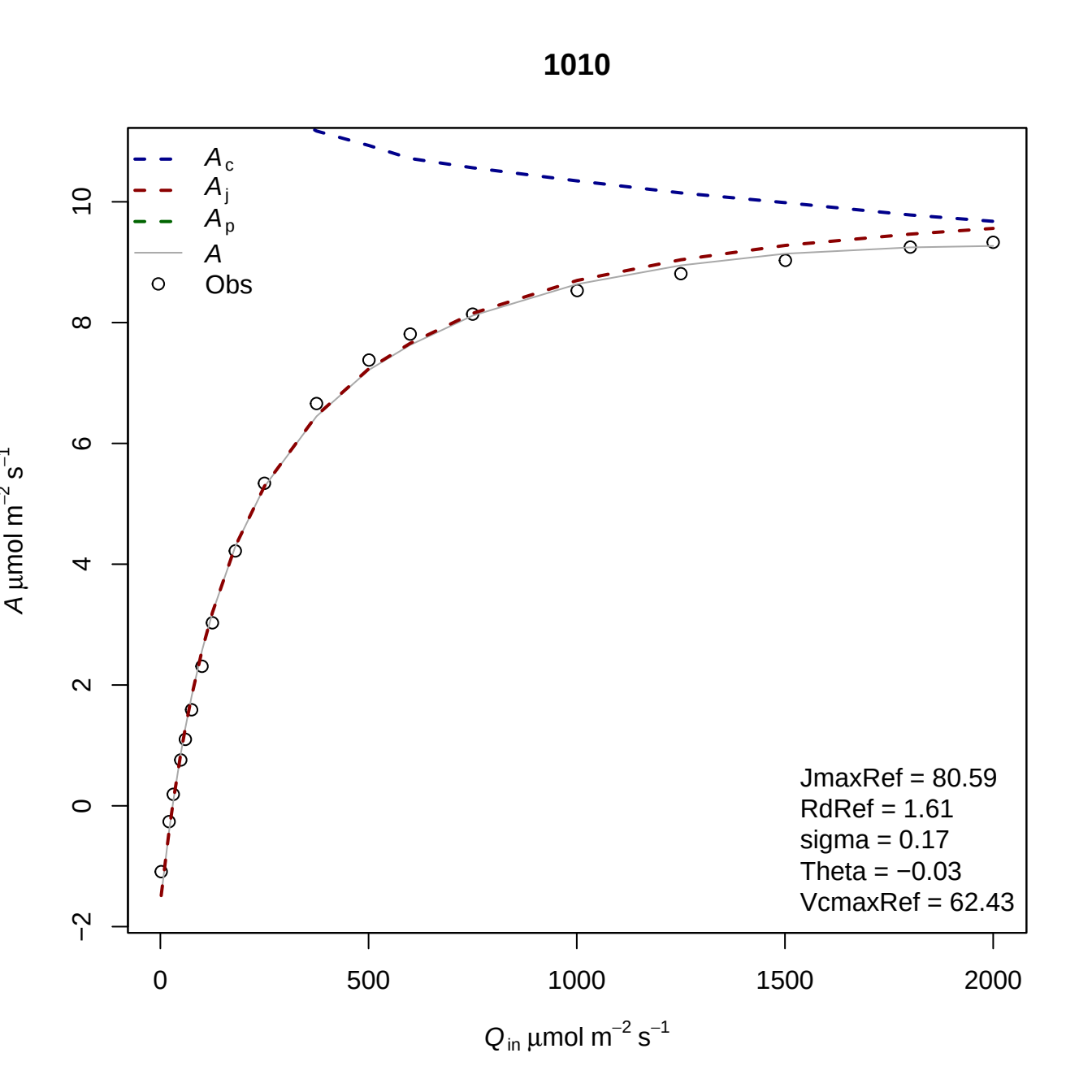

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 184.94
RdRef = 2.2
sigma = 0.17
Theta = 0.53
VcmaxRef = 1961335.91
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

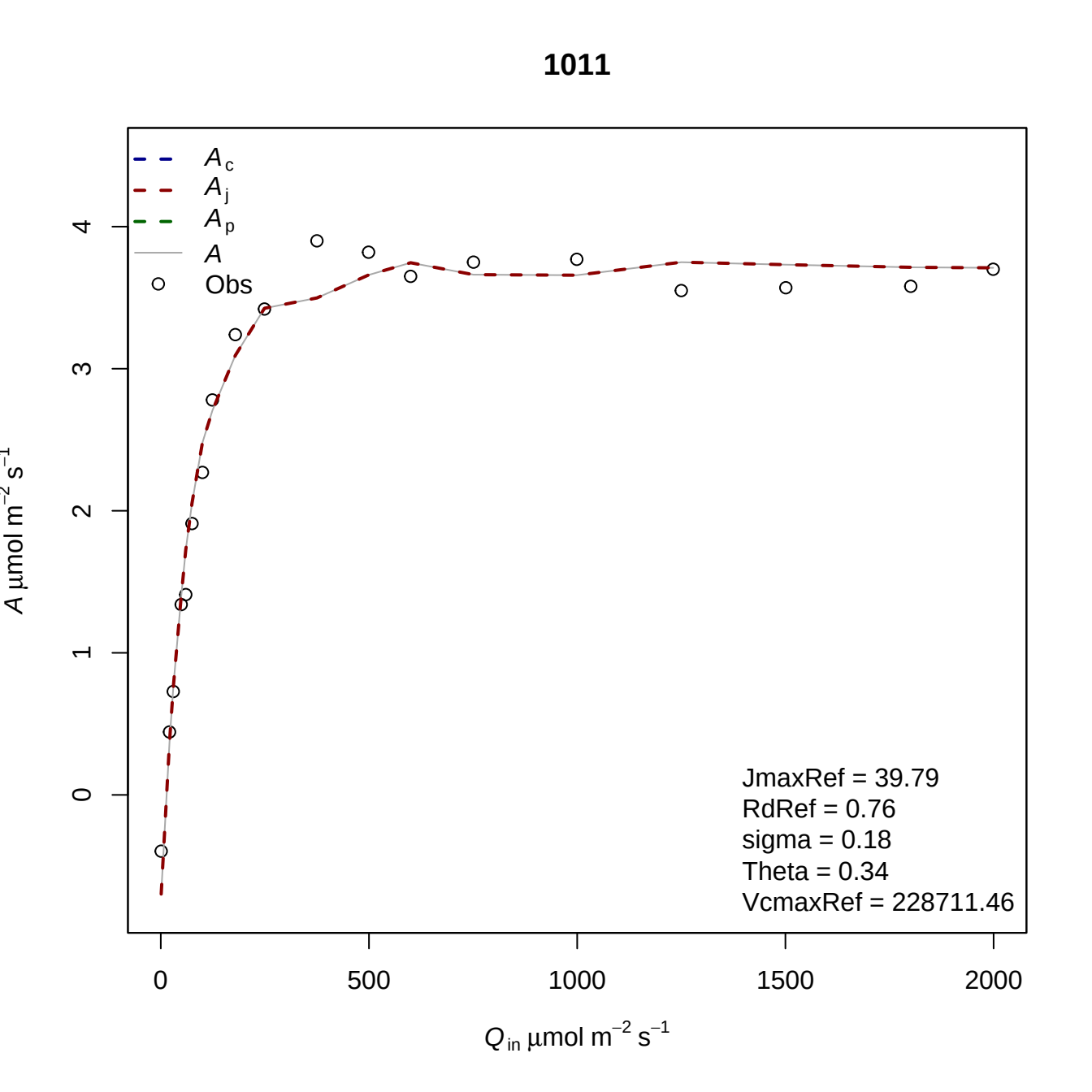

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 109.94
RdRef = 2.38
sigma = 0.19
Theta = 0.46
VcmaxRef = 101.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

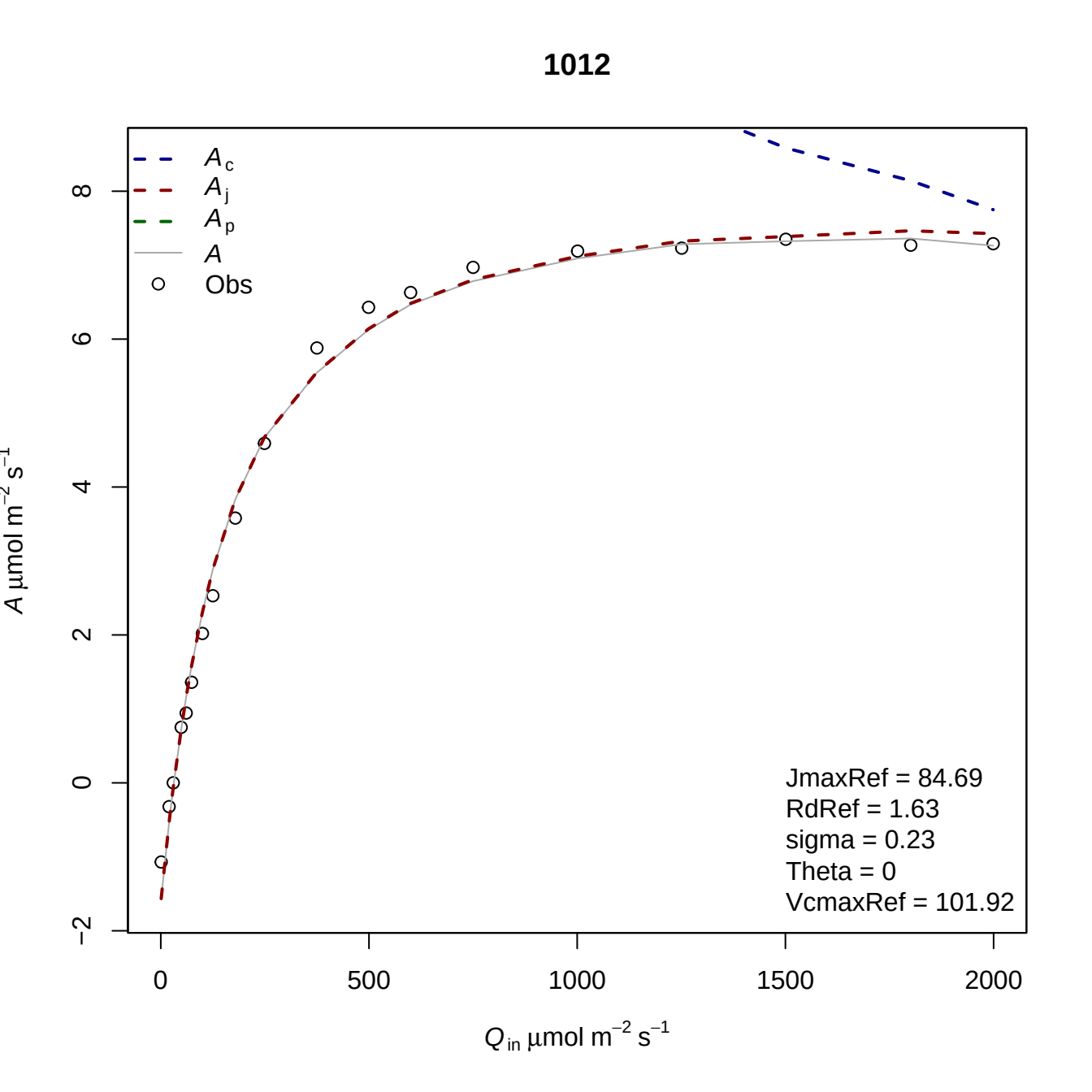

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 61.97
RdRef = 1.51
sigma = 0.1
Theta = 0.31
VcmaxRef = 63.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

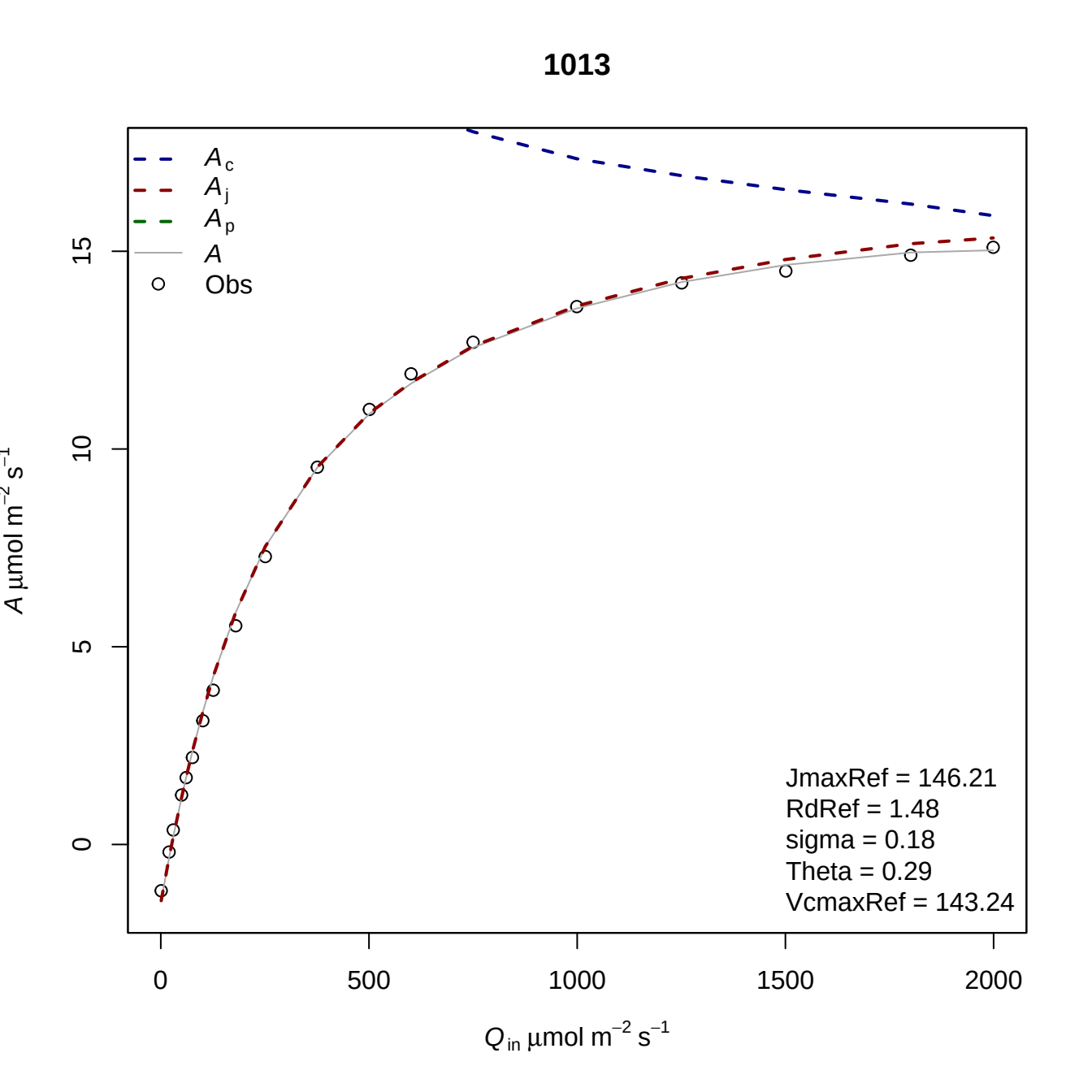

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 76.09
RdRef = 1.4
sigma = 0.14
Theta = −0.45
VcmaxRef = 100134.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

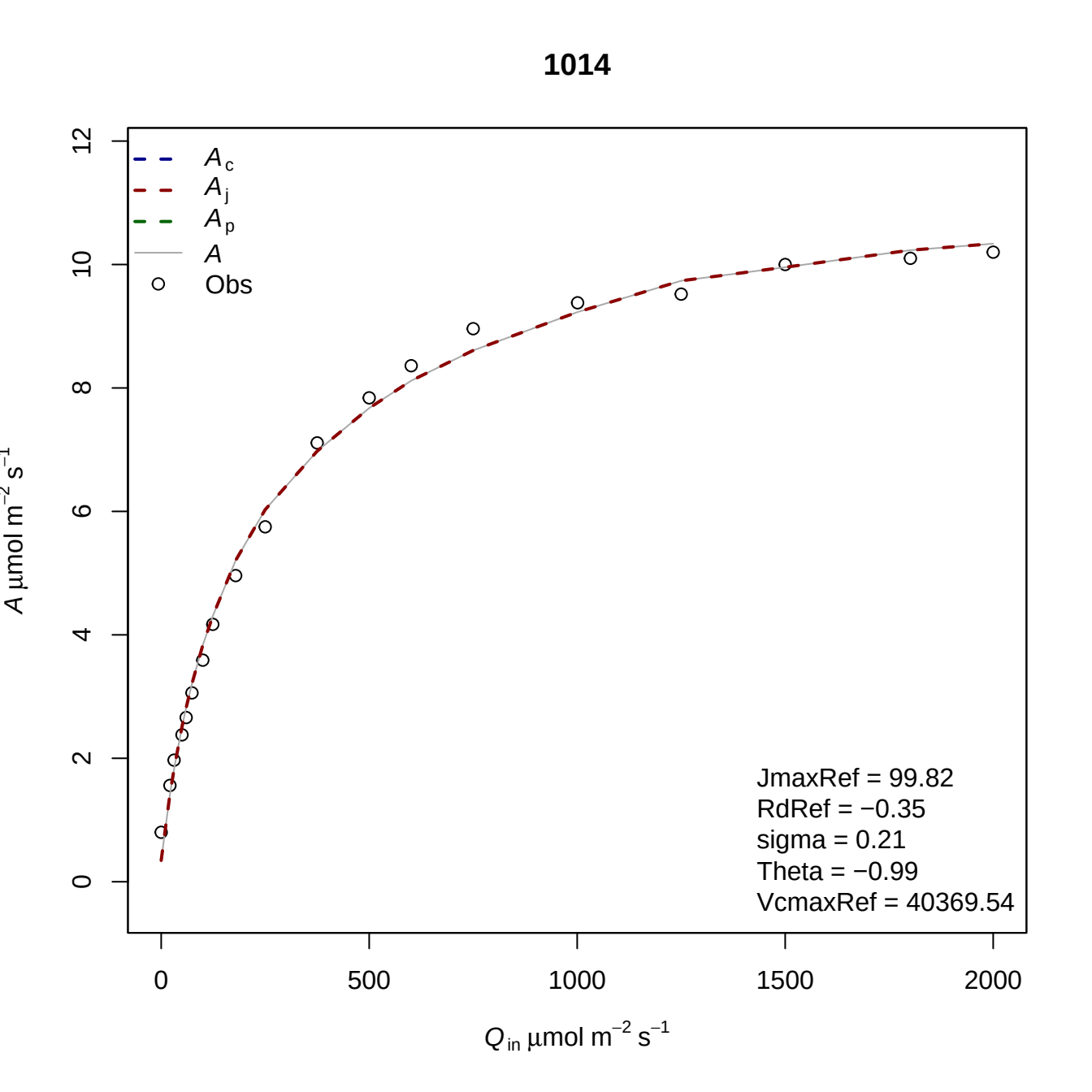

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 82.84
RdRef = 2.19
sigma = 0.15
Theta = 0.01
VcmaxRef = 121394.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

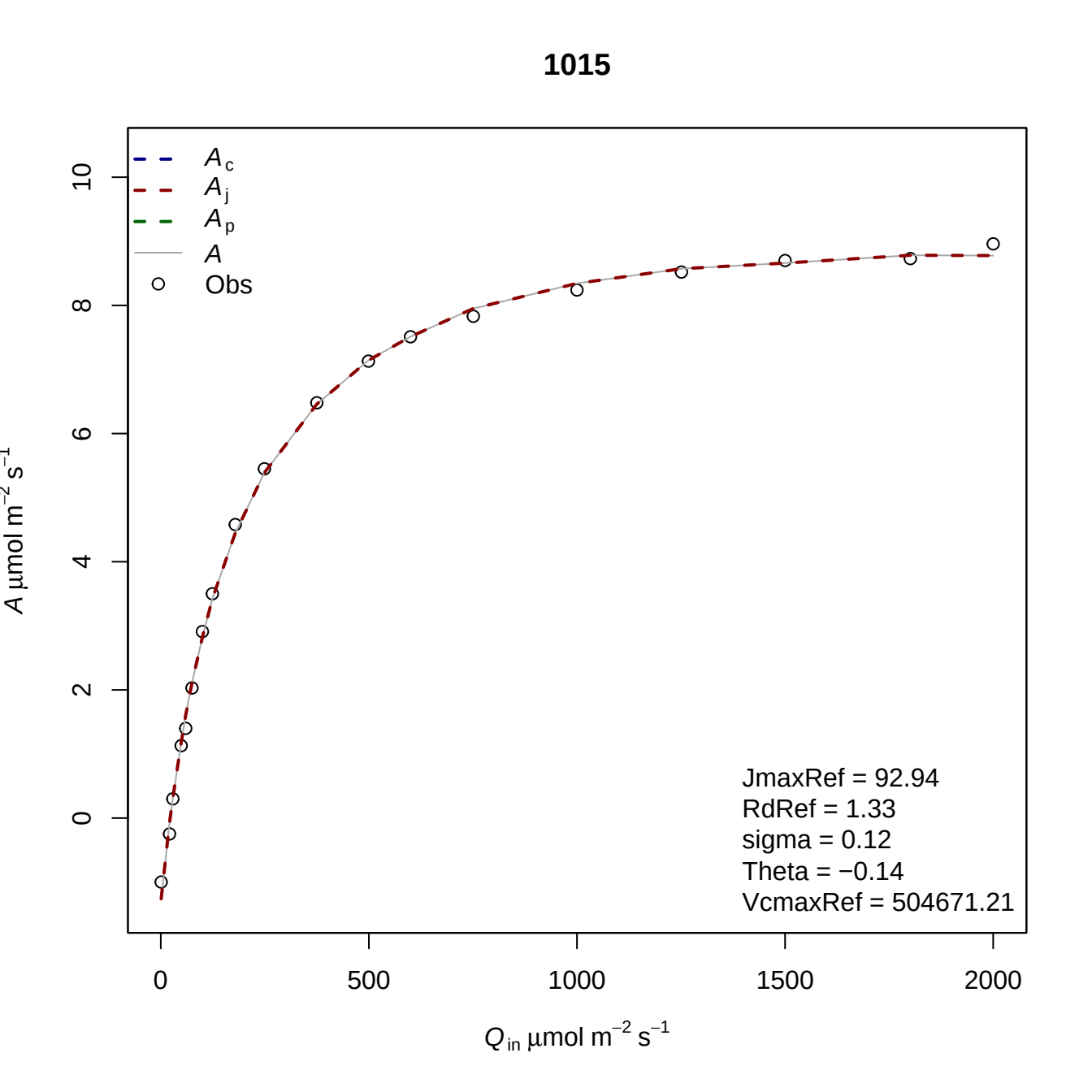

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 80.59
RdRef = 1.61
sigma = 0.17
Theta = −0.03
VcmaxRef = 62.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

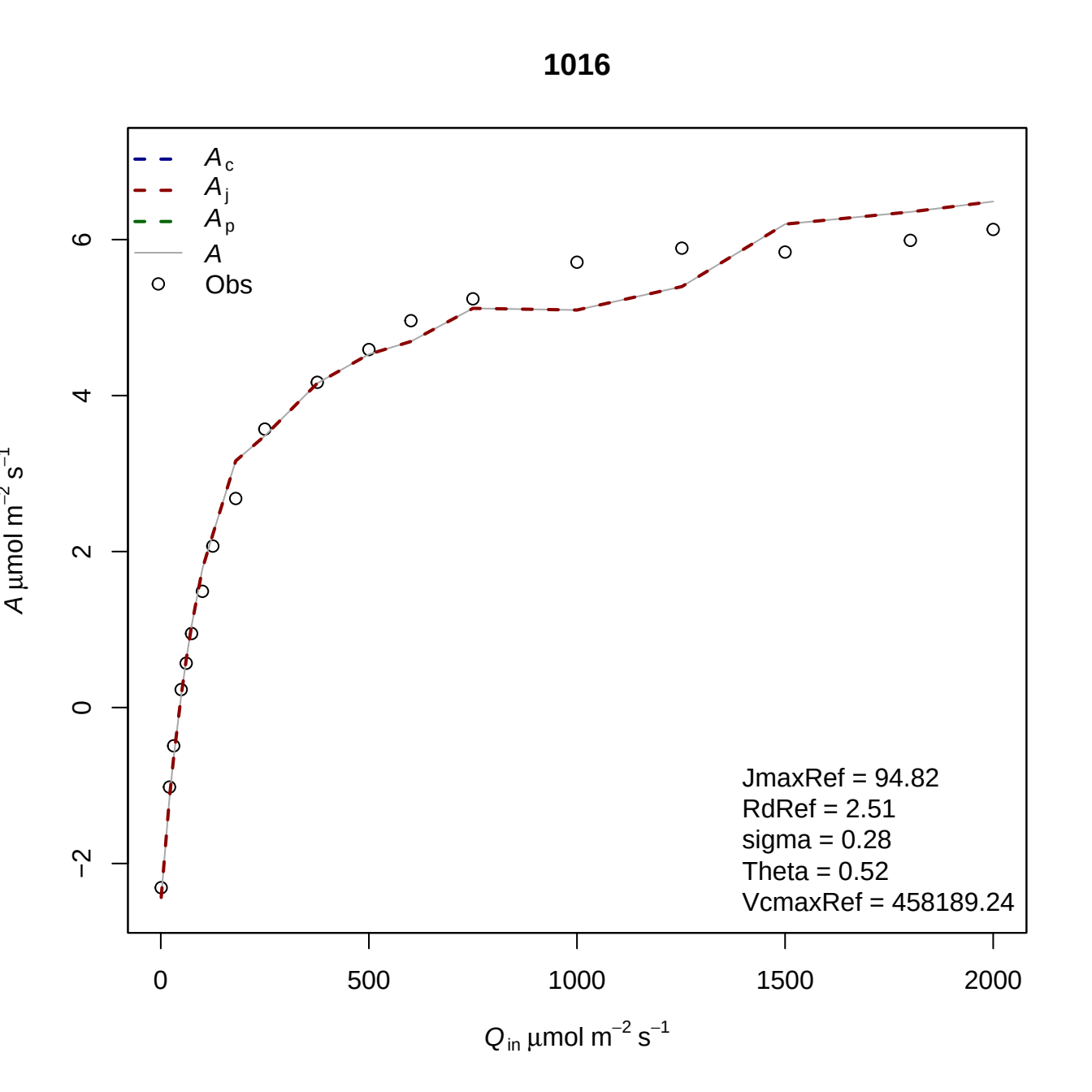

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 39.79
RdRef = 0.76
sigma = 0.18
Theta = 0.34
VcmaxRef = 228711.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

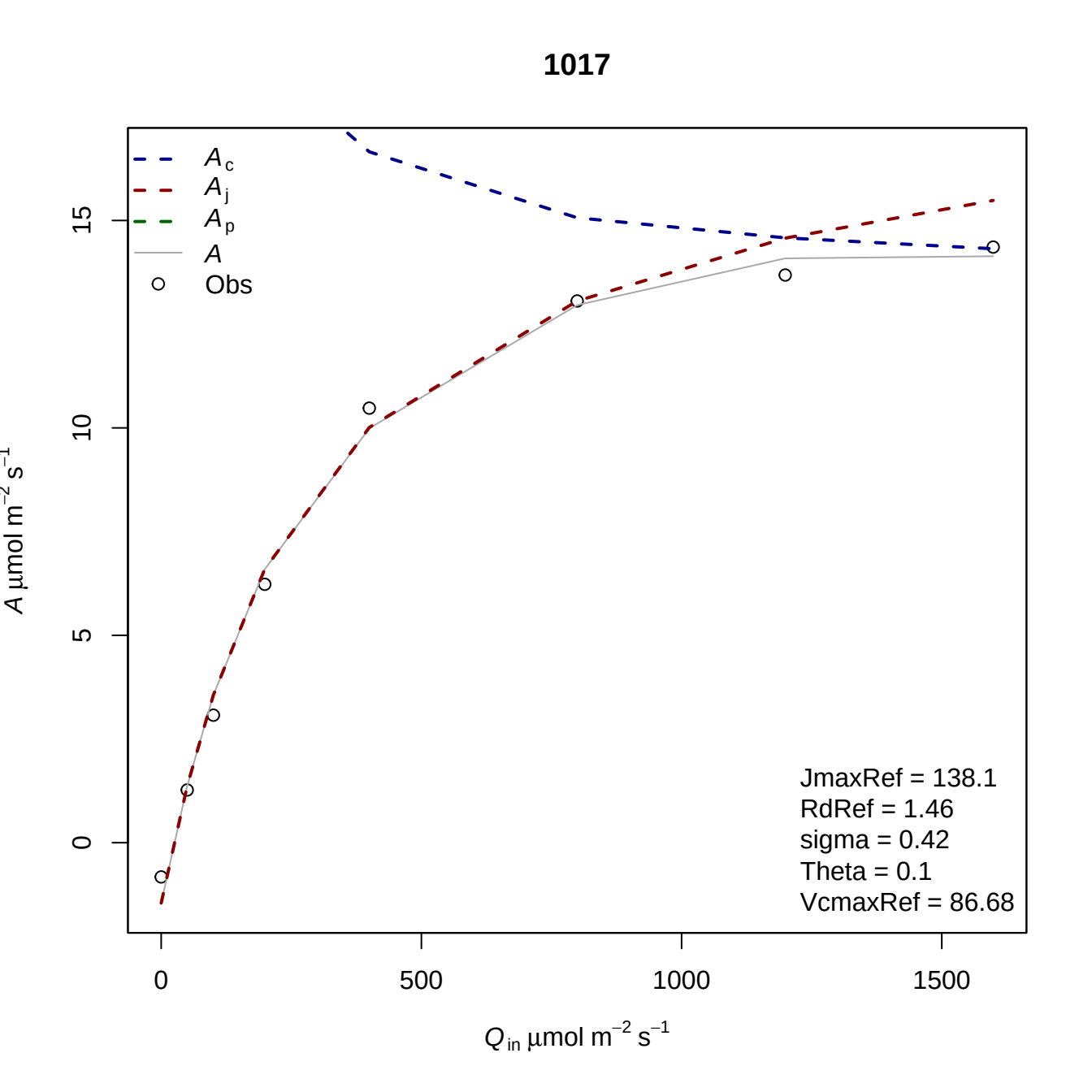

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 84.69
RdRef = 1.63
sigma = 0.23
Theta = 0
VcmaxRef = 101.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

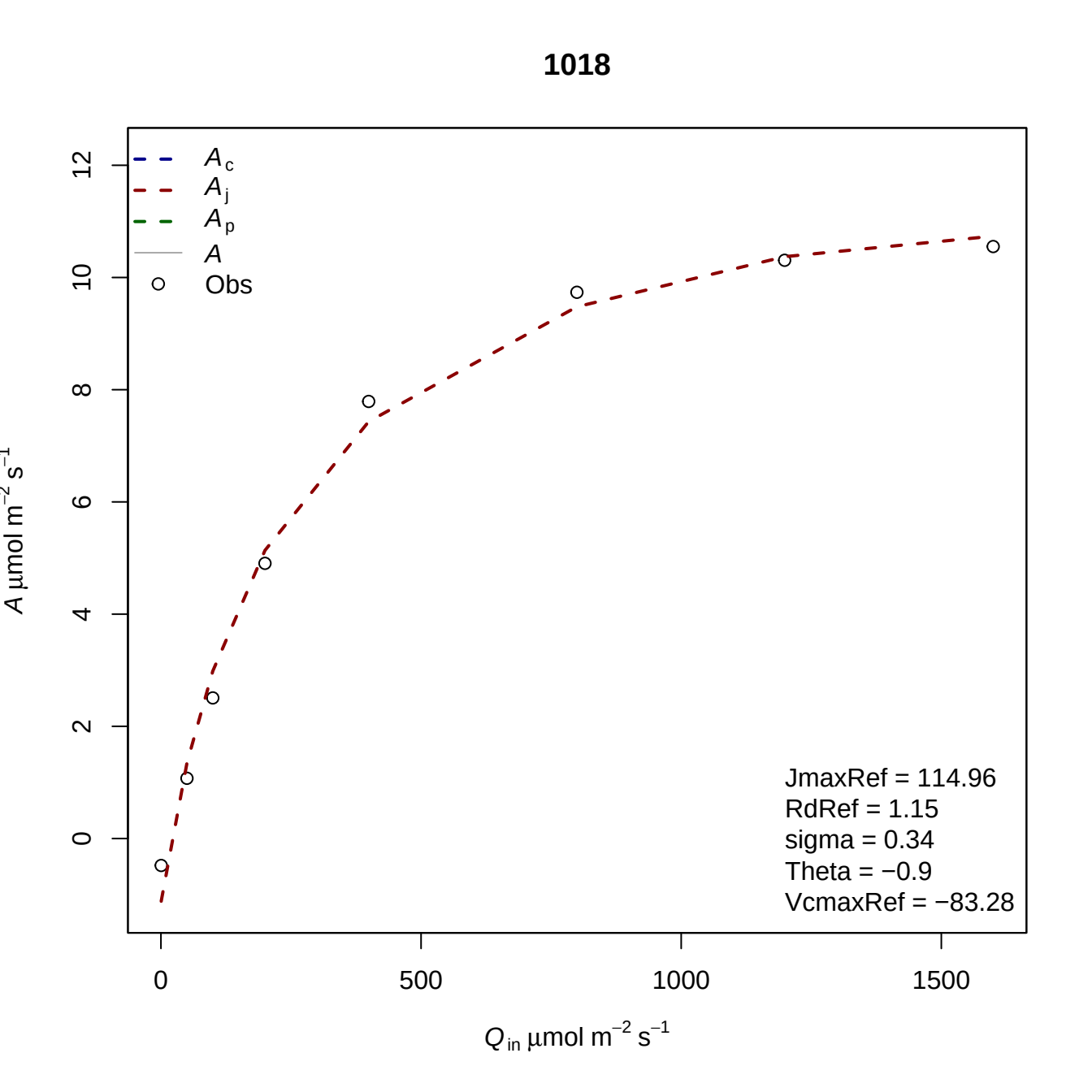

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 146.21
RdRef = 1.48
sigma = 0.18
Theta = 0.29
VcmaxRef = 143.24
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

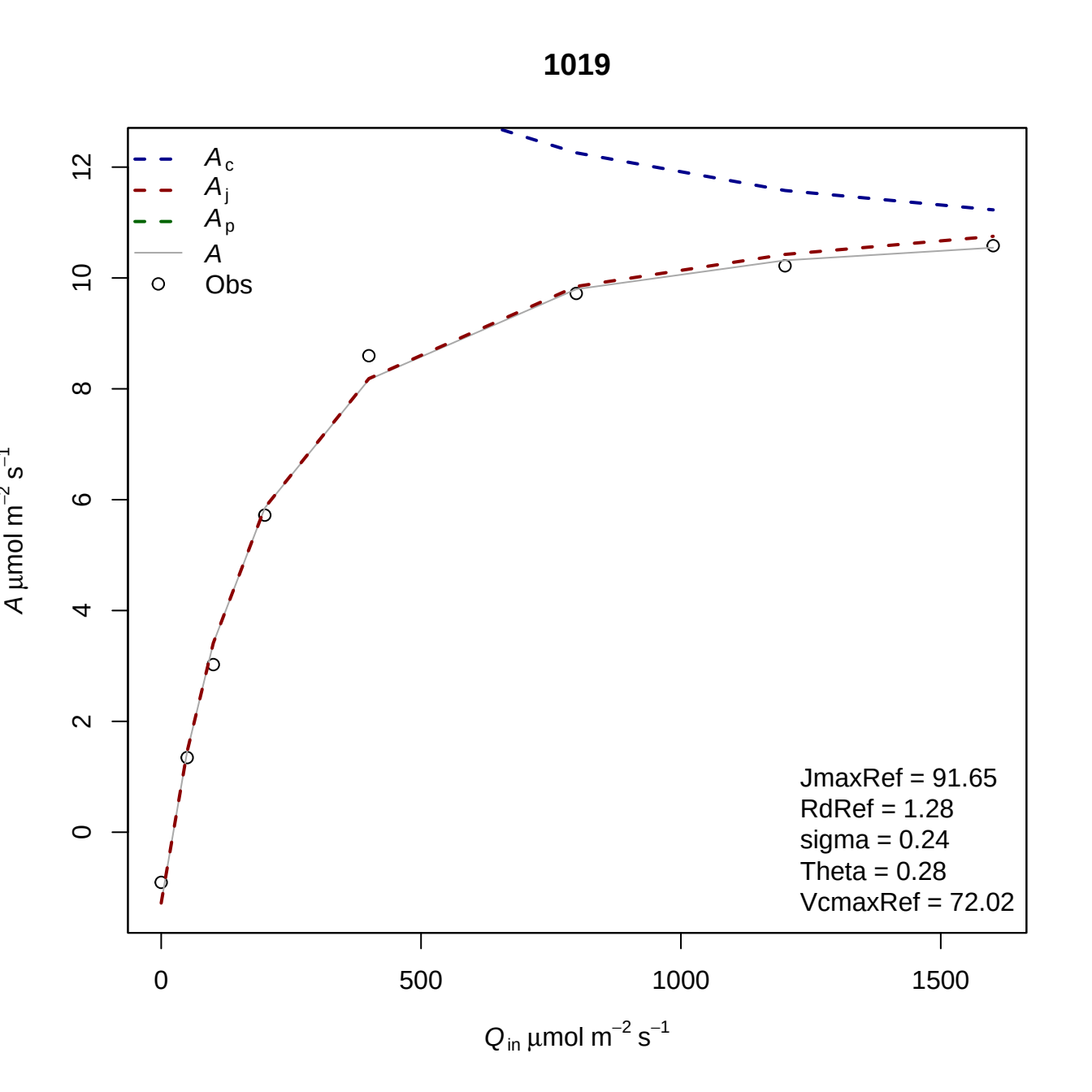

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 99.82
RdRef = −0.35
sigma = 0.21
Theta = −0.99
VcmaxRef = 40369.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

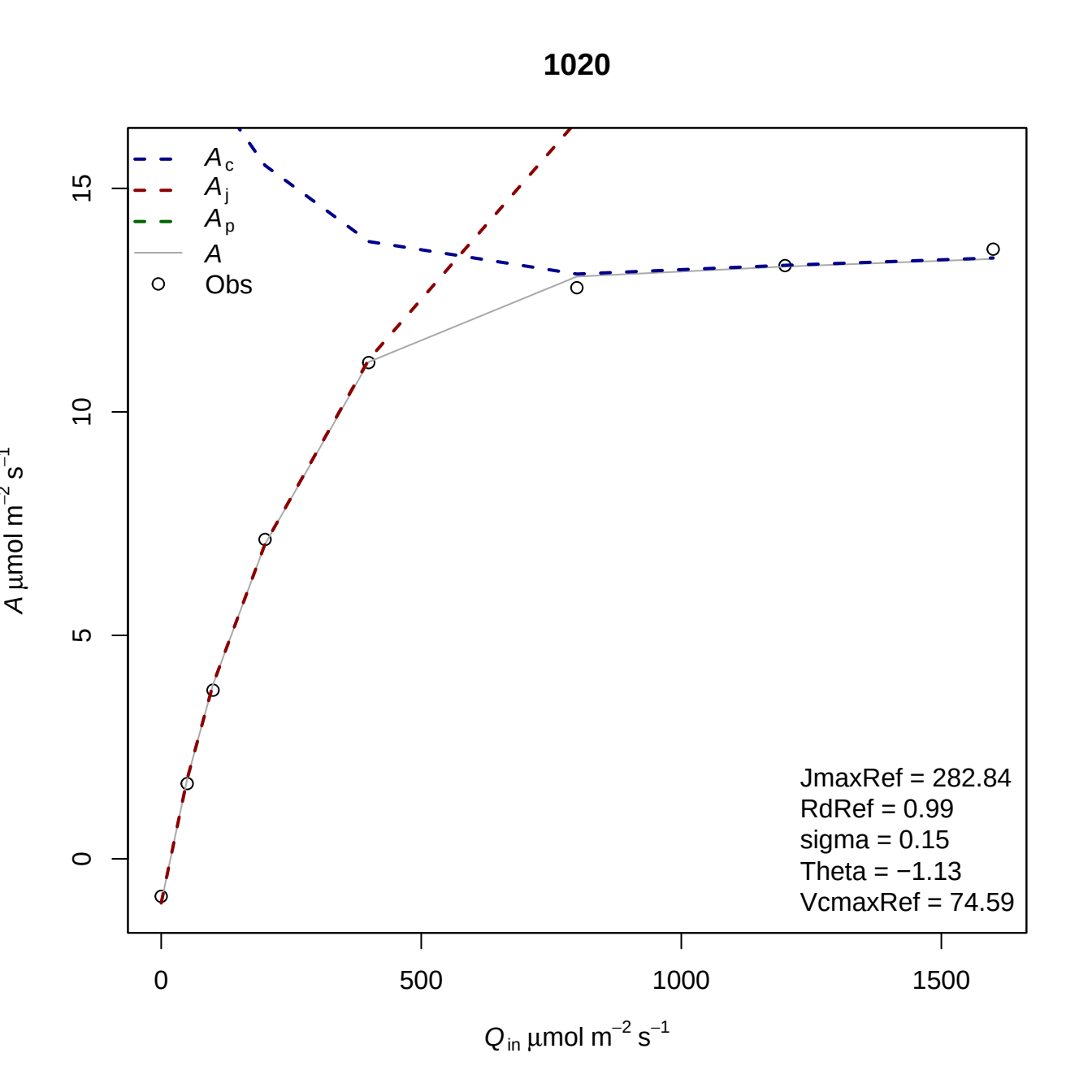

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 92.94
RdRef = 1.33
sigma = 0.12
Theta = −0.14
VcmaxRef = 504671.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

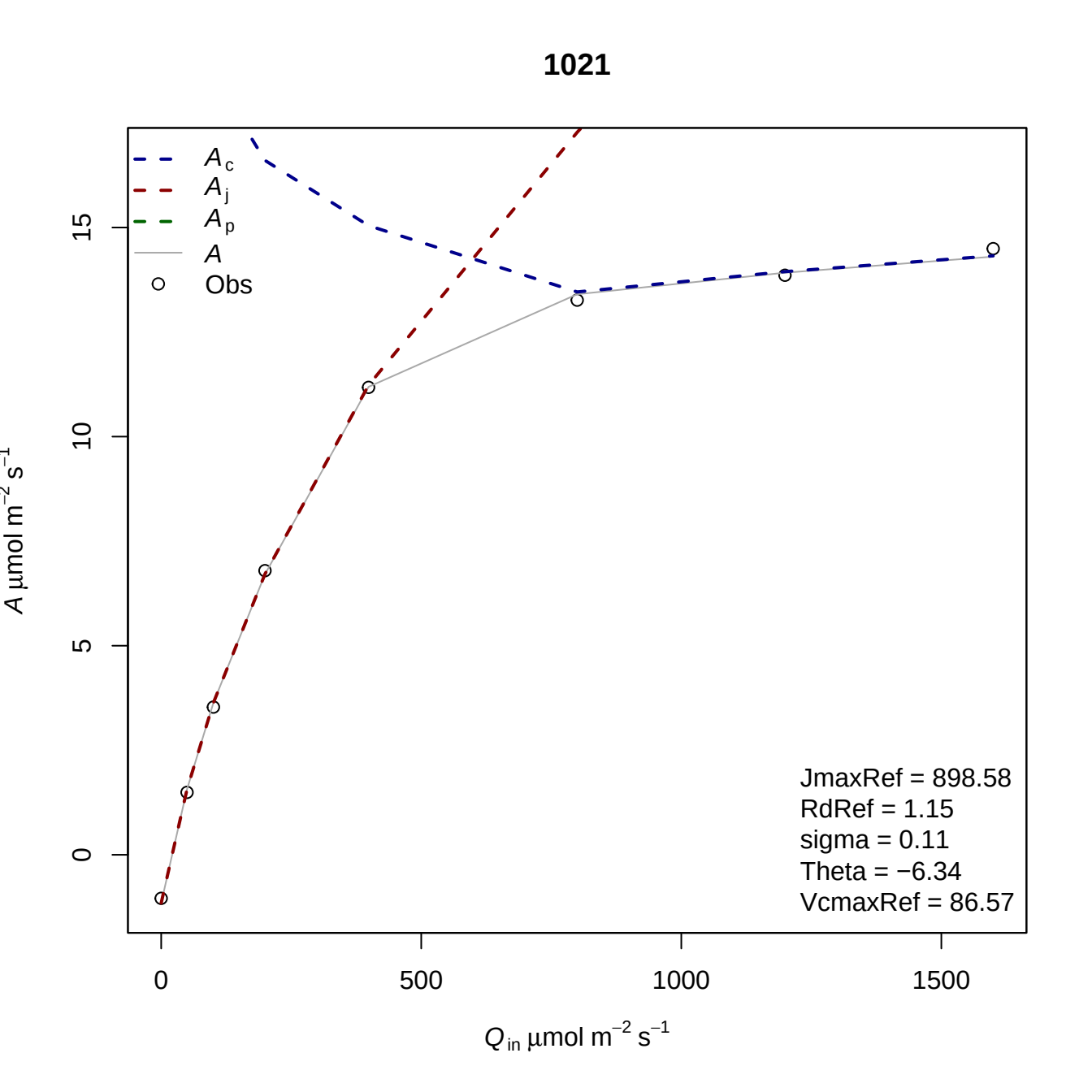

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 94.82
RdRef = 2.51
sigma = 0.28
Theta = 0.52
VcmaxRef = 458189.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

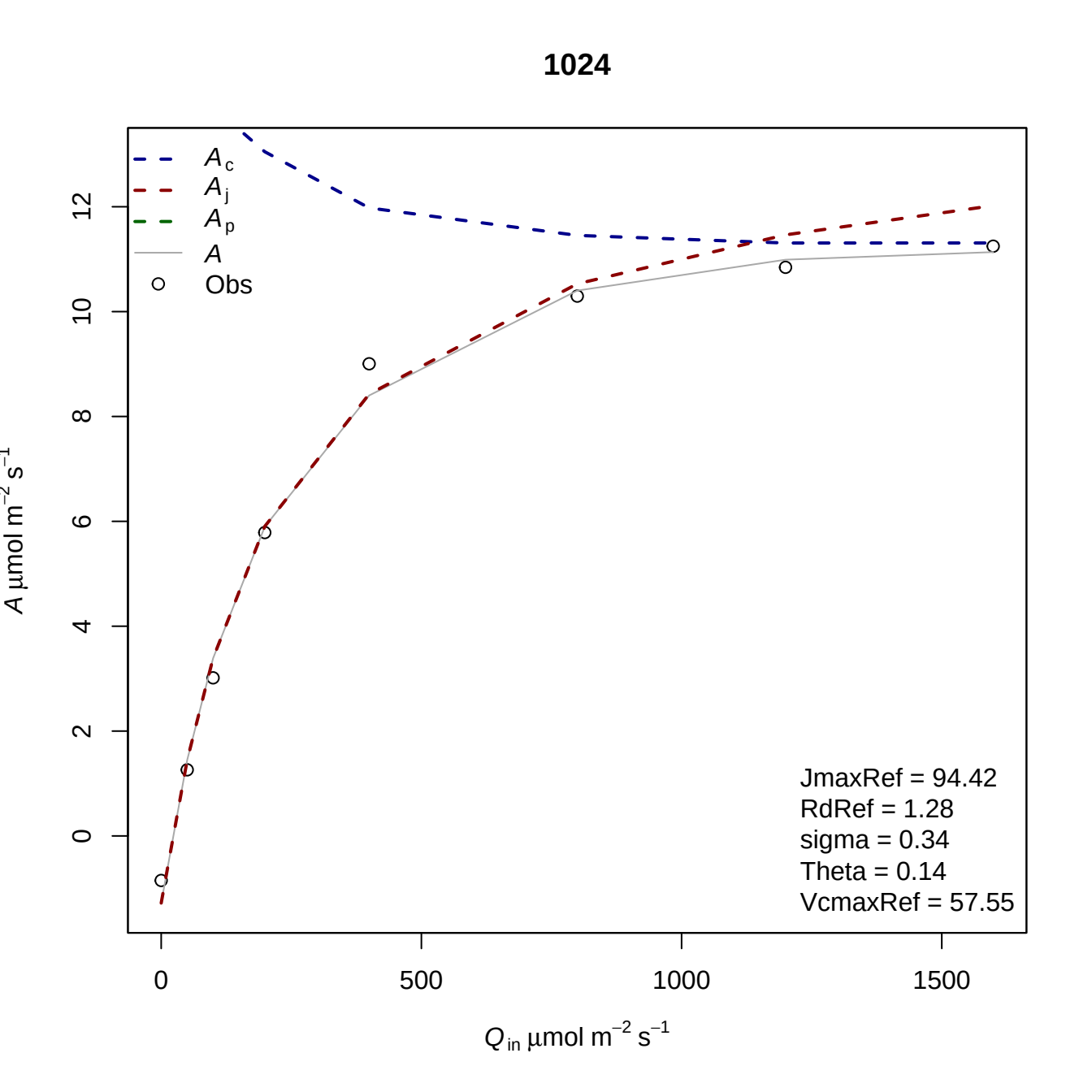

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 138.1
RdRef = 1.46
sigma = 0.42
Theta = 0.1
VcmaxRef = 86.68
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

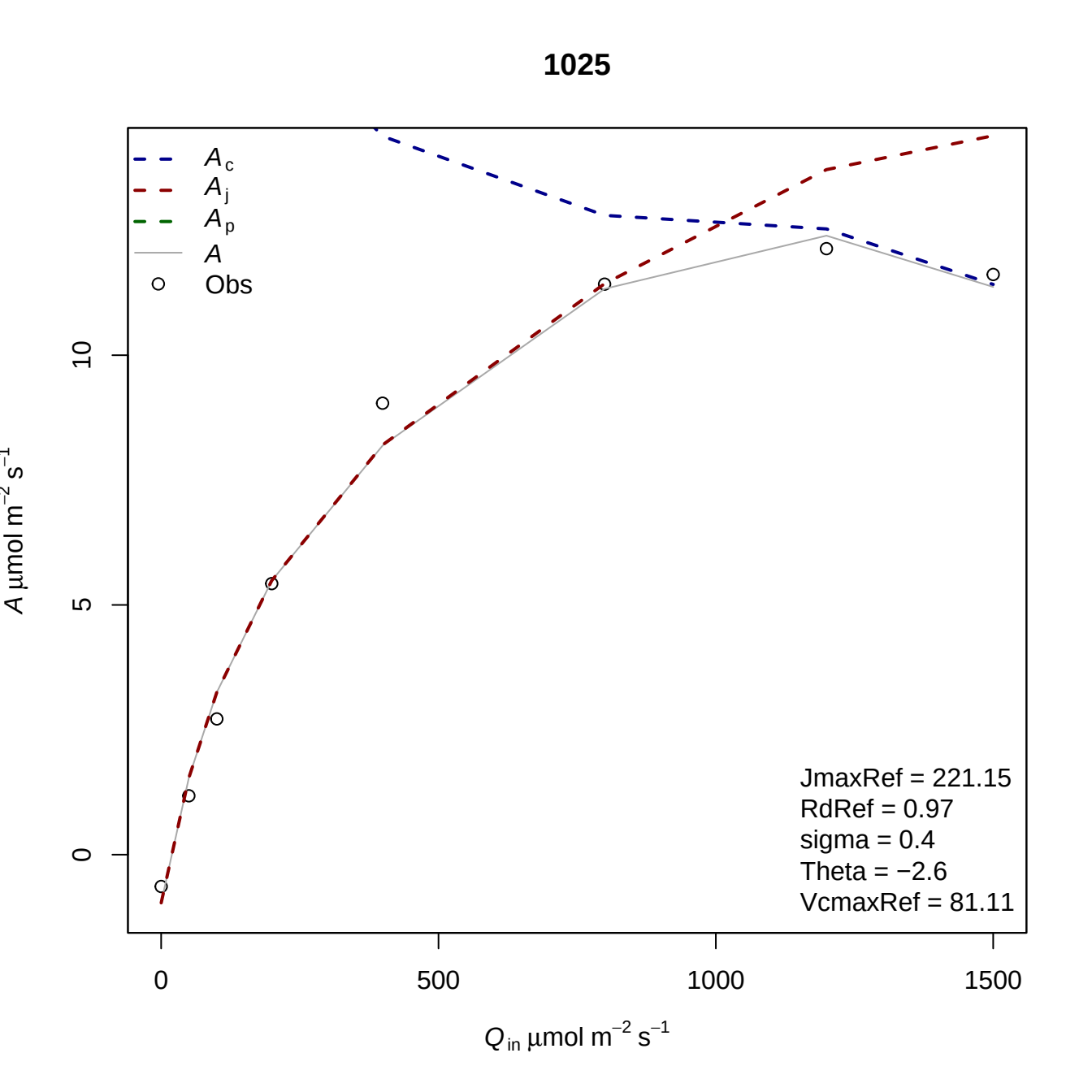

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 114.96
RdRef = 1.15
sigma = 0.34
Theta = −0.9
VcmaxRef = −83.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

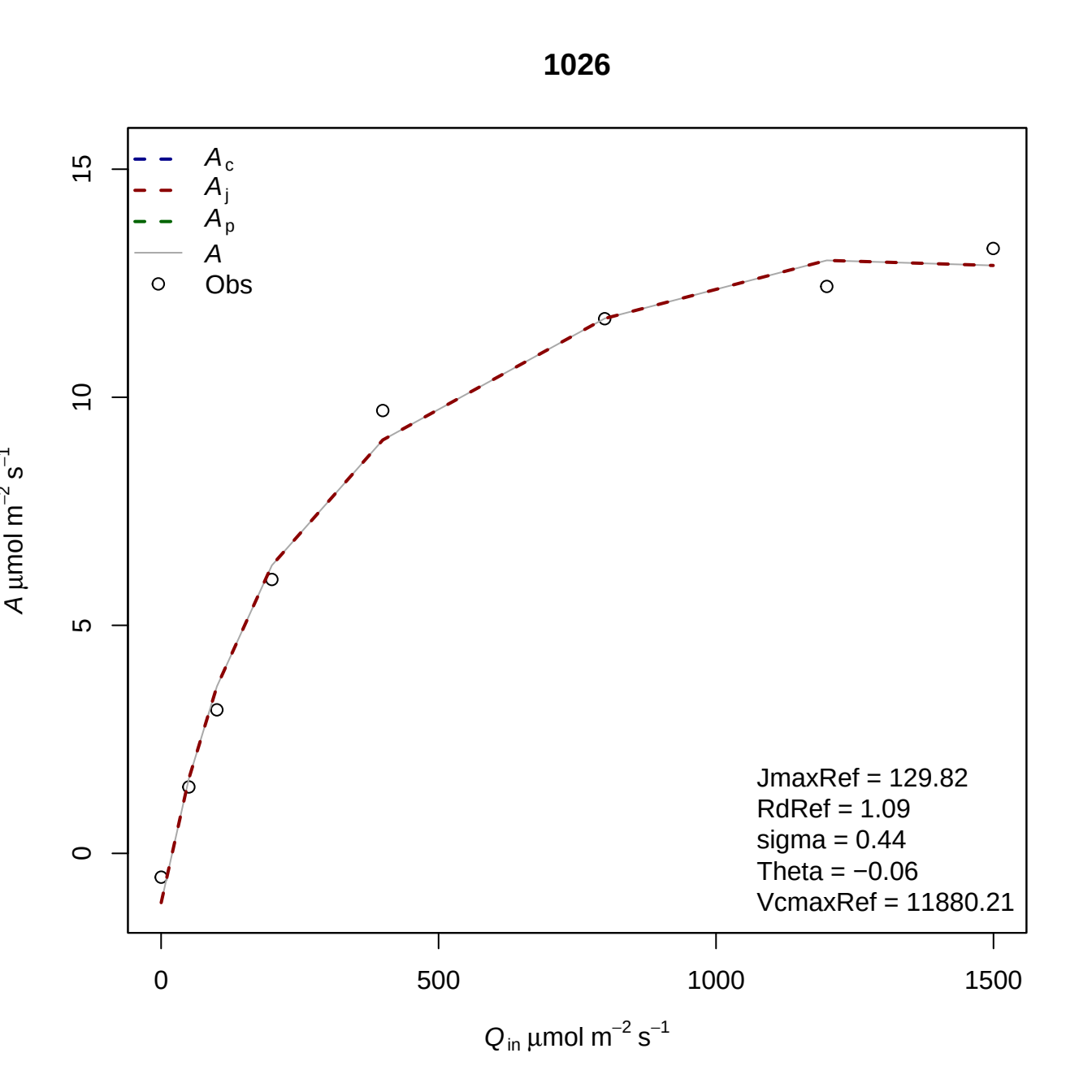

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 91.65
RdRef = 1.28
sigma = 0.24
Theta = 0.28
VcmaxRef = 72.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

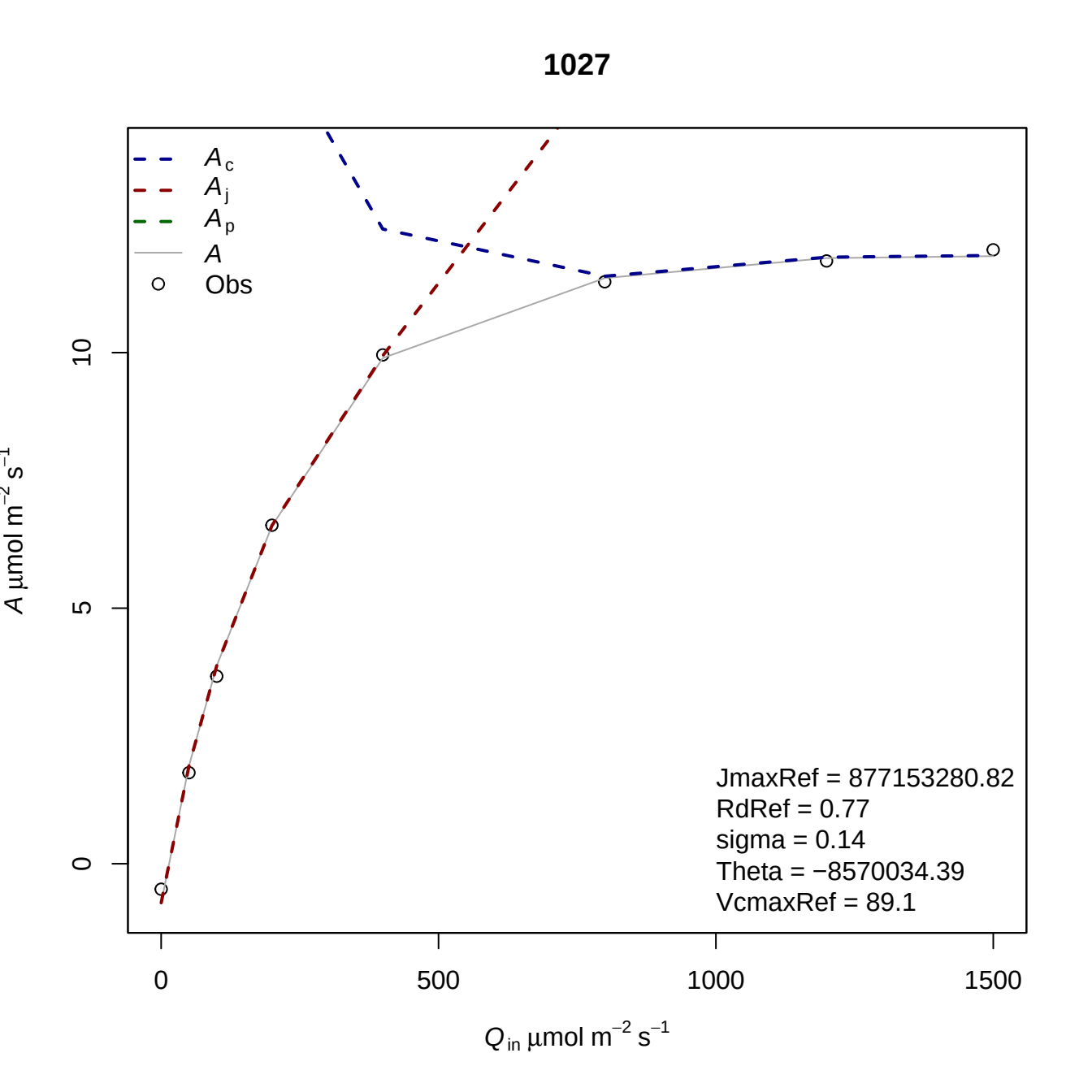

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 282.84
RdRef = 0.99
sigma = 0.15
Theta = −1.13
VcmaxRef = 74.59
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

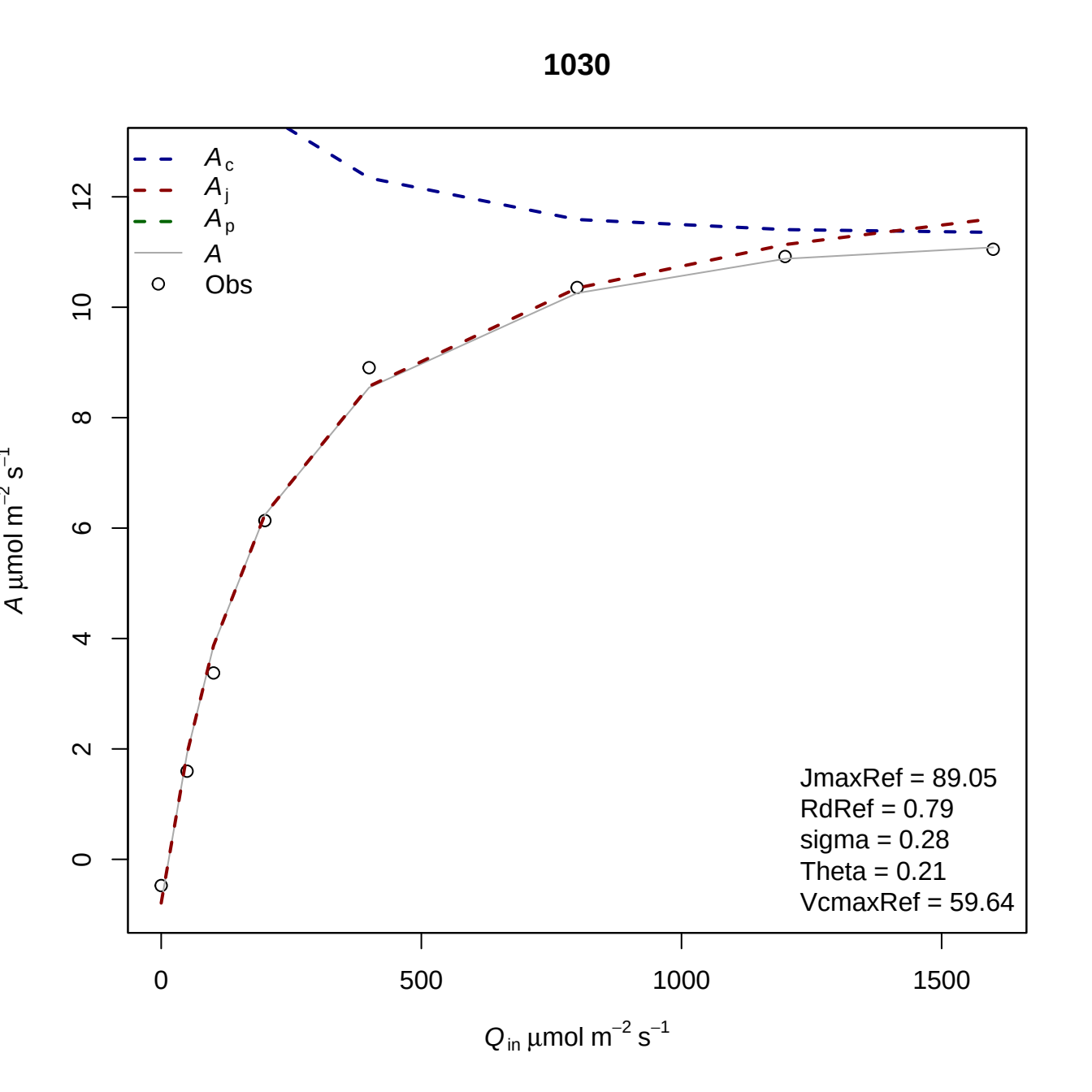

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 898.58
RdRef = 1.15
sigma = 0.11
Theta = −6.34
VcmaxRef = 86.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

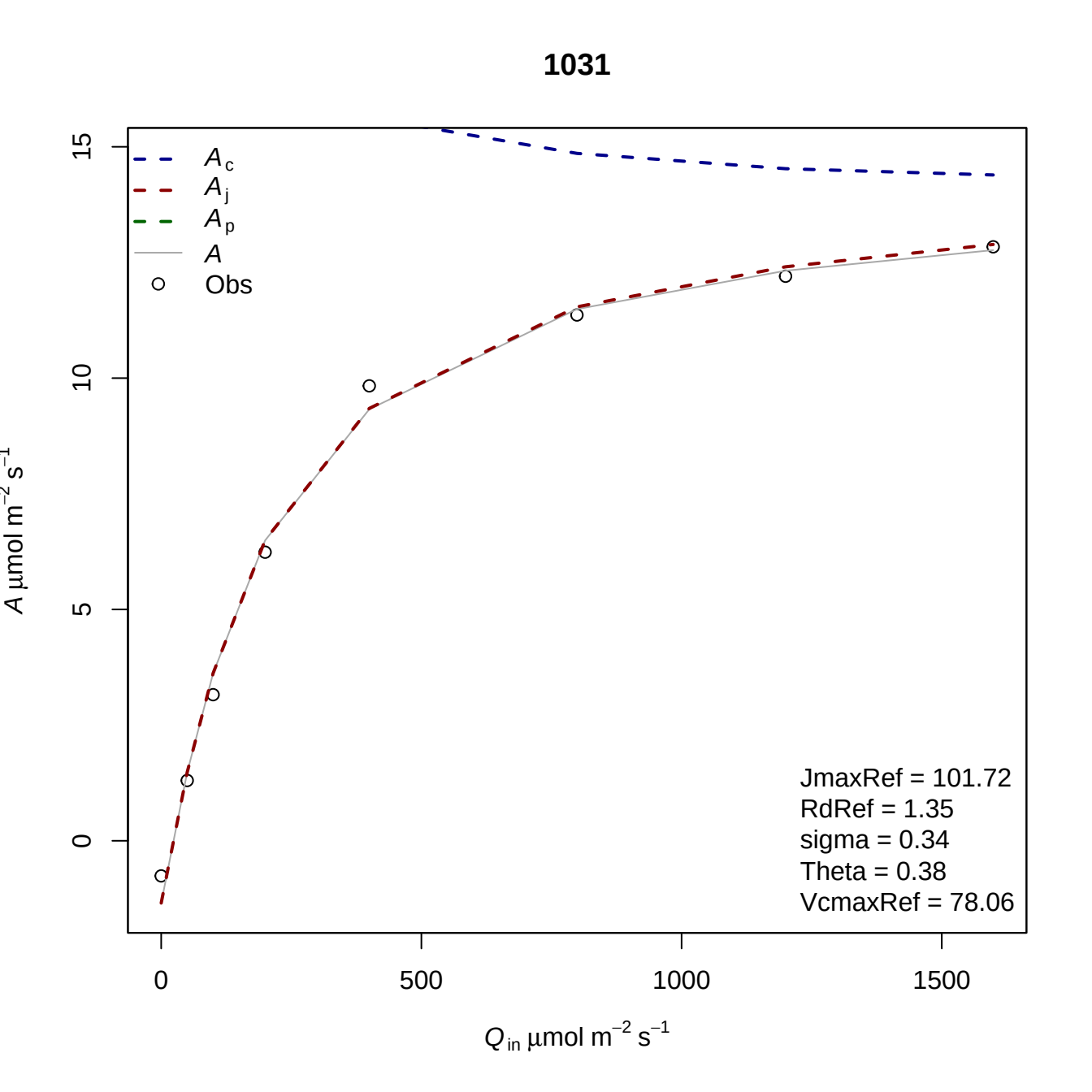

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 94.42
RdRef = 1.28
sigma = 0.34
Theta = 0.14
VcmaxRef = 57.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

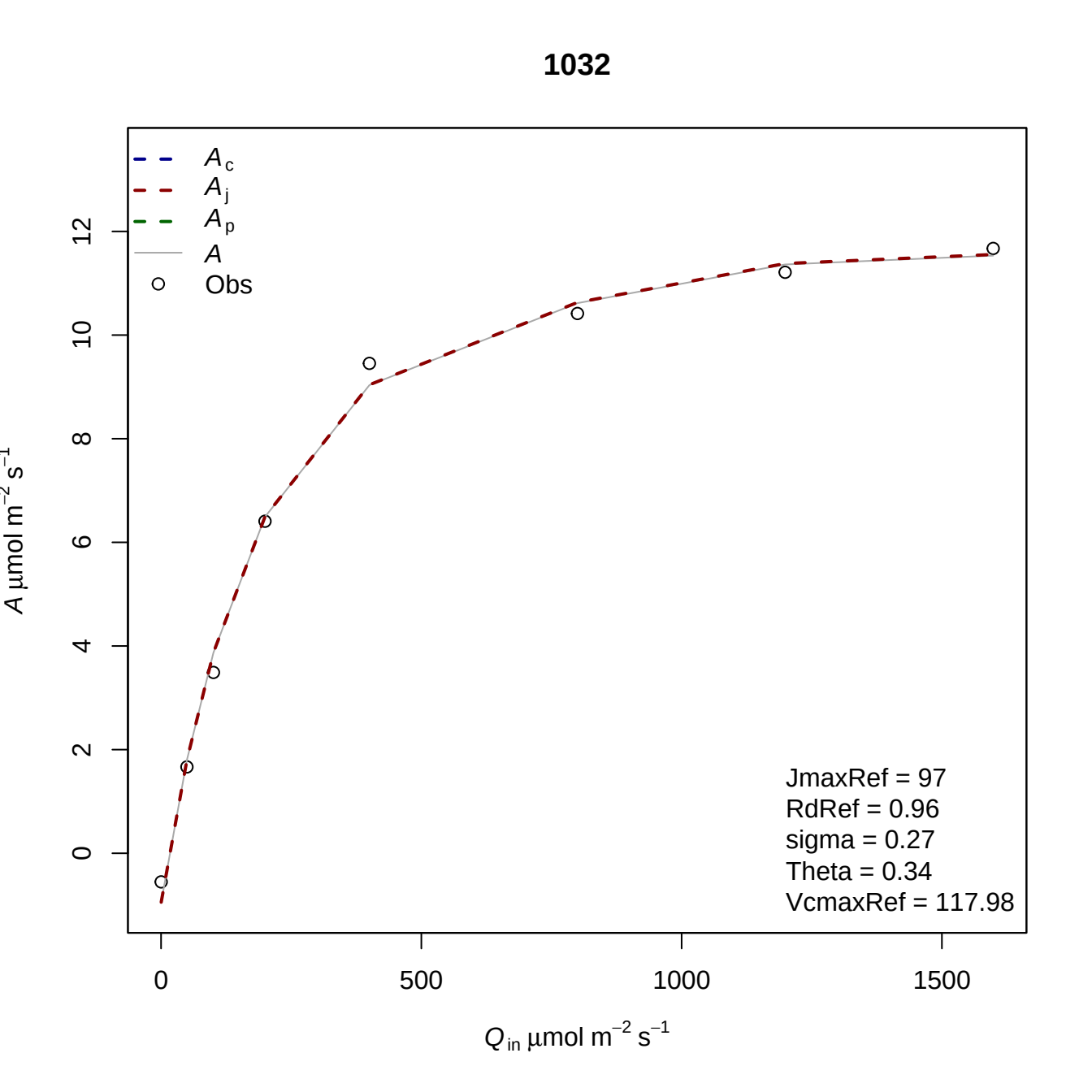

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 221.15
RdRef = 0.97
sigma = 0.4
Theta = −2.6
VcmaxRef = 81.11
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

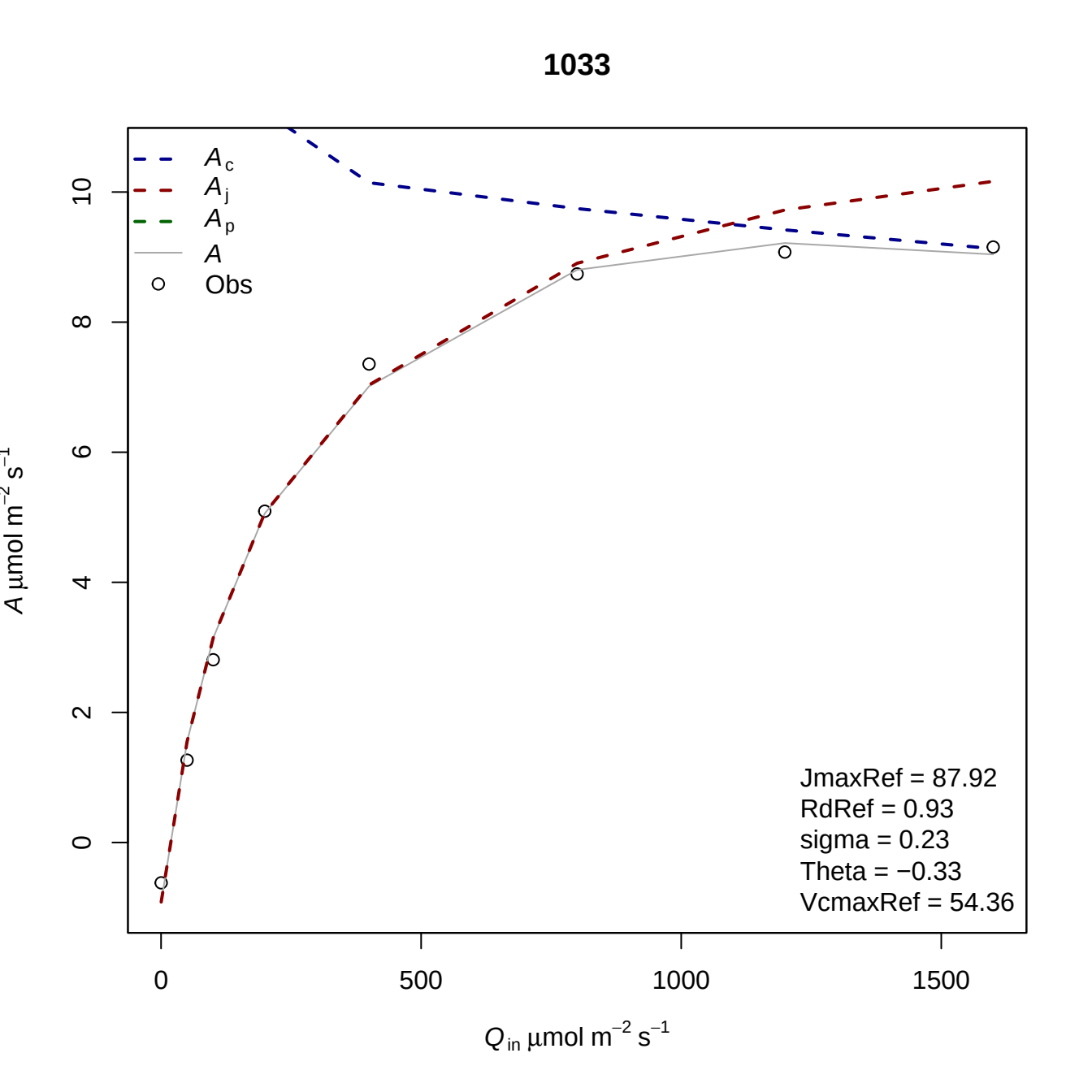

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 129.82
RdRef = 1.09
sigma = 0.44
Theta = −0.06
VcmaxRef = 11880.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

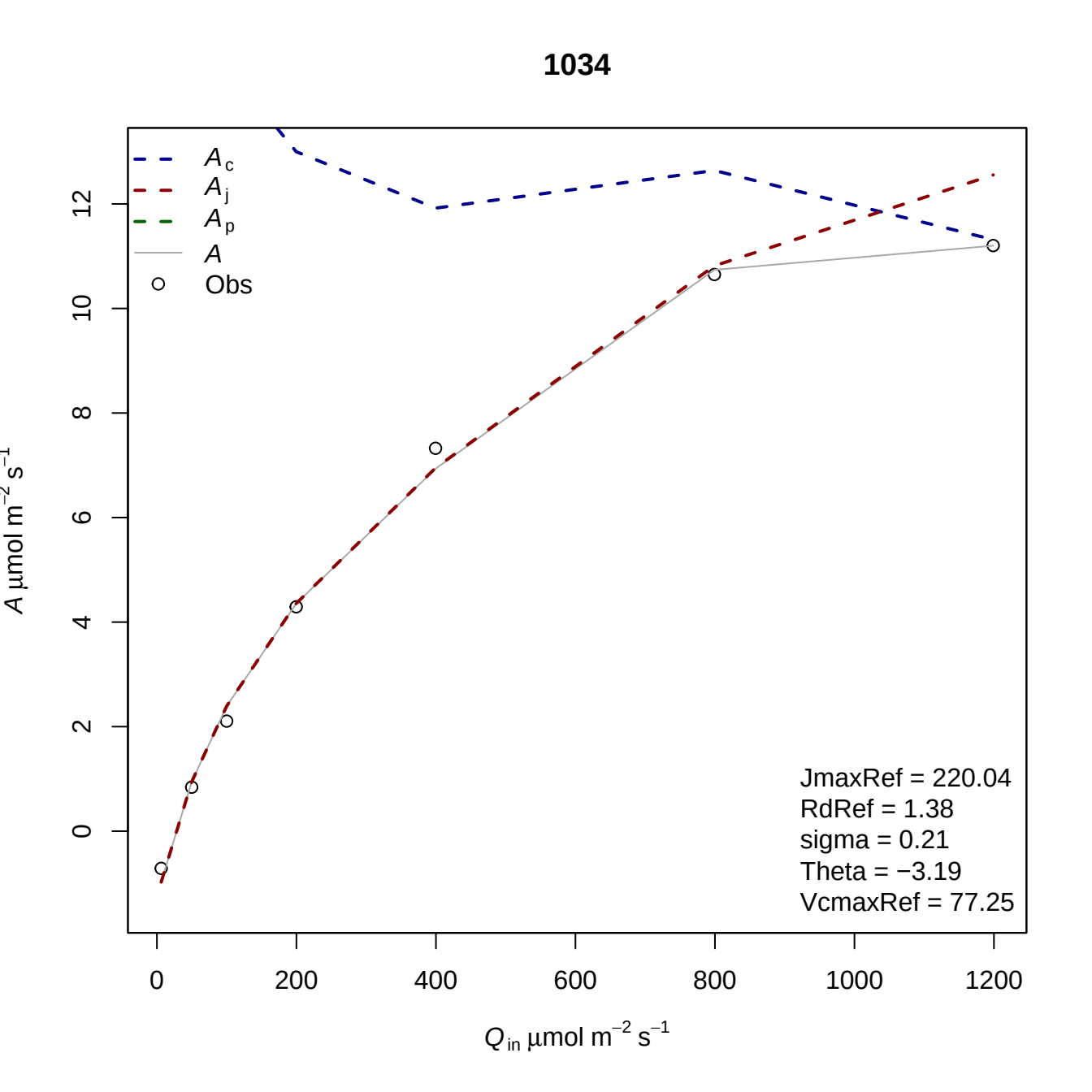

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 877153280.82
RdRef = 0.77
sigma = 0.14
Theta = −8570034.39
VcmaxRef = 89.1
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

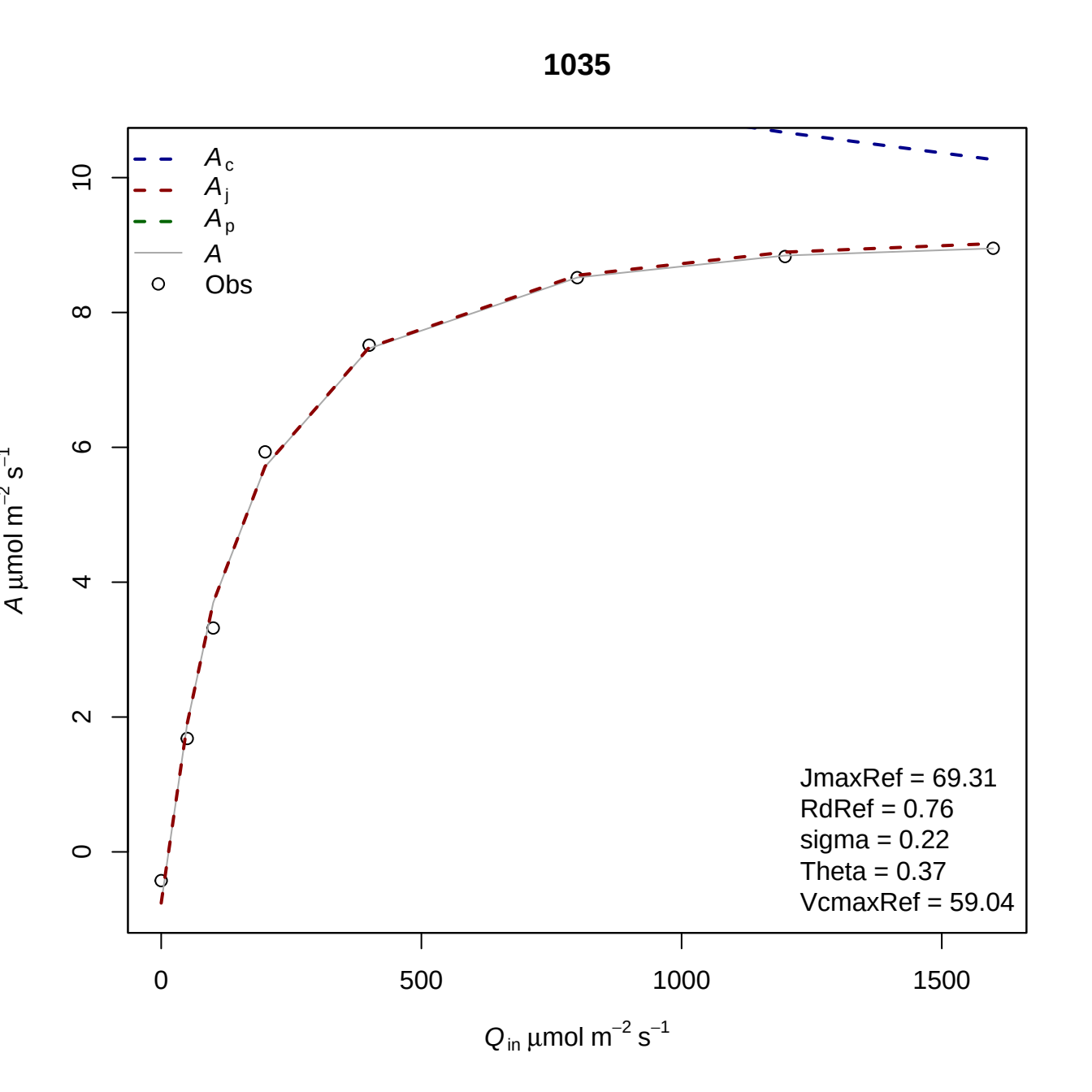

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 89.05
RdRef = 0.79
sigma = 0.28
Theta = 0.21
VcmaxRef = 59.64
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

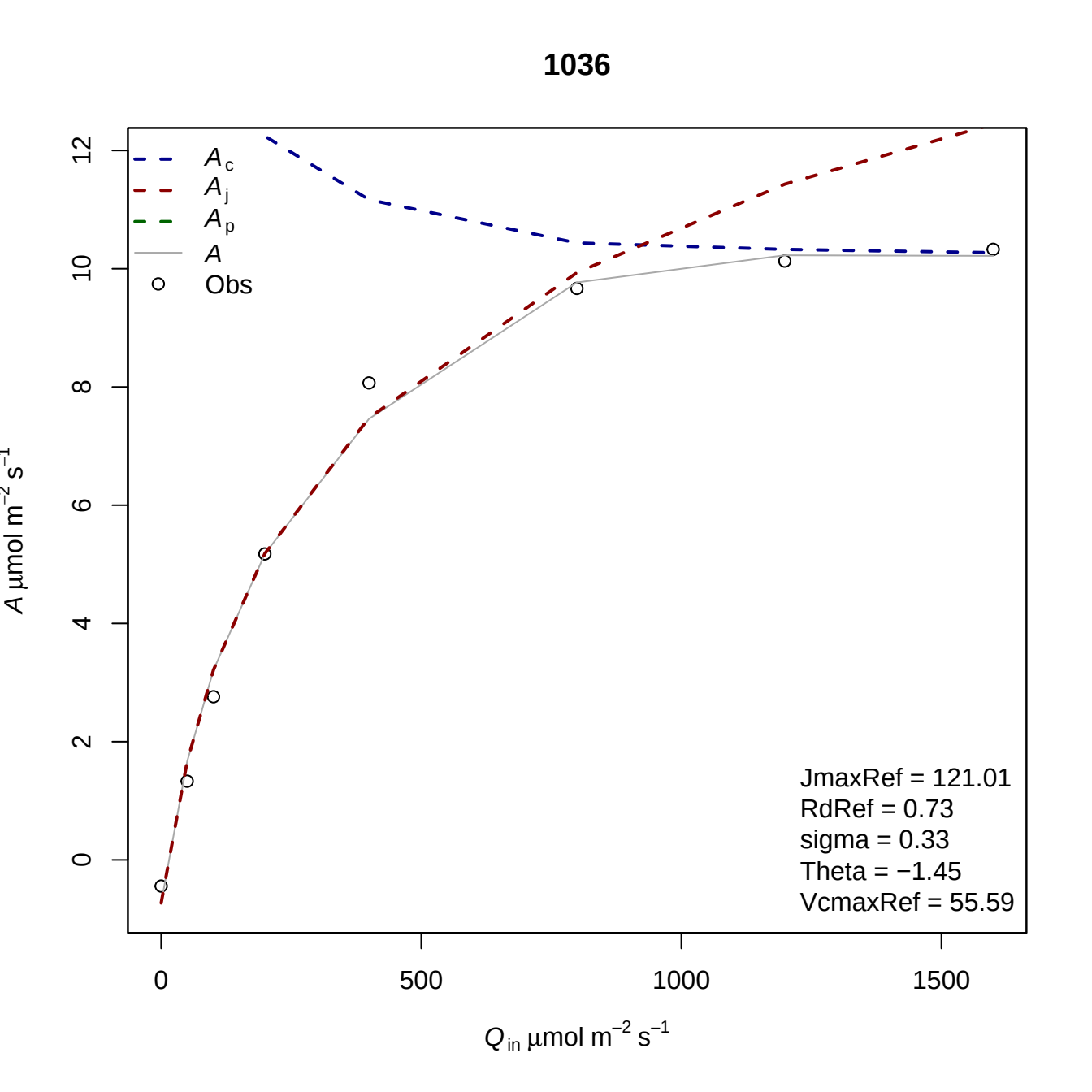

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 101.72
RdRef = 1.35
sigma = 0.34
Theta = 0.38
VcmaxRef = 78.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

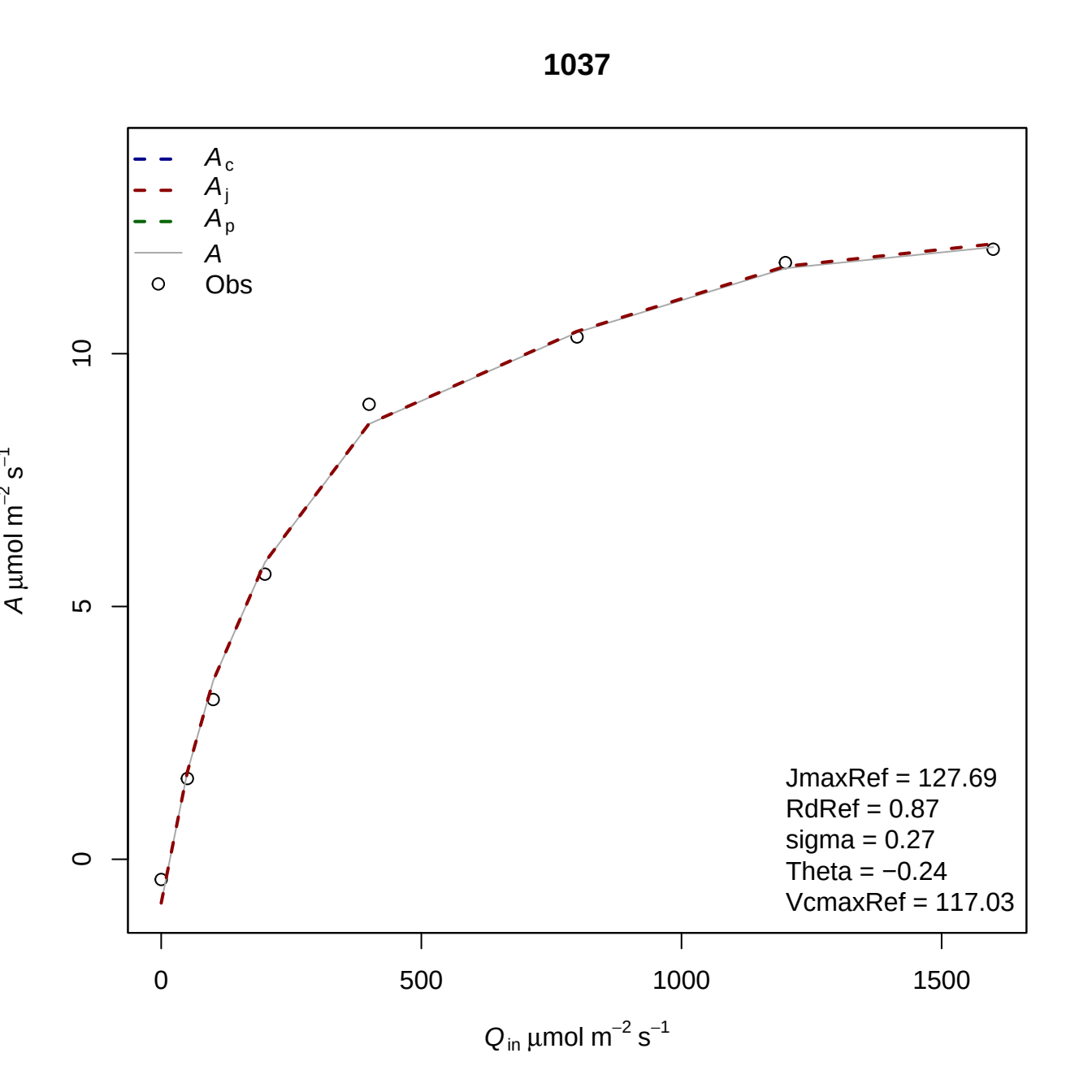

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 97
RdRef = 0.96
sigma = 0.27
Theta = 0.34
VcmaxRef = 117.98
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

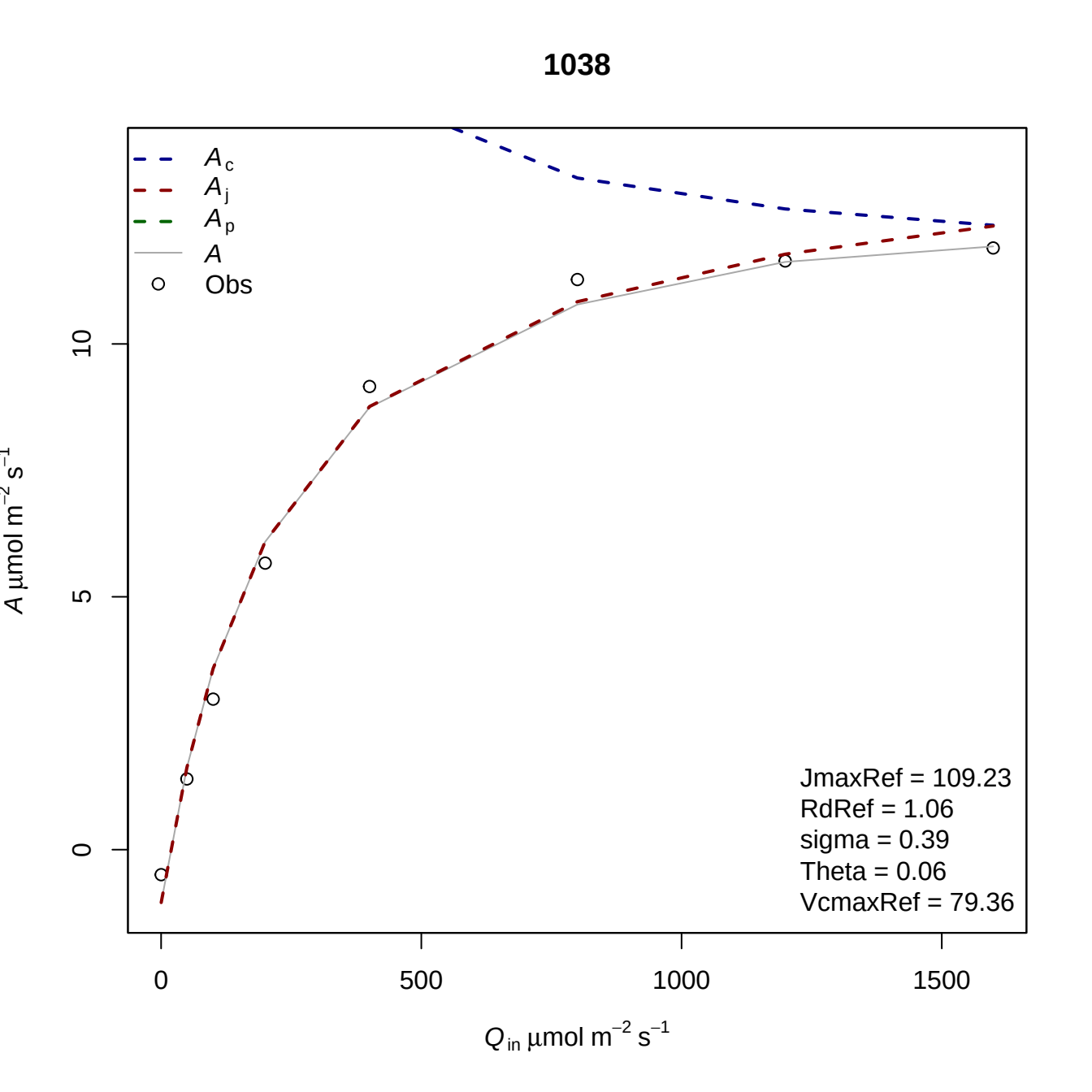

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 87.92
RdRef = 0.93
sigma = 0.23
Theta = −0.33
VcmaxRef = 54.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

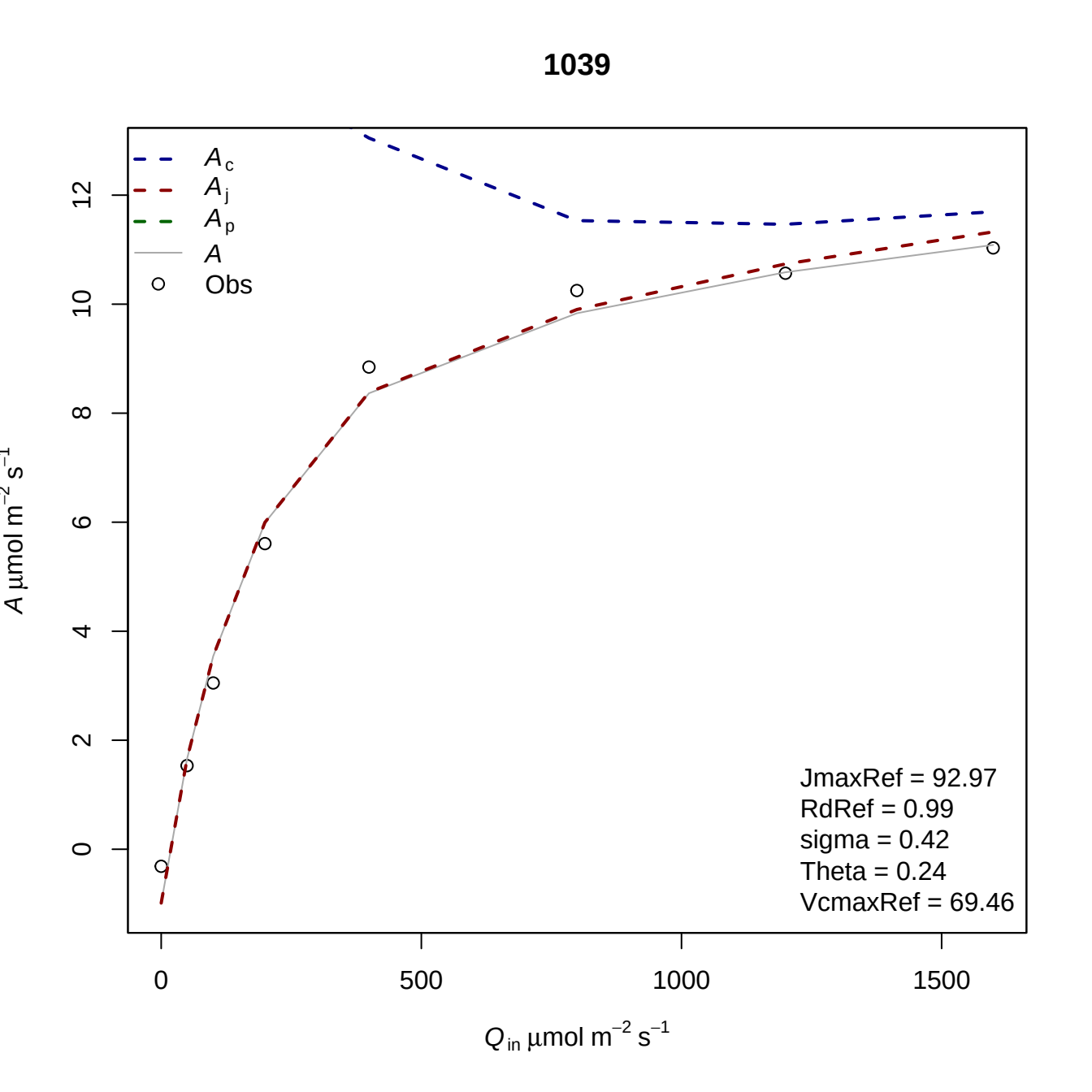

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 220.04
RdRef = 1.38
sigma = 0.21
Theta = −3.19
VcmaxRef = 77.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

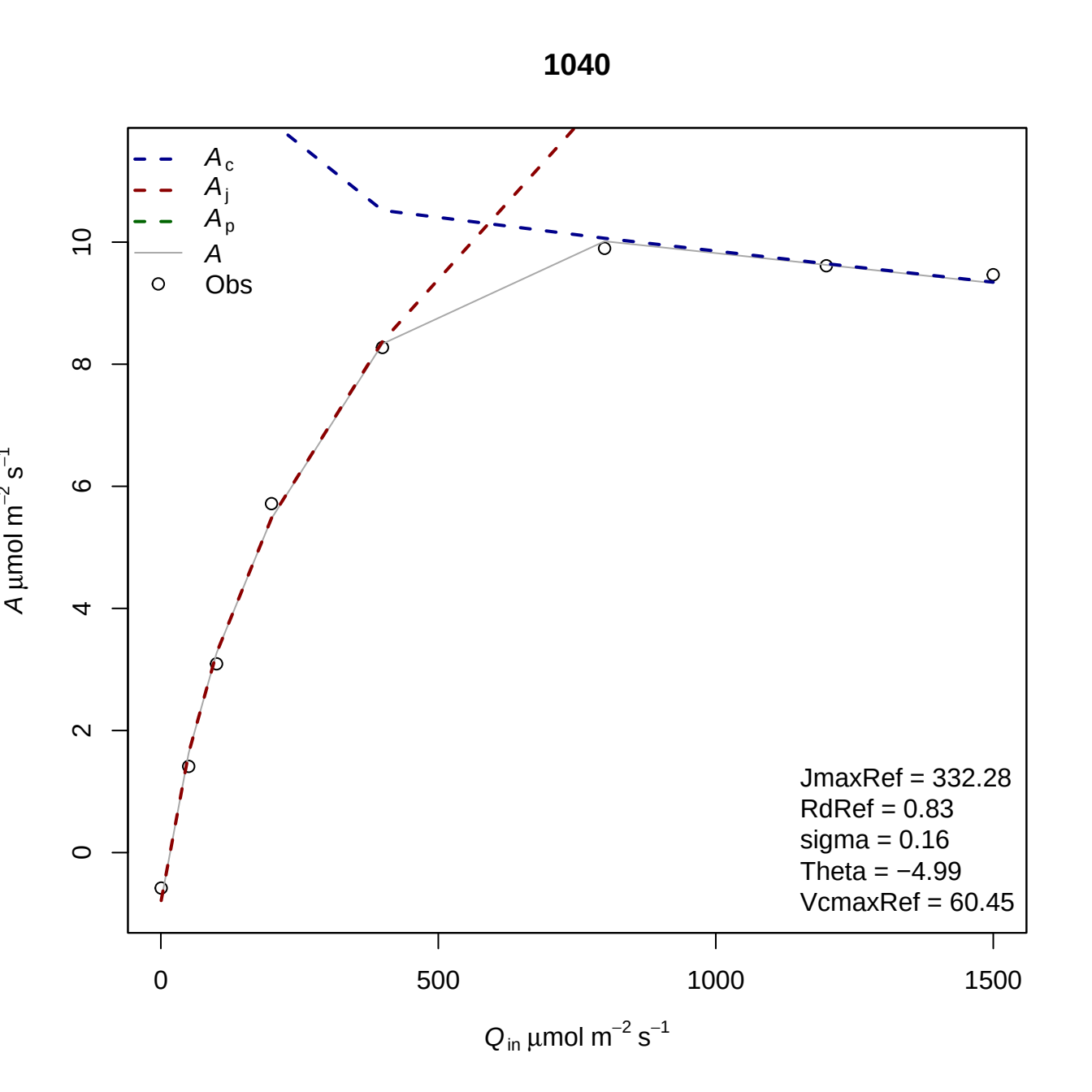

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 69.31
RdRef = 0.76
sigma = 0.22
Theta = 0.37
VcmaxRef = 59.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

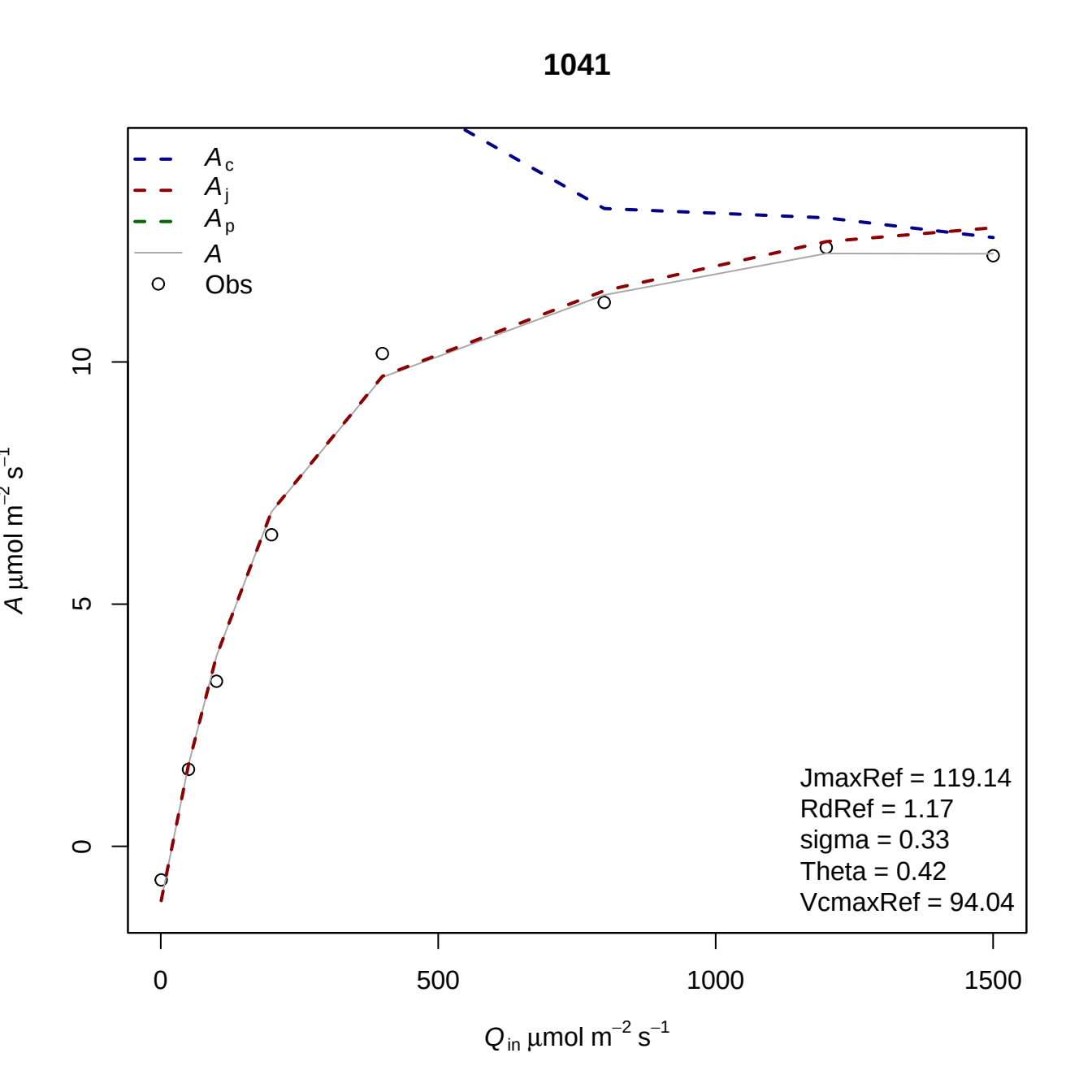

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 121.01
RdRef = 0.73
sigma = 0.33
Theta = −1.45
VcmaxRef = 55.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

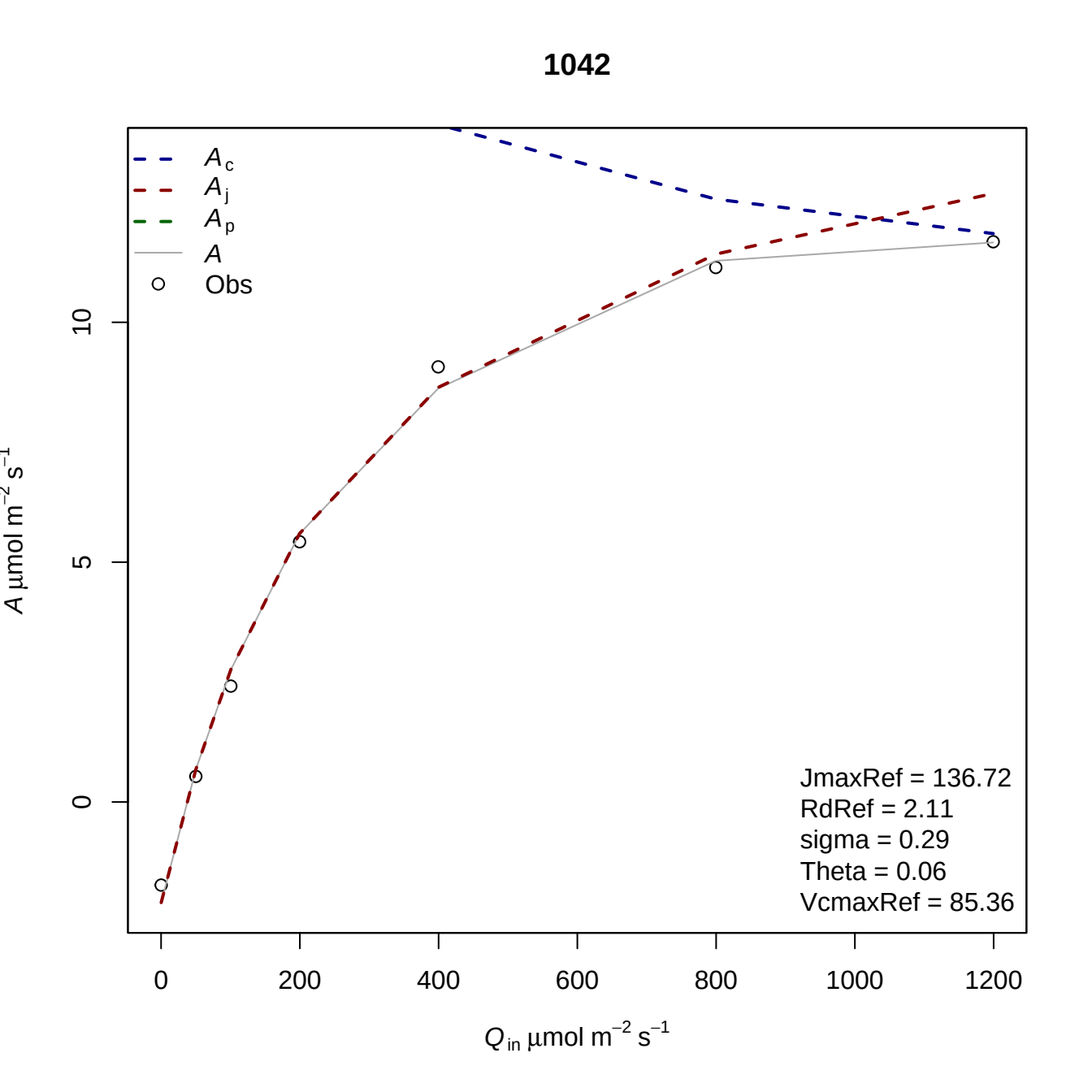

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 127.69
RdRef = 0.87
sigma = 0.27
Theta = −0.24
VcmaxRef = 117.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

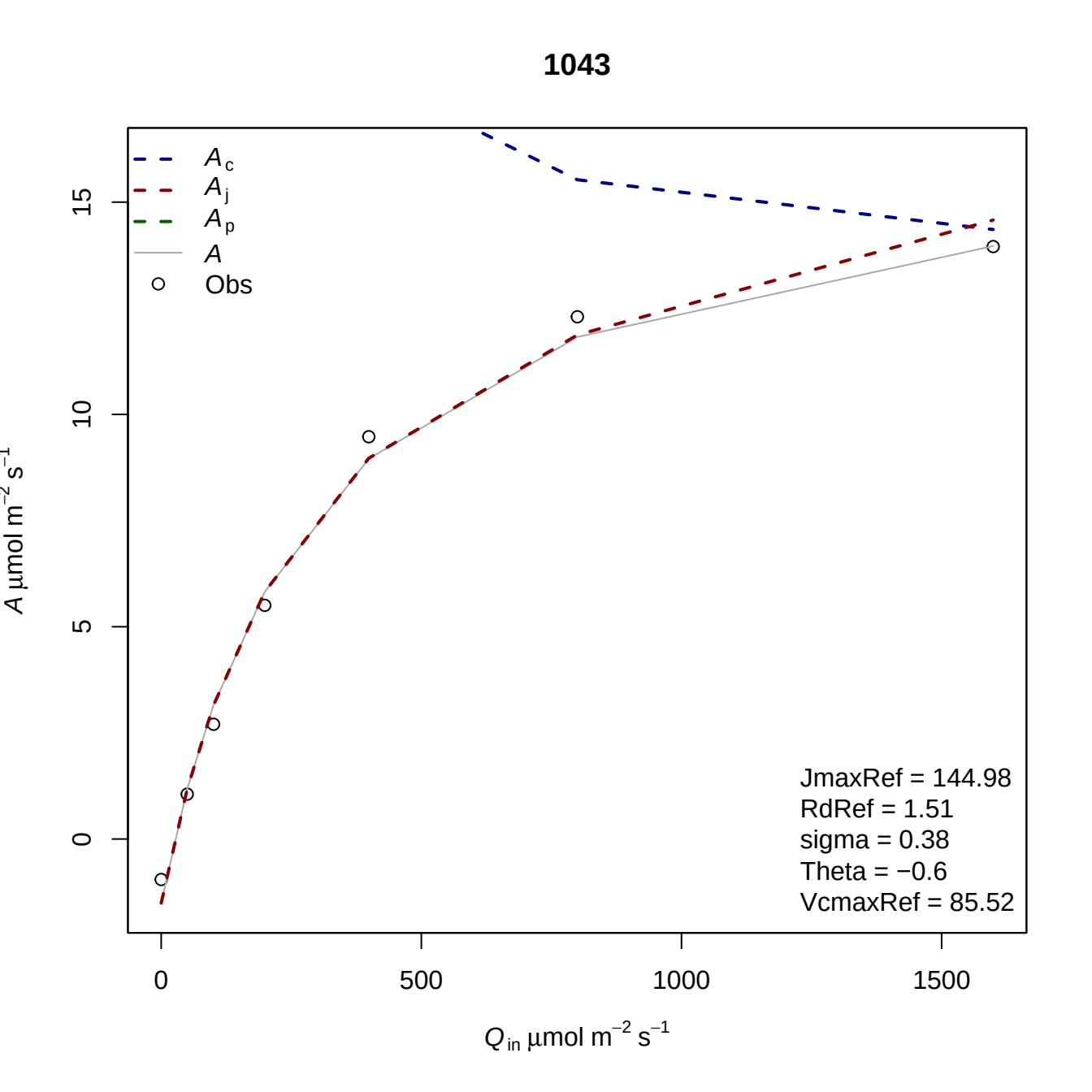

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 109.23
RdRef = 1.06
sigma = 0.39
Theta = 0.06
VcmaxRef = 79.36
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

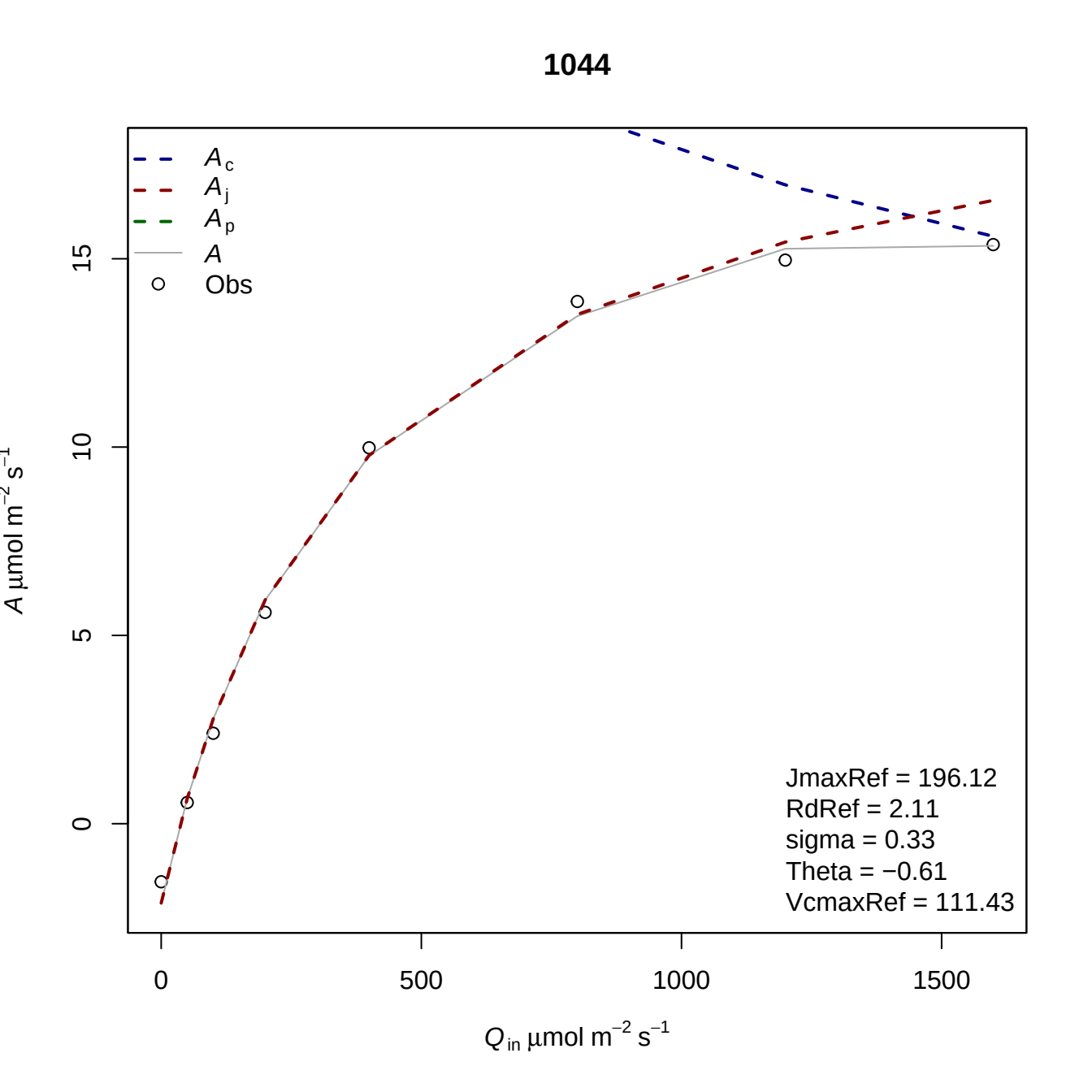

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 92.97
RdRef = 0.99
sigma = 0.42
Theta = 0.24
VcmaxRef = 69.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

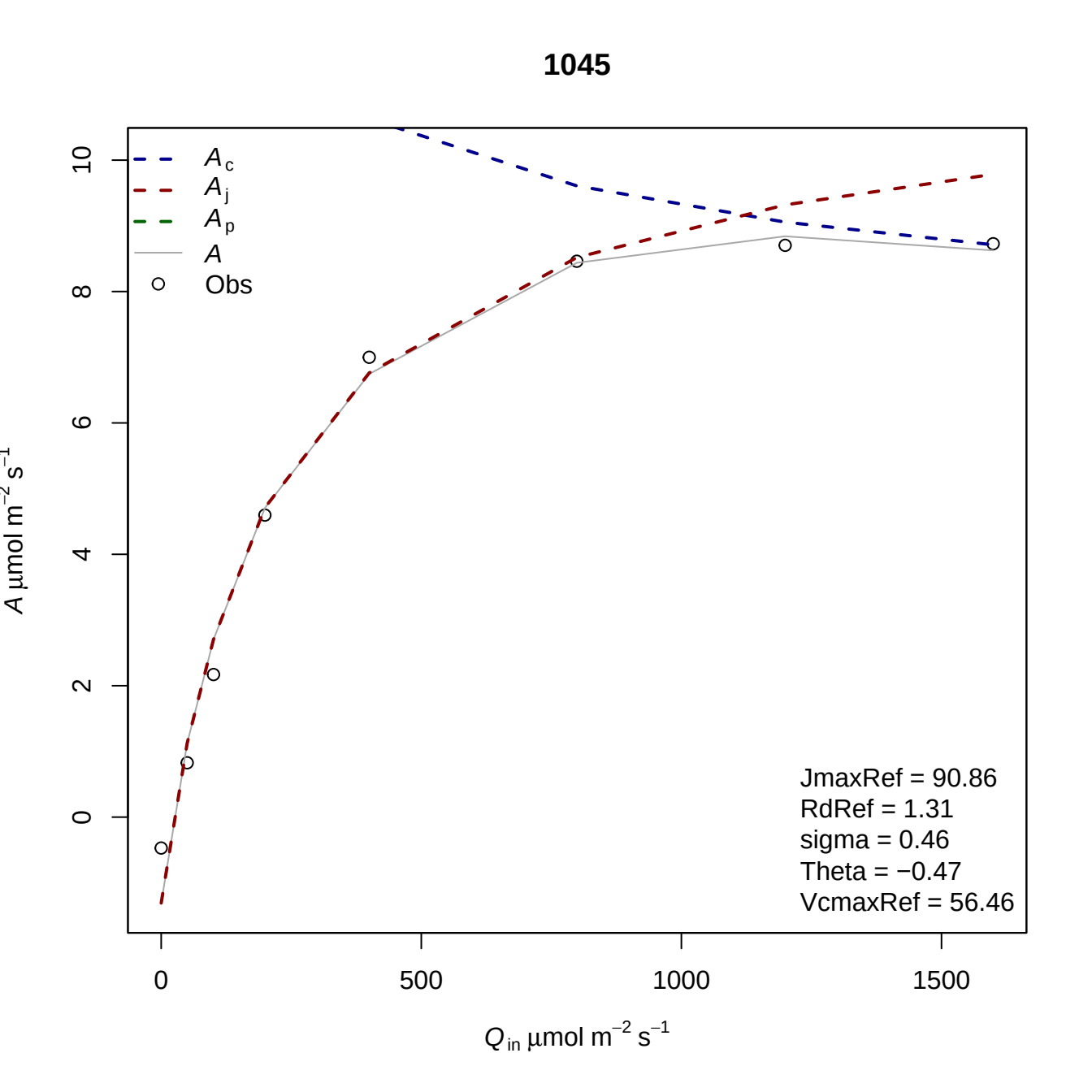

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 332.28
RdRef = 0.83
sigma = 0.16
Theta = −4.99
VcmaxRef = 60.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

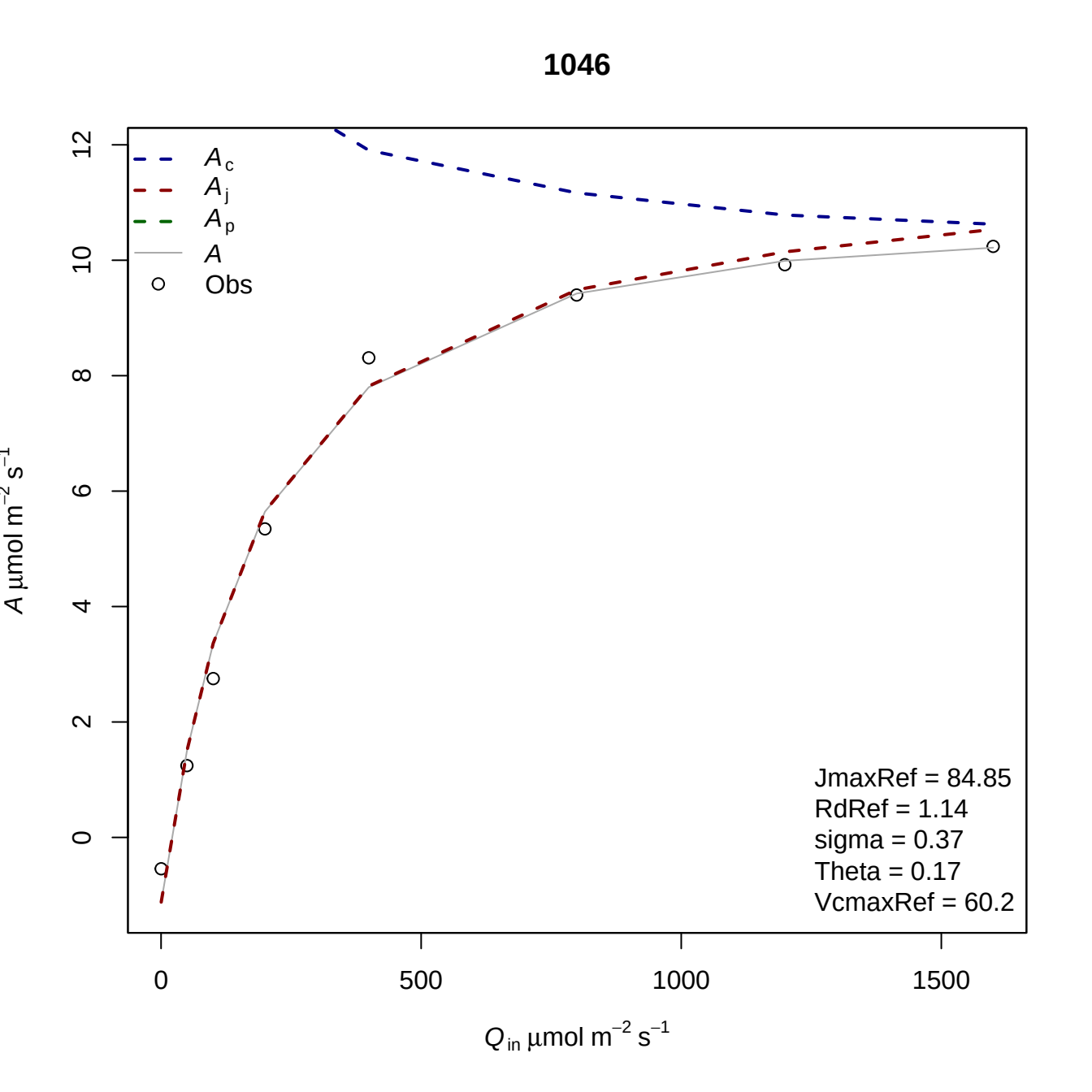

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 119.14
RdRef = 1.17
sigma = 0.33
Theta = 0.42
VcmaxRef = 94.04
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

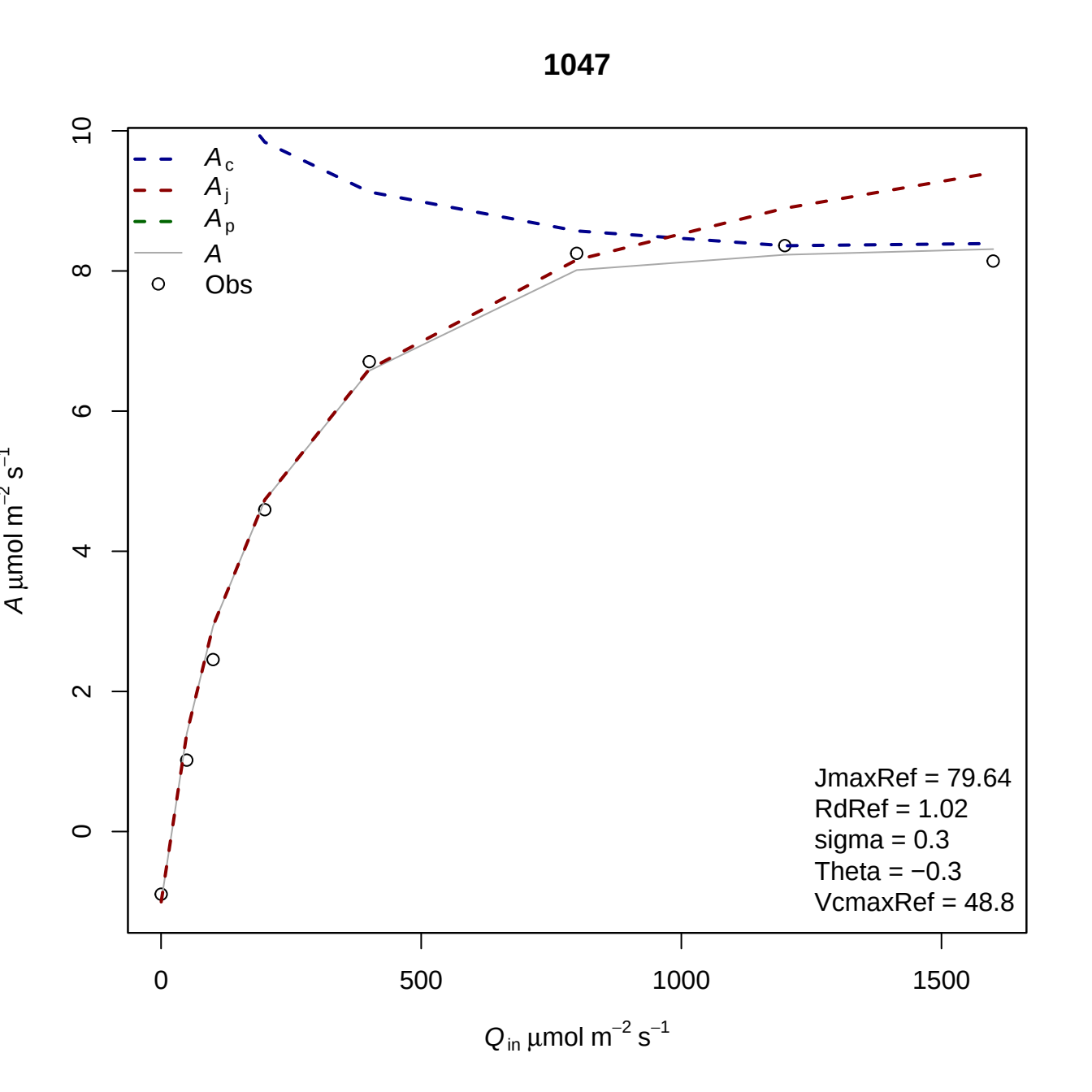

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 136.72
RdRef = 2.11
sigma = 0.29
Theta = 0.06
VcmaxRef = 85.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

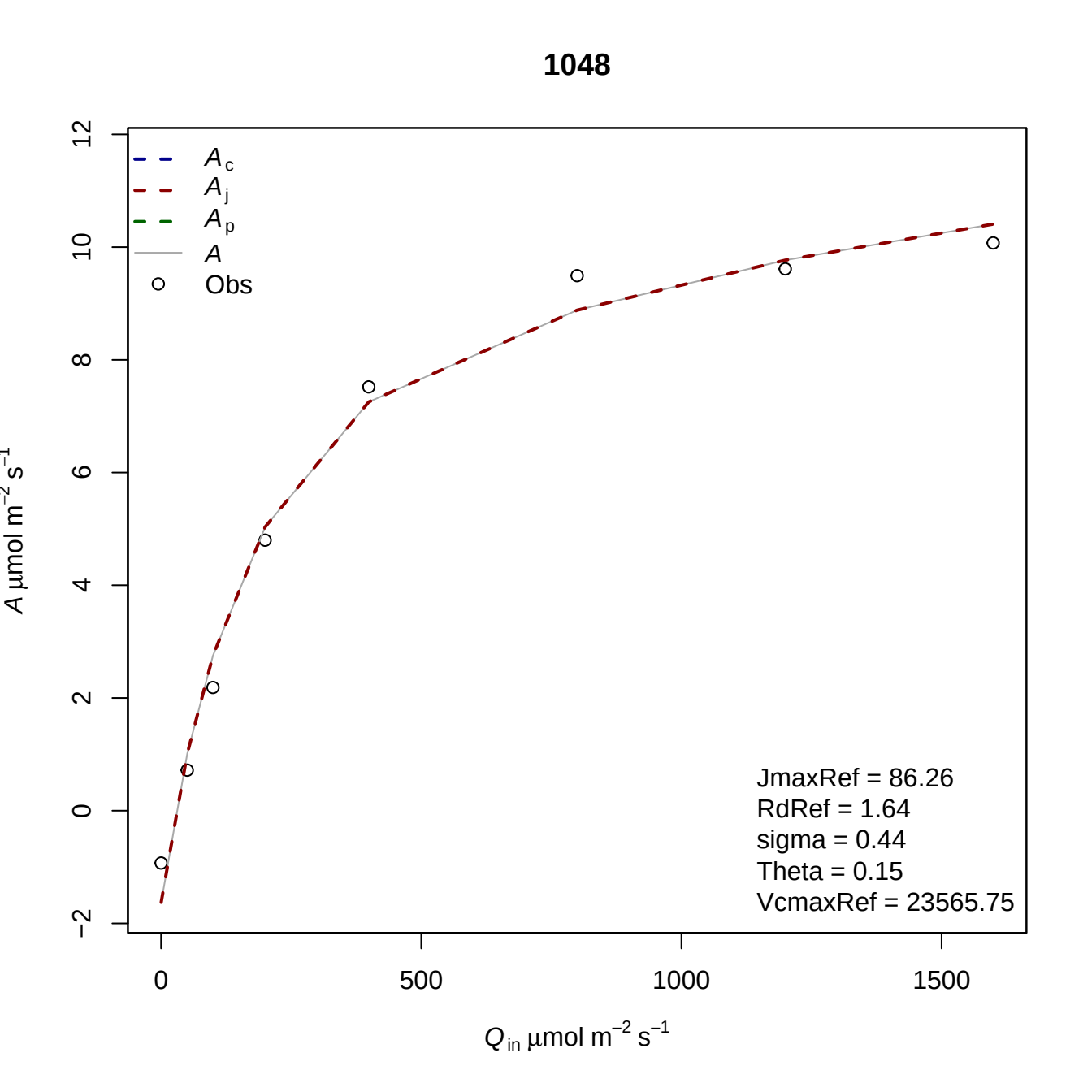

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 144.98
RdRef = 1.51
sigma = 0.38
Theta = −0.6
VcmaxRef = 85.52
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

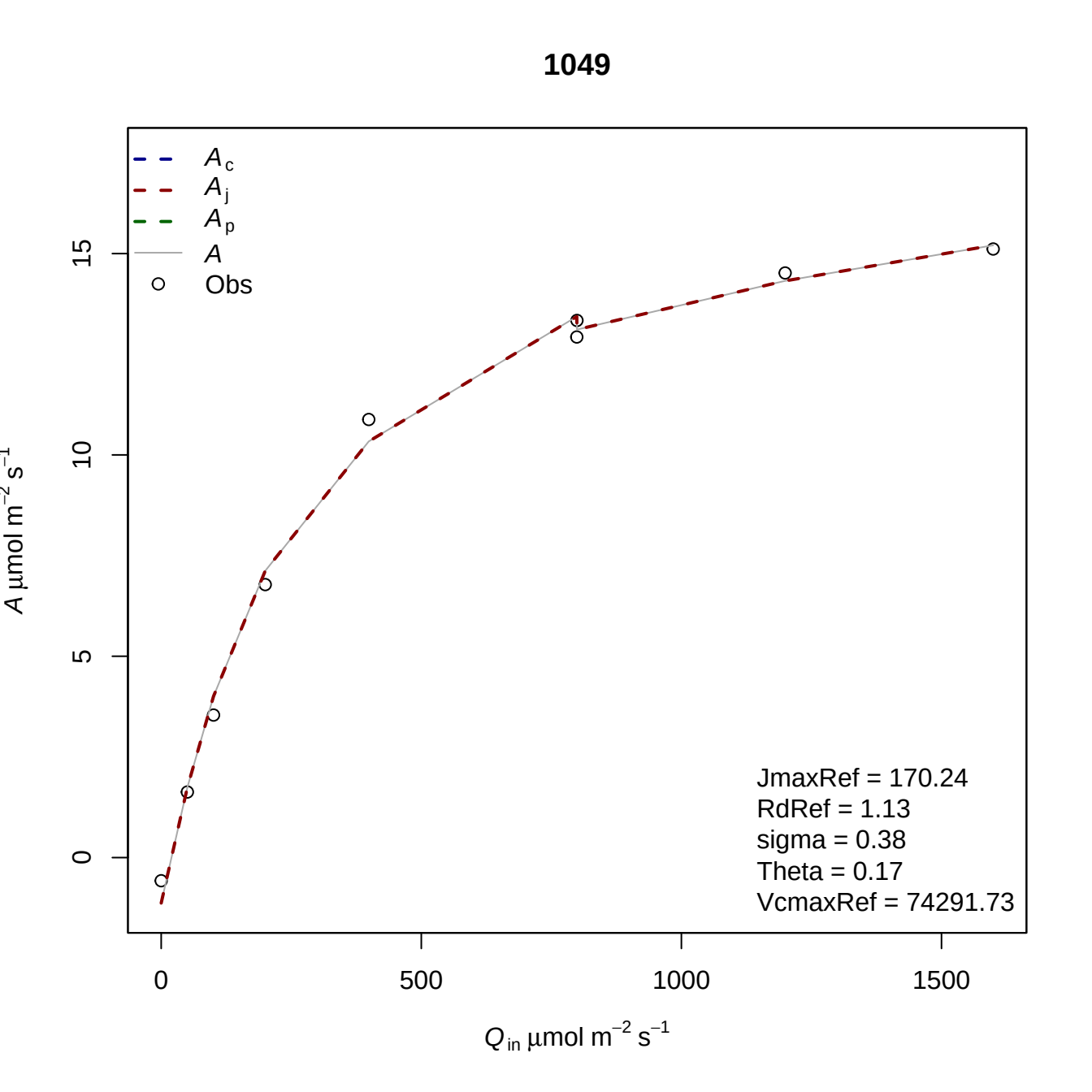

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 196.12
RdRef = 2.11
sigma = 0.33
Theta = −0.61
VcmaxRef = 111.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

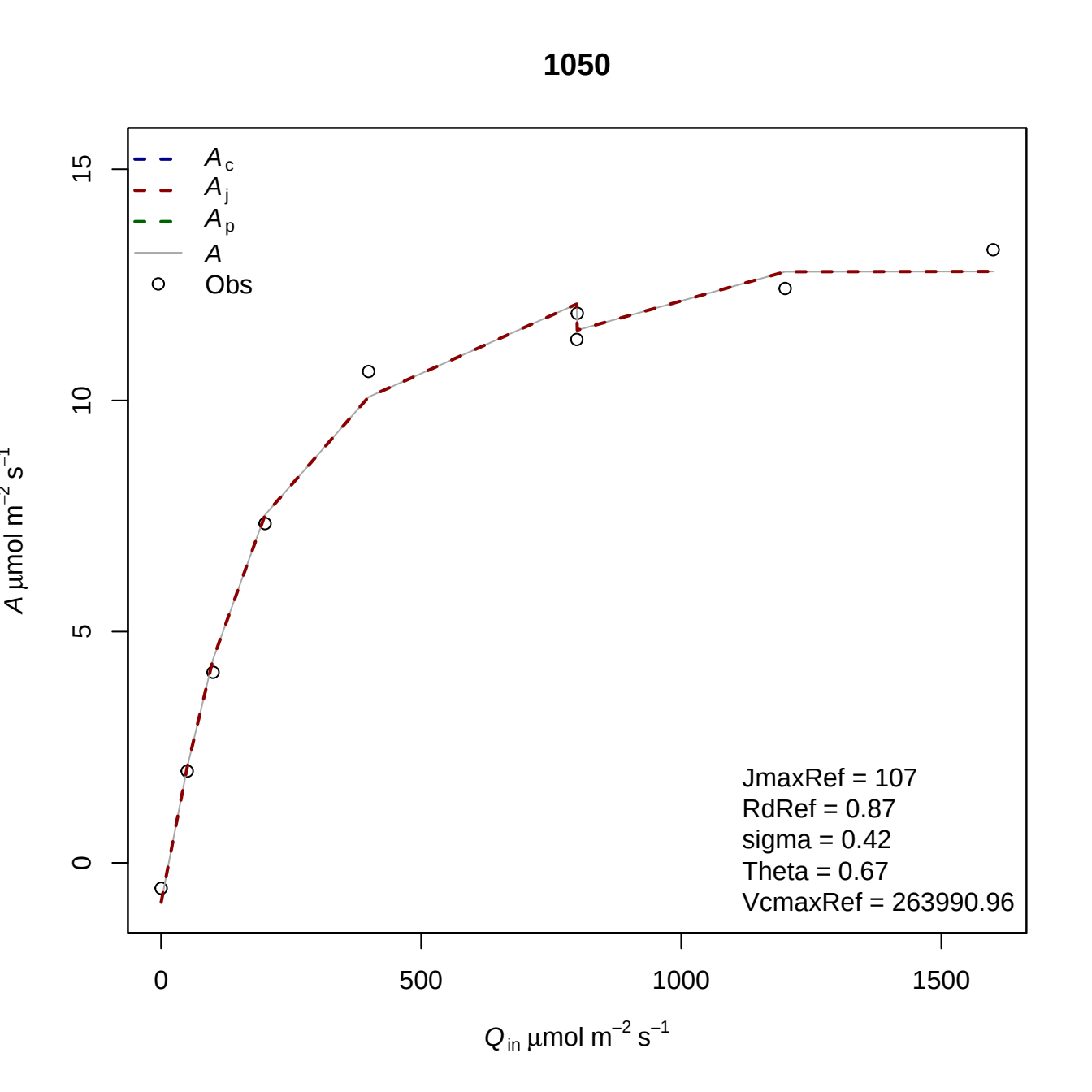

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 90.86
RdRef = 1.31
sigma = 0.46
Theta = −0.47
VcmaxRef = 56.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

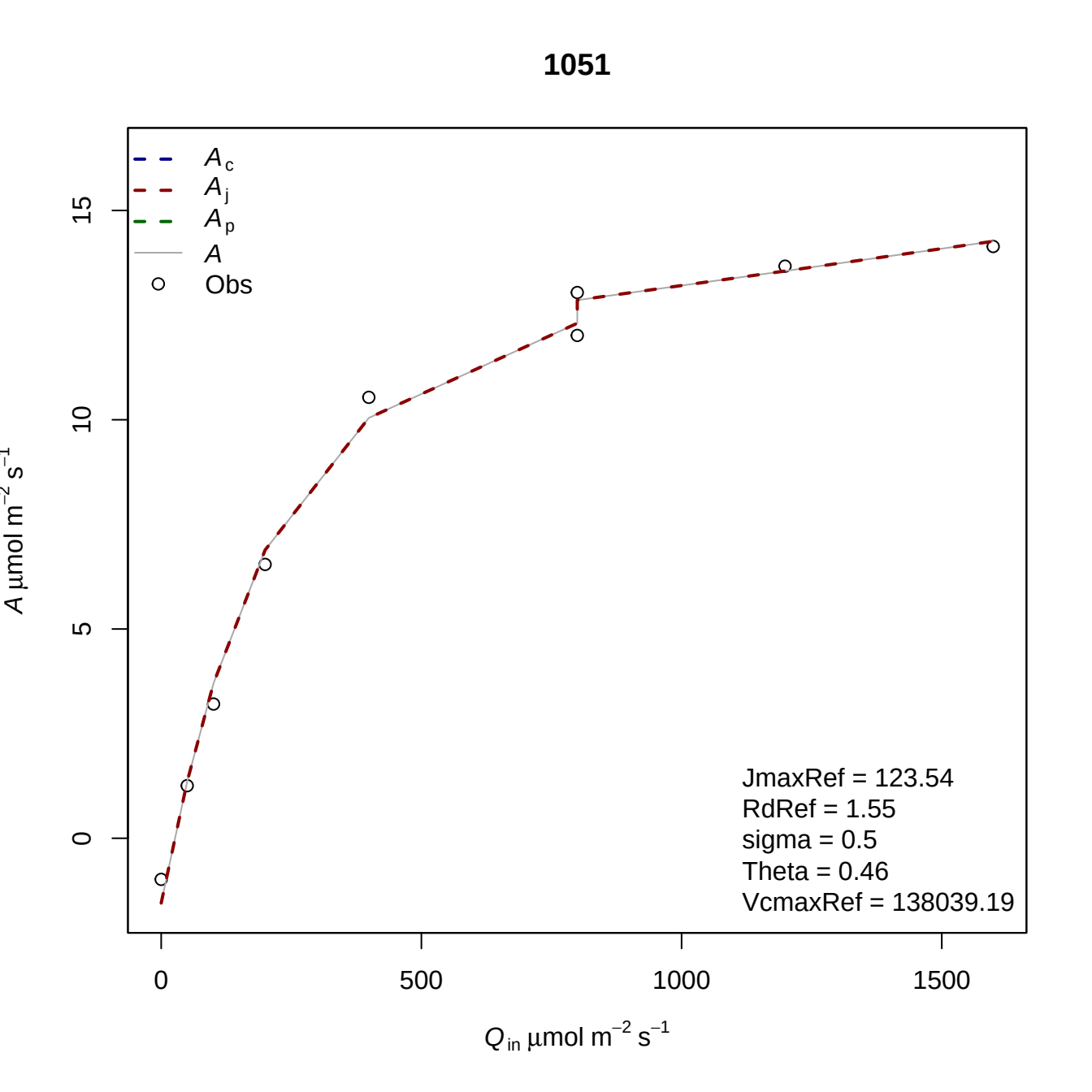

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 84.85
RdRef = 1.14
sigma = 0.37
Theta = 0.17
VcmaxRef = 60.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

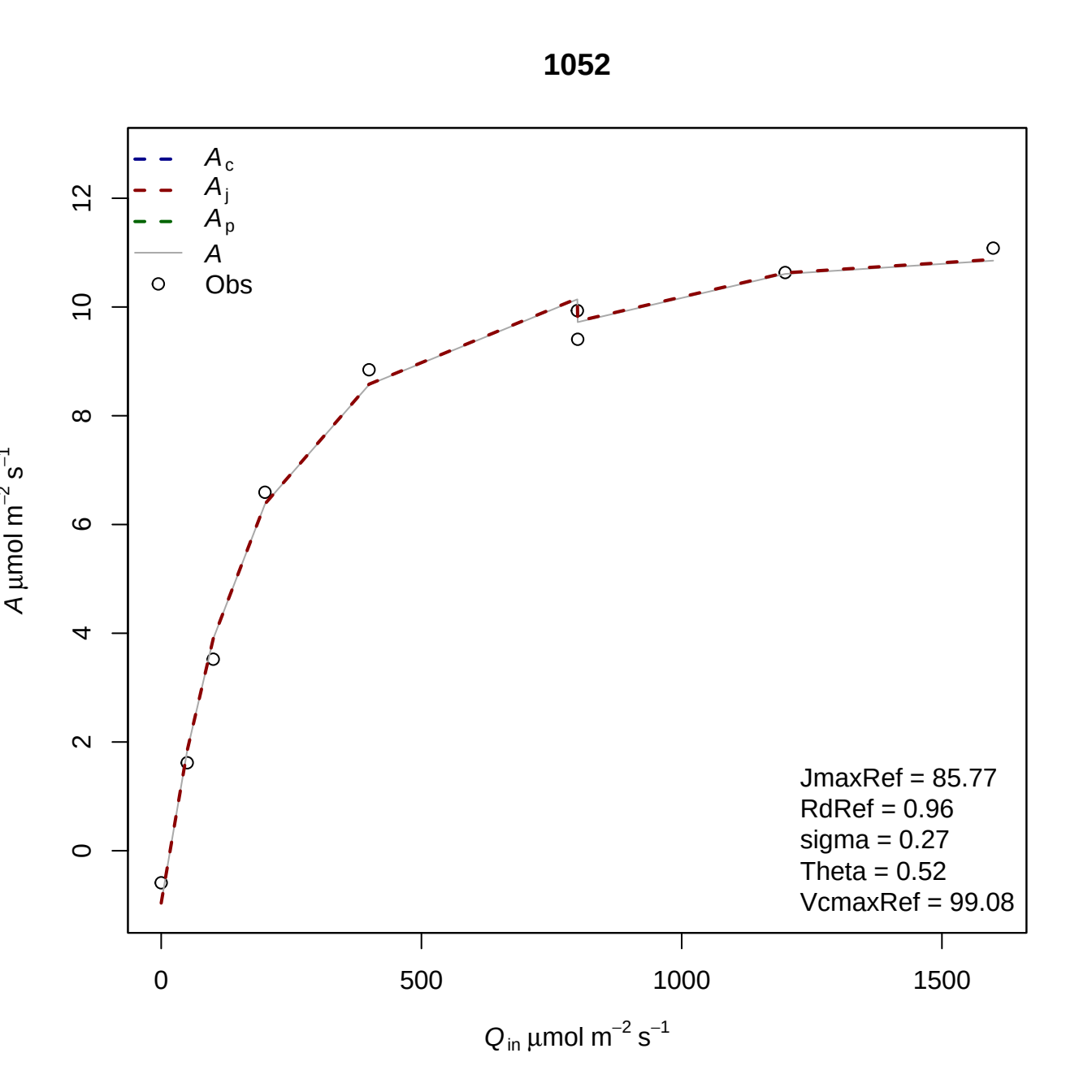

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 79.64
RdRef = 1.02
sigma = 0.3
Theta = −0.3
VcmaxRef = 48.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

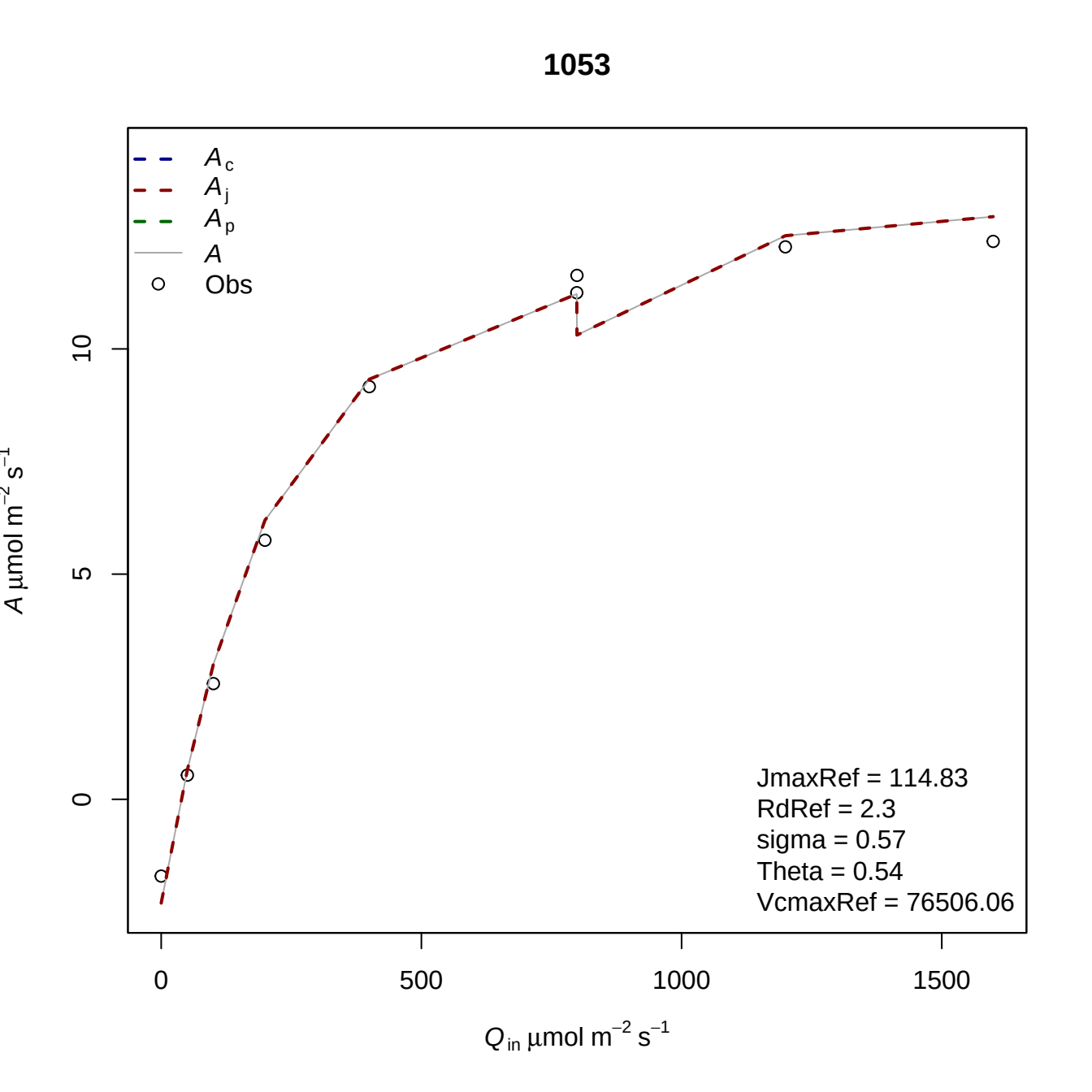

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 86.26
RdRef = 1.64
sigma = 0.44
Theta = 0.15
VcmaxRef = 23565.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

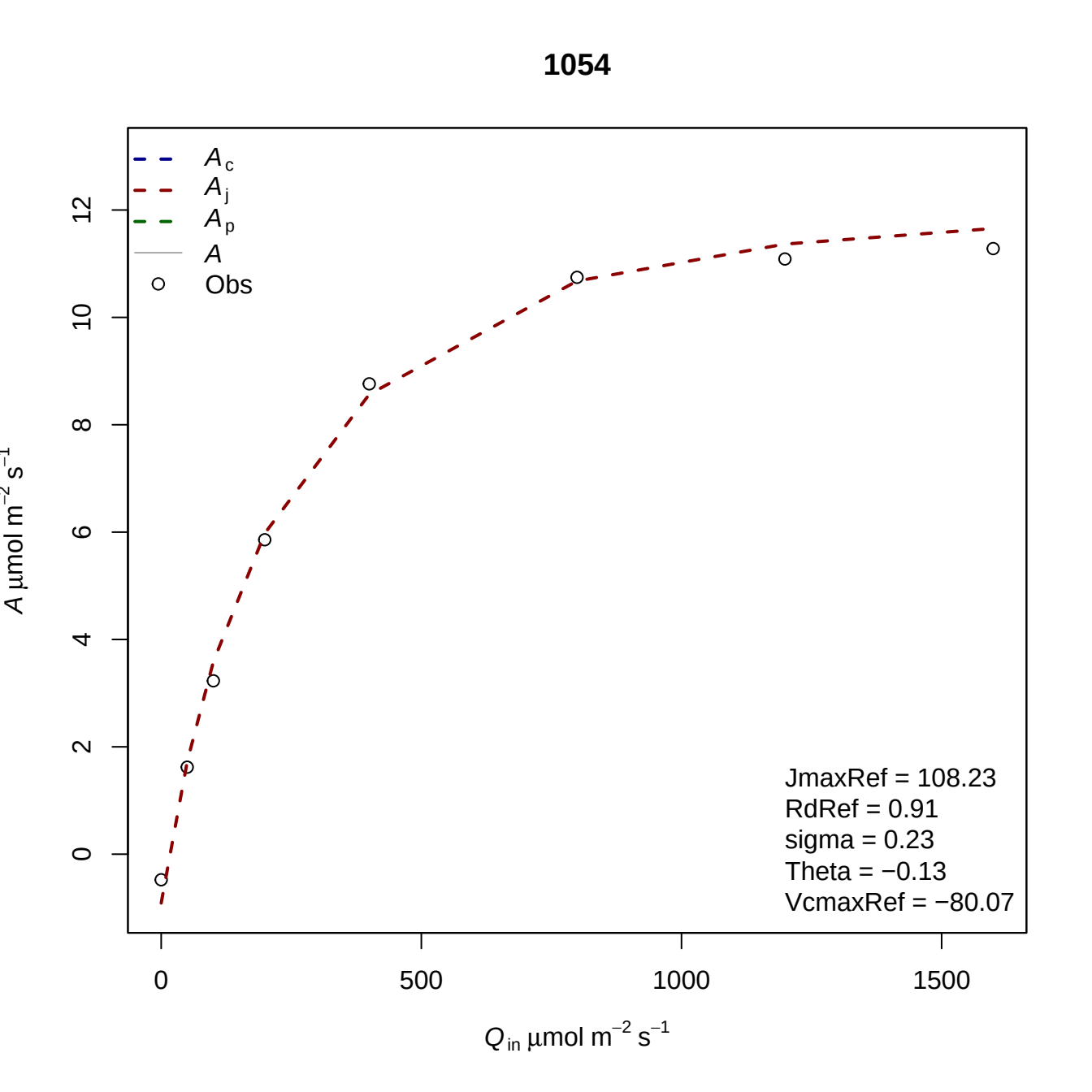

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 170.24
RdRef = 1.13
sigma = 0.38
Theta = 0.17
VcmaxRef = 74291.73
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

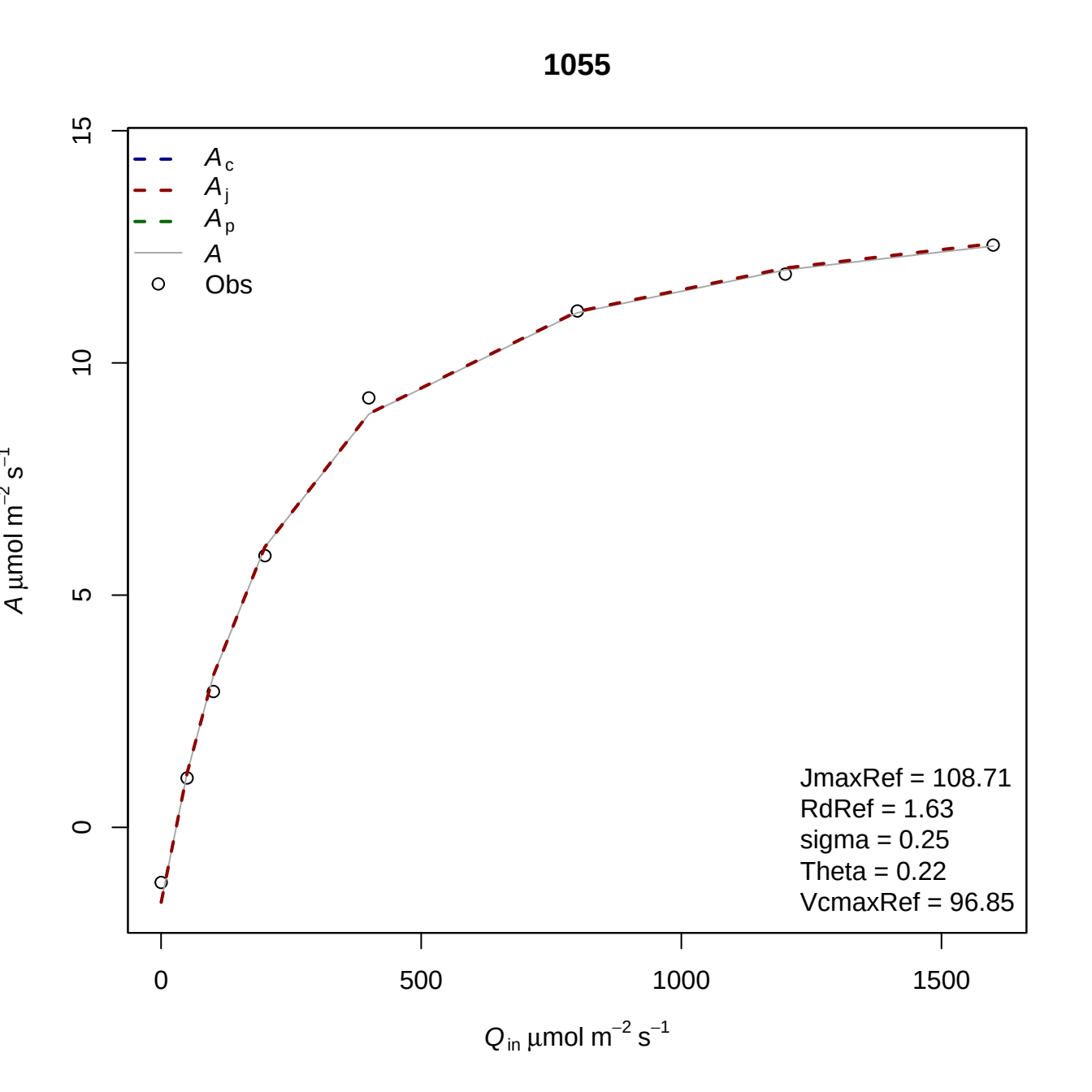

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 107
RdRef = 0.87
sigma = 0.42
Theta = 0.67
VcmaxRef = 263990.96
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

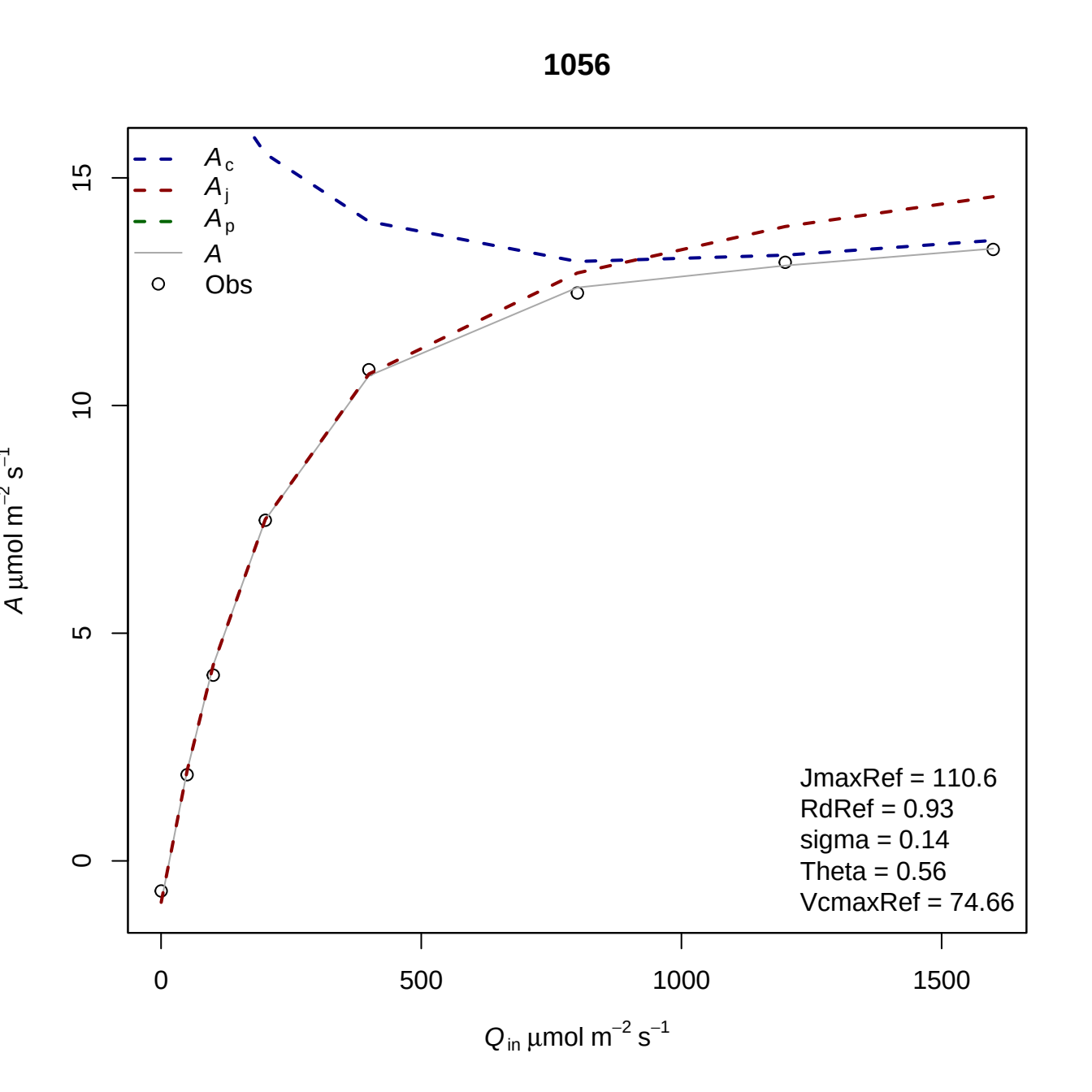

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 123.54
RdRef = 1.55
sigma = 0.5
Theta = 0.46
VcmaxRef = 138039.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

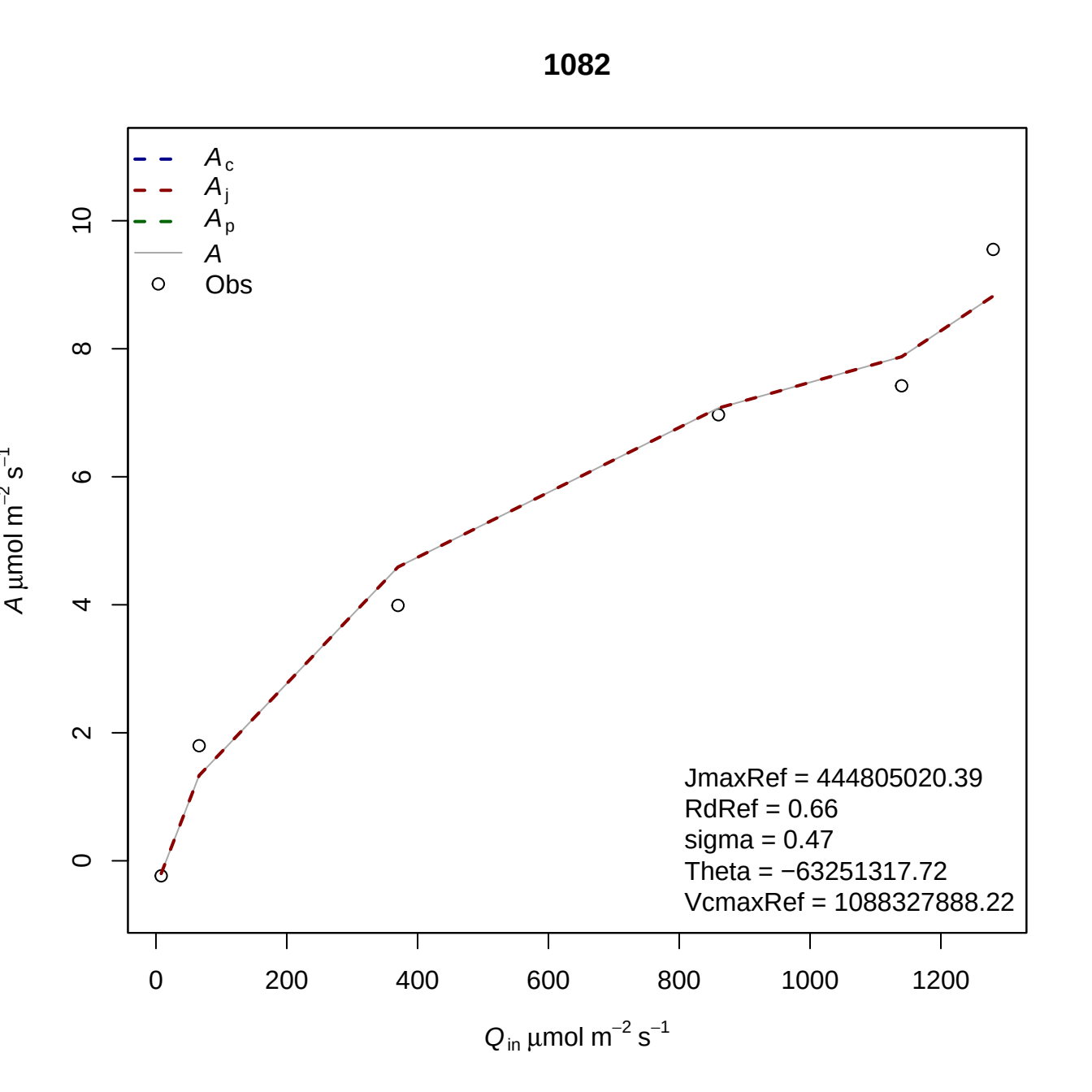

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 85.77
RdRef = 0.96
sigma = 0.27
Theta = 0.52
VcmaxRef = 99.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

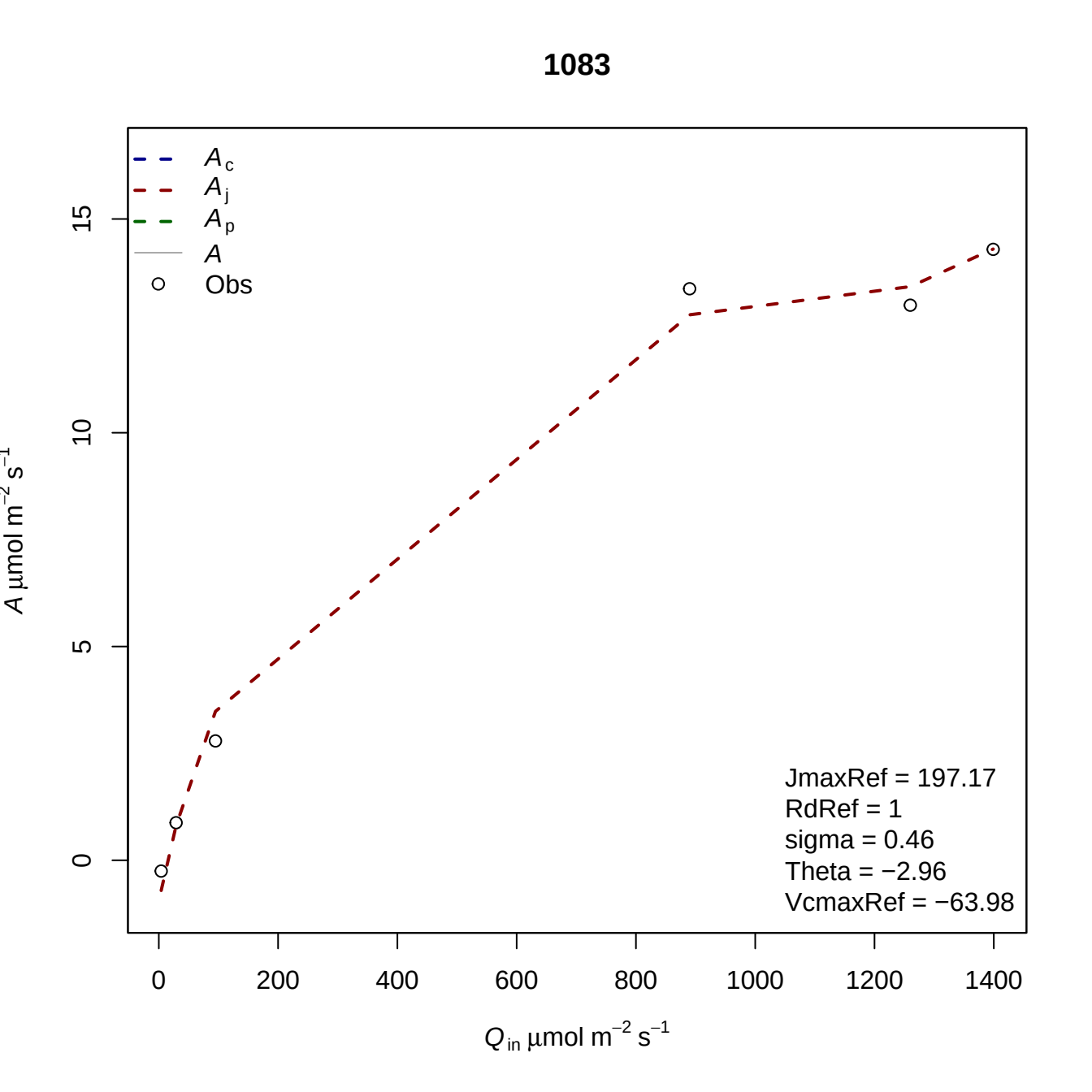

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 114.83
RdRef = 2.3
sigma = 0.57
Theta = 0.54
VcmaxRef = 76506.06
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

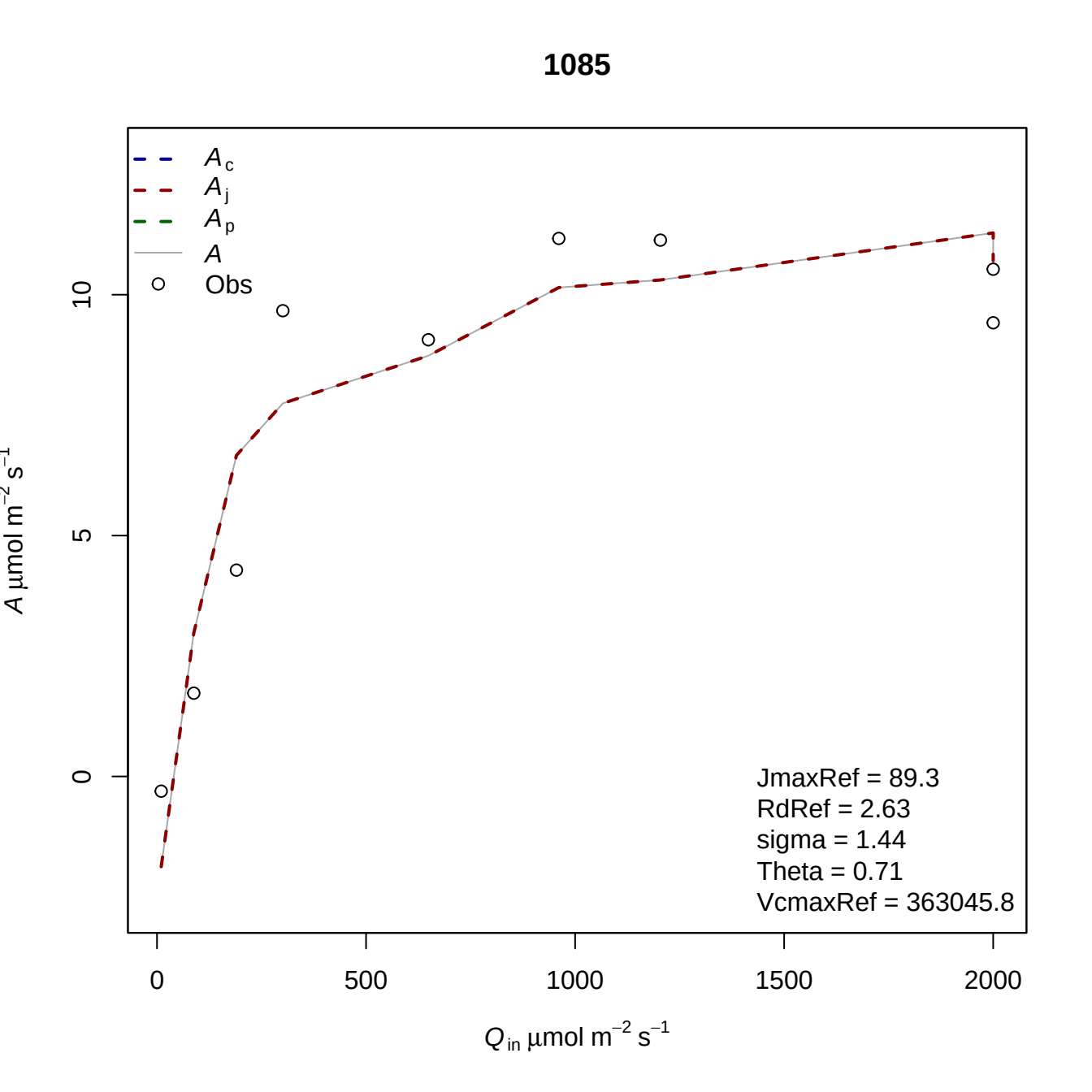

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 108.23
RdRef = 0.91
sigma = 0.23
Theta = −0.13
VcmaxRef = −80.07
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

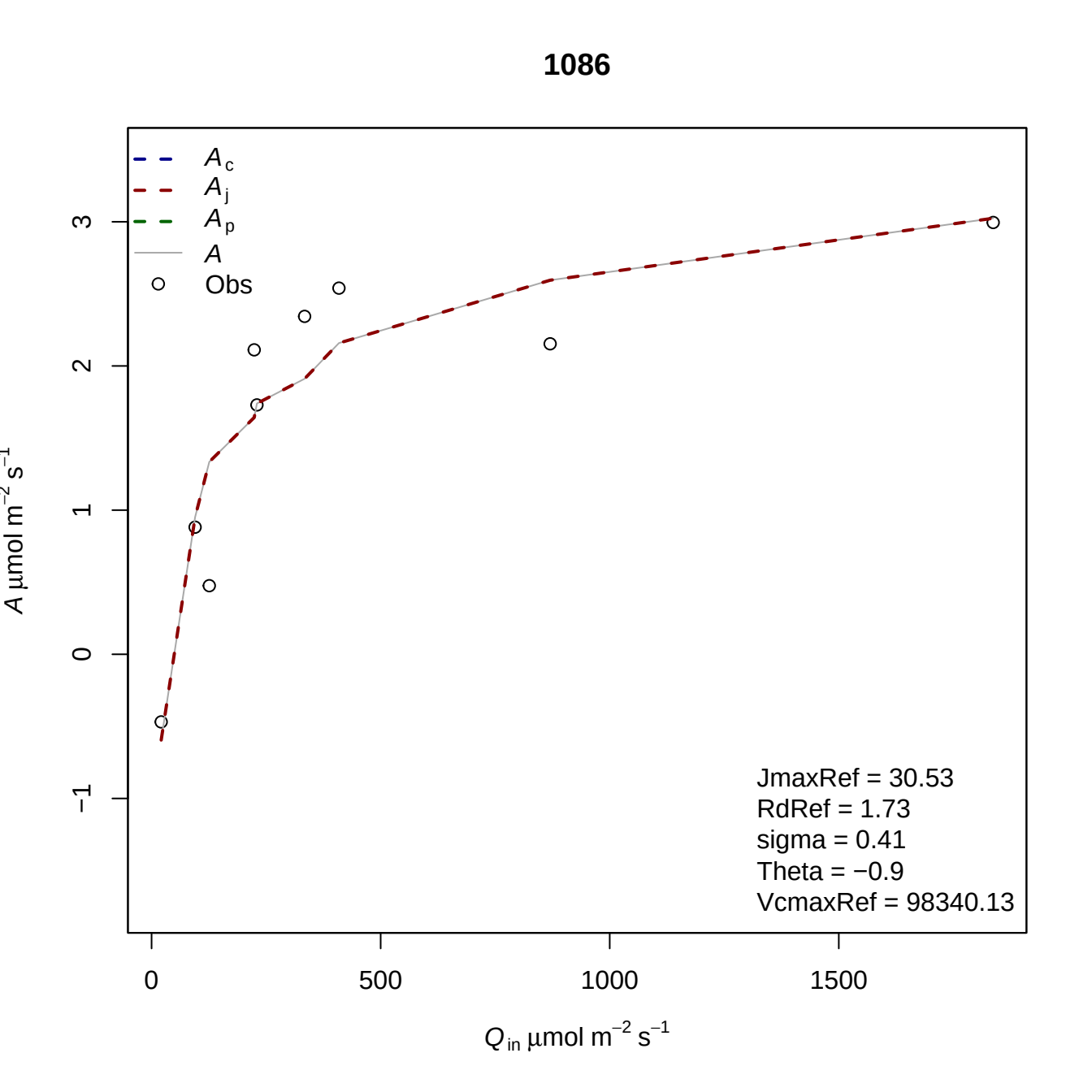

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 108.71
RdRef = 1.63
sigma = 0.25
Theta = 0.22
VcmaxRef = 96.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

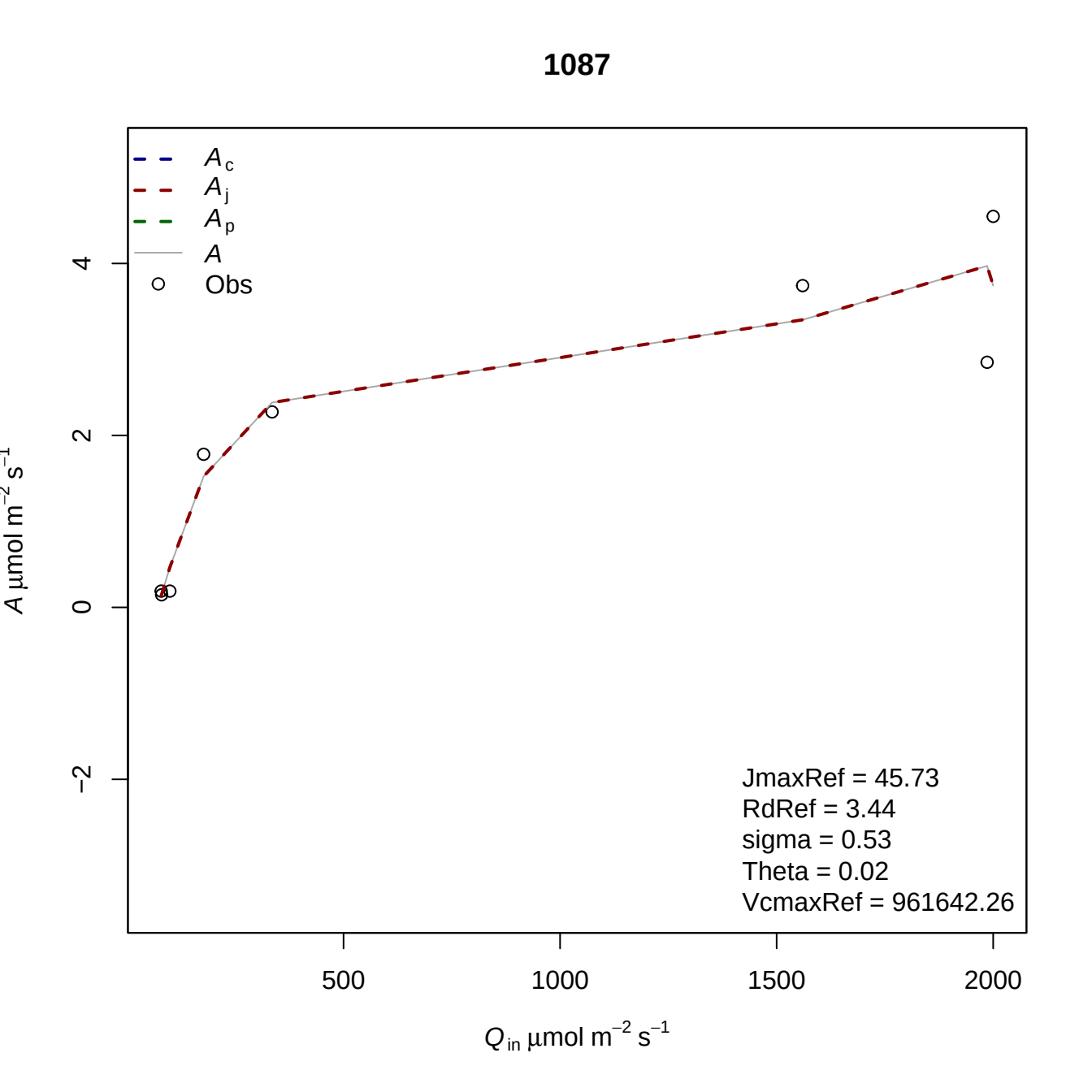

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 110.6
RdRef = 0.93
sigma = 0.14
Theta = 0.56
VcmaxRef = 74.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

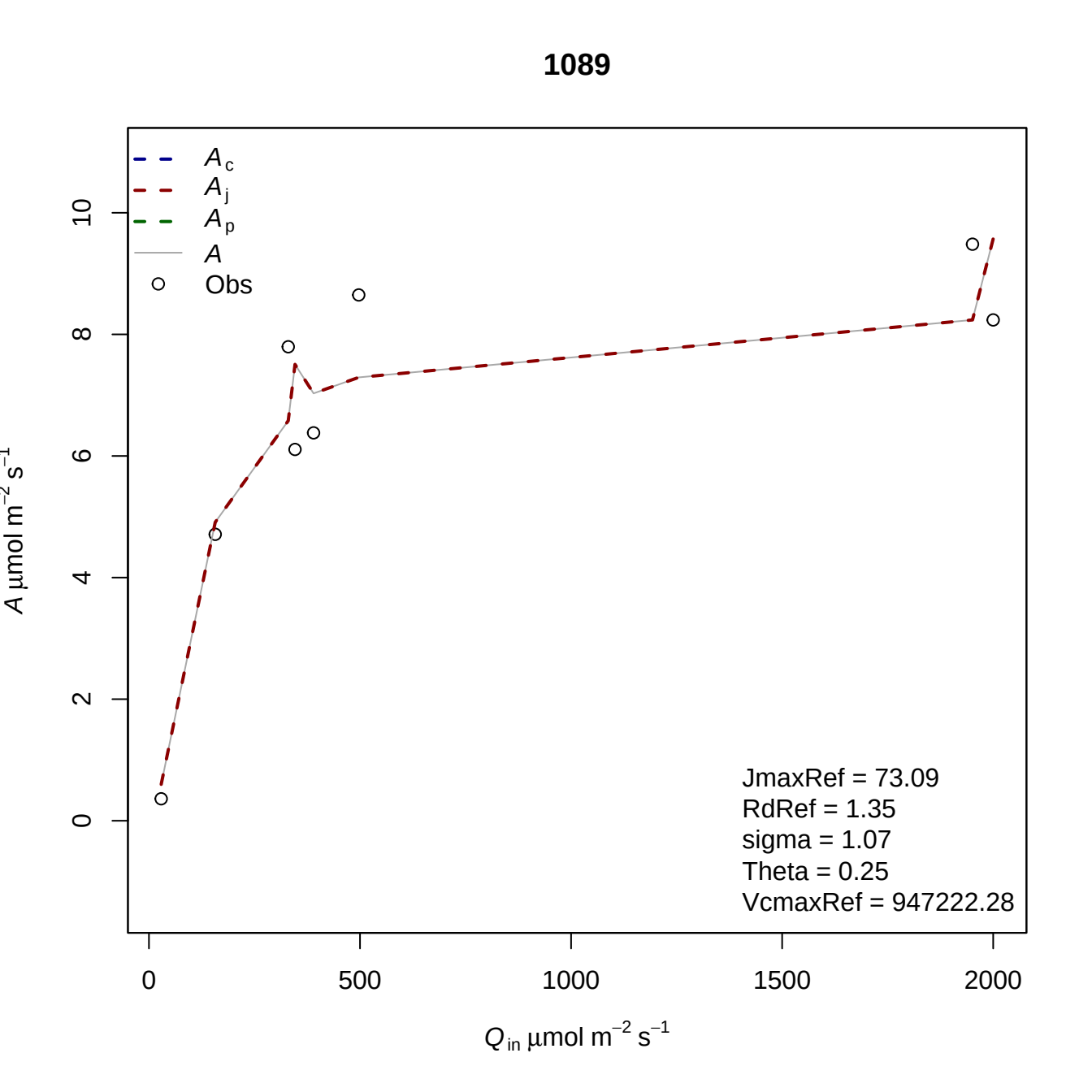

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 444805020.39
RdRef = 0.66
sigma = 0.47
Theta = −63251317.72
VcmaxRef = 1088327888.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

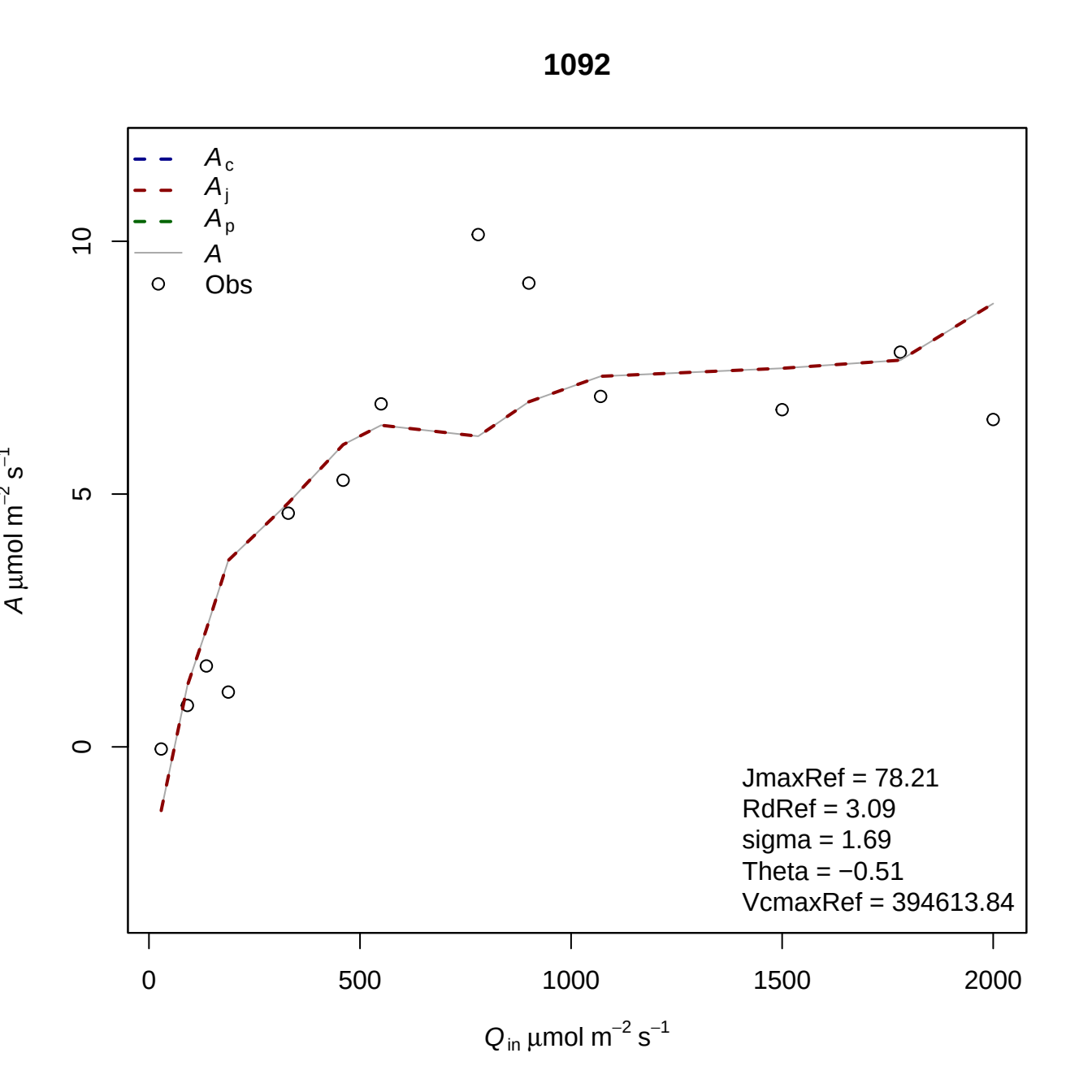

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 197.17
RdRef = 1
sigma = 0.46
Theta = −2.96
VcmaxRef = −63.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

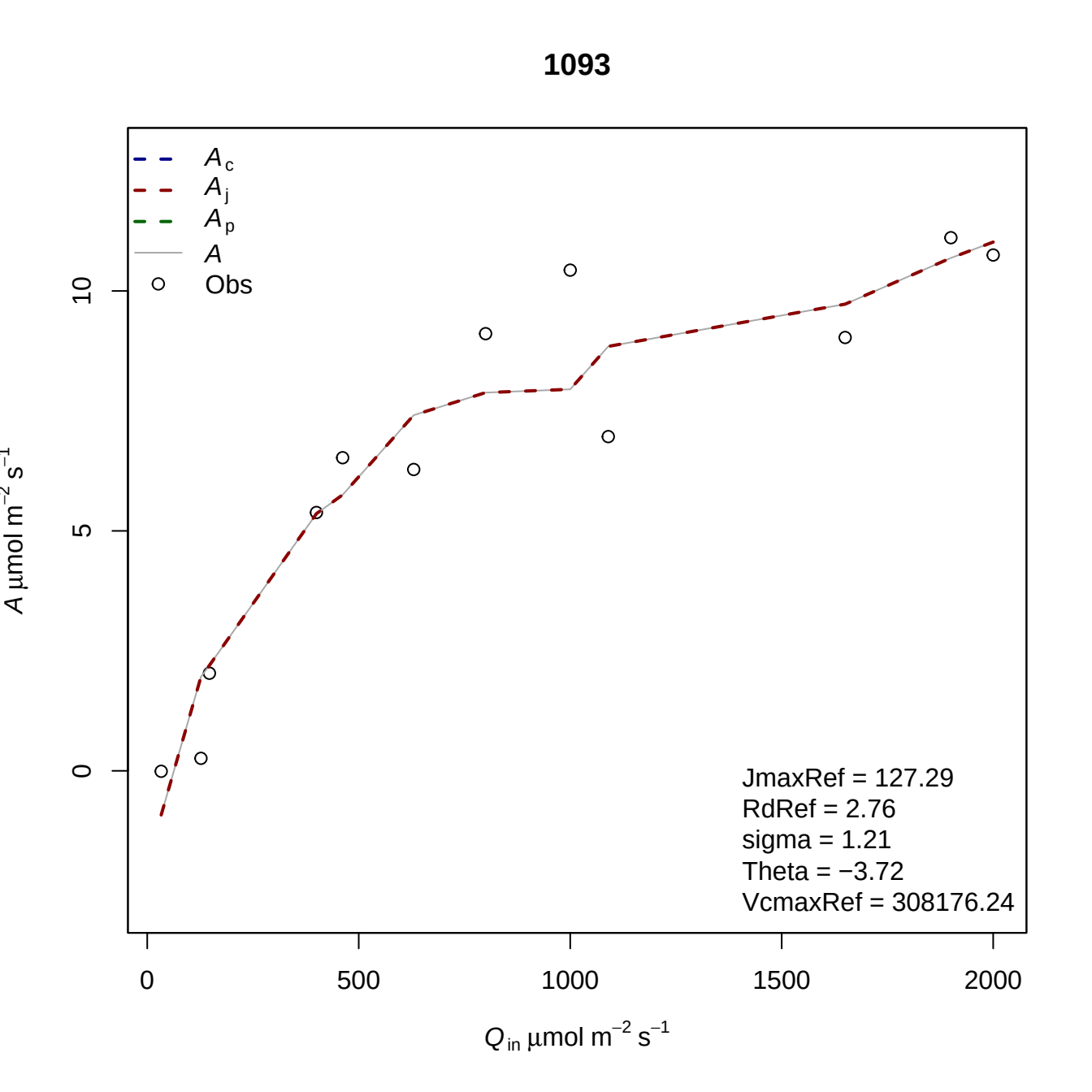

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 89.3
RdRef = 2.63
sigma = 1.44
Theta = 0.71
VcmaxRef = 363045.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

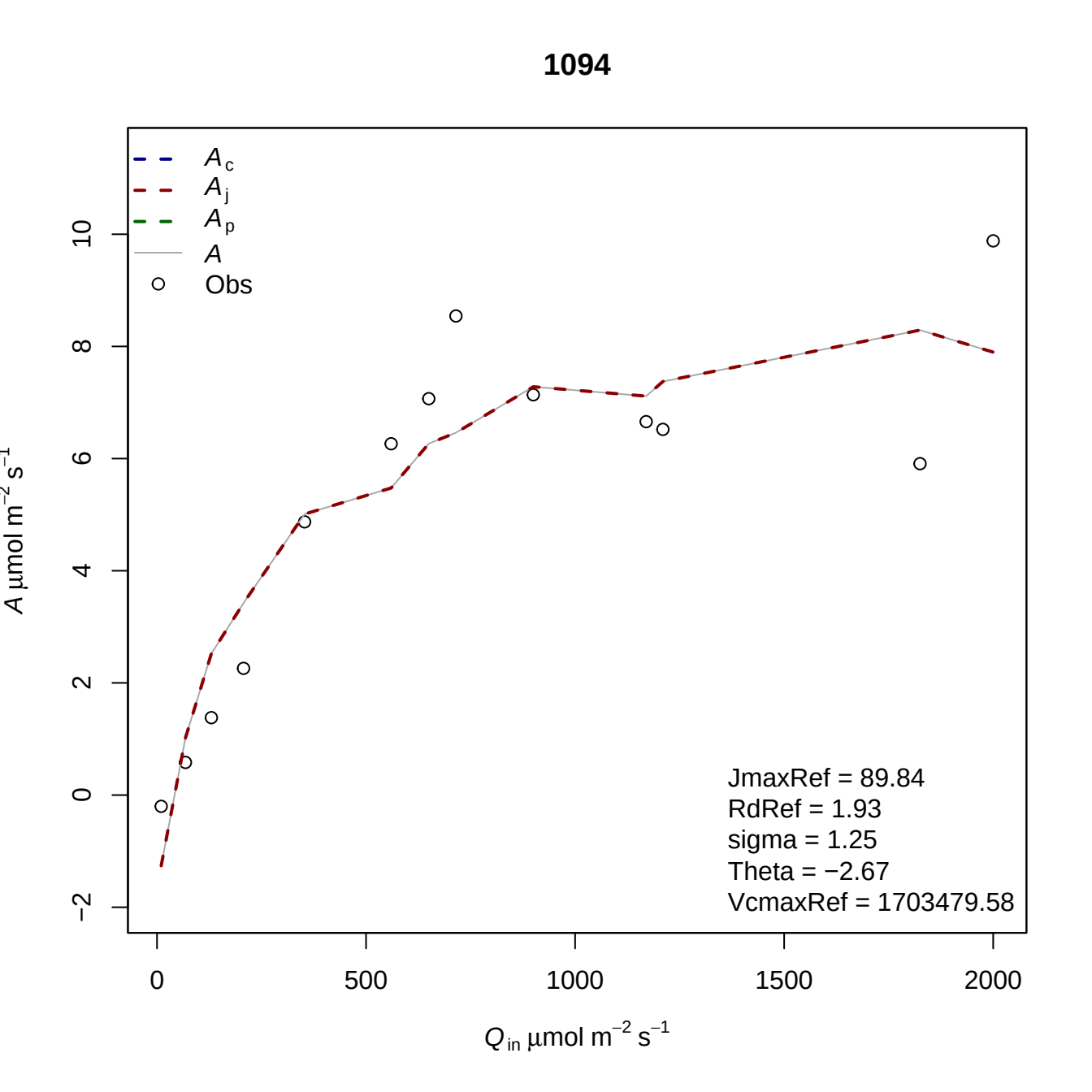

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 30.53
RdRef = 1.73
sigma = 0.41
Theta = −0.9
VcmaxRef = 98340.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

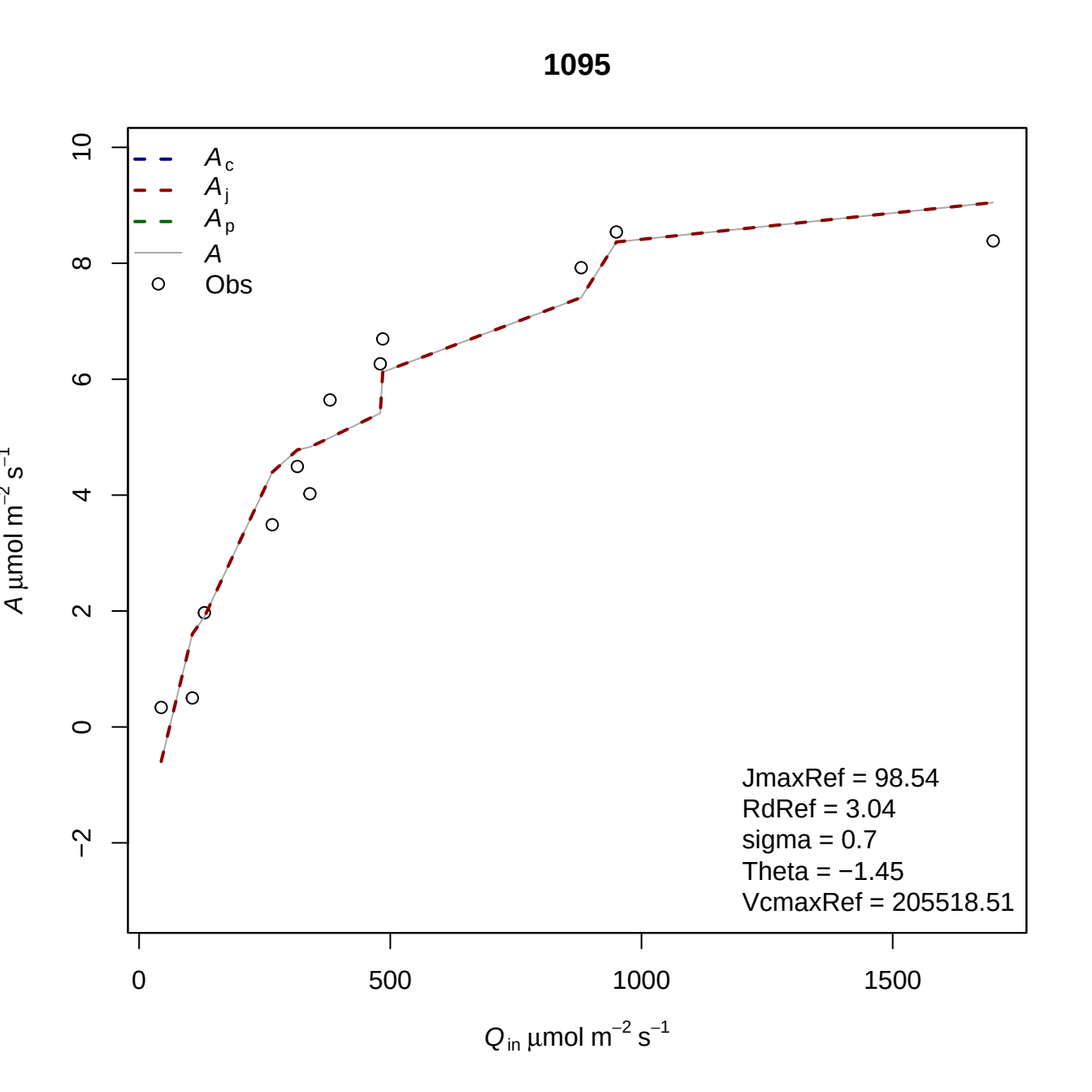

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 45.73
RdRef = 3.44
sigma = 0.53
Theta = 0.02
VcmaxRef = 961642.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
500
1000
1500
2000

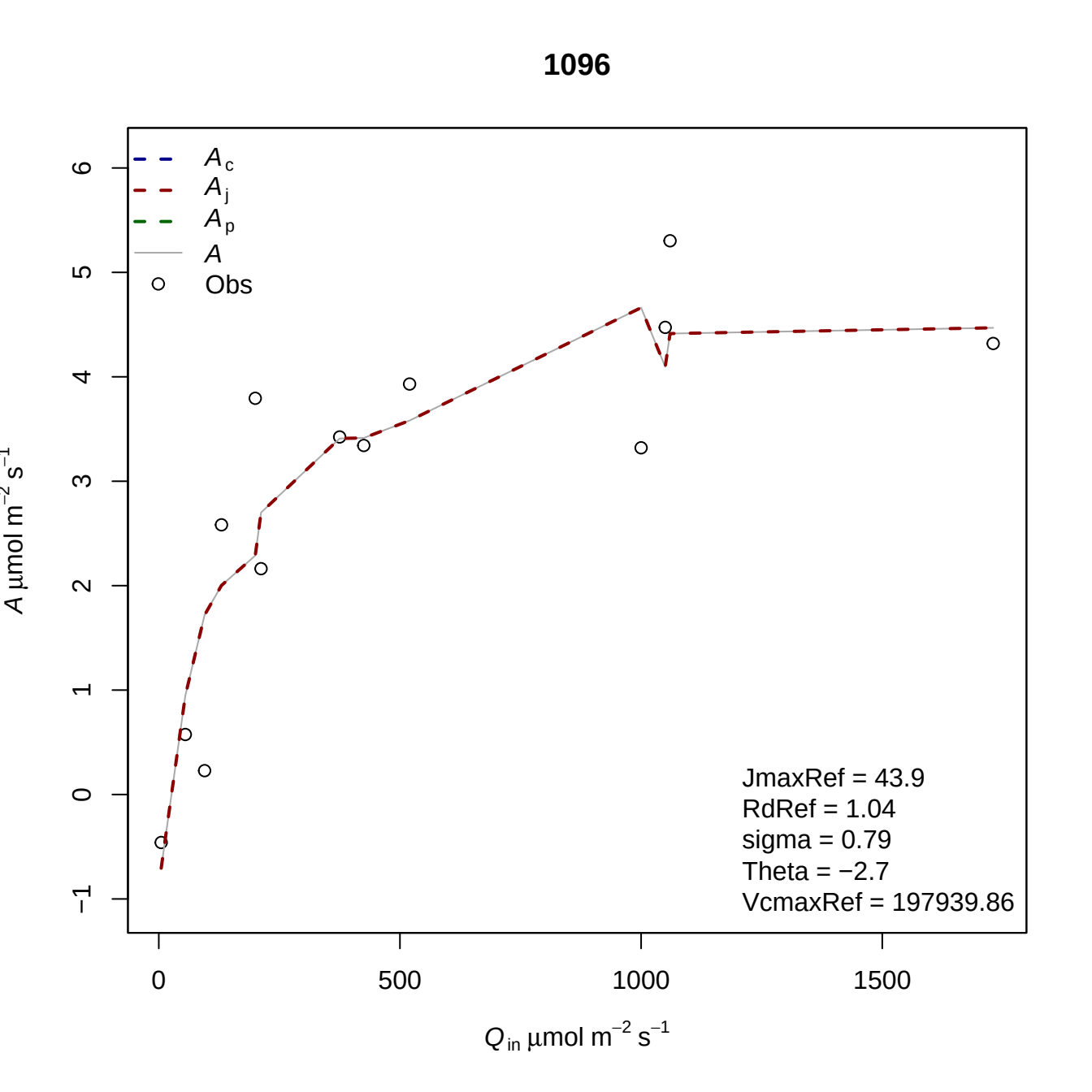

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 73.09
RdRef = 1.35
sigma = 1.07
Theta = 0.25
VcmaxRef = 947222.28
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

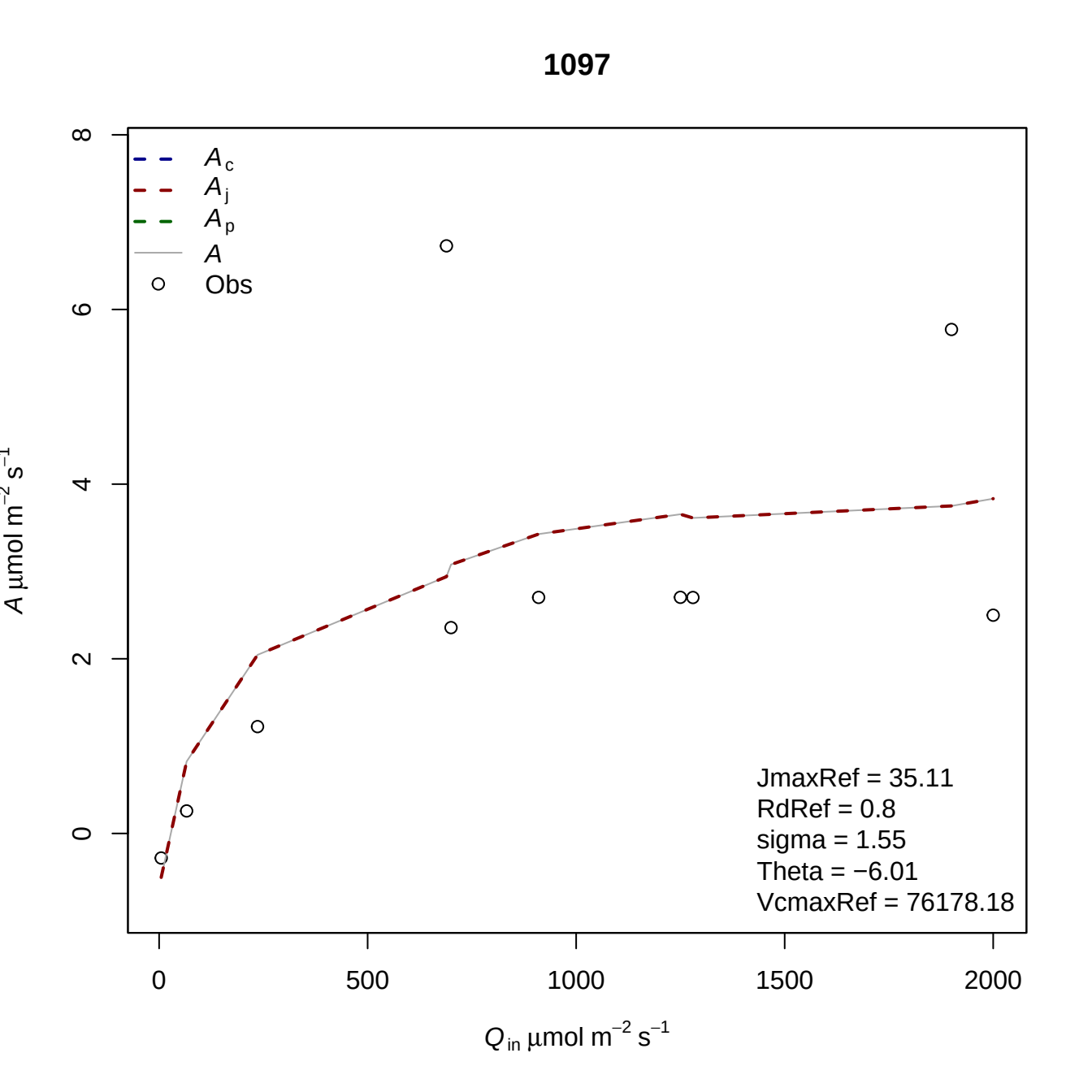

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 78.21
RdRef = 3.09
sigma = 1.69
Theta = −0.51
VcmaxRef = 394613.84
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

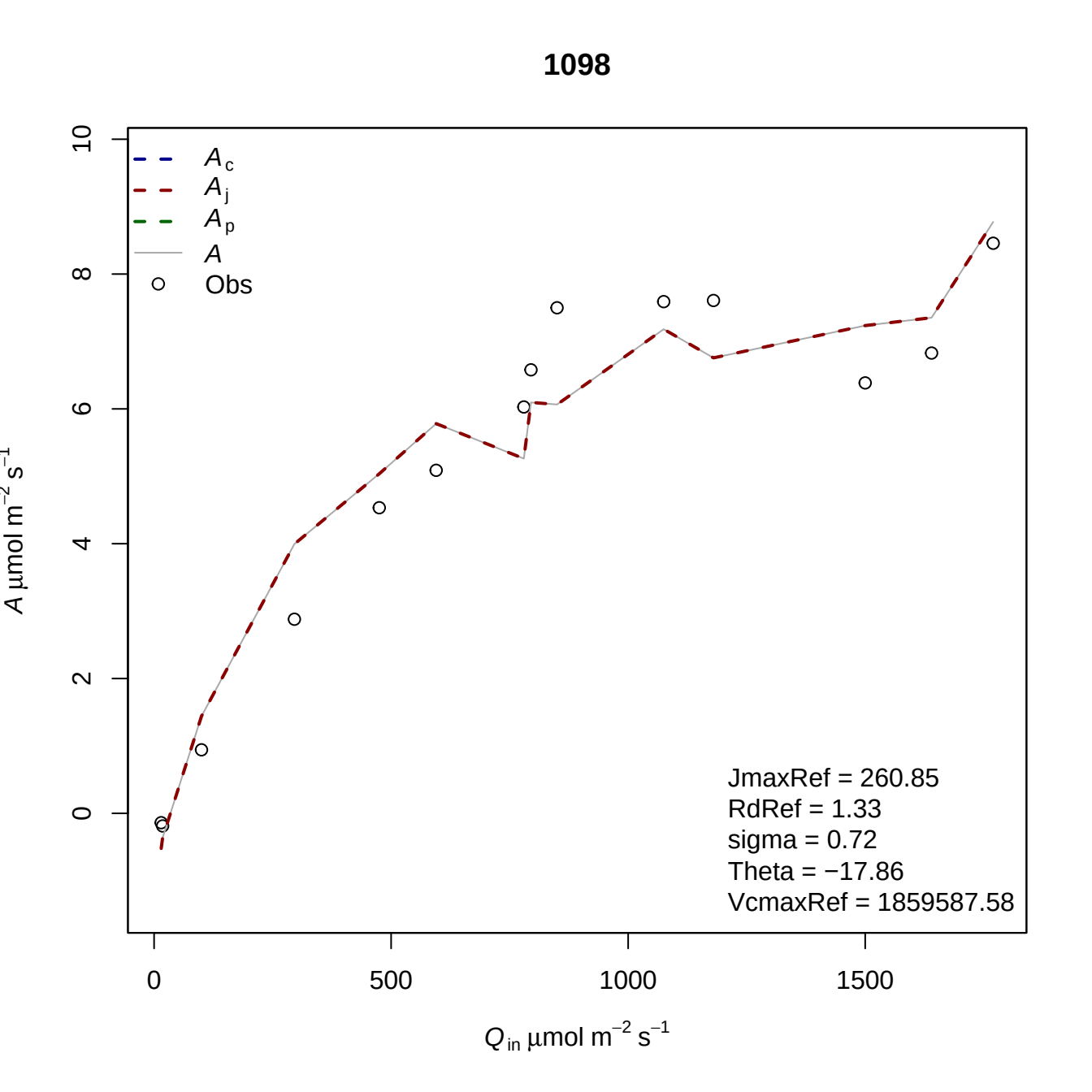

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 127.29
RdRef = 2.76
sigma = 1.21
Theta = −3.72
VcmaxRef = 308176.24
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

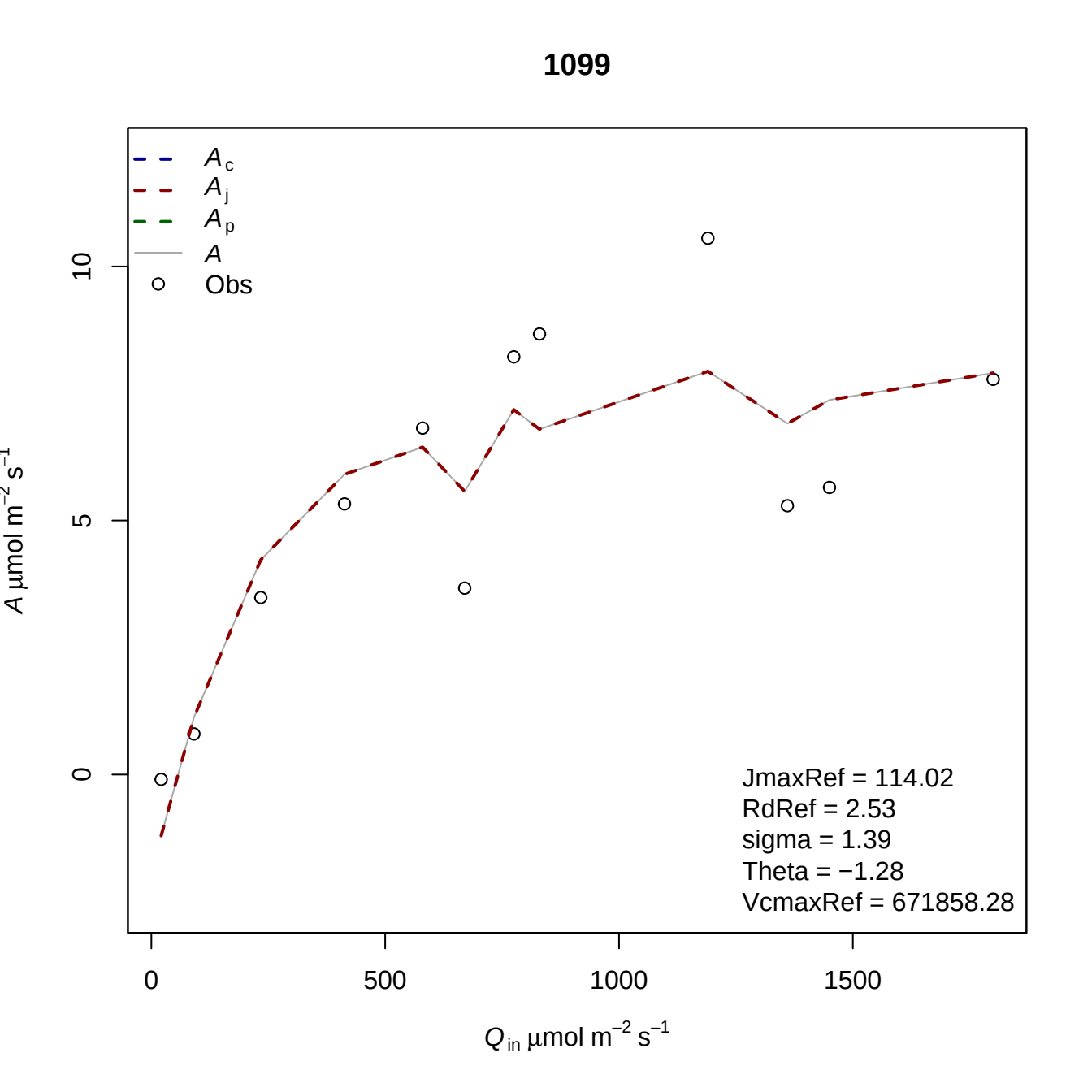

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 89.84
RdRef = 1.93
sigma = 1.25
Theta = −2.67
VcmaxRef = 1703479.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

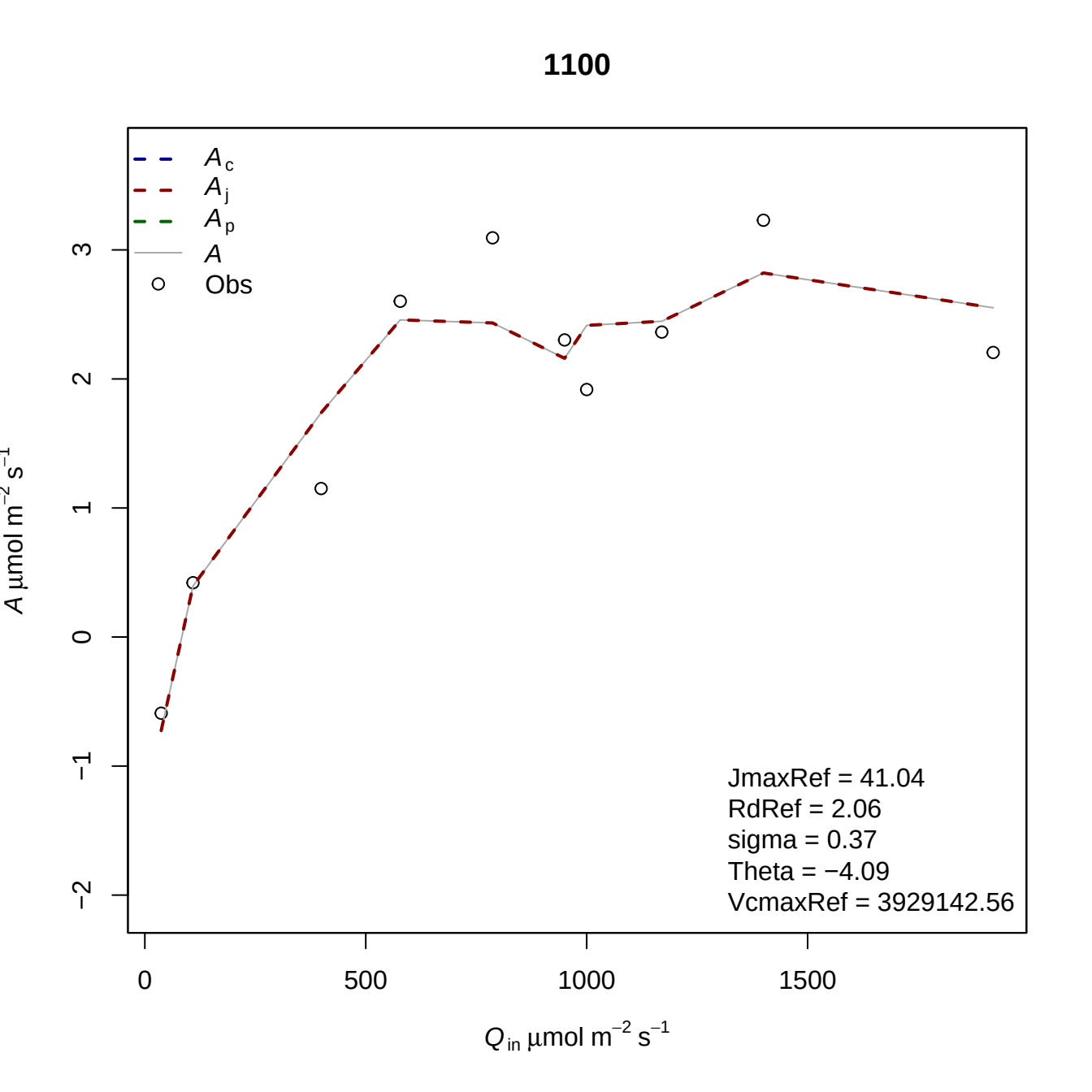

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 98.54
RdRef = 3.04
sigma = 0.7
Theta = −1.45
VcmaxRef = 205518.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

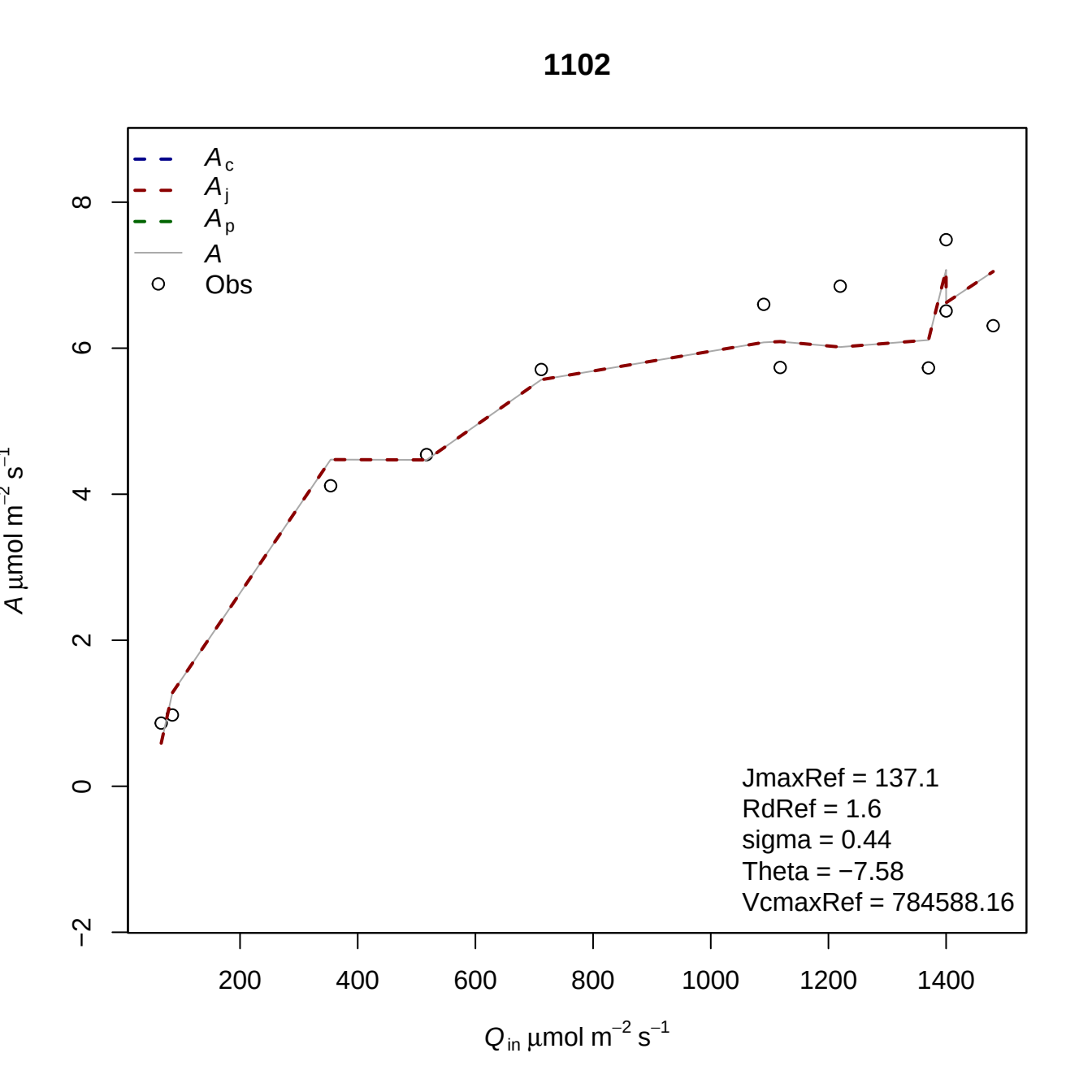

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 43.9
RdRef = 1.04
sigma = 0.79
Theta = −2.7
VcmaxRef = 197939.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

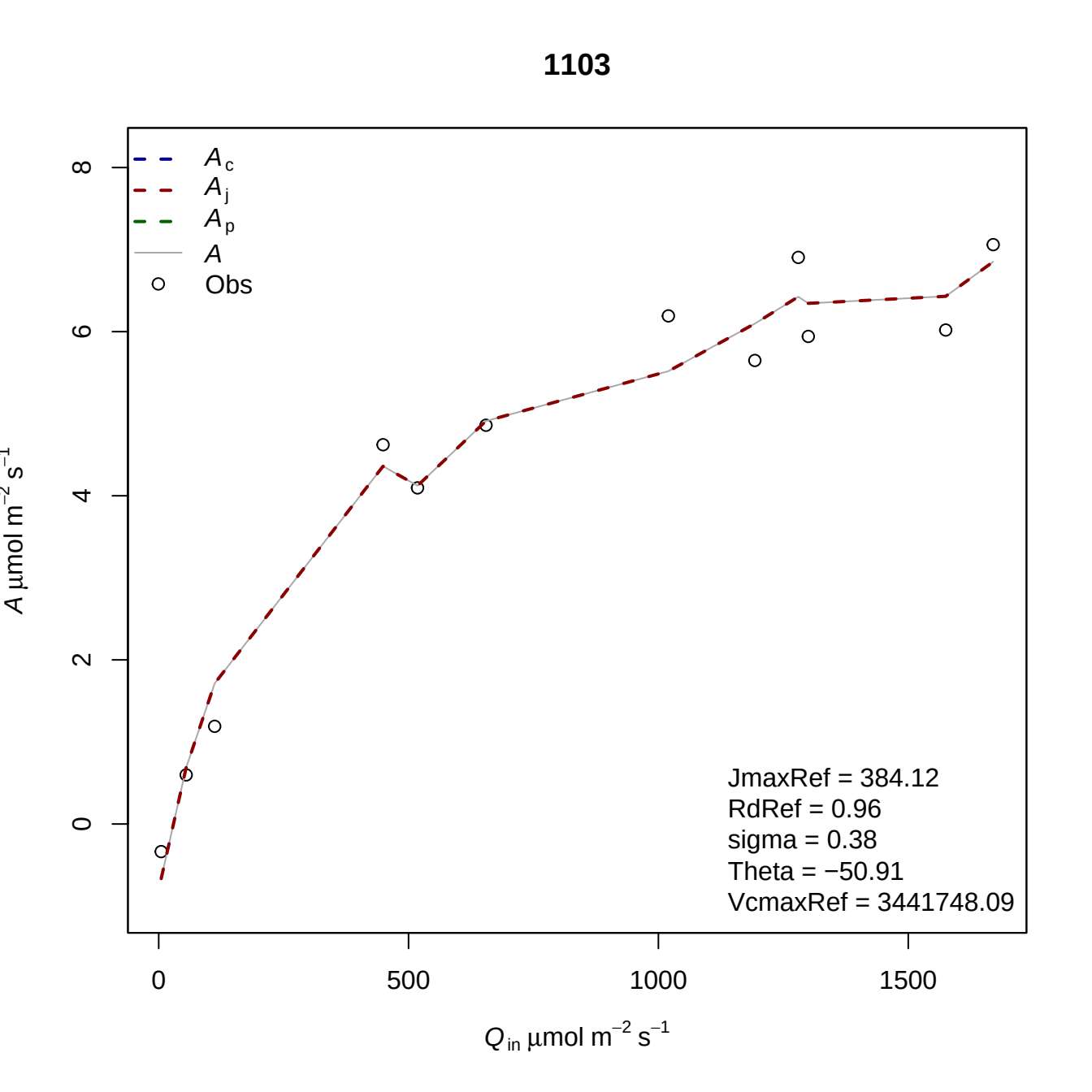

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 35.11
RdRef = 0.8
sigma = 1.55
Theta = −6.01
VcmaxRef = 76178.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

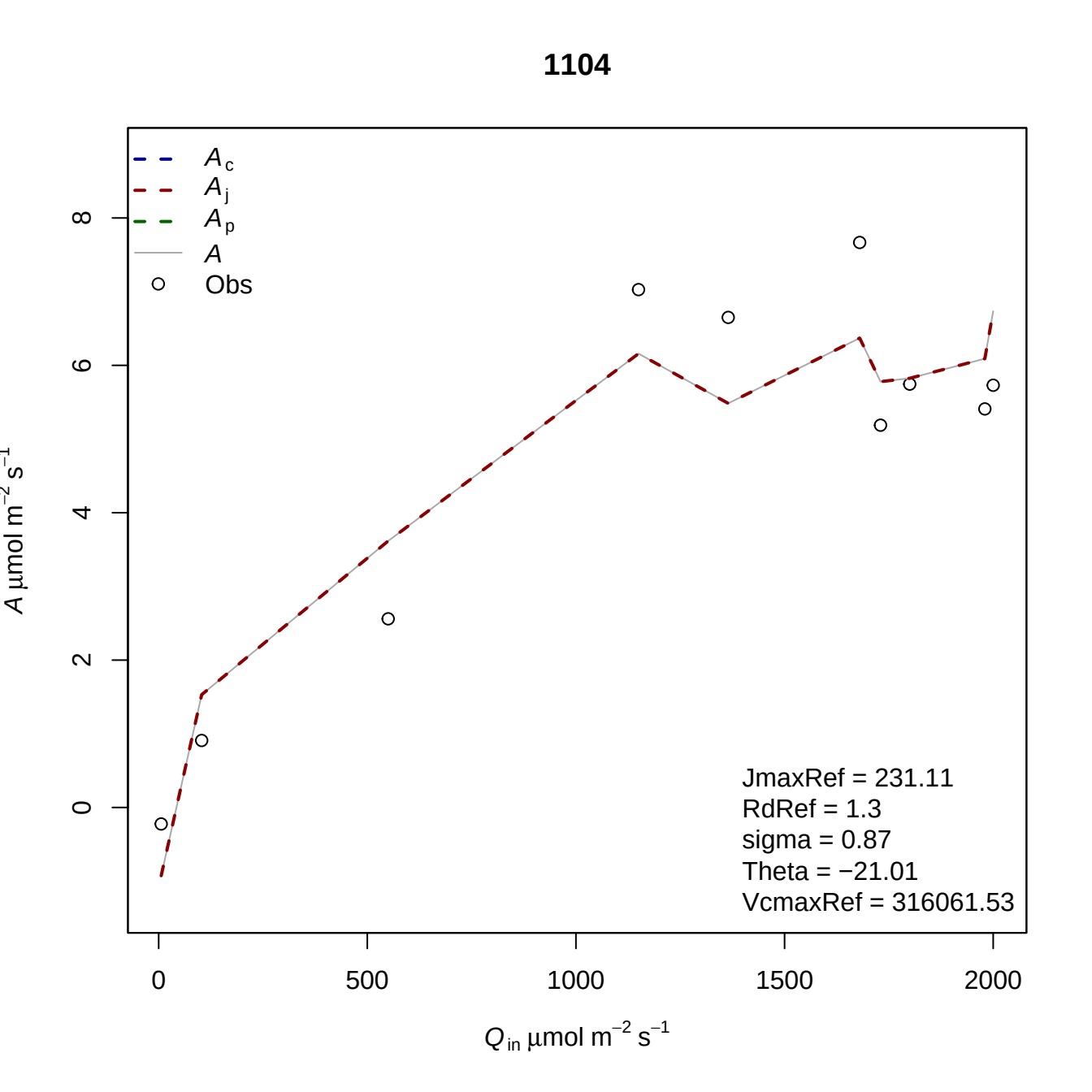

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 260.85
RdRef = 1.33
sigma = 0.72
Theta = −17.86
VcmaxRef = 1859587.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

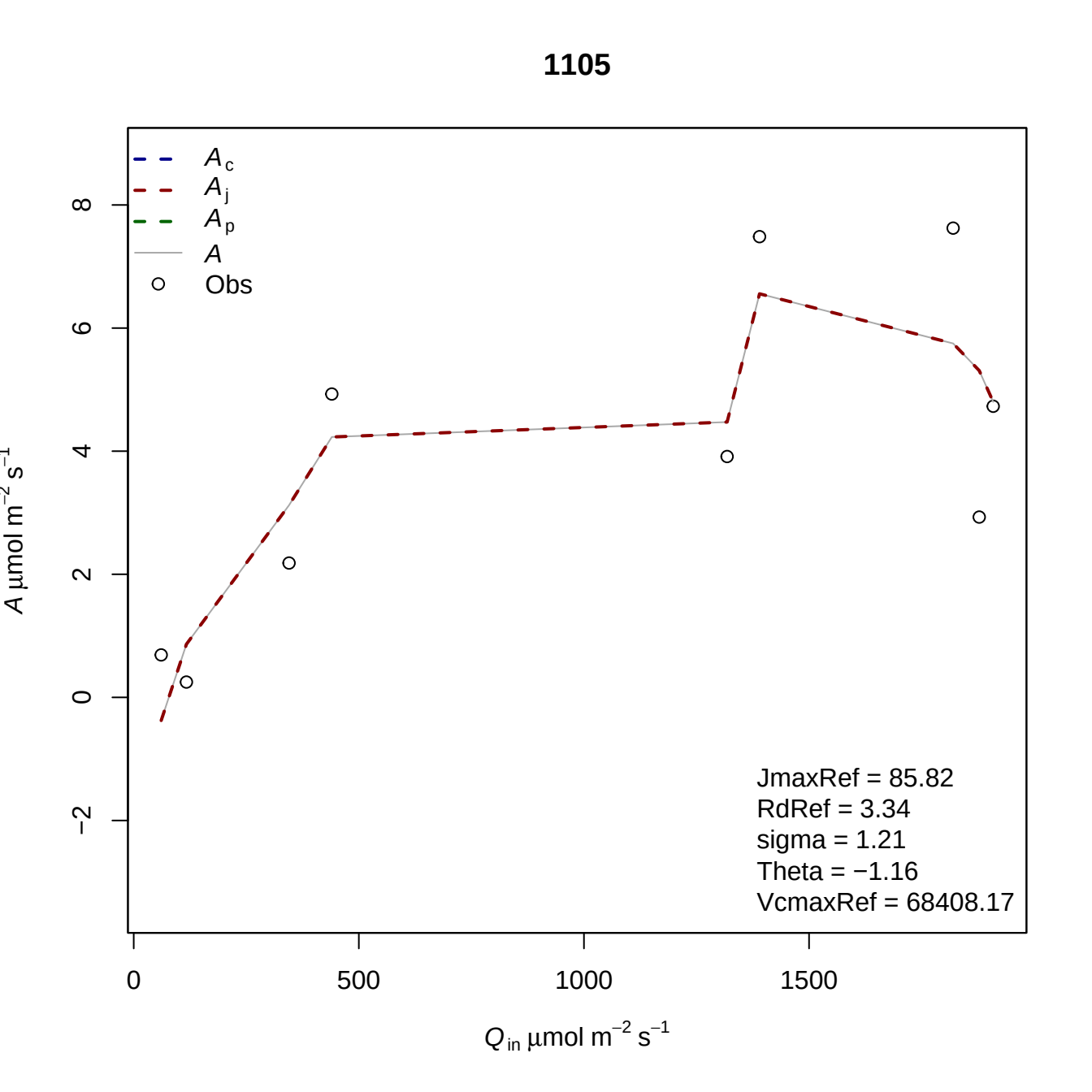

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 114.02
RdRef = 2.53
sigma = 1.39
Theta = −1.28
VcmaxRef = 671858.28
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

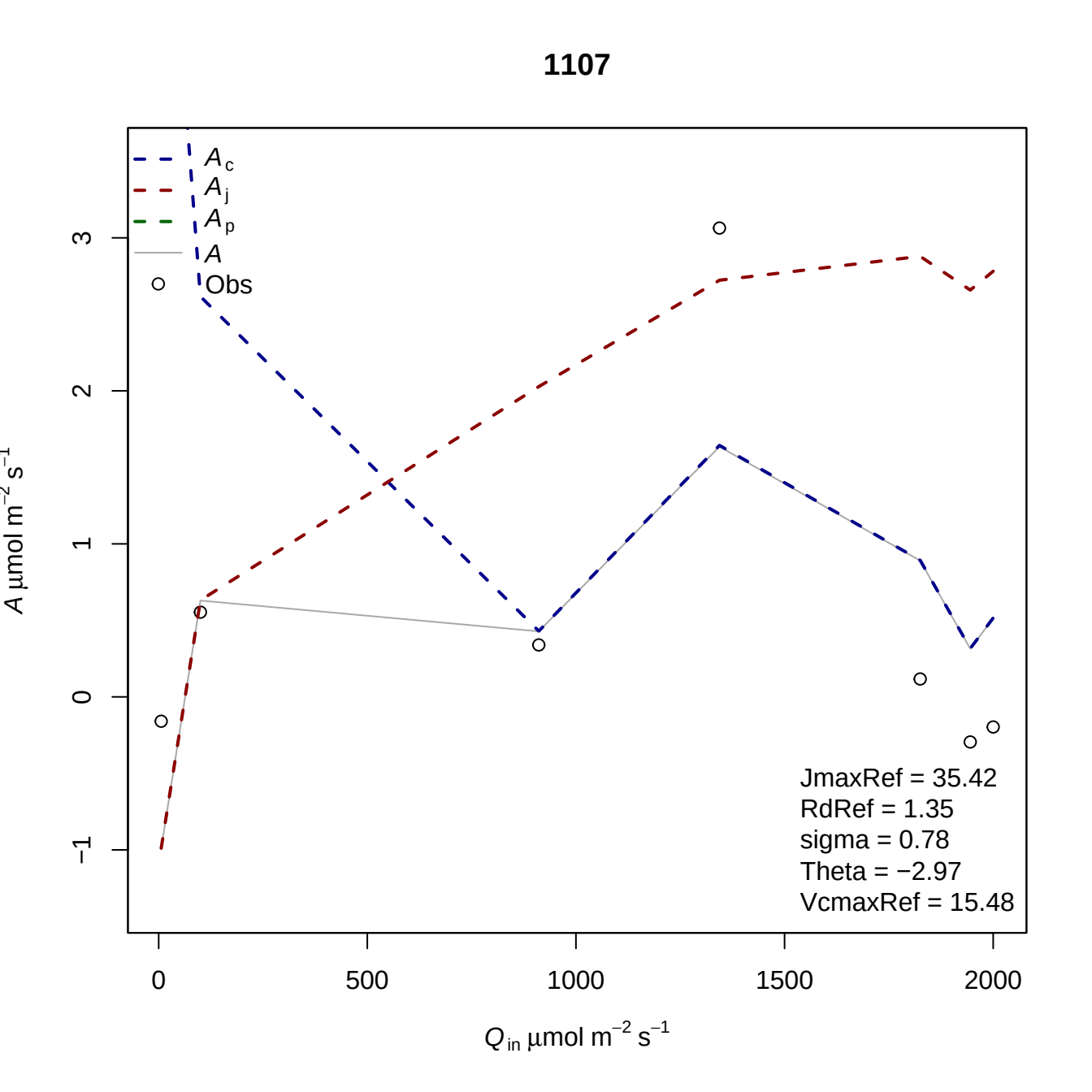

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 41.04
RdRef = 2.06
sigma = 0.37
Theta = −4.09
VcmaxRef = 3929142.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
500
1000
1500

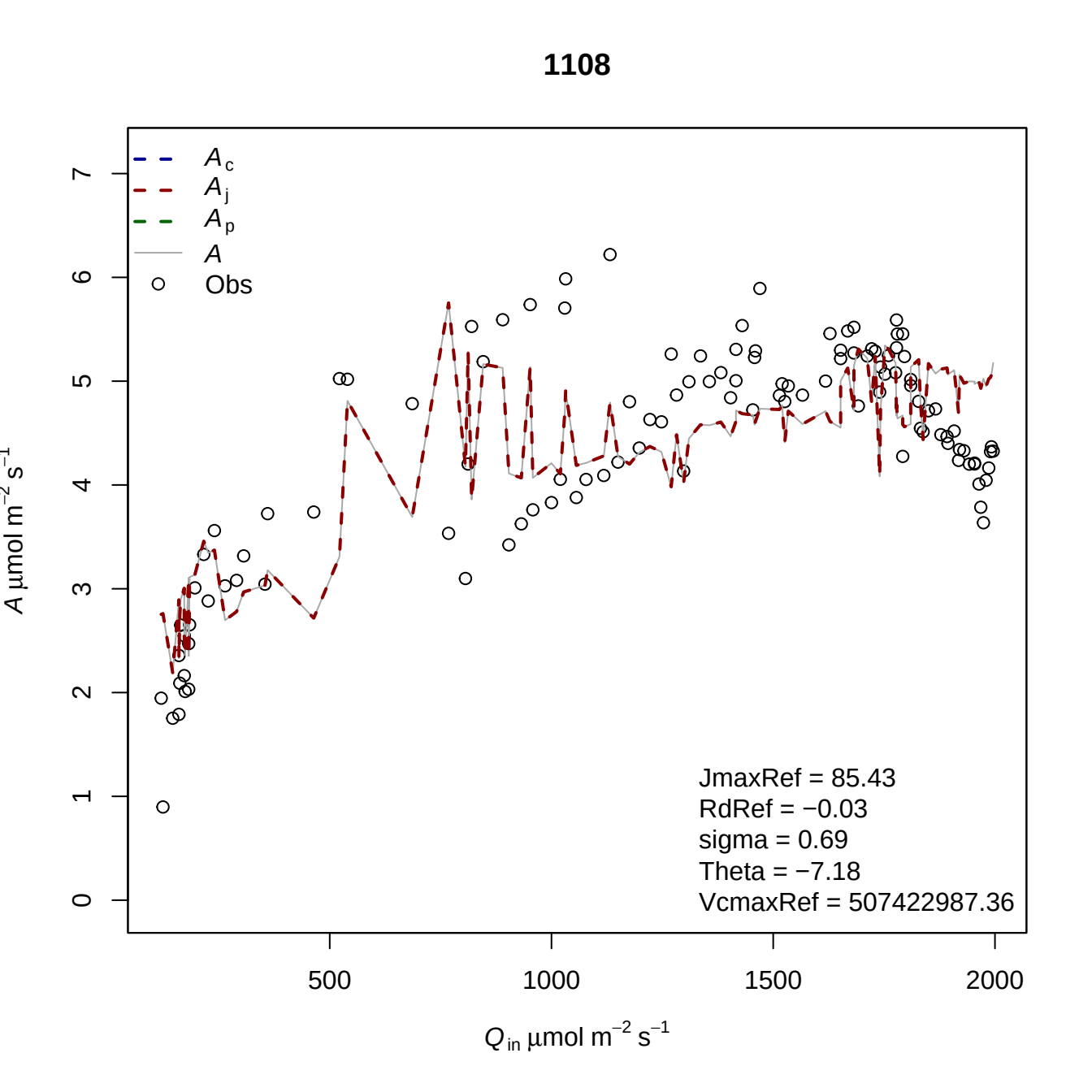

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 137.1
RdRef = 1.6
sigma = 0.44
Theta = −7.58
VcmaxRef = 784588.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
200
400
600
800
1000
1200
1400

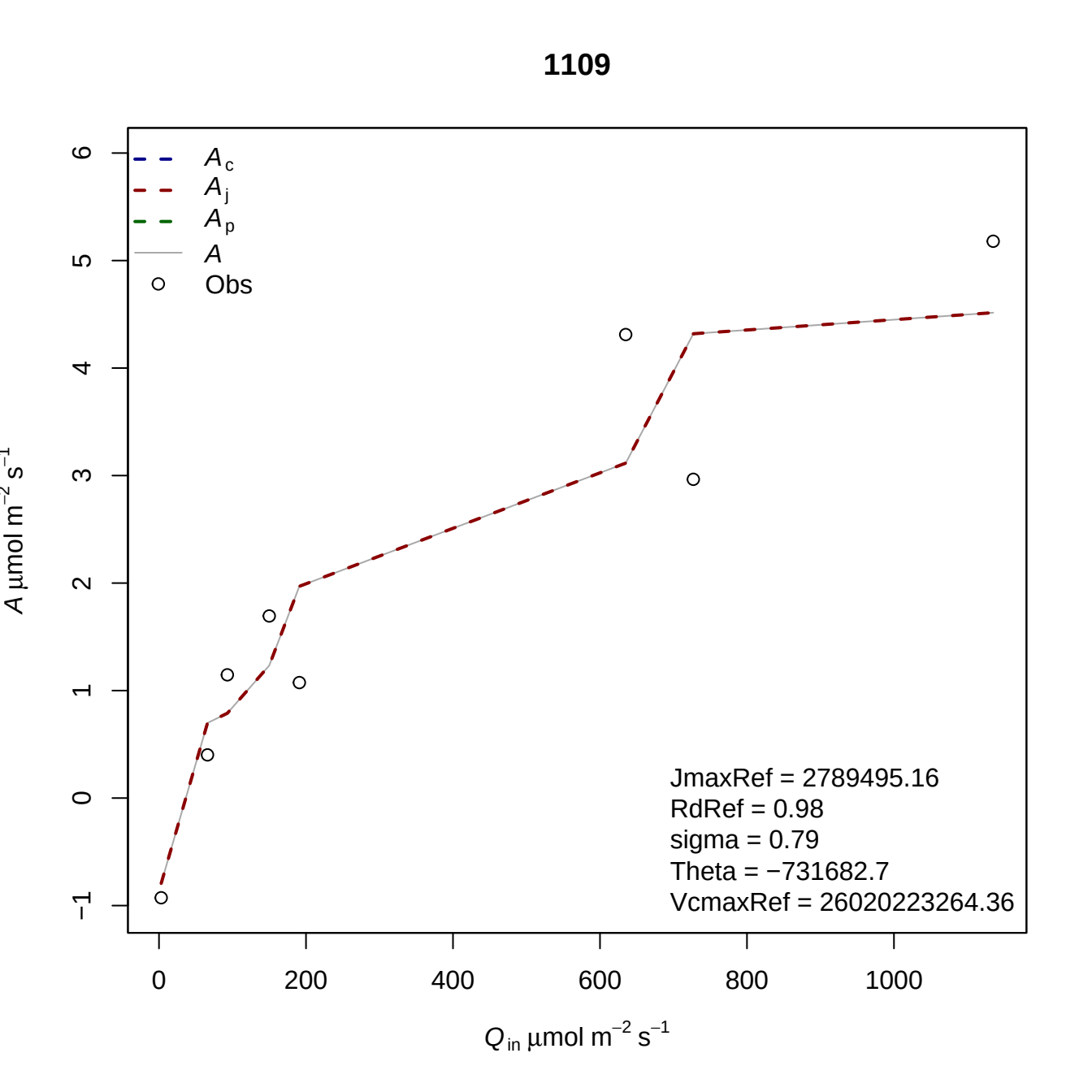

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 384.12
RdRef = 0.96
sigma = 0.38
Theta = −50.91
VcmaxRef = 3441748.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

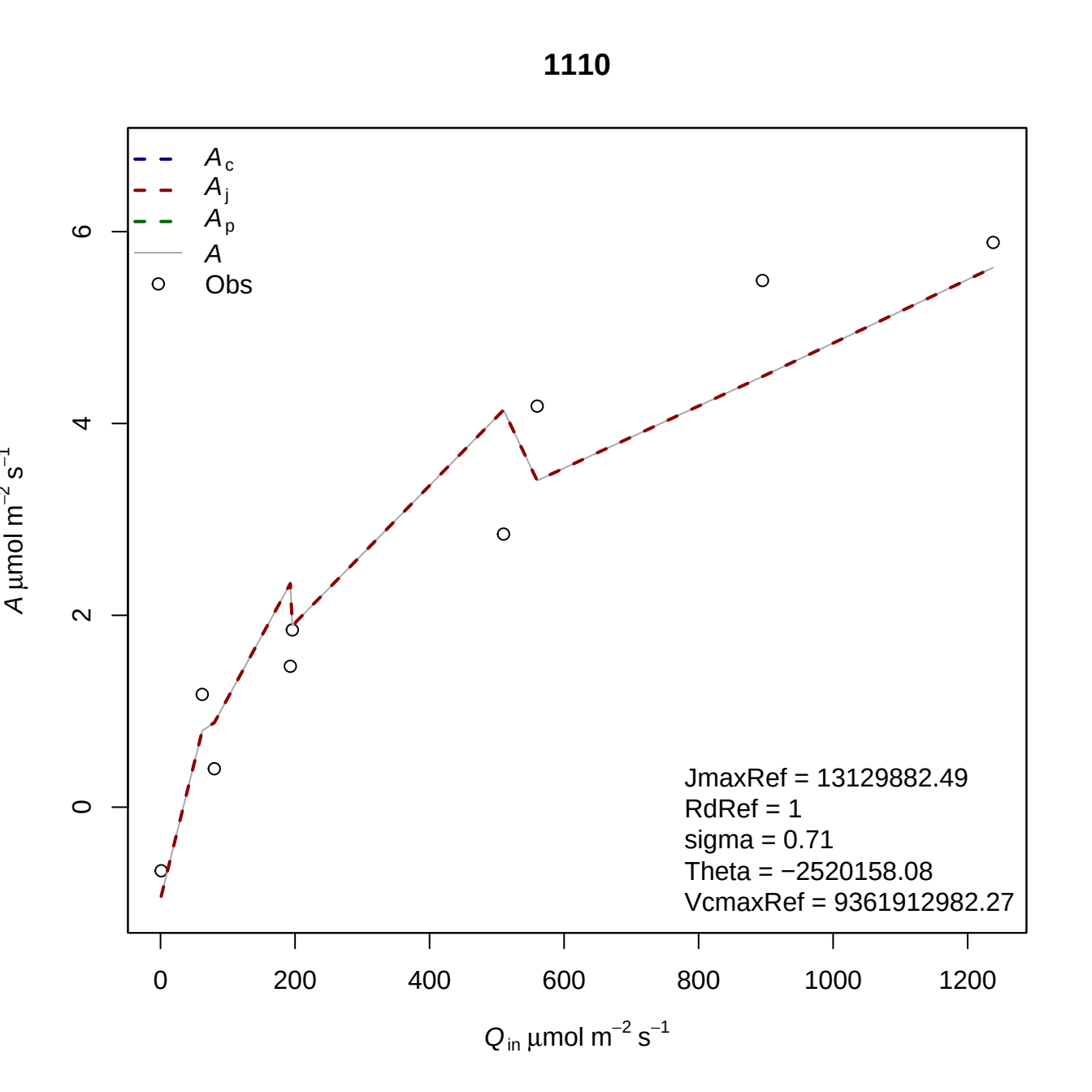

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 231.11
RdRef = 1.3
sigma = 0.87
Theta = −21.01
VcmaxRef = 316061.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

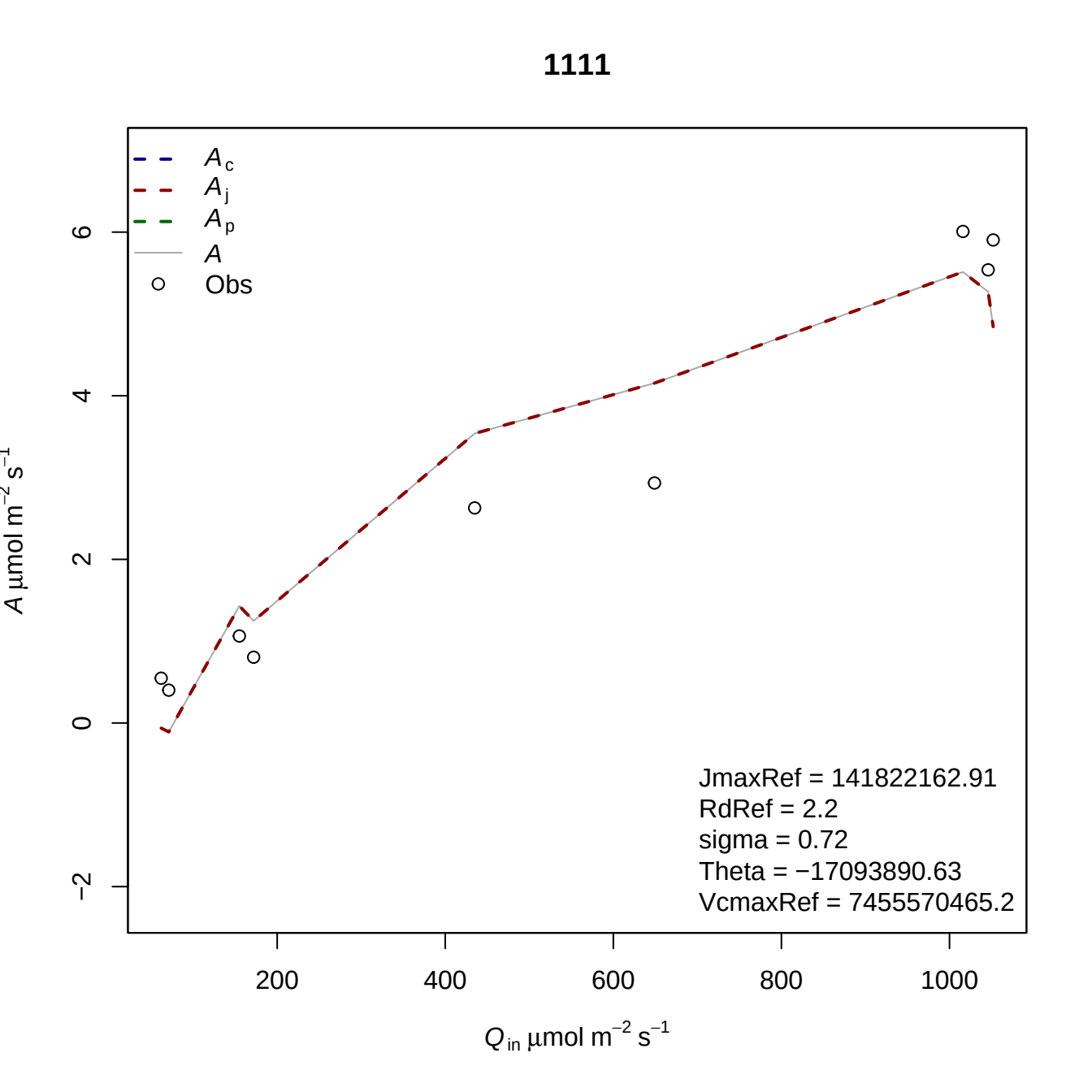

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 85.82
RdRef = 3.34
sigma = 1.21
Theta = −1.16
VcmaxRef = 68408.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

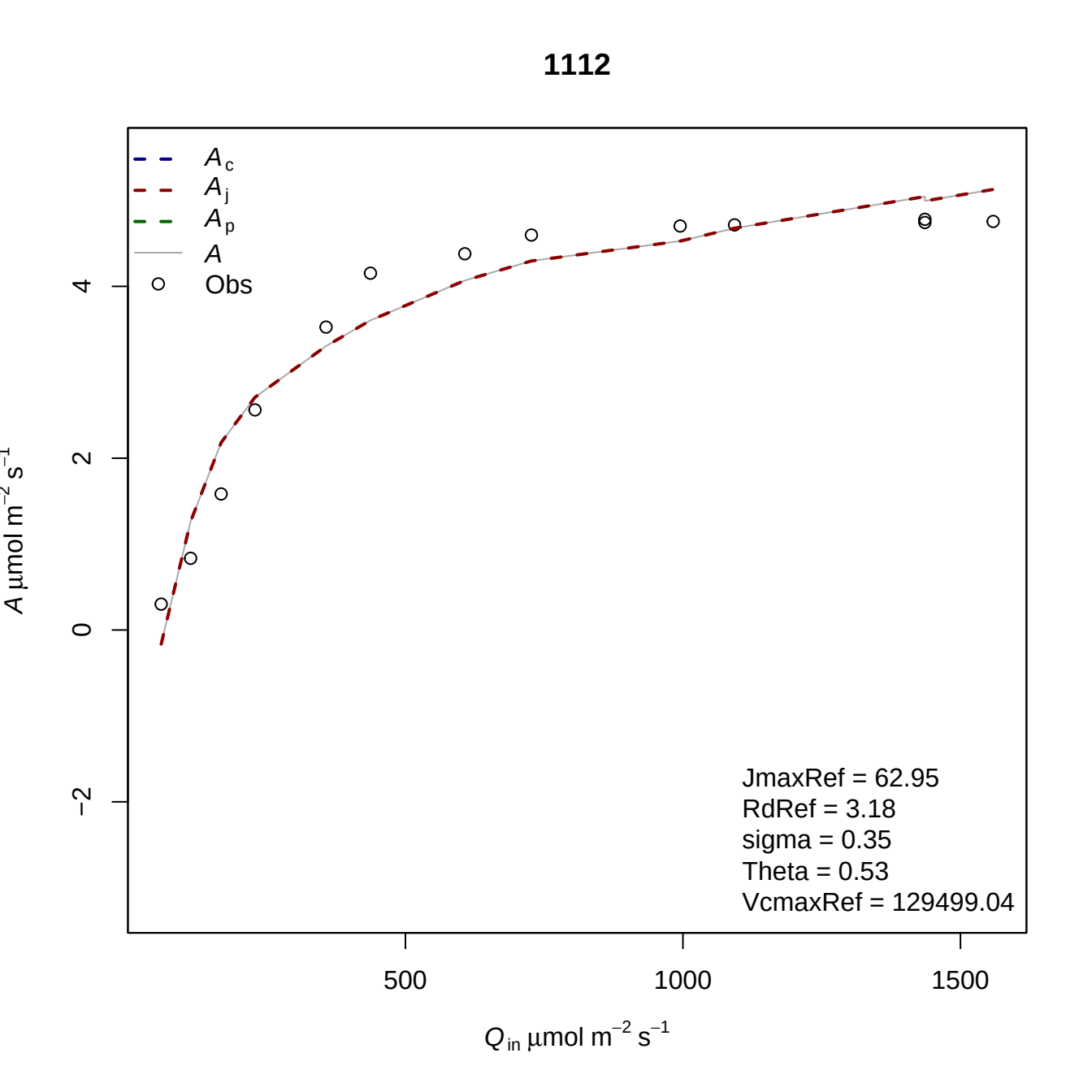

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 35.42
RdRef = 1.35
sigma = 0.78
Theta = −2.97
VcmaxRef = 15.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

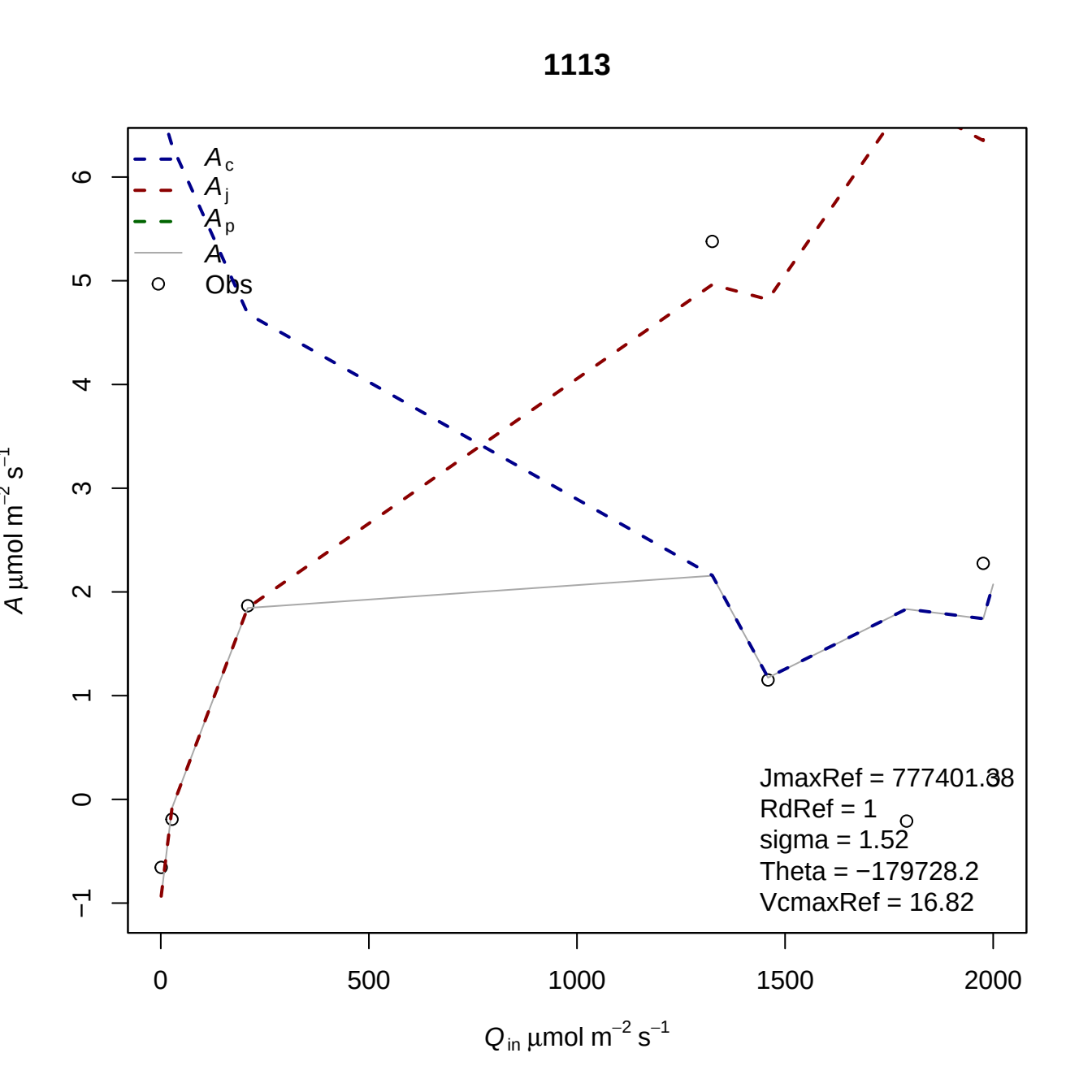

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 85.43
RdRef = −0.03
sigma = 0.69
Theta = −7.18
VcmaxRef = 507422987.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
7
500
1000
1500
2000

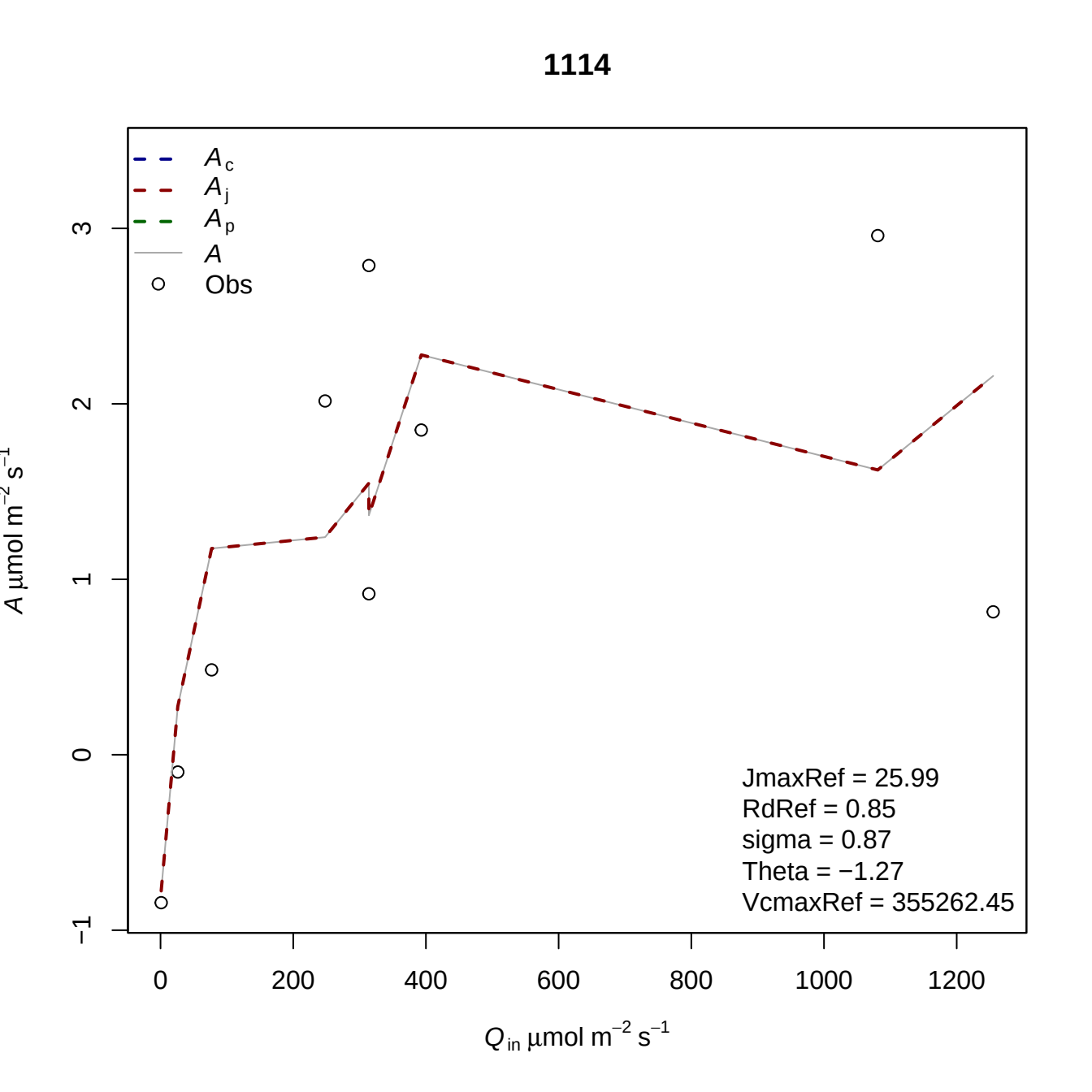

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 2789495.16
RdRef = 0.98
sigma = 0.79
Theta = −731682.7
VcmaxRef = 26020223264.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

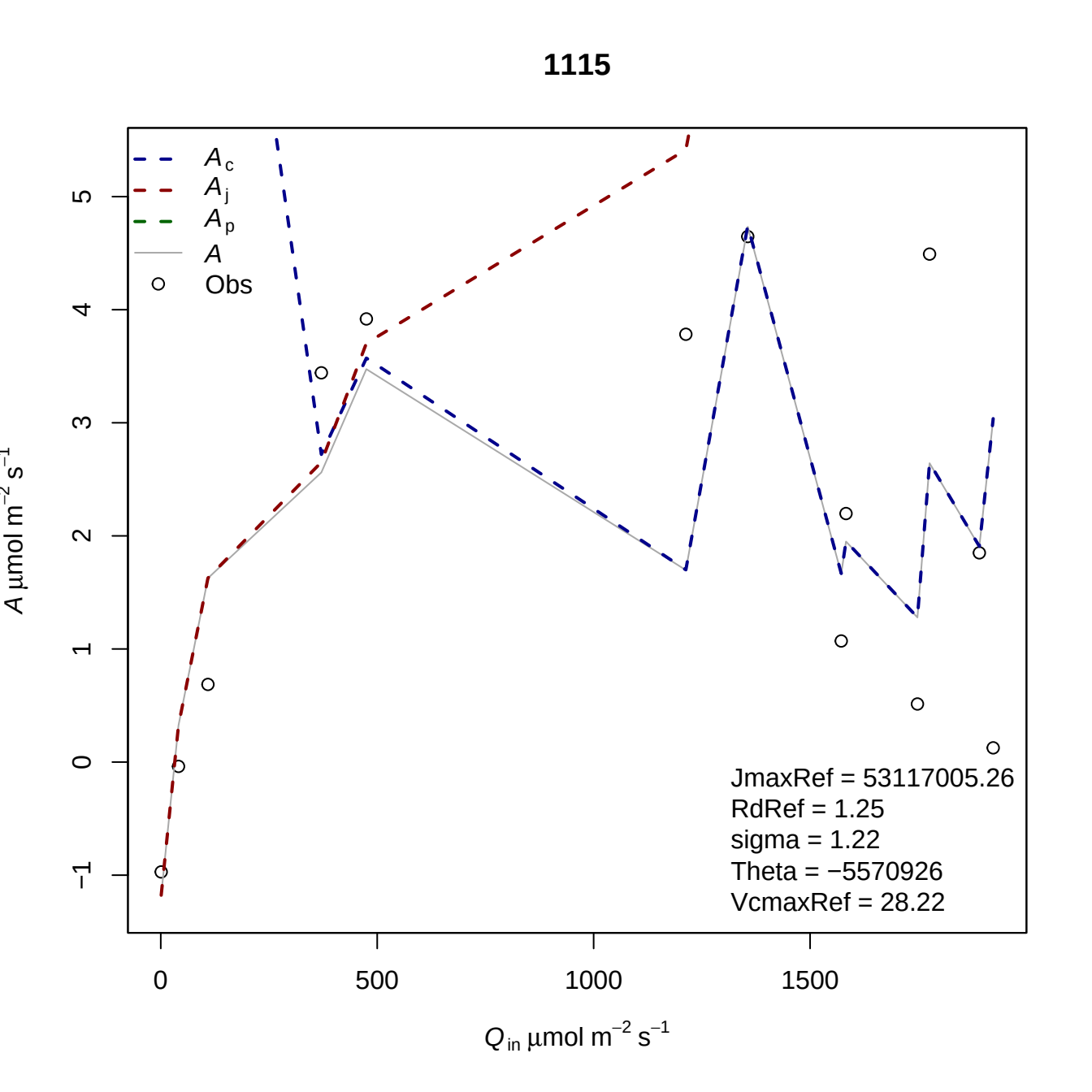

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 13129882.49
RdRef = 1
sigma = 0.71
Theta = −2520158.08
VcmaxRef = 9361912982.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000
1200

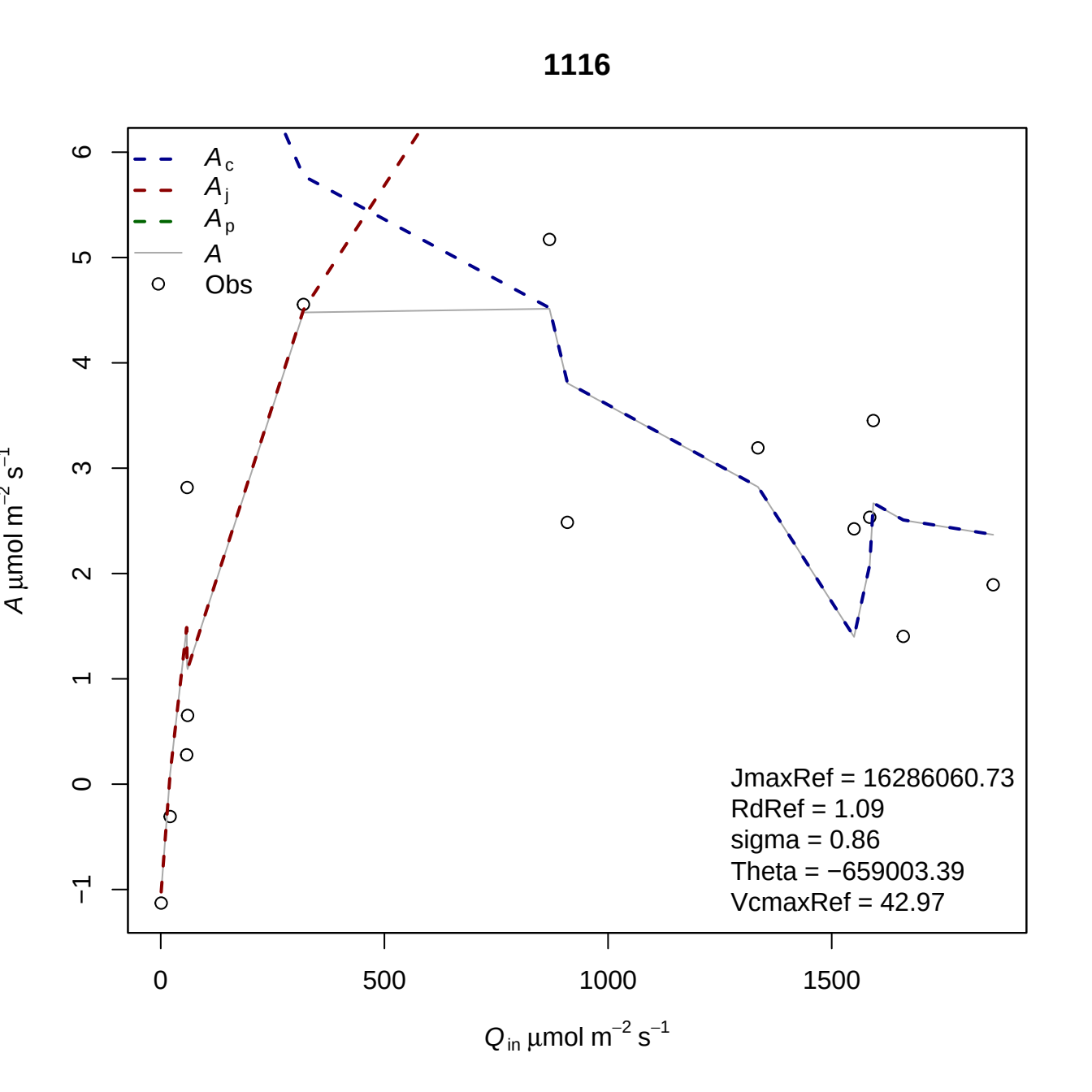

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 141822162.91
RdRef = 2.2
sigma = 0.72
Theta = −17093890.63
VcmaxRef = 7455570465.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
200
400
600
800
1000

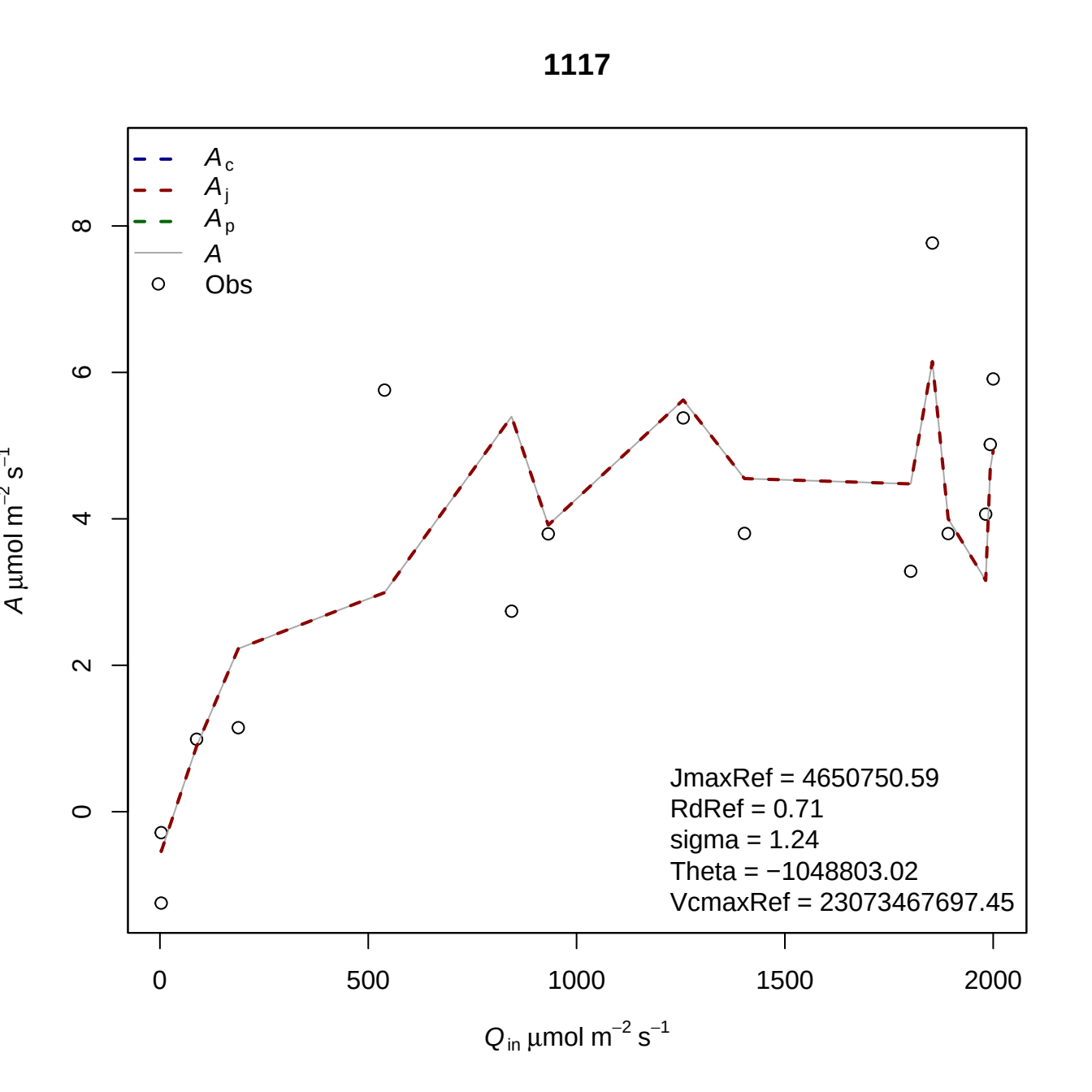

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 62.95
RdRef = 3.18
sigma = 0.35
Theta = 0.53
VcmaxRef = 129499.04
−2
0
2
4
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

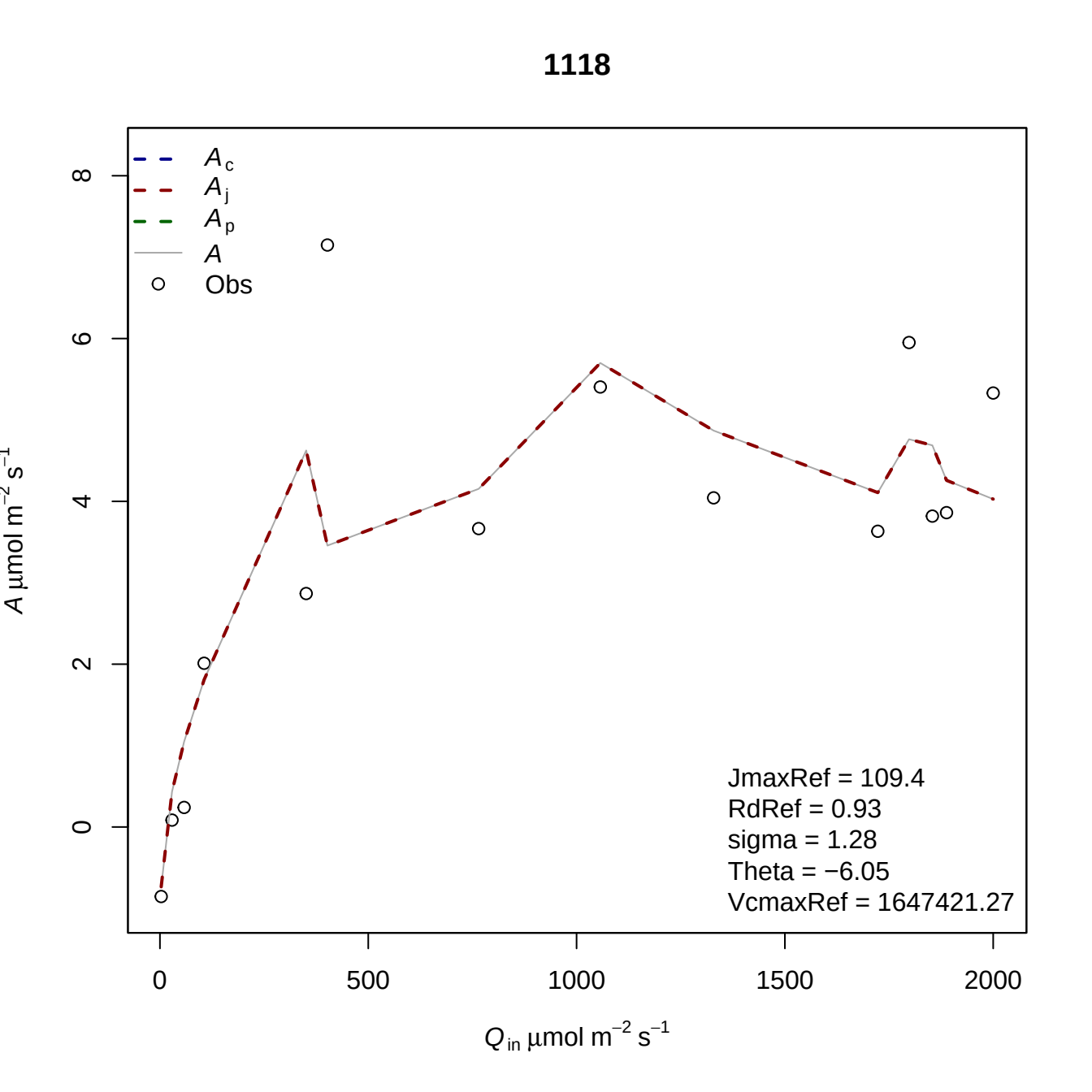

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 777401.38
RdRef = 1
sigma = 1.52
Theta = −179728.2
VcmaxRef = 16.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

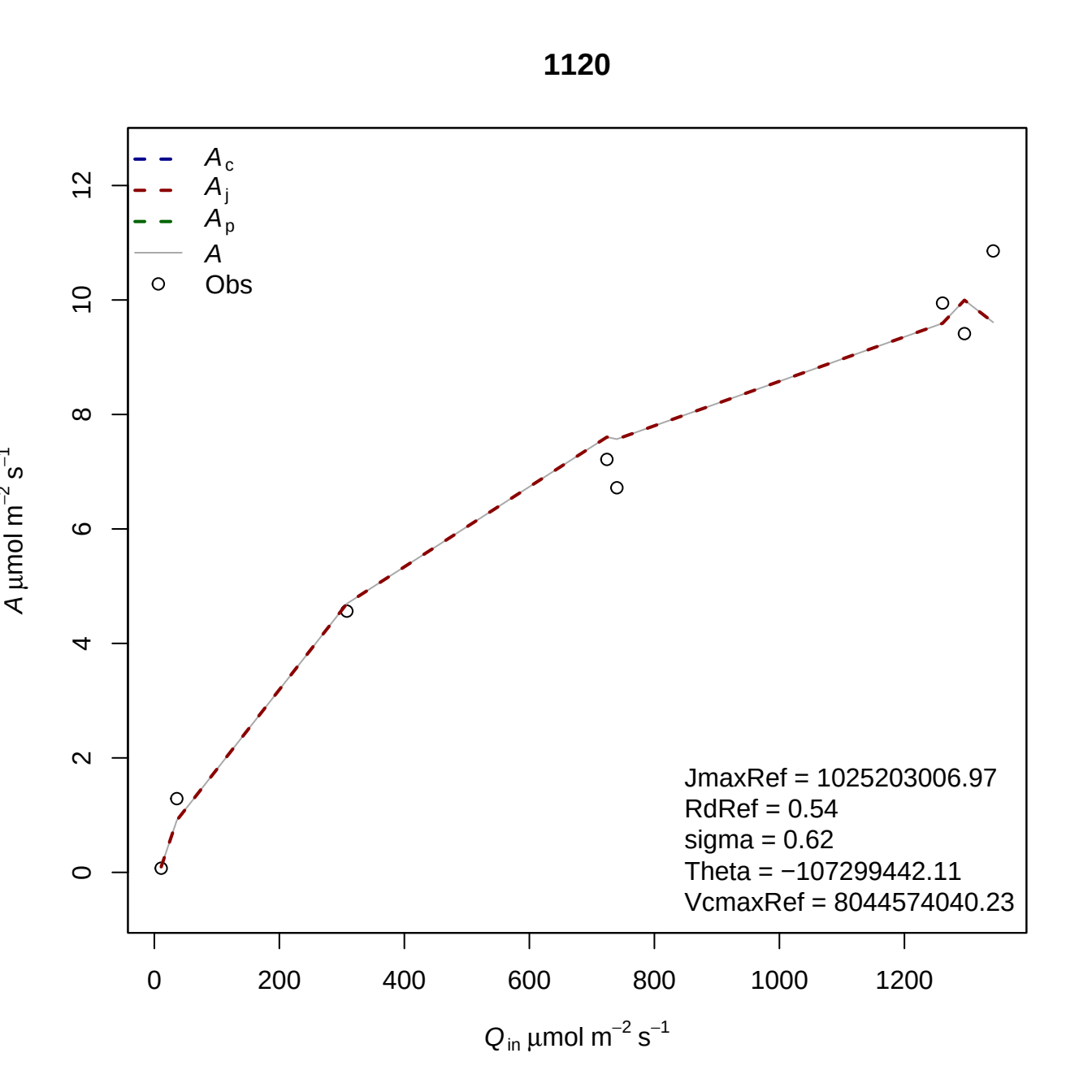

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 25.99
RdRef = 0.85
sigma = 0.87
Theta = −1.27
VcmaxRef = 355262.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

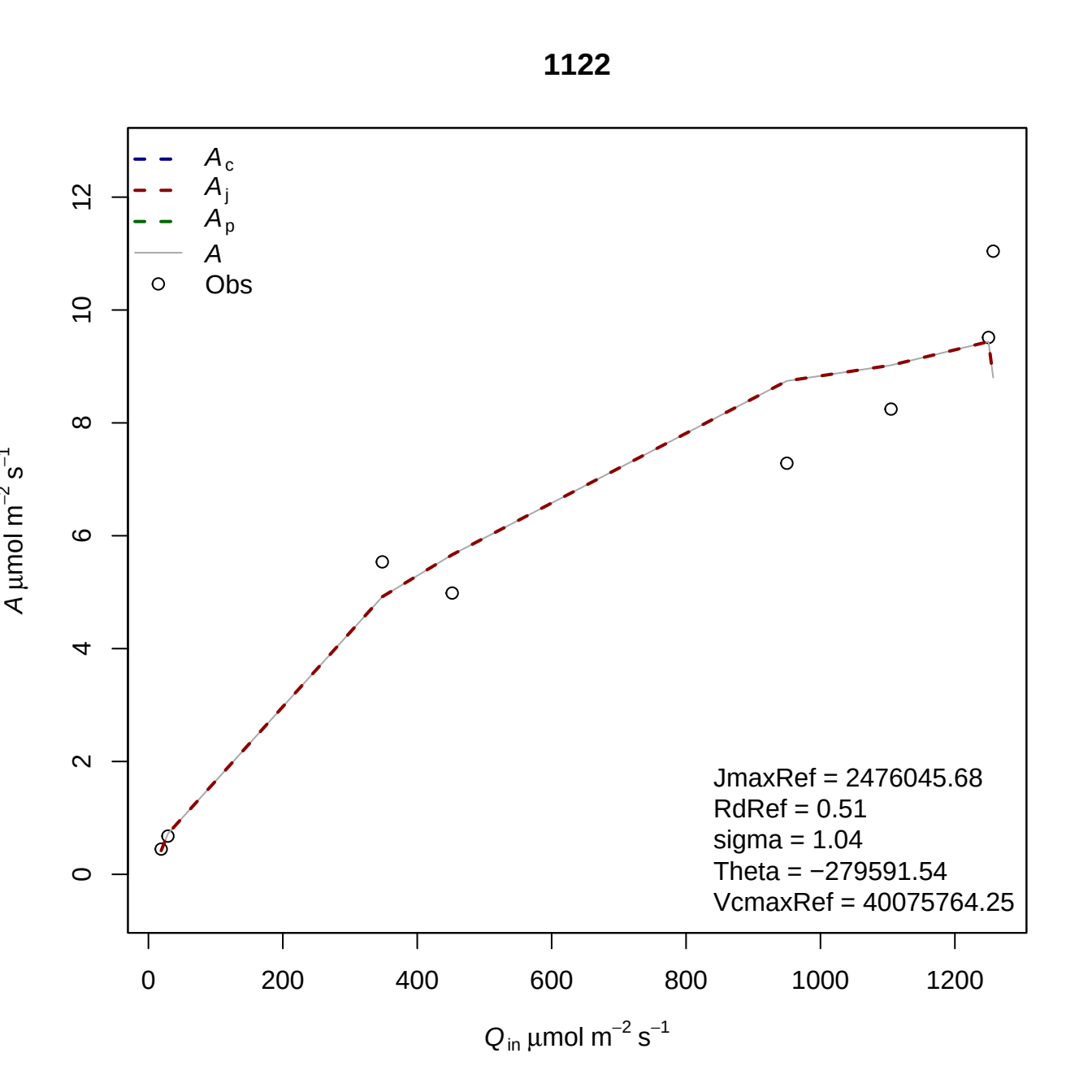

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 53117005.26
RdRef = 1.25
sigma = 1.22
Theta = −5570926
VcmaxRef = 28.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

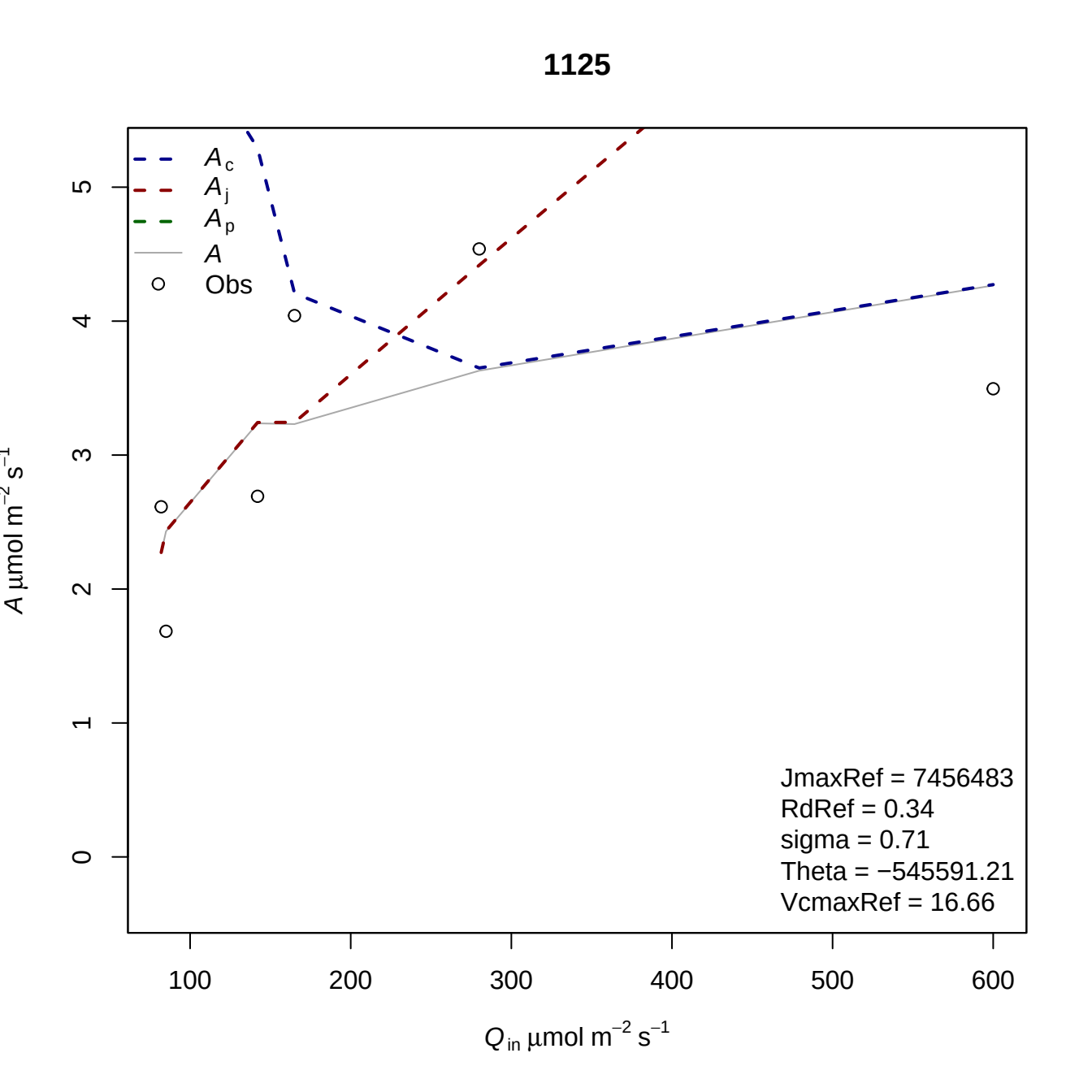

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 16286060.73
RdRef = 1.09
sigma = 0.86
Theta = −659003.39
VcmaxRef = 42.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

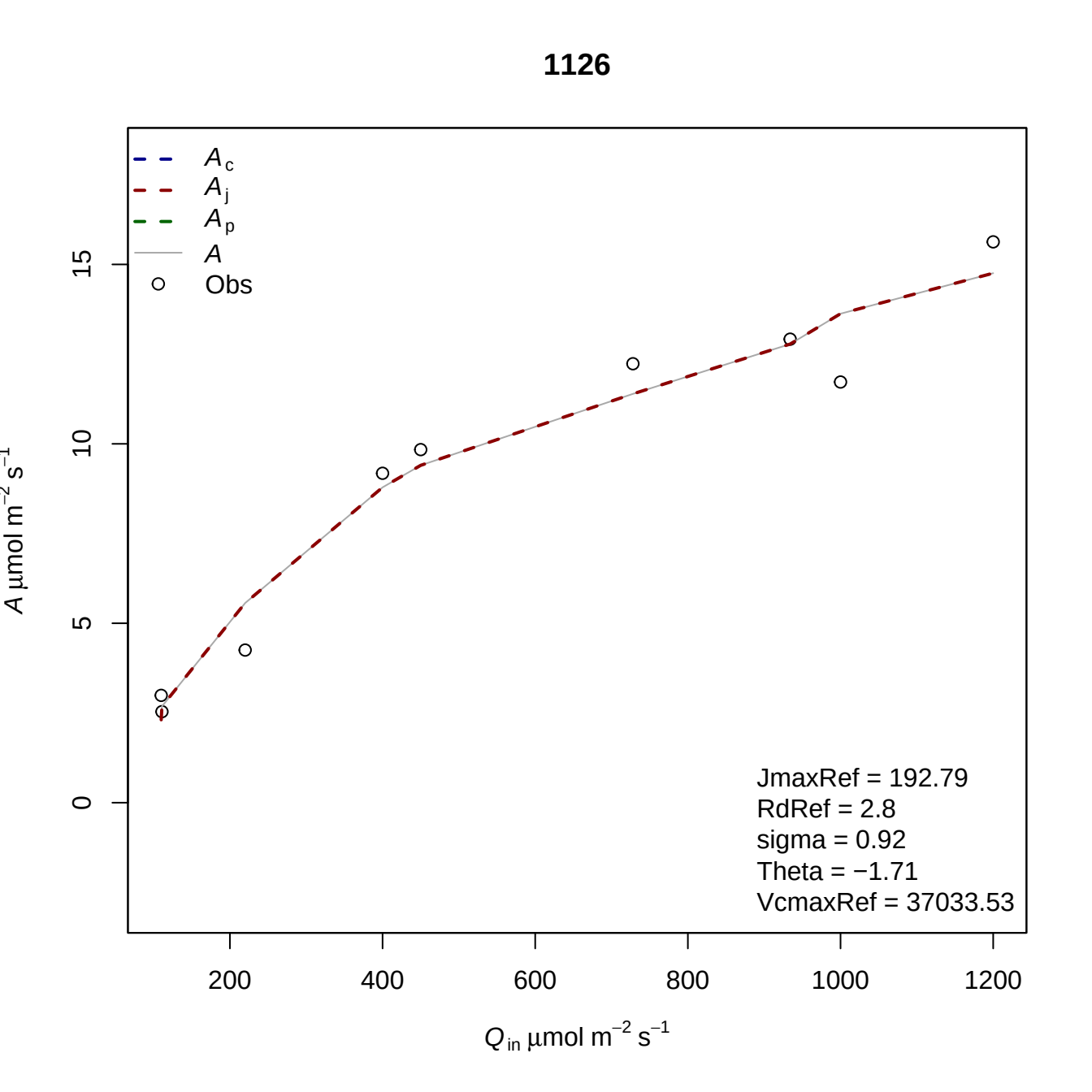

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 4650750.59
RdRef = 0.71
sigma = 1.24
Theta = −1048803.02
VcmaxRef = 23073467697.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

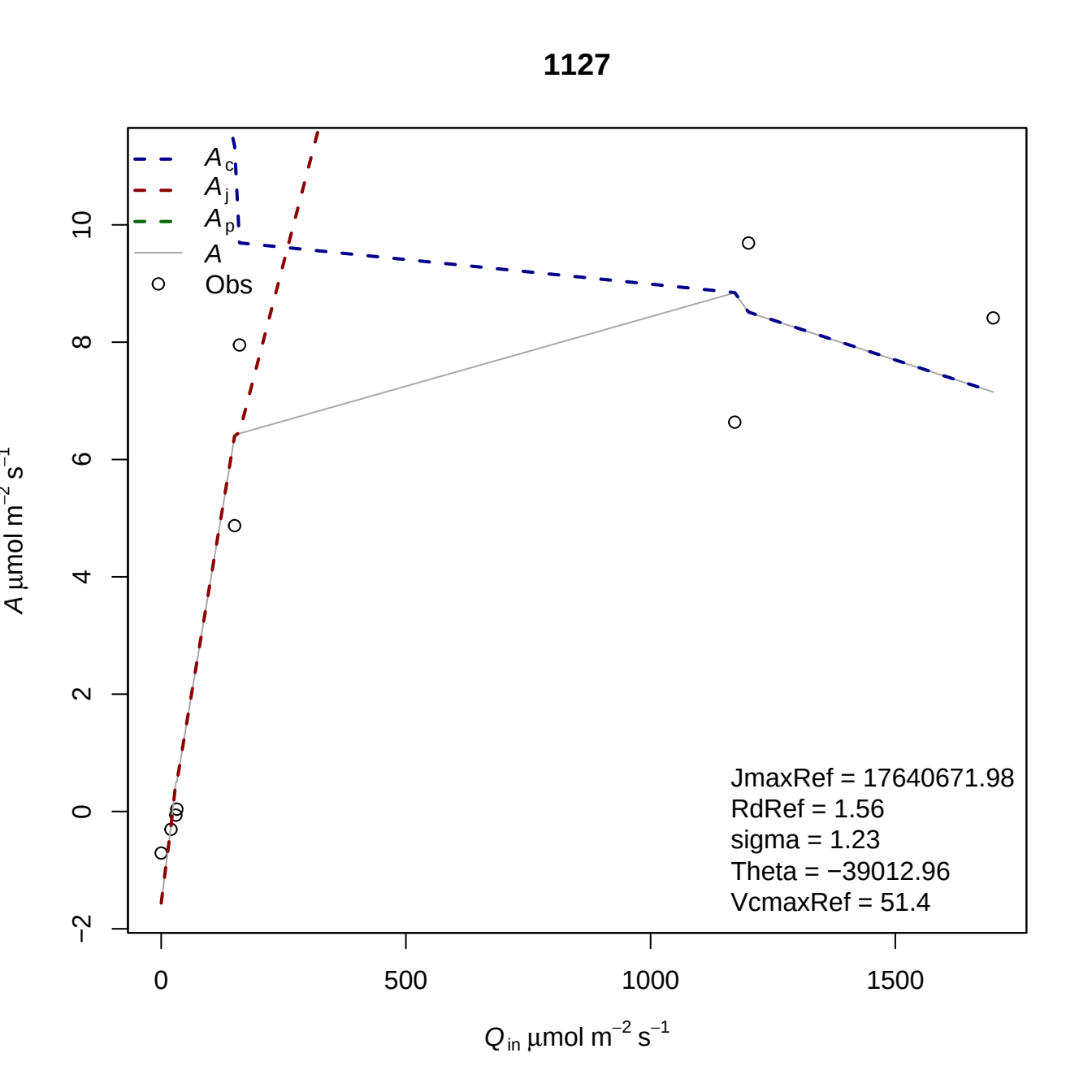

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 109.4
RdRef = 0.93
sigma = 1.28
Theta = −6.05
VcmaxRef = 1647421.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

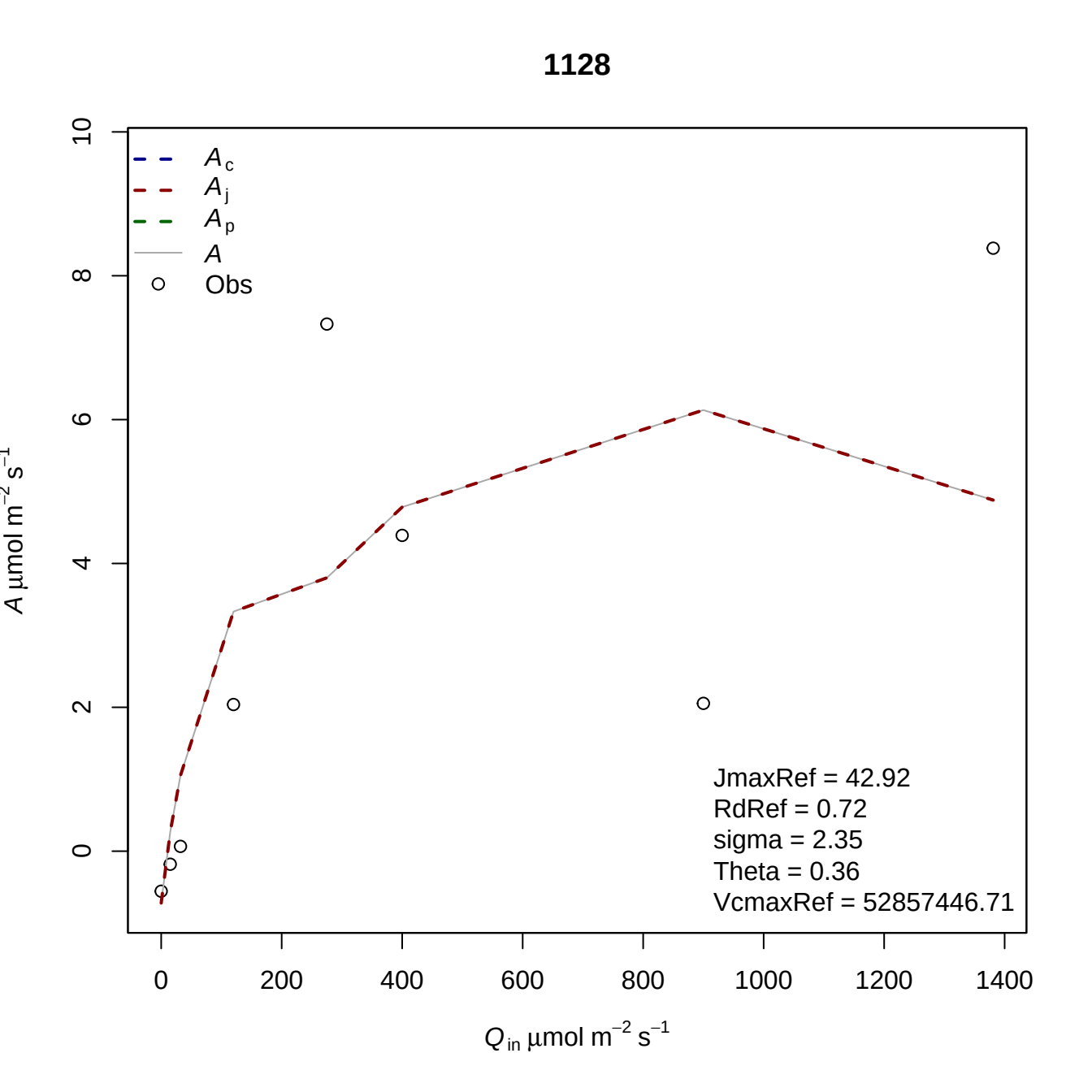

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 1025203006.97
RdRef = 0.54
sigma = 0.62
Theta = −107299442.11
VcmaxRef = 8044574040.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

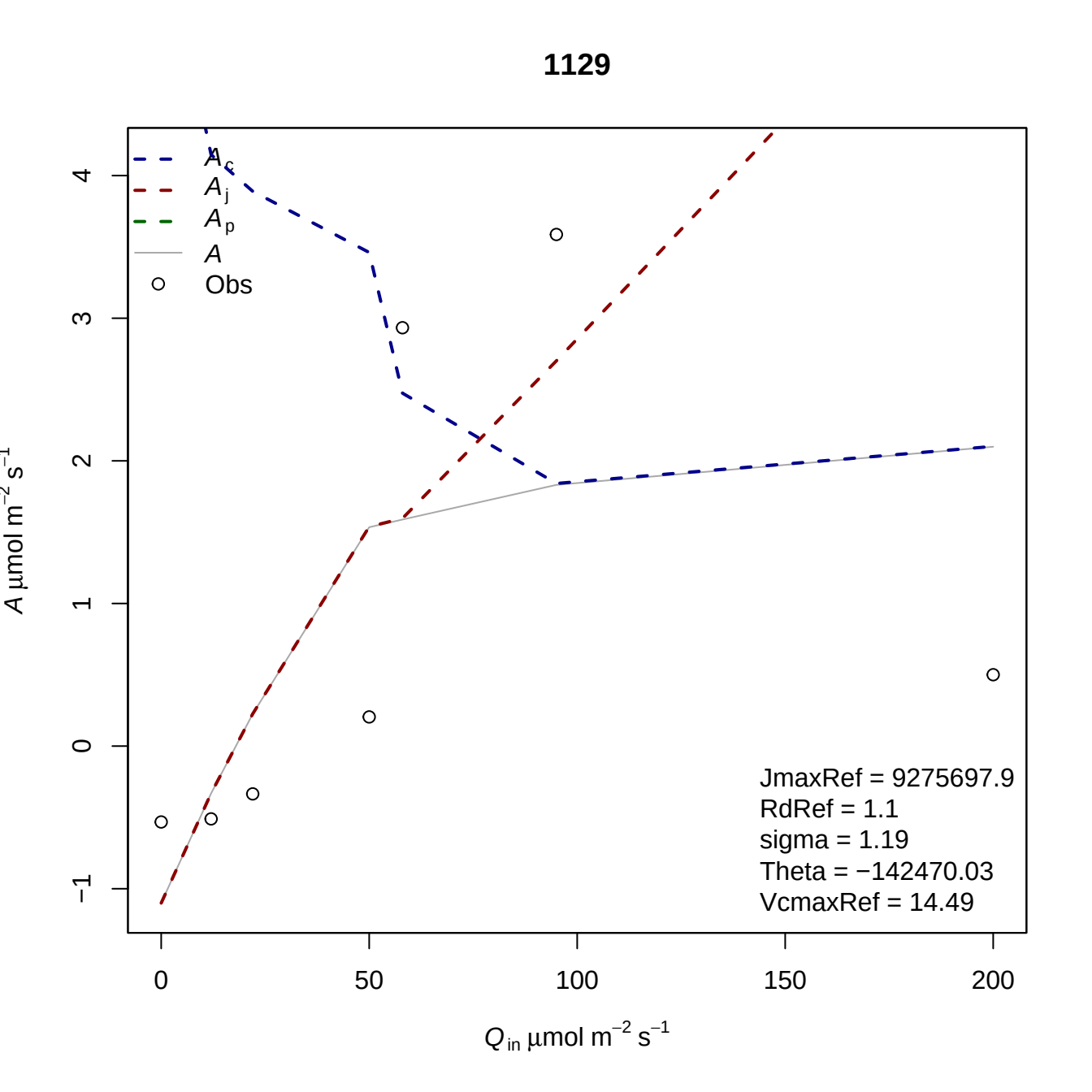

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 2476045.68
RdRef = 0.51
sigma = 1.04
Theta = −279591.54
VcmaxRef = 40075764.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

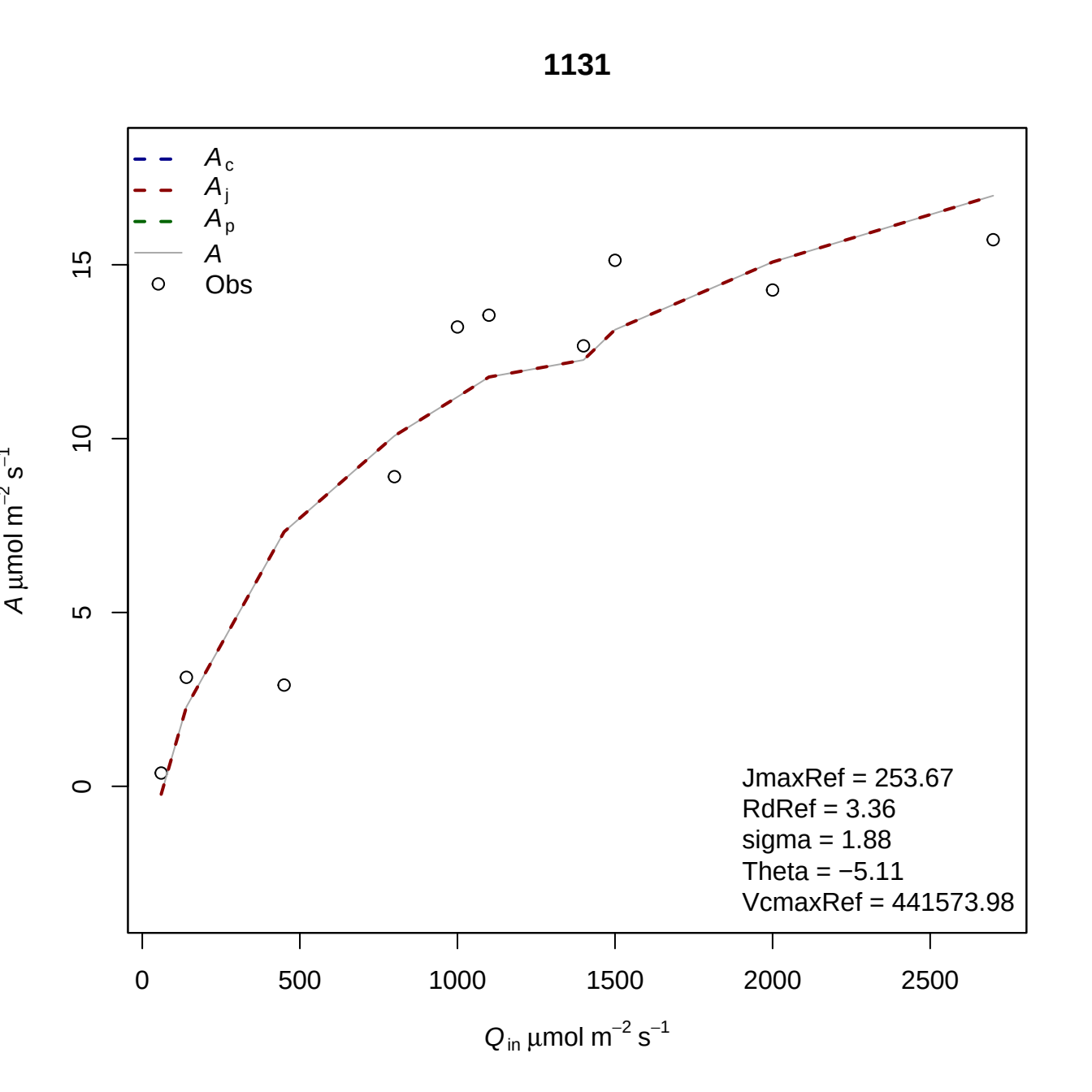

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 7456483
RdRef = 0.34
sigma = 0.71
Theta = −545591.21
VcmaxRef = 16.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
100
200
300
400
500
600

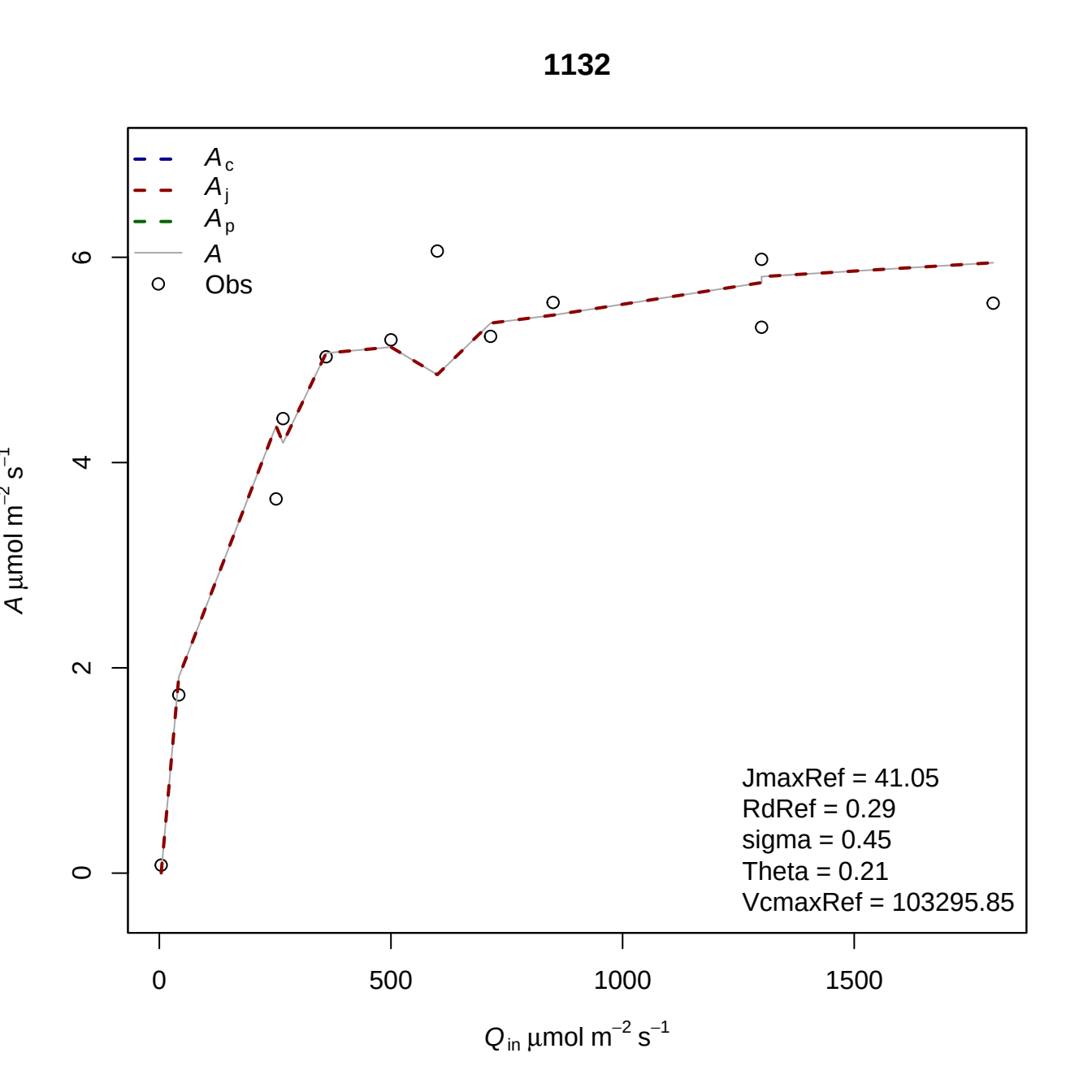

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 192.79
RdRef = 2.8
sigma = 0.92
Theta = −1.71
VcmaxRef = 37033.53
0
5
10
15
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

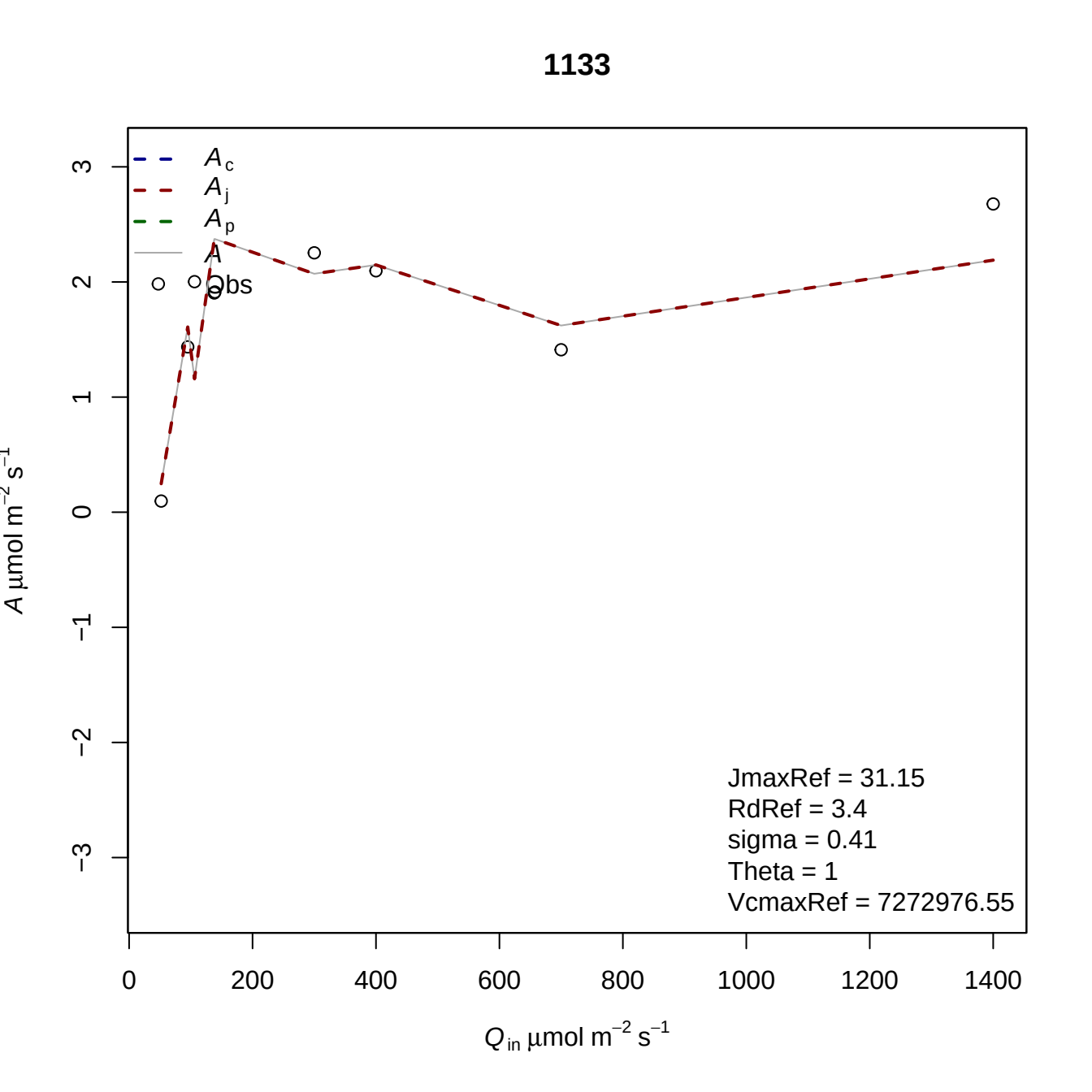

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 17640671.98
RdRef = 1.56
sigma = 1.23
Theta = −39012.96
VcmaxRef = 51.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

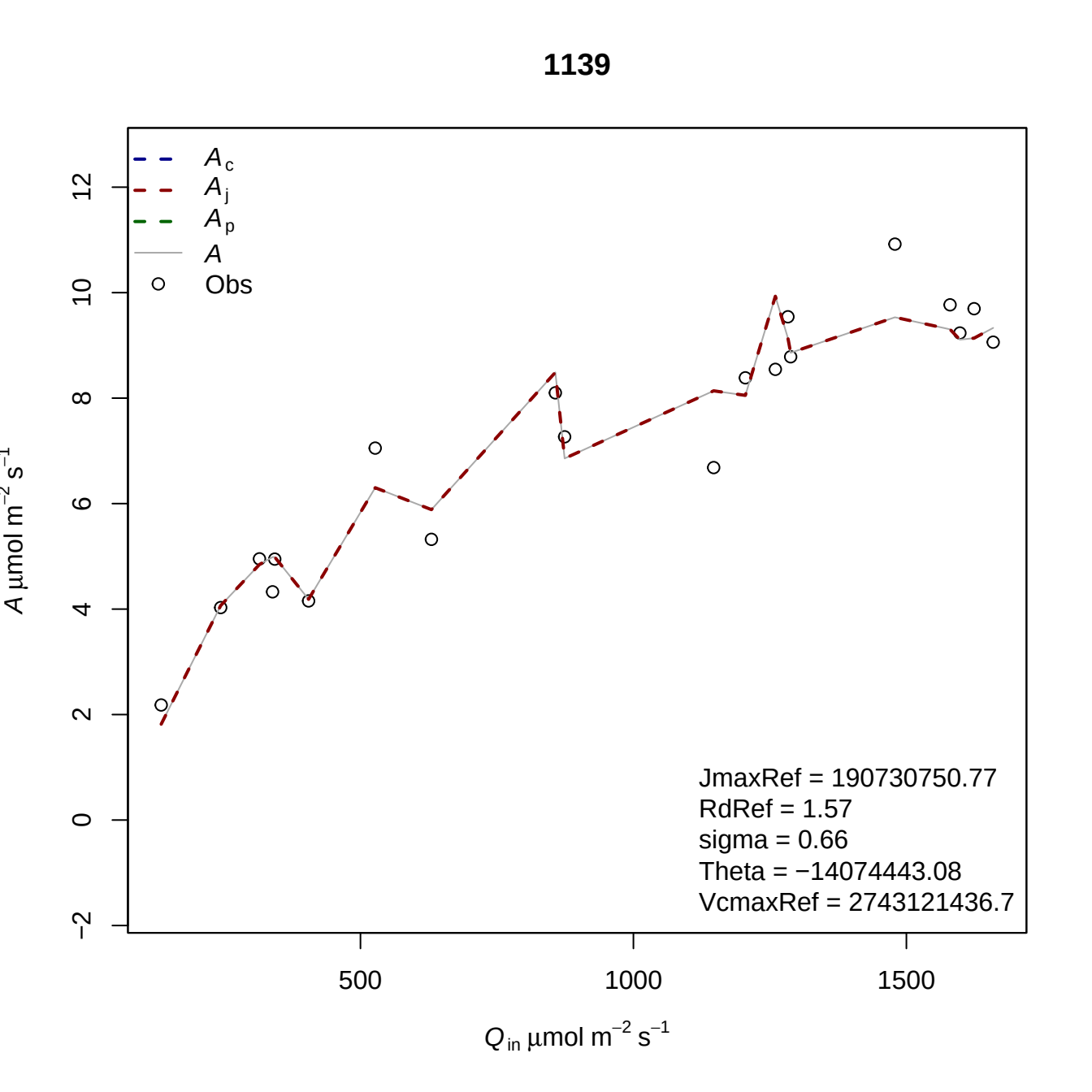

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 42.92
RdRef = 0.72
sigma = 2.35
Theta = 0.36
VcmaxRef = 52857446.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

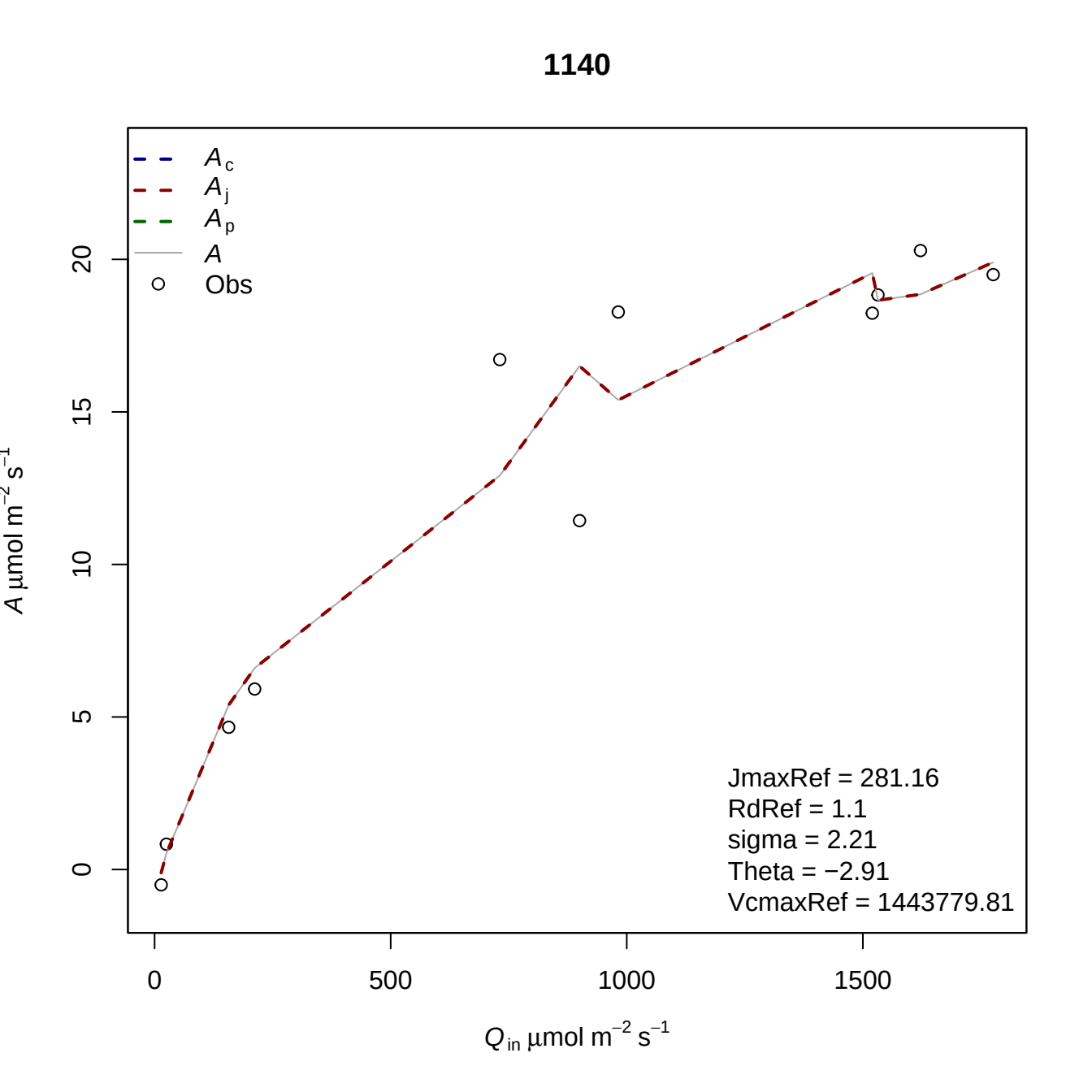

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 9275697.9
RdRef = 1.1
sigma = 1.19
Theta = −142470.03
VcmaxRef = 14.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

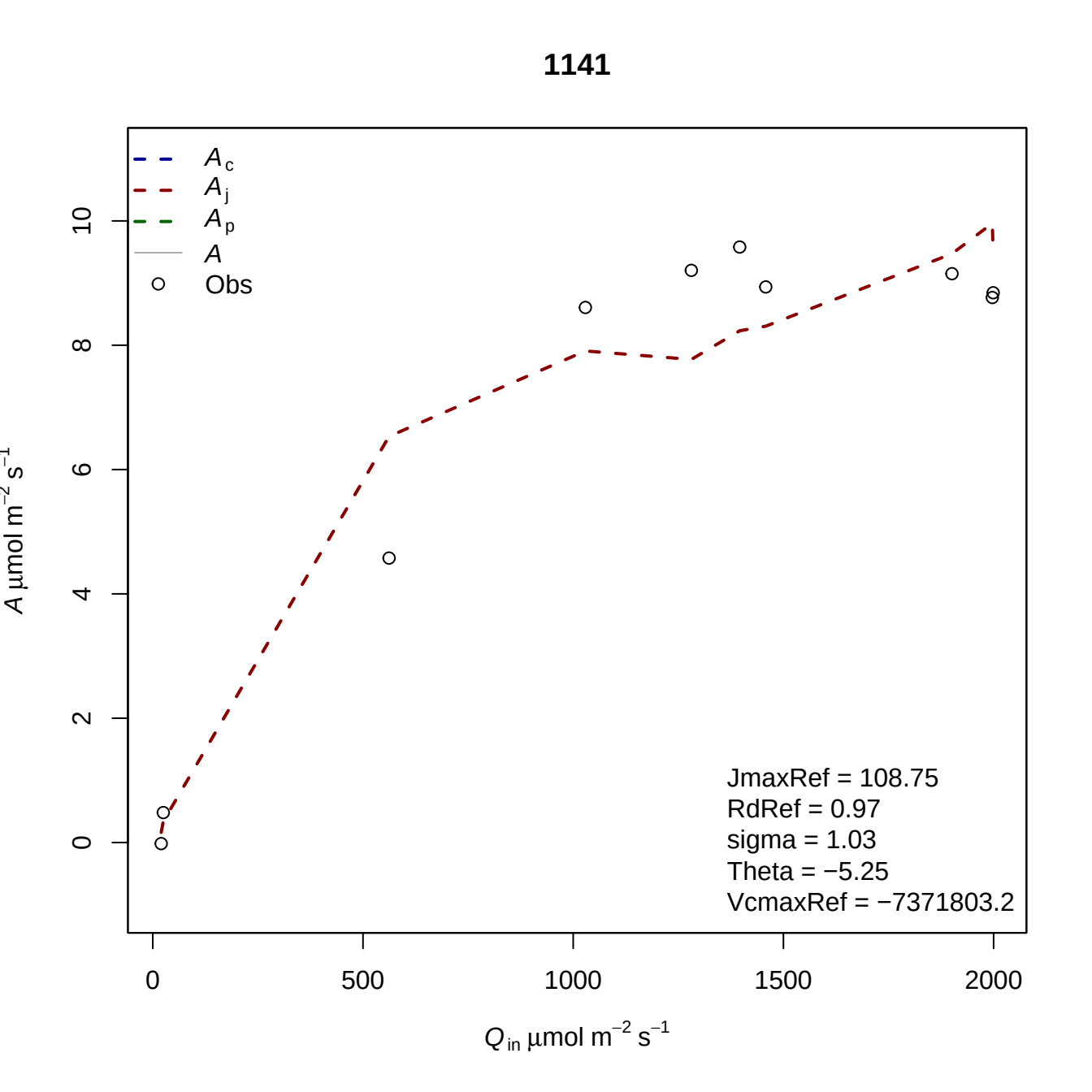

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 253.67
RdRef = 3.36
sigma = 1.88
Theta = −5.11
VcmaxRef = 441573.98
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
2500
Q_{in} µmol m^{-2} s^{-1}

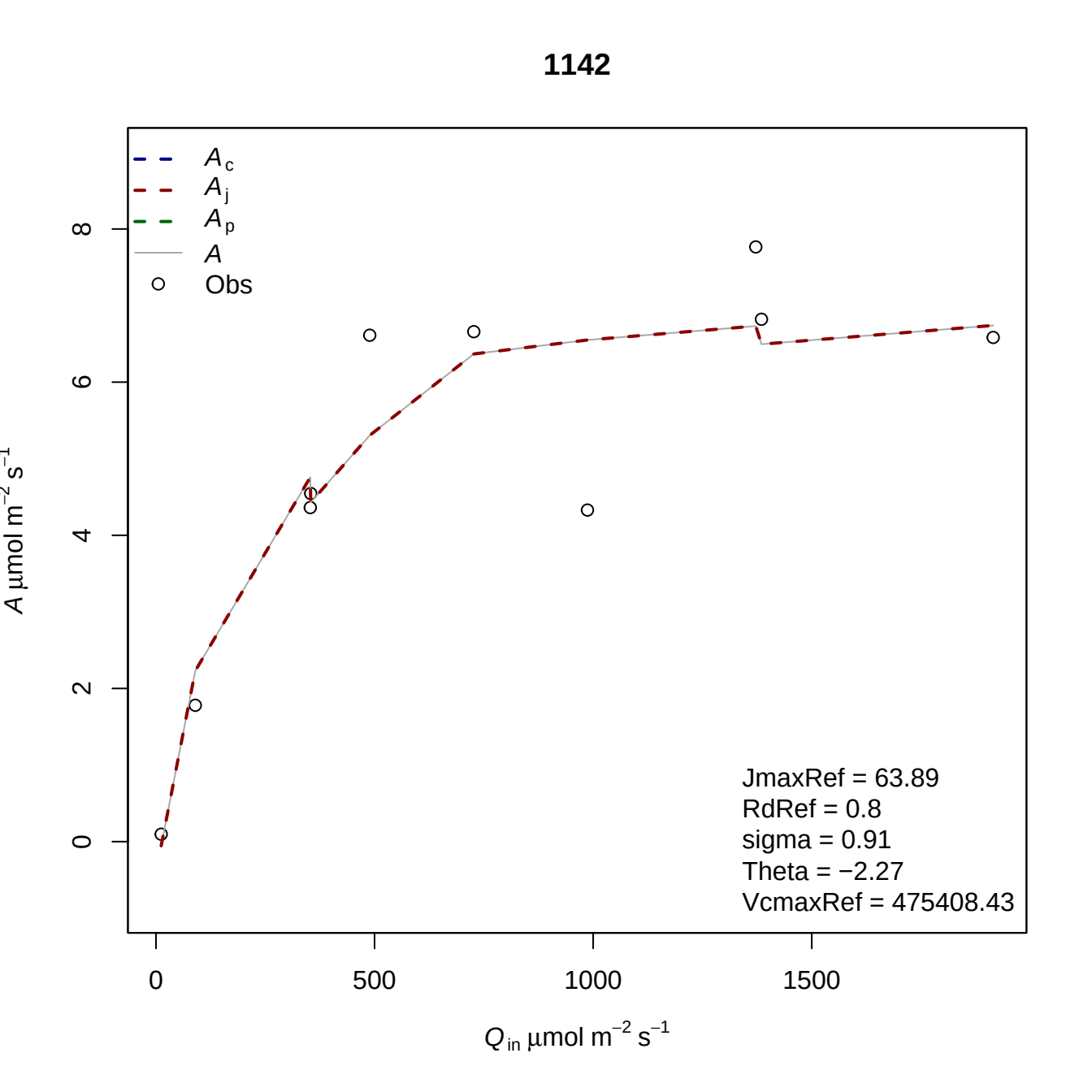

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 41.05
RdRef = 0.29
sigma = 0.45
Theta = 0.21
VcmaxRef = 103295.85
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

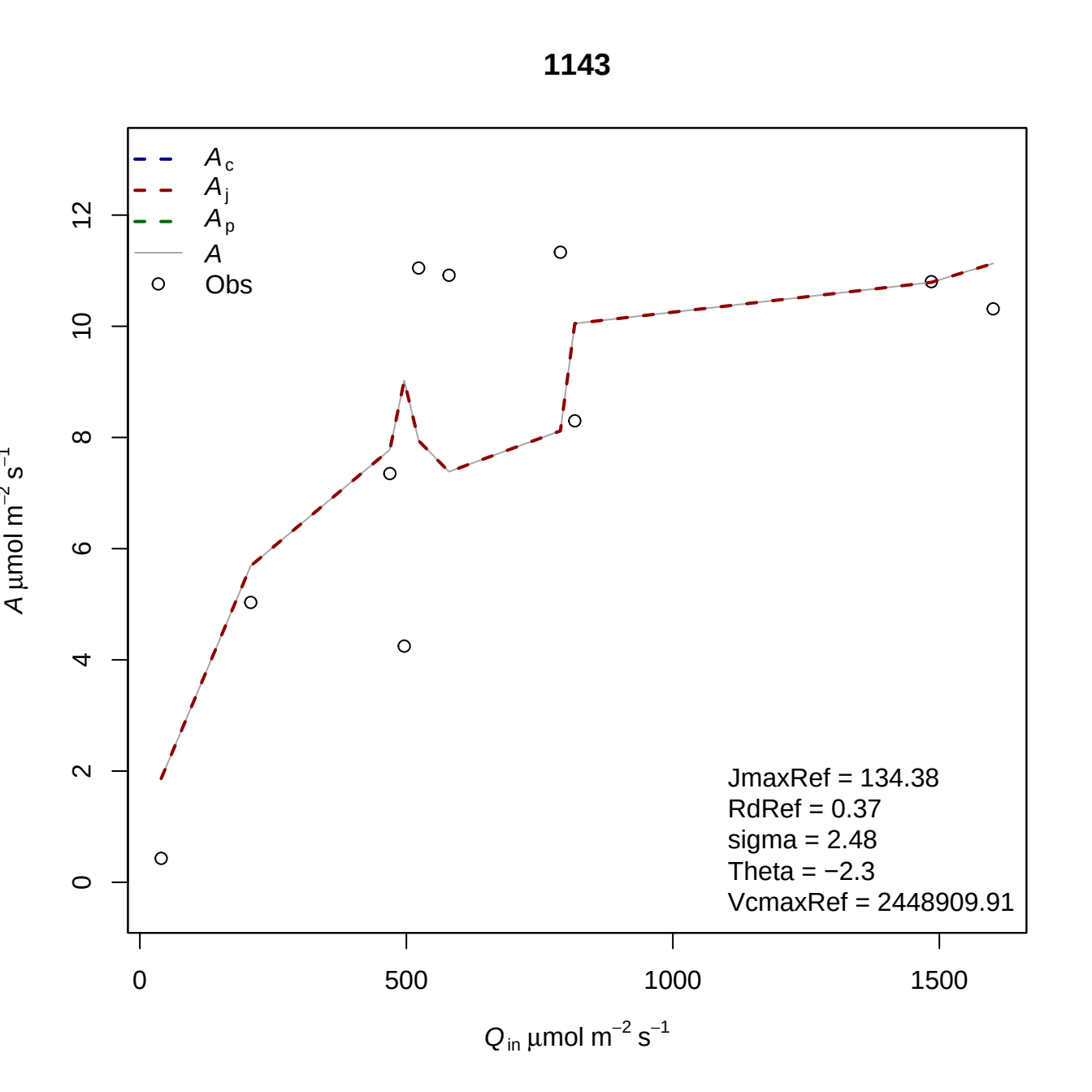

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 31.15
RdRef = 3.4
sigma = 0.41
Theta = 1
VcmaxRef = 7272976.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
−3
0
200
400
600
800
1000
1200
1400

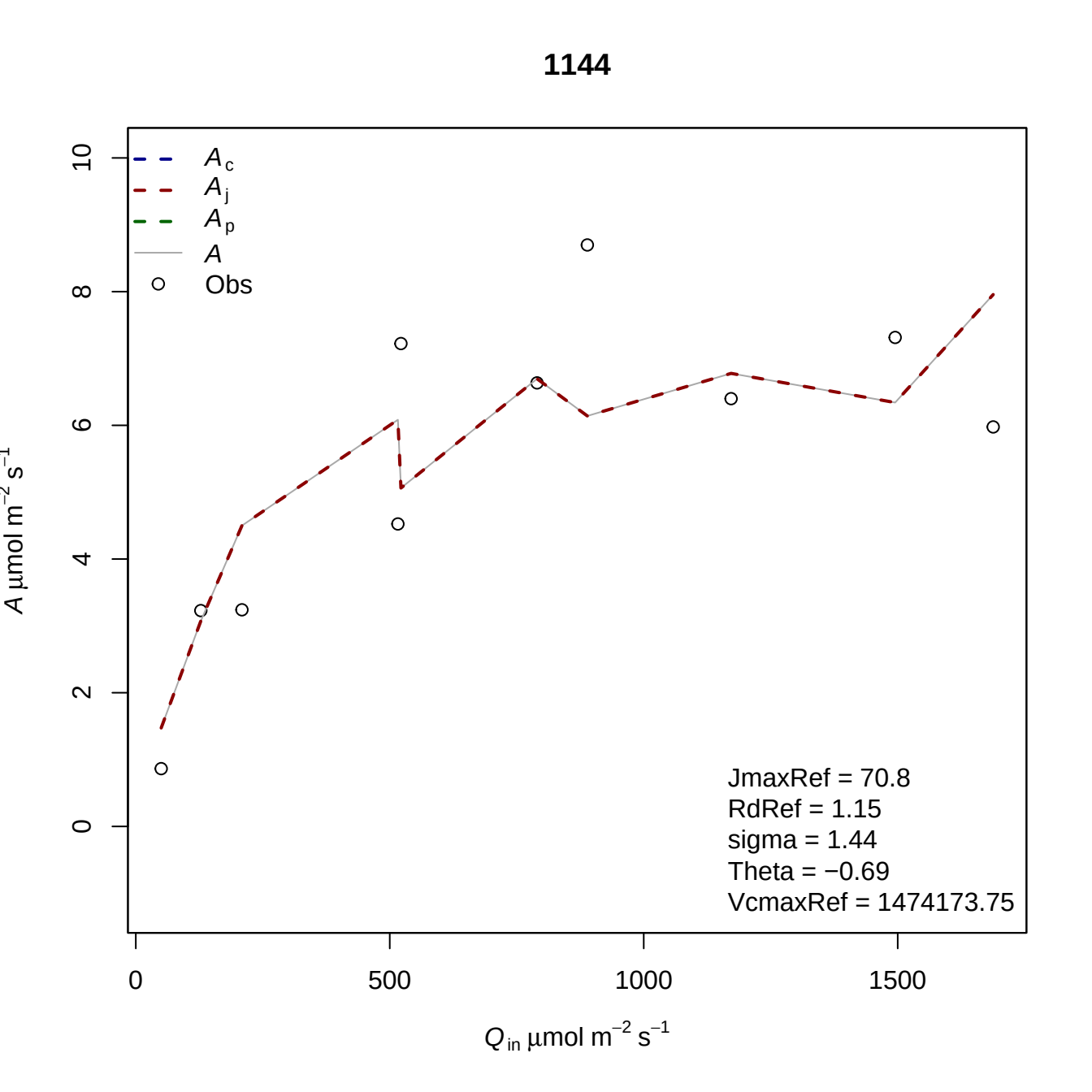

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 190730750.77
RdRef = 1.57
sigma = 0.66
Theta = −14074443.08
VcmaxRef = 2743121436.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
500
1000
1500

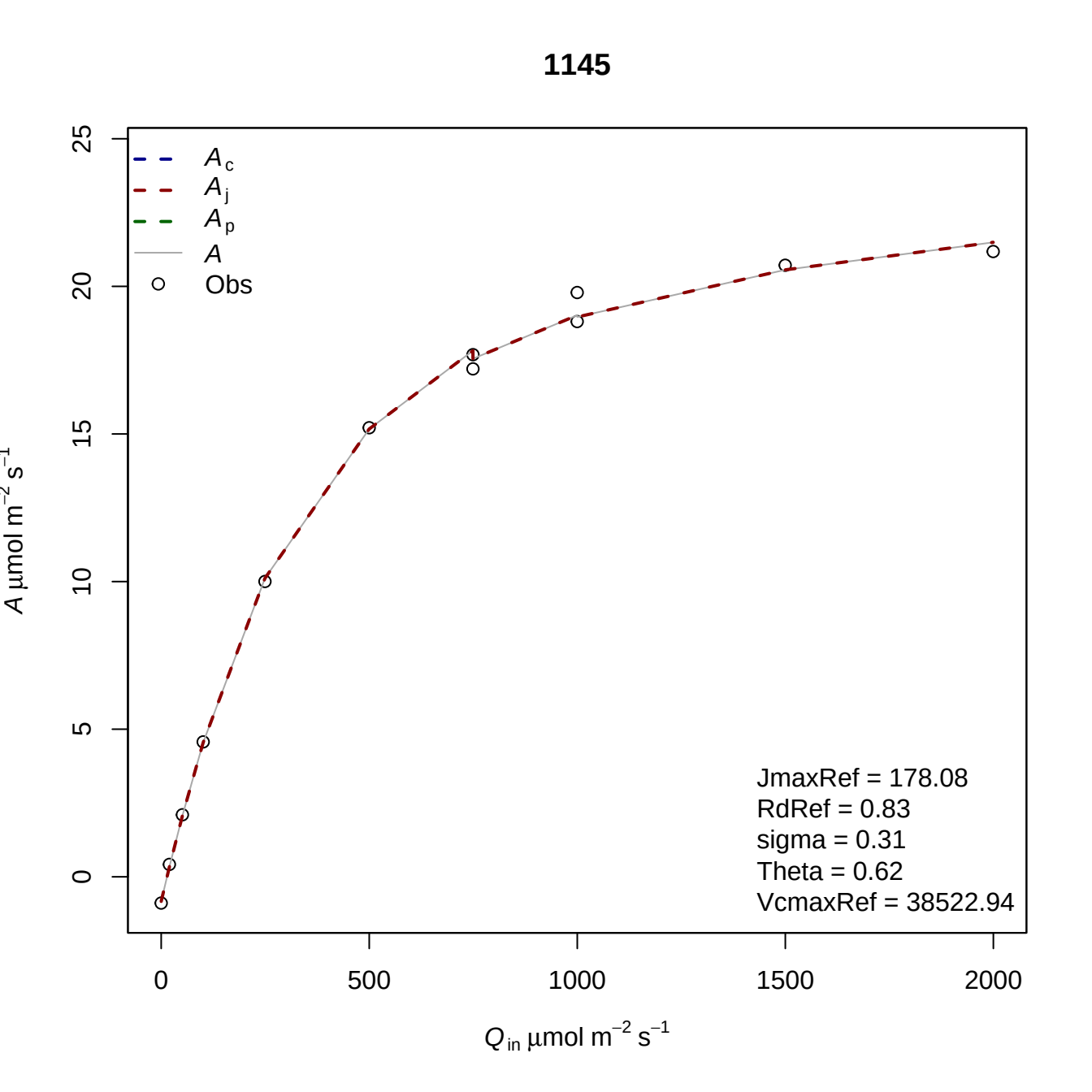

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 281.16
RdRef = 1.1
sigma = 2.21
Theta = −2.91
VcmaxRef = 1443779.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

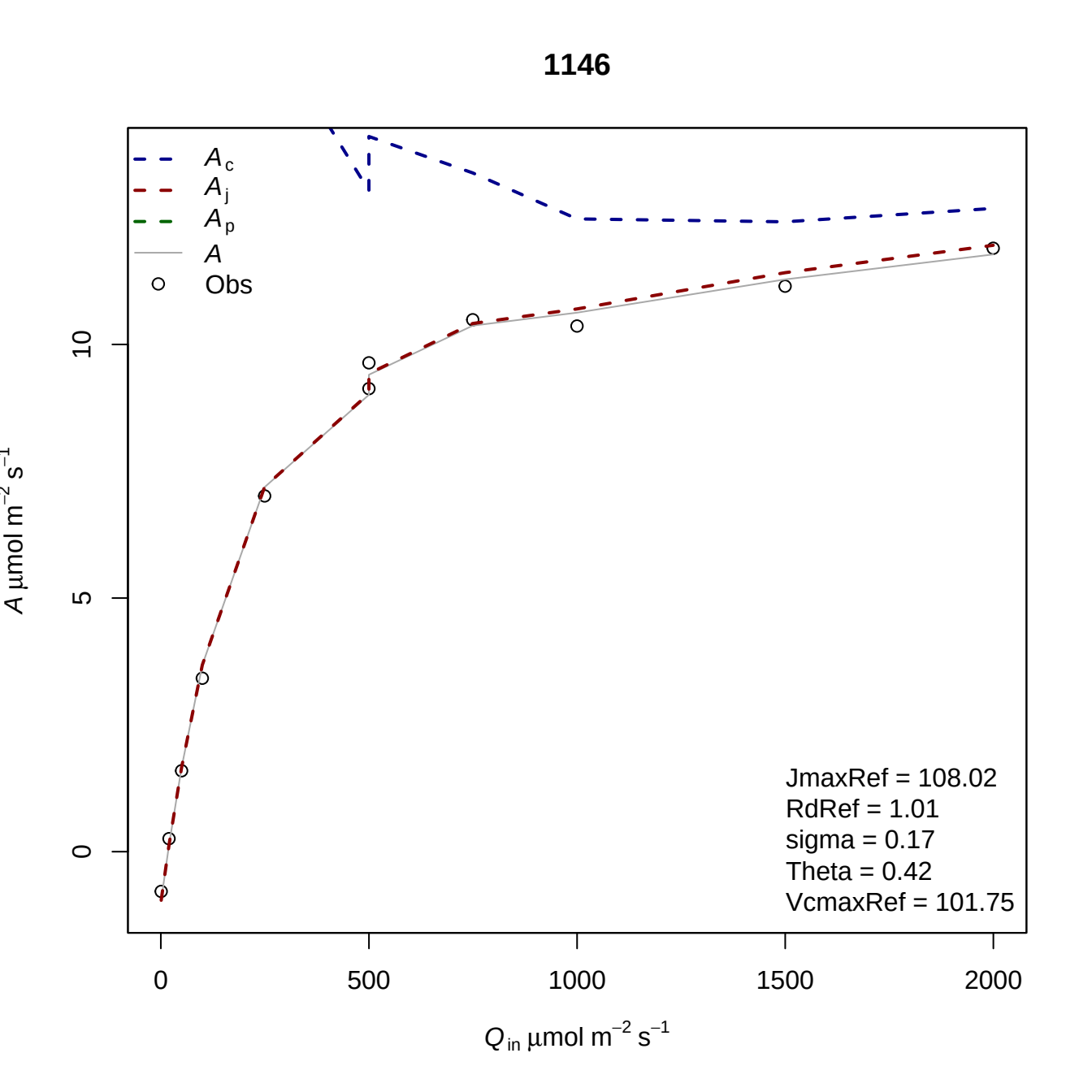

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 108.75
RdRef = 0.97
sigma = 1.03
Theta = −5.25
VcmaxRef = −7371803.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

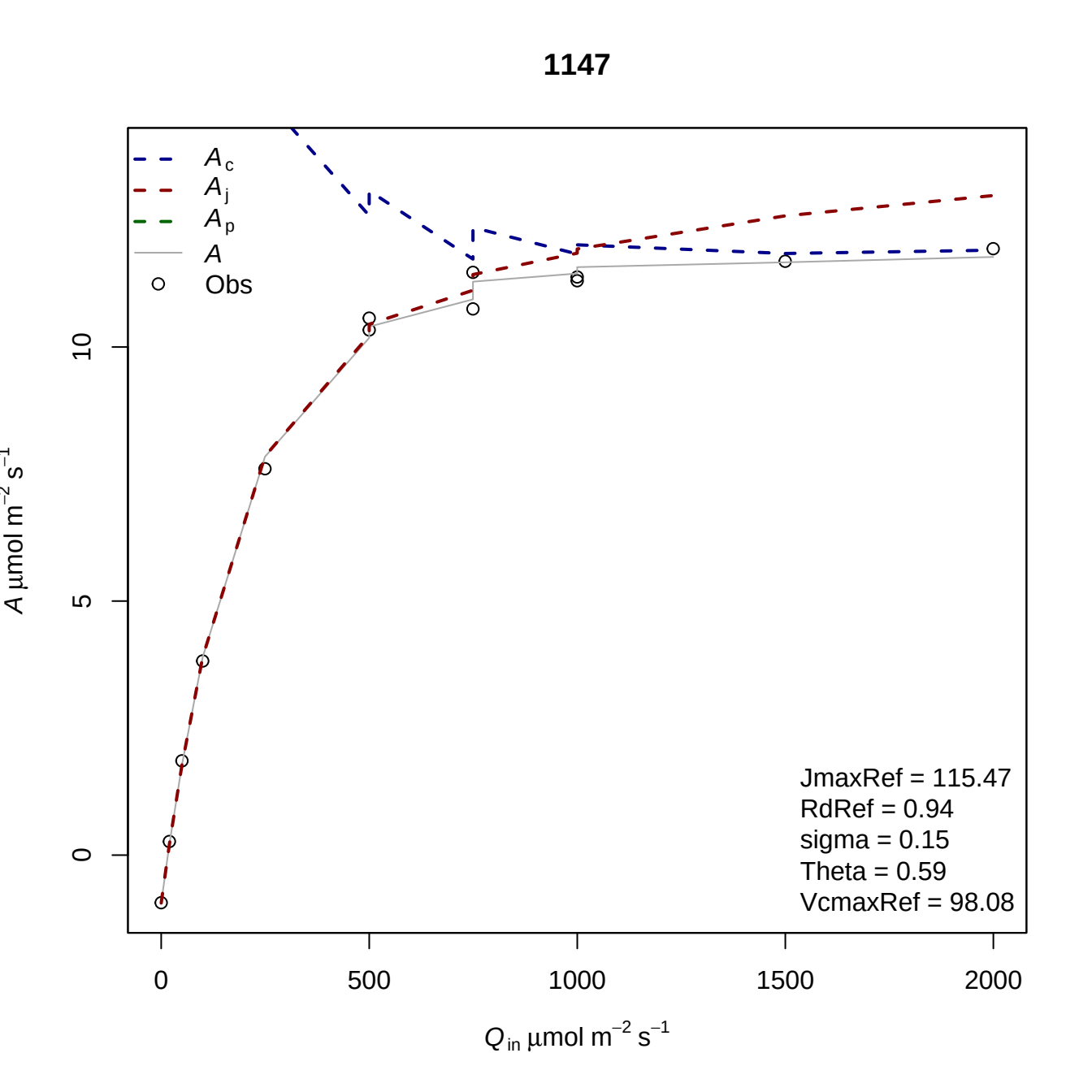

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 63.89
RdRef = 0.8
sigma = 0.91
Theta = −2.27
VcmaxRef = 475408.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

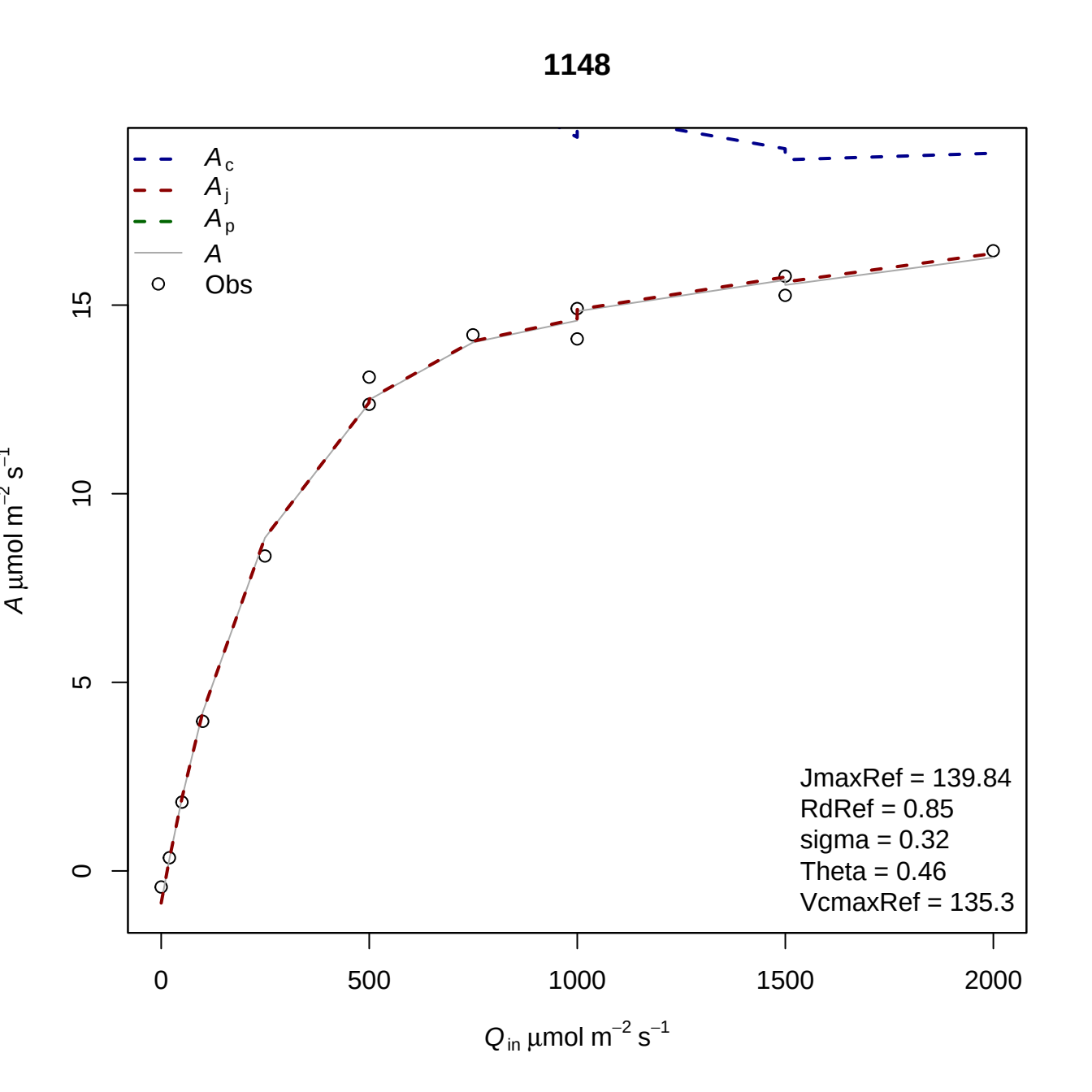

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 134.38
RdRef = 0.37
sigma = 2.48
Theta = −2.3
VcmaxRef = 2448909.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

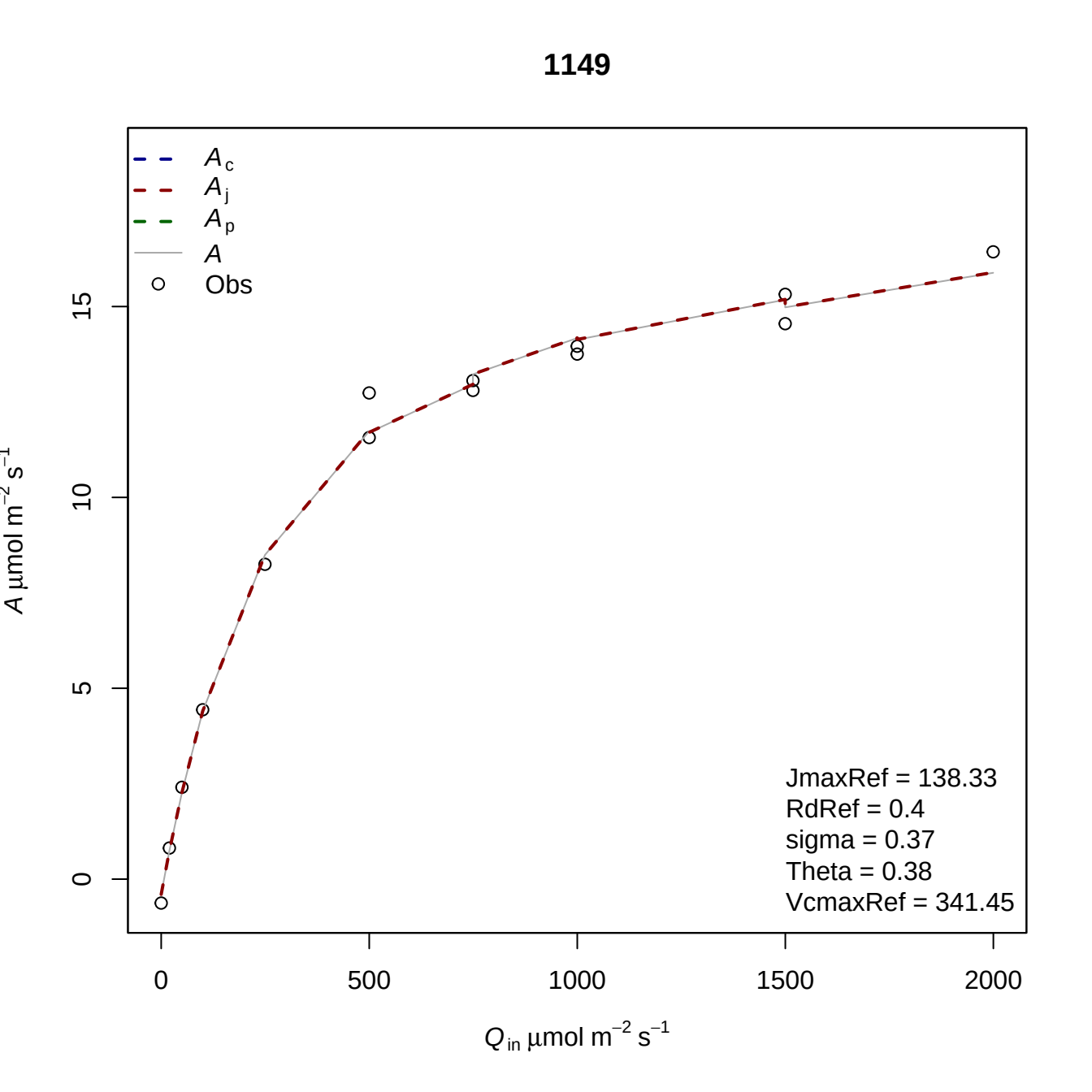

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 70.8
RdRef = 1.15
sigma = 1.44
Theta = −0.69
VcmaxRef = 1474173.75
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

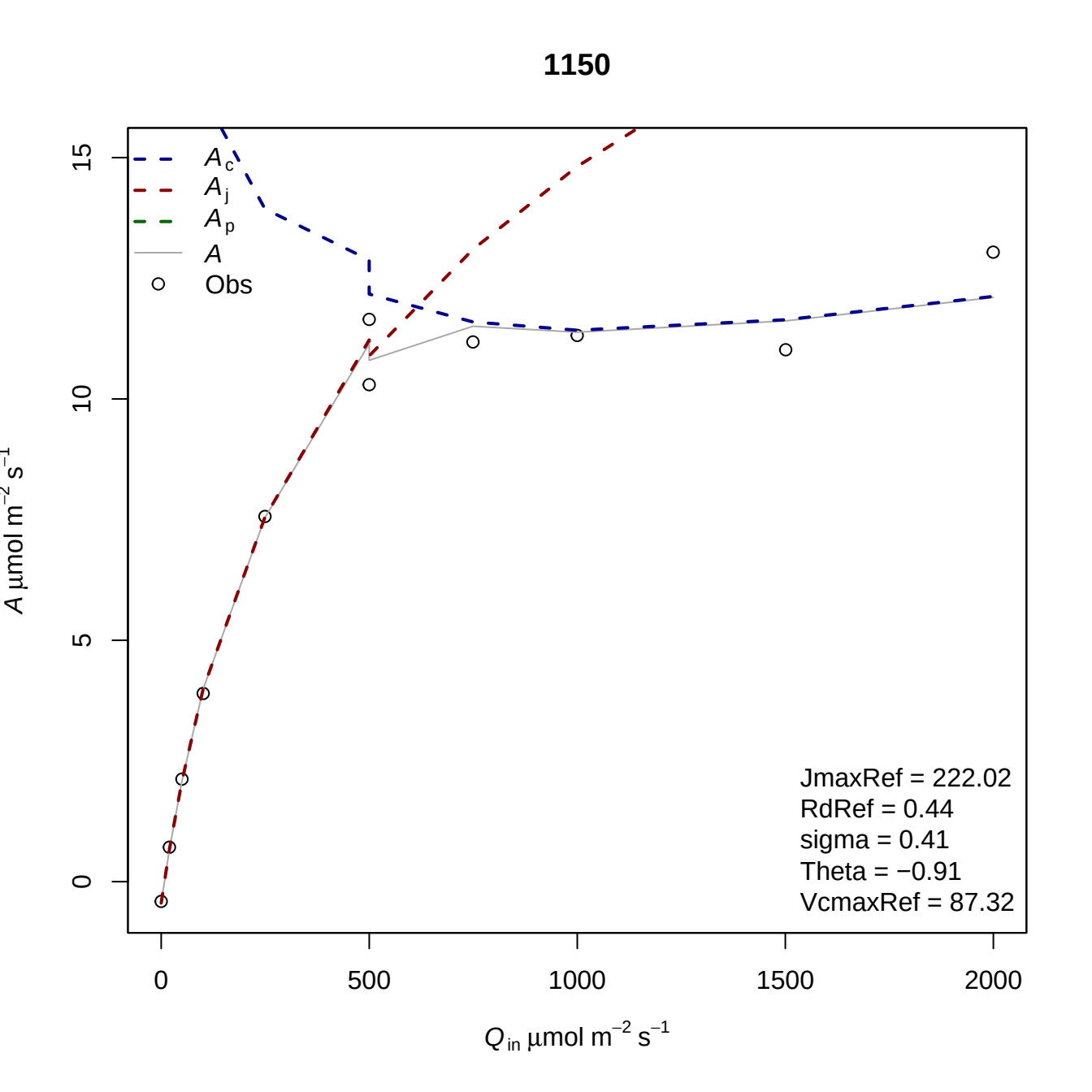

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 178.08
RdRef = 0.83
sigma = 0.31
Theta = 0.62
VcmaxRef = 38522.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

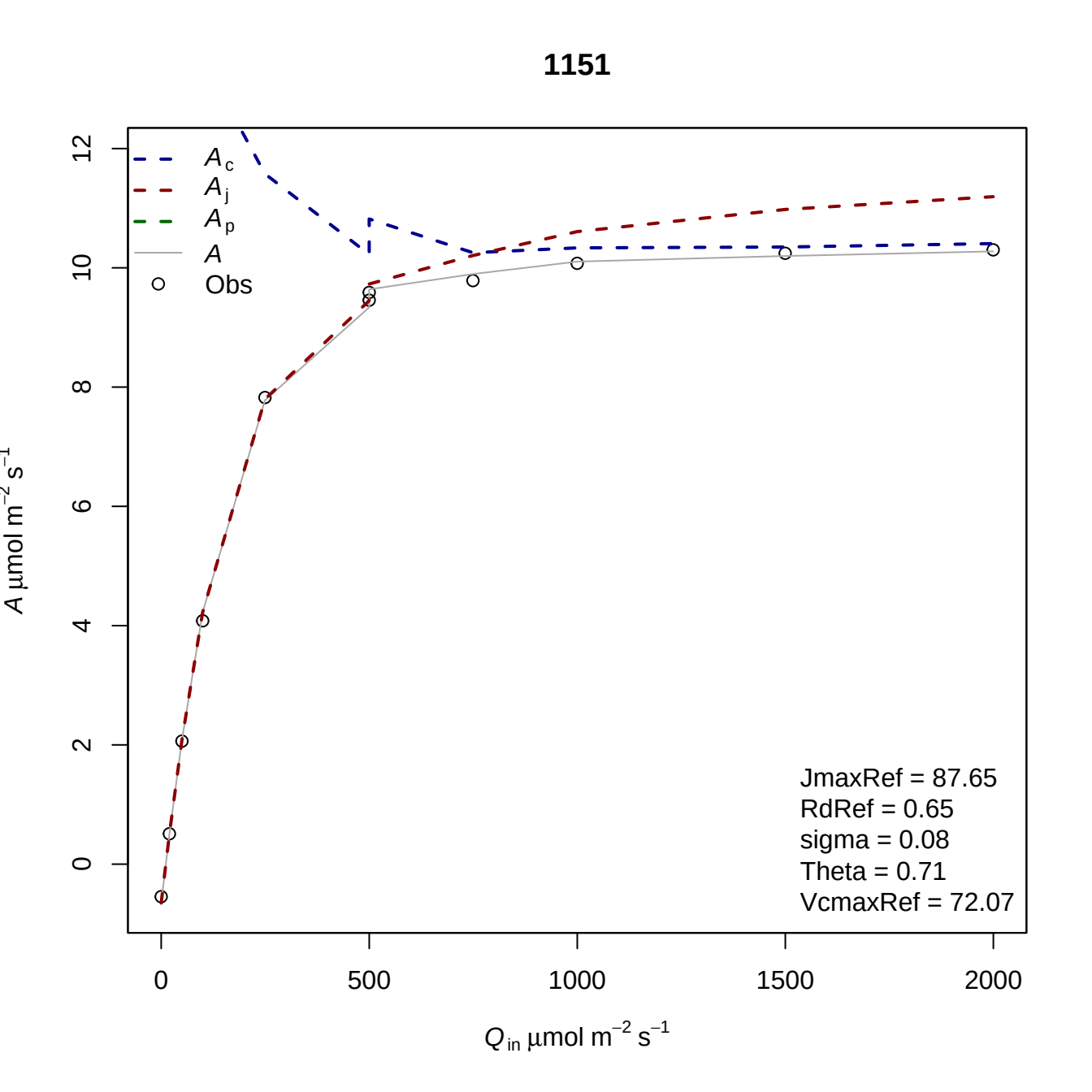

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 108.02
RdRef = 1.01
sigma = 0.17
Theta = 0.42
VcmaxRef = 101.75
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

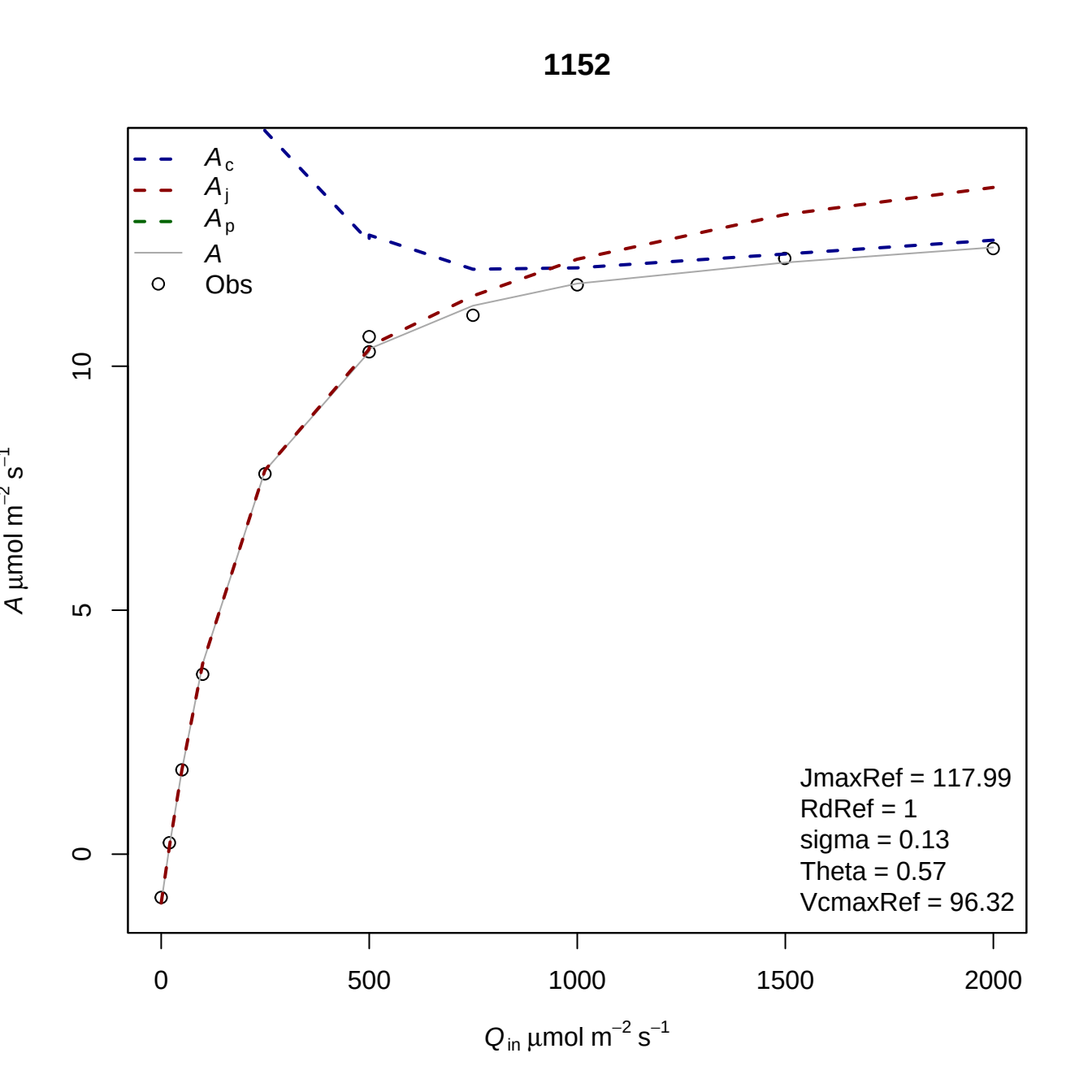

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 115.47
RdRef = 0.94
sigma = 0.15
Theta = 0.59
VcmaxRef = 98.08
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

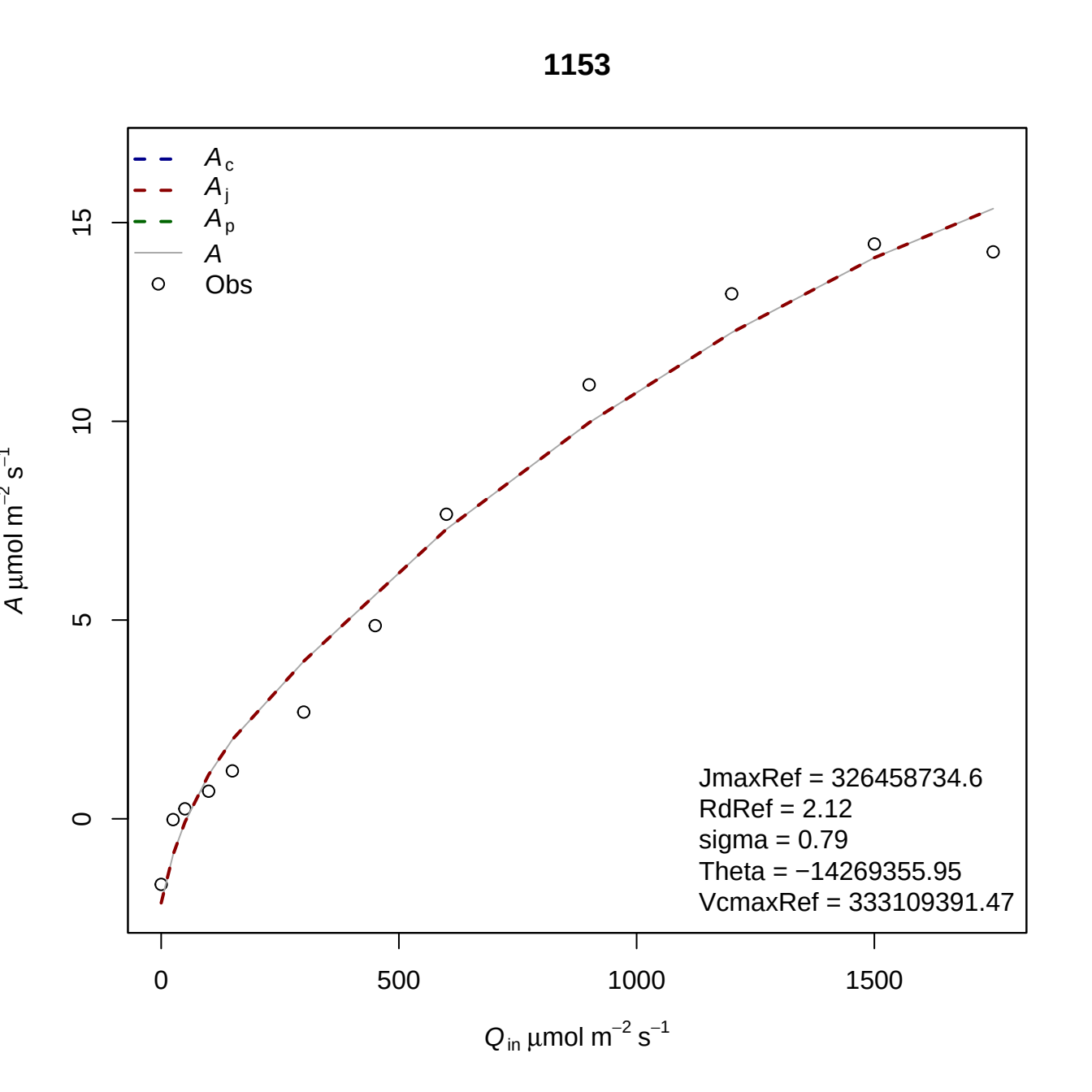

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 139.84
RdRef = 0.85
sigma = 0.32
Theta = 0.46
VcmaxRef = 135.3
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

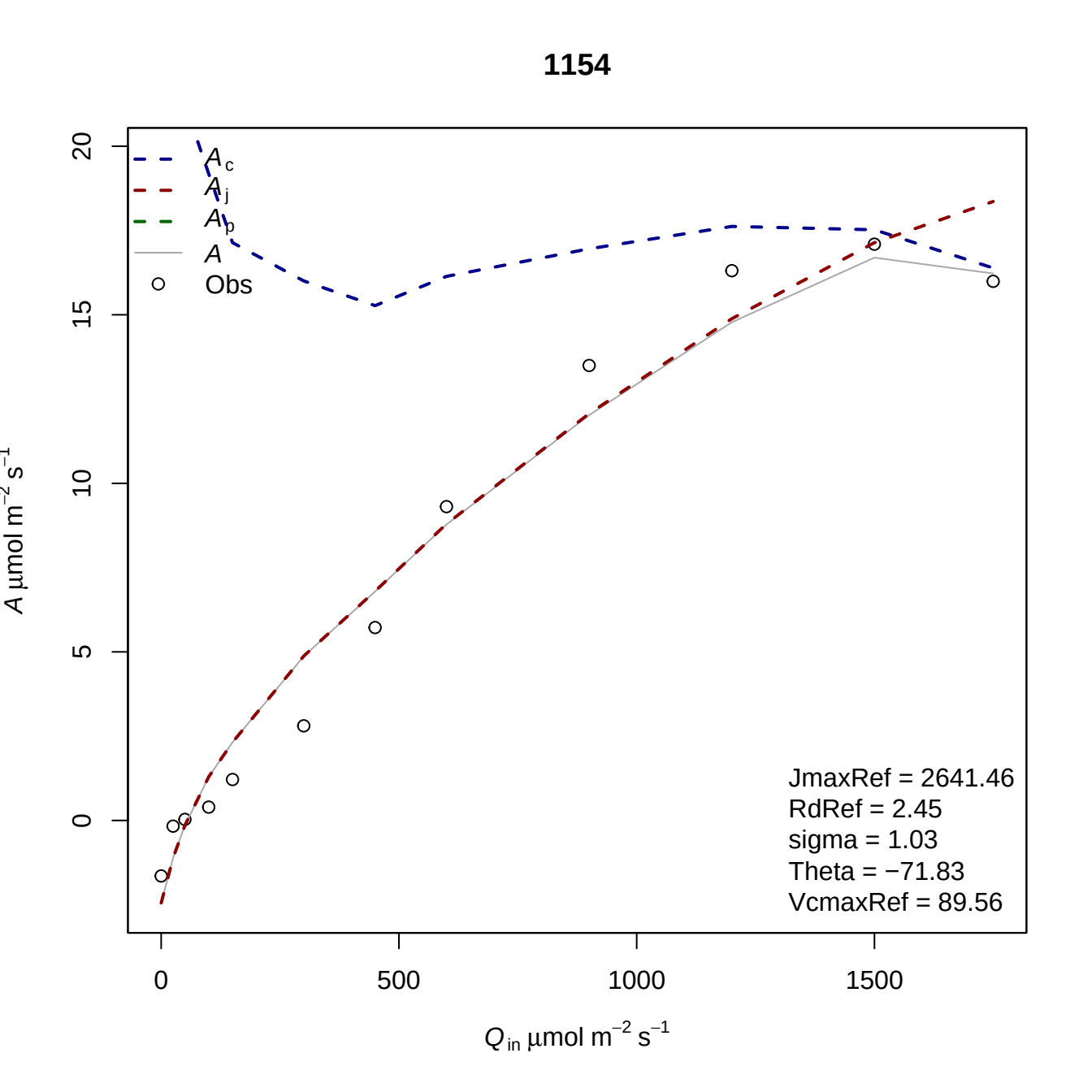

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 138.33
RdRef = 0.4
sigma = 0.37
Theta = 0.38
VcmaxRef = 341.45
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

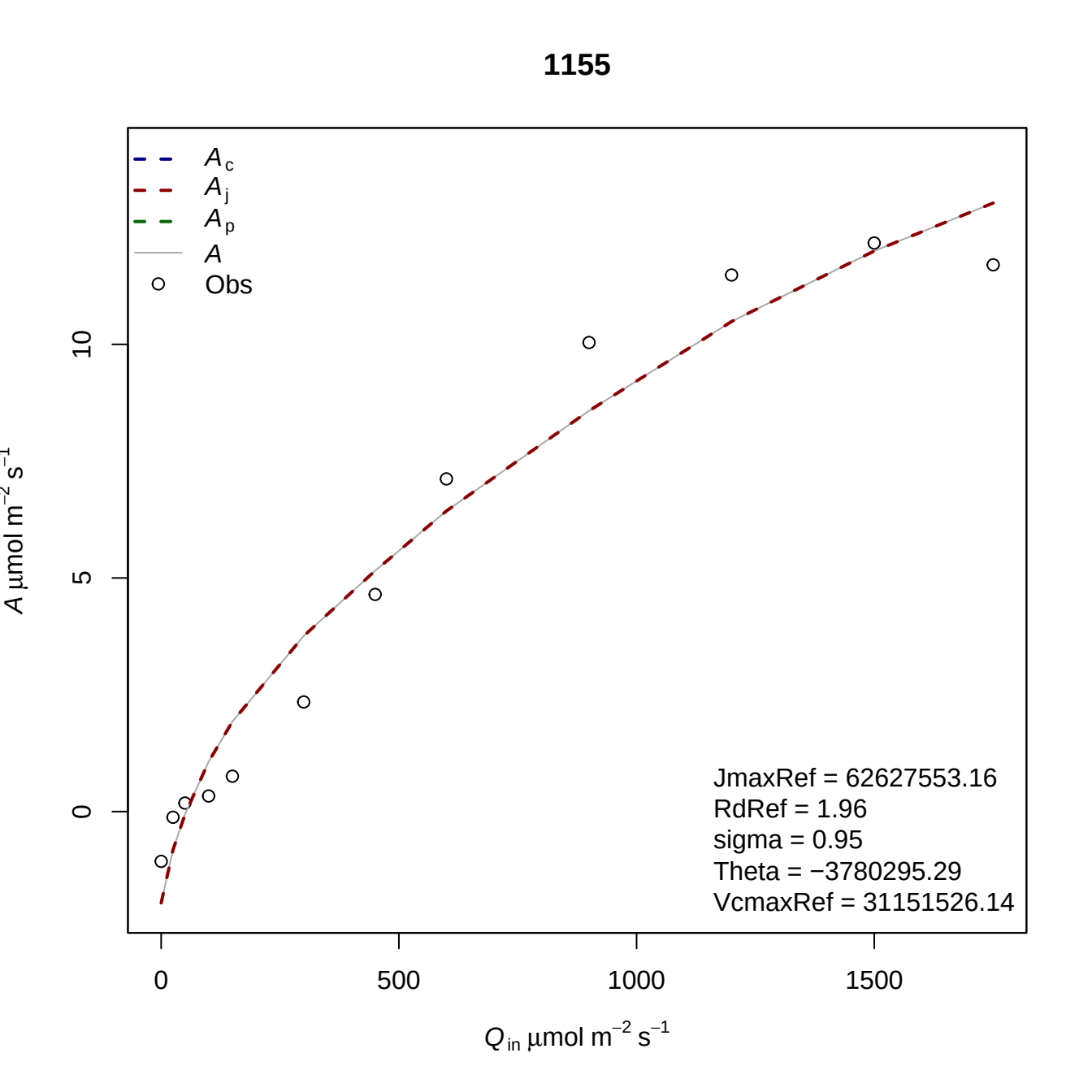

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 222.02
RdRef = 0.44
sigma = 0.41
Theta = −0.91
VcmaxRef = 87.32
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

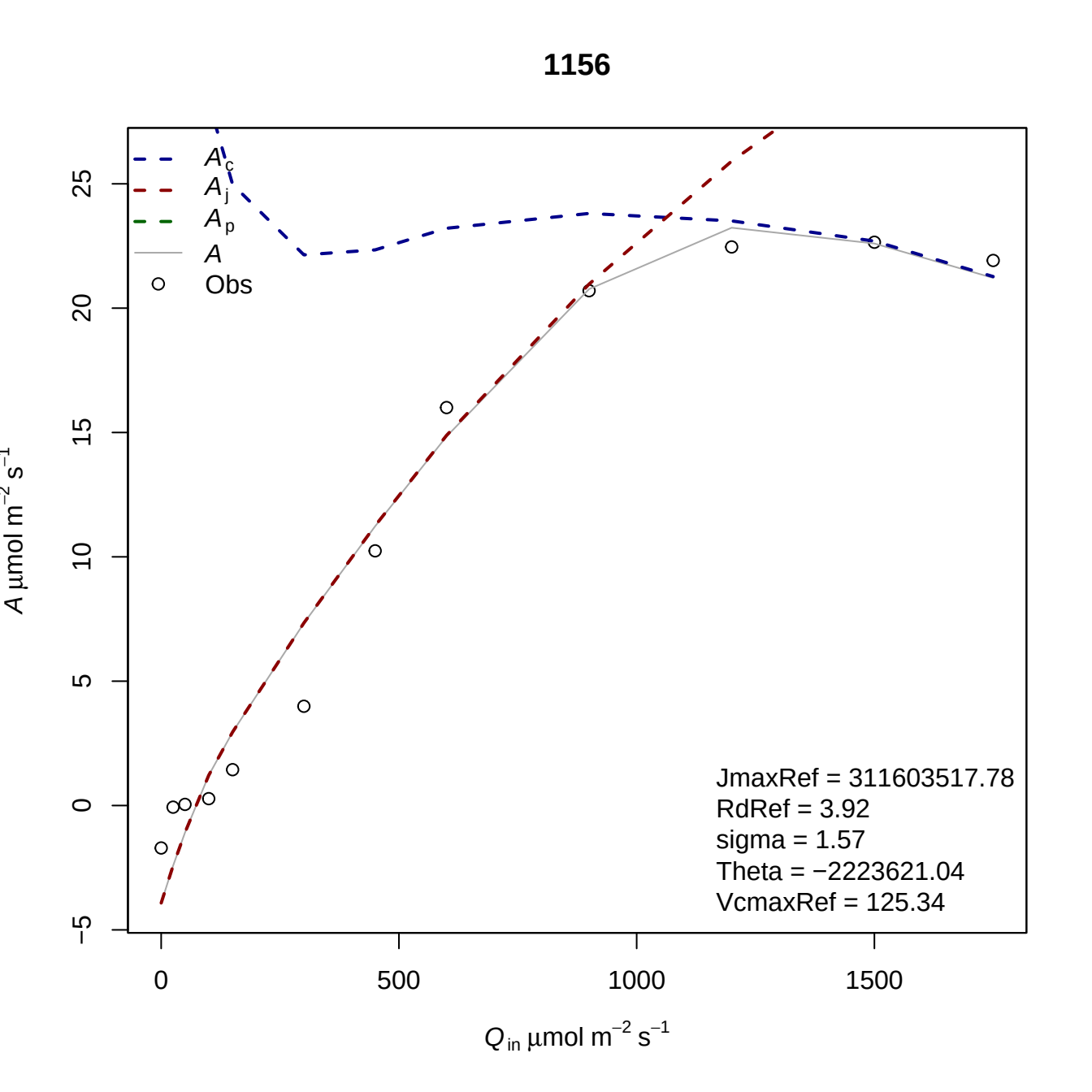

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 87.65
RdRef = 0.65
sigma = 0.08
Theta = 0.71
VcmaxRef = 72.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

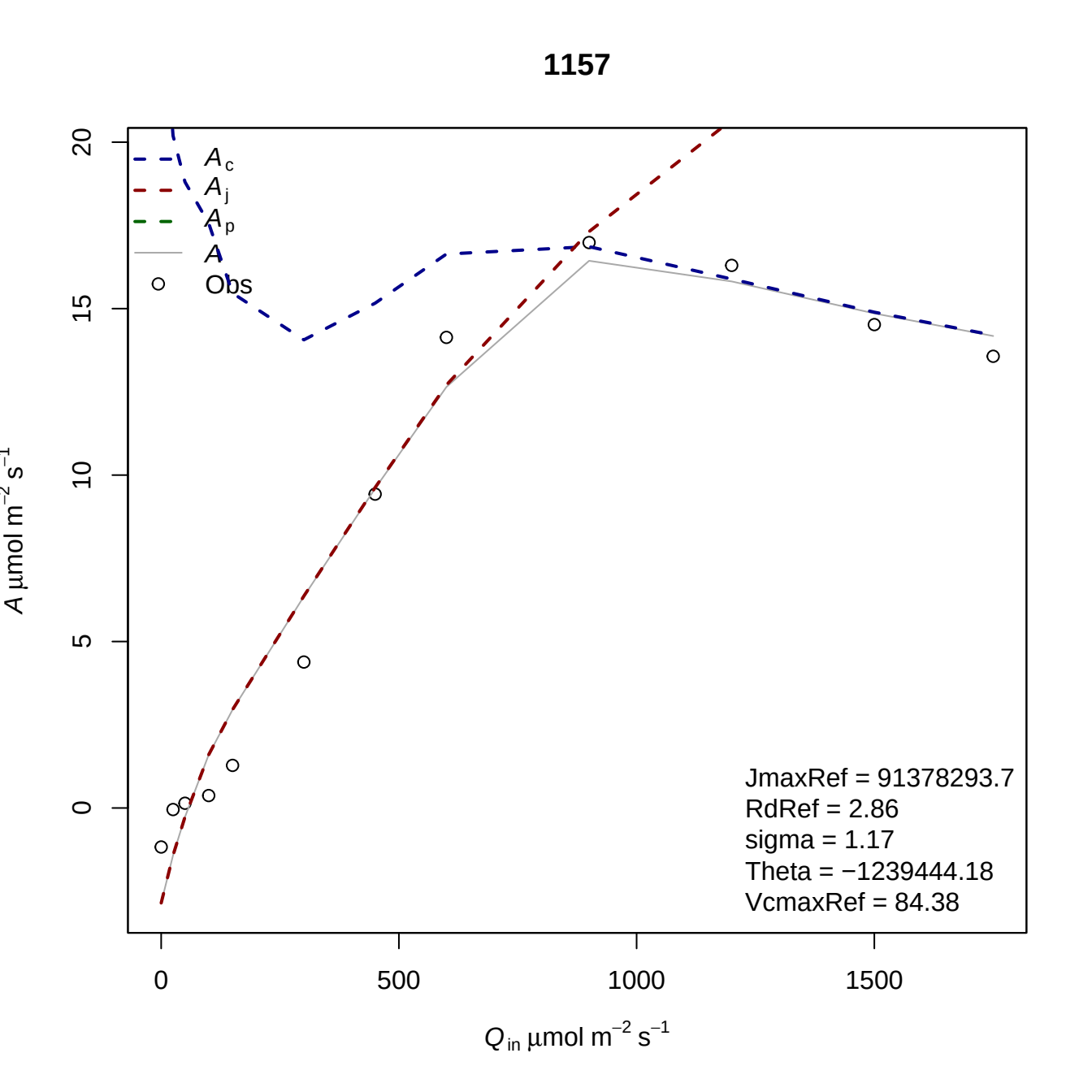

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 117.99
RdRef = 1
sigma = 0.13
Theta = 0.57
VcmaxRef = 96.32
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

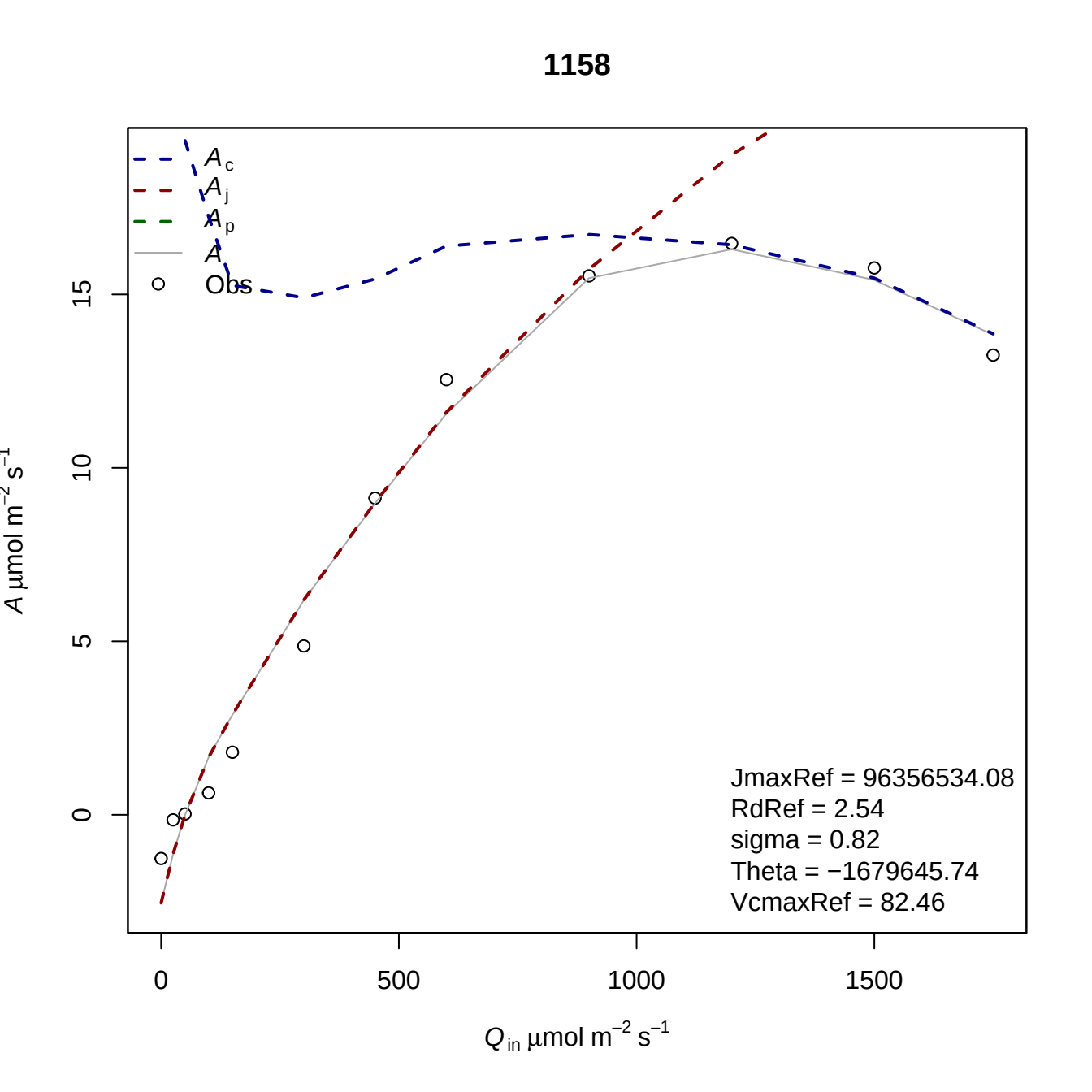

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 326458734.6
RdRef = 2.12
sigma = 0.79
Theta = −14269355.95
VcmaxRef = 333109391.47
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

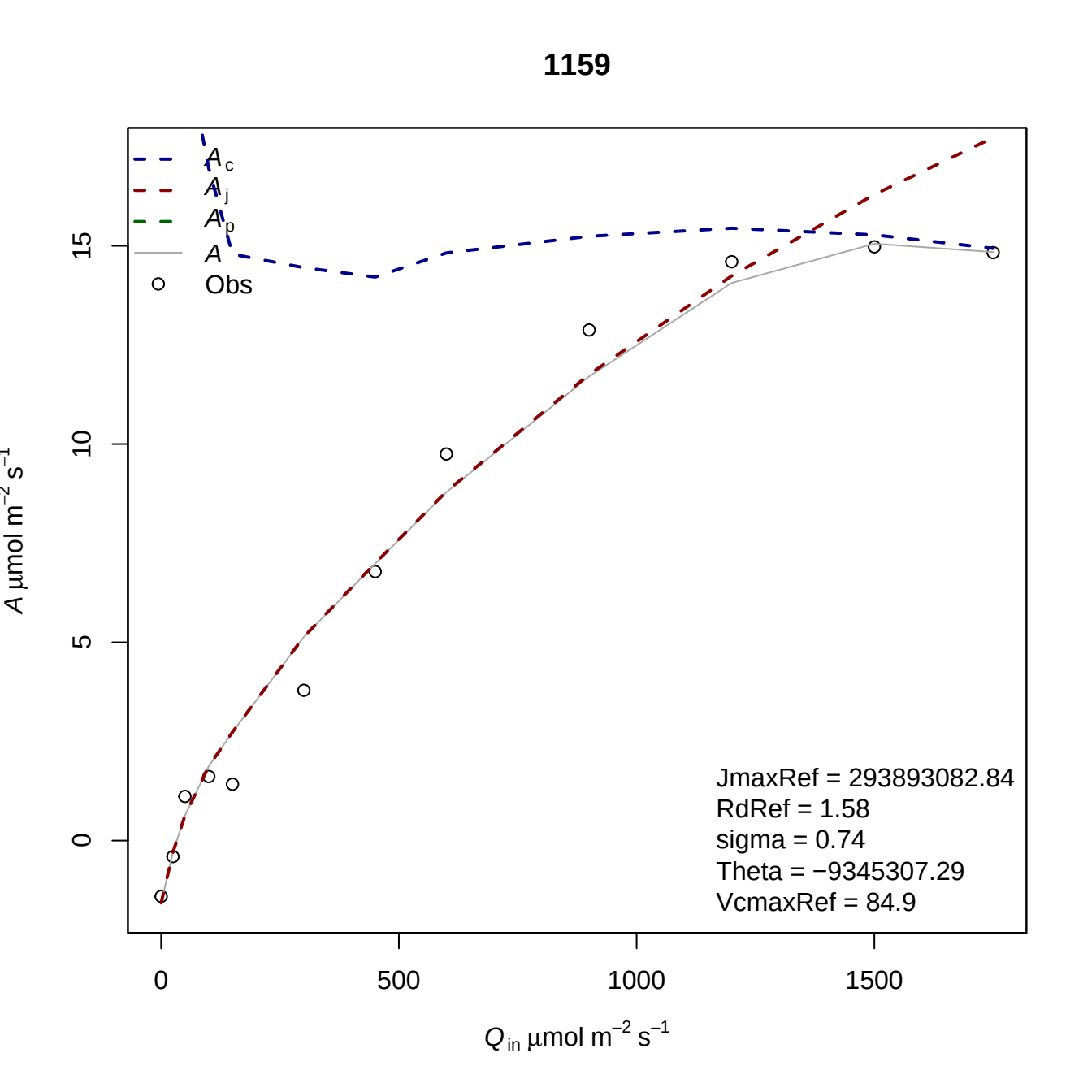

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 2641.46
RdRef = 2.45
sigma = 1.03
Theta = −71.83
VcmaxRef = 89.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

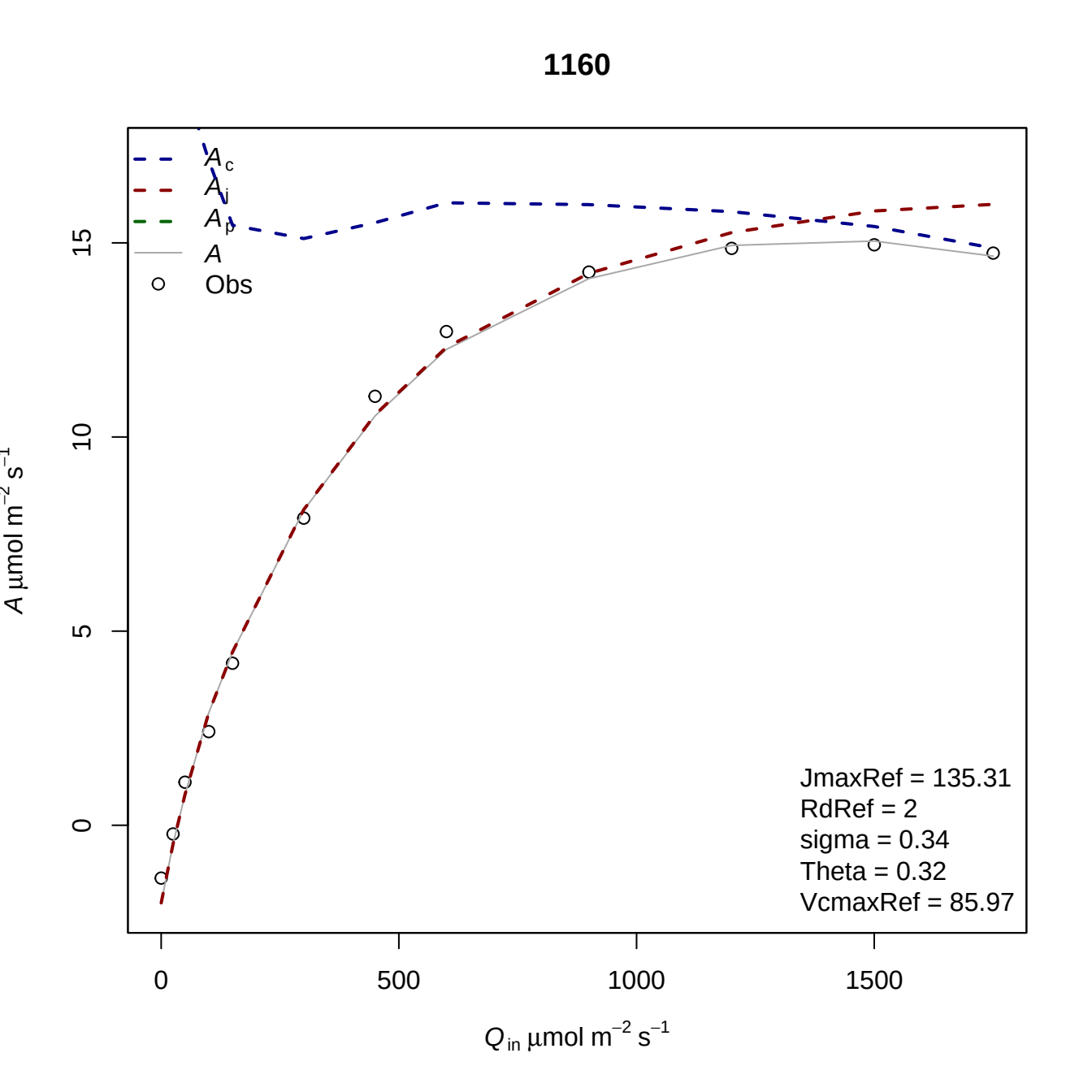

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 62627553.16
RdRef = 1.96
sigma = 0.95
Theta = −3780295.29
VcmaxRef = 31151526.14
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

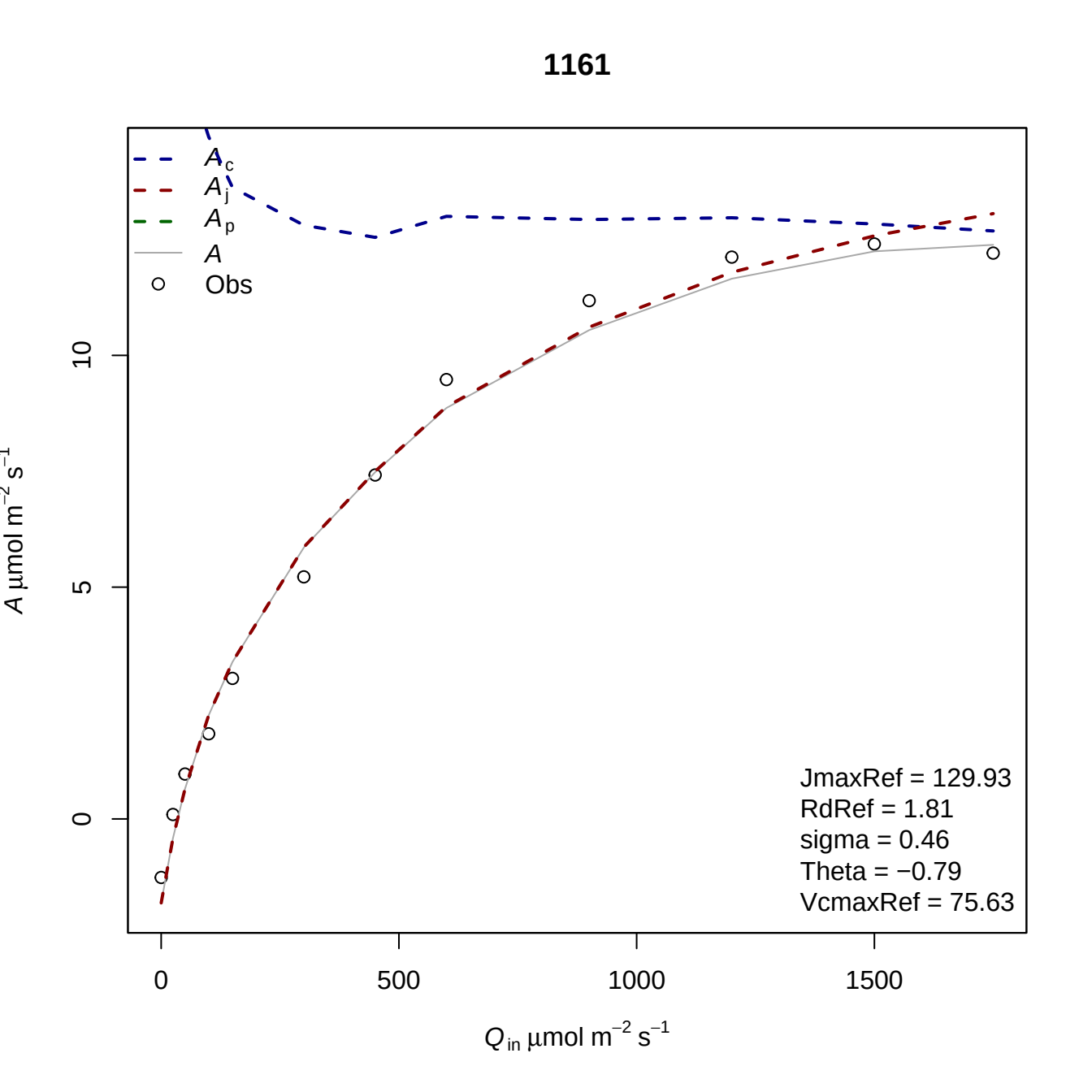

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 311603517.78
RdRef = 3.92
sigma = 1.57
Theta = −2223621.04
VcmaxRef = 125.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

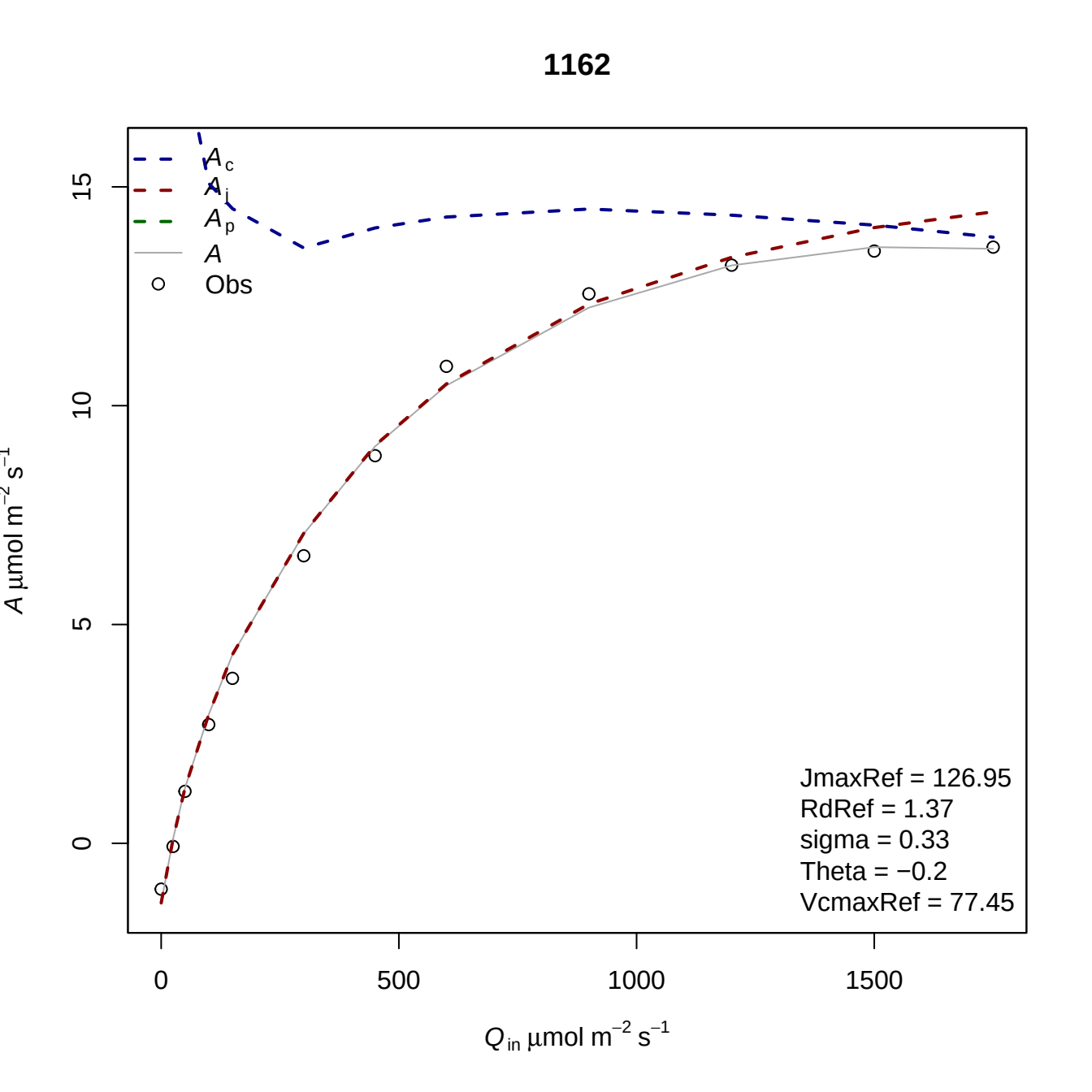

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 91378293.7
RdRef = 2.86
sigma = 1.17
Theta = −1239444.18
VcmaxRef = 84.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

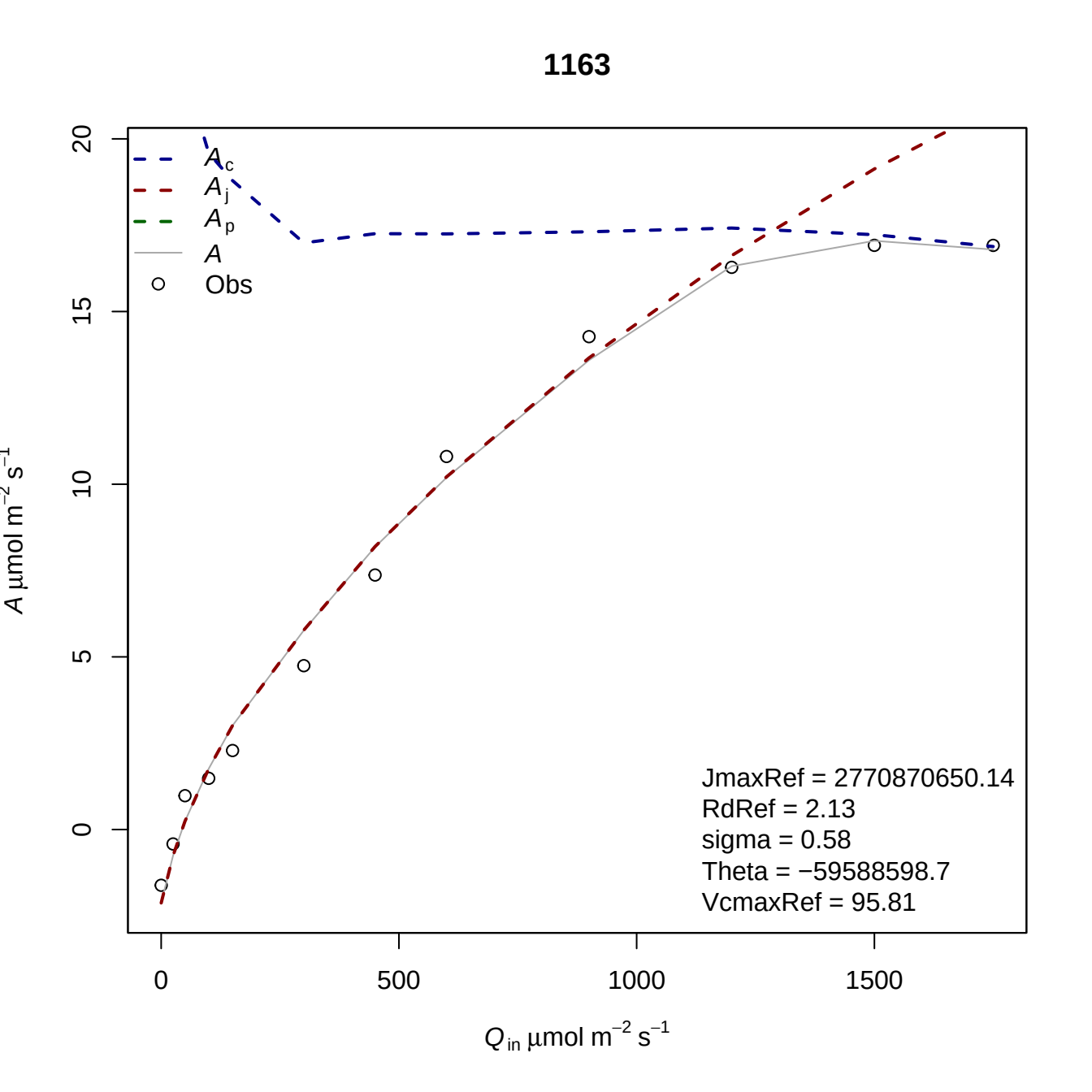

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 96356534.08
RdRef = 2.54
sigma = 0.82
Theta = −1679645.74
VcmaxRef = 82.46
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

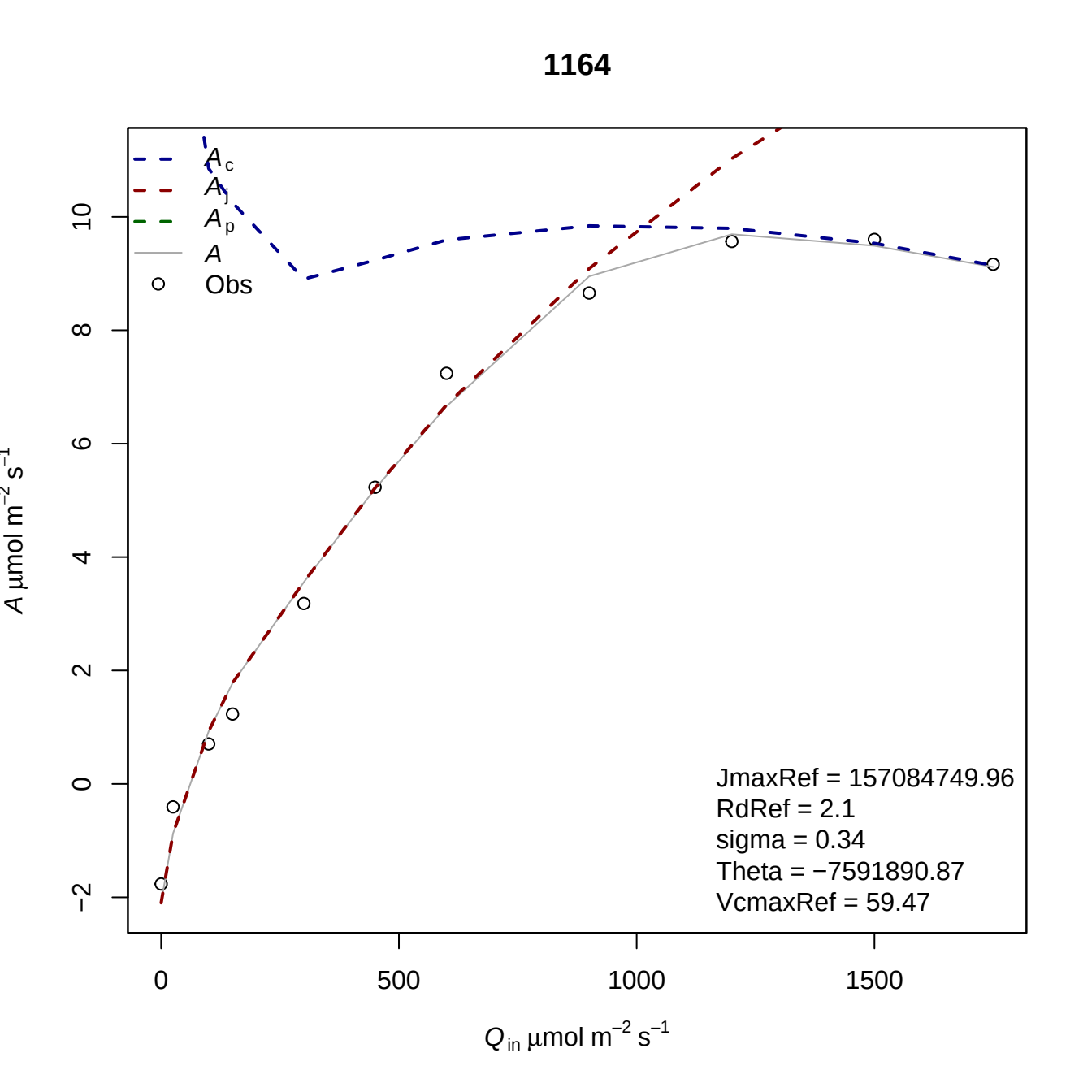

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 293893082.84
RdRef = 1.58
sigma = 0.74
Theta = −9345307.29
VcmaxRef = 84.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

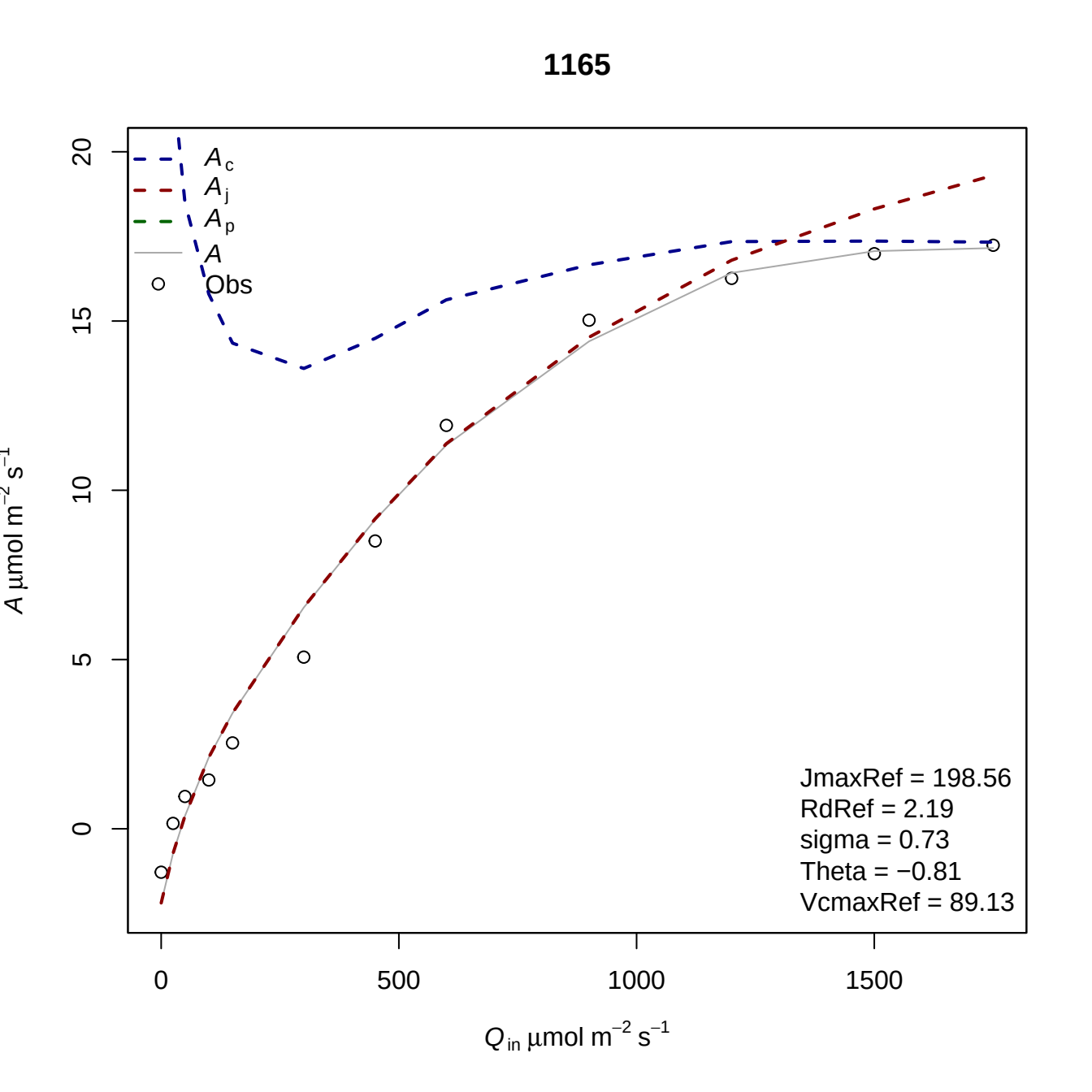

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 135.31
RdRef = 2
sigma = 0.34
Theta = 0.32
VcmaxRef = 85.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

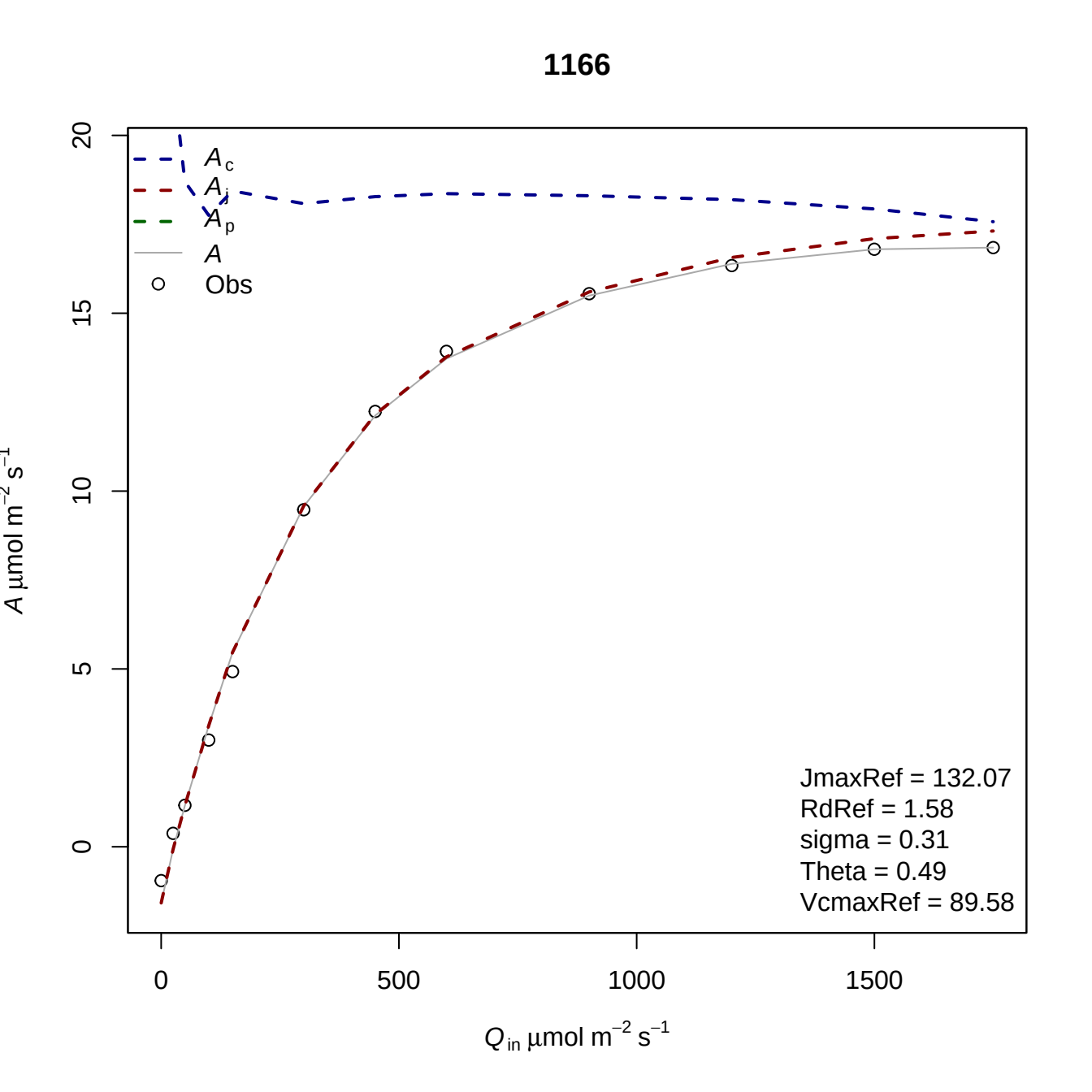

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 129.93
RdRef = 1.81
sigma = 0.46
Theta = −0.79
VcmaxRef = 75.63
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

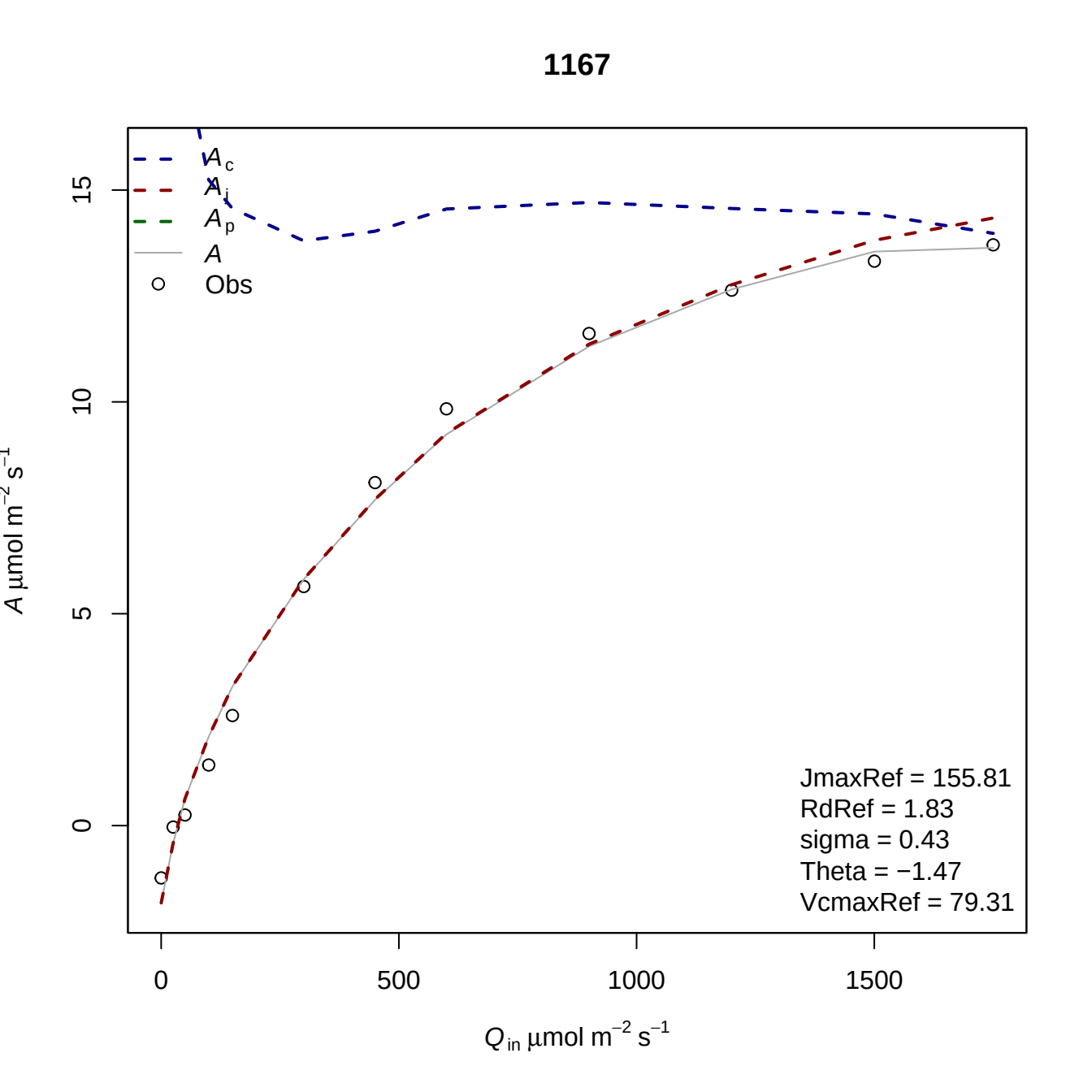

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 126.95
RdRef = 1.37
sigma = 0.33
Theta = −0.2
VcmaxRef = 77.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

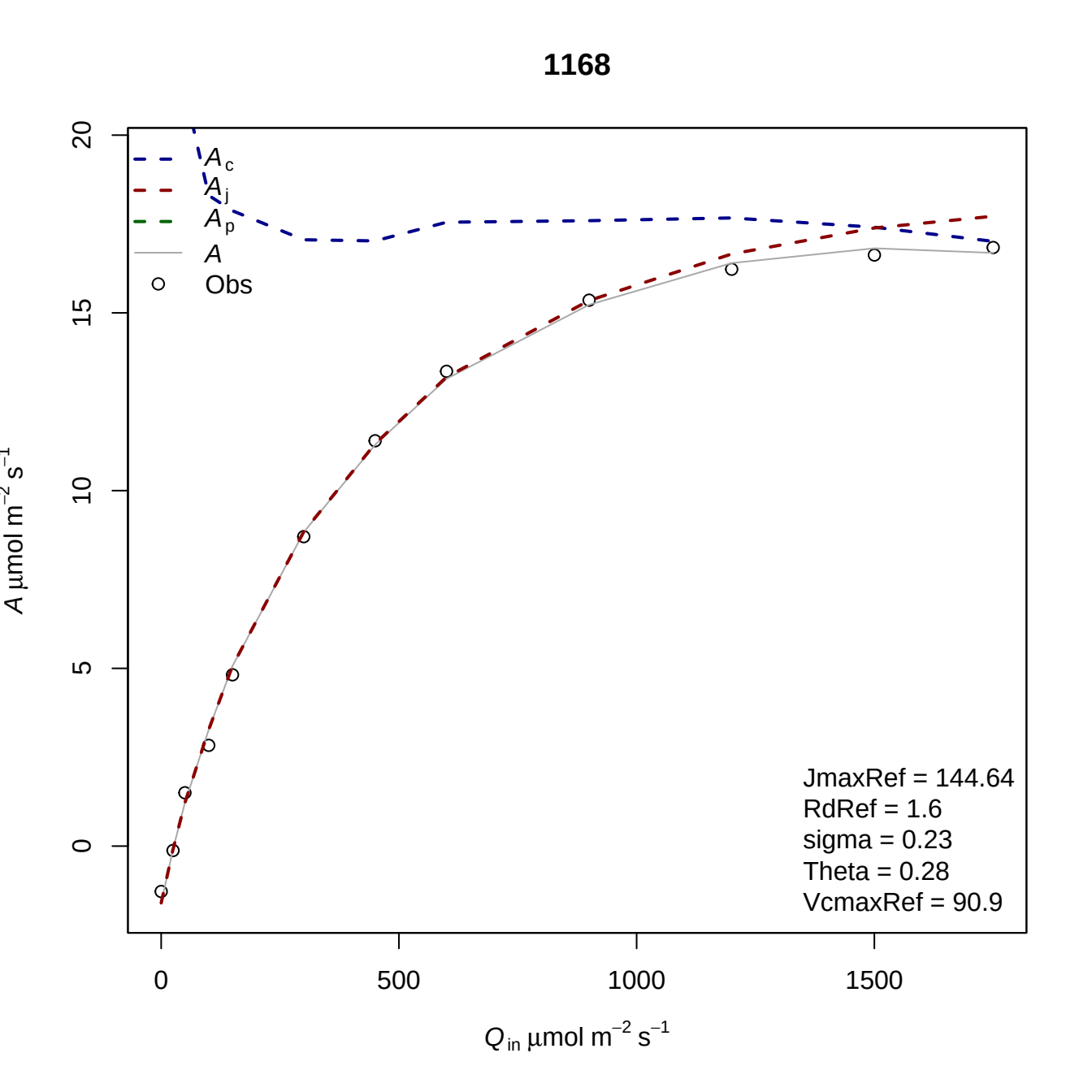

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 2770870650.14
RdRef = 2.13
sigma = 0.58
Theta = −59588598.7
VcmaxRef = 95.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

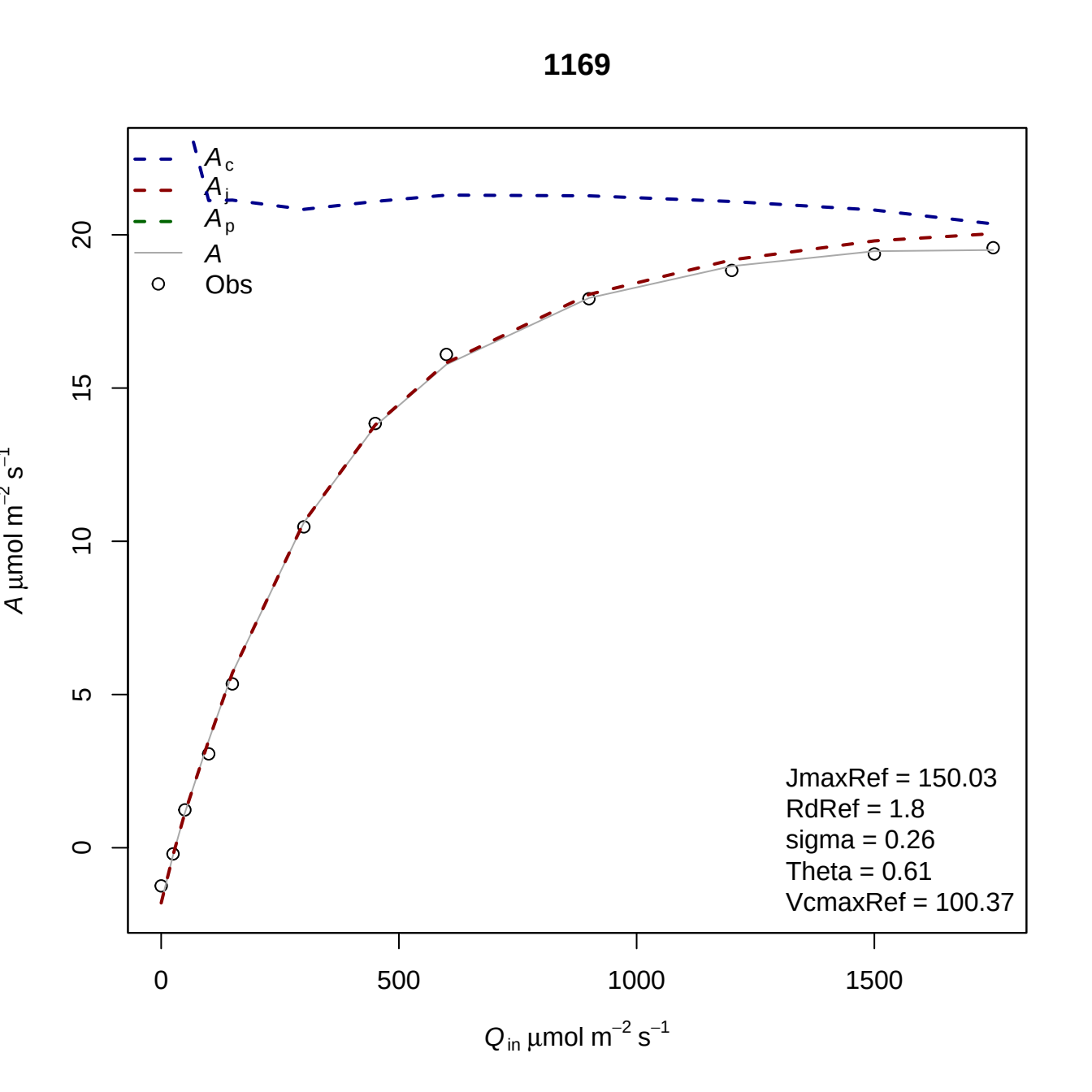

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 157084749.96
RdRef = 2.1
sigma = 0.34
Theta = −7591890.87
VcmaxRef = 59.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

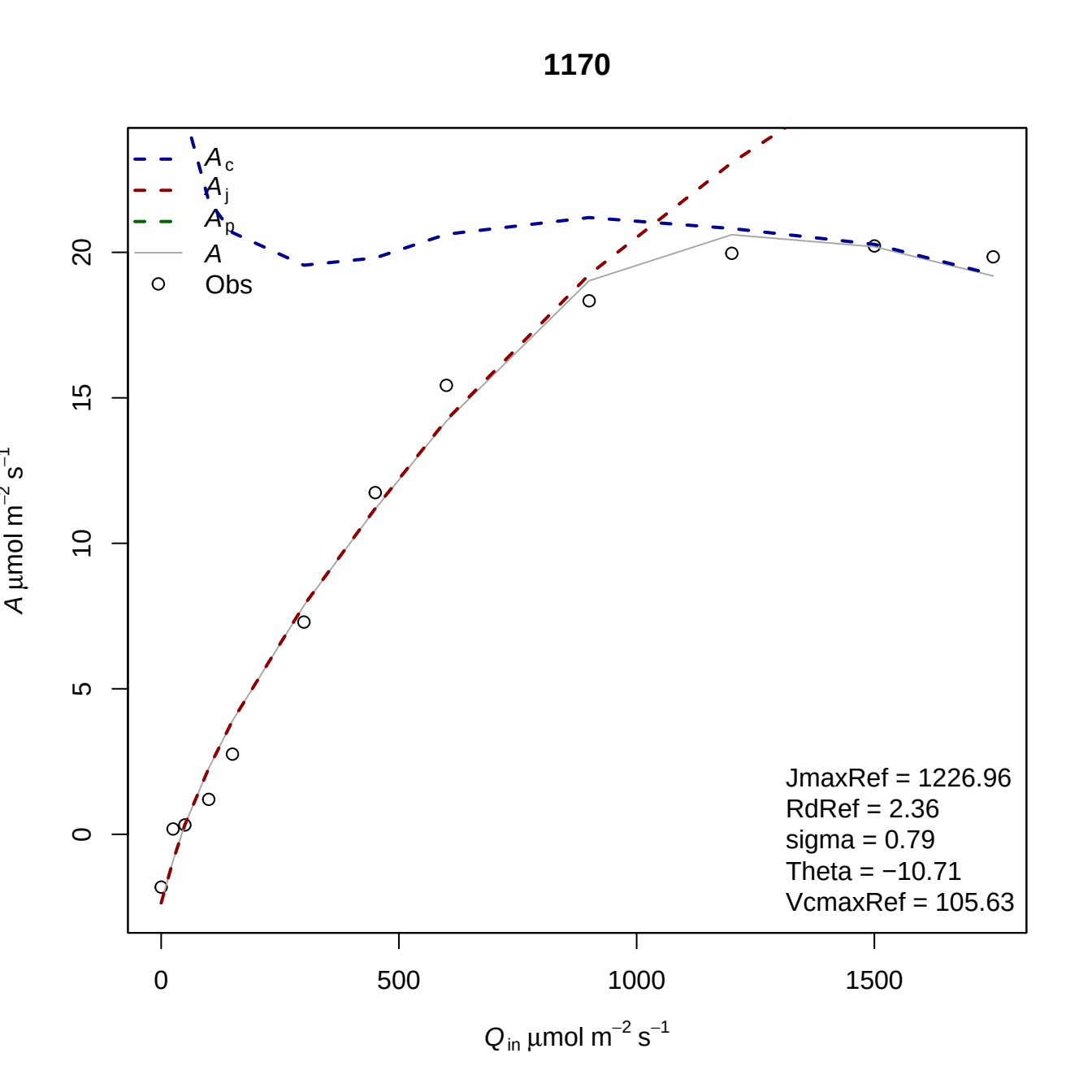

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 198.56
RdRef = 2.19
sigma = 0.73
Theta = −0.81
VcmaxRef = 89.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

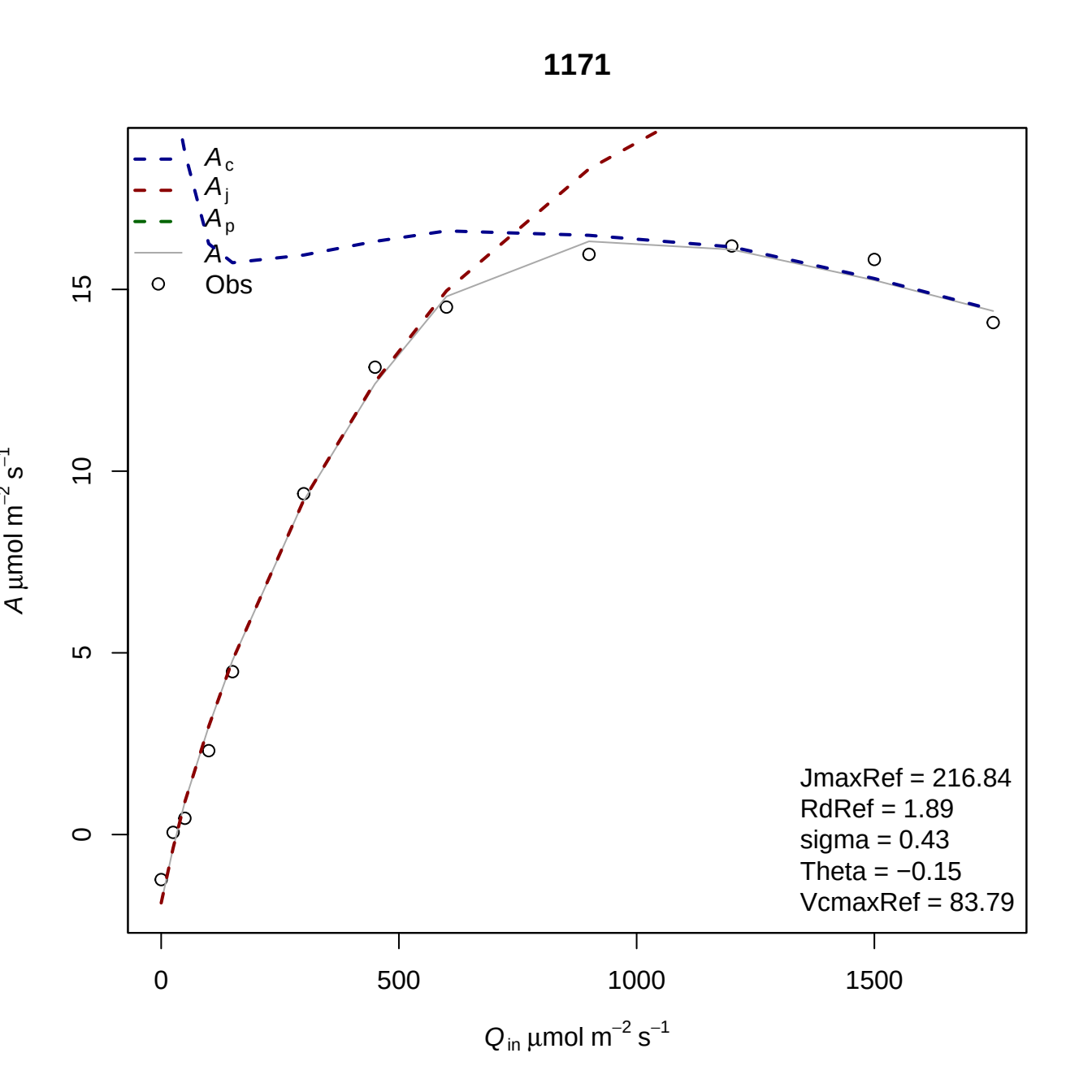

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 132.07
RdRef = 1.58
sigma = 0.31
Theta = 0.49
VcmaxRef = 89.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

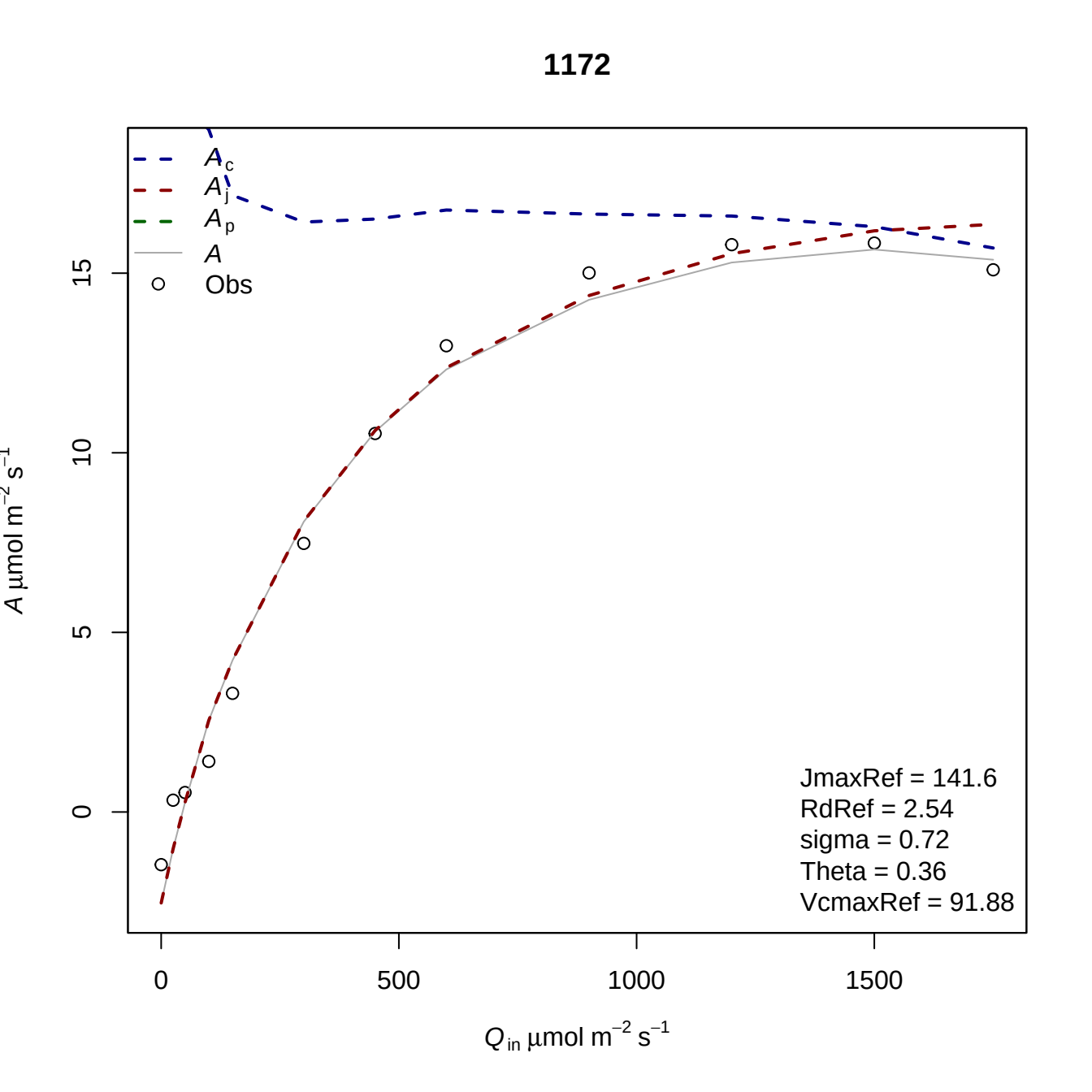

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 155.81
RdRef = 1.83
sigma = 0.43
Theta = −1.47
VcmaxRef = 79.31
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

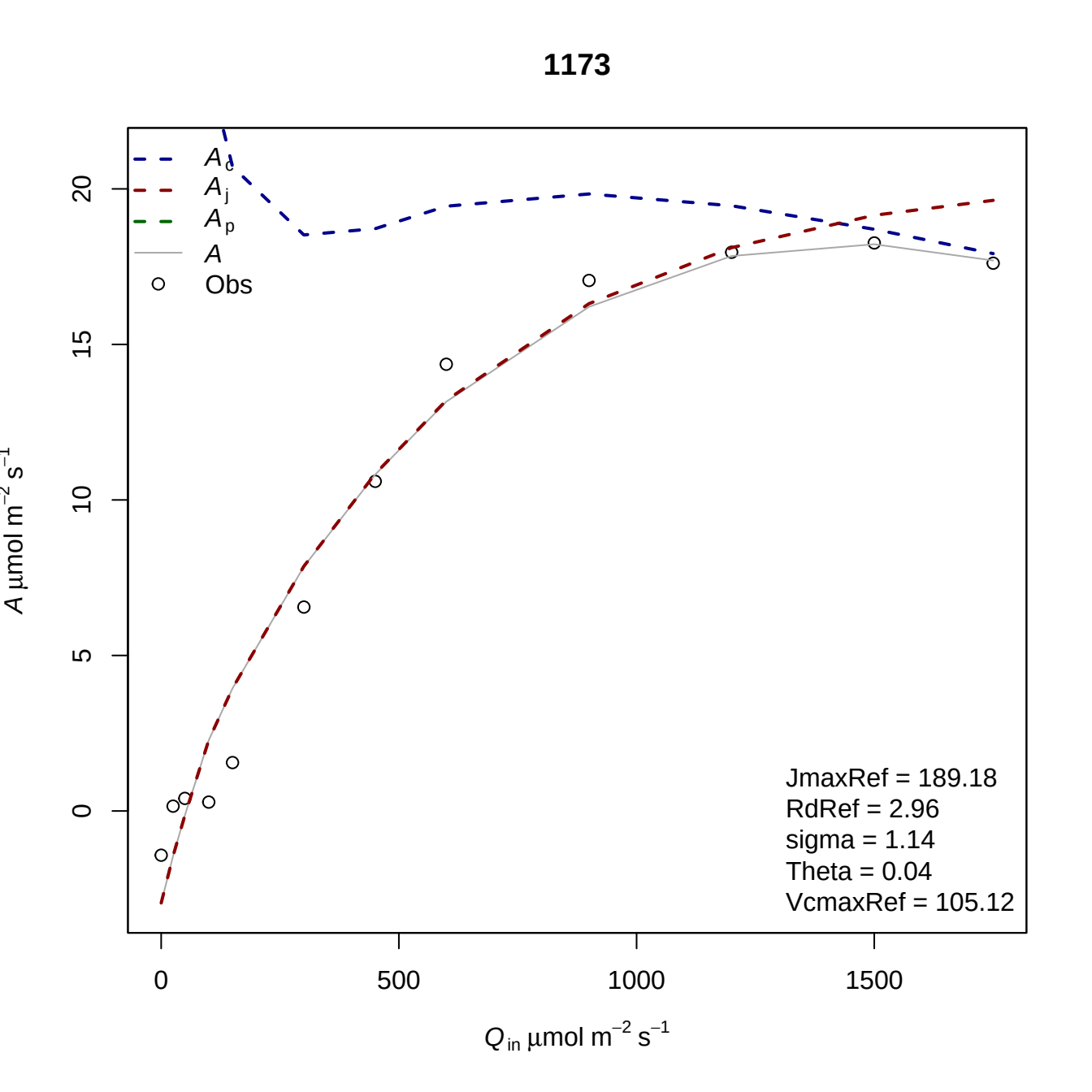

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 144.64
RdRef = 1.6
sigma = 0.23
Theta = 0.28
VcmaxRef = 90.9
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

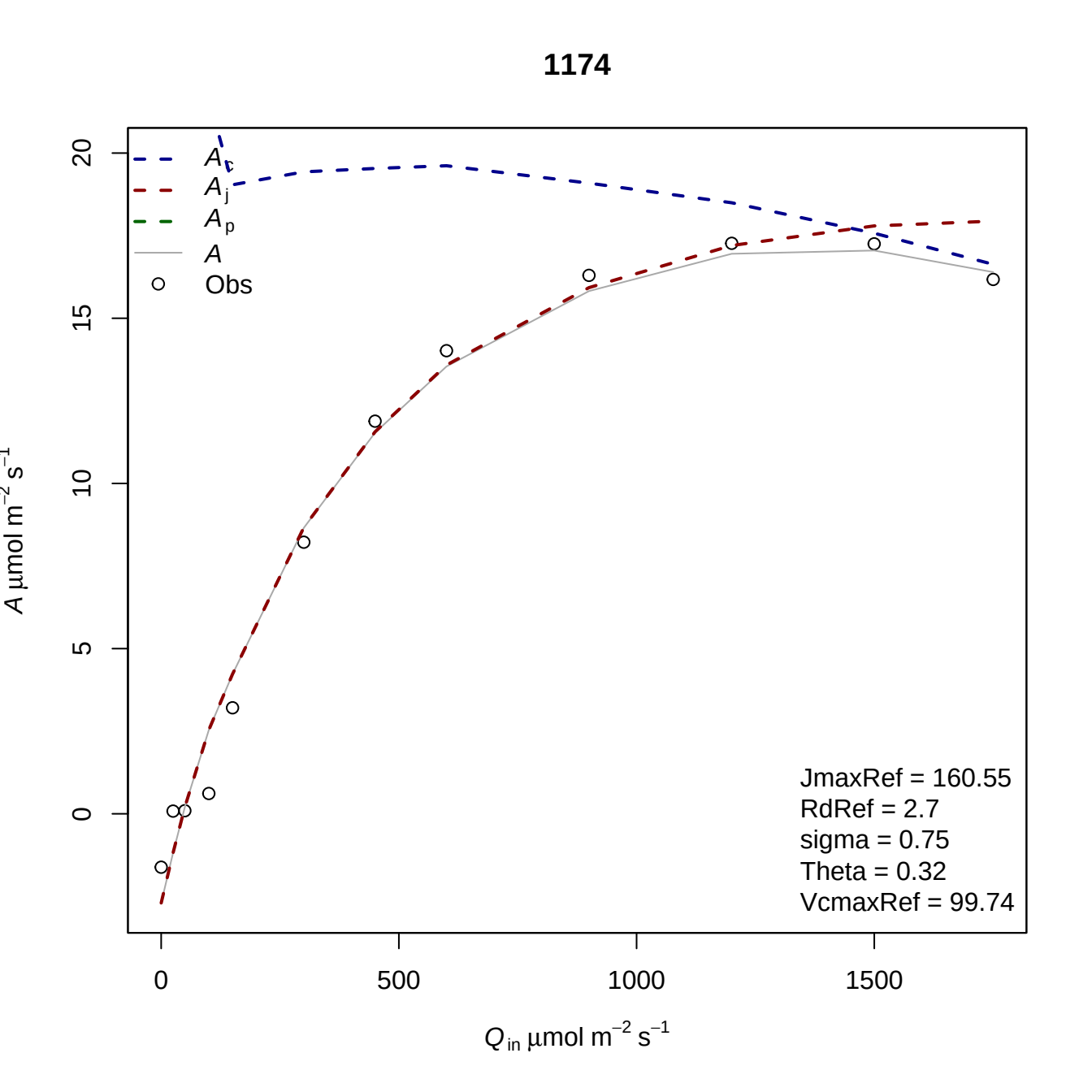

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 150.03
RdRef = 1.8
sigma = 0.26
Theta = 0.61
VcmaxRef = 100.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

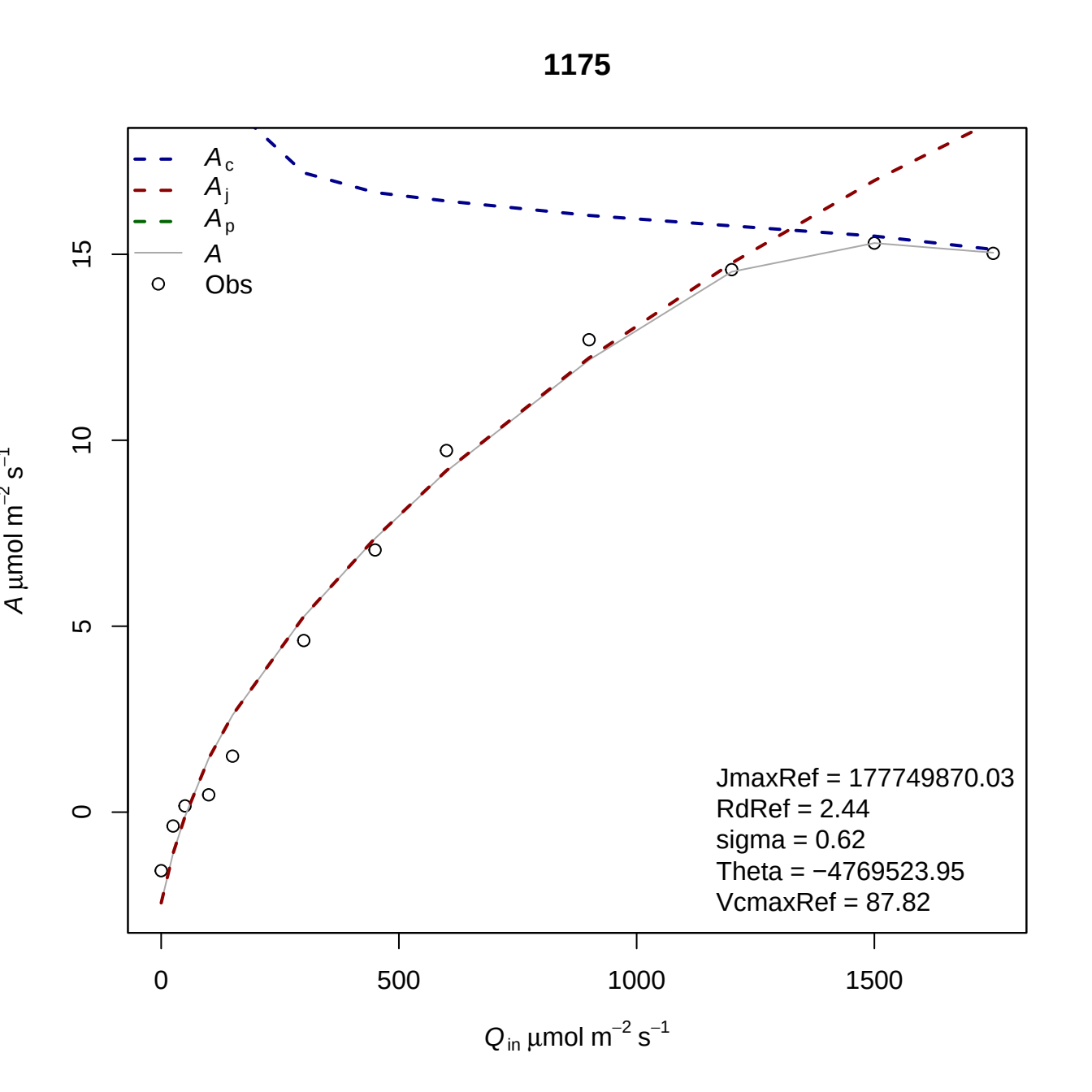

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 1226.96
RdRef = 2.36
sigma = 0.79
Theta = −10.71
VcmaxRef = 105.63
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

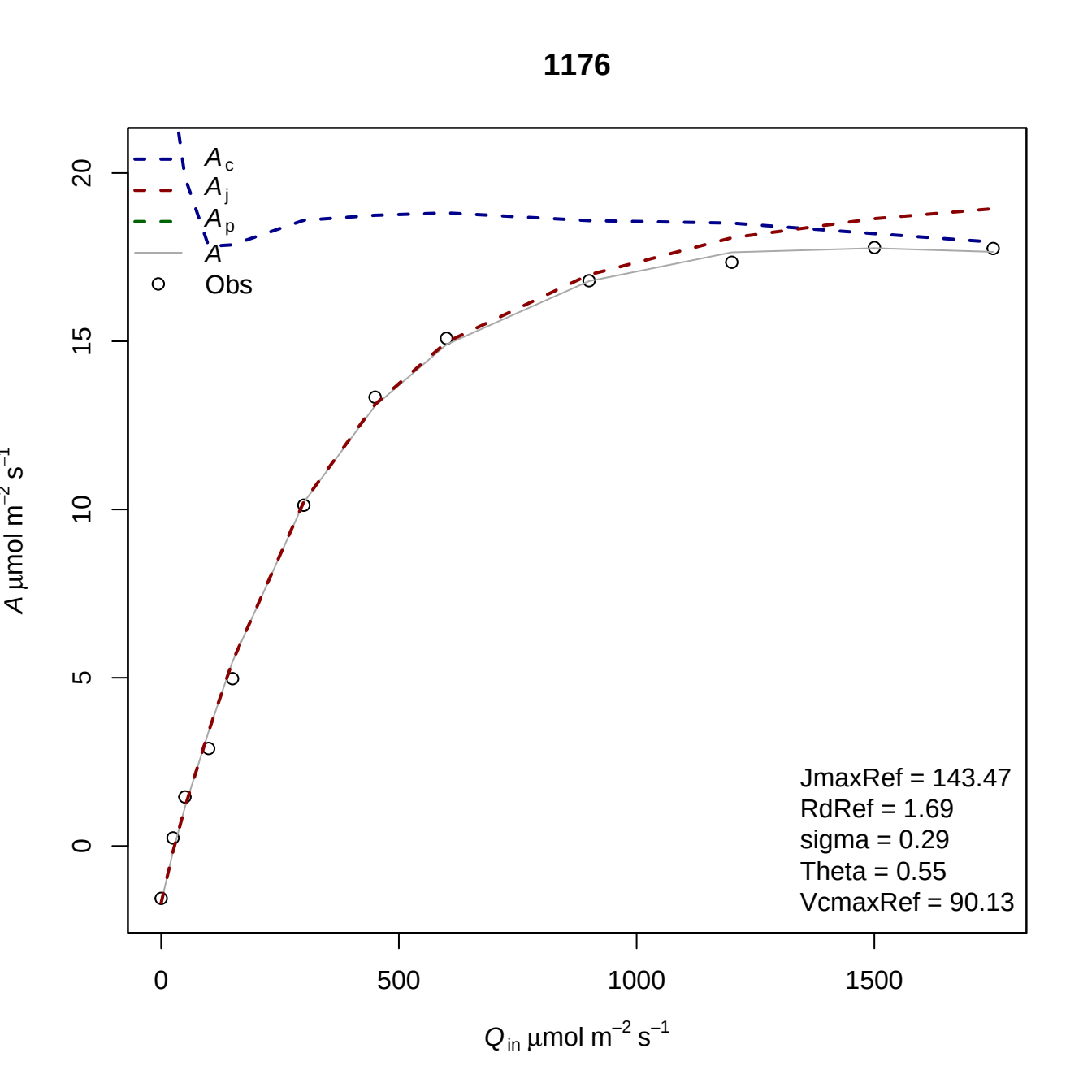

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 216.84
RdRef = 1.89
sigma = 0.43
Theta = −0.15
VcmaxRef = 83.79
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

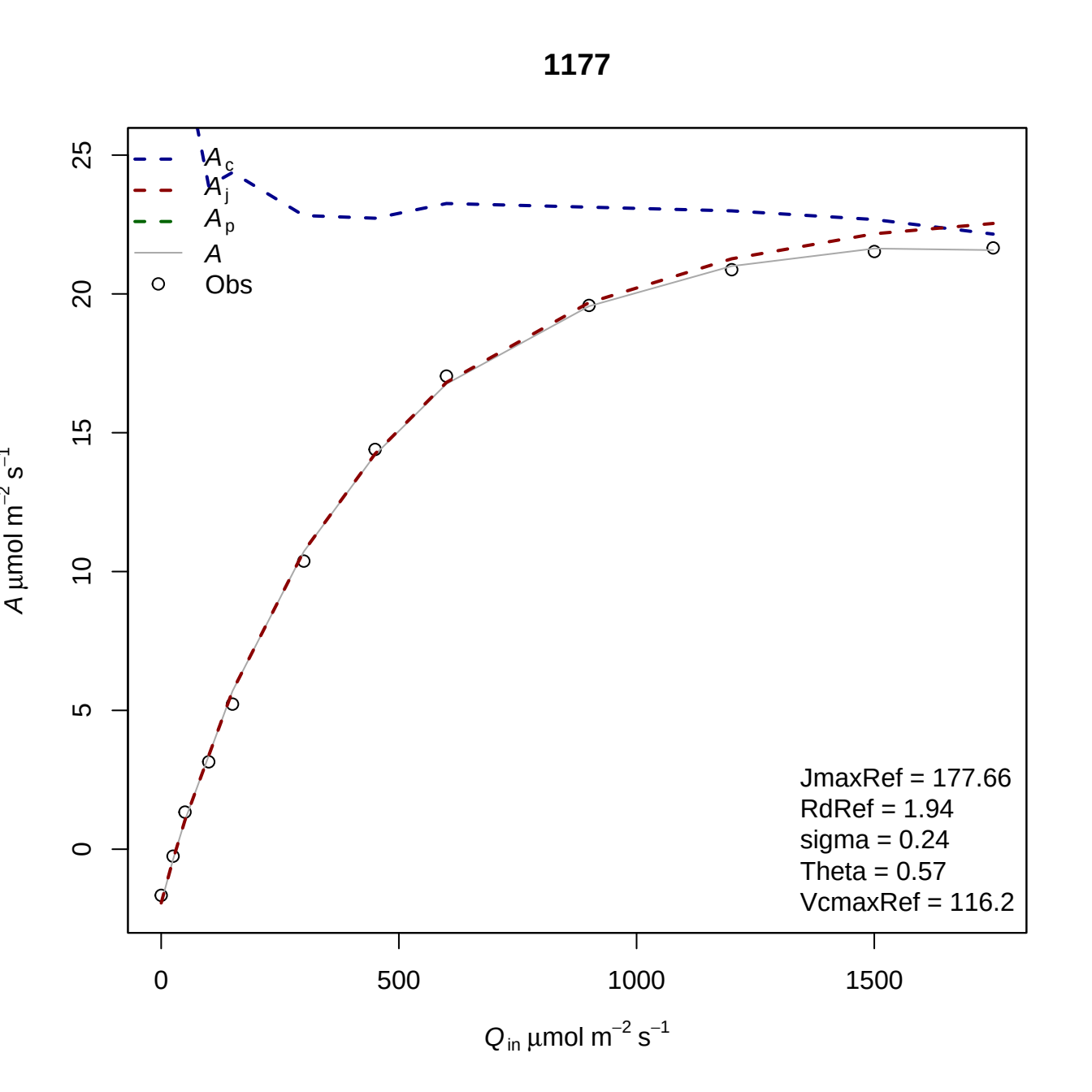

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 141.6
RdRef = 2.54
sigma = 0.72
Theta = 0.36
VcmaxRef = 91.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

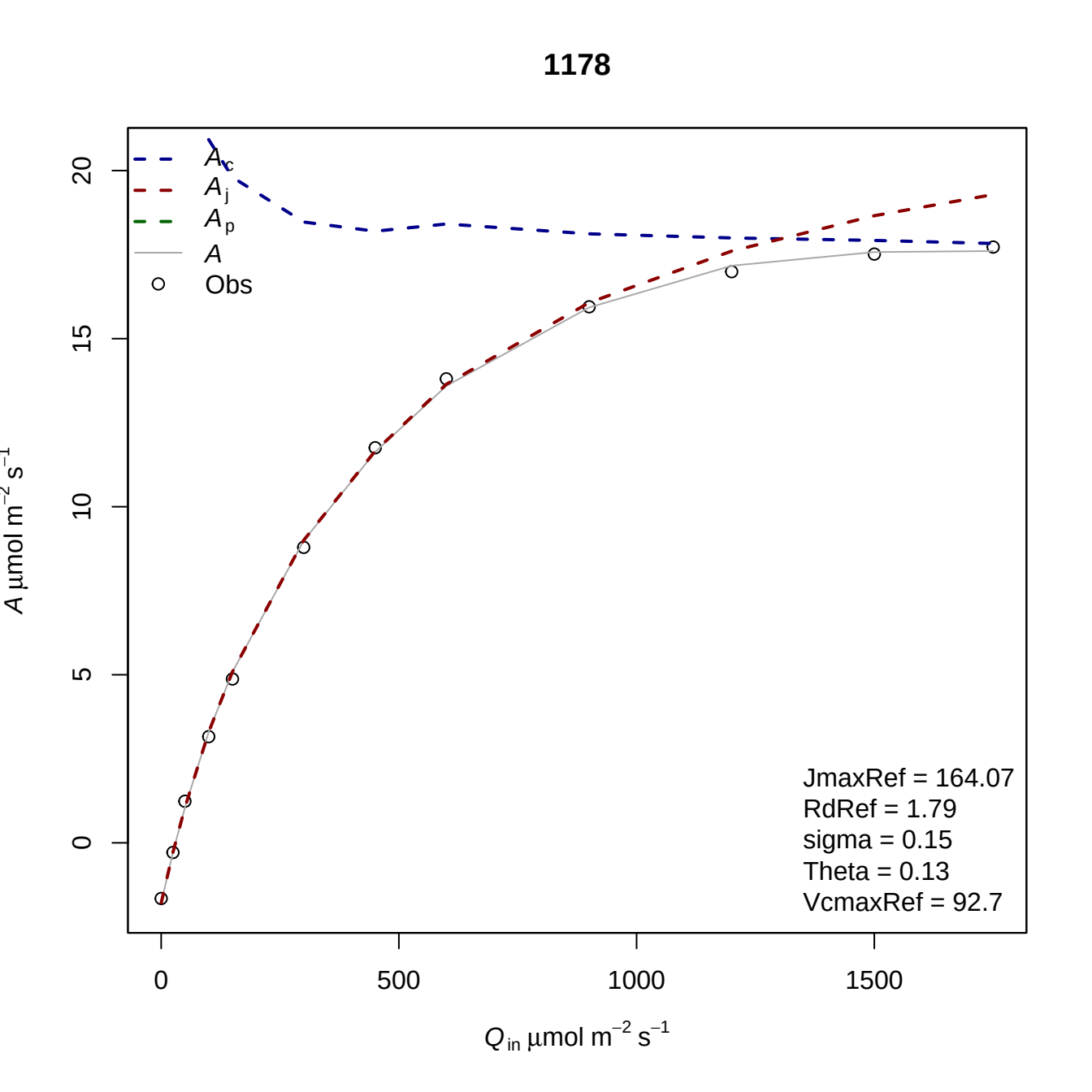

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 189.18
RdRef = 2.96
sigma = 1.14
Theta = 0.04
VcmaxRef = 105.12
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

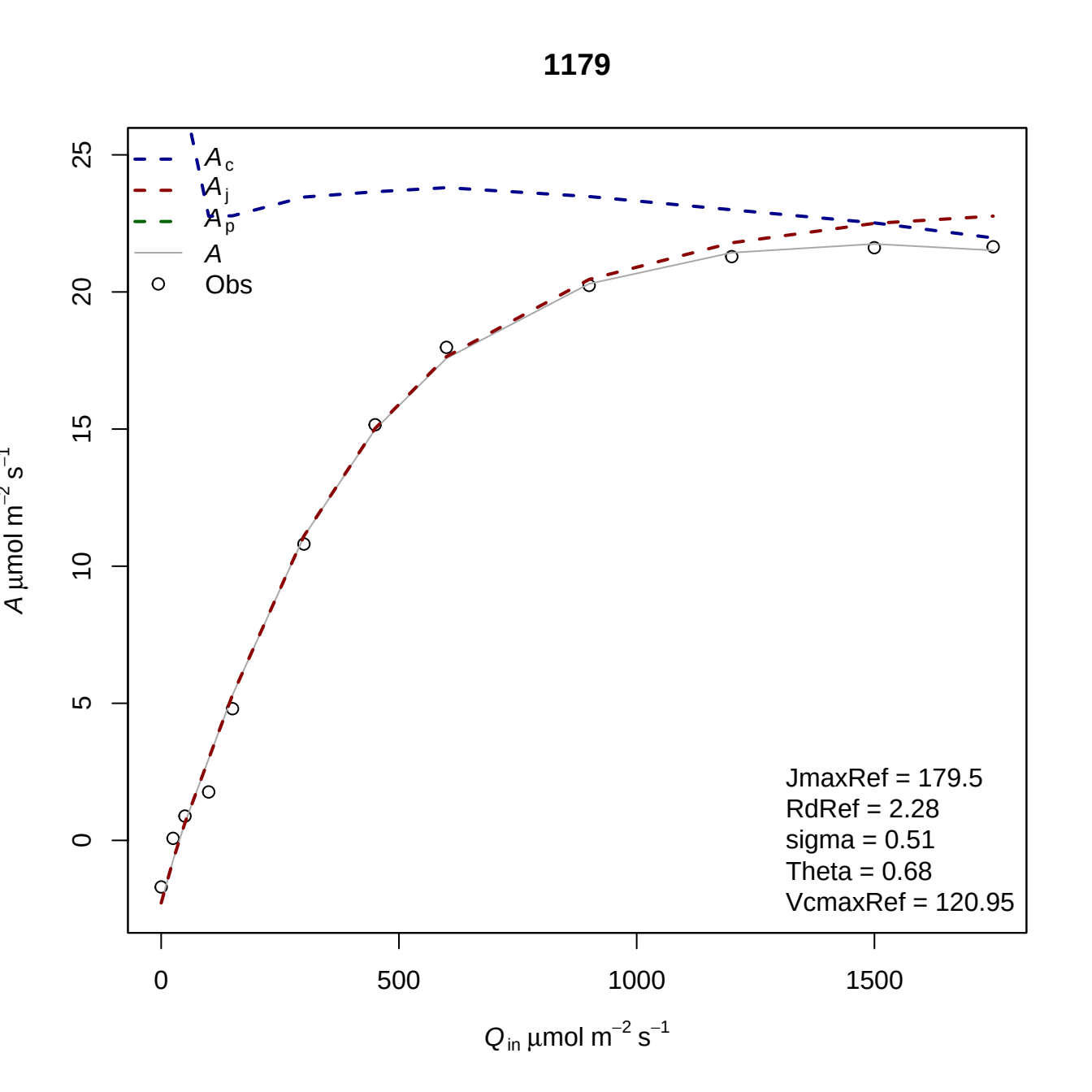

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 160.55
RdRef = 2.7
sigma = 0.75
Theta = 0.32
VcmaxRef = 99.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

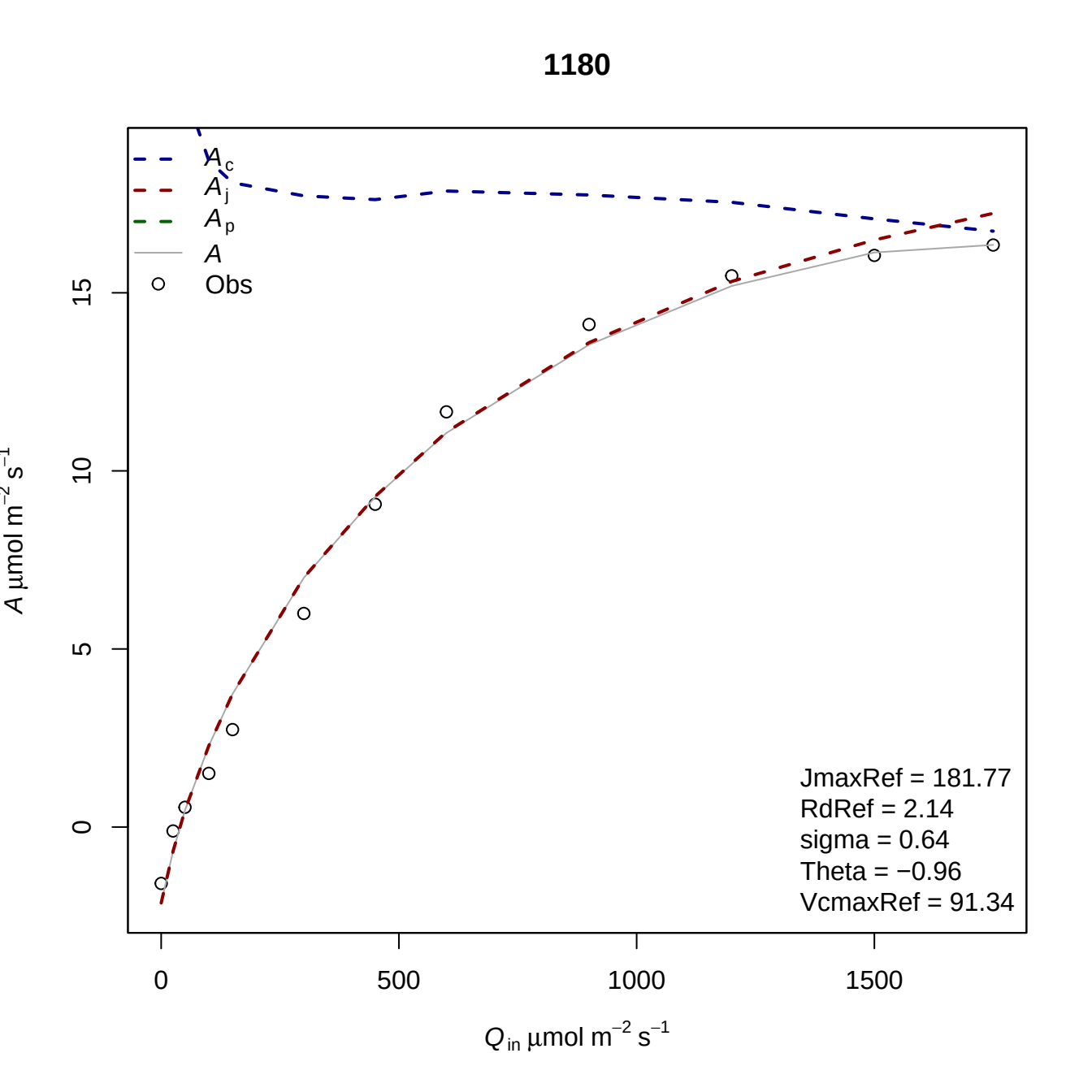

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 177749870.03
RdRef = 2.44
sigma = 0.62
Theta = −4769523.95
VcmaxRef = 87.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

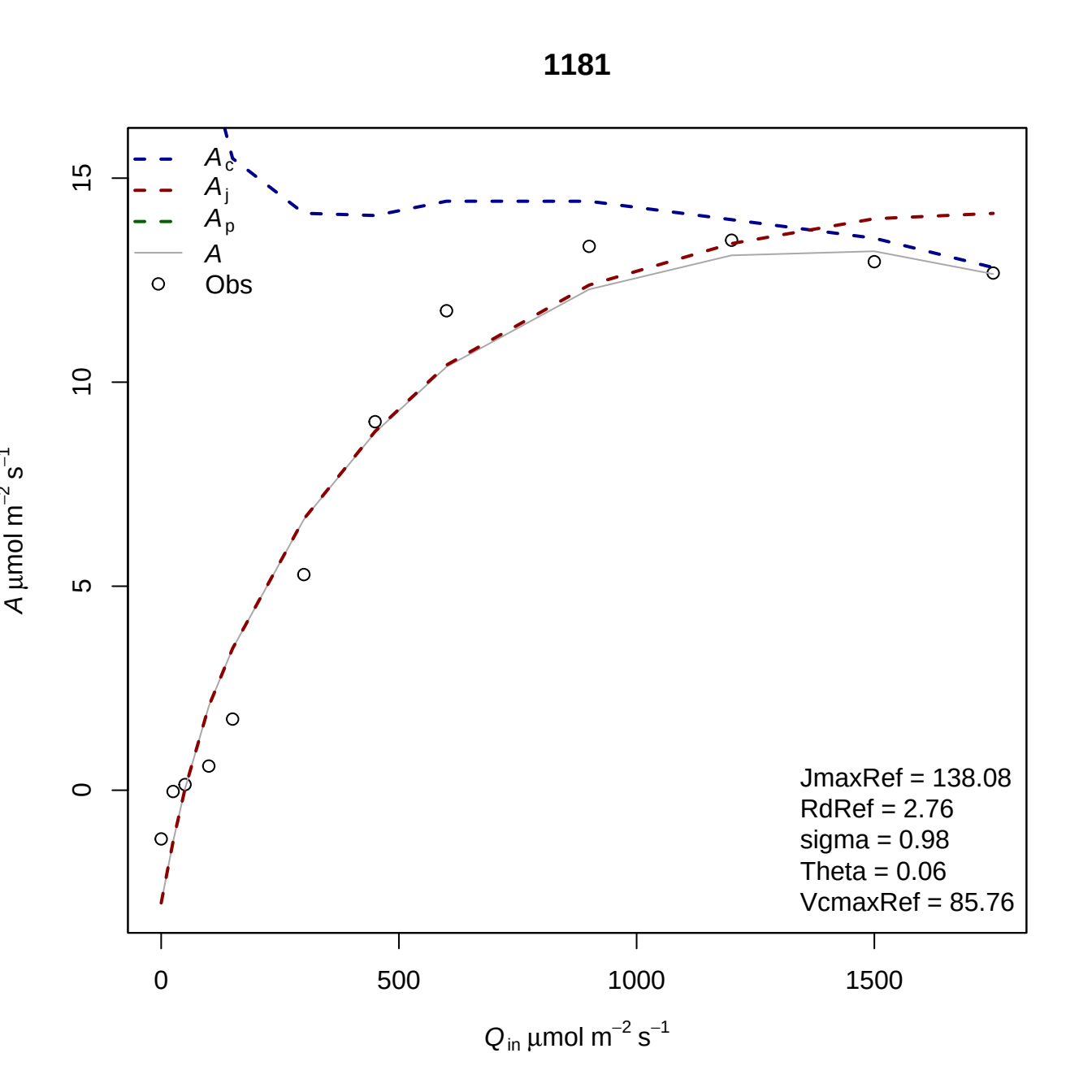

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 143.47
RdRef = 1.69
sigma = 0.29
Theta = 0.55
VcmaxRef = 90.13
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

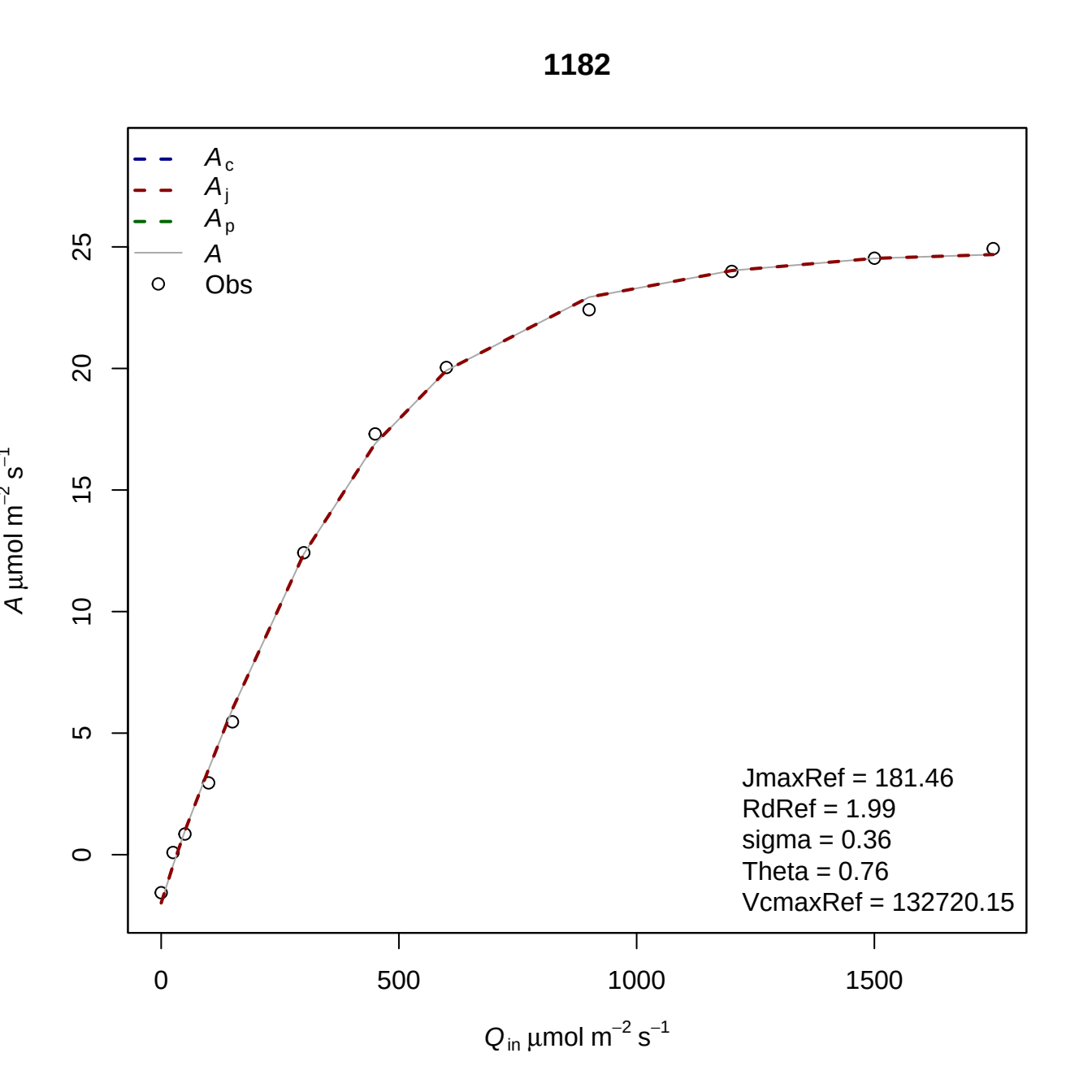

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 177.66
RdRef = 1.94
sigma = 0.24
Theta = 0.57
VcmaxRef = 116.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

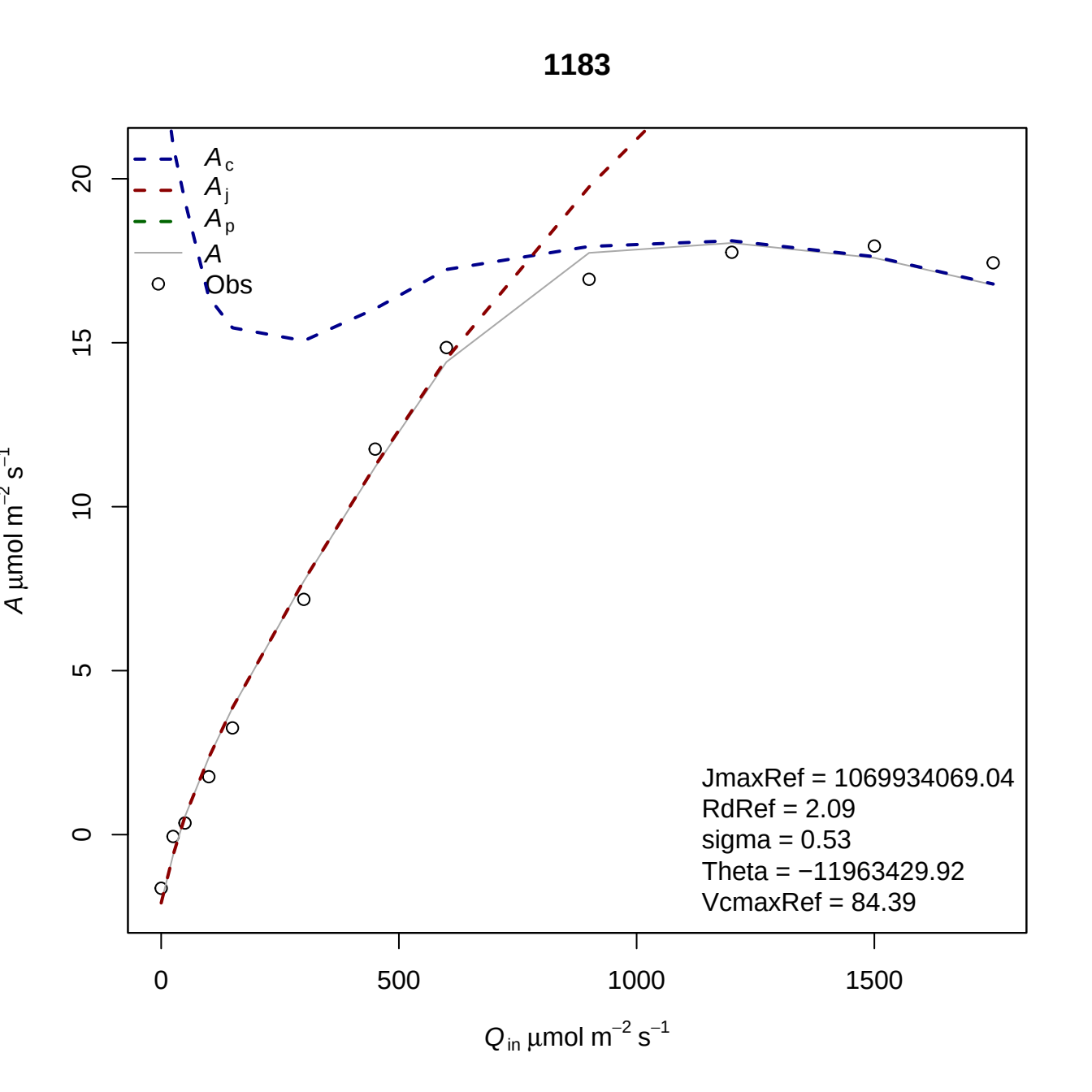

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 164.07
RdRef = 1.79
sigma = 0.15
Theta = 0.13
VcmaxRef = 92.7
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

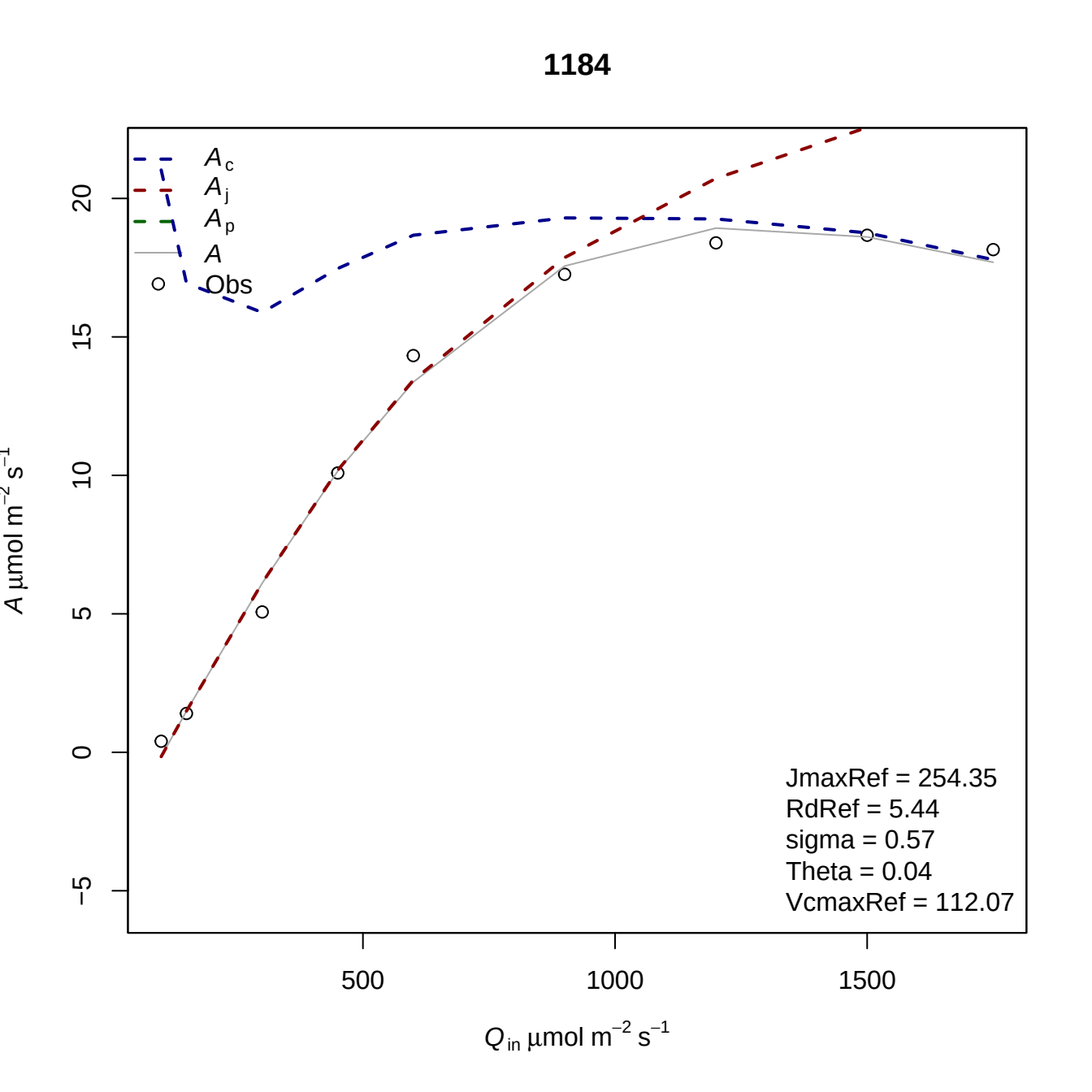

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 179.5
RdRef = 2.28
sigma = 0.51
Theta = 0.68
VcmaxRef = 120.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

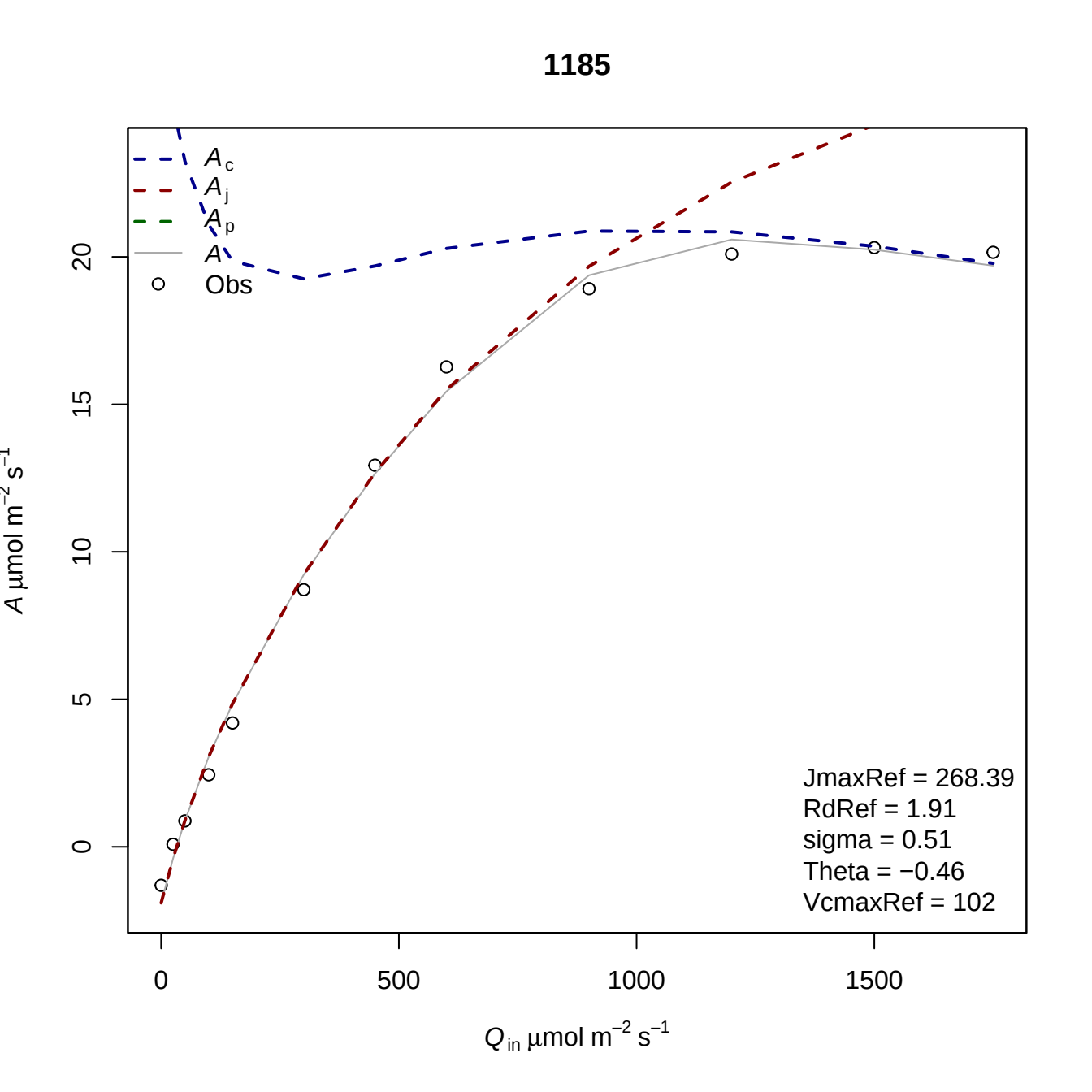

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 181.77
RdRef = 2.14
sigma = 0.64
Theta = −0.96
VcmaxRef = 91.34
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

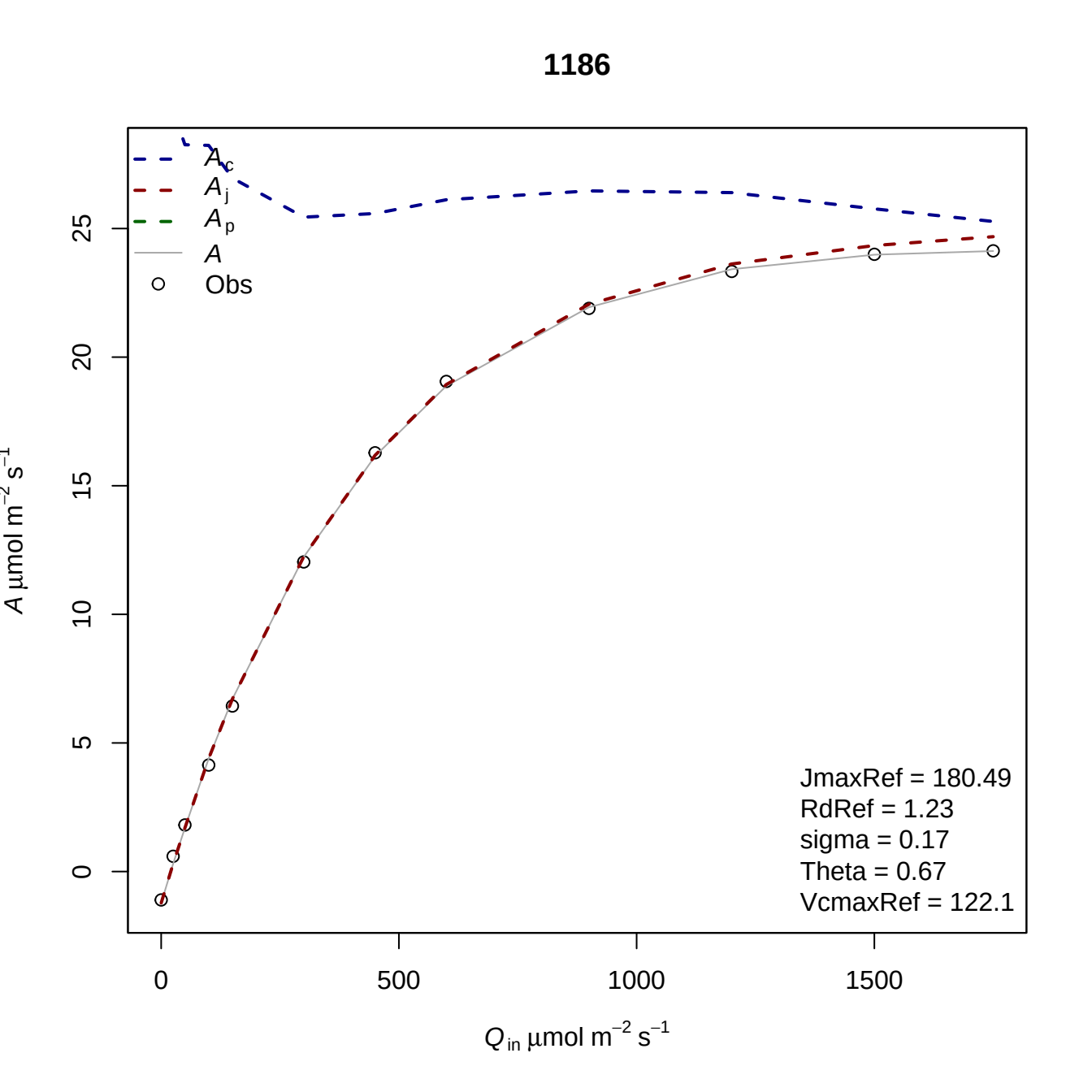

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 138.08
RdRef = 2.76
sigma = 0.98
Theta = 0.06
VcmaxRef = 85.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

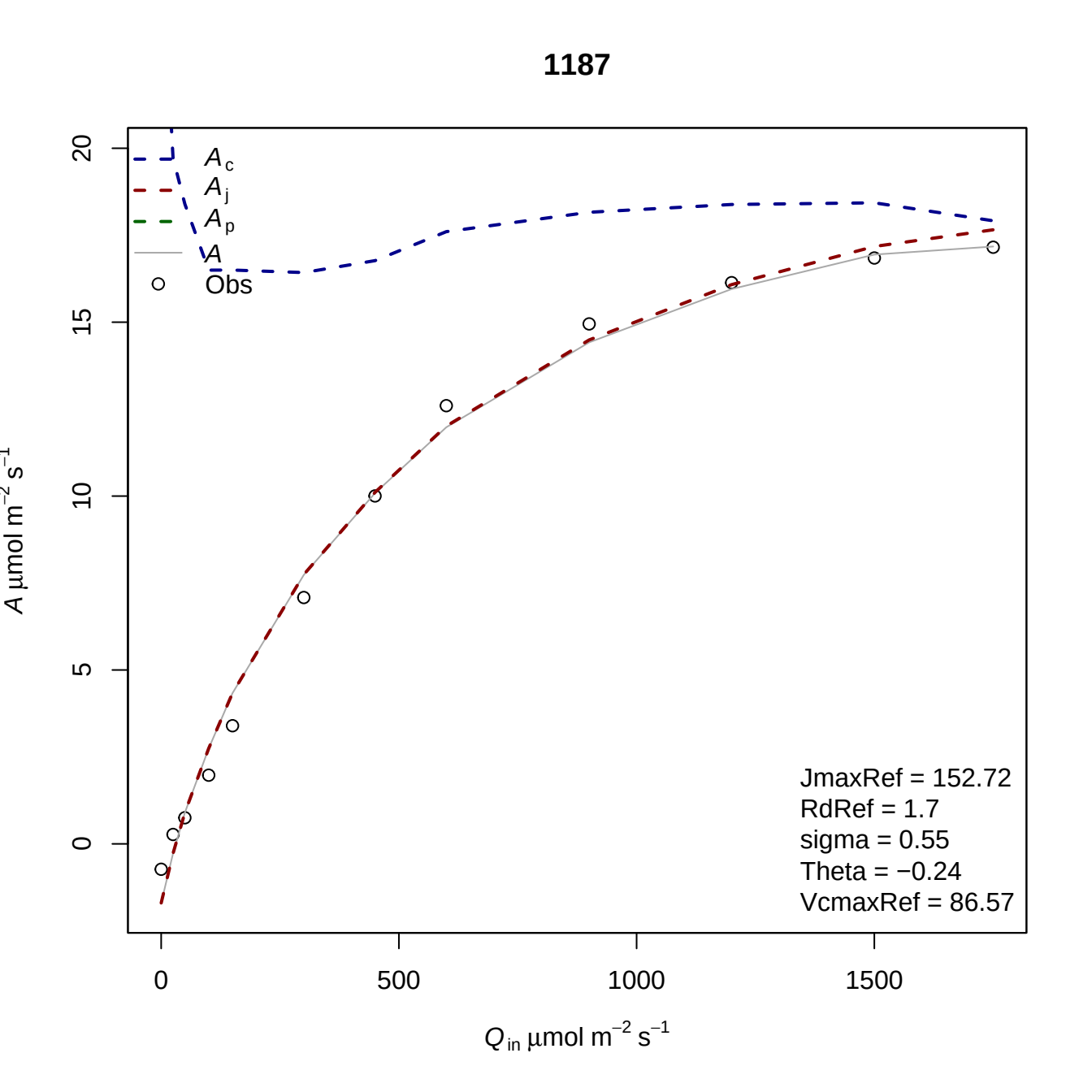

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 181.46
RdRef = 1.99
sigma = 0.36
Theta = 0.76
VcmaxRef = 132720.15
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

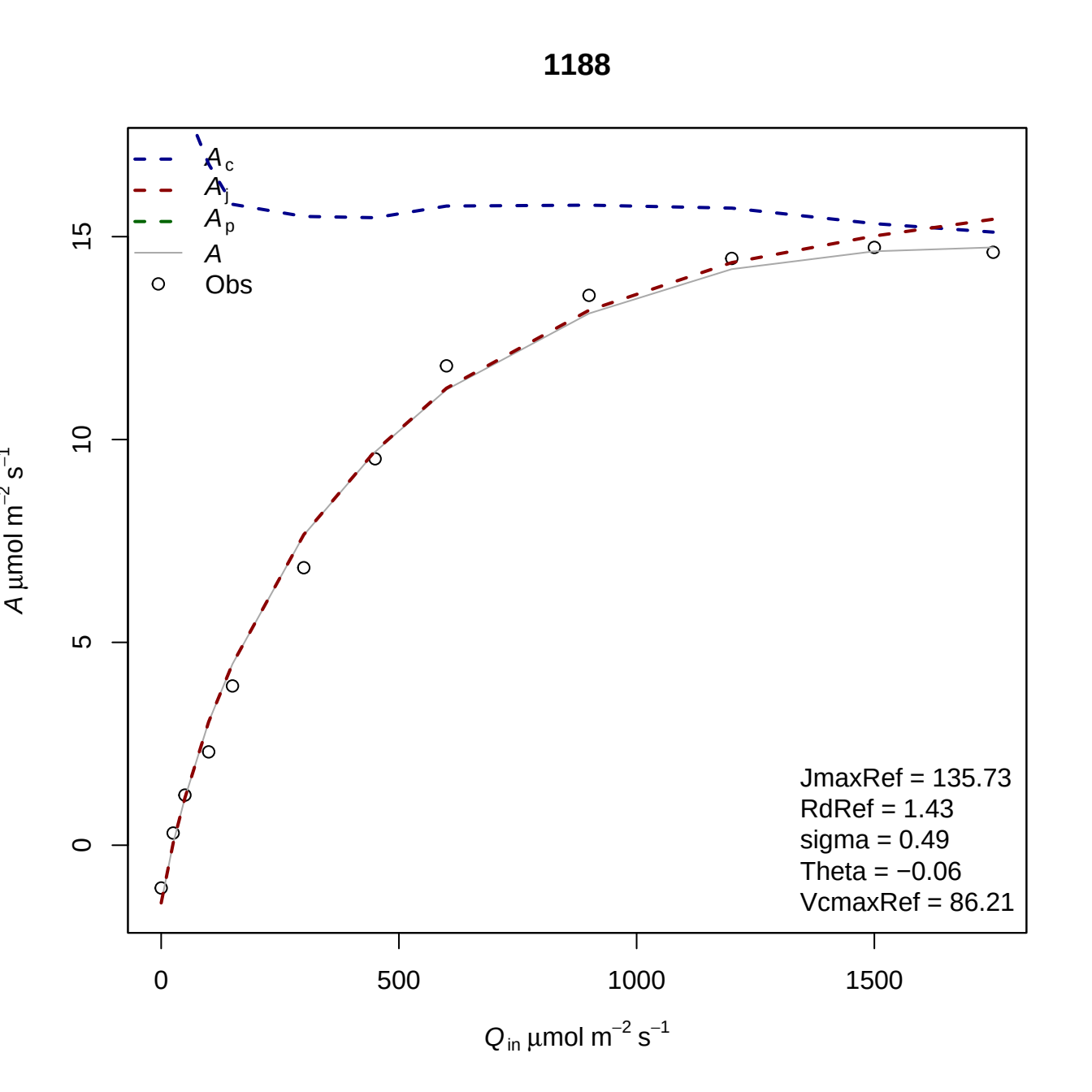

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 1069934069.04
RdRef = 2.09
sigma = 0.53
Theta = −11963429.92
VcmaxRef = 84.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

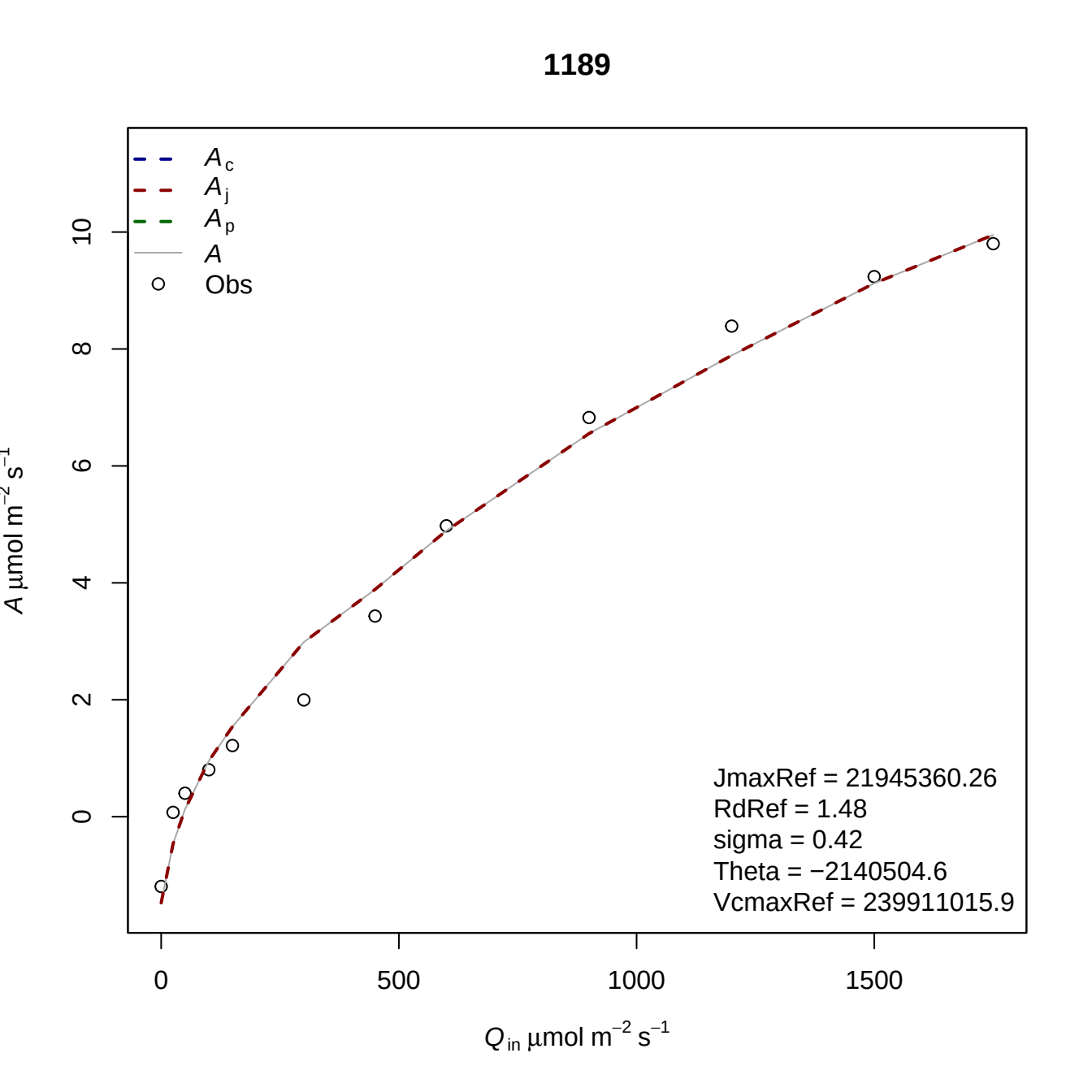

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 254.35
RdRef = 5.44
sigma = 0.57
Theta = 0.04
VcmaxRef = 112.07
20
15
10
5
0
−5
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

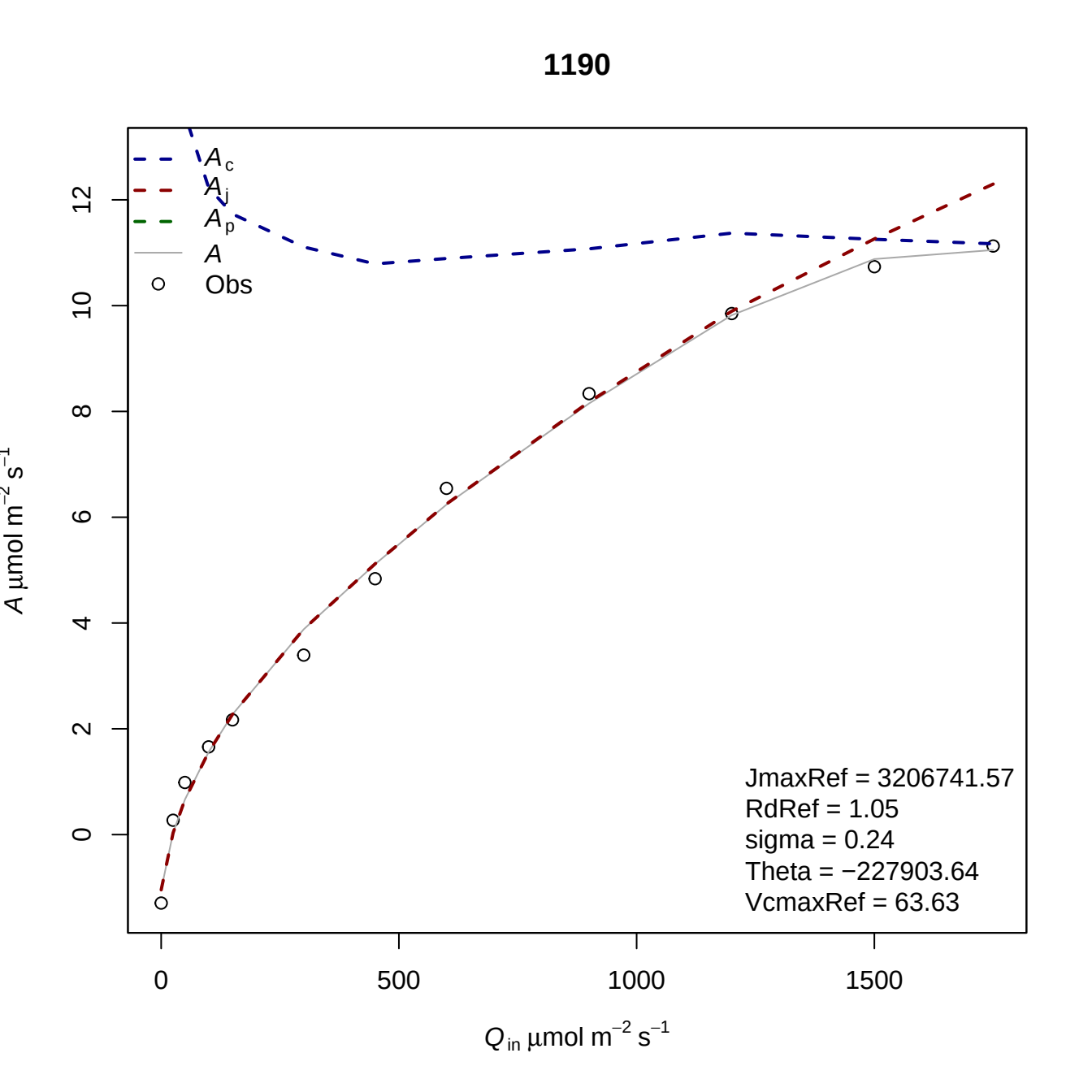

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 268.39
RdRef = 1.91
sigma = 0.51
Theta = −0.46
VcmaxRef = 102
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

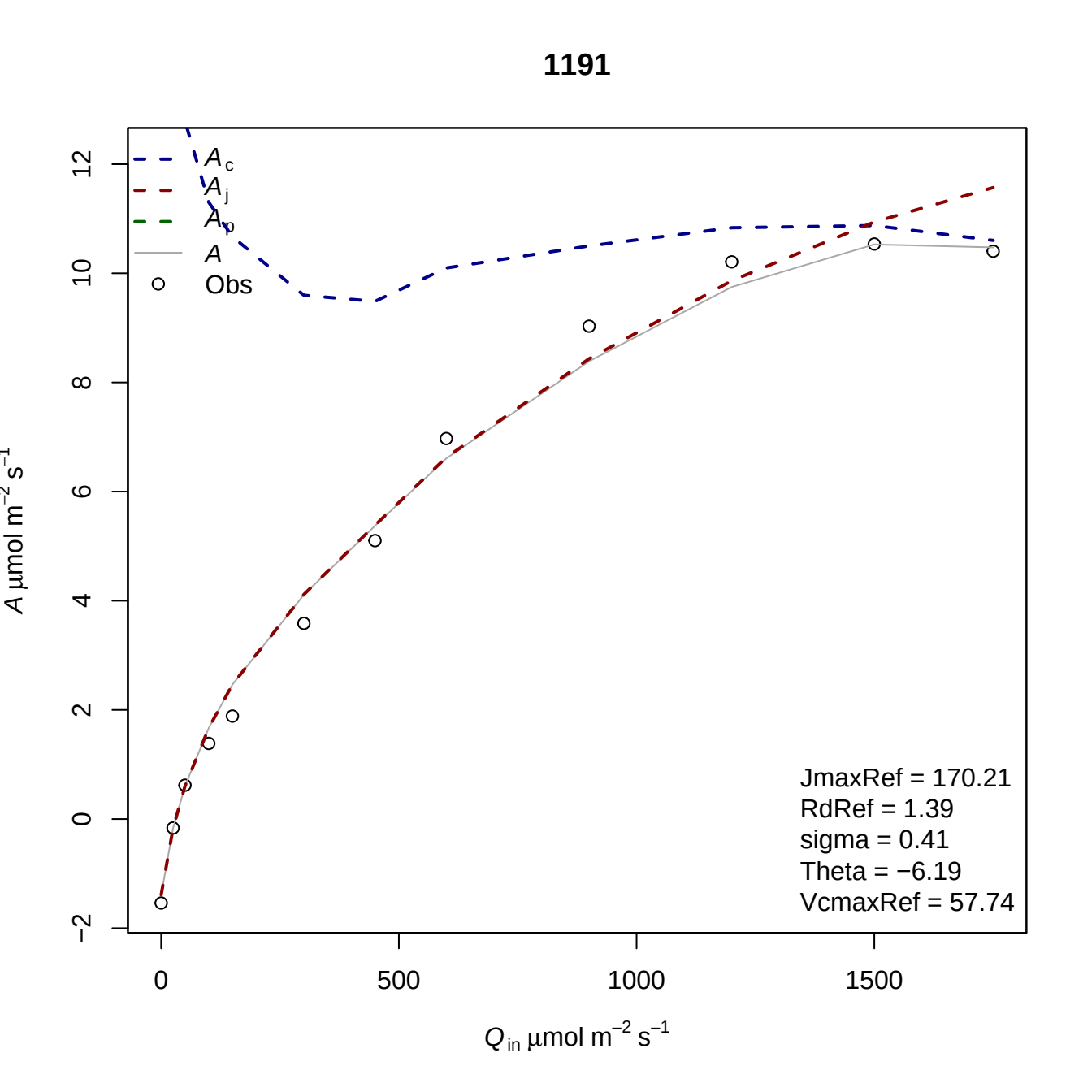

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 180.49
RdRef = 1.23
sigma = 0.17
Theta = 0.67
VcmaxRef = 122.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

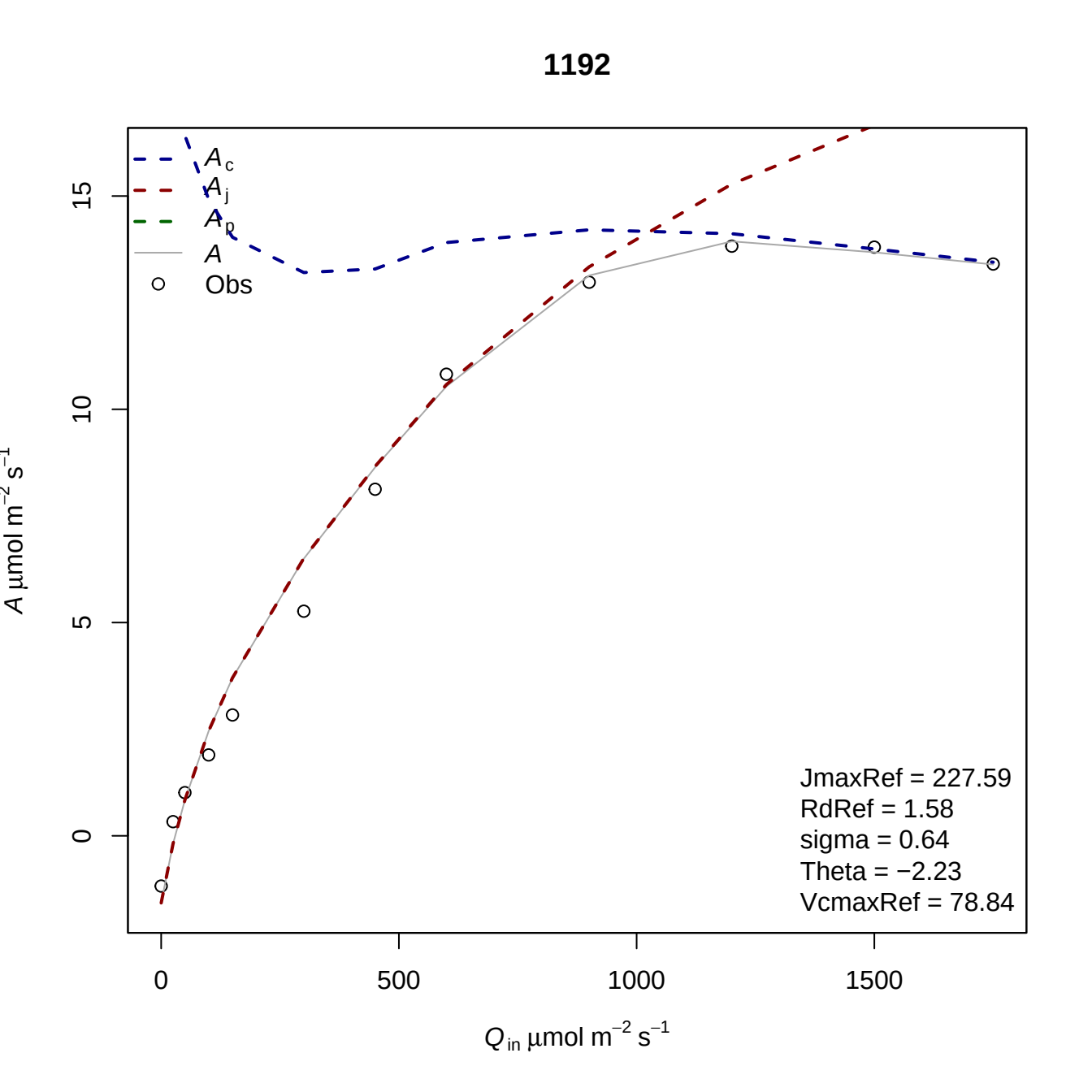

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 152.72
RdRef = 1.7
sigma = 0.55
Theta = −0.24
VcmaxRef = 86.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

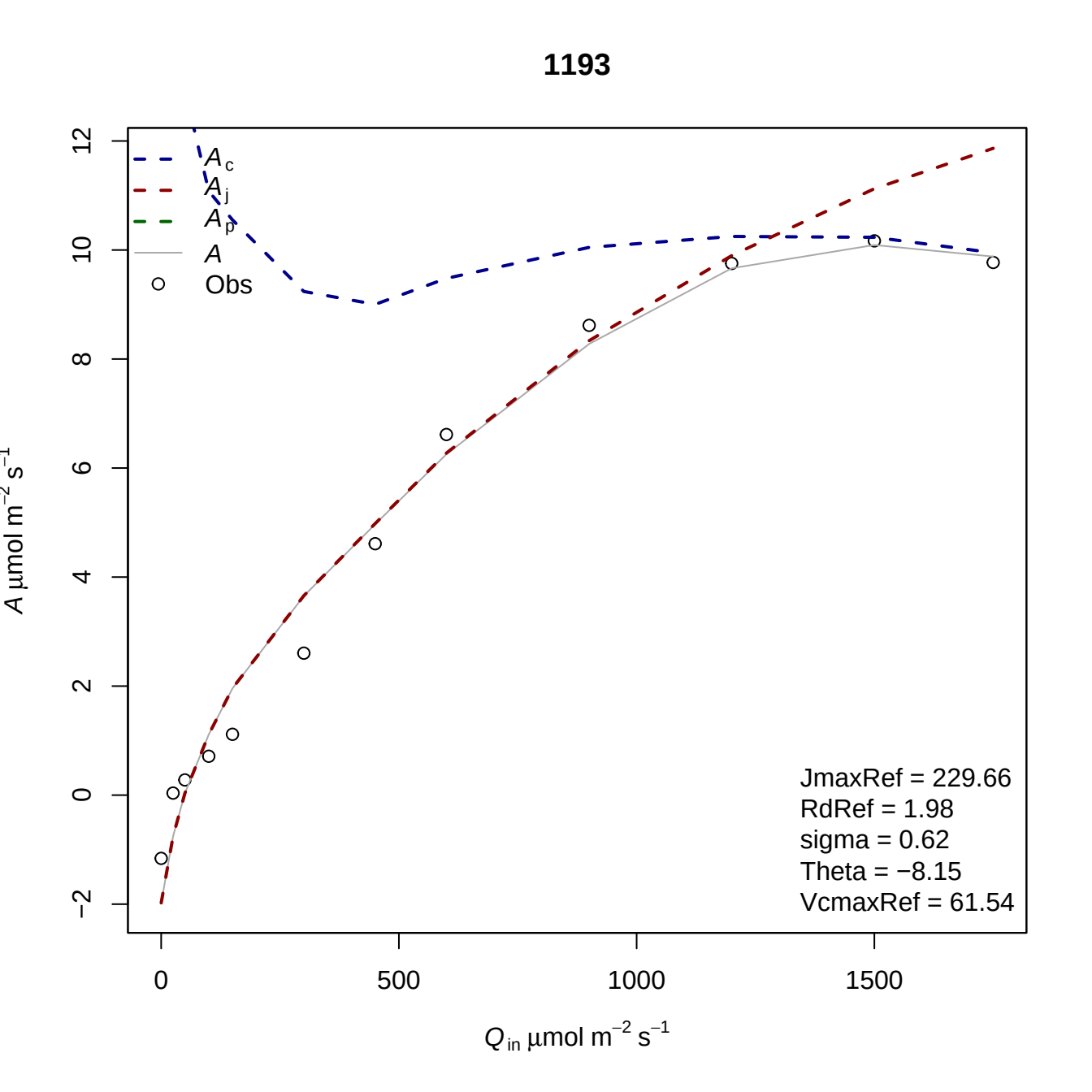

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 135.73
RdRef = 1.43
sigma = 0.49
Theta = −0.06
VcmaxRef = 86.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

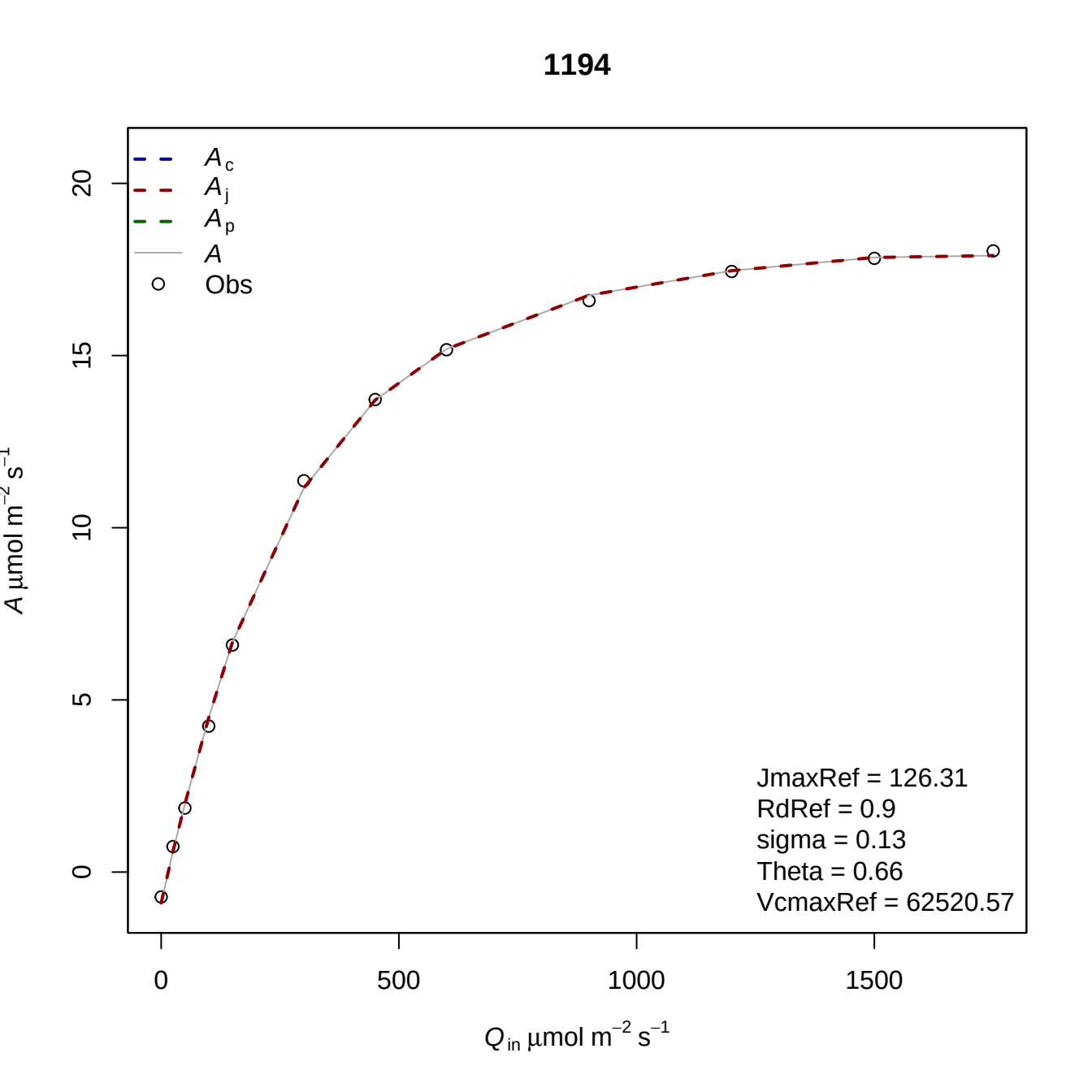

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 21945360.26
RdRef = 1.48
sigma = 0.42
Theta = −2140504.6
VcmaxRef = 239911015.9
A µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

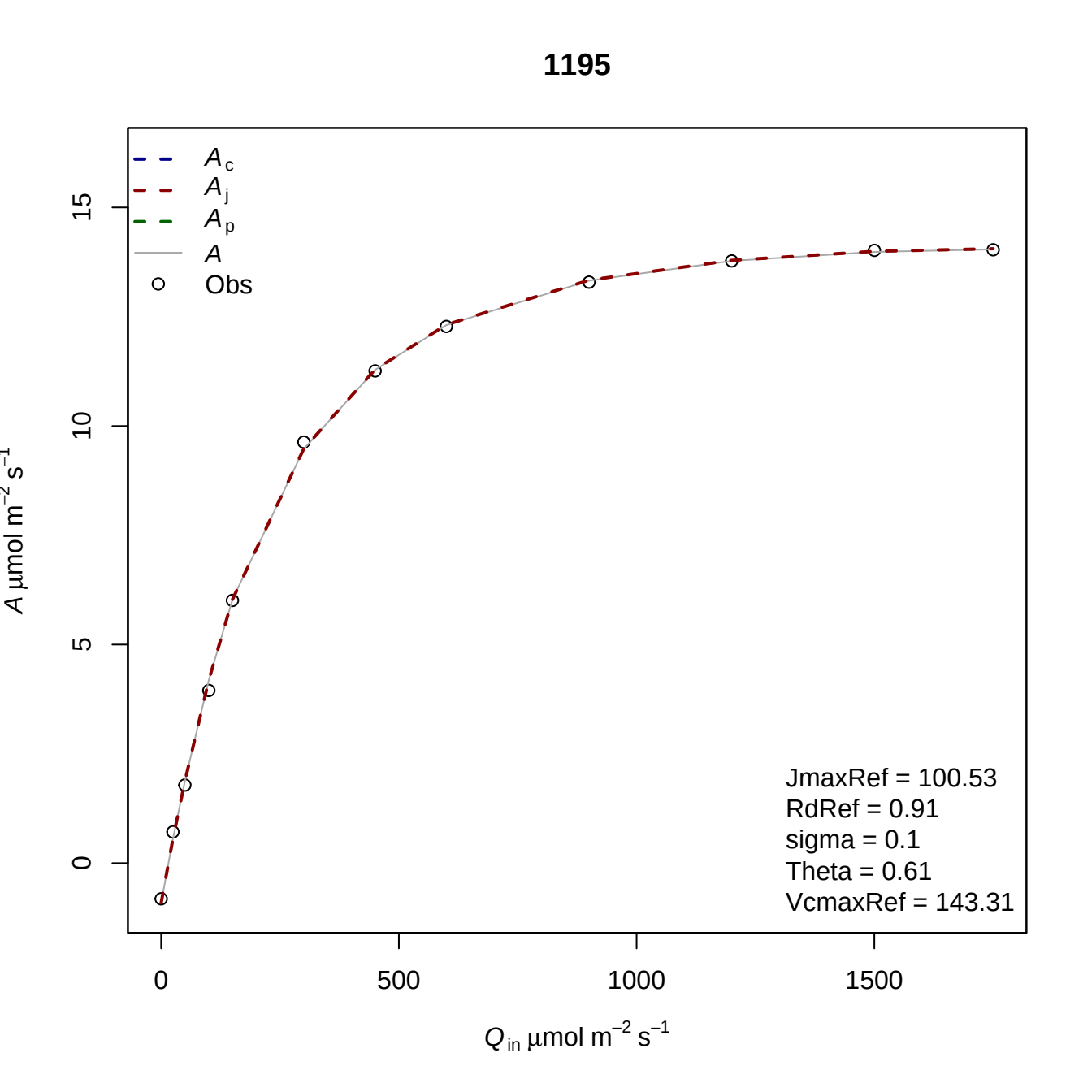

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 3206741.57
RdRef = 1.05
sigma = 0.24
Theta = −227903.64
VcmaxRef = 63.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

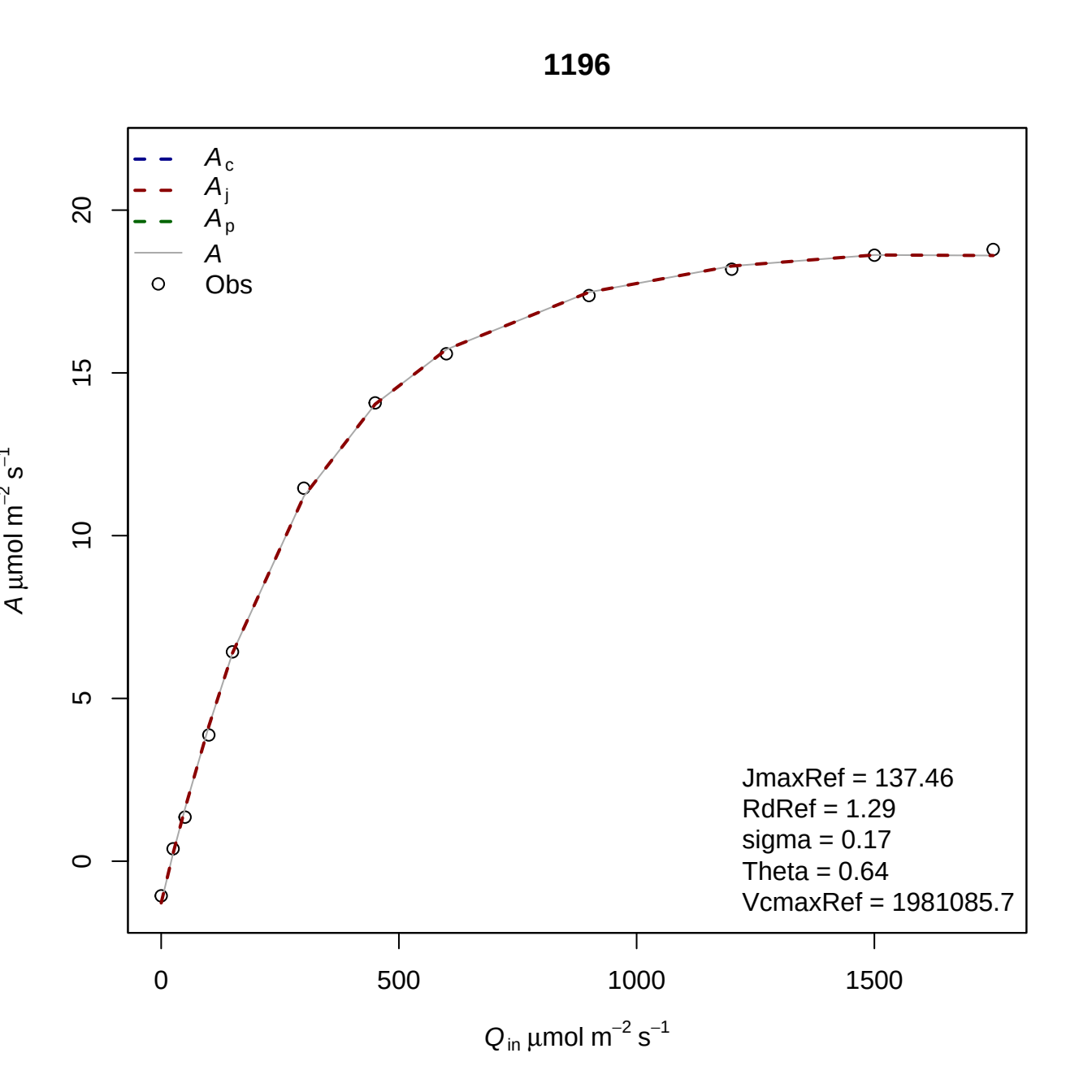

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 170.21
RdRef = 1.39
sigma = 0.41
Theta = −6.19
VcmaxRef = 57.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

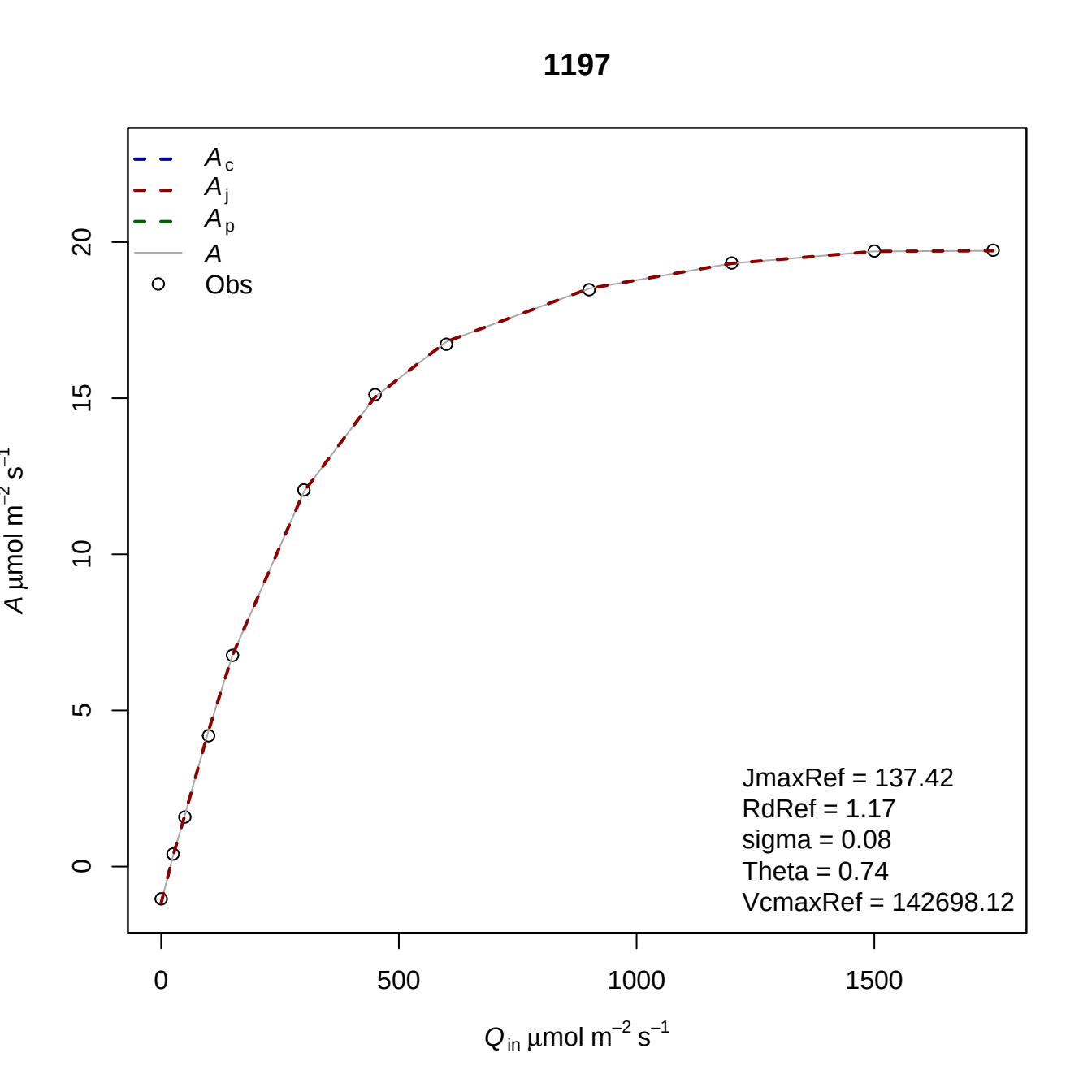

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 227.59
RdRef = 1.58
sigma = 0.64
Theta = −2.23
VcmaxRef = 78.84
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

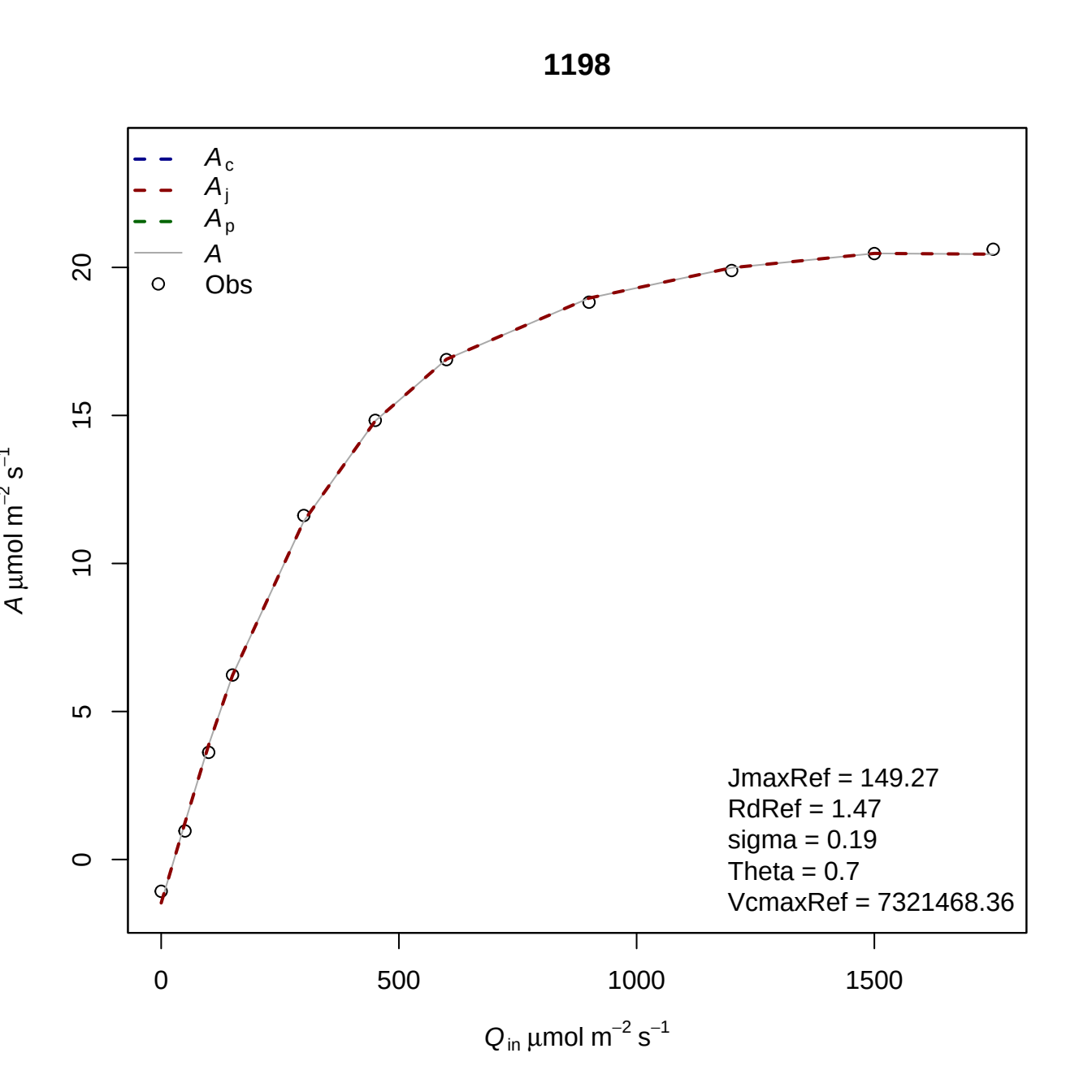

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 229.66
RdRef = 1.98
sigma = 0.62
Theta = −8.15
VcmaxRef = 61.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

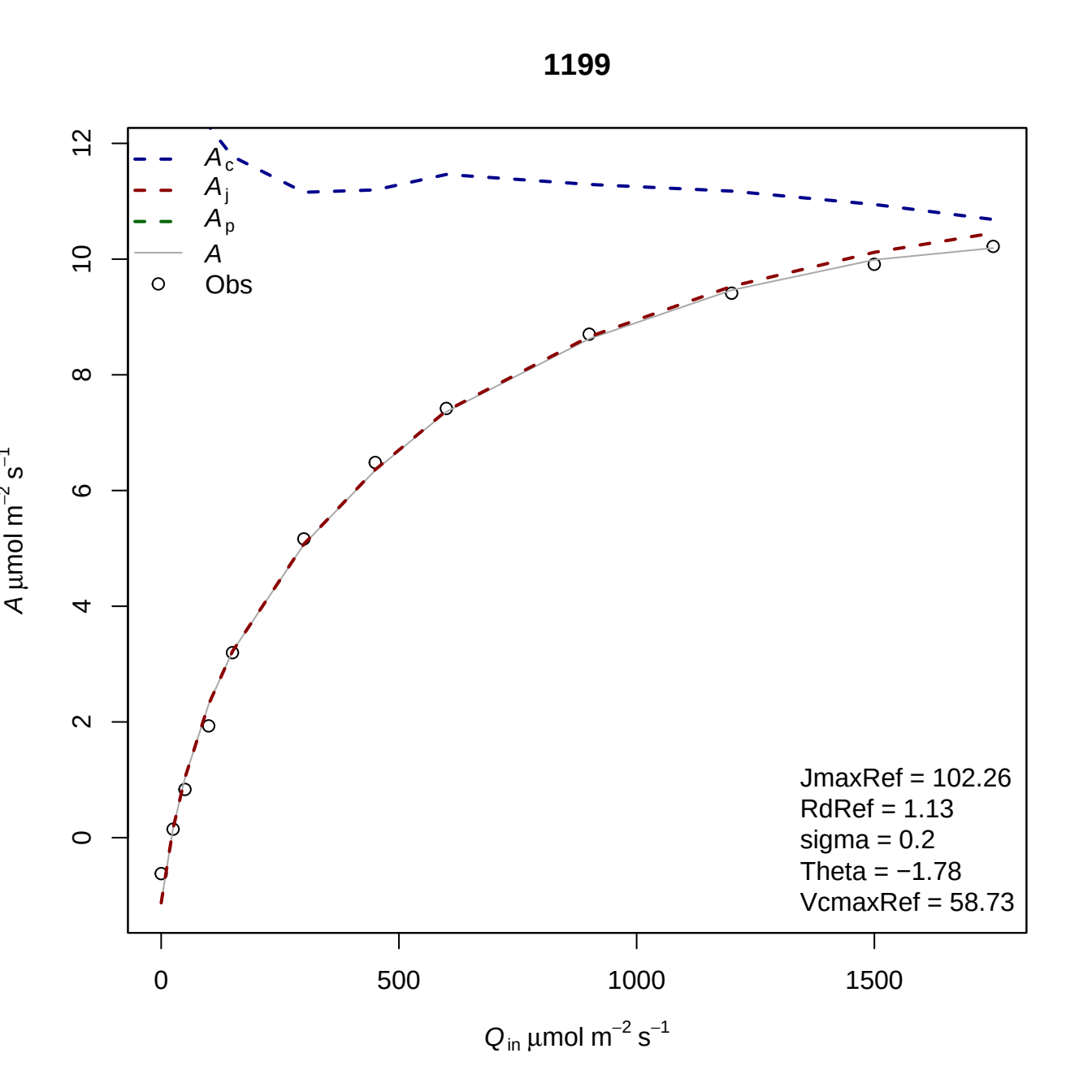

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 126.31
RdRef = 0.9
sigma = 0.13
Theta = 0.66
VcmaxRef = 62520.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

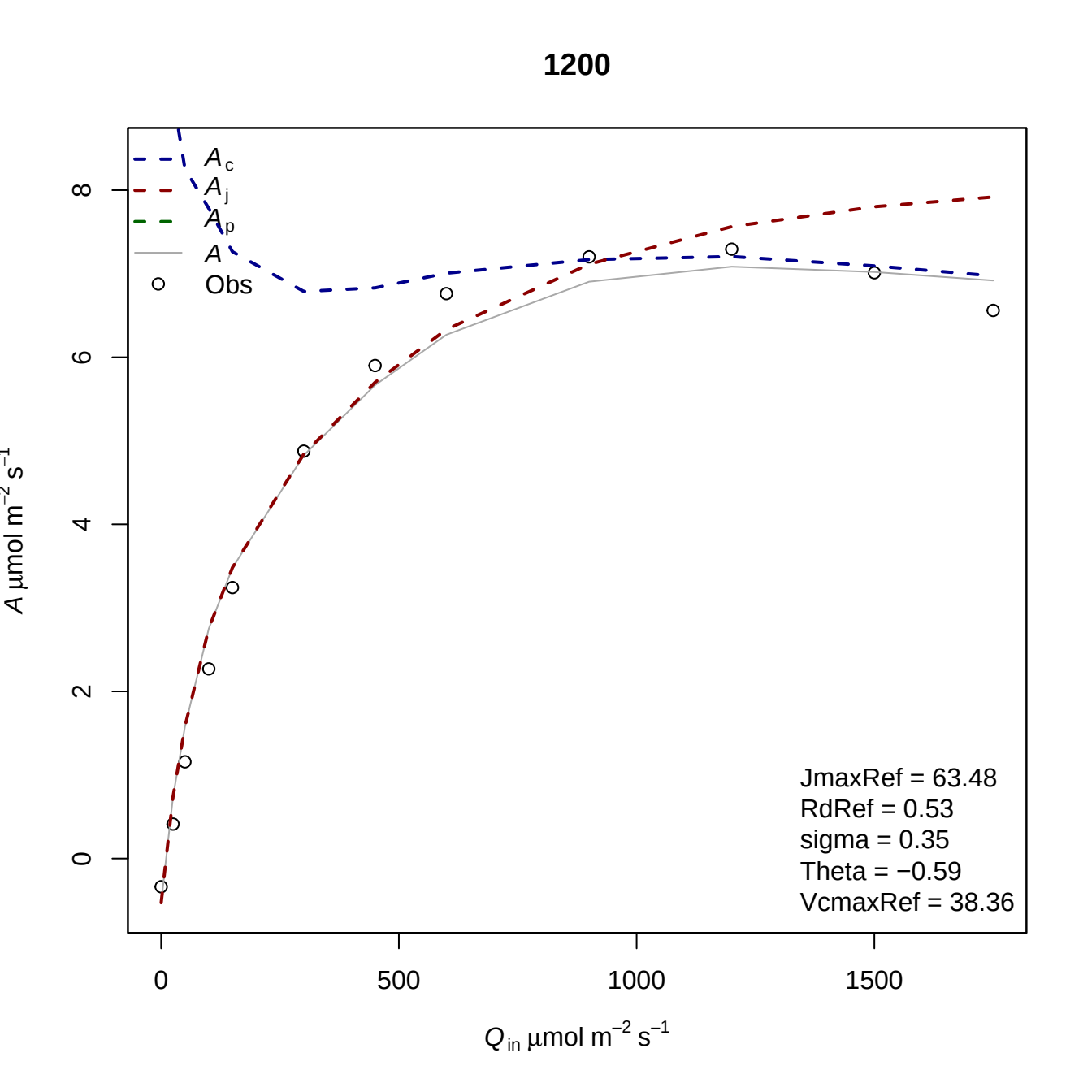

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 100.53
RdRef = 0.91
sigma = 0.1
Theta = 0.61
VcmaxRef = 143.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

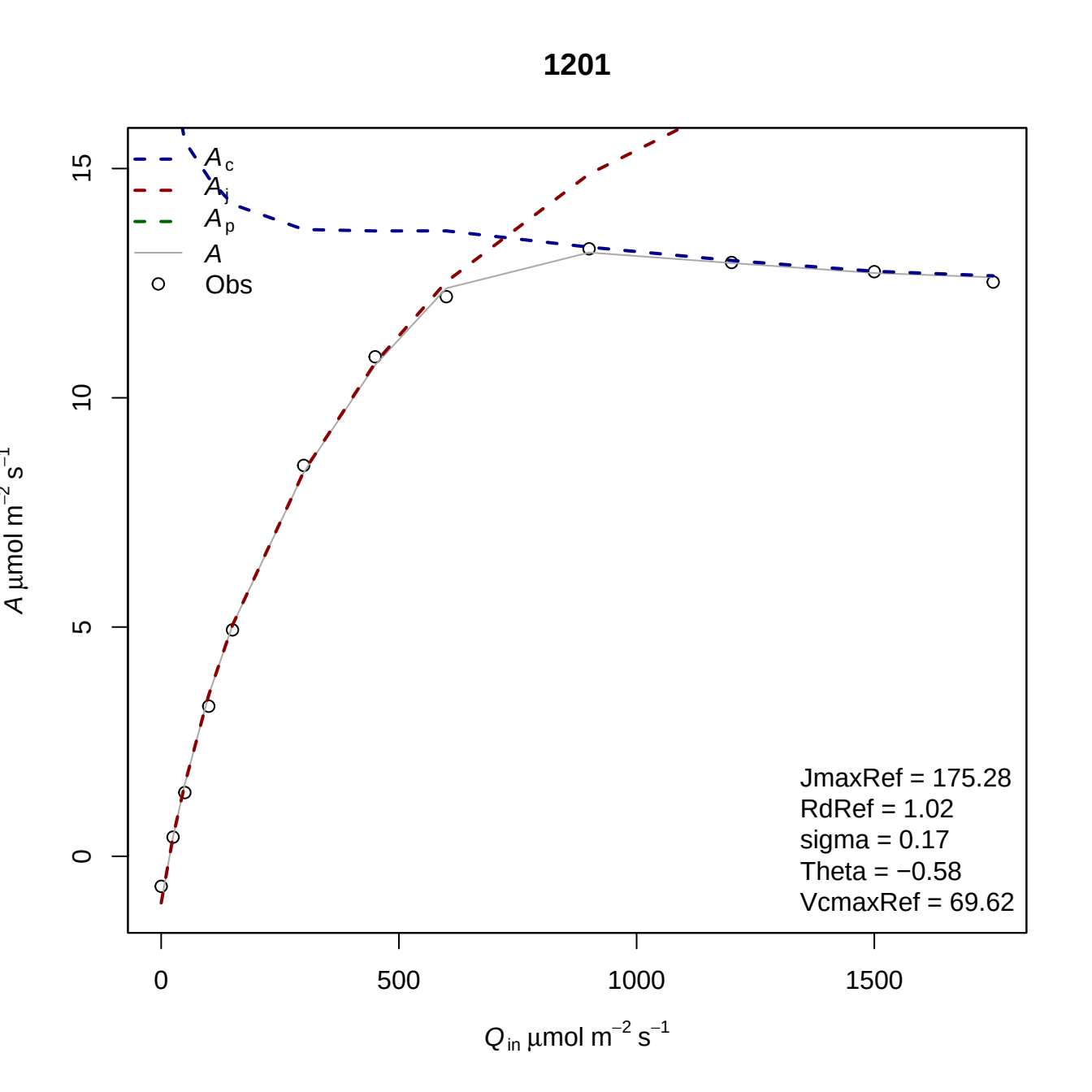

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 137.46
RdRef = 1.29
sigma = 0.17
Theta = 0.64
VcmaxRef = 1981085.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

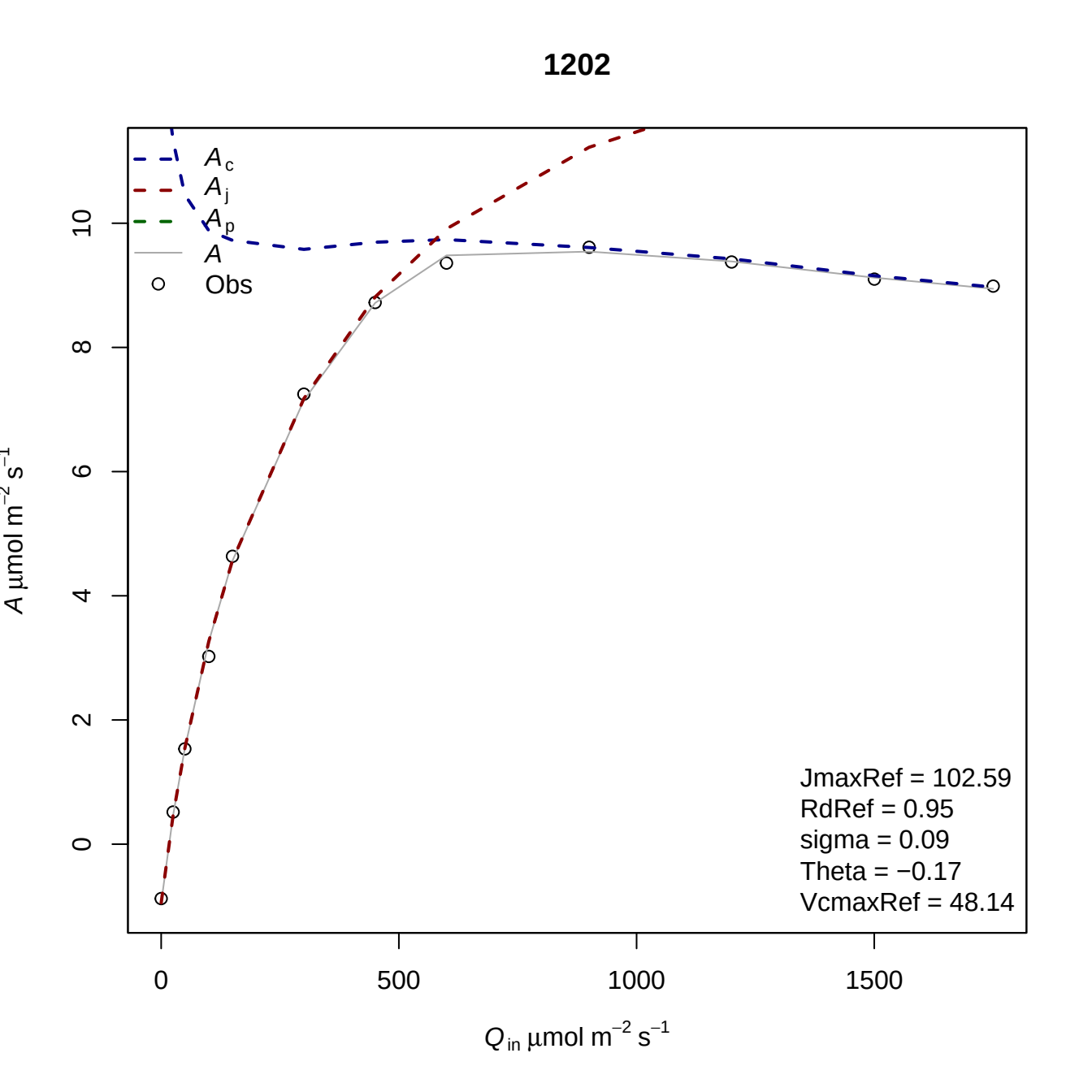

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 137.42
RdRef = 1.17
sigma = 0.08
Theta = 0.74
VcmaxRef = 142698.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

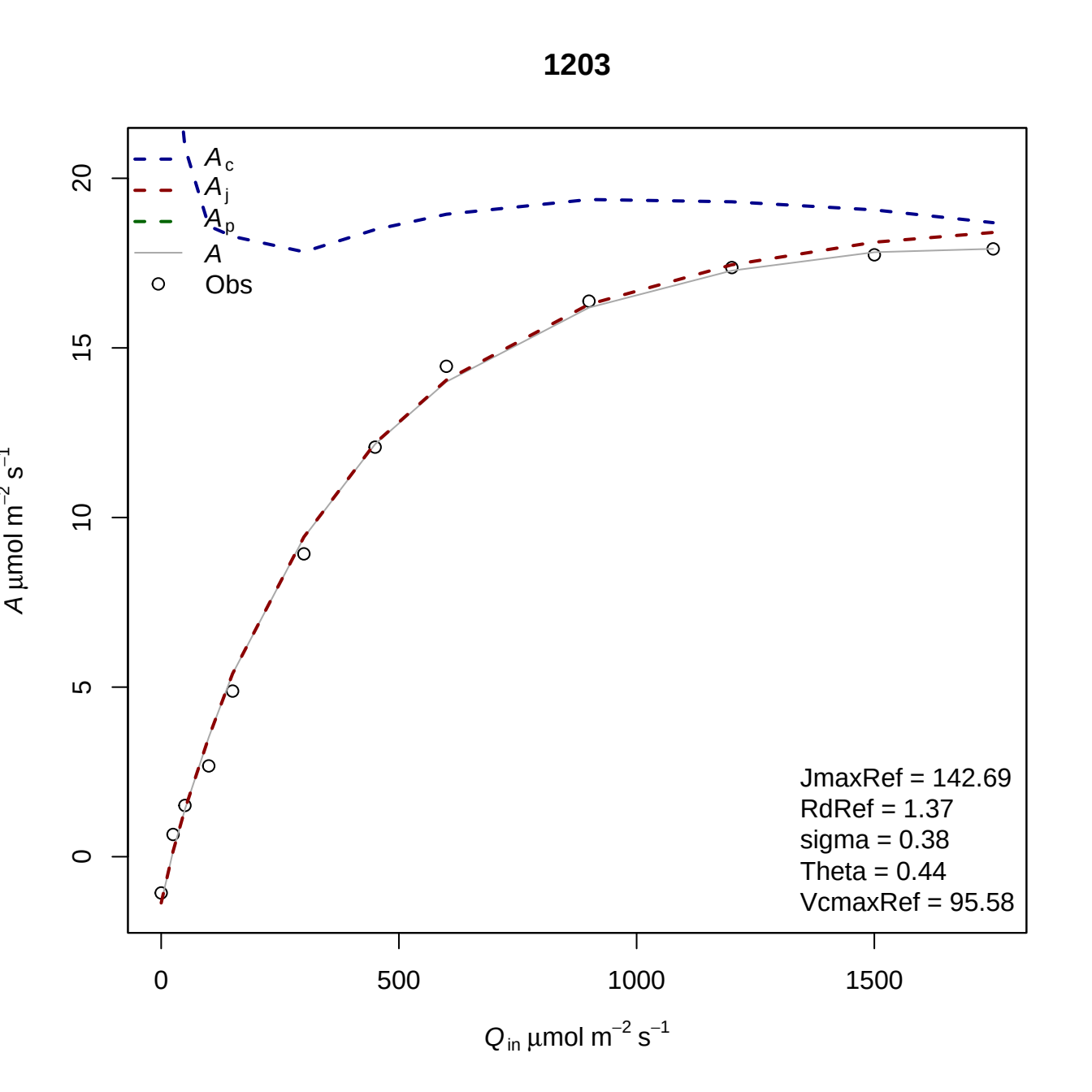

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 149.27
RdRef = 1.47
sigma = 0.19
Theta = 0.7
VcmaxRef = 7321468.36
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

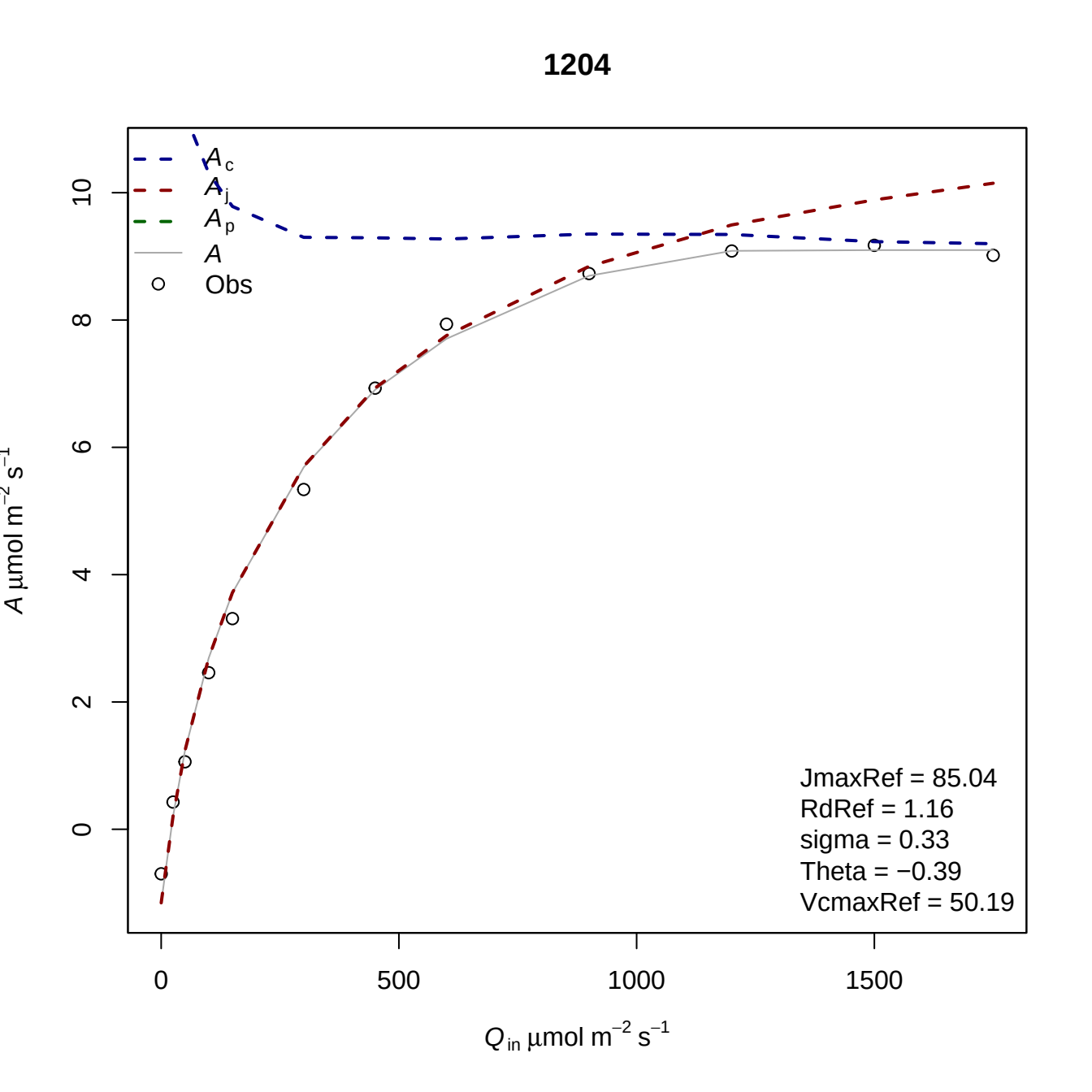

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 102.26
RdRef = 1.13
sigma = 0.2
Theta = −1.78
VcmaxRef = 58.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

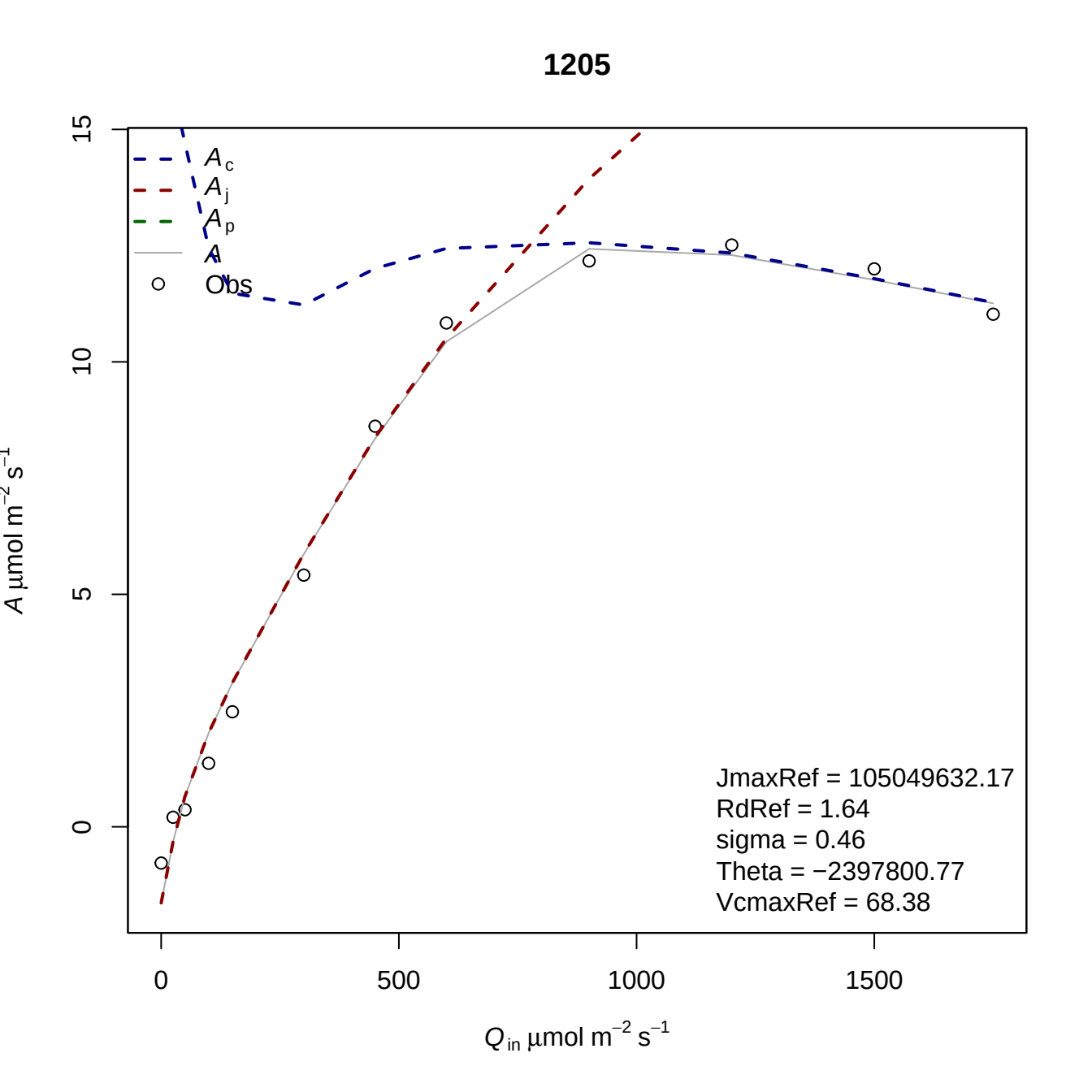

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 63.48
RdRef = 0.53
sigma = 0.35
Theta = −0.59
VcmaxRef = 38.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

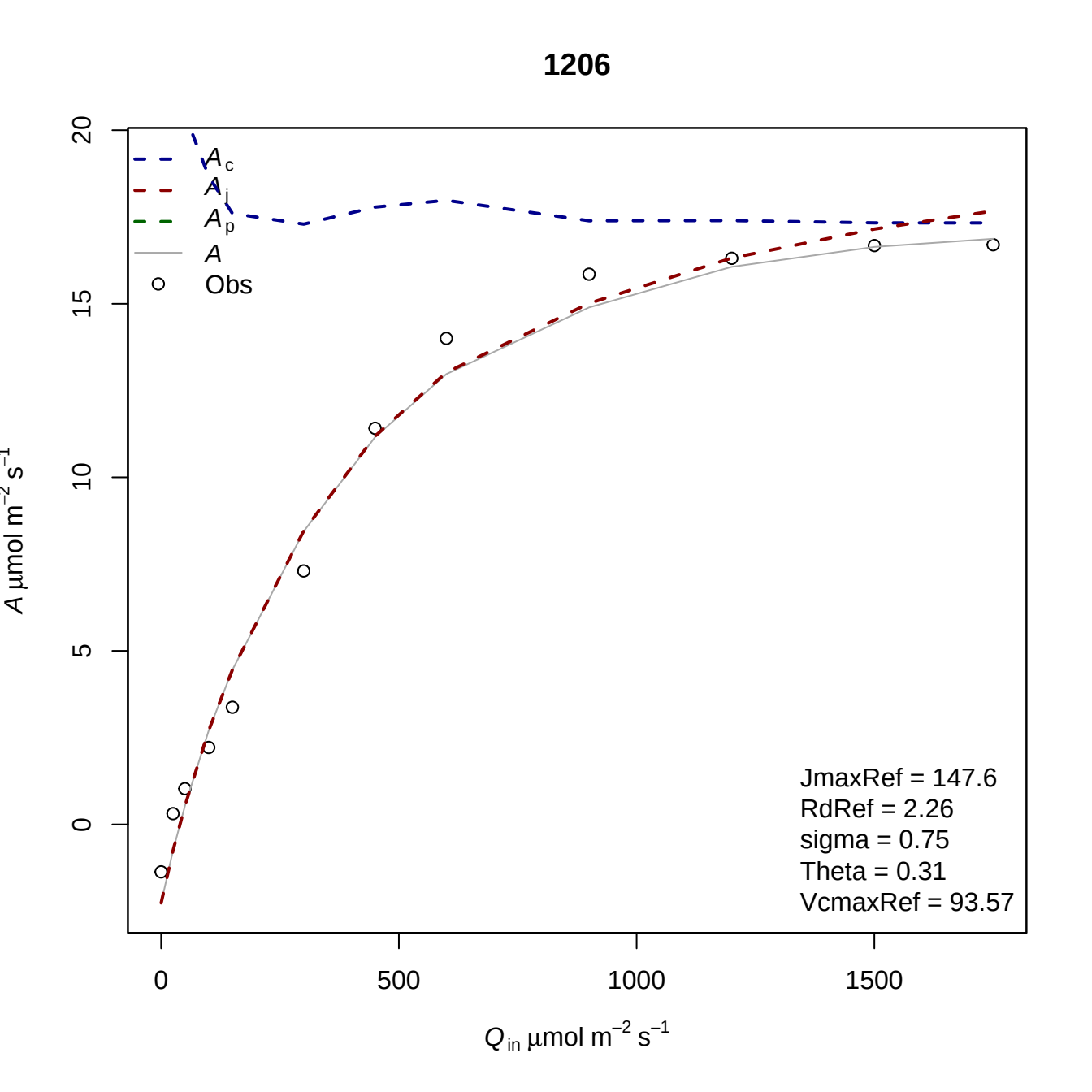

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 175.28
RdRef = 1.02
sigma = 0.17
Theta = −0.58
VcmaxRef = 69.62
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

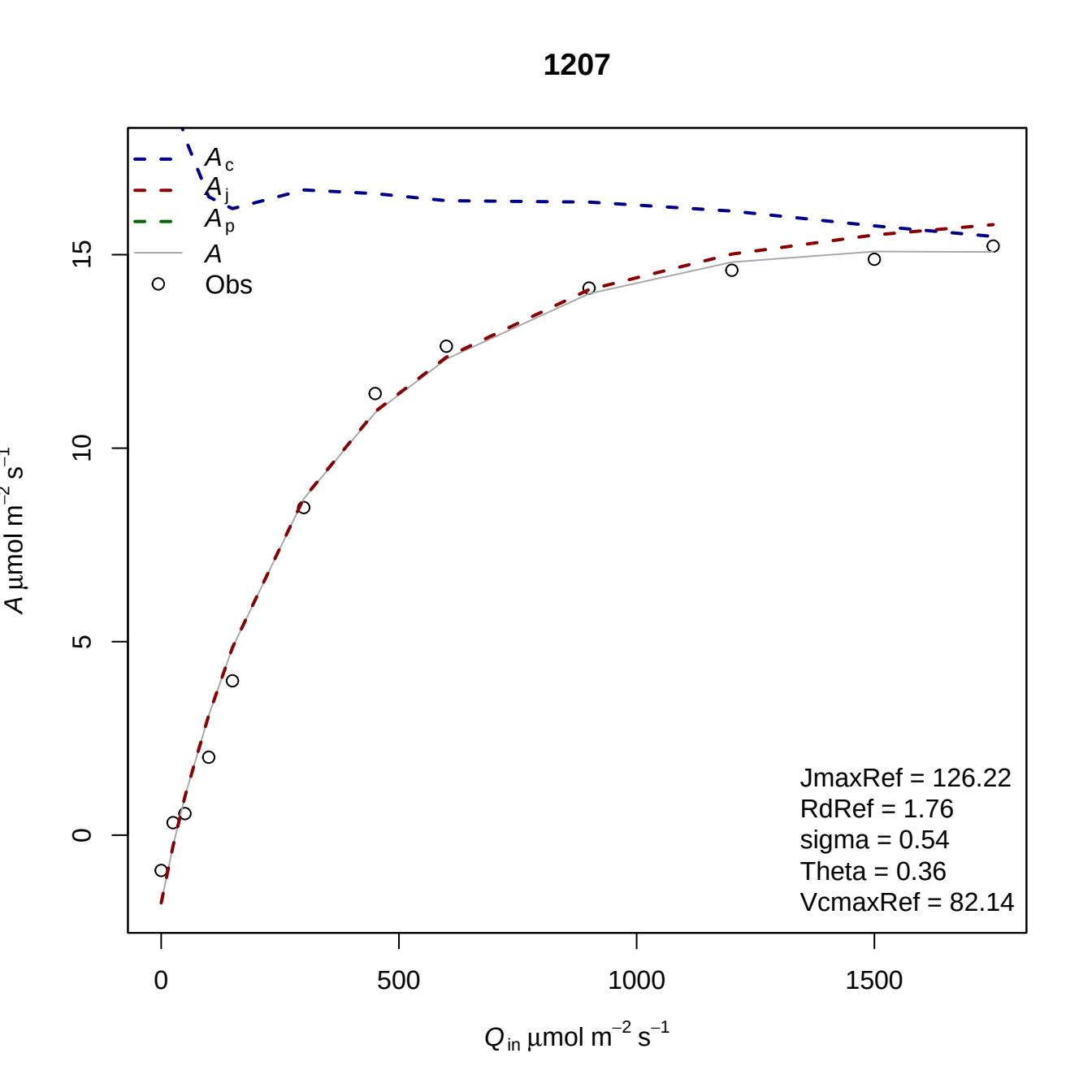

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 102.59
RdRef = 0.95
sigma = 0.09
Theta = −0.17
VcmaxRef = 48.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

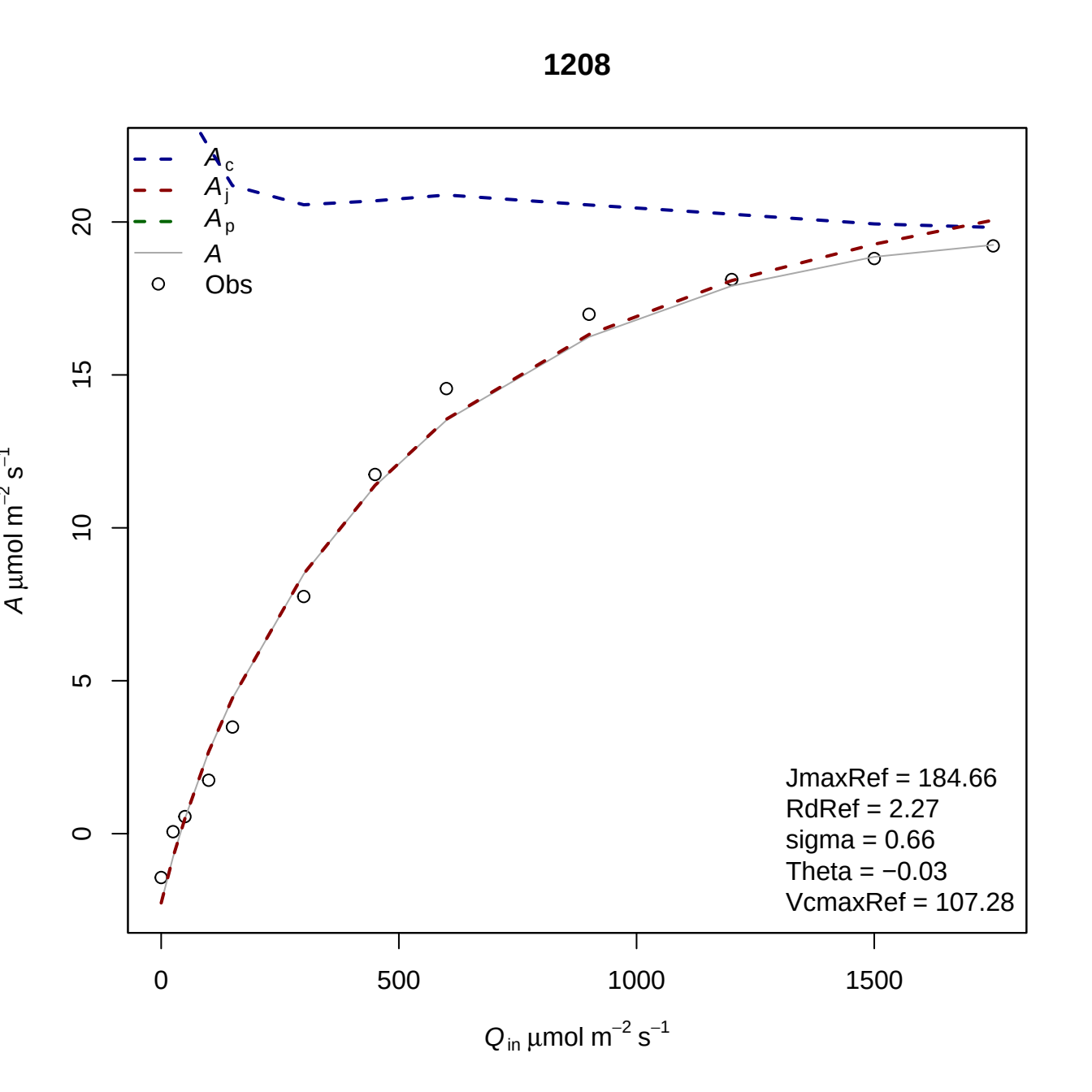

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 142.69
RdRef = 1.37
sigma = 0.38
Theta = 0.44
VcmaxRef = 95.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

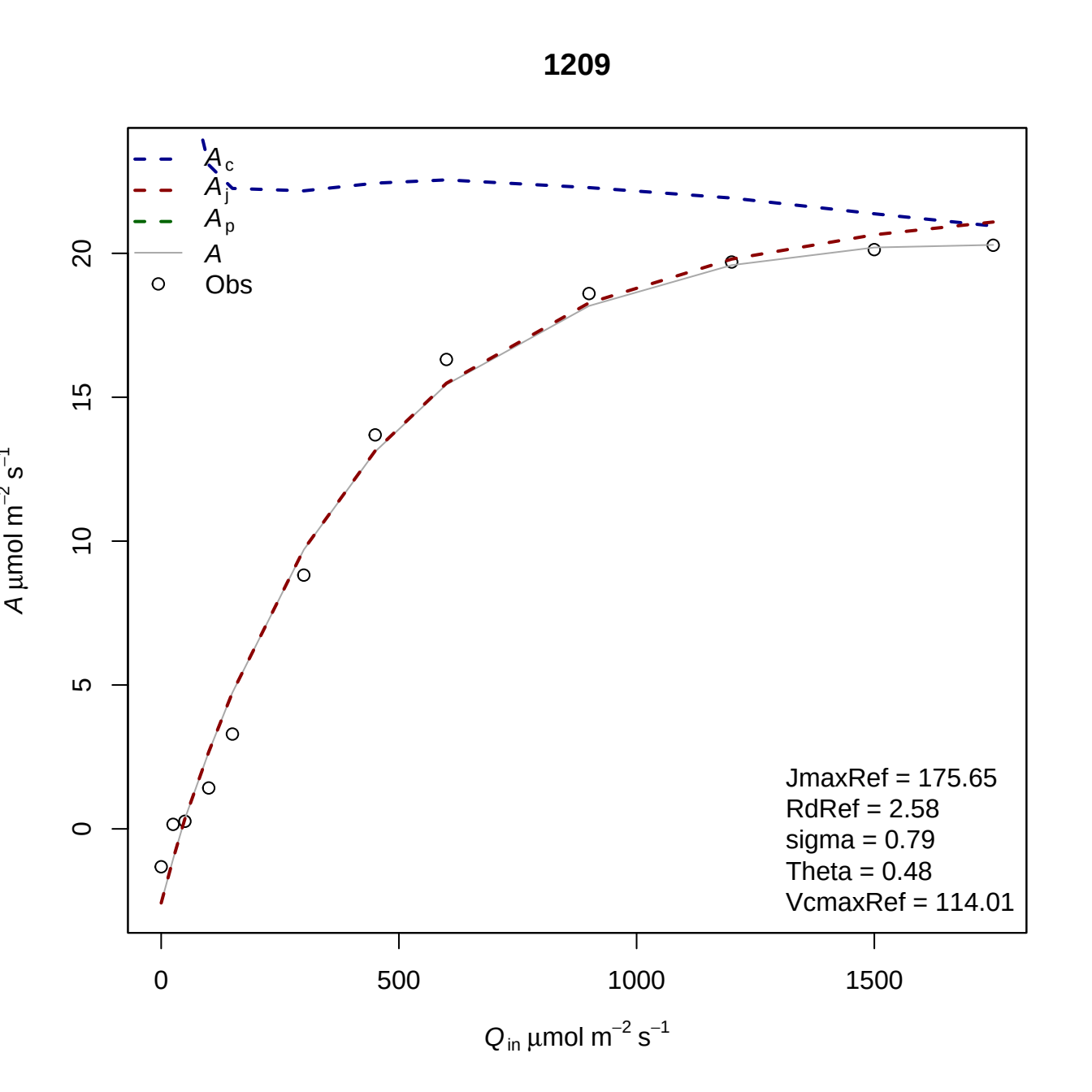

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 85.04
RdRef = 1.16
sigma = 0.33
Theta = −0.39
VcmaxRef = 50.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

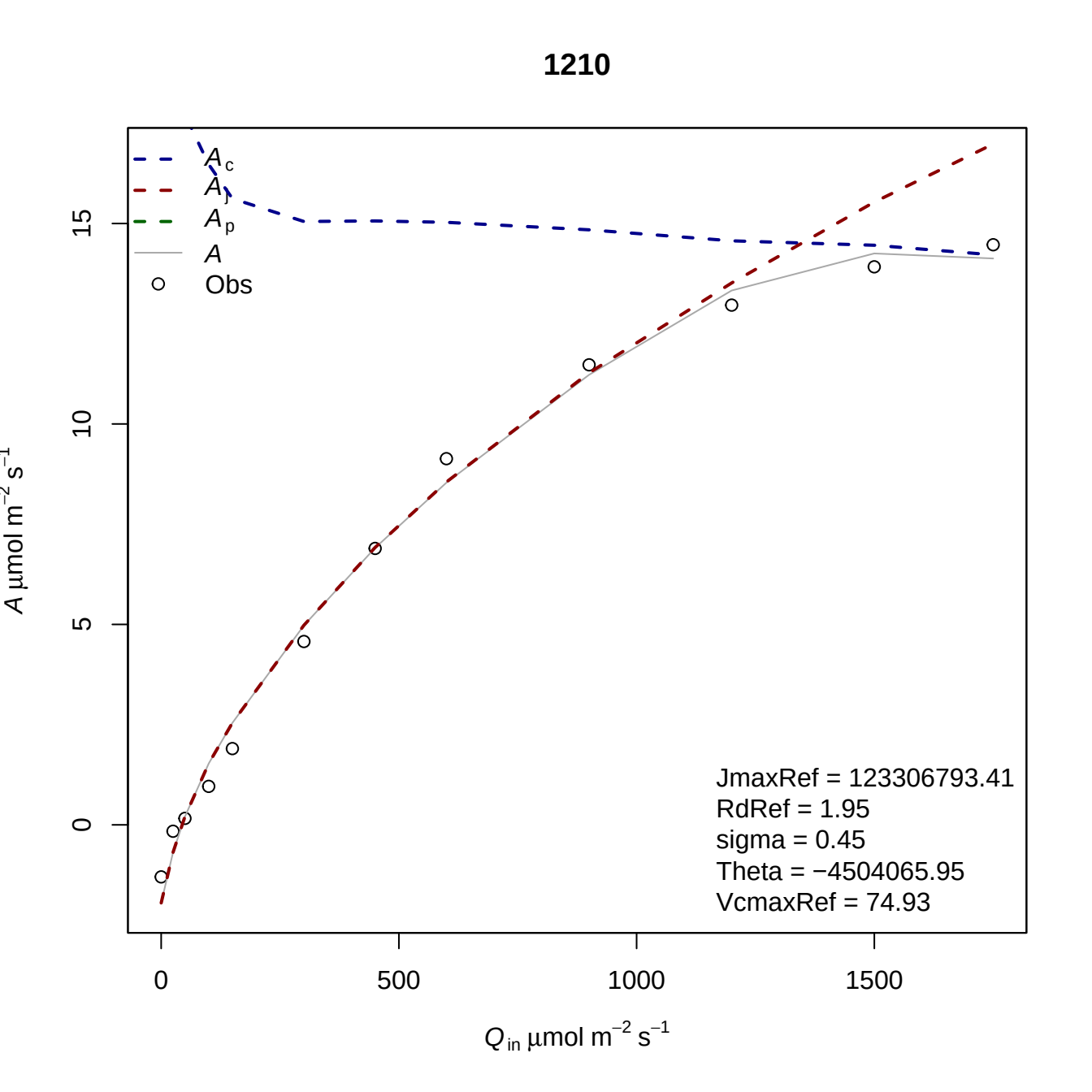

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 105049632.17
RdRef = 1.64
sigma = 0.46
Theta = −2397800.77
VcmaxRef = 68.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

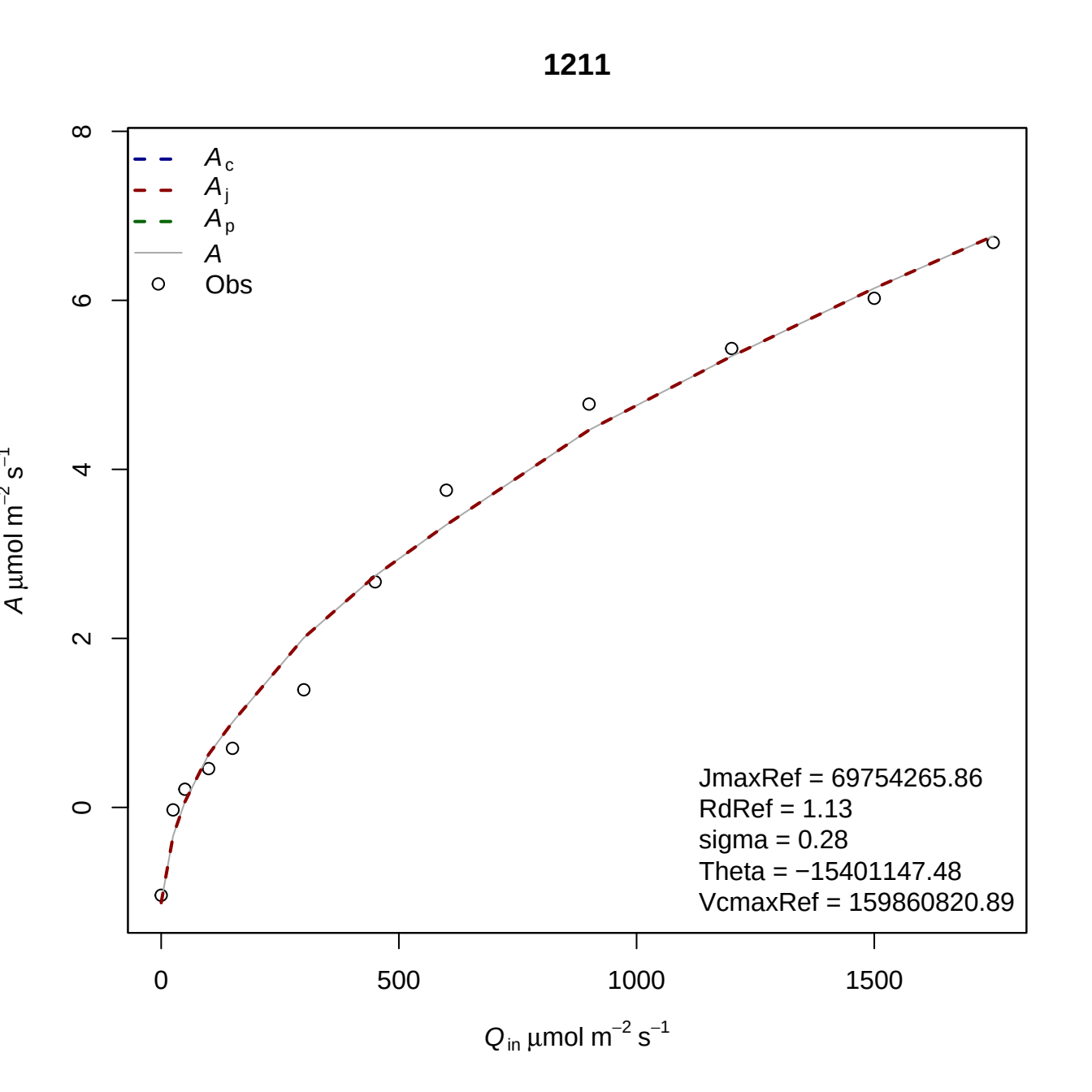

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 147.6
RdRef = 2.26
sigma = 0.75
Theta = 0.31
VcmaxRef = 93.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

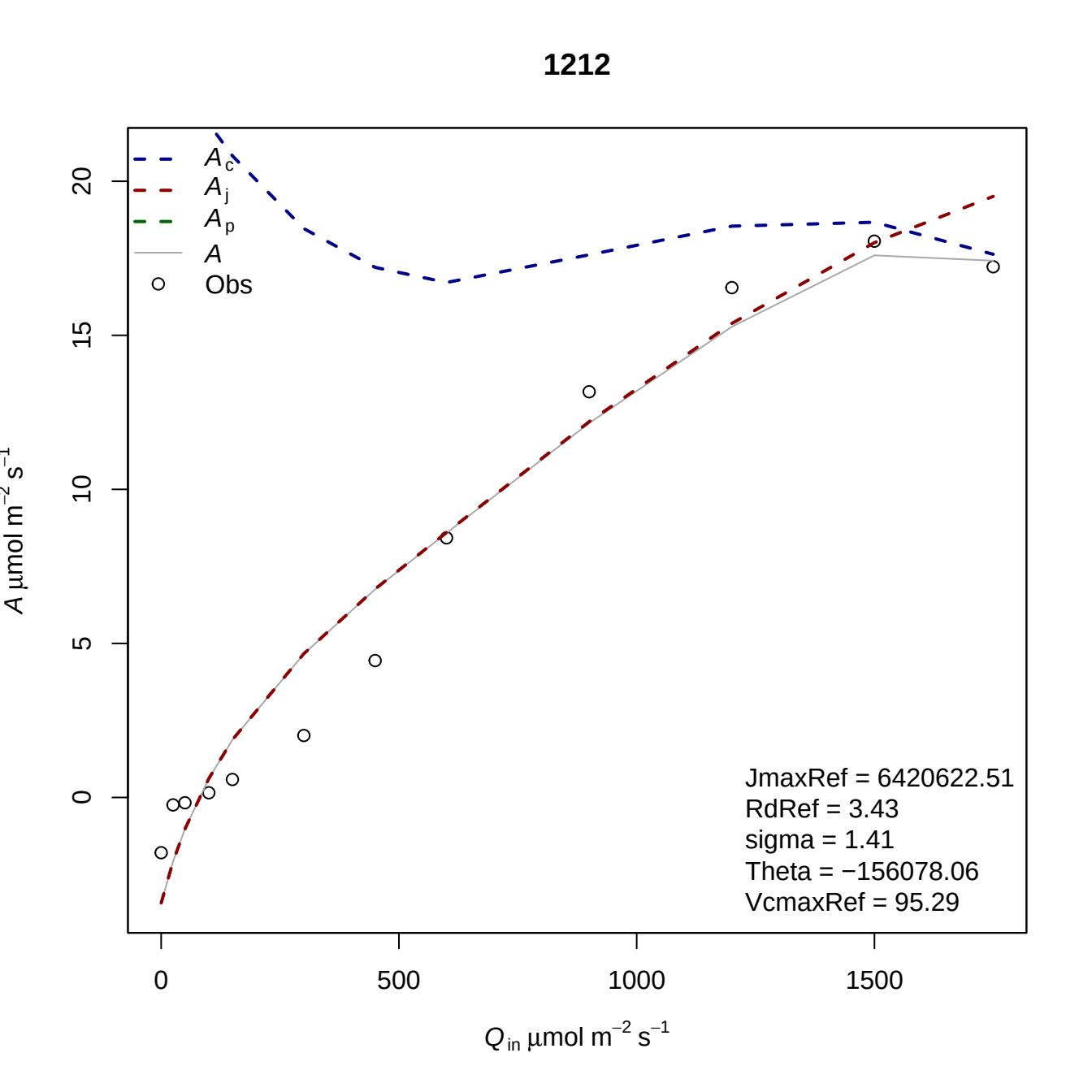

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 126.22
RdRef = 1.76
sigma = 0.54
Theta = 0.36
VcmaxRef = 82.14
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

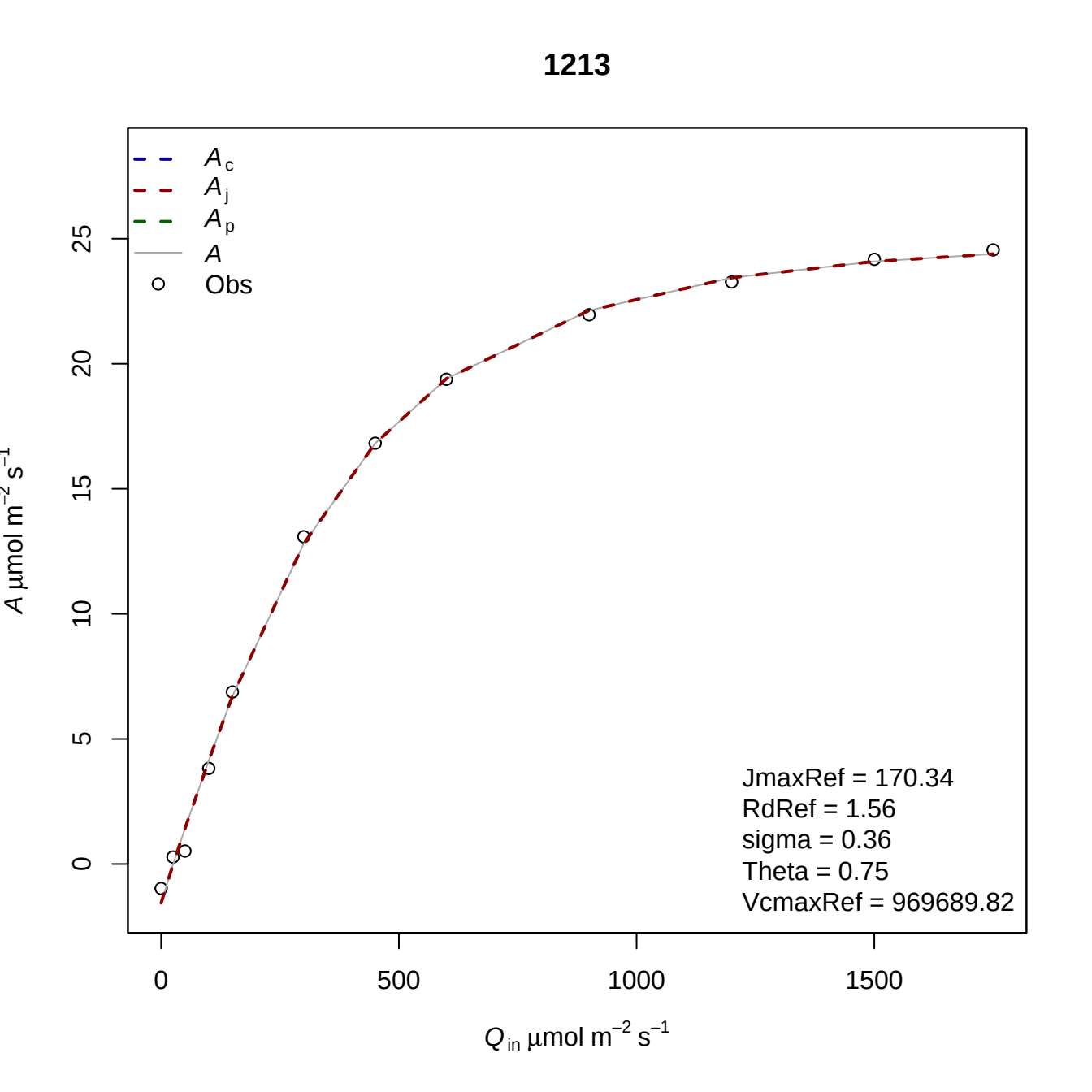

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 184.66
RdRef = 2.27
sigma = 0.66
Theta = −0.03
VcmaxRef = 107.28
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

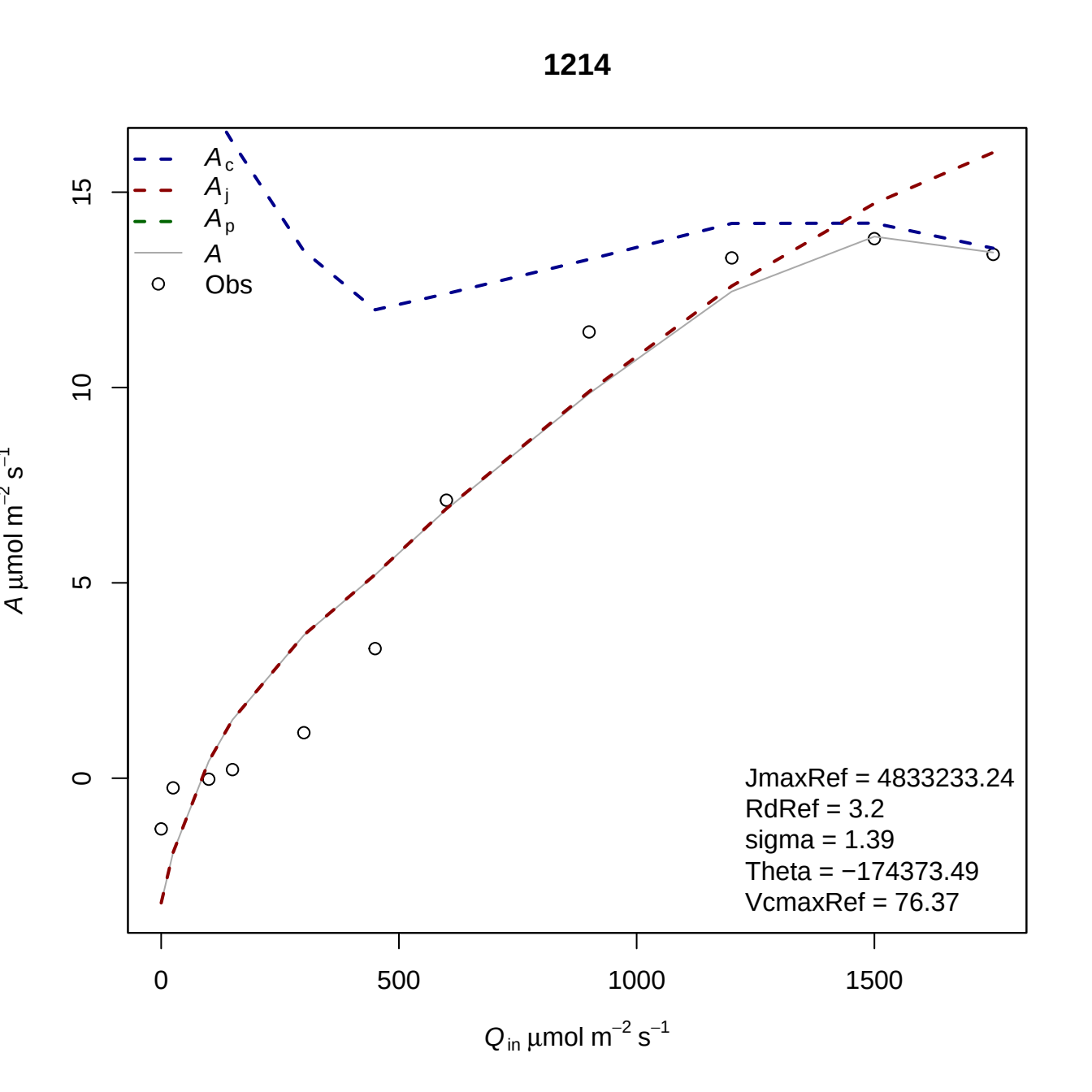

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 175.65
RdRef = 2.58
sigma = 0.79
Theta = 0.48
VcmaxRef = 114.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

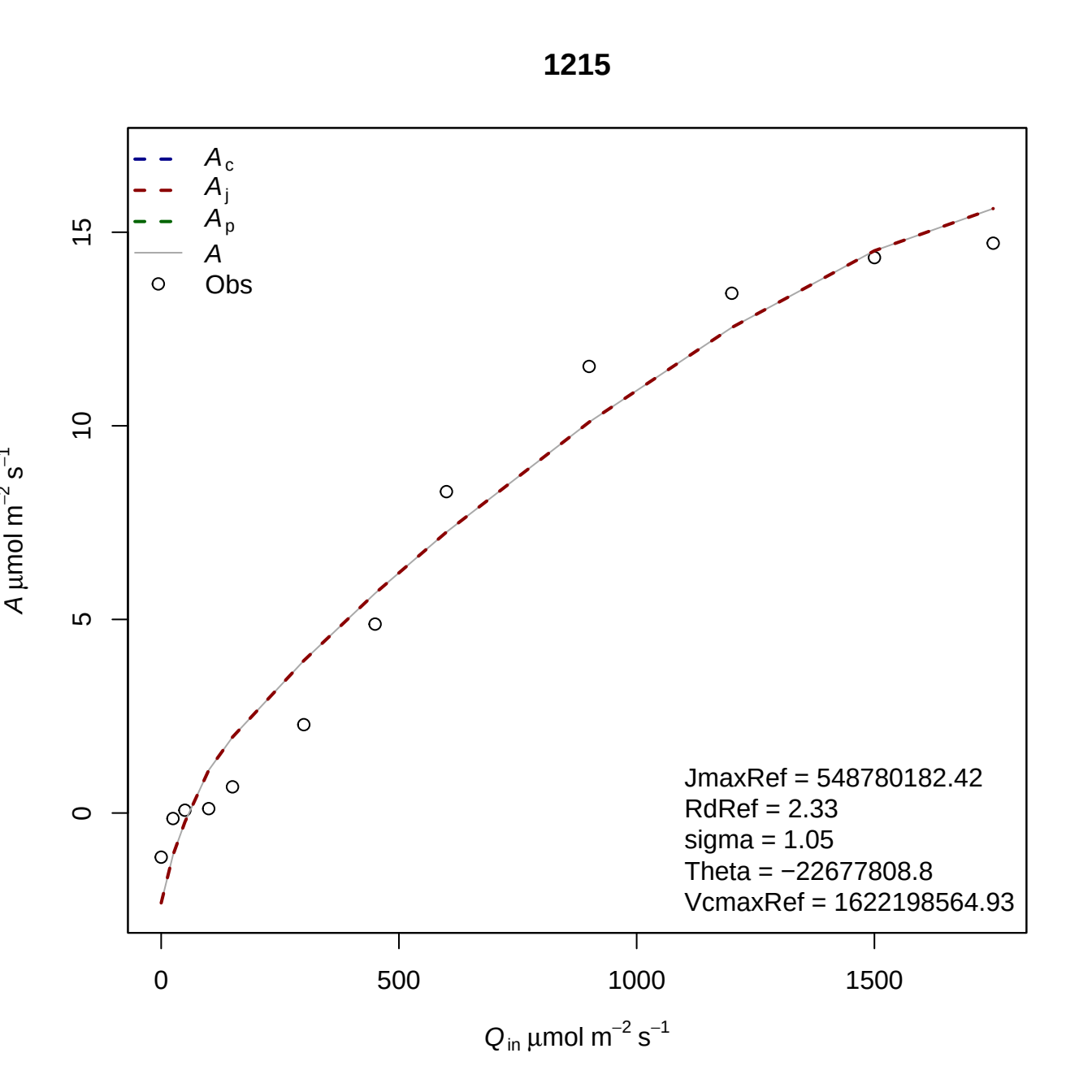

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 123306793.41
RdRef = 1.95
sigma = 0.45
Theta = −4504065.95
VcmaxRef = 74.93
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

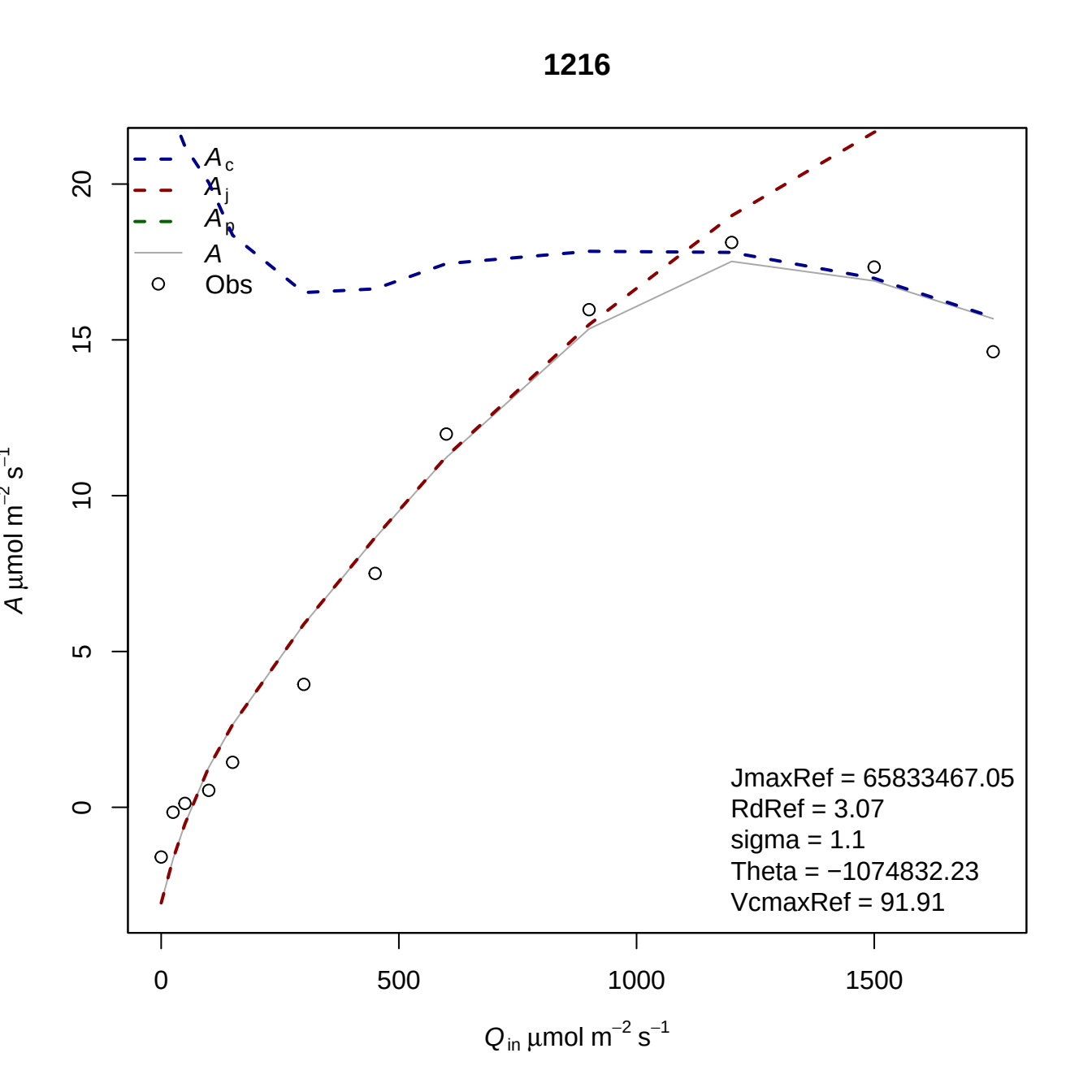

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 69754265.86
RdRef = 1.13
sigma = 0.28
Theta = −15401147.48
VcmaxRef = 159860820.89
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

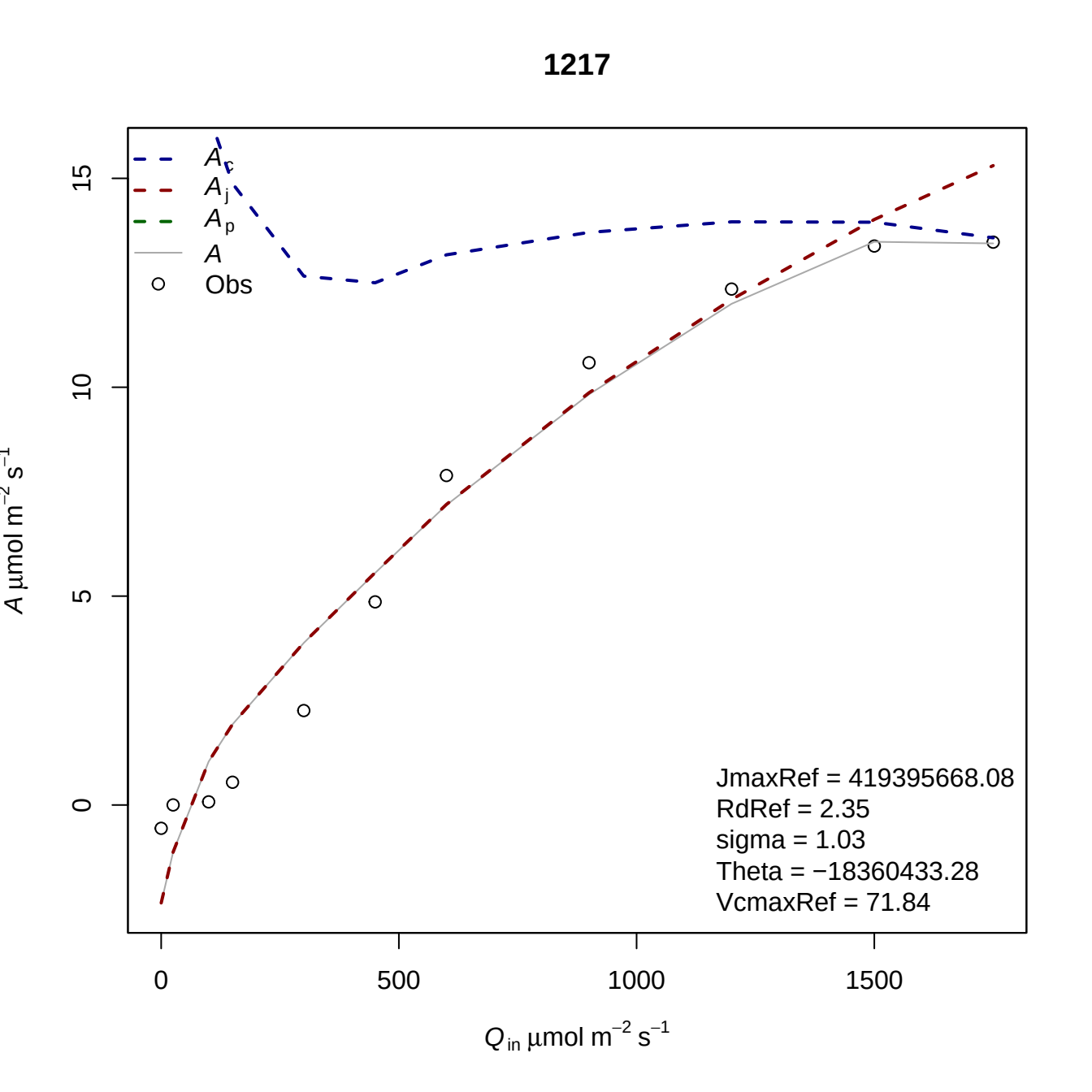

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 6420622.51
RdRef = 3.43
sigma = 1.41
Theta = −156078.06
VcmaxRef = 95.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

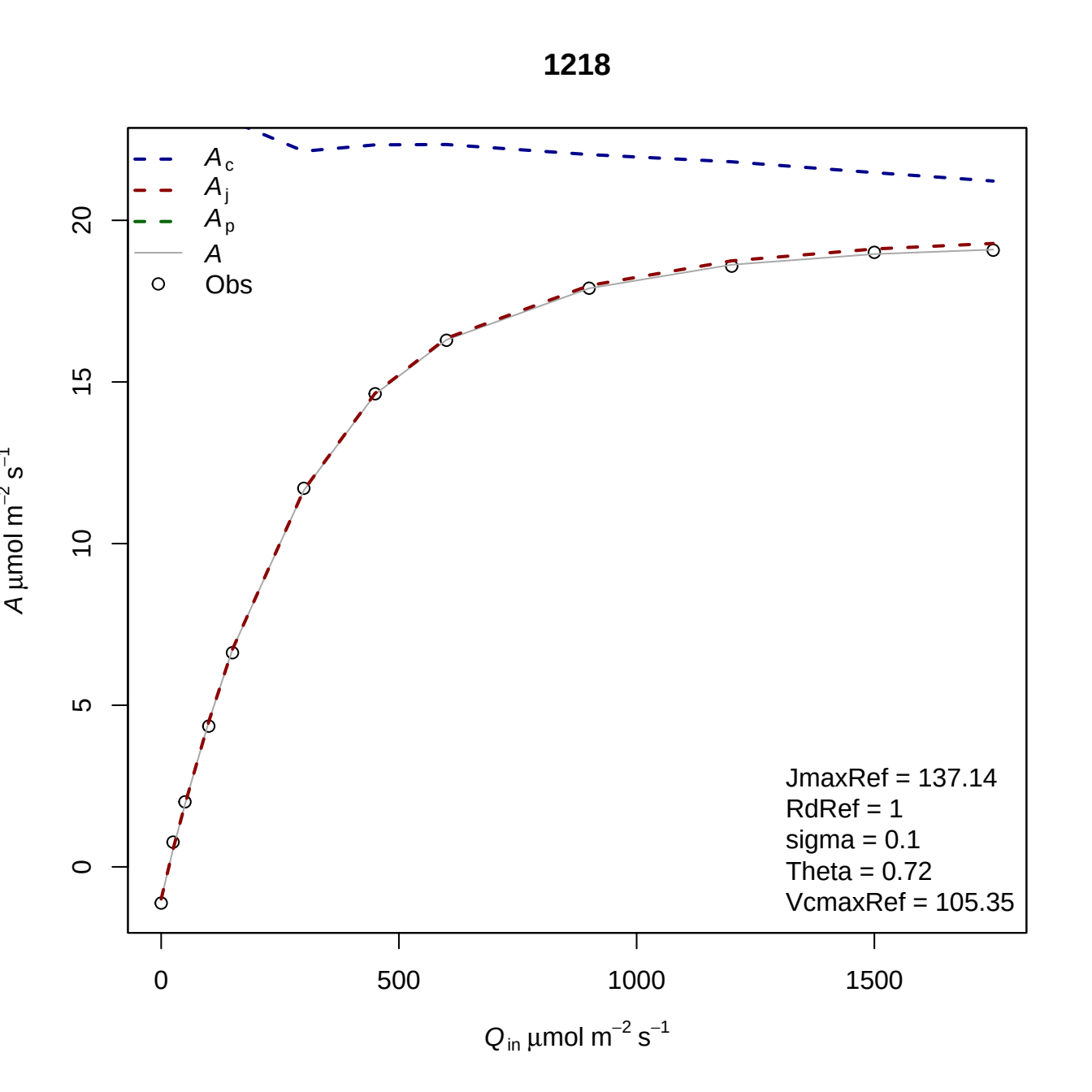

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 170.34
RdRef = 1.56
sigma = 0.36
Theta = 0.75
VcmaxRef = 969689.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

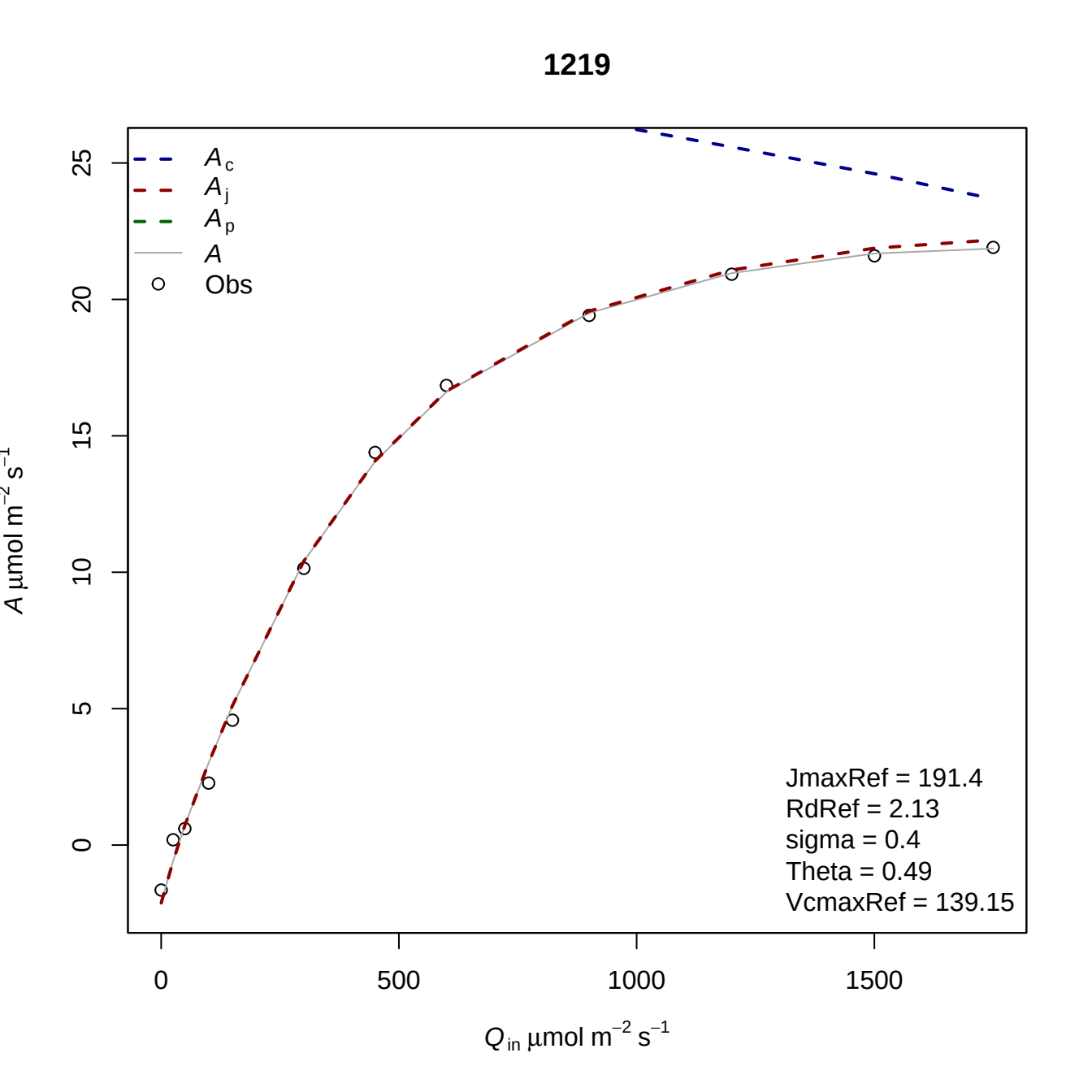

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 4833233.24
RdRef = 3.2
sigma = 1.39
Theta = −174373.49
VcmaxRef = 76.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

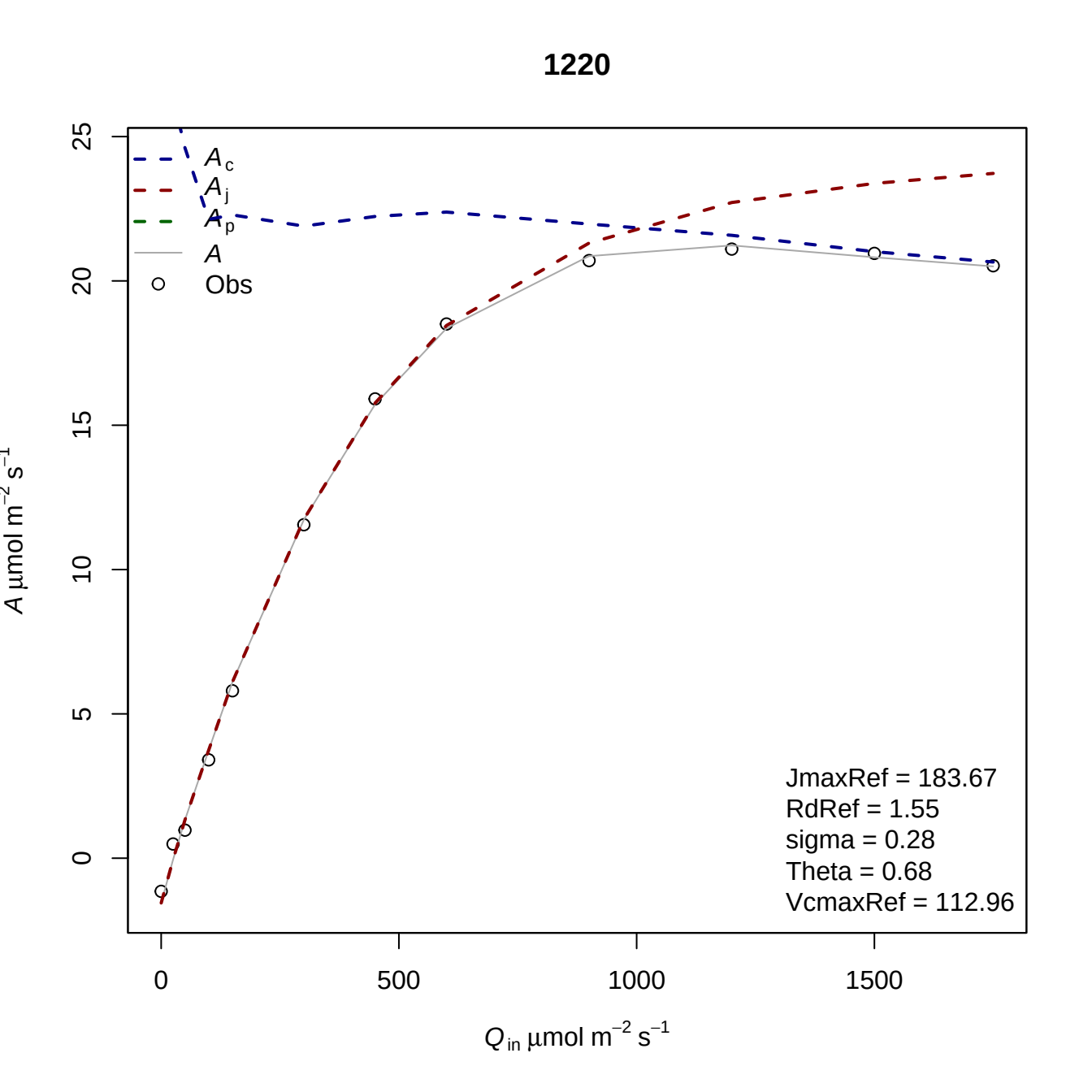

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 548780182.42
RdRef = 2.33
sigma = 1.05
Theta = −22677808.8
VcmaxRef = 1622198564.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

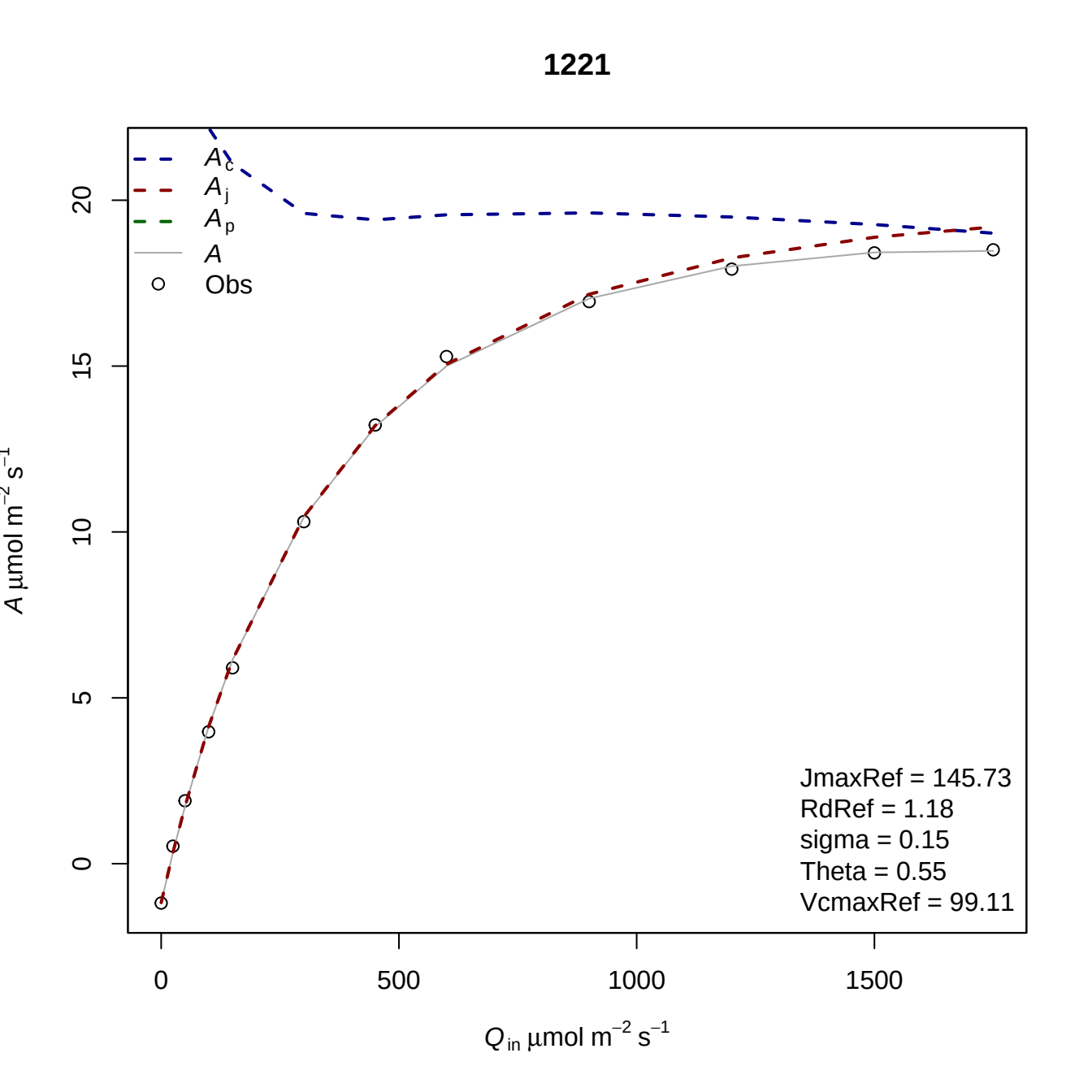

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 65833467.05
RdRef = 3.07
sigma = 1.1
Theta = −1074832.23
VcmaxRef = 91.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

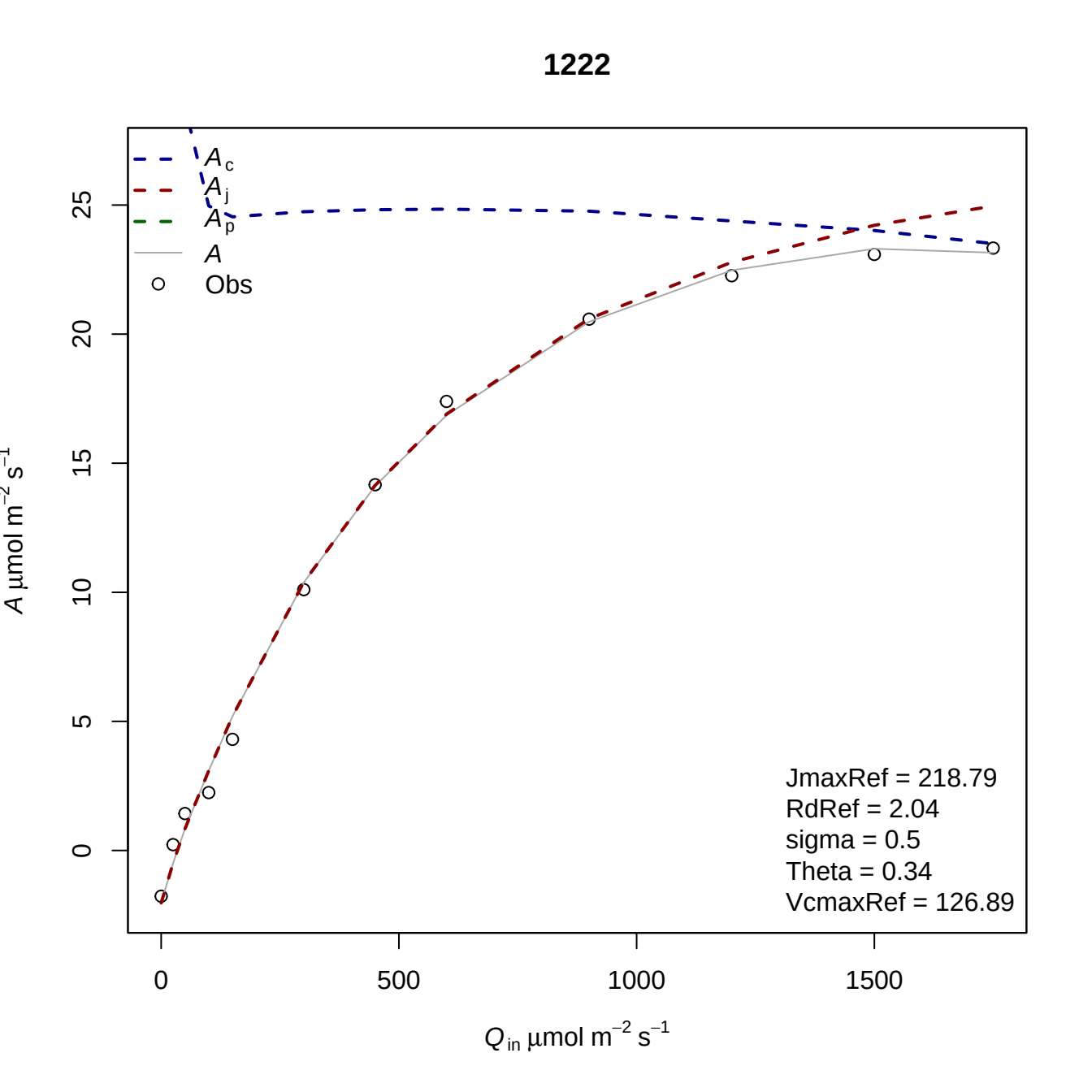

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 419395668.08
RdRef = 2.35
sigma = 1.03
Theta = −18360433.28
VcmaxRef = 71.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

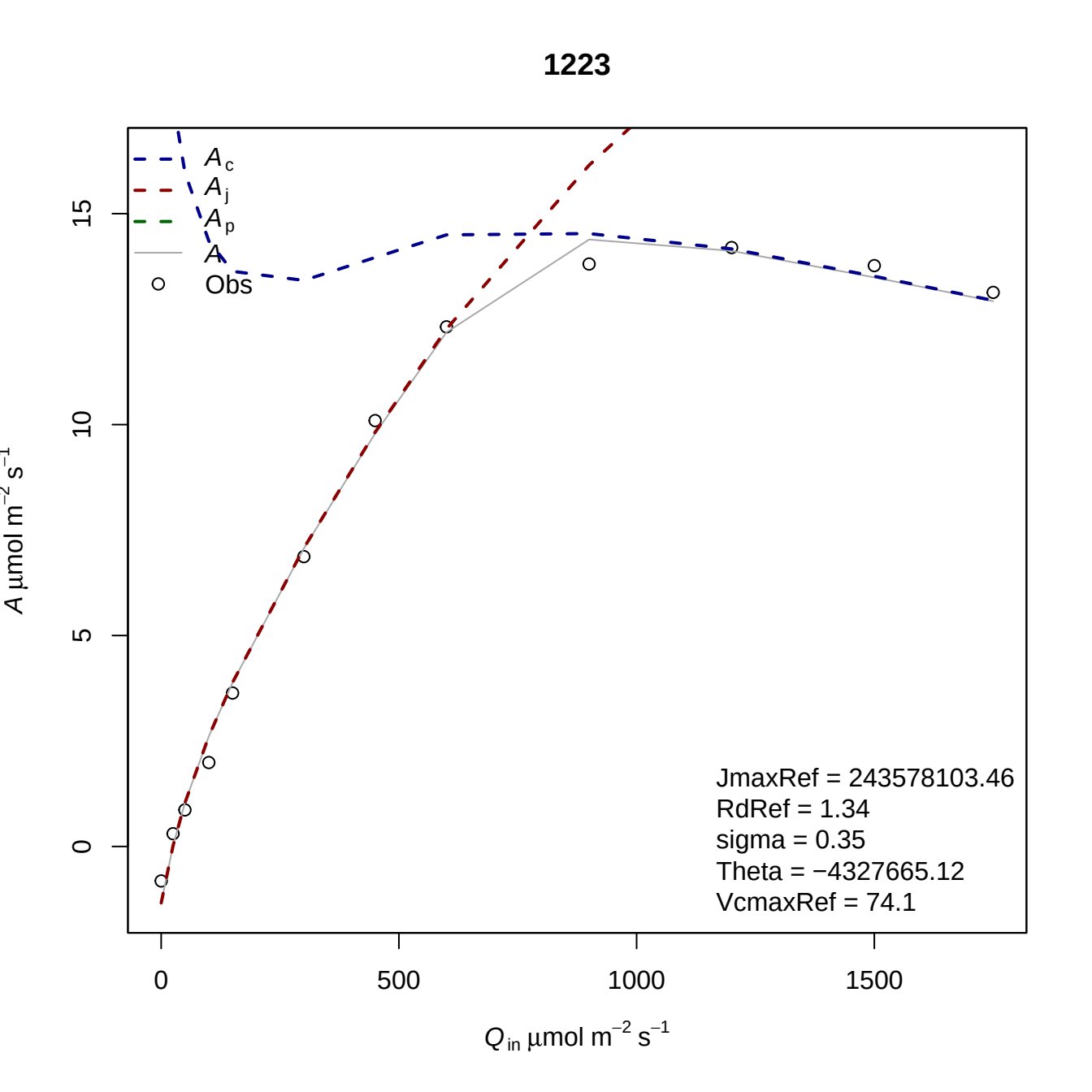

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 137.14
RdRef = 1
sigma = 0.1
Theta = 0.72
VcmaxRef = 105.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

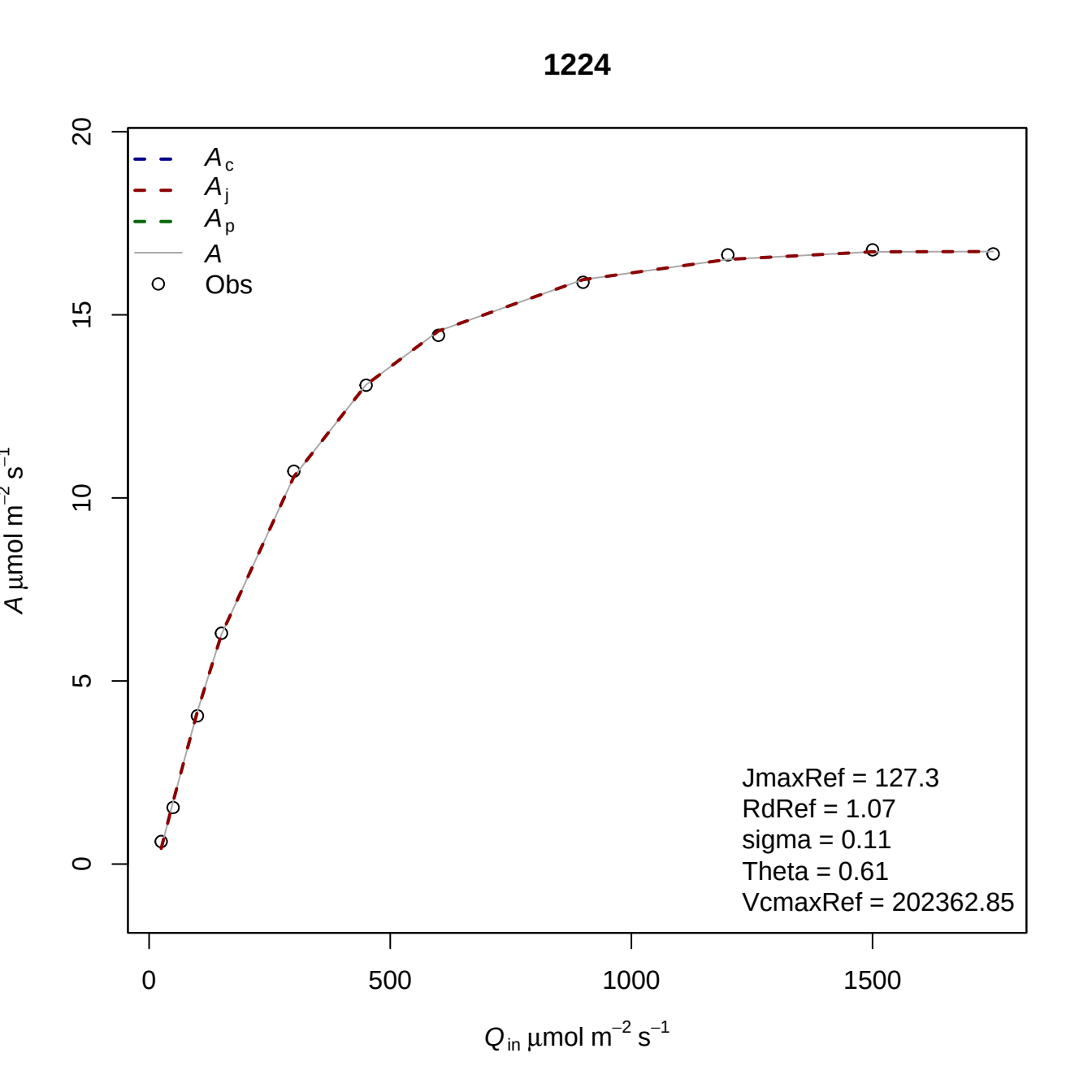

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 191.4
RdRef = 2.13
sigma = 0.4
Theta = 0.49
VcmaxRef = 139.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

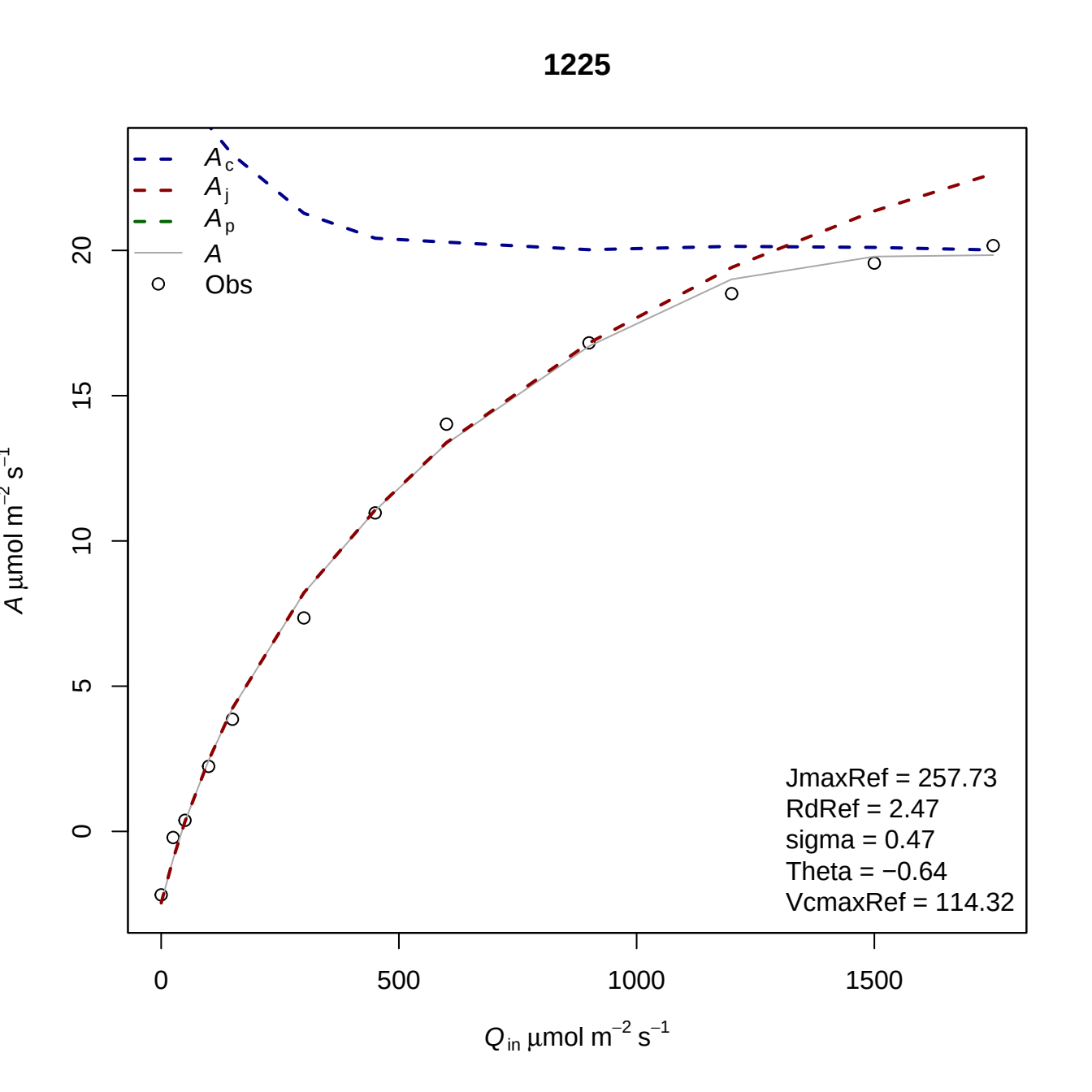

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 183.67
RdRef = 1.55
sigma = 0.28
Theta = 0.68
VcmaxRef = 112.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

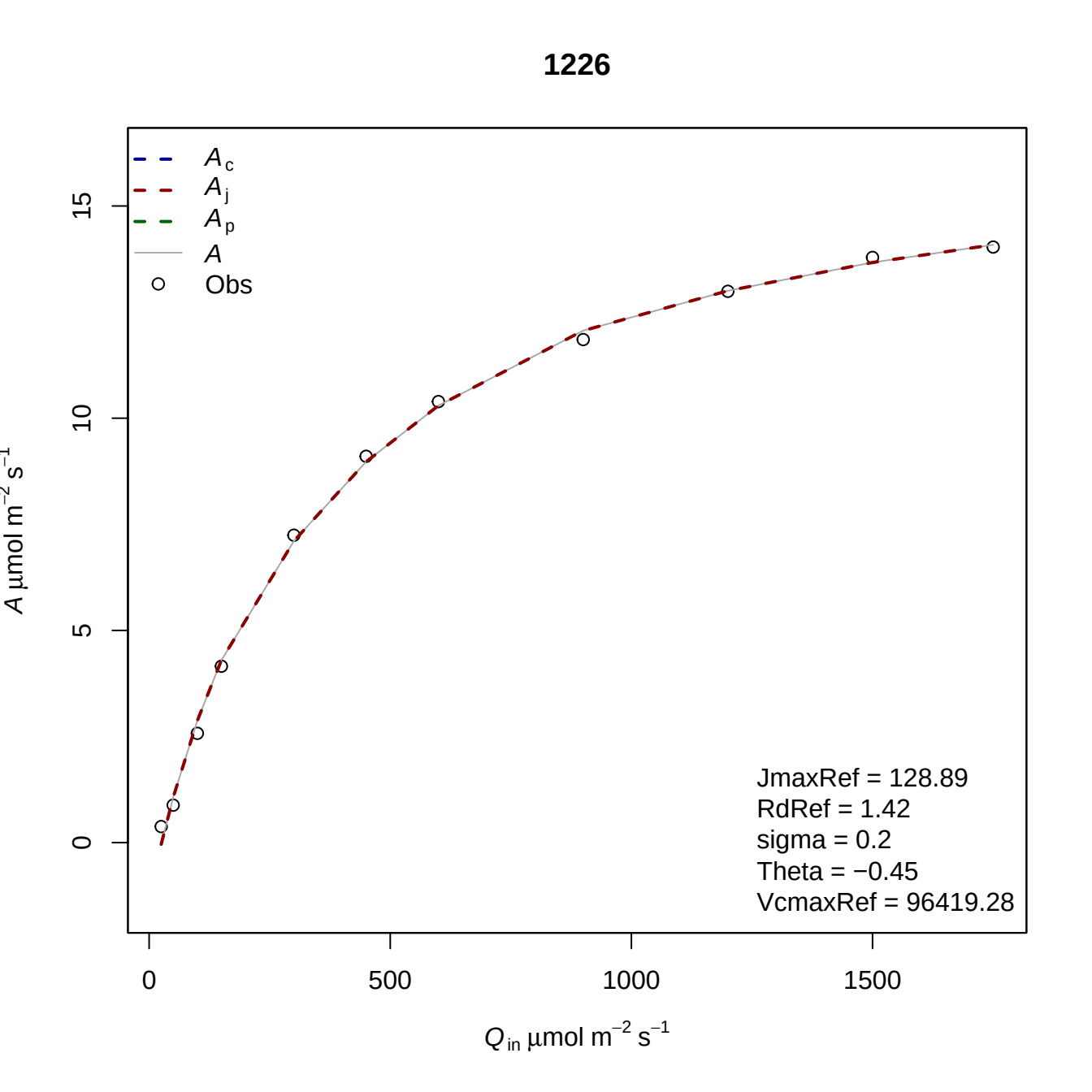

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 145.73
RdRef = 1.18
sigma = 0.15
Theta = 0.55
VcmaxRef = 99.11
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

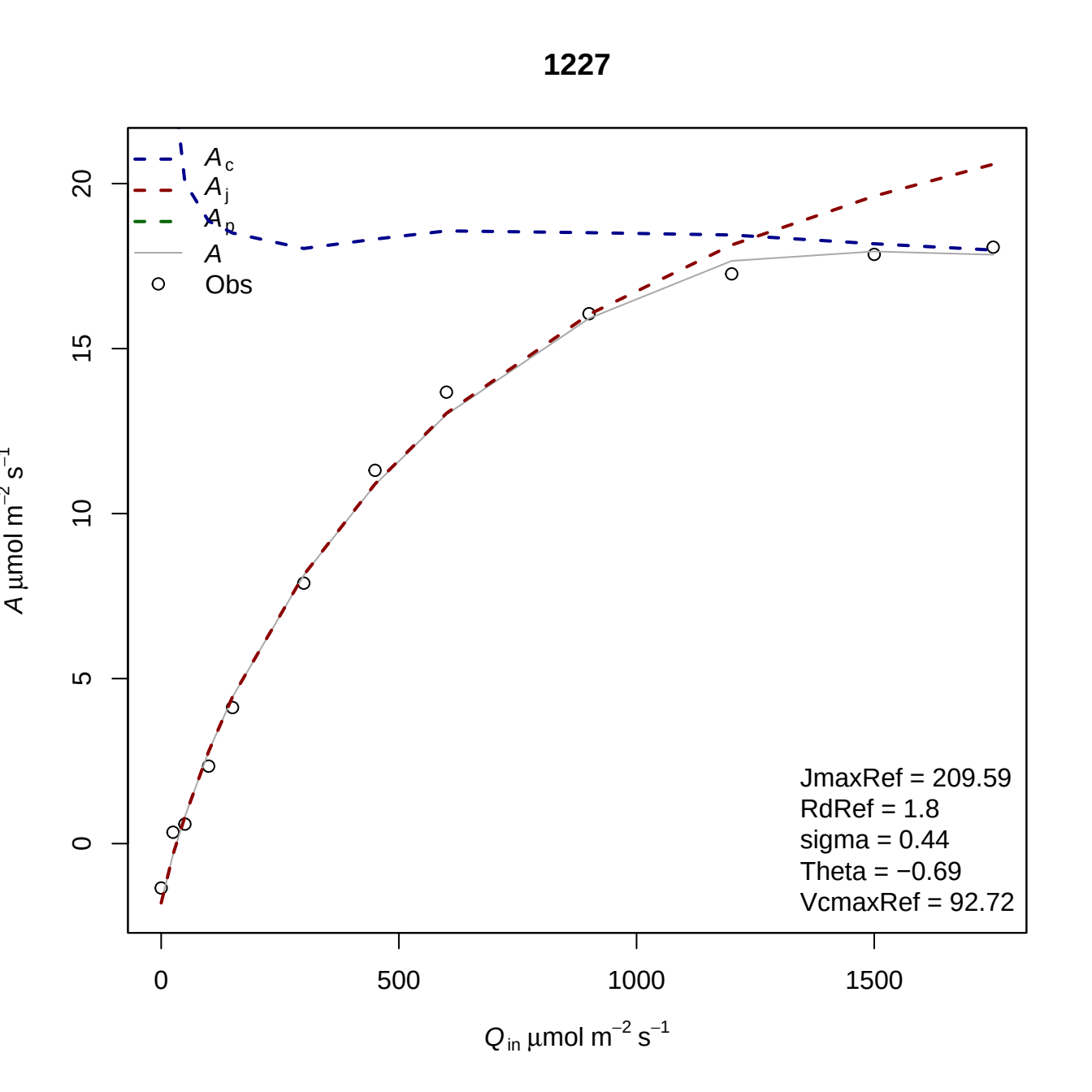

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 218.79
RdRef = 2.04
sigma = 0.5
Theta = 0.34
VcmaxRef = 126.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

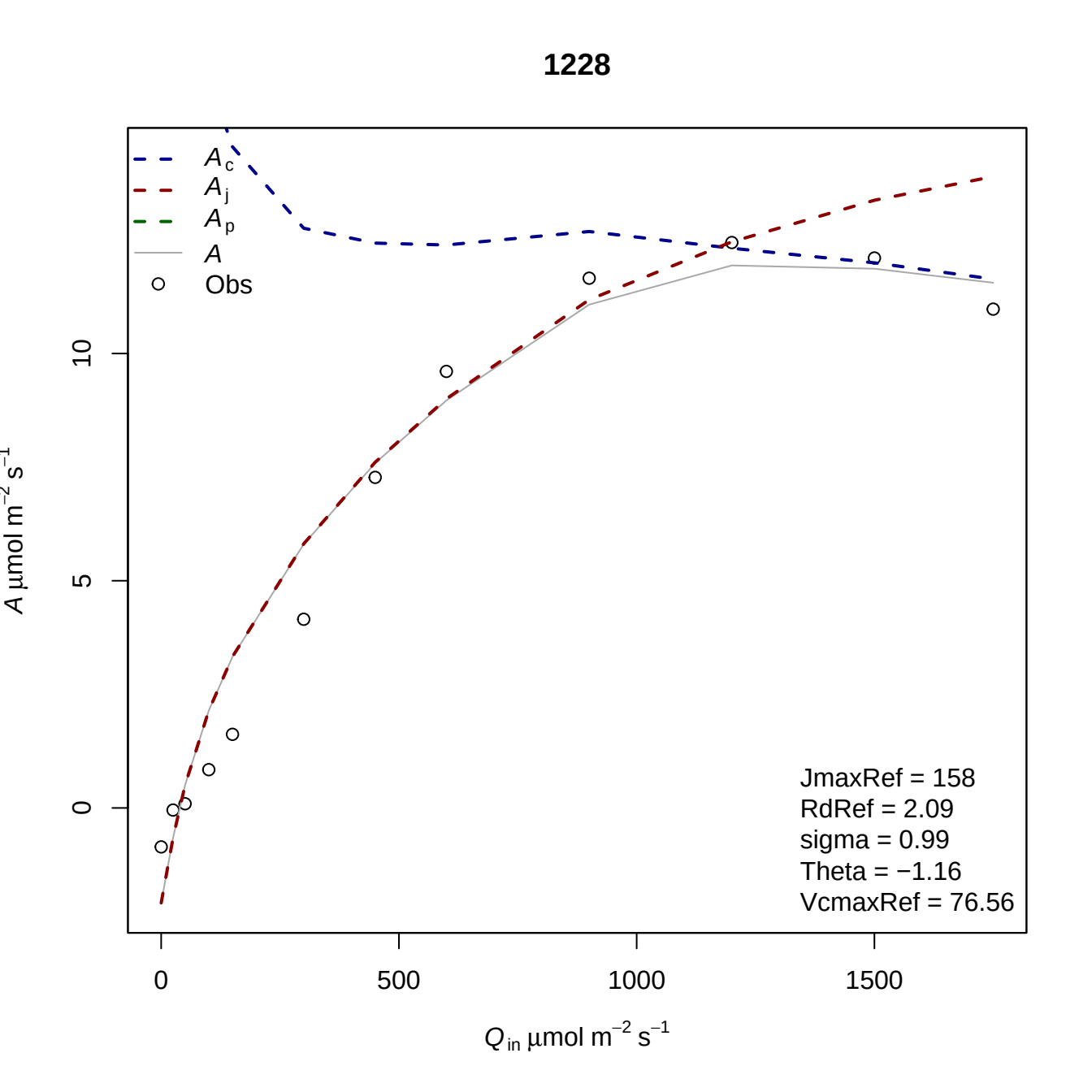

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 243578103.46
RdRef = 1.34
sigma = 0.35
Theta = −4327665.12
VcmaxRef = 74.1
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

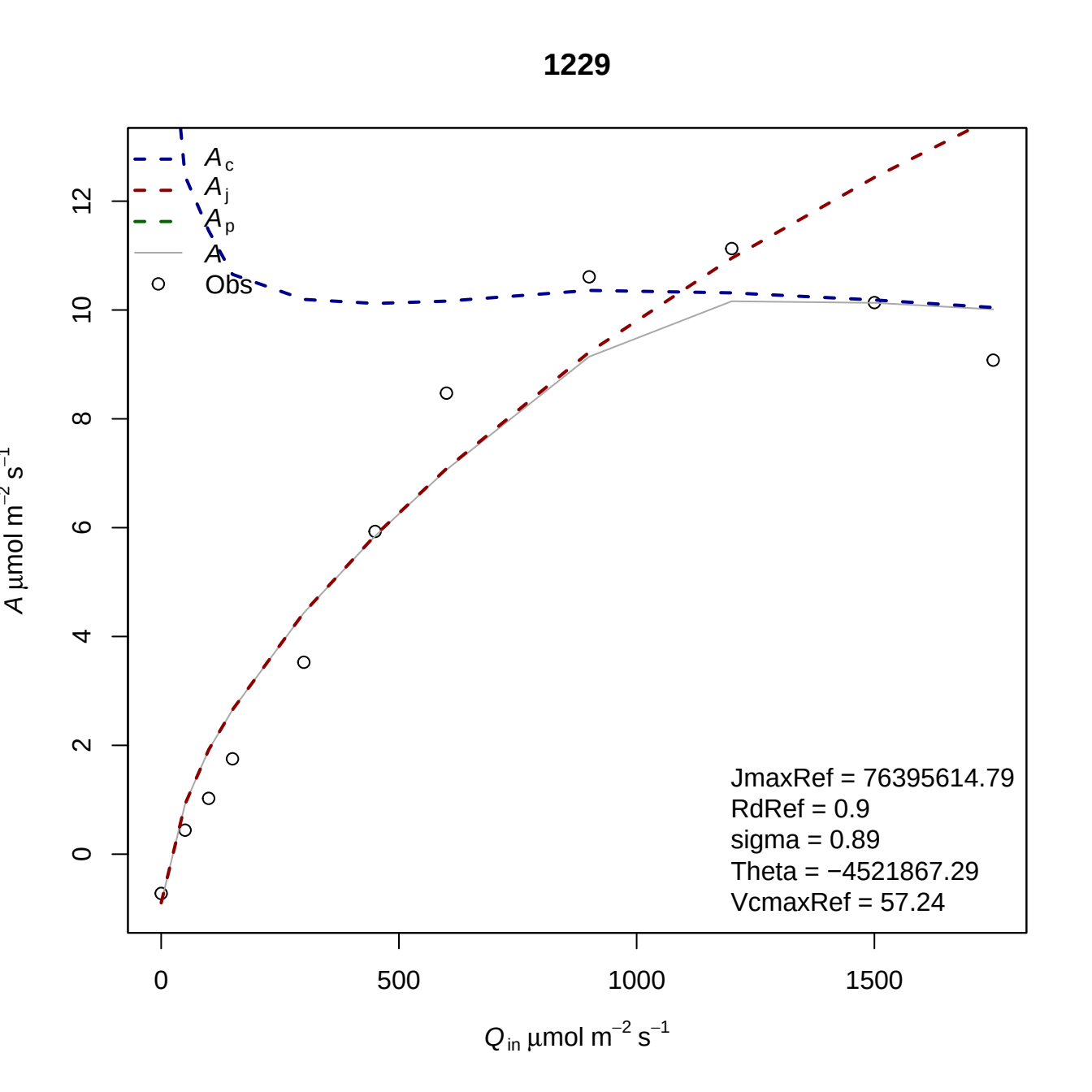

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 127.3
RdRef = 1.07
sigma = 0.11
Theta = 0.61
VcmaxRef = 202362.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

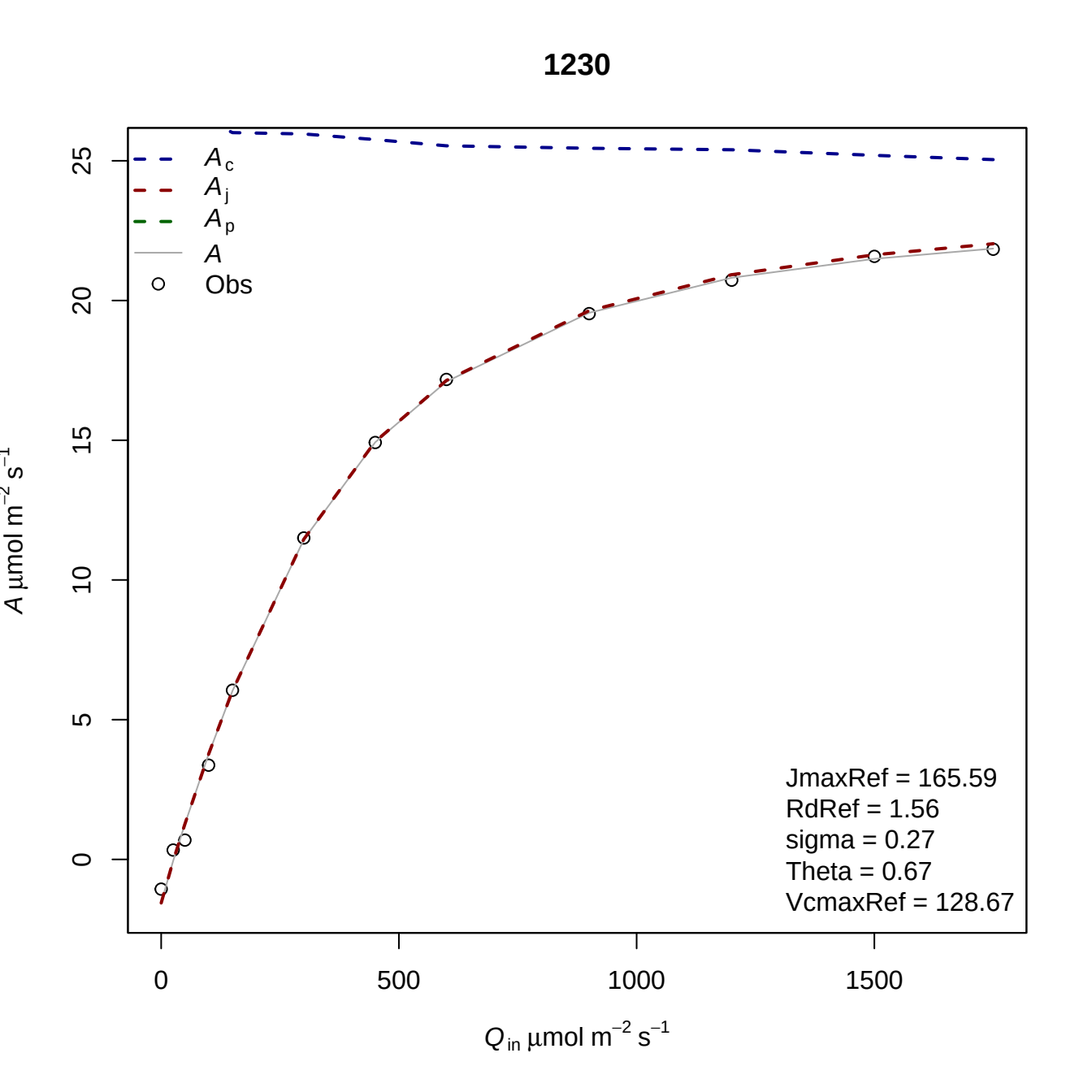

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 257.73
RdRef = 2.47
sigma = 0.47
Theta = −0.64
VcmaxRef = 114.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

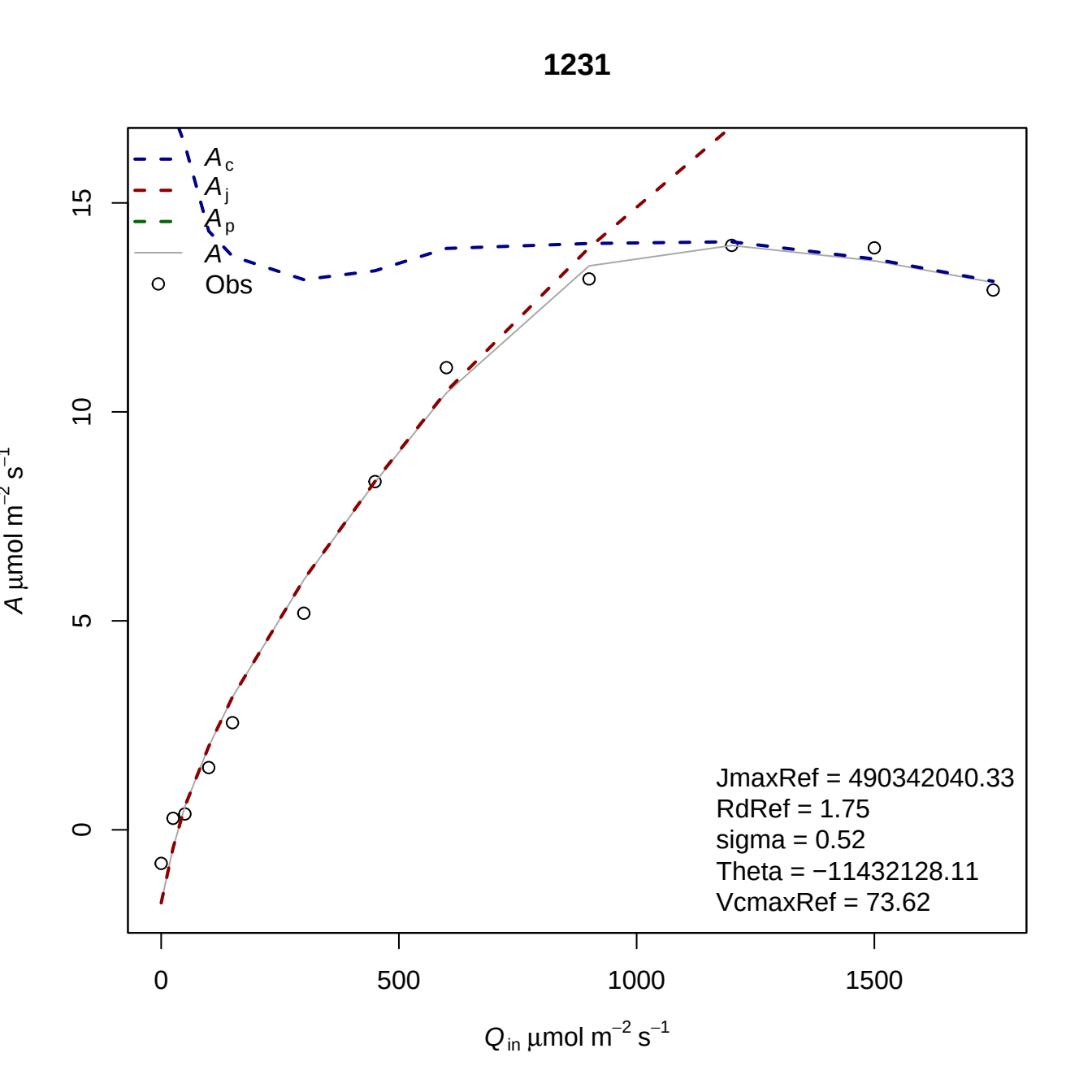

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 128.89
RdRef = 1.42
sigma = 0.2
Theta = −0.45
VcmaxRef = 96419.28
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

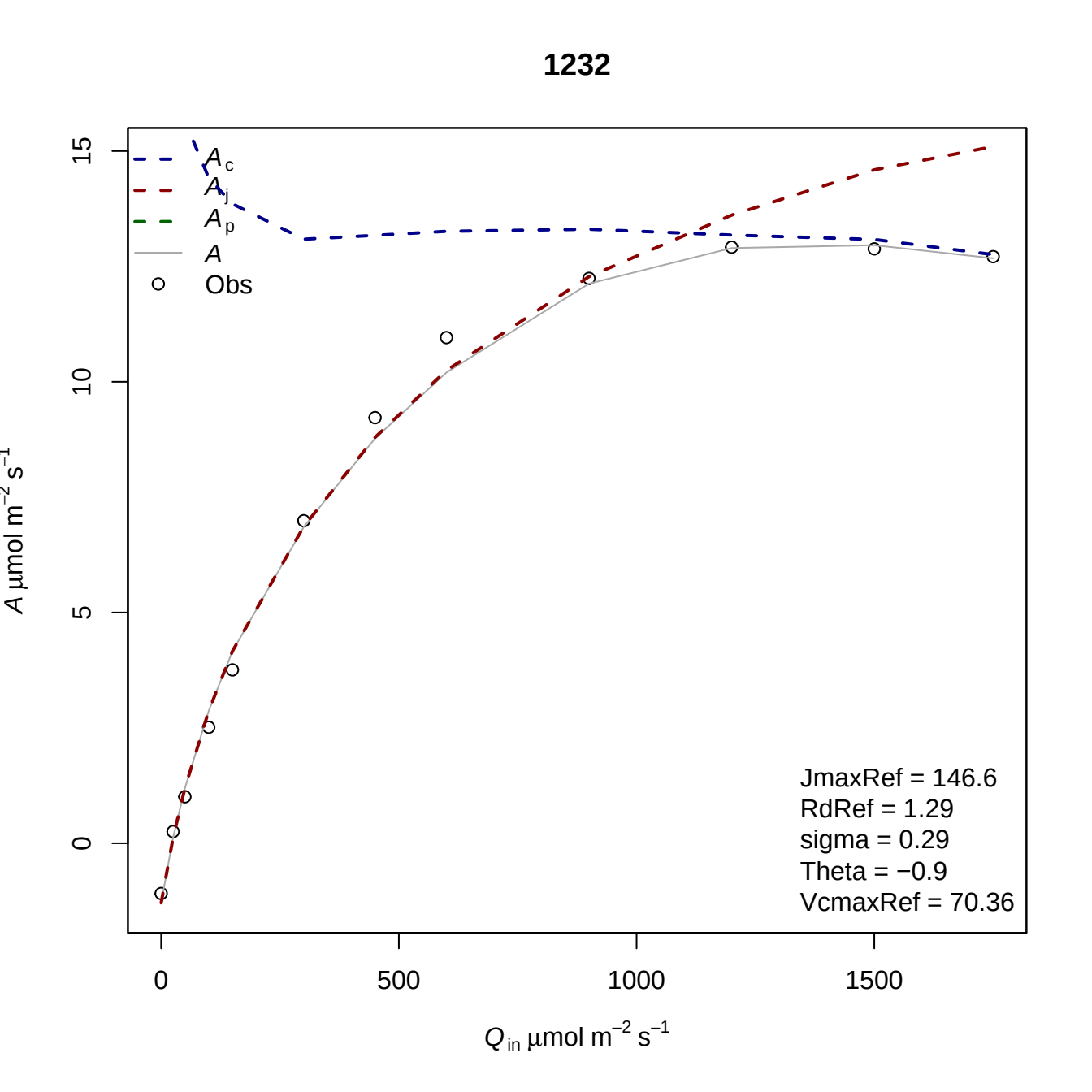

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 209.59
RdRef = 1.8
sigma = 0.44
Theta = −0.69
VcmaxRef = 92.72
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

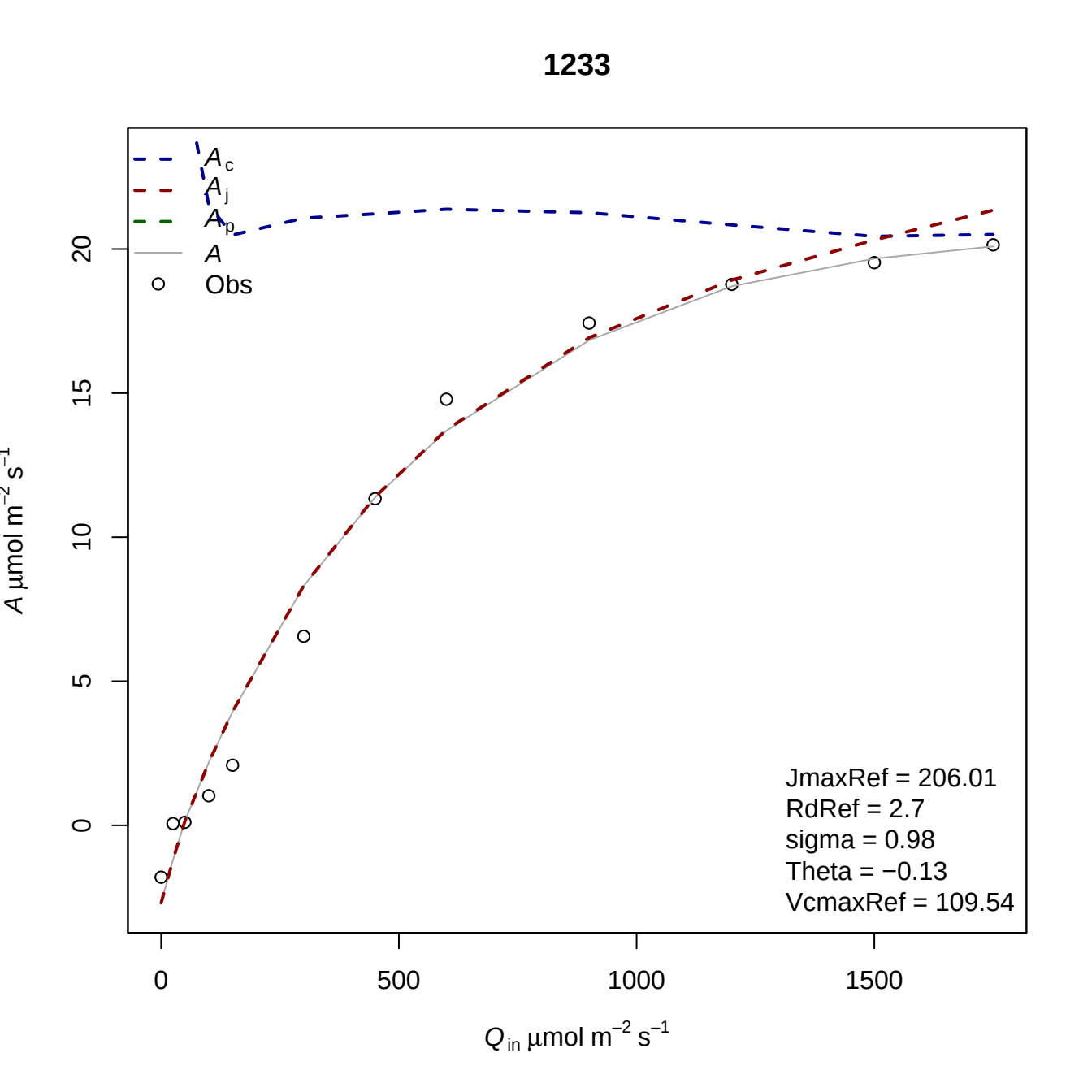

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 158
RdRef = 2.09
sigma = 0.99
Theta = −1.16
VcmaxRef = 76.56
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

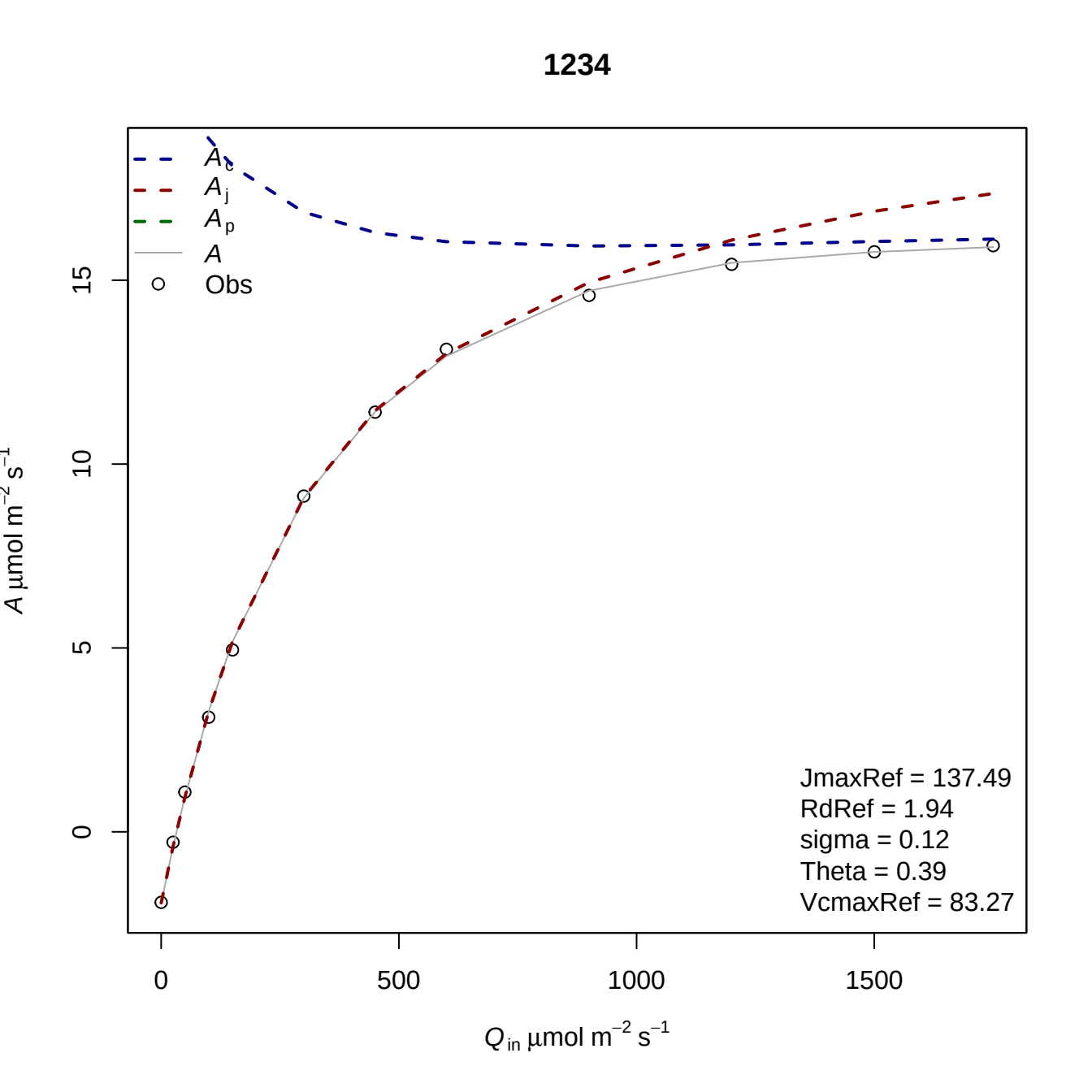

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 76395614.79
RdRef = 0.9
sigma = 0.89
Theta = −4521867.29
VcmaxRef = 57.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

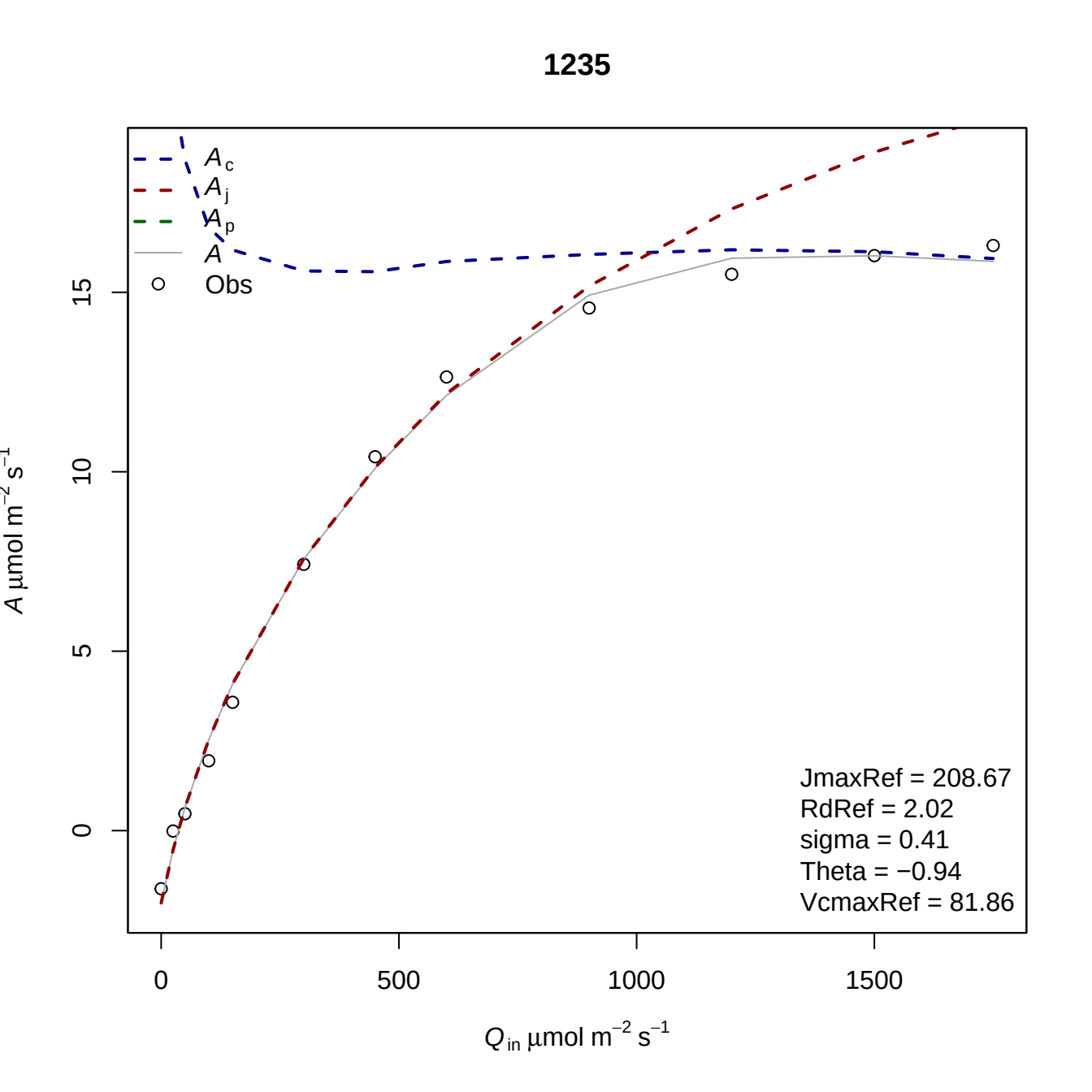

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 165.59
RdRef = 1.56
sigma = 0.27
Theta = 0.67
VcmaxRef = 128.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

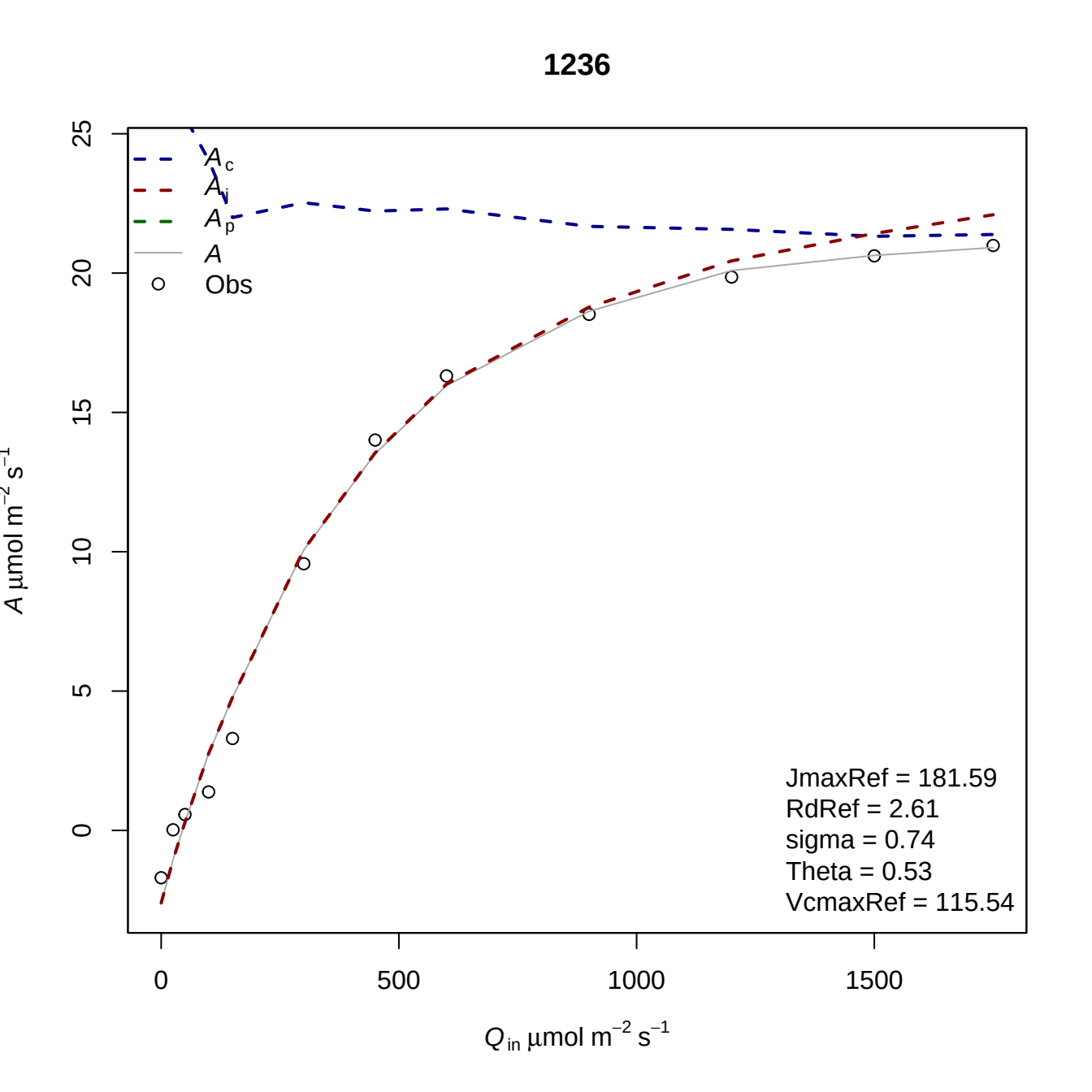

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 490342040.33
RdRef = 1.75
sigma = 0.52
Theta = −11432128.11
VcmaxRef = 73.62
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

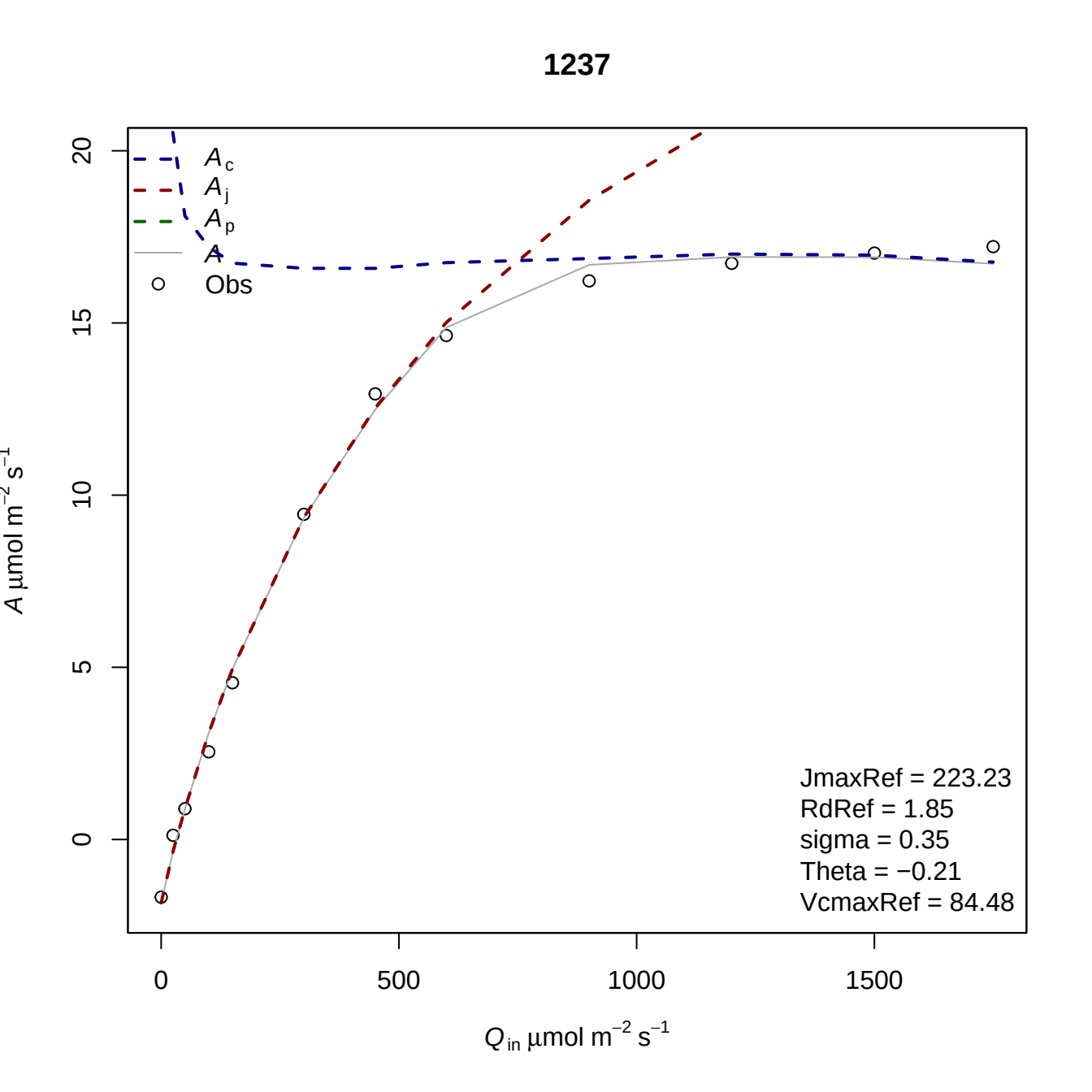

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 146.6
RdRef = 1.29
sigma = 0.29
Theta = −0.9
VcmaxRef = 70.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

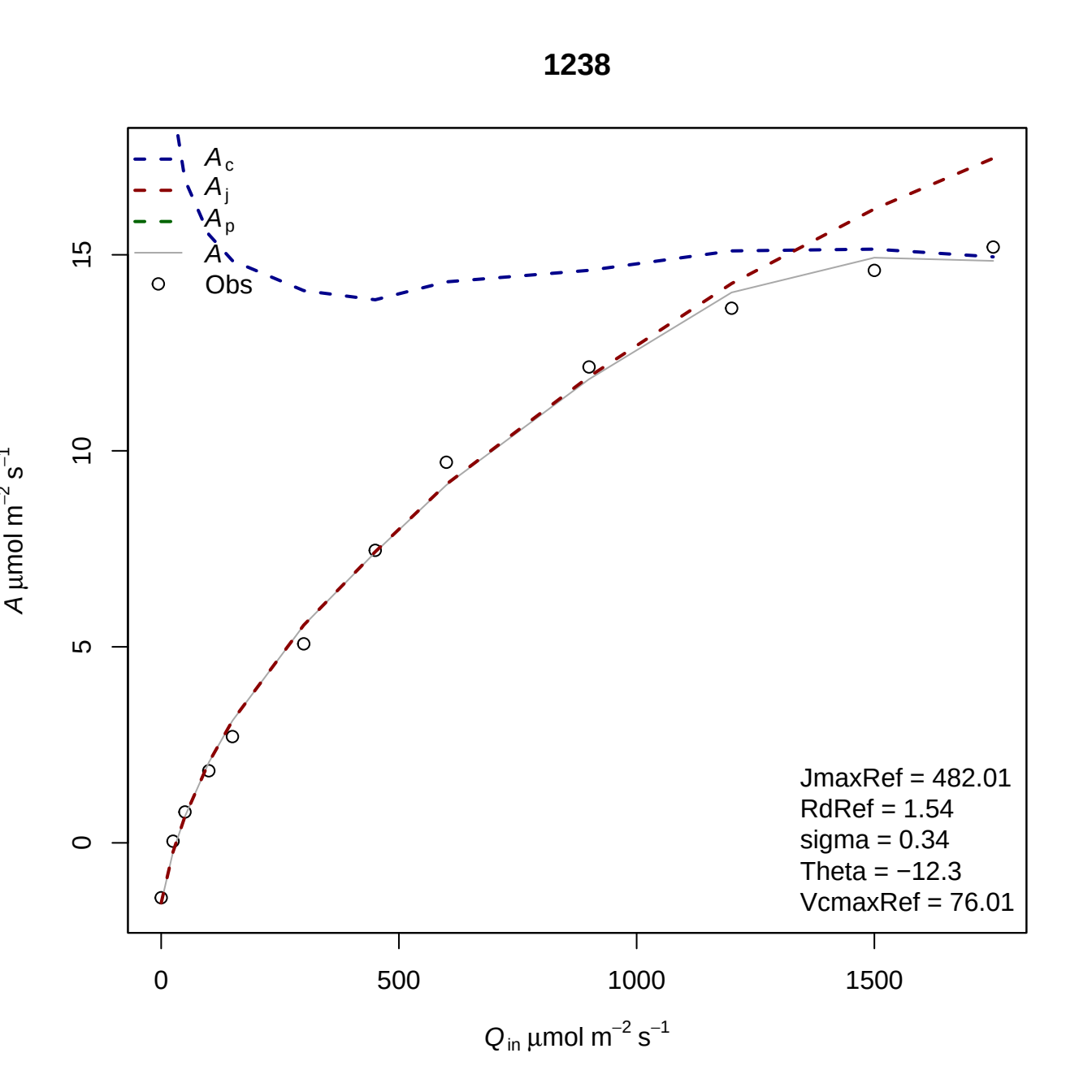

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 206.01
RdRef = 2.7
sigma = 0.98
Theta = −0.13
VcmaxRef = 109.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

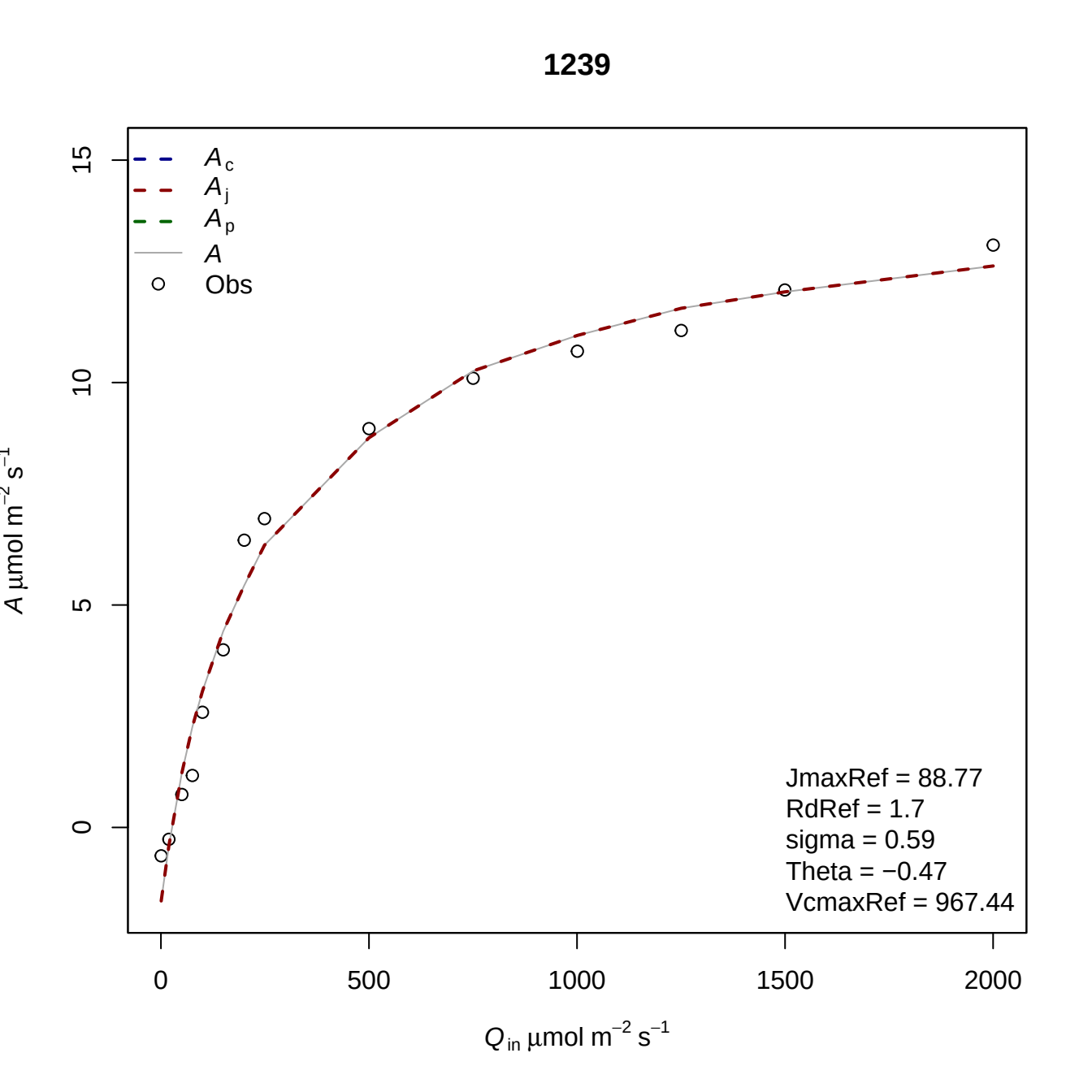

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 137.49
RdRef = 1.94
sigma = 0.12
Theta = 0.39
VcmaxRef = 83.27
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

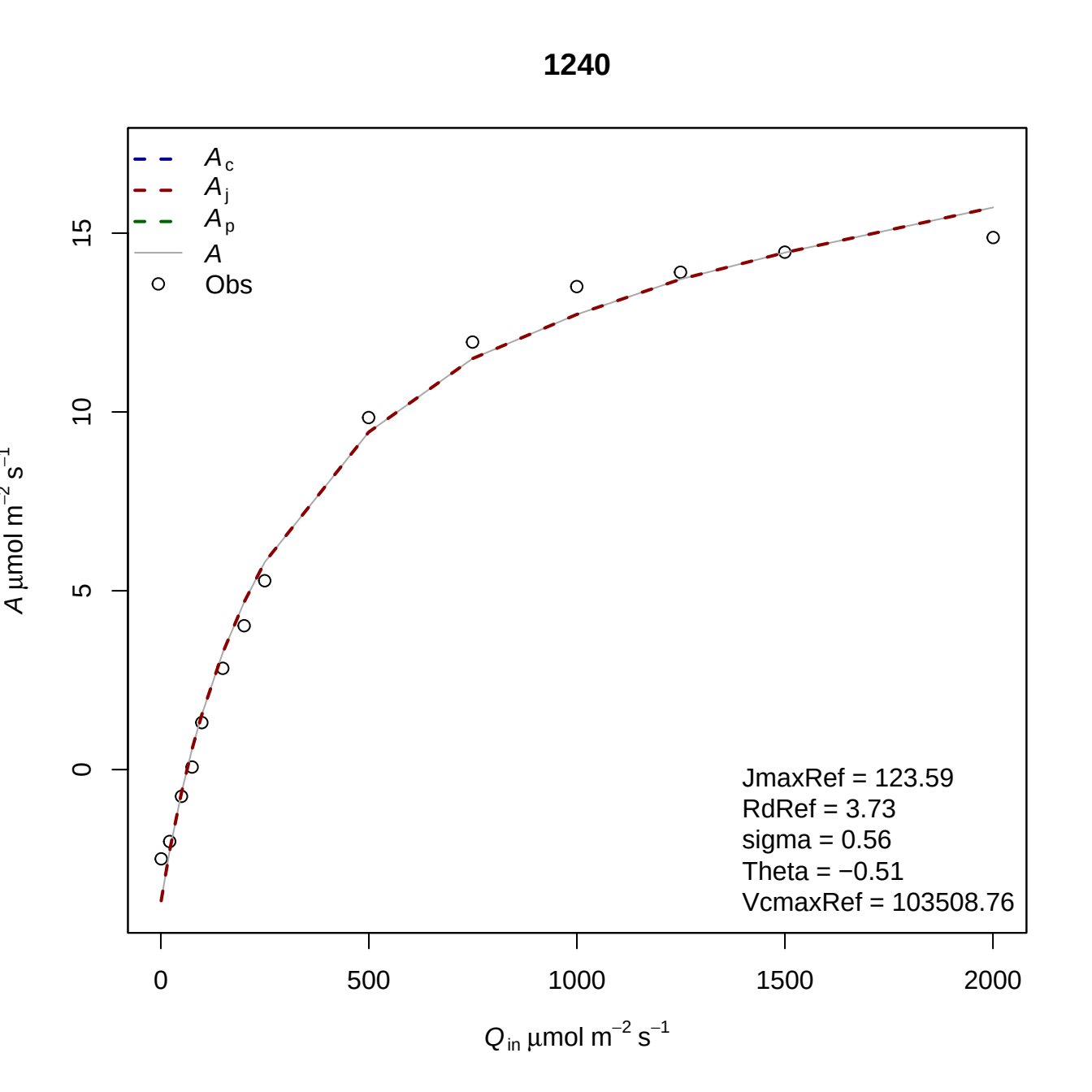

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 208.67
RdRef = 2.02
sigma = 0.41
Theta = −0.94
VcmaxRef = 81.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

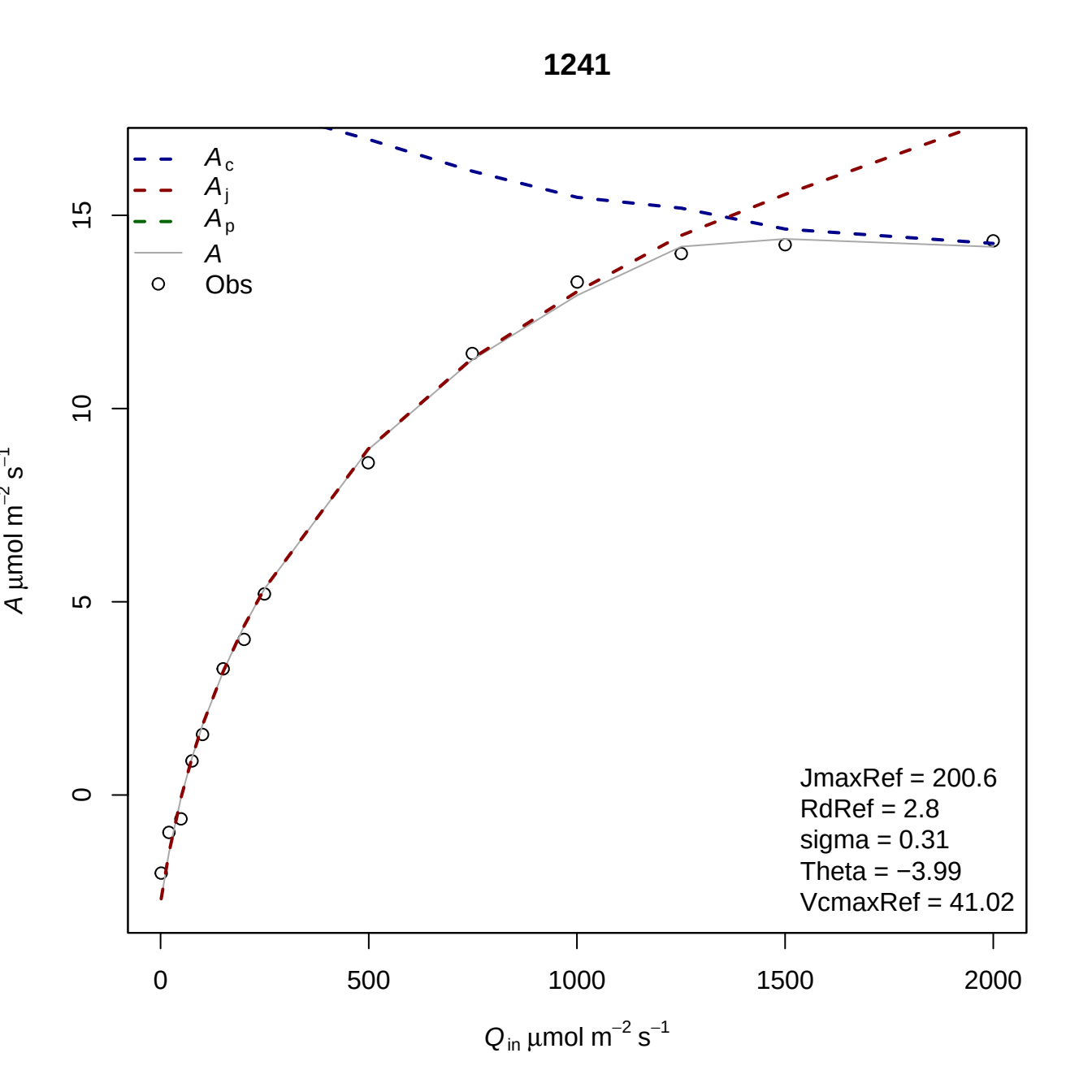

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 181.59
RdRef = 2.61
sigma = 0.74
Theta = 0.53
VcmaxRef = 115.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

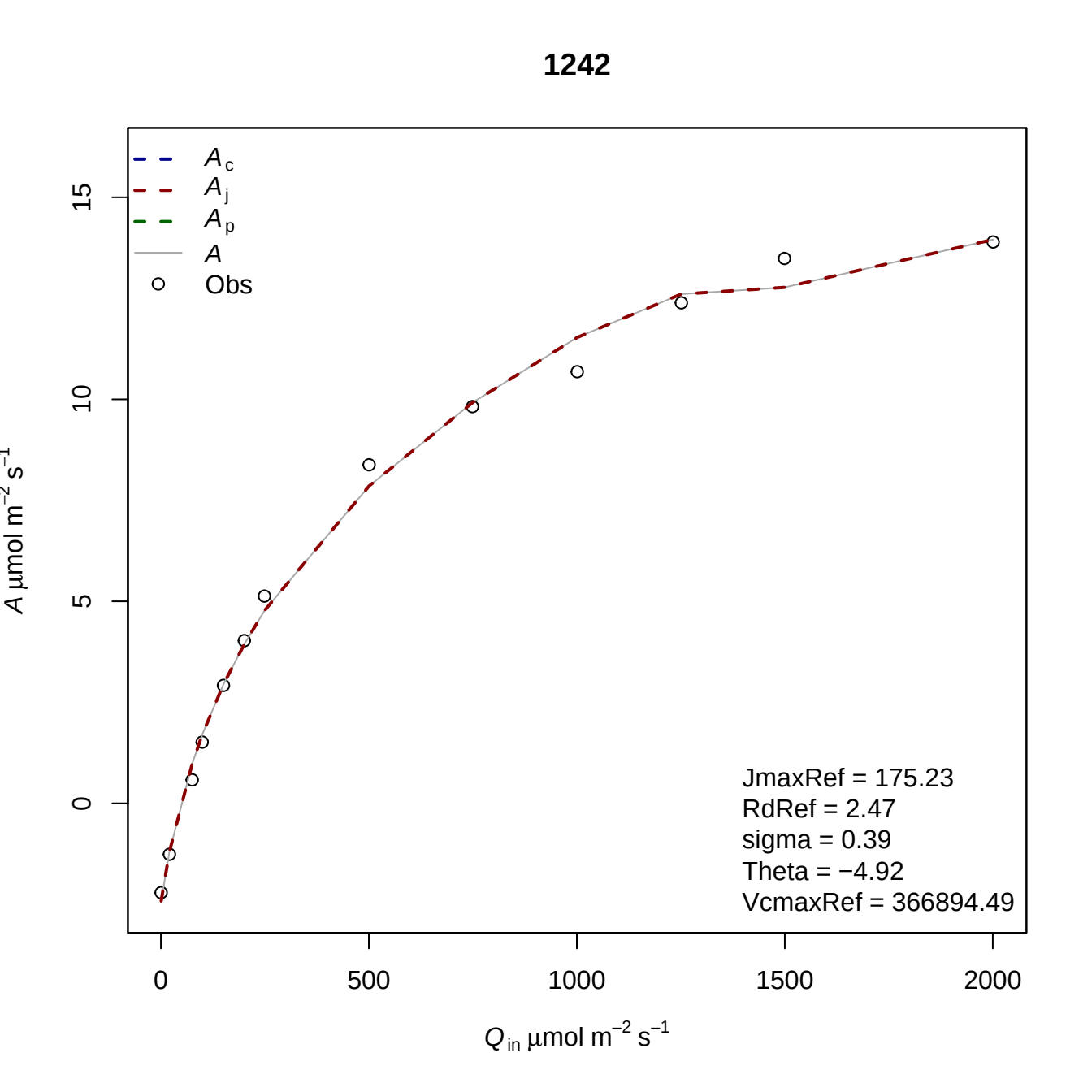

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 223.23
RdRef = 1.85
sigma = 0.35
Theta = −0.21
VcmaxRef = 84.48
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

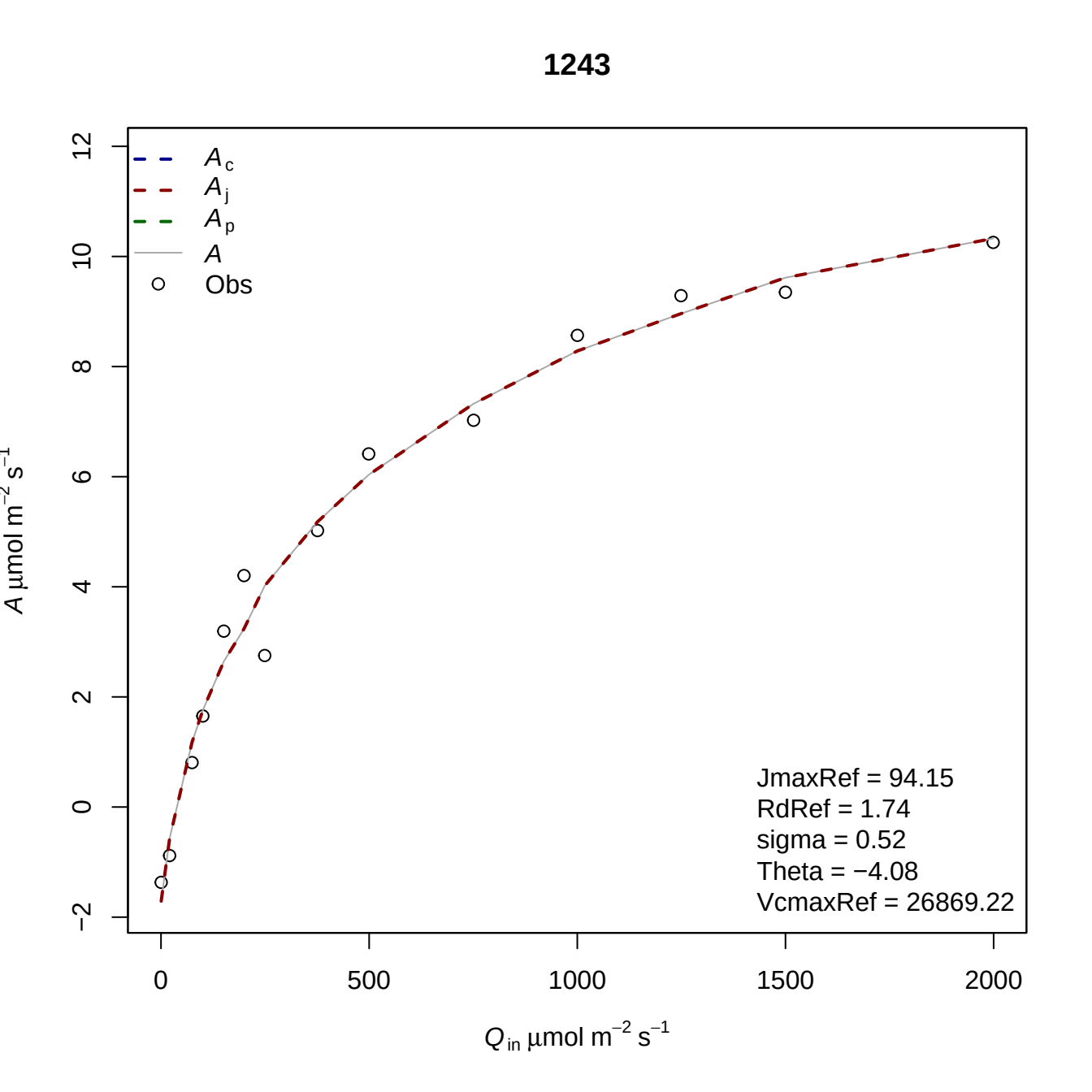

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 482.01
RdRef = 1.54
sigma = 0.34
Theta = −12.3
VcmaxRef = 76.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

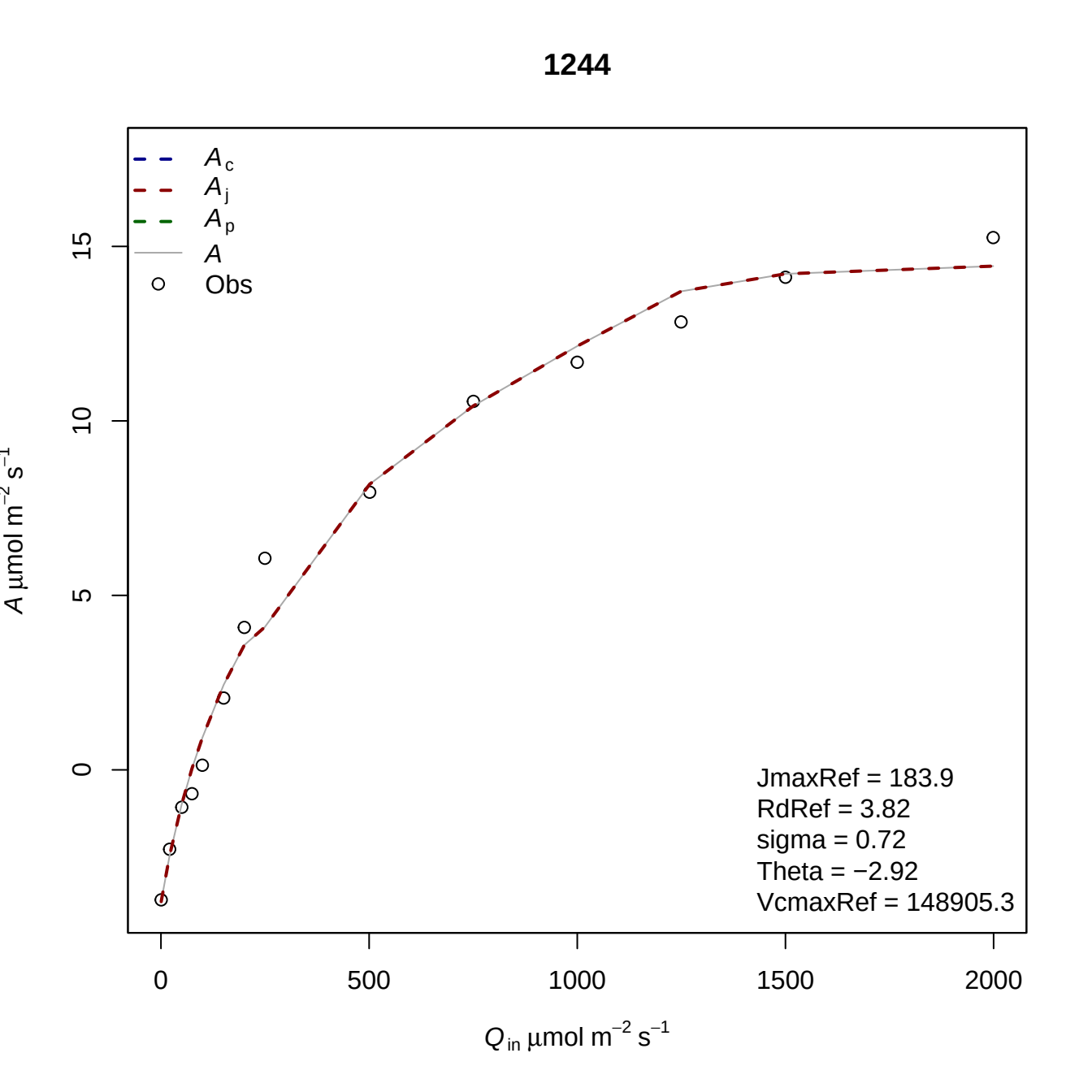

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 88.77
RdRef = 1.7
sigma = 0.59
Theta = −0.47
VcmaxRef = 967.44
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

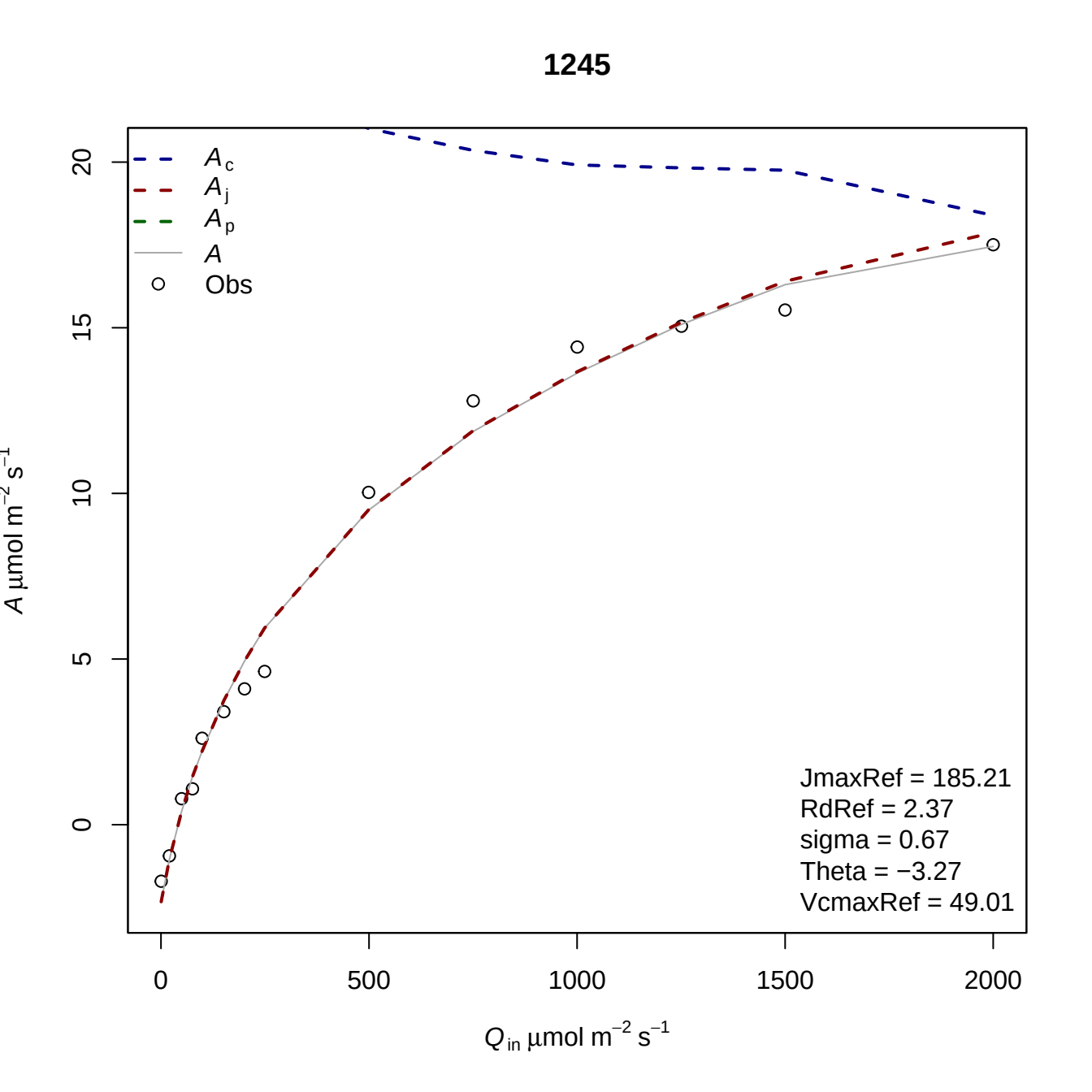

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 123.59
RdRef = 3.73
sigma = 0.56
Theta = −0.51
VcmaxRef = 103508.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

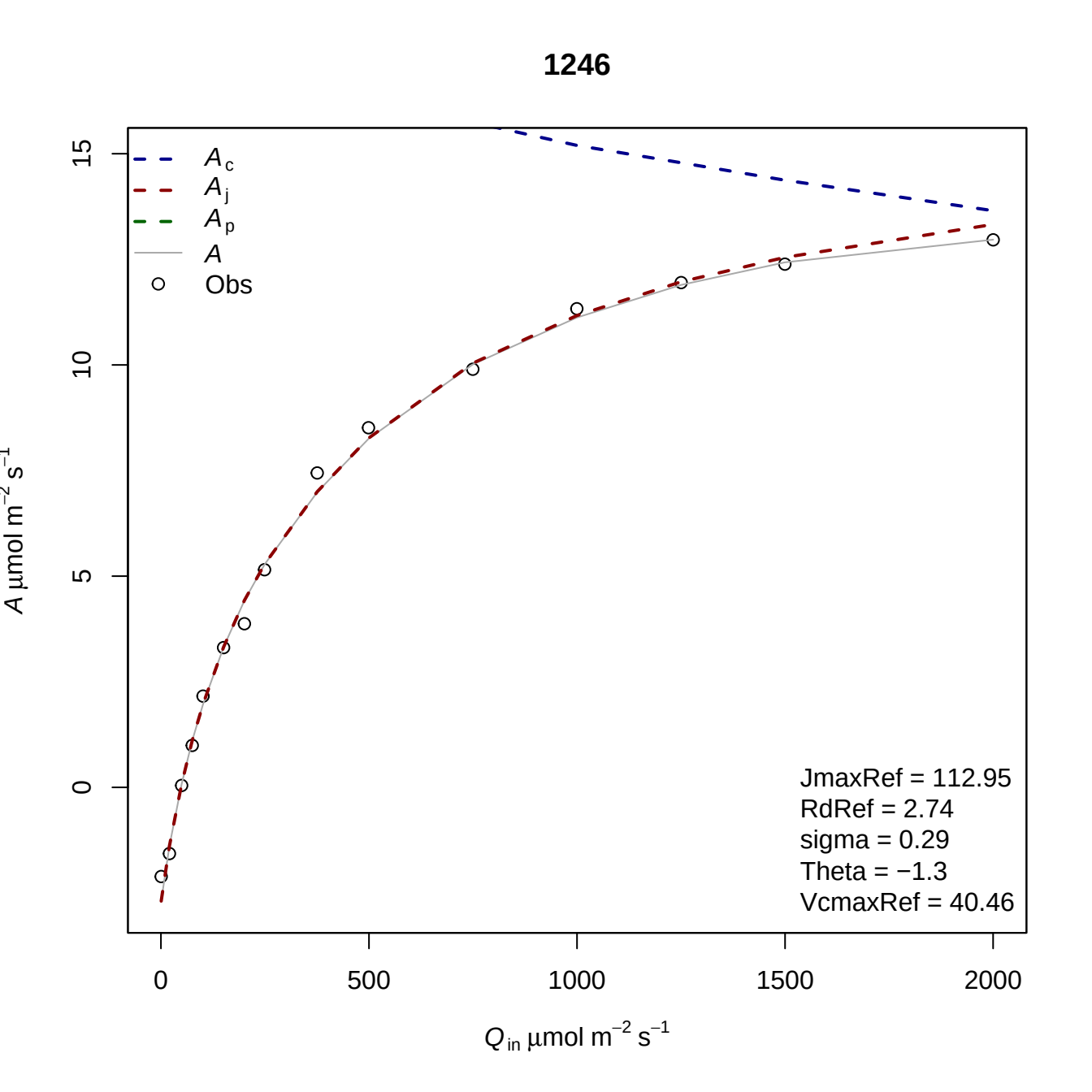

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 200.6
RdRef = 2.8
sigma = 0.31
Theta = −3.99
VcmaxRef = 41.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

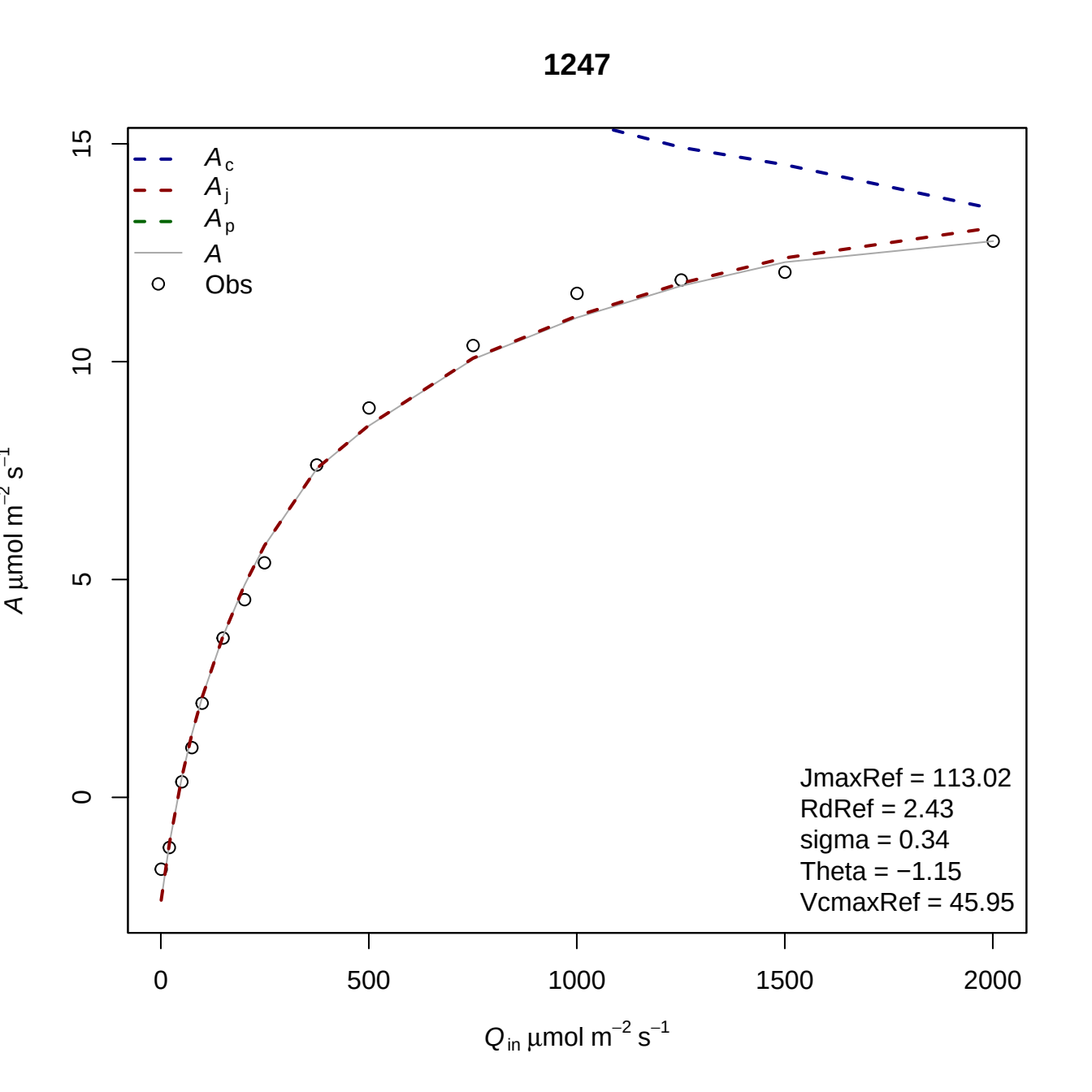

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 175.23
RdRef = 2.47
sigma = 0.39
Theta = −4.92
VcmaxRef = 366894.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

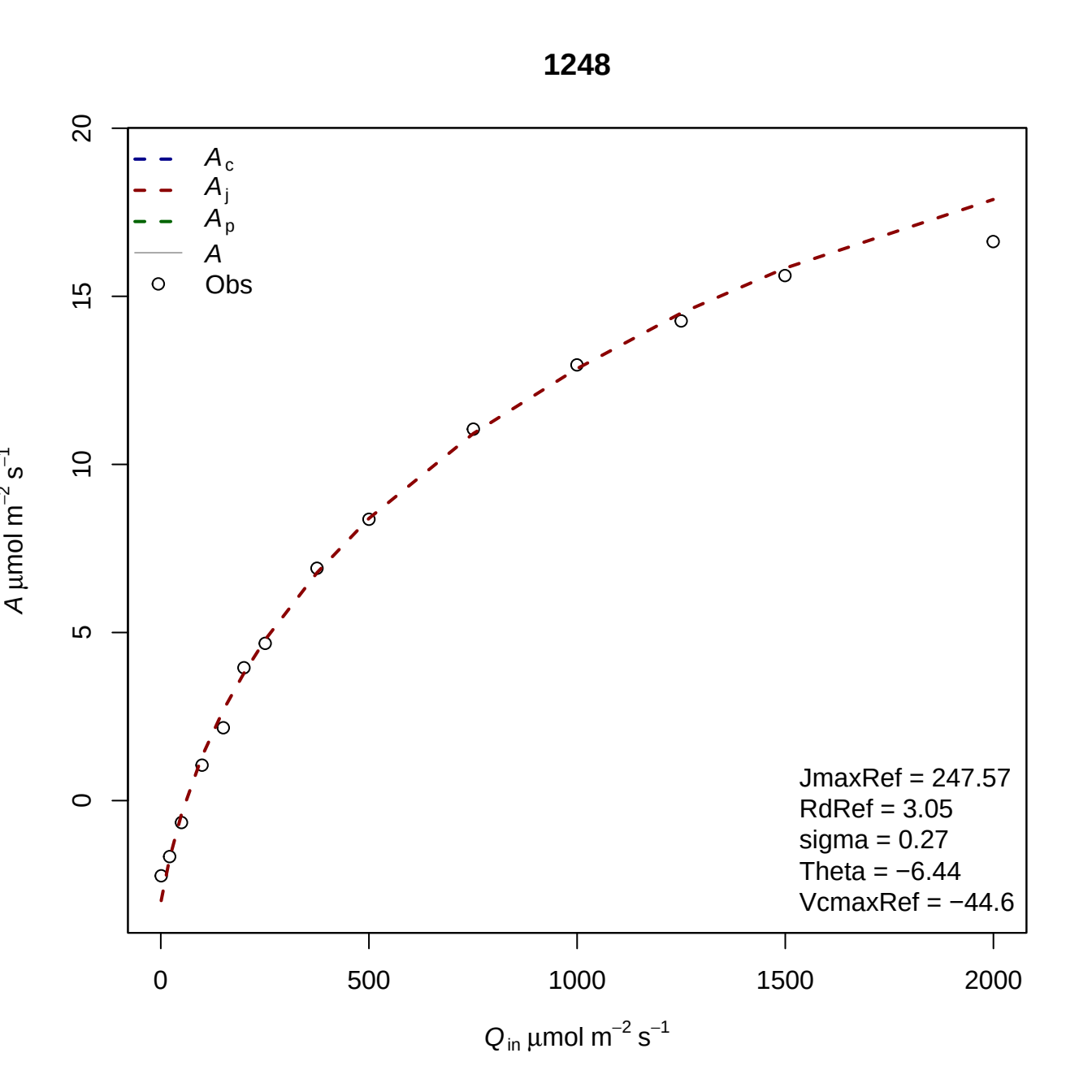

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 94.15
RdRef = 1.74
sigma = 0.52
Theta = −4.08
VcmaxRef = 26869.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

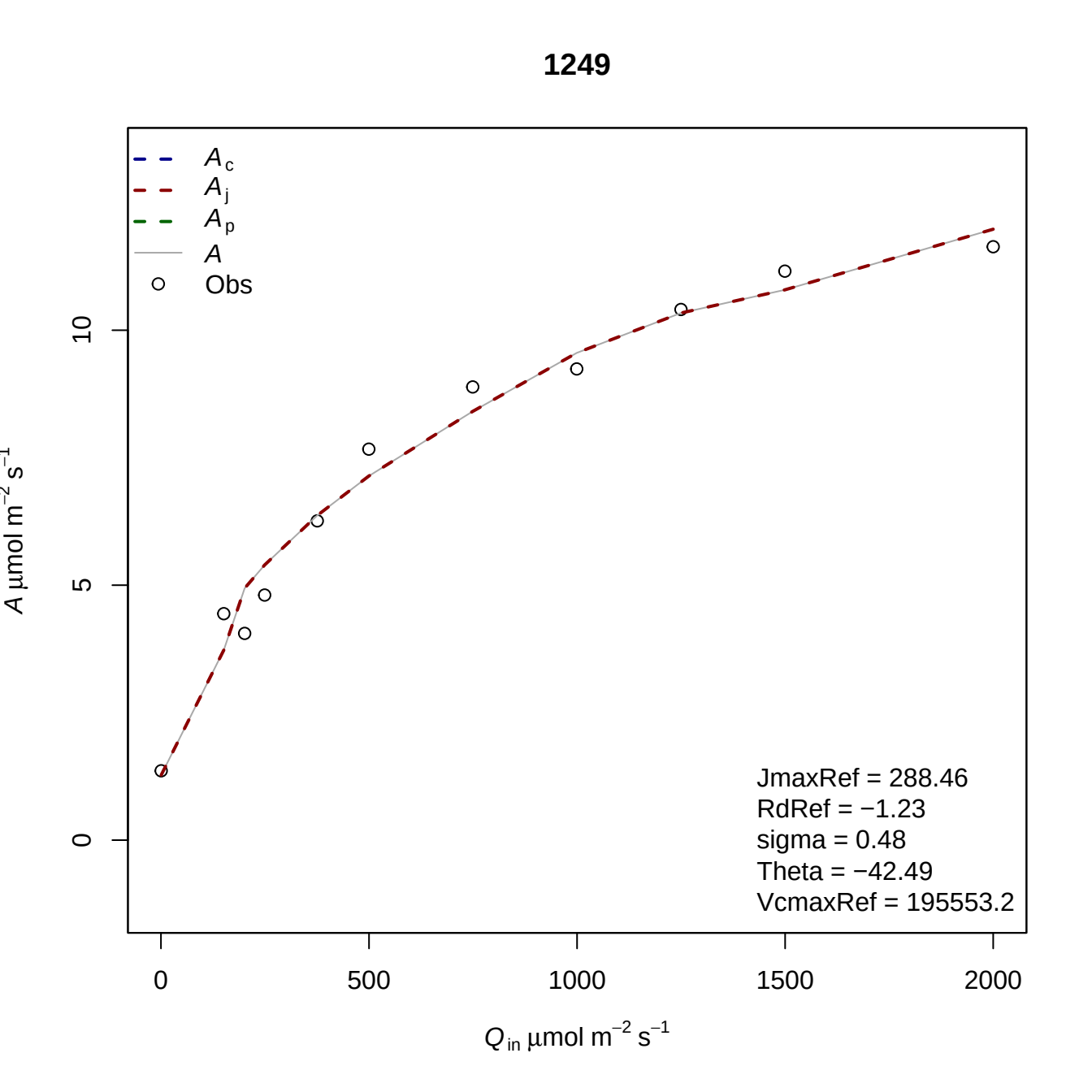

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 183.9
RdRef = 3.82
sigma = 0.72
Theta = −2.92
VcmaxRef = 148905.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

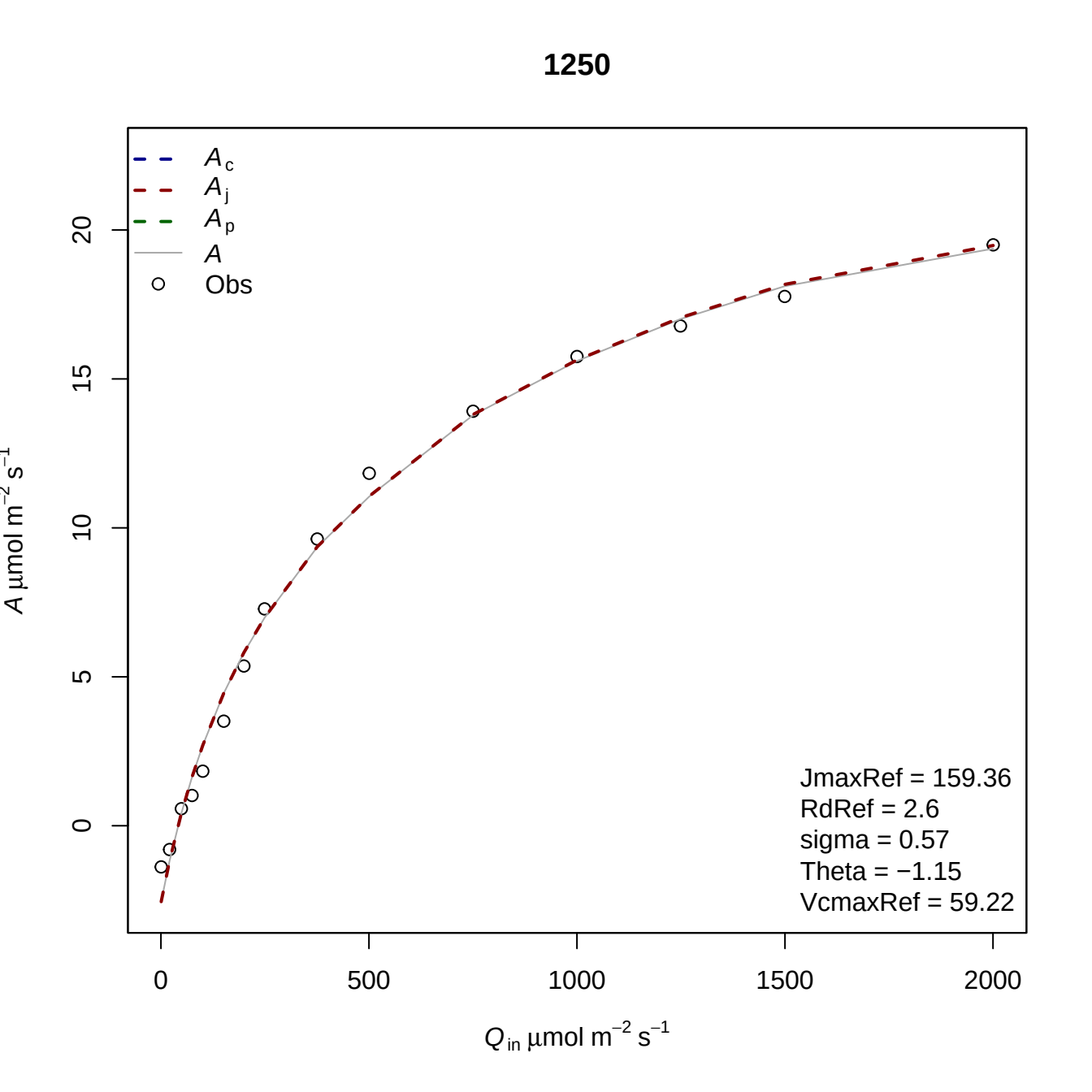

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 185.21
RdRef = 2.37
sigma = 0.67
Theta = −3.27
VcmaxRef = 49.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

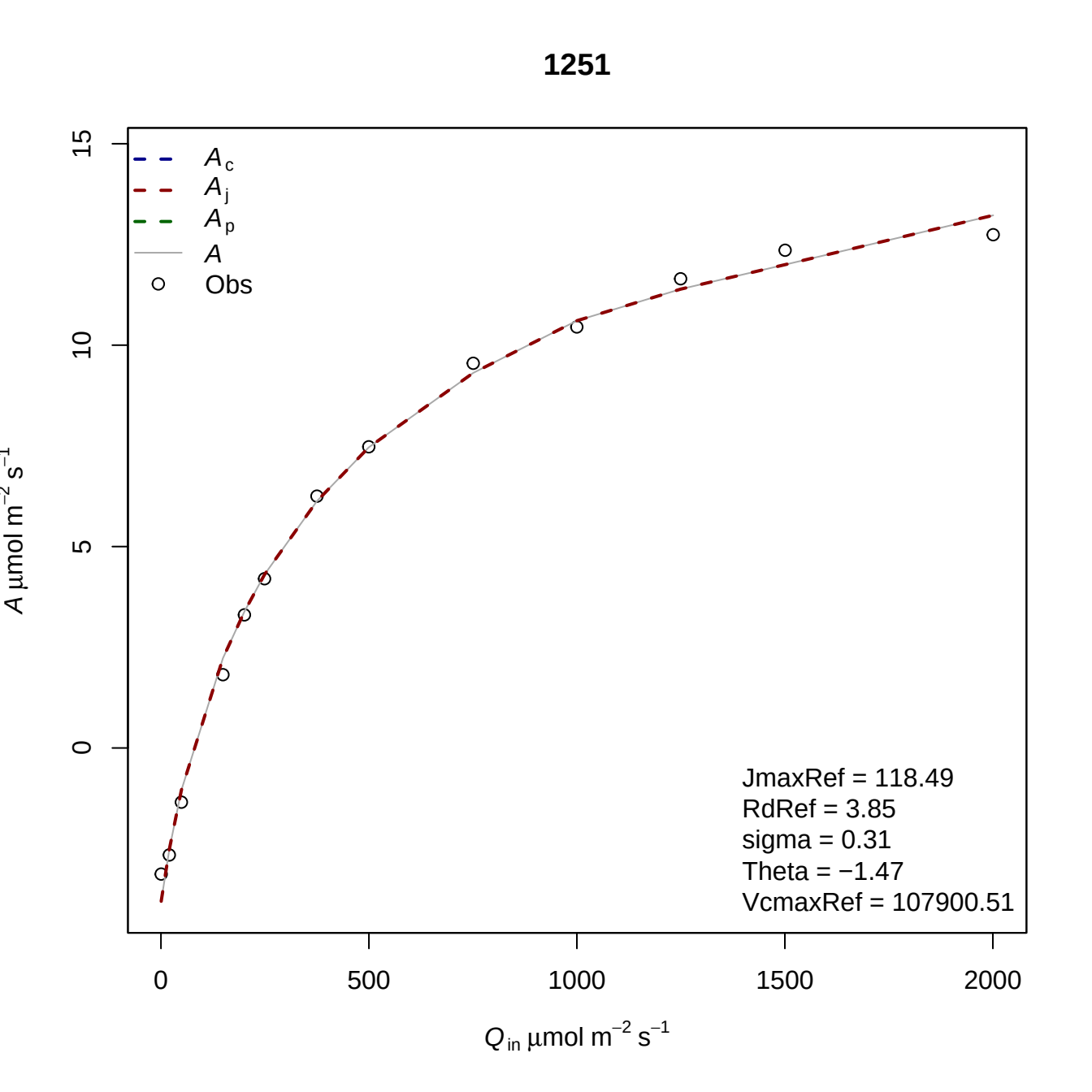

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 112.95
RdRef = 2.74
sigma = 0.29
Theta = −1.3
VcmaxRef = 40.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

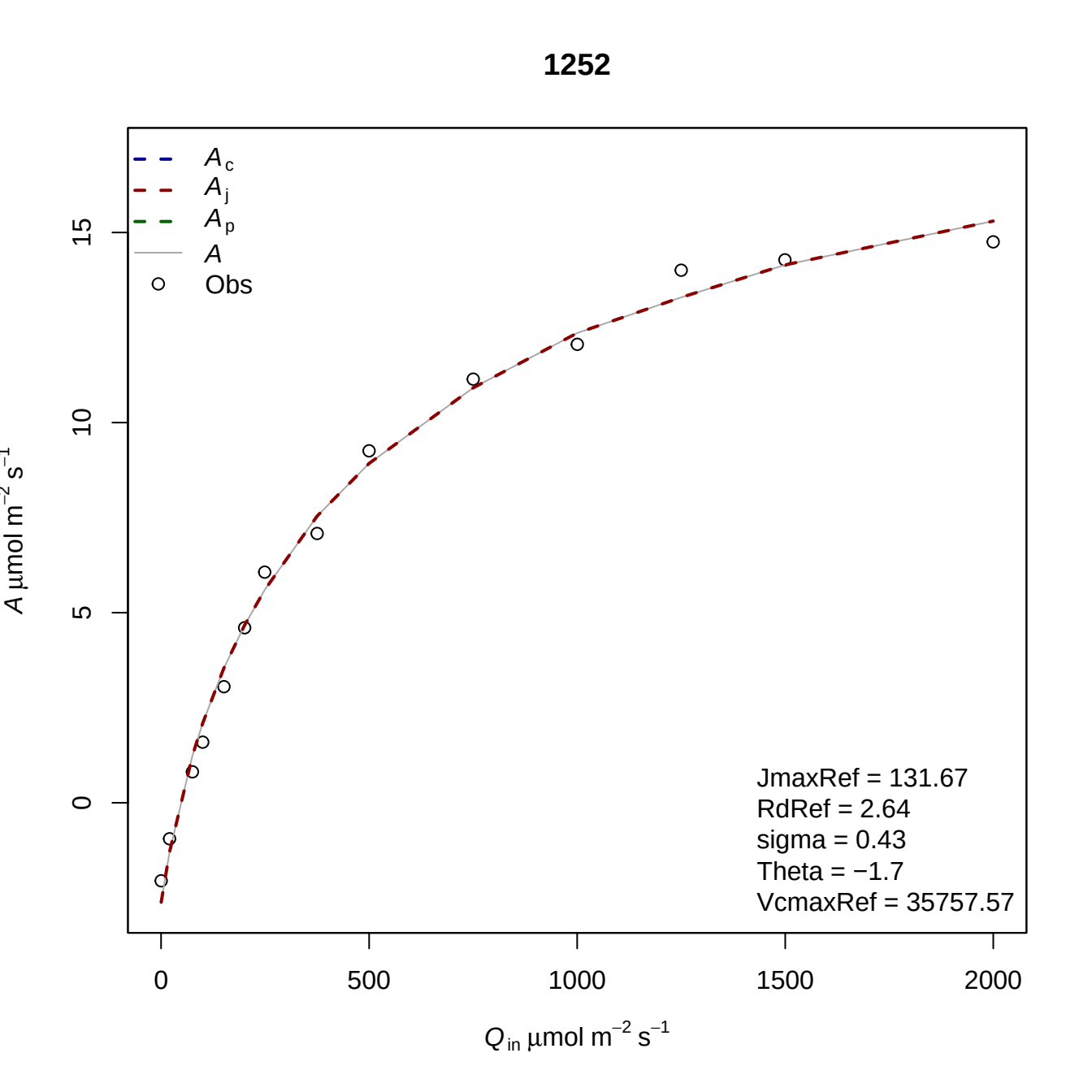

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 113.02
RdRef = 2.43
sigma = 0.34
Theta = −1.15
VcmaxRef = 45.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

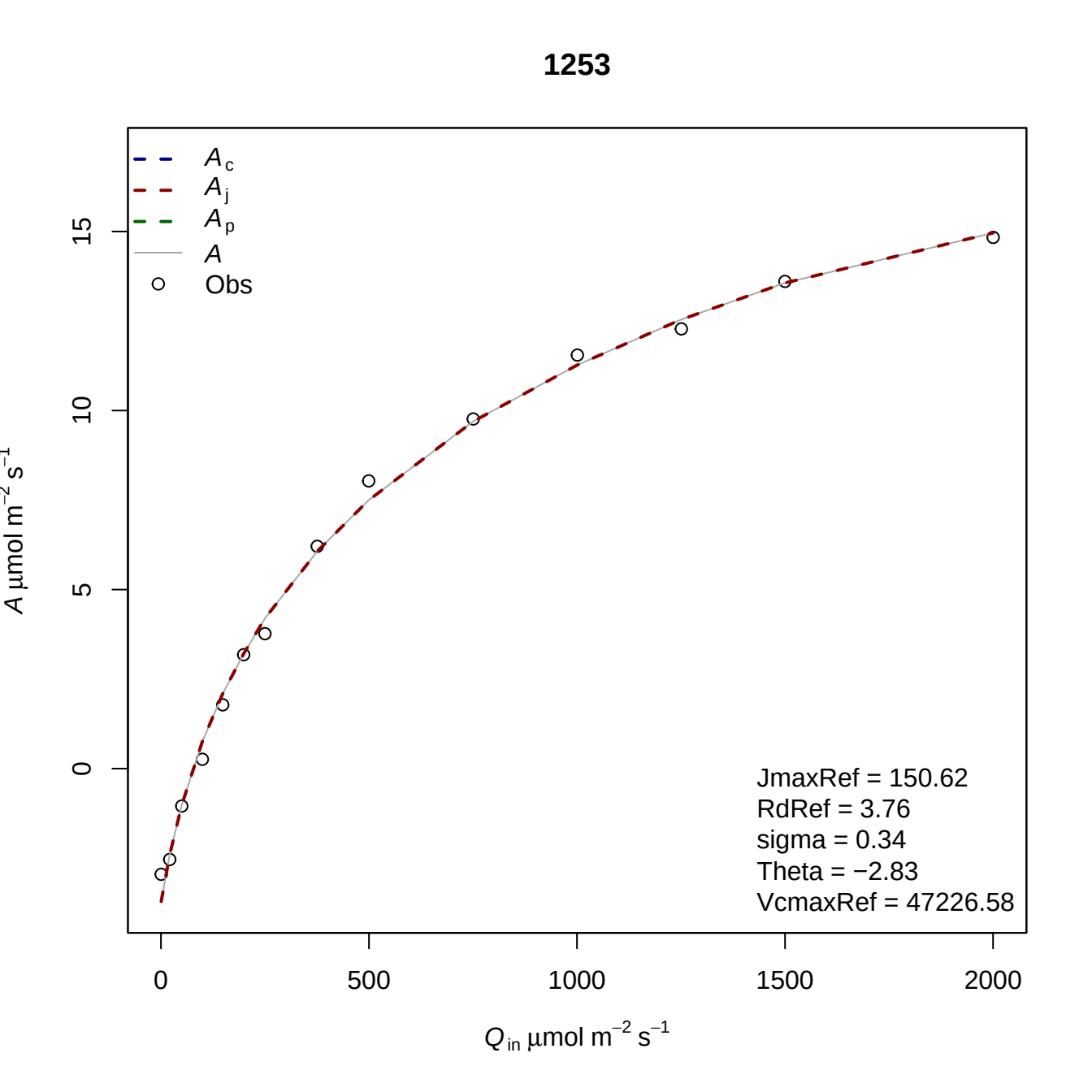

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 247.57
RdRef = 3.05
sigma = 0.27
Theta = −6.44
VcmaxRef = −44.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

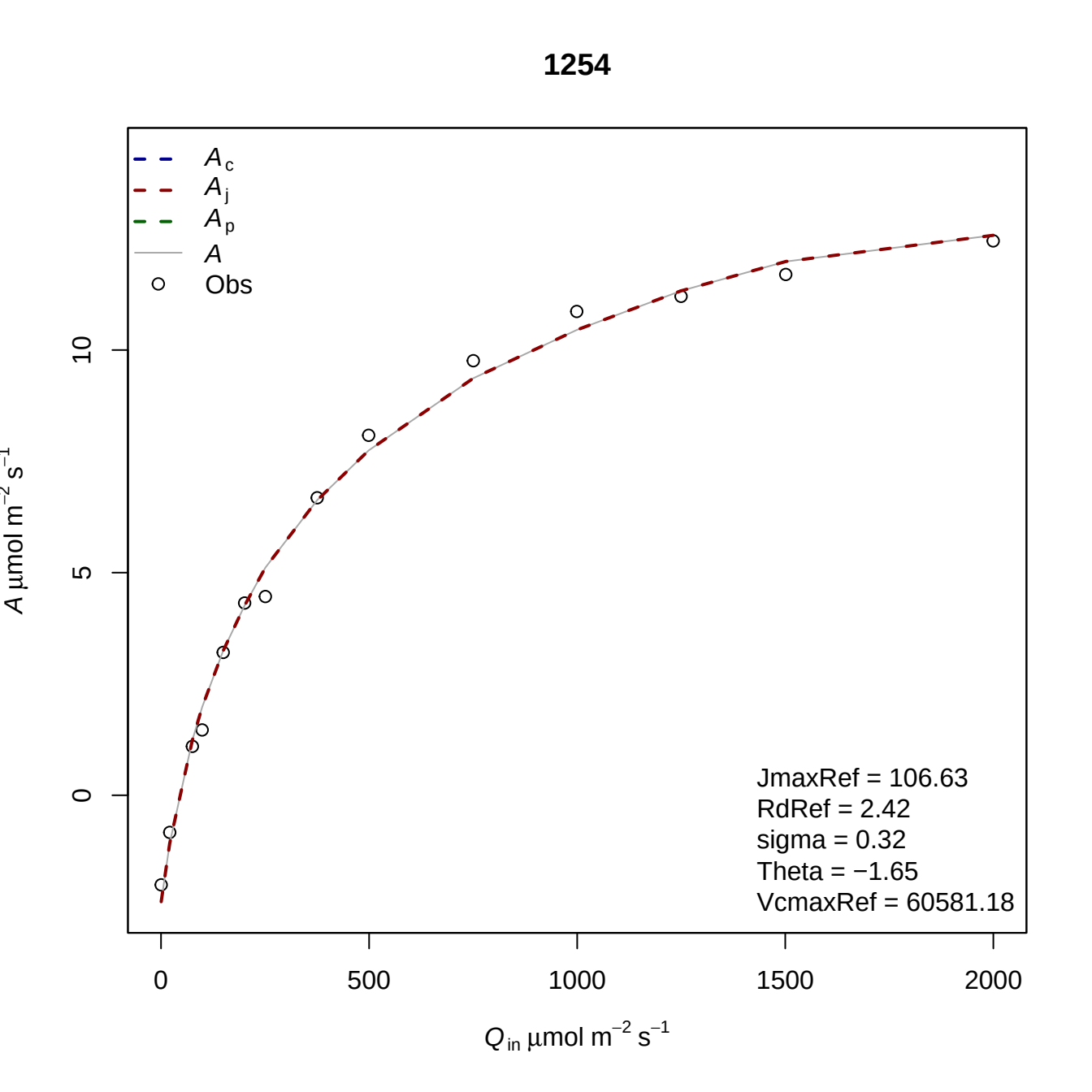

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 288.46
RdRef = −1.23
sigma = 0.48
Theta = −42.49
VcmaxRef = 195553.2
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

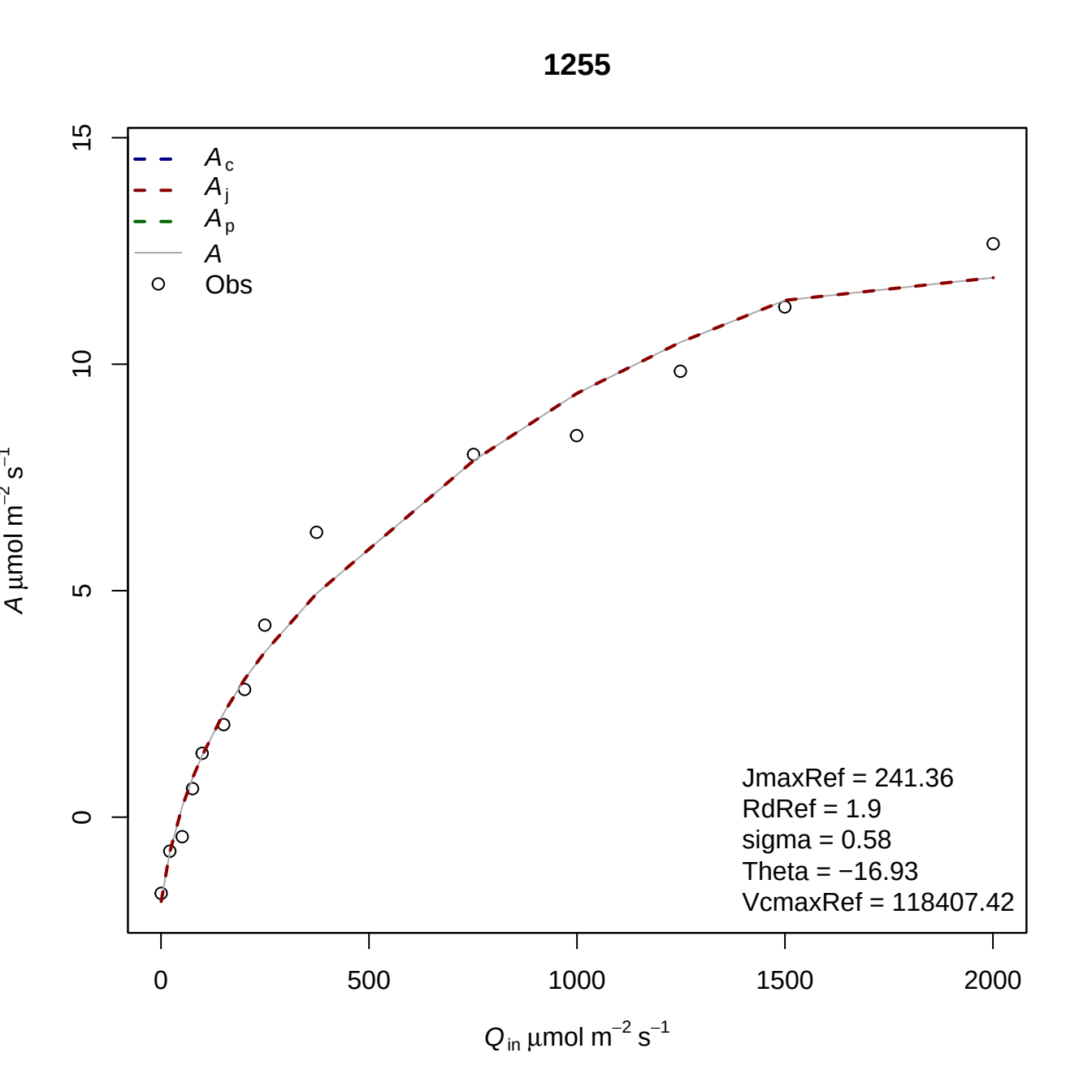

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 159.36
RdRef = 2.6
sigma = 0.57
Theta = −1.15
VcmaxRef = 59.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

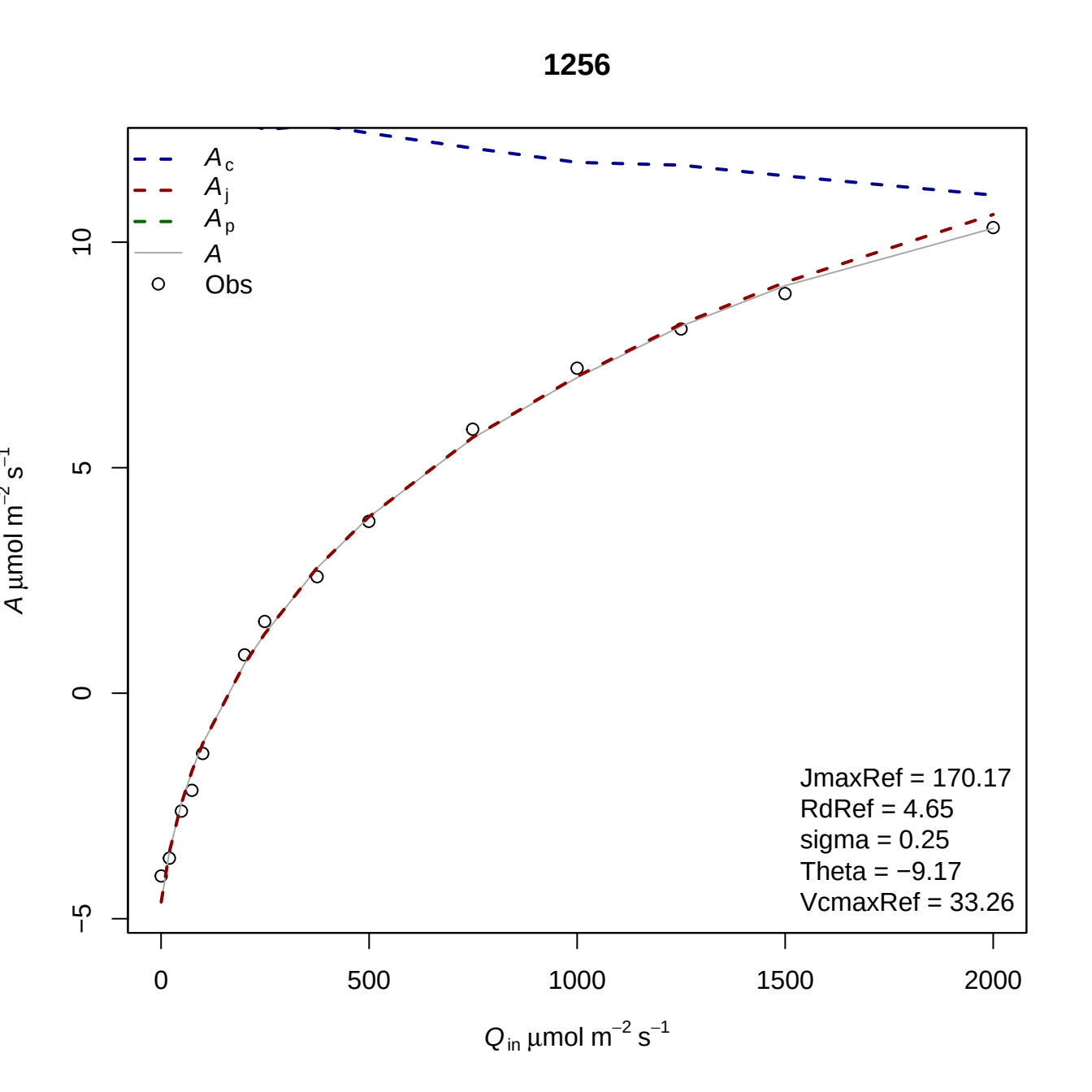

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 118.49
RdRef = 3.85
sigma = 0.31
Theta = −1.47
VcmaxRef = 107900.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

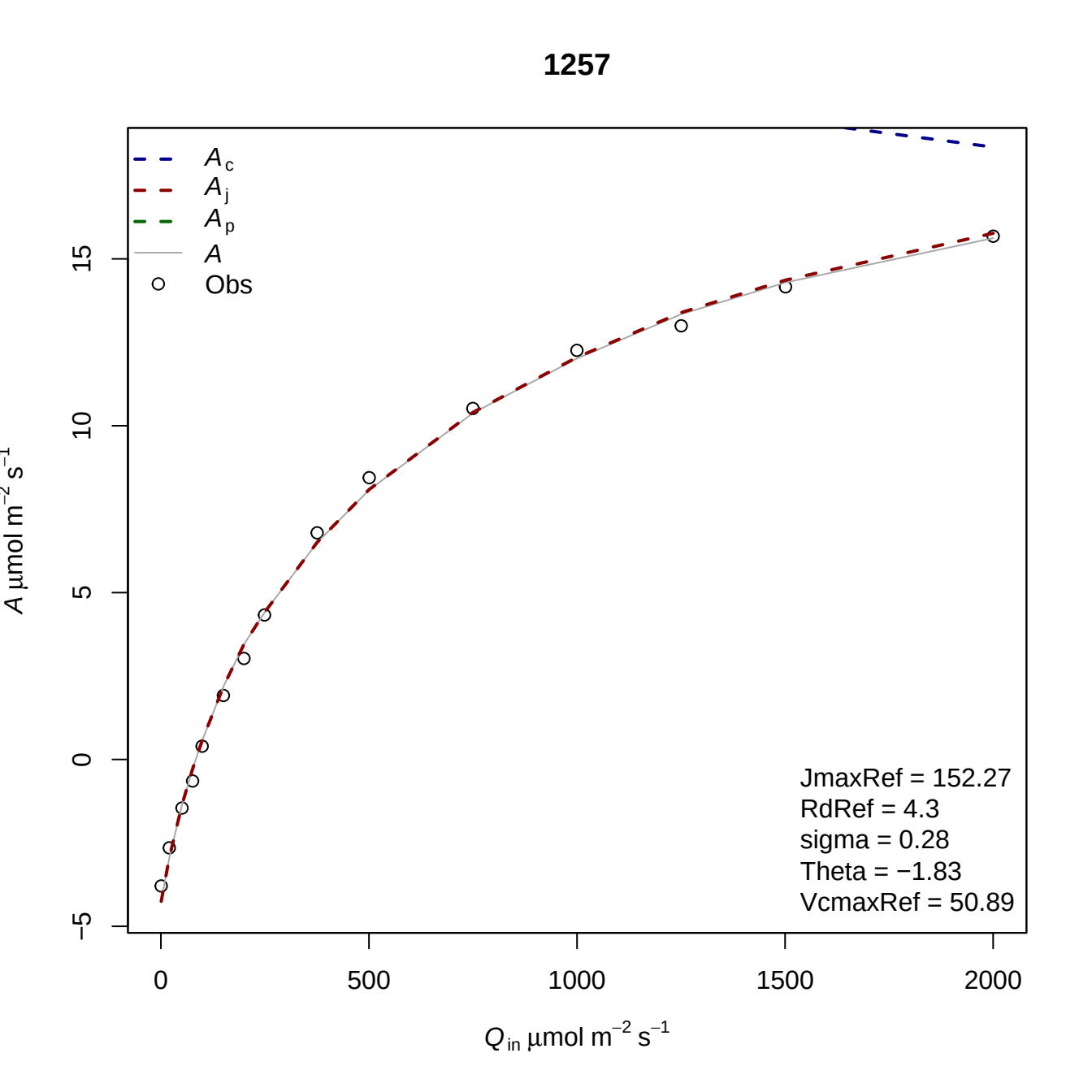

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 131.67
RdRef = 2.64
sigma = 0.43
Theta = −1.7
VcmaxRef = 35757.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

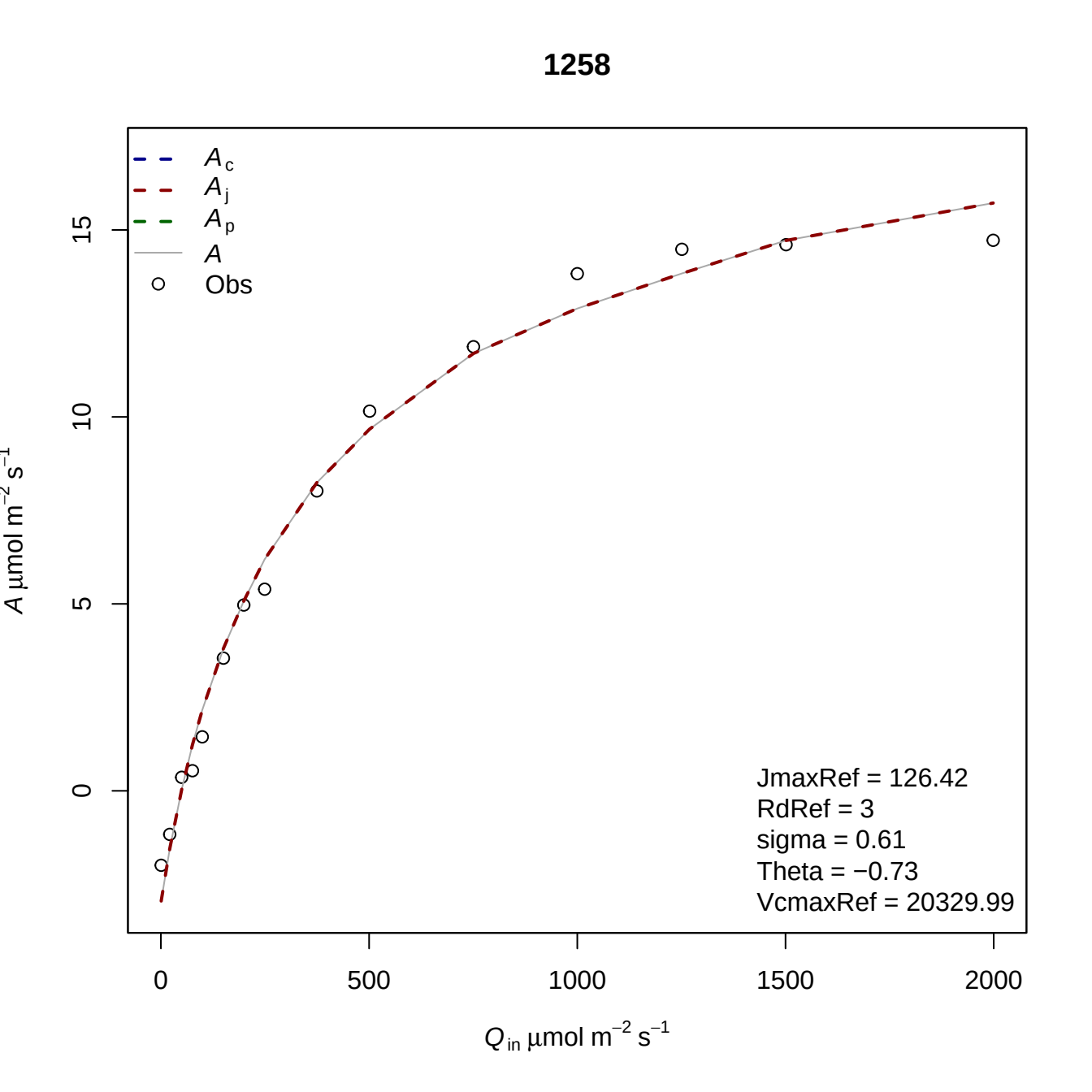

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 150.62
RdRef = 3.76
sigma = 0.34
Theta = −2.83
VcmaxRef = 47226.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

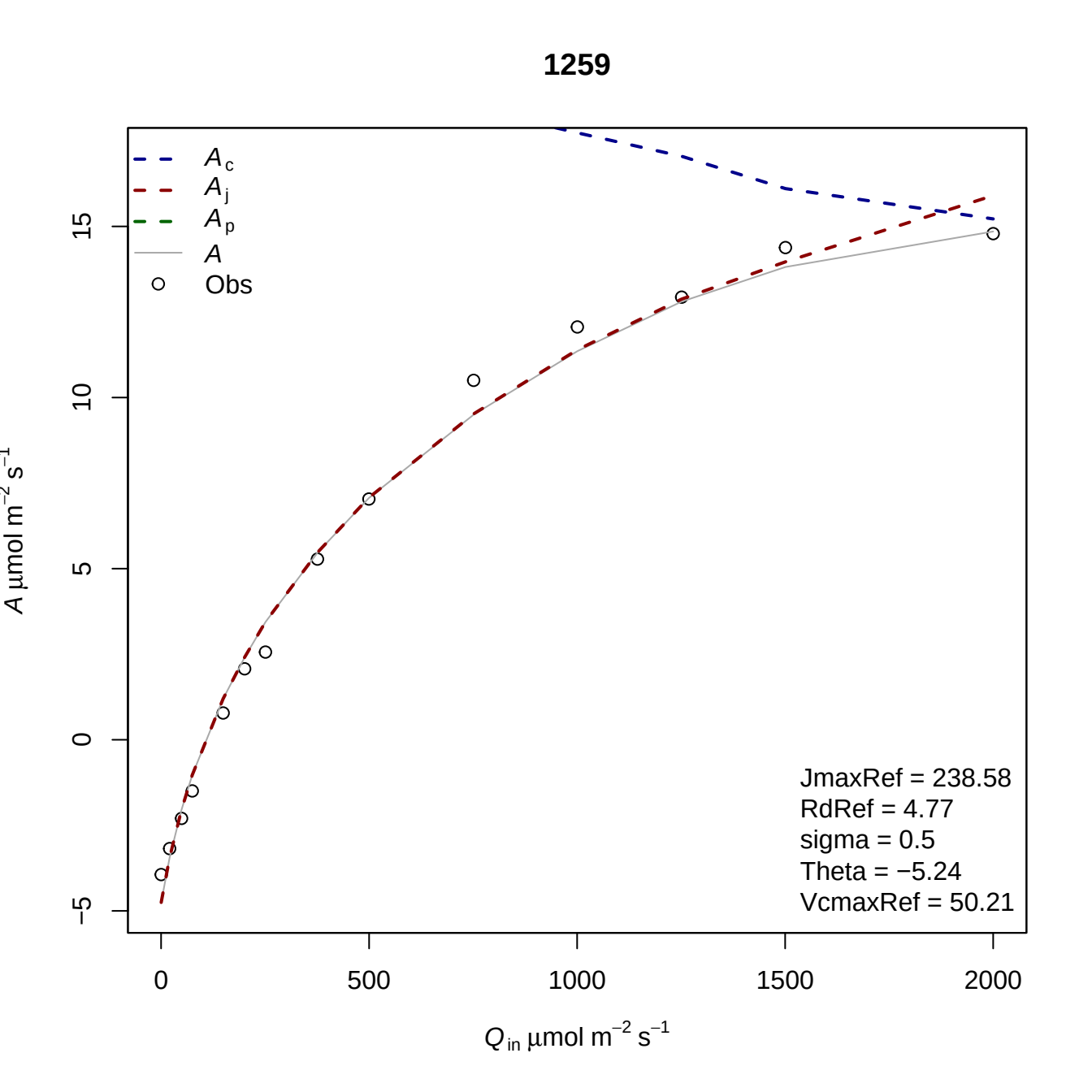

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 106.63
RdRef = 2.42
sigma = 0.32
Theta = −1.65
VcmaxRef = 60581.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

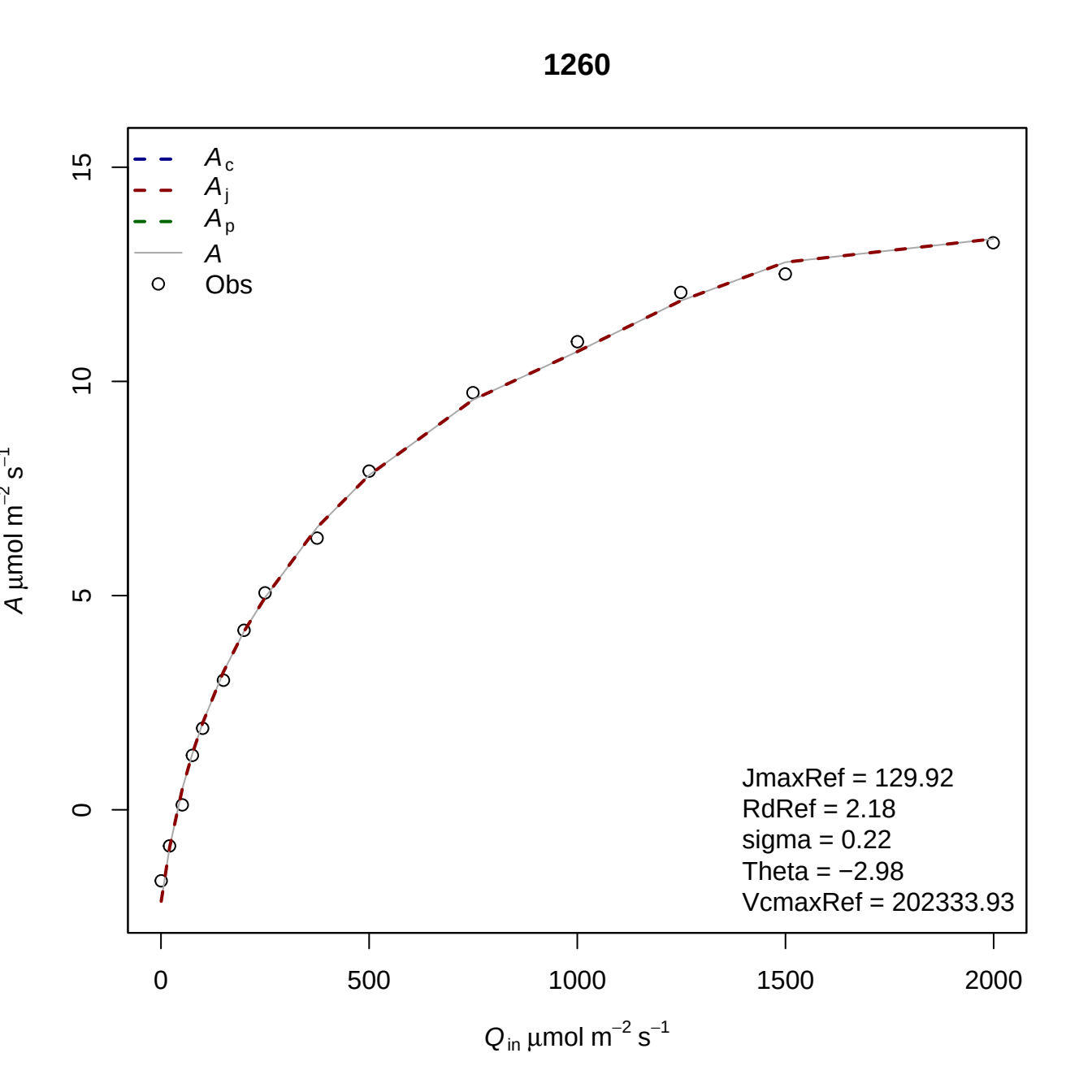

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 241.36
RdRef = 1.9
sigma = 0.58
Theta = −16.93
VcmaxRef = 118407.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

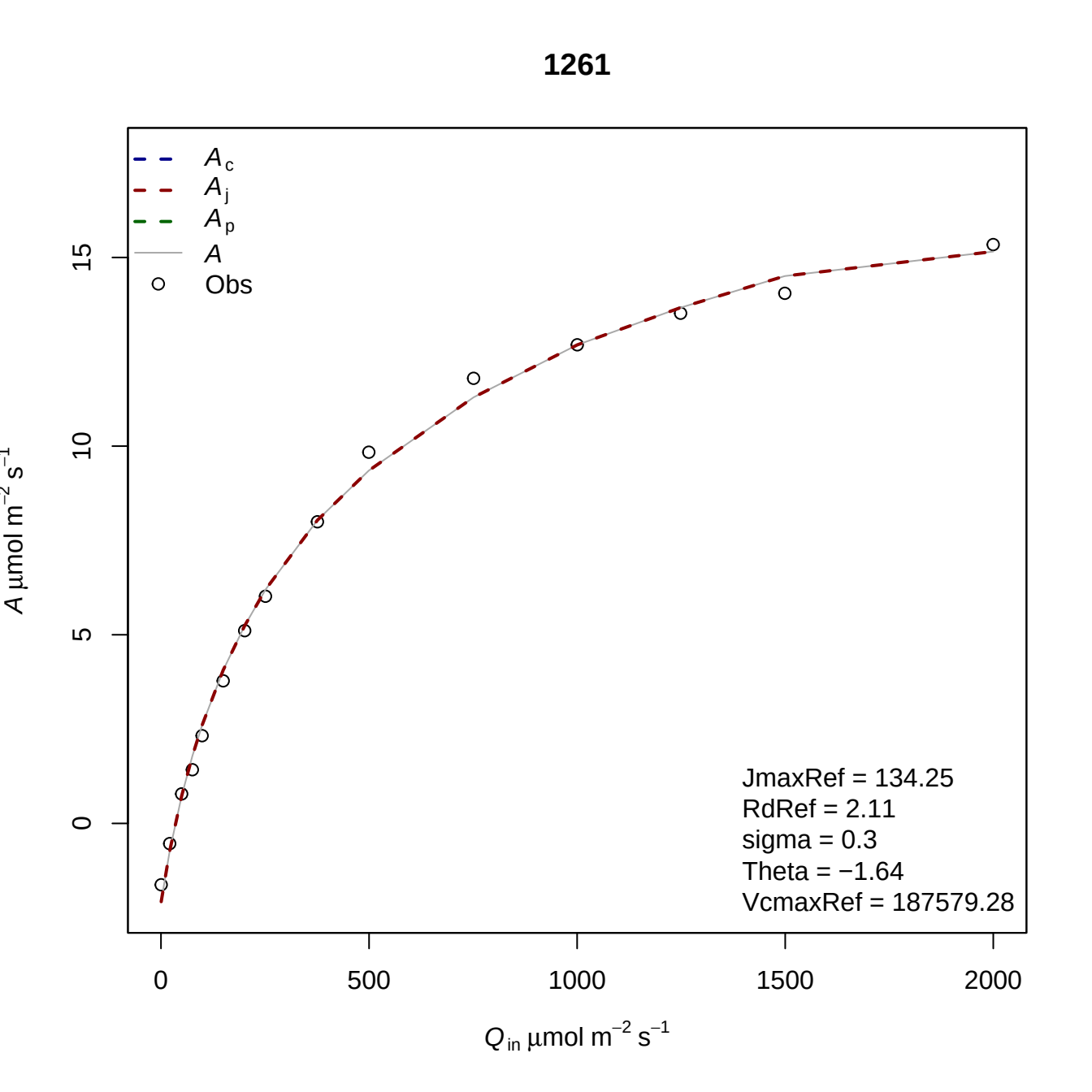

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 170.17
RdRef = 4.65
sigma = 0.25
Theta = −9.17
VcmaxRef = 33.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

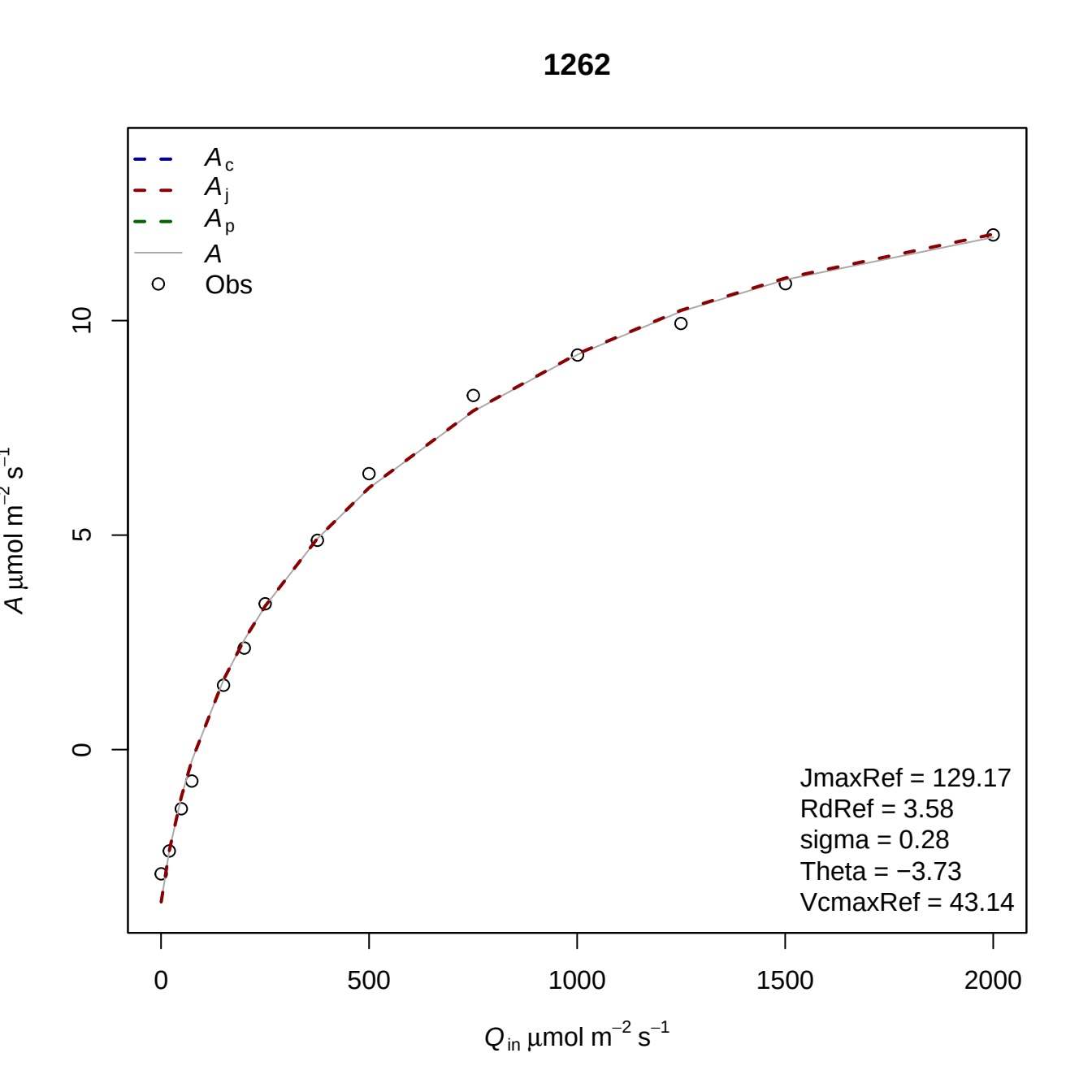

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 152.27
RdRef = 4.3
sigma = 0.28
Theta = −1.83
VcmaxRef = 50.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

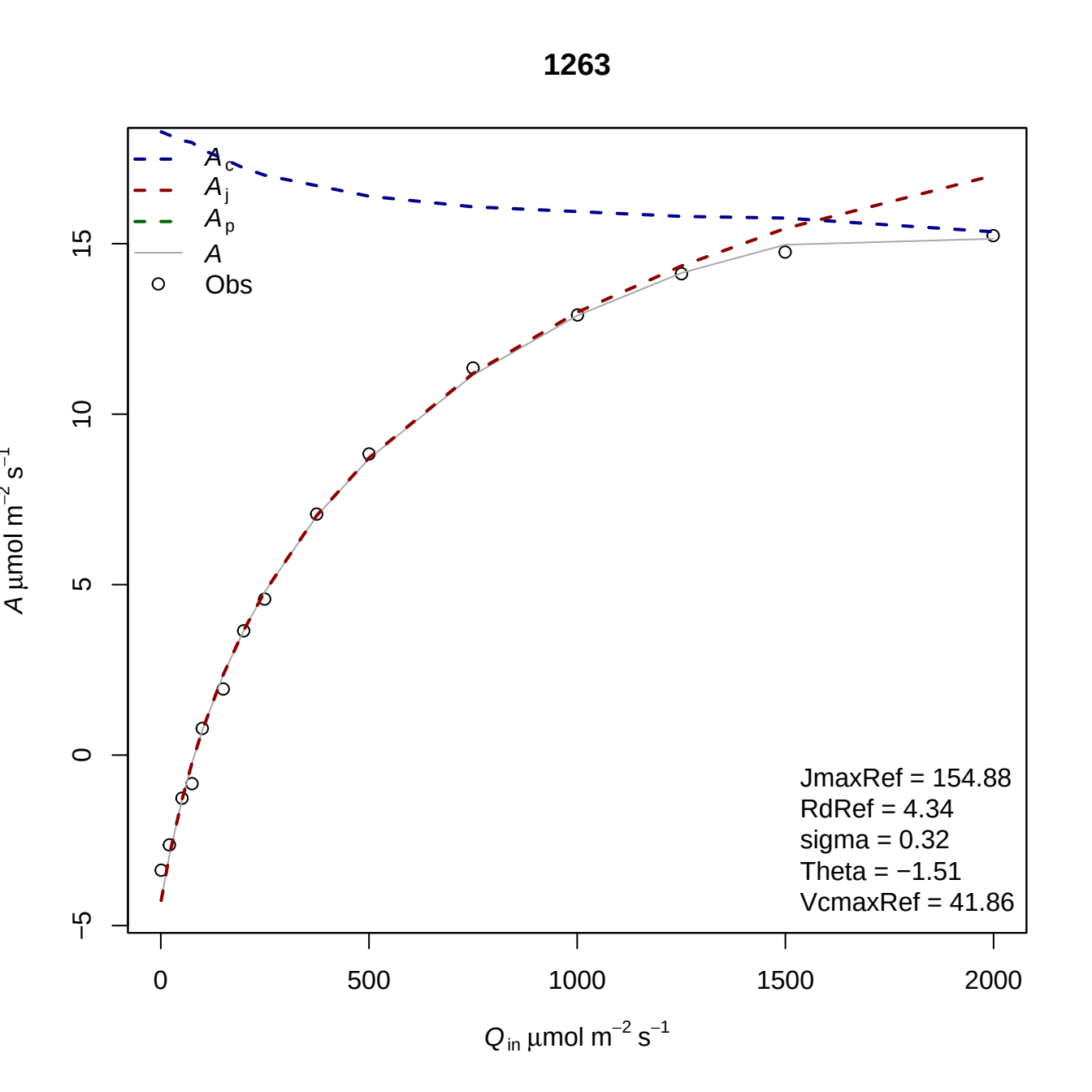

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 126.42
RdRef = 3
sigma = 0.61
Theta = −0.73
VcmaxRef = 20329.99
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

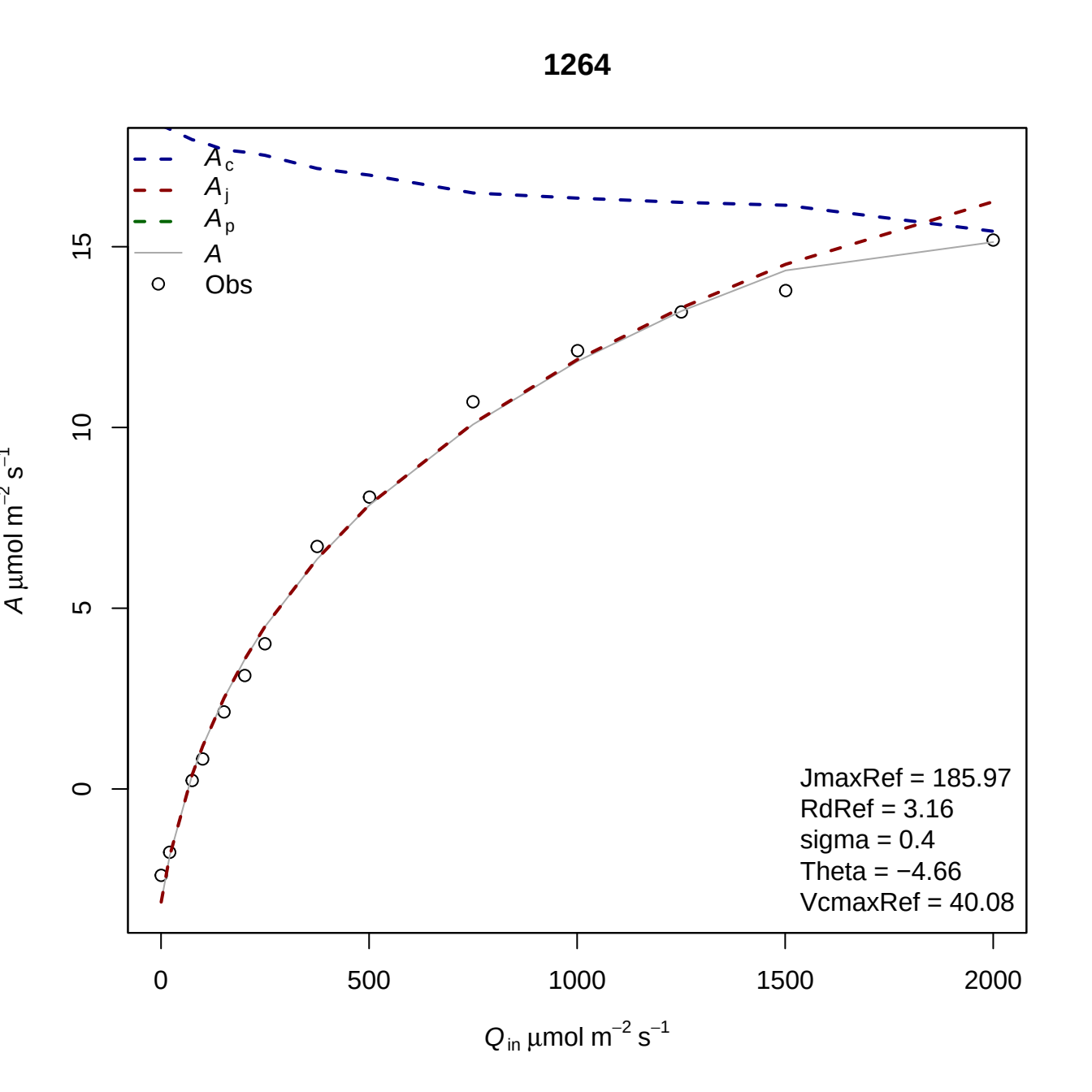

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 238.58
RdRef = 4.77
sigma = 0.5
Theta = −5.24
VcmaxRef = 50.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

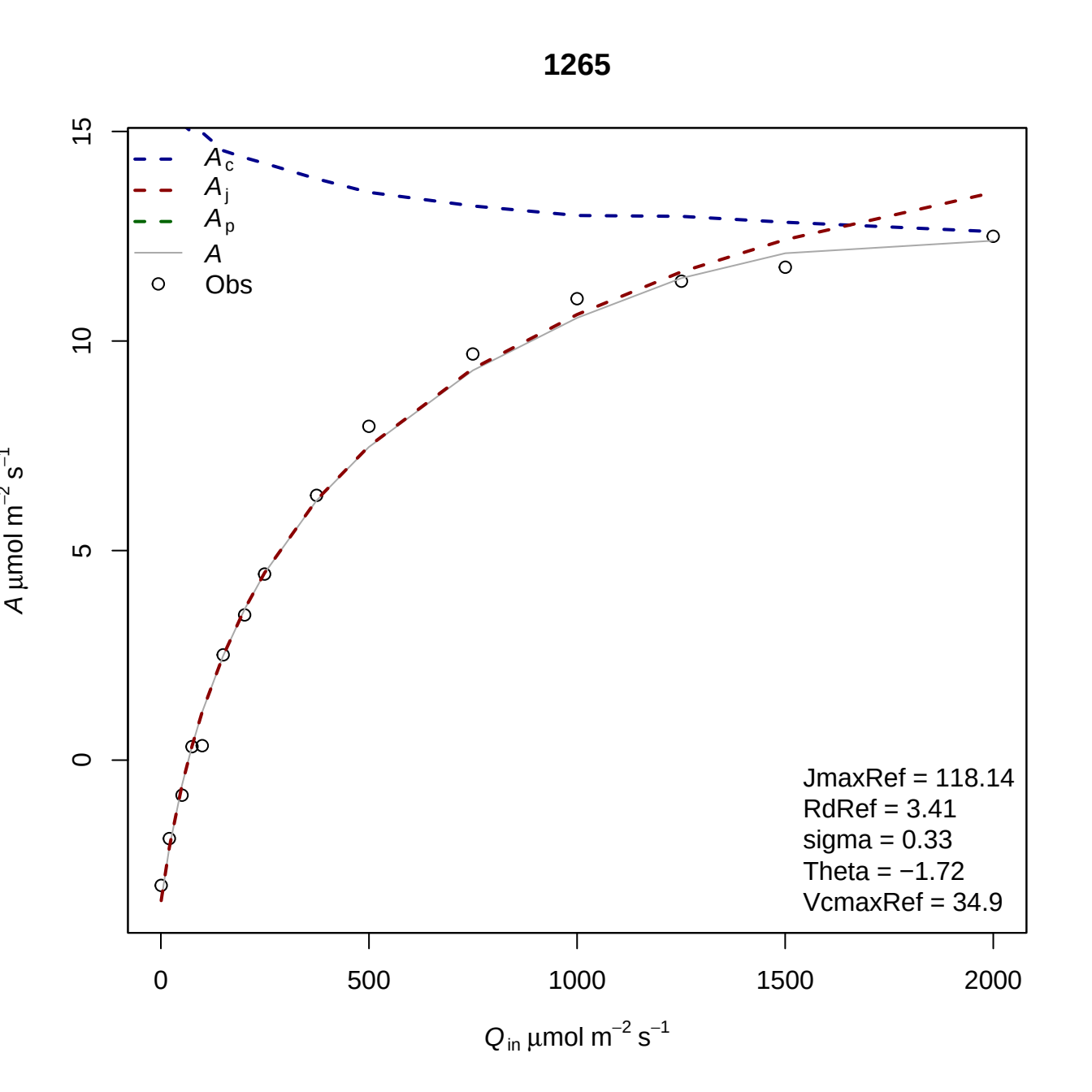

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 129.92
RdRef = 2.18
sigma = 0.22
Theta = −2.98
VcmaxRef = 202333.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

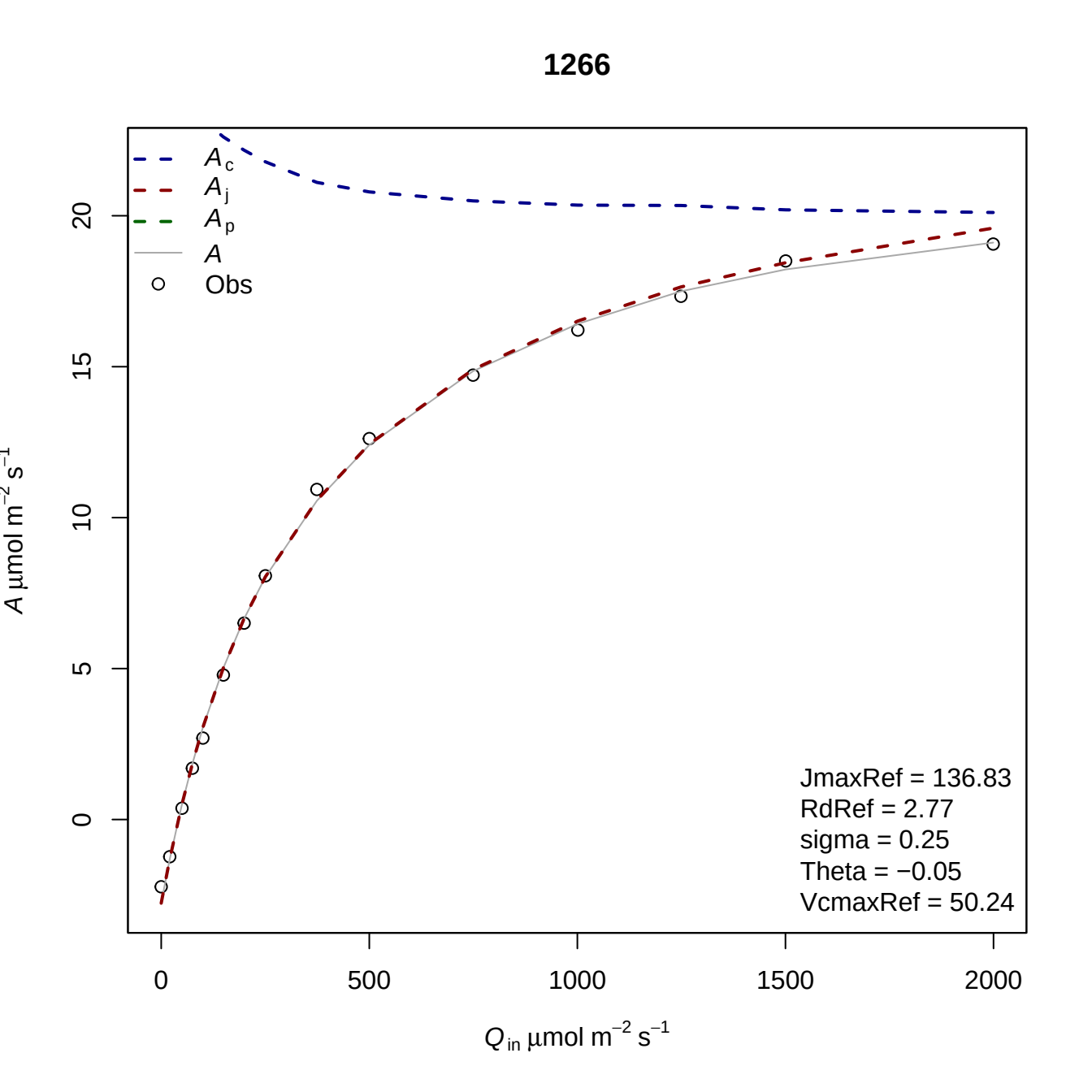

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 134.25
RdRef = 2.11
sigma = 0.3
Theta = −1.64
VcmaxRef = 187579.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

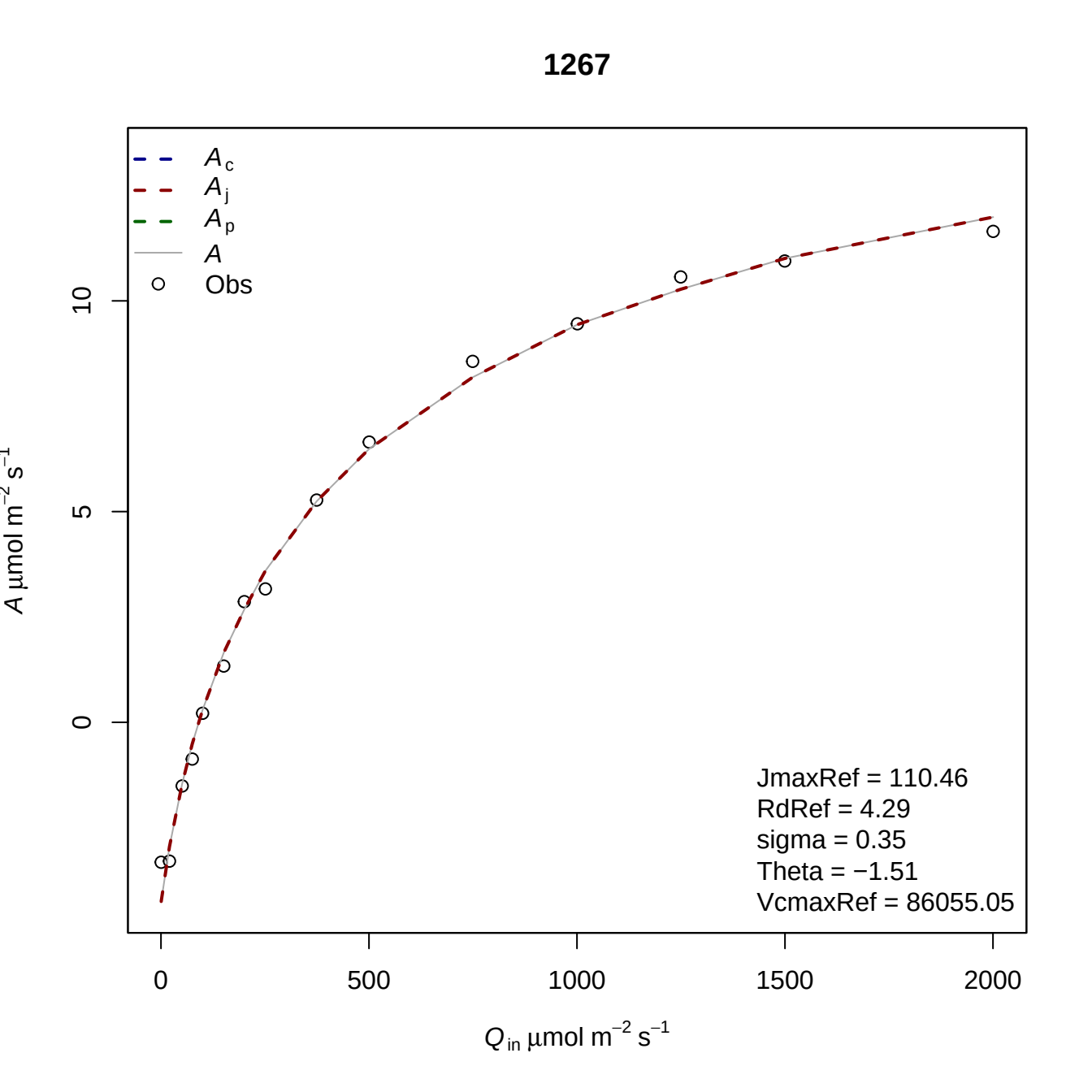

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 129.17
RdRef = 3.58
sigma = 0.28
Theta = −3.73
VcmaxRef = 43.14
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

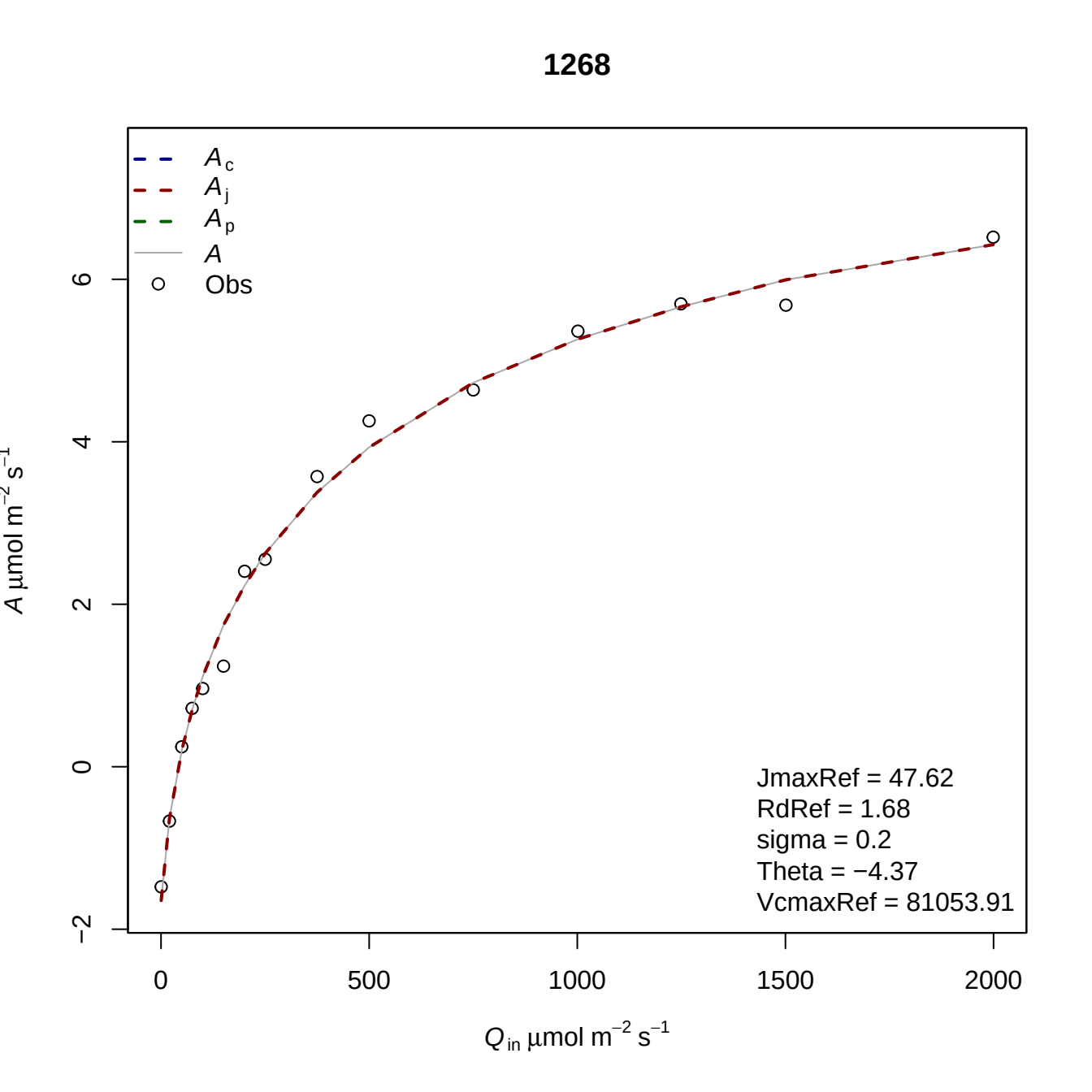

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 154.88
RdRef = 4.34
sigma = 0.32
Theta = −1.51
VcmaxRef = 41.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

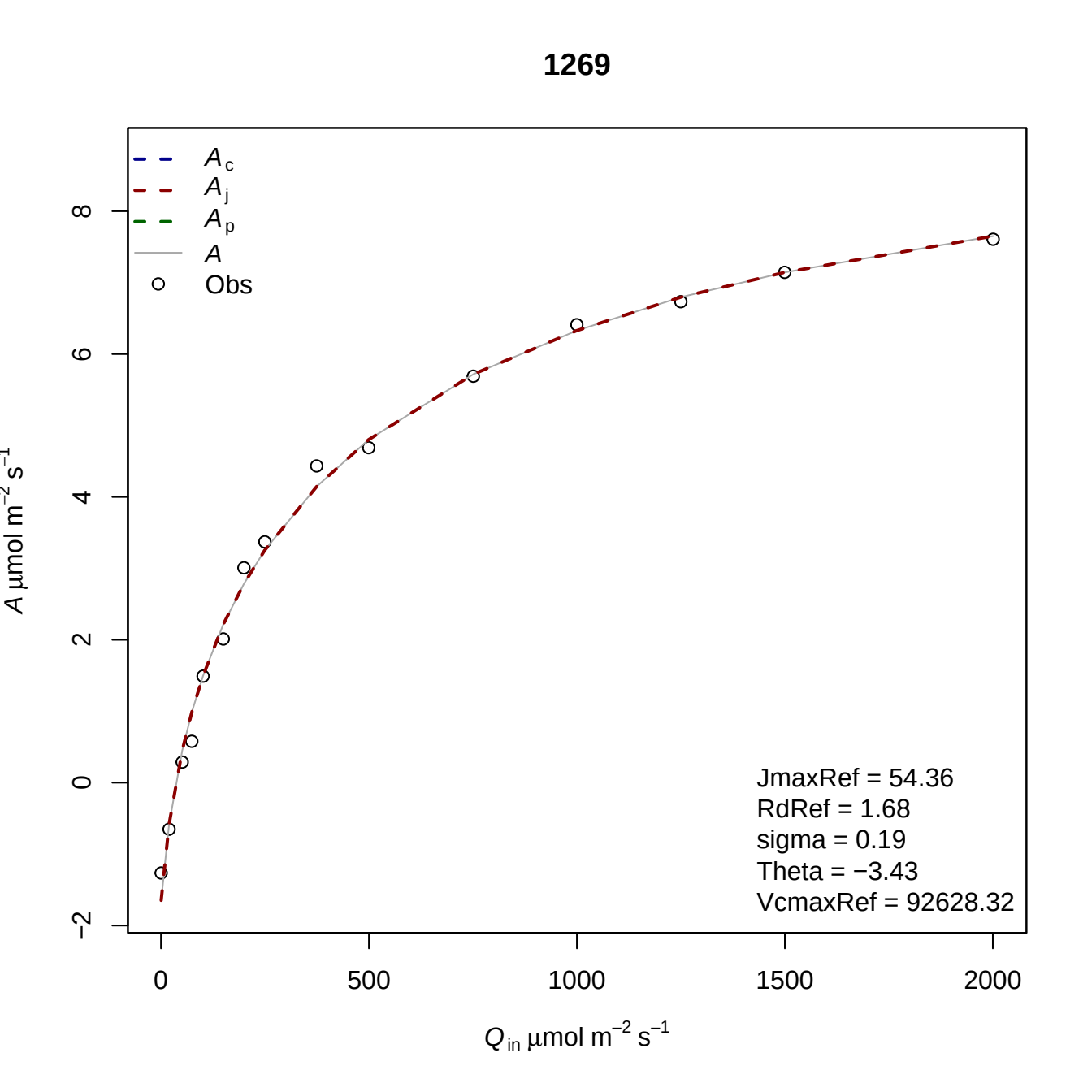

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 185.97
RdRef = 3.16
sigma = 0.4
Theta = −4.66
VcmaxRef = 40.08
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

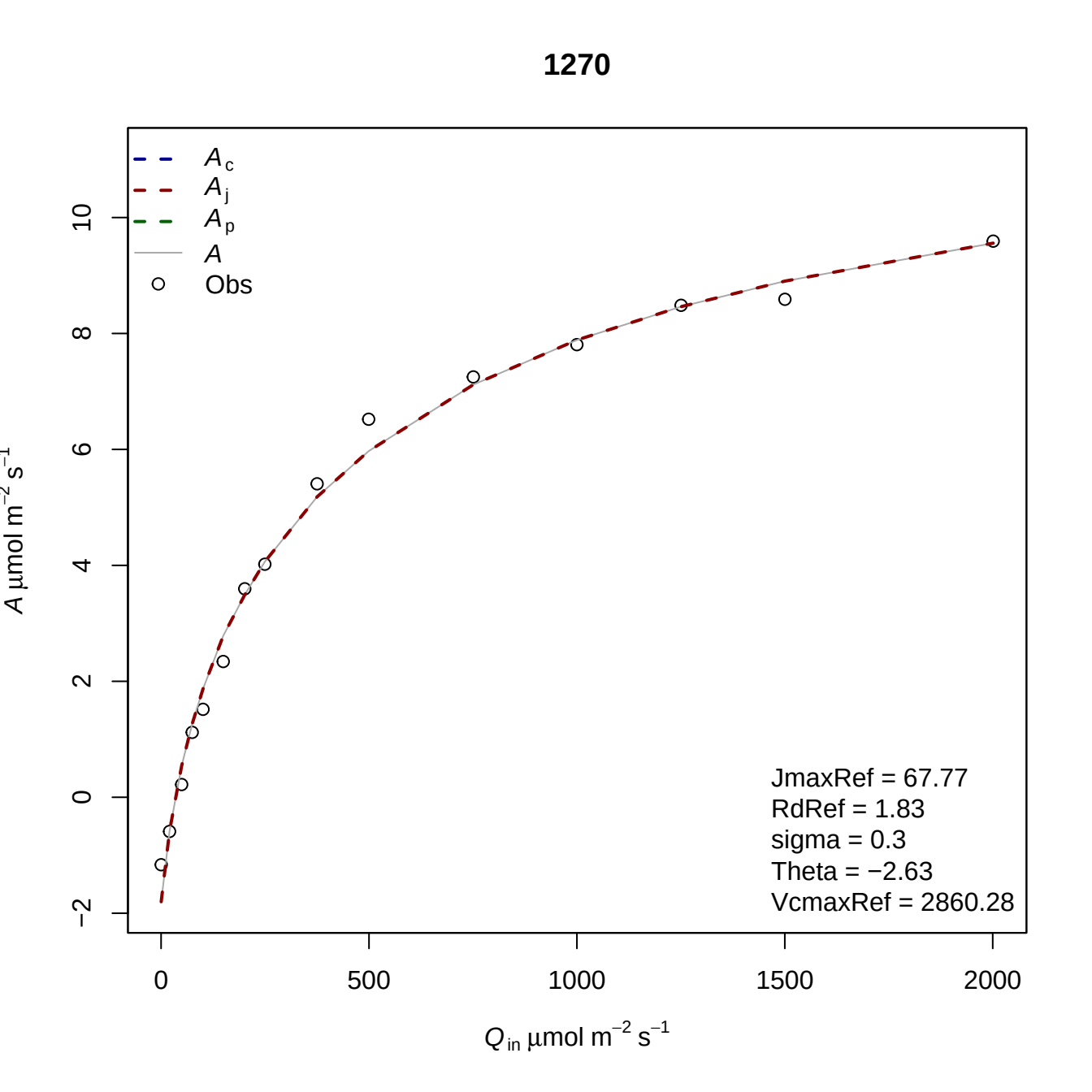

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 118.14
RdRef = 3.41
sigma = 0.33
Theta = −1.72
VcmaxRef = 34.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

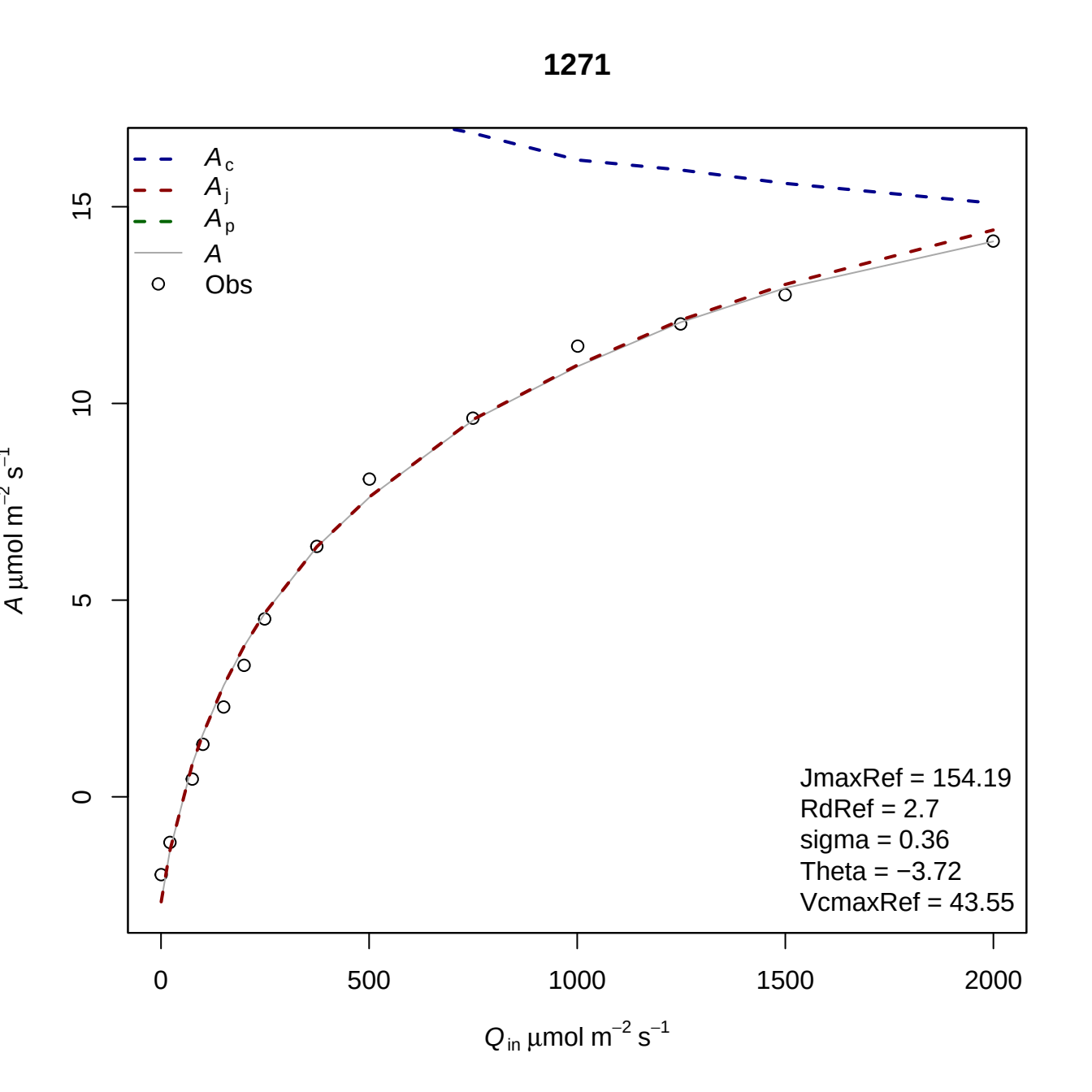

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 136.83
RdRef = 2.77
sigma = 0.25
Theta = −0.05
VcmaxRef = 50.24
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

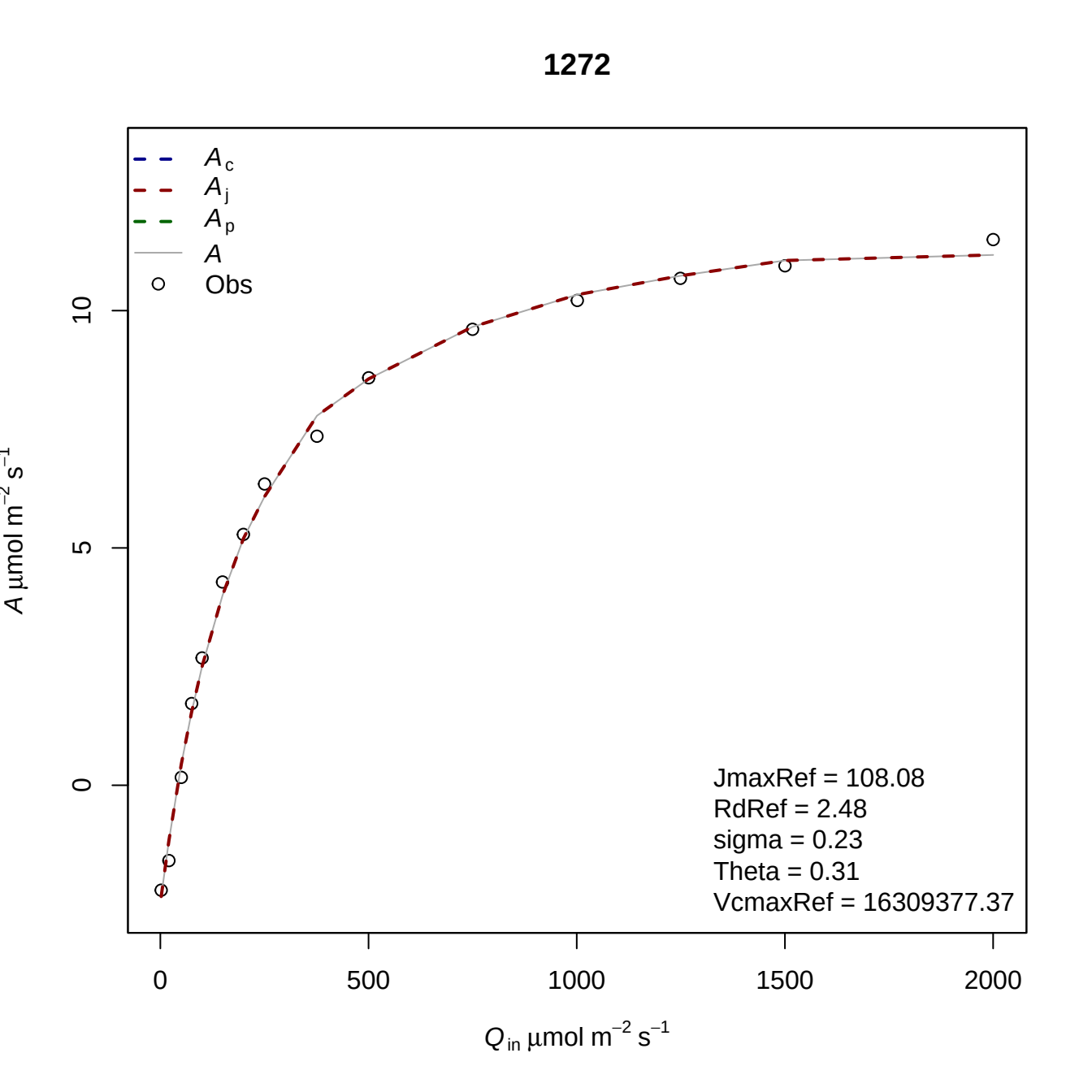

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 110.46
RdRef = 4.29
sigma = 0.35
Theta = −1.51
VcmaxRef = 86055.05
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

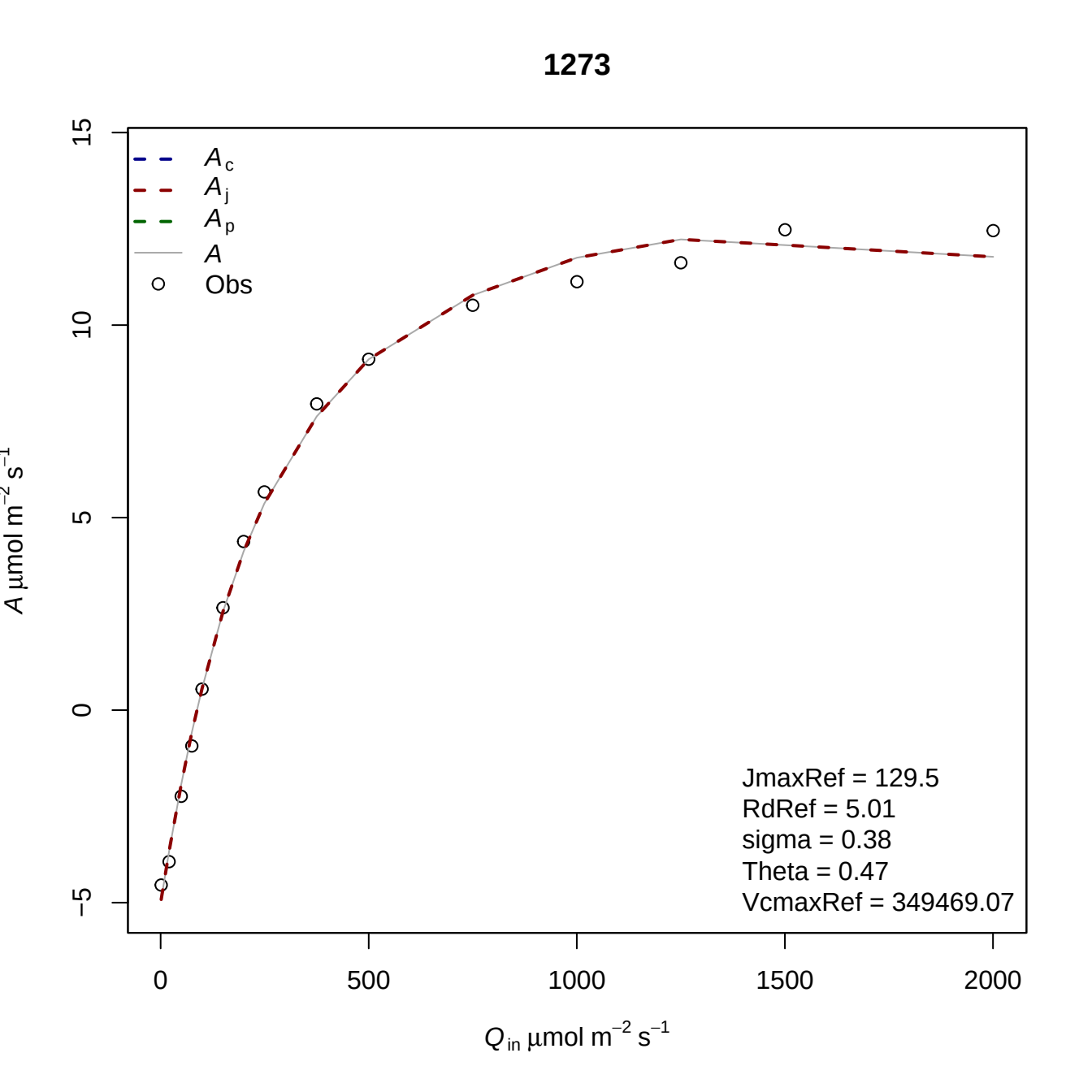

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 47.62
RdRef = 1.68
sigma = 0.2
Theta = −4.37
VcmaxRef = 81053.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

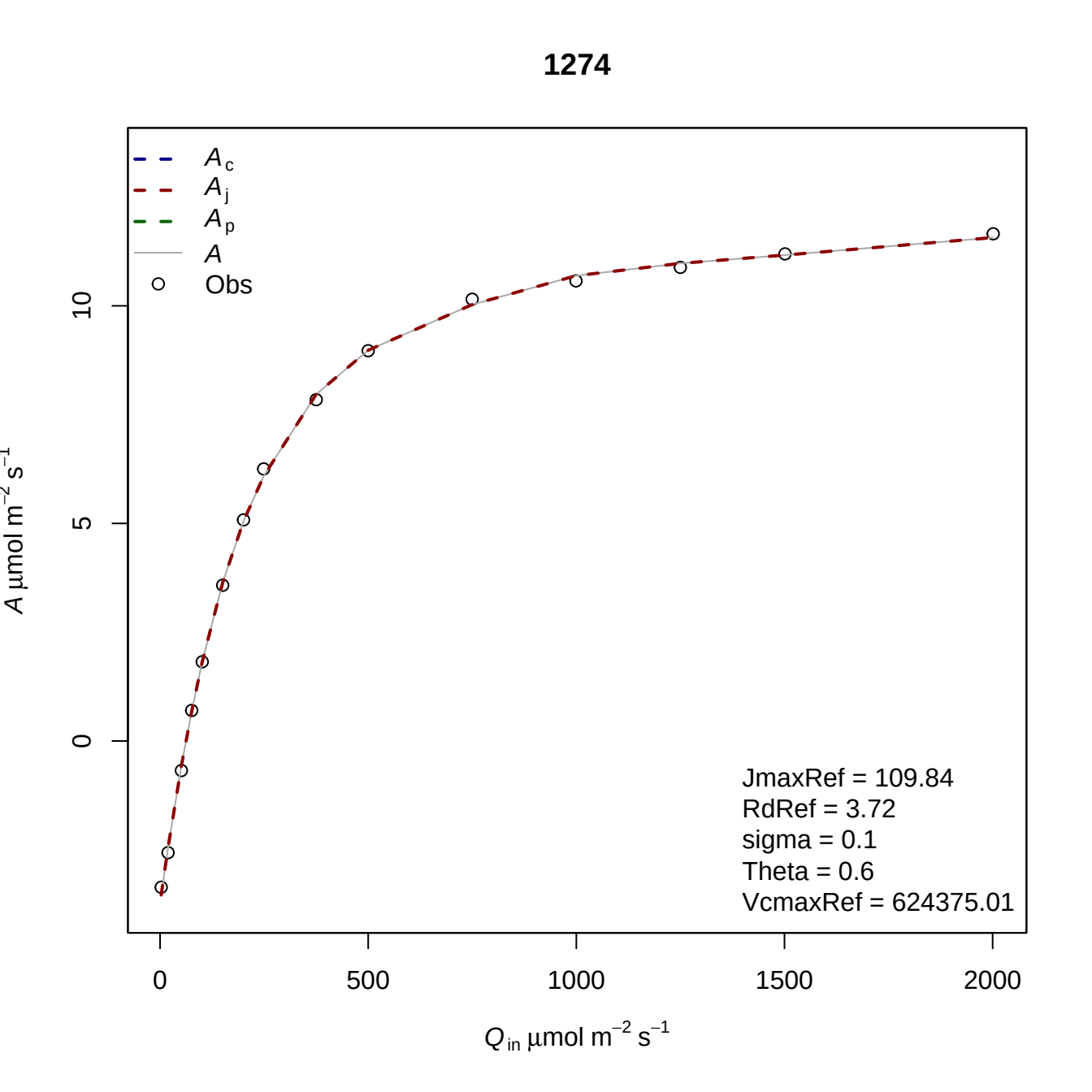

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 54.36
RdRef = 1.68
sigma = 0.19
Theta = −3.43
VcmaxRef = 92628.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

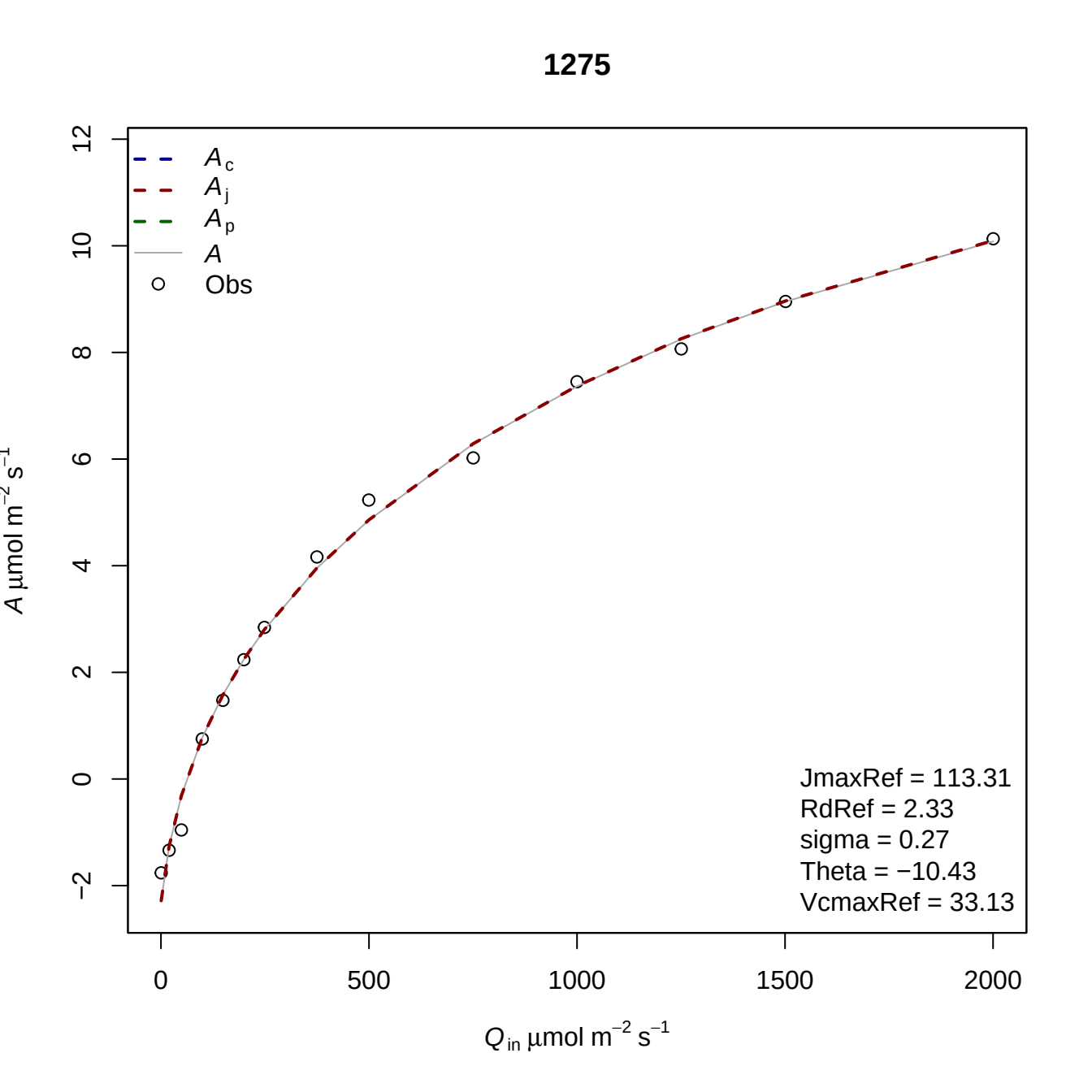

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 67.77
RdRef = 1.83
sigma = 0.3
Theta = −2.63
VcmaxRef = 2860.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

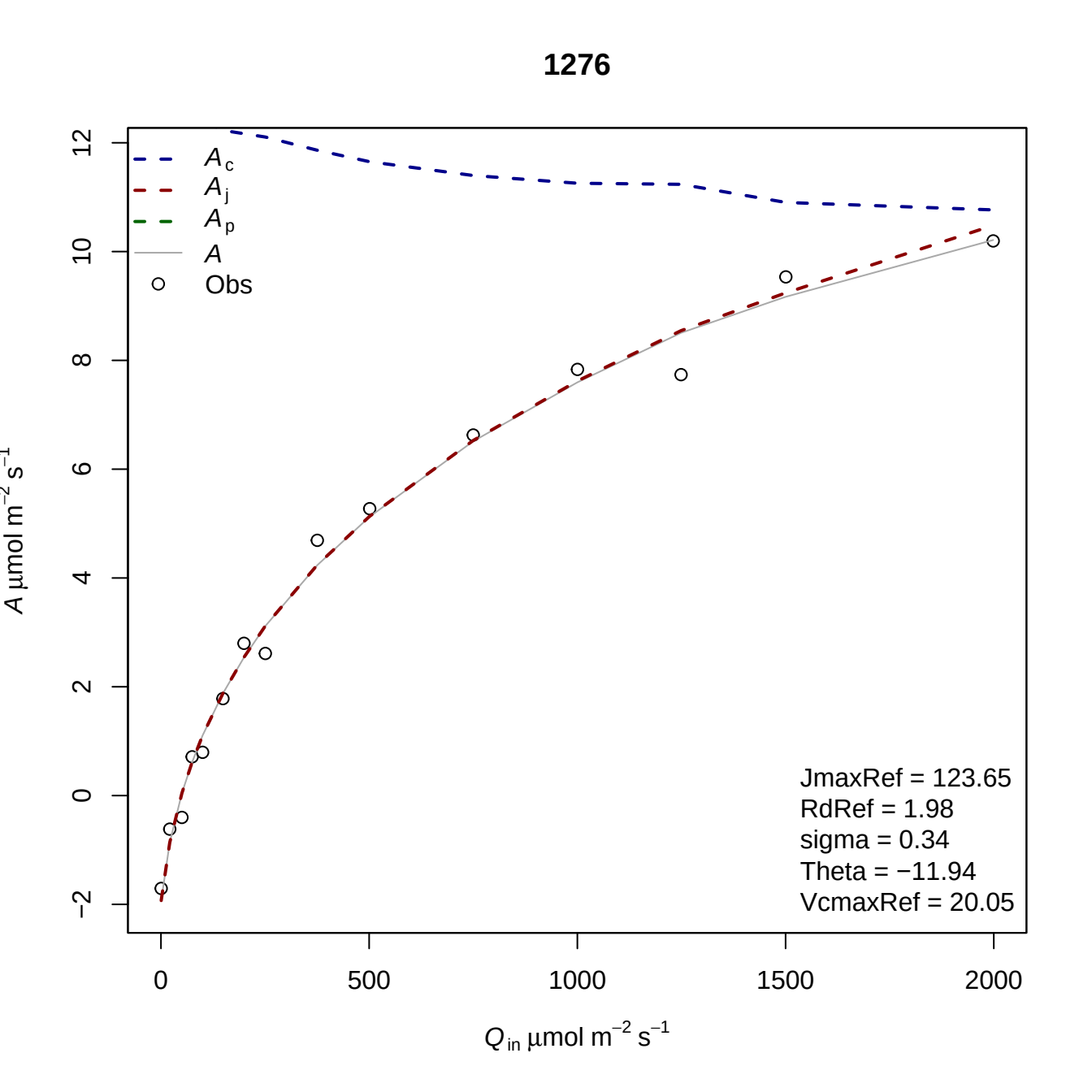

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 154.19
RdRef = 2.7
sigma = 0.36
Theta = −3.72
VcmaxRef = 43.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

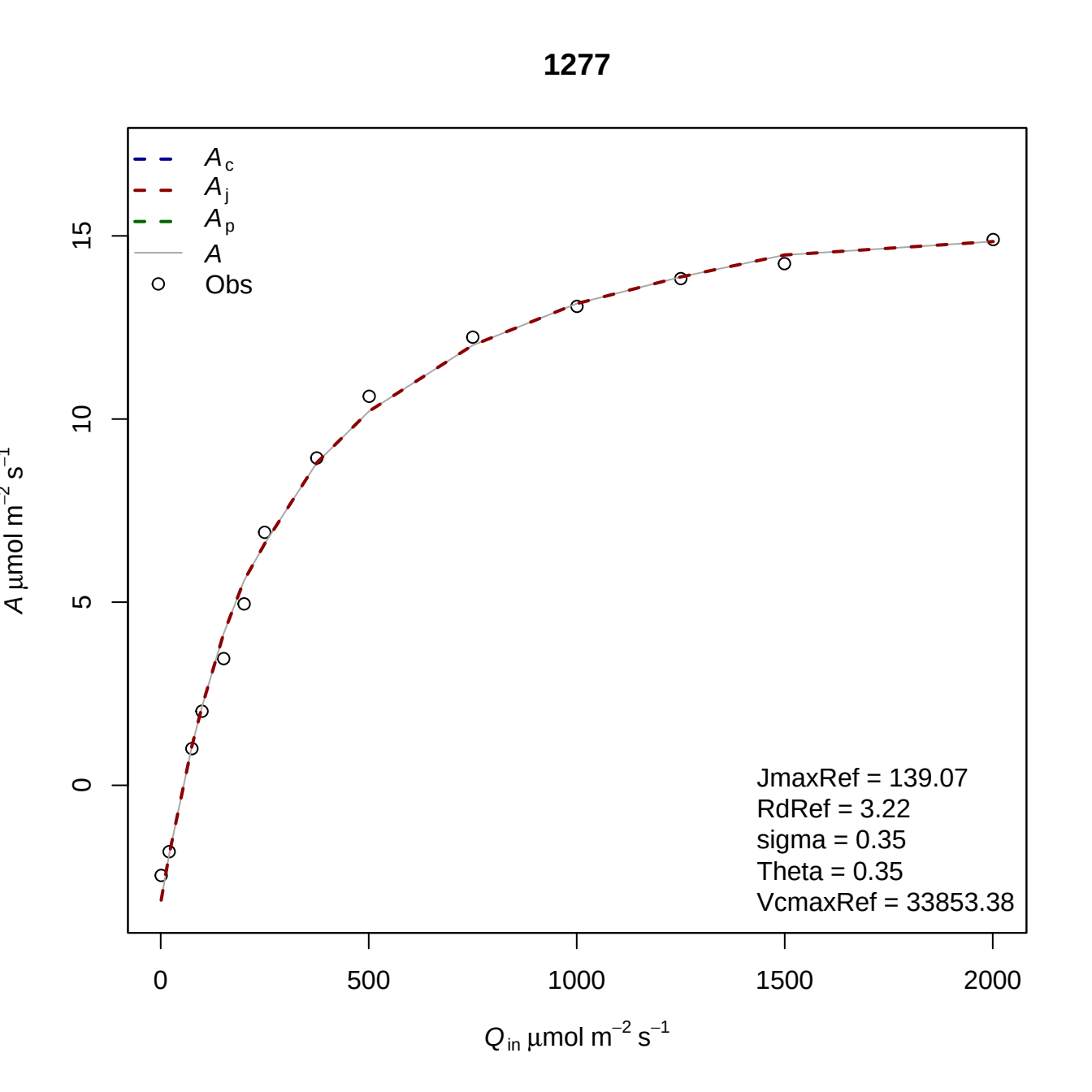

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 108.08
RdRef = 2.48
sigma = 0.23
Theta = 0.31
VcmaxRef = 16309377.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

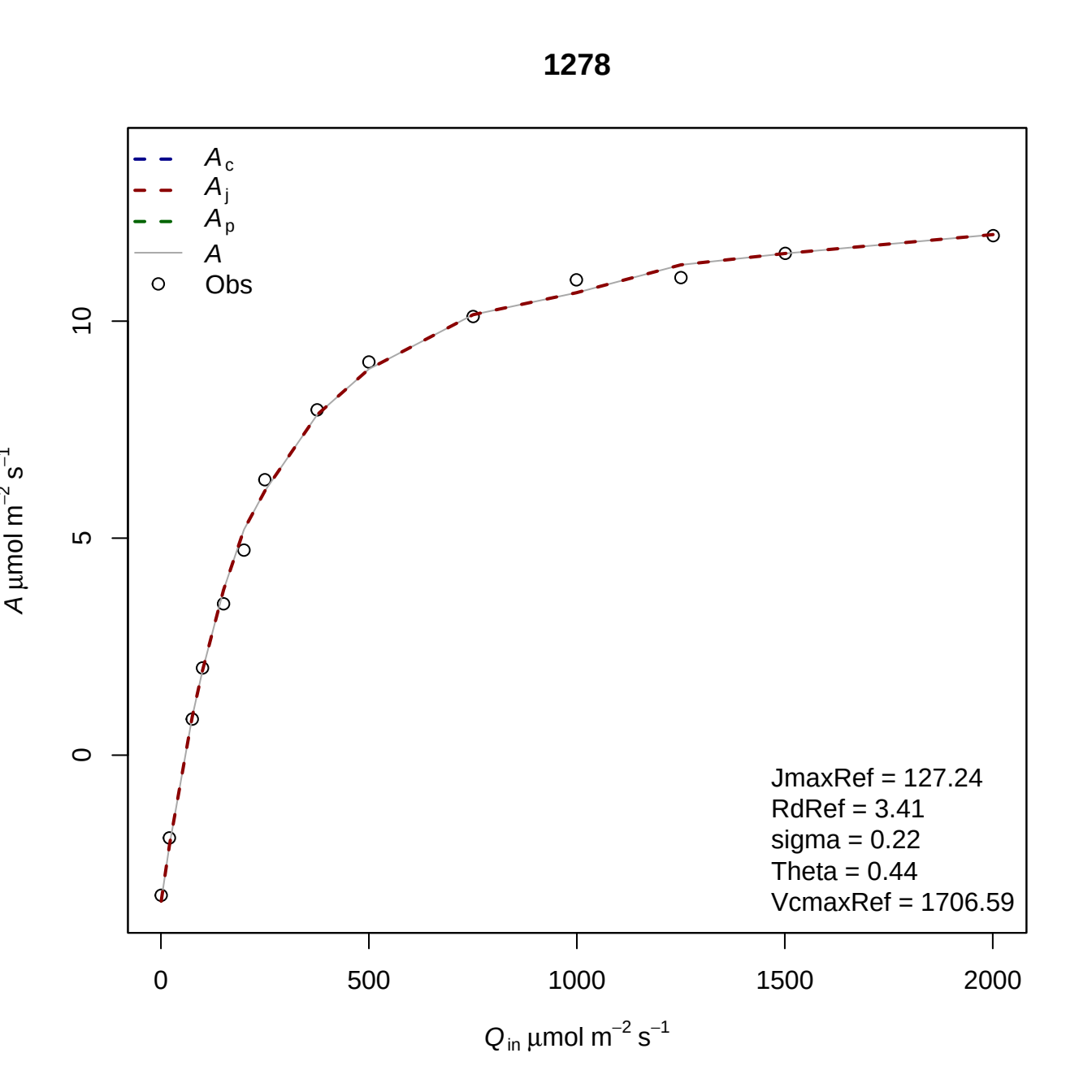

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 129.5
RdRef = 5.01
sigma = 0.38
Theta = 0.47
VcmaxRef = 349469.07
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

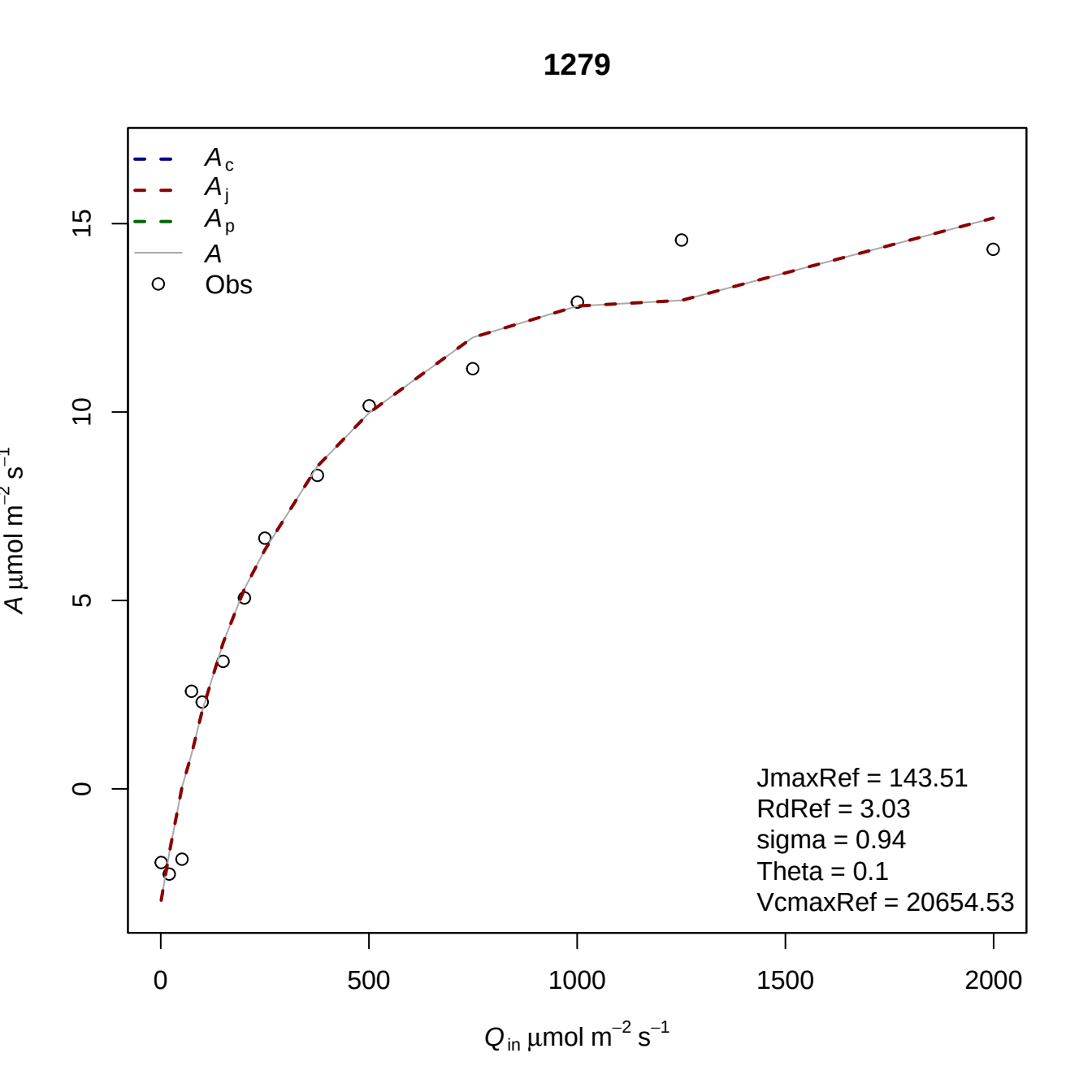

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 109.84
RdRef = 3.72
sigma = 0.1
Theta = 0.6
VcmaxRef = 624375.01
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

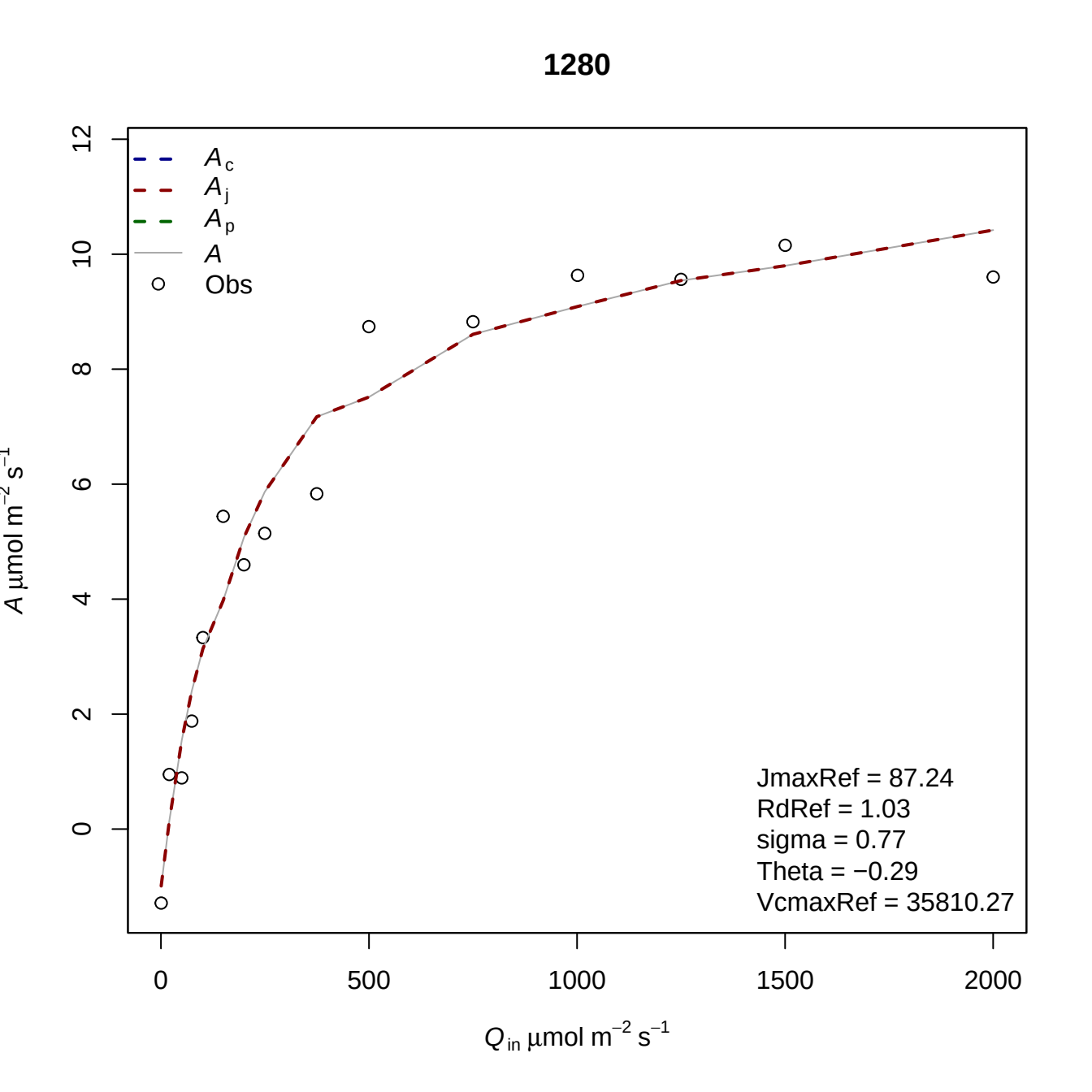

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 113.31
RdRef = 2.33
sigma = 0.27
Theta = −10.43
VcmaxRef = 33.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

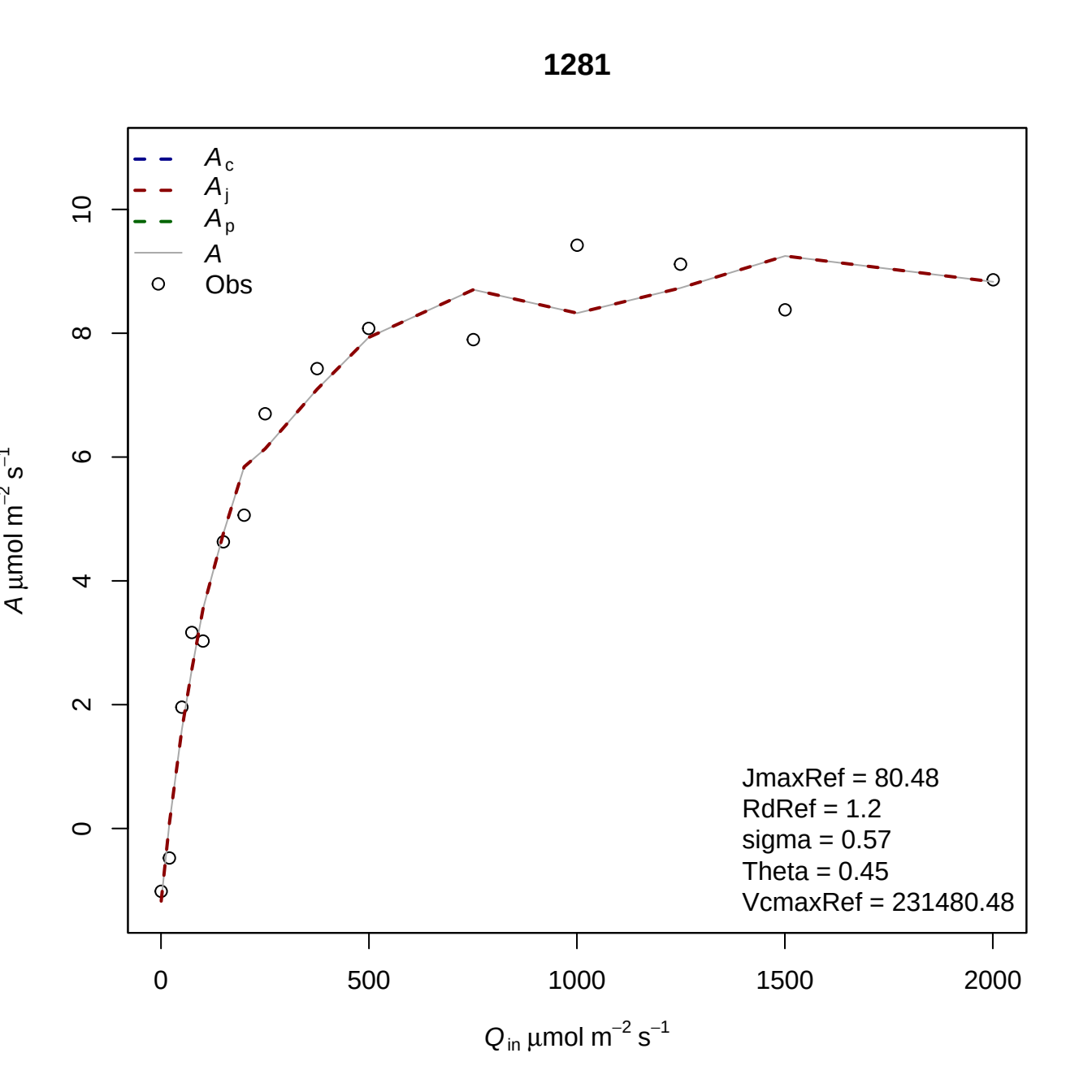

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 123.65
RdRef = 1.98
sigma = 0.34
Theta = −11.94
VcmaxRef = 20.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

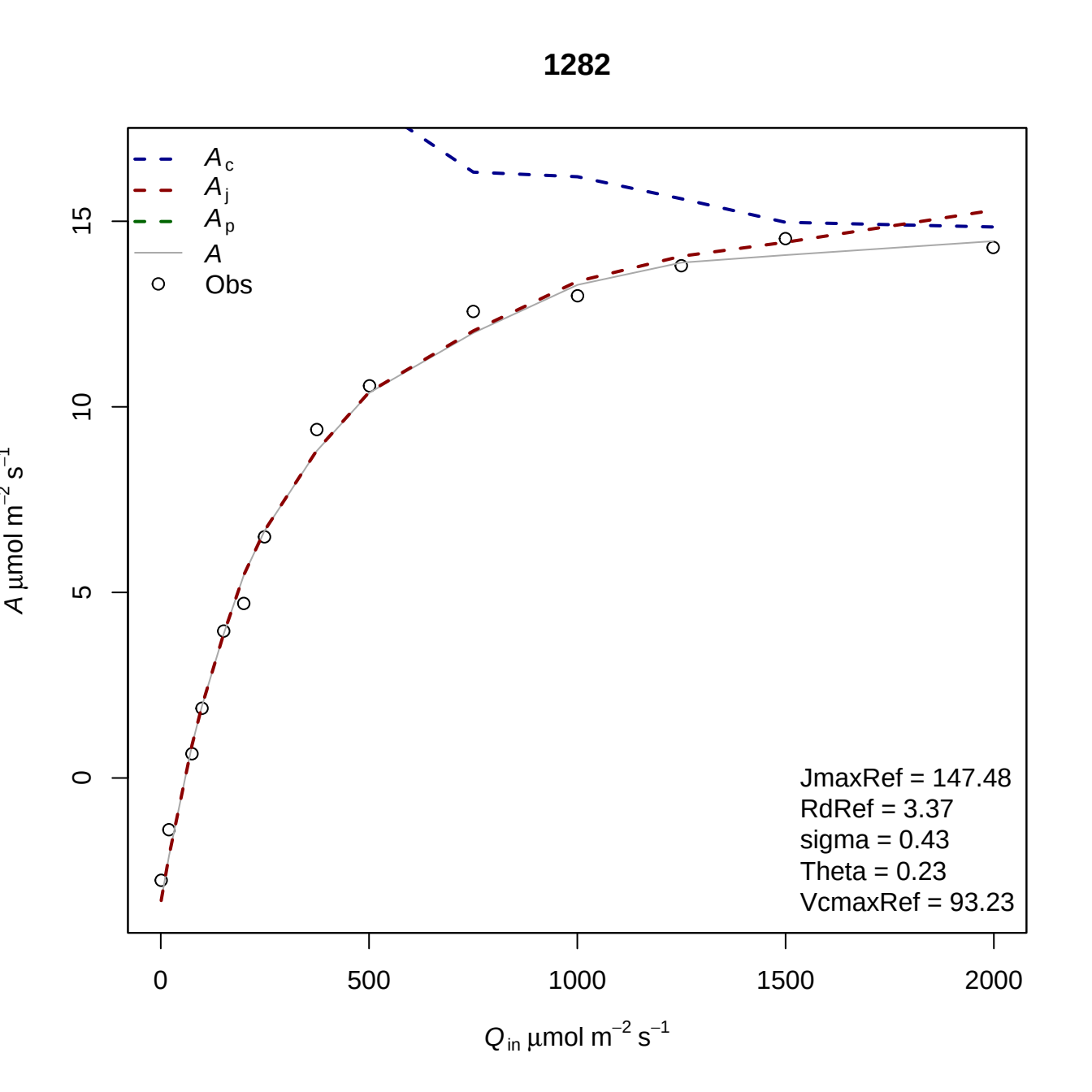

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 139.07
RdRef = 3.22
sigma = 0.35
Theta = 0.35
VcmaxRef = 33853.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

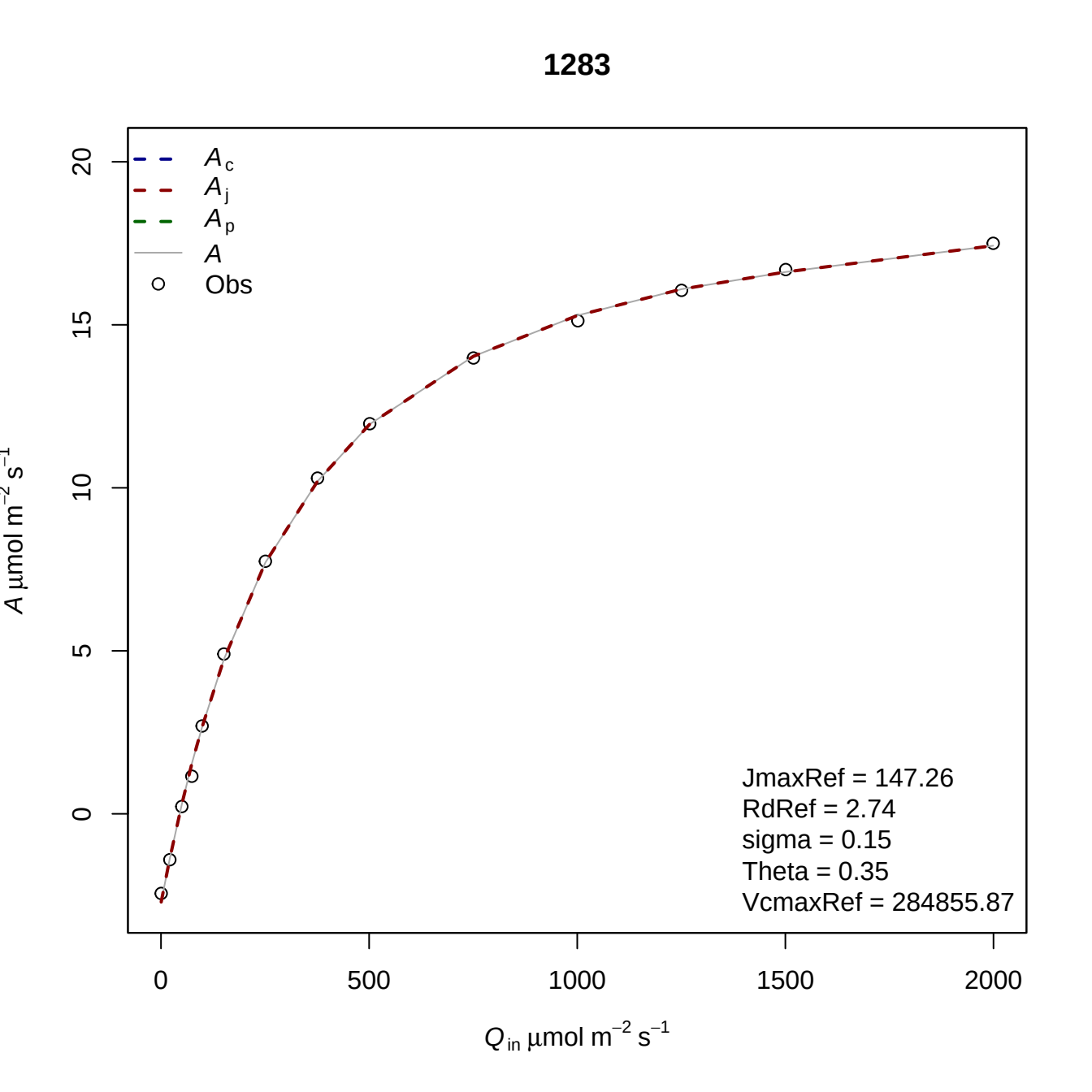

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 127.24
RdRef = 3.41
sigma = 0.22
Theta = 0.44
VcmaxRef = 1706.59
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

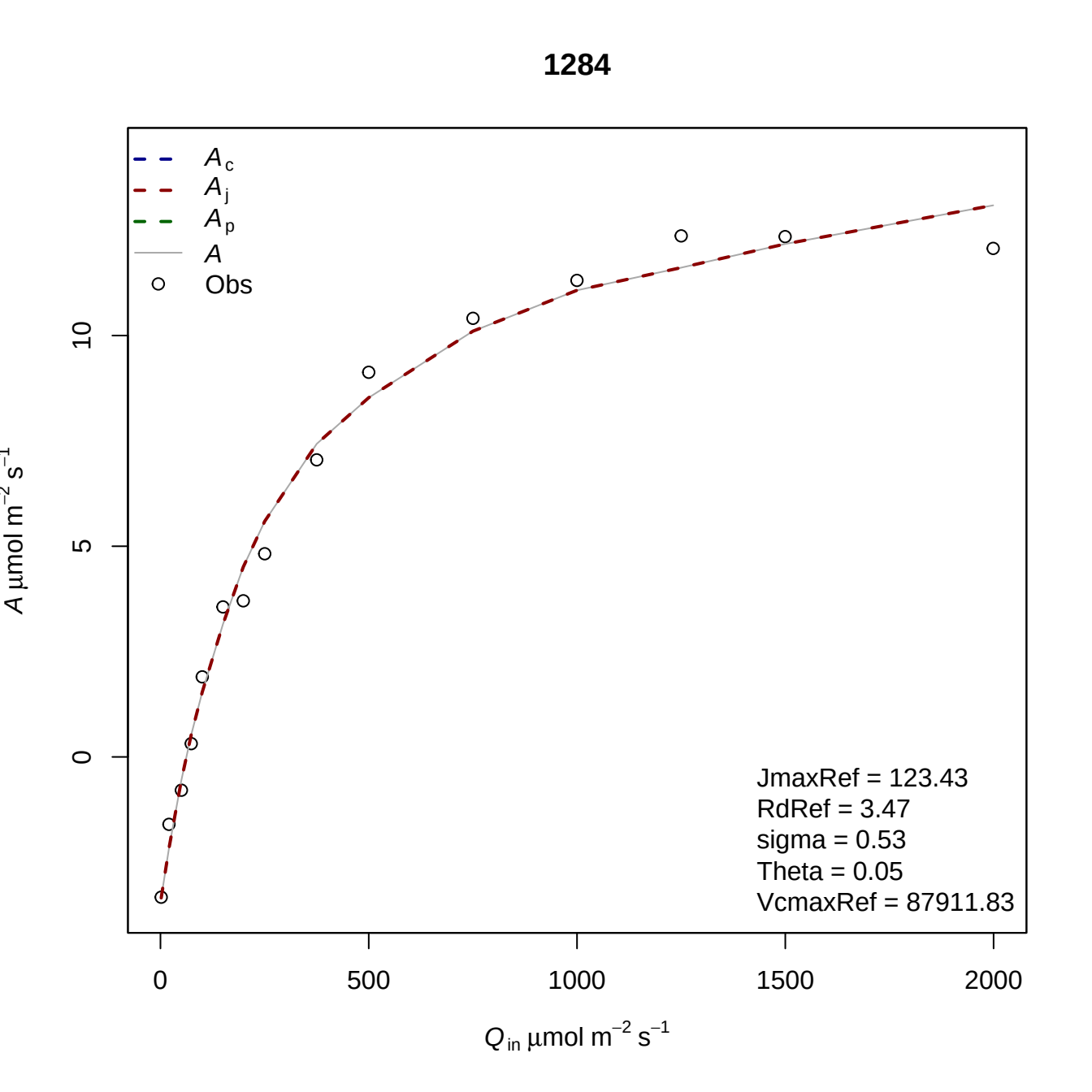

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 143.51
RdRef = 3.03
sigma = 0.94
Theta = 0.1
VcmaxRef = 20654.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

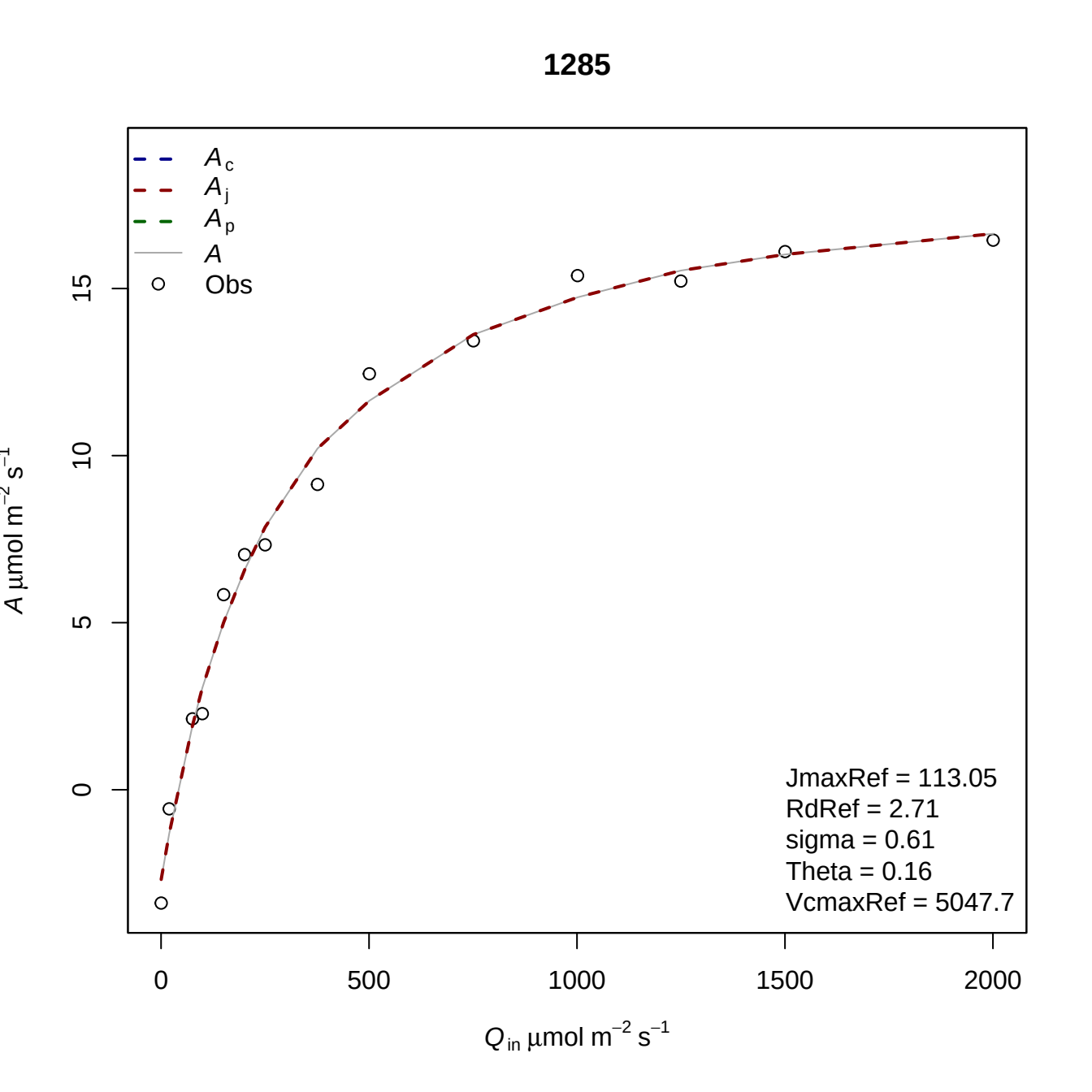

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 87.24
RdRef = 1.03
sigma = 0.77
Theta = −0.29
VcmaxRef = 35810.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

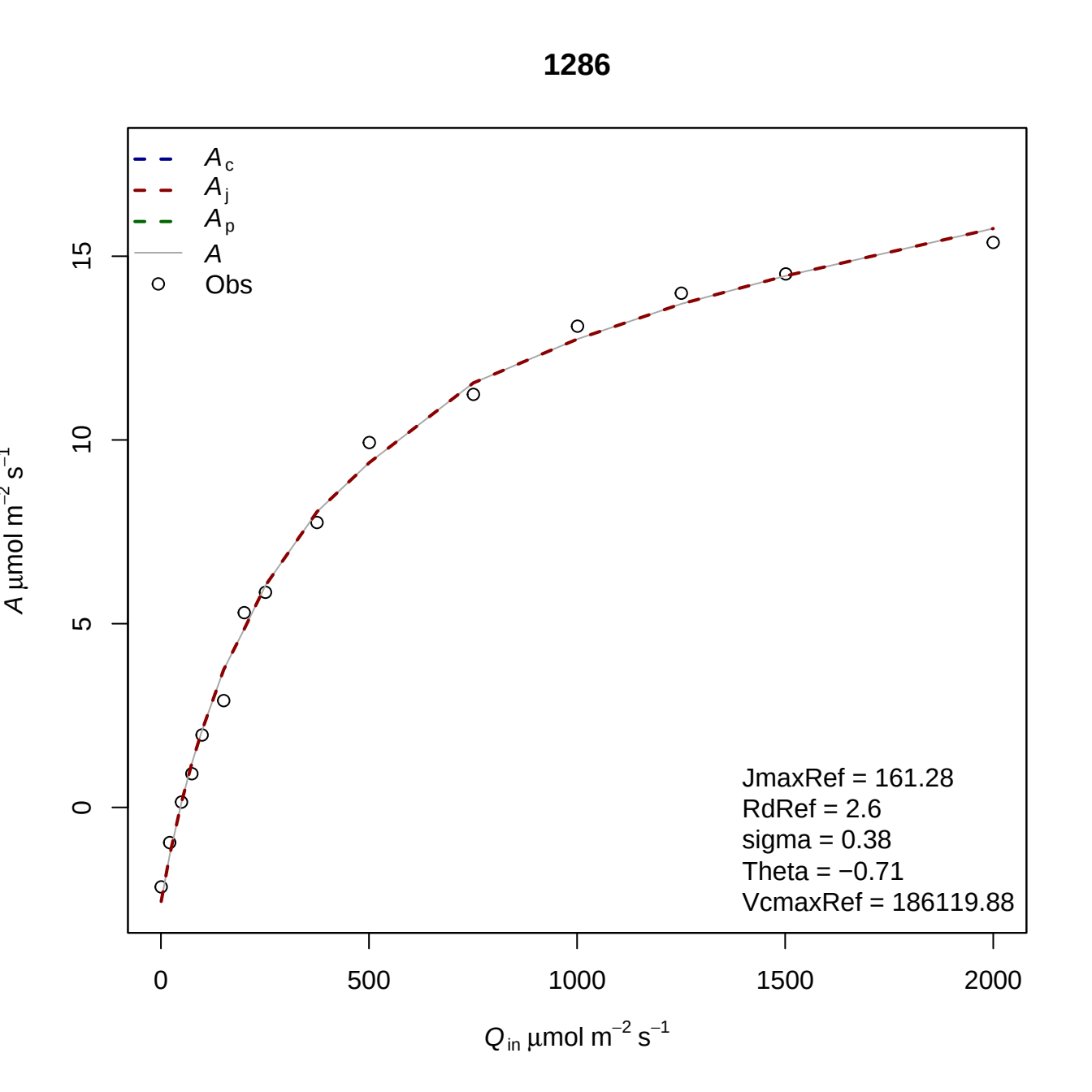

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 80.48
RdRef = 1.2
sigma = 0.57
Theta = 0.45
VcmaxRef = 231480.48
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

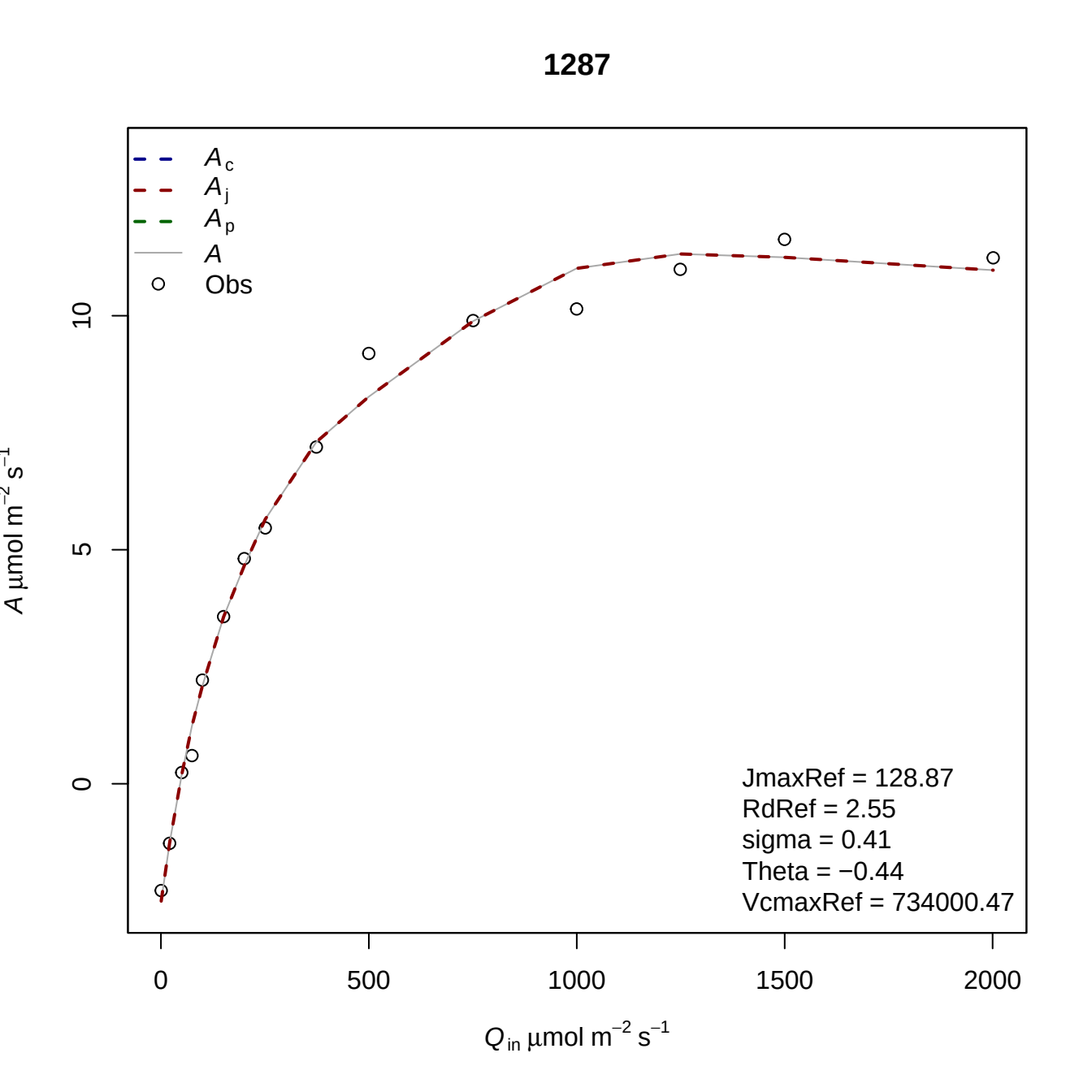

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 147.48
RdRef = 3.37
sigma = 0.43
Theta = 0.23
VcmaxRef = 93.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

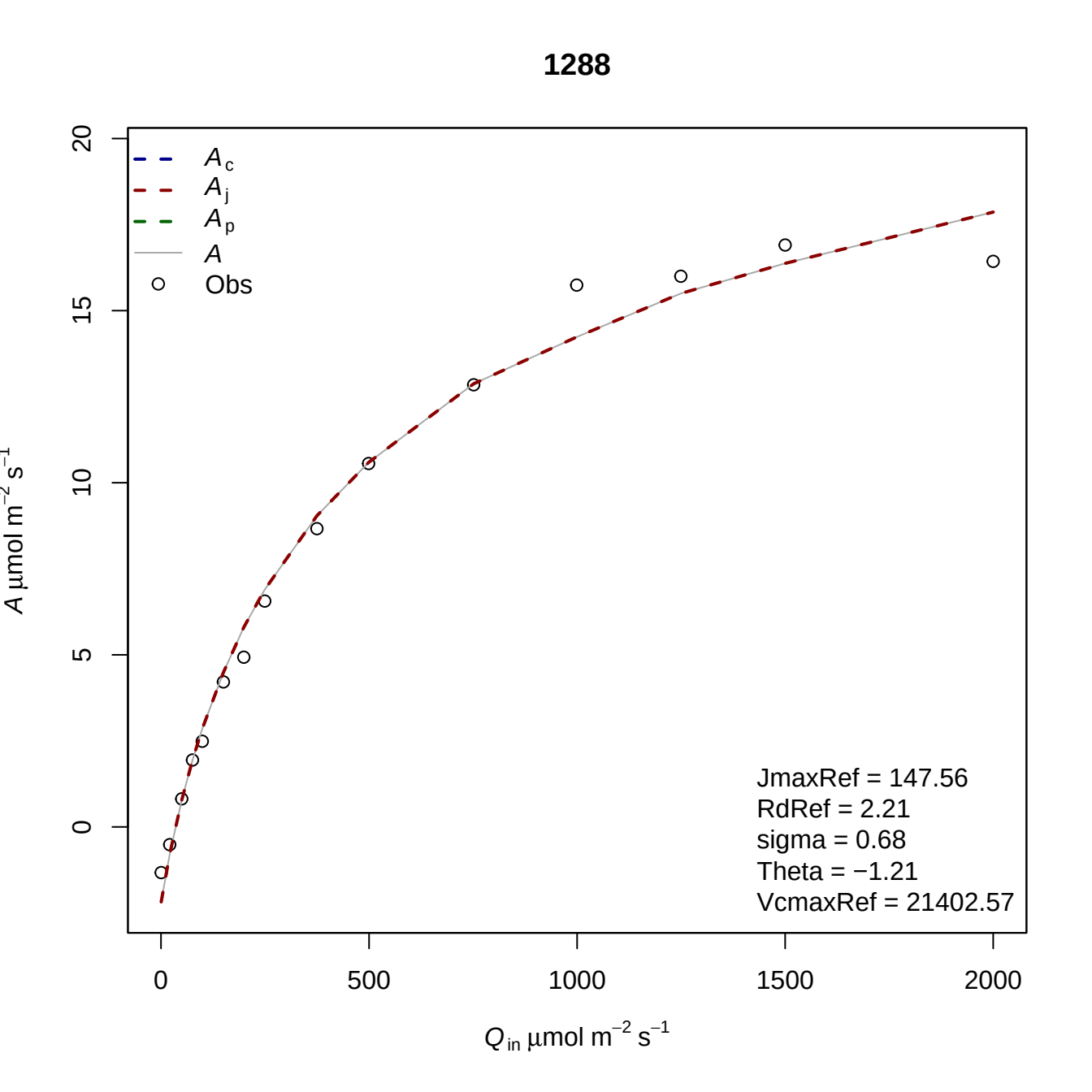

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 147.26
RdRef = 2.74
sigma = 0.15
Theta = 0.35
VcmaxRef = 284855.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

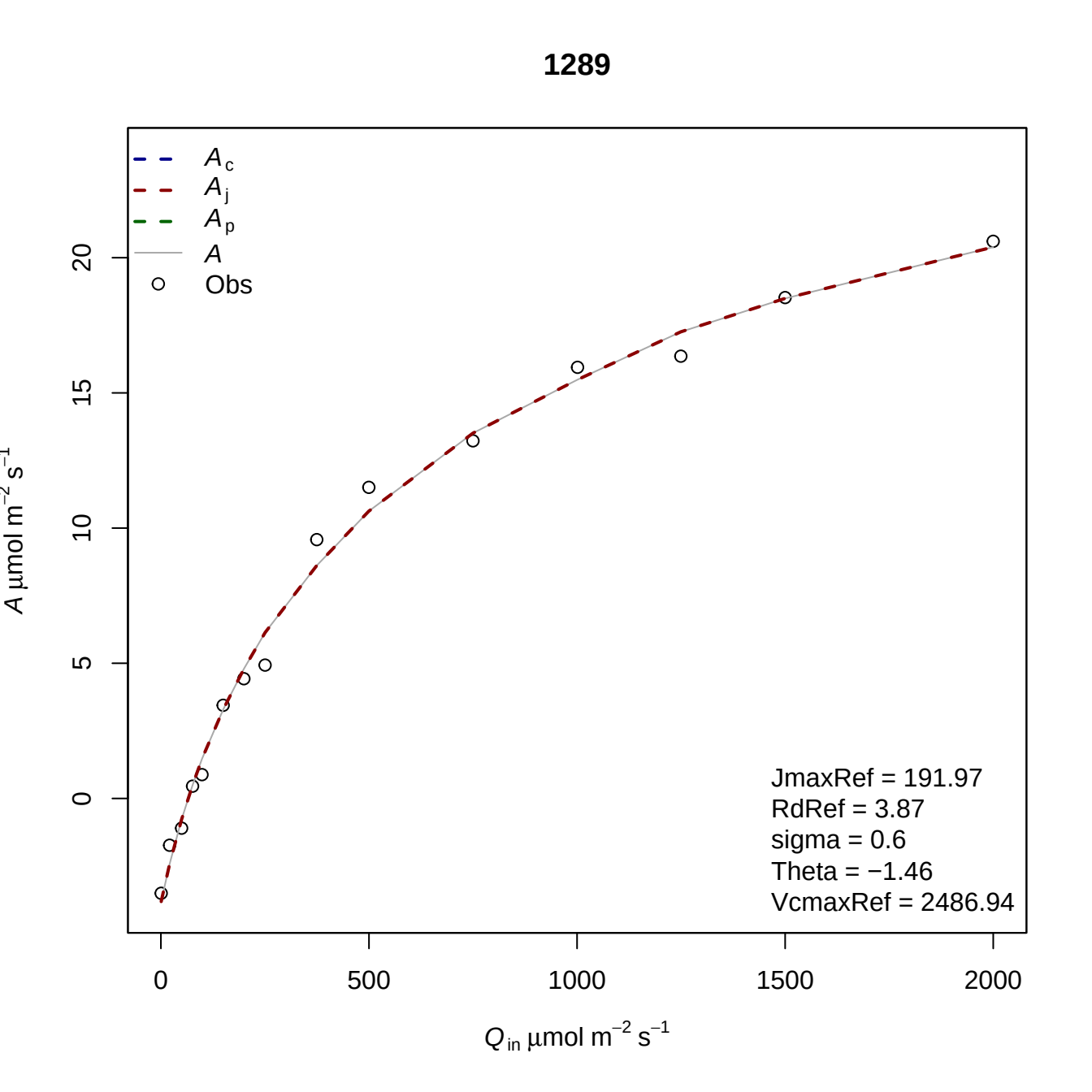

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 123.43
RdRef = 3.47
sigma = 0.53
Theta = 0.05
VcmaxRef = 87911.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

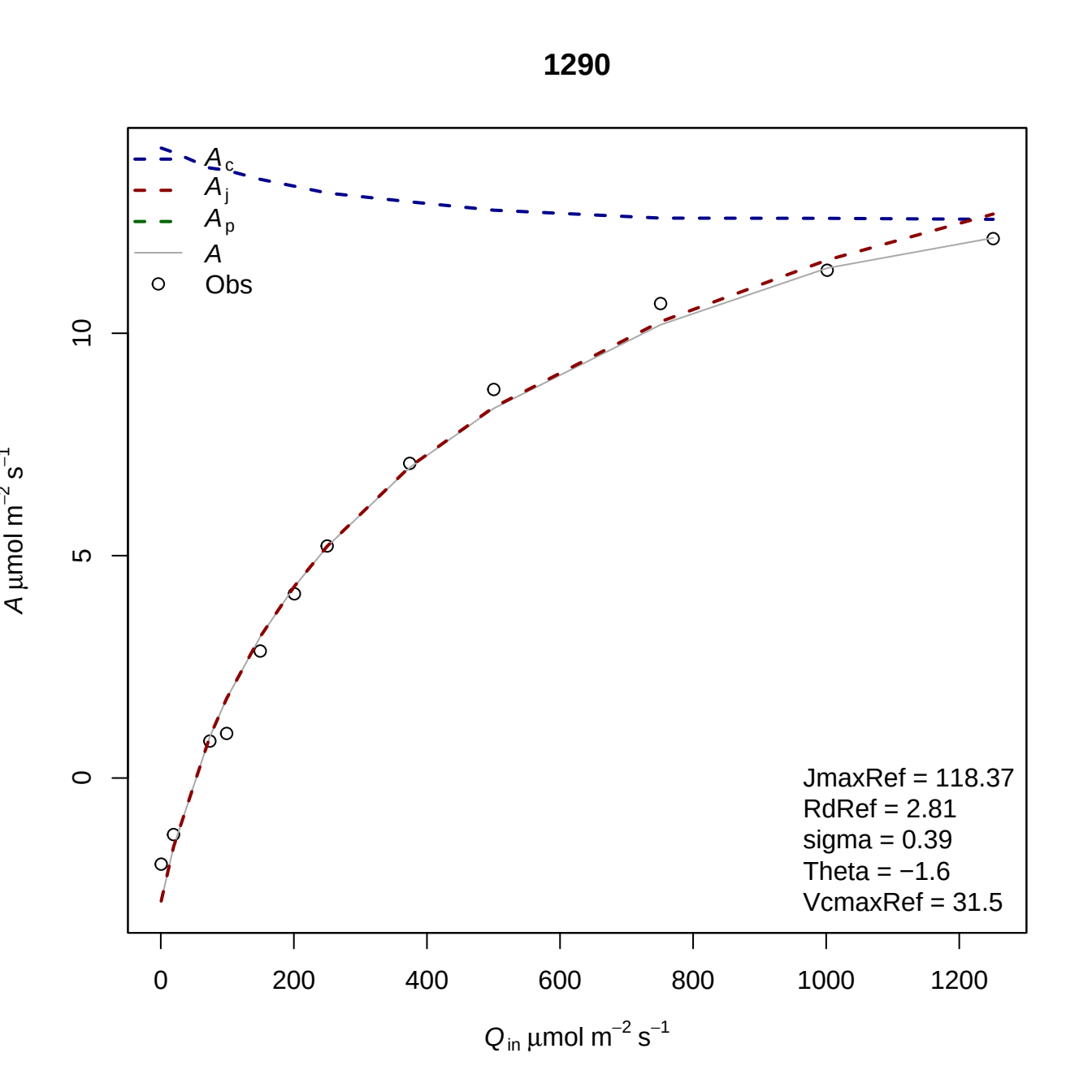

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 113.05
RdRef = 2.71
sigma = 0.61
Theta = 0.16
VcmaxRef = 5047.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

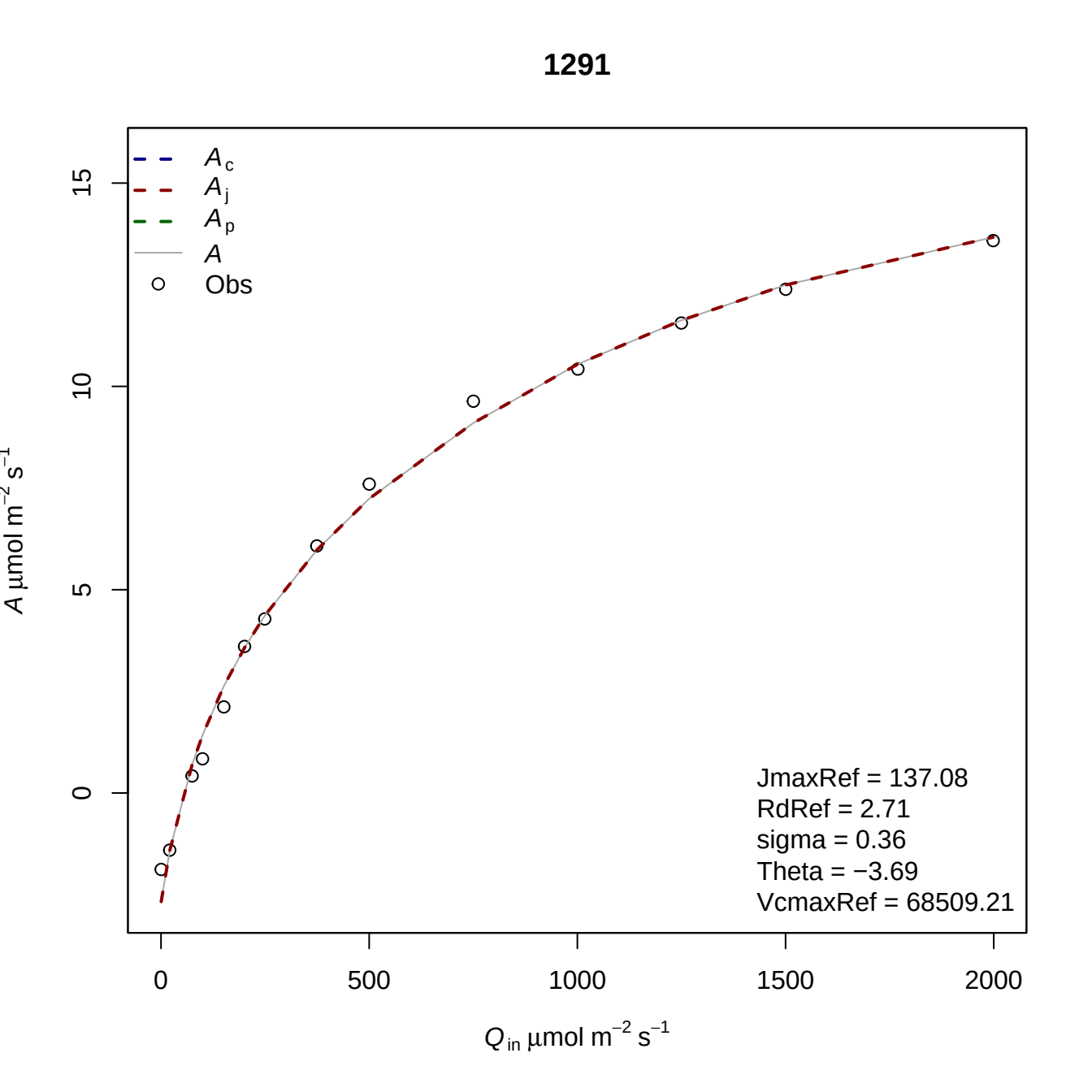

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 161.28
RdRef = 2.6
sigma = 0.38
Theta = −0.71
VcmaxRef = 186119.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

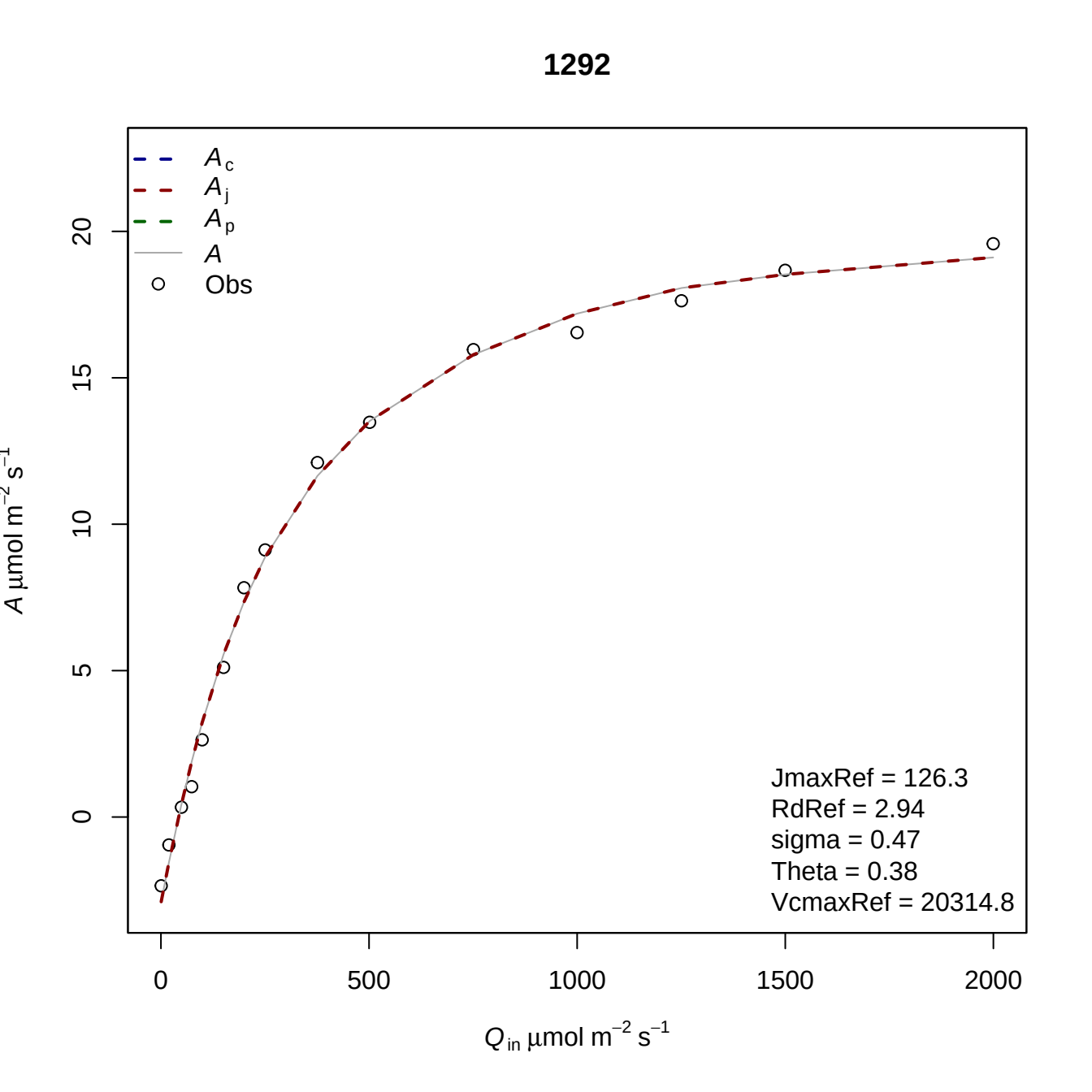

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 128.87
RdRef = 2.55
sigma = 0.41
Theta = −0.44
VcmaxRef = 734000.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

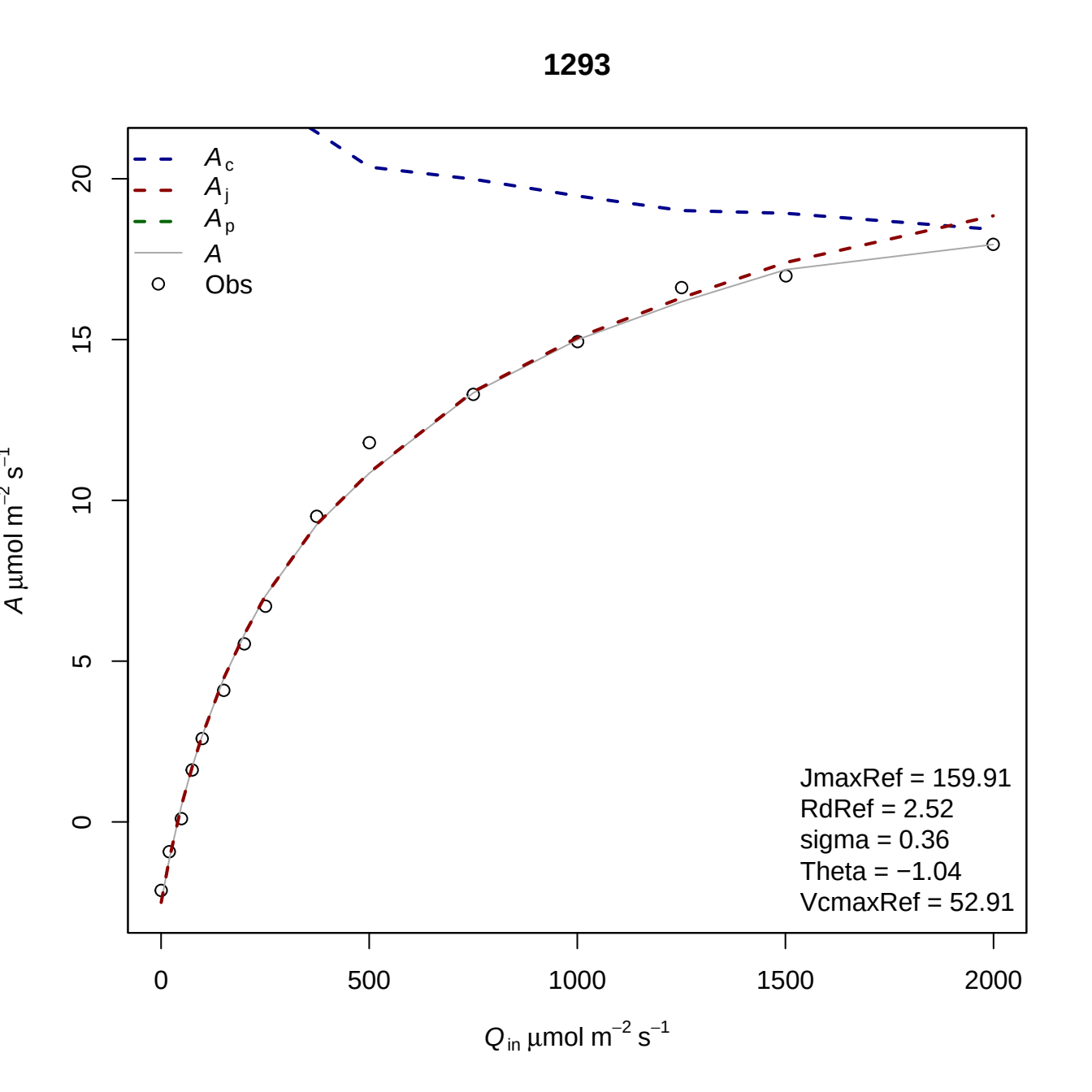

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 147.56
RdRef = 2.21
sigma = 0.68
Theta = −1.21
VcmaxRef = 21402.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

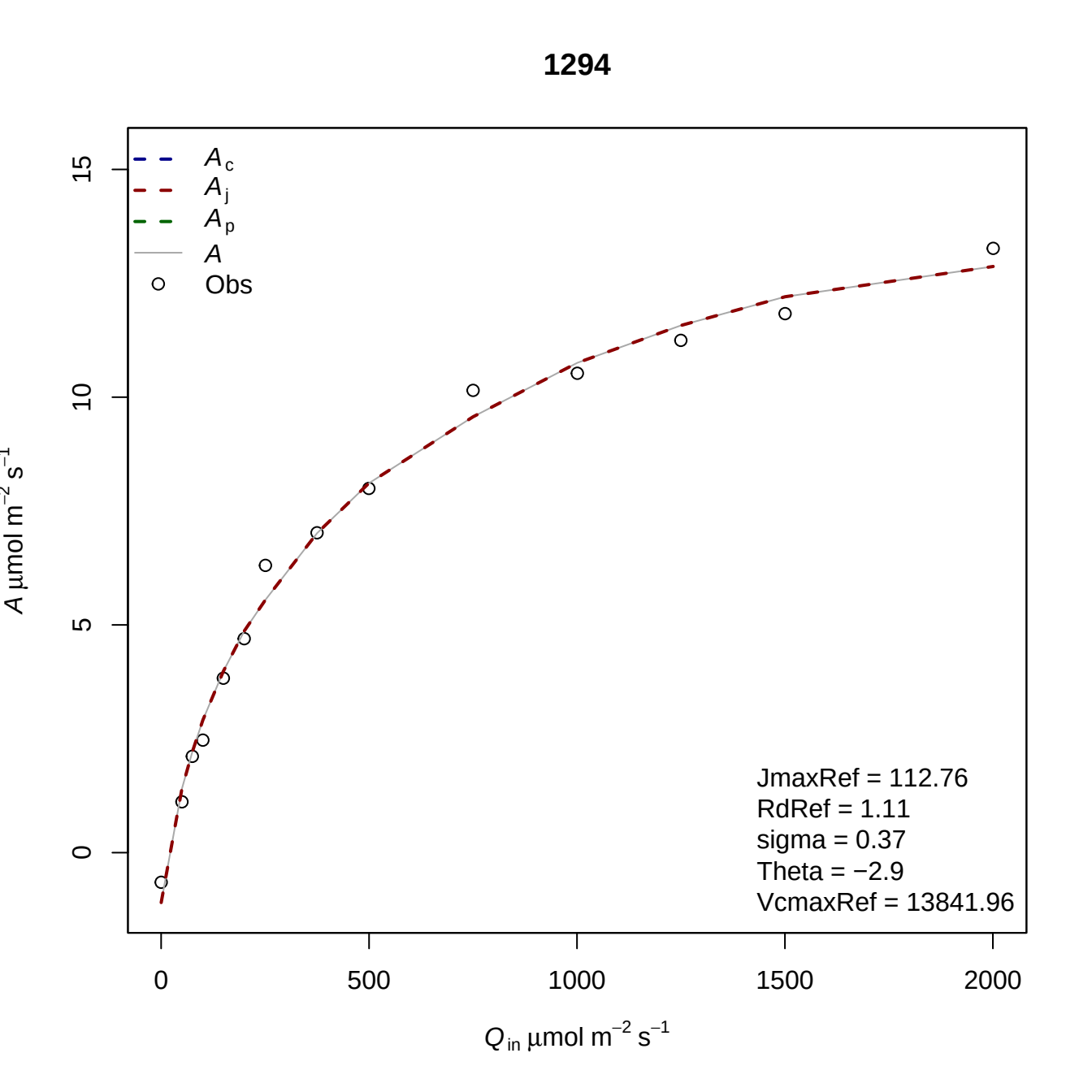

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 191.97
RdRef = 3.87
sigma = 0.6
Theta = −1.46
VcmaxRef = 2486.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

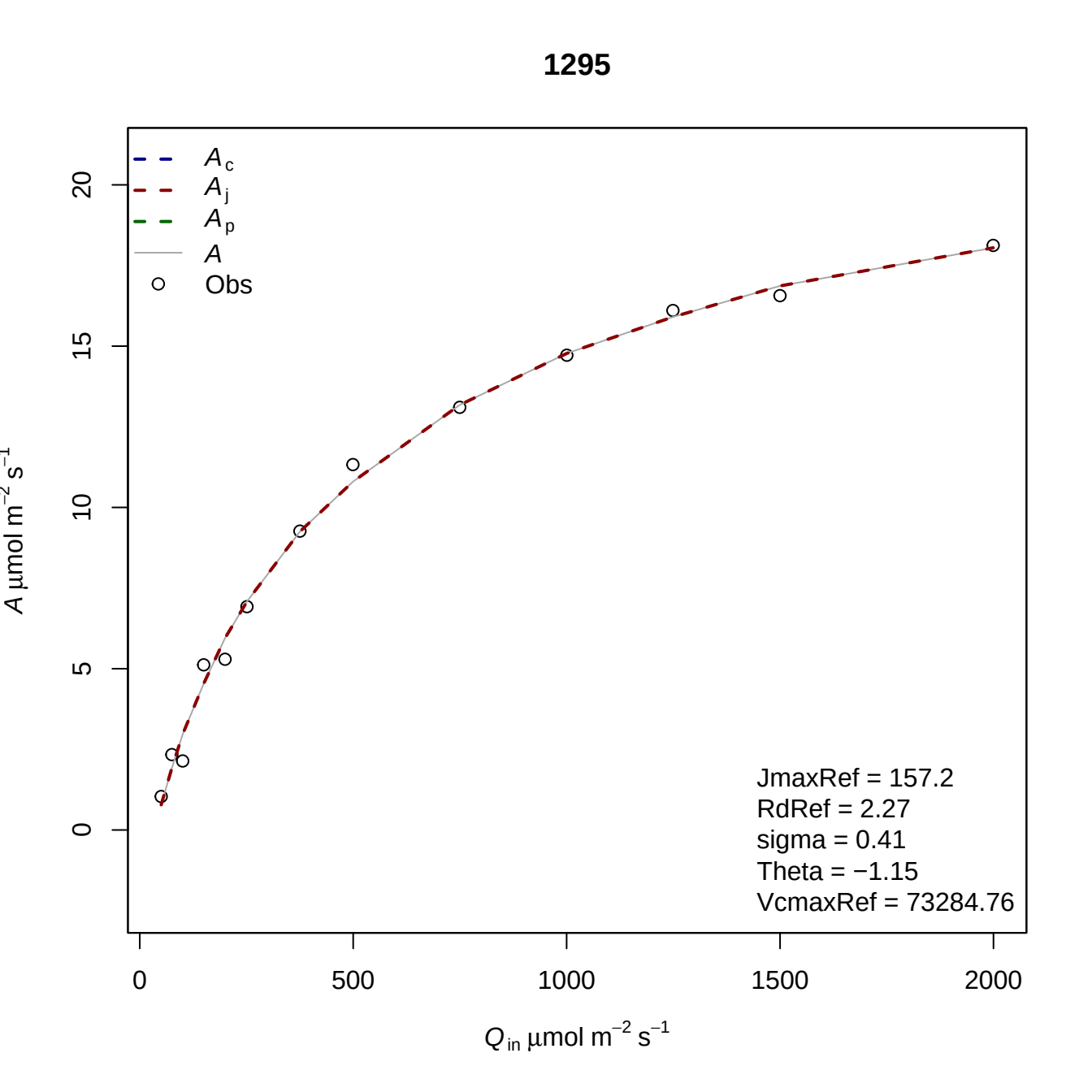

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 118.37
RdRef = 2.81
sigma = 0.39
Theta = −1.6
VcmaxRef = 31.5
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

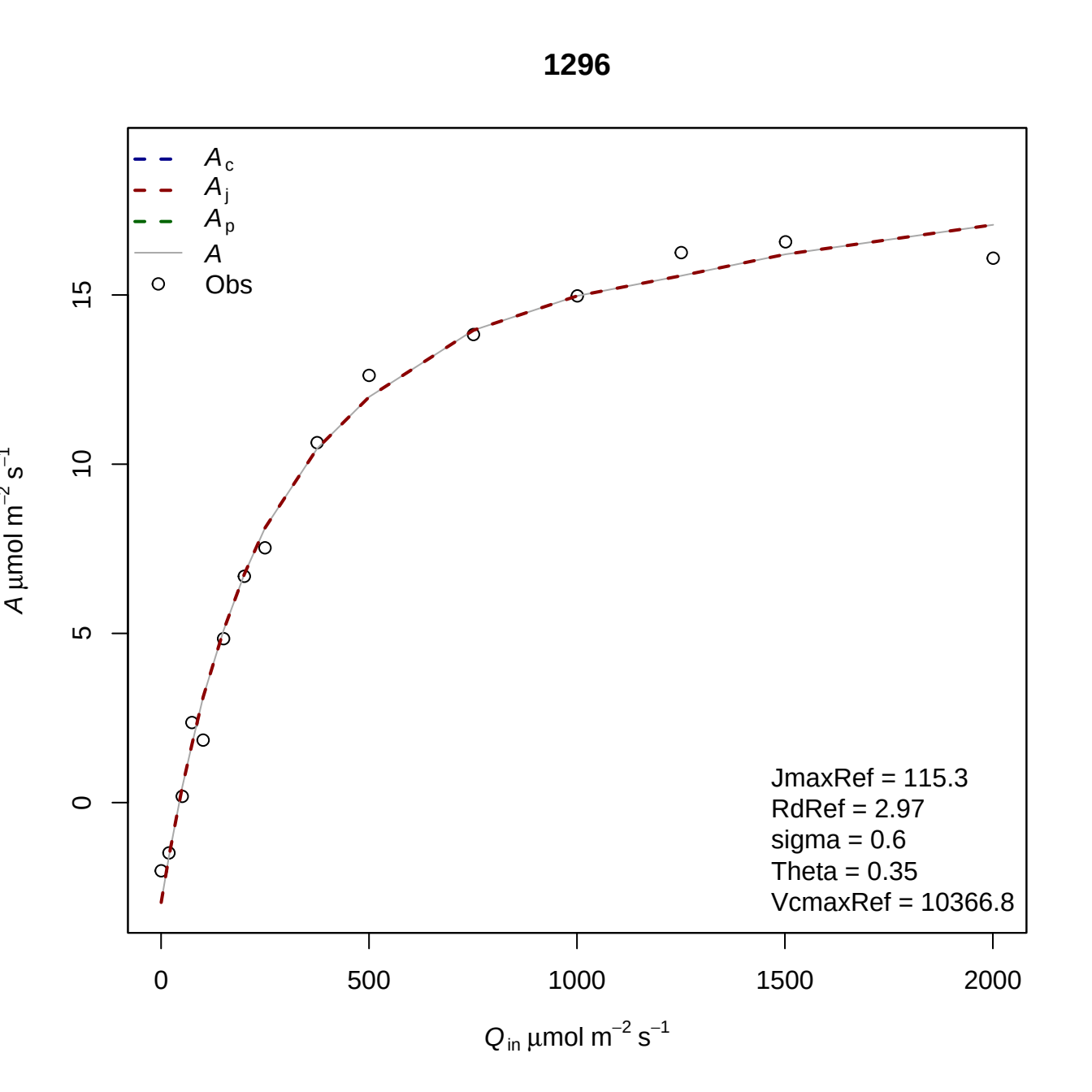

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 137.08
RdRef = 2.71
sigma = 0.36
Theta = −3.69
VcmaxRef = 68509.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

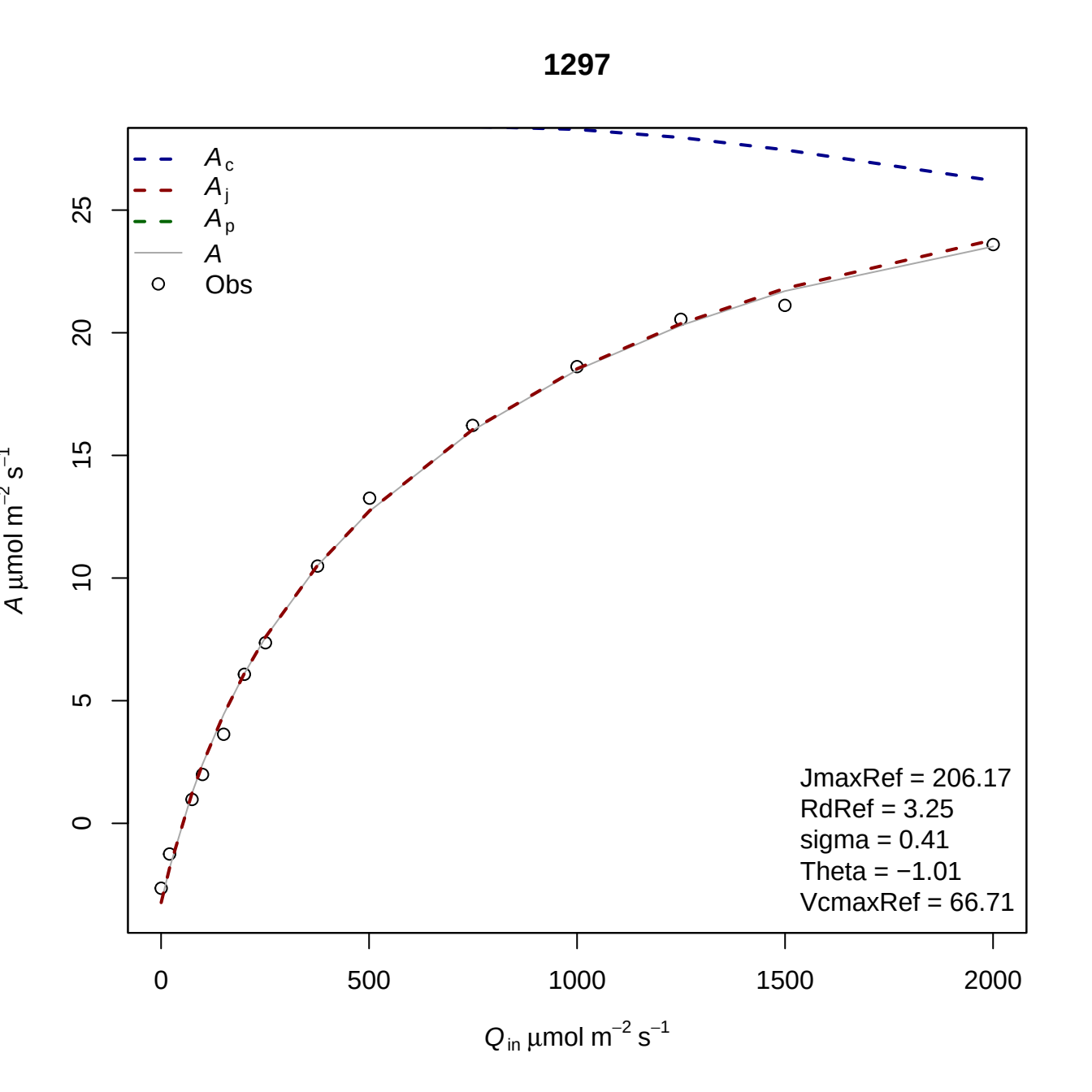

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 126.3
RdRef = 2.94
sigma = 0.47
Theta = 0.38
VcmaxRef = 20314.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

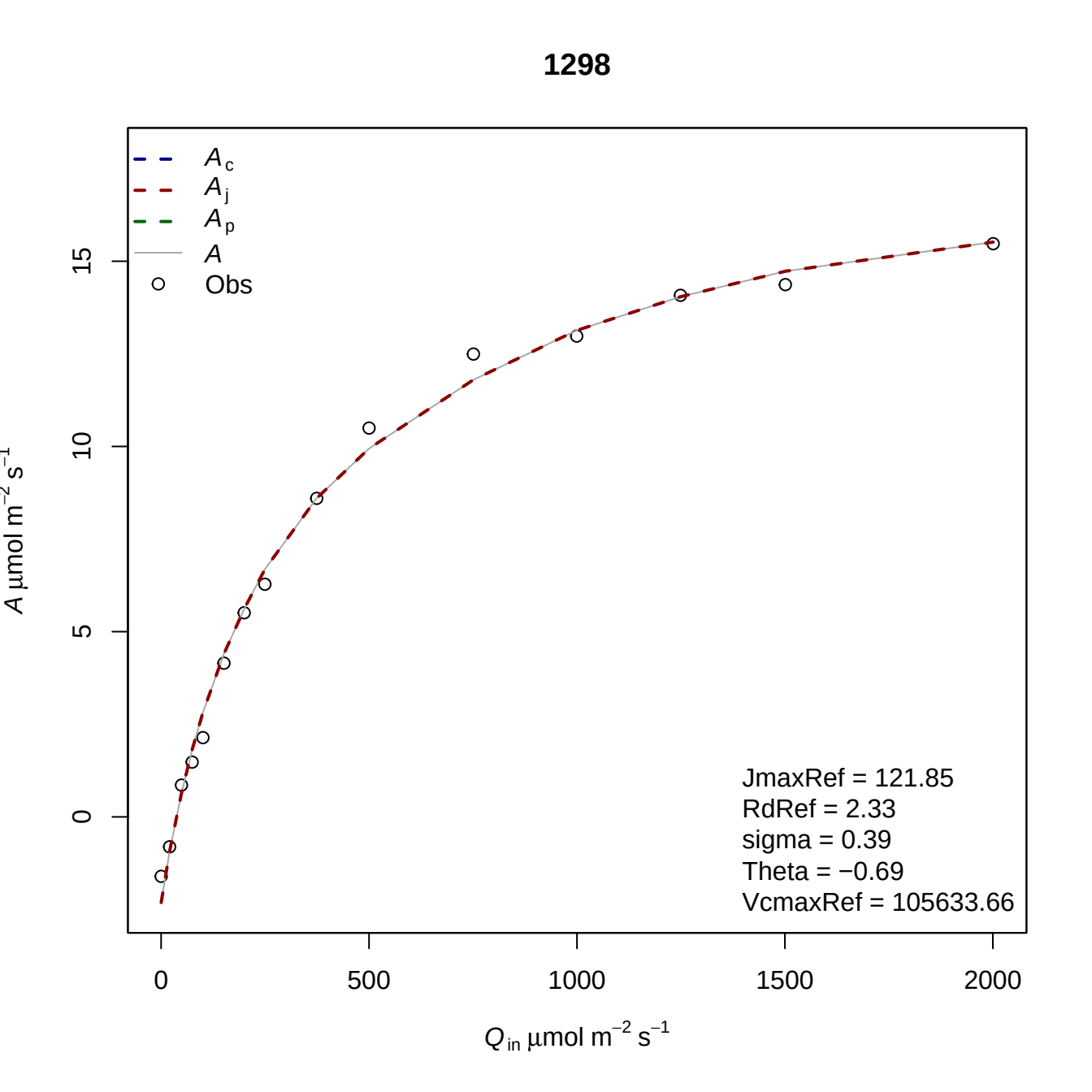

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 159.91
RdRef = 2.52
sigma = 0.36
Theta = −1.04
VcmaxRef = 52.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

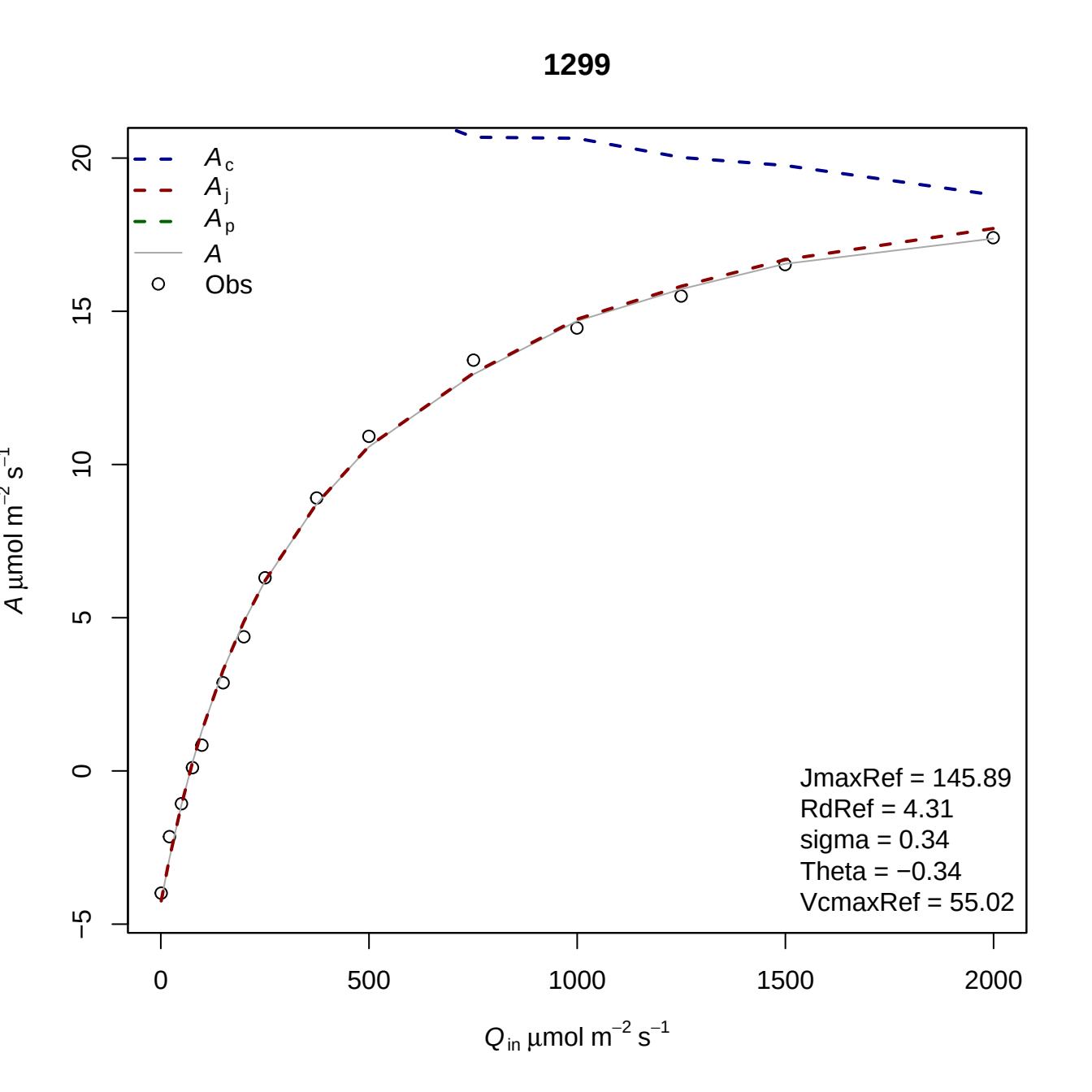

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 112.76
RdRef = 1.11
sigma = 0.37
Theta = −2.9
VcmaxRef = 13841.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

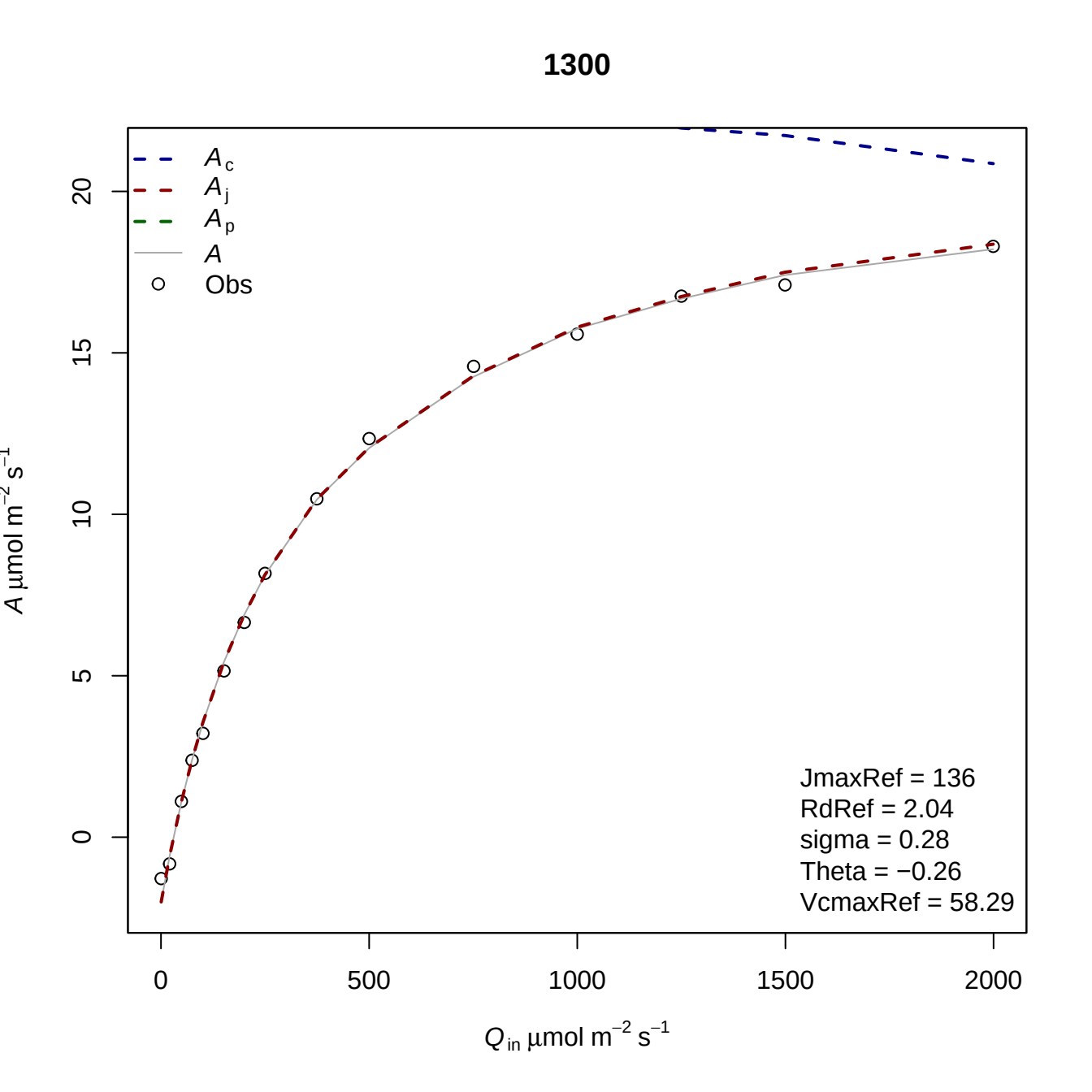

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 157.2
RdRef = 2.27
sigma = 0.41
Theta = −1.15
VcmaxRef = 73284.76
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

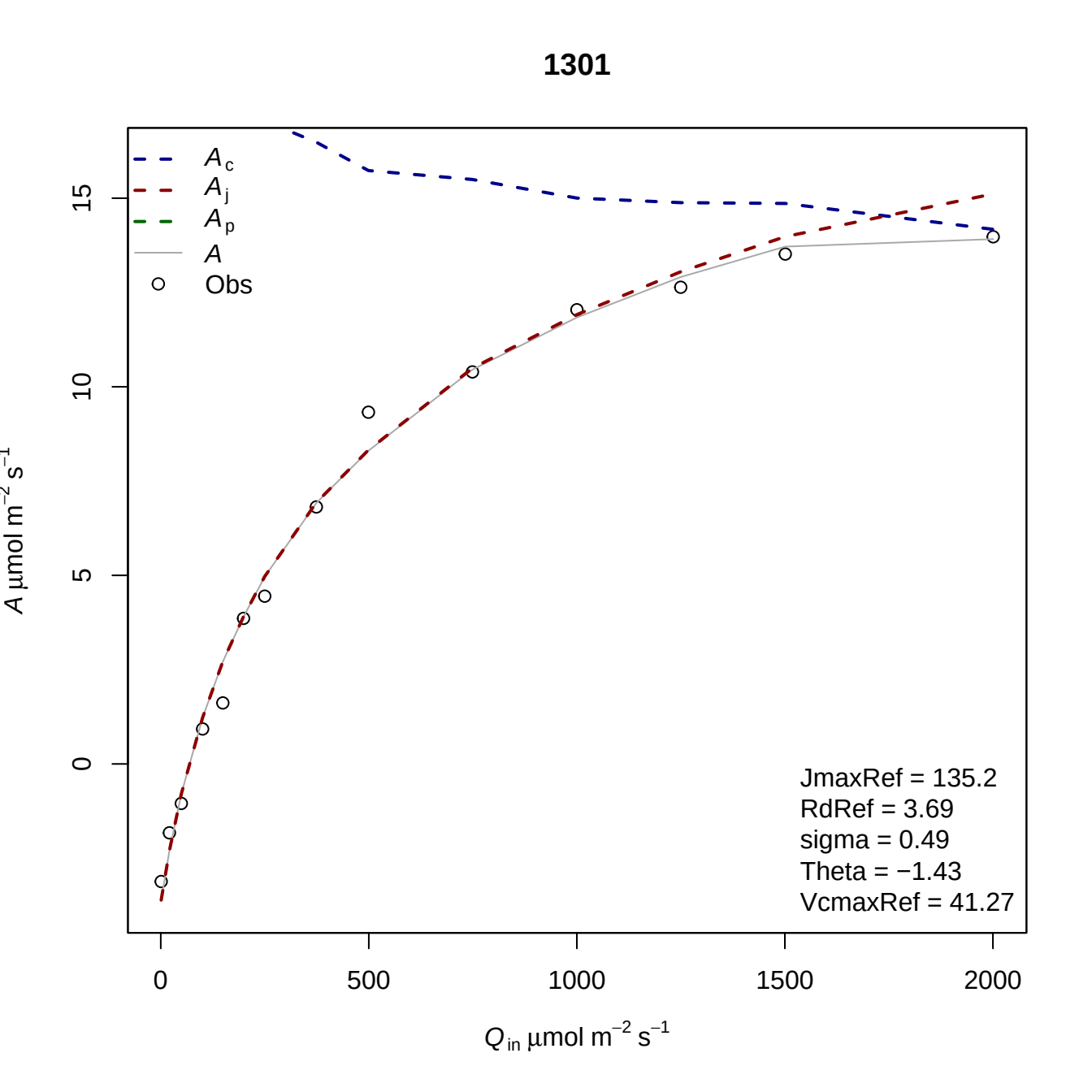

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 115.3
RdRef = 2.97
sigma = 0.6
Theta = 0.35
VcmaxRef = 10366.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

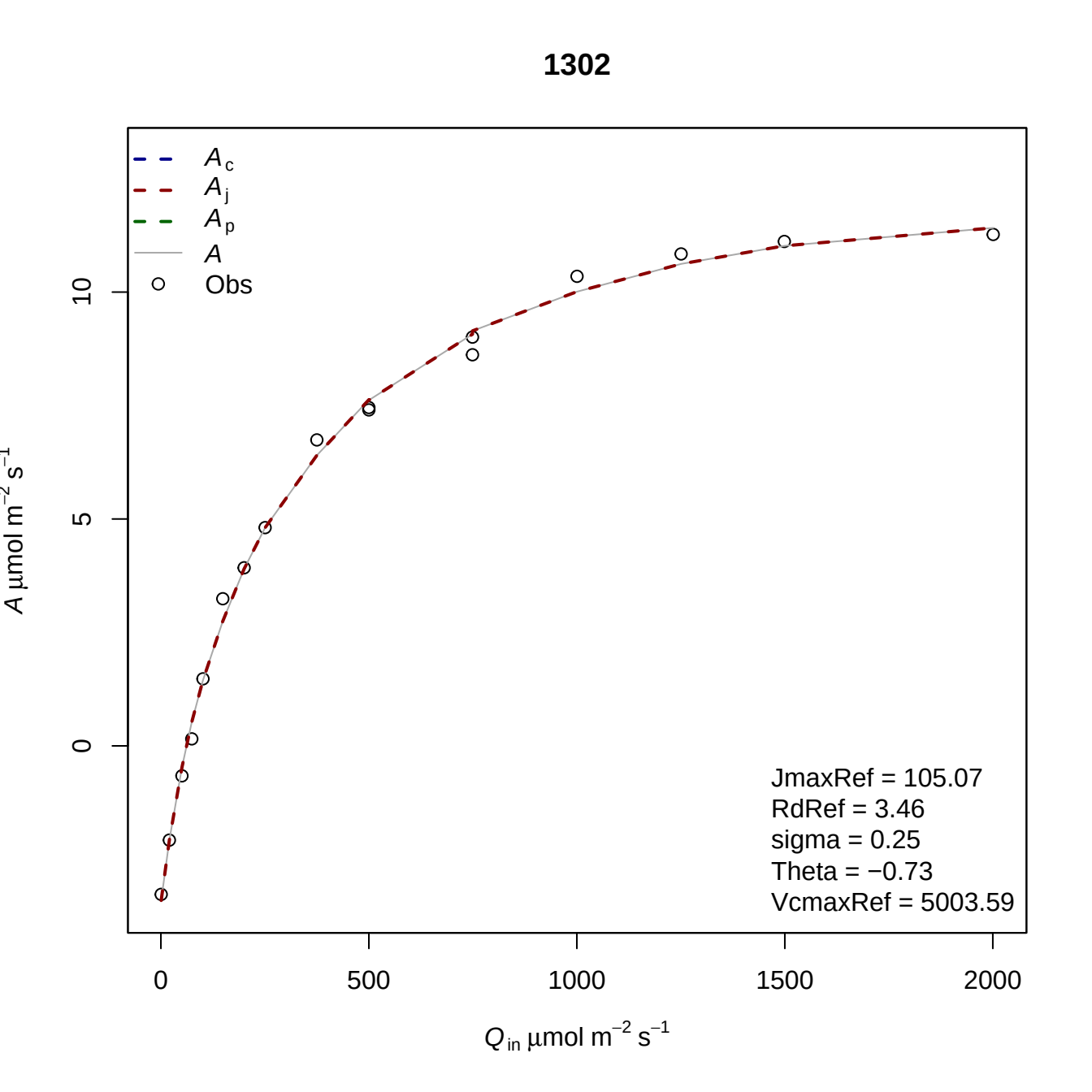

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 206.17
RdRef = 3.25
sigma = 0.41
Theta = −1.01
VcmaxRef = 66.71
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

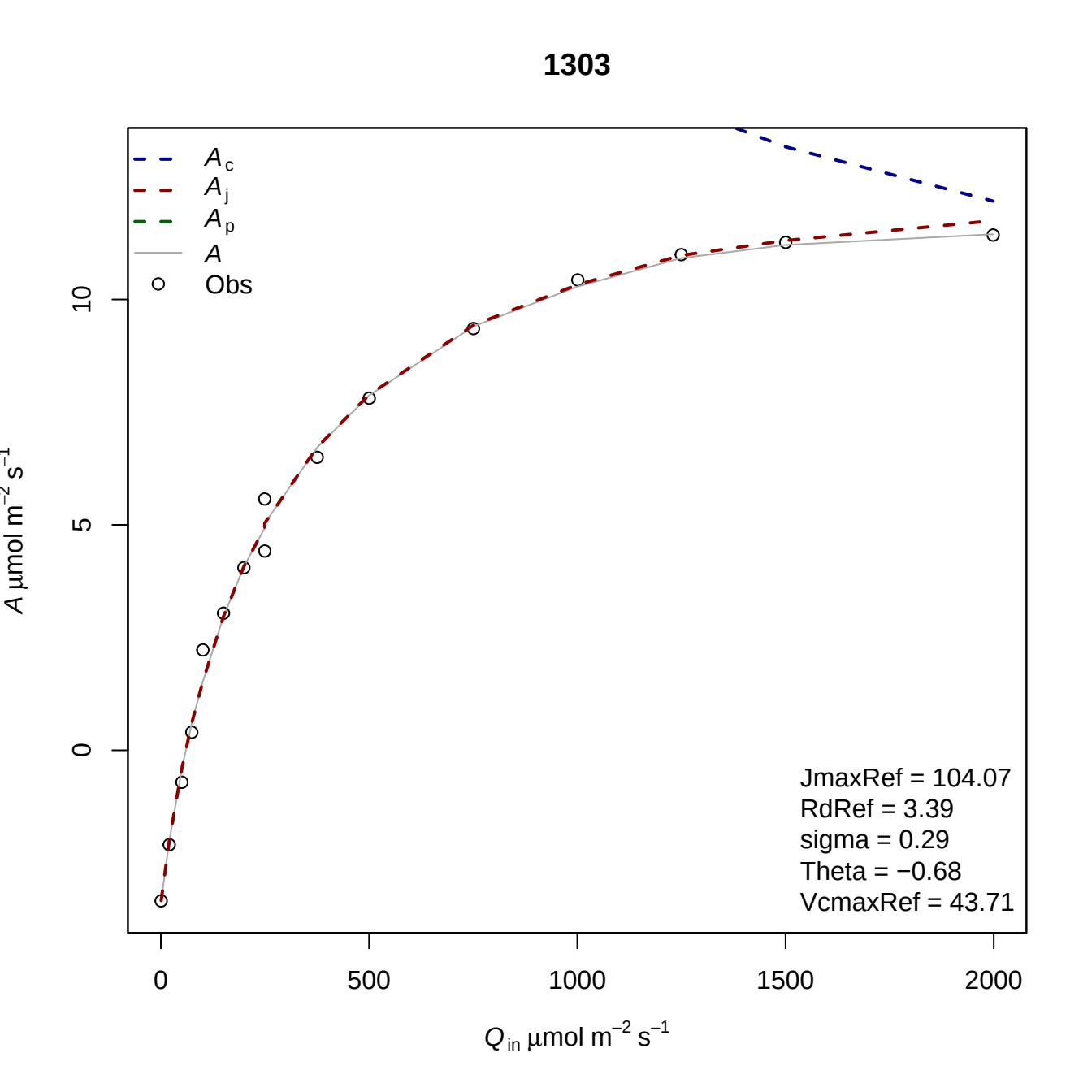

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 121.85
RdRef = 2.33
sigma = 0.39
Theta = −0.69
VcmaxRef = 105633.66
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

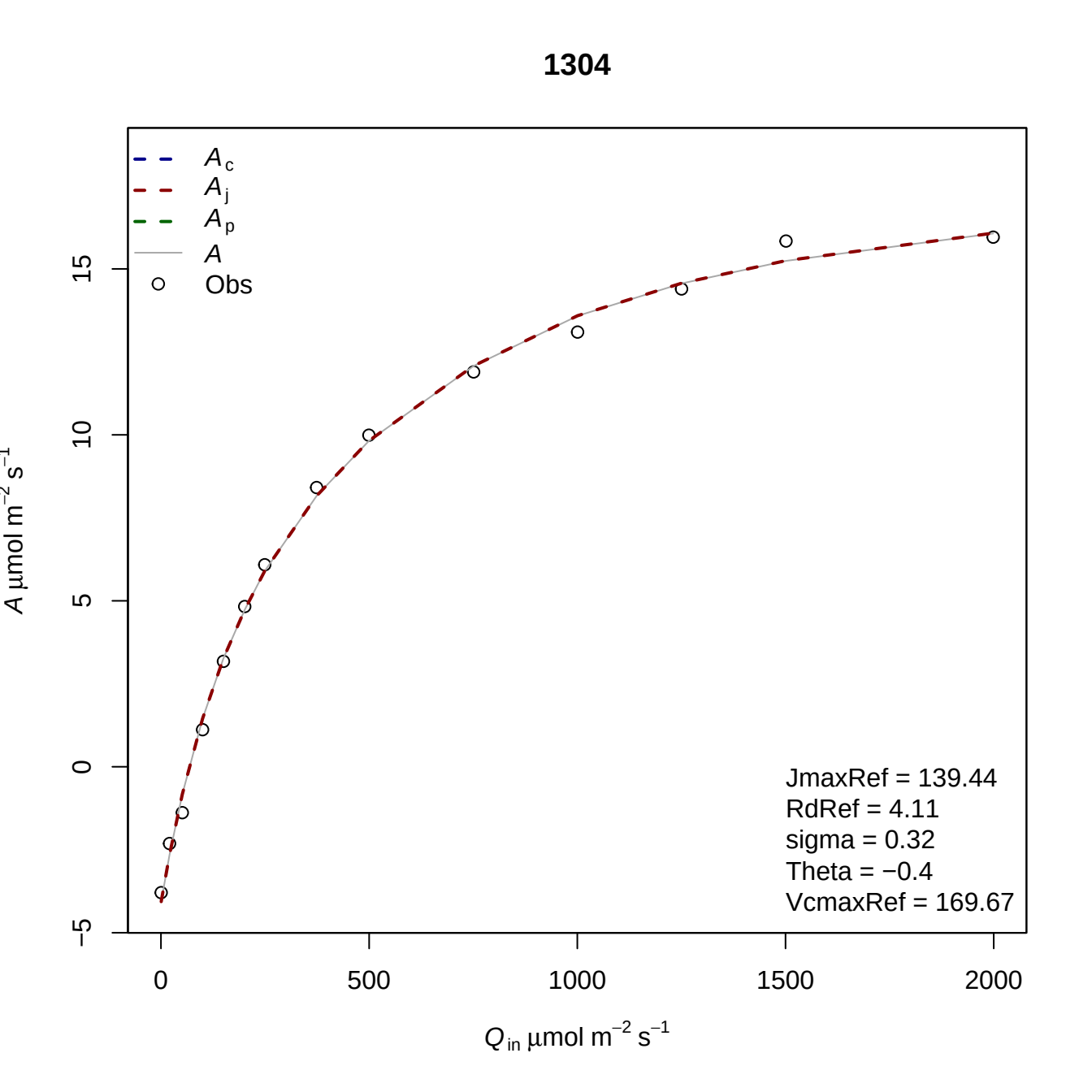

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 145.89
RdRef = 4.31
sigma = 0.34
Theta = −0.34
VcmaxRef = 55.02
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

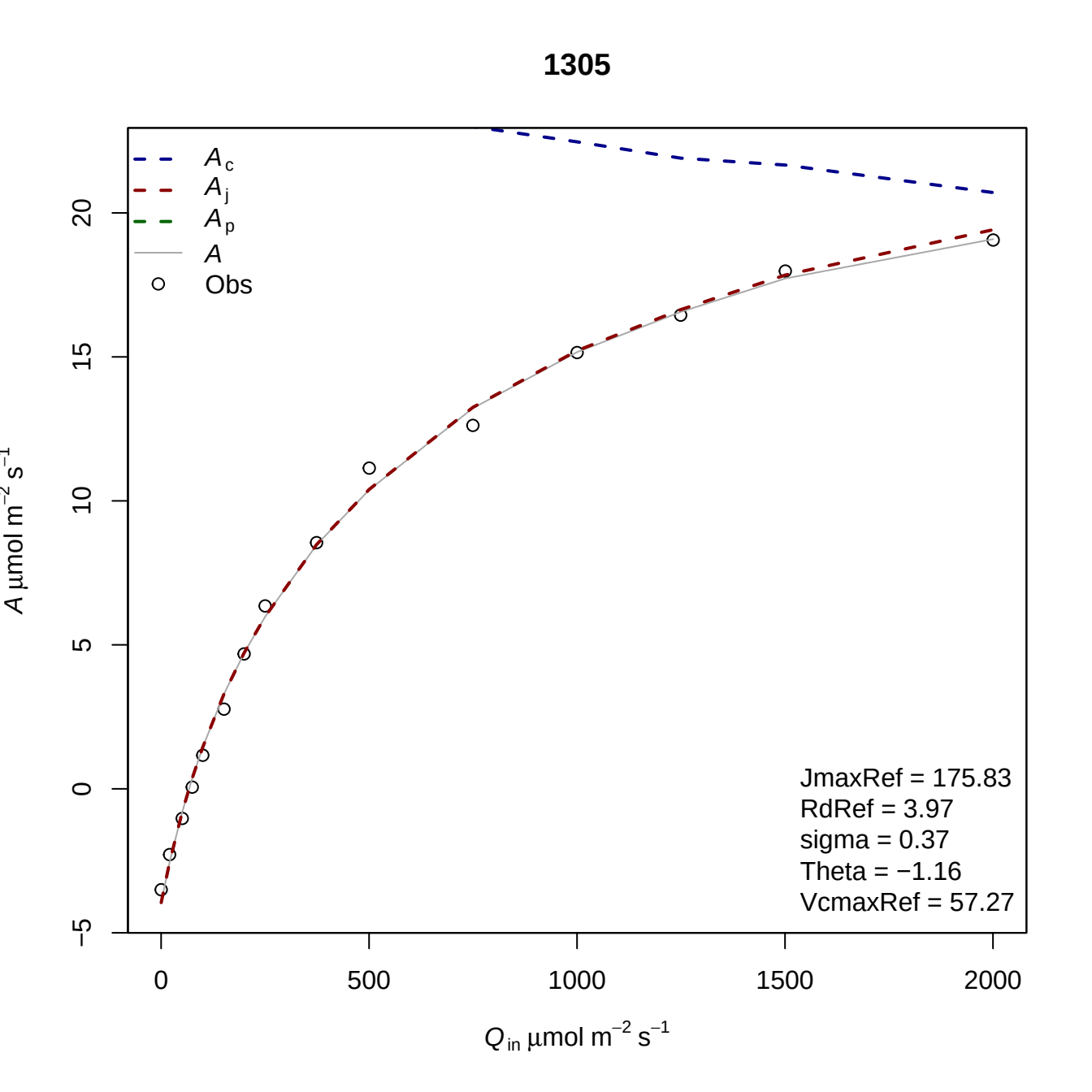

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 136
RdRef = 2.04
sigma = 0.28
Theta = −0.26
VcmaxRef = 58.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

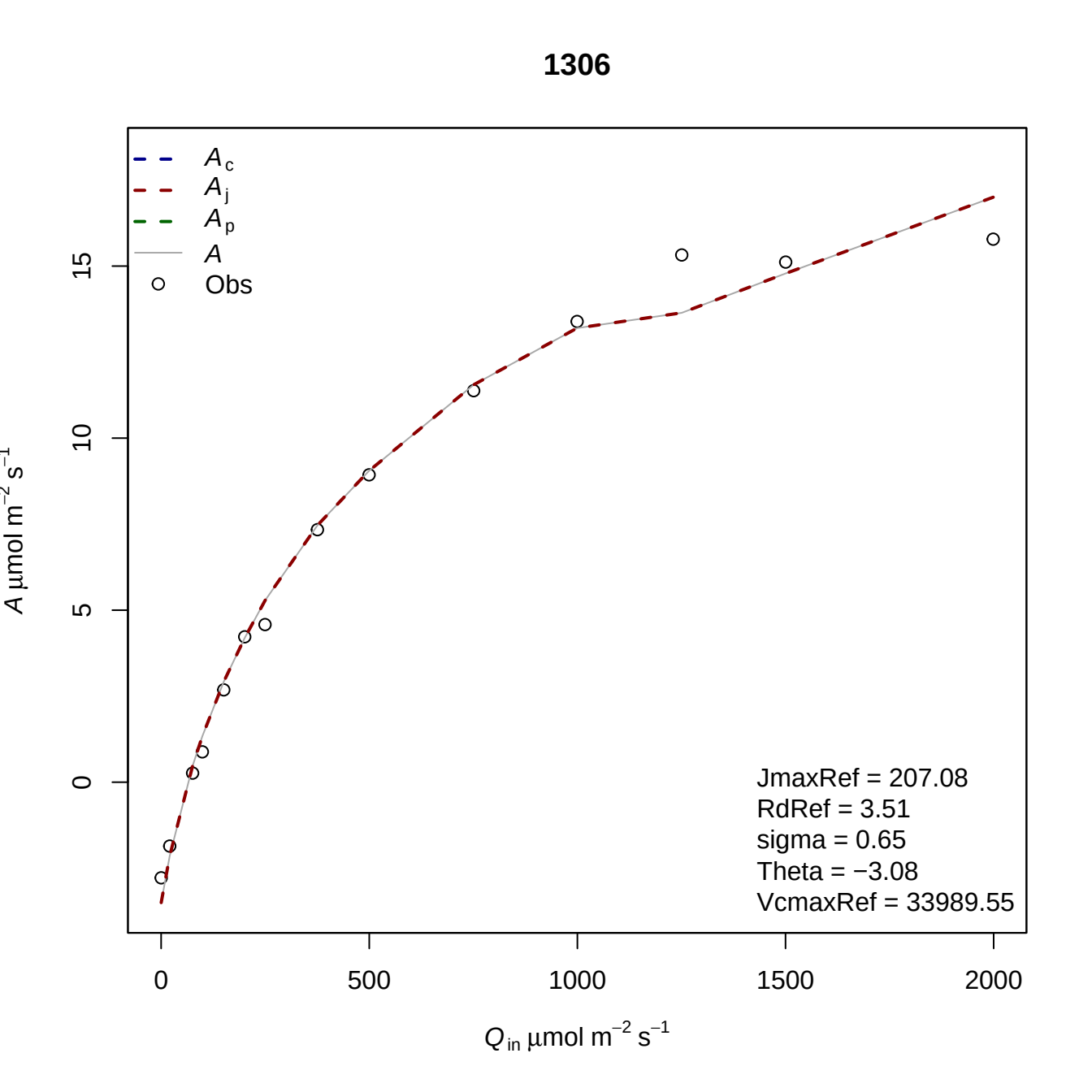

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 135.2
RdRef = 3.69
sigma = 0.49
Theta = −1.43
VcmaxRef = 41.27
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

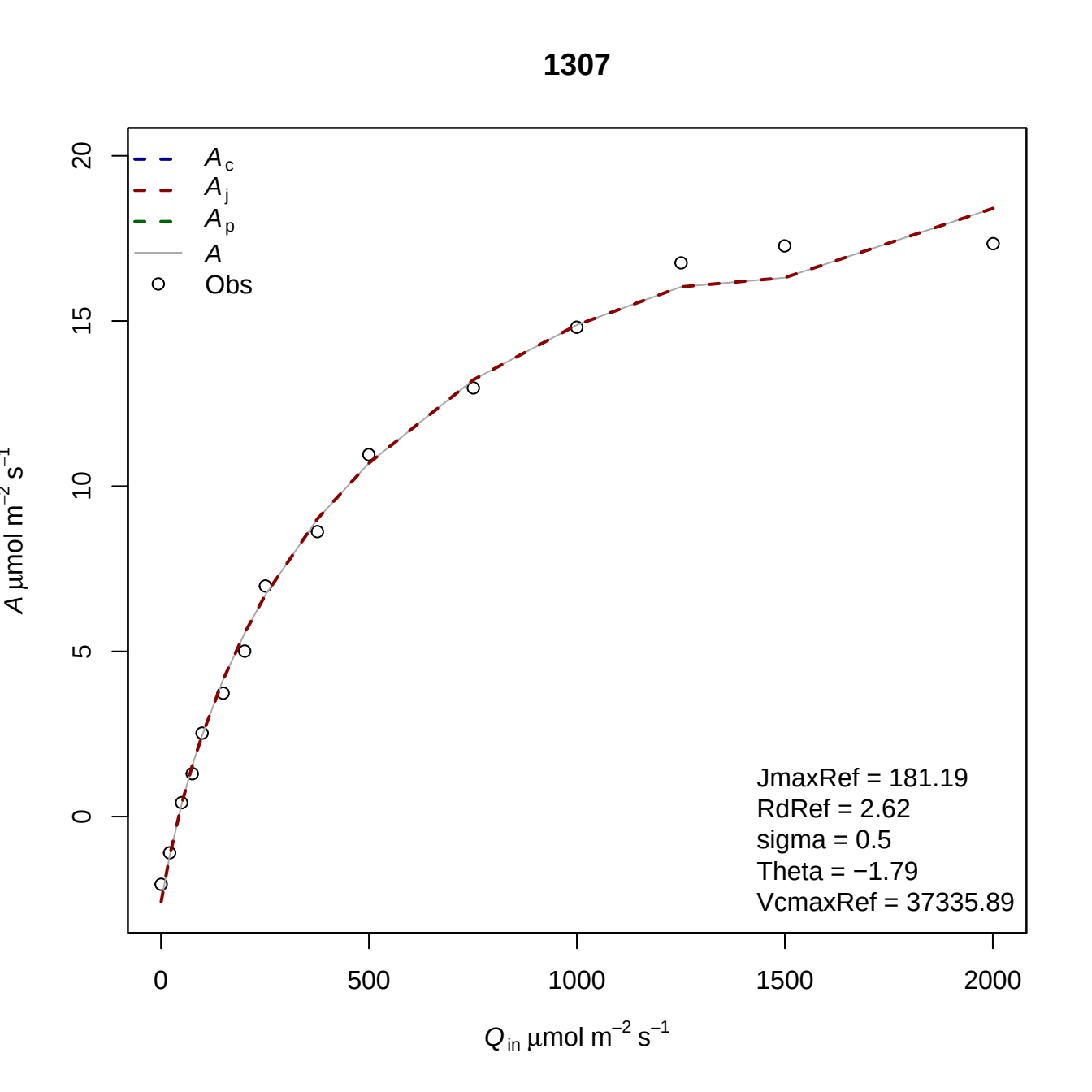

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 105.07
RdRef = 3.46
sigma = 0.25
Theta = −0.73
VcmaxRef = 5003.59
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

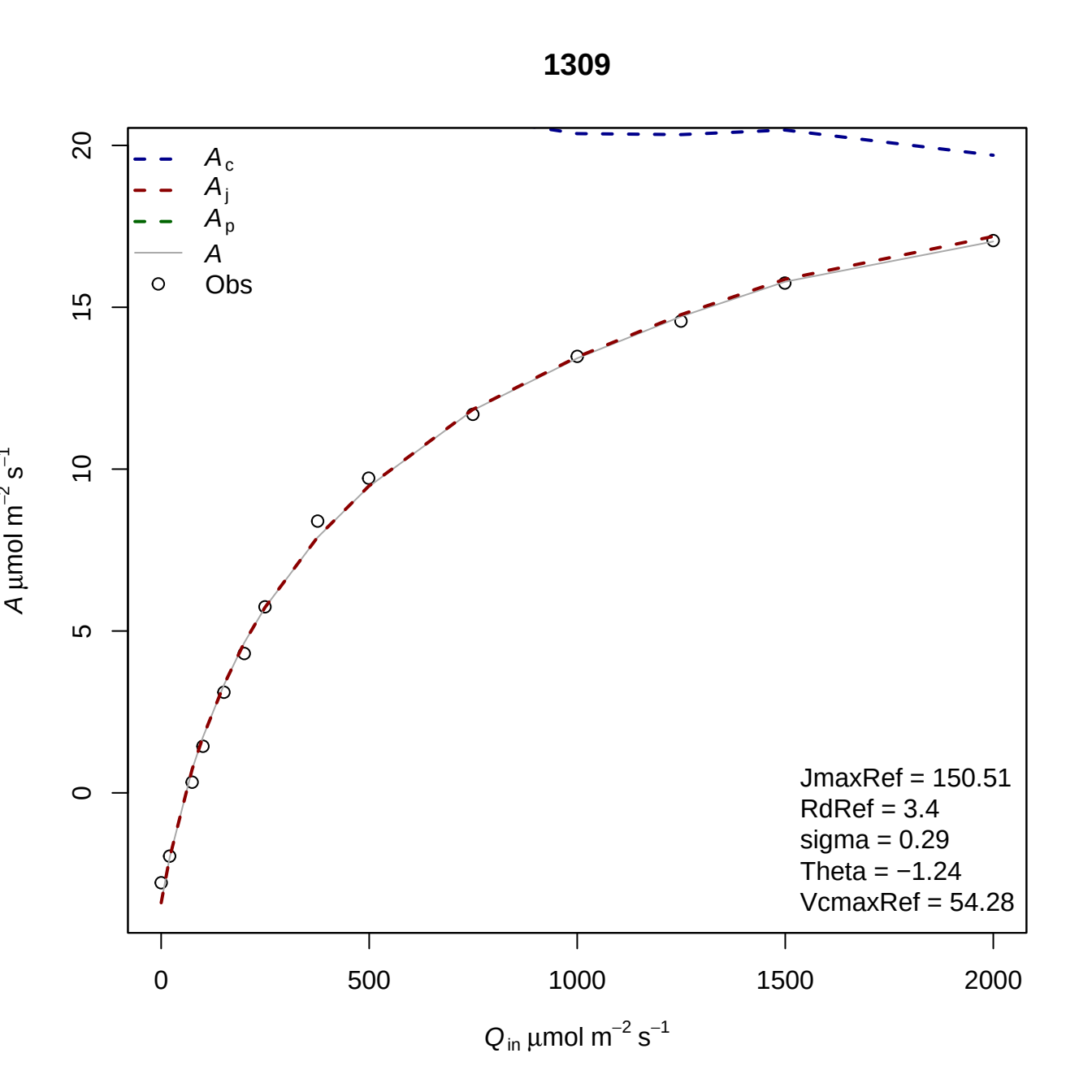

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 104.07
RdRef = 3.39
sigma = 0.29
Theta = −0.68
VcmaxRef = 43.71
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

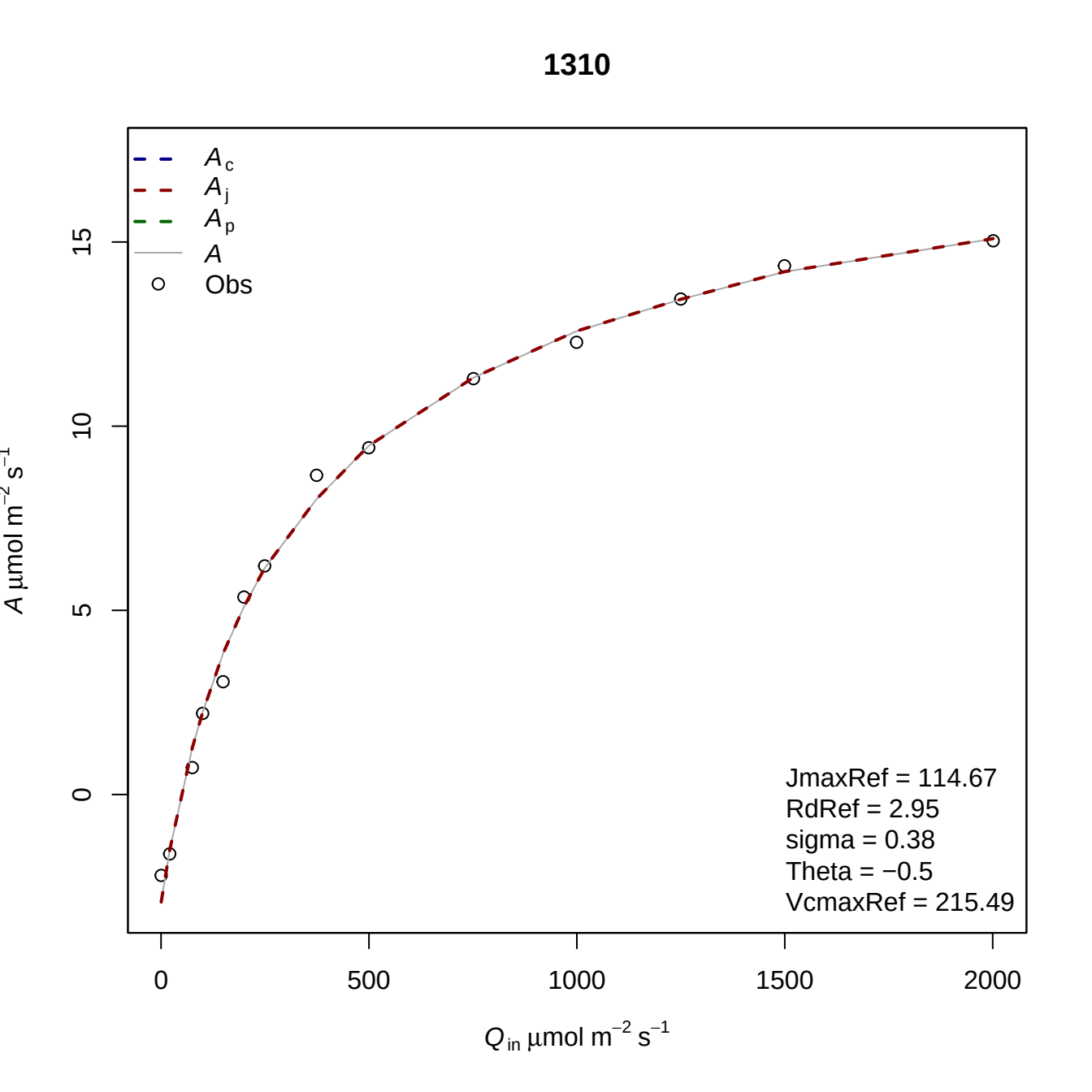

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 139.44
RdRef = 4.11
sigma = 0.32
Theta = −0.4
VcmaxRef = 169.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

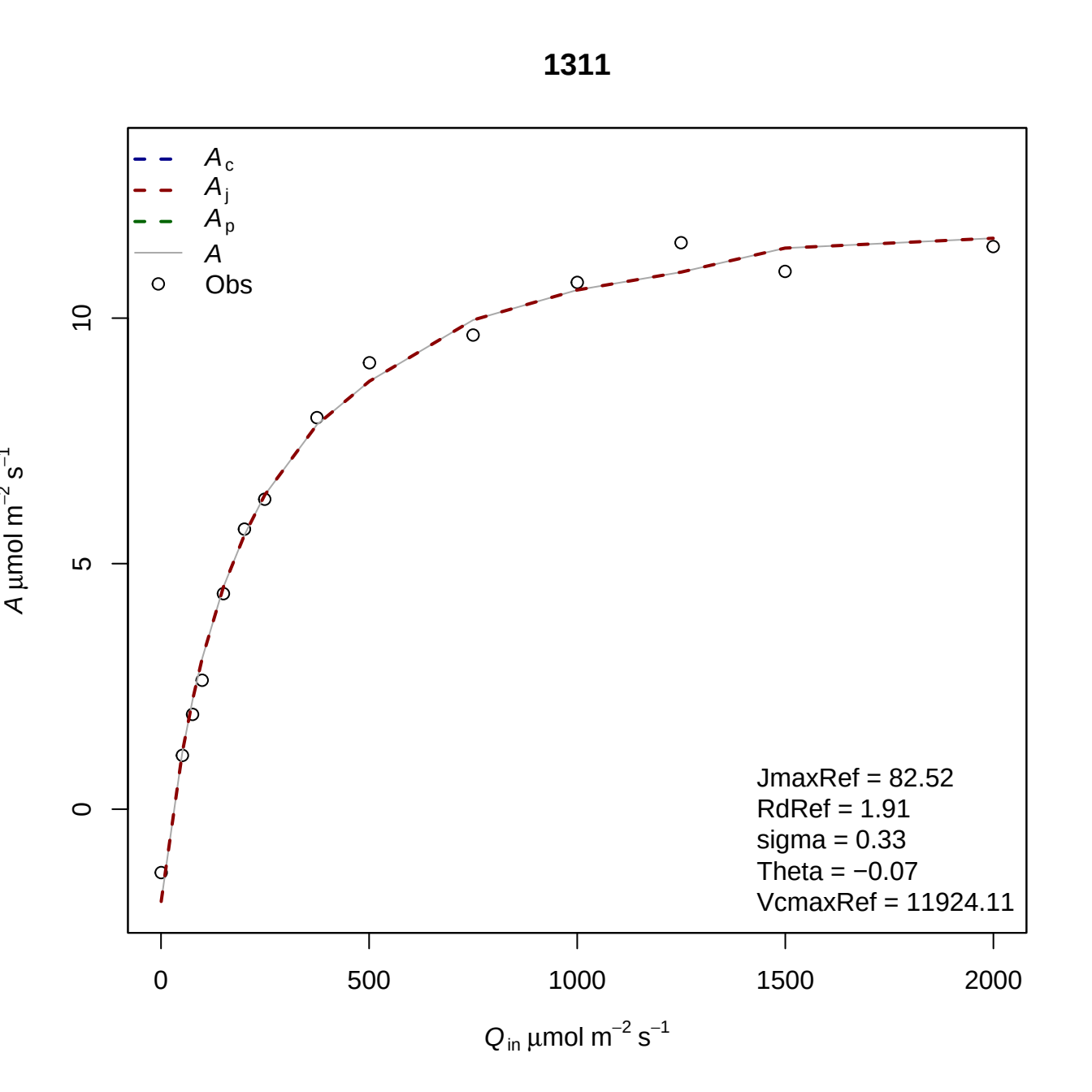

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 175.83
RdRef = 3.97
sigma = 0.37
Theta = −1.16
VcmaxRef = 57.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

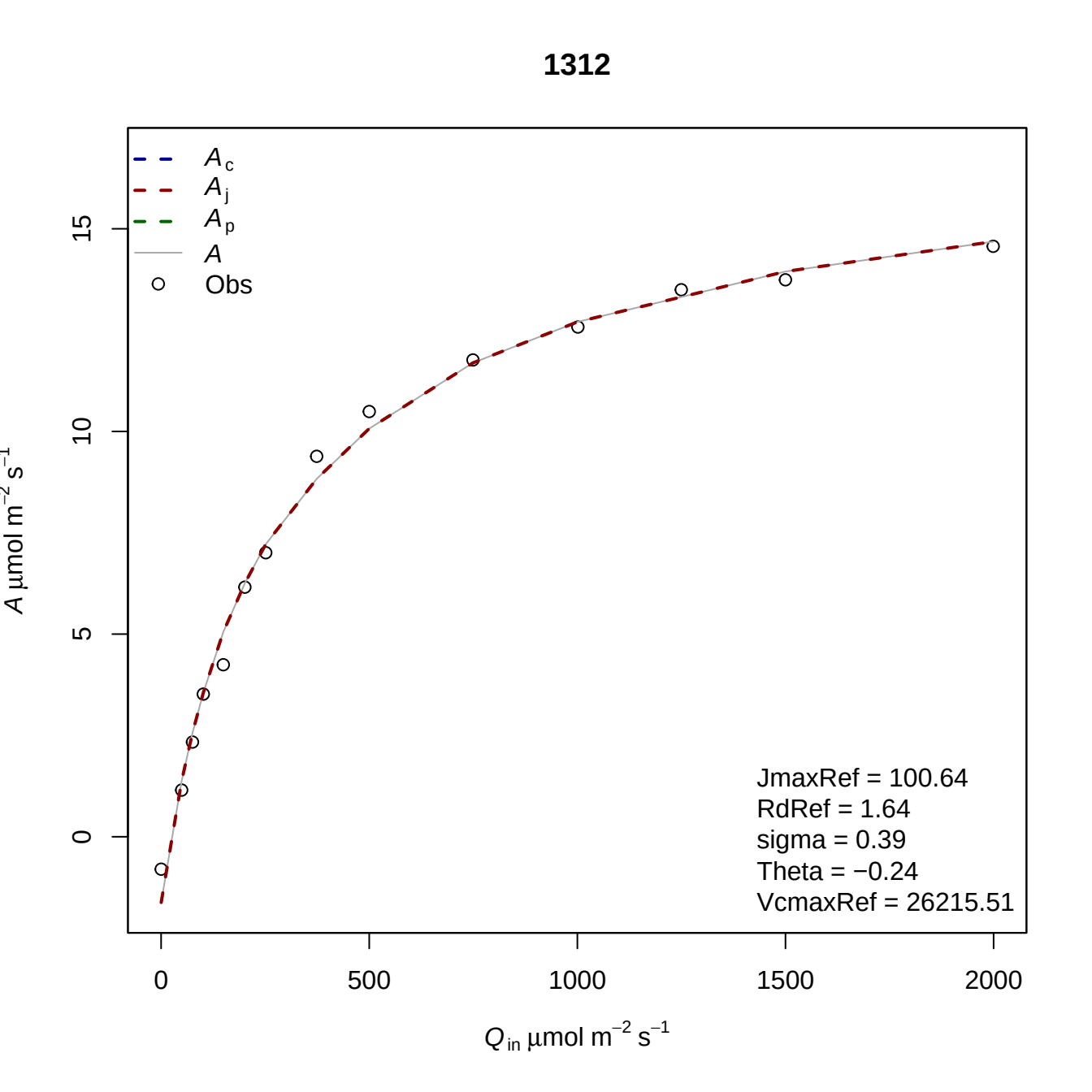

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 207.08
RdRef = 3.51
sigma = 0.65
Theta = −3.08
VcmaxRef = 33989.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

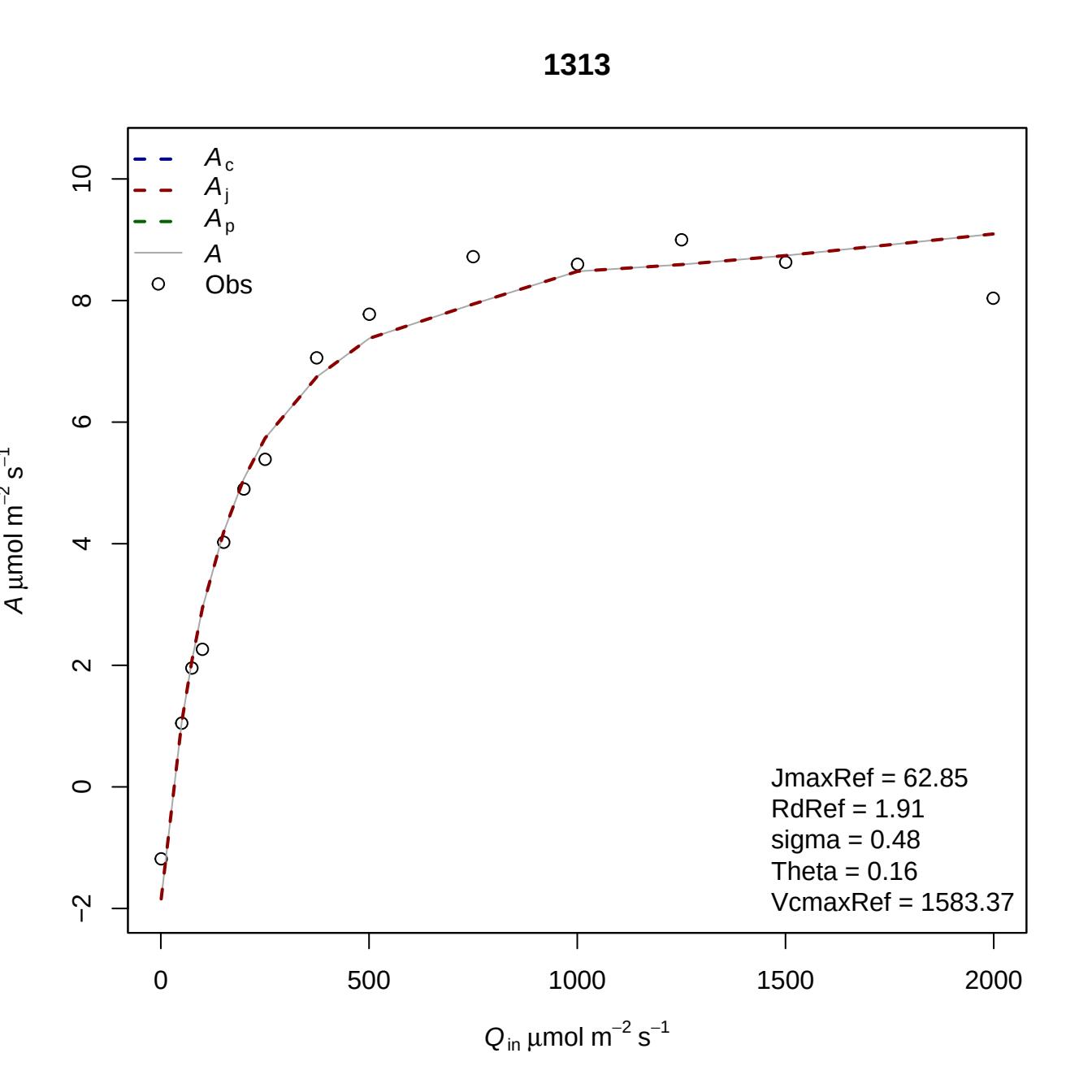

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 181.19
RdRef = 2.62
sigma = 0.5
Theta = −1.79
VcmaxRef = 37335.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

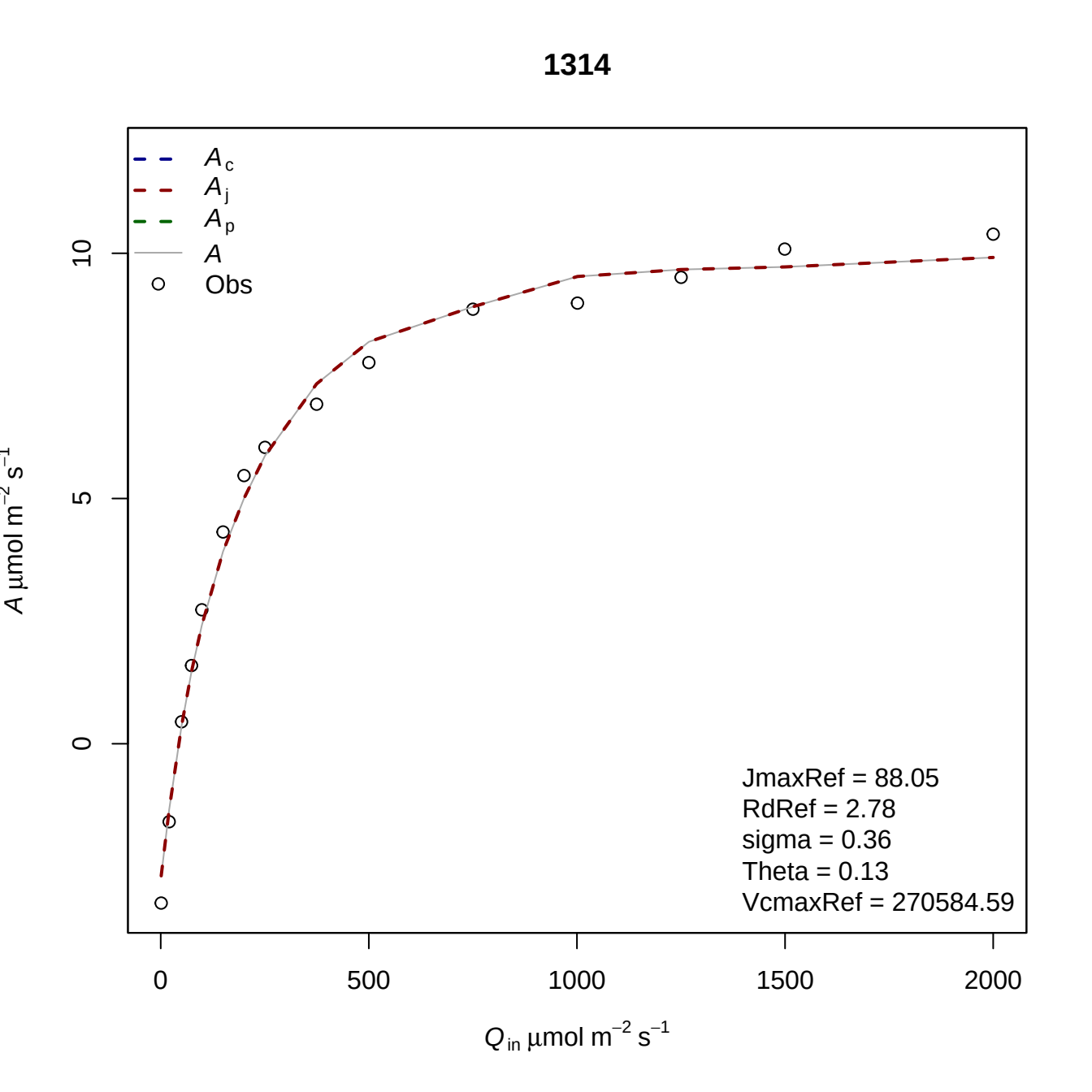

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 150.51
RdRef = 3.4
sigma = 0.29
Theta = −1.24
VcmaxRef = 54.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

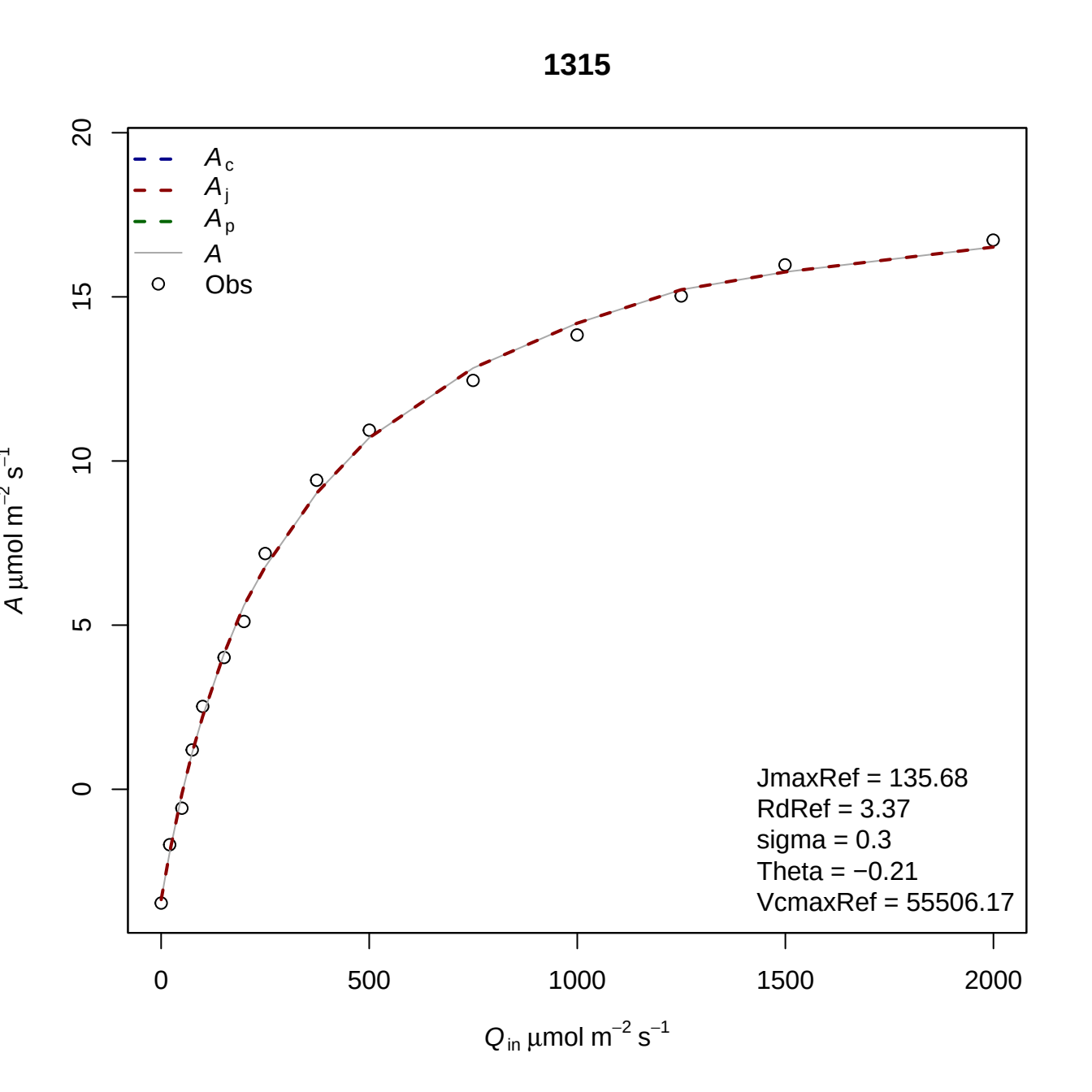

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 114.67
RdRef = 2.95
sigma = 0.38
Theta = −0.5
VcmaxRef = 215.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

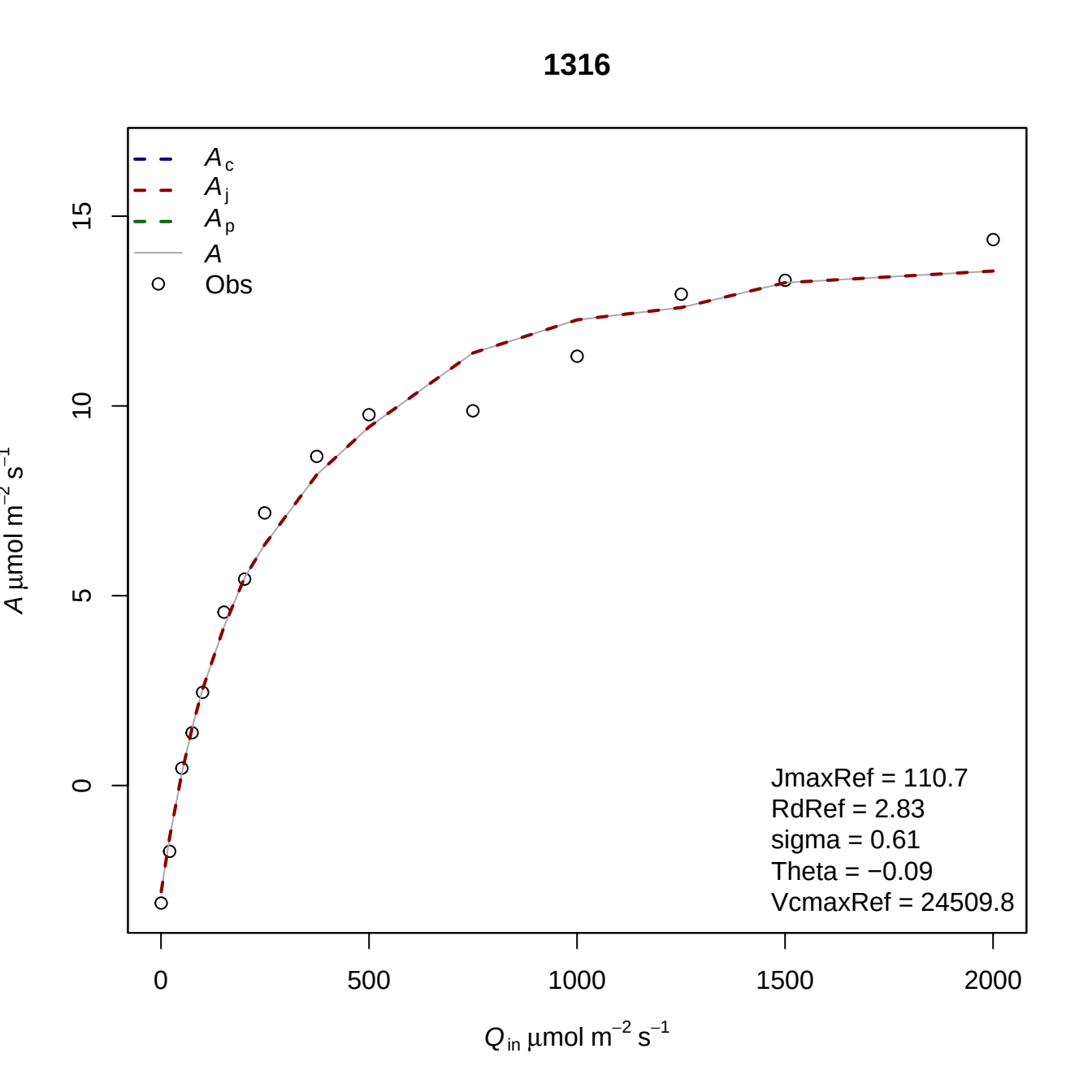

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 82.52
RdRef = 1.91
sigma = 0.33
Theta = −0.07
VcmaxRef = 11924.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

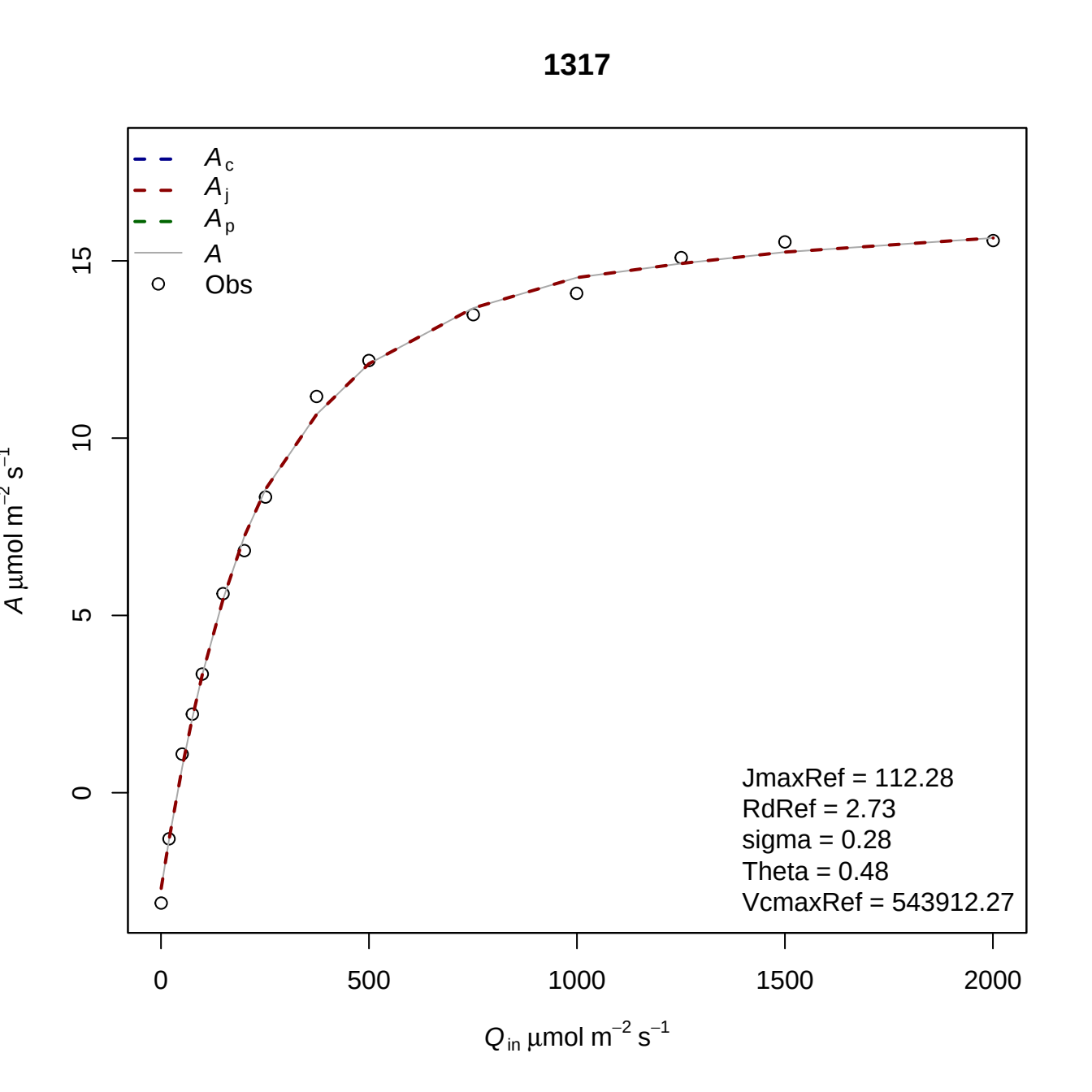

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 100.64
RdRef = 1.64
sigma = 0.39
Theta = −0.24
VcmaxRef = 26215.51
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

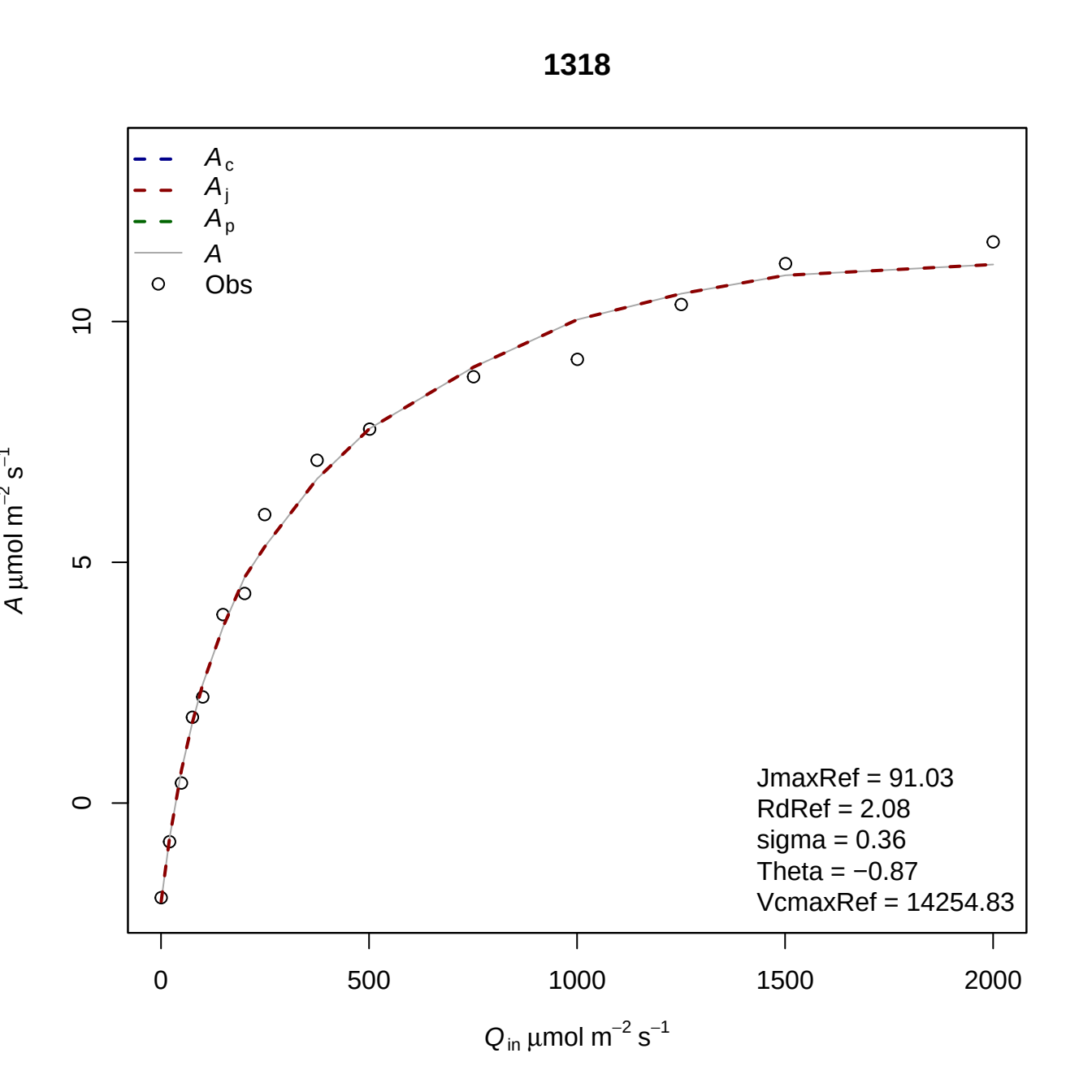

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 62.85
RdRef = 1.91
sigma = 0.48
Theta = 0.16
VcmaxRef = 1583.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

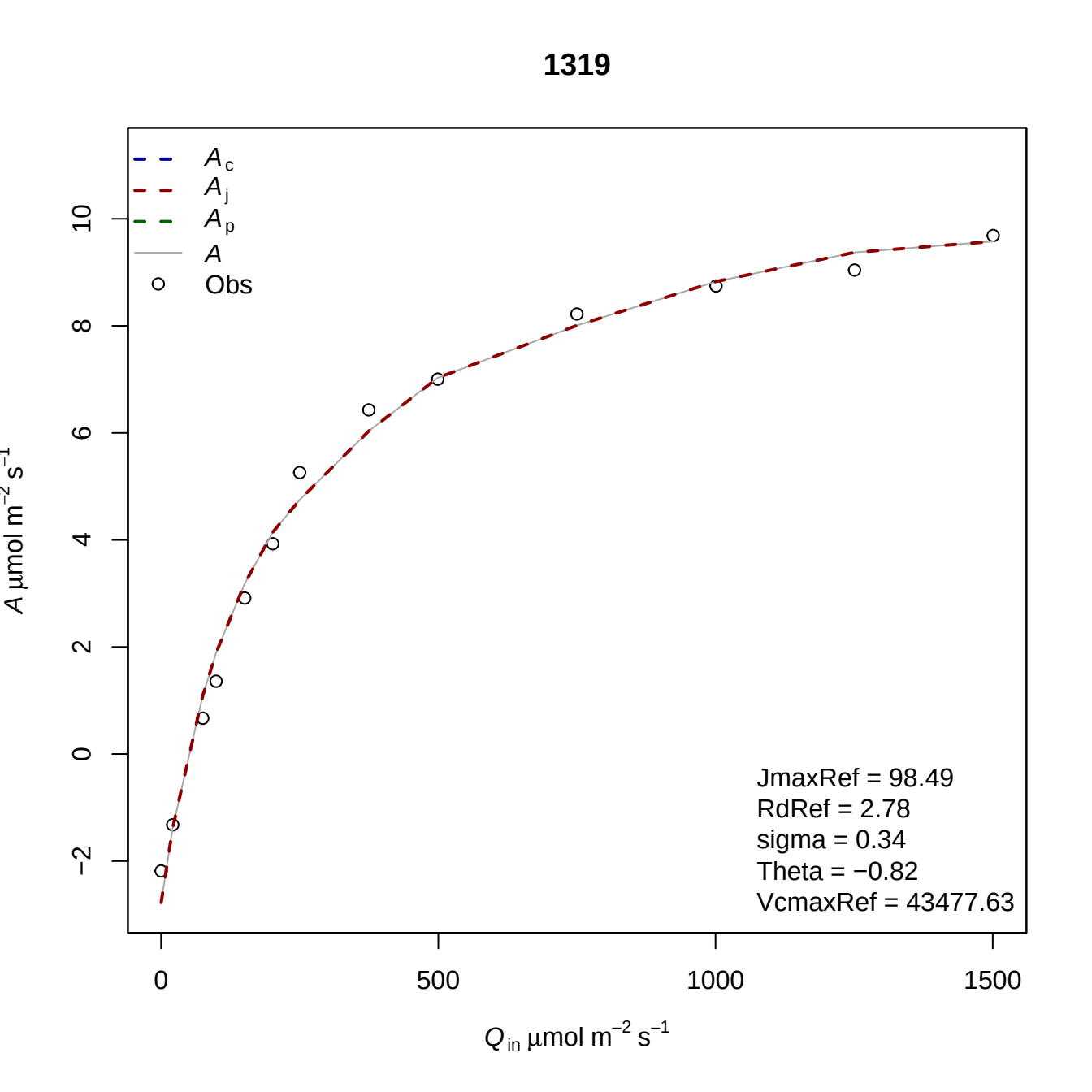

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 88.05
RdRef = 2.78
sigma = 0.36
Theta = 0.13
VcmaxRef = 270584.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

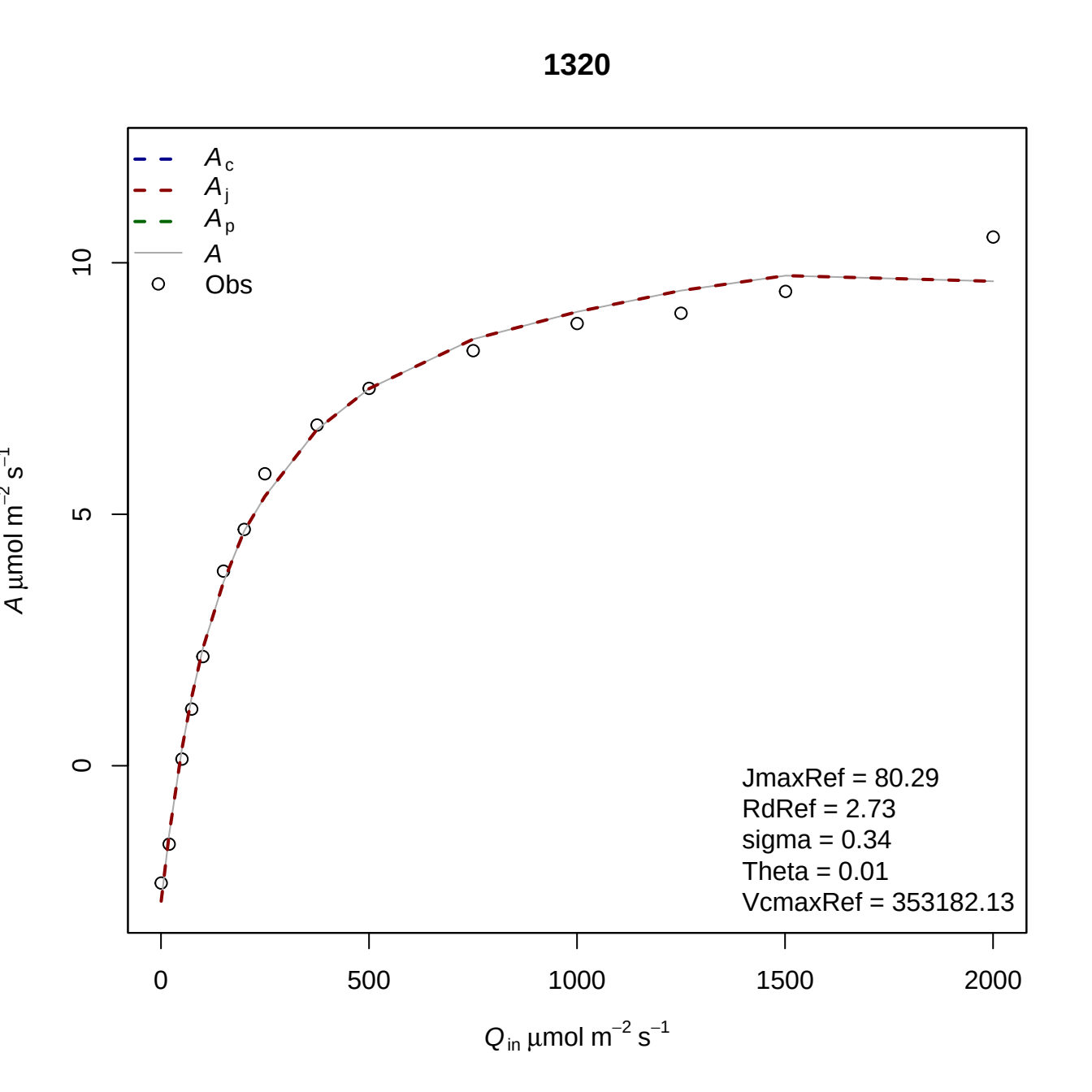

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 135.68
RdRef = 3.37
sigma = 0.3
Theta = −0.21
VcmaxRef = 55506.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

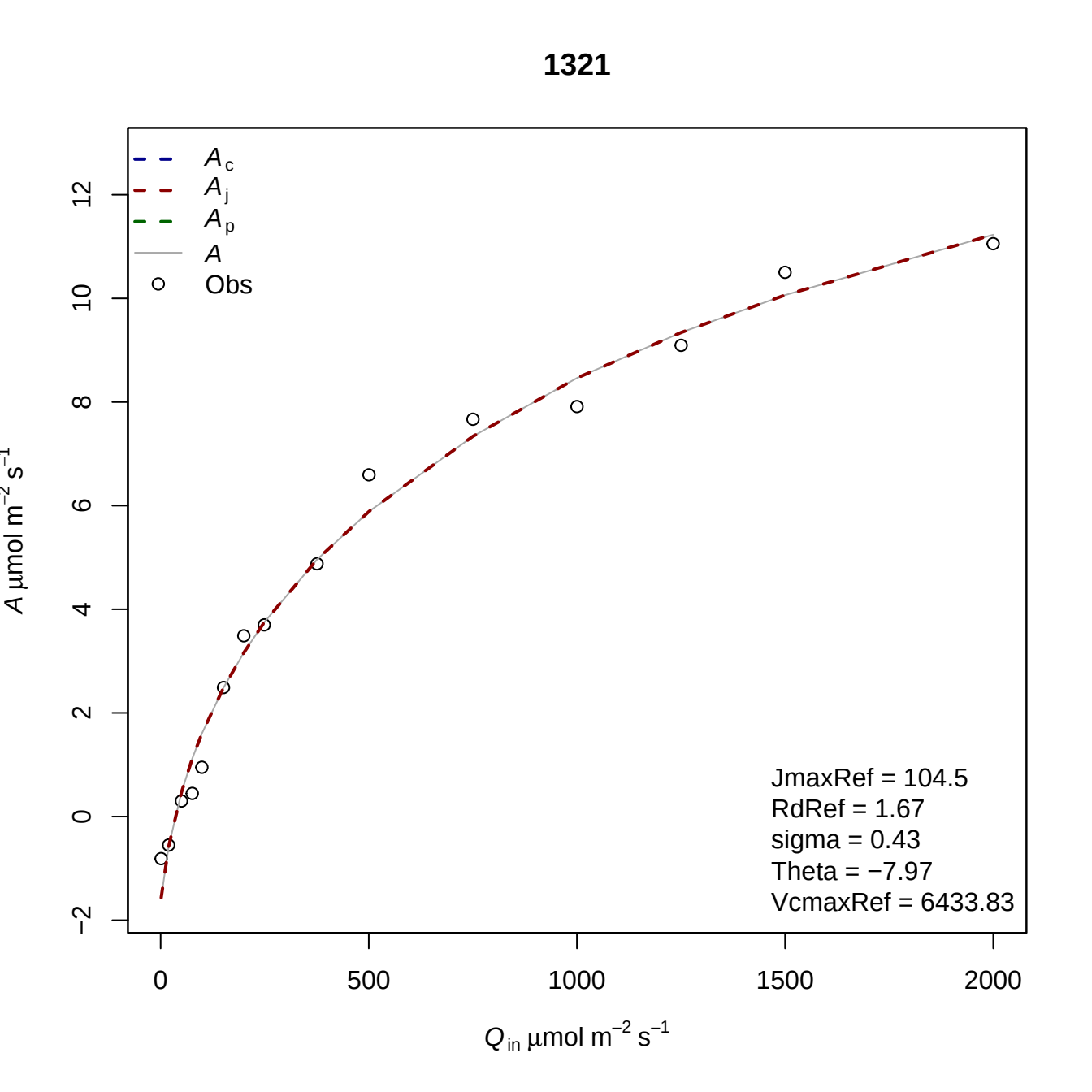

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 110.7
RdRef = 2.83
sigma = 0.61
Theta = −0.09
VcmaxRef = 24509.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

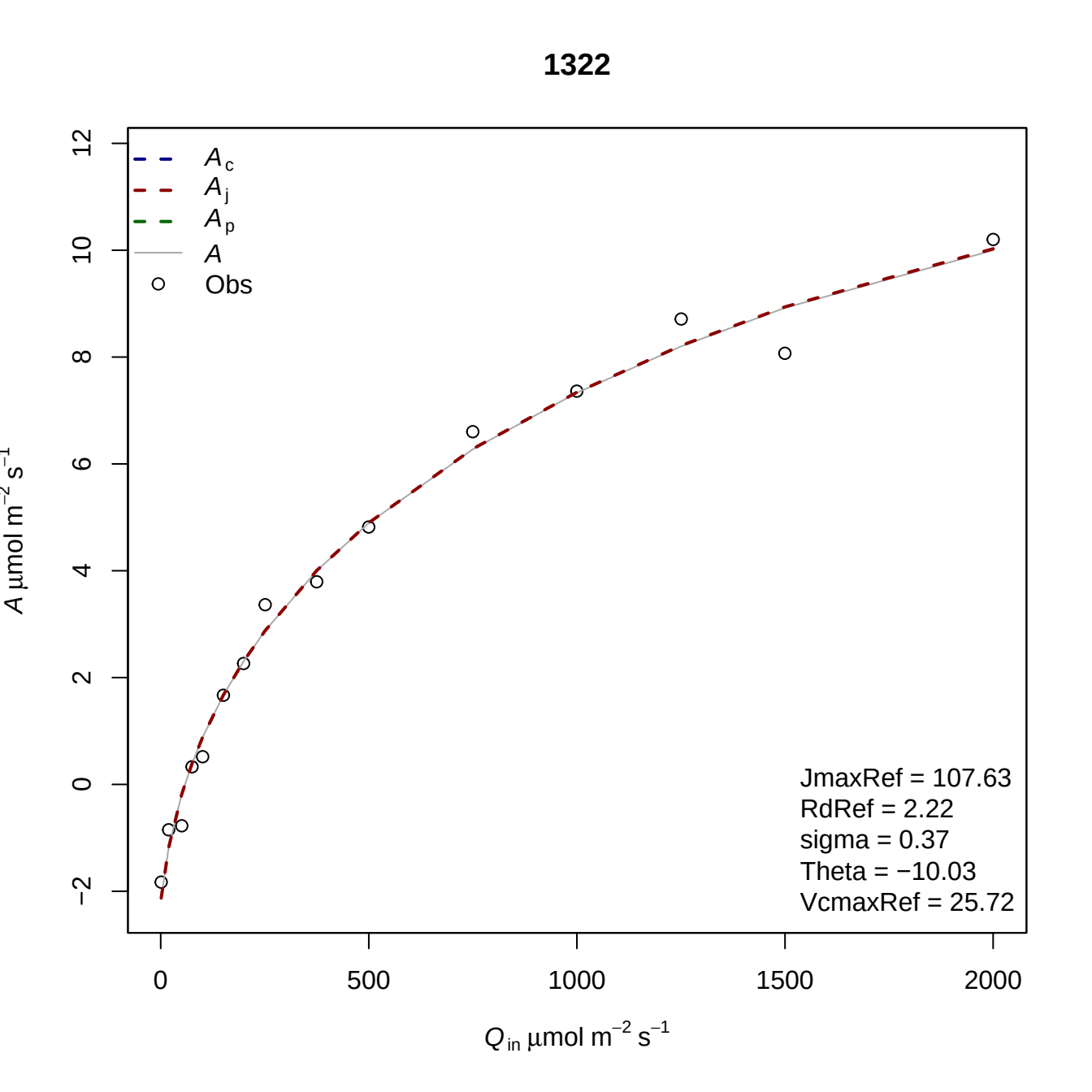

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 112.28
RdRef = 2.73
sigma = 0.28
Theta = 0.48
VcmaxRef = 543912.27
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

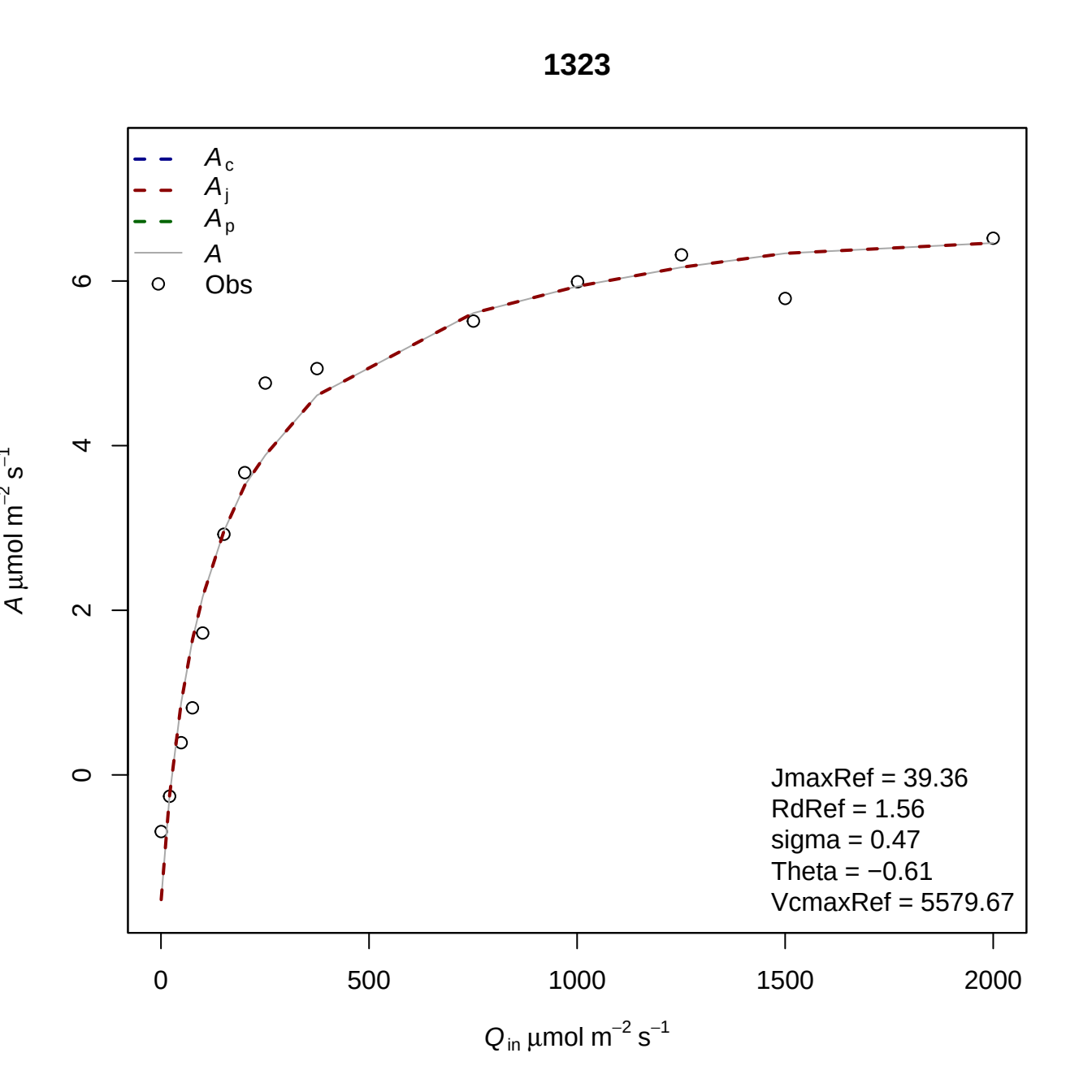

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 91.03
RdRef = 2.08
sigma = 0.36
Theta = −0.87
VcmaxRef = 14254.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

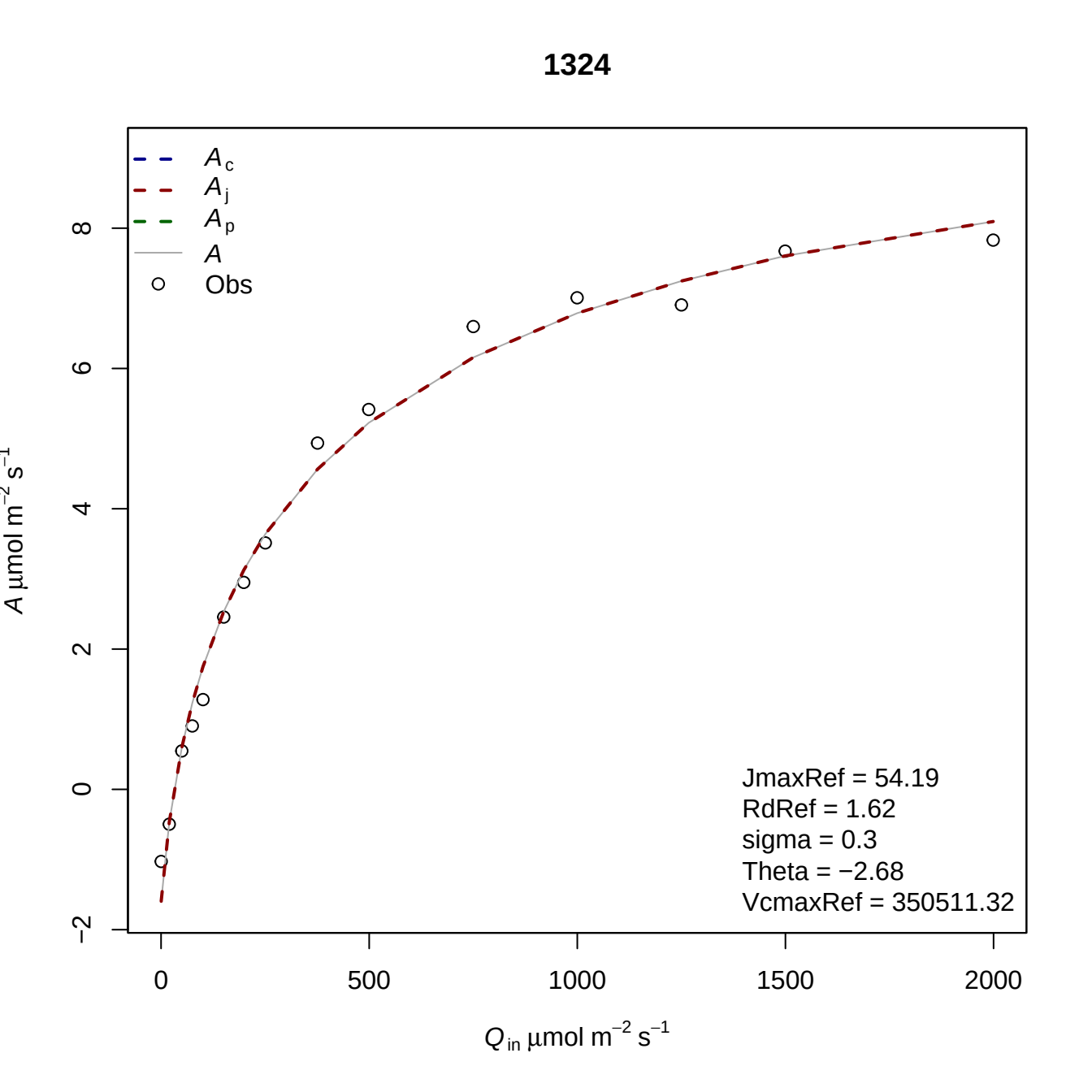

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 98.49
RdRef = 2.78
sigma = 0.34
Theta = −0.82
VcmaxRef = 43477.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

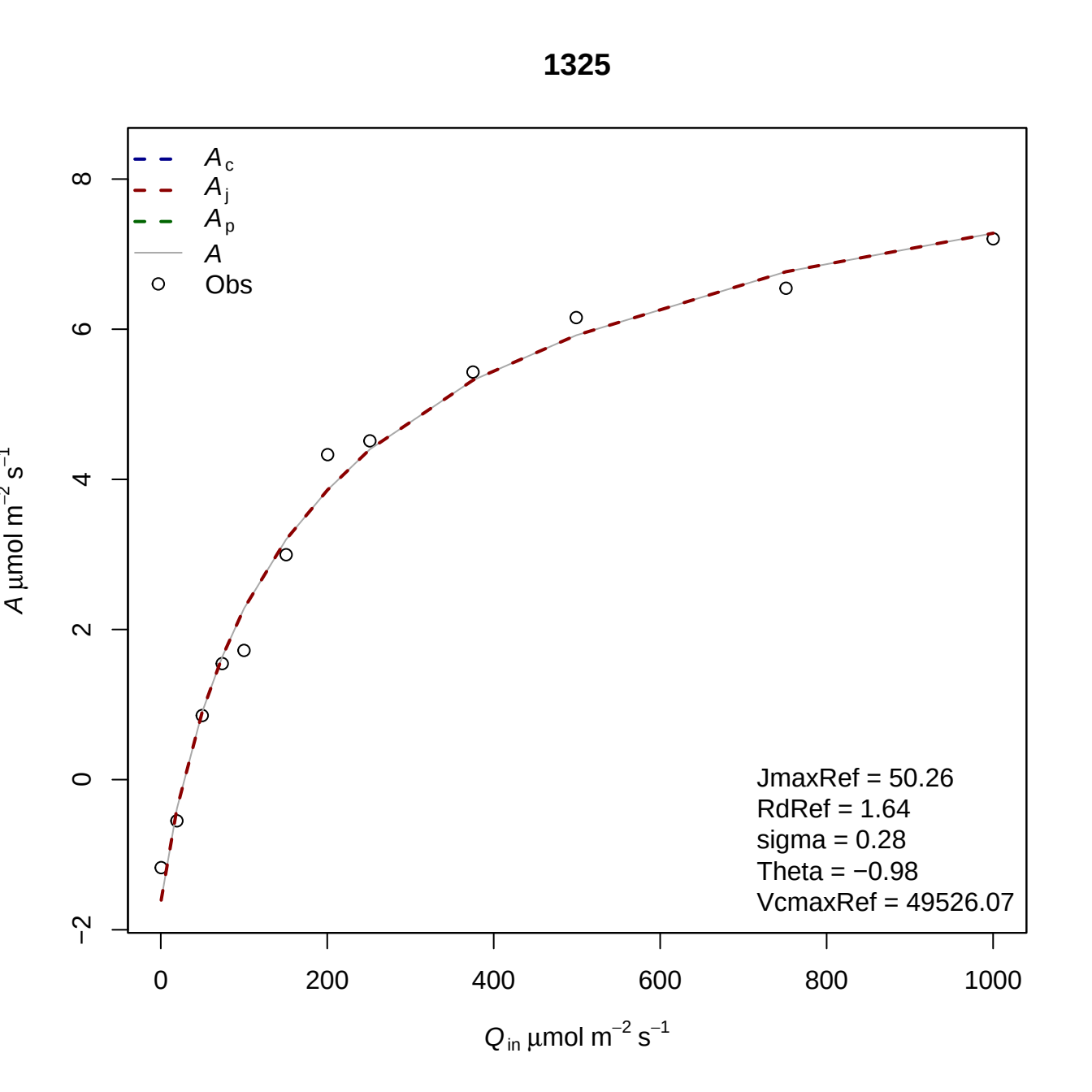

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 80.29
RdRef = 2.73
sigma = 0.34
Theta = 0.01
VcmaxRef = 353182.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

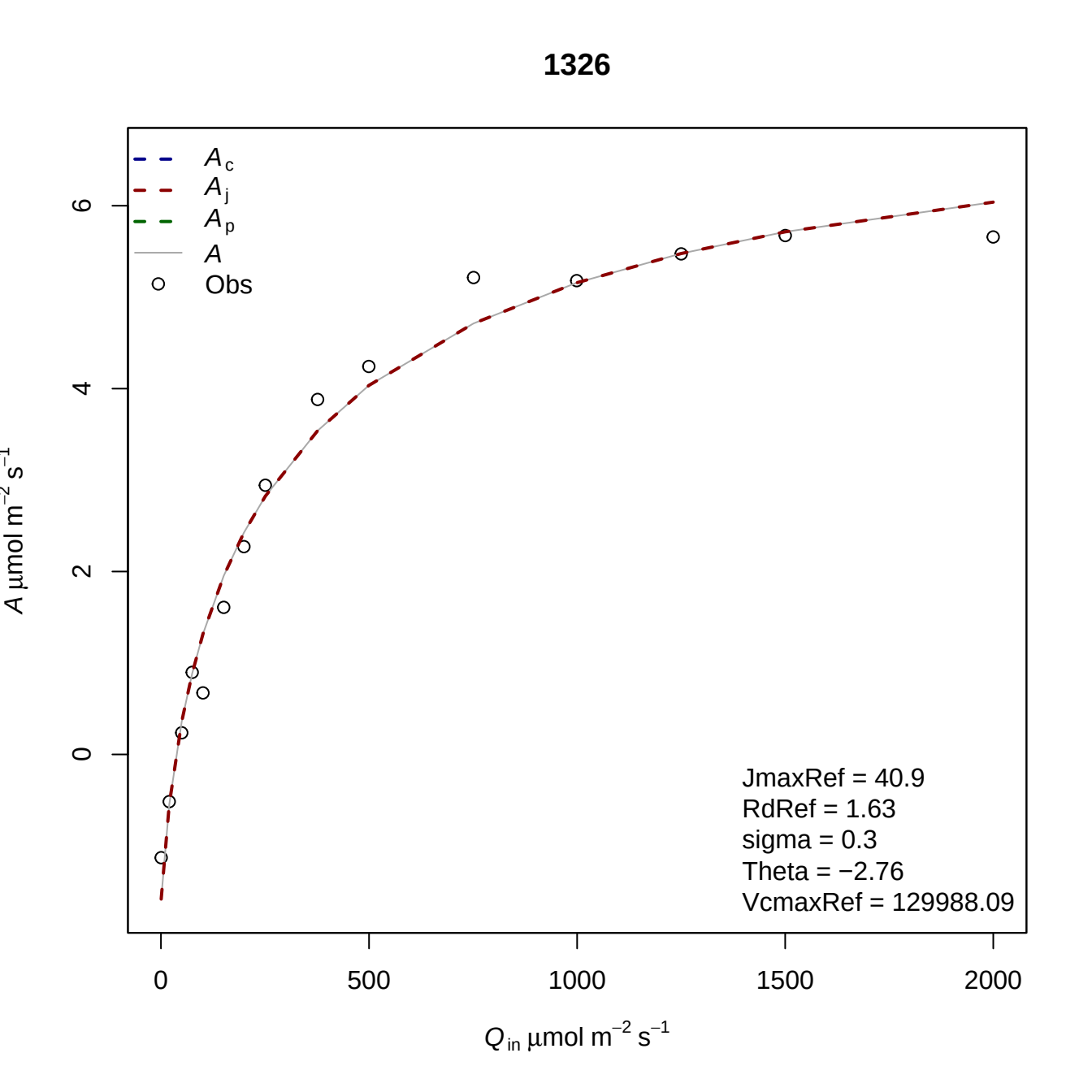

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 104.5
RdRef = 1.67
sigma = 0.43
Theta = −7.97
VcmaxRef = 6433.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

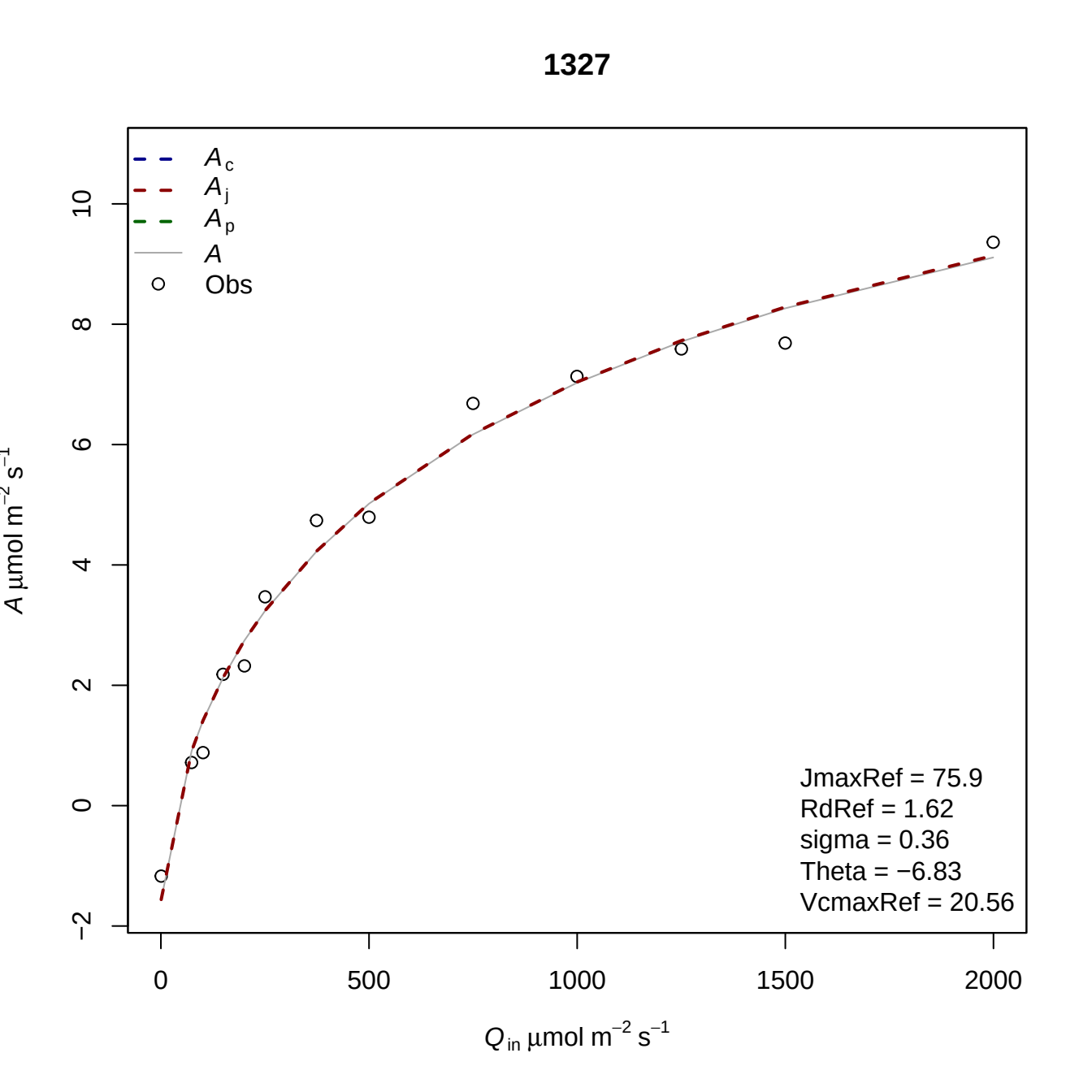

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 107.63
RdRef = 2.22
sigma = 0.37
Theta = −10.03
VcmaxRef = 25.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

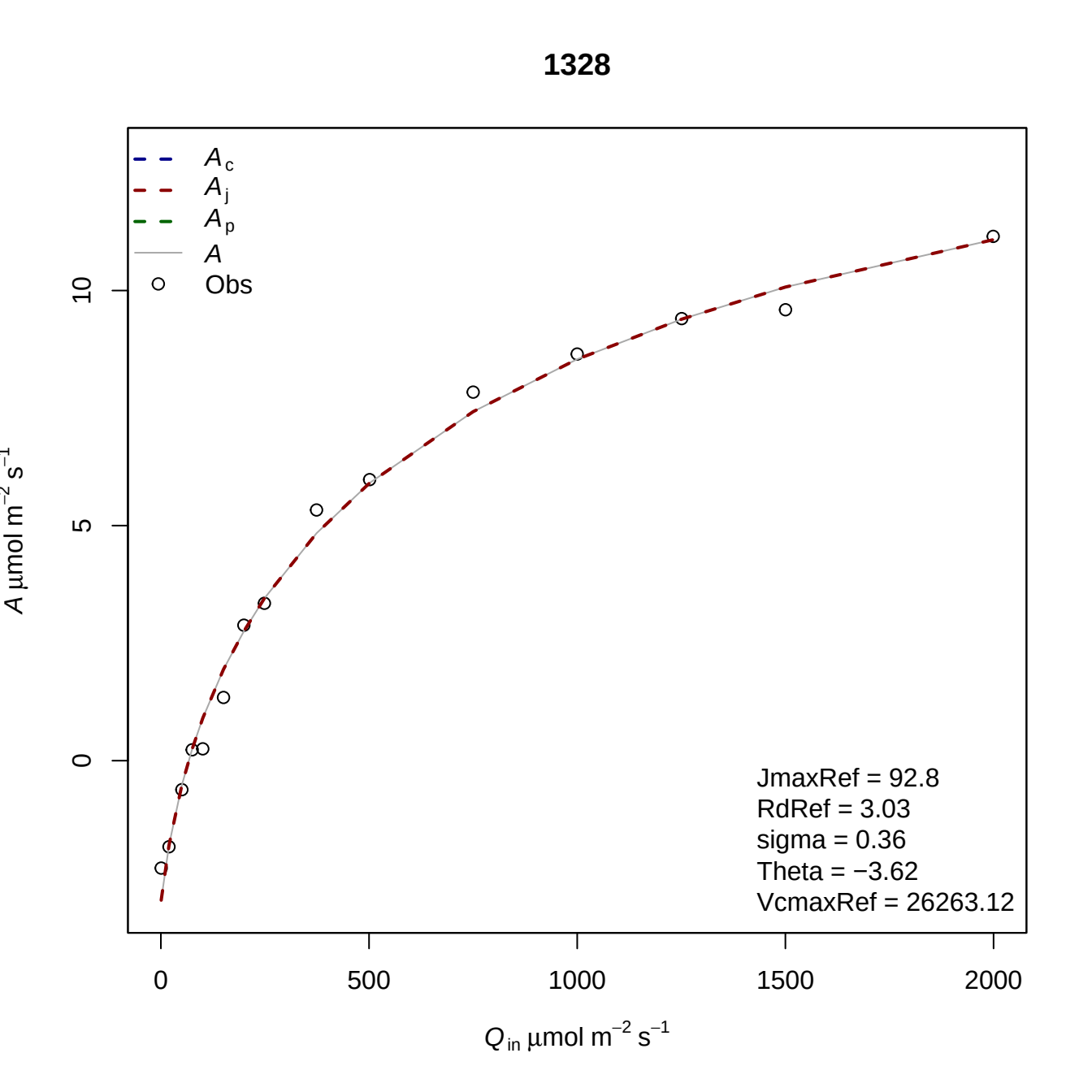

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 39.36
RdRef = 1.56
sigma = 0.47
Theta = −0.61
VcmaxRef = 5579.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

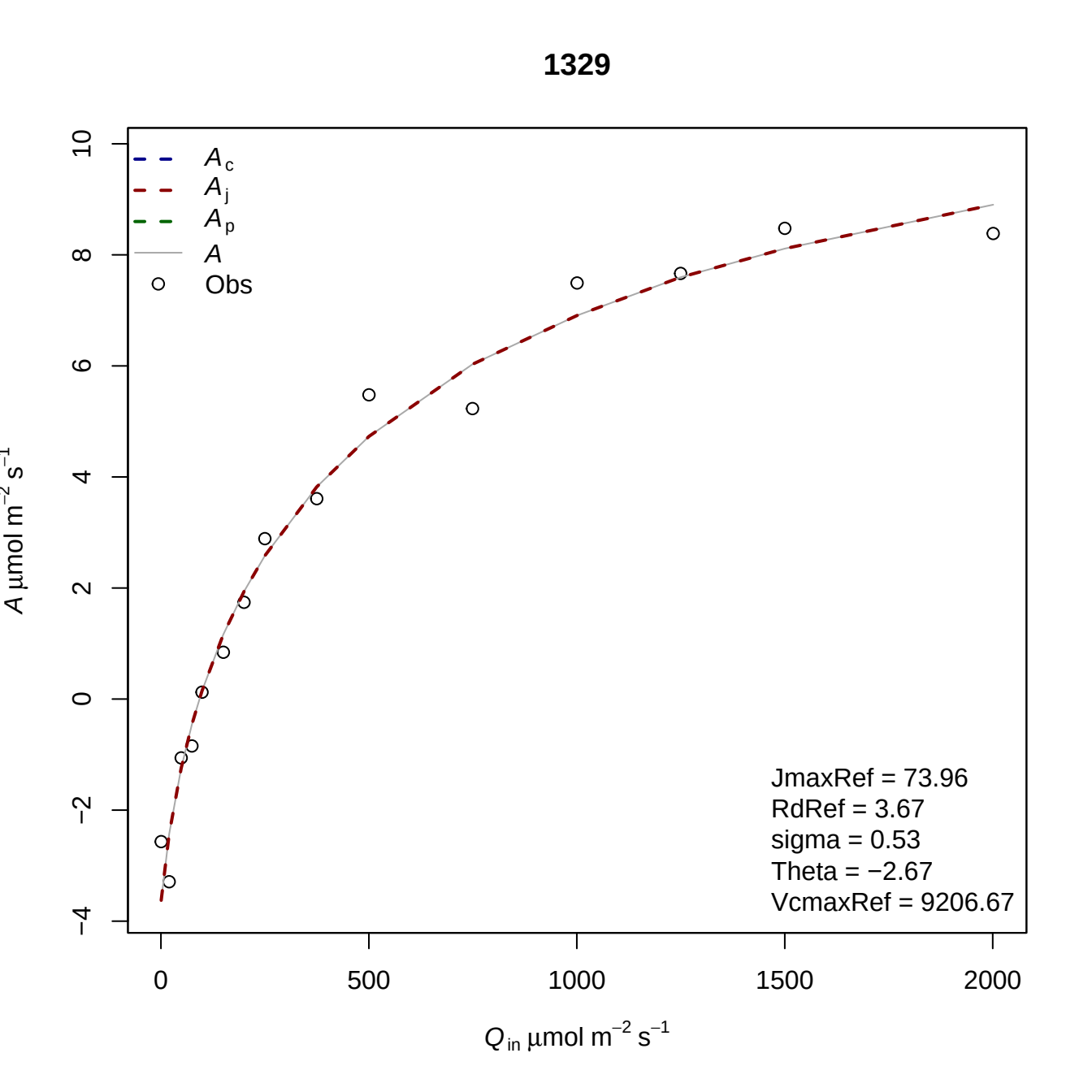

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 54.19
RdRef = 1.62
sigma = 0.3
Theta = −2.68
VcmaxRef = 350511.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

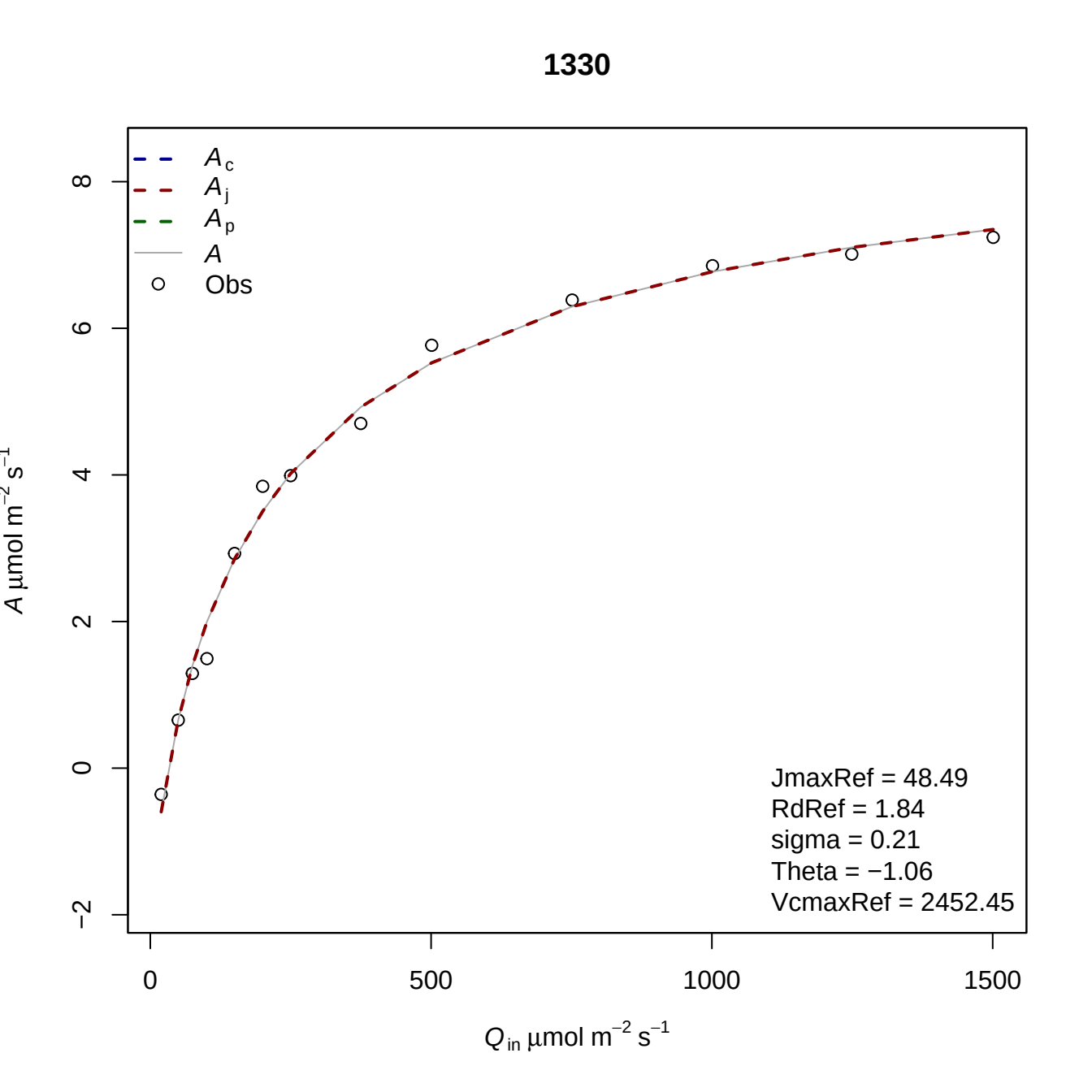

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 50.26
RdRef = 1.64
sigma = 0.28
Theta = −0.98
VcmaxRef = 49526.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000

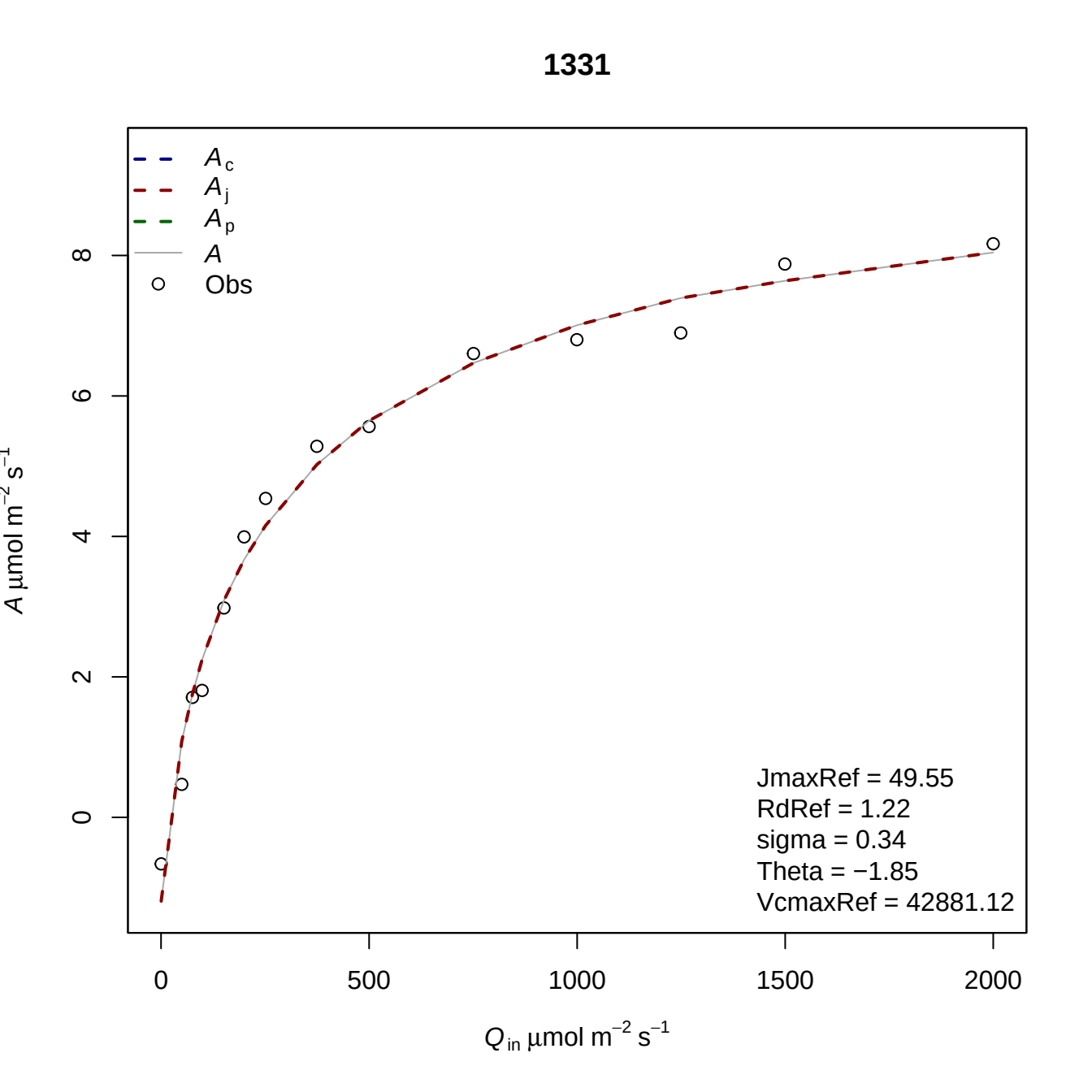

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 40.9
RdRef = 1.63
sigma = 0.3
Theta = −2.76
VcmaxRef = 129988.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

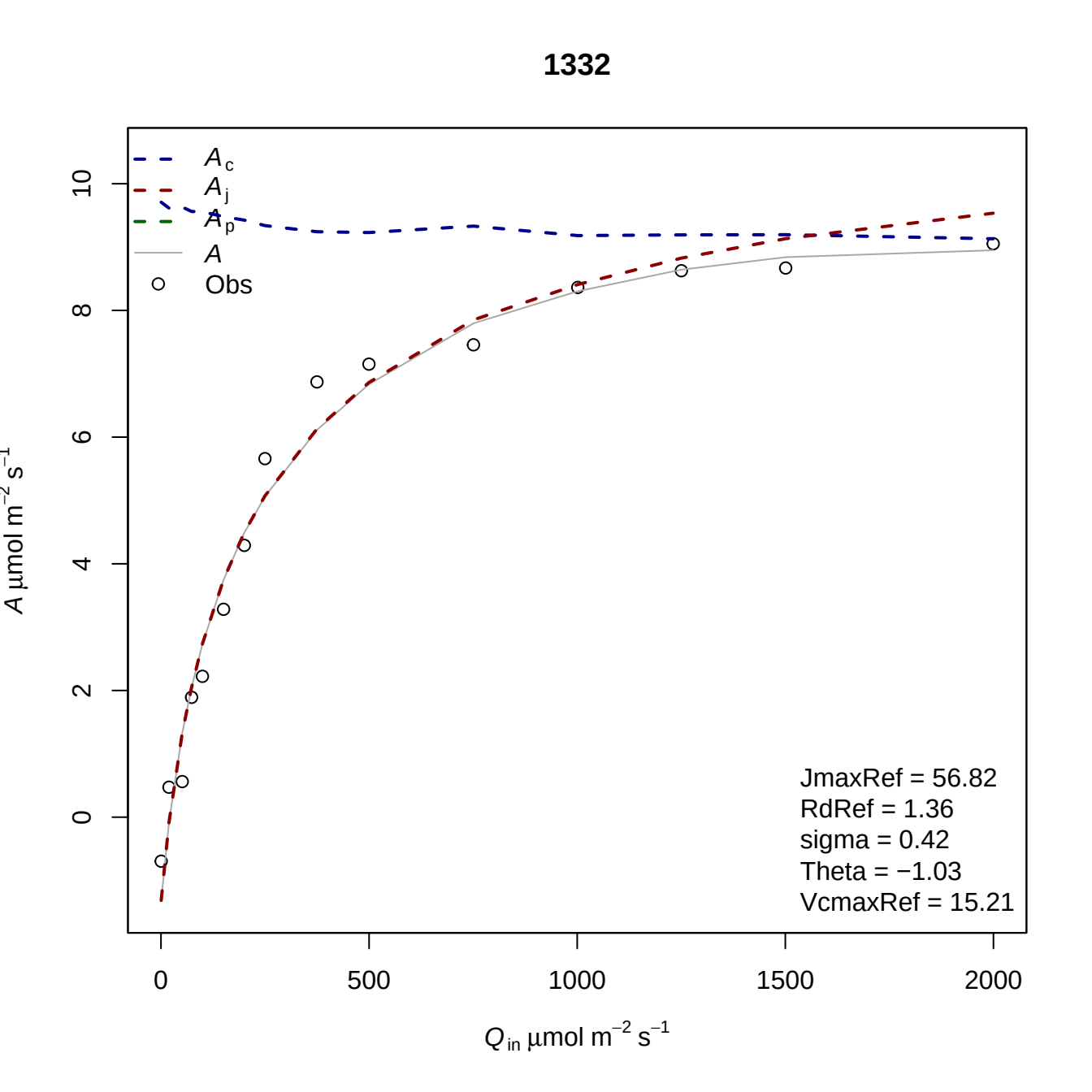

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 75.9
RdRef = 1.62
sigma = 0.36
Theta = −6.83
VcmaxRef = 20.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

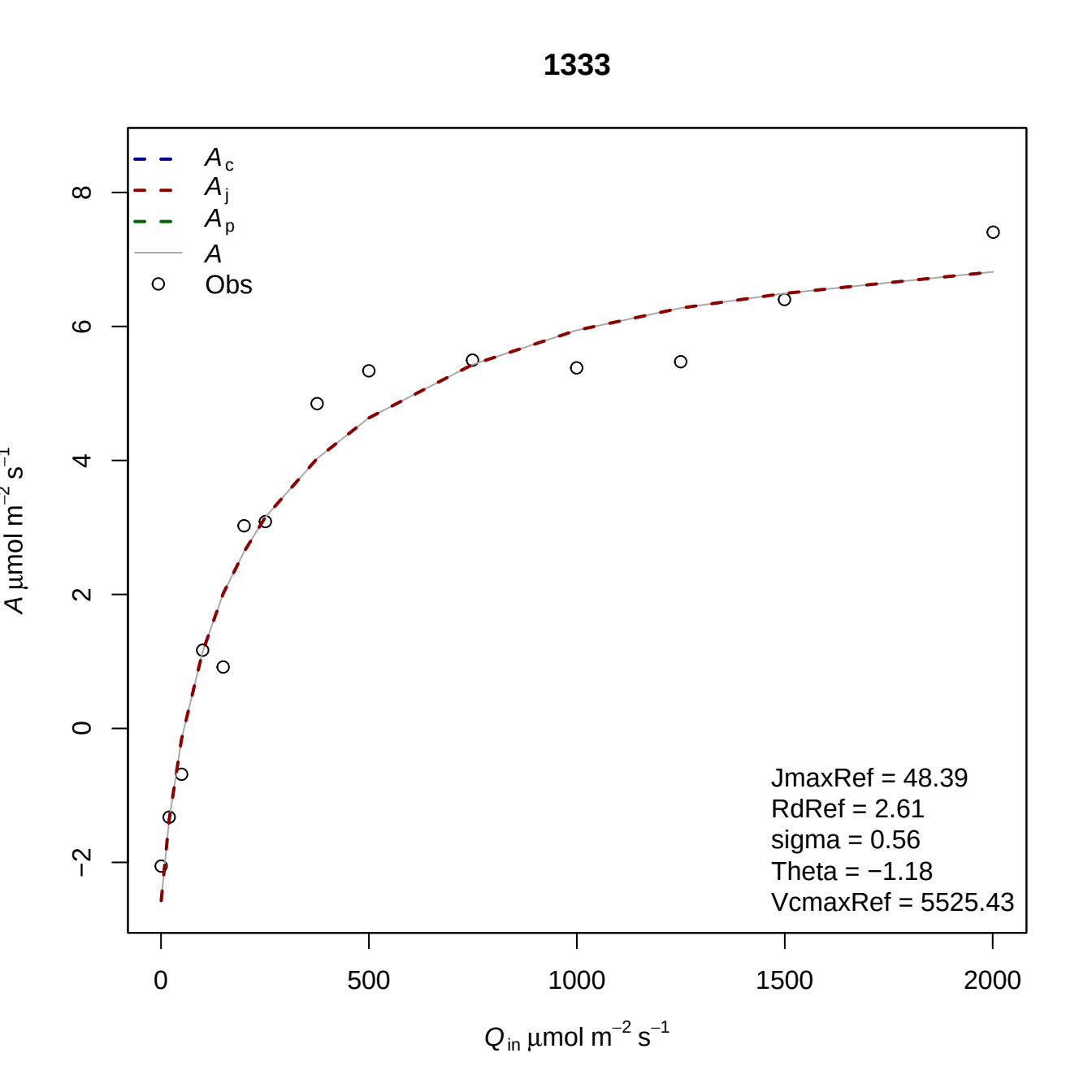

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 92.8
RdRef = 3.03
sigma = 0.36
Theta = −3.62
VcmaxRef = 26263.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

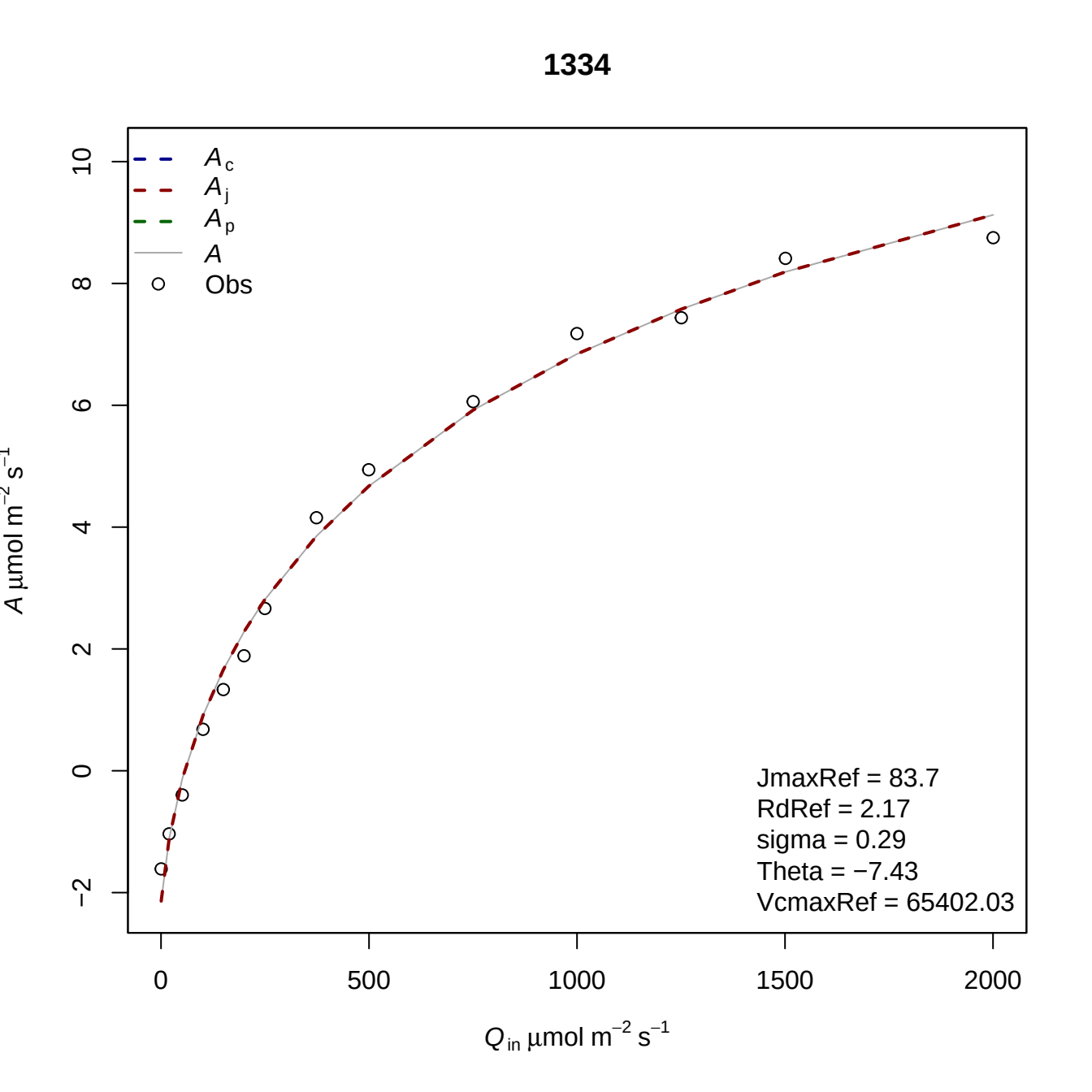

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 73.96
RdRef = 3.67
sigma = 0.53
Theta = −2.67
VcmaxRef = 9206.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

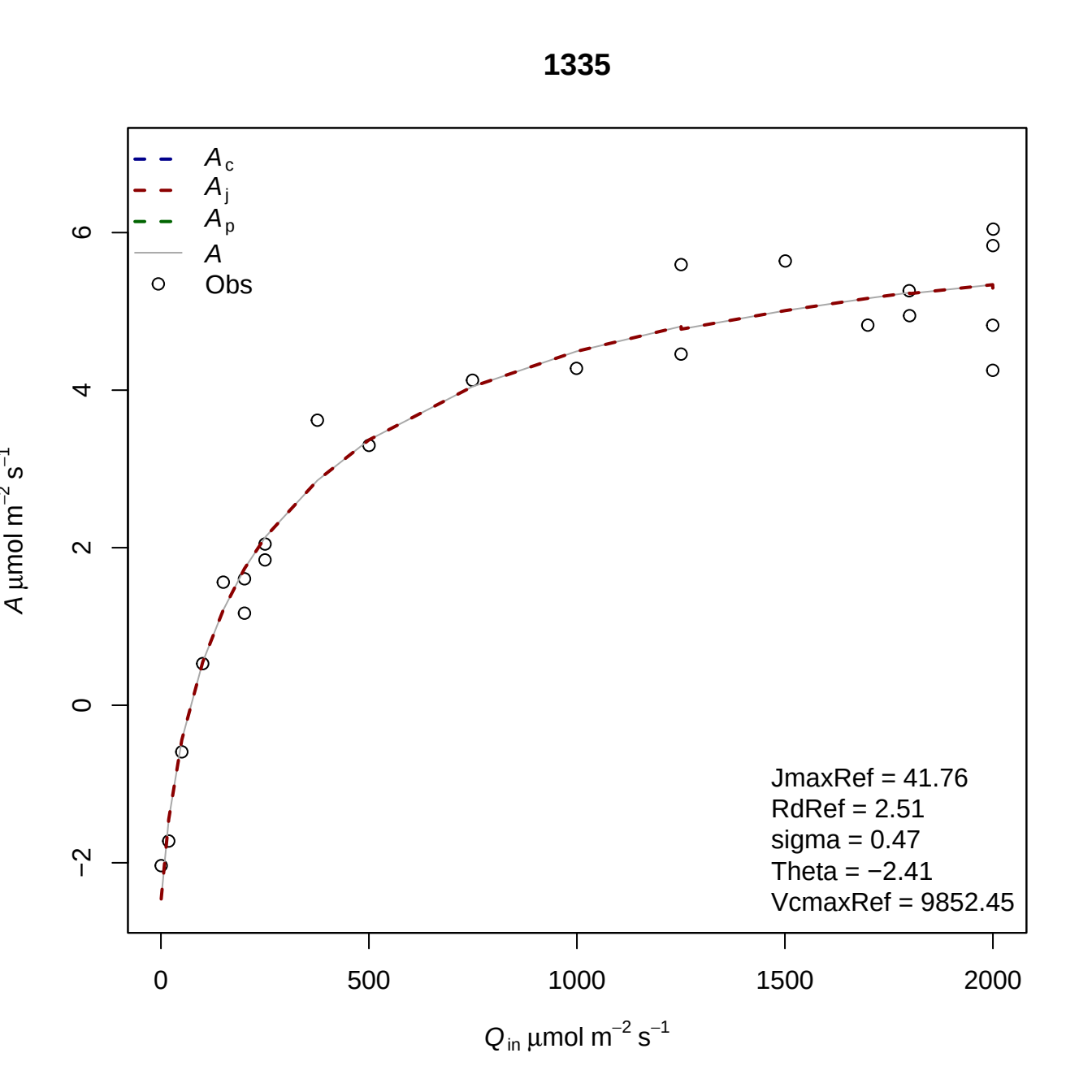

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 48.49
RdRef = 1.84
sigma = 0.21
Theta = −1.06
VcmaxRef = 2452.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

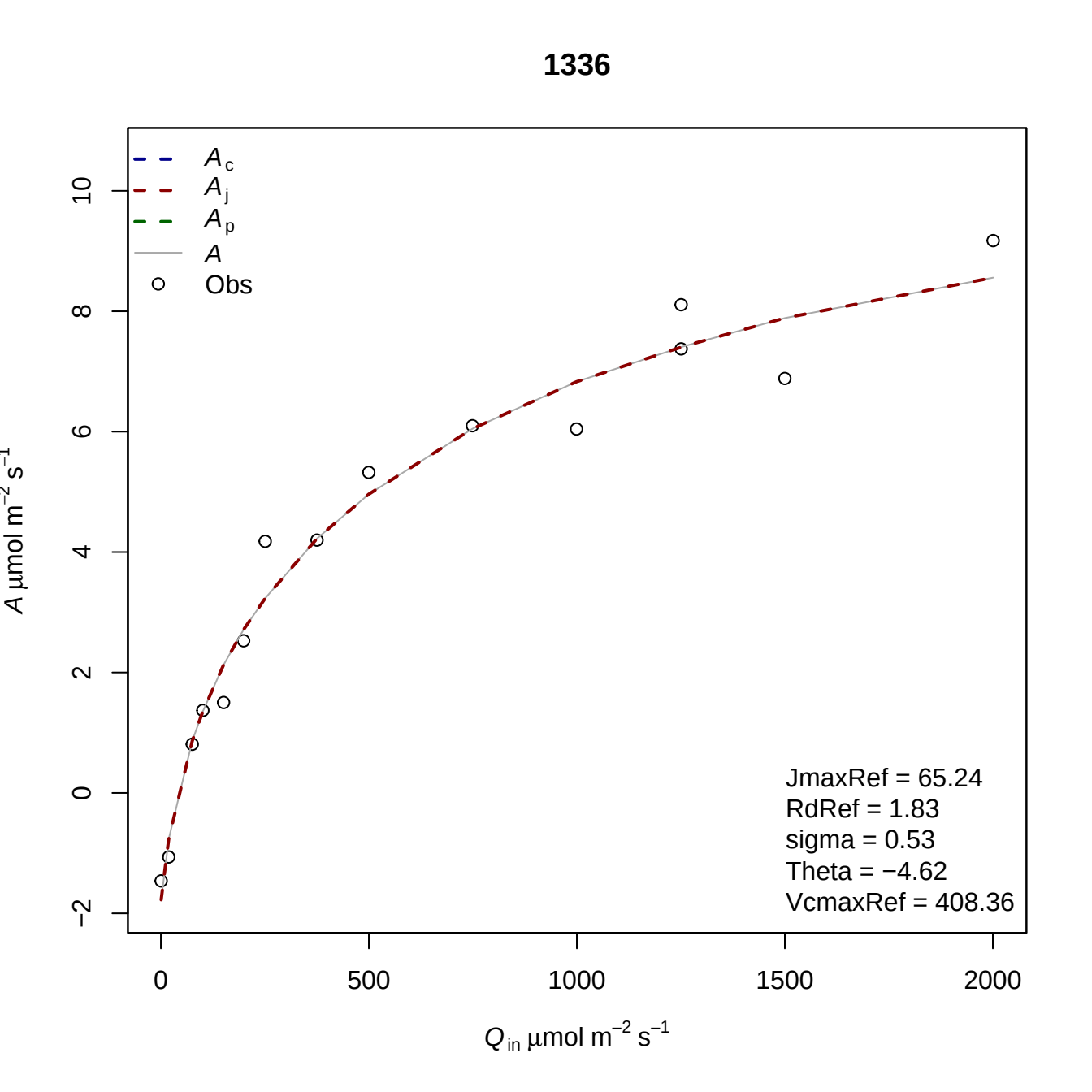

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 49.55
RdRef = 1.22
sigma = 0.34
Theta = −1.85
VcmaxRef = 42881.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

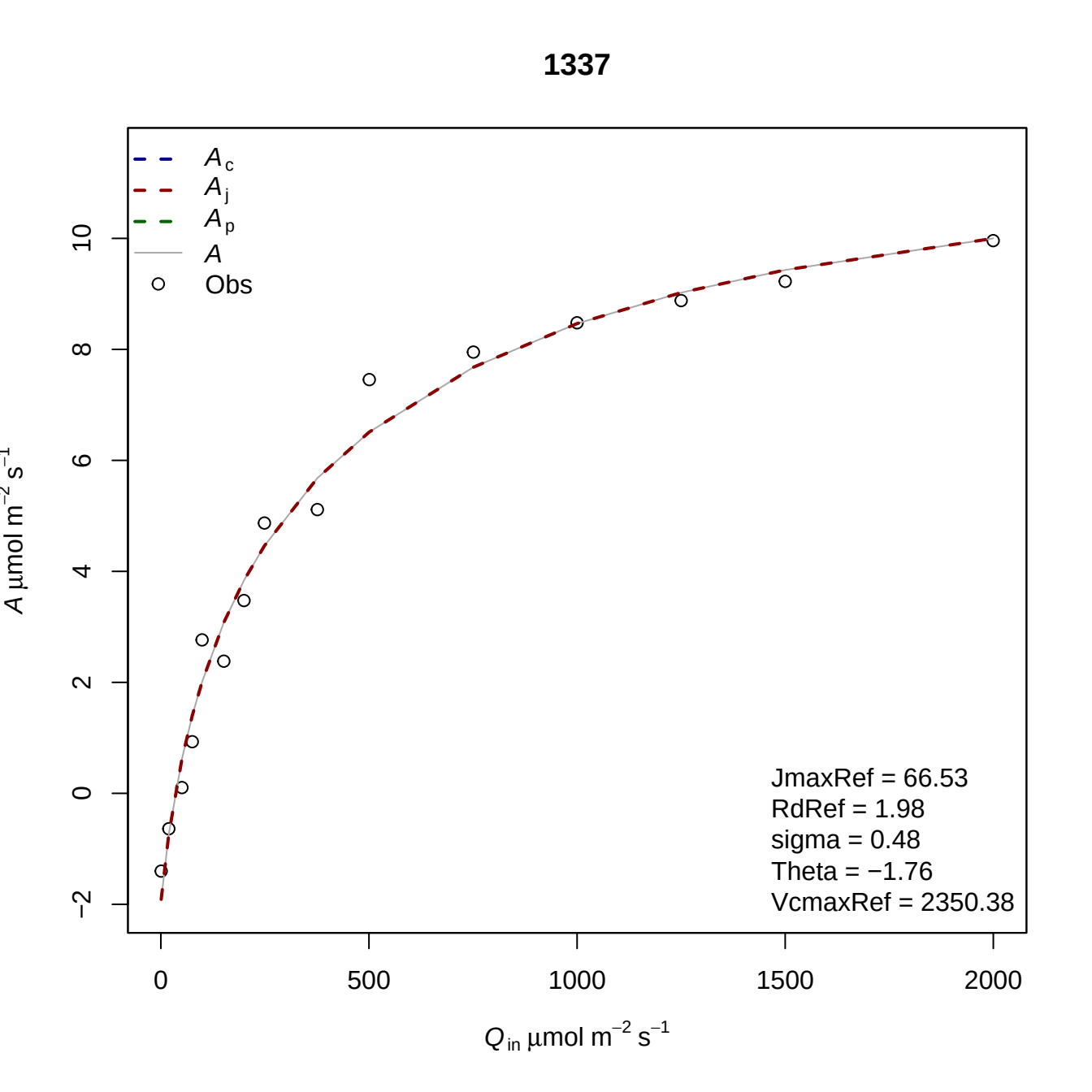

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 56.82
RdRef = 1.36
sigma = 0.42
Theta = −1.03
VcmaxRef = 15.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

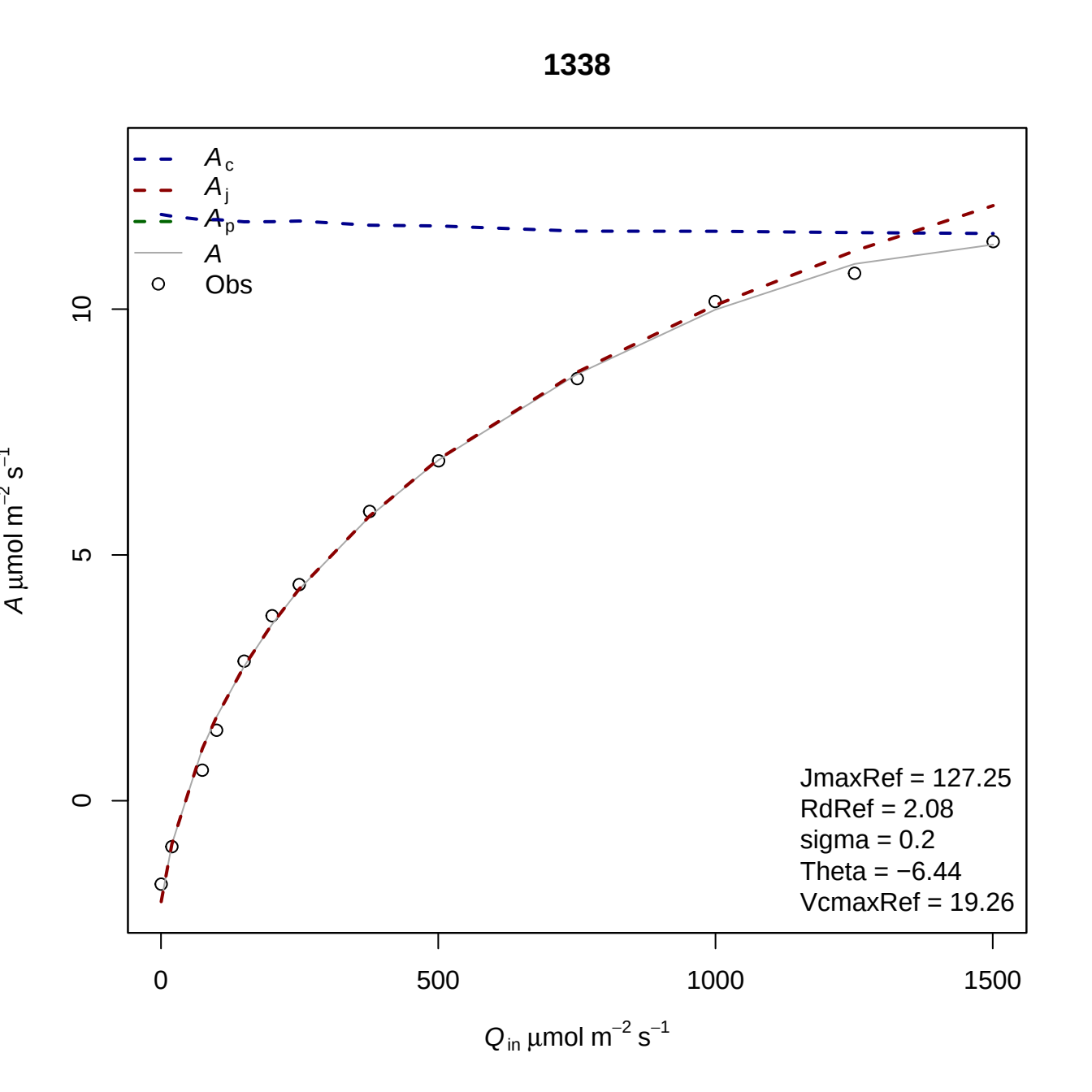

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 2.61
sigma = 0.56
Theta = −1.18
VcmaxRef = 5525.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

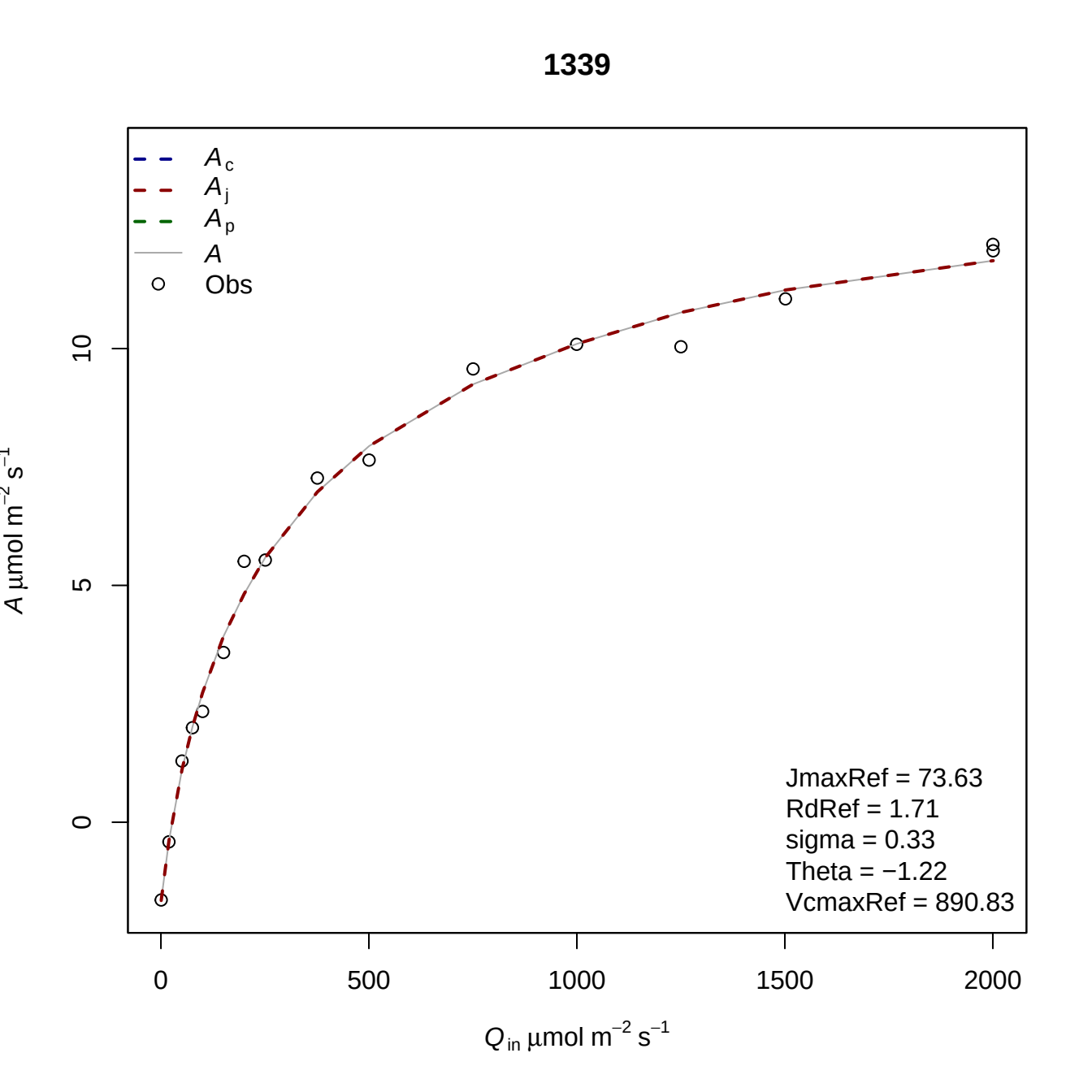

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 83.7
RdRef = 2.17
sigma = 0.29
Theta = −7.43
VcmaxRef = 65402.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

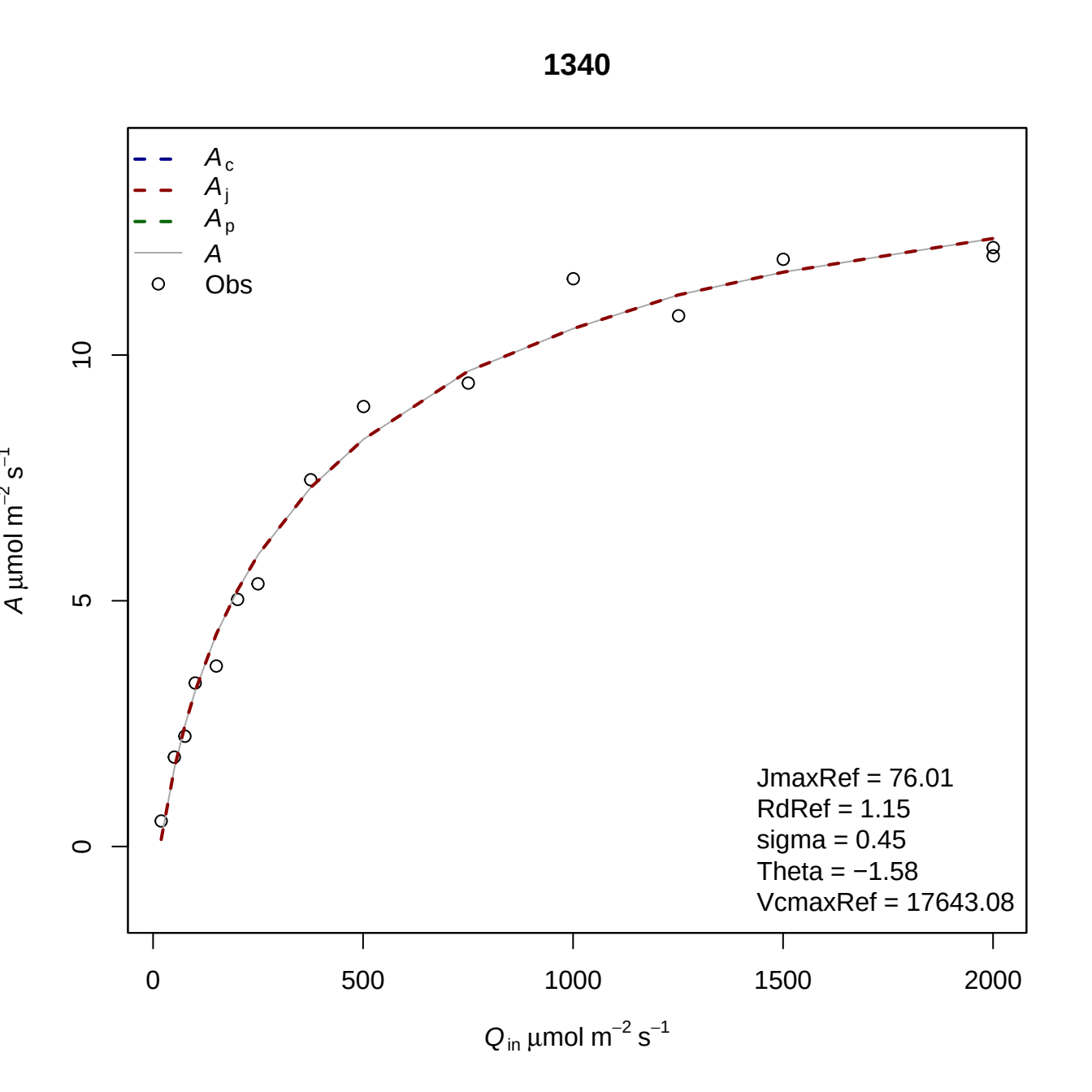

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 41.76
RdRef = 2.51
sigma = 0.47
Theta = −2.41
VcmaxRef = 9852.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

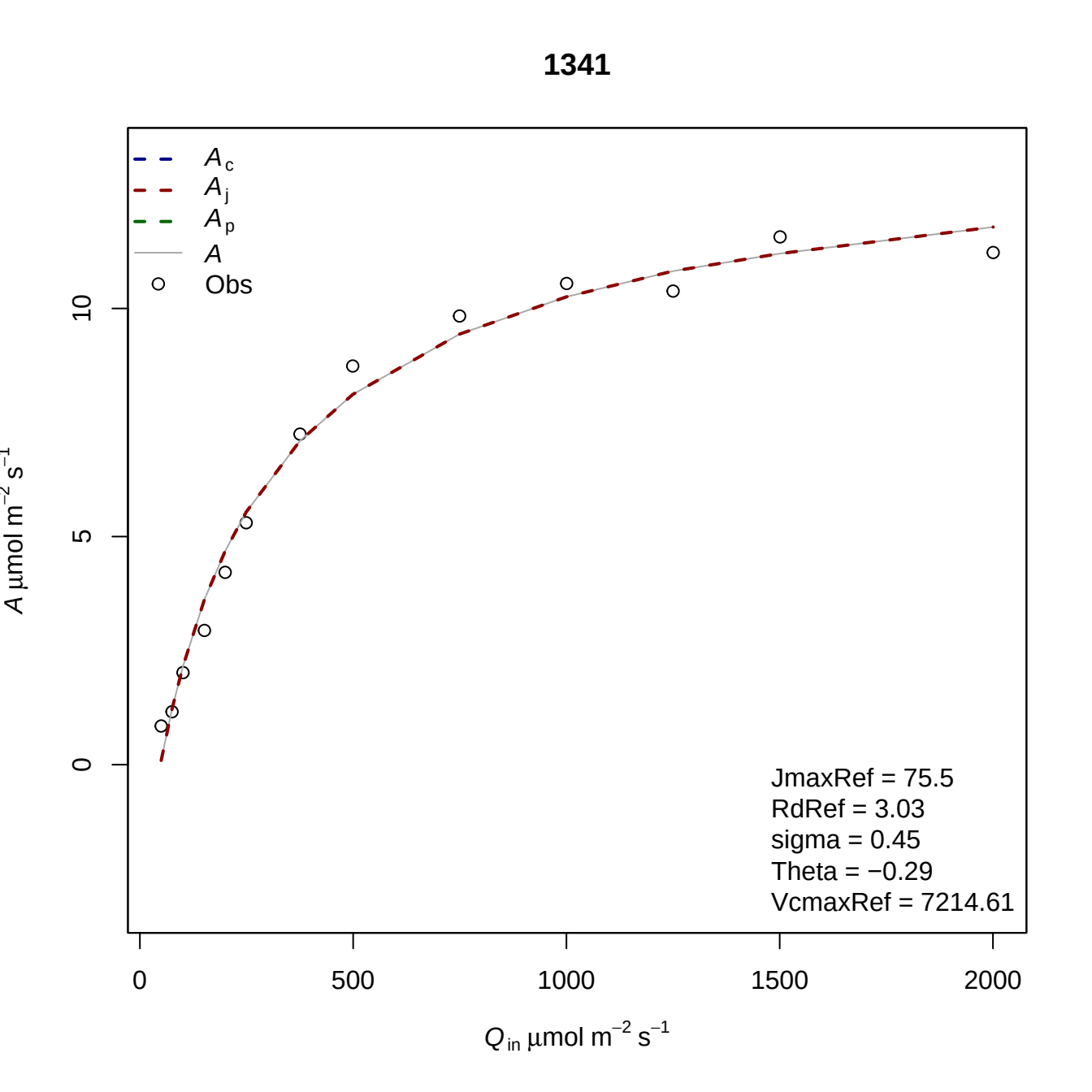

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 65.24
RdRef = 1.83
sigma = 0.53
Theta = −4.62
VcmaxRef = 408.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

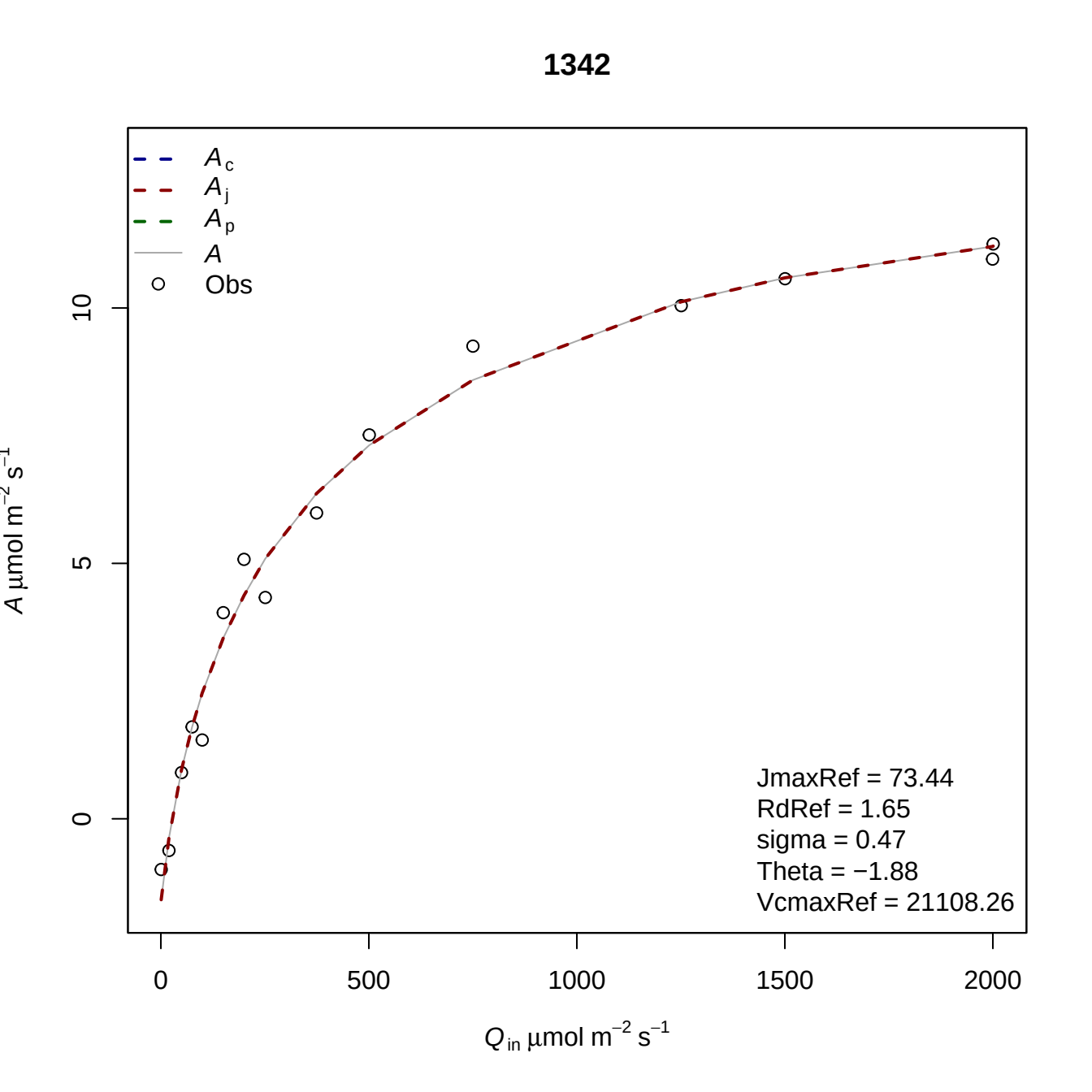

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 66.53
RdRef = 1.98
sigma = 0.48
Theta = −1.76
VcmaxRef = 2350.38
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

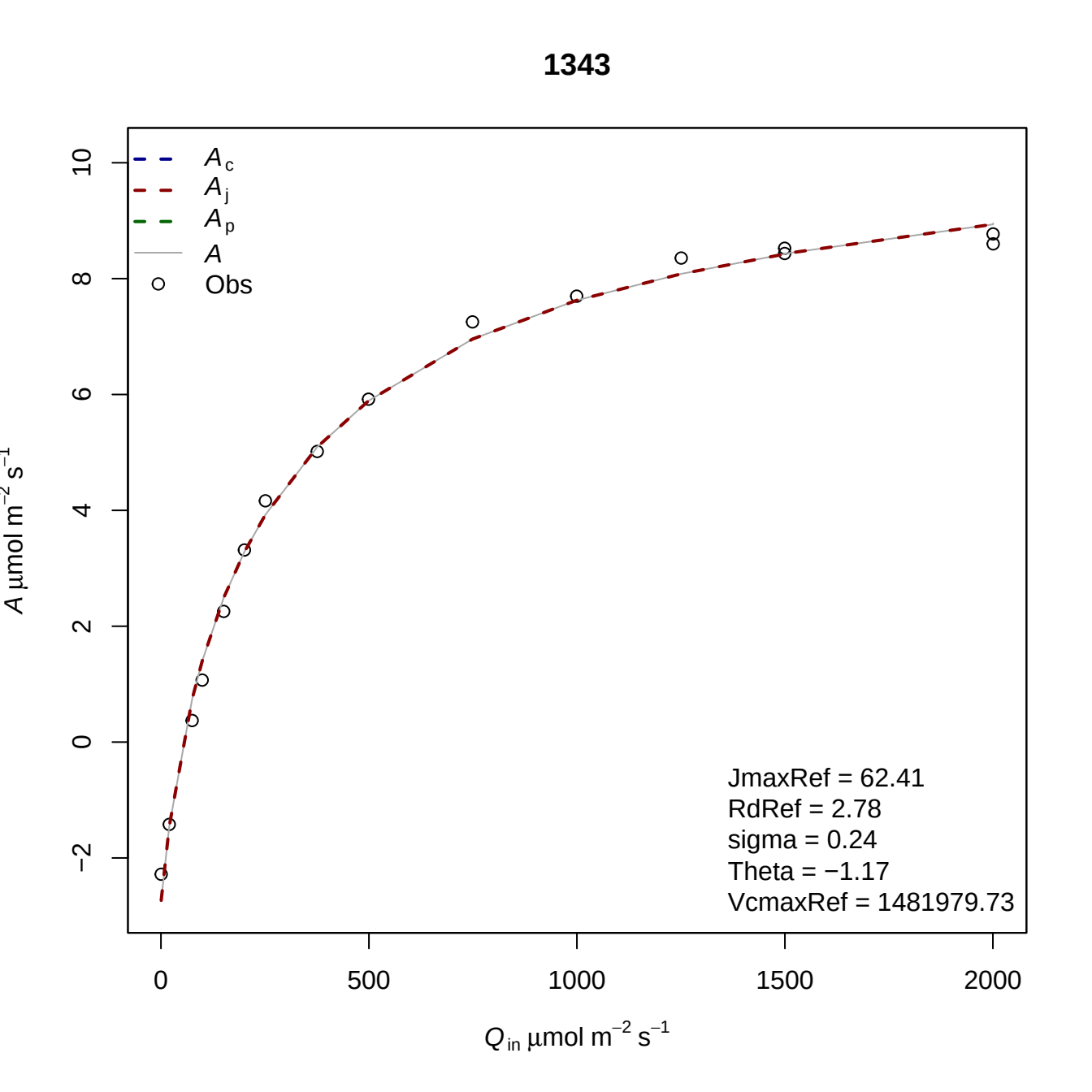

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 127.25
RdRef = 2.08
sigma = 0.2
Theta = −6.44
VcmaxRef = 19.26
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

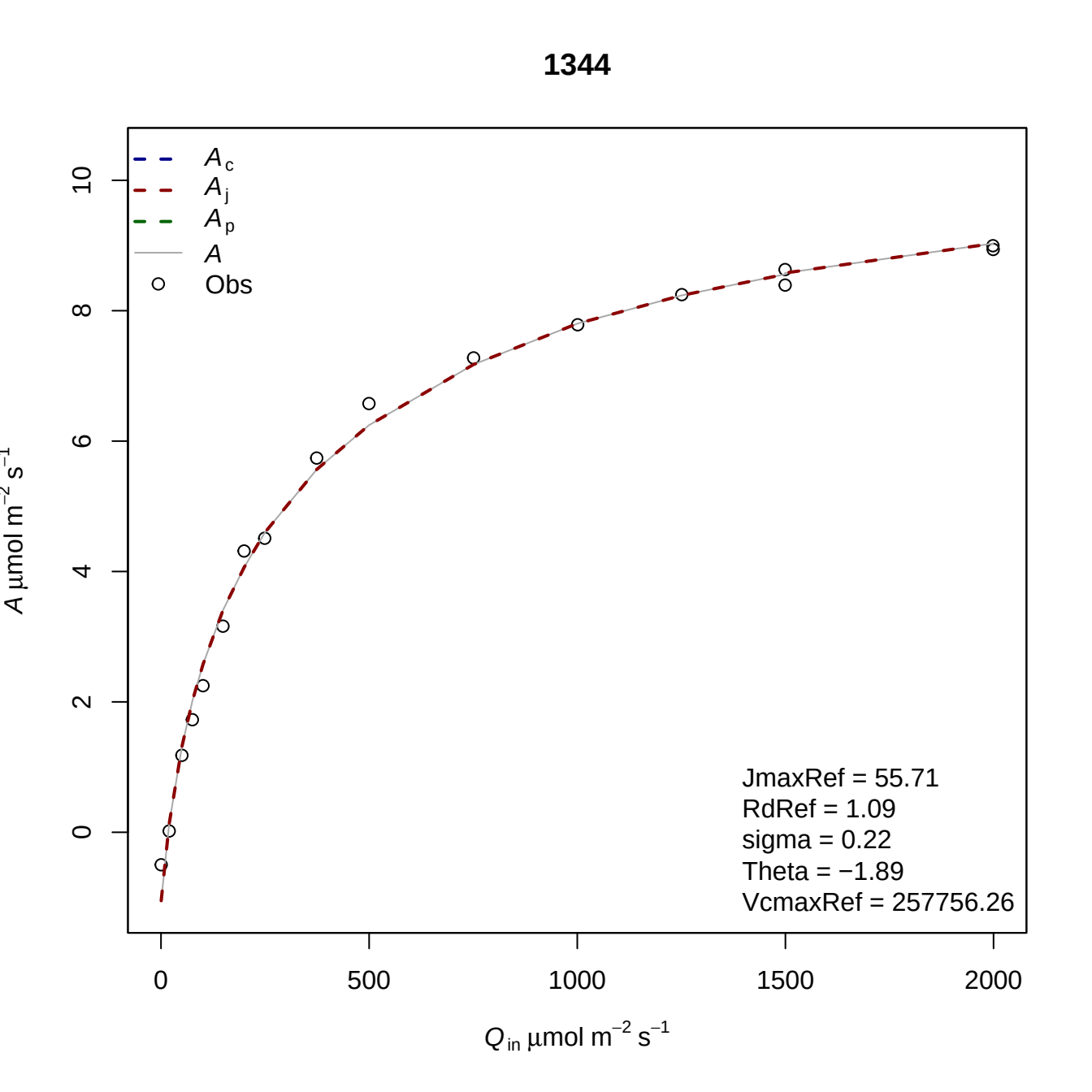

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 73.63
RdRef = 1.71
sigma = 0.33
Theta = −1.22
VcmaxRef = 890.83
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

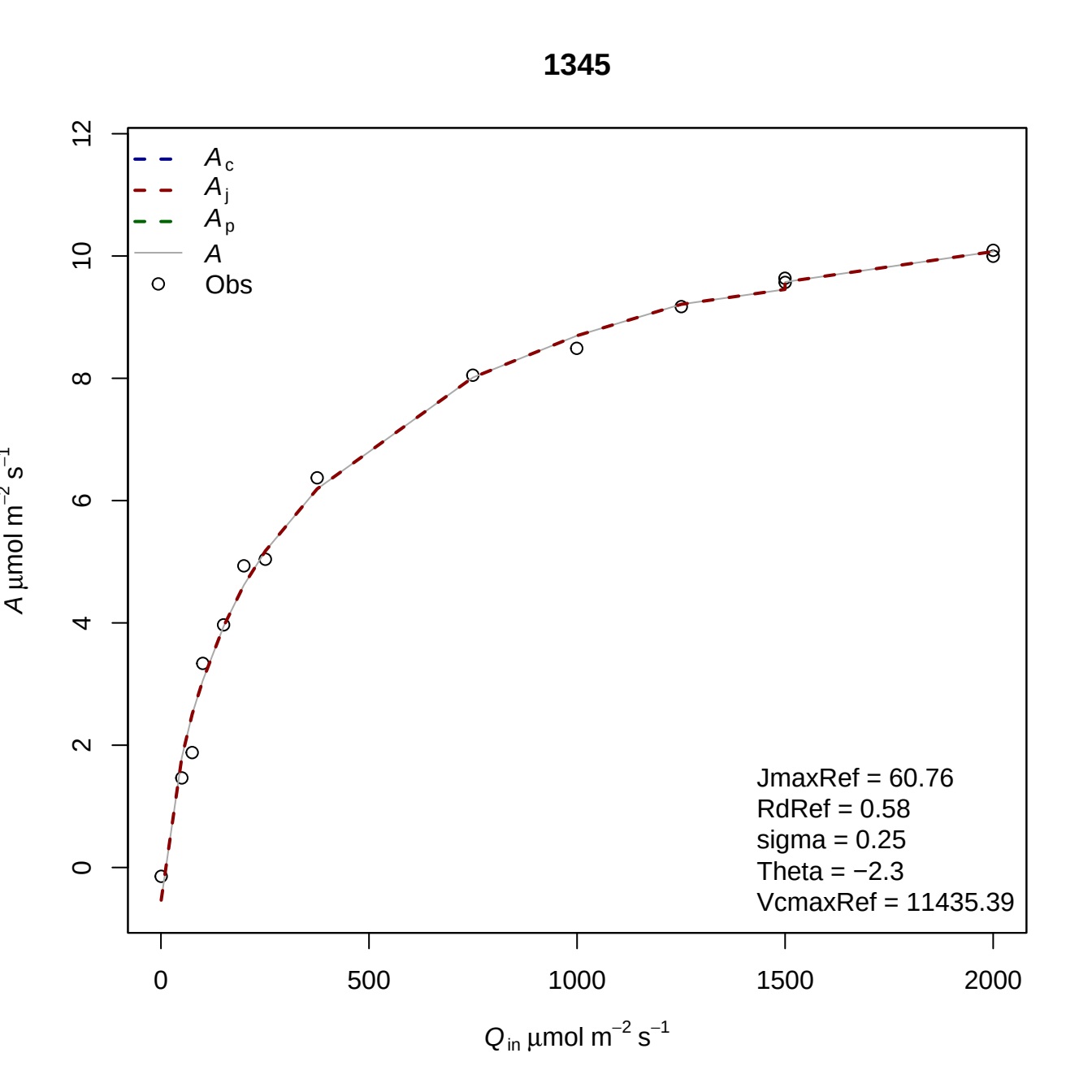

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 76.01
RdRef = 1.15
sigma = 0.45
Theta = −1.58
VcmaxRef = 17643.08
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

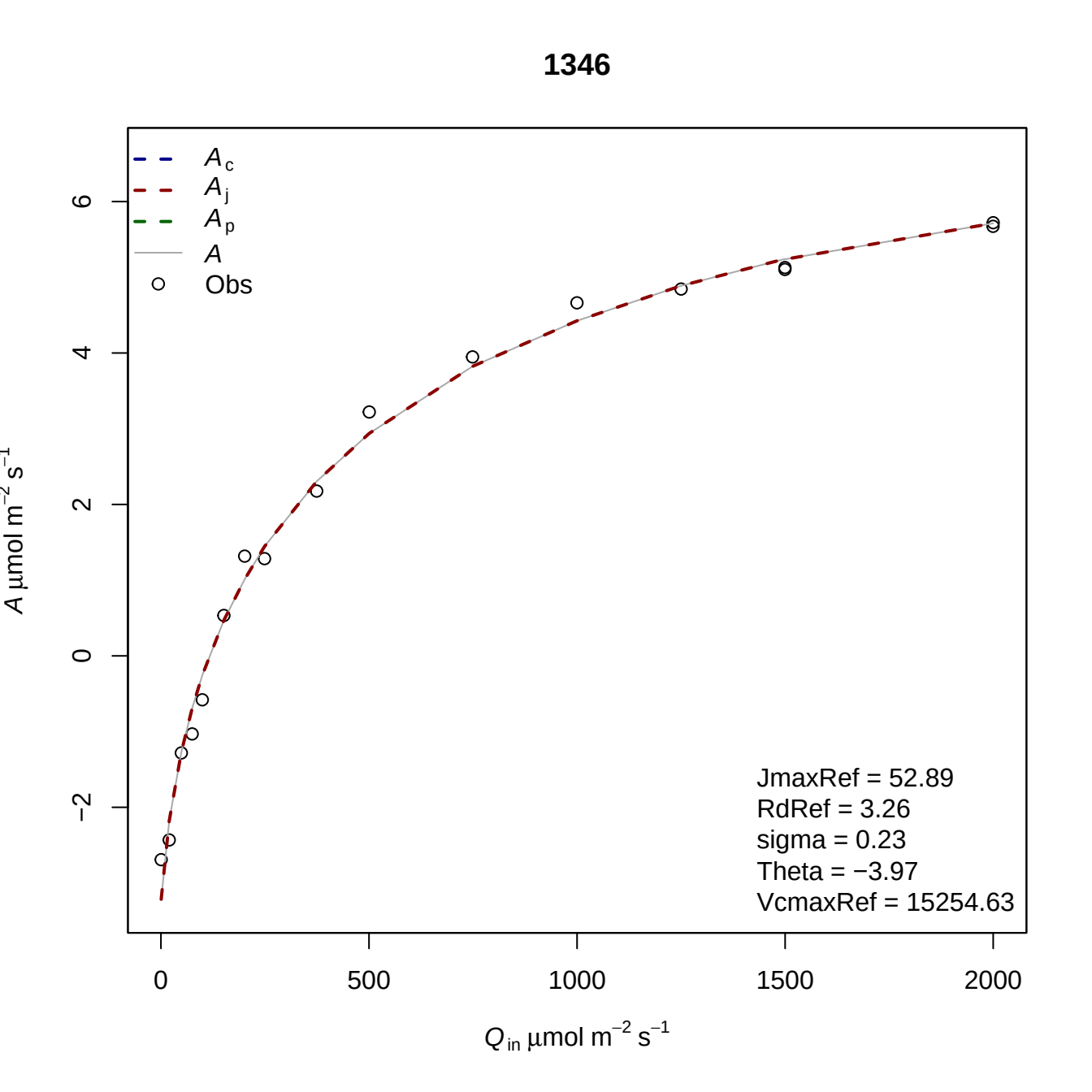

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 75.5
RdRef = 3.03
sigma = 0.45
Theta = −0.29
VcmaxRef = 7214.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

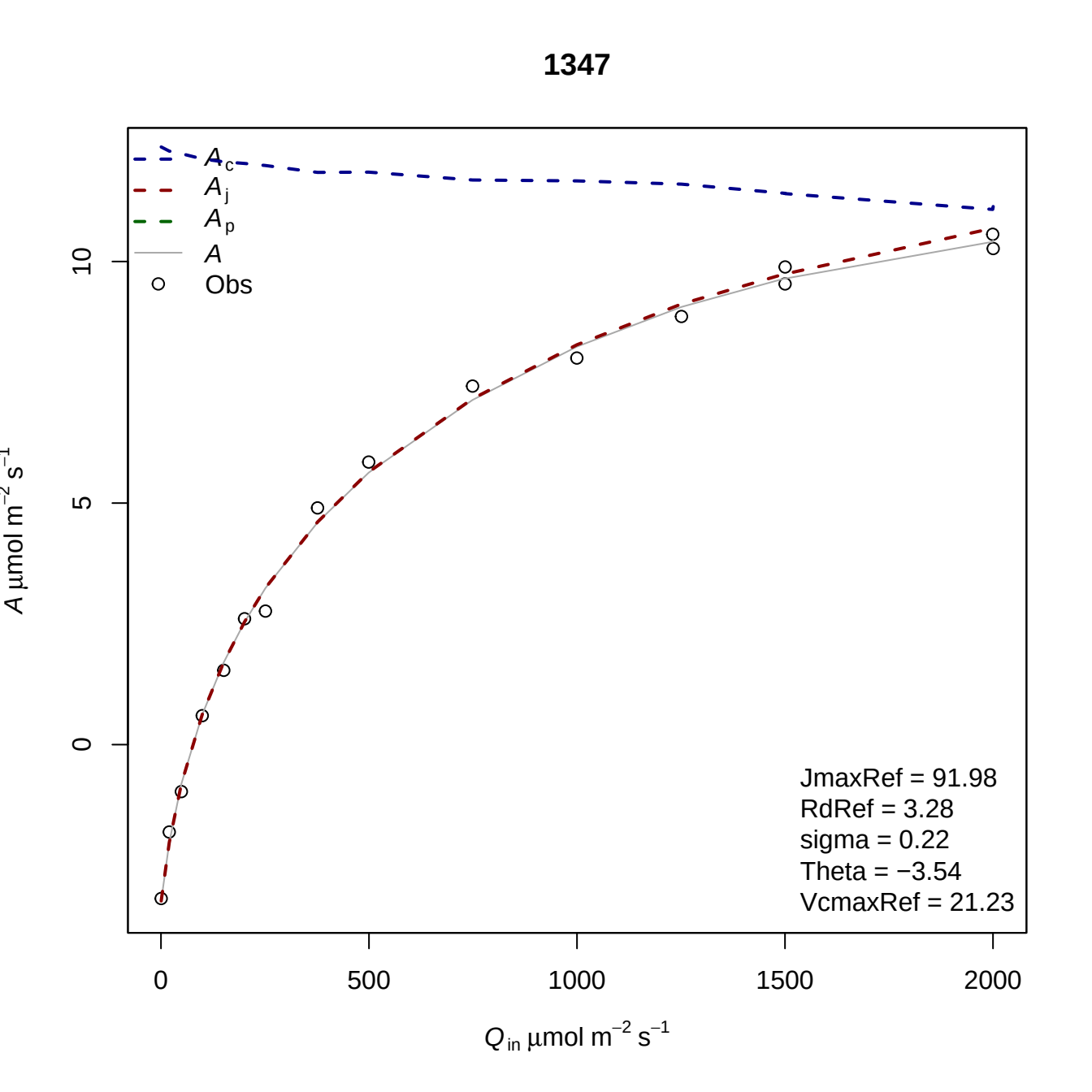

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 73.44
RdRef = 1.65
sigma = 0.47
Theta = −1.88
VcmaxRef = 21108.26
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

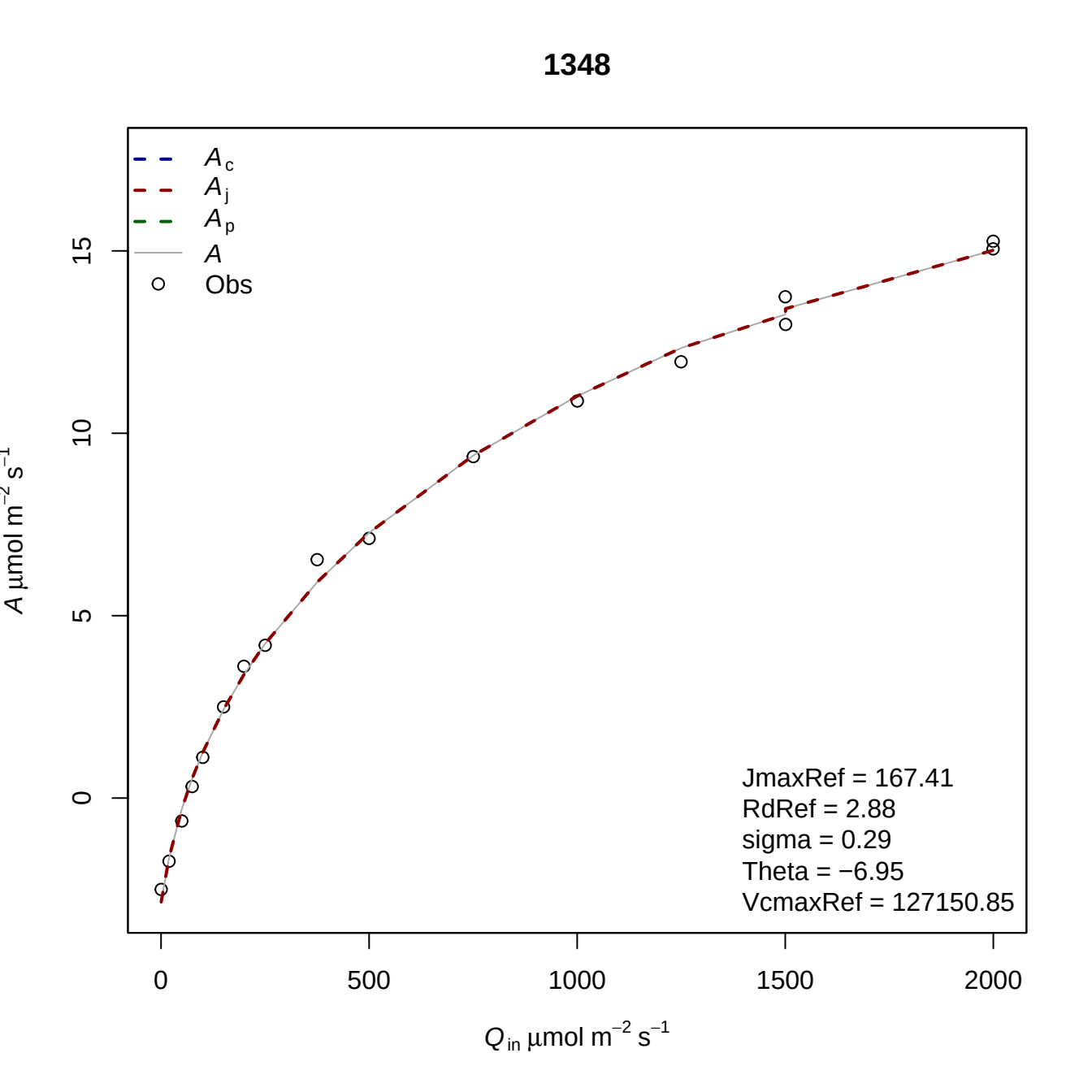

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 62.41
RdRef = 2.78
sigma = 0.24
Theta = −1.17
VcmaxRef = 1481979.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

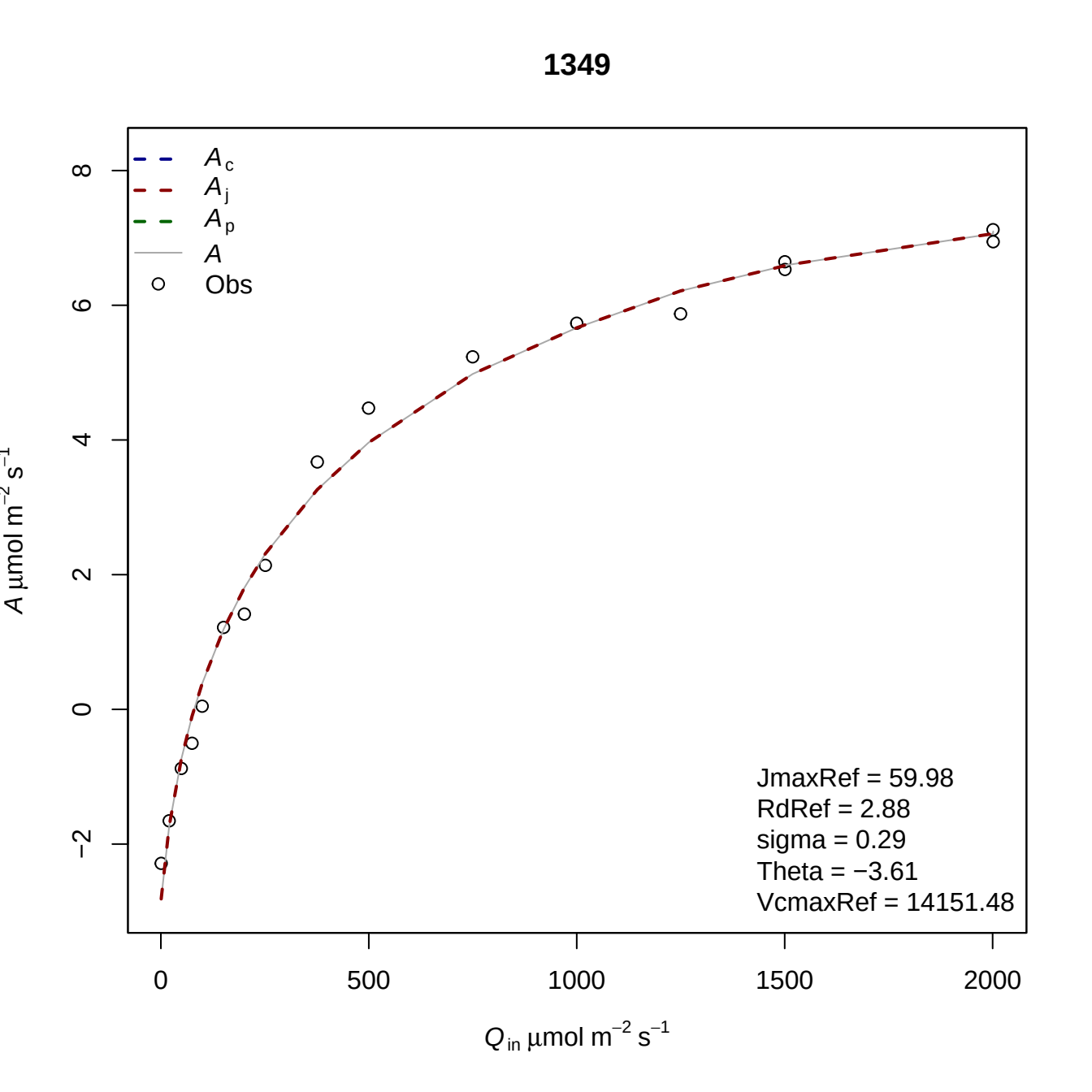

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 55.71
RdRef = 1.09
sigma = 0.22
Theta = −1.89
VcmaxRef = 257756.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

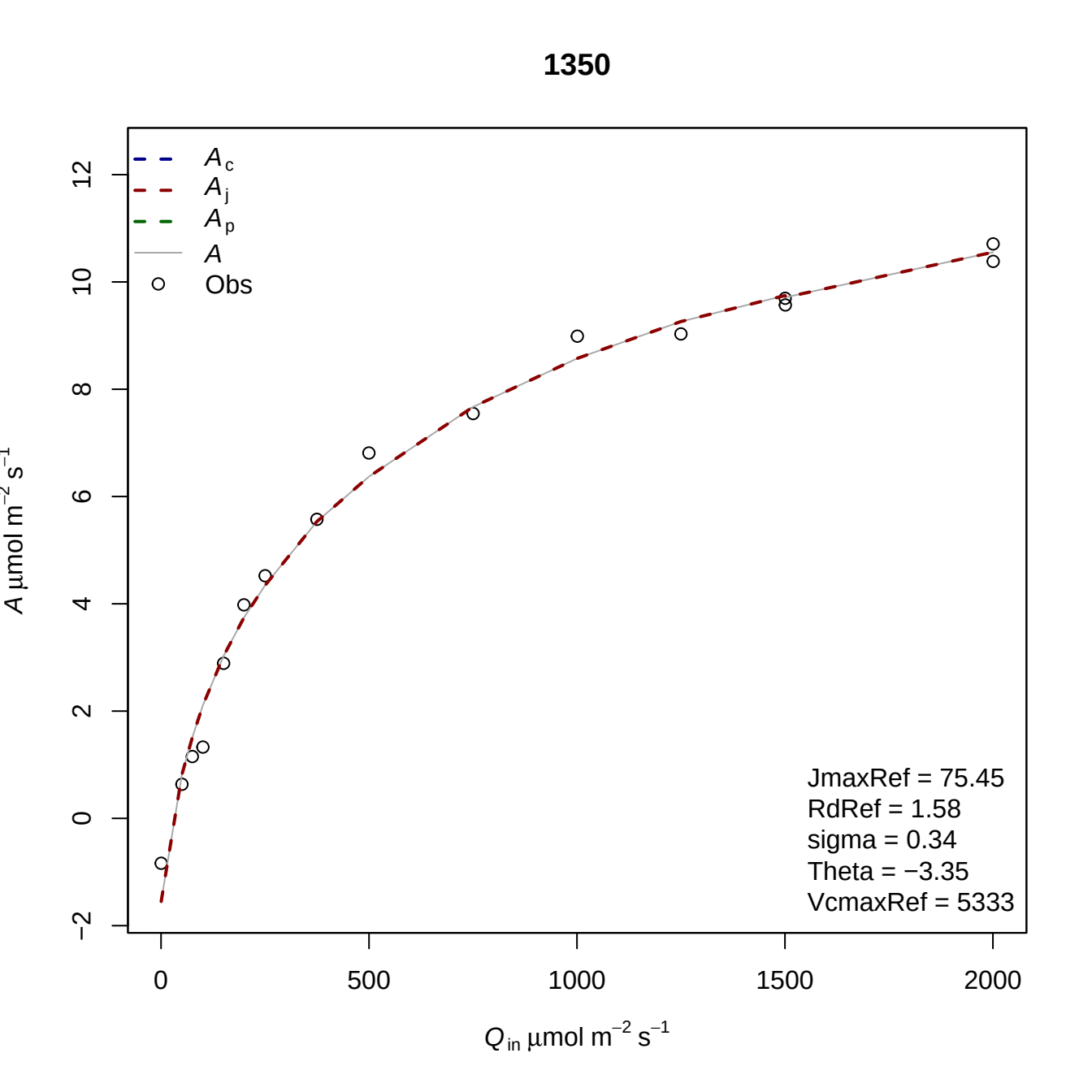

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 60.76
RdRef = 0.58
sigma = 0.25
Theta = −2.3
VcmaxRef = 11435.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

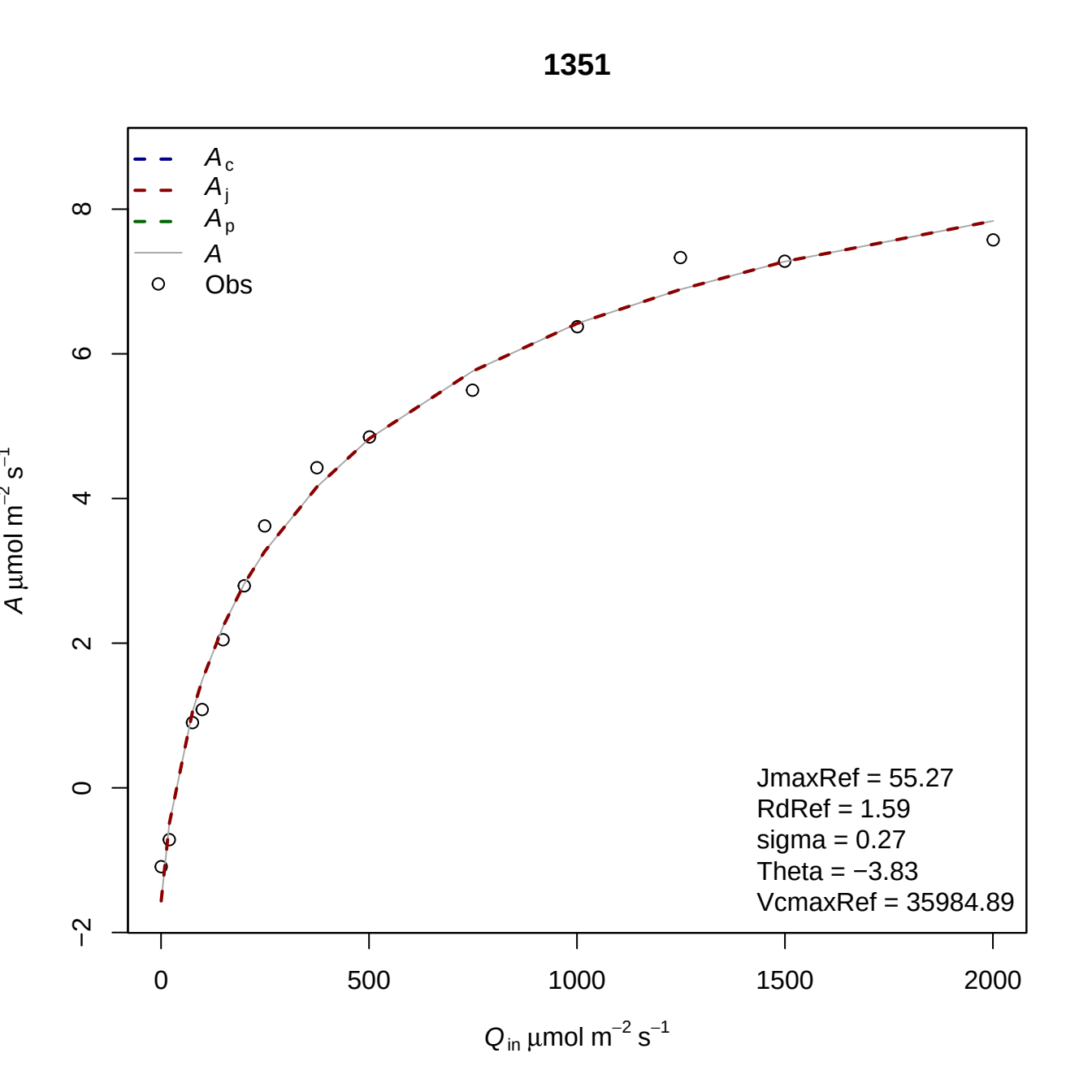

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 52.89
RdRef = 3.26
sigma = 0.23
Theta = −3.97
VcmaxRef = 15254.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

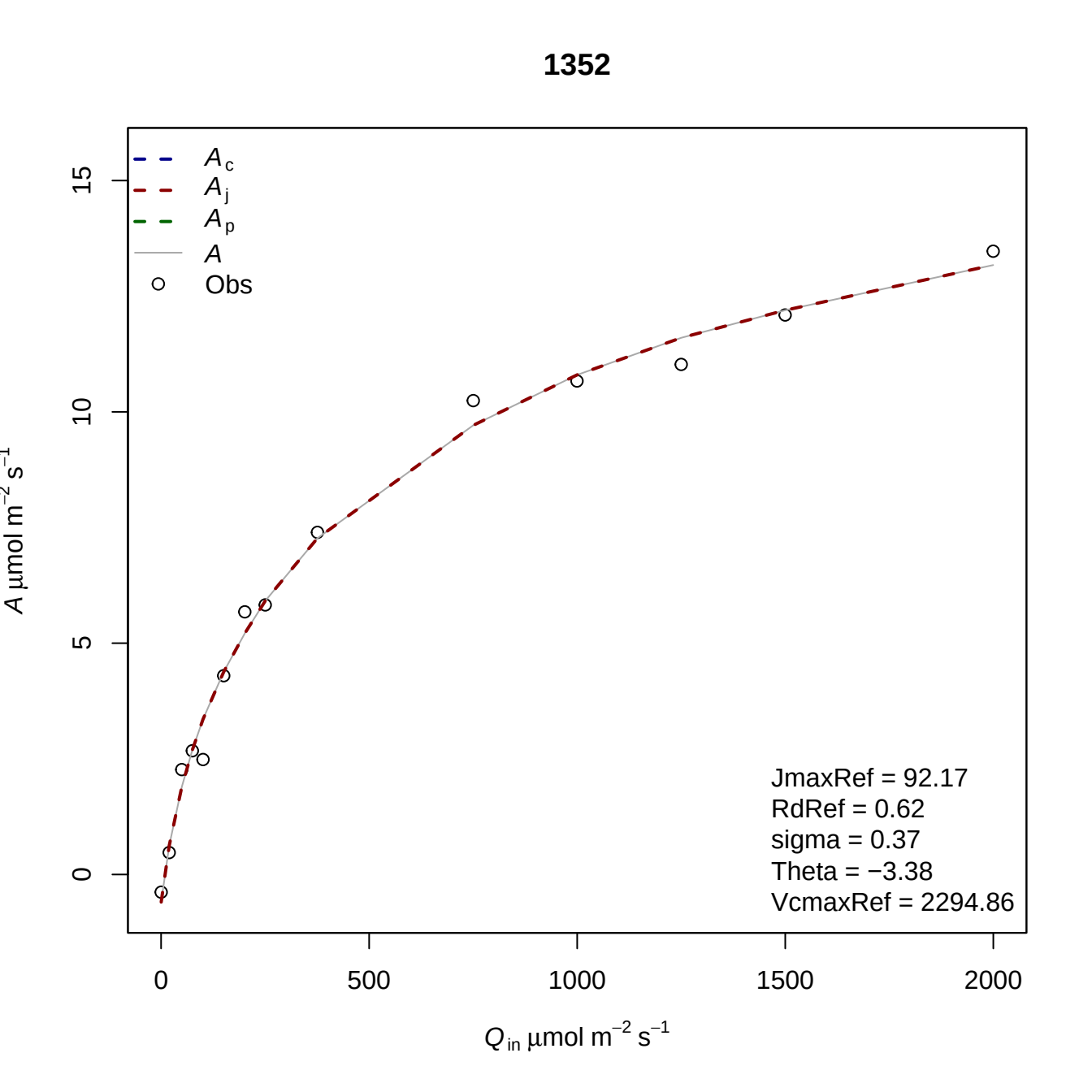

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 91.98
RdRef = 3.28
sigma = 0.22
Theta = −3.54
VcmaxRef = 21.23
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

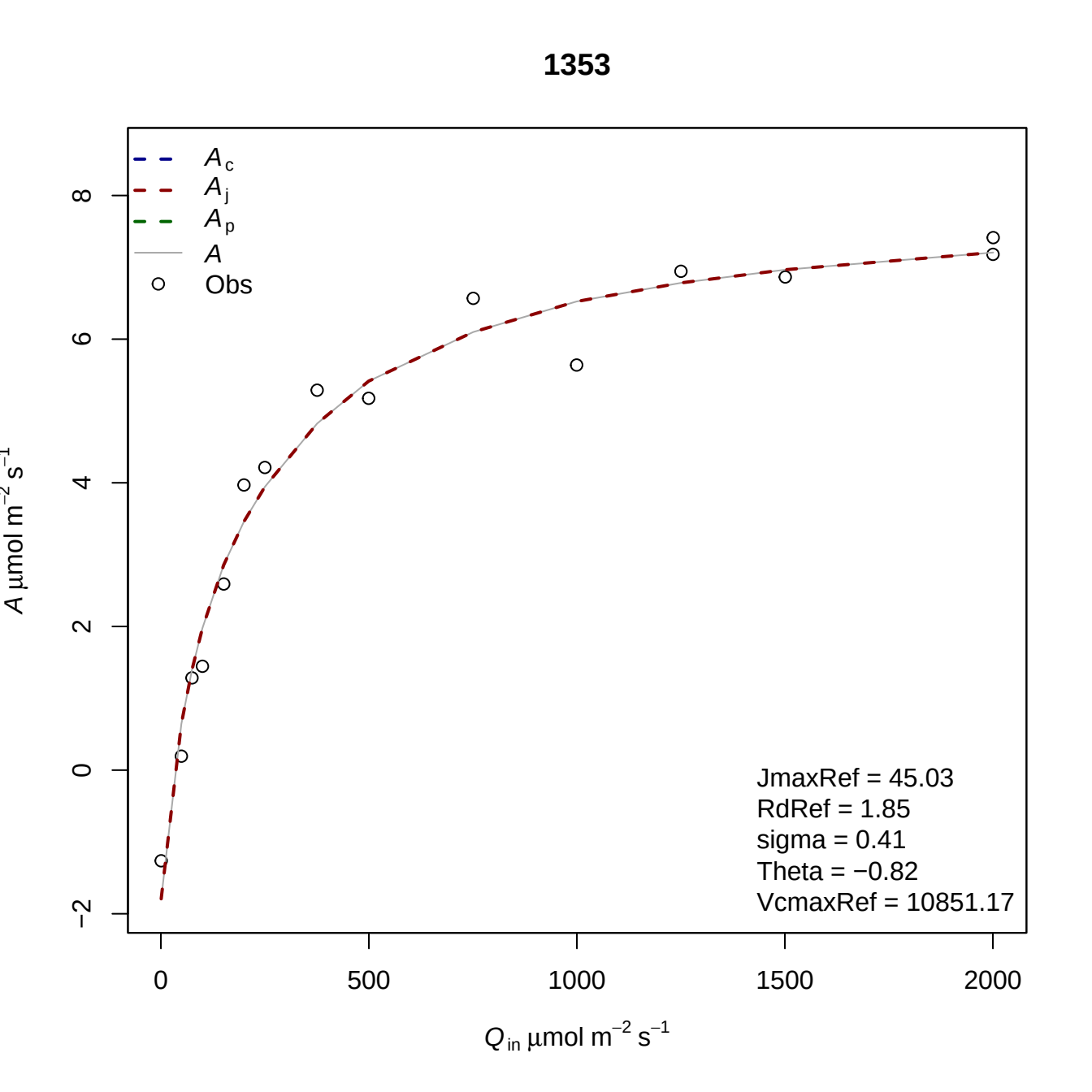

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 167.41
RdRef = 2.88
sigma = 0.29
Theta = −6.95
VcmaxRef = 127150.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

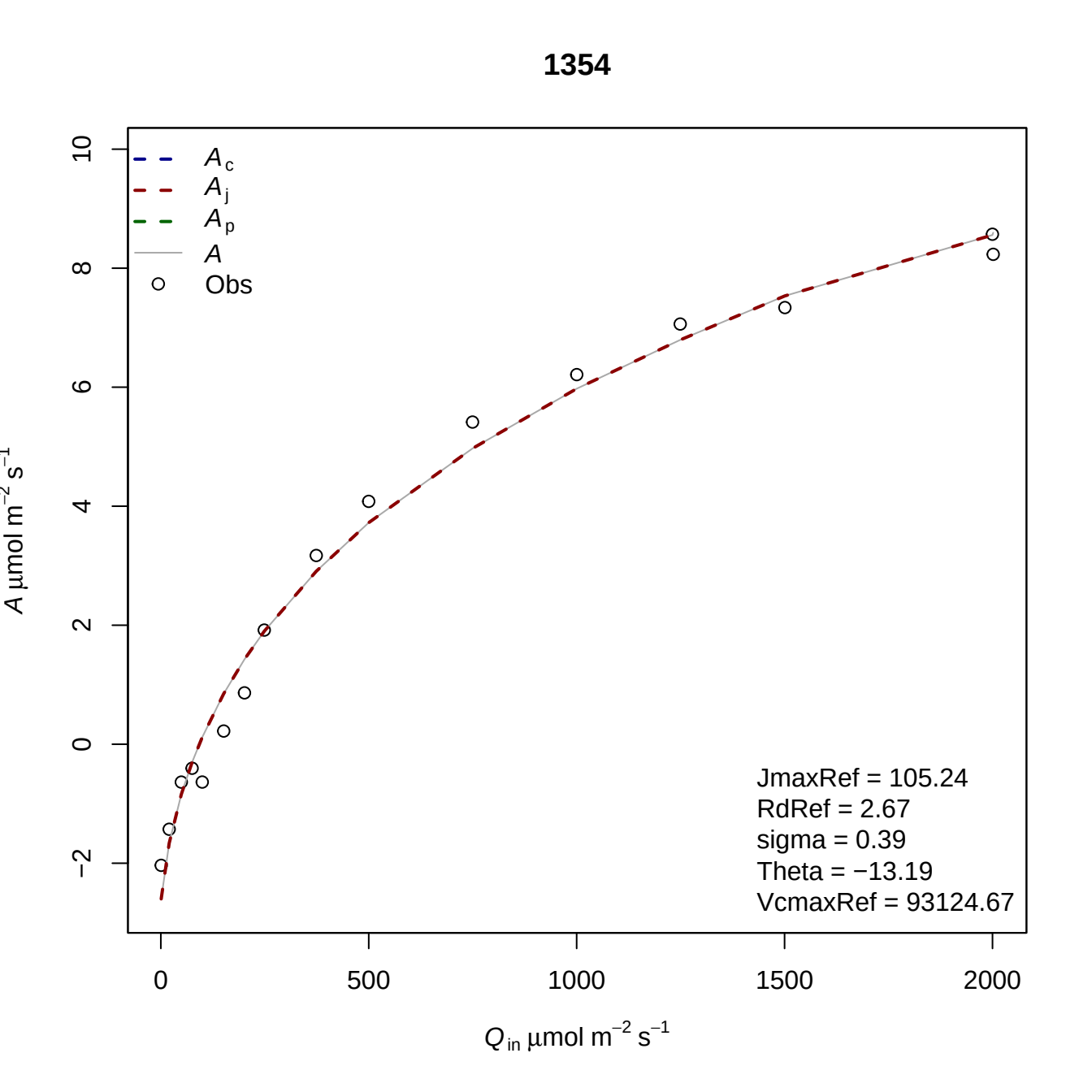

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 59.98
RdRef = 2.88
sigma = 0.29
Theta = −3.61
VcmaxRef = 14151.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

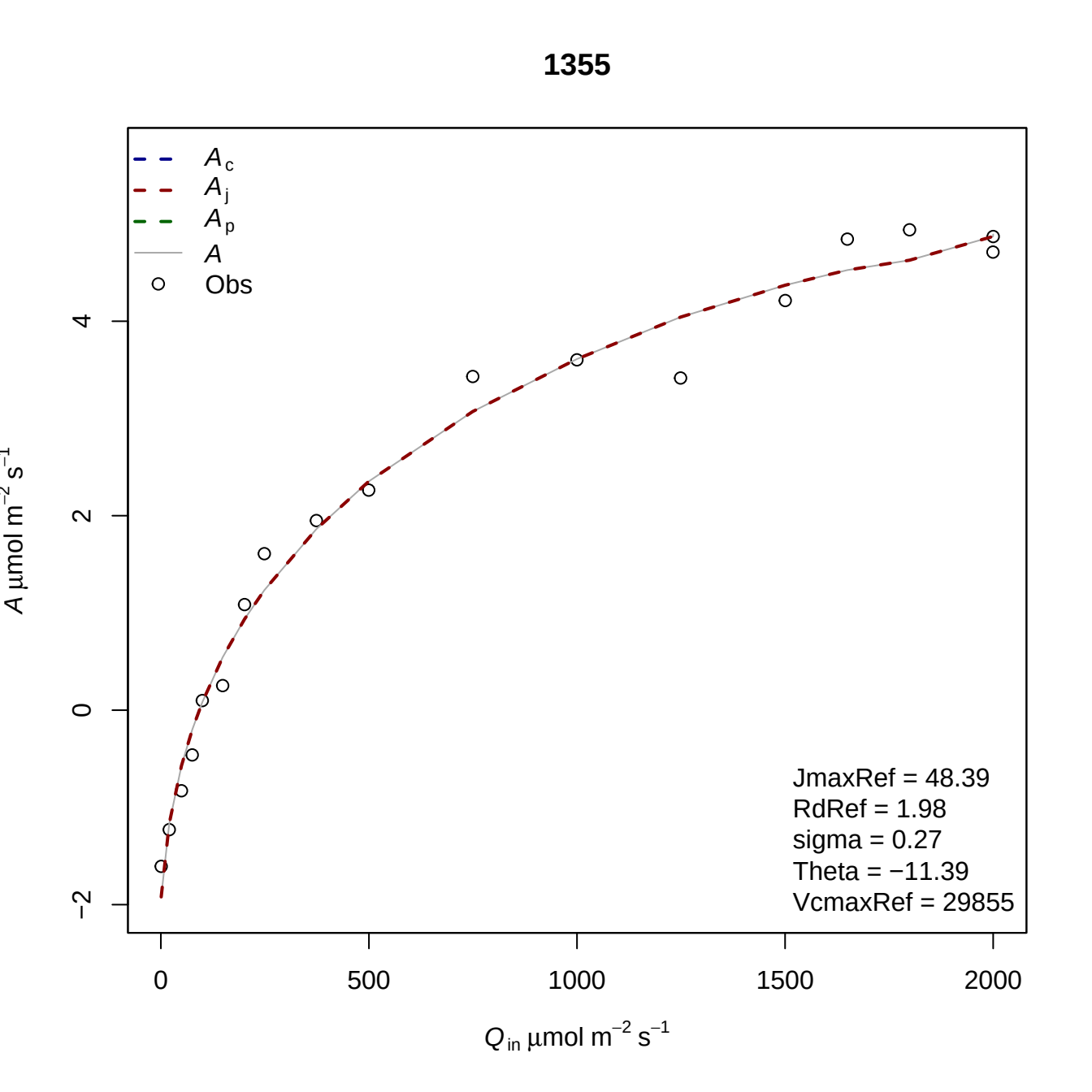

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 75.45
RdRef = 1.58
sigma = 0.34
Theta = −3.35
VcmaxRef = 5333
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500
2000

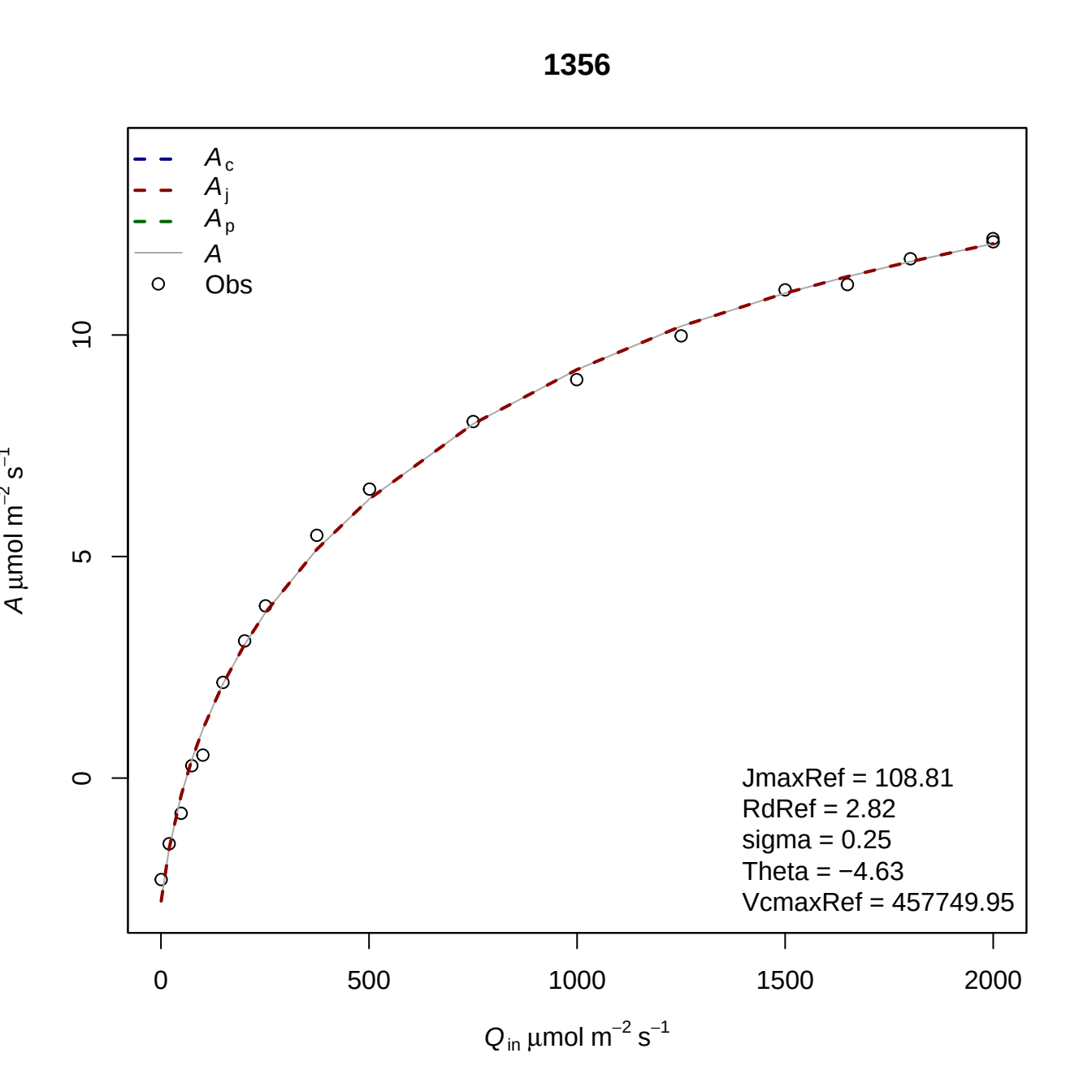

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 55.27
RdRef = 1.59
sigma = 0.27
Theta = −3.83
VcmaxRef = 35984.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

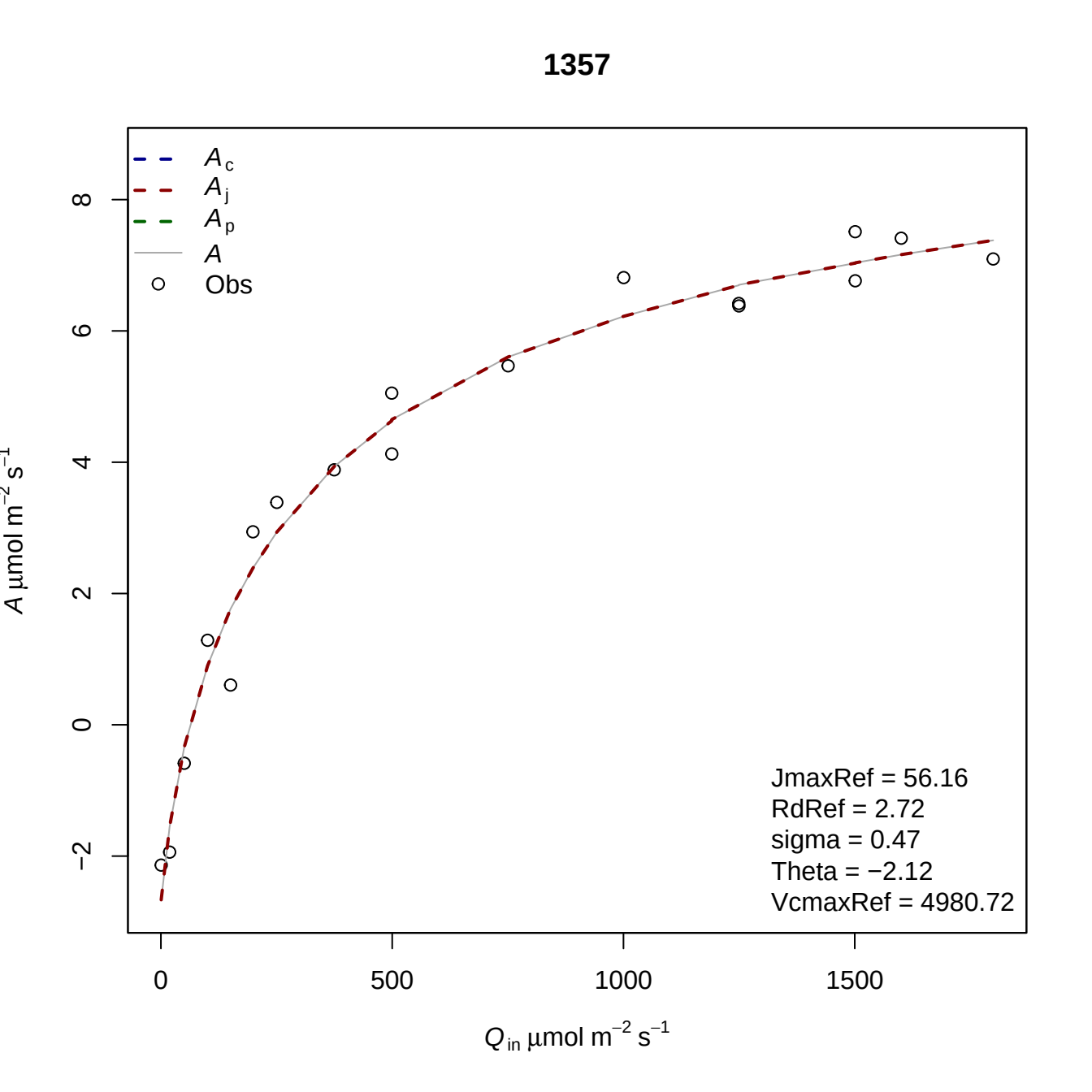

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 92.17
RdRef = 0.62
sigma = 0.37
Theta = −3.38
VcmaxRef = 2294.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

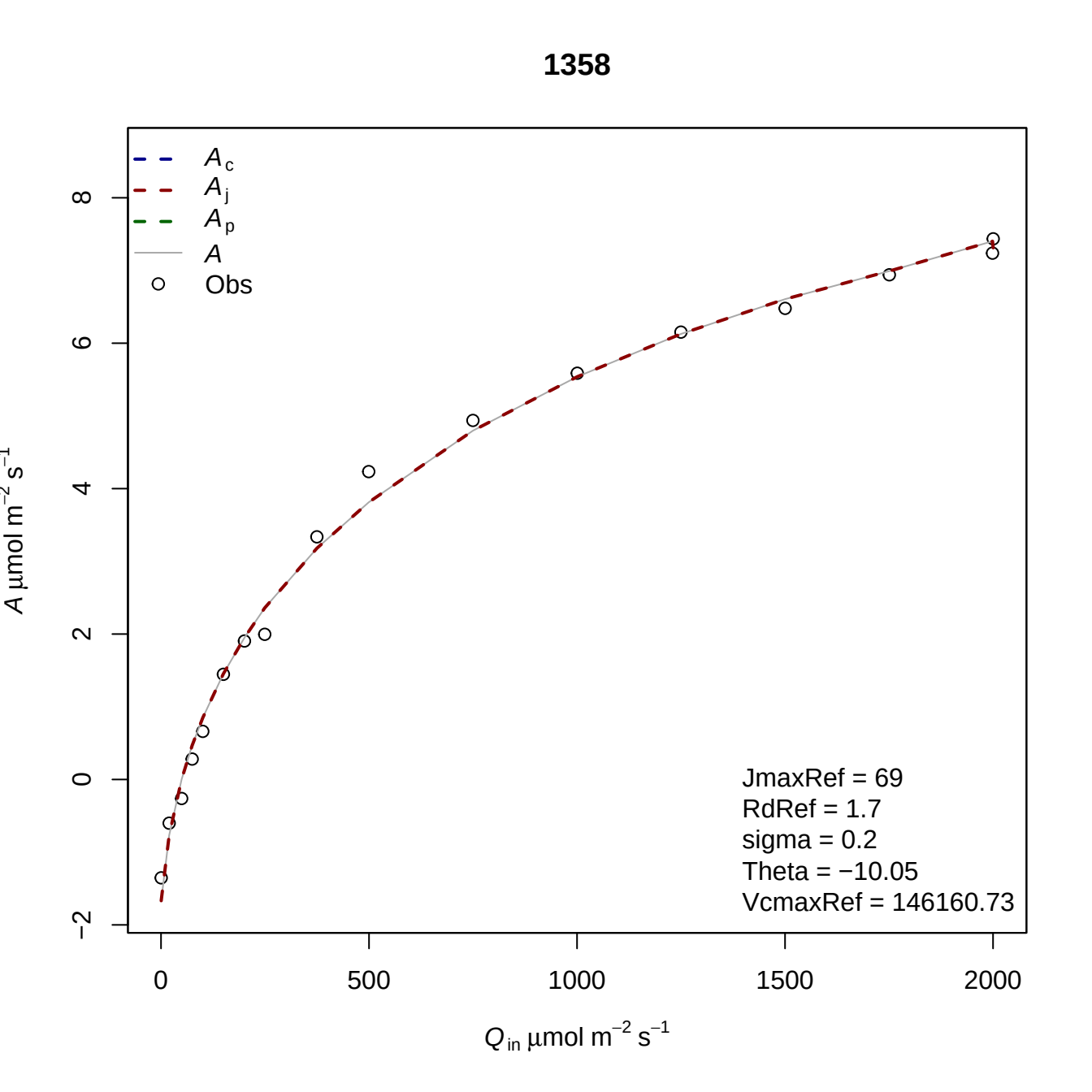

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 45.03
RdRef = 1.85
sigma = 0.41
Theta = −0.82
VcmaxRef = 10851.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

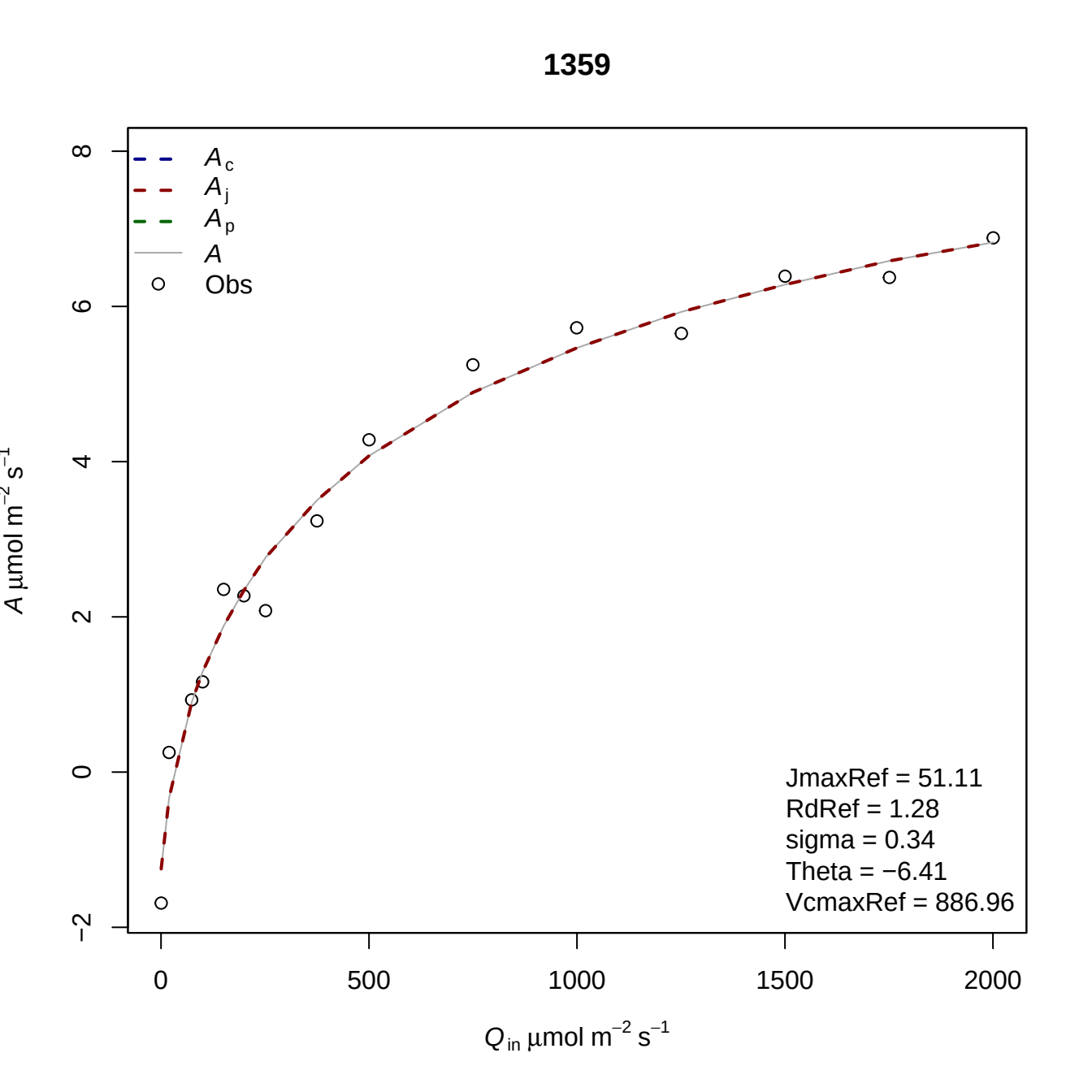

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 105.24
RdRef = 2.67
sigma = 0.39
Theta = −13.19
VcmaxRef = 93124.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

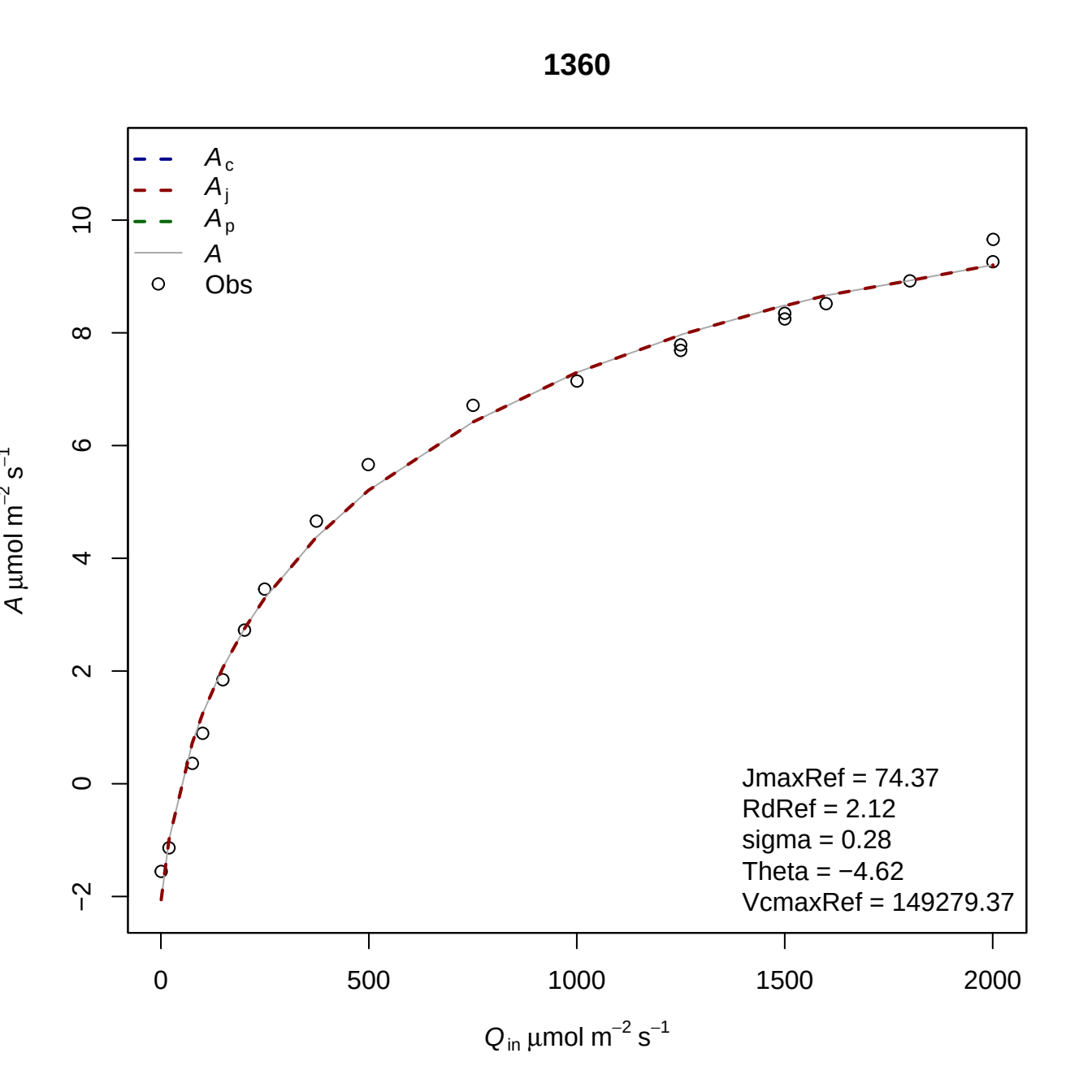

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 1.98
sigma = 0.27
Theta = −11.39
VcmaxRef = 29855
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

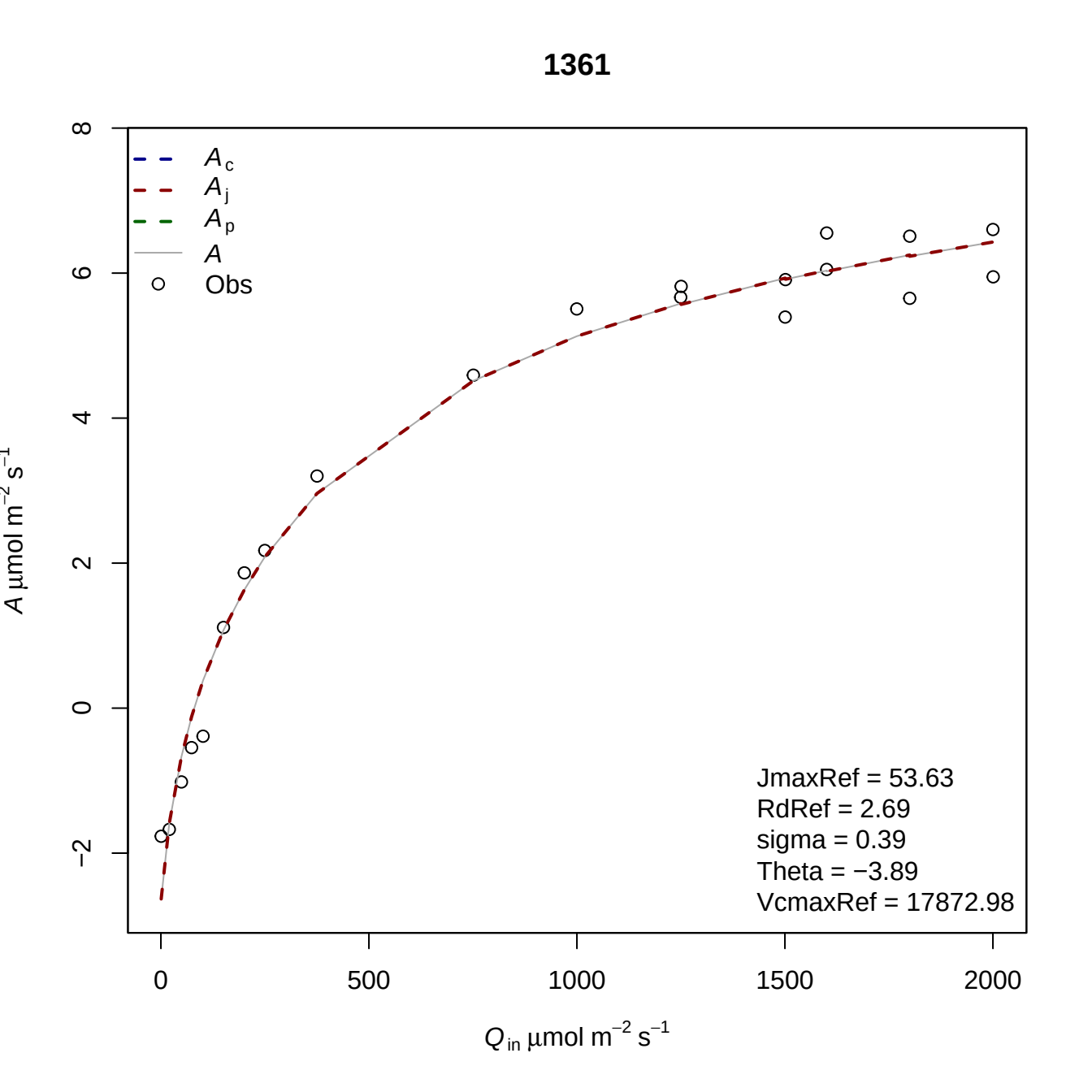

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 108.81
RdRef = 2.82
sigma = 0.25
Theta = −4.63
VcmaxRef = 457749.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

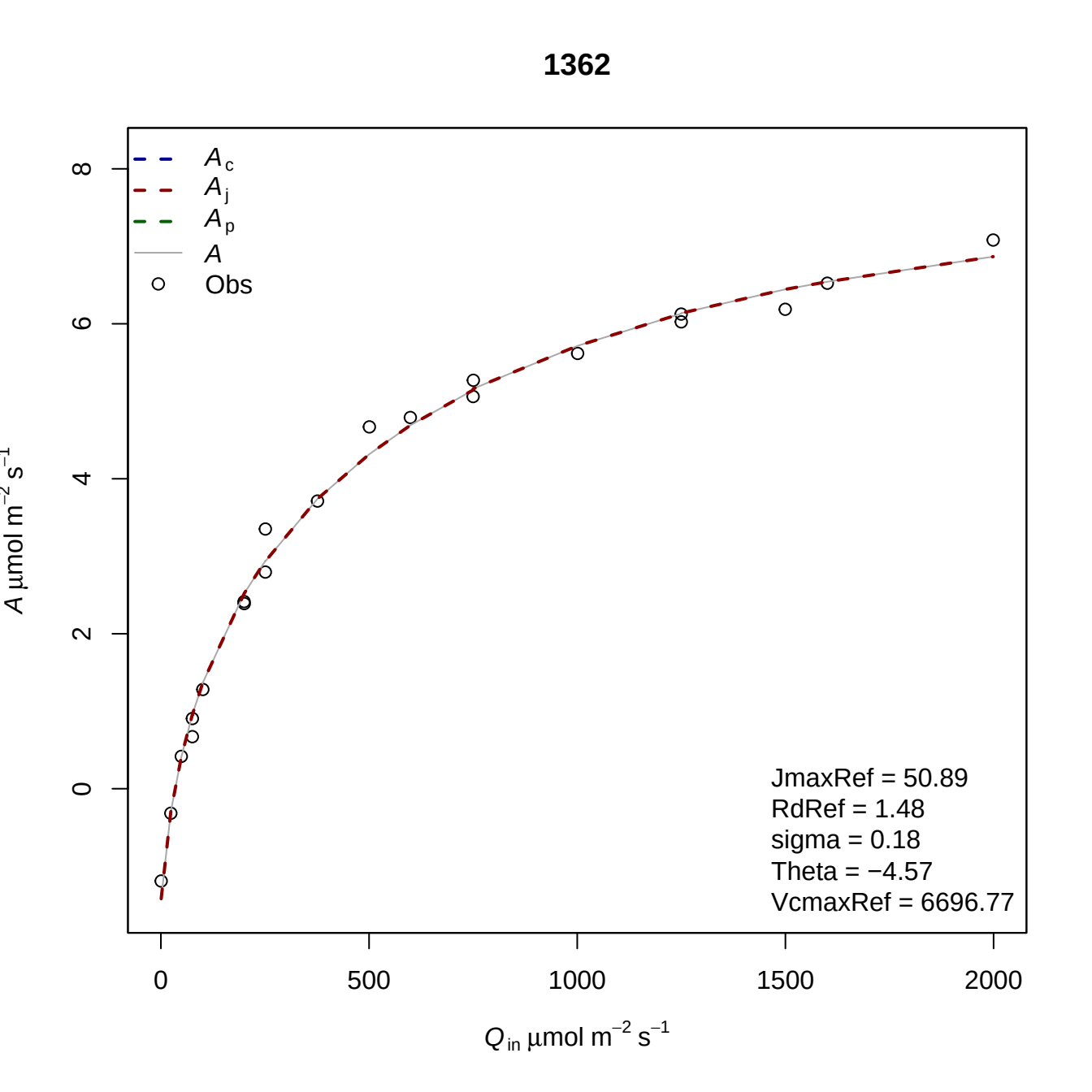

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 56.16
RdRef = 2.72
sigma = 0.47
Theta = −2.12
VcmaxRef = 4980.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

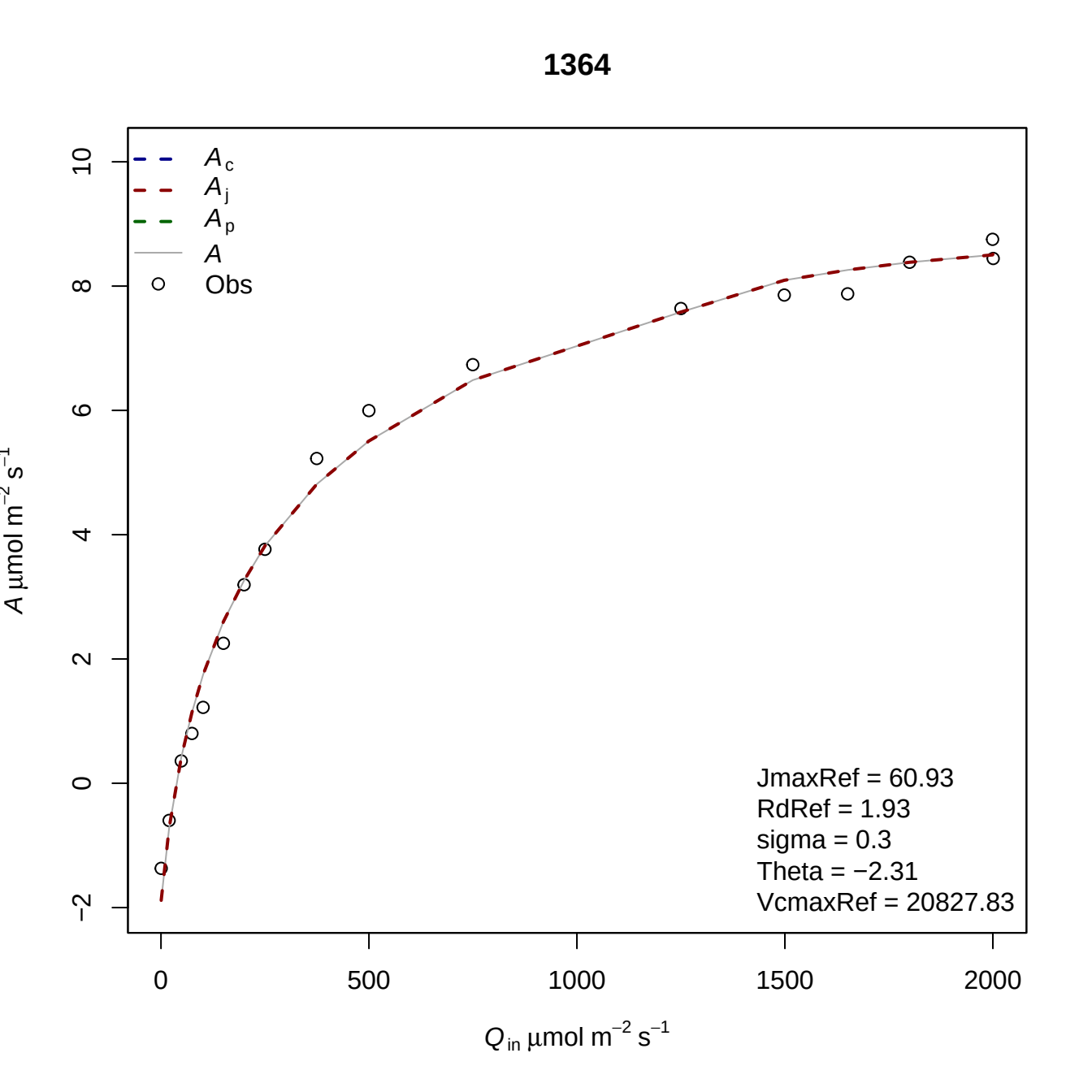

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 69
RdRef = 1.7
sigma = 0.2
Theta = −10.05
VcmaxRef = 146160.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

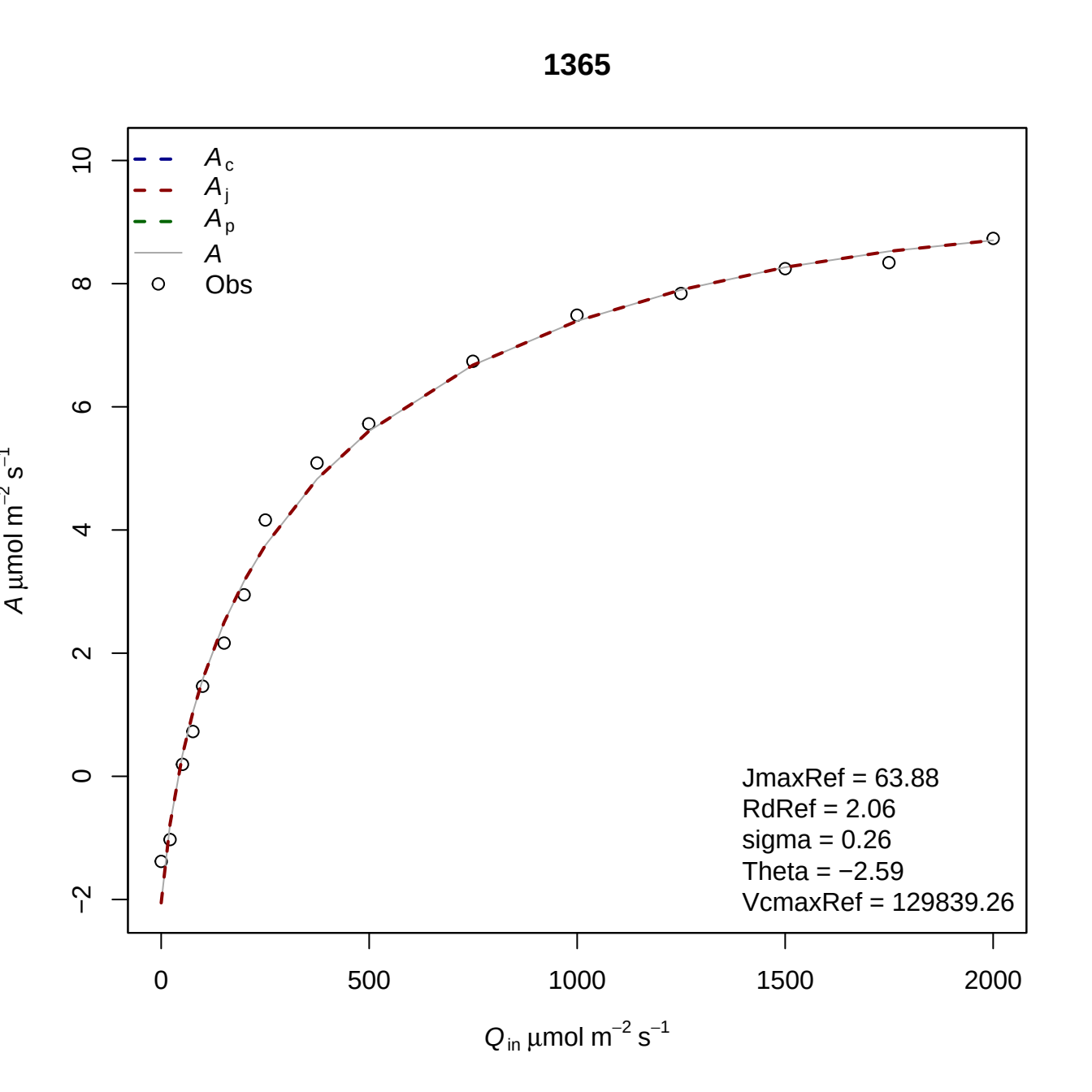

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 51.11
RdRef = 1.28
sigma = 0.34
Theta = −6.41
VcmaxRef = 886.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

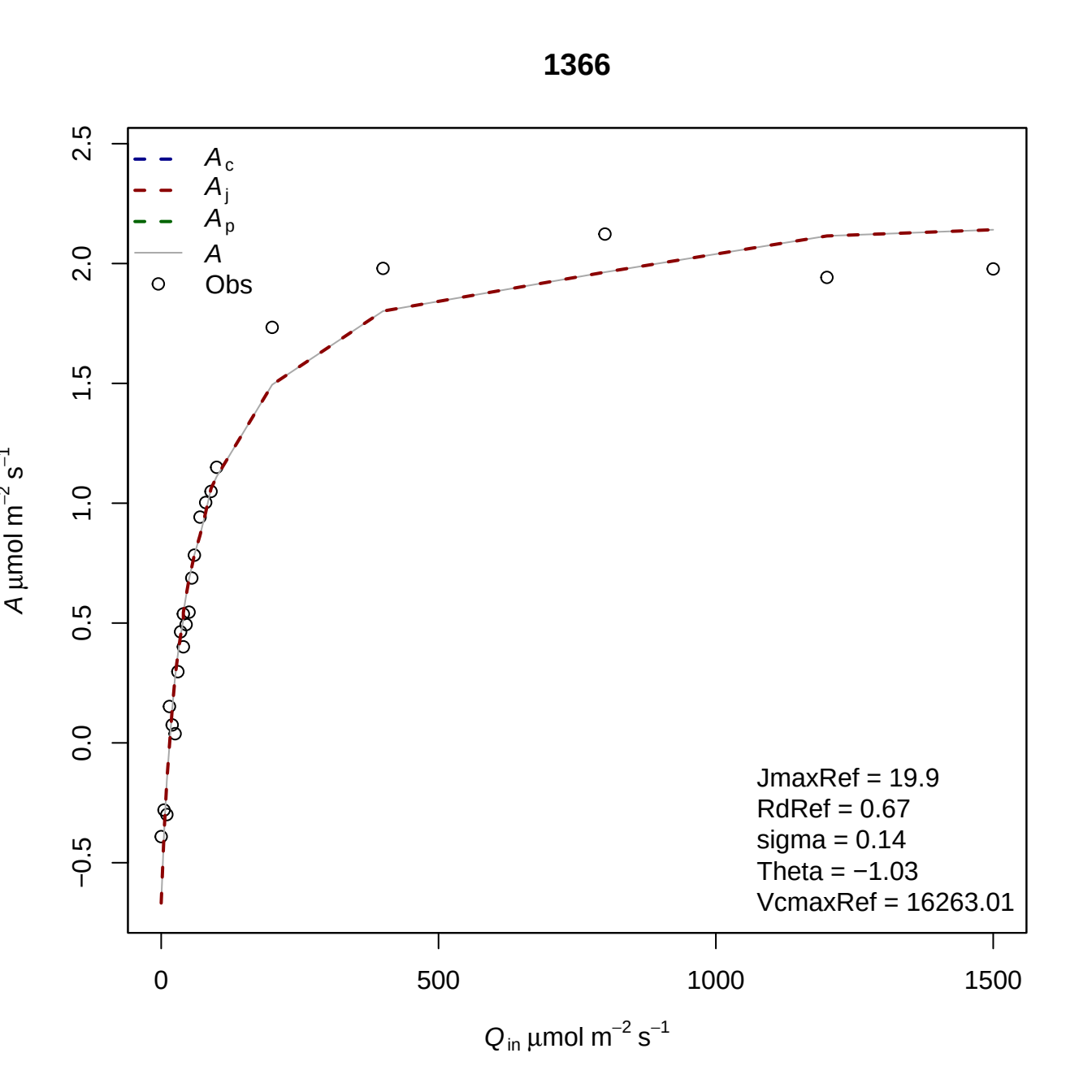

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 74.37
RdRef = 2.12
sigma = 0.28
Theta = −4.62
VcmaxRef = 149279.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

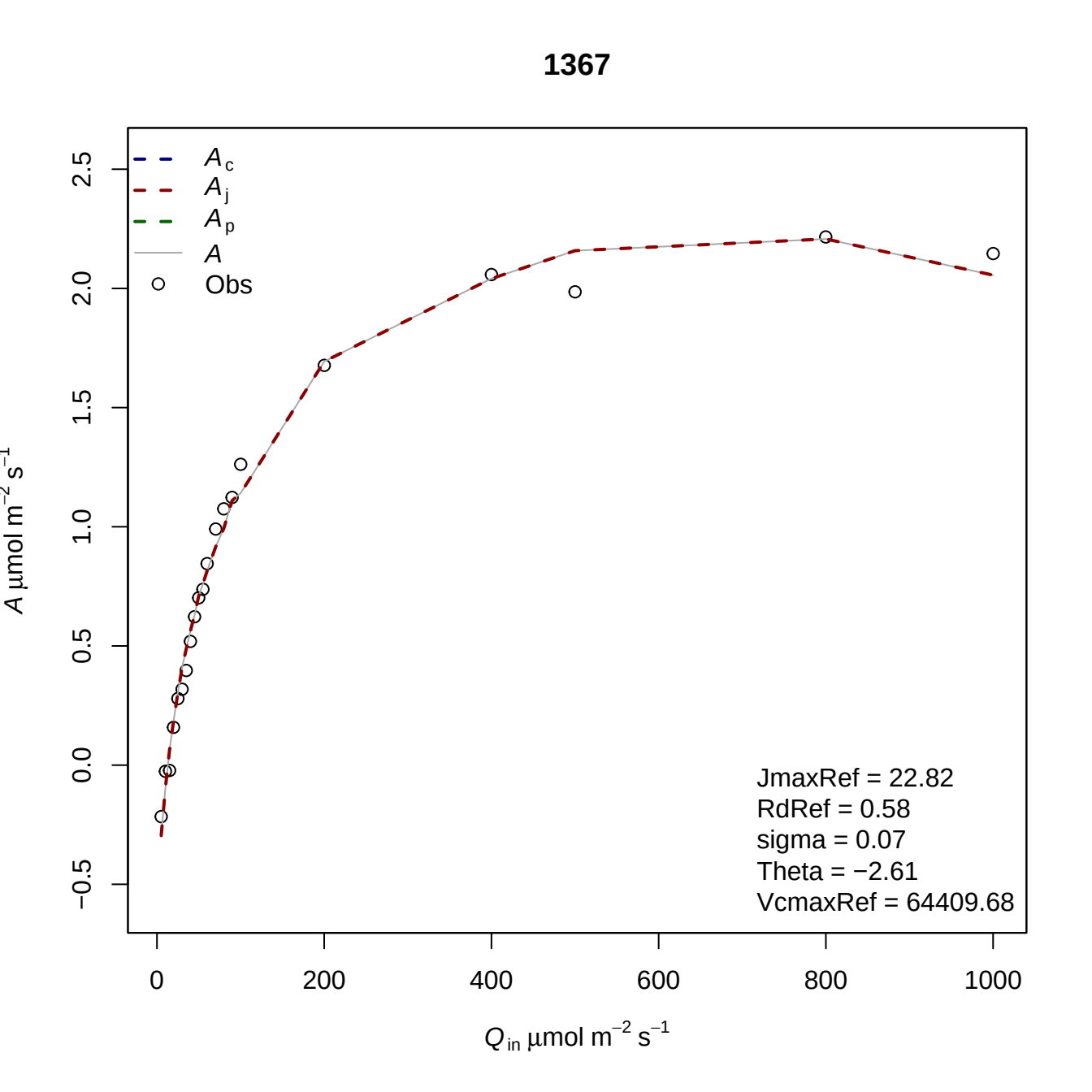

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 53.63
RdRef = 2.69
sigma = 0.39
Theta = −3.89
VcmaxRef = 17872.98
8
6
4
2
0
−2
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

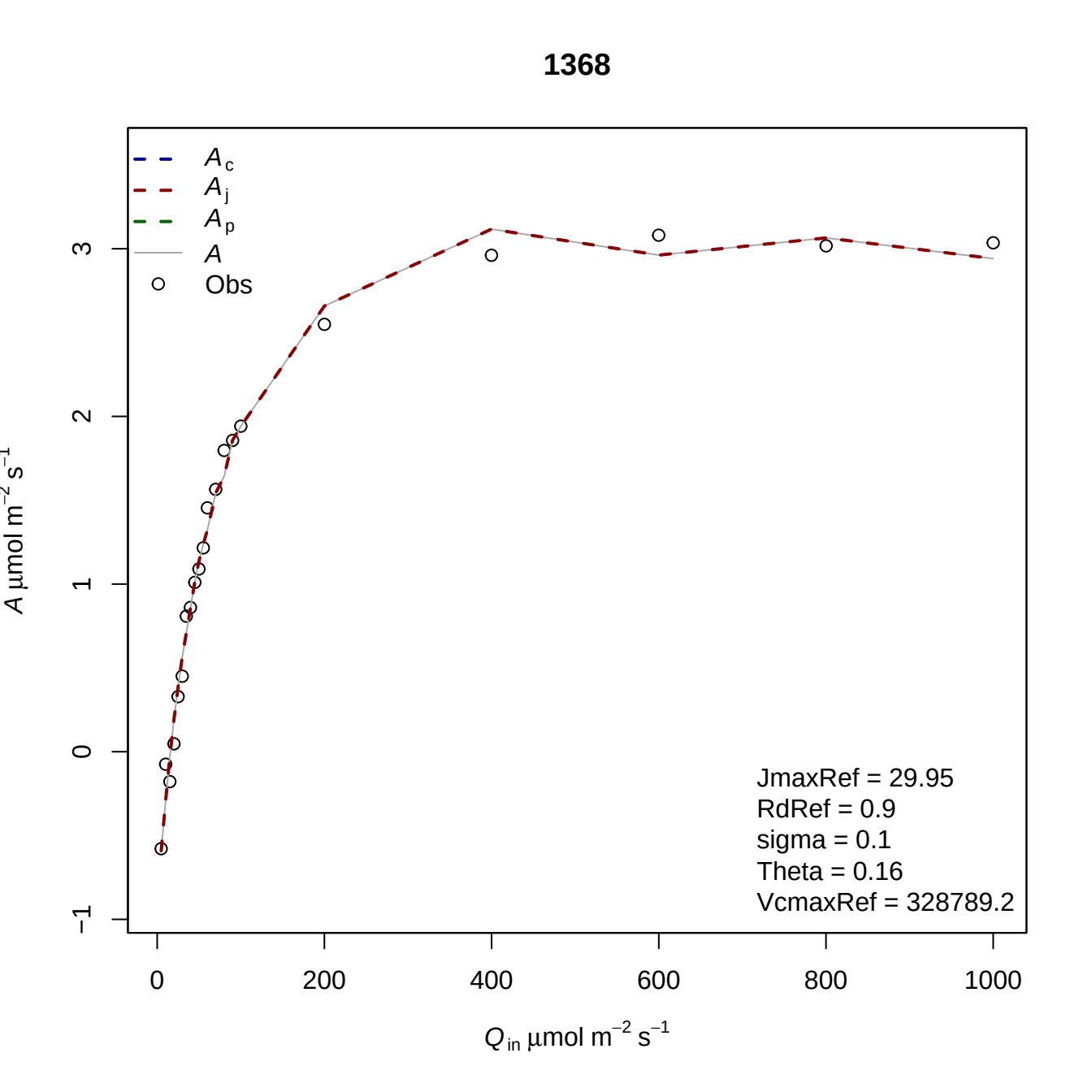

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 50.89
RdRef = 1.48
sigma = 0.18
Theta = −4.57
VcmaxRef = 6696.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

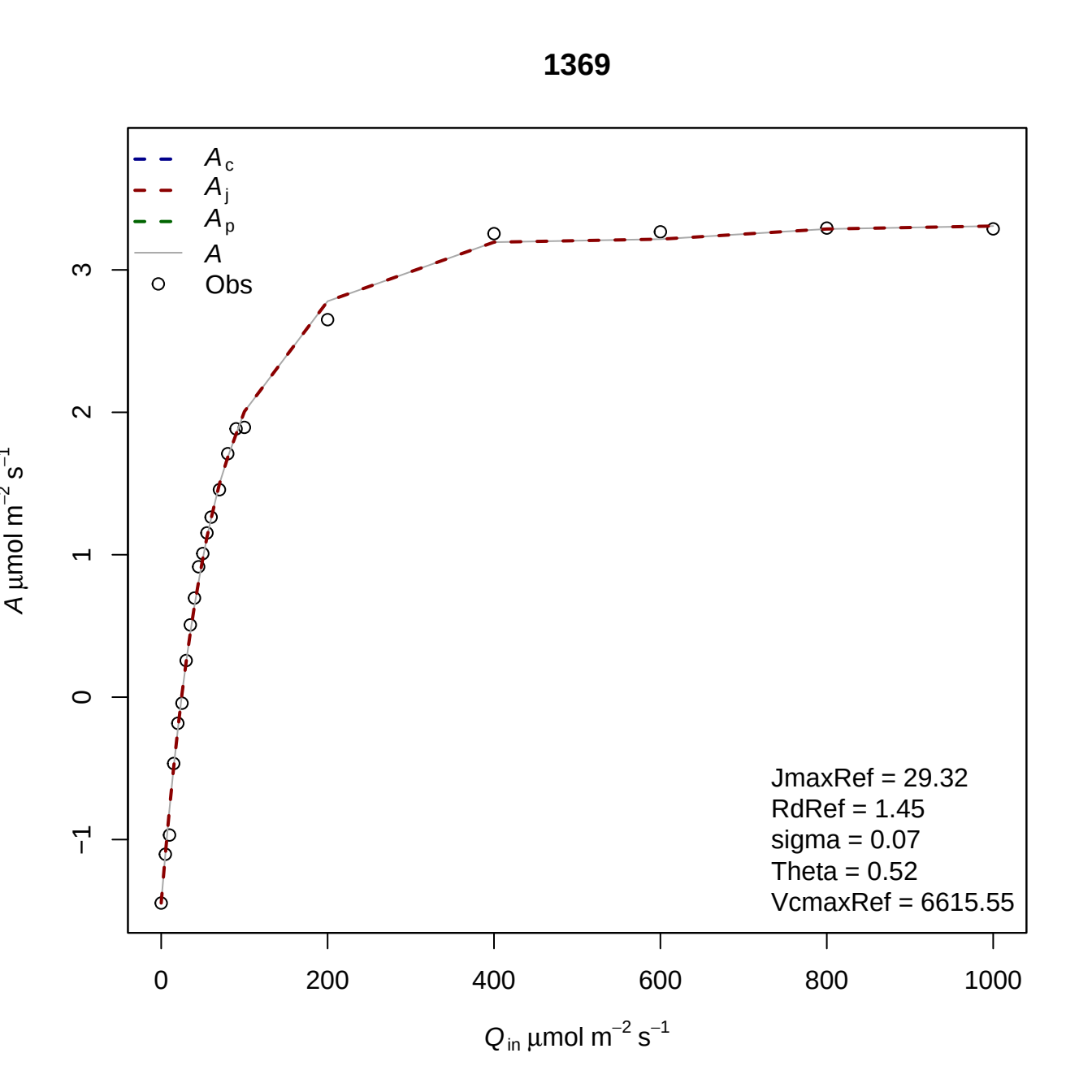

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 60.93
RdRef = 1.93
sigma = 0.3
Theta = −2.31
VcmaxRef = 20827.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

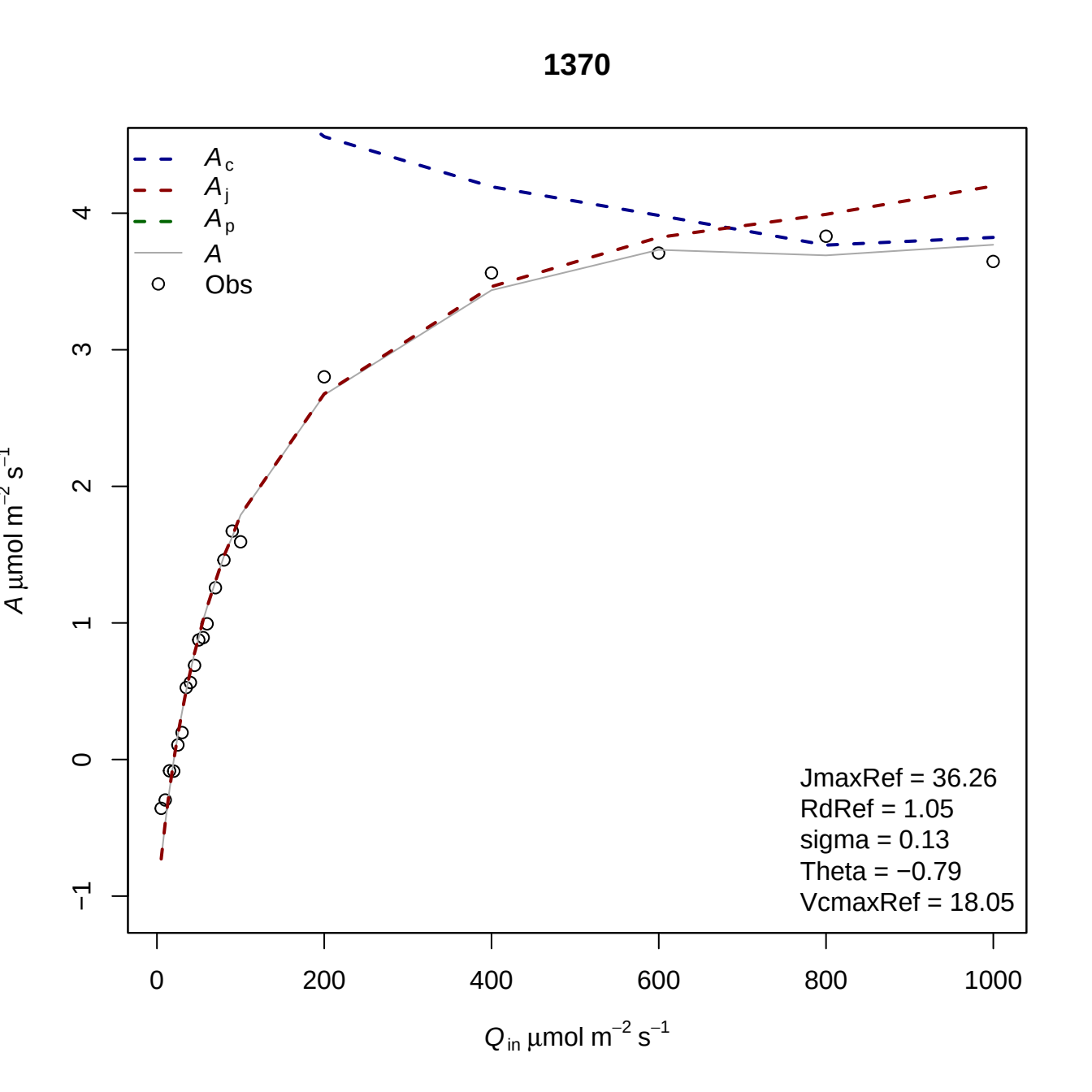

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 63.88
RdRef = 2.06
sigma = 0.26
Theta = −2.59
VcmaxRef = 129839.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

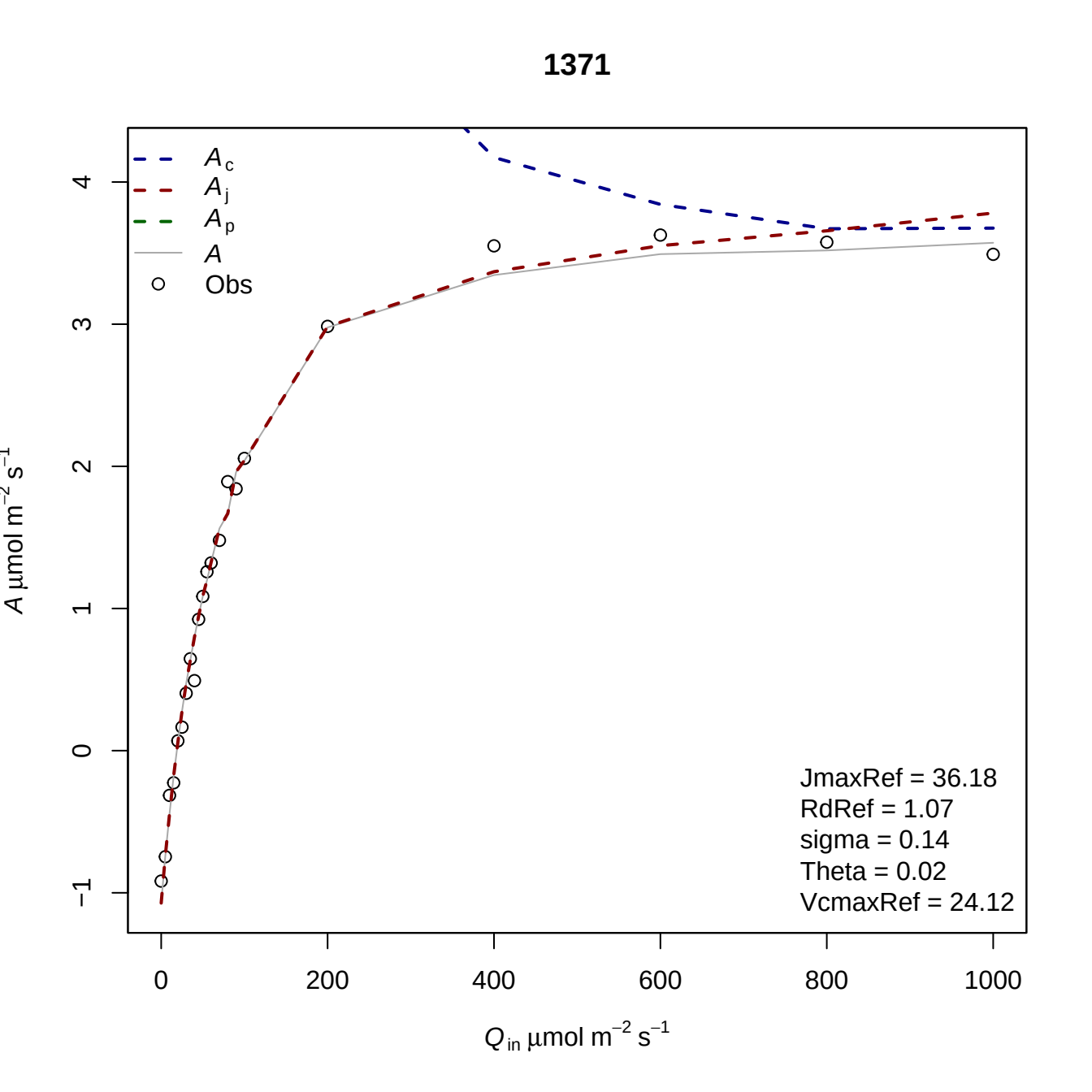

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 19.9
RdRef = 0.67
sigma = 0.14
Theta = −1.03
VcmaxRef = 16263.01
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

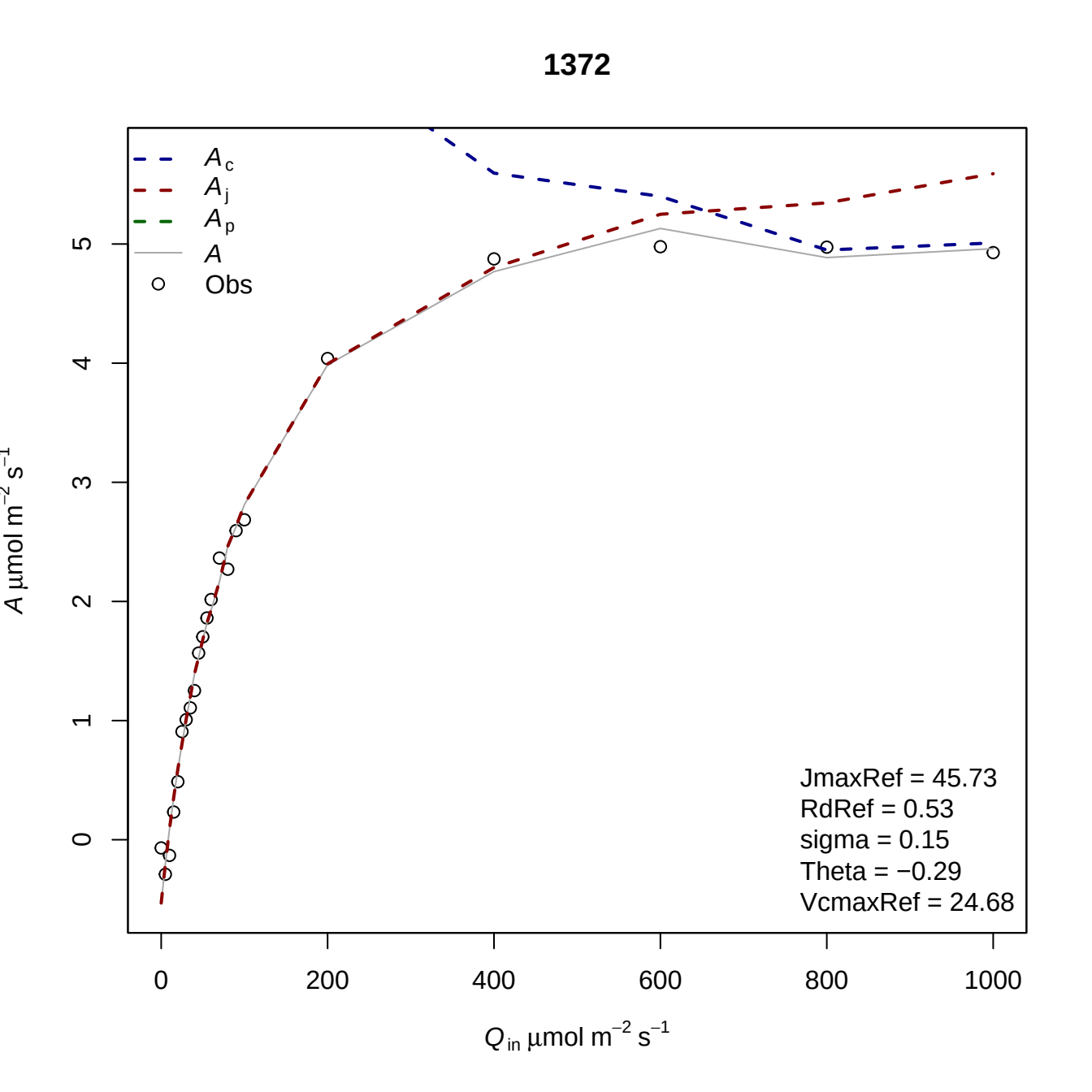

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 22.82
RdRef = 0.58
sigma = 0.07
Theta = −2.61
VcmaxRef = 64409.68
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

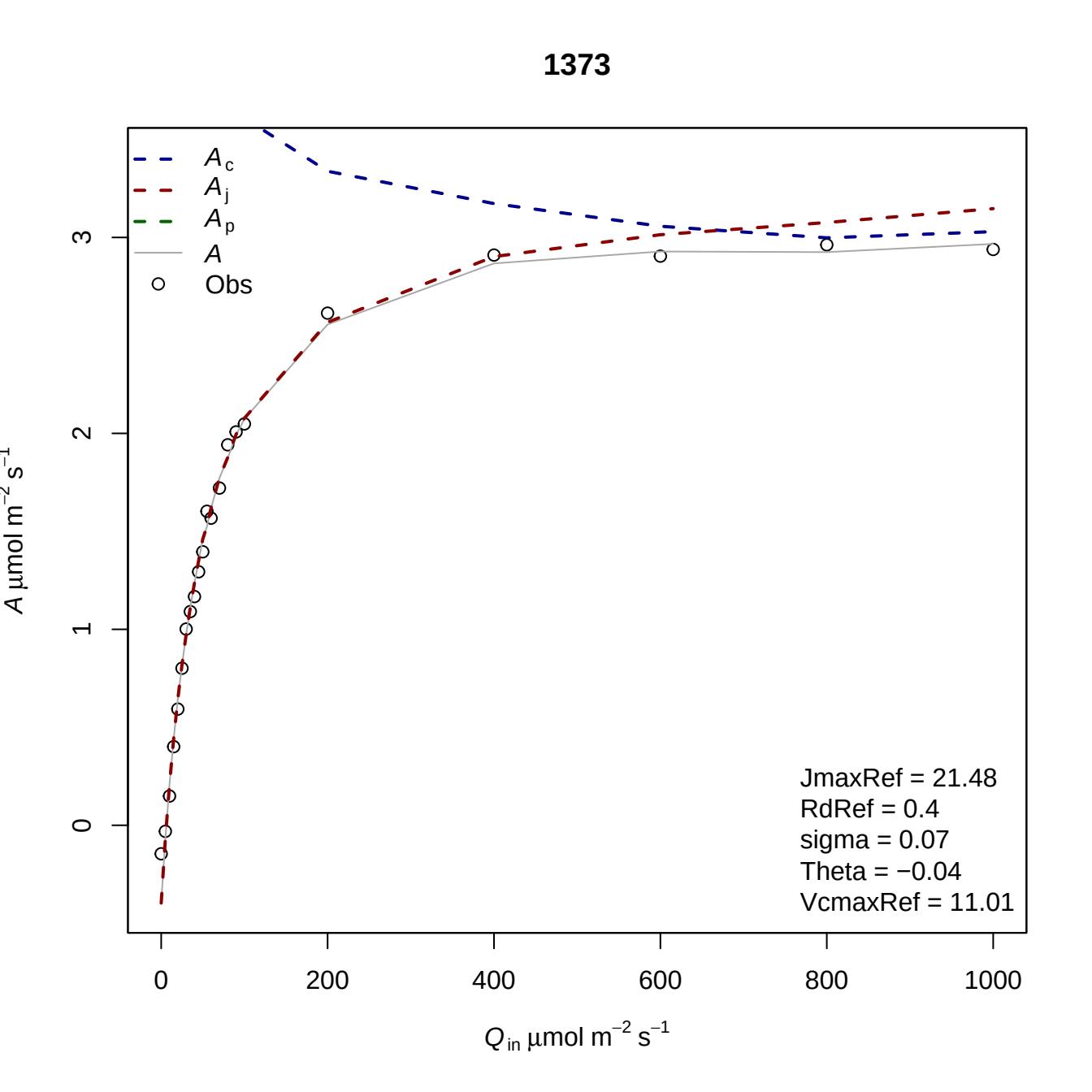

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 29.95
RdRef = 0.9
sigma = 0.1
Theta = 0.16
VcmaxRef = 328789.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

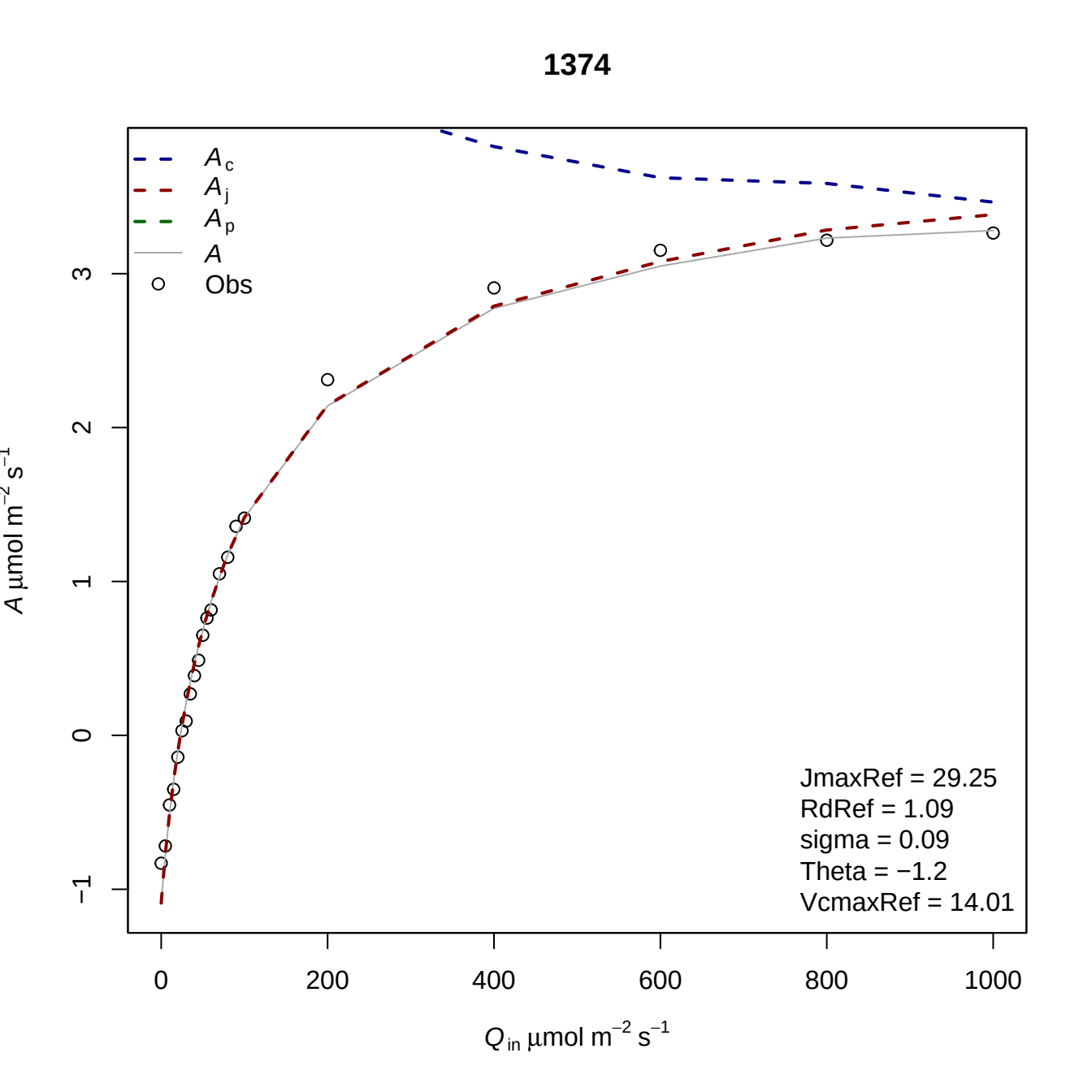

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 29.32
RdRef = 1.45
sigma = 0.07
Theta = 0.52
VcmaxRef = 6615.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

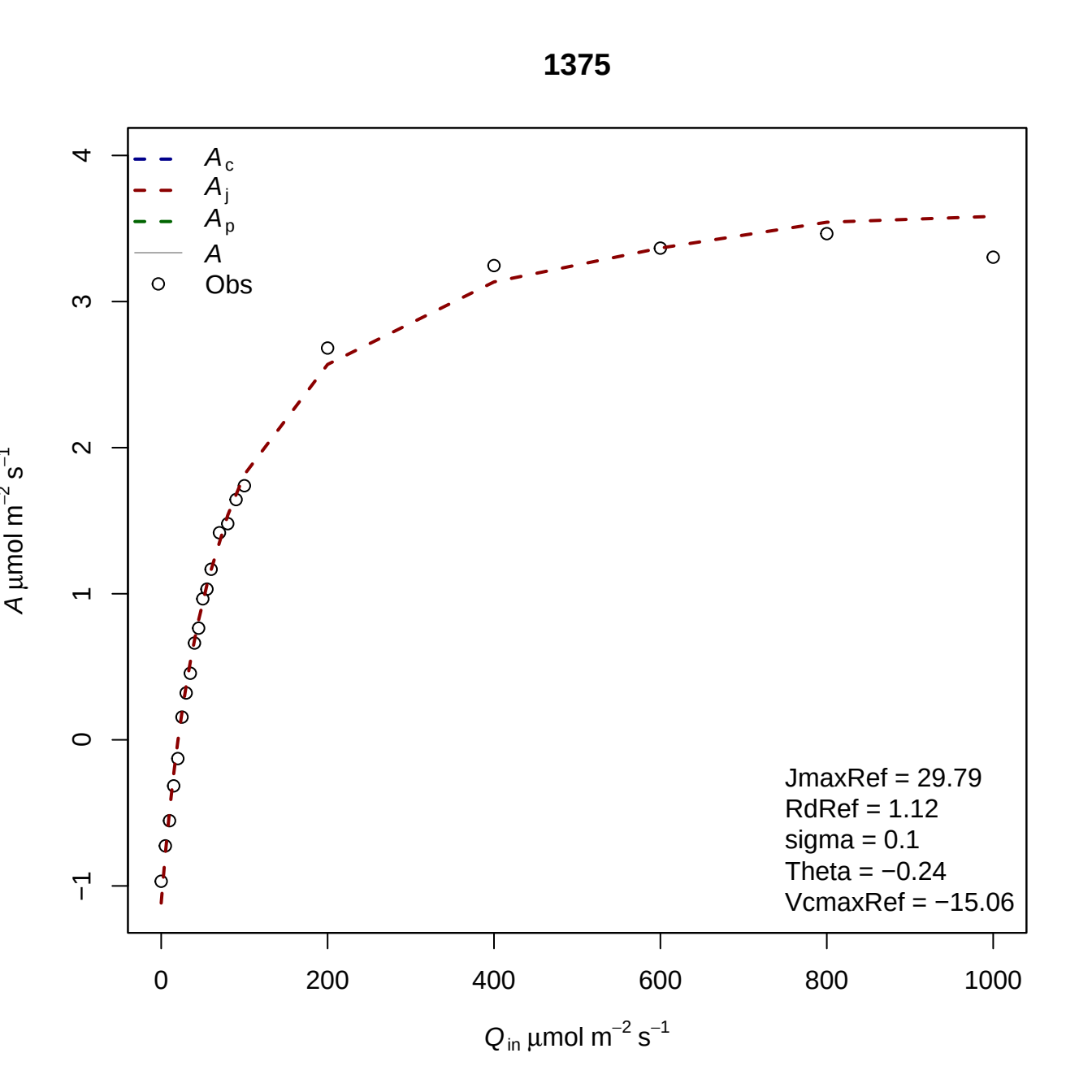

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 36.26
RdRef = 1.05
sigma = 0.13
Theta = −0.79
VcmaxRef = 18.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

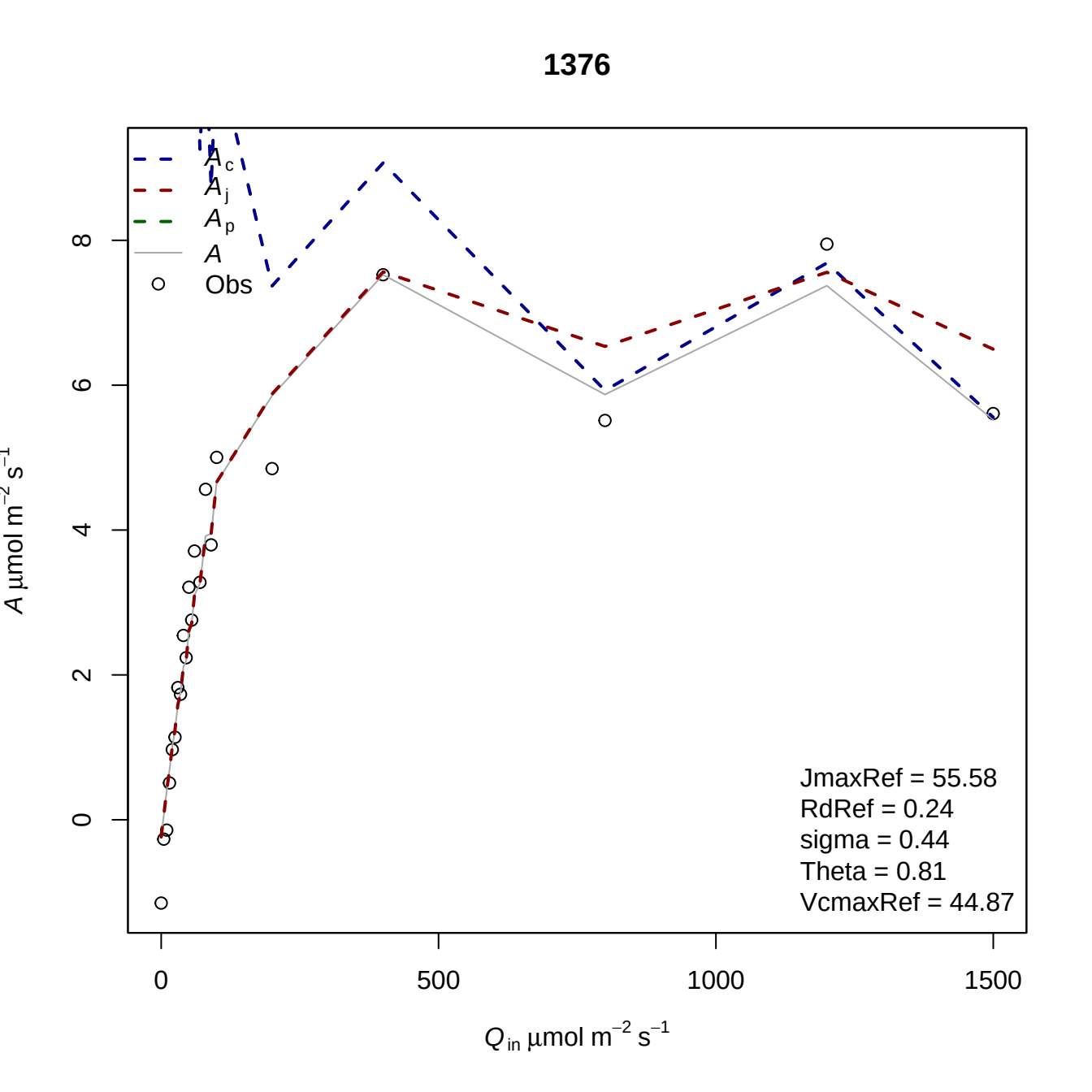

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 36.18
RdRef = 1.07
sigma = 0.14
Theta = 0.02
VcmaxRef = 24.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

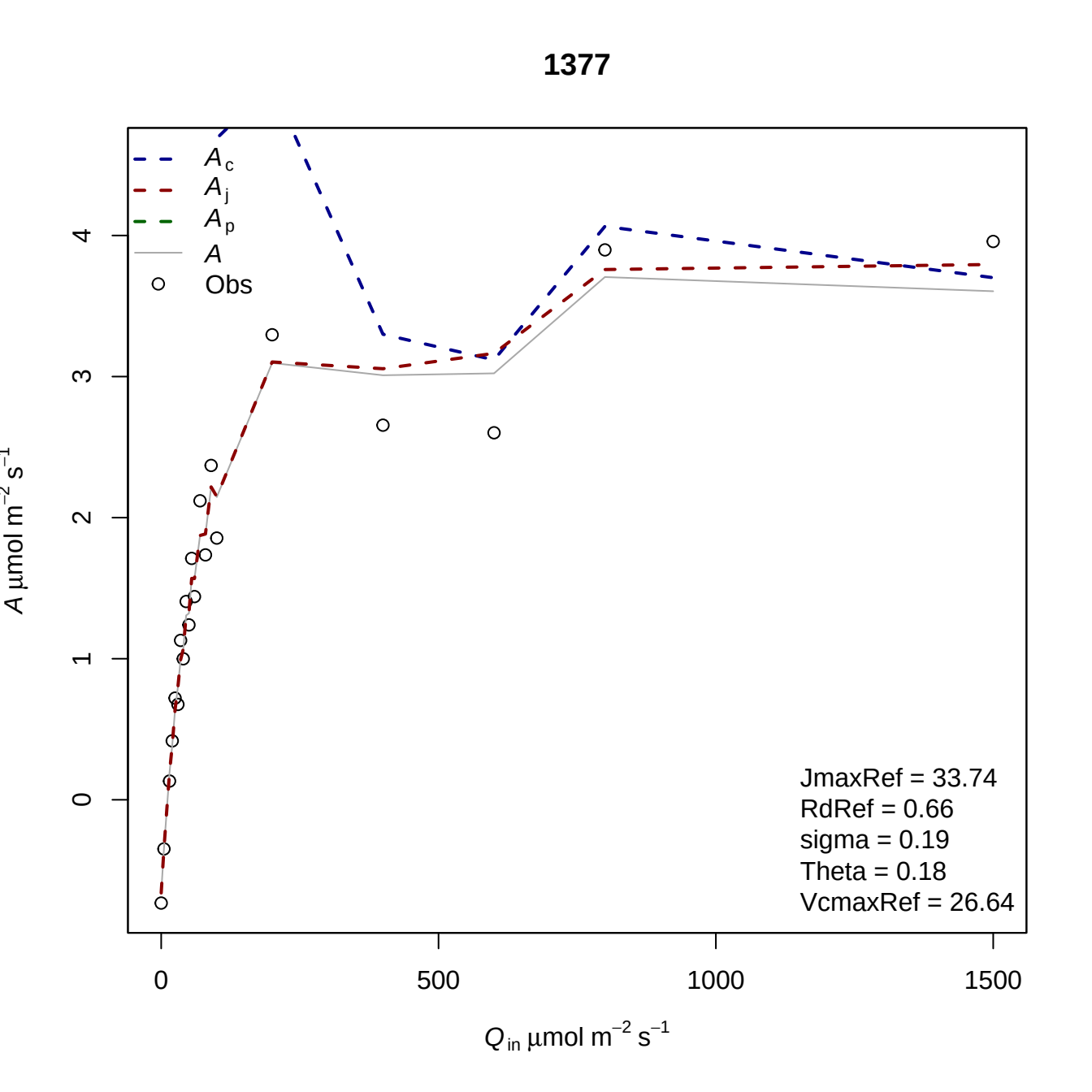

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 45.73
RdRef = 0.53
sigma = 0.15
Theta = −0.29
VcmaxRef = 24.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

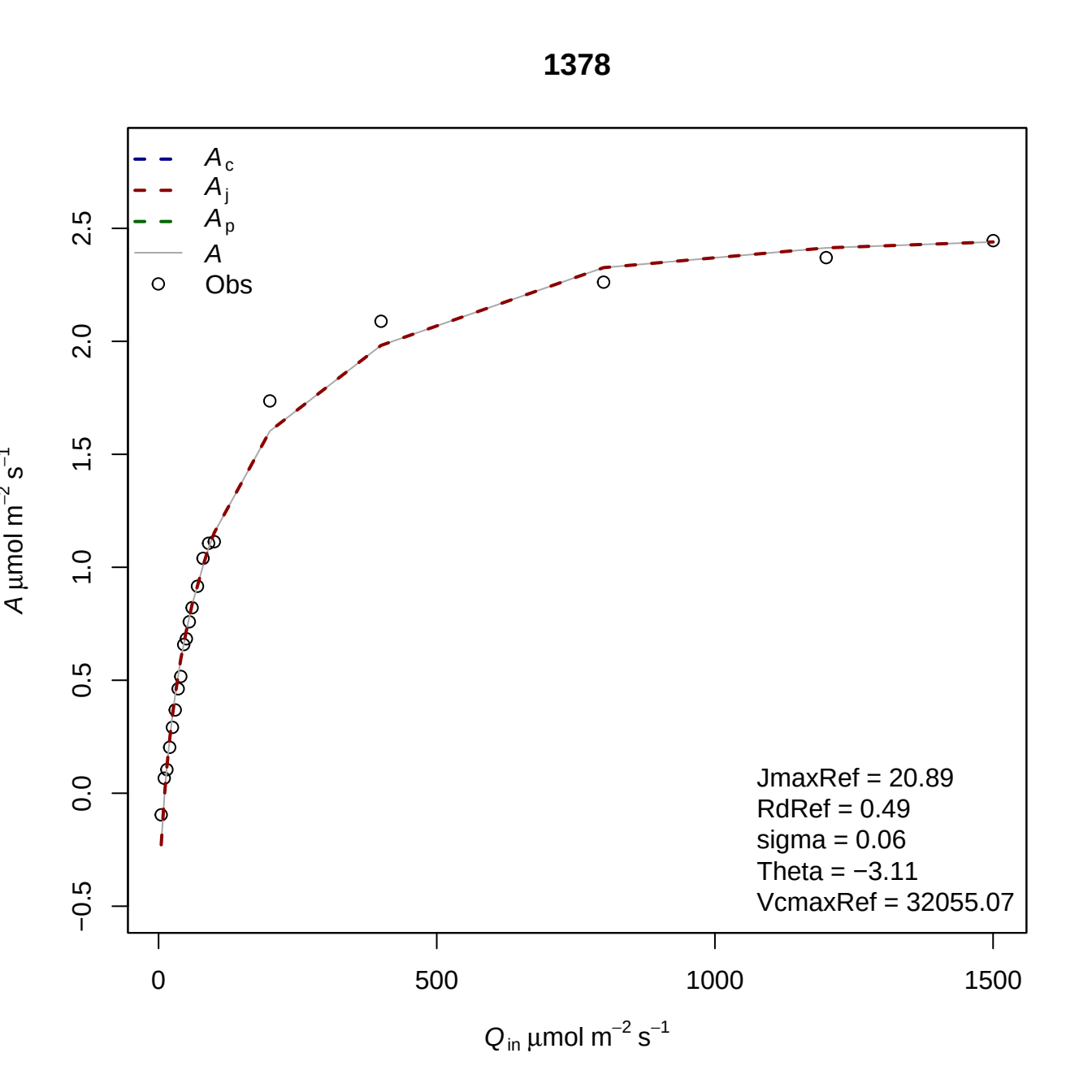

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 21.48
RdRef = 0.4
sigma = 0.07
Theta = −0.04
VcmaxRef = 11.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

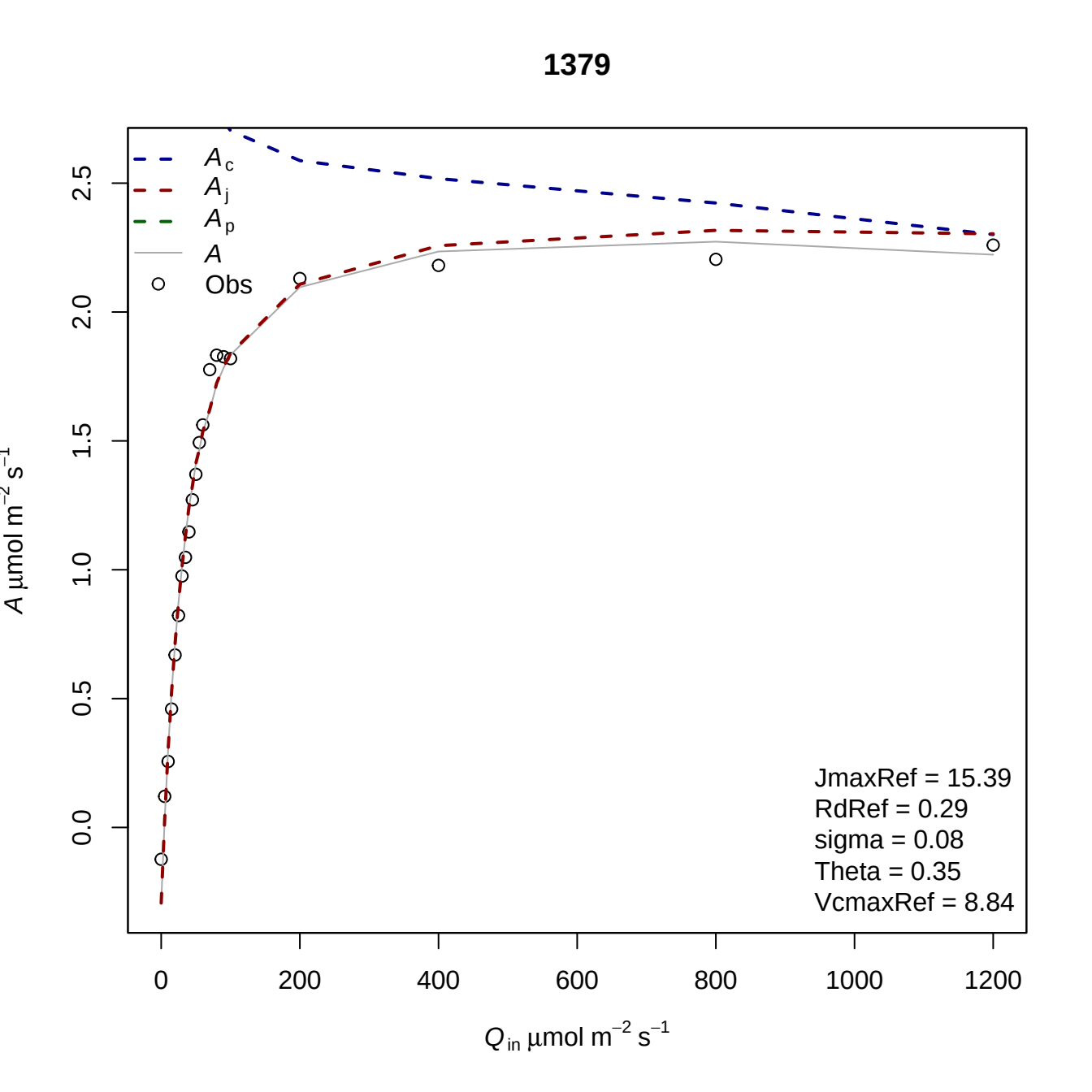

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 29.25
RdRef = 1.09
sigma = 0.09
Theta = −1.2
VcmaxRef = 14.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

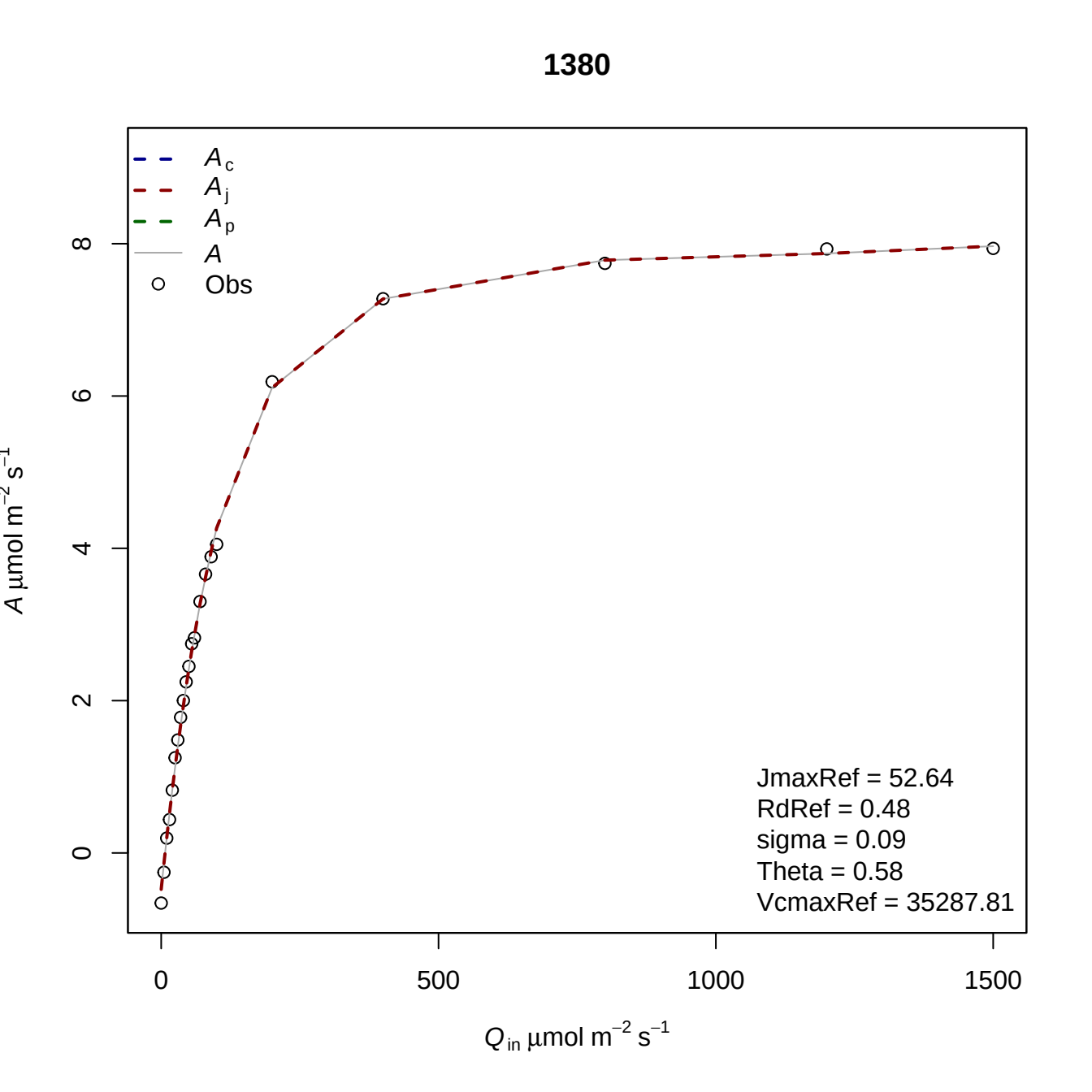

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 29.79
RdRef = 1.12
sigma = 0.1
Theta = −0.24
VcmaxRef = −15.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

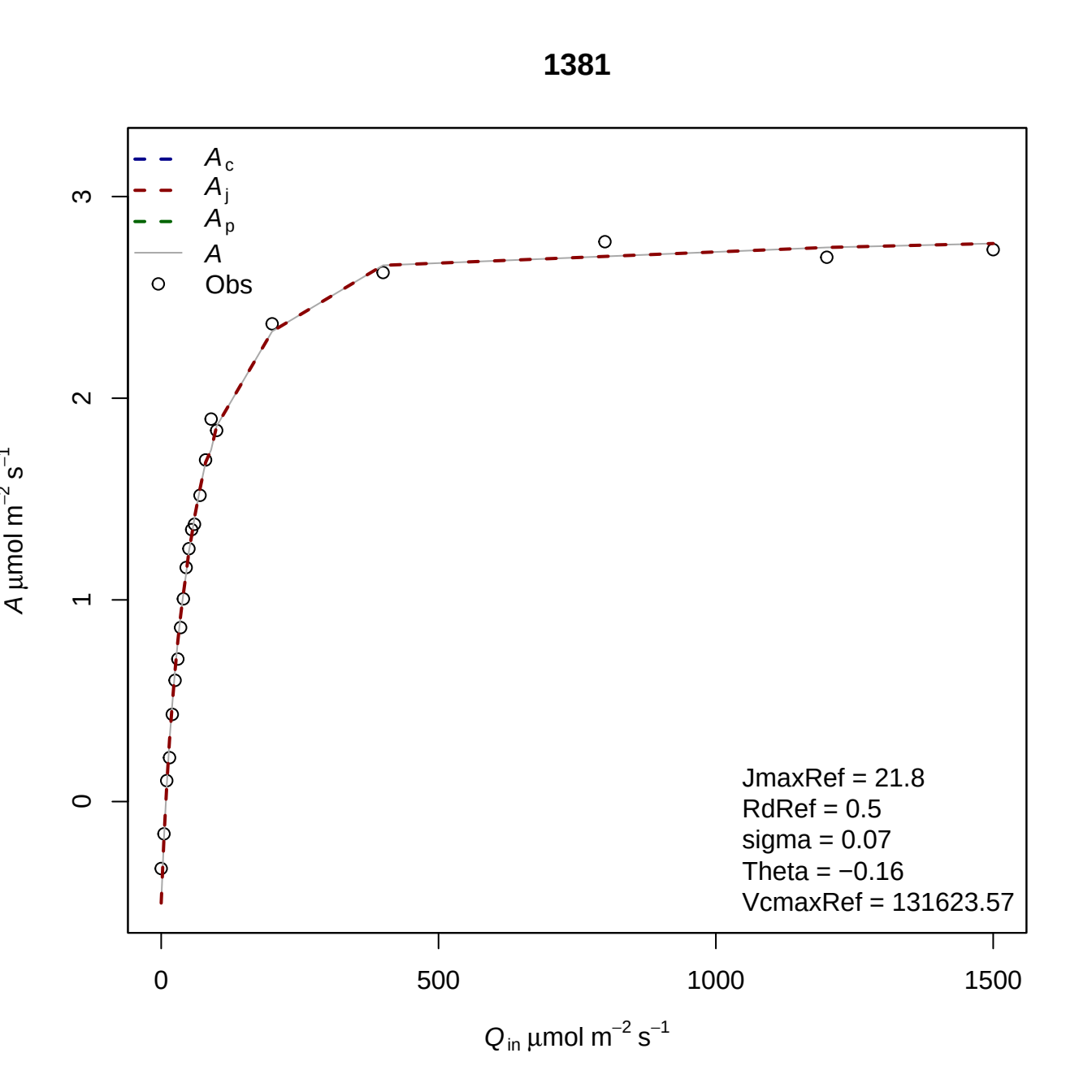

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 55.58
RdRef = 0.24
sigma = 0.44
Theta = 0.81
VcmaxRef = 44.87
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

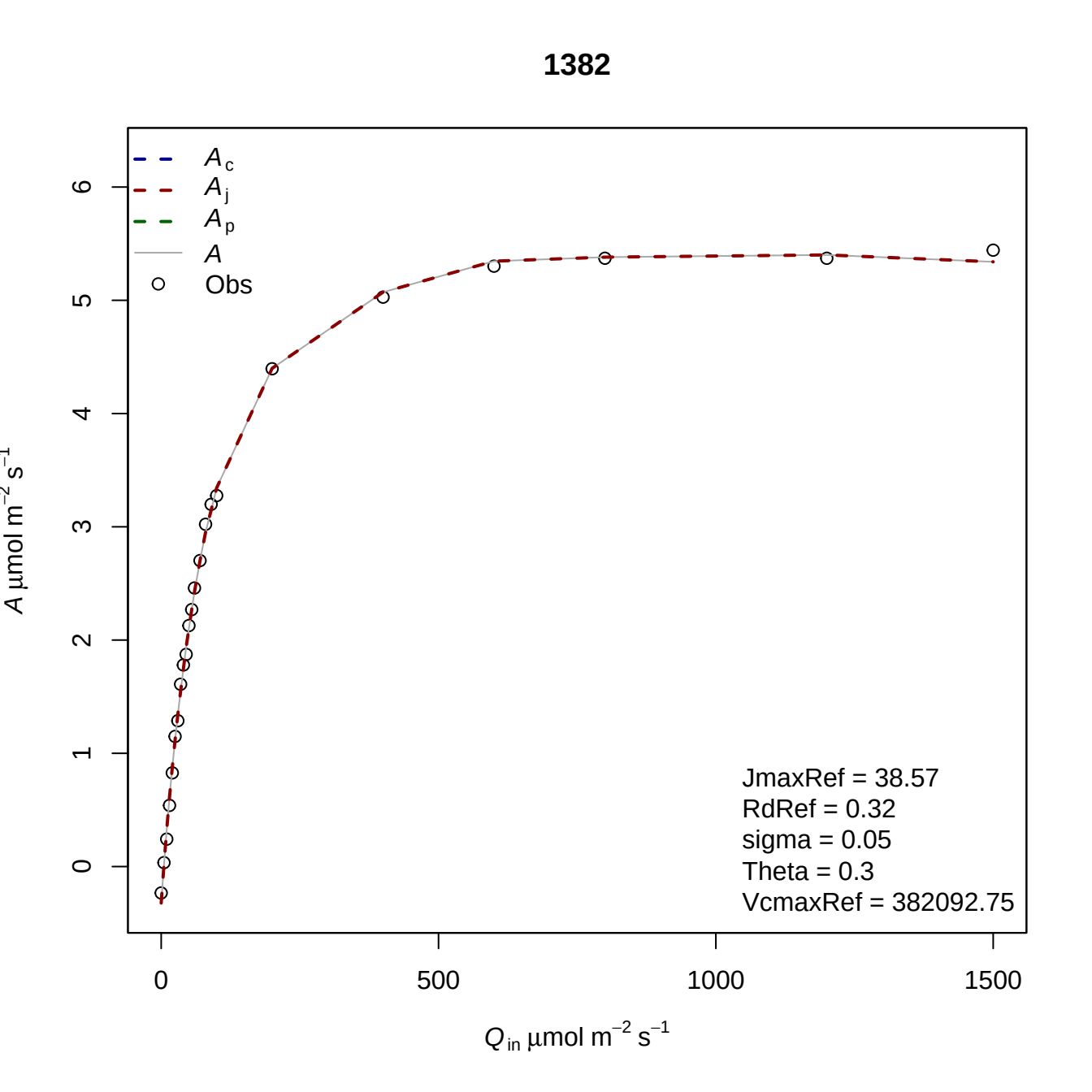

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 33.74
RdRef = 0.66
sigma = 0.19
Theta = 0.18
VcmaxRef = 26.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

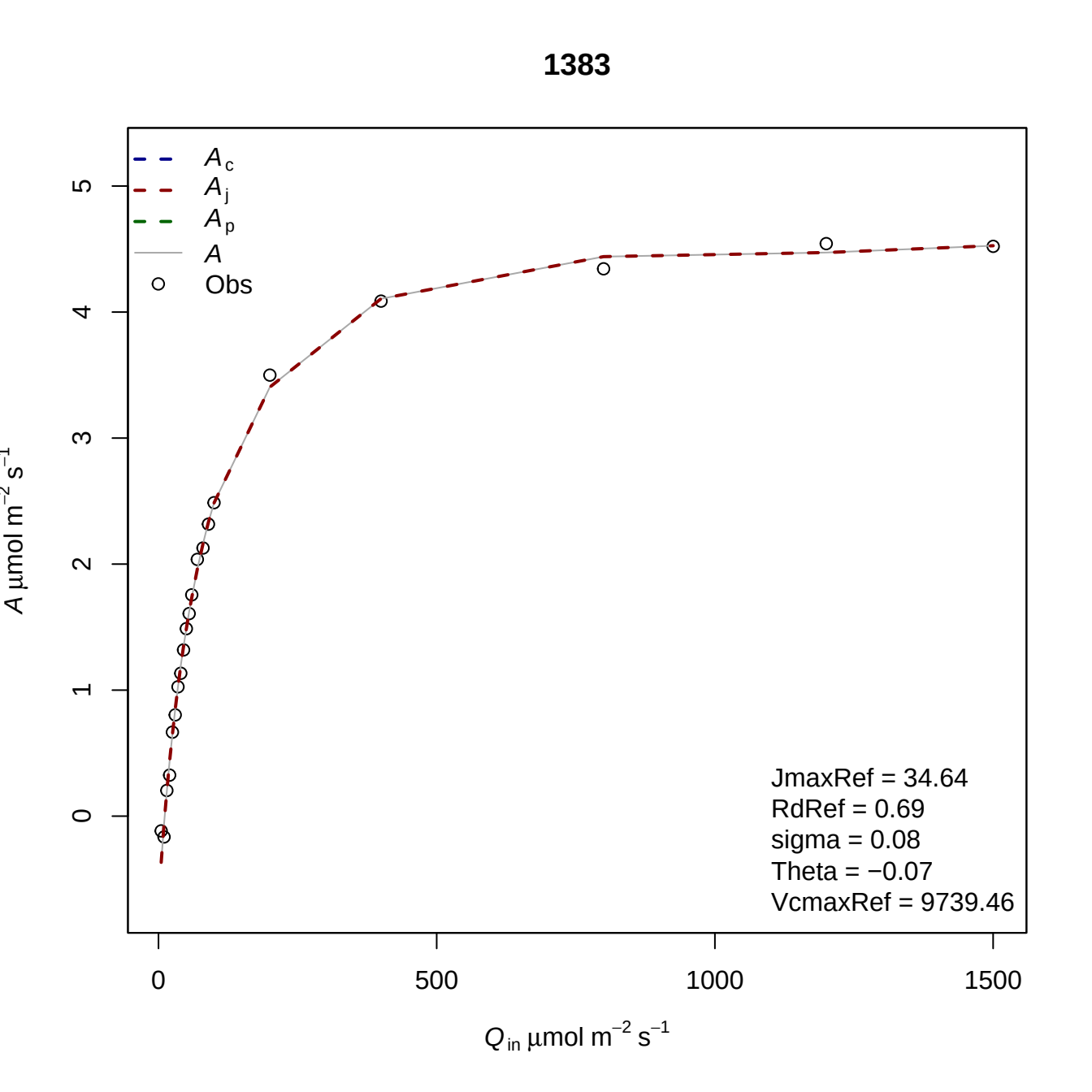

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 20.89
RdRef = 0.49
sigma = 0.06
Theta = −3.11
VcmaxRef = 32055.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

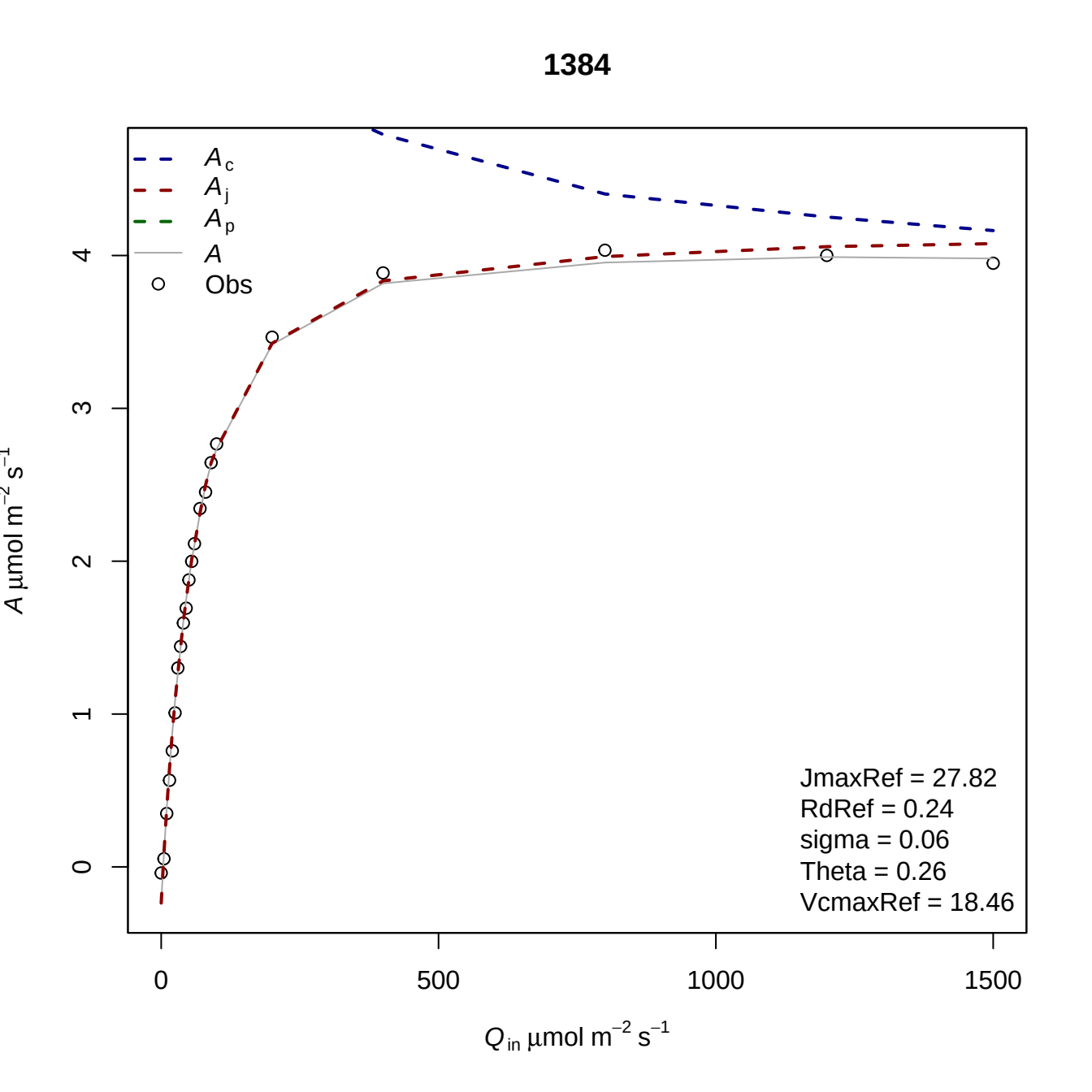

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 15.39
RdRef = 0.29
sigma = 0.08
Theta = 0.35
VcmaxRef = 8.84
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

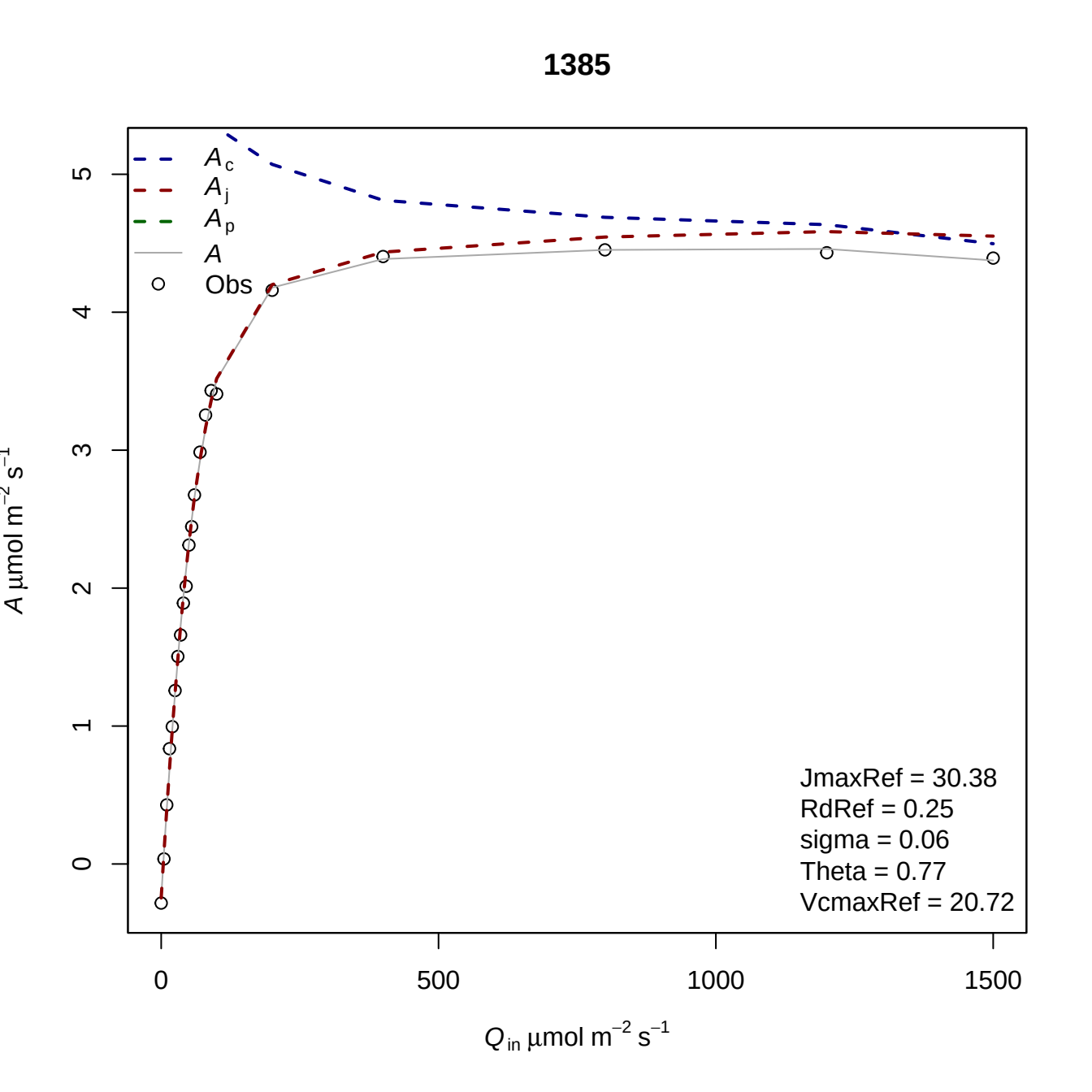

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 52.64
RdRef = 0.48
sigma = 0.09
Theta = 0.58
VcmaxRef = 35287.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

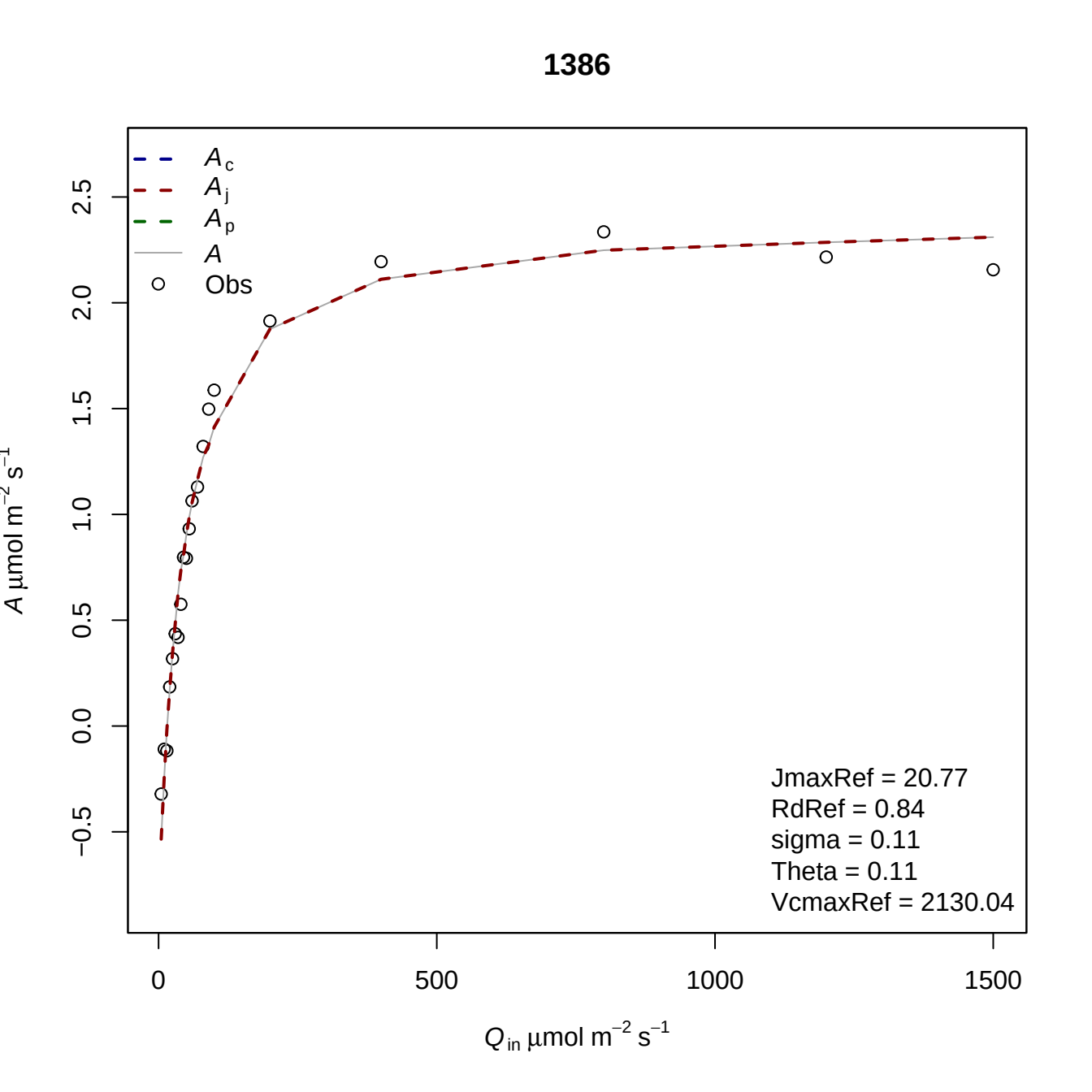

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 21.8
RdRef = 0.5
sigma = 0.07
Theta = −0.16
VcmaxRef = 131623.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

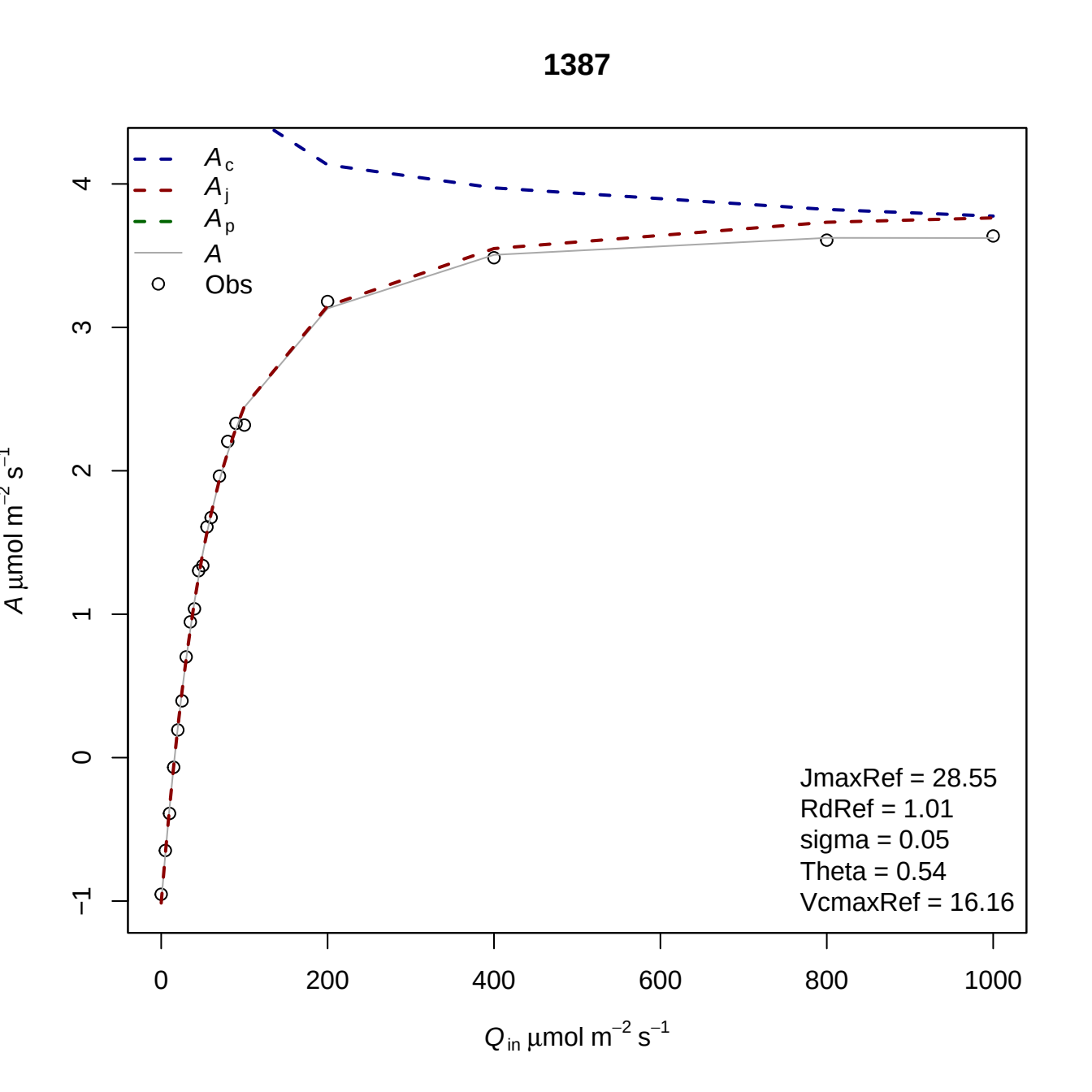

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 38.57
RdRef = 0.32
sigma = 0.05
Theta = 0.3
VcmaxRef = 382092.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

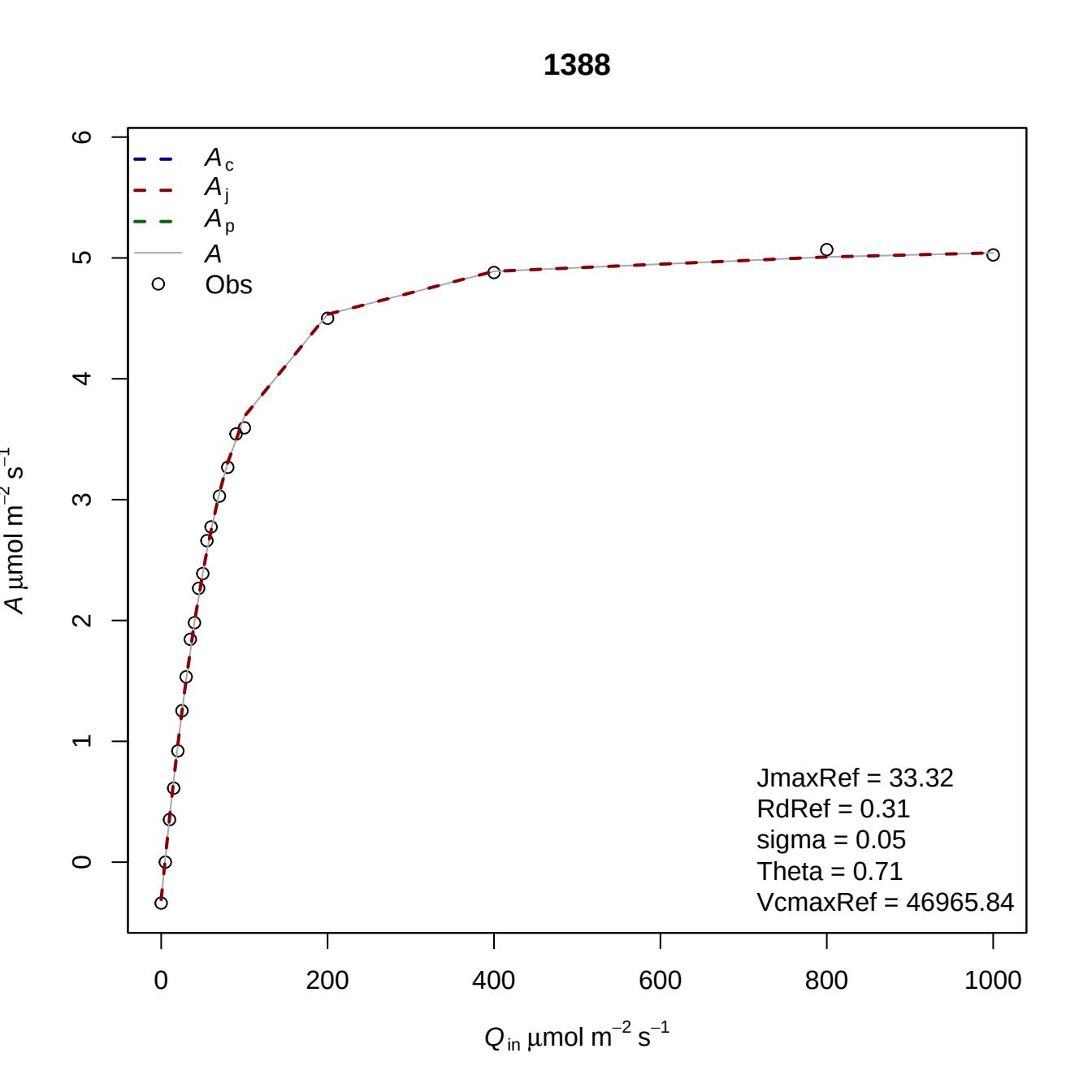

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 34.64
RdRef = 0.69
sigma = 0.08
Theta = −0.07
VcmaxRef = 9739.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

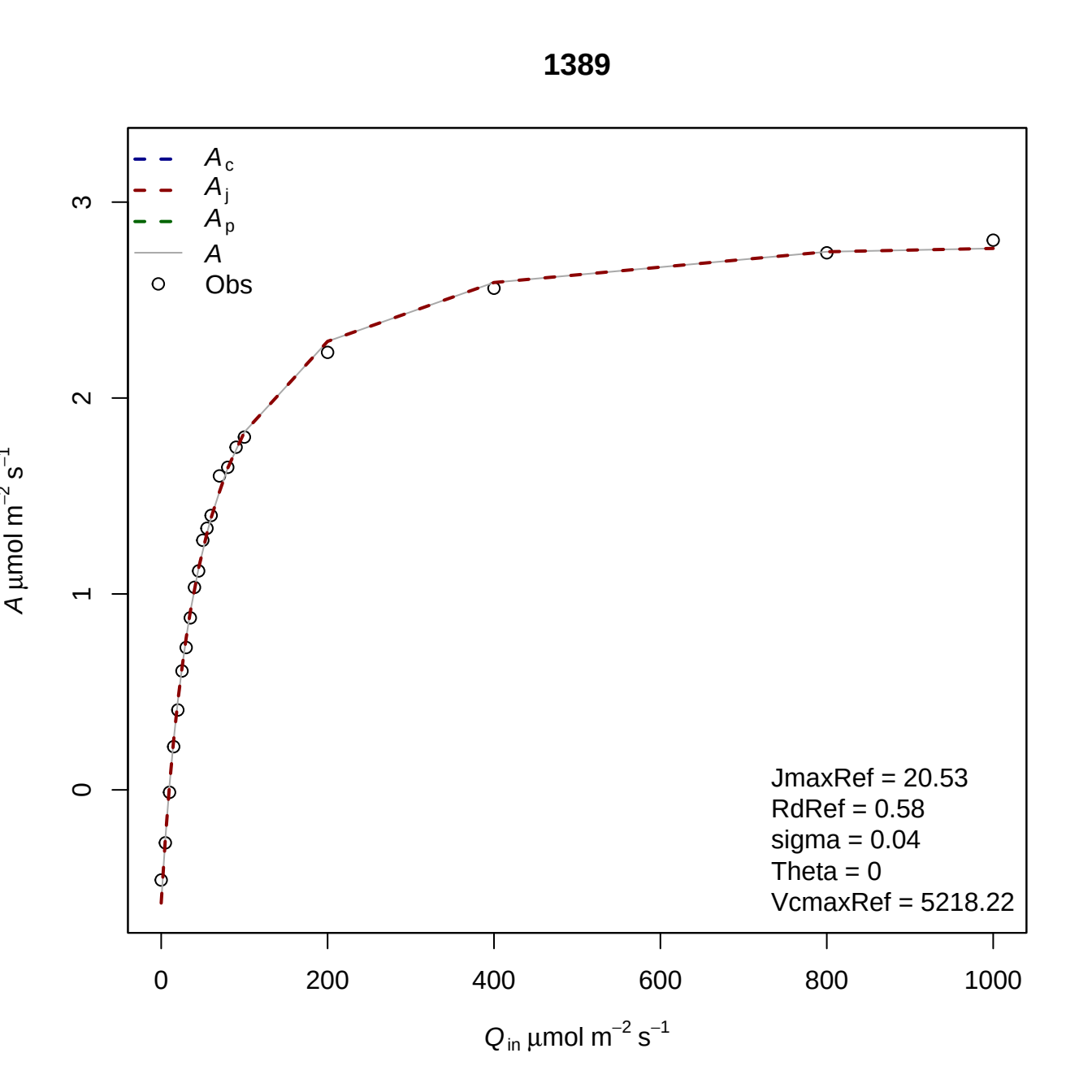

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 27.82
RdRef = 0.24
sigma = 0.06
Theta = 0.26
VcmaxRef = 18.46
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

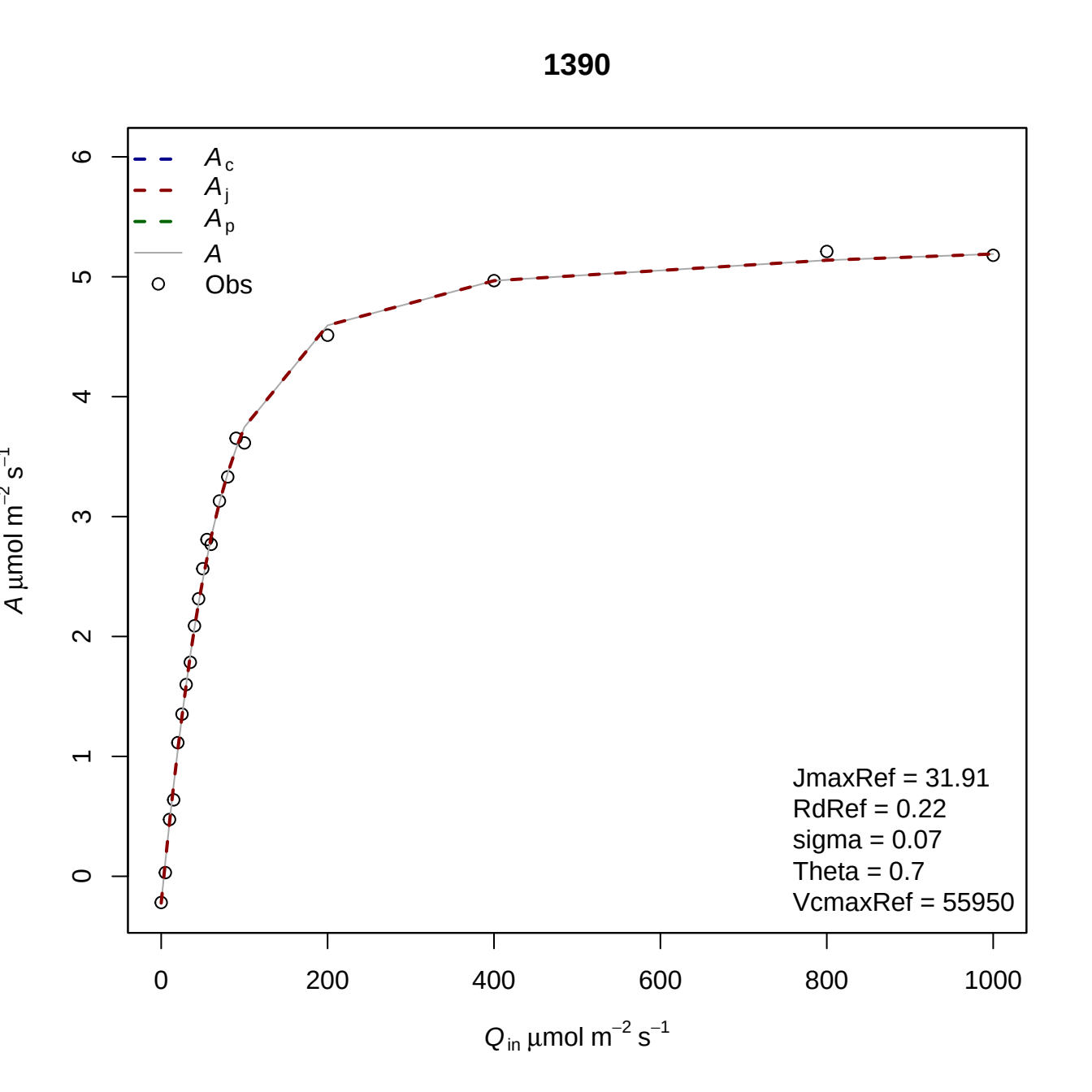

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 30.38
RdRef = 0.25
sigma = 0.06
Theta = 0.77
VcmaxRef = 20.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

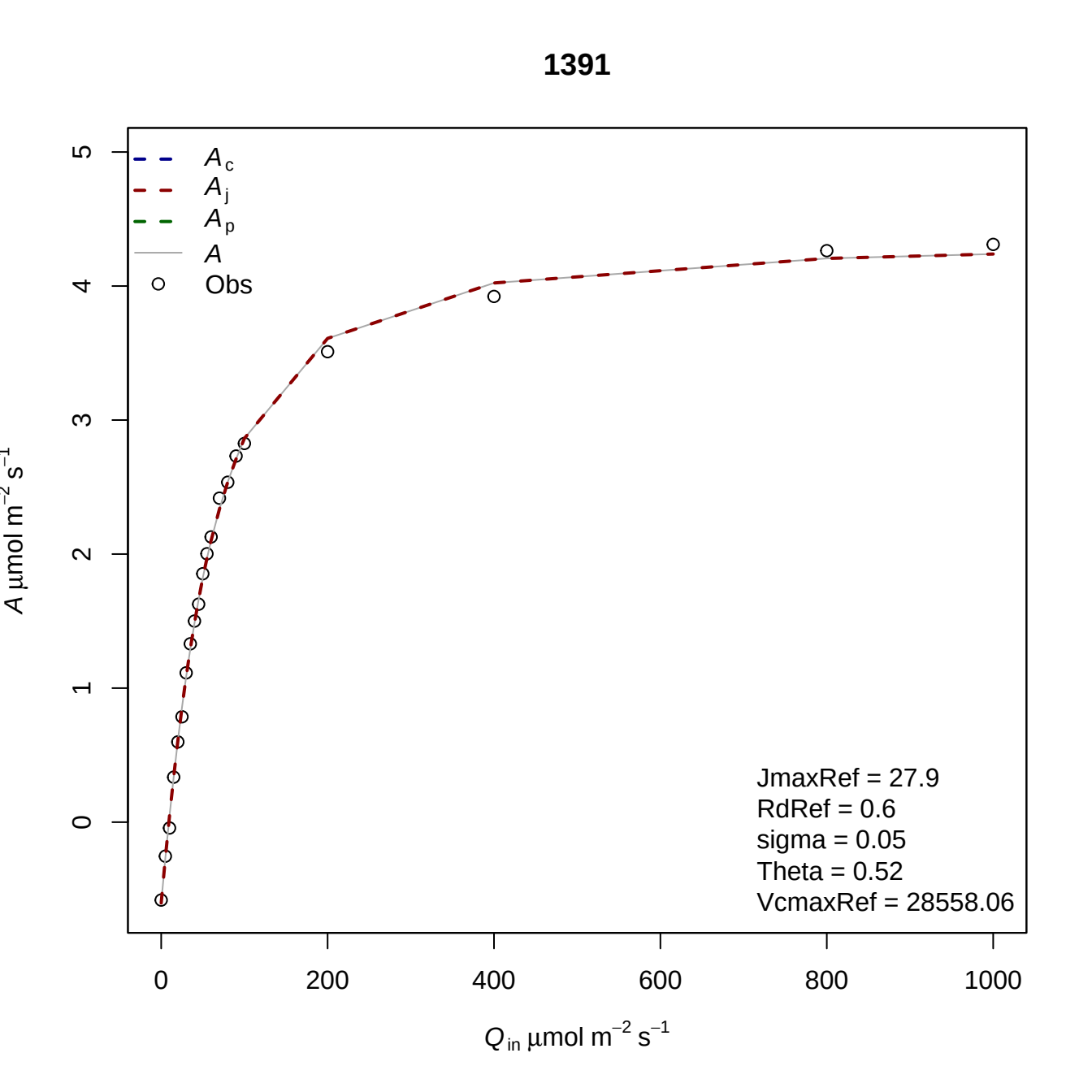

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 20.77
RdRef = 0.84
sigma = 0.11
Theta = 0.11
VcmaxRef = 2130.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

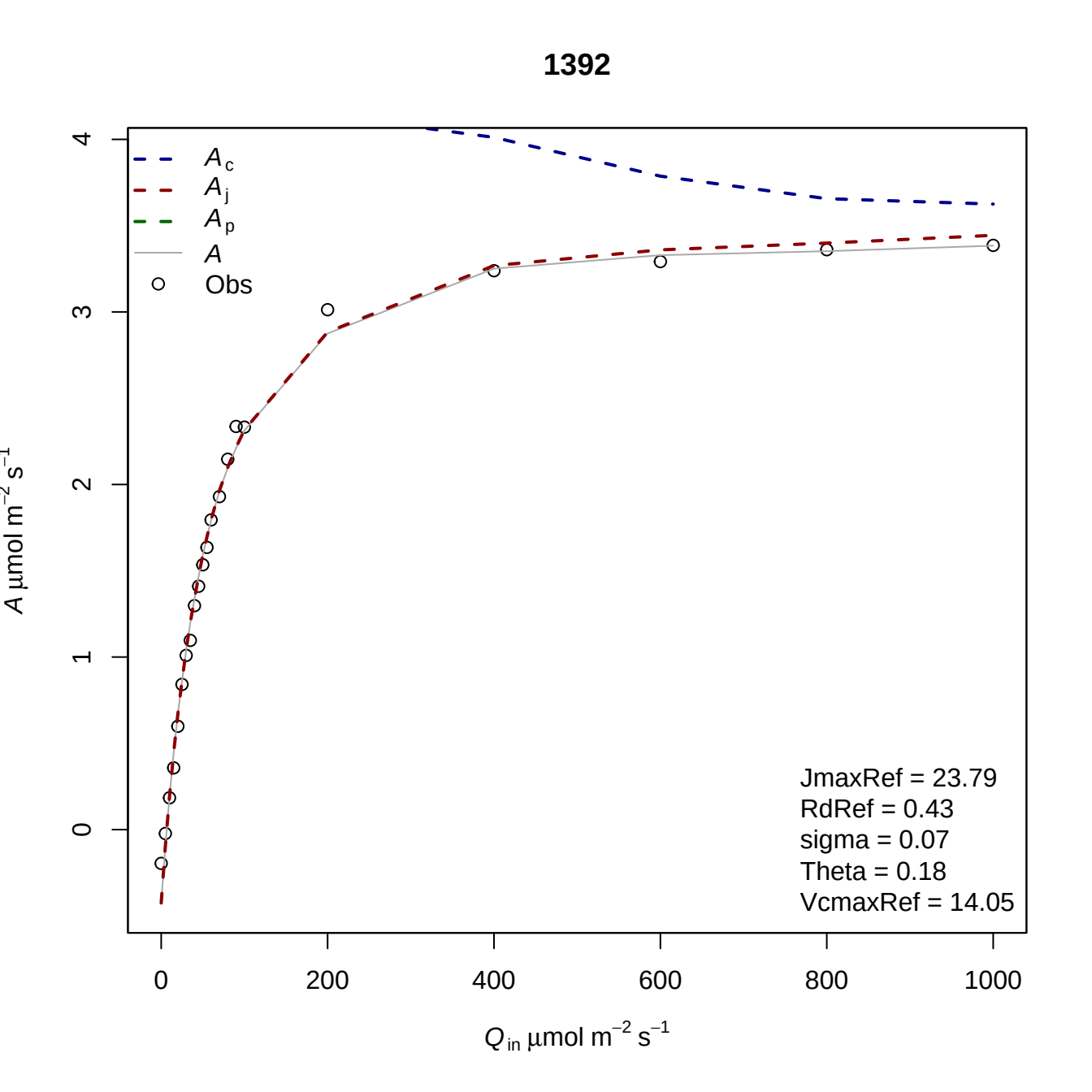

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 28.55
RdRef = 1.01
sigma = 0.05
Theta = 0.54
VcmaxRef = 16.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

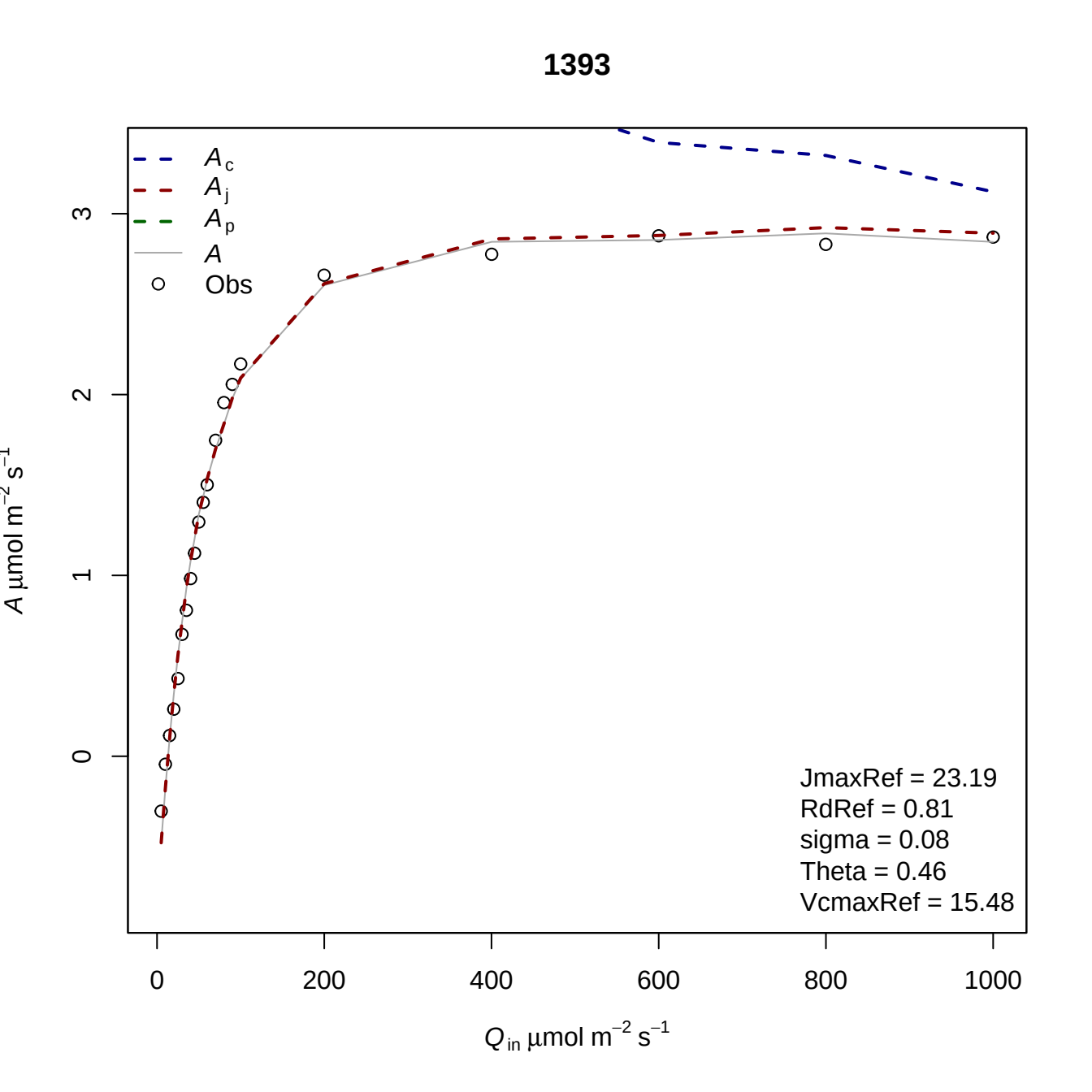

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 33.32
RdRef = 0.31
sigma = 0.05
Theta = 0.71
VcmaxRef = 46965.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

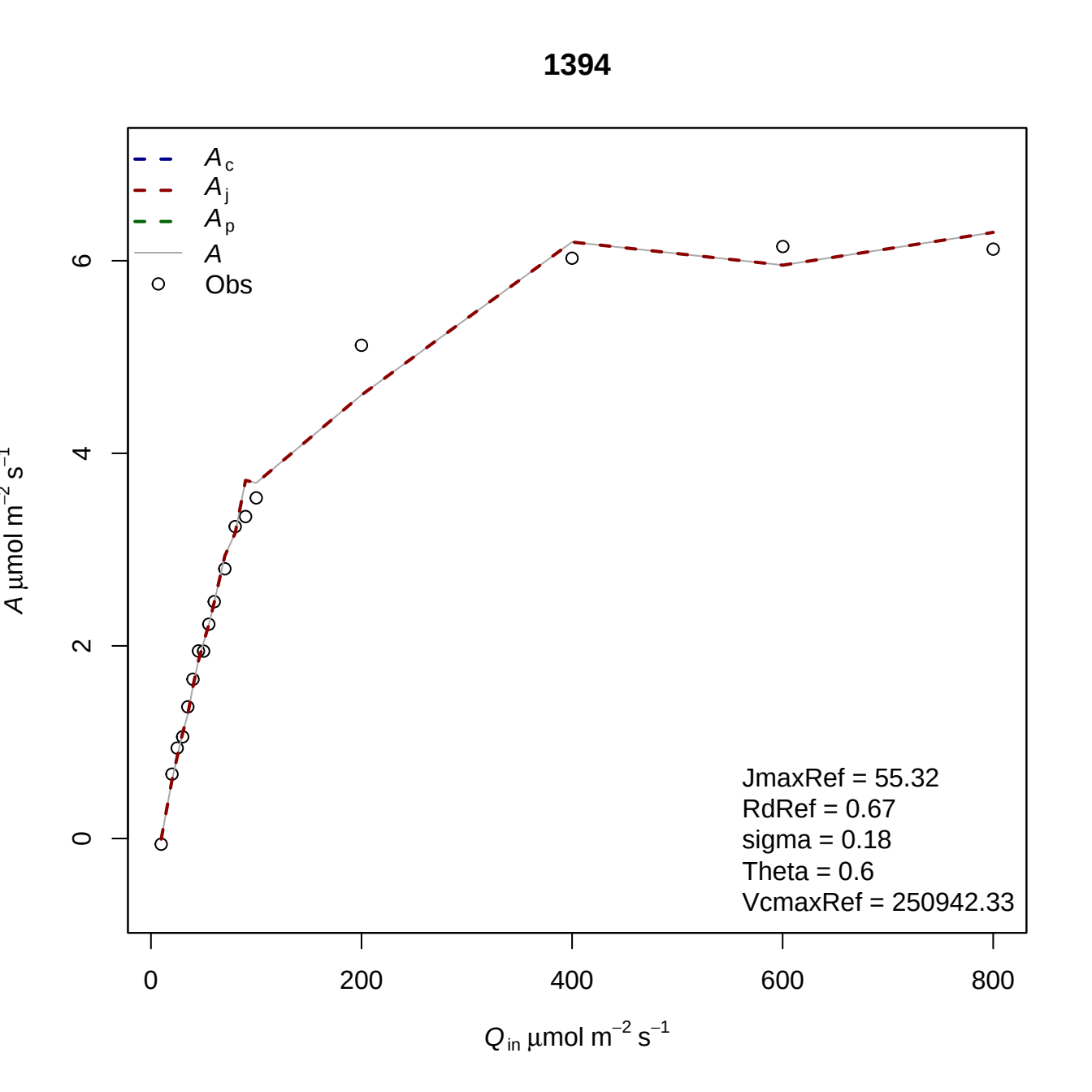

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 20.53
RdRef = 0.58
sigma = 0.04
Theta = 0
VcmaxRef = 5218.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

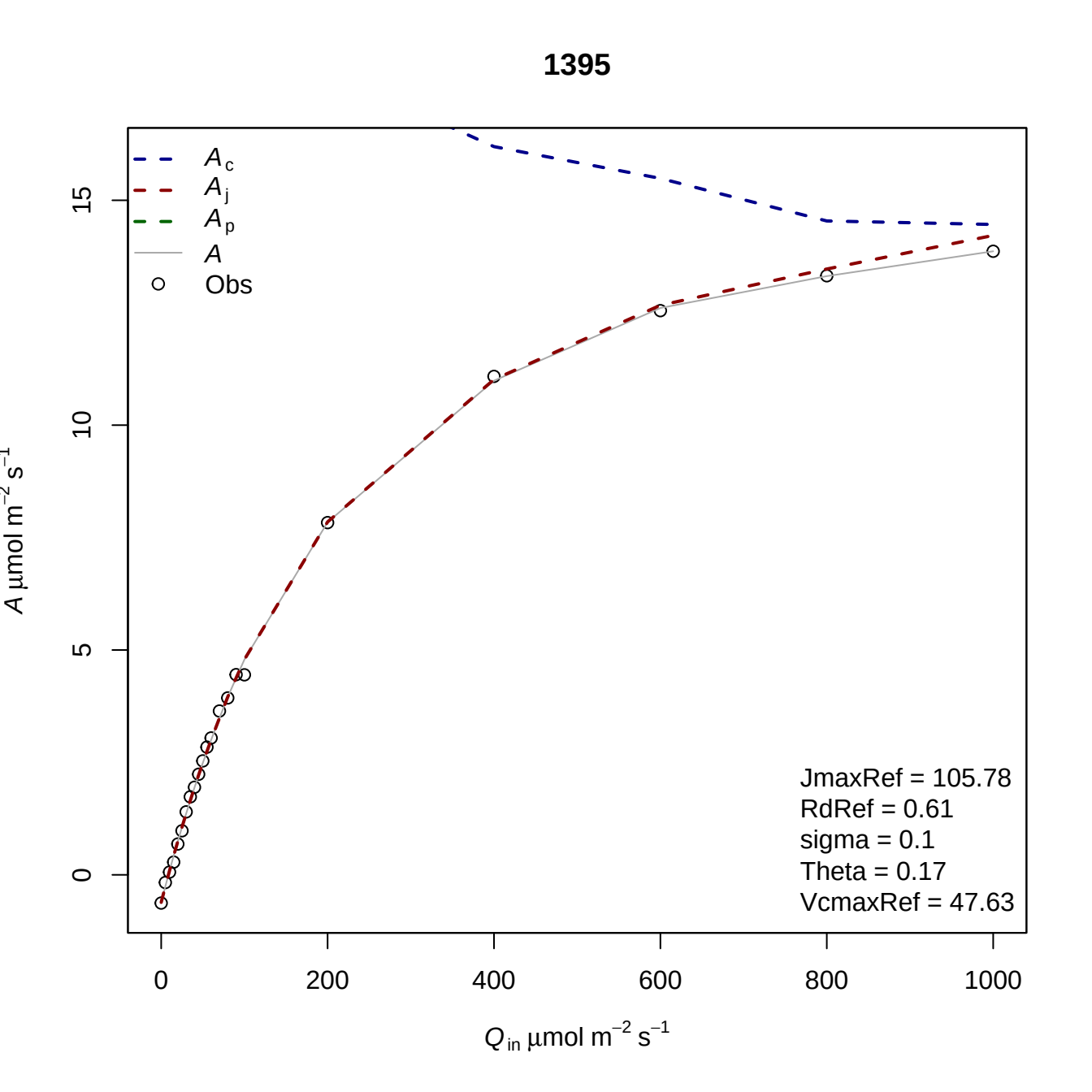

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 31.91
RdRef = 0.22
sigma = 0.07
Theta = 0.7
VcmaxRef = 55950
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

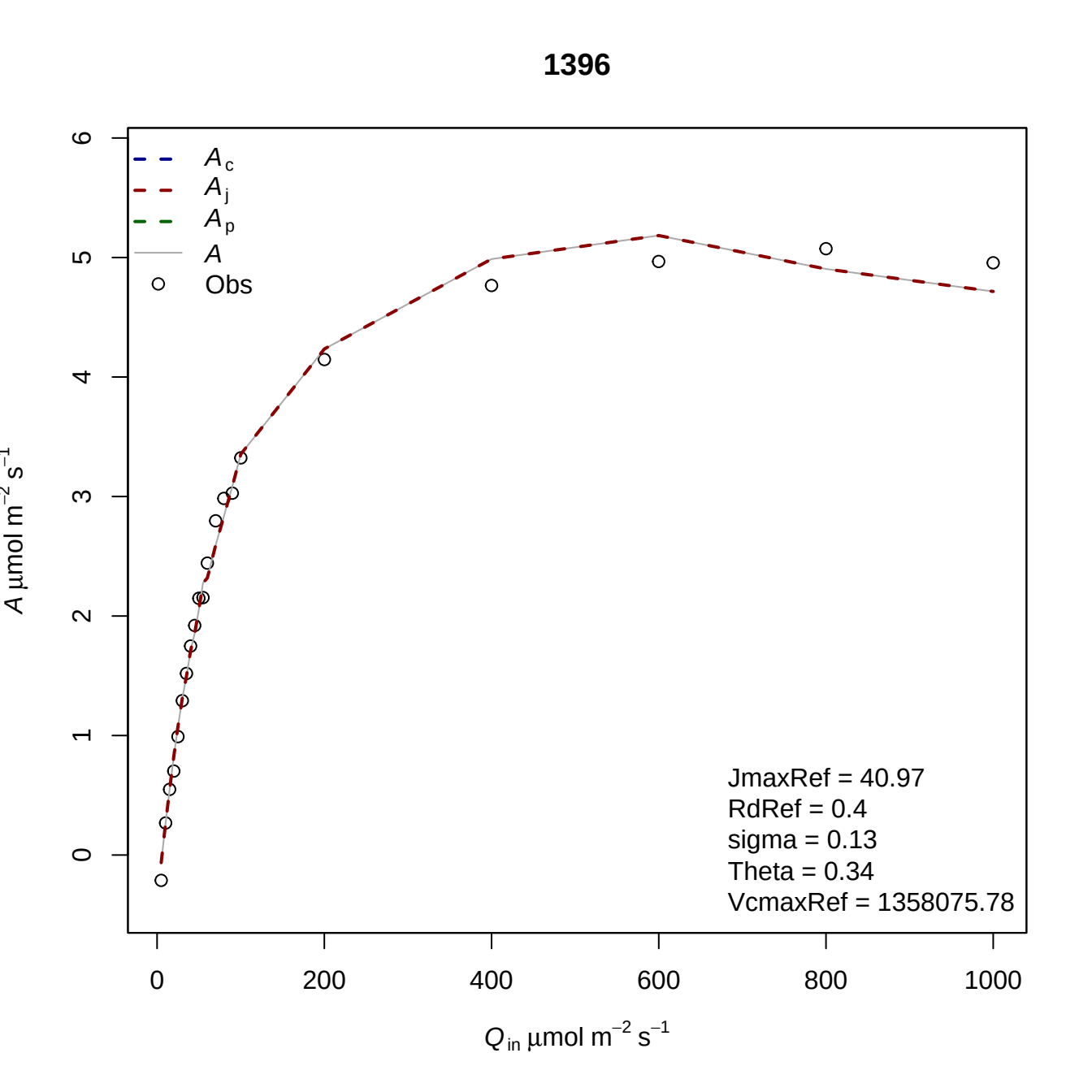

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 27.9
RdRef = 0.6
sigma = 0.05
Theta = 0.52
VcmaxRef = 28558.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

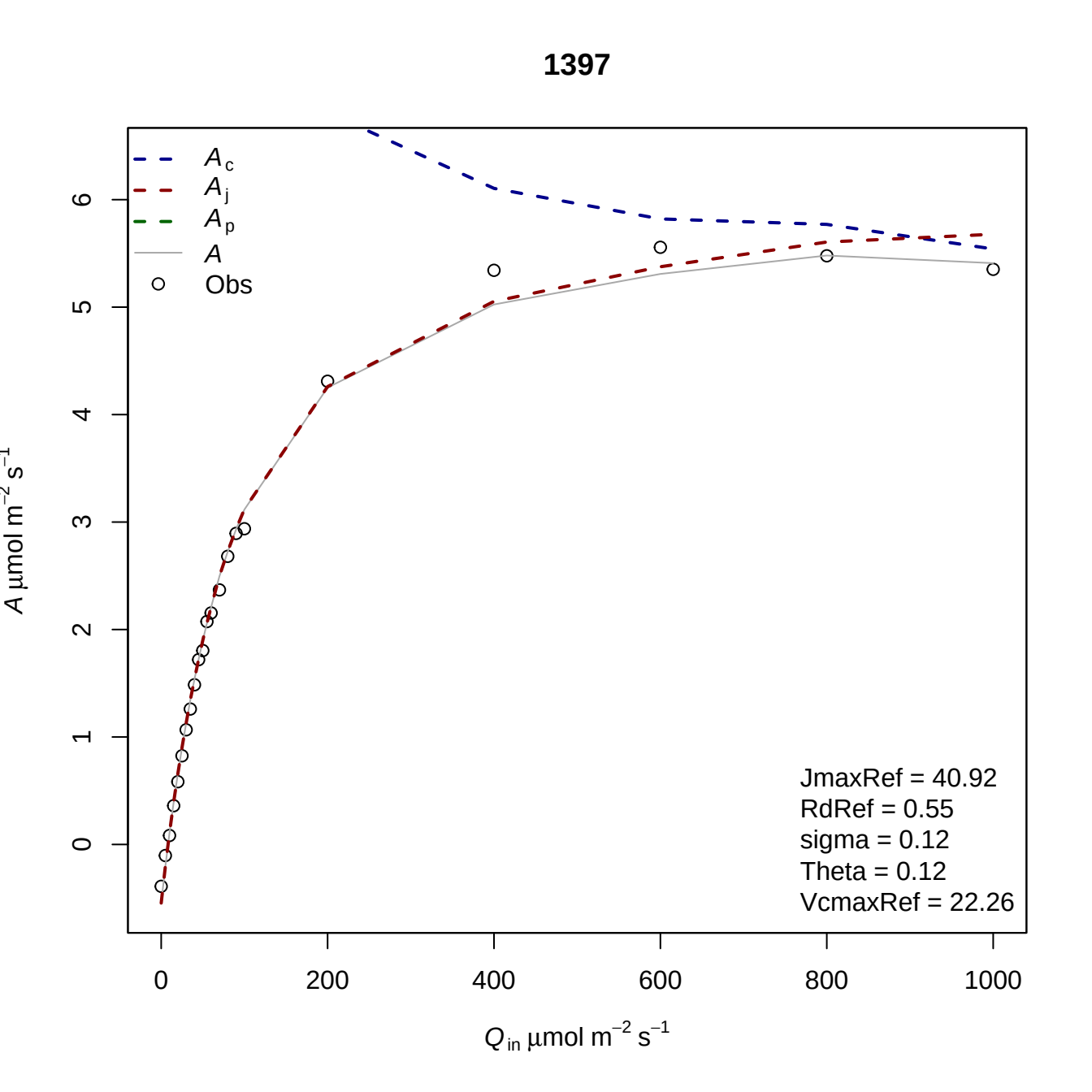

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 23.79
RdRef = 0.43
sigma = 0.07
Theta = 0.18
VcmaxRef = 14.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

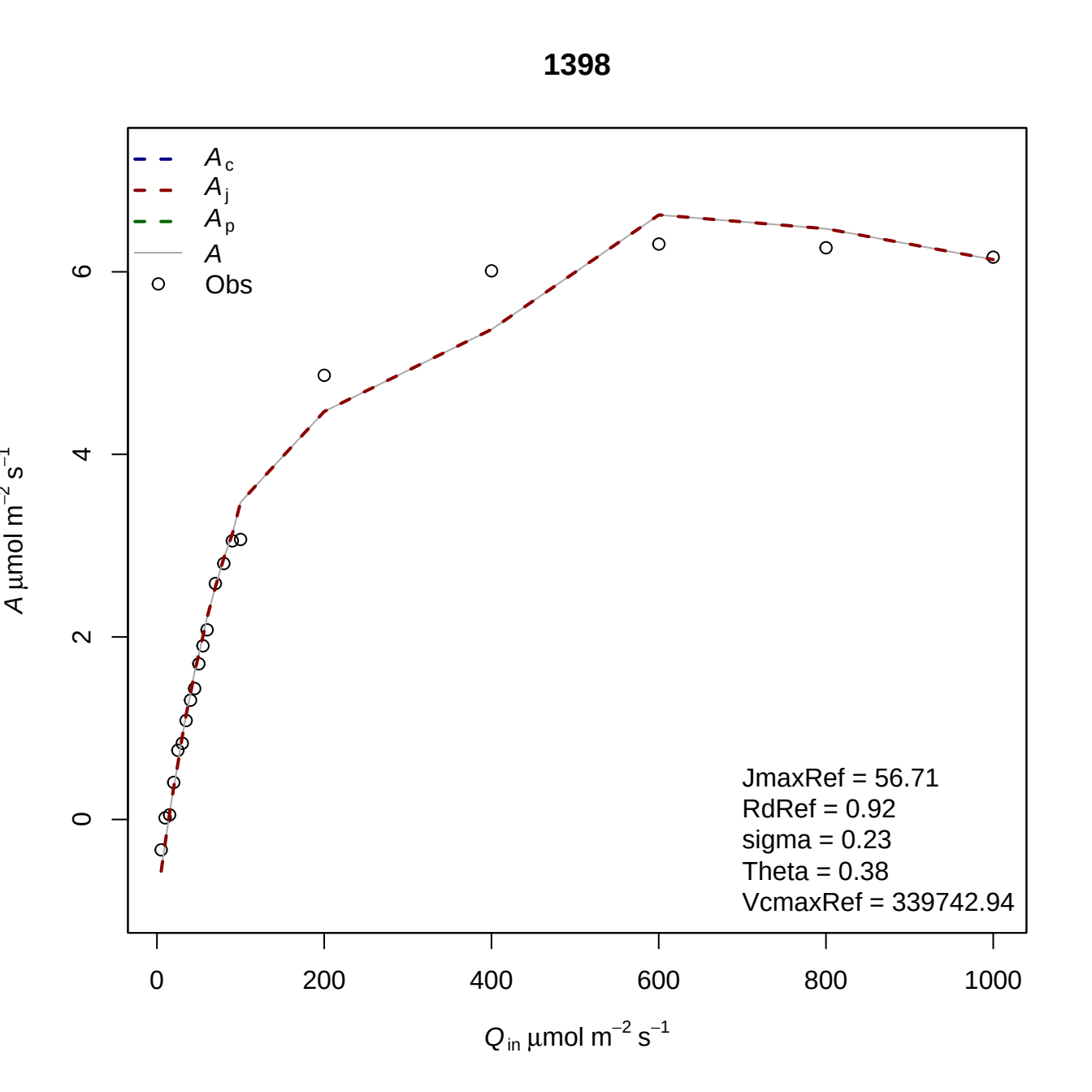

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 23.19
RdRef = 0.81
sigma = 0.08
Theta = 0.46
VcmaxRef = 15.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

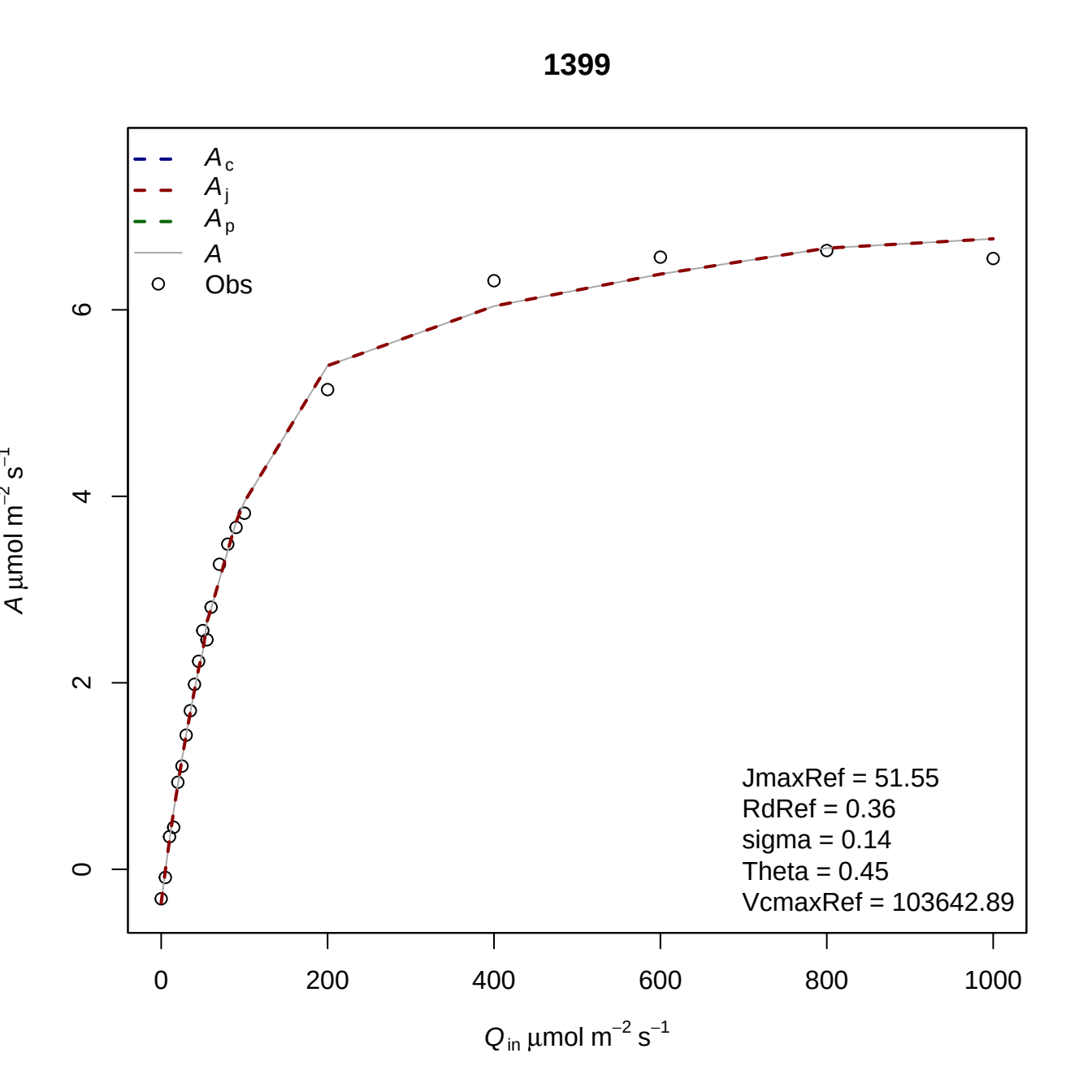

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 55.32
RdRef = 0.67
sigma = 0.18
Theta = 0.6
VcmaxRef = 250942.33
0
2
4
6
0
200
400
600
800
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

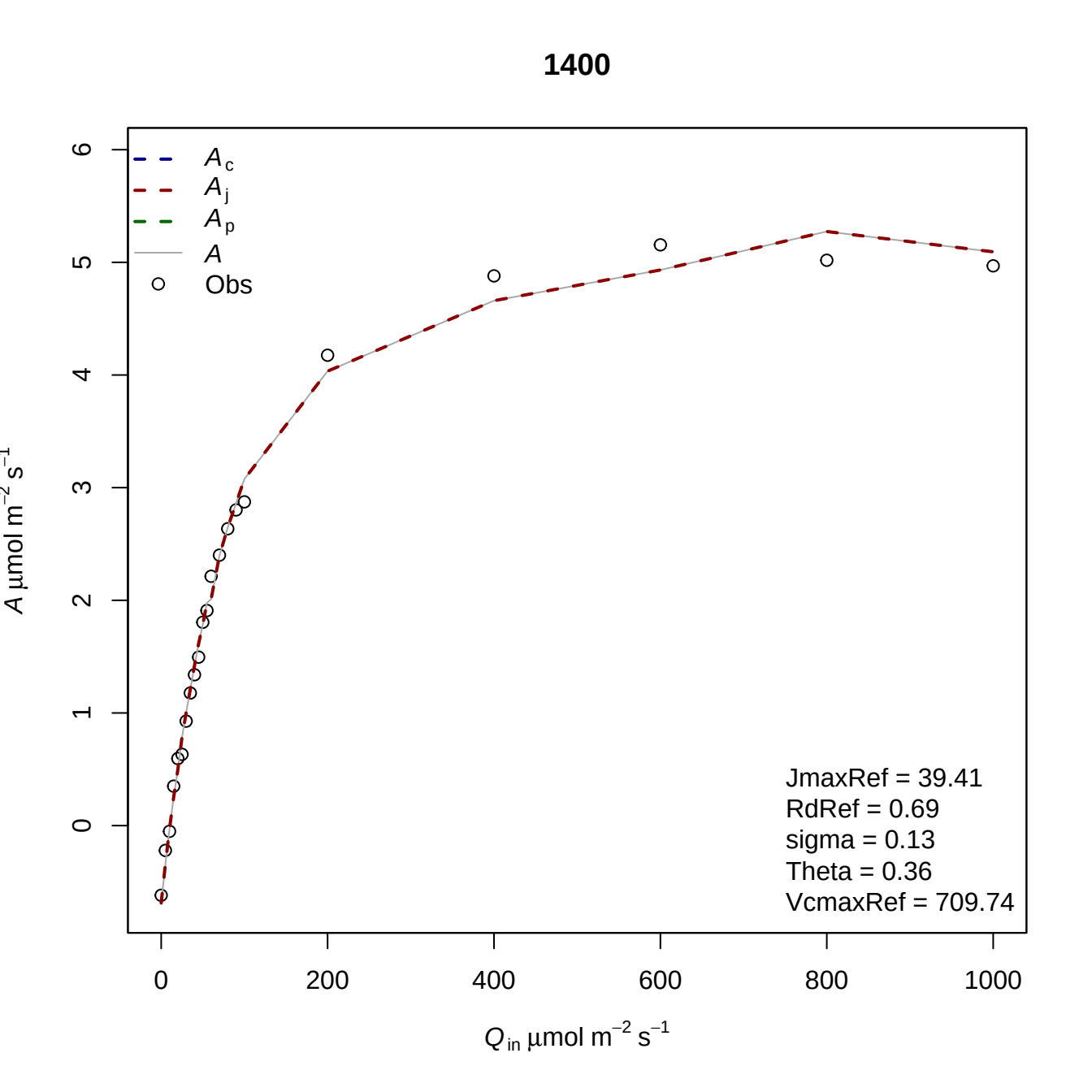

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 105.78
RdRef = 0.61
sigma = 0.1
Theta = 0.17
VcmaxRef = 47.63
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

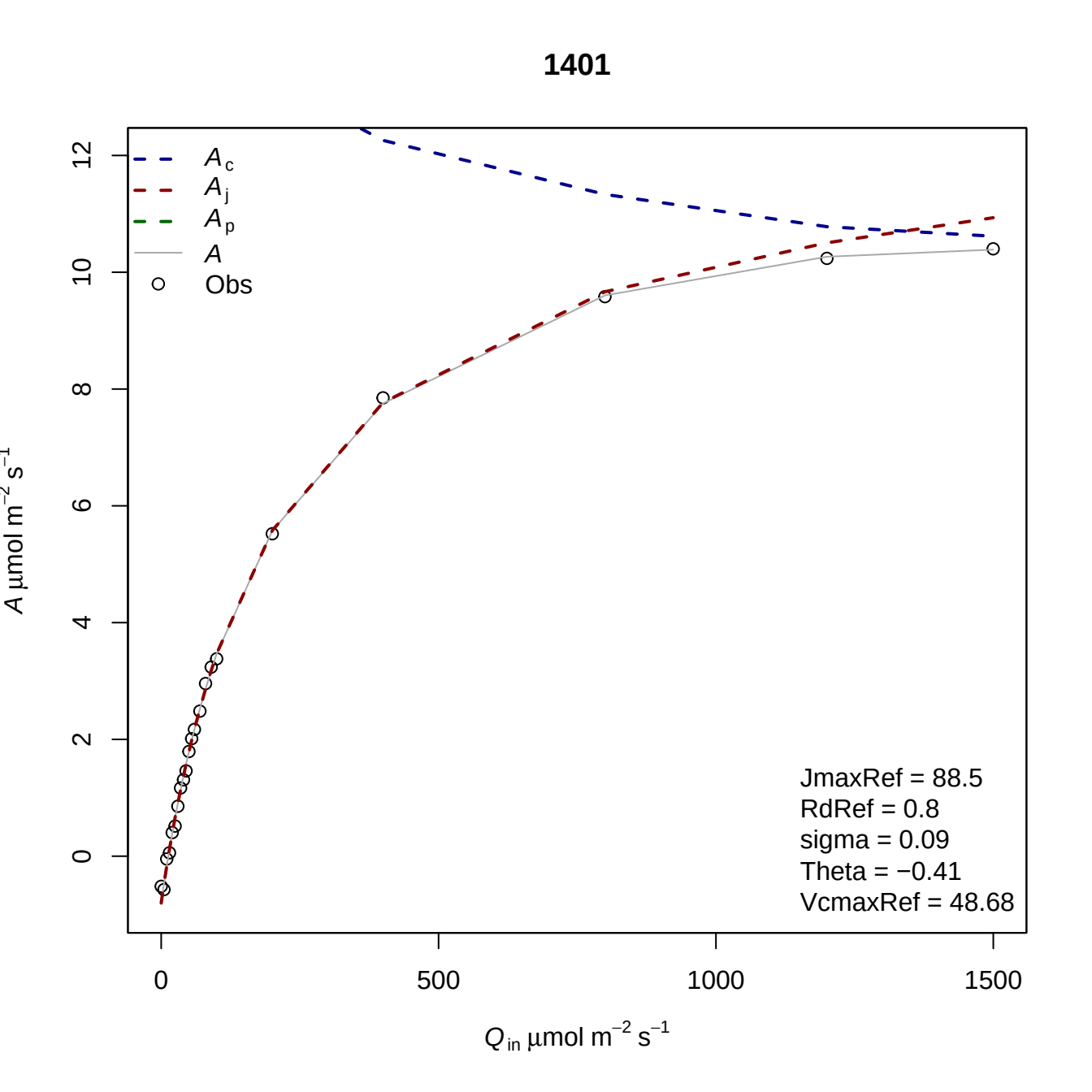

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 40.97
RdRef = 0.4
sigma = 0.13
Theta = 0.34
VcmaxRef = 1358075.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

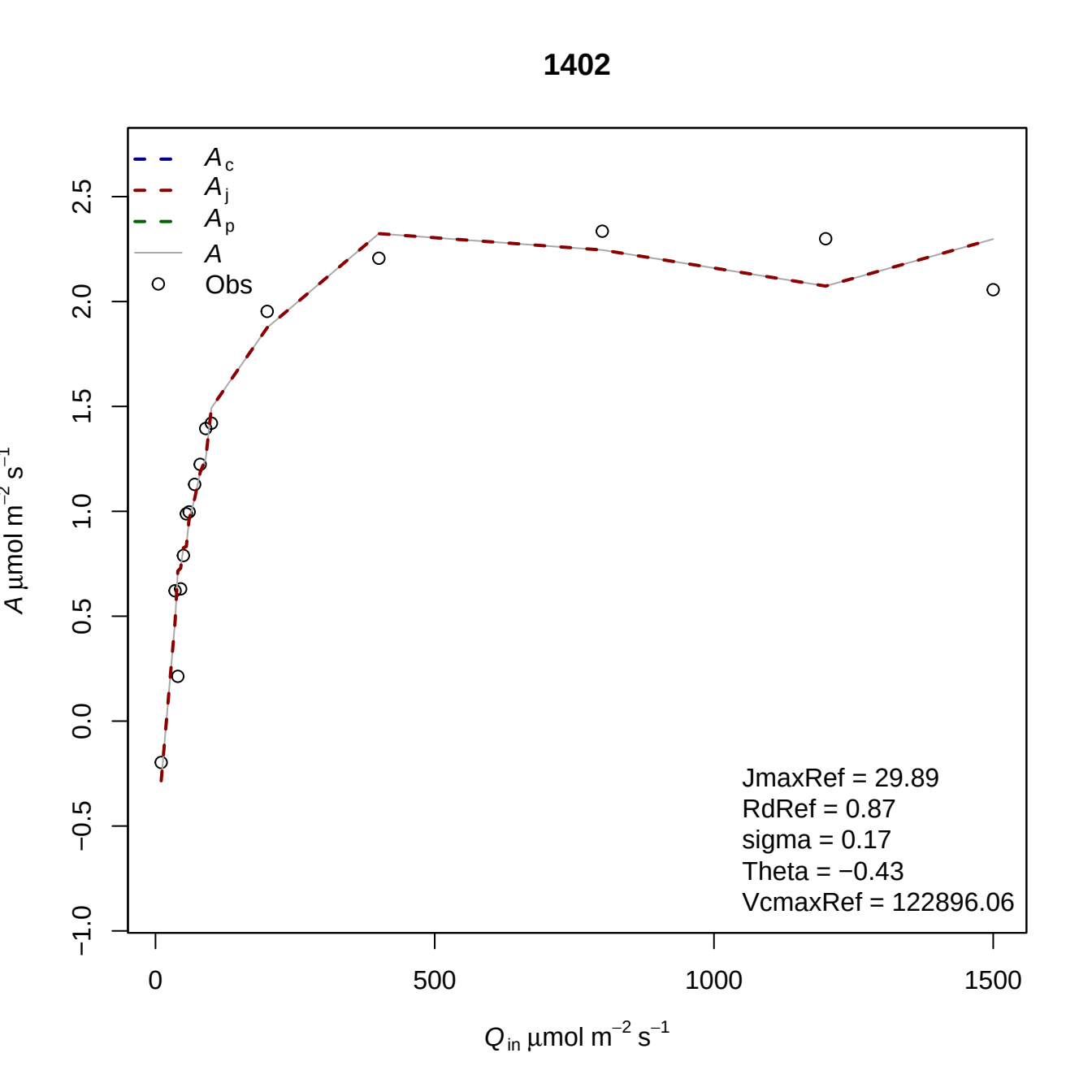

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 40.92
RdRef = 0.55
sigma = 0.12
Theta = 0.12
VcmaxRef = 22.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

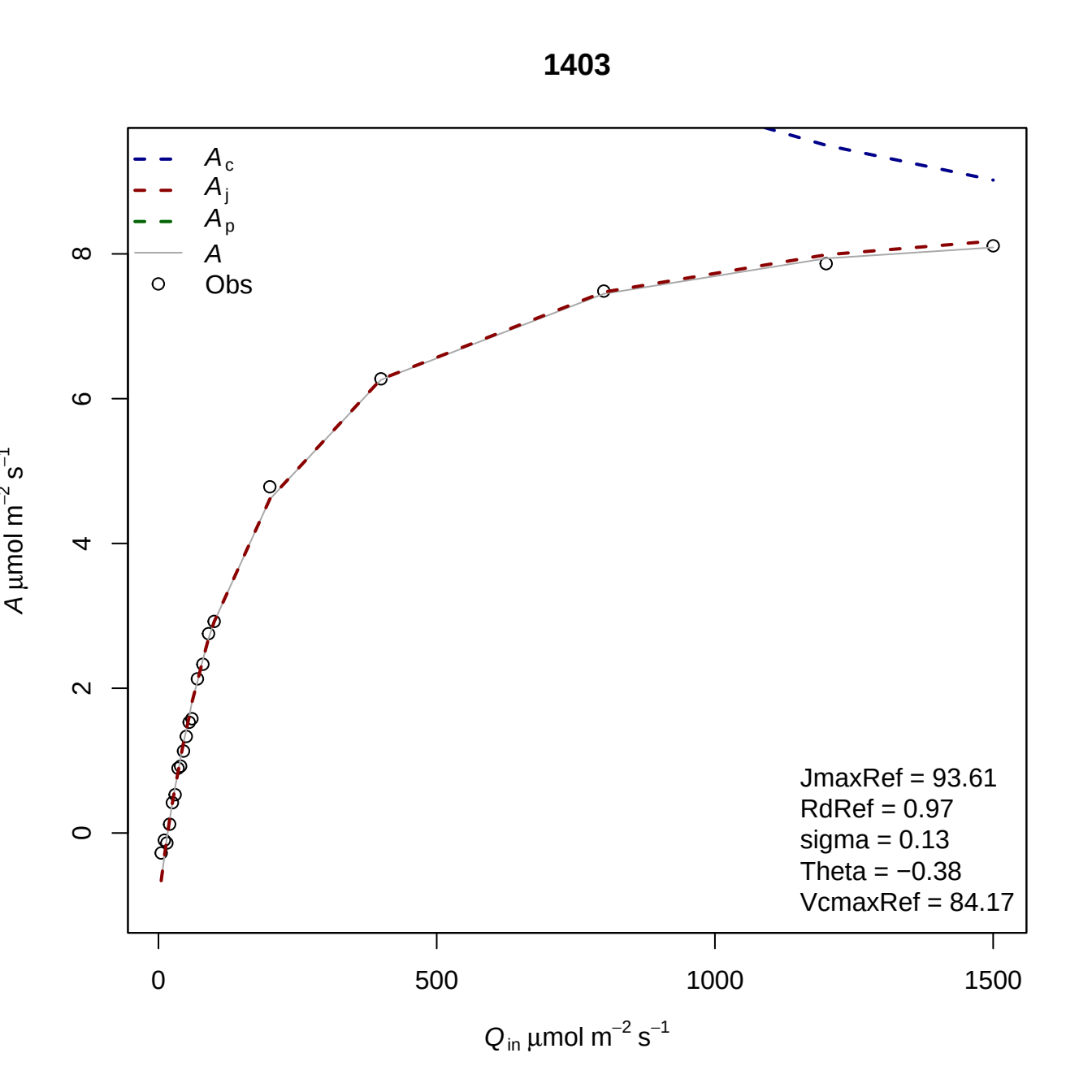

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 56.71
RdRef = 0.92
sigma = 0.23
Theta = 0.38
VcmaxRef = 339742.94
0
2
4
6
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

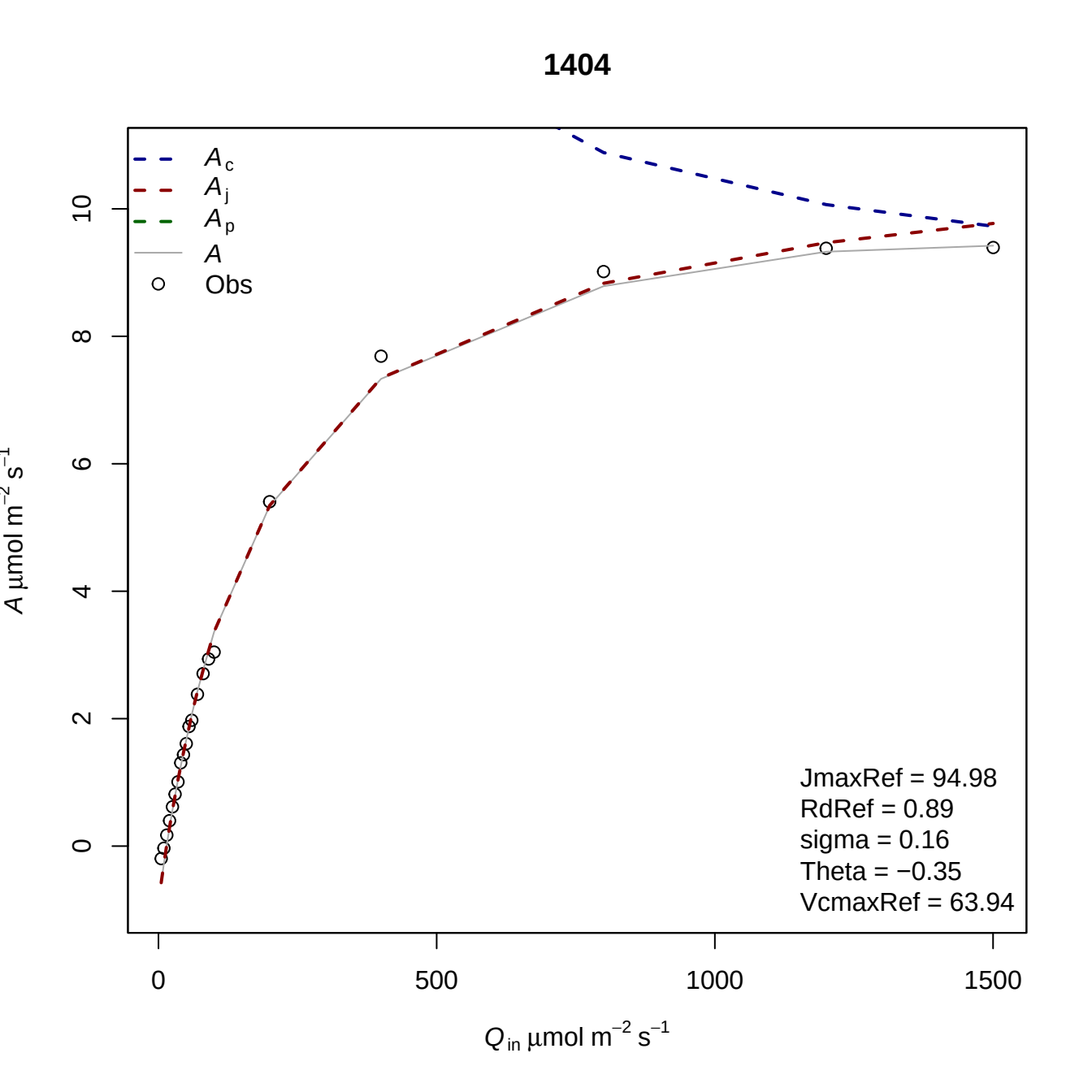

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 51.55
RdRef = 0.36
sigma = 0.14
Theta = 0.45
VcmaxRef = 103642.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

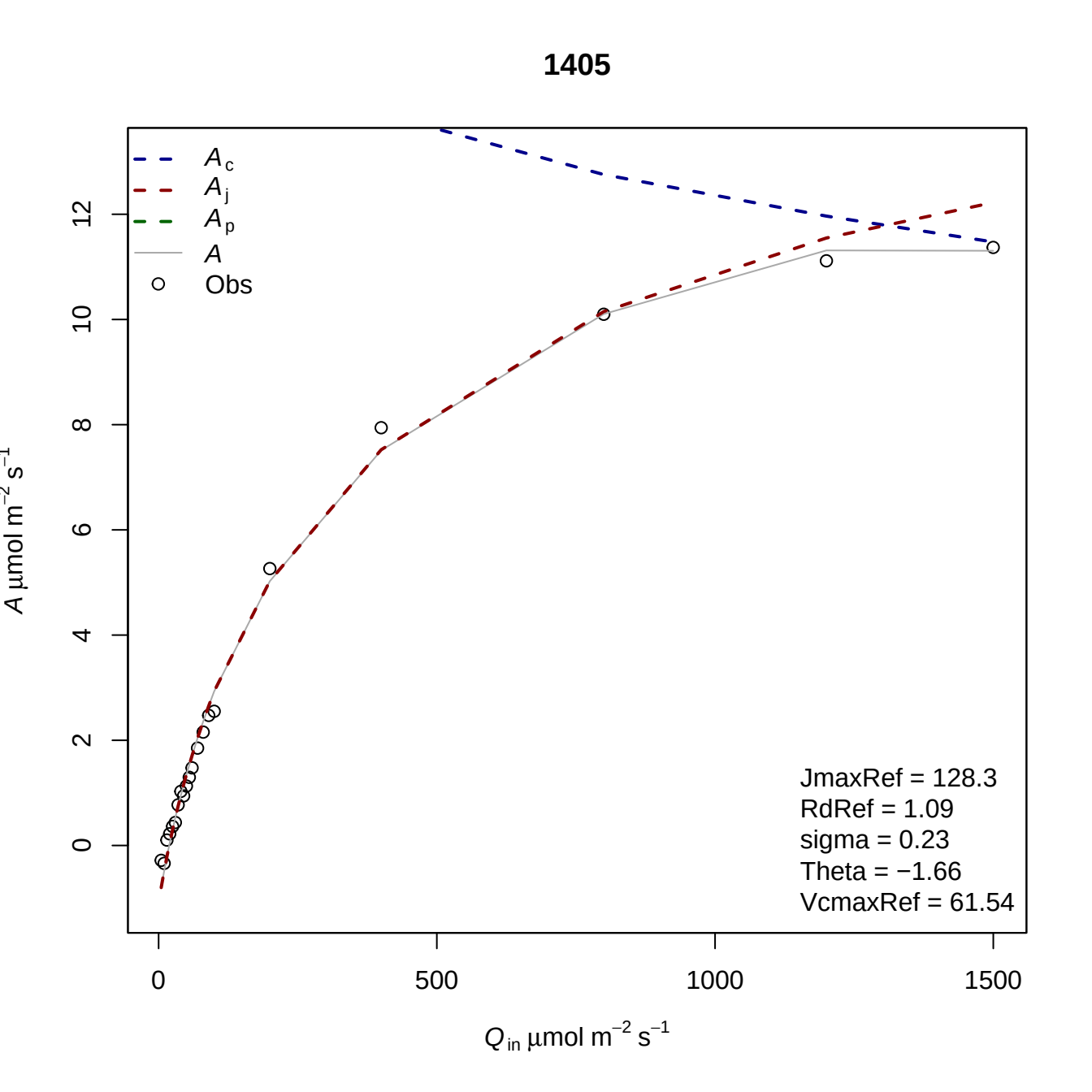

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 39.41
RdRef = 0.69
sigma = 0.13
Theta = 0.36
VcmaxRef = 709.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

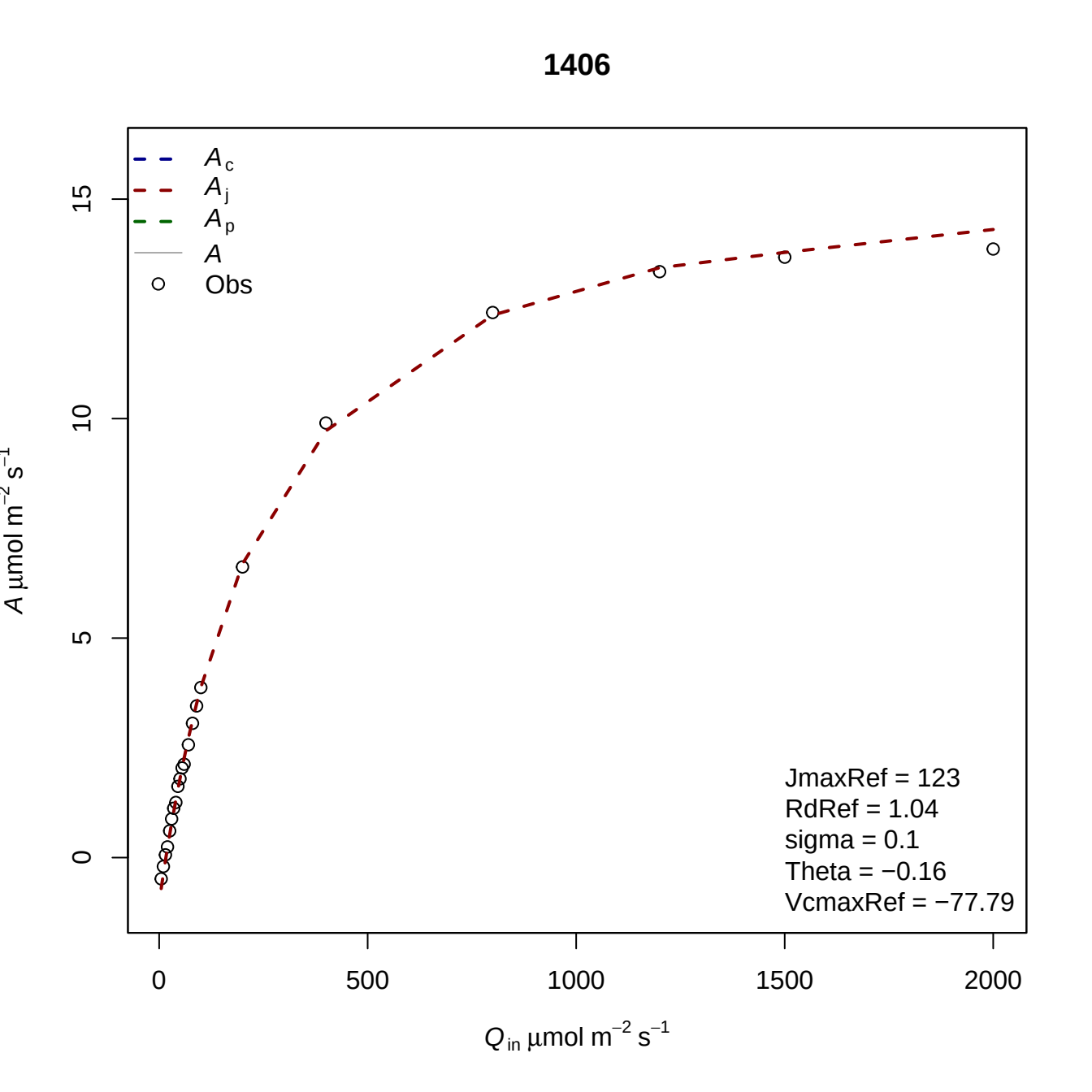

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 88.5
RdRef = 0.8
sigma = 0.09
Theta = −0.41
VcmaxRef = 48.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

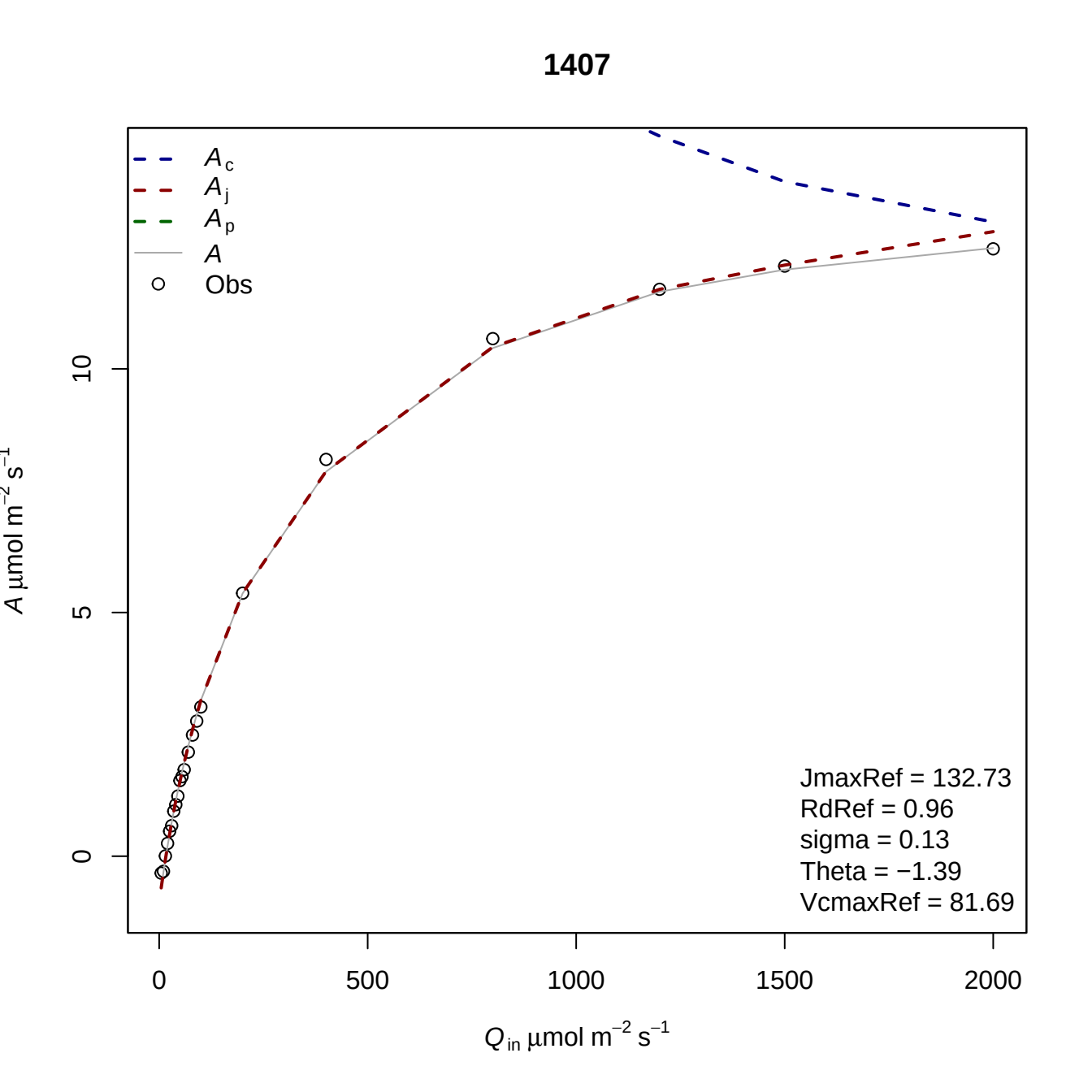

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 29.89
RdRef = 0.87
sigma = 0.17
Theta = −0.43
VcmaxRef = 122896.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500

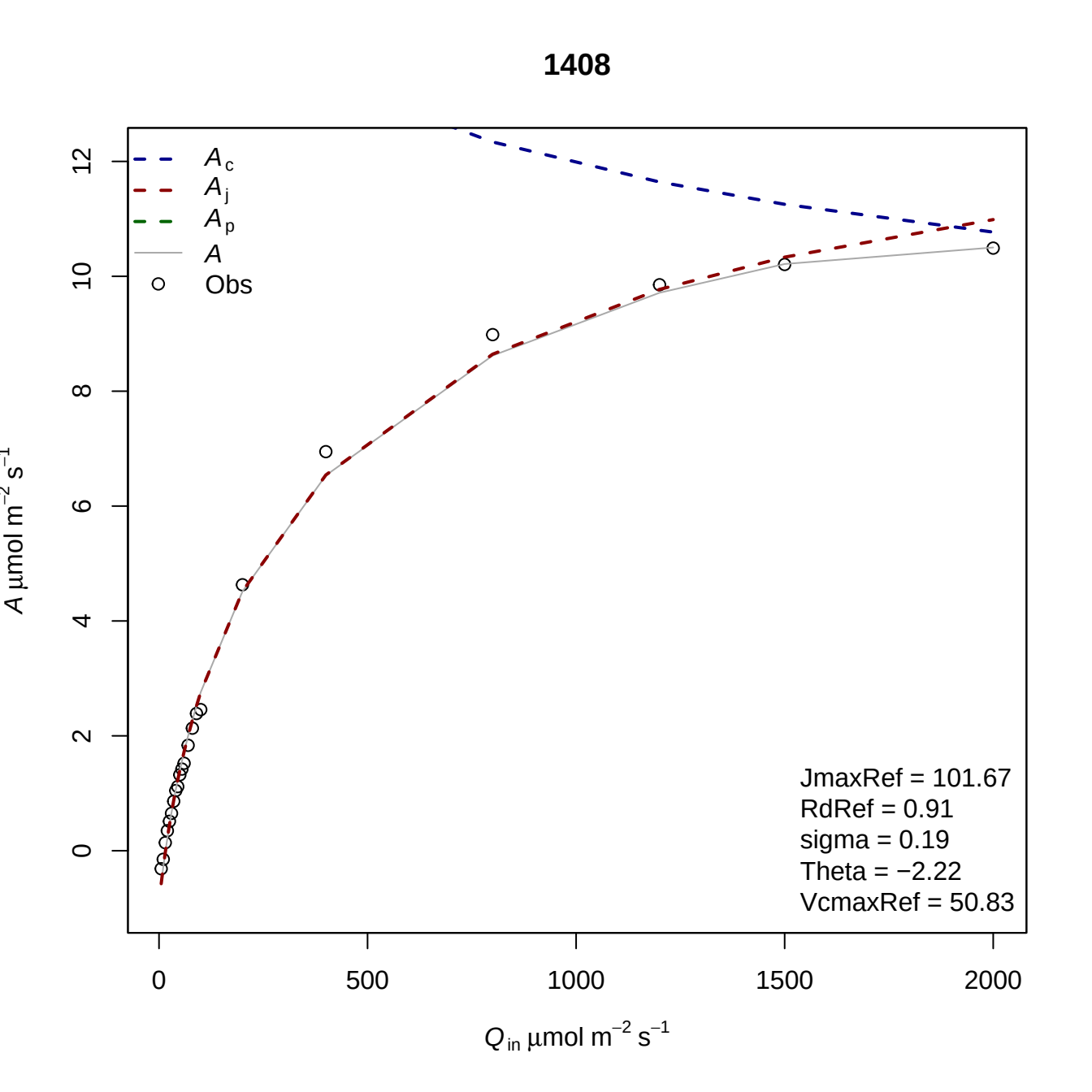

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 93.61
RdRef = 0.97
sigma = 0.13
Theta = −0.38
VcmaxRef = 84.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

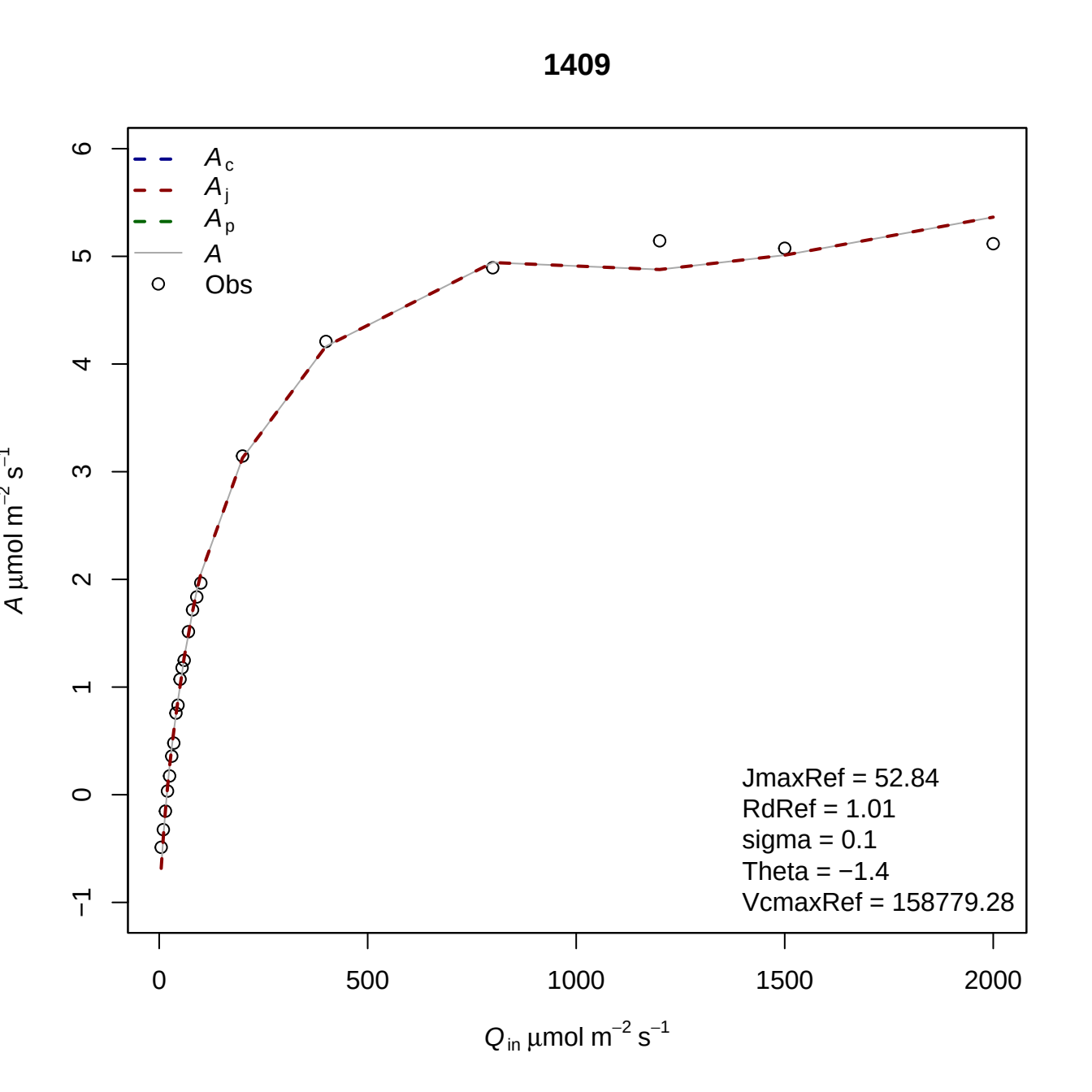

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 94.98
RdRef = 0.89
sigma = 0.16
Theta = −0.35
VcmaxRef = 63.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

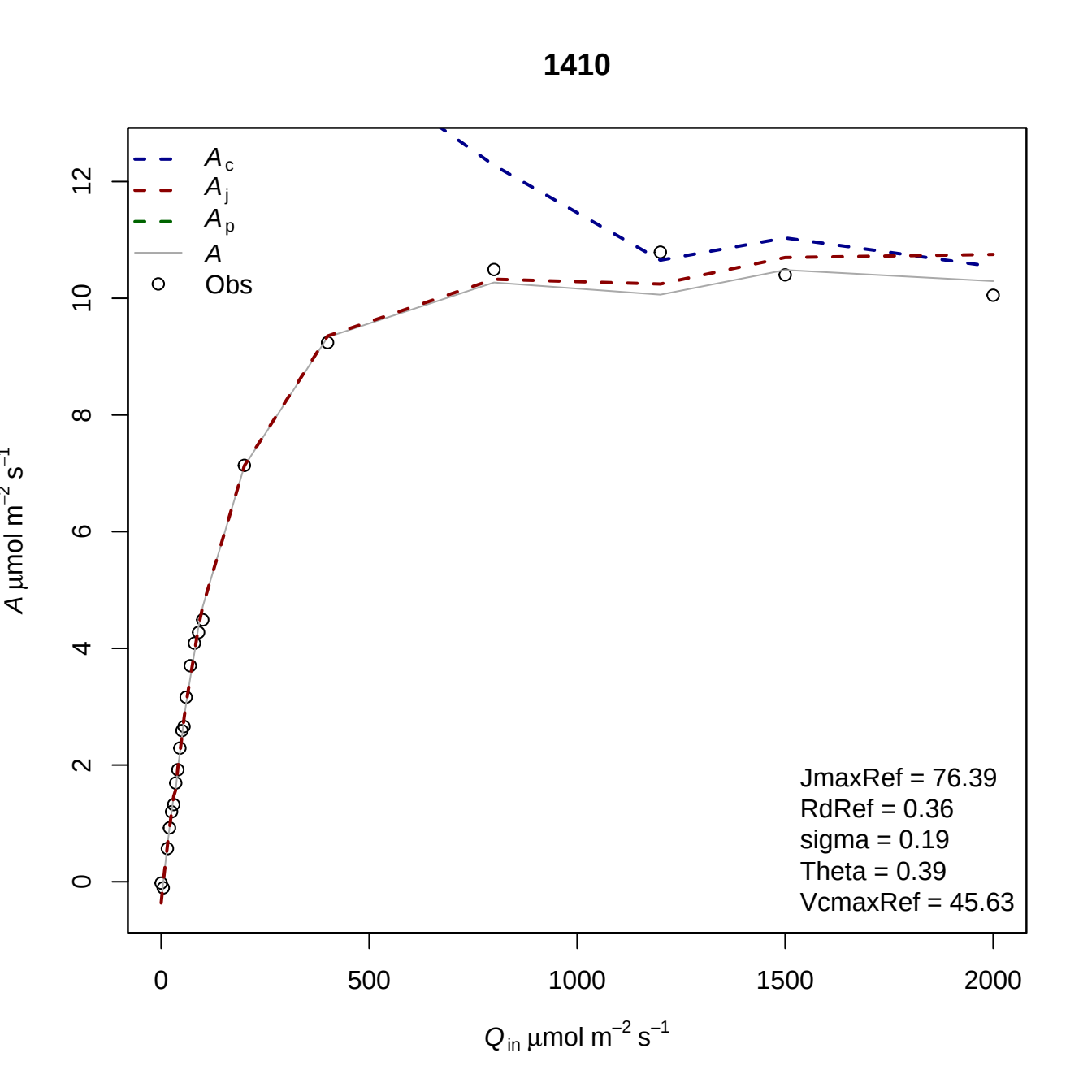

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 128.3
RdRef = 1.09
sigma = 0.23
Theta = −1.66
VcmaxRef = 61.54
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

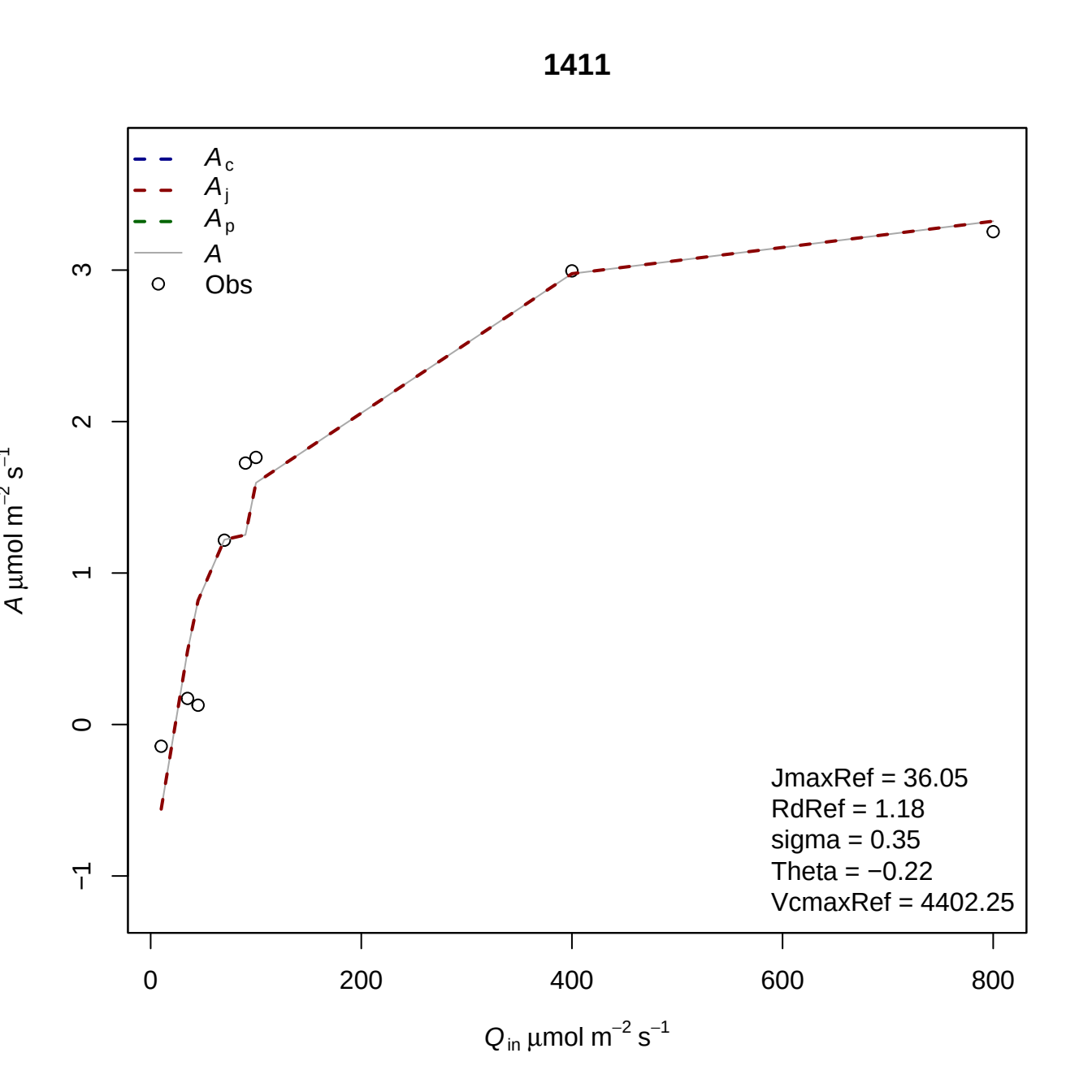

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 123
RdRef = 1.04
sigma = 0.1
Theta = −0.16
VcmaxRef = −77.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

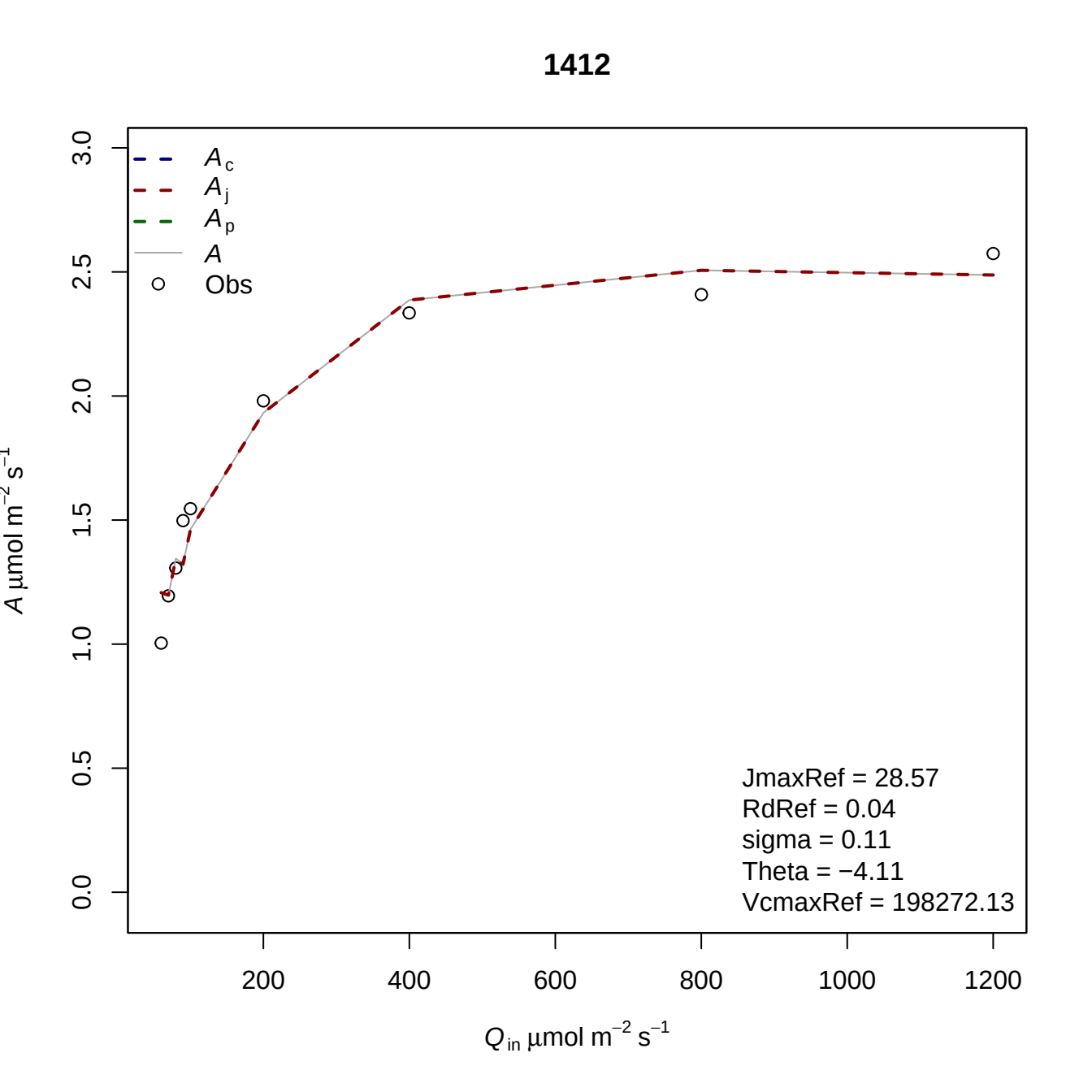

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 132.73
RdRef = 0.96
sigma = 0.13
Theta = −1.39
VcmaxRef = 81.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

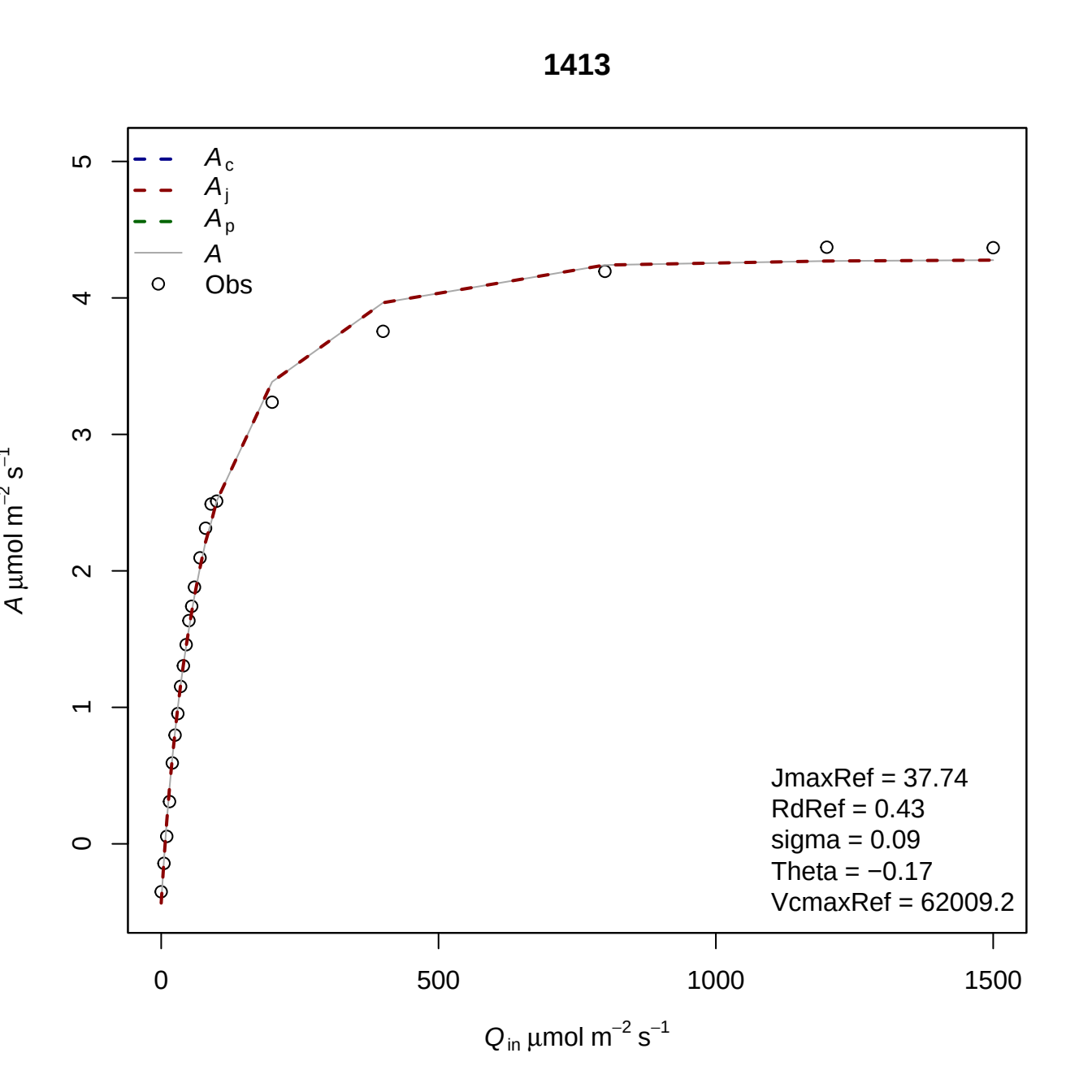

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 101.67
RdRef = 0.91
sigma = 0.19
Theta = −2.22
VcmaxRef = 50.83
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

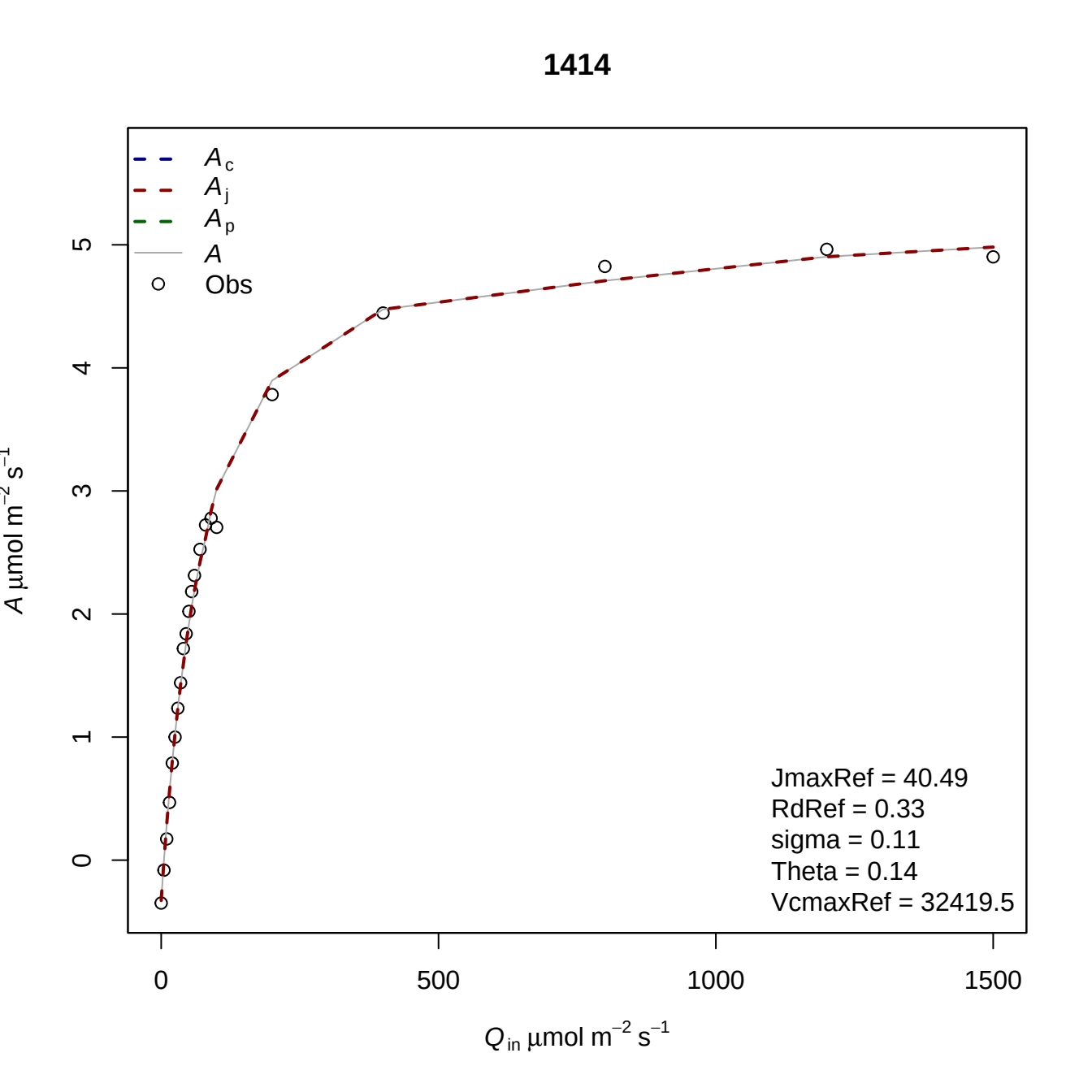

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 52.84
RdRef = 1.01
sigma = 0.1
Theta = −1.4
VcmaxRef = 158779.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

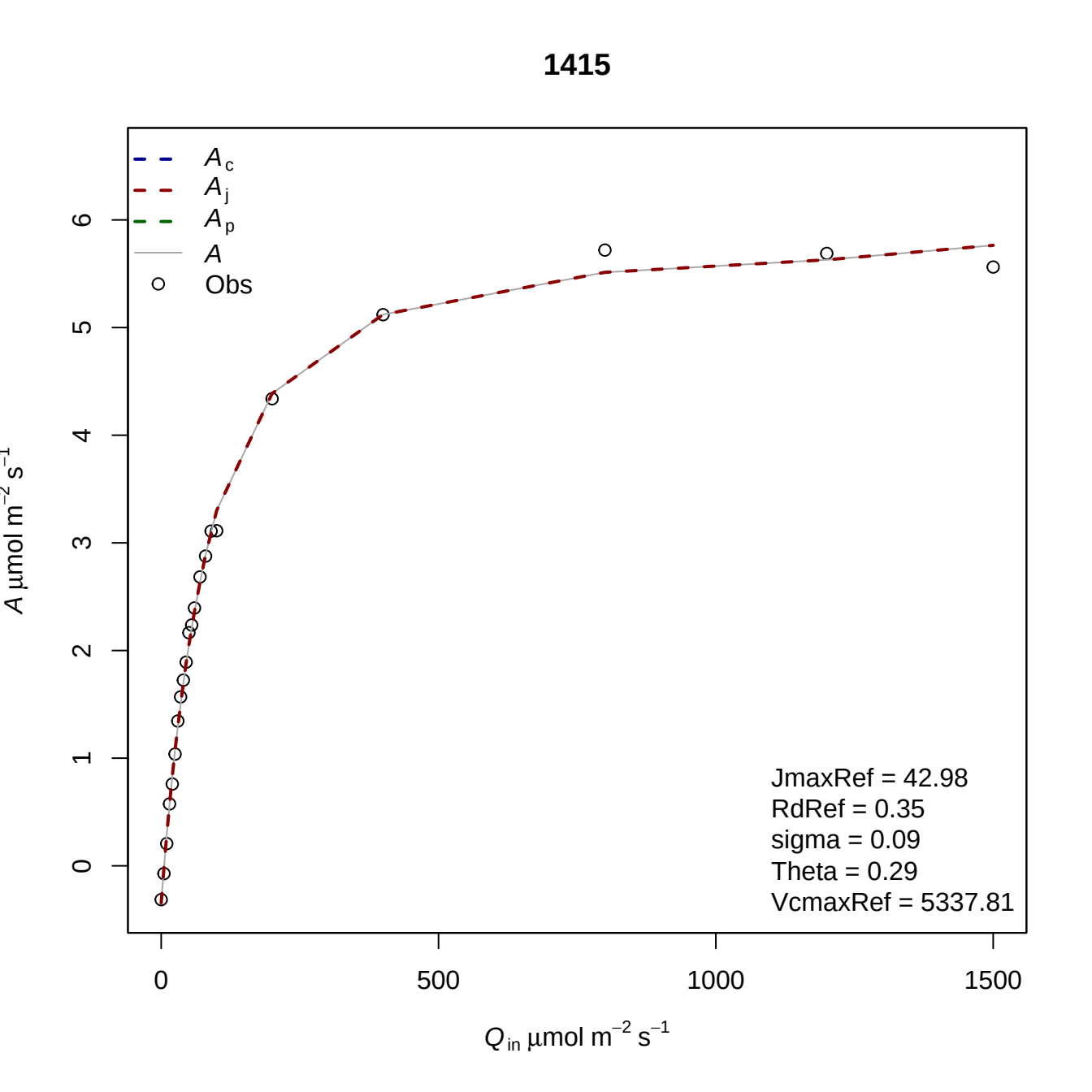

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 76.39
RdRef = 0.36
sigma = 0.19
Theta = 0.39
VcmaxRef = 45.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

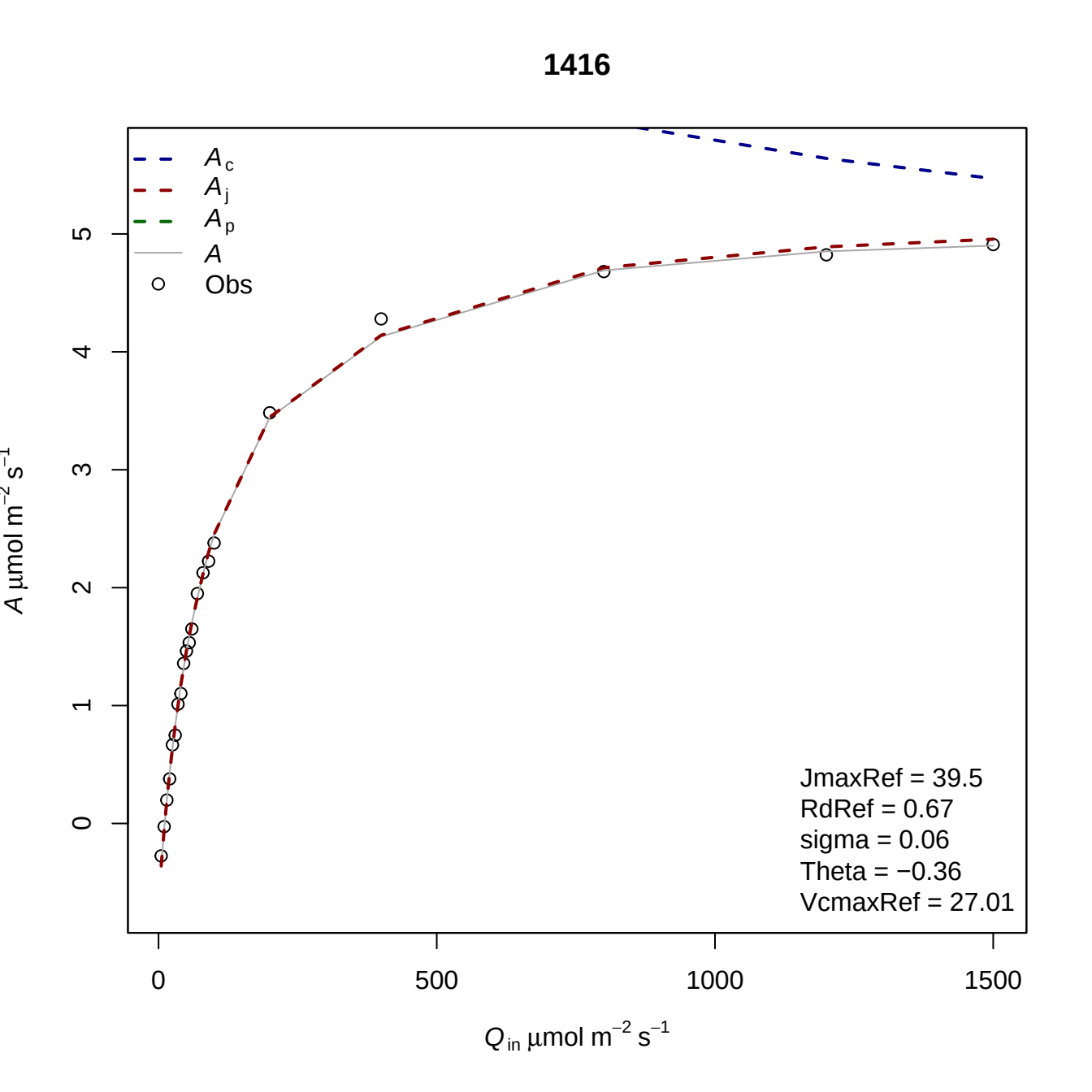

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 36.05
RdRef = 1.18
sigma = 0.35
Theta = −0.22
VcmaxRef = 4402.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

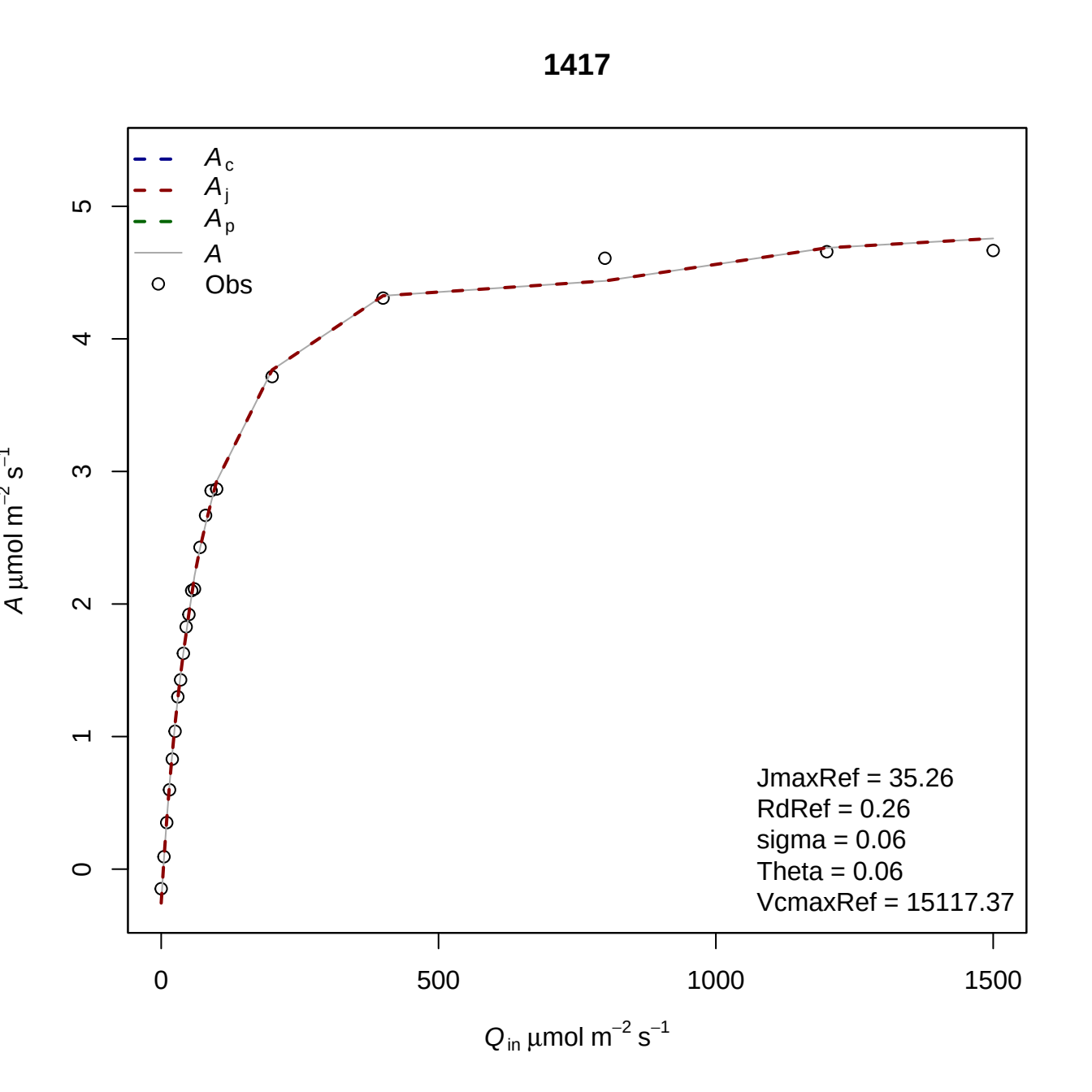

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 28.57
RdRef = 0.04
sigma = 0.11
Theta = −4.11
VcmaxRef = 198272.13
0.0
0.5
1.0
1.5
2.0
2.5
3.0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

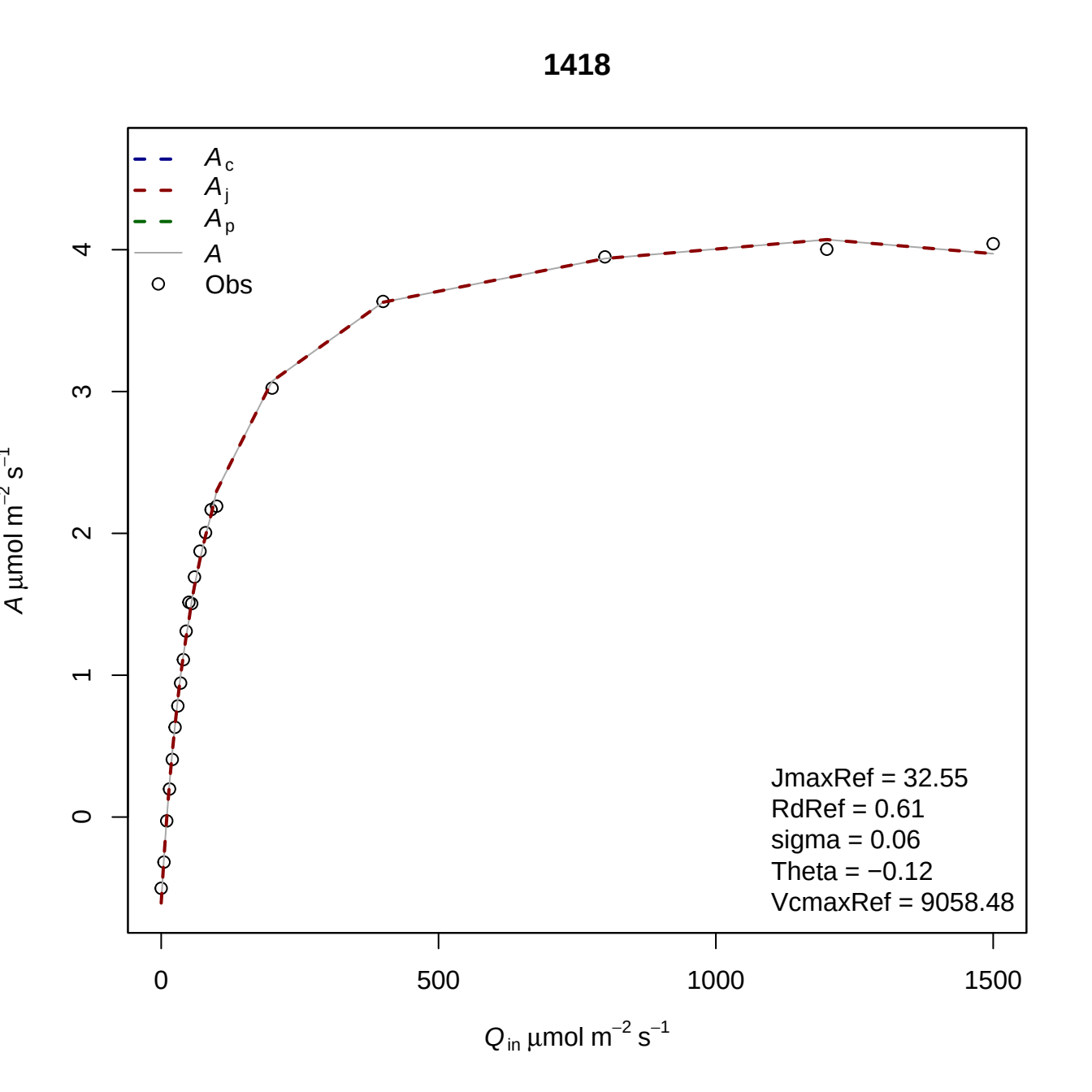

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 37.74
RdRef = 0.43
sigma = 0.09
Theta = −0.17
VcmaxRef = 62009.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

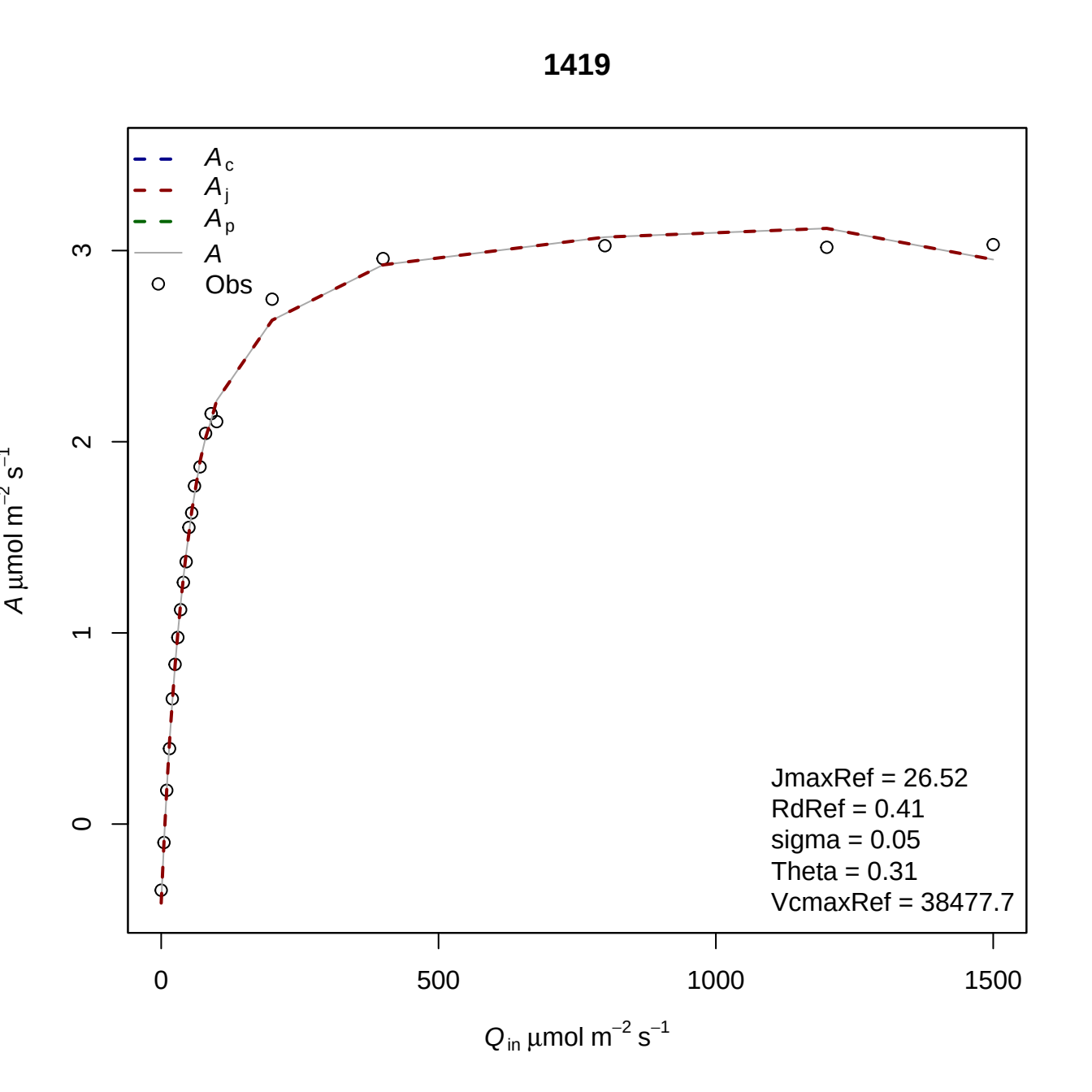

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 40.49
RdRef = 0.33
sigma = 0.11
Theta = 0.14
VcmaxRef = 32419.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

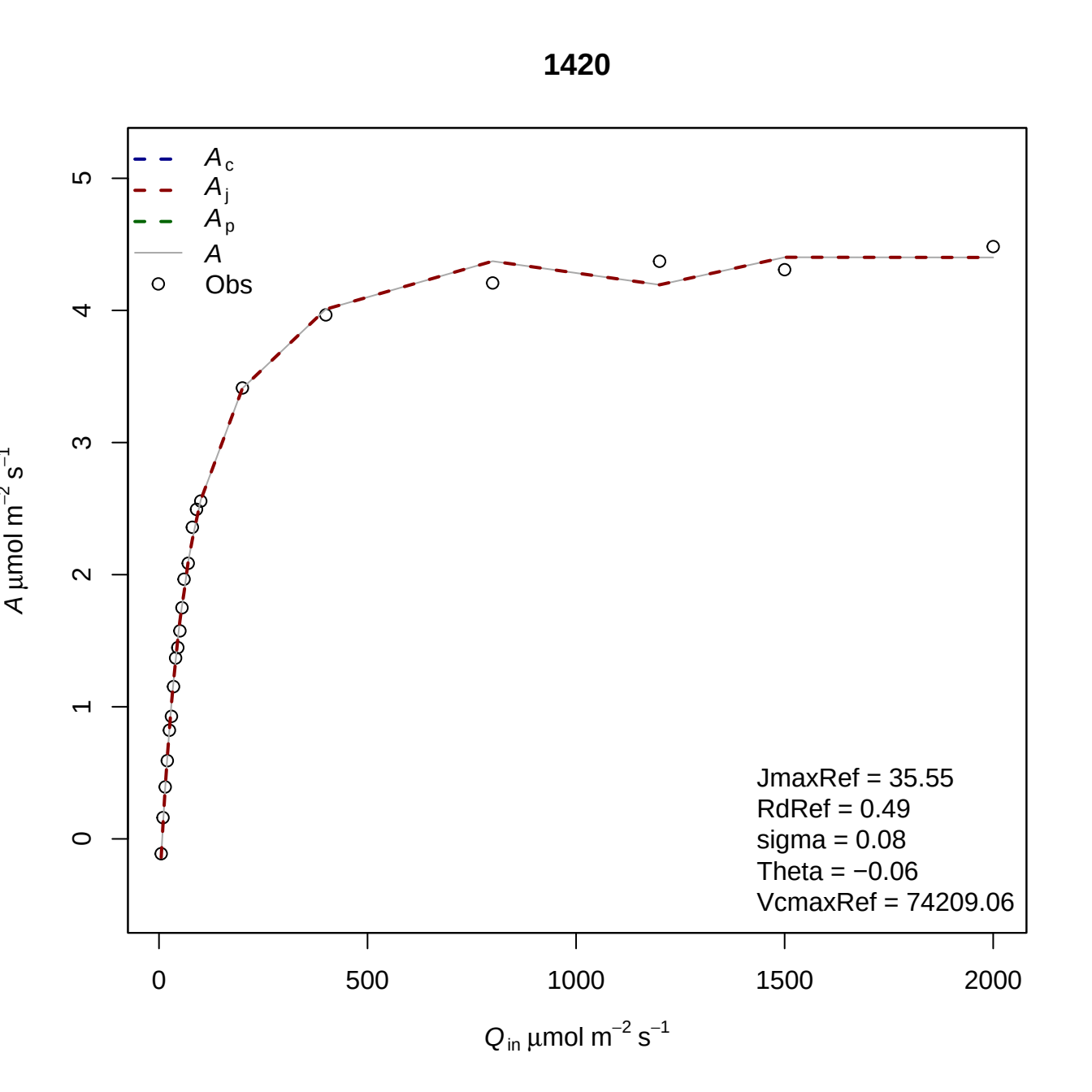

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 42.98
RdRef = 0.35
sigma = 0.09
Theta = 0.29
VcmaxRef = 5337.81
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

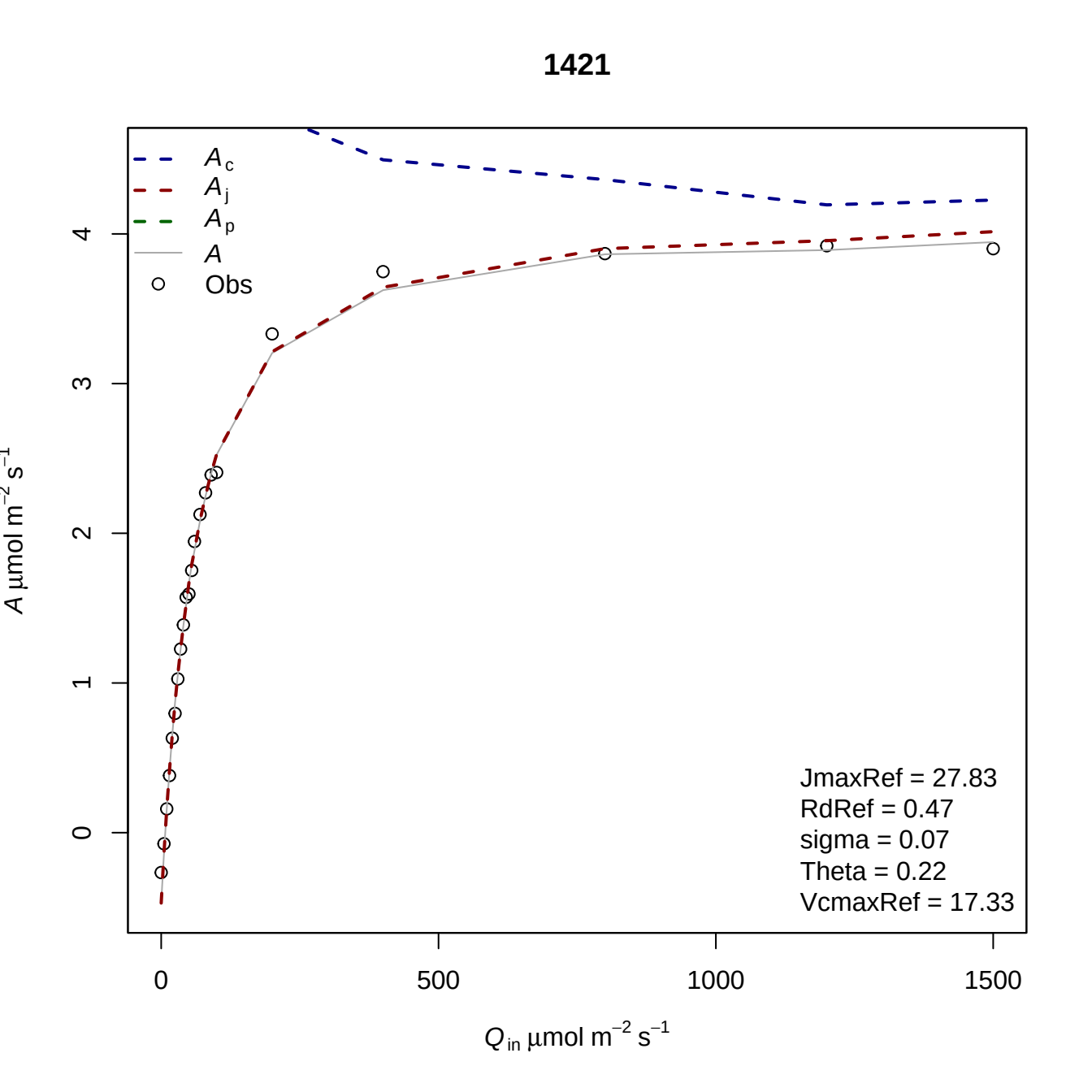

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 39.5
RdRef = 0.67
sigma = 0.06
Theta = −0.36
VcmaxRef = 27.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

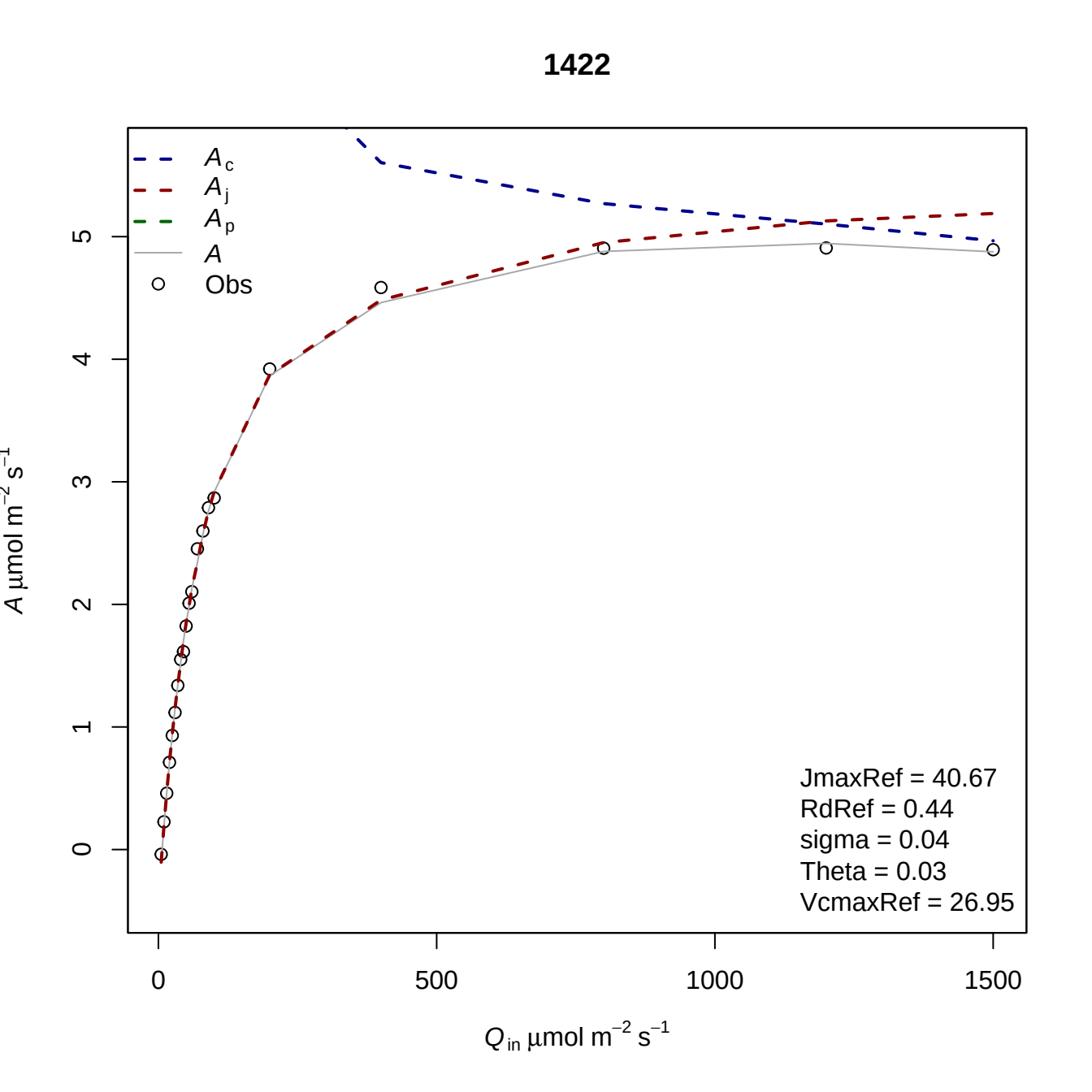

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 35.26
RdRef = 0.26
sigma = 0.06
Theta = 0.06
VcmaxRef = 15117.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

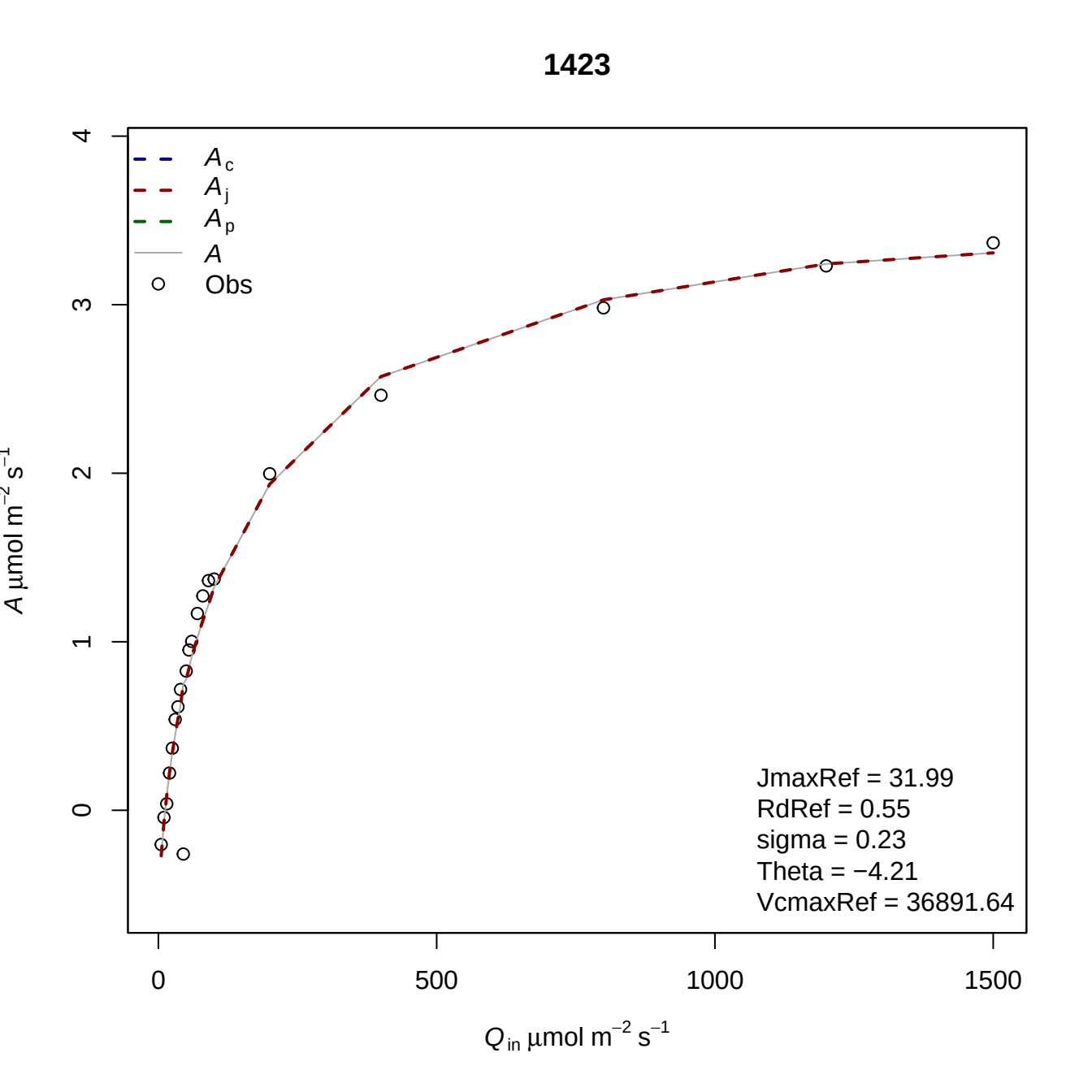

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 32.55
RdRef = 0.61
sigma = 0.06
Theta = −0.12
VcmaxRef = 9058.48
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

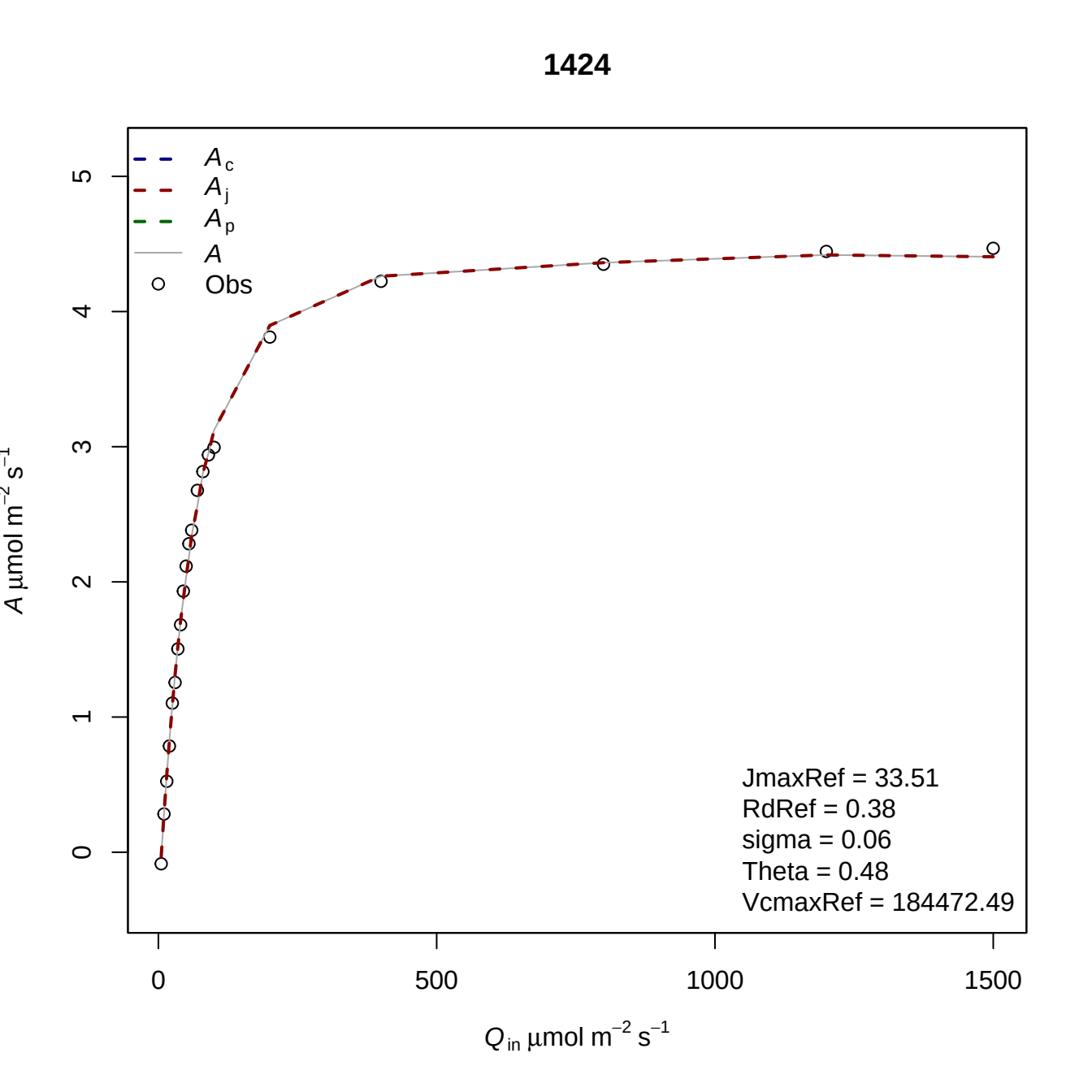

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 26.52
RdRef = 0.41
sigma = 0.05
Theta = 0.31
VcmaxRef = 38477.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

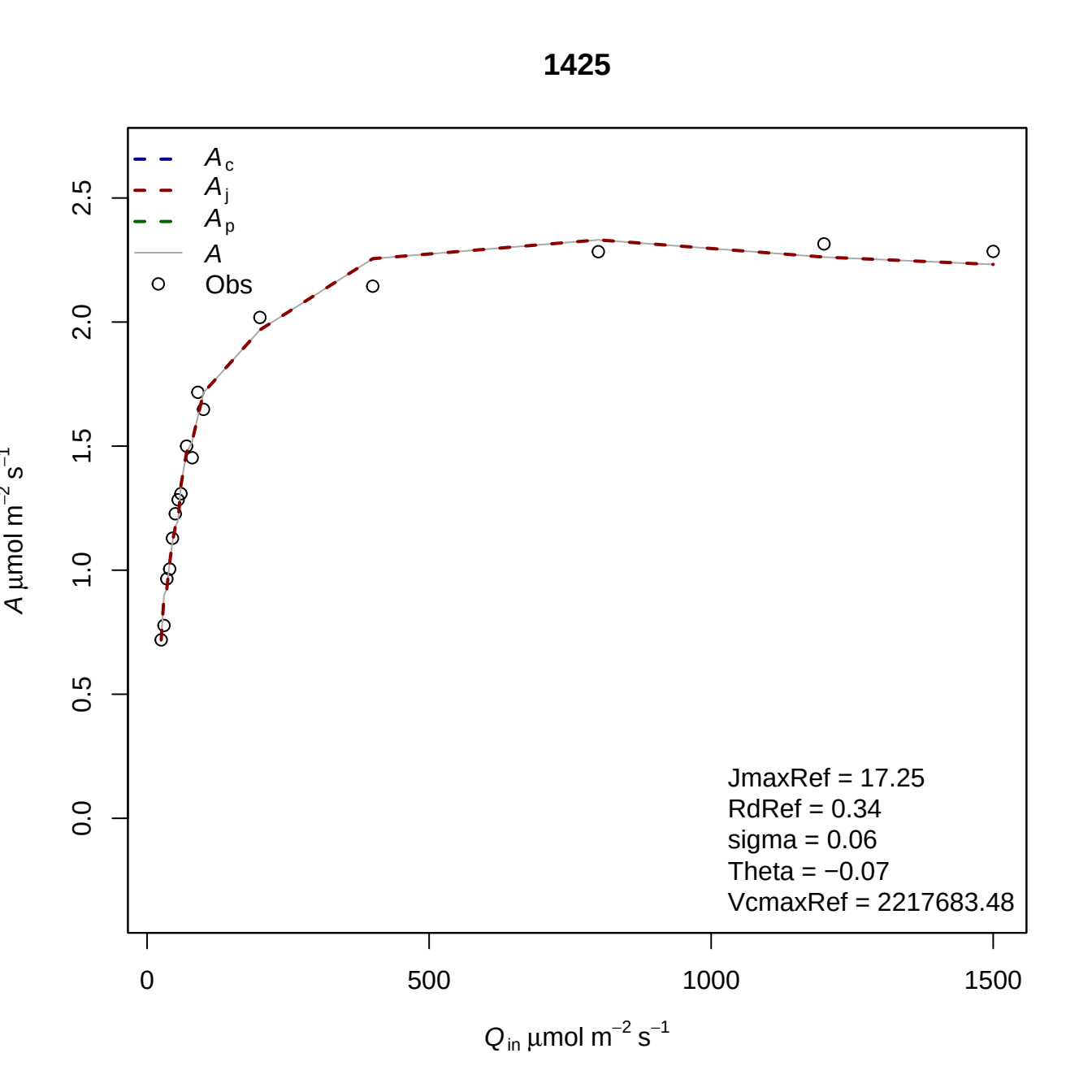

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 35.55
RdRef = 0.49
sigma = 0.08
Theta = −0.06
VcmaxRef = 74209.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500
2000

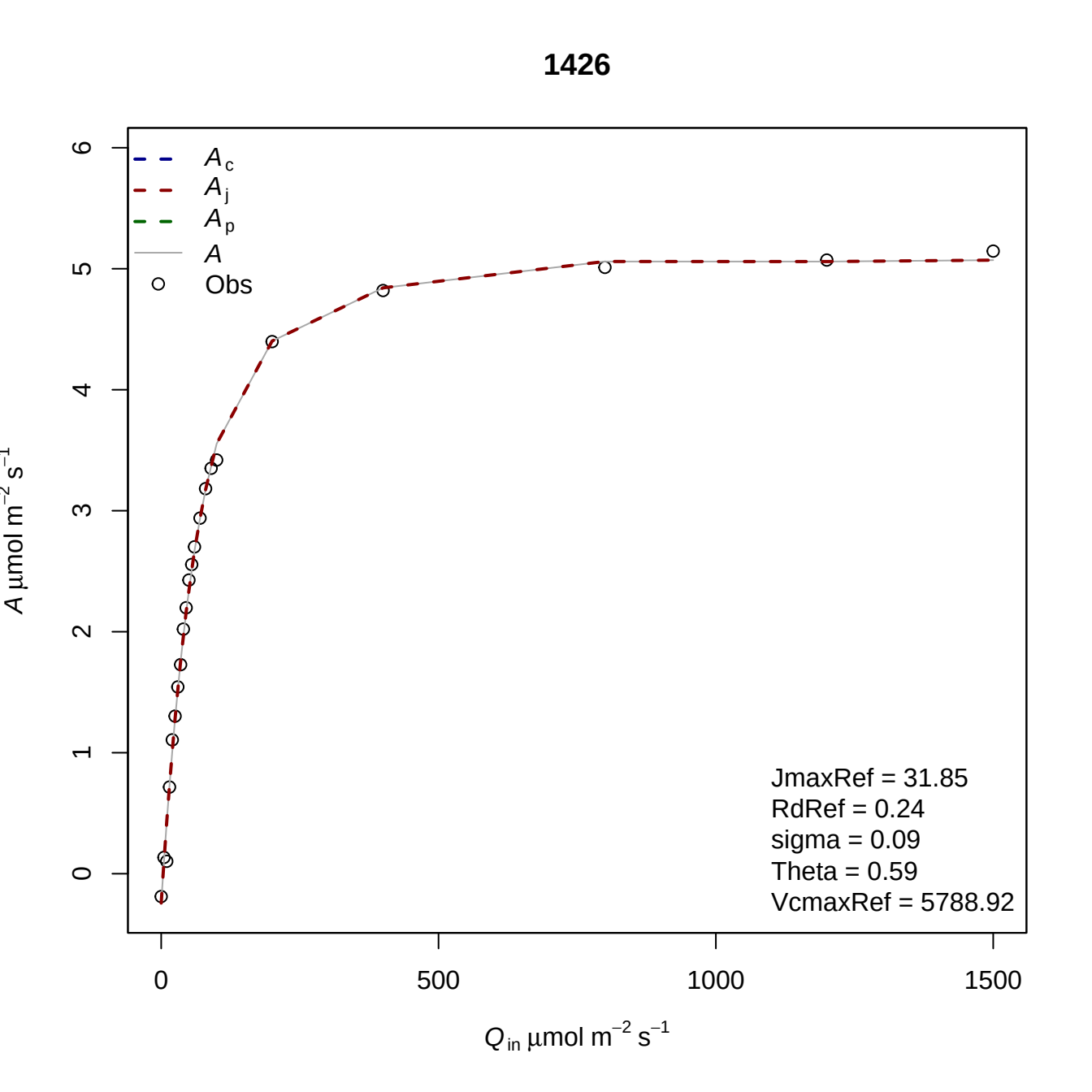

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 27.83
RdRef = 0.47
sigma = 0.07
Theta = 0.22
VcmaxRef = 17.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

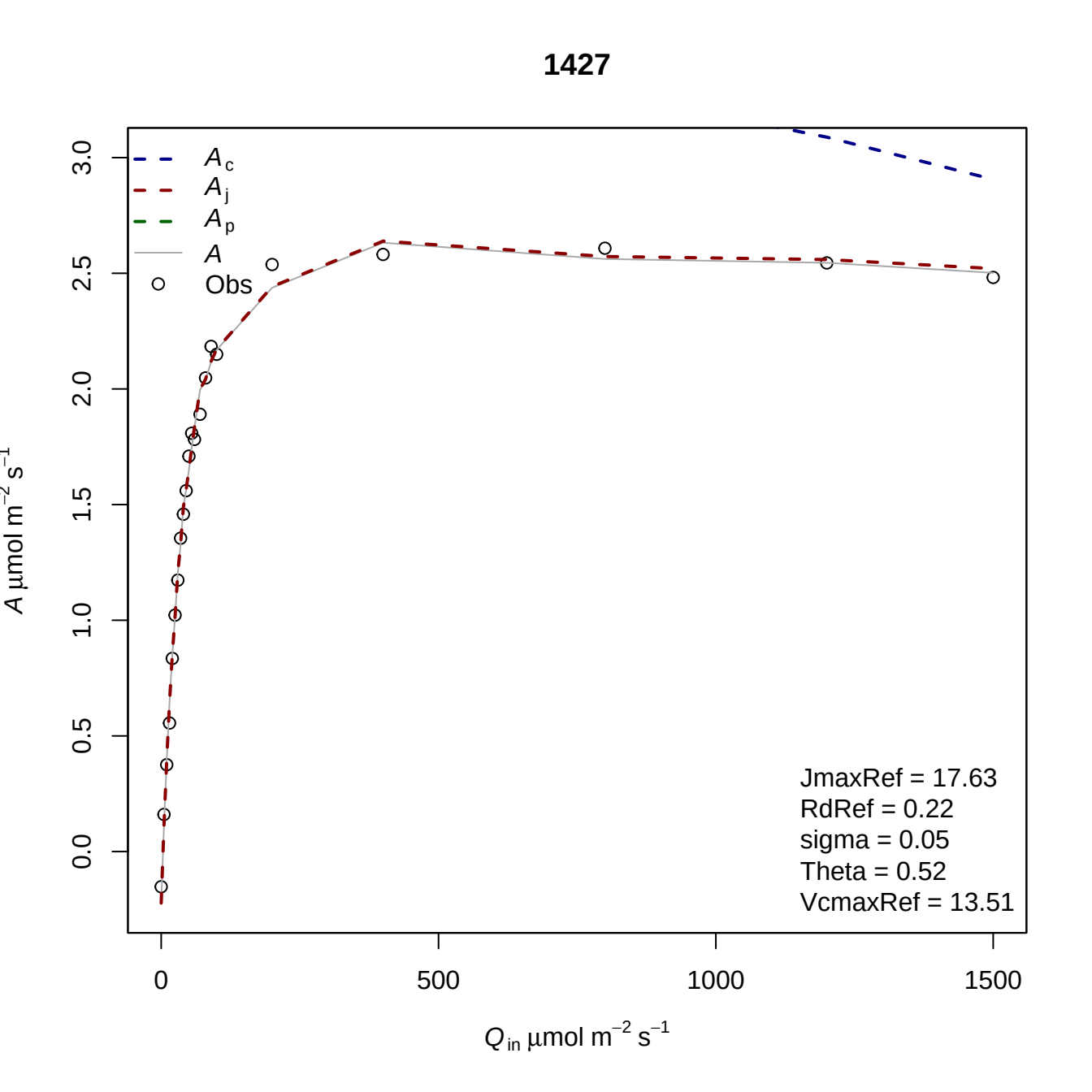

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 40.67
RdRef = 0.44
sigma = 0.04
Theta = 0.03
VcmaxRef = 26.95
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

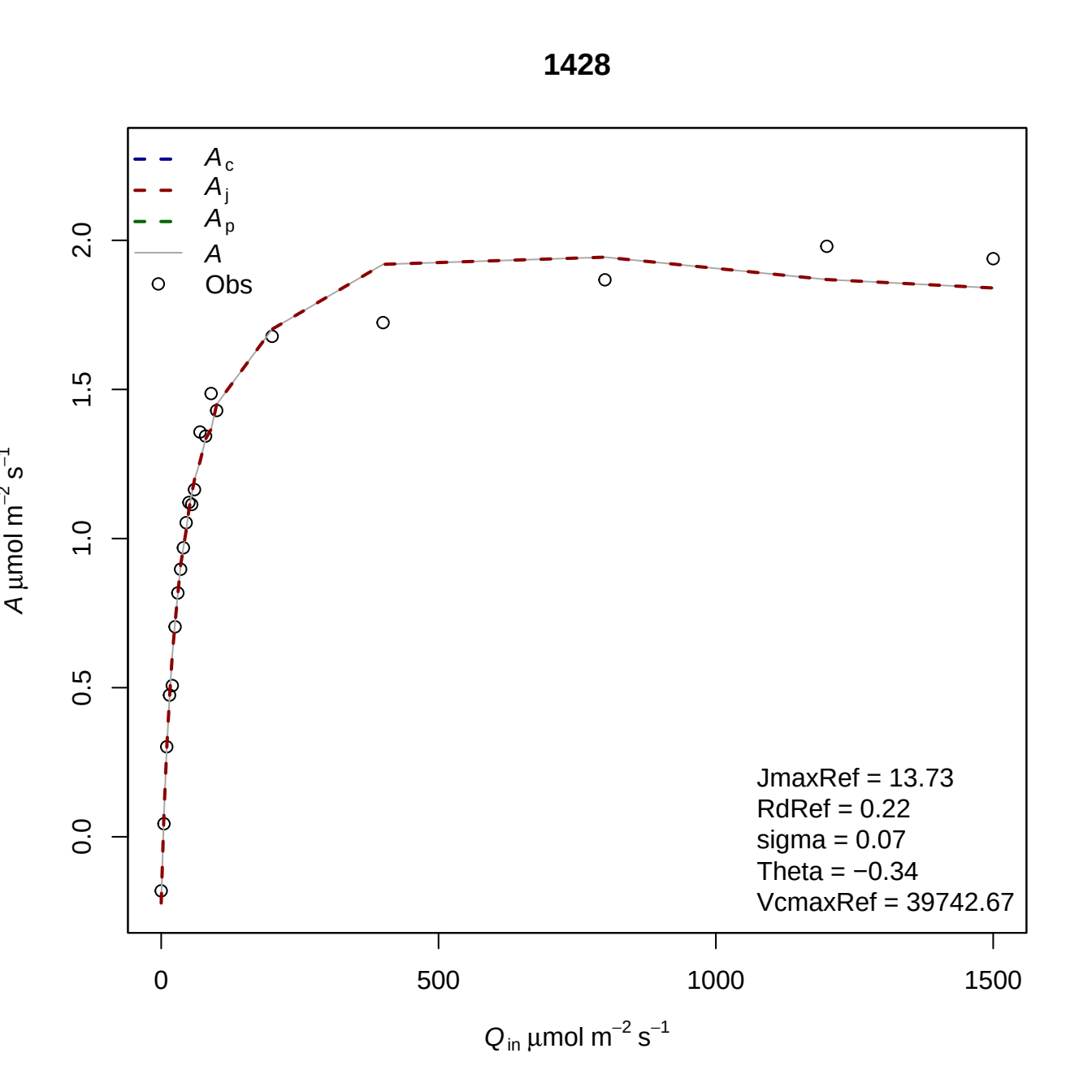

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 31.99
RdRef = 0.55
sigma = 0.23
Theta = −4.21
VcmaxRef = 36891.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

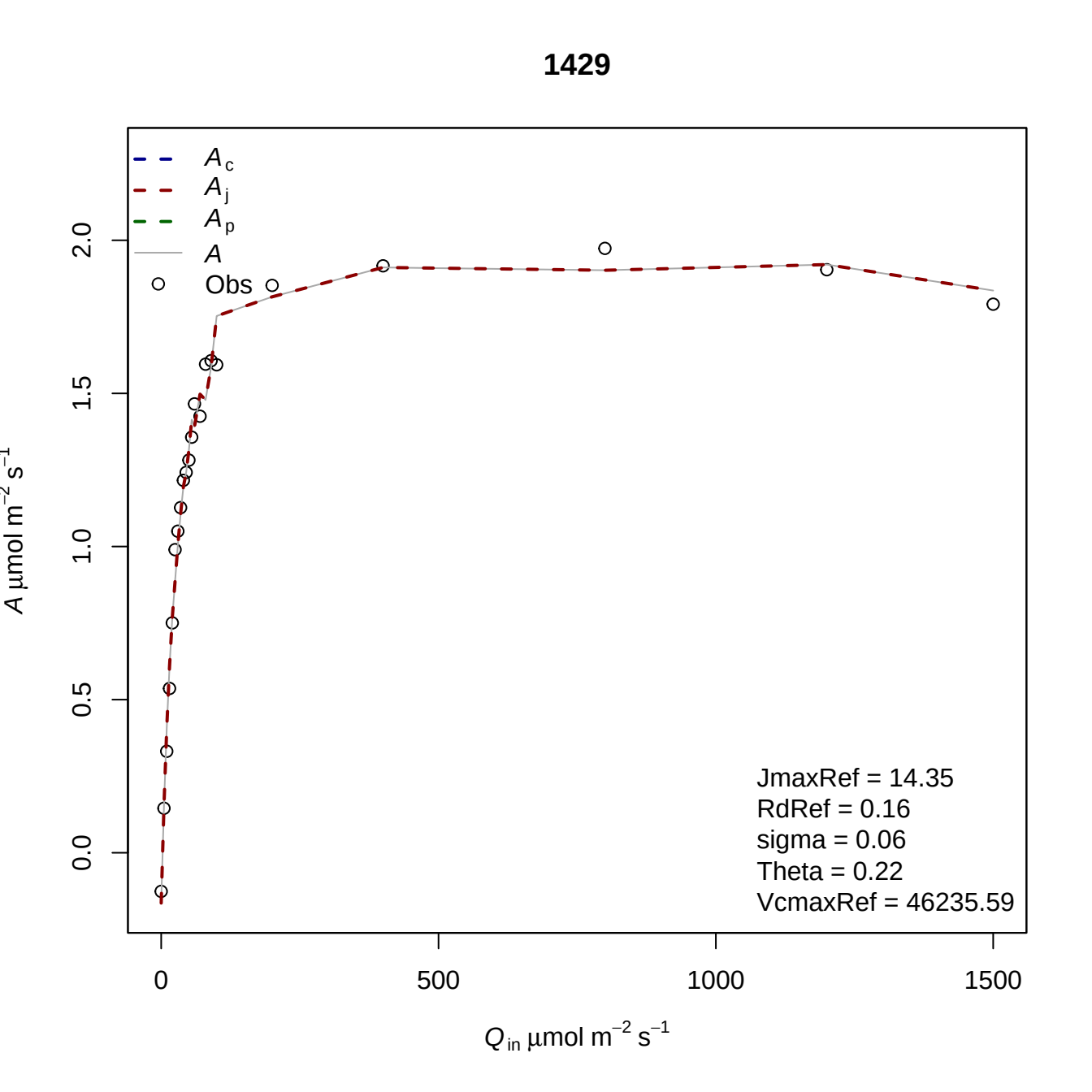

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 33.51
RdRef = 0.38
sigma = 0.06
Theta = 0.48
VcmaxRef = 184472.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

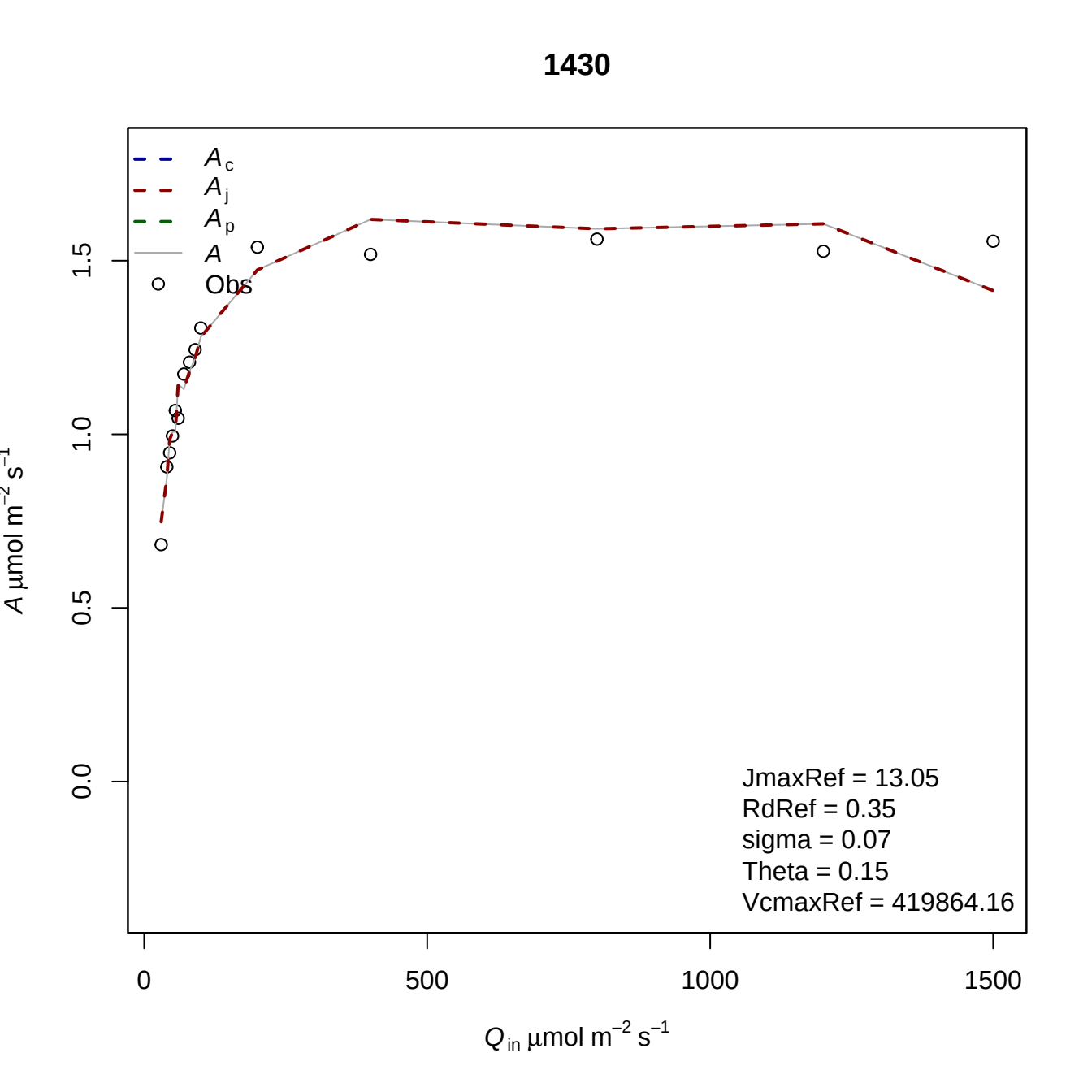

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 17.25
RdRef = 0.34
sigma = 0.06
Theta = −0.07
VcmaxRef = 2217683.48
0.0
0.5
1.0
1.5
2.0
2.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

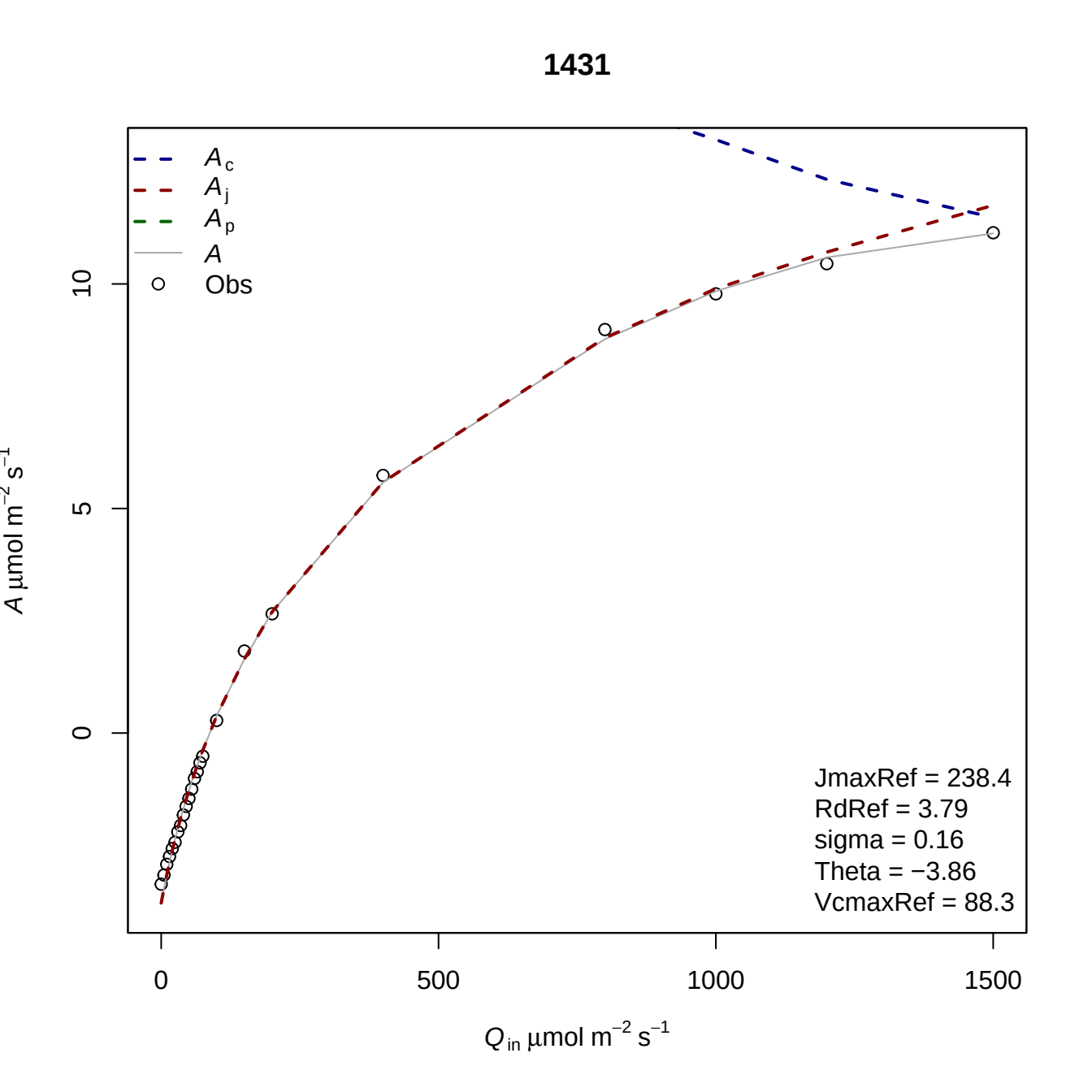

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 31.85
RdRef = 0.24
sigma = 0.09
Theta = 0.59
VcmaxRef = 5788.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

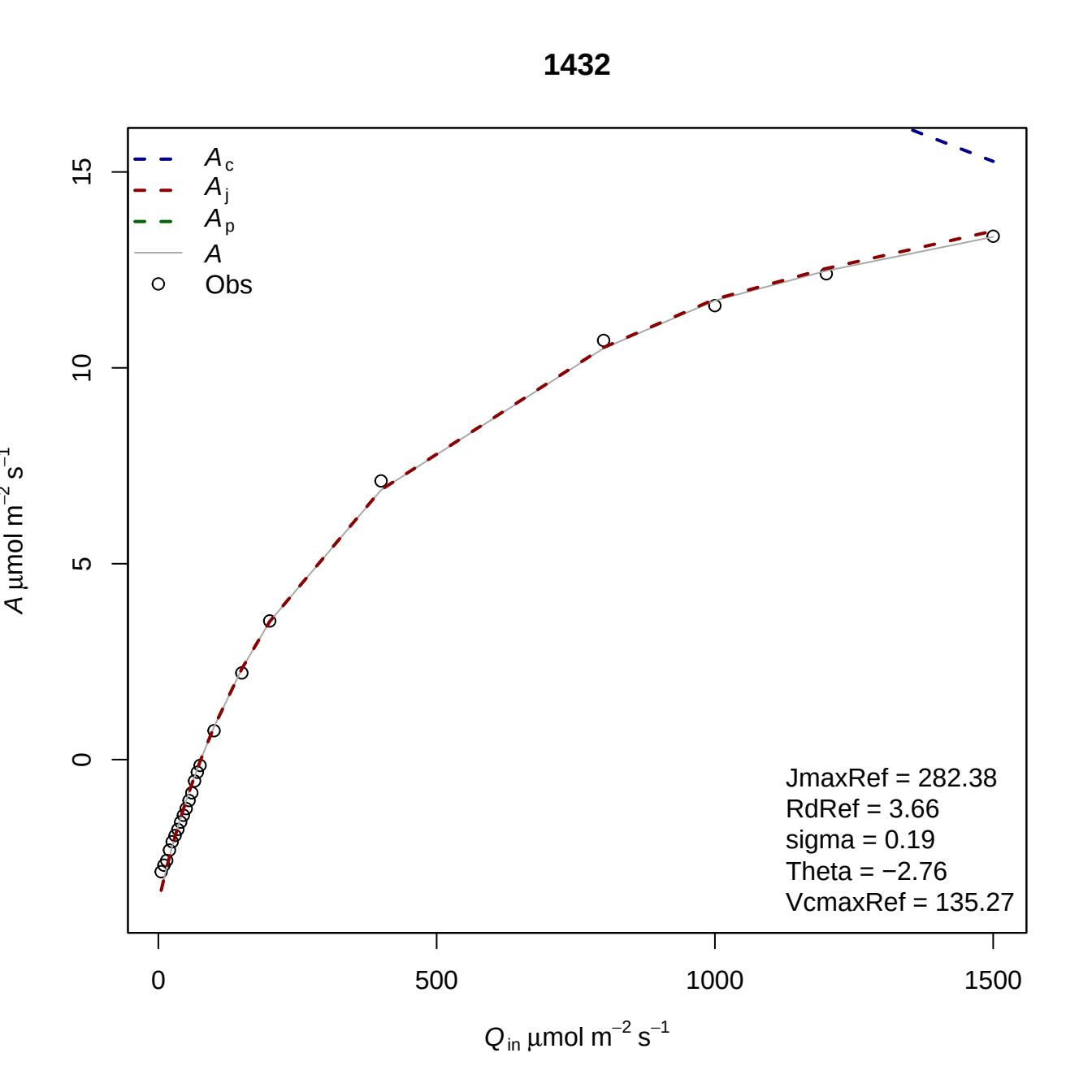

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 17.63
RdRef = 0.22
sigma = 0.05
Theta = 0.52
VcmaxRef = 13.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

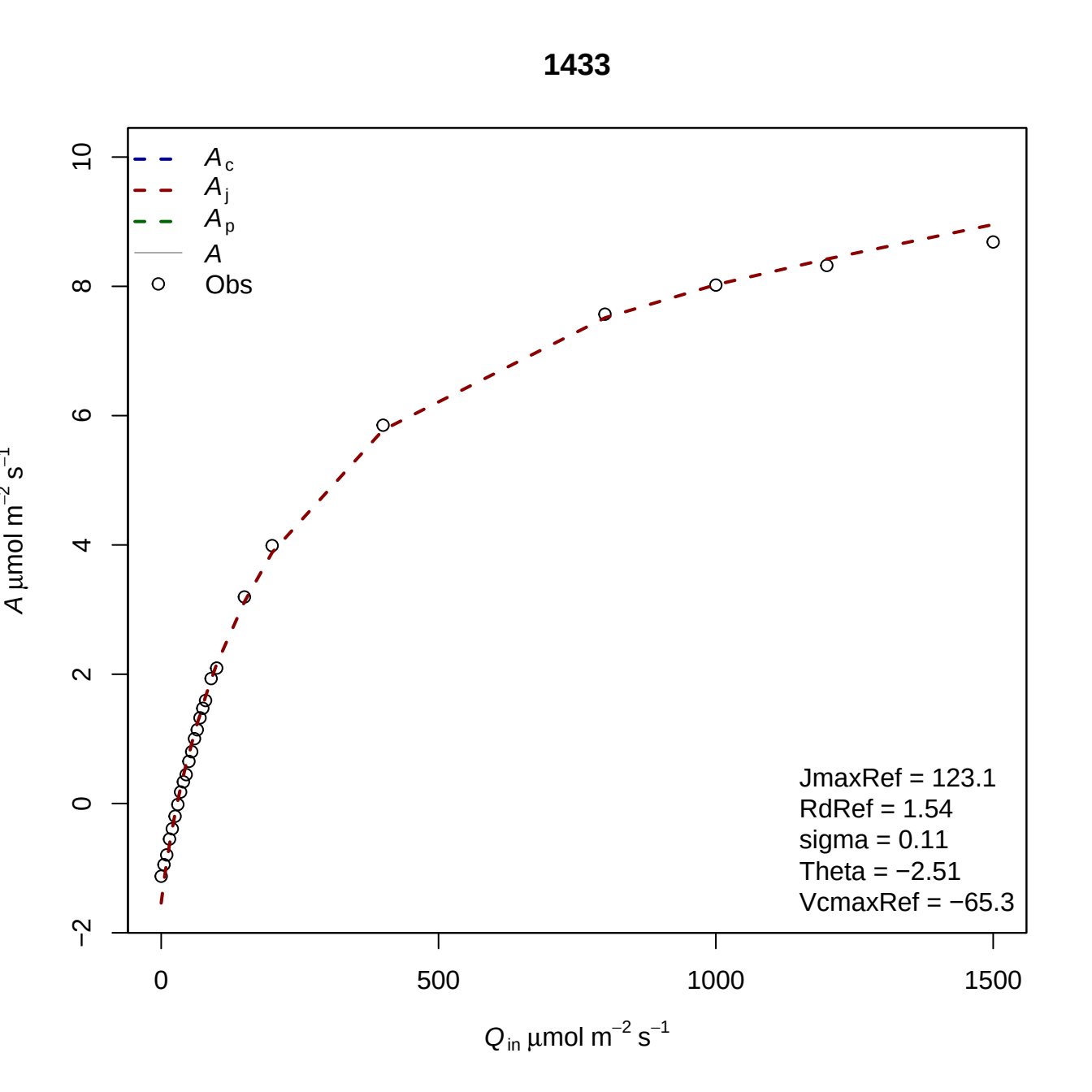

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.73
RdRef = 0.22
sigma = 0.07
Theta = −0.34
VcmaxRef = 39742.67
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

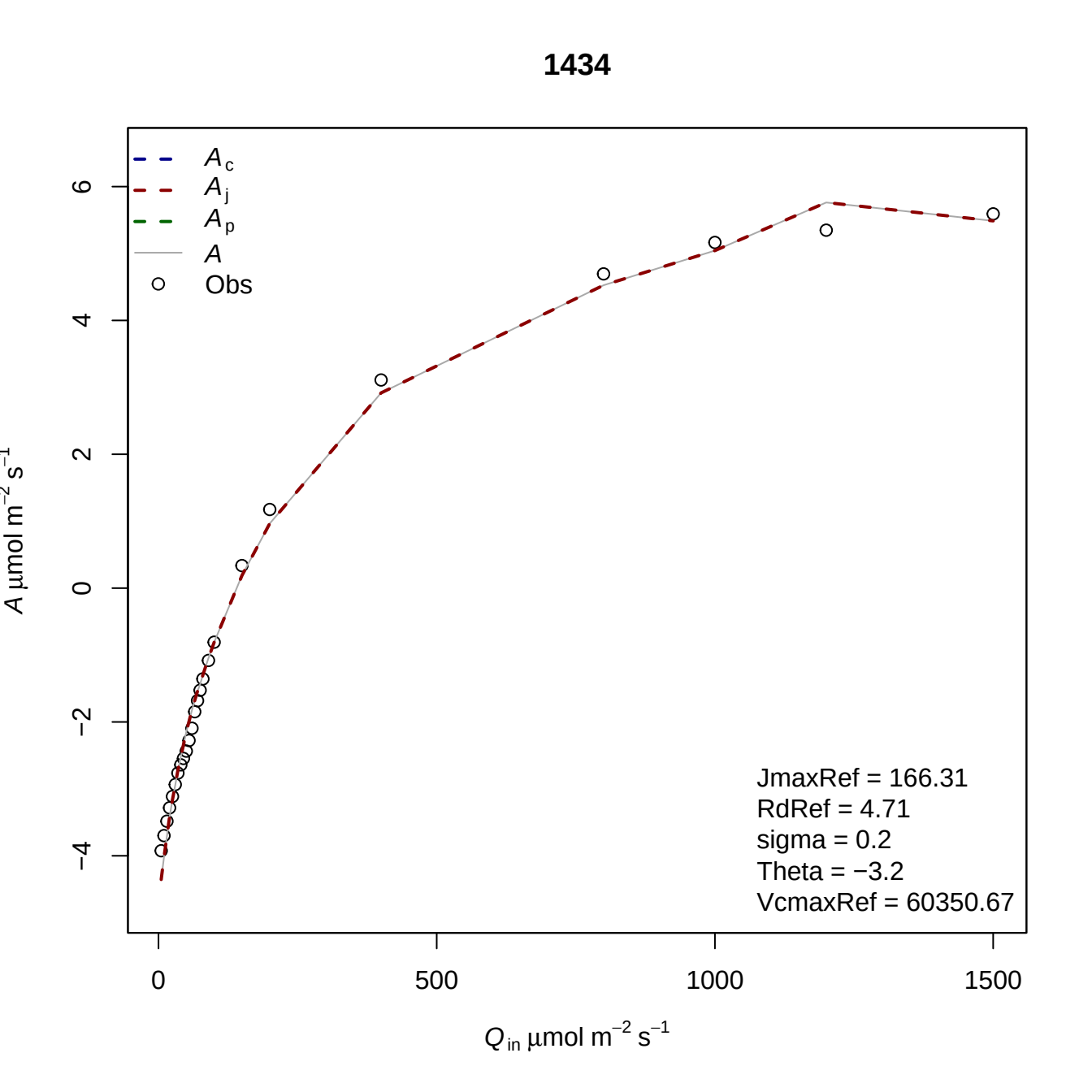

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 14.35
RdRef = 0.16
sigma = 0.06
Theta = 0.22
VcmaxRef = 46235.59
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

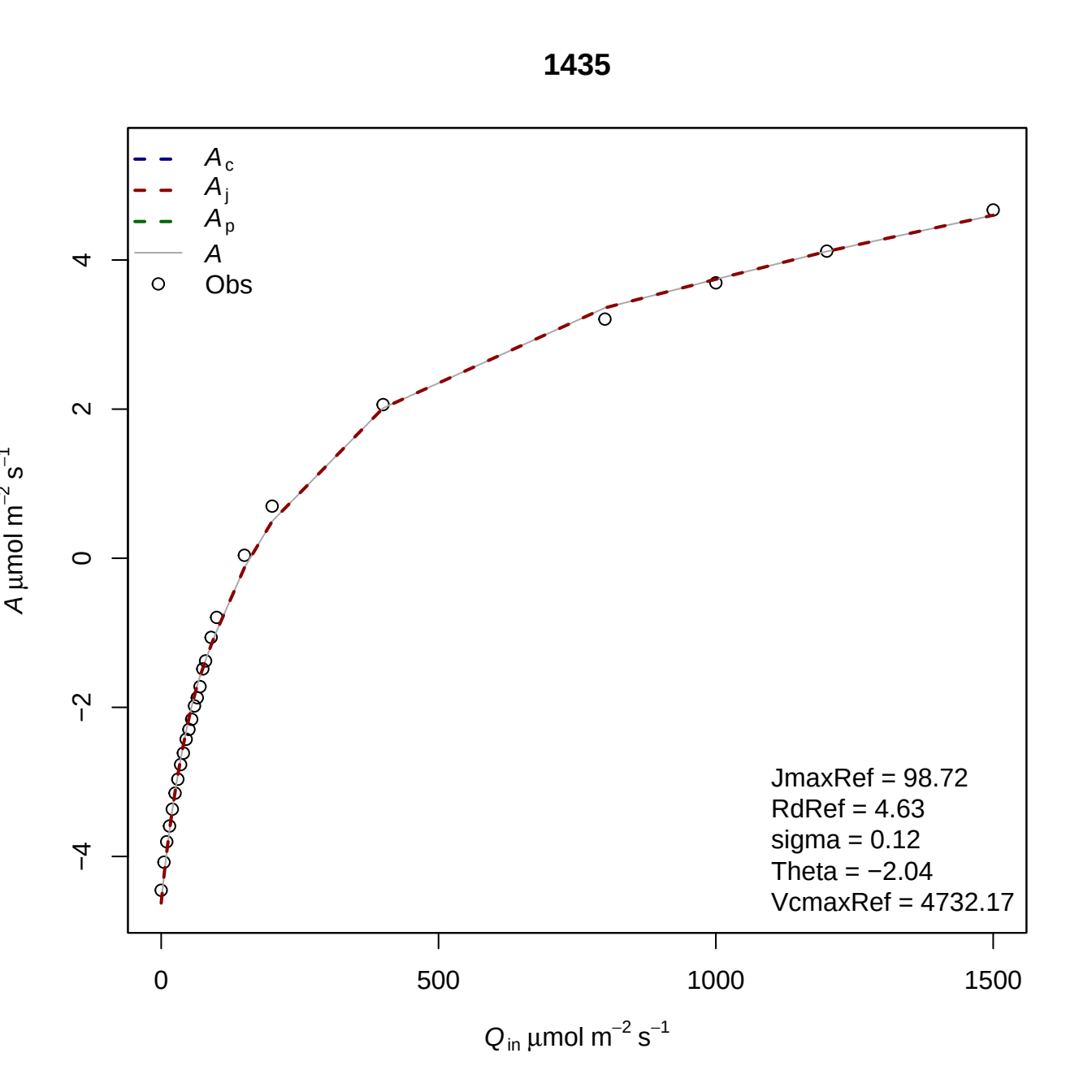

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 13.05
RdRef = 0.35
sigma = 0.07
Theta = 0.15
VcmaxRef = 419864.16
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

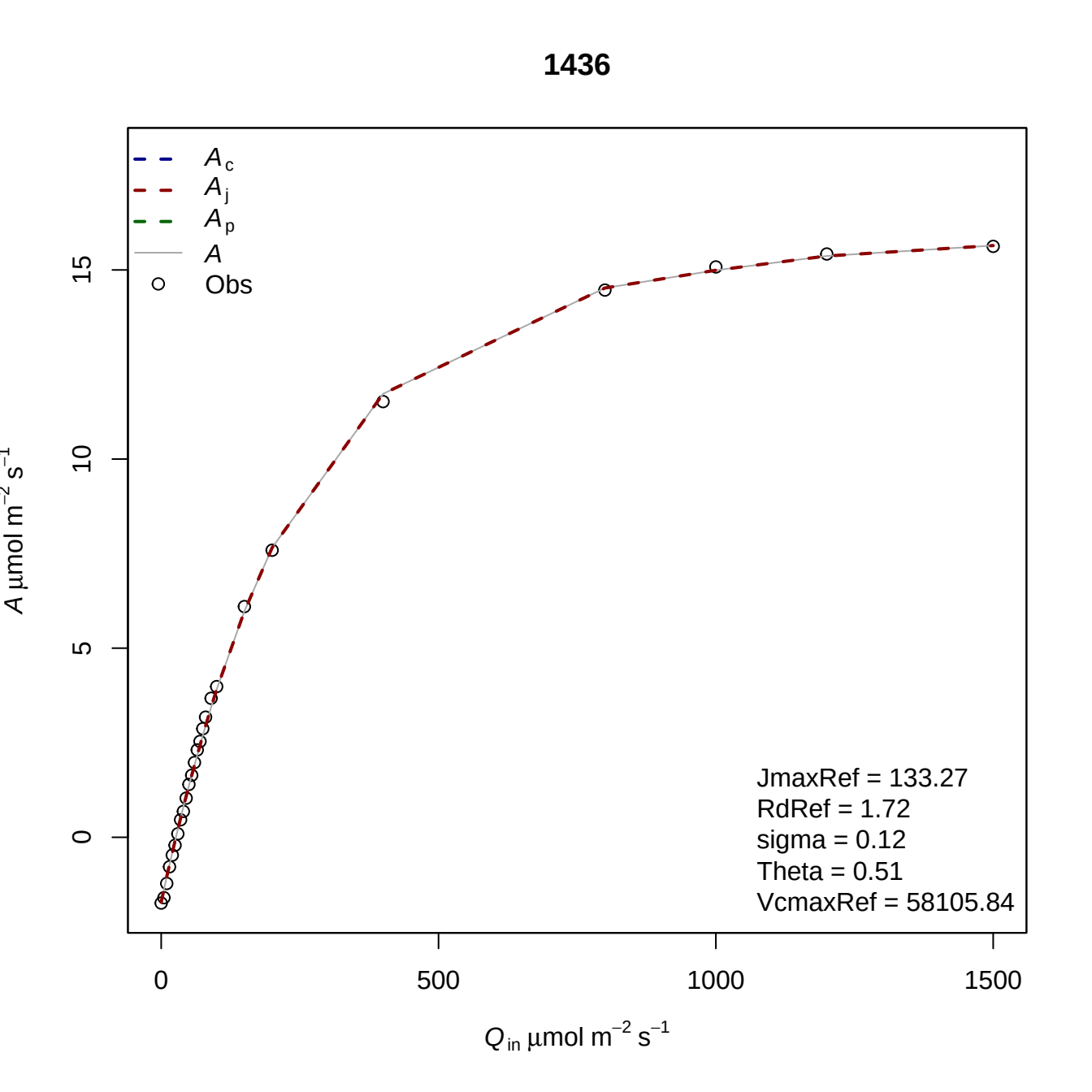

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 238.4
RdRef = 3.79
sigma = 0.16
Theta = −3.86
VcmaxRef = 88.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

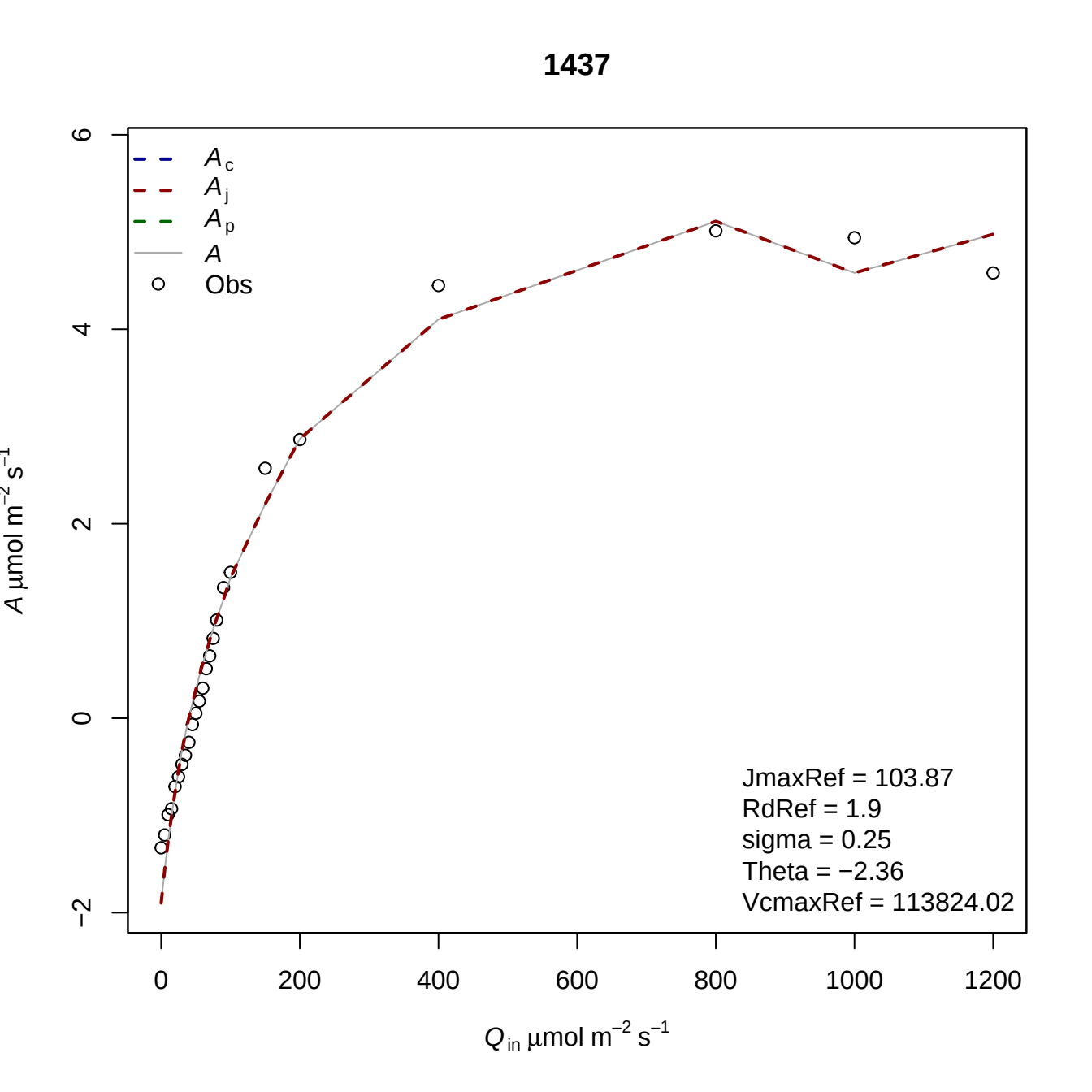

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 282.38
RdRef = 3.66
sigma = 0.19
Theta = −2.76
VcmaxRef = 135.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

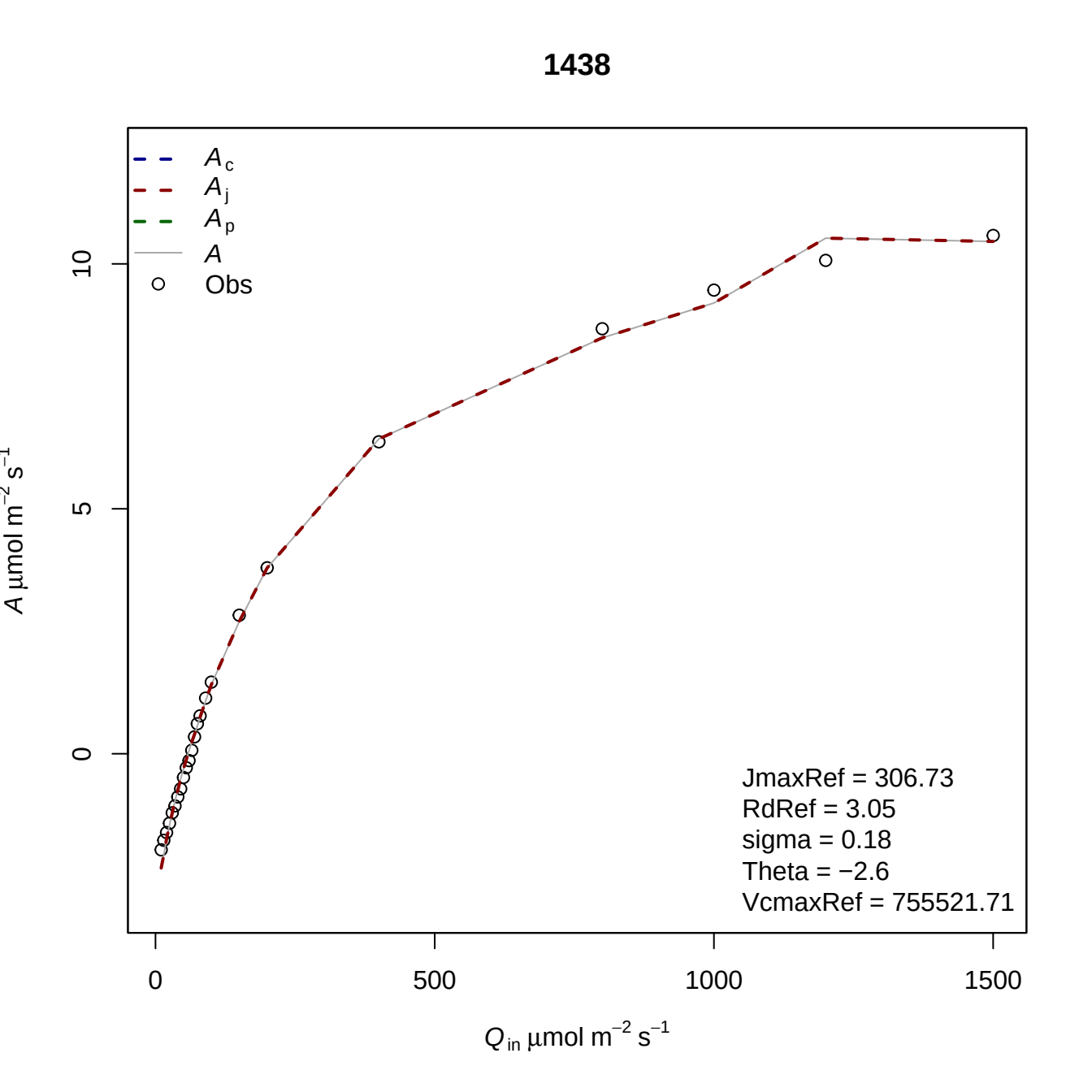

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 123.1
RdRef = 1.54
sigma = 0.11
Theta = −2.51
VcmaxRef = −65.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

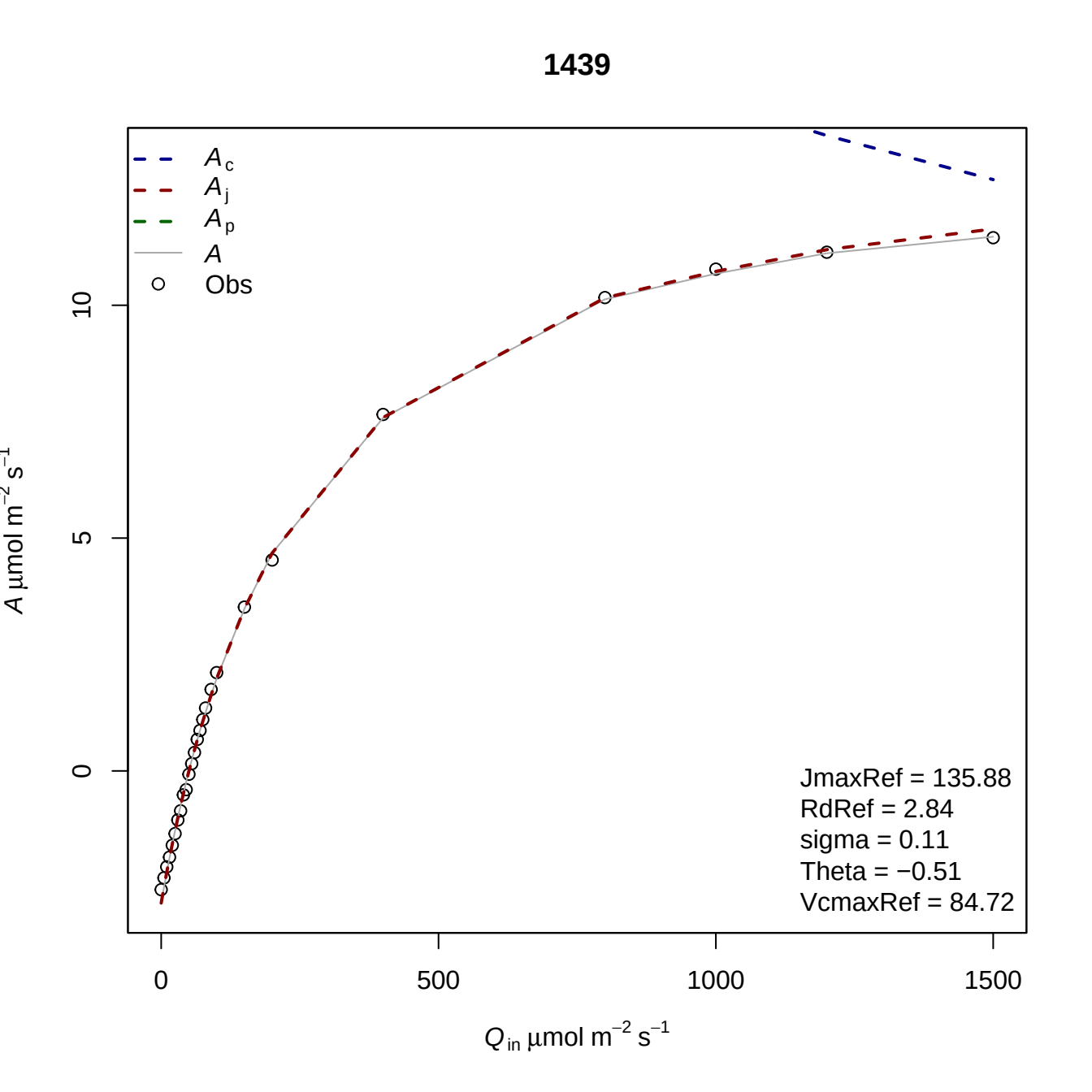

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 166.31
RdRef = 4.71
sigma = 0.2
Theta = −3.2
VcmaxRef = 60350.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

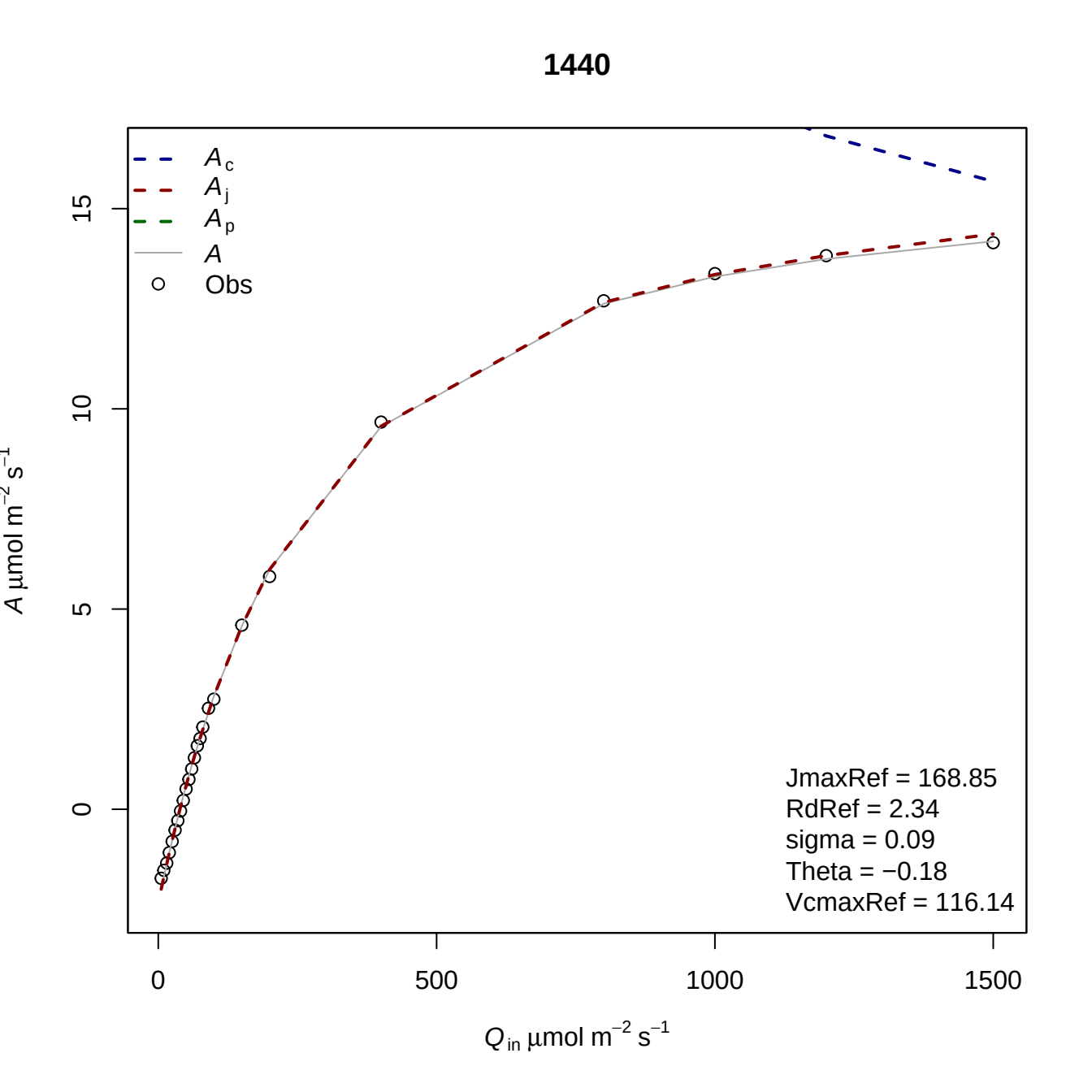

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 98.72
RdRef = 4.63
sigma = 0.12
Theta = −2.04
VcmaxRef = 4732.17
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

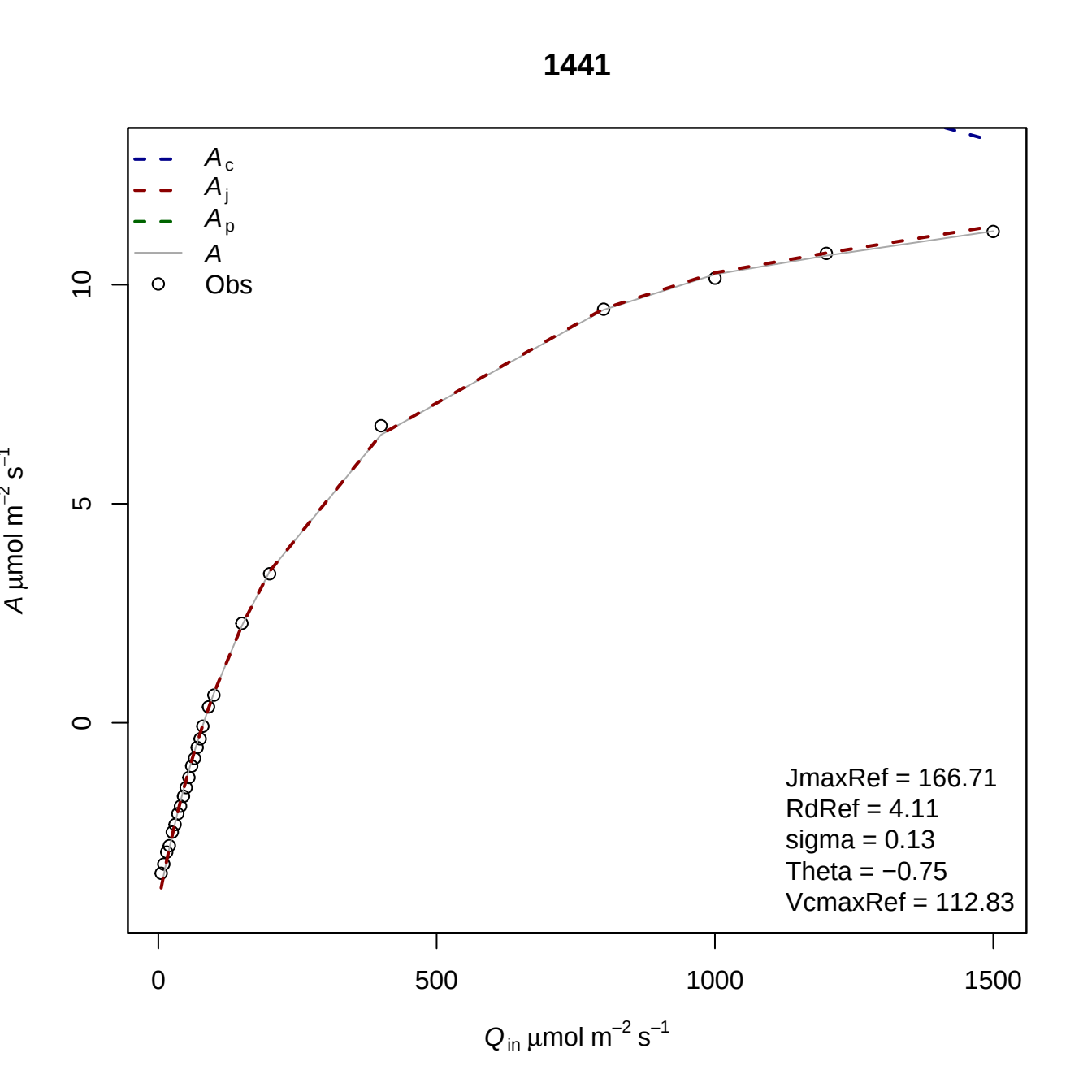

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 133.27
RdRef = 1.72
sigma = 0.12
Theta = 0.51
VcmaxRef = 58105.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

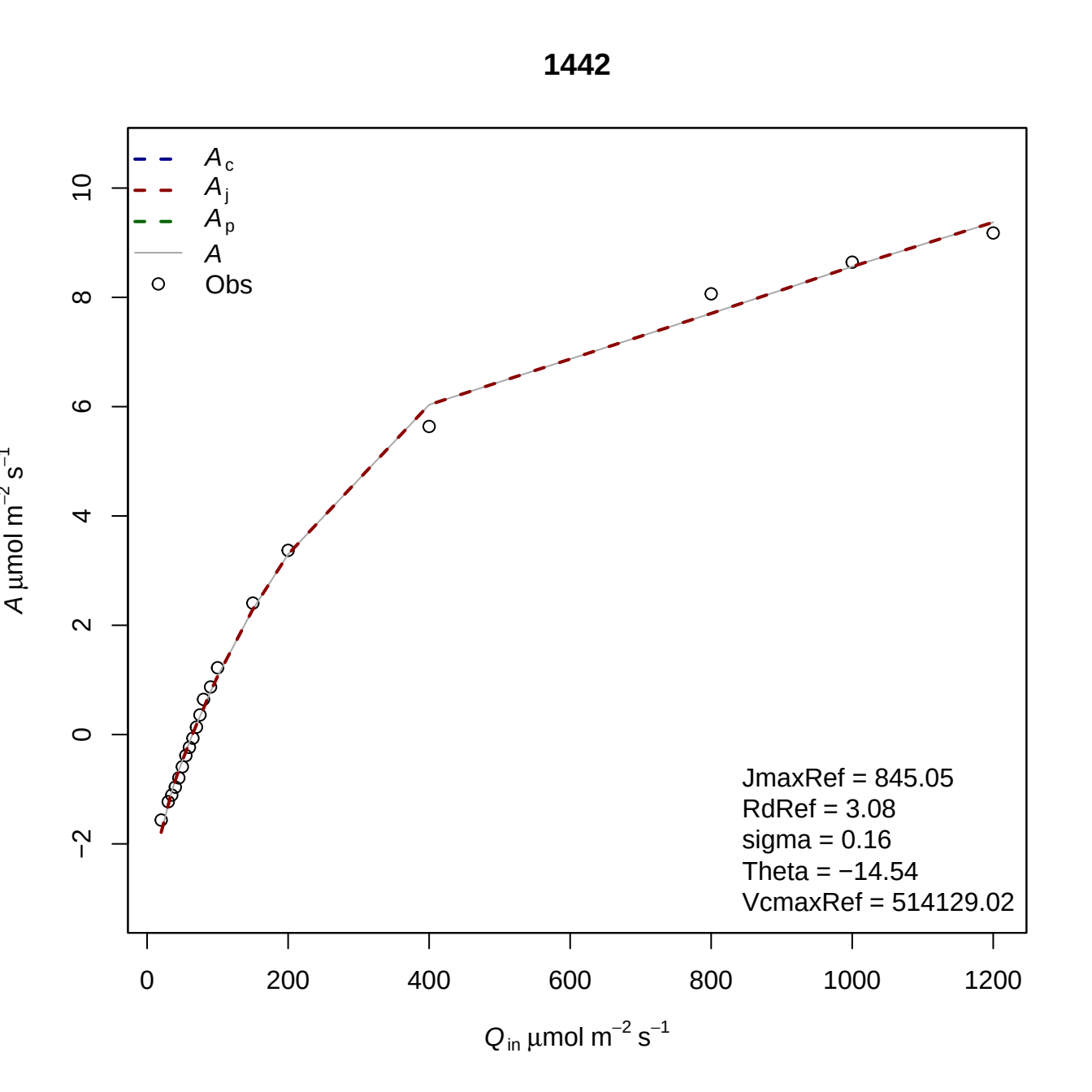

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 103.87
RdRef = 1.9
sigma = 0.25
Theta = −2.36
VcmaxRef = 113824.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

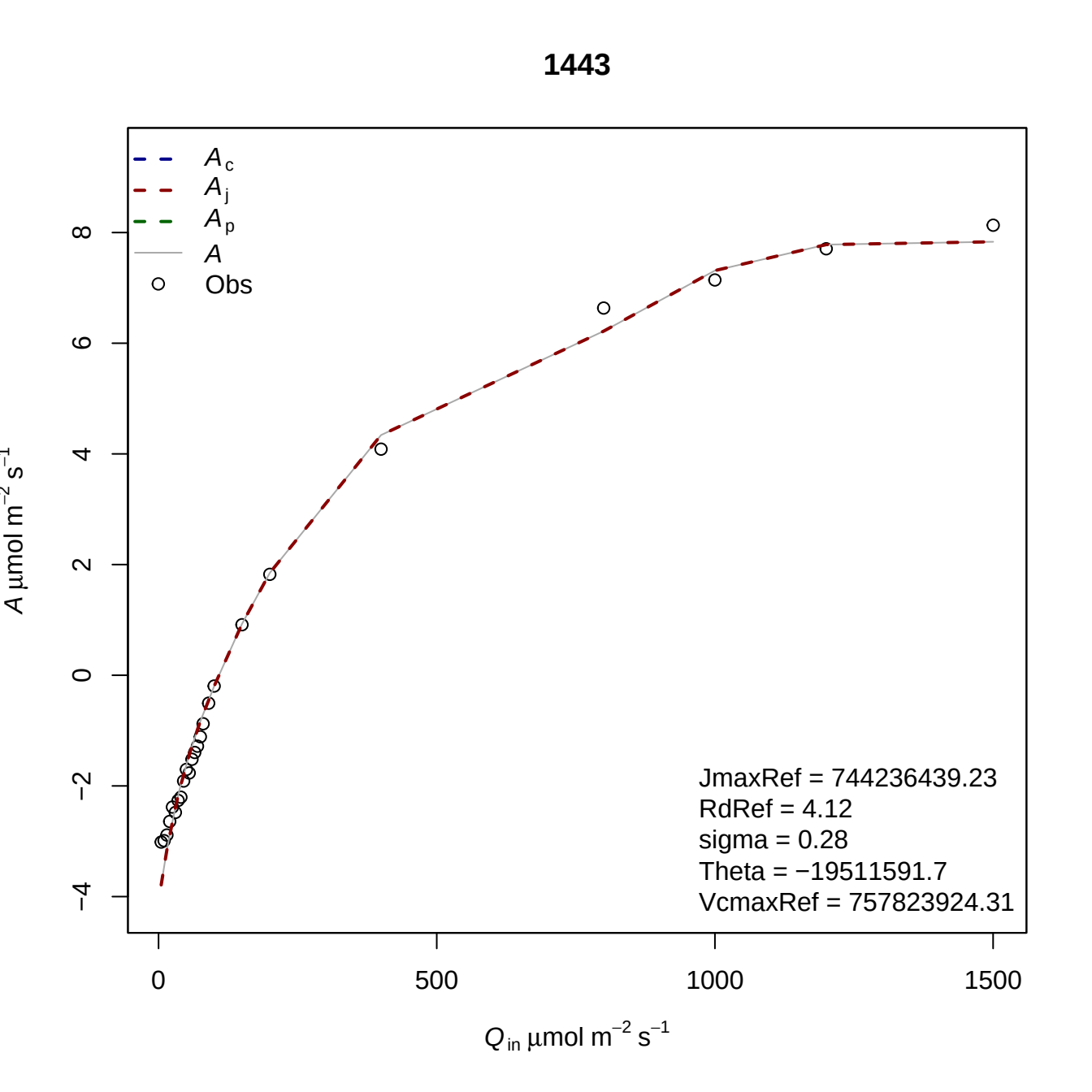

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 306.73
RdRef = 3.05
sigma = 0.18
Theta = −2.6
VcmaxRef = 755521.71
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

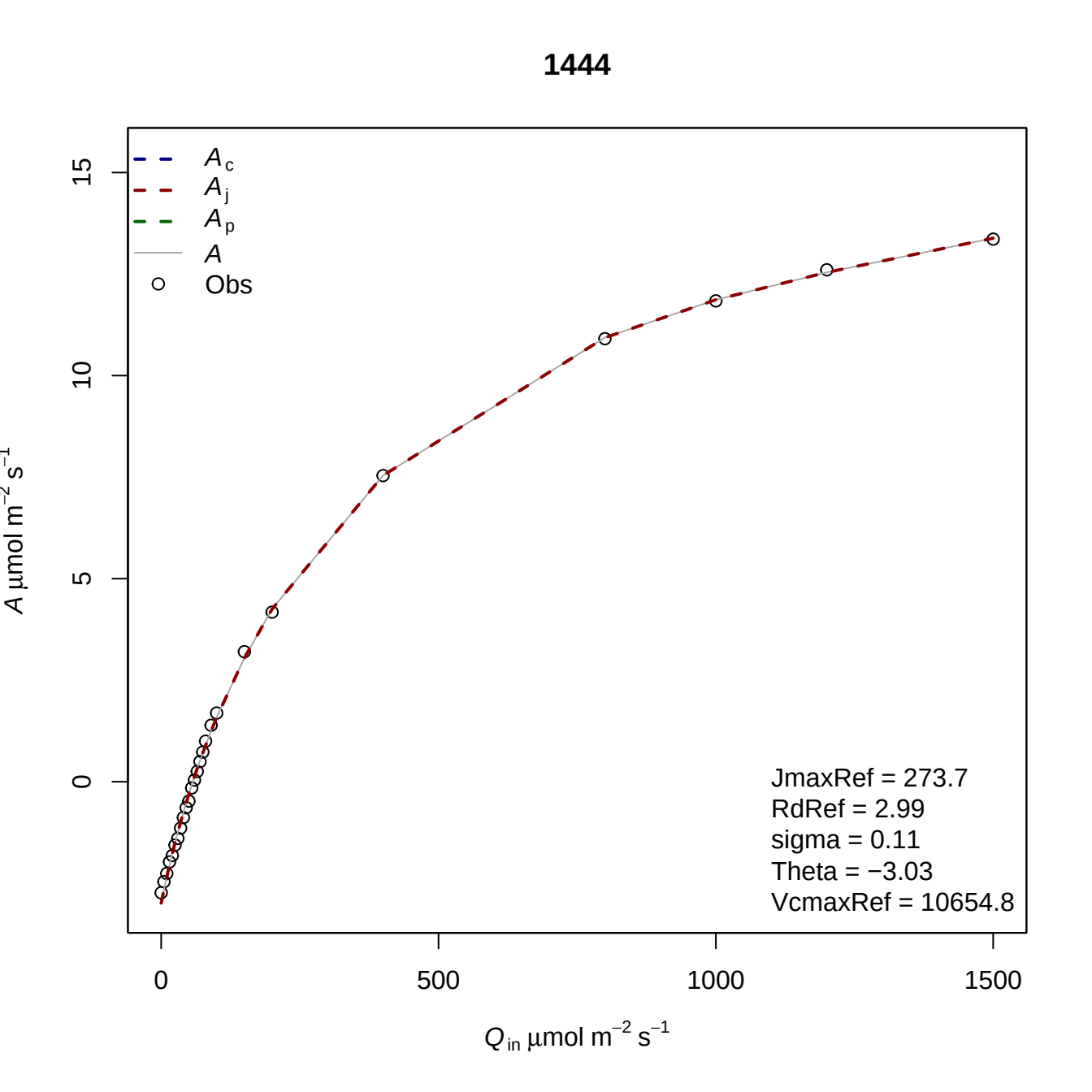

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 135.88
RdRef = 2.84
sigma = 0.11
Theta = −0.51
VcmaxRef = 84.72
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

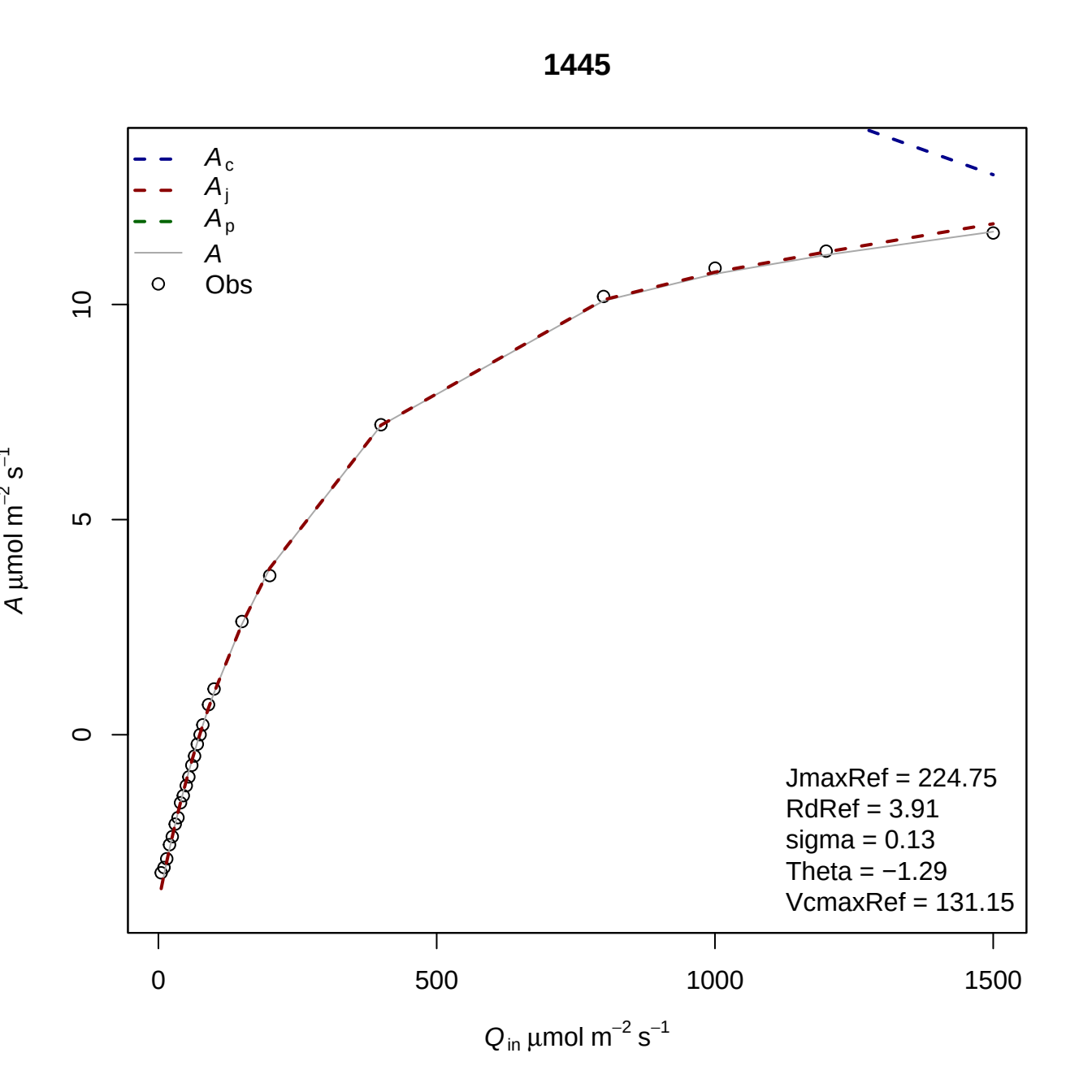

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 168.85
RdRef = 2.34
sigma = 0.09
Theta = −0.18
VcmaxRef = 116.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

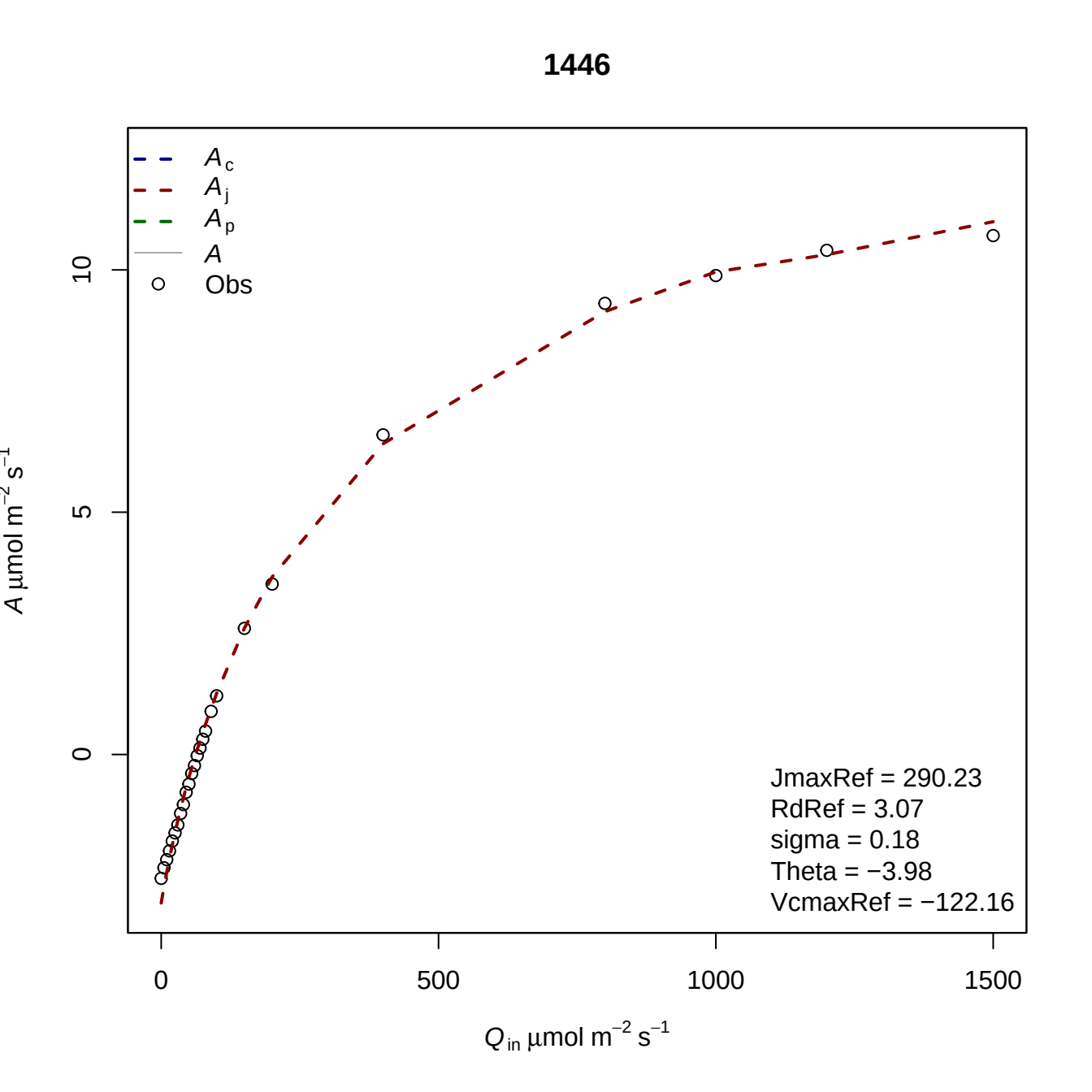

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 166.71
RdRef = 4.11
sigma = 0.13
Theta = −0.75
VcmaxRef = 112.83
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

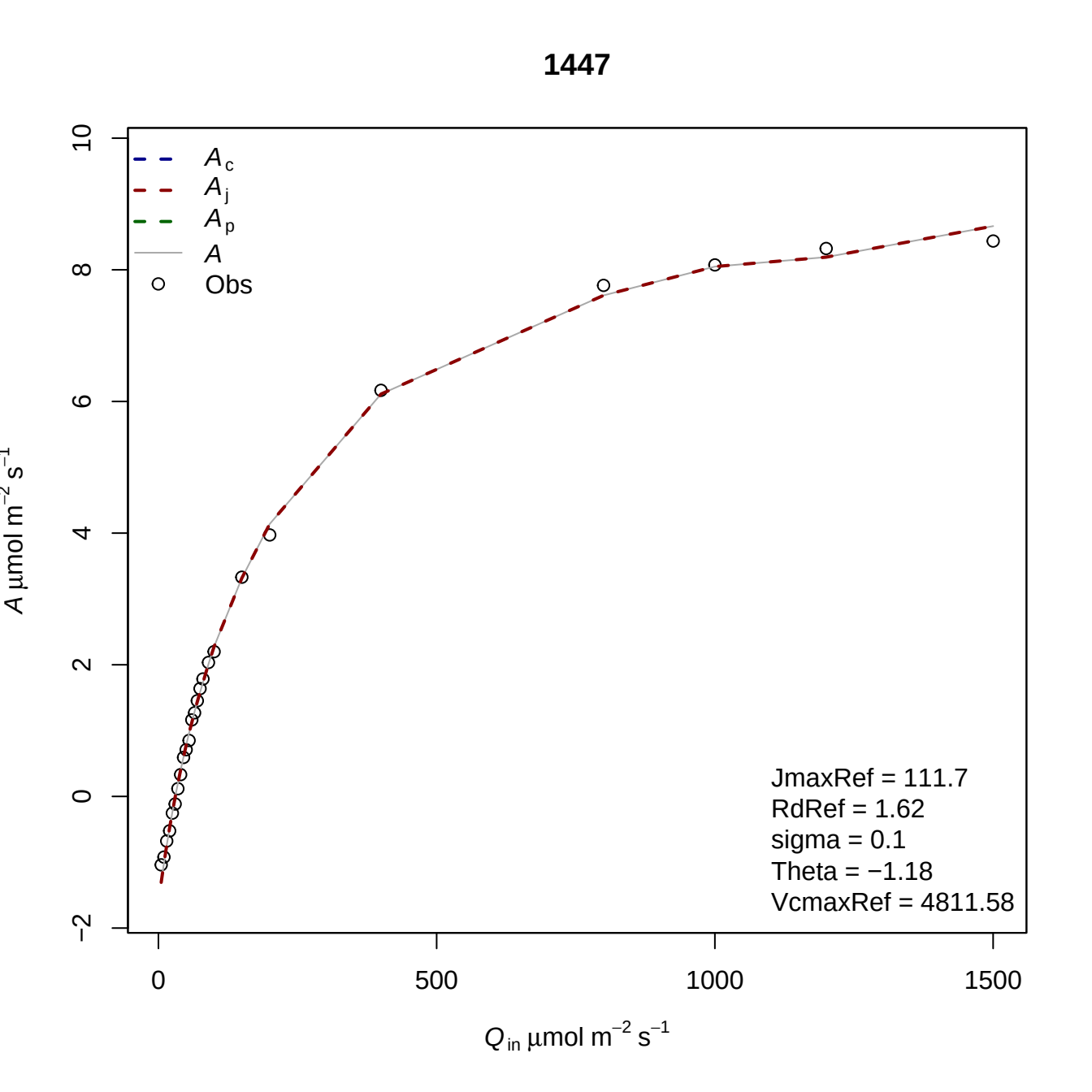

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 845.05
RdRef = 3.08
sigma = 0.16
Theta = −14.54
VcmaxRef = 514129.02
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

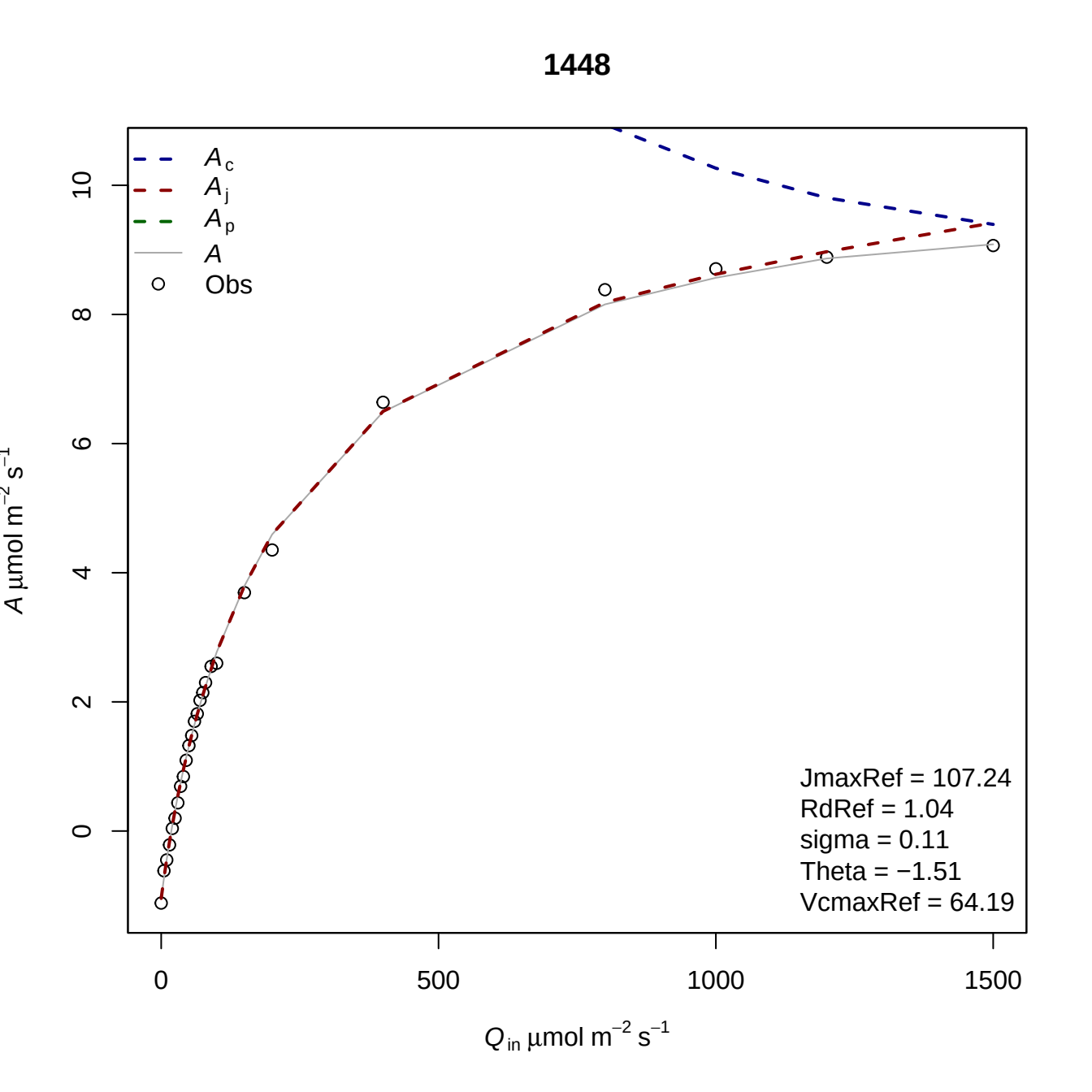

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 744236439.23
RdRef = 4.12
sigma = 0.28
Theta = −19511591.7
VcmaxRef = 757823924.31
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

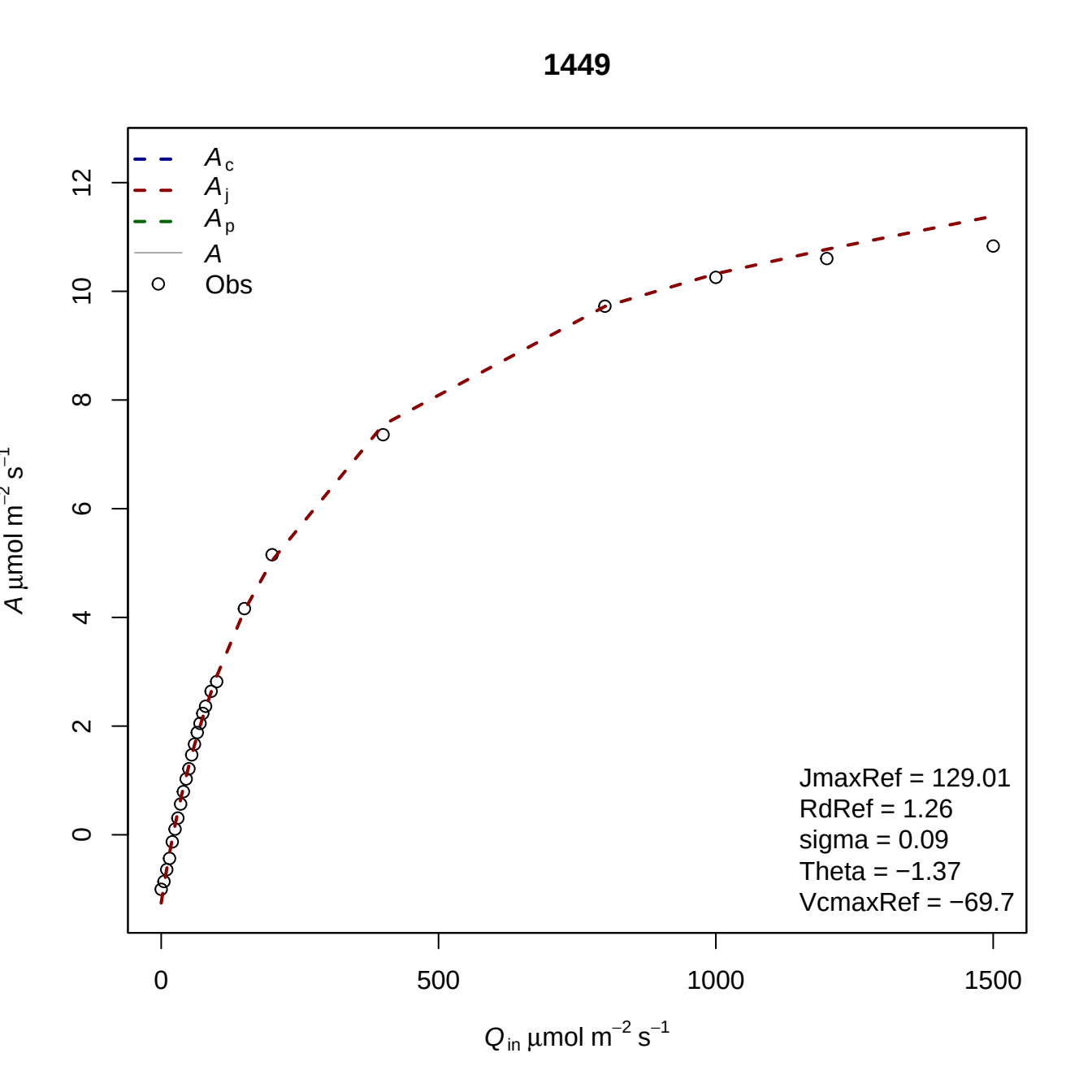

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 273.7
RdRef = 2.99
sigma = 0.11
Theta = −3.03
VcmaxRef = 10654.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

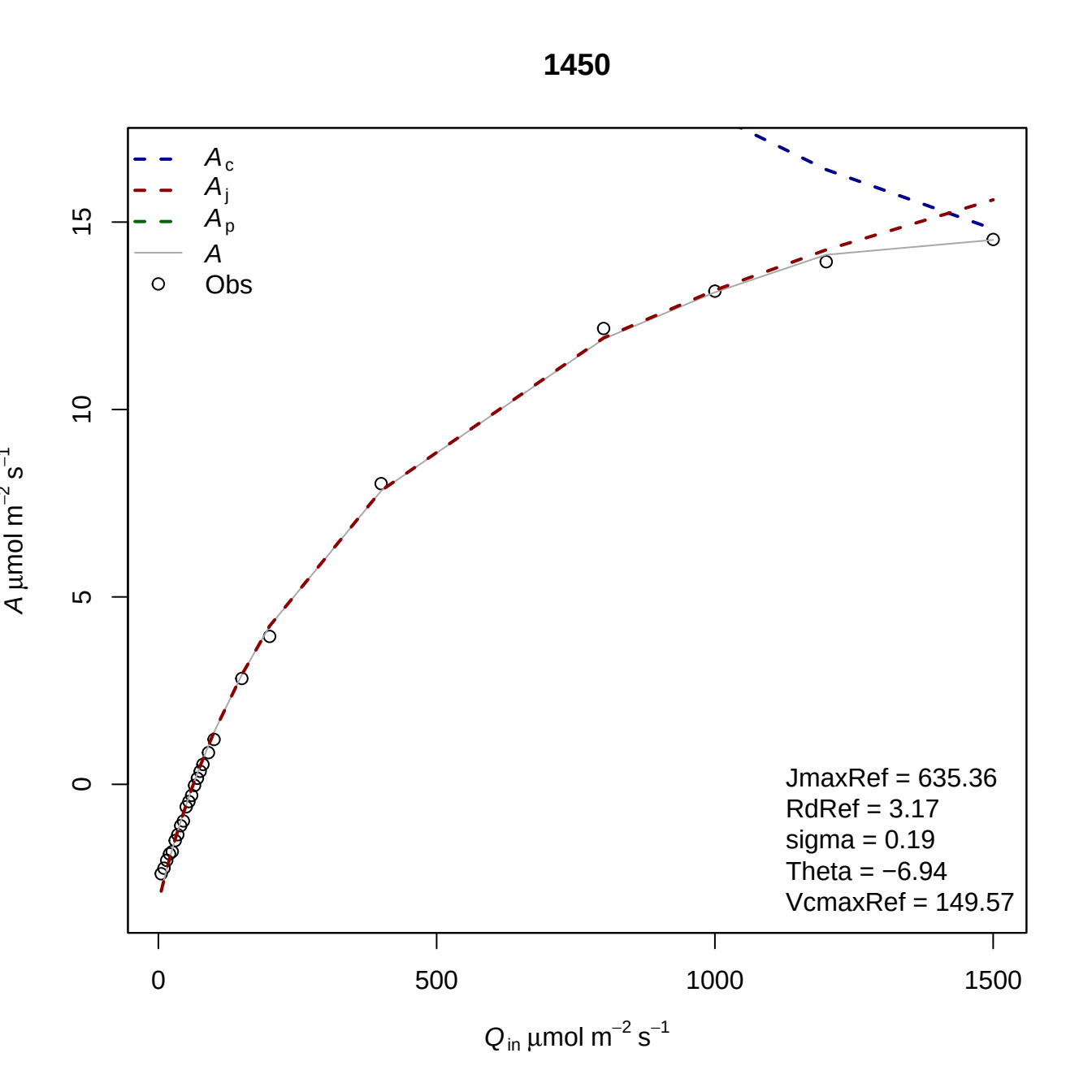

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 224.75
RdRef = 3.91
sigma = 0.13
Theta = −1.29
VcmaxRef = 131.15
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

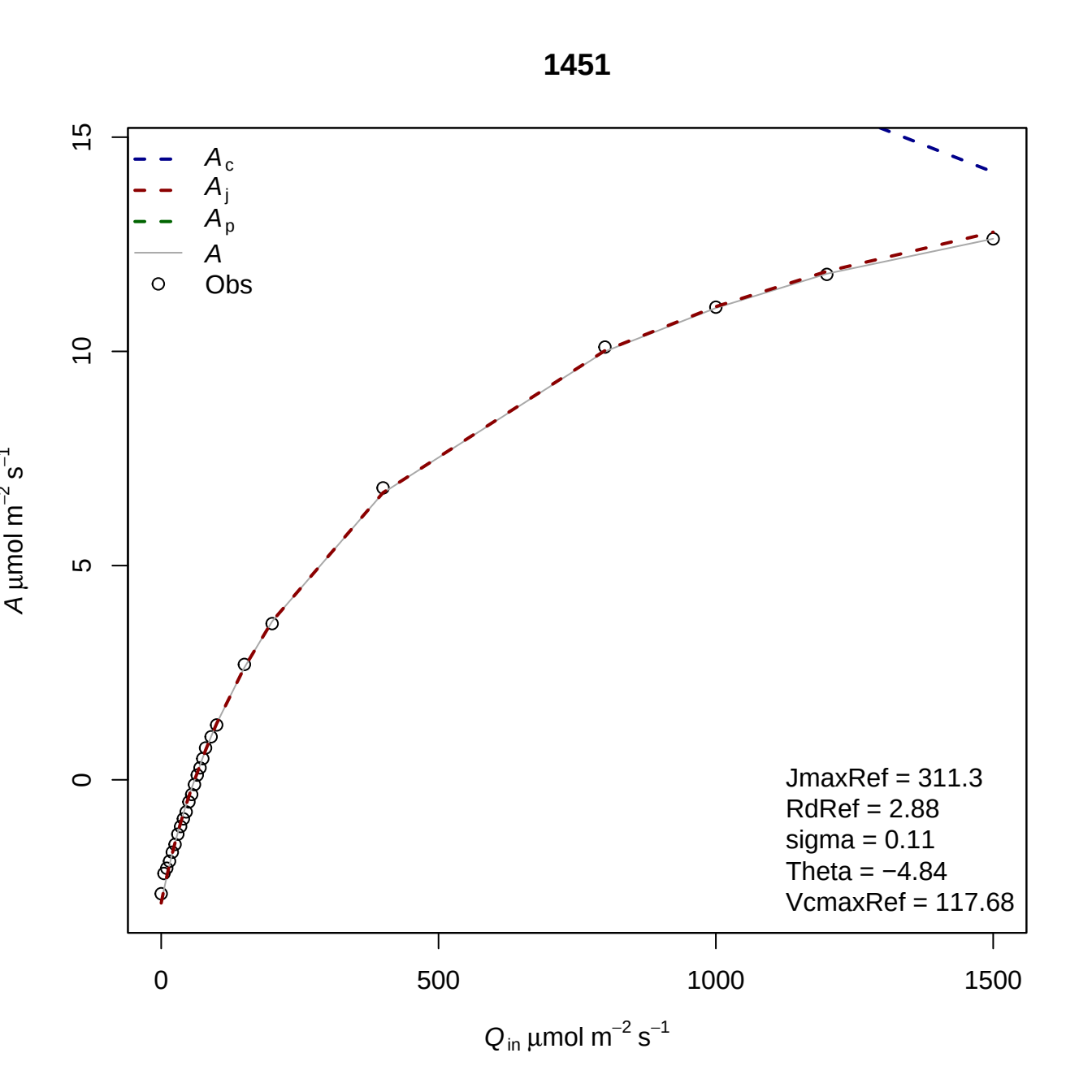

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 290.23
RdRef = 3.07
sigma = 0.18
Theta = −3.98
VcmaxRef = −122.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

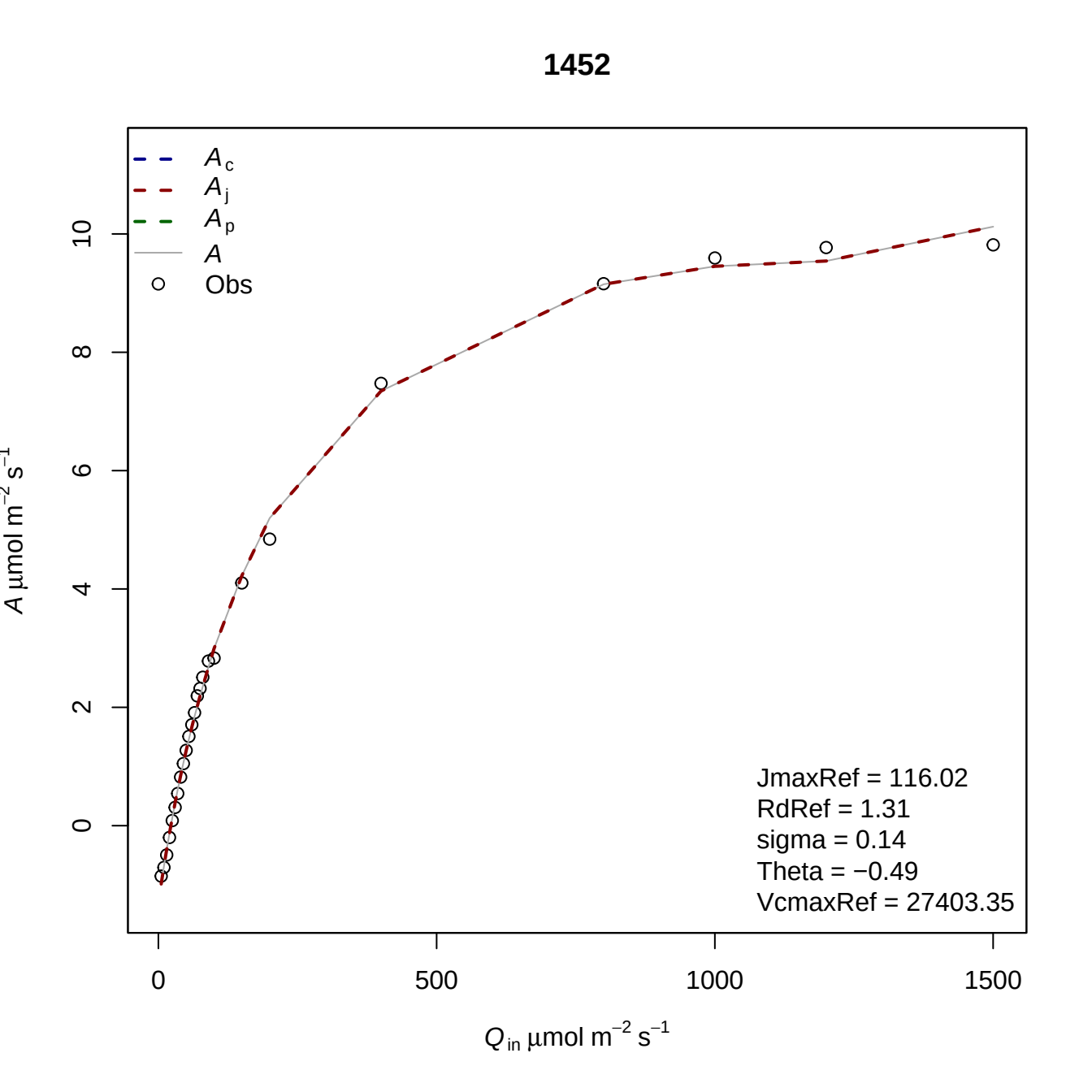

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 111.7
RdRef = 1.62
sigma = 0.1
Theta = −1.18
VcmaxRef = 4811.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

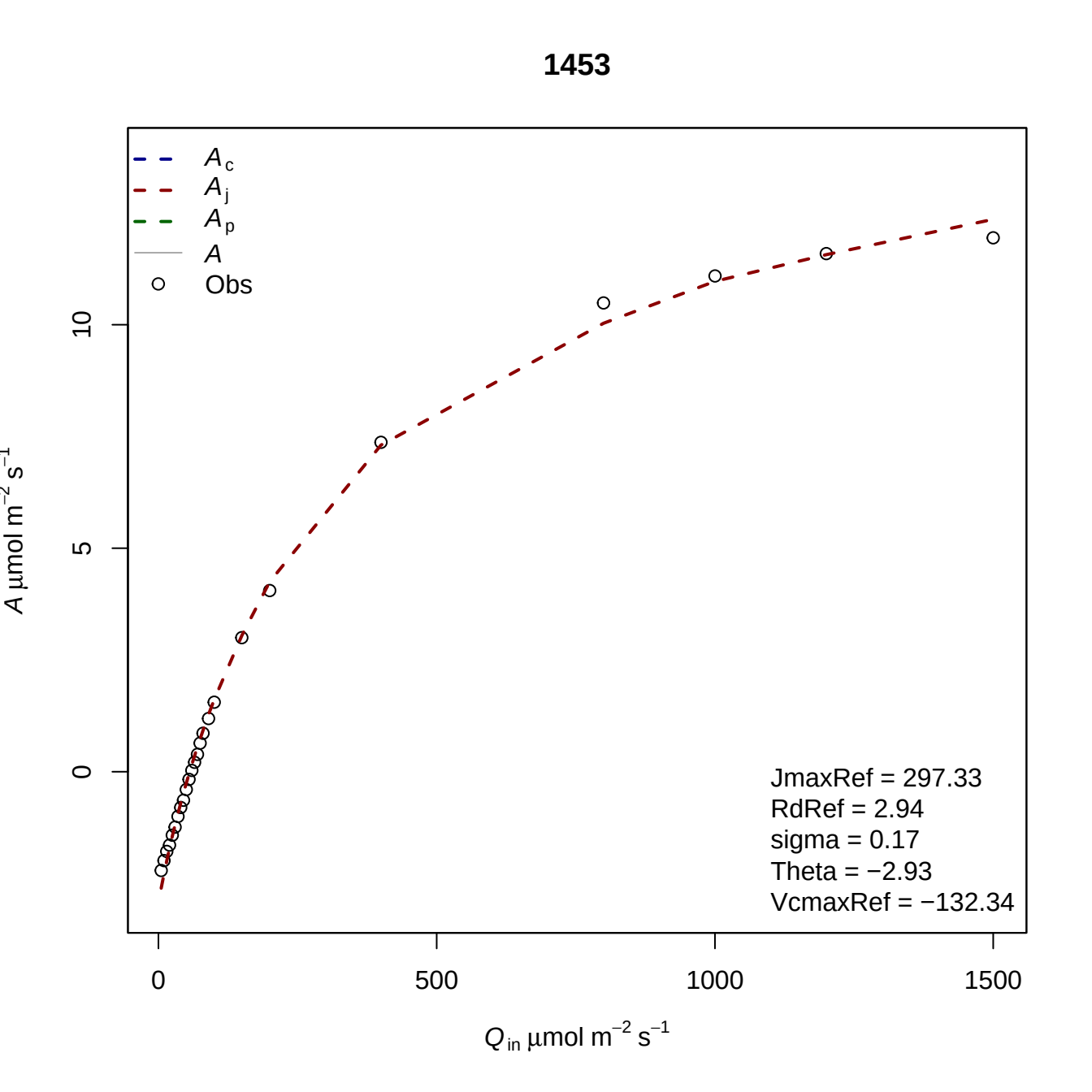

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 107.24
RdRef = 1.04
sigma = 0.11
Theta = −1.51
VcmaxRef = 64.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

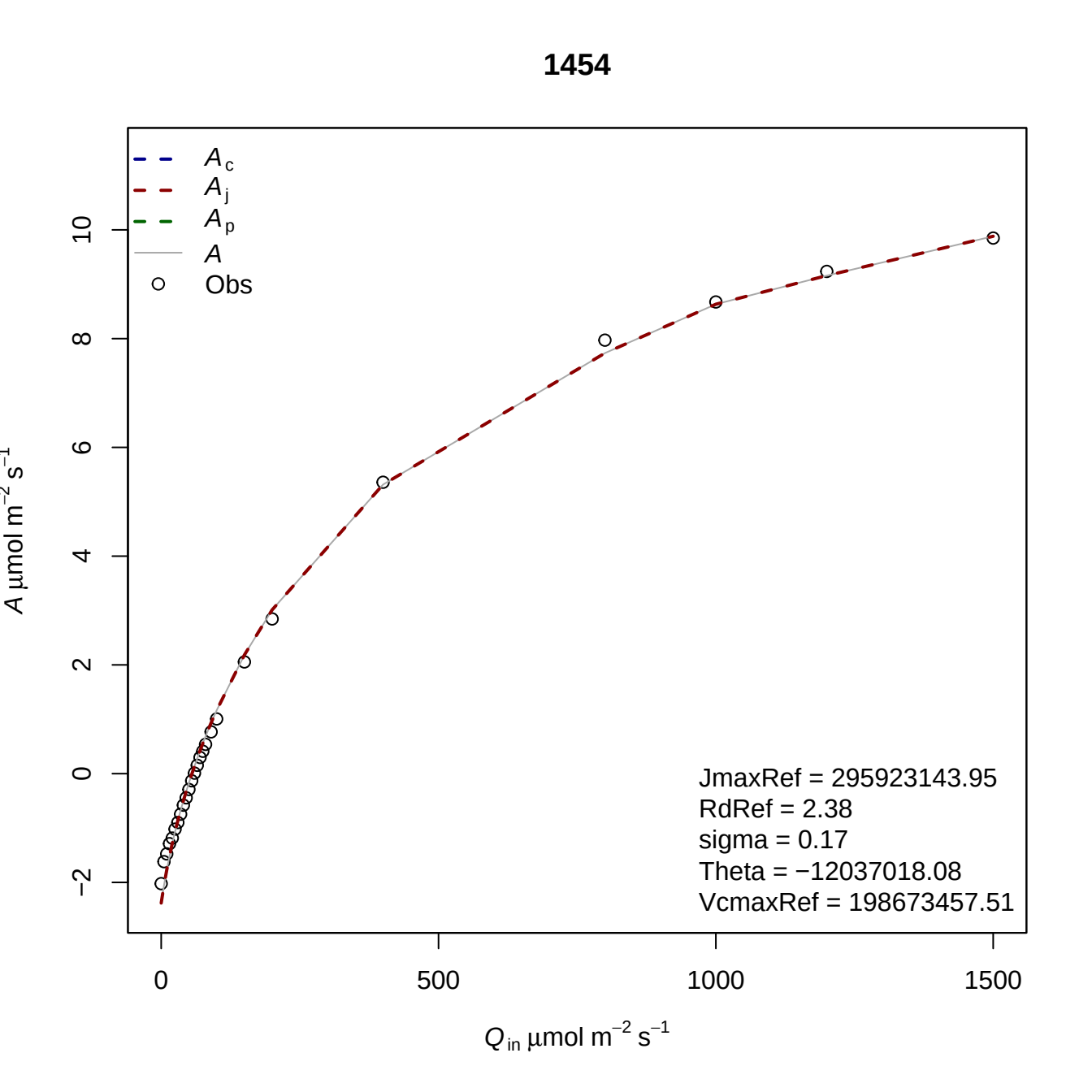

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 129.01
RdRef = 1.26
sigma = 0.09
Theta = −1.37
VcmaxRef = −69.7
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

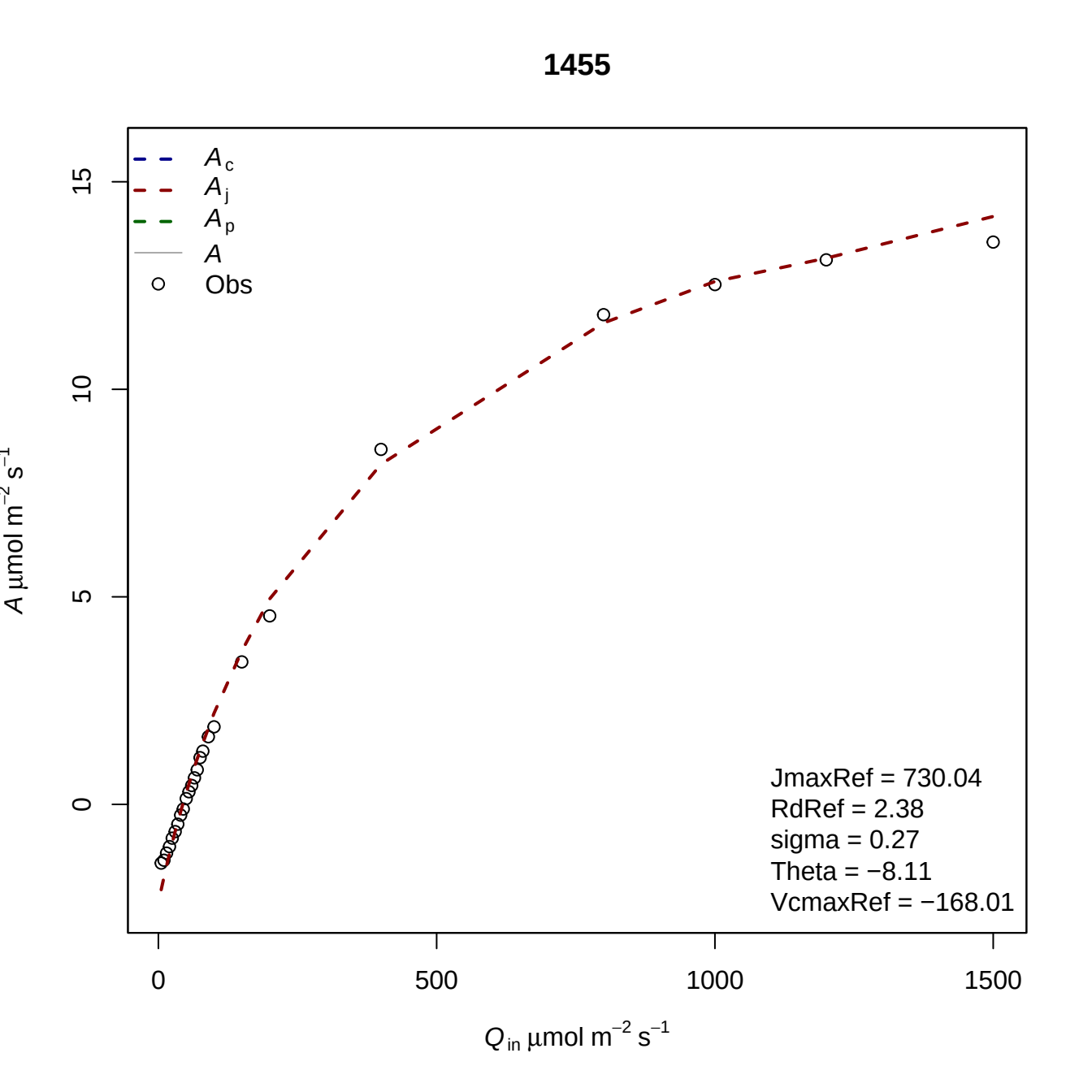

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 635.36
RdRef = 3.17
sigma = 0.19
Theta = −6.94
VcmaxRef = 149.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

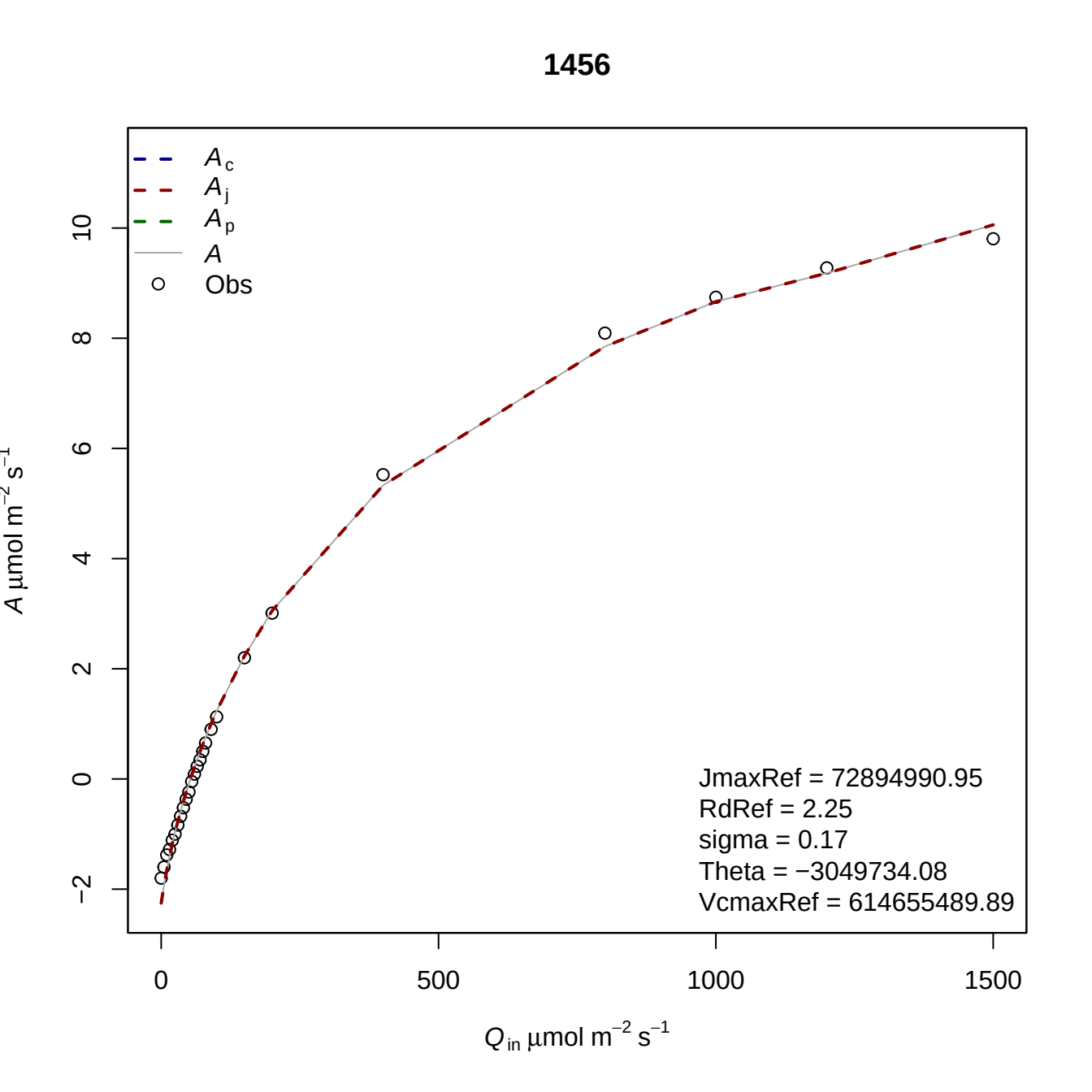

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 311.3
RdRef = 2.88
sigma = 0.11
Theta = −4.84
VcmaxRef = 117.68
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

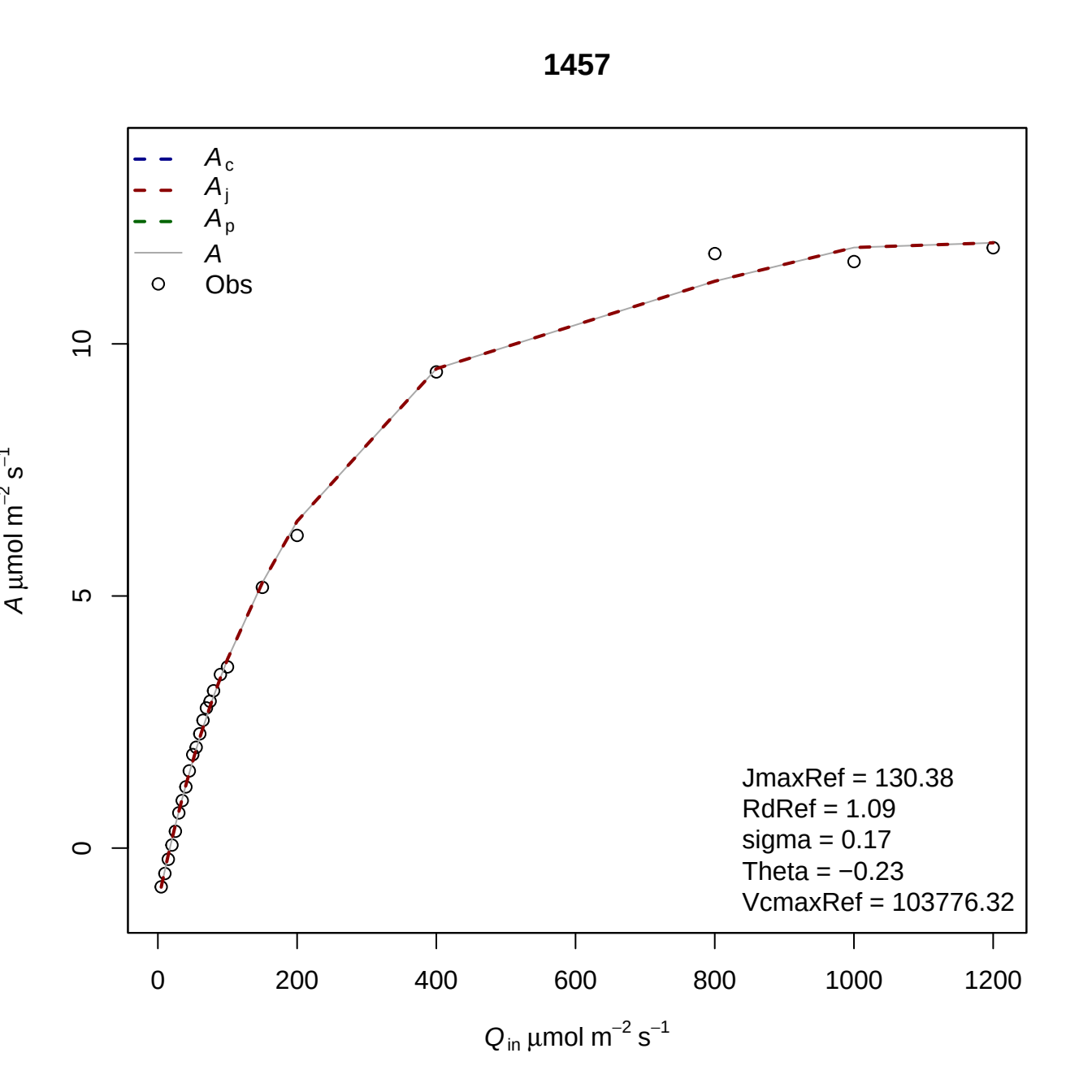

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 116.02
RdRef = 1.31
sigma = 0.14
Theta = −0.49
VcmaxRef = 27403.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

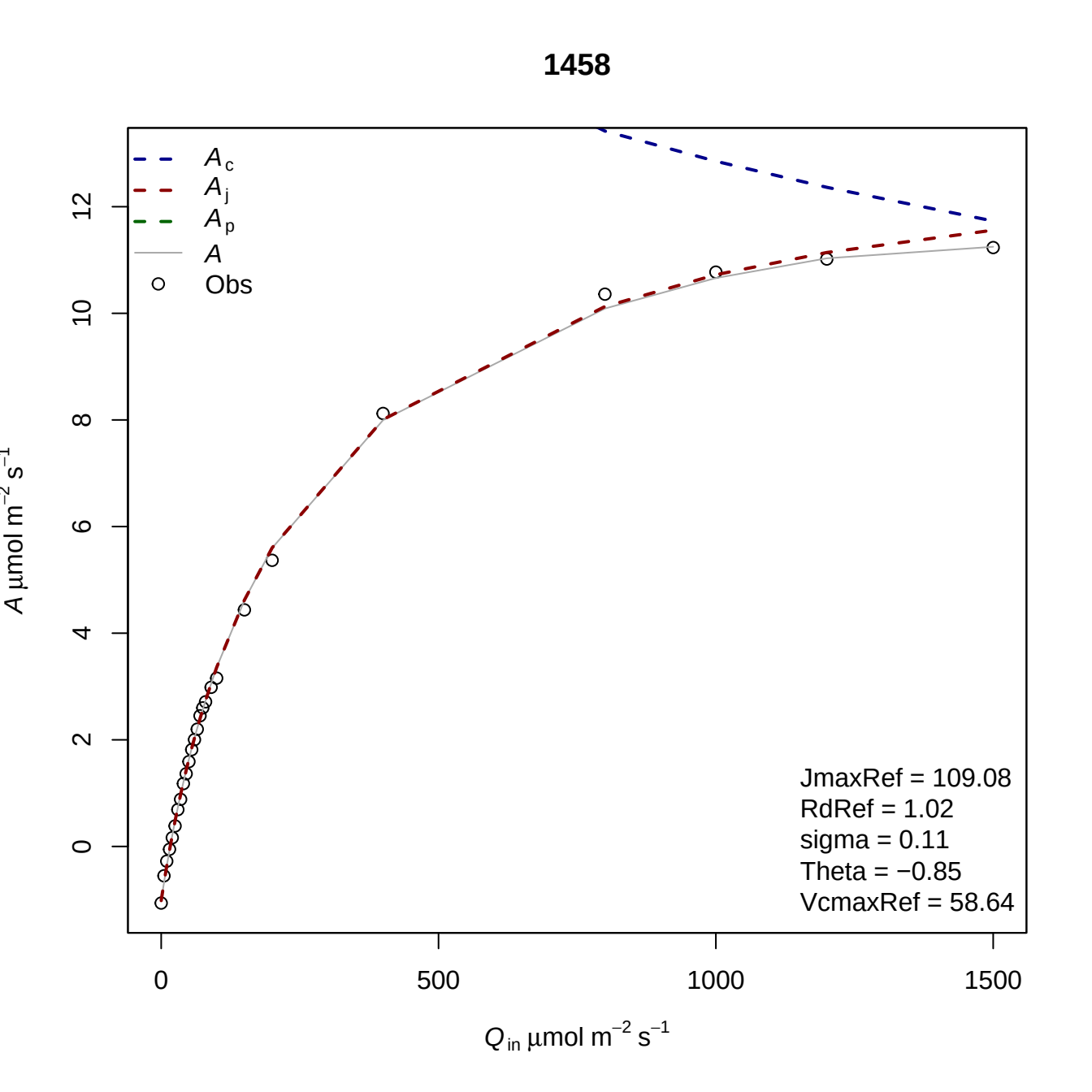

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 297.33
RdRef = 2.94
sigma = 0.17
Theta = −2.93
VcmaxRef = −132.34
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

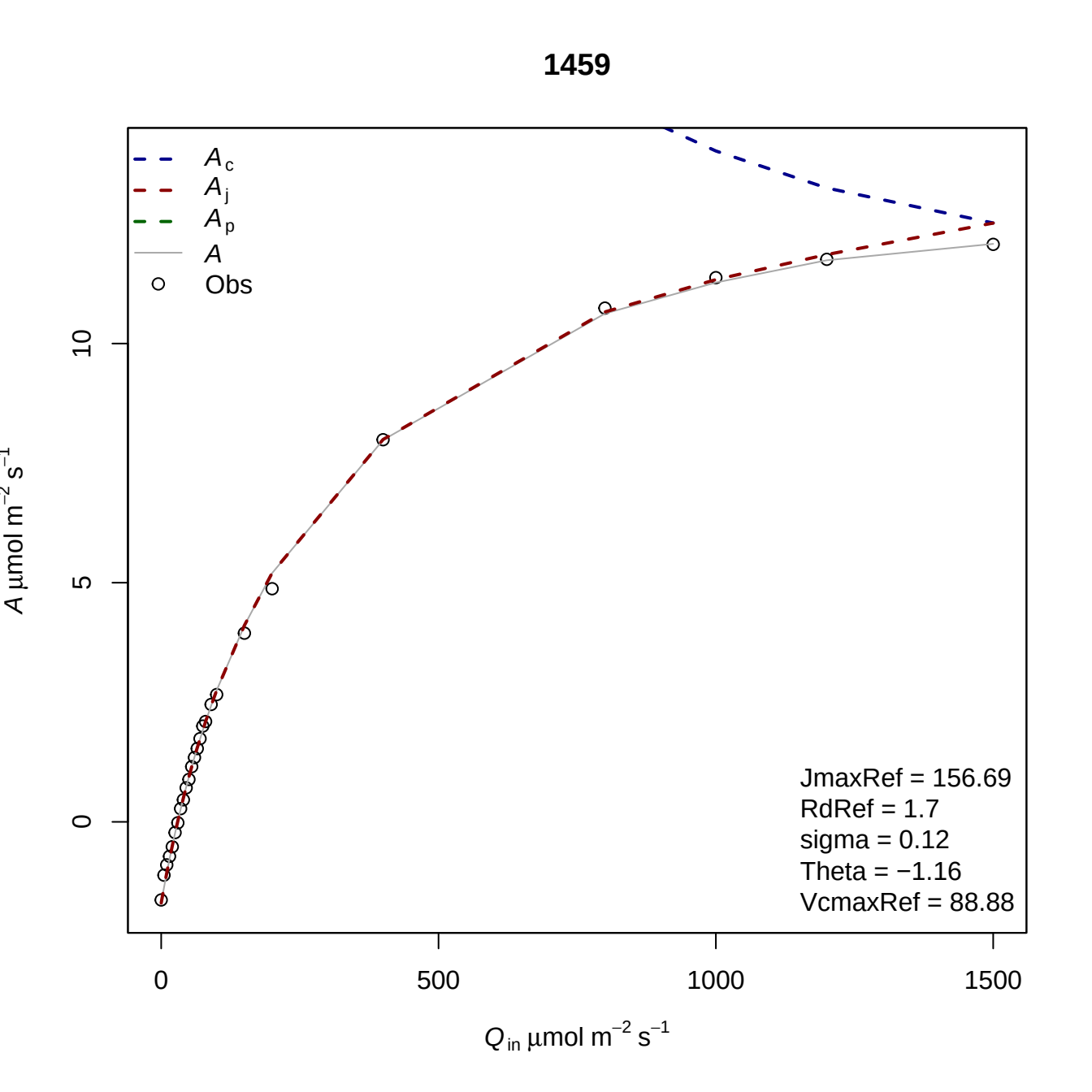

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 295923143.95
RdRef = 2.38
sigma = 0.17
Theta = −12037018.08
VcmaxRef = 198673457.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

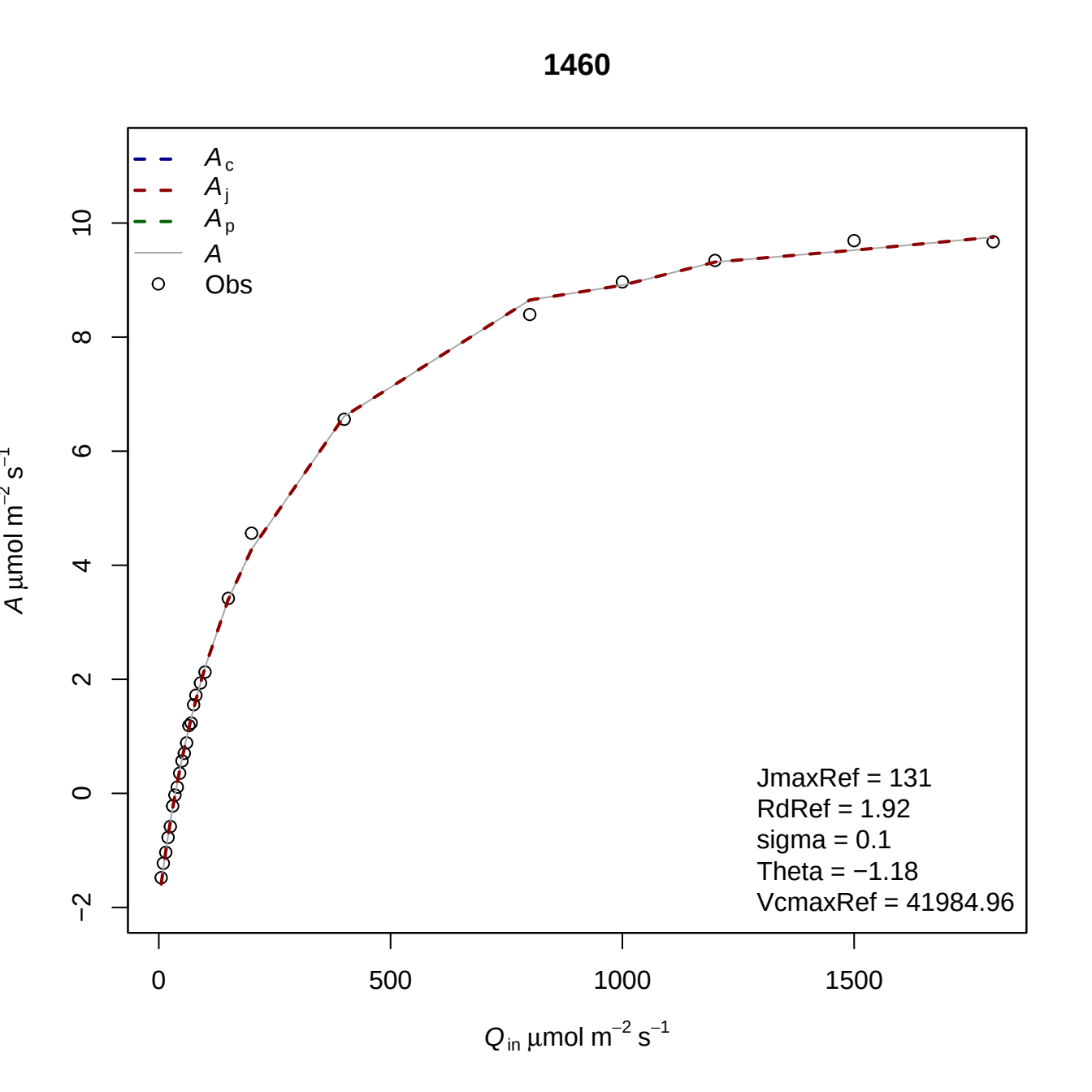

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 730.04
RdRef = 2.38
sigma = 0.27
Theta = −8.11
VcmaxRef = −168.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

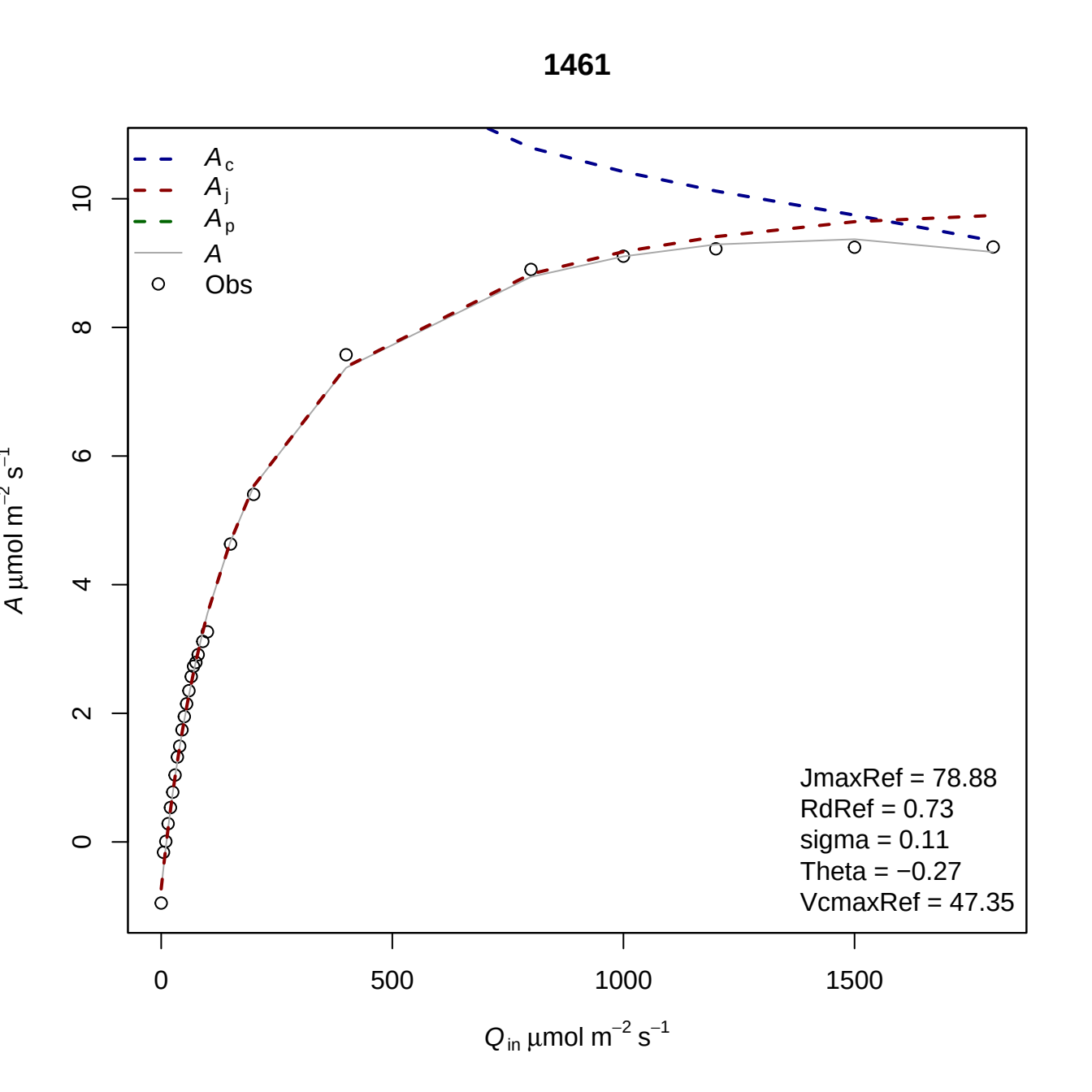

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 72894990.95
RdRef = 2.25
sigma = 0.17
Theta = −3049734.08
VcmaxRef = 614655489.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

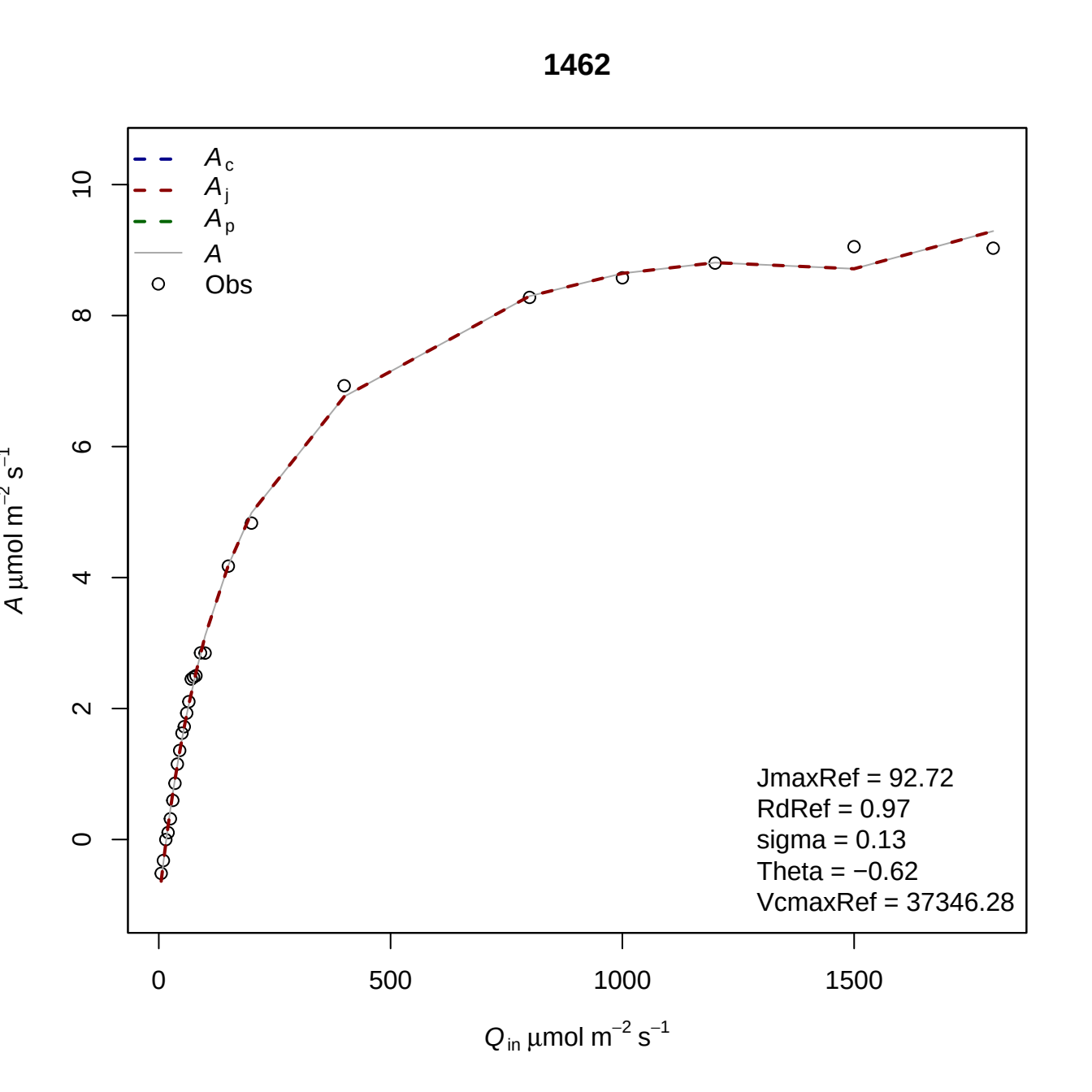

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 130.38
RdRef = 1.09
sigma = 0.17
Theta = −0.23
VcmaxRef = 103776.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

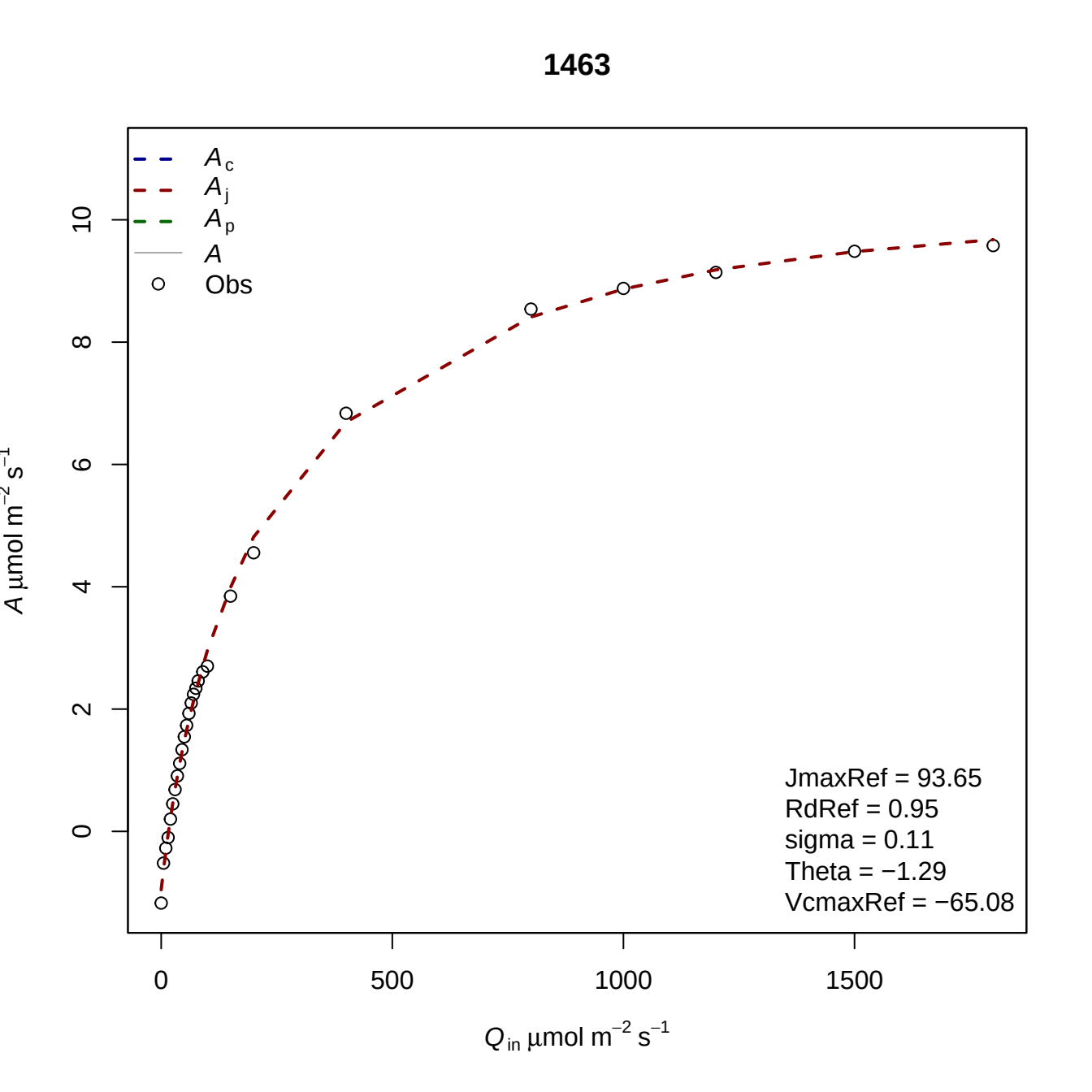

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 109.08
RdRef = 1.02
sigma = 0.11
Theta = −0.85
VcmaxRef = 58.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

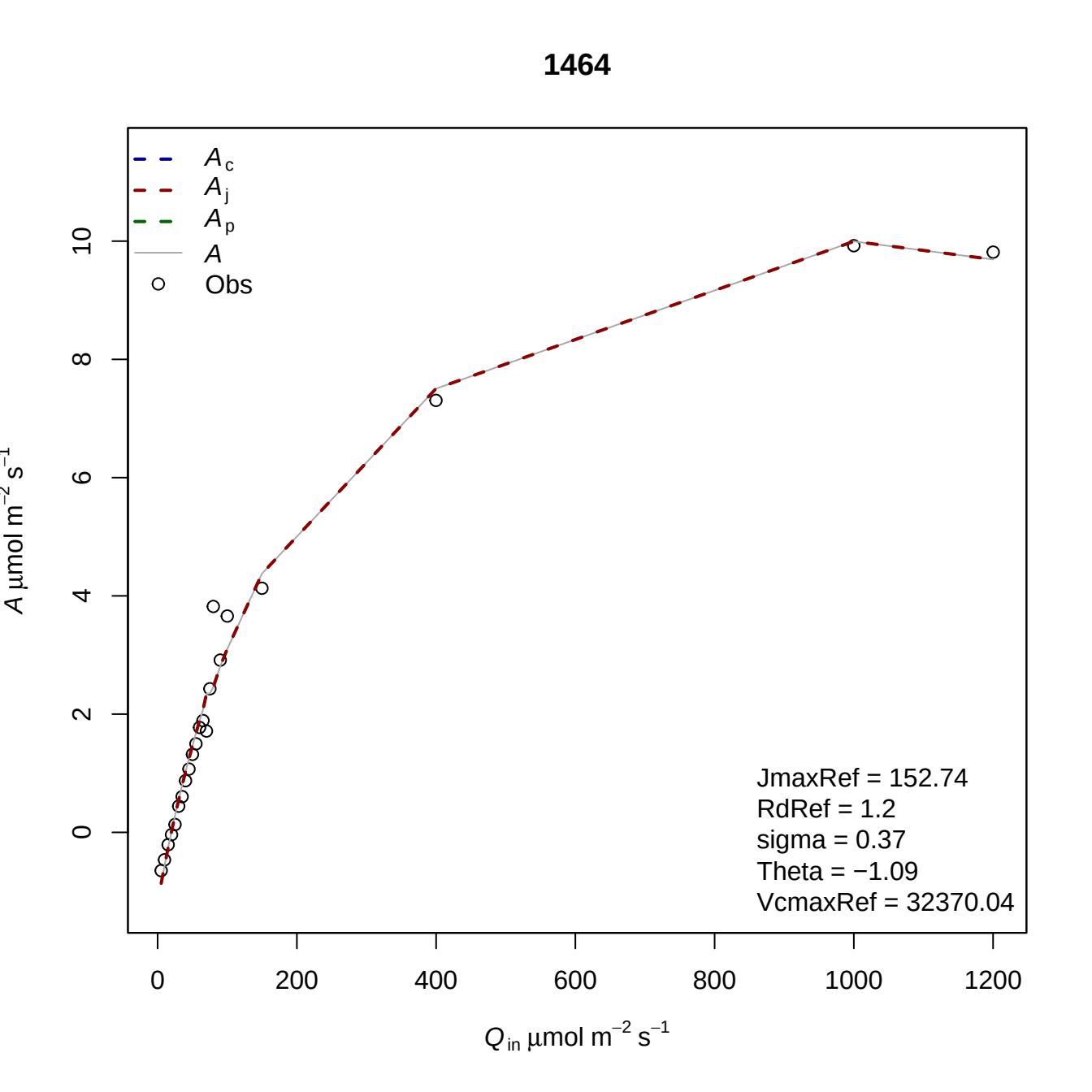

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 156.69
RdRef = 1.7
sigma = 0.12
Theta = −1.16
VcmaxRef = 88.88
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

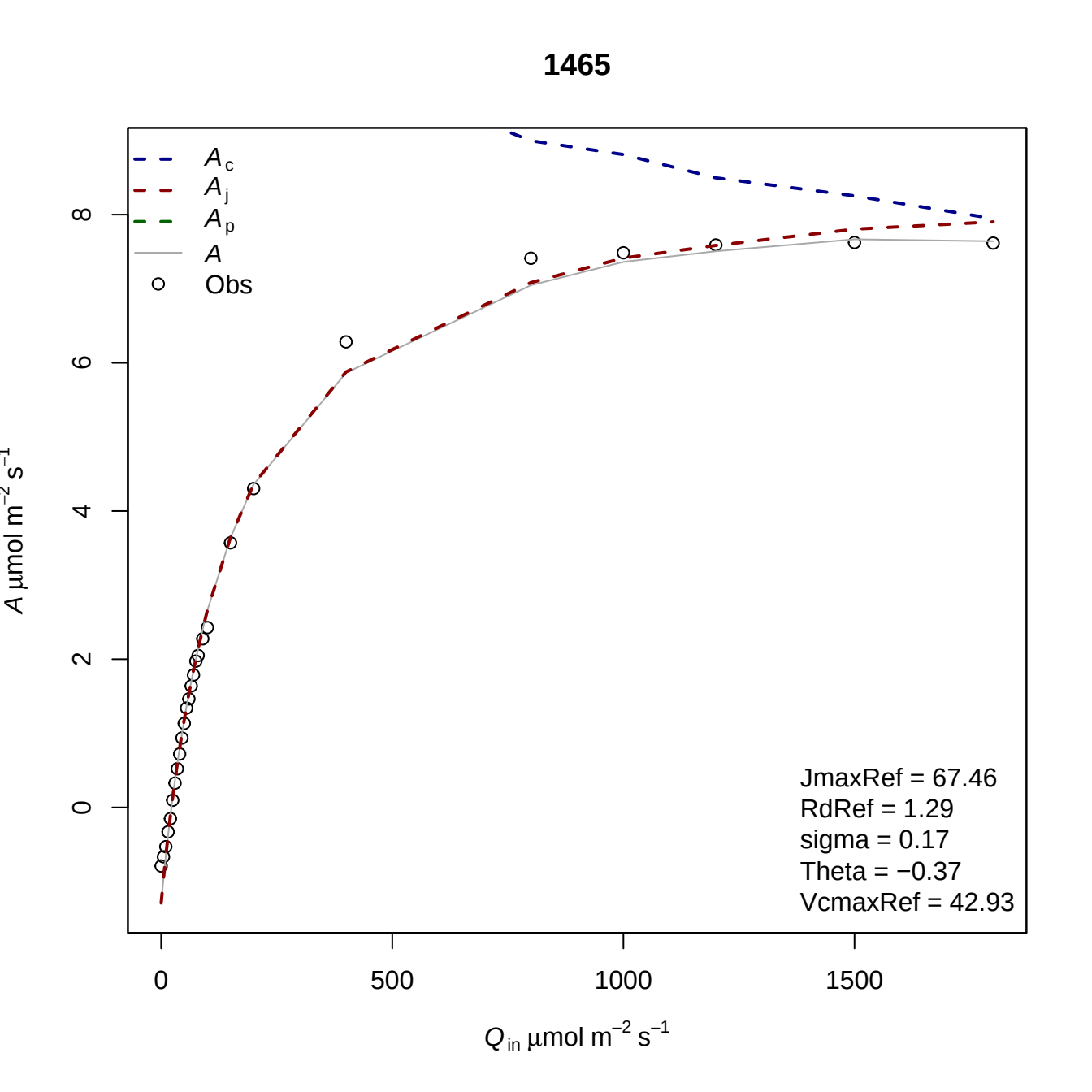

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 131
RdRef = 1.92
sigma = 0.1
Theta = −1.18
VcmaxRef = 41984.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

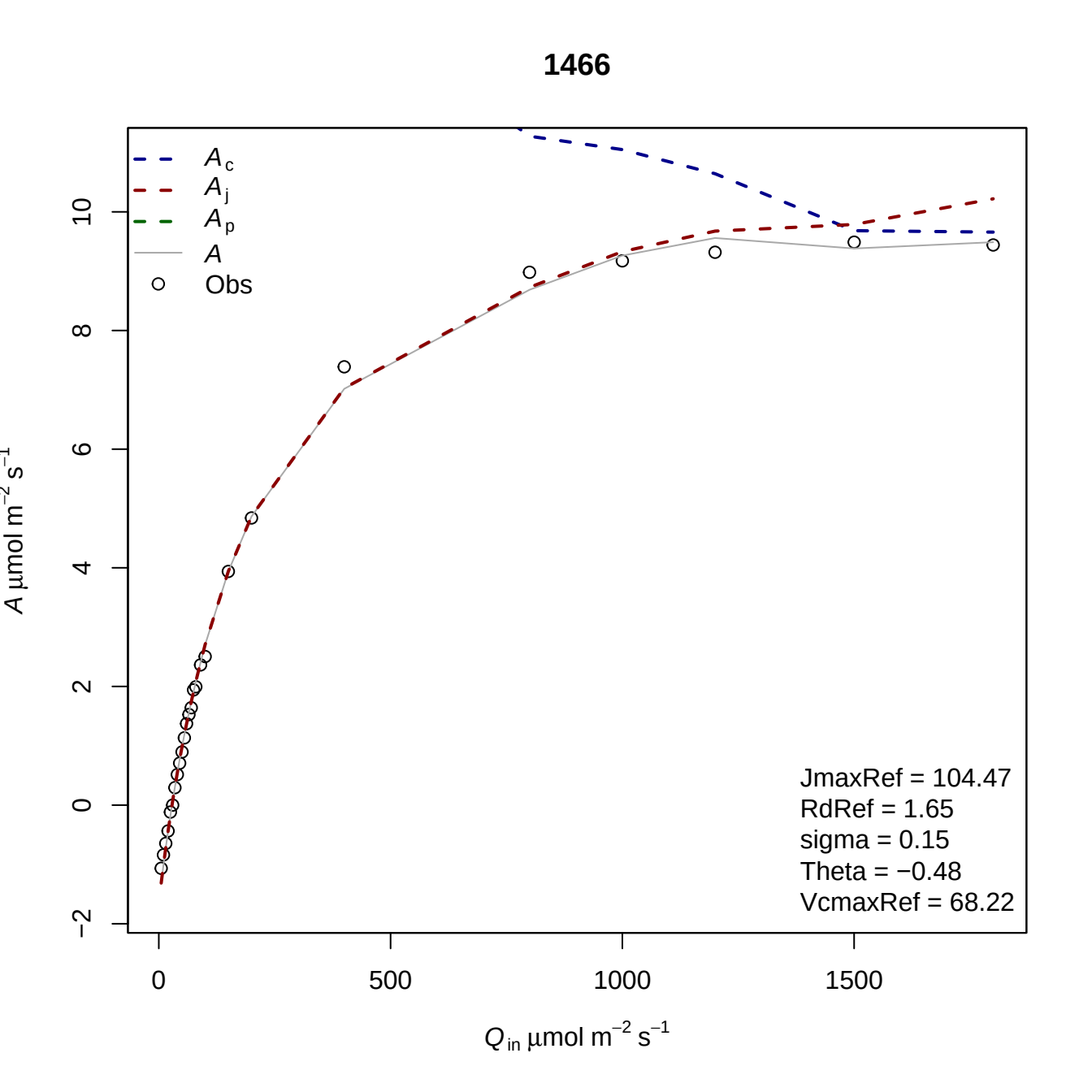

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 78.88
RdRef = 0.73
sigma = 0.11
Theta = −0.27
VcmaxRef = 47.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

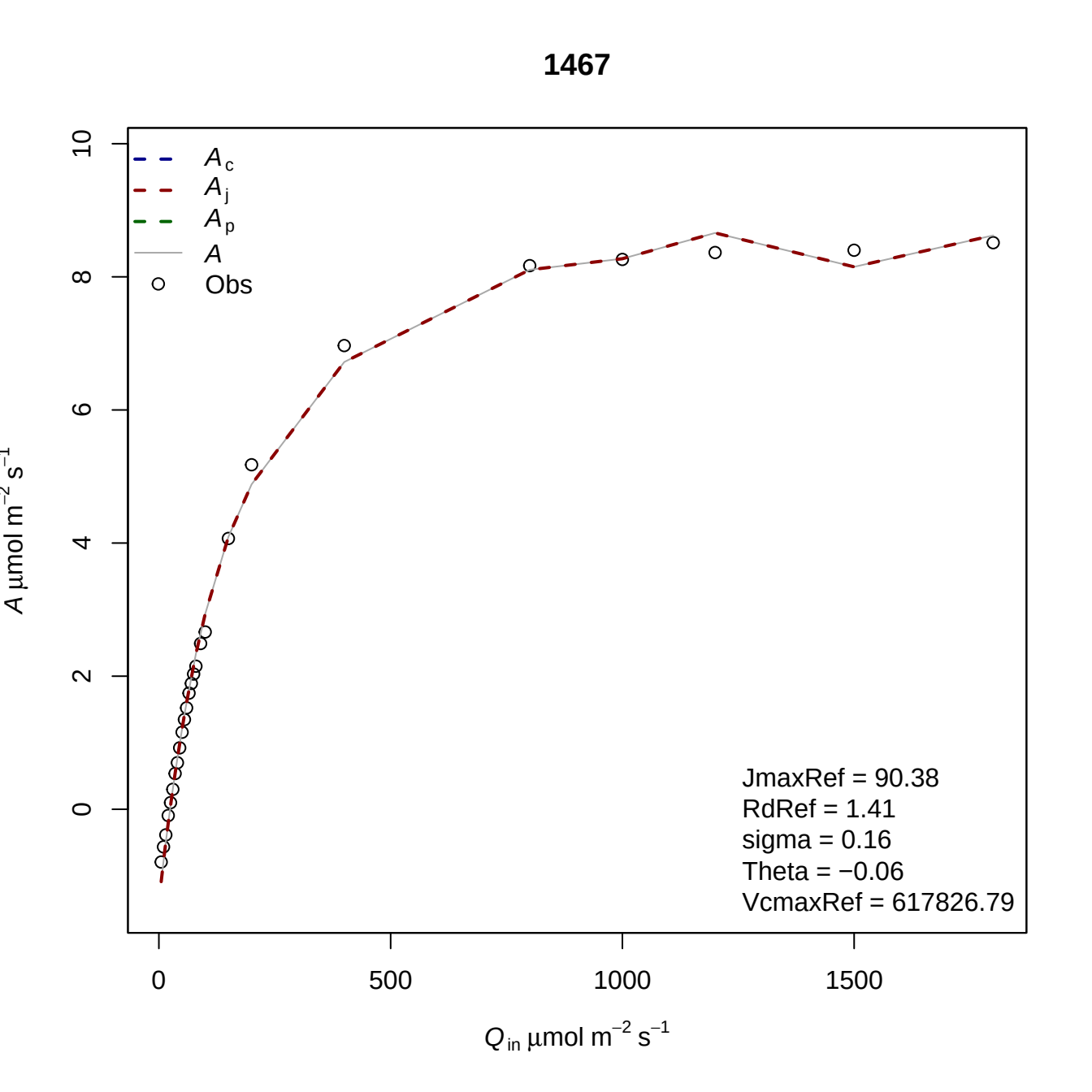

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 92.72
RdRef = 0.97
sigma = 0.13
Theta = −0.62
VcmaxRef = 37346.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

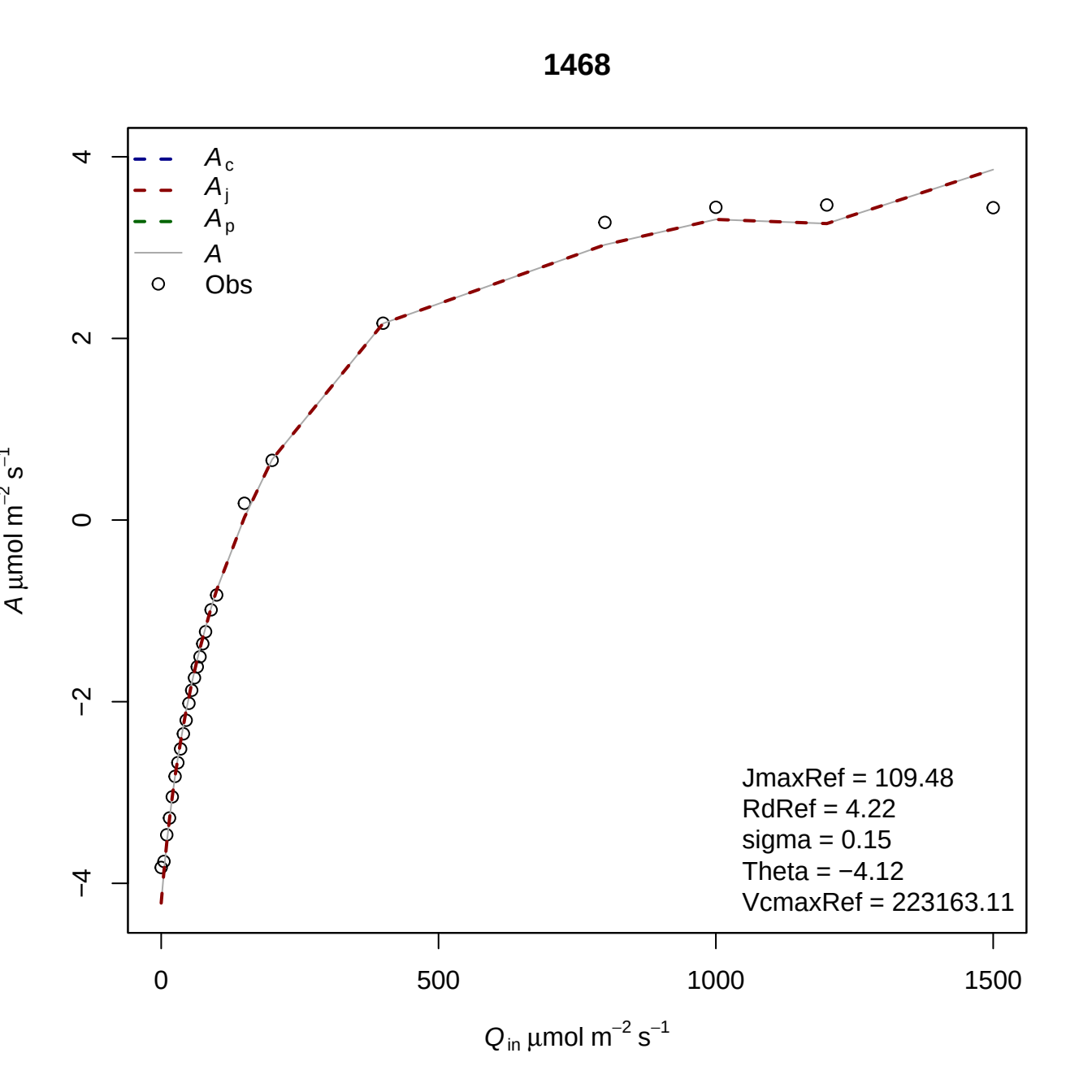

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 93.65
RdRef = 0.95
sigma = 0.11
Theta = −1.29
VcmaxRef = −65.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

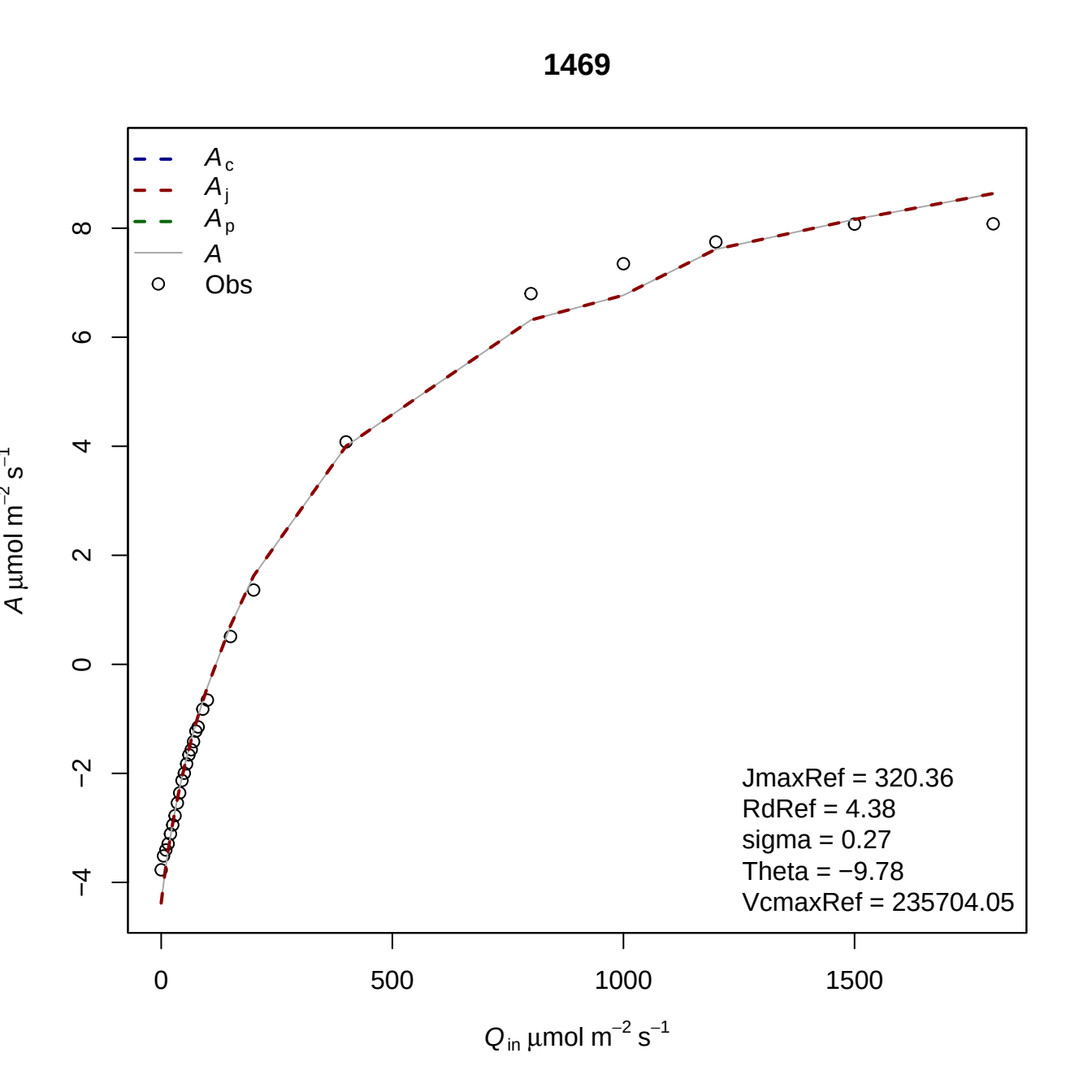

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 152.74
RdRef = 1.2
sigma = 0.37
Theta = −1.09
VcmaxRef = 32370.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

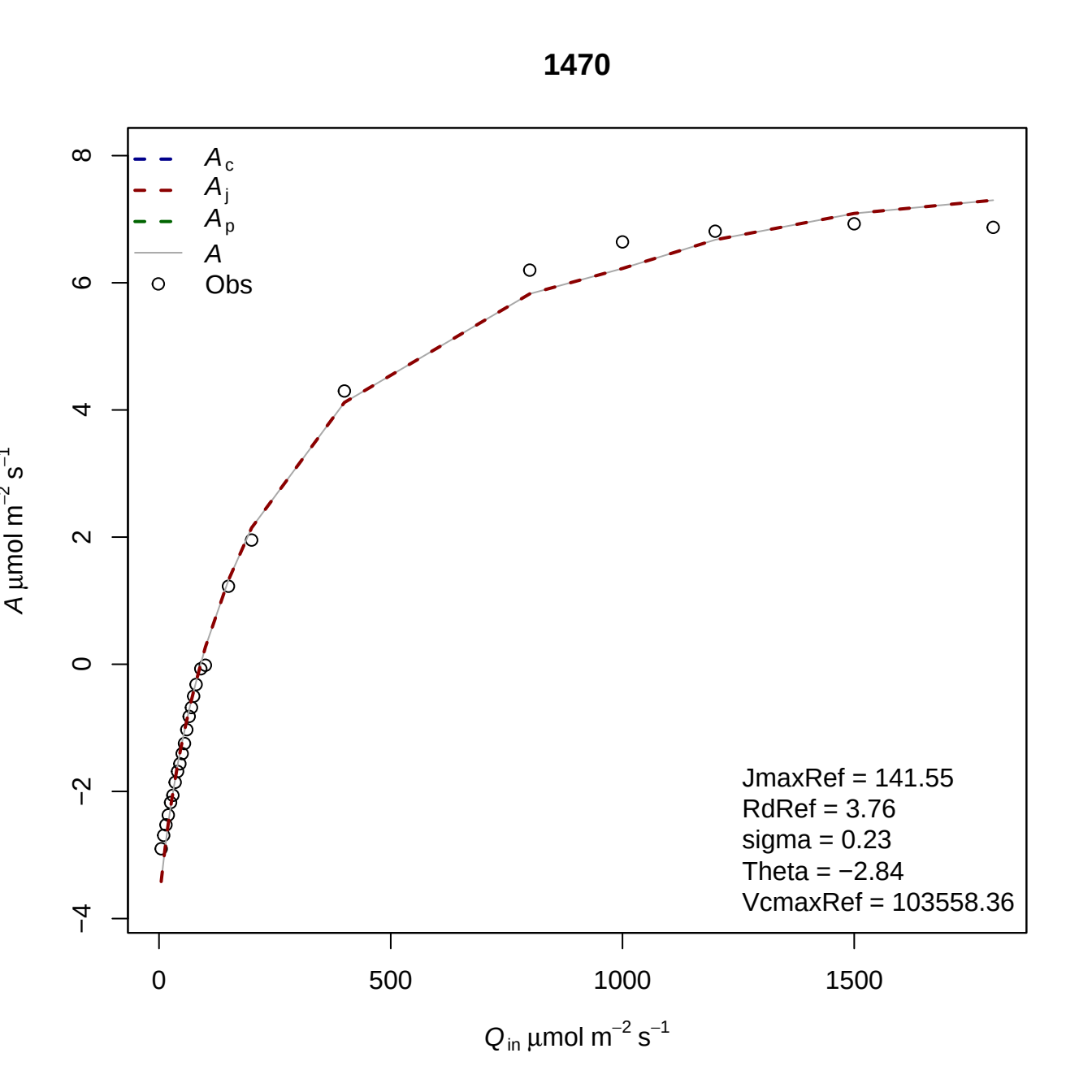

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 67.46
RdRef = 1.29
sigma = 0.17
Theta = −0.37
VcmaxRef = 42.93
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

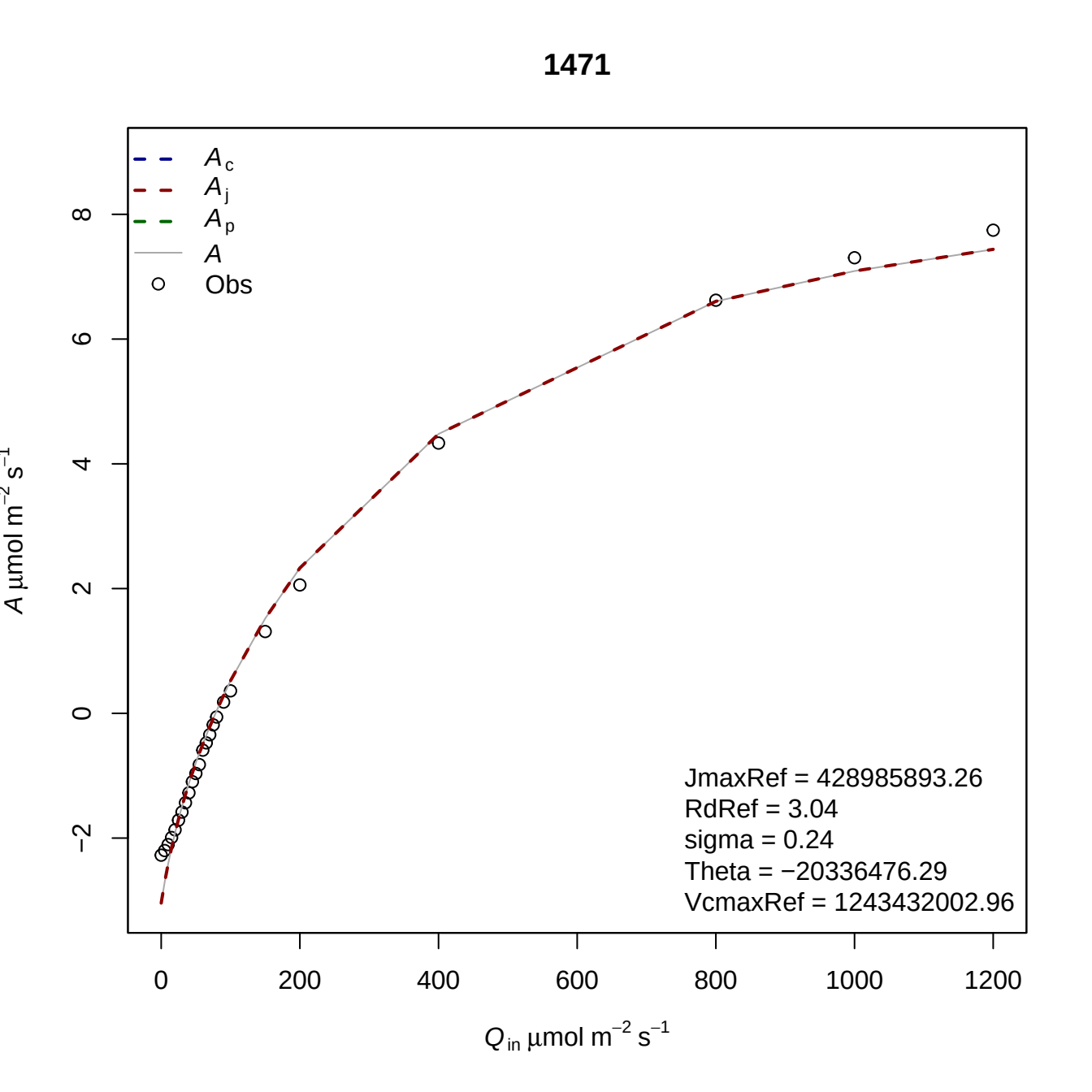

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 104.47
RdRef = 1.65
sigma = 0.15
Theta = −0.48
VcmaxRef = 68.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

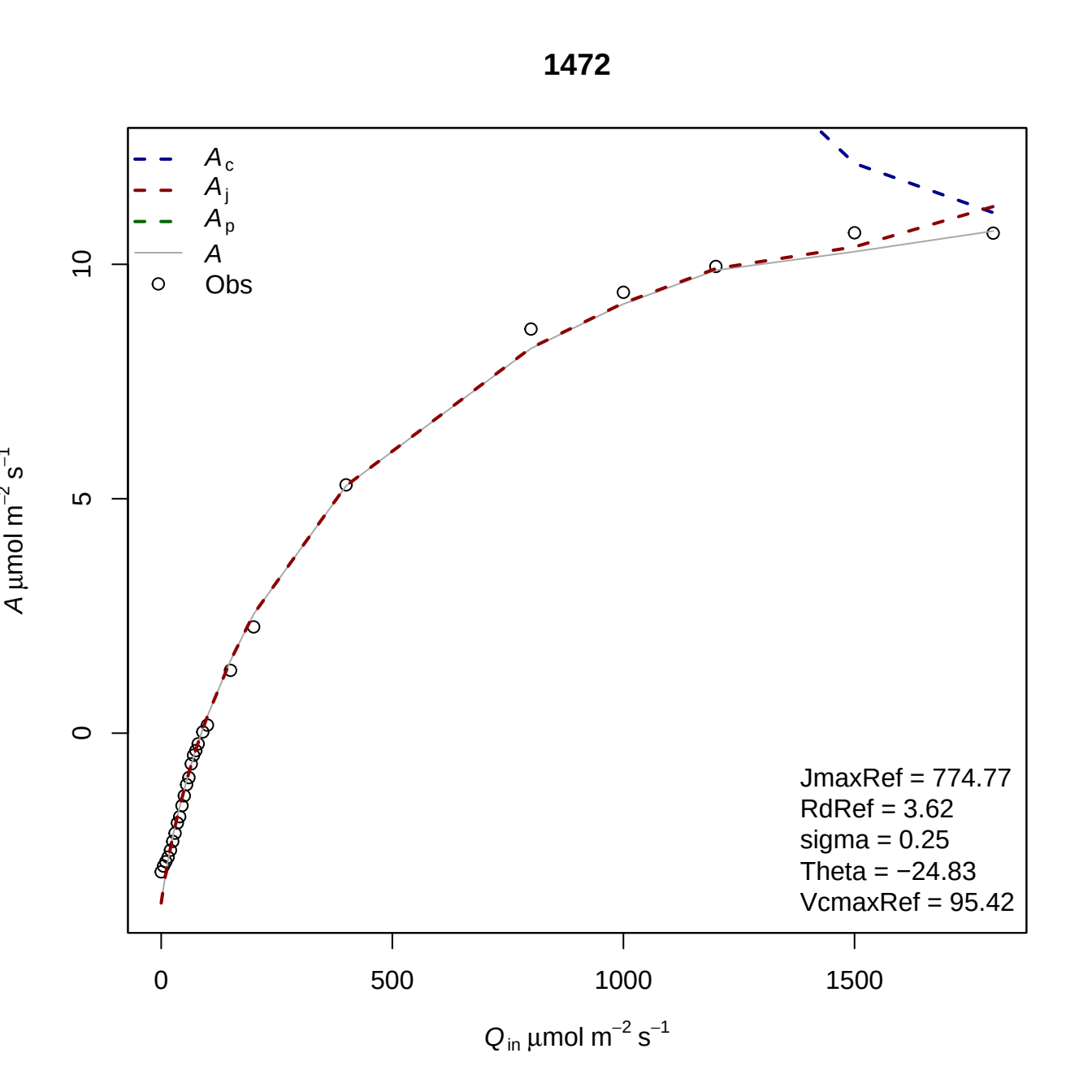

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 90.38
RdRef = 1.41
sigma = 0.16
Theta = −0.06
VcmaxRef = 617826.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

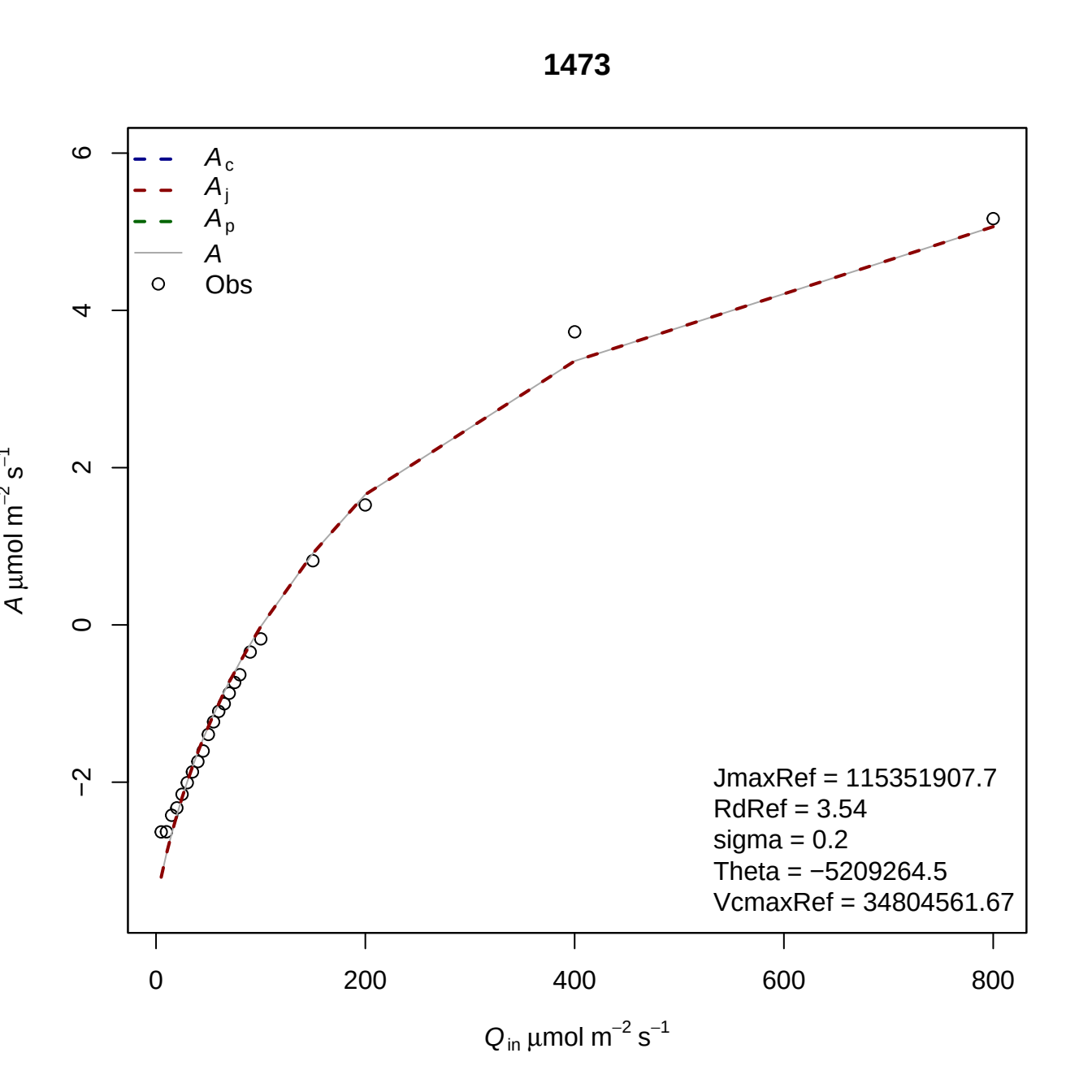

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 109.48
RdRef = 4.22
sigma = 0.15
Theta = −4.12
VcmaxRef = 223163.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

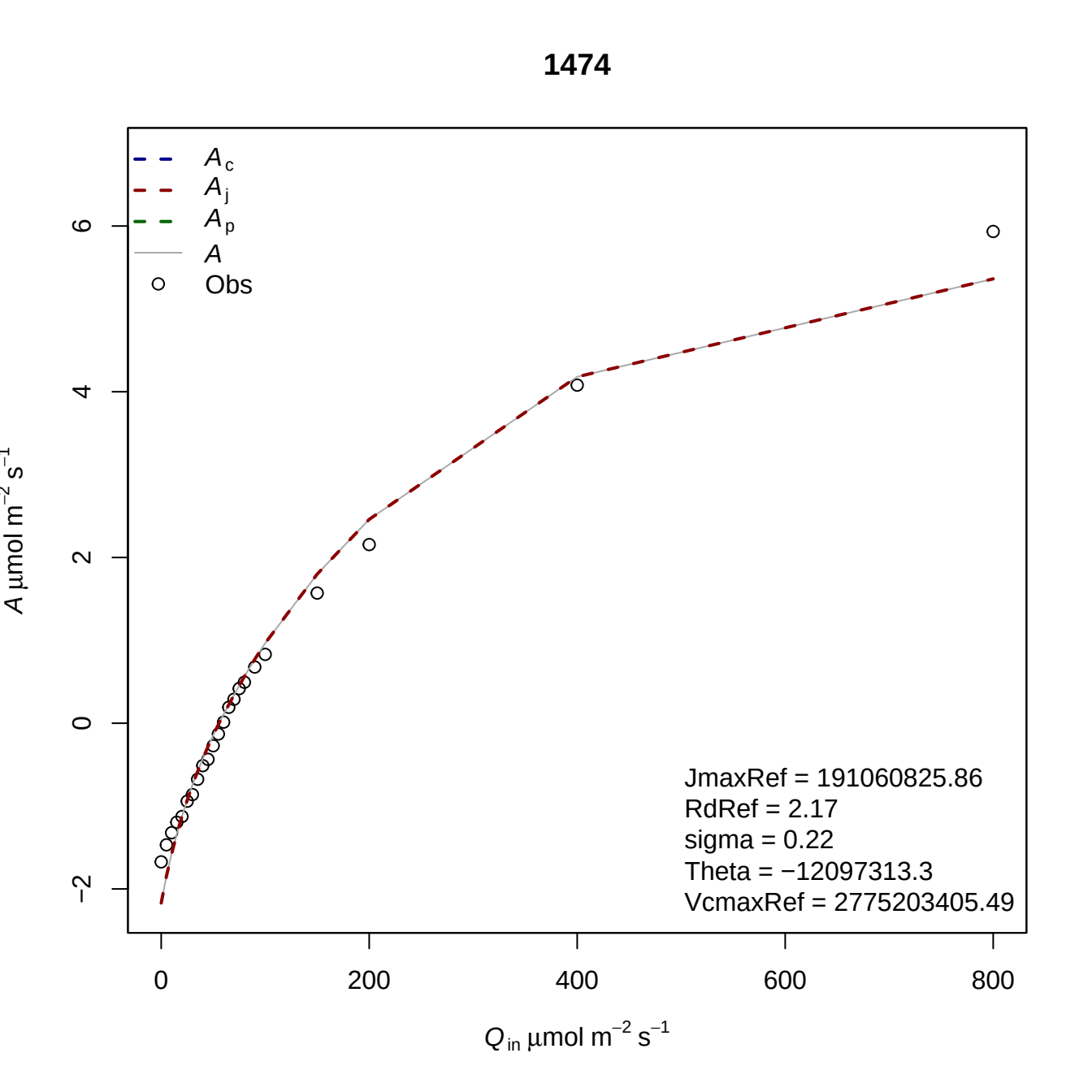

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 320.36
RdRef = 4.38
sigma = 0.27
Theta = −9.78
VcmaxRef = 235704.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

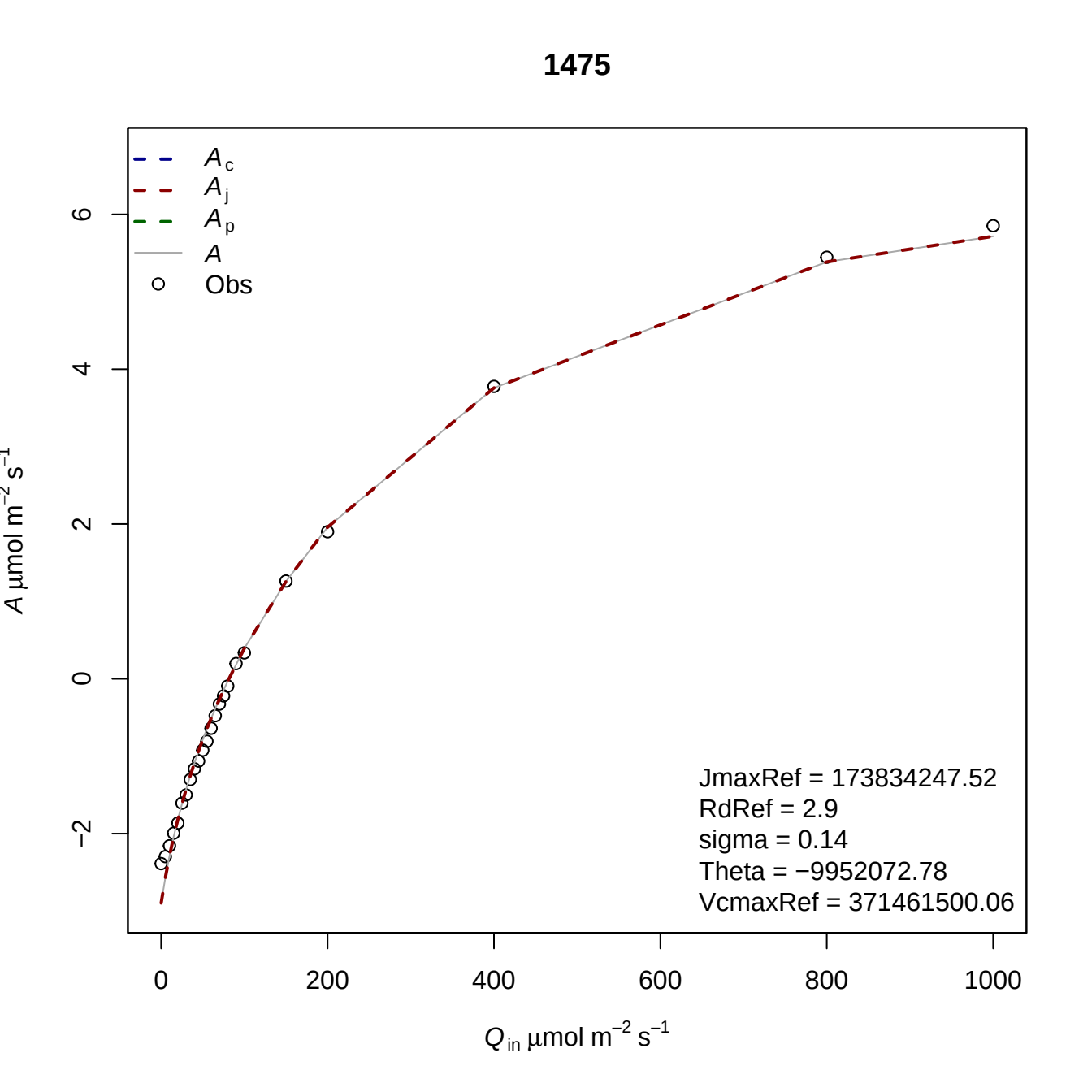

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 141.55
RdRef = 3.76
sigma = 0.23
Theta = −2.84
VcmaxRef = 103558.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

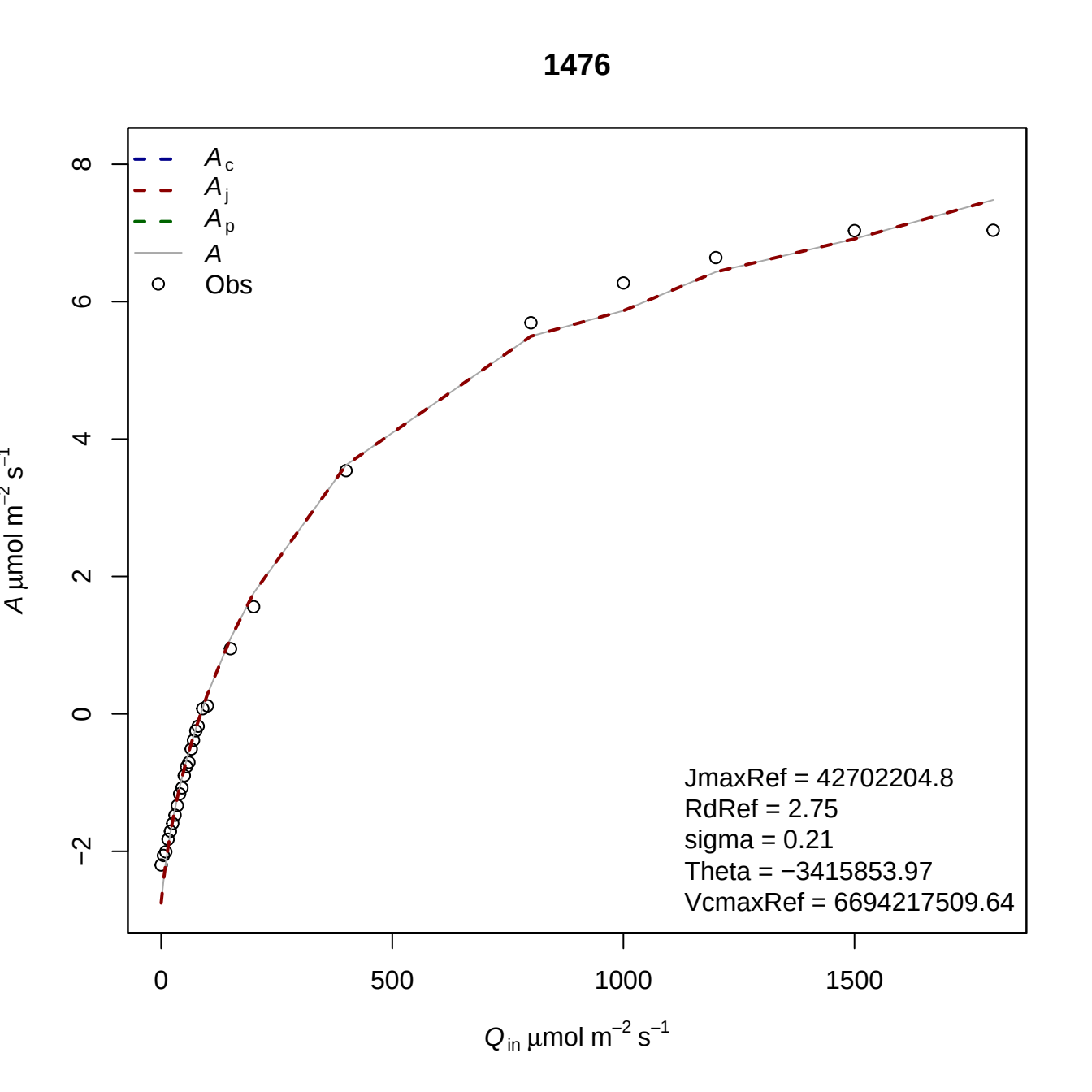

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 428985893.26
RdRef = 3.04
sigma = 0.24
Theta = −20336476.29
VcmaxRef = 1243432002.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

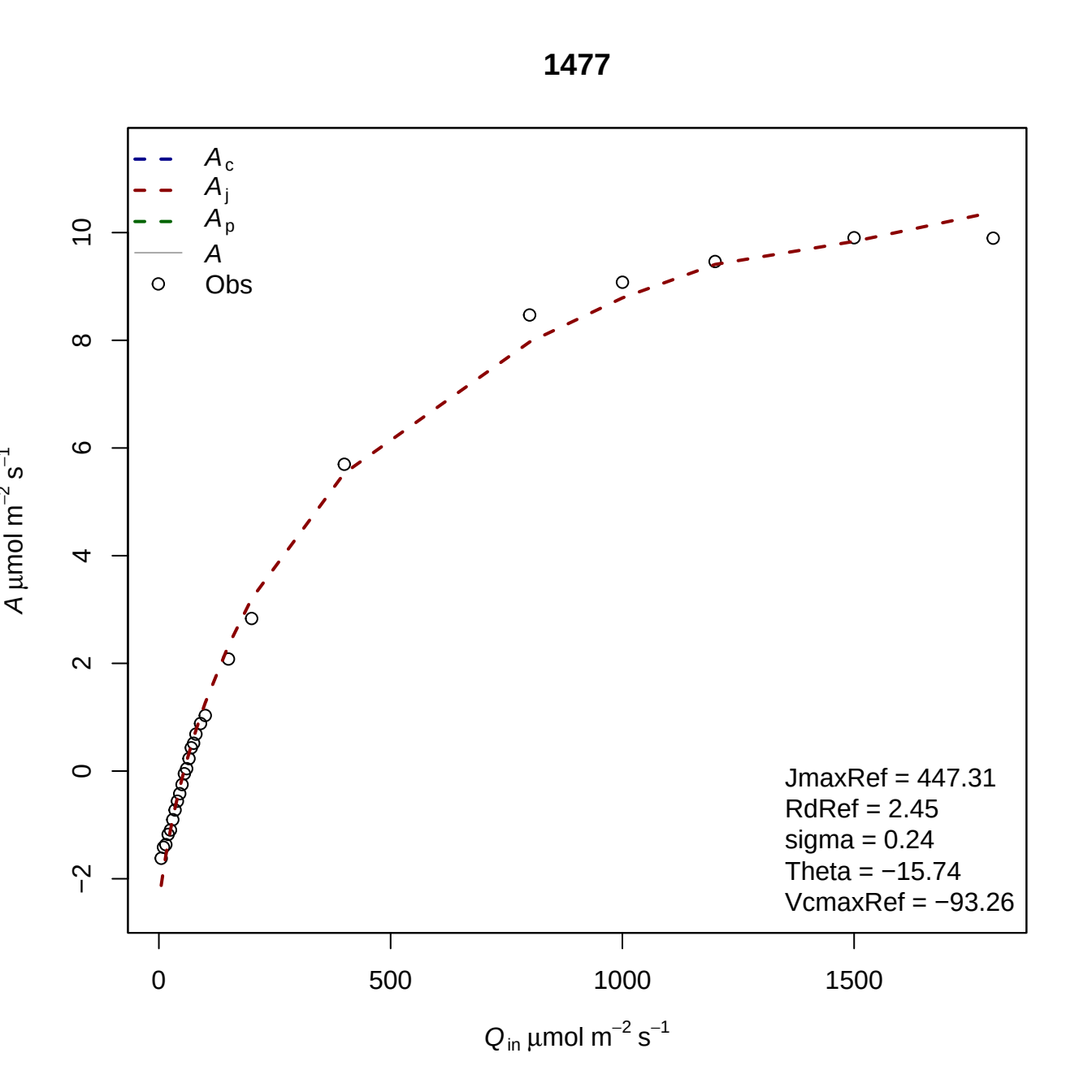

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 774.77
RdRef = 3.62
sigma = 0.25
Theta = −24.83
VcmaxRef = 95.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

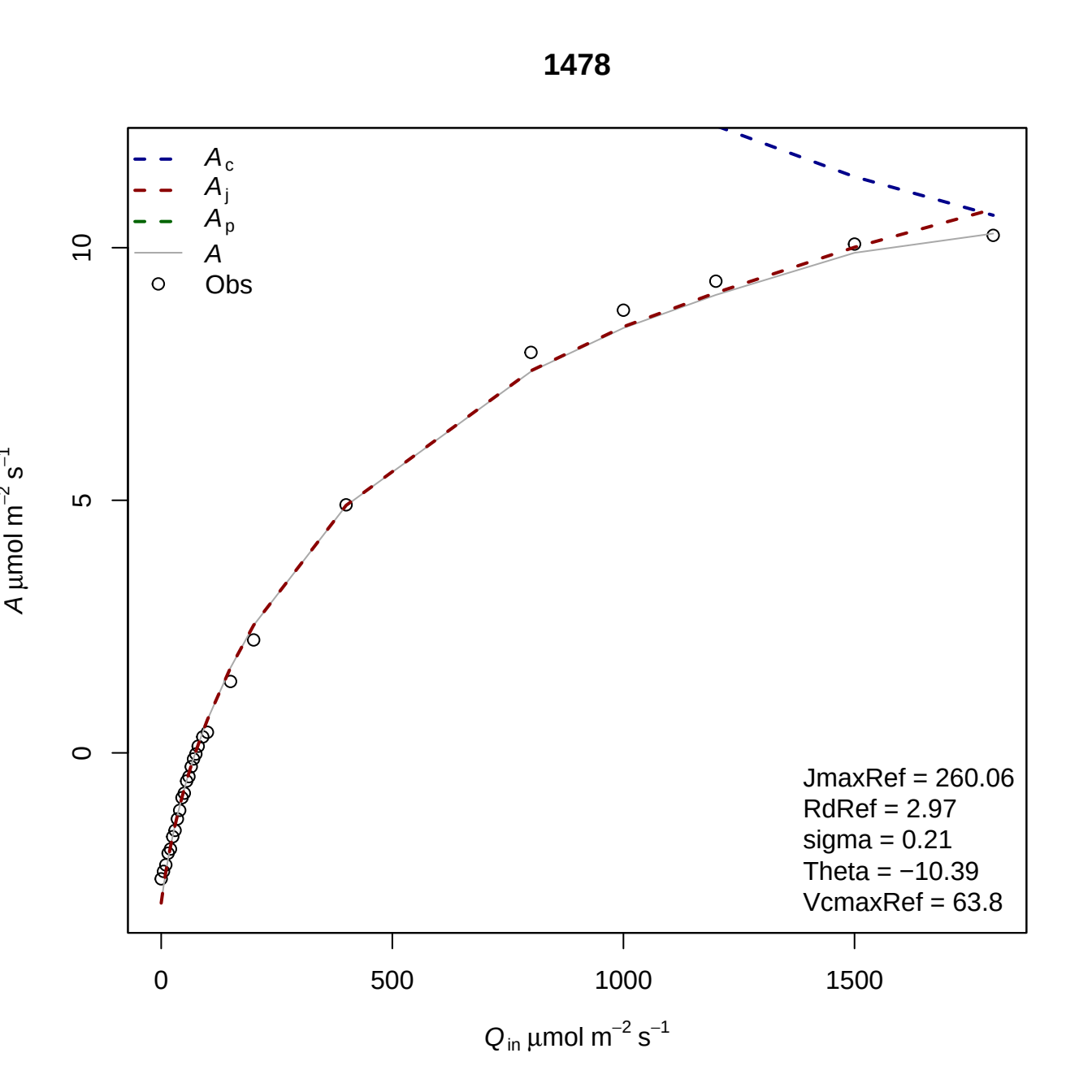

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 115351907.7
RdRef = 3.54
sigma = 0.2
Theta = −5209264.5
VcmaxRef = 34804561.67
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800

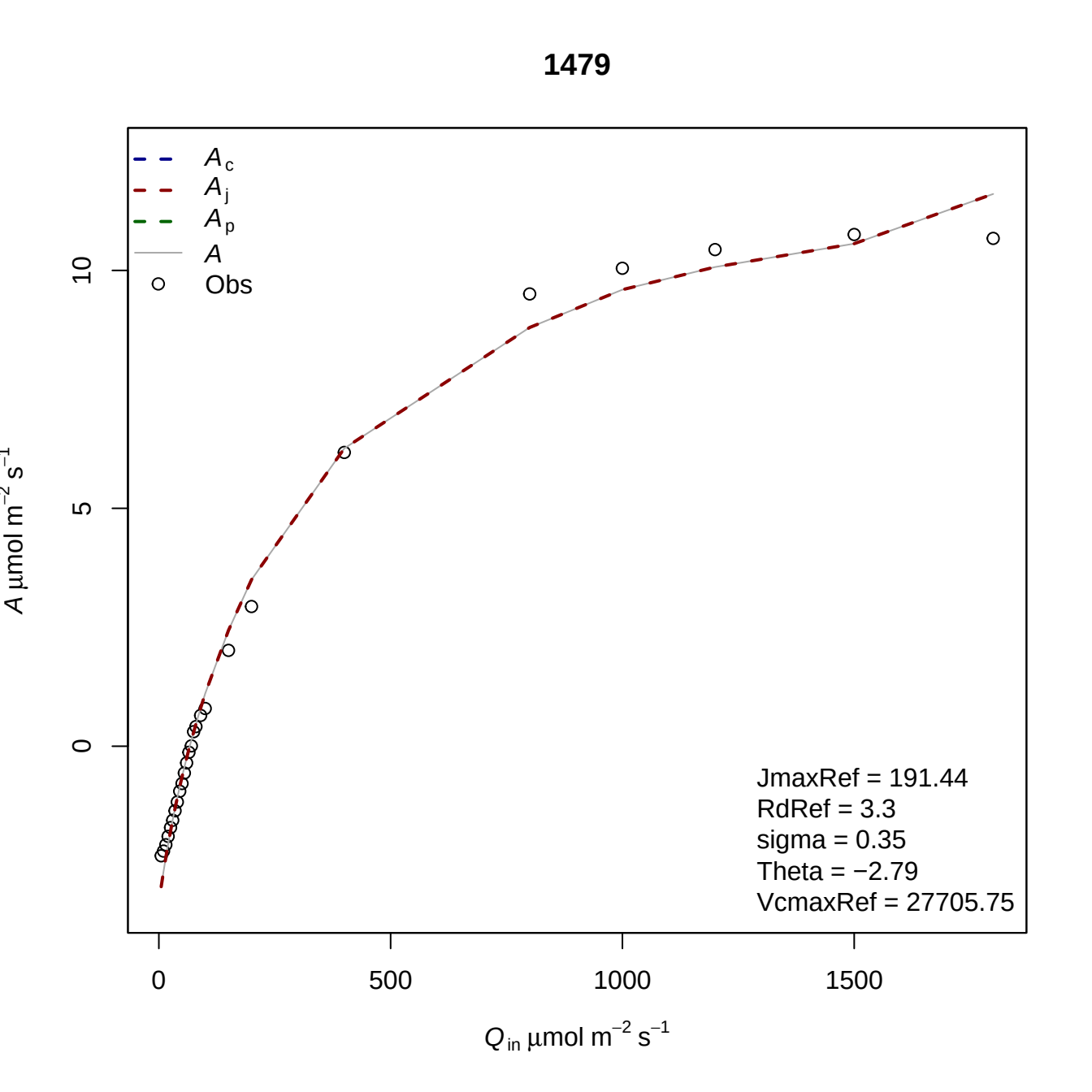

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 191060825.86
RdRef = 2.17
sigma = 0.22
Theta = −12097313.3
VcmaxRef = 2775203405.49
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

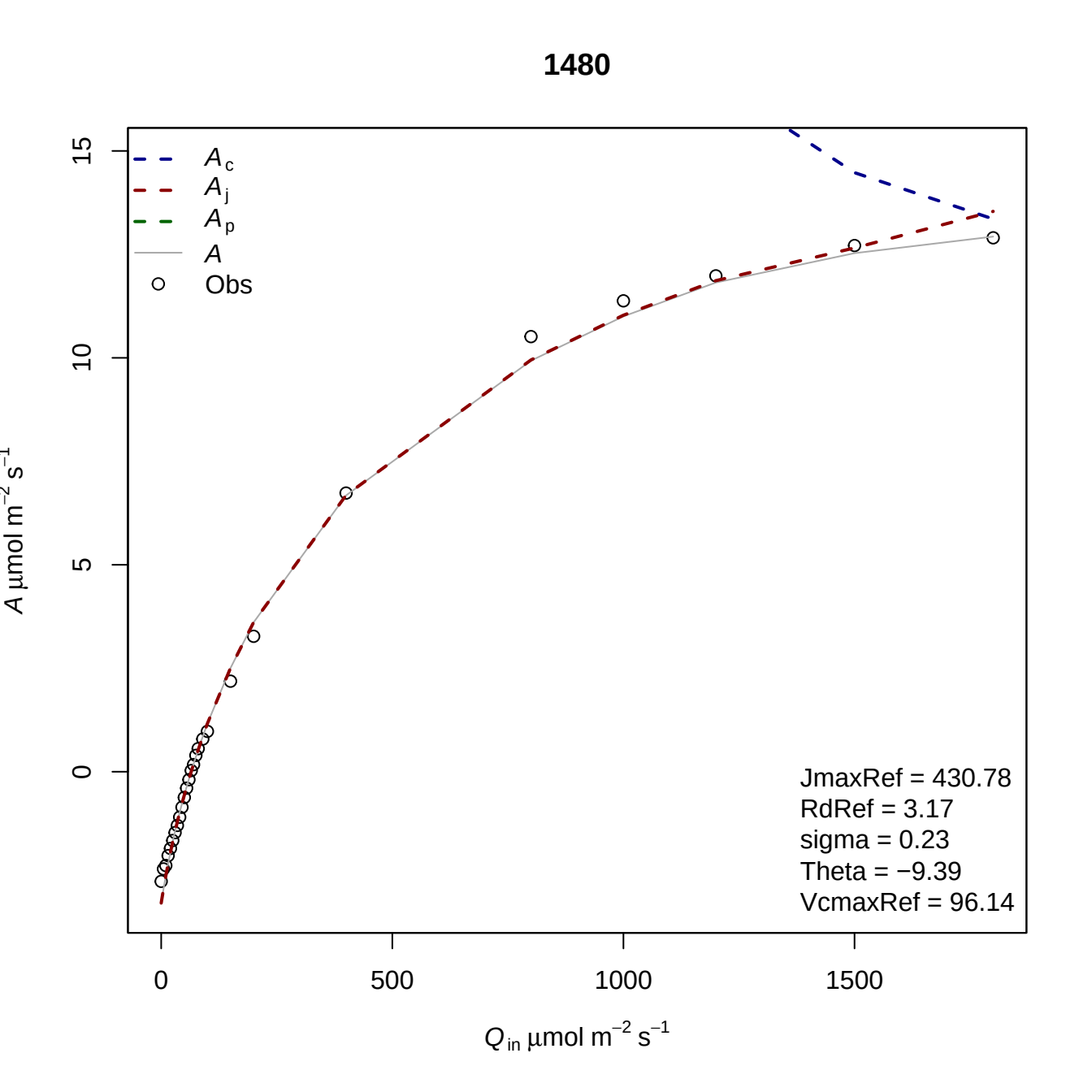

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 173834247.52
RdRef = 2.9
sigma = 0.14
Theta = −9952072.78
VcmaxRef = 371461500.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

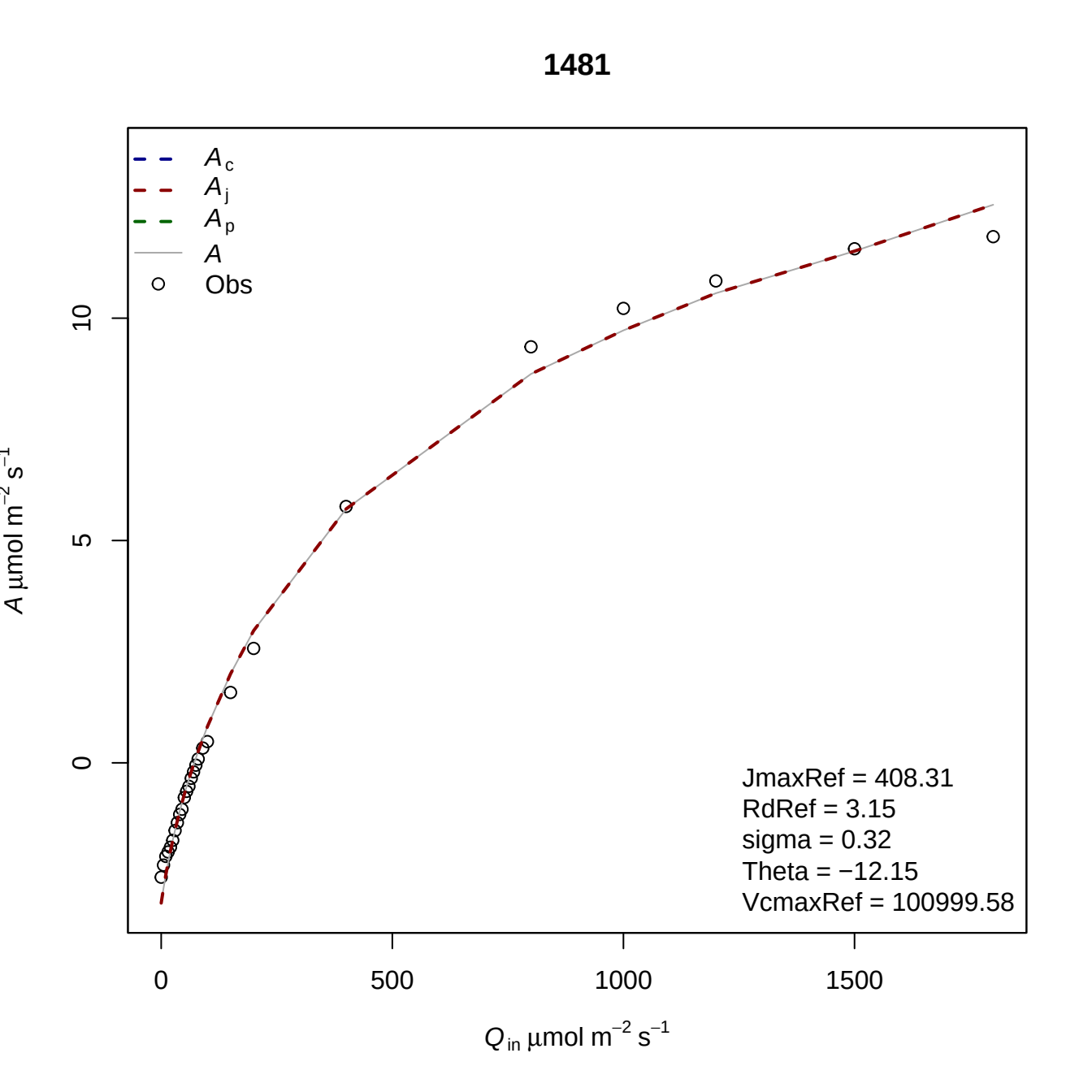

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 42702204.8
RdRef = 2.75
sigma = 0.21
Theta = −3415853.97
VcmaxRef = 6694217509.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

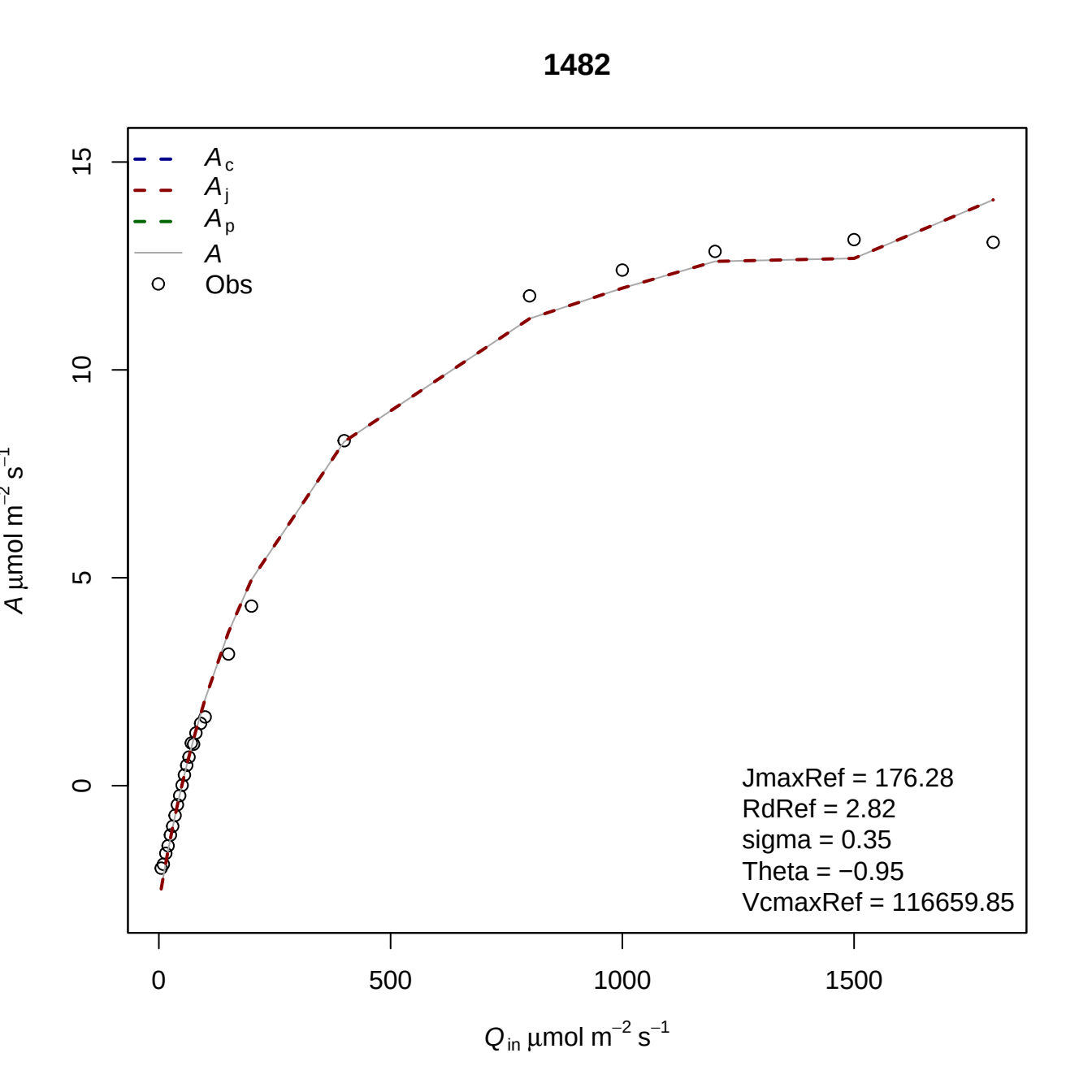

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 447.31
RdRef = 2.45
sigma = 0.24
Theta = −15.74
VcmaxRef = −93.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

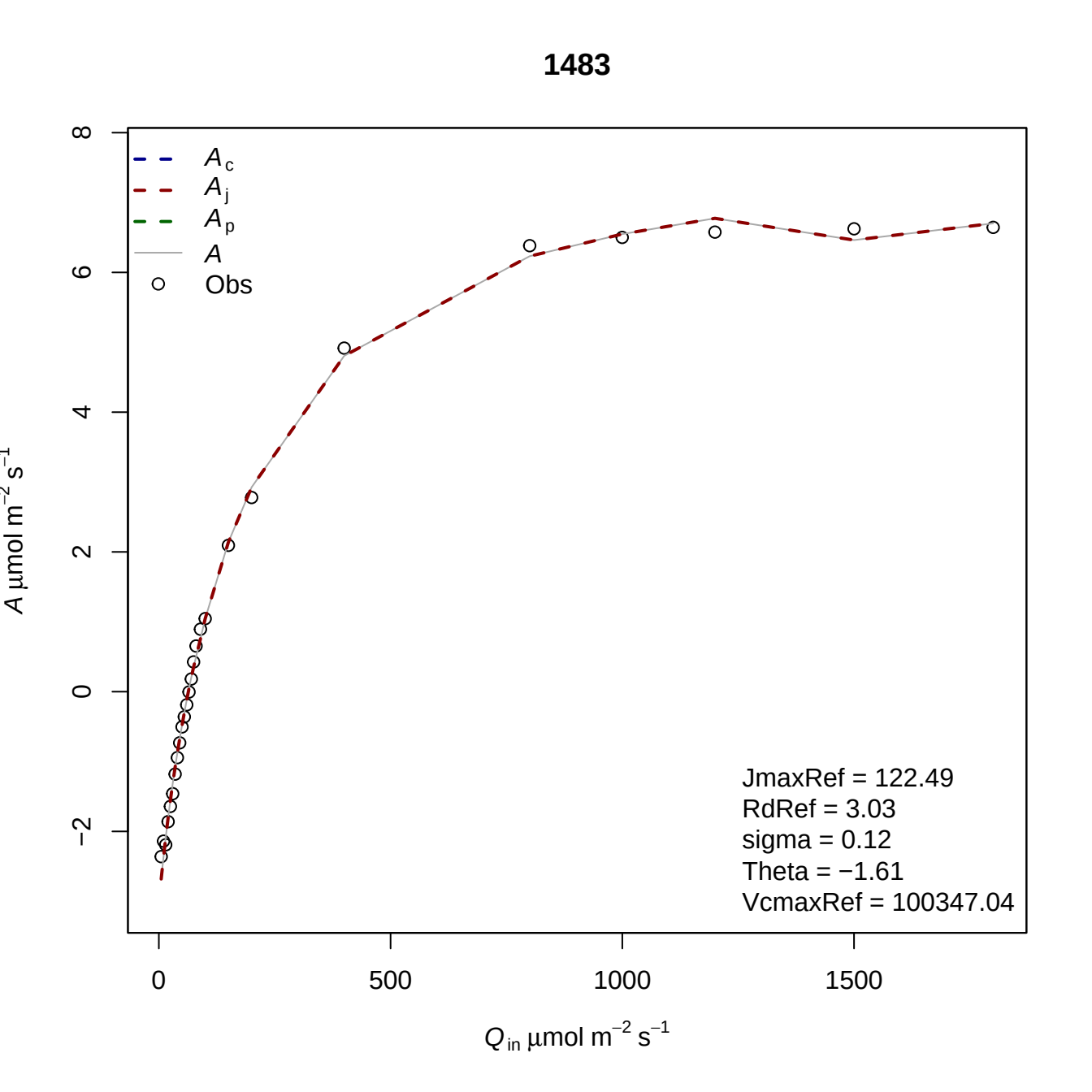

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 260.06
RdRef = 2.97
sigma = 0.21
Theta = −10.39
VcmaxRef = 63.8
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

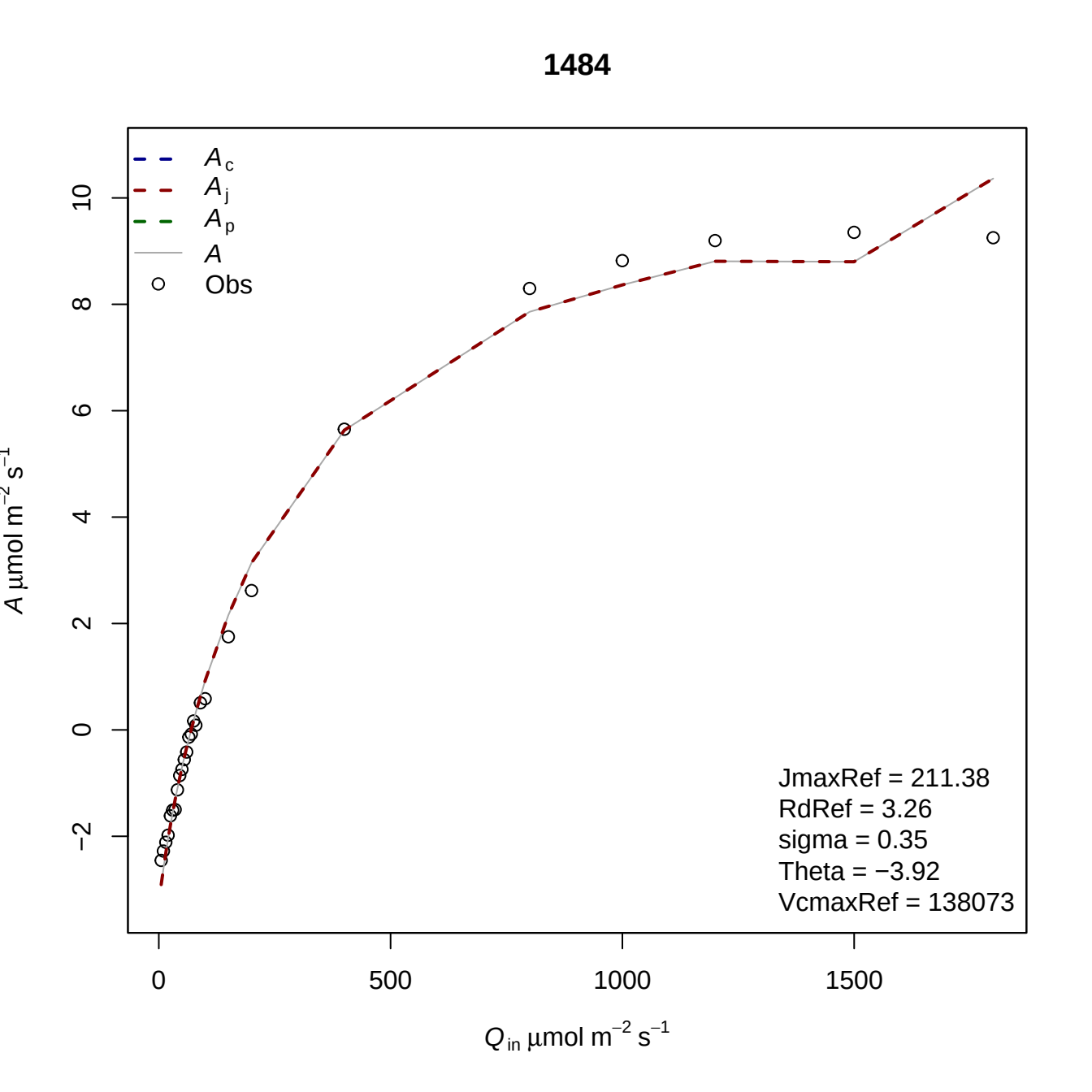

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 191.44
RdRef = 3.3
sigma = 0.35
Theta = −2.79
VcmaxRef = 27705.75
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

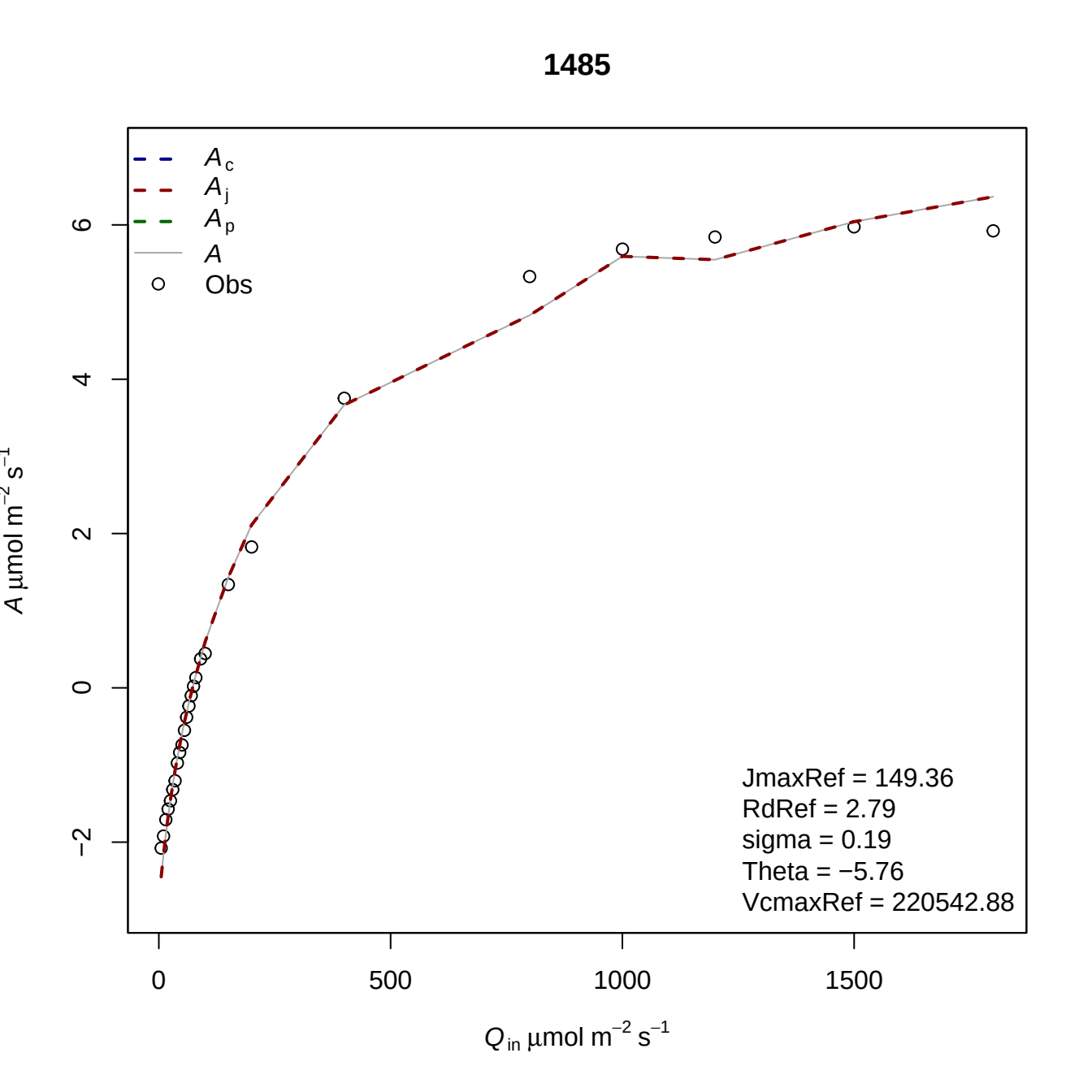

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 430.78
RdRef = 3.17
sigma = 0.23
Theta = −9.39
VcmaxRef = 96.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

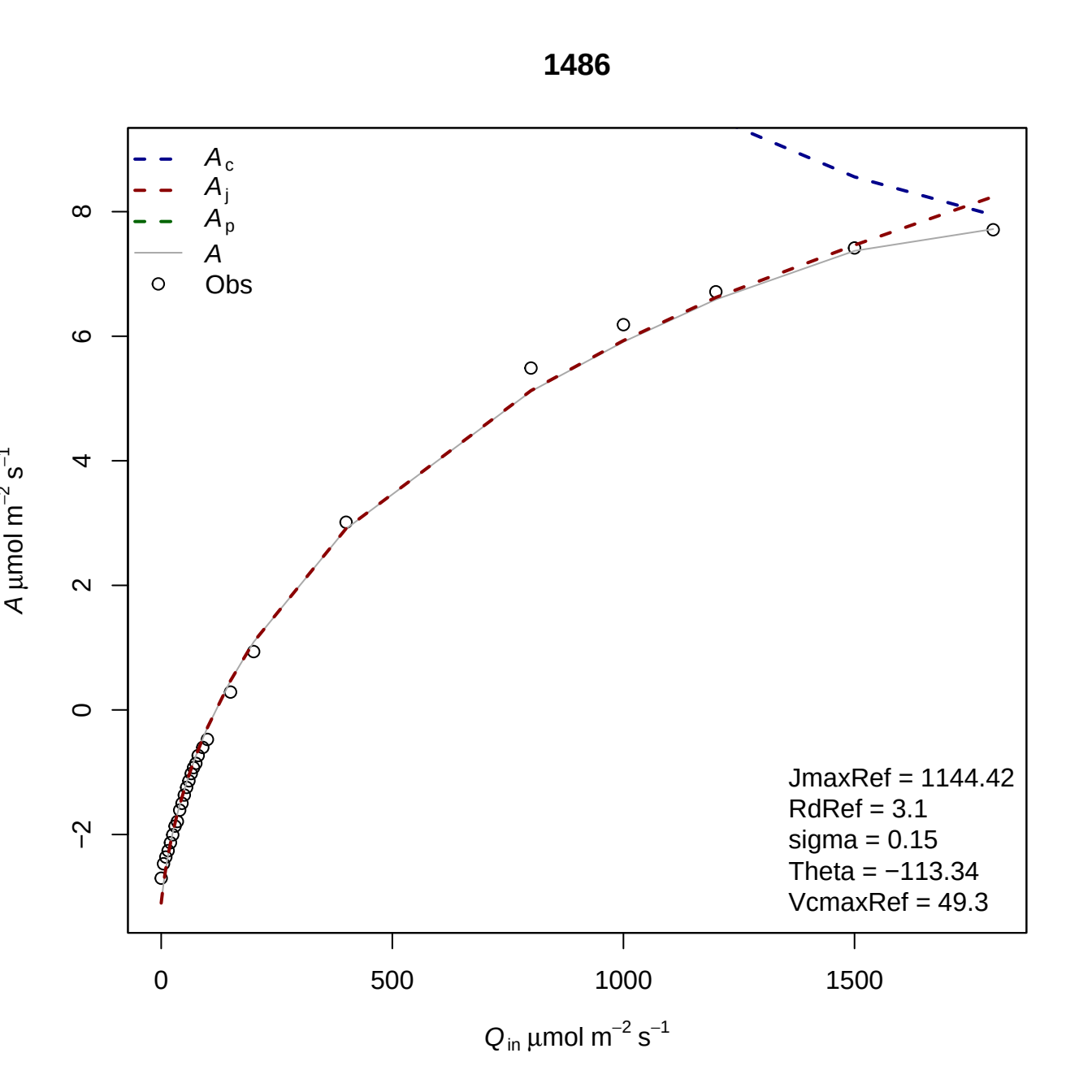

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 408.31
RdRef = 3.15
sigma = 0.32
Theta = −12.15
VcmaxRef = 100999.58
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

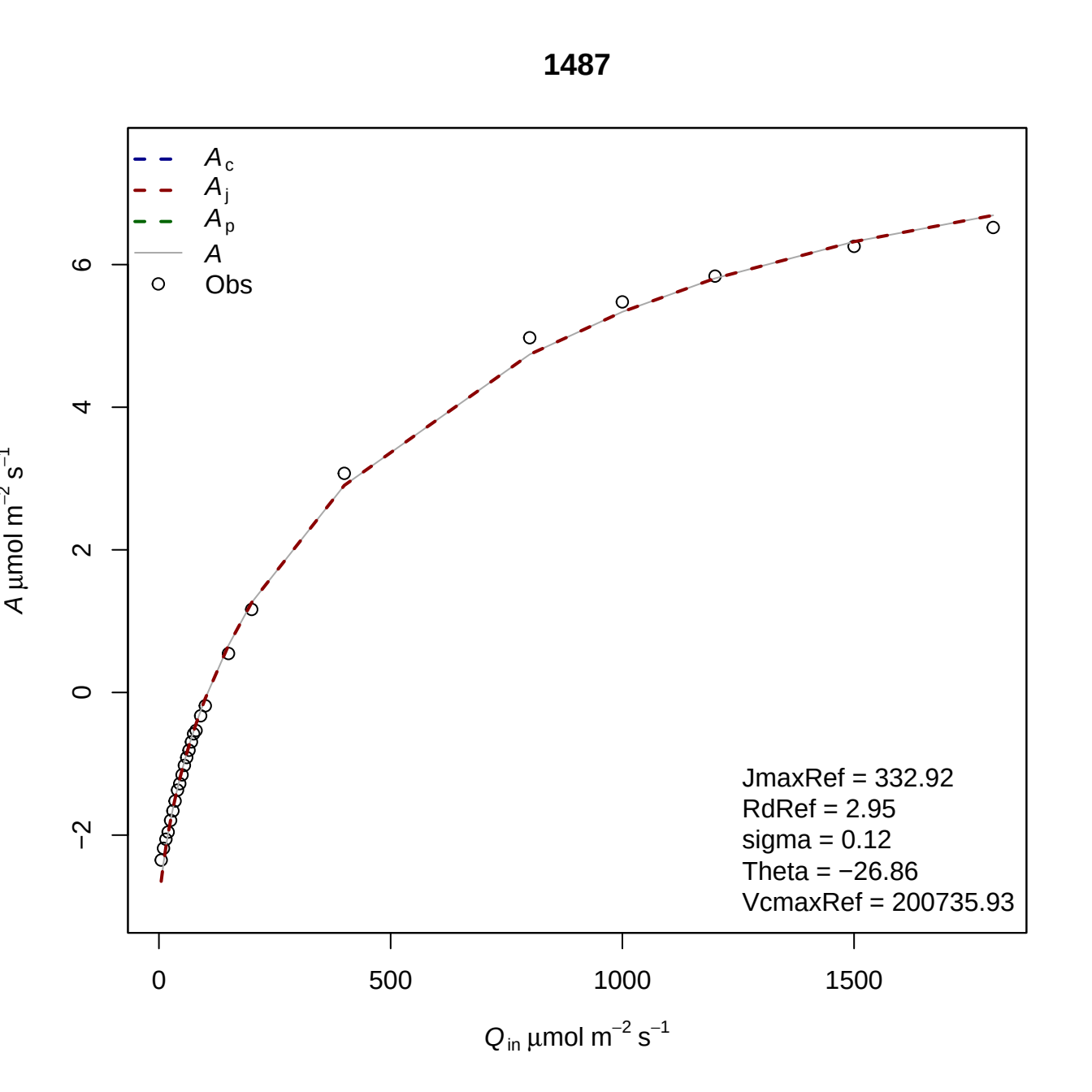

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 176.28
RdRef = 2.82
sigma = 0.35
Theta = −0.95
VcmaxRef = 116659.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

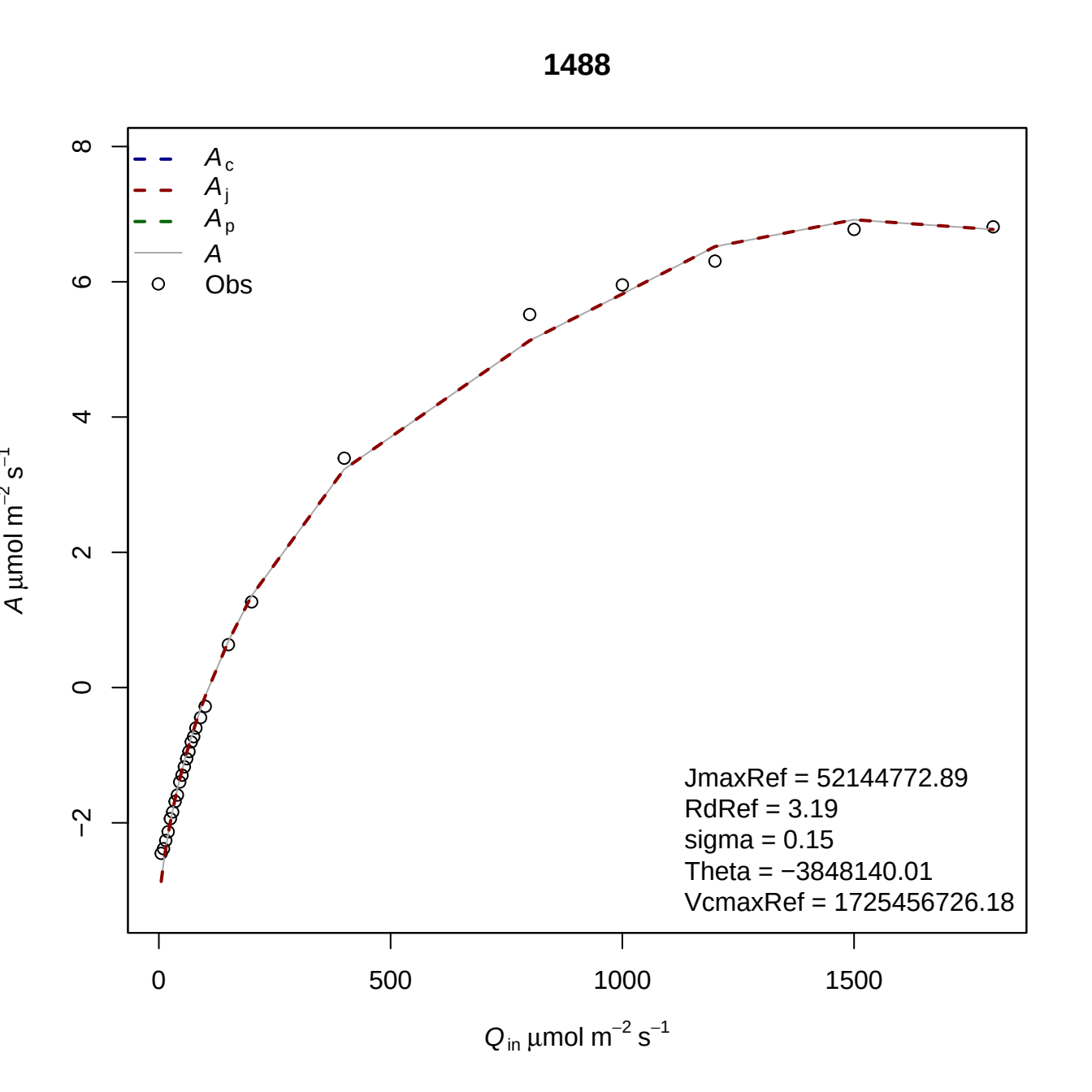

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 122.49
RdRef = 3.03
sigma = 0.12
Theta = −1.61
VcmaxRef = 100347.04
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

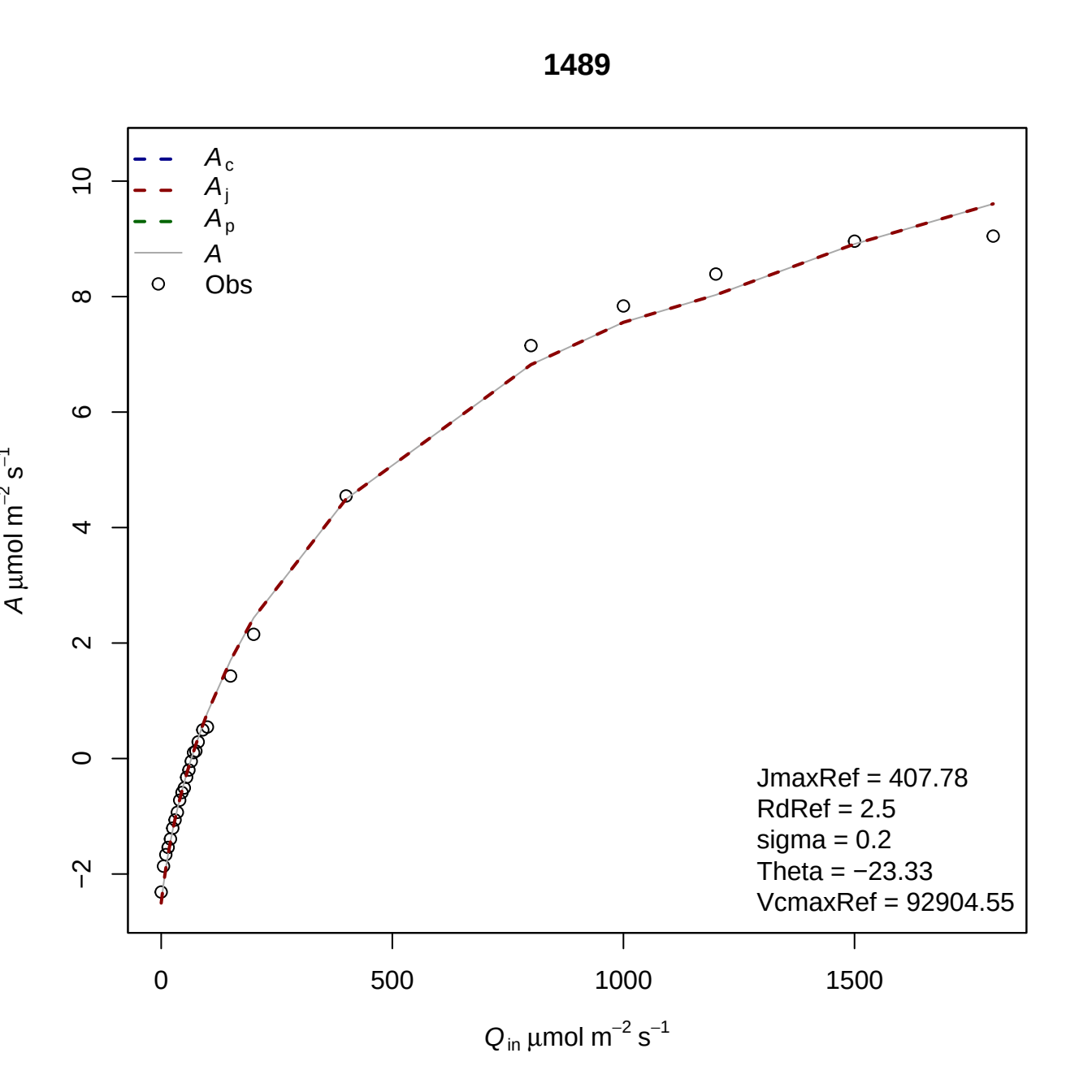

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 211.38
RdRef = 3.26
sigma = 0.35
Theta = −3.92
VcmaxRef = 138073
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

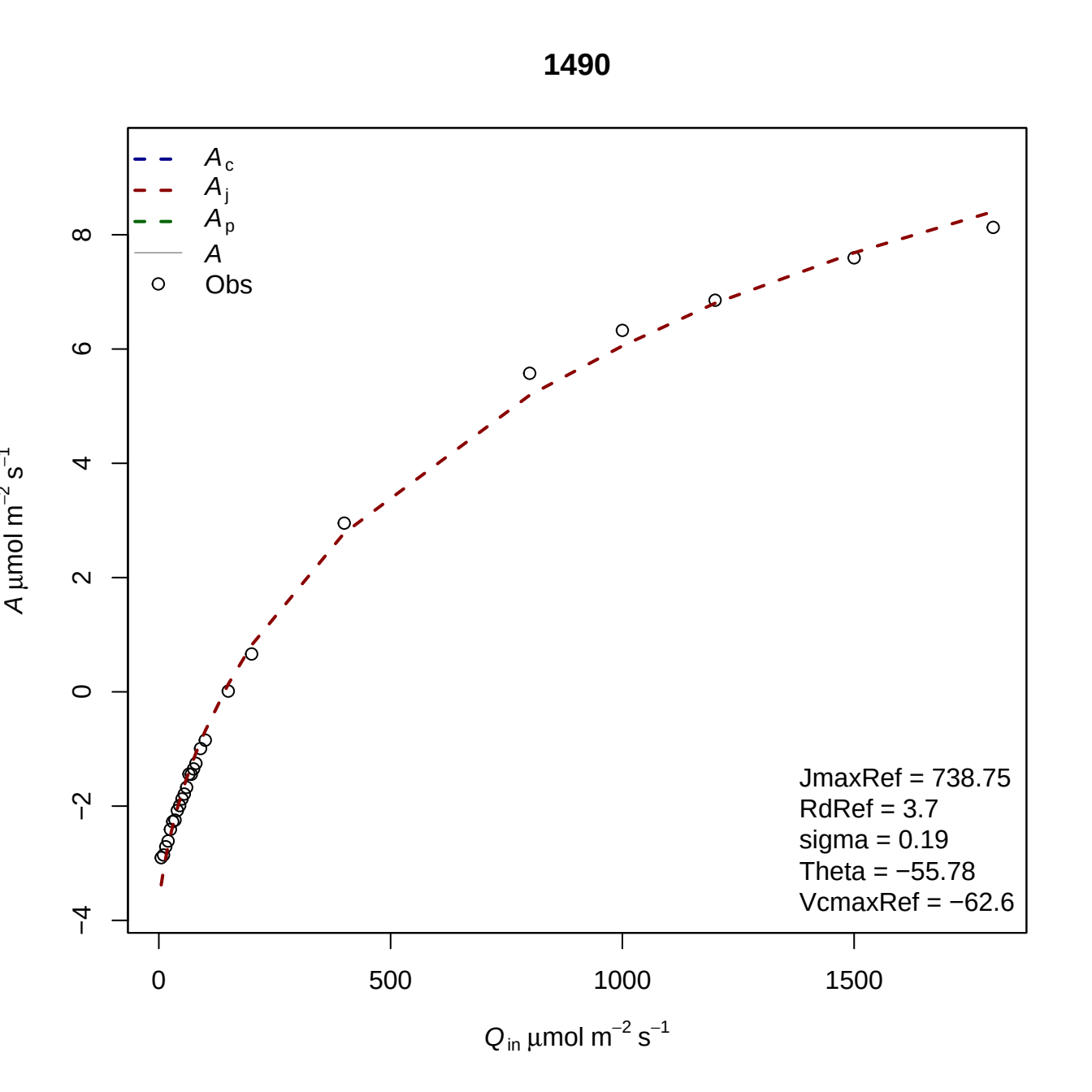

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 149.36
RdRef = 2.79
sigma = 0.19
Theta = −5.76
VcmaxRef = 220542.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

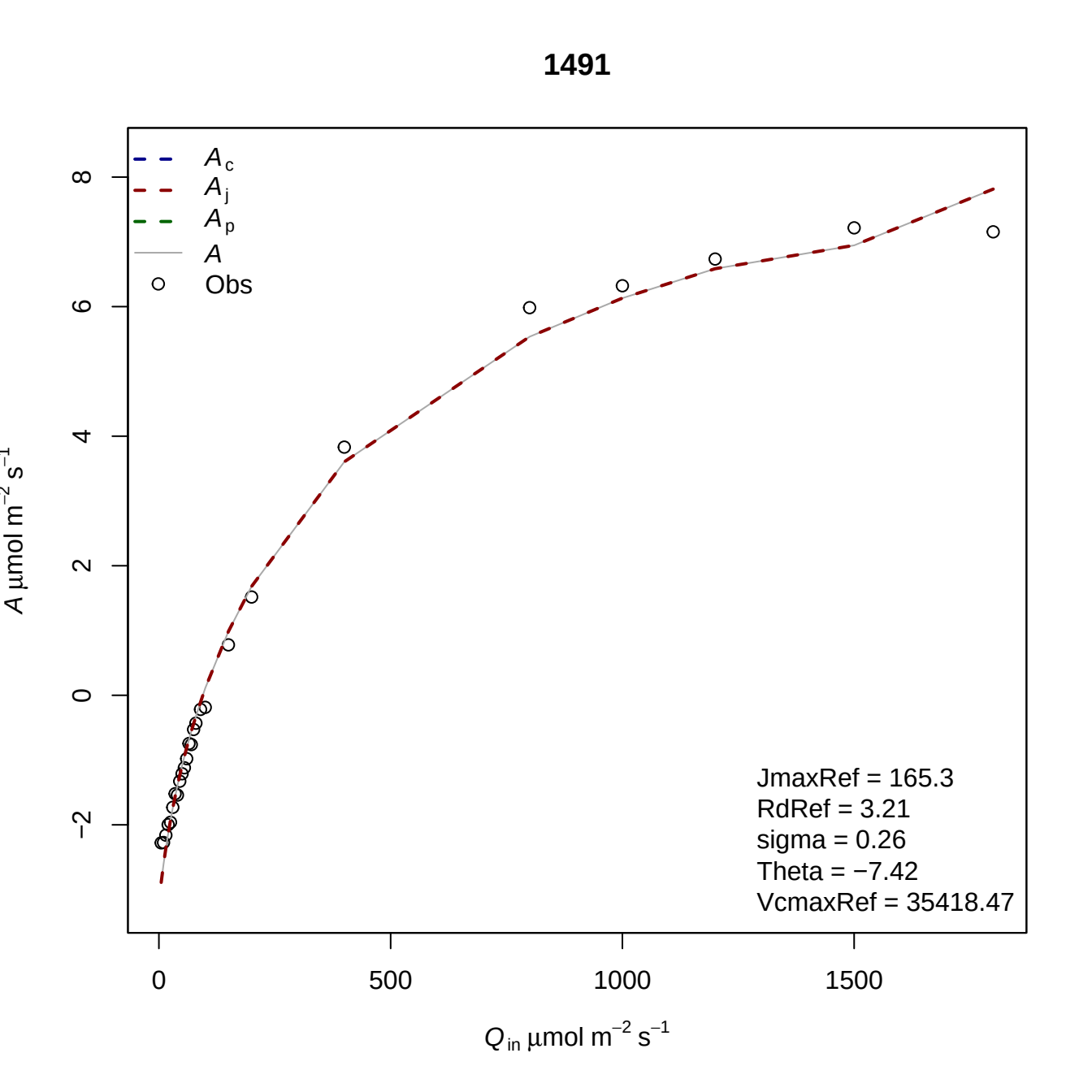

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 1144.42
RdRef = 3.1
sigma = 0.15
Theta = −113.34
VcmaxRef = 49.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

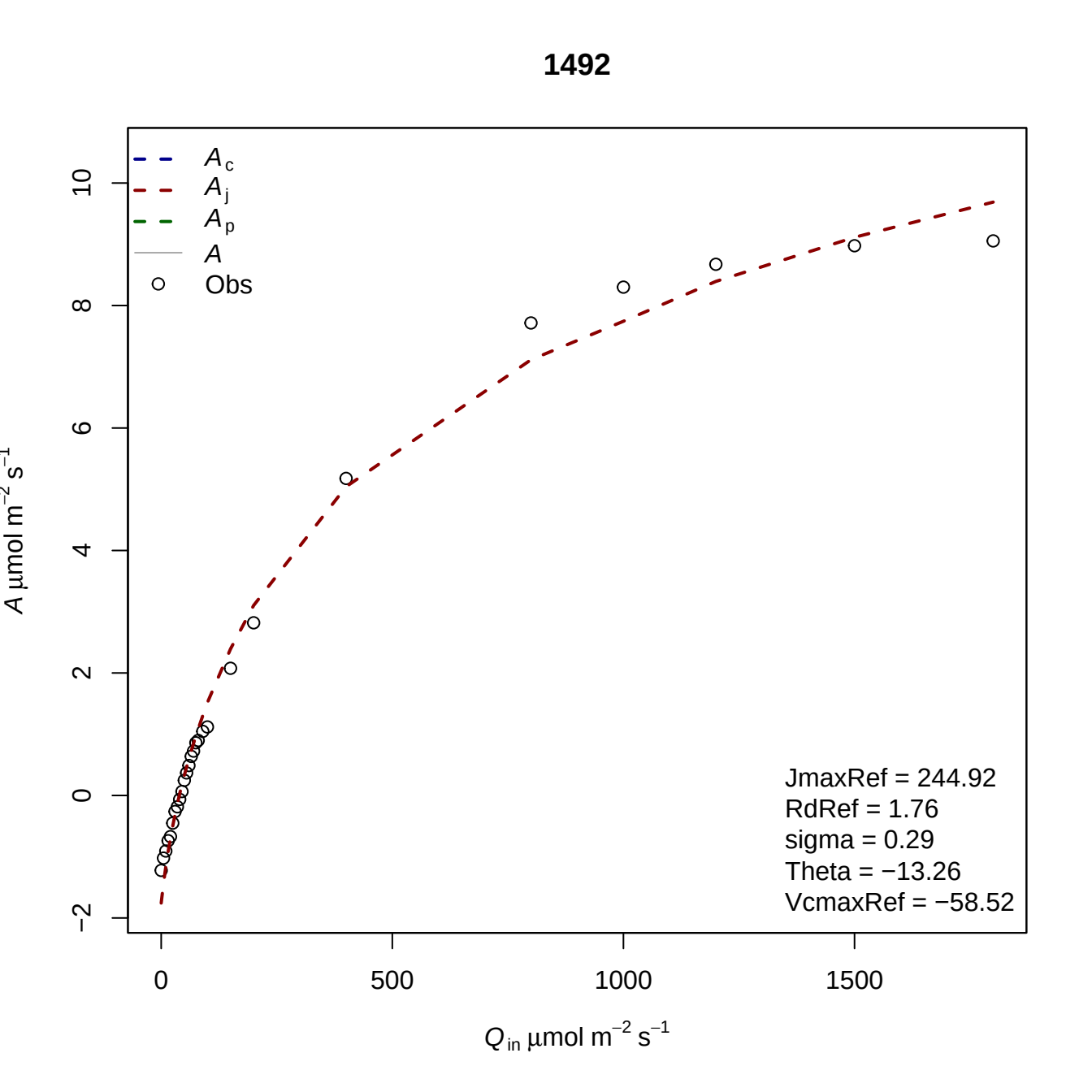

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 332.92
RdRef = 2.95
sigma = 0.12
Theta = −26.86
VcmaxRef = 200735.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

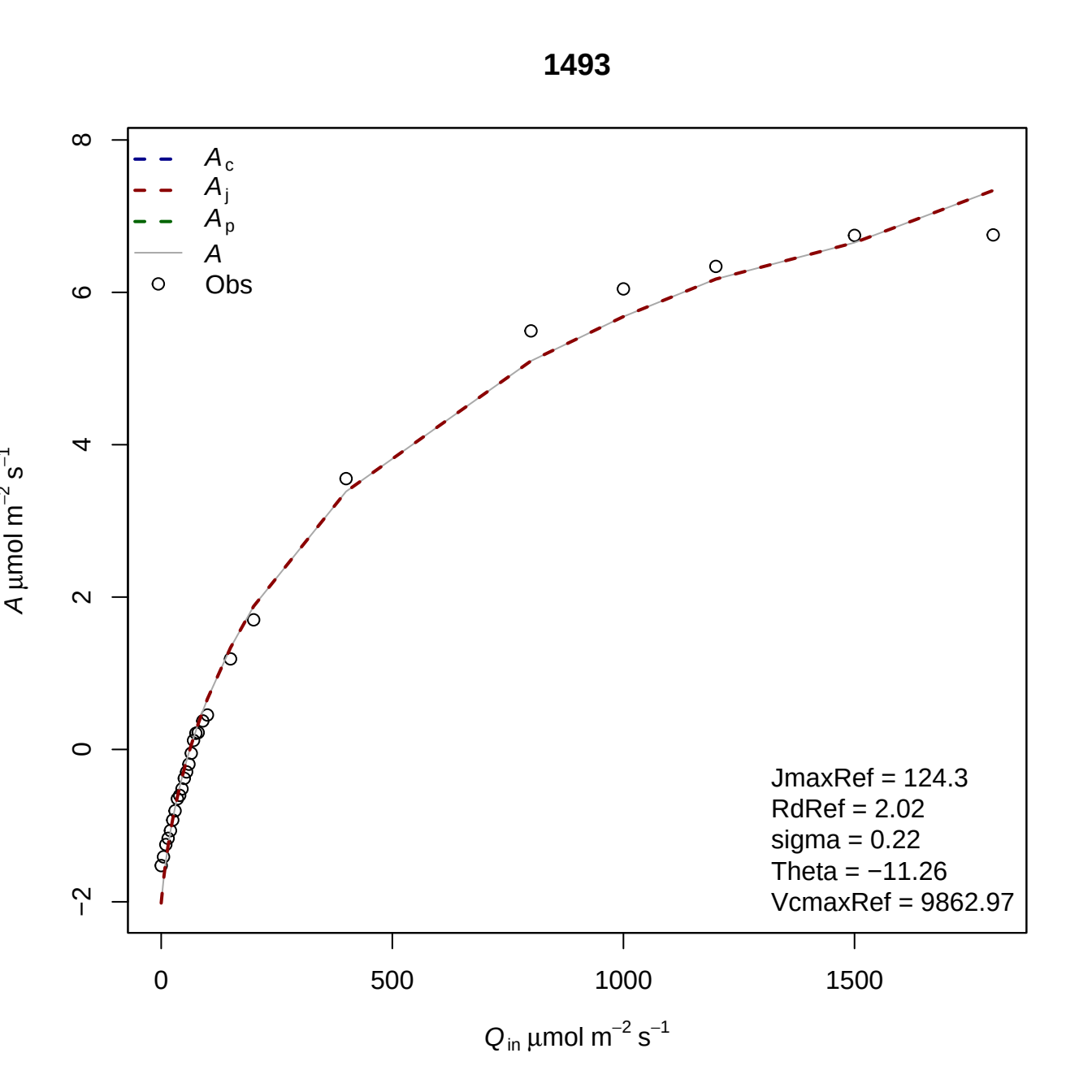

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 52144772.89
RdRef = 3.19
sigma = 0.15
Theta = −3848140.01
VcmaxRef = 1725456726.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

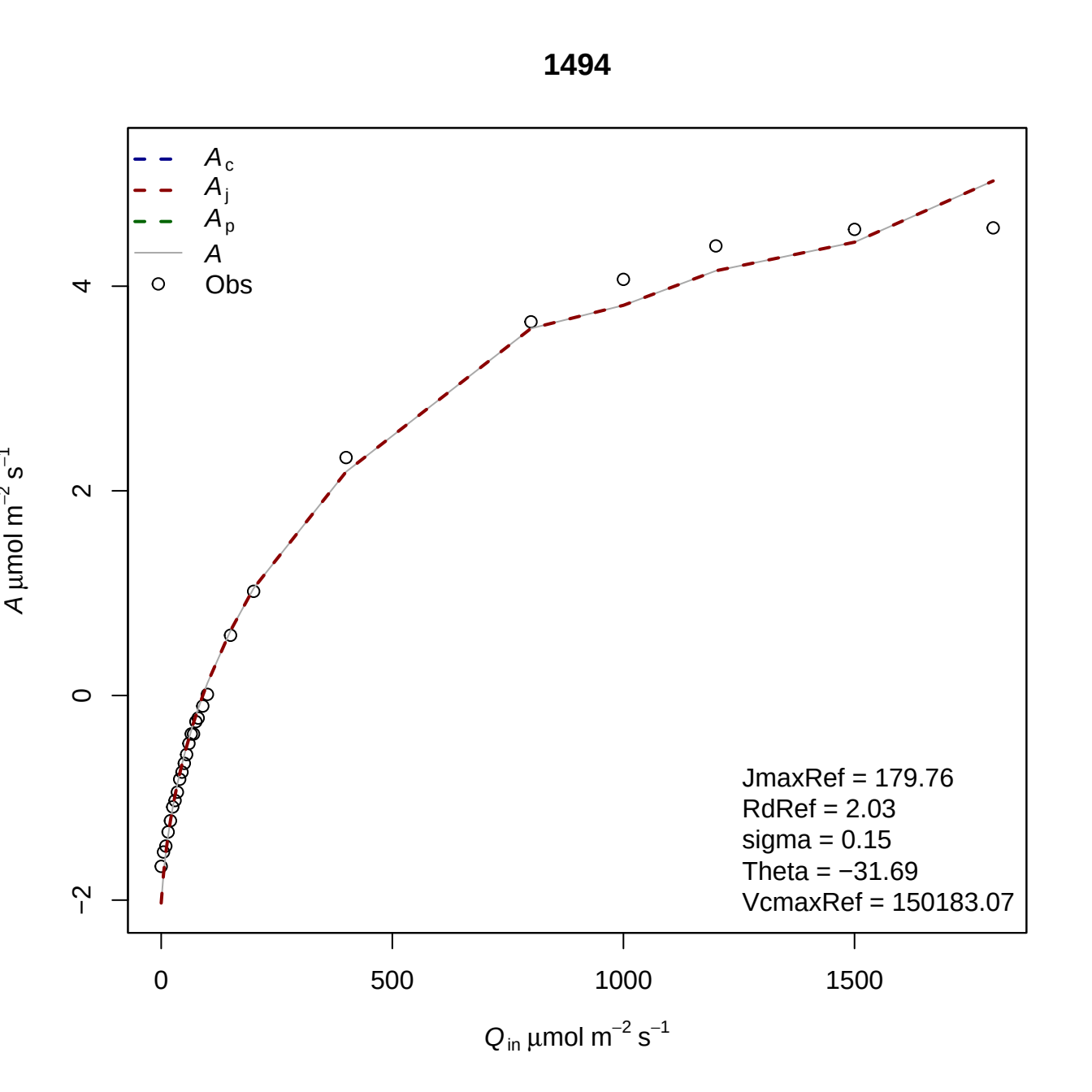

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 407.78
RdRef = 2.5
sigma = 0.2
Theta = −23.33
VcmaxRef = 92904.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

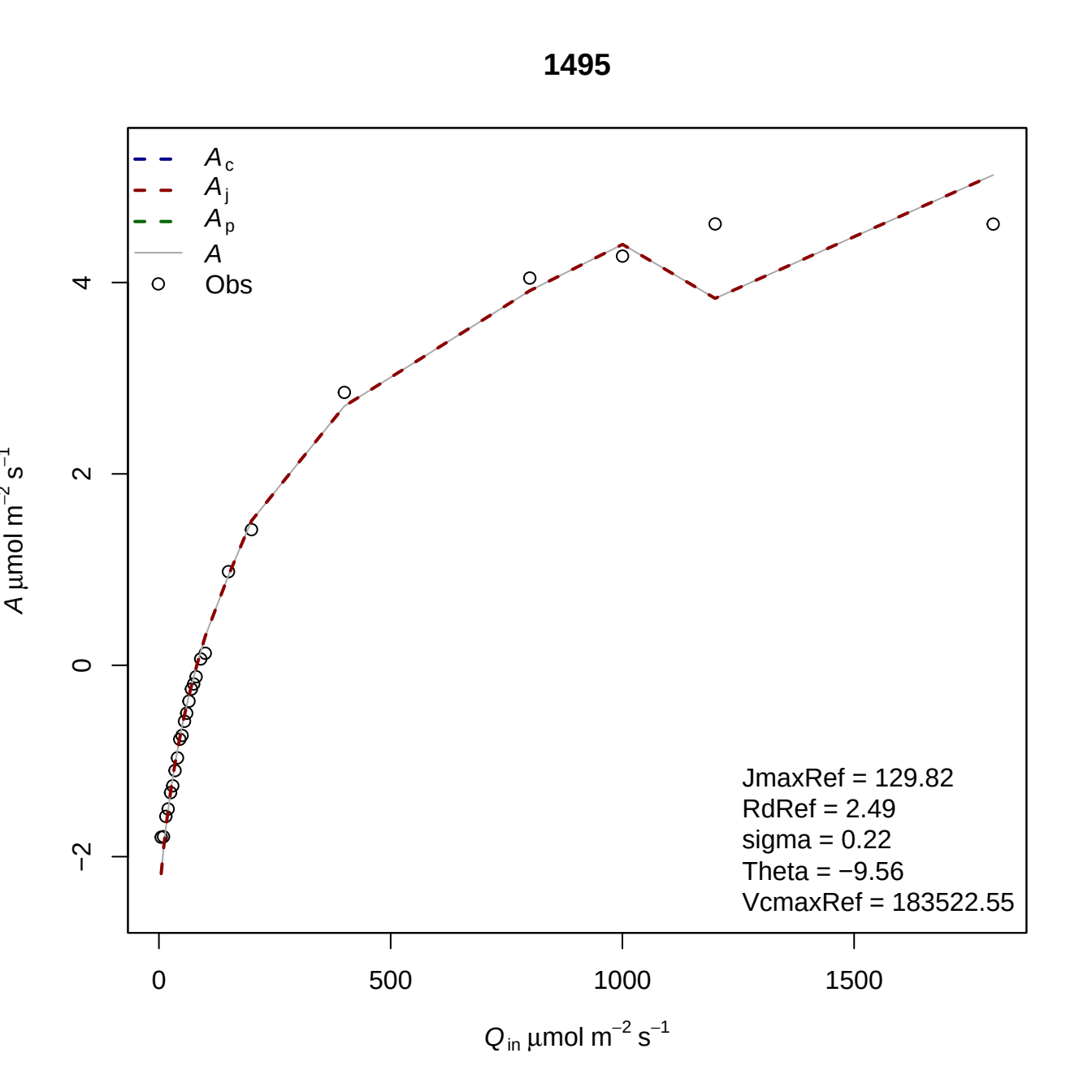

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 738.75
RdRef = 3.7
sigma = 0.19
Theta = −55.78
VcmaxRef = −62.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
500
1000
1500

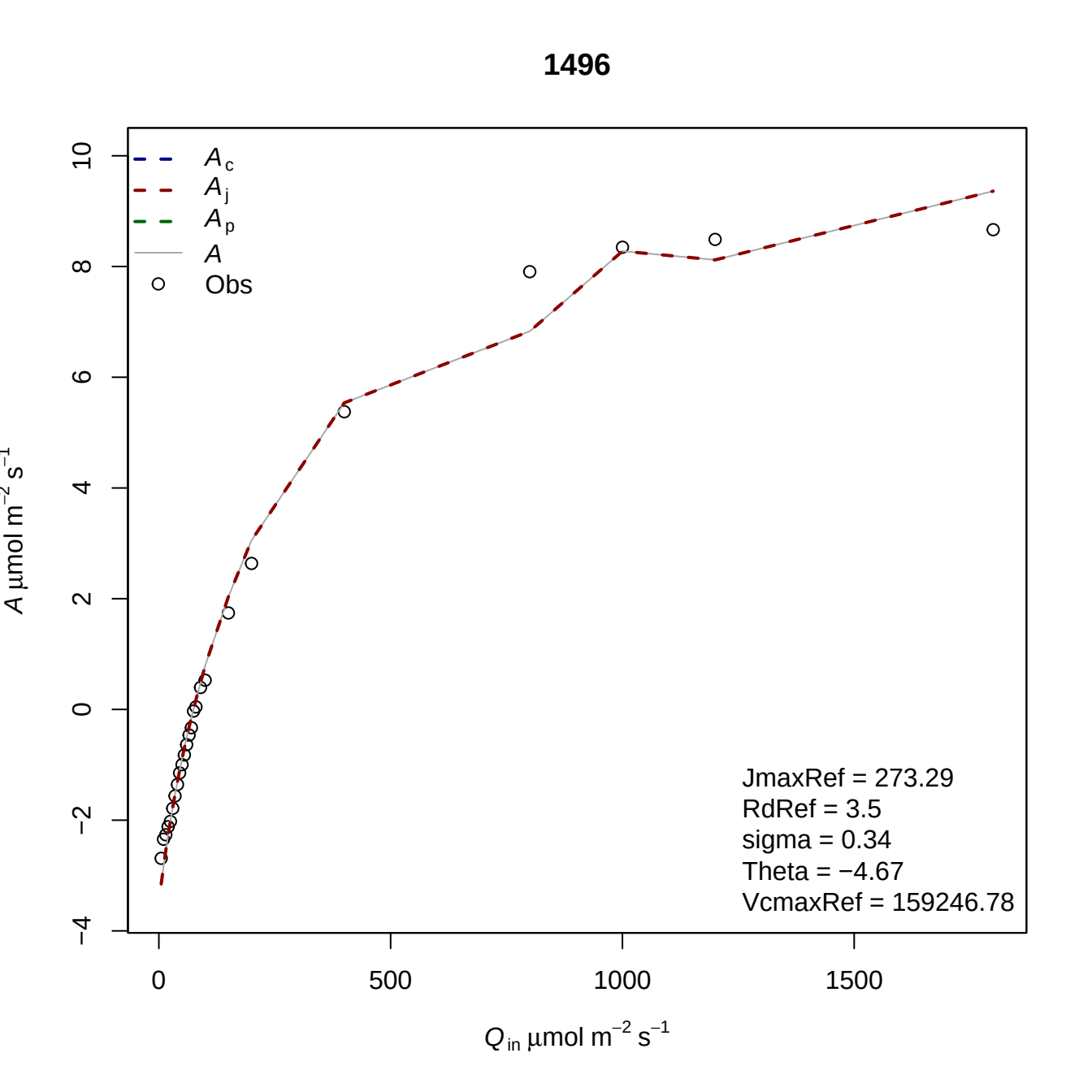

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 165.3
RdRef = 3.21
sigma = 0.26
Theta = −7.42
VcmaxRef = 35418.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

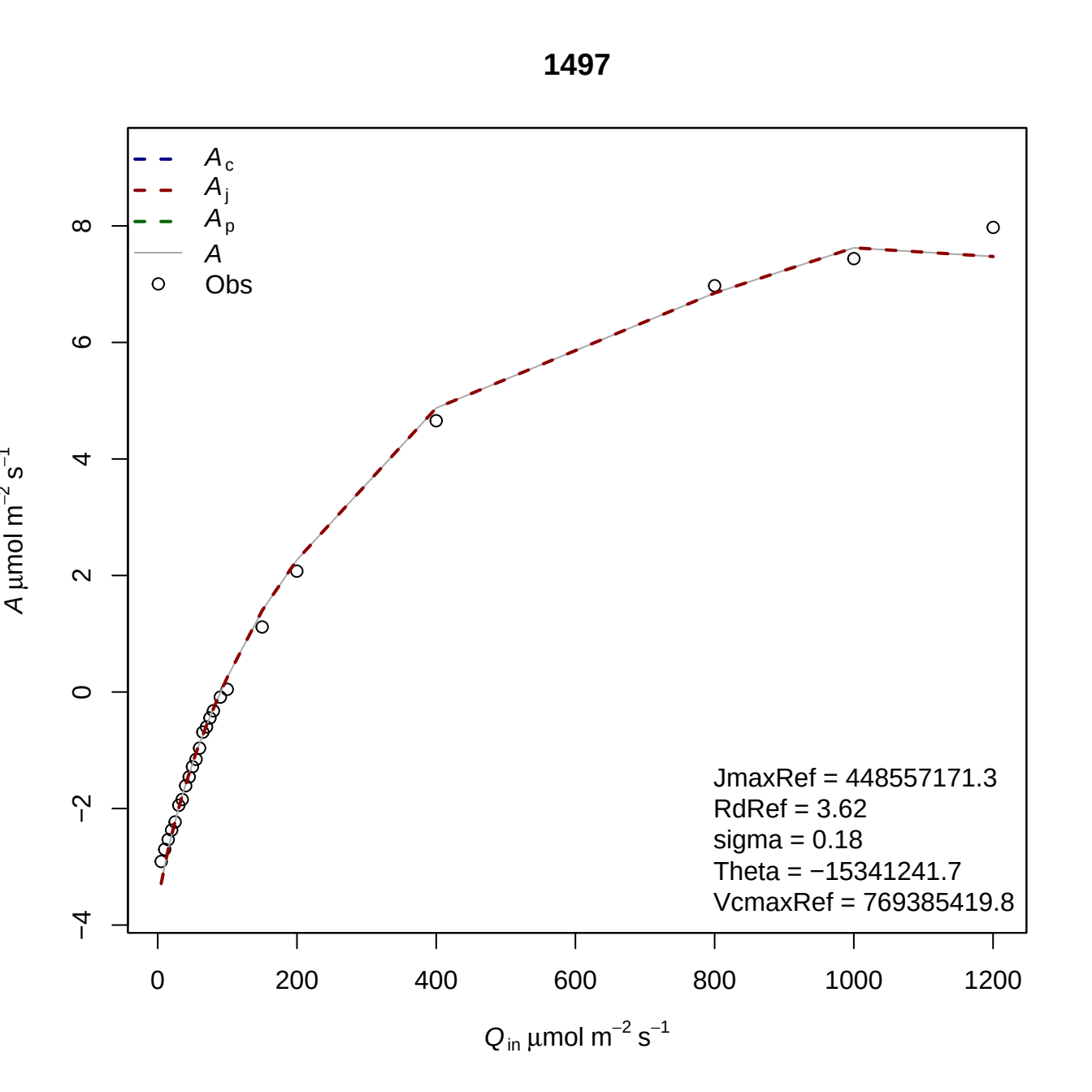

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 244.92
RdRef = 1.76
sigma = 0.29
Theta = −13.26
VcmaxRef = −58.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

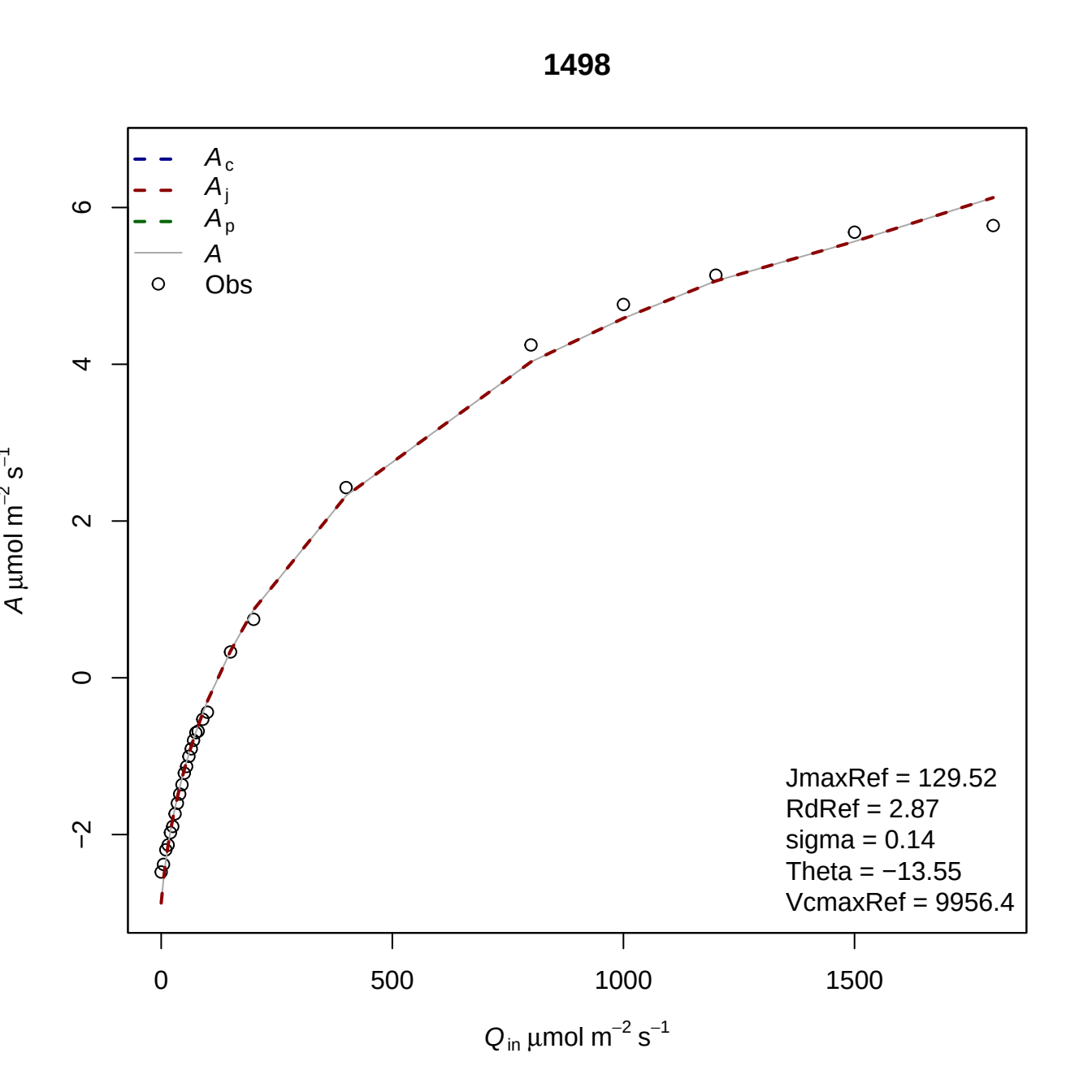

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 124.3
RdRef = 2.02
sigma = 0.22
Theta = −11.26
VcmaxRef = 9862.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

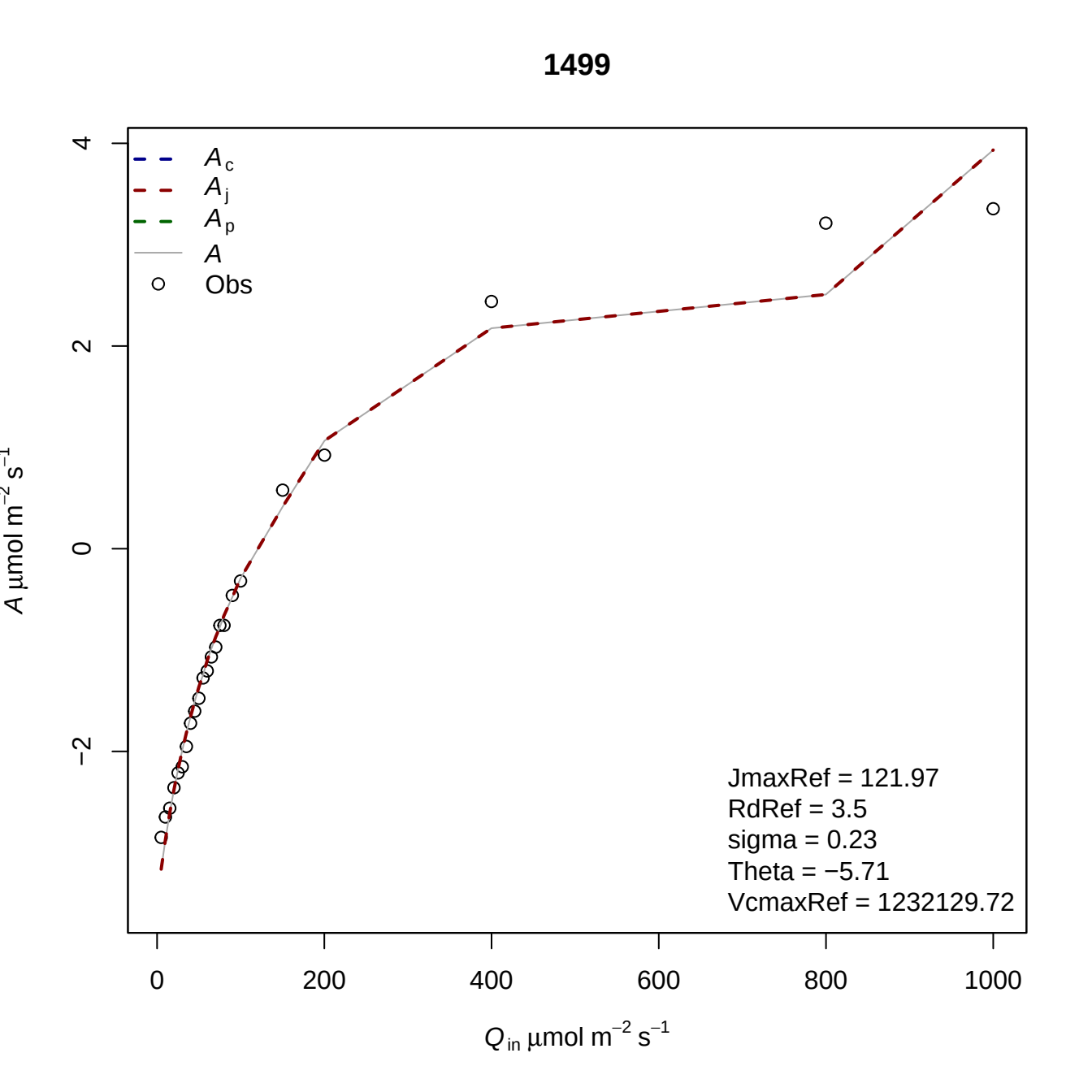

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 179.76
RdRef = 2.03
sigma = 0.15
Theta = −31.69
VcmaxRef = 150183.07
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

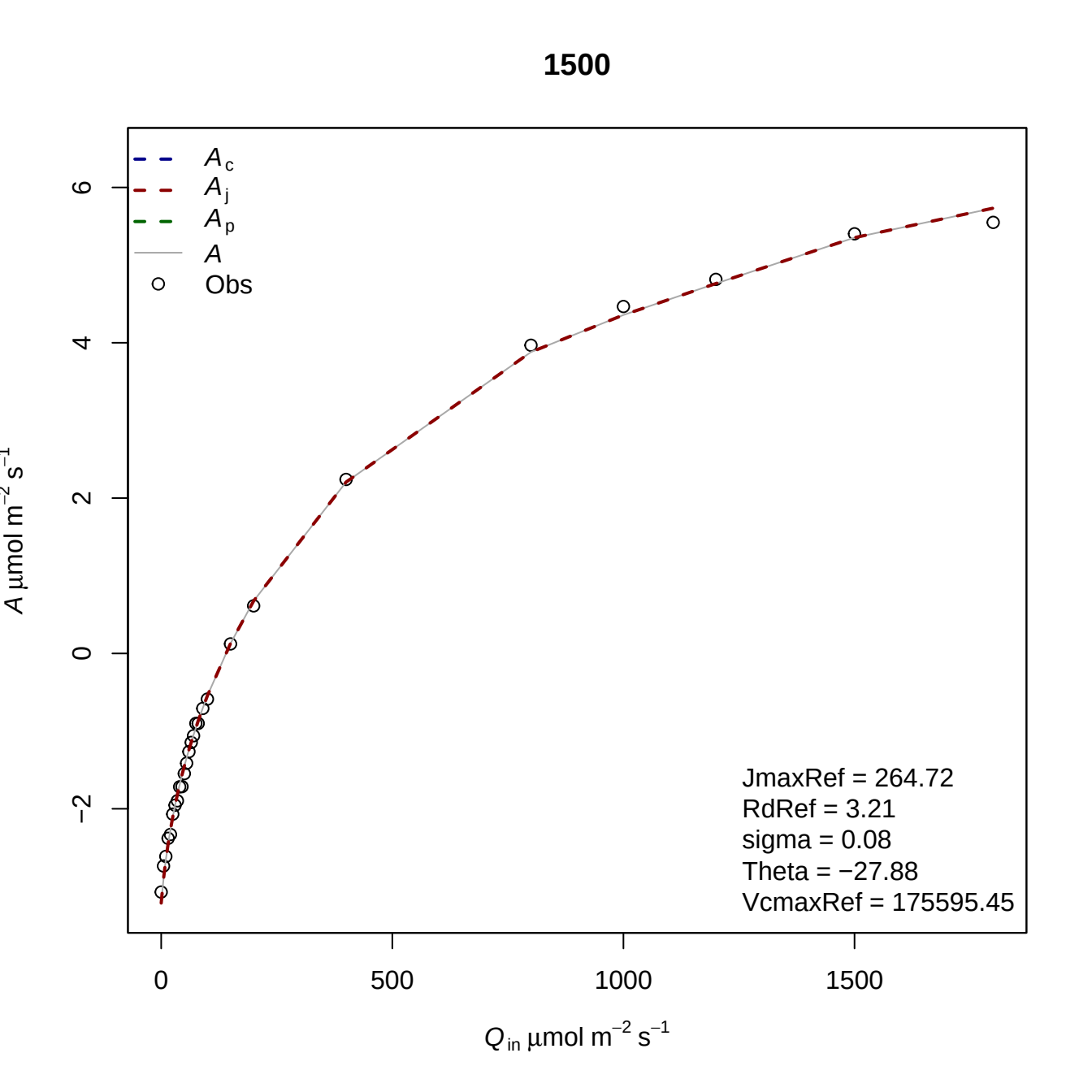

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 129.82
RdRef = 2.49
sigma = 0.22
Theta = −9.56
VcmaxRef = 183522.55
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

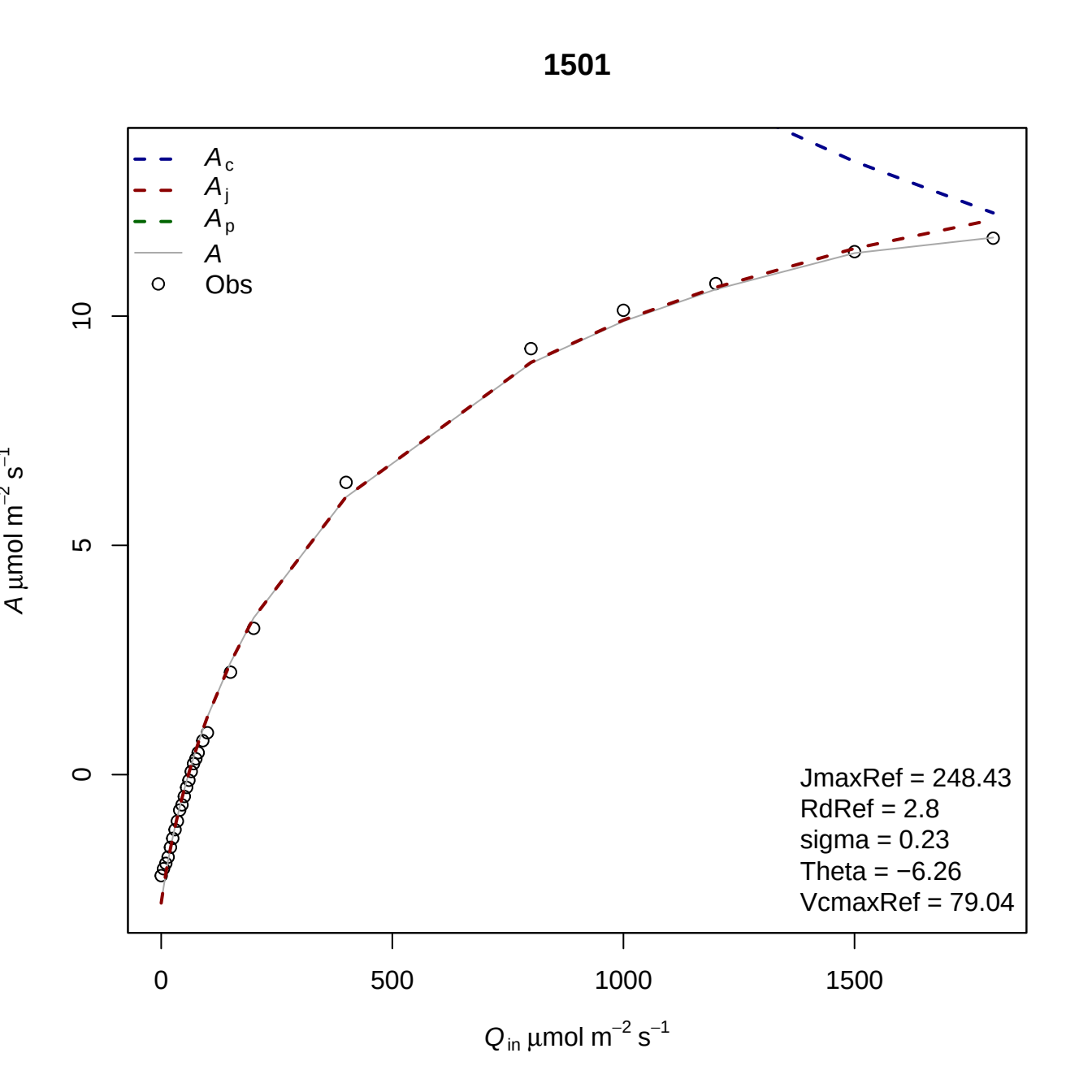

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 273.29
RdRef = 3.5
sigma = 0.34
Theta = −4.67
VcmaxRef = 159246.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

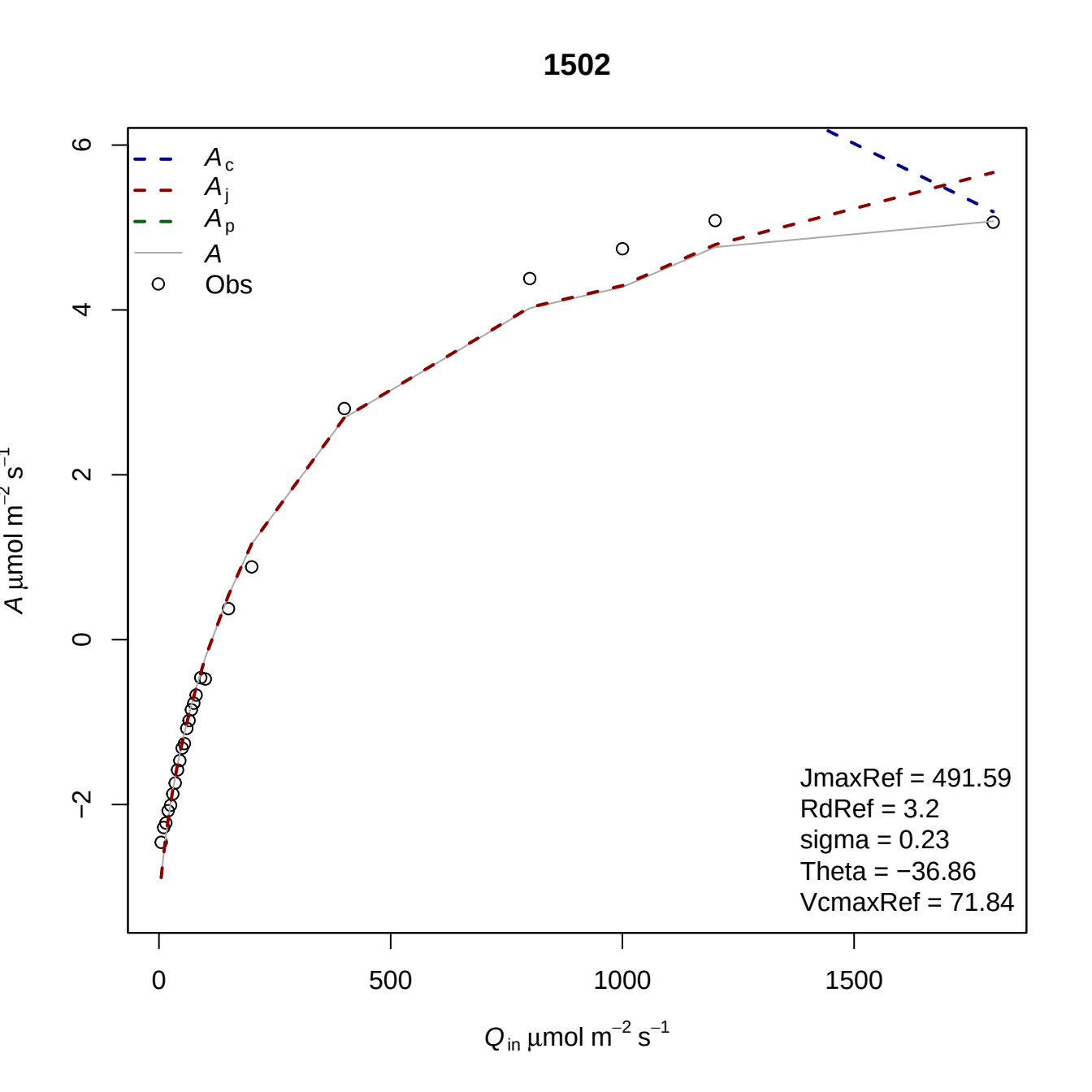

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 448557171.3
RdRef = 3.62
sigma = 0.18
Theta = −15341241.7
VcmaxRef = 769385419.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

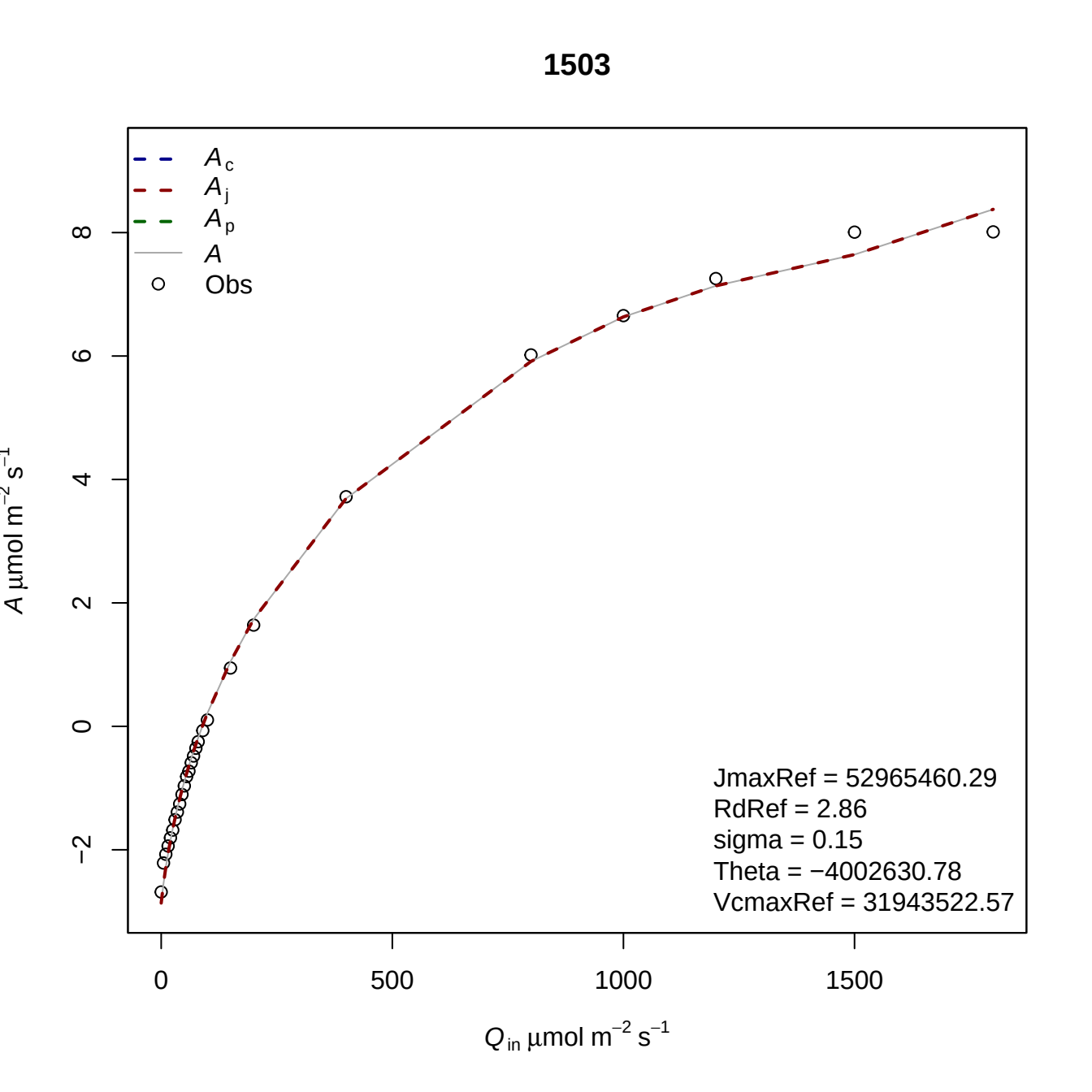

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 129.52
RdRef = 2.87
sigma = 0.14
Theta = −13.55
VcmaxRef = 9956.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

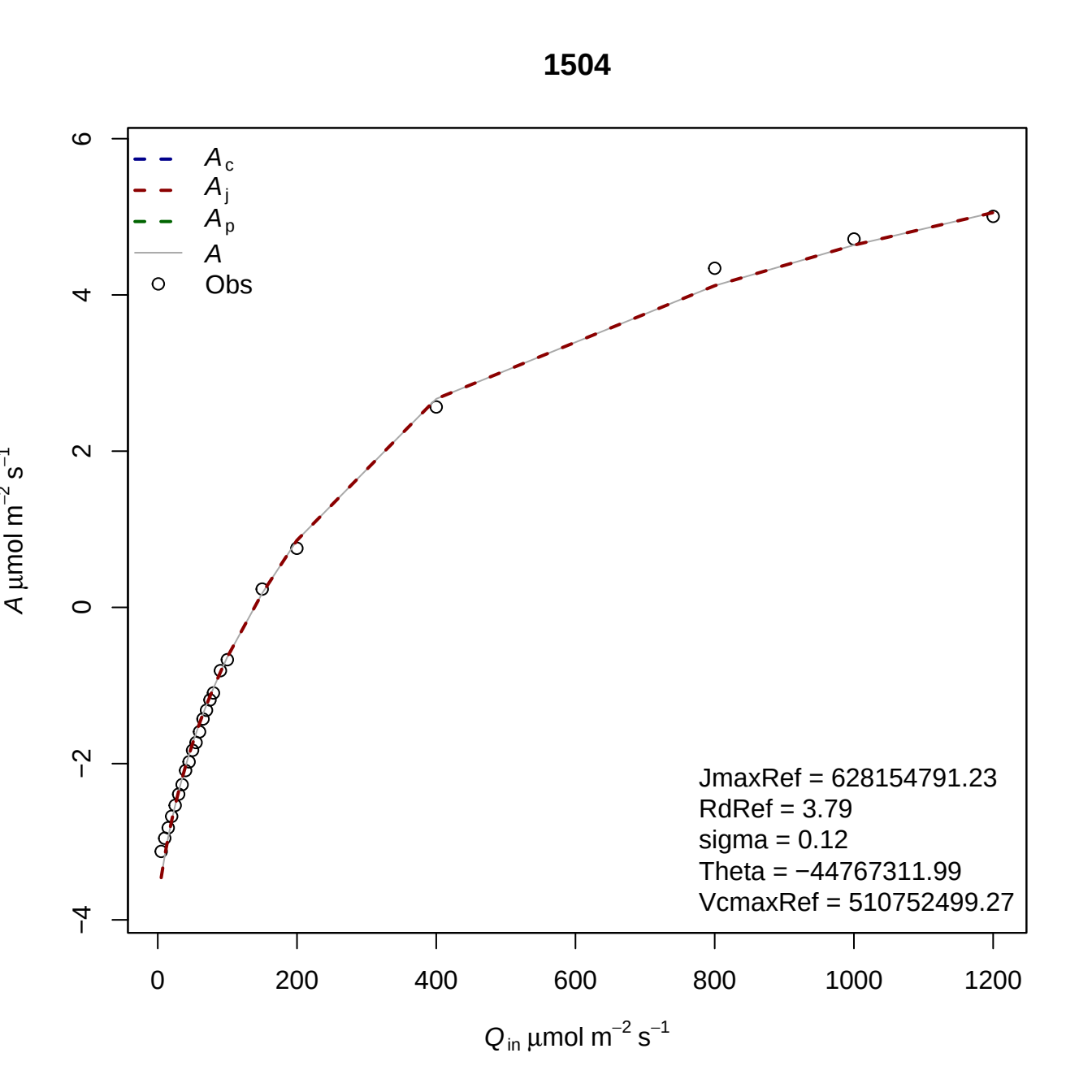

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 121.97
RdRef = 3.5
sigma = 0.23
Theta = −5.71
VcmaxRef = 1232129.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000

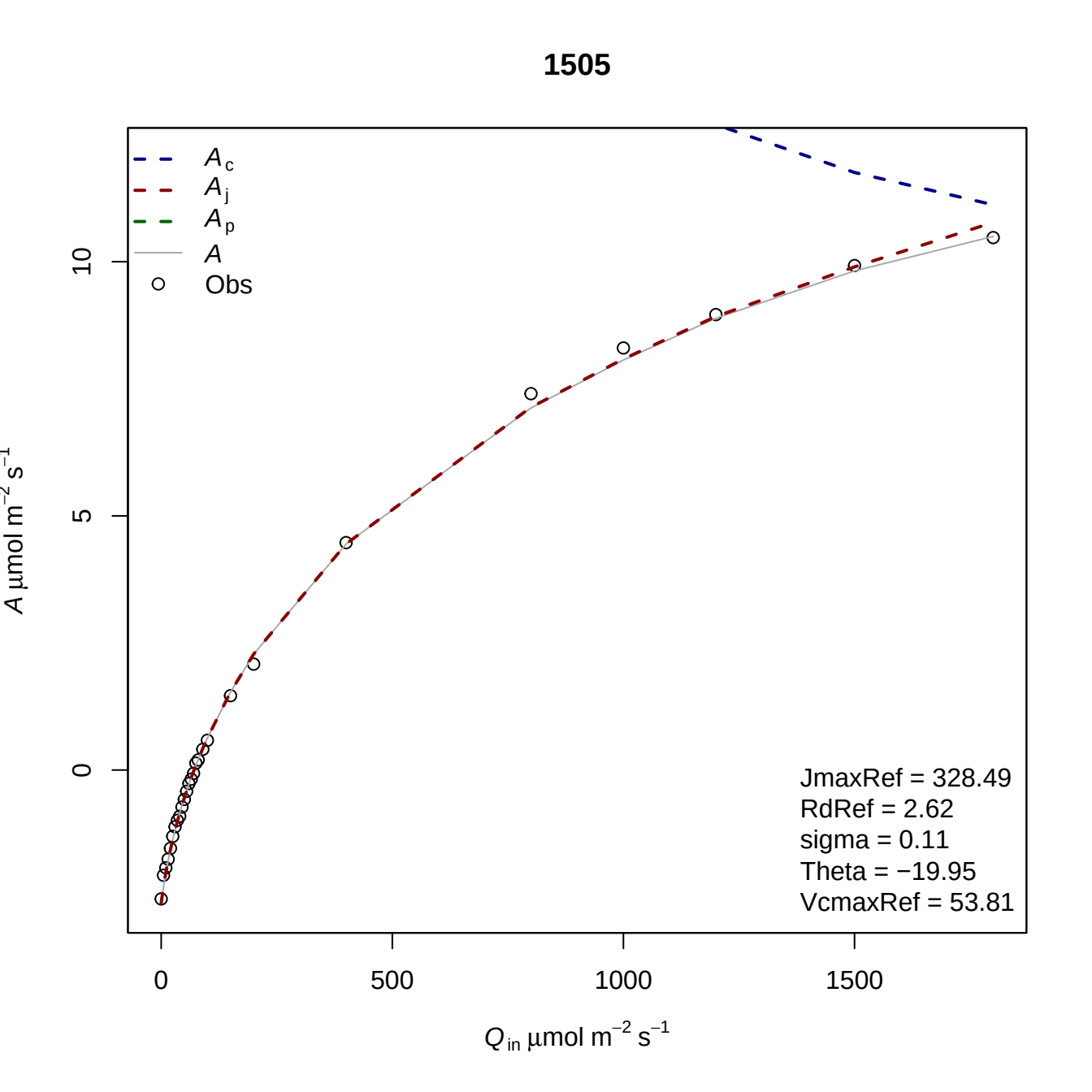

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 264.72
RdRef = 3.21
sigma = 0.08
Theta = −27.88
VcmaxRef = 175595.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

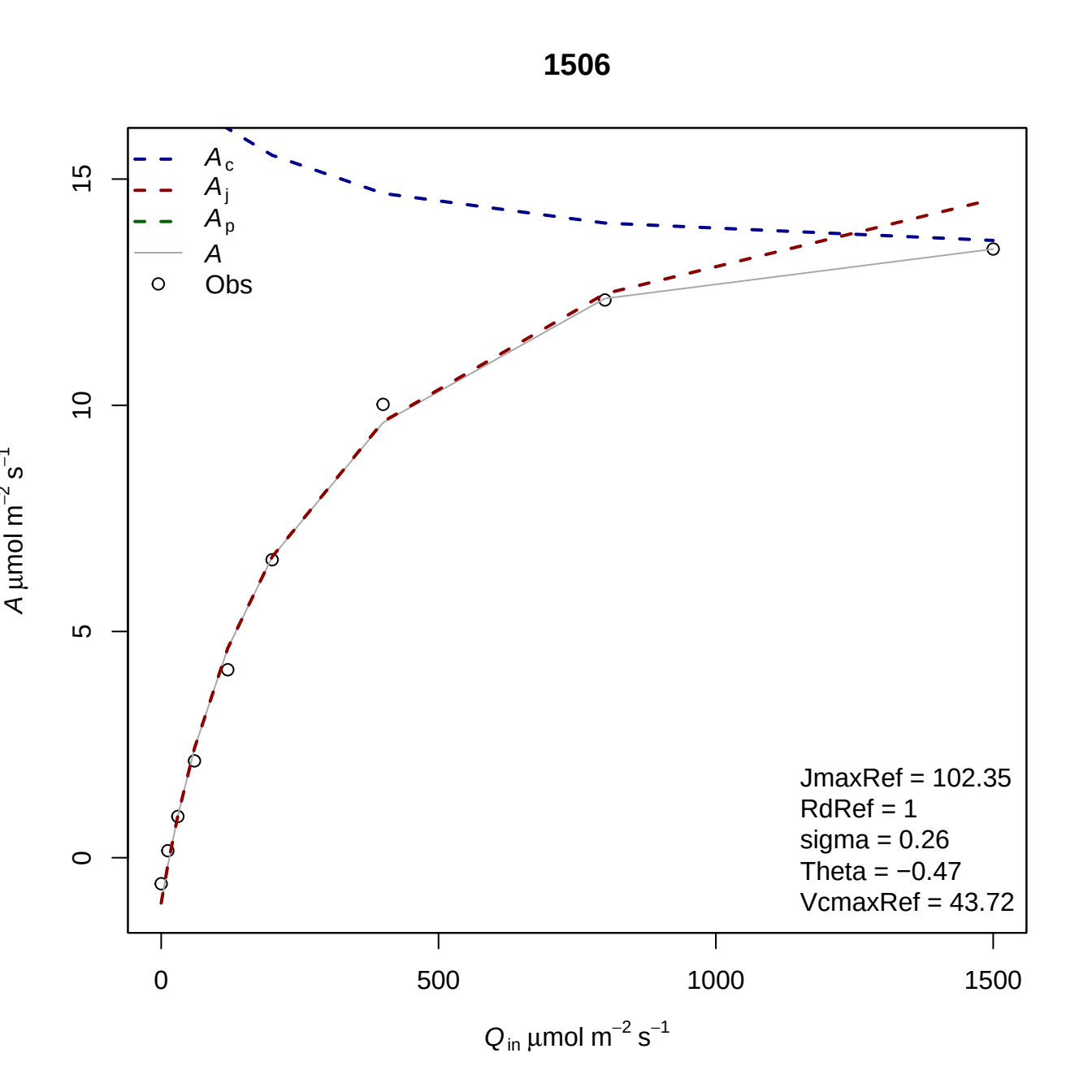

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 248.43
RdRef = 2.8
sigma = 0.23
Theta = −6.26
VcmaxRef = 79.04
0
5
10
A µmol m−2 s−1
0
500
1000
1500
Qin µmol m−2 s−1

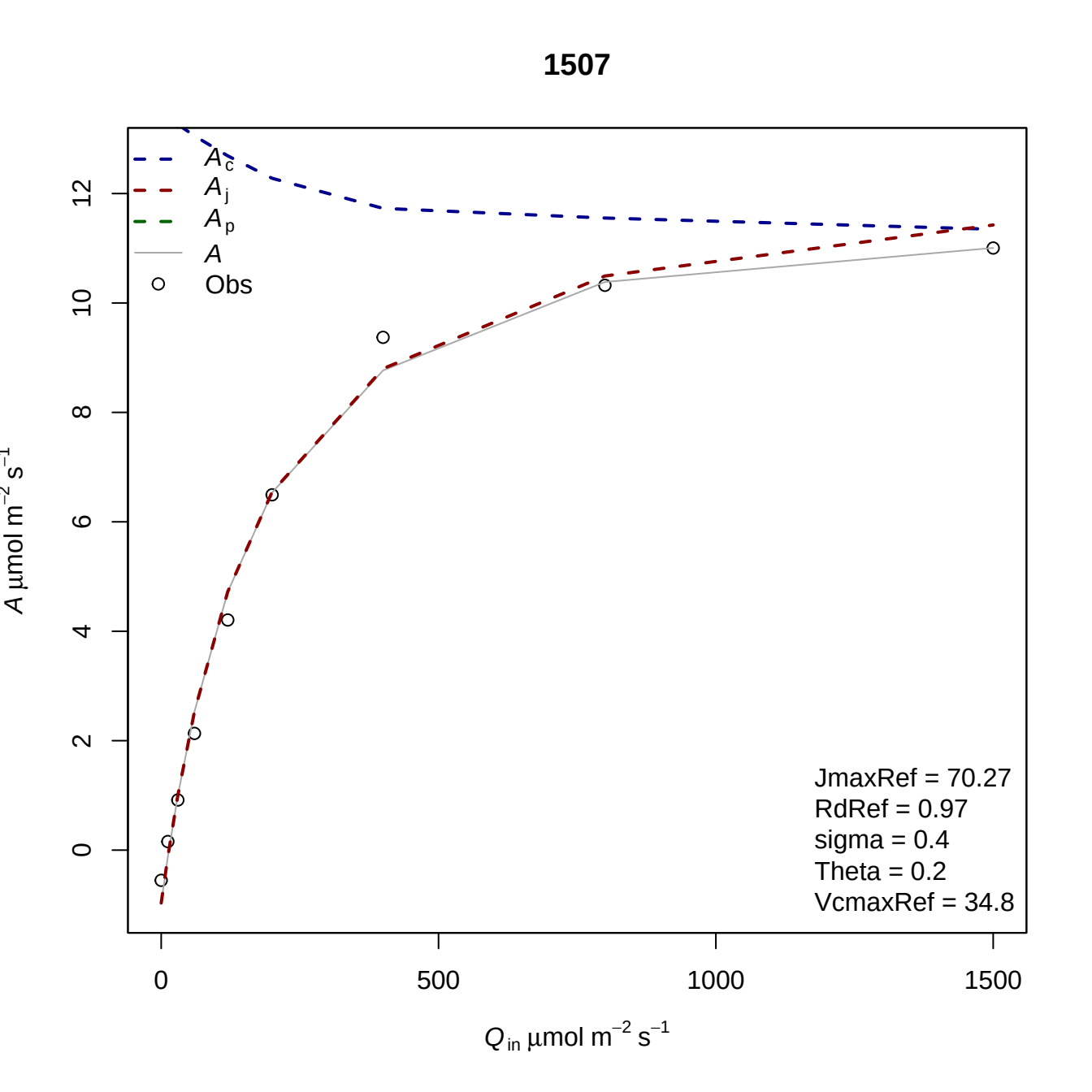

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 491.59
RdRef = 3.2
sigma = 0.23
Theta = −36.86
VcmaxRef = 71.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

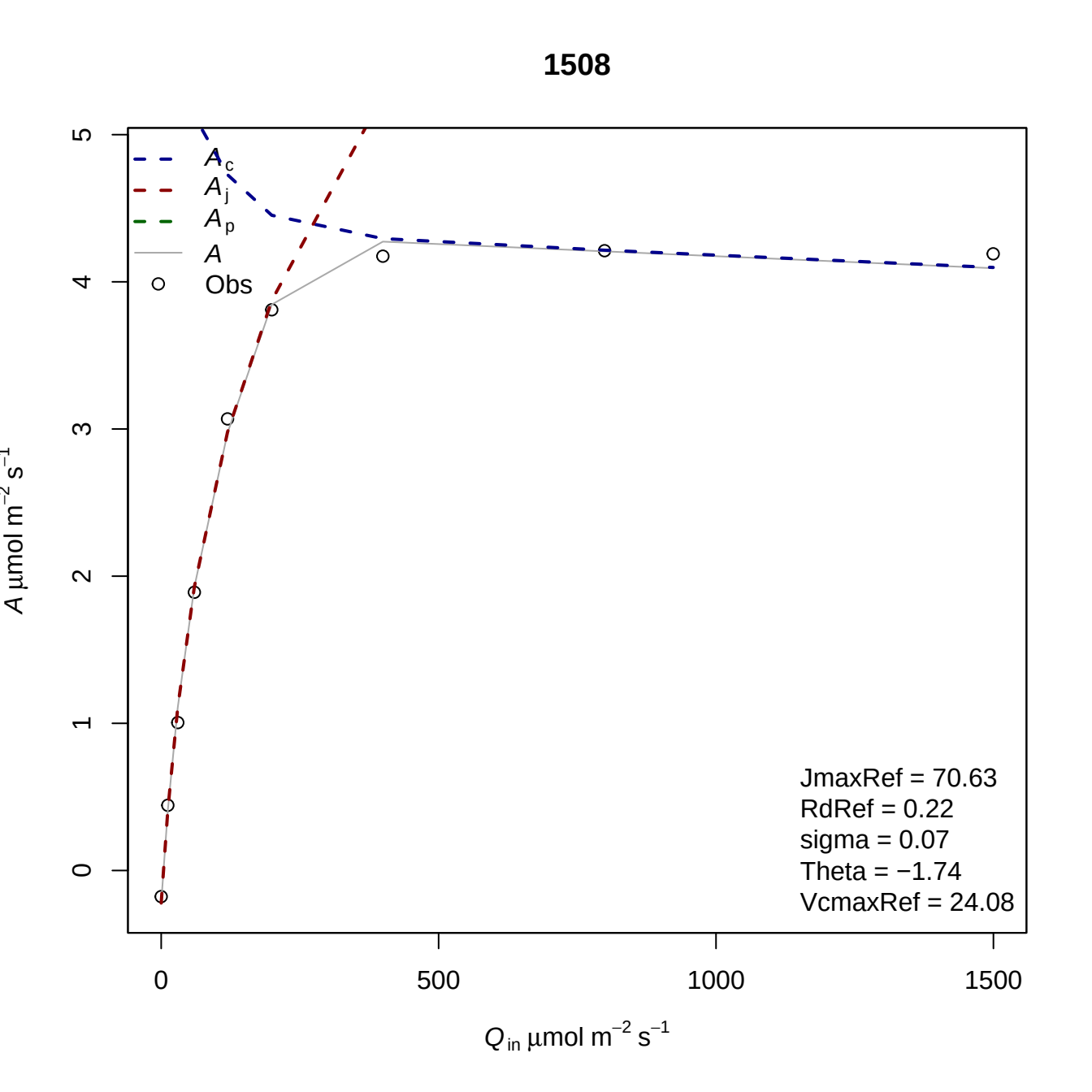

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 52965460.29
RdRef = 2.86
sigma = 0.15
Theta = −4002630.78
VcmaxRef = 31943522.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

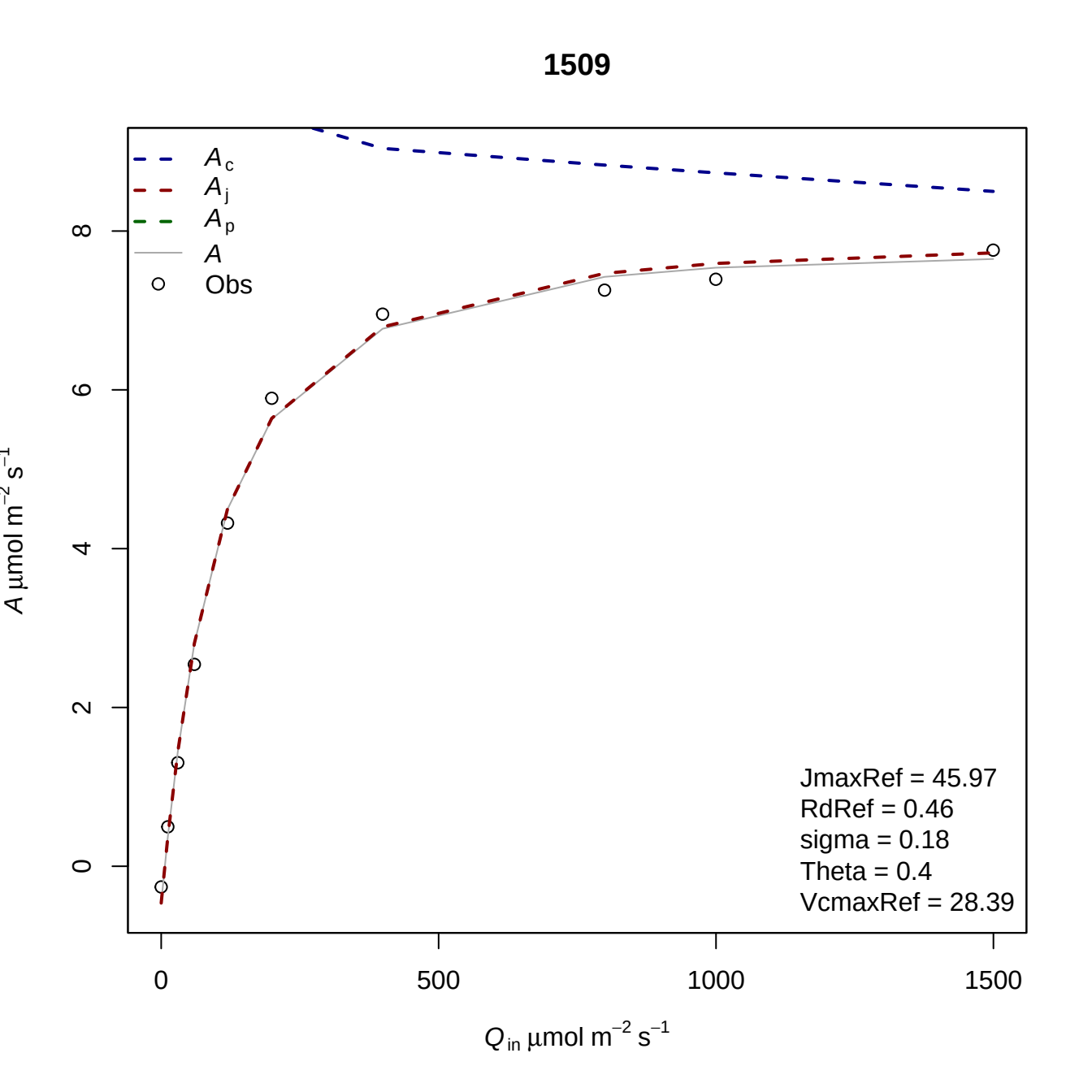

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 628154791.23
RdRef = 3.79
sigma = 0.12
Theta = −44767311.99
VcmaxRef = 510752499.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

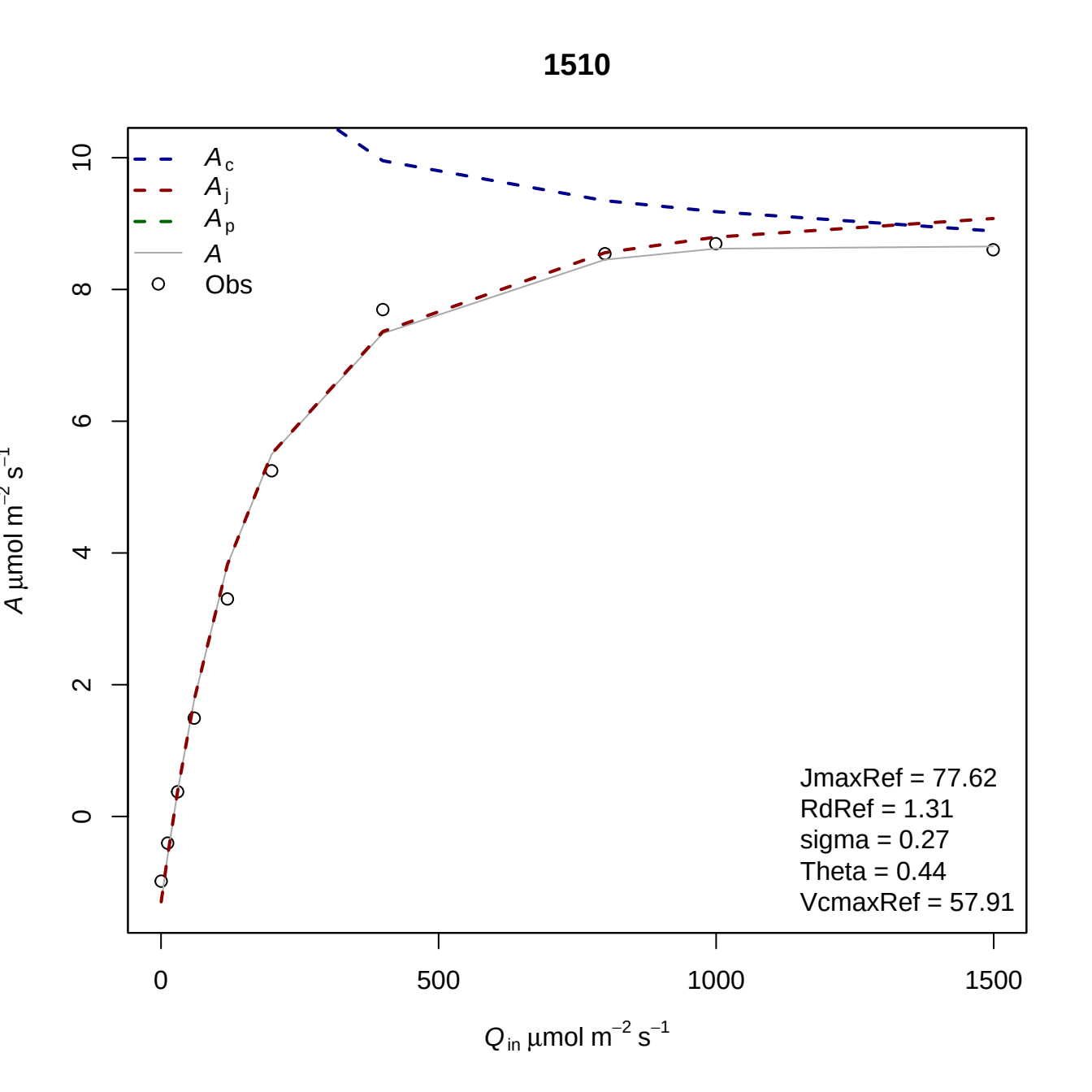

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 328.49
RdRef = 2.62
sigma = 0.11
Theta = −19.95
VcmaxRef = 53.81
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

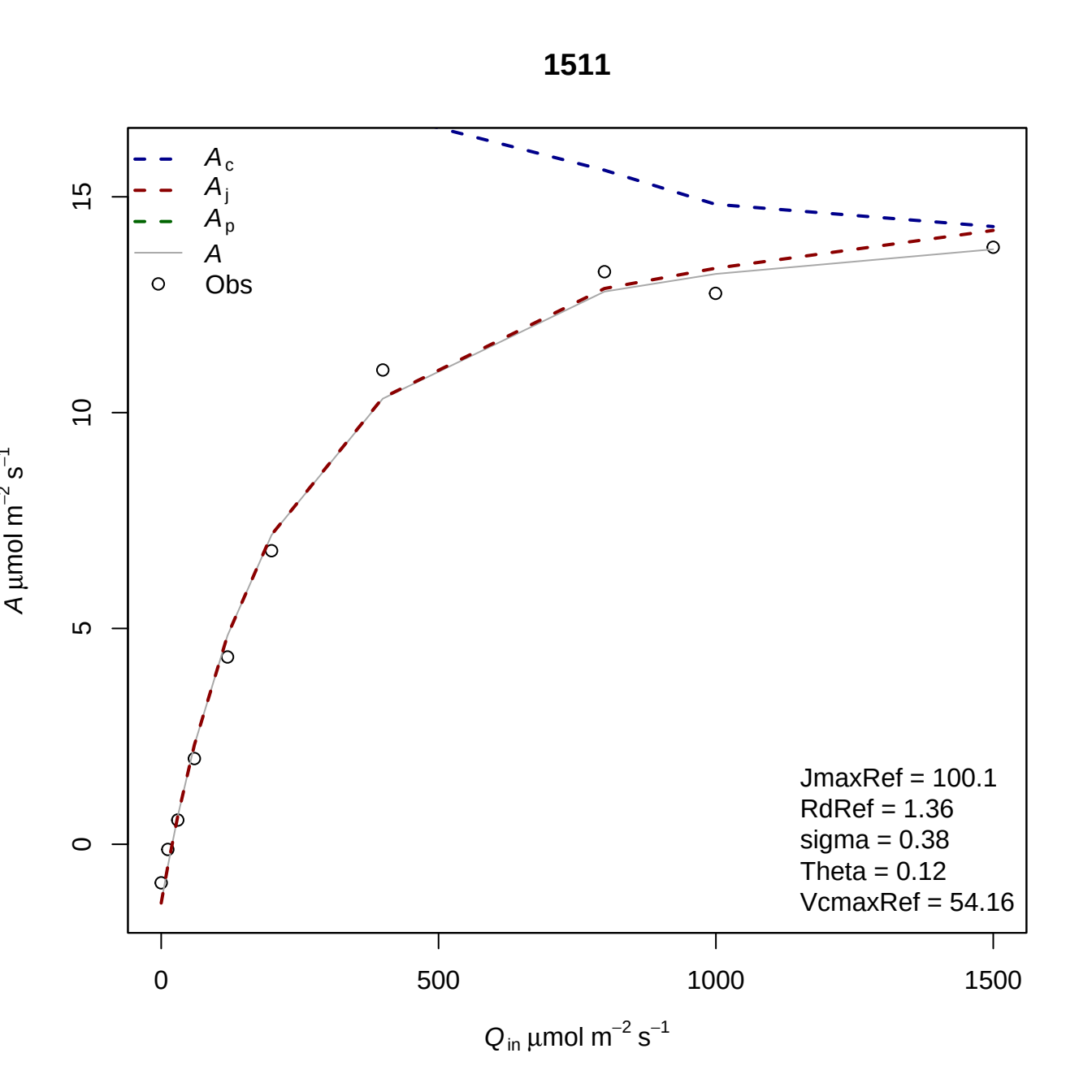

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 102.35
RdRef = 1
sigma = 0.26
Theta = −0.47
VcmaxRef = 43.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

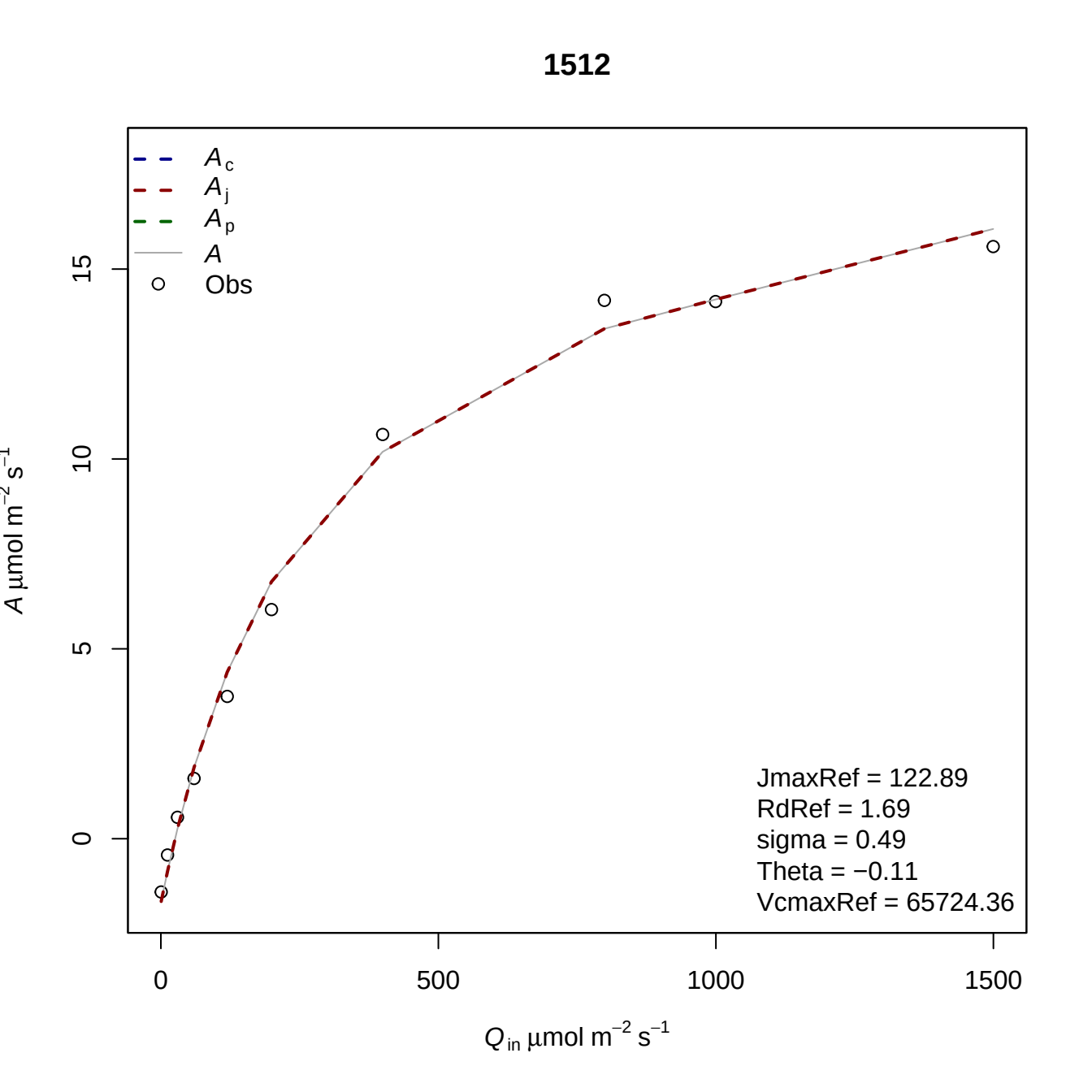

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 70.27
RdRef = 0.97
sigma = 0.4
Theta = 0.2
VcmaxRef = 34.8
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

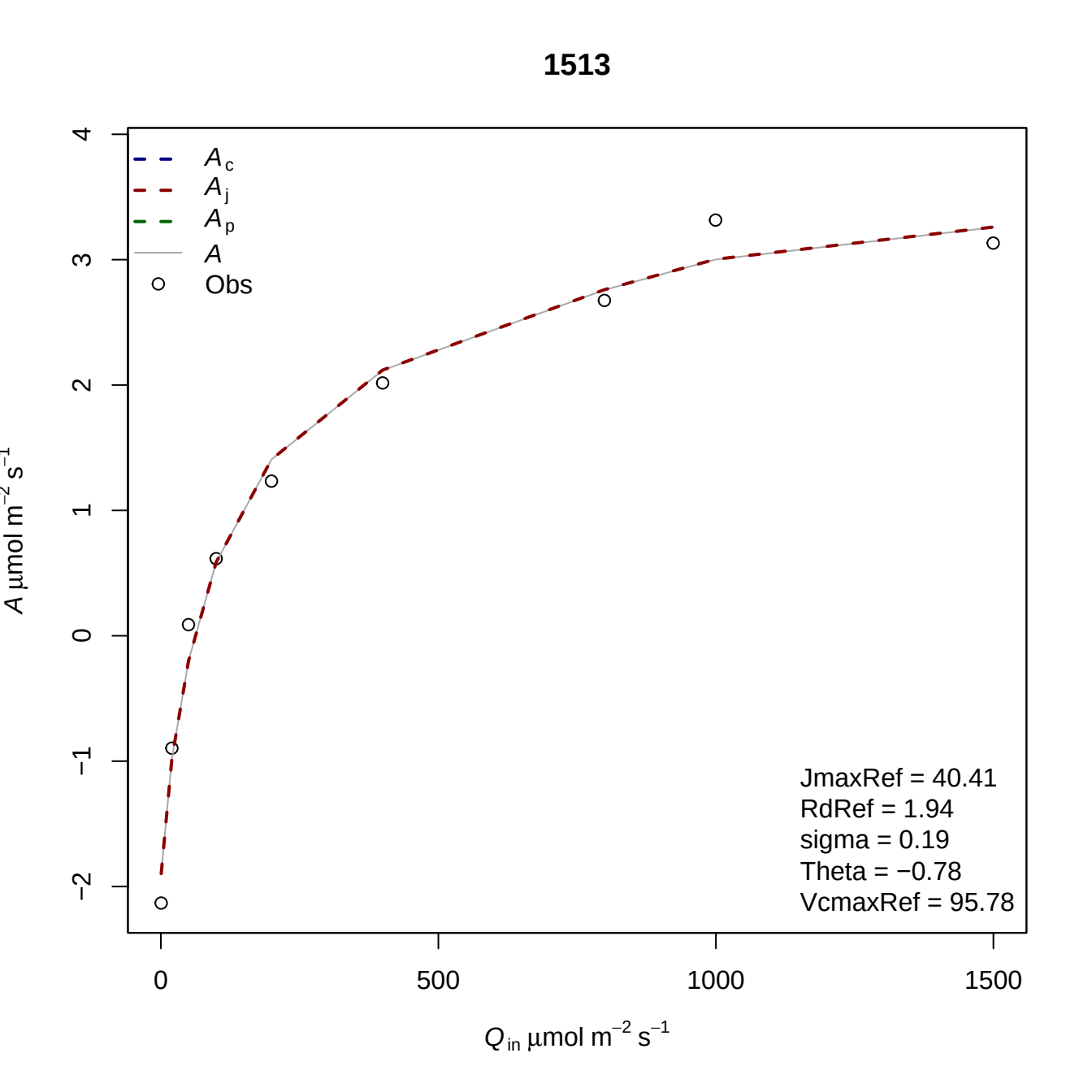

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 70.63
RdRef = 0.22
sigma = 0.07
Theta = −1.74
VcmaxRef = 24.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

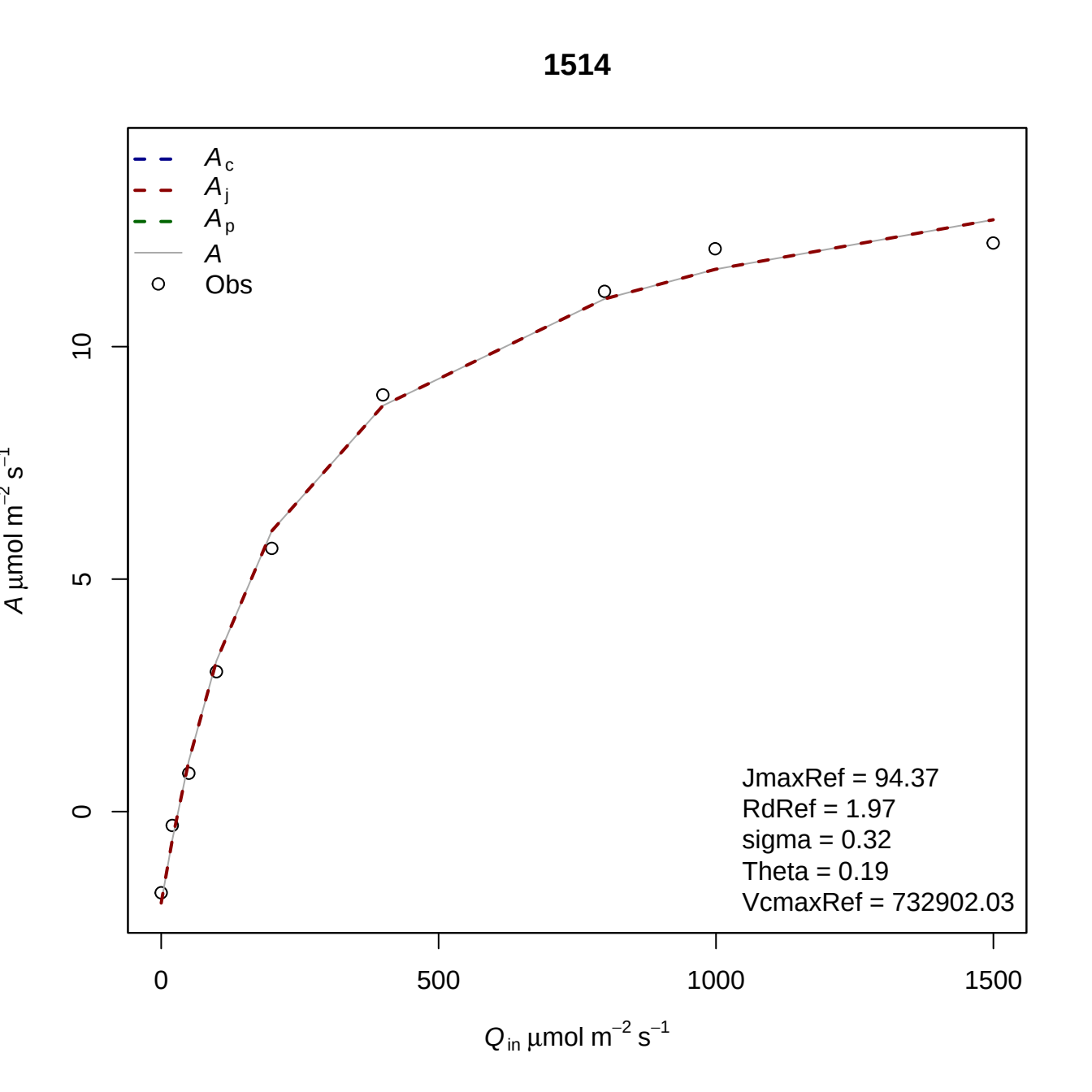

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 45.97
RdRef = 0.46
sigma = 0.18
Theta = 0.4
VcmaxRef = 28.39
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

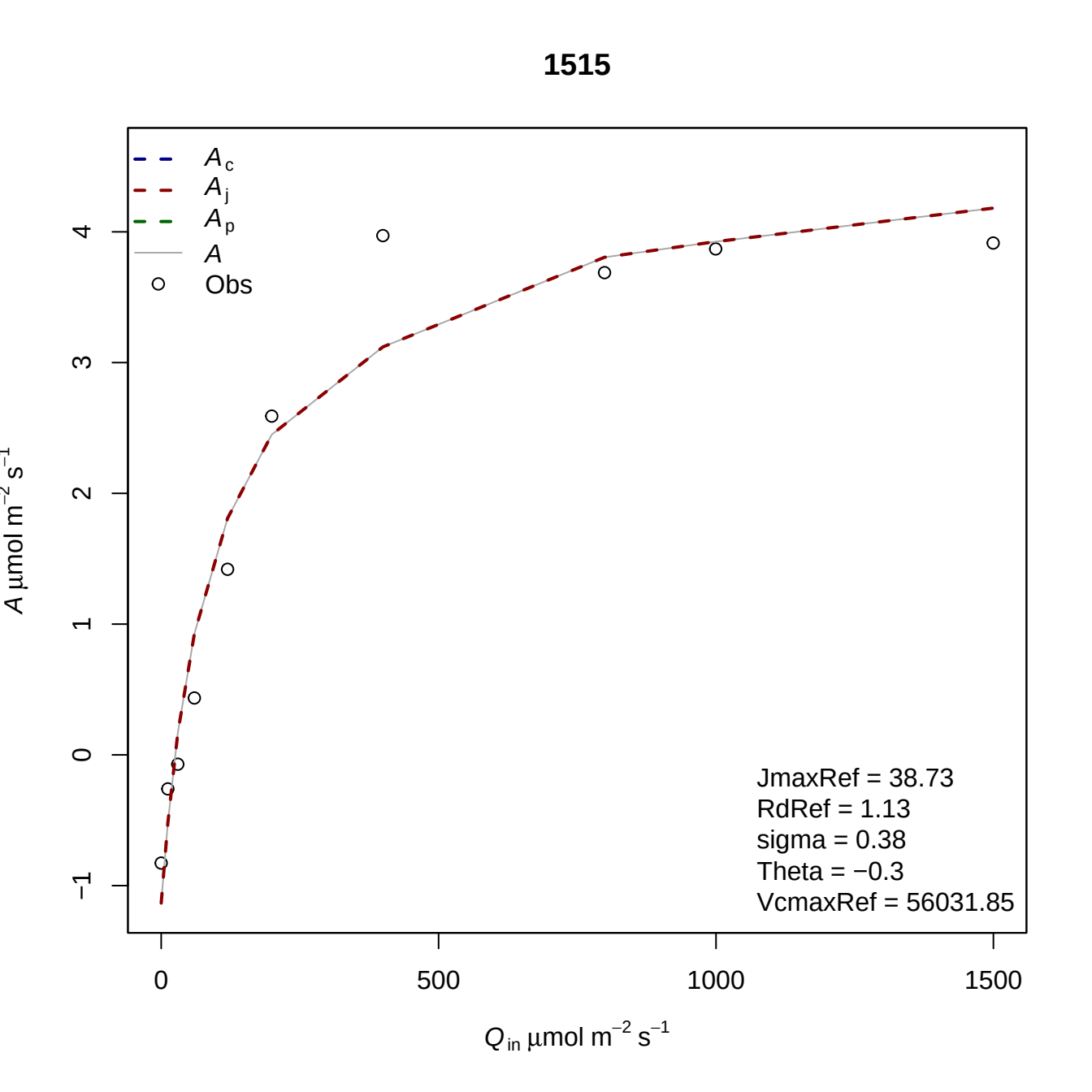

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 77.62
RdRef = 1.31
sigma = 0.27
Theta = 0.44
VcmaxRef = 57.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

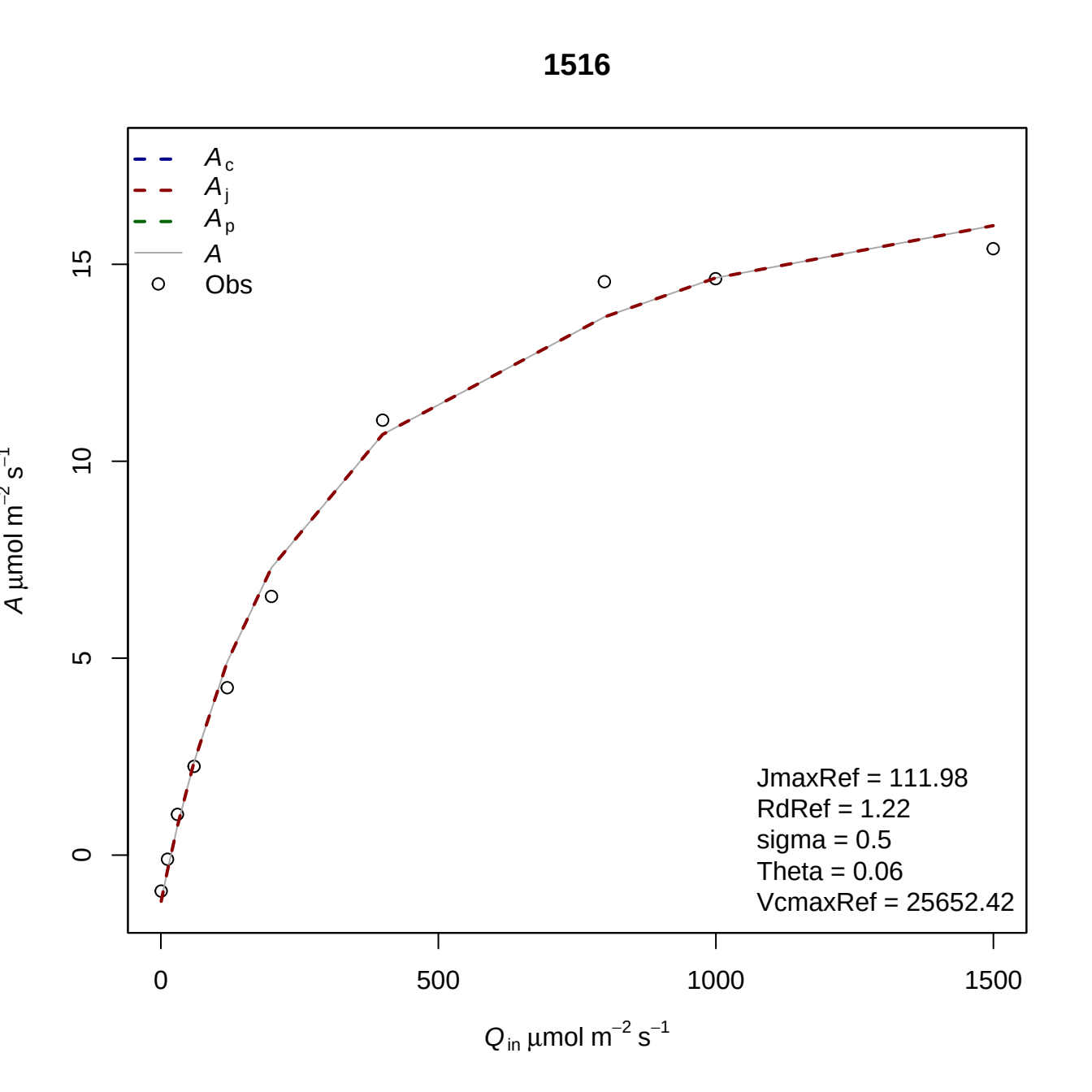

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 100.1
RdRef = 1.36
sigma = 0.38
Theta = 0.12
VcmaxRef = 54.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

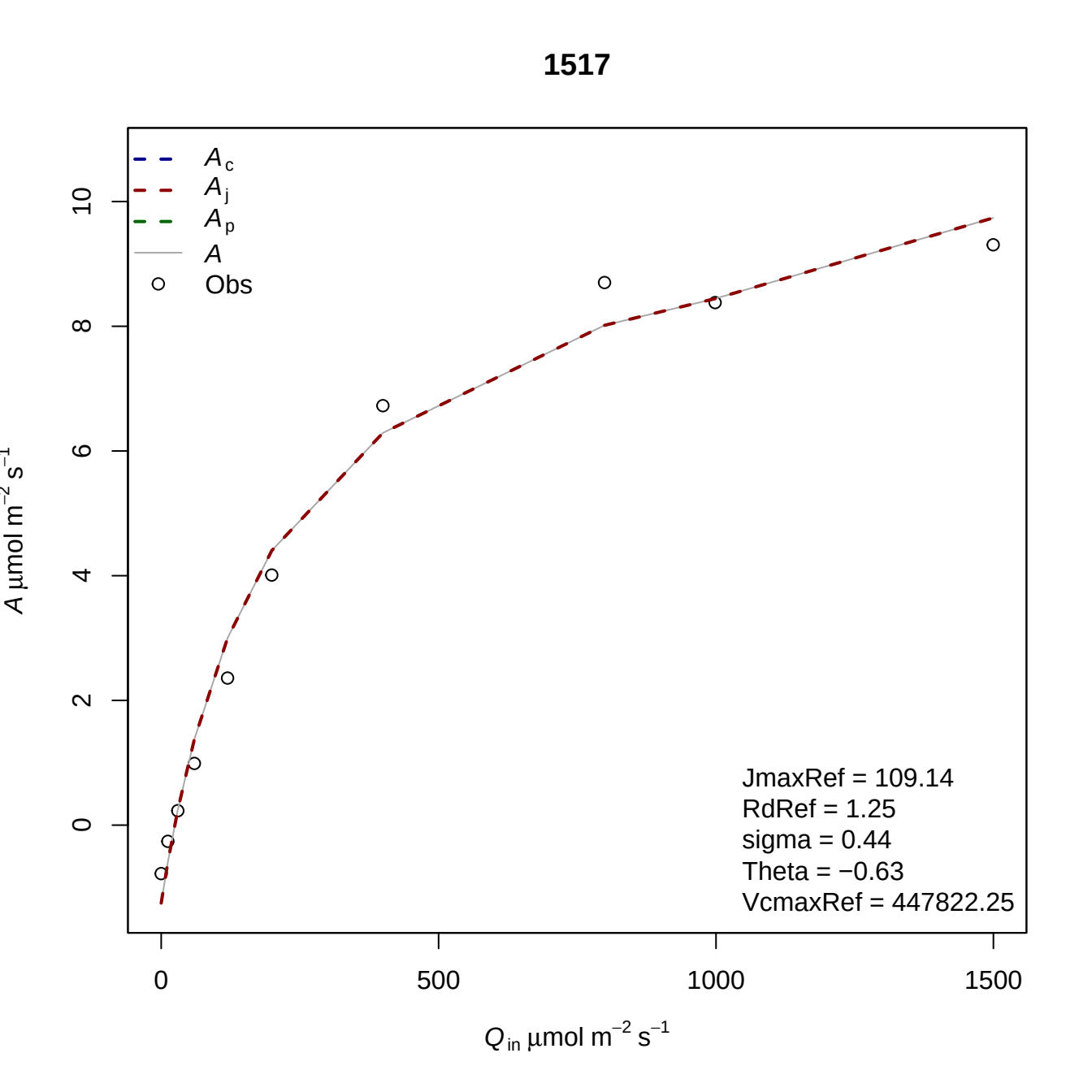

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 122.89
RdRef = 1.69
sigma = 0.49
Theta = −0.11
VcmaxRef = 65724.36
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

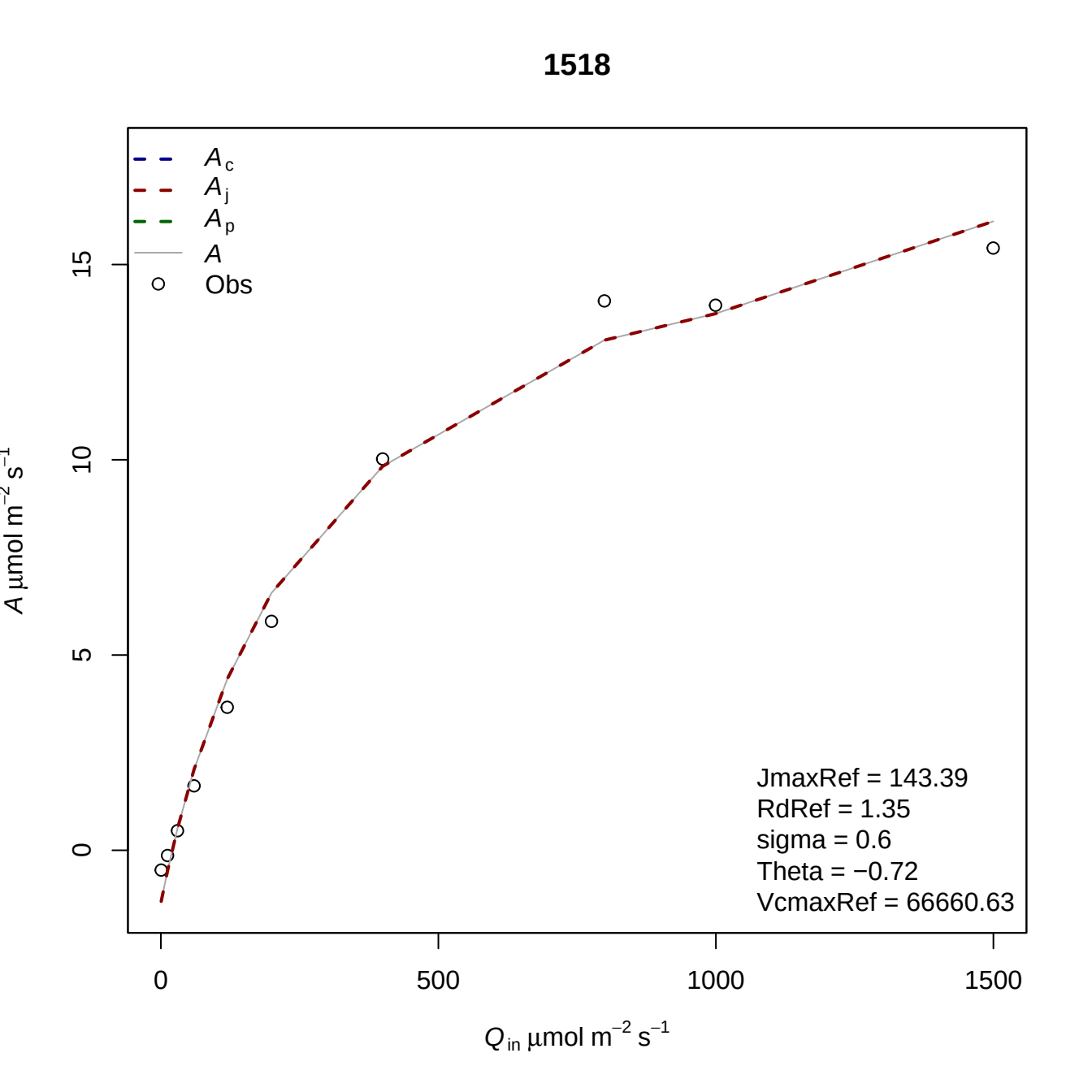

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 40.41
RdRef = 1.94
sigma = 0.19
Theta = −0.78
VcmaxRef = 95.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

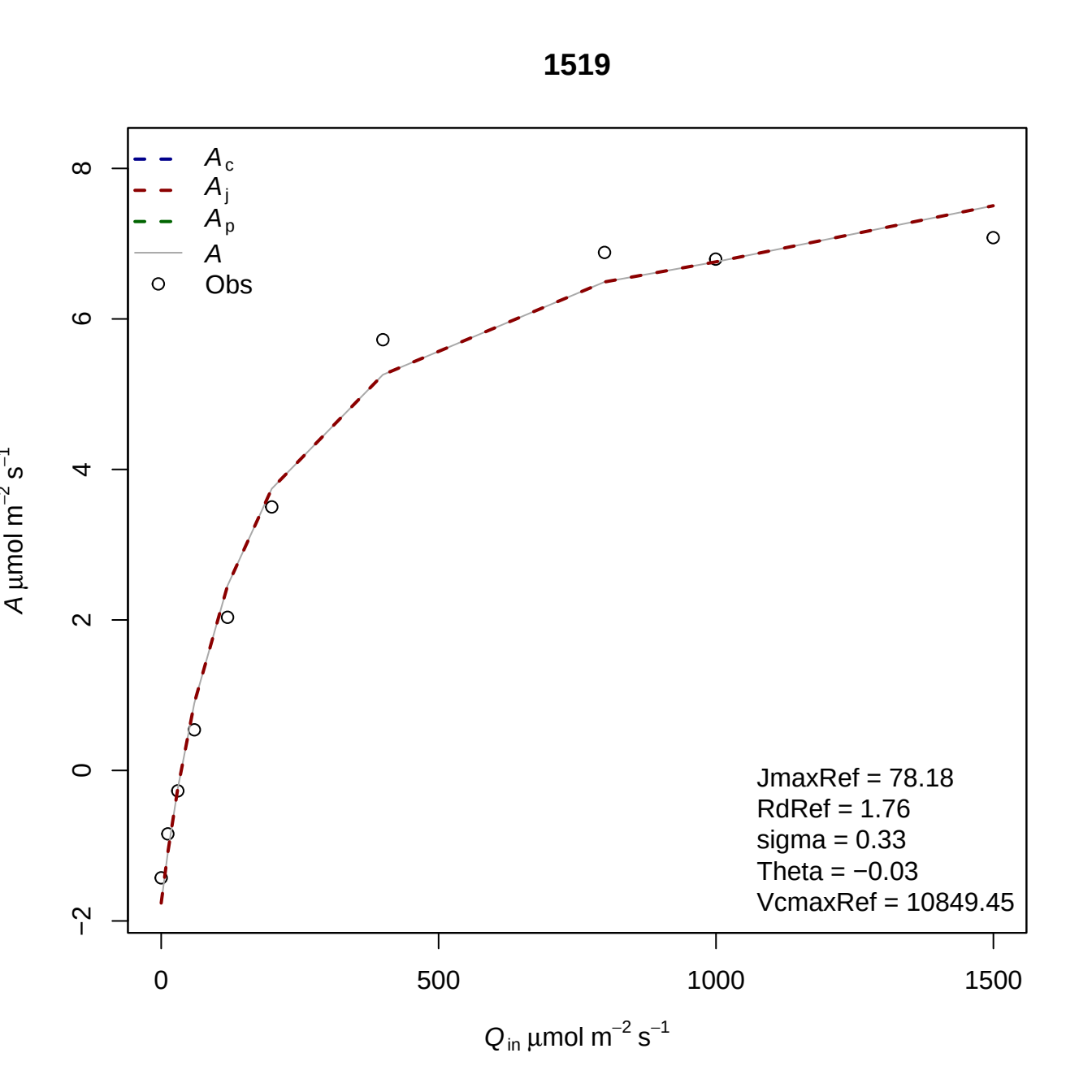

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 94.37
RdRef = 1.97
sigma = 0.32
Theta = 0.19
VcmaxRef = 732902.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

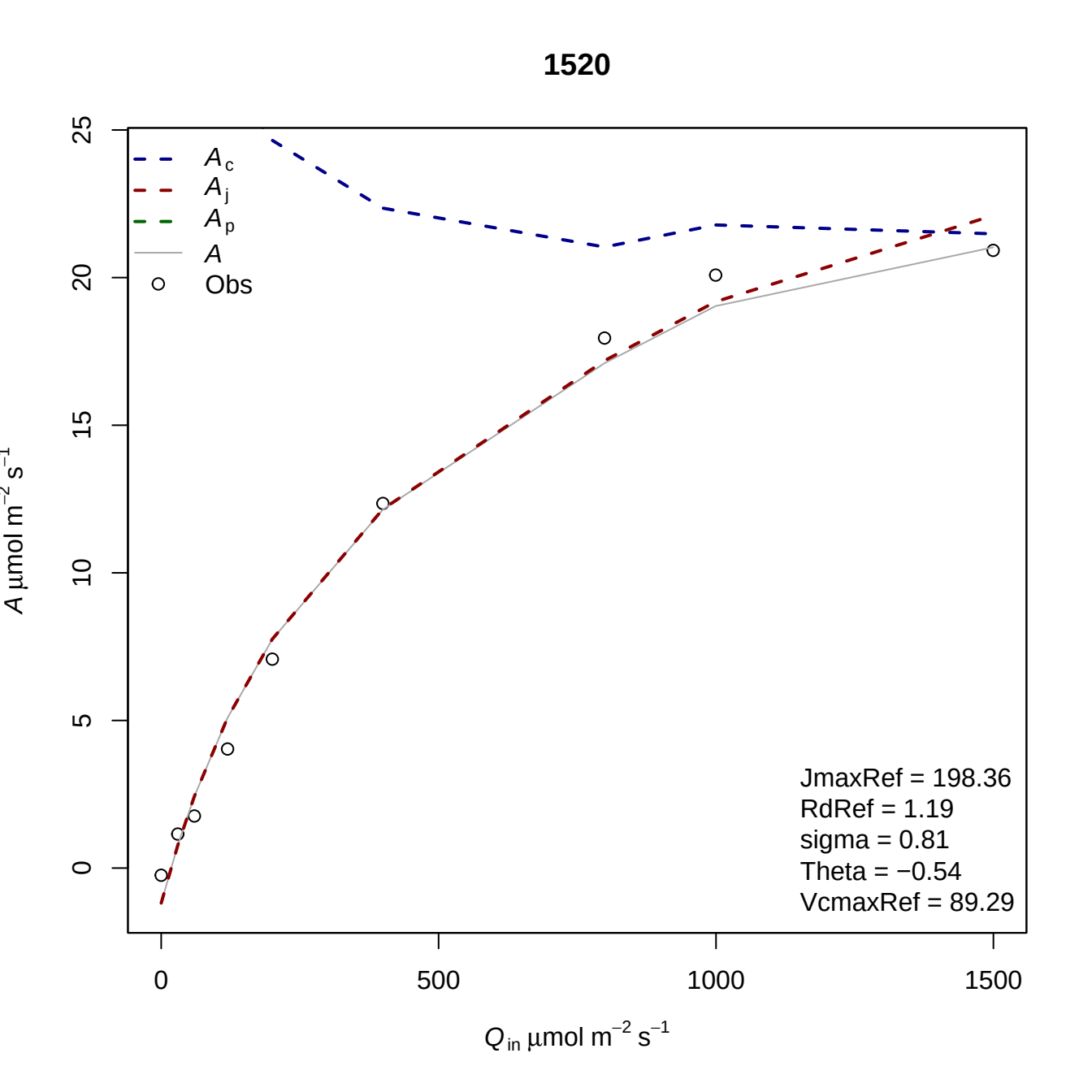

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 38.73
RdRef = 1.13
sigma = 0.38
Theta = −0.3
VcmaxRef = 56031.85
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

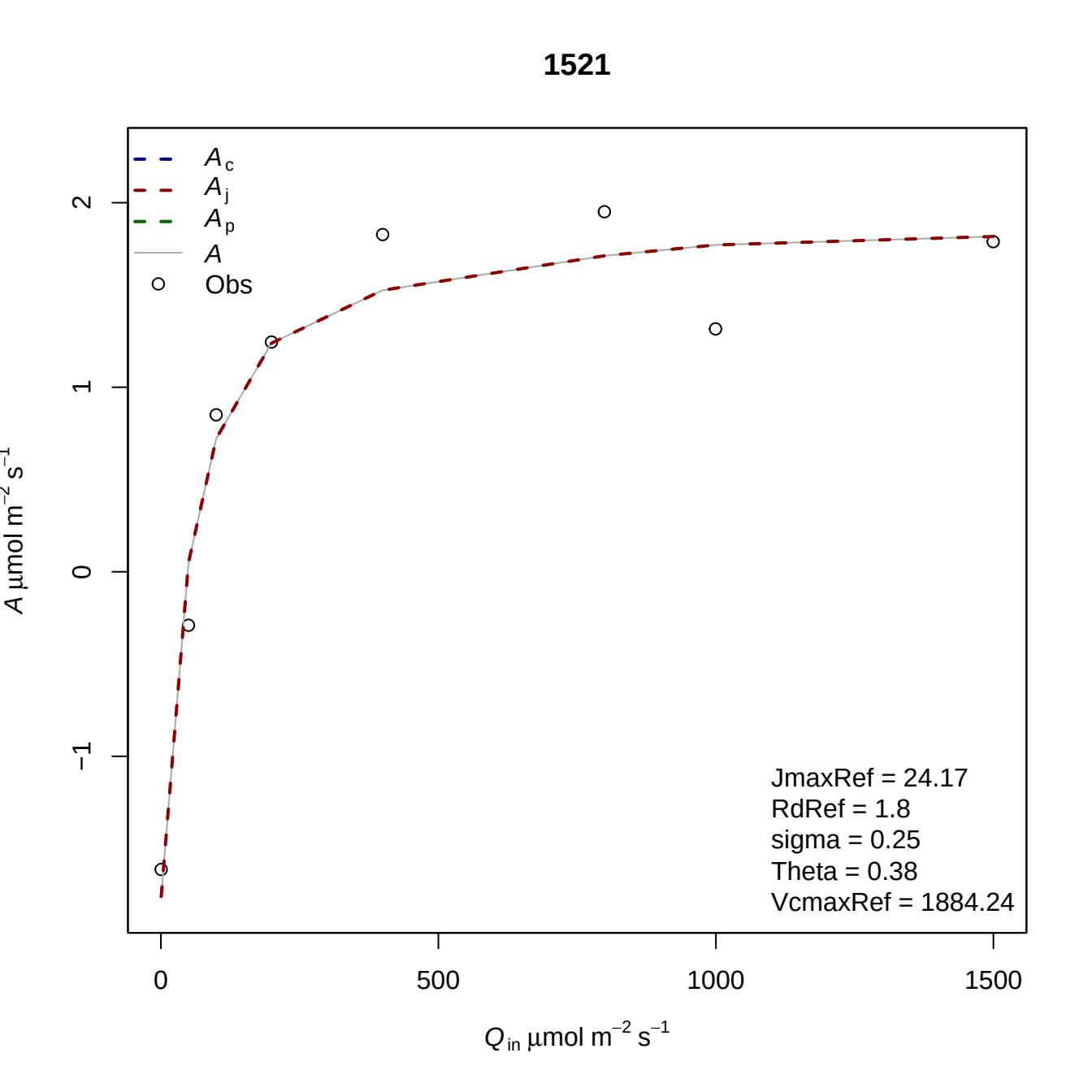

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 111.98
RdRef = 1.22
sigma = 0.5
Theta = 0.06
VcmaxRef = 25652.42
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

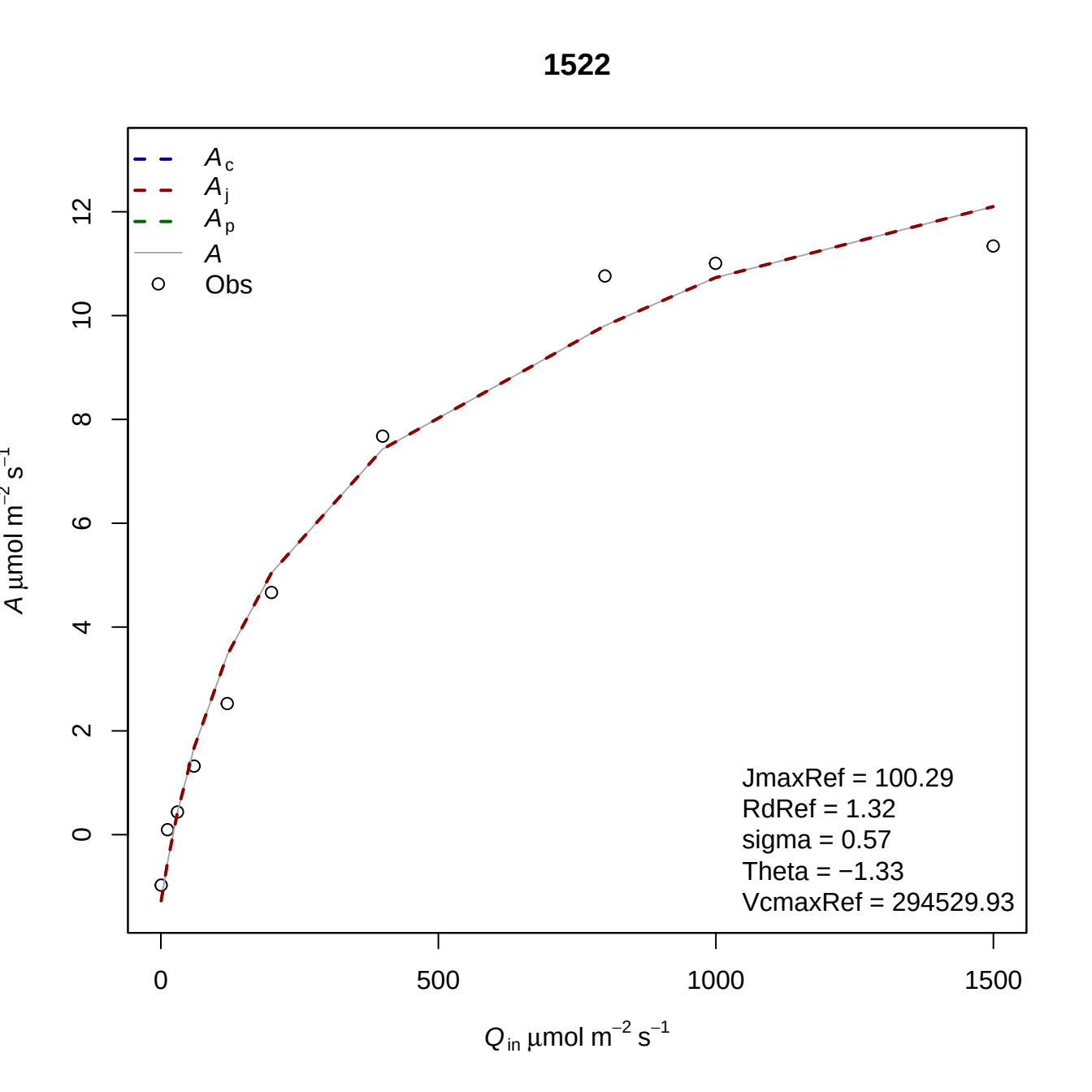

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 109.14
RdRef = 1.25
sigma = 0.44
Theta = −0.63
VcmaxRef = 447822.25
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

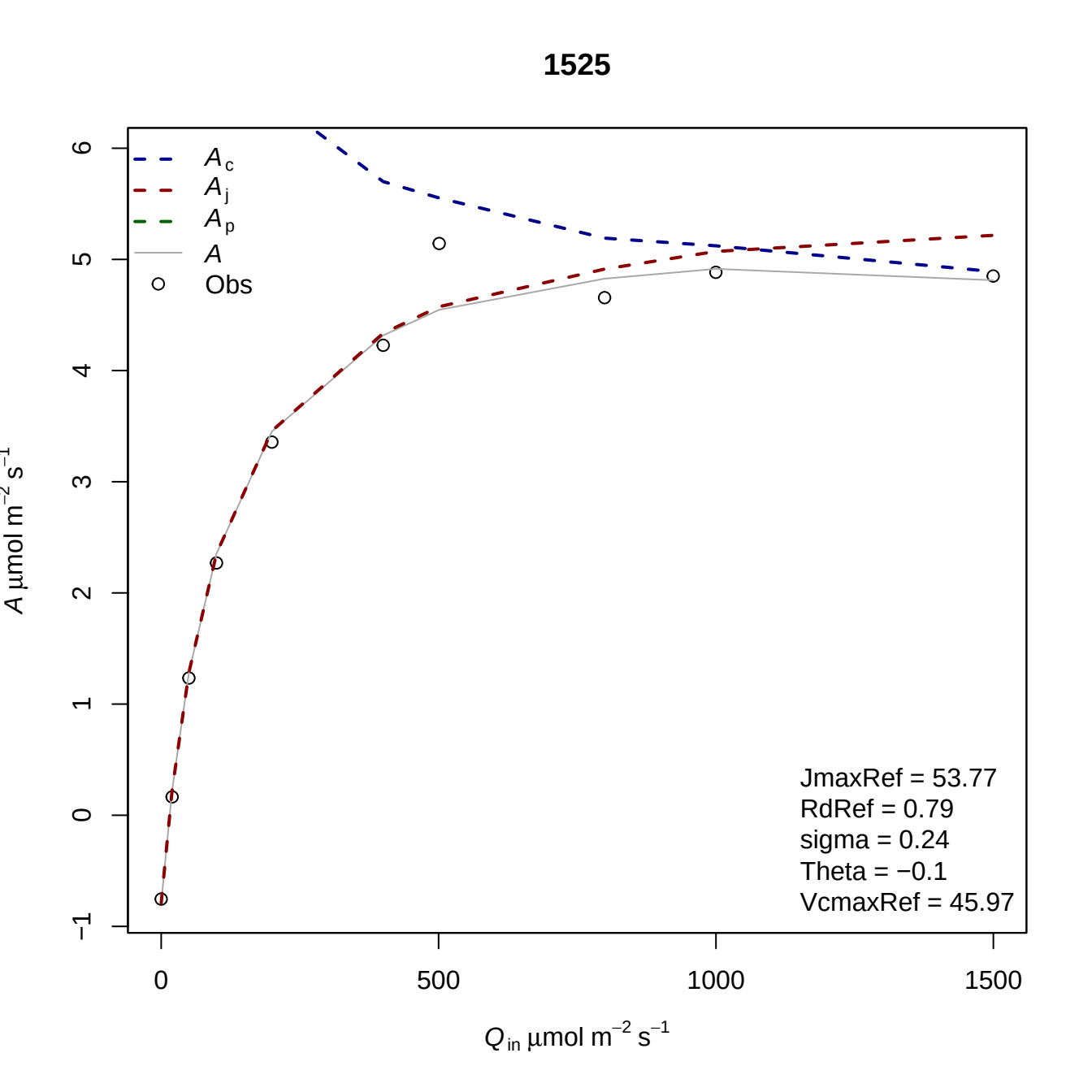

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 143.39
RdRef = 1.35
sigma = 0.6
Theta = −0.72
VcmaxRef = 66660.63
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

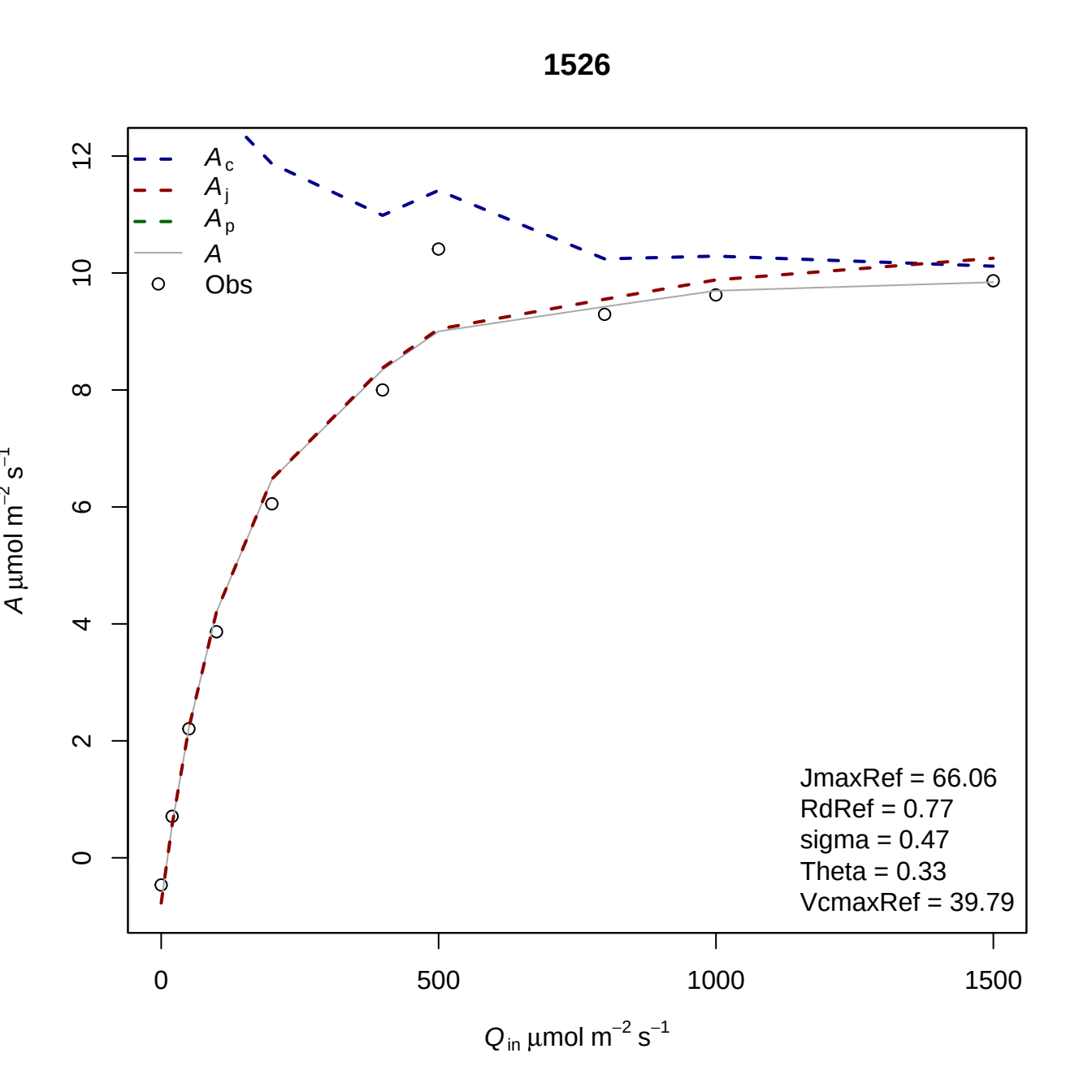

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 78.18
RdRef = 1.76
sigma = 0.33
Theta = −0.03
VcmaxRef = 10849.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

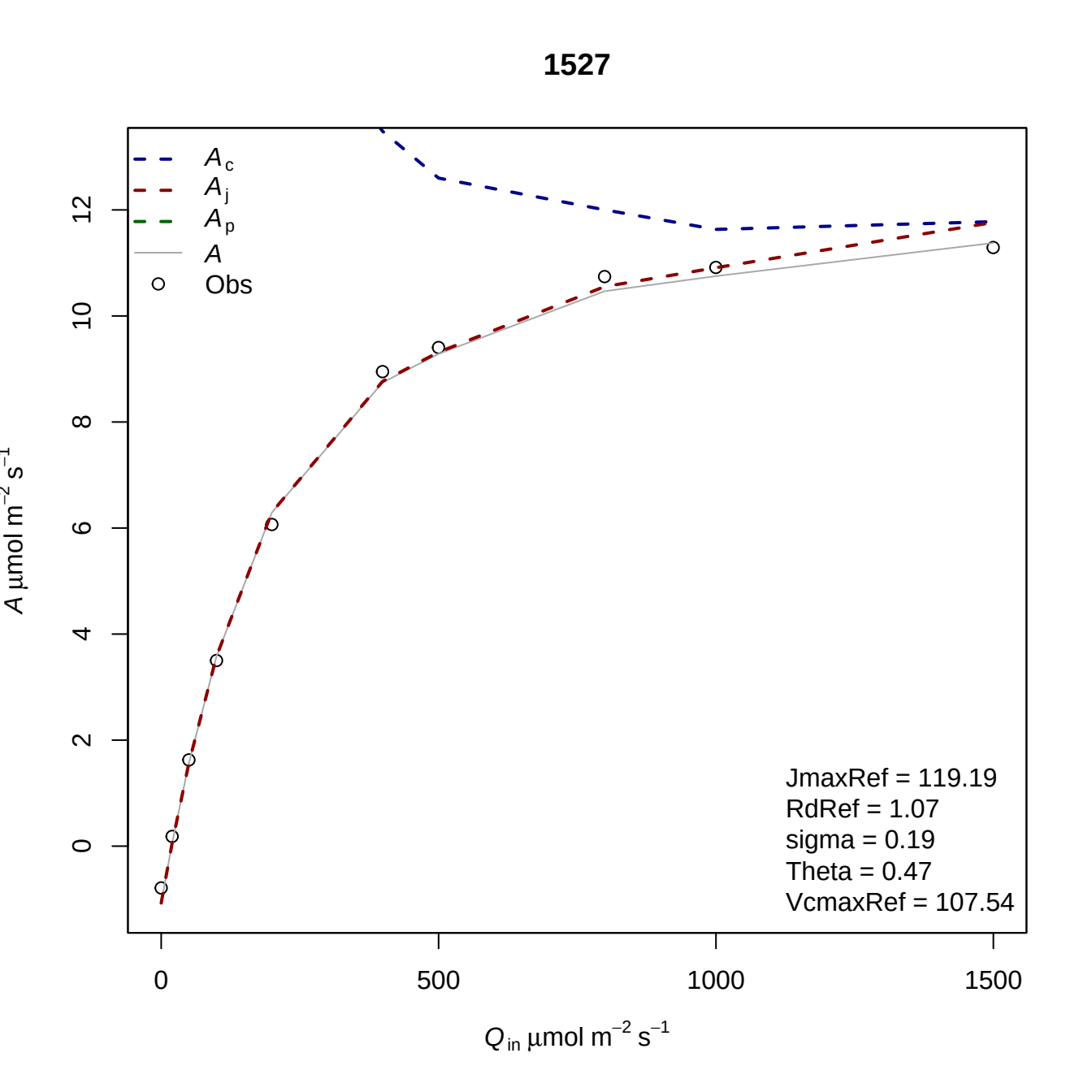

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 198.36
RdRef = 1.19
sigma = 0.81
Theta = −0.54
VcmaxRef = 89.29
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

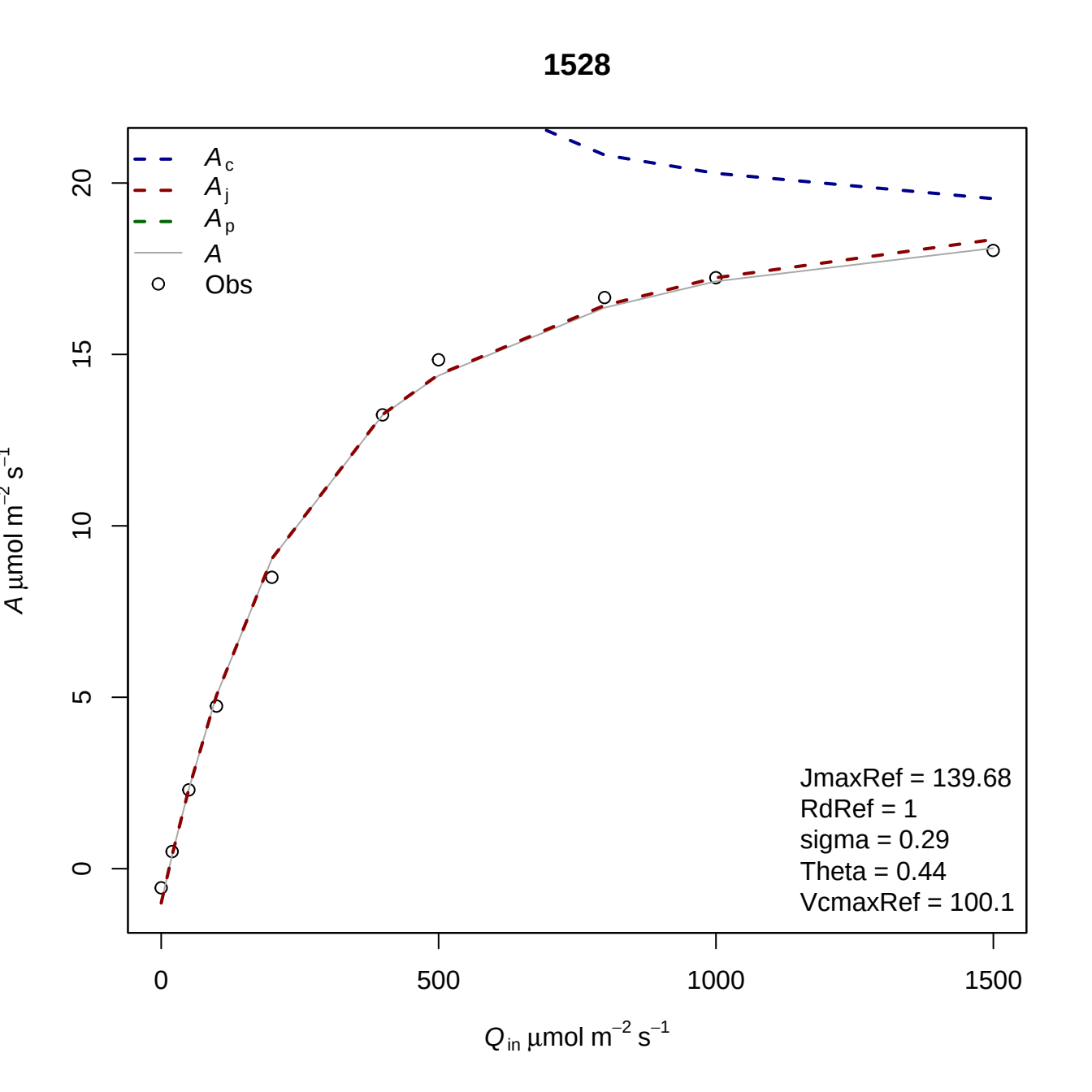

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 24.17
RdRef = 1.8
sigma = 0.25
Theta = 0.38
VcmaxRef = 1884.24
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

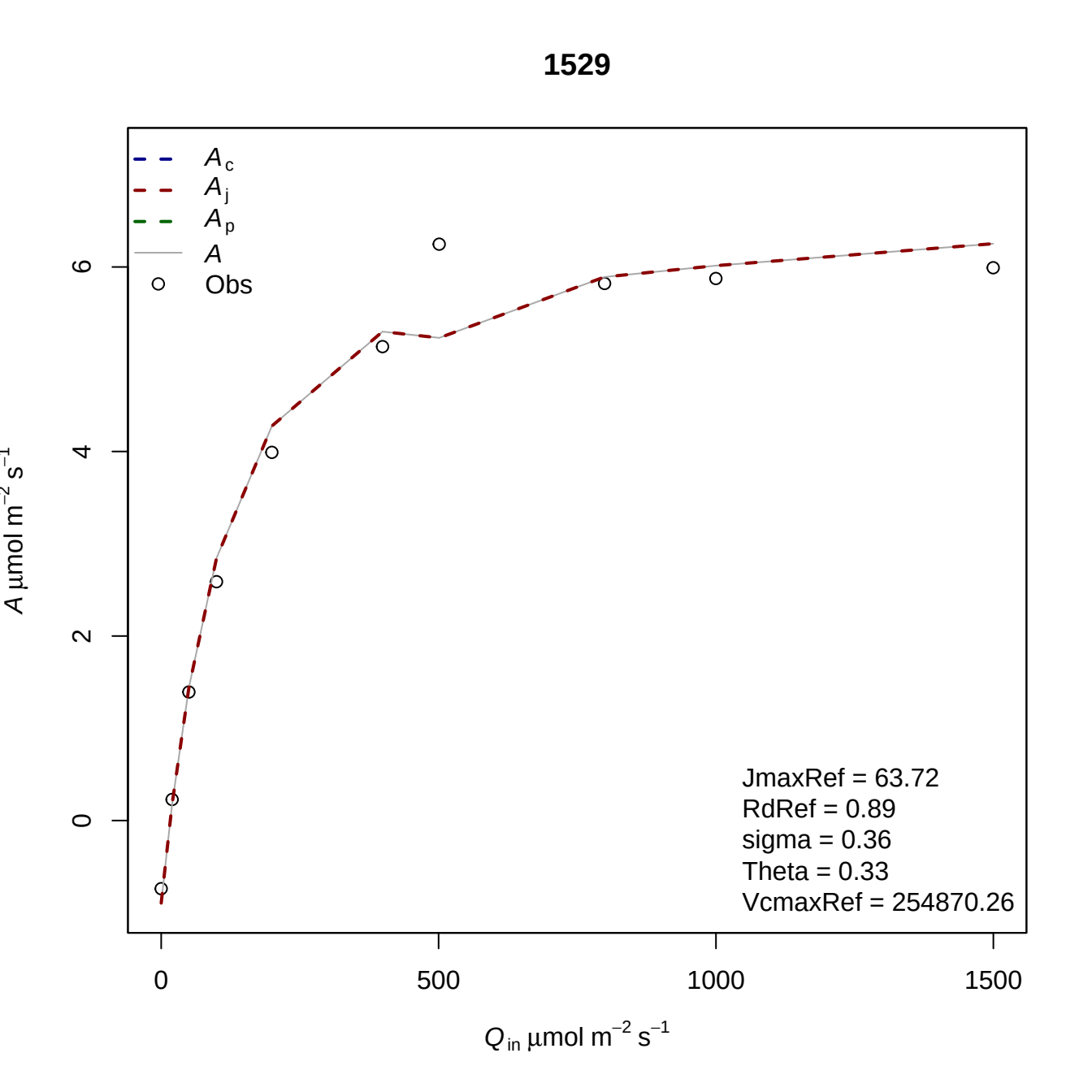

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 100.29
RdRef = 1.32
sigma = 0.57
Theta = −1.33
VcmaxRef = 294529.93
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

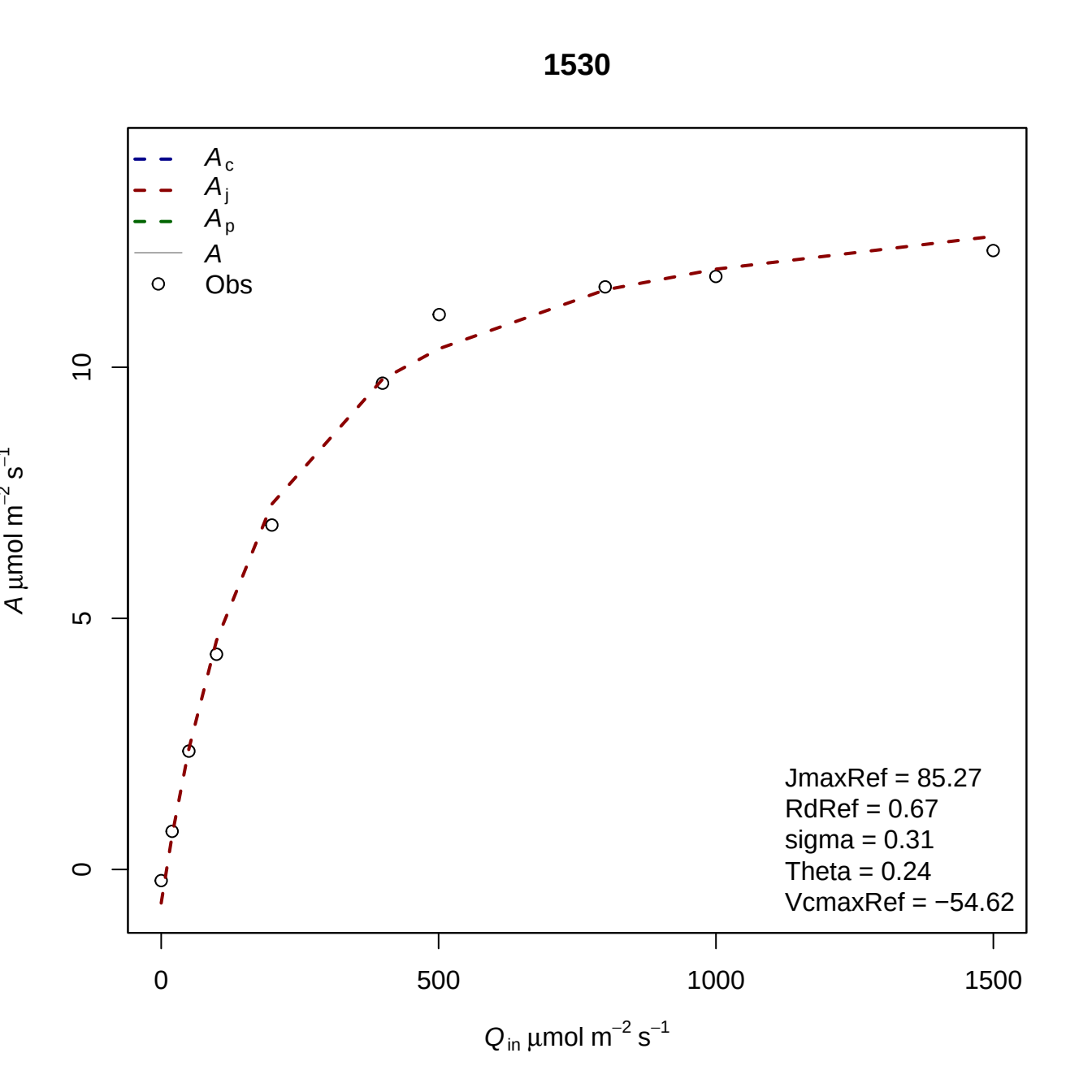

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 53.77
RdRef = 0.79
sigma = 0.24
Theta = −0.1
VcmaxRef = 45.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

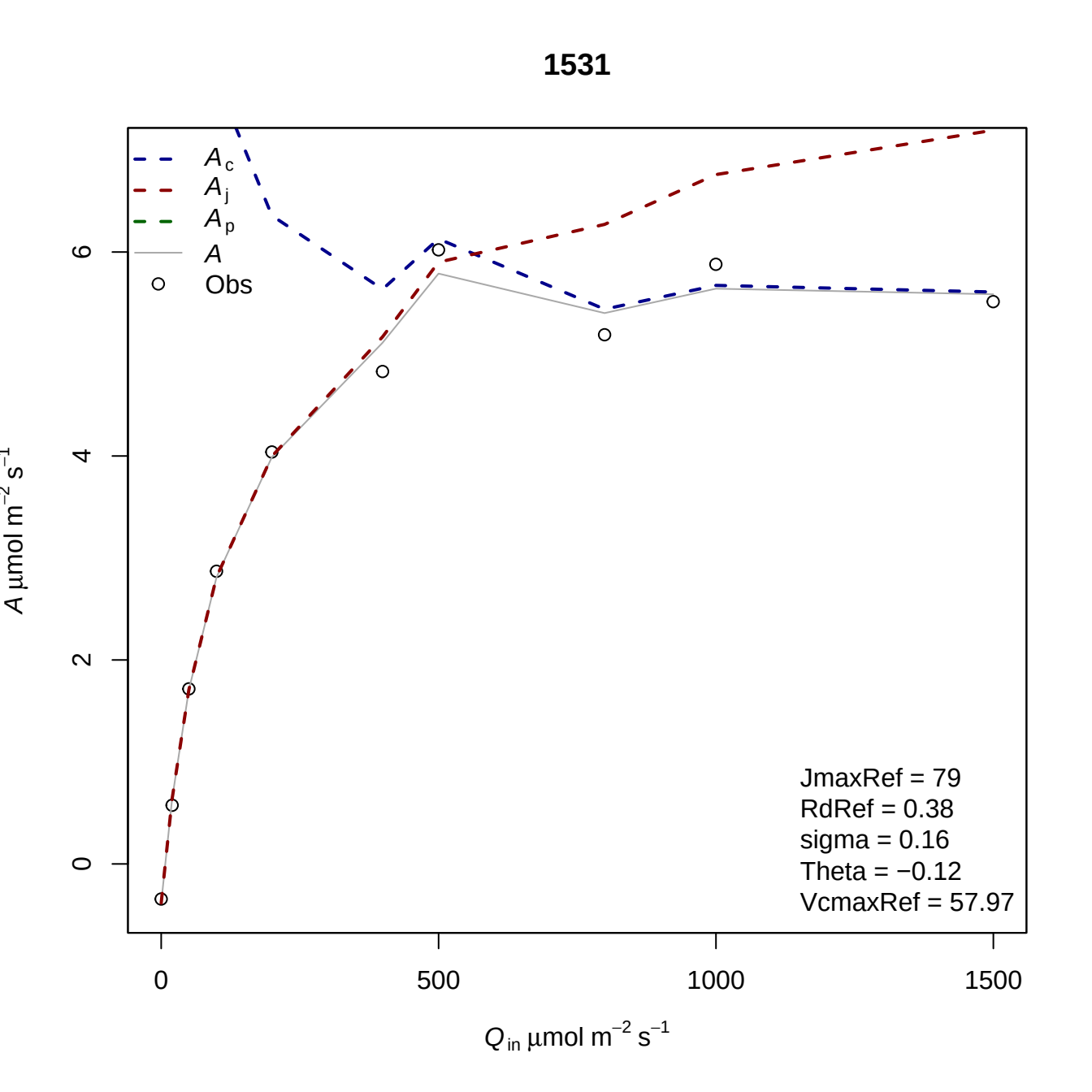

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 66.06
RdRef = 0.77
sigma = 0.47
Theta = 0.33
VcmaxRef = 39.79
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

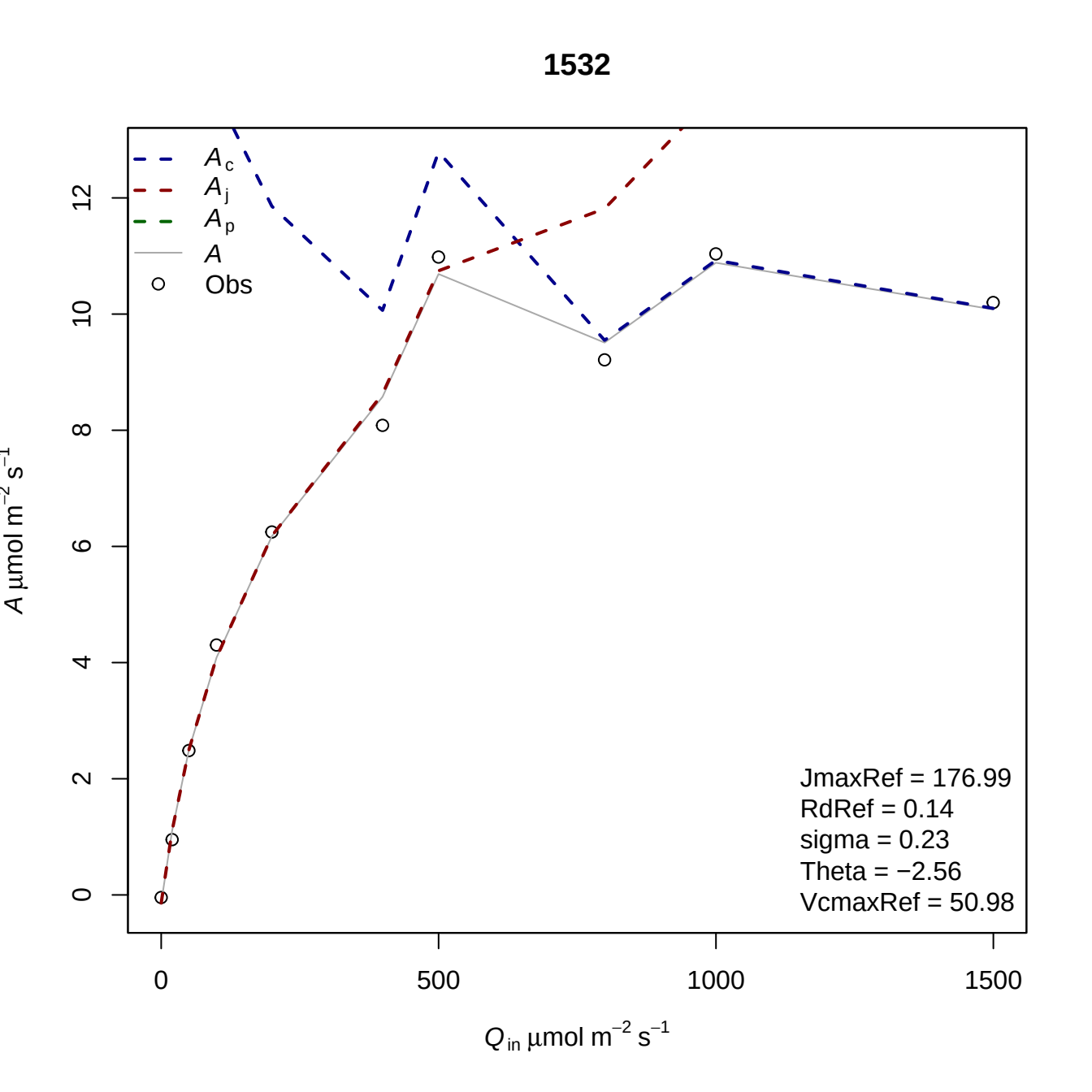

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 119.19
RdRef = 1.07
sigma = 0.19
Theta = 0.47
VcmaxRef = 107.54
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

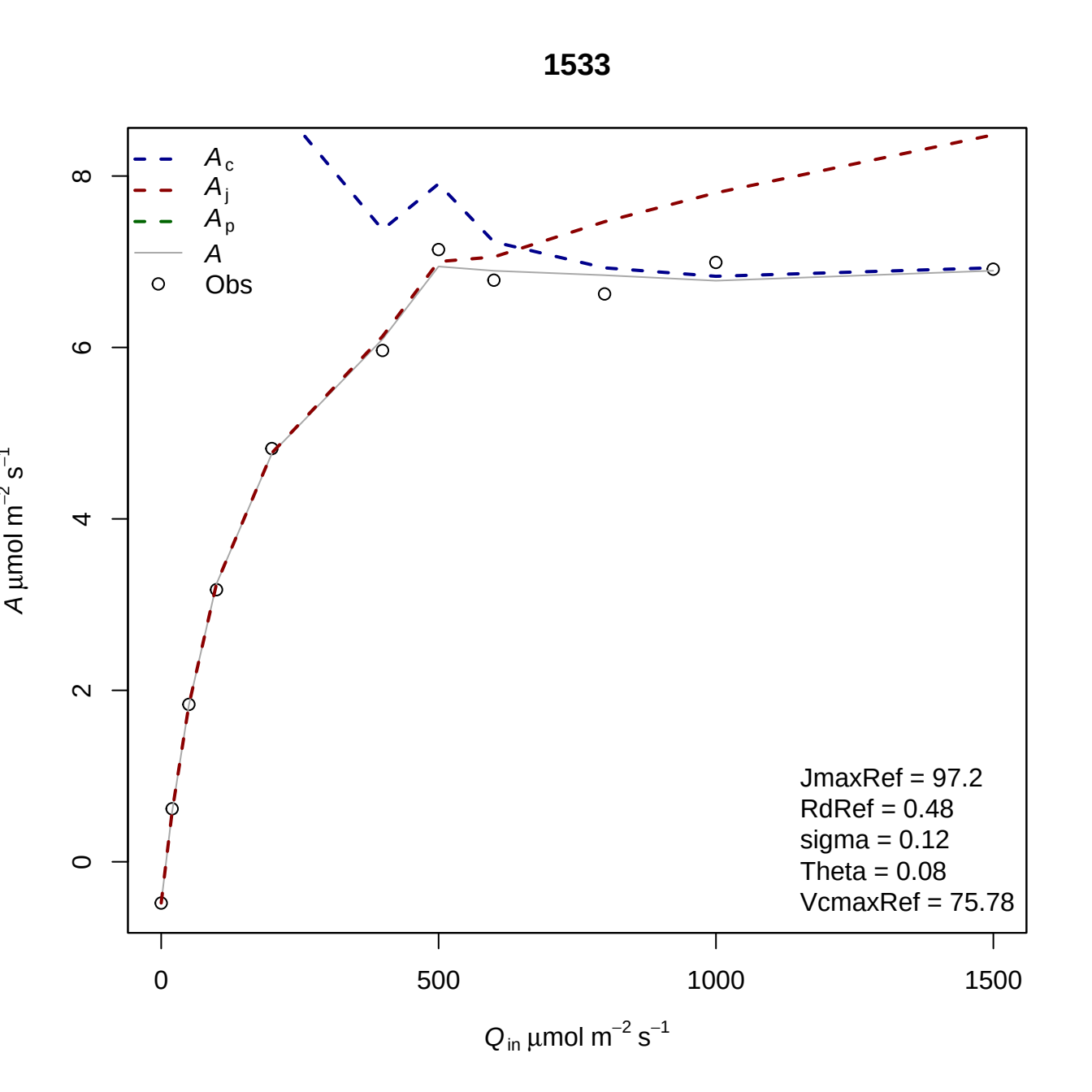

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 139.68
RdRef = 1
sigma = 0.29
Theta = 0.44
VcmaxRef = 100.1
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

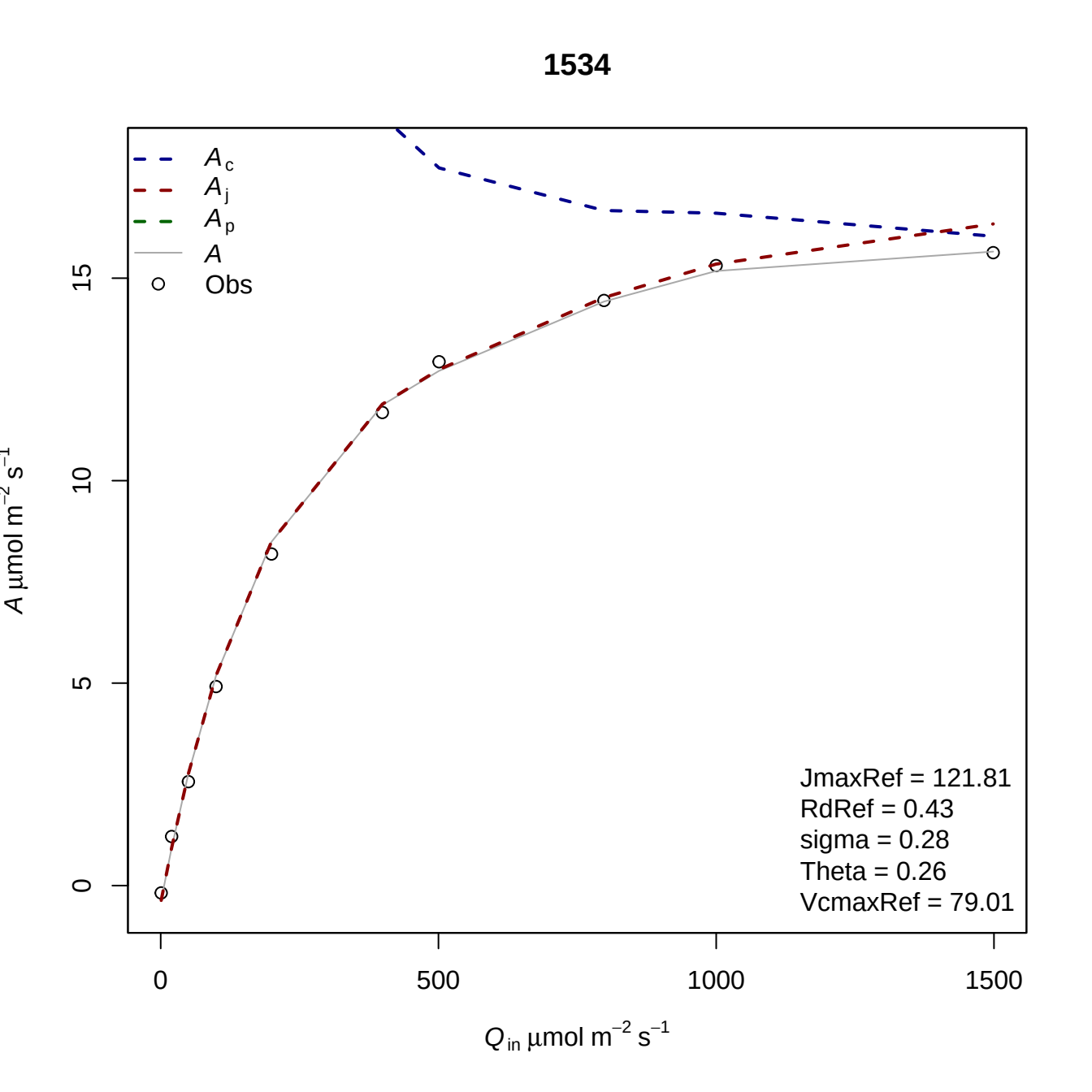

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 63.72
RdRef = 0.89
sigma = 0.36
Theta = 0.33
VcmaxRef = 254870.26
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

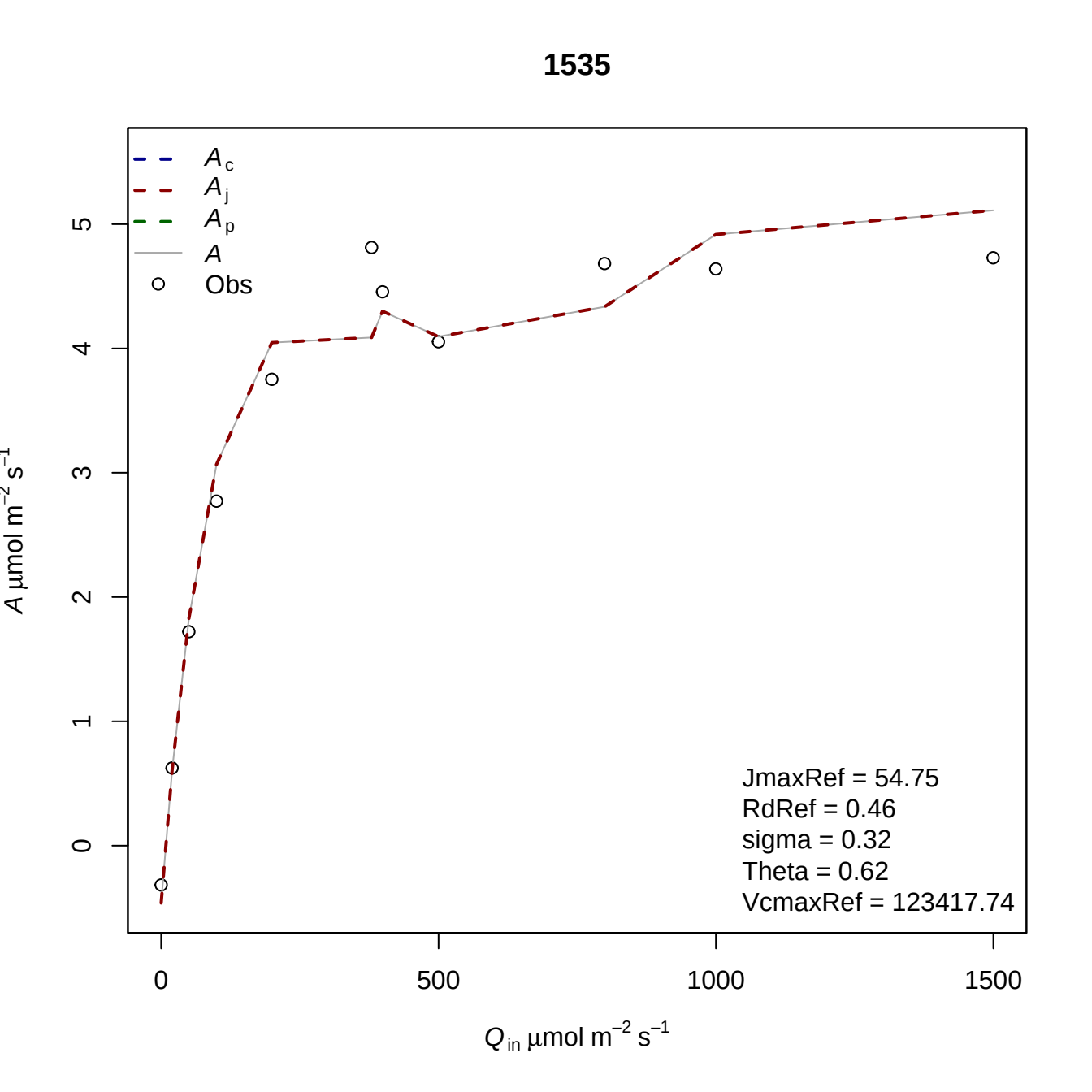

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 85.27
RdRef = 0.67
sigma = 0.31
Theta = 0.24
VcmaxRef = −54.62
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

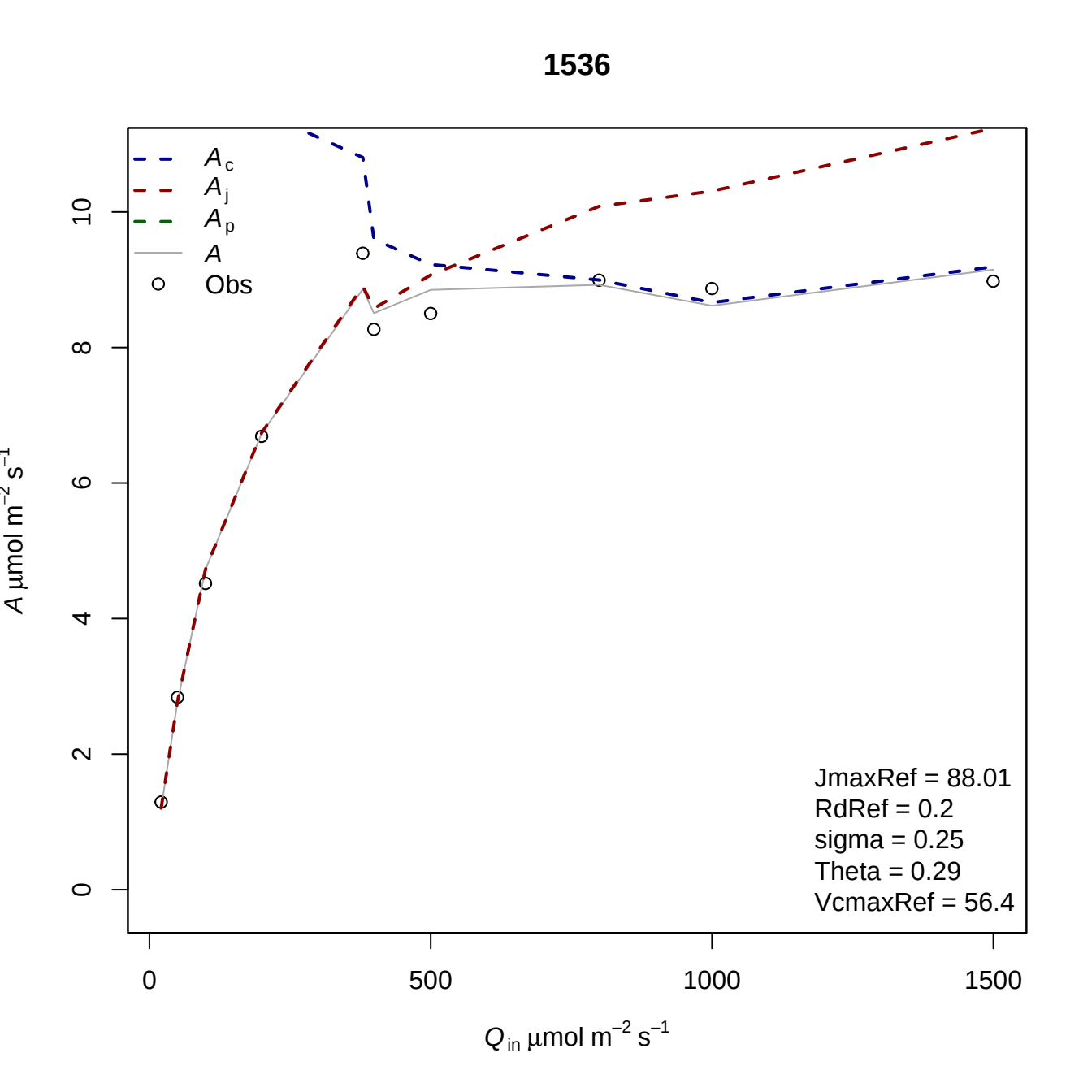

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 79
RdRef = 0.38
sigma = 0.16
Theta = −0.12
VcmaxRef = 57.97
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

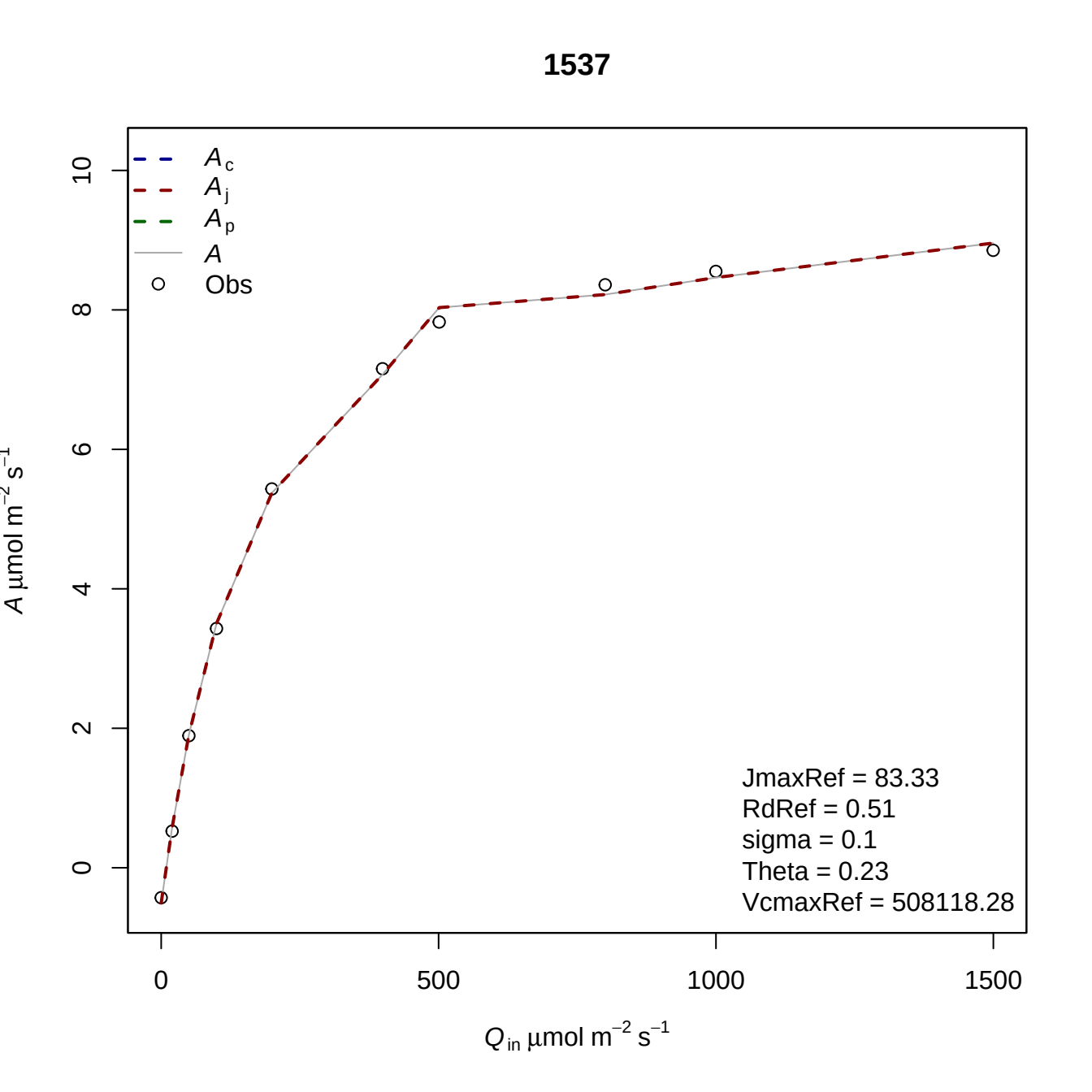

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 176.99
RdRef = 0.14
sigma = 0.23
Theta = −2.56
VcmaxRef = 50.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

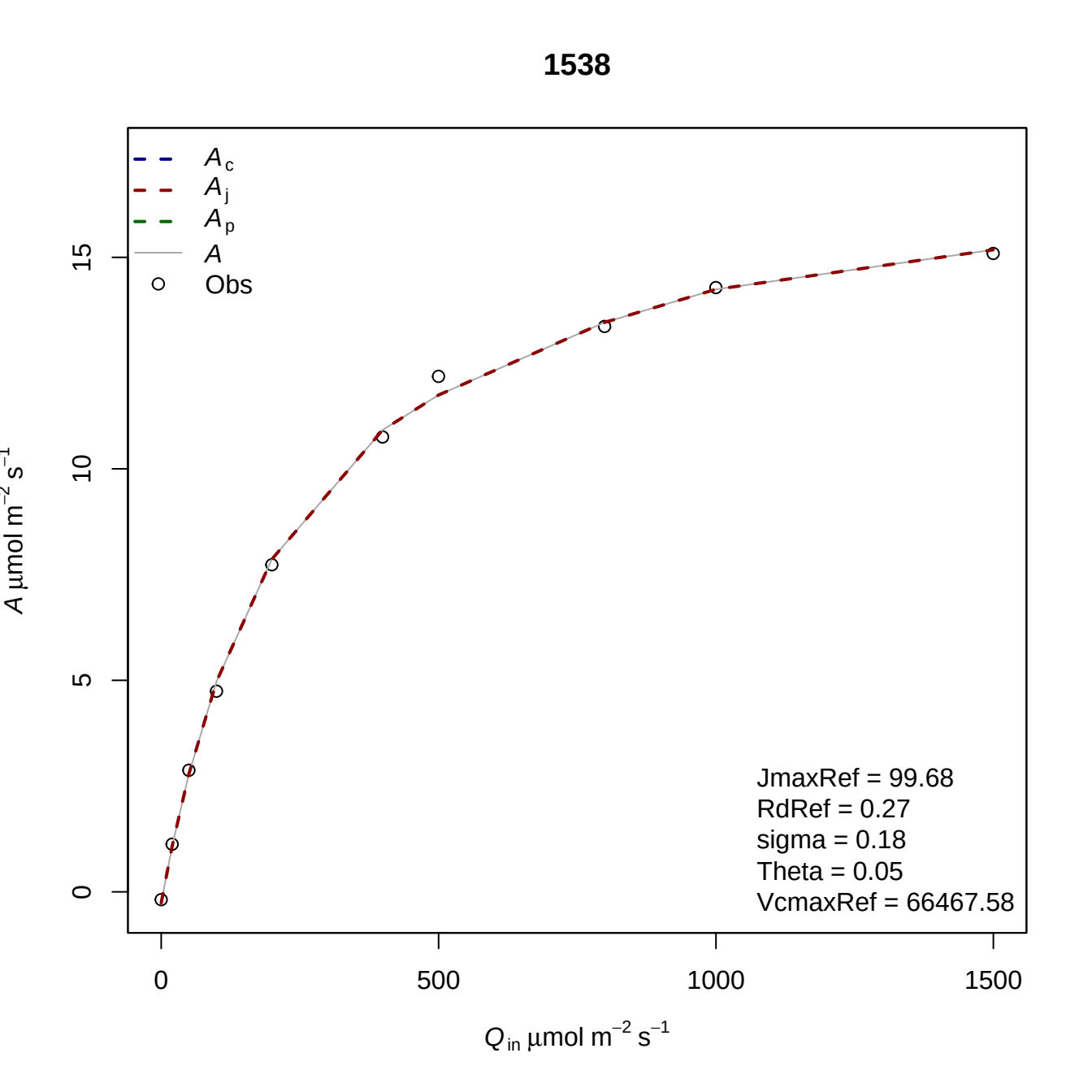

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 97.2
RdRef = 0.48
sigma = 0.12
Theta = 0.08
VcmaxRef = 75.78
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

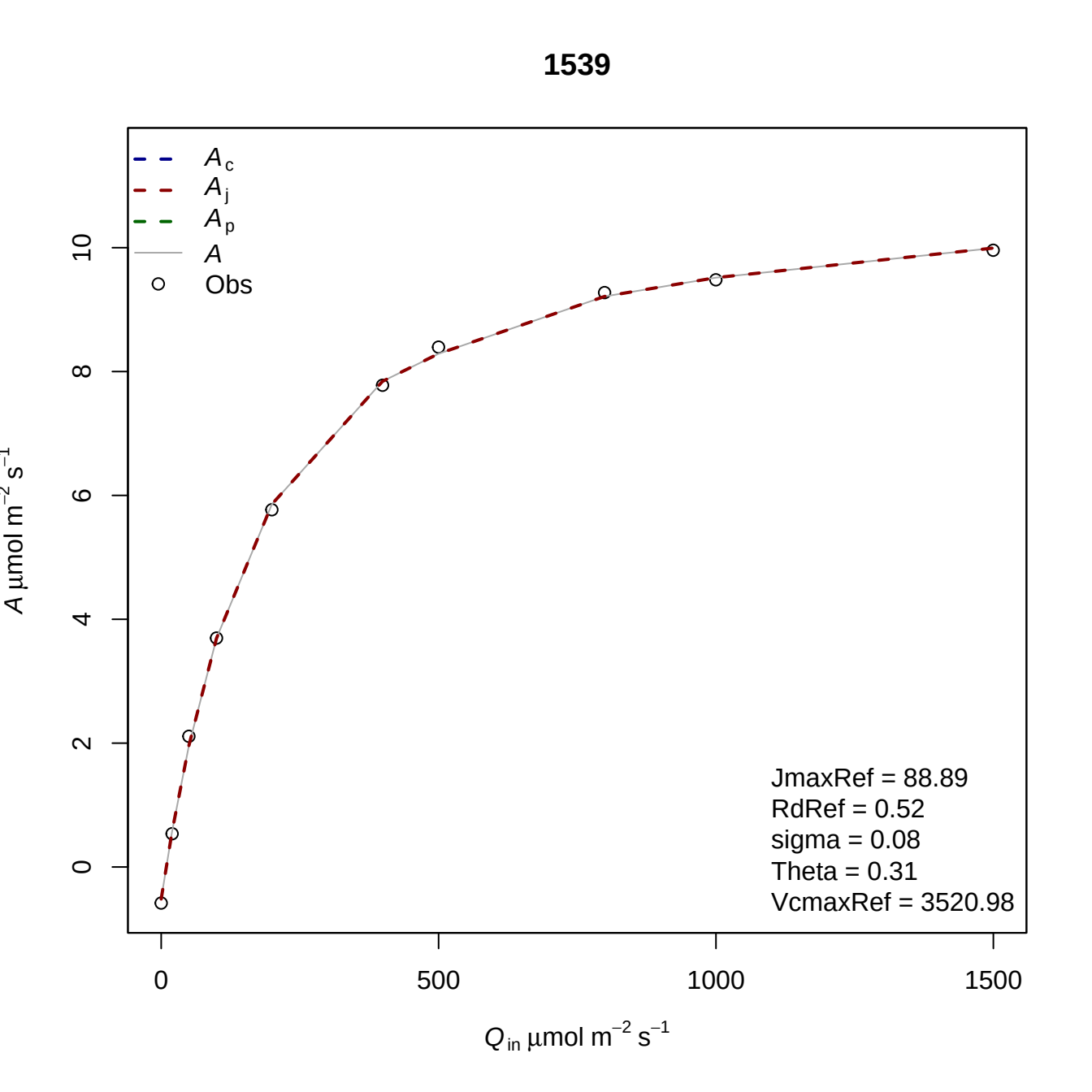

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 121.81
RdRef = 0.43
sigma = 0.28
Theta = 0.26
VcmaxRef = 79.01
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

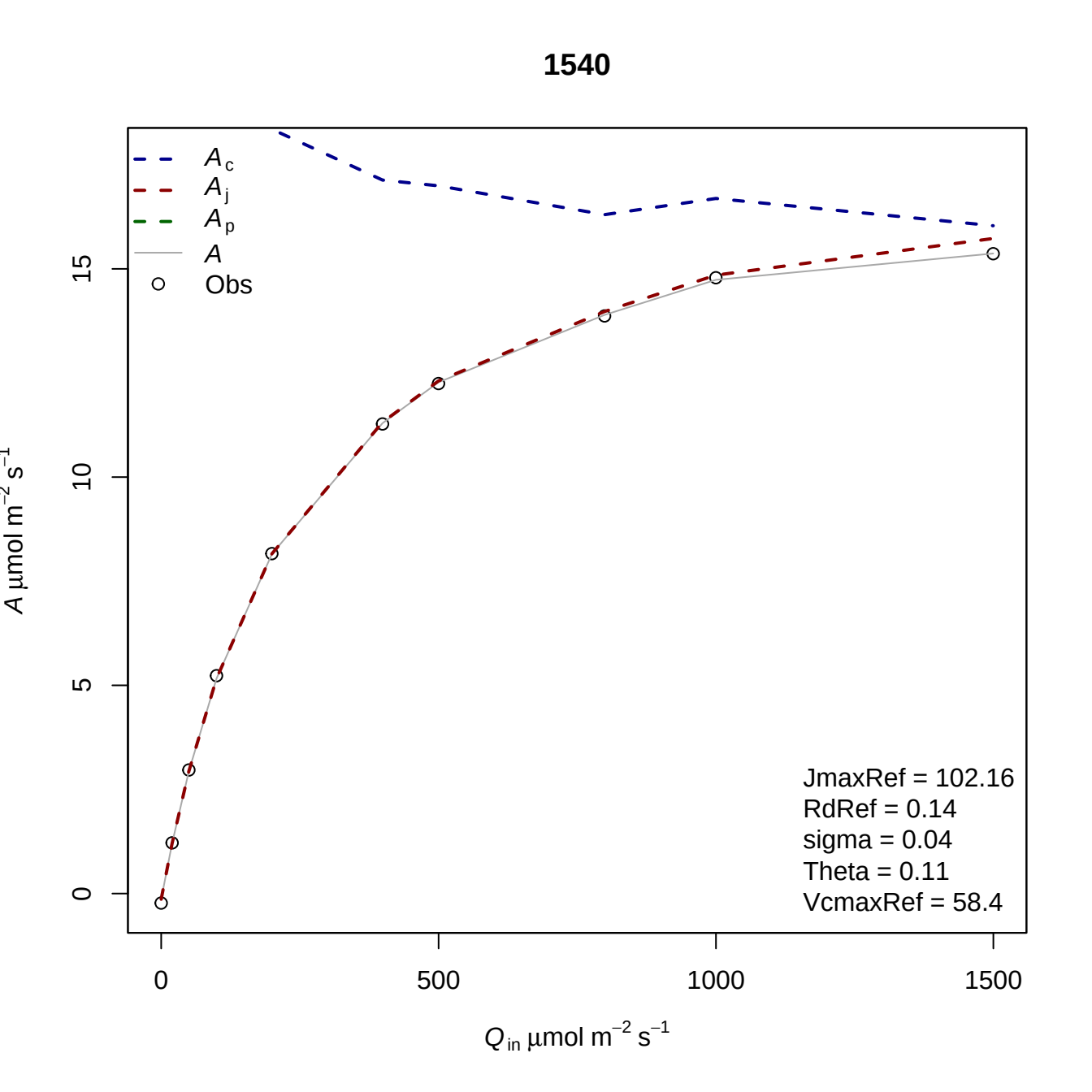

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 54.75
RdRef = 0.46
sigma = 0.32
Theta = 0.62
VcmaxRef = 123417.74
0
1
2
3
4
5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

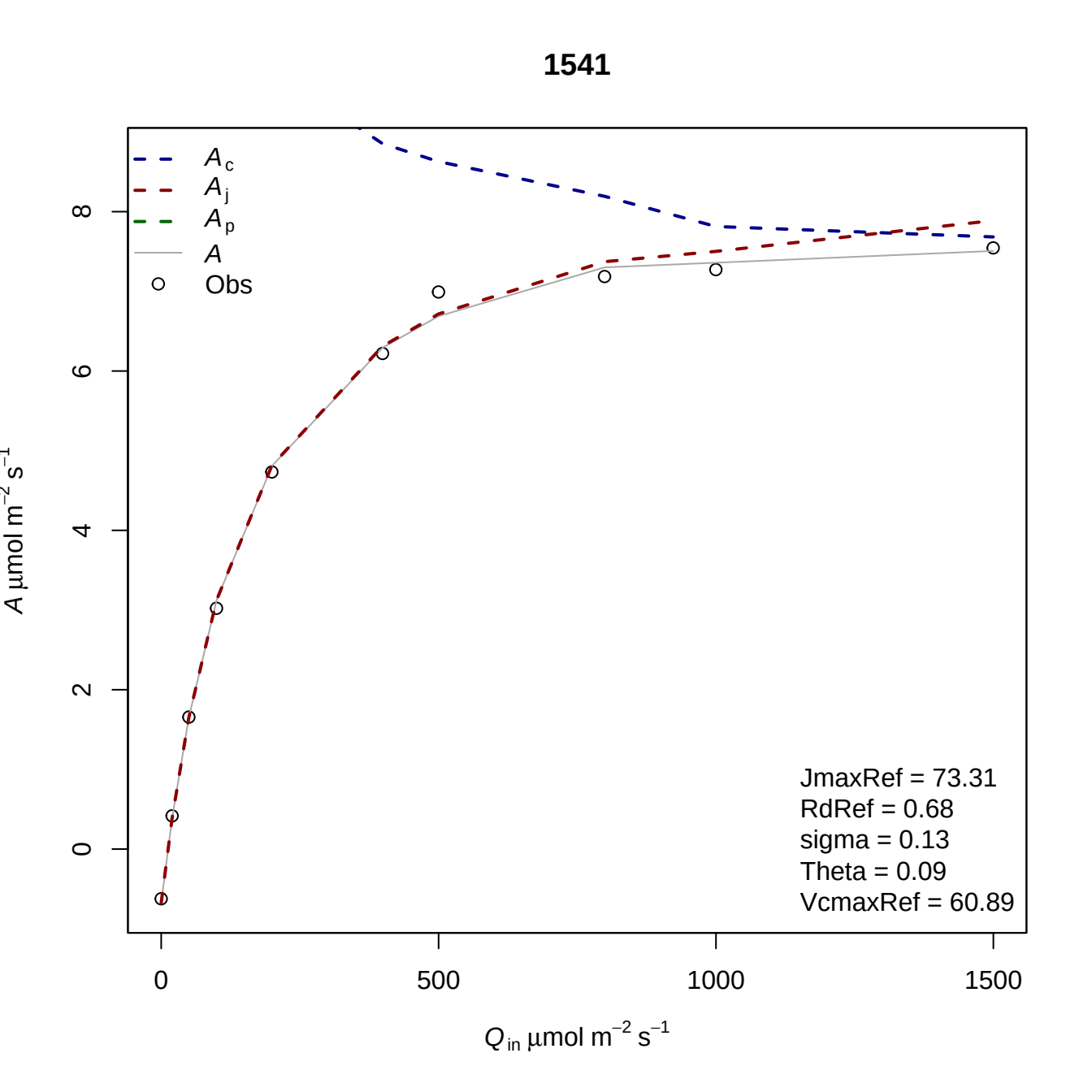

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 88.01
RdRef = 0.2
sigma = 0.25
Theta = 0.29
VcmaxRef = 56.4
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

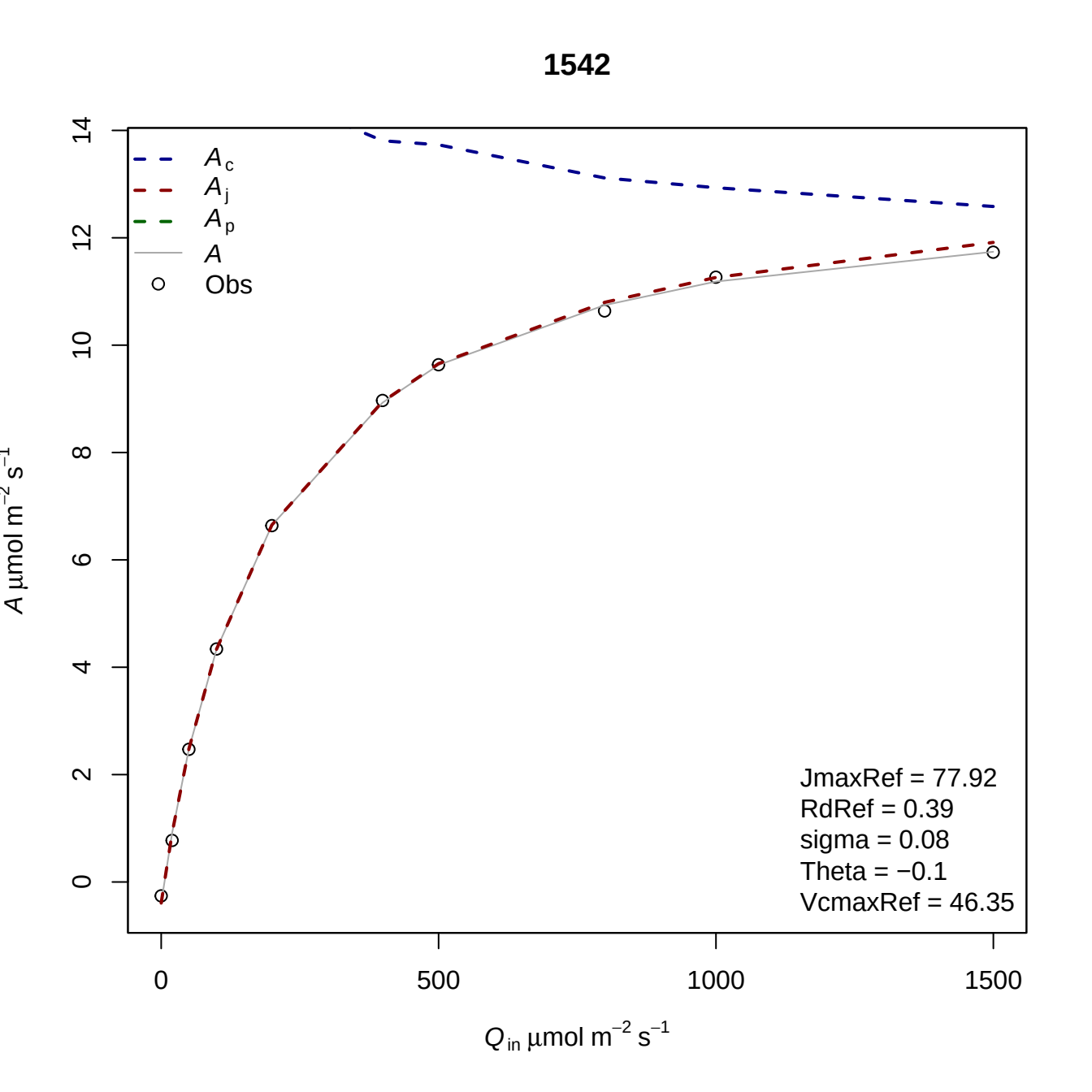

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 83.33
RdRef = 0.51
sigma = 0.1
Theta = 0.23
VcmaxRef = 508118.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

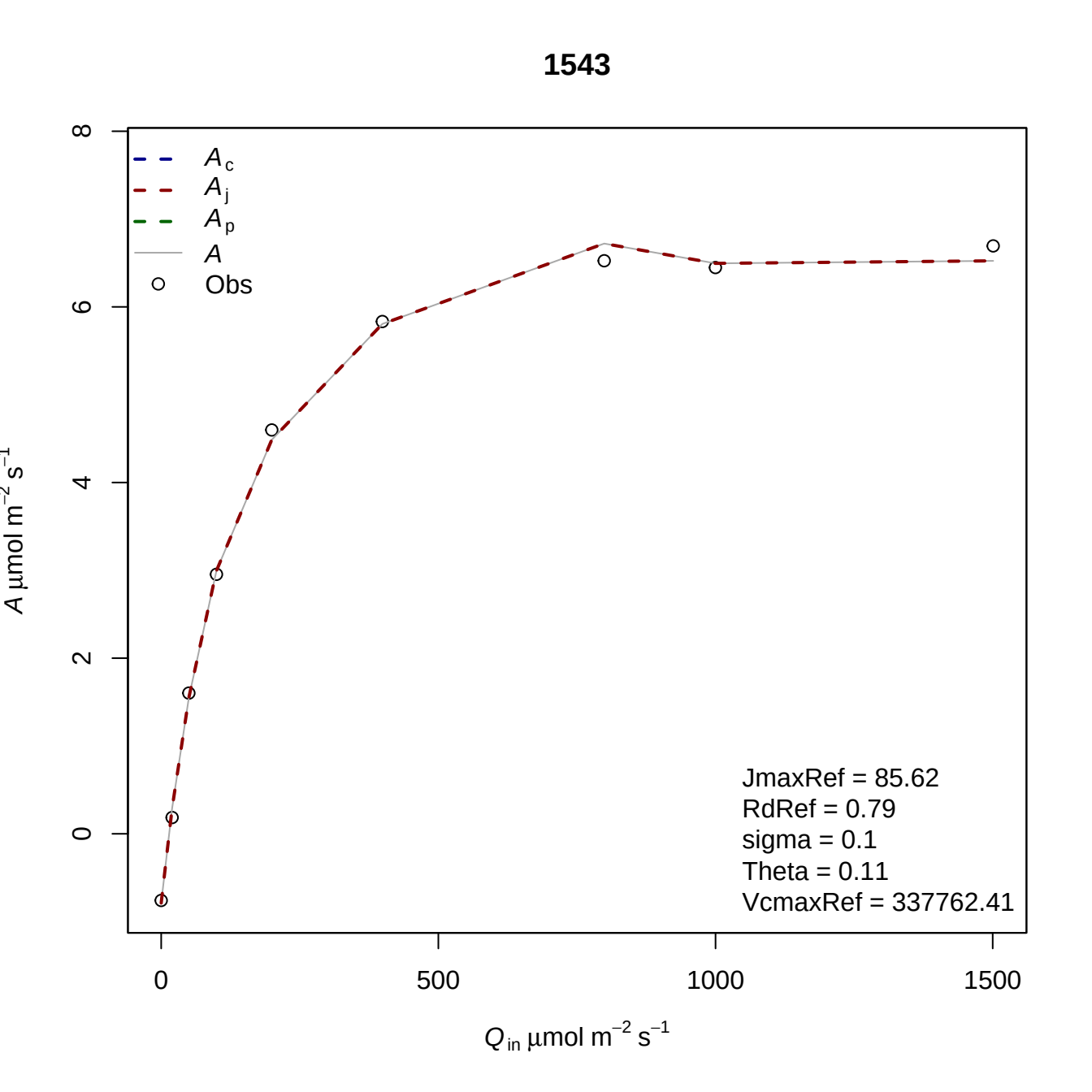

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 99.68
RdRef = 0.27
sigma = 0.18
Theta = 0.05
VcmaxRef = 66467.58
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

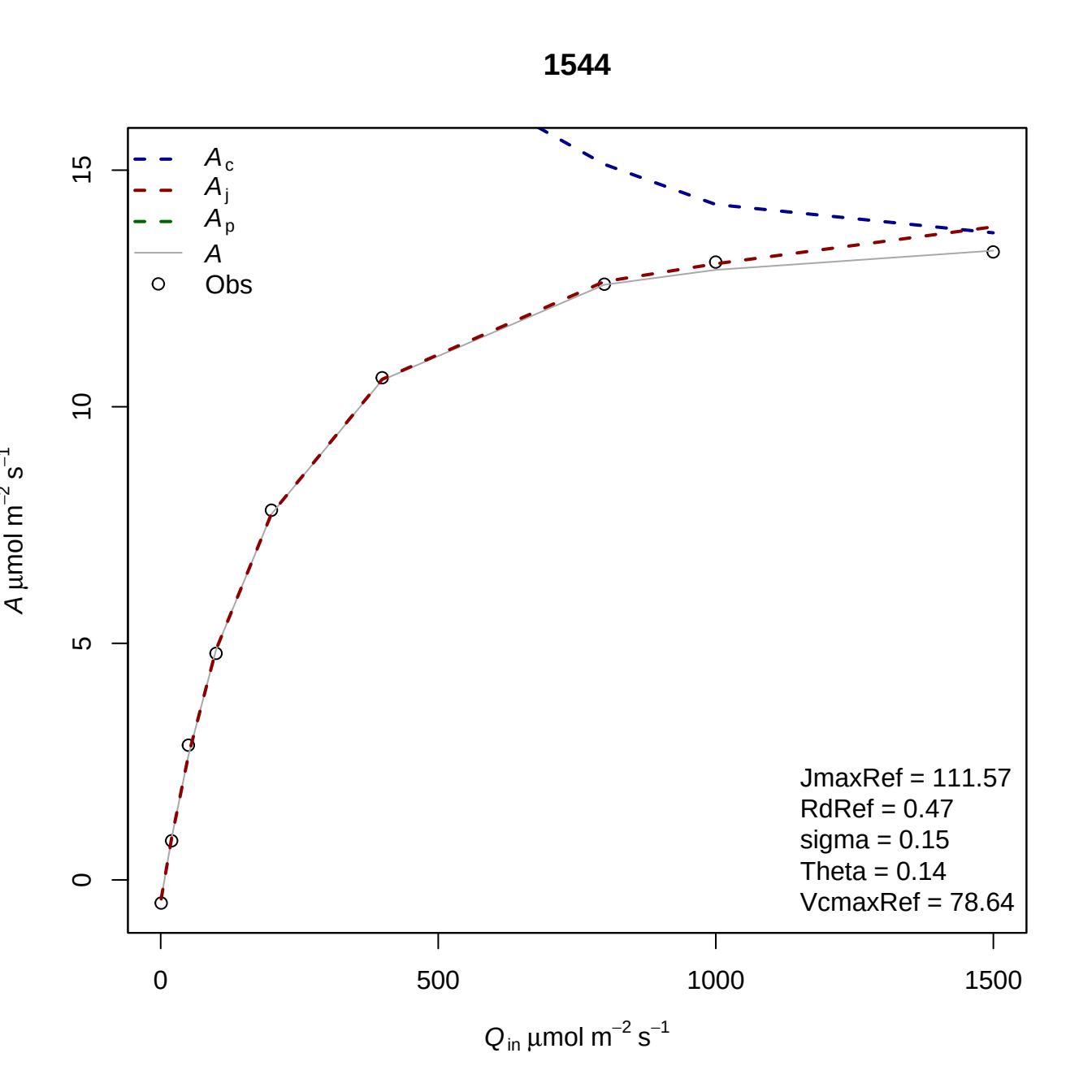

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 88.89
RdRef = 0.52
sigma = 0.08
Theta = 0.31
VcmaxRef = 3520.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

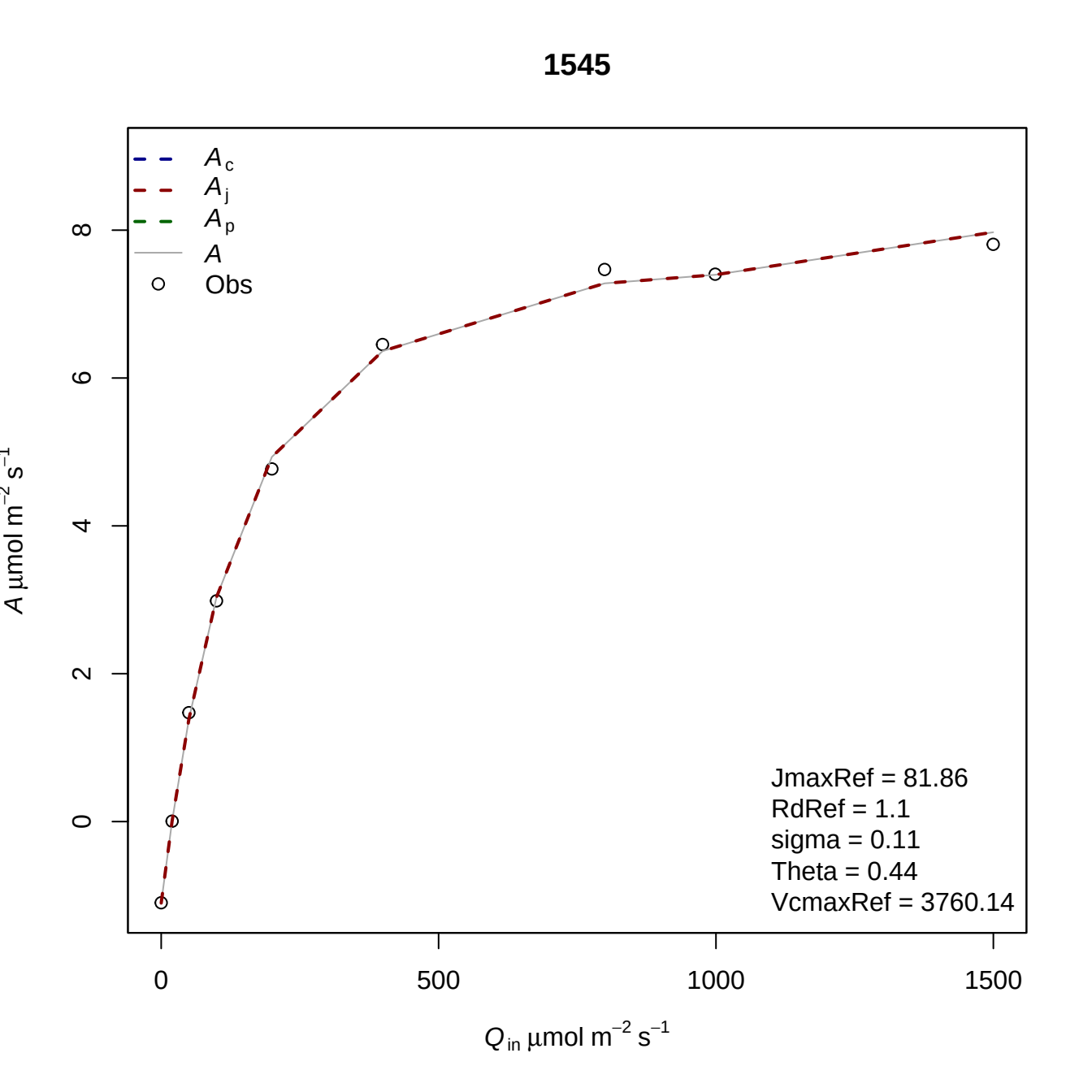

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 102.16
RdRef = 0.14
sigma = 0.04
Theta = 0.11
VcmaxRef = 58.4
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

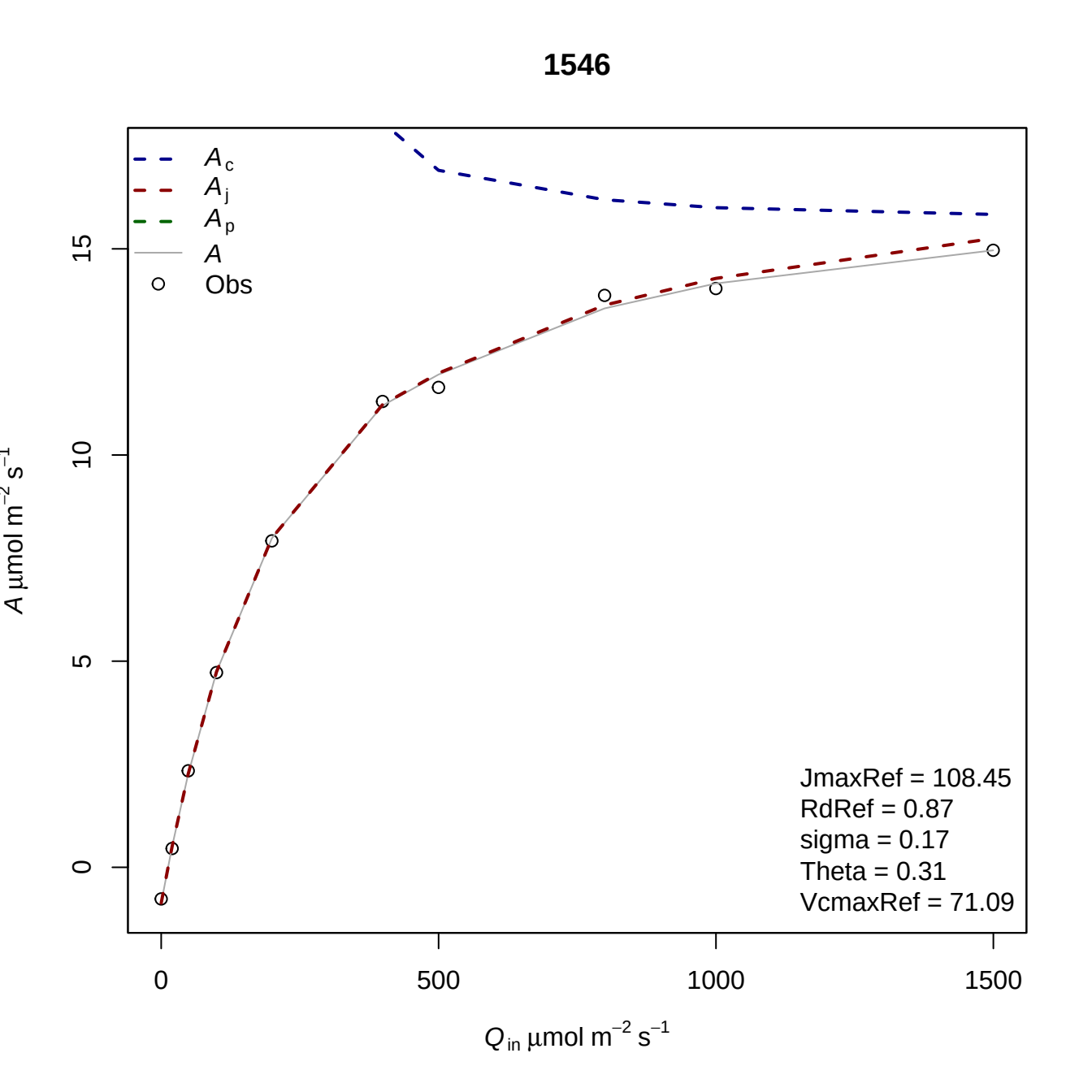

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 73.31
RdRef = 0.68
sigma = 0.13
Theta = 0.09
VcmaxRef = 60.89
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

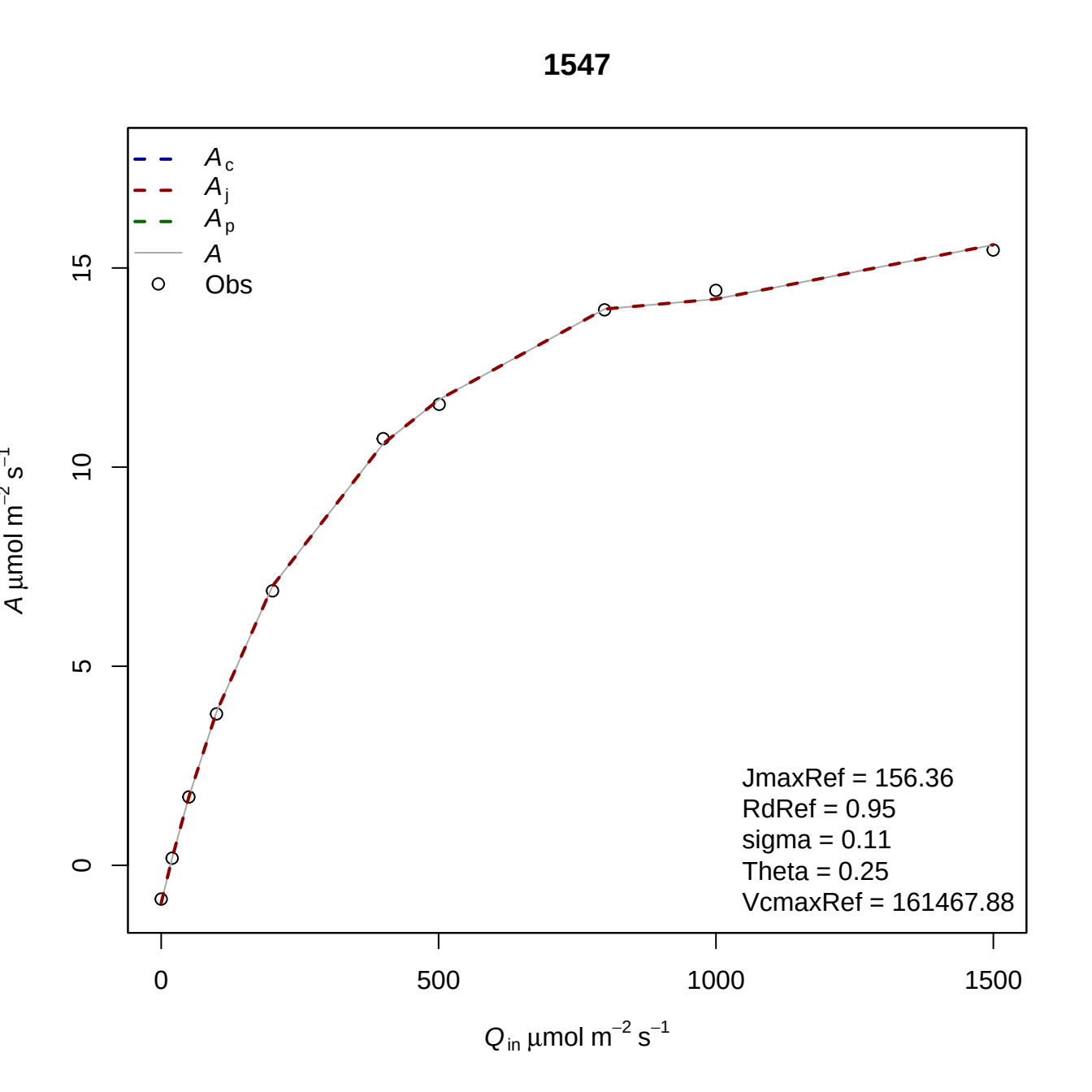

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 77.92
RdRef = 0.39
sigma = 0.08
Theta = −0.1
VcmaxRef = 46.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

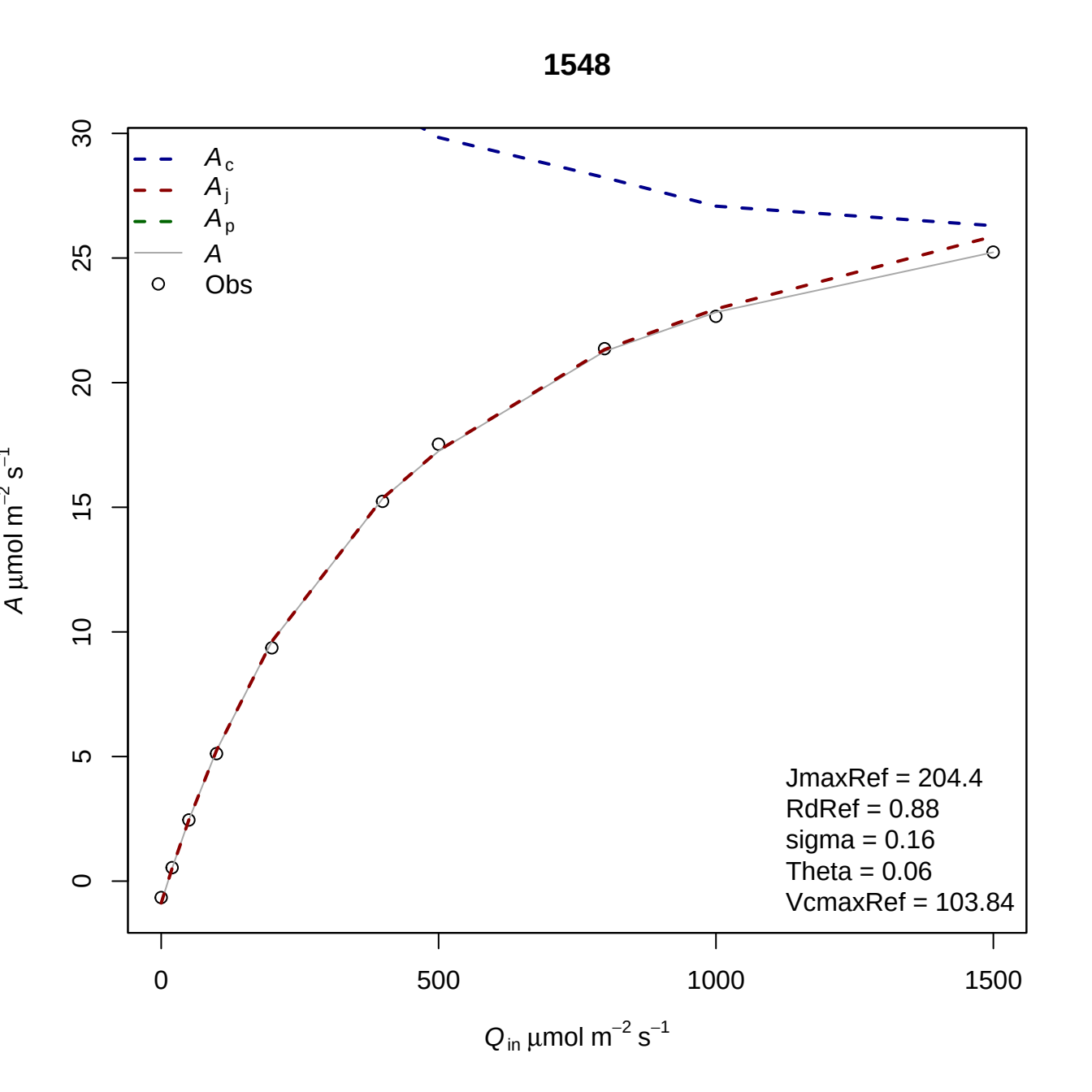

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 85.62
RdRef = 0.79
sigma = 0.1
Theta = 0.11
VcmaxRef = 337762.41
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

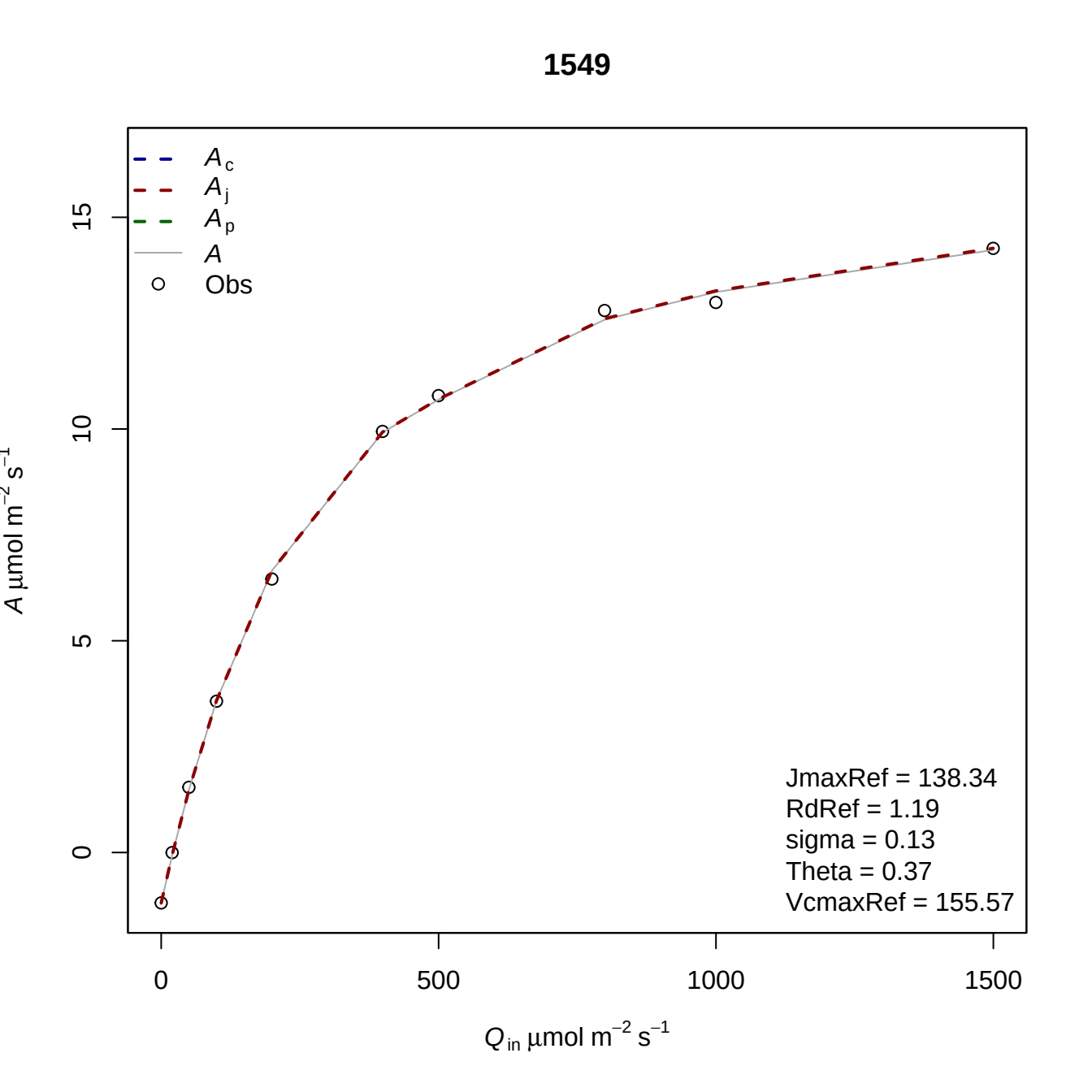

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 111.57
RdRef = 0.47
sigma = 0.15
Theta = 0.14
VcmaxRef = 78.64
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

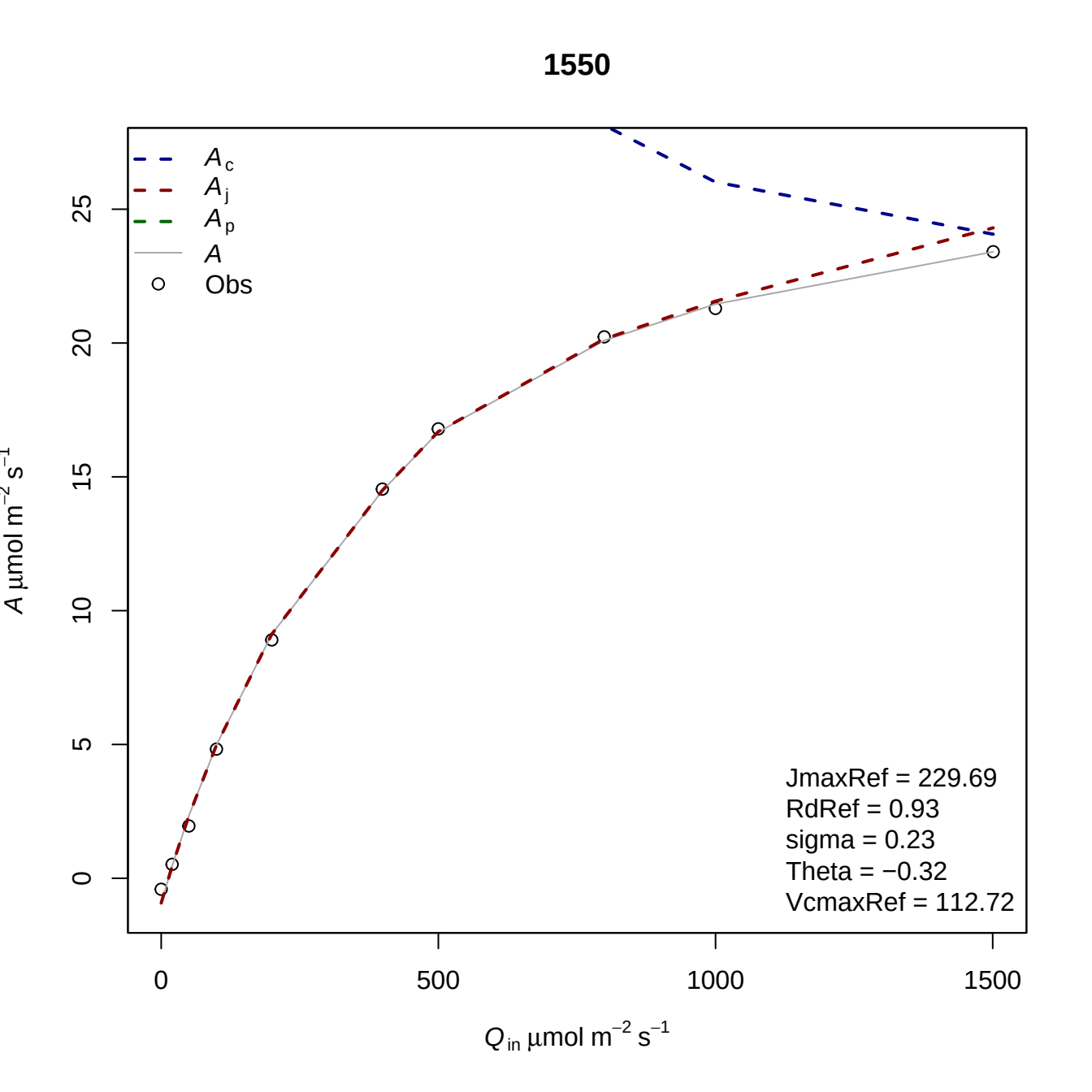

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 81.86
RdRef = 1.1
sigma = 0.11
Theta = 0.44
VcmaxRef = 3760.14
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

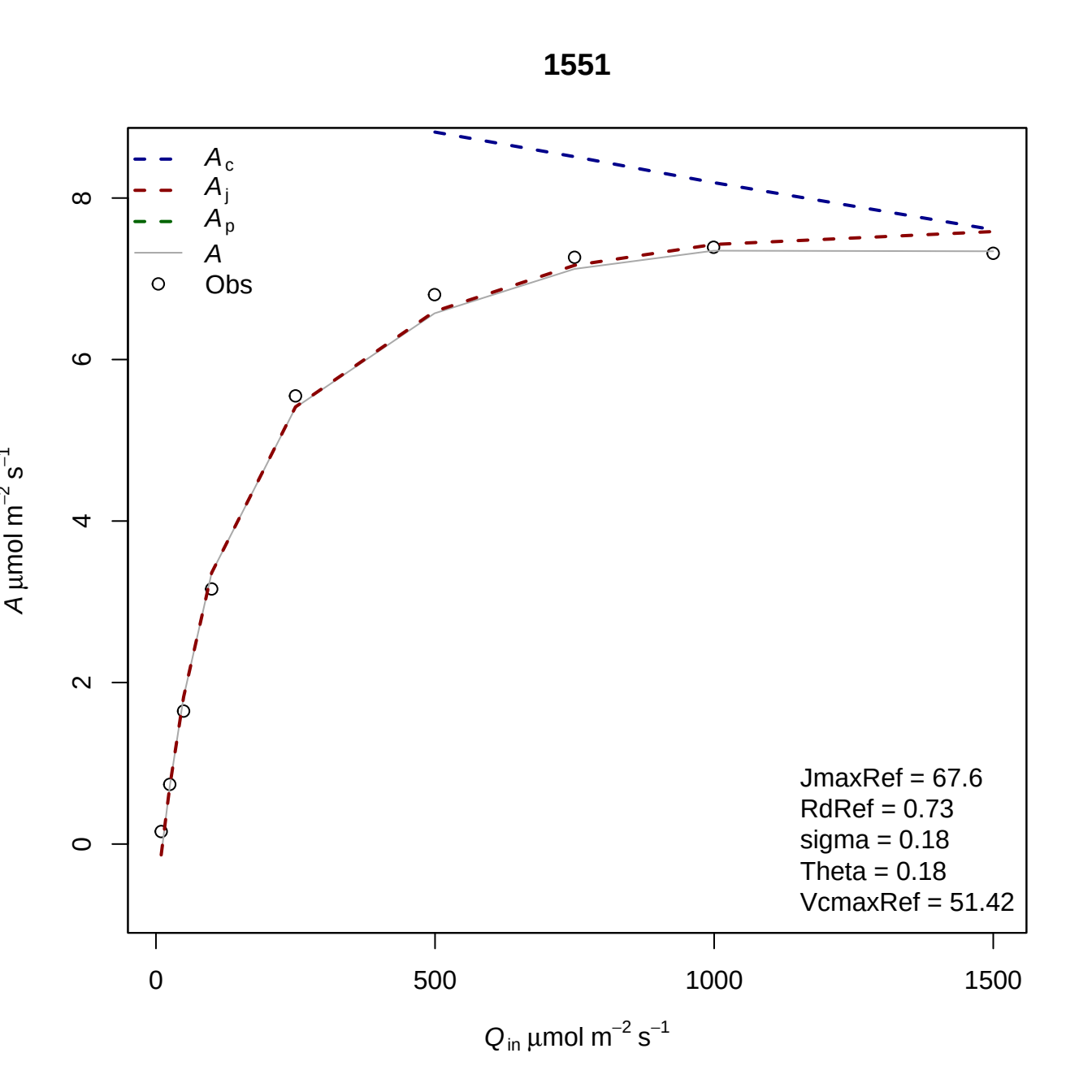

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 108.45
RdRef = 0.87
sigma = 0.17
Theta = 0.31
VcmaxRef = 71.09
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

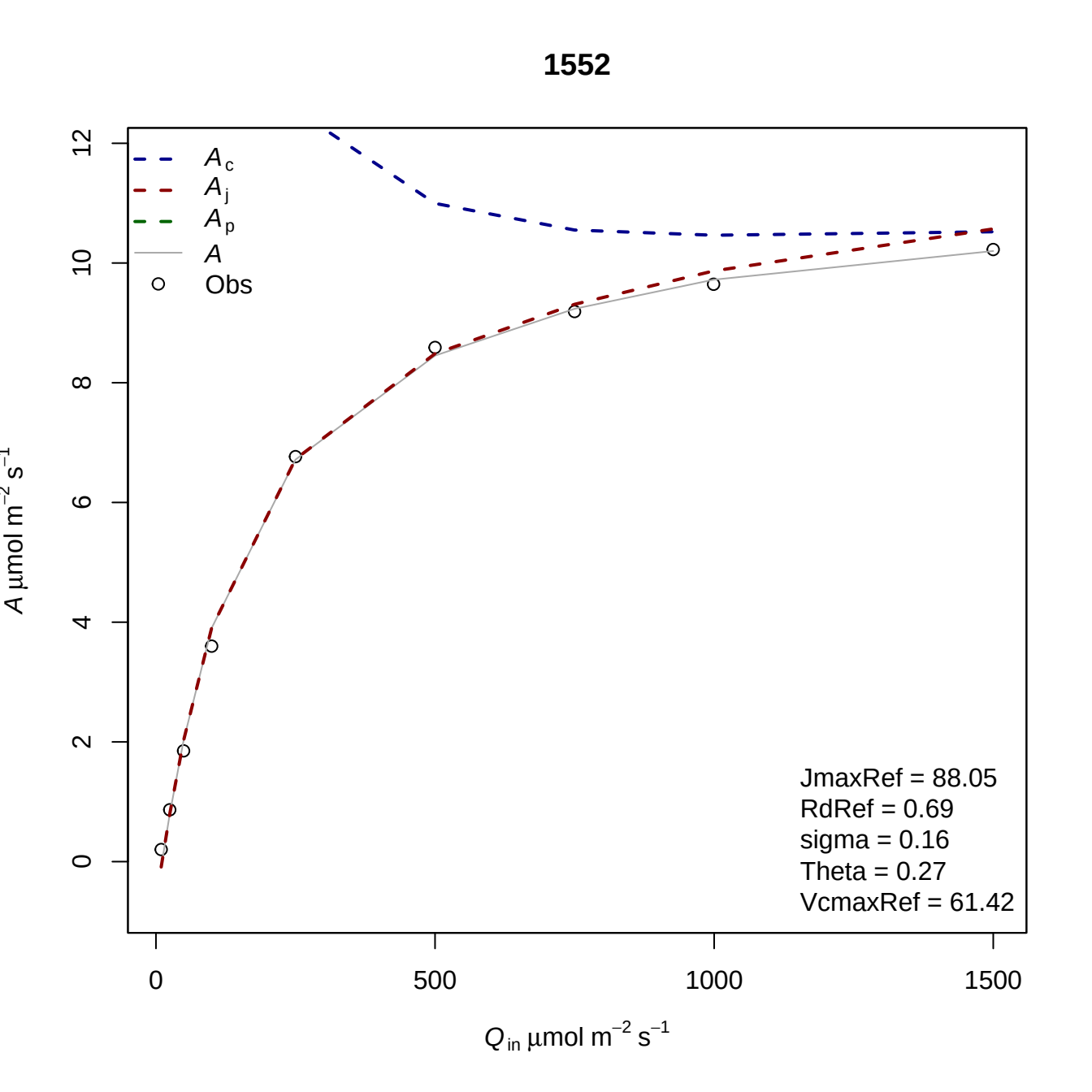

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 156.36
RdRef = 0.95
sigma = 0.11
Theta = 0.25
VcmaxRef = 161467.88
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

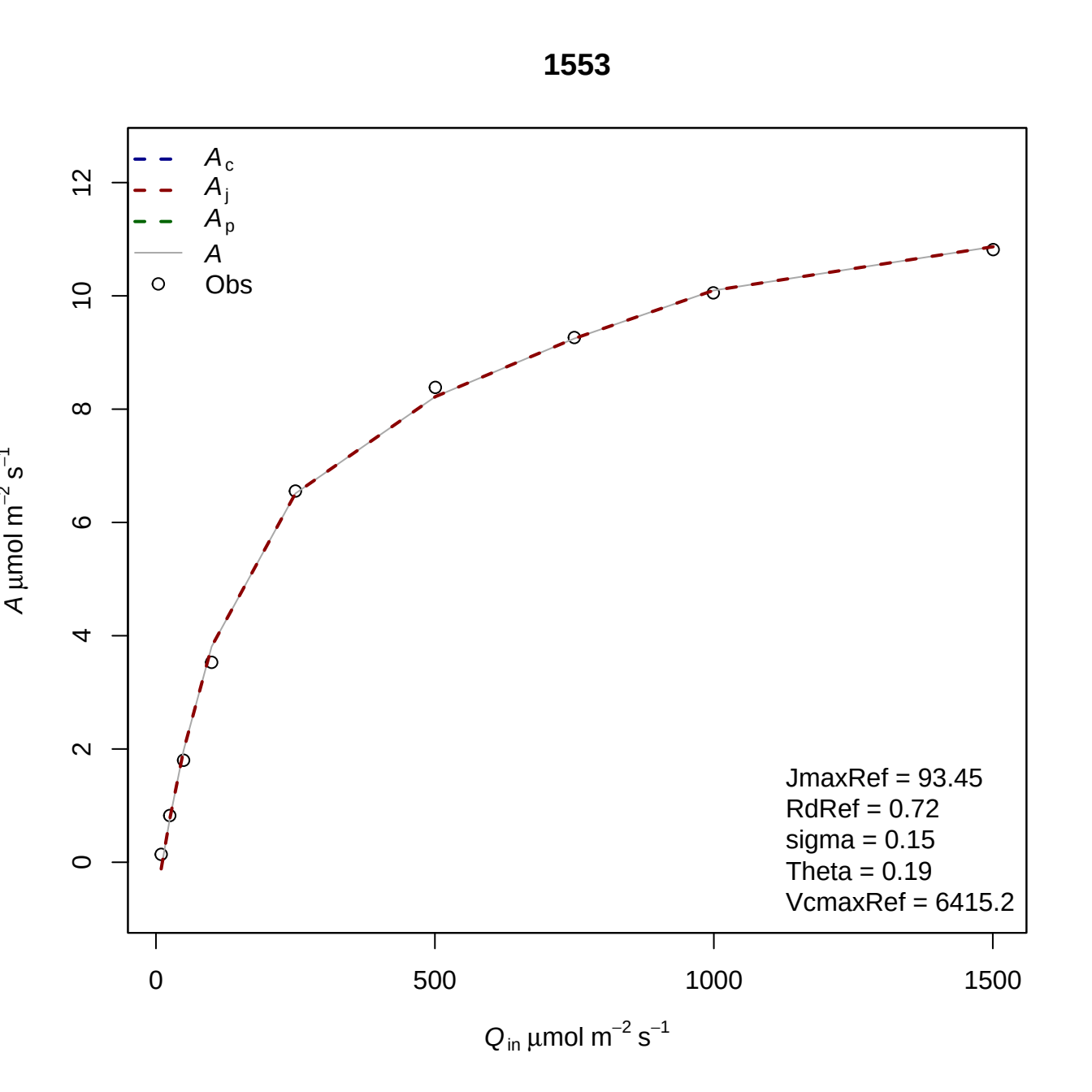

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 204.4
RdRef = 0.88
sigma = 0.16
Theta = 0.06
VcmaxRef = 103.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

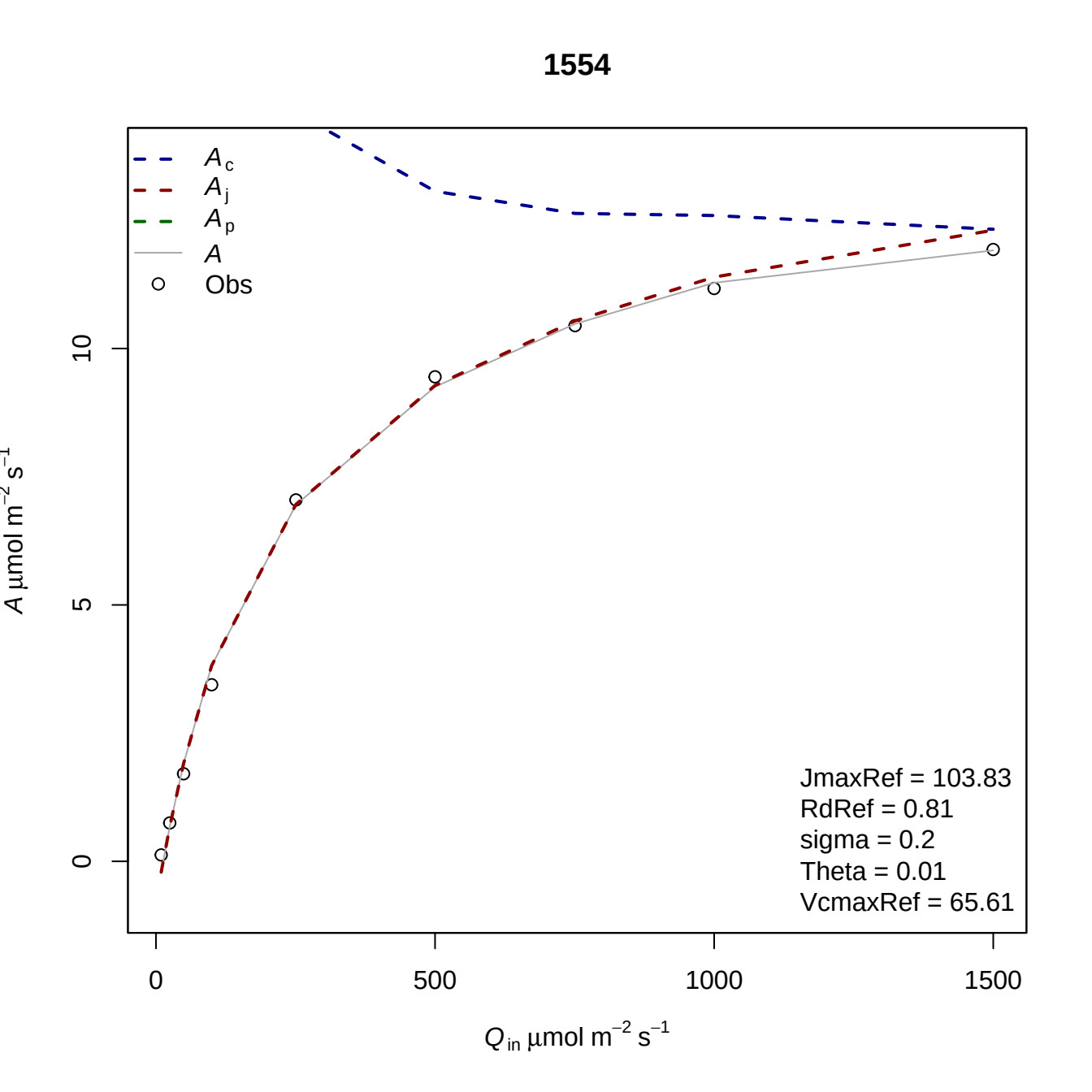

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 138.34
RdRef = 1.19
sigma = 0.13
Theta = 0.37
VcmaxRef = 155.57
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

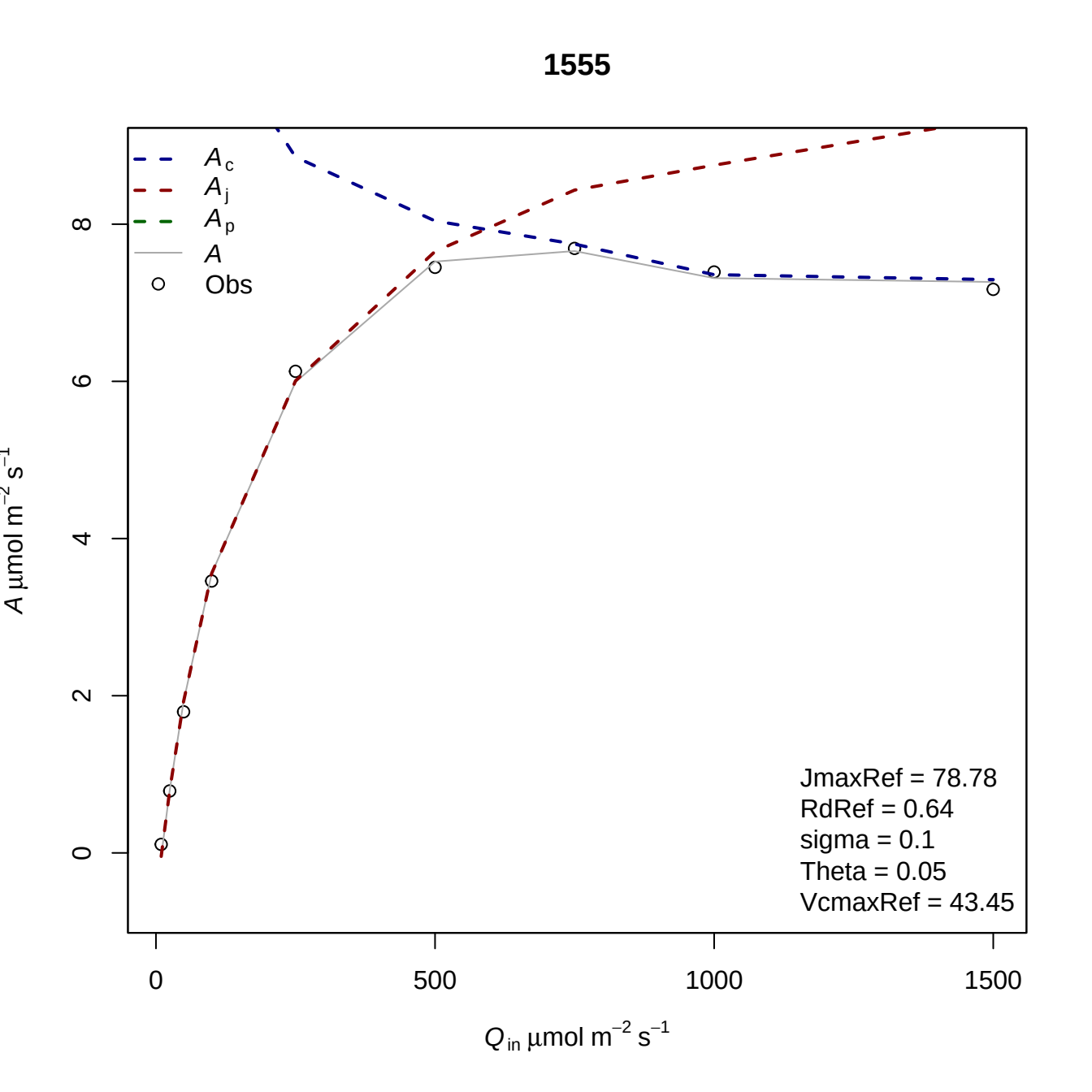

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 229.69
RdRef = 0.93
sigma = 0.23
Theta = −0.32
VcmaxRef = 112.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

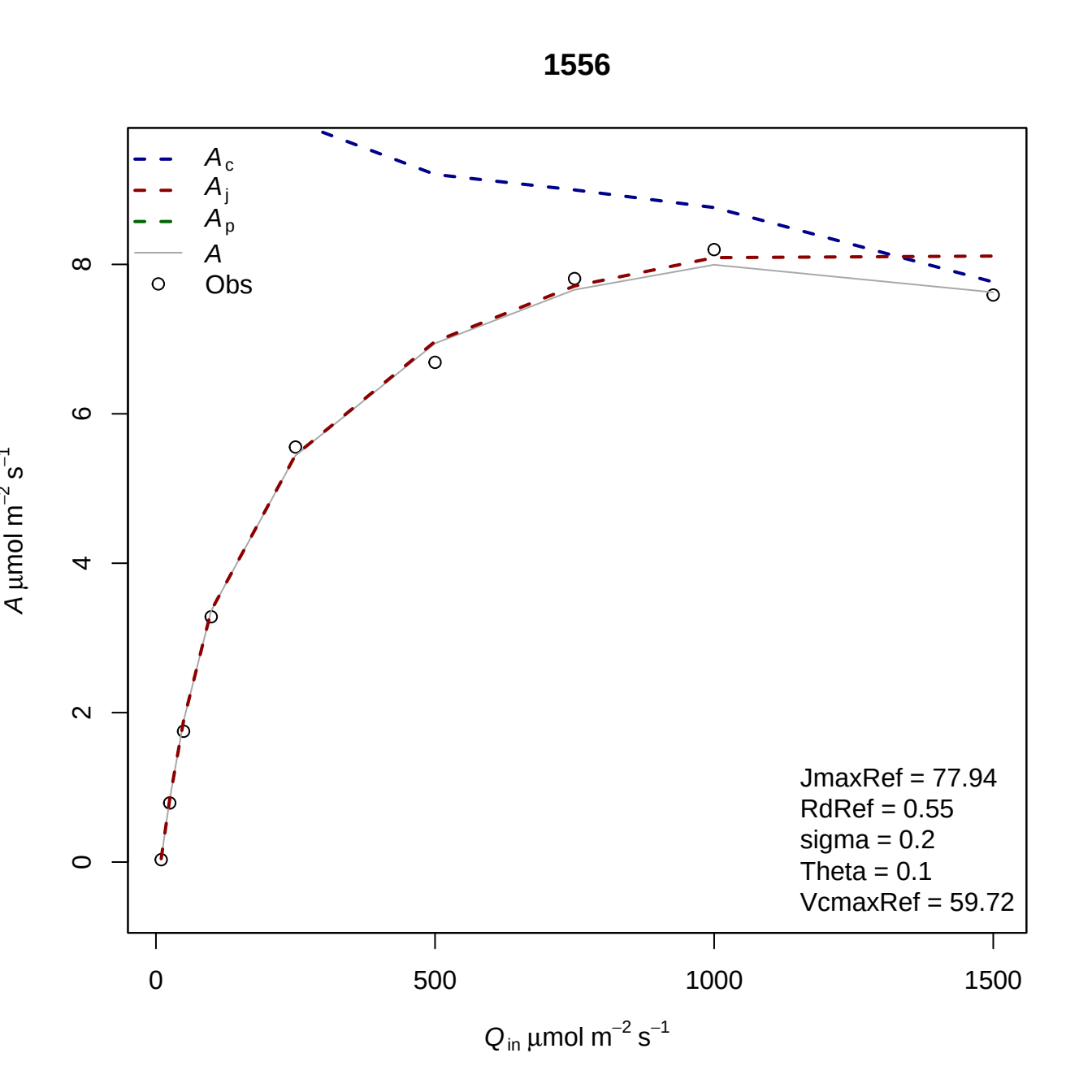

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 67.6
RdRef = 0.73
sigma = 0.18
Theta = 0.18
VcmaxRef = 51.42
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

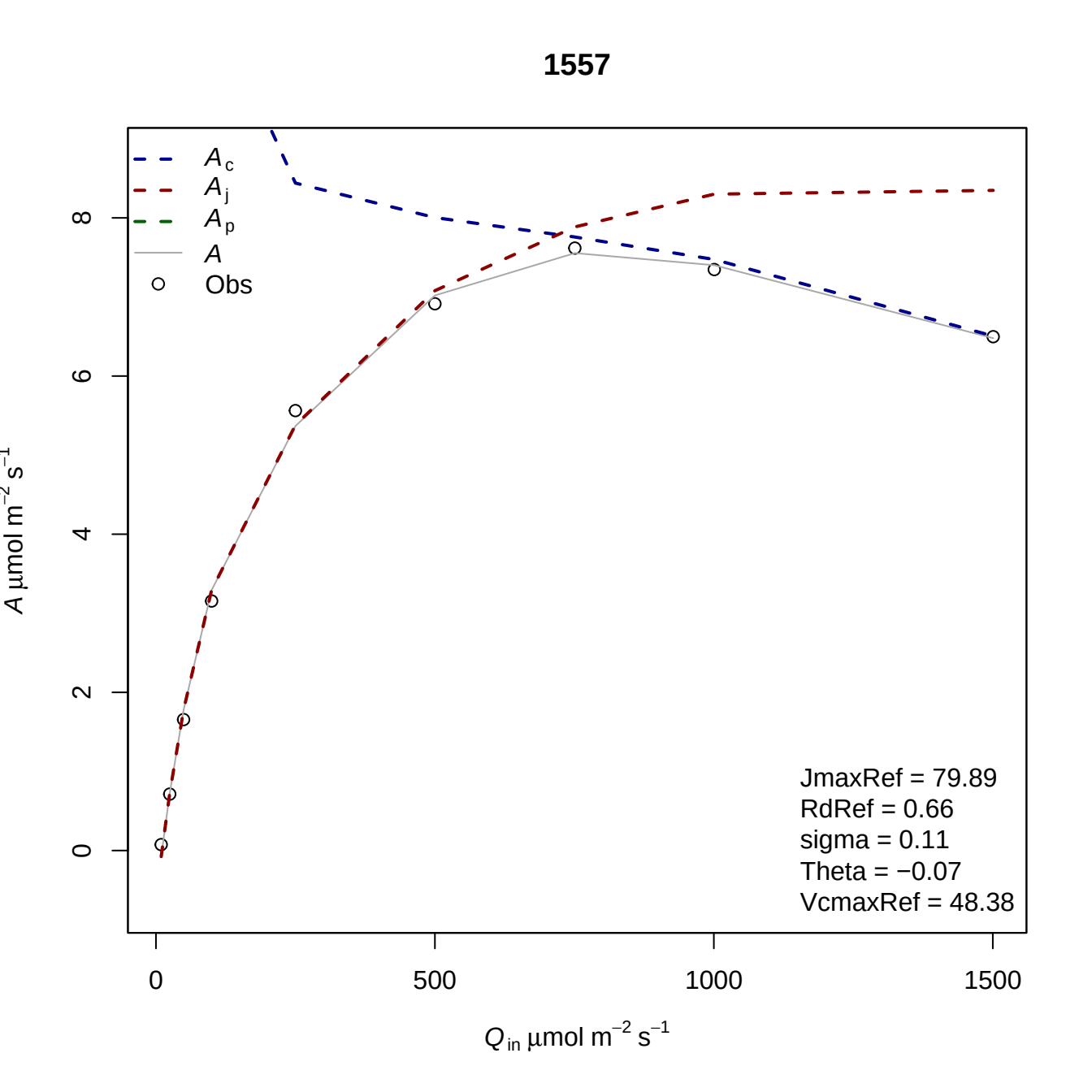

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 88.05
RdRef = 0.69
sigma = 0.16
Theta = 0.27
VcmaxRef = 61.42
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

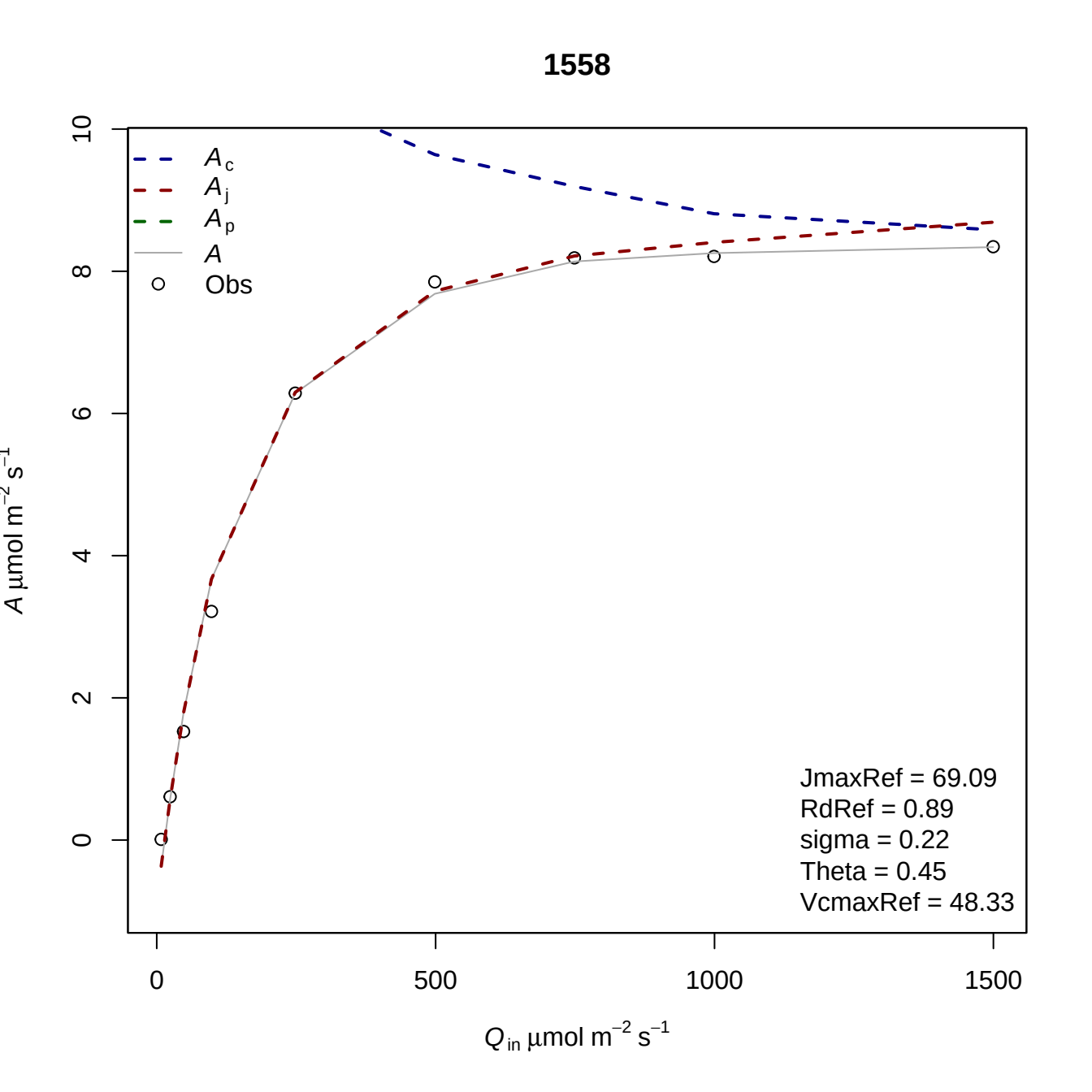

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 93.45
RdRef = 0.72
sigma = 0.15
Theta = 0.19
VcmaxRef = 6415.2
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

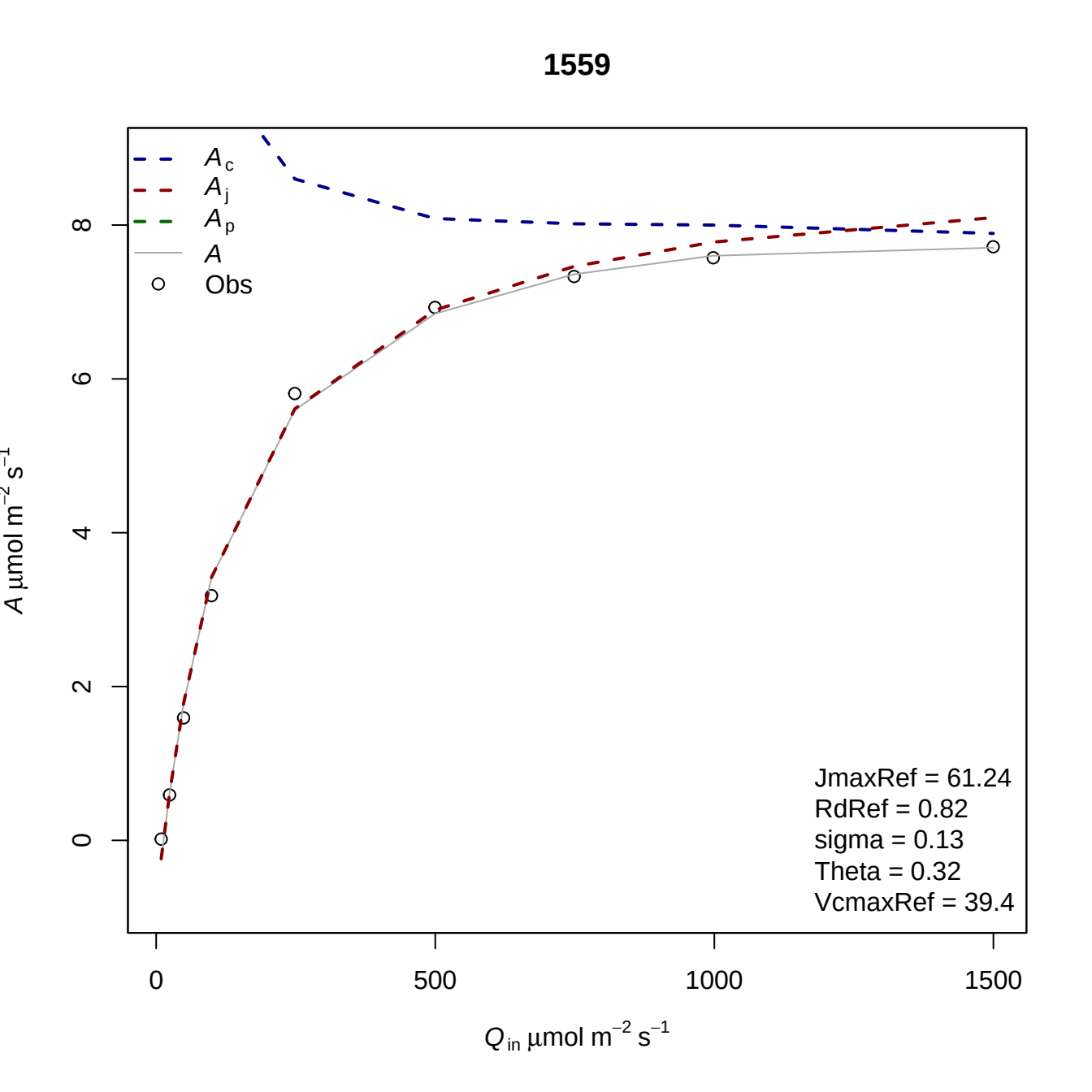

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 103.83
RdRef = 0.81
sigma = 0.2
Theta = 0.01
VcmaxRef = 65.61
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

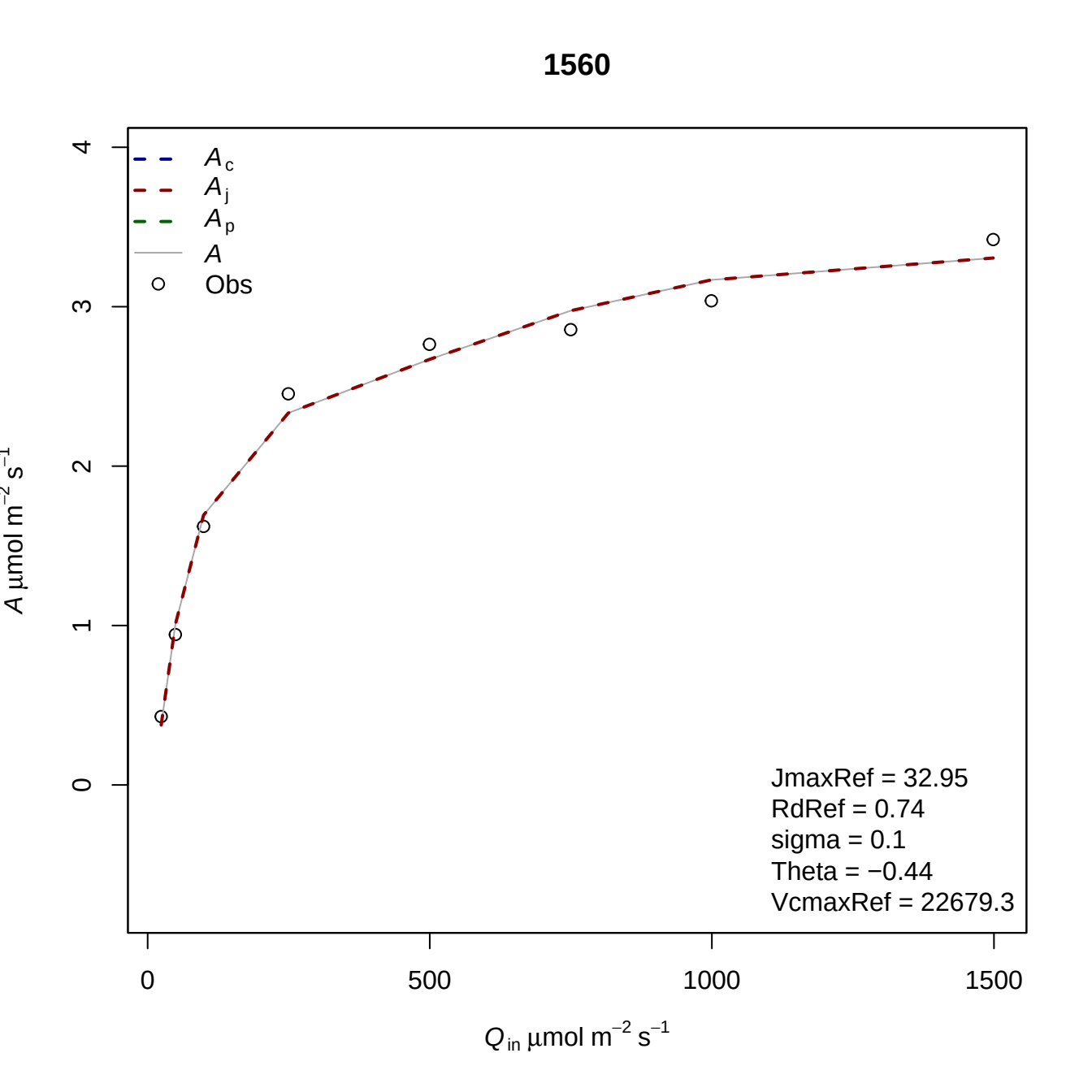

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 78.78
RdRef = 0.64
sigma = 0.1
Theta = 0.05
VcmaxRef = 43.45
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

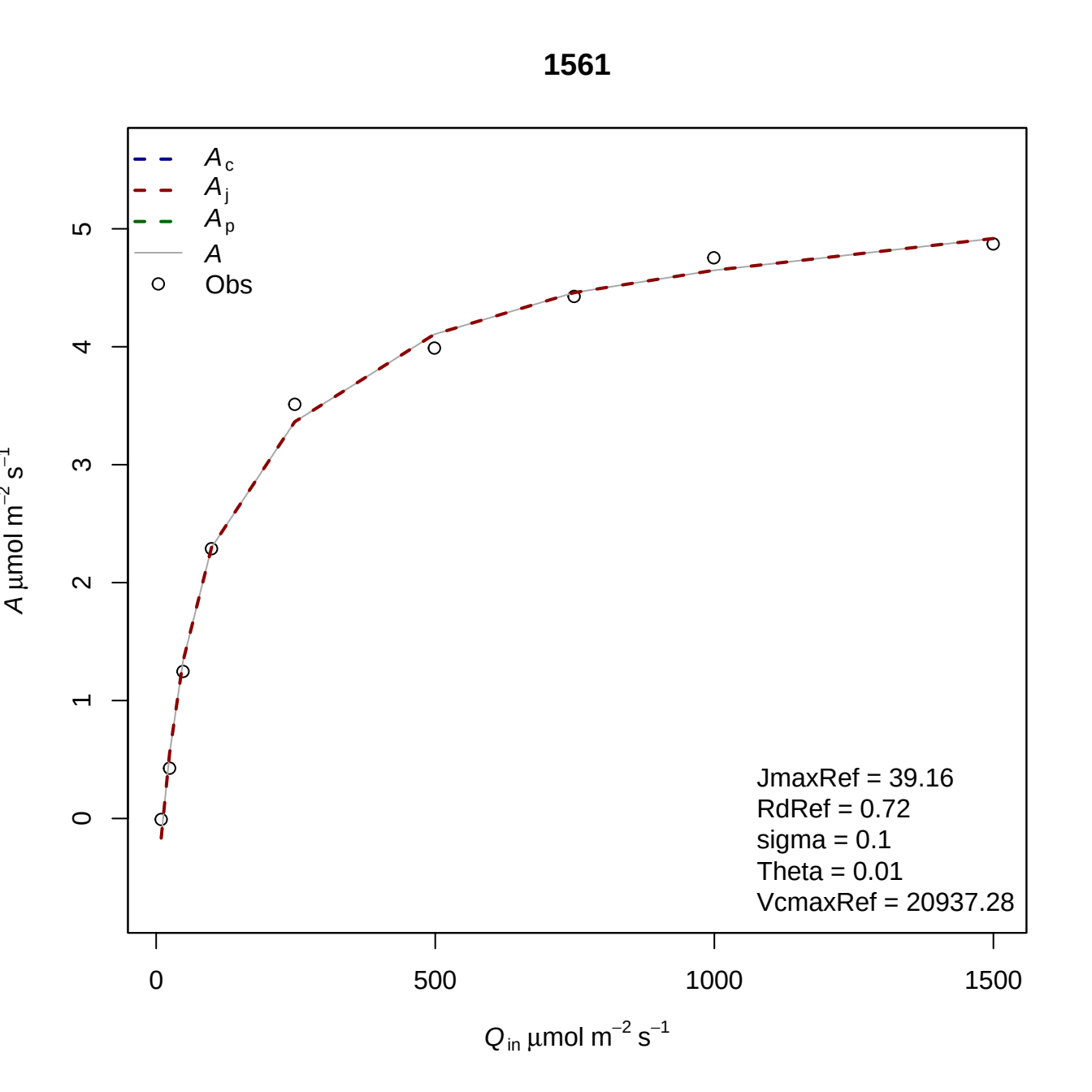

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 77.94
RdRef = 0.55
sigma = 0.2
Theta = 0.1
VcmaxRef = 59.72
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

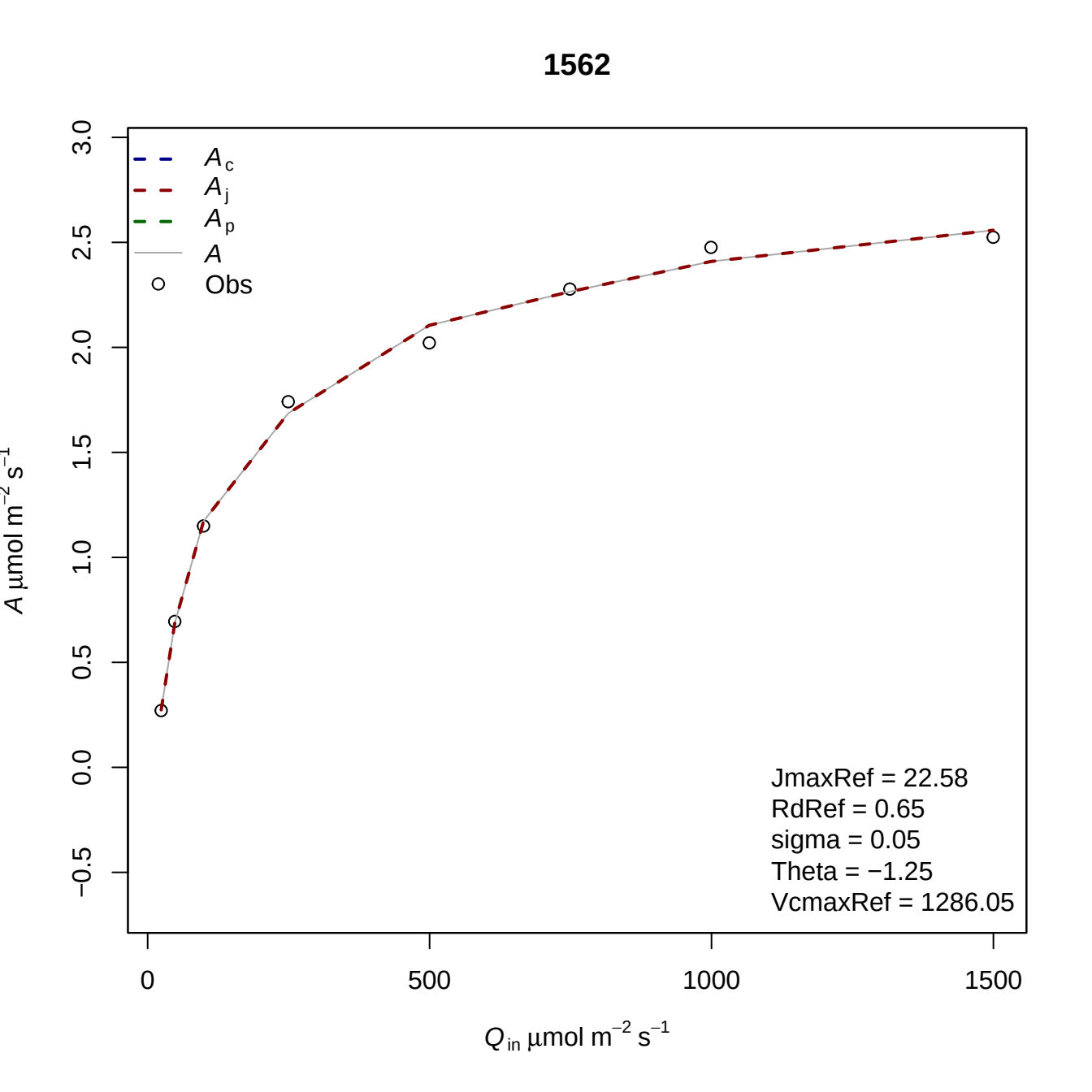

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 79.89
RdRef = 0.66
sigma = 0.11
Theta = −0.07
VcmaxRef = 48.38
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

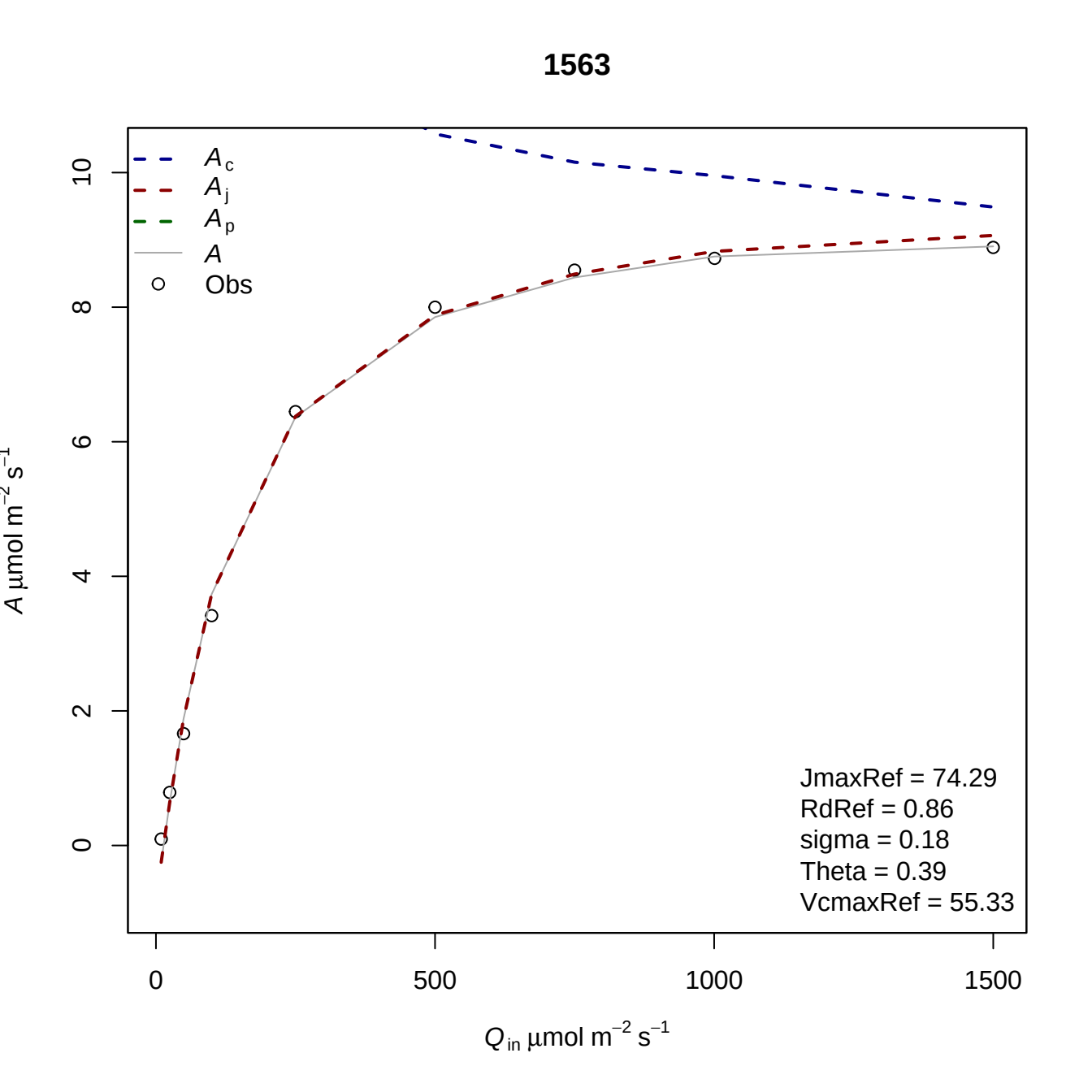

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 69.09
RdRef = 0.89
sigma = 0.22
Theta = 0.45
VcmaxRef = 48.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

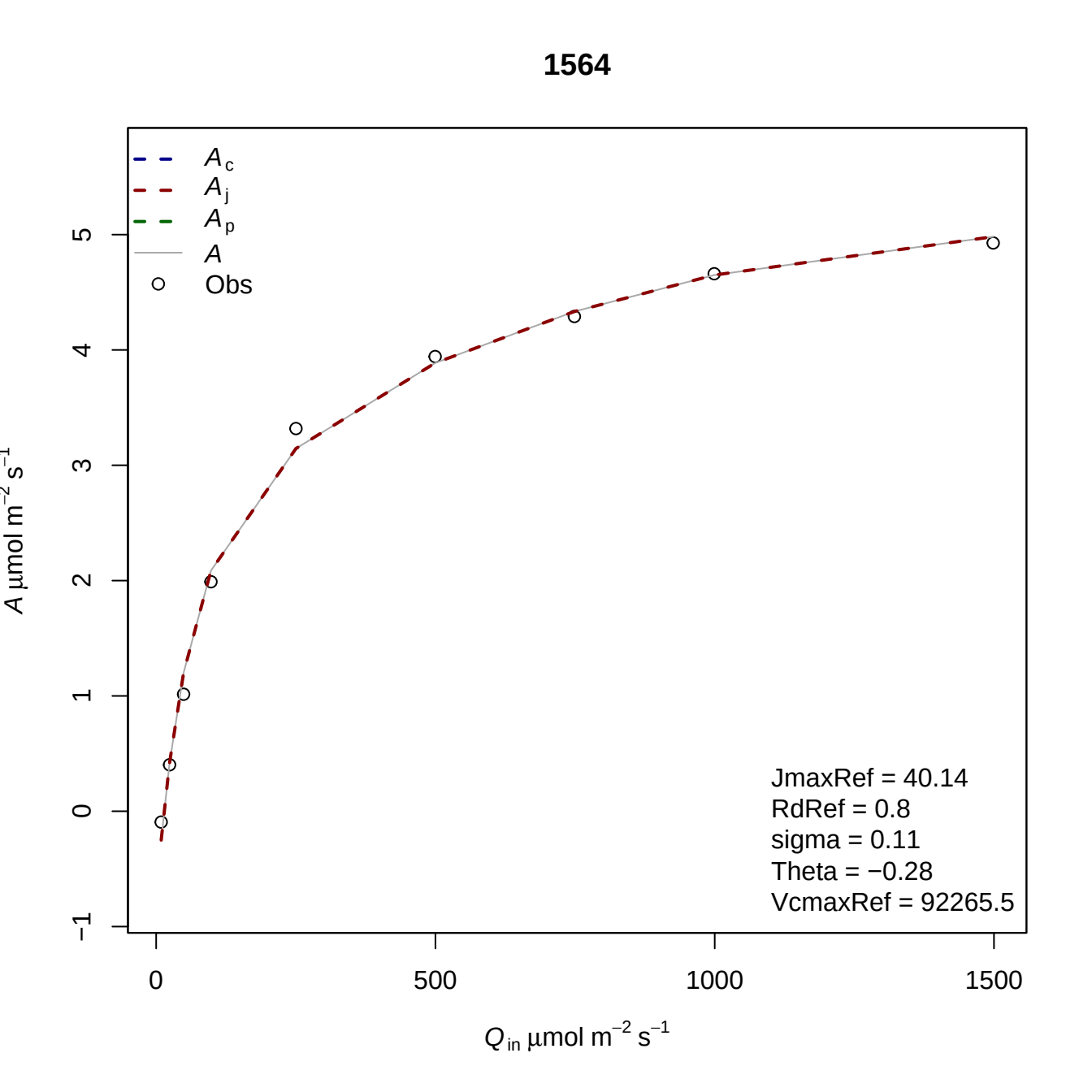

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 61.24
RdRef = 0.82
sigma = 0.13
Theta = 0.32
VcmaxRef = 39.4
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

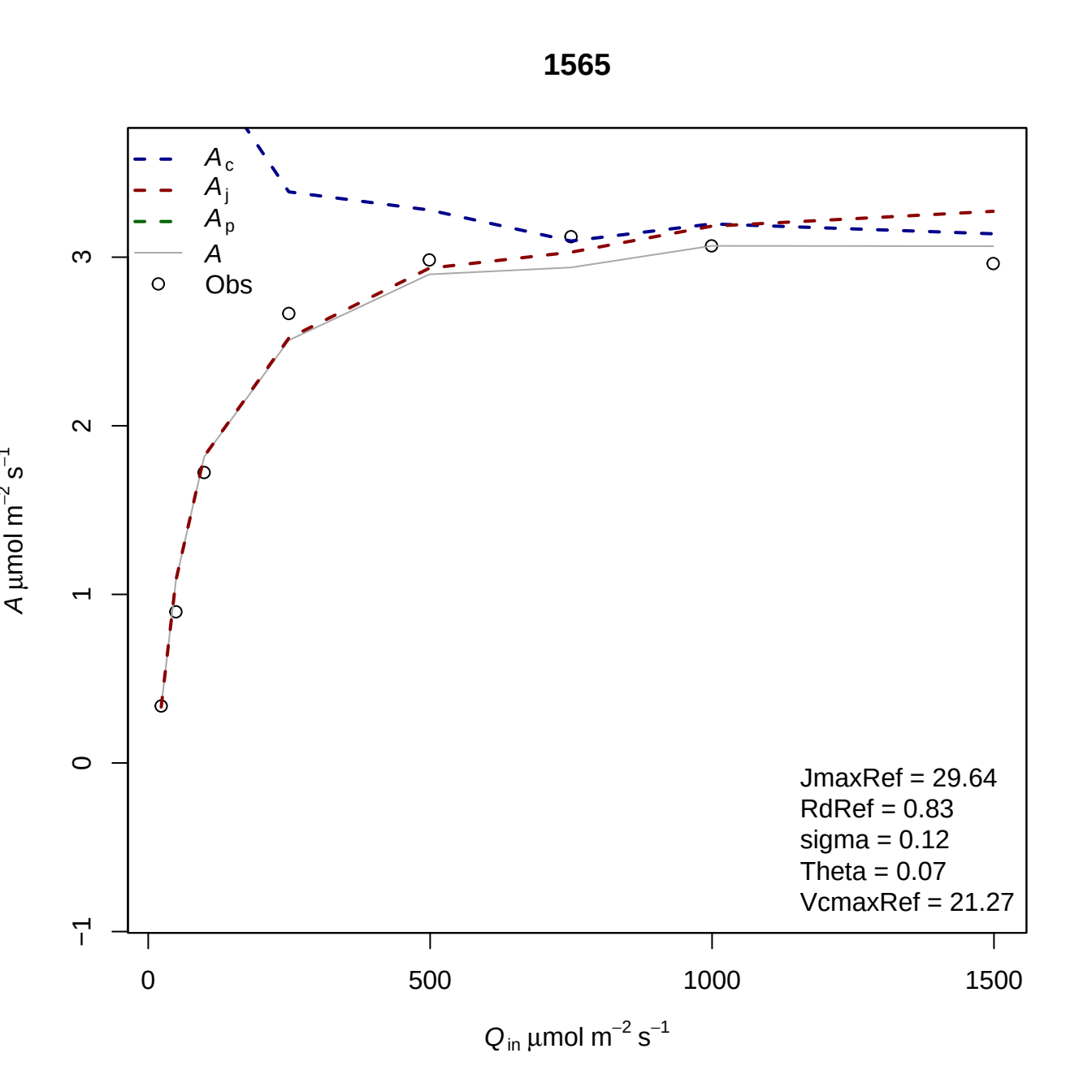

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 32.95
RdRef = 0.74
sigma = 0.1
Theta = −0.44
VcmaxRef = 22679.3
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

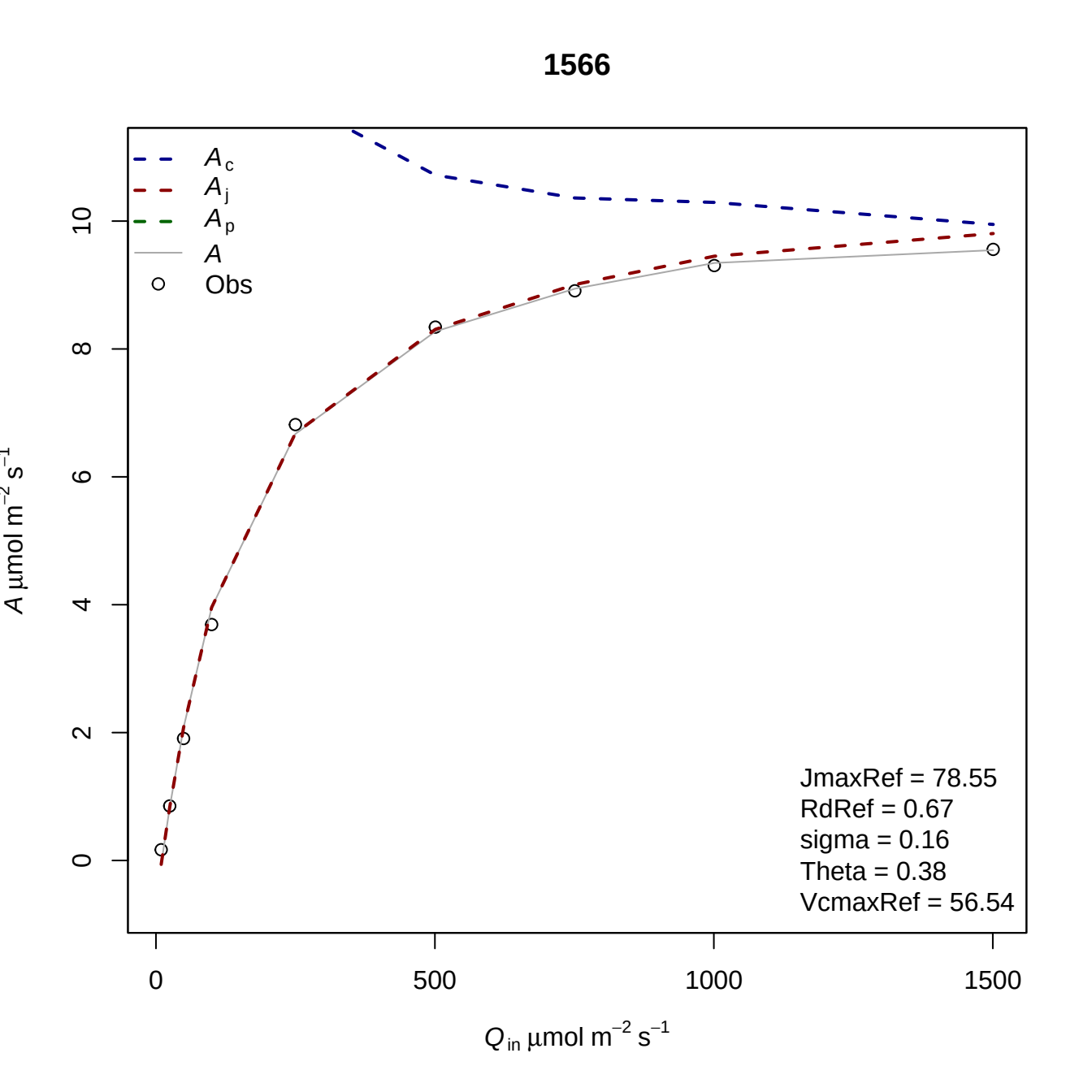

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 39.16
RdRef = 0.72
sigma = 0.1
Theta = 0.01
VcmaxRef = 20937.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

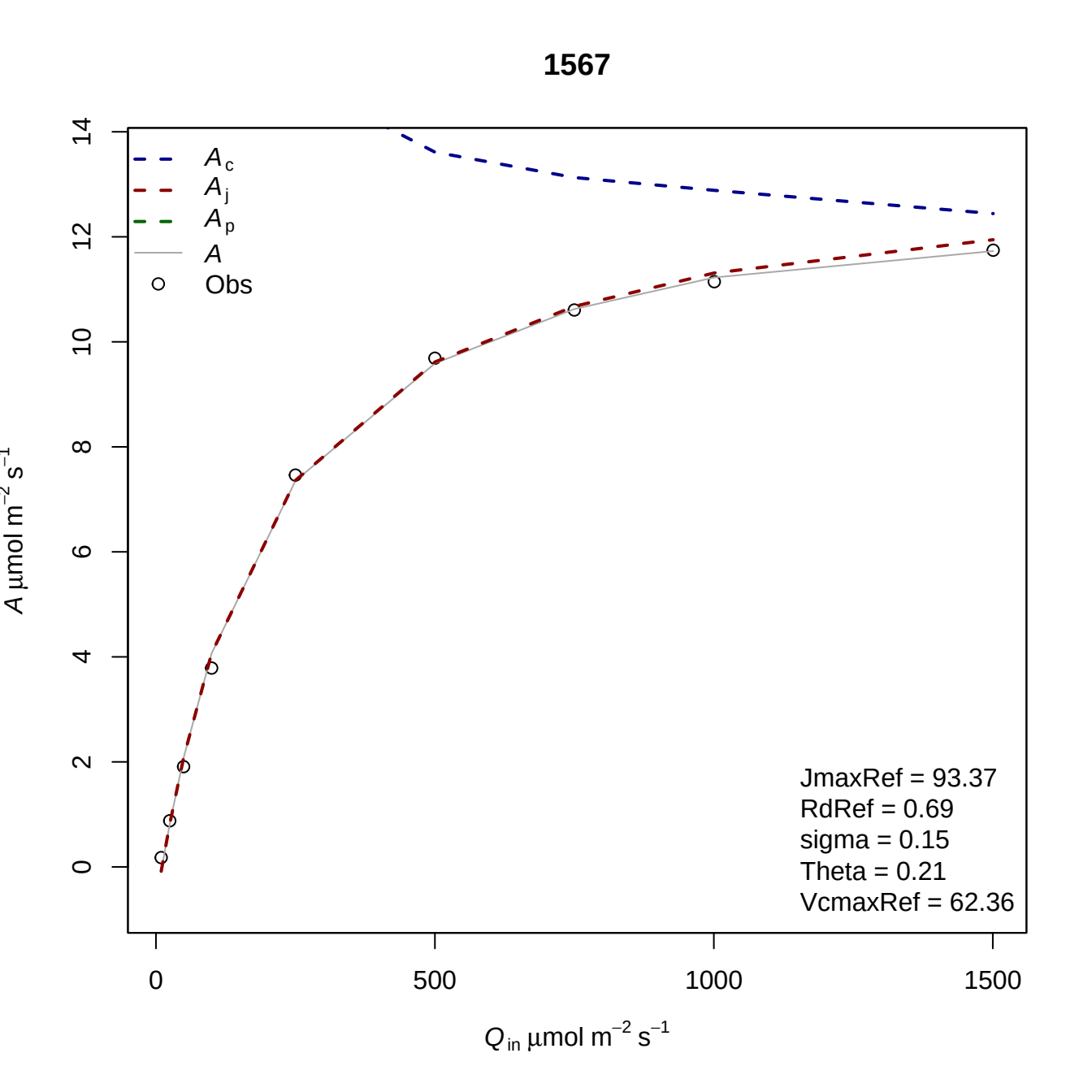

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 22.58
RdRef = 0.65
sigma = 0.05
Theta = −1.25
VcmaxRef = 1286.05
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

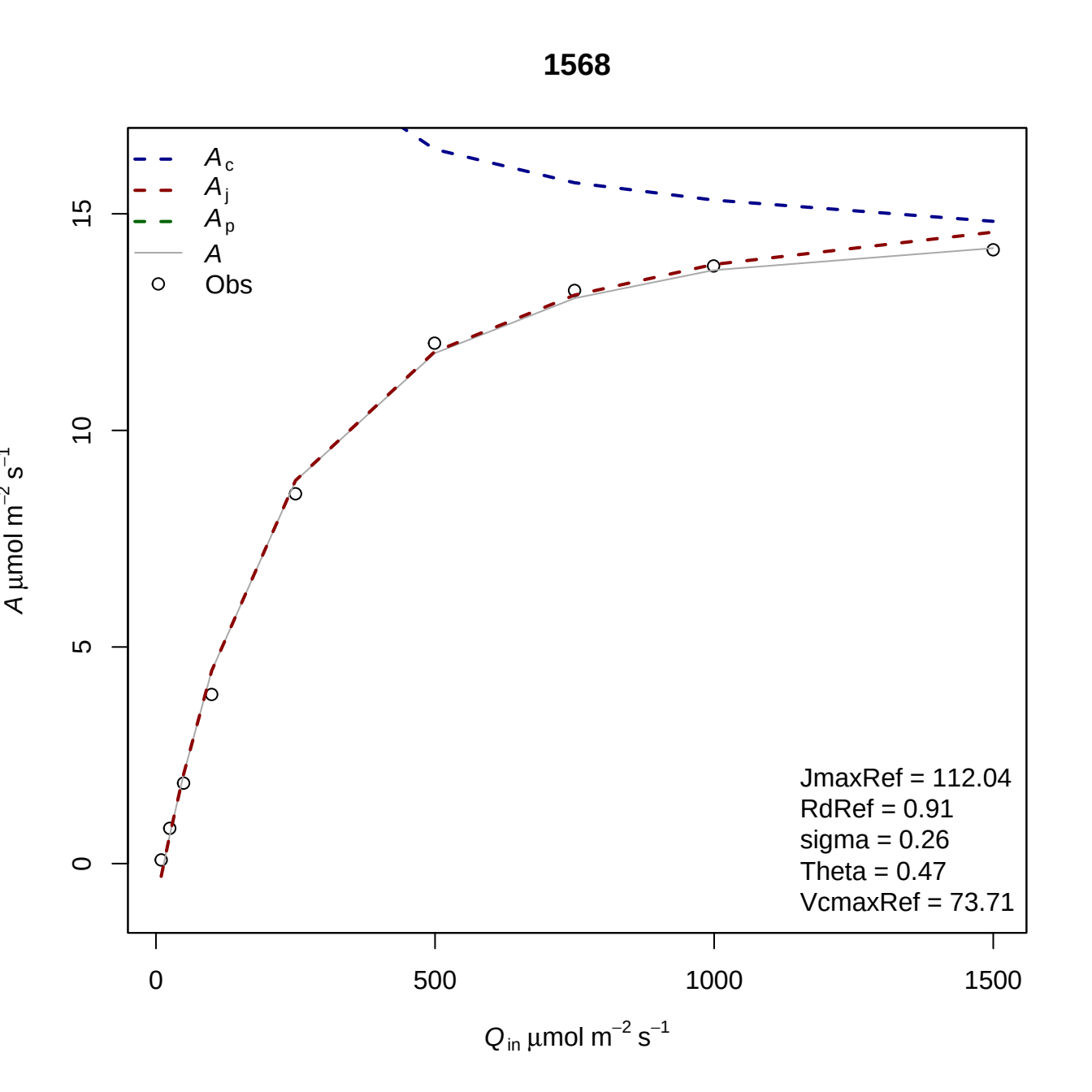

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 74.29
RdRef = 0.86
sigma = 0.18
Theta = 0.39
VcmaxRef = 55.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

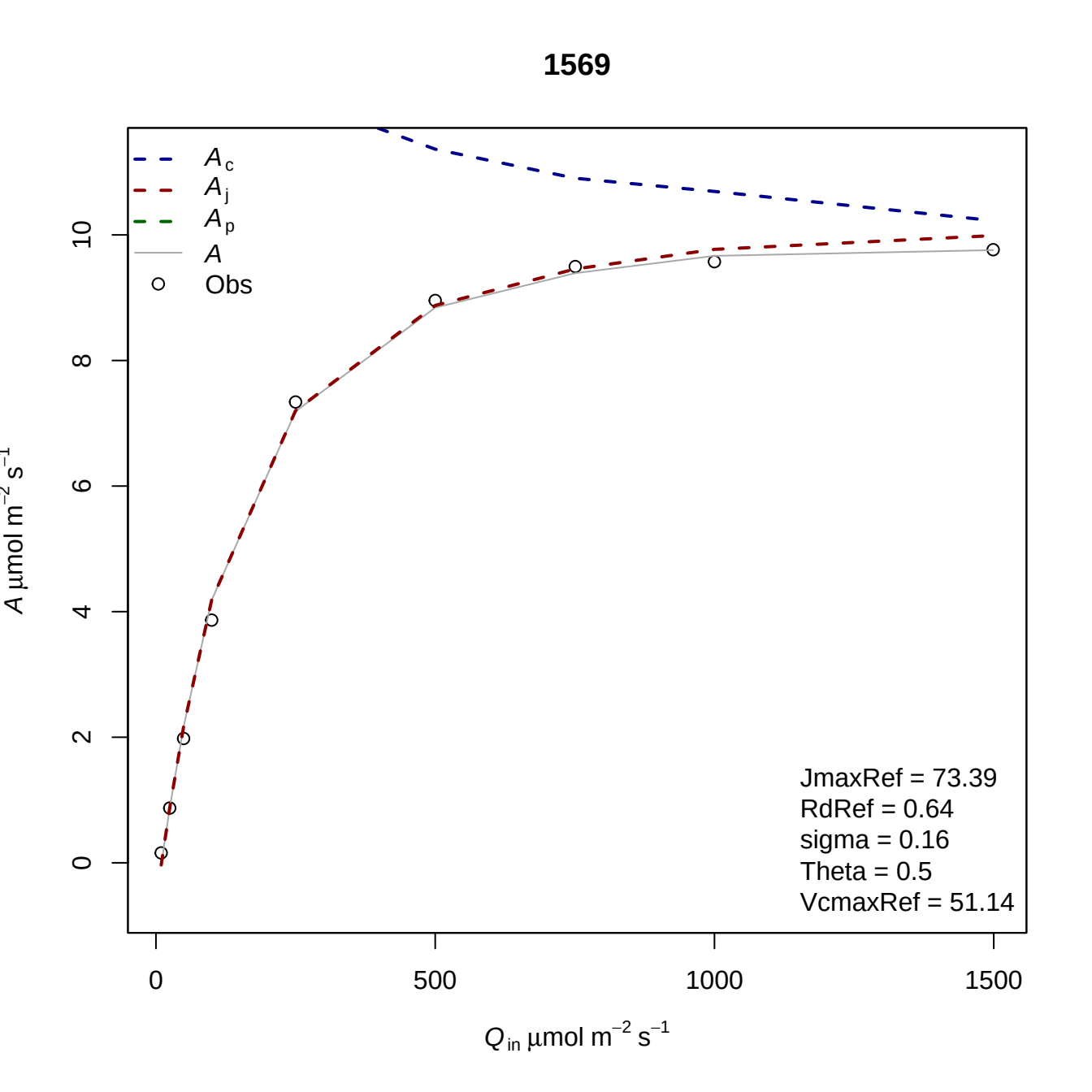

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 40.14
RdRef = 0.8
sigma = 0.11
Theta = −0.28
VcmaxRef = 92265.5
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

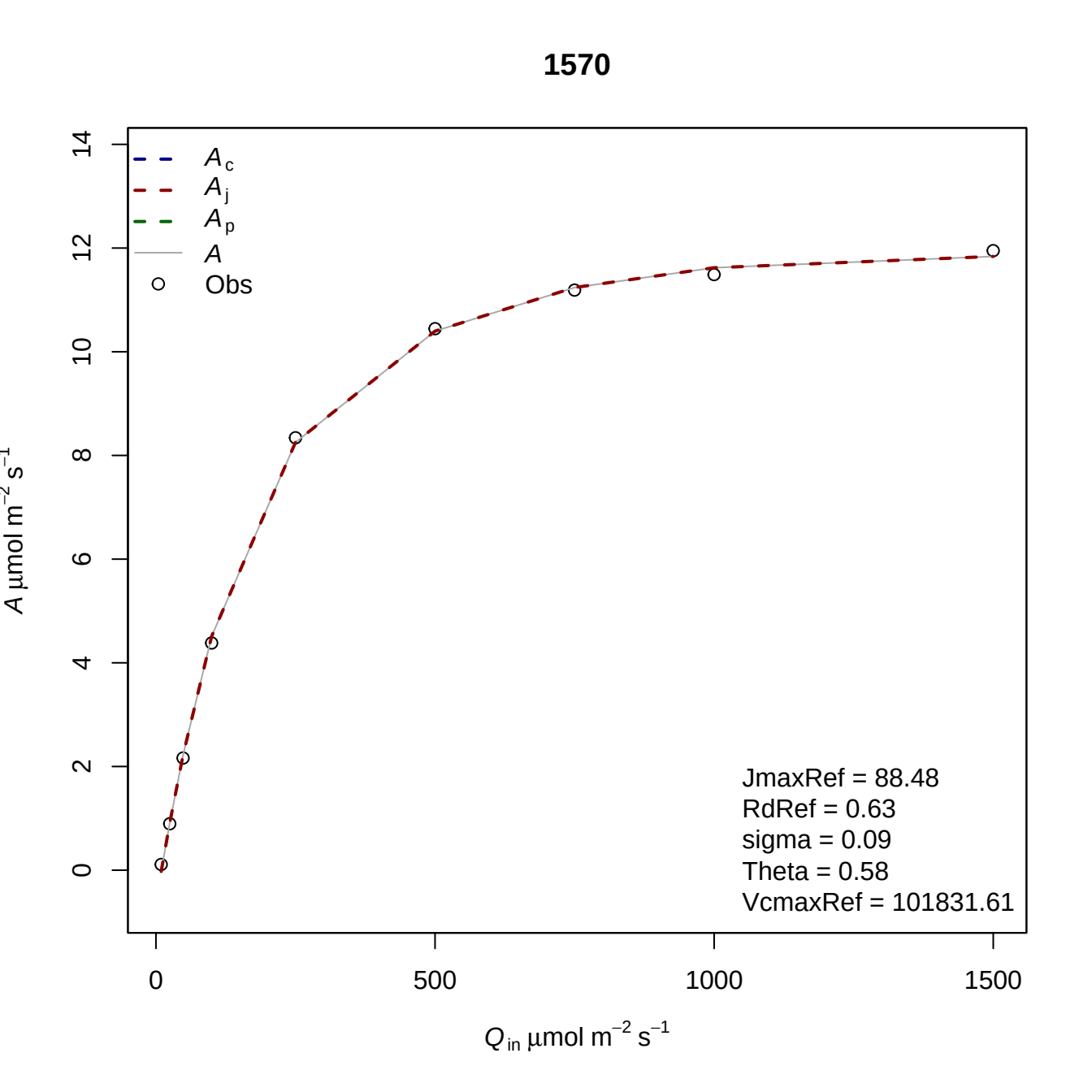

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 29.64
RdRef = 0.83
sigma = 0.12
Theta = 0.07
VcmaxRef = 21.27
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

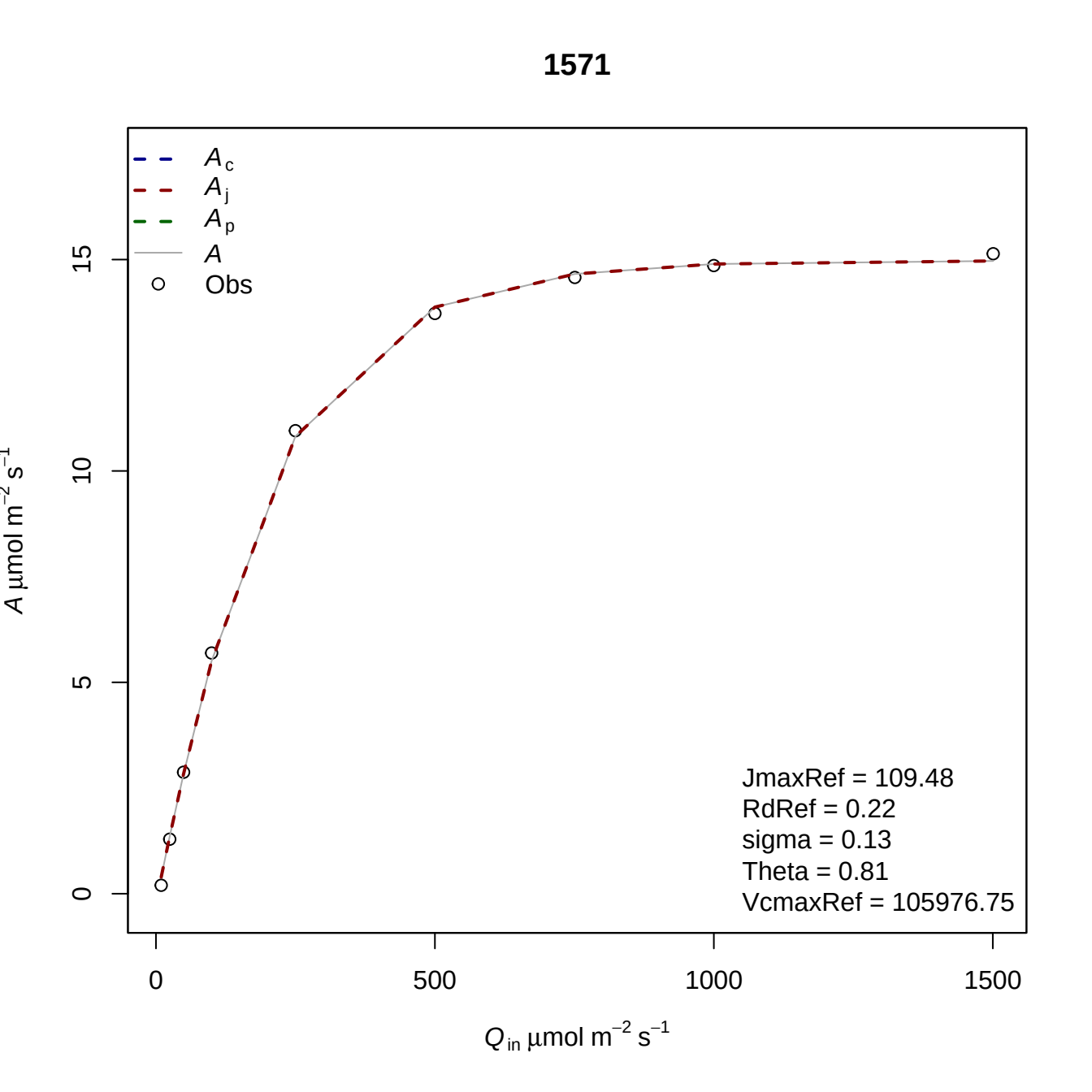

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 78.55
RdRef = 0.67
sigma = 0.16
Theta = 0.38
VcmaxRef = 56.54
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

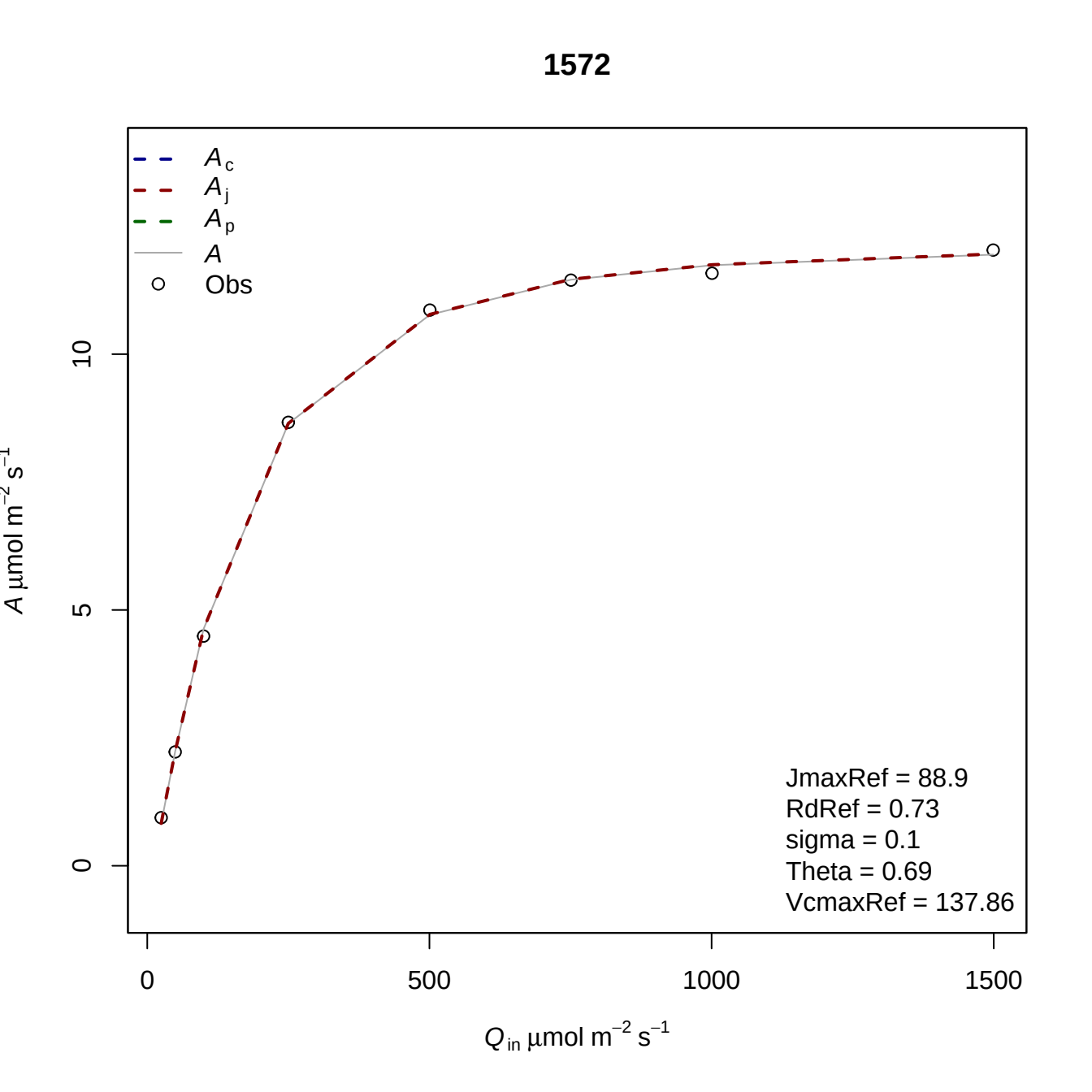

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 93.37
RdRef = 0.69
sigma = 0.15
Theta = 0.21
VcmaxRef = 62.36
0
2
4
6
8
10
12
14
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

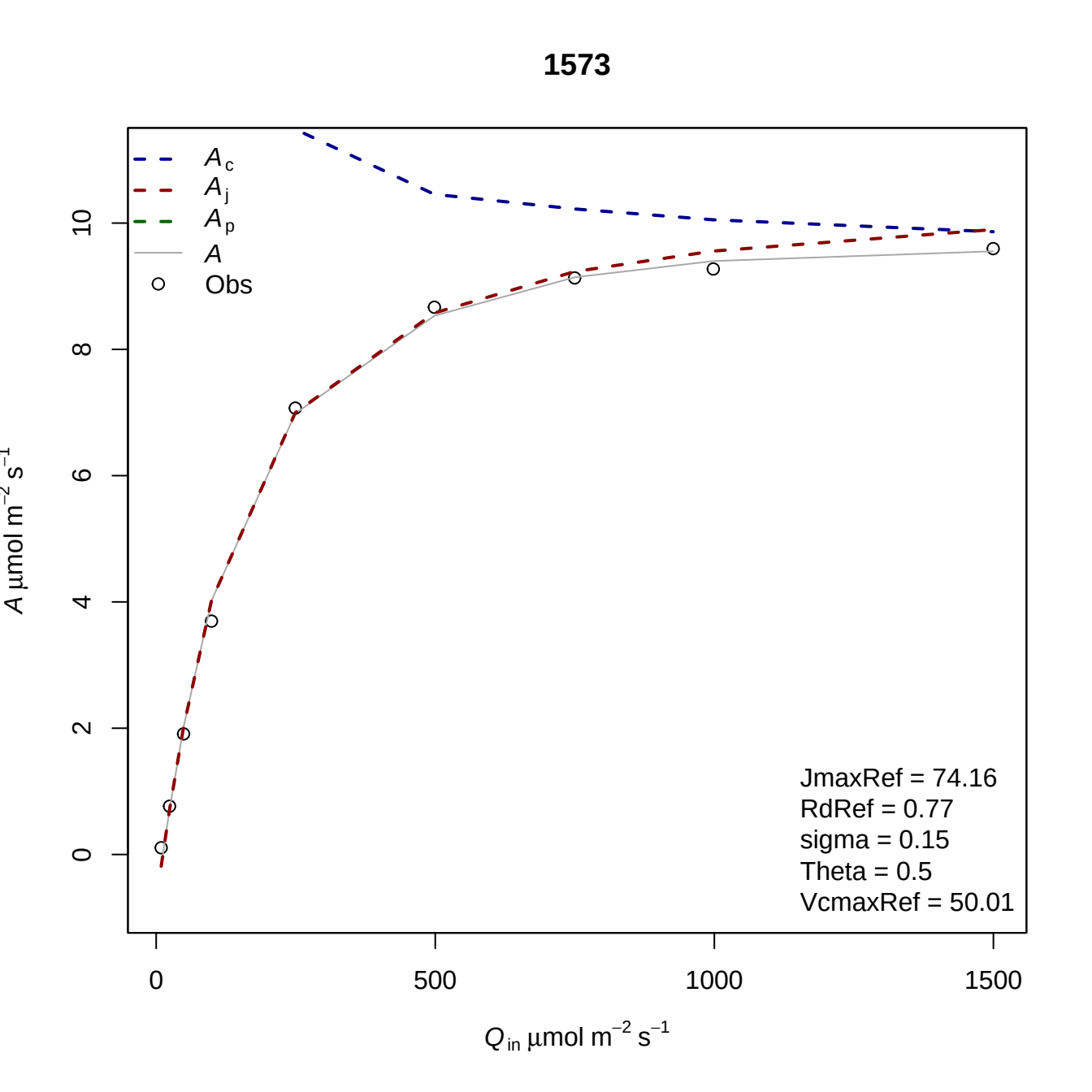

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 112.04
RdRef = 0.91
sigma = 0.26
Theta = 0.47
VcmaxRef = 73.71
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

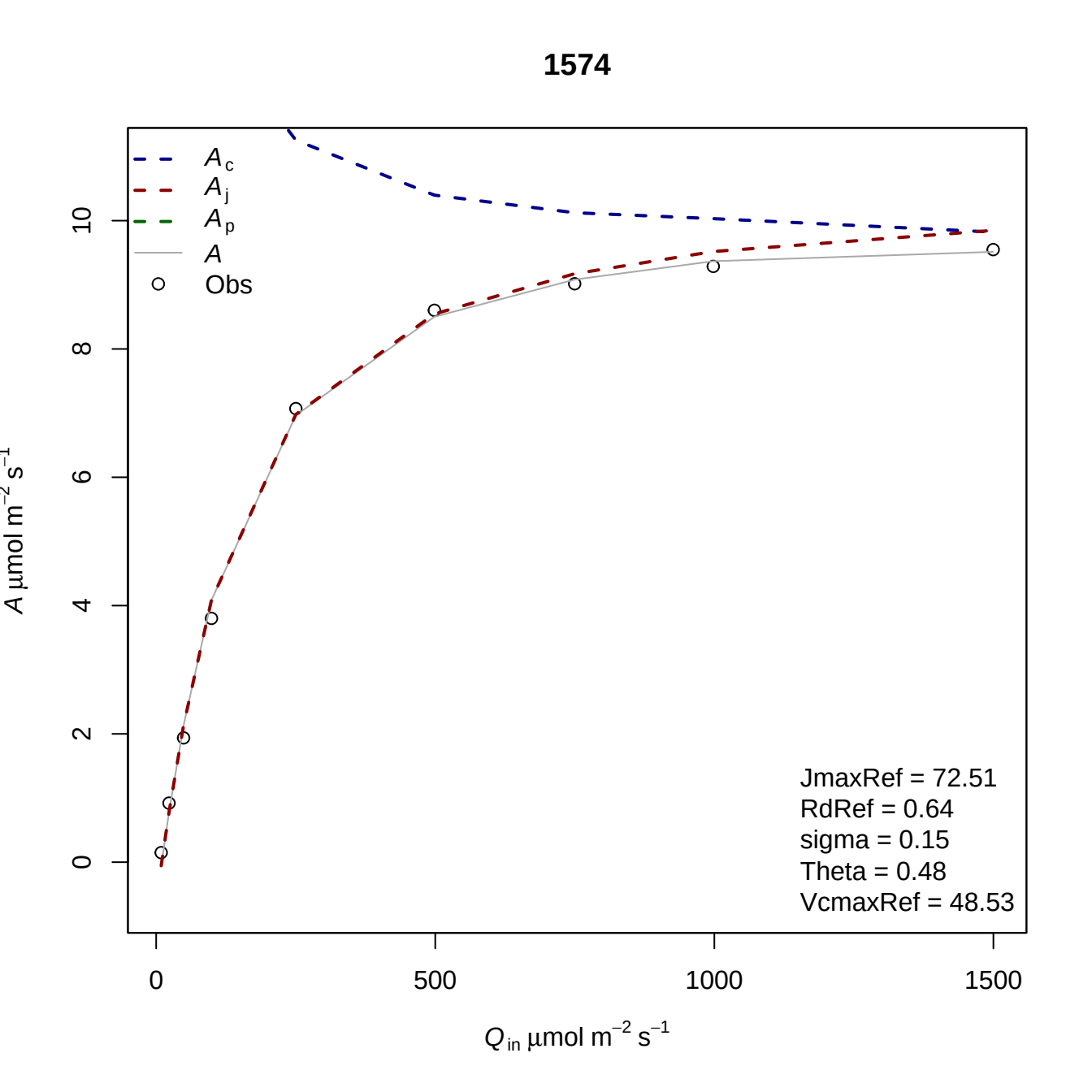

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 73.39
RdRef = 0.64
sigma = 0.16
Theta = 0.5
VcmaxRef = 51.14
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

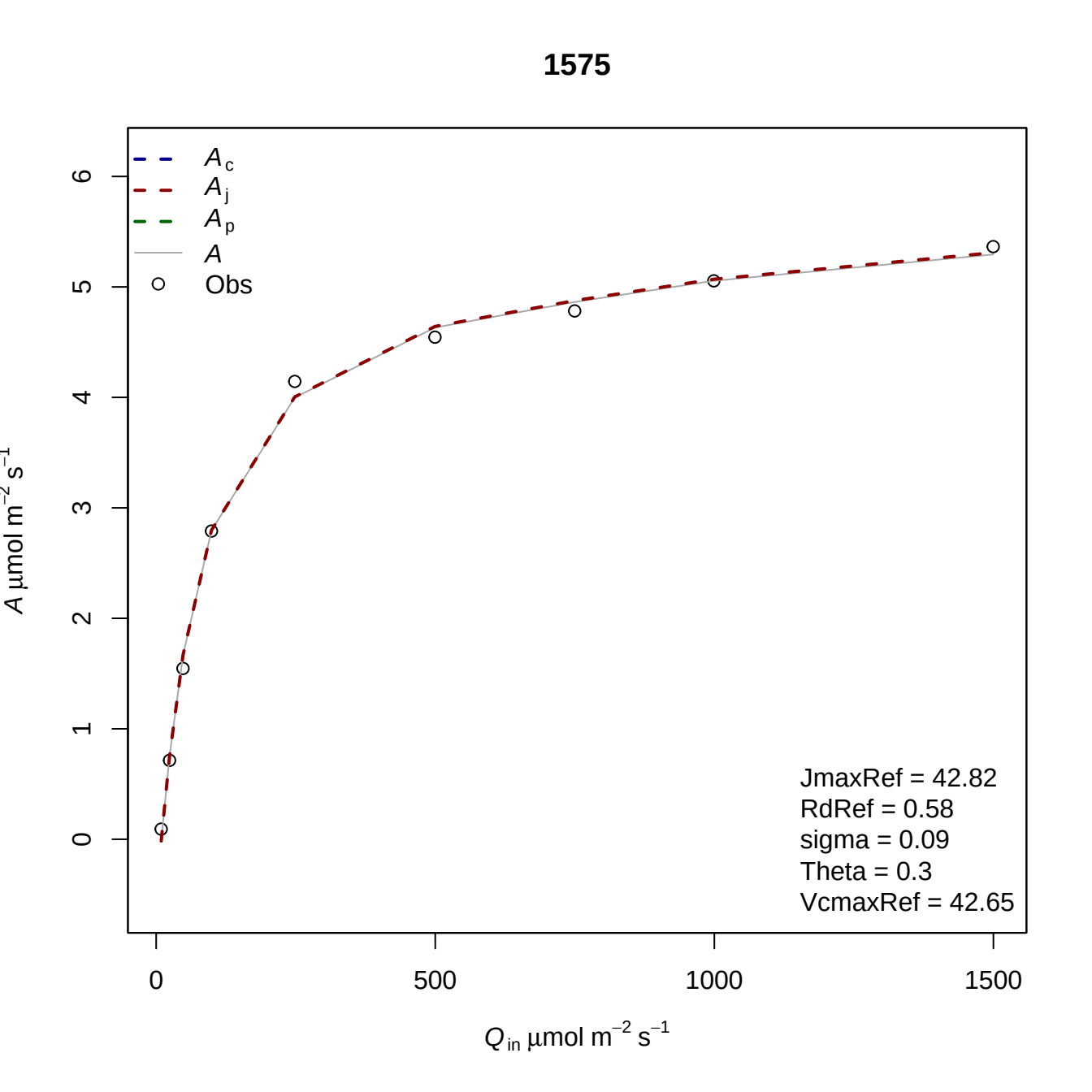

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 88.48
RdRef = 0.63
sigma = 0.09
Theta = 0.58
VcmaxRef = 101831.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

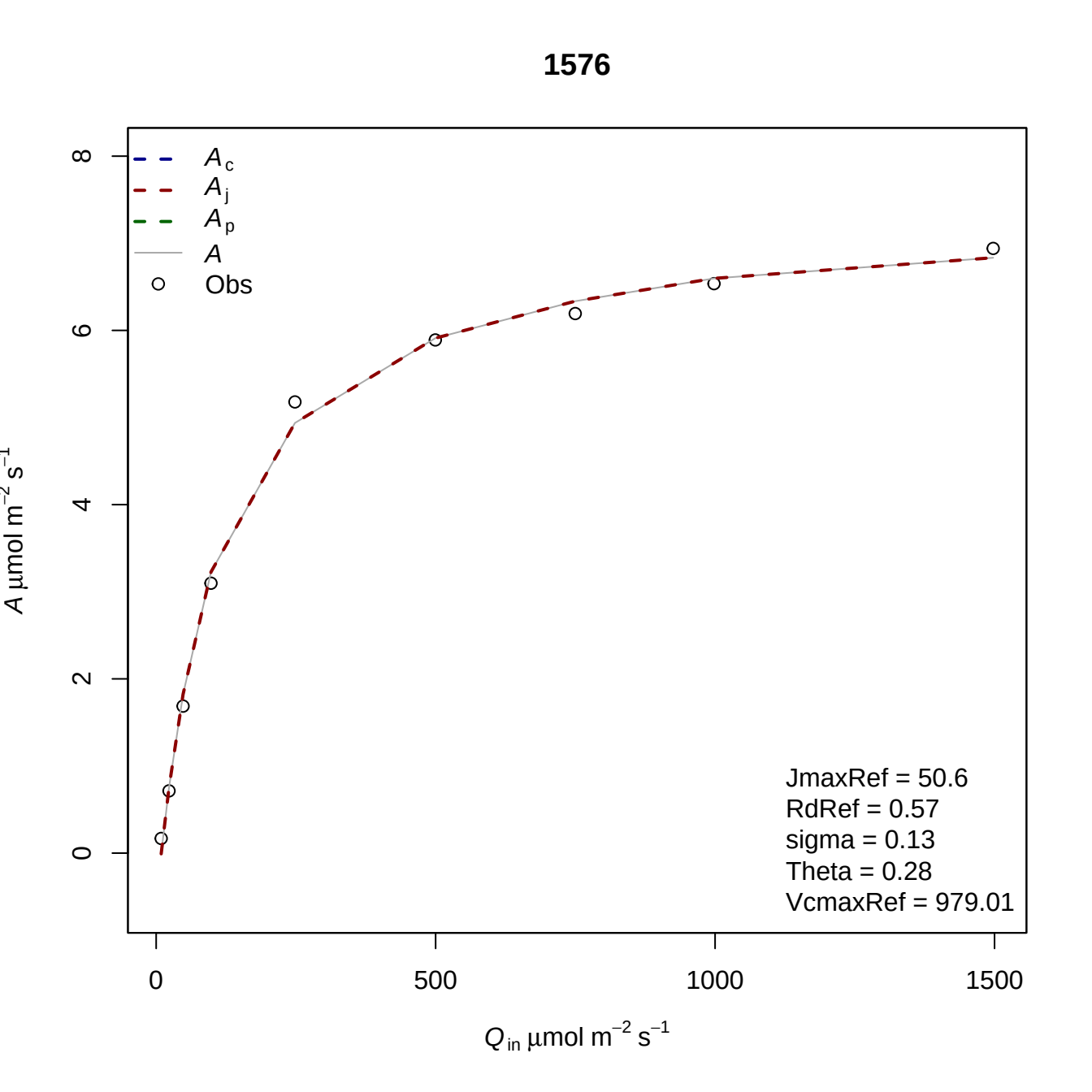

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 109.48
RdRef = 0.22
sigma = 0.13
Theta = 0.81
VcmaxRef = 105976.75
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

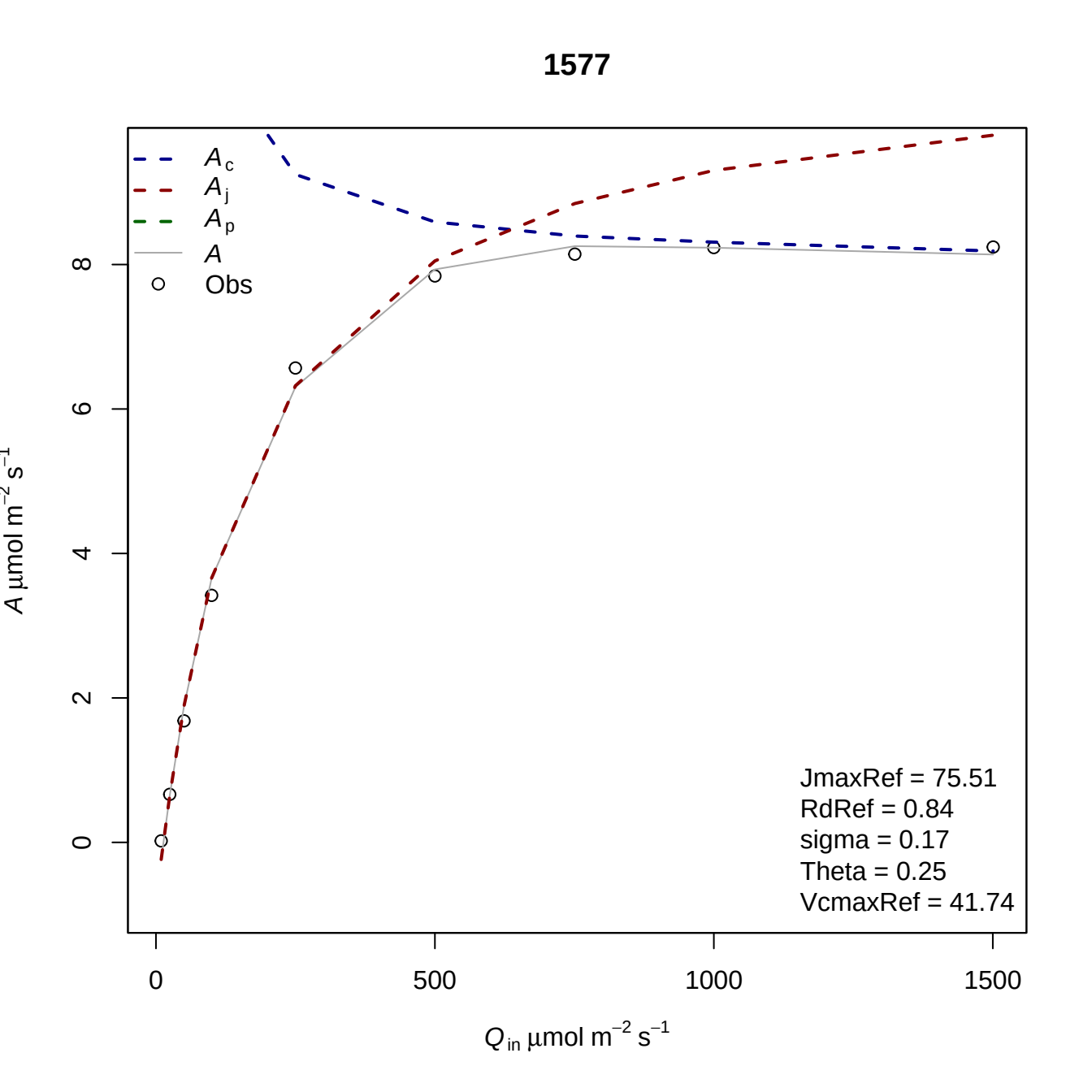

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 88.9
RdRef = 0.73
sigma = 0.1
Theta = 0.69
VcmaxRef = 137.86
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

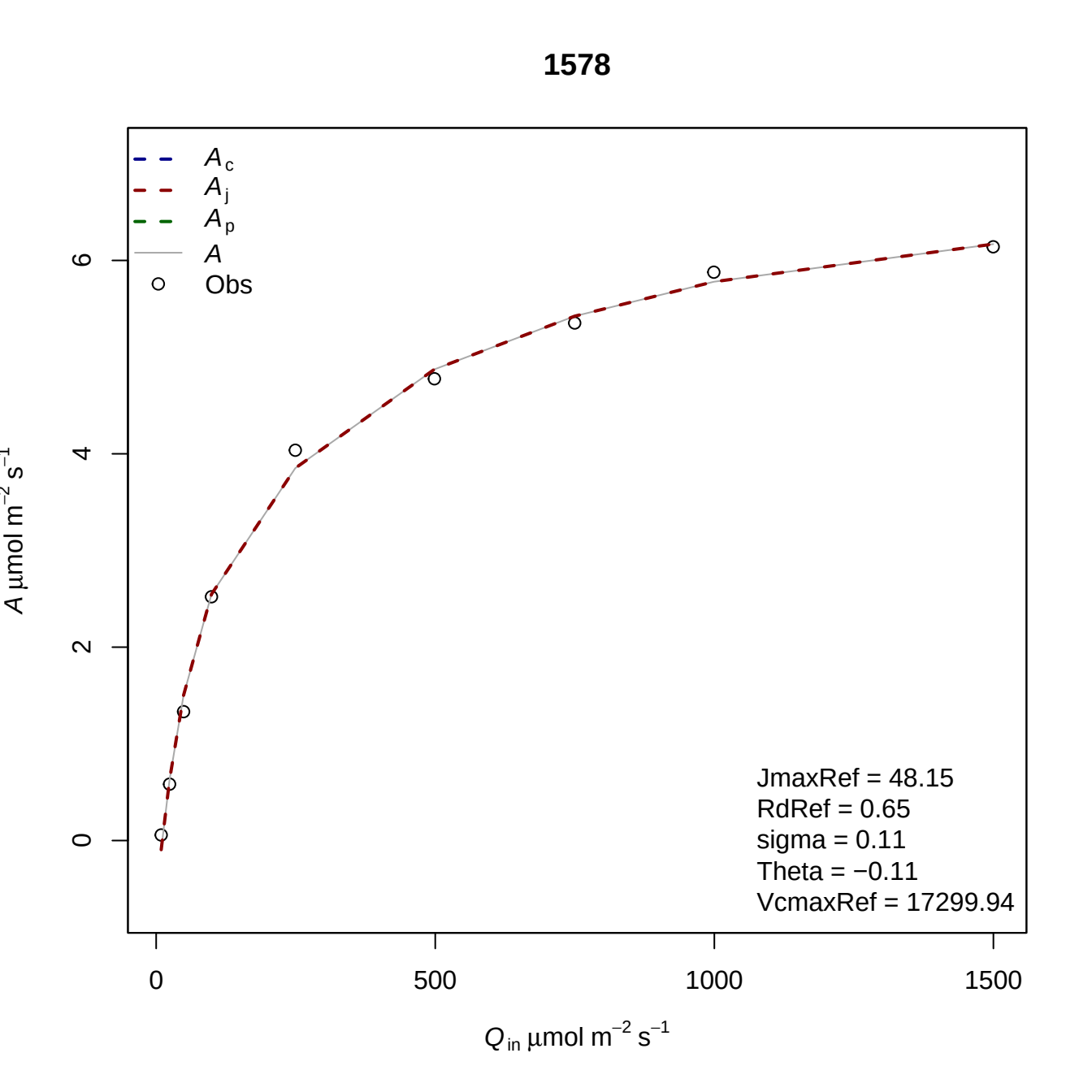

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 74.16
RdRef = 0.77
sigma = 0.15
Theta = 0.5
VcmaxRef = 50.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

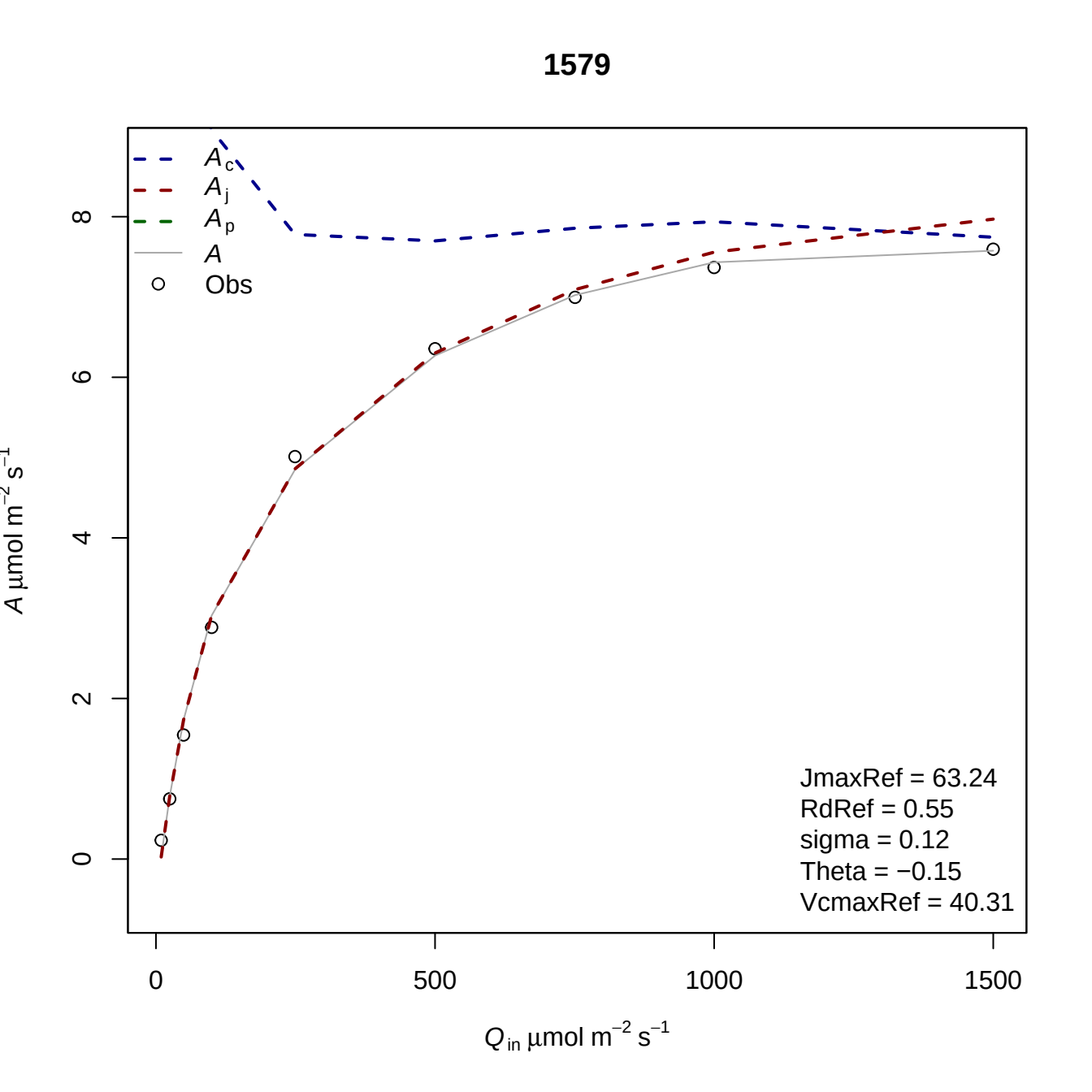

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 72.51
RdRef = 0.64
sigma = 0.15
Theta = 0.48
VcmaxRef = 48.53
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

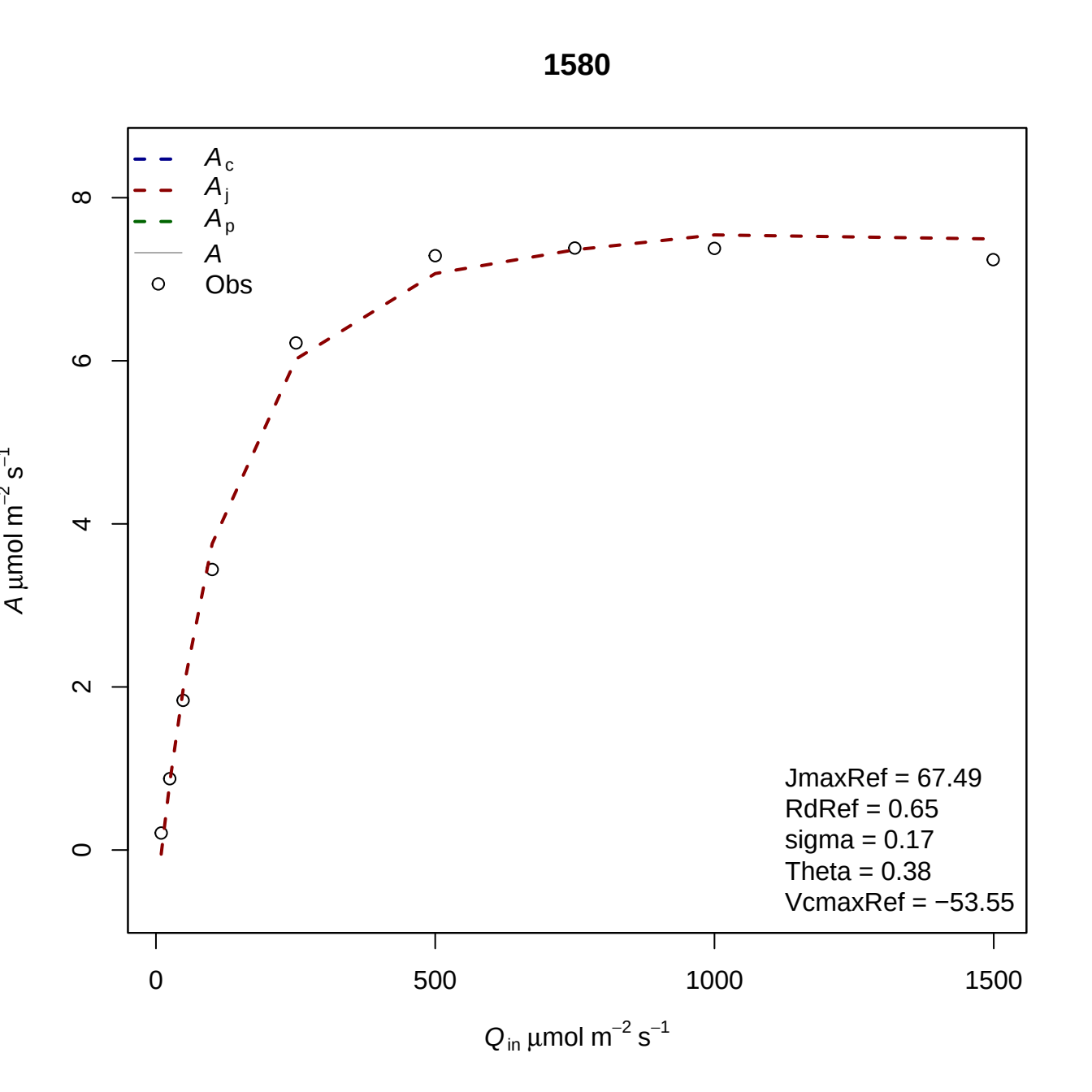

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 42.82
RdRef = 0.58
sigma = 0.09
Theta = 0.3
VcmaxRef = 42.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

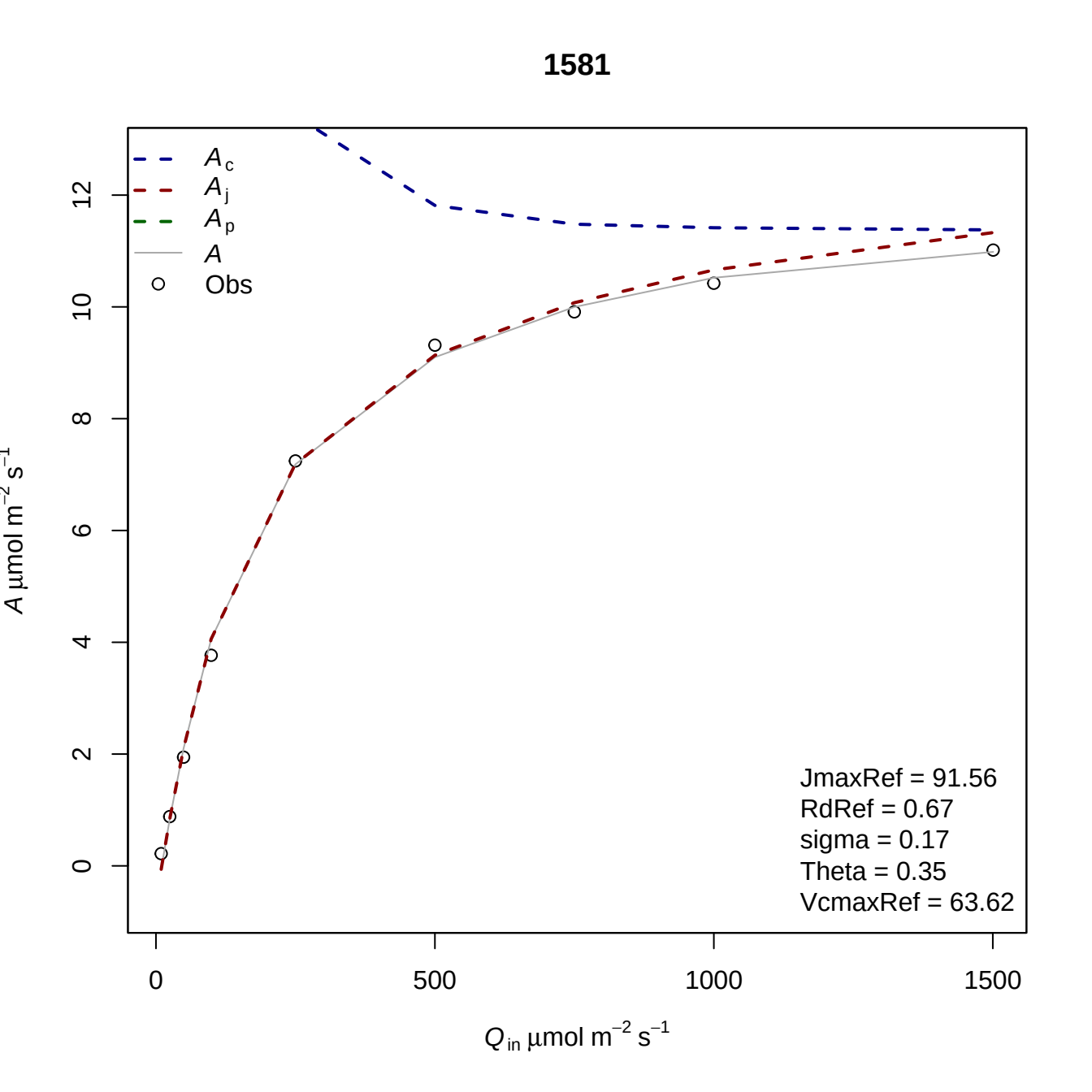

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 50.6
RdRef = 0.57
sigma = 0.13
Theta = 0.28
VcmaxRef = 979.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

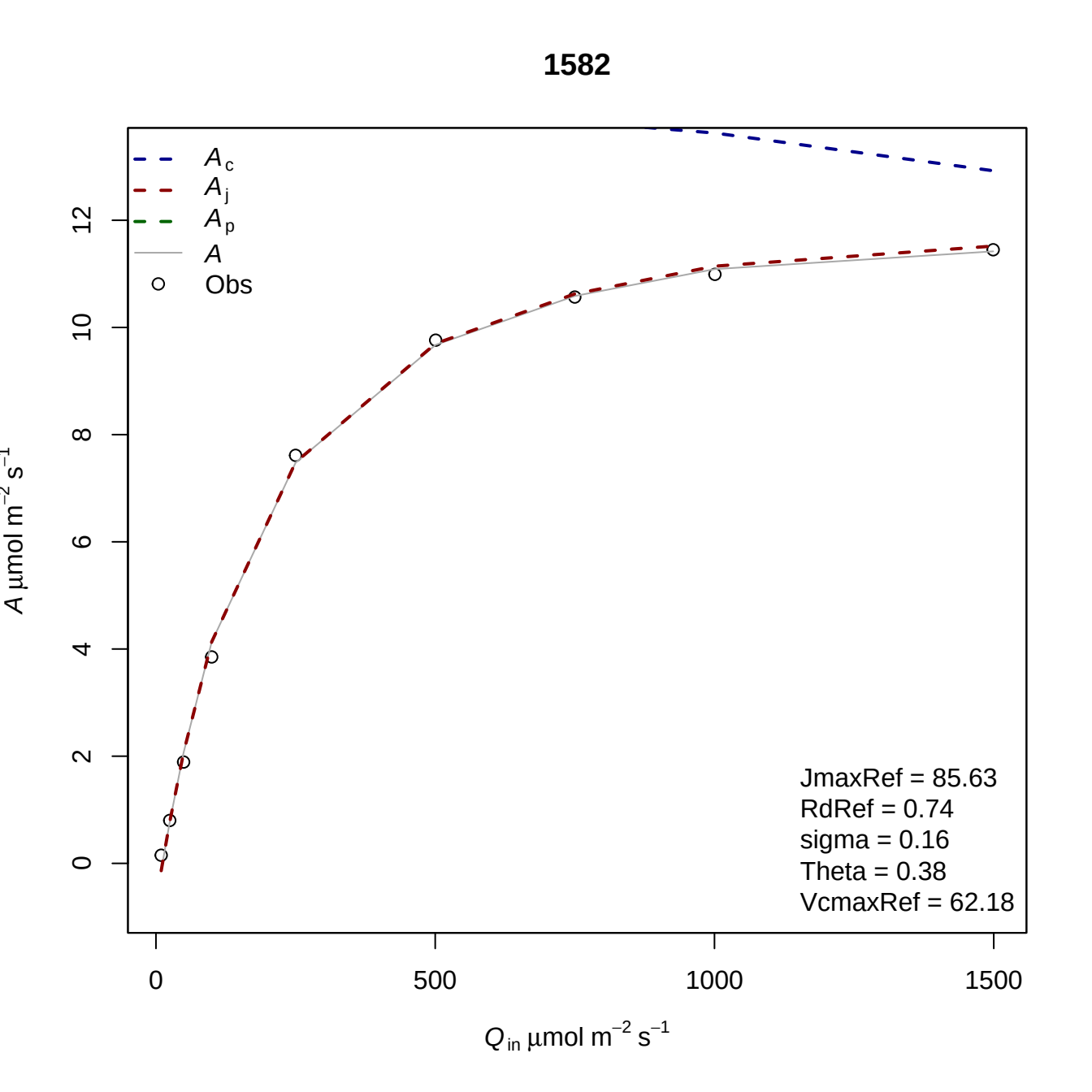

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 75.51
RdRef = 0.84
sigma = 0.17
Theta = 0.25
VcmaxRef = 41.74
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

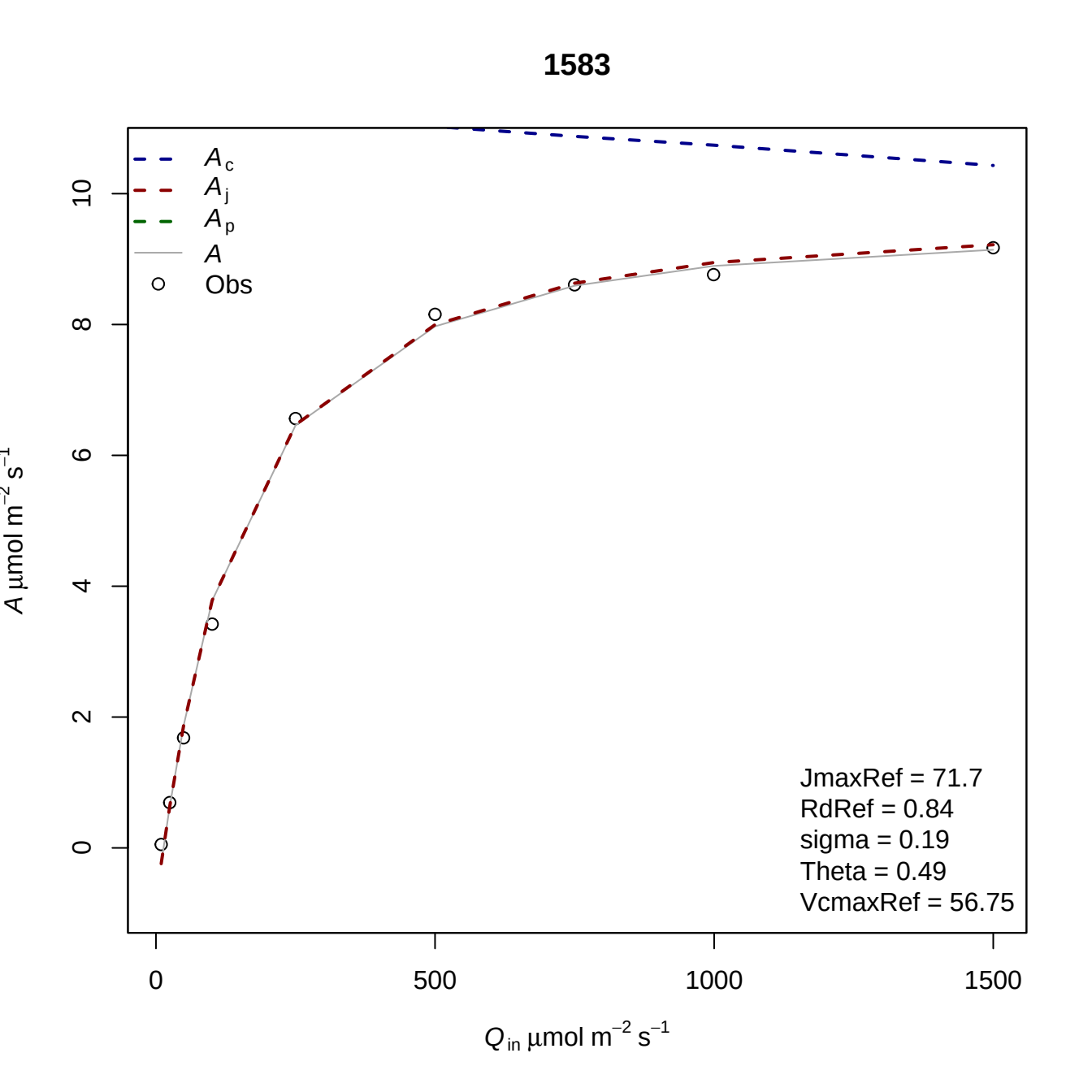

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 48.15
RdRef = 0.65
sigma = 0.11
Theta = −0.11
VcmaxRef = 17299.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

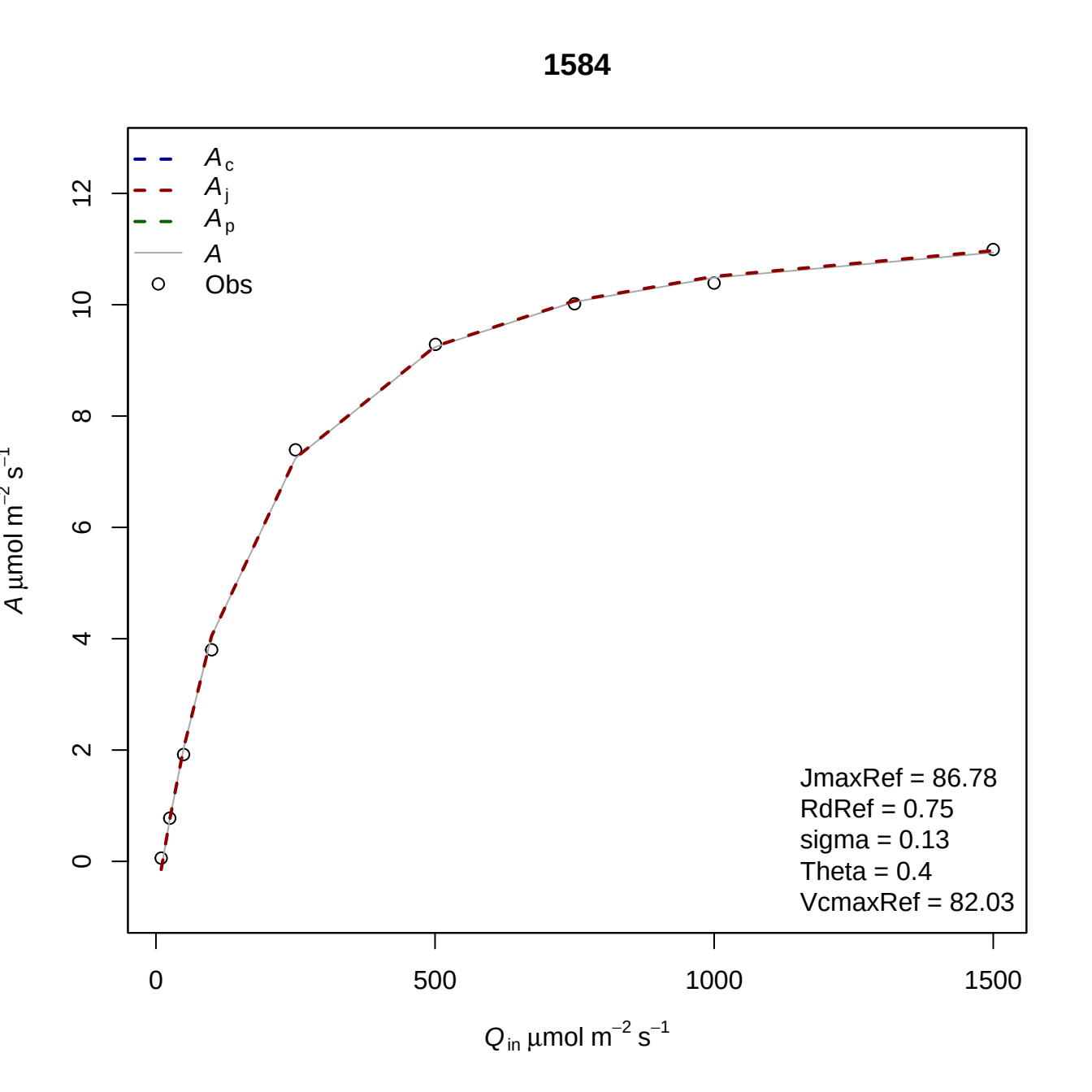

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 63.24
RdRef = 0.55
sigma = 0.12
Theta = −0.15
VcmaxRef = 40.31
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

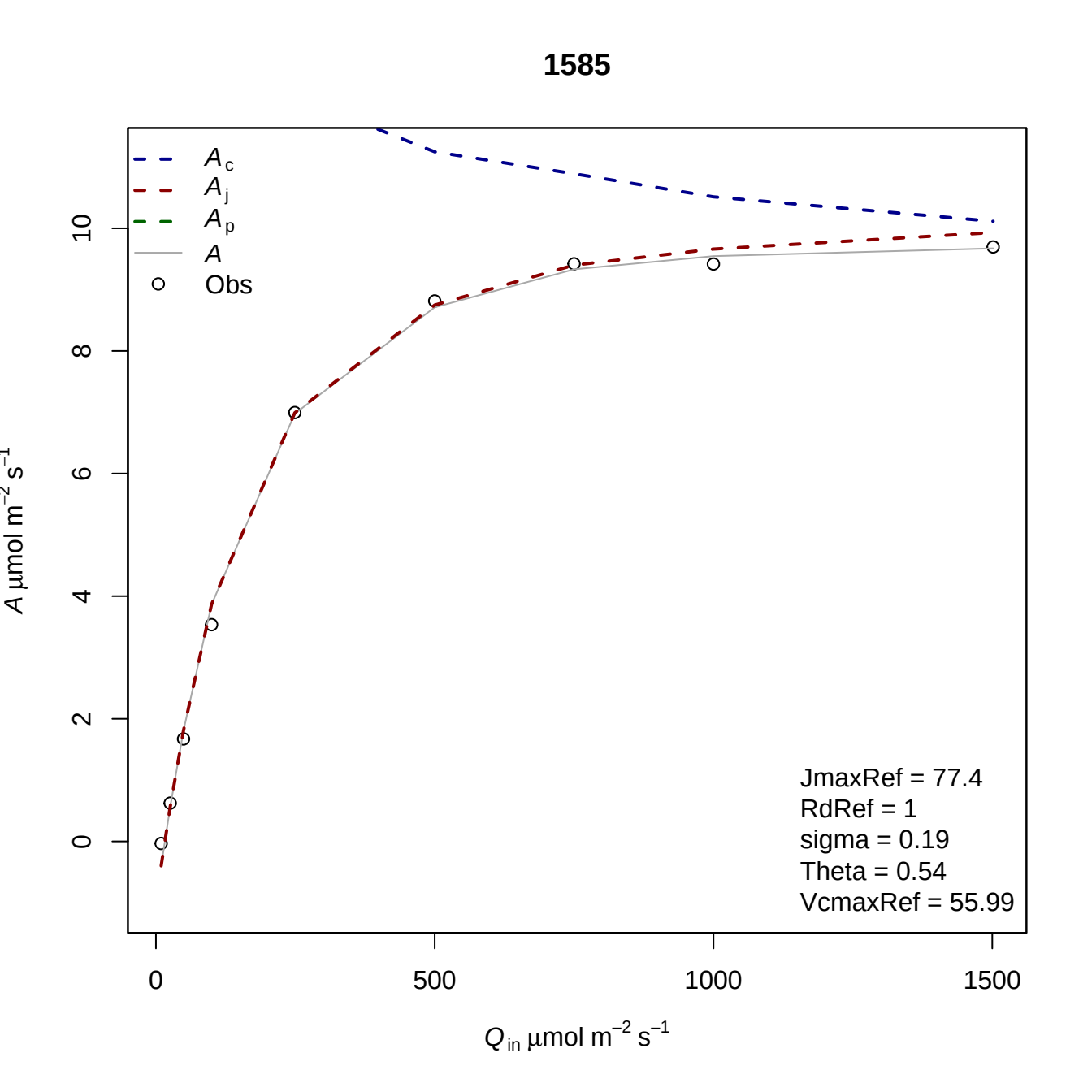

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 67.49
RdRef = 0.65
sigma = 0.17
Theta = 0.38
VcmaxRef = −53.55
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

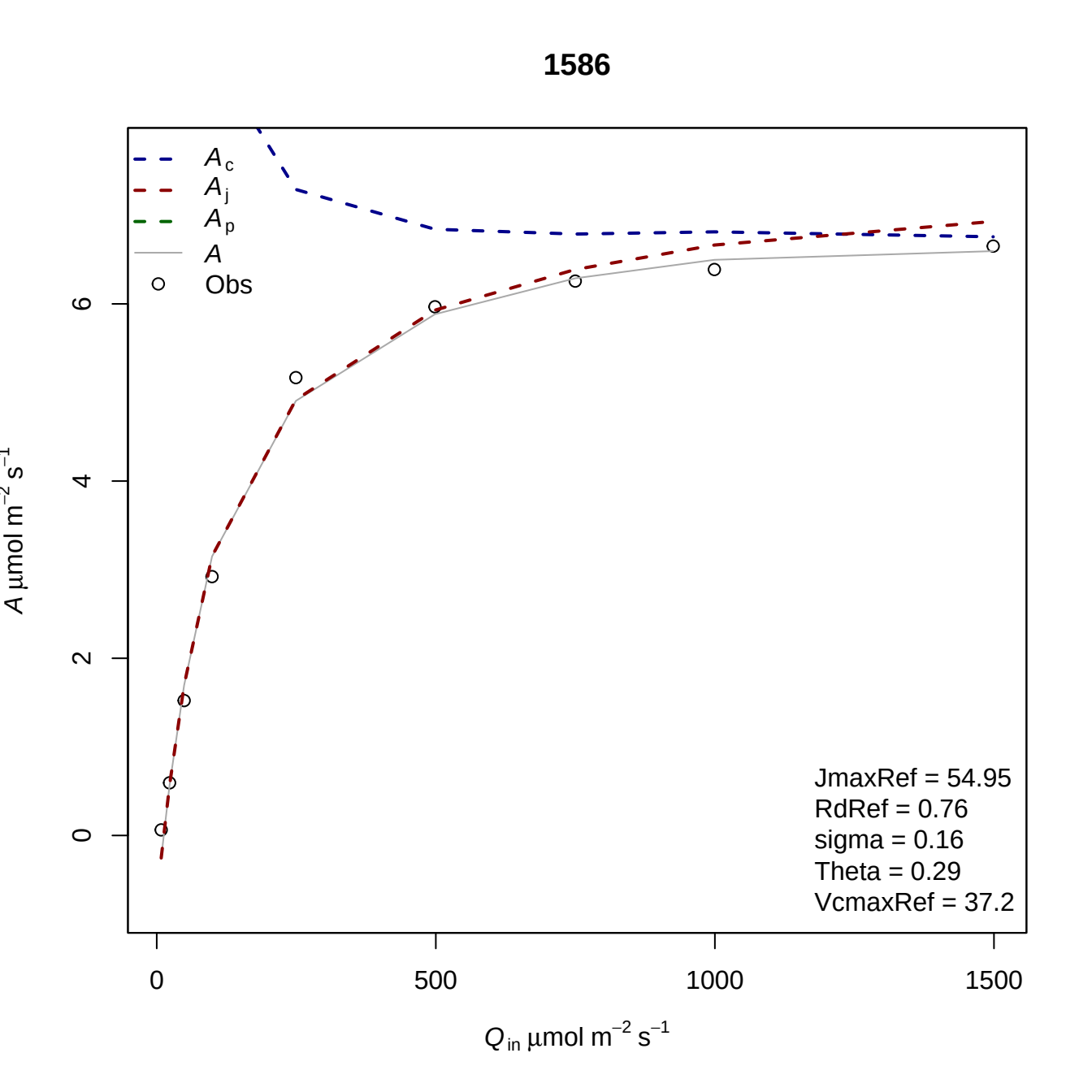

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 91.56
RdRef = 0.67
sigma = 0.17
Theta = 0.35
VcmaxRef = 63.62
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

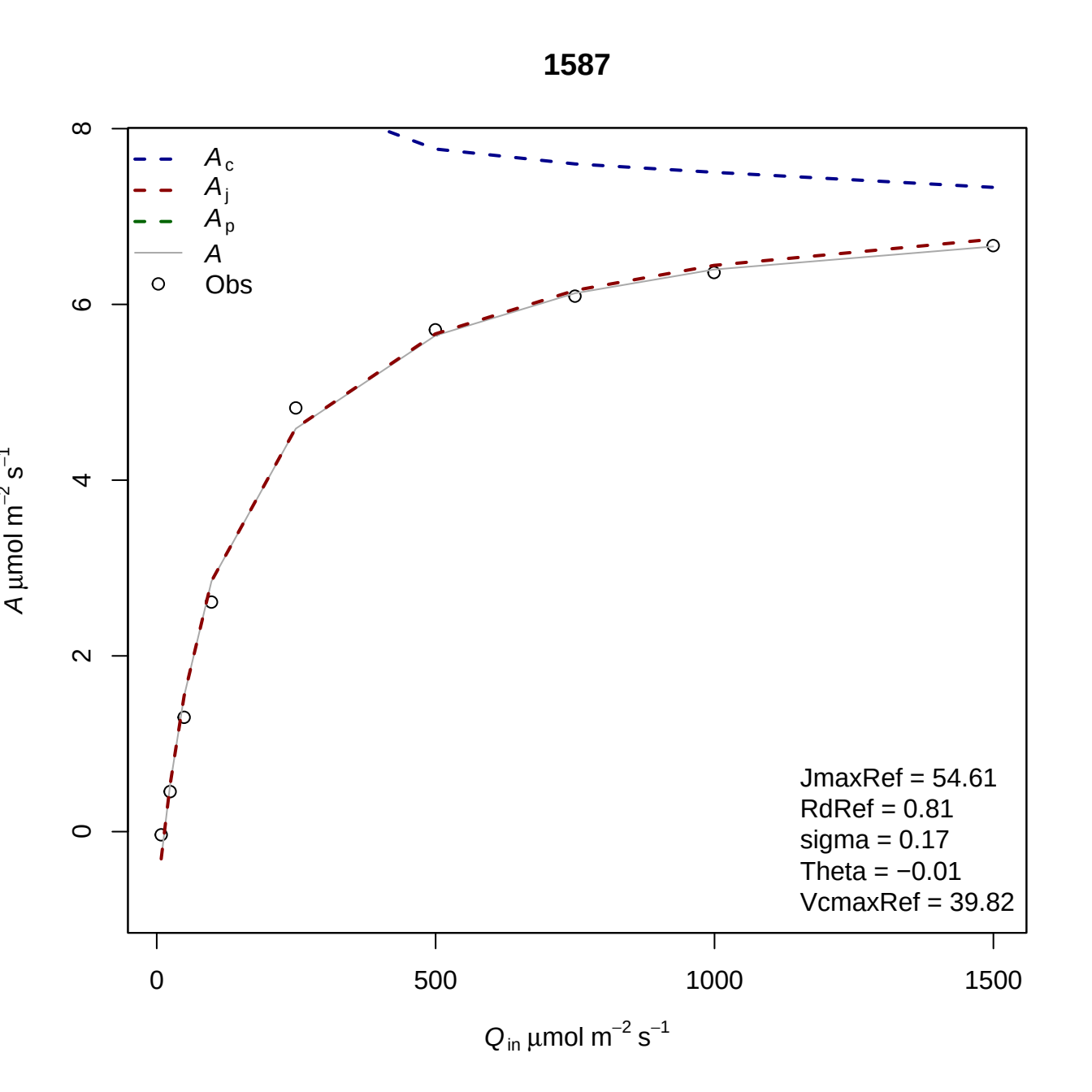

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 85.63
RdRef = 0.74
sigma = 0.16
Theta = 0.38
VcmaxRef = 62.18
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

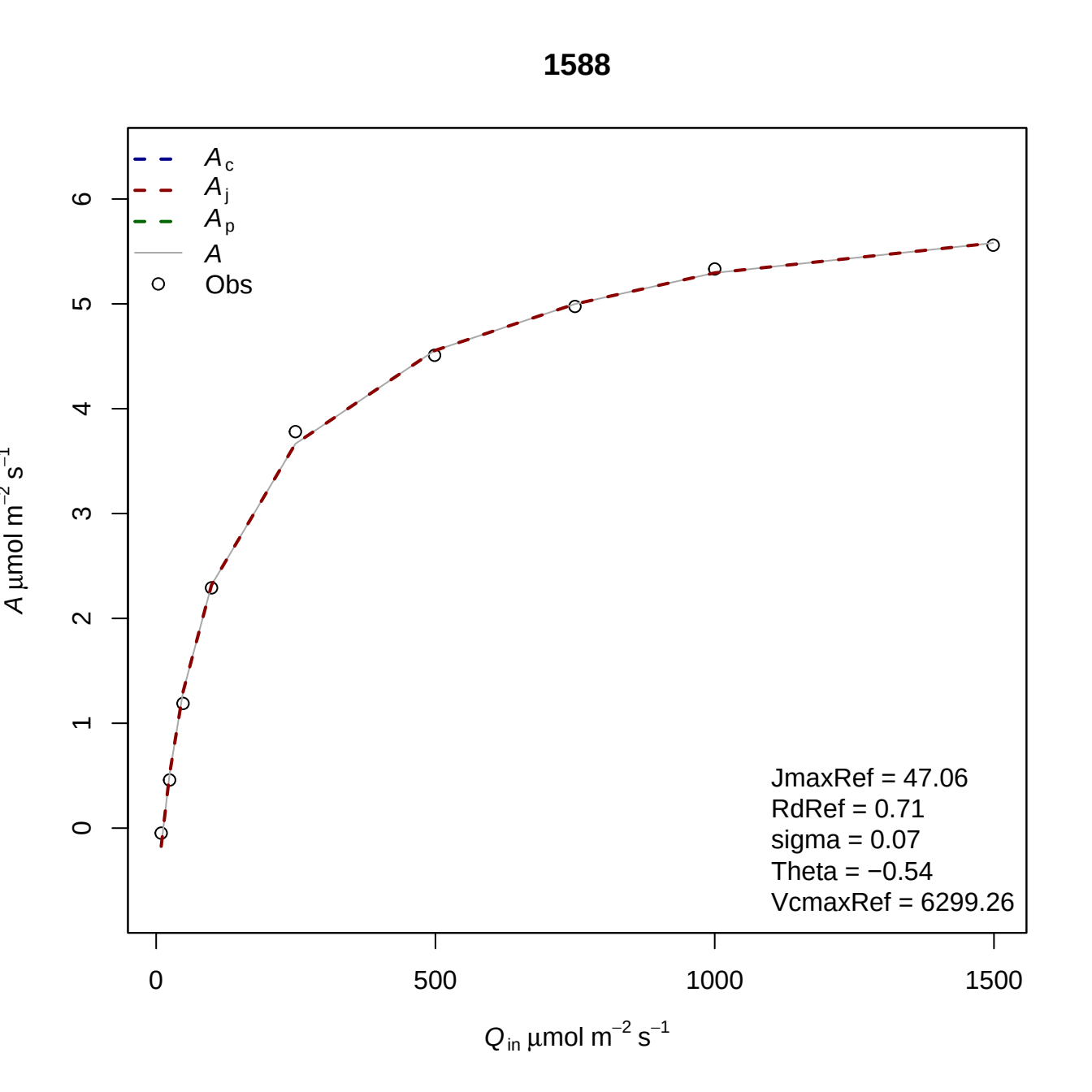

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 71.7
RdRef = 0.84
sigma = 0.19
Theta = 0.49
VcmaxRef = 56.75
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

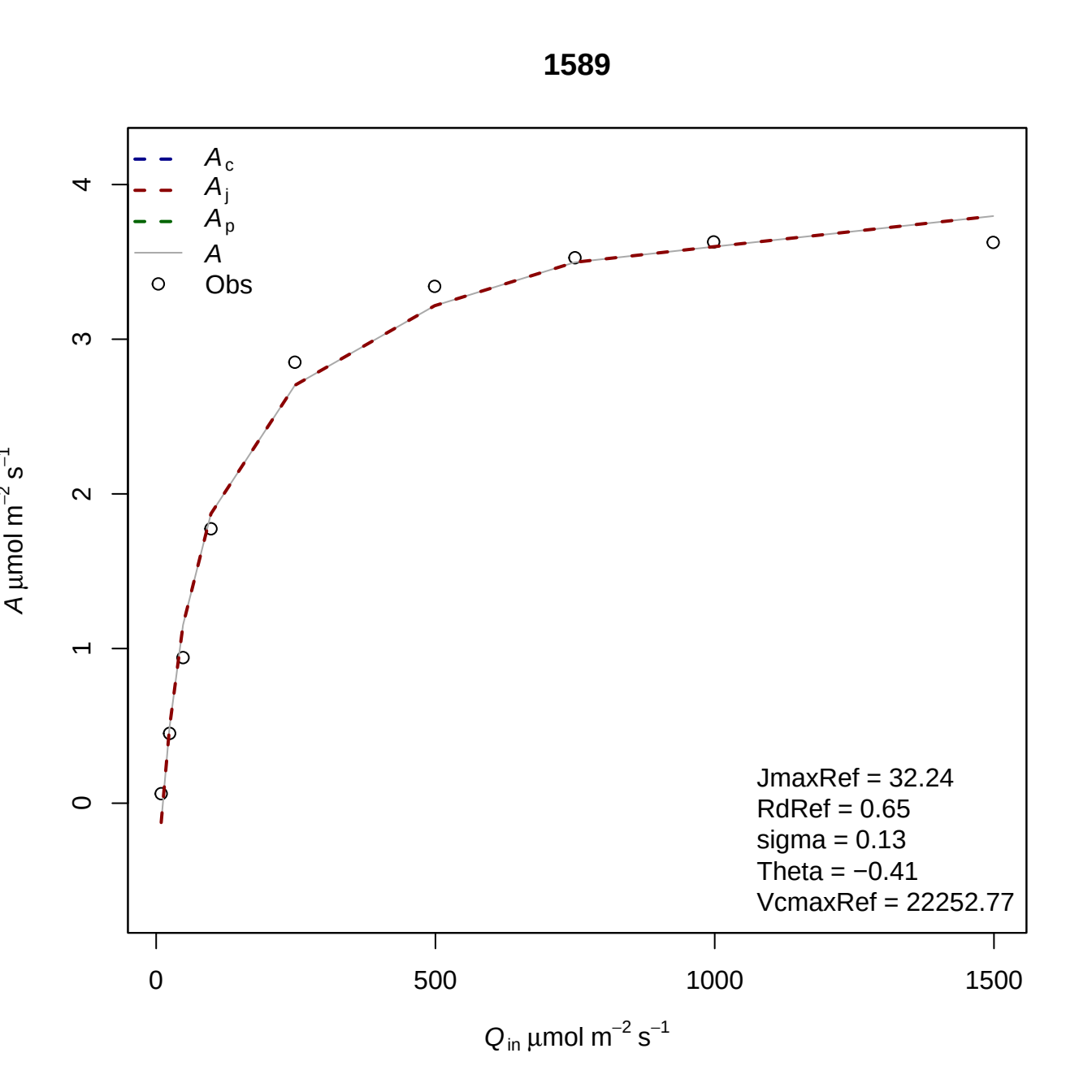

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 86.78
RdRef = 0.75
sigma = 0.13
Theta = 0.4
VcmaxRef = 82.03
0
2
4
6
8
10
12
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

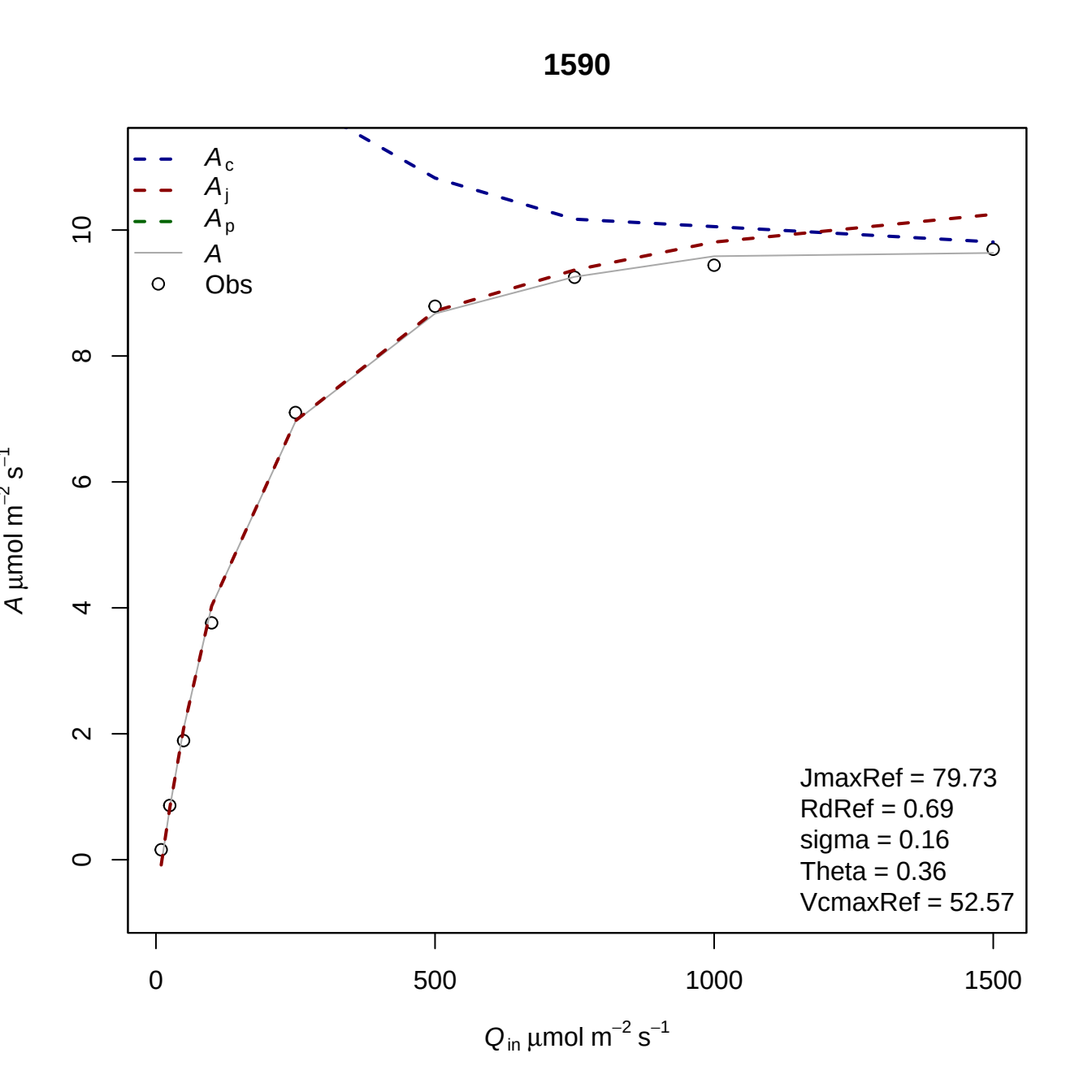

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 77.4
RdRef = 1
sigma = 0.19
Theta = 0.54
VcmaxRef = 55.99
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

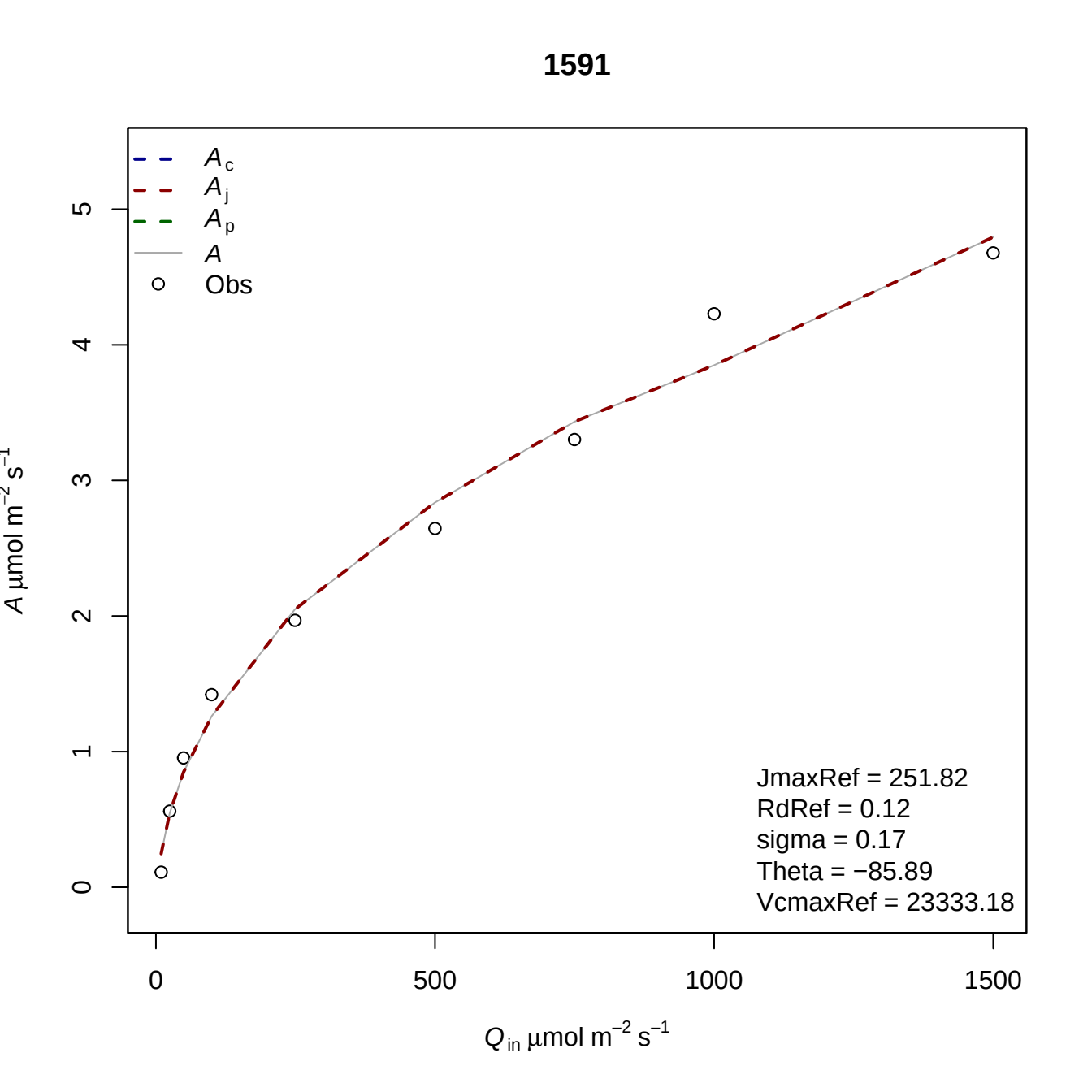

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 54.95
RdRef = 0.76
sigma = 0.16
Theta = 0.29
VcmaxRef = 37.2
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

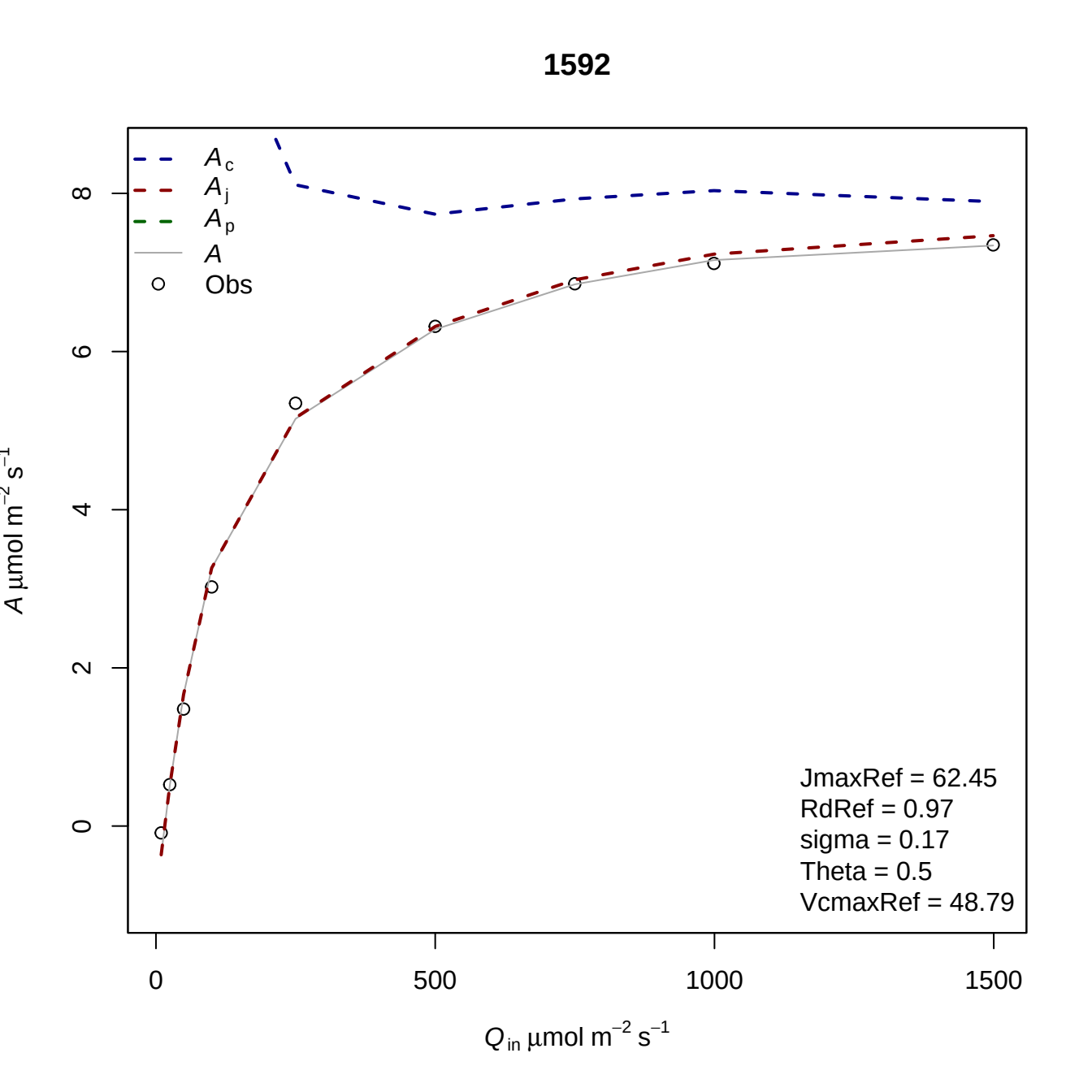

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 54.61
RdRef = 0.81
sigma = 0.17
Theta = −0.01
VcmaxRef = 39.82
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

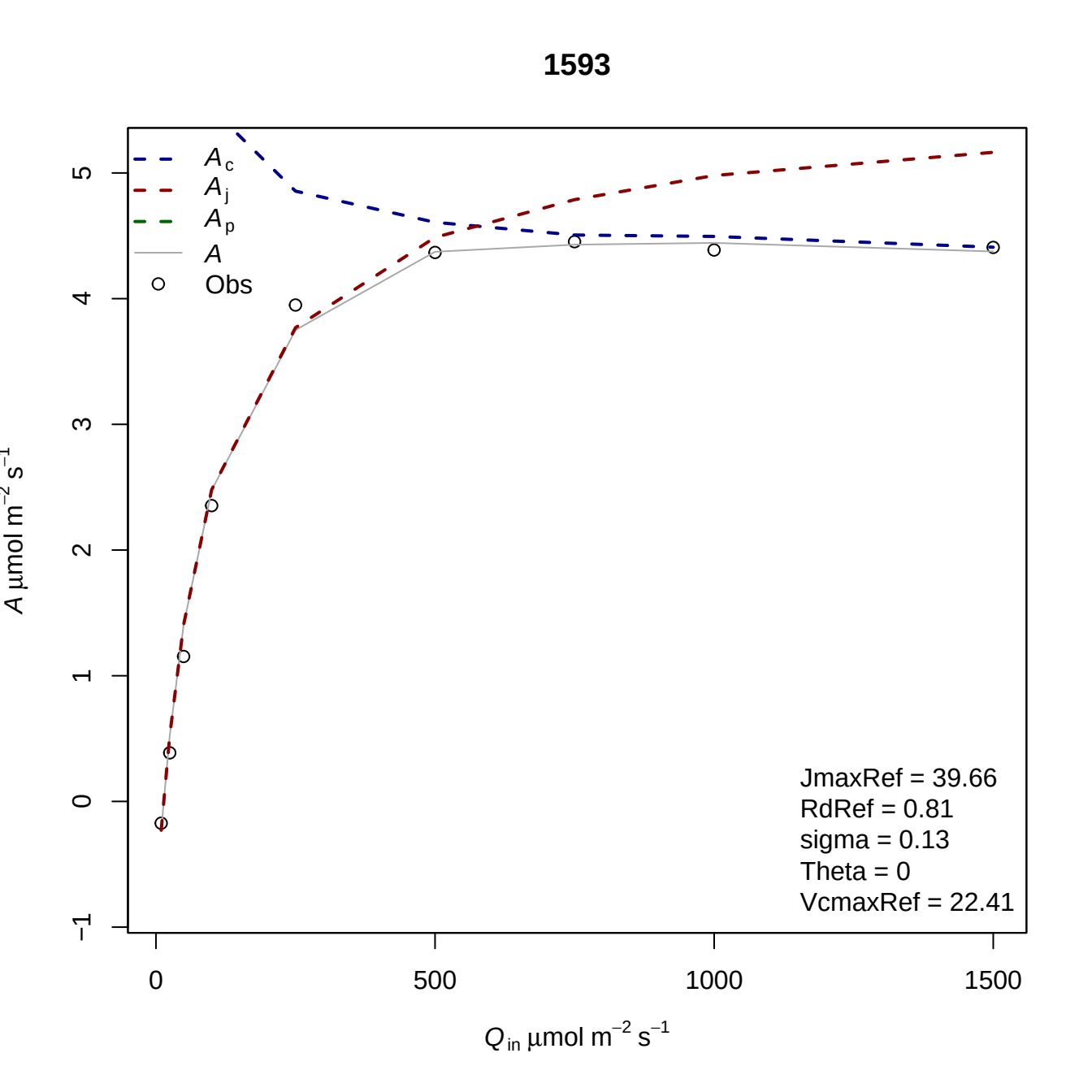

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 47.06
RdRef = 0.71
sigma = 0.07
Theta = −0.54
VcmaxRef = 6299.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

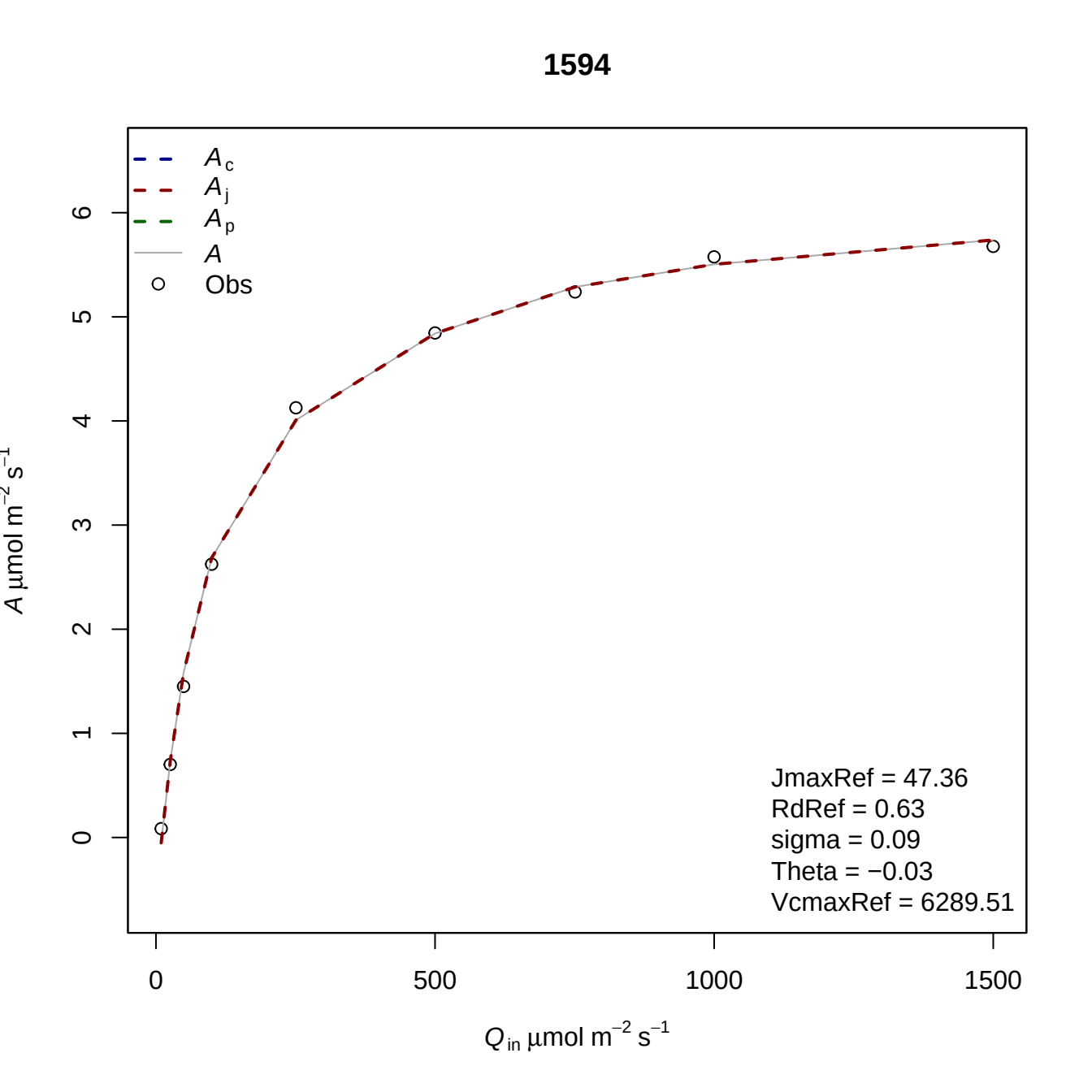

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 32.24
RdRef = 0.65
sigma = 0.13
Theta = −0.41
VcmaxRef = 22252.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

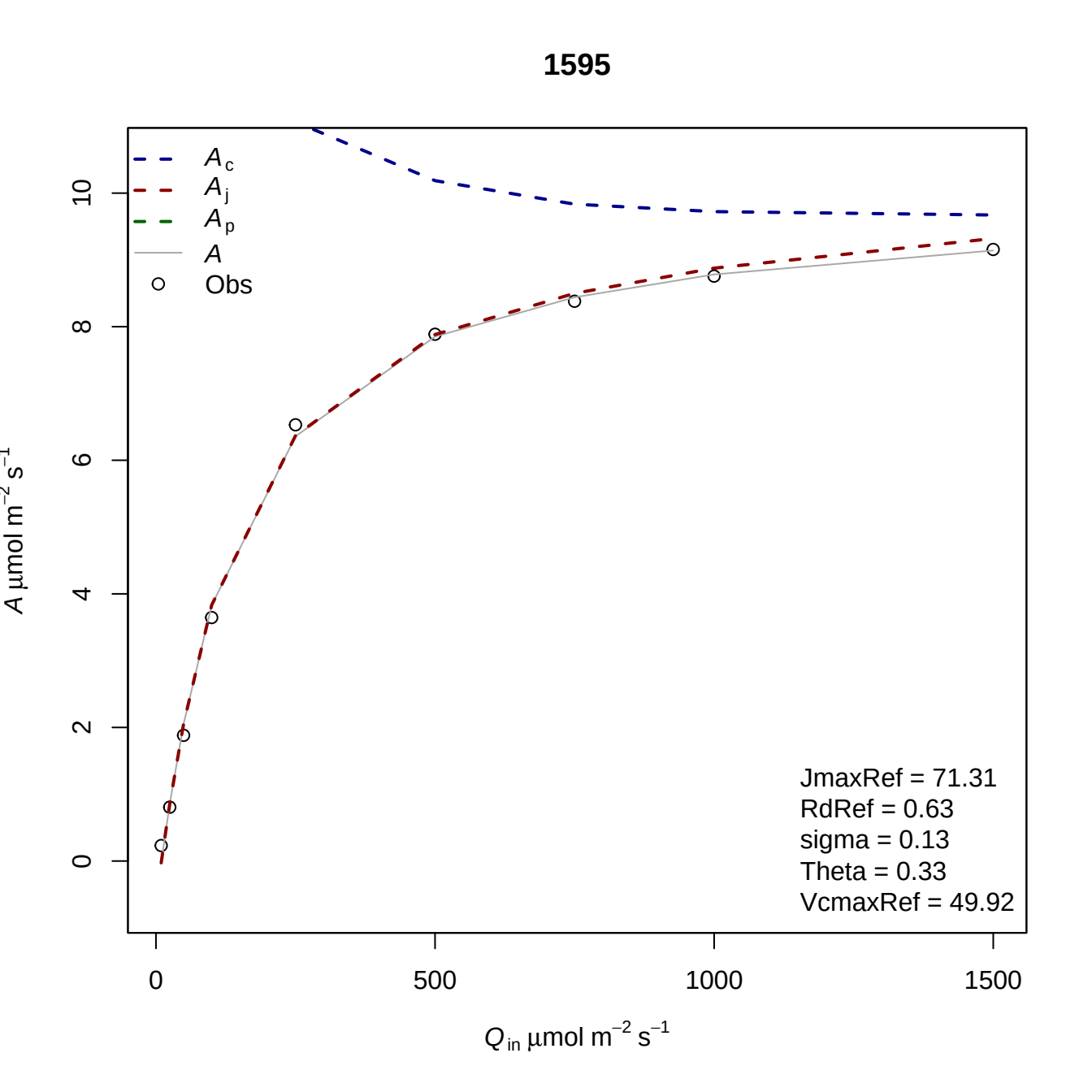

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 79.73
RdRef = 0.69
sigma = 0.16
Theta = 0.36
VcmaxRef = 52.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

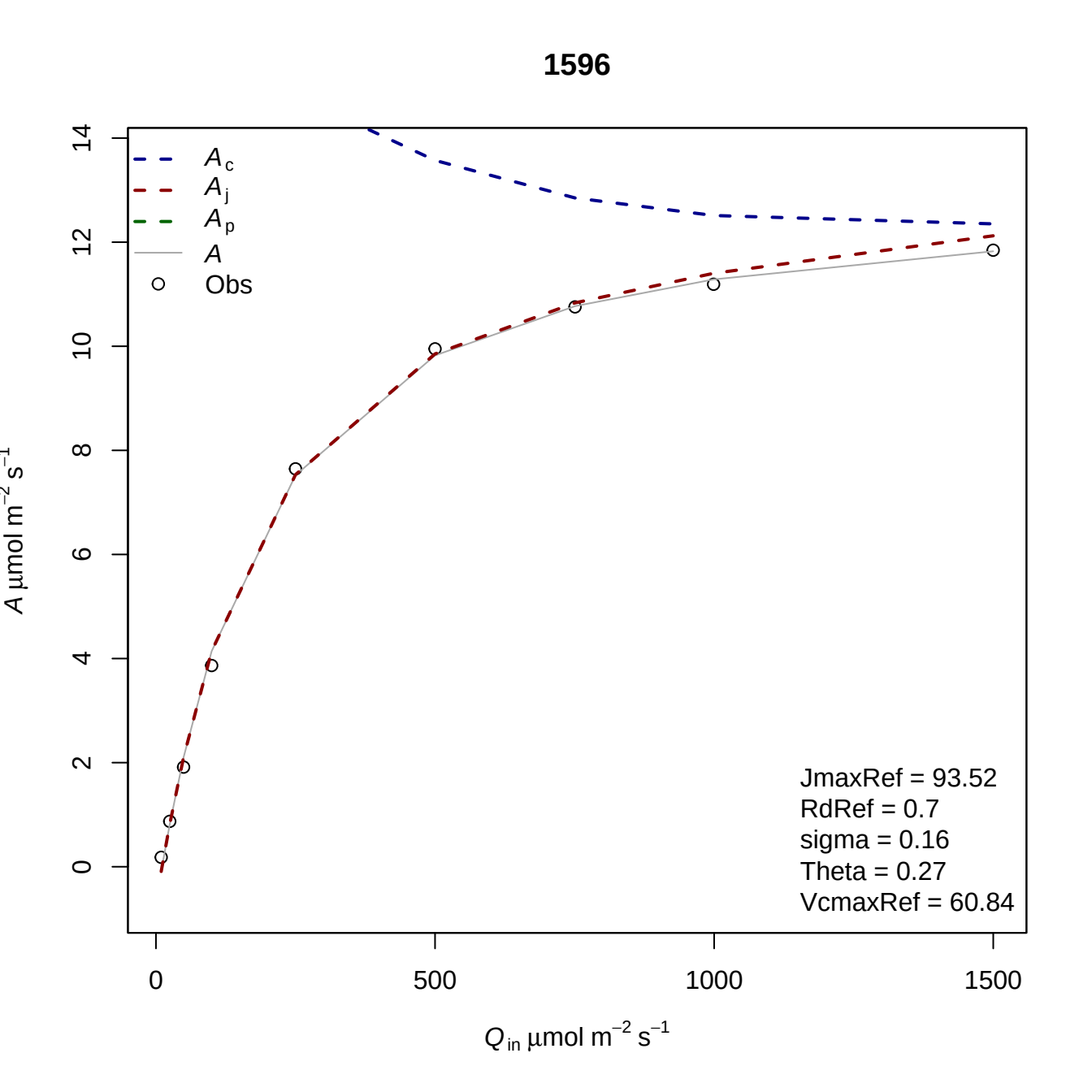

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 251.82
RdRef = 0.12
sigma = 0.17
Theta = −85.89
VcmaxRef = 23333.18
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

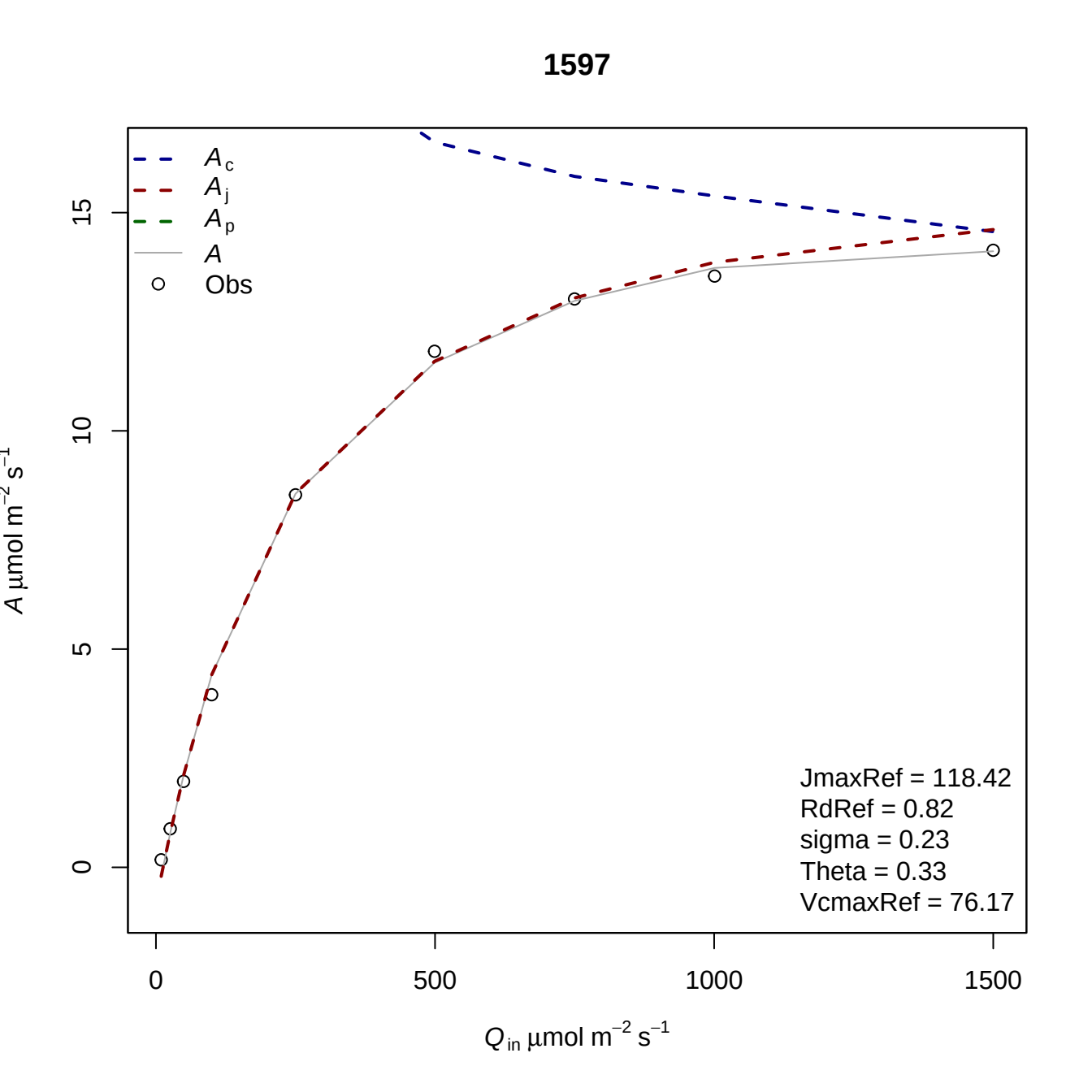

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 62.45
RdRef = 0.97
sigma = 0.17
Theta = 0.5
VcmaxRef = 48.79
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

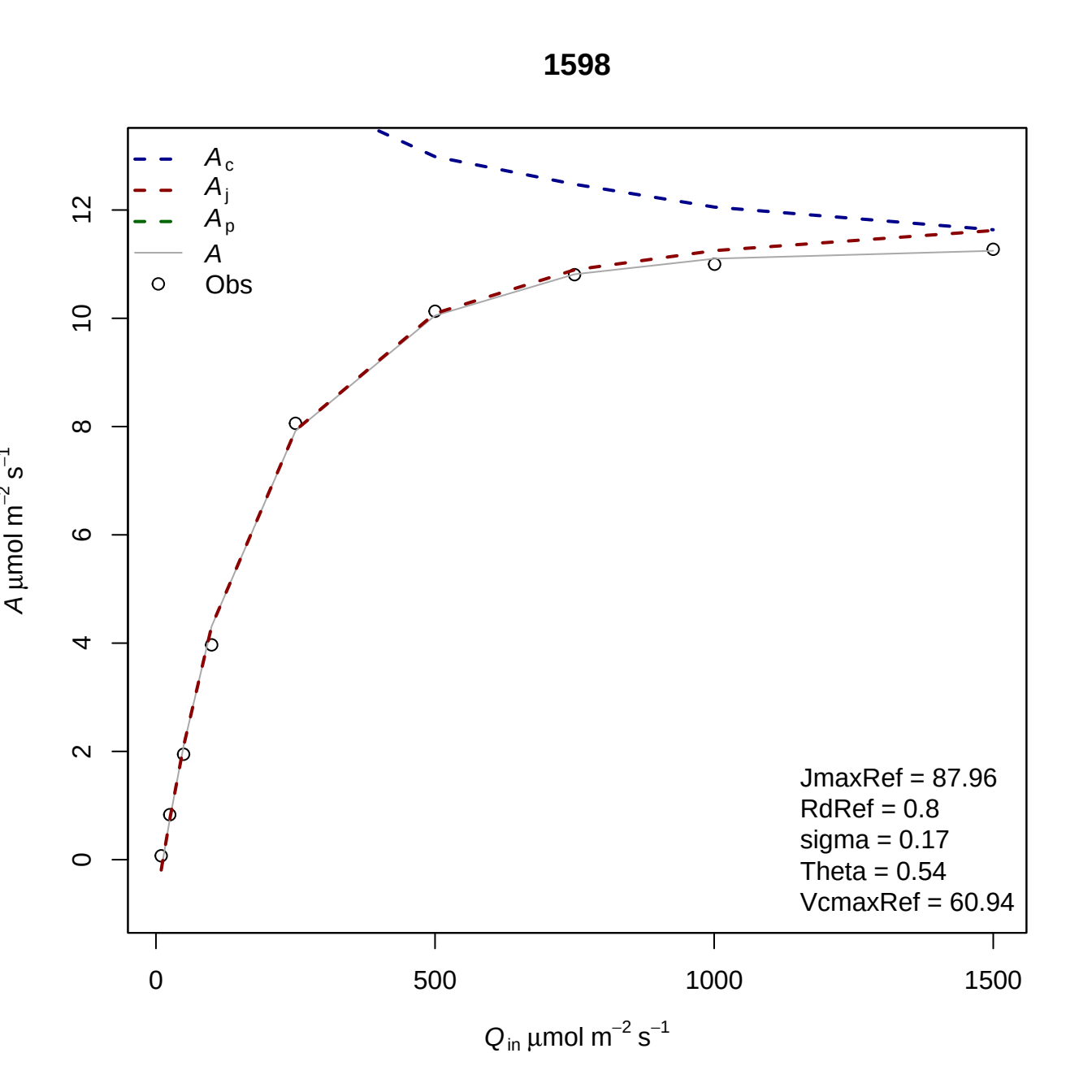

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 39.66
RdRef = 0.81
sigma = 0.13
Theta = 0
VcmaxRef = 22.41
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

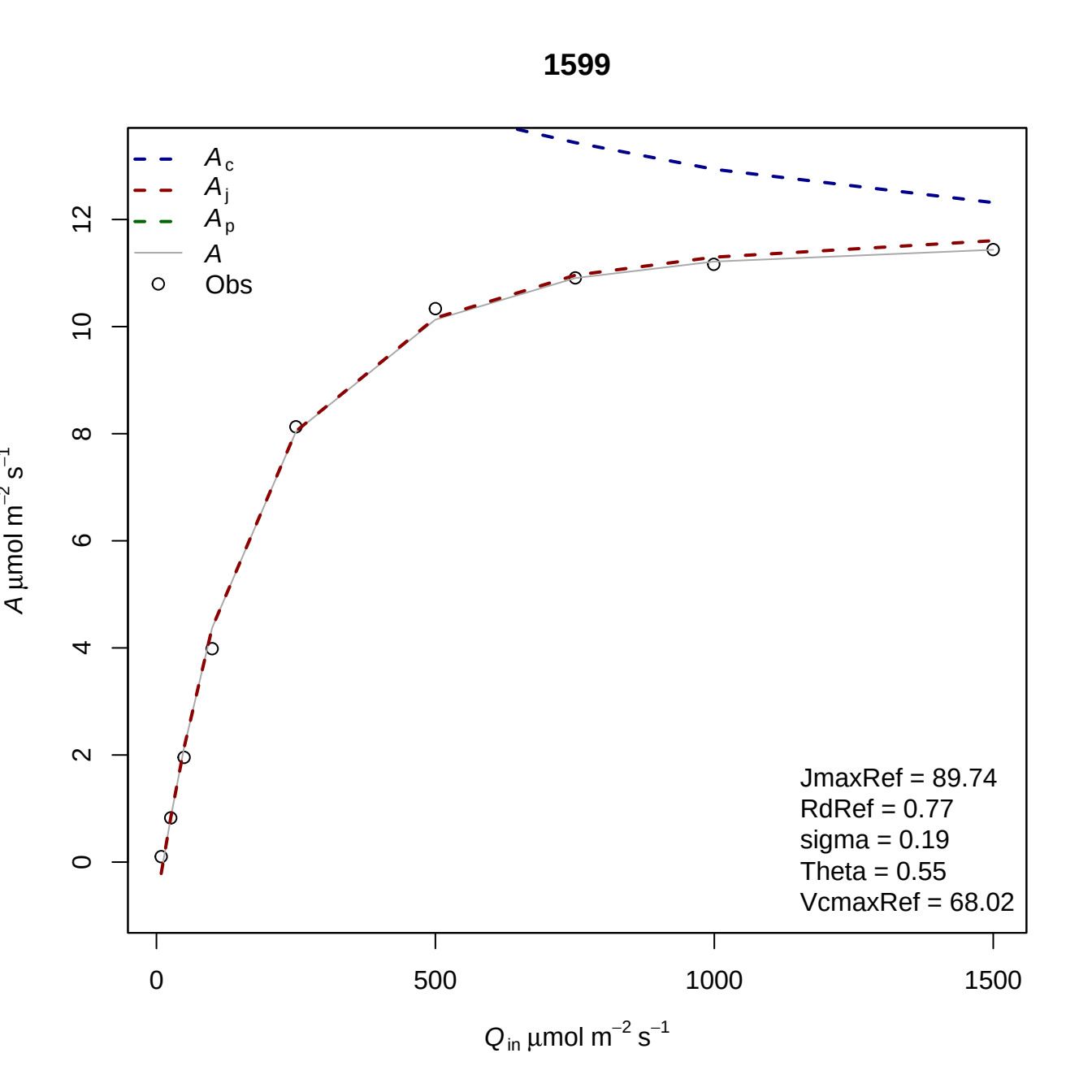

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 47.36
RdRef = 0.63
sigma = 0.09
Theta = −0.03
VcmaxRef = 6289.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

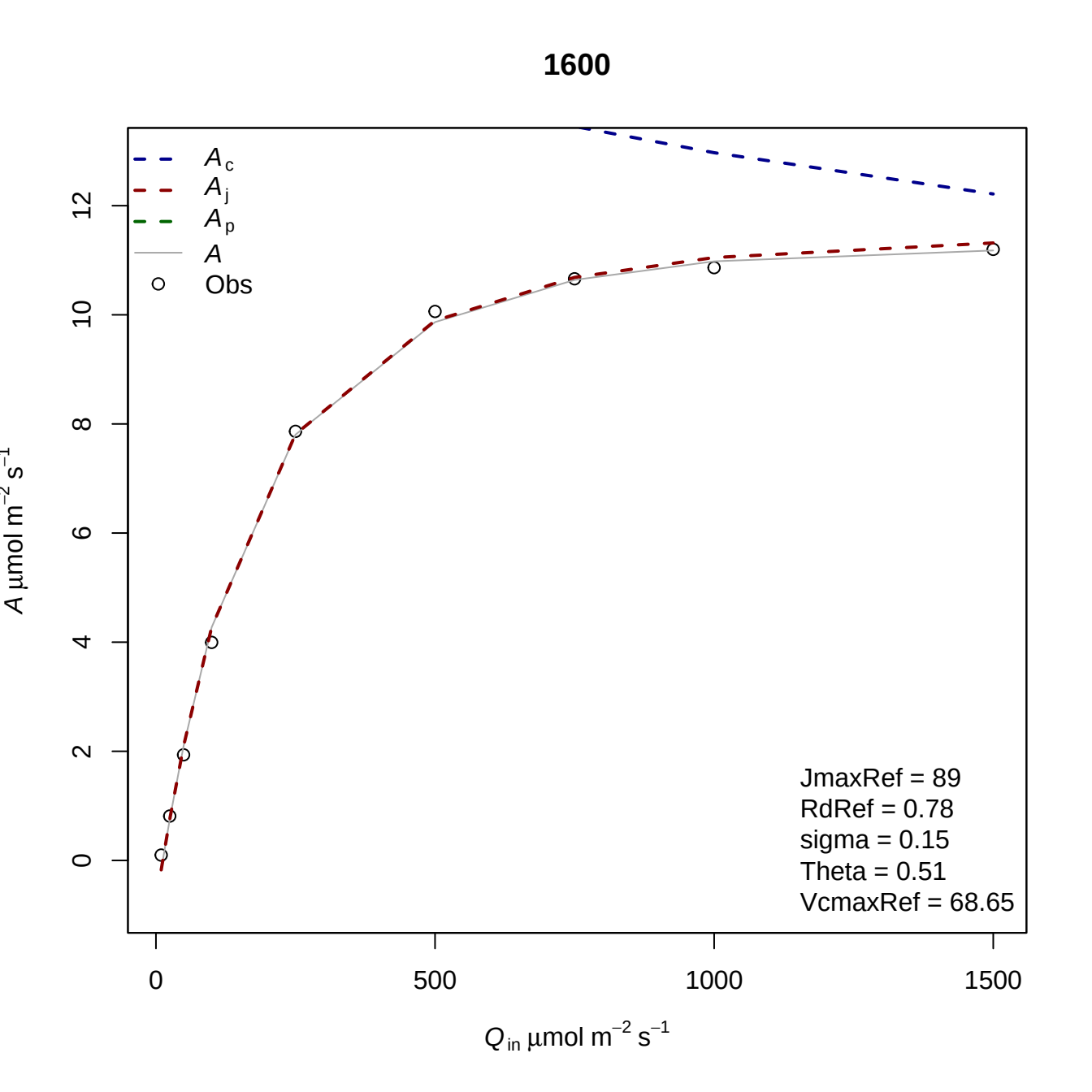

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 71.31
RdRef = 0.63
sigma = 0.13
Theta = 0.33
VcmaxRef = 49.92
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

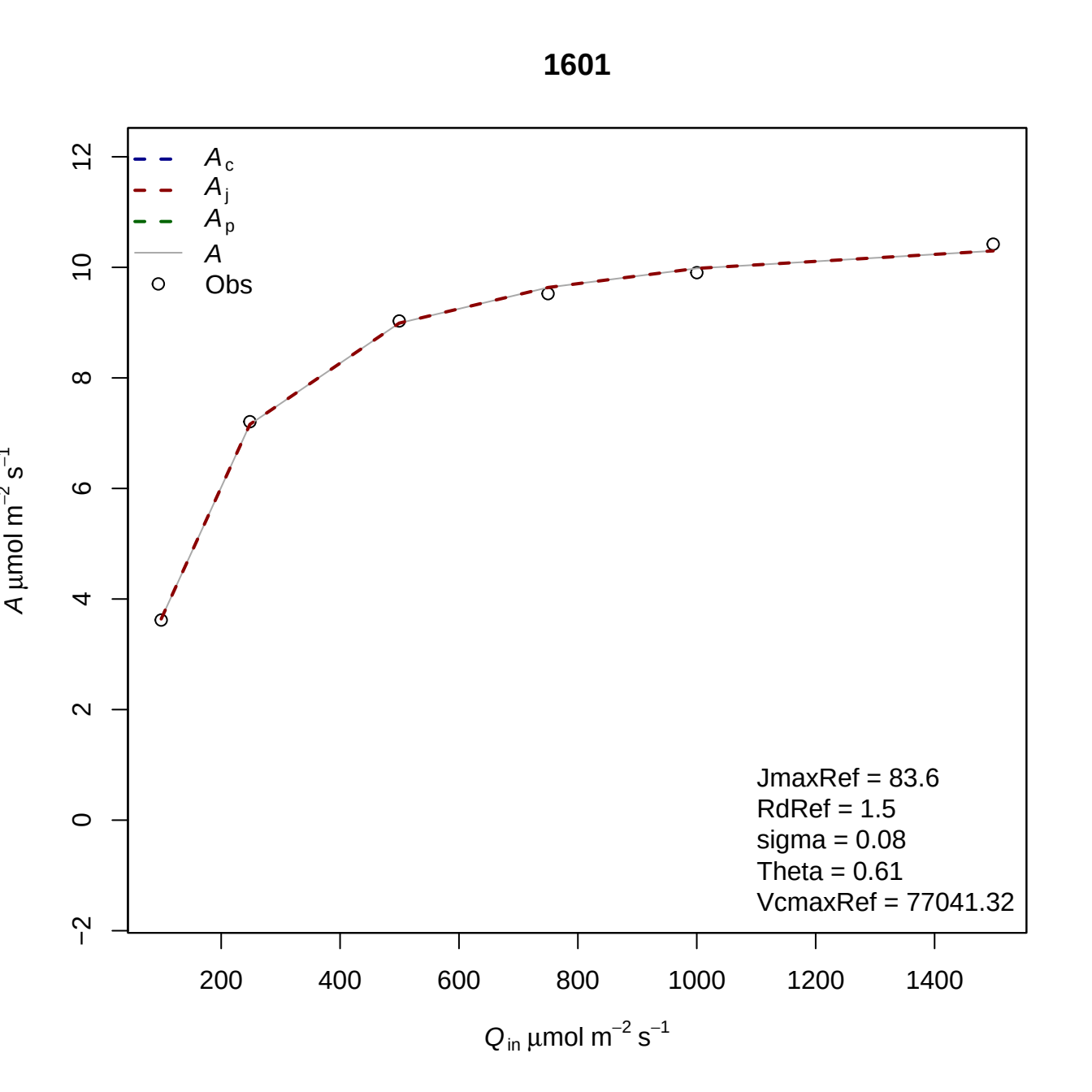

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 93.52
RdRef = 0.7
sigma = 0.16
Theta = 0.27
VcmaxRef = 60.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
14
0
500
1000
1500

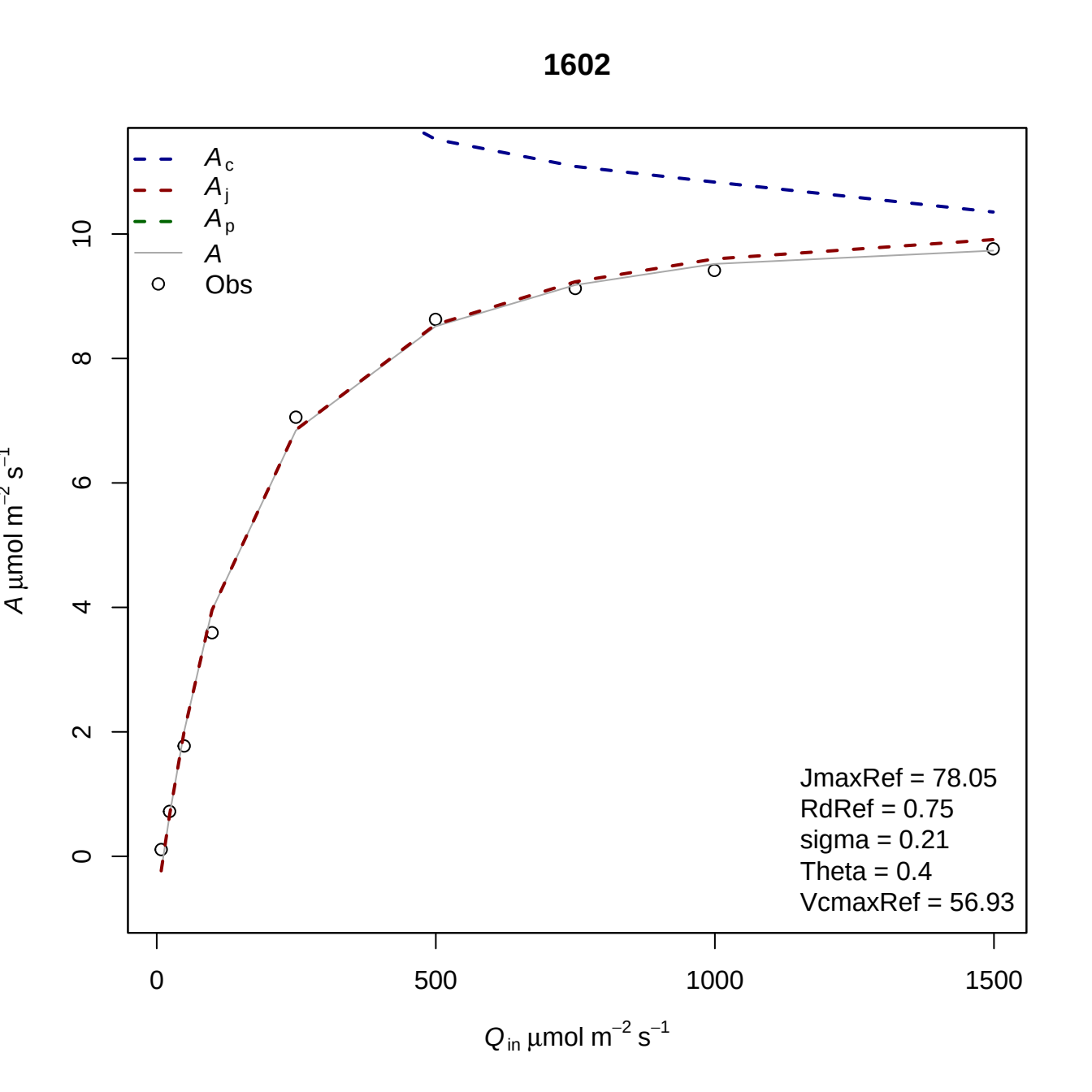

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 118.42
RdRef = 0.82
sigma = 0.23
Theta = 0.33
VcmaxRef = 76.17
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

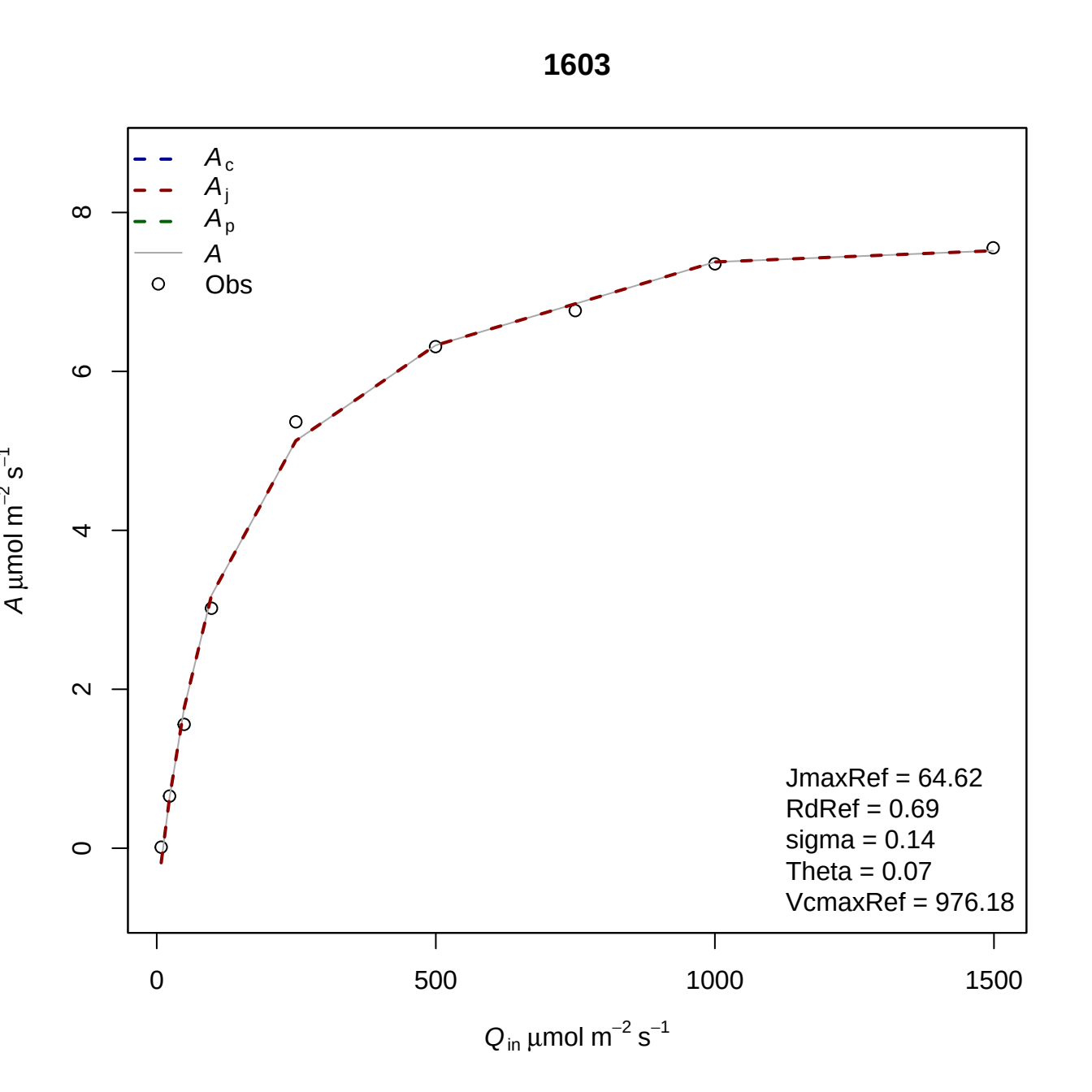

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 87.96
RdRef = 0.8
sigma = 0.17
Theta = 0.54
VcmaxRef = 60.94
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

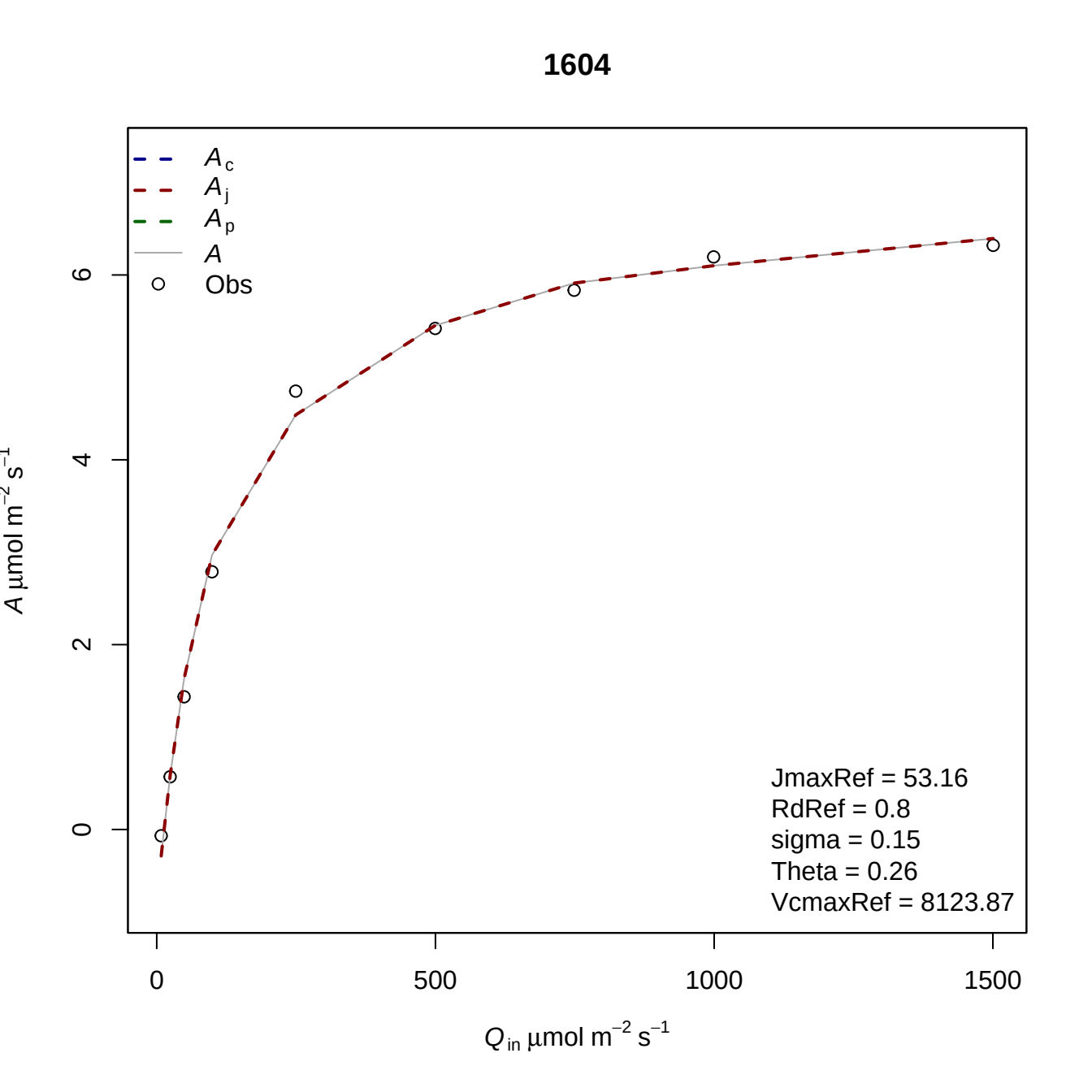

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 89.74
RdRef = 0.77
sigma = 0.19
Theta = 0.55
VcmaxRef = 68.02
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

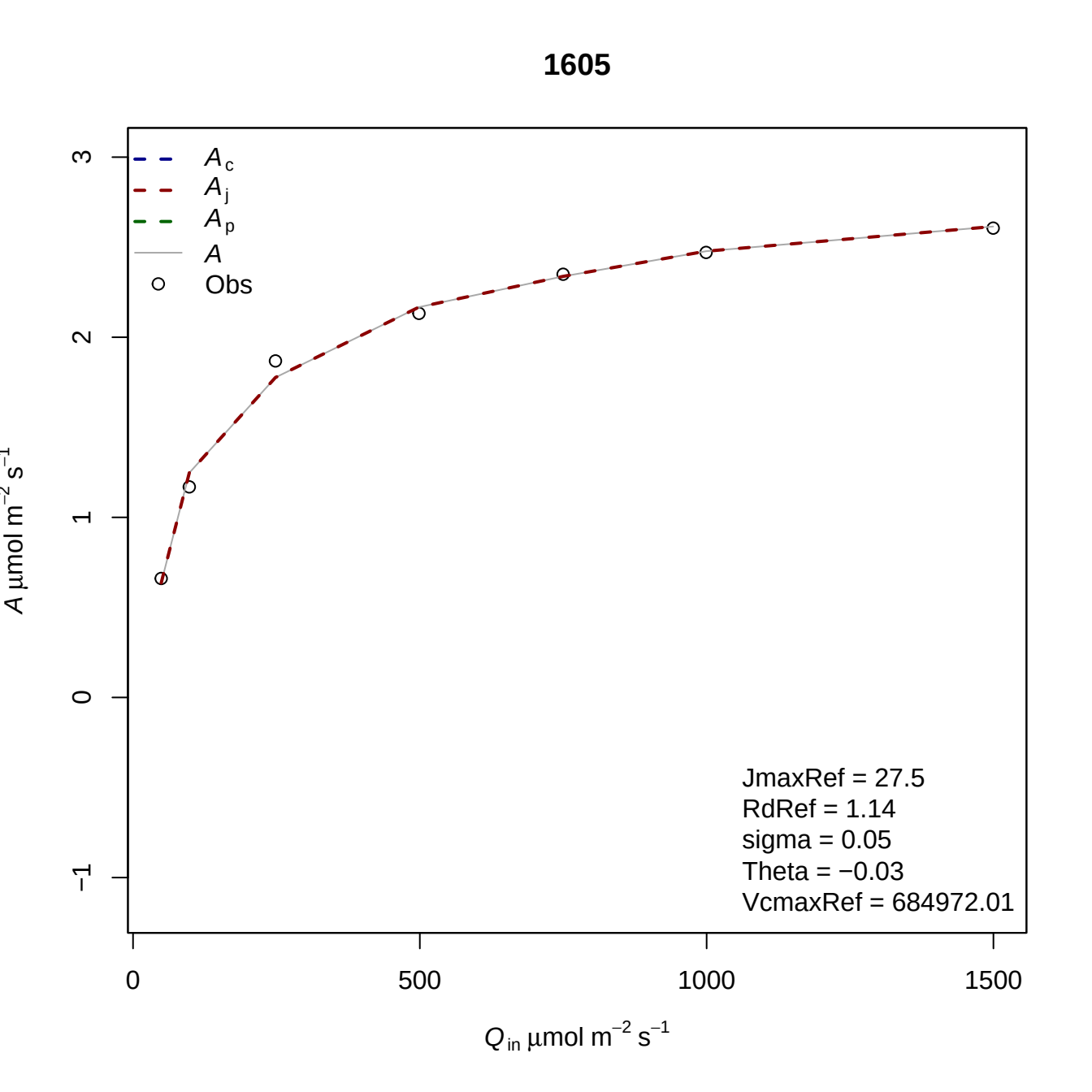

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 89
RdRef = 0.78
sigma = 0.15
Theta = 0.51
VcmaxRef = 68.65
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

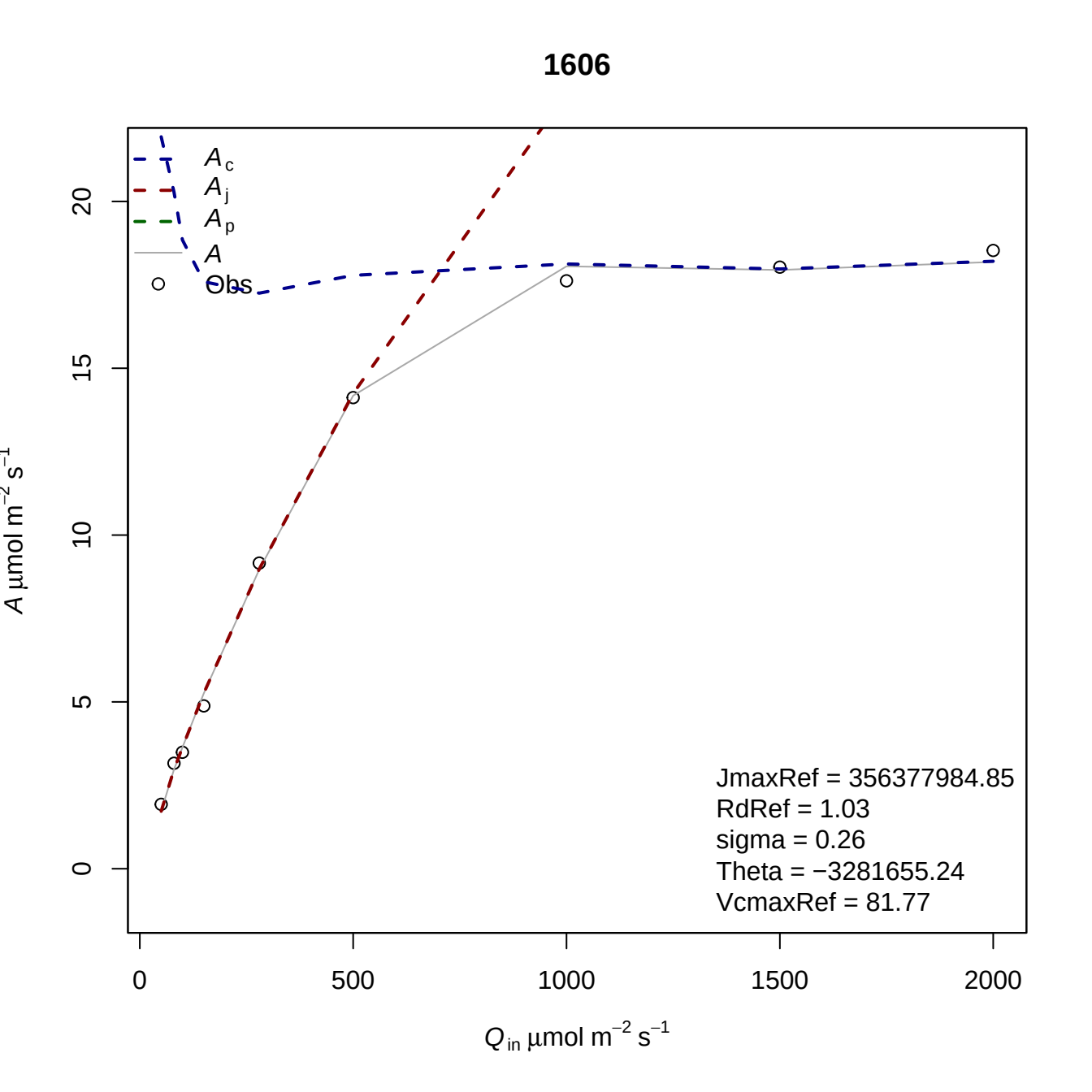

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 83.6
RdRef = 1.5
sigma = 0.08
Theta = 0.61
VcmaxRef = 77041.32
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
200
400
600
800
1000
1200
1400

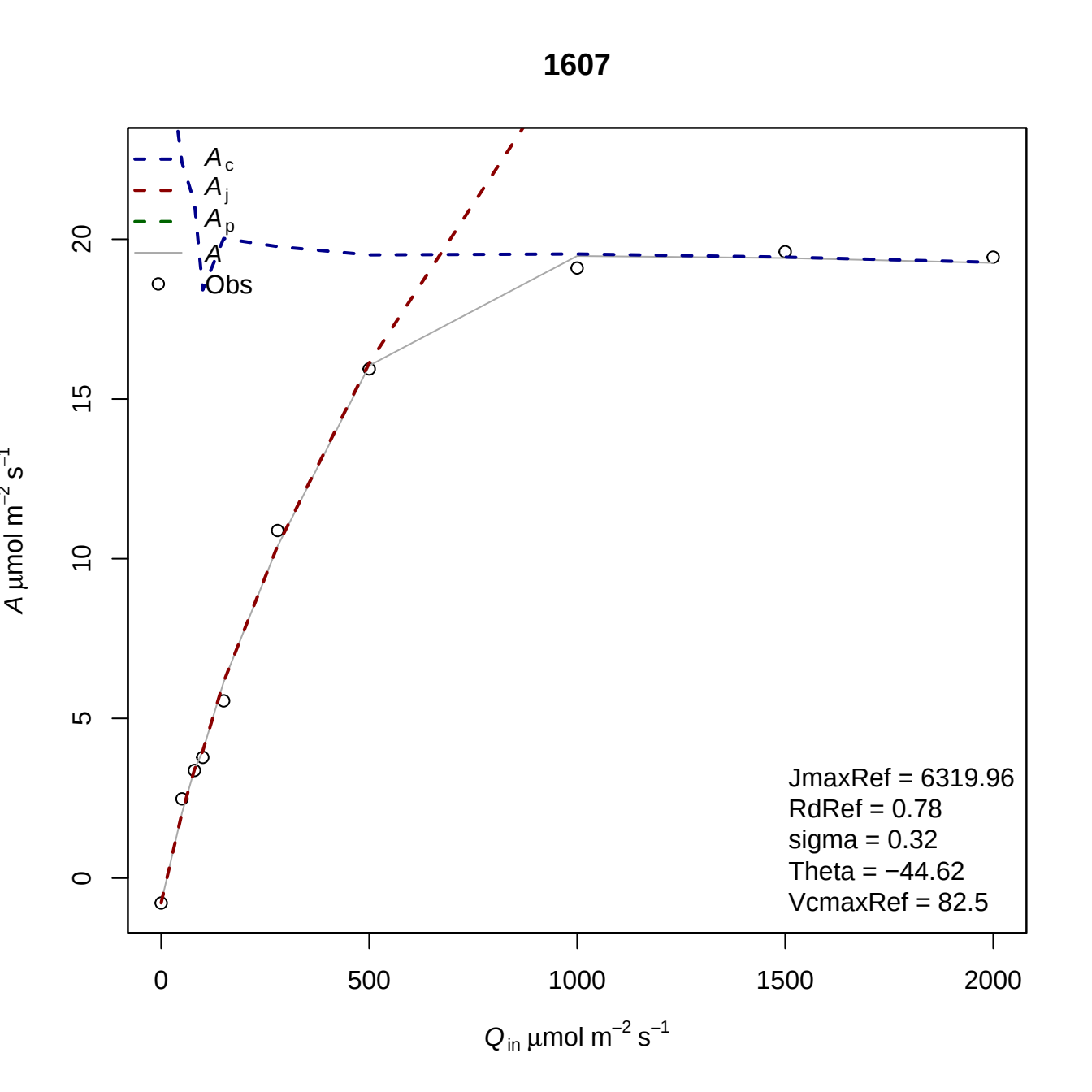

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 78.05
RdRef = 0.75
sigma = 0.21
Theta = 0.4
VcmaxRef = 56.93
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

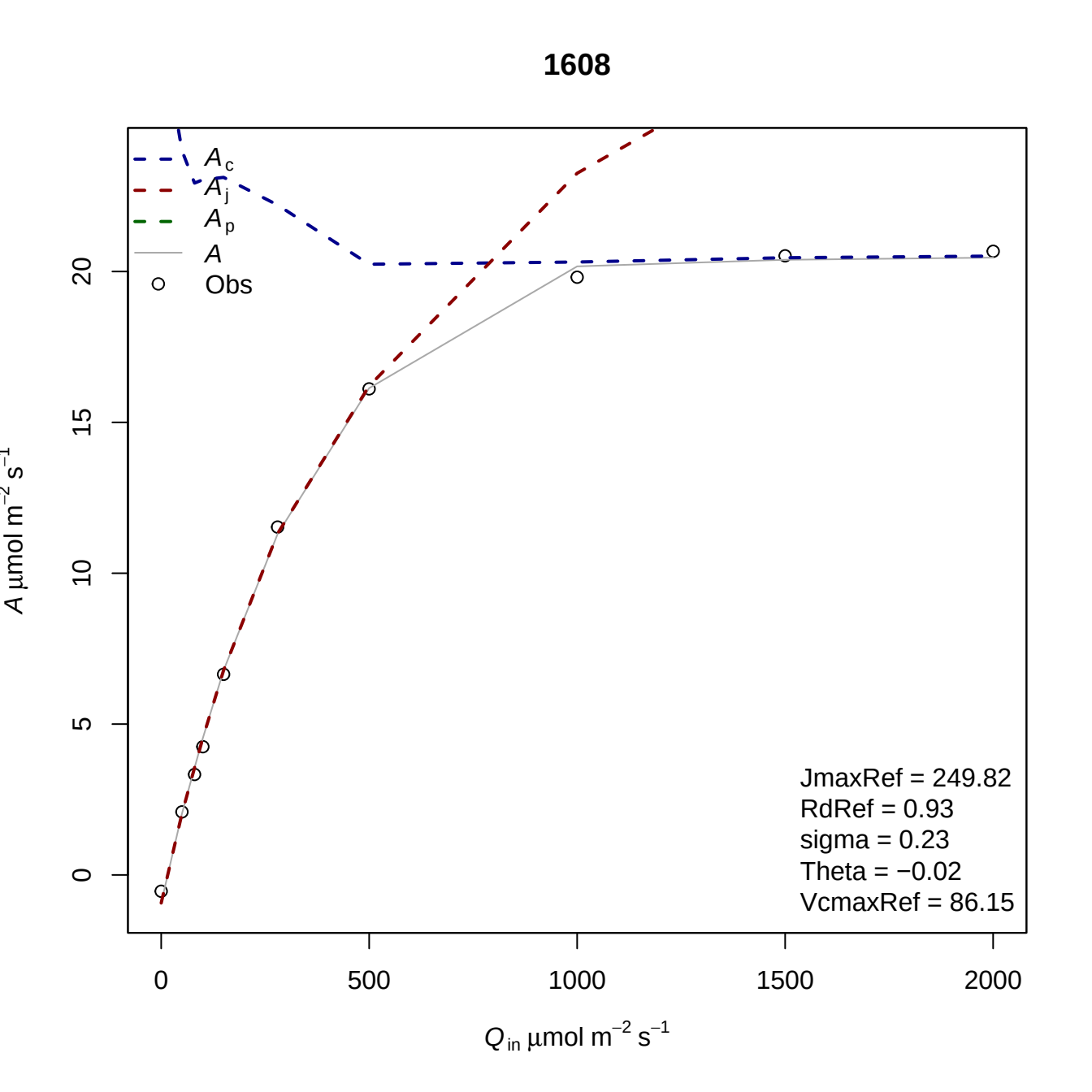

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 64.62
RdRef = 0.69
sigma = 0.14
Theta = 0.07
VcmaxRef = 976.18
A µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

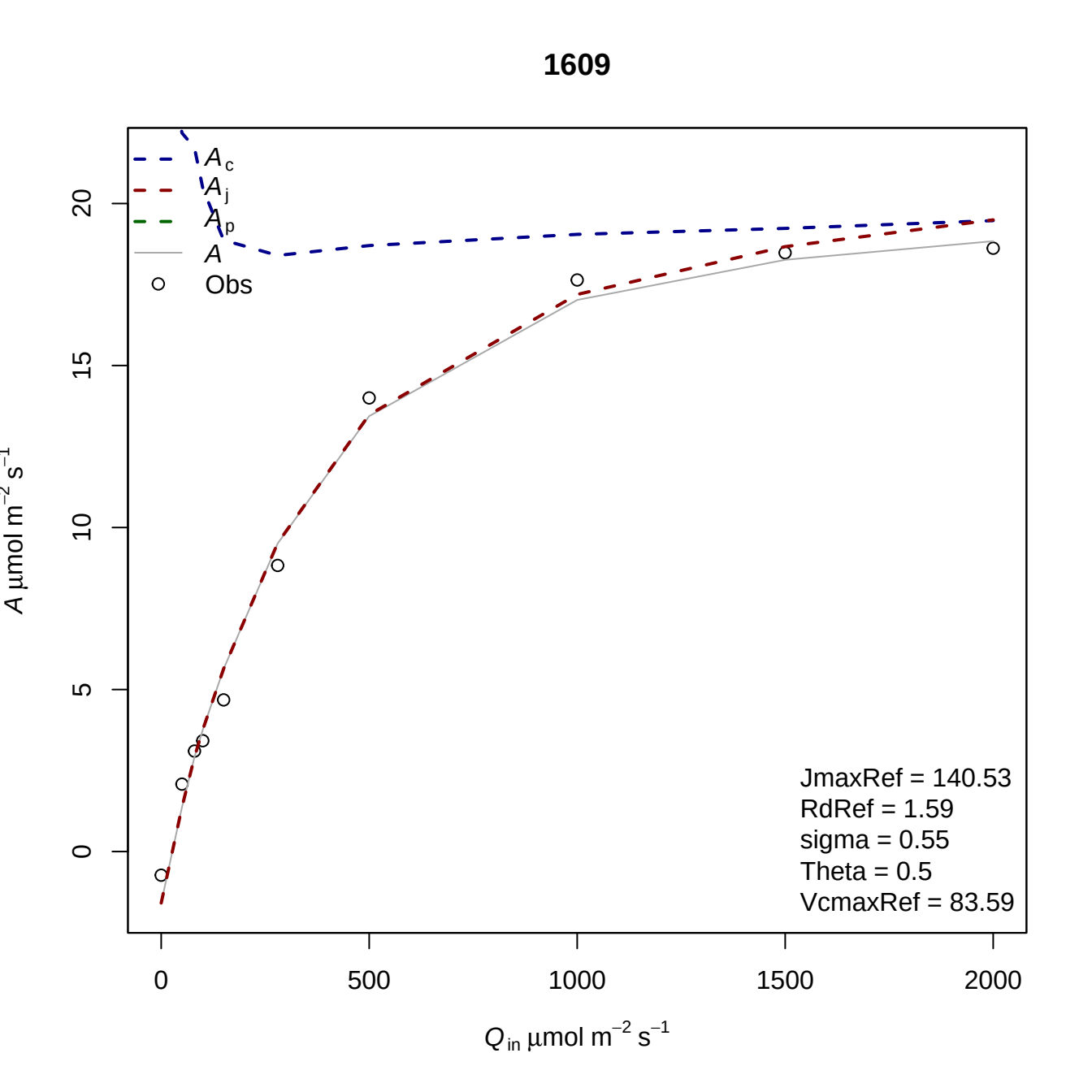

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 53.16
RdRef = 0.8
sigma = 0.15
Theta = 0.26
VcmaxRef = 8123.87
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

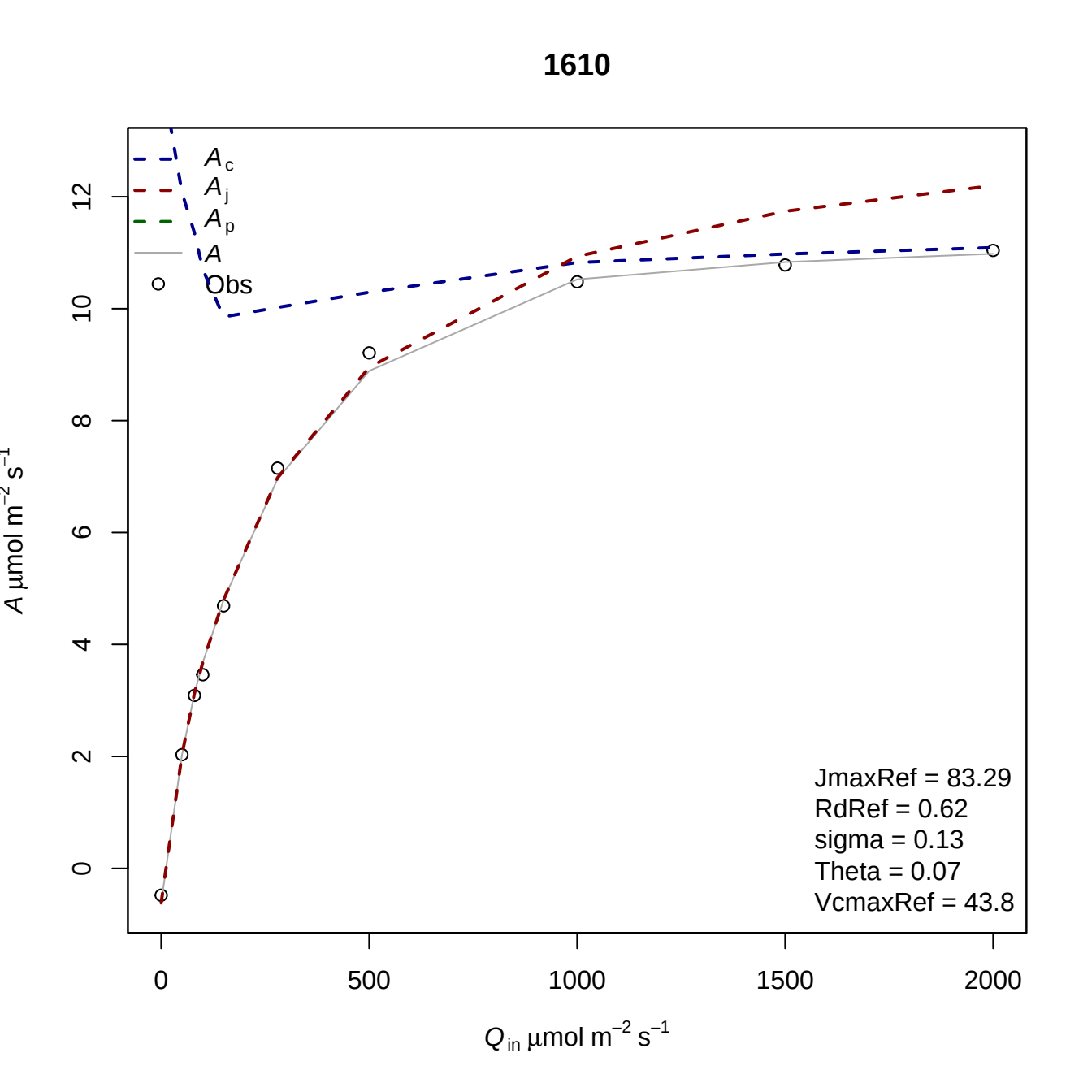

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 27.5
RdRef = 1.14
sigma = 0.05
Theta = −0.03
VcmaxRef = 684972.01
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

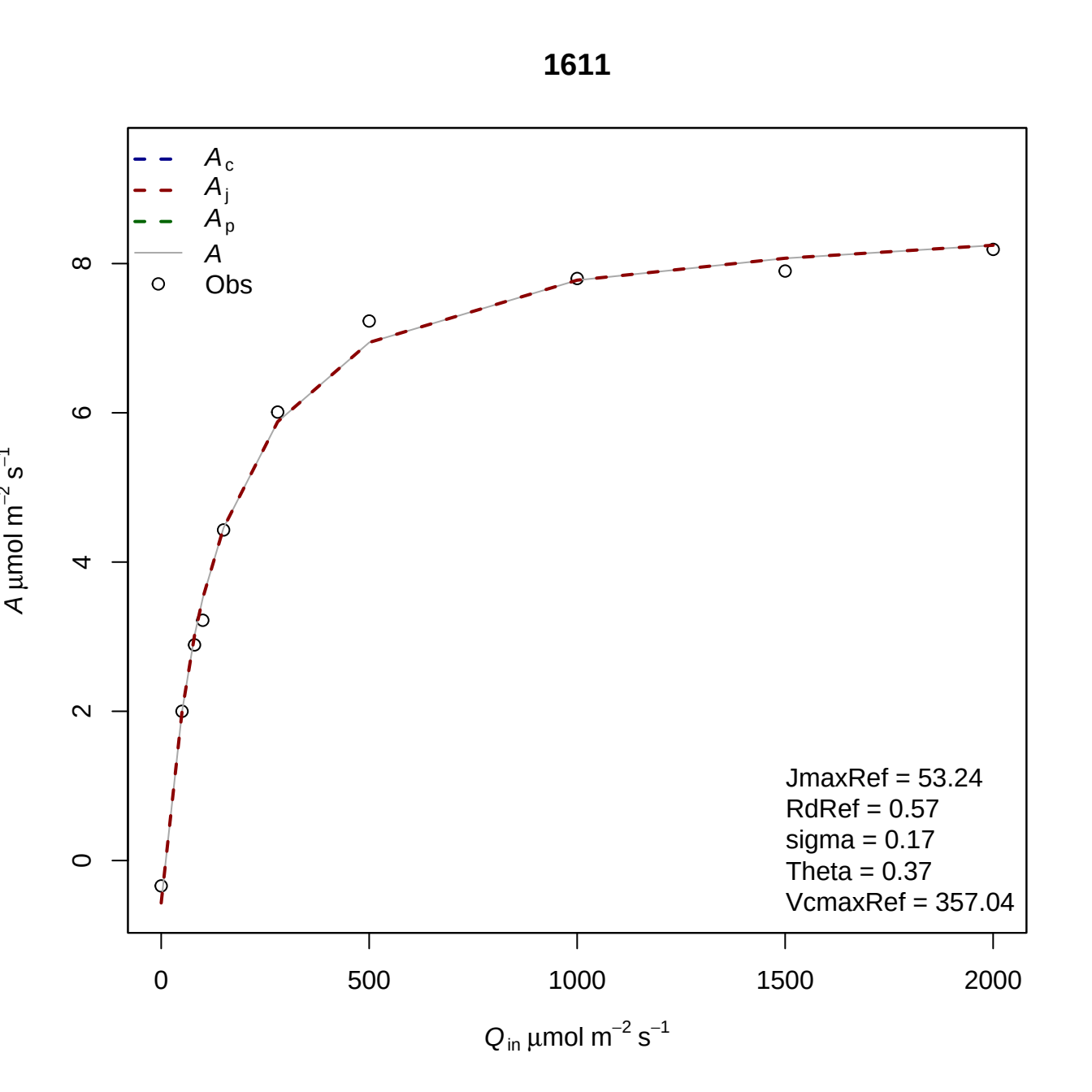

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 356377984.85
RdRef = 1.03
sigma = 0.26
Theta = −3281655.24
VcmaxRef = 81.77
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

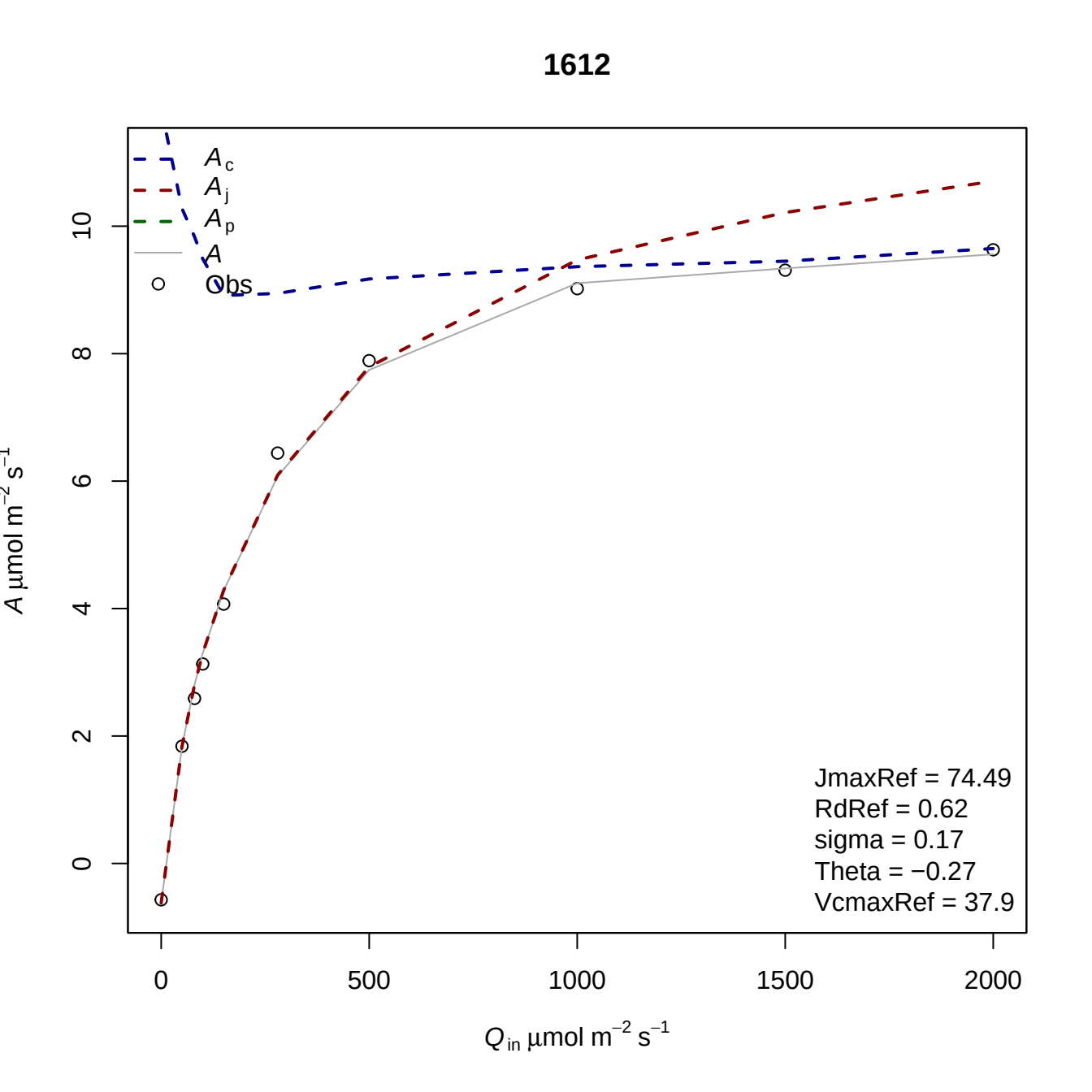

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 6319.96
RdRef = 0.78
sigma = 0.32
Theta = −44.62
VcmaxRef = 82.5
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

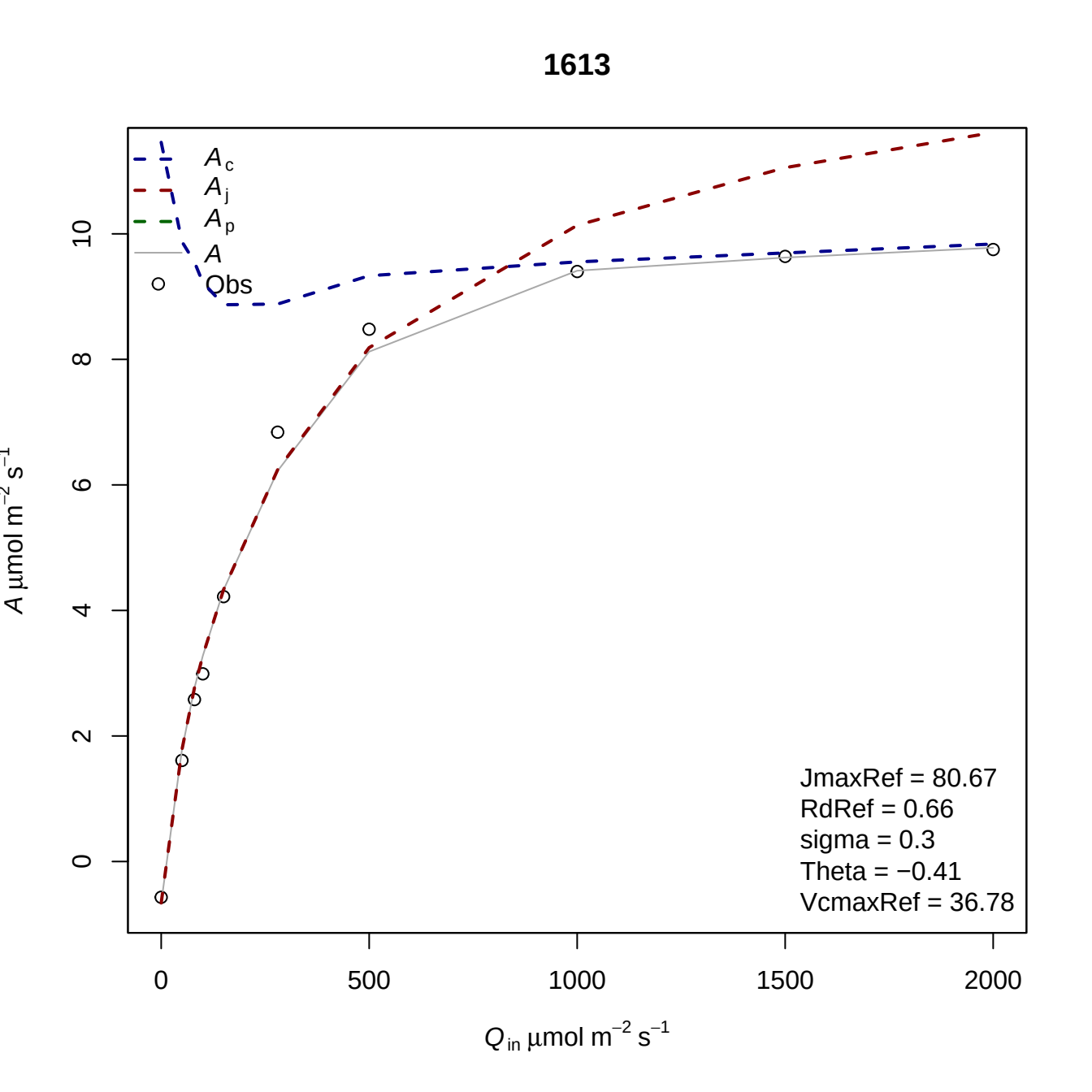

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 249.82
RdRef = 0.93
sigma = 0.23
Theta = −0.02
VcmaxRef = 86.15
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

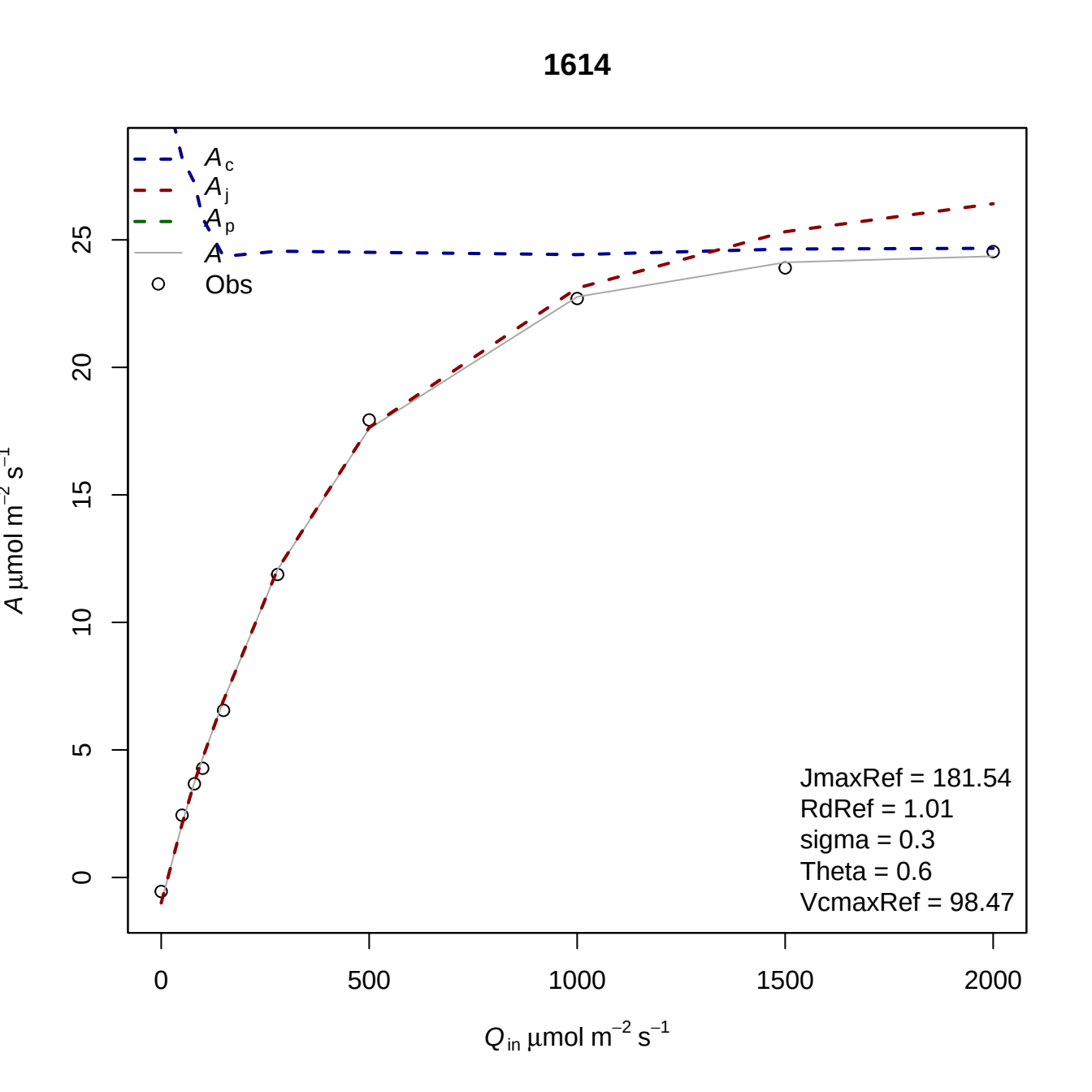

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 140.53
RdRef = 1.59
sigma = 0.55
Theta = 0.5
VcmaxRef = 83.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

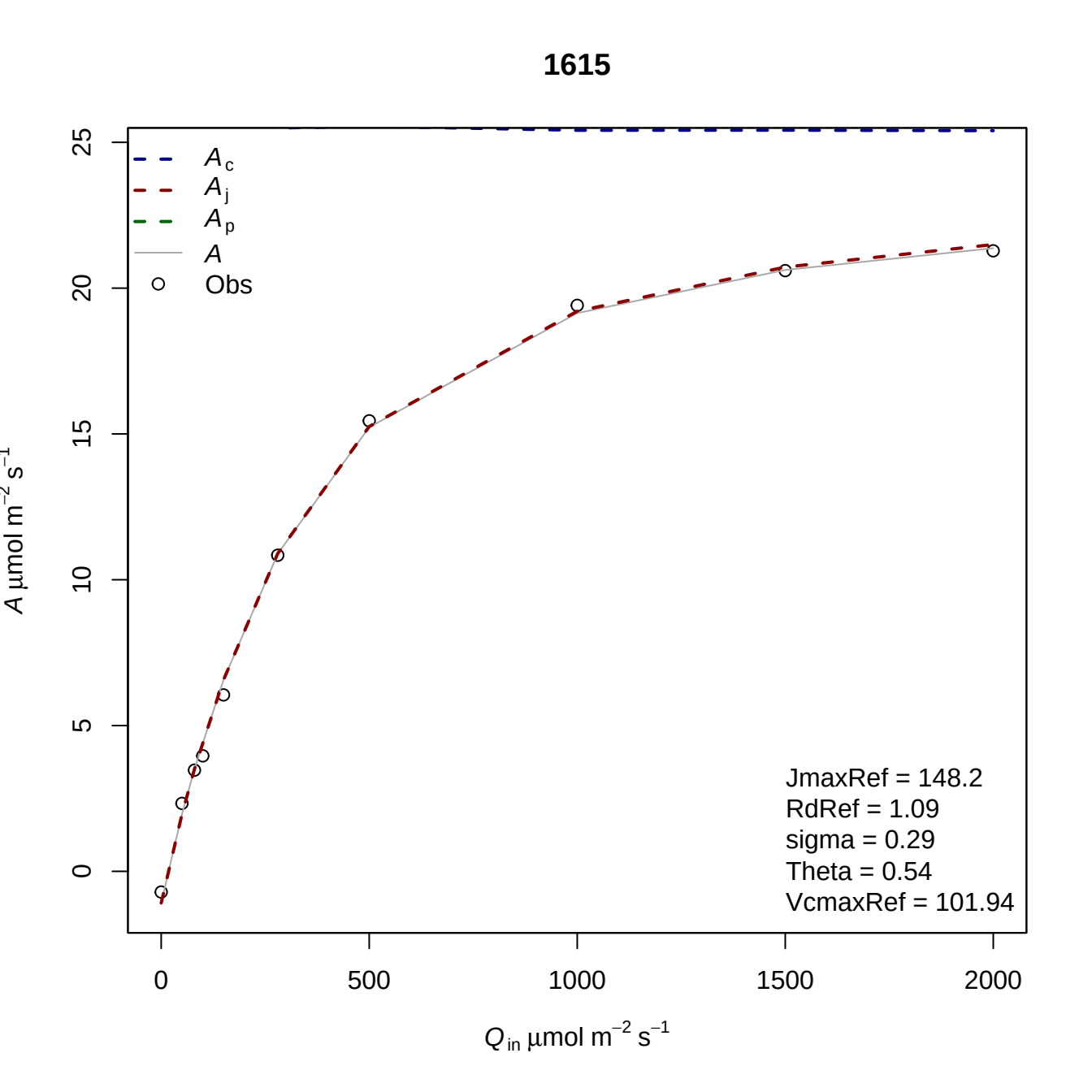

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 83.29
RdRef = 0.62
sigma = 0.13
Theta = 0.07
VcmaxRef = 43.8
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

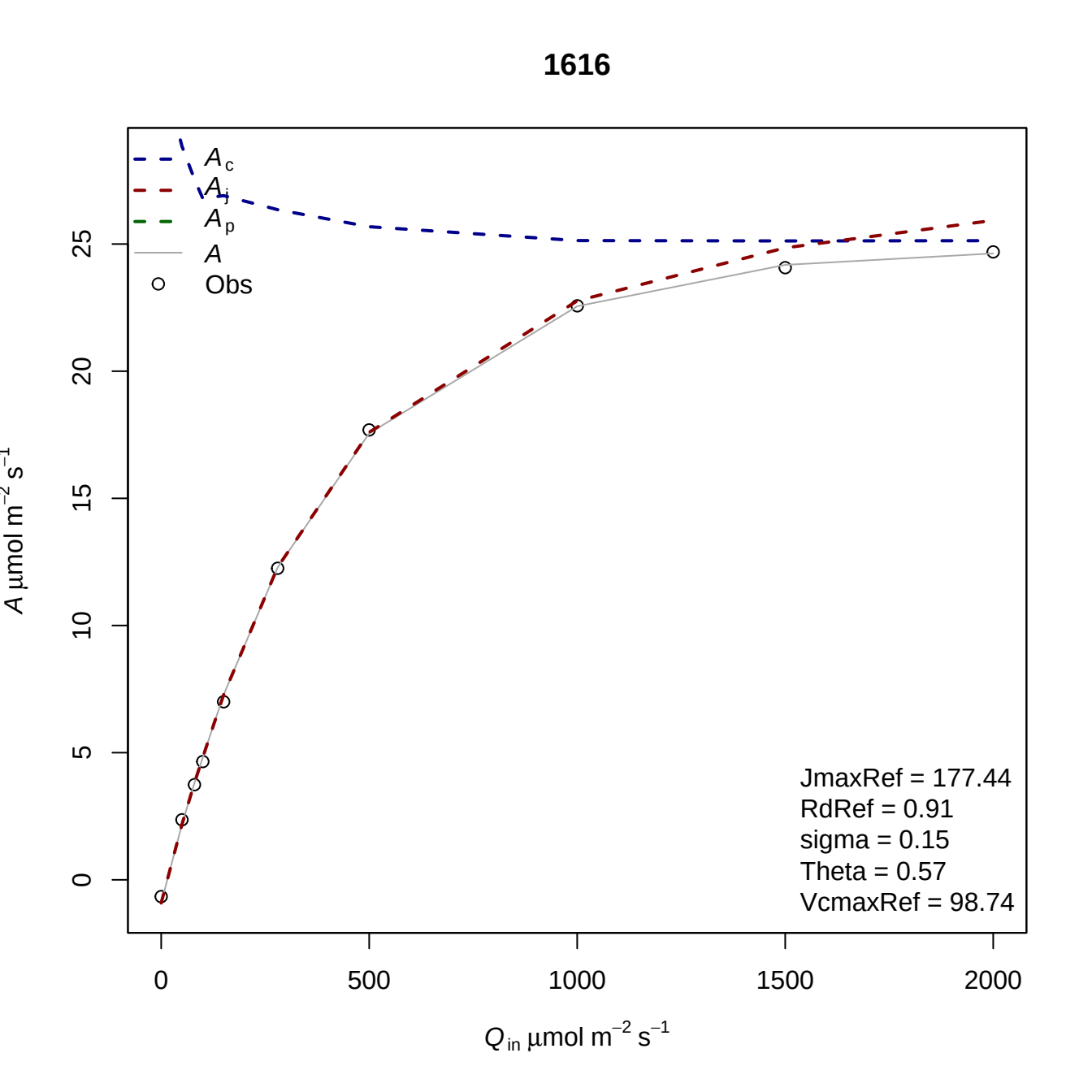

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 53.24
RdRef = 0.57
sigma = 0.17
Theta = 0.37
VcmaxRef = 357.04
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

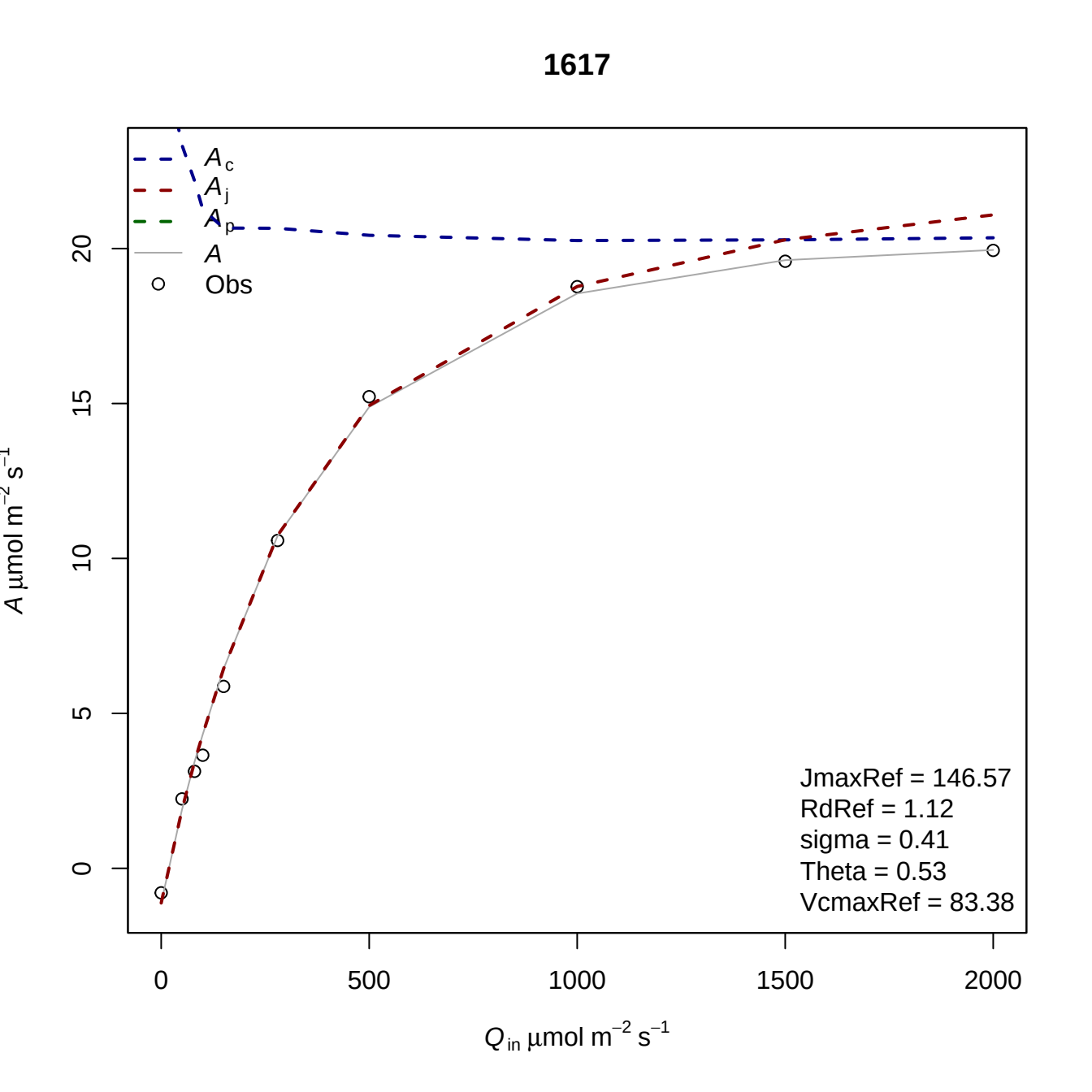

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 74.49
RdRef = 0.62
sigma = 0.17
Theta = −0.27
VcmaxRef = 37.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

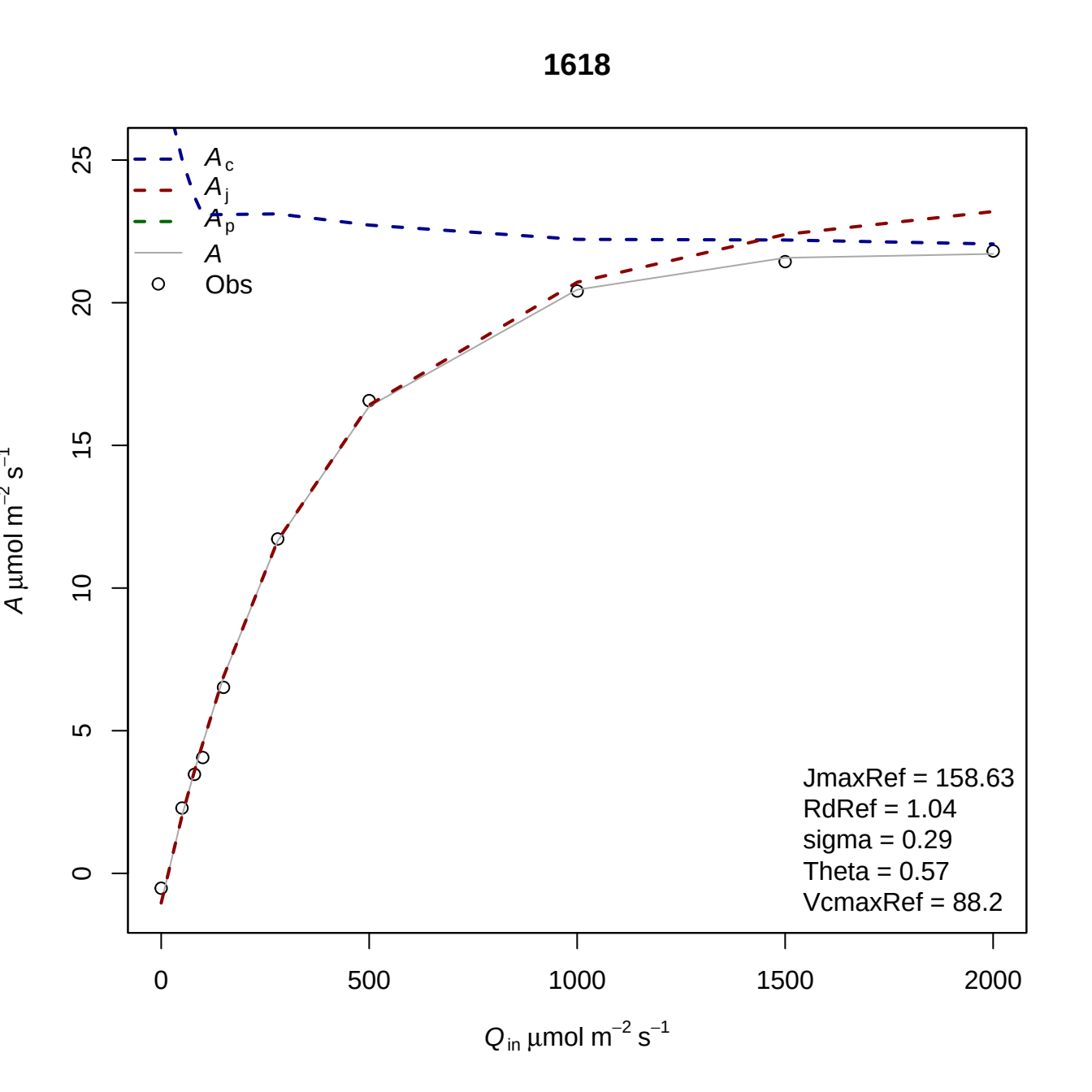

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 80.67
RdRef = 0.66
sigma = 0.3
Theta = −0.41
VcmaxRef = 36.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

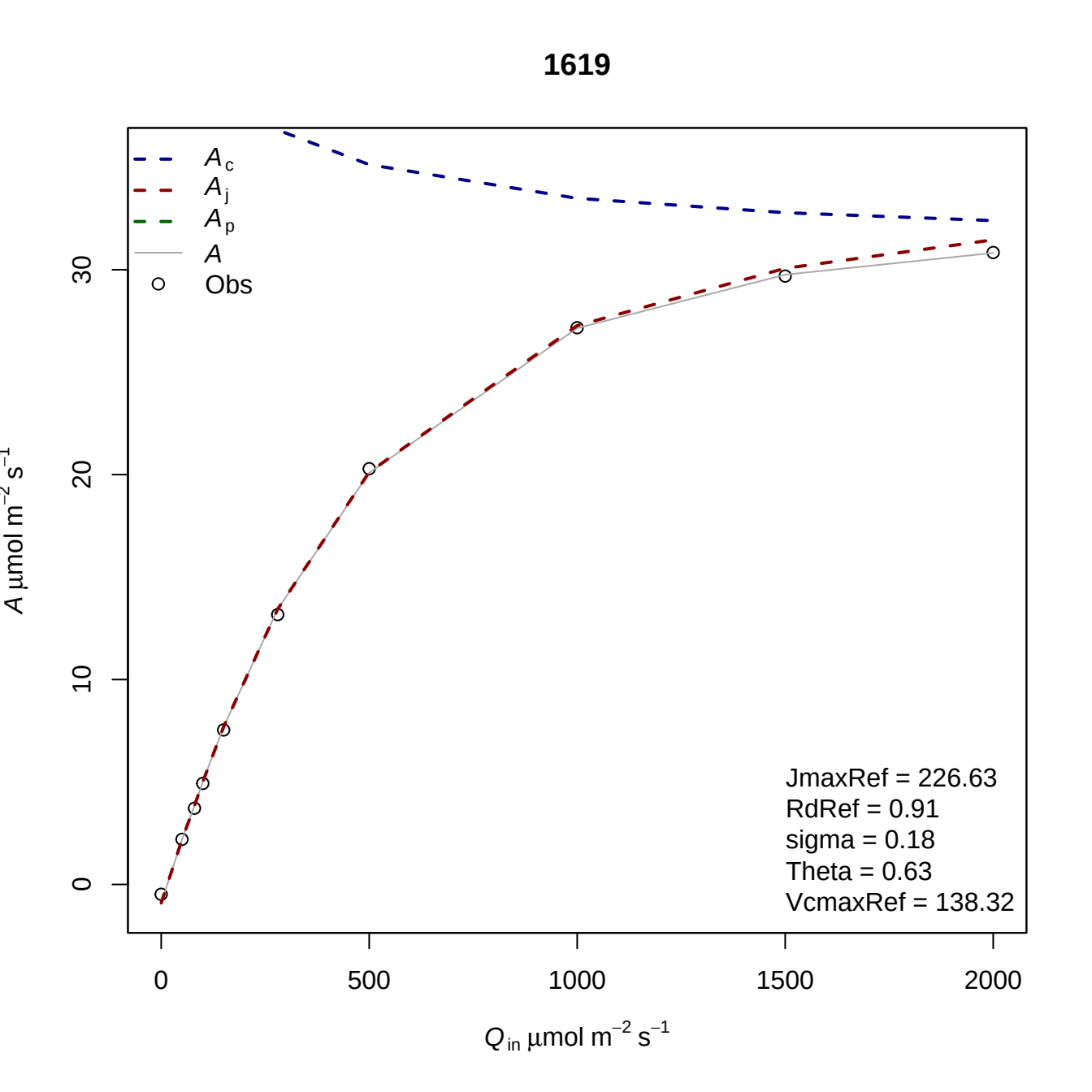

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 181.54
RdRef = 1.01
sigma = 0.3
Theta = 0.6
VcmaxRef = 98.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

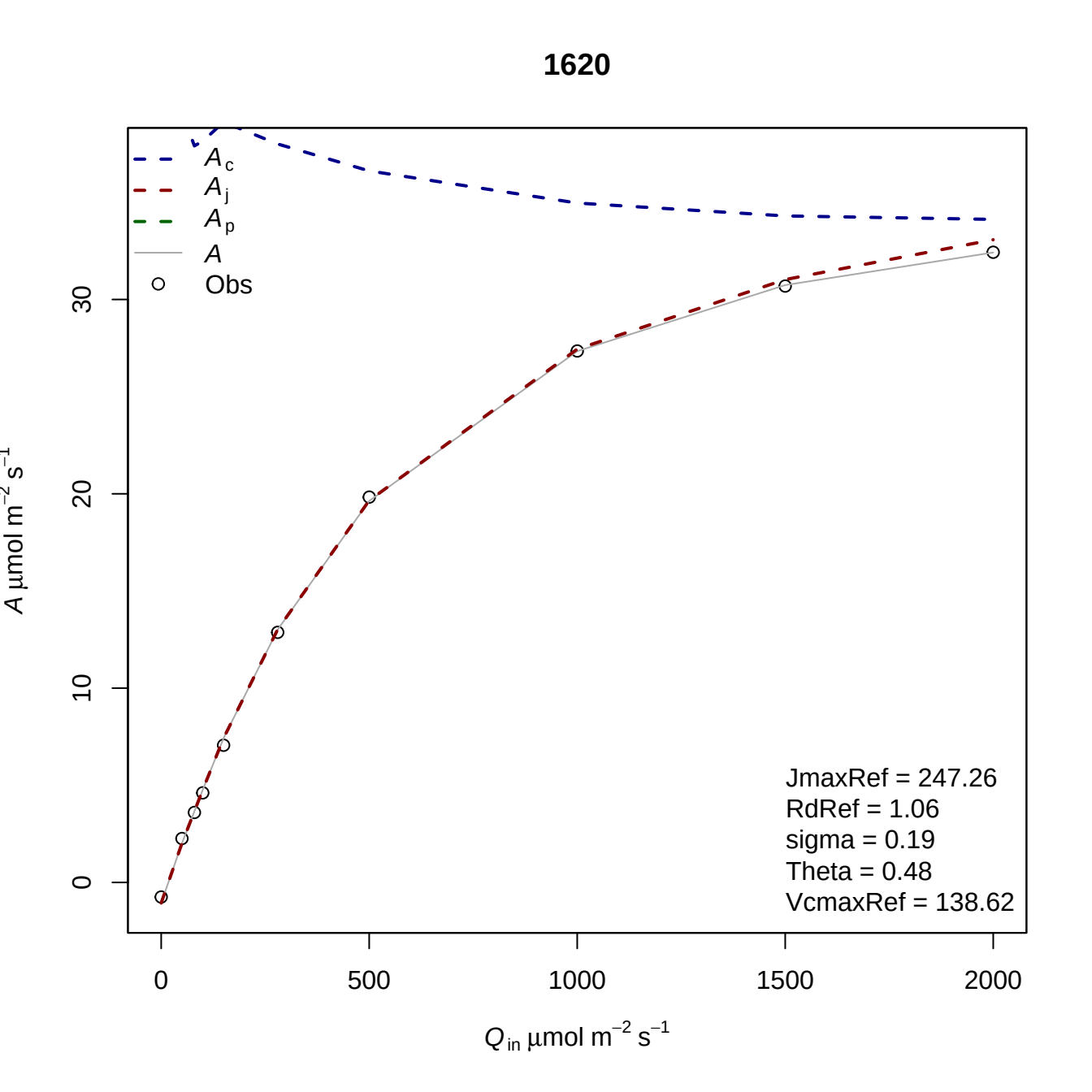

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 148.2
RdRef = 1.09
sigma = 0.29
Theta = 0.54
VcmaxRef = 101.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

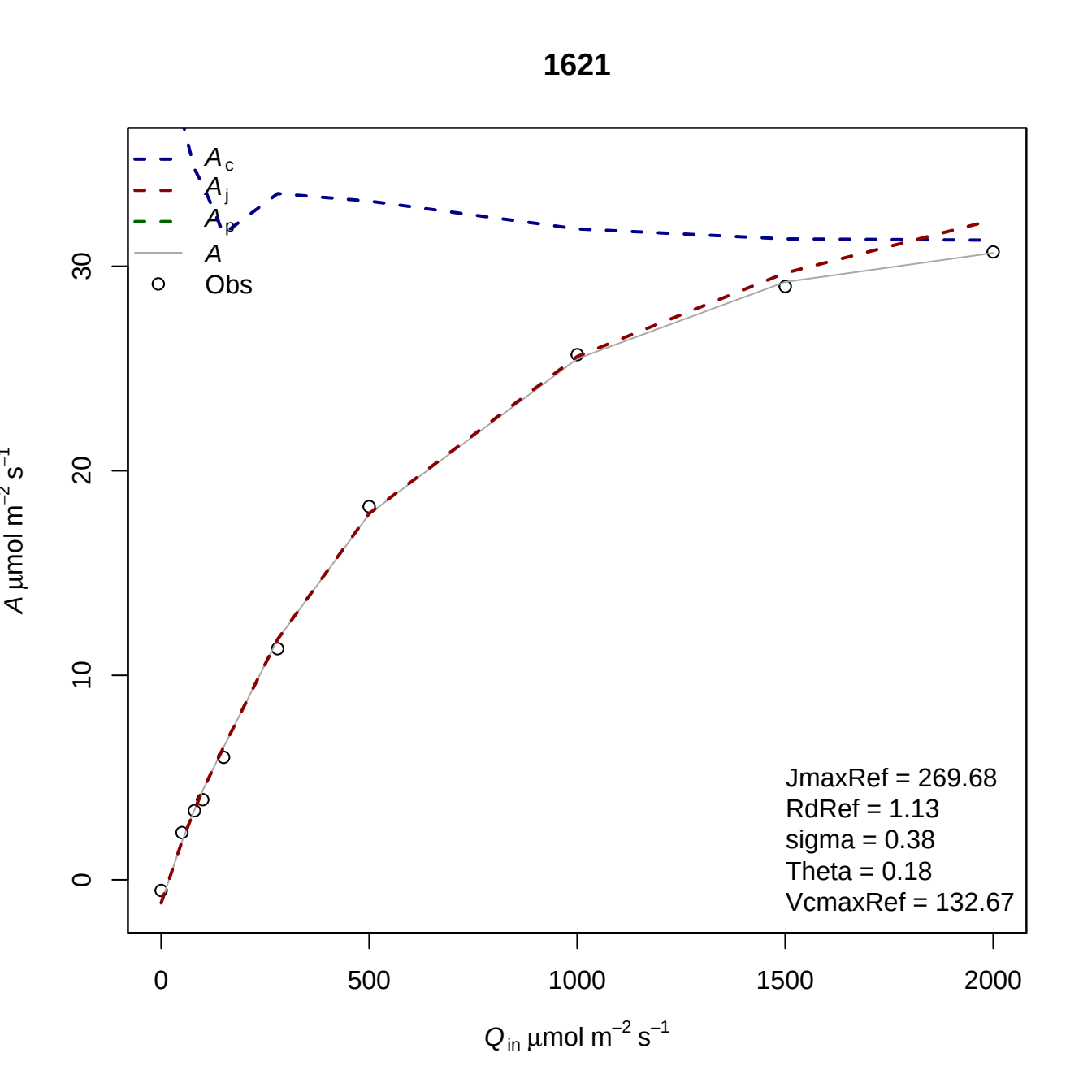

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 177.44
RdRef = 0.91
sigma = 0.15
Theta = 0.57
VcmaxRef = 98.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

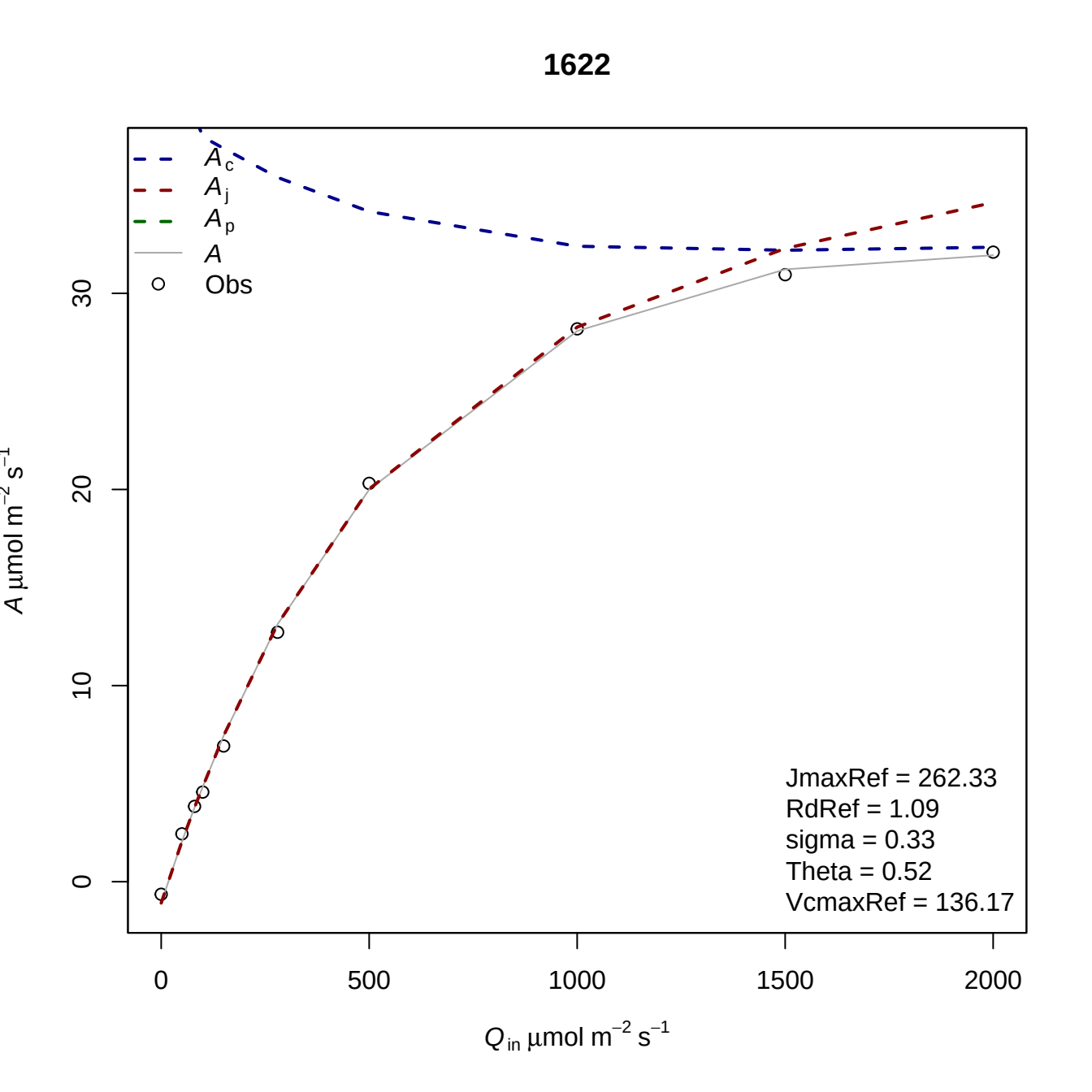

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 146.57
RdRef = 1.12
sigma = 0.41
Theta = 0.53
VcmaxRef = 83.38
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

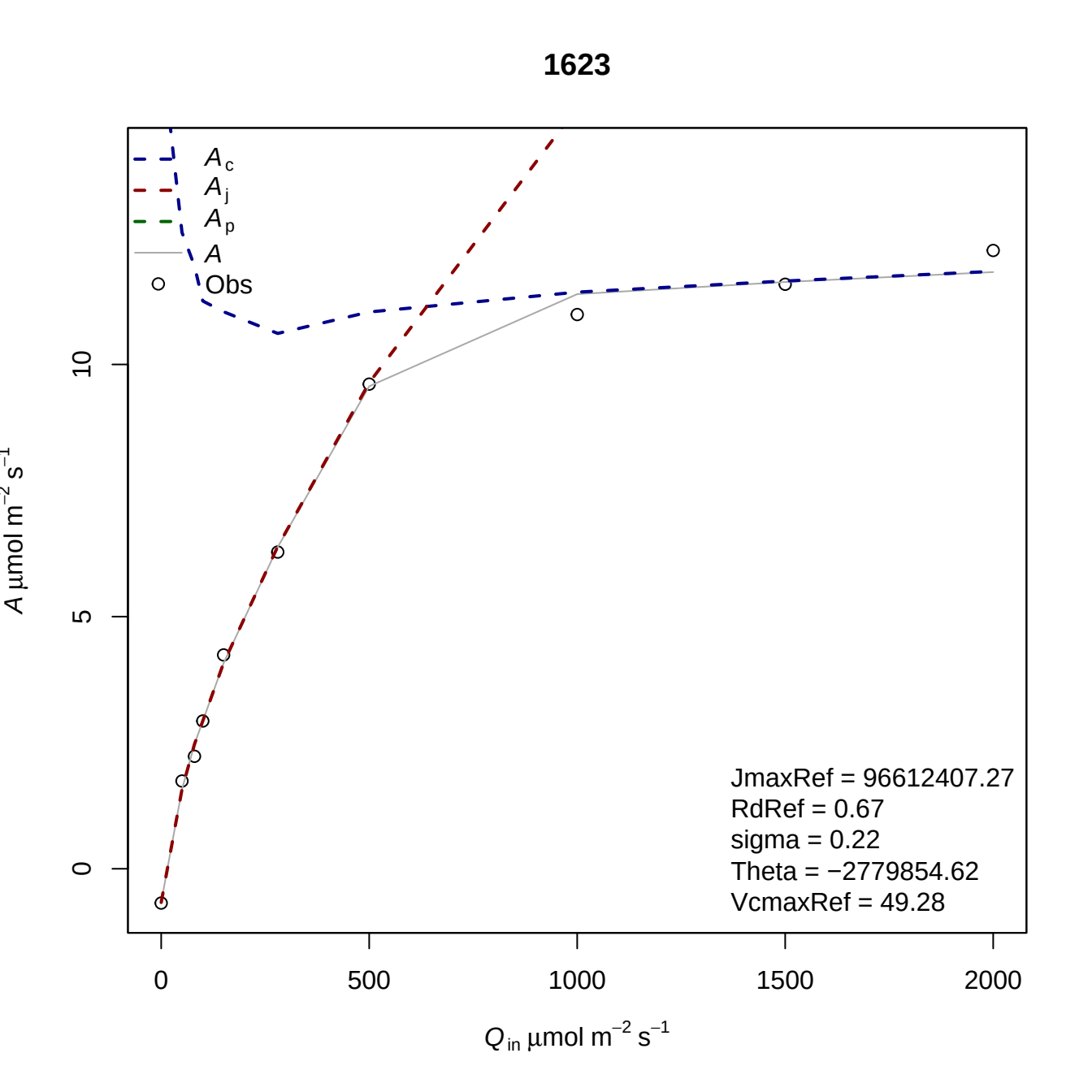

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 158.63
RdRef = 1.04
sigma = 0.29
Theta = 0.57
VcmaxRef = 88.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

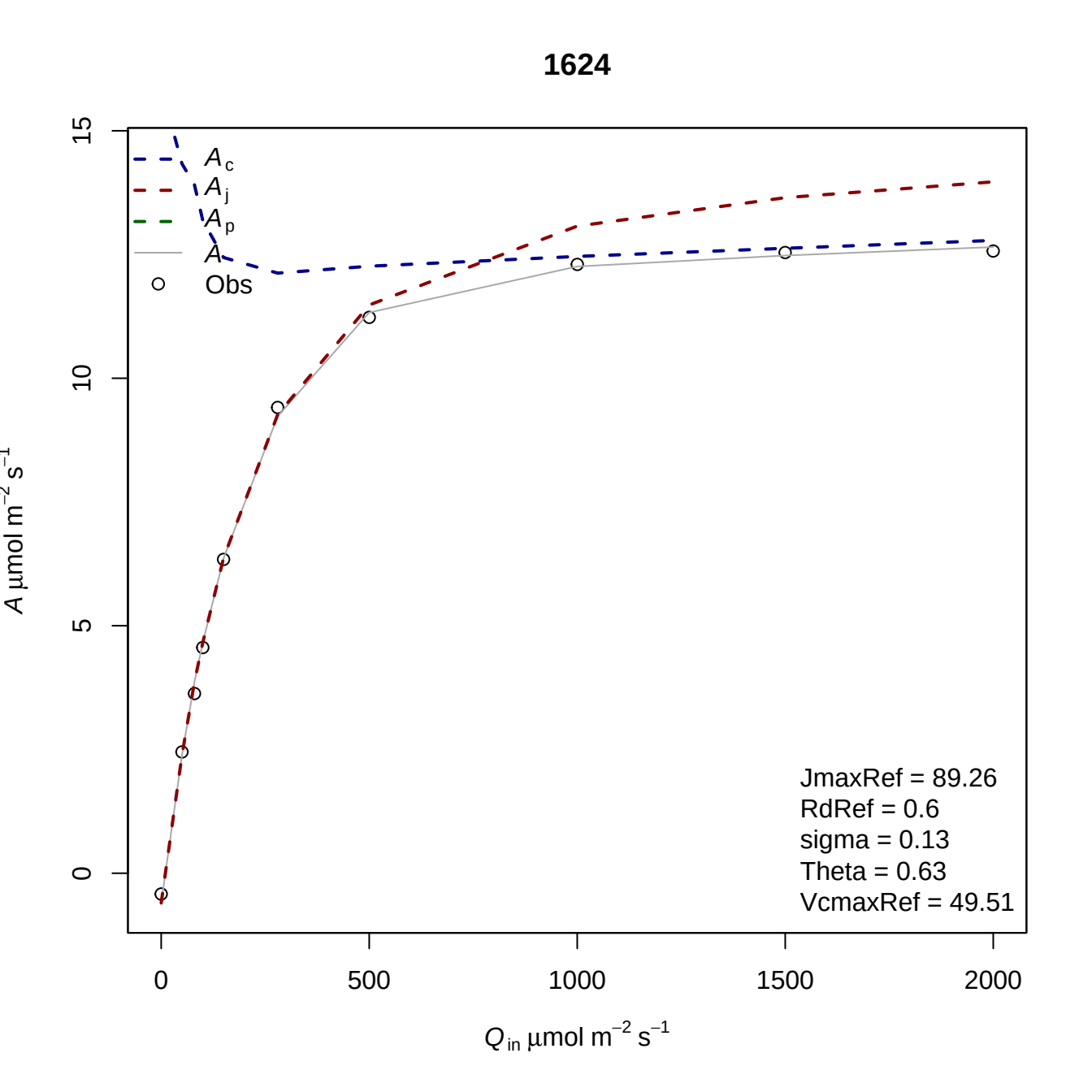

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 226.63
RdRef = 0.91
sigma = 0.18
Theta = 0.63
VcmaxRef = 138.32
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

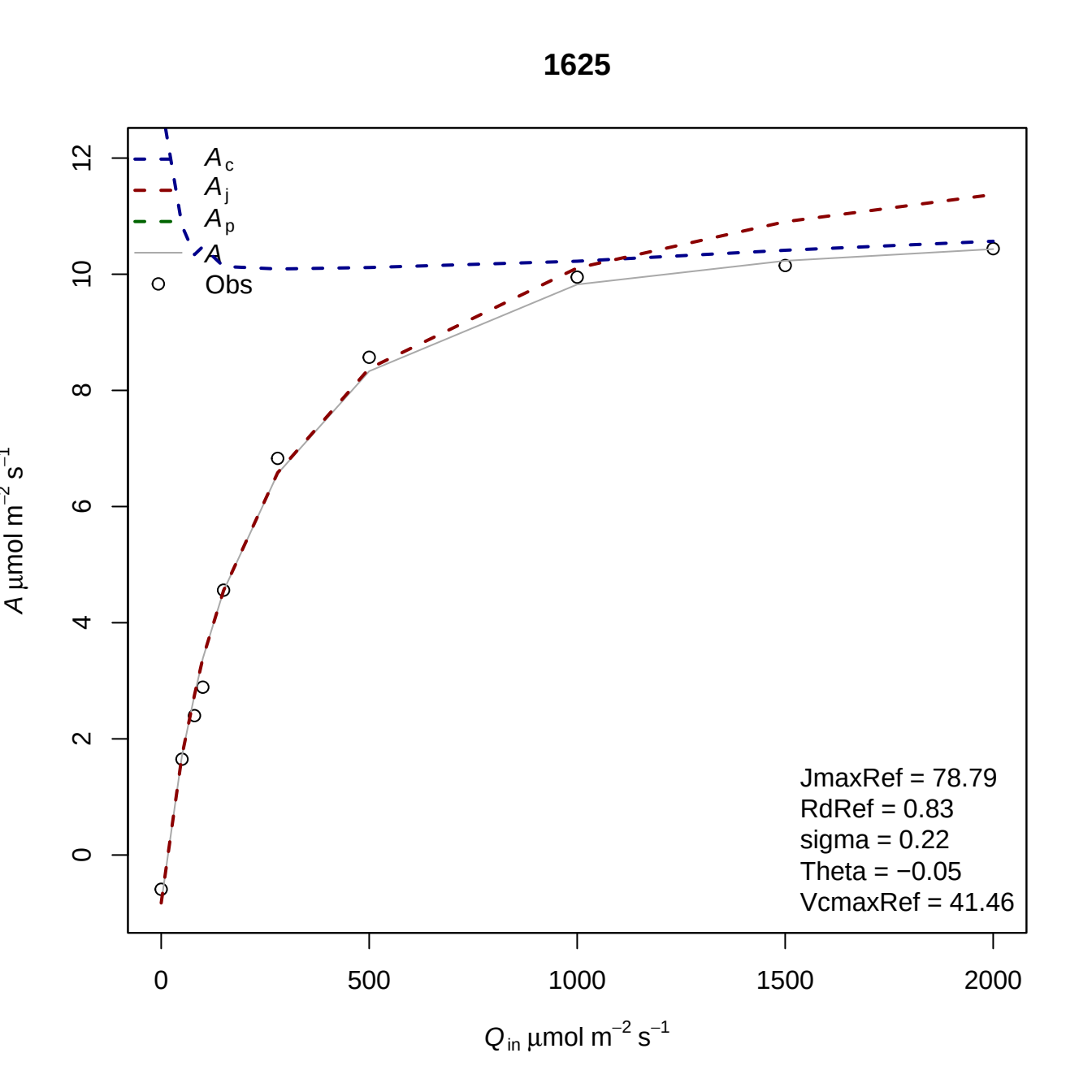

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 247.26
RdRef = 1.06
sigma = 0.19
Theta = 0.48
VcmaxRef = 138.62
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

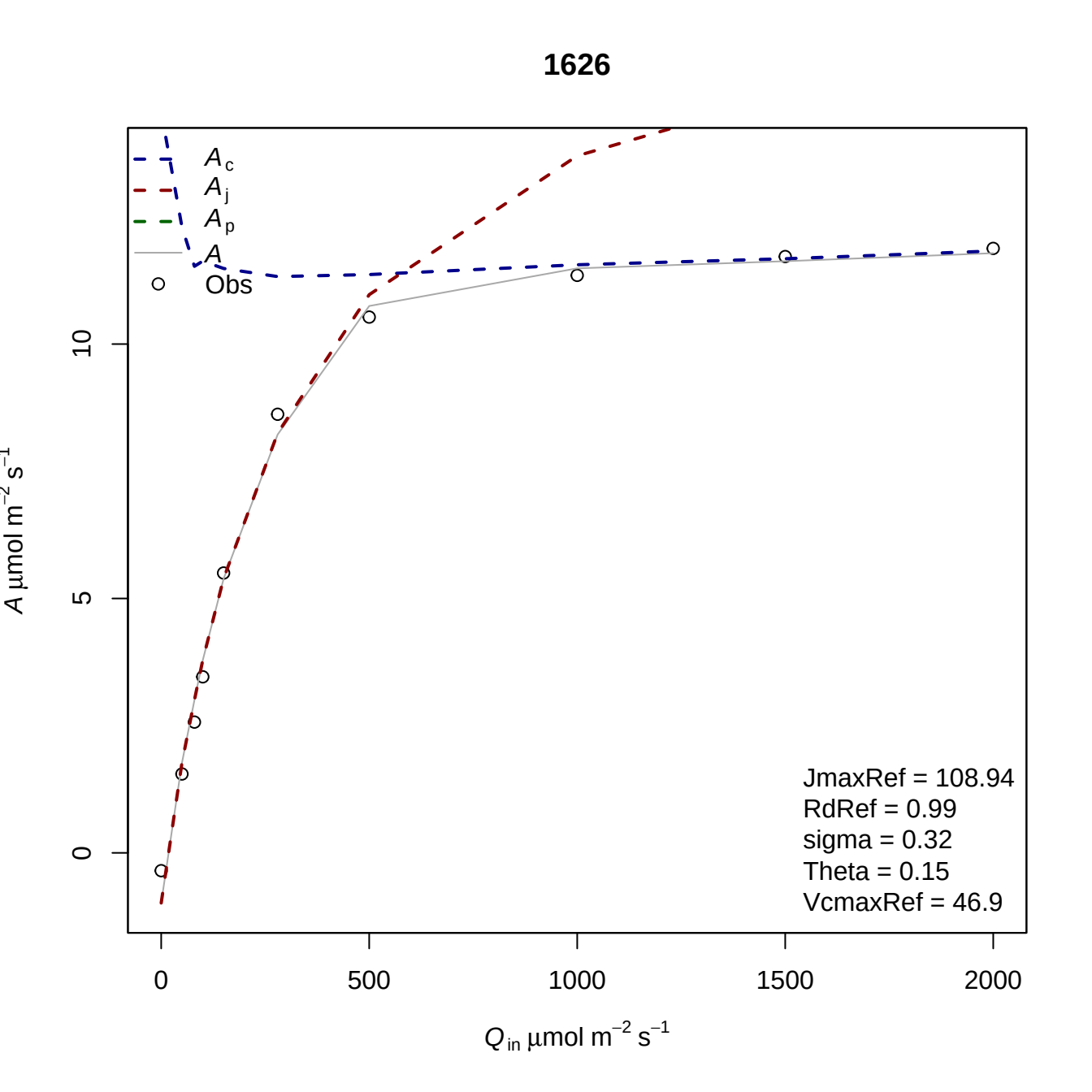

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 269.68
RdRef = 1.13
sigma = 0.38
Theta = 0.18
VcmaxRef = 132.67
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

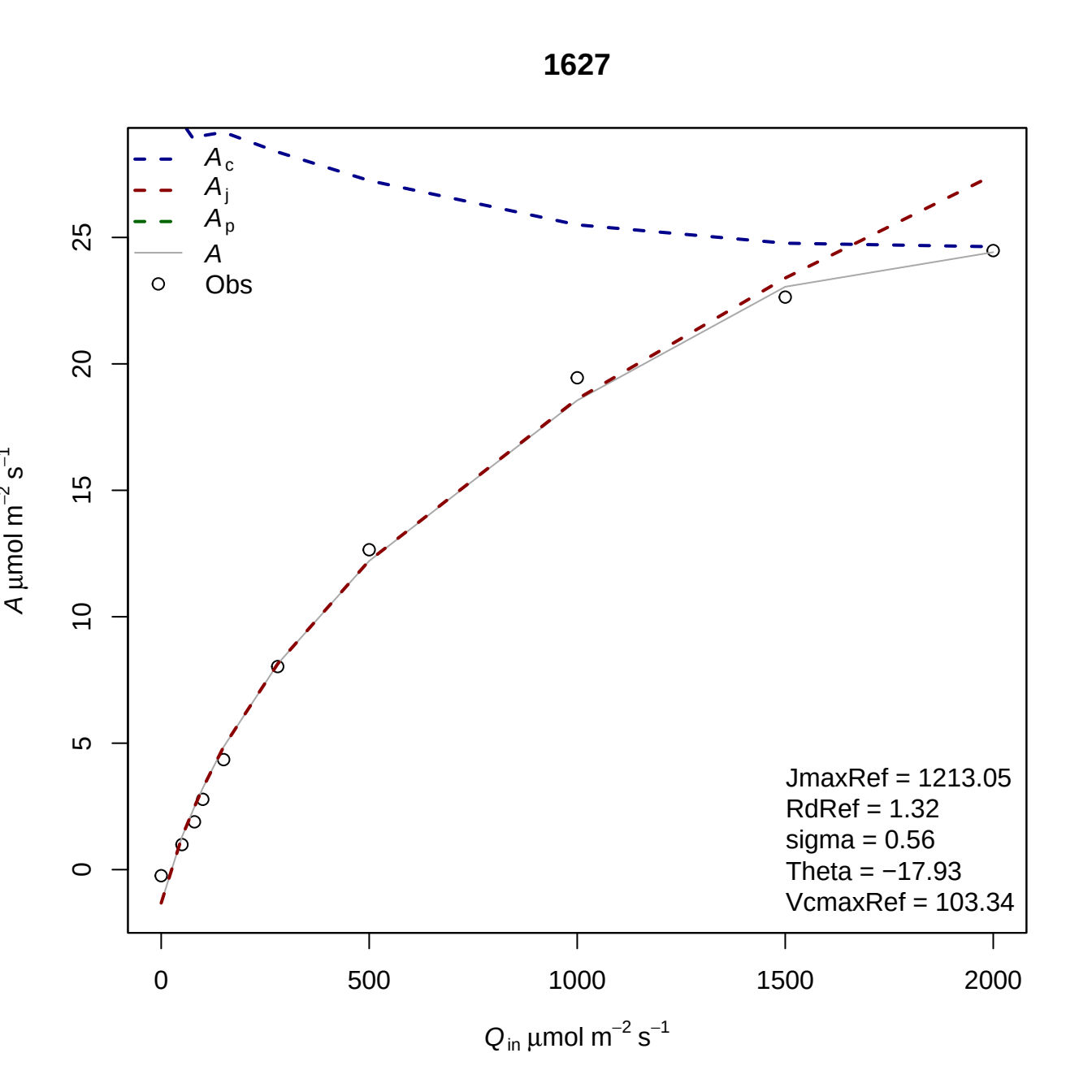

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 262.33
RdRef = 1.09
sigma = 0.33
Theta = 0.52
VcmaxRef = 136.17
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

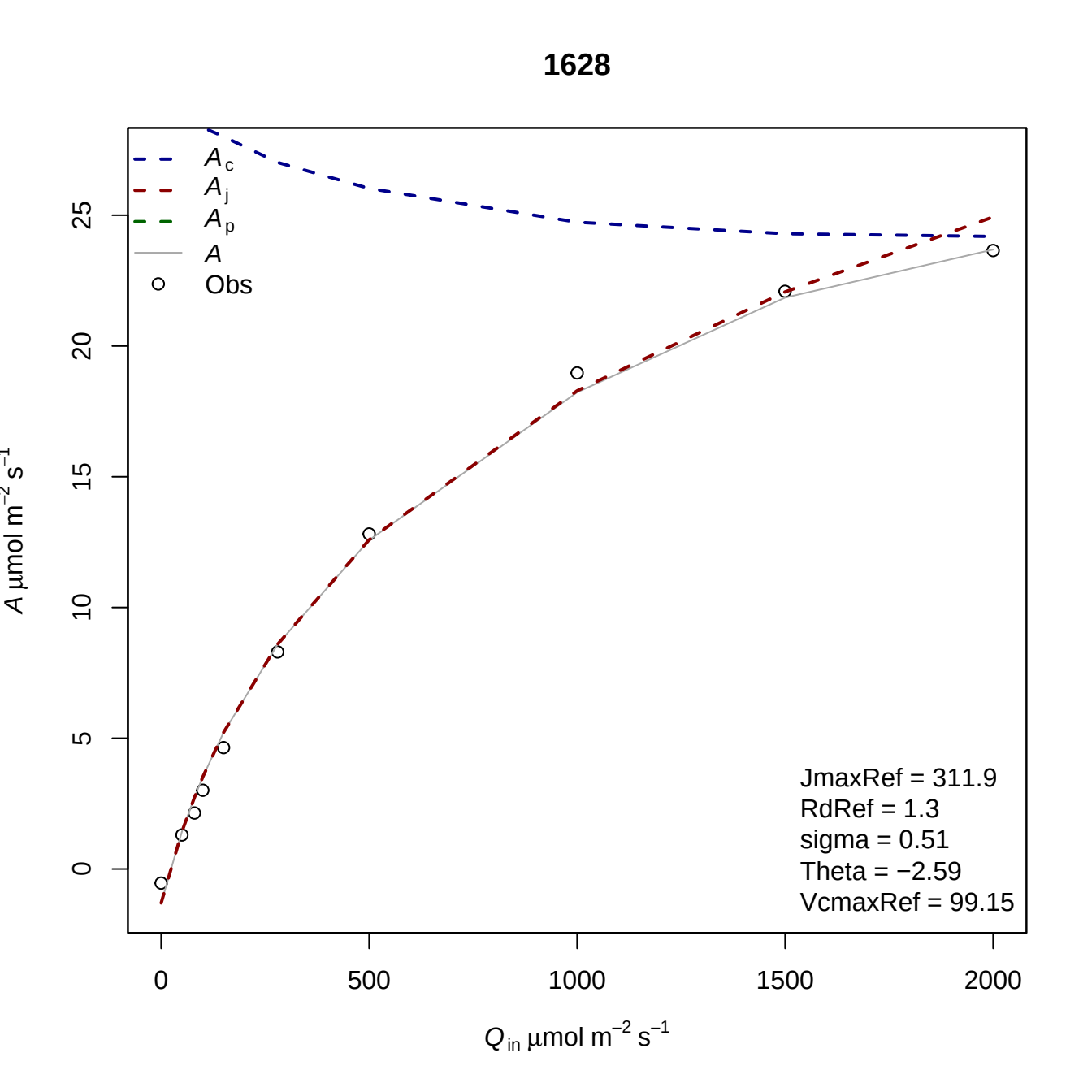

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 96612407.27
RdRef = 0.67
sigma = 0.22
Theta = −2779854.62
VcmaxRef = 49.28
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

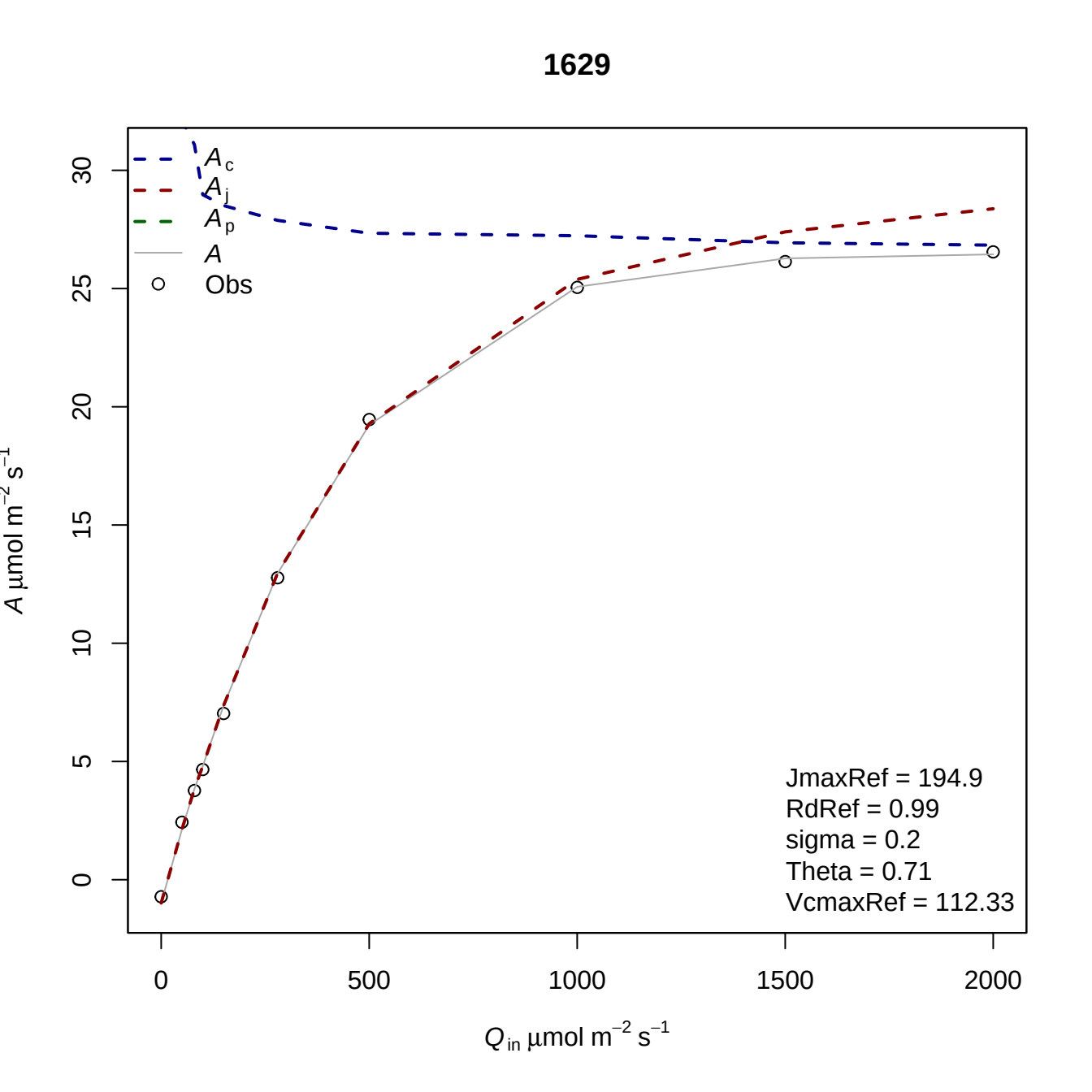

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 89.26
RdRef = 0.6
sigma = 0.13
Theta = 0.63
VcmaxRef = 49.51
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

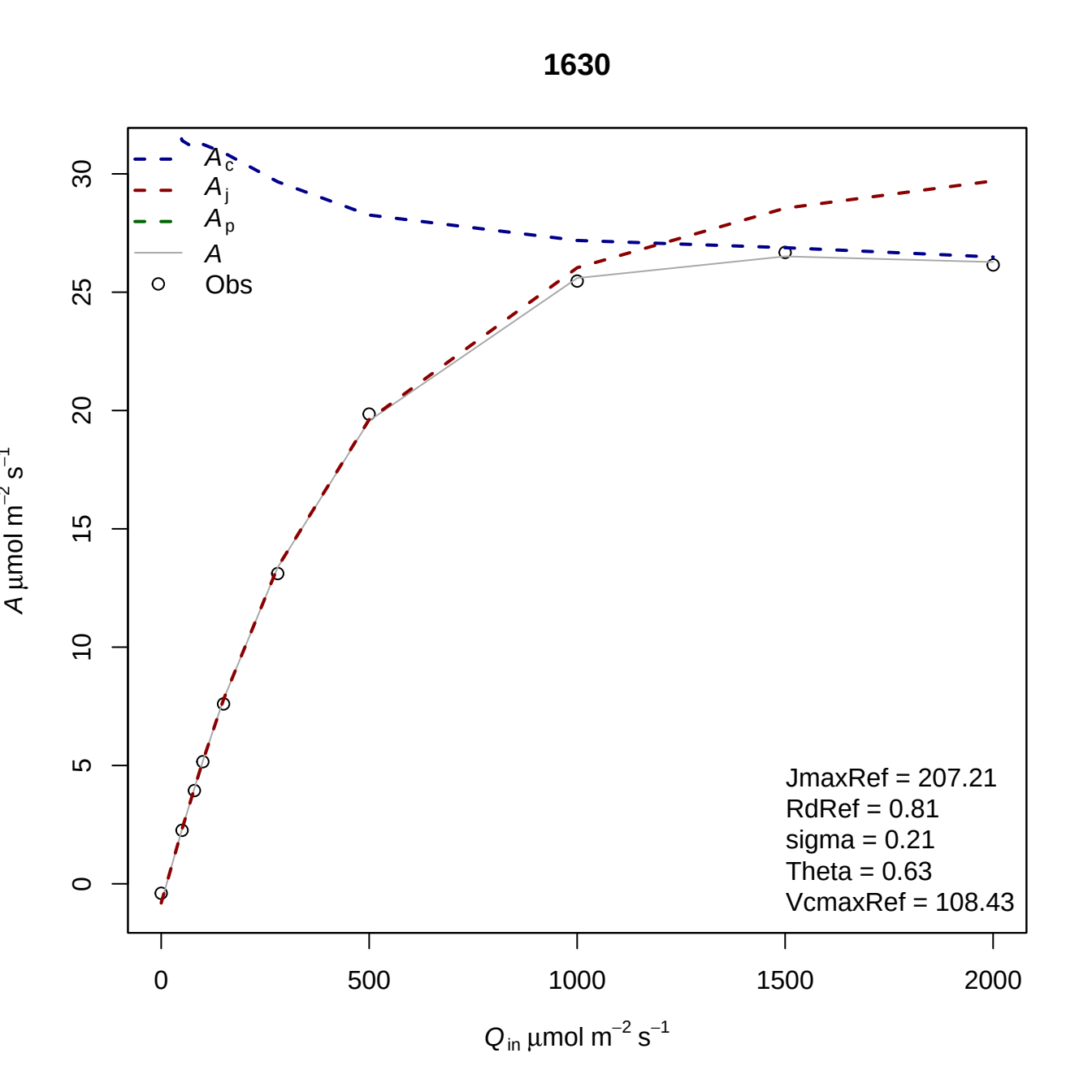

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 78.79
RdRef = 0.83
sigma = 0.22
Theta = −0.05
VcmaxRef = 41.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

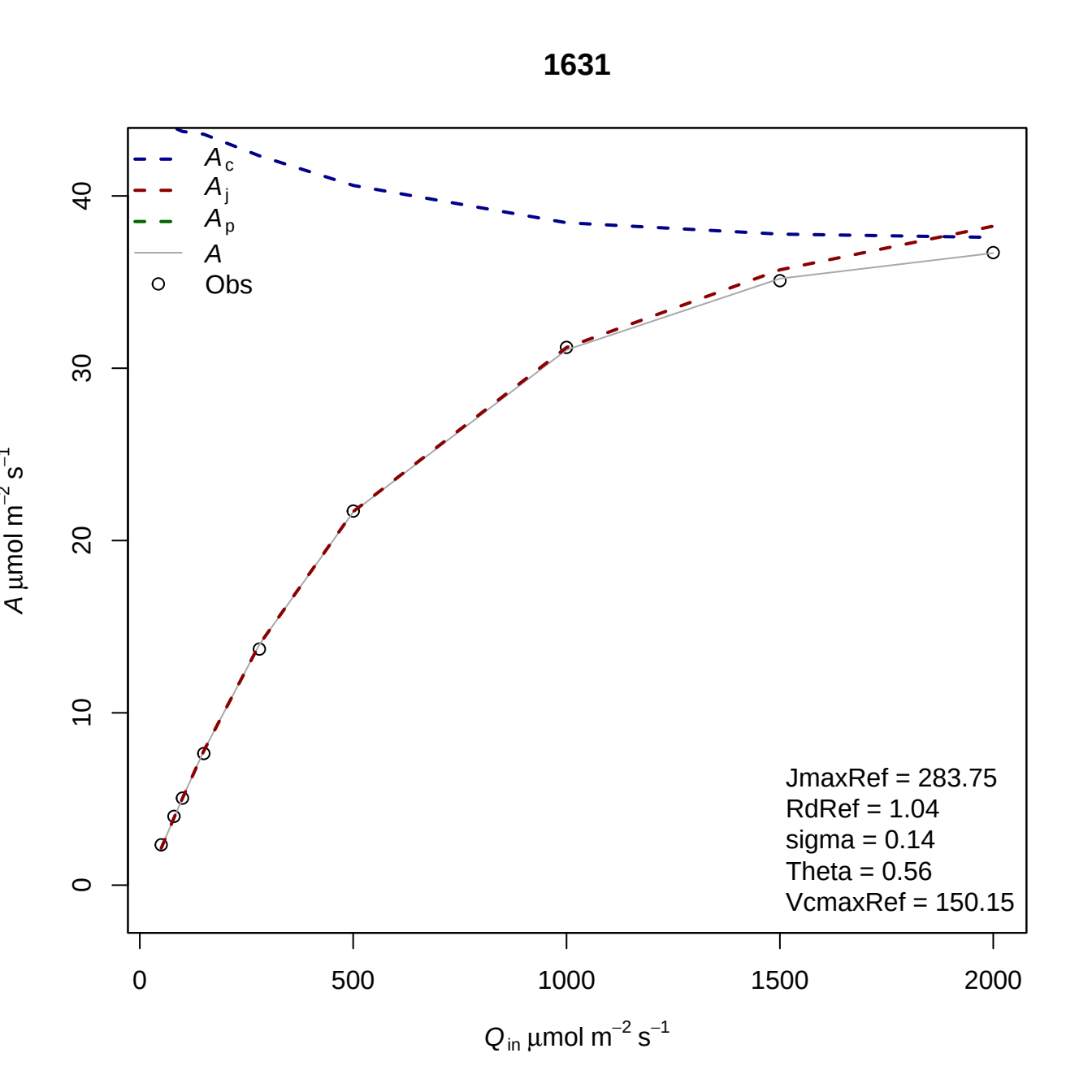

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 108.94
RdRef = 0.99
sigma = 0.32
Theta = 0.15
VcmaxRef = 46.9
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

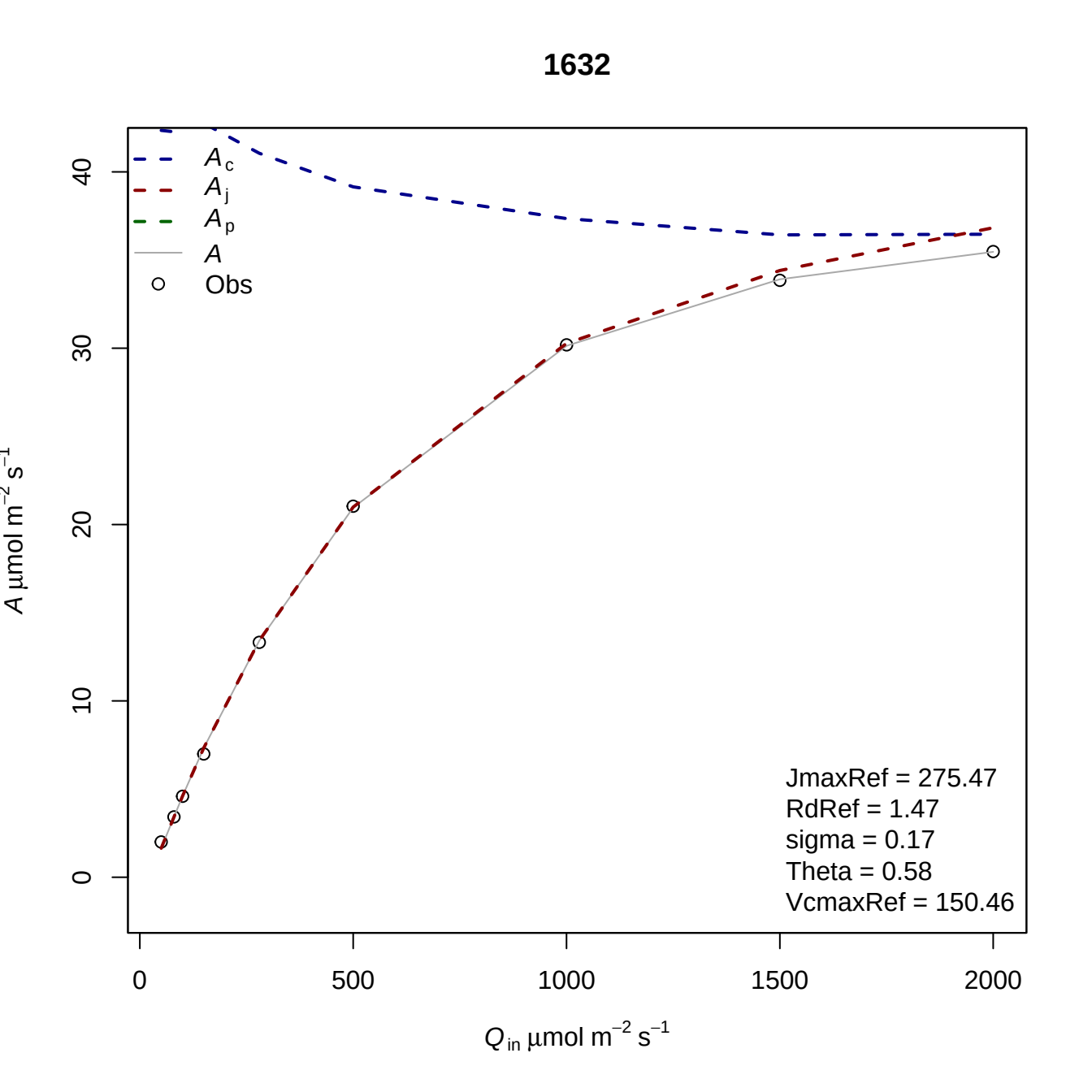

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 1213.05
RdRef = 1.32
sigma = 0.56
Theta = −17.93
VcmaxRef = 103.34
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

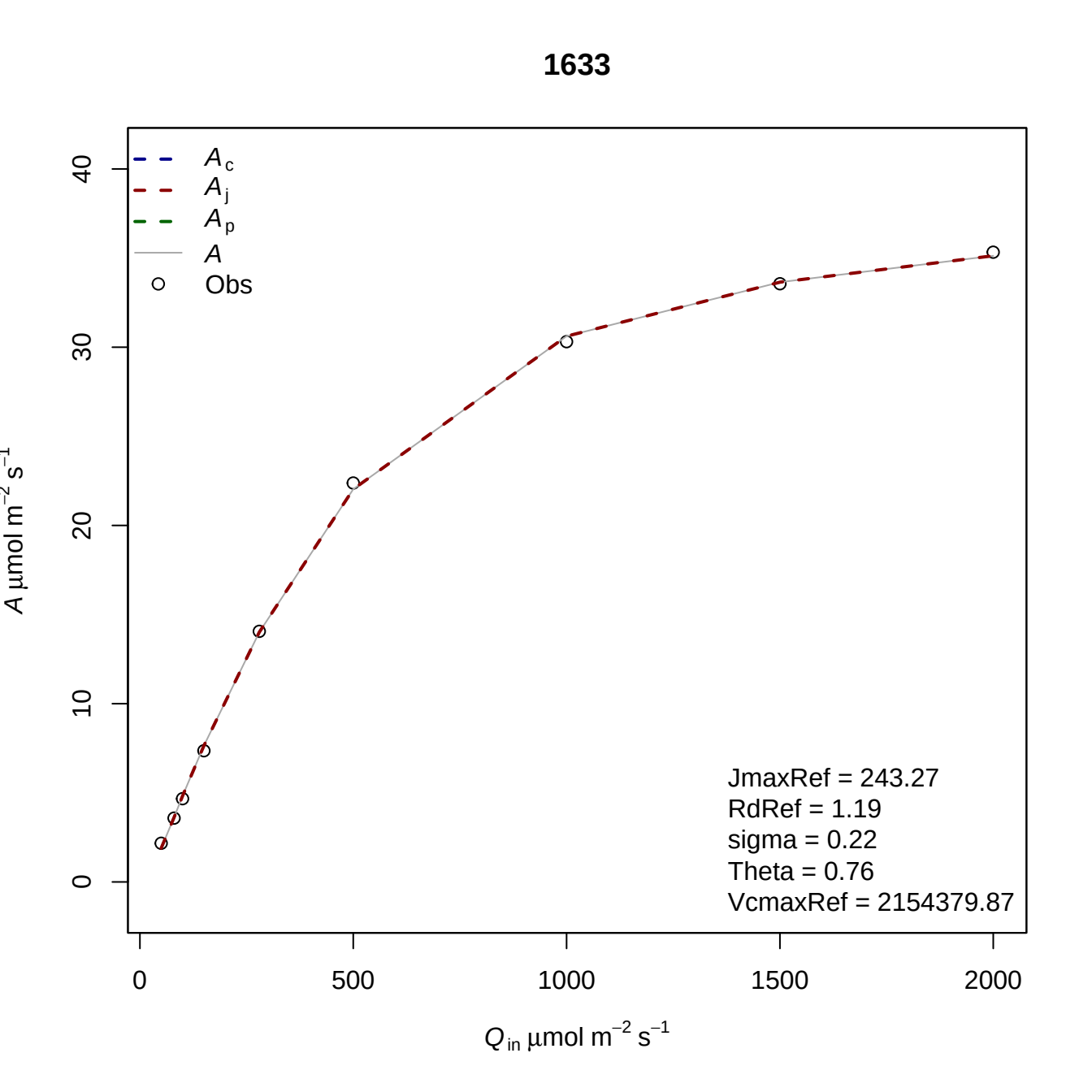

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 311.9
RdRef = 1.3
sigma = 0.51
Theta = −2.59
VcmaxRef = 99.15
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

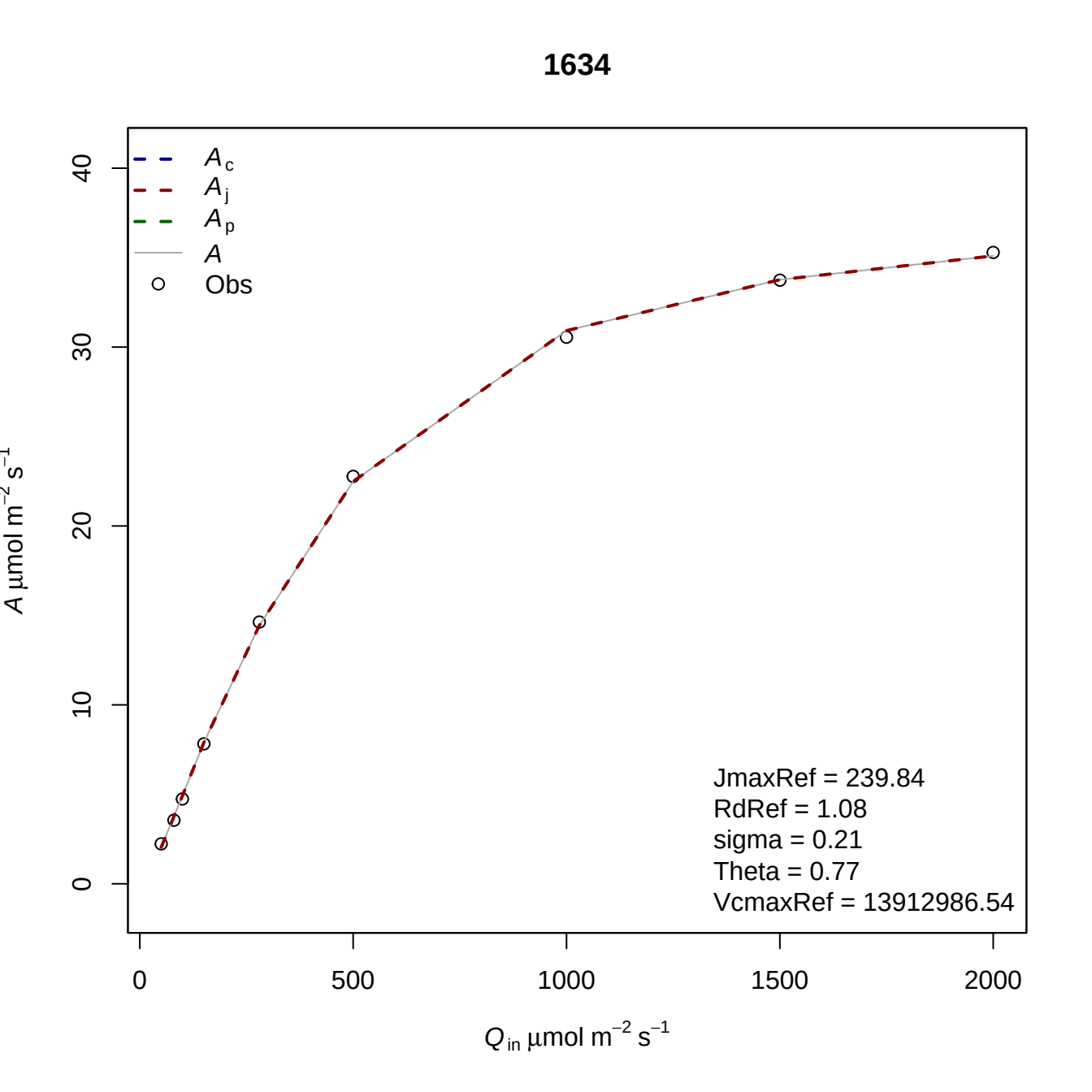

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 194.9
RdRef = 0.99
sigma = 0.2
Theta = 0.71
VcmaxRef = 112.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

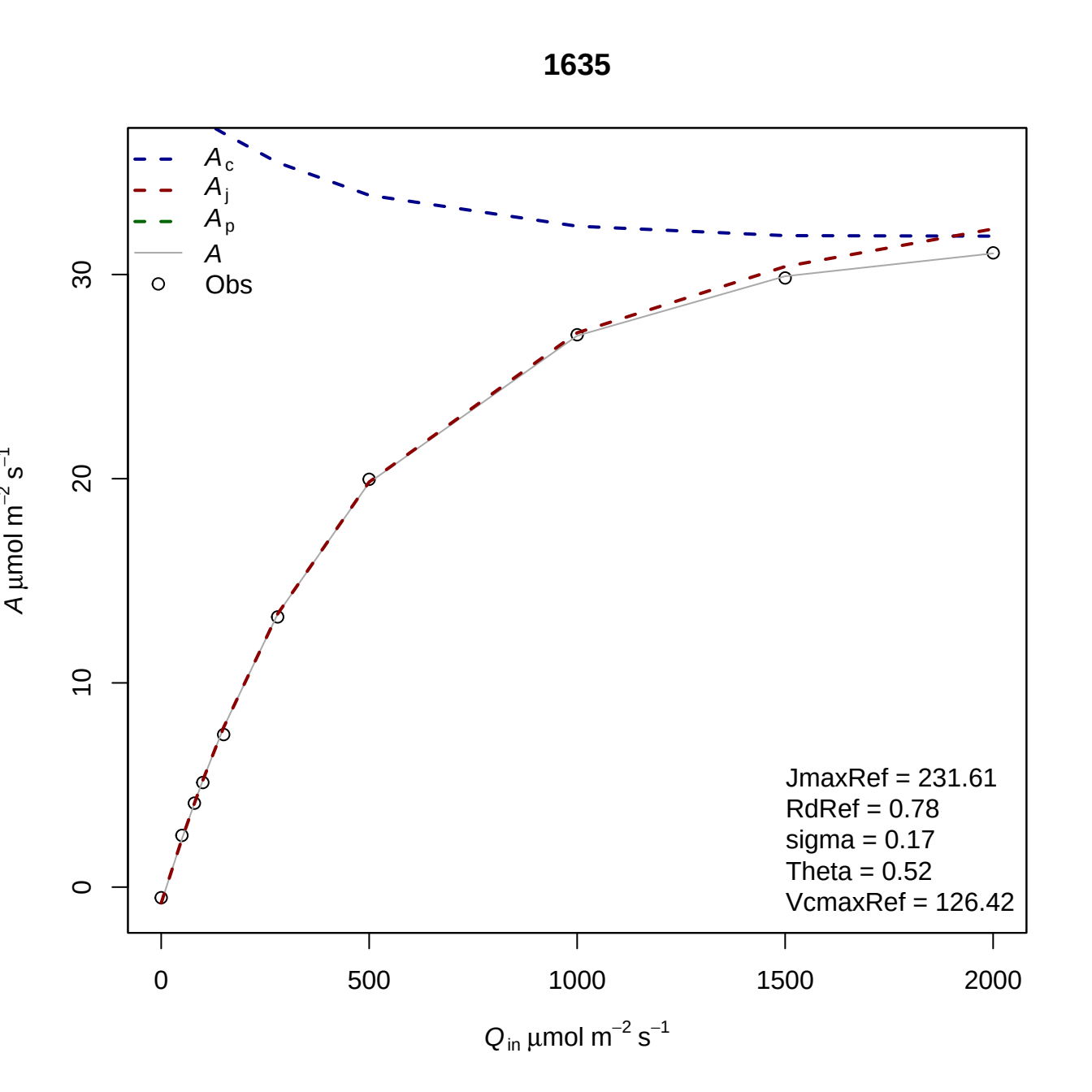

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 207.21
RdRef = 0.81
sigma = 0.21
Theta = 0.63
VcmaxRef = 108.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

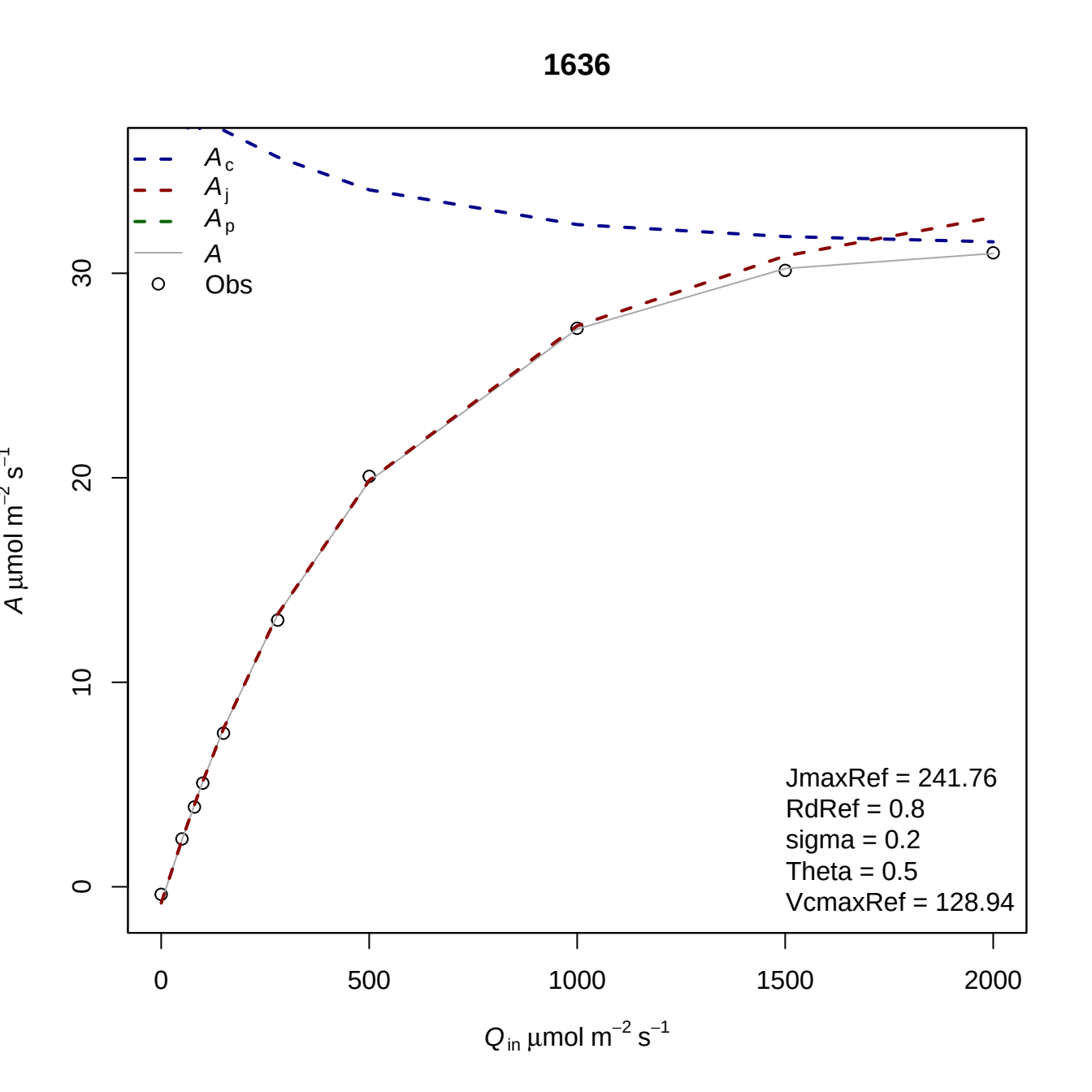

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 283.75
RdRef = 1.04
sigma = 0.14
Theta = 0.56
VcmaxRef = 150.15
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

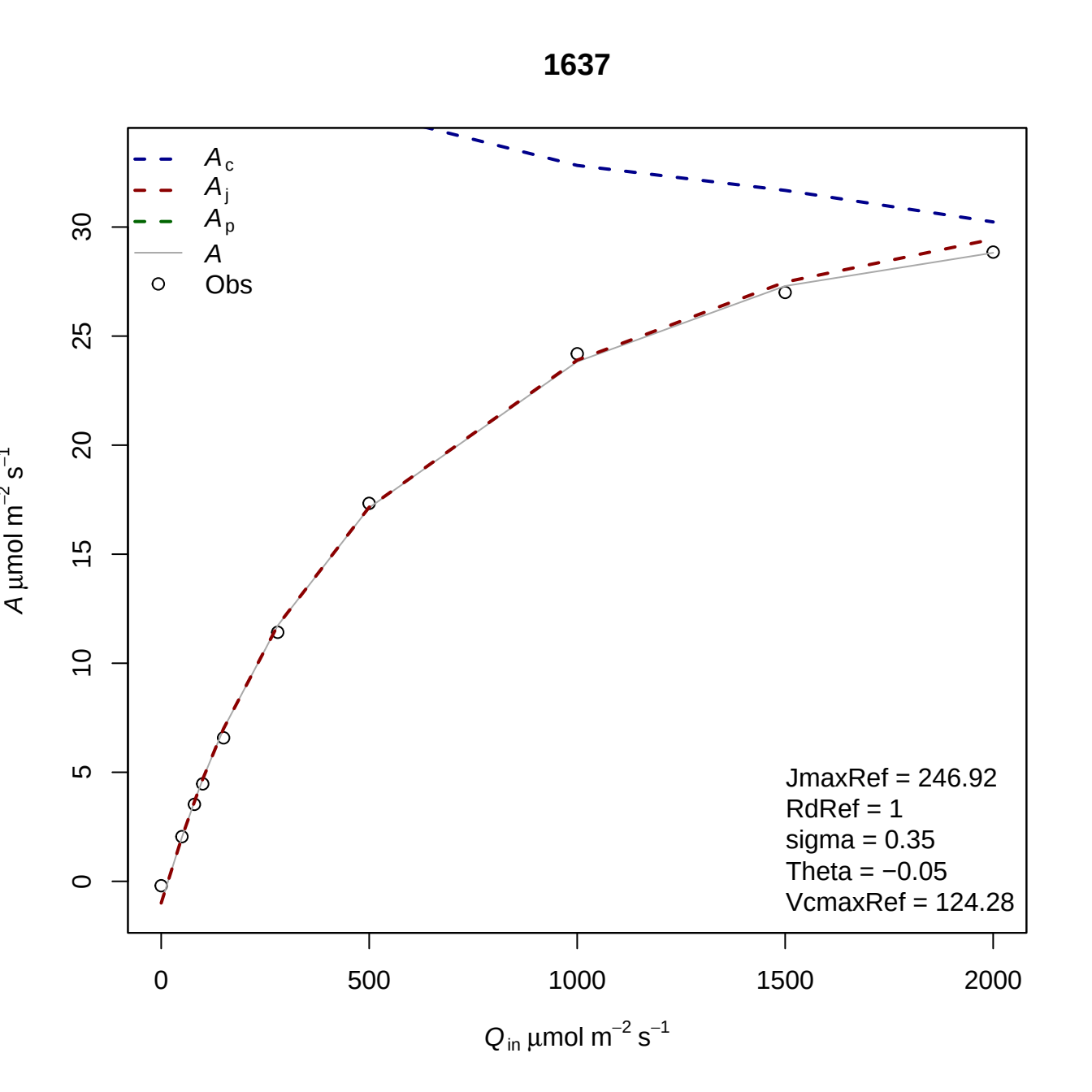

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 275.47
RdRef = 1.47
sigma = 0.17
Theta = 0.58
VcmaxRef = 150.46
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

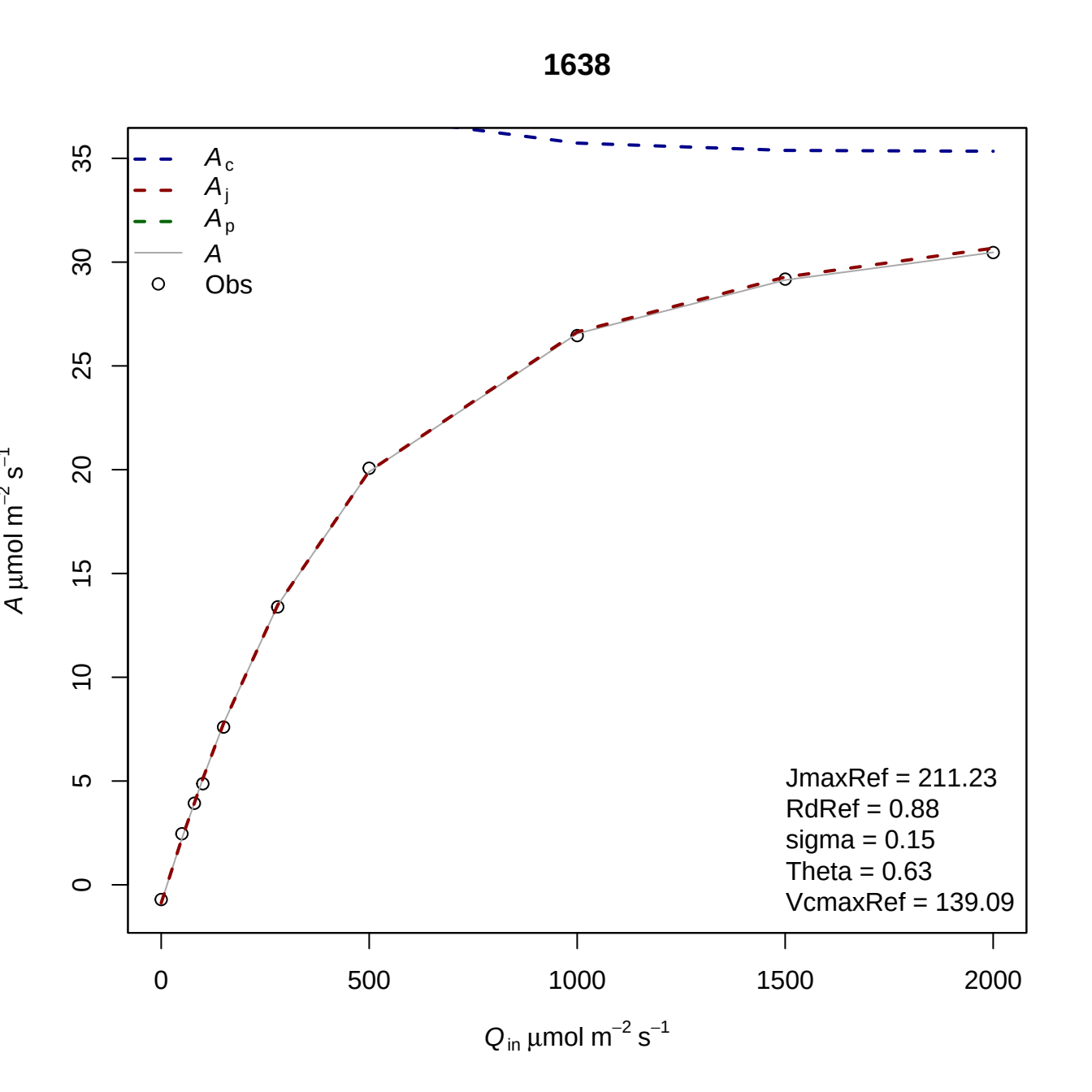

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 243.27
RdRef = 1.19
sigma = 0.22
Theta = 0.76
VcmaxRef = 2154379.87
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

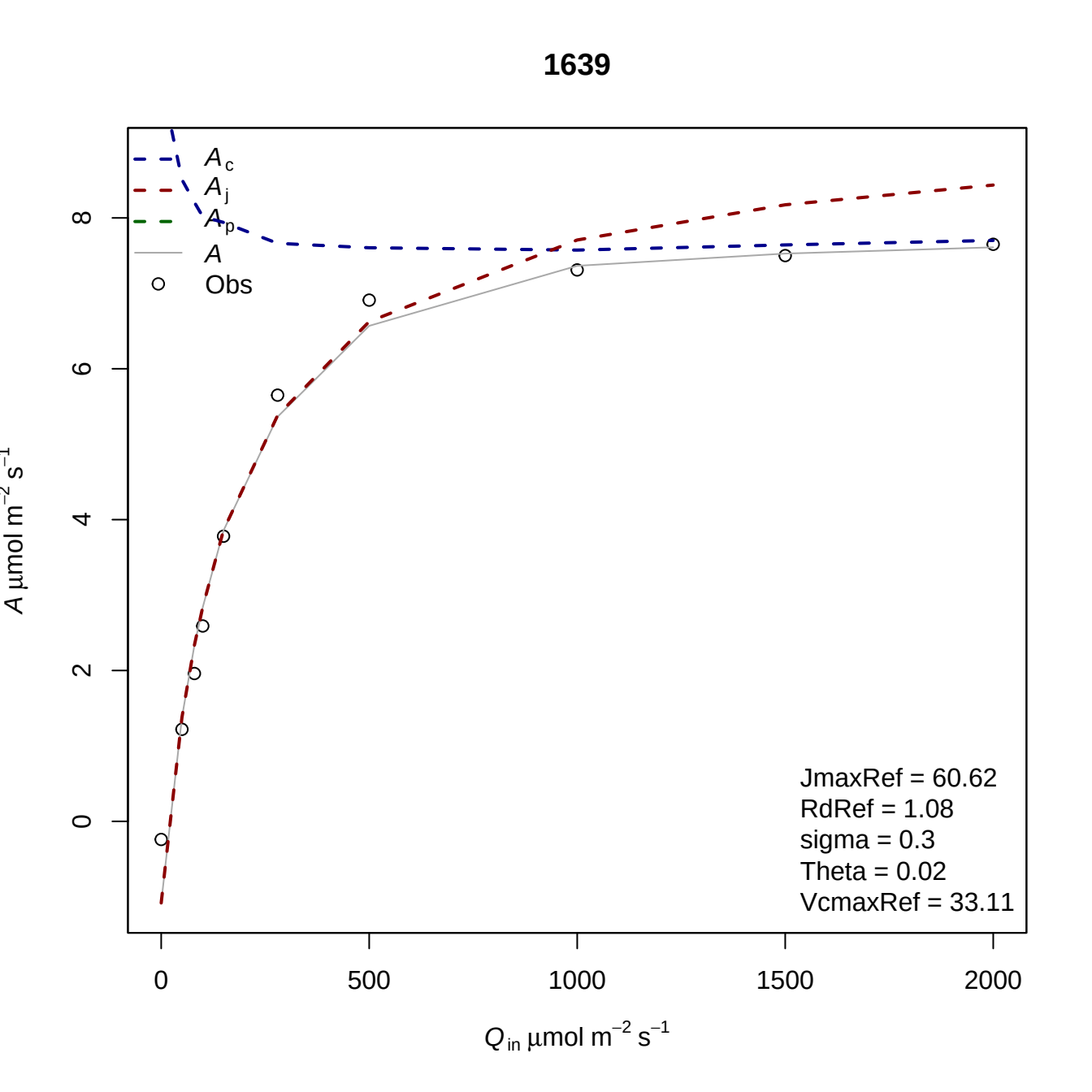

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 239.84
RdRef = 1.08
sigma = 0.21
Theta = 0.77
VcmaxRef = 13912986.54
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

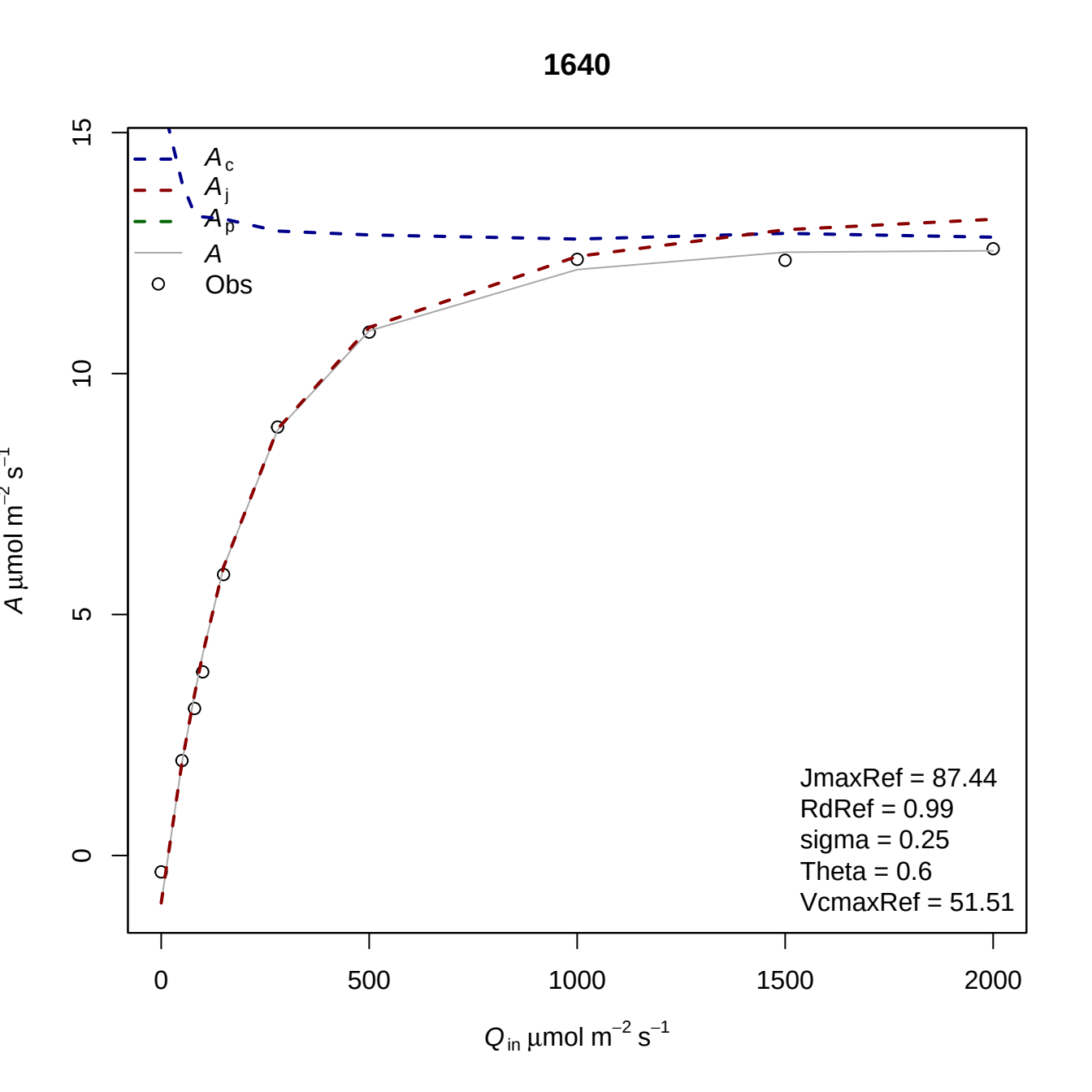

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 231.61
RdRef = 0.78
sigma = 0.17
Theta = 0.52
VcmaxRef = 126.42
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

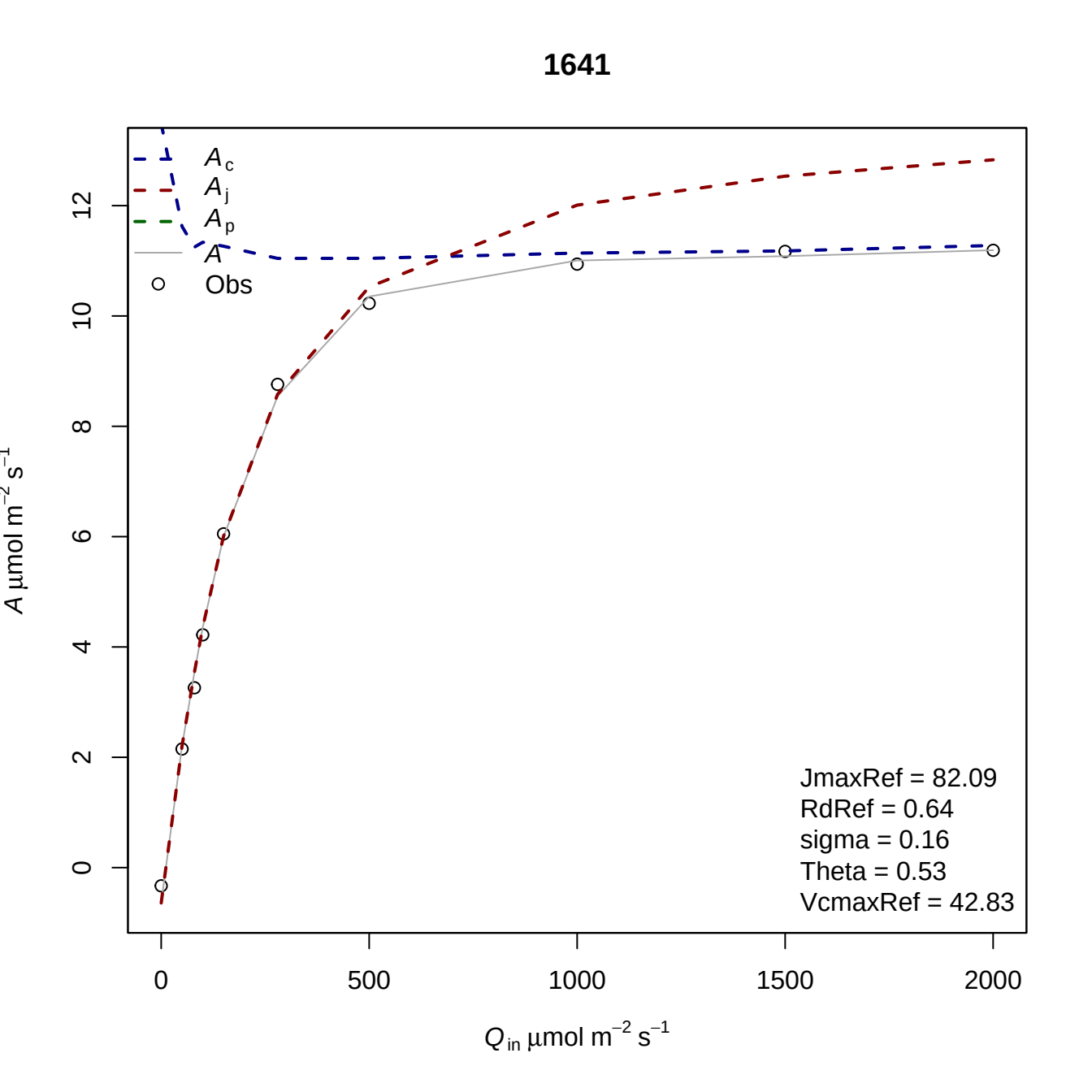

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 241.76
RdRef = 0.8
sigma = 0.2
Theta = 0.5
VcmaxRef = 128.94
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

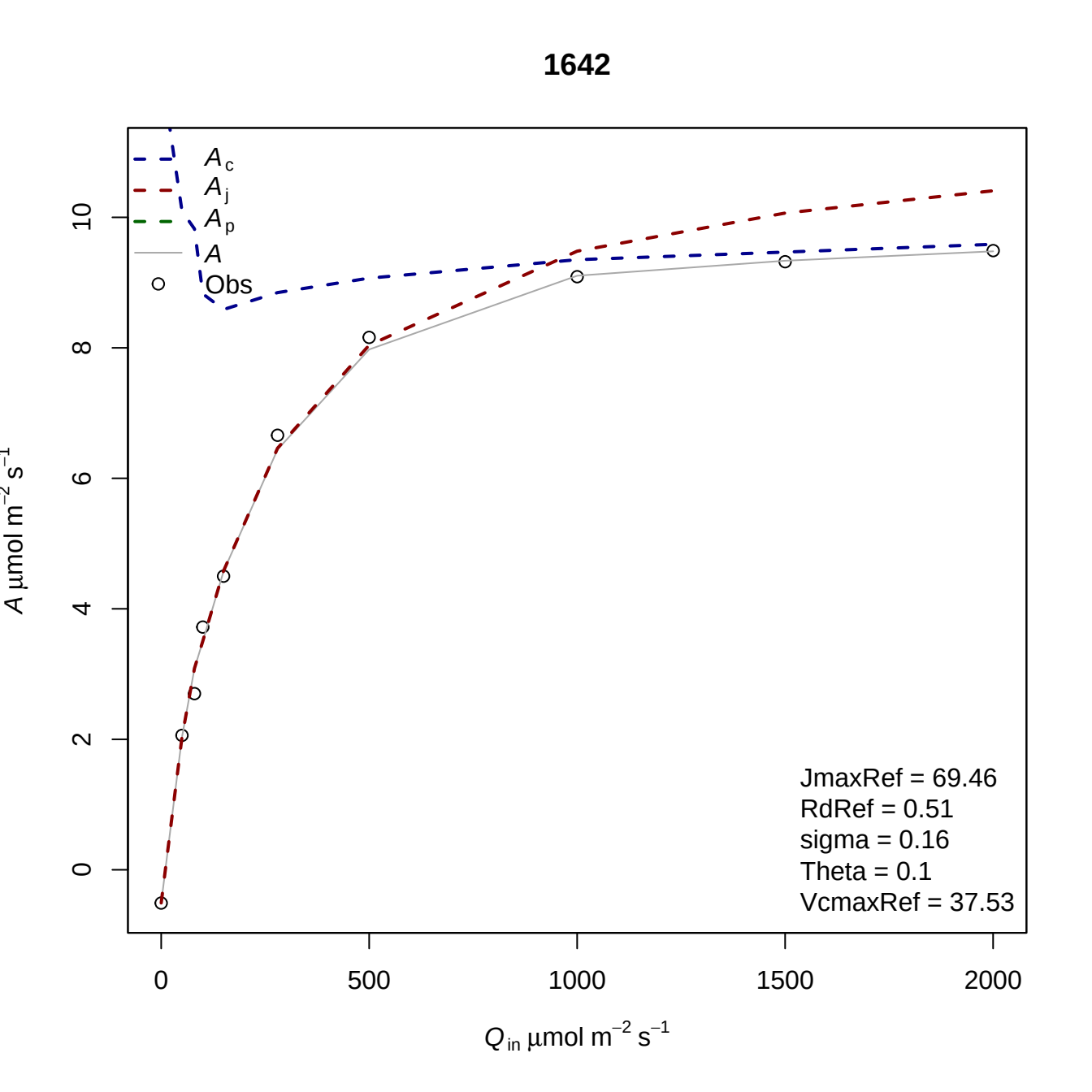

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 246.92
RdRef = 1
sigma = 0.35
Theta = −0.05
VcmaxRef = 124.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

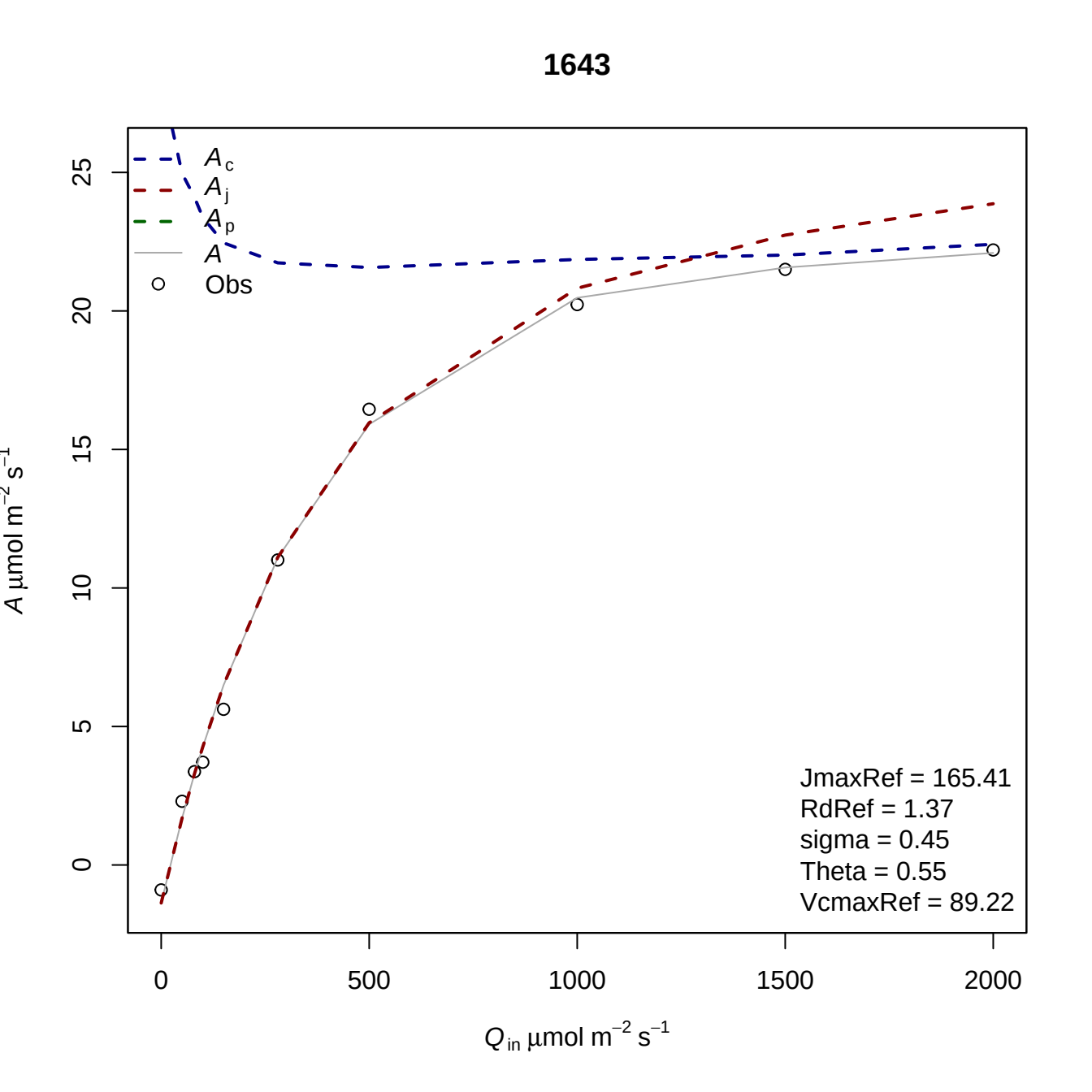

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 211.23
RdRef = 0.88
sigma = 0.15
Theta = 0.63
VcmaxRef = 139.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

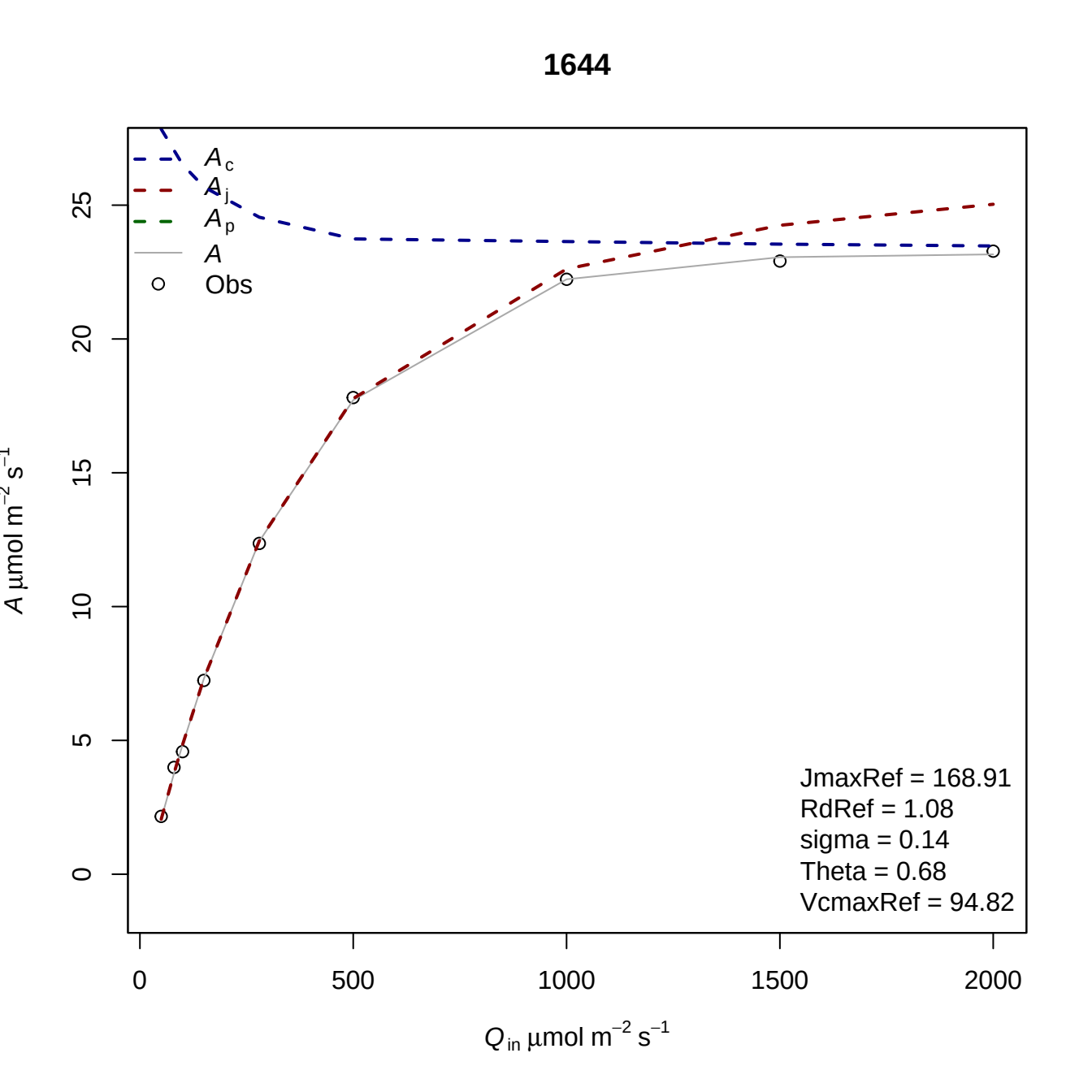

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 60.62
RdRef = 1.08
sigma = 0.3
Theta = 0.02
VcmaxRef = 33.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

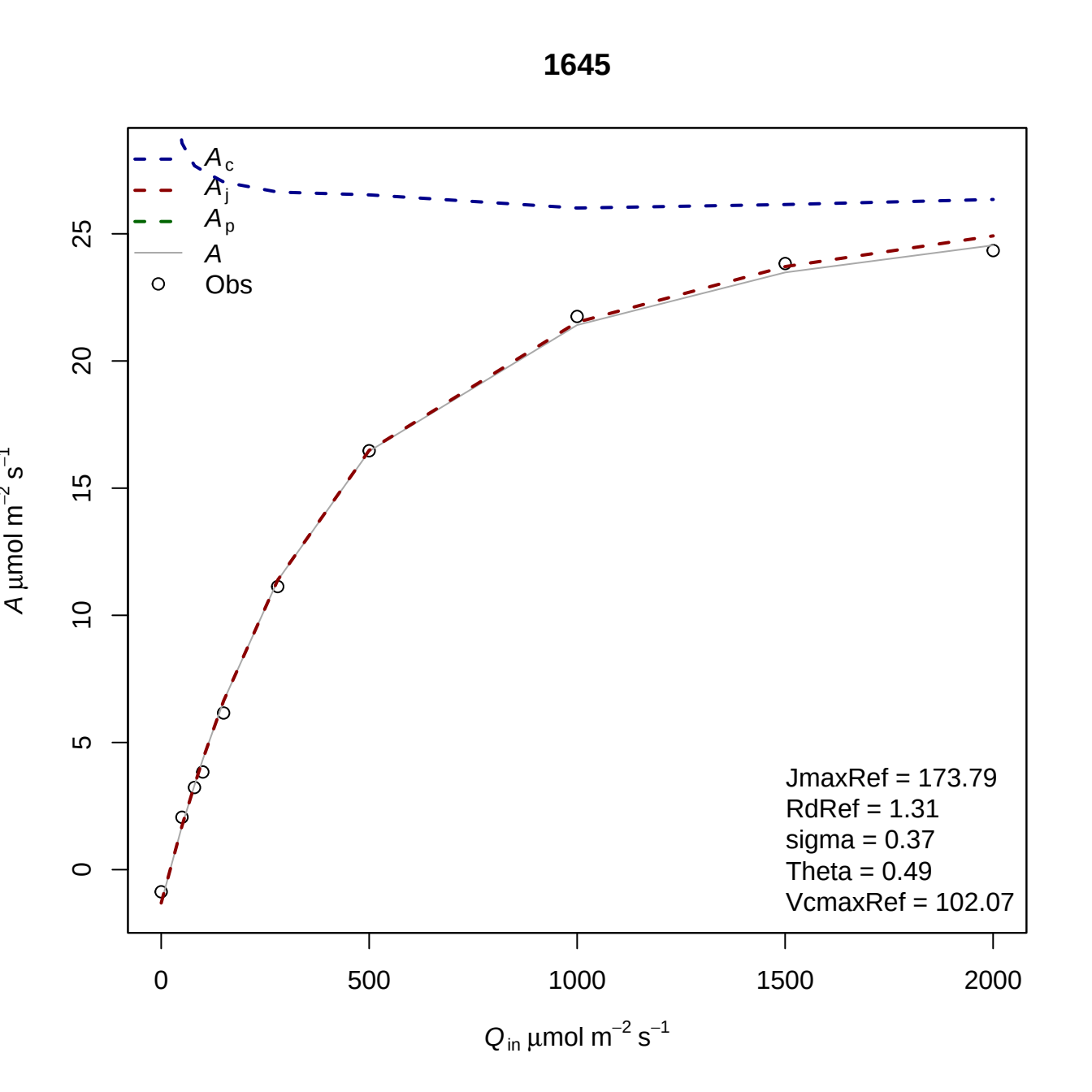

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 87.44
RdRef = 0.99
sigma = 0.25
Theta = 0.6
VcmaxRef = 51.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

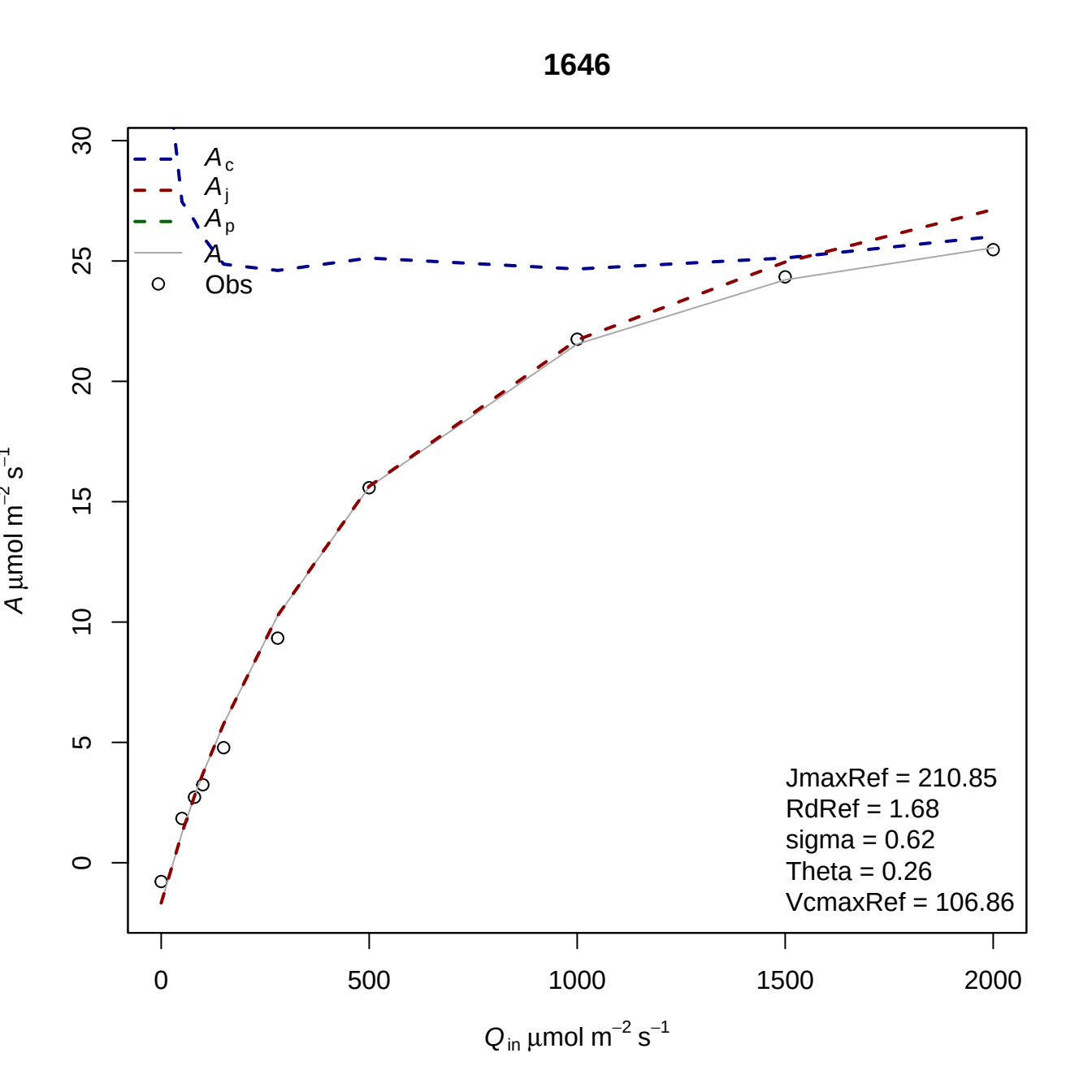

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 82.09
RdRef = 0.64
sigma = 0.16
Theta = 0.53
VcmaxRef = 42.83
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

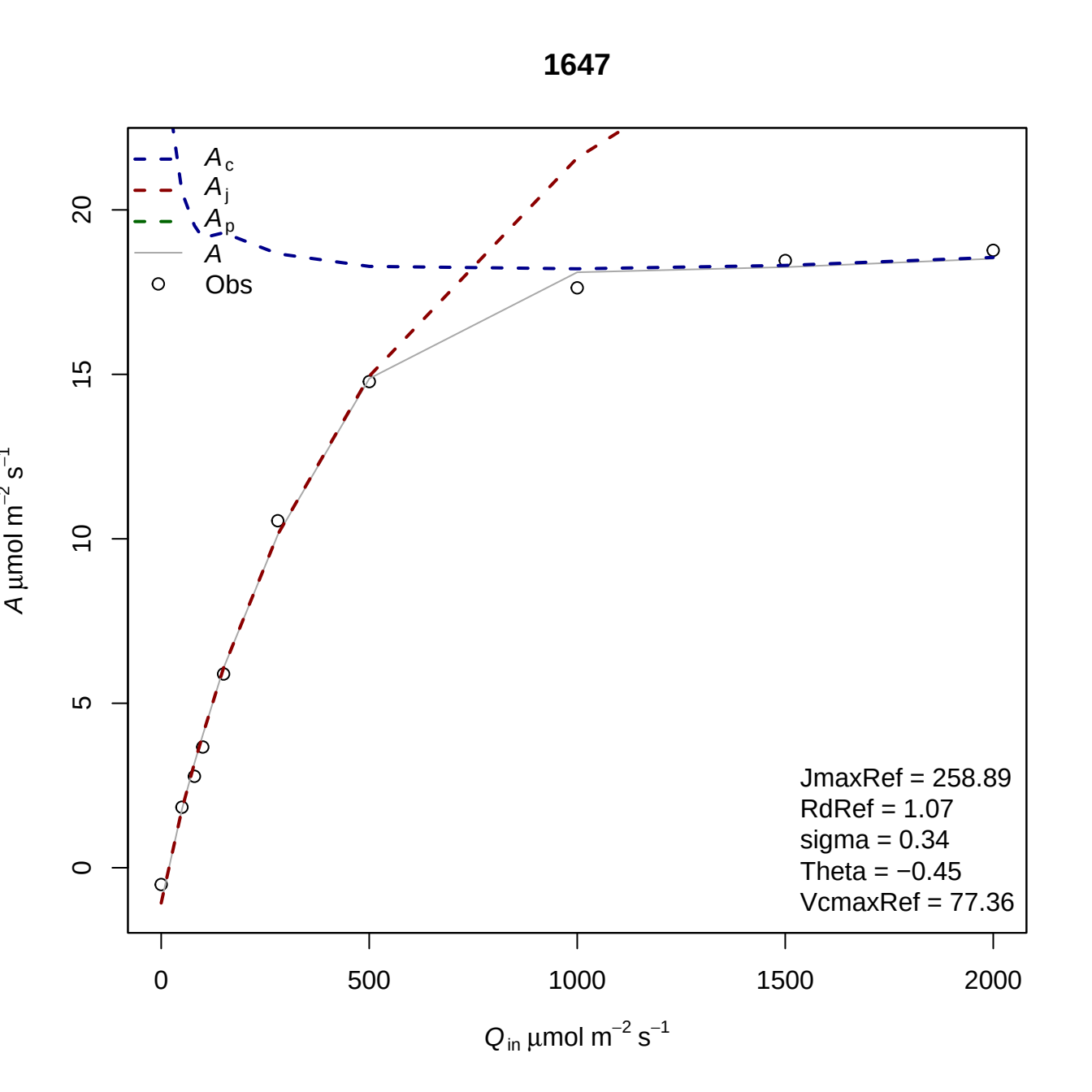

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 69.46
RdRef = 0.51
sigma = 0.16
Theta = 0.1
VcmaxRef = 37.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

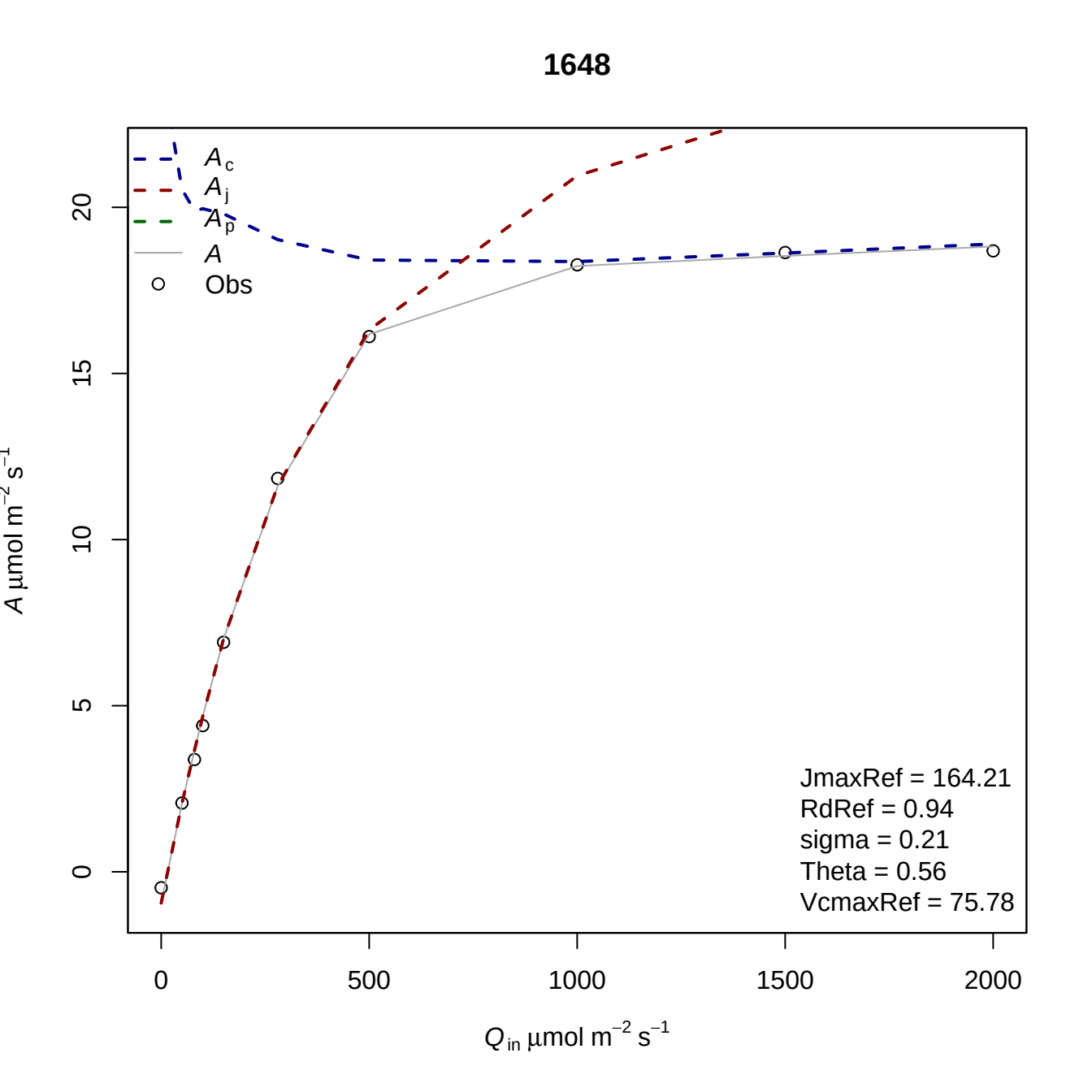

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 165.41
RdRef = 1.37
sigma = 0.45
Theta = 0.55
VcmaxRef = 89.22
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

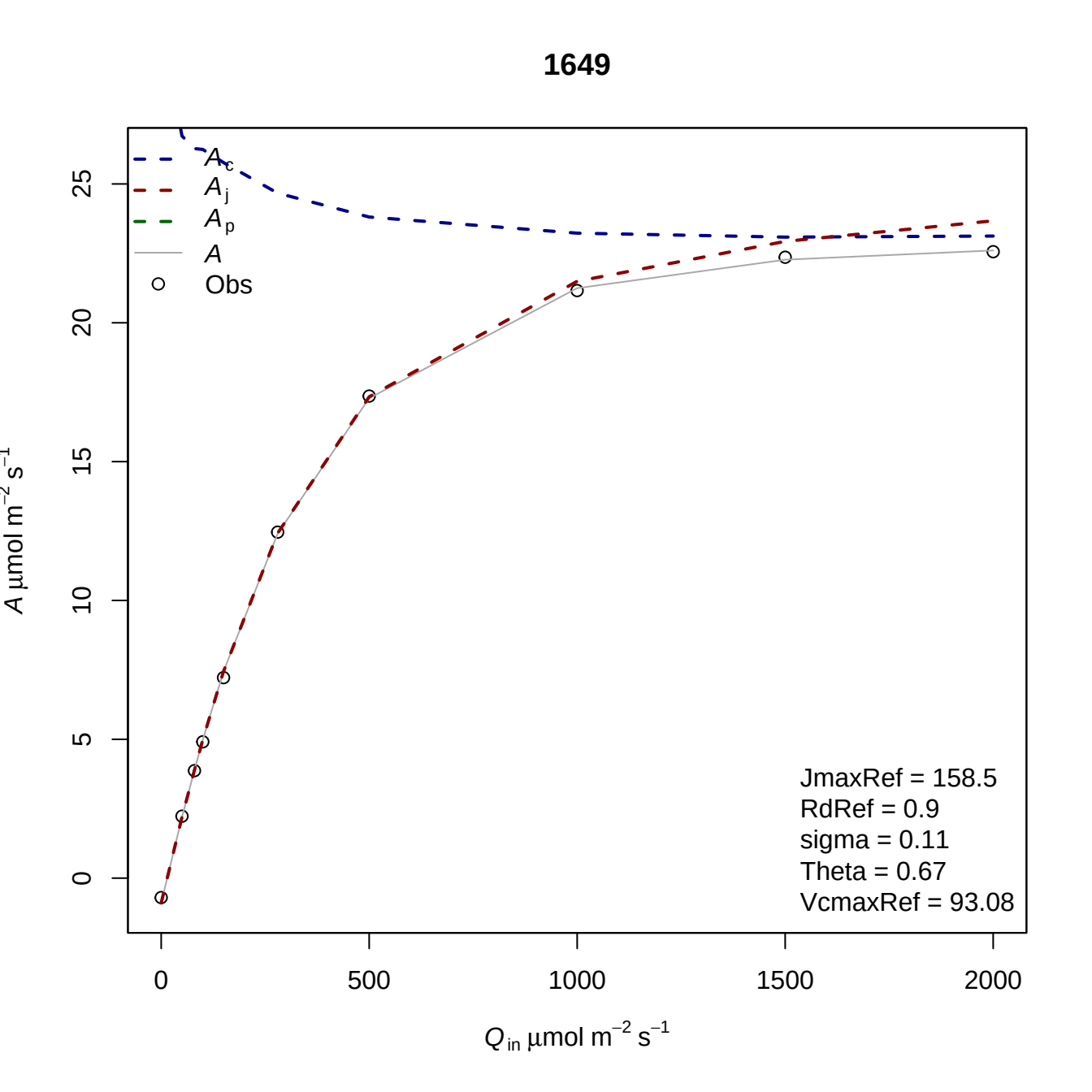

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 168.91
RdRef = 1.08
sigma = 0.14
Theta = 0.68
VcmaxRef = 94.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

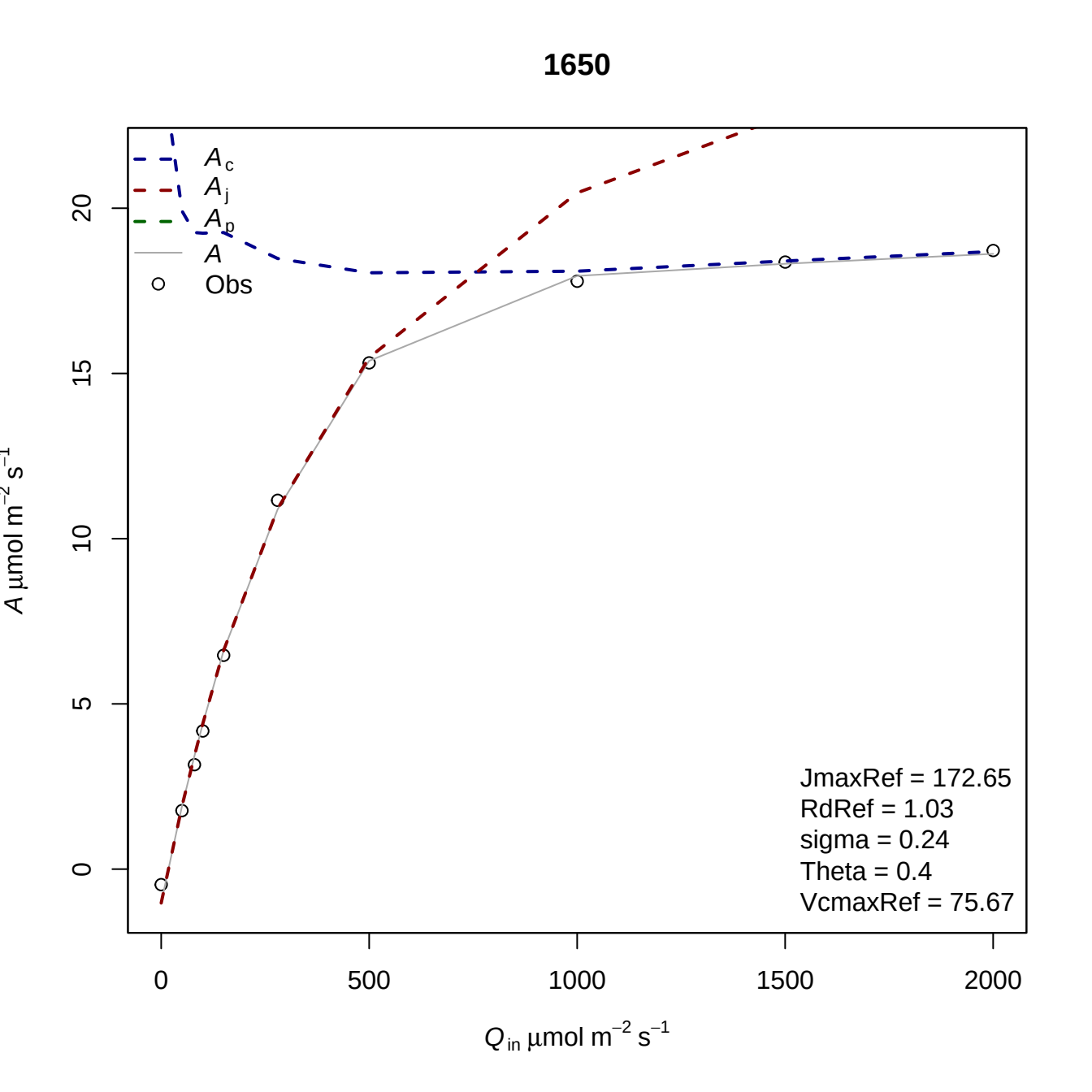

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 173.79
RdRef = 1.31
sigma = 0.37
Theta = 0.49
VcmaxRef = 102.07
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

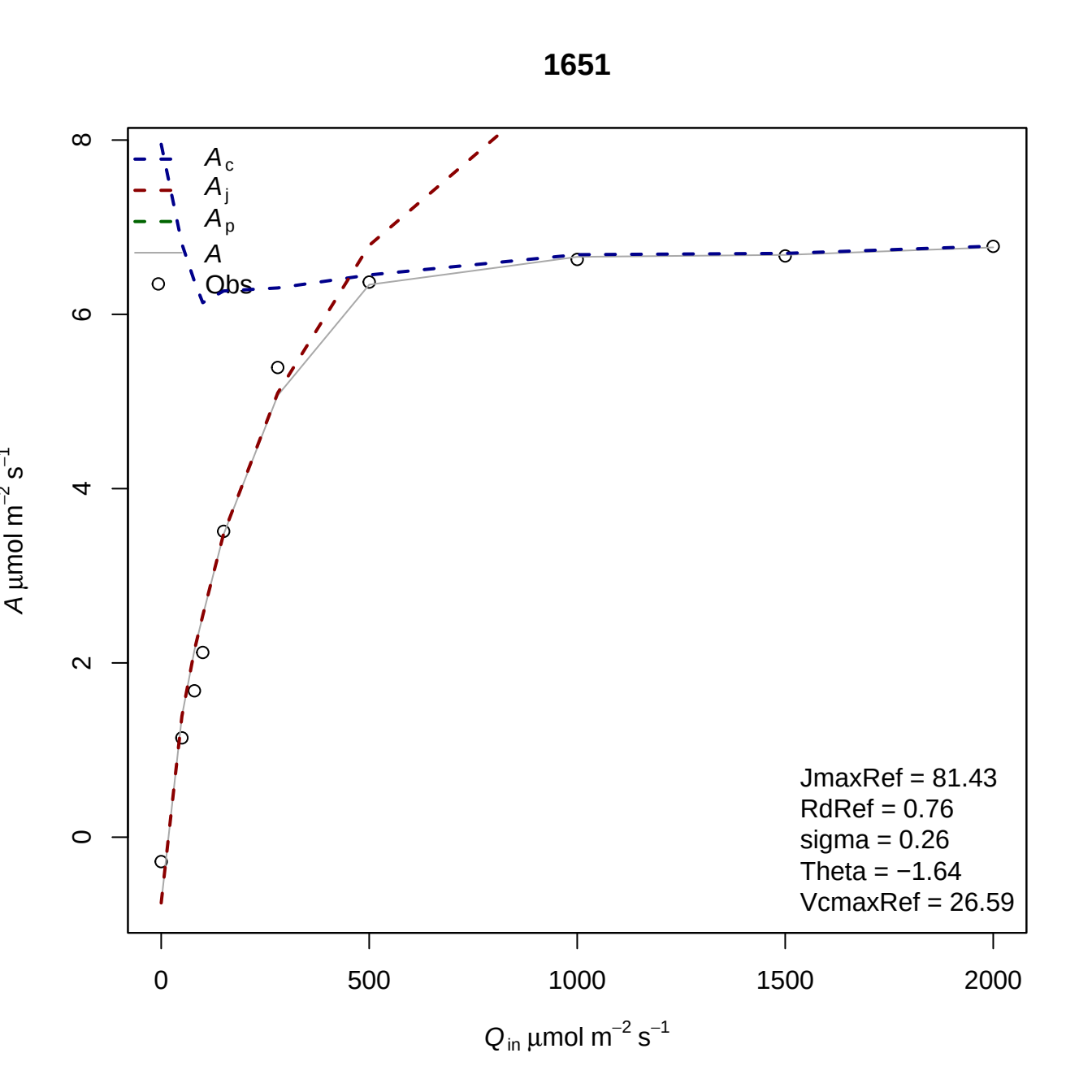

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 210.85
RdRef = 1.68
sigma = 0.62
Theta = 0.26
VcmaxRef = 106.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

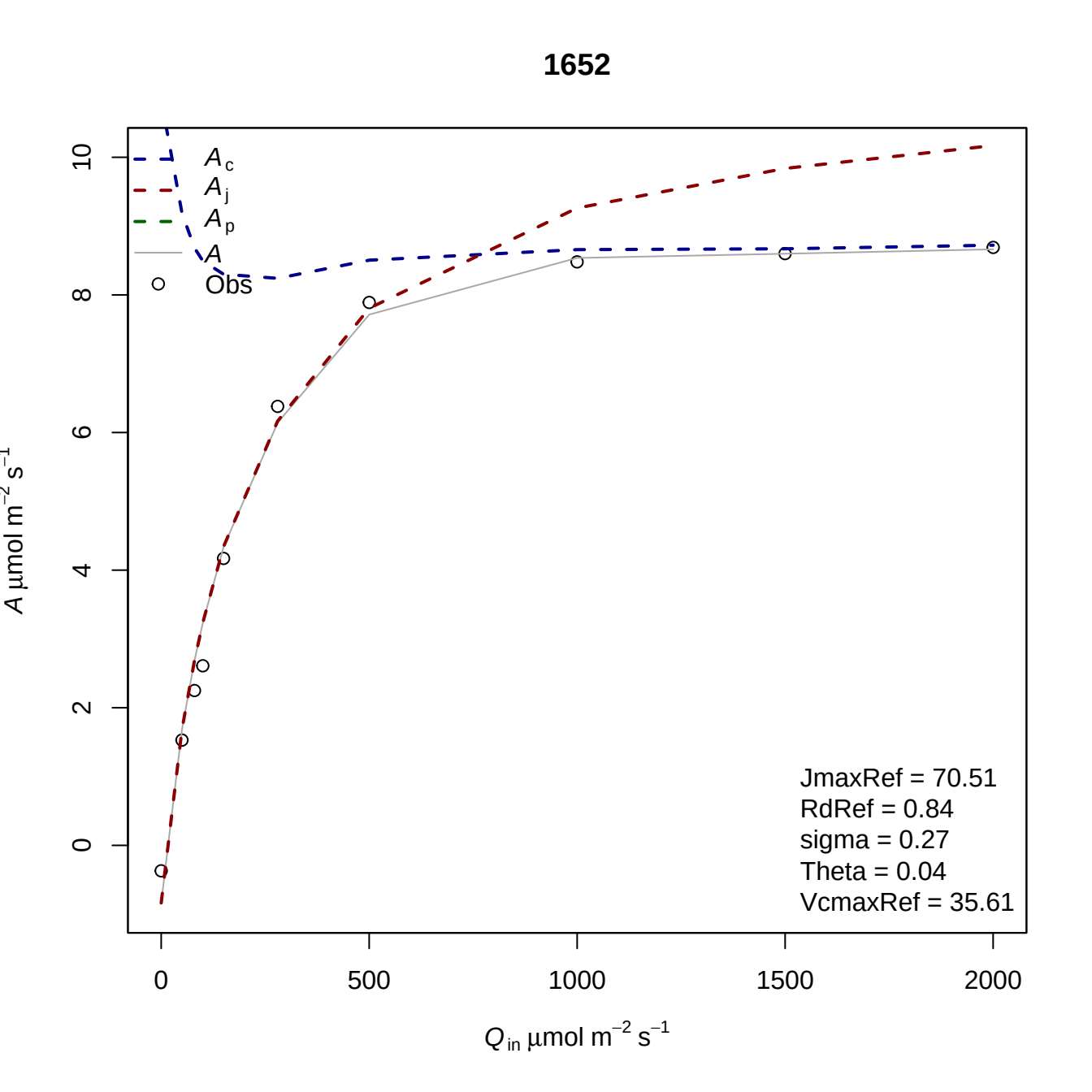

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 258.89
RdRef = 1.07
sigma = 0.34
Theta = −0.45
VcmaxRef = 77.36
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

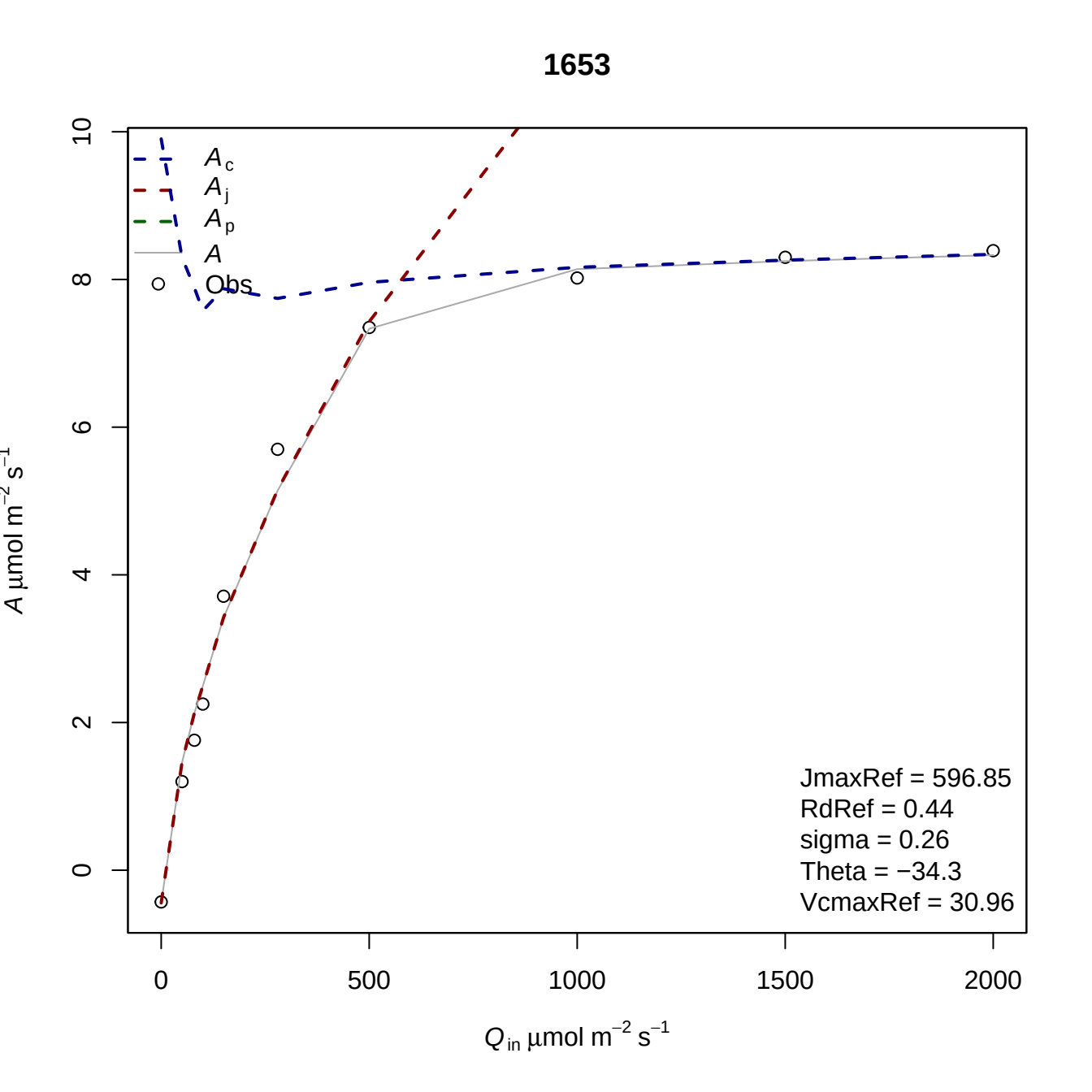

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 164.21
RdRef = 0.94
sigma = 0.21
Theta = 0.56
VcmaxRef = 75.78
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

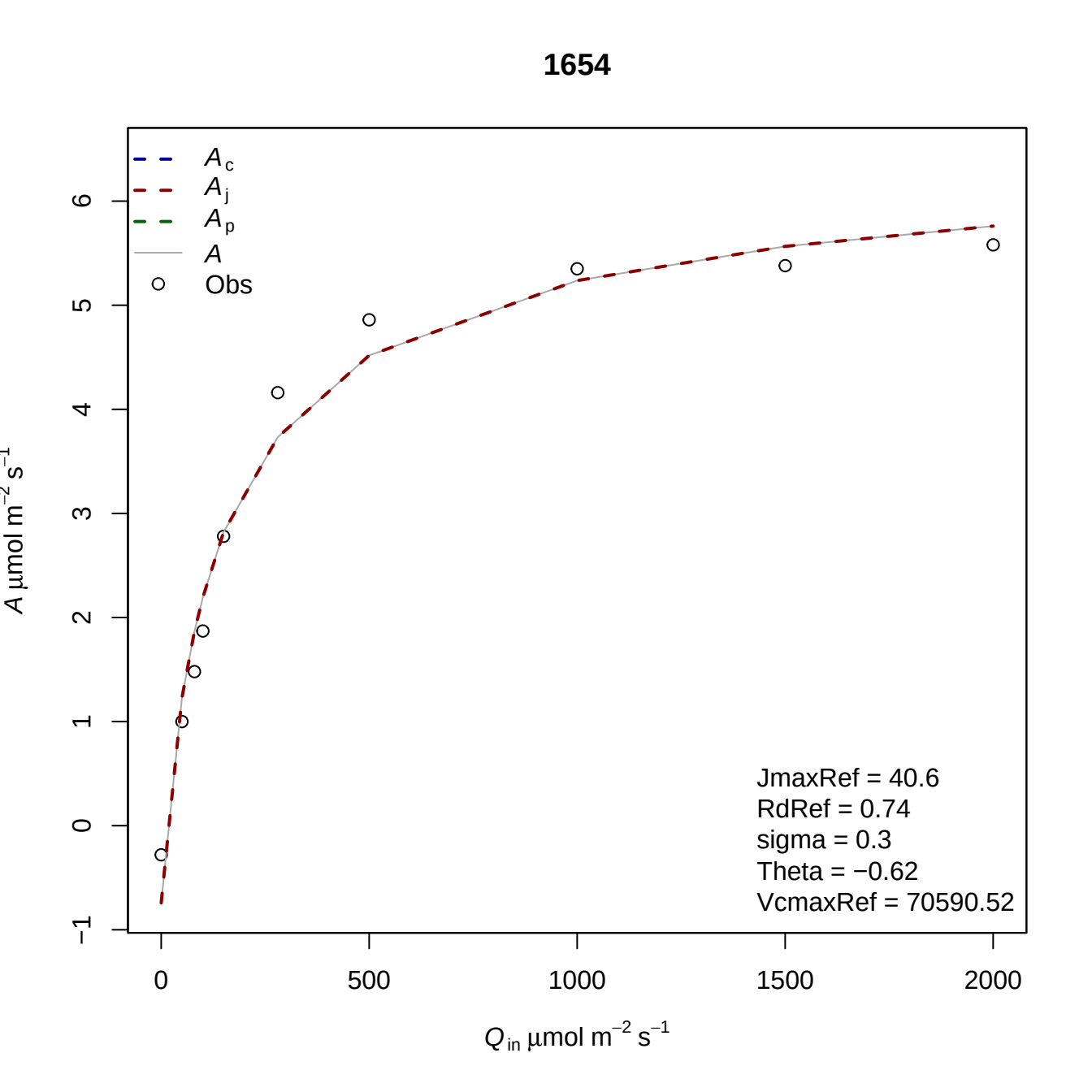

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 158.5
RdRef = 0.9
sigma = 0.11
Theta = 0.67
VcmaxRef = 93.08
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

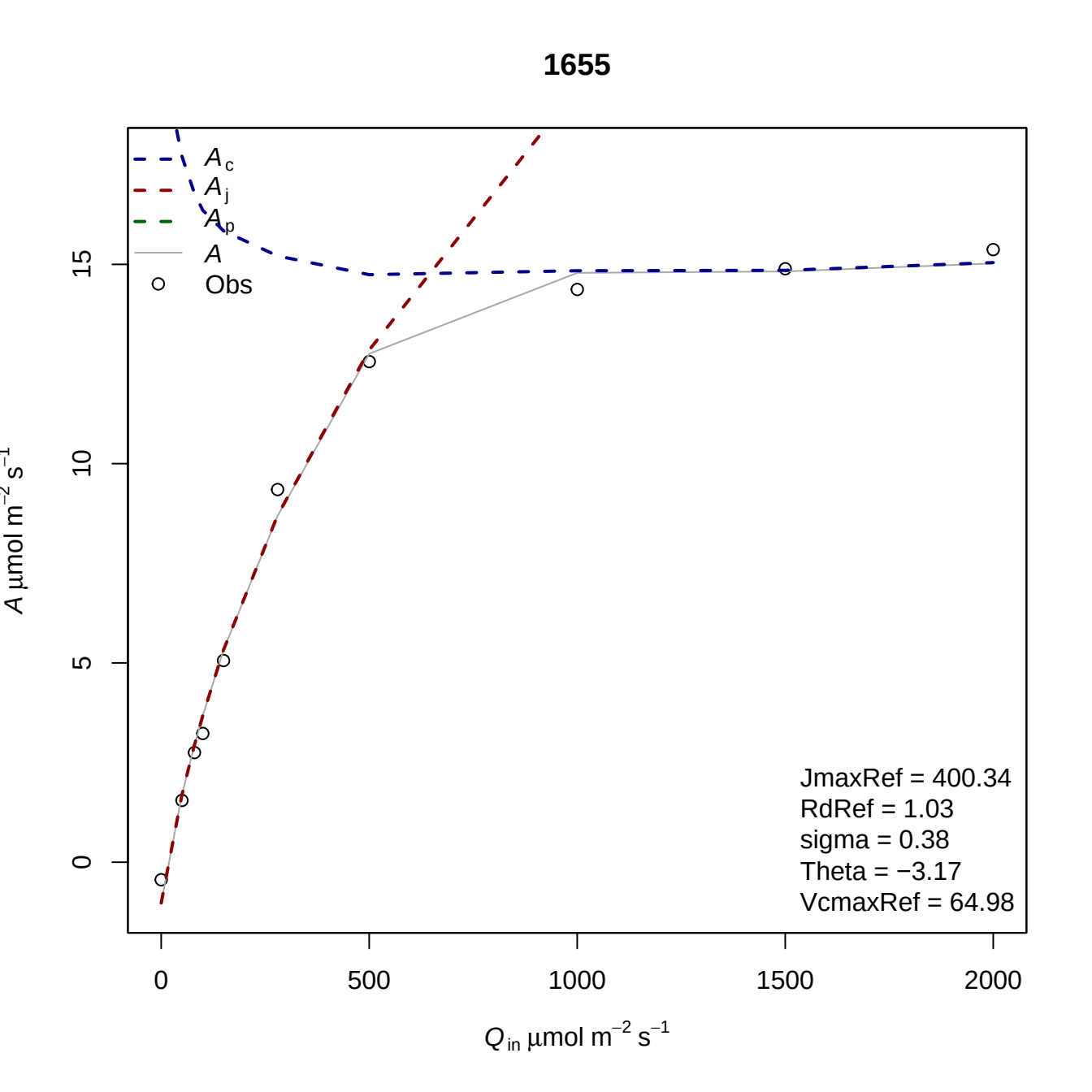

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 172.65
RdRef = 1.03
sigma = 0.24
Theta = 0.4
VcmaxRef = 75.67
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

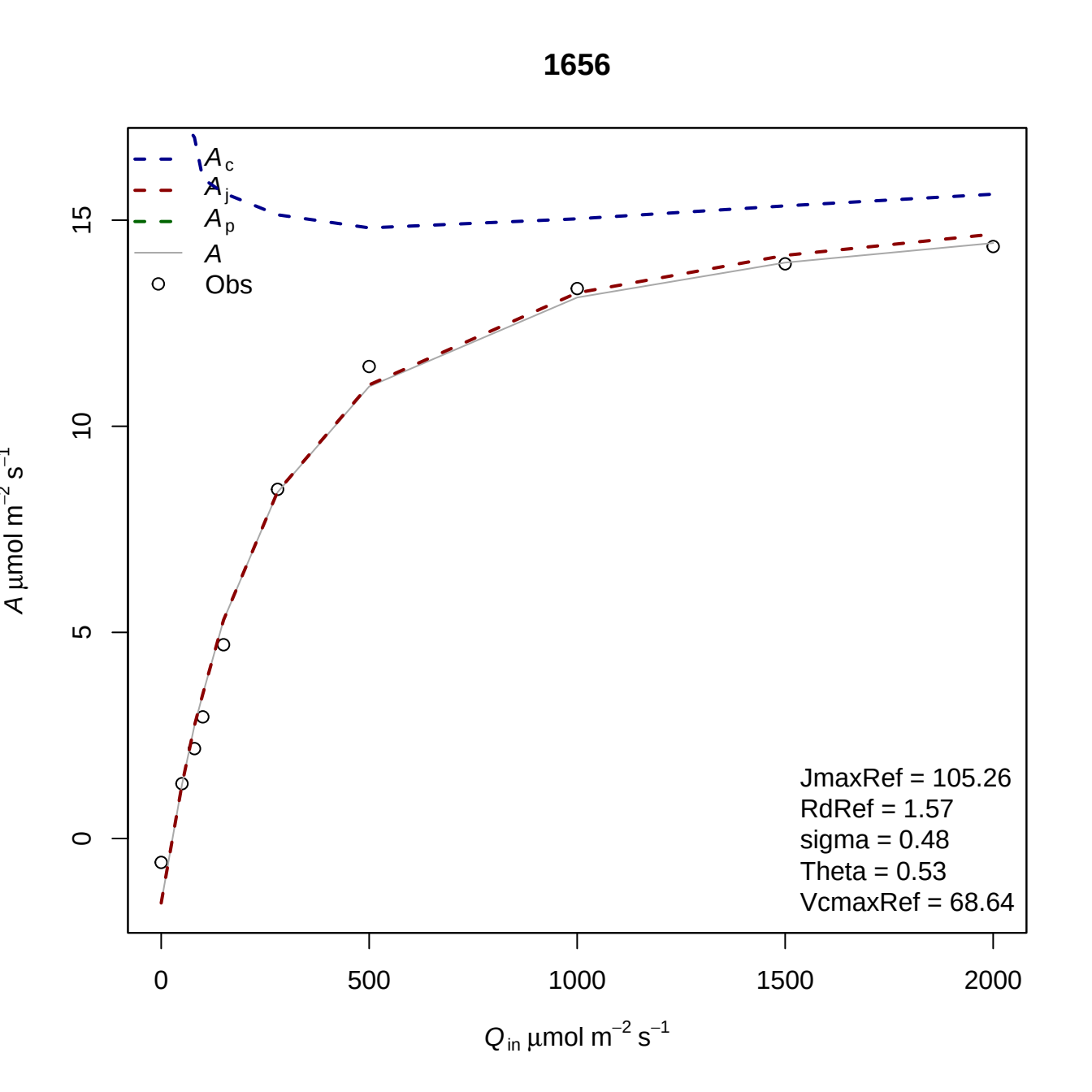

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 81.43
RdRef = 0.76
sigma = 0.26
Theta = −1.64
VcmaxRef = 26.59
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

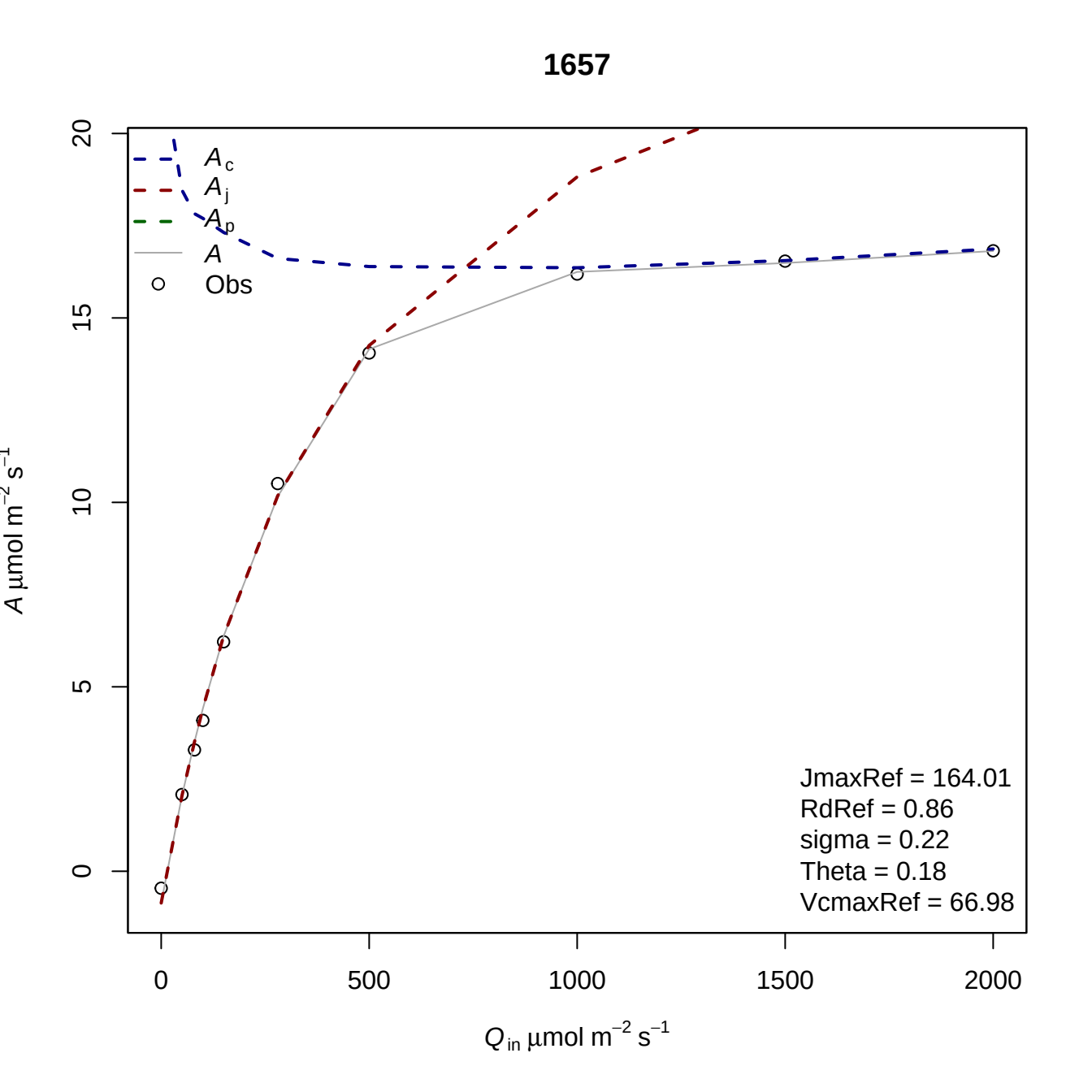

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 70.51
RdRef = 0.84
sigma = 0.27
Theta = 0.04
VcmaxRef = 35.61
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

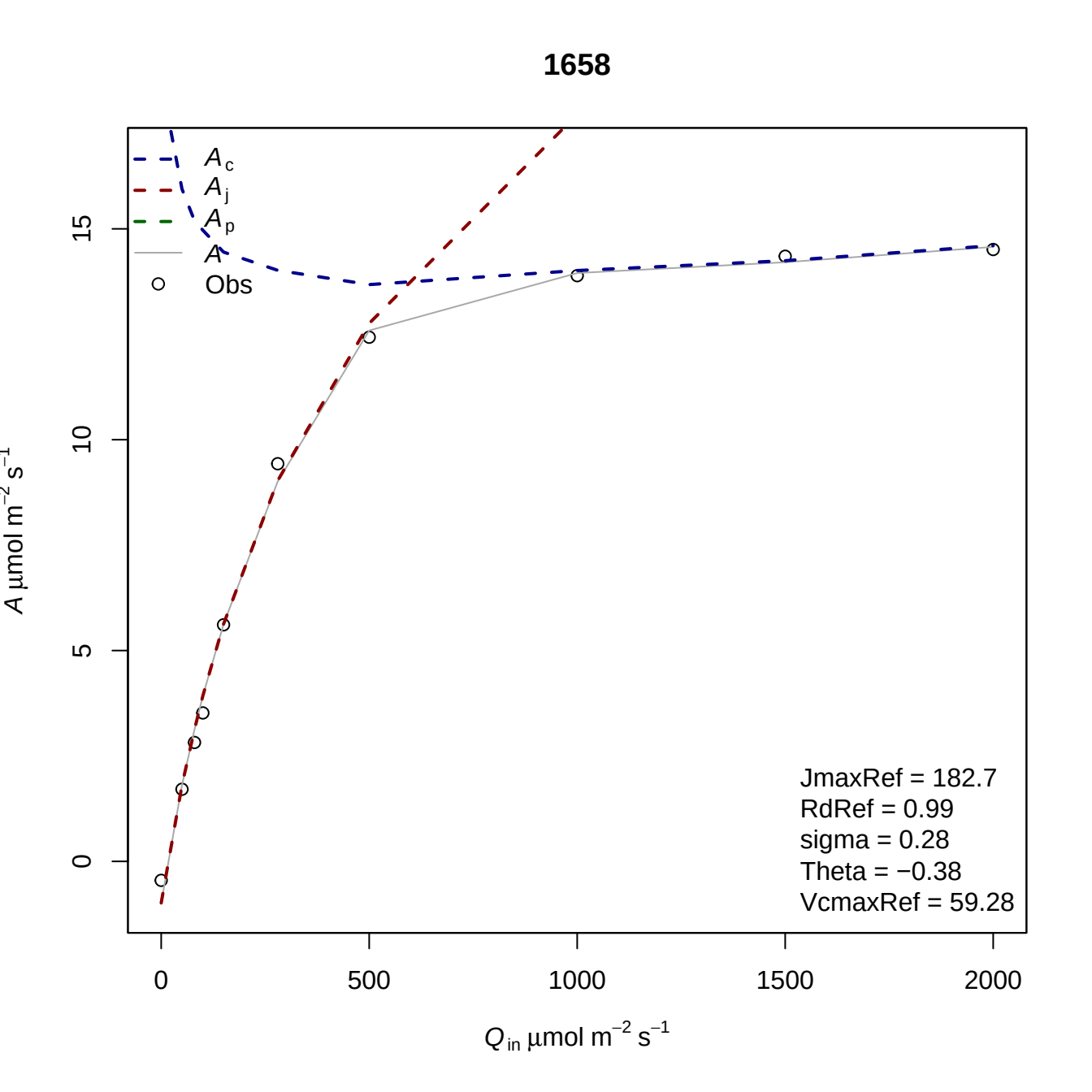

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 596.85
RdRef = 0.44
sigma = 0.26
Theta = −34.3
VcmaxRef = 30.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

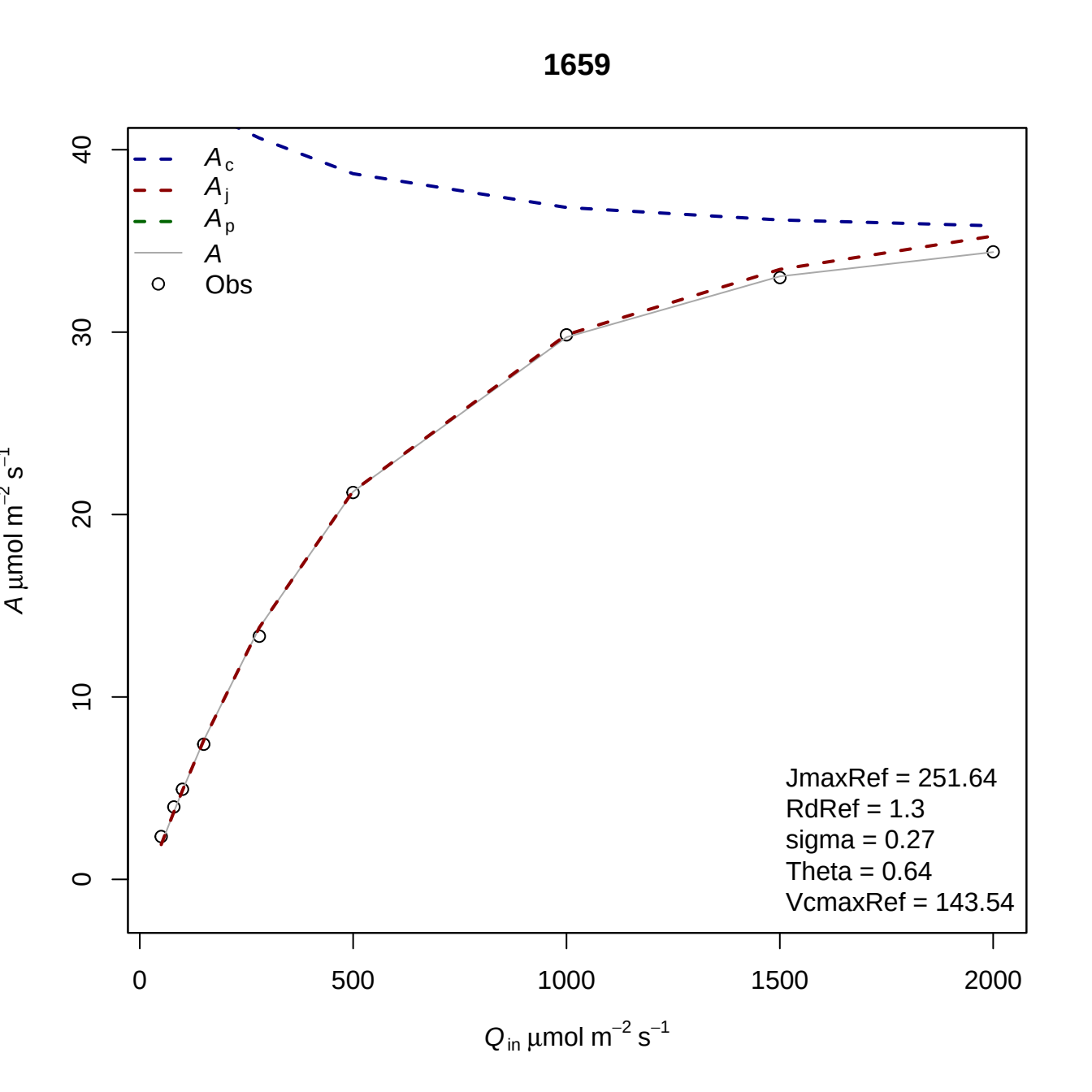

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.6
RdRef = 0.74
sigma = 0.3
Theta = −0.62
VcmaxRef = 70590.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

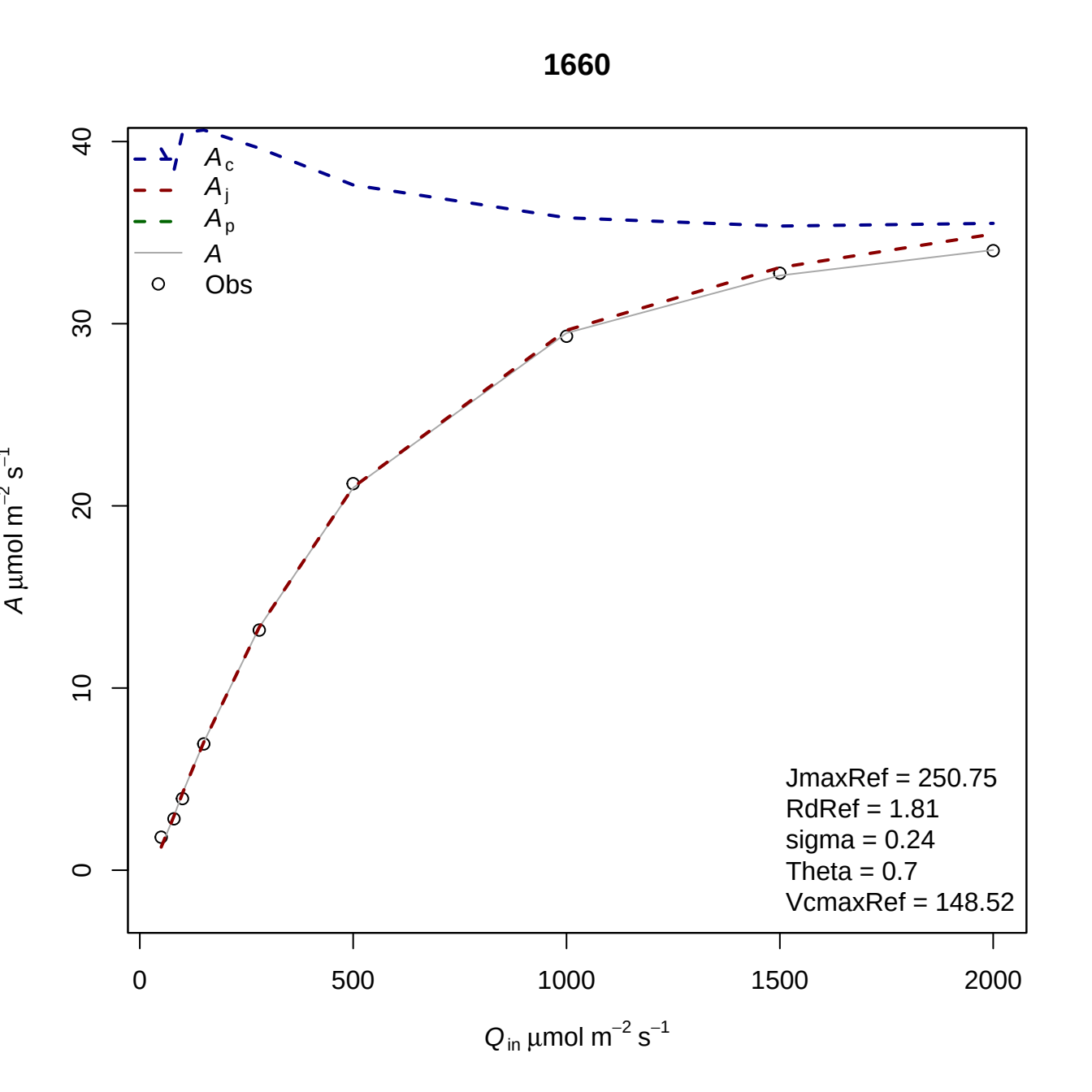

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 400.34
RdRef = 1.03
sigma = 0.38
Theta = −3.17
VcmaxRef = 64.98
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

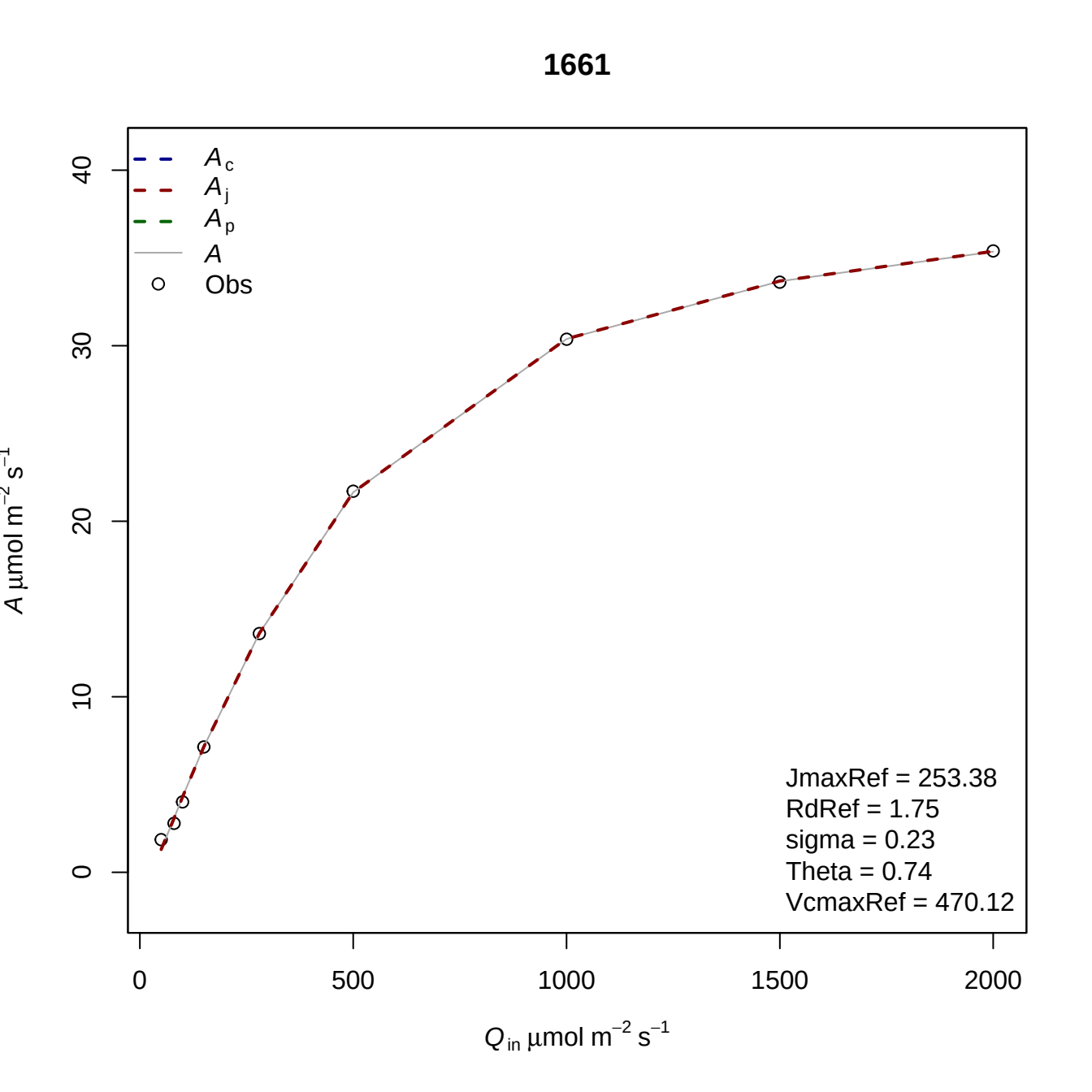

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 105.26
RdRef = 1.57
sigma = 0.48
Theta = 0.53
VcmaxRef = 68.64
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

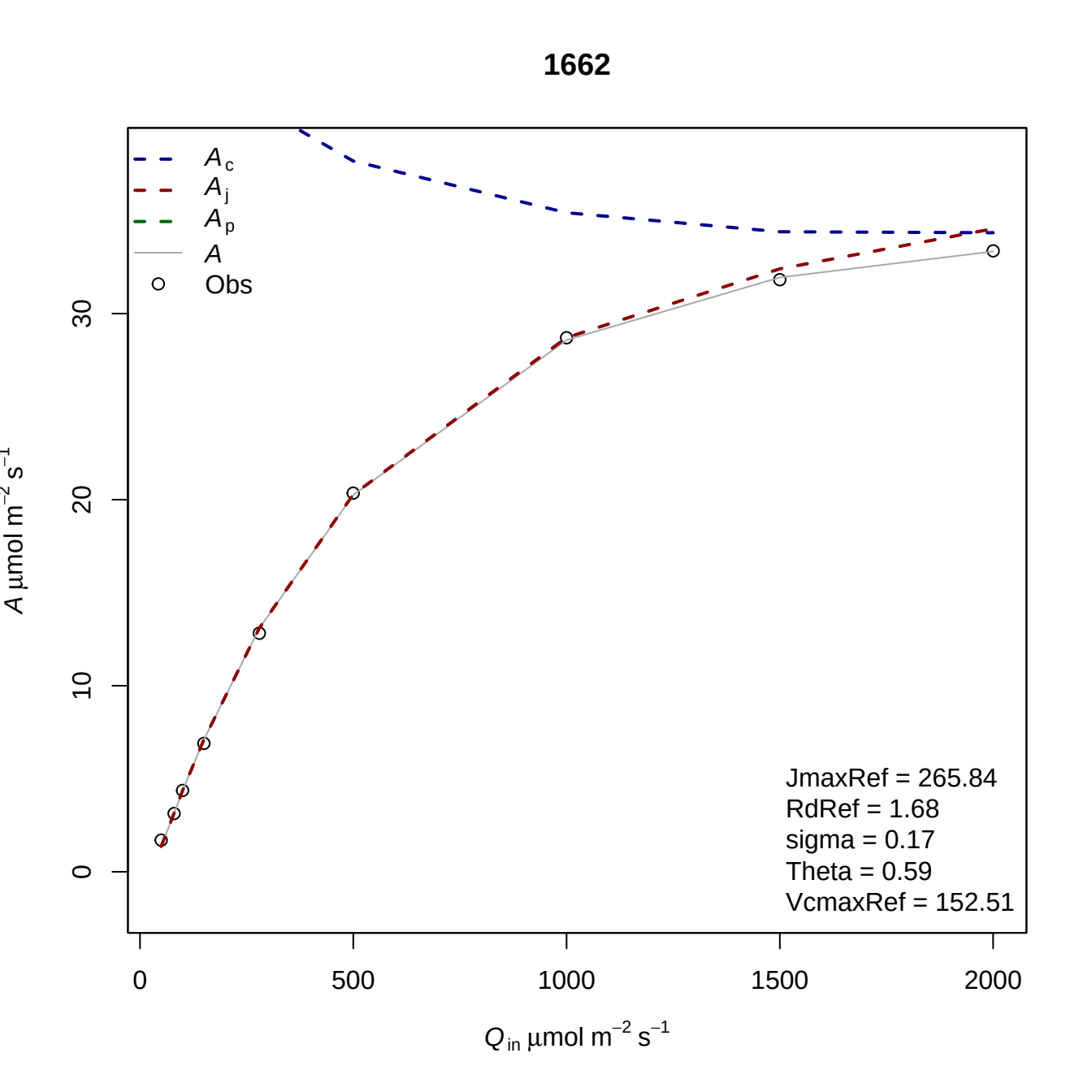

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 164.01
RdRef = 0.86
sigma = 0.22
Theta = 0.18
VcmaxRef = 66.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 182.7
RdRef = 0.99
sigma = 0.28
Theta = −0.38
VcmaxRef = 59.28
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 251.64
RdRef = 1.3
sigma = 0.27
Theta = 0.64
VcmaxRef = 143.54
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 250.75
RdRef = 1.81
sigma = 0.24
Theta = 0.7
VcmaxRef = 148.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 253.38
RdRef = 1.75
sigma = 0.23
Theta = 0.74
VcmaxRef = 470.12
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 265.84
RdRef = 1.68
sigma = 0.17
Theta = 0.59
VcmaxRef = 152.51
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}